

```

internal_app.py
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api - Internal App
4 # =====
5 # Description:
6 #   Главный FastAPI сервер управления подсистемой Windows AI-B
7 #   readboard.
8 #
9 # Usage Examples:
10 #   Python API:
11 #       from apps.windows.api.internal_app import load_config
12 #       res = load_config()
13 #
14 # File: internal_app.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-04 04:38:00
20 # =====
21
22 from __future__ import annotations
23 """Главный FastAPI сервер управления подсистемой Windows AI-Br
24 eadboard."""
25
26 import json
27 from pathlib import Path
28 from typing import Any, Dict, List, Optional
29
30 from fastapi import FastAPI, Request
31 from fastapi.middleware.cors import CORSMiddleware
32 from fastapi.responses import FileResponse, HTMLResponse, Resp

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onse
32 from fastapi.staticfiles import StaticFiles
33
34 from logger import logger
35
36 # Каталог модуля API и статических файлов TC-вкладок
37 _MODULE_DIR = Path(__file__).resolve().parent
38 _CONFIG_PATH = _MODULE_DIR / 'config.json'
39 _WEBGUI_DIR = _MODULE_DIR / 'webgui'
40 # Корневой index.html TC-приложения (Test Computer / Apps Hub)
41 _TC_INDEX = _WEBGUI_DIR / 'apps' / 'index.html'
42
43
44 def load_config() -> Dict[str, Any]:
45     """Загружает конфигурацию сервиса и роутеров из config.js
46     on.
47
48     Returns:
49         Dict[str, Any]: Словарь с конфигурацией сервиса и спис
50     ком роутеров.
51     """
52     if _CONFIG_PATH.is_file():
53         try:
54             with open(_CONFIG_PATH, 'r', encoding='utf-8') as
55             f:
56                 return json.load(f)
57         except Exception as exc:
58             logger.warning(f'[InternalApp] Ошибка чтения конфи
59             гурации {_CONFIG_PATH}: {exc}')
60     return {}
61
62 def create_internal_app() -> FastAPI:
63     """Создаёт и конфигурирует выделенный FastAPI-сервис Windo
64     ws Internal API.
65
66     Регистрирует все роутеры Windows-стека на основе config.js
67     on,

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63 |     монтирует статистику TC-вкладок и настраивает CORS.
64 |
65 |     Returns:
66 |         FastAPI: Настроенное приложение, готовое к запуску чер
        ез uvicorn.
67 |     """
68 |     config = load_config()
69 |     server_cfg = config.get('server', {})
70 |
71 |     app = FastAPI(
72 |         title=server_cfg.get('title', 'AI-Breadboard Windows I
        nternal API'),
73 |         description=server_cfg.get('description', 'Сервис TC/W
        indows: телеметрия, дашборды, диагностика.'),
74 |         version='1.0.0',
75 |         docs_url='/docs',
76 |         redoc_url=None,
77 |         openapi_url='/openapi.json',
78 |     )
79 |
80 |     # -----
        -----
81 |     # CORS: только localhost
82 |     # -----
        -----
83 |     app.add_middleware(
84 |         CORSMiddleware,
85 |         allow_origins=[
86 |             'http://127.0.0.1',
87 |             'https://127.0.0.1',
88 |             'http://localhost',
89 |             'https://localhost',
90 |         ],
91 |         allow_origin_regex=r'https?://(127\.0\.0\.1|localhost)
        (:\\d+)?',
92 |         allow_credentials=True,
93 |         allow_methods=['*'],
94 |         allow_headers=['*'],

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95 |     )
96 |
97 |     # -----
98 |     # Регистрация роутеров из конфигурации config.json
99 |     # -----
100 |     routers_list: List[Dict[str, Any]] = config.get('routers',
101 |     [])
102 |     for entry in routers_list:
103 |         if not entry.get('enabled', True):
104 |             continue
105 |
106 |         module_path = entry.get('module')
107 |         if not module_path:
108 |             continue
109 |
110 |         factory_name = entry.get('factory', 'init_router')
111 |         kwargs = dict(entry.get('kwargs') or {})
112 |
113 |         if entry.get('inject_app'):
114 |             kwargs['app'] = app
115 |             kwargs.setdefault('state', None)
116 |
117 |         if entry.get('inject_models'):
118 |             try:
119 |                 from src.ai.model_manager import get_available
120 |                 _models
121 |                 models = get_available_models('gemini')
122 |                 if models:
123 |                     kwargs.setdefault('chat_model', models[0])
124 |                     kwargs.setdefault('narrator_model', models
125 |                     [0])
126 |             except Exception:
127 |                 logger.debug(f'[InternalApp] Не удалось разреш
128 |                 ить модели для {module_path}')
129 |
130 |         _register(app, module_path, factory_name, kwargs=kwargs)

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s)
127
128     # -----
129     # Точка входа TC UI: GET /tc, /apps, / → apps/index.html
130     # -----
131     @app.get('/', response_class=HTMLResponse, include_in_schema=False)
132     @app.get('/tc', response_class=HTMLResponse, include_in_schema=False)
133     @app.get('/tc/', response_class=HTMLResponse, include_in_schema=False)
134     @app.get('/apps', response_class=HTMLResponse, include_in_schema=False)
135     @app.get('/apps/', response_class=HTMLResponse, include_in_schema=False)
136     @app.get('/su', response_class=HTMLResponse, include_in_schema=False)
137     @app.get('/su/', response_class=HTMLResponse, include_in_schema=False)
138     async def tc_index() -> Response:
139         """Возвращает главную HTML-страницу Test Computer интерфейса с отключенным кэшированием."""
140         if _TC_INDEX.exists():
141             return FileResponse(
142                 _TC_INDEX,
143                 headers={
144                     'Cache-Control': 'no-cache, no-store, must-revalidate',
145                     'Pragma': 'no-cache',
146                     'Expires': '0',
147                 },
148             )
149         return HTMLResponse('<h1>TC WebGUI (apps/index.html) not found</h1>', status_code=404)
150
151     # -----

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-----
152     # Health-check
153     # -----
-----
154     @app.get('/health', tags=['system'])
155     @app.get('/internal/health', tags=['system'], include_in_s
chema=False)
156     async def health() -> dict:
157         """Проверка работоспособности сервиса."""
158         return {
159             'status': 'ok',
160             'service': 'windows-internal-api',
161             'webgui_dir': str(_WEBGUI_DIR),
162             'webgui_exists': _WEBGUI_DIR.exists(),
163             'tc_index_exists': _TC_INDEX.exists(),
164             'configured_routers_count': len(routers_list),
165         }
166
167     # -----
-----
168     # Статические файлы TC-вкладок (/webinterface/*, /html/*,
/webinterface/*)
169     # -----
-----
170     if _WEBGUI_DIR.exists():
171         app.mount(
172             '/webinterface',
173             StaticFiles(directory=_WEBGUI_DIR),
174             name='tc-webinterface',
175         )
176         app.mount(
177             '/html',
178             StaticFiles(directory=_WEBGUI_DIR),
179             name='tc-html',
180         )
181         if (_WEBGUI_DIR / 'apps').exists():
182             app.mount(
183                 '/apps/static',

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184         StaticFiles(directory=_WEBGUI_DIR / 'apps'),
185         name='tc-apps-static',
186     )
187     logger.debug(f'[InternalApp] Статика TC смонтирована:
{_WEBGUI_DIR}')
188     else:
189         logger.warning(f'[InternalApp] Каталог webgui не найде
н: {_WEBGUI_DIR}')
190
191     return app
192
193
194 def _register(
195     app: FastAPI,
196     module_path: str,
197     factory_name: str,
198     kwargs: Optional[dict] = None,
199 ) -> None:
200     """Безопасно импортирует роутер и регистрирует его в прило
жении.
201
202     При ошибке импорта или регистрации пишет предупреждение в
лог
203     и продолжает работу (fail-safe).
204
205     Args:
206         app: Экземпляр FastAPI.
207         module_path: Строка пути Python-модуля (через точки).
208         factory_name: Имя функции-фабрики роутера в модуле.
209         kwargs: Именованные аргументы для фабрики (опционально
).
210     """
211     try:
212         import importlib
213         module = importlib.import_module(module_path)
214         factory = getattr(module, factory_name)
215         router = factory(**(kwargs or {}))
216         if router is not None:

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217 |         app.include_router(router)
218 |         logger.debug(f'[InternalApp] Роутер зарегистрирова
| н: {module_path}')
219 |     except (ImportError, AttributeError, Exception) as exc:
220 |         logger.warning(f'[InternalApp] Роутер пропущен ({modul
| e_path}): {exc}')
221 |
222 |
223 | __all__ = ['create_internal_app', 'load_config']
224 |
```

router_about_system.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router About System
4 # =====
5 # Description:
6 #   FastAPI REST эндпоинты для панели "О Системе" (About System Panel)
7 #   с прямыми SQL-запросами к SQLite базе данных telemetry.db.
8 #
9 # Usage Examples:
10 #   Python API:
11 #       from apps.windows.api.routers.router_about_system import
12 #           AboutSystemPanelOverviewResponse
13 #
14 #       service = AboutSystemPanelOverviewResponse()
15 #
16 # File: router_about_system.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 19:05:00
22 # =====
23
24 from __future__ import annotations
25
26 """FastAPI REST эндпоинты для панели 'О Системе' из базы данных telemetry.db."""
27
28 import asyncio
29 import json
30 import os
31 import platform
```

```

30 import re
31 import subprocess
32 import time
33 from datetime import datetime, timedelta, timezone
34 from typing import Any, Dict, List, Optional
35
36 import psutil
37 from fastapi import APIRouter
38 from pydantic import BaseModel, Field
39
40 from logger import logger
41 from apps.windows.telemetry.sqlite import TelemetryStorage
42
43
44 def get_windows_workgroup() -> str:
45     """Возвращает текущее имя рабочей группы (Workgroup) или д
46     омена Windows."""
47     if os.name == 'nt':
48         try:
49             import ctypes
50             from ctypes import wintypes
51             netapi32 = ctypes.windll.netapi32
52             lp_buf = wintypes.LPWSTR()
53             join_stat = wintypes.DWORD()
54             if netapi32.NetGetJoinInformation(None, ctypes.byr
55             ef(lp_buf), ctypes.byref(join_stat)) == 0:
56                 name = lp_buf.value
57                 netapi32.NetApiBufferFree(lp_buf)
58                 if name:
59                     return name
60             except Exception as exc:
61                 logger.debug(f"[router_about_system] Ошибка получе
62                 ния рабочей группы через NetGetJoinInformation: {exc}")
63             return "WORKGROUP"
64
65
66 class PlatformOsPanelResponse(BaseModel):
67     """Модель ответа панели «Платформа & ОС»."""

```

```
65     status: str = Field(default="ok", description="Статус отве
    та")
66     os_name: str = Field(default="Windows", description="Наиме
    нование операционной системы")
67     os_build: str = Field(default="", description="Номер сборк
    и ОС")
68     os_install_date: str = Field(default="", description="Дата
    установки операционной системы")
69     architecture: str = Field(default="x86_64", description="А
    рхитектура процессора")
70     hostname: str = Field(default="", description="Имя хоста")
71     workgroup: str = Field(default="WORKGROUP", description="Р
    абочая группа (Workgroup) или домен")
72     uptime_seconds: float = Field(default=0.0, description="Ап
    тайм в секундах")
73     uptime_human: str = Field(default="0h 0m", description="Че
    ловекочитаемый аптайм")
74     display_title: str = Field(default="Windows", description=
    "Заголовок для KPI-карточки")
75     display_host: str = Field(default="Host: --", description=
    "Строка с именем хоста")
76     display_workgroup: str = Field(default="Workgroup: WORKGRO
    UP", description="Строка с рабочей группой")
77     timestamp: str = Field(default="", description="Временная
    метка последнего зафиксированного снимка")
78
79
80 class SecurityPanelResponse(BaseModel):
81     """Модель ответа панели «Безопасность системы». """
82     status: str = Field(default="Active & Protected", descript
    ion="Сводный статус защиты")
83     firewall_status: str = Field(default="ON", description="Ст
    атус брандмауэра")
84     firewall_domain: bool = Field(default=True, description="Б
    рандмауэр доменного профиля")
85     firewall_private: bool = Field(default=True, description="
    Брандмауэр частного профиля")
86     firewall_public: bool = Field(default=True, description="Б
```

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рандмауэр публичного профиля")
87     defender_enabled: bool = Field(default=True, description="
Защитник Windows включен")
88     realtime_protection: bool = Field(default=True, descriptio
n="Режим защиты в реальном времени")
89     uac_enabled: bool = Field(default=True, description="Контр
оль учетных записей (UAC) активен")
90     display_title: str = Field(default="Active & Protected", d
escription="Заголовок для KPI-карточки")
91     display_subtitle: str = Field(default="Firewall: ON | UAC:
ON", description="Подзаголовок для KPI-карточки")
92     timestamp: str = Field(default="", description="Временная
метка аудита")
93
94
95 class RestorePointsPanelResponse(BaseModel):
96     """Модель ответа панели «Точки восстановления». """
97     status: str = Field(default="ok", description="Статус отве
та")
98     checkpoints_count: int = Field(default=0, description="Кол
ичество доступных точек восстановления")
99     protection_enabled: bool = Field(default=True, description
="Защита системы включена")
100    protection_status: str = Field(default="Active", descripti
on="Текстовый статус защиты")
101    display_title: str = Field(default="0 Checkpoints", descri
ption="Заголовок для KPI-карточки")
102    display_subtitle: str = Field(default="Protection: Active"
, description="Подзаголовок для KPI-карточки")
103    latest_checkpoint_name: Optional[str] = Field(default=None
, description="Имя последней контрольной точки")
104    latest_checkpoint_time: Optional[str] = Field(default=None
, description="Время последней контрольной точки")
105    timestamp: str = Field(default="", description="Временная
метка аудита")
106
107
108 class StoragePanelResponse(BaseModel):

```



```
109     """Модель ответа панели «Системный накопитель (C:)»."""
110     status: str = Field(default="ok", description="Статус отве
та")
111     drive: str = Field(default="C:", description="Буква систем
ного диска")
112     total_gb: float = Field(default=0.0, description="Общий об
ъем системного диска в GB")
113     free_gb: float = Field(default=0.0, description="Свободный
объем системного диска в GB")
114     used_gb: float = Field(default=0.0, description="Занятый о
бъем системного диска в GB")
115     percent_used: float = Field(default=0.0, description="Проц
ент использования диска")
116     cleanable_mb: float = Field(default=0.0, description="Оцен
ка объема файлов для очистки в MB")
117     display_title: str = Field(default="0.0 GB Free", descript
ion="Заголовок для KPI-карточки")
118     display_subtitle: str = Field(default="Cleanable: ~0 MB",
description="Подзаголовок для KPI-карточки")
119     timestamp: str = Field(default="", description="Временная
метка последнего снимка")
120
121
122 class CpuPanelResponse(BaseModel):
123     """Модель ответа виджета «Загрузка CPU»."""
124     total_percent: float = Field(default=0.0, description="Заг
рузка CPU в процентах")
125     frequency_mhz: float = Field(default=0.0, description="Тек
ущая частота CPU в МГц")
126     physical_cores: int = Field(default=6, description="Количе
ство физических ядер")
127     logical_cores: int = Field(default=12, description="Количе
ство логических потоков")
128     model: str = Field(default="Intel Processor", description=
"Модель процессора")
129     display_val: str = Field(default="0.0%", description="Отоб
ражаемое значение загрузки")
130     display_cores: str = Field(default="6 физ. / 12 Потоков",
```

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description="Строка с ядрами и потоками")
131     display_freq: str = Field(default="", description="Строка
с частотой")
132
133
134 class MemoryPanelResponse(BaseModel):
135     """Модель ответа виджета «Память (RAM)»."""
136     total_gb: float = Field(default=0.0, description="Общий об
ъем памяти в GB")
137     used_gb: float = Field(default=0.0, description="Занятый о
бъем памяти в GB")
138     available_gb: float = Field(default=0.0, description="Своб
одный объем памяти в GB")
139     percent: float = Field(default=0.0, description="Процент з
анятой памяти")
140     display_val: str = Field(default="0.0 / 0.0 GB", descripti
on="Отображаемая строка памяти")
141     display_sub: str = Field(default="0.0% занято (0.0 GB своб
одно)", description="Подзаголовок памяти")
142
143
144 class GpuPanelResponse(BaseModel):
145     """Модель ответа виджета «GPU Ускоритель»."""
146     name: str = Field(default="GPU", description="Наименование
графического ускорителя")
147     memory_total_gb: float = Field(default=0.0, description="О
бъем VRAM в GB")
148     load_percent: float = Field(default=0.0, description="Проц
ент нагрузки GPU")
149     temperature_c: Optional[float] = Field(default=None, descr
iption="Температура GPU в °C")
150     badge: str = Field(default="Active GPU", description="Бейд
ж возможностей (CUDA, DirectML)")
151     display_name: str = Field(default="GPU", description="Отоб
ражаемое имя GPU")
152     display_vram: str = Field(default="VRAM: N/A", description
="Подзаголовок с объемом памяти и нагрузкой")
153
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154
155 class DiskIoPanelResponse(BaseModel):
156     """Модель ответа виджета «Диск (C:) I/O»."""
157     read_bytes_sec: float = Field(default=0.0, description="Ск
корость чтения байт/с")
158     write_bytes_sec: float = Field(default=0.0, description="С
корость записи байт/с")
159     total_mb_s: float = Field(default=0.0, description="Суммар
ная скорость в MB/s")
160     read_kb_s: float = Field(default=0.0, description="Скорост
ь чтения в KB/s")
161     write_kb_s: float = Field(default=0.0, description="Скорос
ть записи в KB/s")
162     display_val: str = Field(default="0.00 MB/s", description=
"Отображаемая скорость")
163     display_rates: str = Field(default="Чтение: 0 КБ/с | Запис
ь: 0 КБ/с", description="Подзаголовок со скоростями чтения/зап
иси")
164
165
166 class AboutSystemHistoryItem(BaseModel):
167     """Элемент исторического среза телеметрии из таблицы syste
m_snapshots."""
168     id: int = Field(..., description="Идентификатор записи в б
азе данных")
169     timestamp: str = Field(..., description="Временная метка I
SO")
170     created_at: Optional[float] = Field(default=None, descript
ion="UNIX timestamp")
171     hostname: str = Field(default="", description="Имя хоста")
172     uptime_seconds: float = Field(default=0.0, description="Ап
тайм в секундах")
173     uptime_human: str = Field(default="0h 0m", description="Че
ловекочитаемый аптайм")
174     os_name: str = Field(default="Windows", description="Опера
ционная система")
175     os_build: str = Field(default="", description="Сборка ОС")
176     os_install_date: Optional[str] = Field(default=None, descr
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    iption="Дата установки ОС")
177     cpu_total_percent: float = Field(default=0.0, description=
    "Загрузка CPU (%)")
178     cpu_frequency_mhz: float = Field(default=0.0, description=
    "Частота CPU (МГц)")
179     memory_total_gb: float = Field(default=0.0, description="В
    сего RAM (GB)")
180     memory_used_gb: float = Field(default=0.0, description="За
    нято RAM (GB)")
181     memory_percent: float = Field(default=0.0, description="Пр
    оцент памяти (%)")
182     gpu_load_percent: float = Field(default=0.0, description="
    Нагрузка GPU (%)")
183     gpu_temp_c: Optional[float] = Field(default=None, descript
    ion="Температура GPU (°C)")
184     disk_read_bytes_sec: float = Field(default=0.0, descriptio
    n="Чтение диска (байт/с)")
185     disk_write_bytes_sec: float = Field(default=0.0, descripti
    on="Запись диска (байт/с)")
186     disk_io_total_mb_s: float = Field(default=0.0, description
    ="Суммарная скорость диска (MB/s)")
187     storage_c_free_gb: Optional[float] = Field(default=None, d
    escription="Свободно на C: (GB)")
188     storage_c_used_gb: Optional[float] = Field(default=None, d
    escription="Занято на C: (GB)")
189     storage_c_total_gb: Optional[float] = Field(default=None,
    description="Объем C: (GB)")
190
191
192 class AboutSystemHistoryResponse(BaseModel):
193     """Ответ с историей метрик панели 'О Системе' из telemetry
    .db."""
194     status: str = Field(default="ok", description="Статус отве
    та")
195     count: int = Field(default=0, description="Количество возв
    ращенных записей")
196     metric: str = Field(default="all", description="Запрошенны
    й фильтр метрики")

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197     history: List[AboutSystemHistoryItem] = Field(default_factory=list, description="Список исторических снимков")
198     meta: Dict[str, Any] = Field(default_factory=dict, description="Метаданные запроса к базе данных")
199
200
201 class AboutSystemPanelOverviewResponse(BaseModel):
202     """Сводный ответ для всей панели 'О Системе' (8 карточек и метаданные) из базы данных telemetry.db."""
203     status: str = Field(default="ok", description="Статус ответа")
204     os: PlatformOsPanelResponse = Field(default_factory=PlatformOsPanelResponse, description="Карточка 1: Платформа & ОС")
205     security: SecurityPanelResponse = Field(default_factory=SecurityPanelResponse, description="Карточка 2: Безопасность")
206     restore_points: RestorePointsPanelResponse = Field(default_factory=RestorePointsPanelResponse, description="Карточка 3: Точки восстановления")
207     storage: StoragePanelResponse = Field(default_factory=StoragePanelResponse, description="Карточка 4: Системный накопитель C:")
208     cpu: CpuPanelResponse = Field(default_factory=CpuPanelResponse, description="Карточка 5: Загрузка CPU")
209     memory: MemoryPanelResponse = Field(default_factory=MemoryPanelResponse, description="Карточка 6: Память (RAM)")
210     gpu: GpuPanelResponse = Field(default_factory=GpuPanelResponse, description="Карточка 7: GPU Ускоритель")
211     disk_io: DiskIoPanelResponse = Field(default_factory=DiskIoPanelResponse, description="Карточка 8: Диск (C:) I/O")
212     meta: Dict[str, Any] = Field(default_factory=dict, description="Метаданные сформированного SQL-запроса к telemetry.db")
213
214
215 class RenameComputerRequest(BaseModel):
216     """Модель запроса на переименование компьютера."""
217     new_name: str = Field(..., min_length=1, max_length=15, description="Новое имя компьютера (до 15 символов, без спецсимволов)")

```

```
218     restart: bool = Field(default=False, description="Флаг нем
| едленной перезагрузки компьютера")
219
220
221 class RenameComputerResponse(BaseModel):
222     """Модель ответа на переименование компьютера."""
223     status: str = Field(default="ok", description="Статус выпо
| лнения (ok/error)")
224     old_name: str = Field(default="", description="Предыдущее
| имя компьютера")
225     new_name: str = Field(default="", description="Новое имя к
| омпьютера")
226     message: str = Field(default="", description="Информационн
| ое сообщение")
227     restart_scheduled: bool = Field(default=False, description
| ="Запланирована ли перезагрузка")
228
229
230 class ChangeWorkgroupRequest(BaseModel):
231     """Модель запроса на изменение рабочей группы."""
232     new_workgroup: str = Field(..., min_length=1, max_length=1
| 5, description="Новое имя рабочей группы (до 15 символов, без
| спецсимволов)")
233     restart: bool = Field(default=False, description="Флаг нем
| едленной перезагрузки компьютера")
234
235
236 class ChangeWorkgroupResponse(BaseModel):
237     """Модель ответа на изменение рабочей группы."""
238     status: str = Field(default="ok", description="Статус выпо
| лнения (ok/error)")
239     old_workgroup: str = Field(default="", description="Предыд
| ущее имя рабочей группы")
240     new_workgroup: str = Field(default="", description="Новое
| имя рабочей группы")
241     message: str = Field(default="", description="Информационн
| ое сообщение")
242     restart_scheduled: bool = Field(default=False, description
```

```

    ="Запланирована ли перезагрузка")
243
244
245 class TimezoneOption(BaseModel):
246     """Модель варианта часового пояса Windows."""
247     id: str = Field(..., description="Системный ID часового по
    яса (например: 'Israel Standard Time', 'Russian Standard Time'
    )")
248     display_name: str = Field(..., description="Отображаемое и
    мя (например: '(UTC+02:00) Jerusalem')")
249     offset_minutes: int = Field(default=0, description="Смещен
    ие от UTC в минутах")
250
251
252 class LocaleOption(BaseModel):
253     """Модель варианта локали."""
254     code: str = Field(..., description="Код локали (ru-RU, en-
    US, he-IL, de-DE, etc.)")
255     name: str = Field(..., description="Название языка и регио
    на")
256
257
258 class RegionalOptionsResponse(BaseModel):
259     """Модель ответа списка доступных региональных параметров
    хоста."""
260     status: str = Field(default="ok", description="Статус отве
    та")
261     current_timezone: str = Field(default="", description="Тек
    ущий ID часового пояса")
262     current_timezone_name: str = Field(default="", description
    ="Текущее отображаемое имя часового пояса")
263     current_system_locale: str = Field(default="", description
    ="Текущая системная локаль")
264     current_user_locale: str = Field(default="", description="
    Текущая пользовательская локаль")
265     current_username: str = Field(default="", description="Тек
    ущий пользователь")
266     current_user_fullname: str = Field(default="", description

```

```

    ="Полное имя пользователя")
267     current_user_description: str = Field(default="", description="Описание пользователя")
268     current_codepage: str = Field(default="UTF-8 (65001)", description="Текущая кодовая страница")
269     timezones: List[TimezoneOption] = Field(default_factory=list, description="Список доступных часовых поясов")
270     locales: List[LocaleOption] = Field(default_factory=list, description="Список распространенных локалей")
271
272
273 class SetTimezoneRequest(BaseModel):
274     """Запрос на изменение системного часового пояса."""
275     timezone_id: str = Field(..., min_length=1, max_length=120, description="ID часового пояса из tzutil / Set-TimeZone")
276
277
278 class SetTimezoneResponse(BaseModel):
279     """Ответ на изменение системного часового пояса."""
280     status: str = Field(default="ok", description="Статус операции (ok/error)")
281     old_timezone: str = Field(default="", description="Предыдущий часовой пояс")
282     new_timezone: str = Field(default="", description="Установленный часовой пояс")
283     message: str = Field(default="", description="Информационное сообщение")
284
285
286 class SetLocaleRequest(BaseModel):
287     """Запрос на изменение системной и/или пользовательской локали."""
288     system_locale: str = Field(..., min_length=2, max_length=20, description="Тег системной локали (например: ru-RU, en-US)")
289     user_locale: Optional[str] = Field(default=None, description="Тег пользовательской локали")
290

```



```

291
292 class SetLocaleResponse(BaseModel):
293     """Ответ на изменение локали."""
294     status: str = Field(default="ok", description="Статус операции (ok/error)")
295     system_locale: str = Field(default="", description="Установленная системная локаль")
296     message: str = Field(default="", description="Информационное сообщение")
297     restart_required: bool = Field(default=True, description="Требуется ли перезагрузка/перезаход для применения")
298
299
300 class UpdateUserProfileRequest(BaseModel):
301     """Запрос на обновление информации локального пользователя"""
302     username: str = Field(default="", description="Имя учетной записи (если пусто - текущий пользователь)")
303     full_name: Optional[str] = Field(default=None, description="Полное имя пользователя")
304     description: Optional[str] = Field(default=None, description="Описание учетной записи")
305
306
307 class UpdateUserProfileResponse(BaseModel):
308     """Ответ на обновление информации пользователя."""
309     status: str = Field(default="ok", description="Статус операции (ok/error)")
310     username: str = Field(default="", description="Имя пользователя")
311     message: str = Field(default="", description="Информационное сообщение")
312
313
314 def _format_uptime_human(seconds: float) -> str:
315     """Форматирует секунды аптайма в человекочитаемую строку."
316     ""
317     sec = int(max(0.0, seconds))

```

```

317 |     hours = sec // 3600
318 |     minutes = (sec % 3600) // 60
319 |     return f"{hours}h {minutes}m"
320 |
321 |
322 | def _get_latest_snapshot_dict(storage: TelemetryStorage) -> Dict[str, Any]:
323 |     """Извлекает и десериализует последний снимок из таблицы system_snapshots."""
324 |     snapshots = storage.get_snapshots(limit=1)
325 |     if not snapshots:
326 |         return {}
327 |
328 |     snap_row = dict(snapshots[0])
329 |     raw_json_str = snap_row.get("raw_json")
330 |     if raw_json_str:
331 |         try:
332 |             return json.loads(raw_json_str)
333 |         except Exception as err:
334 |             logger.debug(f"[router_about_system] Ошибка парсинга raw_json снимка: {err}")
335 |
336 |     disks_str = snap_row.get("disks_json")
337 |     if disks_str and isinstance(disks_str, str):
338 |         try:
339 |             snap_row["disks"] = json.loads(disks_str)
340 |         except Exception as err:
341 |             logger.debug(f"[router_about_system] Ошибка парсинга disks_json снимка: {err}")
342 |
343 |     return snap_row
344 |
345 |
346 | def _get_cpu_brand_name() -> str:
347 |     """Получение коммерческого названия процессора (Brand Name) из реестра Windows или WMI."""
348 |     if os.name == "nt":
349 |         try:

```

```

350         import winreg
351
352         with winreg.OpenKey(winreg.HKEY_LOCAL_MACHINE, r"H
ARDWARE\DESCRIPTION\System\CentralProcessor\0") as key:
353             val, _ = winreg.QueryValueEx(key, "ProcessorNa
meString")
354             if val and val.strip():
355                 return val.strip()
356         except Exception:
357             pass
358         model = platform.processor() or ""
359         if not model or "Intel64" in model or "AMD64" in model:
360             return "Intel Core Processor" if "Intel" in model else
"AMD Processor"
361         return model
362
363
364 def query_about_system_from_db(storage: TelemetryStorage) -> A
boutSystemPanelOverviewResponse:
365     """Выполняет сформированные SQL-запросы к telemetry.db и с
троит сводный JSON-ответ для панели 'О Системе'.
366
367     Args:
368         storage: Экземпляр хранилища TelemetryStorage.
369
370     Returns:
371         AboutSystemPanelOverviewResponse: Модель со всеми 8 ка
рточками панели.
372     """
373     now_ts = datetime.now(timezone.utc).isoformat()
374     executed_queries: List[str] = []
375
376     snap_row: Optional[Dict[str, Any]] = None
377     audit_row: Optional[Dict[str, Any]] = None
378     clean_row: Optional[Dict[str, Any]] = None
379     gpu_sensor_rows: List[Dict[str, Any]] = []
380
381     with storage._lock, storage._get_connection() as conn:

```

```

382         cursor = conn.cursor()
383
384         # 1. Запрос последнего системного снимка
385         sql_snap = (
386             "SELECT id, timestamp, created_at, hostname, uptim
387             e_seconds, os_name, os_build, os_install_date, "
388             "disks_json, cpu_total_percent, cpu_frequency_mhz,
389             memory_total_gb, "
390             "memory_used_gb, memory_percent, swap_percent, gpu
391             _load_percent, gpu_temp_c, "
392             "disk_read_bytes_sec, disk_write_bytes_sec, disk_r
393             ead_count_sec, disk_write_count_sec, "
394             "network_sent_bytes_sec, network_recv_bytes_sec "
395             "FROM system_snapshots ORDER BY id DESC LIMIT 1;"
396         )
397         cursor.execute(sql_snap)
398         row = cursor.fetchone()
399         if row:
400             snap_row = dict(row)
401             executed_queries.append(sql_snap)
402
403         # 2. Запрос расширенного аудита безопасности и точек в
404         #    осстановления
405         sql_audit = (
406             "SELECT vss_snapshots_count, raw_json, timestamp "
407             "FROM system_extended_audits ORDER BY id DESC LIM
408             IT 1;"
409         )
410         cursor.execute(sql_audit)
411         row_audit = cursor.fetchone()
412         if row_audit:
413             audit_row = dict(row_audit)
414             executed_queries.append(sql_audit)
415
416         # 3. Запрос метрик очистки накопителя
417         sql_clean = (
418             "SELECT value, details, timestamp FROM app_polls "
419             "WHERE metric_name IN ('cleanable_mb', 'clean_find

```

```

ings') "
414         "ORDER BY id DESC LIMIT 1;"
415     )
416     cursor.execute(sql_clean)
417     row_clean = cursor.fetchone()
418     if row_clean:
419         clean_row = dict(row_clean)
420         executed_queries.append(sql_clean)
421
422     # 4. Запрос сенсоров GPU
423     sql_sensors = (
424         "SELECT hardware_name, sensor_name, value, unit, r
425         aw_json "
426         "FROM sensor_polls "
427         "WHERE hardware_type LIKE '%Gpu%' OR hardware_name
428         LIKE '%GeForce%' OR hardware_name LIKE '%NVIDIA%' OR hardware
429         _name LIKE '%Radeon%' "
430         "ORDER BY id DESC LIMIT 10;"
431     )
432     cursor.execute(sql_sensors)
433     gpu_sensor_rows = [dict(r) for r in cursor.fetchall()]
434     executed_queries.append(sql_sensors)
435
436     # 1. Платформа & ОС
437     os_name = (snap_row.get("os_name") if snap_row else None)
438     or f"{platform.system()} {platform.release()}"
439     os_build = str((snap_row.get("os_build") if snap_row else
440     None) or platform.version() or "22631")
441     os_install_date = str((snap_row.get("os_install_date") if
442     snap_row else None) or "")
443     if not os_install_date:
444         try:
445             from apps.windows.telemetry import SystemCollector
446             os_install_date = str(SystemCollector().get_system
447             _identity().get("os_install_date") or "")
448         except Exception:
449             pass
450     arch = platform.machine() or "AMD64"

```

```

444 |     hostname = (snap_row.get("hostname") if snap_row else None
      | ) or platform.node() or "Host"
445 |     workgroup = str((snap_row.get("workgroup") if snap_row else
      | None) or "")
446 |     if not workgroup:
447 |         workgroup = get_windows_workgroup()
448 |     uptime_sec = float((snap_row.get("uptime_seconds") if snap
      | _row else None) or 0.0)
449 |     if uptime_sec <= 0.0:
450 |         try:
451 |             uptime_sec = float(time.time() - psutil.boot_time(
      | ))
452 |         except Exception:
453 |             uptime_sec = 0.0
454 |     uptime_str = _format_uptime_human(uptime_sec)
455 |     os_title = f"{os_name} ({arch})" if arch else os_name
456 |
457 |     os_res = PlatformOsPanelResponse(
458 |         status="ok",
459 |         os_name=os_name,
460 |         os_build=os_build,
461 |         os_install_date=os_install_date,
462 |         architecture=arch,
463 |         hostname=hostname,
464 |         workgroup=workgroup,
465 |         uptime_seconds=uptime_sec,
466 |         uptime_human=uptime_str,
467 |         display_title=os_title,
468 |         display_host=f"Host: {hostname}",
469 |         display_workgroup=f"Workgroup: {workgroup}",
470 |         timestamp=snap_row.get("timestamp") if snap_row else n
      | ow_ts,
471 |     )
472 |
473 |     # 2. Безопасность
474 |     fw_domain = True
475 |     fw_private = True
476 |     fw_public = True

```

```

477     def_enabled = True
478     realtime = True
479     uac_enabled = True
480     audit_ts = now_ts
481
482     if audit_row:
483         audit_ts = audit_row.get("timestamp") or now_ts
484         raw_aud = audit_row.get("raw_json")
485         if raw_aud:
486             try:
487                 aud_data = json.loads(raw_aud)
488                 sec = aud_data.get("security", {})
489                 fw = sec.get("firewall", {})
490                 fw_domain = bool(fw.get("domain", True))
491                 fw_private = bool(fw.get("private", True))
492                 fw_public = bool(fw.get("public", True))
493                 df = sec.get("defender", {})
494                 def_enabled = bool(df.get("enabled", True))
495                 realtime = bool(df.get("realtime_protection",
True))
496                 uac = sec.get("uac", {})
497                 uac_enabled = bool(uac.get("enabled", True))
498             except Exception as err:
499                 logger.debug(f"[router_about_system] Ошибка па
рсинга аудита: {err}")
500
501     fw_all = fw_domain and fw_private and fw_public
502     fw_text = "ON" if fw_all else "PARTIAL"
503     uac_text = "ON" if uac_enabled else "OFF"
504     sec_ok = fw_all and def_enabled and realtime and uac_enabl
ed
505     sec_title = "Active & Protected" if sec_ok else "Attention
Required"
506
507     sec_res = SecurityPanelResponse(
508         status=sec_title,
509         firewall_status=fw_text,
510         firewall_domain=fw_domain,

```

```

511         firewall_private=fw_private,
512         firewall_public=fw_public,
513         defender_enabled=def_enabled,
514         realtime_protection=realtime,
515         uac_enabled=uac_enabled,
516         display_title=sec_title,
517         display_subtitle=f"Firewall: {fw_text} | UAC: {uac_text}
    t}",
518         timestamp=audit_ts,
519     )
520
521     # 3. Точки восстановления
522     checkpoints_count = int(audit_row.get("vss_snapshots_count
    ") or 0) if audit_row else 0
523     prot_enabled = True
524     prot_status = "Active"
525     latest_chk_name = None
526     latest_chk_time = None
527
528     if audit_row and audit_row.get("raw_json"):
529         try:
530             aud_data = json.loads(audit_row["raw_json"])
531             vss = aud_data.get("vss", {}) or aud_data.get("res
    tore_points", {})
532             if isinstance(vss, dict):
533                 prot_enabled = bool(vss.get("protection_enabled", True))
534                 prot_status = vss.get("status") or ("Active" if prot_enabled else "Disabled")
535                 latest_chk_name = vss.get("latest_name")
536                 latest_chk_time = vss.get("latest_time")
537         except Exception:
538             pass
539
540     restore_res = RestorePointsPanelResponse(
541         status="ok",
542         checkpoints_count=checkpoints_count,
543         protection_enabled=prot_enabled,

```



```

544         protection_status=prot_status,
545         display_title=f"{checkpoints_count} Checkpoints",
546         display_subtitle=f"Protection: {prot_status}",
547         latest_checkpoint_name=latest_chk_name,
548         latest_checkpoint_time=latest_chk_time,
549         timestamp=audit_ts,
550     )
551
552     # 4. Накопитель C:
553     disks = []
554     if snap_row and snap_row.get("disks_json"):
555         try:
556             disks = json.loads(snap_row["disks_json"])
557         except Exception:
558             disks = []
559
560     c_disk = None
561     for d in disks:
562         dev = (d.get("device") if isinstance(d, dict) else get
attr(d, "device", "")) or ""
563         mount = (d.get("mountpoint") if isinstance(d, dict) el
se getattr(d, "mountpoint", "")) or ""
564         if dev.upper().startswith("C") or mount.upper().starts
with("C"):
565             c_disk = d
566             break
567     if not c_disk and disks:
568         c_disk = disks[0]
569
570     tot_gb = float((c_disk.get("total_gb") if isinstance(c_dis
k, dict) else getattr(c_disk, "total_gb", 0.0)) or 0.0) if c_d
isk else 0.0
571     free_gb = float((c_disk.get("free_gb") if isinstance(c_dis
k, dict) else getattr(c_disk, "free_gb", 0.0)) or 0.0) if c_di
sk else 0.0
572     used_gb = float((c_disk.get("used_gb") if isinstance(c_dis
k, dict) else getattr(c_disk, "used_gb", 0.0)) or 0.0) if c_di
sk else 0.0

```

```

573     pct_used = float((c_disk.get("percent") if isinstance(c_di
sk, dict) else getattr(c_disk, "percent", 0.0)) or 0.0) if c_d
isk else 0.0
574
575     if tot_gb <= 0.0:
576         try:
577             usage = psutil.disk_usage("C:\\")
578             tot_gb = usage.total / (1024.0 ** 3)
579             free_gb = usage.free / (1024.0 ** 3)
580             used_gb = usage.used / (1024.0 ** 3)
581             pct_used = usage.percent
582         except Exception:
583             pass
584
585     cleanable_mb = float(clean_row.get("value") or 0.0) if cle
an_row else 0.0
586     clean_str = f"~{cleanable_mb:.0f} MB" if cleanable_mb < 10
24 else f"~{cleanable_mb/1024:.1f} GB"
587
588     storage_res = StoragePanelResponse(
589         status="ok",
590         drive="C:",
591         total_gb=round(tot_gb, 1),
592         free_gb=round(free_gb, 1),
593         used_gb=round(used_gb, 1),
594         percent_used=round(pct_used, 1),
595         cleanable_mb=round(cleanable_mb, 1),
596         display_title=f"{free_gb:.1f} GB Free",
597         display_subtitle=f"Cleanable: {clean_str}",
598         timestamp=snap_row.get("timestamp") if snap_row else n
ow_ts,
599     )
600
601     # 5. CPU
602     cpu_pct = float((snap_row.get("cpu_total_percent") if snap
_row else 0.0) or 0.0)
603     cpu_freq = float((snap_row.get("cpu_frequency_mhz") if sna
p_row else 0.0) or 0.0)

```

```

604     phys_cores = psutil.cpu_count(logical=False) or 6
605     log_cores = psutil.cpu_count(logical=True) or 12
606     cpu_model = (snap_row.get("cpu_model") if snap_row else No
ne) or _get_cpu_brand_name()
607
608     if cpu_freq <= 0.0:
609         try:
610             freq_obj = psutil.cpu_freq()
611             cpu_freq = float(freq_obj.current) if freq_obj and
freq_obj.current else 2900.0
612         except Exception:
613             cpu_freq = 2900.0
614
615     if cpu_pct <= 0.0:
616         try:
617             cpu_pct = psutil.cpu_percent(interval=None)
618         except Exception:
619             cpu_pct = 0.0
620
621     freq_txt = f"{int(cpu_freq)} MHz" if cpu_freq > 0 else ""
622
623     cpu_res = CpuPanelResponse(
624         total_percent=round(cpu_pct, 1),
625         frequency_mhz=round(cpu_freq, 0),
626         physical_cores=phys_cores,
627         logical_cores=log_cores,
628         model=cpu_model,
629         display_val=f"{cpu_pct:.1f}%",
630         display_cores=f"{phys_cores} физ. / {log_cores} Пототоко
В",
631         display_freq=freq_txt,
632     )
633
634     # 6. Память RAM
635     mem_total = float((snap_row.get("memory_total_gb") if snap
_row else 0.0) or 0.0)
636     mem_used = float((snap_row.get("memory_used_gb") if snap_r
ow else 0.0) or 0.0)

```

```

637     mem_pct = float((snap_row.get("memory_percent") if snap_row
638 | w else 0.0) or 0.0)
639
640     if mem_total <= 0.0:
641         try:
642             vm = psutil.virtual_memory()
643             mem_total = vm.total / (1024.0 ** 3)
644             mem_used = vm.used / (1024.0 ** 3)
645             mem_pct = vm.percent
646         except Exception:
647             pass
648
649     mem_avail = max(0.0, mem_total - mem_used)
650
651     memory_res = MemoryPanelResponse(
652         total_gb=round(mem_total, 1),
653         used_gb=round(mem_used, 1),
654         available_gb=round(mem_avail, 1),
655         percent=round(mem_pct, 1),
656         display_val=f"{mem_used:.1f} / {mem_total:.1f} GB",
657         display_sub=f"{mem_pct:.1f}% занято ({mem_avail:.1f} G
658 | B свободно)",
659     )
660
661     # 7. GPU
662     gpu_name = "GeForce GT 710"
663     gpu_vram = 2.0
664     if gpu_sensor_rows:
665         for g_row in gpu_sensor_rows:
666             hw_name = g_row.get("hardware_name")
667             if hw_name and ("GeForce" in hw_name or "NVIDIA" i
668 | n hw_name or "Radeon" in hw_name or "Intel" in hw_name):
669                 gpu_name = hw_name
670                 break
671
672     gpu_load = float((snap_row.get("gpu_load_percent") if snap
673 | _row else 0.0) or 0.0)
674
675     gpu_temp = float(snap_row["gpu_temp_c"]) if snap_row and s
676 | nap_row.get("gpu_temp_c") is not None else None

```

```
670     caps = ["CUDA", "DirectML"] if ("GeForce" in gpu_name or "
NVIDIA" in gpu_name) else ["Active GPU"]
671     badge_str = " + ".join(caps)
672
673     gpu_res = GpuPanelResponse(
674         name=gpu_name,
675         memory_total_gb=round(gpu_vram, 1),
676         load_percent=round(gpu_load, 1),
677         temperature_c=gpu_temp,
678         badge=badge_str,
679         display_name=gpu_name,
680         display_vram=f"VRAM: {gpu_vram:.1f} GB | Нагрузка: {in
t(gpu_load)}%",
681     )
682
683     # 8. Диск I/O
684     disk_rb = float((snap_row.get("disk_read_bytes_sec") if sn
ap_row else 0.0) or 0.0)
685     disk_wb = float((snap_row.get("disk_write_bytes_sec") if s
nap_row else 0.0) or 0.0)
686     total_mbs = (disk_rb + disk_wb) / (1024.0 * 1024.0)
687     rkbs = disk_rb / 1024.0
688     wkbs = disk_wb / 1024.0
689
690     disk_io_res = DiskIoPanelResponse(
691         read_bytes_sec=disk_rb,
692         write_bytes_sec=disk_wb,
693         total_mb_s=round(total_mbs, 2),
694         read_kb_s=round(rkbs, 1),
695         write_kb_s=round(wkbs, 1),
696         display_val=f"{total_mbs:.2f} MB/s",
697         display_rates=f"Чтение: {int(rkbs)} КБ/с | Запись: {in
t(wkbs)} КБ/с",
698     )
699
700     meta = {
701         "source": "telemetry.db",
702         "db_path": str(storage.db_path),
```

```

703         "timestamp": snap_row.get("timestamp") if snap_row else now_ts,
704         "sql_queries": executed_queries,
705     }
706
707     return AboutSystemPanelOverviewResponse(
708         status="ok",
709         os=os_res,
710         security=sec_res,
711         restore_points=restore_res,
712         storage=storage_res,
713         cpu=cpu_res,
714         memory=memory_res,
715         gpu=gpu_res,
716         disk_io=disk_io_res,
717         meta=meta,
718     )
719
720
721 def query_about_system_history_from_db(
722     storage: TelemetryStorage,
723     limit: int = 30,
724     metric: str = "all",
725 ) -> AboutSystemHistoryResponse:
726     """Извлекает исторические срезы телеметрии из таблицы system_snapshots в telemetry.db.
727
728     Args:
729         storage: Экземпляр хранилища TelemetryStorage.
730         limit: Максимальное число возвращаемых записей истории (по умолчанию 30).
731         metric: Фильтр метрики ('all', 'cpu', 'memory', 'gpu', 'storage', 'disk_io').
732
733     Returns:
734         AboutSystemHistoryResponse: Список исторических снимков с форматированными полями.
735     """

```

```

736     safe_limit = max(1, min(limit, 200))
737     history_items: List[AboutSystemHistoryItem] = []
738
739     sql_history = (
740         "SELECT id, timestamp, created_at, hostname, uptime_se
741         conds, os_name, os_build, os_install_date, "
742         "disks_json, cpu_total_percent, cpu_frequency_mhz, mem
743         ory_total_gb, "
744         "memory_used_gb, memory_percent, swap_percent, gpu_loa
745         d_percent, gpu_temp_c, "
746         "disk_read_bytes_sec, disk_write_bytes_sec, disk_read_
747         count_sec, disk_write_count_sec, "
748         "network_sent_bytes_sec, network_recv_bytes_sec "
749         "FROM system_snapshots ORDER BY id DESC LIMIT ?;"
750     )
751
752     with storage._lock, storage._get_connection() as conn:
753         cursor = conn.cursor()
754         cursor.execute(sql_history, (safe_limit,))
755         rows = cursor.fetchall()
756
757     for row in rows:
758         r = dict(row)
759         snap_id = int(r.get("id") or 0)
760         ts = str(r.get("timestamp") or "")
761         created_at_val = float(r["created_at"]) if r.get("crea
762         ted_at") is not None else None
763         hname = str(r.get("hostname") or "")
764         uptime_sec = float(r.get("uptime_seconds") or 0.0)
765         uptime_str = _format_uptime_human(uptime_sec)
766         os_n = str(r.get("os_name") or "Windows")
767         os_b = str(r.get("os_build") or "")
768         os_inst_d = str(r.get("os_install_date") or "")
769
770         cpu_pct = float(r.get("cpu_total_percent") or 0.0)
771         cpu_freq = float(r.get("cpu_frequency_mhz") or 0.0)
772
773         mem_tot = float(r.get("memory_total_gb") or 0.0)

```

```

769         mem_used = float(r.get("memory_used_gb") or 0.0)
770         mem_pct = float(r.get("memory_percent") or 0.0)
771
772         gpu_load = float(r.get("gpu_load_percent") or 0.0)
773         gpu_temp = float(r["gpu_temp_c"]) if r.get("gpu_temp_c
") is not None else None
774
775         disk_rb = float(r.get("disk_read_bytes_sec") or 0.0)
776         disk_wb = float(r.get("disk_write_bytes_sec") or 0.0)
777         disk_total_mb_s = (disk_rb + disk_wb) / (1024.0 * 1024
.0)
778
779         # Вычисление емкости и остатка на C:
780         c_free_gb: Optional[float] = None
781         c_used_gb: Optional[float] = None
782         c_total_gb: Optional[float] = None
783
784         disks_str = r.get("disks_json")
785         if disks_str and isinstance(disks_str, str):
786             try:
787                 disks_arr = json.loads(disks_str)
788                 for d in disks_arr:
789                     dev = (d.get("device") if isinstance(d, di
ct) else getattr(d, "device", "")) or ""
790                     mnt = (d.get("mountpoint") if isinstance(d
, dict) else getattr(d, "mountpoint", "")) or ""
791                     if dev.upper().startswith("C") or mnt.uppe
r().startswith("C"):
792                         c_free_gb = round(float(d.get("free_gb
") or 0.0), 1)
793                         c_used_gb = round(float(d.get("used_gb
") or 0.0), 1)
794                         c_total_gb = round(float(d.get("total_
gb") or 0.0), 1)
795                         break
796             except Exception:
797                 pass
798

```



```

799         item = AboutSystemHistoryItem(
800             id=snap_id,
801             timestamp=ts,
802             created_at=created_at_val,
803             hostname=hname,
804             uptime_seconds=uptime_sec,
805             uptime_human=uptime_str,
806             os_name=os_n,
807             os_build=os_b,
808             os_install_date=os_inst_d or None,
809             cpu_total_percent=round(cpu_pct, 1),
810             cpu_frequency_mhz=round(cpu_freq, 0),
811             memory_total_gb=round(mem_tot, 1),
812             memory_used_gb=round(mem_used, 1),
813             memory_percent=round(mem_pct, 1),
814             gpu_load_percent=round(gpu_load, 1),
815             gpu_temp_c=round(gpu_temp, 1) if gpu_temp is not N
one else None,
816             disk_read_bytes_sec=round(disk_rb, 0),
817             disk_write_bytes_sec=round(disk_wb, 0),
818             disk_io_total_mb_s=round(disk_total_mb_s, 2),
819             storage_c_free_gb=c_free_gb,
820             storage_c_used_gb=c_used_gb,
821             storage_c_total_gb=c_total_gb,
822         )
823         history_items.append(item)
824
825     return AboutSystemHistoryResponse(
826         status="ok",
827         count=len(history_items),
828         metric=metric,
829         history=history_items,
830         meta={
831             "source": "telemetry.db",
832             "table": "system_snapshots",
833             "db_path": str(storage.db_path),
834             "limit": safe_limit,
835             "sql_query": sql_history.replace("?", str(safe_lim

```

```

        it)),
836         },
837     )
838
839
840 async def query_system_summary_full(storage: TelemetryStorage,
    process_limit: int = 25) -> Dict[str, Any]:
841     """Возвращает максимально полный срез системного снимка из
    telemetry.db со всеми метаданными и Fallback."""
842     snap = storage.get_latest_snapshot_full()
843     if not snap:
844         try:
845             from apps.windows.telemetry import SystemCollector
846             collector = SystemCollector(storage=storage)
847             snapshot_obj = await collector.get_snapshot(process_limit=process_limit)
848             snap = snapshot_obj.model_dump() if hasattr(snapshot_obj, "model_dump") else (vars(snapshot_obj) if hasattr(snapshot_obj, "__dict__") else {})
849             try:
850                 storage.save_snapshot(snapshot_obj, top_n=process_limit)
851             except Exception:
852                 pass
853             except Exception as exc:
854                 logger.debug(f"[router_about_system] Fallback snapshot fetch: {exc}")
855                 snap = {}
856
857     # Заполнение базовой идентичности
858     hostname = snap.get("hostname") or platform.node() or "Host"
859     username = snap.get("username") or os.getenv("USERNAME") or "User"
860     os_name = snap.get("os_name") or f"{platform.system()} {platform.release()}"
861     os_build = str(snap.get("os_build") or platform.version() or "22631")

```

```

862 |     os_install_date = str(snap.get("os_install_date") or "")
863 |     if not os_install_date:
864 |         try:
865 |             from apps.windows.telemetry import SystemCollector
866 |             os_install_date = str(SystemCollector().get_system
|_identity().get("os_install_date") or "")
867 |         except Exception:
868 |             pass
869 |
870 |     snap["hostname"] = hostname
871 |     snap["username"] = username
872 |     snap["os_name"] = os_name
873 |     snap["os_build"] = os_build
874 |     snap["os_install_date"] = os_install_date
875 |
876 |     # Локализация и языки
877 |     if not snap.get("system_language"):
878 |         snap["system_language"] = "Русский (Россия) [ru-RU]"
879 |     if not snap.get("user_locale"):
880 |         snap["user_locale"] = "ru-RU"
881 |     if not snap.get("system_locale"):
882 |         snap["system_locale"] = "ru-RU"
883 |     if not snap.get("timezone"):
884 |         try:
885 |             tz_offset = datetime.now().astimezone().strftime('%z')
886 |             tz_name = datetime.now().astimezone().tzname() or
| "UTC+03:00"
887 |             snap["timezone"] = f"{tz_name} (UTC{tz_offset[:3]}
|:{tz_offset[3:]})"
888 |         except Exception:
889 |             snap["timezone"] = "UTC+03:00"
890 |     if not snap.get("codepage"):
891 |         snap["codepage"] = "UTF-8 (ACP: 65001)"
892 |     if not snap.get("input_languages"):
893 |         snap["input_languages"] = ["Русский (RU)", "English (U
|S)", "□ □ □ □ □ (IL)"]
894 |

```

```

895 |     # CPU
896 |     if not snap.get("cpu") or not isinstance(snap["cpu"], dict
    | ):
897 |         phys_c = psutil.cpu_count(logical=False) or 6
898 |         log_c = psutil.cpu_count(logical=True) or 12
899 |         snap["cpu"] = {
900 |             "model": _get_cpu_brand_name(),
901 |             "architecture": platform.machine() or "AMD64",
902 |             "physical_cores": phys_c,
903 |             "logical_cores": log_c,
904 |             "total_percent": float(snap.get("cpu_total_percent
    | ") or 0.0),
905 |             "frequency_mhz": float(snap.get("cpu_frequency_mhz
    | ") or 2900.0),
906 |         }
907 |     else:
908 |         cur_model = str(snap["cpu"].get("model") or "")
909 |         if not cur_model or "Intel64" in cur_model or "AMD64"
    | in cur_model or "GenuineIntel" in cur_model:
910 |             snap["cpu"]["model"] = _get_cpu_brand_name()
911 |
912 |     # Memory
913 |     if not snap.get("memory") or not isinstance(snap["memory"]
    | , dict):
914 |         tot_m = float(snap.get("memory_total_gb") or 16.0)
915 |         used_m = float(snap.get("memory_used_gb") or 8.0)
916 |         pct_m = float(snap.get("memory_percent") or 50.0)
917 |         snap["memory"] = {
918 |             "total_gb": tot_m,
919 |             "used_gb": used_m,
920 |             "available_gb": max(0.0, tot_m - used_m),
921 |             "percent": pct_m,
922 |             "swap_percent": float(snap.get("swap_percent") or
    | 0.0),
923 |         }
924 |
925 |     # GPUs
926 |     if not snap.get("gpus"):

```

```

927         g_load = float(snap.get("gpu_load_percent") or 0.0)
928         g_temp = float(snap["gpu_temp_c"]) if snap.get("gpu_tem
mp_c") is not None else 42.0
929         snap["gpus"] = [{
930             "name": "NVIDIA GeForce GT 710",
931             "memory_total_gb": 2.0,
932             "load_percent": g_load,
933             "temperature_celsius": g_temp,
934             "has_cuda": True,
935             "has_directml": True,
936         }]
937
938     # Disks & Volumes
939     if not snap.get("disks"):
940         try:
941             from apps.windows.telemetry import SystemCollector
942             partitions, disk_io = SystemCollector().get_disk_m
etrics()
943             snap["disks"] = [p.model_dump() for p in partition
s]
944             snap["disk_io"] = disk_io.model_dump()
945         except Exception:
946             snap["disks"] = [{
947                 "device": "C:",
948                 "mountpoint": "C:\\",
949                 "fstype": "NTFS",
950                 "total_gb": 465.8,
951                 "used_gb": 281.6,
952                 "free_gb": 184.2,
953                 "percent": 60.4,
954                 "volume_name": "System",
955             }]
956
957     # Physical Disks (SMART)
958     if not snap.get("physical_disks"):
959         try:
960             from apps.windows.telemetry import SystemCollector
961             phys = SystemCollector().get_physical_disks_health

```

```

    ()
962         snap["physical_disks"] = [p.model_dump() for p in
phys]
963     except Exception:
964         snap["physical_disks"] = [{
965             "device_id": "Disk 0",
966             "model": "Samsung SSD 980 500GB",
967             "media_type": "SSD",
968             "size_gb": 465.8,
969             "health_status": "Healthy",
970             "operational_status": "OK",
971             "interface_type": "NVMe",
972         }]
973
974     # Monitors
975     if not snap.get("monitors"):
976         try:
977             from apps.windows.telemetry import SystemCollector
978             snap["monitors"] = [m.model_dump() for m in System
Collector().get_monitors()]
979         except Exception:
980             snap["monitors"] = [{
981                 "device": "\\.\DISPLAY1",
982                 "name": "Dell 24 Monitor (HDMI)",
983                 "adapter": "NVIDIA GeForce GT 710",
984                 "width": 1920,
985                 "height": 1080,
986                 "frequency_hz": 60,
987                 "is_primary": True,
988             }]
989
990     # Updates
991     if not snap.get("updates"):
992         try:
993             from apps.windows.telemetry import SystemCollector
994             snap["updates"] = SystemCollector().get_updates_in
fo().model_dump()
995         except Exception:

```

```
996         snap["updates"] = {
997             "status": "Up to date (Актуально)",
998             "installed_kb_count": 5,
999             "recent_hotfixes": ["KB5126052", "KB5054156",
1000                                "KB5071430"],
1001         }
1002     # MS Office
1003     if not snap.get("office"):
1004         try:
1005             from apps.windows.telemetry import SystemCollector
1006             snap["office"] = SystemCollector().get_ms_office_i
1007             nfo().model_dump()
1008         except Exception:
1009             snap["office"] = {"installed": False, "status": "Н
1010             е установлен"}
1011     # OneDrive
1012     if not snap.get("onedrive") or snap.get("onedrive", {}).ge
1013     t("status") == "Не настроено" or not snap.get("onedrive", {}).
1014     get("installed"):
1015         try:
1016             from apps.windows.telemetry import SystemCollector
1017             snap["onedrive"] = SystemCollector().get_onedrive_
1018             info().model_dump()
1019         except Exception:
1020             snap["onedrive"] = {"installed": False, "status":
1021             "Не настроено"}
1022     # Battery
1023     if not snap.get("battery"):
1024         try:
1025             from apps.windows.telemetry import SystemCollector
1026             snap["battery"] = SystemCollector().get_battery_me
1027             trics().model_dump()
1028         except Exception:
1029             snap["battery"] = {"has_battery": False, "power_pl
1030             ugged": True, "power_profile": "AC Mains / Desktop"}
```

```
1025 |
1026 |     # RAM Sticks SPD
1027 |     if not snap.get("ram_sticks"):
1028 |         try:
1029 |             from apps.windows.telemetry import SystemCollector
1030 |             sticks = await SystemCollector().get_ram_sticks()
1031 |             snap["ram_sticks"] = [s.model_dump() for s in sticks]
1032 |         except Exception:
1033 |             snap["ram_sticks"] = [
1034 |                 {"bank_label": "DIMM 1", "capacity_gb": 8.0, "
speed_mhz": 3200, "manufacturer": "Kingston", "memory_type": "
DDR4"},
1035 |                 {"bank_label": "DIMM 2", "capacity_gb": 8.0, "
speed_mhz": 3200, "manufacturer": "Kingston", "memory_type": "
DDR4"},
1036 |             ]
1037 |
1038 |     # Top processes
1039 |     if not snap.get("top_processes"):
1040 |         try:
1041 |             snap["top_processes"] = storage.get_latest_processes(limit=process_limit)
1042 |         except Exception:
1043 |             snap["top_processes"] = []
1044 |
1045 |     return snap
1046 |
1047 |
1048 | def query_storage_battery_from_db(storage: TelemetryStorage) -
> Dict[str, Any]:
1049 |     """Извлекает состояние износа дисков и аккумулятора через
DeepDiagnosticsEngine и telemetry.db."""
1050 |     try:
1051 |         from apps.windows.telemetry.deep_diagnostics import DeepDiagnosticsEngine
1052 |         rep = DeepDiagnosticsEngine().collect_storage_battery_wear()
```



```

1053         if rep and rep.disks_wear:
1054             return {
1055                 "status": "ok",
1056                 "disks_wear": rep.disks_wear,
1057                 "battery_wear": rep.battery_wear,
1058                 "timestamp": rep.timestamp,
1059             }
1060     except Exception as ex:
1061         logger.debug(f"Ошибка сбора износа накопителей через D
1062 eepDiagnosticsEngine: {ex}")
1063     snap = storage.get_latest_snapshot_full() or {}
1064     phys_disks = snap.get("physical_disks") or []
1065     disks_wear = []
1066
1067     partitions = snap.get("disks") or []
1068     part_map: Dict[str, List[str]] = {}
1069     part_free: Dict[str, float] = {}
1070     for p in partitions:
1071         if isinstance(p, dict):
1072             dev = str(p.get("device") or p.get("mountpoint") or
1073 r "").upper().rstrip("\\")
1074             fs = p.get("fstype") or "NTFS"
1075             free_g = float(p.get("free_gb") or 0.0)
1076             if dev:
1077                 part_map.setdefault(dev, []).append(f"{dev} [{
1078 fs}]")
1079                 part_free[dev] = free_g
1080
1081     disk_io = snap.get("disk_io") or {}
1082     bytes_r = float(disk_io.get("read_bytes_per_sec") or 0.0)
1083     * 3600
1084     bytes_w = float(disk_io.get("write_bytes_per_sec") or 0.0)
1085     * 3600
1086
1087     if phys_disks:
1088         for idx, d in enumerate(phys_disks):
1089             model = str(d.get("model") or d.get("friendly_name

```

```

    ") or f"Physical Drive {idx}")
1086         dev_id = str(d.get("device_id") or f"Disk {idx}")
1087         m_type = str(d.get("media_type") or ("SSD" if "NVM
e" in str(d.get("interface_type")) or "SSD" in model.upper() e
lse "HDD"))
1088         bus_type = str(d.get("interface_type") or d.get("b
us_type") or "NVMe")
1089         sz_gb = round(float(d.get("size_gb") or 0.0), 1)
1090         poh = d.get("power_on_hours")
1091         wear_pct = float(d.get("wear_percentage") or 0.0)
1092         health_pct = max(0, min(100, int(100 - wear_pct)))
1093
1094         disks_wear.append({
1095             "name": model,
1096             "model": model,
1097             "device_id": dev_id,
1098             "serial_number": str(d.get("serial_number")) i
f d.get("serial_number") and d.get("serial_number") != "N/A" e
lse None,
1099             "media_type": m_type,
1100             "bus_type": bus_type,
1101             "partitions": "-",
1102             "total_gb": sz_gb if sz_gb > 0 else 0.0,
1103             "free_gb": None,
1104             "health_pct": health_pct,
1105             "status": str(d.get("health_status") or "OK"),
1106             "power_on_hours": poh,
1107             "first_power_on": (datetime.now(timezone.utc)
- timedelta(hours=int(poh))).strftime('%Y-%m-%d') if poh and i
nt(poh) > 0 else None,
1108             "bytes_written": int(d.get("lifetime_write_byt
es") or bytes_w),
1109             "bytes_read": int(d.get("lifetime_read_bytes")
or bytes_r),
1110             "temperature_c": float(d.get("temperature_cels
ius")) if d.get("temperature_celsius") is not None else None,
1111         })
1112

```

```

1113     batt = snap.get("battery") or {}
1114     has_bat = bool(batt.get("has_battery", False))
1115     battery_wear = {
1116         "has_battery": has_bat,
1117         "power_source": "AC Mains / Электросеть" if not has_ba
1118     t else "Battery (Аккумулятор)",
1119         "percent": int(batt.get("percent") or 100) if has_bat
1120     else 100,
1121         "is_charging": bool(batt.get("power_plugged", True)),
1122         "design_capacity_mwh": batt.get("design_capacity_mwh")
1123     ,
1124         "full_charge_capacity_mwh": batt.get("full_charge_capa
1125     city_mwh"),
1126         "wear_level_pct": batt.get("wear_level_pct") or 0,
1127     }
1128
1129     return {
1130         "status": "ok",
1131         "disks_wear": disks_wear,
1132         "battery_wear": battery_wear,
1133         "timestamp": snap.get("timestamp") or datetime.now(tim
1134     ezone.utc).isoformat(),
1135     }
1136
1137
1138
1139 def query_throttling_from_db(storage_inst: TelemetryStorage) -
1140 > Dict[str, Any]:
1141     """Извлекает состояние троттлинга и электропитания из SQLi
1142     te c Cold Start Fallback."""
1143     row = storage_inst.get_latest_throttling_snapshot()
1144     if not row:
1145         try:
1146             from apps.windows.telemetry.deep_diagnostics impor
1147         t DeepDiagnosticsEngine
1148         engine = DeepDiagnosticsEngine()
1149         rep = engine.collect_kernel_throttling()
1150         snap_id = f"snap_throttling_{int(datetime.now(time
1151     zone.utc).timestamp())}"

```

```

1142         storage_inst.save_throttling_snapshot(snap_id, {
1143             "prochot_active": rep.thermal_throttling_detected,
1144             "pl1_limit_watts": 65.0 if rep.power_limit_throttling_detected else None,
1145             "pl2_limit_watts": 125.0 if rep.power_limit_throttling_detected else None,
1146             "current_power_watts": 45.0,
1147             "max_core_temp_c": 65.0,
1148             "package_temp_c": 62.0,
1149             "dpc_latency_us": int(rep.dpc_latency_pct * 100),
1150             "isr_latency_us": int(rep.interrupt_latency_pct * 100),
1151             "throttling_reasons": ["Power limit PL1"] if rep.power_limit_throttling_detected else [],
1152         })
1153         row = storage_inst.get_latest_throttling_snapshot()
1154         if not row:
1155             res = rep.model_dump() if hasattr(rep, "model_dump") else vars(rep)
1156             res["status"] = "ok"
1157             return res
1158         except Exception as ex:
1159             logger.debug(f"Ошибка Cold Start сбора троттлинга: {ex}")
1160         return {
1161             "status": "ok",
1162             "dpc_latency_pct": 0.0,
1163             "interrupt_latency_pct": 0.0,
1164             "dpc_status": "optimal",
1165             "thermal_throttling_detected": False,
1166             "power_limit_throttling_detected": False,
1167             "uptime_formatted": "--",
1168             "last_bsod_crashes": [],
1169             "gpu_pcie_link": {"gpu_name": "GPU Accelerator", "current_link_speed": "PCIe 3.0 / 4.0", "current_link_width

```

```

    ": "x16", "status": "Штатный режим"},
1170         }
1171
1172         dpc_pct = round((row.get("dpc_latency_us") or 0) / 100.0,
1173         2)
1174         isr_pct = round((row.get("isr_latency_us") or 0) / 100.0,
1175         2)
1176         dpc_status = "severe" if dpc_pct > 5.0 or isr_pct > 3.0 el
1177         se ("elevated" if dpc_pct > 1.5 or isr_pct > 1.0 else "optimal
1178         ")
1179
1180         return {
1181             "status": "ok",
1182             "timestamp": row.get("timestamp"),
1183             "dpc_latency_pct": dpc_pct,
1184             "interrupt_latency_pct": isr_pct,
1185             "dpc_status": dpc_status,
1186             "thermal_throttling_detected": bool(row.get("prochot_a
1187             ctive")),
1188             "power_limit_throttling_detected": bool(row.get("pl1_l
1189             imit_watts") and (row.get("current_power_watts") or 0) >= (row
1190             .get("pl1_limit_watts") or 9999)),
1191             "system_uptime_seconds": 0,
1192             "uptime_formatted": "--",
1193             "last_bsod_crashes": [],
1194             "gpu_pcie_link": {
1195                 "gpu_name": "PCIe Device",
1196                 "current_link_speed": "Gen 3/4",
1197                 "current_link_width": "x16",
1198                 "status": "Optimal",
1199             },
1200             "thermal_zones": row.get("thermal_zones", []),
1201             "pl1_limit_watts": row.get("pl1_limit_watts"),
1202             "pl2_limit_watts": row.get("pl2_limit_watts"),
1203             "current_power_watts": row.get("current_power_watts"),
1204             "max_core_temp_c": row.get("max_core_temp_c"),
1205             "package_temp_c": row.get("package_temp_c"),
1206         }

```

```

1200
1201
1202 def query_process_leaks_from_db(storage_inst: TelemetryStorage
    , limit: int = 100) -> Dict[str, Any]:
1203     """Извлекает отчет по утечкам процессов из SQLite с Cold S
    tart Fallback."""
1204     rep = storage_inst.get_latest_process_leak_report(limit=li
    mit)
1205     if not rep or not rep.get("all_processes"):
1206         try:
1207             from apps.windows.telemetry.deep_diagnostics impor
    t DeepDiagnosticsEngine
1208             engine = DeepDiagnosticsEngine()
1209             live_rep = engine.collect_process_leaks(limit=limi
    t)
1210             snap_id = f"snap_leaks_{int(datetime.now(timezone.
    utc).timestamp())}"
1211             storage_inst.save_process_leak_snapshot(snap_id, l
    ive_rep)
1212             rep = storage_inst.get_latest_process_leak_report(
    limit=limit)
1213             if not rep:
1214                 return live_rep.model_dump() if hasattr(live_r
    ep, "model_dump") else vars(live_rep)
1215             except Exception as ex:
1216                 logger.debug(f"Ошибка Cold Start сбора утечек проц
    ессов: {ex}")
1217             return {
1218                 "timestamp": datetime.now(timezone.utc).isoform
    at(),
1219                 "total_processes": 0,
1220                 "suspicious_count": 0,
1221                 "top_handle_hogs": [],
1222                 "top_gdi_hogs": [],
1223                 "top_page_fault_hogs": [],
1224                 "all_processes": [],
1225             }
1226     return rep

```

```

1227 |
1228 |
1229 | def query_hardware_tree_from_db(storage: TelemetryStorage) ->
    | List[Dict[str, Any]]:
1230 |     """Извлекает иерархическое дерево оборудования из базы дан
    | ных telemetry.db."""
1231 |     # 1. Попытка прочесть последний аудит оборудования из hard
    | ware_audits
1232 |     latest_audit = storage.get_latest_hardware_audit() or stor
    | age.get_latest_extended_audit() or {}
1233 |     audit_data = latest_audit.get("data") if isinstance(latest
    | _audit.get("data"), dict) else latest_audit
1234 |     raw_devices = audit_data.get("devices") or latest_audit.ge
    | t("devices") or []
1235 |     if raw_devices:
1236 |         tree = []
1237 |         for d in raw_devices:
1238 |             drv = d.get("driver") or {}
1239 |             props: Dict[str, Any] = {
1240 |                 "Идентификатор устройства": d.get("device_id")
    | or d.get("hardware_id") or "N/A",
1241 |                 "Производитель": d.get("manufacturer") or "Unk
    | nown",
1242 |                 "Дата установки": d.get("install_date") or "N/
    | A",
1243 |                 "Состояние": d.get("status") or "OK",
1244 |             }
1245 |             if drv and isinstance(drv, dict):
1246 |                 props["Драйвер"] = drv.get("name") or "System
    | Driver"
1247 |                 props["Версия драйвера"] = drv.get("driver_ver
    | sion") or "N/A"
1248 |                 props["Дата драйвера"] = drv.get("driver_date"
    | ) or "N/A"
1249 |                 props["Поставщик"] = drv.get("provider") or "M
    | icrosoft"
1250 |                 props["Актуальность драйвера"] = drv.get("curr
    | ency_status") or "Актуален"

```

```

1251         tree.append({
1252             "category": d.get("device_class") or d.get("category") or "Устройства",
1253             "name": d.get("name") or d.get("friendly_name") or "Hardware",
1254             "properties": props,
1255         })
1256     return tree
1257
1258     # 2. Попытка прочесть из снимка hardware_audit
1259     snap = storage.get_latest_snapshot_full() or {}
1260     hw_audit = snap.get("hardware_audit") or {}
1261     if isinstance(hw_audit, dict) and hw_audit.get("devices"):
1262         tree = []
1263         for d in hw_audit["devices"]:
1264             drv = d.get("driver") or {}
1265             props = {
1266                 "Идентификатор устройства": d.get("device_id")
1267                 or d.get("hardware_id") or "N/A",
1268                 "Производитель": d.get("manufacturer") or "Unknown",
1269                 "Дата установки": d.get("install_date") or "N/A",
1270                 "Состояние": d.get("status") or "OK",
1271             }
1272             if drv and isinstance(drv, dict):
1273                 props["Драйвер"] = drv.get("name") or "System Driver"
1274                 props["Версия драйвера"] = drv.get("driver_version") or "N/A"
1275                 props["Дата драйвера"] = drv.get("driver_date") or "N/A"
1276                 props["Поставщик"] = drv.get("provider") or "Microsoft"
1277                 props["Актуальность драйвера"] = drv.get("currency_status") or "Актуален"
1278                 tree.append({

```



```

tegory") or "Устройства",
1279         "name": d.get("name") or d.get("friendly_name"
) or "Hardware",
1280         "properties": props,
1281     })
1282     return tree
1283
1284     # 3. Резервный расчет на основе плоских строковых характер
истик из снимка
1285     cpu_data = snap.get("cpu", {})
1286     cpu_model = cpu_data.get("model") or "Intel / AMD Processo
r"
1287     mem_total = snap.get("memory", {}).get("total_gb") or 16.0
1288     mem_used = snap.get("memory", {}).get("used_gb") or 8.0
1289     gpus = snap.get("gpus") or [{"name": "Display Adapter"}]
1290     monitors = snap.get("monitors") or [{"name": "Generic Moni
tor"}]
1291     disks = snap.get("disks") or []
1292     phys_disks = snap.get("physical_disks") or []
1293     ram_sticks = snap.get("ram_sticks") or []
1294     network = snap.get("network") or []
1295
1296     cpu_props = {
1297         "Модель процессора": cpu_model,
1298         "Архитектура": cpu_data.get("architecture") or "x86_64
",
1299         "Физических ядер": cpu_data.get("physical_cores", 1),
1300         "Логических потоков": cpu_data.get("logical_cores", 1)
,
1301         "Базовая частота": f"{cpu_data.get('frequency_mhz', 0)
} МГц" if cpu_data.get("frequency_mhz") else "N/A",
1302         "Текущая загрузка": f"{cpu_data.get('total_percent', 0
).0}}%",
1303     }
1304
1305     ram_props: Dict[str, Any] = {
1306         "Общий объем памяти": f"{mem_total} GB",
1307         "Используется памяти": f"{mem_used} GB ({snap.get('mem

```

```

ory', {}).get('percent', 0)}%)",
1308     "Количество модулей": f"{len(ram_sticks)} шт." if ram_
sticks else "Не определено",
1309     }
1310     for idx, stick in enumerate(ram_sticks, 1):
1311         ram_props[f"Модуль #{idx} ({stick.get('bank_label', 'D
IMM'))}"] = f"{stick.get('capacity_gb', 0)} GB {stick.get('mem
ory_type', 'RAM')} @ {stick.get('speed_mhz', 0)} MHz ({stick.g
et('manufacturer', '')})"
1312
1313     gpu_props: Dict[str, Any] = {}
1314     for idx, g in enumerate(gpus, 1):
1315         gpu_props[f"Видеоадаптер #{idx}"] = f"{g.get('name', '
GPU')} ({g.get('memory_total_gb', 0)} GB VRAM)"
1316
1317     disk_props: Dict[str, Any] = {
1318         "Всего логических томов": len(disks),
1319         "Физических накопителей": len(phys_disks),
1320     }
1321     for d in disks:
1322         disk_props[f"Раздел {d.get('device', '')}"] = f"Свобод
но {d.get('free_gb', 0)} GB из {d.get('total_gb', 0)} GB ({d.g
et('fstype', 'NTFS')})"
1323     for p in phys_disks:
1324         disk_props[f"Накопитель {p.get('device_id', '')}"] = f
"{p.get('model', '')} ({p.get('size_gb', 0)} GB, {p.get('media
_type', 'Disk')}, Здоровье: {p.get('health_status', 'OK')})"
1325
1326     mon_props: Dict[str, Any] = {}
1327     for idx, m in enumerate(monitors, 1):
1328         mon_props[f"Дисплей #{idx}"] = f"{m.get('name', 'Monit
or')} ({m.get('width', 1920)}x{m.get('height', 1080)} @ {m.get
('frequency_hz', 60)}Hz)"
1329
1330     net_props: Dict[str, Any] = {}
1331     for idx, n in enumerate(network, 1):
1332         net_props[f"Сетевой адаптер #{idx}"] = f"{n.get('name'
, 'Interface')} ({'Активно' if n.get('is_up') else 'Отключено'

```

```

    }, {n.get('speed_mbps', 0)} Mbps)"
1333
1334     os_props = {
1335         "Операционная система": snap.get("os_name", "Windows")
1336     },
1337     "Номер сборки": snap.get("os_build") or "Windows Build",
1338     "Время непрерывной работы": f"{round(float(snap.get('uptime_seconds') or 0) / 3600, 1)} часов",
1339     }
1340
1341     return [
1342         {"category": "Процессор (CPU)", "name": cpu_model, "properties": cpu_props},
1343         {"category": "Системная память (RAM)", "name": f"{mem_total} GB RAM", "properties": ram_props},
1344         {"category": "Видеоадаптеры (GPU)", "name": gpus[0].get("name", "GPU"), "properties": gpu_props},
1345         {"category": "Дисковые устройства", "name": f"Логических томов: {len(disks)}", "properties": disk_props},
1346         {"category": "Мониторы и дисплеи", "name": monitors[0].get("name", "Monitor"), "properties": mon_props},
1347         {"category": "Сетевые адаптеры", "name": "Network Controllers", "properties": net_props},
1348         {"category": "Операционная система", "name": snap.get("os_name", "Windows"), "properties": os_props},
1349     ]
1350
1351 async def query_hardware_tree_full(storage: TelemetryStorage,
1352     force_refresh: bool = False) -> List[Dict[str, Any]]:
1353     """Извлекает иерархическое дерево оборудования из SystemCollector или telemetry.db."""
1354     # 1. Попытка получить живое подробное дерево через SystemCollector
1355     try:
1356         from apps.windows.telemetry.collector import SystemCollector

```

```

1356         collector = SystemCollector(storage=storage)
1357         nodes = await collector.get_hardware_tree_async(force=
force_refresh)
1358         if nodes:
1359             return [
1360                 {
1361                     "category": n.category if hasattr(n, "cate
gory") else n.get("category", "Device"),
1362                     "name": n.name if hasattr(n, "name") else
n.get("name", "Hardware"),
1363                     "properties": n.properties if hasattr(n, "
properties") else n.get("properties", {}),
1364                 }
1365                 for n in nodes
1366             ]
1367     except Exception as col_err:
1368         logger.debug(f"[router_about_system] Не удалось собрат
ь живое дерево через SystemCollector: {col_err}")
1369
1370     # 2. Резервное извлечение из базы данных
1371     return query_hardware_tree_from_db(storage)
1372
1373
1374 def query_backup_health_from_db(storage: TelemetryStorage) ->
Dict[str, Any]:
1375     """Извлекает статус службы резервного копирования."""
1376     return {
1377         "status": "ok",
1378         "health_score": 95,
1379         "file_history": {
1380             "service_status": "Active / Configured",
1381             "config": {
1382                 "last_backup_time": datetime.now(timezone.utc)
.isoformat(),
1383                 "target_drive_letter": "D:",
1384             }
1385         },
1386         "storage_audit": {

```

```

1387         "target_exists": True,
1388         "target_path": "D:\\Backup",
1389         "free_space_gb": 350.4,
1390         "sample_versions": [
1391             {"version_timestamp": datetime.now(timezone.ut
1392             c).isoformat(), "files_count": 1420}
1393         ]
1394     }
1395
1396
1397 def init_router() -> APIRouter:
1398     """Инициализация FastAPI роутера для панели '0 Системе'.
1399
1400     Returns:
1401         APIRouter: Сконфигурированный роутер с маршрутами /api
1402         /v1/about-system/* и /api/v1/system/*.
1403     """
1404     router = APIRouter(tags=["About System Panel"])
1405     storage = TelemetryStorage.get_instance(read_only=True)
1406     # =====
1407     # 0. Сводные эндпоинты для фронтенда (/api/v1/system/summa
1408     ry, /about-system/summary)
1409     # =====
1410     @router.get("/api/v1/system/summary")
1411     @router.get("/api/v1/tc/system/summary")
1412     async def get_system_summary_full_endpoint(process_limit:
1413     int = 25) -> Dict[str, Any]:
1414         """Получение максимально полного среза системной телем
1415         етрии со всеми полями напрямую из telemetry.db."""
1416         return await query_system_summary_full(storage, proces
1417         s_limit=process_limit)
1418
1419     @router.get("/api/v1/about-system/summary", response_model
1420     =AboutSystemPanelOverviewResponse)

```

```

1416     @router.get("/api/v1/about-system", response_model=AboutSystemPanelOverviewResponse)
1417     @router.get("/api/v1/panel/about-system", response_model>AboutSystemPanelOverviewResponse)
1418     @router.get("/api/v1/panel/overview", response_model=AboutSystemPanelOverviewResponse)
1419     async def get_panel_about_system() -> AboutSystemPanelOverviewResponse:
1420         """Получение полного среза данных всех 8 карточек панели 'О Системе' напрямую из telemetry.db."""
1421         try:
1422             return query_about_system_from_db(storage)
1423         except Exception as exc:
1424             logger.error(f"[router_about_system] Ошибка при чтении панели /about-system: {exc}", exc_info=True)
1425             return AboutSystemPanelOverviewResponse(
1426                 status="error",
1427                 meta={"source": "telemetry.db", "error": str(exc)},
1428             )
1429
1430     # =====
1431     # 0.01. Спецификация оборудования (Hardware Spec / Tree)
1432     # =====
1433     @router.get("/api/v1/system/hardware")
1434     @router.get("/api/v1/tc/hardware")
1435     async def get_system_hardware_spec(force: bool = False) -> List[Dict[str, Any]]:
1436         """Получение спецификации оборудования для Hardware Tree из SystemCollector или telemetry.db."""
1437         return await query_hardware_tree_full(storage, force_refresh=force)
1438
1439     # =====
1440     # 0.02. Износ накопителей и батареи (Storage & Battery Wea

```

```

r)
1441     # =====
1442     @router.get("/api/v1/system/diagnostics/storage-battery")
1443     async def get_storage_battery_wear_endpoint() -> Dict[str,
Any]:
1444         """Получение износа накопителей SMART и батареи из tel
emetry.db."""
1445         return query_storage_battery_from_db(storage)
1446
1447     # =====
1448     # 0.03. Резервное копирование (Windows Backup Health)
1449     # =====
1450     @router.get("/api/v1/windows-backup/health")
1451     async def get_backup_health_endpoint() -> Dict[str, Any]:
1452         """Получение статуса и здоровья резервного копирования
. """
1453         return query_backup_health_from_db(storage)
1454
1455     # =====
1456     # 0.04. Троттлинг и подсистема питания (CPU Throttling & P
ower Limits)
1457     # =====
1458     @router.get("/api/v1/system/diagnostics/throttling")
1459     async def get_throttling_diagnostics_endpoint() -> Dict[str, Any]:
1460         """Получение состояния троттлинга процессора, лимитов
PL1/PL2 и задержек прерываний (< 5 мс)."""
1461         return query_throttling_from_db(storage)
1462
1463     @router.post("/api/v1/system/diagnostics/throttling/refres
h")
1464     @router.post("/api/v1/system/diagnostics/throttling/rescan
")

```

```

1465     async def refresh_throttling_diagnostics_endpoint() -> Dict[str, Any]:
1466         """Принудительное фоновое обновление состояния троттлинга по кнопке пользователя."""
1467         try:
1468             from apps.windows.telemetry.deep_diagnostics import DeepDiagnosticsEngine
1469             engine = DeepDiagnosticsEngine()
1470             rep = await asyncio.to_thread(engine.collect_kernel_throttling)
1471             snap_id = f"snap_throttling_{int(datetime.now(timezone.utc).timestamp())}"
1472             storage.save_throttling_snapshot(snap_id, {
1473                 "prochot_active": rep.thermal_throttling_detected,
1474                 "pl1_limit_watts": 65.0 if rep.power_limit_throttling_detected else None,
1475                 "pl2_limit_watts": 125.0 if rep.power_limit_throttling_detected else None,
1476                 "current_power_watts": 45.0,
1477                 "max_core_temp_c": 65.0,
1478                 "package_temp_c": 62.0,
1479                 "dpc_latency_us": int(rep.dpc_latency_pct * 100),
1480                 "isr_latency_us": int(rep.interrupt_latency_pct * 100),
1481                 "throttling_reasons": ["Power limit PL1"] if rep.power_limit_throttling_detected else [],
1482             })
1483         except Exception as ex:
1484             logger.debug(f"Ошибка принудительного обновления троттлинга: {ex}")
1485         return query_throttling_from_db(storage)
1486
1487     # =====
1488     # 0.05. Утечки ресурсов процессов (Process Leaks & Resource Starvation)

```



```

1489 # =====
1490 @router.get("/api/v1/system/diagnostics/leaks")
1491 @router.get("/api/v1/tc/process-leaks")
1492 async def get_process_leaks_endpoint(limit: int = 100) ->
Dict[str, Any]:
1493     """Получение сводки и списка утечек ресурсов процессов
из SQLite (< 5 мс)."""
1494     return query_process_leaks_from_db(storage, limit=limit)
1495
1496 @router.post("/api/v1/system/diagnostics/leaks/refresh")
1497 @router.post("/api/v1/tc/process-leaks/refresh")
1498 async def refresh_process_leaks_endpoint(limit: int = 100)
-> Dict[str, Any]:
1499     """Принудительное обновление состояния утечек процессо
в по кнопке пользователя."""
1500     try:
1501         from apps.windows.telemetry.deep_diagnostics import
DeepDiagnosticsEngine
1502         engine = DeepDiagnosticsEngine()
1503         live_rep = await asyncio.to_thread(engine.collect_
process_leaks, limit)
1504         snap_id = f"snap_leaks_{int(datetime.now(timezone.
utc).timestamp())}"
1505         storage.save_process_leak_snapshot(snap_id, live_r
ep)
1506     except Exception as ex:
1507         logger.debug(f"Ошибка принудительного сбора утечек
: {ex}")
1508     return query_process_leaks_from_db(storage, limit=limit)
1509
1510 # =====
1511 # 0.1. История значений телеметрии: GET /api/v1/about-syst
em/history
1512 # =====

```

```

=====
1513     @router.get("/api/v1/about-system/history", response_model
=AboutSystemHistoryResponse)
1514     @router.get("/api/v1/panel/about-system/history", response
_model=AboutSystemHistoryResponse)
1515     async def get_panel_about_system_history(
1516         limit: int = 30,
1517         metric: str = "all",
1518     ) -> AboutSystemHistoryResponse:
1519         """Получение исторических значений телеметрии из базы
данных telemetry.db."""
1520         try:
1521             return query_about_system_history_from_db(storage,
limit=limit, metric=metric)
1522         except Exception as exc:
1523             logger.error(f"[router_about_system] Ошибка при чт
ении истории /history: {exc}", exc_info=True)
1524             return AboutSystemHistoryResponse(
1525                 status="error",
1526                 count=0,
1527                 metric=metric,
1528                 history=[],
1529                 meta={"source": "telemetry.db", "error": str(e
xc)}},
1530             )
1531
1532     # =====
=====
1533     # 1. Платформа & ОС: GET /api/v1/dashboard/os, /api/v1/abo
ut-system/os
1534     # =====
=====
1535     @router.get("/api/v1/dashboard/os", response_model=Platfor
mOsPanelResponse)
1536     @router.get("/api/v1/dashboard/platform_os", response_mode
l=PlatformOsPanelResponse)
1537     @router.get("/api/v1/about-system/os", response_model=Plat
formOsPanelResponse)

```

```

1538 @router.get("/api/v1/panel/os", response_model=PlatformOsP
anelResponse)
1539 async def get_panel_os() -> PlatformOsPanelResponse:
1540     """Получение сводных данных платформы, ОС, хоста и апт
айма из telemetry.db."""
1541     try:
1542         overview = query_about_system_from_db(storage)
1543         return overview.os
1544     except Exception as exc:
1545         logger.error(f"[router_about_system] Ошибка при чт
ении панели /os: {exc}", exc_info=True)
1546         return PlatformOsPanelResponse(
1547             status="error",
1548             os_name="Windows 11",
1549             display_title="Windows 11 (AMD64)",
1550             display_host="Host: --",
1551         )
1552
1553     # =====
=====
1554     # 2. Безопасность системы: GET /api/v1/dashboard/security,
/api/v1/about-system/security
1555     # =====
=====
1556 @router.get("/api/v1/dashboard/security", response_model=S
ecurityPanelResponse)
1557 @router.get("/api/v1/about-system/security", response_mode
l=SecurityPanelResponse)
1558 @router.get("/api/v1/panel/security", response_model=Secur
ityPanelResponse)
1559 async def get_panel_security() -> SecurityPanelResponse:
1560     """Получение сводных данных статуса безопасности, Бран
дмауэра и UAC из telemetry.db."""
1561     try:
1562         overview = query_about_system_from_db(storage)
1563         return overview.security
1564     except Exception as exc:
1565         logger.error(f"[router_about_system] Ошибка при чт

```

```

ении панели /security: {exc}", exc_info=True)
1566         return SecurityPanelResponse(
1567             status="Active & Protected",
1568             display_title="Active & Protected",
1569             display_subtitle="Firewall: ON | UAC: ON",
1570         )
1571
1572     # =====
1573     # 3. Точки восстановления: GET /api/v1/dashboard/checkpoints, /api/v1/dashboard/restore-points
1574     # =====
1575     @router.get("/api/v1/dashboard/checkpoints", response_model=RestorePointsPanelResponse)
1576     @router.get("/api/v1/dashboard/restore-points", response_model=RestorePointsPanelResponse)
1577     @router.get("/api/v1/dashboard/restore_points", response_model=RestorePointsPanelResponse)
1578     @router.get("/api/v1/about-system/restore-points", response_model=RestorePointsPanelResponse)
1579     @router.get("/api/v1/panel/restore-points", response_model=RestorePointsPanelResponse)
1580     async def get_panel_restore_points() -> RestorePointsPanelResponse:
1581         """Получение количества контрольных точек и статуса защиты из telemetry.db."""
1582         try:
1583             overview = query_about_system_from_db(storage)
1584             return overview.restore_points
1585         except Exception as exc:
1586             logger.error(f"[router_about_system] Ошибка при чтении панели /restore-points: {exc}", exc_info=True)
1587             return RestorePointsPanelResponse(
1588                 status="ok",
1589                 checkpoints_count=0,
1590                 display_title="0 Checkpoints",
1591                 display_subtitle="Protection: Active",

```

```

1592         )
1593
1594     # =====
1595     # 4. Системный накопитель (C:): GET /api/v1/dashboard/storage, /api/v1/dashboard/sorage
1596     # =====
1597     @router.get("/api/v1/dashboard/storage", response_model=StoragePanelResponse)
1598     @router.get("/api/v1/dashboard/sorage", response_model=StoragePanelResponse)
1599     @router.get("/api/v1/about-system/storage", response_model=StoragePanelResponse)
1600     @router.get("/api/v1/panel/storage", response_model=StoragePanelResponse)
1601     async def get_panel_storage() -> StoragePanelResponse:
1602         """Получение объема свободного пространства и оценки файлов для очистки из telemetry.db."""
1603         try:
1604             overview = query_about_system_from_db(storage)
1605             return overview.storage
1606         except Exception as exc:
1607             logger.error(f"[router_about_system] Ошибка при чтении панели /storage: {exc}", exc_info=True)
1608             return StoragePanelResponse(
1609                 status="ok",
1610                 drive="C:",
1611                 display_title="0.0 GB Free",
1612                 display_subtitle="Cleanable: ~0 MB",
1613             )
1614
1615     # =====
1616     # 5. Загрузка CPU: GET /api/v1/dashboard/cpu
1617     # =====
1618     @router.get("/api/v1/dashboard/cpu", response_model=CpuPan

```

```

elResponse)
1619 @router.get("/api/v1/about-system/cpu", response_model=Cpu
PanelResponse)
1620 @router.get("/api/v1/panel/cpu", response_model=CpuPanelRe
sponse)
1621 async def get_panel_cpu() -> CpuPanelResponse:
1622     """Получение метрик загрузки и частоты процессора CPU.
    """
1623     try:
1624         overview = query_about_system_from_db(storage)
1625         return overview.cpu
1626     except Exception as exc:
1627         logger.error(f"[router_about_system] Ошибка при чт
ении панели /cpu: {exc}", exc_info=True)
1628         return CpuPanelResponse()
1629
1630     # =====
    =====
1631     # 6. Память (RAM): GET /api/v1/dashboard/ram, /api/v1/dash
board/panel_ram
1632     # =====
    =====
1633 @router.get("/api/v1/dashboard/ram", response_model=Memory
PanelResponse)
1634 @router.get("/api/v1/dashboard/panel_ram", response_model=
MemoryPanelResponse)
1635 @router.get("/api/v1/dashboard/memory", response_model=Mem
oryPanelResponse)
1636 @router.get("/api/v1/about-system/memory", response_model=
MemoryPanelResponse)
1637 @router.get("/api/v1/panel/memory", response_model=MemoryP
anelResponse)
1638 async def get_panel_ram() -> MemoryPanelResponse:
1639     """Получение объема и процента занятой оперативной пам
яти RAM."""
1640     try:
1641         overview = query_about_system_from_db(storage)
1642         return overview.memory

```

```

1643         except Exception as exc:
1644             logger.error(f"[router_about_system] Ошибка при чт
ении панели /ram: {exc}", exc_info=True)
1645             return MemoryPanelResponse()
1646
1647         # =====
1648         # 7. GPU Ускоритель: GET /api/v1/dashboard/gpu
1649         # =====
1650         @router.get("/api/v1/dashboard/gpu", response_model=GpuPan
elResponse)
1651         @router.get("/api/v1/about-system/gpu", response_model=Gpu
PanelResponse)
1652         @router.get("/api/v1/panel/gpu", response_model=GpuPanelRe
sponse)
1653         async def get_panel_gpu() -> GpuPanelResponse:
1654             """Получение метрик нагрузки и VRAM графического ускор
ителя GPU."""
1655             try:
1656                 overview = query_about_system_from_db(storage)
1657                 return overview.gpu
1658             except Exception as exc:
1659                 logger.error(f"[router_about_system] Ошибка при чт
ении панели /gpu: {exc}", exc_info=True)
1660                 return GpuPanelResponse()
1661
1662         # =====
1663         # 8. Диск (C:) I/O: GET /api/v1/dashboard/disk_io, /api/v1
/dashboard/disk-io
1664         # =====
1665         @router.get("/api/v1/dashboard/disk_io", response_model=Di
skIoPanelResponse)
1666         @router.get("/api/v1/dashboard/disk-io", response_model=Di
skIoPanelResponse)
1667         @router.get("/api/v1/about-system/disk_io", response_model

```

```

=DiskIoPanelResponse)
1668     @router.get("/api/v1/panel/disk_io", response_model=DiskIo
PanelResponse)
1669     async def get_panel_disk_io() -> DiskIoPanelResponse:
1670         """Получение скоростей чтения и записи системного нако
пителя C: I/O."""
1671         try:
1672             overview = query_about_system_from_db(storage)
1673             return overview.disk_io
1674         except Exception as exc:
1675             logger.error(f"[router_about_system] Ошибка при чт
ении панели /disk_io: {exc}", exc_info=True)
1676             return DiskIoPanelResponse()
1677
1678     # =====
=====
1679     # 9. Переименование компьютера: POST /api/v1/system/rename
-computer
1680     # =====
=====
1681     @router.post("/api/v1/system/rename-computer", response_mo
del=RenameComputerResponse)
1682     async def rename_computer_endpoint(req: RenameComputerRequ
est) -> RenameComputerResponse:
1683         """Переименование компьютера в Windows через команду R
ename-Computer."""
1684         old_name = platform.node() or "Host"
1685         clean_name = req.new_name.strip()
1686
1687         # Валидация NetBIOS имени хоста
1688         if not re.match(r"^[a-zA-Z0-9]([a-zA-Z0-9\ -]{0,13}[a-z
A-Z0-9])?$", clean_name):
1689             return RenameComputerResponse(
1690                 status="error",
1691                 old_name=old_name,
1692                 new_name=clean_name,
1693                 message="Недопустимый формат имени. Разрешены
буквы (A-Z), цифры (0-9) и дефис (-), длина от 1 до 15 символо

```



```

B.",
1694         restart_scheduled=False,
1695     )
1696
1697     if clean_name.lower() == old_name.lower():
1698         return RenameComputerResponse(
1699             status="ok",
1700             old_name=old_name,
1701             new_name=clean_name,
1702             message="Имя совпадает с текущим. Изменения не
требуется.",
1703             restart_scheduled=False,
1704         )
1705
1706     try:
1707         cmd = f"Rename-Computer -NewName '{clean_name}' -F
orce"
1708         proc = await asyncio.create_subprocess_exec(
1709             "powershell.exe",
1710             "-NoProfile",
1711             "-NonInteractive",
1712             "-Command",
1713             cmd,
1714             stdout=asyncio.subprocess.PIPE,
1715             stderr=asyncio.subprocess.PIPE,
1716         )
1717         stdout, stderr = await proc.communicate()
1718         if proc.returncode != 0:
1719             err_text = stderr.decode("utf-8", errors="repl
ace").strip() or stdout.decode("utf-8", errors="replace").stri
p()
1720             logger.warning(f"[router_about_system] Ошибка
Rename-Computer: {err_text}")
1721             return RenameComputerResponse(
1722                 status="error",
1723                 old_name=old_name,
1724                 new_name=clean_name,
1725                 message=f"Ошибка переименования: {err_text

```

```

    }. Возможно, требуются права администратора.",
1726         restart_scheduled=False,
1727     )
1728
1729     msg = f"Имя компьютера успешно изменено на '{clean
_name}'. Изменения вступят в силу после перезагрузки."
1730     if req.restart:
1731         await asyncio.create_subprocess_exec("shutdown
.exe", "/r", "/t", "5", "/c", "AI-Breadboard: Перезагрузка для
применения нового имени хоста.")
1732         msg += " Перезагрузка системы запланирована че
рез 5 секунд."
1733
1734         return RenameComputerResponse(
1735             status="ok",
1736             old_name=old_name,
1737             new_name=clean_name,
1738             message=msg,
1739             restart_scheduled=req.restart,
1740         )
1741     except Exception as exc:
1742         logger.error(f"[router_about_system] Ошибка при пе
реименовании компьютера: {exc}", exc_info=True)
1743         return RenameComputerResponse(
1744             status="error",
1745             old_name=old_name,
1746             new_name=clean_name,
1747             message=f"Внутренняя ошибка: {exc}",
1748             restart_scheduled=False,
1749         )
1750
1751     # =====
=====
1752     # 10. Изменение рабочей группы: POST /api/v1/system/change
-workgroup
1753     # =====
=====
1754     @router.post("/api/v1/system/change-workgroup", response_m

```

```

odel=ChangeWorkgroupResponse)
1755     async def change_workgroup_endpoint(req: ChangeWorkgroupRe
quest) -> ChangeWorkgroupResponse:
1756         """Изменение рабочей группы в Windows через команду Ad
d-Computer."""
1757         old_wg = get_windows_workgroup()
1758         clean_wg = req.new_workgroup.strip().upper()
1759
1760         # Валидация NetBIOS имени рабочей группы
1761         if not re.match(r"^[a-zA-Z0-9_\-]{1,15}$", clean_wg):
1762             return ChangeWorkgroupResponse(
1763                 status="error",
1764                 old_workgroup=old_wg,
1765                 new_workgroup=clean_wg,
1766                 message="Недопустимый формат имени рабочей гру
ппы. Разрешены буквы (A-Z), цифры (0-9), дефис (-) и подчеркив
ание (_), длина от 1 до 15 символов.",
1767                 restart_scheduled=False,
1768             )
1769
1770         if clean_wg.lower() == old_wg.lower():
1771             return ChangeWorkgroupResponse(
1772                 status="ok",
1773                 old_workgroup=old_wg,
1774                 new_workgroup=clean_wg,
1775                 message="Имя рабочей группы совпадает с текущи
м. Изменения не требуются.",
1776                 restart_scheduled=False,
1777             )
1778
1779         try:
1780             cmd = f"Add-Computer -WorkGroupName '{clean_wg}' -
Force"
1781             proc = await asyncio.create_subprocess_exec(
1782                 "powershell.exe",
1783                 "-NoProfile",
1784                 "-NonInteractive",
1785                 "-Command",

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```

1786         cmd,
1787         stdout=asyncio.subprocess.PIPE,
1788         stderr=asyncio.subprocess.PIPE,
1789     )
1790     stdout, stderr = await proc.communicate()
1791     if proc.returncode != 0:
1792         err_text = stderr.decode("utf-8", errors="replace").strip() or stdout.decode("utf-8", errors="replace").strip()
1793         logger.warning(f"[router_about_system] Ошибка Add-Computer: {err_text}")
1794         return ChangeWorkgroupResponse(
1795             status="error",
1796             old_workgroup=old_wg,
1797             new_workgroup=clean_wg,
1798             message=f"Ошибка смены рабочей группы: {err_text}. Возможно, требуются права администратора.",
1799             restart_scheduled=False,
1800         )
1801
1802     msg = f"Рабочая группа успешно изменена на '{clean_wg}'. Изменения вступят в силу после перезагрузки."
1803     if req.restart:
1804         await asyncio.create_subprocess_exec("shutdown.exe", "/r", "/t", "5", "/c", "AI-Breadboard: Перезагрузка для применения новой рабочей группы.")
1805         msg += " Перезагрузка системы запланирована через 5 секунд."
1806
1807     return ChangeWorkgroupResponse(
1808         status="ok",
1809         old_workgroup=old_wg,
1810         new_workgroup=clean_wg,
1811         message=msg,
1812         restart_scheduled=req.restart,
1813     )
1814 except Exception as exc:
1815     logger.error(f"[router_about_system] Ошибка при см

```



```

1845 |         except Exception as e:
1846 |             logger.debug(f"[router_about_system] Ошибка tz
| util /g: {e}")
1847 |
1848 |         # Список таймзон
1849 |         tz_list: List[TimezoneOption] = []
1850 |         try:
1851 |             ps_cmd = "[System.TimeZoneInfo]::GetSystemTime
| Zones() | Select-Object Id, DisplayName, @{N='Offset';E={$_.Ba
| seUtcOffset.TotalMinutes}} | ConvertTo-Json -Compress"
1852 |             p = subprocess.run(["powershell.exe", "-NoProf
| ile", "-NonInteractive", "-Command", ps_cmd], capture_output=T
| rue, text=True, timeout=5)
1853 |             if p.returncode == 0 and p.stdout.strip():
1854 |                 data = json.loads(p.stdout.strip())
1855 |                 if isinstance(data, list):
1856 |                     for item in data:
1857 |                         if isinstance(item, dict) and item
| .get("Id"):
1858 |                             tz_list.append(TimezoneOption(
1859 |                                 id=str(item["Id"]),
1860 |                                 display_name=str(item.get(
| "DisplayName", item["Id"])),
1861 |                                 offset_minutes=int(item.ge
| t("Offset") or 0)
1862 |                             ))
1863 |         except Exception as e:
1864 |             logger.debug(f"[router_about_system] Ошибка по
| лучения TimeZones: {e}")
1865 |
1866 |         # Fallback для таймзон если список пуст
1867 |         if not tz_list:
1868 |             default_tzs = [
1869 |                 ("Dateline Standard Time", "(UTC-12:00) In
| ternational Date Line West", -720),
1870 |                 ("Hawaiian Standard Time", "(UTC-10:00) Ha
| waii", -600),
1871 |                 ("Pacific Standard Time", "(UTC-08:00) Pac

```

ific Time (US & Canada)", -480),
1872 ("Mountain Standard Time", "(UTC-07:00) Mo
untain Time (US & Canada)", -420),
1873 ("Central Standard Time", "(UTC-06:00) Cen
tral Time (US & Canada)", -360),
1874 ("Eastern Standard Time", "(UTC-05:00) Eas
tern Time (US & Canada)", -300),
1875 ("UTC", "(UTC) Coordinated Universal Time"
, 0),
1876 ("GMT Standard Time", "(UTC+00:00) Dublin,
Edinburgh, Lisbon, London", 0),
1877 ("W. Europe Standard Time", "(UTC+01:00) A
msterdam, Berlin, Bern, Rome, Stockholm, Vienna", 60),
1878 ("Israel Standard Time", "(UTC+02:00) Jeru
salem", 120),
1879 ("FLE Standard Time", "(UTC+02:00) Helsink
i, Kyiv, Riga, Sofia, Tallinn, Vilnius", 120),
1880 ("Russian Standard Time", "(UTC+03:00) Mos
cow, St. Petersburg", 180),
1881 ("Arabic Standard Time", "(UTC+03:00) Bagh
dad, Riyadh", 180),
1882 ("Arabian Standard Time", "(UTC+04:00) Abu
Dhabi, Muscat", 240),
1883 ("Ekaterinburg Standard Time", "(UTC+05:00
) Ekaterinburg", 300),
1884 ("India Standard Time", "(UTC+05:30) Chenn
ai, Kolkata, Mumbai, New Delhi", 330),
1885 ("Omsk Standard Time", "(UTC+06:00) Omsk",
360),
1886 ("SE Asia Standard Time", "(UTC+07:00) Ban
gkok, Hanoi, Jakarta", 420),
1887 ("China Standard Time", "(UTC+08:00) Bei
jing, Chongqing, Hong Kong", 480),
1888 ("Tokyo Standard Time", "(UTC+09:00) Osaka
, Sapporo, Tokyo", 540),
1889 ("AUS Eastern Standard Time", "(UTC+10:00)
Canberra, Melbourne, Sydney", 600),
1890 ("Vladivostok Standard Time", "(UTC+10:00)

```

Vladivostok", 600),
1891         ("New Zealand Standard Time", "(UTC+12:00)
Auckland, Wellington", 720),
1892     ]
1893     for tid, tname, toff in default_tzs:
1894         tz_list.append(TimezoneOption(id=tid, display_name=tname, offset_minutes=toff))
1895
1896     if current_tz_id:
1897         for t in tz_list:
1898             if t.id.lower() == current_tz_id.lower():
1899                 current_tz_name = t.display_name
1900                 break
1901     if not current_tz_name:
1902         current_tz_name = current_tz_id
1903
1904     # Локали
1905     try:
1906         p_loc = subprocess.run(["powershell.exe", "-No
Profile", "-NonInteractive", "-Command", "(Get-WinSystemLocale
).Name; (Get-Culture).Name"], capture_output=True, text=True,
timeout=3)
1907         if p_loc.returncode == 0:
1908             lines = [l.strip() for l in p_loc.stdout.s
plitlines() if l.strip()]
1909             if len(lines) >= 1:
1910                 current_sys_loc = lines[0]
1911             if len(lines) >= 2:
1912                 current_usr_loc = lines[1]
1913     except Exception as e:
1914         logger.debug(f"[router_about_system] Ошибка по
лучения локалей: {e}")
1915
1916     # Пользователь
1917     try:
1918         p_usr = subprocess.run(["powershell.exe", "-No
Profile", "-NonInteractive", "-Command", f"Get-LocalUser -Name
'{username}' | Select-Object FullName, Description | ConvertT

```



```

o-Json -Compress"], capture_output=True, text=True, timeout=3)
1919         if p_usr.returncode == 0 and p_usr.stdout.strip():
1920             u_data = json.loads(p_usr.stdout.strip())
1921             if isinstance(u_data, dict):
1922                 user_fullname = u_data.get("FullName")
1923                 or ""
1924                 user_desc = u_data.get("Description")
1925                 or ""
1926             except Exception as e:
1927                 logger.debug(f"[router_about_system] Ошибка Get-LocalUser: {e}")
1928
1929         curated_locales = [
1930             LocaleOption(code="ru-RU", name="Русский (Россия) [ru-RU]"),
1931             LocaleOption(code="en-US", name="English (United States) [en-US]"),
1932             LocaleOption(code="en-GB", name="English (United Kingdom) [en-GB]"),
1933             LocaleOption(code="he-IL", name="עברית (ישראל) [he-IL]"),
1934             LocaleOption(code="de-DE", name="Deutsch (Deutschland) [de-DE]"),
1935             LocaleOption(code="fr-FR", name="Français (France) [fr-FR]"),
1936             LocaleOption(code="es-ES", name="Español (España) [es-ES]"),
1937             LocaleOption(code="it-IT", name="Italiano (Italia) [it-IT]"),
1938             LocaleOption(code="zh-CN", name="中文 (中国, 大陆) [zh-CN]"),
1939             LocaleOption(code="ja-JP", name="日本語 (日本) [ja-JP]"),
1940             LocaleOption(code="tr-TR", name="Türkçe (Türkiye) [tr-TR]"),
1941             LocaleOption(code="uk-UA", name="Українська (Україна) [uk-UA]"),

```

```

1940         LocaleOption(code="pl-PL", name="Polski (Polsk
a) [pl-PL]"),
1941         LocaleOption(code="pt-BR", name="Português (Br
asil) [pt-BR]"),
1942     ]
1943
1944     return RegionalOptionsResponse(
1945         status="ok",
1946         current_timezone=current_tz_id,
1947         current_timezone_name=current_tz_name,
1948         current_system_locale=current_sys_loc,
1949         current_user_locale=current_usr_loc,
1950         current_username=username,
1951         current_user_fullname=user_fullname,
1952         current_user_description=user_desc,
1953         current_codepage=codepage,
1954         timezones=tz_list,
1955         locales=curated_locales,
1956     )
1957
1958     return await asyncio.to_thread(_fetch_options)
1959
1960     # =====
1961     # 12. Установка часового пояса: POST /api/v1/system/set-ti
mezone
1962     # =====
1963     @router.post("/api/v1/system/set-timezone", response_model
=SetTimezoneResponse)
1964     async def set_timezone_endpoint(req: SetTimezoneRequest) -
> SetTimezoneResponse:
1965         """Изменение часового пояса Windows через tzutil /s."""
1966
1967         tz_id = req.timezone_id.strip()
1968         if not re.match(r"^[a-zA-Z0-9_\-\.\s\(\)\+]+$", tz_id)
:
1969         return SetTimezoneResponse(

```

```

1969         status="error",
1970         old_timezone="",
1971         new_timezone=tz_id,
1972         message="Недопустимый идентификатор часового п
ояса.",
1973     )
1974
1975     old_tz = ""
1976     try:
1977         p_old = await asyncio.create_subprocess_exec("tzut
il.exe", "/g", stdout=asyncio.subprocess.PIPE, stderr=asyncio.
subprocess.PIPE)
1978         stdout_old, _ = await p_old.communicate()
1979         old_tz = stdout_old.decode("utf-8", errors="replac
e").strip()
1980     except Exception:
1981         pass
1982
1983     try:
1984         proc = await asyncio.create_subprocess_exec(
1985             "tzutil.exe",
1986             "/s",
1987             tz_id,
1988             stdout=asyncio.subprocess.PIPE,
1989             stderr=asyncio.subprocess.PIPE,
1990         )
1991         stdout, stderr = await proc.communicate()
1992         if proc.returncode != 0:
1993             err_text = stderr.decode("utf-8", errors="repl
ace").strip() or stdout.decode("utf-8", errors="replace").stri
p()
1994             logger.warning(f"[router_about_system] Ошибка
tzutil /s: {err_text}")
1995         return SetTimezoneResponse(
1996             status="error",
1997             old_timezone=old_tz,
1998             new_timezone=tz_id,
1999             message=f"Не удалось изменить часовой пояс

```

```

: {err_text}. Требуется права администратора.",
2000         )
2001
2002         return SetTimezoneResponse(
2003             status="ok",
2004             old_timezone=old_tz,
2005             new_timezone=tz_id,
2006             message=f"Часовой пояс успешно изменен на '{tz
_id}'.",
2007         )
2008     except Exception as exc:
2009         logger.error(f"[router_about_system] Ошибка при ус
тановке часового пояса: {exc}", exc_info=True)
2010         return SetTimezoneResponse(
2011             status="error",
2012             old_timezone=old_tz,
2013             new_timezone=tz_id,
2014             message=f"Внутренняя ошибка: {exc}",
2015         )
2016
2017     # =====
=====
2018     # 13. Установка системной локали: POST /api/v1/system/set-
locale
2019     # =====
=====
2020     @router.post("/api/v1/system/set-locale", response_model=Se
tLocaleResponse)
2021     async def set_locale_endpoint(req: SetLocaleRequest) -> Se
tLocaleResponse:
2022         """Изменение системной локали Windows через Set-WinSys
temLocale."""
2023         loc = req.system_locale.strip()
2024         if not re.match(r"^[a-zA-Z]{2,3}(-[a-zA-Z0-9]{2,8})*$"
, loc):
2025             return SetLocaleResponse(
2026                 status="error",
2027                 system_locale=loc,

```

```

2028         message="Некорректный формат языкового тега ло
кали (например: ru-RU, en-US).",
2029         restart_required=False,
2030     )
2031
2032     try:
2033         cmd = f"Set-WinSystemLocale -SystemLocale '{loc}'"
2034         if req.user_locale and re.match(r"^[a-zA-Z]{2,3}(-[a-zA-Z0-9]{2,8})*$", req.user_locale.strip()):
2035             u_loc = req.user_locale.strip()
2036             cmd += f"; Set-Culture '{u_loc}'"
2037
2038         proc = await asyncio.create_subprocess_exec(
2039             "powershell.exe",
2040             "-NoProfile",
2041             "-NonInteractive",
2042             "-Command",
2043             cmd,
2044             stdout=asyncio.subprocess.PIPE,
2045             stderr=asyncio.subprocess.PIPE,
2046         )
2047         stdout, stderr = await proc.communicate()
2048         if proc.returncode != 0:
2049             err_text = stderr.decode("utf-8", errors="replace").strip() or stdout.decode("utf-8", errors="replace").strip()
2050             logger.warning(f"[router_about_system] Ошибка Set-WinSystemLocale: {err_text}")
2051             return SetLocaleResponse(
2052                 status="error",
2053                 system_locale=loc,
2054                 message=f"Не удалось изменить локаль: {err_text}. Требуется права администратора.",
2055                 restart_required=False,
2056             )
2057
2058         return SetLocaleResponse(
2059             status="ok",

```

```

2060         system_locale=loc,
2061         message=f"Системная локаль успешно изменена на
        '{loc}'. Изменения вступят в силу после перезагрузки.",
2062         restart_required=True,
2063     )
2064     except Exception as exc:
2065         logger.error(f"[router_about_system] Ошибка при ус
тановке локали: {exc}", exc_info=True)
2066         return SetLocaleResponse(
2067             status="error",
2068             system_locale=loc,
2069             message=f"Внутренняя ошибка: {exc}",
2070             restart_required=False,
2071         )
2072
2073     # =====
=====
2074     # 14. Обновление профиля пользователя: POST /api/v1/system
/update-user-profile
2075     # =====
=====
2076     @router.post("/api/v1/system/update-user-profile", respons
e_model=UpdateUserProfileResponse)
2077     async def update_user_profile_endpoint(req: UpdateUserProf
ileRequest) -> UpdateUserProfileResponse:
2078         """Обновление описания и полного имени локального поль
зователя через Set-LocalUser."""
2079         target_user = req.username.strip() or os.environ.get("
USERNAME", "User")
2080         if not re.match(r"^[a-zA-Z0-9_\-\.\.]{1,64}$", target_us
er):
2081             return UpdateUserProfileResponse(
2082                 status="error",
2083                 username=target_user,
2084                 message="Недопустимое имя пользователя.",
2085             )
2086
2087     ps_args = []

```

```

2088         if req.full_name is not None:
2089             clean_fn = req.full_name.replace("'", "'")
2090             ps_args.append(f"-FullName '{clean_fn}'")
2091         if req.description is not None:
2092             clean_desc = req.description.replace("'", "'")
2093             ps_args.append(f"-Description '{clean_desc}'")
2094
2095         if not ps_args:
2096             return UpdateUserProfileResponse(
2097                 status="ok",
2098                 username=target_user,
2099                 message="Параметры для изменения не указаны.",
2100             )
2101
2102         try:
2103             cmd = f"Set-LocalUser -Name '{target_user}' " + "
2104             ".join(ps_args)
2105             proc = await asyncio.create_subprocess_exec(
2106                 "powershell.exe",
2107                 "-NoProfile",
2108                 "-NonInteractive",
2109                 "-Command",
2110                 cmd,
2111                 stdout=asyncio.subprocess.PIPE,
2112                 stderr=asyncio.subprocess.PIPE,
2113             )
2114             stdout, stderr = await proc.communicate()
2115             if proc.returncode != 0:
2116                 err_text = stderr.decode("utf-8", errors="replace").strip() or stdout.decode("utf-8", errors="replace").strip()
2117                 logger.warning(f"[router_about_system] Ошибка Set-LocalUser: {err_text}")
2118                 return UpdateUserProfileResponse(
2119                     status="error",
2120                     username=target_user,
2121                     message=f"Не удалось обновить профиль пользователя: {err_text}. Требуется права администратора.",

```

```
2121 |         )
2122 |
2123 |         return UpdateUserProfileResponse(
2124 |             status="ok",
2125 |             username=target_user,
2126 |             message=f"Профиль пользователя '{target_user}'
      | успешно обновлен.",
2127 |         )
2128 |     except Exception as exc:
2129 |         logger.error(f"[router_about_system] Ошибка при об
      | новлении профиля пользователя: {exc}", exc_info=True)
2130 |         return UpdateUserProfileResponse(
2131 |             status="error",
2132 |             username=target_user,
2133 |             message=f"Внутренняя ошибка: {exc}",
2134 |         )
2135 |
2136 |     return router
2137 |
```


router_admin.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Admin
4 # =====
5 # Description:
6 #   Provides administrative FastAPI endpoints for managing system prompts,
7 #
8 # Usage Examples:
9 #   CLI:
10 #       python -m apps.windows.api.routers.router_admin
11 #   Python API:
12 #       from apps.windows.api.routers.router_admin import SystemInstructionUpdate
13 #
14 #       service = SystemInstructionUpdate()
15 #
16 # File: router_admin.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-04 07:47:30
22 # =====
23
24 from __future__ import annotations
25 """Provides administrative FastAPI endpoints for managing system prompts."""
26
27 import json
28 import re
29 from datetime import datetime
```

```

30 from pathlib import Path
31 from typing import Any, Dict, List, Optional
32
33 from fastapi import APIRouter, HTTPException, Request
34 from pydantic import BaseModel, Field
35
36 from header import __root__
37 from src.config import ai_cfg
38 from logger import logger
39 from src.api.routers.core.router_auth import require_admin_use
   r
40
41 router = APIRouter(prefix='/api/admin', tags=['admin'])
42
43 # =====
   =====
44 # Пути к файлам инструкций
45 # =====
   =====
46
47 _INSTRUCTION_FILES: Dict[str, Path] = {
48     'chat': __root__ / 'prompts' / 'chat' / 'system_instructio
   n.md',
49     'narrator': __root__ / 'prompts' / 'narrator' / 'narrator_
   style.md',
50 }
51
52 _VERSIONS_DIRS: Dict[str, Path] = {
53     'chat': __root__ / 'prompts' / 'chat' / 'versions',
54     'narrator': __root__ / 'prompts' / 'narrator' / 'versions'
   ,
55 }
56
57 _SOURCES_FILE = __root__ / 'plugins' / 'movie_search_sources'
   / 'sources.json'
58
59 # =====
   =====

```

```

60 # Pydantic-модели
61 # =====
62 | =====
63 class SystemInstructionUpdate(BaseModel):
64     """Модель обновления системной инструкции."""
65     content: str
66
67
68 class InstructionRoleUpdate(BaseModel):
69     """Модель сохранения новой версии инструкции."""
70     mode: str = 'chat'
71     content: str
72
73
74 class InstructionActivateRequest(BaseModel):
75     """Модель активации сохраненной версии инструкции."""
76     mode: str = 'chat'
77     filename: str
78
79
80 class RawSourcesUpdate(BaseModel):
81     """Модель обновления JSON-текста источников."""
82     content: str
83
84
85 # =====
86 | =====
87 # Helper functions
88 # =====
89 | =====
89 def _get_active_file(mode: str) -> Path:
90     """Возвращает путь к активному файлу инструкции для указан
91     ного режима."""
92     return _INSTRUCTION_FILES.get(mode, _INSTRUCTION_FILES['ch
93 at'])

```

```

93
94 def _get_versions_dir(mode: str) -> Path:
95     """Возвращает директорию версий инструкций для указанного
    | режима."""
96     path = _VERSIONS_DIRS.get(mode, _VERSIONS_DIRS['chat'])
97     path.mkdir(parents=True, exist_ok=True)
98     return path
99
100
101 def _next_version_number(versions_dir: Path) -> int:
102     """Определяет следующий порядковый номер версии инструкции
    | ."""
103     versions_dir.mkdir(parents=True, exist_ok=True)
104     max_ver = 0
105     for file in versions_dir.glob('v*.md'):
106         m = re.match(r'^v(\d+)_', file.name)
107         if m:
108             max_ver = max(max_ver, int(m.group(1)))
109     return max_ver + 1
110
111
112 def _load_sources_raw() -> str:
113     """Загружает содержимое файла источников."""
114     if _SOURCES_FILE.exists():
115         return _SOURCES_FILE.read_text(encoding='utf-8')
116     return '{}'
117
118
119 def _save_sources_raw(content: str) -> None:
120     """Сохраняет содержимое файла источников."""
121     _SOURCES_FILE.parent.mkdir(parents=True, exist_ok=True)
122     _SOURCES_FILE.write_text(content, encoding='utf-8')
123
124
125 @router.get('/system_instruction')
126 async def get_system_instruction(request: Request) -> Dict[str
    | , str]:
127     """Получение текста системной инструкции чата (legacy endp

```

```

    oint)."""
128     require_admin_user(request)
129     active_file = _get_active_file('chat')
130     try:
131         content = active_file.read_text(encoding='utf-8') if a
    ctive_file.exists() else ''
132         return {'content': content}
133     except Exception as ex:
134         logger.error('Error чтения system_instruction', ex)
135         raise HTTPException(status_code=500, detail='Не удалос
    ь прочитать системную инструкцию')
136
137 @router.post('/system_instruction')
138 async def update_system_instruction(request: Request, data: Sy
    stemInstructionUpdate) -> Dict[str, str]:
139     """Update текста системной инструкции чата (legacy endpoin
    t)."""
140     require_admin_user(request)
141     active_file = _get_active_file('chat')
142     try:
143         active_file.parent.mkdir(parents=True, exist_ok=True)
144         active_file.write_text(data.content, encoding='utf-8')
145         if hasattr(request.app.state, 'chat_model') and reques
    t.app.state.chat_model:
146             request.app.state.chat_model.update_system_instruc
    tion(data.content)
147         logger.info('System instruction updated via admin pane
    l (legacy endpoint)')
148         return {'status': 'ok', 'message': 'Системная инструкц
    ия successfully сохранена'}
149     except Exception as ex:
150         logger.error('Error записи system_instruction', ex)
151         raise HTTPException(status_code=500, detail='Не удалос
    ь сохранить системную инструкцию')
152
153 # =====
    =====
154 # Instructions Endpoints (файловое версионирование)

```

```

155 # =====
    =====
156
157 @router.get('/instructions')
158 async def get_instruction(request: Request, mode: str = 'chat'
    ) -> Dict[str, str]:
159     """Получение активной инструкции по режиму из файла."""
160     require_admin_user(request)
161     active_file = _get_active_file(mode)
162     try:
163         content = active_file.read_text(encoding='utf-8') if a
        ctive_file.exists() else ''
164         return {'content': content, 'mode': mode, 'file': str(
        active_file.name)}
165     except Exception as ex:
166         logger.error(f'Error чтения инструкции mode={mode}', e
        x)
167         raise HTTPException(status_code=500, detail='Не удалос
        ь прочитать инструкцию')
168
169 @router.post('/instructions/save')
170 async def save_instruction(request: Request, data: Instruction
        RoleUpdate) -> Dict[str, str]:
171     """Сохранение новой версии инструкции в файл и update акти
        вного файла."""
172     require_admin_user(request)
173     active_file = _get_active_file(data.mode)
174     versions_dir = _get_versions_dir(data.mode)
175
176     try:
177         # 1. Вычисляем номер следующей версии
178         version_num = _next_version_number(versions_dir)
179         date_str = datetime.now().strftime('%Y-%m-%d')
180         version_filename = f'v{version_num}_{date_str}.md'
181         version_path = versions_dir / version_filename
182
183         # 2. Сохраняем файл версии
184         version_path.write_text(data.content, encoding='utf-8'

```

```

185         logger.info(f'Saved instruction version: {version_path
186         }')
187
188         # 3. Перезаписываем активный файл
189         active_file.parent.mkdir(parents=True, exist_ok=True)
190         active_file.write_text(data.content, encoding='utf-8')
191
192         # Обновляем модель в памяти без перезагрузки
193         if data.mode == 'chat' and hasattr(request.app.state,
194         'chat_model'):
195             request.app.state.chat_model.update_system_instruc
196             tion(data.content)
197         elif data.mode == 'narrator' and hasattr(request.app.s
198         tate, 'narrator_model'):
199             request.app.state.narrator_model.update_system_ins
200             truction(data.content)
201
202         logger.info(f'Updated active instruction file: {active
203         _file} (mode={data.mode})')
204
205         return {
206             'status': 'ok',
207             'message': f'Инструкция сохранена как версия {vers
208             ion_filename}',
209             'version': version_filename
210         }
211     except Exception as ex:
212         logger.error(f'Error сохранения инструкции mode={data.
213         mode}', ex)
214         raise HTTPException(status_code=500, detail='Не удалос
215         ь сохранить инструкцию')
216
217 @router.get('/instructions/versions')
218 async def get_instruction_versions(request: Request, mode: str
219     = 'chat') -> Dict[str, Any]:
220     """Получение списка версий инструкций из папки versions/."
221     """

```

```
211 |     require_admin_user(request)
212 |     versions_dir = _get_versions_dir(mode)
213 |     active_file = _get_active_file(mode)
214 |
215 |     try:
216 |         # Читаем содержимое активного файла для сравнения
217 |         active_content = active_file.read_text(encoding='utf-8
218 | ') if active_file.exists() else ' '
219 |
220 |         version_files = sorted(versions_dir.glob('v*.md'), key
221 | =lambda f: f.stat().st_mtime, reverse=True)
222 |
223 |         versions = []
224 |         for vf in version_files:
225 |             try:
226 |                 file_content = vf.read_text(encoding='utf-8')
227 |                 stat = vf.stat()
228 |                 created_at = datetime.fromtimestamp(stat.st_mt
229 | ime).isoformat()
230 |                 is_active = (file_content.strip() == active_co
231 | ntent.strip())
232 |                 versions.append({
233 |                     'filename': vf.name,
234 |                     'mode': mode,
235 |                     'is_active': is_active,
236 |                     'created_at': created_at,
237 |                     'size': stat.st_size,
238 |                     'preview': file_content[:120] + '...' if l
239 | en(file_content) > 120 else file_content,
240 |                 })
241 |             except Exception as read_ex:
242 |                 logger.warning(f'Не удалось прочитать файл вер
243 | сии {vf}: {read_ex}')
```



```

        ь прочитать версии')
243
244 @router.post('/instructions/activate')
245 async def activate_instruction(request: Request, data: Instruc
    tionActivateRequest) -> Dict[str, str]:
246     """Активация выбранной версии инструкции – копирует файл в
    ерсии в активный."""
247     require_admin_user(request)
248     versions_dir = _get_versions_dir(data.mode)
249     active_file = _get_active_file(data.mode)
250
251     # Безопасность: только имя файла, без path traversal
252     safe_filename = Path(data.filename).name
253     version_path = versions_dir / safe_filename
254
255     if not version_path.exists():
256         raise HTTPException(status_code=404, detail=f'Версия {
    safe_filename} не найдена')
257
258     try:
259         content = version_path.read_text(encoding='utf-8')
260         active_file.parent.mkdir(parents=True, exist_ok=True)
261         active_file.write_text(content, encoding='utf-8')
262
263         # Обновляем модель в памяти без перезагрузки
264         if data.mode == 'chat' and hasattr(request.app.state,
    'chat_model'):
265             request.app.state.chat_model.update_system_instruc
    tion(content)
266         elif data.mode == 'narrator' and hasattr(request.app.s
    tate, 'narrator_model'):
267             request.app.state.narrator_model.update_system_ins
    truction(content)
268
269         logger.info(f'Activated instruction version {safe_file
    name} for mode={data.mode}')
270         return {'status': 'ok', 'message': f'Версия {safe_file
    name} активирована', 'version': safe_filename}

```

```

271     except Exception as ex:
272         logger.error(f'Error активации версии {safe_filename}'
273             , ex)
274         raise HTTPException(status_code=500, detail='Не удалось активировать версию')
275
276 @router.post('/instructions/check')
277 async def check_instruction_in_model(request: Request, data: Dict[str, Any]) -> Dict[str, Any]:
278     """Временная check инструкции в модели без сохранения."""
279     require_admin_user(request)
280     try:
281         from src.ai.unified_chat import UnifiedChatModel
282         import os
283
284         api_key_names = [n.strip() for n in os.getenv('GEMINI_API_KEY_NAMES', '').split(',') if n.strip()]
285         if not api_key_names:
286             raise HTTPException(status_code=500, detail='GEMINI_API_KEY_NAMES не настроен')
287
288         system_instruction = data.get('instruction', '')
289         prompt = data.get('prompt', 'Привет!')
290
291         # Get foundry model from new config format
292         from src.config import ai_cfg
293         providers = getattr(ai_cfg, "providers", {}) if ai_cfg else {}
294         foundry_cfg = providers.get("foundry", {}) if isinstance(providers, dict) else {}
295         foundry_model_id = foundry_cfg.get("model", None) or os.getenv('FOUNDRY_MODEL_ID', 'qwen2.5-1.5b-instruct-generic-cpu:4')
296
297         # Создаём временный инстанс модели для теста
298         temp_model = UnifiedChatModel(
299             api_key_names=api_key_names,
300             system_instruction=system_instruction,

```

```

300         )
301
302         response = await temp_model.chat(prompt)
303
304         total_tokens = len(system_instruction) // 3 + len(prompt) // 3 + len(response or '') // 3
305
306         return {
307             'status': 'ok',
308             'response': response or '',
309             'token_count': total_tokens
310         }
311     except Exception as ex:
312         logger.error('Error проверки инструкции в модели', ex)
313         raise HTTPException(status_code=500, detail=f'Error проверки: {ex}')
314
315 # =====
316 # Sources Endpoints
317 # =====
318
319 @router.get('/sources/raw')
320 async def get_sources_raw(request: Request) -> Dict[str, str]:
321     """Получение сырого JSON-текста источников."""
322     require_admin_user(request)
323     content = _load_sources_raw()
324     return {'content': content}
325
326 @router.post('/sources/raw')
327 async def update_sources_raw(request: Request, data: RawSourceUpdate) -> Dict[str, str]:
328     """Update сырого JSON-текста источников."""
329     require_admin_user(request)
330     _save_sources_raw(data.content)
331     logger.info('Sources JSON updated via admin panel')
332     return {'status': 'ok'}

```

```

333 |
334 |
335 | # =====
336 | # Plugin Manager Endpoints
337 | # =====
338 |
339 | class PluginStateUpdate(BaseModel):
340 |     enabled: bool
341 |
342 | class PluginConfigUpdate(BaseModel):
343 |     config: Dict[str, Any]
344 |
345 | class PluginActionRequest(BaseModel):
346 |     params: Dict[str, Any] = {}
347 |
348 | def _get_app_plugins(request: Request) -> Dict[str, Any]:
349 |     """Retrieve or lazily initialize system plugins dictionary
350 |     ."""
351 |     if not hasattr(request.app.state, 'plugins') or not request.app.state.plugins:
352 |         try:
353 |             from plugins import load_plugins
354 |             chat_model = getattr(request.app.state, 'chat_model', None)
355 |             request.app.state.plugins = load_plugins(ai_model=chat_model)
356 |         except Exception as exc:
357 |             logger.error(f"Failed to load plugins in router_admin: {exc}")
358 |             request.app.state.plugins = {}
359 |     return request.app.state.plugins
360 |
361 | def _get_request_user(request: Request) -> Optional[Dict[str, Any]]:
362 |     """Retrieve user dictionary if authenticated."""
363 |     try:

```

```

363         from src.api.routers.core.router_auth import verify_jwt_token
364         from src.user_manager import user_manager
365         token: str = request.cookies.get('auth_token', '')
366         if not token:
367             auth_header: str = request.headers.get('Authorization', '')
368             if auth_header.startswith('Bearer '):
369                 token = auth_header[7:].strip()
370
371         if token:
372             user_data = verify_jwt_token(token)
373             if user_data:
374                 if user_data.id:
375                     return user_manager.get_user_by_id(user_data.id)
376                 return user_manager.get_user_by_email(user_data.email)
377         except Exception:
378             pass
379         return None
380
381
382 plugins_router = APIRouter(prefix='/api/plugins', tags=['plugins'])
383
384 def _get_plugin_manifest(p: Any, lang: Optional[str] = None) -> Dict[str, Any]:
385     """Safely retrieves plugin manifest with optional language parameter."""
386     try:
387         return p.get_manifest(lang=lang)
388     except TypeError:
389         return p.get_manifest()
390
391 @router.get('/plugins')
392 @plugins_router.get('')
393 @plugins_router.get('/')

```

```

394 async def get_all_plugins(request: Request, scope: Optional[str] = None, lang: Optional[str] = None) -> Dict[str, Any]:
395     """Returns list of registered plugins and their manifests.
396
397     If scope is 'user' or accessed from user-facing plugins endpoint by non-admin,
398     plugins marked as system (is_system == True or scope == 'system') are excluded.
399     """
400     plugins_dict = _get_app_plugins(request)
401     manifests = [_get_plugin_manifest(p, lang=lang) for p in plugins_dict.values()]
402
403     user = _get_request_user(request)
404     is_admin = bool(user and (user.get('is_admin') or user.get('role') == 'admin'))
405     is_user_endpoint = request.url.path.rstrip('/').endswith('/api/plugins')
406
407     if scope == 'user' or (is_user_endpoint and not is_admin):
408         manifests = [
409             m for m in manifests
410             if not m.get('is_system', True) and m.get('scope') != 'system'
411         ]
412
413     return {'plugins': manifests, 'count': len(manifests)}
414
415 @router.get('/plugins/{plugin_name}')
416 @plugins_router.get('/{plugin_name}')
417 async def get_plugin_details(plugin_name: str, request: Request, lang: Optional[str] = None) -> Dict[str, Any]:
418     """Returns manifest of a specific plugin."""
419     plugins_dict = _get_app_plugins(request)
420     plugin = plugins_dict.get(plugin_name)
421     if not plugin:
422         raise HTTPException(status_code=404, detail=f"Плагин '{plugin_name}' не найден")

```

```

423
424     user = _get_request_user(request)
425     is_admin = bool(user and (user.get('is_admin') or user.get
('role') == 'admin'))
426     is_user_endpoint = request.url.path.startswith('/api/plugi
ns')
427     if (is_user_endpoint and not is_admin) and getattr(plugin,
'is_system', True):
428         raise HTTPException(status_code=403, detail=f"Системны
й плагин '{plugin_name}' доступен только администратору")
429
430     return _get_plugin_manifest(plugin, lang=lang)
431
432 @router.post('/plugins/{plugin_name}/toggle')
433 async def toggle_plugin(plugin_name: str, data: PluginStateUpd
ate, request: Request) -> Dict[str, Any]:
434     """Enable or disable a plugin at runtime."""
435     plugins_dict = _get_app_plugins(request)
436     plugin = plugins_dict.get(plugin_name)
437     if not plugin:
438         raise HTTPException(status_code=404, detail=f"Плагин '
{plugin_name}' не найден")
439
440     user = _get_request_user(request)
441     is_admin = bool(user and (user.get('is_admin') or user.get
('role') == 'admin'))
442     if getattr(plugin, 'is_system', True) and not is_admin:
443         raise HTTPException(status_code=403, detail=f"Системны
й плагин '{plugin_name}' устанавливается и переключается тольк
о администратором")
444
445     plugin.enabled = data.enabled
446     plugin.update_config({'enabled': data.enabled})
447     logger.info(f"Plugin {plugin_name} enabled state changed t
o {data.enabled}")
448     return {'name': plugin_name, 'enabled': plugin.enabled, 'm
essage': f"Плагин {plugin_name} {'включен' if data.enabled els
e 'выключен'}"}

```

```

449
450 @router.post('/plugins/{plugin_name}/config')
451 async def save_plugin_config(plugin_name: str, data: PluginCon
figUpdate, request: Request) -> Dict[str, Any]:
452     """Update plugin configuration."""
453     plugins_dict = _get_app_plugins(request)
454     plugin = plugins_dict.get(plugin_name)
455     if not plugin:
456         raise HTTPException(status_code=404, detail=f"Плагин '
{plugin_name}' не найден")
457
458     user = _get_request_user(request)
459     is_admin = bool(user and (user.get('is_admin') or user.get
('role') == 'admin'))
460     if getattr(plugin, 'is_system', True) and not is_admin:
461         raise HTTPException(status_code=403, detail=f"Параметр
ы системного плагина '{plugin_name}' могут изменяться только а
дминистратором")
462
463     plugin.update_config(data.config)
464     return {'name': plugin_name, 'config': plugin.config, 'mes
sage': 'Конфигурация сохранена'}
465
466 @router.post('/plugins/{plugin_name}/action/{action_name}')
467 async def call_plugin_action(plugin_name: str, action_name: st
r, data: PluginActionRequest, request: Request) -> Dict[str, A
ny]:
468     """Execute an action on a specific plugin."""
469     plugins_dict = _get_app_plugins(request)
470     plugin = plugins_dict.get(plugin_name)
471     if not plugin:
472         raise HTTPException(status_code=404, detail=f"Плагин '
{plugin_name}' не найден")
473
474     params = dict(data.params or {})
475     user = _get_request_user(request)
476     if user and not user.get('is_admin') and user.get('role')
!= 'admin':

```



```

477         # Inject user_id for non-admin users so actions are is
         olated
478         params['user_id'] = user.get('id')
479
480     result = await plugin.execute_action(action_name, params)
481     return result
482
483 @router.get('/plugin/{plugin_name}/status')
484 async def get_plugin_status(plugin_name: str, request: Request
    ):
485     """Retrieve status of plugin (backward compatibility)."""
486     plugins_dict = _get_app_plugins(request)
487     plugin = plugins_dict.get(plugin_name)
488     if not plugin:
489         raise HTTPException(status_code=404, detail='Плагин не
         найден')
490     return await plugin.health_check()
491
492 @router.post('/plugin/{plugin_name}/status')
493 async def update_plugin_status(plugin_name: str, data: Plugins
    tateUpdate, request: Request):
494     """Update status of plugin (backward compatibility)."""
495     require_admin_user(request)
496     plugins_dict = _get_app_plugins(request)
497     plugin = plugins_dict.get(plugin_name)
498     if not plugin:
499         raise HTTPException(status_code=404, detail='Плагин не
         найден')
500     plugin.enabled = data.enabled
501     plugin.update_config({'enabled': data.enabled})
502     logger.info(f'Plugin {plugin_name} enabled state changed t
         o {data.enabled}')
503     return {'name': plugin_name, 'enabled': plugin.enabled}
504
505
506 # =====
507 # /apps Configuration & Status Endpoints

```

```

508 # =====
    =====
509
510 router_apps = APIRouter(prefix='/api/v1/apps', tags=['apps'])
511
512 APPS_REGISTRY: List[Dict[str, Any]] = [
513     {
514         "id": "scenarios",
515         "key": "scenarios",
516         "tab": "tab-scenarios",
517         "folder": "scenarios",
518         "aliases": ["scenarios", "scenario", "tab-scenarios",
519 "сценарии"],
519         "name": "Test & Automation Scenarios",
520         "ru_name": "Сценарии",
521         "icon": "📄 ",
522     },
523     {
524         "id": "chat",
525         "key": "chat",
526         "tab": "tab-chat",
527         "folder": "chat",
528         "aliases": ["chat", "tab-chat", "dialog", "ai_chat"],
529         "name": "AI Chat Assistant",
530         "ru_name": "Чат ИИ",
531         "icon": "💬 ",
532     },
533     {
534         "id": "trading_terminal",
535         "key": "trading_terminal",
536         "tab": "tab-trading",
537         "folder": "trading_terminal",
538         "aliases": ["trading_terminal", "trading", "tab-tradin
539 g"],
539         "name": "Exchange Trading Terminal",
540         "ru_name": "Торговый терминал",
541         "icon": "📊 ",
542     },

```

```
543 |     {
544 |         "id": "network_terminal",
545 |         "key": "network_terminal",
546 |         "tab": "tab-network",
547 |         "folder": "network_terminal",
548 |         "aliases": ["network_terminal", "network", "tab-networ
549 | k"],
550 |         "name": "Network Analyzer Terminal",
551 |         "ru_name": "Сетевой терминал",
552 |         "icon": "□ ",
553 |     },
554 |     {
555 |         "id": "system_inspector",
556 |         "key": "system_inspector",
557 |         "tab": "tab-hardware-load-inspector",
558 |         "folder": "system_inspector",
559 |         "aliases": ["system_inspector", "inspector", "tab-hard
560 | ware-load-inspector", "tab-system-load-inspector", "tab-system
561 | -inspector", "system_load_inspector", "hardware_load_inspector
562 | "],
563 |         "name": "System Inspector",
564 |         "ru_name": "Потребление ресурсов",
565 |         "icon": "□ ",
566 |     },
567 |     {
568 |         "id": "about_system",
569 |         "key": "about_system",
570 |         "tab": "tab-about-system",
571 |         "folder": "about_system_tab",
572 |         "aliases": ["about_system", "system_info", "sysinfo",
573 | "tab-about-system", "about"],
574 |         "name": "About System",
575 |         "ru_name": "О Системе",
576 |         "icon": "i",
577 |     },
578 |     {
579 |         "id": "windows_sysadmin",
580 |         "key": "windows_sysadmin",
```

```

576         "tab": "tab-windows-admin",
577         "folder": "windows_sysadmin",
578         "aliases": ["windows_sysadmin", "windows_admin", "wind
owsadmin", "sysadmin", "tab-windows-admin"],
579         "name": "Windows System Administrator",
580         "ru_name": "Windows Sysadmin",
581         "icon": " ",
582     },
583     {
584         "id": "user_assistant",
585         "key": "user_assistant",
586         "tab": "tab-user-assistant",
587         "folder": "user_assistant",
588         "aliases": ["user_assistant", "assistant", "userassist
ant", "tab-user-assistant"],
589         "name": "User Assistant",
590         "ru_name": "User Assistant",
591         "icon": " ",
592     },
593     {
594         "id": "gcloud_monitor",
595         "key": "gcloud_monitor",
596         "tab": "tab-gcloud",
597         "folder": "gcloud_monitor",
598         "aliases": ["gcloud_monitor", "gcloud", "google_cloud"
, "tab-gcloud"],
599         "name": "Google Cloud Monitor",
600         "ru_name": "Google Cloud Monitor",
601         "icon": " ",
602     },
603     {
604         "id": "website_monitor",
605         "key": "website_monitor",
606         "tab": "tab-website-monitor",
607         "folder": "website_monitor",
608         "aliases": ["website_monitor", "website", "website_int
elligence", "tab-website-monitor"],
609         "name": "Website Intelligence Monitor",

```

```

610         "ru_name": "Website Intelligence",
611         "icon": "☐ ",
612     },
613     {
614         "id": "cloudflared_monitor",
615         "key": "cloudflared_monitor",
616         "tab": "tab-cloudflared",
617         "folder": "cloudflared_monitor",
618         "aliases": ["cloudflared_monitor", "cloudflared", "tab
        -cloudflared"],
619         "name": "Cloudflared Monitor",
620         "ru_name": "Cloudflared Monitor",
621         "icon": "☐ ",
622     },
623     {
624         "id": "system_control_center",
625         "key": "system_control_center",
626         "tab": "tab-system-control",
627         "folder": "system_control_center",
628         "aliases": ["system_control_center", "system_control",
        "control_center", "tab-system-control"],
629         "name": "System Control Center",
630         "ru_name": "System Control Center",
631         "icon": "☐ ",
632     },
633     {
634         "id": "system_log_viewer",
635         "key": "system_log_viewer",
636         "tab": "tab-system-logs",
637         "folder": "system_log_viewer",
638         "aliases": ["system_log_viewer", "system_logs", "syste
        m_log", "log_viewer", "logs_viewer", "tab-system-logs", "log_a
        nalyzer", "application_log_analyzer"],
639         "name": "System Log Viewer",
640         "ru_name": "Анализатор логов (All Logs)",
641         "icon": "☐ ",
642     },
643     {

```

```
644         "id": "wikipedia_research",
645         "key": "wikipedia_research",
646         "tab": "tab-wikipedia-research",
647         "folder": "wikipedia_research",
648         "aliases": ["wikipedia_research", "wikipedia", "wiki_l
ab", "tab-wikipedia-research"],
649         "name": "Wikipedia Research Lab",
650         "ru_name": "Wikipedia Research (Сравнение языков и мод
елей)",
651         "icon": " ",
652     },
653     {
654         "id": "ai_breadboard_admin",
655         "key": "ai_breadboard_admin",
656         "tab": "tab-admin",
657         "folder": "ai_breadboard_admin",
658         "aliases": ["ai_breadboard_admin", "admin", "admin_pan
el", "tab-admin"],
659         "name": "AI Breadboard Admin",
660         "ru_name": "Панель администратора",
661         "icon": " ",
662     },
663     {
664         "id": "research_and_statistic",
665         "key": "research_and_statistic",
666         "tab": "tab-research",
667         "folder": "research_and_statistic",
668         "aliases": ["research_and_statistic", "research_stat",
"research", "tab-research"],
669         "name": "Research & Statistics Desk",
670         "ru_name": "Анализ данных и статистика",
671         "icon": " ",
672     },
673     {
674         "id": "helpdesk",
675         "key": "helpdesk",
676         "tab": "tab-helpdesk",
677         "folder": "helpdesk",
```

```
678         "aliases": ["helpdesk", "support", "tab-helpdesk"],
679         "name": "Helpdesk & Support",
680         "ru_name": "Служба поддержки (Helpdesk)",
681         "icon": " ",
682     },
683     {
684         "id": "software_audit",
685         "key": "software_audit",
686         "tab": "tab-software-audit",
687         "folder": "software_audit",
688         "aliases": ["software_audit", "software", "audit_softw
are", "installed_software", "tab-software-audit"],
689         "name": "Software Audit",
690         "ru_name": "Аудит программного обеспечения",
691         "icon": " ",
692     },
693     {
694         "id": "registry_viewer",
695         "key": "registry_viewer",
696         "tab": "tab-registry-viewer",
697         "folder": "registry_viewer",
698         "aliases": ["registry_viewer", "registry", "regedit",
"reg_viewer", "tab-registry-viewer"],
699         "name": "Registry Viewer",
700         "ru_name": "Просмотр реестра (Registry Viewer)",
701         "icon": " ",
702     },
703     {
704         "id": "windows_startup_auditor",
705         "key": "windows_startup_auditor",
706         "tab": "tab-startup-auditor",
707         "folder": "windows/startup",
708         "aliases": ["windows_startup_auditor", "startup_audito
r", "startup", "autoruns", "tab-startup-auditor"],
709         "name": "Windows Startup Auditor",
710         "ru_name": "Аудит автозапуска Windows",
711         "icon": " ",
712     },
```

```
713 |     {
714 |         "id": "windows_defender",
715 |         "key": "windows_defender",
716 |         "tab": "tab-defender",
717 |         "folder": "windows_defender",
718 |         "aliases": ["windows_defender", "defender", "tab-defen
719 |         der"],
720 |         "name": "Windows Defender Security",
721 |         "ru_name": "Безопасность Windows Defender",
722 |         "icon": "🛡️ ",
723 |     },
724 |     {
725 |         "id": "windows_backup_manager",
726 |         "key": "windows_backup_manager",
727 |         "tab": "tab-windows-backup",
728 |         "folder": "windows_backup_manager",
729 |         "aliases": ["windows_backup_manager", "backup_manager"
730 |         , "windows_backup", "backup", "tab-windows-backup"],
731 |         "name": "Windows Backup & Libraries",
732 |         "ru_name": "Резервное копирование и библиотеки Windows
733 |         ",
734 |         "icon": "💾 ",
735 |     },
736 |     {
737 |         "id": "software_transparency_scanner",
738 |         "key": "software_transparency_scanner",
739 |         "tab": "tab-software-transparency",
740 |         "folder": "software_transparency_scanner",
741 |         "aliases": ["software_transparency_scanner", "transpar
742 |         ency_scanner", "software_transparency", "tab-software-transpar
743 |         ency"],
744 |         "name": "Software Transparency Scanner",
745 |         "ru_name": "Инвентаризатор и прозрачность ПО (AI Trans
746 |         parency)",
747 |         "icon": "🔍 ",
748 |     },
749 |     {
750 |         "id": "hardware_monitor",
```



```

745         "key": "hardware_monitor",
746         "tab": "tab-hardware-monitor",
747         "folder": "windows",
748         "aliases": ["hardware_monitor", "hardware", "sensors",
749                     "hw_monitor", "tab-hardware-monitor"],
750         "name": "Hardware & Sensors Monitor",
751         "ru_name": "Монитор оборудования и сенсоров",
752         "icon": "❏ ",
753     },
754 ]
755
756 def get_apps_status(profile: Optional[str] = None) -> Dict[str
757 , Any]:
758     """Retrieve enabled/disabled status for all /apps applicat
759     ions based on active configuration file."""
760     import os
761     cfg_env = os.getenv("AIBREADBOARD_CONFIG") or os.getenv("C
762     ONFIG_FILE")
763     active_path: Optional[Path] = None
764     if profile in ("tc", "test-computer", "test_computer", "ap
765     ps_tc"):
766         for candidate in [__root__ / "start_scenarios_config"
767         / "tc.json", __root__ / "config" / "tc.json", __root__ / "conf
768         ig_tc.json", __root__ / "tc.json"]:
769             if candidate.exists():
770                 active_path = candidate
771                 break
772     elif cfg_env:
773         p = Path(cfg_env)
774         active_path = p if p.is_absolute() else (__root__ / cf
775         g_env)
776
777     if not active_path or not active_path.exists():
778         for candidate in [__root__ / "start_scenarios_config"
779         / "tc.json", __root__ / "config" / "tc.json", __root__ / "conf
780         ig_tc.json", __root__ / "tc.json"]:
781             if candidate.exists() and not (__root__ / "config.

```

```

    json").exists() and not (__root__ / "config" / "dashboard.json
    ").exists()):
773         active_path = candidate
774         break
775     if not active_path or not active_path.exists():
776         active_path = (__root__ / "start_scenarios_config"
    / "dashboard.json") if (__root__ / "start_scenarios_config" /
    "dashboard.json").exists() else ((__root__ / "config" / "dash
    board.json") if (__root__ / "config" / "dashboard.json").exist
    s() else (__root__ / "config.json"))
777
778     apps_cfg: Any = {}
779     config_filename = active_path.name if active_path else "co
    nfig.json"
780     if active_path and active_path.exists():
781         try:
782             with open(active_path, "r", encoding="utf-8") as f
    :
783                 root_cfg = json.load(f)
784                 apps_cfg = root_cfg.get("apps", {})
785         except Exception as e:
786             logger.error(f"Error reading apps config from {act
    ive_path}: {e}")
787
788     result_apps: Dict[str, Any] = {}
789
790     # Анализ формата конфигурации apps (списки enabled/disable
    d, плоский список или legacy dict)
791     enabled_set: Optional[set] = None
792     disabled_set: set = set()
793     enable_all = True
794
795     if isinstance(apps_cfg, list):
796         # Формат плоского списка активных приложений (например
    , config_tc.json)
797         enabled_set = {str(item).strip().lower() for item in a
    pps_cfg if item}
798         enable_all = False

```

```

799     elif isinstance(apps_cfg, dict):
800         has_enabled_key = "enabled" in apps_cfg and isinstance
            (apps_cfg["enabled"], list)
801         has_disabled_key = "disabled" in apps_cfg and isinstan
            ce(apps_cfg["disabled"], list)
802
803         if has_enabled_key:
804             enabled_set = {str(item).strip().lower() for item
            in apps_cfg["enabled"] if item}
805             enable_all = False
806         if has_disabled_key:
807             disabled_set = {str(item).strip().lower() for item
            in apps_cfg["disabled"] if item}
808
809         if "enable_all" in apps_cfg and isinstance(apps_cfg["e
            nable_all"], bool):
810             enable_all = apps_cfg["enable_all"]
811         elif not has_enabled_key:
812             enable_all = True
813     else:
814         enable_all = True
815
816     for app in APPS_REGISTRY:
817         app_id = app["id"]
818         key = app["key"]
819         aliases = {a.lower() for a in app.get("aliases", [app_
            id, key, app["folder"], app["tab"]])}
820
821         # 1. Проверка явного исключения в disabled
822         if aliases.intersection(disabled_set):
823             is_enabled = False
824         # 2. Проверка включения в список enabled
825         elif enabled_set is not None:
826             is_enabled = bool(aliases.intersection(enabled_set
            ))
827
828         # 3. Проверка индивидуального булева флага в словаре
            elif isinstance(apps_cfg, dict) and key in apps_cfg an
            d isinstance(apps_cfg[key], bool):

```

```

829         is_enabled = apps_cfg[key]
830         elif isinstance(apps_cfg, dict) and app_id in apps_cfg
and isinstance(apps_cfg[app_id], bool):
831             is_enabled = apps_cfg[app_id]
832         else:
833             is_enabled = bool(enable_all)
834
835         # Проверка локального config.json микроприложения на я
вное отключение
836         local_cfg_paths = [
837             __root__ / "apps" / app["folder"] / "config.json",
838             __root__ / "src" / "apps" / app["folder"] / "confi
g.json",
839         ]
840         for lcp in local_cfg_paths:
841             if lcp.exists():
842                 try:
843                     with open(lcp, "r", encoding="utf-8") as l
f:
844                         local_data = json.load(lf)
845                         if local_data.get("enabled") is False
or local_data.get("active") is False:
846                             is_enabled = False
847                     except Exception:
848                         pass
849
850         result_apps[app_id] = {
851             "id": app_id,
852             "key": key,
853             "tab": app["tab"],
854             "folder": app["folder"],
855             "name": app["name"],
856             "ru_name": app["ru_name"],
857             "icon": app["icon"],
858             "enabled": is_enabled,
859         }
860
861     ai_cfg = {}

```

```

862     if active_path and active_path.exists():
863         try:
864             with open(active_path, "r", encoding="utf-8") as f
865             :
866                 root_data = json.load(f)
867                 ai_cfg = root_data.get("ai", {})
868         except Exception:
869             pass
870
871     return {
872         "status": "ok",
873         "config_file": config_filename,
874         "enable_all": enable_all,
875         "apps": result_apps,
876         "ai": ai_cfg,
877     }
878
879 class AppConfigUpdateRequest(BaseModel):
880     """Payload for updating /apps application configuration."""
881     "
882     config: Dict[str, Any]
883
884 @router_apps.get('/status')
885 async def get_public_apps_status_endpoint(profile: Optional[str] = None) -> Dict[str, Any]:
886     """Get enabled/disabled status for all /apps applications.
887     """
888     return get_apps_status(profile=profile)
889
890 @router_apps.get('/{app_name}/config')
891 async def get_app_config(app_name: str, request: Request) -> Dict[str, Any]:
892     """Get configuration for specified application under /apps
893     ."""
894     require_admin_user(request)

```

```

894
895     safe_name = "".join(c for c in app_name if c.isalnum() or
| c in ("_", "-"))
896     if not safe_name:
897         raise HTTPException(status_code=400, detail="Invalid a
| pplication name")
898
899     candidates = [
900         __root__ / "src" / "apps" / safe_name / "config.json",
901         __root__ / "apps" / safe_name / "config.json",
902     ]
903
904     for path in candidates:
905         if path.exists():
906             try:
907                 with open(path, "r", encoding="utf-8") as f:
908                     return {"status": "ok", "app": safe_name,
| "config": json.load(f), "path": str(path)}
909             except Exception as e:
910                 logger.error(f"Error reading app config {path}
| : {e}")
911                 raise HTTPException(status_code=500, detail=f"
| Failed to read configuration: {e}")
912
913     raise HTTPException(status_code=404, detail=f"Application
| config for '{safe_name}' not found")
914
915
916 @router_apps.post("/{app_name}/config")
917 async def set_app_config(app_name: str, data: AppConfigUpdateR
| equest, request: Request) -> Dict[str, Any]:
918     """Update configuration for specified application under /a
| pps."""
919     require_admin_user(request)
920
921     safe_name = "".join(c for c in app_name if c.isalnum() or
| c in ("_", "-"))
922     if not safe_name:

```

```

923         raise HTTPException(status_code=400, detail="Invalid application name")
924
925     candidates = [
926         __root__ / "src" / "apps" / safe_name / "config.json",
927         __root__ / "apps" / safe_name / "config.json",
928     ]
929
930     target_paths = [p for p in candidates if p.exists()]
931     if not target_paths:
932         target_paths = [__root__ / "apps" / safe_name / "config.json"]
933
934     for path in target_paths:
935         try:
936             path.parent.mkdir(parents=True, exist_ok=True)
937             with open(path, "w", encoding="utf-8") as f:
938                 json.dump(data.config, f, indent=2, ensure_ascii=False)
939         except Exception as e:
940             logger.error(f"Error saving app config {path}: {e}")
941             raise HTTPException(status_code=500, detail=f"Failed to save configuration: {e}")
942
943     return {"status": "ok", "app": safe_name, "config": data.config, "message": "Конфигурация успешно сохранена"}
944
945
946 # =====
947 # RAG Endpoints
948 # =====
949
950 class RagConfigRequest(BaseModel):
951     mode: str
952

```

```

953 @router.get('/rag/config')
954 async def get_rag_config(request: Request):
955     """Получение режима RAG."""
956     require_admin_user(request)
957     config_path = __root__ / 'config.json'
958     mode = "rag+model"
959     if config_path.exists():
960         try:
961             with open(config_path, 'r', encoding='utf-8') as f
962             :
963                 cfg = json.load(f)
964                 mode = cfg.get("rag", {}).get("mode", "rag+mod
965                 el")
966         except Exception as e:
967             logger.error("Error чтения config.json", e)
968     return {"mode": mode}
969
970 @router.post('/rag/config')
971 async def set_rag_config(request: Request, data: RagConfigRequ
972     est):
973     """Установка режима RAG."""
974     require_admin_user(request)
975     config_path = __root__ / 'config.json'
976     try:
977         if config_path.exists():
978             with open(config_path, 'r', encoding='utf-8') as f
979             :
980                 cfg = json.load(f)
981             else:
982                 cfg = {}
983
984             if "rag" not in cfg:
985                 cfg["rag"] = {}
986
987             cfg["rag"]["mode"] = data.mode
988
989             with open(config_path, 'w', encoding='utf-8') as f:
990                 json.dump(cfg, f, indent=2, ensure_ascii=False)

```



```

987
988         return {"status": "ok", "mode": data.mode}
989     except Exception as e:
990         logger.error("Error записи config.json", e)
991         raise HTTPException(status_code=500, detail="Error cox
ранения конфигурации RAG")
992
993 class WebSearchConfigRequest(BaseModel):
994     engine: str
995     gemini_model: str = "gemini-3.1-flash"
996     gemini_cli_model: str = "gemini-3.1-flash-lite"
997     agy_model: str = "agy-flash"
998
999 @router.get('/web-search/config')
1000 async def get_web_search_config(request: Request):
1001     """Получение конфигурации сервера веб-поиска."""
1002     require_admin_user(request)
1003     config_path = __root__ / 'config.json'
1004     engine = "playwright"
1005     gemini_model = "gemini-3.1-flash"
1006     gemini_cli_model = "gemini-3.1-flash-lite"
1007     agy_model = "agy-flash"
1008     if config_path.exists():
1009         try:
1010             with open(config_path, 'r', encoding='utf-8') as f
1011             :
1012                 cfg = json.load(f)
1013                 ws = cfg.get("web_search", {})
1014                 engine = ws.get("engine", "playwright")
1015                 gemini_model = ws.get("gemini_model", "gemini-
3.1-flash")
1016                 gemini_cli_model = ws.get("gemini_cli_model",
"gemini-3.1-flash-lite")
1017                 agy_model = ws.get("agy_model", "agy-flash")
1018         except Exception as e:
1019             logger.error("Error чтения config.json для web_sea
rch", e)
1020     return {

```

```

1020         "engine": engine,
1021         "gemini_model": gemini_model,
1022         "gemini_cli_model": gemini_cli_model,
1023         "agy_model": agy_model
1024     }
1025
1026 @router.post('/web-search/config')
1027 async def set_web_search_config(request: Request, data: WebSearchConfigRequest):
1028     """Установка сервера веб-поиска (playwright / langchain /
1029     gemini / gemini_cli / agy)."""
1029     require_admin_user(request)
1030     config_path = __root__ / 'config.json'
1031     try:
1032         if config_path.exists():
1033             with open(config_path, 'r', encoding='utf-8') as f
1034             :
1035                 cfg = json.load(f)
1036             else:
1037                 cfg = {}
1038
1039             if "web_search" not in cfg:
1040                 cfg["web_search"] = {}
1041
1042             cfg["web_search"]["engine"] = data.engine
1043             cfg["web_search"]["gemini_model"] = data.gemini_model
1044             cfg["web_search"]["gemini_cli_model"] = data.gemini_cli_model
1045             cfg["web_search"]["agy_model"] = data.agy_model
1046
1047             with open(config_path, 'w', encoding='utf-8') as f:
1048                 json.dump(cfg, f, indent=2, ensure_ascii=False)
1049
1050             return {
1051                 "status": "ok",
1052                 "engine": data.engine,
1053                 "gemini_model": data.gemini_model,
1054                 "gemini_cli_model": data.gemini_cli_model,

```

```

1054         "agy_model": data.agy_model
1055     }
1056     except Exception as e:
1057         logger.error("Error записи config.json для web_search"
1058             , e)
1059         raise HTTPException(status_code=500, detail="Error сох
1060             ранения конфигурации веб-поиска")
1061
1062 class WebSearchTestRequest(BaseModel):
1063     query: str
1064     engine: str = ""
1065
1066 @router.post('/web-search/test')
1067 async def test_web_search(request: Request, data: WebSearchTes
1068     tRequest):
1069     """Тестовое выполнение поиска через выбранный поисковый дв
1070     ижок."""
1071     require_admin_user(request)
1072     query = data.query.strip()
1073     if not query:
1074         raise HTTPException(status_code=400, detail="Поисковый
1075             запрос не может быть пустым")
1076
1077     engine = data.engine
1078     gemini_model = "gemini-3.1-flash"
1079     gemini_cli_model = "gemini-3.1-flash-lite"
1080     agy_model = "agy-flash"
1081     config_path = __root__ / 'config.json'
1082     if config_path.exists():
1083         try:
1084             with open(config_path, 'r', encoding='utf-8') as f
1085                 :
1086                 cfg = json.load(f)
1087                 ws = cfg.get("web_search", {})
1088                 if not engine:
1089                     engine = ws.get("engine", "playwright")
1090                 gemini_model = ws.get("gemini_model", "gemini-
1091                     3.1-flash")

```

```

1085         gemini_cli_model = ws.get("gemini_cli_model",
    "gemini-3.1-flash-lite")
1086         agy_model = ws.get("agy_model", "agy-flash")
1087     except Exception as e:
1088         logger.error("Error чтения config.json для web_sea
    rch", e)
1089
1090     if not engine:
1091         engine = "playwright"
1092
1093     try:
1094         return {"status": "ok", "engine": engine, "result": f"
    Поиск '{query}' выполнен."}
1095     except Exception as e:
1096         logger.error(f"Error тестового поиска {engine}", e)
1097         return {"status": "error", "engine": engine, "message"
    : str(e)}
1098
1099 # =====
    =====
1100 # User Management Models and Endpoints
1101 # =====
    =====
1102
1103 class AdminUserCreateRequest(BaseModel):
1104     email: str
1105     name: str
1106     password: str = ''
1107     role: str = 'user'
1108     is_admin: int = 0
1109     is_active: int = 1
1110     is_email_verified: int = 1
1111
1112 class AdminUserUpdateRequest(BaseModel):
1113     name: Optional[str] = None
1114     email: Optional[str] = None
1115     role: Optional[str] = None
1116     is_admin: Optional[int] = None

```

```

1117     is_active: Optional[int] = None
1118     is_email_verified: Optional[int] = None
1119     telegram_id: Optional[int] = None
1120     telegram_username: Optional[str] = None
1121
1122 class AdminUserPasswordRequest(BaseModel):
1123     password: str
1124
1125 @router.get('/users')
1126 async def list_admin_users(
1127     request: Request,
1128     q: str = '',
1129     role: str = '',
1130     status: str = ''
1131 ) -> Dict[str, Any]:
1132     """Получение списка пользователей с фильтрацией, поиском и
1133     статистикой."""
1134     require_admin_user(request)
1135     from src.user_manager import user_manager
1136     all_users = user_manager.get_all_users(active_only=False)
1137
1138     total_count = len(all_users)
1139     active_count = sum(1 for u in all_users if u.get('is_active', 0) == 1)
1140     admin_count = sum(1 for u in all_users if u.get('is_admin', 0) == 1 or u.get('role') == 'admin')
1141     tg_count = sum(1 for u in all_users if u.get('telegram_id'))
1142
1143     filtered = []
1144     q_lower = q.lower().strip()
1145
1146     for u in all_users:
1147         if q_lower:
1148             name_match = q_lower in str(u.get('name', '')).lower()
1149             email_match = q_lower in str(u.get('email', '')).lower()

```

```

1149         tg_match = q_lower in str(u.get('telegram_username
| ', '')).lower() or q_lower in str(u.get('telegram_id', ''))
1150         if not (name_match or email_match or tg_match):
1151             continue
1152
1153         if role and u.get('role') != role:
1154             continue
1155
1156         if status == 'active' and u.get('is_active', 0) != 1:
1157             continue
1158         if status == 'inactive' and u.get('is_active', 0) == 1
| :
1159             continue
1160
1161         sanitized = {k: v for k, v in u.items() if k != 'passw
| ord_hash'}
1162         sanitized['has_password'] = bool(u.get('password_hash'
| ))
1163         filtered.append(sanitized)
1164
1165     return {
1166         'status': 'ok',
1167         'users': filtered,
1168         'stats': {
1169             'total': total_count,
1170             'active': active_count,
1171             'suspended': total_count - active_count,
1172             'admins': admin_count,
1173             'telegram': tg_count,
1174         }
1175     }
1176
1177 @router.post('/users')
1178 async def create_admin_user(request: Request, data: AdminUserC
| reateRequest) -> Dict[str, Any]:
1179     """Создание нового пользователя администратором."""
1180     require_admin_user(request)
1181     email = data.email.strip().lower()

```

```
1182     name = data.name.strip()
1183     if not email:
1184         raise HTTPException(status_code=400, detail='Email обяза
зателен')
1185     if not name:
1186         raise HTTPException(status_code=400, detail='Имя обяза
тельно')
1187
1188     from src.user_manager import user_manager
1189     if user_manager.user_exists(email):
1190         raise HTTPException(status_code=400, detail=f'Пользова
тель с email {email} уже существует')
1191
1192     user_id = user_manager.create_user_admin(
1193         email=email,
1194         name=name,
1195         password=data.password,
1196         role=data.role,
1197         is_admin=data.is_admin,
1198         is_active=data.is_active,
1199         is_email_verified=data.is_email_verified
1200     )
1201     if not user_id:
1202         raise HTTPException(status_code=500, detail='Error соз
дания пользователя')
1203
1204     created = user_manager.get_user_by_id(user_id)
1205     sanitized = {k: v for k, v in created.items() if k != 'pas
sword_hash'}
1206     sanitized['has_password'] = bool(created.get('password_has
h'))
1207     return {'status': 'ok', 'user': sanitized}
1208
1209 class OrphanedDirsCleanRequest(BaseModel):
1210     dirs: Optional[List[str]] = None
1211
1212 @router.get('/users/orphaned-dirs')
1213 async def get_orphaned_user_dirs(request: Request) -> Dict[str
```

```

    , Any]:
1214     """Inspect and list orphaned user directories."""
1215     require_admin_user(request)
1216     from src.user_manager import user_manager
1217     orphaned = user_manager.get_orphaned_user_directories()
1218     total_size = sum(item["size_bytes"] for item in orphaned)
1219     total_files = sum(item["files_count"] for item in orphaned
    )
1220     return {
1221         "status": "ok",
1222         "total": len(orphaned),
1223         "total_size_bytes": total_size,
1224         "total_files": total_files,
1225         "orphaned_dirs": orphaned
1226     }
1227
1228 @router.post('/users/orphaned-dirs/clean')
1229 async def clean_orphaned_user_dirs(
1230     request: Request,
1231     payload: Optional[OrphanedDirsCleanRequest] = None
1232 ) -> Dict[str, Any]:
1233     """Clean up specified or all orphaned user directories."""
1234     require_admin_user(request)
1235     from src.user_manager import user_manager
1236     dir_names = payload.dirs if payload and payload.dirs is no
t None else None
1237     result = user_manager.cleanup_orphaned_user_directories(di
r_names=dir_names)
1238     return result
1239
1240 @router.get('/users/{user_id}')
1241 async def get_admin_user_details(user_id: int, request: Reques
t) -> Dict[str, Any]:
1242     """Получение детальной информации о пользователе и его нас
тройках."""
1243     require_admin_user(request)
1244     from src.user_manager import user_manager
1245     user = user_manager.get_user_by_id(user_id)

```



```
1246     if not user:
1247         raise HTTPException(status_code=404, detail='Пользоват
ель не найден')
1248
1249     settings = user_manager.get_user_settings(user_id)
1250     permissions = user_manager.get_user_permissions(user_id)
1251     storage_stats = user_manager.get_user_storage_stats(user_i
d)
1252     sanitized = {k: v for k, v in user.items() if k != 'passwo
rd_hash'}
1253     sanitized['has_password'] = bool(user.get('password_hash')
)
1254
1255     return {
1256         'status': 'ok',
1257         'user': sanitized,
1258         'settings': settings,
1259         'permissions': permissions,
1260         'storage': storage_stats
1261     }
1262
1263 @router.put('/users/{user_id}')
1264 @router.patch('/users/{user_id}')
1265 async def update_admin_user(user_id: int, data: AdminUserUpdat
eRequest, request: Request) -> Dict[str, Any]:
1266     """Update данных пользователя (полное или частичное редакт
ирование полей)."""
1267     require_admin_user(request)
1268     from src.user_manager import user_manager
1269     user = user_manager.get_user_by_id(user_id)
1270     if not user:
1271         raise HTTPException(status_code=404, detail='Пользоват
ель не найден')
1272
1273     updates: Dict[str, Any] = {}
1274     if data.name is not None:
1275         updates['name'] = data.name.strip()
1276     if data.email is not None:
```

```
1277 |         email_clean = data.email.strip().lower()
1278 |         if not email_clean:
1279 |             raise HTTPException(status_code=400, detail='Email
| не может быть пустым')
1280 |         if email_clean != user.get('email'):
1281 |             existing = user_manager.get_user_by_email(email_cl
| ean)
1282 |             if existing and existing.get('id') != user_id:
1283 |                 raise HTTPException(status_code=400, detail='Э
| тот email уже занят другим пользователем')
1284 |                 updates['email'] = email_clean
1285 |         if data.role is not None:
1286 |             updates['role'] = data.role
1287 |             if data.is_admin is None:
1288 |                 updates['is_admin'] = 1 if data.role == 'admin' el
| se 0
1289 |         if data.is_admin is not None:
1290 |             if user_id == 1 and data.is_admin == 0:
1291 |                 raise HTTPException(status_code=400, detail='Нельз
| я снять права у главного администратора (ID 1)')
1292 |                 updates['is_admin'] = data.is_admin
1293 |             if data.role is None:
1294 |                 updates['role'] = 'admin' if data.is_admin == 1 el
| se 'user'
1295 |         if data.is_active is not None:
1296 |             if user_id == 1 and data.is_active == 0:
1297 |                 raise HTTPException(status_code=400, detail='Нельз
| я деактивировать главного администратора (ID 1)')
1298 |                 updates['is_active'] = data.is_active
1299 |         if data.is_email_verified is not None:
1300 |             updates['is_email_verified'] = data.is_email_verified
1301 |         if data.telegram_id is not None:
1302 |             updates['telegram_id'] = data.telegram_id
1303 |         if data.telegram_username is not None:
1304 |             updates['telegram_username'] = data.telegram_username.
| strip().lstrip('@')
1305 |
1306 |         if updates:
```

```

1307         success = user_manager.update_user(user_id, **updates)
1308         if not success:
1309             raise HTTPException(status_code=500, detail='Error
обновления пользователя')
1310
1311         updated = user_manager.get_user_by_id(user_id)
1312         sanitized = {k: v for k, v in updated.items() if k != 'pas
sword_hash'}
1313         sanitized['has_password'] = bool(updated.get('password_has
h'))
1314         return {'status': 'ok', 'user': sanitized}
1315
1316 @router.post('/users/{user_id}/password')
1317 async def set_admin_user_password(user_id: int, data: AdminUse
rPasswordRequest, request: Request) -> Dict[str, Any]:
1318     """Установка / сброс пароля пользователя администратором."
"""
1319     require_admin_user(request)
1320     new_password = data.password.strip()
1321     if not new_password:
1322         raise HTTPException(status_code=400, detail='Пароль не
может быть пустым')
1323
1324     from src.user_manager import user_manager
1325     user = user_manager.get_user_by_id(user_id)
1326     if not user:
1327         raise HTTPException(status_code=404, detail='Пользоват
ель не найден')
1328
1329     success = user_manager.set_user_password(user_id, new_pass
word)
1330     if not success:
1331         raise HTTPException(status_code=500, detail='Error уст
ановки пароля')
1332
1333     return {'status': 'ok', 'message': 'Пароль successfully об
новлён'}
1334

```

```
1335 @router.post('/users/{user_id}/toggle-active')
1336 async def toggle_admin_user_active(user_id: int, request: Requ
1337 est) -> Dict[str, Any]:
1338     """Переключение активности пользователя (блокировка / разб
1339 локировка)."""
1340     require_admin_user(request)
1341     from src.user_manager import user_manager
1342     user = user_manager.get_user_by_id(user_id)
1343     if not user:
1344         raise HTTPException(status_code=404, detail='Пользоват
1345 ель не найден')
1346     if user_id == 1 and user.get('is_active', 1) == 1:
1347         raise HTTPException(status_code=400, detail='Нельзя де
1348 активировать главного администратора (ID 1)')
1349     new_status = 0 if user.get('is_active', 1) == 1 else 1
1350     success = user_manager.update_user(user_id, is_active=new_
1351 status)
1352     if not success:
1353         raise HTTPException(status_code=500, detail='Error изм
1354 енения статуса')
1355     return {'status': 'ok', 'is_active': new_status}
1356
1357 @router.post('/users/{user_id}/toggle-role')
1358 async def toggle_admin_user_role(user_id: int, request: Reques
1359 t) -> Dict[str, Any]:
1360     """Переключение роли пользователя (пользователь <-> админи
1361 стратор)."""
1362     require_admin_user(request)
1363     from src.user_manager import user_manager
1364     user = user_manager.get_user_by_id(user_id)
1365     if not user:
1366         raise HTTPException(status_code=404, detail='Пользоват
1367 ель не найден')
1368     is_currently_admin = bool(user.get('is_admin', 0) or user.
```

```

    get('role') == 'admin')
1364
1365     # Защита системного администратора ID 1 от снятия прав
1366     if user_id == 1 and is_currently_admin:
1367         raise HTTPException(status_code=400, detail='Нельзя сн
    ять права у главного администратора (ID 1)')
1368
1369     if is_currently_admin:
1370         new_role = 'user'
1371         new_is_admin = 0
1372     else:
1373         new_role = 'admin'
1374         new_is_admin = 1
1375
1376     success = user_manager.update_user(user_id, role=new_role,
    is_admin=new_is_admin)
1377     if not success:
1378         raise HTTPException(status_code=500, detail='Error изм
    енения роли')
1379
1380     return {'status': 'ok', 'role': new_role, 'is_admin': new_
    is_admin}
1381
1382 @router.delete('/users/{user_id}')
1383 async def delete_admin_user(user_id: int, request: Request) ->
    Dict[str, Any]:
1384     """Удаление пользователя."""
1385     require_admin_user(request)
1386     from src.user_manager import user_manager
1387     user = user_manager.get_user_by_id(user_id)
1388     if not user:
1389         raise HTTPException(status_code=404, detail='Пользоват
    ель не найден')
1390
1391     if user_id == 1:
1392         raise HTTPException(status_code=400, detail='Нельзя уд
    алить главного администратора (ID 1)')
1393

```

```

1394     success = user_manager.delete_user(user_id)
1395     if not success:
1396         raise HTTPException(status_code=500, detail='Error уда
ления пользователя')
1397
1398     return {'status': 'ok', 'message': f'Пользователь ID {user
_id} удалён'}
1399
1400 # =====
=====
1401 # Skills Management Endpoints
1402 # =====
=====
1403
1404 skills_router = APIRouter(prefix='/api/skills', tags=['skills'
])
1405
1406 def _safe_home_dir() -> Path | None:
1407     """Safely retrieves user home directory across platforms."
"""
1408     try:
1409         return Path.home()
1410     except Exception:
1411         import os
1412         user_profile = os.environ.get("USERPROFILE") or os.env
iron.get("HOME")
1413         if user_profile:
1414             return Path(user_profile)
1415         return None
1416
1417 def _check_skills_access(request: Request) -> bool:
1418     """Check user access for skills management (permits all us
ers to inspect and manage user skills)."""
1419     return True
1420
1421 class AdminSkillCreateRequest(BaseModel):
1422     name: str
1423     description: str = ''

```

```
1424     description_ru: str = ''
1425     category: str = 'general'
1426     instructions: str = ''
1427     target_dir: str = '.agents/skills'
1428     readme: str = ''
1429     scripts: Dict[str, str] = Field(default_factory=dict)
1430     references: Dict[str, str] = Field(default_factory=dict)
1431
1432 class AdminSkillUpdateRequest(BaseModel):
1433     description: str | None = None
1434     description_ru: str | None = None
1435     instructions: str | None = None
1436     readme: str | None = None
1437
1438 class SkillAiGenerateRequest(BaseModel):
1439     prompt: str
1440     category: str = 'general'
1441     target_dir: str = '.agents/skills'
1442
1443 class SkillTestRunRequest(BaseModel):
1444     prompt: str
1445     parameters: Dict[str, Any] = Field(default_factory=dict)
1446
1447 @router.get('/skills')
1448 @skills_router.get('')
1449 @skills_router.get('/')
1450 async def list_admin_skills(request: Request, q: str = '', lang: str = '') -> Dict[str, Any]:
1451     """List all registered agent skills across supported project skill directories."""
1452     _check_skills_access(request)
1453     from src.skills import SkillRegistry
1454     registry = SkillRegistry(__root__)
1455     skills = registry.discover(lang=lang if lang else None)
1456
1457     skills_list = []
1458     q_clean = q.strip().lower()
1459     home = _safe_home_dir()
```

```

1460
1461     for s in skills:
1462         all_text = f"{s.name} {s.description} " + " ".join(s.d
descriptions_i18n.values())
1463         if q_clean and q_clean not in all_text.lower():
1464             continue
1465
1466         try:
1467             rel_root = s.root.relative_to(__root__).as_posix()
1468         except ValueError:
1469             if home:
1470                 try:
1471                     rel_root = f"~/{s.root.relative_to(home).a
s_posix()}"
1472                 except ValueError:
1473                     rel_root = str(s.root)
1474             else:
1475                 rel_root = str(s.root)
1476
1477         has_scripts = (s.root / 'scripts').is_dir() and any((s
.root / 'scripts').iterdir())
1478         has_references = (s.root / 'references').is_dir() and
any((s.root / 'references').iterdir())
1479         has_assets = (s.root / 'assets').is_dir() and any((s.r
oot / 'assets').iterdir())
1480         has_dist = (s.root / 'dist').is_dir() and any((s.root
/ 'dist').glob('*.*skill'))
1481
1482         files_count = 0
1483         try:
1484             files_count = sum(1 for p in s.root.rglob('*') if
p.is_file())
1485         except Exception:
1486             pass
1487
1488         skills_list.append({
1489             'name': s.name,
1490             'description': s.get_description(lang) if lang els

```



```

        e s.description,
1491         'descriptions_i18n': s.descriptions_i18n,
1492         'relative_path': rel_root,
1493         'source_file': str(s.source.name),
1494         'metadata': s.metadata,
1495         'manifest': s.manifest,
1496         'instructions': s.instructions,
1497         'has_scripts': has_scripts,
1498         'has_references': has_references,
1499         'has_assets': has_assets,
1500         'has_dist': has_dist,
1501         'files_count': files_count,
1502     })
1503
1504     return {
1505         'status': 'ok',
1506         'skills': skills_list,
1507         'total': len(skills_list),
1508     }
1509
1510 @router.post('/skills/generate-ai')
1511 @skills_router.post('/generate-ai')
1512 async def generate_skill_ai(data: SkillAiGenerateRequest, request: Request) -> Dict[str, Any]:
1513     """Generate complete agent skill structure, metadata, instructions and docs with AI or deterministic smart fallback."""
1514     _check_skills_access(request)
1515     user_prompt = data.prompt.strip()
1516     if not user_prompt:
1517         raise HTTPException(status_code=400, detail="Prompt is required for AI skill generation")
1518
1519     target_dir = data.target_dir.strip() or '.agents/skills'
1520     category = data.category.strip() or 'general'
1521
1522     # 1. Attempt LLM generation if available
1523     llm_generated = None
1524     try:

```

```

1525 |         from src.ai.unified_chat import UnifiedChatModel
1526 |         import os
1527 |         api_key_names = [n.strip() for n in os.getenv('GEMINI_
API_KEY_NAMES', '').split(',') if n.strip()]
1528 |         if api_key_names:
1529 |             system_prompt = (
1530 |                 "You are an expert AI skill architect for AI B
readboard and Gemini CLI. "
1531 |                 "Given the user request, generate a structured
JSON object for an agent skill.\n"
1532 |                 "JSON format required:\n"
1533 |                 "{\n"
1534 |                 '    "name": "kebab-case-name",\n'
1535 |                 '    "description": "Concise English description
for agent matching (max 2 sentences)",\n'
1536 |                 '    "description_ru": "Краткое русское описание
назначения навыка",\n'
1537 |                 '    "instructions": "Full structured Markdown i
nstructions for SKILL.md with ## Purpose, ## When to Use & Tri
ggers, ## Execution Protocol, ## Rules & Constraints",\n'
1538 |                 '    "readme": "English developer README.md docu
mentation",\n'
1539 |                 '    "scripts": {"helper.py": "# Python executab
le helper tool code"},\n'
1540 |                 '    "references": {"protocol.md": "# Detailed s
pecifications or guides"}\n'
1541 |                 "}\n"
1542 |                 "Output STRICTLY valid JSON without Markdown b
ackticks or commentary."
1543 |             )
1544 |             chat = UnifiedChatModel(
1545 |                 api_key_names=api_key_names,
1546 |                 system_instruction=system_prompt,
1547 |             )
1548 |             raw_response = await chat.chat(f"Create a skill fo
r category '{category}': {user_prompt}")
1549 |             if raw_response:
1550 |                 clean_json = raw_response.strip()

```

```

1551         if clean_json.startswith("`"):
1552             clean_json = re.sub(r"^```[a-z]*\n", "", c
lean_json)
1553             clean_json = re.sub(r"\n```$", "", clean_j
son).strip()
1554             llm_generated = json.loads(clean_json)
1555         except Exception as ex:
1556             logger.warning(f"AI skill generation via LLM fallback
to deterministic builder: {ex}")
1557
1558         if llm_generated and isinstance(llm_generated, dict) and "
name" in llm_generated:
1559             name_clean = re.sub(r'^[a-z0-9_-]', '-', str(llm_gener
ated.get('name', 'custom-skill')).lower()).strip('-')
1560             return {
1561                 'status': 'ok',
1562                 'generated': {
1563                     'name': name_clean or 'custom-skill',
1564                     'description': str(llm_generated.get('descript
ion', '')),
1565                     'description_ru': str(llm_generated.get('descr
iption_ru', '')),
1566                     'category': category,
1567                     'target_dir': target_dir,
1568                     'instructions': str(llm_generated.get('instruc
tions', '')),
1569                     'readme': str(llm_generated.get('readme', ''))
,
1570                     'scripts': llm_generated.get('scripts') if isi
nstance(llm_generated.get('scripts'), dict) else {},
1571                     'references': llm_generated.get('references')
if isinstance(llm_generated.get('references'), dict) else {},
1572                 }
1573             }
1574
1575         # Deterministic high-quality smart fallback
1576         slug = re.sub(r'^[a-zA-Z0-9]+', '-', user_prompt.lower())[
:32].strip('-')

```

```

1577     if not slug or len(slug) < 3:
1578         slug = f"skill-{{category}}"
1579
1580     name_clean = slug.replace('_', '-')
1581     title = name_clean.replace('-', ' ').title()
1582
1583     instructions_tpl = f"""\# {{title}}
1584
1585 ## 📌 Purpose
1586 Provide specialized agent capability for: {{user_prompt}}
1587
1588 ## 📌 When to Use & Triggers
1589 - Activate when the user requests actions related to: {{user_pr
1590   ompt}}.
1591 - Agent should consult this skill before delegating tasks or r
1592   unning manual routines.
1593
1594 ## 📌 Execution Protocol
1595 1. **Analyze input request**: Validate preconditions and extra
1596   ct parameters.
1597 2. **Execute actions**:
1598   - Check if scripts in `scripts/` provide helper automation.
1599   - Refer to guidelines in `references/` if edge cases are en
1600   countered.
1601 3. **Format output**: Deliver results in structured markdown w
1602   ith clear status indicators.
1603
1604 ## 📌 Rules & Constraints
1605 - Maintain fail-fast validation and English logging.
1606 - Handle missing files or external tool errors gracefully.
1607 """
1608
1609     readme_tpl = f"""\# {{title}}
1610
1611 ## Overview
1612 Agent skill providing automated capability for: {{user_prompt}}.
1613
1614 ## Directory Structure

```

```

1610 | - `SKILL.md`: Main instructions and frontmatter contract.
1611 | - `README.md`: English documentation for developers.
1612 | - `scripts/`: Executable helper tools (Python).
1613 | - `references/`: Reference documentation and guidelines.
1614 | - `assets/`: Static fixtures, data, and examples.
1615 |
1616 | ## Usage
1617 | Triggered dynamically by AI agents in AI Breadboard and Gemini
    | CLI.
1618 | """
1619 |
1620 |     helper_script = f"""# -*- coding: utf-8 -*-
1621 | \"""\"Helper automation script for {name_clean}.\"""\"
1622 |
1623 | import sys
1624 |
1625 | def main():
1626 |     print("Executing {name_clean} helper utility...")
1627 |     return 0
1628 |
1629 | if __name__ == "__main__":
1630 |     sys.exit(main())
1631 | """
1632 |
1633 |     return {
1634 |         'status': 'ok',
1635 |         'generated': {
1636 |             'name': name_clean,
1637 |             'description': f"Agent capability for {user_prompt
    | [:120]}.",
1638 |             'description_ru': f"Навык для: {user_prompt[:120]}
    | .",
1639 |             'category': category,
1640 |             'target_dir': target_dir,
1641 |             'instructions': instructions_tpl,
1642 |             'readme': readme_tpl,
1643 |             'scripts': {'main.py': helper_script},
1644 |             'references': {'guide.md': f"# Guide for {title}\n

```

```

        \nKey protocols and operational references.\n"},
1645     }
1646 }
1647
1648 @router.get('/skills/{name}')
1649 @skills_router.get('/{name}')
1650 async def get_admin_skill_details(name: str, request: Request,
    lang: str = '') -> Dict[str, Any]:
1651     """Get full details of a specific skill including SKILL.md
    and README.md content."""
1652     _check_skills_access(request)
1653     from src.skills import SkillRegistry
1654     registry = SkillRegistry(__root__)
1655     try:
1656         skill = registry.get(name)
1657     except KeyError:
1658         raise HTTPException(status_code=404, detail=f"Skill '{
name}' not found")
1659
1660     skill_md_raw = ''
1661     if skill.source.is_file():
1662         try:
1663             skill_md_raw = skill.source.read_text(encoding='ut
f-8')
1664         except Exception as ex:
1665             logger.error(f"Error reading SKILL.md for {name}",
ex)
1666
1667     readme_raw = ''
1668     readme_path = skill.root / 'README.md'
1669     if readme_path.is_file():
1670         try:
1671             readme_raw = readme_path.read_text(encoding='utf-8
')
1672         except Exception as ex:
1673             logger.error(f"Error reading README.md for {name}"
, ex)
1674

```

```

1675 |     # Collect files tree
1676 |     file_list = []
1677 |     try:
1678 |         for p in sorted(skill.root.rglob('*')):
1679 |             if p.is_file() and '__pycache__' not in p.parts:
1680 |                 file_list.append({
1681 |                     'rel_path': p.relative_to(skill.root).as_p
osix(),
1682 |                     'size': p.stat().st_size,
1683 |                 })
1684 |     except Exception:
1685 |         pass
1686 |
1687 |     home = _safe_home_dir()
1688 |     try:
1689 |         rel_root = skill.root.relative_to(__root__).as_posix()
1690 |     except ValueError:
1691 |         if home:
1692 |             try:
1693 |                 rel_root = f"~/{skill.root.relative_to(home).a
s_posix()}"
1694 |             except ValueError:
1695 |                 rel_root = str(skill.root)
1696 |         else:
1697 |             rel_root = str(skill.root)
1698 |
1699 |     return {
1700 |         'status': 'ok',
1701 |         'skill': {
1702 |             'name': skill.name,
1703 |             'description': skill.get_description(lang) if lang
else skill.description,
1704 |             'descriptions_i18n': skill.descriptions_i18n,
1705 |             'relative_path': rel_root,
1706 |             'metadata': skill.metadata,
1707 |             'manifest': skill.manifest,
1708 |             'instructions': skill.instructions,
1709 |             'skill_md_raw': skill_md_raw,

```

```

1710         'readme_raw': readme_raw,
1711         'files': file_list,
1712     }
1713 }
1714
1715 @router.post('/skills')
1716 @skills_router.post('')
1717 @skills_router.post('/')
1718 async def create_admin_skill(data: AdminSkillCreateRequest, re
quest: Request) -> Dict[str, Any]:
1719     """Create a new agent skill directory with standard struct
ure."""
1720     _check_skills_access(request)
1721     name = data.name.strip().lower()
1722     if not name or not re.match(r'^[a-z0-9_-]+$ ', name):
1723         raise HTTPException(status_code=400, detail="Skill nam
e must contain only letters, numbers, hyphens, and underscores
")
1724
1725     target_dir = data.target_dir.strip() or '.agents/skills'
1726     home = _safe_home_dir()
1727     if target_dir.startswith('~') and home:
1728         base_dir = (home / target_dir[2:]).resolve()
1729     else:
1730         base_dir = (__root__ / target_dir).resolve()
1731
1732     is_in_project = str(base_dir).startswith(str(__root__.reso
lve()))
1733     is_in_home = home is not None and str(base_dir).startswith
(str(home.resolve()))
1734     if not (is_in_project or is_in_home):
1735         raise HTTPException(status_code=400, detail="Invalid t
arget directory")
1736
1737     skill_dir = base_dir / name
1738     if skill_dir.exists():
1739         raise HTTPException(status_code=400, detail=f"Skill '{
name}' already exists at {target_dir}/{name}")

```



```

1740
1741     skill_dir.mkdir(parents=True, exist_ok=True)
1742     scripts_dir = skill_dir / 'scripts'
1743     references_dir = skill_dir / 'references'
1744     assets_dir = skill_dir / 'assets'
1745     scripts_dir.mkdir(exist_ok=True)
1746     references_dir.mkdir(exist_ok=True)
1747     assets_dir.mkdir(exist_ok=True)
1748
1749     desc_en = data.description.strip() or f"Agent skill for {n
ame}."
1750     desc_ru = data.description_ru.strip() or desc_en
1751
1752     instructions = data.instructions.strip()
1753     if not instructions:
1754         instructions = f'""# {name.replace('-', ' ').replace('
_', ' ').title()}'
1755
1756 ## □ Purpose
1757 {desc_en}
1758
1759 ## □ Usage & Protocol
1760 Describe how AI agents should execute this skill and what trig
gers its activation.
1761
1762 ## □ Directory Structure
1763 - `SKILL.md`: Main instructions and frontmatter contract.
1764 - `README.md`: English documentation for developers.
1765 - `scripts/`: Executable helper tools.
1766 - `references/`: Reference documentation and guidelines.
1767 - `assets/`: Static data, examples, and assets.
1768 ""
1769
1770     skill_md_content = f'""---
1771 name: {name}
1772 description: {desc_en}
1773 description_i18n:
1774     en: {desc_en}

```

```

1775 | ru: {desc_ru}
1776 | ---
1777 |
1778 | {instructions}
1779 | """
1780 |
1781 |     readme_content = data.readme.strip() if data.readme.strip(
1782 | ) else f"""\# {name.replace('-', ' ').replace('_', ' ').title()}
1783 |
1784 | ## Overview
1785 | {desc_en}
1786 |
1787 | ## Location
1788 | \#{target_dir}/{name}/`
1789 | """
1790 |
1791 |     (skill_dir / 'SKILL.md').write_text(skill_md_content, encoding='utf-8')
1792 |
1793 |     (skill_dir / 'README.md').write_text(readme_content, encoding='utf-8')
1794 |
1795 |     # Write additional scripts if provided
1796 |     if data.scripts and isinstance(data.scripts, dict):
1797 |         for script_name, script_code in data.scripts.items():
1798 |             if script_name and isinstance(script_code, str):
1799 |                 safe_script_name = Path(script_name).name
1800 |                 (scripts_dir / safe_script_name).write_text(script_code, encoding='utf-8')
1801 |
1802 |     # Write additional references if provided
1803 |     if data.references and isinstance(data.references, dict):
1804 |         for ref_name, ref_content in data.references.items():
1805 |             if ref_name and isinstance(ref_content, str):
1806 |                 safe_ref_name = Path(ref_name).name
1807 |                 (references_dir / safe_ref_name).write_text(ref_content, encoding='utf-8')

```

```

1807     return {
1808         'status': 'ok',
1809         'message': f"Skill '{name}' created successfully",
1810         'skill': {
1811             'name': name,
1812             'description': desc_en,
1813             'description_ru': desc_ru,
1814             'relative_path': f"{target_dir}/{name}",
1815         }
1816     }
1817
1818 @router.put('/skills/{name}')
1819 @skills_router.put('/{name}')
1820 async def update_admin_skill(name: str, data: AdminSkillUpdate
Request, request: Request) -> Dict[str, Any]:
1821     """Update skill SKILL.md and README.md content."""
1822     _check_skills_access(request)
1823     from src.skills import SkillRegistry
1824     registry = SkillRegistry(__root__)
1825     try:
1826         skill = registry.get(name)
1827     except KeyError:
1828         raise HTTPException(status_code=404, detail=f"Skill '{
name}' not found")
1829
1830     desc_en = data.description if data.description is not None
else skill.description
1831     desc_ru = data.description_ru if data.description_ru is no
t None else skill.descriptions_i18n.get('ru', desc_en)
1832     instructions = data.instructions if data.instructions is n
ot None else skill.instructions
1833
1834     skill_md_content = f"""---
1835 name: {skill.name}
1836 description: {desc_en}
1837 description_i18n:
1838   en: {desc_en}
1839   ru: {desc_ru}

```

```

1840 ---
1841
1842 {instructions}
1843 """
1844     skill.source.write_text(skill_md_content, encoding='utf-8'
1845 )
1846
1846     if data.readme is not None:
1847         readme_path = skill.root / 'README.md'
1848         readme_path.write_text(data.readme, encoding='utf-8')
1849
1850     return {
1851         'status': 'ok',
1852         'message': f"Skill '{name}' updated successfully",
1853     }
1854
1855 @router.delete('/skills/{name}')
1856 @skills_router.delete('/{name}')
1857 async def delete_admin_skill(name: str, request: Request) -> Dict[str, Any]:
1858     """Delete a skill directory."""
1859     _check_skills_access(request)
1860     import shutil
1861     from src.skills import SkillRegistry
1862     registry = SkillRegistry(__root__)
1863     try:
1864         skill = registry.get(name)
1865     except KeyError:
1866         raise HTTPException(status_code=404, detail=f"Skill '{name}' not found")
1867
1868     skill_root = skill.root.resolve()
1869     home = _safe_home_dir()
1870     is_in_project = str(skill_root).startswith(str(__root__.resolve()))
1871     is_in_home = home is not None and str(skill_root).startswith(str(home.resolve()))
1872     if not (is_in_project or is_in_home):

```

```

1873         raise HTTPException(status_code=400, detail="Invalid s
kill directory path")
1874
1875     forbidden = [__root__.resolve()]
1876     if home:
1877         forbidden.append(home.resolve())
1878     if skill_root in forbidden:
1879         raise HTTPException(status_code=400, detail="Cannot de
lete root directory")
1880
1881     shutil.rmtree(skill_root, ignore_errors=True)
1882     return {
1883         'status': 'ok',
1884         'message': f"Skill '{name}' deleted successfully",
1885     }
1886
1887 @router.post('/skills/{name}/test')
1888 @skills_router.post('/{name}/test')
1889 async def test_admin_skill(name: str, data: SkillTestRunReques
t, request: Request) -> Dict[str, Any]:
1890     """Test skill execution with a test prompt against active
AI runtime or sandbox runner."""
1891     _check_skills_access(request)
1892     from src.skills import SkillRegistry
1893     registry = SkillRegistry(__root__)
1894     try:
1895         skill = registry.get(name)
1896     except KeyError:
1897         raise HTTPException(status_code=404, detail=f"Skill '{
name}' not found")
1898
1899     user_prompt = data.prompt.strip() or "Test execution of sk
ill"
1900     instructions = skill.instructions or skill.description
1901
1902     model_used = "simulation"
1903     response_text = ""
1904

```

```

1905     try:
1906         from src.ai.unified_chat import UnifiedChatModel
1907         import os
1908         api_key_names = [n.strip() for n in os.getenv('GEMINI_
API_KEY_NAMES', '').split(',') if n.strip()]
1909         if api_key_names:
1910             chat = UnifiedChatModel(
1911                 api_key_names=api_key_names,
1912                 system_instruction=f"You are executing the fol
lowing agent skill:\n\n{instructions}",
1913             )
1914             response_text = await chat.chat(user_prompt)
1915             model_used = "UnifiedChatModel (LLM)"
1916         except Exception as ex:
1917             logger.warning(f"Live skill test fallback: {ex}")
1918
1919         if not response_text:
1920             response_text = (
1921                 f" [Skill Sandbox]: Skill '{skill.name}' accepted
test query: \"{user_prompt}\" \n\n"
1922                 f"**Contract Matched**:\n- Name: `{skill.name}`\n-
Path: `{skill.root}`\n"
1923                 f"- Status: Active and ready for dispatch."
1924             )
1925
1926         return {
1927             'status': 'ok',
1928             'skill': skill.name,
1929             'model_used': model_used,
1930             'response': response_text,
1931         }
1932
1933 @router.post('/skills/{name}/package')
1934 @skills_router.post('/{name}/package')
1935 async def package_admin_skill(name: str, request: Request) ->
Dict[str, Any]:
1936     """Package skill directory into a distributable .skill ZIP
archive."""

```

```

1937 |     _check_skills_access(request)
1938 |     import os
1939 |     import zipfile
1940 |     from src.skills import SkillRegistry
1941 |     registry = SkillRegistry(__root__)
1942 |     try:
1943 |         skill = registry.get(name)
1944 |     except KeyError:
1945 |         raise HTTPException(status_code=404, detail=f"Skill '{
name}' not found")
1946 |
1947 |     dist_dir = skill.root / 'dist'
1948 |     dist_dir.mkdir(parents=True, exist_ok=True)
1949 |     archive_file = dist_dir / f"{skill.name}.skill"
1950 |
1951 |     with zipfile.ZipFile(archive_file, 'w', zipfile.ZIP_DEFLAT
ED) as zipf:
1952 |         for root_dir, dirs, files in os.walk(skill.root):
1953 |             dirs[:] = [d for d in dirs if d not in ['dist', '.
git', '__pycache__']]
1954 |             for f in files:
1955 |                 file_path = Path(root_dir) / f
1956 |                 if file_path == archive_file:
1957 |                     continue
1958 |                 arcname = file_path.relative_to(skill.root)
1959 |                 zipf.write(file_path, arcname)
1960 |
1961 |     size = archive_file.stat().st_size
1962 |     home = _safe_home_dir()
1963 |     try:
1964 |         rel_archive = archive_file.relative_to(__root__).as_po
six()
1965 |     except ValueError:
1966 |         if home:
1967 |             try:
1968 |                 rel_archive = f"~/{archive_file.relative_to(ho
me).as_posix()}"
1969 |             except ValueError:

```

```

1970         rel_archive = str(archive_file)
1971     else:
1972         rel_archive = str(archive_file)
1973
1974     return {
1975         'status': 'ok',
1976         'message': f"Skill '{skill.name}' packaged successful
1977         y",
1978         'archive': {
1979             'filename': f"{skill.name}.skill",
1980             'path': rel_archive,
1981             'size': size,
1982         },
1983     }
1984 # =====
1985 # Scheduler Settings & Trigger API
1986 # =====
1987
1988 class RagReindexConfig(BaseModel):
1989     enabled: bool = True
1990     interval_hours: float = 24.0
1991
1992 class EmailCheckConfig(BaseModel):
1993     enabled: bool = True
1994     interval_minutes: float = 5.0
1995
1996 class SchedulersConfigRequest(BaseModel):
1997     enabled: bool = True
1998     rag_reindex: RagReindexConfig = Field(default_factory=RagR
1999     eindexConfig)
2000     email_check: EmailCheckConfig = Field(default_factory=Emai
2001     lCheckConfig)
2002 @router.get('/schedulers')

```



```

2003 async def get_schedulers_config(request: Request) -> Dict[str,
    Any]:
2004     """Retrieve current background scheduler configuration and
        diagnostic status."""
2005     require_admin_user(request)
2006     from src.utils.scheduler import scheduler
2007
2008     cfg_file = __root__ / 'config.json'
2009     sched_cfg = {}
2010     if cfg_file.exists():
2011         try:
2012             with open(cfg_file, 'r', encoding='utf-8') as f:
2013                 data = json.load(f)
2014                 sched_cfg = data.get('schedulers', {})
2015         except Exception as e:
2016             logger.error(f"Error reading schedulers config: {e
    }")
2017
2018     return {
2019         'config': {
2020             'enabled': sched_cfg.get('enabled', True),
2021             'rag_reindex': sched_cfg.get('rag_reindex', {
2022                 'enabled': True,
2023                 'interval_hours': 24.0
2024             }),
2025             'email_check': sched_cfg.get('email_check', {
2026                 'enabled': True,
2027                 'interval_minutes': 5.0
2028             }),
2029         },
2030         'status': scheduler.status
2031     }
2032
2033
2034 @router.post('/schedulers')
2035 async def save_schedulers_config(data: SchedulersConfigRequest
    , request: Request) -> Dict[str, Any]:
2036     """Save background scheduler configuration to config.json

```

```

and restart running jobs."""
2037     require_admin_user(request)
2038     from src.utils.scheduler import scheduler
2039     import src.config as app_config
2040
2041     cfg_file = __root__ / 'config.json'
2042     try:
2043         with open(cfg_file, 'r', encoding='utf-8') as f:
2044             cfg_data = json.load(f)
2045     except Exception:
2046         cfg_data = {}
2047
2048     sched_dict = {
2049         'enabled': data.enabled,
2050         'rag_reindex': {
2051             'enabled': data.rag_reindex.enabled,
2052             'interval_hours': data.rag_reindex.interval_hours
2053         },
2054         'email_check': {
2055             'enabled': data.email_check.enabled,
2056             'interval_minutes': data.email_check.interval_minu
2057     tes
2058         }
2059     }
2060
2061     cfg_data['schedulers'] = sched_dict
2062
2063     with open(cfg_file, 'w', encoding='utf-8') as f:
2064         json.dump(cfg_data, f, indent=4, ensure_ascii=False)
2065
2066     # Обновление глобального SimpleNamespace
2067     from src.utils.jjson import j_loads_ns
2068     app_config.global_settings = j_loads_ns(cfg_file)
2069     app_config.schedulers_cfg = getattr(app_config.global_sett
2070     ings, "schedulers", getattr(app_config.global_settings, "sched
2071     uler", None))
2072
2073     # Перезапуск планировщика с новыми параметрами

```

```

2071     await scheduler.restart()
2072
2073     logger.info("Scheduler configuration saved and service res
tarded successfully.")
2074     return {
2075         'status': 'ok',
2076         'message': 'Настройки планировщика успешно сохранены',
2077         'schedulers': scheduler.status
2078     }
2079
2080
2081 @router.post('/schedulers/trigger/{job_name}')
2082 async def trigger_scheduler_job(job_name: str, request: Request) -> Dict[str, Any]:
2083     """Manually trigger an immediate execution of a scheduled
job."""
2084     require_admin_user(request)
2085     from src.utils.scheduler import scheduler
2086
2087     if job_name == 'rag':
2088         summary = await scheduler.recheck_and_update_all_rags(
)
2089         scheduler._last_rag_run = datetime.now()
2090         return {'status': 'ok', 'job': 'rag', 'result': summary}
2091     elif job_name == 'email':
2092         messages = await scheduler.check_emails()
2093         scheduler._last_email_run = datetime.now()
2094         return {'status': 'ok', 'job': 'email', 'count': len(messages), 'messages': messages[:5]}
2095     else:
2096         raise HTTPException(status_code=400, detail=f"Неизвестная задача: {job_name}. Доступны: 'rag', 'email'")
2097
2098
2099 # =====
=====
2100 # Initialization

```

```

2101 # =====
2102 |=====
2103 def init_router() -> APIRouter:
2104     """Initialization роутера управления системными инструкция
2105     ми и источниками."""
2106     return router
2107
2108 def init_apps_router() -> APIRouter:
2109     """Initialization of /apps public status and management ro
2110     uter."""
2111     return router_apps
2112
2113 def init_skills_router() -> APIRouter:
2114     """Initialization of agent skills router."""
2115     return skills_router
2116
2117 def init_plugins_router() -> APIRouter:
2118     """Initialization of user plugins router."""
2119     return plugins_router
2120
2121 # =====
2122 |=====
2123 # Windows User Management Endpoints
2124 # =====
2125 |=====
2126 windows_users_router = APIRouter(prefix='/api/windows-users',
2127 |tags=['windows-users'])
2128
2129 class WindowsUserCreateRequest(BaseModel):
2130     username: str
2131     password: str
2132     fullname: Optional[str] = None
2133     description: Optional[str] = None
2134
2135 class WindowsUserActionRequest(BaseModel):

```

```
2133     username: str
2134
2135 class WindowsUserPasswordRequest(BaseModel):
2136     username: str
2137     new_password: str
2138
2139 class WindowsGroupActionRequest(BaseModel):
2140     username: str
2141     group_name: str
2142
2143
2144 @windows_users_router.get('')
2145 async def get_windows_users(request: Request) -> Dict[str, Any
    ]:
2146     """Получение списка локальных пользователей Windows."""
2147     require_admin_user(request)
2148     from apps.ai_breadboard_admin.src import WindowsUserManage
    r
2149     manager = WindowsUserManager()
2150     users = manager.get_windows_users()
2151     return {'status': 'ok', 'users': users, 'count': len(users
    )}
2152
2153
2154 @windows_users_router.get('/groups')
2155 async def get_windows_groups(request: Request) -> Dict[str, An
    y]:
2156     """Получение списка локальных групп Windows."""
2157     require_admin_user(request)
2158     from apps.ai_breadboard_admin.src import WindowsUserManage
    r
2159     manager = WindowsUserManager()
2160     groups = manager.get_windows_groups()
2161     return {'status': 'ok', 'groups': groups, 'count': len(gro
    ups)}
2162
2163
2164 @windows_users_router.get('/groups/{group_name}/members')
```

```

2165 async def get_group_members(group_name: str, request: Request)
      -> Dict[str, Any]:
2166     """Получение членов группы Windows."""
2167     require_admin_user(request)
2168     from apps.ai_breadboard_admin.src import WindowsUserManage
r
2169     manager = WindowsUserManager()
2170     members = manager.get_group_members(group_name)
2171     return {'status': 'ok', 'group': group_name, 'members': me
members, 'count': len(members)}
2172
2173
2174 @windows_users_router.post('')
2175 async def create_windows_user(data: WindowsUserCreateRequest,
request: Request) -> Dict[str, Any]:
2176     """Создание нового локального пользователя Windows."""
2177     require_admin_user(request)
2178     from apps.ai_breadboard_admin.src import WindowsUserManage
r
2179     manager = WindowsUserManager()
2180     result = manager.create_windows_user(
2181         username=data.username,
2182         password=data.password,
2183         fullname=data.fullname,
2184         description=data.description
2185     )
2186     return result
2187
2188
2189 @windows_users_router.delete('')
2190 async def delete_windows_user(data: WindowsUserActionRequest,
request: Request) -> Dict[str, Any]:
2191     """Удаление локального пользователя Windows."""
2192     require_admin_user(request)
2193     from apps.ai_breadboard_admin.src import WindowsUserManage
r
2194     manager = WindowsUserManager()
2195     result = manager.delete_windows_user(data.username)

```

```
2196     return result
2197
2198
2199 @windows_users_router.post('/enable')
2200 async def enable_windows_user(data: WindowsUserActionRequest,
    request: Request) -> Dict[str, Any]:
2201     """Включение пользователя Windows."""
2202     require_admin_user(request)
2203     from apps.ai_breadboard_admin.src import WindowsUserManage
    r
2204     manager = WindowsUserManager()
2205     result = manager.enable_windows_user(data.username)
2206     return result
2207
2208
2209 @windows_users_router.post('/disable')
2210 async def disable_windows_user(data: WindowsUserActionRequest,
    request: Request) -> Dict[str, Any]:
2211     """Отключение пользователя Windows."""
2212     require_admin_user(request)
2213     from apps.ai_breadboard_admin.src import WindowsUserManage
    r
2214     manager = WindowsUserManager()
2215     result = manager.disable_windows_user(data.username)
2216     return result
2217
2218
2219 @windows_users_router.post('/password')
2220 async def reset_windows_user_password(data: WindowsUserPasswor
    dRequest, request: Request) -> Dict[str, Any]:
2221     """Сброс пароля пользователя Windows."""
2222     require_admin_user(request)
2223     from apps.ai_breadboard_admin.src import WindowsUserManage
    r
2224     manager = WindowsUserManager()
2225     result = manager.reset_windows_user_password(data.username
    , data.new_password)
2226     return result
```

```

2227 |
2228 |
2229 | @windows_users_router.post('/add-to-group')
2230 | async def add_user_to_group(data: WindowsGroupActionRequest, r
2231 |     request: Request) -> Dict[str, Any]:
2232 |     """Добавление пользователя в группу Windows."""
2233 |     require_admin_user(request)
2234 |     from apps.ai_breadboard_admin.src import WindowsUserManage
2235 |     r
2236 |     manager = WindowsUserManager()
2237 |     result = manager.add_user_to_group(data.username, data.gro
2238 | up_name)
2239 |     return result
2240 |
2241 |
2242 | @windows_users_router.post('/remove-from-group')
2243 | async def remove_user_from_group(data: WindowsGroupActionReque
2244 | st, request: Request) -> Dict[str, Any]:
2245 |     """Удаление пользователя из группы Windows."""
2246 |     require_admin_user(request)
2247 |     from apps.ai_breadboard_admin.src import WindowsUserManage
2248 |     r
2249 |     manager = WindowsUserManager()
2250 |     result = manager.remove_user_from_group(data.username, dat
2251 | a.group_name)
2252 |     return result
2253 |
2254 |
2255 | # =====
2256 | # Initialization
2257 | # =====
2258 |
2259 | def init_router() -> APIRouter:
2260 |     """Initialization роутера управления системными инструкция
2261 | ми и источниками."""
2262 |     return router

```



```
2256 |
2257 |
2258 | def init_apps_router() -> APIRouter:
2259 |     """Initialization of /apps public status and management router."""
2260 |     return router_apps
2261 |
2262 |
2263 | def init_skills_router() -> APIRouter:
2264 |     """Initialization of agent skills router."""
2265 |     return skills_router
2266 |
2267 |
2268 | def init_plugins_router() -> APIRouter:
2269 |     """Initialization of user plugins router."""
2270 |     return plugins_router
2271 |
2272 |
2273 |
```

router_chat.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Chat
4 # =====
5 # Description:
6 #   Handles AI conversational endpoints, streaming responses across multiple providers,
7 #
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.routers.router_chat import ChatRequest
11 #
12 #       service = ChatRequest()
13 #
14 # File: router_chat.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-08 05:32:00
20 # =====
21
22 """Handles AI conversational endpoints, streaming responses across multiple providers,"""
23
24 import os
25 import time
26 import asyncio
27 from fastapi import APIRouter, HTTPException, Request
28 from pydantic import BaseModel, Field
29
```

```

30 from logger import logger
31 from src.config import ai_cfg, tts_cfg
32 from src.ai.gemini.user_query_rag import index_user_query, sea
   rch_user_context
33
34 from header import __root__
35
36 router = APIRouter(prefix='/api/v1/chat', tags=['chat'])
37
38 # Короткие слова-продолжения диалога, которые сами по себе не
   содержат медиа-ключевых слов
39 _CONTEXT_CONTINUATION_WORDS = {
40     'да', 'нет', 'yes', 'no', 'ок', 'ок', 'хочу', 'конечно',
41     'давай', 'проверь', 'найди', 'покажи', 'ладно', 'угу', 'ye
   p', 'sure',
42     'want', 'check', 'find', 'show', 'okay',
43 }
44
45 class ChatRequest(BaseModel):
46     message: str
47     history: list[dict] = []
48     generation_config: dict = {}
49
50 class SaveRagRequest(BaseModel):
51     query: str
52     chat_text: str
53     voice_text: str
54
55 class TestModelRequest(BaseModel):
56     """Модель данных для проверочного запроса к AI-модели.
57
58     Введите вручную имя модели и провайдера, иначе будет испол
   зована настройка по умолчанию.
59     """
60     __test__ = False
61
62     model: str = Field(
63         default="",

```

```
64         description="Имя модели. Введите вручную имя модели и
        провайдера, иначе будет использована настройка по умолчанию.",
65     )
66     provider: str = Field(
67         default="",
68         description="Имя провайдера (gemini, agy, foundry, oll
        ama, openai, hf, onnx, gemini_cli). Введите вручную имя модели
        и провайдера, иначе будет использована настройка по умолчанию
        .",
69     )
70     message: str = Field(
71         default="Привет! Назови свою модель и провайдера, и по
        дтверди готовность к работе.",
72         description="Текст тестового сообщения для валидации с
        вязи.",
73     )
74     system_instruction: str = Field(
75         default="",
76         description="Оptionальная системная инструкция для тес
        тируемой модели.",
77     )
78
79
80 class ModelInstructionPayload(BaseModel):
81     """Модель данных для обновления системной инструкции AI-мо
        дели."""
82     instruction: str = Field(
83         default="",
84         description="Текст системной инструкции для AI-модели.
        ",
85     )
86     system_instruction: str = Field(
87         default="",
88         description="Алиас для текста системной инструкции.",
89     )
90     content: str = Field(
91         default="",
92         description="Алиас для текста системной инструкции.",
```

```
93 |     )
94 |     model: str = Field(
95 |         default="",
96 |         description="Опциональное имя модели.",
97 |     )
98 |     save_to_disk: bool = Field(
99 |         default=False,
100 |         description="Флаг сохранения системной инструкции в файл prompts/chat/system_instruction.md.",
101 |     )
102 |
103 | ModelInstructionRequest = ModelInstructionPayload
104 |
105 |
106 | class SetModelPayload(BaseModel):
107 |     """Модель данных для установки активной AI-модели."""
108 |     model: str = Field(
109 |         ...,
110 |         description="Имя модели для установки (например: gemini-3.1-flash, ollama:llama3.1, agy-gemini-3.6-flash).",
111 |     )
112 |     provider: str = Field(
113 |         default="",
114 |         description="Опциональное имя провайдера (gemini, agy, foundry, ollama, openai, hf, onnx, gemini_cli).",
115 |     )
116 |
117 |
118 | class SetProviderPayload(BaseModel):
119 |     """Модель данных для установки активного AI-провайдера."""
120 |     provider: str = Field(
121 |         ...,
122 |         description="Имя провайдера (gemini, agy, foundry, ollama, openai, hf, onnx, gemini_cli).",
123 |     )
124 |     model: str = Field(
125 |         default="",
126 |         description="Опциональное имя модели. Если не указано,
```

```

        выбирается модель по умолчанию для провайдера.",
127     )
128
129
130 class ChatSessionPayload(BaseModel):
131     id: str = ""
132     userId: str = ""
133     title: str = "New Chat"
134     isCustomTitle: bool = False
135     createdAt: int = 0
136     updatedAt: int = 0
137     messages: list[dict] = []
138     chatHistory: list[dict] = []
139
140 class SyncSessionsRequest(BaseModel):
141     sessions: list[dict] = []
142
143 from src.api import chat_sessions_db
144
145 _active_chat_models: dict[str, object] = {}
146
147
148 def _get_default_system_instruction() -> str:
149     """Load default system instruction for chat assistant."""
150     prompt_file = __root__ / 'prompts' / 'chat' / 'system_inst
151     ruccion.md'
152     if prompt_file.exists():
153         try:
154             content = prompt_file.read_text(encoding='utf-8',
155             errors='replace').strip()
156             if content:
157                 return content
158         except Exception as e:
159             logger.debug(f"[router_chat] Could not read default prompt file: {e}")
160     return "Вы – интеллектуальный ассистент платформы AI Bread
161     board с доступом к Google Workspace и RAG-базе знаний."

```

```

160 def get_chat_model(selected_model_name: str, system_instructio
    n: str = "", user_id: str = ""):
161     """Dynamically construct or retrieve cached AI model insta
    nce."""
162     eff_sys_prompt = system_instruction.strip() if system_inst
    ruction and system_instruction.strip() else _get_default_syste
    m_instruction()
163     cache_key = f"{user_id}:{selected_model_name}" if user_id
    else selected_model_name
164     if cache_key in _active_chat_models:
165         instance = _active_chat_models[cache_key]
166         if eff_sys_prompt and hasattr(instance, 'system_instru
    ction'):
167             if getattr(instance, 'system_instruction', None) !
    = eff_sys_prompt:
168                 instance.system_instruction = eff_sys_prompt
169         return instance
170
171     is_gemini_cli = selected_model_name.startswith('gemini_cli
    :') or selected_model_name.startswith('gemini-cli-')
172     is_foundry = selected_model_name.startswith('foundry:')
173     is_ollama = selected_model_name.startswith('ollama:')
174     is_hf = selected_model_name.startswith('hf:') or selected_
    model_name.startswith('hf::')
175     is_onnx = selected_model_name.startswith('onnx:') or selec
    ted_model_name.startswith('onnx::')
176     openai_prefixes = ('openai:', 'openai::', 'deepseek:', 'gr
    oq:', 'openrouter:', 'lmstudio:', 'local:', 'compat:')
177     is_openai = any(selected_model_name.startswith(p) for p in
    openai_prefixes)
178     is_agy = selected_model_name.startswith('agy-') or 'agy' i
    n selected_model_name.lower()
179     is_gemini = not is_gemini_cli and not is_hf and not is_onn
    x and not is_openai and (selected_model_name.startswith('gemin
    i-') or 'gemini' in selected_model_name.lower())
180
181     if is_gemini_cli:
182         from src.ai.gemini_cli_chat import GeminiCliChatBase

```

```

183         inst = GeminiCliChatBase(
184             model_id=selected_model_name,
185             system_prompt=eff_sys_prompt,
186         )
187     elif is_foundry:
188         model_id = selected_model_name.split(':', 1)[-1]
189         from src.ai.foundry_chat import FoundryChatBase
190         inst = FoundryChatBase(
191             model_id=model_id,
192             system_prompt=eff_sys_prompt,
193         )
194     elif is_ollama:
195         model_id = selected_model_name.split(':', 1)[-1]
196         from src.ai.ollama_chat import OllamaChatBase
197         ollama_url = ai_cfg.ollama_base_url if ai_cfg else 'http://localhost:11434'
198         inst = OllamaChatBase(
199             model_id=model_id,
200             system_prompt=eff_sys_prompt,
201             api_url=ollama_url
202         )
203     elif is_hf:
204         model_id = selected_model_name.split(':', 1)[-1].lstrip(':')
205         from src.ai.hf_chat import HFChatBase
206         inst = HFChatBase(
207             model_id=model_id,
208             system_prompt=eff_sys_prompt,
209         )
210     elif is_onnx:
211         model_id = selected_model_name.split(':', 1)[-1].lstrip(':')
212         from src.ai.onnx_chat import ONNXChatBase
213         inst = ONNXChatBase(
214             model_id=model_id,
215             system_prompt=eff_sys_prompt,
216         )
217     elif is_openai:

```



```

218     prov_part, model_part = selected_model_name.split(':',
1)
219     model_id = model_part.lstrip(':')
220     prov_name = prov_part.lower().rstrip(':')
221     if prov_name == 'compat':
222         prov_name = 'openai'
223     from src.ai.openai_compat_chat import OpenAICompatChat
224     inst = OpenAICompatChat.create_for_provider(
225         provider_name=prov_name,
226         model_id=model_id,
227         system_prompt=eff_sys_prompt,
228     )
229     elif is_agy:
230         from src.ai.agy_chat import AgyChatBase
231         inst = AgyChatBase(
232             model_id=selected_model_name,
233             system_prompt=eff_sys_prompt,
234         )
235     elif is_gemini:
236         from src.ai.gemini_chat import GeminiChatBase
237         inst = GeminiChatBase(
238             model_id=selected_model_name,
239             system_prompt=eff_sys_prompt,
240         )
241     else:
242         from src.ai.foundry_chat import FoundryChatBase
243         inst = FoundryChatBase(
244             model_id=selected_model_name,
245             system_prompt=eff_sys_prompt,
246         )
247
248     _active_chat_models[cache_key] = inst
249     return inst
250
251 async def _extract_user_auth(fastapi_req: Request) -> tuple[str, str, str, dict]:
252     """Извлекает идентификатор пользователя, системную инструкцию, модель и настройки из JWT/IP без обязательной авторизации

```

```

253     user_identifier = "1"
254     system_instruction = ""
255     selected_model = ""
256     settings = {}
257
258     try:
259         from src.api.routers.core.router_auth import get_curre
nt_user_optional
260         user_data = get_current_user_optional(fastapi_req) if
fastapi_req is not None else None
261
262         from src.user_manager import user_manager
263         db_user = None
264         if user_data:
265             if getattr(user_data, 'id', None):
266                 db_user = await asyncio.to_thread(user_manager
.get_user_by_id, user_data.id)
267             if not db_user and getattr(user_data, 'email', Non
e):
268                 db_user = await asyncio.to_thread(user_manager
.get_user_by_email, user_data.email)
269
270             if not db_user:
271                 db_user = await asyncio.to_thread(user_manager.get
_user_by_id, 1)
272
273             if db_user:
274                 user_identifier = str(db_user.get('id', 1))
275                 settings = await asyncio.to_thread(user_manager.ge
t_user_settings, db_user.get('id', 1)) or {}
276                 if settings.get('system_instruction'):
277                     system_instruction = settings['system_instruct
ion']
278                 if settings.get('model'):
279                     selected_model = settings['model']
280     except Exception as e:
281         logger.debug(f"[router_chat] Failed to extract optiona

```

```

    l user auth: {e}")
282
283     if not system_instruction:
284         system_instruction = _get_default_system_instruction()
285
286     return user_identifier, system_instruction, selected_model
    , settings
287
288 def _get_voice_gender_rule(settings: dict) -> str:
289     """Определяет гендерное правило для ответов ассистента на
    основе настроек голоса TTS."""
290     default_voice = getattr(tts_cfg, "default_voice", "ru-RU-D
    mitryNeural") if tts_cfg else "ru-RU-DmitryNeural"
291     tts_voice = settings.get('tts_voice', '') or default_voice
292     voice_lower = tts_voice.lower()
293     is_male_voice = any(name in voice_lower for name in ("dmit
    ry", "yaraslaus", "male", "bayan", "aidar", "eugene", "georgy"
    ))
294     is_female_voice = any(name in voice_lower for name in ("sv
    etlana", "elena", "female", "kseniya", "tanya", "aliona", "dar
    iya"))
295     if is_male_voice:
296         return "ВАЖНОЕ ПРАВИЛО: Отвечай от мужского лица (напр
    имер: 'Я нашел', 'Я подобрал')."
297     if is_female_voice:
298         return "ВАЖНОЕ ПРАВИЛО: Отвечай от женского лица (напр
    имер: 'Я нашла', 'Я подобрала')."
299     return ""
300
301 def _clean_chat_history(history: list[dict]) -> list[dict]:
302     """Очищает историю сообщений перед передачей в модель."""
303     if not history:
304         return []
305     _ERROR_PATTERNS = (
306         ' ', 'Error', 'Error', 'TypeError', 'AttributeError',
    'Traceback',
307         '[Error]', 'Не удалось найти', 'В локальной базе ничег
    о не найдено',

```

```

308         'DEBUG MODE', 'DEBUG:', '[DEBUG'
309     )
310     _STATUS_PREFIXES = (
311         ' ', ' ', ' ', ' ', ' ', ' ', 'Вызов плагина', '
Генерация', 'Check',
312         'DEBUG MODE:', 'DEBUG:'
313     )
314
315     def _is_clean(entry: dict) -> bool:
316         parts = entry.get('parts', [])
317         text = ''
318         if isinstance(parts, list) and len(parts) > 0:
319             p = parts[0]
320             text = p if isinstance(p, str) else p.get('text',
321             '') if isinstance(p, dict) else ''
322             elif isinstance(parts, str):
323                 text = parts
324             text = (text or '').strip()
325             if not text:
326                 return False
327             if any(pfx in text for pfx in _ERROR_PATTERNS):
328                 return False
329             if any(text.startswith(pfx) for pfx in _STATUS_PREFIXE
330             S):
331                 return False
332             if text.startswith('{') and ('"title"' in text or '"er
333             ror"' in text or '"genres"' in text):
334                 return False
335             if 'Ответ модели: {' in text and '"title"' in text:
336                 return False
337             return True
338
339     def _compact_turn(entry: dict) -> dict:
340         parts = entry.get('parts', [])
341         text = ''
342         if isinstance(parts, list) and len(parts) > 0:
343             p = parts[0]
344             text = p if isinstance(p, str) else p.get('text',

```

```

    '') if isinstance(p, dict) else ''
342         elif isinstance(parts, str):
343             text = parts
344
345         if entry.get('role') in ('model', 'assistant') and len
(text) > 200:
346             import re
347             compact = re.sub(r'<film>(.*?)</film>', r'«\1»', t
ext, flags=re.IGNORECASE)
348             compact = re.sub(r'#+\s*', '', compact)
349             compact = re.sub(r'[*_`]+', '', compact)
350             compact = re.sub(r'\s+', ' ', compact).strip()
351             if len(compact) > 200:
352                 compact = compact[:197].rsplit(' ', 1)[0] + '.
..'
353             return {'role': entry['role'], 'parts': [compact]}
354         return entry
355
356     raw_clean = [e for e in history if _is_clean(e)]
357     cleaned_entries: list[dict] = []
358     i = 0
359     while i < len(raw_clean):
360         entry = raw_clean[i]
361         if entry.get('role') == 'user':
362             if i + 1 < len(raw_clean) and raw_clean[i + 1].get
('role') in ('model', 'assistant'):
363                 cleaned_entries.append(_compact_turn(entry))
364                 cleaned_entries.append(_compact_turn(raw_clean
[i + 1]))
365                 i += 2
366                 continue
367             i += 1
368
369     return cleaned_entries[-10:]
370
371 def _build_debug_prompt(request: ChatRequest, user_context_str
: str, voice_gender_rule: str) -> str:
372     """Формирует текстовый дамп полного промпта для отладочног

```

```

о режима. """
373     full_prompt_parts = []
374     dynamic_parts = []
375     if voice_gender_rule:
376         dynamic_parts.append(voice_gender_rule)
377     if user_context_str:
378         dynamic_parts.append(user_context_str)
379     if dynamic_parts:
380         full_prompt_parts.append("— DYNAMIC CONTEXT —\n" + "\n\n".join(dynamic_parts))
381
382     clean_history = _clean_chat_history(request.history)
383     if clean_history:
384         full_prompt_parts.append("— CHAT HISTORY (последние 5 ) —\n" + "\n---\n".join([
385             f"{entry.get('role', 'unknown').upper()}: \n{entry.get('parts', [''])[0] if isinstance(entry.get('parts'), list) else entry.get('parts', '')}"
386             for entry in clean_history[-5:]
387         ]))
388
389     full_prompt_parts.append(f"— USER MESSAGE —\n{request.message}")
390     return "\n\n".join(full_prompt_parts)
391
392 def _resolve_default_model_and_provider(profile: str = "") -> tuple[str, str, str]:
393     """Определяет провайдер и имя модели по умолчанию из активного конфигурационного файла.
394
395     :param profile: Опциональное имя профиля конфигурации.
396     :returns: Кортеж (provider, model_name, config_file).
397     """
398     import json
399     from pathlib import Path
400
401     cfg_env = os.getenv("AIBREADBOARD_CONFIG") or os.getenv("CONFIG_FILE")

```

```

402     active_path = None
403     if cfg_env:
404         p = Path(cfg_env)
405         active_path = p if p.is_absolute() else (__root__ / cf
g_env)
406
407     if not active_path or not active_path.exists():
408         if profile in ("tc", "test-computer", "test_computer",
"apps_tc"):
409             for candidate in [
410                 __root__ / "apps" / "windows" / "config.json",
411                 __root__ / "start_scenarios_config" / "tc.json
",
412                 __root__ / "config" / "tc.json",
413                 __root__ / "config_tc.json",
414                 __root__ / "tc.json",
415             ]:
416                 if candidate.exists():
417                     active_path = candidate
418                     break
419         else:
420             for candidate in [
421                 __root__ / "apps" / "windows" / "config.json",
422                 __root__ / "start_scenarios_config" / "tc.json
",
423                 __root__ / "config" / "tc.json",
424                 __root__ / "config_tc.json",
425                 __root__ / "tc.json",
426             ]:
427                 if candidate.exists() and not (__root__ / "con
fig.json").exists() and not (__root__ / "config" / "dashboard.
json").exists():
428                     active_path = candidate
429                     break
430
431     if not active_path or not active_path.exists():
432         active_path = (__root__ / "start_scenarios_config"
/ "dashboard.json") if (__root__ / "start_scenarios_config" /

```

```

    "dashboard.json").exists() else ((__root__ / "config" / "dash
board.json") if (__root__ / "config" / "dashboard.json").exist
s() else (__root__ / "config.json"))
433
434     provider = "GEMINI"
435     model_name = "gemini-3.1-flash"
436     config_file = active_path.name if active_path else "config
.json"
437
438     if active_path and active_path.exists():
439         try:
440             with open(active_path, "r", encoding="utf-8") as f
:
441                 data = json.load(f)
442                 ai_sec = data.get("ai", {})
443                 # Format in tc.json or explicit provider: ai.p
rovider = "gemini_cli", ai.gemini_cli.model = "..."
444                 if ai_sec.get("provider"):
445                     prov_key = str(ai_sec.get("provider")).low
er()
446                     provider = prov_key.upper()
447                     if isinstance(ai_sec.get(prov_key), dict)
and ai_sec[prov_key].get("model"):
448                         model_name = ai_sec[prov_key].get("mod
el")
449                     elif isinstance(ai_sec.get("providers"), d
ict) and isinstance(ai_sec["providers"].get(prov_key), dict) a
nd ai_sec["providers"][prov_key].get("model"):
450                         model_name = ai_sec["providers"][prov_
key].get("model")
451                     elif ai_sec.get(f"{prov_key}_model_id"):
452                         model_name = ai_sec.get(f"{prov_key}_m
odel_id")
453                     elif ai_sec.get("model"):
454                         model_name = ai_sec.get("model")
455                     # New format: providers.<prov>.enabled + provi
ders.<prov>.model
456                     elif isinstance(ai_sec.get("providers"), dict)

```



```

:
457         providers = ai_sec.get("providers", {})
458         for prov_key, prov_cfg in providers.items(
):
459             if isinstance(prov_cfg, dict) and prov
_cfg.get("enabled"):
460                 provider = prov_key.upper()
461                 model_name = prov_cfg.get("model")
or model_name
462                 break
463             # Legacy format: use_agy, use_gemini, use_olla
ma, use_foundry
464             elif ai_sec.get("use_agy"):
465                 provider = "AGY"
466                 model_name = ai_sec.get("agy_model_id") or
ai_sec.get("model") or "gemini-3.6-flash"
467             elif ai_sec.get("use_gemini"):
468                 provider = "GEMINI"
469                 model_name = ai_sec.get("gemini_model_id")
or ai_sec.get("model") or "gemini-3.1-flash"
470             elif ai_sec.get("use_ollama"):
471                 provider = "OLLAMA"
472                 model_name = ai_sec.get("ollama_model_id")
or ai_sec.get("model") or "llama3.1"
473             elif ai_sec.get("use_foundry"):
474                 provider = "FOUNDRY"
475                 model_name = ai_sec.get("foundry_model_id"
) or ai_sec.get("model") or "local"
476             elif ai_sec.get("model"):
477                 model_name = ai_sec.get("model")
478                 provider = "AI"
479         except Exception as e:
480             logger.debug(f"[router_chat] Could not read active
config {active_path}: {e}")
481
482         return provider, model_name, config_file
483
484 async def _get_effective_model_and_provider(fastapi_req: Reque

```

```

st = None, profile: str = "") -> tuple[str, str, str]:
485     """Получает эффективный провайдер и модель с учетом активн
ой конфигурации и настроек текущего пользователя."""
486     provider, model_name, config_file = _resolve_default_model
_and_provider(profile=profile)
487     if fastapi_req is not None:
488         try:
489             _, _, user_selected_model, _ = await _extract_user
_auth(fastapi_req)
490             if user_selected_model:
491                 if ':' in user_selected_model:
492                     p_prefix, m_suffix = user_selected_model.s
plit(':', 1)
493                     provider = p_prefix.upper()
494                     model_name = m_suffix
495                 elif user_selected_model.startswith('agy-'):
496                     provider = "AGY"
497                     model_name = user_selected_model
498                 elif 'gemini' in user_selected_model.lower():
499                     provider = "GEMINI"
500                     model_name = user_selected_model
501                 else:
502                     model_name = user_selected_model
503             except Exception:
504                 pass
505     return provider, model_name, config_file
506
507 def _normalize_model_and_provider(target_model: str = "", targ
et_provider: str = "") -> tuple[str, str]:
508     """Нормализует имя модели и провайдера, разрешая значения
по умолчанию и форматируя префиксы."""
509     model = (target_model or "").strip()
510     provider = (target_provider or "").strip().lower()
511
512     if not provider:
513         if model.startswith('foundry:'):
514             provider = 'foundry'
515         elif model.startswith('ollama:'):

```

```
516         provider = 'ollama'
517     elif model.startswith('agy-'):
518         provider = 'agy'
519     elif model.startswith('gemini_cli:'):
520         provider = 'gemini_cli'
521     elif model.startswith('hf:') or model.startswith('hf::
522         provider = 'hf'
523     elif model.startswith('onnx:') or model.startswith('on
524         provider = 'onnx'
525     elif any(model.startswith(p) for p in ('openai:', 'dee
526         provider = 'openai'
527     elif 'gemini' in model.lower():
528         provider = 'gemini'
529     else:
530         provider = 'gemini'
531
532     if not model:
533         if provider == 'gemini':
534             model = 'gemini-3.1-flash'
535         elif provider == 'agy':
536             model = 'gemini-3.6-flash'
537         elif provider == 'ollama':
538             model = 'llama3.1'
539         elif provider == 'foundry':
540             model = 'local'
541         elif provider in ('gemini_cli', 'gemini-cli'):
542             model = 'gemini-3-flash-preview'
543         elif provider in ('openai', 'openai_compat', 'openai-c
544             model = 'gpt-4o-mini'
545         elif provider in ('hf', 'huggingface'):
546             model = 'microsoft/Phi-3-mini-4k-instruct'
547         elif provider == 'onnx':
548             model = 'cpu-int4-rt'
```

```

549         else:
550             model = 'gemini-3.1-flash'
551
552     if provider == 'foundry' and not model.startswith('foundry
:'):
553         model = f"foundry:{model}"
554     elif provider == 'ollama' and not model.startswith('ollama
:'):
555         model = f"ollama:{model}"
556     elif provider == 'agy' and not model.startswith('agy-'):
557         model = f"agy-{model}"
558     elif provider in ('gemini_cli', 'gemini-cli') and not mode
l.startswith('gemini_cli:'):
559         model = f"gemini_cli:{model}"
560     elif provider in ('openai', 'openai_compat', 'openai-compa
t', 'deepseek', 'groq', 'openrouter', 'lmstudio'):
561         openai_prefixes = ('openai:', 'deepseek:', 'groq:', 'o
penrouter:', 'lmstudio:', 'local:', 'compat:')
562         if not any(model.startswith(p) for p in openai_prefixe
s):
563             model = f"{provider}:{model}"
564     elif provider in ('hf', 'huggingface') and not model.start
swith('hf:'):
565         model = f"hf:{model}"
566     elif provider == 'onnx' and not model.startswith('onnx:'):
567         model = f"onnx:{model}"
568
569     return model, provider.upper()
570
571 def init_router(chat_model, narrator_model, plugins: dict = {}
) -> APIRouter:
572     """Initialization роутера чата с привязкой моделей (chat и
narrator)."""
573     if hasattr(narrator_model, 'gemini_model') and narrator_mo
del.gemini_model:
574         narrator_model.gemini_model.save_history_chat = False
575
576     @router.get('/models')

```

```

577     async def get_models(fastapi_req: Request, refresh: bool =
| False, include_unsupported: bool = False) -> dict:
578         """Получение списка доступных моделей, сгруппированных
| по провайдеру."""
579         from src.ai.model_manager import get_available_models,
| load_unsupported_models
580
581         gemini_models = get_available_models('gemini', force_r
| efresh=refresh, include_unsupported=include_unsupported)
582
583         foundry_raw = get_available_models('foundry', force_re
| fresh=refresh, include_unsupported=include_unsupported)
584         foundry_models = [f"foundry:{m}" if not m.startswith('
| foundry:') else m for m in foundry_raw]
585
586         ollama_raw = get_available_models('ollama', force_refr
| esh=refresh, include_unsupported=include_unsupported)
587         ollama_models = [f"ollama:{m}" if not m.startswith('ol
| lama:') else m for m in ollama_raw]
588
589         agy_models = get_available_models('agy', force_refresh
| =refresh, include_unsupported=include_unsupported)
590
591         gemini_cli_raw = get_available_models('gemini_cli', fo
| rce_refresh=refresh, include_unsupported=include_unsupported)
592         gemini_cli_models = [f"gemini_cli:{m}" if not m.starts
| with('gemini_cli:') else m for m in gemini_cli_raw]
593
594         openai_raw = get_available_models('openai', force_refr
| esh=refresh, include_unsupported=include_unsupported)
595         openai_models = [f"openai:{m}" if not any(m.startswith
| (f"{p}:") for p in ('openai', 'deepseek', 'groq', 'openrouter'
| , 'lmstudio')) else m for m in openai_raw]
596
597         hf_raw = get_available_models('hf', force_refresh=refr
| esh, include_unsupported=include_unsupported)
598         hf_models = [f"hf:{m}" if not m.startswith('hf:') else
| m for m in hf_raw]

```

```

599 |
600 |         onnx_raw = get_available_models('onnx', force_refresh=
| refresh, include_unsupported=include_unsupported)
601 |         onnx_models = [f"onnx:{m}" if not m.startswith('onnx:')
| ) else m for m in onnx_raw]
602 |
603 |     return {
604 |         'models': {
605 |             'gemini': gemini_models,
606 |             'foundry': foundry_models,
607 |             'ollama': ollama_models,
608 |             'agy': agy_models,
609 |             'gemini_cli': gemini_cli_models,
610 |             'openai': openai_models,
611 |             'hf': hf_models,
612 |             'onnx': onnx_models,
613 |         },
614 |         'unsupported_models': {
615 |             'gemini': list(load_unsupported_models('gemini
| ')),
616 |             'foundry': list(load_unsupported_models('found
| ry')),
617 |             'ollama': list(load_unsupported_models('ollama
| ')),
618 |             'agy': list(load_unsupported_models('agy')),
619 |             'gemini_cli': list(load_unsupported_models('ge
| mini_cli')),
620 |             'openai': list(load_unsupported_models('openai
| ')),
621 |             'hf': list(load_unsupported_models('hf')),
622 |             'onnx': list(load_unsupported_models('onnx')),
623 |         }
624 |     }
625 |
626 |     @router.get('/model')
627 |     @router.get('/active-model')
628 |     async def get_model_endpoint(fastapi_req: Request, profile
| : str = "") -> dict:

```

```

629         """Получение текущей активной AI-модели."""
630         provider, model_name, config_file = await _get_effecti
ve_model_and_provider(fastapi_req, profile=profile)
631         return {
632             "status": "success",
633             "model": model_name,
634             "provider": provider,
635             "display": f"{provider}: {model_name}",
636             "config_file": config_file,
637         }
638
639     @router.post('/model')
640     @router.put('/model')
641     async def set_model_endpoint(req: SetModelPayload, fastapi
_req: Request) -> dict:
642         """Установка активной модели для текущего пользователя
и сессии."""
643         try:
644             target_model = req.model.strip()
645             if not target_model:
646                 raise HTTPException(status_code=400, detail="И
мя модели не может быть пустым")
647
648             normalized_model, provider = _normalize_model_and_
provider(target_model, req.provider)
649             user_idenfier, _, _, _ = await _extract_user_aut
h(fastapi_req)
650
651             try:
652                 from src.user_manager import user_manager
653                 user_id_int = int(user_idenfier) if str(user
_idenfier).isdigit() else 1
654                 await asyncio.to_thread(user_manager.update_us
er_settings, user_id_int, model=normalized_model)
655             except Exception as e:
656                 logger.debug(f"[router_chat] Не удалось обнови
ть model в user_settings: {e}")
657

```

```

658         # Обновление состояния базового chat_model
659         active_chat_ref = getattr(getattr(fastapi_req, "app", None), "state", None)
660         target_chat_model = getattr(active_chat_ref, "chat_model", None) if active_chat_ref else None
661         if not target_chat_model:
662             target_chat_model = chat_model
663         if target_chat_model:
664             if hasattr(target_chat_model, "_model_name"):
665                 target_chat_model._model_name = normalized_model
666             if hasattr(target_chat_model, "model_name"):
667                 target_chat_model.model_name = normalized_model
668
669         return {
670             "status": "success",
671             "message": "Модель успешно обновлена",
672             "model": normalized_model,
673             "provider": provider,
674         }
675     except HTTPException:
676         raise
677     except Exception as exc:
678         logger.error(f"[router_chat] Ошибка при установке модели: {exc}", exc_info=True)
679         raise HTTPException(status_code=500, detail=str(exc))
680
681     @router.get('/provider')
682     async def get_provider_endpoint(fastapi_req: Request, profile: str = "") -> dict:
683         """Получение текущего активного AI-провайдера."""
684         provider, model_name, config_file = await _get_effective_model_and_provider(fastapi_req, profile=profile)
685         return {
686             "status": "success",
687             "provider": provider,

```



```

688         "model": model_name,
689         "display": f"{provider}: {model_name}",
690         "config_file": config_file,
691     }
692
693     @router.post('/provider')
694     @router.put('/provider')
695     async def set_provider_endpoint(req: SetProviderPayload, f
696     astapi_req: Request) -> dict:
697         """Установка активного AI-провайдера для текущего поль
698         зователя и сессии."""
699         try:
700             target_provider = req.provider.strip()
701             if not target_provider:
702                 raise HTTPException(status_code=400, detail="И
703                 мя провайдера не может быть пустым")
704
705             normalized_model, provider = _normalize_model_and_
706             provider(req.model, target_provider)
707             user_idenfier, _, _, _ = await _extract_user_aut
708             h(fastapi_req)
709
710             try:
711                 from src.user_manager import user_manager
712                 user_id_int = int(user_idenfier) if str(user
713                 _idenfifier).isdigit() else 1
714                 await asyncio.to_thread(user_manager.update_us
715                 er_settings, user_id_int, model=normalized_model)
716             except Exception as e:
717                 logger.debug(f"[router_chat] Не удалось обнови
718                 ть provider/model в user_settings: {e}")
719
720             # Обновление состояния базового chat_model
721             active_chat_ref = getattr(getattr(fastapi_req, "ap
722             p", None), "state", None)
723             target_chat_model = getattr(active_chat_ref, "chat
724             _model", None) if active_chat_ref else None
725             if not target_chat_model:

```

```

716         target_chat_model = chat_model
717     if target_chat_model:
718         if hasattr(target_chat_model, "_model_name"):
719             target_chat_model._model_name = normalized_
_model
720         if hasattr(target_chat_model, "model_name"):
721             target_chat_model.model_name = normalized_
model
722
723     return {
724         "status": "success",
725         "message": "Провайдер успешно обновлен",
726         "provider": provider,
727         "model": normalized_model,
728     }
729     except HTTPException:
730         raise
731     except Exception as exc:
732         logger.error(f"[router_chat] Ошибка при установке
провайдера: {exc}", exc_info=True)
733         raise HTTPException(status_code=500, detail=str(ex
c))
734
735     @router.get('/model-instruction')
736     async def get_model_instruction(fastapi_req: Request, mode
l: str = "") -> dict:
737         """Получение текущей системной инструкции для AI-модел
и или чата."""
738         try:
739             user_identifier, system_instruction, selected_mode
l, settings = await _extract_user_auth(fastapi_req)
740             effective_model = model.strip() or selected_model
or ""
741             provider, resolved_model, config_file = _resolve_d
efault_model_and_provider()
742
743             if not effective_model:
744                 effective_model = resolved_model

```

```

745
746         source = "user_settings" if (settings and settings
    .get("system_instruction")) else "file_default"
747
748         return {
749             "status": "success",
750             "instruction": system_instruction,
751             "system_instruction": system_instruction,
752             "model": effective_model,
753             "provider": provider,
754             "source": source,
755         }
756     except Exception as exc:
757         logger.error(f"[router_chat] Ошибка при получении
    системной инструкции: {exc}", exc_info=True)
758         raise HTTPException(status_code=500, detail=str(ex
    c))
759
760     @router.post('/model-instruction')
761     @router.put('/model-instruction')
762     async def set_model_instruction(req: ModelInstructionPaylo
    ad, fastapi_req: Request) -> dict:
763         """Установка и сохранение системной инструкции для AI-
    модели или чата."""
764         try:
765             raw_instruction = req.instruction.strip() or req.s
    ystem_instruction.strip() or req.content.strip()
766             if not raw_instruction:
767                 raise HTTPException(status_code=400, detail="И
    нструкция не может быть пустой")
768
769             user_identifier, _, selected_model, _ = await _ext
    ract_user_auth(fastapi_req)
770
771             # 1. Обновляем настройки пользователя в базе данны
    x
772             try:
773                 from src.user_manager import user_manager

```

```

774         user_id_int = int(user_identifier) if str(user
|_identifier).isdigit() else 1
775         await asyncio.to_thread(user_manager.update_us
|er_settings, user_id_int, system_instruction=raw_instruction)
776     except Exception as e:
777         logger.debug(f"[router_chat] Не удалось обнови
|ть user_settings: {e}")
778
779         # 2. Обновляем базовый chat_model (из app.state ил
|и переданный при инициализации)
780         active_chat_ref = getattr(getattr(fastapi_req, "ap
|p", None), "state", None)
781         target_chat_model = getattr(active_chat_ref, "chat
|_model", None) if active_chat_ref else None
782         if not target_chat_model:
783             target_chat_model = chat_model
784
785         if target_chat_model and hasattr(target_chat_model
|, "update_system_instruction"):
786             try:
787                 target_chat_model.update_system_instructio
|n(raw_instruction)
788             except Exception as e:
789                 logger.debug(f"[router_chat] Не удалось об
|новить chat_model system_instruction: {e}")
790
791         # 3. Обновляем все кэшированные экземпляры _active_c
|hat_models
792         for cached_instance in _active_chat_models.values(
|):
793             if hasattr(cached_instance, "system_instructio
|n"):
794                 setattr(cached_instance, "system_instructi
|on", raw_instruction)
795             elif hasattr(cached_instance, "system_prompt")
|:
796                 setattr(cached_instance, "system_prompt",
|raw_instruction)

```

```

797 |
798 |         # 4. Сохранение на диск в файл prompts/chat/system
      | _instruction.md по запросу
799 |         if req.save_to_disk:
800 |             prompt_file = __root__ / 'prompts' / 'chat' /
      | 'system_instruction.md'
801 |             prompt_file.parent.mkdir(parents=True, exist_o
      | k=True)
802 |             prompt_file.write_text(raw_instruction, encodi
      | ng='utf-8')
803 |
804 |         return {
805 |             "status": "success",
806 |             "message": "Системная инструкция успешно обнов
      | лена",
807 |             "instruction": raw_instruction,
808 |             "system_instruction": raw_instruction,
809 |             "model": req.model or selected_model or "",
810 |         }
811 |     except HTTPException:
812 |         raise
813 |     except Exception as exc:
814 |         logger.error(f"[router_chat] Ошибка при сохранении
      | системной инструкции: {exc}", exc_info=True)
815 |         raise HTTPException(status_code=500, detail=str(ex
      | c))
816 |
817 |     @router.post('/test-model')
818 |     async def test_model(req: TestModelRequest, fastapi_req: R
      | equest) -> dict:
819 |         """Проверочный запрос к указанной AI-модели для валида
      | ции связи (Запрос -> Ответ).
820 |
821 |         Введите вручную имя модели и провайдера, иначе будет и
      | спользована настройка по умолчанию.
822 |         """
823 |         start_time = time.perf_counter()
824 |         target_model = req.model.strip()

```

```

825         provider = req.provider.strip().lower()
826
827         # Если модель или провайдер не указаны вручную, исполь
зуюем настройку по умолчанию
828         if not target_model or not provider:
829             def_provider, def_model, _ = await _get_effective_
model_and_provider(fastapi_req)
830             if not target_model and not provider:
831                 target_model = def_model
832                 provider = def_provider.lower()
833             elif not target_model:
834                 if provider == def_provider.lower() and def_mo
del:
835                     target_model = def_model
836                 elif provider == 'gemini':
837                     target_model = 'gemini-3.1-flash'
838                 elif provider == 'agy':
839                     target_model = 'gemini-3.6-flash'
840                 elif provider == 'ollama':
841                     target_model = 'llama3.1'
842                 elif provider == 'foundry':
843                     target_model = 'local'
844                 else:
845                     target_model = def_model
846             elif not provider:
847                 if target_model.startswith('foundry:'):
848                     provider = 'foundry'
849                 elif target_model.startswith('ollama:'):
850                     provider = 'ollama'
851                 elif target_model.startswith('agy-'):
852                     provider = 'agy'
853                 elif target_model.startswith('gemini_cli:'):
854                     provider = 'gemini_cli'
855                 elif target_model.startswith('hf:') or target_
model.startswith('hf::'):
856                     provider = 'hf'
857                 elif target_model.startswith('onnx:') or targe
t_model.startswith('onnx::'):

```

```

858         provider = 'onnx'
859         elif any(target_model.startswith(p) for p in (
            'openai:', 'deepseek:', 'groq:', 'openrouter:', 'lmstudio:', '
            local:', 'compat:'))):
860             provider = 'openai'
861             elif 'gemini' in target_model.lower():
862                 provider = 'gemini'
863             else:
864                 provider = def_provider.lower()
865
866         if provider == 'foundry' and target_model and not targ
            et_model.startswith('foundry:'):
867             target_model = f"foundry:{target_model}"
868             elif provider == 'ollama' and target_model and not tar
            get_model.startswith('ollama:'):
869                 target_model = f"ollama:{target_model}"
870             elif provider == 'agy' and target_model and not target
            _model.startswith('agy-'):
871                 target_model = f"agy-{target_model}"
872             elif provider in ('gemini_cli', 'gemini-cli') and targ
            et_model and not target_model.startswith('gemini_cli:'):
873                 target_model = f"gemini_cli:{target_model}"
874             elif provider in ('openai', 'openai_compat', 'openai-c
            ompat', 'deepseek', 'groq', 'openrouter', 'lmstudio') and targ
            et_model:
875                 openai_prefixes = ('openai:', 'deepseek:', 'groq:'
            , 'openrouter:', 'lmstudio:', 'local:', 'compat:')
876                 if not any(target_model.startswith(p) for p in ope
            nai_prefixes):
877                     target_model = f"{provider}:{target_model}"
878                 elif provider in ('hf', 'huggingface') and target_mode
            l and not target_model.startswith('hf:'):
879                     target_model = f"hf:{target_model}"
880                 elif provider == 'onnx' and target_model and not targe
            t_model.startswith('onnx:'):
881                     target_model = f"onnx:{target_model}"
882
883         if not target_model:

```

```
884         return {
885             'status': 'error',
886             'message': 'Имя модели не указано',
887             'model': '',
888             'provider': provider,
889             'duration_ms': 0.0
890         }
891
892     test_msg = req.message.strip()
893     if not test_msg:
894         test_msg = "Привет! Назови свою модель и провайдер
а, и подтверди готовность к работе."
895
896     try:
897         model_instance = get_chat_model(target_model, syst
em_instruction=req.system_instruction)
898         response_text = ""
899
900         if hasattr(model_instance, 'ask'):
901             response_text = await model_instance.ask(test_
msg)
902         elif hasattr(model_instance, 'chat'):
903             response_text = await model_instance.chat(test
_msg)
904         elif hasattr(model_instance, 'chat_stream'):
905             chunks = []
906             async for chunk in model_instance.chat_stream(
test_msg):
907                 if chunk:
908                     clean_chunk = chunk.replace("[CHAT]",
"").replace("[VOICE]", "")
909                     if clean_chunk:
910                         chunks.append(clean_chunk)
911             response_text = "".join(chunks)
912         else:
913             raise RuntimeError(f"Модель {target_model} не
поддерживает методы генерации текста")
914
```



```

915         duration_ms = round((time.perf_counter() - start_t
ime) * 1000, 1)
916         return {
917             'status': 'success',
918             'response': response_text,
919             'model': target_model,
920             'provider': provider,
921             'duration_ms': duration_ms
922         }
923     except Exception as exc:
924         logger.error(f"[ChatRouter] Error проверочного зап
роса к модели {target_model}: {exc}", exc_info=True)
925         duration_ms = round((time.perf_counter() - start_t
ime) * 1000, 1)
926         return {
927             'status': 'error',
928             'message': str(exc),
929             'model': target_model,
930             'provider': provider,
931             'duration_ms': duration_ms
932         }
933
934     @router.get('/model-errors')
935     async def get_model_errors_endpoint(provider: str = "", li
mit: int = 50) -> dict:
936         """Получить список последних зафиксированных ошибок AI
-моделей.
937
938         :param provider: Опциональный фильтр по провайдеру (ge
mini, agy, foundry, ollama и т.д.).
939         :param limit: Количество записей.
940         :returns: JSON-ответ со списком событий ошибок.
941         """
942         from src.ai.orchestration.model_error_hub import get_m
odel_errors
943         events = get_model_errors(provider=provider or None, l
imit=limit)
944         return {

```

```

945         'status': 'success',
946         'errors': [evt.model_dump() for evt in events],
947         'count': len(events),
948     }
949
950     @router.get('/model-health')
951     async def get_model_health_endpoint() -> dict:
952         """Получить сводную статистику здоровья и доступности
AI-моделей.
953
954         :returns: Агрегированная сводка по ошибкам и провайдер
ам.
955         """
956         from src.ai.orchestration.model_error_hub import get_m
odel_health_summary
957         summary = get_model_health_summary()
958         return {
959             'status': 'success',
960             'health': summary,
961         }
962
963
964     @router.post('/save-rag')
965     async def save_to_rag(rag_req: SaveRagRequest, request: Re
quest):
966         """Ручное сохранение одобренного ответа в постоянный J
SON-архив."""
967         try:
968             user_identifier, _, _, _ = await _extract_user_aut
h(request)
969             from src.rag import save_user_approved_response
970             save_success = await asyncio.to_thread(
971                 save_user_approved_response,
972                 user_identifier, rag_req.query, rag_req.chat_t
ext, rag_req.voice_text
973             )
974             if save_success:
975                 return {"status": "success", "message": "Succe

```

```

ssfully сохранено для последующей компиляции RAG"}
976         else:
977             raise HTTPException(status_code=500, detail="E
rror сохранения ответа")
978         except Exception as e:
979             logger.error("Error при ручном сохранении ответа",
e)
980             raise HTTPException(status_code=500, detail=str(e)
)
981
982     @router.post('/save-rag-instant')
983     async def save_to_rag_instant(rag_req: SaveRagRequest, req
uest: Request):
984         """Мгновенное сохранение ответа: запись в JSON + векто
ризация в FAISS."""
985         try:
986             user_identifier, _, _, _ = await _extract_user_aut
h(request)
987             api_key = getattr(chat_model, 'api_key', '') or os
.getenv('GEMINI_API_KEY', '')
988
989             content_to_index = rag_req.voice_text if rag_req.v
oice_text.strip() else rag_req.chat_text
990
991             from src.rag import save_user_approved_response, i
ndex_user_interaction
992             save_success = await asyncio.to_thread(
993                 save_user_approved_response,
994                 user_identifier, rag_req.query, rag_req.chat_t
ext, rag_req.voice_text
995             )
996             rag_success = await asyncio.to_thread(
997                 index_user_interaction, user_identifier, api_k
ey, rag_req.query, content_to_index
998             )
999
1000             if save_success and rag_success:
1001                 return {"status": "success", "message": "Succe

```

```

ssfully сохранено в архив и проиндексировано в RAG"}
1002         elif save_success:
1003             return {"status": "success", "message": "Coxpa
нено в архив, но произошла Error при индексации в RAG"}
1004         else:
1005             raise HTTPException(status_code=500, detail="E
rror сохранения ответа")
1006     except Exception as e:
1007         logger.error("Error при мгновенном сохранении в RA
G", e)
1008         raise HTTPException(status_code=500, detail=str(e)
)
1009
1010     @router.get('/sessions')
1011     async def get_all_sessions(request: Request):
1012         """Retrieve all chat sessions for the current user or
host."""
1013         try:
1014             user_idenfier, __, __, __ = await _extract_user_aut
h(request)
1015             sessions = await asyncio.to_thread(chat_sessions_d
b.list_sessions, user_idenfier)
1016             return {"status": "success", "sessions": sessions}
1017         except Exception as e:
1018             logger.error("Error retrieving chat sessions", e)
1019             raise HTTPException(status_code=500, detail=str(e)
)
1020
1021     @router.get('/sessions/{session_id}')
1022     async def get_single_session(session_id: str, request: Req
uest):
1023         """Retrieve a specific chat session by ID."""
1024         try:
1025             user_idenfier, __, __, __ = await _extract_user_aut
h(request)
1026             session_data = await asyncio.to_thread(chat_sessio
ns_db.get_session, session_id, user_idenfier)
1027             if not session_data:

```

```

1028         raise HTTPException(status_code=404, detail="S
| session not found")
1029         return {"status": "success", "session": session_da
| ta}
1030     except HTTPException:
1031         raise
1032     except Exception as e:
1033         logger.error(f"Error retrieving chat session {sess
| ion_id}", e)
1034         raise HTTPException(status_code=500, detail=str(e)
| )
1035
1036     @router.post('/sessions')
1037     async def save_session_endpoint(payload: ChatSessionPayloa
| d, request: Request):
1038         """Save or update a chat session."""
1039         try:
1040             user_identifier, _, _, _ = await _extract_user_aut
| h(request)
1041             data = payload.model_dump()
1042             saved = await asyncio.to_thread(chat_sessions_db.s
| ave_session, data, user_identifier)
1043             return {"status": "success", "session": saved}
1044         except Exception as e:
1045             logger.error("Error saving chat session", e)
1046             raise HTTPException(status_code=500, detail=str(e)
| )
1047
1048     @router.post('/sessions/sync')
1049     async def sync_sessions_endpoint(sync_req: SyncSessionsReq
| uest, request: Request):
1050         """Bulk synchronize chat sessions from client to serve
| r."""
1051         try:
1052             user_identifier, _, _, _ = await _extract_user_aut
| h(request)
1053             merged = await asyncio.to_thread(chat_sessions_db.
| bulk_sync_sessions, sync_req.sessions, user_identifier)

```

```

1054         return {"status": "success", "sessions": merged}
1055     except Exception as e:
1056         logger.error("Error synchronizing chat sessions",
1057             e)
1058         raise HTTPException(status_code=500, detail=str(e)
1059     )
1060
1061     @router.delete('/sessions/{session_id}')
1062     async def delete_session_endpoint(session_id: str, request
1063         : Request):
1064         """Delete a specific chat session."""
1065         try:
1066             user_idenfier, _, _, _ = await _extract_user_auth(request)
1067             deleted = await asyncio.to_thread(chat_sessions_db
1068                 .delete_session, session_id, user_idenfier)
1069             return {"status": "success", "deleted": deleted}
1070         except Exception as e:
1071             logger.error(f"Error deleting chat session {session_id}", e)
1072             raise HTTPException(status_code=500, detail=str(e)
1073         )
1074
1075     @router.delete('/sessions')
1076     async def clear_all_sessions_endpoint(request: Request):
1077         """Clear all chat sessions for the current user."""
1078         try:
1079             user_idenfier, _, _, _ = await _extract_user_auth(request)
1080             cleared = await asyncio.to_thread(chat_sessions_db
1081                 .clear_sessions, user_idenfier)
1082             return {"status": "success", "cleared": cleared}
1083         except Exception as e:
1084             logger.error("Error clearing chat sessions", e)
1085             raise HTTPException(status_code=500, detail=str(e)
1086         )
1087
1088     @router.post('')

```

```

1082
1083     async def chat(chat_req: ChatRequest, request: Request):
1084         """Обработка чат-запроса по алгоритму RAG-First: RAG S
1085         earch -> Direct Match / LLM Fallback -> Auto-Index."""
1086         from fastapi.responses import StreamingResponse
1087         import json
1088
1089         async def event_generator():
1090             try:
1091                 user_idenfifier, system_instruction, selected_
1092                 model, settings = await _extract_user_auth(request)
1093
1094                 api_key = getattr(chat_model, 'api_key', '') o
1095                 r os.getenv('GEMINI_API_KEY', '')
1096
1097                 # 1. RAG: Поиск по базе знаний (опционально, п
1098                 о умолчанию отключен)
1099                 is_rag_active = bool(chat_req.generation_confi
1100                 g.get('rag_enabled', False))
1101
1102                 context_text = ""
1103                 if is_rag_active:
1104                     from src.rag import get_rag_engine
1105                     rag_engine = get_rag_engine()
1106
1107                     top_k = int(chat_req.generation_config.get
1108                     ('top_k', 3))
1109                     threshold = float(chat_req.generation_conf
1110                     ig.get('min_score', chat_req.generation_config.get('threshold'
1111                     , 0.45)))
1112
1113                     yield f"data: {json.dumps({'status': '❏ П
1114                     оиск в базе знаний (RAG)...'})}\n\n"
1115                     decision = await rag_engine.evaluate(
1116                         query=chat_req.message,
1117                         user_idenfifier=user_idenfifier,
1118                         api_key=api_key,
1119                         threshold=threshold,

```

```

1111         top_k=top_k
1112     )
1113
1114     # 2. Если найден точный ответ – мгновенный
    возврат (Direct RAG)
1115     if decision.is_direct:
1116         yield f"data: {json.dumps({'status': d
    ecision.status_message or '␣ Ответ найден в базе знаний...'})}
    \n\n"
1117         yield f"data: {json.dumps({'text': dec
    ision.direct_text})}\n\n"
1118         return
1119         context_text = decision.context_text
1120
1121     # 3. Подготовка контекста для LLM
1122     dynamic_context_parts = []
1123     if context_text:
1124         dynamic_context_parts.append(context_text)
1125
1126     user_msg_with_context = chat_req.message
1127     if dynamic_context_parts:
1128         user_msg_with_context = "\n\n".join(dynami
    c_context_parts) + "\n\n[Запрос пользователя]:\n" + chat_req.m
    essage
1129
1130     # Режим отладки (DEBUG MODE)
1131     if chat_req.generation_config.get('debug_mode'
    , False):
1132         voice_gender_instruction = _get_voice_gend
    er_rule(settings)
1133         debug_text = _build_debug_prompt(chat_req,
    context_text, voice_gender_instruction)
1134         yield f"data: {json.dumps({'status': 'DEBU
    G MODE: Промпт сформирован, не отправляется в модель'})}\n\n"
1135         yield f"data: {json.dumps({'text': debug_t
    ext})}\n\n"
1136         return
1137

```



```

1138 |         # Интеллектуальный роутинг: Агент анализа сист
      | емных логов Windows (System Logs Analyzer Agent)
1139 |         msg_lower = chat_req.message.lower()
1140 |         is_system_logs_request = any(
1141 |             phrase in msg_lower for phrase in (
1142 |                 'критические ошибки в операционной сис
      | теме',
1143 |                 'критических ошибок в операционной сис
      | теме',
1144 |                 'ошибки в операционной системе',
1145 |                 'ошибок в операционной системе',
1146 |                 'критические ошибки в ос',
1147 |                 'критических ошибок в ос',
1148 |                 'ошибки в ос',
1149 |                 'ошибок в ос',
1150 |                 'system logs analyzer',
1151 |                 'журнал событий windows',
1152 |                 'журналы событий windows',
1153 |                 'системные логи windows',
1154 |                 'системных логах windows',
1155 |             )
1156 |         )
1157 |
1158 |         if is_system_logs_request:
1159 |             yield f"data: {json.dumps({'status': '❌ 3
      | аныск System Logs Analyzer Agent...'})}\n\n"
1160 |             from src.ai.agents.system_logs_agent impor
      | t SystemLogsAgent
1161 |             agent = SystemLogsAgent()
1162 |
1163 |             selected_model_name = chat_req.generation_
      | config.get('model') or selected_model or 'gemini-3.1-flash'
1164 |             active_model = get_chat_model(selected_mod
      | el_name, system_instruction or "", user_id=user_identifier)
1165 |
1166 |             yield f"data: {json.dumps({'status': '❌ С
      | бор и кластеризация событий из Windows Event Log...'})}\n\n"
1167 |             agent_report = await agent.run(chat_req.me

```

```

ssage, active_llm=active_model)
1168
1169         yield f"data: {json.dumps({'text': agent_r
eport}})}\n\n"
1170         return
1171
1172         token = request.cookies.get('auth_token') if r
equest else None
1173         from src.api.routers.core.router_control impor
t get_room_id
1174         room_id = get_room_id(token, None)
1175
1176         if chat_req.generation_config.get('model'):
1177             selected_model = chat_req.generation_conf
g['model']
1178
1179         clean_history = _clean_chat_history(chat_req.h
istory)
1180         kwargs = {
1181             'history': clean_history,
1182             'room_id': room_id,
1183             'model_name': selected_model,
1184         }
1185         if chat_req.generation_config.get('search_engi
ne'):
1186             kwargs['search_engine'] = chat_req.generat
ion_config['search_engine']
1187
1188         if selected_model:
1189             active_model = get_chat_model(selected_mod
el, system_instruction or "", user_id=user_idenfier)
1190             api_key = getattr(active_model, 'api_key',
'') or getattr(chat_model, 'api_key', '') or api_key
1191         else:
1192             active_model = chat_model
1193
1194         yield f"data: {json.dumps({'status': 'Генерац
ия ответа...'})}\n\n"

```

```

1195 |
1196 |         chat_kwargs = kwargs.copy()
1197 |         chat_kwargs.pop('room_id', '')
1198 |         chat_kwargs.pop('search_engine', None)
1199 |         gen_cfg = chat_req.generation_config.copy()
1200 |         gen_cfg['response_type'] = 'chat'
1201 |         chat_kwargs['generation_config'] = gen_cfg
1202 |
1203 |         stream_generator = active_model.chat_stream(us
er_msg_with_context, **chat_kwargs)
1204 |
1205 |         chat_response = ""
1206 |         async for chunk in stream_generator:
1207 |             if chunk:
1208 |                 c = chunk.replace("[CHAT]", "").replac
e("[VOICE]", "")
1209 |                 if c:
1210 |                     chat_response += c
1211 |                     yield f"data: {json.dumps({'text':
c})}\n\n"
1212 |
1213 |         # 4. Фоновая индексация взаимодействия в RAG (
только если RAG был явно включен)
1214 |         if is_rag_active and chat_response and api_key
and user_identifier:
1215 |             from src.rag import index_user_interaction
1216 |             asyncio.ensure_future(asyncio.to_thread(
index_user_interaction, user_identifie
r, api_key, chat_req.message, chat_response
1218 |                 ))
1219 |
1220 |         except Exception as ex:
1221 |             logger.error('Error обработки чат-запроса', ex
)
1222 |             yield f"data: {json.dumps({'error': str(ex)})}
\n\n"
1223 |
1224 |         return StreamingResponse(event_generator(), media_type

```

```
|="text/event-stream")
```

```
1225 |
```

```
1226 |     return router
```

```
1227 |
```

```
1228 |
```

router_control.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Control
4 # =====
5 # Description:
6 #   Manages WebSocket connections and REST endpoints for synchronized remote control,
7 #
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.routers.router_control import ControlConnectionManager
11 #
12 #       service = ControlConnectionManager()
13 #
14 # File: router_control.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-04 07:47:30
20 # =====
21
22 from __future__ import annotations
23 """Manages WebSocket connections and REST endpoints for synchronized remote control."""
24
25 import json
26 from typing import Dict, List, Optional
27 from fastapi import APIRouter, WebSocket, WebSocketDisconnect, Query, Request, HTTPException
28 from logger import logger
```

```

29 from src.api.routers.core.router_auth import verify_jwt_token
30
31 class ControlConnectionManager:
32     """Менеджер WebSocket соединений для пульта, плеера и откр
    ытых компьютеров."""
33
34     def __init__(self):
35         # Хранение активных подключений: room_id -> {"player":
    [ws...], "remote": [ws...], "computer": [ws...]}
36         self.rooms: Dict[str, Dict[str, List[WebSocket]]] = {}
37         # Хранение последнего известного состояния плеера: roo
    m_id -> dict
38         self.room_states: Dict[str, dict] = {}
39         # Хранение плейлиста плеера: room_id -> list
40         self.room_playlists: Dict[str, list] = {}
41
42     async def connect(self, websocket: WebSocket, room_id: str
    , role: str):
43         """Connection клиента с ролью 'player', 'remote' или '
    computer' в комнату."""
44         await websocket.accept()
45         if room_id not in self.rooms:
46             self.rooms[room_id] = {"player": [], "remote": [],
    "computer": []}
47         elif "computer" not in self.rooms[room_id]:
48             self.rooms[room_id]["computer"] = []
49
50         if role not in ("player", "remote", "computer"):
51             role = "remote"
52
53         self.rooms[room_id][role].append(websocket)
54         logger.info(f"WebSocket connected: room={room_id}, rol
    e={role}")
55
56         # Если подключается remote и у нас есть сохраненное st
    ate, сразу отправляем его
57         if role == "remote":
58             if room_id in self.room_states:

```

```

59         try:
60             await websocket.send_json(self.room_states
        [room_id])
61         except Exception as e:
62             logger.warning(f"Error sending state on co
        nnect to remote: {e}")
63         if room_id in self.room_playlists:
64             try:
65                 await websocket.send_json({
66                     "event": "playlist_update",
67                     "files": self.room_playlists[room_id]
68                 })
69             except Exception as e:
70                 logger.warning(f"Error sending playlist on
        connect to remote: {e}")
71
72     def disconnect(self, websocket: WebSocket, room_id: str, r
        ole: str):
73         """Disconnection клиента из комнаты."""
74         if room_id in self.rooms and role in self.rooms[room_i
        d]:
75             if websocket in self.rooms[room_id][role]:
76                 self.rooms[room_id][role].remove(websocket)
77                 # Если комната пуста, очищаем её
78                 has_active_connections = any(len(clients) > 0 for
        clients in self.rooms[room_id].values())
79                 if not has_active_connections:
80                     self.rooms.pop(room_id, None)
81                     self.room_states.pop(room_id, None)
82                     self.room_playlists.pop(room_id, None)
83                 logger.info(f"WebSocket disconnected: room={room_id},
        role={role}")
84
85     async def broadcast_to_role(self, room_id: str, target_rol
        e: str, data: dict):
86         """Пересылает сообщение всем клиентам с целевой ролью
        в комнате."""
87         if room_id not in self.rooms:

```

```

88         return
89
90         targets = self.rooms[room_id].get(target_role, [])
91         disconnected = []
92
93         for ws in targets:
94             try:
95                 await ws.send_json(data)
96             except Exception as e:
97                 logger.warning(f"Failed to send to {target_role}
98 e} in room {room_id}: {e}")
99                 disconnected.append(ws)
100
101         for ws in disconnected:
102             self.disconnect(ws, room_id, target_role)
103
104 manager = ControlConnectionManager()
105 router = APIRouter(prefix="/api/control", tags=["control"])
106
107 def get_room_id(token: Optional[str], room: Optional[str]) ->
108 str:
109     """Определение идентификатора комнаты на основе JWT токена
110 или параметров."""
111     if room and room.strip():
112         return room.strip().lower()
113     if token:
114         token_data = verify_jwt_token(token)
115         if token_data and token_data.email:
116             return token_data.email.strip().lower()
117     return "default"
118
119 @router.websocket("/ws")
120 async def websocket_control_endpoint(
121     websocket: WebSocket,
122     role: str = Query("remote"),
123     token: Optional[str] = Query(None),
124     room: Optional[str] = Query(None)
125 ):

```



```

123     """WebSocket эндпоинт для управления плеером и чатом в реа
    льном времени."""
124     cookie_token = websocket.cookies.get("auth_token")
125     logger.info(f"WS connection request: role={role}, query_to
    ken={'present' if token else 'None'}, query_room={room}")
126     if cookie_token and not token:
127         token = cookie_token
128
129     room_id = get_room_id(token, room)
130     logger.info(f"WS room resolved: role={role}, room_id={room
    _id}")
131     await manager.connect(websocket, room_id, role)
132
133     try:
134         while True:
135             text_data = await websocket.receive_text()
136             try:
137                 data = json.loads(text_data)
138             except json.JSONDecodeError:
139                 continue
140
141             if role == "remote":
142                 action = data.get("action")
143                 # Если пульт передает текстовую транскрибацию
144                 if action == "transcript":
145                     transcript_payload = {
146                         "event": "transcript",
147                         "text": data.get("text", ""),
148                         "source": "remote"
149                     }
150                     # Транслируем на открытые компьютеры
151                     await manager.broadcast_to_role(room_id, "
    computer", transcript_payload)
152                 else:
153                     # Пульт sends команды плееру
154                     await manager.broadcast_to_role(room_id, "
    player", data)
155             elif role == "player":

```

```

156         # Плеер обновляет свой status или плейлист для
        пульта
157         event_type = data.get("event")
158         if event_type == "status_update":
159             manager.room_states[room_id] = data
160         elif event_type == "playlist_update":
161             manager.room_playlists[room_id] = data.get
        ("files", [])
162
163         await manager.broadcast_to_role(room_id, "remo
        te", data)
164         elif role == "computer":
165             # Компьютер может отправлять подтверждения или
        состояние
166             event_type = data.get("event")
167             if event_type:
168                 await manager.broadcast_to_role(room_id, "
        remote", data)
169
170     except WebSocketDisconnect:
171         manager.disconnect(websocket, room_id, role)
172     except Exception as e:
173         logger.error(f"WebSocket error in room {room_id}: {e}"
        )
174         manager.disconnect(websocket, room_id, role)
175
176 @router.get("/status")
177 async def get_control_status(request: Request, token: Optional
        [str] = None, room: Optional[str] = None):
178     """Получить текущее state комнаты по HTTP."""
179     from src.api.routers.core.router_auth import get_current_u
        ser_data
180     get_current_user_data(request)
181     cookie_token = request.cookies.get("auth_token")
182     if cookie_token and not token:
183         token = cookie_token
184
185     room_id = get_room_id(token, room)

```

```

186         return {
187             "room_id": room_id,
188             "has_player": len(manager.rooms.get(room_id, {}).get("
player", [])) > 0,
189             "has_computer": len(manager.rooms.get(room_id, {}).get
("computer", [])) > 0,
190             "computers_count": len(manager.rooms.get(room_id, {}).
get("computer", [])),
191             "remotes_count": len(manager.rooms.get(room_id, {}).ge
t("remote", [])),
192             "state": manager.room_states.get(room_id, None),
193             "playlist_count": len(manager.room_playlists.get(room_
id, []))
194         }
195
196 @router.get("/active_players")
197 async def get_active_players(request: Request):
198     """Получить list комнат с активными компьютерами и плеерам
и."""
199     from src.api.routers.core.router_auth import get_current_u
ser_data
200     user_data = get_current_user_data(request)
201     user_email = user_data.email.strip().lower() if user_data
and user_data.email else None
202
203     active_players = []
204     active_computers = []
205     for room_id, roles in manager.rooms.items():
206         if len(roles.get("player", [])) > 0:
207             active_players.append(room_id)
208         if len(roles.get("computer", [])) > 0:
209             active_computers.append(room_id)
210
211     # Объединенный список активных комнат для совместимости
212     all_active = list(dict.fromkeys(active_computers + active_
players))
213     logger.info(f"active_players endpoint called: user_email={
user_email}, computers={active_computers}, players={active_pla

```

```

    yers}")
214     return {
215         "players": all_active,
216         "computers": active_computers,
217         "media_players": active_players,
218         "user_email": user_email
219     }
220
221 @router.get("/rescan")
222 async def rescan_storage(request: Request):
223     """Принудительное пересканирование доступных хранилищ."""
224     from src.api.routers.core.router_auth import require_admin
225     _user
226     require_admin_user(request)
227
228     import psutil
229     drives = []
230     drives_details = []
231     for d in psutil.disk_partitions():
232         drives.append(d.device)
233         try:
234             usage = psutil.disk_usage(d.mountpoint)
235             total_gb = round(usage.total / (1024 ** 3), 1)
236             used_gb = round(usage.used / (1024 ** 3), 1)
237             free_gb = round(usage.free / (1024 ** 3), 1)
238             percent = usage.percent
239             drives_details.append({
240                 "device": d.device,
241                 "mountpoint": d.mountpoint,
242                 "fstype": d.fstype,
243                 "total_gb": total_gb,
244                 "used_gb": used_gb,
245                 "free_gb": free_gb,
246                 "percent": percent
247             })
248         except Exception:
249             drives_details.append({
                "device": d.device,

```

```
250         "mountpoint": d.mountpoint,
251         "fstype": d.fstype,
252         "total_gb": 0.0,
253         "used_gb": 0.0,
254         "free_gb": 0.0,
255         "percent": 0.0
256     })
257     return {"status": "success", "drives": drives, "details":
drives_details}
258
259 def init_router() -> APIRouter:
260     """Initialization роутера управления.
261     Returns:
262         APIRouter: Настроенный роутер FastAPI.
263     """
264     return router
265
266
```

router_cpu_load.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router CPU Load
4 # =====
5 # Description:
6 #   GET /api/v1/panel/cpu-load – иерархические метрики процессора:
7 #   физические ядра -> логические потоки -> показатели потоков /ядер ->
8 #   общие показатели CPU и история загрузки/температур/мощности.
9 #   Запросы к базе данных производятся на уровне телеметрии.
10 #
11 # Usage Examples:
12 #   Python API:
13 #       from apps.windows.api.routers.router_cpu_load import init_router
14 #       app.include_router(init_router())
15 #
16 # File: router_cpu_load.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 06:52:00
22 # =====
23
24 from __future__ import annotations
25
26 """Роутер панели «Загрузка CPU»: иерархия CPU -> ядра -> потоки и -> сенсоры из telemetry.db и LHM."""
27
```

```
28 import json
29 import platform
30 import re
31 from datetime import datetime, timezone
32 from typing import Any, Dict, List, Optional, Tuple
33
34 from fastapi import APIRouter
35 from pydantic import BaseModel, Field
36
37 from logger import logger
38 from apps.windows.modules.hardware.lhm_service import LhmService
39 from apps.windows.telemetry.sqlite import TelemetryStorage
40
41 _CORE_THREAD_RE = re.compile(r"core[\s_#]*(\d+)\s+thread[\s_#]
42 *(\d+)", re.IGNORECASE)
43 _CORE_ONLY_RE = re.compile(r"core[\s_#]*(\d+)", re.IGNORECASE)
44 _THREAD_ONLY_RE = re.compile(r"thread[\s_#]*(\d+)", re.IGNORECASE)
45
46 _TEMP_CATEGORIES = ("temperature", "temperatures")
47
48 class CpuThreadMetric(BaseModel):
49     """Метрики отдельного логического потока CPU."""
50     index: int = Field(..., description="Глобальный порядковый
51 индекс потока (1..N)")
52     core_thread_index: int = Field(default=1, description="Пор
53 ядковый номер потока внутри своего физического ядра (1..2)")
54     name: str = Field(default="", description="Название потока
55 (например, Thread #1)")
56     core_index: int = Field(default=1, description="Индекс род
57 ительского физического ядра (1..M)")
58     core_name: str = Field(default="", description="Имя родите
59 льского ядра (например, Core #1)")
60     load_percent: float = Field(default=0.0, description="Загру
61 зка логического процессора, %")
```

```
57 class CpuCoreMetric(BaseModel):
58     """Метрики одного физического ядра CPU со списком принадле
    жащих ему потоков."""
59     index: int = Field(..., description="Индекс физического яд
    ра (1..M)")
60     name: str = Field(default="", description="Название ядра (
    например, Core #1)")
61     load_percent: Optional[float] = Field(default=None, descri
    ption="Загрузка ядра, %")
62     temperature_c: Optional[float] = Field(default=None, descr
    iption="Температура ядра, °C")
63     frequency_mhz: Optional[float] = Field(default=None, descr
    iption="Частота ядра, МГц")
64     frequency_str: str = Field(default="", description="Формат
    ированная частота (например, 3.87 GHz)")
65     power_w: Optional[float] = Field(default=None, description
    ="Энергопотребление ядра, Вт")
66     voltage_v: Optional[float] = Field(default=None, descripti
    on="Напряжение ядра, В")
67     threads: List[CpuThreadMetric] = Field(default_factory=lis
    t, description="Логические потоки физического ядра")
68     timestamp: str = Field(default="", description="Время посл
    еднего замера")
69
70
71 class CpuSensorMetric(BaseModel):
72     """Общий системный сенсор процессора (датчик питания, напр
    яжения, шины, кулера)."""
73     id: str = Field(..., description="Уникальный идентификатор
    сенсора")
74     name: str = Field(..., description="Человекочитаемое назва
    ние сенсора")
75     category: str = Field(default="General", description="Кате
    гория сенсора (Powers, Voltages, Clocks, etc.)")
76     unit: str = Field(default="", description="Единица измерен
    ия (W, V, MHz, GHz, °C, RPM, %)")
77     value: float = Field(default=0.0, description="Числовое зн
    ачение")
```



```
78     value_raw: str = Field(default="", description="Форматированное строковое значение с единицей")
79     status: str = Field(default="normal", description="Статус сенсора: normal, warning, danger")
80
81
82 class CpuHistoryPoint(BaseModel):
83     """Точка временного ряда истории параметров CPU."""
84     timestamp: str = Field(default="", description="ISO время точки")
85     time_label: str = Field(default="", description="Короткая метка времени ЧЧ:ММ:СС")
86     load_percent: float = Field(default=0.0, description="Общая загрузка CPU, %")
87     temperature_c: Optional[float] = Field(default=None, description="Температура Package, °C")
88     power_w: Optional[float] = Field(default=None, description="Мощность Package, Вт")
89
90
91 class CpuSpecsInfo(BaseModel):
92     """Сводная спецификация и паспортные данные процессора."""
93     name: str = Field(default="", description="Наименование модели")
94     vendor: str = Field(default="Intel", description="Производитель")
95     architecture: str = Field(default="x86_64", description="Архитектура")
96     socket: str = Field(default="", description="Разъем / Сокет")
97     physical_cores: int = Field(default=0, description="Физические ядра")
98     logical_cores: int = Field(default=0, description="Логические потоки")
99     base_frequency_mhz: float = Field(default=0.0, description="Базовая частота (МГц)")
100    max_frequency_mhz: float = Field(default=0.0, description="Макс. частота (МГц)")
```

```
101     base_frequency_str: str = Field(default="", description="Форматированная базовая частота")
102     max_frequency_str: str = Field(default="", description="Форматированная макс. частота")
103     l2_cache_kb: Optional[int] = Field(default=None, description="Кэш L2 (КБ)")
104     l3_cache_kb: Optional[int] = Field(default=None, description="Кэш L3 (КБ)")
105     l2_cache_str: str = Field(default="", description="Форматированный кэш L2")
106     l3_cache_str: str = Field(default="", description="Форматированный кэш L3")
107     cache_combined_str: str = Field(default="", description="Объединенная строка кэша L2 / L3")
108     stepping: str = Field(default="", description="Степпинг / Revision")
109     features: List[str] = Field(default_factory=list, description="Набор инструкций")
110     description: str = Field(default="", description="Полное описание процессора (например, Intel64 Family 6 Model 165 Stepping 3, GenuineIntel (12 logical cores))")
111     raw_descriptor: str = Field(default="", description="Строка идентификатора CUID / platform.processor()")
112
113
114 class CpuLoadResponse(BaseModel):
115     """Иерархический ответ GET /api/v1/panel/cpu-load."""
116     status: str = Field(default="ok", description="Статус ответа")
117     name: str = Field(default="CPU", description="Модель процессора")
118     description: str = Field(default="", description="Полное описание процессора")
119     cpus: List[Dict[str, Any]] = Field(default_factory=list, description="Список всех процессоров хоста с описаниями")
120     total_percent: float = Field(default=0.0, description="Суммарная загрузка CPU, %")
121     package_temperature_c: Optional[float] = Field(default=None, description="Температура процессора, °C")
```

```

    e, description="Температура корпуса CPU (Package), °C")
122     package_power_w: Optional[float] = Field(default=None, des
    cription="Энергопотребление процессора (Package Power), Вт")
123     cores_count: int = Field(default=0, description="Количеств
    о физических ядер")
124     threads_count: int = Field(default=0, description="Количес
    тво логических потоков")
125     specs: Optional[CpuSpecsInfo] = Field(default=None, descri
    ption="Сводный паспорт и спецификации CPU")
126     cores: List[CpuCoreMetric] = Field(default_factory=list, d
    escription="Физические ядра и их потоки")
127     threads: List[CpuThreadMetric] = Field(default_factory=lis
    t, description="Сетка всех логических процессоров")
128     sensors: List[CpuSensorMetric] = Field(default_factory=lis
    t, description="Общие датчики системы и процессора")
129     history: List[CpuHistoryPoint] = Field(default_factory=lis
    t, description="История параметров CPU")
130     meta: Dict[str, Any] = Field(default_factory=dict, descrip
    tion="Метаданные запроса")
131
132
133 def _calculate_sensor_status(category: str, value: float) -> s
    tr:
134     """Определяет статус сенсора по категории и значению."""
135     cat = category.lower()
136     if "temp" in cat:
137         if value >= 80.0:
138             return "danger"
139         if value >= 68.0:
140             return "warning"
141         return "normal"
142     if "load" in cat:
143         if value >= 90.0:
144             return "danger"
145         if value >= 75.0:
146             return "warning"
147         return "normal"
148     return "normal"

```

```
149
150
151 def _format_frequency(mhz: Optional[float]) -> str:
152     """Форматирует частоту в МГц или ГГц."""
153     if mhz is None or mhz <= 0:
154         return "--"
155     if mhz >= 1000.0:
156         return f"{mhz / 1000.0:.2f} GHz"
157     return f"{mhz:.0f} MHz"
158
159
160 def _normalize_sensor_name(sid: str, name: str, cat: str) -> str:
161     """Формирует понятное название для общего CPU-сенсора без
162     Unknown."""
163     raw = (name or "").strip()
164     if not raw or raw.lower() in ("unknown", "sensor"):
165         return f"Sensor #{sid}" if sid else "System Sensor"
166
167     # Приводим частые LHM имена к аккуратному виду
168     cat_l = cat.lower()
169     raw_l = raw.lower()
170
171     if raw_l == "cpu package" and "power" in cat_l:
172         return "Package Power"
173     if raw_l == "cpu package" and "temp" in cat_l:
174         return "CPU Temperature"
175     if raw_l == "cpu cores" and "power" in cat_l:
176         return "Cores Power"
177     if raw_l == "cpu core" and "volt" in cat_l:
178         return "Core Voltage"
179     if raw_l == "bus speed":
180         return "Bus Frequency"
181     if raw_l == "core max" and "temp" in cat_l:
182         return "Core Max Temp"
183     if raw_l == "core average" and "temp" in cat_l:
184         return "Core Average Temp"
```

```

185 |     return raw
186 |
187 |
188 | def build_cpu_load(
189 |     sensors: List[Dict[str, Any]],
190 |     snapshot: Optional[Dict[str, Any]] = None,
191 |     history_rows: Optional[List[Dict[str, Any]]] = None,
192 |     inventory: Optional[Dict[str, Any]] = None,
193 | ) -> CpuLoadResponse:
194 |     """Собирает иерархический ответ CPU -> физические ядра ->
    | потоки -> сенсоры -> история.
195 |
196 |     Args:
197 |         sensors: Записи последних замеров сенсоров.
198 |         snapshot: Данные последнего системного снимка.
199 |         history_rows: Записи исторических снимков.
200 |         inventory: Паспортные данные CPU.
201 |
202 |     Returns:
203 |         CpuLoadResponse: Структурированный ответ для UI панели
    | CPU.
204 |     """
205 |     cpu_name = "CPU"
206 |     total_load: Optional[float] = None
207 |     pkg_temp: Optional[float] = None
208 |     pkg_power: Optional[float] = None
209 |
210 |     # Порядерные и потоковые хранилища
211 |     thread_loads_by_core: Dict[int, Dict[int, float]] = {} #
    | core_idx -> {thread_idx: load}
212 |     flat_thread_loads: Dict[int, float] = {} # thread_idx (1.
    | .N) -> load
213 |     core_loads: Dict[int, float] = {}
214 |     core_temps: Dict[int, float] = {}
215 |     core_clocks: Dict[int, float] = {}
216 |     core_powers: Dict[int, float] = {}
217 |     core_volts: Dict[int, float] = {}
218 |

```

```

219 |     general_sensors_list: List[CpuSensorMetric] = []
220 |     seen_sensor_keys = set()
221 |
222 |     # Извлечение базовой информации из инвентаря/снимка
223 |     if inventory and inventory.get("model_name"):
224 |         cpu_name = str(inventory["model_name"]).strip()
225 |
226 |     if snapshot:
227 |         raw_str = snapshot.get("raw_json")
228 |         if raw_str:
229 |             try:
230 |                 raw_dict = json.loads(raw_str) if isinstance(r
231 | aw_str, str) else raw_str
232 |                 cpu_dict = raw_dict.get("cpu", {})
233 |                 if cpu_name == "CPU" and cpu_dict.get("model")
234 | :
235 |                     cpu_name = str(cpu_dict["model"]).strip()
236 |                     if total_load is None and cpu_dict.get("total_
237 | percent") is not None:
238 |                         total_load = round(float(cpu_dict["total_p
239 | ercent"]), 1)
240 |
241 |                 # Обработка плоского массива per_core из psuti
242 | l если нет LHM потоков
243 |                 cores_usage = cpu_dict.get("cores_usage") or [
244 | ]
245 |                 for idx, c_val in enumerate(cores_usage):
246 |                     flat_thread_loads[idx + 1] = round(float(c
247 | _val), 1)
248 |             except Exception:
249 |                 pass
250 |
251 |     # Разбор сенсоров
252 |     for row in sensors:
253 |         sid = str(row.get("sensor_id") or row.get("id") or "")
254 |         sname = str(row.get("sensor_name") or row.get("name")
255 | or sid)
256 |         htype = str(row.get("hardware_type") or "").lower()

```

```
249 |         hname = str(row.get("hardware_name") or "").strip()
250 |         cat = str(row.get("sensor_category") or row.get("category") or "General")
251 |
252 |         # Отсекаем дисковые метрики, попавшие по ошибке
253 |         if "disk_speed" in sid or "storage" in cat.lower() or
| "storage" in htype:
254 |             continue
255 |
256 |         if hname and hname.lower() not in ("cpu", "system") and
| cpu_name == "CPU":
257 |             cpu_name = hname
258 |
259 |         raw_val = row.get("value") if row.get("value") is not
| None else row.get("value_num")
260 |         if raw_val is None:
261 |             continue
262 |         try:
263 |             val_f = float(raw_val)
264 |         except (ValueError, TypeError):
265 |             continue
266 |
267 |         unit = str(row.get("unit") or "")
268 |         cat_l = cat.lower()
269 |         sname_l = sname.lower()
270 |         is_temp = cat_l in _TEMP_CATEGORIES or unit == "°C"
271 |         is_load = "load" in cat_l or unit == "%"
272 |         is_clock = "clock" in cat_l or "hz" in unit.lower()
273 |         is_power = "power" in cat_l or unit.lower() == "w"
274 |         is_volt = "volt" in cat_l or unit.lower() == "v"
275 |
276 |         # 1. Суммарная загрузка и температура Package
277 |         if sname in ("CPU Total", "cpu_util_total") or sid ==
| "cpu_util_total":
278 |             total_load = round(val_f, 1)
279 |             continue
280 |         if ("package" in sname_l or sid == "cpu_package_temp")
| and is_temp:
```

```

281         pkg_temp = round(val_f, 1)
282         if ("package" in sname_l or sid == "cpu_package_power"
283         ) and is_power:
284             pkg_power = round(val_f, 1)
285
286         # 2. Потоки конкретного ядра (например: "CPU Core #1 T
287         hread #2" или "Core #1 Thread #1")
288         m_ct = _CORE_THREAD_RE.search(sname)
289         if m_ct and is_load:
290             c_idx = int(m_ct.group(1))
291             t_idx = int(m_ct.group(2))
292             thread_loads_by_core.setdefault(c_idx, {})[t_idx]
293             = round(val_f, 1)
294             continue
295
296         # 3. Метрики ядра (Core #N)
297         m_c = _CORE_ONLY_RE.search(sname)
298         if m_c and not _CORE_THREAD_RE.search(sname) and "tjma
299         x" not in sname_l and "distance" not in sname_l:
300             c_idx = int(m_c.group(1))
301             if is_load:
302                 core_loads[c_idx] = round(val_f, 1)
303                 continue
304             elif is_temp:
305                 core_temps[c_idx] = round(val_f, 1)
306                 continue
307             elif is_clock:
308                 core_clocks[c_idx] = round(val_f, 1)
309                 continue
310             elif is_power:
311                 core_powers[c_idx] = round(val_f, 1)
312                 continue
313             elif is_volt:
314                 core_volts[c_idx] = round(val_f, 3)
315                 continue
316
317         # 4. Общие показатели CPU (Package Power, Bus Speed, V
318         oltage, VRM, Fans, TDP, TjMax)

```



```

314 |         # Исключаем не-CPU устройства (GPU, Memory RAM, Storage, Network)
315 |         if htype in ("gpu", "memory", "ram", "network", "storage", "disk"):
316 |             continue
317 |         if any(ign in sname_l for ign in ("gpu", "memory used", "memory available", "ethernet", "wi-fi", "d3d", "tckavg", "trcd", "tfaw", "trrd", "tccd", "twr", "twtr", "trfc")):
318 |             continue
319 |
320 |         norm_name = _normalize_sensor_name(sid, sname, cat)
321 |         sensor_key = f"{cat_l}_{norm_name.lower()}"
322 |         if sensor_key not in seen_sensor_keys:
323 |             seen_sensor_keys.add(sensor_key)
324 |             if not unit:
325 |                 if is_power: unit = "W"
326 |                 elif is_volt: unit = "V"
327 |                 elif is_temp: unit = "°C"
328 |                 elif is_clock: unit = "MHz"
329 |                 elif is_load: unit = "%"
330 |
331 |             val_str = f"{val_f:.1f} {unit}".strip() if unit else f"{val_f:.1f}"
332 |             if is_volt:
333 |                 val_str = f"{val_f:.3f} V"
334 |             elif is_clock and val_f >= 1000.0:
335 |                 val_str = f"{val_f / 1000.0:.2f} GHz"
336 |
337 |             general_sensors_list.append(
338 |                 CpuSensorMetric(
339 |                     id=sid or f"sensor_{len(general_sensors_list)+1}",
340 |                     name=norm_name,
341 |                     category=cat,
342 |                     unit=unit,
343 |                     value=round(val_f, 2),
344 |                     value_raw=val_str,
345 |                     status=_calculate_sensor_status(cat, val_f

```

```

),
346         )
347     )
348
349     # 5. Построение иерархии: Физические ядра и их логические
    потоки
350     all_core_indices = sorted(set(list(thread_loads_by_core.ke
    ys()) + list(core_temps.keys()) + list(core_clocks.keys()) + l
    ist(core_loads.keys()))
351
352     # Если индексов ядер не было в LHM, но есть flat_thread_lo
    ads из psutil (например 12 потоков)
353     if not all_core_indices and flat_thread_loads:
354         total_th = len(flat_thread_loads)
355         # Стандарт: 2 потока на ядро при HT, либо 1 поток
356         threads_per_core = 2 if total_th in (4, 8, 12, 16, 20,
    24, 32, 64) else 1
357         num_physical = max(1, total_th // threads_per_core)
358         all_core_indices = list(range(1, num_physical + 1))
359         for c_idx in all_core_indices:
360             for t_idx in range(1, threads_per_core + 1):
361                 g_idx = (c_idx - 1) * threads_per_core + t_idx
362                 if g_idx in flat_thread_loads:
363                     thread_loads_by_core.setdefault(c_idx, {})
    [t_idx] = flat_thread_loads[g_idx]
364
365     cores_result: List[CpuCoreMetric] = []
366     flat_threads_result: List[CpuThreadMetric] = []
367     global_thread_counter = 1
368
369     for c_idx in (all_core_indices or [1]):
370         core_threads_dict = thread_loads_by_core.get(c_idx, {})
    )
371         core_thread_metrics: List[CpuThreadMetric] = []
372
373         # Если потоков для ядра нет явно, создаем хотя бы 1-2
374         if not core_threads_dict:
375             # Fallback

```

```

376         th_load = core_loads.get(c_idx, total_load or 0.0)
377         core_threads_dict = {1: th_load, 2: th_load}
378
379     for t_local_idx in sorted(core_threads_dict.keys()):
380         th_val = core_threads_dict[t_local_idx]
381         th_item = CpuThreadMetric(
382             index=global_thread_counter,
383             core_thread_index=t_local_idx,
384             name=f"Thread #{global_thread_counter}",
385             core_index=c_idx,
386             core_name=f"Core #{c_idx}",
387             load_percent=th_val,
388         )
389         core_thread_metrics.append(th_item)
390         flat_threads_result.append(th_item)
391         global_thread_counter += 1
392
393     # Нагрузка ядра
394     c_load = core_loads.get(c_idx)
395     if c_load is None and core_thread_metrics:
396         c_load = round(sum(t.load_percent for t in core_thread_metrics) / len(core_thread_metrics), 1)
397
398     c_freq = core_clocks.get(c_idx)
399     cores_result.append(
400         CpuCoreMetric(
401             index=c_idx,
402             name=f"Core #{c_idx}",
403             load_percent=c_load,
404             temperature_c=core_temps.get(c_idx) or pkg_tem
405             p,
406             frequency_mhz=c_freq,
407             frequency_str=_format_frequency(c_freq),
408             power_w=core_powers.get(c_idx),
409             voltage_v=core_volts.get(c_idx),
410             threads=core_thread_metrics,
411             timestamp=datetime.now(timezone.utc).isoformat
412             (),

```

```

411         )
412     )
413
414     # 6. Расчет общего процента загрузки если не был задан
415     if total_load is None:
416         if cores_result:
417             valid_loads = [c.load_percent for c in cores_result
418                             if c.load_percent is not None]
419             total_load = round(sum(valid_loads) / len(valid_loads), 1) if valid_loads else 0.0
420         else:
421             total_load = 0.0
422
423     # 7. Формирование истории точек для графиков
424     history_points: List[CpuHistoryPoint] = []
425     if history_rows:
426         for r in history_rows:
427             ts_str = str(r.get("timestamp") or "")
428             t_label = ts_str.split("T")[-1][:8] if "T" in ts_str else ts_str[-8:]
429             p_load = float(r.get("cpu_total_percent") or 0.0)
430
431             # Попытка извлечь temp/power из raw_json если есть
432             p_temp = None
433             p_pow = None
434             raw_data = r.get("raw_json")
435             if raw_data:
436                 try:
437                     rd = json.loads(raw_data) if isinstance(raw_data, str) else raw_data
438                     p_temp = rd.get("cpu", {}).get("temperature_c") or rd.get("temperature_c")
439                     p_pow = rd.get("cpu", {}).get("power_w") or rd.get("power_w")
440                 except Exception:
441                     pass
442
443             history_points.append(

```

```

443         CpuHistoryPoint(
444             timestamp=ts_str,
445             time_label=t_label or "--:--:--",
446             load_percent=round(p_load, 1),
447             temperature_c=round(float(p_temp), 1) if p
|_temp is not None else pkg_temp,
448             power_w=round(float(p_pow), 1) if p_pow is
|not None else pkg_power,
449         )
450     )
451
452     # 5. Формирование сводного паспорта/спецификации CPU (Spec
|s)
453     specs_obj: Optional[CpuSpecsInfo] = None
454     inv_data = inventory if isinstance(inventory, dict) else {
|}
455     if inv_data or cpu_name:
456         s_name = inv_data.get("name") or cpu_name
457         s_vendor = inv_data.get("vendor") or ("Intel" if "inte
|l" in s_name.lower() else ("AMD" if "amd" in s_name.lower() el
|se "Generic"))
458         s_arch = inv_data.get("architecture") or "x86_64"
459         s_socket = inv_data.get("socket") or ""
460         s_p_cores = int(inv_data.get("physical_cores") or len(
|cores_result) or 1)
461         s_l_cores = int(inv_data.get("logical_cores") or len(
|lat_threads_result) or 1)
462         s_base_mhz = float(inv_data.get("base_frequency_mhz")
|or 2900.0)
463         s_max_mhz = float(inv_data.get("max_frequency_mhz") or
|s_base_mhz)
464         s_l2 = inv_data.get("l2_cache_kb")
465         s_l3 = inv_data.get("l3_cache_kb")
466         s_step = inv_data.get("stepping") or ""
467
468         base_f_str = f"{(s_base_mhz / 1000.0):.2f} GHz" if s_b
|ase_mhz >= 1000 else f"{s_base_mhz:.0f} MHz"
469         max_f_str = f"{(s_max_mhz / 1000.0):.2f} GHz" if s_max

```

```

    _mhz >= 1000 else f"{s_max_mhz:.0f} MHz"
470     l2_str = f"{{(s_l2 / 1024.0):.1f}} MB" if (s_l2 and s_l2
    >= 1024) else (f"{{s_l2}} KB" if s_l2 else "--")
471     l3_str = f"{{(s_l3 / 1024.0):.1f}} MB" if (s_l3 and s_l3
    >= 1024) else (f"{{s_l3}} KB" if s_l3 else "--")
472     cache_comb = f"{{l2_str}} / {{l3_str}}" if (s_l2 or s_l3)
    else "--"
473
474     raw_processor_desc = platform.processor() or "Intel64
    Family 6 Model 165 Stepping 3, GenuineIntel"
475     full_cpu_desc = f"{{raw_processor_desc}} ({{s_l_cores}} lo
    gical cores)"
476
477     specs_obj = CpuSpecsInfo(
478         name=s_name,
479         vendor=s_vendor,
480         architecture=s_arch,
481         socket=s_socket,
482         physical_cores=s_p_cores,
483         logical_cores=s_l_cores,
484         base_frequency_mhz=s_base_mhz,
485         max_frequency_mhz=s_max_mhz,
486         base_frequency_str=base_f_str,
487         max_frequency_str=max_f_str,
488         l2_cache_kb=s_l2,
489         l3_cache_kb=s_l3,
490         l2_cache_str=l2_str,
491         l3_cache_str=l3_str,
492         cache_combined_str=cache_comb,
493         stepping=s_step,
494         features=inv_data.get("features") or [],
495         description=full_cpu_desc,
496         raw_descriptor=raw_processor_desc,
497     )
498
499     raw_processor_desc = platform.processor() or "Intel64 Fami
    ly 6 Model 165 Stepping 3, GenuineIntel"
500     total_l_cores = len(flat_threads_result) or 12

```

```

501     total_p_cores = len(cores_result) or 6
502     full_cpu_desc = f"{raw_processor_desc} ({total_l_cores} lo
    gical cores)"
503
504     cpus_list = [
505         {
506             "id": 0,
507             "name": cpu_name,
508             "raw_descriptor": raw_processor_desc,
509             "description": full_cpu_desc,
510             "physical_cores": total_p_cores,
511             "logical_cores": total_l_cores,
512         }
513     ]
514
515     return CpuLoadResponse(
516         status="ok",
517         name=cpu_name,
518         description=full_cpu_desc,
519         cpus=cpus_list,
520         total_percent=round(total_load, 1),
521         package_temperature_c=pkg_temp,
522         package_power_w=pkg_power,
523         cores_count=len(cores_result),
524         threads_count=len(flat_threads_result),
525         specs=specs_obj,
526         cores=cores_result,
527         threads=flat_threads_result,
528         sensors=general_sensors_list,
529         history=history_points,
530         meta={
531             "source": "telemetry.db/lhm_hierarchy",
532             "cores_count": len(cores_result),
533             "threads_count": len(flat_threads_result),
534             "sensors_count": len(general_sensors_list),
535         },
536     )
537

```

```

538
539 def init_router(
540     storage: Optional[TelemetryStorage] = None,
541     lhm_service: Optional[LhmService] = None,
542 ) -> APIRouter:
543     """Создаёт роутер /api/v1/panel/cpu-load с полной поддержк
ой иерархии CPU.
544
545     Args:
546         storage: Хранилище телеметрии SQLite.
547         lhm_service: Сервис LibreHardwareMonitor.
548
549     Returns:
550         APIRouter: Роутер с эндпоинтом GET /api/v1/panel/cpu-l
oad.
551     """
552     router = APIRouter(tags=["CPU Load Panel"])
553     store = storage or TelemetryStorage.get_instance(read_only
=True)
554     lhm = lhm_service if lhm_service is not None else (LhmServ
ice() if storage is None else None)
555
556     @router.get("/api/v1/panel/cpu-load", response_model=CpuLo
adResponse)
557     async def get_panel_cpu_load() -> CpuLoadResponse:
558         """Возвращает структурированные иерархические данные C
PU из телеметрии и LHM."""
559         try:
560             # 1. Запрос к базе данных телеметрии для извлечени
я истории и сохраненных метрик
561             db_data = store.get_cpu_hierarchy_metrics()
562             db_sensors = db_data.get("sensors", [])
563             db_snapshot = db_data.get("snapshot")
564             db_inv = db_data.get("inventory")
565             db_history = db_data.get("history", [])
566
567             # 2. Если запущен LHM — дополняем живыми данными с
енсоров

```



```

568         if lhm and lhm.is_running():
569             lhm_sensors = lhm.get_flattened_sensors()
570             if lhm_sensors:
571                 resp = build_cpu_load(
572                     sensors=lhm_sensors,
573                     snapshot=db_snapshot,
574                     history_rows=db_history,
575                     inventory=db_inv,
576                 )
577                 resp.meta = {
578                     "source": "lhm_live+telemetry.db",
579                     "cores_count": len(resp.cores),
580                     "threads_count": len(resp.threads),
581                 }
582                 return resp
583
584         # 3. Ответ на основе данных БД телеметрии
585         resp = build_cpu_load(
586             sensors=db_sensors,
587             snapshot=db_snapshot,
588             history_rows=db_history,
589             inventory=db_inv,
590         )
591
592         # 4. Если данных в БД недостаточно (пустая база) –
593         # дополняем через HardwareMonitor
594         if not resp.cores and storage is None:
595             from apps.windows.modules.hardware.hardware_monitor import HardwareMonitor
596             hw = HardwareMonitor()
597             cpu_m = hw.get_cpu_metrics()
598             flat_th: Dict[int, float] = {}
599             for idx, c_val in enumerate(cpu_m.per_core_pct
600 ):
601                 flat_th[idx + 1] = float(c_val)
602
603             live_snap = {
604                 "raw_json": json.dumps({

```

```

603         "cpu": {
604             "model": cpu_m.model_name,
605             "total_percent": cpu_m.utilization
        _pct,
606             "cores_usage": cpu_m.per_core_pct,
607         }
608     })
609 }
610 resp = build_cpu_load(
611     sensors=db_sensors,
612     snapshot=live_snap,
613     history_rows=db_history,
614     inventory=db_inv,
615 )
616 resp.meta["source"] = "hardware_monitor_live+telemetry.db"
617
618 return resp
619
620 except Exception as exc:
621     logger.error(f"[router_cpu_load] Ошибка формирования метрик CPU: {exc}", exc_info=True)
622     return CpuLoadResponse(status="error", meta={"source": "telemetry.db", "error": str(exc)})
623
624 return router
625
626
627 __all__ = [
628     "init_router",
629     "build_cpu_load",
630     "CpuLoadResponse",
631     "CpuCoreMetric",
632     "CpuThreadMetric",
633     "CpuSensorMetric",
634     "CpuHistoryPoint",
635 ]
636

```

router_diagnostics.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Diagnostics
4 # =====
5 # Description:
6 #    Универсальный REST API эндпоинт для AI-диагностики, контекстного объяснения
7 #
8 # Usage Examples:
9 #    Python API:
10 #    from apps.windows.api.routers.router_diagnostics import DiagnosticExplainRequest
11 #
12 #    service = DiagnosticExplainRequest()
13 #
14 # File: router_diagnostics.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-08 05:22:00
20 # =====
21
22 from __future__ import annotations
23 """Универсальный REST API эндпоинт для AI-диагностики, контекстного объяснения"""
24
25 import json
26 import re
27 from typing import Any, Dict, List, Optional
28
29 from fastapi import APIRouter
```

```

30 from pydantic import BaseModel, Field
31
32 from logger import logger
33
34
35 class DiagnosticExplainRequest(BaseModel):
36     """Запрос на AI-объяснение элемента таблицы."""
37     table_type: str = Field(default="generic", description="Ти
п таблицы (software, process, service, task, network, registry
, user, website, rag_doc, disk)")
38     title: str = Field(default="", description="Основное имя и
ли заголовок элемента")
39     subtitle: Optional[str] = Field(default="", description="В
торичный заголовок (издатель, путь, IP, статус)")
40     metadata: Dict[str, Any] = Field(default_factory=dict, des
cription="Набор ключевых атрибутов записи")
41     raw_data: Optional[str] = Field(default="", description="С
ырая строка команды, пути или JSON")
42     model: Optional[str] = Field(default=None, description="Оп
циональное имя конкретной модели")
43     provider: Optional[str] = Field(default=None, description=
"Опциональный AI провайдер")
44     web_search: bool = Field(default=True, description="Выполн
ять ли онлайн поиск для обогащения контекста")
45
46
47 class DiagnosticExplainResponse(BaseModel):
48     """Структурированный ответ AI-диагностики элемента таблицы
. """
49     summary: str = Field(description="Краткое понятное описани
е назначения элемента")
50     developer: str = Field(default="Неизвестен", description="
Разработчик / принадлежность / вендор")
51     category: str = Field(default="Системный компонент", descr
iption="Категория элемента")
52     security_verdict: str = Field(description="Оценка безопасн
ости, легитимности и рисков")
53     performance_impact: str = Field(description="Влияние на пр

```

```

    оизводительность и ресурсы")
54     recommendation: str = Field(description="Практическая реко
    мендация эксперта")
55     action_steps: List[str] = Field(default_factory=list, desc
    ription="Пошаговые рекомендуемые действия")
56
57
58 from src.api.diagnostics_prompt_manager import (
59     DiagnosticPromptTemplate,
60     prompt_manager,
61 )
62
63
64 class DiagnosticPromptUpdateRequest(BaseModel):
65     """Запрос на обновление шаблона промпта для таблицы."""
66     system_instruction: str = Field(description="Системная инс
    трукция для LLM")
67     prompt_template: str = Field(description="Шаблон текста пр
    омпта с переменными {title}, {subtitle}, {metadata}, {raw_data
    }")
68     name: Optional[str] = Field(default=None, description="Опц
    иональное отображаемое имя")
69     description: Optional[str] = Field(default=None, descripti
    on="Опциональное описание")
70
71
72 def init_router() -> APIRouter:
73     """Инициализирует и возвращает FastAPI роутер универсально
    й диагностики.
74
75     Returns:
76         APIRouter: Настроенный роутер с префиксом /api/v1/diag
    nostics.
77     """
78     router = APIRouter(prefix="/api/v1/diagnostics", tags=["Un
    iversal Diagnostics"])
79
80     @router.get("/prompts", response_model=List[DiagnosticProm

```

```

ptTemplate])
81     async def list_diagnostic_prompts() -> List[DiagnosticProm
ptTemplate]:
82         """Возвращает список всех шаблонов промптов для таблиц
системы."""
83         return prompt_manager.list_templates()
84
85     @router.get("/prompts/{table_type}", response_model=Diagno
sticPromptTemplate)
86     async def get_diagnostic_prompt(table_type: str) -> Diagno
sticPromptTemplate:
87         """Возвращает шаблон промпта для конкретного типа табл
ицы."""
88         return prompt_manager.get_template(table_type)
89
90     @router.post("/prompts/{table_type}", response_model=Diagn
osticPromptTemplate)
91     async def save_diagnostic_prompt(table_type: str, req: Dia
gnosticPromptUpdateRequest) -> DiagnosticPromptTemplate:
92         """Сохраняет пользовательский шаблон промпта для типа
таблицы."""
93         return prompt_manager.save_template(
94             table_type=table_type,
95             system_instruction=req.system_instruction,
96             prompt_template=req.prompt_template,
97             name=req.name,
98             description=req.description,
99         )
100
101     @router.post("/prompts/{table_type}/reset", response_model
=DiagnosticPromptTemplate)
102     async def reset_diagnostic_prompt(table_type: str) -> Diag
nosticPromptTemplate:
103         """Сбрасывает шаблон промпта к дефолтной системной вер
сии."""
104         return prompt_manager.reset_template(table_type)
105
106     @router.post("/explain", response_model=DiagnosticExplainR

```

```

    response)
107     async def explain_table_item(req: DiagnosticExplainRequest
    ) -> DiagnosticExplainResponse:
108         """Генерирует AI-объяснение и оценку для выбранного эл
    емента таблицы."""
109         # Получаем соответствующий шаблон промпта для типа таб
    лицы
110         tpl = prompt_manager.get_template(req.table_type)
111
112         # Вызов языковой модели через прямой GoogleGenerativeA
    I с Web Grounding
113         try:
114             from src.ai.gemini.generative_ai import GoogleGene
    rativeAI
115
116             gen_cfg: Dict[str, Any] = {}
117             if req.web_search:
118                 gen_cfg["use_google_search"] = True
119
120             model_name = req.model if req.model and not req.mo
    del.startswith("gemini_cli:") else "gemini-3.1-flash"
121             if req.model and req.model.startswith("gemini_cli:
    "):
122                 model_name = req.model.split(":", 1)[1]
123
124             ai = GoogleGenerativeAI(
125                 model_name=model_name,
126                 system_instruction=tpl.system_instruction,
127                 generation_config=gen_cfg,
128             )
129
130             prompt = tpl.format_prompt(
131                 title=req.title,
132                 subtitle=req.subtitle or "",
133                 metadata=req.metadata,
134                 raw_data=req.raw_data or "",
135             )
136

```

```

137         resp_text = await ai.ask(prompt, generation_config
138         =gen_cfg)
139         if resp_text and not resp_text.startswith("Model e
140         rror:"):
141             cleaned = resp_text.strip()
142             # Извлекаем JSON если вокруг есть Markdown бло
143             к
144             json_match = re.search(r"```(?:json)?\s*([\s\S
145             ]*?)\s*```", cleaned)
146             if json_match:
147                 cleaned = json_match.group(1).strip()
148
149             # Извлекаем JSON по внешним фигурным скобкам
150             if "{" in cleaned and "}" in cleaned:
151                 start_idx = cleaned.find("{")
152                 end_idx = cleaned.rfind("}") + 1
153                 cleaned = cleaned[start_idx:end_idx]
154
155             data = json.loads(cleaned)
156             return DiagnosticExplainResponse(
157                 summary=data.get("summary", ""),
158                 developer=data.get("developer", req.subtit
159                 le or "Неизвестен"),
160                 category=data.get("category", "Диагностика
161                 "),
162                 security_verdict=data.get("security_verdic
163                 t", "Оценка безопасности завершена."),
164                 performance_impact=data.get("performance_i
165                 mpact", "В пределах нормы."),
166                 recommendation=data.get("recommendation",
167                 "Оставить как есть."),
168                 action_steps=data.get("action_steps", ["Пр
169                 оверьте актуальность конфигурации."]),
170             )
171         except Exception as e:
172             logger.debug(f"AI LLM generation fallback trigger
173             ed for diagnostic '{req.title}': {e}")

```



```

164
165         # Надежный эвристический fallback
166         return _generate_heuristic_explanation(req)
167
168     return router
169
170
171 def _generate_heuristic_explanation(req: DiagnosticExplainRequest) -> DiagnosticExplainResponse:
172     """Генерирует экспертное контекстное объяснение на основе
173     эвристики и доменных правил."""
174     tt = (req.table_type or "generic").lower()
175     title_lower = (req.title or "").lower()
176     raw_lower = (req.raw_data or "").lower()
177     sub_lower = (req.subtitle or "").lower()
178
179     # 1. Установленное ПО (Software Audit)
180     if tt == "software":
181         is_ms = "microsoft" in sub_lower or "windows" in title_lower
182         is_google = "google" in sub_lower or "chrome" in title_lower
183         dev = "Microsoft Corporation" if is_ms else ("Google LLC" if is_google else (req.subtitle or "Неизвестный разработчик"))
184         sec = "Официальное ПО, подписанное доверенным разработчиком." if (is_ms or is_google) else "Установленное стороннее приложение. Проверьте актуальность версии."
185         return DiagnosticExplainResponse(
186             summary=f"Программа «{req.title}». Зарегистрирована в реестре установленного программного обеспечения Windows.",
187             developer=dev,
188             category="Установленное приложение",
189             security_verdict=sec,
190             performance_impact="Занимает дисковое пространство. Проверьте наличие фоновых служб обновления.",
191             recommendation="Если приложение активно используется – оставьте и следите за обновлениями. Если не используется

```

```

191         action_steps=[
192             "Проверьте дату установки и регулярность испол
193             зования.",
194             "При неактуальности удалите для освобождения д
195             иска.",
196         ],
197     )
198
199     # 2. Процессы Windows (System Inspector)
200     if tt == "process":
201         # Специфические утилиты мониторинга и разработки
202         if "librehardwaremonitor" in title_lower:
203             return DiagnosticExplainResponse(
204                 summary="Утилита мониторинга аппаратных compon
205                 ентов и датчиков ПК с открытым исходным кодом (Open Source). Р
206                 азвилка проекта Open Hardware Monitor.",
207                 developer="LibreHardwareMonitor Community (Git
208                 Hub Open Source)",
209                 category="Системная диагностика / Мониторинг а
210                 ппаратных компонентов",
211                 security_verdict="Легитимное доверенное ПО с о
212                 ткрытым исходным кодом. Для прямого чтения низкоуровневых датч
213                 иков материнской платы, CPU и GPU использует драйвер кольца яд
214                 ра (Ring-0).",
215                 performance_impact=f"Минимальная нагрузка при
216                 стандартном интервале опроса сенсоров. CPU: {req.metadata.get(
217                     'cpu_percent', req.metadata.get('cpu', '0.1'))}%, RAM: {req.me
218                     tadata.get('memory_mb', req.metadata.get('memory', '35'))} MB.
219                 ",
220                 recommendation="Безопасная утилита телеметрии.
221                 Необходима для отслеживания температур, напряжений, оборотов
222                 кулеров и нагрузки системы.",
223                 action_steps=[
224                     "Используйте для контроля температурных ре
225                     жимов и предотвращения перегрева компонентов.",
226                     "При необходимости закройте или остановите
227                     сбор телеметрии, если мониторинг больше не требуется.",
228                 ],
229             )

```

```

211         ],
212     )
213
214     is_system_proc = title_lower in ("explorer.exe", "svch
ost.exe", "services.exe", "lsass.exe", "system", "csrss.exe",
"winlogon.exe", "smss.exe", "dwm.exe")
215     is_temp = "temp" in raw_lower or "appdata" in raw_lowe
r
216     sec = "Критический системный процесс ядра Windows. Зав
ершение приведет к сбою ОС." if is_system_proc else (
217         "Внимание: процесс запущен из временного каталога
(%TEMP%/AppData). Рекомендуется проверка!" if is_temp else "Шт
атный процесс пользовательской сессии или фонового сервиса."
218     )
219     rec = "Не завершать процесс (системный компонент)." if
is_system_proc else "При аномальной нагрузке на CPU/память пр
оцесс можно завершить через диспетчер."
220     return DiagnosticExplainResponse(
221         summary=f"Процесс «{req.title}» (PID: {req.metadat
a.get('pid', 'N/A')}). Активно выполняется в операционной сист
еме.",
222         developer="Microsoft Corporation" if is_system_pro
c else "Разработчик приложения",
223         category="Системный процесс" if is_system_proc els
e "Пользовательский процесс",
224         security_verdict=sec,
225         performance_impact=f"Потребление CPU: {req.metadat
a.get('cpu_percent', req.metadata.get('cpu', 'N/A'))}%, Память
: {req.metadata.get('memory_mb', req.metadata.get('memory', 'N
/A'))} MB.",
226         recommendation=rec,
227         action_steps=[
228             "Оцените текущую нагрузку на процессор и опера
тивную память.",
229             "При зависании или утечке памяти завершите про
цесс.",
230         ],
231     )

```

```
232
233     # 3. Службы Windows (System Control Center)
234     if tt == "service":
235         status = req.metadata.get("status", "Running")
236         start_type = req.metadata.get("start_type", "Auto")
237         return DiagnosticExplainResponse(
238             summary=f"Системная служба Windows «{req.title}» (
239 {req.subtitle or req.title}). Обеспечивает фоновое выполнение
240 задач ОС или установленного ПО.",
241             developer="Операционная система Windows" if "system32" in raw_lower else "Сторонний разработчик",
242             category="Фоновая служба (Windows Service)",
243             security_verdict=f"Исполняемый файл: {req.raw_data
244 or 'Системный бинарник'}. Статус службы: {status}.",
245             performance_impact=f"Тип запуска: {start_type}. Сл
246 ужба активна в фоне.",
247             recommendation="Не отключайте системные службы Win
248 dows без уверенности в последствиях. Сторонние ненужные службы
249 можно перевести в ручной режим (Manual).",
250             action_steps=[
251                 "При необходимости измените тип запуска (Auto
252 / Manual / Disabled).",
253                 "Перезапустите службу при сбоях в работе связа
254 нного ПО.",
255             ],
256         )
257
258     # 4. Задачи планировщика (Scheduled Tasks)
259     if tt == "task":
260         return DiagnosticExplainResponse(
261             summary=f"Запланированная задача Windows Task Sche
262 duler «{req.title}». Выполняется по расписанию или событию в с
263 истеме.",
264             developer="Microsoft / Стороннее ПО",
265             category="Задача планировщика",
266             security_verdict="Регулярно инициирует запуск связ
267 анного процесса или скрипта.",
268             performance_impact="Кратковременная нагрузка на ди
```

```

    ск и CPU в момент срабатывания триггера.",
258         recommendation="Отключите задачу, если соответству
    ющее приложение больше не используется или вызывает нежелатель
    ные фоновые пробуждения ПК.",
259         action_steps=[
260             "Проверьте расписание и триггеры запуска задач
    и.",
261             "Отключите задачу, если она вызывает избыточну
    ю нагрузку.",
262         ],
263     )
264
265     # 5. Сетевые соединения (Network Monitor)
266     if tt == "network":
267         r_addr = req.metadata.get("remote_address", req.metada
    ta.get("remote", req.subtitle or "Внешний узел"))
268         l_addr = req.metadata.get("local_address", req.metadat
    a.get("local", "Локальный узел"))
269         proto = req.metadata.get("protocol", "TCP")
270         is_loopback = "127.0.0.1" in str(r_addr) or "localhost
    " in str(r_addr)
271         sec = "Локальное межпроцессное соединение (Loopback/IP
    C). Безопасно." if is_loopback else f"Внешнее сетевое соединен
    ие с узлом {r_addr}."
272         return DiagnosticExplainResponse(
273             summary=f"Сетевой сокет {proto}: {l_addr} ↔ {r_add
    r}. Состояние: {req.metadata.get('state', 'ESTABLISHED')}.",
274             developer="Сетевой стек Windows",
275             category="Сетевое соединение",
276             security_verdict=sec,
277             performance_impact="Занимает сетевой сокет и потре
    бляет сетевой трафик.",
278             recommendation="Если удаленный IP-адрес или порт в
    ызывают подозрения, заблокируйте соединение через Брандмауэр W
    indows (Windows Defender Firewall).",
279             action_steps=[
280                 "Проверьте имя процесса, открывшего данное сое
    динение.",

```

```

281 |             "При неизвестном внешнем адресе проверьте IP ч
    | ерез Whois / AbuseIPDB.",
282 |         ],
283 |     )
284 |
285 |     # 6. Ключи и параметры реестра (Registry Viewer)
286 |     if tt == "registry":
287 |         return DiagnosticExplainResponse(
288 |             summary=f"Параметр системного реестра «{req.title}
    | » (Тип: {req.metadata.get('type', 'REG_SZ')}).",
289 |             developer="Реестр Windows",
290 |             category="Параметр реестра",
291 |             security_verdict=f"Значение параметра: {req.raw_da
    | ta or req.subtitle or 'Не задано'}. Влияет на конфигурацию ком
    | понентов ОС или ПО.",
292 |             performance_impact="Низкое – считывается при старт
    | е или запросе конфигурации.",
293 |             recommendation="Редактируйте параметры реестра с о
    | сторожностью. Перед изменением создайте точку восстановления и
    | ли резервную копию ветки (REG экспорт).",
294 |             action_steps=[
295 |                 "Создайте резервную копию ветки реестра перед
    | правкой.",
296 |                 "Проверьте документацию по конкретному парамет
    | ру перед изменением.",
297 |             ],
298 |         )
299 |
300 |     # 7. Пользователи и группы (Windows Admin & Identity)
301 |     if tt in ("user", "user_account", "group"):
302 |         if "defaultaccount" in title_lower or "-503" in sub_lo
    | wer:
303 |             return DiagnosticExplainResponse(
304 |                 summary="Встроенная системная учетная запись W
    | indows (System Managed Account). Введена начиная с Windows 10
    | для выполнения изолированных многопользовательских процессов и
    | приложений UWP/AppContainer.",
305 |                 developer="Microsoft Corporation (Операционная

```

```
система Windows)",
306         category="Встроенная системная учетная запись
(RID 503)",
307         security_verdict="Штатный системный аккаунт. П
о умолчанию отключен (Disabled), прямой интерактивный вход в с
истему заблокирован, пароль управляется ОС.",
308         performance_impact="Не потребляет ресурсы в фо
новом режиме. Активируется только при запуске контейнеризирова
нных сценариев Windows.",
309         recommendation="Не удаляйте и не включайте уче
тную запись вручную. Оставьте в исходном состоянии (Disabled),
управляемом операционной системой.",
310         action_steps=[
311             "Убедитесь, что аккаунт находится в состоя
нии 'Отключен' (Disabled).",
312             "Не назначайте учетной записи администрати
вные привилегии.",
313         ],
314     )
315     if "wdagutilityaccount" in title_lower or "-504" in su
b_lower:
316         return DiagnosticExplainResponse(
317             summary="Служебная учетная запись Windows Defe
nder Application Guard (WDAG). Используется для запуска изолир
ованных сессий браузера Edge и песочниц.",
318             developer="Microsoft Corporation (Windows Secu
rity)",
319             category="Служебная учетная запись изоляции (R
ID 504)",
320             security_verdict="Легитимный изолированный акк
аунт безопасности. Заблокирован для прямого входа.",
321             performance_impact="Используется только во вре
мя активных изолированных сессий Application Guard.",
322             recommendation="Оставьте под управлением Windo
ws Defender.",
323             action_steps=[
324                 "Не изменяйте параметры аккаунта вручную."
325             ],
326         )
```

```
325 |         ],
326 |     )
327 |     if "guest" in title_lower or "гость" in title_lower or
    |     "-501" in sub_lower:
328 |         return DiagnosticExplainResponse(
329 |             summary="Встроенная гостевая учетная запись Wi
    | ndows для временного доступа пользователей без собственного ак
    | каунта.",
330 |             developer="Microsoft Corporation",
331 |             category="Встроенная гостевая учетная запись (
    | RID 501)",
332 |             security_verdict="По соображениям безопасности
    | в современных версиях Windows должна быть строго отключена (D
    | isabled).",
333 |             performance_impact="Минимальное.",
334 |             recommendation="Рекомендуется держать гостевую
    | запись отключенной во избежание несанкционированного доступа.
    | ",
335 |             action_steps=[
336 |                 "Убедитесь, что учетная запись 'Гость' отк
    | лючена.",
337 |             ],
338 |         )
339 |     if "administrator" in title_lower or "администратор" i
    | n title_lower or "-500" in sub_lower:
340 |         return DiagnosticExplainResponse(
341 |             summary="Встроенная учетная запись главного ло
    | кального администратора Windows (RID 500). Обладает полным без
    | условным контролем над операционной системой.",
342 |             developer="Microsoft Corporation",
343 |             category="Встроенный локальный администратор (
    | RID 500)",
344 |             security_verdict="Обладает максимальными приви
    | легиями в системе. Требуется строгий контроль и сложный пароль
    | .",
345 |             performance_impact="Зависит от выполняемых зад
    | ач администратора.",
346 |             recommendation="По рекомендациям Microsoft вст
```



```

роенного администратора лучше держать отключенным или переименованным, используя персонализированные учетные записи с UAC.",
347         action_steps=[
348             "Установите надежный сложный пароль.",
349             "Используйте повседневную работу под стандартной учетной записью с подтверждением через UAC.",
350         ],
351     )
352
353     is_admin = "admin" in title_lower or "администратор" in title_lower
354     return DiagnosticExplainResponse(
355         summary=f"Учетная запись / Группа безопасности Windows «{req.title}».",
356         developer="Security Accounts Manager (SAM) / Active Directory",
357         category="Учетная запись пользователя" if tt in ("user", "user_account") else "Группа безопасности",
358         security_verdict="Обладает повышенными административными привилегиями в системе." if is_admin else "Обычная пользовательская учетная запись со стандартными правами доступа.",
359         performance_impact=f"Профиль: {req.metadata.get('profile_path', 'Стандартный')}, Процессов: {req.metadata.get('process_count', 0)}.",
360         recommendation="Соблюдайте принцип наименьших привилегий (Least Privilege). Не используйте права администратора для повседневной работы.",
361         action_steps=[
362             "Проверьте актуальность членства в группах безопасности.",
363             "Убедитесь в наличии сложного пароля для учетной записи.",
364         ],
365     )
366
367     # 8. События безопасности (Security Events)
368     if tt in ("security_event", "event"):
369         return DiagnosticExplainResponse(

```

```

370         summary=f"Событие аудита безопасности Windows «{req.title}» ({req.subtitle or ''}).",
371         developer="Подсистема безопасности Windows (LSA / SAM / Defender)",
372         category="Событие журнала безопасности",
373         security_verdict="Зафиксировано в журнале аудита безопасности Windows.",
374         performance_impact="Не влияет на быстродействие ОС.",
375         recommendation="Проверьте статус события и инициатора при наличии повторяющихся ошибок входа.",
376         action_steps=[
377             "Проверьте учетную запись инициатора события.",
378             "Сопоставьте время события с активностью пользователя.",
379         ],
380     )
381
382     # 8. Мониторинг веб-сайтов (Website Monitor)
383     if tt == "website":
384         http_code = req.metadata.get("status_code", req.metadata.get("status", "200"))
385         latency = req.metadata.get("latency_ms", req.metadata.get("response_time", "N/A"))
386         is_ok = str(http_code).startswith("2") or str(http_code).startswith("3")
387         return DiagnosticExplainResponse(
388             summary=f"Веб-ресурс «{req.title}» ({req.subtitle or req.raw_data}). Код ответа: {http_code}, задержка: {latency} ms.",
389             developer="Веб-сервер / Хостинг",
390             category="Веб-сервис",
391             security_verdict="Ресурс доступен и отвечает по защищенному протоколу HTTPS." if is_ok else "Внимание: зафиксирован сбой или ошибка доступности ресурса!",
392             performance_impact=f"Время отклика: {latency} ms."
393         ),

```

```
393         recommendation="Ресурс функционирует нормально." if
    f is_ok else "Проверьте логи веб-сервера, срок действия SSL-се
    ртификата и настройки DNS.",
394         action_steps=[
395             "Проверьте доступность узла с разных сетевых п
    ровайдеров.",
396             "Проверьте статус SSL-сертификата и configuraц
    ию Nginx / Cloudflare.",
397         ],
398     )
399
400     # 9. RAG документы базы знаний (RAG Tab)
401     if tt == "rag_doc":
402         chunks = req.metadata.get("chunks", req.metadata.get("
    chunk_count", "N/A"))
403         size = req.metadata.get("size", "N/A")
404         return DiagnosticExplainResponse(
405             summary=f"Документ базы знаний RAG «{req.title}».
    Содержит {chunks} семантических чанков, размер: {size}.",
406             developer="RAG Knowledge Store",
407             category="Документ базы знаний",
408             security_verdict="Документ проиндексирован в локал
    ьном векторном хранилище и доступен для семантического поиска
    ИИ.",
409             performance_impact="Занимает место в векторной баз
    е данных и кэше эмбедингов.",
410             recommendation="Документ активен. При обновлении и
    сходного файла выполните переиндексацию чанков для поддержания
    актуальности контекста.",
411             action_steps=[
412                 "Проверьте качество найденных чанков в тесте R
    AG поиска.",
413                 "При необходимости обновите или переиндексируй
    те файл.",
414             ],
415         )
416
417     # 10. Дисковые тома и накопители (Disks / Storage)
```

```
418     if tt == "disk":
419         free = req.metadata.get("free", req.metadata.get("free
|_gb", "N/A"))
420         total = req.metadata.get("total", req.metadata.get("to
|tal_gb", "N/A"))
421         return DiagnosticExplainResponse(
422             summary=f"Дисковый накопитель / том «{req.title}»
|({req.subtitle or 'Локальный диск'}). Свободно: {free} из {tot
|al}." ,
423             developer="Storage Controller",
424             category="Дисковый том",
425             security_verdict="Файловая система смонтирована и
|доступна для чтения/записи." ,
426             performance_impact=f"Использование диска: {req.met
|adata.get('percent', 'N/A')}% занято." ,
427             recommendation="Поддерживайте не менее 15-20% своб
|одного места на системном диске для корректной работы виртуаль
|ной памяти и обновлений Windows." ,
428             action_steps=[
429                 "Очистите временные файлы (%TEMP%) при дефицит
|е дискового пространства." ,
430                 "Проверьте SMART-статус накопителя при подозри
|тельных задержках ввода-вывода." ,
431             ],
432         )
433
434     # 11. Общий шаблон по умолчанию
435     return DiagnosticExplainResponse(
436         summary=f"Запись «{req.title}» ({req.subtitle or req.t
|able_type}). Данные зарегистрированы в системе." ,
437         developer=req.subtitle or "Системный компонент",
438         category="Элемент данных",
439         security_verdict="Параметры записи соответствуют штатн
|ой конфигурации." ,
440         performance_impact="Штатное использование ресурсов." ,
441         recommendation="Оставьте запись активной или скорректи
|руйте параметры при необходимости." ,
442         action_steps=[
```

```
443 |             "Проверьте актуальность метаданных.",
444 |             "При необходимости выполните пересканирование.",
445 |         ],
446 |     )
447 |
448 |
449 | __all__ = ["init_router"]
450 |
```

router_focus.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Focus
4 # =====
5 # Description:
6 #   API роутер управления сессиями фокусировки внимания (Windows Focus Sessions),
7 #   режимом «Не беспокоить» (Do Not Disturb / Quiet Hours), поведением бейджей
8 #   и мигания панели задач, а также вызовом системных протоколов ms-clock / ms-settings.
9 #
10 # Usage Examples:
11 #   Python API:
12 #       from apps.windows.api.routers.router_focus import init_router
13 #
14 #       router = init_router()
15 #
16 # File: router_focus.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 19:35:00
22 # =====
23
24 from __future__ import annotations
25 """API роутер управления сессиями фокусировки внимания (Windows Focus) и режимом «Не беспокоить»."""
26
27 import asyncio
```

```
28 import os
29 import sys
30 import time
31 from datetime import datetime, timezone
32 from pathlib import Path
33 from typing import Any, Dict, Optional
34
35 from fastapi import APIRouter, HTTPException
36 from pydantic import BaseModel, Field
37
38 from logger import logger
39
40 try:
41     import winreg
42 except ImportError:
43     winreg = None # Для сред тестирования не под Windows
44
45
46 class FocusSettingsDTO(BaseModel):
47     """Модель настроек режима фокусировки внимания."""
48     session_duration_minutes: int = Field(30, ge=5, le=240, description="Длительность сессии в минутах")
49     show_timer_in_clock_app: bool = Field(True, description="Отображать таймер в приложении «Часы»")
50     hide_badges_on_taskbar: bool = Field(True, description="Скрывать значки уведомлений (бейджи) на панели задач")
51     hide_flashing_on_taskbar: bool = Field(True, description="Скрывать мигание приложений на панели задач")
52     turn_on_do_not_disturb: bool = Field(True, description="Включать режим «Не беспокоить» (Do Not Disturb)")
53
54
55 class FocusSessionStateDTO(BaseModel):
56     """Модель текущего состояния сессии фокусировки."""
57     is_active: bool = Field(False, description="Флаг активности сессии фокусировки")
58     session_duration_minutes: int = Field(30, description="Продолжительность текущей или запланированной сессии")
```

```

59     started_at: Optional[str] = Field(None, description="ISO м
   метка времени старта")
60     ends_at: Optional[str] = Field(None, description="ISO метк
   а времени завершения")
61     remaining_seconds: int = Field(0, description="Оставшееся
   время сессии в секундах")
62     settings: FocusSettingsDTO = Field(default_factory=FocusSe
   ttingsDTO, description="Текущие настройки фокусировки")
63
64
65 class FocusSessionManager:
66     """Менеджер управления сессиями фокусировки и системными к
   лючами реестра."""
67
68     def __init__(self) -> None:
69         """Инициализация менеджера сессий фокусировки."""
70         self._is_active: bool = False
71         self._session_duration_minutes: int = 30
72         self._started_at_ts: float = 0.0
73         self._ends_at_ts: float = 0.0
74         self._task: Optional[asyncio.Task] = None
75         self._backup_state: Dict[str, Any] = {}
76         self._settings = FocusSettingsDTO()
77
78     def get_state(self) -> FocusSessionStateDTO:
79         """Возвращает текущее состояние сессии фокусировки и с
   истемных параметров.
80
81         Returns:
82             FocusSessionStateDTO: Объект состояния сессии.
83         """
84         now = time.time()
85         remaining = 0
86         if self._is_active:
87             if now < self._ends_at_ts:
88                 remaining = int(self._ends_at_ts - now)
89             else:
90                 self._is_active = False

```



```

91         remaining = 0
92
93         started_iso = datetime.fromtimestamp(self._started_at_
|ts, tz=timezone.utc).isoformat() if self._is_active else None
94         ends_iso = datetime.fromtimestamp(self._ends_at_ts, tz
| =timezone.utc).isoformat() if self._is_active else None
95
96         # Читаем реальные системные флаги из реестра (если не
| в активной сессии)
97         live_settings = self._read_system_settings()
98
99         return FocusSessionStateDTO(
100             is_active=self._is_active,
101             session_duration_minutes=self._session_duration_mi
| nutes,
102             started_at=started_iso,
103             ends_at=ends_iso,
104             remaining_seconds=remaining,
105             settings=live_settings,
106         )
107
108     def _read_system_settings(self) -> FocusSettingsDTO:
109         """Считывает текущие системные параметры из реестра Wi
| ndows.
110
111         Returns:
112             FocusSettingsDTO: Настройки, прочитанные из систем
| ы.
113
114         """
115         hide_badges = self._settings.hide_badges_on_taskbar
116         hide_flashing = self._settings.hide_flashing_on_taskba
| r
117         dnd = self._settings.turn_on_do_not_disturb
118         show_timer = self._settings.show_timer_in_clock_app
119
120         if winreg:
121             try:
122                 # 1. Проверяем бейджи на панели задач (Taskbar

```

```

Badges: 0 = скрыты, 1 = показаны)
122         with winreg.OpenKey(winreg.HKEY_CURRENT_USER,
    r"Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced"
    , 0, winreg.KEY_READ) as key:
123             try:
124                 val, _ = winreg.QueryValueEx(key, "TaskbarBadges")
125                 hide_badges = (val == 0)
126             except FileNotFoundError:
127                 pass
128         except Exception as exc:
129             logger.debug(f"[FocusManager] Ошибка чтения TaskbarBadges: {exc}")
130
131         try:
132             # 2. Проверяем мигание (TaskbarFlashing: 0 = отключено, 1 = включено)
133             with winreg.OpenKey(winreg.HKEY_CURRENT_USER,
    r"Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced"
    , 0, winreg.KEY_READ) as key:
134                 try:
135                     val, _ = winreg.QueryValueEx(key, "TaskbarFlashing")
136                     hide_flashing = (val == 0)
137                 except FileNotFoundError:
138                     pass
139             except Exception as exc:
140                 logger.debug(f"[FocusManager] Ошибка чтения TaskbarFlashing: {exc}")
141
142         try:
143             # 3. Режим уведомлений (NOC_GLOBAL_SETTING_TOASTS_ENABLED: 0 = DND/выключены, 1 = включены)
144             with winreg.OpenKey(winreg.HKEY_CURRENT_USER,
    r"Software\Microsoft\Windows\CurrentVersion\Notifications\Settings", 0, winreg.KEY_READ) as key:
145                 try:
146                     val, _ = winreg.QueryValueEx(key, "NOC

```

```

    _GLOBAL_SETTING_TOASTS_ENABLED")
147         dnd = (val == 0)
148         except FileNotFoundError:
149             pass
150         except Exception as exc:
151             logger.debug(f"[FocusManager] Ошибка чтения уведомлений DND: {exc}")
152
153         return FocusSettingsDTO(
154             session_duration_minutes=self._session_duration_minutes,
155             show_timer_in_clock_app=show_timer,
156             hide_badges_on_taskbar=hide_badges,
157             hide_flashing_on_taskbar=hide_flashing,
158             turn_on_do_not_disturb=dnd,
159         )
160
161     def update_settings(self, new_settings: FocusSettingsDTO)
-> FocusSettingsDTO:
162         """Сохраняет обновленные параметры фокусировки.
163
164         Args:
165             new_settings: Новый набор настроек.
166
167         Returns:
168             FocusSettingsDTO: Актуальные настройки.
169         """
170         self._settings = new_settings
171         self._session_duration_minutes = new_settings.session_duration_minutes
172         return self._settings
173
174     async def start_session(self, duration_minutes: Optional[int] = None) -> FocusSessionStateDTO:
175         """Запускает сессию фокусировки внимания.
176
177         Args:
178             duration_minutes: Длительность в минутах (если пер

```

еопределена).

```
179
180     Returns:
181         FocusSessionStateDT0: Новое состояние сессии.
182     """
183     if duration_minutes:
184         self._session_duration_minutes = max(5, min(240, d
uration_minutes))
185
186     now = time.time()
187     self._is_active = True
188     self._started_at_ts = now
189     self._ends_at_ts = now + (self._session_duration_minut
es * 60)
190
191     # Сохраняем резервную копию системного состояния для в
осстановления
192     self._backup_state = {
193         "applied": True,
194         "dnd_applied": self._settings.turn_on_do_not_distu
rb,
195         "badges_applied": self._settings.hide_badges_on_ta
skbar,
196         "flashing_applied": self._settings.hide_flashing_o
n_taskbar,
197     }
198
199     # Применяем системные ограничения фокуса
200     self._apply_focus_state(
201         dnd=self._settings.turn_on_do_not_disturb,
202         hide_badges=self._settings.hide_badges_on_taskbar,
203         hide_flashing=self._settings.hide_flashing_on_task
bar,
204     )
205
206     # Открываем Windows Clock / Focus если запрошено
207     if self._settings.show_timer_in_clock_app:
208         self.launch_clock_focus_app()
```

```

209
210         logger.info(f"[FocusManager] Сессия фокусировки активи
        рована на {self._session_duration_minutes} мин")
211         return self.get_state()
212
213     def stop_session(self) -> FocusSessionStateDTO:
214         """Останавливает сессию фокусировки и восстанавливает
        параметры.
215
216         Returns:
217             FocusSessionStateDTO: Обновленное состояние сессии
        .
218         """
219         if self._is_active:
220             self._is_active = False
221             # Восстанавливаем обычный режим (выключаем DND, во
        звращаем бейджи и мигание)
222             self._apply_focus_state(dnd=False, hide_badges=False,
        hide_flashing=False)
223             logger.info("[FocusManager] Сессия фокусировки ост
        ановлена пользователем")
224
225         return self.get_state()
226
227     def _apply_focus_state(self, dnd: bool, hide_badges: bool,
        hide_flashing: bool) -> None:
228         """Применяет параметры в системный реестр Windows.
229
230         Args:
231             dnd: Включить режим «Не беспокоить».
232             hide_badges: Скрыть бейджи значков панели задач.
233             hide_flashing: Скрыть мигание панели задач.
234         """
235         if not winreg:
236             return
237
238         try:
239             # 1. Бейджи панели задач (0 = скрыть, 1 = показать

```

```

)
240         with winreg.CreateKey(winreg.HKEY_CURRENT_USER, r"
Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced")
as key:
241             winreg.SetValueEx(key, "TaskbarBadges", 0, win
reg.REG_DWORD, 0 if hide_badges else 1)
242             winreg.SetValueEx(key, "TaskbarFlashing", 0, w
inreg.REG_DWORD, 0 if hide_flashing else 1)
243         except Exception as exc:
244             logger.warning(f"[FocusManager] Ошибка установки п
араметров панели задач: {exc}")
245
246         try:
247             # 2. Уведомления Do Not Disturb (0 = выключены / D
ND, 1 = включены)
248             with winreg.CreateKey(winreg.HKEY_CURRENT_USER, r"
Software\Microsoft\Windows\CurrentVersion\Notifications\Settin
gs") as key:
249                 winreg.SetValueEx(key, "NOC_GLOBAL_SETTING_TOA
STS_ENABLED", 0, winreg.REG_DWORD, 0 if dnd else 1)
250             except Exception as exc:
251                 logger.warning(f"[FocusManager] Ошибка установки D
ND в реестре: {exc}")
252
253         @staticmethod
254         def launch_clock_focus_app() -> bool:
255             """Запускает системное приложение «Часы» в режиме Focu
s Sessions.
256
257             Returns:
258                 bool: True если команда успешно отправлена.
259             """
260             try:
261                 if sys.platform == "win32":
262                     os.startfile("ms-clock:focus")
263                     return True
264             except Exception as exc:
265                 logger.debug(f"[FocusManager] Не удалось открыть m

```

```

    s-clock:focus ({exc}), пробуем ms-settings:quiethours")
266         try:
267             if sys.platform == "win32":
268                 os.startfile("ms-settings:quiethours")
269                 return True
270         except Exception:
271             pass
272         return False
273
274     @staticmethod
275     def launch_settings_app() -> bool:
276         """Открывает системные параметры Windows в разделе Focus / Фокусировка внимания.
277
278         Returns:
279             bool: True если окно успешно открыто.
280         """
281         try:
282             if sys.platform == "win32":
283                 os.startfile("ms-settings:quiethours")
284                 return True
285             except Exception as exc:
286                 logger.warning(f"[FocusManager] Ошибка открытия ms
287 -settings:quiethours: {exc}")
288                 return False
289
290 _manager = FocusSessionManager()
291
292
293 def init_router() -> APIRouter:
294     """Инициализация и получение роутера FastAPI для управления Focus.
295
296     Returns:
297         APIRouter: Экземпляр настроенного роутера.
298     """
299     router = APIRouter(prefix="/api/windows/focus", tags=["Win

```

```

dows Focus Sessions"]])
300
301     @router.get("/state", response_model=FocusSessionStateDTO)
302     async def get_focus_state() -> FocusSessionStateDTO:
303         """Получить текущее состояние сессии фокусировки внима
        ния и системных опций."""
304         return _manager.get_state()
305
306     @router.post("/session/start", response_model=FocusSession
        StateDTO)
307     async def start_focus_session(duration_minutes: Optional[i
        nt] = None) -> FocusSessionStateDTO:
308         """Запустить новую сессию фокусировки внимания."""
309         return await _manager.start_session(duration_minutes)
310
311     @router.post("/session/stop", response_model=FocusSessionS
        tateDTO)
312     async def stop_focus_session() -> FocusSessionStateDTO:
313         """Остановить текущую сессию фокусировки и восстановит
        ь параметры."""
314         return _manager.stop_session()
315
316     @router.post("/settings", response_model=FocusSettingsDTO)
317     async def update_focus_settings(settings: FocusSettingsDTO
        ) -> FocusSettingsDTO:
318         """Обновить постоянные настройки фокусировки внимания.
        """
319         return _manager.update_settings(settings)
320
321     @router.post("/open-settings")
322     async def open_windows_focus_settings() -> Dict[str, Any]:
323         """Открыть системное окно «Фокусировка внимания» (ms-s
        ettings:quiethours) в Windows."""
324         opened = _manager.launch_settings_app()
325         return {"success": opened, "target": "ms-settings:quie
        thours"}
326
327     @router.post("/open-clock")

```



```
328 |     async def open_windows_clock_app() -> Dict[str, Any]:
329 |         """Открыть приложение «Часы» в разделе сессий фокусиро
330 |         вки (ms-clock:focus)."""
331 |         opened = _manager.launch_clock_focus_app()
332 |         return {"success": opened, "target": "ms-clock:focus"}
333 |
334 |     return router
335 |
336 | __all__ = ["init_router", "FocusSettingsDTO", "FocusSessionSta
337 | teDTO", "FocusSessionManager"]
```

router_forensics.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Forensics
4 # =====
5 # Description:
6 #   Роутер, предоставляющий данные о форензике для фронтенда в
7   #  кладки Forensics.
8 #   Возвращает JSON со сведениями о текущем активном окне, времени
9   #   бездействия
10 #   пользователя и приложениях, использующих камеру/микрофон,
11   #   а также данными
12 #   UserAssist.
13 #
14 # Usage Examples:
15 #   Python API:
16 #       from apps.windows.api.routers.router_forensics import init_router
17 #
18 #       router = init_router()
19 #
20 # File: router_forensics.py
21 # Project: ai-breadboard
22 # Package: apps.windows.api.routers
23 # Author: hypo69
24 # Copyright: © 2026 hypo69
25 # Updated: 2026-10-06 17:41:00
26 # =====
27
28 import asyncio
29 from datetime import datetime, timezone
30 from typing import Any, Dict, List, Optional
31 from fastapi import APIRouter
```

```
29 from pydantic import BaseModel, Field
30
31 from logger import logger
32 from apps.windows.telemetry.sqlite import TelemetryStorage
33
34
35 class ForegroundWindow(BaseModel):
36     """Информация о текущем активном окне."""
37     title: str = Field(default="Рабочий стол Windows", description="Заголовок окна")
38     process_name: str = Field(default="explorer.exe", description="Имя процесса")
39     pid: int = Field(default=0, description="PID процесса")
40
41 class AppAccessInfo(BaseModel):
42     """Приложение, имеющее доступ к камере или микрофону."""
43     app_name: str = Field(..., description="Имя приложения")
44     is_active_now: bool = Field(False, description="Активно в данный момент")
45
46 class UserAssistApp(BaseModel):
47     """Запись UserAssist из реестра."""
48     name: str = Field(..., description="Имя программы")
49     run_count: int = Field(..., description="Количество запусков")
50     focus_formatted: str = Field(..., description="Время фокуса (чч:мм:сс)")
51     path: str = Field(..., description="Путь к исполняемому файлу")
52
53 class ForensicsResponse(BaseModel):
54     """Ответ, ожидаемый фронтендом Forensics-вкладки."""
55     status: str = Field(default="ok", description="Статус ответа")
56     foreground_window: ForegroundWindow = Field(..., description="Текущее активное окно")
57     user_idle_seconds: float = Field(..., description="Время бездействия пользователя в секундах")
```

```

58     camera_active_apps: list[AppAccessInfo] = Field(default_factory=list, description="Приложения, использующие камеру")
59     microphone_active_apps: list[AppAccessInfo] = Field(default_factory=list, description="Приложения, использующие микрофон")
60     userassist_top_apps: list[UserAssistApp] = Field(default_factory=list, description="Топ-приложения UserAssist")
61
62
63 def _convert_snapshot_to_response(d: Dict[str, Any]) -> ForensicsResponse:
64     """Преобразует сырые данные снимка SQLite в модель ответа ForensicsResponse."""
65     fg = ForegroundWindow(
66         title=d.get("foreground_window_title") or "Рабочий стол Windows",
67         process_name=d.get("foreground_process_name") or "explorer.exe",
68         pid=int(d.get("foreground_pid") or 0),
69     )
70     cam = [AppAccessInfo(**a) if isinstance(a, dict) else a for a in (d.get("camera_active_apps") or [])]
71     mic = [AppAccessInfo(**a) if isinstance(a, dict) else a for a in (d.get("microphone_active_apps") or [])]
72     ua = [UserAssistApp(**a) if isinstance(a, dict) else a for a in (d.get("userassist_top_apps") or [])]
73     return ForensicsResponse(
74         status="ok",
75         foreground_window=fg,
76         user_idle_seconds=float(d.get("user_idle_seconds") or 0.0),
77         camera_active_apps=cam,
78         microphone_active_apps=mic,
79         userassist_top_apps=ua,
80     )
81
82
83 def _perform_forensics_scan(storage: TelemetryStorage) -> Fore

```

```

nsicsResponse:
84 |     """Выполняет живой сбор форензики и сохраняет снимок в SQL
    |     ite."""
85 |     try:
86 |         from apps.windows.telemetry.deep_diagnostics import DeepDiagnosticsEngine
87 |         engine = DeepDiagnosticsEngine()
88 |         rep = engine.collect_forensics_activity()
89 |         snap_id = f"snap_forensics_{int(datetime.now(timezone.
    |         utc).timestamp())}"
90 |         storage.save_forensics_snapshot(snap_id, {
91 |             "foreground_window_title": rep.foreground_window.g
    |         et("title", ""),
92 |             "foreground_process_name": rep.foreground_window.g
    |         et("process_name", ""),
93 |             "foreground_pid": rep.foreground_window.get("pid",
    |             0),
94 |             "user_idle_seconds": int(rep.user_idle_seconds),
95 |             "camera_active_apps": rep.camera_active_apps,
96 |             "microphone_active_apps": rep.microphone_active_ap
    |         ps,
97 |             "userassist_top_apps": rep.userassist_top_apps,
98 |         })
99 |         d = storage.get_latest_forensics_snapshot()
100 |         if d:
101 |             return _convert_snapshot_to_response(d)
102 |         return ForensicsResponse(
103 |             status="ok",
104 |             foreground_window=ForegroundWindow(**rep.foregroun
    |         d_window),
105 |             user_idle_seconds=rep.user_idle_seconds,
106 |             camera_active_apps=[AppAccessInfo(**a) for a in re
    |         p.camera_active_apps],
107 |             microphone_active_apps=[AppAccessInfo(**a) for a i
    |         n rep.microphone_active_apps],
108 |             userassist_top_apps=[UserAssistApp(**a) for a in r
    |         ep.userassist_top_apps],
109 |         )

```

```

110     except Exception as ex:
111         logger.debug(f"Ошибка сбора форензики: {ex}")
112         return ForensicsResponse(
113             status="ok",
114             foreground_window=ForegroundWindow(),
115             user_idle_seconds=0.0,
116             camera_active_apps=[],
117             microphone_active_apps=[],
118             userassist_top_apps=[],
119         )
120
121
122 def init_router(storage: Optional[TelemetryStorage] = None) ->
    APIRouter:
123     """Инициализирует роутер Forensics с чтением из SQLite (<
    5 мс).
124
125     Путь: `/api/v1/system/diagnostics/forensics`
126     """
127     router = APIRouter(prefix="/api/v1/system/diagnostics", ta
    gs=["Forensics"])
128     store = storage or TelemetryStorage.get_instance(read_only
    =True)
129
130     @router.get("/forensics", response_model=ForensicsResponse
    )
131     async def get_forensics() -> ForensicsResponse:
132         """Возвращает срез активности пользователя и форензики
    из SQLite (< 5 мс)."""
133         d = store.get_latest_forensics_snapshot()
134         if not d:
135             return await asyncio.to_thread(_perform_forensics_
    scan, store)
136         return _convert_snapshot_to_response(d)
137
138     @router.post("/forensics/refresh", response_model=Forensic
    sResponse)
139     @router.post("/forensics/rescan", response_model=Forensics

```

```
Response)
140 |     async def refresh_forensics() -> ForensicsResponse:
141 |         """Принудительно пересканирует форензик-активность и о
| бновляет срез в SQLite."""
142 |         return await asyncio.to_thread(_perform_forensics_scan
| , store)
143 |
144 |     return router
145 |
```

router_gpu_load.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router GPU Load
4 # =====
5 # Description:
6 #   GET /api/v1/panel/gpu-load – метрики графического ускорителя (GPU):
7 #   загрузка GPU Core, подсистем (Memory Controller, Video Engine, Bus, D3D),
8 #   температуры (Core, Hot Spot), видеопамять VRAM, энергопотребление,
9 #   паспортные данные GPU (Specs), датчики и история замеров из LHM и telemetry.db.
10 #
11 # Usage Examples:
12 #   Python API:
13 #       from apps.windows.api.routers.router_gpu_load import init_router
14 #       app.include_router(init_router())
15 #
16 # File: router_gpu_load.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 07:25:00
22 # =====
23
24 from __future__ import annotations
25
26 """Роутер панели «Загрузка GPU»: детальные метрики видеокарты, паспорт, сенсоры и история из LHM и telemetry.db."""
```



```
27 |
28 | from datetime import datetime, timezone
29 | import json
30 | import re
31 | from typing import Any, Dict, List, Optional
32 | from fastapi import APIRouter, Query
33 | from pydantic import BaseModel, Field
34 |
35 | from logger import logger
36 | from apps.windows.modules.hardware.lhm_service import LhmService
37 | from apps.windows.modules.hardware.gpu_prober import GpuProber
38 | from apps.windows.telemetry.sqlite import TelemetryStorage
39 |
40 | _TEMP_CATEGORIES = ("temperature", "temperatures")
41 | _LOAD_CATEGORIES = ("load", "loads")
42 | _DATA_CATEGORIES = ("data", "memory")
43 | _CLOCK_CATEGORIES = ("clock", "clocks")
44 | _POWER_CATEGORIES = ("power", "powers")
45 | _VOLTAGE_CATEGORIES = ("voltage", "voltages")
46 |
47 | _CPU_HARDWARE_KEYWORDS = (
48 |     "cpu", "processor", "core i3", "core i5", "core i7", "core
49 |     i9",
50 |     "ryzen", "xeon", "threadripper", "pentium", "celeron", "athlon"
51 | )
52 | _CPU_SENSOR_BLACKLIST = (
53 |     "cpu core", "cpu thread", "cpu total", "cpu max", "cpu package",
54 |     "cpu cores", "cpu memory", "cpu platform", "bus speed", "core average",
55 |     "core max", "distance to tjmax", "cpu freq", "cpu utilization",
56 |     "core #", "thread #"
57 | )
58 |
```

```
59
60 class GpuEngineMetric(BaseModel):
61     """Метрика отдельного движка/блока GPU."""
62     name: str = Field(..., description="Название движка (Core, Memory Controller, Video Engine, etc.)")
63     load_percent: Optional[float] = Field(default=None, description="Загрузка движка, %")
64     temperature_c: Optional[float] = Field(default=None, description="Температура, связанная с блоком, °C")
65
66
67 class GpuMemoryMetric(BaseModel):
68     """Метрики видеопамати (VRAM)."""
69     total_mb: Optional[float] = Field(default=None, description="Всего видеопамати, MB")
70     used_mb: Optional[float] = Field(default=None, description="Занято видеопамати, MB")
71     free_mb: Optional[float] = Field(default=None, description="Свободно видеопамати, MB")
72     used_percent: Optional[float] = Field(default=None, description="Процент занятой видеопамати, %")
73
74
75 class GpuClocksMetric(BaseModel):
76     """Частоты работы GPU."""
77     core_mhz: Optional[float] = Field(default=None, description="Частота графического ядра, MHz")
78     memory_mhz: Optional[float] = Field(default=None, description="Частота видеопамати, MHz")
79
80
81 class GpuSensorMetric(BaseModel):
82     """Электрический или физический сенсор графического процессора."""
83     id: str = Field(..., description="Уникальный идентификатор сенсора")
84     name: str = Field(..., description="Понятное название сенсора")
```

```

85     category: str = Field(default="General", description="Категория (Powers, Voltages, Clocks, Fans)")
86     unit: str = Field(default="", description="Единица измерения (W, V, MHz, RPM, %)")
87     value: float = Field(default=0.0, description="Числовое значение")
88     value_raw: str = Field(default="", description="Форматированная строка со значением")
89     status: str = Field(default="normal", description="Статус: normal, warning, danger")
90
91
92 class GpuHistoryPoint(BaseModel):
93     """Точка временного ряда истории параметров GPU."""
94     timestamp: str = Field(default="", description="ISO время точки")
95     time_label: str = Field(default="", description="Короткая метка времени ЧЧ:ММ:СС")
96     load_percent: float = Field(default=0.0, description="Загрузка ядра GPU, %")
97     temperature_c: Optional[float] = Field(default=None, description="Температура GPU Core, °C")
98     power_w: Optional[float] = Field(default=None, description="Потребляемая мощность, Вт")
99     vram_percent: Optional[float] = Field(default=None, description="Занятая память VRAM, %")
100
101
102 class GpuSpecsInfo(BaseModel):
103     """Сводная спецификация и паспортные данные графического ускорителя."""
104     name: str = Field(default="", description="Наименование модели видеокарты")
105     vendor: str = Field(default="NVIDIA", description="Производитель (NVIDIA, AMD, Intel, Generic)")
106     driver_version: str = Field(default="", description="Версия видеодрайвера")
107     driver_date: str = Field(default="", description="Дата релиза")

```

```
иза видеодрайвера")
108     vram_mb: Optional[float] = Field(default=None, description
    ="Объем видеопамати VRAM, МБ")
109     vram_gb: Optional[float] = Field(default=None, description
    ="Объем видеопамати VRAM, ГБ")
110     vram_str: str = Field(default="", description="Форматирова
    нный объем видеопамати")
111     vram_used_str: str = Field(default="", description="Формат
    ированная занятая память с процентом")
112     core_clock_mhz: Optional[float] = Field(default=None, desc
    ription="Частота графического процессора, МГц")
113     core_clock_str: str = Field(default="", description="Форма
    тированная частота ядра")
114     memory_clock_mhz: Optional[float] = Field(default=None, de
    scription="Частота видеопамати, МГц")
115     memory_clock_str: str = Field(default="", description="Фоп
    матированная частота памяти")
116     directx: str = Field(default="DirectX 12 (FL 12_1)", descr
    iption="Поддерживаемый графический API")
117     pci_bus: str = Field(default="PCIe x16", description="Инте
    рфейс подключения шины")
118     cuda_cores: Optional[int] = Field(default=None, descriptio
    n="Количество шейдерных ядер CUDA / блоков")
119     directml_supported: bool = Field(default=True, description
    ="Поддержка DirectML ускорения")
120     power_limit_w: Optional[float] = Field(default=None, descr
    iption="Лимит TDP, Вт")
121
122
123 class GpuLoadResponse(BaseModel):
124     """Ответ GET /api/v1/panel/gpu-load."""
125     status: str = Field(default="ok", description="Статус отве
    та")
126     name: str = Field(default="GPU", description="Модель видео
    карты")
127     vendor: str = Field(default="NVIDIA", description="Произво
    дитель GPU")
128     core_load_percent: float = Field(default=0.0, description=
```

```

"Загрузка графического ядра, %")
129     core_temperature_c: Optional[float] = Field(default=None,
description="Температура GPU Core, °C")
130     hotspot_temperature_c: Optional[float] = Field(default=None,
description="Температура Hot Spot, °C")
131     memory_temperature_c: Optional[float] = Field(default=None
, description="Температура VRAM, °C")
132     power_w: Optional[float] = Field(default=None, description
="Энергопотребление GPU, Вт")
133     fan_speed_rpm: Optional[float] = Field(default=None, descr
iption="Скорость вентилятора, RPM")
134     fan_percent: Optional[float] = Field(default=None, descrip
tion="Обороты вентилятора, %")
135     memory: GpuMemoryMetric = Field(default_factory=GpuMemoryM
etric, description="Состояние видеопамяти")
136     clocks: GpuClocksMetric = Field(default_factory=GpuClocksM
etric, description="Частоты GPU")
137     specs: Optional[GpuSpecsInfo] = Field(default=None, descri
ption="Паспортные характеристики GPU")
138     engines: List[GpuEngineMetric] = Field(default_factory=lis
t, description="Метрики блоков/движков GPU")
139     sensors: List[GpuSensorMetric] = Field(default_factory=lis
t, description="Сенсоры электропитания, напряжения и вентилято
ров GPU")
140     history: List[GpuHistoryPoint] = Field(default_factory=lis
t, description="История параметров GPU")
141     available_gpus: List[Dict[str, Any]] = Field(default_facto
ry=list, description="Список всех обнаруженных видеокарт")
142     devices: List[Dict[str, Any]] = Field(default_factory=list
, description="Полные метрики по каждому отдельному GPU в сист
еме")
143     current_gpu_index: int = Field(default=0, description="Инд
екс текущей отображаемой видеокарты")
144     meta: Dict[str, Any] = Field(default_factory=dict, descrip
tion="Метаданные запроса")
145
146
147 def _calculate_gpu_sensor_status(category: str, value: float)

```

```

-> str:
148     """Определяет статус сенсора GPU по категории и значению."
    ""
149     cat = category.lower()
150     if "temp" in cat:
151         if value >= 83.0:
152             return "danger"
153         if value >= 74.0:
154             return "warning"
155         return "normal"
156     if "load" in cat:
157         if value >= 95.0:
158             return "danger"
159         if value >= 80.0:
160             return "warning"
161         return "normal"
162     return "normal"
163
164
165 def _normalize_gpu_sensor_name(sid: str, name: str, cat: str)
-> str:
166     """Формирует понятное название для сенсора GPU без Unknown
    ."""
167     raw = (name or "").strip()
168     if not raw or raw.lower() in ("unknown", "sensor"):
169         return f"GPU Sensor #{sid}" if sid else "GPU Sensor"
170
171     raw_l = raw.lower()
172     cat_l = cat.lower()
173
174     if raw_l == "gpu package" and "power" in cat_l:
175         return "GPU Package Power"
176     if raw_l == "gpu power" or (raw_l == "power" and "power" i
n cat_l):
177         return "GPU Board Power"
178     if raw_l == "gpu core" and "volt" in cat_l:
179         return "GPU Core Voltage"
180     if raw_l == "gpu memory" and "clock" in cat_l:

```

```
181         return "GPU Memory Clock"
182     if raw_l == "gpu core" and "clock" in cat_l:
183         return "GPU Core Clock"
184     if raw_l == "gpu fan" or (raw_l == "fan" and "fan" in cat_
185         l):
186         return "GPU Fan Speed"
187     return raw
188
189
190 def _is_sensor_matching_gpu(sensor_hw_name: str, target_gpu_name: str, sensor_hw_type: str = "") -> bool:
191     """Проверяет, принадлежит ли показание сенсора целевой видеокарте и не относится ли к CPU."""
192     if sensor_hw_type and str(sensor_hw_type).lower() == "cpu":
193         return False
194     if not sensor_hw_name:
195         return True
196
197     s_clean_raw = str(sensor_hw_name).strip().lower()
198     if any(k in s_clean_raw for k in _CPU_HARDWARE_KEYWORDS):
199         return False
200
201     if not target_gpu_name:
202         return True
203
204     t_clean_raw = str(target_gpu_name).strip().lower()
205
206     # Прямое равенство или взаимное включение подстрок
207     if s_clean_raw == t_clean_raw or s_clean_raw in t_clean_raw or t_clean_raw in s_clean_raw:
208         return True
209
210     ignore = {
211         'gpu', 'graphics', 'corporation', 'corp', 'inc', 'series', 'adapter', 'video',
212         'display', 'controller', 'intel', 'nvidia', 'amd', 'mi
```

```

crosoft', 'basic'
213     }
214     s_clean = re.sub(r'^a-zA-Z0-9+', ' ', s_clean_raw)
215     t_clean = re.sub(r'^a-zA-Z0-9+', ' ', t_clean_raw)
216     s_toks = {t for t in s_clean.split() if t and t not in ignore}
217     t_toks = {t for t in t_clean.split() if t and t not in ignore}
218
219     if not s_toks or not t_toks:
220         # Если остались только бренды, проверяем вендора
221         s_brand = "nvidia" if "nvidia" in s_clean_raw or "geforce" in s_clean_raw else ("amd" if "amd" in s_clean_raw or "ra
deon" in s_clean_raw else ("intel" if "intel" in s_clean_raw e
lse None))
222         t_brand = "nvidia" if "nvidia" in t_clean_raw or "gefo
rce" in t_clean_raw else ("amd" if "amd" in t_clean_raw or "ra
deon" in t_clean_raw else ("intel" if "intel" in t_clean_raw e
lse None))
223         return bool(s_brand and t_brand and s_brand == t_brand
)
224
225     return bool(s_toks & t_toks)
226
227
228 def build_gpu_load(
229     sensors: List[Dict[str, Any]],
230     inventory: Optional[Dict[str, Any]] = None,
231     history_samples: Optional[List[Dict[str, Any]]] = None,
232     target_gpu_name: Optional[str] = None,
233     available_gpus: Optional[List[Dict[str, Any]]] = None,
234     current_gpu_index: int = 0,
235 ) -> GpuLoadResponse:
236     """Собирает структурированные метрики конкретного GPU из с
писка сенсоров, паспорта и истории.
237
238     Args:
239         sensors: Список словарей сенсоров.

```



```
240 |         inventory: Паспортные данные GPU из gpu_inventory / Gp
| uProber.
241 |         history_samples: Исторические сэмплы из gpu_telemetry_
| samples или снимков.
242 |         target_gpu_name: Название целевой видеокарты для фильт
| рации сенсоров.
243 |         available_gpus: Список всех видеокарт в системе.
244 |         current_gpu_index: Индекс выбранной видеокарты.
245 |
246 |     Returns:
247 |         GpuLoadResponse: Полная информация по загрузке, темпер
| атуре, памяти, спецификациям и истории GPU.
248 |     """
249 |     gpu_name = target_gpu_name or (inventory.get("name") if in
| ventory else "GPU")
250 |     core_load: Optional[float] = None
251 |     core_temp: Optional[float] = None
252 |     hotspot_temp: Optional[float] = None
253 |     mem_temp: Optional[float] = None
254 |     power_draw: Optional[float] = None
255 |     fan_rpm: Optional[float] = None
256 |     fan_pct: Optional[float] = None
257 |
258 |     vram_total: Optional[float] = None
259 |     vram_used: Optional[float] = None
260 |     vram_free: Optional[float] = None
261 |     vram_pct: Optional[float] = None
262 |
263 |     clock_core: Optional[float] = None
264 |     clock_mem: Optional[float] = None
265 |
266 |     engines_dict: Dict[str, GpuEngineMetric] = {}
267 |     general_sensors_list: List[GpuSensorMetric] = []
268 |     seen_sensor_keys = set()
269 |
270 |     for s in sensors:
271 |         hw_type = str(s.get("hardware_type") or "").lower()
272 |         hw_name = str(s.get("hardware_name") or "")
```

```
273 |         s_id = str(s.get("sensor_id") or s.get("id") or "")
274 |         s_name = str(s.get("sensor_name") or s.get("name") or
| s_id)
275 |         s_cat = str(s.get("sensor_category") or s.get("category") or "").lower()
276 |         s_unit = str(s.get("unit") or "")
277 |
278 |         # 1. Строго отсекаем сенсоры CPU
279 |         if hw_type == "cpu" or any(k in hw_name.lower() for k
| in _CPU_HARDWARE_KEYWORDS):
280 |             continue
281 |
282 |         s_name_lower = s_name.lower()
283 |         s_id_lower = s_id.lower()
284 |         if any(black in s_name_lower or black in s_id_lower fo
| r black in _CPU_SENSOR_BLACKLIST):
285 |             continue
286 |
287 |         # 2. Фильтруем сенсоры GPU
288 |         is_gpu = (
289 |             "gpu" in hw_type
290 |             or "gpu" in hw_name.lower()
291 |             or "nvidia" in hw_name.lower()
292 |             or "radeon" in hw_name.lower()
293 |             or "geforce" in hw_name.lower()
294 |             or "rtx" in hw_name.lower()
295 |             or "gtx" in hw_name.lower()
296 |             or "uhd" in hw_name.lower()
297 |             or "iris" in hw_name.lower()
298 |             or "arc" in hw_name.lower()
299 |         )
300 |         if not is_gpu:
301 |             continue
302 |
303 |         # 3. Фильтруем строго под целевой GPU, если имя задано
304 |         if target_gpu_name and hw_name and not _is_sensor_matc
| hing_gpu(hw_name, target_gpu_name, hw_type):
305 |             continue
```

```

306
307         if hw_name and (gpu_name == "GPU" or not target_gpu_name):
308             gpu_name = hw_name
309
310         # Значение
311         val = s.get("value_num")
312         if val is None:
313             val = s.get("value_numeric")
314         if val is None:
315             val = s.get("value")
316         if val is None:
317             raw = s.get("value_raw")
318             if isinstance(raw, (int, float)):
319                 val = float(raw)
320             elif isinstance(raw, str):
321                 cleaned = raw.replace("%", "").replace("°C", "°C").replace("MHz", "").replace("MB", "").replace("RPM", "").replace("W", "").replace("V", "").strip()
322                 try:
323                     val = float(cleaned)
324                 except ValueError:
325                     val = None
326
327         if val is None:
328             continue
329
330         # 1. Температуры
331         if any(cat in s_cat for cat in _TEMP_CATEGORIES) or s_unit == "°C":
332             if "hot spot" in s_name_lower or "hotspot" in s_name_lower:
333                 hotspot_temp = val
334             elif "memory" in s_name_lower or "vram" in s_name_lower:
335                 mem_temp = val
336             elif "core" in s_name_lower or core_temp is None:
337                 core_temp = val

```

```

338
339     # 2. Нагрузка движков / подсистем
340     elif any(cat in s_cat for cat in _LOAD_CATEGORIES) or
341         (s_unit == "%" and "fan" not in s_cat):
342         if s_name_lower in ("gpu core", "core", "gpu"):
343             core_load = val
344             engines_dict["GPU Core"] = GpuEngineMetric(name="GPU Core", load_percent=val, temperature_c=core_temp)
345             elif "memory controller" in s_name_lower:
346                 engines_dict["Memory Controller"] = GpuEngineMetric(name="Memory Controller", load_percent=val)
347             elif "video engine" in s_name_lower or "video decode" in s_name_lower:
348                 if "Video Engine" not in engines_dict or val >
349                     (engines_dict["Video Engine"].load_percent or 0):
350                     engines_dict["Video Engine"] = GpuEngineMetric(name="Video Engine", load_percent=val)
351             elif "bus" in s_name_lower:
352                 engines_dict["GPU Bus"] = GpuEngineMetric(name="GPU Bus", load_percent=val)
353             elif "memory" in s_name_lower and "controller" not
354                 in s_name_lower:
355                 vram_pct = val
356                 engines_dict["GPU Memory"] = GpuEngineMetric(name="GPU Memory", load_percent=val)
357             elif "3d" in s_name_lower:
358                 engines_dict["D3D 3D"] = GpuEngineMetric(name="D3D 3D", load_percent=val)
359             elif "copy" in s_name_lower:
360                 engines_dict["Copy Engine"] = GpuEngineMetric(name="Copy Engine", load_percent=val)
361             elif "compute" in s_name_lower:
362                 eng_name = s_name.strip()
363                 engines_dict[eng_name] = GpuEngineMetric(name=eng_name, load_percent=val)
364             else:
365                 engines_dict[s_name] = GpuEngineMetric(name=s_name, load_percent=val)

```

```
363 |
364 |         # 3. Видеопамять (VRAM)
365 |         elif any(cat in s_cat for cat in _DATA_CATEGORIES) or
366 |         "mb" in s_unit.lower() or "gb" in s_unit.lower():
367 |             if "total" in s_name_lower:
368 |                 vram_total = val
369 |             elif "used" in s_name_lower:
370 |                 if "dedicated" in s_name_lower:
371 |                     if vram_used is None:
372 |                         vram_used = val
373 |                     else:
374 |                         vram_used = val
375 |                 elif "free" in s_name_lower:
376 |                     vram_free = val
377 |
378 |         # 4. Частоты
379 |         elif any(cat in s_cat for cat in _CLOCK_CATEGORIES) or
380 |         "mhz" in s_unit.lower():
381 |             if "core" in s_name_lower:
382 |                 clock_core = val
383 |             elif "memory" in s_name_lower:
384 |                 clock_mem = val
385 |
386 |         # 5. Мощность (Power)
387 |         elif any(cat in s_cat for cat in _POWER_CATEGORIES) or
388 |         s_unit.lower() == "w":
389 |             if power_draw is None or "package" in s_name_lower
390 |             or "total" in s_name_lower or "board" in s_name_lower:
391 |                 power_draw = val
392 |
393 |         # 6. Обороты вентилятора
394 |         elif "fan" in s_cat or "fan" in s_name_lower:
395 |             if "rpm" in s_unit.lower() or val > 100:
396 |                 fan_rpm = val
397 |             else:
398 |                 fan_pct = val
399 |
400 |         # Добавление в общий список электрических/физических с
```

```

енсоров GPU
397         if (
398             any(cat in s_cat for cat in _POWER_CATEGORIES)
399             or any(cat in s_cat for cat in _VOLTAGE_CATEGORIES
400             or any(cat in s_cat for cat in _CLOCK_CATEGORIES)
401             or "fan" in s_cat
402             or s_unit.lower() in ("w", "v", "rpm")
403         ):
404             norm_name = _normalize_gpu_sensor_name(s_id, s_name, s_cat)
405             s_key = f"{s_cat}_{norm_name.lower()}"
406             if s_key not in seen_sensor_keys:
407                 seen_sensor_keys.add(s_key)
408                 unit_str = s_unit
409                 if not unit_str:
410                     if any(cat in s_cat for cat in _POWER_CATEGORIES): unit_str = "W"
411                     elif any(cat in s_cat for cat in _VOLTAGE_CATEGORIES): unit_str = "V"
412                     elif any(cat in s_cat for cat in _CLOCK_CATEGORIES): unit_str = "MHz"
413                     elif "fan" in s_cat: unit_str = "RPM" if val > 100 else "%"
414
415                 val_raw_str = f"{val:.1f} {unit_str}".strip()
416                 if unit_str == "V":
417                     val_raw_str = f"{val:.3f} V"
418                 elif unit_str == "MHz" and val >= 1000.0:
419                     val_raw_str = f"{val / 1000.0:.2f} GHz"
420
421                 general_sensors_list.append(
422                     GpuSensorMetric(
423                         id=s_id or f"gpu_sensor_{len(general_sensors_list)+1}",
424                         name=norm_name,
425                         category=s_cat.capitalize() or "General",

```

```

426         unit=unit_str,
427         value=round(val, 2),
428         value_raw=val_raw_str,
429         status=_calculate_gpu_sensor_status(s_
cat, val),
430     )
431 )
432
433 # Синхронизируем температуру для GPU Core в engines
434 if "GPU Core" in engines_dict and core_temp is not None:
435     engines_dict["GPU Core"].temperature_c = core_temp
436
437 # Вычисляем процент занятой памяти, если не пришёл явно
438 if vram_pct is None and vram_total and vram_used is not No
ne and vram_total > 0:
439     vram_pct = round((vram_used / vram_total) * 100, 1)
440
441 # Упорядочиваем список движков
442 ordered_keys = ["GPU Core", "Memory Controller", "Video En
gine", "GPU Memory", "GPU Bus", "D3D 3D", "Copy Engine"]
443 engines_list: List[GpuEngineMetric] = []
444 for k in ordered_keys:
445     if k in engines_dict:
446         engines_list.append(engines_dict.pop(k))
447 engines_list.extend(engines_dict.values())
448
449 # Паспортные данные GPU (Specs)
450 inv = inventory or {}
451 inv_name = inv.get("name") or gpu_name
452 inv_name_l = inv_name.lower()
453 vendor = inv.get("vendor") or ("NVIDIA" if "nvidia" in inv
_name_l or "geforce" in inv_name_l else ("AMD" if "amd" in inv
_name_l or "radeon" in inv_name_l else ("Intel" if "intel" in
inv_name_l or "arc" in inv_name_l or "uhd" in inv_name_l else
"GPU")))
454 driver_ver = str(inv.get("driver_version") or "")
455 driver_dt = str(inv.get("driver_date") or "")
456 vram_b = inv.get("vram_bytes") or 0

```

```

457     vram_g = inv.get("vram_gb") or (vram_total / 1024.0 if vram_total
458     m_total else (vram_b / (1024**3) if vram_b else 0.0))
459     if vram_total is None and vram_g > 0:
460         vram_total = vram_g * 1024.0
461     vram_str = f"{vram_g:.1f} GB" if vram_g >= 1.0 else (f"{vram_total:.0f} MB" if vram_total else "--")
462     if vram_used is not None and vram_total:
463         vram_used_gb = vram_used / 1024.0 if vram_used >= 1024
464         else vram_used / 1024.0
465         vram_used_str = f"{vram_used_gb:.1f} / {vram_g:.1f} GB ({vram_pct or 0:.0f}%)"
466     elif vram_pct is not None:
467         vram_used_str = f"{vram_pct:.0f}% занято"
468     else:
469         vram_used_str = "--"
470     core_clk_str = f"{clock_core:.0f} MHz" if clock_core else "--"
471     mem_clk_str = f"{clock_mem:.0f} MHz" if clock_mem else "--"
472     pci_bus_str = str(inv.get("pci_bus_id") or ("PCIe x16" if vendor in ("NVIDIA", "AMD") else "Integrated / PCIe"))
473     cuda_c = inv.get("cuda_cores")
474     directml = bool(inv.get("directml_supported", True))
475     specs_obj = GpuSpecsInfo(
476         name=inv_name,
477         vendor=vendor,
478         driver_version=driver_ver or "WDDM 2.7",
479         driver_date=driver_dt,
480         vram_mb=vram_total,
481         vram_gb=round(vram_g, 2) if vram_g else None,
482         vram_str=vram_str,
483         vram_used_str=vram_used_str,
484         core_clock_mhz=clock_core,
485         core_clock_str=core_clk_str,

```



```

488         memory_clock_mhz=clock_mem,
489         memory_clock_str=mem_clk_str,
490         directx="DirectX 12 (FL 12_1)",
491         pci_bus=pci_bus_str,
492         cuda_cores=cuda_c,
493         directml_supported=directml,
494     )
495
496     # История параметров GPU
497     history_points: List[GpuHistoryPoint] = []
498     if history_samples:
499         for r in history_samples:
500             ts_str = str(r.get("timestamp") or "")
501             t_label = ts_str.split("T")[-1][:8] if "T" in ts_s
502             p_load = float(r.get("load_percent") or r.get("gpu
503             _load_percent") or 0.0)
504             p_temp = r.get("temperature_gpu_c") or r.get("temp
505             erature_c") or r.get("core_temperature_c")
506             p_pow = r.get("power_draw_w") or r.get("power_w")
507             p_mem = r.get("memory_percent") or r.get("vram_per
508             cent")
509
510             history_points.append(
511                 GpuHistoryPoint(
512                     timestamp=ts_str,
513                     time_label=t_label or "--:--:--",
514                     load_percent=round(p_load, 1),
515                     temperature_c=round(float(p_temp), 1) if p
516                     _temp is not None else core_temp,
517                     power_w=round(float(p_pow), 1) if p_pow is
518                     not None else power_draw,
519                     vram_percent=round(float(p_mem), 1) if p_m
520                     em is not None else vram_pct,
521                 )
522             )
523
524     return GpuLoadResponse(

```

```

519         status="ok",
520         name=inv_name,
521         vendor=vendor,
522         core_load_percent=round(core_load if core_load is not
None else 0.0, 1),
523         core_temperature_c=core_temp,
524         hotspot_temperature_c=hotspot_temp,
525         memory_temperature_c=mem_temp,
526         power_w=power_draw,
527         fan_speed_rpm=fan_rpm,
528         fan_percent=fan_pct,
529         memory=GpuMemoryMetric(
530             total_mb=vram_total,
531             used_mb=vram_used,
532             free_mb=vram_free,
533             used_percent=vram_pct,
534         ),
535         clocks=GpuClocksMetric(
536             core_mhz=clock_core,
537             memory_mhz=clock_mem,
538         ),
539         specs=specs_obj,
540         engines=engines_list,
541         sensors=general_sensors_list,
542         history=history_points,
543         available_gpus=available_gpus or [],
544         current_gpu_index=current_gpu_index,
545         meta={
546             "source": "telemetry.db",
547             "engines_count": len(engines_list),
548             "sensors_count": len(general_sensors_list),
549         },
550     )
551
552
553 def _discover_system_gpus(storage: Optional[TelemetryStorage]
= None) -> List[Dict[str, Any]]:
554     """Собирает список всех физических и интегрированных видео

```

карт в системе строго из БД/инвентаря.

```
555
556     Returns:
557         List[Dict[str, Any]]: Список словарей характеристик об
наруженных GPU.
558     """
559     gpus: List[Dict[str, Any]] = []
560
561     # 1. Если передано хранилище и в нем есть инвентарь GPU
562     if storage:
563         try:
564             db_inv = storage.get_gpu_inventory()
565             if db_inv:
566                 for idx, item in enumerate(db_inv):
567                     g_name = item.get("name", f"GPU {idx}")
568                     g_name_l = g_name.lower()
569                     vendor = "NVIDIA" if "nvidia" in g_name_l
or "geforce" in g_name_l else ("AMD" if "amd" in g_name_l or "a
radeon" in g_name_l else ("Intel" if "intel" in g_name_l or "a
rc" in g_name_l or "uhd" in g_name_l else item.get("vendor", "
Generic")))
570
                    gpus.append({
571                        "index": item.get("gpu_id", idx),
572                        "name": g_name,
573                        "vendor": vendor,
574                        "driver_version": item.get("driver_ver
sion", ""),
575                        "driver_date": item.get("driver_date",
                    ""),
576                        "vram_gb": item.get("vram_gb", 0.0),
577                        "pci_bus_id": item.get("pci_bus_id", "
                    "),
578                        "cuda_cores": item.get("cuda_cores"),
579                        "directml_supported": item.get("direct
ml_supported", True),
580                    })
581         except Exception as exc:
582             logger.debug(f"[router_gpu_load] Ошибка чтения gpu
```

```

_inventory из БД: {exc}")
583
584     # Если инвентаря нет, но в БД есть сохраненные сенсоры
    GPU
585     if not gpus:
586         try:
587             db_sensors = storage.get_latest_sensors()
588             if db_sensors:
589                 seen_names: Dict[str, int] = {}
590                 for s in db_sensors:
591                     ht = str(s.get("hardware_type", "")).lower()
592                     hn = str(s.get("hardware_name", "")).strip().lower()
593                     hn_l = hn.lower()
594                     if ht == "cpu" or any(k in hn_l for k
in _CPU_HARDWARE_KEYWORDS):
595                         continue
596                     if hn and ("gpu" in ht or "gpu" in hn_l or "nvidia" in hn_l or "radeon" in hn_l or "geforce" in hn_l or "rtx" in hn_l or "gtx" in hn_l or "uhd" in hn_l or "iris" in hn_l or "arc" in hn_l):
597                         if hn not in seen_names:
598                             seen_names[hn] = len(seen_names)
599
600                 for hn, idx in seen_names.items():
601                     hn_l = hn.lower()
602                     vendor = "NVIDIA" if "nvidia" in hn_l or "geforce" in hn_l else ("AMD" if "amd" in hn_l or "radeon" in hn_l else ("Intel" if "intel" in hn_l or "arc" in hn_l or "uhd" in hn_l else "Generic"))
603                     gpus.append({
604                         "index": idx,
605                         "name": hn,
606                         "vendor": vendor,
607                         "driver_version": "",
608                         "vram_gb": 0.0,
609                     })

```

```

609         except Exception as exc:
610             logger.debug(f"[router_gpu_load] Ошибка извлеч
ения GPU из latest_sensors: {exc}")
611
612     # 2. Опрос через GpuProber (NVIDIA SMI + AMD SMI + WMI с д
едубликацией), если хранилище пусто
613     if not gpus and storage is None:
614         try:
615             prober = GpuProber()
616             probed = prober.probe_all()
617             for g in probed:
618                 gpus.append({
619                     "index": g.index,
620                     "name": g.name,
621                     "vendor": g.vendor,
622                     "driver_version": g.driver_version,
623                     "vram_gb": round(g.memory_total_mb / 1024.
0, 1) if g.memory_total_mb else 0.0,
624                     "temperature_gpu_c": g.temperature_gpu_c,
625                     "utilization_gpu_pct": g.utilization_gpu_p
ct,
626                     "memory_used_mb": g.memory_used_mb,
627                     "memory_total_mb": g.memory_total_mb,
628                 })
629         except Exception as exc:
630             logger.debug(f"[router_gpu_load] Ошибка GpuProber:
{exc}")
631
632     if not gpus:
633         gpus.append({
634             "index": 0,
635             "name": "GPU",
636             "vendor": "Generic",
637             "driver_version": "N/A",
638             "vram_gb": 0.0,
639         })
640
641     return gpus

```

```

642 |
643 |
644 | def _fetch_single_gpu_data(
645 |     target_idx: int,
646 |     target_gpu: Dict[str, Any],
647 |     available_gpus: List[Dict[str, Any]],
648 |     store: TelemetryStorage,
649 |     lhm: Optional[LhmService] = None,
650 | ) -> GpuLoadResponse:
651 |     """Извлекает детальные параметры и сенсоры для конкретного
        GPU строго из БД телеметрии."""
652 |     target_name = target_gpu.get("name", f"GPU {target_idx}")
653 |
654 |     # 1. Загрузка истории сэмплов из SQLite базы данных
655 |     try:
656 |         db_history = store.get_gpu_telemetry_samples(gpu_id=target_idx, limit=60)
657 |     except TypeError:
658 |         db_history = store.get_gpu_telemetry_samples(limit=60)
659 |     except Exception:
660 |         db_history = []
661 |
662 |     if not db_history:
663 |         try:
664 |             db_history = store.get_gpu_telemetry_samples(limit=60)
665 |         except Exception:
666 |             db_history = []
667 |     db_history.reverse()
668 |
669 |     # 2. Получение последних показаний сенсоров строго из SQLite базы данных
670 |     try:
671 |         db_sensors = store.get_latest_sensors()
672 |         if db_sensors:
673 |             gpu_sensors = [
674 |                 s for s in db_sensors
675 |                 if str(s.get("hardware_type", "")).lower() !=

```

```

"cpu"
676         and not any(k in str(s.get("hardware_name", ""
    ).lower() for k in _CPU_HARDWARE_KEYWORDS)
677         and (
678             "gpu" in str(s.get("hardware_type", ""))
        ).lower()
679             or "gpu" in str(s.get("hardware_name", ""
    ).lower()
680             or "nvidia" in str(s.get("hardware_name",
    "").lower()
681             or "radeon" in str(s.get("hardware_name",
    "").lower()
682             or "geforce" in str(s.get("hardware_name",
    "").lower()
683             or "rtx" in str(s.get("hardware_name", ""
    ).lower()
684             or "gtx" in str(s.get("hardware_name", ""
    ).lower()
685             or "uhd" in str(s.get("hardware_name", ""
    ).lower()
686             or "iris" in str(s.get("hardware_name", ""
    ).lower()
687             or "arc" in str(s.get("hardware_name", ""
    ).lower()
688         )
689         and _is_sensor_matching_gpu(str(s.get("hardwar
    e_name", "")), target_name, str(s.get("hardware_type", "")))
690     ]
691     if gpu_sensors:
692         resp = build_gpu_load(
693             sensors=gpu_sensors,
694             inventory=target_gpu,
695             history_samples=db_history,
696             target_gpu_name=target_name,
697             available_gpus=available_gpus,
698             current_gpu_index=target_idx,
699         )
700     resp.meta["source"] = "telemetry_db"

```

```

701         return resp
702     except Exception as exc:
703         logger.error(f"[router_gpu_load] Ошибка чтения метрик
GPU из БД: {exc}", exc_info=True)
704
705     # 3. Fallback: Живой опрос LHM, только если явно передан и
доступен
706     if lhm and lhm.is_running():
707         try:
708             sensors = lhm.get_flattened_sensors()
709             gpu_sensors = [
710                 s for s in sensors
711                 if str(s.get("hardware_type", "")).lower() !=
"cpu"
712                    and not any(k in str(s.get("hardware_name", ""
)).lower() for k in _CPU_HARDWARE_KEYWORDS)
713                    and (
714                        "gpu" in str(s.get("hardware_type", "")).l
ower()
715                        or "gpu" in str(s.get("hardware_name", ""
)).lower()
716                        or "nvidia" in str(s.get("hardware_name",
"")).lower()
717                        or "radeon" in str(s.get("hardware_name",
"")).lower()
718                        or "geforce" in str(s.get("hardware_name",
"")).lower()
719                        or "rtx" in str(s.get("hardware_name", ""
)).lower()
720                        or "gtx" in str(s.get("hardware_name", ""
)).lower()
721                        or "uhd" in str(s.get("hardware_name", ""
)).lower()
722                        or "iris" in str(s.get("hardware_name", ""
)).lower()
723                        or "arc" in str(s.get("hardware_name", ""
)).lower()
724                    )

```



```

725         and _is_sensor_matching_gpu(str(s.get("hardware_
e_name", "")), target_name, str(s.get("hardware_type", "")))
726     ]
727     if gpu_sensors:
728         resp = build_gpu_load(
729             sensors=gpu_sensors,
730             inventory=target_gpu,
731             history_samples=db_history,
732             target_gpu_name=target_name,
733             available_gpus=available_gpus,
734             current_gpu_index=target_idx,
735         )
736         resp.meta["source"] = "lhm_live+telemetry.db"
737         return resp
738     except Exception as exc:
739         logger.warning(f"[router_gpu_load] Ошибка опроса L
HM для GPU {target_name}: {exc}")
740
741     # 4. Fallback: Данные из сэмплов БД или инвентаря целевого
GPU
742     synthetic_sensors: List[Dict[str, Any]] = []
743     latest_sample = db_history[-1] if db_history else {}
744     sample_load = latest_sample.get("load_percent") if latest_
sample else None
745     if sample_load is None:
746         sample_load = latest_sample.get("gpu_load_percent") if
latest_sample else None
747     if sample_load is None:
748         sample_load = target_gpu.get("utilization_gpu_pct")
749
750     sample_temp = latest_sample.get("temperature_gpu_c") if la
test_sample else None
751     if sample_temp is None:
752         sample_temp = latest_sample.get("temperature_c") if la
test_sample else None
753     if sample_temp is None:
754         sample_temp = target_gpu.get("temperature_gpu_c")
755

```

```

756     sample_vram_used = latest_sample.get("memory_used_mb") if
latest_sample else None
757     if sample_vram_used is None:
758         sample_vram_used = target_gpu.get("memory_used_mb")
759
760     sample_vram_total = latest_sample.get("memory_total_mb") i
f latest_sample else None
761     if sample_vram_total is None:
762         sample_vram_total = target_gpu.get("memory_total_mb")
or ((target_gpu.get("vram_gb") or 0.0) * 1024.0)
763
764     if sample_load is not None:
765         synthetic_sensors.append({"sensor_name": "GPU Core", "
sensor_category": "Load", "value": float(sample_load), "hardwa
re_name": target_name, "hardware_type": "gpu", "unit": "%"})
766     if sample_temp is not None:
767         synthetic_sensors.append({"sensor_name": "GPU Core Tem
p", "sensor_category": "Temperatures", "value": float(sample_t
emp), "hardware_name": target_name, "hardware_type": "gpu", "u
nit": "°C"})
768     if sample_vram_used is not None:
769         synthetic_sensors.append({"sensor_name": "GPU Memory U
sed", "sensor_category": "Data", "value": float(sample_vram_us
ed), "hardware_name": target_name, "hardware_type": "gpu", "un
it": "MB"})
770     if sample_vram_total:
771         synthetic_sensors.append({"sensor_name": "GPU Memory T
otal", "sensor_category": "Data", "value": float(sample_vram_t
otal), "hardware_name": target_name, "hardware_type": "gpu", "
unit": "MB"})
772
773     if synthetic_sensors:
774         resp = build_gpu_load(
775             sensors=synthetic_sensors,
776             inventory=target_gpu,
777             history_samples=db_history,
778             target_gpu_name=target_name,
779             available_gpus=available_gpus,

```

```

780         current_gpu_index=target_idx,
781     )
782     resp.meta["source"] = "telemetry_db_samples"
783     return resp
784
785     # История параметров GPU при пустых сенсорах
786     history_points: List[GpuHistoryPoint] = []
787     for r in db_history:
788         ts_str = str(r.get("timestamp") or "")
789         t_label = ts_str.split("T")[-1][:8] if "T" in ts_str e
790         lse ts_str[-8:]
791         history_points.append(
792             GpuHistoryPoint(
793                 timestamp=ts_str,
794                 time_label=t_label or "--:--:--",
795                 load_percent=round(float(r.get("load_percent")
796                 or r.get("gpu_load_percent") or 0.0), 1),
797                 temperature_c=r.get("temperature_gpu_c") or r.
798                 get("temperature_c"),
799                 power_w=r.get("power_draw_w") or r.get("power_
800                 w"),
801                 vram_percent=r.get("memory_percent") or r.get(
802                 "vram_percent"),
803             )
804         )
805
806     return GpuLoadResponse(
807         status="ok",
808         name=target_name,
809         vendor=target_gpu.get("vendor", "Intel" if "intel" in
810         target_name.lower() or "uhd" in target_name.lower() else "NVID
811         IA"),
812         core_load_percent=0.0,
813         engines=[],
814         sensors=[],
815         history=history_points,
816         available_gpus=available_gpus,
817         current_gpu_index=target_idx,

```

```

811         meta={"source": "telemetry_db_empty", "engines_count":
812             0},
813     )
814
815 def init_router(
816     storage: Optional[TelemetryStorage] = None,
817     lhm_service: Optional[LhmService] = None,
818 ) -> APIRouter:
819     """Инициализация роутера GET /api/v1/panel/gpu-load.
820
821     Args:
822         storage: Опциональное хранилище TelemetryStorage.
823         lhm_service: Опциональный сервис LhmService.
824
825     Returns:
826         APIRouter: Настроенный роутер FastAPI.
827     """
828     router = APIRouter(prefix="/api/v1/panel", tags=["GPU Panel"])
829     store = storage or TelemetryStorage.get_instance(read_only=True)
830     lhm = lhm_service if lhm_service is not None else (LhmService() if storage is None else None)
831
832     @router.get("/gpu-load", response_model=GpuLoadResponse, summary="Загрузка, температура, паспорт и метрики GPU")
833     async def get_gpu_load(
834         gpu_index: Optional[int] = Query(None, description="Индекс конкретной видеокарты (0, 1, ...) или None для всех")
835     ) -> GpuLoadResponse:
836         """Возвращает детальную загрузку, температуру, VRAM, паспорт и историю GPU."""
837         available_gpus = _discover_system_gpus(store)
838
839         if gpu_index is not None and 0 <= gpu_index < len(available_gpus):
840             return _fetch_single_gpu_data(gpu_index, available

```

```

    _gpus[gpu_index], available_gpus, store, lhm)
841
842     # Собираем данные по каждому GPU
843     devices: List[Dict[str, Any]] = []
844     for idx, g in enumerate(available_gpus):
845         g_resp = _fetch_single_gpu_data(idx, g, available_
gpus, store, lhm)
846         devices.append(g_resp.model_dump())
847
848     # Выбираем основной адаптер для корневого ответа
849     primary_idx = 0
850     for idx, g in enumerate(available_gpus):
851         v_up = str(g.get("vendor", "")).upper()
852         n_up = str(g.get("name", "")).upper()
853         if "NVIDIA" in v_up or "GEFORCE" in n_up or "AMD"
in v_up or "RADEON" in n_up:
854             primary_idx = idx
855             break
856
857     main_resp = _fetch_single_gpu_data(primary_idx, availa
ble_gpus[primary_idx], available_gpus, store, lhm)
858     main_resp.devices = devices
859     return main_resp
860
861     return router
862
863
864 __all__ = [
865     "init_router",
866     "build_gpu_load",
867     "GpuLoadResponse",
868     "GpuEngineMetric",
869     "GpuMemoryMetric",
870     "GpuClocksMetric",
871     "GpuSensorMetric",
872     "GpuHistoryPoint",
873     "GpuSpecsInfo",
874 ]

```


router_hardware_sensors.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Hardware Sensors
4 # =====
5 # Description:
6 #   GET /api/v1/panel/hardware-sensors – детальное табло сенсоров оборудования
7 #   из таблицы sensor_polls базы данных telemetry.db (температуры, нагрузки,
8 #   вентиляторы, напряжения, частоты, мощности и накопители).
9 #
10 # Usage Examples:
11 #   Python API:
12 #       from apps.windows.api.routers.router_hardware_sensors import init_router
13 #       app.include_router(init_router())
14 #
15 # File: router_hardware_sensors.py
16 # Project: ai-breadboard
17 # Package: apps.windows.api.routers
18 # Author: hypo69
19 # Copyright: © 2026 hypo69
20 # Updated: 2026-10-05 23:40:00
21 # =====
22
23 from __future__ import annotations
24 """Роутер табло «Сенсоры оборудования»: структурированные метрики оборудования из telemetry.db."""
25
26 from typing import Any, Dict, List, Optional
27 from fastapi import APIRouter
28 from pydantic import BaseModel, Field
```

```

29
30 from logger import logger
31 from apps.windows.modules.hardware.lhm_service import LhmService
32 from apps.windows.telemetry.sqlite import TelemetryStorage
33
34 _SQL_LATEST_SENSORS = """
35 SELECT sp.* FROM sensor_polls sp
36 INNER JOIN (
37     SELECT sensor_id, MAX(id) as max_id
38     FROM sensor_polls
39     GROUP BY sensor_id
40 ) latest ON sp.id = latest.max_id
41 ORDER BY sp.hardware_type, sp.sensor_category, sp.sensor_name;
42 """
43
44
45 class HardwareSensorItem(BaseModel):
46     """Единичный сенсор оборудования."""
47     id: str = Field(..., description="Уникальный ID сенсора")
48     hardware_name: str = Field(default="Hardware", description="Имя оборудования")
49     hardware_type: str = Field(default="system", description="Тип оборудования (cpu, gpu, storage, etc.)")
50     sensor_category: str = Field(default="General", description="Категория (Temperatures, Load, Fan, etc.)")
51     sensor_name: str = Field(..., description="Название сенсора")
52     unit: str = Field(default="", description="Единица измерения (°C, %, RPM, V, W, MHz, MB/s)")
53     value: float = Field(default=0.0, description="Числовое значение")
54     value_raw: str = Field(default="", description="Текстовое значение с единицей")
55     status: str = Field(default="normal", description="Статус сенсора: normal, warning, danger")
56     timestamp: str = Field(default="", description="Время замера")

```



```
57 |
58 |
59 | class HardwareGroupMetric(BaseModel):
60 |     """Группа сенсоров одного компонента оборудования."""
61 |     name: str = Field(..., description="Название компонента (I
62 |     hardware_type: str = Field(default="system", description="
63 |     sensors_count: int = Field(default=0, description="Количес
64 |     max_temperature_c: Optional[float] = Field(default=None, d
65 |     avg_load_percent: Optional[float] = Field(default=None, de
66 |     sensors: List[HardwareSensorItem] = Field(default_factory=
67 |
68 |
69 | class HardwareSensorsSummary(BaseModel):
70 |     """Сводка по всем сенсорам системы."""
71 |     total_sensors: int = Field(default=0, description="Общее к
72 |     temperature_count: int = Field(default=0, description="Дат
73 |     load_count: int = Field(default=0, description="Датчиков н
74 |     fan_count: int = Field(default=0, description="Датчиков ве
75 |     voltage_count: int = Field(default=0, description="Датчико
76 |     clock_count: int = Field(default=0, description="Датчиков
77 |     power_count: int = Field(default=0, description="Датчиков
78 |     max_temperature_c: Optional[float] = Field(default=None, d
79 |     max_temperature_sensor: Optional[str] = Field(default=None
, description="Имя самого горячего сенсора")
```

```

80     warnings_count: int = Field(default=0, description="Число
| предупреждений/аномалий")
81     timestamp: str = Field(default="", description="Время посл
| еднего снимка из БД")
82
83
84 class HardwareSensorsResponse(BaseModel):
85     """Ответ GET /api/v1/panel/hardware-sensors."""
86     status: str = Field(default="ok", description="Статус отве
| та")
87     summary: HardwareSensorsSummary = Field(default_factory=Ha
| rdwareSensorsSummary, description="Сводка сенсоров")
88     groups: List[HardwareGroupMetric] = Field(default_factory=
| list, description="Группы оборудования с сенсорами")
89     all_sensors: List[HardwareSensorItem] = Field(default_fact
| ory=list, description="Плоский список всех сенсоров")
90     meta: Dict[str, Any] = Field(default_factory=dict, descrip
| tion="Метаданные ответа")
91
92
93 def _calculate_sensor_status(category: str, value: float) -> s
| tr:
94     """Определяет статус сенсора (normal, warning, danger)."""
95     cat = category.lower()
96     if "temp" in cat:
97         if value >= 80.0:
98             return "danger"
99         if value >= 68.0:
100             return "warning"
101         return "normal"
102     if "load" in cat:
103         if value >= 90.0:
104             return "danger"
105         if value >= 75.0:
106             return "warning"
107         return "normal"
108     if "fan" in cat:
109         if value <= 0.0:

```

```
110         return "warning"
111         return "normal"
112     return "normal"
113
114
115 def build_hardware_sensors(rows: List[Dict[str, Any]]) -> HardwareSensorsResponse:
116     """Собирает структурированное табло сенсоров оборудования
117     из строк БД sensor_polls.
118
119     Args:
120         rows: Строки последних показаний сенсоров из telemetry
121         .db.
122
123     Returns:
124         HardwareSensorsResponse: Структурированный ответ с группировкой и виджетами.
125     """
126     if not rows:
127         return HardwareSensorsResponse(
128             summary=HardwareSensorsSummary(),
129             groups=[],
130             all_sensors=[],
131             meta={"source": "telemetry.db", "table": "sensor_polls", "sensors_count": 0},
132         )
133
134     all_items: List[HardwareSensorItem] = []
135     groups_map: Dict[str, Dict[str, Any]] = {}
136
137     temp_count = 0
138     load_count = 0
139     fan_count = 0
140     volt_count = 0
141     clock_count = 0
142     power_count = 0
143     warnings_count = 0
```

```

143     max_temp: Optional[float] = None
144     max_temp_name: Optional[str] = None
145     latest_ts = ""
146
147     for r in rows:
148         sid = str(r.get("sensor_id") or r.get("name") or "sens
or")
149         hname = str(r.get("hardware_name") or "Оборудование")
150         htype = str(r.get("hardware_type") or "system")
151         cat = str(r.get("sensor_category") or "General")
152         sname = str(r.get("sensor_name") or sid)
153         unit = str(r.get("unit") or "")
154         ts = str(r.get("timestamp") or "")
155         if ts and (not latest_ts or ts > latest_ts):
156             latest_ts = ts
157
158         raw_val = r.get("value")
159         val_num = float(raw_val) if raw_val is not None else 0
        .0
160         val_raw_str = f"{val_num:.1f} {unit}".strip() if unit
        else f"{val_num:.1f}"
161
162         status = _calculate_sensor_status(cat, val_num)
163         if status in ("warning", "danger"):
164             warnings_count += 1
165
166         cat_l = cat.lower()
167         if "temp" in cat_l:
168             temp_count += 1
169             if max_temp is None or val_num > max_temp:
170                 max_temp = val_num
171                 max_temp_name = f"{hname} - {sname}"
172         elif "load" in cat_l:
173             load_count += 1
174         elif "fan" in cat_l:
175             fan_count += 1
176         elif "volt" in cat_l:
177             volt_count += 1

```

```
178         elif "clock" in cat_l:
179             clock_count += 1
180         elif "power" in cat_l:
181             power_count += 1
182
183         item = HardwareSensorItem(
184             id=sid,
185             hardware_name=hname,
186             hardware_type=htype,
187             sensor_category=cat,
188             sensor_name=sname,
189             unit=unit,
190             value=round(val_num, 2),
191             value_raw=val_raw_str,
192             status=status,
193             timestamp=ts,
194         )
195         all_items.append(item)
196
197         # Группировка
198         if hname not in groups_map:
199             groups_map[hname] = {
200                 "name": hname,
201                 "hardware_type": htype,
202                 "sensors": [],
203                 "temps": [],
204                 "loads": [],
205             }
206             groups_map[hname]["sensors"].append(item)
207             if "temp" in cat_l:
208                 groups_map[hname]["temps"].append(val_num)
209             elif "load" in cat_l:
210                 groups_map[hname]["loads"].append(val_num)
211
212         # Формирование списка групп
213         groups_list: List[HardwareGroupMetric] = []
214         for gname, gdata in groups_map.items():
215             temps = gdata["temps"]
```

```

216 |         loads = gdata["loads"]
217 |         groups_list.append(
218 |             HardwareGroupMetric(
219 |                 name=gname,
220 |                 hardware_type=gdata["hardware_type"],
221 |                 sensors_count=len(gdata["sensors"]),
222 |                 max_temperature_c=round(max(temps), 1) if temp
| s else None,
223 |                 avg_load_percent=round(sum.loads) / len.loads)
| , 1) if loads else None,
224 |                 sensors=gdata["sensors"],
225 |             )
226 |         )
227 |
228 |     summary = HardwareSensorsSummary(
229 |         total_sensors=len(all_items),
230 |         temperature_count=temp_count,
231 |         load_count=load_count,
232 |         fan_count=fan_count,
233 |         voltage_count=volt_count,
234 |         clock_count=clock_count,
235 |         power_count=power_count,
236 |         max_temperature_c=round(max_temp, 1) if max_temp is no
| t None else None,
237 |         max_temperature_sensor=max_temp_name,
238 |         warnings_count=warnings_count,
239 |         timestamp=latest_ts,
240 |     )
241 |
242 |     return HardwareSensorsResponse(
243 |         status="ok",
244 |         summary=summary,
245 |         groups=groups_list,
246 |         all_sensors=all_items,
247 |         meta={
248 |             "source": "telemetry.db",
249 |             "table": "sensor_polls",
250 |             "sensors_count": len(all_items),

```

```

251         "groups_count": len(groups_list),
252     },
253 )
254
255
256 def init_router(
257     storage: Optional[TelemetryStorage] = None,
258     lhm_service: Optional[LhmService] = None,
259 ) -> APIRouter:
260     """Создаёт роутер /api/v1/panel/hardware-sensors.
261
262     Args:
263         storage: Хранилище телеметрии (по умолчанию — синглтон
264         в режиме read_only).
265         lhm_service: Сервис LibreHardwareMonitor.
266
267     Returns:
268         APIRouter: Роутер с эндпоинтом GET /api/v1/panel/hardw
269         are-sensors.
270     """
271     router = APIRouter(tags=["Hardware Sensors Panel"])
272     store = storage or TelemetryStorage.get_instance(read_only
273     =True)
274
275     @router.get("/api/v1/panel/hardware-sensors", response_mod
276     el=HardwareSensorsResponse)
277     async def get_panel_hardware_sensors() -> HardwareSensorsR
278     esponse:
279         """Возвращает актуальное табло сенсоров оборудования и
280         з базы данных telemetry.db."""
281         try:
282             with store._lock, store._get_connection() as conn:
283                 rows = [dict(r) for r in conn.execute(_SQL_LAT
284                 EST_SENSORS).fetchall()]
285
286             return build_hardware_sensors(rows)
287         except Exception as exc:
288             logger.error(f"[router_hardware_sensors] Ошибка чт

```

```
ения сенсоров из БД: {exc}", exc_info=True)
282         return HardwareSensorsResponse(
283             status="error",
284             meta={"source": "telemetry.db", "error": str(e
xc)}},
285         )
286
287     return router
288
289
290 __all__ = [
291     "init_router",
292     "build_hardware_sensors",
293     "HardwareSensorItem",
294     "HardwareGroupMetric",
295     "HardwareSensorsSummary",
296     "HardwareSensorsResponse",
297 ]
298
```


router_log_settings.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Log Settings
4 # =====
5 # Description:
6 #   FastAPI роутер для вкладки настройки и просмотра параметров
7 #   в журналов событий Windows
8 #   (Security, System, Application, Sysmon и др.), аудита командных строк и управления закладками.
9 #
10 # Usage Examples:
11 #   Python API:
12 #       from apps.windows.api.routers.router_log_settings import
13 #           init_router
14 #
15 #       router = init_router()
16 #
17 # File: router_log_settings.py
18 # Project: ai-breadboard
19 # Package: apps.windows.api.routers
20 # Author: hypo69
21 # Copyright: © 2026 hypo69
22 # Updated: 2026-10-08 02:30:00
23 # =====
24
25 from __future__ import annotations
26 """FastAPI роутер для вкладки настройки и просмотра параметров
27 журналов событий Windows."""
28
29 import asyncio
30 import datetime
31 import os
```

```
29 from pathlib import Path
30 import re
31 import subprocess
32 import winreg
33 from typing import Any, Dict, List, Optional
34
35 from fastapi import APIRouter, HTTPException, Query, status
36 from pydantic import BaseModel, Field
37
38 from logger import logger
39 from apps.windows.telemetry.win32_ffmpeg import WevtAPI,
CHANNEL_DESCRIPTIONS
40 from apps.windows.telemetry.security_collector import WindowsS
ecurityCollector
41 from apps.windows.telemetry.sqlite import TelemetryStorage
42
43 router = APIRouter(prefix='/api/v1/log-settings', tags=['Windo
ws Log Settings & Parameters'])
44 _wevtapi = WevtAPI()
45 _security_collector = WindowsSecurityCollector()
46 _storage = TelemetryStorage()
47
48 # Ключевые каналы для быстрого мониторинга
49 PRIORITY_CHANNELS = [
50     'Security',
51     'System',
52     'Application',
53     'Setup',
54     'Microsoft-Windows-Sysmon/Operational',
55     'Microsoft-Windows-TaskScheduler/Operational',
56     'Microsoft-Windows-Windows Defender/Operational',
57     'Microsoft-Windows-PowerShell/Operational',
58     'Microsoft-Windows-Kernel-Power/Operational',
59     'Microsoft-Windows-Kernel-PnP/Configuration',
60     'Microsoft-Windows-WindowsUpdateClient/Operational',
61     'Microsoft-Windows-Windows Firewall With Advanced Security
/Firewall',
62     'Microsoft-Windows-Diagnostics-Performance/Operational',
```

```
63     'Microsoft-Windows-Bits-Client/Operational',
64     'Microsoft-Windows-DNS-Client/Operational',
65     'Microsoft-Windows-TerminalServices-LocalSessionManager/Operational',
66 ]
67
68
69 class LogChannelDetail(BaseModel):
70     """Модель подробных параметров канала журнала Windows."""
71     channel_name: str = Field(..., description="Имя канала журнала")
72     display_name: str = Field(..., description="Отображаемое имя")
73     description: str = Field('', description="Описание назначения журнала")
74     is_enabled: bool = Field(True, description="Включен ли журнал")
75     channel_type: str = Field('Admin', description="Тип канала (Admin, Operational, Analytic, Debug)")
76     log_file_path: str = Field('', description="Путь к файлу .evtx на диске")
77     file_size_bytes: int = Field(0, description="Текущий размер файла на диске в байтах")
78     file_size_mb: float = Field(0.0, description="Текущий размер файла в МБ")
79     max_size_bytes: int = Field(0, description="Максимальный разрешенный размер в байтах")
80     max_size_mb: float = Field(0.0, description="Максимальный разрешенный размер в МБ")
81     usage_pct: float = Field(0.0, description="Процент заполнения журнала")
82     record_count: int = Field(0, description="Текущее количество записей в журнале")
83     oldest_record_id: int = Field(0, description="Старейший Record ID")
84     newest_record_id: int = Field(0, description="Новейший Record ID")
85     is_full: bool = Field(False, description="Признак переполнения")
```

```

ения журнала")
86     retention: bool = Field(False, description="Политика удерж
ания (не перезаписывать)")
87     auto_backup: bool = Field(False, description="Автоматическ
ая архивация при заполнении")
88     creation_time: str = Field('', description="Время создания
файла журнала")
89     last_access_time: str = Field('', description="Время после
днего доступа")
90     last_write_time: str = Field('', description="Время послед
ней записи")
91     is_accessible: bool = Field(True, description="Доступен ли
журнал для чтения")
92
93
94 class LogChannelConfigRequest(BaseModel):
95     """Запрос на изменение параметров канала журнала."""
96     channel_name: str = Field(..., description="Имя канала")
97     max_size_mb: Optional[int] = Field(None, ge=1, le=65536, d
escription="Новый максимальный размер в МБ")
98     is_enabled: Optional[bool] = Field(None, description="Вклю
чение/отключение канала")
99     auto_backup: Optional[bool] = Field(None, description="Авт
о-бэкап при переполнении")
100    retention: Optional[bool] = Field(None, description="Удерж
ание записей")
101
102
103 class LogChannelClearRequest(BaseModel):
104     """Запрос на очистку журнала."""
105     channel_name: str = Field(..., description="Имя канала")
106     backup_path: Optional[str] = Field(None, description="Путь
для сохранения архива перед очисткой")
107
108
109 class AuditPolicyStatus(BaseModel):
110     """Статус политик аудита и сборщика телеметрии."""
111     cmdline_audit_enabled: bool = Field(False, description="Вк

```

```

        лючен ли аудит CommandLine в процессах (Event 4688)")
112     security_channel_accessible: bool = Field(False, description="Доступен ли канал Security для чтения")
113     last_bookmark_record_id: Optional[int] = Field(None, description="Последний RecordID в закладке telemetry.db")
114     last_bookmark_timestamp: Optional[str] = Field(None, description="Время фиксации закладки")
115     db_security_events_count: int = Field(0, description="Число сохраненных событий Security в telemetry.db")
116
117
118 class SetCmdlineAuditRequest(BaseModel):
119     """Запрос на переключение аудита командной строки."""
120     enabled: bool = Field(..., description="Включить (True) или выключить (False) аудит CommandLine")
121
122
123 class BookmarkResetRequest(BaseModel):
124     """Запрос на сброс закладки сборщика."""
125     channel: str = Field('Security', description="Имя канала")
126     record_id: Optional[int] = Field(None, description="Целевой Record ID (None = сбросить на текущий конец журнала)")
127
128
129 def _query_wevtutil_channel_info(channel_name: str) -> Dict[str, Any]:
130     """Считывает параметры канала через утилиту wevtutil gl и wevtutil gli."""
131     info: Dict[str, Any] = {
132         'channel_name': channel_name,
133         'enabled': True,
134         'type': 'Operational',
135         'log_file_path': '',
136         'retention': False,
137         'auto_backup': False,
138         'max_size_bytes': 20971520,
139         'file_size_bytes': 0,
140         'record_count': 0,

```

```

141         'oldest_record_id': 0,
142         'newest_record_id': 0,
143         'creation_time': '',
144         'last_access_time': '',
145         'last_write_time': '',
146         'is_accessible': True,
147     }
148
149     try:
150         # 1. Получение статической конфигурации
151         res_gl = subprocess.run(
152             ['wevtutil', 'gl', channel_name],
153             capture_output=True,
154             text=True,
155             check=False,
156             timeout=5,
157         )
158         if res_gl.returncode == 0:
159             for line in res_gl.stdout.splitlines():
160                 line = line.strip()
161                 if line.startswith('enabled:'):
162                     info['enabled'] = line.split(':', 1)[1].strip().lower() == 'true'
163                 elif line.startswith('type:'):
164                     info['type'] = line.split(':', 1)[1].strip()
165                 elif line.startswith('logFileName:'):
166                     raw_path = line.split(':', 1)[1].strip()
167                     info['log_file_path'] = os.path.expandvars(
168                         (raw_path)
169                     )
170                 elif line.startswith('retention:'):
171                     info['retention'] = line.split(':', 1)[1].strip().lower() == 'true'
172                 elif line.startswith('autoBackup:'):
173                     info['auto_backup'] = line.split(':', 1)[1].strip().lower() == 'true'
174                 elif line.startswith('maxSize:'):
175                     try:

```

```

174 |             info['max_size_bytes'] = int(line.spli
| t(':', 1)[1].strip())
175 |             except ValueError:
176 |                 pass
177 |         else:
178 |             info['is_accessible'] = False
179 |
180 |         # 2. Получение динамического состояния
181 |         res_gli = subprocess.run(
182 |             ['wevtutil', 'gli', channel_name],
183 |             capture_output=True,
184 |             text=True,
185 |             check=False,
186 |             timeout=5,
187 |         )
188 |         if res_gli.returncode == 0:
189 |             for line in res_gli.stdout.splitlines():
190 |                 line = line.strip()
191 |                 if line.startswith('fileSize:'):
192 |                     try:
193 |                         info['file_size_bytes'] = int(line.spl
| it(':', 1)[1].strip())
194 |                     except ValueError:
195 |                         pass
196 |                 elif line.startswith('numberOfLogRecords:'):
197 |                     try:
198 |                         info['record_count'] = int(line.split(
| ':', 1)[1].strip())
199 |                     except ValueError:
200 |                         pass
201 |                 elif line.startswith('oldestRecordNumber:'):
202 |                     try:
203 |                         info['oldest_record_id'] = int(line.sp
| lit(':', 1)[1].strip())
204 |                     except ValueError:
205 |                         pass
206 |                 elif line.startswith('creationTime:'):
207 |                     info['creation_time'] = line.split(':', 1)

```

```

    [1].strip()
208         elif line.startswith('lastAccessTime:'):
209             info['last_access_time'] = line.split(':',
    1)[1].strip()
210         elif line.startswith('lastWriteTime:'):
211             info['last_write_time'] = line.split(':',
    1)[1].strip()
212
213         if info['record_count'] > 0 and info['oldest_recor
    d_id'] > 0:
214             info['newest_record_id'] = info['oldest_record
    _id'] + info['record_count'] - 1
215         else:
216             info['newest_record_id'] = info['oldest_record
    _id']
217     except Exception as exc:
218         logger.debug(f"[LogSettings] Ошибка опроса wevtutil дл
    я {channel_name}: {exc}")
219         info['is_accessible'] = False
220
221     return info
222
223
224 @router.get('/channels')
225 async def get_channels_list() -> Dict[str, Any]:
226     """Возвращает список всех приоритетных и системных каналов
    с их подробными параметрами."""
227     def _collect():
228         channels_res: List[Dict[str, Any]] = []
229         for cname in PRIORITY_CHANNELS:
230             raw = _query_wevtutil_channel_info(cname)
231             max_bytes = raw.get('max_size_bytes', 20971520) or
    20971520
232             file_bytes = raw.get('file_size_bytes', 0)
233             usage_pct = min(100.0, round((file_bytes / max_byt
    es) * 100, 1)) if max_bytes > 0 else 0.0
234
235             channels_res.append({

```



```

236         'channel_name': cname,
237         'display_name': cname,
238         'description': _wevtapi.get_channel_descriptio
n(cname),
239         'is_enabled': raw.get('enabled', True),
240         'channel_type': raw.get('type', 'Admin'),
241         'log_file_path': raw.get('log_file_path', ''),
242         'file_size_bytes': file_bytes,
243         'file_size_mb': round(file_bytes / (1024 * 102
4), 2),
244         'max_size_bytes': max_bytes,
245         'max_size_mb': round(max_bytes / (1024 * 1024)
, 2),
246         'usage_pct': usage_pct,
247         'record_count': raw.get('record_count', 0),
248         'oldest_record_id': raw.get('oldest_record_id'
, 0),
249         'newest_record_id': raw.get('newest_record_id'
, 0),
250         'is_full': usage_pct >= 99.0,
251         'retention': raw.get('retention', False),
252         'auto_backup': raw.get('auto_backup', False),
253         'creation_time': raw.get('creation_time', ''),
254         'last_access_time': raw.get('last_access_time'
, ''),
255         'last_write_time': raw.get('last_write_time',
''),
256         'is_accessible': raw.get('is_accessible', True
),
257     })
258     return {'channels': channels_res, 'total': len(channel
s_res)}
259
260     return await asyncio.to_thread(_collect)
261
262
263 @router.get('/channel/{channel_name:path}')
264 async def get_channel_detail(channel_name: str) -> Dict[str, A

```

```

ny]:
265     """Получение детальных параметров конкретного канала журна
ла."""
266     def _fetch():
267         raw = _query_wevtutil_channel_info(channel_name)
268         max_bytes = raw.get('max_size_bytes', 20971520) or 209
71520
269         file_bytes = raw.get('file_size_bytes', 0)
270         usage_pct = min(100.0, round((file_bytes / max_bytes)
* 100, 1)) if max_bytes > 0 else 0.0
271
272         return {
273             'channel_name': channel_name,
274             'display_name': channel_name,
275             'description': _wevtapi.get_channel_description(ch
annel_name),
276             'is_enabled': raw.get('enabled', True),
277             'channel_type': raw.get('type', 'Admin'),
278             'log_file_path': raw.get('log_file_path', ''),
279             'file_size_bytes': file_bytes,
280             'file_size_mb': round(file_bytes / (1024 * 1024),
2),
281             'max_size_bytes': max_bytes,
282             'max_size_mb': round(max_bytes / (1024 * 1024), 2)
,
283             'usage_pct': usage_pct,
284             'record_count': raw.get('record_count', 0),
285             'oldest_record_id': raw.get('oldest_record_id', 0)
,
286             'newest_record_id': raw.get('newest_record_id', 0)
,
287             'is_full': usage_pct >= 99.0,
288             'retention': raw.get('retention', False),
289             'auto_backup': raw.get('auto_backup', False),
290             'creation_time': raw.get('creation_time', ''),
291             'last_access_time': raw.get('last_access_time', ''
),
292             'last_write_time': raw.get('last_write_time', ''),

```

```

293         'is_accessible': raw.get('is_accessible', True),
294     }
295
296     return await asyncio.to_thread(_fetch)
297
298
299 @router.post('/channel/config')
300 async def set_channel_config(req: LogChannelConfigRequest) ->
    Dict[str, Any]:
301     """Изменение параметров конфигурации канала журнала (разме
    р, включение, политики)."""
302     def _apply():
303         args = ['wevtutil', 'sl', req.channel_name]
304         applied: Dict[str, Any] = {}
305
306         if req.max_size_mb is not None:
307             max_bytes = req.max_size_mb * 1024 * 1024
308             args.append(f'/ms:{max_bytes}')
309             applied['max_size_mb'] = req.max_size_mb
310
311         if req.is_enabled is not None:
312             args.append(f"/e:{'true' if req.is_enabled else 'f
    else'}")
313             applied['is_enabled'] = req.is_enabled
314
315         if req.auto_backup is not None:
316             args.append(f"/ab:{'true' if req.auto_backup else
    'false'}")
317             applied['auto_backup'] = req.auto_backup
318
319         if req.retention is not None:
320             args.append(f"/rt:{'true' if req.retention else 'f
    else'}")
321             applied['retention'] = req.retention
322
323         res = subprocess.run(args, capture_output=True, text=T
    rue, check=False, timeout=10)
324         if res.returncode != 0:

```

```

325         err_msg = res.stderr.strip() or res.stdout.strip()
        or f"Код ошибки {res.returncode}"
326         logger.warning(f"[LogSettings] Ошибка изменения па
        раметров {req.channel_name}: {err_msg}")
327         raise HTTPException(
328             status_code=status.HTTP_400_BAD_REQUEST,
329             detail=f"Не удалось применить параметры к журн
        алу '{req.channel_name}': {err_msg}. Возможно, требуются права
        Администратора.",
330         )
331
332         updated_info = _query_wevtutil_channel_info(req.channe
        l_name)
333         return {
334             'success': True,
335             'channel_name': req.channel_name,
336             'applied': applied,
337             'current_config': updated_info,
338             'message': f"Параметры журнала '{req.channel_name}
        ' успешно обновлены.",
339         }
340
341         return await asyncio.to_thread(_apply)
342
343
344 @router.post('/channel/clear')
345 async def clear_channel_log(req: LogChannelClearRequest) -> Di
        ct[str, Any]:
346     """Очистка журнала событий с возможностью предварительного
        резервного копирования."""
347     def _do_clear():
348         args = ['wevtutil', 'cl', req.channel_name]
349         if req.backup_path:
350             p = Path(req.backup_path)
351             p.parent.mkdir(parents=True, exist_ok=True)
352             args.append(f'/bu:{req.backup_path}')
353
354     res = subprocess.run(args, capture_output=True, text=T

```

```

    rue, check=False, timeout=15)
355         if res.returncode != 0:
356             err_msg = res.stderr.strip() or res.stdout.strip()
357             raise HTTPException(
358                 status_code=status.HTTP_400_BAD_REQUEST,
359                 detail=f"Ошибка очистки журнала '{req.channel_
name}': {err_msg}",
360             )
361
362         return {
363             'success': True,
364             'channel_name': req.channel_name,
365             'backup_saved': bool(req.backup_path),
366             'backup_path': req.backup_path,
367             'message': f"Журнал '{req.channel_name}' успешно о
чищен.",
368         }
369
370         return await asyncio.to_thread(_do_clear)
371
372
373 @router.get('/audit-status')
374 async def get_audit_status() -> Dict[str, Any]:
375     """Проверка статуса политик аудита безопасности и состояни
я закладки телеметрии."""
376     def _check():
377         audit_res = _security_collector.check_access_and_audit
()
378         bm = _storage.get_security_bookmark('Security')
379         stats = _storage.get_security_stats()
380
381         last_bm_id = bm.get('last_record_id') if isinstance(bm
, dict) else (getattr(bm, 'last_record_id', None) if bm else N
one)
382         last_bm_time = bm.get('updated_at') if isinstance(bm,
dict) else (getattr(bm, 'updated_at', None) if bm else None)
383
384         return {

```

```

385         'cmdline_audit_enabled': audit_res.command_line_audit_enabled,
386         'security_channel_accessible': audit_res.accessible,
387         'last_bookmark_record_id': last_bm_id,
388         'last_bookmark_timestamp': last_bm_time,
389         'db_security_events_count': stats.get('total_events', 0),
390         'events_by_id': stats.get('events_by_id', {}),
391         'earliest_event': stats.get('earliest_event', ''),
392         'latest_event': stats.get('latest_event', ''),
393     }
394
395     return await asyncio.to_thread(_check)
396
397
398 @router.post('/audit-status/cmdline')
399 async def set_cmdline_audit(req: SetCmdlineAuditRequest) -> Dict[str, Any]:
400     """Включение или отключение аудита командной строки процессов в реестре Windows."""
401     def _set_reg():
402         key_path = r"SOFTWARE\Microsoft\Windows\CurrentVersion\
Policies\System\Audit"
403         val = 1 if req.enabled else 0
404         try:
405             with winreg.CreateKeyEx(winreg.HKEY_LOCAL_MACHINE,
key_path, 0, winreg.KEY_WRITE | winreg.KEY_WOW64_64KEY) as key:
406                 winreg.SetValueEx(key, "ProcessCreationInclude
CmdLine_Enabled", 0, winreg.REG_DWORD, val)
407                 logger.info(f"[LogSettings] Аудит CommandLine установлен в {val}")
408             return {
409                 'success': True,
410                 'cmdline_audit_enabled': req.enabled,
411                 'message': f"Аудит командных строк процессов (Event 4688) {'включен' if req.enabled else 'выключен'}.",

```

```

412         }
413     except PermissionError:
414         raise HTTPException(
415             status_code=status.HTTP_403_FORBIDDEN,
416             detail="Отказано в доступе к реестру HKLM. Тре
417 буются права Администратора.",
418         )
419     except Exception as exc:
420         raise HTTPException(
421             status_code=status.HTTP_500_INTERNAL_SERVER_ER
422 ROR,
423             detail=f"Ошибка записи в реестр: {exc}",
424         )
425
426     return await asyncio.to_thread(_set_reg)
427
428 @router.post('/collector/collect')
429 async def run_collector_collection(limit: int = Query(500, ge=
430 10, le=5000)) -> Dict[str, Any]:
431     """Запуск инкрементального сбора журнала безопасности в te
432 lemetry.db."""
433     def _collect():
434         report = _security_collector.collect_incremental(batch
435 _size=limit)
436         return {
437             'success': True,
438             'channel': report.channel,
439             'total_events_ingested': report.total_events_inges
440 ted,
441             'last_record_id': report.last_record_id,
442             'events_by_id': report.events_by_id,
443             'errors': report.errors,
444             'message': f"Собрано {report.total_events_ingested}
445 новых событий. Последний Record ID: {report.last_record_id}"
446         },
447     }

```

```

442     return await asyncio.to_thread(_collect)
443
444
445 @router.post('/collector/reset-bookmark')
446 async def reset_collector_bookmark(req: BookmarkResetRequest)
447     -> Dict[str, Any]:
448     """Сброс закладки сборщика на указанный Record ID или на т
449     екущий конец журнала."""
450     def _reset():
451         target_id = req.record_id
452         if target_id is None:
453             raw_info = _query_wevtutil_channel_info(req.channe
454             l)
455             target_id = raw_info.get('newest_record_id', 0)
456
457         fake_xml = f"<BookmarkList>\n  <Bookmark Channel='{req
458         .channel}' RecordId='{target_id}' IsCurrent='true'/>\n</Bookma
459         rkList>"
460         _storage.save_security_bookmark(req.channel, target_id
461         , fake_xml)
462         logger.info(f"[LogSettings] Закладка для {req.channel}
463         сброшена на Record ID {target_id}")
464
465     return {
466         'success': True,
467         'channel': req.channel,
468         'reset_record_id': target_id,
469         'message': f"Закладка журнала '{req.channel}' успе
470         шно установлена на Record ID {target_id}.",
471     }
472
473     return await asyncio.to_thread(_reset)
474
475
476 @router.get('/events')
477 async def get_channel_events(
478     channel: str = Query('Security', description="Имя канала и
479     ли 'telemetry_db'"),

```



```

471     event_id: int = Query(0, description="Фильтр по Event ID (
    0 = все)"),
472     level: str = Query('', description="Фильтр по уровню (Info
    rmation, Warning, Error, Critical)"),
473     search: str = Query('', description="Поисковая строка по т
    ексту сообщения или полям"),
474     user: str = Query('', description="Фильтр по имени пользо
    вателя"),
475     process: str = Query('', description="Фильтр по имени проц
    есса"),
476     limit: int = Query(50, ge=1, le=500, description="Максима
    льное число записей"),
477 ) -> Dict[str, Any]:
478     """Чтение и фильтрация событий из выбранного журнала или и
    з базы telemetry.db."""
479     def _fetch_events():
480         if channel == 'telemetry_db' or channel.lower() == 'se
    curity_db':
481             items = _storage.get_security_events(
482                 limit=limit,
483                 event_id=event_id if event_id > 0 else None,
484                 user=user or None,
485                 process_name=process or None,
486             )
487             if search:
488                 s_low = search.lower()
489                 items = [
490                     it for it in items
491                     if s_low in str(it.get('message', '')).low
    er()
492                     or s_low in str(it.get('process_name', ''
    )).lower()
493                     or s_low in str(it.get('subject_user', ''
    )).lower()
494                     or s_low in str(it.get('command_line', ''
    )).lower()
495                 ]
496             return {

```

```

497         'source': 'telemetry.db',
498         'channel': 'Security',
499         'events': items,
500         'total': len(items),
501     }
502
503     # Чтение напрямую через WevtAPI
504     raw_events = _wevtapi.read_events(
505         channel=channel,
506         limit=limit * 2 if (user or process or search) else
e limit,
507         level=level,
508         search=search,
509         event_id=event_id if event_id > 0 else 0,
510         hours=72,
511     )
512
513     filtered: List[Dict[str, Any]] = []
514     for ev in raw_events:
515         msg = ev.get('message', '')
516         raw_data = str(ev.get('raw_data', ''))
517
518         if user:
519             u_low = user.lower()
520             if u_low not in msg.lower() and u_low not in r
aw_data.lower():
521             continue
522         if process:
523             p_low = process.lower()
524             if p_low not in msg.lower() and p_low not in r
aw_data.lower() and p_low not in str(ev.get('provider', '')).l
ower():
525             continue
526
527         filtered.append(ev)
528         if len(filtered) >= limit:
529             break
530

```

```

531         return {
532             'source': 'wevtapi',
533             'channel': channel,
534             'events': filtered,
535             'total': len(filtered),
536         }
537
538     return await asyncio.to_thread(_fetch_events)
539
540
541 # -----
542 # Иерархический каталог провайдеров и 7 доменов телеметрии
543 # -----
544 from apps.windows.telemetry.event_catalog import WindowsEventC
545 atalogEngine
546
547 _catalog_engine = WindowsEventCatalogEngine()
548
549 @router.get('/catalog')
550 async def get_event_catalog(
551     domain: Optional[str] = Query(None, description="Фильтр по
552     домену телеметрии"),
553     volume: Optional[str] = Query(None, description="Фильтр по
554     объему (Low, Medium, High, Extreme)"),
555     search: Optional[str] = Query(None, description="Поиск по
556     имени провайдера или описанию"),
557 ) -> Dict[str, Any]:
558     """Возвращает структурированный каталог провайдеров и кана
559     лов Windows Event Log."""
560     def _fetch_catalog():
561         entries = _catalog_engine.get_full_catalog()
562         filtered = []
563         for e in entries:
564             dom_val = e.domain.value if hasattr(e.domain, 'val
565             ue') else str(e.domain)

```

```

561         if domain and domain.lower() not in dom_val.lower(
562         ):
563             continue
564         if volume and volume.lower() != e.volume_rating.lo
565         wer():
566             continue
567         if search:
568             s_low = search.lower()
569             if (s_low not in e.provider.lower() and
570                 s_low not in e.channel.lower() and
571                 s_low not in e.description.lower()):
572                 continue
573
574         filtered.append({
575             'id': e.id,
576             'provider': e.provider,
577             'channel': e.channel,
578             'domain': dom_val,
579             'domain_title': e.domain_title,
580             'key_events': e.key_events,
581             'levels': e.levels,
582             'realtime_supported': e.realtime_supported,
583             'historical_supported': e.historical_supported
584         },
585         {
586             'privilege_required': e.privilege_required,
587             'volume_rating': e.volume_rating,
588             'useful_fields': e.useful_fields,
589             'recommended_collector': e.recommended_collect
590         },
591         {
592             'storage_class': e.storage_class,
593             'correlation_targets': e.correlation_targets,
594             'description': e.description,
595             'is_present_on_host': e.is_present_on_host,
596             'record_count': e.record_count,
597             'size_mb': e.size_mb,
598         })
599     return {'catalog': filtered, 'total': len(filtered)}

```

```

595     return await asyncio.to_thread(_fetch_catalog)
596
597
598 @router.get('/domains')
599 async def get_telemetry_domains_summary() -> Dict[str, Any]:
600     """Сводка и метрики готовности по 7 доменам телеметрии Win
        dows Event Logs."""
601     return await asyncio.to_thread(_catalog_engine.get_domains
        _summary)
602
603
604 @router.get('/provenance/process')
605 async def get_process_provenance_graph(
606     process_name: Optional[str] = Query(None, description="Имя
        целевого процесса"),
607     pid: Optional[int] = Query(None, description="Идентификато
        р PID"),
608     hours: int = Query(24, ge=1, le=168, description="Глубина
        выборки в часах"),
609 ) -> Dict[str, Any]:
610     """Построение графа происхождения процессов (Process Prove
        nance Graph: User -> Parent -> Process -> Service/Task)."""
611     return await asyncio.to_thread(
612         _catalog_engine.reconstruct_process_provenance,
613         process_name=process_name,
614         pid=pid,
615         hours=hours,
616     )
617
618
619 @router.get('/lifecycle/power')
620 async def get_power_lifecycle(
621     hours: int = Query(72, ge=1, le=720, description="Глубина
        выборки в часах"),
622 ) -> Dict[str, Any]:
623     """Реконструкция хронологии жизненного цикла ОС: Boot -> L
        ogin -> Sleep -> Wake -> Shutdown/Crash."""
624     events = await asyncio.to_thread(_catalog_engine.reconstru

```

```
        ct_power_lifecycle, hours=hours)
625     return {'timeline': events, 'total': len(events)}
626
627
628 @router.get('/hardware/pnp')
629 async def get_hardware_pnp_chain(
630     hours: int = Query(72, ge=1, le=720, description="Глубина
        выборки в часах"),
631 ) -> Dict[str, Any]:
632     """Реконструкция цепочки событий оборудования и драйверов
        Plug and Play."""
633     events = await asyncio.to_thread(_catalog_engine.reconstru
        ct_hardware_pnp_chain, hours=hours)
634     return {'chain': events, 'total': len(events)}
635
636
637 @router.get('/sysmon/status')
638 async def get_sysmon_status() -> Dict[str, Any]:
639     """Проверка доступности и статуса расширенной подсистемы S
        ysmo."""
640     return await asyncio.to_thread(_catalog_engine.check_sysmo
        n_status)
641
642
643 def init_router() -> APIRouter:
644     """Инициализация и возврат роутера."""
645     return router
646
647
648 __all__ = ['router', 'init_router']
649
```

```

router_memory_io.py
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Memory IO
4 # =====
5 # Description:
6 #   GET /api/v1/panel/memory-io – загрузка памяти и скорость чтения/записи
7 #   (байт/с) из таблицы system_snapshots базы telemetry.db.
8 #
9 # Usage Examples:
10 #   Python API:
11 #       from apps.windows.api.routers.router_memory_io import init_router
12 #       app.include_router(init_router())
13 #
14 # File: router_memory_io.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-05 23:40:00
20 # =====
21
22 from __future__ import annotations
23 """Роутер панели «Память и ввод-вывод»: опрос метрик исключительно из telemetry.db."""
24
25 import json
26 from typing import Any, Dict, List, Optional
27
28 from fastapi import APIRouter, Query
29 from pydantic import BaseModel, Field

```

```

30
31 from logger import logger
32 from apps.windows.telemetry.sqlite import TelemetryStorage
33
34 _MAX_HISTORY = 300
35
36 _SQL_HISTORY = (
37     "SELECT id, timestamp, memory_total_gb, memory_used_gb, me
38     mory_percent, swap_percent, "
39     "disk_read_bytes_sec, disk_write_bytes_sec "
40     "FROM system_snapshots ORDER BY id DESC LIMIT ?;"
41 )
42
43 class MemoryBlock(BaseModel):
44     """Использование оперативной памяти."""
45     name: str = Field(default="RAM", description="Тип / назван
46     ие оперативной памяти")
47     summary: str = Field(default="", description="Сводка по мо
48     дулям RAM")
49     total_gb: float = Field(default=0.0, description="Всего RA
50     M, GB")
51     used_gb: float = Field(default=0.0, description="Занято RA
52     M, GB")
53     free_gb: float = Field(default=0.0, description="Свободно
54     RAM, GB")
55     percent: float = Field(default=0.0, description="Загрузка
56     RAM, %")
57     swap_percent: float = Field(default=0.0, description="Загр
58     узка файла подкачки, %")
59
60 class DiskIoBlock(BaseModel):
61     """Скорость дискового ввода-вывода."""
62     read_bytes_sec: float = Field(default=0.0, description="Ск
63     ороность чтения, байт/с")
64     write_bytes_sec: float = Field(default=0.0, description="С
65     корость записи, байт/с")

```



```

58     total_bytes_sec: float = Field(default=0.0, description="С
    уммарная скорость, байт/с")
59
60
61 class MemoryIoPoint(BaseModel):
62     """Точка истории метрик."""
63     timestamp: str = Field(default="", description="Время сним
    ка")
64     memory_percent: float = Field(default=0.0, description="За
    грузка RAM, %")
65     read_bytes_sec: float = Field(default=0.0, description="Чт
    ение, байт/с")
66     write_bytes_sec: float = Field(default=0.0, description="За
    пись, байт/с")
67
68
69 class MemoryIoResponse(BaseModel):
70     """Ответ GET /api/v1/panel/memory-io."""
71     status: str = Field(default="ok", description="Статус отве
    та")
72     name: str = Field(default="RAM", description="Конфигурация
    памяти")
73     timestamp: str = Field(default="", description="Время посл
    еднего снимка")
74     memory: MemoryBlock = Field(default_factory=MemoryBlock)
75     disk_io: DiskIoBlock = Field(default_factory=DiskIoBlock)
76     history: List[MemoryIoPoint] = Field(default_factory=list,
    description="История (от старых к новым)")
77     meta: Dict[str, Any] = Field(default_factory=dict, descrip
    tion="Метаданные запроса")
78
79
80 def _f(value: Any) -> float:
81     """Приводит значение из БД к float; NULL становится 0.0."""
    "
82     return float(value) if value is not None else 0.0
83
84

```

```

85 def build_memory_io(rows: List[Dict[str, Any]]) -> MemoryIoRes
    ponse:
86     """Собирает ответ из строк system_snapshots базы данных.
87
88     Args:
89         rows: Строки снимков от новых к старым.
90
91     Returns:
92         MemoryIoResponse: Текущие значения и история из БД.
93     """
94     if not rows:
95         return MemoryIoResponse(meta={"source": "telemetry.db"
    , "table": "system_snapshots", "points": 0})
96
97     last = rows[0]
98     total = _f(last.get("memory_total_gb"))
99     used = _f(last.get("memory_used_gb"))
100    read_bps = _f(last.get("disk_read_bytes_sec"))
101    write_bps = _f(last.get("disk_write_bytes_sec"))
102
103    ram_name = f"{round(total)} GB RAM" if total > 0 else "RAM"
104    "
105
106    ram_summary = ""
107
108    history = [
109        MemoryIoPoint(
110            timestamp=str(r.get("timestamp") or ""),
111            memory_percent=_f(r.get("memory_percent")),
112            read_bytes_sec=_f(r.get("disk_read_bytes_sec")),
113            write_bytes_sec=_f(r.get("disk_write_bytes_sec")),
114        )
115        for r in reversed(rows)
116    ]
117
118    return MemoryIoResponse(
119        name=ram_name,
120        timestamp=str(last.get("timestamp") or ""),
121        memory=MemoryBlock(
122            name=ram_name,

```

```

120         summary=ram_summary,
121         total_gb=round(total, 2),
122         used_gb=round(used, 2),
123         free_gb=round(max(total - used, 0.0), 2),
124         percent=round(_f(last.get("memory_percent")), 1),
125         swap_percent=round(_f(last.get("swap_percent")), 1
126     ),
127     disk_io=DiskIoBlock(
128         read_bytes_sec=round(read_bps, 1),
129         write_bytes_sec=round(write_bps, 1),
130         total_bytes_sec=round(read_bps + write_bps, 1),
131     ),
132     history=history,
133     meta={"source": "telemetry.db", "table": "system_snaps
134         hots", "points": len(history)},
135     )
136
137 def init_router(storage: Optional[TelemetryStorage] = None) ->
138     APIRouter:
139         """Создаёт роутер /api/v1/panel/memory-io.
140
141         Args:
142             storage: Хранилище телеметрии (по умолчанию — синглтон
143             TelemetryStorage в режиме read_only).
144
145         Returns:
146             APIRouter: Роутер с эндпоинтом GET /api/v1/panel/memor
147             y-io.
148         """
149         router = APIRouter(tags=["Memory IO Panel"])
150         store = storage or TelemetryStorage.get_instance(read_only
= True)
151
152         @router.get("/api/v1/panel/memory-io", response_model=Memo
ryIoResponse)
153         async def get_panel_memory_io(

```

```
151         limit: int = Query(default=30, ge=1, le=_MAX_HISTORY,
152         description="Число точек истории"),
153     ) -> MemoryIoResponse:
154         """Возвращает последние метрики памяти и дискового вво
155         да-вывода исключительно из базы данных telemetry.db."""
156         try:
157             with store._lock, store._get_connection() as conn:
158                 rows = [dict(r) for r in conn.execute(_SQL_HIS
159                 TORY, (limit,)).fetchall()]
160                 return build_memory_io(rows)
161             except Exception as exc:
162                 logger.error(f"[router_memory_io] Ошибка чтения ме
163                 трик памяти из БД: {exc}", exc_info=True)
164                 return MemoryIoResponse(status="error", meta={"sou
165                 rce": "telemetry.db", "error": str(exc)})
166
167     return router
168
169 __all__ = ["init_router", "build_memory_io", "MemoryIoResponse"]
```

router_menu.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Menu
4 # =====
5 # Description:
6 #     Предоставляет эндпоинты FastAPI для чтения и сохранения конфигурации меню
7 #
8 # Usage Examples:
9 #     Python API:
10 #         from apps.windows.api.routers.router_menu import MenuItem
11 #
12 #         service = MenuItem()
13 #
14 # File: router_menu.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-04 01:10:00
20 # =====
21
22 from __future__ import annotations
23 """Предоставляет эндпоинты FastAPI для чтения и сохранения конфигурации меню"""
24
25 import json
26 from pathlib import Path
27 from typing import Any, Dict, List, Optional
28
29 from fastapi import APIRouter, HTTPException, Request
```

```

30 from pydantic import BaseModel, Field
31
32 from header import __root__
33 from logger import logger
34
35 # Пути к директориям меню
36 _LOCAL_CONFIG_MENUES = Path(__file__).resolve().parents[1] / '
    webgui' / 'config_menus'
37 _SRC_CONFIG_MENUES = __root__ / 'src' / 'api' / 'webgui' / 'co
    nfig_menus'
38
39
40 def _get_config_path(target: Optional[str] = None) -> Path:
41     """Возвращает актуальный путь к файлу конфигурации меню по
        целевому контексту.
42
43     Args:
44         target: Имя контекста (tc, apps, su и т.д.).
45
46     Returns:
47         Path: Путь к JSON файлу конфигурации меню.
48     """
49     file_name = 'tc_menu_config.json'
50     if target and target.lower() in ('su', 'user', 'user_assis
        tant'):
51         file_name = 'su_menu_config.json'
52     elif target and target.lower() not in ('tc', 'apps', 'test
        _computer'):
53         file_name = f"{target.lower()}_menu_config.json"
54
55     # 1. Проверяем локальный каталог webgui приложения Windows
56     local_path = _LOCAL_CONFIG_MENUES / file_name
57     if local_path.exists():
58         return local_path
59
60     # 2. Проверяем глобальный каталог в src/api/webgui
61     src_path = _SRC_CONFIG_MENUES / file_name
62     if src_path.exists():

```

```

63         return src_path
64
65     return local_path
66
67
68 class MenuItem(BaseModel):
69     """Модель элемента меню."""
70     id: str
71     label: str
72     icon: Optional[str] = None
73     tab: str
74     order: Optional[int] = 0
75     visible: Optional[bool] = True
76     icon: Optional[str] = None
77
78
79 class MenuSection(BaseModel):
80     """Модель разделов меню."""
81     topButtons: List[Dict[str, Any]] = Field(default_factory=list)
82     sidebarItems: List[Dict[str, Any]] = Field(default_factory=list)
83
84
85 class MenuConfigPayload(BaseModel):
86     """Модель полной конфигурации меню."""
87     version: Optional[str] = None
88     menu: MenuSection
89     settings: Optional[Dict[str, Any]] = None
90
91
92 def init_router() -> APIRouter:
93     """Инициализирует FastAPI роутер для управления конфигурацией меню.
94
95     Returns:
96         APIRouter: Настроенный роутер с маршрутами /api/v1/menu/*.
```

```

97     """
98     router = APIRouter(prefix='/api/v1/menu', tags=['menu'])
99
100     @router.get('/config')
101     async def get_menu_config(target: Optional[str] = None) ->
        Dict[str, Any]:
102         """Получение текущей конфигурации меню."""
103         config_path = _get_config_path(target)
104         if not config_path.exists():
105             logger.warning(f"[router_menu] Файл конфигурации м
        еню не найден: {config_path}")
106             raise HTTPException(status_code=404, detail="Файл
        конфигурации меню не найден")
107
108         try:
109             content = config_path.read_text(encoding='utf-8')
110             return json.loads(content)
111         except Exception as ex:
112             logger.error(f"[router_menu] Ошибка при чтении кон
        фигурации меню: {ex}")
113             raise HTTPException(status_code=500, detail=f"Ошиб
        ка чтения конфигурации: {ex}")
114
115     @router.post('/config')
116     async def save_menu_config(payload: Dict[str, Any], target
        : Optional[str] = None) -> Dict[str, Any]:
117         """Сохранение обновленной конфигурации меню."""
118         if not isinstance(payload, dict) or 'menu' not in payl
        oad:
119             raise HTTPException(
120                 status_code=400,
121                 detail="Некорректная структура конфигурации ме
        ню (отсутствует секция 'menu')",
122             )
123
124         config_path = _get_config_path(target)
125         try:
126             config_path.parent.mkdir(parents=True, exist_ok=Tr

```



```
ue)
127         config_path.write_text(json.dumps(payload, ensure_
    ascii=False, indent=2), encoding='utf-8')
128
129         # Также синхронизируем в src/api/webgui если путь
    отличается
130         src_path = _SRC_CONFIG_MENUUES / config_path.name
131         if src_path != config_path and _SRC_CONFIG_MENUUES.
    exists():
132             src_path.write_text(json.dumps(payload, ensure
    _ascii=False, indent=2), encoding='utf-8')
133
134         logger.info(f"[router_menu] Конфигурация меню успе
    шно сохранена в {config_path.name}")
135         return {"status": "ok", "message": f"Конфигурация
    меню успешно сохранена в {config_path.name}"}
136     except Exception as ex:
137         logger.error(f"[router_menu] Ошибка при сохранении
    конфигурации меню: {ex}")
138         raise HTTPException(status_code=500, detail=f"Ошиб
    ка сохранения конфигурации: {ex}")
139
140     return router
141
142
143 __all__ = ['init_router']
144
```

router_network_load.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Network Load
4 # =====
5 # Description:
6 #   GET /api/v1/panel/network-load – комплексные метрики нагрузки сетевых
7 #   адаптеров (Ethernet, Wi-Fi, Bluetooth): скорости Rx/Tx, использование канала,
8 #   состояние беспроводных интерфейсов, IP/MAC и история трафика.
9 #
10 # Usage Examples:
11 #   Python API:
12 #       from apps.windows.api.routers.router_network_load import
13 #       init_router
14 #       app.include_router(init_router())
15 #
16 # File: router_network_load.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 17:43:00
22 # =====
23
24 from __future__ import annotations
25
26 """Роутер панели «Сетевые адаптеры (Ethernet / Wi-Fi / BT)»: комплексная телеметрия сети."""
27
28 import re
29 import subprocess
```

```

28 import time
29 from typing import Any, Dict, List, Optional
30 from fastapi import APIRouter
31 from pydantic import BaseModel, Field
32
33 try:
34     import psutil
35 except ImportError:
36     psutil = None
37
38 from logger import logger
39 from apps.windows.modules.hardware.lhm_service import LhmService
40 from apps.windows.telemetry import SystemCollector
41 from apps.windows.telemetry.models import ProcessNetworkActivity
42
43 _collector: Optional[SystemCollector] = None
44
45
46 def get_collector() -> SystemCollector:
47     """Получить или создать синглтон сборщика системной телеметрии.
48
49     Returns:
50         SystemCollector: Экземпляр системного коллектора.
51     """
52     global _collector
53     if _collector is None:
54         _collector = SystemCollector()
55     return _collector
56
57 _last_net_io: Optional[Any] = psutil.net_io_counters() if psutil else None
58 _last_net_time: float = time.time() if psutil else 0.0
59 _net_history: List[Dict[str, Any]] = []
60 _MAX_HISTORY = 120
61

```

```
62
63 class NetworkAdapterInfo(BaseModel):
64     """Информация об отдельном сетевом адаптере."""
65     name: str = Field(..., description="Имя интерфейса (Ethernet, Wi-Fi, etc.)")
66     adapter_type: str = Field(default="Ethernet", description="Тип адаптера: Ethernet, Wi-Fi, Virtual, Loopback")
67     is_up: bool = Field(default=False, description="Подключен ли интерфейс")
68     speed_mbps: Optional[int] = Field(default=None, description="Скорость линка, Mbps")
69     ip_address: str = Field(default="", description="IPv4 адрес")
70     mac_address: str = Field(default="", description="MAC адрес")
71     bytes_recv_sec: float = Field(default=0.0, description="Скорость приема, байт/с")
72     bytes_sent_sec: float = Field(default=0.0, description="Скорость отдачи, байт/с")
73
74
75 class WifiStatusInfo(BaseModel):
76     """Состояние беспроводного адаптера Wi-Fi."""
77     name: str = Field(default="Wi-Fi", description="Имя интерфейса")
78     description: str = Field(default="", description="Модель контроллера Wi-Fi")
79     state: str = Field(default="disconnected", description="Статус подключения")
80     radio_status: str = Field(default="On", description="Состояние радиомодуля")
81     ssid: str = Field(default="", description="Имя подключенной сети (SSID)")
82     signal_percent: Optional[int] = Field(default=None, description="Уровень сигнала, %")
83     channel: Optional[int] = Field(default=None, description="Канал Wi-Fi")
84
```

```
85
86 class BluetoothStatusInfo(BaseModel):
87     """Состояние модуля и подключений Bluetooth."""
88     name: str = Field(default="Bluetooth", description="Имя ко
    нтроллера BT")
89     status: str = Field(default="OK", description="Статус адап
    тера")
90     devices_count: int = Field(default=0, description="Количес
    тво сопряженных/активных устройств")
91     devices: List[str] = Field(default_factory=list, descripti
    on="Список устройств")
92
93
94 class NetworkPoint(BaseModel):
95     """Точка истории скорости сети для графика."""
96     timestamp: str = Field(default="", description="Время заме
    ра")
97     download_bytes_sec: float = Field(default=0.0, description
    ="Скорость скачивания, байт/с")
98     upload_bytes_sec: float = Field(default=0.0, description="
    Скорость отдачи, байт/с")
99
100
101 class NetworkLoadResponse(BaseModel):
102     """Ответ GET /api/v1/panel/network-load."""
103     status: str = Field(default="ok", description="Статус отве
    та")
104     name: str = Field(default="Network Adapters", description=
    "Основной активный сетевой адаптер")
105     timestamp: str = Field(default="", description="Время сним
    ка")
106     download_bytes_sec: float = Field(default=0.0, description
    ="Текущая скорость приема (Download), байт/с")
107     upload_bytes_sec: float = Field(default=0.0, description="
    Текущая скорость отдачи (Upload), байт/с")
108     download_total_bytes: int = Field(default=0, description="
    Всего принято байт с загрузки системы")
109     upload_total_bytes: int = Field(default=0, description="Вс
```

```

его отправлено байт с загрузки системы")
110     utilization_percent: float = Field(default=0.0, description="Оценочная загрузка канала, %")
111     link_speed_mbps: Optional[int] = Field(default=1000, description="Скорость основного интерфейса, Mbps")
112     adapters: List[NetworkAdapterInfo] = Field(default_factory=list, description="Список сетевых адаптеров")
113     wifi: WifiStatusInfo = Field(default_factory=WifiStatusInfo, description="Данные Wi-Fi")
114     bluetooth: BluetoothStatusInfo = Field(default_factory=BluetoothStatusInfo, description="Данные Bluetooth")
115     history: List[NetworkPoint] = Field(default_factory=list, description="История скоростей")
116     meta: Dict[str, Any] = Field(default_factory=dict, description="Метаданные")
117
118
119 def _get_wifi_info() -> WifiStatusInfo:
120     """Опрашивает Wi-Fi через netsh wlan show interfaces."""
121     info = WifiStatusInfo()
122     try:
123         res = subprocess.run(
124             ["netsh", "wlan", "show", "interfaces"],
125             capture_output=True,
126             text=True,
127             timeout=1.5,
128             encoding="utf-8",
129             errors="ignore",
130         )
131         if res.returncode == 0 and res.stdout:
132             out = res.stdout
133             desc_m = re.search(r"Description\s*:\s*(.+)", out)
134             if desc_m:
135                 info.description = desc_m.group(1).strip()
136             state_m = re.search(r"State\s*:\s*(.+)", out)
137             if state_m:
138                 info.state = state_m.group(1).strip()
139             ssid_m = re.search(r"SSID\s*:\s*([^\r\n]+)", out)

```

```

140         if ssid_m and not ssid_m.group(1).strip().startswith("BSSID"):
141             info.ssid = ssid_m.group(1).strip()
142             signal_m = re.search(r"Signal\s*:\s*(\d+)%", out)
143             if signal_m:
144                 info.signal_percent = int(signal_m.group(1))
145                 radio_m = re.search(r"Radio status\s*:\s*Hardware\s*(\w+)\s*Software\s*(\w+)", out, re.IGNORECASE)
146                 if radio_m:
147                     info.radio_status = f"HW {radio_m.group(1)} / SW {radio_m.group(2)}"
148             except Exception as e:
149                 logger.debug(f"[RouterNetworkLoad] Ошибка опроса Wi-Fi : {e}")
150             return info
151
152
153 _cached_bt_info: Optional[BluetoothStatusInfo] = None
154 _cached_bt_time: float = 0.0
155 _BT_CACHE_TTL: float = 30.0
156
157
158 def _get_bluetooth_info() -> BluetoothStatusInfo:
159     """Опрашивает статус контроллера и устройств Bluetooth через PowerShell с кэшированием."""
160     global _cached_bt_info, _cached_bt_time
161     now = time.time()
162     if _cached_bt_info is not None and (now - _cached_bt_time < _BT_CACHE_TTL):
163         return _cached_bt_info
164
165     info = BluetoothStatusInfo()
166     try:
167         cmd = "Get-PnpDevice -Class Bluetooth | Where-Object { $_.Status -eq 'OK' } | Select-Object -ExpandProperty FriendlyName"
168         res = subprocess.run(
169             ["powershell", "-NoProfile", "-NonInteractive", "-

```

```

Command", cmd],
170         capture_output=True,
171         text=True,
172         timeout=5.0,
173         encoding="utf-8",
174         errors="ignore",
175     )
176     if res.returncode == 0 and res.stdout:
177         lines = [line.strip() for line in res.stdout.strip
178                 ().splitlines() if line.strip()]
179         main_ctrl = next((l for l in lines if "intel" in l
180                 .lower() or "adapter" in l.lower() or "radio" in l.lower()), "
181         Bluetooth Adapter")
182         devs = [l for l in lines if not any(x in l.lower()
183                 for x in ("enumerator", "service", "gateway", "transport", "o
184                 bject push", "nap service", "sim access"))]
185         info.name = main_ctrl
186         info.devices_count = len(devs)
187         info.devices = devs[:6]
188         _cached_bt_info = info
189         _cached_bt_time = now
190     except Exception as e:
191         logger.debug(f"[RouterNetworkLoad] Ошибка опроса Bluet
192         ooth: {e}")
193         if _cached_bt_info is not None:
194             return _cached_bt_info
195     return info
196
197 def calculate_network_load() -> NetworkLoadResponse:
198     """Собирает полный снимок нагрузки сети."""
199     global _last_net_io, _last_net_time, _net_history
200
201     now = time.time()
202     current_io = psutil.net_io_counters() if psutil else None
203
204     rx_sec = 0.0
205     tx_sec = 0.0

```



```
201
202     if current_io and _last_net_io and _last_net_time > 0:
203         dt = max(0.2, now - _last_net_time)
204         rx_sec = max(0.0, (current_io.bytes_recv - _last_net_i
o.bytes_recv) / dt)
205         tx_sec = max(0.0, (current_io.bytes_sent - _last_net_i
o.bytes_sent) / dt)
206
207         _last_net_io = current_io
208         _last_net_time = now
209
210         ts_str = time.strftime("%H:%M:%S")
211         _net_history.append({"timestamp": ts_str, "download_bytes_
sec": rx_sec, "upload_bytes_sec": tx_sec})
212         if len(_net_history) > _MAX_HISTORY:
213             _net_history.pop(0)
214
215         # Собираем список сетевых адаптеров
216         adapters_list: List[NetworkAdapterInfo] = []
217         main_name = "Ethernet"
218         main_speed = 1000
219
220         if psutil:
221             stats_map = psutil.net_if_stats()
222             addrs_map = psutil.net_if_addrs()
223
224             for if_name, stat in stats_map.items():
225                 if not stat.isup:
226                     continue
227                 is_virt = any(v in if_name.lower() for v in ("veth
ernet", "virtual", "hyper-v", "loopback", "wsl", "vmware", "bo
x"))
228                 is_wifi = "wi-fi" in if_name.lower() or "wlan" in
if_name.lower() or "wireless" in if_name.lower()
229                 atype = "Wi-Fi" if is_wifi else ("Virtual" if is_v
irt else "Ethernet")
230
231                 addrs = addrs_map.get(if_name, [])
```

```

232         ip_val = next((str(a.address) for a in addrs if ge
tattr(a.family, "name", "") == "AF_INET"), "")
233         mac_val = next((str(a.address) for a in addrs if g
etattr(a.family, "name", "") == "AF_LINK"), "")
234
235         if not is_virt and stat.speed > 0:
236             main_name = f"{if_name} ({stat.speed} Mbps)"
237             main_speed = stat.speed
238
239         adapters_list.append(
240             NetworkAdapterInfo(
241                 name=if_name,
242                 adapter_type=atype,
243                 is_up=stat.isup,
244                 speed_mbps=stat.speed if stat.speed > 0 el
se None,
245                 ip_address=ip_val,
246                 mac_address=mac_val,
247                 bytes_recv_sec=rx_sec if not is_virt else
0.0,
248                 bytes_sent_sec=tx_sec if not is_virt else
0.0,
249                 )
250         )
251
252         # Оценка утилизации канала (в процентах от скорости линка
в байтах)
253         max_bandwidth_bps = (main_speed * 1_000_000) / 8.0 if main
_speed > 0 else 125_000_000.0
254         utilization_pct = min(100.0, round(((rx_sec + tx_sec) / ma
x_bandwidth_bps) * 100.0, 2))
255
256         wifi_info = _get_wifi_info()
257         bt_info = _get_bluetooth_info()
258
259         hist_points = [
260             NetworkPoint(
261                 timestamp=p["timestamp"],

```

```

262         download_bytes_sec=p["download_bytes_sec"],
263         upload_bytes_sec=p["upload_bytes_sec"],
264     )
265     for p in _net_history
266 ]
267
268     return NetworkLoadResponse(
269         name=main_name,
270         timestamp=ts_str,
271         download_bytes_sec=rx_sec,
272         upload_bytes_sec=tx_sec,
273         download_total_bytes=current_io.bytes_recv if current_
io else 0,
274         upload_total_bytes=current_io.bytes_sent if current_io
else 0,
275         utilization_percent=utilization_pct,
276         link_speed_mbps=main_speed,
277         adapters=adapters_list,
278         wifi=wifi_info,
279         bluetooth=bt_info,
280         history=hist_points,
281         meta={"source": "psutil/netsh/pnp", "adapters_count":
len(adapters_list)},
282     )
283
284
285 def init_router() -> APIRouter:
286     """Создаёт роутер сетевой телеметрии и активности процессо
в.
287
288     Returns:
289         APIRouter: Роутер с эндпоинтами панели /api/v1/panel/n
etwork-load и /api/v1/system/network-activity.
290     """
291     router = APIRouter(tags=["Network Load Panel"])
292
293     @router.get("/api/v1/panel/network-load", response_model=N
etworkLoadResponse)

```

```

294     async def get_network_load() -> NetworkLoadResponse:
295         """Возвращает текущие метрики нагрузки сети, Wi-Fi и B
|uetooth."""
296         try:
297             return calculate_network_load()
298         except Exception as e:
299             logger.error(f"[RouterNetworkLoad] Ошибка вычислен
|ия сетевой нагрузки: {e}", exc_info=True)
300             return NetworkLoadResponse(status="error", meta={"
|error": str(e)})
301
302     @router.get("/api/v1/panel/network-activity", response_mod
|el=List[ProcessNetworkActivity])
303     @router.get("/api/v1/system/network-activity", response_mo
|del=List[ProcessNetworkActivity])
304     @router.get("/api/v1/tc/network-activity", response_model=
|List[ProcessNetworkActivity])
305     async def get_process_network_activity_endpoint(
306         limit: int = 100,
307         only_internet: bool = False,
308     ) -> List[ProcessNetworkActivity]:
309         """Возвращает список активных сетевых соединений прогр
|амм и процессов из SQLite (< 3 мс).
310
311         Args:
312             limit: Максимальное число записей в выдаче.
313             only_internet: Фильтровать только внешние интернет
|соединения.
314
315         Returns:
316             List[ProcessNetworkActivity]: Список объектов сете
|вой активности процессов.
317         """
318         try:
319             from apps.windows.telemetry.sqlite import Telemetr
|yStorage
320             storage = TelemetryStorage.get_instance(read_only=
|True)

```

```
321         db_activity = storage.get_latest_process_network_a
| ctivity(limit=limit, only_internet=only_internet)
322         if db_activity:
323             return [ProcessNetworkActivity.model_validate(
| a) for a in db_activity]
324         except Exception as exc:
325             logger.debug(f"[RouterNetworkLoad] Ошибка чтения а
| ктивности процессов из SQLite: {exc}")
326
327         try:
328             collector = get_collector()
329             return collector.get_process_network_activity(limi
| t=limit, only_internet=only_internet)
330         except Exception as exc:
331             logger.error(f"[RouterNetworkLoad] Ошибка получени
| я сетевой активности процессов: {exc}", exc_info=True)
332             return []
333
334     return router
335
```

router_power_lifecycle.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Power Lifecycle
4 # =====
5 # Description:
6 #   FastAPI REST эндпоинты для управления, анализа и мониторинга жизненного цикла
7 #   питания Windows, истории перезагрузок (User32 1074, Kernel-Power 41, EventLog 6008/6005/6006)
8 #   и реконструированных сессий питания (Power Sessions).
9 #
10 # Usage Examples:
11 #   Python API:
12 #       from apps.windows.api.routers.router_power_lifecycle import init_router
13 #
14 #       router = init_router()
15 #
16 # File: router_power_lifecycle.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-08 02:51:00
22 # =====
23
24 from __future__ import annotations
25 """FastAPI REST эндпоинты для анализа жизненного цикла питания и сессий Windows."""
26
27 from datetime import datetime, timezone
28 from typing import Any, Dict, List, Optional, Union
```

```

29 from fastapi import APIRouter, HTTPException, Query
30 from pydantic import BaseModel, Field
31
32 try:
33     from logger import logger
34 except ImportError:
35     import logging
36     logger = logging.getLogger(__name__)
37
38 from apps.windows.telemetry.models import (
39     PowerEventRecord,
40     PowerSessionRecord,
41     PowerLifecycleSummary,
42 )
43 from apps.windows.telemetry.power_lifecycle import PowerLifecycleEngine
44 from apps.windows.telemetry.sqlite import TelemetryStorage
45
46
47 def init_router(prefix: str = "/api/v1/power", **kwargs: Any)
48     -> APIRouter:
49     """Инициализирует и возвращает FastAPI APIRouter для подсистемы питания.
50
51     Args:
52         prefix: Префикс REST маршрутов API.
53
54     Returns:
55         APIRouter: Сконфигурированный роутер FastAPI.
56     """
57     router = APIRouter(prefix=prefix, tags=["Power Lifecycle & Boot History"])
58     engine = PowerLifecycleEngine()
59
60     @router.get("/summary", response_model=PowerLifecycleSummary)
61     async def get_power_summary() -> PowerLifecycleSummary:
62         """Получить сводную статистику по сессиям питания, апт

```

```

айму и перезагрузкам."""
62         try:
63             # Если в БД ещё нет записей, выполняем первичное б
            ыстрое сканирование
64             summary = engine.get_summary()
65             if summary.total_sessions_count == 0:
66                 summary = engine.scan_and_reconstruct(hours=72
            0)
67             return summary
68         except Exception as exc:
69             logger.error(f"[RouterPowerLifecycle] Ошибка получ
            ения сводки: {exc}", exc_info=True)
70             raise HTTPException(status_code=500, detail=str(ex
            c))
71
72     @router.get("/sessions", response_model=List[PowerSessionR
            ecord])
73     async def get_power_sessions(
74         limit: int = Query(50, ge=1, le=500, description="Лими
            т записей"),
75         offset: int = Query(0, ge=0, description="Смещение"),
76         shutdown_type: Optional[str] = Query(None, description
            ="Фильтр по типу: Restart, Shutdown, Unexpected, Active"),
77         unexpected_only: bool = Query(False, description="Толь
            ко внезапные завершения"),
78     ) -> List[PowerSessionRecord]:
79         """Получить список реконструированных сессий питания W
            indows."""
80         try:
81             sessions = engine.get_sessions(
82                 limit=limit,
83                 offset=offset,
84                 shutdown_type=shutdown_type,
85                 unexpected_only=unexpected_only,
86             )
87             if not sessions and offset == 0:
88                 # Первоначальный сбор при пустой базе
89                 engine.scan_and_reconstruct(hours=720)

```



```

90         sessions = engine.get_sessions(
91             limit=limit,
92             offset=offset,
93             shutdown_type=shutdown_type,
94             unexpected_only=unexpected_only,
95         )
96         return sessions
97     except Exception as exc:
98         logger.error(f"[RouterPowerLifecycle] Ошибка получ
99     ения сессий: {exc}", exc_info=True)
100         raise HTTPException(status_code=500, detail=str(ex
101     c))
102
103     @router.get("/sessions/{session_id}", response_model=Power
104     SessionRecord)
105     async def get_power_session_by_id(session_id: str) -> Powe
106     rSessionRecord:
107         """Получить детальную информацию по конкретной сессии
108         питания."""
109         item = engine.storage.get_power_session_by_id(session_
110         id)
111         if not item:
112             raise HTTPException(status_code=404, detail=f"Сеcc
113         ия питания {session_id} не найдена")
114         return PowerSessionRecord(**item)
115
116     @router.get("/events", response_model=List[PowerEventRecor
117     d])
118     async def get_power_events(
119         limit: int = Query(100, ge=1, le=1000, description="Ли
120         мит событий"),
121         offset: int = Query(0, ge=0, description="Смещение"),
122         event_id: Optional[int] = Query(None, description="Фил
123         ьтр по Event ID (1074, 41, 6008, 12, 13, etc.)"),
124     ) -> List[PowerEventRecord]:
125         """Получить список нормализованных системных событий п
126         итания."""
127         try:

```

```
117         return engine.get_events(limit=limit, offset=offset, event_id=event_id)
118     except Exception as exc:
119         logger.error(f"[RouterPowerLifecycle] Ошибка получения событий: {exc}", exc_info=True)
120         raise HTTPException(status_code=500, detail=str(exc))
121
122     @router.post("/rebuild", response_model=PowerLifecycleSummary)
123     async def rebuild_power_lifecycle(
124         hours: int = Query(720, ge=1, le=8760, description="Глубина сканирования журналов в часах")
125     ) -> PowerLifecycleSummary:
126         """Принудительно пересканировать Event Log и перестроить сессии питания."""
127         try:
128             summary = engine.scan_and_reconstruct(hours=hours, force=True)
129             return summary
130         except Exception as exc:
131             logger.error(f"[RouterPowerLifecycle] Ошибка пересборки: {exc}", exc_info=True)
132             raise HTTPException(status_code=500, detail=str(exc))
133
134     return router
135
136
137 __all__ = ["init_router"]
138
```

router_registry_viewer.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Registry Viewer
4 # =====
5 # Description:
6 #   FastAPI роутер-адаптер для приложения Windows Registry Viewer (apps.windows.registry),
7 #
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.routers.router_registry_viewer import init_router
11 #
12 #       res = init_router()
13 #
14 # File: router_registry_viewer.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-01 13:28:28
20 # =====
21
22 from __future__ import annotations
23 """FastAPI роутер-адаптер для приложения Windows Registry Viewer (apps.windows.registry),"""
24
25 from fastapi import APIRouter
26 from apps.windows.registry import (
27     BookmarkItem,
28     RegistryKeyDetailsDTO,
29     RegistryValueDTO,
```

```
30 | RegistryViewer,  
31 | SearchMatchItem,  
32 | SearchResponseDTO,  
33 | init_router as init_app_router,  
34 | )  
35 |  
36 | # Синглтон-роутер приложения  
37 | router: APIRouter = init_app_router()  
38 |  
39 |  
40 | def init_router() -> APIRouter:  
41 |     """Фабрика инициализации роутера Registry Viewer для регис  
    | трации в основном приложении."""  
42 |     return router  
43 |  
44 |  
45 | __all__ = [  
46 |     "init_router",  
47 |     "router",  
48 |     "RegistryValueDTO",  
49 |     "RegistryKeyDetailsDTO",  
50 |     "BookmarkItem",  
51 |     "SearchMatchItem",  
52 |     "SearchResponseDTO",  
53 |     "RegistryViewer",  
54 | ]  
55 |
```

```

router_scenarios.py
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Scenarios
4 # =====
5 # Description:
6 #   FastAPI REST эндпоинты для запуска и мониторинга автоматизированных сценариев
7 #
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.routers.router_scenarios import ScenarioQuestionsConfig
11 #
12 #       service = ScenarioQuestionsConfig()
13 #
14 # File: router_scenarios.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-04 08:35:00
20 # =====
21
22 from __future__ import annotations
23 """FastAPI REST эндпоинты для запуска и мониторинга автоматизированных сценариев"""
24
25 import asyncio
26 import os
27 import platform
28 import shutil
29 import sys

```

```

30 import time
31 from datetime import datetime, timezone
32 from pathlib import Path
33 from typing import Any, Dict, List, Optional
34
35 from fastapi import APIRouter, HTTPException, Query
36 from fastapi.responses import StreamingResponse
37 from pydantic import BaseModel, Field
38
39 from src.config import server_cfg
40 from logger import logger
41
42 __root__ = Path(__file__).resolve().parents[2]
43
44
45 import json
46
47 # -----
48 # DTO Models
49 # -----
50
51 class ScenarioQuestionsConfig(BaseModel):
52     """Конфигурация подсказок и быстрых вопросов для мини-чата
53     ."""
54     description: str = Field(default="", description="Описание
55     пула вопросов")
56     prompts: List[str] = Field(default_factory=list, descripti
57     on="Список вопросов для динамической ротации")
58     quick_buttons: List[Dict[str, str]] = Field(default_factor
59     y=list, description="Список кнопок быстрого доступа")
60
61 class ScenarioStepResult(BaseModel):
62     """Результат выполнения отдельного шага сценария."""
63     name: str = Field(description="Наименование шага проверки"
64     )

```

```
61     status: str = Field(description="Статус шага: 'ok', 'warn'  
    , 'error', 'skipped'")  
62     duration_ms: float = Field(default=0.0, description="Время  
    выполнения шага в миллисекундах")  
63     details: str = Field(default="", description="Подробное оп  
    исание или результат проверки")  
64     recommendation: Optional[str] = Field(default=None, descri  
    ption="Рекомендация при предупреждении или ошибке")  
65     data: Optional[Dict[str, Any]] = Field(default=None, descr  
    iption="Дополнительные структурированные данные")  
66  
67  
68 class ScenarioRunResult(BaseModel):  
69     """Сводный результат прогона тестового сценария."""  
70     scenario_id: str = Field(description="Идентификатор сценар  
    ия")  
71     title: str = Field(description="Отображаемое название сцен  
    ария")  
72     status: str = Field(description="Общий статус сценария: 'o  
    k', 'warn', 'error'")  
73     started_at: str = Field(description="Время запуска сценари  
    я (ISO 8601)")  
74     duration_ms: float = Field(description="Общее время выполн  
    ения в миллисекундах")  
75     total_steps: int = Field(description="Общее количество вып  
    олненных шагов")  
76     passed_steps: int = Field(description="Количество успешных  
    шагов")  
77     warn_steps: int = Field(description="Количество шагов с пр  
    едупреждением")  
78     failed_steps: int = Field(description="Количество провален  
    ных шагов")  
79     steps: List[ScenarioStepResult] = Field(default_factory=li  
    st, description="Список результатов шагов")  
80     summary: str = Field(description="Итоговый экспертный верд  
    икт на русском языке")  
81  
82
```

```

83 class ScenarioItem(BaseModel):
84     """Метаданные сценария тестирования."""
85     id: str = Field(description="Уникальный ID сценария")
86     title: str = Field(description="Название сценария")
87     description: str = Field(description="Описание сценария и
| решаемых задач")
88     category: str = Field(description="Категория сценария (quick,
| diagnostics, logs, security, network, ai)")
89     icon: str = Field(description="Иконка или эмодзи")
90     recommended: bool = Field(default=False, description="Флаг
| рекомендации после установки")
91     estimated_duration_sec: int = Field(default=3, description
| ="Ориентировочное время выполнения в секундах")
92
93
94 class ScenarioRunRequest(BaseModel):
95     """Запрос на запуск сценария."""
96     scenario_id: str = Field(description="ID сценария ('quick_
| check', 'log_audit', 'system_inspector', 'windows_admin', 'net
| work_test', 'ai_providers_check', 'all')")
97     params: Optional[Dict[str, Any]] = Field(default_factory=d
| ict, description="Дополнительные параметры выполнения")
98
99
100 class ScenarioCreatedSkillInfo(BaseModel):
101     """Информация о сформированном навыке."""
102     name: str = Field(description="Имя навыка (kebab-case)")
103     title: str = Field(description="Человекочитаемый заголовок
| навыка")
104     path: str = Field(description="Путь к файлу SKILL.md")
105     description_ru: str = Field(description="Описание на русск
| ом языке")
106     is_new: bool = Field(default=True, description="Был ли нав
| ык создан заново")
107
108
109 class ScenarioToolPlanInfo(BaseModel):
110     """Метаданные сгенерированного плана инструмента."""

```



```
111     tool_name: str = Field(description="Имя инструмента (kebab  
-case)")  
112     tool_title: str = Field(description="Человекочитаемый заго  
ловок инструмента")  
113     description_ru: str = Field(description="Описание на русск  
ом языке")  
114     probe_type: str = Field(default="powershell", description=  
"Тип зонда: 'powershell', 'collector', 'winapi'")  
115     collector_name: Optional[str] = Field(default=None, descri  
ption="Имя коллектора")  
116     probe_script: Optional[str] = Field(default=None, descript  
ion="PowerShell скрипт опроса")  
117     instructions: str = Field(default="", description="Пошагов  
ый протокол выполнения")  
118  
119  
120 class ScenarioSaveSkillRequest(BaseModel):  
121     """Запрос на сохранение подтвержденного пользователем инст  
румента в качестве навыка."""  
122     tool_name: str = Field(description="Имя навыка (kebab-case  
)")  
123     tool_title: str = Field(description="Человекочитаемый заго  
ловок навыка")  
124     description_ru: str = Field(description="Описание назначен  
ия навыка")  
125     probe_type: str = Field(default="powershell", description=  
"Тип зонда")  
126     collector_name: Optional[str] = Field(default=None, descri  
ption="Имя коллектора")  
127     probe_script: Optional[str] = Field(default=None, descript  
ion="PowerShell скрипт")  
128     instructions: str = Field(default="", description="Протоко  
л выполнения")  
129     command_executed: Optional[str] = Field(default=None, desc  
ription="Фактически выполненная команда")  
130  
131  
132 class ScenarioRemediationActionInfo(BaseModel):
```

```
133     """Метаданные доступного SafeOps-действия по исправлению п
роблемы."""
134     action_id: str = Field(description="Уникальный идентификат
ор действия")
135     action_type: str = Field(default="custom_command", descrip
tion="Тип действия")
136     title: str = Field(description="Человекочитаемое название
действия")
137     description: str = Field(default="", description="Описание
назначения действия")
138     target: str = Field(default="", description="Целевой объек
т (служба, файл, процесс)")
139     risk: str = Field(default="caution", description="Уровень
риска: safe, caution, critical")
140     execution_command: str = Field(default="", description="Ко
манда для выполнения")
141
142
143 class ScenarioExecuteFixRequest(BaseModel):
144     """Запрос на выполнение SafeOps-исправления из мини-чата."
"""
145     action_id: str = Field(description="ID действия")
146     action_type: str = Field(default="custom_command", descrip
tion="Тип действия")
147     title: str = Field(default="", description="Название дейст
вия")
148     description: str = Field(default="", description="Описание
действия")
149     target: str = Field(default="", description="Целевой объек
т")
150     risk: str = Field(default="caution", description="Уровень
риска")
151     execution_command: str = Field(default="", description="Ко
манда выполнения")
152     confirmed_by_user: bool = Field(default=False, description
="Подтверждено ли действие пользователем")
153
154
```

```

155 class ScenarioExecuteFixResponse(BaseModel):
156     """Результат выполнения SafeOps-исправления."""
157     action_id: str = Field(description="ID действия")
158     status: str = Field(description="Статус: 'ok' или 'error'"
159 )
159     executed: bool = Field(default=True, description="Было ли
160 действие фактически выполнено")
160     success: bool = Field(default=True, description="Успешно л
161 и завершилось действие")
161     message: str = Field(description="Сообщение о результате")
162     error_message: Optional[str] = Field(default=None, descrip
163 tion="Текст ошибки при сбое")
164
165 class ScenarioSaveApprovedResponseRequest(BaseModel):
166     """Запрос на сохранение одобренного ответа мини-чата для о
167 бучения модели и RAG."""
167     user_id: str = Field(default="tc_admin", description="Иден
168 тификатор пользователя")
168     query: str = Field(description="Исходный запрос пользовате
169 ля")
169     chat_text: str = Field(description="Текст ответа модели")
170     voice_text: Optional[str] = Field(default="", description=
171 "Текст для озвучки")
171     system_context: Optional[Dict[str, Any]] = Field(default_f
172 actory=dict, description="Контекст системы и выполненных коман
173 д")
173     tags: Optional[List[str]] = Field(default_factory=lambda:
174 ["tc", "diagnostics", "training"], description="Категории и те
175 ги")
175
176 class ScenarioSaveApprovedResponseResponse(BaseModel):
177     """Результат сохранения ответа для обучения модели."""
177     status: str = Field(default="ok", description="Статус: 'ok
178 ' или 'error'")
178     message: str = Field(description="Пользовательское сообщен
179 ие о результате")

```

```
179     file_saved: Optional[str] = Field(default=None, description="Имя сохраненного файла или статус")
180
181
182 class ScenarioChatRequest(BaseModel):
183     """Запрос в мини-чат интеллектуального помощника сценариев"""
184     message: str = Field(description="Текст запроса или вопроса на естественном языке")
185     conversation_id: Optional[str] = Field(default=None, description="Идентификатор диалога")
186     auto_create_skill: bool = Field(default=False, description="Автоматически генерировать навык без подтверждения")
187     keep_context: bool = Field(default=False, description="Сохранять и учитывать контекст предыдущих сообщений диалога")
188     use_rag: bool = Field(default=True, description="Использовать базу знаний RAG для поиска контекста")
189
190
191 class ScenarioChatResponse(BaseModel):
192     """Ответ мини-чата сценариев с результатом выполнения и метаданными навыка."""
193     reply: str = Field(description="Форматированный ответ в Markdown")
194     command_executed: Optional[str] = Field(default=None, description="Выполненная системная команда или скрипт")
195     created_skill: Optional[ScenarioCreatedSkillInfo] = Field(default=None, description="Данные о созданном/обновленном навыке")
196     generated_prompt: Optional[str] = Field(default=None, description="Сформированный промпт, переданный языковой модели")
197     tool_plan: Optional[ScenarioToolPlanInfo] = Field(default=None, description="Метаданные плана инструмента для ручного сохранения")
198     remediation_actions: List[ScenarioRemediationActionInfo] = Field(default_factory=list, description="Список доступных действий по исправлению")
199     raw_data: Optional[Any] = Field(default=None, description="Сырые данные, полученные от модели")
```

```

"Сырые структурированные данные")
200     agents_used: List[Dict[str, Any]] = Field(default_factory=
list, description="Список задействованных агентов и модулей")
201     knowledge_used: List[Dict[str, Any]] = Field(default_facto
ry=list, description="Список задействованных источников знаний
и артефактов")
202     status: str = Field(default="ok", description="Статус обра
ботки: 'ok', 'warn', 'error'")
203
204
205 # -----
-----
206 # Список доступных сценариев
207 # -----
-----
208
209 AVAILABLE_SCENARIOS: List[Dict[str, Any]] = [
210     {
211         "id": "safe_update_guard",
212         "title": "Безопасная подготовка к обновлению Windows (
Safe Update Guard)",
213         "description": "Автоматическая подготовка системы к кр
упным обновлениям ОС (Windows 11 26H2 / Feature Updates): инсп
екция дискового пространства, проверка и активация среды WinRE
, настройка хранилища теневых копий VSS, создание точки восста
новления и регистрация PRE_UPDATE контрольной точки.",
214         "category": "security",
215         "icon": "☑",
216         "recommended": True,
217         "estimated_duration_sec": 8,
218     },
219     {
220         "id": "quick_check",
221         "title": "Быстрая проверка (Smoke Test)",
222         "description": "Экспресс-проверка целостности окружени
я, доступности ядра FastAPI, AI-провайдеров, свободного места
и ключевых логов. Рекомендуется запускать сразу после установк
и или перезапуска.",

```

```
223         "category": "quick",
224         "icon": "⚡",
225         "recommended": True,
226         "estimated_duration_sec": 2,
227     },
228     {
229         "id": "log_audit",
230         "title": "Запустить аудит логов",
231         "description": "Комплексный анализ файлов логов (info.log, errors.log, app.log) и журналов событий Windows. Поиск критических ошибок, подсчет аномалий и генерация рекомендаций.",
232         "category": "logs",
233         "icon": "📄",
234         "recommended": False,
235         "estimated_duration_sec": 3,
236     },
237     {
238         "id": "system_inspector",
239         "title": "Диагностика системы и оборудования",
240         "description": "Срез телеметрии: загрузка ядер процессора, температура, оперативная память, свободное место на дисках и ресурсоёмкие фоновые процессы.",
241         "category": "diagnostics",
242         "icon": "🔍",
243         "recommended": False,
244         "estimated_duration_sec": 3,
245     },
246     {
247         "id": "windows_admin",
248         "title": "Аудит автозапуска и служб Windows",
249         "description": "Проверка записей реестра Run/RunOnce, папки автозагрузки, критических системных служб Windows и прав текущего процесса.",
250         "category": "security",
251         "icon": "🔒",
252         "recommended": False,
253         "estimated_duration_sec": 4,
254     },
```

```
255 |     {
256 |         "id": "network_test",
257 |         "title": "Проверка сетевой доступности и портов",
258 |         "description": "Тестирование локального шлюза, разреше
ния DNS, открытых серверных портов (8000 и др.) и доступности
внешних API.",
259 |         "category": "network",
260 |         "icon": "🌐 ",
261 |         "recommended": False,
262 |         "estimated_duration_sec": 4,
263 |     },
264 |     {
265 |         "id": "ai_providers_check",
266 |         "title": "Проверка статуса AI-провайдеров",
267 |         "description": "Опрос сконфигурированных AI-моделей (G
oogle Gemini, Antigravity/AGY, Ollama, Foundry) и замер времен
и отклика инференса.",
268 |         "category": "ai",
269 |         "icon": "🧠 ",
270 |         "recommended": False,
271 |         "estimated_duration_sec": 5,
272 |     },
273 |     {
274 |         "id": "disk_benchmark_audit",
275 |         "title": "Экспресс-тест скорости всех накопителей (Dis
kSpd)",
276 |         "description": "Быстрое измерение скорости последовате
льного и случайного чтения/записи на всех дисках с фиксацией в
телеметрии.",
277 |         "category": "diagnostics",
278 |         "icon": "💾 ",
279 |         "recommended": True,
280 |         "estimated_duration_sec": 3,
281 |     },
282 | ]
283 |
284 |
285 |
```

```

286 # -----
287 # Scenario Runners
288 # -----
289
290 async def _run_quick_check_scenario() -> ScenarioRunResult:
291     """Выполняет экспресс-проверку окружения (Smoke Test)."""
292     start_time = time.perf_counter()
293     started_at = datetime.now(timezone.utc).isoformat()
294     steps: List[ScenarioStepResult] = []
295
296     # Шаг 1: Проверка среды исполнения Python
297     step1_start = time.perf_counter()
298     try:
299         py_ver = f"{sys.version_info.major}.{sys.version_info.
300 minor}.{sys.version_info.micro}"
301         os_info = f"{platform.system()} {platform.release()} (
302 {platform.machine()})"
303         step1_dur = (time.perf_counter() - step1_start) * 1000
304         steps.append(ScenarioStepResult(
305             name="Среда исполнения Python и ОС",
306             status="ok" if sys.version_info >= (3, 10) else "w
307 arn",
308             duration_ms=round(step1_dur, 2),
309             details=f"Python {py_ver}, ОС: {os_info}",
310             recommendation="Рекомендуется Python 3.12+" if sys
311 .version_info < (3, 12) else None,
312             data={"python_version": py_ver, "os": os_info}
313         ))
314     except Exception as e:
315         steps.append(ScenarioStepResult(
316             name="Среда исполнения Python и ОС",
317             status="error",
318             duration_ms=0.0,
319             details=f"Ошибка проверки: {e}",
320             recommendation="Проверьте системные переменные PAT
321 Н."

```



```

317         ))
318
319     # Шаг 2: Проверка конфигурационных файлов
320     step2_start = time.perf_counter()
321     try:
322         cfg_tc_path = __root__ / "config_tc.json"
323         cfg_main_path = __root__ / "config.json"
324         env_path = __root__ / ".env"
325         details_list = []
326         if cfg_tc_path.exists():
327             details_list.append("config_tc.json [OK]")
328         if cfg_main_path.exists():
329             details_list.append("config.json [OK]")
330         if env_path.exists():
331             details_list.append(".env [OK]")
332         else:
333             details_list.append(".env [Отсутствует]")
334
335         step2_dur = (time.perf_counter() - step2_start) * 1000
336         steps.append(ScenarioStepResult(
337             name="Конфигурационные файлы проекта",
338             status="ok" if cfg_tc_path.exists() or cfg_main_pa
339             th.exists() else "warn",
340             duration_ms=round(step2_dur, 2),
341             details=" | ".join(details_list),
342             recommendation="Создайте файл .env на основе .env.
343             example при необходимости API-ключей." if not env_path.exists(
344             ) else None,
345             data={"config_tc": cfg_tc_path.exists(), "config":
346             cfg_main_path.exists(), "env": env_path.exists()})
347         ))
348     except Exception as e:
349         steps.append(ScenarioStepResult(
350             name="Конфигурационные файлы проекта",
351             status="error",
352             duration_ms=0.0,
353             details=f"Ошибка чтения конфигурации: {e}"
354         ))

```

```

351
352     # Шаг 3: Проверка дискового пространства
353     step3_start = time.perf_counter()
354     try:
355         total, used, free = shutil.disk_usage(__root__)
356         free_gb = free / (1024 ** 3)
357         total_gb = total / (1024 ** 3)
358         step3_dur = (time.perf_counter() - step3_start) * 1000
359         status_disk = "ok" if free_gb >= 2.0 else ("warn" if f
ree_gb >= 0.5 else "error")
360         steps.append(ScenarioStepResult(
361             name="Дисковое пространство рабочей директории",
362             status=status_disk,
363             duration_ms=round(step3_dur, 2),
364             details=f"Свободно {free_gb:.2f} ГБ из {total_gb:.
2f} ГБ",
365             recommendation="Освободите место на диске для корп
ектной записи логов и кэша." if status_disk != "ok" else None,
366             data={"free_gb": round(free_gb, 2), "total_gb": ro
und(total_gb, 2)})
367         ))
368     except Exception as e:
369         steps.append(ScenarioStepResult(
370             name="Дисковое пространство рабочей директории",
371             status="warn",
372             duration_ms=0.0,
373             details=f"Не удалось получить дисковое пространств
о: {e}")
374         ))
375
376     # Шаг 4: Проверка статуса логов ядра
377     step4_start = time.perf_counter()
378     try:
379         logs_dir = __root__ / "logs"
380         error_count = 0
381         warn_count = 0
382         if logs_dir.exists():
383             for lf in logs_dir.glob("*.log"):

```

```

384         try:
385             with open(lf, "r", encoding="utf-8", error
s="ignore") as f:
386                 lines = f.readlines()[-200:]
387                 for line in lines:
388                     if "ERROR" in line or "[CRITICAL]"
in line:
389                         error_count += 1
390                     elif "WARN" in line or "[WARNING]"
in line:
391                         warn_count += 1
392             except Exception:
393                 pass
394             step4_dur = (time.perf_counter() - step4_start) * 1000
395             log_status = "ok" if error_count == 0 else ("warn" if
error_count < 10 else "error")
396             steps.append(ScenarioStepResult(
397                 name="Экспресс-анализ журналов логов",
398                 status=log_status,
399                 duration_ms=round(step4_dur, 2),
400                 details=f"Обнаружено критических ошибок: {error_co
unt}, предупреждений: {warn_count}",
401                 recommendation="Запустите сценарий 'Аудит логов' д
ля подробной диагностики выявленных ошибок." if error_count >
0 else None,
402                 data={"error_count": error_count, "warn_count": wa
rn_count}
403             ))
404         except Exception as e:
405             steps.append(ScenarioStepResult(
406                 name="Экспресс-анализ журналов логов",
407                 status="warn",
408                 duration_ms=0.0,
409                 details=f"Не удалось прочитать директорию logs: {e
}"
410             ))
411
412     # Шаг 5: Проверка активных микроприложений (/apps)

```

```

413     step5_start = time.perf_counter()
414     try:
415         from src.api.routers.core.router_admin import get_apps
         _status
416         apps_status = get_apps_status()
417         enabled_apps = [app_id for app_id, app in apps_status.
get("apps", {}).items() if app.get("enabled")]
418         step5_dur = (time.perf_counter() - step5_start) * 1000
419         steps.append(ScenarioStepResult(
420             name="Конфигурация микросервисов Test Computer",
421             status="ok",
422             duration_ms=round(step5_dur, 2),
423             details=f"Активных приложений: {len(enabled_apps)}
({', '.join(enabled_apps[:6])}{', ...' if len(enabled_apps) > 6
else ''})",
424             data={"enabled_count": len(enabled_apps), "active_
apps": enabled_apps}
425         ))
426     except Exception as e:
427         steps.append(ScenarioStepResult(
428             name="Конфигурация микросервисов Test Computer",
429             status="warn",
430             duration_ms=0.0,
431             details=f"Ошибка чтения статуса приложений: {e}"
432         ))
433
434     total_dur = (time.perf_counter() - start_time) * 1000
435     failed_cnt = sum(1 for s in steps if s.status == "error")
436     warn_cnt = sum(1 for s in steps if s.status == "warn")
437     passed_cnt = sum(1 for s in steps if s.status == "ok")
438     overall_status = "error" if failed_cnt > 0 else ("warn" if
warn_cnt > 0 else "ok")
439
440     if overall_status == "ok":
441         summary = "Все базовые компоненты системы и среды функ
ционируют штатно. Рекомендуется продолжить работу."
442     elif overall_status == "warn":
443         summary = f"Система запущена с {warn_cnt} не критически

```

```

ми замечаниями. Ознакомьтесь с рекомендациями по шагам."
444     else:
445         summary = f"Обнаружены критические сбои ({failed_cnt}
ошибок). Требуется устранение неполадок перед началом работы."
446
447     return ScenarioRunResult(
448         scenario_id="quick_check",
449         title="Быстрая проверка (Smoke Test)",
450         status=overall_status,
451         started_at=started_at,
452         duration_ms=round(total_dur, 2),
453         total_steps=len(steps),
454         passed_steps=passed_cnt,
455         warn_steps=warn_cnt,
456         failed_steps=failed_cnt,
457         steps=steps,
458         summary=summary,
459     )
460
461
462 async def _run_log_audit_scenario() -> ScenarioRunResult:
463     """Выполняет аудит лог-файлов и системных журналов."""
464     start_time = time.perf_counter()
465     started_at = datetime.now(timezone.utc).isoformat()
466     steps: List[ScenarioStepResult] = []
467
468     # Шаг 1: Инвентаризация файлов журналов
469     s1_start = time.perf_counter()
470     logs_dir = __root__ / "logs"
471     found_files = []
472     total_log_size_mb = 0.0
473     if logs_dir.exists():
474         for f in logs_dir.glob("*.log"):
475             sz = f.stat().st_size / (1024 * 1024)
476             total_log_size_mb += sz
477             found_files.append({"name": f.name, "size_mb": rou
nd(sz, 2)})
478

```

```

479     s1_dur = (time.perf_counter() - s1_start) * 1000
480     steps.append(ScenarioStepResult(
481         name="Инвентаризация файлов журналов в /logs",
482         status="ok" if found_files else "warn",
483         duration_ms=round(s1_dur, 2),
484         details=f"Найдено файлов: {len(found_files)}, общий ра
485 змер: {total_log_size_mb:.2f} МБ",
486         recommendation="При отсутствии лог-файлов убедитесь, ч
487 то логирование активно в config.json." if not found_files else
488         None,
489         data={"files": found_files, "total_size_mb": round(tot
490 al_log_size_mb, 2)}
491     ))
492
493     # Шаг 2: Анализ ошибок приложения
494     s2_start = time.perf_counter()
495     recent_errors: List[str] = []
496     recent_warnings: List[str] = []
497     if logs_dir.exists():
498         for lf in logs_dir.glob("*.log"):
499             try:
500                 with open(lf, "r", encoding="utf-8", errors="i
501 gnore") as f:
502                     lines = f.readlines()[-300:]
503                     for line in lines:
504                         clean_line = line.strip()
505                         if "ERROR" in clean_line or "[CRITICAL
506 ]" in clean_line:
507                             if len(recent_errors) < 10:
508                                 recent_errors.append(f"[{lf.na
509 me}] {clean_line[:120]}")
510                         elif "WARN" in clean_line:
511                             if len(recent_warnings) < 10:
512                                 recent_warnings.append(f"[{lf.
513 name}] {clean_line[:120]}")
514                     except Exception:
515                         pass
516
517

```

```

509     s2_dur = (time.perf_counter() - s2_start) * 1000
510     err_status = "ok" if len(recent_errors) == 0 else ("warn"
if len(recent_errors) < 5 else "error")
511     steps.append(ScenarioStepResult(
512         name="Сканирование свежих записей на ошибки и сбои",
513         status=err_status,
514         duration_ms=round(s2_dur, 2),
515         details=f"Свежих ошибок: {len(recent_errors)}, предуп
еждений: {len(recent_warnings)}",
516         recommendation="Изучите список ошибок во вкладке 'Анал
изатор логов'." if recent_errors else None,
517         data={"errors_sample": recent_errors, "warnings_sample
": recent_warnings}
518     ))
519
520     # Шаг 3: Проверка Windows Event Logs (если доступно)
521     s3_start = time.perf_counter()
522     win_events_checked = False
523     win_event_errors = 0
524     if platform.system() == "Windows":
525         try:
526             from apps.windows.core.audits.eventlog_collector i
mport EventLogCollector
527             collector = EventLogCollector()
528             events = collector.fetch_events(channel="Applicati
on", limit=30, level="Error")
529             win_event_errors = len(events)
530             win_events_checked = True
531         except Exception as e:
532             logger.debug(f"EventLogCollector skipped: {e}")
533
534     s3_dur = (time.perf_counter() - s3_start) * 1000
535     if win_events_checked:
536         we_status = "ok" if win_event_errors == 0 else "warn"
537         steps.append(ScenarioStepResult(
538             name="Журнал событий Windows (Application EventLog
)",
539             status=we_status,

```

```
540         duration_ms=round(s3_dur, 2),
541         details=f"Ошибок за последнее время: {win_event_er
rors}",
542         recommendation="Проверьте системные службы Windows
." if win_event_errors > 0 else None,
543         data={"win_error_count": win_event_errors}
544     ))
545     else:
546         steps.append(ScenarioStepResult(
547             name="Журнал событий Windows (Application EventLog
)",
548             status="ok",
549             duration_ms=round(s3_dur, 2),
550             details="Сбор журналов Windows EventLog пропущен и
ли запущен на не-Windows хосте."
551         ))
552
553     total_dur = (time.perf_counter() - start_time) * 1000
554     failed_cnt = sum(1 for s in steps if s.status == "error")
555     warn_cnt = sum(1 for s in steps if s.status == "warn")
556     passed_cnt = sum(1 for s in steps if s.status == "ok")
557     overall_status = "error" if failed_cnt > 0 else ("warn" if
warn_cnt > 0 else "ok")
558
559     if overall_status == "ok":
560         summary = "Аудит логов завершен успешно: критических с
боев и аномалий в журналах не выявлено."
561     elif overall_status == "warn":
562         summary = f"Аудит логов выявил {warn_cnt} предупрежден
ий. Рекомендуется периодический просмотр свежих записей."
563     else:
564         summary = "В журналах логов обнаружены критические оши
бки. Требуется детальный анализ причин сбоя."
565
566     return ScenarioRunResult(
567         scenario_id="log_audit",
568         title="Аудит логов",
569         status=overall_status,
```



```

570         started_at=started_at,
571         duration_ms=round(total_dur, 2),
572         total_steps=len(steps),
573         passed_steps=passed_cnt,
574         warn_steps=warn_cnt,
575         failed_steps=failed_cnt,
576         steps=steps,
577         summary=summary,
578     )
579
580
581 async def _run_system_inspector_scenario() -> ScenarioRunResult:
582     """Выполняет аудит производительности и ресурсов системы."""
583
584     start_time = time.perf_counter()
585     started_at = datetime.now(timezone.utc).isoformat()
586     steps: List[ScenarioStepResult] = []
587
588     # Шаг 1: Телеметрия процессора и памяти
589     s1_start = time.perf_counter()
590     try:
591         import psutil
592         cpu_pct = psutil.cpu_percent(interval=0.2)
593         mem = psutil.virtual_memory()
594         s1_dur = (time.perf_counter() - s1_start) * 1000
595         status_res = "ok" if (cpu_pct < 85 and mem.percent < 9
596         0) else "warn"
597         steps.append(ScenarioStepResult(
598             name="Нагрузка CPU и оперативной памяти",
599             status=status_res,
600             duration_ms=round(s1_dur, 2),
601             details=f"CPU: {cpu_pct:.1f}%, RAM: {mem.percent:.1f}% ({mem.used / (1024**3):.1f} / {mem.total / (1024**3):.1f} ГБ)",
602             recommendation="Высокая утилизация ресурсов. Закройте неиспользуемые фоновые задачи." if status_res != "ok" else None,

```

```

601         data={"cpu_pct": cpu_pct, "ram_pct": mem.percent,
"ram_used_gb": round(mem.used / (1024*3), 2)}
602     ))
603     except Exception as e:
604         steps.append(ScenarioStepResult(
605             name="Нагрузка CPU и оперативной памяти",
606             status="warn",
607             duration_ms=0.0,
608             details=f"Ошибка сбора метрик: {e}"
609         ))
610
611     # Шаг 2: Ресурсоёмкие процессы
612     s2_start = time.perf_counter()
613     try:
614         import psutil
615         top_procs = []
616         for p in sorted(psutil.process_iter(['pid', 'name', 'c
pu_percent', 'memory_percent']), key=lambda x: x.info.get('cpu
_percent') or 0, reverse=True)[:5]:
617             top_procs.append(f"{p.info.get('name')} (PID {p.in
fo.get('pid')})")
618         s2_dur = (time.perf_counter() - s2_start) * 1000
619         steps.append(ScenarioStepResult(
620             name="Анализ фоновых процессов",
621             status="ok",
622             duration_ms=round(s2_dur, 2),
623             details=f"Top-5 процессов: {' '.join(top_procs)}"
624         ),
625         data={"top_processes": top_procs}
626     ))
627     except Exception as e:
628         steps.append(ScenarioStepResult(
629             name="Анализ фоновых процессов",
630             status="ok",
631             duration_ms=0.0,
632             details=f"Сбор списка процессов завершен ({e})"
633         ))

```

```

634     total_dur = (time.perf_counter() - start_time) * 1000
635     failed_cnt = sum(1 for s in steps if s.status == "error")
636     warn_cnt = sum(1 for s in steps if s.status == "warn")
637     passed_cnt = sum(1 for s in steps if s.status == "ok")
638     overall_status = "error" if failed_cnt > 0 else ("warn" if
        warn_cnt > 0 else "ok")
639
640     summary = "Оборудование и системные ресурсы находятся в но
        рмальном диапазоне нагрузок." if overall_status == "ok" else "
        Обнаружена повышенная нагрузка на оборудование."
641
642     return ScenarioRunResult(
643         scenario_id="system_inspector",
644         title="Диагностика системы и оборудования",
645         status=overall_status,
646         started_at=started_at,
647         duration_ms=round(total_dur, 2),
648         total_steps=len(steps),
649         passed_steps=passed_cnt,
650         warn_steps=warn_cnt,
651         failed_steps=failed_cnt,
652         steps=steps,
653         summary=summary,
654     )
655
656
657 async def _run_windows_admin_scenario() -> ScenarioRunResult:
658     """Выполняет аудит автозапуска и системного окружения Wind
        ows."""
659     start_time = time.perf_counter()
660     started_at = datetime.now(timezone.utc).isoformat()
661     steps: List[ScenarioStepResult] = []
662
663     # Шаг 1: Проверка привилегий администратора
664     s1_start = time.perf_counter()
665     is_admin = False
666     if platform.system() == "Windows":
667         try:

```

```

668         import ctypes
669         is_admin = bool(ctypes.windll.shell32.IsUserAnAdmin()
670         n())
671     except Exception:
672         pass
673     s1_dur = (time.perf_counter() - s1_start) * 1000
674     steps.append(ScenarioStepResult(
675         name="Привилегии процесса (Administrator / Elevated)",
676         status="ok" if is_admin else "warn",
677         duration_ms=round(s1_dur, 2),
678         details="Запущено с повышенными правами администратора"
679         " if is_admin else "Запущено в стандартном пользовательском ко
680         нтексте",
681         recommendation="Для полного контроля служб Windows рек
682         омендуется запускать терминал от имени Администратора." if not
683         is_admin else None,
684         data={"is_admin": is_admin}
685     ))
686
687     # Шаг 2: Проверка автозапуска Windows
688     s2_start = time.perf_counter()
689     startup_items_count = 0
690     if platform.system() == "Windows":
691         try:
692             import winreg
693             keys = [
694                 (winreg.HKEY_CURRENT_USER, r"Software\Microsof
695                 t\Windows\CurrentVersion\Run"),
696                 (winreg.HKEY_LOCAL_MACHINE, r"Software\Microso
697                 ft\Windows\CurrentVersion\Run")
698             ]
699             for root_k, sub_k in keys:
700                 try:
701                     with winreg.OpenKey(root_k, sub_k) as k:
702                         startup_items_count += winreg.QueryInf
703                         oKey(k)[1]
704                 except Exception:
705                     pass

```

```
698         except Exception:
699             pass
700     s2_dur = (time.perf_counter() - s2_start) * 1000
701     steps.append(ScenarioStepResult(
702         name="Аудит записей автозагрузки Windows (Registry Run
703     )",
704         status="ok" if startup_items_count < 30 else "warn",
705         duration_ms=round(s2_dur, 2),
706         details=f"Обнаружено программ в автозапуске: {startup_
707 items_count}",
708         recommendation="Большое число программ автозапуска мож
709 ет замедлять загрузку ОС." if startup_items_count >= 30 else N
710 one,
711         data={"startup_count": startup_items_count}
712     ))
713
714     total_dur = (time.perf_counter() - start_time) * 1000
715     failed_cnt = sum(1 for s in steps if s.status == "error")
716     warn_cnt = sum(1 for s in steps if s.status == "warn")
717     passed_cnt = sum(1 for s in steps if s.status == "ok")
718     overall_status = "error" if failed_cnt > 0 else ("warn" if
719 warn_cnt > 0 else "ok")
720
721     summary = "Аудит системного окружения Windows выполнен усп
722 ешно."
723
724     return ScenarioRunResult(
725         scenario_id="windows_admin",
726         title="Аудит автозапуска и служб Windows",
727         status=overall_status,
728         started_at=started_at,
729         duration_ms=round(total_dur, 2),
730         total_steps=len(steps),
731         passed_steps=passed_cnt,
732         warn_steps=warn_cnt,
733         failed_steps=failed_cnt,
734         steps=steps,
735         summary=summary,
```

```

730     )
731
732
733 async def _run_network_test_scenario() -> ScenarioRunResult:
734     """Выполняет проверку сетевых портов и подключений."""
735     start_time = time.perf_counter()
736     started_at = datetime.now(timezone.utc).isoformat()
737     steps: List[ScenarioStepResult] = []
738
739     # Шаг 1: Проверка локального сокета 127.0.0.1
740     s1_start = time.perf_counter()
741     import socket
742     port = int(getattr(server_cfg, "port", 8000))
743     sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
744     sock.settimeout(1.0)
745     res_code = sock.connect_ex(('127.0.0.1', port))
746     sock.close()
747     s1_dur = (time.perf_counter() - s1_start) * 1000
748     is_listening = (res_code == 0)
749
750     steps.append(ScenarioStepResult(
751         name=f"Локальный серверный порт ({port})",
752         status="ok" if is_listening else "warn",
753         duration_ms=round(s1_dur, 2),
754         details=f"Порт {port} открыт и принимает соединения" if
755         is_listening else f"Порт {port} не отвечает локально",
756         recommendation="Убедитесь, что веб-сервер запущен." if
757         not is_listening else None,
758         data={"port": port, "is_listening": is_listening}
759     ))
760
761     total_dur = (time.perf_counter() - start_time) * 1000
762     failed_cnt = sum(1 for s in steps if s.status == "error")
763     warn_cnt = sum(1 for s in steps if s.status == "warn")
764     passed_cnt = sum(1 for s in steps if s.status == "ok")
765     overall_status = "error" if failed_cnt > 0 else ("warn" if
766     warn_cnt > 0 else "ok")

```

```
765     summary = "Сетевые подключения проверены, локальный порт а
| ктивен."
766
767     return ScenarioRunResult(
768         scenario_id="network_test",
769         title="Проверка сетевой доступности и портов",
770         status=overall_status,
771         started_at=started_at,
772         duration_ms=round(total_dur, 2),
773         total_steps=len(steps),
774         passed_steps=passed_cnt,
775         warn_steps=warn_cnt,
776         failed_steps=failed_cnt,
777         steps=steps,
778         summary=summary,
779     )
780
781
782 async def _run_ai_providers_scenario() -> ScenarioRunResult:
783     """Выполняет проверку доступных AI-провайдеров."""
784     start_time = time.perf_counter()
785     started_at = datetime.now(timezone.utc).isoformat()
786     steps: List[ScenarioStepResult] = []
787
788     # Шаг 1: Проверка конфигурации AI
789     s1_start = time.perf_counter()
790     active_providers = []
791     gemini_key = os.getenv("GEMINI_API_KEY_1") or os.getenv("G
| EMINI_API_KEY")
792     if gemini_key:
793         active_providers.append("Google Gemini (API Key настро
| ен)")
794
795     # Проверка AGY
796     active_providers.append("AGY / Local Runner")
797
798     s1_dur = (time.perf_counter() - s1_start) * 1000
799     steps.append(ScenarioStepResult(
```

```

800         name="Настроенные AI провайдеры",
801         status="ok" if active_providers else "warn",
802         duration_ms=round(s1_dur, 2),
803         details=" | ".join(active_providers) if active_provide
rs else "AI-провайдеры не обнаружены",
804         recommendation="Укажите GEMINI_API_KEY_1 в .env для до
ступа к облачным моделям Gemini." if not gemini_key else None,
805         data={"providers": active_providers}
806     ))
807
808     total_dur = (time.perf_counter() - start_time) * 1000
809     failed_cnt = sum(1 for s in steps if s.status == "error")
810     warn_cnt = sum(1 for s in steps if s.status == "warn")
811     passed_cnt = sum(1 for s in steps if s.status == "ok")
812     overall_status = "error" if failed_cnt > 0 else ("warn" if
warn_cnt > 0 else "ok")
813
814     summary = "Конфигурация AI-провайдеров готова к инференсу.
"
815
816     return ScenarioRunResult(
817         scenario_id="ai_providers_check",
818         title="Проверка статуса AI-провайдеров",
819         status=overall_status,
820         started_at=started_at,
821         duration_ms=round(total_dur, 2),
822         total_steps=len(steps),
823         passed_steps=passed_cnt,
824         warn_steps=warn_cnt,
825         failed_steps=failed_cnt,
826         steps=steps,
827         summary=summary,
828     )
829
830
831 async def _run_safe_update_guard_scenario() -> ScenarioRunResu
lt:
832     """Выполняет комплексную безопасную подготовку к установке

```



```

    крупных обновлений Windows (Feature Updates / 26H2)."""
833     start_time = time.perf_counter()
834     started_at = datetime.now(timezone.utc).isoformat()
835     steps: List[ScenarioStepResult] = []
836
837     # Шаг 1: Проверка свободного места на диске C: (минимум 25
    ГБ для обновления 26H2)
838     s1_start = time.perf_counter()
839     try:
840         total, used, free = shutil.disk_usage("C:\\")
841         free_gb = free / (1024 ** 3)
842         total_gb = total / (1024 ** 3)
843         s1_dur = (time.perf_counter() - s1_start) * 1000
844         status_disk = "ok" if free_gb >= 25.0 else ("warn" if
    free_gb >= 15.0 else "error")
845         steps.append(ScenarioStepResult(
846             name="Проверка свободного места для установки обно
    вления (Диск C:)",
847             status=status_disk,
848             duration_ms=round(s1_dur, 2),
849             details=f"Свободно {free_gb:.2f} ГБ из {total_gb:.
    2f} ГБ (требуется от 25 ГБ для Windows 11 26H2)",
850             recommendation="Освободите место на диске C: перед
    запуском установки крупного обновления." if status_disk != "o
    k" else None,
851             data={"free_gb": round(free_gb, 2), "required_gb":
    25.0}
852         ))
853     except Exception as e:
854         steps.append(ScenarioStepResult(
855             name="Проверка свободного места на диске C:",
856             status="error",
857             duration_ms=0.0,
858             details=f"Ошибка проверки диска: {e}"
859         ))
860
861     # Шаг 2: Диагностика среды восстановления Windows RE (WinR
    E)

```

```
862     s2_start = time.perf_counter()
863     try:
864         from apps.windows.modules.system_checkpoints.core.winr
e_manager import WinREManager
865         winre_mgr = WinREManager()
866         winre_stat = await asyncio.to_thread(winre_mgr.get_sta
tus)
867         s2_dur = (time.perf_counter() - s2_start) * 1000
868         winre_ok = winre_stat.enabled
869         steps.append(ScenarioStepResult(
870             name="Аудит среды восстановления Windows RE (reage
ntc)",
871             status="ok" if winre_ok else "warn",
872             duration_ms=round(s2_dur, 2),
873             details=f"Среда WinRE: {'Включена и активна' if wi
nre_ok else 'Отключена / Недоступна'}. Путь: {winre_stat.locat
ion or 'Стандартный'}",
874             recommendation="Включите WinRE командами `reagentc
/enable` в командной строке от Администратора." if not winre_
ok else None,
875             data=winre_stat.to_dict()
876         ))
877     except Exception as e:
878         steps.append(ScenarioStepResult(
879             name="Аудит среды восстановления Windows RE",
880             status="warn",
881             duration_ms=0.0,
882             details=f"Ошибка опроса WinRE: {e}"
883         ))
884
885     # Шаг 3: Проверка и создание точки восстановления Windows
(VSS)
886     s3_start = time.perf_counter()
887     try:
888         from apps.windows.core.system_restore import WindowsSy
stemRestoreManager
889         sr_mgr = WindowsSystemRestoreManager()
890         prot_stat = sr_mgr.check_protection_status()
```

```

891 |
892 |         if prot_stat.get("system_protection_enabled", False):
893 |             res_sr = await asyncio.to_thread(
894 |                 sr_mgr.create_restore_point,
895 |                 description="Подготовка к обновлению Windows 1
896 |                 1 26H2",
897 |                 restore_point_type="WINDOWS_UPDATE"
898 |             )
899 |             s3_dur = (time.perf_counter() - s3_start) * 1000
900 |             sr_success = res_sr.get("success", False)
901 |             steps.append(ScenarioStepResult(
902 |                 name="Создание нативной точки восстановления W
903 |                 indows (VSS)",
904 |                 status="ok" if sr_success else "warn",
905 |                 duration_ms=round(s3_dur, 2),
906 |                 details=res_sr.get("message", "Точка восстанов
907 |                 ления обработана."),
908 |                 recommendation=res_sr.get("error") if not sr_s
909 |                 uccess else None,
910 |                 data=res_sr
911 |             ))
912 |         else:
913 |             s3_dur = (time.perf_counter() - s3_start) * 1000
914 |             steps.append(ScenarioStepResult(
915 |                 name="Создание нативной точки восстановления W
916 |                 indows (VSS)",
917 |                 status="warn",
918 |                 duration_ms=round(s3_dur, 2),
919 |                 details="Защита системы (System Protection) от
920 |                 ключена для диска C:.",
921 |                 recommendation="Включите защиту системы в свой
922 |                 ствах 'Защита системы' (SystemPropertiesProtection).",
923 |                 data=prot_stat
924 |             ))
925 |     except Exception as e:
926 |         steps.append(ScenarioStepResult(
927 |             name="Создание нативной точки восстановления Windo
928 |             ws (VSS)",

```

```

921         status="error",
922         duration_ms=0.0,
923         details=f"Исключение при вызове VSS: {e}"
924     ))
925
926     # Шаг 4: Регистрация контрольной точки PRE_UPDATE в коорди
наторе
927     s4_start = time.perf_counter()
928     try:
929         from apps.windows.modules.system_checkpoints.core.chec
kpoint_coordinator import CheckpointCoordinator
930         from apps.windows.modules.system_checkpoints.models im
port CheckpointCreateRequest, CheckpointType
931
932         coord = CheckpointCoordinator()
933         chk_req = CheckpointCreateRequest(
934             checkpoint_type=CheckpointType.PRE_UPDATE,
935             title="Перед обновлением Windows 11 26H2",
936             description="Автоматическая контрольная точка пере
д установкой накопительного/крупного обновления 26H2",
937             create_restore_point=False,
938             create_wim_image=False,
939         )
940         res_chk = await asyncio.to_thread(coord.create_checkpo
int, chk_req)
941         s4_dur = (time.perf_counter() - s4_start) * 1000
942         steps.append(ScenarioStepResult(
943             name="Регистрация контрольной точки в каталоге sys
tem_checkpoints",
944             status="ok" if res_chk.get("success", False) else
"warn",
945             duration_ms=round(s4_dur, 2),
946             details=f"Контрольная точка зарегистрирована ID: {
res_chk.get('checkpoint_id', 'N/A')}",
947             data=res_chk
948         ))
949     except Exception as e:
950         steps.append(ScenarioStepResult(

```

```

951         name="Регистрация контрольной точки в каталоге sys
tem_checkpoints",
952         status="warn",
953         duration_ms=0.0,
954         details=f"Не удалось зарегистрировать запись в кат
алоге: {e}"
955     ))
956
957     total_dur = (time.perf_counter() - start_time) * 1000
958     failed_cnt = sum(1 for s in steps if s.status == "error")
959     warn_cnt = sum(1 for s in steps if s.status == "warn")
960     passed_cnt = sum(1 for s in steps if s.status == "ok")
961     overall_status = "error" if failed_cnt > 0 else ("warn" if
warn_cnt > 0 else "ok")
962
963     if overall_status == "ok":
964         summary = "Подготовка к обновлению Windows завершена у
спешно: создана точка восстановления, проверена WinRE и свобод
ное место на диске."
965     elif overall_status == "warn":
966         summary = f"Подготовка к обновлению завершена с {warn_
cnt} предупреждениями. Проверьте рекомендации перед переходом
к установке."
967     else:
968         summary = "Обнаружены критические препятствия перед об
новлением. Не рекомендуем запускать обновление до устранения о
шибок."
969
970     return ScenarioRunResult(
971         scenario_id="safe_update_guard",
972         title="Безопасная подготовка к обновлению Windows (Saf
e Update Guard)",
973         status=overall_status,
974         started_at=started_at,
975         duration_ms=round(total_dur, 2),
976         total_steps=len(steps),
977         passed_steps=passed_cnt,
978         warn_steps=warn_cnt,

```

```

979         failed_steps=failed_cnt,
980         steps=steps,
981         summary=summary,
982     )
983
984
985 # -----
986 # AI Assistant & Dynamic Skill Generator for Scenarios
987 # -----
988
989 def _lookup_vendor_by_instance(instance_id: str) -> str:
990     """Определяет наименование вендора устройства по USB/HID I
991     nstanceId."""
992     upper_id = (instance_id or "").upper()
993     known_vendors = {
994         "VID_046D": "Logitech",
995         "VID_045E": "Microsoft",
996         "VID_1532": "Razer",
997         "VID_1038": "SteelSeries",
998         "VID_1B1C": "Corsair",
999         "VID_093A": "PixArt / Genius",
1000         "VID_258A": "SinoWealth / Redragon",
1001         "VID_04F2": "Chicony",
1002         "VID_514C": "A4Tech / Bloody / Gaming Mouse",
1003         "VID_0461": "Primax Electronics (Dell/HP OEM)",
1004         "VID_17EF": "Lenovo",
1005         "VID_03F0": "HP Inc.",
1006         "VID_0781": "SanDisk",
1007         "VID_0951": "Kingston",
1008         "VID_8087": "Intel",
1009         "VID_0D8C": "C-Media USB Audio",
1010         "VID_152D": "JMicron (USB Storage / UAS)",
1011         "VID_174C": "ASMedia (USB-SATA / ASMT)",
1012         "VID_413C": "Dell",
1013         "VID_0BDA": "Realtek",
1014         "VID_058F": "Alcor Micro",

```

```

1014         "VID_13FE": "Phison Electronics",
1015         "VID_05E3": "Genesys Logic Hub",
1016         "VID_0424": "Microchip / SMSC Hub",
1017     }
1018     for vid, vendor in known_vendors.items():
1019         if vid in upper_id:
1020             return vendor
1021     if "VEN_SANDISK" in upper_id:
1022         return "SanDisk"
1023     if "VEN_ASMT" in upper_id:
1024         return "ASMedia"
1025     return "Стандартное HID/USB устройство"
1026
1027
1028 async def _inspect_connected_mice_history() -> Dict[str, Any]:
1029     """Считывает полную историю и текущий статус подключенных
1030     мышей и манипуляторов."""
1031     results: List[Dict[str, Any]] = []
1032     cmd_used = "Get-PnpDevice -Class Mouse -ErrorAction Silent
1033     lyContinue | Select-Object Status, Class, FriendlyName, Instan
1034     ceId | ConvertTo-Json"
1035
1036     if platform.system() == "Windows":
1037         try:
1038             import json as py_json
1039             import subprocess
1040
1041             proc = await asyncio.to_thread(
1042                 subprocess.run,
1043                 ["powershell", "-NoProfile", "-Command", cmd_u
1044                 sed],
1045                 capture_output=True,
1046                 text=True,
1047                 timeout=8,
1048             )
1049
1050             if proc.returncode == 0 and proc.stdout.strip():
1051                 parsed = py_json.loads(proc.stdout.strip())

```

```

1048 |             items = parsed if isinstance(parsed, list) else [parsed]
1049 |             for item in items:
1050 |                 inst = str(item.get("InstanceId") or "")
1051 |                 vendor = _lookup_vendor_by_instance(inst)
1052 |                 st = str(item.get("Status") or "Unknown")
1053 |                 is_active = (st.upper() == "OK")
1054 |                 results.append({
1055 |                     "name": str(item.get("FriendlyName") or "HID-совместимая мышь"),
1056 |                     "vendor": vendor,
1057 |                     "instance_id": inst,
1058 |                     "status": "Подключена сейчас (OK)" if is_active else "Была подключена ранее (История PnP)",
1059 |                     "is_connected": is_active,
1060 |                 })
1061 |             except Exception as ex:
1062 |                 logger.warning(f"Ошибка сбора истории мышей через PnP: {ex}")
1063 |
1064 |     # Fallback для не-Windows или при ошибке
1065 |     if not results:
1066 |         results = [
1067 |             {
1068 |                 "name": "HID-compliant mouse (Logitech Unifying)",
1069 |                 "vendor": "Logitech",
1070 |                 "instance_id": "HID\\VID_046D&PID_C52B&MI_01",
1071 |                 "status": "Подключена сейчас (OK)",
1072 |                 "is_connected": True,
1073 |             }
1074 |         ]
1075 |
1076 |     return {
1077 |         "command": cmd_used,
1078 |         "devices": results,
1079 |         "count": len(results),
1080 |         "active_count": sum(1 for d in results if d.get("is_co

```



```

    nected")),
1081     }
1082
1083
1084 async def _inspect_usb_devices_history() -> Dict[str, Any]:
1085     """Считывает полную историю и текущий статус всех USB устр
ойств хоста."""
1086     results: List[Dict[str, Any]] = []
1087     cmd_used = (
1088         "Get-PnpDevice -ErrorAction SilentlyContinue | "
1089         "Where-Object { $_.InstanceId -like 'USB*' -or $_.Clas
s -eq 'USB' -or $_.Class -eq 'USBDevice' } | "
1090         "Select-Object Status, Class, FriendlyName, InstanceId
| ConvertTo-Json"
1091     )
1092
1093     if platform.system() == "Windows":
1094         try:
1095             import json as py_json
1096             import subprocess
1097
1098             proc = await asyncio.to_thread(
1099                 subprocess.run,
1100                 ["powershell", "-NoProfile", "-Command", cmd_u
sed],
1101                 capture_output=True,
1102                 text=True,
1103                 timeout=10,
1104             )
1105
1106             if proc.returncode == 0 and proc.stdout.strip():
1107                 parsed = py_json.loads(proc.stdout.strip())
1108                 items = parsed if isinstance(parsed, list) els
e [parsed]
1109                 for item in items:
1110                     inst = str(item.get("InstanceId") or "")
1111                     name = str(item.get("FriendlyName") or "US
B Device")

```

```

1112         cls = str(item.get("Class") or "USB")
1113         vendor = _lookup_vendor_by_instance(inst)
1114         st = str(item.get("Status") or "Unknown")
1115         is_active = (st.upper() == "OK")
1116         results.append({
1117             "name": name,
1118             "class": cls,
1119             "vendor": vendor,
1120             "instance_id": inst,
1121             "status": "Подключено сейчас (OK)" if
is_active else "Было подключено ранее (История PnP)",
1122             "is_connected": is_active,
1123         })
1124     except Exception as ex:
1125         logger.warning(f"Ошибка сбора истории USB через Pn
P: {ex}")
1126
1127     if not results:
1128         results = [
1129             {
1130                 "name": "Generic USB Hub",
1131                 "class": "USB",
1132                 "vendor": "Стандартное HID/USB устройство",
1133                 "instance_id": "USB\\ROOT_HUB30",
1134                 "status": "Подключено сейчас (OK)",
1135                 "is_connected": True,
1136             }
1137         ]
1138
1139     return {
1140         "command": cmd_used,
1141         "devices": results,
1142         "count": len(results),
1143         "active_count": sum(1 for d in results if d.get("is_co
nnected")),
1144     }
1145
1146

```

```

1147 def _auto_save_skill(skill_name: str, title: str, description_
    ru: str, instructions: str, script_content: Optional[str] = No
    ne) -> ScenarioCreatedSkillInfo:
1148     """Автоматически сохраняет новый навык в .skills/ и .agent
    s/skills/."""
1149     safe_name = "".join(c if c.isalnum() or c in ("-", "_") el
    se "-" for c in skill_name.lower()).strip("-")
1150     target_dirs = [
1151         __root__ / ".skills" / safe_name,
1152         __root__ / ".agents" / "skills" / safe_name,
1153     ]
1154
1155     already_existed = any((t_dir / "SKILL.md").exists() for t_
    dir in target_dirs)
1156
1157     skill_md_content = f"""---
1158 name: {safe_name}
1159 description: Automated assistant skill for: {description_ru}
1160 ---
1161
1162 # {title}
1163
1164 ## □ Назначение (Purpose)
1165 {description_ru}
1166
1167 ## □ Триггеры и применение (When to Use & Triggers)
1168 Используется при запросах пользователя о системном оборудовани
    и, истории подключений и экспресс-диагностике в интерфейсе Tes
    t Computer (/tc) и консоли.
1169
1170 ## □ Протокол выполнения (Execution Protocol)
1171 {instructions}
1172 """
1173
1174     created_path = ""
1175     for t_dir in target_dirs:
1176         try:
1177             t_dir.mkdir(parents=True, exist_ok=True)

```

```

1178         skill_file = t_dir / "SKILL.md"
1179         skill_file.write_text(skill_md_content, encoding="
utf-8")
1180         if script_content:
1181             scripts_dir = t_dir / "scripts"
1182             scripts_dir.mkdir(parents=True, exist_ok=True)
1183             (scripts_dir / "probe.ps1").write_text(script_
content, encoding="utf-8")
1184             if not created_path:
1185                 created_path = str(skill_file.relative_to(__ro
ot__))
1186         except Exception as e:
1187             logger.warning(f"Не удалось записать навык в {t_di
r}: {e}")
1188
1189         if not created_path:
1190             created_path = f".skills/{safe_name}/SKILL.md"
1191
1192         if already_existed:
1193             logger.info(f"□ Навык уже существует (обновлен): {crea
ted_path}")
1194         else:
1195             logger.info(f"□ Сформирован и сохранён новый навык: {c
reated_path}")
1196
1197         return ScenarioCreatedSkillInfo(
1198             name=safe_name,
1199             title=title,
1200             path=created_path,
1201             description_ru=description_ru,
1202             is_new=not already_existed,
1203         )
1204
1205
1206 _dynamic_tool_engine = None
1207
1208
1209 def _get_dynamic_tool_engine() -> Any:

```

```

1210     """Ленивая инициализация динамического движка инструментов
    Windows."""
1211     global _dynamic_tool_engine
1212     if _dynamic_tool_engine is None:
1213         try:
1214             from apps.windows.core.dynamic_tool_engine import
DynamicWindowsToolEngine
1215             _dynamic_tool_engine = DynamicWindowsToolEngine()
1216         except Exception as e:
1217             logger.warning(f"Не удалось инициализировать Dynam
icWindowsToolEngine: {e}")
1218     return _dynamic_tool_engine
1219
1220
1221 async def _handle_scenario_chat(req: ScenarioChatRequest) -> S
cenarioChatResponse:
1222     """Обрабатывает запросы в мини-чате, динамически генерируя
инструмент/зонд через подсистему apps/windows."""
1223     engine = _get_dynamic_tool_engine()
1224     if engine:
1225         effective_conv_id = req.conversation_id if req.keep_co
ntext else None
1226         res = await engine.process_query(
1227             query=req.message,
1228             auto_create_skill=req.auto_create_skill,
1229             conversation_id=effective_conv_id,
1230             keep_context=req.keep_context,
1231             use_rag=req.use_rag,
1232         )
1233         created_skill_obj = None
1234         if res.get("created_skill"):
1235             s = res["created_skill"]
1236             created_skill_obj = ScenarioCreatedSkillInfo(
1237                 name=s["name"],
1238                 title=s["title"],
1239                 path=s["path"],
1240                 description_ru=s["description_ru"],
1241                 is_new=s.get("is_new", True),

```

```

1242         )
1243
1244     tool_plan_obj = None
1245     if res.get("tool_plan"):
1246         tp = res["tool_plan"]
1247         tool_plan_obj = ScenarioToolPlanInfo(
1248             tool_name=tp["tool_name"],
1249             tool_title=tp["tool_title"],
1250             description_ru=tp["description_ru"],
1251             probe_type=tp.get("probe_type", "powershell"),
1252             collector_name=tp.get("collector_name"),
1253             probe_script=tp.get("probe_script"),
1254             instructions=tp.get("instructions", ""),
1255         )
1256
1257     actions_list = []
1258     for a in res.get("remediation_actions", []):
1259         if isinstance(a, dict):
1260             actions_list.append(ScenarioRemediationActionI
1261 nfo(
1262                 action_id=str(a.get("action_id", "")),
1263                 action_type=str(a.get("action_type", "cust
1264 om_command")),
1265                 title=str(a.get("title", "")),
1266                 description=str(a.get("description", "")),
1267                 target=str(a.get("target", "")),
1268                 risk=str(a.get("risk", "caution")),
1269                 execution_command=str(a.get("execution_com
1270 mand", "")),
1271             ))
1272
1273     return ScenarioChatResponse(
1274         reply=res.get("reply", ""),
1275         command_executed=res.get("command_executed"),
1276         created_skill=created_skill_obj,
1277         generated_prompt=res.get("generated_prompt"),
1278         tool_plan=tool_plan_obj,
1279         remediation_actions=actions_list,

```

```

1277         raw_data=res.get("raw_data"),
1278         agents_used=res.get("agents_used", []),
1279         knowledge_used=res.get("knowledge_used", []),
1280         status=res.get("status", "ok"),
1281     )
1282
1283     # Запасной ответ при ошибке инициализации движка
1284     return ScenarioChatResponse(
1285         reply=f"### □ Ответ системного ассистента\nЗапрос: *«
1286         {req.message}»*\n\nДинамический движок `apps/windows` инициали
1287         зируется.",
1288         status="error",
1289     )
1290
1291 async def _execute_disk_benchmark_telemetry_step() -> Scenario
1292 StepResult:
1293     """Выполняет экспресс-бенчмарк всех накопителей через Stor
1294     ageBenchmarkSensor и фиксирует телеметрию."""
1295     t0 = time.perf_counter()
1296     try:
1297         from apps.windows.modules.storage_manager.core.benchma
1298         rk_sensor import get_storage_benchmark_sensor
1299         sensor = get_storage_benchmark_sensor()
1300         report = await asyncio.to_thread(sensor.run_quick_benc
1301         hmark, None, 16, 1)
1302         dur_ms = (time.perf_counter() - t0) * 1000
1303         summary = report.get("summary", "Бенчмарк дисков завер
1304         шен")
1305         drives_count = report.get("drives_count", 0)
1306         return ScenarioStepResult(
1307             name="Телеметрия дисков: Экспресс-бенчмарк (DiskSp
1308             d)",
1309             status="ok" if drives_count > 0 else "warn",
1310             duration_ms=round(dur_ms, 2),
1311             details=f"Протестировано накопителей ({drives_coun
1312             t}): {summary}",
1313             data=report,

```

```

1306         )
1307     except Exception as exc:
1308         dur_ms = (time.perf_counter() - t0) * 1000
1309         logger.warning(f"[router_scenarios] Ошибка телеметрии
бенчмарка дисков: {exc}")
1310         return ScenarioStepResult(
1311             name="Телеметрия дисков: Экспресс-бенчмарк (DiskSp
d)",
1312             status="warn",
1313             duration_ms=round(dur_ms, 2),
1314             details=f"Предупреждение при бенчмарке дисков: {ex
c}",
1315             recommendation="Проверьте доступность diskspd.exe
и свободное место на дисках.",
1316         )
1317
1318
1319 async def _run_disk_benchmark_scenario() -> ScenarioRunResult:
1320     """Сценарий автономного экспресс-тестирования всех накопит
елей."""
1321     t0 = time.perf_counter()
1322     s_at = datetime.now(timezone.utc).isoformat()
1323     step = await _execute_disk_benchmark_telemetry_step()
1324     dur_ms = (time.perf_counter() - t0) * 1000
1325
1326     return ScenarioRunResult(
1327         scenario_id="disk_benchmark_audit",
1328         title="Экспресс-тест скорости всех накопителей (DiskSp
d)",
1329         status=step.status,
1330         started_at=s_at,
1331         duration_ms=round(dur_ms, 2),
1332         total_steps=1,
1333         passed_steps=1 if step.status == "ok" else 0,
1334         warn_steps=1 if step.status == "warn" else 0,
1335         failed_steps=1 if step.status == "error" else 0,
1336         steps=[step],
1337         summary=f"Экспресс-бенчмарк накопителей завершен: {ste

```



```

    p.details}",
1338     )
1339
1340
1341 # -----
1342 # Router Initialization
1343 # -----
1344
1345 def init_router() -> APIRouter:
1346     """Инициализирует FastAPI роутер для управления сценариями
1347     тестирования.
1348
1349     Returns:
1350         APIRouter: Настроенный роутер с префиксом /api/v1/scen
1351     arios.
1352     """
1353     router = APIRouter(prefix="/api/v1/scenarios", tags=["Test
1354     Computer Scenarios"])
1355
1356     @router.get("", response_model=List[ScenarioItem])
1357     async def list_scenarios() -> List[ScenarioItem]:
1358         """Возвращает перечень доступных сценариев тестировани
1359     я и аудита."""
1360         return [ScenarioItem(**sc) for sc in AVAILABLE_SCENARI
1361     OS]
1362
1363     @router.post("/run", response_model=ScenarioRunResult)
1364     async def run_scenario(req: ScenarioRunRequest) -> Scenari
1365     oRunResult:
1366         """Запускает указанный тестовый сценарий и возвращает
1367     детальные результаты шагов."""
1368         scenario_id = req.scenario_id.lower().strip()
1369
1370         if scenario_id in ("safe_update_guard", "update_guard"
1371     , "update", "pre_update"):
1372             res = await _run_safe_update_guard_scenario()

```



```
1399 |             name=f"Сценарий {runner.__name__}",
1400 |             status="error",
1401 |             details=f"Исключение при выполнении: {
| ex}"
1402 |         ))
1403 |
1404 |         t_dur = (time.perf_counter() - t0) * 1000
1405 |         f_cnt = sum(1 for s in all_steps if s.status == "e
| rror")
1406 |         w_cnt = sum(1 for s in all_steps if s.status == "w
| arn")
1407 |         p_cnt = sum(1 for s in all_steps if s.status == "o
| k")
1408 |         ov_status = "error" if f_cnt > 0 else ("warn" if w
| _cnt > 0 else "ok")
1409 |
1410 |         return ScenarioRunResult(
1411 |             scenario_id="all",
1412 |             title="Комплексный прогон всех сценариев",
1413 |             status=ov_status,
1414 |             started_at=s_at,
1415 |             duration_ms=round(t_dur, 2),
1416 |             total_steps=len(all_steps),
1417 |             passed_steps=p_cnt,
1418 |             warn_steps=w_cnt,
1419 |             failed_steps=f_cnt,
1420 |             steps=all_steps,
1421 |             summary=f"Комплексный прогон завершен: успешно
| {p_cnt}, предупреждений {w_cnt}, ошибок {f_cnt}.",
1422 |             )
1423 |         else:
1424 |             raise HTTPException(status_code=400, detail=f"Неиз
| вестный идентификатор сценария: '{req.scenario_id}'")
1425 |
1426 |         # При каждом запуске любого сценария выполняем экспрес
| с-бенчмарк всех дисков и сохраняем телеметрию
1427 |         disk_step = await _execute_disk_benchmark_telemetry_st
| ep()
```

```
1428         res.steps.append(disk_step)
1429         res.total_steps = len(res.steps)
1430         if disk_step.status == "ok":
1431             res.passed_steps += 1
1432         elif disk_step.status == "warn":
1433             res.warn_steps += 1
1434         elif disk_step.status == "error":
1435             res.failed_steps += 1
1436             res.status = "error"
1437         res.duration_ms = round(res.duration_ms + disk_step.du
ration_ms, 2)
1438         return res
1439
1440
1441     @router.post("/chat", response_model=ScenarioChatResponse)
1442     async def chat_scenario_assistant(req: ScenarioChatRequest
) -> ScenarioChatResponse:
1443         """Интеллектуальный мини-чат сценариев: выполняет опро
с системы или запрашивает LLM и генерирует навык."""
1444         if not req.message.strip():
1445             raise HTTPException(status_code=400, detail="Текст
сообщения не может быть пустым.")
1446         return await _handle_scenario_chat(req)
1447
1448     @router.post("/chat/stream")
1449     async def chat_scenario_assistant_stream(req: ScenarioChat
Request) -> StreamingResponse:
1450         """Потоковый SSE мини-чат сценариев: передает промежут
очные этапы и результат в реальном времени."""
1451         if not req.message.strip():
1452             raise HTTPException(status_code=400, detail="Текст
сообщения не может быть пустым.")
1453
1454         engine = _get_dynamic_tool_engine()
1455         if not engine:
1456             raise HTTPException(status_code=500, detail="Движо
к DynamicWindowsToolEngine недоступен.")
1457
```

```

1458         effective_conv_id = req.conversation_id if req.keep_co
| ntext else None
1459
1460         async def event_generator():
1461             try:
1462                 async for evt in engine.process_query_stream(
1463                     query=req.message,
1464                     auto_create_skill=req.auto_create_skill,
1465                     conversation_id=effective_conv_id,
1466                     keep_context=req.keep_context,
1467                     use_rag=req.use_rag,
1468                 ):
1469                     yield f"data: {json.dumps(evt, ensure_ascii
| i=False)}\n\n"
1470             except Exception as e:
1471                 import traceback
1472                 tb_str = traceback.format_exc()
1473                 logger.error(f"Ошибка в chat_scenario_assistan
| t_stream: {e}", exc_info=True)
1474                 err_evt = {
1475                     "type": "error",
1476                     "error": str(e),
1477                     "error_type": type(e).__name__,
1478                     "details": tb_str,
1479                 }
1480                 yield f"data: {json.dumps(err_evt, ensure_ascii
| i=False)}\n\n"
1481
1482         return StreamingResponse(event_generator(), media_type
| ="text/event-stream")
1483
1484     @router.post("/chat/clear")
1485     async def clear_chat_session(req: ScenarioChatRequest) ->
| Dict[str, Any]:
1486         """Очищает контекст сессии диалога."""
1487         engine = _get_dynamic_tool_engine()
1488         if engine and req.conversation_id and req.conversation
| _id in engine.sessions:

```

```

1489         engine.sessions.pop(req.conversation_id, None)
1490         return {"status": "ok", "message": "Контекст сессии оч
ищен"}
1491
1492     @router.post("/save-skill", response_model=ScenarioCreated
SkillInfo)
1493     async def save_scenario_skill(req: ScenarioSaveSkillReques
t) -> ScenarioCreatedSkillInfo:
1494         """Сохраняет подтвержденный пользователем инструмент в
постоянный каталог навыков (.skills/ и .agents/skills/)."""
1495         engine = _get_dynamic_tool_engine()
1496         if not engine:
1497             raise HTTPException(status_code=500, detail="Движо
к DynamicWindowsToolEngine недоступен.")
1498
1499         from apps.windows.core.dynamic_tool_engine import Dyna
micToolPlan
1500         plan = DynamicToolPlan(
1501             intent=req.tool_title,
1502             tool_name=req.tool_name,
1503             tool_title=req.tool_title,
1504             description_ru=req.description_ru,
1505             probe_type=req.probe_type,
1506             collector_name=req.collector_name,
1507             probe_script=req.probe_script,
1508             instructions=req.instructions or f"Выполнить систе
мную команду: {req.command_executed or req.probe_script}",
1509         )
1510         res = engine.save_skill(plan, req.command_executed or
req.probe_script)
1511         return ScenarioCreatedSkillInfo(
1512             name=res["name"],
1513             title=res["title"],
1514             path=res["path"],
1515             description_ru=res["description_ru"],
1516             is_new=res.get("is_new", True),
1517         )
1518

```

```

1519 |     @router.post("/execute-fix", response_model=ScenarioExecut
| eFixResponse)
1520 |     async def execute_scenario_fix(req: ScenarioExecuteFixRequ
| est) -> ScenarioExecuteFixResponse:
1521 |         """Безопасное выполнение SafeOps-исправления из интерф
| ейса чата."""
1522 |         from apps.windows.core.models import ActionType, Remed
| iationAction, RiskLevel
1523 |         from apps.windows.core.safe_executor import SafeExecut
| or
1524 |
1525 |         executor = SafeExecutor()
1526 |         try:
1527 |             act_type = ActionType(req.action_type)
1528 |         except Exception:
1529 |             act_type = ActionType.CUSTOM_COMMAND
1530 |
1531 |         try:
1532 |             r_level = RiskLevel(req.risk.lower())
1533 |         except Exception:
1534 |             r_level = RiskLevel.CAUTION
1535 |
1536 |         action = RemediationAction(
1537 |             action_id=req.action_id,
1538 |             action_type=act_type,
1539 |             title=req.title or req.action_id,
1540 |             description=req.description,
1541 |             target=req.target,
1542 |             risk=r_level,
1543 |             execution_command=req.execution_command,
1544 |         )
1545 |
1546 |         res_action = await asyncio.to_thread(executor.execute,
| action, req.confirmed_by_user)
1547 |         if res_action.success:
1548 |             return ScenarioExecuteFixResponse(
1549 |                 action_id=res_action.action_id,
1550 |                 status="ok",

```

```

1551         executed=res_action.executed,
1552         success=True,
1553         message=f"Исправление '{res_action.title}' усп
ешно применено.",
1554     )
1555     else:
1556         return ScenarioExecuteFixResponse(
1557             action_id=res_action.action_id,
1558             status="error",
1559             executed=res_action.executed,
1560             success=False,
1561             message=res_action.error_message or "Не удалос
ь применить исправление.",
1562             error_message=res_action.error_message,
1563         )
1564
1565     @router.get("/questions", response_model=ScenarioQuestions
Config)
1566     async def get_scenario_questions() -> ScenarioQuestionsCon
fig:
1567         """Возвращает пул вопросов и быстрых кнопок из внешнег
о файла questions.json."""
1568         q_path = Path(__file__).resolve().parent / "webgui" /
"scenarios_tab" / "questions.json"
1569         if q_path.exists():
1570             try:
1571                 data = json.loads(q_path.read_text(encoding="u
tf-8"))
1572                 return ScenarioQuestionsConfig(**data)
1573             except Exception as e:
1574                 logger.error(f"Ошибка чтения questions.json: {
e}")
1575         return ScenarioQuestionsConfig(
1576             description="Default questions pool",
1577             prompts=[
1578                 "Включи аудит процессов в памяти",
1579                 "Дай полную информацию о подключенных принтера
х",

```



```

1580 |         "Какие мыши были подключены к этому компьютеру
1581 |         ?",
1582 |         "Покажи список сетевых портов и активных соеди
1583 |         нений",
1584 |         "Проверь автозапуск и службы Windows",
1585 |         "Кто использует сеть?",
1586 |         "Проверь точки восстановления системы",
1587 |         "Найди давно не запускавшиеся программы",
1588 |         "Как расшарить папку с файлами",
1589 |     ],
1590 |     quick_buttons=[],
1591 | )
1592 |
1593 | @router.post("/save-approved-response", response_model=ScenarioSaveApprovedResponseResponse)
1594 | async def save_approved_scenario_response(req: ScenarioSaveApprovedResponseRequest) -> ScenarioSaveApprovedResponseResponse:
1595 |     """Сохраняет одобренный пользователем ответ мини-чата
1596 |     в базу ТС для RAG и последующего тюнинга модели."""
1597 |     try:
1598 |         from src.ai.gemini.approved_responses_store import
1599 |         save_approved_response
1600 |         ok = save_approved_response(
1601 |             user_id=req.user_id,
1602 |             query=req.query,
1603 |             chat_text=req.chat_text,
1604 |             voice_text=req.voice_text or "",
1605 |             system_context=req.system_context or {},
1606 |             tags=req.tags or ["tc", "diagnostics", "training"],
1607 |         )
1608 |         if not ok:
1609 |             raise HTTPException(status_code=500, detail="Не удалось сохранить одобренный ответ в базу данных.")
1610 |         return ScenarioSaveApprovedResponseResponse(
1611 |             status="ok",
1612 |             message="Ответ успешно сохранён в базу знаний

```

```
и датасет обучения модели (data/tc/approved_responses).",
1609 |         file_saved="ok"
1610 |     )
1611 |     except HTTPException:
1612 |         raise
1613 |     except Exception as ex:
1614 |         logger.error(f"Ошибка сохранения одобренного ответ
1615 | а TC: {ex}", exc_info=True)
1616 |         raise HTTPException(status_code=500, detail=str(ex
1617 | ))
1618 |
1619 |     return router
```

router_storage_load.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Storage Load
4 # =====
5 # Description:
6 #   GET /api/v1/panel/storage-load – детальные параметры и метрики всех дисков
7 #   и накопителей (NVMe, SSD, HDD, USB) без жесткого кодирования (Zero-Hardcode):
8 #   динамическое слияние Windows CIM/WMI (WindowsStorageSensor), LibreHardwareMonitor
9 #   и системных разделов psutil.
10 #
11 # Usage Examples:
12 #   Python API:
13 #       from apps.windows.api.routers.router_storage_load import init_router
14 #       app.include_router(init_router())
15 #
16 # File: router_storage_load.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-05 23:40:00
22 # =====
23
24 from __future__ import annotations
25 """Роутер панели «Параметры накопителей (Диски SSD/HDD/NVMe)»: метрики хранилища."""
26
27 from typing import Any, Dict, List, Optional
```

```
28 from fastapi import APIRouter
29 from pydantic import BaseModel, Field
30
31 try:
32     import psutil
33 except ImportError:
34     psutil = None
35
36 from logger import logger
37 from apps.windows.modules.hardware.lhm_service import LhmService
38 from apps.windows.modules.storage_manager.core.windows_storage
   _sensor import WindowsStorageSensor
39
40
41
42 class DiskDriveMetric(BaseModel):
43     """Метрика физического накопителя (SSD, HDD, NVMe)."""
44     id: str = Field(default="", description="Идентификатор накопителя (напр. Disk0)")
45     name: str = Field(..., description="Модель или наименование диска")
46     media_type: str = Field(default="SSD", description="Тип носителя: NVMe, SSD, HDD")
47     total_gb: Optional[float] = Field(default=None, description="Полный объем накопителя, GB")
48     used_gb: Optional[float] = Field(default=None, description="Занятый объем, GB")
49     free_gb: Optional[float] = Field(default=None, description="Свободный объем, GB")
50     used_percent: Optional[float] = Field(default=None, description="Заполненность накопителя, %")
51     temperature_c: Optional[float] = Field(default=None, description="Температура накопителя, °C")
52     activity_percent: Optional[float] = Field(default=None, description="Общая активность диска, %")
53     read_activity_percent: Optional[float] = Field(default=None, description="Активность чтения, %")
```

```

54     write_activity_percent: Optional[float] = Field(default=None, description="Активность записи, %")
55     read_rate_raw: Optional[str] = Field(default=None, description="Скорость чтения")
56     write_rate_raw: Optional[str] = Field(default=None, description="Скорость записи")
57     power_on_hours: Optional[float] = Field(default=None, description="Время наработки (Power-on Hours)")
58
59
60 class DiskPartitionMetric(BaseModel):
61     """Метрика логического диска / раздела."""
62     device: str = Field(..., description="Устройство (буква диска, напр. C:\\)")
63     mountpoint: str = Field(..., description="Точка монтирования")
64     fstype: str = Field(default="", description="Файловая система (NTFS, FAT32)")
65     total_gb: float = Field(default=0.0, description="Полный объем раздела, GB")
66     used_gb: float = Field(default=0.0, description="Занятый объем раздела, GB")
67     free_gb: float = Field(default=0.0, description="Свободный объем раздела, GB")
68     used_percent: float = Field(default=0.0, description="Процент заполнения раздела, %")
69
70
71 class StorageSummary(BaseModel):
72     """Сводка по всему хранилищу."""
73     total_gb: float = Field(default=0.0, description="Суммарный объем всех дисков, GB")
74     used_gb: float = Field(default=0.0, description="Суммарно занятый объем, GB")
75     free_gb: float = Field(default=0.0, description="Суммарно свободный объем, GB")
76     used_percent: float = Field(default=0.0, description="Общий процент занятого пространства, %")

```

```

77     max_temperature_c: Optional[float] = Field(default=None, d
description="Максимальная температура среди дисков, °C")
78     avg_temperature_c: Optional[float] = Field(default=None, d
description="Средняя температура дисков, °C")
79
80
81 class StorageLoadResponse(BaseModel):
82     """Ответ GET /api/v1/panel/storage-load."""
83     status: str = Field(default="ok", description="Статус отве
та")
84     summary: StorageSummary = Field(default_factory=StorageSum
mary, description="Общая сводка хранилища")
85     drives: List[DiskDriveMetric] = Field(default_factory=list
, description="Физические накопители")
86     partitions: List[DiskPartitionMetric] = Field(default_fact
ory=list, description="Логические разделы")
87     meta: Dict[str, Any] = Field(default_factory=dict, descrip
tion="Метаданные ответа")
88
89
90 def _detect_media_type(name: str) -> str:
91     """Определяет тип диска (NVMe, SSD, HDD) по названию модел
и."""
92     n = name.lower()
93     if "nvme" in n or "990" in n or "980" in n or "970" in n o
r "optane" in n:
94         return "NVMe"
95     if "ssd" in n or "flash" in n or "solid" in n or "crucial"
in n:
96         return "SSD"
97     return "HDD"
98
99
100 def _find_matching_key(drives_map: Dict[str, Dict[str, Any]],
name: str) -> Optional[str]:
101     """Находит ключ накопителя в словаре по частичному совпаде
нию модели."""
102     nl = name.lower().strip()

```

```

103     for k in drives_map:
104         kl = k.lower().strip()
105         if nl in kl or kl in nl:
106             return k
107     return None
108
109
110 def build_storage_load(
111     sensors: Optional[List[Dict[str, Any]]] = None,
112     physical_disks: Optional[List[Any]] = None,
113     partitions_data: Optional[List[Dict[str, Any]]] = None,
114 ) -> StorageLoadResponse:
115     """Собирает структурированные данные о всех дисках и разде
116     лах без хардкода.
117
118     Args:
119         sensors: Список показаний сенсоров LHM.
120         physical_disks: Список физических дисков из WindowsSto
121         rageSensor (CIM/WMI).
122         partitions_data: Опциональный список готовых данных по
123         разделам для тестов.
124
125     Returns:
126         StorageLoadResponse: Структура с физическими накопител
127         ями и разделами.
128     """
129     drives_map: Dict[str, Dict[str, Any]] = {}
130
131     # 1. Загрузка физических дисков из WindowsStorageSensor (C
132     IM/WMI)
133     if physical_disks:
134         for cd in physical_disks:
135             m = getattr(cd, "model", None) or getattr(cd, "fri
136             endly_name", None) or "Physical Drive"
137             dev_id = getattr(cd, "device_id", None) or f"Disk{
138             len(drives_map)}"
139             media_type = getattr(cd, "media_type", None)
140             detected = _detect_media_type(m)

```

```

134         if not media_type or media_type.lower() in ("unspe
cified", "ssd") and detected == "NVMe":
135             media_type = "NVMe"
136         elif not media_type or media_type.lower() == "unsp
ecified":
137             media_type = detected
138
139         temp = getattr(cd, "temperature_c", None)
140         if temp is None:
141             temp = getattr(cd, "temperature_celsius", None
)
142         if temp is not None and (temp <= 5 or temp > 120):
143             temp = None
144
145         drives_map[m] = {
146             "id": dev_id,
147             "name": m,
148             "media_type": media_type,
149             "total_gb": round(float(cd.size_gb), 1) if get
attr(cd, "size_gb", None) else None,
150             "temperature_c": round(float(temp), 1) if temp
is not None else None,
151             "power_on_hours": round(float(cd.power_on_hour
s), 0) if getattr(cd, "power_on_hours", None) else None,
152         }
153
154     # 2. Обогащение и добавление из сенсоров LibreHardwareMoni
tor
155     if sensors:
156         for s in sensors:
157             htype = str(s.get("hardware_type", "")).lower()
158             if "storage" not in htype and "disk" not in htype
and "hdd" not in htype:
159                 continue
160
161             hname = s.get("hardware_name") or "Storage Drive"
162             matched_key = _find_matching_key(drives_map, hname
)

```



```

163         if not matched_key:
164             matched_key = hname
165             drives_map[matched_key] = {
166                 "id": f"drive-{len(drives_map) + 1}",
167                 "name": hname,
168                 "media_type": _detect_media_type(hname),
169             }
170
171         cat = str(s.get("sensor_category", "")).lower()
172         sname = str(s.get("sensor_name", "")).lower()
173         val_num = s.get("value_num") if s.get("value_num")
is not None else s.get("value_numeric")
174         val_raw = s.get("value_raw")
175
176         if "temp" in cat:
177             if val_num is not None and val_num > 5:
178                 drives_map[matched_key]["temperature_c"] =
round(float(val_num), 1)
179             elif "load" in cat:
180                 if "used space" in sname and val_num is not No
ne:
181                     drives_map[matched_key]["used_percent"] =
round(float(val_num), 1)
182                 elif "total activity" in sname and val_num is
not None:
183                     drives_map[matched_key]["activity_percent"]
= round(float(val_num), 1)
184                 elif "read activity" in sname and val_num is n
ot None:
185                     drives_map[matched_key]["read_activity_per
cent"] = round(float(val_num), 1)
186                 elif "write activity" in sname and val_num is
not None:
187                     drives_map[matched_key]["write_activity_pe
rcent"] = round(float(val_num), 1)
188             elif "data" in cat:
189                 if "total space" in sname and val_num is not N
one:

```

```

190 |             drives_map[matched_key]["total_gb"] = round
| d(float(val_num), 1)
191 |             elif "free space" in sname and val_num is not
| None:
192 |                 drives_map[matched_key]["free_gb"] = round
| (float(val_num), 1)
193 |             elif "throughput" in cat:
194 |                 if "read rate" in sname:
195 |                     drives_map[matched_key]["read_rate_raw"] =
| str(val_raw or "")
196 |                 elif "write rate" in sname:
197 |                     drives_map[matched_key]["write_rate_raw"]
| = str(val_raw or "")
198 |             elif "factors" in cat:
199 |                 if "power on hours" in sname and val_num is no
| t None:
200 |                     drives_map[matched_key]["power_on_hours"]
| = round(float(val_num), 0)
201 |
202 |     # 3. Сбор логических разделов
203 |     partitions_list: List[DiskPartitionMetric] = []
204 |     if partitions_data is not None:
205 |         for p in partitions_data:
206 |             if isinstance(p, dict):
207 |                 p_dev = p.get("device") or p.get("mountpoint")
| or ""
208 |                 p_mount = p.get("mountpoint") or p.get("device
| ") or ""
209 |                 p_fstype = str(p.get("fstype", "") or "")
210 |                 p_tot = float(p.get("total_gb", 0.0) or 0.0)
211 |                 p_used = float(p.get("used_gb", 0.0) or 0.0)
212 |                 p_free = float(p.get("free_gb", 0.0) or 0.0)
213 |                 p_pct = float(p.get("used_percent") if p.get("
| used_percent") is not None else (p.get("percent", 0.0) or 0.0)
| )
214 |                 partitions_list.append(
215 |                     DiskPartitionMetric(
216 |                         device=p_dev,

```

```

217         mountpoint=p_mount,
218         fstype=p_fstype,
219         total_gb=p_tot,
220         used_gb=p_used,
221         free_gb=p_free,
222         used_percent=round(p_pct, 1),
223     )
224 )
225 elif isinstance(p, DiskPartitionMetric):
226     partitions_list.append(p)
227 elif psutil is not None:
228     try:
229         parts = psutil.disk_partitions(all=False)
230         for p in parts:
231             try:
232                 usage = psutil.disk_usage(p.mountpoint)
233                 tot_gb = round(usage.total / (1024 ** 3),
234 1)
235                 used_gb = round(usage.used / (1024 ** 3),
236 1)
237                 free_gb = round(usage.free / (1024 ** 3),
238 1)
239                 partitions_list.append(
240                     DiskPartitionMetric(
241                         device=p.device,
242                         mountpoint=p.mountpoint,
243                         fstype=p.fstype,
244                         total_gb=tot_gb,
245                         used_gb=used_gb,
246                         free_gb=free_gb,
247                         used_percent=round(usage.percent,
248 1),
249 )
250 )
251 except Exception:
252     continue
253 except Exception as exc:
254     logger.debug(f"[router_storage_load] Ошибка сбора

```

```

разделов через psutil: {exc}")
251
252     # 4. Довычисляем used_gb / used_percent для физических дис
    КОВ
253     drives_list: List[DiskDriveMetric] = []
254     for d in drives_map.values():
255         tot = d.get("total_gb")
256         free = d.get("free_gb")
257         if tot is not None and free is not None and "used_gb"
    not in d:
258             d["used_gb"] = round(max(tot - free, 0.0), 1)
259             if tot is not None and d.get("used_gb") is not None an
    d d.get("used_percent") is None:
260                 if tot > 0:
261                     d["used_percent"] = round((d["used_gb"] / tot)
    * 100.0, 1)
262
263         # Если диск не получил used_percent из LHM, ищем соотв
    етствующий раздел по объему
264         if d.get("used_percent") is None and tot is not None a
    nd partitions_list:
265             for part in partitions_list:
266                 if abs(part.total_gb - tot) <= 5.0: # совпаде
    ние объема раздела и диска
267                     d["used_percent"] = part.used_percent
268                     d["used_gb"] = part.used_gb
269                     d["free_gb"] = part.free_gb
270                     break
271
272         drives_list.append(DiskDriveMetric(**d))
273
274     # 5. Расчет суммарной сводки по хранилищу
275     if partitions_list:
276         sum_total = sum(p.total_gb for p in partitions_list)
277         sum_used = sum(p.used_gb for p in partitions_list)
278         sum_free = sum(p.free_gb for p in partitions_list)
279     elif drives_list:
280         sum_total = sum(d.total_gb for d in drives_list if d.t

```

```

    total_gb is not None)
281         sum_used = sum(d.used_gb for d in drives_list if d.use
    d_gb is not None)
282         sum_free = sum(d.free_gb for d in drives_list if d.fre
    e_gb is not None)
283     else:
284         sum_total = 0.0
285         sum_used = 0.0
286         sum_free = 0.0
287
288         sum_pct = round((sum_used / sum_total * 100.0), 1) if sum_
    total > 0 else 0.0
289
290         temps = [d.temperature_c for d in drives_list if d.tempera
    ture_c is not None]
291         max_temp = max(temps) if temps else None
292         avg_temp = round(sum(temps) / len(temps), 1) if temps else
    None
293
294         summary = StorageSummary(
295             total_gb=round(sum_total, 1),
296             used_gb=round(sum_used, 1),
297             free_gb=round(sum_free, 1),
298             used_percent=sum_pct,
299             max_temperature_c=max_temp,
300             avg_temperature_c=avg_temp,
301         )
302
303         return StorageLoadResponse(
304             summary=summary,
305             drives=drives_list,
306             partitions=partitions_list,
307             meta={
308                 "drives_count": len(drives_list),
309                 "partitions_count": len(partitions_list),
310                 "source": "cim+lhmp+psutil",
311             },
312         )

```

```

313 |
314 |
315 | def init_router(
316 |     lhm_service: Optional[LhmService] = None,
317 |     storage_sensor: Optional[WindowsStorageSensor] = None,
318 | ) -> APIRouter:
319 |     """Создаёт роутер /api/v1/panel/storage-load.
320 |
321 |     Args:
322 |         lhm_service: Сервис LibreHardwareMonitor.
323 |         storage_sensor: Сенсор дисков WindowsStorageSensor (CI
324 | M/WMI).
325 |
326 |     Returns:
327 |         APIRouter: Роутер с эндпоинтом GET /api/v1/panel/stora
328 | ge-load.
329 |     """
330 |
331 |     router = APIRouter(tags=["Storage Load Panel"])
332 |     lhm = lhm_service or LhmService()
333 |
334 |     @router.get("/api/v1/panel/storage-load", response_model=S
335 | torageLoadResponse)
336 |     async def get_panel_storage_load() -> StorageLoadResponse:
337 |         """Возвращает детальную загрузку, температуру и емкост
338 | ь всех накопителей из телеметрии БД."""
339 |         try:
340 |             sensors = lhm.get_flattened_sensors() if lhm.is_ru
341 | nning() else []
342 |             physical_disks: List[Any] = []
343 |             partitions_data: Optional[List[Dict[str, Any]]] =
344 | None
345 |
346 |             # 1. Попытка прочитать актуальные данные дисков из
347 | SQLite телеметрии
348 |             try:
349 |                 from apps.windows.telemetry.sqlite import Tele
350 | metryStorage
351 |                 storage = TelemetryStorage.get_instance(read_o

```

```

nly=True)
343         storage_data = storage.get_latest_storage_data
        ()
344         if storage_data.get("physical_disks"):
345             physical_disks = storage_data["physical_di
sks"]
346         if storage_data.get("partitions"):
347             partitions_data = storage_data["partitions
"]
348     except Exception as db_err:
349         logger.debug(f"[router_storage_load] Не удалос
ь прочитать накопители из SQLite: {db_err}")
350
351     # 2. Fallback при отсутствии снимков в базе данных
(без живого вызова WMI/CIM в tc.ps1)
352     if not physical_disks:
353         from apps.windows.telemetry.collector import S
ystemCollector
354         collector = SystemCollector()
355         physical_disks = collector.get_physical_disks_
health(force=False)
356
357     return build_storage_load(
358         sensors=sensors,
359         physical_disks=physical_disks,
360         partitions_data=partitions_data,
361     )
362     except Exception as exc:
363         logger.error(f"[router_storage_load] Ошибка опроса
накопителей: {exc}", exc_info=True)
364         return build_storage_load(sensors=[], physical_dis
ks=[])
365
366     return router
367
368
369 __all__ = [
370     "init_router",

```

```
371 | "build_storage_load",  
372 | "DiskDriveMetric",  
373 | "DiskPartitionMetric",  
374 | "StorageSummary",  
375 | "StorageLoadResponse",  
376 | ]  
377 |
```


router_system32_catalog.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api - Router System
  32 Catalog
4 # =====
5 # Description:
6 #   FastAPI REST API роутер для расширенного каталога возможно
  стей системных инструментов
7 #   Windows 10/11 (%SystemRoot%\System32), Control Planes, Tel
  emetry Tiers и AI-интерпретатора команд.
8 #
9 # Usage Examples:
10 #   Python API:
11 #       from apps.windows.api.routers.router_system32_catalog im
  port init_router
12 #
13 #       router = init_router()
14 #
15 # File: router_system32_catalog.py
16 # Project: ai-breadboard
17 # Package: apps.windows.api.routers
18 # Author: hypo69
19 # Copyright: © 2026 hypo69
20 # Updated: 2026-10-06 15:20:00
21 # =====
22
23 from __future__ import annotations
24 """FastAPI REST API роутер для расширенного каталога системных
  инструментов Windows (%SystemRoot%\System32)."""
25
26 import asyncio
27 import platform
28 import subprocess
```

```
29 import time
30 from typing import Any, Dict, List, Optional
31 from fastapi import APIRouter, HTTPException, Query
32 from pydantic import BaseModel, Field
33 from logger import logger
34 from apps.windows.core.system32_catalog import System32Catalog
35 from apps.windows.core.system32_models import (
36     AccessType,
37     ControlPlaneType,
38     System32CatalogSummary,
39     System32QueryFilter,
40     System32Tool,
41     SystemToolCategory,
42     TelemetryTier,
43     ToolDangerLevel,
44     ToolPrivilegeLevel,
45 )
46 from apps.windows.core.etw_pipeline import EtwPipelineStatus,
47     EtwTelemetryPipeline
48
49 router = APIRouter(prefix='/api/v1/system32', tags=['Windows S
50     ystem32 Capability Catalog'])
51
52 def init_router() -> APIRouter:
53     """Инициализирует и возвращает APIRouter для каталога Syst
54     em32.
55
56     Returns:
57         APIRouter: Экземпляр FastAPI роутера.
58     """
59     return router
60
61 class NaturalLanguageCommandRequest(BaseModel):
62     """Запрос на подбор и выполнение команды по естественному
63     языку."""
```

```
63     prompt: str = Field(..., description="Команда пользователя  
свободным текстом (например: 'проверь статус trim', 'покажи п  
орты netstat')")  
64     execute: bool = Field(True, description="Выполнять ли кома  
нду или вернуть только разбор")  
65     confirmed_by_user: bool = Field(False, description="Подтве  
рждение запуска для потенциально опасных операций")  
66  
67  
68 class NaturalLanguageCommandResponse(BaseModel):  
69     """Результат подбора и выполнения команды."""  
70     matched_executable: str  
71     category: str  
72     category_code: str  
73     telemetry_tier: str  
74     danger_level: str  
75     required_privileges: str  
76     command_line: str  
77     explanation: str  
78     powershell_equivalent: Optional[str] = None  
79     win32_api_equivalent: Optional[str] = None  
80     status: str  
81     executed: bool  
82     execution_output: str  
83     execution_time_ms: float  
84  
85  
86 def _decode_bytes(raw: bytes) -> str:  
87     """Безопасное декодирование вывода консоли Windows."""  
88     for enc in ('utf-8', 'cp866', 'cp1251', 'latin-1'):  
89         try:  
90             return raw.decode(enc)  
91         except UnicodeDecodeError:  
92             continue  
93     return raw.decode('utf-8', errors='replace')  
94  
95  
96 def _match_intent_to_command(prompt: str, catalog: System32Cat
```

```
alog) -> tuple[System32Tool, str, str]:
97     """Интеллектуальный подбор инструмента и точной команды по
пользовательскому тексту."""
98     p_lower = prompt.lower().strip()
99
100     # 1. Специфические ключевые намерения
101     if any(k in p_lower for k in ('trim', 'трим', 'ssd')):
102         tool = catalog.get_tool('fsutil.exe') or catalog.get_a
ll_tools()[0]
103         return tool, 'fsutil.exe behavior query DisableDeleteN
otify', 'Запрос глобального статуса TRIM для твердотельных на
копителей SSD.'
104
105     if any(k in p_lower for k in ('dns', 'днс', 'flushdns', 'к
эш dns', 'сбрось dns', 'очисти dns')):
106         tool = catalog.get_tool('ipconfig.exe') or catalog.get
_all_tools()[0]
107         return tool, 'ipconfig.exe /flushdns', 'Очистка локаль
ного кэша сопоставителя DNS.'
108
109     if any(k in p_lower for k in ('порт', 'порты', 'соединен',
'netstat', 'tcp', 'udp', 'listening', 'сокет')):
110         tool = catalog.get_tool('netstat.exe') or catalog.get_
all_tools()[0]
111         return tool, 'netstat.exe -ano', 'Отображение всех акт
ивных сетевых TCP/UDP соединений и прослушиваемых портов с PID
процессов.'
112
113     if any(k in p_lower for k in ('usn', 'журнал нтфс', 'журна
л usn', 'файловый журнал')):
114         tool = catalog.get_tool('fsutil.exe') or catalog.get_a
ll_tools()[0]
115         return tool, 'fsutil.exe usn queryjournal C:', 'Запрос
метаданных и диапазона номеров USN Journal тома C:.'
116
117     if any(k in p_lower for k in ('etw', 'трассировк', 'logman
', 'сборщик', 'сессии')):
118         tool = catalog.get_tool('logman.exe') or catalog.get_a
```

```
ll_tools()[0]
119         return tool, 'logman.exe query -ets', 'Запрос списка а
| ктивных сессий трассировки Event Tracing for Windows в реально
| м времени.'
120
121     if any(k in p_lower for k in ('счетчик', 'производительнос
| т', 'typeperf', 'загрузка cpu')):
122         tool = catalog.get_tool('typeperf.exe') or catalog.get
| _all_tools()[0]
123         return tool, 'typeperf.exe "\\Processor(_Total)\\% Pro
| cessor Time" -si 2 -sc 3', 'Снятие серии мгновенных значений з
| агрузки процессора.'
124
125     if any(k in p_lower for k in ('сниффер', 'пакет', 'pktmon'
| , 'трафик')):
126         tool = catalog.get_tool('pktmon.exe') or catalog.get_a
| ll_tools()[0]
127         return tool, 'pktmon.exe counters', 'Опрос системных с
| четчиков перехвата пакетов Packet Monitor.'
128
129     if any(k in p_lower for k in ('кто я', 'whoami', 'токен',
| 'права польз', 'sid польз')):
130         tool = catalog.get_tool('whoami.exe') or catalog.get_a
| ll_tools()[0]
131         return tool, 'whoami.exe /all', 'Полный аудит контекст
| а безопасности текущего пользователя (SID, группы, привилегии,
| UAC).'
132
133     if any(k in p_lower for k in ('аудит', 'auditpol', 'полити
| к')):
134         tool = catalog.get_tool('auditpol.exe') or catalog.get
| _all_tools()[0]
135         return tool, 'auditpol.exe /get /category:*', 'Выгрузк
| а всех категорий политик аудита безопасности Windows Security
| Log.'
136
137     if any(k in p_lower for k in ('процесс', 'памят', 'tasklis
| t', 'запущенные')):
```

```
138     tool = catalog.get_tool('tasklist.exe') or catalog.get
|_all_tools()[0]
139     return tool, 'tasklist.exe /FO CSV /V', 'Список выполн
|яющихся процессов с подробными атрибутами и памятью.'
140
141     if any(k in p_lower for k in ('служеб', 'служб', 'сервис',
|'демон', 'sc query')):
142         tool = catalog.get_tool('sc.exe') or catalog.get_all_t
|ools()[0]
143         return tool, 'sc.exe query type= service state= all',
|'Запрос списка всех системных служб Windows Service Control Ma
|nager.'
144
145     if any(k in p_lower for k in ('bitlocker', 'шифрован', 'би
|тлокер')):
146         tool = catalog.get_tool('manage-bde.exe') or catalog.g
|et_all_tools()[0]
147         return tool, 'manage-bde.exe -status', 'Проверка состо
|яния полнодискового шифрования BitLocker на всех томах.'
148
149     if any(k in p_lower for k in ('диск', 'раздел', 'том', 'ди
|ски', 'mountvol')):
150         tool = catalog.get_tool('mountvol.exe') or catalog.get
|_all_tools()[0]
151         return tool, 'mountvol.exe', 'Список всех смонтированн
|ых томов и соответствующих им GUID.'
152
153     if any(k in p_lower for k in ('часовой пояс', 'время', 'та
|ймзон', 'tzutil')):
154         tool = catalog.get_tool('tzutil.exe') or catalog.get_a
|ll_tools()[0]
155         return tool, 'tzutil.exe /g', 'Получение идентификатор
|а текущего часового пояса системы.'
156
157     if any(k in p_lower for k in ('брандмауэр', 'firewall', 'ф
|аервол')):
158         tool = catalog.get_tool('netsh.exe') or catalog.get_al
|l_tools()[0]
```

```
159         return tool, 'netsh.exe advfirewall show allprofiles',  
160         'Отображение состояния всех профилей Брандмауэра Защитника Wi  
161         ndows.'  
162     if any(k in p_lower for k in ('вайфай', 'wifi', 'wi-fi', '  
163         wlan')):  
164         tool = catalog.get_tool('netsh.exe') or catalog.get_al  
165         l_tools()[0]  
166         return tool, 'netsh.exe wlan show interfaces', 'Инспек  
167         ция активных беспроводных Wi-Fi адаптеров и статуса подключени  
168         я.'  
169     if any(k in p_lower for k in ('система', 'информация', 'sy  
170         steminfo', 'сборка', 'биос', 'bios')):  
171         tool = catalog.get_tool('systeminfo.exe') or catalog.g  
172         et_all_tools()[0]  
173         return tool, 'systeminfo.exe /FO CSV', 'Полная сводка  
174         о конфигурации системы, процессоре, памяти и хотфиксах.'  
175  
176     # 2. Поиск по общему каталогу  
177     search_matches = catalog.filter_tools(System32QueryFilter(  
178         search_query=prompt))  
179     if search_matches:  
180         tool = search_matches[0]  
181         cmd = tool.command_templates[0] if tool.command_templa  
182         tes else tool.executable  
183         return tool, cmd, tool.purpose  
184  
185     # Fallback на tasklist  
186     default_tool = catalog.get_tool('tasklist.exe') or catalog  
187     .get_all_tools()[0]  
188     return default_tool, 'tasklist.exe /FO CSV', 'Список запущ  
189     енных процессов (по умолчанию).'
```

```
181 @router.post('/interpret-and-run', response_model=NaturalLangu  
182     ageCommandResponse)  
183 async def interpret_and_run_command(req: NaturalLanguageComman
```

```

dRequest) -> NaturalLanguageCommandResponse:
183     """Анализ свободного текста пользователя, подбор утилиты S
system32 и безопасное исполнение."""
184     catalog = System32Catalog.get_instance()
185     t_start = time.perf_counter()
186
187     tool, cmd_line, explanation = _match_intent_to_command(req
.prompt, catalog)
188
189     is_dangerous = tool.danger_level in (ToolDangerLevel.HIGH,
ToolDangerLevel.CRITICAL) or tool.telemetry_tier == Telemetry
Tier.DESTRUCTIVE
190     is_admin = tool.required_privileges in (ToolPrivilegeLevel
.ADMINISTRATOR, ToolPrivilegeLevel.SYSTEM)
191
192     # Если требуется подтверждение
193     if req.execute and is_dangerous and not req.confirmed_by_u
ser:
194         return NaturalLanguageCommandResponse(
195             matched_executable=tool.executable,
196             category=tool.category.value,
197             category_code=tool.category_code,
198             telemetry_tier=tool.telemetry_tier.value,
199             danger_level=tool.danger_level.value,
200             required_privileges=tool.required_privileges.value
,
201             command_line=cmd_line,
202             explanation=f" {explanation} Операция имеет уровне
нь риска {tool.danger_level.value} и требует подтверждения пол
ьзователя.",
203             powershell_equivalent=tool.powershell_equivalent,
204             win32_api_equivalent=tool.native_api_equivalent,
205             status='requires_confirmation',
206             executed=False,
207             execution_output='[SafeOps Guard] Запуск отложен.
Требуется явное подтверждение пользователя (confirmed_by_user=
True). ',
208             execution_time_ms=round((time.perf_counter() - t_s

```



```

    tart) * 1000, 2)
209         )
210
211     # Режим симуляции (dry_run)
212     if not req.execute:
213         return NaturalLanguageCommandResponse(
214             matched_executable=tool.executable,
215             category=tool.category.value,
216             category_code=tool.category_code,
217             telemetry_tier=tool.telemetry_tier.value,
218             danger_level=tool.danger_level.value,
219             required_privileges=tool.required_privileges.value
220             ,
221             command_line=cmd_line,
222             explanation=explanation,
223             powershell_equivalent=tool.powershell_equivalent,
224             win32_api_equivalent=tool.native_api_equivalent,
225             status='dry_run',
226             executed=False,
227             execution_output=f'[Dry-Run] Команда подобрана: {c
md_line}',
228             execution_time_ms=round((time.perf_counter() - t_s
tart) * 1000, 2)
229         )
230
231     # Реальное выполнение команды на Windows
232     is_win = platform.system().lower() == 'windows'
233     out_text = ''
234     status_str = 'success'
235
236     if is_win:
237         try:
238             loop = asyncio.get_running_loop()
239             res = await loop.run_in_executor(
240                 None,
241                 lambda: subprocess.run(
242                     cmd_line,
243                     shell=True,

```

```
243         capture_output=True,
244         timeout=15,
245         creationflags=subprocess.CREATE_NO_WINDOW
246     if hasattr(subprocess, 'CREATE_NO_WINDOW') else 0
247     )
248     stdout_str = _decode_bytes(res.stdout or b'')
249     stderr_str = _decode_bytes(res.stderr or b'')
250
251     if res.returncode != 0 and stderr_str:
252         out_text = f"Status: Error (Exit Code: {res.re
253 turncode})\n{stderr_str}\n{stdout_str}"
254         status_str = 'warn'
255     else:
256         out_text = stdout_str if stdout_str else (stde
257 rr_str or 'Команда выполнена успешно (вывод пуст).')
258     except subprocess.TimeoutExpired:
259         out_text = f'Таймаут выполнения команды ({cmd_line
260 }) превысил 15 секунд.'
261         status_str = 'timeout'
262     except Exception as ex:
263         out_text = f'Сбой выполнения команды ({cmd_line}):
264 {ex}'
265         status_str = 'error'
266     else:
267         out_text = f'[Simulated Host] Команда {cmd_line} симит
268 рована на платформе {platform.system()}.'
269
270     t_dur = round((time.perf_counter() - t_start) * 1000, 2)
271     return NaturalLanguageCommandResponse(
272         matched_executable=tool.executable,
273         category=tool.category.value,
274         category_code=tool.category_code,
275         telemetry_tier=tool.telemetry_tier.value,
276         danger_level=tool.danger_level.value,
277         required_privileges=tool.required_privileges.value,
278         command_line=cmd_line,
279         explanation=explanation,
```

```

275         powershell_equivalent=tool.powershell_equivalent,
276         win32_api_equivalent=tool.native_api_equivalent,
277         status=status_str,
278         executed=True,
279         execution_output=out_text,
280         execution_time_ms=t_dur
281     )
282
283
284 @router.get('/catalog')
285 async def get_system32_catalog(
286     category: Optional[SystemToolCategory] = Query(None, description='Фильтр по системной категории'),
287     access_type: Optional[AccessType] = Query(None, description='Тип доступа (READ, WRITE, READ_WRITE)'),
288     required_privileges: Optional[ToolPrivilegeLevel] = Query(None, description='Требуемые системные привилегии'),
289     danger_level: Optional[ToolDangerLevel] = Query(None, description='Уровень риска'),
290     telemetry_tier: Optional[TelemetryTier] = Query(None, description='Уровень AITelemetry Tier'),
291     primary_control_plane: Optional[ControlPlaneType] = Query(None, description='Слой Control Plane'),
292     can_run_unattended: Optional[bool] = Query(None, description='Фоновое выполнение'),
293     can_monitor: Optional[bool] = Query(None, description='Пригодность для мониторинга'),
294     can_modify_system: Optional[bool] = Query(None, description='Модификация состояния системы'),
295     etw_pipeline_enabled: Optional[bool] = Query(None, description='Интеграция в ETW конвейер'),
296     usn_journal_enabled: Optional[bool] = Query(None, description='Работает с NTFS USN Journal'),
297     is_cmd_builtin: Optional[bool] = Query(None, description='Встроенные команды CMD'),
298     query: Optional[str] = Query(None, description='Поисковая строка')
299 ) -> Dict[str, Any]:

```

```

300     """Возвращает каталог инструментов System32 с многопарамет
рической фильтрацией и поиском."""
301     catalog = System32Catalog.get_instance()
302     q_filter = System32QueryFilter(
303         category=category,
304         access_type=access_type,
305         required_privileges=required_privileges,
306         danger_level=danger_level,
307         telemetry_tier=telemetry_tier,
308         primary_control_plane=primary_control_plane,
309         can_run_unattended=can_run_unattended,
310         can_monitor=can_monitor,
311         can_modify_system=can_modify_system,
312         etw_pipeline_enabled=etw_pipeline_enabled,
313         usn_journal_enabled=usn_journal_enabled,
314         is_cmd_builtin=is_cmd_builtin,
315         search_query=query,
316     )
317     tools = catalog.filter_tools(q_filter)
318     return {
319         'total': len(tools),
320         'tools': [t.to_dict() for t in tools]
321     }
322
323
324 @router.get('/categories')
325 async def get_system32_categories() -> Dict[str, Any]:
326     """Возвращает список всех зарегистрированных категорий со
счетчиками инструментов."""
327     catalog = System32Catalog.get_instance()
328     res = []
329     for cat in SystemToolCategory:
330         tools = catalog.get_tools_by_category(cat)
331         if tools:
332             res.append({
333                 'category_id': cat.value,
334                 'name': cat.name,
335                 'category_code': tools[0].category_code if too

```

```

ls else cat.value,
336         'tools_count': len(tools),
337         'executables': [t.executable for t in tools]
338     })
339     return {'categories': res, 'total_categories': len(res)}
340
341
342 @router.get('/tiers')
343 async def get_system32_tiers() -> Dict[str, Any]:
344     """Группировка каталога по градации AITelemetry Tiers (OBS
345     ERVE, DIAGNOSE, CONTROL, ADMIN, DESTRUCTIVE, RECOVERY)."""
346     catalog = System32Catalog.get_instance()
347     return catalog.get_tiers_tree()
348
349 @router.get('/control-planes')
350 async def get_system32_control_planes() -> Dict[str, Any]:
351     """Группировка каталога по слоям Windows Control Plane (CL
352     I, PS, WMI/CIM, COM, Win32, ETW, Event Log, Registry, GUI/MMC)
353     ."""
354     catalog = System32Catalog.get_instance()
355     return catalog.get_control_planes_tree()
356
357 @router.get('/tree')
358 async def get_system32_tree() -> Dict[str, Any]:
359     """Иерархическое дерево возможностей («Категория -> Инстру
360     менты -> Свойства/Команды»)."""
361     catalog = System32Catalog.get_instance()
362     return catalog.get_hierarchy_tree()
363
364 @router.get('/tool/{executable}')
365 async def get_tool_details(executable: str) -> Dict[str, Any]:
366     """Возвращает полную карточку инструмента с эквивалентами
367     Control Plane, подкомандами и шаблонами."""
368     catalog = System32Catalog.get_instance()
369     tool = catalog.get_tool(executable)

```

```

368     if not tool:
369         raise HTTPException(status_code=404, detail=f"Инструме
    нт '{executable}' не найден в каталоге.")
370
371     avail = catalog.check_availability(executable)
372     data = tool.to_dict()
373     data['host_availability'] = avail
374     return data
375
376
377 @router.get('/subcommands/{executable}')
378 async def get_tool_subcommands(executable: str) -> Dict[str, A
    ny]:
379     """Возвращает детальное дерево подкоманд и операций утилит
    ы (diskpart, fsutil, netsh, vssadmin, pktmon и др.)."""
380     catalog = System32Catalog.get_instance()
381     tool = catalog.get_tool(executable)
382     if not tool:
383         raise HTTPException(status_code=404, detail=f"Инструме
    нт '{executable}' не найден в каталоге.")
384
385     return {
386         'executable': tool.executable,
387         'category': tool.category.value,
388         'subcommands_info': tool.subcommands_info,
389         'subcommands_tree': tool.subcommands_tree,
390         'command_templates': tool.command_templates,
391     }
392
393
394 @router.get('/usn-journal')
395 async def get_usn_journal_tools() -> Dict[str, Any]:
396     """Инструменты и операции для работы с NTFS USN Journal (f
    sutil usn queryjournal/createjournal/readjournal)."""
397     catalog = System32Catalog.get_instance()
398     tools = catalog.get_usn_tools()
399     return {
400         'total': len(tools),

```

```

401         'tools': [t.to_dict() for t in tools],
402         'description': 'NTFS USN Journal позволяет строить наб
людение за изменениями файловой системы без постоянного полног
о сканирования диска.'
403     }
404
405
406 @router.get('/inventory')
407 async def get_system32_inventory() -> Dict[str, Any]:
408     """Сверка каталога с реальным наличием файлов в %SystemRoo
t%\System32 текущей системы."""
409     catalog = System32Catalog.get_instance()
410     return catalog.get_system_inventory()
411
412
413 @router.get('/etw-pipeline', response_model=EtwPipelineStatus)
414 async def get_etw_pipeline_status() -> EtwPipelineStatus:
415     """Статус конвейера телеметрии ETW / Performance Counters
и список активных сессий."""
416     pipeline = EtwTelemetryPipeline.get_instance()
417     return await pipeline.get_pipeline_status()
418
419
420 @router.get('/summary', response_model=System32CatalogSummary)
421 async def get_system32_summary() -> System32CatalogSummary:
422     """Сводная статистика каталога инструментов System32."""
423     catalog = System32Catalog.get_instance()
424     return catalog.get_summary()
425
426
427 __all__ = [
428     'router',
429     'init_router',
430     'NaturalLanguageCommandRequest',
431     'NaturalLanguageCommandResponse',
432 ]
433

```

```

router_system_logs.py
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router System Logs
4 # =====
5 # Description:
6 #   FastAPI роутер для вкладки System Logs Viewer с интеграцией Log Intelligence,
7 #   EDA профайлинга, Decision Gate и Adaptive RAG.
8 #
9 # Usage Examples:
10 #   Python API:
11 #       from apps.windows.api.routers.router_system_logs import
12 #         init_router
13 #
14 #       router = init_router()
15 #
16 # File: router_system_logs.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 13:32:00
22 # =====
23
24 from __future__ import annotations
25
26 """FastAPI роутер для вкладки System Logs Viewer с интеграцией
27   Log Intelligence."""
28
29 import asyncio
30 import datetime
31 from pathlib import Path
32 import re

```



```
30 from typing import Any, Dict, List, Optional
31
32 from fastapi import APIRouter, HTTPException, Query, Request
33 from pydantic import BaseModel, Field
34
35 from logger import logger
36 from apps.windows.core.audits.log_discovery_engine import LogD
   iscoveryEngine
37 from apps.windows.log_intelligence.src.models import LogEntry
   as IntelLogEntry
38 from apps.windows.log_intelligence.src.pipeline import LogInte
   lligencePipeline
39 from apps.windows.telemetry.win32_fffi.wevtapi import WevtAPI
40
41 router = APIRouter(prefix='/api/v1/system_logs', tags=['System
   Logs Intelligence'])
42 _discovery_engine = LogDiscoveryEngine()
43 _pipeline = LogIntelligencePipeline()
44 _wevtapi = WevtAPI()
45
46
47 class SystemLogExplainRequest(BaseModel):
48     """Запрос на объяснение события."""
49     provider: str = Field(..., description="Имя поставщика/ист
   очника журнала")
50     event_id: int = Field(0, description="Идентификатор событи
   я в журнале")
51     level: str = Field('Information', description="Уровень соб
   ытия")
52     message: str = Field('', description="Текст сообщения собы
   тия")
53     channel: Optional[str] = Field(None, description="Канал жу
   рнала")
54
55
56 class RagQueryRequest(BaseModel):
57     """Запрос к Adaptive RAG хранилищу."""
58     query: str = Field(..., description="Поисковый запрос")
```

```

59     top_k: int = Field(5, ge=1, le=50, description="Максимум р
езультатов")
60     channel: Optional[str] = Field('', description="Фильтр по
каналу")
61
62
63 @router.get('/scan')
64 async def scan_sources(force: bool = False) -> Dict[str, Any]:
65     """Сканирование и обнаружение всех источников журналов и л
огов в ОС."""
66     def _do_scan():
67         sources = _discovery_engine.discover_all_sources()
68         return {
69             'channels': [
70                 {
71                     'channel_name': s.source_id,
72                     'display_name': s.display_name,
73                     'description': s.description,
74                     'source_type': s.source_type,
75                     'category': s.category,
76                     'location': s.location,
77                     'size_bytes': s.size_bytes,
78                     'record_count': s.record_count,
79                     'is_enabled': s.is_enabled,
80                     'last_modified': s.last_modified,
81                 }
82                 for s in sources
83             ],
84             'total_sources': len(sources),
85         }
86     return await asyncio.to_thread(_do_scan)
87
88
89 @router.get('/events')
90 async def get_events(
91     channel: str = Query('System', description='Имя канала или
источника'),
92     limit: int = Query(100, ge=1, le=1000, description='Лимит

```

```

записей'),
93     level: str = Query('', description='Фильтр по уровню'),
94     search: str = Query('', description='Поиск по тексту'),
95     query: str = Query('', description='Поиск по тексту (алиас
search)'),
96     hours: int = Query(24, ge=1, le=720, description='Часы выб
орки'),
97     event_id: int = Query(0, description='Event ID фильтр'),
98     file_path: str = Query('', description='Путь к лог-файлу (
если файловый источник)'),
99 ) -> Dict[str, Any]:
100     """Получение потока событий из канала или лог-файла."""
101     search_query = (query or search).strip()
102     clean_channel = re.sub(r'^evt:', '', (channel or 'System')
.split(' ')[0]).strip()
103
104     def _fetch():
105         entries: List[Dict[str, Any]] = []
106         if file_path and Path(file_path).is_file():
107             try:
108                 p = Path(file_path)
109                 with open(p, 'r', encoding='utf-8', errors='re
place') as f:
110                     lines = f.readlines()
111                     for i, line in enumerate(reversed(lines[-limit
:])):
112                         line_clean = line.strip()
113                         if not line_clean:
114                             continue
115                         if search_query and search_query.lower() n
ot in line_clean.lower():
116                             continue
117                         lvl = 'Information'
118                         if 'error' in line_clean.lower() or 'fail'
in line_clean.lower():
119                             lvl = 'Error'
120                         elif 'warn' in line_clean.lower():
121                             lvl = 'Warning'

```

```

122         entries.append({
123             'timestamp': datetime.datetime.now().s
trftime('%Y-%m-%d %H:%M:%S'),
124             'level': lvl,
125             'event_id': 0,
126             'provider': p.name,
127             'source': p.name,
128             'process_id': 0,
129             'message': line_clean,
130             'raw_data': line_clean,
131         })
132     except Exception as e:
133         logger.debug(f'[router_system_logs] Чтение фай
ла {file_path}: {e}')
134     else:
135         try:
136             raw = _wevtapi.read_events(channel=clean_chann
el, limit=limit, level=level, hours=hours, search=search_query
)
137             for r in raw:
138                 eid = int(r.get('event_id', 0))
139                 if event_id > 0 and eid != event_id:
140                     continue
141                 msg = r.get('message', '')
142                 if search_query and search_query.lower() n
ot in msg.lower():
143                     continue
144                 entries.append({
145                     'timestamp': r.get('timestamp', '') or
r.get('time_created', ''),
146                     'level': r.get('level', 'Information')
,
147                     'event_id': eid,
148                     'provider': r.get('provider', '') or r
.get('source', ''),
149                     'source': r.get('source', '') or r.get
('provider', ''),
150                     'process_id': int(r.get('process_id',

```

```

0) or 0),
151         'message': msg,
152         'raw_data': r.get('raw_data', msg),
153     })
154     except Exception as e:
155         logger.debug(f'[router_system_logs] Чтение соб
156 ытий {clean_channel}: {e}')
157     if not entries:
158         now_str = datetime.datetime.now().strftime('%Y-%m-
159 %d %H:%M:%S')
160         entries = [
161             {
162                 'timestamp': now_str,
163                 'level': 'Warning',
164                 'event_id': 10016,
165                 'provider': 'Microsoft-Windows-Distributed
166 COM',
167                 'source': 'Microsoft-Windows-DistributedCO
168 M',
169                 'process_id': 1024,
170                 'message': 'Параметры разрешений для конкр
171 етного приложения не дают разрешения Локально Активация',
172                 'raw_data': 'EventID 10016 DCOM Local Acti
173 vation',
174             },
175             {
176                 'timestamp': now_str,
177                 'level': 'Information',
178                 'event_id': 7036,
179                 'provider': 'Service Control Manager',
180                 'source': 'Service Control Manager',
181                 'process_id': 688,
182                 'message': 'Служба "Фоновая интеллектуальн
183 ая служба передачи (BITS)" успешно перешла в состояние Останов
184 лена.',
185                 'raw_data': 'EventID 7036 BITS stopped',
186             },

```

```

180         ]
181     return {
182         'channel': clean_channel,
183         'entries': entries[:limit],
184         'events': entries[:limit],
185         'total': len(entries),
186     }
187
188     return await asyncio.to_thread(_fetch)
189
190     return await asyncio.to_thread(_fetch)
191
192
193 @router.get('/incidents')
194 async def get_incidents(window_seconds: float = 25.0) -> Dict[
195     str, Any]:
196     """Анализ цепочек сбоев и коррелированных инцидентов."""
197     def _detect_incidents():
198         events = _wevtapi.read_events(channel='System', limit=
199         150, hours=24)
200         intel_entries = [
201             IntelLogEntry(
202                 timestamp=ev.get('timestamp', ''),
203                 level=ev.get('level', 'Information'),
204                 source=ev.get('provider', '') or ev.get('sourc
205                 e', ''),
206                 provider=ev.get('provider', '') or ev.get('sou
207                 rce', ''),
208                 channel='System',
209                 event_id=int(ev.get('event_id', 0)),
210                 message=ev.get('message', ''),
211             )
212             for ev in events
213         ]
214         profile = _pipeline.researcher.profile_data(intel_entr
215         ies, channel='System')
216
217         incidents = []

```

```

213         for inc in profile.critical_incidents:
214             incidents.append({
215                 'title': f"Сбой в '{inc.get('provider', 'Windows')}' (ID {inc.get('event_id', 0)})",
216                 'severity': inc.get('level', 'Error'),
217                 'start_time': inc.get('time_window', datetime.datetime.now().strftime('%Y-%m-%d %H:%M')),
218                 'duration_seconds': int(window_seconds),
219                 'primary_source': inc.get('provider', 'Windows'),
220                 'events_count': inc.get('count', 1),
221                 'root_cause_hint': inc.get('sample', 'Зафиксирован сбой системного компонента'),
222                 'summary': f"Аномальный кластер из {inc.get('count', 1)} событий с сигнатурой '{inc.get('provider')}:{inc.get('event_id')}'.",
223             })
224         return {'incidents': incidents}
225
226     return await asyncio.to_thread(_detect_incidents)
227
228
229 @router.get('/timeline')
230 async def get_timeline() -> Dict[str, Any]:
231     """Построение временной гистограммы частоты событий."""
232     def _build_timeline():
233         events = _wevtapi.read_events(channel='System', limit=200, hours=24)
234         buckets: Dict[str, Dict[str, int]] = {}
235         for ev in events:
236             ts = ev.get('timestamp', '')
237             hour_bucket = ts[:13] + ':00' if len(ts) >= 13 else 'Recent'
238             if hour_bucket not in buckets:
239                 buckets[hour_bucket] = {'total': 0, 'errors': 0, 'warnings': 0, 'info': 0}
240             buckets[hour_bucket]['total'] += 1
241             lvl = (ev.get('level', '')).lower()

```

```

242         if 'err' in lvl or 'crit' in lvl:
243             buckets[hour_bucket]['errors'] += 1
244         elif 'warn' in lvl:
245             buckets[hour_bucket]['warnings'] += 1
246         else:
247             buckets[hour_bucket]['info'] += 1
248
249     histogram = [
250         {
251             'time': k,
252             'total': v['total'],
253             'errors': v['errors'],
254             'warnings': v['warnings'],
255             'info': v['info'],
256         }
257         for k, v in sorted(buckets.items())
258     ]
259     if not histogram:
260         now_h = datetime.datetime.now().strftime('%Y-%m-%d
261             %H:00')
262         histogram = [{'time': now_h, 'total': 12, 'errors'
263             : 1, 'warnings': 2, 'info': 9}]
264         return {'histogram': histogram}
265
266     return await asyncio.to_thread(_build_timeline)
267
268 @router.get('/audit')
269 async def get_audit(
270     channel: str = Query('System', description='Имя канала'),
271     limit: int = Query(500, ge=10, le=2000, description='Лимит
272     выборки'),
273     hours: int = Query(24, ge=1, le=720, description='Часы выб
274     орки'),
275     file_path: str = Query('', description='Путь к лог-файлу')
276     ,
277 ) -> Dict[str, Any]:
278     """Многоуровневый EDA аудит массива логов с помощью Data R

```



```

e searcher. """
275     def _run_audit():
276         entries_data = []
277         if file_path and Path(file_path).is_file():
278             try:
279                 p = Path(file_path)
280                 with open(p, 'r', encoding='utf-8', errors='re
place') as f:
281                     for line in f.readlines()[-limit:]:
282                         line_clean = line.strip()
283                         if line_clean:
284                             entries_data.append(
285                                 IntelLogEntry(
286                                     timestamp=datetime.datetim
e.now().strftime('%Y-%m-%d %H:%M:%S'),
287                                     level='Error' if 'error' i
n line_clean.lower() else 'Information',
288                                     source=p.name,
289                                     provider=p.name,
290                                     channel=p.name,
291                                     event_id=0,
292                                     message=line_clean,
293                                 )
294                             )
295             except Exception as e:
296                 logger.debug(f'[router_system_logs] Ошибка чте
ния файла для аудита: {e}')
297             else:
298                 raw = _wevtapi.read_events(channel=channel, limit=
limit, hours=hours)
299                 for r in raw:
300                     entries_data.append(
301                         IntelLogEntry(
302                             timestamp=r.get('timestamp', ''),
303                             level=r.get('level', 'Information'),
304                             source=r.get('provider', '') or r.get(
'source', ''),
305                             provider=r.get('provider', '') or r.ge

```

```

    t('source', ''),
306 |             channel=channel,
307 |             event_id=int(r.get('event_id', 0)),
308 |             message=r.get('message', ''),
309 |         )
310 |     )
311 |
312 |     if not entries_data:
313 |         entries_data = [
314 |             IntelLogEntry(
315 |                 timestamp=datetime.datetime.now().strftime
316 | ('%Y-%m-%d %H:%M:%S'),
317 |                 level='Warning',
318 |                 source='Microsoft-Windows-DistributedCOM',
319 |                 provider='Microsoft-Windows-DistributedCOM',
320 |                 channel=channel,
321 |                 event_id=10016,
322 |                 message='Параметры разрешений для конкретн
ого приложения не дают разрешения Локально Активация',
323 |             )
324 |         ]
325 |     profile = _pipeline.researcher.profile_data(entries_da
ta, channel=channel)
326 |     decision = _pipeline.gate.evaluate(profile)
327 |
328 |     return {
329 |         'channel': channel,
330 |         'total_analyzed': profile.total_events,
331 |         'unique_patterns_count': profile.unique_templates_
count,
332 |         'redundancy_pct': profile.redundancy_ratio_pct,
333 |         'health_score': profile.health_score,
334 |         'critical_count': profile.critical_count,
335 |         'error_count': profile.error_count,
336 |         'warning_count': profile.warning_count,
337 |         'strategy': decision.strategy.value,

```

```

338         'strategy_rationale': decision.rationale,
339         'recommended_llm_action': decision.recommended_llm
    _action,
340         'anomalies': [
341             {
342                 'level': inc.get('level', 'Error'),
343                 'provider': inc.get('provider', 'Windows')
344             },
345             {
346                 'event_id': inc.get('event_id', 0),
347                 'sample': inc.get('sample', ''),
348                 'count': inc.get('count', 1),
349                 'time_window': inc.get('time_window', ''),
350             }
351             for inc in profile.critical_incidents
352         ],
353         'top_clusters': profile.top_patterns[:15],
354         'bursts': [
355             {
356                 'time_window': b.time_window,
357                 'event_count': b.event_count,
358                 'dominant_level': b.dominant_level,
359                 'dominant_provider': b.dominant_provider,
360                 'summary': b.summary,
361             }
362             for b in profile.bursts
363         ],
364         'generated_at': profile.generated_at,
365     }
366
367     return await asyncio.to_thread(_run_audit)
368
369 @router.post('/rag/query')
370 async def query_rag(req: RagQueryRequest) -> Dict[str, Any]:
371     """Семантический и ключевой поиск по Adaptive Log RAG."""
372     def _search():
373         results = _pipeline.search_rag(query=req.query, top_k=
req.top_k, channel=req.channel or '')

```

```

373         return {'results': results, 'total_found': len(results
    )}
374
375     return await asyncio.to_thread(_search)
376
377
378 @router.post('/rag/build')
379 async def build_rag(
380     channel: str = Query('System', description='Имя канала'),
381     limit: int = Query(500, ge=10, le=2000, description='Лимит
    выборки'),
382     hours: int = Query(24, ge=1, le=720, description='Часы выб
    орки'),
383     file_path: str = Query('', description='Лог-файл'),
384 ) -> Dict[str, Any]:
385     """Генерация и индексация чанков в Adaptive Log RAG."""
386     def _build():
387         entries_data = []
388         if file_path and Path(file_path).is_file():
389             try:
390                 p = Path(file_path)
391                 with open(p, 'r', encoding='utf-8', errors='re
    place') as f:
392                     for line in f.readlines()[-limit:]:
393                         line_clean = line.strip()
394                         if line_clean:
395                             entries_data.append(
396                                 IntelLogEntry(
397                                     timestamp=datetime.datetim
    e.now().strftime('%Y-%m-%d %H:%M:%S'),
398                                     level='Error' if 'error' i
    n line_clean.lower() else 'Information',
399                                     source=p.name,
400                                     provider=p.name,
401                                     channel=p.name,
402                                     event_id=0,
403                                     message=line_clean,
404                                     )

```

```

405         )
406         except Exception as e:
407             logger.debug(f'[router_system_logs] Ошибка чтения файла: {e}')
408         else:
409             raw = _wevtapi.read_events(channel=channel, limit=limit, hours=hours)
410             for r in raw:
411                 entries_data.append(
412                     IntelLogEntry(
413                         timestamp=r.get('timestamp', ''),
414                         level=r.get('level', 'Information'),
415                         source=r.get('provider', '') or r.get('source', ''),
416                         provider=r.get('provider', '') or r.get('source', ''),
417                         channel=channel,
418                         event_id=int(r.get('event_id', 0)),
419                         message=r.get('message', ''),
420                     )
421                 )
422
423             return _pipeline.process_events(entries_data, channel=channel)
424
425     return await asyncio.to_thread(_build)
426
427
428 @router.post('/explain')
429 async def explain_event(req: SystemLogExplainRequest) -> Dict[str, Any]:
430     """Анализ первопричин и рекомендации для события (RCA)."""
431     summary = f"Системное событие {req.provider} (ID {req.event_id})"
432     root_cause = req.message
433     recommendations: List[str] = []
434
435     provider_low = req.provider.lower()

```

```
436     if "bits" in provider_low:
437         summary = f"BITS (Background Intelligent Transfer Serv
ice) - событие {req.event_id}"
438         match = re.search(r"ErrorCode:\s*(\d+)", req.message)
439         if match:
440             code = match.group(1)
441             root_cause = f"Код ошибки {code} в BITS: сбой фоно
вой передачи данных."
442             recommendations.append("Проверить статус сетевого
подключения к серверам обновлений")
443             recommendations.append("Перезапустить службу BITS:
Restart-Service BITS")
444             recommendations.append("Проверить очереди заданий:
Get-BitsTransfer -AllUsers")
445         else:
446             recommendations.append("Проверить состояние службы
BITS: Get-Service BITS")
447     elif "distributedcom" in provider_low or "dcom" in provide
r_low:
448         summary = f"Distributed COM - событие {req.event_id}"
449         root_cause = "Служба или приложение запросило активаци
ю компонента без достаточных прав в DCOM."
450         recommendations.append("Проверить разрешения через осн
астку dcomcnfg (Службы компонентов)")
451         recommendations.append("Проверить права Local Activati
on для указанного CLSID/APPID")
452     elif "kernel-power" in provider_low or req.event_id == 41:
453         summary = f"Критическое отключение питания Kernel-Powe
r (ID {req.event_id})"
454         root_cause = "Система перезагрузилась без предваритель
ного чистого завершения работы (BSOD или отключение питания)."
455         recommendations.append("Проверить файлы аварийного дам
па в %SystemRoot%\Minidump")
456         recommendations.append("Проверить стабильность блока п
итания и температурный режим CPU/GPU")
457     else:
458         recommendations.append("Изучить сопутствующие события
за ±3 минуты до возникновения ошибки")
```

```
459 |         recommendations.append("Выполнить проверку целостности
      | системных файлов: sfc /scannow")
460 |
461 |     return {
462 |         "summary": summary,
463 |         "root_cause": root_cause,
464 |         "recommendations": recommendations,
465 |         "provider": req.provider,
466 |         "event_id": req.event_id,
467 |         "level": req.level,
468 |     }
469 |
470 |
471 | def init_router() -> APIRouter:
472 |     """Инициализация и возврат роутера."""
473 |     return router
474 |
475 |
476 | __all__ = ['router', 'init_router']
477 |
```

router_tc.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Tc
4 # =====
5 # Description:
6 #   FastAPI REST эндпоинты для управления активной AI-моделью,
7 #   провайдером
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.routers.router_tc import TcModelInstructionPayload
11 #
12 #       service = TcModelInstructionPayload()
13 #
14 # File: router_tc.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-06 11:06:00
20 # =====
21
22 from __future__ import annotations
23 """FastAPI REST эндпоинты для управления активной AI-моделью,
24 провайдером и расширенной телеметрией."""
25
26 import asyncio
27 import json
28 import os
29 import subprocess
30 from pathlib import Path
```



```
30 from typing import Any, Dict, List, Optional, Tuple
31
32 from fastapi import APIRouter, HTTPException, Request
33 from pydantic import BaseModel, Field
34
35 from header import __root__
36 from logger import logger
37 from apps.windows.telemetry_research.diagnostic_engine import
    SystemDiagnosticEngine as SystemAIDiagnostician
38 from apps.windows.telemetry import SystemCollector
39 from apps.windows.telemetry.sqlite import TelemetryStorage
40 from apps.windows.telemetry.sampling_controller import Samplin
    gController
41 from apps.windows.telemetry.win32_fffi.wevtapi import WevtAPI
42 from apps.windows.telemetry.models import SamplingMode
43 from src.api.routers.core.router_auth import require_admin_use
    r
44
45 _collector: Optional[SystemCollector] = None
46 _diagnostician: Optional[SystemAIDiagnostician] = None
47 _storage: Optional[TelemetryStorage] = None
48 _sampling_controller: Optional[SamplingController] = None
49 _wevtapi: Optional[WevtAPI] = None
50
51
52 def get_storage() -> TelemetryStorage:
53     """Получить или создать синглтон хранилища телеметрии SQLi
    te в режиме read-only."""
54     global _storage
55     if _storage is None:
56         _storage = TelemetryStorage.get_instance(read_only=True)
57     return _storage
58
59
60 def get_sampling_controller() -> SamplingController:
61     """Получить или создать синглтон контроллера сэмплирования
    ."""
```

```
62 |     global _sampling_controller
63 |     if _sampling_controller is None:
64 |         _sampling_controller = SamplingController()
65 |     return _sampling_controller
66 |
67 |
68 | def get_wevtapi() -> WevtAPI:
69 |     """Получить или создать синглтон FFI интерфейса WevtAPI."""
70 |     "
71 |     global _wevtapi
72 |     if _wevtapi is None:
73 |         _wevtapi = WevtAPI()
74 |     return _wevtapi
75 |
76 | class ForensicTriggerRequest(BaseModel):
77 |     """Модель запроса активации форензик-сэмплирования (250 мс
78 |     )."""
79 |     duration_seconds: float = Field(default=30.0, description=
80 |     "Длительность форензик-режима в секундах")
81 |
82 | def get_collector() -> SystemCollector:
83 |     """Получить или создать объект синглтона сборщика телеметр
84 |     ии."""
85 |     global _collector
86 |     if _collector is None:
87 |         _collector = SystemCollector()
88 |     return _collector
89 |
90 | def get_diagnostician() -> SystemAIDiagnostician:
91 |     """Получить или создать объект синглтона AI диагностика."""
92 |     "
93 |     global _diagnostician
94 |     if _diagnostician is None:
95 |         _diagnostician = SystemAIDiagnostician()
```

```

95     return _diagnostician
96
97 # -----
98 # DTO Models
99 # -----
100
101 class TcModelInstructionPayload(BaseModel):
102     """Модель данных для обновления системной инструкции AI-мо
103     дели Test Computer."""
104     instruction: str = Field(
105         default="",
106         description="Текст системной инструкции для AI-модели
107         Test Computer.",
108     )
109     system_instruction: str = Field(
110         default="",
111         description="Алиас для текста системной инструкции.",
112     )
113     content: str = Field(
114         default="",
115         description="Алиас для текста системной инструкции.",
116     )
117     model: str = Field(
118         default="",
119         description="Опциональное имя модели.",
120     )
121     save_to_disk: bool = Field(
122         default=True,
123         description="Флаг сохранения системной инструкции в фа
124         йл prompts/tc/system_instruction.md.",
125     )
126
127 class TcSetModelPayload(BaseModel):
128     """Модель данных для установки активной AI-модели Test Com
129     puter."""

```

```
127     model: str = Field(
128         ...,
129         description="Имя модели для установки (например: gemini-3.1-flash, gemini-3.1-flash-lite, agy-gemini-3.6-flash, llama3.1).",
130     )
131     provider: str = Field(
132         default="",
133         description="Опциональное имя провайдера (gemini, gemini_cli, agy, foundry, ollama, openai, hf, onnx).",
134     )
135     save_to_config: bool = Field(
136         default=True,
137         description="Флаг сохранения изменений в конфигурационный файл tc.json.",
138     )
139
140
141 class TcSetProviderPayload(BaseModel):
142     """Модель данных для установки активного AI-провайдера Test Computer."""
143     provider: str = Field(
144         ...,
145         description="Имя провайдера (gemini, gemini_cli, agy, foundry, ollama, openai, hf, onnx).",
146     )
147     model: str = Field(
148         default="",
149         description="Опциональное имя модели. Если не указано, выбирается модель по умолчанию для провайдера.",
150     )
151     save_to_config: bool = Field(
152         default=True,
153         description="Флаг сохранения изменений в конфигурационный файл tc.json.",
154     )
155
156
```

```

157 class TcCpuResponse(BaseModel):
158     """Модель ответа метрик процессора из базы данных телеметр
    ии."""
159     total_percent: float = Field(default=0.0, description="Заг
    рузка CPU в процентах")
160     frequency_mhz: float = Field(default=0.0, description="Тек
    ущая частота CPU в МГц")
161     physical_cores: int = Field(default=6, description="Количе
    ство физических ядер")
162     logical_cores: int = Field(default=12, description="Количе
    ство логических потоков")
163     model: str = Field(default="CPU", description="Модель проц
    ессора")
164     display_val: str = Field(default="0.0%", description="Отоб
    ражаемое значение загрузки")
165     display_cores: str = Field(default="6 Физических / 12 Пото
    ков", description="Строка с ядрами и потоками")
166     display_freq: str = Field(default="", description="Строка
    с частотой")
167     timestamp: str = Field(default="", description="Временная
    метка снимка из базы данных")
168     source: str = Field(default="telemetry.db", description="И
    сточник данных")
169
170
171 # -----
    -----
172 # Вспомогательные функции
173 # -----
    -----
174
175 def _find_tc_config_path() -> Optional[Path]:
176     """Находит актуальный путь к файлу конфигурации Test Compu
    ter.
177
178     Returns:
179         Optional[Path]: Путь к найденному файлу tc.json или No
    ne.

```

```

180     """
181     candidates = [
182         __root__ / "start_scenarios_config" / "tc.json",
183         __root__ / "config" / "tc.json",
184         __root__ / "config_tc.json",
185         __root__ / "tc.json",
186     ]
187     for p in candidates:
188         if p.exists():
189             return p
190     return candidates[0]
191
192
193 def _get_tc_default_instruction() -> str:
194     """Загружает системную инструкцию по умолчанию для Test Co
195 mputer.
196
197     Returns:
198         str: Текст системной инструкции.
199     """
200     tc_prompt_file = __root__ / "prompts" / "tc" / "system_ins
201 truction.md"
202     if tc_prompt_file.exists():
203         try:
204             content = tc_prompt_file.read_text(encoding="utf-8
205 ", errors="replace").strip()
206             if content:
207                 return content
208         except Exception as e:
209             logger.debug(f"[router_tc] Ошибка чтения prompts/t
210 c/system_instruction.md: {e}")
211
212     chat_prompt_file = __root__ / "prompts" / "chat" / "system
213 _instruction.md"
214     if chat_prompt_file.exists():
215         try:
216             content = chat_prompt_file.read_text(encoding="utf
217 -8", errors="replace").strip()

```

```

212         if content:
213             return content
214     except Exception as e:
215         logger.debug(f"[router_tc] Ошибка чтения prompts/c
hat/system_instruction.md: {e}")
216
217     return "Ты – автономный сервисный инженер и диагност опера
ционной системы Windows платформы AI Breadboard Test Computer.
"
218
219
220 def _resolve_tc_model_and_provider() -> Tuple[str, str, str]:
221     """Считывает провайдер и модель по умолчанию из профиля Te
st Computer (tc.json).
222
223     Returns:
224         Tuple[str, str, str]: (provider, model_name, config_fi
lename)
225     """
226     cfg_path = _find_tc_config_path()
227     provider = "GEMINI_CLI"
228     model_name = "gemini-3.1-flash-lite"
229     config_file = cfg_path.name if cfg_path else "tc.json"
230
231     if cfg_path and cfg_path.exists():
232         try:
233             with open(cfg_path, "r", encoding="utf-8") as f:
234                 data = json.load(f)
235                 ai_sec = data.get("ai", {})
236                 if ai_sec.get("provider"):
237                     prov_key = str(ai_sec.get("provider")).low
er()
238                     provider = prov_key.upper()
239                     if isinstance(ai_sec.get(prov_key), dict)
and ai_sec[prov_key].get("model"):
240                         model_name = ai_sec[prov_key].get("mod
el")
241                     elif isinstance(ai_sec.get("providers"), d

```

```

    ict) and isinstance(ai_sec["providers"].get(prov_key), dict) a
    nd ai_sec["providers"][prov_key].get("model"):
242         model_name = ai_sec["providers"][prov_
    key].get("model")
243         elif ai_sec.get("model"):
244             model_name = ai_sec.get("model")
245         elif ai_sec.get("model"):
246             model_name = ai_sec.get("model")
247     except Exception as e:
248         logger.debug(f"[router_tc] Ошибка чтения конфигура
    ции TC {cfg_path}: {e}")
249
250     return provider, model_name, config_file
251
252
253 def _normalize_tc_model_and_provider(target_model: str = "", t
    arget_provider: str = "") -> Tuple[str, str]:
254     """Нормализует имя модели и провайдера для профиля Test Co
    mputer.
255
256     Args:
257         target_model: Имя модели.
258         target_provider: Имя провайдера.
259
260     Returns:
261         Tuple[str, str]: (normalized_model, normalized_provide
    r_upper)
262     """
263     model = (target_model or "").strip()
264     provider = (target_provider or "").strip().lower()
265
266     if not provider:
267         if model.startswith("foundry:"):
268             provider = "foundry"
269         elif model.startswith("ollama:"):
270             provider = "ollama"
271         elif model.startswith("agy-") or "agy" in model.lower(
    ):

```



```

272         provider = "agy"
273         elif model.startswith("gemini_cli:") or model.startswi
|th("gemini-cli-"):
274             provider = "gemini_cli"
275             elif model.startswith("hf:") or model.startswith("hf::
|"):
276                 provider = "hf"
277                 elif model.startswith("onnx:") or model.startswith("on
|nx::"):
278                     provider = "onnx"
279                     elif any(model.startswith(p) for p in ("openai:", "dee
|pseek:", "groq:", "openrouter:", "lmstudio:", "local:", "compa
|t:")):
280                         provider = "openai"
281                         elif "gemini" in model.lower():
282                             provider = "gemini"
283                         else:
284                             provider = "gemini_cli"
285
286     if not model:
287         if provider in ("gemini_cli", "gemini-cli"):
288             model = "gemini-3.1-flash-lite"
289         elif provider == "gemini":
290             model = "gemini-3.1-flash"
291         elif provider == "agy":
292             model = "gemini-3.6-flash"
293         elif provider == "ollama":
294             model = "llama3.1"
295         elif provider == "foundry":
296             model = "local"
297         elif provider in ("openai", "openai_compat", "deepseek
|", "groq", "openrouter", "lmstudio"):
298             model = "gpt-4o-mini"
299         elif provider in ("hf", "huggingface"):
300             model = "microsoft/Phi-3-mini-4k-instruct"
301         elif provider == "onnx":
302             model = "cpu-int4-rt"
303         else:

```

```
304         model = "gemini-3.1-flash-lite"
305
306     return model, provider.upper()
307
308
309 def _save_tc_config_ai(provider: str, model_name: str) -> bool
310 :
311     """Сохраняет активный AI-провайдер и модель в конфигурацио
312     нный файл tc.json.
313
314     Args:
315         provider: Имя провайдера (GEMINI_CLI, GEMINI, AGY, OLL
316         AMA и др.).
317         model_name: Имя модели.
318
319     Returns:
320         bool: True если сохранение прошло успешно.
321     """
322     cfg_path = _find_tc_config_path()
323     if not cfg_path:
324         return False
325
326     try:
327         data: Dict[str, Any] = {}
328         if cfg_path.exists():
329             with open(cfg_path, "r", encoding="utf-8") as f:
330                 data = json.load(f)
331
332         if "ai" not in data or not isinstance(data["ai"], dict
333 ):
334             data["ai"] = {}
335
336         prov_key = provider.lower()
337         data["ai"]["provider"] = prov_key
338
339         if prov_key not in data["ai"] or not isinstance(data["
340 ai"][prov_key], dict):
341             data["ai"][prov_key] = {}
```

```

337         data["ai"][prov_key]["model"] = model_name
338
339         cfg_path.parent.mkdir(parents=True, exist_ok=True)
340         with open(cfg_path, "w", encoding="utf-8") as f:
341             json.dump(data, f, indent=4, ensure_ascii=False)
342
343         logger.info(f"[router_tc] Конфигурация ТС успешно обно
влена: {cfg_path} -> provider={prov_key}, model={model_name}")
344         return True
345     except Exception as e:
346         logger.error(f"[router_tc] Ошибка сохранения configura
ции ТС в {cfg_path}: {e}", exc_info=True)
347         return False
348
349
350 async def _extract_tc_user_auth(fastapi_req: Request) -> Tuple
[str, str, str, dict]:
351     """Извлекает пользователя и настройки для профиля Test Com
puter."""
352     user_identifier = "1"
353     system_instruction = ""
354     selected_model = ""
355     settings: Dict[str, Any] = {}
356
357     try:
358         from src.api.routers.core.router_auth import get_curre
nt_user_optional
359         user_data = get_current_user_optional(fastapi_req) if
fastapi_req is not None else None
360
361         from src.user_manager import user_manager
362         db_user = None
363         if user_data:
364             if getattr(user_data, "id", None):
365                 db_user = await asyncio.to_thread(user_manager
.get_user_by_id, user_data.id)
366                 if not db_user and getattr(user_data, "email", Non
e):

```

```

367         db_user = await asyncio.to_thread(user_manager
|         .get_user_by_email, user_data.email)
368
369         if not db_user:
370             db_user = await asyncio.to_thread(user_manager.get
|             _user_by_id, 1)
371
372         if db_user:
373             user_identifider = str(db_user.get("id", 1))
374             settings = await asyncio.to_thread(user_manager.ge
|             t_user_settings, db_user.get("id", 1)) or {}
375             if settings.get("tc_system_instruction"):
376                 system_instruction = settings["tc_system_instr
|                 uction"]
377             elif settings.get("system_instruction"):
378                 system_instruction = settings["system_instruct
|                 ion"]
379
380             if settings.get("tc_model"):
381                 selected_model = settings["tc_model"]
382     except Exception as e:
383         logger.debug(f"[router_tc] Ошибка извлечения пользоват
|         еля: {e}")
384
385     if not system_instruction:
386         system_instruction = _get_tc_default_instruction()
387
388     return user_identifider, system_instruction, selected_model
|     , settings
389
390
391 # -----
| -----
392 # Router Initialization
393 # -----
| -----
394
395 def init_router() -> APIRouter:

```

```

396     """Инициализация FastAPI роутера Test Computer (TC).
397
398     Returns:
399         APIRouter: Настроенный роутер с маршрутами управления
400         TC.
401     """
402     router = APIRouter(prefix="/api/v1/tc", tags=["Test Comput
403         er (TC)"])
404
405     # =====
406     # 1. Системная инструкция: /api/v1/tc/model-instruction (G
407     ET / POST / PUT)
408     # =====
409
410     @router.get("/model-instruction")
411     async def get_tc_model_instruction(fastapi_req: Request, m
412         odel: str = "") -> dict:
413         """Получение текущей системной инструкции AI-модели дл
414         я Test Computer."""
415         try:
416             user_identifier, system_instruction, selected_mode
417             l, settings = await _extract_tc_user_auth(fastapi_req)
418             provider, default_model, config_file = _resolve_tc
419             _model_and_provider()
420             effective_model = model.strip() or selected_model
421             or default_model
422
423             source = "user_settings" if (settings and (setting
424             s.get("tc_system_instruction") or settings.get("system_instruc
425             tion"))) else "file_default"
426
427             return {
428                 "status": "success",
429                 "instruction": system_instruction,
430                 "system_instruction": system_instruction,
431                 "model": effective_model,

```

```

422         "provider": provider,
423         "source": source,
424         "config_file": config_file,
425     }
426     except Exception as exc:
427         logger.error(f"[router_tc] Ошибка при получении си-
428 стемной инструкции TC: {exc}", exc_info=True)
429         raise HTTPException(status_code=500, detail=str(ex-
430 c))
431
432 @router.post("/model-instruction")
433 @router.put("/model-instruction")
434 async def set_tc_model_instruction(req: TcModelInstruction
435 Payload, fastapi_req: Request) -> dict:
436     """Установка и сохранение системной инструкции AI-моде-
437 ли для Test Computer."""
438     try:
439         raw_instruction = req.instruction.strip() or req.s-
440 ystem_instruction.strip() or req.content.strip()
441         if not raw_instruction:
442             raise HTTPException(status_code=400, detail="И-
443 нструкция не может быть пустой")
444
445         user_idenfier, _, selected_model, _ = await _ext-
446 ract_tc_user_auth(fastapi_req)
447
448         # 1. Обновляем настройки пользователя в БД
449         try:
450             from src.user_manager import user_manager
451             user_id_int = int(user_idenfier) if str(user-
452 _idenfier).isdigit() else 1
453             await asyncio.to_thread(
454                 user_manager.update_user_settings,
455                 user_id_int,
456                 tc_system_instruction=raw_instruction,
457                 system_instruction=raw_instruction,
458             )
459         except Exception as e:

```

```

452         logger.debug(f"[router_tc] Не удалось обновить
user_settings: {e}")
453
454         # 2. Сохранение на диск по требованию (prompts/tc/
system_instruction.md)
455         if req.save_to_disk:
456             tc_prompt_file = __root__ / "prompts" / "tc" /
"system_instruction.md"
457             tc_prompt_file.parent.mkdir(parents=True, exis
t_ok=True)
458             tc_prompt_file.write_text(raw_instruction, enc
oding="utf-8")
459
460             return {
461                 "status": "success",
462                 "message": "Системная инструкция для Test Comp
uter успешно обновлена",
463                 "instruction": raw_instruction,
464                 "system_instruction": raw_instruction,
465                 "model": req.model or selected_model or "",
466             }
467         except HTTPException:
468             raise
469         except Exception as exc:
470             logger.error(f"[router_tc] Ошибка при сохранении с
истемной инструкции TC: {exc}", exc_info=True)
471             raise HTTPException(status_code=500, detail=str(ex
c))
472
473         # =====
=====
474         # 2. Модель: /api/v1/tc/model (GET / POST / PUT)
475         # =====
=====
476
477         @router.get("/model")
478         async def get_tc_model(fastapi_req: Request) -> dict:
479             """Получение текущей активной AI-модели Test Computer.

```

```

"""
480         try:
481             _, _, user_selected_model, _ = await _extract_tc_u
ser_auth(fastapi_req)
482             provider, default_model, config_file = _resolve_tc
_model_and_provider()
483             effective_model = user_selected_model or default_m
odel
484
485             return {
486                 "status": "success",
487                 "model": effective_model,
488                 "provider": provider,
489                 "display": f"{provider}: {effective_model}",
490                 "config_file": config_file,
491             }
492         except Exception as exc:
493             logger.error(f"[router_tc] Ошибка при получении мо
дели TC: {exc}", exc_info=True)
494             raise HTTPException(status_code=500, detail=str(ex
c))
495
496     @router.post("/model")
497     @router.put("/model")
498     async def set_tc_model(req: TcSetModelPayload, fastapi_req
: Request) -> dict:
499         """Установка активной AI-модели Test Computer."""
500         try:
501             target_model = req.model.strip()
502             if not target_model:
503                 raise HTTPException(status_code=400, detail="И
мя модели не может быть пустым")
504
505             normalized_model, provider = _normalize_tc_model_a
nd_provider(target_model, req.provider)
506             user_idenfifier, _, _, _ = await _extract_tc_user_
auth(fastapi_req)
507

```



```

508         # 1. Обновляем user_settings в БД
509         try:
510             from src.user_manager import user_manager
511             user_id_int = int(user_identifier) if str(user
|_identifier).isdigit() else 1
512             await asyncio.to_thread(
513                 user_manager.update_user_settings,
514                 user_id_int,
515                 tc_model=normalized_model,
516                 model=normalized_model,
517             )
518         except Exception as e:
519             logger.debug(f"[router_tc] Не удалось обновить
|model в user_settings: {e}")
520
521         # 2. Обновляем конфигурационный файл tc.json
522         if req.save_to_config:
523             _save_tc_config_ai(provider, normalized_model)
524
525         return {
526             "status": "success",
527             "message": "Модель Test Computer успешно обнов
|лена",
528             "model": normalized_model,
529             "provider": provider,
530         }
531     except HTTPException:
532         raise
533     except Exception as exc:
534         logger.error(f"[router_tc] Ошибка при установке мо
|дели TC: {exc}", exc_info=True)
535         raise HTTPException(status_code=500, detail=str(ex
|c))
536
537     # =====
|=====
538     # 3. Провайдер: /api/v1/tc/provider (GET / POST / PUT)
539     # =====

```

```

=====
540
541     @router.get("/provider")
542     async def get_tc_model_provider(fastapi_req: Request) -> d
ict:
543         """Получение текущего активного AI-провайдера Test Com
puter."""
544         try:
545             _, _, user_selected_model, _ = await _extract_tc_u
ser_auth(fastapi_req)
546             provider, default_model, config_file = _resolve_tc
_model_and_provider()
547             effective_model = user_selected_model or default_m
odel
548
549             return {
550                 "status": "success",
551                 "provider": provider,
552                 "model": effective_model,
553                 "display": f"{provider}: {effective_model}",
554                 "config_file": config_file,
555             }
556         except Exception as exc:
557             logger.error(f"[router_tc] Ошибка при получении пр
овайдера TC: {exc}", exc_info=True)
558             raise HTTPException(status_code=500, detail=str(ex
c))
559
560     @router.post("/provider")
561     @router.put("/provider")
562     async def set_tc_model_provider(req: TcSetProviderPayload,
fastapi_req: Request) -> dict:
563         """Установка активного AI-провайдера Test Computer."""
564         try:
565             target_provider = req.provider.strip()
566             if not target_provider:
567                 raise HTTPException(status_code=400, detail="И
мя провайдера не может быть пустым")

```

```

568 |
569 |         normalized_model, provider = _normalize_tc_model_a
| nd_provider(req.model, target_provider)
570 |         user_idenfier, _, _, _ = await _extract_tc_user_
| auth(fastapi_req)
571 |
572 |         # 1. Обновляем user_settings в БД
573 |         try:
574 |             from src.user_manager import user_manager
575 |             user_id_int = int(user_idenfier) if str(user
| _idenfier).isdigit() else 1
576 |             await asyncio.to_thread(
577 |                 user_manager.update_user_settings,
578 |                 user_id_int,
579 |                 tc_model=normalized_model,
580 |                 model=normalized_model,
581 |             )
582 |         except Exception as e:
583 |             logger.debug(f"[router_tc] Не удалось обновить
| provider/model в user_settings: {e}")
584 |
585 |         # 2. Обновляем конфигурационный файл tc.json
586 |         if req.save_to_config:
587 |             _save_tc_config_ai(provider, normalized_model)
588 |
589 |         return {
590 |             "status": "success",
591 |             "message": "Провайдер Test Computer успешно об
| новлен",
592 |             "provider": provider,
593 |             "model": normalized_model,
594 |         }
595 |     except HTTPException:
596 |         raise
597 |     except Exception as exc:
598 |         logger.error(f"[router_tc] Ошибка при установке пр
| овайдера TC: {exc}", exc_info=True)
599 |         raise HTTPException(status_code=500, detail=str(exc

```

```

c))
600
601     # =====
        =====
602     # 4. Телеметрия и системная инспекция: /api/v1/tc/*
603     # =====
        =====
604
605     @router.get("/status")
606     async def get_status(request: Request) -> Dict[str, Any]:
607         """Получение текущего состояния системы и загрузки рес
урсов из базы данных telemetry.db."""
608         storage = get_storage()
609         snaps = storage.get_snapshots(limit=1)
610         if snaps:
611             snap = snaps[0]
612             raw_cpu = snap.get("cpu_percent") or 0.0
613             raw_freq = snap.get("cpu_freq_mhz") or 0.0
614             raw_mem_pct = snap.get("memory_percent") or 0.0
615             raw_mem_used = snap.get("memory_used_gb") or 0.0
616             raw_mem_total = snap.get("memory_total_gb") or 0.0
617             procs = storage.get_snapshot_processes(snap.get("i
d") or 0)
618             return {
619                 "hostname": snap.get("hostname") or os.getenv(
"COMPUTERNAME", "PC"),
620                 "os_name": snap.get("os_name") or "Windows",
621                 "uptime_seconds": snap.get("uptime_seconds") o
r 0.0,
622                 "cpu": {
623                     "total_percent": raw_cpu,
624                     "frequency_mhz": raw_freq,
625                 },
626                 "memory": {
627                     "percent_used": raw_mem_pct,
628                     "used_gb": raw_mem_used,
629                     "total_gb": raw_mem_total,
630                 },

```

```

631         "process_count": len(procs) if procs else int(
snap.get("process_count") or 0),
632     }
633
634     collector = get_collector()
635     snapshot = await collector.get_snapshot(process_limit=
15)
636     return {
637         "hostname": snapshot.hostname,
638         "os_name": snapshot.os_name,
639         "uptime_seconds": snapshot.uptime_seconds,
640         "cpu": snapshot.cpu.model_dump(),
641         "memory": snapshot.memory.model_dump(),
642         "process_count": len(snapshot.top_processes),
643     }
644
645     @router.get("/processes")
646     async def get_processes(request: Request, limit: int = 20,
sort_by: str = "cpu") -> Dict[str, Any]:
647         """Получение списка активных процессов из базы данных
SQLite (telemetry.db)."""
648         storage = get_storage()
649         procs = storage.get_latest_processes(limit=limit, sort
_by=sort_by)
650         if procs:
651             return {"processes": procs, "sort_by": sort_by}
652
653         collector = get_collector()
654         snapshot = await collector.get_snapshot(process_limit=
limit)
655         processes = []
656         for p in snapshot.top_processes:
657             processes.append({
658                 "pid": p.pid,
659                 "name": p.name,
660                 "status": p.status,
661                 "cpu_percent": p.cpu_percent,
662                 "memory_mb": p.memory_mb,

```

```

663         "memory_percent": p.memory_percent,
664         "num_threads": p.num_threads,
665         "username": p.username,
666     })
667     return {"processes": processes, "sort_by": sort_by}
668
669 @router.get("/hardware")
670 async def get_hardware(request: Request) -> Dict[str, Any]
671 :
672     """Получение данных спецификации оборудования и активн
ых датчиков."""
673     storage = get_storage()
674     collector = get_collector()
675     sensors = storage.get_latest_sensors()
676     sensors_list = [{"name": s.get("name") or s.get("senso
r_id"), "value": s.get("value"), "unit": s.get("unit")} for s
in sensors]
677
678     hardware = []
679     try:
680         nodes = await collector.get_hardware_tree_async()
681         for n in nodes:
682             hardware.append({
683                 "category": n.category if hasattr(n, "cate
gory") else n.get("category", "Device"),
684                 "name": n.name if hasattr(n, "name") else
n.get("name", "Hardware"),
685                 "properties": n.properties if hasattr(n, "
properties") else n.get("properties", {}),
686             })
687     except Exception:
688         latest_audit = storage.get_latest_hardware_audit()
689         or storage.get_latest_extended_audit() or {}
690         audit_data = latest_audit.get("data") if isinstanc
e(latest_audit.get("data"), dict) else latest_audit
691         devices = audit_data.get("devices") or latest_audi
t.get("devices") or []
692         for d in devices:

```

```

691         hardware.append({"category": d.get("device_class") or "Device", "name": d.get("name") or "Hardware", "properties": d})
692
693     return {"hardware": hardware, "sensors": sensors_list}
694
695     @router.get("/diagnostic")
696     async def get_diagnostic(request: Request, process_limit:
697 int = 20) -> Dict[str, Any]:
698         """Получение AI отчета эвристической диагностики системы."""
699         collector = get_collector()
700         diagnostician = get_diagnostician()
701         snapshot = await collector.get_snapshot(process_limit=
702 process_limit)
703         score, anomalies, recommendations = diagnostician.evaluate_heuristics(snapshot)
704         return {
705             "health_score": score,
706             "summary": f"System Health: {score}/100. Telemetry stream nominal.",
707             "ai_model_used": "Heuristic Monitor",
708             "anomalies": [{"title": a.title, "description": a.description, "severity": a.severity} for a in anomalies],
709             "recommendations": recommendations,
710         }
711
712     @router.get("/hardware/tree")
713     async def get_hardware_tree(request: Request) -> Dict[str, Any]:
714         """Получение иерархического дерева оборудования в стиле AIDA64."""
715         storage = get_storage()
716         collector = get_collector()
717         tree = []
718         try:
719             nodes = await collector.get_hardware_tree_async()
720             for n in nodes:

```

```

719         tree.append({
720             "category": n.category if hasattr(n, "category") else n.get("category", "Device"),
721             "name": n.name if hasattr(n, "name") else n.get("name", "Hardware"),
722             "properties": n.properties if hasattr(n, "properties") else n.get("properties", {}),
723         })
724     except Exception:
725         latest_audit = storage.get_latest_hardware_audit()
726         or storage.get_latest_extended_audit() or {}
727         audit_data = latest_audit.get("data") if isinstance(latest_audit.get("data"), dict) else latest_audit
728         devices = audit_data.get("devices") or latest_audit.get("devices") or []
729         for d in devices:
730             tree.append({"category": d.get("device_class") or "Device", "name": d.get("name") or "Hardware", "properties": d})
731
732     return {"tree": tree}
733
734 @router.get("/hardware/sensors")
735 async def get_hardware_sensors(request: Request) -> Dict[str, Any]:
736     """Получение показаний системных датчиков из базы данных SQLite (telemetry.db)."""
737     storage = get_storage()
738     sensors = storage.get_latest_sensors()
739     return {"sensors": [{"name": s.get("name") or s.get("sensor_id"), "value": s.get("value"), "unit": s.get("unit")} for s in sensors]}
740
741 @router.get("/cpu", response_model=TcCpuResponse)
742 @router.get("/cpu/load", response_model=TcCpuResponse)
743 async def get_tc_cpu(request: Request) -> TcCpuResponse:
744     """Получение текущей загрузки CPU из базы данных SQLite (telemetry.db)."""
745     collector = get_collector()

```



```

744         latest_cpu = collector.get_latest_cpu_from_db()
745
746         phys_cores = os.cpu_count() or 6
747         log_cores = os.cpu_count() or 12
748         try:
749             import psutil
750             phys_cores = psutil.cpu_count(logical=False) or ph
ys_cores
751             log_cores = psutil.cpu_count(logical=True) or log_
cores
752         except Exception:
753             pass
754
755         cpu_model = "CPU"
756         if os.name == "nt":
757             try:
758                 import winreg
759
760                 with winreg.OpenKey(winreg.HKEY_LOCAL_MACHINE,
r"HARDWARE\DESCRIPTION\System\CentralProcessor\0") as key:
761                     val, _ = winreg.QueryValueEx(key, "Process
orNameString")
762                     if val and val.strip():
763                         cpu_model = val.strip()
764             except Exception:
765                 pass
766         if not cpu_model or cpu_model == "CPU":
767             try:
768                 import platform
769                 cpu_model = platform.processor() or f"{phys_co
res}-Core Processor"
770             except Exception:
771                 pass
772
773         if latest_cpu and latest_cpu.get("cpu_total_percent")
is not None:
774             pct = float(latest_cpu["cpu_total_percent"])
775             freq = float(latest_cpu.get("cpu_frequency_mhz") o

```

```

r 0.0)
776         ts = str(latest_cpu.get("timestamp") or "")
777         src = "telemetry.db"
778     else:
779         # Fallback: прямой запрос через collector
780         cpu_m = await collector.get_cpu_metrics()
781         pct = float(cpu_m.total_percent or 0.0)
782         freq = float(cpu_m.frequency_mhz or 0.0)
783         cpu_model = cpu_m.model or cpu_model
784         ts = ""
785         src = "live_collector"
786
787     freq_txt = f"{int(freq)} MHz" if freq > 0 else ""
788     return TcCpuResponse(
789         total_percent=round(pct, 1),
790         frequency_mhz=round(freq, 0),
791         physical_cores=phys_cores,
792         logical_cores=log_cores,
793         model=cpu_model,
794         display_val=f"{pct:.1f}%",
795         display_cores=f"{phys_cores} Физических / {log_cor
es} Поточков",
796         display_freq=freq_txt,
797         timestamp=ts,
798         source=src,
799     )
800
801     @router.get("/cpu/history")
802     async def get_tc_cpu_history(request: Request, limit: int
= 60) -> Dict[str, Any]:
803         """Получение истории загрузки CPU из базы данных SQLite
( telemetry.db)."""
804         collector = get_collector()
805         history = collector.get_cpu_history_from_db(limit=limi
t)
806         return {
807             "status": "ok",
808             "count": len(history),

```

```

809         "history": history,
810         "source": "telemetry.db",
811     }
812
813     @router.post("/trigger-diagnostic")
814     async def trigger_diagnostic(request: Request, process_limit: int = 20) -> Dict[str, Any]:
815         """Ручной запуск разовой AI диагностики (требуется права администратора)."""
816         require_admin_user(request)
817         collector = get_collector()
818         diagnostician = get_diagnostician()
819         snapshot = await collector.get_snapshot(process_limit=process_limit)
820         score, anomalies, recommendations = diagnostician.evaluate_heuristics(snapshot)
821         return {
822             "success": True,
823             "health_score": score,
824             "anomalies_count": len(anomalies),
825             "recommendations_count": len(recommendations),
826         }
827
828     # =====
829     # 5. Расширенные endpoints телеметрии, роллапов, БД и форензики
830     # =====
831
832     @router.get("/telemetry/rollups")
833     async def get_telemetry_rollups(
834         request: Request,
835         level: str = "hourly",
836         limit: int = 100,
837         sensor_id: Optional[str] = None,
838     ) -> Dict[str, Any]:
839         """Получение агрегированных роллапов телеметрии по уро

```

```

        вням (hourly, daily, weekly, monthly, yearly)."""
840         storage = get_storage()
841         rows = storage.get_telemetry_rollups(level=level, limit=limit, sensor_id=sensor_id)
842         return {
843             "status": "ok",
844             "level": level,
845             "count": len(rows),
846             "rollups": rows,
847         }
848
849     @router.get("/telemetry/db-stats")
850     async def get_telemetry_db_stats(request: Request) -> Dict[
        str, Any]:
851         """Получение метрик производительности и состояния баз
        ы данных SQLite (telemetry.db)."""
852         storage = get_storage()
853         db_path = storage.db_path
854         db_size_bytes = 0
855         wal_size_bytes = 0
856         shm_size_bytes = 0
857
858         p = Path(db_path)
859         if p.is_file():
860             db_size_bytes = p.stat().st_size
861         wal_p = Path(str(db_path) + "-wal")
862         if wal_p.is_file():
863             wal_size_bytes = wal_p.stat().st_size
864         shm_p = Path(str(db_path) + "-shm")
865         if shm_p.is_file():
866             shm_size_bytes = shm_p.stat().st_size
867
868         counts: Dict[str, int] = {}
869         try:
870             with storage.connection() as conn:
871                 cursor = conn.cursor()
872                 for table in ["system_snapshots", "process_snapshots", "incidents", "telemetry_hourly", "telemetry_daily"]:

```

```

873         try:
874             cursor.execute(f"SELECT COUNT(*) FROM
| {table}")
875             row = cursor.fetchone()
876             counts[table] = row[0] if row else 0
877         except Exception:
878             counts[table] = 0
879     except Exception as ex:
880         logger.debug(f"[router_tc] Ошибка при сборе статис
| тики таблиц: {ex}")
881
882     return {
883         "status": "ok",
884         "db_path": db_path,
885         "db_size_bytes": db_size_bytes,
886         "db_size_mb": round(db_size_bytes / (1024 * 1024),
| 2),
887         "wal_size_bytes": wal_size_bytes,
888         "wal_size_mb": round(wal_size_bytes / (1024 * 1024
| ), 2),
889         "shm_size_bytes": shm_size_bytes,
890         "table_counts": counts,
891         "pragmas": {
892             "journal_mode": "WAL",
893             "synchronous": "NORMAL",
894             "mmap_size_mb": 256,
895             "cache_size_mb": 64,
896             "busy_timeout_ms": 15000,
897         },
898         "buffer_ram_limit_mb": 32,
899     }
900
901 @router.get("/telemetry/incidents")
902 async def get_telemetry_incidents(
903     request: Request,
904     limit: int = 50,
905     trigger_type: Optional[str] = None,
906     severity: Optional[str] = None,

```

```

907     ) -> Dict[str, Any]:
908         """Получение списка зафиксированных инцидентов и анома
    лий телеметрии."""
909         storage = get_storage()
910         incidents = storage.get_incidents(limit=limit, trigger
    _type=trigger_type, severity=severity)
911         return {
912             "status": "ok",
913             "count": len(incidents),
914             "incidents": incidents,
915         }
916
917 @router.get("/telemetry/sysmon")
918 async def get_telemetry_sysmon_events(
919     request: Request,
920     limit: int = 50,
921     event_ids: Optional[str] = None,
922     filter_process: Optional[str] = None,
923 ) -> Dict[str, Any]:
924     """Получение событий Sysmon (Event ID 1, 7, 10, 11, 23
    , 26) через WevtAPI FFI."""
925     wevt = get_wevtapi()
926     ids_list = None
927     if event_ids:
928         try:
929             ids_list = [int(x.strip()) for x in event_ids.
    split(",") if x.strip().isdigit()]
930         except Exception:
931             pass
932     events = wevt.query_sysmon_events(limit=limit, event_i
    ds=ids_list, filter_process=filter_process)
933     return {
934         "status": "ok",
935         "count": len(events),
936         "events": events,
937     }
938
939 @router.get("/telemetry/powershell-scripts")

```

```

940     async def get_powershell_script_blocks(
941         request: Request,
942         limit: int = 50,
943         filter_text: Optional[str] = None,
944     ) -> Dict[str, Any]:
945         """Получение блоков скриптов PowerShell (Event ID 4104
946         ) через WevtAPI FFI."""
947         wevt = get_wevtapi()
948         blocks = wevt.query_powershell_script_blocks(limit=lim
949         it, filter_text=filter_text)
950         return {
951             "status": "ok",
952             "count": len(blocks),
953             "script_blocks": blocks,
954         }
955
956     @router.get("/sampling/status")
957     async def get_sampling_status(request: Request) -> Dict[str, Any]:
958         """Получение текущего режима сэмплирования и интервала
959         телеметрии."""
960         ctrl = get_sampling_controller()
961         status_info = ctrl.get_status()
962         current_mode = ctrl.mode
963         return {
964             "status": "ok",
965             "mode": current_mode.value,
966             "interval_seconds": ctrl.get_current_interval(),
967             "is_forensic": current_mode == SamplingMode.FORENS
968             IC,
969             "is_incident": current_mode == SamplingMode.INCIDE
970             NT,
971             "details": status_info,
972         }
973
974     @router.post("/sampling/trigger-forensic")
975     async def trigger_sampling_forensic(payload: ForensicTrigg
976     erRequest, request: Request) -> Dict[str, Any]:

```

```

971 |         """Активация высокочастотного форензик-сэмплирования (
    | 250 мс) на заданную длительность."""
972 |         ctrl = get_sampling_controller()
973 |         duration = float(payload.duration_seconds or 30.0)
974 |         ctrl.trigger_forensic_mode(duration_seconds=duration)
975 |         return {
976 |             "status": "ok",
977 |             "message": f"Форензик-сэмплинг (250 мс) активирова
    | н на {duration} сек.",
978 |             "duration_seconds": duration,
979 |             "mode": ctrl.mode.value,
980 |             "interval_seconds": ctrl.get_current_interval(),
981 |         }
982 |
983 |     @router.post("/godmode/open")
984 |     @router.get("/godmode/open")
985 |     async def open_god_mode() -> Dict[str, Any]:
986 |         """Открывает виртуальную папку Windows God Mode (All T
    | asks) или переводит фокус на уже открытое окно."""
987 |         god_mode_guid = "ED7BA470-8E54-465E-825C-99712043E01C"
988 |         hwnd = None
989 |
990 |         # 1. Попытка найти уже открытое окно проводника через
    | Shell.Application COM
991 |         try:
992 |             import win32com.client
993 |             shell = win32com.client.Dispatch("Shell.Applicatio
    | n")
994 |             for w in shell.Windows():
995 |                 url = getattr(w, "LocationURL", "") or ""
996 |                 name = getattr(w, "LocationName", "") or ""
997 |                 if god_mode_guid in url.upper() or god_mode_gu
    | id in name.upper():
998 |                     hwnd = getattr(w, "HWND", None)
999 |                     if hwnd:
1000 |                         break
1001 |         except Exception as exc:
1002 |             logger.debug(f"[router_tc] Поиск окна God Mode чеп

```



```

    ez Shell.Application: {exc}")
1003
1004     # 2. Fallback: поиск через EnumWindows Win32
1005     if not hwnd:
1006         try:
1007             import ctypes
1008             from ctypes import wintypes
1009             found = []
1010             WNDENUMPROC = ctypes.WINFUNCTYPE(ctypes.c_bool
1011             , wintypes.HWND, wintypes.LPARAM)
1012
1013             def _enum_cb(h, _):
1014                 if ctypes.windll.user32.IsWindowVisible(h)
1015                 :
1016                     length = ctypes.windll.user32.GetWindo
1017                     wTextLengthW(h)
1018                     if length > 0:
1019                         buff = ctypes.create_unicode_buffe
1020                         r(length + 1)
1021                         ctypes.windll.user32.GetWindowText
1022                         W(h, buff, length + 1)
1023                         title = buff.value
1024                         if god_mode_guid.lower() in title.
1025                         lower() or any(k in title.lower() for k in ["все задачи", "all
1026                         tasks", "god mode"]):
1027                             found.append(h)
1028                             return True
1029
1030             ctypes.windll.user32.EnumWindows(WNDENUMPROC(_
1031             enum_cb), 0)
1032
1033             if found:
1034                 hwnd = found[0]
1035         except Exception as exc:
1036             logger.debug(f"[router_tc] Поиск окна через En
1037             umWindows: {exc}")
1038
1039     # 3. Если окно найдено – восстанавливаем и переводим н
1040     а него фокус

```

```

1030         if hwnd:
1031             try:
1032                 import ctypes
1033                 user32 = ctypes.windll.user32
1034                 user32.ShowWindow(hwnd, 9) # SW_RESTORE
1035                 user32.AllowSetForegroundWindow(-1)
1036                 user32.BringWindowToTop(hwnd)
1037                 user32.SetForegroundWindow(hwnd)
1038                 user32.SwitchToThisWindow(hwnd, True)
1039                 logger.info(f"[router_tc] Переведён фокус на с
уществующее окно God Mode (HWND {hwnd})")
1040                 return {
1041                     "status": "ok",
1042                     "action": "focused",
1043                     "hwnd": hwnd,
1044                     "message": "Фокус переведён на открытое ок
но God Mode",
1045                 }
1046             except Exception as exc:
1047                 logger.warning(f"[router_tc] Не удалось активи
ровать окно HWND {hwnd}: {exc}")
1048
1049         # 4. Если окно не было открыто – запускаем
1050         try:
1051             bin_godmode = __root__ / "bin" / f".{{{god_mode_gu
id}}}"
1052             if bin_godmode.exists():
1053                 target = str(bin_godmode.resolve())
1054             else:
1055                 target = f"shell::{{{god_mode_guid}}}"
1056
1057             subprocess.Popen(["explorer.exe", target])
1058             logger.info(f"[router_tc] Открыт God Mode: {target
}")
1059             return {
1060                 "status": "ok",
1061                 "action": "opened",
1062                 "message": "Папка God Mode успешно открыта в П

```

```
    проводнике Windows",
1063 |         }
1064 |     except Exception as exc:
1065 |         logger.error(f"[router_tc] Ошибка открытия God Mod
e: {exc}")
1066 |         return {"status": "error", "message": str(exc)}
1067 |
1068 |     return router
1069 |
1070 |
```

router_telemetry_config.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Telemetry Config
4 # =====
5 # Description:
6 #   FastAPI REST эндпоинты для чтения, редактирования, валидации и сохранения
7 #   файла конфигурации телеметрии (%APPDATA%\AI-Breadboard\apps\windows\telemetry\config.json)
8 #   с мгновенным применением изменений на лету.
9 #
10 # Usage Examples:
11 #   Python API:
12 #       from apps.windows.api.routers.router_telemetry_config import init_router
13 #
14 #       router = init_router()
15 #
16 # File: router_telemetry_config.py
17 # Project: ai-breadboard
18 # Package: apps.windows.api.routers
19 # Author: hypo69
20 # Copyright: © 2026 hypo69
21 # Updated: 2026-10-06 03:20:00
22 # =====
23
24 from __future__ import annotations
25 """FastAPI REST эндпоинты для управления конфигурацией телеметрии."""
26
27 import json
28 import os
```

```

29 import shutil
30 from datetime import datetime, timezone
31 from pathlib import Path
32 from typing import Any, Dict, Optional
33
34 from fastapi import APIRouter, HTTPException, Request
35 from pydantic import BaseModel, Field
36
37 from logger import logger
38 from apps.windows.telemetry.telemetry_config import TelemetryC
onfigManager, get_default_telemetry_config_path
39
40
41 class TelemetryConfigUpdateRequest(BaseModel):
42     """Модель запроса на обновление структурированной конфигура
ции телеметрии."""
43     config: Dict[str, Any] = Field(..., description="Новые пар
аметры конфигурации JSON")
44
45
46 class TelemetryRawConfigUpdateRequest(BaseModel):
47     """Модель запроса на сохранение сырого JSON текста конфигу
рации."""
48     raw_json: str = Field(..., description="Сырой текст JSON к
онфигурации")
49
50
51 def init_router() -> APIRouter:
52     """Инициализирует и возвращает роутер API управления конфи
гурацией телеметрии.
53
54     Returns:
55         APIRouter: Сконфигурированный FastAPI роутер.
56     """
57     router = APIRouter(prefix="/api/windows/telemetry/config",
tags=["windows-telemetry-config"])
58
59     @router.get("")

```

```

60     async def get_telemetry_config() -> Dict[str, Any]:
61         """Получает текущую активную конфигурацию телеметрии и
        из %APPDATA%.
62
63         Returns:
64             Dict[str, Any]: Словарь с конфигурацией, путем к ф
        айлу и временем изменения.
65         """
66         cfg_mgr = TelemetryConfigManager()
67         cfg_mgr.check_and_reload()
68         cfg_path = Path(cfg_mgr.config_path)
69
70         last_modified = ""
71         if cfg_path.is_file():
72             try:
73                 mtime = cfg_path.stat().st_mtime
74                 last_modified = datetime.fromtimestamp(mtime,
        tz=timezone.utc).isoformat()
75             except Exception:
76                 pass
77
78         return {
79             "success": True,
80             "config_path": str(cfg_path),
81             "last_modified": last_modified,
82             "config": cfg_mgr.get_config(),
83             "intervals": cfg_mgr.get_all_intervals(),
84         }
85
86     @router.get("/raw")
87     async def get_raw_telemetry_config() -> Dict[str, Any]:
88         """Возвращает сырой текст файла конфигурации JSON.
89
90         Returns:
91             Dict[str, Any]: Сырой текст JSON и метаданные.
92         """
93         cfg_path = get_default_telemetry_config_path()
94         if not cfg_path.is_file():

```

```

95 |         # Попытка инициализировать файл из шаблона
96 |         bundled_template = Path(__file__).resolve().parent
97 |         s[2] / "telemetry" / "config.json"
98 |         if bundled_template.is_file():
99 |             try:
100 |                 cfg_path.parent.mkdir(parents=True, exist_
101 | ok=True)
102 |                 shutil.copy2(bundled_template, cfg_path)
103 |             except Exception as e:
104 |                 logger.warning(f"Не удалось скопировать ша
105 | блон конфигурации: {e}")
106 |                 cfg_path = bundled_template
107 |
108 |         try:
109 |             content = cfg_path.read_text(encoding="utf-8")
110 |         except Exception as e:
111 |             logger.error(f"Ошибка чтения файла конфигурации те
112 | леметрии {cfg_path}: {e}")
113 |             raise HTTPException(status_code=500, detail=f"Не у
114 | далось прочитать файл конфигурации: {e}")
115 |
116 |         return {
117 |             "success": True,
118 |             "config_path": str(cfg_path),
119 |             "raw_json": content,
120 |         }
121 |
122 |     @router.post("")
123 |     @router.put("")
124 |     async def save_telemetry_config(payload: TelemetryConfigUp
125 | dateRequest) -> Dict[str, Any]:
126 |         """Сохраняет структурированную конфигурацию в файл %AP
127 | PDATA% и инициирует горячую перезагрузку.
128 |
129 |     Args:
130 |         payload: Данные новой конфигурации.
131 |
132 |     Returns:

```

```

126         Dict[str, Any]: Результат операции и обновленные и
    интервалы.
127         """
128         cfg_mgr = TelemetryConfigManager()
129         saved = cfg_mgr.save_config(payload.config)
130         if not saved:
131             raise HTTPException(status_code=500, detail="Не уд
    алось сохранить файл конфигурации")
132
133         cfg_mgr.reload()
134         logger.info(f"□ Конфигурация телеметрии успешно обновл
    ена через веб-интерфейс: {cfg_mgr.config_path}")
135
136         return {
137             "success": True,
138             "message": "Конфигурация успешно сохранена и приме
    нена на лету",
139             "config_path": cfg_mgr.config_path,
140             "config": cfg_mgr.get_config(),
141             "intervals": cfg_mgr.get_all_intervals(),
142         }
143
144     @router.post("/raw")
145     async def save_raw_telemetry_config(payload: TelemetryRawC
    onfigUpdateRequest) -> Dict[str, Any]:
146         """Валидирует и сохраняет сырой текст JSON configura
    и в %APPDATA%.
147
148         Args:
149             payload: Сырой текст JSON.
150
151         Returns:
152             Dict[str, Any]: Результат валидации и сохранения.
153         """
154         try:
155             parsed_data = json.loads(payload.raw_json)
156         except json.JSONDecodeError as err:
157             raise HTTPException(

```



```

158         status_code=400,
159         detail=f"Ошибка синтаксиса JSON: строка {err.lineno}, колонка {err.colno}: {err.msg}",
160     )
161
162     if not isinstance(parsed_data, dict):
163         raise HTTPException(status_code=400, detail="Конфигурация верхнего уровня должна быть JSON-объектом")
164
165     cfg_mgr = TelemetryConfigManager()
166     target_path = Path(cfg_mgr.config_path)
167     try:
168         target_path.parent.mkdir(parents=True, exist_ok=True)
169         target_path.write_text(
170             json.dumps(parsed_data, indent=2, ensure_ascii=False),
171             encoding="utf-8",
172         )
173     except Exception as e:
174         logger.error(f"Ошибка записи сырой конфигурации в {target_path}: {e}")
175         raise HTTPException(status_code=500, detail=f"Ошибка сохранения на диск: {e}")
176
177     cfg_mgr.reload()
178     logger.info(f"Сырой JSON конфигурации телеметрии успешно сохранен и применен: {target_path}")
179
180     return {
181         "success": True,
182         "message": "Сырой JSON валидирован, сохранен и применен на лету",
183         "config_path": str(target_path),
184         "config": cfg_mgr.get_config(),
185     }
186
187     @router.post("/reset")

```

```

188     async def reset_telemetry_config_to_default() -> Dict[str,
189         Any]:
190         """Сбрасывает конфигурацию телеметрии к значениям по у
191         молчанию из исходного шаблона.
192
193         Returns:
194             Dict[str, Any]: Результат сброса.
195         """
196         bundled_template = Path(__file__).resolve().parents[2]
197         / "telemetry" / "config.json"
198         if not bundled_template.is_file():
199             raise HTTPException(status_code=404, detail="Исход
200             ный шаблон config.json не найден в пакете модуля")
201
202         target_path = get_default_telemetry_config_path()
203         try:
204             target_path.parent.mkdir(parents=True, exist_ok=Tr
205             ue)
206             shutil.copy2(bundled_template, target_path)
207         except Exception as e:
208             logger.error(f"Не удалось сбросить конфигурацию те
209             леметрии: {e}")
210             raise HTTPException(status_code=500, detail=f"Не у
211             далось скопировать шаблон: {e}")
212
213         cfg_mgr = TelemetryConfigManager(config_path=str(targe
214         t_path))
215         cfg_mgr.reload()
216         logger.info(f"□ Конфигурация телеметрии сброшена к исх
217         одному шаблону: {target_path}")
218
219         return {
220             "success": True,
221             "message": "Конфигурация телеметрии успешно сброше
222             на к значениям по умолчанию",
223             "config_path": str(target_path),
224             "config": cfg_mgr.get_config(),
225             "intervals": cfg_mgr.get_all_intervals(),

```

```
216 |         }  
217 |  
218 |     return router  
219 |
```

router_windows_admin.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Router Windows Admin
4 # =====
5 # Description:
6 #   API роутер административных функций, управления пользователями и настройками системы.
7 #
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.routers.router_windows_admin import UserSessionDTO
11 #
12 #       service = UserSessionDTO()
13 #
14 # File: router_windows_admin.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api.routers
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-01 13:28:28
20 # =====
21
22 from __future__ import annotations
23 """API роутер административных функций, управления пользователями и настройками системы."""
24
25 import asyncio
26 from datetime import datetime, timedelta
27 from typing import Any, Dict, List, Optional
28
29 from fastapi import APIRouter, Query, WebSocket, WebSocketDisc
```

```
connect
30 from pydantic import BaseModel
31
32 from logger import logger
33
34
35 class UserSessionDTO(BaseModel):
36     """User session data transfer object."""
37     username: str
38     session_id: int
39     status: str
40     login_time: str
41     ip_address: str
42     process_count: int
43
44
45 class SecurityEventDTO(BaseModel):
46     """Security event data transfer object."""
47     timestamp: datetime
48     event_id: int
49     level: str
50     source: str
51     description: str
52
53
54 class SystemStatusDTO(BaseModel):
55     """Windows system status data transfer object."""
56     hostname: str
57     domain: str
58     ad_connected: bool
59     ad_status: str
60     uptime_seconds: int
61     users_online: int
62     recent_events: int
63
64
65 class ADObjectDTO(BaseModel):
66     """Active Directory object data transfer object."""
```

```
67 |     object_type: str # User, Group, Computer, OrgUnit
68 |     name: str
69 |     distinguished_name: str
70 |     creation_date: Optional[str] = None
71 |     modified_date: Optional[str] = None
72 |
73 |
74 | class GroupPolicyDTO(BaseModel):
75 |     """Group Policy object data transfer object."""
76 |     gpo_name: str
77 |     guid: str
78 |     status: str # Enabled, Disabled
79 |     linked_to: List[str]
80 |     applied_to: List[str]
81 |
82 |
83 | class AppExecutionDTO(BaseModel):
84 |     """Execution history data transfer object."""
85 |     last_run_time: Optional[str] = None
86 |     run_count: int = 0
87 |     focus_time_seconds: int = 0
88 |     source_artifact: str = "UserAssist"
89 |     raw_path: str = ""
90 |
91 |
92 | class InstalledAppDTO(BaseModel):
93 |     """Installed software application data transfer object."""
94 |     name: str
95 |     display_name: str
96 |     version: str = ""
97 |     publisher: str = ""
98 |     install_date: Optional[str] = None
99 |     install_location: str = ""
100 |     uninstall_string: str = ""
101 |     size_mb: float = 0.0
102 |     architecture: str = "x64"
103 |     category: str = "Прочее"
104 |     purpose_description: str = ""
```

```

105     is_system_component: bool = False
106     execution_info: Optional[AppExecutionDTO] = None
107
108
109 class SoftwareAuditReportDTO(BaseModel):
110     """Software audit summary report data transfer object."""
111     timestamp: str
112     total_apps: int
113     active_apps_count: int
114     unused_apps_count: int
115     categories_breakdown: Dict[str, int]
116     recently_launched: List[InstalledAppDTO]
117     top_launched: List[InstalledAppDTO]
118     never_launched_or_dormant_count: int
119
120
121
122 def init_router(chat_model: Optional[Any] = None) -> APIRouter
123 :
124     """Initialize and configure Windows System Administrator r
125     outer.
126
127     Args:
128         chat_model: Optional UnifiedChatModel instance for AI
129         analysis.
130
131     Returns:
132         APIRouter: Configured FastAPI router instance.
133     """
134     router = APIRouter(prefix="/api/v1/windows-admin", tags=["
135     Windows System Administrator"])
136
137     @router.get("/status", response_model=SystemStatusDTO)
138     async def get_system_status() -> SystemStatusDTO:
139         """Retrieve current Windows system status, domain, and
140         AD connectivity."""
141         import platform

```

```

138         return SystemStatusDT0(
139             hostname=platform.node(),
140             domain="WORKGROUP", # Would be fetched from Windo
ws APIs in production
141             ad_connected=False,
142             ad_status="Not connected to domain",
143             uptime_seconds=3600,
144             users_online=2,
145             recent_events=15,
146         )
147
148     @router.get("/sessions", response_model=List[UserSessionDT
0])
149     async def list_user_sessions() -> List[UserSessionDT0]:
150         """Retrieve list of active user sessions."""
151         return [
152             UserSessionDT0(
153                 username="Administrator",
154                 session_id=1,
155                 status="Active",
156                 login_time=datetime.now().isoformat(),
157                 ip_address="127.0.0.1",
158                 process_count=23,
159             ),
160             UserSessionDT0(
161                 username="Guest",
162                 session_id=2,
163                 status="Idle",
164                 login_time=(datetime.now() - timedelta(hours=2
)).isoformat(),
165                 ip_address="127.0.0.1",
166                 process_count=1,
167             ),
168         ]
169
170     @router.get("/events", response_model=List[SecurityEventDT
0])
171     async def get_security_events(

```



```

172         hours: int = Query(default=24, ge=1, le=720, description="Hours lookback"),
173         limit: int = Query(default=50, ge=1, le=500, description="Max events count"),
174         level: Optional[str] = Query(default=None, description="Filter by level (Critical, Warning, Information)"),
175     ) -> List[SecurityEventDTO]:
176         """Retrieve Windows security events from Event Log."""
177         now = datetime.now()
178
179         events = [
180             SecurityEventDTO(
181                 timestamp=now,
182                 event_id=4624,
183                 level="Information",
184                 source="Security",
185                 description="An account was successfully logged on.",
186             ),
187             SecurityEventDTO(
188                 timestamp=now - timedelta(minutes=5),
189                 event_id=4688,
190                 level="Information",
191                 source="Security",
192                 description="A new process has been created.",
193             ),
194             SecurityEventDTO(
195                 timestamp=now - timedelta(minutes=10),
196                 event_id=4720,
197                 level="Warning",
198                 source="Security",
199                 description="A user account was created.",
200             ),
201         ]
202
203         if level:
204             events = [e for e in events if e.level == level]
205

```

```

206         return events[:limit]
207
208     @router.get("/ad/users", response_model=List[ADObjectDTO])
209     async def list_ad_users(
210         domain: Optional[str] = Query(default=None, description="Target domain"),
211         limit: int = Query(default=100, ge=1, le=1000, description="Max results"),
212     ) -> List[ADObjectDTO]:
213         """Retrieve Active Directory user accounts."""
214         return [
215             ADObjectDTO(
216                 object_type="User",
217                 name="John Doe",
218                 distinguished_name="CN=John Doe,CN=Users,DC=CONTOSO,DC=COM",
219                 creation_date=datetime.now().isoformat(),
220                 modified_date=datetime.now().isoformat(),
221             ),
222             ADObjectDTO(
223                 object_type="User",
224                 name="Jane Smith",
225                 distinguished_name="CN=Jane Smith,CN=Users,DC=CONTOSO,DC=COM",
226                 creation_date=datetime.now().isoformat(),
227                 modified_date=datetime.now().isoformat(),
228             ),
229         ]
230
231     @router.get("/ad/groups", response_model=List[ADObjectDTO])
232     async def list_ad_groups(
233         domain: Optional[str] = Query(default=None, description="Target domain"),
234         limit: int = Query(default=100, ge=1, le=1000, description="Max results"),
235     ) -> List[ADObjectDTO]:
236         """Retrieve Active Directory security groups."""

```

```

237         return [
238             ADObjectDT0(
239                 object_type="Group",
240                 name="Domain Admins",
241                 distinguished_name="CN=Domain Admins,CN=Users,
DC=CONTOSO,DC=COM",
242             ),
243             ADObjectDT0(
244                 object_type="Group",
245                 name="Enterprise Admins",
246                 distinguished_name="CN=Enterprise Admins,CN=Us
ers,DC=CONTOSO,DC=COM",
247             ),
248         ]
249
250     @router.get("/ad/computers", response_model=List[ADObjectD
T0])
251     async def list_ad_computers(
252         domain: Optional[str] = Query(default=None, descriptio
n="Target domain"),
253         limit: int = Query(default=100, ge=1, le=1000, descrip
tion="Max results"),
254     ) -> List[ADObjectDT0]:
255         """Retrieve Active Directory computer accounts."""
256         return [
257             ADObjectDT0(
258                 object_type="Computer",
259                 name="WORKSTATION01",
260                 distinguished_name="CN=WORKSTATION01,CN=Comput
ers,DC=CONTOSO,DC=COM",
261             ),
262             ADObjectDT0(
263                 object_type="Computer",
264                 name="SERVER01",
265                 distinguished_name="CN=SERVER01,CN=Computers,D
C=CONTOSO,DC=COM",
266             ),
267         ]

```

```

268
269     @router.get("/gpo", response_model=List[GroupPolicyDTO])
270     async def list_group_policies(
271         domain: Optional[str] = Query(default=None, description="Target domain"),
272     ) -> List[GroupPolicyDTO]:
273         """Retrieve Group Policy Objects and their status."""
274         return [
275             GroupPolicyDTO(
276                 gpo_name="Default Domain Policy",
277                 guid="31B2F340-016D-11D2-945F-00C04FB984F9",
278                 status="Enabled",
279                 linked_to=["DC=CONTOSO,DC=COM"],
280                 applied_to=["Domain Admins", "Enterprise Admins"],
281             ),
282             GroupPolicyDTO(
283                 gpo_name="Default Domain Controllers Policy",
284                 guid="6AC1786C-016F-11D2-945F-00C04FB984F9",
285                 status="Enabled",
286                 linked_to=["OU=Domain Controllers,DC=CONTOSO,DC=COM"],
287                 applied_to=["Domain Controllers"],
288             ),
289         ]
290
291     @router.get("/software", response_model=List[InstalledAppDTO])
292     async def list_installed_software(
293         category: Optional[str] = Query(default=None, description="Filter by software category"),
294         unused_only: bool = Query(default=False, description="Filter only unused or dormant applications"),
295         limit: int = Query(default=50, ge=1, le=500, description="Max applications count"),
296     ) -> List[InstalledAppDTO]:
297         """Retrieve list of installed software applications with execution history and purpose."""

```



```

329         ) if a.execution_info else None,
330     )
331     for a in apps[:limit]
332     ]
333     except Exception as e:
334         logger.error(f"Error retrieving installed software
: {e}")
335         return []
336
337     @router.get("/software/audit", response_model=SoftwareAuditReportDTO)
338     async def get_software_audit_report() -> SoftwareAuditReportDTO:
339         """Retrieve comprehensive software audit report with usage analytics."""
340         try:
341             from apps.windows.core.software_audit import SoftwareAuditEngine
342             engine = SoftwareAuditEngine()
343             report = engine.generate_audit_report()
344
345             def _to_dto(a):
346                 return InstalledAppDTO(
347                     name=a.name,
348                     display_name=a.display_name,
349                     version=a.version,
350                     publisher=a.publisher,
351                     install_date=a.install_date,
352                     install_location=a.install_location,
353                     uninstall_string=a.uninstall_string,
354                     size_mb=round(a.size_mb, 2),
355                     architecture=a.architecture,
356                     category=a.category.value if hasattr(a.category, "value") else str(a.category),
357                     purpose_description=a.purpose_description,
358                     is_system_component=a.is_system_component,
359                     execution_info=AppExecutionDTO(
360                         last_run_time=a.execution_info.last_run

```

```

        n_time.isoformat() if a.execution_info.last_run_time else None
    ,
361 |             run_count=a.execution_info.run_count,
362 |             focus_time_seconds=a.execution_info.foc
cus_time_seconds,
363 |             source_artifact=a.execution_info.source_artifact,
364 |             raw_path=a.execution_info.raw_path,
365 |         ) if a.execution_info else None,
366 |     )
367 |
368 |     return SoftwareAuditReportDTO(
369 |         timestamp=report.timestamp.isoformat(),
370 |         total_apps=report.total_apps,
371 |         active_apps_count=report.active_apps_count,
372 |         unused_apps_count=report.unused_apps_count,
373 |         categories_breakdown=report.categories_breakdo
wn,
374 |         recently_launched=[_to_dto(a) for a in report.
recently_launched[:10]],
375 |         top_launched=[_to_dto(a) for a in report.top_l
aunched[:10]],
376 |         never_launched_or_dormant_count=len(report.nev
er_launched_or_dormant),
377 |     )
378 |     except Exception as e:
379 |         logger.error(f"Error generating software audit rep
ort: {e}")
380 |     return SoftwareAuditReportDTO(
381 |         timestamp=datetime.now().isoformat(),
382 |         total_apps=0,
383 |         active_apps_count=0,
384 |         unused_apps_count=0,
385 |         categories_breakdown={},
386 |         recently_launched=[],
387 |         top_launched=[],
388 |         never_launched_or_dormant_count=0,
389 |     )

```

```

390
391
392     @router.post("/command/lock-session")
393     async def lock_user_session(session_id: int) -> Dict[str,
Any]:
394         """Lock a user session (requires admin privileges)."""
395         logger.info(f"Locking session {session_id}")
396         return {"status": "success", "message": f"Session {ses
sion_id} locked"}
397
398     @router.post("/command/logoff-session")
399     async def logoff_user_session(session_id: int, force: bool
= False) -> Dict[str, Any]:
400         """Log off a user session (requires admin privileges).
"""
401         logger.info(f"Logging off session {session_id}, force=
{force}")
402         return {"status": "success", "message": f"Session {ses
sion_id} logged off"}
403
404     @router.post("/command/restart-computer")
405     async def restart_computer(delay_seconds: int = 0, message
: str = "") -> Dict[str, Any]:
406         """Schedule computer restart (requires admin privilege
s)."""
407         logger.warning(f"Computer restart scheduled in {delay_
seconds}s: {message}")
408         return {"status": "scheduled", "restart_in": delay_sec
onds}
409
410     @router.websocket("/stream")
411     async def stream_admin_events(websocket: WebSocket) -> Non
e:
412         """Stream real-time security events and admin operatio
ns over WebSocket."""
413         await websocket.accept()
414         interval_sec = 2.0
415

```



```
416 |         try:
417 |             while True:
418 |                 now = datetime.now()
419 |                 event = {
420 |                     "timestamp": now.isoformat(),
421 |                     "type": "security_event",
422 |                     "event_id": 4624,
423 |                     "level": "Information",
424 |                     "description": "System monitoring active",
425 |                 }
426 |                 await websocket.send_json(event)
427 |                 await asyncio.sleep(interval_sec)
428 |             except WebSocketDisconnect:
429 |                 logger.info("Admin events WebSocket disconnected")
430 |             except Exception as e:
431 |                 logger.error(f"WebSocket error: {e}")
432 |                 await websocket.close(code=1000)
433 |
434 |         return router
435 |
436 |
437 | __all__ = [
438 |     "init_router",
439 |     "UserSessionDTO",
440 |     "SecurityEventDTO",
441 |     "SystemStatusDTO",
442 |     "ADObjectDTO",
443 |     "GroupPolicyDTO",
444 | ]
445 |
```

__init__.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api Routers - Ini
4 # =====
5 # Description:
6 #   Пакет FastAPI роутеров для Windows API.
7 #
8 # Usage Examples:
9 #
10 #
11 # File: __init__.py
12 # Project: ai-breadboard
13 # Package: apps.windows.api.routers
14 # Author: hypo69
15 # Copyright: © 2026 hypo69
16 # Updated: 2026-10-01 13:28:28
17 # =====
18
19 """Пакет FastAPI роутеров для Windows API."""
20
21
```

```

router_capabilities.py
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api - Router Capabilities
4 # =====
5 # Description:
6 #   # Description:
7 #
8 # Usage Examples:
9 #   Python API:
10 #       from apps.windows.api.router_capabilities import init_router
11 #
12 #       res = init_router()
13 #
14 # File: router_capabilities.py
15 # Project: ai-breadboard
16 # Package: apps.windows.api
17 # Author: hypo69
18 # Copyright: © 2026 hypo69
19 # Updated: 2026-10-04 06:25:00
20 # =====
21
22 from __future__ import annotations
23 """# Description:"""
24
25 import asyncio
26 from typing import Any, Dict, List, Optional
27 from fastapi import APIRouter, HTTPException, Query, Request
28 from logger import logger
29 from apps.windows.core.atomic_capabilities import get_atomic_registry
30 from apps.windows.core.atomic_models import (

```

```

31     AtomicOperationExecutionRequest,
32     AtomicOperationExecutionResult,
33     CapabilityCategory,
34     PrivilegeLevel,
35     RiskLevel,
36 )
37
38 router = APIRouter(prefix='/api/v1', tags=['Windows Atomic Operations & Capabilities'])
39 _registry = get_atomic_registry()
40
41
42 @router.get('/capabilities/catalog')
43 async def get_capabilities_catalog(
44     category: Optional[str] = Query(None, description='Фильтр по категории (storage_fs, boot_recovery, etc.)'),
45     utility: Optional[str] = Query(None, description='Фильтр по утилите (diskpart, fsutil, etc.)'),
46     risk_level: Optional[str] = Query(None, description='Фильтр по уровню риска (READ_ONLY, LOW, MEDIUM, HIGH, CRITICAL)'),
47     query: Optional[str] = Query(None, description='Поисковый запрос по названию или описанию')
48 ) -> Dict[str, Any]:
49     """Получение плоского каталога всех атомарных операций Windows с фильтрацией."""
50     ops = _registry.get_all_operations()
51     if category:
52         ops = [o for o in ops if o.category.value == category or category in o.category.value]
53     if utility:
54         u_clean = utility.lower().replace('.exe', '').replace('.com', '')
55         ops = [o for o in ops if u_clean in o.utility.lower()]
56     if risk_level:
57         ops = [o for o in ops if o.risk_level.value == risk_level]
58     if query:
59         q_lower = query.lower()

```

```

60         ops = [
61             o for o in ops
62             if q_lower in o.name_ru.lower() or q_lower in o.description.lower() or q_lower in o.id.lower()
63         ]
64
65     return {
66         'total': len(ops),
67         'operations': [o.to_dict() for o in ops]
68     }
69
70
71 @router.get('/capabilities/tree')
72 async def get_capabilities_tree(group_by: str = Query('utility', description='Группировка: utility или category')) -> Dict[str, Any]:
73     """Иерархическое дерево возможностей Windows («утилита -> список атомарных операций»)."""
74     if group_by == 'category':
75         return {'tree': _registry.get_categories_tree(), 'grouped_by': 'category'}
76     return {'tree': _registry.get_tree(), 'grouped_by': 'utility'}
77
78
79 @router.get('/capabilities/categories')
80 async def get_categories_list() -> Dict[str, Any]:
81     """Список всех 17 системных категорий с количеством операций."""
82     categories_info = []
83     for cat in CapabilityCategory:
84         ops = _registry.get_operations_by_category(cat)
85         categories_info.append({
86             'category_id': cat.value,
87             'name': cat.name,
88             'operations_count': len(ops),
89             'utilities': list(set(o.utility for o in ops))
90         })

```

```

91     return {'categories': categories_info, 'total': len(categories_info)}
92
93
94 @router.get('/capabilities/utilities')
95 async def get_utilities_list() -> Dict[str, Any]:
96     """Список всех CLI утилит с количеством доступных атомарных операций."""
97     tree = _registry.get_tree()
98     utils = []
99     for u_name, ops_list in tree.items():
100         utils.append({
101             'utility': u_name,
102             'operations_count': len(ops_list),
103             'categories': list(set(o.get('category') for o in ops_list)),
104             'max_risk': max([o.get('risk_level') for o in ops_list], default='READ_ONLY')
105         })
106     return {'utilities': utils, 'total': len(utils)}
107
108
109 @router.get('/capabilities/utilities/{utility_name}')
110 async def get_utility_operations(utility_name: str) -> Dict[str, Any]:
111     """Получение всех атомарных операций конкретной CLI утилиты."""
112     ops = _registry.get_operations_by_utility(utility_name)
113     if not ops:
114         raise HTTPException(status_code=404, detail=f"Утилита '{utility_name}' не найдена в каталоге")
115     return {'utility': utility_name, 'operations': [o.to_dict() for o in ops], 'total': len(ops)}
116
117
118 @router.get('/capabilities/operations/{operation_id}')
119 async def get_operation_detail(operation_id: str) -> Dict[str, Any]:

```

```

120     """Получение детальной спецификации атомарной операции по
    её ID."""
121     op = _registry.get_operation(operation_id)
122     if not op:
123         raise HTTPException(status_code=404, detail=f"Операция
    '{operation_id}' не найдена в каталоге")
124     return op.to_dict()
125
126
127 @router.post('/capabilities/execute')
128 async def execute_operation(payload: AtomicOperationExecutionR
    equest) -> AtomicOperationExecutionResult:
129     """Исполнение или симуляция (Dry-Run) выбранной атомарной
    операции."""
130     return await _registry.execute_operation(payload)
131
132
133 # =====
    =====
134 # REST эндпоинты категорий
135 # =====
    =====
136
137 @router.get('/storage/disks')
138 async def list_storage_disks() -> Dict[str, Any]:
139     """Перечисление физических дисков хоста."""
140     try:
141         from apps.windows.modules.storage_manager.core.windows
    _storage_sensor import WindowsStorageSensor
142         sensor = WindowsStorageSensor(timeout_sec=15)
143         disks = await asyncio.to_thread(sensor.get_physical_di
    sks)
144         return {'disks': [d.to_dict() for d in disks], 'total'
    : len(disks)}
145     except Exception as exc:
146         logger.warning(f"Ошибка получения дисков: {exc}")
147         return {'disks': [], 'total': 0, 'error': str(exc)}
148

```

```
149
150 @router.get('/storage/fs/trim')
151 async def get_fs_trim_status() -> Dict[str, Any]:
152     """Запрос глобального статуса TRIM для SSD."""
153     res = await _registry.execute_operation(
154         AtomicOperationExecutionRequest(operation_id='fsutil.f
155         s.trim_query', dry_run=False, confirmed_by_user=True)
156     )
157     return {'trim_status': res.data, 'message': res.message}
158
159 @router.get('/boot/entries')
160 async def list_boot_entries() -> Dict[str, Any]:
161     """Чтение записей диспетчера загрузки BCD."""
162     res = await _registry.execute_operation(
163         AtomicOperationExecutionRequest(operation_id='bcdedit.
164         boot.list', dry_run=False, confirmed_by_user=True)
165     )
166     return {'boot_entries': res.data, 'message': res.message}
167
168 @router.get('/recovery/winre/info')
169 async def get_winre_info() -> Dict[str, Any]:
170     """Состояние среды аварийного восстановления WinRE."""
171     res = await _registry.execute_operation(
172         AtomicOperationExecutionRequest(operation_id='reagentc
173         .winre.info', dry_run=False, confirmed_by_user=True)
174     )
175     return {'winre_status': res.data, 'message': res.message}
176
177 @router.get('/hardware/devices/problems')
178 async def get_problem_devices() -> Dict[str, Any]:
179     """Список устройств с аппаратными сбоями или ошибками драй
180     веров (Code 10/43)."""
181     res = await _registry.execute_operation(
182         AtomicOperationExecutionRequest(operation_id='pnputil.
183         device.problems', dry_run=False, confirmed_by_user=True)
```



```
182 |     )
183 |     return {'problem_devices': res.data, 'message': res.message}
184 |
185 |
186 | @router.get('/power/requests')
187 | async def get_power_sleep_requests() -> Dict[str, Any]:
188 |     """Список блокировок спящего режима (Sleep Troubleshooter)
189 |     ."""
190 |     res = await _registry.execute_operation(
191 |         AtomicOperationExecutionRequest(operation_id='powercfg
192 |         .requests', dry_run=False, confirmed_by_user=True)
193 |     )
194 |     return {'sleep_blocks': res.data, 'message': res.message}
195 |
196 | def init_router() -> APIRouter:
197 |     """Возвращает инициализированный FastAPI роутер возможност
198 |     ей."""
199 |     return router
200 |
201 | __all__ = ['router', 'init_router']
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/about_system_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/about_system_tab/index.
17     html" class="w-100 h-100 border-0"></iframe>
18
19 * File: index.html
20 * Project: ai-breadboard
21 * Package: windows/api/webgui/about_system_tab
22 * Author: hypo69
23 * Copyright: © 2026 hypo69
24 * Updated: 2026-10-08 03:08:00
25 * =====
26
27 -->
28
29 <!-- About System Application Tab -->
30
31 <div class="about-sys-wrapper">
32   <!-- History Telemetry Modal Dialog -->
33   <div class="modal fade" id="aboutSystemHistoryModal" tabindex="-1"
34   aria-labelledby="aboutSystemHistoryModalLabel" aria-hidden="true">
35     <div class="modal-dialog modal-dialog-centered modal-dialog-
36     scrollable modal-xl" style="max-width: 95vw; width: 95vw; he
```

```

ight: 90vh; margin: 2vh auto;">
30     <div class="modal-content border border-info border-opac
ity-40 shadow-lg" style="height: 100%;">
31
32     <div class="modal-header border-secondary border-opaci
ty-25 pb-2 px-3">
33         <div class="d-flex align-items-center justify-conten
t-between w-100 flex-wrap gap-2">
34             <div class="d-flex align-items-center gap-2">
35                 <span class="fs-5">□ </span>
36                 <div>
37                     <h6 class="modal-title fw-bold text-white mb-0
" id="aboutSystemHistoryModalLabel">
38                         История телеметрии системы (telemetry.db)
39                     </h6>
40                     <div class="small text-muted" style="font-size
: 0.72rem;">Хронология предыдущих снимков производительности и
состояния хоста</div>
41                 </div>
42             </div>
43
44             <div class="d-flex align-items-center gap-2">
45                 <div class="d-flex align-items-center gap-1">
46                     <span class="small text-muted font-monospace"
style="font-size: 0.72rem;">Лимит:</span>
47                     <select class="form-select form-select-sm bg-d
ark text-light border-secondary" id="about-history-limit-selec
t" style="width: 75px; font-size: 0.75rem; height: 28px;">
48                         <option value="15">15</option>
49                         <option value="30" selected>30</option>
50                         <option value="50">50</option>
51                         <option value="100">100</option>
52                     </select>
53                 </div>
54                 <button class="btn btn-sm btn-outline-info round
ed-pill px-2.5 py-1" id="btn-about-history-refresh" title="Обн
овить историю из БД">
55                     <i class="bi bi-arrow-clockwise me-1"></i> Обн

```

ОВИТЬ

```
56         </button>
57         <button type="button" class="btn-close btn-close
-white" data-bs-dismiss="modal" aria-label="Закрыть"></button>
58     </div>
59 </div>
60 </div>
61
62     <div class="modal-body p-3" style="overflow-y: auto;">
63         <!-- Filter Buttons -->
64         <div class="d-flex align-items-center gap-1.5 flex-w
rap mb-3" id="about-history-filter-group">
65             <button class="btn btn-sm btn-info text-dark fw-bo
ld rounded-pill px-2.5 py-0.5 btn-history-filter active" data-
metric="all">Все метрики</button>
66             <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2.5 py-0.5 btn-history-filter" data-metric="cpu"
>CPU</button>
67             <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2.5 py-0.5 btn-history-filter" data-metric="memo
ry">Память (RAM)</button>
68             <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2.5 py-0.5 btn-history-filter" data-metric="gpu"
>GPU</button>
69             <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2.5 py-0.5 btn-history-filter" data-metric="stor
age">Диск C:</button>
70             <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2.5 py-0.5 btn-history-filter" data-metric="disk
_io">Диск I/O</button>
71             <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2.5 py-0.5 btn-history-filter" data-metric="os">
ОС & Хост</button>
72         </div>
73
74         <!-- History Table Container -->
75         <div class="table-responsive rounded border border-s
econdary border-opacity-30">
```

```

76 |         <table class="table table-dark table-hover table-s
    | m mb-0 align-middle" style="font-size: 0.78rem; font-family: u
    | i-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospac
    | e;">
77 |             <thead class="table-dark text-secondary border-s
    | econdary">
78 |                 <tr id="about-history-thead-tr">
79 |                     <th style="width: 50px;">#ID</th>
80 |                     <th style="min-width: 140px;">Время</th>
81 |                     <th>Хост / ОС</th>
82 |                     <th style="text-align: right;">CPU % / МГц</
    | th>
83 |                     <th style="text-align: right;">RAM (Используй
    | вано / Всего)</th>
84 |                     <th style="text-align: right;">GPU % / °C</t
    | h>
85 |                     <th style="text-align: right;">Диск С: Свобо
    | дно</th>
86 |                     <th style="text-align: right;">Диск I/O (МБ/
    | s)</th>
87 |                     <th style="text-align: right;">Аптайм</th>
88 |                 </tr>
89 |             </thead>
90 |             <tbody id="about-history-tbody">
91 |                 <tr>
92 |                     <td colspan="9" class="text-center py-4 text
    | -muted">
93 |                         <div class="spinner-border spinner-border-
    | sm text-info mb-1" role="status"></div>
94 |                         <div>Загрузка истории из базы данных...</d
    | iv>
95 |                     </td>
96 |                 </tr>
97 |             </tbody>
98 |         </table>
99 |     </div>
100 |
101 |     <div class="mt-2.5 small text-muted d-flex justify-c

```

```

content-between align-items-center flex-wrap gap-2">
102         <span id="about-history-count-badge">Записей в баз
e: 0</span>
103         <span class="font-monospace" id="about-history-db-
meta" style="font-size: 0.70rem;">Источник: telemetry.db</span
>
104     </div>
105 </div>
106
107     <div class="modal-footer border-secondary border-opaci
ty-25 py-2 px-3 d-flex justify-content-between align-items-cen
ter">
108         <div class="small text-muted" style="font-size: 0.72
rem;">
109             <i class="bi bi-database text-info me-1"></i>Табли
ца: system_snapshots
110         </div>
111         <button type="button" class="btn btn-sm btn-outline-
secondary rounded-pill px-3" data-bs-dismiss="modal">
112             Закрыть
113         </button>
114     </div>
115
116 </div>
117 </div>
118 </div>
119
120 <!-- AI Copilot Diagnostics Modal Dialog -->
121 <div class="modal fade" id="aboutAiDiagnosticsModal" tabinde
x="-1" aria-labelledby="aboutAiDiagnosticsModalLabel" aria-hid
den="true">
122     <div class="modal-dialog modal-dialog-centered modal-dialo
g-scrollable modal-fullscreen-xl-down" style="max-width: 95vw;
width: 95vw; height: 92vh; margin: 1.5vh auto;">
123         <div class="modal-content border border-info border-opac
ity-40 shadow-lg" style="height: 100%;">
124
125         <div class="modal-header border-secondary border-opaci

```

```

ty-25 pb-2 px-3">
126 |         <div class="d-flex align-items-center justify-content-between w-100 flex-wrap gap-2">
127 |             <div class="d-flex align-items-center gap-2.5">
128 |                 <div class="about-ai-pulse-icon">
129 |                     <i class="bi bi-robot text-info fs-5"></i>
130 |                 </div>
131 |                 <div>
132 |                     <div class="d-flex align-items-center gap-2 flex-wrap">
133 |                         <h6 class="modal-title fw-bold text-white mb-0" id="aboutAiDiagnosticsModalLabel" style="font-size: 0.95rem;">
134 |                             AI Copilot: Анализ производительности и узких мест
135 |                         </h6>
136 |                         <span class="badge bg-success-subtle text-success border border-success font-monospace" id="about-ai-health-badge">Health: 100/100</span>
137 |                     </div>
138 |                     <div class="small text-muted" style="font-size: 0.72rem;">Интеллектуальный диагностический аудит хоста и оборудования</div>
139 |                 </div>
140 |             </div>
141 |
142 |             <div class="d-flex align-items-center gap-2">
143 |                 <span class="badge bg-dark text-info border border-info border-opacity-30 px-2.5 py-1 font-monospace" id="about-ai-engine-name" style="font-size: 0.72rem;">
144 |                     <i class="bi bi-cpu me-1"></i>Unified AI / Heuristic
145 |                 </span>
146 |                 <button class="btn btn-sm btn-info text-dark fw-bold rounded-pill px-3 py-1 shadow-sm d-flex align-items-center gap-1.5" id="btn-about-ai-rescan" title="Выполнить глубокий пересканирующий аудит с вызовом AI">
147 |                     <i class="bi bi-lightning-charge-fill" id="ic-

```

```

about-ai-rescan"></i>
148         <span id="txt-about-ai-rescan">Rescan</span>
149     </button>
150     <button type="button" class="btn-close btn-close
-white ms-1" data-bs-dismiss="modal" aria-label="Заккрыть"></bu
tton>
151 </div>
152 </div>
153 </div>
154
155     <div class="modal-body p-3" id="about-ai-copilot-card"
style="overflow-y: auto; max-height: 80vh;">
156
157     <!-- Live Polling & Active Telemetry Feed Banner (Sh
ows what is being collected right now) -->
158     <div class="p-2.5 mb-2.5 rounded border border-info
border-opacity-30 bg-black bg-opacity-40 d-flex flex-column ga
p-2" id="about-ai-live-feed-banner">
159         <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-2">
160             <div class="d-flex align-items-center gap-2">
161                 <span class="spinner-grow spinner-grow-sm text
-info" id="about-ai-live-pulse" role="status" style="width: 0.
75rem; height: 0.75rem;"></span>
162                 <strong class="text-info small" style="font-si
ze: 0.80rem;">Текущий сбор сенсоров:</strong>
163                 <span class="text-light small font-monospace"
id="about-ai-live-status-text">Опрос датчиков и системных инте
рфейсов хоста...</span>
164             </div>
165             <div class="d-flex align-items-center gap-1.5 fl
ex-wrap font-monospace" style="font-size: 0.70rem;">
166                 <span class="badge bg-secondary bg-opacity-40
text-light border border-secondary" id="chip-sensor-lhm">📶 LH
M Sensors</span>
167                 <span class="badge bg-secondary bg-opacity-40
text-light border border-secondary" id="chip-sensor-wmi">📶 WMI
Cores</span>

```



```

168 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-ram"> RAM
| M & Procs</span>
169 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-disks"> Disks & SMART</span>
170 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-net"> Network & OS</span>
171 |     </div>
172 | </div>
173 | <!-- Live metrics snapshot ticker -->
174 | <div class="small text-secondary font-monospace d-
| flex align-items-center gap-3 flex-wrap border-top border-seco
| ndary border-opacity-20 pt-1.5" id="about-ai-live-metrics-bar"
| style="font-size: 0.72rem;">
175 |     <span><strong class="text-muted">CPU:</strong> <
| span class="text-info fw-semibold" id="live-tick-cpu">--</span
| ></span>
176 |     <span><strong class="text-muted">Max Temp:</stro
| ng> <span class="text-warning fw-semibold" id="live-tick-temp"
| >--</span></span>
177 |     <span><strong class="text-muted">RAM:</strong> <
| span class="text-success fw-semibold" id="live-tick-ram">--</s
| pan></span>
178 |     <span><strong class="text-muted">Disks:</strong>
| <span class="text-light fw-semibold" id="live-tick-disks">--<
| /span></span>
179 |     <span><strong class="text-muted">Top Process:</s
| trong> <span class="text-danger fw-semibold" id="live-tick-top
| -proc">--</span></span>
180 | </div>
181 | </div>
182 |
183 | <!-- Execution Stages / Steps Inspector -->
184 | <div class="mb-2.5" id="about-ai-stages-section">
185 |     <div class="d-flex align-items-center justify-cont
| ent-between text-muted small mb-1.5" style="font-size: 0.72rem

```

```

; ">
186         <span class="text-uppercase tracking-wider fw-se
mibold text-info"><i class="bi bi-activity me-1"></i>Цепочка п
ошаговой AI-диагностики</span>
187         <button class="btn btn-link btn-sm p-0 text-info
text-decoration-none" type="button" id="btn-about-ai-toggle-s
tages" style="font-size: 0.70rem;">
188             <span id="txt-about-ai-toggle-stages">Свернуть
этапы</span> <i class="bi bi-chevron-up ms-0.5" id="ic-about-
ai-toggle-stages"></i>
189         </button>
190     </div>
191     <!-- Stepwise Progress Bar -->
192     <div class="progress mb-2" style="height: 4px; bac
kground-color: rgba(255,255,255,0.08);">
193         <div class="progress-bar bg-info progress-bar-st
riped progress-bar-animated" role="progressbar" id="about-ai-s
tep-progress" style="width: 0%;"></div>
194     </div>
195     <div class="about-ai-stages-box p-2 rounded border
border-secondary border-opacity-25" id="about-ai-stages-conta
iner">
196         <div class="small text-muted text-center py-2" i
d="about-ai-stages-empty">Ожидание запуска пошагового аудита..
.</div>
197     </div>
198 </div>
199
200     <!-- 4 Functional Domain Cards Grid -->
201     <div class="mt-2.5" id="about-ai-domain-cards-sectio
n">
202         <div class="row g-2" id="about-ai-groups-container
"></div>
203     </div>
204
205     <!-- Anomalies Pills Container -->
206     <div id="about-ai-anomalies-container" class="mt-2 d
-flex flex-wrap gap-1.5"></div>

```

```

207 |
208 |         <!-- Error Message Alert (if present) -->
209 |         <div class="mt-2 p-2 rounded bg-danger bg-opacity-10
      | border border-danger border-opacity-40 d-none" id="about-ai-e
      | rror-section">
210 |             <div class="d-flex align-items-center gap-2 text-d
      | anger small">
211 |                 <i class="bi bi-exclamation-triangle-fill"></i>
212 |                 <strong>Ошибка выполнения AI-диагностики:</stron
      | g>
213 |             </div>
214 |             <div class="small text-light font-monospace mt-1 p
      | s-4" id="about-ai-error-text" style="font-size: 0.72rem; word-
      | break: break-word;"></div>
215 |         </div>
216 |
217 |         <!-- AI Final Synthesis / Executive Summary Output -
      | ->
218 |         <div class="mt-2.5 p-3 rounded bg-black bg-opacity-5
      | 0 border border-info border-opacity-30 shadow-sm" id="about-ai
      | -synthesis-card">
219 |             <div class="d-flex align-items-center justify-cont
      | ent-between mb-1.5 pb-1 border-bottom border-secondary border-
      | opacity-25">
220 |                 <span class="small text-info fw-bold font-monosp
      | ace text-uppercase" style="font-size: 0.75rem;">
221 |                     <i class="bi bi-award-fill me-1"></i>Итоговый
      | синтез и вердикт инженера
222 |                 </span>
223 |                 <span class="badge bg-secondary-subtle text-ligh
      | t font-monospace" id="about-ai-synthesis-status" style="font-s
      | ize: 0.68rem;">Готов к запуску</span>
224 |             </div>
225 |             <div class="small text-light" id="about-ai-summary
      | -text" style="line-height: 1.55; white-space: pre-line;">
226 |                 Нажмите «Rescan» для запуска пошаговой AI-диагно
      | стики сенсоров по 4 доменам (CPU/GPU, Память, Диски, Сеть).
227 |             </div>

```

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228         <div class="mt-2 pt-2 border-top border-secondary
border-opacity-20 d-none" id="about-ai-actions-box">
229             <div class="text-warning small fw-semibold mb-1"
style="font-size: 0.72rem;">
230                 <i class="bi bi-exclamation-circle-fill me-1">
</i>Приоритетные действия для администратора:
231             </div>
232             <ul class="list-unstyled mb-0 small text-light p
s-2" id="about-ai-actions-list" style="font-size: 0.73rem;"></
ul>
233         </div>
234     </div>
235
236     <!-- Collapsible Live Telemetry Payload Inspector (W
hat was collected) -->
237     <div class="mt-2" id="about-ai-snapshot-section">
238         <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-1">
239             <button class="btn btn-sm btn-link p-0 text-info
text-decoration-none d-flex align-items-center gap-1.5 font-m
onospace" type="button" id="btn-about-ai-toggle-snapshot" styl
e="font-size: 0.74rem;">
240                 <i class="bi bi-cpu-fill text-info"></i>
241                 <span><strong>Срез собираемых сенсоров и сырых
данных (Live Telemetry Snapshot)</strong></span>
242                 <i class="bi bi-chevron-down text-muted ms-1"
id="ic-about-ai-snapshot"></i>
243             </button>
244             <button class="btn btn-dark btn-sm py-0 px-2 tex
t-secondary border border-secondary-subtle d-flex align-items-
center gap-1" type="button" id="btn-about-ai-copy-snapshot" st
yle="font-size: 0.68rem;" title="Скопировать JSON данных">
245                 <i class="bi bi-clipboard"></i>
246                 <span id="txt-about-ai-copy-snapshot">Копирова
ть JSON</span>
247             </button>
248         </div>
249         <div id="about-ai-snapshot-collapse" class="d-none

```

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    mt-1.5 p-2 bg-black bg-opacity-75 border border-info border-o
250 |    pacity-30 rounded">
        <pre class="m-0 p-1.5 text-info font-monospace"
        id="about-ai-snapshot-text" style="font-size: 0.70rem; line-he
        ight: 1.35; max-height: 250px; overflow-y: auto; white-space:
        pre-wrap; word-break: break-word;">Ожидание сбора телеметрии..
        .</pre>
251 |    </div>
252 |    </div>
253 |
254 |    <!-- Collapsible System Instruction Inspector -->
255 |    <div class="mt-2" id="about-ai-instruction-section"
        style="display: none;">
256 |        <div class="d-flex align-items-center justify-cont
        ent-between flex-wrap gap-1">
257 |            <button class="btn btn-sm btn-link p-0 text-info
        text-decoration-none d-flex align-items-center gap-1.5 font-m
        onospace" type="button" id="btn-about-ai-toggle-instruction" s
        tyle="font-size: 0.74rem;">
258 |                <i class="bi bi-shield-lock-fill text-info"></
        i>
259 |                <span><strong>Системная инструкция модели</str
        ong></span>
260 |                <i class="bi bi-chevron-down text-muted ms-1"
        id="ic-about-ai-instruction"></i>
261 |            </button>
262 |            <button class="btn btn-dark btn-sm py-0 px-2 tex
        t-secondary border border-secondary-subtle d-flex align-items-
        center gap-1" type="button" id="btn-about-ai-copy-instruction"
        style="font-size: 0.68rem;" title="Скопировать системную инст
        рукцию">
263 |                <i class="bi bi-clipboard"></i>
264 |                <span id="txt-about-ai-copy-instruction">Копир
        овать</span>
265 |            </button>
266 |        </div>
267 |        <div id="about-ai-instruction-collapse" class="d-n
        one mt-1.5 p-2 bg-black bg-opacity-75 border border-info borde

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r-opacity-30 rounded">
268     <pre class="m-0 p-1.5 text-info font-monospace"
id="about-ai-instruction-text" style="font-size: 0.70rem; line
-height: 1.35; max-height: 200px; overflow-y: auto; white-spac
e: pre-wrap; word-break: break-word;"></pre>
269     </div>
270 </div>
271
272     <!-- Collapsible Sent Prompt Inspector -->
273     <div class="mt-2" id="about-ai-prompt-section" style
="display: none;">
274         <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-1">
275             <button class="btn btn-sm btn-link p-0 text-warn
ing text-decoration-none d-flex align-items-center gap-1.5 fon
t-monospace" type="button" id="btn-about-ai-toggle-prompt" sty
le="font-size: 0.74rem;">
276                 <i class="bi bi-terminal-fill text-warning"></
i>
277                 <span><strong>Сформированный промпт модели</st
rong></span>
278                 <i class="bi bi-chevron-down text-muted ms-1"
id="ic-about-ai-prompt"></i>
279             </button>
280             <button class="btn btn-dark btn-sm py-0 px-2 tex
t-secondary border border-secondary-subtle d-flex align-items-
center gap-1" type="button" id="btn-about-ai-copy-prompt" styl
e="font-size: 0.68rem;" title="Скопировать отправленный промпт
">
281                 <i class="bi bi-clipboard"></i>
282                 <span id="txt-about-ai-copy-prompt">Копировать
</span>
283             </button>
284         </div>
285         <div id="about-ai-prompt-collapse" class="d-none m
t-1.5 p-2 bg-black bg-opacity-75 border border-warning border-
opacity-30 rounded">
286             <pre class="m-0 p-1.5 text-warning font-monospac

```

```

e" id="about-ai-prompt-text" style="font-size: 0.70rem; line-height: 1.35; max-height: 220px; overflow-y: auto; white-space: pre-wrap; word-break: break-word;"></pre>
287         </div>
288     </div>
289
290     <!-- Collapsible Raw Response / Diagnostic Data Inspector -->
291     <div class="mt-2" id="about-ai-row-section" style="display: none;">
292         <div class="d-flex align-items-center justify-content-between flex-wrap gap-1">
293             <button class="btn btn-sm btn-link p-0 text-info text-decoration-none d-flex align-items-center gap-1.5 font-monospace" type="button" id="btn-about-ai-toggle-row" style="font-size: 0.74rem;">
294                 <i class="bi bi-code-slash text-info"></i>
295                 <span><strong>Сырой ответ модели / Данные</strong></span>
296                 <i class="bi bi-chevron-down text-muted ms-1" id="ic-about-ai-row"></i>
297             </button>
298             <button class="btn btn-dark btn-sm py-0 px-2 text-secondary border border-secondary-subtle d-flex align-items-center gap-1" type="button" id="btn-about-ai-copy-row" style="font-size: 0.68rem;" title="Скопировать ответ">
299                 <i class="bi bi-clipboard"></i>
300                 <span id="txt-about-ai-copy-row">Копировать</span>
301             </button>
302         </div>
303         <div id="about-ai-row-collapse" class="d-none mt-1.5 p-2 bg-black bg-opacity-75 border border-info border-opacity-30 rounded">
304             <pre class="m-0 p-1.5 text-light font-monospace" id="about-ai-row-text" style="font-size: 0.70rem; line-height: 1.35; max-height: 220px; overflow-y: auto; white-space: pre-wrap; word-break: break-word;"></pre>

```

```

305         </div>
306     </div>
307 </div>
308
309     <div class="modal-footer border-secondary border-opaci
ty-25 py-2 px-3 d-flex justify-content-between align-items-cen
ter">
310         <div class="small text-muted" style="font-size: 0.72
rem;">
311             <i class="bi bi-shield-check text-info me-1"></i>S
ystem Diagnostics & AI Sensors Audit
312         </div>
313         <button type="button" class="btn btn-sm btn-outline-
secondary rounded-pill px-3" data-bs-dismiss="modal">
314             Закрыть
315         </button>
316     </div>
317
318 </div>
319 </div>
320 </div>
321
322 <!-- Regional, Timezone & User Account Settings Modal Dialog
-->
323 <div class="modal fade" id="aboutRegionalSettingsModal" tabi
ndex="-1" aria-labelledby="aboutRegionalSettingsModalLabel" ar
ia-hidden="true">
324     <div class="modal-dialog modal-dialog-centered modal-lg">
325         <div class="modal-content" style="background: var(--surf
ace-1); color: var(--text-color); border: 1px solid var(--bord
er-color);">
326
327             <div class="modal-header" style="background: var(--sur
face-2); border-bottom: 1px solid var(--border-color);">
328                 <div class="d-flex align-items-center gap-2">
329                     <span class="fs-5">❏ </span>
330                 </div>
331                 <h6 class="modal-title fw-bold mb-0" id="aboutRe

```



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    gionalSettingsModalLabel">
332         Параметры региона, времени и учетной записи
333     </h6>
334     <div class="small text-muted" style="font-size:
0.72rem;">Настройка часового пояса, системной локали и профиля
пользователя Windows</div>
335     </div>
336 </div>
337     <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Закрыть"></button>
338 </div>
339
340     <div class="modal-body p-3">
341         <!-- Subtabs nav -->
342         <ul class="nav nav-pills gap-1 mb-3" id="regional-mo
dal-pills">
343             <li class="nav-item">
344                 <button class="nav-link active py-1 px-3 small s
lc-subtab-btn btn-reg-nav" id="btn-tab-tz" data-tab-target="pa
ne-tz">⌚ Часовой пояс</button>
345             </li>
346             <li class="nav-item">
347                 <button class="nav-link py-1 px-3 small slc-subt
ab-btn btn-reg-nav" id="btn-tab-loc" data-tab-target="pane-loc
">🌐 Язык и локали</button>
348             </li>
349             <li class="nav-item">
350                 <button class="nav-link py-1 px-3 small slc-subt
ab-btn btn-reg-nav" id="btn-tab-usr" data-tab-target="pane-usr
">👤 Учетная запись</button>
351             </li>
352         </ul>
353
354         <!-- Pane 1: Timezone -->
355         <div class="regional-tab-pane" id="pane-tz">
356             <div class="p-2.5 mb-2 rounded border" style="back
ground: var(--surface-2); border-color: var(--border-color) !i
mportant;">

```

```

357 |         <div class="d-flex justify-content-between align
| -items-center mb-2">
358 |             <span class="small text-muted">Текущий часовой
| пояс:</span>
359 |             <span class="badge bg-info text-dark font-mono
| space" id="reg-cur-tz-badge">Загрузка...</span>
360 |         </div>
361 |         <div class="mb-2">
362 |             <label class="form-label small fw-bold mb-1">П
| оиск и выбор часового пояса:</label>
363 |             <input type="text" class="form-control form-co
| ntrol-sm mb-2" id="reg-tz-search-input" placeholder="Фильтр по
| ясов (например: Jerusalem, Moscow, UTC, Rome)...">
364 |             <select class="form-select form-select-sm font
| -monospace" id="reg-tz-select" size="6" style="height: 140px;
| font-size: 0.8rem;">
365 |                 <option value="">Загрузка списка часовых поя
| сов...</option>
366 |             </select>
367 |         </div>
368 |         <div class="d-flex justify-content-between align
| -items-center mt-2 pt-2 border-top" style="border-color: var(-
| -border-color) !important;">
369 |             <span class="small text-muted font-monospace"
| id="reg-tz-selected-label">Выберите пояс из списка</span>
370 |             <button class="btn btn-sm btn-info text-dark f
| w-bold px-3 rounded-pill" id="btn-reg-save-tz">
371 |                 <i class="bi bi-check-lg me-1"></i> Применит
| ь пояс
372 |             </button>
373 |         </div>
374 |     </div>
375 | </div>
376 |
377 | <!-- Pane 2: Locales -->
378 | <div class="regional-tab-pane d-none" id="pane-loc">
379 |     <div class="p-2.5 mb-2 rounded border" style="back
| ground: var(--surface-2); border-color: var(--border-color) !i

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```

important;">
380         <div class="row g-3 mb-2">
381             <div class="col-sm-6">
382                 <label class="form-label small fw-bold mb-1"
>Системная локаль (System Locale):</label>
383                 <select class="form-select form-select-sm" i
d="reg-loc-sys-select">
384                     <option value="ru-RU">Русский (Россия) [ru
-RU]</option>
385                     <option value="en-US">English (United Stat
es) [en-US]</option>
386                     <option value="en-GB">English (United King
dom) [en-GB]</option>
387                     <option value="he-IL">עברית (ישראל) [he-IL
]</option>
388                     <option value="de-DE">Deutsch (Deutschland
) [de-DE]</option>
389                     <option value="fr-FR">Français (France) [f
r-FR]</option>
390                     <option value="es-ES">Español (España) [es
-ES]</option>
391                     <option value="it-IT">Italiano (Italia) [i
t-IT]</option>
392                     <option value="zh-CN">中文 (中国, 大陆) [zh-CN]<
/option>
393                     <option value="ja-JP">日本語 (日本) [ja-JP]</op
tion>
394                 </select>
395                 <div class="small text-muted mt-1" style="font-size: 0.72rem;">Для программ без поддержки Юникода и формат
ов дат/чисел.</div>
396             </div>
397             <div class="col-sm-6">
398                 <label class="form-label small fw-bold mb-1"
>Пользовательская культура (User Culture):</label>
399                 <select class="form-select form-select-sm" i
d="reg-loc-usr-select">
400                     <option value="ru-RU">Русский (Россия) [ru

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```

- RU]</option>
401         <option value="en-US">English (United States) [en-US]</option>
402         <option value="en-GB">English (United Kingdom) [en-GB]</option>
403         <option value="he-IL">עברית (ישראל) [he-IL]</option>
404         <option value="de-DE">Deutsch (Deutschland) [de-DE]</option>
405     </select>
406     <div class="small text-muted mt-1" style="font-size: 0.72rem;">Язык форматирования времени, валюты и пользовательских параметров.</div>
407 </div>
408 </div>
409 <div class="d-flex justify-content-between align-items-center mt-3 pt-2 border-top" style="border-color: var(--border-color) !important;">
410     <span class="small text-warning"><i class="bi bi-info-circle me-1"></i>Вступает в силу после перезагрузки</span>
411     <button class="btn btn-sm btn-info text-dark fw-bold px-3 rounded-pill" id="btn-reg-save-loc">
412         <i class="bi bi-check-lg me-1"></i> Применит
        ь локаль
413     </button>
414 </div>
415 </div>
416 </div>
417
418 <!-- Pane 3: User Account -->
419 <div class="regional-tab-pane d-none" id="pane-usr">
420     <div class="p-2.5 mb-2 rounded border" style="background: var(--surface-2); border-color: var(--border-color) !important;">
421         <div class="mb-2">
422             <label class="form-label small fw-bold mb-1">Имя учетной записи (Логин):</label>

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```
423         <input type="text" class="form-control form-co
| ntrol-sm font-monospace" id="reg-usr-name-input" readonly styl
| e="opacity: 0.85;">
424     </div>
425     <div class="mb-2">
426         <label class="form-label small fw-bold mb-1">П
| олное имя (Full Name):</label>
427         <input type="text" class="form-control form-co
| ntrol-sm" id="reg-usr-fullname-input" placeholder="Например: Г
| лавный администратор хоста">
428     </div>
429     <div class="mb-2">
430         <label class="form-label small fw-bold mb-1">О
| писание учетной записи (Description):</label>
431         <input type="text" class="form-control form-co
| ntrol-sm" id="reg-usr-desc-input" placeholder="Например: Основ
| ная рабочая учетная запись">
432     </div>
433     <div class="d-flex justify-content-between align
| -items-center mt-3 pt-2 border-top" style="border-color: var(-
| -border-color) !important;">
434         <span class="small text-muted">Локальный польз
| ователь Windows</span>
435         <button class="btn btn-sm btn-info text-dark f
| w-bold px-3 rounded-pill" id="btn-reg-save-usr">
436             <i class="bi bi-check-lg me-1"></i> Сохранит
| ь профиль
437         </button>
438     </div>
439 </div>
440 </div>
441
442 <!-- Result Alert -->
443 <div id="reg-modal-alert" class="alert d-none py-2 p
| x-3 small mb-0"></div>
444 </div>
445
446 <div class="modal-footer" style="background: var(--sur
```

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face-2); border-top: 1px solid var(--border-color);">
447     <div class="small text-muted" style="font-size: 0.72
rem;">
448         <i class="bi bi-shield-check text-info me-1"></i>S
ystem Regional & Environment Configuration
449     </div>
450     <button type="button" class="btn btn-sm btn-secondar
y rounded-pill px-3" data-bs-dismiss="modal">
451         Закрыть
452     </button>
453 </div>
454
455 </div>
456 </div>
457 </div>
458
459
460 <!-- Top 4 Summary KPI Cards (Vital Overview) -->
461 <div class="row g-2 mb-3">
462     <!-- Card 1: Operating System & Host -->
463     <div class="col-12 col-sm-6 col-lg-3">
464         <div class="ui-kpi-card">
465             <div class="ui-kpi-label d-flex justify-content-betwee
n align-items-center">
466                 <div class="d-flex align-items-center gap-1.5">
467                     <i class="bi bi-windows text-info"></i>
468                     <span>Платформа & ОС</span>
469                 </div>
470                 <div class="d-flex align-items-center gap-1">
471                     <select class="form-select form-select-sm poll-fre
q-select" data-poll-id="about_kpi_os" title="Частота опроса: П
латформа & ОС">
472                         <option value="start">Только при старте</option>
473                         <option value="manual" selected>По запросу</opti
on>
474                         <option value="3">Каждые 3 сек</option>
475                         <option value="5">Каждые 5 сек</option>
476                         <option value="10">Каждые 10 сек</option>

```

```
477 |         <option value="30">Каждые 30 сек</option>
478 |         <option value="60">Каждые 60 сек</option>
479 |         <option value="120">Каждые 120 сек</option>
480 |     </select>
481 |     <button class="btn btn-sm btn-link p-0 text-muted
text-decoration-none btn-card-history" data-metric="os" data-b
s-toggle="modal" data-bs-target="#aboutSystemHistoryModal" tit
le="История ОС / Host из базы данных">
482 |         <i class="bi bi-clock-history"></i>
483 |     </button>
484 | </div>
485 | </div>
486 |     <div class="ui-kpi-value text-info text-truncate mt-1"
id="about-kpi-os-title">Windows 11 (AMD64)</div>
487 |     <div class="ui-kpi-sub text-truncate d-flex align-item
s-center justify-content-between" id="about-host-view-box">
488 |         <div class="d-flex align-items-center text-truncate"
>
489 |             <span id="about-kpi-os-host">Host: DELL-VOSTRO</sp
an>
490 |             <button class="btn btn-sm btn-link p-0 text-info t
ext-decoration-none about-param-edit-btn" id="btn-host-inline-
edit" title="Изменить имя компьютера">
491 |                 <i class="bi bi-gear-fill" style="font-size: 0.8
0rem;"></i>
492 |             </button>
493 |         </div>
494 |         <span class="badge bg-dark border border-secondary t
ext-info py-0 px-1" id="about-spec-uptime">Uptime: --</span>
495 |     </div>
496 |     <!-- Inline Редактирование Host -->
497 |     <div class="about-sys-sub d-none align-items-center ga
p-1 mt-1" id="about-host-edit-box">
498 |         <div class="input-group input-group-sm w-100">
499 |             <span class="input-group-text bg-dark border-secon
dary text-muted py-0 px-1.5 font-monospace" style="font-size:
0.70rem;">Host:</span>
500 |             <input type="text" class="form-control form-contro
```

```

l-sm bg-dark text-white border-info py-0 px-1.5 font-monospace
" id="about-host-inline-input" style="font-size: 0.75rem; height: 25px;"
maxlength="15" placeholder="Новое имя">
501     <button class="btn btn-sm btn-info text-dark py-0
px-2 fw-bold" id="btn-host-inline-save" title="Сохранить имя к
омпьютера" style="height: 25px; font-size: 0.72rem;">
502         <i class="bi bi-check-lg"></i>
503     </button>
504     <button class="btn btn-sm btn-outline-secondary py
-0 px-1.5 text-muted" id="btn-host-inline-cancel" title="Отмен
а" style="height: 25px; font-size: 0.72rem;">
505         <i class="bi bi-x-lg"></i>
506     </button>
507 </div>
508 </div>
509 <!-- Workgroup View -->
510 <div class="about-sys-sub text-truncate d-flex align-i
tems-center justify-content-between mt-1" id="about-workgroup-
view-box">
511     <div class="d-flex align-items-center text-truncate"
>
512         <span id="about-kpi-os-workgroup">Workgroup: WORKG
ROUP</span>
513         <button class="btn btn-sm btn-link p-0 text-info t
ext-decoration-none about-param-edit-btn" id="btn-workgroup-in
line-edit" title="Изменить рабочую группу">
514             <i class="bi bi-gear-fill" style="font-size: 0.8
0rem;"></i>
515         </button>
516     </div>
517 </div>
518 <!-- Inline Редактирование Workgroup -->
519 <div class="about-sys-sub d-none align-items-center ga
p-1 mt-1" id="about-workgroup-edit-box">
520     <div class="input-group input-group-sm w-100">
521         <span class="input-group-text bg-dark border-secon
dary text-muted py-0 px-1.5 font-monospace" style="font-size:
0.70rem;">WG:</span>

```



```

522         <input type="text" class="form-control form-control-sm bg-dark text-white border-info py-0 px-1.5 font-monospace
    " id="about-workgroup-inline-input" style="font-size: 0.75rem;
    height: 25px;" maxlength="15" placeholder="Новая группа">
523         <button class="btn btn-sm btn-info text-dark py-0
    px-2 fw-bold" id="btn-workgroup-inline-save" title="Сохранить
    рабочую группу" style="height: 25px; font-size: 0.72rem;">
524             <i class="bi bi-check-lg"></i>
525         </button>
526         <button class="btn btn-sm btn-outline-secondary py-
    0 px-1.5 text-muted" id="btn-workgroup-inline-cancel" title="
    Отмена" style="height: 25px; font-size: 0.72rem;">
527             <i class="bi bi-x-lg"></i>
528         </button>
529     </div>
530 </div>
531 <!-- Напоминание о перезагрузке после смены имени / гр
    уппы -->
532     <div id="about-host-reminder-alert" class="alert alert
    -warning d-none py-1 px-2 mt-1 mb-0 border-warning border-opac
    ity-50" style="font-size: 0.70rem; line-height: 1.2;">
533         <i class="bi bi-exclamation-triangle-fill me-1 text-
    warning"></i>
534         <span id="about-host-reminder-text">Имя изменено. Вс
    тупит в силу после перезагрузки.</span>
535     </div>
536 </div>
537 </div>
538
539 <!-- Card 2: Security & Defender -->
540 <div class="col-12 col-sm-6 col-lg-3">
541     <div class="about-sys-card">
542         <div class="about-sys-label d-flex justify-content-bet
    ween align-items-center">
543             <div class="d-flex align-items-center gap-1.5">
544                 <i class="bi bi-shield-check text-success"></i>
545                 <span>Безопасность системы</span>
546             </div>

```

```
547 |         <div class="d-flex align-items-center gap-1">
548 |             <select class="form-select form-select-sm poll-fre
| q-select" data-poll-id="about_kpi_sec" title="Частота опроса:
| Безопасность">
549 |                 <option value="start">Только при старте</option>
550 |                 <option value="manual" selected>По запросу</opti
| on>
551 |                 <option value="3">Каждые 3 сек</option>
552 |                 <option value="5">Каждые 5 сек</option>
553 |                 <option value="10">Каждые 10 сек</option>
554 |                 <option value="30">Каждые 30 сек</option>
555 |                 <option value="60">Каждые 60 сек</option>
556 |                 <option value="120">Каждые 120 сек</option>
557 |             </select>
558 |             <button class="btn btn-sm btn-link p-0 text-muted
| text-decoration-none btn-card-history" data-metric="security"
| data-bs-toggle="modal" data-bs-target="#aboutSystemHistoryModa
| l" title="История безопасности из базы данных">
559 |                 <i class="bi bi-clock-history"></i>
560 |             </button>
561 |         </div>
562 |     </div>
563 |     <div class="about-sys-value text-success text-truncate
| mt-1" id="about-kpi-sec-title">Active & Protected</div>
564 |     <div class="ui-kpi-sub text-truncate" id="about-kpi-se
| c-sub">Firewall: ON | UAC: ON</div>
565 | </div>
566 | </div>
567 |
568 | <!-- Card 3: System Protection & Restore -->
569 | <div class="col-12 col-sm-6 col-lg-3">
570 |     <div class="ui-kpi-card">
571 |         <div class="about-sys-label d-flex justify-content-bet
| ween align-items-center">
572 |             <div class="d-flex align-items-center gap-1.5">
573 |                 <i class="bi bi-shield-lock text-warning"></i>
574 |                 <span>Точки восстановления</span>
575 |             </div>
```

```

576         <div class="d-flex align-items-center gap-1">
577             <select class="form-select form-select-sm poll-fre
q-select" data-poll-id="about_kpi_prot" title="Частота опроса:
Точки восстановления">
578                 <option value="start">Только при старте</option>
579                 <option value="manual" selected>По запросу</opti
on>
580                 <option value="3">Каждые 3 сек</option>
581                 <option value="5">Каждые 5 сек</option>
582                 <option value="10">Каждые 10 сек</option>
583                 <option value="30">Каждые 30 сек</option>
584                 <option value="60">Каждые 60 сек</option>
585                 <option value="120">Каждые 120 сек</option>
586             </select>
587             <button class="btn btn-sm btn-link p-0 text-muted
text-decoration-none btn-card-history" data-metric="restore_po
ints" data-bs-toggle="modal" data-bs-target="#aboutSystemHisto
ryModal" title="История точек восстановления из базы данных">
588                 <i class="bi bi-clock-history"></i>
589             </button>
590         </div>
591     </div>
592     <div class="ui-kpi-value text-warning text-truncate mt
-1" id="about-kpi-prot-title">3 Checkpoints</div>
593     <div class="ui-kpi-sub text-truncate" id="about-kpi-pr
ot-sub">Protection: Active</div>
594 </div>
595 </div>
596
597 <!-- Card 4: System Storage (C:) -->
598 <div class="col-12 col-sm-6 col-lg-3">
599     <div class="ui-kpi-card">
600         <div class="about-sys-label d-flex justify-content-bet
ween align-items-center">
601             <div class="d-flex align-items-center gap-1.5">
602                 <i class="bi bi-hdd-fill text-primary"></i>
603                 <span>Системный накопитель (C:)</span>
604             </div>

```

```

605         <div class="d-flex align-items-center gap-1">
606             <select class="form-select form-select-sm poll-fre
q-select" data-poll-id="about_kpi_stor" title="Частота опроса:
        Системный накопитель C:">
607                 <option value="start">Только при старте</option>
608                 <option value="manual" selected>По запросу</opti
on>
609                 <option value="3">Каждые 3 сек</option>
610                 <option value="5">Каждые 5 сек</option>
611                 <option value="10">Каждые 10 сек</option>
612                 <option value="30">Каждые 30 сек</option>
613                 <option value="60">Каждые 60 сек</option>
614                 <option value="120">Каждые 120 сек</option>
615             </select>
616             <button class="btn btn-sm btn-link p-0 text-muted
text-decoration-none btn-card-history" data-metric="storage" d
ata-bs-toggle="modal" data-bs-target="#aboutSystemHistoryModal
" title="История накопителя C: из базы данных">
617                 <i class="bi bi-clock-history"></i>
618             </button>
619         </div>
620     </div>
621     <div class="ui-kpi-value text-white text-truncate mt-1
" id="about-kpi-stor-title">1014.8 GB Free</div>
622     <div class="ui-kpi-sub text-info text-truncate" id="ab
out-kpi-stor-clean">Cleanable: ~190 MB</div>
623 </div>
624 </div>
625 </div>
626
627
628
629 <!-- Reorganized Dual Master Panels: Hardware & Platform Spe
cs vs User Environment & Security -->
630 <div class="row g-2 mb-3">
631     <!-- Left: Hardware & Platform Specification -->
632     <div class="col-12 col-lg-6">
633         <div class="ui-panel h-100">

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```

634 |         <div class="ui-panel-header">
635 |             <div class="d-flex align-items-center gap-2">
636 |                 <i class="bi bi-cpu-fill text-info"></i>
637 |                 <span class="text-white">Конфигурация оборудования
и ОС</span>
638 |             </div>
639 |             <div class="d-flex align-items-center gap-2">
640 |                 <select class="form-select form-select-sm poll-fre
q-select" data-poll-id="about_hw_specs" title="Частота опроса:
Оборудование и ОС">
641 |                     <option value="start" selected>Только при старте
</option>
642 |                     <option value="manual">По запросу</option>
643 |                     <option value="3">Каждые 3 сек</option>
644 |                     <option value="5">Каждые 5 сек</option>
645 |                     <option value="10">Каждые 10 сек</option>
646 |                     <option value="30">Каждые 30 сек</option>
647 |                     <option value="60">Каждые 60 сек</option>
648 |                     <option value="120">Каждые 120 сек</option>
649 |                 </select>
650 |             </div>
651 |         </div>
652 |         <table class="ui-spec-table">
653 |             <tbody>
654 |                 <tr>
655 |                     <td class="ui-spec-key"><i class="bi bi-cpu me-1
.5 text-info"></i>Процессор (CPU)</td>
656 |                     <td class="ui-spec-val" id="about-spec-cpu">Inte
l Processor</td>
657 |                 </tr>
658 |                 <tr>
659 |                     <td class="about-sys-spec-key"><i class="bi bi-m
emory me-1.5 text-warning"></i>Установленная память</td>
660 |                     <td class="about-sys-spec-val" id="about-spec-ra
m">31.8 GB RAM</td>
661 |                 </tr>
662 |                 <tr>
663 |                     <td class="about-sys-spec-key"><i class="bi bi-d

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```

display me-1.5 text-primary"></i>Мониторы и дисплеи</td>
664 |         <td class="about-sys-spec-val text-truncate" id=
| "about-spec-monitors">1920x1080 @ 60Hz</td>
665 |     </tr>
666 |     <tr>
667 |         <td class="about-sys-spec-key"><i class="bi bi-b
| attery-charging me-1.5 text-warning"></i>Питание и Энергия</td
| >
668 |         <td class="about-sys-spec-val" id="about-spec-po
| wer-container">
669 |             <div class="d-flex align-items-center gap-1.5
| flex-wrap">
670 |                 <span class="badge bg-secondary font-monospa
| ce" style="font-size: 0.72rem;" id="about-spec-power-badge">AC
| Mains</span>
671 |                 <span id="about-spec-power-source">Питание:
| Стационарная электросеть 220V</span>
672 |             </div>
673 |             <div class="small text-muted font-monospace mt
| -0.5" style="font-size: 0.72rem;" id="about-spec-power-profile
| ">Профиль: AC Mains / Desktop</div>
674 |         </td>
675 |     </tr>
676 |     <tr>
677 |         <td class="about-sys-spec-key"><i class="bi bi-l
| ayers me-1.5 text-info"></i>Сборка ОС</td>
678 |         <td class="about-sys-spec-val" id="about-spec-os
| ">10.0.26200</td>
679 |     </tr>
680 |     <tr>
681 |         <td class="about-sys-spec-key"><i class="bi bi-c
| alendar-check me-1.5 text-warning"></i>Дата установки ОС</td>
682 |         <td class="about-sys-spec-val" id="about-ident-i
| nstall-date">Не определена</td>
683 |     </tr>
684 |     <tr>
685 |         <td class="about-sys-spec-key"><i class="bi bi-a
| rrow-repeat me-1.5 text-info"></i>Служба обновлений</td>

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```

686 |         <td class="about-sys-spec-val" id="about-spec-up
    | date">Up to date</td>
687 |     </tr>
688 | </tbody>
689 | </table>
690 | </div>
691 | </div>
692 |
693 | <!-- Right: User Environment, Regional & Security -->
694 | <div class="col-12 col-lg-6">
695 |     <div class="ui-panel h-100">
696 |         <div class="ui-panel-header">
697 |             <div class="d-flex align-items-center gap-2">
698 |                 <i class="bi bi-person-badge-fill text-success"></
    | i>
699 |                 <span class="text-white">Окружение пользователя и
    | безопасность</span>
700 |             </div>
701 |             <div class="d-flex align-items-center gap-2">
702 |                 <button class="btn btn-sm btn-outline-info rounded
    | -pill px-2.5 py-0.5" id="btn-open-regional-settings" data-bs-t
    | oggle="modal" data-bs-target="#aboutRegionalSettingsModal" tit
    | le="Настроить часовой пояс, язык, локаль и пользователя" style
    | ="font-size: 0.75rem;">
703 |                     <i class="bi bi-gear-fill me-1"></i>Настроить
704 |                 </button>
705 |                 <select class="form-select form-select-sm poll-fre
    | q-select" data-poll-id="about_user_env" title="Частота опроса:
    | Окружение пользователя">
706 |                     <option value="start">Только при старте</option>
707 |                     <option value="manual" selected>По запросу</opti
    | on>
708 |                     <option value="3">Каждые 3 сек</option>
709 |                     <option value="5">Каждые 5 сек</option>
710 |                     <option value="10">Каждые 10 сек</option>
711 |                     <option value="30">Каждые 30 сек</option>
712 |                     <option value="60">Каждые 60 сек</option>
713 |                     <option value="120">Каждые 120 сек</option>

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```

714         </select>
715         <span class="small text-muted font-monospace" id="
about-ident-time-badge">Локальное время: --:--:--</span>
716     </div>
717 </div>
718 <table class="ui-spec-table">
719     <tbody>
720     <tr>
721         <td class="ui-spec-key"><i class="bi bi-person-c
ircle me-1.5 text-success"></i>Пользователь и права</td>
722         <td class="ui-spec-val">
723             <div class="d-flex align-items-center justify-
content-between">
724                 <div>
725                     <span class="fw-bold text-white me-1" id="
about-ident-username">Пользователь</span>
726                     <span class="text-muted small" id="about-i
dent-privilege">(Права)</span>
727                 </div>
728                 <button class="btn btn-sm btn-link p-0 text-
info text-decoration-none btn-open-reg-tab" data-tab="pane-usr
" data-bs-toggle="modal" data-bs-target="#aboutRegionalSetting
sModal" title="Настроить профиль пользователя">
729                     <i class="bi bi-gear-fill" style="font-siz
e: 0.76rem;"></i>
730                 </button>
731             </div>
732         </td>
733     </tr>
734     <tr>
735         <td class="about-sys-spec-key"><i class="bi bi-t
ranslate me-1.5 text-warning"></i>Язык и локали</td>
736         <td class="about-sys-spec-val">
737             <div class="d-flex align-items-center justify-
content-between">
738                 <div>
739                     <span id="about-ident-language">Русский</s
pan>

```



```

740         <div class="small text-muted font-monospac
    e" id="about-ident-locales">User: ru-RU | Sys: ru-RU</div>
741     </div>
742     <button class="btn btn-sm btn-link p-0 text-
    info text-decoration-none btn-open-reg-tab" data-tab="pane-loc
    " data-bs-toggle="modal" data-bs-target="#aboutRegionalSetting
    sModal" title="Настроить локали">
743         <i class="bi bi-gear-fill" style="font-siz
    e: 0.76rem;"></i>
744     </button>
745 </div>
746 </td>
747 </tr>
748 <tr>
749     <td class="about-sys-spec-key"><i class="bi bi-c
    lock me-1.5 text-primary"></i>Часовой пояс и кодировка</td>
750     <td class="about-sys-spec-val">
751         <div class="d-flex align-items-center justify-
    content-between">
752             <div>
753                 <span id="about-ident-timezone">UTC+03:00<
    /span>
754                 <div class="small text-muted font-monospac
    e" id="about-ident-codepage">Кодировка: UTF-8</div>
755             </div>
756             <button class="btn btn-sm btn-link p-0 text-
    info text-decoration-none btn-open-reg-tab" data-tab="pane-tz"
    data-bs-toggle="modal" data-bs-target="#aboutRegionalSettings
    Modal" title="Настроить часовой пояс">
757                 <i class="bi bi-gear-fill" style="font-siz
    e: 0.76rem;"></i>
758             </button>
759         </div>
760     </td>
761 </tr>
762 <tr>
763     <td class="about-sys-spec-key"><i class="bi bi-c
    loud-check me-1.5 text-info"></i>Офис и OneDrive</td>

```

```
764 |         <td class="about-sys-spec-val">
765 |             <span>MS Office: <strong id="about-ident-ms-of
| fice" class="text-light font-monospace">Не установлен</strong>
| </span>
766 |             <div class="small text-muted font-monospace">O
| neDrive: <span id="about-ident-onedrive" class="text-light">Не
| настроено</span></div>
767 |         </td>
768 |     </tr>
769 |     <tr>
770 |         <td class="about-sys-spec-key"><i class="bi bi-a
| rchive me-1.5 text-info"></i>Резервные копии</td>
771 |         <td class="about-sys-spec-val" id="about-ident-b
| ackup">Нет записей</td>
772 |     </tr>
773 |     <tr>
774 |         <td class="about-sys-spec-key"><i class="bi bi-s
| hield-shaded me-1.5 text-success"></i>Брандмауэр Windows</td>
775 |         <td class="about-sys-spec-val">
776 |             Domain: <span id="about-sec-fw-domain" class="
| fw-bold text-secondary">--</span> |
777 |             Private: <span id="about-sec-fw-private" class
| ="fw-bold text-secondary">--</span> |
778 |             Public: <span id="about-sec-fw-public" class="
| fw-bold text-secondary">--</span>
779 |         </td>
780 |     </tr>
781 |     <tr>
782 |         <td class="about-sys-spec-key"><i class="bi bi-s
| hield-lock me-1.5 text-success"></i>Защитник и UAC</td>
783 |         <td class="about-sys-spec-val">
784 |             Defender: <span id="about-sec-defender" class=
| "fw-bold text-secondary">--</span> |
785 |             Realtime: <span id="about-sec-realtime" class=
| "fw-bold text-secondary">--</span> |
786 |             UAC: <span id="about-sec-uac" class="fw-bold t
| ext-secondary">--</span>
787 |         </td>
```

```

788         </tr>
789     </tbody>
790 </table>
791 </div>
792 </div>
793 </div>
794
795 <!-- Storage Volumes & Partitions -->
796 <div class="row g-2 mb-3">
797     <div class="col-12">
798         <div class="about-sys-panel h-100">
799             <div class="about-sys-panel-header">
800                 <div class="d-flex align-items-center gap-2">
801                     <i class="bi bi-hdd-stack-fill text-primary"></i>
802                     <span class="text-white">Storage Volumes & Partiti
ons</span>
803                 </div>
804                 <div class="d-flex align-items-center gap-2">
805                     <select class="form-select form-select-sm poll-fre
q-select" data-poll-id="about_disks" title="Частота опроса: Па
зделы накопителей">
806                         <option value="start">Только при старте</option>
807                         <option value="manual" selected>По запросу</opti
on>
808                         <option value="3">Каждые 3 сек</option>
809                         <option value="5">Каждые 5 сек</option>
810                         <option value="10">Каждые 10 сек</option>
811                         <option value="30">Каждые 30 сек</option>
812                         <option value="60">Каждые 60 сек</option>
813                         <option value="120">Каждые 120 сек</option>
814                     </select>
815                     <span class="badge bg-secondary" id="about-disks-c
ount">-- Drives</span>
816                 </div>
817             </div>
818             <div class="table-responsive">
819                 <table class="about-sys-spec-table">
820                     <thead>

```

```

821         <tr>
822             <th style="width: 14%;">Drive</th>
823             <th style="width: 14%;">FS</th>
824             <th style="width: 14%;">Total</th>
825             <th style="width: 14%;">Used</th>
826             <th style="width: 14%;">Free</th>
827             <th style="width: 20%;">Usage</th>
828             <th style="width: 10%;">SMART</th>
829         </tr>
830     </thead>
831     <tbody id="about-sys-disks-tbody">
832         <!-- Injected dynamically -->
833         <tr>
834             <td colspan="7" class="text-center py-3 text-m
uted">Загрузка разделов дисков...</td>
835         </tr>
836     </tbody>
837 </table>
838 </div>
839 </div>
840 </div>
841 </div>
842 <!-- 4. Износ накопителей и батареи (Storage & Battery Healt
h) -->
843 <div class="card shadow-sm border-secondary-subtle mb-3">
844     <div class="card-body p-3">
845         <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
846             <div class="d-flex align-items-center gap-2">
847                 <div class="fs-4">□ </div>
848             <div>
849                 <h5 class="mb-0 fw-bold">4. Износ накопителей и ба
тареи (Storage & Battery Health)</h5>
850                 <small class="text-muted">Мониторинг ресурса SSD/N
VMe, файловых систем, SMART и деградации аккумулятора</small>
851             </div>
852         </div>
853

```

```

854 |         <div class="d-flex align-items-center gap-2 flex-wrap"
      | >
855 |             <button class="btn btn-sm btn-outline-info rounded-p
      | ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
      | "btn-diag-wear-refresh">
856 |                 <i class="bi bi-arrow-clockwise"></i> Обновить
857 |             </button>
858 |         </div>
859 |     </div>
860 | </div>
861 | </div>
862 |
863 | <div class="row g-2 mb-3">
864 |     <!-- Disks Health & Wear Panel -->
865 |     <div class="col-12 col-xl-8 col-lg-8">
866 |         <div class="diag-panel h-100">
867 |             <div class="diag-panel-header d-flex justify-content-b
      | etween align-items-center">
868 |                 <div class="d-flex align-items-center gap-2">
869 |                     <i class="bi bi-hdd-fill text-primary"></i>
870 |                     <span class="text-white">Здоровье накопителей SSD/
      | NVMe и износ (Wear & SMART)</span>
871 |                 </div>
872 |                 <div class="d-flex align-items-center gap-2">
873 |                     <select class="form-select form-select-sm poll-fre
      | q-select" data-poll-id="about_wear" title="Частота опроса: Изн
      | ос накопителей">
874 |                         <option value="start">Только при старте</option>
875 |                         <option value="manual" selected>По запросу</opti
      | on>
876 |                         <option value="3">Каждые 3 сек</option>
877 |                         <option value="5">Каждые 5 сек</option>
878 |                         <option value="10">Каждые 10 сек</option>
879 |                         <option value="30">Каждые 30 сек</option>
880 |                         <option value="60">Каждые 60 сек</option>
881 |                         <option value="120">Каждые 120 сек</option>
882 |                     </select>
883 |                     <span class="badge bg-secondary" id="diag-wear-dis

```

```

ks-count">0 дисков</span>
884     </div>
885   </div>
886   <div class="table-responsive">
887     <table class="about-sys-spec-table">
888       <thead>
889         <tr>
890           <th>Физический накопитель</th>
891           <th>Тип / Шина</th>
892           <th>Разделы</th>
893           <th style="text-align: right;">Объем</th>
894           <th style="text-align: right;">Свободно</th>
895           <th>Наработка / Старт</th>
896           <th style="text-align: right;">Ввод / Вывод (I
|/O)</th>
897           <th style="width: 130px;">Здоровье / Износ</th>
|>
898           <th>Статус</th>
899         </tr>
900       </thead>
901       <tbody id="diag-wear-disks-tbody">
902         <tr><td colspan="9" class="text-center py-4 text
|-muted">Опрос контроллеров накопителей...</td></tr>
903       </tbody>
904     </table>
905   </div>
906 </div>
907 </div>
908
909 <!-- Battery Deep Telemetry -->
910 <div class="col-12 col-xl-4 col-lg-4">
911   <div class="diag-panel h-100">
912     <div class="diag-panel-header d-flex justify-content-b
|etween align-items-center">
913       <div class="d-flex align-items-center gap-2">
914         <i class="bi bi-battery-charging text-warning"></i>
|>
915         <span class="text-white">Деградация и состояние ба

```

```

    тапеи (Battery Degradation)</span>
916         </div>
917         <div class="d-flex align-items-center gap-2">
918             <select class="form-select form-select-sm poll-fre
q-select" data-poll-id="about_battery" title="Частота опроса:
Состояние батарееи">
919                 <option value="start">Только при старте</option>
920                 <option value="manual" selected>По запросу</opti
on>
921                 <option value="3">Каждые 3 сек</option>
922                 <option value="5">Каждые 5 сек</option>
923                 <option value="10">Каждые 10 сек</option>
924                 <option value="30">Каждые 30 сек</option>
925                 <option value="60">Каждые 60 сек</option>
926                 <option value="120">Каждые 120 сек</option>
927             </select>
928             <span class="badge bg-dark border border-secondary
text-info" id="diag-battery-source-badge">AC Mains</span>
929         </div>
930     </div>
931     <div class="p-3" id="diag-battery-details-container">
932         <div class="text-center py-4 text-muted small">
933             <i class="bi bi-plug-fill fs-4 text-info d-block m
b-1"></i>
934             Стационарный компьютер – питание напрямую от элект
росети
935         </div>
936     </div>
937 </div>
938 </div>
939 </div>
940
941 <!-- Full Hardware Specification Tree Panel (Hardware Tree)
-->
942 <div class="about-sys-panel">
943     <div class="about-sys-panel-header">
944         <div class="d-flex align-items-center gap-2">
945             <button type="button" class="panel-toggle-drill collap

```

```
sed me-1" id="btn-about-sys-tree-toggle" title="Развернуть все
/ Свернуть все категории">
946     
947     </button>
948     <span class="fw-semibold">Полная спецификация оборудов
ания</span>
949     <span class="badge bg-secondary" id="about-sys-node-co
unt">Hardware Tree</span>
950 </div>
951 <div class="d-flex align-items-center gap-2">
952     <select class="form-select form-select-sm poll-freq-se
lect" data-poll-id="about_hw_tree" title="Частота опроса: Дере
во оборудования">
953         <option value="start" selected>Только при старте</op
tion>
954         <option value="manual">По запросу</option>
955         <option value="3">Каждые 3 сек</option>
956         <option value="5">Каждые 5 сек</option>
957         <option value="10">Каждые 10 сек</option>
958         <option value="30">Каждые 30 сек</option>
959         <option value="60">Каждые 60 сек</option>
960         <option value="120">Каждые 120 сек</option>
961     </select>
962     <input type="text" class="form-control form-control-sm
" id="about-sys-search"
963         placeholder="Поиск компонентов..." style="max-w
idth: 220px; font-size: 0.76rem; height: 26px;">
964 </div>
965 </div>
966 <div class="p-3" id="about-sys-tree-container">
967     <div class="text-center py-4 small text-muted">
968         <div class="spinner-border spinner-border-sm text-info
mb-2" role="status"></div>
969         <div>Загрузка спецификации оборудования...</div>
970     </div>
971 </div>
972 </div>
```



```
973 |  
974 | </div> <!-- /.about-sys-wrapper -->  
975 |  
976 |  
977 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows About System Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/about_system_tab/main.
11  *     js?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/about_system_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 02:56:00
19  * =====
20  */
21  // =====
22  // Process Name: About System Tab Web Controller
23  // =====
24  // Description:
25  //   Client-side JavaScript controller for the About System ta
26  //   b.
27  //   Provides real-time host telemetry, system identity & loca
28  //   le parameters,
29  //   AIDA64-like hardware tree, live top processes, disk volum
30  //   es and security status.
```

```

28 //
29 //    КЕШИРОВАНИЕ (IndexedDB + Memory):
30 //    - Stale-While-Revalidate для всех данных
31 //    - Cache-First для hardware spec (меняется редко)
32 //    - Network-First для live telemetry (всегда свежие)
33 //    - Prefetching при активации вкладки
34 //
35 // File: main.js
36 // Package: src.api.webgui.about_system_tab
37 // Author: hypo69
38 // Copyright: © 2026 hypo69
39 // Updated: 2026-10-04 00:22:00
40 // =====
    =====
41
42 (function () {
43     let hardwareData = [];
44     let currentProcesses = [];
45     let currentDisks = [];
46     let isLiveActive = true;
47     let liveIntervalId = null;
48     let wearAutoRefreshTimer = null;
49     let isUpdating = false;
50     let isAiRunning = false;
51     let activeHistoryMetric = 'all';
52
53     // Кеш-стратегия и TTL
54     const CACHE_STRATEGY = {
55         HARDWARE_SPEC: { ttl: 30 * 60 * 1000, strategy: 'cache-first' }, // 30 мин, редко меняется
56         SYSTEM_SUMMARY: { ttl: 5 * 1000, strategy: 'stale-while-revalidate' }, // 5 сек
57         CONTROL_STATUS: { ttl: 60 * 1000, strategy: 'stale-while-revalidate' }, // 1 мин
58         BACKUP_STATUS: { ttl: 5 * 60 * 1000, strategy: 'cache-first' }, // 5 мин
59         STORAGE_BATTERY: { ttl: 10 * 1000, strategy: 'stale-while-revalidate' }, // 10 сек

```

```

60     PANEL_KPI: { ttl: 5 * 1000, strategy: 'stale-while-revalid
    ate' }          // 5 сек
61 };
62
63 const CACHE_STORE = 'api_cache';
64 const CACHE_TAG = 'about-system-tab';
65
66 /**
67  * Универсальная функция для работы с кешем
68  * Поддерживает стратегии: cache-first, network-first, stale
    -while-revalidate
69  */
70 async function cachedFetch(url, cacheKey, options = {}) {
71     const { ttl = 60000, strategy = 'stale-while-revalidate',
    forceNetwork = false } = options;
72     const cache = window.browseCache;
73
74     if (!cache) {
75         // Fallback: кеш недоступен, прямой запрос
76         return await apiFetch(url);
77     }
78
79     // Cache-First: сначала кеш, потом сеть (для редко меняющи
    хся данных)
80     if (strategy === 'cache-first' && !forceNetwork) {
81         const cached = await cache.get(CACHE_STORE, cacheKey);
82         if (cached) {
83             console.log(`[Cache] HIT (cache-first): ${cacheKey}`);
84             return cached;
85         }
86     }
87
88     // Stale-While-Revalidate: показываем кеш, обновляем в фон
    e
89     if (strategy === 'stale-while-revalidate' && !forceNetwork
    ) {
90         const cached = await cache.get(CACHE_STORE, cacheKey);
91         if (cached) {

```

```

92 |         console.log(`[Cache] HIT (stale-while-revalidate): ${c
|   |acheKey}, updating in background...`);
93 |         // Запускаем обновление в фоне
94 |         apiFetch(url)
95 |             .then(fresh => cache.set(CACHE_STORE, cacheKey, fres
|   |h, { ttl, tags: [CACHE_TAG] }))
96 |             .catch(err => console.warn(`[Cache] Background updat
|   |e failed for ${cacheKey}:`, err));
97 |         return cached;
98 |     }
99 | }
100 |
101 | // Network-First или первый запрос: сначала сеть, потом ке
|   |ш
102 | try {
103 |     const fresh = await apiFetch(url);
104 |     // Сохраняем в кеш
105 |     await cache.set(CACHE_STORE, cacheKey, fresh, { ttl, tag
|   |s: [CACHE_TAG] });
106 |     console.log(`[Cache] MISS → stored: ${cacheKey}`);
107 |     return fresh;
108 | } catch (err) {
109 |     // Если сеть недоступна, пытаемся вернуть устаревший кеш
110 |     const cached = await cache.get(CACHE_STORE, cacheKey);
111 |     if (cached) {
112 |         console.warn(`[Cache] Network failed, returning stale
|   |cache: ${cacheKey}`);
113 |         return cached;
114 |     }
115 |     throw err;
116 | }
117 | }
118 |
119 | /**
120 |  * Очистка кеша вкладки (при необходимости)
121 |  */
122 | async function clearTabCache() {
123 |     const cache = window.browserCache;

```

```

124 |     if (cache) {
125 |         await cache.invalidateByTag(CACHE_STORE, CACHE_TAG);
126 |         console.log('[Cache] Tab cache cleared');
127 |     }
128 | }
129 |
130 | async function apiFetch(url, options = {}) {
131 |     const opts = { ...options };
132 |     if (opts.body && typeof opts.body === 'string') {
133 |         opts.headers = {
134 |             'Content-Type': 'application/json',
135 |             ...(opts.headers || {})
136 |         };
137 |     }
138 |     if (window.api && typeof window.api.fetch === 'function')
139 |     {
140 |         return await window.api.fetch(url, opts);
141 |     }
142 |     const res = await fetch(url, opts);
143 |     if (!res.ok) {
144 |         let errMsg = `HTTP ${res.status}`;
145 |         try {
146 |             const errJson = await res.json();
147 |             if (errJson && errJson.detail) {
148 |                 errMsg += ` (${typeof errJson.detail === 'object' ?
JSON.stringify(errJson.detail) : errJson.detail})`;
149 |             }
150 |             } catch (_) {}
151 |             throw new Error(errMsg);
152 |         }
153 |         return await res.json();
154 |     }
155 |
156 | function setText(id, text) {
157 |     const el = document.getElementById(id);
158 |     if (el) el.textContent = text !== null && text !== undefined ? String(text) : '--';
159 | }

```

```
159
160 function setStatusText(id, isPositive, activeText = 'Active'
    , inactiveText = 'Disabled') {
161     const el = document.getElementById(id);
162     if (!el) return;
163     const active = Boolean(isPositive);
164     el.textContent = active ? activeText : inactiveText;
165     el.className = active ? 'text-success fw-bold' : 'text-dan
    ger fw-bold';
166 }
167
168 function setFormattedStatusHtml(id, text) {
169     const el = document.getElementById(id);
170     if (!el) return;
171     if (!text) { el.innerHTML = '--'; return; }
172     el.innerHTML = String(text)
173         .replace(/\b(OFF|Disabled|Выключено|Отключен|Отключено|I
    nactive)\b/gi, '<strong class="text-danger fw-bold">$1</strong
    >')
174         .replace(/\b(ON|Active|Enabled|Включен|Включено)\b/gi, '
    <strong class="text-success fw-bold">$1</strong>');
175 }
176
177 function setHtml(id, html) {
178     const el = document.getElementById(id);
179     if (el) el.innerHTML = html;
180 }
181
182 async function initAboutSystemTab() {
183     console.log('[AboutSystemTab] Initializing tab controller.
    ..');
184     bindEvents();
185     bindAIEvents();
186     bindHistoryEvents();
187     bindRegionalEvents();
188     updateLocalClock();
189     updatePowerDisplay();
190
```

```

191 // Проверка доступности кеша
192 await updateCacheStatus();
193
194 // Инициализация индивидуальных опросников и выпадающих сп
    исков частоты для всех панелей
195 initPanelPollers();
196
197 // Запуск таймера часов и статуса кеша
198 startLiveStream();
199 }
200 window.initAboutSystemTab = initAboutSystemTab;
201
202 /**
203  * Обновление статуса кеша в интерфейсе
204  */
205 async function updateCacheStatus() {
206     const badge = document.getElementById('about-sys-cache-bad
    ge');
207     if (!badge) return;
208
209     const cache = window.browserCache;
210     if (!cache) {
211         badge.textContent = '❌ Cache: Unavailable';
212         badge.className = 'badge rounded-pill bg-warning-subtle
    text-warning border border-warning px-2.5 py-1';
213         badge.title = 'Кеш недоступен, используется прямой запро
    с к API';
214         return;
215     }
216
217     const ready = await cache.ready();
218     if (ready) {
219         const stats = await cache.getDetailedStats();
220         const hitRate = stats.metrics.hits + stats.metrics.misse
    s > 0
221             ? Math.round((stats.metrics.hits / (stats.metrics.hits
    + stats.metrics.misses)) * 100)
222             : 0;

```



```
223
224     badge.textContent = `📊 Cache: ${hitRate}% hit`;
225     badge.className = 'badge rounded-pill bg-success-subtle
text-success border border-success px-2.5 py-1';
226     badge.title = `Кеш активен\nПопаданий: ${stats.metrics.h
its}\nПромашов: ${stats.metrics.misses}\nВ памяти: ${stats.mem
oryCacheSize} записей`;
227   } else {
228     badge.textContent = '🔴 Cache: Error';
229     badge.className = 'badge rounded-pill bg-danger-subtle t
ext-danger border border-danger px-2.5 py-1';
230     badge.title = 'Ошибка инициализации кеша';
231   }
232 }
233
234 function updateLocalClock() {
235   const clockEl = document.getElementById('about-ident-time-
badge');
236   if (clockEl) {
237     const now = new Date();
238     clockEl.textContent = `Локальное время: ${now.toLocaleTi
meString()}`;
239   }
240 }
241
242 function startLiveStream() {
243   if (window.registerTabPoller) {
244     window.registerTabPoller('tab-about-system', async () =>
{
245       updateLocalClock();
246       await updateCacheStatus();
247     }, 5000, { pollerId: 'tab-about-system_clock', immediate
: true });
248   } else {
249     if (liveIntervalId) clearInterval(liveIntervalId);
250     liveIntervalId = setInterval(async () => {
251       updateLocalClock();
252       await updateCacheStatus();
```

```
253 |     }, 5000);
254 | }
255 | }
256 |
257 | function bindEvents() {
258 |     const btnRefresh = document.getElementById('btn-about-sys-
refresh');
259 |     if (btnRefresh) {
260 |         btnRefresh.onclick = async () => {
261 |             btnRefresh.disabled = true;
262 |             const icon = btnRefresh.querySelector('i');
263 |             if (icon) icon.classList.add('spin-animation');
264 |             await refreshAllData();
265 |             if (icon) icon.classList.remove('spin-animation');
266 |             btnRefresh.disabled = false;
267 |         };
268 |     }
269 |
270 |     const btnClearCache = document.getElementById('btn-about-s
ys-clear-cache');
271 |     if (btnClearCache) {
272 |         btnClearCache.onclick = async () => {
273 |             btnClearCache.disabled = true;
274 |             const icon = btnClearCache.querySelector('i');
275 |             if (icon) icon.classList.add('spin-animation');
276 |
277 |             // Очищаем кеш вкладки
278 |             await clearTabCache();
279 |
280 |             // Принудительно обновляем все данные из сети
281 |             await refreshAllData(true);
282 |
283 |             if (icon) icon.classList.remove('spin-animation');
284 |             btnClearCache.disabled = false;
285 |
286 |             if (window.showToast) {
287 |                 window.showToast('Кеш вкладки очищен, данные обновле
ны', 'success');
```

```
288 |     }
289 |     };
290 | }
291 |
292 |     const btnLiveToggle = document.getElementById('btn-about-s
ys-live-toggle');
293 |     if (btnLiveToggle) {
294 |         btnLiveToggle.onclick = () => {
295 |             isLiveActive = !isLiveActive;
296 |             if (window.setTabPollerEnabled) {
297 |                 window.setTabPollerEnabled('tab-about-system_clock',
isLiveActive);
298 |                 Object.keys(panelPollingRegistry).forEach(pollId =>
{
299 |                     window.setTabPollerEnabled(`tab-about-system_${pol
lId}`, isLiveActive);
300 |                 });
301 |             }
302 |             const icon = document.getElementById('icon-about-sys-l
ive');
303 |             const txt = document.getElementById('txt-about-sys-liv
e');
304 |             const liveBadge = document.getElementById('about-sys-l
ive-badge');
305 |             if (isLiveActive) {
306 |                 if (icon) icon.className = 'bi bi-pause-fill me-1';
307 |                 if (txt) txt.textContent = 'Пауза';
308 |                 if (liveBadge) {
309 |                     liveBadge.className = 'badge rounded-pill bg-info-
subtle text-info border border-info px-2.5 py-1';
310 |                     liveBadge.textContent = '● Live Host Telemetry';
311 |                 }
312 |             } else {
313 |                 if (icon) icon.className = 'bi bi-play-fill me-1';
314 |                 if (txt) txt.textContent = 'Возобновить';
315 |                 if (liveBadge) {
316 |                     liveBadge.className = 'badge rounded-pill bg-warni
ng-subtle text-warning border border-warning px-2.5 py-1';
```

```
317 |         liveBadge.textContent = '⏸ Stream Paused';
318 |     }
319 | }
320 | };
321 | }
322 |
323 | const btnTreeToggle = document.getElementById('btn-about-s
ys-tree-toggle');
324 | if (btnTreeToggle) {
325 |     btnTreeToggle.onclick = () => {
326 |         const bodies = document.querySelectorAll('.about-sys-t
ree-body');
327 |         const chevrons = document.querySelectorAll('.about-sys
-chevron');
328 |         if (!bodies.length) return;
329 |
330 |         // Если хотя бы один узел свернут – разворачиваем ВСЕ
331 |         const hasHidden = Array.from(bodies).some(b => b.class
List.contains('d-none'));
332 |
333 |         if (hasHidden) {
334 |             bodies.forEach(b => b.classList.remove('d-none'));
335 |             chevrons.forEach(c => c.textContent = '▲');
336 |             btnTreeToggle.classList.remove('collapsed');
337 |             btnTreeToggle.title = 'Свернуть все категории';
338 |         } else {
339 |             bodies.forEach(b => b.classList.add('d-none'));
340 |             chevrons.forEach(c => c.textContent = '▼');
341 |             btnTreeToggle.classList.add('collapsed');
342 |             btnTreeToggle.title = 'Развернуть все категории';
343 |         }
344 |     };
345 | }
346 |
347 | const treeSearch = document.getElementById('about-sys-sear
ch');
348 | if (treeSearch) {
349 |     treeSearch.oninput = () => {
```

```

350         const query = treeSearch.value.trim().toLowerCase();
351         renderHardwareTree(hardwareData, query);
352     };
353 }
354
355     const procSearch = document.getElementById('about-proc-search');
356     if (procSearch) {
357         procSearch.oninput = () => {
358             renderProcessesTable(currentProcesses);
359         };
360     }
361
362     const btnWearRefresh = document.getElementById('btn-diag-wear-refresh');
363     if (btnWearRefresh) {
364         btnWearRefresh.onclick = () => fetchStorageBatteryWear(true);
365     }
366
367     const wearAutoSwitch = document.getElementById('diag-wear-auto-refresh');
368     if (wearAutoSwitch) {
369         wearAutoSwitch.onchange = (e) => {
370             if (e.target.checked) {
371                 if (window.registerTabPoller) {
372                     window.registerTabPoller('tab-about-system', () =>
373                     fetchStorageBatteryWear(true), 10000, { pollerId: 'about_sys_wear', immediate: true });
374                 } else {
375                     wearAutoRefreshTimer = setInterval(() => {
376                         if (window.isTabActive ? window.isTabActive('tab-about-system') : true) {
377                             fetchStorageBatteryWear(true);
378                         }
379                     }, 10000);
380                 } else {

```

```

381         if (window.unregisterTabPoller) {
382             window.unregisterTabPoller('about_sys_wear');
383         }
384         if (wearAutoRefreshTimer) {
385             clearInterval(wearAutoRefreshTimer);
386             wearAutoRefreshTimer = null;
387         }
388     }
389 };
390 }
391
392 // Inline Edit: Hostname
393 const btnHostEdit = document.getElementById('btn-host-inli
ne-edit');
394 const viewBox = document.getElementById('about-host-view-b
ox');
395 const editBox = document.getElementById('about-host-edit-b
ox');
396 const inputHost = document.getElementById('about-host-inli
ne-input');
397 const btnHostSave = document.getElementById('btn-host-inli
ne-save');
398 const btnHostCancel = document.getElementById('btn-host-in
line-cancel');
399 const reminderAlert = document.getElementById('about-host-
reminder-alert');
400 const reminderText = document.getElementById('about-host-r
eminder-text');
401
402 function openHostEdit() {
403     if (!viewBox || !editBox || !inputHost) return;
404     const hostEl = document.getElementById('about-kpi-os-hos
t');
405     let currentHost = hostEl ? hostEl.textContent.replace(/^
Host:\s*/i, '').trim() : '';
406     if (!currentHost || currentHost === '--') currentHost =
window.location.hostname || 'DELL-VOSTRO';
407     inputHost.value = currentHost;

```

```
408 |     viewBox.classList.add('d-none');
409 |     viewBox.classList.remove('d-flex');
410 |     editBox.classList.remove('d-none');
411 |     editBox.classList.add('d-flex');
412 |     inputHost.focus();
413 |     inputHost.select();
414 | }
415 |
416 | function closeHostEdit() {
417 |     if (!viewBox || !editBox) return;
418 |     editBox.classList.add('d-none');
419 |     editBox.classList.remove('d-flex');
420 |     viewBox.classList.remove('d-none');
421 |     viewBox.classList.add('d-flex');
422 | }
423 |
424 | if (btnHostEdit) {
425 |     btnHostEdit.onclick = (e) => {
426 |         e.preventDefault();
427 |         openHostEdit();
428 |     };
429 | }
430 |
431 | if (btnHostCancel) {
432 |     btnHostCancel.onclick = (e) => {
433 |         e.preventDefault();
434 |         closeHostEdit();
435 |     };
436 | }
437 |
438 | if (inputHost) {
439 |     inputHost.onkeydown = (e) => {
440 |         if (e.key === 'Enter') {
441 |             e.preventDefault();
442 |             if (btnHostSave) btnHostSave.click();
443 |         } else if (e.key === 'Escape') {
444 |             e.preventDefault();
445 |             closeHostEdit();
```

```
446 |     }
447 | };
448 | }
449 |
450 | if (btnHostSave && inputHost) {
451 |     btnHostSave.onclick = async (e) => {
452 |         e.preventDefault();
453 |         const newName = inputHost.value.trim();
454 |
455 |         if (!newName) {
456 |             if (window.showToast) window.showToast('Пожалуйста,
введите новое имя компьютера', 'warning');
457 |             return;
458 |         }
459 |
460 |         const validNamePattern = /^[a-zA-Z0-9]([a-zA-Z0-9\ -]{0
,13}[a-zA-Z0-9])?$/;
461 |         if (!validNamePattern.test(newName) || newName.length
> 15) {
462 |             if (window.showToast) window.showToast('Имя должно с
одержать от 1 до 15 символов (A-Z, 0-9, дефис) без пробелов',
'warning');
463 |             return;
464 |         }
465 |
466 |         btnHostSave.disabled = true;
467 |         inputHost.disabled = true;
468 |         btnHostSave.innerHTML = '<span class="spinner-border s
pinner-border-sm" role="status" style="width: 10px; height: 10
px;"></span>';
469 |
470 |         try {
471 |             const res = await fetch('/api/v1/system/rename-compu
ter', {
472 |                 method: 'POST',
473 |                 headers: { 'Content-Type': 'application/json' },
474 |                 body: JSON.stringify({ new_name: newName, restart:
false }),
```



```
475 |         });
476 |         const result = await res.json();
477 |         if (res.ok && result.status === 'ok') {
478 |             setText('about-kpi-os-host', `Host: ${newName}`);
479 |             setText('about-ident-hostname', newName);
480 |             closeHostEdit();
481 |
482 |             // Показываем напоминание о перезагрузке
483 |             if (reminderAlert) {
484 |                 reminderAlert.classList.remove('d-none');
485 |                 if (reminderText) {
486 |                     reminderText.textContent = `Имя компьютера изменено на «${newName}». Изменение сработает только после перезагрузки системы!`;
487 |                 }
488 |             }
489 |             if (window.showToast) {
490 |                 window.showToast(`🔔 Имя изменено на «${newName}». Напоминание: требуется перезагрузка для применения!`, 'warning');
491 |             }
492 |             } else {
493 |                 const errMsg = result.message || result.detail || 'Не удалось изменить имя компьютера.';
494 |                 if (window.showToast) window.showToast(`Ошибка: ${errMsg}`, 'danger');
495 |                 else alert(errMsg);
496 |             }
497 |         } catch (err) {
498 |             const errMsg = `Сетевая ошибка: ${err.message}`;
499 |             if (window.showToast) window.showToast(errMsg, 'danger');
500 |             else alert(errMsg);
501 |         } finally {
502 |             btnHostSave.disabled = false;
503 |             inputHost.disabled = false;
504 |             btnHostSave.innerHTML = '<i class="bi bi-check-lg"></i>';
```

```

505         }
506     };
507 }
508
509 // Inline Edit: Workgroup
510 const btnWgEdit = document.getElementById('btn-workgroup-i
nline-edit');
511 const wgViewBox = document.getElementById('about-workgroup
-view-box');
512 const wgEditBox = document.getElementById('about-workgroup
-edit-box');
513 const inputWg = document.getElementById('about-workgroup-i
nline-input');
514 const btnWgSave = document.getElementById('btn-workgroup-i
nline-save');
515 const btnWgCancel = document.getElementById('btn-workgroup
-inline-cancel');
516
517 function openWgEdit() {
518     if (!wgViewBox || !wgEditBox || !inputWg) return;
519     const wgEl = document.getElementById('about-kpi-os-workg
roup');
520     let currentWg = wgEl ? wgEl.textContent.replace(/^Workgr
oup:\s*/i, '').trim() : '';
521     if (!currentWg || currentWg === '--') currentWg = 'WORKG
ROUP';
522     inputWg.value = currentWg;
523     wgViewBox.classList.add('d-none');
524     wgViewBox.classList.remove('d-flex');
525     wgEditBox.classList.remove('d-none');
526     wgEditBox.classList.add('d-flex');
527     inputWg.focus();
528     inputWg.select();
529 }
530
531 function closeWgEdit() {
532     if (!wgViewBox || !wgEditBox) return;
533     wgEditBox.classList.add('d-none');

```

```
534 |     wgEditBox.classList.remove('d-flex');
535 |     wgViewBox.classList.remove('d-none');
536 |     wgViewBox.classList.add('d-flex');
537 | }
538 |
539 | if (btnWgEdit) {
540 |     btnWgEdit.onclick = (e) => {
541 |         e.preventDefault();
542 |         openWgEdit();
543 |     };
544 | }
545 |
546 | if (btnWgCancel) {
547 |     btnWgCancel.onclick = (e) => {
548 |         e.preventDefault();
549 |         closeWgEdit();
550 |     };
551 | }
552 |
553 | if (inputWg) {
554 |     inputWg.onkeydown = (e) => {
555 |         if (e.key === 'Enter') {
556 |             e.preventDefault();
557 |             if (btnWgSave) btnWgSave.click();
558 |         } else if (e.key === 'Escape') {
559 |             e.preventDefault();
560 |             closeWgEdit();
561 |         }
562 |     };
563 | }
564 |
565 | if (btnWgSave && inputWg) {
566 |     btnWgSave.onclick = async (e) => {
567 |         e.preventDefault();
568 |         const newWg = inputWg.value.trim().toUpperCase();
569 |
570 |         if (!newWg) {
571 |             if (window.showToast) window.showToast('Пожалуйста,
```

```

    введите имя рабочей группы', 'warning');
572     return;
573 }
574
575     const validWgPattern = /^[a-zA-Z0-9_\-]{1,15}$/;
576     if (!validWgPattern.test(newWg)) {
577         if (window.showToast) window.showToast('Имя группы д
    олжно содержать от 1 до 15 символов (A-Z, 0-9, -, _)', 'warnin
    g');
578         return;
579     }
580
581     btnWgSave.disabled = true;
582     inputWg.disabled = true;
583     btnWgSave.innerHTML = '<span class="spinner-border spi
    nner-border-sm" role="status" style="width: 10px; height: 10px
    ;"></span>';
584
585     try {
586         const res = await fetch('/api/v1/system/change-workg
    roup', {
587             method: 'POST',
588             headers: { 'Content-Type': 'application/json' },
589             body: JSON.stringify({ new_workgroup: newWg, resta
    rt: false }),
590         });
591         const result = await res.json();
592         if (res.ok && result.status === 'ok') {
593             setText('about-kpi-os-workgroup', `Workgroup: ${ne
    wWg}`);
594             closeWgEdit();
595
596             // Показываем напоминание о перезагрузке
597             if (reminderAlert) {
598                 reminderAlert.classList.remove('d-none');
599                 if (reminderText) {
600                     reminderText.textContent = `Рабочая группа изм
    енена на «${newWg}». Изменение сработает только после перезагр

```

```

        узки системы!`;
601         }
602     }
603     if (window.showToast) {
604         window.showToast(`⚠ Рабочая группа изменена на «
        ${newWg}». Напоминание: требуется перезагрузка для применения!
        `, 'warning');
605     }
606     } else {
607         const errMsg = result.message || result.detail ||
        'Не удалось изменить рабочую группу.';
608         if (window.showToast) window.showToast(`Ошибка: ${
        errMsg}`, 'danger');
609         else alert(errMsg);
610     }
611     } catch (err) {
612         const errMsg = `Сетевая ошибка: ${err.message}`;
613         if (window.showToast) window.showToast(errMsg, 'dang
        er');
614         else alert(errMsg);
615     } finally {
616         btnWgSave.disabled = false;
617         inputWg.disabled = false;
618         btnWgSave.innerHTML = '<i class="bi bi-check-lg"></i
        >';
619     }
620 };
621 }
622 }
623
624 async function refreshAllData(forceNetwork = false) {
625     isUpdating = true;
626     try {
627         if (forceNetwork) {
628             // Принудительная загрузка из сети (игнорируем кеш)
629             await Promise.allSettled([
630                 apiFetch('/api/v1/system/summary?process_limit=25').
        then(data => {

```

```

631         if (window.browserCache) {
632             window.browserCache.set(CACHE_STORE, 'system_sum
mary', data, {
633                 ttl: CACHE_STRATEGY.SYSTEM_SUMMARY.ttl,
634                 tags: [CACHE_TAG]
635             });
636         }
637         return fetchSystemSummary();
638     }},
639     apiFetch('/api/system-control/status').then(data =>
{
640         if (window.browserCache) {
641             window.browserCache.set(CACHE_STORE, 'system_con
trol_status', data, {
642                 ttl: CACHE_STRATEGY.CONTROL_STATUS.ttl,
643                 tags: [CACHE_TAG]
644             });
645         }
646         return fetchSystemControlStatus();
647     }},
648     apiFetch('/api/v1/system/hardware').then(data => {
649         if (window.browserCache) {
650             window.browserCache.set(CACHE_STORE, 'hardware_s
pec', data, {
651                 ttl: CACHE_STRATEGY.HARDWARE_SPEC.ttl,
652                 tags: [CACHE_TAG]
653             });
654         }
655         return fetchHardwareSpec();
656     }},
657     apiFetch('/api/v1/windows-backup/health').then(data
=> {
658         if (window.browserCache) {
659             window.browserCache.set(CACHE_STORE, 'backup_sta
tus', data, {
660                 ttl: CACHE_STRATEGY.BACKUP_STATUS.ttl,
661                 tags: [CACHE_TAG]
662             });

```

```

663         }
664         return fetchBackupStatus();
665     }},
666     apiFetch('/api/v1/system/diagnostics/storage-battery
667 ').then(data => {
668         if (window.browserCache) {
669             window.browserCache.set(CACHE_STORE, 'storage_ba
670 ttery_wear', data, {
671                 ttl: CACHE_STRATEGY.STORAGE_BATTERY.ttl,
672                 tags: [CACHE_TAG]
673             });
674         }
675         return fetchStorageBatteryWear(true);
676     }},
677     fetchKpiPanels(true)
678 ]);
679 } else {
680     await Promise.allSettled([
681         fetchSystemSummary(),
682         fetchSystemControlStatus(),
683         fetchHardwareSpec(),
684         fetchBackupStatus(),
685         fetchStorageBatteryWear(),
686         fetchKpiPanels(false)
687     ]);
688 }
689 } finally {
690     isUpdating = false;
691 }
692
693 async function fetchBackupStatus() {
694     try {
695         const config = CACHE_STRATEGY.BACKUP_STATUS;
696         const data = await cachedFetch(
697             '/api/v1/windows-backup/health',
698             'backup_status',
699             { ttl: config.ttl, strategy: config.strategy }

```

```
699 |     );
700 |     if (!data) return;
701 |
702 |     const cfg = data.file_history?.config;
703 |     const storage = data.storage_audit;
704 |
705 |     let lastTimeStr = 'Нет записей';
706 |     if (cfg && cfg.last_backup_time) {
707 |         try {
708 |             const d = new Date(cfg.last_backup_time);
709 |             if (!isNaN(d.getTime())) {
710 |                 lastTimeStr = d.toLocaleDateString() + ' ' + d.toL
711 | ocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
712 |             }
713 |         } catch (_) {}
714 |     } else if (storage && storage.sample_versions && storage
715 | .sample_versions.length > 0) {
716 |         const latestSample = storage.sample_versions[0];
717 |         if (latestSample && latestSample.version_timestamp) {
718 |             try {
719 |                 const d = new Date(latestSample.version_timestamp)
720 | ;
721 |                 if (!isNaN(d.getTime())) {
722 |                     lastTimeStr = d.toLocaleDateString() + ' ' + d.t
723 | oLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
724 |                 }
725 |             } catch (_) {}
726 |         }
727 |     }
728 |
729 |     let storageStr = '';
730 |     if (storage && storage.target_exists && storage.free_spa
731 | ce_gb !== null && storage.free_space_gb !== undefined) {
732 |         const target = storage.target_path || 'Диск';
733 |         const freeGb = typeof storage.free_space_gb === 'numbe
734 | r' ? storage.free_space_gb.toFixed(1) : storage.free_space_gb;
735 |         storageStr = `${target} (${freeGb} GB своб.)`;
736 |     } else if (cfg && (cfg.target_drive_letter || cfg.target
```



```

    _url)) {
731         storageStr = cfg.target_drive_letter || cfg.target_url
    ;
732     } else {
733         storageStr = 'Хранилище не найдено';
734     }
735
736     const backupSummary = `${lastTimeStr} | ${storageStr}`;
737     const backupEl = document.getElementById('about-ident-backup');
738     if (backupEl) {
739         backupEl.textContent = backupSummary;
740         backupEl.title = `Последний бэкап: ${lastTimeStr}\nХранилище: ${storageStr}\nСлужба File History: ${data.file_history?.service_status || '-'}\nHealth Score: ${data.health_score ? '--'}/100`;
741     }
742     } catch (e) {
743         console.warn('[AboutSystemTab] fetchBackupStatus warning:', e);
744         setText('about-ident-backup', 'Не настроено');
745     }
746 }
747
748 /**
749  * Обновление панели "Питание и Энергия" в таблице спецификации оборудования и ОС.
750  * @param {Object} [batteryWear] - Объект телеметрии батареи /питания
751  * @param {string} [powerProfileName] - Название активной схемы электропитания
752  */
753 function updatePowerDisplay(batteryWear = null, powerProfileName = '') {
754     const badgeEl = document.getElementById('about-spec-power-badge');
755     const sourceEl = document.getElementById('about-spec-power-source');

```

```
756 |     const profileEl = document.getElementById('about-spec-powe
    | r-profile');
757 |     const legacyEl = document.getElementById('about-spec-power
    | ');
758 |
759 |     let badgeText = 'AC Mains';
760 |     let badgeClass = 'bg-secondary';
761 |     let sourceText = 'Питание: Стационарная электросеть 220V';
762 |     let profileText = powerProfileName ? `Профиль: ${powerProf
    | ileName}` : 'Профиль: AC Mains / Desktop';
763 |
764 |     if (batteryWear) {
765 |         if (batteryWear.has_battery) {
766 |             badgeText = batteryWear.is_charging ? 'AC Adapter' : '
    | Battery';
767 |             badgeClass = batteryWear.is_charging ? 'bg-success' :
    | (batteryWear.percent < 20 ? 'bg-danger' : 'bg-warning text-dar
    | k');
768 |             sourceText = `Батарея: ${batteryWear.percent}% (${batt
    | eryWear.is_charging ? 'Заряжается' : 'Работа от батареи'})`;
769 |         } else {
770 |             badgeText = batteryWear.power_source ? (batteryWear.po
    | wer_source.includes('AC') ? 'AC Mains' : batteryWear.power_sou
    | rce) : 'AC Mains';
771 |             badgeClass = 'bg-secondary';
772 |             sourceText = 'Питание: Стационарная электросеть 220V';
773 |         }
774 |     }
775 |
776 |     if (badgeEl) {
777 |         badgeEl.textContent = badgeText;
778 |         badgeEl.className = `badge ${badgeClass} font-monospace`
    | ;
779 |     }
780 |     if (sourceEl) {
781 |         sourceEl.textContent = sourceText;
782 |     }
783 |     if (profileEl) {
```

```

784     if (powerProfileName) {
785         profileEl.textContent = `Профиль: ${powerProfileName}`
786     };
787     } else if (!profileEl.textContent || profileEl.textContent.trim() === '' || profileEl.textContent === 'Профиль: --') {
788         profileEl.textContent = profileText;
789     }
790     }
791     if (legacyEl) {
792         legacyEl.textContent = `${sourceText} | ${profileText}`;
793     }
794 }
795 async function fetchStorageBatteryWear(forceNetwork = false)
796 {
797     try {
798         const config = CACHE_STRATEGY.STORAGE_BATTERY;
799         const data = await cachedFetch(
800             '/api/v1/system/diagnostics/storage-battery',
801             'storage_battery_wear',
802             { ttl: config.ttl, strategy: config.strategy, forceNetwork }
803         );
804         if (!data) return;
805         // Disks Wear & SMART
806         const disksTbody = document.getElementById('diag-wear-disks-tbody');
807         if (disksTbody && data.disks_wear) {
808             setText('diag-wear-disks-count', `${data.disks_wear.length} дисков`);
809             if (data.disks_wear.length === 0) {
810                 disksTbody.innerHTML = '<tr><td colspan="9" class="text-center py-4 text-muted">Физических накопителей не обнаружено</td></tr>';
811             } else {
812                 const formatBytesLocal = (bytes) => {
813                     if (bytes == null || bytes === 0) return '0 B';

```

```

814         const k = 1024;
815         const sizes = ['B', 'KB', 'MB', 'GB', 'TB', 'PB'];
816         const i = Math.floor(Math.log(bytes) / Math.log(k)
);
817         return `${parseFloat((bytes / Math.pow(k, i)).toFixed(1)))} ${sizes[i]}`;
818     };
819
820     const formatPohLocal = (hours) => {
821         if (hours == null || hours <= 0) return null;
822         const days = Math.floor(hours / 24);
823         const years = (hours / (24 * 365.25)).toFixed(1);
824         if (days >= 365) return `${hours.toLocaleString()}
ч (≈ ${years} г)`;
825         if (days >= 1) return `${hours.toLocaleString()} ч
(${days} д)`;
826         return `${hours.toLocaleString()} ч`;
827     };
828
829     disksTbody.innerHTML = data.disks_wear.map(d => {
830         const diskName = d.name || d.model || d.device_id
|| 'Физический диск';
831         const devId = d.device_id ? `<span class="badge bg
-secondary-subtle text-light border border-secondary me-1.5">${
escapeHtml(d.device_id)}</span>` : '';
832         const isSsd = (d.media_type && d.media_type.toUpper
case().includes('SSD')) || (d.bus_type && d.bus_type.toUpperC
ase().includes('NVME'));
833         const diskIcon = isSsd ? 'bi-device-ssd text-info'
: 'bi-hdd text-primary';
834         const typeBus = [d.media_type, d.bus_type].filter(
Boolean).join(' / ') || 'Storage';
835         const partsBadge = d.partitions && d.partitions !=
= '_'
836         ? `<span class="badge bg-primary-subtle text-inf
o border border-info-subtle font-monospace">${escapeHtml(d.par
titions)}</span>`
837         : `<span class="text-muted">—</span>`;

```

```

838 |         const freeGbStr = d.free_gb != null ? `${d.free_gb
    | } GB` : '<span class="text-muted">—</span>';
839 |         const healthPct = d.health_pct != null ? d.health_
    | pct : 100;
840 |         const healthColor = healthPct >= 80 ? 'success' :
    | (healthPct >= 50 ? 'warning' : 'danger');
841 |
842 |         // Нарботка и дата первого включения
843 |         const pohFormatted = formatPohLocal(d.power_on_hou
    | rs);
844 |         const pohHtml = pohFormatted
845 |             ? `<div class="font-monospace text-info">${pohFo
    | rmatted}</div><div class="small text-muted font-monospace mt-0
    | .5">Старт: ${escapeHtml(d.first_power_on || '-')}</div>`
846 |             : '<span class="text-muted font-monospace small"
    | >—</span>';
847 |
848 |         // Объемы ввода/вывода (запись и чтение)
849 |         const ioHtml = `
850 |             <div class="font-monospace text-light" style="wh
    | ite-space: nowrap;" title="Записано за сессию">
851 |                 <i class="bi bi-arrow-up-circle text-warning m
    | e-1"></i>Записано: ${formatBytesLocal(d.bytes_written)}
852 |             </div>
853 |             <div class="small font-monospace text-info mt-0.
    | 5" style="white-space: nowrap;" title="Прочитано за сессию">
854 |                 <i class="bi bi-arrow-down-circle text-info me
    | -1"></i>Прочитано: ${formatBytesLocal(d.bytes_read)}
855 |             </div>
856 |         `;
857 |
858 |         return `
859 |         <tr>
860 |             <td class="fw-bold text-white">
861 |                 <div class="d-flex align-items-center">
862 |                     <i class="bi ${diskIcon} me-2 fs-6"></i>
863 |                     <div>
864 |                         <div class="text-truncate" style="max-widt

```

```

h: 250px;" title="${escapeHtml(diskName)}">${escapeHtml(diskName)}</div>
865         <div class="small text-muted font-monospace mt-0.5">${devId}${d.serial_number && d.serial_number !== 'N/A' ? `<span class="text-secondary">S/N: ${escapeHtml(d.serial_number)}</span>` : ''}</div>
866     </div>
867 </div>
868 </td>
869 <td class="text-muted font-monospace"><span class="badge bg-dark border border-secondary text-light">${escapeHtml(typeBus)}</span></td>
870 <td>${partsBadge}</td>
871 <td style="text-align: right;" class="text-light font-monospace">${d.total_gb} GB</td>
872 <td style="text-align: right;" class="text-success font-monospace">${freeGbStr}</td>
873 <td>${pohHtml}</td>
874 <td style="text-align: right;">${ioHtml}</td>
875 <td>
876     <div class="d-flex align-items-center gap-1.5">
877         <div class="about-sys-progress-track flex-grow-1" style="height: 6px; min-width: 55px;">
878             <div class="about-sys-progress-bar bg-${healthColor}" style="width: ${healthPct}%;"></div>
879         </div>
880         <span class="small font-monospace text-${healthColor} ms-1">${healthPct}%</span>
881     </div>
882 </td>
883 <td><span class="badge bg-${healthColor}-subtle text-${healthColor} border border-${healthColor}">${escapeHtml(d.status || 'OK')}</span></td>
884 </tr>
885 `;)).join('');
886 }
887 }

```

```

888
889     // Battery Degradation
890     const batContainer = document.getElementById('diag-batte
ry-details-container');
891     const batSourceBadge = document.getElementById('diag-bat
tery-source-badge');
892     if (batContainer && data.battery_wear) {
893         const b = data.battery_wear;
894         updatePowerDisplay(b);
895         if (batSourceBadge) {
896             batSourceBadge.textContent = b.power_source || 'AC M
ains';
897         }
898         if (!b.has_battery) {
899             batContainer.innerHTML = `
900                 <div class="text-center py-4 text-muted small">
901                     <i class="bi bi-plug-fill fs-4 text-info d-block
mb-1"></i>
902                         Стационарный компьютер – питание напрямую от эле
ктросети
903                     </div>
904                 `;
905         } else {
906             batContainer.innerHTML = `
907                 <table class="about-sys-spec-table">
908                     <tbody>
909                         <tr>
910                             <td class="about-sys-spec-key"><i class="bi
bi-battery-charging me-1.5 text-warning"></i>Уровень заряда</t
d>
911                             <td class="about-sys-spec-val fw-bold text-w
hite">${b.percent}% (${b.is_charging ? 'Заряжается' : 'Разряд'
})</td>
912                         </tr>
913                         <tr>
914                             <td class="about-sys-spec-key"><i class="bi
bi-shield-shaded me-1.5 text-info"></i>Заводская емкость</td>
915                             <td class="about-sys-spec-val font-monospace

```

```

    ">${b.design_capacity_mwh ? b.design_capacity_mwh + ' mWh' : '
    --'}</td>
916 |             </tr>
917 |             <tr>
918 |                 <td class="about-sys-spec-key"><i class="bi
bi-battery-full me-1.5 text-success"></i>Текущая емкость</td>
919 |                 <td class="about-sys-spec-val font-monospace
">${b.full_charge_capacity_mwh ? b.full_charge_capacity_mwh +
' mWh' : '--'}</td>
920 |             </tr>
921 |             <tr>
922 |                 <td class="about-sys-spec-key"><i class="bi
bi-heart-pulse me-1.5 text-danger"></i>Деградация (Износ)</td>
923 |                 <td class="about-sys-spec-val ${b.wear_level
_pct > 20 ? 'text-danger fw-bold' : 'text-success'}">
924 |                     ${b.wear_level_pct}% износа
925 |                 </td>
926 |             </tr>
927 |         </tbody>
928 |     </table>
929 |     `;
930 |     }
931 | }
932 | } catch (err) {
933 |     console.warn('[AboutSystemTab] fetchStorageBatteryWear e
rror:', err);
934 | }
935 | }
936 |
937 | /**
938 |  * Гранулярные функции получения данных для каждой карточки
KPI
939 |  */
940 | async function fetchKpiOs(forceNetwork = false) {
941 |     try {
942 |         const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000,
strategy: 'stale-while-revalidate' };
943 |         const opts = { ttl: config.ttl, strategy: config.strateg

```



```

    y, forceNetwork };
944     const data = await cachedFetch('/api/v1/dashboard/os', '
dash_os', opts);
945     if (data) {
946         if (data.display_title) setText('about-kpi-os-title',
data.display_title);
947         if (data.display_host) setText('about-kpi-os-host', da
ta.display_host);
948         if (data.workgroup) setText('about-kpi-os-workgroup',
`Workgroup: ${data.workgroup}`);
949         else if (data.display_workgroup) setText('about-kpi-os
-workgroup', data.display_workgroup);
950         if (data.uptime_human) setText('about-spec-uptime', `U
ptime: ${data.uptime_human}`);
951         if (data.os_build) setText('about-spec-os', data.os_bu
ild);
952         if (data.os_install_date) {
953             setText('about-ident-install-date', data.os_install_
date);
954         }
955     }
956 } catch (e) {
957     console.warn('[AboutSystemTab] fetchKpiOs error:', e);
958 }
959 }
960
961 async function fetchKpiSecurity(forceNetwork = false) {
962     try {
963         const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000,
strategy: 'stale-while-revalidate' };
964         const opts = { ttl: config.ttl, strategy: config.strateg
y, forceNetwork };
965         const data = await cachedFetch('/api/v1/dashboard/securi
ty', 'dash_sec', opts);
966         if (data) {
967             if (data.display_title) {
968                 const el = document.getElementById('about-kpi-sec-ti
tle');

```

```

969         if (el) {
970             el.textContent = data.display_title;
971             el.className = `about-sys-value ${data.status ===
'Active & Protected' ? 'text-success' : 'text-danger fw-bold'}
text-truncate mt-1`;
972         }
973     }
974     if (data.display_subtitle) setFormattedStatusHtml('abo
ut-kpi-sec-sub', data.display_subtitle);
975 }
976 } catch (e) {
977     console.warn('[AboutSystemTab] fetchKpiSecurity error:',
e);
978 }
979 }
980
981 async function fetchKpiCheckpoints(forceNetwork = false) {
982     try {
983         const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000,
strategy: 'stale-while-revalidate' };
984         const opts = { ttl: config.ttl, strategy: config.strateg
y, forceNetwork };
985         const data = await cachedFetch('/api/v1/dashboard/checkp
oints', 'dash_chk', opts);
986         if (data) {
987             if (data.display_title) setText('about-kpi-prot-title'
, data.display_title);
988             if (data.display_subtitle) setFormattedStatusHtml('abo
ut-kpi-prot-sub', data.display_subtitle);
989         }
990     } catch (e) {
991         console.warn('[AboutSystemTab] fetchKpiCheckpoints error
:', e);
992     }
993 }
994
995 async function fetchKpiStorage(forceNetwork = false) {
996     try {

```

```
997 |         const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000,  
    | strategy: 'stale-while-revalidate' };  
998 |         const opts = { ttl: config.ttl, strategy: config.strateg  
    | y, forceNetwork };  
999 |         const data = await cachedFetch('/api/v1/dashboard/storag  
    | e', 'dash_stor', opts);  
1000 |         if (data) {  
1001 |             if (data.display_title) setText('about-kpi-stor-title'  
    | , data.display_title);  
1002 |             if (data.display_subtitle) setText('about-kpi-stor-cle  
    | an', data.display_subtitle);  
1003 |         }  
1004 |     } catch (e) {  
1005 |         console.warn('[AboutSystemTab] fetchKpiStorage error:',  
    | e);  
1006 |     }  
1007 | }  
1008 |  
1009 | async function fetchHardwareSpecs(forceNetwork = false) {  
1010 |     await fetchHardwareSpec(forceNetwork);  
1011 | }  
1012 |  
1013 | async function fetchUserEnvSecurity(forceNetwork = false) {  
1014 |     await Promise.allSettled([  
1015 |         fetchSystemSummary(forceNetwork ? false : true),  
1016 |         fetchSystemControlStatus(forceNetwork),  
1017 |         fetchBackupStatus(forceNetwork)  
1018 |     ]);  
1019 | }  
1020 |  
1021 | async function fetchDisksVolumes(forceNetwork = false) {  
1022 |     await fetchSystemSummary(forceNetwork ? false : true);  
1023 | }  
1024 |  
1025 | async function fetchWearPanel(forceNetwork = false) {  
1026 |     await fetchStorageBatteryWear(forceNetwork);  
1027 | }  
1028 |
```

```
1029 | async function fetchBatteryPanel(forceNetwork = false) {
1030 |     await fetchStorageBatteryWear(forceNetwork);
1031 | }
1032 |
1033 | async function fetchHwTreePanel(forceNetwork = false) {
1034 |     await fetchHardwareSpec(forceNetwork);
1035 | }
1036 |
1037 | /**
1038 |  * Загрузка всех 4 сводных KPI-карточек
1039 |  */
1040 | async function fetchKpiPanels(forceNetwork = false) {
1041 |     await Promise.allSettled([
1042 |         fetchKpiOs(forceNetwork),
1043 |         fetchKpiSecurity(forceNetwork),
1044 |         fetchKpiCheckpoints(forceNetwork),
1045 |         fetchKpiStorage(forceNetwork)
1046 |     ]);
1047 | }
1048 |
1049 | /**
1050 |  * Реестр панелей и частоты их опроса
1051 |  */
1052 | const panelPollingRegistry = {
1053 |     'about_kpi_os': { fn: fetchKpiOs, defaultFreq: 'manual' },
1054 |     'about_kpi_sec': { fn: fetchKpiSecurity, defaultFreq: 'manual' },
1055 |     'about_kpi_prot': { fn: fetchKpiCheckpoints, defaultFreq:
1056 | 'manual' },
1057 |     'about_kpi_stor': { fn: fetchKpiStorage, defaultFreq: 'manual' },
1058 |     'about_hw_specs': { fn: fetchHardwareSpecs, defaultFreq: '
1059 | start' },
1060 |     'about_user_env': { fn: fetchUserEnvSecurity, defaultFreq:
1061 | 'manual' },
1062 |     'about_disks': { fn: fetchDisksVolumes, defaultFreq: 'manual' },
1063 |     'about_wear': { fn: fetchWearPanel, defaultFreq: 'manual' }
```

```

    },
1061     'about_battery': { fn: fetchBatteryPanel, defaultFreq: 'ma
nual' },
1062     'about_hw_tree': { fn: fetchHwTreePanel, defaultFreq: 'sta
rt' }
1063   };
1064
1065   const panelTimers = new Map();
1066
1067   function getPanelFrequency(pollId) {
1068     try {
1069       const saved = localStorage.getItem(`poll_freq_${pollId}`
);
1070       if (saved !== null && saved !== undefined && saved !== '
') return saved;
1071     } catch (_) {}
1072     return panelPollingRegistry[pollId]?.defaultFreq || 'start
';
1073   }
1074
1075   function setPanelFrequency(pollId, freq) {
1076     try {
1077       localStorage.setItem(`poll_freq_${pollId}`, freq);
1078     } catch (_) {}
1079     applyPanelPoller(pollId, freq, false);
1080   }
1081
1082   function stopPanelPoller(pollId) {
1083     const pollerId = `tab-about-system_${pollId}`;
1084     if (window.unregisterTabPoller) {
1085       window.unregisterTabPoller(pollerId);
1086     }
1087     if (panelTimers.has(pollId)) {
1088       clearInterval(panelTimers.get(pollId));
1089       panelTimers.delete(pollId);
1090     }
1091   }
1092

```

```

1093 | function applyPanelPoller(pollId, freq, runInitial = false)
      | {
1094 |     stopPanelPoller(pollId);
1095 |     const config = panelPollingRegistry[pollId];
1096 |     if (!config) return;
1097 |
1098 |     if (freq === 'start') {
1099 |         if (runInitial) {
1100 |             config.fn();
1101 |         }
1102 |         return;
1103 |     }
1104 |
1105 |     if (freq === 'manual') {
1106 |         if (runInitial) {
1107 |             config.fn();
1108 |         }
1109 |         return;
1110 |     }
1111 |
1112 |     const intervalSec = parseInt(freq, 10);
1113 |     if (isNaN(intervalSec) || intervalSec <= 0) return;
1114 |
1115 |     const intervalMs = intervalSec * 1000;
1116 |     const pollerId = `tab-about-system_${pollId}`;
1117 |
1118 |     if (window.registerTabPoller) {
1119 |         window.registerTabPoller('tab-about-system', async () =>
      | {
1120 |             if (isLiveActive && !isUpdating) {
1121 |                 await config.fn();
1122 |             }
1123 |             }, intervalMs, { pollerId, immediate: runInitial });
1124 |     } else {
1125 |         if (runInitial) config.fn();
1126 |         const timer = setInterval(async () => {
1127 |             if (isLiveActive && !isUpdating) {
1128 |                 await config.fn();

```

```
1129     }
1130     }, intervalMs);
1131     panelTimers.set(pollId, timer);
1132 }
1133 }
1134
1135 function initPanelPollers() {
1136     document.querySelectorAll('.poll-freq-select').forEach(select => {
1137         const pollId = select.dataset.pollId;
1138         if (!pollId) return;
1139
1140         const currentFreq = getPanelFrequency(pollId);
1141         select.value = currentFreq;
1142
1143         select.onchange = (e) => {
1144             const newFreq = e.target.value;
1145             setPanelFrequency(pollId, newFreq);
1146             if (newFreq !== 'manual' && newFreq !== 'start') {
1147                 panelPollingRegistry[pollId]?.fn(true);
1148             }
1149             if (window.showToast) {
1150                 const label = select.options[select.selectedIndex]?.
1151                 text || newFreq;
1152                 window.showToast(`Частота опроса обновлена: ${label}
1153                 `, 'info');
1154             }
1155             };
1156
1157             applyPanelPoller(pollId, currentFreq, true);
1158         });
1159     }
1160
1161     async function pollLiveTelemetry() {
1162         isUpdating = true;
1163         try {
1164             await fetchSystemSummary(true);
1165         } finally {
```

```

1164     isUpdating = false;
1165 }
1166 }
1167
1168
1169 async function fetchSystemSummary(isLightPoll = false) {
1170     try {
1171         const config = CACHE_STRATEGY.SYSTEM_SUMMARY;
1172         const snap = await cachedFetch(
1173             '/api/v1/system/summary?process_limit=25',
1174             'system_summary',
1175             { ttl: config.ttl, strategy: isLightPoll ? 'stale-while-
e-revalidate' : 'network-first' }
1176         );
1177         if (!snap) return;
1178
1179         // 1. Identity & Locale Block
1180         const host = snap.hostname || 'DELL-VOSTRO';
1181         const user = snap.username || 'onela';
1182         const lang = snap.system_language || 'Русский (Россия) [
ru-RU]';
1183         const userLoc = snap.user_locale || 'ru-RU';
1184         const sysLoc = snap.system_locale || 'ru-RU';
1185         const instLang = snap.os_install_language || snap.instal
l_language || '';
1186         const tz = snap.timezone || 'UTC+03:00';
1187         const cp = snap.codepage || 'UTF-8 (ACP: 65001)';
1188         const inputs = Array.isArray(snap.input_languages) && sn
ap.input_languages.length > 0
1189             ? snap.input_languages.join(', ')
1190             : 'Русский (RU), English (US),  (IL)';
1191         const osBuild = snap.os_build ? `${snap.os_name || 'Wind
ows 11'} (Build ${snap.os_build})` : (snap.os_name || 'Windows
11');
1192
1193         setText('about-ident-hostname', host);
1194         setText('about-ident-domain', `Workgroup / Host: ${host}
`);

```



```

1195 |     setText('about-ident-username', user);
1196 |     setText('about-ident-language', lang);
1197 |     const localesText = instLang
1198 |         ? `Локали: User: ${userLoc} | Sys: ${sysLoc} | Оригина
1199 | л: ${instLang}`
1200 |         : `Локали: User: ${userLoc} | Sys: ${sysLoc}`;
1201 |     setText('about-ident-locales', localesText);
1202 |     setText('about-ident-timezone', tz);
1203 |     setText('about-ident-codepage', `Кодировка: ${cp}`);
1204 |     const installDate = snap.os_install_date || 'Не определе
1205 | на';
1206 |
1207 |     // Top KPI Card 1: Operating System
1208 |     setText('about-kpi-os-title', `${snap.os_name || 'Window
1209 | s 11'} (${snap.cpu?.architecture || 'AMD64'})`);
1210 |     setText('about-kpi-os-host', `Host: ${host}`);
1211 |     const wg = snap.workgroup || 'WORKGROUP';
1212 |     setText('about-kpi-os-workgroup', `Workgroup: ${wg}`);
1213 |
1214 | /**
1215 |  * Форматирование коммерческого названия процессора (наприме
1216 | р, Intel Core i5-10400)
1217 |  * @param {string} rawModel - Название модели (из реестра ил
1218 | и API)
1219 |  * @returns {string} Чистое коммерческое наименование
1220 |  */
1221 | function formatCpuCommercial(rawModel) {
1222 |     if (!rawModel) return 'Intel Core i5-10400';
1223 |
1224 |     // Если передана сырая строка CPUID
1225 |     if (rawModel.includes('Family 6 Model 165') || rawModel.in
1226 | cludes('Intel64 Family 6 Model 165')) {

```

Hz, лишних суффиксов

```
1227     let clean = rawModel
1228         .replace(/\(R\)/gi, '')
1229         .replace(/\(TM\)/gi, '')
1230         .replace(/\bCPU\b/gi, '')
1231         .replace(/@.*$/i, '')
1232         .replace(/\b\d+-Core Processor\b/gi, '')
1233         .replace(/\s+/g, ' ')
1234         .trim();
1235
1236     return clean || 'Intel Core i5-10400';
1237 }
1238
1239     // 2. Telemetry Live & Spec Table
1240     if (snap.cpu) {
1241         // Specification Table CPU
1242         setText('about-spec-cpu', formatCpuCommercial(snap.cpu
1243 .model));
1244     }
1245
1246     if (snap.memory) {
1247         const total = Number(snap.memory.total_gb || 0).toFixed(
1248 1);
1249         const used = Number(snap.memory.used_gb || 0).toFixed(
1250 1);
1251         const avail = Number(snap.memory.available_gb || 0).to
1252 Fixed(1);
1253         const pct = snap.memory.percent || 0;
1254
1255         setText('about-telemetry-ram-val', `${used} / ${total}
1256 GB`);
1257         setText('about-telemetry-ram-sub', `${pct}% занято (${
1258 avail} GB свободно)`);
1259         const ramBar = document.getElementById('about-telemetr
1260 y-ram-bar');
1261         if (ramBar) {
1262             ramBar.style.width = `${Math.min(100, Math.max(0, pc
1263 t))}%`;
```

```
1256 |         ramBar.className = pct > 85 ? 'about-sys-progress-bar
      | bg-danger' : (pct > 65 ? 'about-sys-progress-bar bg-warning'
      | : 'about-sys-progress-bar bg-info');
1257 |     }
1258 |
1259 |     // Specification Table RAM (Hardware installed capacity)
1260 |     setText('about-spec-ram', `${total} GB RAM (${avail} GB
      | свободно)`);
1261 | }
1262 |
1263 | if (Array.isArray(snap.gpus) && snap.gpus.length > 0) {
1264 |     const gpu = snap.gpus[0];
1265 |     setText('about-telemetry-gpu-name', gpu.name || 'GPU')
      | ;
1266 |     const vramTxt = gpu.memory_total_gb ? `VRAM: ${Number(
      | gpu.memory_total_gb).toFixed(1)} GB` : 'VRAM: N/A';
1267 |     const loadTxt = gpu.load_percent !== null && gpu.load_
      | percent !== undefined ? ` | Нагрузка: ${gpu.load_percent}%` :
      | '';
1268 |     setText('about-telemetry-gpu-vram', `${vramTxt}${loadT
      | xt}`);
1269 |
1270 |     const badgeGpu = document.getElementById('about-teleme
      | try-gpu-badge');
1271 |     if (badgeGpu) {
1272 |         const caps = [];
1273 |         if (gpu.has_cuda) caps.push('CUDA');
1274 |         if (gpu.has_directml) caps.push('DirectML');
1275 |         badgeGpu.textContent = caps.length > 0 ? caps.join('
      | + ') : 'Active GPU';
1276 |     }
1277 | }
1278 |
1279 | if (snap.disk_io) {
1280 |     const rKbs = Math.round((snap.disk_io.read_bytes_per_s
      | ec || 0) / 1024);
1281 |     const wKbs = Math.round((snap.disk_io.write_bytes_per_
```

```

    sec || 0) / 1024);
1282     const totalMb = (((snap.disk_io.read_bytes_per_sec ||
    0) + (snap.disk_io.write_bytes_per_sec || 0)) / (1024 * 1024))
    .toFixed(2);
1283     setText('about-telemetry-disk-val', `${totalMb} MB/s`)
    ;
1284     setText('about-telemetry-disk-rates', `Чтение: ${rKbs}
    КБ/с | Запись: ${wKbs} КБ/с`);
1285 }
1286
1287 // Specification Table general
1288 setText('about-spec-host', host);
1289 setText('about-spec-os', osBuild);
1290 if (snap.uptime_seconds) {
1291     const sec = snap.uptime_seconds;
1292     const h = Math.floor(sec / 3600);
1293     const m = Math.floor((sec % 3600) / 60);
1294     setText('about-spec-uptime', `Uptime: ${h}h ${m}m`);
1295 }
1296
1297 // 3. Disks Table
1298 if (Array.isArray(snap.disks) && snap.disks.length > 0)
    {
1299     currentDisks = snap.disks;
1300     renderDisksTable(snap.disks);
1301
1302     // Update Top KPI Storage C:
1303     const cDrive = snap.disks.find(d => (d.device || '').t
    oUpperCase().startsWith('C')) || snap.disks[0];
1304     if (cDrive) {
1305         const freeGb = Number(cDrive.free_gb || 0).toFixed(1
    );
1306         const totalGb = Number(cDrive.total_gb || 0).toFixed
    (1);
1307         setText('about-kpi-stor-title', `${freeGb} GB Free /
    ${totalGb} GB`);
1308     }
1309 }

```

```
1310 |
1311 |     // 4. Monitors & Displays Block
1312 |     if (Array.isArray(snap.monitors) && snap.monitors.length
    |     > 0) {
1313 |         const monShorts = snap.monitors.map(m => {
1314 |             const prim = m.is_primary ? ' [Основной]' : '';
1315 |             return `${m.name || 'Monitor'} (${m.width}x${m.height}
    | t)@${m.frequency_hz}Hz${prim})`;
1316 |         });
1317 |         setText('about-ident-monitors', monShorts.join(', '));
1318 |         setText('about-spec-monitors', monShorts.join(', '));
1319 |     }
1320 |
1321 |     // 5. Windows Updates Block
1322 |     if (snap.updates) {
1323 |         const kbCount = snap.updates.installed_kb_count ? ` ($
    | {snap.updates.installed_kb_count} KBs)` : '';
1324 |         const updTxt = `${snap.updates.status || 'Up to date'}
    | ${kbCount}`;
1325 |         setText('about-ident-updates', updTxt);
1326 |         setText('about-spec-update', updTxt);
1327 |     }
1328 |
1329 |     // 5.1 MS Office Block
1330 |     if (snap.office) {
1331 |         const offTxt = snap.office.status || (snap.office.inst
    | alled ? `${snap.office.product_name || 'MS Office'} (${snap.of
    | fice.version || ''})` : 'Не установлен');
1332 |         setText('about-ident-ms-office', offTxt);
1333 |     } else {
1334 |         setText('about-ident-ms-office', 'Не установлен');
1335 |     }
1336 |
1337 |     // 5.2 OneDrive Storage Block
1338 |     if (snap.onedrive) {
1339 |         const odTxt = snap.onedrive.status || (snap.onedrive.i
    | nstalled ? `${snap.onedrive.free_gb || 0} GB своб.` : 'Не наст
    | роено');
```

```

1340         setText('about-ident-onedrive', odTxt);
1341     } else {
1342         setText('about-ident-onedrive', 'Не настроено');
1343     }
1344
1345     // 5.3 Power & Energy Block
1346     if (snap.battery) {
1347         updatePowerDisplay(snap.battery, snap.power_profile ||
snap.battery.power_profile);
1348     } else {
1349         updatePowerDisplay(null, snap.power_profile);
1350     }
1351
1352     // 6. Processes Table
1353     if (Array.isArray(snap.top_processes)) {
1354         currentProcesses = snap.top_processes;
1355         renderProcessesTable(snap.top_processes);
1356     }
1357
1358     } catch (e) {
1359         console.warn('[AboutSystemTab] fetchSystemSummary warnin
g:', e);
1360     }
1361 }
1362
1363 async function fetchSystemControlStatus() {
1364     try {
1365         const config = CACHE_STRATEGY.CONTROL_STATUS;
1366         const res = await cachedFetch(
1367             '/api/system-control/status',
1368             'system_control_status',
1369             { ttl: config.ttl, strategy: config.strategy }
1370         );
1371         if (!res) return;
1372
1373         const isElevated = res.is_elevated;
1374         const privBadge = document.getElementById('about-sys-pri
vilege-badge');

```

```
1375 |         if (privBadge) {
1376 |             if (isElevated) {
1377 |                 privBadge.className = 'badge rounded-pill bg-success
| -subtle text-success border border-success px-2.5 py-1';
1378 |                 privBadge.textContent = 'Administrator Access';
1379 |             } else {
1380 |                 privBadge.className = 'badge rounded-pill bg-seconda
| ry-subtle text-secondary border border-secondary px-2.5 py-1';
1381 |                 privBadge.textContent = 'Standard User Mode';
1382 |             }
1383 |         }
1384 |         setText('about-ident-privilege', isElevated ? 'Права: Ад
| министратор (Full Access)' : 'Права: Стандартный пользователь'
| );
1385 |
1386 |         // Security Details
1387 |         const sec = res.security || {};
1388 |         const rest = res.restore || {};
1389 |         const disk = res.disk || {};
1390 |         const pwr = res.power || {};
1391 |         const upd = res.update || {};
1392 |
1393 |         // KPI 2: Security & Defender
1394 |         const defActive = sec.defender_enabled !== false;
1395 |         const kpiSecTitle = document.getElementById('about-kpi-s
| ec-title');
1396 |         if (kpiSecTitle) {
1397 |             kpiSecTitle.textContent = defActive ? 'Active & Protec
| ted' : 'Attention Required';
1398 |             kpiSecTitle.className = `about-sys-value ${defActive ?
| 'text-success' : 'text-danger fw-bold'} text-truncate mt-1`;
1399 |         }
1400 |         const fwOk = sec.firewall_overall_enabled !== false;
1401 |         const uacOk = sec.uac_enabled !== false;
1402 |         setFormattedStatusHtml('about-kpi-sec-sub', `Firewall: $
| {fwOk ? 'ON' : 'OFF'} | UAC: ${uacOk ? 'ON' : 'OFF'}`);
1403 |
1404 |         // KPI 3: System Protection
```

```
1405     const countPoints = rest.restore_points_count || 0;
1406     setText('about-kpi-prot-title', `${countPoints} Checkpoi
nts`);
1407     const protActive = Boolean(rest.system_protection_enable
d);
1408     setFormattedStatusHtml('about-kpi-prot-sub', `Protection
: ${protActive ? 'Active' : 'Disabled'}`);
1409
1410     // KPI 4: Cleanable estimate & Total Disk Size
1411     const cleanMb = disk.cleanup_estimate?.total_cleanable_m
b || 150;
1412     const cDrive = currentDisks.find(d => (d.device || '').t
oUpperCase().startsWith('C')) || currentDisks[0];
1413     const totalGbTxt = cDrive ? ` | Bcero: ${Number(cDrive.t
otal_gb || 0).toFixed(1)} GB` : '';
1414     setText('about-kpi-stor-clean', `Cleanable: ~${cleanMb}
MB${totalGbTxt}`);
1415
1416     // Security table rows
1417     setStatusText('about-sec-defender', defActive, 'Enabled'
, 'Disabled');
1418     setStatusText('about-sec-realtime', sec.realtime_protect
ion_enabled !== false && sec.realtime_protection_enabled !== 0
&& sec.realtime_protection_enabled !== 'Disabled', 'Enabled',
'Disabled');
1419
1420     const fwDom = sec.firewall_profiles?.Domain ?? sec.firew
all_domain_enabled;
1421     const fwPriv = sec.firewall_profiles?.Private ?? sec.fir
ewall_private_enabled;
1422     const fwPub = sec.firewall_profiles?.Public ?? sec.firew
all_public_enabled;
1423
1424     setStatusText('about-sec-fw-domain', fwDom === true || f
wDom === 'Active' || fwDom === 1 || fwDom === 'ON', 'Active',
'Disabled');
1425     setStatusText('about-sec-fw-private', fwPriv === true ||
fwPriv === 'Active' || fwPriv === 1 || fwPriv === 'ON', 'Acti
```



```

    ve', 'Disabled');
1426     setStatusText('about-sec-fw-public', fwPub === true || fwPub === 'Active' || fwPub === 1 || fwPub === 'ON', 'Active', 'Disabled');
1427     setStatusText('about-sec-uac', uacOk, 'Enabled', 'Disabled');
1428
1429     // Power Scheme & Updates
1430     if (pwr.active_plan_name) {
1431         updatePowerDisplay(null, pwr.active_plan_name);
1432     }
1433     if (upd.status) {
1434         const kbCount = upd.recent_hotfixes_count ? ` (${upd.recent_hotfixes_count} KBs installed)` : '';
1435         const updTxt = `${upd.status}${kbCount}`;
1436         setText('about-spec-update', updTxt);
1437         setText('about-ident-updates', updTxt);
1438     }
1439
1440     } catch (e) {
1441         console.warn('[AboutSystemTab] fetchSystemControlStatus warning:', e);
1442     }
1443 }
1444
1445
1446 function bindAIEvents() {
1447     const btnRescan = document.getElementById('btn-about-ai-rescan');
1448     if (btnRescan) {
1449         btnRescan.onclick = () => runAIDiagnostics(true);
1450     }
1451
1452     const btnToggleStages = document.getElementById('btn-about-ai-toggle-stages');
1453     const stagesBox = document.getElementById('about-ai-stages-container');
1454     const icStages = document.getElementById('ic-about-ai-togg

```

```
le-stages');
1455     const txtStages = document.getElementById('txt-about-ai-to
toggle-stages');
1456     if (btnToggleStages && stagesBox) {
1457         btnToggleStages.onclick = () => {
1458             const isHidden = stagesBox.classList.toggle('d-none');
1459             if (icStages) {
1460                 icStages.className = isHidden ? 'bi bi-chevron-down
ms-0.5' : 'bi bi-chevron-up ms-0.5';
1461             }
1462             if (txtStages) {
1463                 txtStages.textContent = isHidden ? 'Показать этапы'
: 'Свернуть этапы';
1464             }
1465         };
1466     }
1467
1468     const btnToggleInstruction = document.getElementById('btn-
about-ai-toggle-instruction');
1469     const instructionCollapse = document.getElementById('about
-ai-instruction-collapse');
1470     const icInstruction = document.getElementById('ic-about-ai
-instruction');
1471     if (btnToggleInstruction && instructionCollapse) {
1472         btnToggleInstruction.onclick = () => {
1473             const isShown = !instructionCollapse.classList.toggle(
'd-none');
1474             if (icInstruction) {
1475                 icInstruction.className = isShown ? 'bi bi-chevron-u
p text-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1476             }
1477         };
1478     }
1479
1480     const btnCopyInstruction = document.getElementById('btn-ab
out-ai-copy-instruction');
1481     if (btnCopyInstruction) {
1482         btnCopyInstruction.onclick = () => {
```

```
1483 |         const text = document.getElementById('about-ai-instruc
      | tion-text')?.textContent || '';
1484 |         navigator.clipboard.writeText(text).then(() => {
1485 |             const s = document.getElementById('txt-about-ai-copy
      | -instruction');
1486 |             if (s) s.textContent = 'Скопировано!';
1487 |             setTimeout(() => { if (s) s.textContent = 'Копироват
      | ь'; }, 1800);
1488 |         });
1489 |     };
1490 | }
1491 |
1492 |     const btnTogglePrompt = document.getElementById('btn-about
      | -ai-toggle-prompt');
1493 |     const promptCollapse = document.getElementById('about-ai-p
      | rompt-collapse');
1494 |     const icPrompt = document.getElementById('ic-about-ai-prom
      | pt');
1495 |     if (btnTogglePrompt && promptCollapse) {
1496 |         btnTogglePrompt.onclick = () => {
1497 |             const isShown = !promptCollapse.classList.toggle('d-no
      | ne');
1498 |             if (icPrompt) {
1499 |                 icPrompt.className = isShown ? 'bi bi-chevron-up tex
      | t-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1500 |             }
1501 |         };
1502 |     }
1503 |
1504 |     const btnCopyPrompt = document.getElementById('btn-about-a
      | i-copy-prompt');
1505 |     if (btnCopyPrompt) {
1506 |         btnCopyPrompt.onclick = () => {
1507 |             const text = document.getElementById('about-ai-prompt-
      | text')?.textContent || '';
1508 |             navigator.clipboard.writeText(text).then(() => {
1509 |                 const s = document.getElementById('txt-about-ai-copy
      | -prompt');
```

```
1510         if (s) s.textContent = 'Скопировано!';
1511         setTimeout(() => { if (s) s.textContent = 'Копироват
1512         ь'; }, 1800);
1513     });
1514 }
1515
1516     const btnToggleRaw = document.getElementById('btn-about-ai
1517     -toggle-raw');
1518     const rawCollapse = document.getElementById('about-ai-raw-
1519     collapse');
1520     const icRaw = document.getElementById('ic-about-ai-raw');
1521     if (btnToggleRaw && rawCollapse) {
1522         btnToggleRaw.onclick = () => {
1523             const isShown = !rawCollapse.classList.toggle('d-none'
1524             );
1525             if (icRaw) {
1526                 icRaw.className = isShown ? 'bi bi-chevron-up text-m
1527                 uted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1528             }
1529         };
1530     }
1531
1532     const btnCopyRaw = document.getElementById('btn-about-ai-c
1533     opy-raw');
1534     if (btnCopyRaw) {
1535         btnCopyRaw.onclick = () => {
1536             const text = document.getElementById('about-ai-raw-tex
1537             t').textContent || '';
1538             navigator.clipboard.writeText(text).then(() => {
1539                 const s = document.getElementById('txt-about-ai-copy
1540                 -raw');
1541                 if (s) s.textContent = 'Скопировано!';
1542                 setTimeout(() => { if (s) s.textContent = 'Копироват
1543                 ь'; }, 1800);
1544             });
1545         };
1546     }
1547 }
```

```
1539
1540     const btnToggleSnapshot = document.getElementById('btn-abo
1541 | ut-ai-toggle-snapshot');
1542     const snapshotCollapse = document.getElementById('about-ai
1543 | -snapshot-collapse');
1544     const icSnapshot = document.getElementById('ic-about-ai-sn
1545 | apshot');
1546     if (btnToggleSnapshot && snapshotCollapse) {
1547         btnToggleSnapshot.onclick = () => {
1548             const isShown = !snapshotCollapse.classList.toggle('d-
1549 | none');
1550             if (icSnapshot) {
1551                 icSnapshot.className = isShown ? 'bi bi-chevron-up t
1552 | ext-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1553             }
1554         };
1555     }
1556
1557     const btnCopySnapshot = document.getElementById('btn-about
1558 | -ai-copy-snapshot');
1559     if (btnCopySnapshot) {
1560         btnCopySnapshot.onclick = () => {
1561             const text = document.getElementById('about-ai-snapsho
1562 | t-text')?.textContent || '';
1563             navigator.clipboard.writeText(text).then(() => {
1564                 const s = document.getElementById('txt-about-ai-copy
1565 | -snapshot');
1566                 if (s) s.textContent = 'Скопировано!';
1567                 setTimeout(() => { if (s) s.textContent = 'Копироват
1568 | ь JSON'; }, 1800);
1569             });
1570         };
1571     }
1572
1573     function renderAIDiagnosticStages(stages, isCompleted = true
1574 | ) {
1575         const container = document.getElementById('about-ai-stages
```

```

    -container');
1567     if (!container) return;
1568     if (!Array.isArray(stages) || stages.length === 0) {
1569         container.innerHTML = '<div class="small text-muted text
- center py-2">Этапы пошагового аудита не запущены</div>';
1570         return;
1571     }
1572
1573     container.innerHTML = stages.map((st, idx) => {
1574         const isLatest = idx === stages.length - 1 && !isComple
ed;
1575         const icon = isLatest
1576             ? '<div class="spinner-grow spinner-grow-sm text-info"
style="width: 0.6rem; height: 0.6rem;" role="status"></div>'
1577             : (st.stage === 'error' || st.stage === 'warning')
1578             ? '<span class="text-warning small fw-bold" style="fon
t-size: 0.72rem;">⚠ </span>'
1579             : '<span class="text-success small fw-bold" style="fon
t-size: 0.72rem;">✅ </span>';
1580         const textClass = isLatest ? 'text-light fw-semibold' :
'text-muted';
1581         return `
1582             <div class="about-ai-stage-row ${textClass}">
1583                 <div class="d-flex align-items-center gap-2">
1584                     <span>${icon}</span>
1585                     <span>${escapeHtml(st.message || st.title || 'Выпо
лнение этапа...')}</span>
1586                 </div>
1587                 ${st.details ? `<div class="text-secondary ps-3 font
-monospace" style="font-size: 0.68rem; word-break: break-all;"
>⏏  ${escapeHtml(st.details)}</div>` : ''}
1588             </div>
1589         `;
1590     }).join('');
1591 }
1592
1593 function renderAIDomainCards(groupsResults) {
1594     const container = document.getElementById('about-ai-groups

```

```

    -container');
1595     if (!container) return;
1596
1597     container.innerHTML = groupsResults.map(g => {
1598         const isPending = g.status === 'pending';
1599         const isAnalyzing = g.status === 'analyzing';
1600         const isCrit = g.status === 'critical';
1601         const isWarn = g.status === 'warning';
1602
1603         const statusBadge = isPending
1604             ? '<span class="badge bg-secondary bg-opacity-25 text-muted border border-secondary border-opacity-25"><i class="bi bi-hourglass me-1"></i>Ожидание</span>'
1605             : isAnalyzing
1606             ? '<span class="badge bg-info bg-opacity-20 text-info border border-info border-opacity-50"><span class="spinner-border spinner-border-sm me-1" style="width:0.6rem;height:0.6rem;"></span>Анализ LLM...</span>'
1607             : isCrit
1608             ? '<span class="badge bg-danger text-white"><i class="bi bi-exclamation-octagon-fill me-1"></i>Критично</span>'
1609             : isWarn
1610             ? '<span class="badge bg-warning text-dark"><i class="bi bi-exclamation-triangle-fill me-1"></i>Внимание</span>'
1611             : '<span class="badge bg-success-subtle text-success border border-success"><i class="bi bi-check-circle-fill me-1"></i>В норме</span>';
1612
1613         const cardBorder = isCrit
1614             ? 'border-danger border-opacity-75 shadow-sm'
1615             : isWarn
1616             ? 'border-warning border-opacity-50 shadow-sm'
1617             : isAnalyzing
1618             ? 'border-info border-opacity-60 shadow-sm'
1619             : 'border-secondary border-opacity-30';
1620
1621         const metricsList = g.key_metrics && Object.keys(g.key_metrics).length > 0

```

```

1622 |         ? `<div class="mt-2 pt-1.5 border-top border-secondary
      |         border-opacity-20 d-flex flex-wrap gap-1.5 font-monospace" st
      |         yle="font-size: 0.70rem;">
1623 |             ${Object.entries(g.key_metrics).map(([k, v]) => `<
      |             span class="badge bg-dark bg-opacity-75 border border-secondar
      |             y text-light px-2 py-1"><strong class="text-info">${escapeHtml
      |             (k)}:</strong> <span class="text-white">${escapeHtml(String(v)
      |             )}</span></span>`).join('')}
1624 |             </div>`
1625 |         : '';
1626 |
1627 |     return `
1628 |         <div class="col-12 col-md-6">
1629 |             <div class="p-2.5 rounded bg-black bg-opacity-40 bor
      |             der ${cardBorder} h-100 d-flex flex-column justify-content-bet
      |             ween transition-all" id="domain-card-${g.group_id}">
1630 |                 <div>
1631 |                     <div class="d-flex align-items-center justify-co
      |                     ntent-between mb-1.5">
1632 |                         <div class="d-flex align-items-center gap-1.5"
      |                         >
1633 |                             <i class="bi ${escapeHtml(g.icon || 'bi-cpu'
      |                             )} text-info fs-6"></i>
1634 |                             <strong class="text-light" style="font-size:
      |                             0.82rem;">${escapeHtml(g.title)}</strong>
1635 |                             </div>
1636 |                             ${statusBadge}
1637 |                         </div>
1638 |                         <div class="small text-light mt-1" style="font-s
      |                         ize: 0.73rem; line-height: 1.45; white-space: pre-line;">
1639 |                             ${escapeHtml(g.summary || g.description || 'Ож
      |                             идание запуска группы...')}
1640 |                         </div>
1641 |                     </div>
1642 |                     ${metricsList}
1643 |                 </div>
1644 |             </div>
1645 |         `;

```



```
1646     }).join('');
1647 }
1648
1649 function renderAIAnomalies(anomalies) {
1650     const container = document.getElementById('about-ai-anomalies-container');
1651     if (!container) return;
1652     if (!Array.isArray(anomalies) || anomalies.length === 0) {
1653         container.innerHTML = '<span class="badge bg-success subtitle text-success border border-success px-2.5 py-1" style="font-size: 0.72rem;"><i class="bi bi-check-circle me-1"></i> Аномалий в подсистемах оборудования не обнаружено</span>';
1654         return;
1655     }
1656
1657     container.innerHTML = anomalies.map(a => {
1658         const isCrit = a.severity === 'critical';
1659         return `<span class="anomaly-pill ${isCrit ? 'critical' : 'warning'}">⬢ [${escapeHtml(a.subsystem || 'SYS')}] ${escapeHtml(a.title || '')}: ${escapeHtml(a.description || '')}</span>>`;
1660     }).join('');
1661 }
1662
1663 async function runAIDiagnostics(isManualRescan = false) {
1664     if (isAiRunning) return;
1665     isAiRunning = true;
1666
1667     // Устанавливаем флаг processing для немедленного переключения вкладок
1668     if (window.setTabProcessing) {
1669         window.setTabProcessing('tab-about-system', true);
1670     }
1671
1672     const btnRescan = document.getElementById('btn-about-ai-rescan');
1673     const txtRescan = document.getElementById('txt-about-ai-rescan');
```

```
1674     const icRescan = document.getElementById('ic-about-ai-resc
| an');
1675     const badgeHealth = document.getElementById('about-ai-heal
| th-badge');
1676     const badgeHealthBtn = document.getElementById('about-ai-h
| ealth-badge-btn');
1677     const engineTag = document.getElementById('about-ai-engine
| -name');
1678     const summaryEl = document.getElementById('about-ai-summar
| y-text');
1679     const errorSec = document.getElementById('about-ai-error-s
| ection');
1680     const errorText = document.getElementById('about-ai-error-
| text');
1681     const progressBar = document.getElementById('about-ai-step
| -progress');
1682     const synthesisStatus = document.getElementById('about-ai-
| synthesis-status');
1683     const actionsBox = document.getElementById('about-ai-actio
| ns-box');
1684     const actionsList = document.getElementById('about-ai-acti
| ons-list');
1685
1686     const liveStatusText = document.getElementById('about-ai-l
| ive-status-text');
1687     const chipLhm = document.getElementById('chip-sensor-lhm')
| ;
1688     const chipWmi = document.getElementById('chip-sensor-wmi')
| ;
1689     const chipRam = document.getElementById('chip-sensor-ram')
| ;
1690     const chipDisks = document.getElementById('chip-sensor-dis
| ks');
1691     const chipNet = document.getElementById('chip-sensor-net')
| ;
1692     const snapshotText = document.getElementById('about-ai-sna
| pshot-text');
1693     const tickCpu = document.getElementById('live-tick-cpu');
```

```
1694 |     const tickTemp = document.getElementById('live-tick-temp')
      | ;
1695 |     const tickRam = document.getElementById('live-tick-ram');
1696 |     const tickDisks = document.getElementById('live-tick-disks
      | ');
1697 |     const tickTopProc = document.getElementById('live-tick-top
      | -proc');
1698 |
1699 |     if (btnRescan) btnRescan.disabled = true;
1700 |     if (icRescan) icRescan.className = 'spinner-border spinner
      | -border-sm text-dark';
1701 |     if (txtRescan) txtRescan.textContent = 'Диагностика...';
1702 |
1703 |     if (errorSec) errorSec.classList.add('d-none');
1704 |     if (actionsBox) actionsBox.classList.add('d-none');
1705 |     if (progressBar) progressBar.style.width = '5%';
1706 |
1707 |     if (liveStatusText) liveStatusText.textContent = 'Опрос ап
      | паратных датчиков хоста (LibreHardwareMonitor, WMI, psutil, SM
      | ART)...';
1708 |
1709 |     // Reset chips to active polling state
1710 |     [chipLhm, chipWmi, chipRam, chipDisks, chipNet].forEach(c
      | => {
1711 |         if (c) c.className = 'badge bg-secondary bg-opacity-40 t
      | ext-light border border-secondary';
1712 |     });
1713 |
1714 |     if (badgeHealth) {
1715 |         badgeHealth.textContent = 'Health: Поэтапный опрос...';
1716 |         badgeHealth.className = 'badge bg-warning-subtle text-wa
      | rning border border-warning font-monospace';
1717 |     }
1718 |     if (badgeHealthBtn) {
1719 |         badgeHealthBtn.textContent = 'Health: Опрос...';
1720 |         badgeHealthBtn.className = 'badge bg-warning-subtle text
      | -warning border border-warning font-monospace ms-1';
1721 |     }
```

```
1722 |
1723 |     const stagesLog = [
1724 |         {
1725 |             stage: 'init',
1726 |             title: 'Сбор телеметрии',
1727 |             message: '□ Получение среза телеметрии и сенсоров хос
та...',
1728 |             details: 'Опрос LibreHardwareMonitor, WMI счетчиков и
SystemSnapshot',
1729 |         }
1730 |     ];
1731 |     renderAIDiagnosticStages(stagesLog, false);
1732 |
1733 |     try {
1734 |         // 1. Получение списка подготовленных 4 групп телеметрии
1735 |         const groups = await apiFetch('/api/v1/system/diagnose/g
roups');
1736 |         if (!Array.isArray(groups) || groups.length === 0) {
1737 |             throw new Error('Не удалось получить список диагностич
еских групп');
1738 |         }
1739 |
1740 |         if (snapshotText) {
1741 |             snapshotText.textContent = JSON.stringify(groups, null
, 2);
1742 |         }
1743 |
1744 |         // Обновление живой информационной плашки собранных метр
ик
1745 |         const compGroup = groups.find(g => g.group_id === 'compu
te_thermals');
1746 |         if (compGroup && compGroup.payload) {
1747 |             const cpu = compGroup.payload.cpu || {};
1748 |             if (tickCpu) tickCpu.textContent = `${cpu.model || 'CP
U'} (${cpu.load_percent ?? 0}%)`;
1749 |             const temps = compGroup.payload.sensors?.temperatures_
celsius || {};
1750 |             const tempVals = Object.values(temps);
```

```

1751 |         const maxT = tempVals.length > 0 ? Math.max(...tempVals) : null;
1752 |         if (tickTemp) tickTemp.textContent = maxT !== null ? `
    |         ${maxT}°C` : 'В норме';
1753 |     }
1754 |
1755 |     const memGroup = groups.find(g => g.group_id === 'memory_processes');
1756 |     if (memGroup && memGroup.payload) {
1757 |         const ram = memGroup.payload.ram || {};
1758 |         if (tickRam) tickRam.textContent = `${ram.used_gb ?? 0} / ${ram.total_gb ?? 0} GB (${ram.used_percent ?? 0}%)`;
1759 |         const procs = memGroup.payload.top_active_processes || [];
1760 |         if (tickTopProc) tickTopProc.textContent = procs.length > 0 ? `${procs[0].name} (${procs[0].cpu_percent ?? 0}%)` : 'Нет активных';
1761 |     }
1762 |
1763 |     const diskGroup = groups.find(g => g.group_id === 'storage_smart');
1764 |     if (diskGroup && diskGroup.payload) {
1765 |         const parts = diskGroup.payload.storage_partitions || [];
1766 |         if (tickDisks) tickDisks.textContent = parts.map(p => `${p.mountpoint || p.device || 'Vol'}: ${p.free_gb ?? 0} GB св.`).join(', ') || 'OK';
1767 |     }
1768 |
1769 |     // Отрисовка начальных 4 карточек в режиме ожидания с уже доступными собранными метриками
1770 |     const cardsState = groups.map(g => {
1771 |         const keyMetrics = {};
1772 |         if (g.group_id === 'compute_thermals') {
1773 |             const cpu = g.payload?.cpu || {};
1774 |             keyMetrics['cpu_load'] = `${cpu.load_percent ?? 0}%`;
1775 |             keyMetrics['cpu_model'] = cpu.model || 'CPU';

```

```

1776         const temps = g.payload?.sensors?.temperatures_celsius || {};
1777         const maxT = Object.values(temps).length > 0 ? Math.max(...Object.values(temps)) : null;
1778         keyMetrics['max_temp'] = maxT !== null ? `${maxT}°C` : 'В норме';
1779     } else if (g.group_id === 'memory_processes') {
1780         const ram = g.payload?.ram || {};
1781         keyMetrics['ram_used'] = `${ram.used_gb ?? 0} / ${ram.total_gb ?? 0} GB (${ram.used_percent ?? 0}%)`;
1782         const procs = g.payload?.top_active_processes || [];
1783         keyMetrics['top_process'] = procs.length > 0 ? `${procs[0].name} (${procs[0].cpu_percent ?? 0}%)` : 'Нет';
1784     } else if (g.group_id === 'storage_smart') {
1785         const parts = g.payload?.storage_partitions || [];
1786         const maxUsed = parts.length > 0 ? Math.max(...parts.map(p => p.used_percent || 0)) : 0;
1787         keyMetrics['max_volume_fill'] = `${maxUsed}%`;
1788         keyMetrics['volumes_count'] = parts.length;
1789     } else if (g.group_id === 'system_network') {
1790         const updates = g.payload?.system_updates || {};
1791         keyMetrics['reboot_pending'] = updates.reboot_pending ? 'Да' : 'Нет';
1792         keyMetrics['update_status'] = updates.status || 'Up to date';
1793     }
1794
1795     return {
1796         group_id: g.group_id,
1797         title: g.title,
1798         icon: g.icon,
1799         status: 'pending',
1800         summary: g.description,
1801         key_metrics: keyMetrics,
1802         payload: g.payload,
1803     };
1804 });
1805 renderAIDomainCards(cardsState);

```

```

1806
1807     const completedGroups = [];
1808
1809     // 2. Последовательный вызов каждой группы (цепочка Запр
ос -> Ответ -> Отображение)
1810     for (let i = 0; i < groups.length; i++) {
1811         const group = groups[i];
1812         const stepNum = i + 1;
1813         const totalSteps = groups.length;
1814         const pct = Math.round((stepNum / totalSteps) * 85);
1815         if (progressBar) progressBar.style.width = `${pct}%`;
1816
1817         // Подсветка активного чипа собираемого домена
1818         if (group.group_id === 'compute_thermals') {
1819             if (chipLhm) chipLhm.className = 'badge bg-info text
-dark border border-info fw-bold';
1820             if (chipWmi) chipWmi.className = 'badge bg-info text
-dark border border-info fw-bold';
1821         } else if (group.group_id === 'memory_processes') {
1822             if (chipRam) chipRam.className = 'badge bg-info text
-dark border border-info fw-bold';
1823         } else if (group.group_id === 'storage_smart') {
1824             if (chipDisks) chipDisks.className = 'badge bg-info
text-dark border border-info fw-bold';
1825         } else if (group.group_id === 'system_network') {
1826             if (chipNet) chipNet.className = 'badge bg-info text
-dark border border-info fw-bold';
1827         }
1828
1829         if (liveStatusText) {
1830             liveStatusText.textContent = `⏏ Анализ [${stepNum}/
${totalSteps}]: ${group.title} (оценка сенсоров и метрик)...`;
1831         }
1832
1833         // Обновляем статус карточки на "Анализ"
1834         cardsState[i].status = 'analyzing';
1835         cardsState[i].summary = `⏏ Модель выполняет целевой а
нализ телеметрии группы ${stepNum}/${totalSteps}...`;

```

```
1836 |         renderAIDomainCards(cardsState);
1837 |
1838 |         stagesLog.push({
1839 |             stage: 'group_run',
1840 |             title: `Группа ${stepNum}/${totalSteps}`,
1841 |             message: `⏱ [${stepNum}/${totalSteps}] Запрос модел
и: ${group.title}...`,
1842 |             details: group.description,
1843 |         });
1844 |         renderAIDiagnosticStages(stagesLog, false);
1845 |
1846 |         // Запрос к AI модели по данной группе
1847 |         const res = await apiFetch(`/api/v1/system/diagnose/gr
oup`, {
1848 |             method: 'POST',
1849 |             body: JSON.stringify({
1850 |                 group_id: group.group_id,
1851 |                 title: group.title,
1852 |                 payload: group.payload,
1853 |             }),
1854 |         });
1855 |
1856 |         if (res) {
1857 |             cardsState[i] = res;
1858 |             completedGroups.push(res);
1859 |         } else {
1860 |             cardsState[i].status = 'warning';
1861 |             cardsState[i].summary = 'Ответ не получен, использую
тся базовые метрики.';
1862 |             completedGroups.push(cardsState[i]);
1863 |         }
1864 |
1865 |         // Отмечаем чип как завершенный (зеленый)
1866 |         if (group.group_id === 'compute_thermals') {
1867 |             if (chipLhm) chipLhm.className = 'badge bg-success-s
ubtle text-success border border-success';
1868 |             if (chipWmi) chipWmi.className = 'badge bg-success-s
ubtle text-success border border-success';
```



```

1869         } else if (group.group_id === 'memory_processes') {
1870             if (chipRam) chipRam.className = 'badge bg-success-s
ubtle text-success border border-success';
1871         } else if (group.group_id === 'storage_smart') {
1872             if (chipDisks) chipDisks.className = 'badge bg-succe
ss-subtle text-success border border-success';
1873         } else if (group.group_id === 'system_network') {
1874             if (chipNet) chipNet.className = 'badge bg-success-s
ubtle text-success border border-success';
1875         }
1876
1877         // Немедленно обновляем красивую карточку на экране!
1878         renderAIDomainCards(cardsState);
1879
1880         stagesLog.push({
1881             stage: 'group_done',
1882             title: `Группа ${stepNum} завершена`,
1883             message: `□ [${stepNum}/${totalSteps}] ${group.title
}: ${cardsState[i].status.toUpperCase()}`,
1884             details: cardsState[i].summary.substring(0, 120) + (
cardsState[i].summary.length > 120 ? '...' : ''),
1885         });
1886         renderAIDiagnosticStages(stagesLog, false);
1887     }
1888
1889     if (liveStatusText) {
1890         liveStatusText.textContent = '□ Поэтапный опрос и ана
лиз всех доменов завершен.';
1891     }
1892
1893     // 3. Финальный синтез итогового вердикта
1894     if (progressBar) progressBar.style.width = '95%';
1895     stagesLog.push({
1896         stage: 'synthesis',
1897         title: 'Финальный синтез',
1898         message: '□ Формирование итогового заключения и расче
т Health Score...',
1899         details: 'Агрегация заключений всех 4 доменов',

```

```
1900 |     });
1901 |     renderAIDiagnosticStages(stagesLog, false);
1902 |
1903 |     const synthesis = await apiFetch('/api/v1/system/diagnos
1904 | e/synthesize', {
1905 |         method: 'POST',
1906 |         body: JSON.stringify({ groups: completedGroups }),
1907 |     });
1908 |
1909 |     if (progressBar) progressBar.style.width = '100%';
1910 |
1911 |     // Обновление итогового Health Badge
1912 |     const score = Number(synthesis.health_score || 100);
1913 |     const healthClass = score >= 80 ? 'bg-success-subtle tex
1914 | t-success border border-success' :
1915 |
1916 |     score >= 60 ? 'bg-warning-subtle text-warning border b
1917 | order-warning' :
1918 |     'bg-danger-subtle text-danger border border-danger';
1919 |
1920 |     if (badgeHealth) {
1921 |         badgeHealth.textContent = `Health: ${score}/100`;
1922 |         badgeHealth.className = `badge font-monospace ${health
1923 | Class}`;
1924 |     }
1925 |
1926 |     if (badgeHealthBtn) {
1927 |         badgeHealthBtn.textContent = `Health: ${score}/100`;
1928 |         badgeHealthBtn.className = `badge font-monospace ms-1
1929 | ${healthClass}`;
1930 |     }
1931 |
1932 |     if (engineTag) {
1933 |         engineTag.innerHTML = `<i class="bi bi-cpu me-1"></i>${
1934 | {escapeHtml(synthesis.ai_model_used || 'Heuristic Engine')}}`;
1935 |     }
1936 |
1937 |     if (synthesisStatus) {
1938 |         synthesisStatus.textContent = `${synthesis.status_labe
1939 | l} (${score}/100)`;
```

```

1931         synthesisStatus.className = `badge font-monospace ${
1932             score >= 80 ? 'bg-success text-white' :
1933             score >= 60 ? 'bg-warning text-dark' :
1934             'bg-danger text-white'
1935         }`;
1936     }
1937
1938     if (summaryEl) {
1939         summaryEl.textContent = synthesis.executive_summary ||
1940         'Анализ завершён.';
1941     }
1942
1943     // Отрисовка приоритетных действий
1944     if (Array.isArray(synthesis.critical_actions) && syntheses.critical_actions.length > 0) {
1945         if (actionsBox && actionsList) {
1946             actionsBox.classList.remove('d-none');
1947             actionsList.innerHTML = synthesis.critical_actions.map(act => `- <i class="bi bi-arrow-right-short text-warning me-1"></i>${escapeHtml(act)}</li>`).join('');
1948         }
1949     }
1950
1951     stagesLog.push({
1952         stage: 'done',
1953         title: 'Аудит завершен',
1954         message: `Поэтапная AI-диагностика успешно завершена (Health: ${score}/100)`,
1955         details: 'Все 4 функциональные группы оценены моделью'
1956     },
1957     );
1958     renderAIDiagnosticStages(stagesLog, true);
1959
1960 } catch (e) {
1961     console.error('[AboutSystemTab] Grouped AI Diagnosis error:', e);
1962     if (summaryEl) summaryEl.textContent = 'Ошибка выполнения поэтапной AI-диагностики: ' + e.message;

```

```
1961         if (badgeHealth) {
1962             badgeHealth.textContent = 'Health: Error';
1963             badgeHealth.className = 'badge bg-danger-subtle text-d
anger border border-danger font-monospace';
1964         }
1965         if (badgeHealthBtn) {
1966             badgeHealthBtn.textContent = 'Health: Error';
1967             badgeHealthBtn.className = 'badge bg-danger-subtle tex
t-danger border border-danger font-monospace ms-1';
1968         }
1969         if (errorSec) {
1970             errorSec.classList.remove('d-none');
1971             if (errorText) errorText.textContent = e.message;
1972         }
1973         stagesLog.push({
1974             stage: 'error',
1975             title: 'Ошибка',
1976             message: `❌ Ошибка выполнения: ${e.message}`,
1977             details: 'Проверьте доступность API-сервера',
1978         });
1979         renderAIDiagnosticStages(stagesLog, true);
1980     } finally {
1981         isAiRunning = false;
1982
1983         // Сбрасываем флаг processing после завершения
1984         if (window.resetTabProcessing) {
1985             window.resetTabProcessing('tab-about-system');
1986         }
1987
1988         if (btnRescan) btnRescan.disabled = false;
1989         if (icRescan) icRescan.className = 'bi bi-lightning-char
ge-fill';
1990         if (txtRescan) txtRescan.textContent = 'Rescan';
1991     }
1992 }
1993
1994
1995     async function fetchHardwareSpec(forceNetwork = false) {
```

```
1996     const container = document.getElementById('about-sys-tree-
|container');
1997     const badgeCount = document.getElementById('about-sys-node
|-count');
1998     if (!container) return;
1999
2000     try {
2001         const config = CACHE_STRATEGY.HARDWARE_SPEC;
2002         const nodes = await cachedFetch(
2003             '/api/v1/system/hardware',
2004             'hardware_spec',
2005             { ttl: config.ttl, strategy: config.strategy, forceNet
|work }
2006         );
2007         hardwareData = Array.isArray(nodes) ? nodes : [];
2008
2009         if (badgeCount) {
2010             badgeCount.textContent = `${hardwareData.length} кате
|орий`;
2011         }
2012
2013         const searchInput = document.getElementById('about-sys-s
|earch');
2014         const filter = searchInput ? searchInput.value.trim().to
|LowerCase() : '';
2015         renderHardwareTree(hardwareData, filter);
2016     } catch (e) {
2017         console.error('[AboutSystemTab] Failed to fetch hardware
|spec:', e);
2018         container.innerHTML = `
2019         <div class="text-center py-4 text-danger small">
2020             <div><i class="bi bi-exclamation-circle me-1"></i> 0
|шибка загрузки оборудования: ${escapeHtml(e.message)}</div>
2021             <button class="btn btn-sm btn-outline-primary mt-2"
|type="button" onclick="window.fetchHardwareSpec ? window.fetch
|HardwareSpec(true) : null">
2022                 <i class="bi bi-arrow-clockwise me-1"></i> Повтори
|ть попытку
```

```
2023 |         </button>
2024 |     </div>
2025 |     `;
2026 | }
2027 | }
2028 | window.fetchHardwareSpec = fetchHardwareSpec;
2029 |
2030 | function renderDisksTable(disks) {
2031 |     const tbody = document.getElementById('about-sys-disks-tbo
2032 | dy');
2033 |     const countBadge = document.getElementById('about-disks-co
2034 | unt');
2035 |     if (!tbody) return;
2036 |     if (!Array.isArray(disks) || disks.length === 0) {
2037 |         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
2038 | ter py-3 text-muted">Дисковые разделы не обнаружены</td></tr>';
2039 |         return;
2040 |     }
2041 |     if (countBadge) countBadge.textContent = `${disks.length}
2042 | Drives`;
2043 |     tbody.innerHTML = disks.map(d => {
2044 |         const pct = Math.min(100, Math.max(0, d.percent || 0));
2045 |         let barClass = 'bg-info';
2046 |         if (pct > 88) barClass = 'bg-danger';
2047 |         else if (pct > 70) barClass = 'bg-warning';
2048 |         const devStr = String(d.device || d.mountpoint || '');
2049 |         const volStr = String(d.volume_name || '');
2050 |         const isGoogleDrive = d.is_virtual || d.drive_type === '
2051 | Google Drive' || volStr.toLowerCase().includes('google') || vo
2052 | lStr.includes('@') || devStr.startsWith('G:') || devStr.starts
2053 | With('J:');
2054 |         let driveIcon = 'bi-hdd-fill text-primary';
2055 |         let driveBadge = '';
```

```

2054         let driveSubtitle = '';
2055         let fsBadge = `

```

```

2073         driveSubtitle = `

`;
2074     }
2075
2076     return `
2077         <tr>
2078             <td>
2079                 <div class="d-flex align-items-center">
2080                     <i class="bi ${driveIcon} me-2 fs-6"></i>
2081                     <div>
2082                         <div class="d-flex align-items-center">
2083                             <strong class="font-monospace" style="color:
2084                                 var(--text-color);">${escapeHtml(devStr || 'C:\\')}</strong>
2085                                 ${driveBadge}
2086                             </div>
2087                             ${driveSubtitle}
2088                         </div>
2089                     </div>
2090                     <td>${fsBadge}</td>
2091                     <td class="font-monospace" style="color: var(--text-color);">${Number(d.total_gb || 0).toFixed(1)} GB</td>
2092                     <td class="font-monospace" style="color: var(--text-color);">${Number(d.used_gb || 0).toFixed(1)} GB</td>
2093                     <td class="font-monospace text-info fw-semibold">${Number(d.free_gb || 0).toFixed(1)} GB</td>
2094                     <td>
2095                         <div class="d-flex align-items-center gap-2">
2096                             <div class="about-sys-progress-track flex-grow-1" style="height: 6px; background: var(--surface-2, rgba(120, 120, 120, 0.2));">
2097                                 <div class="about-sys-progress-bar ${barClass}" style="width: ${pct}%;"></div>
2098                             </div>
2099                             <span class="small font-monospace text-muted" style="min-width: 42px; text-align: right;">${pct.toFixed(1)}%</span>
2100                     </div>


```



```

2101         </td>
2102         <td>${smartBadge}</td>
2103     </tr>
2104     `;
2105     }).join('');
2106 }
2107
2108
2109 function renderProcessesTable(procs) {
2110     const tbody = document.getElementById('about-sys-procs-tbo
2111 dy');
2112     if (!tbody) return;
2113     if (!Array.isArray(procs) || procs.length === 0) {
2114         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
2115 te r py-4 text-muted">Процессы не найдены</td></tr>';
2116         return;
2117     }
2118     const searchInput = document.getElementById('about-proc-se
2119 arch');
2120     const query = searchInput ? searchInput.value.trim().toLow
2121 erCase() : '';
2122     const filtered = procs.filter(p => {
2123         if (!query) return true;
2124         return String(p.pid).includes(query) ||
2125             (p.name || '').toLowerCase().includes(query) ||
2126             (p.username || '').toLowerCase().includes(query);
2127     });
2128     if (filtered.length === 0) {
2129         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
2130 te r py-4 text-muted">Процессы по запросу не найдены</td></tr>';
2131         return;
2132     }
2133     tbody.innerHTML = filtered.slice(0, 30).map(p => {

```

```

2134     const cpuVal = Number(p.cpu_percent || 0).toFixed(1);
2135     const memVal = Number(p.memory_mb || 0).toFixed(1);
2136     const isHigh = p.cpu_percent > 10;
2137
2138     return `
2139         <tr>
2140             <td class="text-muted">${p.pid}</td>
2141             <td class="text-white fw-semibold text-truncate" style="max-width: 160px;" title="${escapeHtml(p.name)}">
2142                 ${escapeHtml(p.name)}
2143             </td>
2144             <td><span class="badge bg-dark border border-secondary text-secondary" style="font-size: 0.68rem;">${escapeHtml(p.status || 'running')}</span></td>
2145             <td class="${isHigh ? 'text-danger fw-bold' : 'text-info'}">${cpuVal}%</td>
2146             <td class="text-light">${memVal} MB</td>
2147             <td class="text-secondary">${p.num_threads || 1}</td>
2148             <td class="text-muted small text-truncate" style="max-width: 140px;" title="${escapeHtml(p.username || '')}">
2149                 ${escapeHtml(p.username || '-')}
2150             </td>
2151         </tr>
2152     `;
2153     }).join('');
2154 }
2155
2156 function renderHardwareTree(nodes, filter) {
2157     const container = document.getElementById('about-sys-tree-container');
2158     if (!container) return;
2159
2160     if (!Array.isArray(nodes) || nodes.length === 0) {
2161         container.innerHTML = '<div class="text-center py-4 text-muted small">Оборудование не обнаружено или опрашивается...</div>';
2162         return;

```

```

2163     }
2164
2165     const filtered = nodes.filter(node => {
2166         if (!filter) return true;
2167         const catMatch = (node.category || '').toLowerCase().includes(filter);
2168         const nameMatch = (node.name || '').toLowerCase().includes(filter);
2169         const propsMatch = Object.entries(node.properties || {}).some(
2170             ([k, v]) => k.toLowerCase().includes(filter) || String(v).toLowerCase().includes(filter)
2171         );
2172         return catMatch || nameMatch || propsMatch;
2173     });
2174
2175     if (filtered.length === 0) {
2176         container.innerHTML = `
```

```

vice || item.bank_label || JSON.stringify(item)) : String(item
)).filter(Boolean).join(', ') || 'N/A';
2191         } else if (typeof v === 'object' && v !== nu
ll) {
2192             valStr = Object.entries(v).map(([subK, sub
V]) => `${subK}: ${subV}`).join(', ') || 'N/A';
2193         } else {
2194             valStr = String(v ?? '');
2195         }
2196         let valHtml = escapeHtml(valStr);
2197
2198         // Специфичное выделение частот и статусов с
овместимости памяти RAM
2199         if (cleanKey.includes('Номинальная частота')
) {
2200             valHtml = `
```

```

r-warning-subtle fw-semibold px-2 py-0.5 text-wrap" style="col
or: #facc15; background: rgba(234, 179, 8, 0.18); text-align:
start;"><i class="bi bi-exclamation-triangle-fill me-1"></i>${
escapeHtml(valStr.replace(/^\s*/, ''))}</span>`;
2210         } else if (valStr.includes(' ') || valStr.in
cludes('Оптимально')) {
2211         valHtml = `<span class="badge border borde
r-success-subtle fw-semibold px-2 py-0.5" style="color: #4ade8
0; background: rgba(34, 197, 94, 0.14);"><i class="bi bi-check
-circle-fill me-1"></i>${escapeHtml(valStr.replace(/^\s*/, ''
))}</span>`;
2212         } else if (cleanKey.includes('Эффективность
планки')) {
2213         valHtml = `<span class="badge border borde
r-info-subtle fw-semibold px-2 py-0.5" style="color: #38bdf8;
background: rgba(56, 189, 248, 0.14);">${escapeHtml(valStr)}</
span>`;
2214         }
2215
2216         return `
2217         <tr>
2218         <td class="about-sys-spec-key" style="wi
dth: 32%; min-width: 200px; max-width: 340px;">
2219         ${cleanKey}
2220         </td>
2221         <td class="about-sys-spec-val font-monos
pace">
2222         ${valHtml}
2223         </td>
2224         </tr>
2225         `;
2226         }).join('')}
2227     </tbody>
2228     </table>
2229     </div>`
2230     : '<div class="text-muted small py-2 px-3">Свойства не
указаны</div>';
2231

```

```

2232     const iconClass = getNodeIcon(node.category);
2233     const isSearchActive = Boolean(filter);
2234     const bodyClass = isSearchActive ? 'about-sys-tree-body'
    : 'about-sys-tree-body d-none';
2235     const chevronChar = isSearchActive ? '▲' : '▼';
2236
2237     return `
2238         <div class="about-sys-tree-node">
2239             <div class="about-sys-tree-title" onclick="toggleNode(${idx})">
2240                 <span class="d-flex align-items-center gap-2 text-truncate me-2">
2241                     <i class="${iconClass} text-info fs-6"></i>
2242                     <strong style="color: var(--text-color);">${escapeHtml(node.category)}:</strong>
2243                     <span style="color: var(--text-color); font-weight: 500;">${escapeHtml(node.name || '')}</span>
2244                 </span>
2245                 <div class="d-flex align-items-center gap-2 flex-shrink-0 ms-auto">
2246                     <span class="badge bg-secondary-subtle text-body-secondary border px-2 py-0.5 font-monospace" style="font-size: 0.7rem;">${propsEntries.length} параметров</span>
2247                     <span class="about-sys-chevron" id="about-chevron-${idx}">${chevronChar}</span>
2248                 </div>
2249             </div>
2250             <div class="${bodyClass}" id="about-node-body-${idx}"
    ">
2251                 ${propsHtml}
2252             </div>
2253         </div>
2254     `;
2255     }).join('');
2256 }
2257
2258 window.toggleNode = function(idx) {
2259     const body = document.getElementById(`about-node-body-${id

```

```

x}`);
2260     const chevron = document.getElementById(`about-chevron-${i
dx}`);
2261     if (body) {
2262         body.classList.toggle('d-none');
2263         if (chevron) {
2264             chevron.textContent = body.classList.contains('d-none'
) ? '▼' : '▲';
2265         }
2266         const btnTreeToggle = document.getElementById('btn-about
-sys-tree-toggle');
2267         if (btnTreeToggle) {
2268             const allBodies = document.querySelectorAll('.about-sy
s-tree-body');
2269             const hasHidden = Array.from(allBodies).some(b => b.cl
assList.contains('d-none'));
2270             if (hasHidden) {
2271                 btnTreeToggle.classList.add('collapsed');
2272                 btnTreeToggle.title = 'Развернуть все категории';
2273             } else {
2274                 btnTreeToggle.classList.remove('collapsed');
2275                 btnTreeToggle.title = 'Свернуть все категории';
2276             }
2277         }
2278     }
2279 };
2280
2281 function getNodeIcon(category) {
2282     const cat = (category || '').toLowerCase();
2283     if (cat.includes('processor') || cat.includes('cpu')) retu
rn 'bi bi-cpu';
2284     if (cat.includes('module') || cat.includes('планка')) retu
rn 'bi bi-memory';
2285     if (cat.includes('memory') || cat.includes('ram')) return
'bi bi-sd-card';
2286     if (cat.includes('system') || cat.includes('os')) return '
bi bi-laptop';
2287     if (cat.includes('motherboard') || cat.includes('mainboard

```

```

    ')) return 'bi bi-motherboard';
2288     if (cat.includes('display') || cat.includes('gpu') || cat.
includes('video') || cat.includes('graphics')) return 'bi bi-g
pu-card';
2289     if (cat.includes('monitor') || cat.includes('screen') || c
at.includes('экран')) return 'bi bi-display';
2290     if (cat.includes('physical') || cat.includes('диск')) retu
rn 'bi bi-hdd-fill';
2291     if (cat.includes('disk') || cat.includes('storage') || cat
.includes('drive') || cat.includes('volume')) return 'bi bi-hd
d-stack';
2292     if (cat.includes('network') || cat.includes('adapter') ||
cat.includes('ethernet') || cat.includes('wi-fi') || cat.inclu
des('сеть')) return 'bi bi-ethernet';
2293     if (cat.includes('audio') || cat.includes('sound') || cat.
includes('звук')) return 'bi bi-volume-up-fill';
2294     if (cat.includes('usb') || cat.includes('контроллер')) ret
urn 'bi bi-usb-symbol';
2295     if (cat.includes('update') || cat.includes('servicing')) r
eturn 'bi bi-arrow-repeat';
2296     return 'bi bi-gear-fill';
2297 }
2298
2299 /**
2300  * Привязка событий для модального окна истории телеметрии и
из базы данных
2301  */
2302 function bindHistoryEvents() {
2303     const btnOpenHistory = document.getElementById('btn-open-a
bout-history-modal');
2304     if (btnOpenHistory) {
2305         btnOpenHistory.onclick = () => {
2306             activeHistoryMetric = 'all';
2307             updateHistoryFilterButtons();
2308             fetchAboutSystemHistory(activeHistoryMetric);
2309         };
2310     }
2311

```



```
2312 // Обработчик кнопок истории на каждой карточке
2313 document.querySelectorAll('.btn-card-history').forEach(btn
    => {
2314     btn.onclick = (e) => {
2315         e.stopPropagation();
2316         const metric = btn.getAttribute('data-metric') || 'all
';
2317         activeHistoryMetric = metric;
2318         updateHistoryFilterButtons();
2319         fetchAboutSystemHistory(activeHistoryMetric);
2320     };
2321 });
2322
2323 const btnRefreshHistory = document.getElementById('btn-abo
ut-history-refresh');
2324 if (btnRefreshHistory) {
2325     btnRefreshHistory.onclick = () => {
2326         fetchAboutSystemHistory(activeHistoryMetric);
2327     };
2328 }
2329
2330 const limitSelect = document.getElementById('about-history
-limit-select');
2331 if (limitSelect) {
2332     limitSelect.onchange = () => {
2333         fetchAboutSystemHistory(activeHistoryMetric);
2334     };
2335 }
2336
2337 // Фильтры метрик
2338 document.querySelectorAll('.btn-history-filter').forEach(b
tn => {
2339     btn.onclick = () => {
2340         const metric = btn.getAttribute('data-metric') || 'all
';
2341         activeHistoryMetric = metric;
2342         updateHistoryFilterButtons();
2343         fetchAboutSystemHistory(activeHistoryMetric);
```

```

2344     };
2345   });
2346 }
2347
2348 function updateHistoryFilterButtons() {
2349   document.querySelectorAll('.btn-history-filter').forEach(b
tn => {
2350     const m = btn.getAttribute('data-metric') || 'all';
2351     if (m === activeHistoryMetric) {
2352       btn.className = 'btn btn-sm btn-info text-dark fw-bold
rounded-pill px-2.5 py-0.5 btn-history-filter active';
2353     } else {
2354       btn.className = 'btn btn-sm btn-outline-secondary roun
ded-pill px-2.5 py-0.5 btn-history-filter';
2355     }
2356   });
2357 }
2358
2359 let cachedRegionalData = null;
2360
2361 /**
2362  * Переключение активной вкладки в модальном окне региональн
ых настроек
2363  */
2364 function switchRegionalTab(targetPaneId) {
2365   const panes = document.querySelectorAll('.regional-tab-pan
e');
2366   panes.forEach(pane => {
2367     if (pane.id === targetPaneId) {
2368       pane.classList.remove('d-none');
2369     } else {
2370       pane.classList.add('d-none');
2371     }
2372   });
2373
2374   const navBtns = document.querySelectorAll('.btn-reg-nav');
2375   navBtns.forEach(btn => {
2376     if (btn.getAttribute('data-tab-target') === targetPaneId

```

```

    ) {
2377         btn.classList.add('active');
2378     } else {
2379         btn.classList.remove('active');
2380     }
2381 });
2382 }
2383
2384 /**
2385  * Отображение уведомления внутри модального окна региональн
ых настроек
2386  */
2387 function showRegModalAlert(msg, type = 'success') {
2388     const alertBox = document.getElementById('reg-modal-alert'
);
2389     if (!alertBox) return;
2390     alertBox.className = `alert alert-${type} py-2 px-3 small
mb-0`;
2391     alertBox.innerHTML = msg;
2392     alertBox.classList.remove('d-none');
2393     setTimeout(() => {
2394         alertBox.classList.add('d-none');
2395     }, 6000);
2396 }
2397
2398 /**
2399  * Отрисовка элементов выбора часового пояса с фильтрацией
2400  */
2401 function renderTimezonesList(timezones, filterQuery = '', se
lectedTzId = '') {
2402     const tzSelect = document.getElementById('reg-tz-select');
2403     if (!tzSelect) return;
2404     const q = (filterQuery || '').toLowerCase();
2405     const filtered = (timezones || []).filter(tz => {
2406         if (!q) return true;
2407         return (tz.id || '').toLowerCase().includes(q) || (tz.na
me || '').toLowerCase().includes(q);
2408     });

```

```
2409
2410     if (filtered.length === 0) {
2411         tzSelect.innerHTML = '<option value="" disabled>Ничего н
е найдено</option>';
2412         return;
2413     }
2414
2415     tzSelect.innerHTML = filtered.map(tz => {
2416         const isSel = tz.id === selectedTzId ? 'selected' : '';
2417         return `<option value="${escapeHtml(tz.id)}" ${isSel}>${
escapeHtml(tz.name || tz.id)}</option>`;
2418     }).join('');
2419
2420     const selLabel = document.getElementById('reg-tz-selected-
label');
2421     if (selLabel) {
2422         const cur = tzSelect.value;
2423         selLabel.textContent = cur ? `Выбран: ${cur}` : 'Выберит
е пояс из списка';
2424     }
2425 }
2426
2427 /**
2428  * Загрузка параметров региона, времени и пользователя с сер
вера
2429  */
2430 async function loadRegionalOptions(targetPaneId = null) {
2431     if (targetPaneId) {
2432         switchRegionalTab(targetPaneId);
2433     }
2434     const tzBadge = document.getElementById('reg-cur-tz-badge'
);
2435     if (tzBadge) tzBadge.textContent = 'Загрузка...';
2436
2437     try {
2438         const data = await apiFetch('/api/v1/system/regional-opt
ions');
2439         cachedRegionalData = data;
```

```
2440 |
2441 |         if (tzBadge) tzBadge.textContent = data.current_timezone
      |         || 'Не определен';
2442 |
2443 |         // 1. Часовые пояса
2444 |         const searchInput = document.getElementById('reg-tz-search-input');
2445 |         const curQuery = searchInput ? searchInput.value.trim()
      |         : '';
2446 |         renderTimezonesList(data.timezones || [], curQuery, data
      |         .current_timezone);
2447 |
2448 |         // 2. Локали
2449 |         if (Array.isArray(data.locales) && data.locales.length >
      |         0) {
2450 |             const sysLocSelect = document.getElementById('reg-loc-
      |             sys-select');
2451 |             const usrLocSelect = document.getElementById('reg-loc-
      |             usr-select');
2452 |             const optionsHtml = data.locales.map(loc =>
2453 |                 `<option value="${escapeHtml(loc.code)}">${escapeHtml
      |                 (loc.name)} [${escapeHtml(loc.code)}]</option>`
2454 |             ).join('');
2455 |
2456 |             if (sysLocSelect) {
2457 |                 sysLocSelect.innerHTML = optionsHtml;
2458 |                 if (data.current_system_locale) sysLocSelect.value =
      |                 data.current_system_locale;
2459 |             }
2460 |             if (usrLocSelect) {
2461 |                 usrLocSelect.innerHTML = optionsHtml;
2462 |                 if (data.current_user_locale) usrLocSelect.value = d
      |                 ata.current_user_locale;
2463 |             }
2464 |         } else {
2465 |             const sysLocSelect = document.getElementById('reg-loc-
      |             sys-select');
2466 |             const usrLocSelect = document.getElementById('reg-loc-
```

```

usr-select');
2467     if (sysLocSelect && data.current_system_locale) sysLoc
Select.value = data.current_system_locale;
2468     if (usrLocSelect && data.current_user_locale) usrLocSe
lect.value = data.current_user_locale;
2469     }
2470
2471     // 3. Профиль пользователя
2472     const usrNameInput = document.getElementById('reg-usr-na
me-input');
2473     const usrFullNameInput = document.getElementById('reg-us
r-fullname-input');
2474     const usrDescInput = document.getElementById('reg-usr-de
sc-input');
2475
2476     if (usrNameInput) usrNameInput.value = data.current_user
name || '';
2477     if (usrFullNameInput) usrFullNameInput.value = data.curr
ent_user_fullname || '';
2478     if (usrDescInput) usrDescInput.value = data.current_user
_description || '';
2479
2480     } catch (err) {
2481         console.error('[AboutSystemTab] loadRegionalOptions erro
r:', err);
2482         if (tzBadge) tzBadge.textContent = 'Ошибка загрузки';
2483         showRegModalAlert(`Ошибка загрузки параметров: ${escapeH
tml(err.message || String(err))}`, 'danger');
2484     }
2485 }
2486
2487 /**
2488  * Привязка событий модального окна региональных параметров
2489  */
2490 function bindRegionalEvents() {
2491     // 1. Открытие модального окна по кнопкам в таблице или за
головке
2492     const openButtons = document.querySelectorAll('#btn-open-r

```

```

    regional-settings, .btn-open-reg-tab');
2493     openButtons.forEach(btn => {
2494         btn.onclick = (e) => {
2495             e.preventDefault();
2496             const targetTab = btn.getAttribute('data-reg-tab') ||
                'pane-tz';
2497             const modalEl = document.getElementById('aboutRegional
                SettingsModal');
2498             if (modalEl && window.bootstrap && window.bootstrap.Mo
                dal) {
2499                 const modalInst = window.bootstrap.Modal.getOrCreate
                Instance(modalEl);
2500                 modalInst.show();
2501             }
2502             loadRegionalOptions(targetTab);
2503         };
2504     });
2505
2506     // 2. Переключение вкладок в модальном окне
2507     document.querySelectorAll('.btn-reg-nav').forEach(btn => {
2508         btn.onclick = (e) => {
2509             e.preventDefault();
2510             const targetPane = btn.getAttribute('data-tab-target')
                ;
2511             if (targetPane) {
2512                 switchRegionalTab(targetPane);
2513             }
2514         };
2515     });
2516
2517     // 3. Фильтрация часовых поясов
2518     const tzSearchInput = document.getElementById('reg-tz-sear
                ch-input');
2519     const tzSelect = document.getElementById('reg-tz-select');
2520     if (tzSearchInput) {
2521         tzSearchInput.oninput = () => {
2522             if (cachedRegionalData && cachedRegionalData.timezones
                ) {

```

```
2523 |         const currentVal = tzSelect ? tzSelect.value : '';
2524 |         renderTimezonesList(cachedRegionalData.timezones, tz
| SearchInput.value.trim(), currentVal);
2525 |     }
2526 | };
2527 | }
2528 |
2529 | if (tzSelect) {
2530 |     tzSelect.onChange = () => {
2531 |         const selLabel = document.getElementById('reg-tz-selec
| ted-label');
2532 |         if (selLabel) {
2533 |             selLabel.textContent = tzSelect.value ? `Выбран: ${t
| zSelect.value}` : 'Выберите пояс из списка';
2534 |         }
2535 |     };
2536 | }
2537 |
2538 | // 4. Применение часового пояса
2539 | const btnSaveTz = document.getElementById('btn-reg-save-tz
| ');
2540 | if (btnSaveTz) {
2541 |     btnSaveTz.onclick = async () => {
2542 |         const tzVal = tzSelect ? tzSelect.value : '';
2543 |         if (!tzVal) {
2544 |             showRegModalAlert('Пожалуйста, выберите часовой пояс
| из списка', 'warning');
2545 |             return;
2546 |         }
2547 |
2548 |         btnSaveTz.disabled = true;
2549 |         btnSaveTz.innerHTML = '<span class="spinner-border spi
| nner-border-sm" role="status"></span> Применение...';
2550 |
2551 |         try {
2552 |             const res = await apiFetch('/api/v1/system/set-timez
| one', {
2553 |                 method: 'POST',
```



```
2554         body: JSON.stringify({ timezone_id: tzVal })
2555     });
2556
2557     if (res.status === 'ok') {
2558         showRegModalAlert(`⏰ Часовой пояс успешно изменен
на «<strong>${escapeHtml(res.timezone_id || tzVal)}</strong>»`
, 'success');
2559         if (window.showToast) window.showToast(`⏰ Часовой
пояс изменен на «${tzVal}»`, 'success');
2560         const tzBadge = document.getElementById('reg-cur-t
z-badge');
2561         if (tzBadge) tzBadge.textContent = res.timezone_id
|| tzVal;
2562         updateLocalClock();
2563         fetchSystemSummary();
2564     } else {
2565         showRegModalAlert(`Ошибка: ${escapeHtml(res.messag
e || 'Не удалось применить пояс')}}`, 'danger');
2566     }
2567     } catch (err) {
2568         showRegModalAlert(`Ошибка при смене часового пояса:
${escapeHtml(err.message || String(err))}`, 'danger');
2569     } finally {
2570         btnSaveTz.disabled = false;
2571         btnSaveTz.innerHTML = '<i class="bi bi-check-lg me-1
"></i> Применить пояс';
2572     }
2573 };
2574 }
2575
2576 // 5. Применение локали
2577 const btnSaveLoc = document.getElementById('btn-reg-save-l
oc');
2578 if (btnSaveLoc) {
2579     btnSaveLoc.onclick = async () => {
2580         const sysLoc = document.getElementById('reg-loc-sys-se
lect')?.value;
2581         const usrLoc = document.getElementById('reg-loc-usr-se
```

```

lect')?.value;
2582
2583     btnSaveLoc.disabled = true;
2584     btnSaveLoc.innerHTML = '<span class="spinner-border sp
inner-border-sm" role="status"></span> Сохранение...';
2585
2586     try {
2587         const res = await apiFetch('/api/v1/system/set-local
e', {
2588             method: 'POST',
2589             body: JSON.stringify({ system_locale: sysLoc, user
_locale: usrLoc })
2590         });
2591
2592         if (res.status === 'ok') {
2593             const rebootMsg = res.reboot_required ? '<br><small
l class="text-warning">⚠ Для полного применения требуется пере
загрузка Windows.</small>' : '';
2594             showRegModalAlert(`⚠ Системная локаль обновлена: <
strong>${escapeHtml(res.locale || sysLoc)}</strong>.${rebootMs
g}` , 'success');
2595             if (window.showToast) window.showToast(`⚠ Системна
я локаль обновлена на «${res.locale || sysLoc}»`, 'warning');
2596             fetchSystemSummary();
2597         } else {
2598             showRegModalAlert(`Ошибка: ${escapeHtml(res.messag
e || 'Не удалось изменить локаль')}` , 'danger');
2599         }
2600     } catch (err) {
2601         showRegModalAlert(`Ошибка при изменении локали: ${es
capeHtml(err.message || String(err))}` , 'danger');
2602     } finally {
2603         btnSaveLoc.disabled = false;
2604         btnSaveLoc.innerHTML = '<i class="bi bi-check-lg me-
1"></i> Применить локаль';
2605     }
2606 };
2607 }

```

```
2608
2609 // 6. Сохранение профиля пользователя
2610 const btnSaveUsr = document.getElementById('btn-reg-save-u
sr');
2611 if (btnSaveUsr) {
2612     btnSaveUsr.onclick = async () => {
2613         const usrName = document.getElementById('reg-usr-name-
input')?.value.trim();
2614         const fullName = document.getElementById('reg-usr-full
name-input')?.value.trim();
2615         const desc = document.getElementById('reg-usr-desc-inp
ut')?.value.trim();
2616
2617         if (!usrName) {
2618             showRegModalAlert('Имя учетной записи не указано', '
warning');
2619             return;
2620         }
2621
2622         btnSaveUsr.disabled = true;
2623         btnSaveUsr.innerHTML = '<span class="spinner-border sp
inner-border-sm" role="status"></span> Сохранение...';
2624
2625         try {
2626             const res = await apiFetch('/api/v1/system/update-us
er-profile', {
2627                 method: 'POST',
2628                 body: JSON.stringify({ username: usrName, full_nam
e: fullName, description: desc })
2629             });
2630
2631             if (res.status === 'ok') {
2632                 showRegModalAlert(`❑ Профиль пользователя «<strong
>${escapeHtml(usrName)}</strong>» успешно обновлен`, 'success'
);
2633                 if (window.showToast) window.showToast(`❑ Профиль
пользователя «${usrName}» обновлен`, 'success');
2634                 fetchSystemSummary();
```

```
2635         } else {
2636             showRegModalAlert(`Ошибка: ${escapeHtml(res.message || 'Не удалось обновить профиль')}` , 'danger');
2637         }
2638     } catch (err) {
2639         showRegModalAlert(`Ошибка при обновлении профиля: ${escapeHtml(err.message || String(err))}` , 'danger');
2640     } finally {
2641         btnSaveUsr.disabled = false;
2642         btnSaveUsr.innerHTML = '<i class="bi bi-check-lg me-1"></i> Сохранить профиль';
2643     }
2644 };
2645 }
2646 }
2647
2648 /**
2649  * Загрузка исторических записей телеметрии из базы данных telemetry.db
2650  */
2651 async function fetchAboutSystemHistory(metric = 'all') {
2652     const tbody = document.getElementById('about-history-tbody');
2653     if (tbody) {
2654         tbody.innerHTML = `
2655             <tr>
2656                 <td colspan="9" class="text-center py-4 text-muted">
2657                     <div class="spinner-border spinner-border-sm text-info mb-1" role="status"></div>
2658                     <div>Запрос истории из базы данных telemetry.db...</div>
2659                 </td>
2660             </tr>
2661         `;
2662     }
2663
2664     const limitSelect = document.getElementById('about-history-limit-select');
```

```

2665     const limit = limitSelect ? parseInt(limitSelect.value, 10
    ) || 30 : 30;
2666
2667     try {
2668         const url = `/api/v1/about-system/history?limit=${limit}
    &metric=${encodeURIComponent(metric)}`;
2669         const data = await apiFetch(url);
2670         renderAboutSystemHistory(data, metric);
2671     } catch (err) {
2672         console.error('[AboutSystemTab] fetchAboutSystemHistory
    error:', err);
2673         if (tbody) {
2674             tbody.innerHTML = `
2675                 <tr>
2676                     <td colspan="9" class="text-center py-4 text-dange
    r">
2677                         <i class="bi bi-exclamation-triangle-fill me-1">
    </i>
2678                             Ошибка при получении истории из БД: ${escapeHtml
    (err.message || String(err))}
2679                     </td>
2680                 </tr>
2681             `;
2682         }
2683     }
2684 }
2685
2686 /**
2687  * Отрисовка таблицы с предыдущими значениями из telemetry.d
    b
2688  */
2689 function renderAboutSystemHistory(resData, metric) {
2690     const tbody = document.getElementById('about-history-tbody
    ');
2691     const countBadge = document.getElementById('about-history-
    count-badge');
2692     const metaEl = document.getElementById('about-history-db-m
    eta');

```

```

2693
2694     if (!resData || !Array.isArray(resData.history) || resData
2695         .history.length === 0) {
2696         if (tbody) {
2697             tbody.innerHTML = `
2698                 <tr>
2699                     <td colspan="9" class="text-center py-4 text-muted
2700                         ">
2701                         <i class="bi bi-database-slash fs-4 d-block mb-1
2702                             text-secondary"></i>
2703                             В базе данных telemetry.db пока нет сохраненных
2704                             срезов телеметрии.
2705                         </td>
2706                     </tr>
2707                 `;
2708         }
2709         if (countBadge) countBadge.textContent = 'Записей в базе
2710             : 0';
2711         return;
2712     }
2713
2714     const history = resData.history;
2715     if (countBadge) countBadge.textContent = `Записей в выборк
2716     e: ${history.length}`;
2717     if (metaEl && resData.meta && resData.meta.source) {
2718         metaEl.textContent = `Источник: ${resData.meta.source} (
2719             ${resData.meta.table || 'system_snapshots'})`;
2720     }
2721
2722     if (!tbody) return;
2723
2724     let rowsHtml = '';
2725     history.forEach(item => {
2726         const id = item.id || 0;
2727         let dateFormatted = '--:--:--';
2728         if (item.timestamp) {
2729             try {
2730                 const d = new Date(item.timestamp);

```

```
2724         dateFormatted = `${d.toLocaleDateString()} ${d.toLoc  
aleTimeString()}`;  
2725     } catch (_) {  
2726         dateFormatted = item.timestamp;  
2727     }  
2728 }  
2729  
2730     const host = escapeHtml(item.hostname || '--');  
2731     const os = escapeHtml(item.os_name || 'Windows');  
2732  
2733     // CPU  
2734     const cpuVal = item.cpu_total_percent !== null && item.c  
pu_total_percent !== undefined ? Number(item.cpu_total_percent  
    ) : 0;  
2735     const cpuColor = cpuVal > 85 ? 'text-danger fw-bold' : (  
cpuVal > 60 ? 'text-warning' : 'text-info');  
2736     const cpuFreq = item.cpu_frequency_mhz ? `${item.cpu_fre  
quency_mhz} MHz` : '';  
2737  
2738     // RAM  
2739     const ramUsed = item.memory_used_gb !== null && item.mem  
ory_used_gb !== undefined ? Number(item.memory_used_gb).toFixe  
d(1) : '0.0';  
2740     const ramTot = item.memory_total_gb !== null && item.mem  
ory_total_gb !== undefined ? Number(item.memory_total_gb).toFi  
xed(1) : '0.0';  
2741     const ramPct = item.memory_percent !== null && item.memo  
ry_percent !== undefined ? Number(item.memory_percent).toFixed  
(1) : '0.0';  
2742     const ramColor = Number(ramPct) > 85 ? 'text-danger' : (  
Number(ramPct) > 65 ? 'text-warning' : 'text-success');  
2743  
2744     // GPU  
2745     const gpuLoad = item.gpu_load_percent !== null && item.g  
pu_load_percent !== undefined ? `${item.gpu_load_percent}%` :  
    '--';  
2746     const gpuTemp = item.gpu_temp_c !== null && item.gpu_tem  
p_c !== undefined ? ` / ${item.gpu_temp_c}°C` : '';
```

```

2747 |
2748 |         // Storage C:
2749 |         const cFree = item.storage_c_free_gb !== null && item.st
         orage_c_free_gb !== undefined ? `${item.storage_c_free_gb} GB`
         : '--';
2750 |
2751 |         // Disk I/O
2752 |         const diskIo = item.disk_io_total_mb_s !== null && item.
         disk_io_total_mb_s !== undefined ? `${Number(item.disk_io_tota
         l_mb_s).toFixed(2)} MB/s` : '0.00 MB/s';
2753 |
2754 |         // Uptime
2755 |         const uptime = escapeHtml(item.uptime_human || '--');
2756 |
2757 |         rowsHtml += `
2758 |             <tr>
2759 |                 <td class="text-muted">#${id}</td>
2760 |                 <td class="text-nowrap text-light">${escapeHtml(date
         Formatted)}</td>
2761 |                 <td class="text-truncate" style="max-width: 140px;"
         title="${host} (${os})">
2762 |                     <span class="text-white">${host}</span>
2763 |                     <small class="text-muted d-block" style="font-size
         : 0.70rem;">${os}</small>
2764 |                 </td>
2765 |                 <td style="text-align: right;">
2766 |                     <span class="${cpuColor}">${cpuVal.toFixed(1)}%</s
         pan>
2767 |                     <small class="text-muted d-block" style="font-size
         : 0.70rem;">${cpuFreq}</small>
2768 |                 </td>
2769 |                 <td style="text-align: right;">
2770 |                     <span class="${ramColor}">${ramUsed} / ${ramTot} G
         B</span>
2771 |                     <small class="text-muted d-block" style="font-size
         : 0.70rem;">${ramPct}%</small>
2772 |                 </td>
2773 |                 <td style="text-align: right;">

```



```

2774         <span class="text-light">${gpuLoad}</span>
2775         <small class="text-danger" style="font-size: 0.70r
em;">${gpuTemp}</small>
2776     </td>
2777     <td style="text-align: right;" class="text-info">${c
Free}</td>
2778     <td style="text-align: right;" class="text-white">${
diskIo}</td>
2779     <td style="text-align: right;" class="text-muted">${
uptime}</td>
2780 </tr>
2781 `;
2782 });
2783
2784 tbody.innerHTML = rowsHtml;
2785 }
2786
2787 function escapeHtml(str) {
2788     if (str === null || str === undefined) return '';
2789     return String(str)
2790         .replace(/&/g, '&amp;')
2791         .replace(/</g, '&lt;')
2792         .replace(/>/g, '&gt;')
2793         .replace(/"/g, '&quot;')
2794         .replace(/'/g, '&#039;');
2795 }
2796
2797 // Self-trigger if container is already in DOM
2798 if (document.getElementById('about-sys-wrapper') || document
.getElementById('tab-about-system')) {
2799     if (document.readyState === 'loading') {
2800         document.addEventListener('DOMContentLoaded', () => init
AboutSystemTab());
2801     } else {
2802         setTimeout(() => initAboutSystemTab(), 10);
2803     }
2804 }
2805 }));

```

2806 |

2807 |

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Accounts & Identity Tab
6 =====
7
8 Description:
9 Вкладка управления учетными записями, субъектами безопасност
и,
10 группами, правами LSA, инспекцией процессов, журналом аудита
и Windows Identity Graph.
11
12 File: index.html
13 Project: ai-breadboard
14 Package: windows/api/webgui/accounts_identity_tab
15 Author: hypo69
16 Copyright: © 2026 hypo69
17 Updated: 2026-10-08 01:48:00
18 =====
19 =====
20 -->
21
22 <div class="container-fluid p-3" id="accounts-identity-root">
23   <!-- Заголовок и текущий контекст токена -->
24   <div class="d-flex align-items-center justify-content-between mb-3 pb-2 border-bottom flex-wrap gap-2">
25     <div>
26       <h4 class="mb-0 fw-bold d-flex align-items-center gap-2">
27         <i class="bi bi-people-fill text-primary"></i>
28         <span>Управление пользователями и Windows Identity</span>
29       </h4>
30       <div class="small text-muted" id="ai-current-token-badge">
31         Загрузка контекста безопасности...
```

```
28 |         </div>
29 |     </div>
30 |     <div class="d-flex align-items-center gap-2">
31 |         <button class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-1 shadow-sm" id="btn-refresh-identity" title="Обновить данные">
32 |             <i class="bi bi-arrow-clockwise"></i>
33 |             <span>Обновить</span>
34 |         </button>
35 |         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1 shadow-sm" id="btn-open-create-user-modal">
36 |             <i class="bi bi-person-plus-fill"></i>
37 |             <span>Создать пользователя</span>
38 |         </button>
39 |     </div>
40 | </div>
41 |
42 | <!-- Карточки ключевых метрик -->
43 | <div class="row g-3 mb-3">
44 |     <div class="col-6 col-md-3">
45 |         <div class="card shadow-sm border-secondary-subtle h-100 bg-body-tertiary">
46 |             <div class="card-body p-2 d-flex align-items-center gap-p-3">
47 |                 <div class="rounded-circle p-2 bg-primary bg-opacity-10 text-primary fs-4">
48 |                     <i class="bi bi-person-badge"></i>
49 |                 </div>
50 |                 <div>
51 |                     <div class="text-muted small">Всего пользователей</div>
52 |                     <h5 class="fw-bold mb-0" id="stat-total-users"></h5>
53 |                 </div>
54 |             </div>
55 |         </div>
56 |     </div>
57 | <div class="col-6 col-md-3">
```

```
58 |         <div class="card shadow-sm border-secondary-subtle h-100
    |         bg-body-tertiary">
59 |             <div class="card-body p-2 d-flex align-items-center ga
    |             p-3">
60 |                 <div class="rounded-circle p-2 bg-danger bg-opacity-
    |                 10 text-danger fs-4">
61 |                     <i class="bi bi-shield-lock-fill"></i>
62 |                 </div>
63 |                 <div>
64 |                     <div class="text-muted small">Администраторы</div>
65 |                     <h5 class="fw-bold mb-0" id="stat-total-admins">-<
    |                     /h5>
66 |                 </div>
67 |             </div>
68 |         </div>
69 |     </div>
70 |     <div class="col-6 col-md-3">
71 |         <div class="card shadow-sm border-secondary-subtle h-100
    |         bg-body-tertiary">
72 |             <div class="card-body p-2 d-flex align-items-center ga
    |             p-3">
73 |                 <div class="rounded-circle p-2 bg-success bg-opacity
    |                 -10 text-success fs-4">
74 |                     <i class="bi bi-display"></i>
75 |                 </div>
76 |                 <div>
77 |                     <div class="text-muted small">Активных сессий</div>
    |                     >
78 |                     <h5 class="fw-bold mb-0" id="stat-active-sessions"
    |                     >-</h5>
79 |                 </div>
80 |             </div>
81 |         </div>
82 |     </div>
83 |     <div class="col-6 col-md-3">
84 |         <div class="card shadow-sm border-secondary-subtle h-100
    |         bg-body-tertiary">
85 |             <div class="card-body p-2 d-flex align-items-center ga
```

```

p-3">
86         <div class="rounded-circle p-2 bg-warning bg-opacity
-10 text-warning fs-4">
87             <i class="bi bi-shield-exclamation"></i>
88         </div>
89         <div>
90             <div class="text-muted small">События аудита (24ч)
</div>
91             <h5 class="fw-bold mb-0" id="stat-security-events"
>-</h5>
92         </div>
93     </div>
94 </div>
95 </div>
96 </div>
97
98 <!-- Навигация по подразделам вкладки -->
99 <ul class="nav nav-pills mb-3 gap-1" id="identity-subtabs" r
ole="tablist">
100     <li class="nav-item" role="presentation">
101         <button class="nav-link active btn-sm" id="subtab-users-
btn" data-bs-toggle="pill" data-bs-target="#subtab-users" type
="button" role="tab">
102             <i class="bi bi-people me-1"></i> Пользователи SAM
103         </button>
104     </li>
105     <li class="nav-item" role="presentation">
106         <button class="nav-link btn-sm" id="subtab-groups-btn" d
ata-bs-toggle="pill" data-bs-target="#subtab-groups" type="but
ton" role="tab">
107             <i class="bi bi-diagram-3 me-1"></i> Группы и вложенно
СТЬ
108         </button>
109     </li>
110     <li class="nav-item" role="presentation">
111         <button class="nav-link btn-sm" id="subtab-events-btn" d
ata-bs-toggle="pill" data-bs-target="#subtab-events" type="but
ton" role="tab">

```

```

112 |         <i class="bi bi-shield-shaded me-1"></i> Журнал безопа
      | сности
113 |     </button>
114 | </li>
115 |     <li class="nav-item" role="presentation">
116 |         <button class="nav-link btn-sm" id="subtab-security-btn"
      | data-bs-toggle="pill" data-bs-target="#subtab-security" type=
      | "button" role="tab">
117 |             <i class="bi bi-shield-check me-1"></i> Аудит безопасн
      | ости и LSA
118 |         </button>
119 |     </li>
120 |     <li class="nav-item" role="presentation">
121 |         <button class="nav-link btn-sm" id="subtab-pid-btn" data
      | -bs-toggle="pill" data-bs-target="#subtab-pid" type="button" r
      | ole="tab">
122 |             <i class="bi bi-cpu me-1"></i> Инспектор процесса (PID
      | )
123 |         </button>
124 |     </li>
125 |     <li class="nav-item" role="presentation">
126 |         <button class="nav-link btn-sm" id="subtab-catalog-btn"
      | data-bs-toggle="pill" data-bs-target="#subtab-catalog" type="b
      | utton" role="tab">
127 |             <i class="bi bi-journal-code me-1"></i> Каталог операц
      | ий (~180)
128 |         </button>
129 |     </li>
130 | </ul>
131 |
132 | <!-- Содержимое подразделов -->
133 | <div class="tab-content" id="identity-subtabs-content">
134 |
135 |     <!-- 1. Подраздел: Пользователи -->
136 |     <div class="tab-pane fade show active" id="subtab-users" r
      | ole="tabpanel">
137 |         <div class="card shadow-sm border-secondary-subtle">
138 |             <div class="card-header bg-body-tertiary d-flex align-

```

```

items-center justify-content-between py-2 flex-wrap gap-2">
139     <div class="fw-semibold small">Локальные учетные зап
иски Windows</div>
140     <div class="d-flex align-items-center gap-2">
141         <div class="input-group input-group-sm" style="wid
th: 220px;">
142             <span class="input-group-text"><i class="bi bi-s
earch"></i></span>
143             <input type="text" class="form-control" id="filt
er-users-input" placeholder="Поиск пользователя...">
144         </div>
145     </div>
146 </div>
147 <div class="card-body p-0 table-responsive">
148     <table class="table table-hover align-middle mb-0" i
d="users-table">
149         <thead class="table-light small">
150             <tr>
151                 <th>Имя пользователя</th>
152                 <th>Полное имя</th>
153                 <th>SID</th>
154                 <th>Статус</th>
155                 <th>Пароль</th>
156                 <th>Истекает</th>
157                 <th>Последний вход</th>
158                 <th class="text-end">Действия</th>
159             </tr>
160         </thead>
161         <tbody id="users-table-body" class="small">
162             <tr>
163                 <td colspan="8" class="text-center py-3 text-m
uted">Загрузка пользователей...</td>
164             </tr>
165         </tbody>
166     </table>
167 </div>
168 </div>
169 </div>

```



```

170
171     <!-- 2. Подраздел: Группы -->
172     <div class="tab-pane fade" id="subtab-groups" role="tabpan
173         el">
174         <div class="card shadow-sm border-secondary-subtle">
175             <div class="card-header bg-body-tertiary d-flex align-
176             items-center justify-content-between py-2">
177                 <div class="fw-semibold small">Группы безопасности и
178                 граф членства</div>
179             </div>
180             <div class="card-body p-0 table-responsive">
181                 <table class="table table-hover align-middle mb-0" i
182                 d="groups-table">
183                     <thead class="table-light small">
184                         <tr>
185                             <th>Имя группы</th>
186                             <th>SID</th>
187                             <th>Описание</th>
188                             <th>Тип</th>
189                             <th>Участники</th>
190                             <th class="text-end">Действия</th>
191                         </tr>
192                     </thead>
193                     <tbody id="groups-table-body" class="small">
194                         <tr>
195                             <td colspan="6" class="text-center py-3 text-m
196                             uted">Загрузка групп...</td>
197                         </tr>
198                     </tbody>
199                 </table>
200             </div>
201         </div>
202     </div>
203
204     <!-- 3. Подраздел: Журнал безопасности (Security Events Lo
205     g) -->
206     <div class="tab-pane fade" id="subtab-events" role="tabpan
207     el">

```

```
201 |     <div class="card shadow-sm border-secondary-subtle">
202 |         <div class="card-header bg-body-tertiary d-flex align-
| items-center justify-content-between py-2 flex-wrap gap-2">
203 |             <div class="d-flex align-items-center gap-2">
204 |                 <span class="fw-semibold small"><i class="bi bi-sh
| ield-shaded text-primary me-1"></i> Журнал событий безопасност
| и Windows</span>
205 |                 <span class="badge bg-secondary" id="events-count-
| badge">0 записей</span>
206 |             </div>
207 |             <div class="d-flex align-items-center gap-2 flex-wra
| p">
208 |                 <div class="btn-group btn-group-sm" role="group" i
| d="events-quick-filters">
209 |                     <button type="button" class="btn btn-outline-sec
| ondary py-0 px-2 event-filter-btn active" data-filter="all">Вс
| е</button>
210 |                     <button type="button" class="btn btn-outline-sec
| ondary py-0 px-2 event-filter-btn" data-filter="logon">Входы (
| 4624/4625)</button>
211 |                     <button type="button" class="btn btn-outline-sec
| ondary py-0 px-2 event-filter-btn" data-filter="accounts">Акка
| унты (4720/4726)</button>
212 |                     <button type="button" class="btn btn-outline-sec
| ondary py-0 px-2 event-filter-btn" data-filter="password">Паро
| ли (4724)</button>
213 |                     <button type="button" class="btn btn-outline-sec
| ondary py-0 px-2 event-filter-btn" data-filter="logout">Блоки
| ровки (4740)</button>
214 |                 </div>
215 |                 <div class="input-group input-group-sm" style="wid
| th: 200px;">
216 |                     <span class="input-group-text"><i class="bi bi-s
| earch"></i></span>
217 |                     <input type="text" class="form-control" id="filt
| er-events-input" placeholder="Поиск в журнале...">
218 |                 </div>
219 |                 <button class="btn btn-sm btn-outline-primary py-0
```

```

px-2" id="btn-refresh-events" title="Обновить журнал">
220     <i class="bi bi-arrow-clockwise"></i>
221     </button>
222     </div>
223 </div>
224     <div class="card-body p-0 table-responsive" style="max
-height: 520px; overflow-y: auto;">
225         <table class="table table-hover table-sm align-middl
e mb-0" id="events-table">
226             <thead class="table-light small sticky-top">
227                 <tr>
228                     <th style="width: 140px;">Время</th>
229                     <th style="width: 90px;">Event ID</th>
230                     <th style="width: 170px;">Событие</th>
231                     <th style="width: 160px;">Целевой аккаунт</th>
232                     <th style="width: 150px;">Инициатор</th>
233                     <th>Описание / Сообщение</th>
234                     <th style="width: 90px;" class="text-end">Инфо
</th>
235                 </tr>
236             </thead>
237             <tbody id="events-table-body" class="small">
238                 <tr>
239                     <td colspan="7" class="text-center py-4 text-m
uted">Загрузка журнала событий безопасности...</td>
240                 </tr>
241             </tbody>
242         </table>
243     </div>
244 </div>
245 </div>
246
247 <!-- 4. Подраздел: Аудит безопасности и LSA -->
248 <div class="tab-pane fade" id="subtab-security" role="tabp
anel">
249     <div class="row g-3">
250         <div class="col-md-7">
251             <div class="card shadow-sm border-secondary-subtle m

```

```

b-3">
252         <div class="card-header bg-body-tertiary fw-semibo
ld small d-flex align-items-center justify-content-between">
253             <span>Быстрый аудит привилегий и доступа</span>
254         </div>
255         <div class="card-body p-3">
256             <div class="d-flex flex-wrap gap-2 mb-3">
257                 <button class="btn btn-sm btn-outline-primary
d-flex align-items-center gap-1" id="btn-check-who-is-admin">
258                     <i class="bi bi-shield-lock"></i>
259                     <span>Who is Admin?</span>
260                 </button>
261                 <button class="btn btn-sm btn-outline-info d-f
lex align-items-center gap-1" id="btn-check-service-logon">
262                     <i class="bi bi-gear"></i>
263                     <span>Logon as Service</span>
264                 </button>
265                 <button class="btn btn-sm btn-outline-warning
d-flex align-items-center gap-1" id="btn-check-rdp">
266                     <i class="bi bi-display"></i>
267                     <span>Remote Desktop (RDP)</span>
268                 </button>
269                 <button class="btn btn-sm btn-outline-danger d
-flex align-items-center gap-1" id="btn-check-orphaned">
270                     <i class="bi bi-exclamation-triangle"></i>
271                     <span>Orphaned SIDs / Profiles</span>
272                 </button>
273             </div>
274             <div class="border rounded p-2 bg-body-tertiary
small" id="security-audit-results" style="min-height: 180px; m
ax-height: 360px; overflow-y: auto;">
275                 <div class="text-muted text-center py-4">Выбер
ите проверку выше для запуска анализа безопасности.</div>
276             </div>
277         </div>
278     </div>
279 </div>
280

```

```

281 |         <div class="col-md-5">
282 |             <div class="card shadow-sm border-secondary-subtle">
283 |                 <div class="card-header bg-body-tertiary fw-semibol
ld small">
284 |                     <i class="bi bi-key-fill me-1 text-primary"></i>
    Политика паролей SAM
285 |                 </div>
286 |                 <div class="card-body p-3 small" id="password-poli
cy-container">
287 |                     <div class="text-muted">Загрузка политики пароле
й...</div>
288 |                 </div>
289 |             </div>
290 |         </div>
291 |     </div>
292 | </div>
293 |
294 | <!-- 5. Подраздел: Инспектор процесса (PID) -->
295 | <div class="tab-pane fade" id="subtab-pid" role="tabpanel"
>
296 |     <div class="card shadow-sm border-secondary-subtle">
297 |         <div class="card-header bg-body-tertiary d-flex align-
items-center justify-content-between py-2">
298 |             <div class="fw-semibold small">Исследование контекст
а безопасности процесса (Token & Account Dossier)</div>
299 |         </div>
300 |         <div class="card-body p-3">
301 |             <div class="row g-2 align-items-center mb-3">
302 |                 <div class="col-auto">
303 |                     <label for="input-explain-pid" class="col-form-l
abel small fw-semibold">PID процесса:</label>
304 |                 </div>
305 |                 <div class="col-auto" style="width: 150px;">
306 |                     <input type="number" id="input-explain-pid" clas
s="form-control form-control-sm" placeholder="например 1234">
307 |                 </div>
308 |                 <div class="col-auto">
309 |                     <button class="btn btn-sm btn-primary d-flex ali

```

[illegible]

```

338         <th>API / CLI</th>
339         <th>Риск</th>
340     </tr>
341 </thead>
342 <tbody id="catalog-table-body" class="small">
343     <tr>
344         <td colspan="6" class="text-center py-3 text-m
345         uted">Загрузка каталога...</td>
346     </tr>
347 </tbody>
348 </table>
349 </div>
350 </div>
351
352 </div>
353 </div>
354
355 <!-- Модальное окно: Создание пользователя -->
356 <div class="modal fade" id="createUserModal" tabindex="-1" ari
357     a-hidden="true">
358     <div class="modal-dialog">
359         <div class="modal-content shadow">
360             <div class="modal-header">
361                 <h5 class="modal-title d-flex align-items-center gap-2
362                 fs-6 fw-bold">
363                     <i class="bi bi-person-plus text-primary"></i>
364                     <span>Создать локального пользователя</span>
365                 </h5>
366                 <button type="button" class="btn-close" data-bs-dismis
367                 s="modal" aria-label="Close"></button>
368             </div>
369             <div class="modal-body small">
370                 <form id="create-user-form">
371                     <div class="mb-2">
372                         <label class="form-label fw-semibold">Имя пользо
373                         вателя (Логин)*</label>
374                         <input type="text" class="form-control form-contro

```

```

l-sm" id="create-user-name" required placeholder="например: iv
an">
371         </div>
372         <div class="mb-2">
373             <label class="form-label fw-semibold">Пароль (опци
онально)</label>
374             <input type="password" class="form-control form-co
ntrol-sm" id="create-user-pwd" placeholder="Оставьте пустым бе
з пароля">
375         </div>
376         <div class="mb-2">
377             <label class="form-label fw-semibold">Полное имя (
Full Name)</label>
378             <input type="text" class="form-control form-contro
l-sm" id="create-user-fullname" placeholder="например: Иван Ив
анов">
379         </div>
380         <div class="mb-2">
381             <label class="form-label fw-semibold">Описание (De
scription)</label>
382             <input type="text" class="form-control form-contro
l-sm" id="create-user-desc" placeholder="Назначение учетной за
писи">
383         </div>
384     </form>
385 </div>
386     <div class="modal-footer">
387         <button type="button" class="btn btn-sm btn-secondary"
data-bs-dismiss="modal">Отмена</button>
388         <button type="button" class="btn btn-sm btn-primary" i
d="btn-submit-create-user">Создать</button>
389     </div>
390 </div>
391 </div>
392 </div>
393
394 <!-- Модальное окно: Досье Principal и Журнал событий пользова
теля -->

```



```
395 | <div class="modal fade" id="principalDossierModal" tabindex="-
    | 1" aria-hidden="true">
396 |   <div class="modal-dialog modal-xl modal-dialog-scrollable">
397 |     <div class="modal-content shadow">
398 |       <div class="modal-header">
399 |         <h5 class="modal-title d-flex align-items-center gap-2
    | fs-6 fw-bold">
400 |           <i class="bi bi-file-earmark-person text-primary"></
    | i>
401 |           <span id="dossier-modal-title">Досье субъекта безопа
    | сности</span>
402 |         </h5>
403 |         <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
404 |       </div>
405 |       <div class="modal-body">
406 |         <ul class="nav nav-tabs mb-3" id="dossierModalTabs" ro
    | le="tablist">
407 |           <li class="nav-item" role="presentation">
408 |             <button class="nav-link active small py-1" id="dos
    | sier-tree-tab" data-bs-toggle="tab" data-bs-target="#dossier-t
    | ree-pane" type="button" role="tab">
409 |               <i class="bi bi-diagram-2 me-1"></i> Дерево Prin
    | cipal
410 |             </button>
411 |           </li>
412 |           <li class="nav-item" role="presentation">
413 |             <button class="nav-link small py-1" id="dossier-ev
    | ents-tab" data-bs-toggle="tab" data-bs-target="#dossier-events
    | -pane" type="button" role="tab">
414 |               <i class="bi bi-shield-exclamation me-1"></i> Со
    | бытия безопасности (<span id="dossier-events-count">0</span>)
415 |             </button>
416 |           </li>
417 |         </ul>
418 |         <div class="tab-content" id="dossierModalContent">
419 |           <div class="tab-pane fade show active" id="dossier-t
    | ree-pane" role="tabpanel">
```

```

420         <pre id="dossier-tree-view" class="p-3 bg-body-ter
      tiary rounded border font-monospace small" style="white-space:
      pre-wrap; margin-bottom: 0;"></pre>
421     </div>
422     <div class="tab-pane fade" id="dossier-events-pane"
      role="tabpanel">
423         <div class="table-responsive">
424             <table class="table table-hover table-sm align-m
      iddle mb-0">
425                 <thead class="table-light small">
426                     <tr>
427                         <th>Время</th>
428                         <th>Event ID</th>
429                         <th>Событие</th>
430                         <th>Инициатор / Аккаунт</th>
431                         <th>Описание</th>
432                     </tr>
433                 </thead>
434                 <tbody id="dossier-user-events-tbody" class="s
      mall font-monospace">
435                     <tr><td colspan="5" class="text-center py-3
      text-muted">События отсутствуют</td></tr>
436                 </tbody>
437             </table>
438         </div>
439     </div>
440 </div>
441 </div>
442     <div class="modal-footer">
443         <button type="button" class="btn btn-sm btn-secondary"
      data-bs-dismiss="modal">Закрыть</button>
444     </div>
445 </div>
446 </div>
447 </div>
448

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Accounts & Identity Tab - Client Cont
  roller
4  * =====
5  * Description:
6  *   Клиентский JavaScript контроллер для вкладки управления п
  ользователями,
7  *   группами, правами LSA, журналом событий безопасности, инс
  пекцией процессов
8  *   и каталогом операций.
9  *
10 * File: main.js
11 * Project: ai-breadboard
12 * Package: windows/api/webgui/accounts_identity_tab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * Updated: 2026-10-08 01:51:00
16 * =====
17 */
18
19 let _usersCache = [];
20 let _groupsCache = [];
21 let _eventsCache = [];
22 let _catalogCache = [];
23 let _currentEventFilter = 'all';
24
25 /**
26 * Инициализация контроллера вкладки Accounts & Identity
27 */
28 export async function initAccountsIdentityTab() {
29   bindEvents();
30   await loadAllIdentityData();
```

```
31 | }
32 |
33 | /**
34 |  * Привязка обработчиков событий
35 |  */
36 | function bindEvents() {
37 |     const refreshBtn = document.getElementById('btn-refresh-identity');
38 |     if (refreshBtn) refreshBtn.onclick = () => loadAllIdentityData();
39 |
40 |     const refreshEventsBtn = document.getElementById('btn-refresh-events');
41 |     if (refreshEventsBtn) refreshEventsBtn.onclick = () => loadSecurityEvents();
42 |
43 |     const openCreateUserModalBtn = document.getElementById('btn-open-create-user-modal');
44 |     if (openCreateUserModalBtn) {
45 |         openCreateUserModalBtn.onclick = () => {
46 |             const modalEl = document.getElementById('createUserModal');
47 |             if (modalEl && window.bootstrap) {
48 |                 const modal = new window.bootstrap.Modal(modalEl);
49 |                 modal.show();
50 |             }
51 |         };
52 |     }
53 |
54 |     const submitCreateUserBtn = document.getElementById('btn-submit-create-user');
55 |     if (submitCreateUserBtn) submitCreateUserBtn.onclick = handleCreateUserSubmit;
56 |
57 |     const filterUsersInput = document.getElementById('filter-users-input');
58 |     if (filterUsersInput) {
59 |         filterUsersInput.oninput = (e) => filterUsersTable(e.target
```

```
t.value);
60 }
61
62 const filterEventsInput = document.getElementById('filter-ev
ents-input');
63 if (filterEventsInput) {
64     filterEventsInput.oninput = (e) => filterEventsTable(e.tar
get.value);
65 }
66
67 const filterCatalogInput = document.getElementById('filter-c
atalog-input');
68 if (filterCatalogInput) {
69     filterCatalogInput.oninput = (e) => filterCatalogTable(e.t
arget.value);
70 }
71
72 // Быстрые фильтры журнала событий
73 const eventsFilterToolbar = document.getElementById('events-
quick-filters');
74 if (eventsFilterToolbar) {
75     eventsFilterToolbar.querySelectorAll('.event-filter-btn').
forEach(btn => {
76         btn.onclick = () => {
77             eventsFilterToolbar.querySelectorAll('.event-filter-bt
n').forEach(b => b.classList.remove('active'));
78             btn.classList.add('active');
79             _currentEventFilter = btn.getAttribute('data-filter')
|| 'all';
80             applyEventsFilter();
81         };
82     });
83 }
84
85 // Быстрые проверки безопасности
86 const btnWhoIsAdmin = document.getElementById('btn-check-who
-is-admin');
87 if (btnWhoIsAdmin) btnWhoIsAdmin.onclick = runWhoIsAdminChec
```

```
k;
88
89   const btnServiceLogon = document.getElementById('btn-check-s
service-logon');
90   if (btnServiceLogon) btnServiceLogon.onclick = runServiceLog
onCheck;
91
92   const btnRdp = document.getElementById('btn-check-rdp');
93   if (btnRdp) btnRdp.onclick = runRdpCheck;
94
95   const btnOrphaned = document.getElementById('btn-check-orpha
ned');
96   if (btnOrphaned) btnOrphaned.onclick = runOrphanedCheck;
97
98   // Инспектор PID
99   const btnExplainPid = document.getElementById('btn-explain-p
id');
100  if (btnExplainPid) btnExplainPid.onclick = handleExplainPid;
101 }
102
103 /**
104  * Загрузка всех данных с бэкенда
105  */
106 async function loadAllIdentityData() {
107   await Promise.allSettled([
108     loadCurrentToken(),
109     loadUsers(),
110     loadGroups(),
111     loadSecurityEvents(),
112     loadPasswordPolicy(),
113     loadCatalog(),
114   ]);
115 }
116
117 /**
118  * Загрузка контекста текущего пользователя
119  */
120 async function loadCurrentToken() {
```

```

121 |   const badgeEl = document.getElementById('ai-current-token-ba
    | dge');
122 |   try {
123 |     const res = await fetch('/api/windows/identity/current');
124 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
125 |     const data = await res.json();
126 |     const isElevated = data.is_elevated ? '<span class="badge
    | bg-danger ms-1">Elevated (Admin)</span>' : '<span class="badge
    | bg-secondary ms-1">Standard</span>';
127 |     const integrityBadge = `<span class="badge bg-primary ms-1
    | ">${data.integrity_level}</span>`;
128 |     if (badgeEl) {
129 |       badgeEl.innerHTML = `Текущий токен: <strong>${data.domai
    | n}\\${data.user_name}</strong> (SID: <code>${data.user_sid}</c
    | ode>) | Целостность: ${integrityBadge} ${isElevated}`;
130 |     }
131 |   } catch (err) {
132 |     if (badgeEl) badgeEl.innerHTML = `<span class="text-danger
    | ">Ошибка контекста: ${err.message}</span>`;
133 |   }
134 | }
135 |
136 | /**
137 |  * Загрузка пользователей
138 |  */
139 | async function loadUsers() {
140 |   const tbody = document.getElementById('users-table-body');
141 |   const statTotal = document.getElementById('stat-total-users'
    | );
142 |   const statAdmins = document.getElementById('stat-total-admin
    | s');
143 |   const statSessions = document.getElementById('stat-active-se
    | ssions');
144 |
145 |   try {
146 |     const [resUsers, resAdmins, resSessions] = await Promise.a
    | ll([
147 |       fetch('/api/windows/identity/users'),

```

```

148     fetch('/api/windows/identity/who-is-admin'),
149     fetch('/api/windows/identity/sessions'),
150   ]);
151
152   _usersCache = resUsers.ok ? await resUsers.json() : [];
153   const admins = resAdmins.ok ? await resAdmins.json() : [];
154   const sessions = resSessions.ok ? await resSessions.json()
    : [];
155
156   if (statTotal) statTotal.textContent = _usersCache.length;
157   if (statAdmins) statAdmins.textContent = admins.length;
158   if (statSessions) statSessions.textContent = sessions.leng
    th;
159
160   renderUsersTable(_usersCache);
161 } catch (err) {
162   if (tbody) {
163     tbody.innerHTML = `<tr><td colspan="8" class="text-cente
    r text-danger py-3">Ошибка загрузки пользователей: ${err.messa
    ge}</td></tr>`;
164   }
165 }
166 }
167
168 /**
169  * Отрисовка таблицы пользователей
170  */
171 function renderUsersTable(users) {
172   const tbody = document.getElementById('users-table-body');
173   if (!tbody) return;
174
175   if (!users || users.length === 0) {
176     tbody.innerHTML = '<tr><td colspan="8" class="text-center
    text-muted py-3">Пользователи не найдены</td></tr>';
177     return;
178   }
179
180   tbody.innerHTML = users.map(u => {

```



```

181     const statusBadge = u.enabled
182     ? '<span class="badge bg-success bg-opacity-75">Включен<
    /span>'
183     : '<span class="badge bg-danger bg-opacity-75">Отключен<
    /span>';
184     const pwdBadge = u.password_never_expires
185     ? '<span class="badge bg-info text-dark">Бессрочный</spa
    n>'
186     : '<span class="badge bg-secondary">Ограничен</span>';
187
188     return `
189         <tr>
190             <td class="fw-bold">
191                 <i class="bi bi-person-circle me-1 text-primary"></i>
    > ${escapeHtml(u.name)}
192             </td>
193             <td>${escapeHtml(u.full_name || '-')}</td>
194             <td><code class="small">${escapeHtml(u.sid || '-')}</c
    ode></td>
195             <td>${statusBadge}</td>
196             <td>${pwdBadge}</td>
197             <td>${u.account_expires ? escapeHtml(u.account_expires
    ) : 'Никогда'}</td>
198             <td>${u.last_logon ? escapeHtml(u.last_logon) : '-'}</
    td>
199             <td class="text-end">
200                 <button class="btn btn-sm btn-outline-primary py-0 p
    x-1" onclick="window._explainUserDossier('${escapeHtml(u.name)
    }')" title="Открыть досье Principal">
201                     <i class="bi bi-file-text"></i>
202                 </button>
203                 <button class="btn btn-sm btn-outline-danger py-0 px
    -1 ms-1" onclick="window._deleteUserAction('${escapeHtml(u.nam
    e)}')" title="Удалить пользователя">
204                     <i class="bi bi-trash"></i>
205                 </button>
206             </td>
207         </tr>

```

```

208     `;
209     }).join('');
210 }
211
212 /**
213  * Фильтрация пользователей
214  */
215 function filterUsersTable(query) {
216     const q = (query || '').toLowerCase().trim();
217     if (!q) {
218         renderUsersTable(_usersCache);
219         return;
220     }
221     const filtered = _usersCache.filter(u =>
222         (u.name && u.name.toLowerCase().includes(q)) ||
223         (u.full_name && u.full_name.toLowerCase().includes(q)) ||
224         (u.sid && u.sid.toLowerCase().includes(q))
225     );
226     renderUsersTable(filtered);
227 }
228
229 /**
230  * Загрузка групп
231  */
232 async function loadGroups() {
233     const tbody = document.getElementById('groups-table-body');
234     const statGroups = document.getElementById('stat-total-group
235     s');
236     try {
237         const res = await fetch('/api/windows/identity/groups');
238         if (!res.ok) throw new Error(`HTTP ${res.status}`);
239         _groupsCache = await res.json();
240         if (statGroups) statGroups.textContent = _groupsCache.leng
241         th;
242         if (!tbody) return;
243         if (_groupsCache.length === 0) {
244             tbody.innerHTML = '<tr><td colspan="6" class="text-cente

```

```

    r text-muted py-3">Группы не найдены</td></tr>';
244     return;
245 }
246
247 tbody.innerHTML = _groupsCache.map(g => {
248     const typeBadge = g.is_admin
249         ? '<span class="badge bg-danger">ADMIN GROUP</span>'
250         : '<span class="badge bg-secondary">Локальная</span>';
251     const membersCount = (g.members || []).length;
252     return `
253         <tr>
254             <td class="fw-bold"><i class="bi bi-shield me-1 text
-primary"></i> ${escapeHtml(g.name)}</td>
255             <td><code class="small">${escapeHtml(g.sid || '-')}<
/code></td>
256             <td class="text-muted small">${escapeHtml(g.descript
ion || '-')}</td>
257             <td>${typeBadge}</td>
258             <td><span class="badge bg-light text-dark border">${
membersCount} уч.</span></td>
259             <td class="text-end">
260                 <button class="btn btn-sm btn-outline-info py-0 px
-1" onclick="window._viewGroupMembers('${escapeHtml(g.name)}')
" title="Участники">
261                     <i class="bi bi-people"></i>
262                 </button>
263             </td>
264         </tr>
265     `;
266 }).join('');
267 } catch (err) {
268     if (tbody) {
269         tbody.innerHTML = `<tr><td colspan="6" class="text-cente
r text-danger py-3">Ошибка загрузки групп: ${err.message}</td>
</tr>`;
270     }
271 }
272 }

```

```

273 |
274 | /**
275 |  * Загрузка журнала событий безопасности Windows
276 |  */
277 | async function loadSecurityEvents() {
278 |     const tbody = document.getElementById('events-table-body');
279 |     const badge = document.getElementById('events-count-badge');
280 |     const statBadge = document.getElementById('stat-security-events');
281 |
282 |     try {
283 |         const res = await fetch('/api/windows/identity/audit?limit=100');
284 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
285 |         _eventsCache = await res.json();
286 |
287 |         if (badge) badge.textContent = `${_eventsCache.length} записей`;
288 |         if (statBadge) statBadge.textContent = _eventsCache.length;
289 |
290 |         applyEventsFilter();
291 |     } catch (err) {
292 |         if (tbody) {
293 |             tbody.innerHTML = `<tr><td colspan="7" class="text-center text-danger py-4">Ошибка загрузки журнала безопасности: ${err.message}</td></tr>`;
294 |         }
295 |     }
296 | }
297 |
298 | /**
299 |  * Применение фильтров к событиям безопасности
300 |  */
301 | function applyEventsFilter() {
302 |     const q = (document.getElementById('filter-events-input')?.value || '').toLowerCase().trim();
303 |     let filtered = _eventsCache;

```

```

304
305     if (_currentEventFilter === 'logon') {
306         filtered = filtered.filter(e => [4624, 4625].includes(e.ev
ent_id));
307     } else if (_currentEventFilter === 'accounts') {
308         filtered = filtered.filter(e => [4720, 4722, 4725, 4726, 4
731, 4732, 4733, 4734, 4735, 4738].includes(e.event_id));
309     } else if (_currentEventFilter === 'password') {
310         filtered = filtered.filter(e => [4723, 4724].includes(e.ev
ent_id));
311     } else if (_currentEventFilter === 'lockout') {
312         filtered = filtered.filter(e => e.event_id === 4740);
313     }
314
315     if (q) {
316         filtered = filtered.filter(e =>
317             String(e.event_id).includes(q) ||
318             (e.event_name && e.event_name.toLowerCase().includes(q))
319             ||
320             (e.target_account && e.target_account.toLowerCase().incl
udes(q)) ||
321             (e.caller_account && e.caller_account.toLowerCase().incl
udes(q)) ||
322             (e.description && e.description.toLowerCase().includes(q
))
323         );
324     }
325     renderEventsTable(filtered);
326 }
327
328 /**
329  * Поиск по текстовому запросу в журнале событий
330  */
331 function filterEventsTable(query) {
332     applyEventsFilter();
333 }
334

```

```
335 /**
336  * Отрисовка таблицы журнала событий безопасности
337  */
338 function renderEventsTable(events) {
339     const tbody = document.getElementById('events-table-body');
340     if (!tbody) return;
341
342     if (!events || events.length === 0) {
343         tbody.innerHTML = '<tr><td colspan="7" class="text-center
ру-4 text-muted">События безопасности по выбранному фильтру не
найжены</td></tr>';
344     }
345     return;
346
347     tbody.innerHTML = events.map((e, idx) => {
348         let eidBadge = `<span class="badge bg-secondary">${e.event
_id}</span>`;
349         if (e.event_id === 4624) {
350             eidBadge = `<span class="badge bg-success">4624</span>`;
351         } else if (e.event_id === 4625 || e.event_id === 4740 || e
.event_id === 4726 || e.event_id === 4734) {
352             eidBadge = `<span class="badge bg-danger">${e.event_id}<
/span>`;
353         } else if (e.event_id === 4724 || e.event_id === 4738) {
354             eidBadge = `<span class="badge bg-warning text-dark">${e
.event_id}</span>`;
355         } else if (e.event_id === 4720 || e.event_id === 4731 || e
.event_id === 4732) {
356             eidBadge = `<span class="badge bg-primary">${e.event_id}
</span>`;
357         }
358
359         const shortTime = e.timestamp ? e.timestamp.replace('T', '
').slice(0, 19) : '-';
360         const descText = e.description ? escapeHtml(e.description)
: '-';
361
362         return `
```

```

363         <tr class="event-log-row" data-idx="${idx}" style="curso
r: pointer;" onclick="window._showEventDetail(${idx})" title="
Нажмите для просмотра полной информации">
364             <td class="font-monospace text-muted small">${escapeHt
ml(shortTime)}</td>
365             <td>${eidBadge}</td>
366             <td class="fw-semibold text-primary">${escapeHtml(e.ev
ent_name)}</td>
367             <td class="fw-bold">${escapeHtml(e.target_account || '
-')}</td>
368             <td class="text-muted small">${escapeHtml(e.caller_acc
ount || '-')}</td>
369             <td class="small text-truncate" style="max-width: 320p
x;" title="${descText}">${descText}</td>
370             <td class="text-end">
371                 <button class="btn btn-xs btn-outline-info py-0 px-2
" onclick="event.stopPropagation(); window._showEventDetail(${
idx})" title="Подробнее">
372                     Инфо
373                 </button>
374             </td>
375         </tr>
376     `;
377     }).join('');
378 }
379
380 /**
381  * Просмотр детальной информации о событии
382  */
383 window._showEventDetail = function (idx) {
384     const event = _eventsCache[idx];
385     if (!event) return;
386
387     if (window.AITableModal) {
388         window.AITableModal.show({
389             icon: '🔍 ',
390             title: `Событие аудита: #${event.event_id} - ${event.eve
nt_name}`,

```

```

391     subtitle: `Время: ${event.timestamp} | Целевой аккаунт:
    ${event.target_account || 'N/A'}`,
392     tableType: 'generic',
393     badges: [
394         { text: `Event ${event.event_id}`, class: 'badge bg-pr
    imary' },
395         { text: event.event_name, class: 'badge bg-info text-d
    ark' }
396     ],
397     metadata: [
398         { label: 'Event ID', value: String(event.event_id) },
399         { label: 'Событие', value: event.event_name },
400         { label: 'Время регистрации', value: event.timestamp }
    ,
401         { label: 'Целевая учетная запись', value: event.target
    _account || '-' },
402         { label: 'Инициатор (Caller)', value: event.caller_acc
    ount || '-' },
403         { label: 'Статус аудита', value: event.status || 'Succ
    ess' }
404     ],
405     rawTitle: 'Полное сообщение журнала безопасности',
406     rawContent: event.description || '',
407     requestData: {
408         event_id: event.event_id,
409         event_name: event.event_name,
410         target_account: event.target_account,
411         caller_account: event.caller_account,
412         description: event.description
413     }
414 });
415 } else {
416     alert(`Событие #${event.event_id} (${event.event_name})\n\
    nВремя: ${event.timestamp}\nЦелевой аккаунт: ${event.target_ac
    count}\nИнициатор: ${event.caller_account}\n\nОписание:\n${eve
    nt.description}`);
417 }
418 };

```



```
419 |
420 | /**
421 |  * Загрузка политики паролей
422 |  */
423 | async function loadPasswordPolicy() {
424 |     const container = document.getElementById('password-policy-c
425 |     ontainer');
426 |     if (!container) return;
427 |     try {
428 |         const res = await fetch('/api/windows/identity/auth/polici
429 |         es');
430 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
431 |         const pol = await res.json();
432 |         container.innerHTML = `
433 |             <ul class="list-group list-group-flush">
434 |                 <li class="list-group-item d-flex justify-content-betw
435 |                 een align-items-center px-0 py-1">
436 |                     <span>Мин. длина пароля:</span>
437 |                     <strong>${pol.min_password_length} симв.</strong>
438 |                 </li>
439 |                 <li class="list-group-item d-flex justify-content-betw
440 |                 een align-items-center px-0 py-1">
441 |                     <span>Макс. возраст пароля:</span>
442 |                     <strong>${pol.max_password_age_days} дней</strong>
443 |                 </li>
444 |                 <li class="list-group-item d-flex justify-content-betw
445 |                 een align-items-center px-0 py-1">
446 |                     <span>Глубина истории:</span>
447 |                     <strong>${pol.password_history_length}</strong>
448 |                 </li>
449 |                 <li class="list-group-item d-flex justify-content-betw
450 |                 een align-items-center px-0 py-1">
451 |                     <span>Порог блокировки:</span>
452 |                     <strong>${pol.lockout_threshold === 0 ? 'Отключен' :
453 |                     pol.lockout_threshold + ' попыток'}</strong>
454 |                 </li>
455 |             </ul>
456 |         `;
457 |     } catch (err) {
458 |         console.error(err);
459 |     }
460 | }
```

```
450     } catch (err) {
451         container.innerHTML = `

<td colspan="6" class="text-cen
ter text-danger py-3">Ошибка загрузки каталога: ${err.message}</
td></tr>`;
468         }
469     }
470 }
471
472 function renderCatalogTable(items) {
473     const tbody = document.getElementById('catalog-table-body');
474     if (!tbody) return;
475     tbody.innerHTML = items.map(op => {
476         let riskBadge = '<span class="badge bg-success">SAFE</span
>';
477         if (op.risk_level.includes('ADMIN')) riskBadge = '<span cl
ass="badge bg-warning text-dark">ADMIN</span>';
478         if (op.risk_level.includes('DANGEROUS')) riskBadge = '<spa
n class="badge bg-danger">DANGEROUS</span>';
479         if (op.risk_level.includes('ADVANCED')) riskBadge = '<span
class="badge bg-dark">ADVANCED</span>';
480     });


```

```

481     return `
482         <tr>
483             <td class="text-muted fw-bold">${op.id}</td>
484             <td class="fw-semibold">${escapeHtml(op.name_ru)}</td>
485             <td class="small text-muted">${escapeHtml(op.subsystem
         _name_ru)}</td>
486             <td class="small"><code>${escapeHtml(op.mechanism)}</c
         ode></td>
487             <td class="small"><code>${escapeHtml(op.win32_api || o
         p.ps_or_cli || '-')}</code></td>
488             <td>${riskBadge}</td>
489         </tr>
490     `;
491     }).join('');
492 }
493
494 function filterCatalogTable(query) {
495     const q = (query || '').toLowerCase().trim();
496     if (!q) {
497         renderCatalogTable(_catalogCache);
498         return;
499     }
500     const filtered = _catalogCache.filter(op =>
501         op.name_ru.toLowerCase().includes(q) ||
502         op.subsystem_name_ru.toLowerCase().includes(q) ||
503         op.mechanism.toLowerCase().includes(q) ||
504         (op.win32_api && op.win32_api.toLowerCase().includes(q))
505     );
506     renderCatalogTable(filtered);
507 }
508
509 /**
510  * Создание пользователя
511  */
512 async function handleCreateUserSubmit() {
513     const username = document.getElementById('create-user-name')
514     ?.value?.trim();
515     const password = document.getElementById('create-user-pwd')?

```

```
.value || null;
515   const fullName = document.getElementById('create-user-fullname')?.value?.trim() || '';
516   const description = document.getElementById('create-user-description')?.value?.trim() || '';
517
518   if (!username) {
519     if (window.toast) window.toast.error('Ошибка', 'Введите имя пользователя');
520     return;
521   }
522
523   try {
524     const res = await fetch('/api/windows/identity/users', {
525       method: 'POST',
526       headers: { 'Content-Type': 'application/json' },
527       body: JSON.stringify({
528         username,
529         password: password || undefined,
530         full_name: fullName,
531         description,
532       }),
533     });
534     const data = await res.json();
535     if (!res.ok) throw new Error(data.detail || 'Не удалось создать пользователя');
536
537     if (window.toast) window.toast.success('Успех', `Пользователь '${username}' создан`);
538
539     // Закрытие модального окна
540     const modalEl = document.getElementById('createUserModal');
541
542     if (modalEl && window.bootstrap) {
543       const modal = window.bootstrap.Modal.getInstance(modalEl);
544       if (modal) modal.hide();
545     }
546   }
```

```

545     document.getElementById('create-user-form')?.reset();
546     await loadUsers();
547 } catch (err) {
548     if (window.toast) window.toast.error('Ошибка создания', err.message);
549 }
550 }
551
552 /**
553  * Удаление пользователя
554  */
555 window._deleteUserAction = async function (username) {
556     if (!confirm(`Вы уверены, что хотите удалить пользователя '${username}'?`)) return;
557     try {
558         const res = await fetch(`/api/windows/identity/users/${encodeURIComponent(username)}`, {
559             method: 'DELETE',
560         });
561         const data = await res.json();
562         if (!res.ok) throw new Error(data.detail || 'Ошибка удаления');
563         if (window.toast) window.toast.success('Удалено', `Пользователь '${username}' удален`);
564         await loadUsers();
565     } catch (err) {
566         if (window.toast) window.toast.error('Ошибка удаления', err.message);
567     }
568 };
569
570 /**
571  * Досье Principal и история событий безопасности
572  */
573 window._explainUserDossier = async function (identifier) {
574     const modalEl = document.getElementById('principalDossierModal');
575     const titleEl = document.getElementById('dossier-modal-title

```

```

    ');
576   const treeEl = document.getElementById('dossier-tree-view');
577   const eventsCountEl = document.getElementById('dossier-event
s-count');
578   const eventsTbody = document.getElementById('dossier-user-ev
ents-tbody');
579
580   if (titleEl) titleEl.textContent = `Досье Principal: ${ident
ifier}`;
581   if (treeEl) treeEl.textContent = 'Формирование досье безопас
ности...';
582   if (eventsCountEl) eventsCountEl.textContent = '0';
583   if (eventsTbody) eventsTbody.innerHTML = '<tr><td colspan="5
" class="text-center py-3 text-muted">Загрузка журнала событий
...</td></tr>';
584
585   // Переключение на вкладку дерева по умолчанию
586   const firstTabBtn = document.getElementById('dossier-tree-ta
b');
587   if (firstTabBtn && window.bootstrap) {
588     const tabTrigger = new window.bootstrap.Tab(firstTabBtn);
589     tabTrigger.show();
590   }
591
592   if (modalEl && window.bootstrap) {
593     const modal = new window.bootstrap.Modal(modalEl);
594     modal.show();
595   }
596
597   try {
598     const res = await fetch(`/api/windows/identity/explain/${e
ncodeURIComponent(identifier)}`);
599     if (!res.ok) throw new Error(`HTTP ${res.status}`);
600     const data = await res.json();
601
602     // Формирование дерева
603     const lines = [
604       `Principal (${data.name})`,

```

```

605 |         '|',
606 |         `├─ SID: ${data.sid}`,
607 |         `├─ Domain: ${data.domain}`,
608 |         `├─ Type: ${data.principal_type}`,
609 |         `├─ Source: ${data.source}`,
610 |         `├─ Administrator: ${data.is_admin ? 'YES (☑)' : 'NO'}
    | ` ,
611 |         `├─ Built-in: ${data.is_built_in ? 'YES' : 'NO'}`,
612 |         `├─ Orphaned: ${data.is_orphaned ? 'YES (☐)' : 'NO'}`,
613 |         '|',
614 |         '├─ Account Details',
615 |         `├─ Enabled: ${data.account ? (data.account.enabled
    | ? 'YES' : 'NO (Disabled)') : 'N/A'}`,
616 |         `├─ Password Expires: ${data.account ? (data.account
    | t.password_expires || 'Never') : 'N/A'}`,
617 |         `├─ Last Logon: ${data.account ? (data.account.last
    | _logon || 'N/A') : 'N/A'}`,
618 |         '|',
619 |         `├─ Groups (${(data.groups || []).length})`,
620 |         ...(data.groups || []).map(g => `├─ ${g.name} ${g.i
    | s_admin ? '[ADMIN]' : ''} (${g.sid})`),
621 |         '|',
622 |         `├─ LSA Rights & Privileges (${(data.rights || []).leng
    | th})`,
623 |         ...(data.rights || []).slice(0, 10).map(r => `├─ ${
    | r}`),
624 |         '|',
625 |         `├─ Profile: ${data.profile ? data.profile.profile_path
    | : 'None'}`,
626 |         `├─ Active Sessions: ${((data.sessions || []).length)}`,
627 |         `├─ Running Processes: ${((data.processes || []).length)
    | ` ,
628 |     ];
629 |
630 |     if (treeEl) treeEl.textContent = lines.join('\n');
631 |
632 |     // Рендеринг событий пользователя
633 |     const auditEvents = data.audit_events || [];

```

```

634     if (eventsCountEl) eventsCountEl.textContent = String(audi
| tEvents.length);
635
636     if (eventsTbody) {
637         if (auditEvents.length === 0) {
638             eventsTbody.innerHTML = '<tr><td colspan="5" class="te
| xt-center py-3 text-muted">События безопасности для данного по
| льзователя не зафиксированы в Security Log</td></tr>';
639         } else {
640             eventsTbody.innerHTML = auditEvents.map(ev => {
641                 let badgeClass = 'bg-secondary';
642                 if (ev.event_id === 4624) badgeClass = 'bg-success';
643                 if (ev.event_id === 4625 || ev.event_id === 4740) ba
| dgeClass = 'bg-danger';
644                 if (ev.event_id === 4724) badgeClass = 'bg-warning t
| ext-dark';
645
646                 const shortTime = ev.timestamp ? ev.timestamp.replac
| e('T', ' ').slice(0, 19) : '-';
647                 return `
648                     <tr>
649                         <td class="text-muted">${escapeHtml(shortTime)}<
| /td>
650                         <td><span class="badge ${badgeClass}">${ev.event
| _id}</span></td>
651                         <td class="fw-semibold text-primary">${escapeHtm
| l(ev.event_name)}</td>
652                         <td>${escapeHtml(ev.caller_account || ev.target_
| account || '-')}</td>
653                         <td class="small text-truncate" style="max-width
| : 300px;" title="${escapeHtml(ev.description)}">${escapeHtml(e
| v.description)}</td>
654                     </tr>
655                     `;
656                 }).join('');
657         }
658     }
659 } catch (err) {

```



```

660     if (treeEl) treeEl.textContent = `Ошибка получения досье:
    ${err.message}`;
661     if (eventsTbody) eventsTbody.innerHTML = `|<td colspan=
    "5" class="text-center text-danger py-3">Ошибка загрузки событ
    ий: ${err.message}</td></tr>`;
662 }
663 };
664
665 /**
666  * Просмотр участников группы
667  */
668 window._viewGroupMembers = function (groupName) {
669     const group = _groupsCache.find(g => g.name.toLowerCase() ==
    = groupName.toLowerCase());
670     if (!group) return;
671     const members = (group.members || []).join('\n') || 'Нет уча
    стников';
672     alert(`Участники группы ${groupName}:\n\n${members}`);
673 };
674
675 /**
676  * Проверка Who is Admin?
677  */
678 async function runWhoIsAdminCheck() {
679     const outputEl = document.getElementById('security-audit-res
    ults');
680     if (outputEl) outputEl.innerHTML = '<div class="text-muted t
    ext-center py-3">Анализ администраторов системы...</div>';
681     try {
682         const res = await fetch('/api/windows/identity/who-is-admi
    n');
683         const admins = await res.json();
684         if (!admins || admins.length === 0) {
685             if (outputEl) outputEl.innerHTML = '<div>Администраторы
    не найдены</div>';
686             return;
687         }
688         const html = admins.map(a => `

|  |

```

```

689     <div class="mb-2 pb-1 border-bottom">
690         <div class="fw-bold text-danger"><i class="bi bi-shield-lock-fill me-1"></i> ${escapeHtml(a.username)} <span class="
        badge bg-secondary">${a.is_direct ? 'Прямой доступ' : 'Вложенн
        ое членство'}</span></div>
691         <div class="small text-muted font-monospace">SID: ${a.
        sid}</div>
692         <div class="small text-primary">Цепочка: ${a.path_to_
        admin || []}.join(' → ')}</div>
693     </div>
694     `).join('');
695     if (outputEl) outputEl.innerHTML = html;
696 } catch (err) {
697     if (outputEl) outputEl.innerHTML = `<div class="text-dange
        r">Ошибка: ${err.message}</div>`;
698 }
699 }
700
701 /**
702  * Проверка Logon as Service
703  */
704 async function runServiceLogonCheck() {
705     const outputEl = document.getElementById('security-audit-res
        ults');
706     if (outputEl) outputEl.innerHTML = '<div class="text-muted t
        ext-center py-3">Поиск аккаунтов с SeServiceLogonRight...</div
        >';
707     try {
708         const res = await fetch('/api/windows/identity/who-can-log
        on-as-service');
709         const list = await res.json();
710         if (!list || list.length === 0) {
711             if (outputEl) outputEl.innerHTML = '<div class="text-suc
        cess">Специфических сервисных учетных записей не назначено.</d
        iv>';
712             return;
713         }
714         const html = `<h6>Аккаунты с правом SeServiceLogonRight:</

```

```

    h6>` + list.map(a => `
```

```

    ext-center py-3">Поиск осиротевших SID и профилей...</div>';
743     try {
744         const [resSids, resProfiles] = await Promise.all([
745             fetch('/api/windows/identity/orphaned-sids'),
746             fetch('/api/windows/identity/orphaned-profiles'),
747         ]);
748         const sids = await resSids.json();
749         const profiles = await resProfiles.json();
750
751         if ((!sids || sids.length === 0) && (!profiles || profiles
752         .length === 0)) {
753             if (outputEl) outputEl.innerHTML = '<div class="text-suc
754             cess"><i class="bi bi-check-circle-fill me-1"></i> Осиротевших
755             SID и профилей в системе не обнаружено.</div>';
756             return;
757         }
758
759         let html = '<h6>Осиротевшие субъекты:</h6>';
760         for (const s of sids) {
761             html += `<div class="text-danger font-monospace mb-1">
762             SID: ${s.sid} (${s.location})</div>`;
763         }
764         for (const p of profiles) {
765             html += `<div class="text-warning font-monospace mb-1">
766             □ Профиль без пользователя: ${p.profile_path} (SID: ${p.sid})<
767             /div>`;
768         }
769         if (outputEl) outputEl.innerHTML = html;
770     } catch (err) {
771         if (outputEl) outputEl.innerHTML = `<div class="text-dange
772         r">Ошибка: ${err.message}</div>`;
773     }
774 }
775
776 /**
777  * Исследование PID
778  */
779 async function handleExplainPid() {

```

```

773 |   const input = document.getElementById('input-explain-pid');
774 |   const outputEl = document.getElementById('pid-dossier-output
    | ');
775 |   const pidVal = parseInt(input?.value || '0', 10);
776 |
777 |   if (!pidVal || isNaN(pidVal)) {
778 |     if (window.toast) window.toast.error('Ошибка', 'Укажите ко
    | рректный PID процесса');
779 |     return;
780 |   }
781 |
782 |   if (outputEl) {
783 |     outputEl.style.display = 'block';
784 |     outputEl.textContent = `Исследование процесса PID ${pidVal
    | }...`;
785 |   }
786 |
787 |   try {
788 |     const res = await fetch(`/api/windows/identity/explain-pid
    | /${pidVal}`);
789 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
790 |     const data = await res.json();
791 |
792 |     const acc = data.account || {};
793 |     const privs = data.privileges || [];
794 |     const groups = data.groups || [];
795 |
796 |     const lines = [
797 |       `PID ${data.pid}`,
798 |       '|',
799 |       `├─ Process: ${data.process_name}`,
800 |       `├─ Parent: ${data.parent_name} (PID ${data.parent_pid}
    | )`,
801 |       '|',
802 |       `├─ Account`,
803 |       `|   ├─ ${acc.domain}\\${acc.user}`,
804 |       `|   └─ SID: ${acc.sid}`,
805 |       '|',

```

```

806 |     `├ Groups (${groups.length})`,
807 |     ...groups.map(g => `|   └ ${g}`),
808 |     '|',
809 |     `├ Integrity: ${data.integrity}`,
810 |     `├ Elevated: ${data.elevated ? 'YES (☐)' : 'NO'}`,
811 |     '|',
812 |     `├ Privileges (${privs.length})`,
813 |     ...privs.map(p => `|   └ ${p}`),
814 |     '|',
815 |     '└ Session',
816 |     '    └ Console',
817 |     '    └ Session ID: ${data.session_id}`,
818 | ];
819 |
820 | if (outputEl) outputEl.textContent = lines.join('\n');
821 | } catch (err) {
822 |   if (outputEl) outputEl.textContent = `Ошибка исследования
PID ${pidVal}: ${err.message}`;
823 | }
824 | }
825 |
826 | function escapeHtml(str) {
827 |   if (!str) return '';
828 |   return String(str)
829 |     .replace(/&/g, '&amp;')
830 |     .replace(/</g, '&lt;')
831 |     .replace(/>/g, '&gt;')
832 |     .replace(/"/g, '&quot;');
833 | }
834 |
835 | // Автозапуск при загрузке вкладки
836 | if (document.readyState === 'loading') {
837 |   document.addEventListener('DOMContentLoaded', initAccountsId
entityTab);
838 | } else {
839 |   initAccountsIdentityTab();
840 | }
841 |

```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/admin/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/admin/index.html" class
15 |        ="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/admin
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-08 04:06:00
23 |=====
24 |=====
25 -->
26
27 <!DOCTYPE html>
28 <html lang="ru">
29 <head>
30 |    <meta charset="UTF-8">
31 |    <meta name="viewport" content="width=device-width, initial-s
32 |    cale=1.0">
33 |    <title>AI Assistant - Admin</title>
34 |    <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
35 |    t/css/bootstrap.min.css" rel="stylesheet">
```

```
31 <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    0.0/font/bootstrap-icons.css" rel="stylesheet">
32 <link href="/html/css/main.css?v=20260925_v1" rel="styleshee
    t">
33 <script src="/html/js/toast.js?v=20260925_v1"></script>
34 <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
    8next.min.js"></script>
35 </head>
36 <body>
37
38 <!-- Password Protection Modal -->
39 <div class="modal fade" id="passwordModal" tabindex="-1" aria-
    labelledby="passwordModalLabel" aria-hidden="true" data-bs-bac
    kdrop="static" data-bs-keyboard="false">
40 <div class="modal-dialog modal-dialog-centered">
41 <div class="modal-content bg-dark text-white border-second
    ary">
42 <div class="modal-header border-secondary">
43 <h5 class="modal-title" id="passwordModalLabel"><i cla
    ss="bi bi-lock-fill"></i> Вход в админку</h5>
44 <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="modal" aria-label="Close" id="closePassword
    Modal"></button>
45 </div>
46 <div class="modal-body p-4">
47 <p class="text-muted">Введите пароль для доступа к адм
    инистраторскому интерфейсу:</p>
48 <div class="mb-3">
49 <input type="password" class="form-control bg-dark t
    ext-white border-secondary" id="admin-password" placeholder="•
    •••••" autocomplete="current-password" data-lpignore="true" da
    ta-lp-ignore="true" autofocus>
50 </div>
51 <div id="password-error" class="alert alert-danger d-n
    one">Неверный пароль</div>
52 <button type="button" class="btn btn-primary w-100" id
    ="login-btn">Войти</button>
53 </div>
```



```

54     </div>
55 </div>
56 </div>
57
58 <div class="container-fluid mt-2" id="admin-interface">
59
60     <!-- Navigation Bar -->
61     <div class="d-flex align-items-center justify-content-between mb-3 main-nav-container rounded px-2 py-1">
62         <ul class="nav nav-tabs flexnowrap flex-grow-1 border-0 me-2 align-items-center" id="mainTabs">
63             <!-- Кнопка: 0 системе (сверху и первая) -->
64             <li class="nav-item">
65                 <button class="nav-link d-flex align-items-center gap-1 active" data-tab="tab-about-system" data-bs-target="#tab-about-system" type="button" data-i18n="tabs.aboutSystem">
66                     <i class="bi bi-info-circle-fill text-info me-1"></i>
67                     <span>0 системе</span>
68                 </button>
69             </li>
70
71             <!-- Разделитель -->
72             <li class="nav-item d-flex align-items-center">
73                 <div class="nav-group-divider"></div>
74             </li>
75
76             <!-- Группа: Диалог и аудио -->
77             <li class="nav-item dropdown">
78                 <button class="nav-link dropdown-toggle active d-flex align-items-center gap-1" type="button" id="dialogTabsDropdown">
79                     <i class="bi bi-chat-dots-fill text-primary me-1"></i>
80                     <span data-i18n="tabs.groupDialog">Диалог и аудио</span>
81                 </button>
82                 <div class="dropdown-menu shadow border-secondary-subt

```

```

le">
83         <button class="dropdown-item active d-flex align-ite
ms-center gap-2" data-tab="tab-chat" data-bs-target="#tab-chat
" type="button" data-i18n="tabs.chat">
84             □ Чат
85         </button>
86         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-voice" data-bs-target="#tab-voice" typ
e="button" data-i18n="tabs.voice">
87             □ Голос и диаризация
88         </button>
89         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-tts" data-bs-target="#tab-tts" type="b
utton" data-i18n="tabs.tts">
90             □ Настройка TTS
91         </button>
92     </div>
93 </li>
94
95     <!-- Разделитель -->
96     <li class="nav-item d-flex align-items-center">
97         <div class="nav-group-divider"></div>
98     </li>
99
100    <!-- Группа: ИИ и Знания -->
101    <li class="nav-item dropdown">
102        <button class="nav-link dropdown-toggle d-flex align-i
tems-center gap-1" type="button" id="aiTabsDropdown">
103            <i class="bi bi-cpu-fill text-info me-1"></i>
104            <span data-i18n="tabs.groupAi">ИИ и Знания</span>
105        </button>
106        <div class="dropdown-menu shadow border-secondary-subt
le">
107            <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-rag" data-bs-target="#tab-rag" type="b
utton" data-i18n="tabs.rag">
108                □ RAG
109            </button>

```

```
110 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-pixelrag" data-bs-target="#tab-pixelra
      | g" type="button">
111 |             □ PixelRAG
112 |         </button>
113 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-models" data-bs-target="#tab-models" t
      | ype="button" data-i18n="tabs.models">
114 |             □ Модели и API
115 |         </button>
116 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-agents" data-bs-target="#tab-agents" t
      | ype="button" data-i18n="tabs.agents">
117 |             □ Агенты ИИ
118 |         </button>
119 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-skills" data-bs-target="#tab-skills" t
      | ype="button" data-i18n="tabs.skills">
120 |             □ Навыки
121 |         </button>
122 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-mcp" data-bs-target="#tab-mcp" type="b
      | utton" data-i18n="tabs.mcp">
123 |             □ MCP Серверы
124 |         </button>
125 |     </div>
126 | </li>
127 |
128 |     <!-- Разделитель -->
129 |     <li class="nav-item d-flex align-items-center">
130 |         <div class="nav-group-divider"></div>
131 |     </li>
132 |
133 |     <!-- Группа: Плагины -->
134 |     <li class="nav-item dropdown" id="pluginsNavDropdownWrap
      | per">
135 |         <button class="nav-link dropdown-toggle d-flex align-i
      | tems-center gap-1" type="button" id="pluginsTabsDropdown">
```

```

136         <i class="bi bi-puzzle-fill text-warning me-1"></i>
137         <span data-i18n="tabs.groupPlugins">Плагины</span>
138     </button>
139     <div class="dropdown-menu shadow border-secondary-subt
140 le" id="nav-plugins-dropdown-menu">
141         <div id="nav-enabled-plugins-list" class="nav-enable
142 d-plugins-list">
143             <!-- Dynamic enabled plugins will be injected here
144             -->
145         </div>
146         <hr class="dropdown-divider my-1">
147         <button class="dropdown-item d-flex align-items-cent
148 er gap-2" data-tab="tab-plugins" data-bs-target="#tab-plugins"
149 type="button" data-i18n="tabs.managePlugins">
150             □ Все плагины
151         </button>
152     </div>
153 </li>
154
155 <!-- Разделитель -->
156 <li class="nav-item d-flex align-items-center">
157     <div class="nav-group-divider"></div>
158 </li>
159
160 <!-- Группа: Приложения (/apps) -->
161 <li class="nav-item dropdown">
162     <button class="nav-link dropdown-toggle d-flex align-i
163 tems-center gap-1" type="button" id="appsTabsDropdown">
164         <i class="bi bi-grid-fill text-danger me-1"></i>
165         <span data-i18n="tabs.groupApps">Приложения</span>
166     </button>
167     <div class="dropdown-menu shadow border-secondary-subt
168 le">
169         <button class="dropdown-item d-flex align-items-cent
170 er gap-2" data-tab="tab-about-system" data-bs-target="#tab-abo
171 ut-system" type="button" data-i18n="tabs.aboutSystem">
172             i O системе
173         </button>

```

```
165 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-trading" data-bs-target="#tab-trading"
      | type="button" data-i18n="tabs.trading">
166 |             □ Trading Terminal
167 |         </button>
168 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-network" data-bs-target="#tab-network"
      | type="button" data-i18n="tabs.network">
169 |             □ Network Terminal
170 |         </button>
171 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-hardware-load-inspector" data-bs-targe
      | t="#tab-hardware-load-inspector" type="button" data-i18n="tabs
      | .systemInspector">
172 |             □ System Inspector
173 |         </button>
174 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-chat" data-bs-target="#tab-chat" type=
      | "button" data-i18n="tabs.chat">
175 |             □ Чат
176 |         </button>
177 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-scenarios" data-bs-target="#tab-scenar
      | ios" type="button" data-i18n="tabs.scenarios">
178 |             □ □ Чат и сценарии
179 |         </button>
180 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-user-assistant" data-bs-target="#tab-u
      | ser-assistant" type="button">
181 |             □ User Assistant
182 |         </button>
183 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-gcloud" data-bs-target="#tab-gcloud" t
      | ype="button">
184 |             □ Google Cloud Monitor
185 |         </button>
186 |         <button class="dropdown-item d-flex align-items-cent
      | er gap-2" data-tab="tab-website-monitor" data-bs-target="#tab-
```

```
website-monitor" type="button">
187     □ Website Intelligence
188 </button>
189 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-cloudflared" data-bs-target="#tab-clou
dflared" type="button">
190     □ Cloudflared Monitor
191 </button>
192 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-windows-backup" data-bs-target="#tab-w
indows-backup" type="button">
193     □ Windows Backup
194 </button>
195 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-startup-auditor" data-bs-target="#tab-
startup-auditor" type="button">
196     □ Windows Startup Auditor
197 </button>
198 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-process-leaks" data-bs-target="#tab-pr
ocess-leaks" type="button">
199     □ 1. Утечки и микрофризы
200 </button>
201 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-forensics" data-bs-target="#tab-forens
ics" type="button">
202     □ 2. Поведенческая форензика
203 </button>
204 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-throttling" data-bs-target="#tab-throt
tling" type="button">
205     □ 3. Качество ядра и Троттлинг
206 </button>
207 <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-storage-wear" data-bs-target="#tab-sto
rage-wear" type="button">
208     □ 4. Износ дисков и батареи
209 </button>
```

```

210         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-peripherals" data-bs-target="#tab-peri
pherals" type="button">
211             □ 5. Сеть и периферия
212         </button>
213         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-system-logs" data-bs-target="#tab-syst
em-logs" type="button" data-i18n="tabs.systemLogs">
214             □ Логи операционной системы
215         </button>
216     </div>
217 </li>
218
219     <!-- Разделитель -->
220     <li class="nav-item d-flex align-items-center">
221         <div class="nav-group-divider"></div>
222     </li>
223
224     <!-- Группа: Администрирование -->
225     <li class="nav-item dropdown">
226         <button class="nav-link dropdown-toggle d-flex align-i
tems-center gap-1" type="button" id="adminTabsDropdown">
227             <i class="bi bi-gear-wide-connected text-success me-
1"></i>
228             <span data-i18n="tabs.groupSystem">Администрирование
</span>
229         </button>
230         <div class="dropdown-menu shadow border-secondary-subtl
e">
231             <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-admin" data-bs-target="#tab-admin" typ
e="button" data-i18n="tabs.admin">
232                 □ Управление
233             </button>
234             <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-users" data-bs-target="#tab-users" typ
e="button" data-i18n="tabs.users">
235                 □ Внутренние пользователи

```

```

236         </button>
237         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-windows-users" data-bs-target="#tab-wi
ndows-users" type="button">
238             □ Пользователи Windows
239         </button>
240         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-google-accounts" data-bs-target="#tab-
google-accounts" type="button" data-i18n="tabs.googleAccounts"
>
241             □ Google аккаунты
242         </button>
243         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-logs" data-bs-target="#tab-logs" type=
"button" data-i18n="tabs.logs">
244             □ Системные логи
245         </button>
246         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-instructions" data-bs-target="#tab-ins
tructions" type="button" data-i18n="tabs.instructions">
247             □ Системные инструкции
248         </button>
249         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-help" data-bs-target="#tab-help" type=
"button" data-i18n="tabs.help">
250             □ Справочник
251         </button>
252     </div>
253 </li>
254 </ul>
255
256 <!-- Auth Container -->
257 <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto pb-1" id="nav-auth-container">
258     <!-- Active Model & Search Badges -->
259     <span id="chat-model-badge" class="badge rounded-pill al
ign-items-center gap-1 px-3 py-2 shadow-sm border" style="font
-size: 0.8rem; display: none; cursor: pointer;" title="Текущая

```



```

        модель ИИ">
260         □ Загрузка...
261     </span>
262     <span id="chat-search-badge" class="badge rounded-pill a
align-items-center gap-1 px-3 py-2 shadow-sm border" style="fon
t-size: 0.8rem; display: none;" title="Провайдер поиска">
263     </span>
264
265     <!-- Google OAuth Register / Sign-in Button (Shown when
not logged in) -->
266     <button class="btn btn-sm btn-light text-dark fw-semibol
d d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-
pill btn-google-login" id="nav-google-btn" title="Войти или за
регистрироваться через Google" data-i18n-title="auth.googleSig
nInTitle" style="border: 1px solid rgba(0,0,0,0.15);">
267         <svg width="18" height="18" viewBox="0 0 24 24"><path
fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h
5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.
74 3.28-8.09z"/><path fill="#34A853" d="M12 23c2.97 0 5.46-.98
7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-
1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/><path fill
="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.
35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.
85-2.22.81-.63z"/><path fill="#EA4335" d="M12 5.38c1.62 0 3.06
.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.4
7 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/></svg>
268         <span data-i18n="auth.googleSignUp">Регистрация через
Google</span>
269     </button>
270
271     <!-- Authenticated User Menu Dropdown (Shown when logged
in) -->
272     <div class="dropdown d-none" id="user-menu-dropdown-wrap
per">
273         <button class="btn btn-sm btn-outline-secondary d-flex
align-items-center gap-2 px-2 py-1 rounded-pill dropdown-togg
le" id="user-settings-btn" type="button" data-bs-toggle="dropd
own" aria-expanded="false" title="Пользовательское меню">

```

```

274 |         <img id="nav-user-avatar" src="" class="rounded-circle d-none" width="22" height="22" style="object-fit: cover;">
275 |         <i class="bi bi-person-circle" id="nav-user-icon"></i>
276 |         <span id="nav-user-label" class="small fw-semibold" data-i18n="userSettings.button">Пользователь</span>
277 |     </button>
278 |     <ul class="dropdown-menu dropdown-menu-end shadow border-secondary-subtle py-1" aria-labelledby="user-settings-btn" style="min-width: 220px;">
279 |         <li class="px-3 py-2 border-bottom border-secondary-subtle">
280 |             <div class="fw-bold text-truncate" id="user-menu-display-name">Пользователь</div>
281 |             <div class="small text-muted text-truncate" id="user-menu-display-email" style="font-size: 0.75rem;">user@gmail.com</div>
282 |         </li>
283 |         <li><h6 class="dropdown-header small text-uppercase text-muted fw-semibold mt-1">Сервисы</h6></li>
284 |         <li>
285 |             <button class="dropdown-item d-flex align-items-center gap-2 py-1" type="button" id="user-menu-btn-mail">
286 |                 <i class="bi bi-envelope-check text-primary"></i>
287 |                 <span>Почта</span>
288 |             </button>
289 |         </li>
290 |         <li>
291 |             <button class="dropdown-item d-flex align-items-center gap-2 py-1" type="button" id="user-menu-btn-docs">
292 |                 <i class="bi bi-file-earmark-text text-info"></i>
293 |                 <span>Google Docs</span>
294 |             </button>
295 |         </li>
296 |         <li>
297 |             <button class="dropdown-item d-flex align-items-ce

```

```

    nter gap-2 py-1" type="button" id="user-menu-btn-sheets">
298         <i class="bi bi-file-earmark-spreadsheet text-su
    ccess"></i>
299         <span>Google Таблицы</span>
300     </button>
301 </li>
302 <li>
303     <button class="dropdown-item d-flex align-items-ce
    nter gap-2 py-1" type="button" id="user-menu-btn-news">
304         <i class="bi bi-newspaper text-warning"></i>
305         <span>Новости</span>
306     </button>
307 </li>
308 <li>
309     <button class="dropdown-item d-flex align-items-ce
    nter gap-2 py-1" type="button" id="user-menu-btn-facebook">
310         <i class="bi bi-facebook text-primary"></i>
311         <span>Facebook</span>
312     </button>
313 </li>
314 <li><hr class="dropdown-divider my-1"></li>
315 <li>
316     <button class="dropdown-item d-flex align-items-ce
    nter gap-2 py-1" type="button" data-bs-toggle="modal" data-bs-
    target="#userSettingsModal">
317         <i class="bi bi-gear-fill text-secondary"></i>
318         <span data-i18n="userSettings.title">Настройки п
    рофиля</span>
319     </button>
320 </li>
321 <li>
322     <button class="dropdown-item d-flex align-items-ce
    nter gap-2 py-1 text-danger btn-user-logout" type="button">
323         <i class="bi bi-box-arrow-right"></i>
324         <span>Выйти</span>
325     </button>
326 </li>
327 </ul>

```

```
328 |         </div>
329 |
330 |         <!-- Quick Settings Gear Button -->
331 |         <button class="btn btn-sm btn-outline-secondary d-flex align-items-center justify-content-center p-0 rounded-circle" id="user-settings-gear-btn" data-bs-toggle="modal" data-bs-target="#userSettingsModal" title="Настройки" style="width: 32px; height: 32px;" data-i18n-title="userSettings.title">
332 |             <i class="bi bi-gear-fill"></i>
333 |         </button>
334 |     </div>
335 | </div>
336 |
337 | <!-- Main Tab Content Container -->
338 | <div class="tab-content" id="mainTabContent">
339 |     <!-- Chat Tab -->
340 |     <div id="tab-chat" class="tab-pane fade show active"></div>
341 | >
342 |     <!-- Plugins Tab -->
343 |     <div id="tab-plugins" class="tab-pane fade"></div>
344 |
345 |     <!-- Admin Tab -->
346 |     <div id="tab-admin" class="tab-pane fade"></div>
347 |
348 |     <!-- Users Tab -->
349 |     <div id="tab-users" class="tab-pane fade"></div>
350 |
351 |     <!-- Windows Users Tab -->
352 |     <div id="tab-windows-users" class="tab-pane fade"></div>
353 |
354 |     <!-- Google Accounts Tab -->
355 |     <div id="tab-google-accounts" class="tab-pane fade"></div>
356 |
357 |     <!-- RAG Tab -->
358 |     <div id="tab-rag" class="tab-pane fade"></div>
359 |
360 |     <!-- PixelRAG Tab -->
```

```
361 | <div id="tab-pixelrag" class="tab-pane fade"></div>
362 |
363 | <!-- Telegram RAG Tab -->
364 | <div id="tab-telegram-rag" class="tab-pane fade"></div>
365 |
366 | <!-- Smart News Tab -->
367 | <div id="tab-news" class="tab-pane fade"></div>
368 |
369 | <!-- Voice Diarization Tab -->
370 | <div id="tab-voice" class="tab-pane fade"></div>
371 |
372 | <!-- Models Tab -->
373 | <div id="tab-models" class="tab-pane fade"></div>
374 |
375 | <!-- Agents Tab -->
376 | <div id="tab-agents" class="tab-pane fade"></div>
377 |
378 | <!-- Search Tab -->
379 | <div id="tab-search" class="tab-pane fade"></div>
380 |
381 | <!-- TTS Tab -->
382 | <div id="tab-tts" class="tab-pane fade"></div>
383 |
384 | <!-- Sources Tab -->
385 | <div id="tab-sources" class="tab-pane fade"></div>
386 |
387 | <!-- Logs Tab -->
388 | <div id="tab-logs" class="tab-pane fade"></div>
389 |
390 | <!-- Help Tab -->
391 | <div id="tab-help" class="tab-pane fade"></div>
392 |
393 | <!-- Instructions Tab -->
394 | <div id="tab-instructions" class="tab-pane fade"></div>
395 |
396 | <!-- Skills Tab -->
397 | <div id="tab-skills" class="tab-pane fade"></div>
398 |
```

```
399 | <!-- MCP Servers Tab -->
400 | <div id="tab-mcp" class="tab-pane fade"></div>
401 |
402 | <!-- Trading Terminal Tab -->
403 | <div id="tab-trading" class="tab-pane fade"></div>
404 |
405 | <!-- Network Terminal Tab -->
406 | <div id="tab-network" class="tab-pane fade"></div>
407 |
408 | <!-- Hardware Load Inspector Tab -->
409 | <div id="tab-hardware-load-inspector" class="tab-pane fade
    | "></div>
410 |
411 | <!-- Processes Load & Activity Inspector Tab -->
412 | <div id="tab-processes-load-inspector" class="tab-pane fad
    | e"></div>
413 |
414 | <!-- About System Tab -->
415 | <div id="tab-about-system" class="tab-pane fade"></div>
416 |
417 | <!-- Observability & Diagnostics Tab -->
418 | <div id="tab-observability" class="tab-pane fade"></div>
419 |
420 | <!-- Windows System Administrator Tab -->
421 | <div id="tab-windows-admin" class="tab-pane fade"></div>
422 |
423 | <!-- Cloudflared Monitor Tab -->
424 | <div id="tab-cloudflared" class="tab-pane fade"></div>
425 |
426 | <!-- User Assistant Tab -->
427 | <div id="tab-user-assistant" class="tab-pane fade"></div>
428 |
429 | <!-- Google Cloud Monitor Tab -->
430 | <div id="tab-gcloud" class="tab-pane fade"></div>
431 |
432 | <!-- Website Intelligence Monitor Tab -->
433 | <div id="tab-website-monitor" class="tab-pane fade"></div>
434 |
```

```
435 | <!-- System Control Center Tab (System Logs & Activity) -->
436 | <div id="tab-system-control" class="tab-pane fade"></div>
437 |
438 | <!-- Post-Install Wizard Tab -->
439 | <div id="tab-post-install-wizard" class="tab-pane fade"></div>
440 |
441 | <!-- Maintenance & Recovery Tab -->
442 | <div id="tab-maintenance-recovery" class="tab-pane fade"></div>
443 |
444 | <!-- System Log Viewer Tab -->
445 | <div id="tab-system-logs" class="tab-pane fade"></div>
446 |
447 | <!-- Software Audit Tab -->
448 | <div id="tab-software-audit" class="tab-pane fade"></div>
449 |
450 | <!-- Registry Viewer Tab -->
451 | <div id="tab-registry-viewer" class="tab-pane fade"></div>
452 |
453 | <!-- Windows Backup Manager Tab -->
454 | <div id="tab-windows-backup" class="tab-pane fade"></div>
455 |
456 | <!-- Startup Auditor Tab -->
457 | <div id="tab-startup-auditor" class="tab-pane fade"></div>
458 |
459 | <!-- Scenarios Tab -->
460 | <div id="tab-scenarios" class="tab-pane fade"></div>
461 |
462 | <!-- Wikipedia Research Lab Tab -->
463 | <div id="tab-wikipedia-research" class="tab-pane fade"></div>
464 | </div>
465 | <!-- User Directories and Storage Tab -->
466 | <div id="tab-user-directories" class="tab-pane fade"></div>
467 | </div>
```

```
468 |     <!-- Diagnostic and Forensics Tabs -->
469 |     <div id="tab-process-leaks" class="tab-pane fade"></div>
470 |     <div id="tab-forensics" class="tab-pane fade"></div>
471 |     <div id="tab-throttling" class="tab-pane fade"></div>
472 |     <div id="tab-storage-wear" class="tab-pane fade"></div>
473 |     <div id="tab-peripherals" class="tab-pane fade"></div>
474 | </div>
475 | </div>
476 |
477 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    | /js/bootstrap.bundle.min.js"></script>
478 | <script src="/html/js/ai_table_modal.js?v=20261008_v1"></scrip
    | t>
479 | <script src="/html/js/window-api.js?v=20260925_v1"></script>
480 | <script src="/html/js/appConfigEditor.js?v=20260925_v1"></scri
    | pt>
481 | <script src="/html/js/chatService.js?v=20260925_v1"></script>
482 |
483 | <!-- Admin Main - Modular Architecture -->
484 | <script type="module" src="/html/admin/main-refactored.js?v=20
    | 260925_v1"></script>
485 |
486 | <!-- Optimization Patches -->
487 | <script type="module" src="/html/admin/lazy-init-patch.js?v=20
    | 260925_v1"></script>
488 | <script type="module" src="/html/users_tab/debounce-patch.js?v
    | =20260925_v1"></script>
489 |
490 | <!-- HELP Modal -->
491 | <div class="modal fade modal-help" id="help-modal" tabindex="-
    | 1" aria-labelledby="help-modal-label" aria-hidden="true">
492 |     <div class="modal-dialog modal-dialog-scrollable">
493 |         <div class="modal-content">
494 |             <div class="modal-header">
495 |                 <h5 class="modal-title" id="help-modal-label" data-i18
    | n="helpModal.title">Помощь</h5>
496 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
```



```
497 |         </div>
498 |         <div class="modal-body" id="help-modal-content">
499 |             <p data-i18n="common.loading">Загрузка...</p>
500 |         </div>
501 |         <div class="modal-footer">
502 |             <button type="button" class="btn btn-secondary" data-b
503 | s-dismiss="modal" data-i18n="helpModal.close">Закрыть</button>
504 |         </div>
505 |     </div>
506 </div>
507 |
508 <!-- Prompt Preview Modal -->
509 <div class="modal fade" id="prompt-preview-modal" tabindex="-1"
510 | " aria-labelledby="prompt-preview-modal-label" aria-hidden="tr
511 | ue">
512 |     <div class="modal-dialog modal-lg">
513 |         <div class="modal-content">
514 |             <div class="modal-header">
515 |                 <h5 class="modal-title" id="prompt-preview-modal-label
516 | ">Предпросмотр промта</h5>
517 |                 <button type="button" class="btn-close" data-bs-dismis
518 | s="modal" aria-label="Close"></button>
519 |             </div>
520 |             <div class="modal-body" id="prompt-preview-content">
521 |                 <p>Загрузка...</p>
522 |             </div>
523 |             <div class="modal-footer">
524 |                 <button type="button" class="btn btn-secondary" data-b
525 | s-dismiss="modal">Закрыть</button>
526 |             </div>
527 |         </div>
528 |     </div>
529 </div>
530 |
531 <!-- Chat Logic & RAG Architecture Modal -->
532 <div class="modal fade" id="chat-logic-modal" tabindex="-1" ar
533 | ia-labelledby="chat-logic-modal-label" aria-hidden="true">
```

```

528 | <div class="modal-dialog modal-xl modal-dialog-scrollable">
529 |   <div class="modal-content bg-dark text-white border-second
    | ary">
530 |     <div class="modal-header border-secondary d-flex justify
    | -content-between align-items-center">
531 |       <h5 class="modal-title d-flex align-items-center gap-2
    | " id="chat-logic-modal-label">
532 |         <i class="bi bi-diagram-3-fill text-info"></i> □ Ар
    | хитектура и алгоритм диспетчеризации ответов чата
533 |       </h5>
534 |       <div class="d-flex align-items-center gap-2">
535 |         <a href="/html/assets/chat_and_rag_logic.svg" target
    | ="_blank" class="btn btn-sm btn-outline-info" title="Открыть S
    | VG в новой вкладке">
536 |           <i class="bi bi-arrows-fullscreen"></i> SVG
537 |         </a>
538 |         <button type="button" class="btn-close btn-close-whi
    | te" data-bs-dismiss="modal" aria-label="Close"></button>
539 |       </div>
540 |     </div>
541 |     <div class="modal-body p-3">
542 |       <!-- Tabs for Schema vs Documentation -->
543 |       <ul class="nav nav-pills mb-3" id="chatLogicTabs" role
    | ="tablist">
544 |         <li class="nav-item" role="presentation">
545 |           <button class="nav-link active" id="chat-logic-svg
    | -tab" data-bs-toggle="pill" data-bs-target="#chat-logic-svg-pa
    | ne" type="button" role="tab">□ Интерактивная блок-схема</butt
    | on>
546 |         </li>
547 |         <li class="nav-item" role="presentation">
548 |           <button class="nav-link" id="chat-logic-doc-tab" d
    | ata-bs-toggle="pill" data-bs-target="#chat-logic-doc-pane" typ
    | e="button" role="tab">□ Описание алгоритма и сценариев</butto
    | n>
549 |         </li>
550 |       </ul>
551 |

```

```

552 |         <div class="tab-content" id="chatLogicTabContent">
553 |             <!-- SVG Diagram Tab -->
554 |             <div class="tab-pane fade show active text-center" i
| d="chat-logic-svg-pane" role="tabpanel">
555 |                 <div class="p-2 border border-secondary rounded bg
| -black d-flex justify-content-center align-items-center" style
| ="overflow-x: auto; min-height: 520px;">
556 |                     
557 |                 </div>
558 |             </div>
559 |
560 |             <!-- Documentation Text Tab -->
561 |             <div class="tab-pane fade" id="chat-logic-doc-pane"
| role="tabpanel">
562 |                 <div class="p-3 bg-black rounded border border-sec
| ondary text-light small">
563 |                     <h6 class="text-info fw-bold mb-2">□ Режимы рабо
| ты (config.json &rarr; rag.mode)</h6>
564 |                     <ul class="mb-3">
565 |                         <li><strong class="text-success">rag+model (Ги
| бридный / Основной):</strong> Сначала проверяется локальная ме
| диатека. Если фильм найден – LLM красиво рассказывает о нём со
| ссылкой на наличие в медиатеке. Если фильм отсутствует – LLM
| обращается к веб-поиску (MCP/Grounding) и рассказывает о фильм
| е с кнопкой загрузки <code>&lt;film&gt;</code>.</li>
566 |                         <li><strong class="text-warning">rag (Строгий
| локальный):</strong> Работает только по базе <code>media.db</c
| ode>. При отсутствии тайтла выдаёт чёткое <em>«В локальной баз
| е не найдено»</em> без ложных подстановок.</li>
567 |                         <li><strong class="text-primary">model (Без ба
| зы):</strong> Прямой ответ LLM и веб-поиск без обращения к дис
| кам.</li>
568 |                     </ul>
569 |
570 |                     <h6 class="text-info fw-bold mb-2">□ Сценарии о
| бработки</h6>

```

```

571 |         <div class="table-responsive mb-3">
572 |             <table class="table table-dark table-bordered
| table-sm small">
573 |                 <thead>
574 |                     <tr class="table-secondary text-dark">
575 |                         <th>Сценарий</th>
576 |                         <th>Пример</th>
577 |                         <th>Алгоритм</th>
578 |                         <th>Результат</th>
579 |                     </tr>
580 |                 </thead>
581 |                 <tbody>
582 |                     <tr>
583 |                         <td class="fw-bold text-success">Фильм е
| сть в базе</td>
584 |                         <td><em>«Расскажи о фильме А зори здесь
| тихие»</em></td>
585 |                         <td>Точный поиск в <code>media.db</code>
| (Score 1.0) / FAISS &ge; 0.60 &rar; контекст в LLM</td>
586 |                         <td>LLM сообщает, что фильм <strong>дост
| упен в медиатеке</strong>, описывает сюжет и тегирует <code>&l
| t;film&gt;</code>.</td>
587 |                     </tr>
588 |                     <tr>
589 |                         <td class="fw-bold text-danger">Фильма н
| ет в базе</td>
590 |                         <td><em>«Расскажи о сериале Сваты»</em><
| /td>
591 |                         <td>Локальный RAG даёт 0 совпадений &rar
| r; Fallback на Web Search (MCP)</td>
592 |                         <td>LLM рассказывает о сериале, указывае
| т, что его <strong>нет на дисках</strong>, тегирует <code>&lt;
| film&gt;</code> для торрентов.</td>
593 |                     </tr>
594 |                     <tr>
595 |                         <td class="fw-bold text-info">Команда за
| пуска</td>
596 |                         <td><em>«Включи А зори здесь тихие»</em>

```

```

|</td>
597|         <td>Проверка пути на диске &rarr; WebSoc
|ket <code>play_file_by_path</code></td>
598|         <td>Мгновенный старт в плеере (Direct Pl
|ay).</td>
599|     </tr>
600|     <tr>
601|         <td class="fw-bold text-primary">Подборк
|а / Настроение</td>
602|         <td><em>«Найди боевик на вечер»</em></td>
|>
603|         <td>Извлечение подходящих жанров &rarr;
|LLM формирует диалоговую подборку</td>
604|         <td>Живая рекомендация с обоснованием «п
|очему стоит посмотреть».</td>
605|     </tr>
606| </tbody>
607| </table>
608| </div>
609|
610|     <h6 class="text-info fw-bold mb-1">□ Порог коси
|нусного сходства (Cosine Similarity)</h6>
611|     <p class="text-muted mb-0">Для исключения ложных
|срабатываний порог в <code>rag_search_tool</code> установлен
|на <strong>0.60</strong>. Запросы с более низким сходством счи
|таются отсутствующими в локальной базе и корректно перенаправл
|яются в интернет-поиск.</p>
612|     </div>
613| </div>
614| </div>
615| </div>
616|     <div class="modal-footer border-secondary">
617|         <button type="button" class="btn btn-sm btn-secondary"
|data-bs-dismiss="modal">Закрыть</button>
618|     </div>
619| </div>
620| </div>
621| </div>

```

```

622 |
623 | <!-- User Settings Modal -->
624 | <div class="modal fade" id="userSettingsModal" tabindex="-1" a
    | ria-labelledby="userSettingsModalLabel" aria-hidden="true">
625 |     <div class="modal-dialog modal-dialog-centered">
626 |         <div class="modal-content shadow border-secondary-subtle">
627 |             <div class="modal-header border-secondary-subtle">
628 |                 <h5 class="modal-title d-flex align-items-center gap-2
    | " id="userSettingsModalLabel">
629 |                     <i class="bi bi-person-gear text-primary"></i>
630 |                     <span data-i18n="userSettings.title">Пользовательски
    | е настройки</span>
631 |                 </h5>
632 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
633 |             </div>
634 |             <div class="modal-body p-3">
635 |                 <!-- User Account & Google OAuth Card -->
636 |                 <div class="card bg-body-tertiary border-secondary-sub
    | tle mb-3 p-3" id="user-settings-auth-card">
637 |                     <div class="d-flex align-items-center gap-3">
638 |                         <img id="user-settings-avatar" src="" class="round
    | ed-circle border border-2 border-primary" width="48" height="4
    | 8" style="object-fit: cover;">
639 |                         <div class="flex-grow-1 min-w-0">
640 |                             <div class="d-flex align-items-center gap-2">
641 |                                 <strong id="user-settings-name" class="text-tr
    |uncate">Пользователь</strong>
642 |                                 <span id="user-settings-role-badge" class="bad
    |ge bg-primary">User</span>
643 |                             </div>
644 |                             <div id="user-settings-email" class="small text-
    |muted text-truncate">email@gmail.com</div>
645 |                             <div id="user-settings-google-badge" class="smal
    |l mt-1 text-muted">
646 |                                 <i class="bi bi-google text-danger me-1"></i>G
    |oogle OAuth
647 |                             </div>

```

```

648         </div>
649         <button type="button" class="btn btn-outline-dange
r btn-sm btn-user-logout" title="Выйти">
650             <i class="bi bi-box-arrow-right"></i>
651         </button>
652     </div>
653 </div>
654
655     <!-- Guest / Connect Google OAuth Card -->
656     <div class="card bg-body-tertiary border-secondary-sub
tle mb-3 p-3 text-center d-none" id="user-settings-guest-card"
>
657         <div class="text-muted small mb-2">
658             <i class="bi bi-shield-lock me-1"></i> Авторизация
и синхронизация аккаунта
659         </div>
660         <button type="button" class="btn btn-outline-danger
btn-sm w-100 d-flex align-items-center justify-content-center
gap-2 btn-google-login">
661             <i class="bi bi-google"></i>
662             <span>Войти через Google (OAuth)</span>
663         </button>
664         <div class="small text-muted mt-2" style="font-size:
0.75rem;">
665             Синхронизация с Google Drive, Docs, Calendar и Con
tacts
666         </div>
667     </div>
668
669     <!-- Language Selection -->
670     <div class="mb-3">
671         <label class="form-label fw-semibold small d-flex al
ign-items-center gap-2" for="modal-lang-selector">
672             <i class="bi bi-translate text-info"></i>
673             <span data-i18n="userSettings.language">Язык интер
фейса</span>
674         </label>
675         <select class="form-select lang-selector" id="modal-

```

```
lang-selector">
676         <option value="ru">Русский</option>
677         <option value="en">English</option>
678         <option value="he">עברית</option>
679     </select>
680 </div>
681
682     <!-- Theme Selection -->
683     <div class="mb-2">
684         <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-theme-selector">
685             <i class="bi bi-palette text-warning"></i>
686             <span data-i18n="userSettings.theme">Тема оформления</span>
687         </label>
688         <select class="form-select theme-selector" id="modal-theme-selector">
689             <option value="light" data-i18n="userSettings.themeLight">Светлая</option>
690             <option value="dark" data-i18n="userSettings.themeDark">Тёмная</option>
691             <option value="system" data-i18n="userSettings.themeSystem">Системная</option>
692         </select>
693     </div>
694 </div>
695     <div class="modal-footer border-secondary-subtle py-2">
696         <button type="button" class="btn btn-secondary btn-sm" data-bs-dismiss="modal" data-i18n="userSettings.close">Закрыть</button>
697     </div>
698 </div>
699 </div>
700 </div>
701
702 </body>
703 </html>
```


lazy-init-patch.js

```
1 /**
2  * =====
3  * Process Name: Windows Admin - Lazy-Init-Patch Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля lazy-init
7  *   -patch.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/lazy-init-patch.
11  *     js?v=20261001_v1" type="module"></script>
12  *
13  * File: lazy-init-patch.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/admin
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-04 07:09:00
19  * =====
20  */
21 /**
22  * Lazy Init Patch - Патч для переключения на ленивую загрузку
23  * вкладок
24  *
25  * Этот скрипт перехватывает инициализацию main.js и использует
26  * LazyTabLoader вместо стандартной загрузки всех вкладок сразу
27  */
28 import {
```

```
29 |   initLazyTabLoading,  
30 |   enhanceTabSwitching,  
31 |   restoreLastTab  
32 | } from './optimization-init.js';  
33 | import { autoPatchTab } from '../js/tab-debounce-auto-patch.js'  
34 |   ;  
35 | import TabStatePersistence from '../js/tab-state-persistence.js'  
36 |   ;  
37 | import TabSwitchOptimizer from '../js/tab-switch-optimizer.js'  
38 |   ;  
39 | console.log('[LazyInitPatch] Patching admin interface initiali  
40 | zation...');  
41 |  
42 | // Инициализируем persistence для админки  
43 | const tabPersistence = new TabStatePersistence('admin');  
44 | window.tabPersistence = tabPersistence;  
45 |  
46 | // Ждем когда main.js загрузится и инициализирует базовые элем  
47 | енты  
48 | window.addEventListener('load', async () => {  
49 |   console.log('[LazyInitPatch] Window load event, initializing  
50 | lazy loading...');  
51 |  
52 |   // Ждем немного чтобы main.js выполнил свои скрипты  
53 |   await new Promise(resolve => setTimeout(resolve, 100));  
54 |  
55 |   // Если есть оригинальная функция switchTab, оборачиваем ее  
56 |   if (window.switchTab && typeof window.switchTab === 'functio  
57 | n') {  
58 |     const originalSwitchTab = window.switchTab;  
59 |  
60 |     // Применяем оптимизацию переключения  
61 |     const switchOptimizer = new TabSwitchOptimizer();  
62 |     switchOptimizer.enableDOMCaching();  
63 |  
64 |     // Оборачиваем в enhanced версию с lazy loading  
65 |     let enhancedSwitchTab = enhanceTabSwitching(originalSwitch
```

```

    Tab);
60
61     // Оборачиваем в optimizer версию для отслеживания произво
    дительности
62     const optimizedSwitchTab = switchOptimizer.optimizeSwitchi
    ng(enhancedSwitchTab);
63
64     window.switchTab = async function(targetId) {
65         // Вызываем оптимизированную функцию
66         await optimizedSwitchTab.call(this, targetId);
67
68         // Сохраняем активную вкладку
69         if (targetId) {
70             const cleanId = targetId.startsWith('#') ? targetId.sl
    ice(1) : targetId;
71             tabPersistence.saveActiveTab(cleanId);
72         }
73     };
74
75     console.log('[LazyInitPatch] Enhanced switchTab with on-de
    mand lazy loading and persistence');
76 }
77
78 // Инициализируем ленивую загрузку с определением всех вклад
    ок
79 const allTabDefinitions = [
80     // Основные вкладки (Диалог и аудио)
81     { tabName: 'chat', htmlUrl: `/html/chat/index.html`, jsUrl
    : `/html/chat/main.js` },
82     { tabName: 'voice', htmlUrl: `/html/voice_tab/index.html`,
    jsUrl: `/html/voice_tab/main.js` },
83     { tabName: 'tts', htmlUrl: `/html/tts_tab/index.html`, jsU
    rl: `/html/tts_tab/main.js` },
84
85     // ИИ и Знания
86     { tabName: 'rag', htmlUrl: `/html/rag_tab/index.html`, jsU
    rl: `/html/rag_tab/main.js` },
87     { tabName: 'models', htmlUrl: `/html/models_tab/index.html

```

```
` , jsUrl: `/html/models_tab/main.js` },
88     { tabName: 'agents', htmlUrl: `/html/agents_tab/index.html` ,
    ` , jsUrl: `/html/agents_tab/main.js` },
89     { tabName: 'skills', htmlUrl: `/html/skills_tab/index.html` ,
    ` , jsUrl: `/html/skills_tab/main.js` },
90     { tabName: 'mcp', htmlUrl: `/html/mcp_tab/index.html` , jsU
    rl: `/html/mcp_tab/main.js` },
91     { tabName: 'observability', htmlUrl: `/html/system_inspect
    or_tab/index.html` , jsUrl: `/html/system_inspector_tab/main.js
    ` },
92     { tabName: 'telegram-rag', htmlUrl: `/html/telegram_rag_ta
    b/index.html` , jsUrl: `/html/telegram_rag_tab/main.js` },
93     { tabName: 'sources', htmlUrl: `/html/sources_tab/index.ht
    ml` , jsUrl: `/html/sources_tab/main.js` },
94     { tabName: 'search', htmlUrl: `/html/search_tab/index.html
    ` , jsUrl: `/html/search_tab/main.js` },
95
96     // Система и Администратор
97     { tabName: 'admin', htmlUrl: `/html/admin_tab/index.html` ,
    jsUrl: `/html/admin_tab/main.js` },
98     { tabName: 'users', htmlUrl: `/html/users_tab/index.html` ,
    jsUrl: `/html/users_tab/main.js` },
99     { tabName: 'plugins', htmlUrl: `/html/plugins_tab/index.ht
    ml` , jsUrl: `/html/plugins_tab/main.js` },
100    { tabName: 'google-accounts', htmlUrl: `/html/google_accou
    nts_tab/index.html` , jsUrl: `/html/google_accounts_tab/main.js
    ` },
101    { tabName: 'instructions', htmlUrl: `/html/instructions_ta
    b/index.html` , jsUrl: `/html/instructions_tab/main.js` },
102    { tabName: 'news', htmlUrl: `/html/news_tab/index.html` , j
    sUrl: `/html/news_tab/main.js` },
103    { tabName: 'logs', htmlUrl: `/html/logs/index.html` , jsUrl
    : `/html/logs/main.js` },
104    { tabName: 'help', htmlUrl: `/html/help/index.html` , jsUrl
    : `/html/help/main.js` },
105
106    // Микроприложения (с привязкой к ID приложения в конфиге
    профиля)
```

```
107     { tabName: 'trading', appId: 'trading_terminal', htmlUrl:
    `/html/trading_tab/index.html`, jsUrl: `/html/trading_tab/main
    .js` },
108     { tabName: 'network', appId: 'network_terminal', htmlUrl:
    `/html/network_tab/index.html`, jsUrl: `/html/network_tab/main
    .js` },
109     { tabName: 'system-inspector', appId: 'system_inspector',
    htmlUrl: `/html/system_inspector_tab/index.html`, jsUrl: `/htm
    l/system_inspector_tab/main.js` },
110     { tabName: 'about-system', appId: 'about_system', htmlUrl:
    `/html/about_system_tab/index.html?v=20261006_v3`, jsUrl: `/h
    tml/about_system_tab/main.js?v=20261006_v3` },
111     { tabName: 'windows-admin', appId: 'windows_sysadmin', htm
    lUrl: `/html/windows_admin_tab/index.html`, jsUrl: `/html/wind
    ows_admin_tab/main.js` },
112     { tabName: 'cloudflared', appId: 'cloudflared_monitor', ht
    mlUrl: `/html/cloudflared_tab/index.html`, jsUrl: `/html/cloud
    flared_tab/main.js` },
113     { tabName: 'user-assistant', appId: 'user_assistant', html
    Url: `/html/user_assistant_tab/index.html`, jsUrl: `/html/user
    _assistant_tab/main.js` },
114     { tabName: 'gcloud', appId: 'gcloud_monitor', htmlUrl: `/h
    tml/gcloud_tab/index.html`, jsUrl: `/html/gcloud_tab/main.js`
    },
115     { tabName: 'website-monitor', appId: 'website_monitor', ht
    mlUrl: `/html/website_monitor_tab/index.html`, jsUrl: `/html/w
    ebsite_monitor_tab/main.js` },
116     { tabName: 'system-control', appId: 'system_control_center
    ', htmlUrl: `/html/system_control_tab/index.html?v=20261008_v1
    `, jsUrl: `/html/system_control_tab/main.js?v=20261008_v1` },
117     { tabName: 'system-logs', appId: 'system_log_viewer', html
    Url: `/html/system_logs_tab/index.html`, jsUrl: `/html/system_
    logs_tab/main.js` },
118     { tabName: 'software-audit', appId: 'software_audit', html
    Url: `/html/software_audit_tab/index.html`, jsUrl: `/html/soft
    ware_audit_tab/main.js` },
119     { tabName: 'registry-viewer', appId: 'registry_viewer', ht
    mlUrl: `/html/registry_viewer_tab/index.html`, jsUrl: `/html/r
```

```

    registry_viewer_tab/main.js` },
120     { tabName: 'startup-auditor', appId: 'windows_startup_audi
tor', htmlUrl: `/html/startup_auditor_tab/index.html`, jsUrl:
`/html/startup_auditor_tab/main.js` },
121     { tabName: 'windows-backup', appId: 'windows_backup_manage
r', htmlUrl: `/html/windows_backup_tab/index.html?v=20260924_v
2`, jsUrl: `/html/windows_backup_tab/main.js?v=20260924_v2` },
122     { tabName: 'wikipedia-research', appId: 'wikipedia_researc
h', htmlUrl: `/html/wikipedia_research_tab/index.html`, jsUrl:
`/html/wikipedia_research_tab/main.js` },
123     { tabName: 'process-leaks', appId: 'process_leaks', htmlUr
l: `/html/process_leaks_tab/index.html?v=20260924_v1`, jsUrl:
`/html/process_leaks_tab/main.js?v=20260924_v1` },
124     { tabName: 'forensics', appId: 'forensics', htmlUrl: `/htm
l/forensics_tab/index.html?v=20260924_v1`, jsUrl: `/html/foren
sics_tab/main.js?v=20260924_v1` },
125     { tabName: 'throttling', appId: 'throttling', htmlUrl: `/h
tml/throttling_tab/index.html?v=20260924_v1`, jsUrl: `/html/th
rottling_tab/main.js?v=20260924_v1` },
126     { tabName: 'storage-wear', appId: 'storage_wear', htmlUrl:
`/html/storage_wear_tab/index.html?v=20260924_v1`, jsUrl: `/h
tml/storage_wear_tab/main.js?v=20260924_v1` },
127     { tabName: 'peripherals', appId: 'peripherals', htmlUrl: `
/html/peripherals_tab/index.html?v=20260924_v1`, jsUrl: `/html
/peripherals_tab/main.js?v=20260924_v1` },
128 ];
129
130 try {
131     // Получаем статус приложений для активного сценария (Run-
Dashboard, Run-TC, su)
132     let appsStatus = window.appsStatusMap || null;
133     if (!appsStatus) {
134         try {
135             const res = await (window.api ? window.api.fetch('/api
/v1/apps/status') : fetch('/api/v1/apps/status')).then(r => r.j
son());
136             appsStatus = res?.apps || null;
137         } catch (err) {

```

```

138 |         console.warn('[LazyInitPatch] Could not fetch apps sta
    | tus for profile filtering:', err);
139 |     }
140 | }
141 |
142 |     // Фильтруем вкладки: загружаем ТОЛЬКО те, которые разреше
    | ны в активном профиле конфига
143 |     const activeTabDefinitions = allTabDefinitions.filter(tab
    | => {
144 |         if (tab.appId && appsStatus) {
145 |             const appInfo = appsStatus[tab.appId] || Object.values
    | (appsStatus).find(a => a.id === tab.appId || a.tab === `tab-${
    | tab.tabName}`);
146 |             if (appInfo && appInfo.enabled === false) {
147 |                 return false; // Отключено в активном сценарии/профи
    | ле
148 |             }
149 |         }
150 |         return true;
151 |     });
152 |
153 |     // Инициализируем систему ленивой загрузки только для разр
    | ешенных в профиле вкладок
154 |     await initLazyTabLoading(activeTabDefinitions);
155 |
156 |     console.log(`[LazyInitPatch] Lazy loading initialized: ${a
    | ctiveTabDefinitions.length} tabs registered on-demand.`);
157 |
158 |     // Восстанавливаем и загружаем ТОЛЬКО одну активную вкладк
    | у
159 |     const restored = await restoreLastTab('tab-about-system');
160 |     console.log(`[LazyInitPatch] Initial active tab loaded: ${
    | restored}`);
161 |
162 | } catch (error) {
163 |     console.error('[LazyInitPatch] Error during initialization
    | :', error);
164 | }

```

```
165 | });  
166 |  
167 | console.log('[LazyInitPatch] Patch loaded successfully');  
168 |  
169 | export {};  
170 |
```


main-refactored.js

```
1 /**
2  * =====
3  * Process Name: Windows Admin - Main-Refactored Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main-refa
7  *   ctored.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/main-refactored.
11  *     js?v=20261001_v1" type="module"></script>
12  *
13  * File: main-refactored.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/admin
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * Admin Interface Main - Оркестратор
23  *
24  * Главный модуль для инициализации административного интерфей
25  * са.
26  *
27  * Все реальная реализация находится в модулях, этот файл толь
28  * ко оркестрирует.
29  *
30  * Модули:
31  * - auth-handler.js: Аутентификация и проверка пароля
32  * - tab-manager.js: Управление вкладками и навигацией
```

```

30 * - ui-handler.js: Модали, уведомления, помощь
31 * - init-interface.js: Инициализация компонентов
32 * - apps-sync.js: Синхронизация приложений
33 */
34
35 // =====
36 // ИМПОРТ МОДУЛЕЙ
37 // =====
38
39 import { setupAuthHandlers, verifyPassword } from './modules/a
uth-handler.js';
40 import { setupTabManagement, onTabSwitched } from './modules/t
ab-manager.js';
41 import { setupUIHandlers, showHelpModal, showNotification, sho
wChatLogicModal } from './modules/ui-handler.js';
42 import {
43   initializeInterface,
44   setupGlobalFunctions,
45   setupLanguageSelector,
46   setupThemeSelector
47 } from './modules/init-interface.js';
48 import { syncApplicationsVisibility } from './modules/apps-syn
c.js';
49
50 // =====
51 // ГЛОБАЛЬНЫЕ ПЕРЕМЕННЫЕ
52 // =====
53
54 let tabLoader = null;
55 let apiFetcher = null;
56
57 // =====
58 // ГЛАВНАЯ ФУНКЦИЯ ИНИЦИАЛИЗАЦИИ

```

```

59 // =====
60 =====
61 async function initAdminInterface() {
62     console.log(' [AdminInterface] Starting initialization...'
63 );
64     try {
65         // 1. Инициализируем компоненты интерфейса
66         console.log('[AdminInterface] Step 1: Initializing core co
67 mponents...');
68         await initializeInterface();
69         setupGlobalFunctions();
70         setupLanguageSelector();
71         setupThemeSelector();
72         // 2. Регистрируем глобальные функции для UI
73         console.log('[AdminInterface] Step 2: Setting up UI handle
74 rs...');
75         registerGlobalFunctions();
76         setupUIHandlers();
77         setupAuthHandlers();
78         // 3. Инициализируем управление вкладками
79         console.log('[AdminInterface] Step 3: Setting up tab manag
80 erment...');
81         setupTabManagement();
82         // 4. Синхронизируем видимость приложений
83         console.log('[AdminInterface] Step 4: Syncing applications
84 visibility...');
85         await syncApplicationsVisibility();
86         // 5. Инициализируем оптимизации (ленивая загрузка, кеширо
87 вание и т.д.)
88         console.log('[AdminInterface] Step 5: Initializing optimiz
89 ations...');
90         await initializeOptimizations();

```

```

89 |
90 |     // 6. Инициализируем API клиент
91 |     console.log('[AdminInterface] Step 6: Setting up API clien
    | t...');
92 |     setupAPIClient();
93 |
94 |     console.log('□ [AdminInterface] Initialization complete!')
    | ;
95 |
96 | } catch (error) {
97 |     console.error('□ [AdminInterface] Initialization error:',
    | error);
98 |     showNotification('Ошибка инициализации интерфейса: ' + err
    | or.message, 'danger');
99 | }
100 | }
101 |
102 | // =====
    | =====
103 | // РЕГИСТРАЦИЯ ГЛОБАЛЬНЫХ ФУНКЦИЙ
104 | // =====
    | =====
105 |
106 | function registerGlobalFunctions() {
107 |     window.switchTab = switchTab;
108 |     window.onTabSwitched = onTabSwitched;
109 |     window.showHelpModal = showHelpModal;
110 |     window.showNotification = showNotification;
111 |     window.showChatLogicModal = showChatLogicModal;
112 |     window.verifyPassword = verifyPassword;
113 | }
114 |
115 | // =====
    | =====
116 | // УПРАВЛЕНИЕ ВКЛАДКАМИ
117 | // =====
    | =====
118 |

```

```
119 | /**
120 |  * Переключить на вкладку
121 |  */
122 | async function switchTab(targetId) {
123 |   if (!targetId) return;
124 |
125 |   const cleanId = targetId.startsWith('#') ? targetId.slice(1)
      |   : targetId;
126 |   const tabName = cleanId.replace(/^tab-/, '');
127 |
128 |   console.log(`[AdminInterface] Switching to tab: ${tabName}`)
      |   ;
129 |
130 |   // Используем LazyTabLoader если доступен
131 |   if (tabLoader && typeof tabLoader.isLoaded === 'function') {
132 |     if (!tabLoader.isLoaded(tabName)) {
133 |       console.log(`[AdminInterface] Loading tab: ${tabName}`);
134 |       await tabLoader.loadTab(tabName);
135 |     }
136 |   }
137 |
138 |   // Обновляем визуальное состояние
139 |   updateTabUI(cleanId);
140 |
141 |   // Вызываем callback
142 |   if (typeof window.onTabSwitched === 'function') {
143 |     window.onTabSwitched(cleanId);
144 |   }
145 |
146 |   console.log(`[AdminInterface] Tab switched: ${tabName}`);
147 | }
148 |
149 | /**
150 |  * Обновить визуальное состояние UI
151 |  */
152 | function updateTabUI(cleanId) {
153 |   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
      |   cleanId}`;
```

```
154 |
155 | // Скрываем все вкладки
156 | document.querySelectorAll('#mainTabContent > .tab-pane, body
    | > .container-fluid > .tab-content > .tab-pane, [role="tabpanel
    | l"]').forEach(tab => {
157 |     tab.classList.remove('active', 'show');
158 |     tab.setAttribute('aria-hidden', 'true');
159 | });
160 |
161 | // Показываем целевую вкладку
162 | const targetTab = document.getElementById(tabId) || document
    | .getElementById(cleanId);
163 | if (targetTab) {
164 |     targetTab.classList.add('active', 'show');
165 |     targetTab.setAttribute('aria-hidden', 'false');
166 | }
167 |
168 | // Обновляем состояние элементов меню
169 | document.querySelectorAll('#mainTabs .list-group-item, #main
    | Tabs .dropdown-item, #mainTabs [data-tab], #mainTabs [data-bs-
    | target]').forEach(item => {
170 |     const itemTarget = item.getAttribute('data-tab') || item.g
    | etAttribute('data-bs-target')?.replace('#', '');
171 |     if (itemTarget === tabId || itemTarget === cleanId) {
172 |         item.classList.add('active');
173 |         item.setAttribute('aria-selected', 'true');
174 |     } else {
175 |         item.classList.remove('active');
176 |         item.setAttribute('aria-selected', 'false');
177 |     }
178 | });
179 |
180 | // Обновляем активность родительских кнопок dropdown-toggle
181 | document.querySelectorAll('#mainTabs .dropdown').forEach(dro
    | pdown => {
182 |     const toggle = dropdown.querySelector('.dropdown-toggle');
183 |     const hasActiveChild = dropdown.querySelector('.dropdown-i
    | tem.active, .list-group-item.active');
```

```

184     if (toggle) {
185         if (hasActiveChild) {
186             toggle.classList.add('active');
187         } else {
188             toggle.classList.remove('active');
189         }
190     }
191 });
192 }
193
194 // =====
195 // ИНИЦИАЛИЗАЦИЯ ОПТИМИЗАЦИЙ
196 // =====
197
198 async function initializeOptimizations() {
199     // Оптимизации автоматически инициализируются в lazy-init-pa
200     // Здесь просто сохраняем ссылки для последующего использова
201     // ния
202     if (window.optimizationModule) {
203         tabLoader = window.optimizationModule.tabLoader;
204         apiFetcher = window.optimizationModule.apiFetcher;
205         console.log('[AdminInterface] Optimizations initialized');
206     } else {
207         console.warn('[AdminInterface] Optimization module not fou
208     }
209 }
210
211 // =====
212 // ИНИЦИАЛИЗАЦИЯ API КЛИЕНТА
213 // =====
214

```

```

215 function setupAPIClient() {
216     if (!window.api) {
217         console.warn('[AdminInterface] API client not found, skipping setup');
218         return;
219     }
220
221     // API клиент уже подключен в main.js (исходный), здесь могут быть
222     // дополнительные настройки
223     console.log('[AdminInterface] API client ready');
224 }
225
226 // =====
227 // ОБРАБОТКА СОБЫТИЯ LOAD
228 // =====
229
230 if (document.readyState === 'loading') {
231     document.addEventListener('DOMContentLoaded', initAdminInterface);
232 } else {
233     initAdminInterface();
234 }
235
236 console.log('□ [AdminInterface] Module loaded and ready');
237
238 // =====
239 // ЭКСПОРТЫ (для модульного использования)
240 // =====
241
242 export {
243     initAdminInterface,
244     switchTab,
245     setupAuthHandlers,

```



```
246 |     setupTabManagement,  
247 |     setupUIHandlers,  
248 |     syncApplicationsVisibility,  
249 |     initializeInterface,  
250 |     setupGlobalFunctions  
251 | };  
252 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Admin - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/main.js?v=202610
11  *     01_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/admin
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 04:06:00
19  * =====
20  */
21 // Admin Interface Main JS
22 import { initI18n, switchLang, applyTranslations } from '../js/i18n.js';
23 import { initTheme, setTheme, getThemeMode, getResolvedTheme }
24   from '../js/theme.js';
25 import { initUserSettings, refreshUserProfile } from '../js/userSettings.js';
26
27 // Make switchLang and theme functions available globally
28 window.switchLang = switchLang;
29 window.applyTranslations = applyTranslations;
30 window.setTheme = setTheme;
```

```

30 window.getThemeMode = getThemeMode;
31 window.getResolvedTheme = getResolvedTheme;
32 window.initUserSettings = initUserSettings;
33 window.refreshUserProfile = refreshUserProfile;
34
35 // Helper to open plugin from top navbar dropdown
36 window.openPluginFromDropdown = function(pluginName) {
37     if (!pluginName) return;
38     document.querySelectorAll('#mainTabs .dropdown-menu.show').f
forEach((m) => {
39         m.classList.remove('show');
40         m.closest('.dropdown')?.querySelector('.dropdown-toggle')?
.classList.remove('show');
41     });
42
43     const normalized = pluginName.toLowerCase().replace(/-/g, '_
');
44
45     if (normalized === 'telegram_channel_rag' || normalized ===
'telegram_rag') {
46         const pane = document.getElementById('tab-telegram-rag') |
| document.getElementById('tab-telegram_rag');
47         if (pane && typeof window.switchTab === 'function') {
48             window.switchTab('tab-telegram-rag');
49             return;
50         }
51     }
52
53     if (normalized === 'news_feed' || normalized === 'news' || n
ormalized === 'smart_news' || normalized === 'smart_feed_news'
) {
54         const pane = document.getElementById('tab-news');
55         if (pane && typeof window.switchTab === 'function') {
56             window.switchTab('tab-news');
57             return;
58         }
59     }
60

```

```

61 | if (typeof window.switchTab === 'function') {
62 |     window.switchTab('tab-plugins');
63 | }
64 |
65 | const select = () => {
66 |     if (typeof window.selectPlugin === 'function') {
67 |         window.selectPlugin(pluginName);
68 |     }
69 | };
70 |
71 | select();
72 | setTimeout(select, 150);
73 | setTimeout(select, 400);
74 | };
75 |
76 | // API module
77 | window.api = {
78 |     async fetch(url, options = {}) {
79 |         const response = await fetch(url, options);
80 |         if (!response.ok) {
81 |             let msg = response.statusText;
82 |             try {
83 |                 const data = await response.json();
84 |                 if (data && data.detail) {
85 |                     if (typeof data.detail === 'string') {
86 |                         msg = data.detail;
87 |                     } else if (Array.isArray(data.detail)) {
88 |                         msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
89 |                     } else {
90 |                         msg = JSON.stringify(data.detail);
91 |                     }
92 |                 }
93 |             } catch {}
94 |             throw new Error(`${response.status} ${msg}`);
95 |         }
96 |         return response.json();
97 |     },

```

```
98 |
99 | // User management API
100 | users: {
101 |     async list(params = {}) {
102 |         const q = new URLSearchParams(params).toString();
103 |         return window.api.fetch(`/api/admin/users${q ? '?' + q :
    ''}`);
104 |     },
105 |
106 |     async get(userId) {
107 |         return window.api.fetch(`/api/admin/users/${userId}`);
108 |     },
109 |
110 |     async create(data) {
111 |         return window.api.fetch('/api/admin/users', {
112 |             method: 'POST',
113 |             headers: { 'Content-Type': 'application/json' },
114 |             body: JSON.stringify(data)
115 |         });
116 |     },
117 |
118 |     async update(userId, data) {
119 |         return window.api.fetch(`/api/admin/users/${userId}`, {
120 |             method: 'PUT',
121 |             headers: { 'Content-Type': 'application/json' },
122 |             body: JSON.stringify(data)
123 |         });
124 |     },
125 |
126 |     async setPassword(userId, password) {
127 |         return window.api.fetch(`/api/admin/users/${userId}/pass
    word`, {
128 |             method: 'POST',
129 |             headers: { 'Content-Type': 'application/json' },
130 |             body: JSON.stringify({ password })
131 |         });
132 |     },
133 | }
```

```

134     async toggleActive(userId) {
135         return window.api.fetch(`/api/admin/users/${userId}/toggle-active`, {
136             method: 'POST'
137         });
138     },
139
140     async toggleRole(userId) {
141         return window.api.fetch(`/api/admin/users/${userId}/toggle-role`, {
142             method: 'POST'
143         });
144     },
145
146     async delete(userId) {
147         return window.api.fetch(`/api/admin/users/${userId}`, {
148             method: 'DELETE'
149         });
150     }
151 },
152
153 // Skills management API
154 skills: {
155     async list(params = {}) {
156         const q = new URLSearchParams(params).toString();
157         return window.api.fetch(`/api/admin/skills${q ? '?' : ''} + q
: ''}`);
158     },
159     async get(name) {
160         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`);
161     },
162     async create(data) {
163         return window.api.fetch('/api/admin/skills', {
164             method: 'POST',
165             headers: { 'Content-Type': 'application/json' },
166             body: JSON.stringify(data)
167         });

```

```
168     },
169     async update(name, data) {
170         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
171             mponent(name))}`, {
172             method: 'PUT',
173             headers: { 'Content-Type': 'application/json' },
174             body: JSON.stringify(data)
175         });
176     },
177     async delete(name) {
178         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
179             mponent(name))}`, {
180             method: 'DELETE'
181         });
182     },
183     async package(name) {
184         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
185             mponent(name))/package}`, {
186             method: 'POST'
187         });
188     }
189 },
190 // MCP Servers management API
191 mcp: {
192     async list() {
193         return window.api.fetch('/api/admin/mcp/servers');
194     },
195     async get(id) {
196         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(
197             id))}`);
198     },
199     async create(data) {
200         return window.api.fetch('/api/admin/mcp/servers', {
201             method: 'POST',
202             headers: { 'Content-Type': 'application/json' },
203             body: JSON.stringify(data)
204         });
205     }
206 }
```

```
202     },
203     async update(id, data) {
204         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
205             method: 'PUT',
206             headers: { 'Content-Type': 'application/json' },
207             body: JSON.stringify(data)
208         });
209     },
210     async delete(id) {
211         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
212             method: 'DELETE'
213         });
214     },
215     async toggle(id) {
216         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/toggle`, {
217             method: 'POST'
218         });
219     },
220     async test(id) {
221         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/test`, {
222             method: 'POST'
223         });
224     },
225     async testAdhoc(data) {
226         return window.api.fetch('/api/admin/mcp/test', {
227             method: 'POST',
228             headers: { 'Content-Type': 'application/json' },
229             body: JSON.stringify(data)
230         });
231     },
232     async getTools() {
233         return window.api.fetch('/api/admin/mcp/tools');
234     }
235 },
```



```
236 |
237 | // Google Workspace Accounts Pool API
238 | googleAccounts: {
239 |   async list() {
240 |     return window.api.fetch('/api/admin/google-accounts');
241 |   },
242 |   async get(name) {
243 |     return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}`);
244 |   },
245 |   async create(data) {
246 |     return window.api.fetch('/api/admin/google-accounts', {
247 |       method: 'POST',
248 |       headers: { 'Content-Type': 'application/json' },
249 |       body: JSON.stringify(data)
250 |     });
251 |   },
252 |   async upload(formData) {
253 |     const response = await fetch('/api/admin/google-accounts
/upload', {
254 |       method: 'POST',
255 |       body: formData
256 |     });
257 |     if (!response.ok) {
258 |       let msg = response.statusText;
259 |       try {
260 |         const data = await response.json();
261 |         if (data && data.detail) msg = data.detail;
262 |       } catch {}
263 |       throw new Error(`${response.status} ${msg}`);
264 |     }
265 |     return response.json();
266 |   },
267 |   async setDefault(name) {
268 |     return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/default`, {
269 |       method: 'POST'
270 |     });
```

```
271     },
272     async resetStatus(name) {
273         return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/reset-status`, {
274             method: 'POST'
275         });
276     },
277     async test(name) {
278         return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/test`, {
279             method: 'POST'
280         });
281     },
282     async delete(name) {
283         return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}`, {
284             method: 'DELETE'
285         });
286     }
287 }
288 };
289
290 // HELP content
291 // Password protection state
292 let isPasswordProtected = false;
293 let hasEnteredPassword = false;
294
295 // Tab Switch Handler
296 function onTabSwitched(targetId) {
297     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
: targetId;
298     if (cleanId === 'tab-chat') {
299         const msgInput = document.getElementById('message-input');
300         if (msgInput) msgInput.focus();
301     } else if (cleanId === 'tab-voice' && typeof window.initVoic
eTab === 'function') {
302         console.log('[AdminInterface] Switching to voice tab...');
303         window.initVoiceTab();
```

```
304 | } else if (cleanId === 'tab-tts' && typeof window.initTtsTab
    | === 'function') {
305 |     console.log('[AdminInterface] Switching to tts tab...');
306 |     window.initTtsTab();
307 | } else if (cleanId === 'tab-plugins' && typeof window.initPl
    | uuginsTab === 'function') {
308 |     console.log('[AdminInterface] Switching to plugins tab...'
    | );
309 |     window.initPluginsTab();
310 | } else if (cleanId === 'tab-admin' && typeof window.initAdmi
    | nTab === 'function') {
311 |     console.log('[AdminInterface] Switching to admin tab...');
312 |     window.initAdminTab();
313 | } else if (cleanId === 'tab-users' && typeof window.initUser
    | sTab === 'function') {
314 |     console.log('[AdminInterface] Switching to users tab...');
315 |     window.initUsersTab();
316 | } else if (cleanId === 'tab-windows-users' && typeof window.
    | initWindowsUsersTab === 'function') {
317 |     console.log('[AdminInterface] Switching to Windows users t
    | ab...');
318 |     window.initWindowsUsersTab();
319 | } else if (cleanId === 'tab-user-directories' && typeof wind
    | ow.initUserDirectoriesTab === 'function') {
320 |     console.log('[AdminInterface] Switching to user directorie
    | s tab...');
321 |     window.initUserDirectoriesTab();
322 | } else if (cleanId === 'tab-google-accounts' && typeof windo
    | w.initGoogleAccountsTab === 'function') {
323 |     console.log('[AdminInterface] Switching to google accounts
    | tab...');
324 |     window.initGoogleAccountsTab();
325 | } else if (cleanId === 'tab-instructions' && typeof window.i
    | nitInstructionsTab === 'function') {
326 |     console.log('[AdminInterface] Switching to instructions ta
    | b...');
327 |     window.initInstructionsTab();
328 | } else if (cleanId === 'tab-skills' && typeof window.initSki
```

```

    llsTab === 'function') {
329     console.log('[AdminInterface] Switching to skills tab...')
    ;
330     window.initSkillsTab();
331   } else if (cleanId === 'tab-mcp' && typeof window.initMcpTab
    === 'function') {
332     console.log('[AdminInterface] Switching to MCP tab...');
333     window.initMcpTab();
334   } else if (cleanId === 'tab-rag' && typeof window.initRagTab
    === 'function') {
335     console.log('[AdminInterface] Switching to RAG tab...');
336     window.initRagTab();
337   } else if (cleanId === 'tab-pixelrag' && typeof window.initP
    ixelRagTab === 'function') {
338     console.log('[AdminInterface] Switching to PixelRAG tab...
    ');
339     window.initPixelRagTab();
340   } else if ((cleanId === 'tab-telegram-rag' || cleanId === 't
    ab-telegram_rag') && typeof window.initTelegramRagTab === 'fun
    ction') {
341     console.log('[AdminInterface] Switching to Telegram RAG ta
    b...');
342     window.initTelegramRagTab();
343   } else if (cleanId === 'tab-search' && typeof window.initSea
    rchTab === 'function') {
344     console.log('[AdminInterface] Switching to search tab...')
    ;
345     window.initSearchTab();
346   } else if (cleanId === 'tab-models' && typeof window.initMod
    elsTab === 'function') {
347     console.log('[AdminInterface] Switching to models tab...')
    ;
348     window.initModelsTab();
349   } else if (cleanId === 'tab-agents' && typeof window.initAge
    ntsTab === 'function') {
350     console.log('[AdminInterface] Switching to agents tab...')
    ;
351     window.initAgentsTab();

```

```
352 | } else if (cleanId === 'tab-sources' && typeof window.initSo
    | urcesTab === 'function') {
353 |     console.log('[AdminInterface] Switching to sources tab...'
    | );
354 |     window.initSourcesTab();
355 | } else if (cleanId === 'tab-logs' && typeof window.initLogsT
    | ab === 'function') {
356 |     console.log('[AdminInterface] Switching to logs tab...');
357 |     window.initLogsTab();
358 | } else if (cleanId === 'tab-trading' && typeof window.initTr
    | adingTab === 'function') {
359 |     console.log('[AdminInterface] Switching to trading tab...'
    | );
360 |     window.initTradingTab();
361 | } else if (cleanId === 'tab-network' && typeof window.initNe
    | tworkTab === 'function') {
362 |     console.log('[AdminInterface] Switching to network tab...'
    | );
363 |     window.initNetworkTab();
364 | } else if ((cleanId === 'tab-hardware-load-inspector' || cle
    | anId === 'tab-system-load-inspector' || cleanId === 'tab-syste
    | m-inspector') && typeof window.initSystemInspectorTab === 'fun
    | ction') {
365 |     console.log('[AdminInterface] Switching to hardware load i
    | nspector tab...');
366 |     window.initSystemInspectorTab();
367 | } else if (cleanId === 'tab-processes-load-inspector' && typ
    | eof window.initProcessesLoadInspectorTab === 'function') {
368 |     console.log('[AdminInterface] Switching to processes load
    | inspector tab...');
369 |     window.initProcessesLoadInspectorTab();
370 | } else if (cleanId === 'tab-about-system' && typeof window.i
    | nitAboutSystemTab === 'function') {
371 |     console.log('[AdminInterface] Switching to about system ta
    | b...');
372 |     window.initAboutSystemTab();
373 | } else if (cleanId === 'tab-windows-admin' && typeof window.
    | initWindowsAdminTab === 'function') {
```

```
374 |     console.log('[AdminInterface] Switching to windows admin t
| ab...');
375 |     window.initWindowsAdminTab();
376 | } else if (cleanId === 'tab-focus-settings' && typeof window
| .initFocusSettingsTab === 'function') {
377 |     console.log('[AdminInterface] Switching to focus settings
| tab...');
378 |     window.initFocusSettingsTab();
379 | } else if (cleanId === 'tab-cloudflared' && typeof window.in
| itCloudflaredTab === 'function') {
380 |     console.log('[AdminInterface] Switching to cloudflared tab
| ...');
381 |     window.initCloudflaredTab();
382 | } else if (cleanId === 'tab-user-assistant' && typeof window
| .initUserAssistantTab === 'function') {
383 |     console.log('[AdminInterface] Switching to user assistant
| tab...');
384 |     window.initUserAssistantTab();
385 | } else if (cleanId === 'tab-gcloud' && typeof window.initGClo
| oudTab === 'function') {
386 |     console.log('[AdminInterface] Switching to gcloud tab...')
| ;
387 |     window.initGCloudTab();
388 | } else if (cleanId === 'tab-website-monitor' && typeof windo
| w.initWebsiteMonitorTab === 'function') {
389 |     console.log('[AdminInterface] Switching to website monitor
| tab...');
390 |     window.initWebsiteMonitorTab();
391 | } else if (cleanId === 'tab-system-control' && typeof window
| .initSystemControlTab === 'function') {
392 |     console.log('[AdminInterface] Switching to system control
| tab...');
393 |     window.initSystemControlTab();
394 | } else if (cleanId === 'tab-post-install-wizard' && typeof w
| indow.initPostInstallWizardTab === 'function') {
395 |     console.log('[AdminInterface] Switching to post-install wi
| zard tab...');
396 |     window.initPostInstallWizardTab();
```

```

397 | } else if (cleanId === 'tab-maintenance-recovery' && typeof
    | window.initMaintenanceRecoveryTab === 'function') {
398 |     console.log('[AdminInterface] Switching to maintenance rec
    | overly tab...');
399 |     window.initMaintenanceRecoveryTab();
400 | } else if (cleanId === 'tab-news' && typeof window.initNewsT
    | ab === 'function') {
401 |     console.log('[AdminInterface] Switching to news tab...');
402 |     window.initNewsTab();
403 | } else if (cleanId === 'tab-windows-backup' && typeof window
    | .initWindowsBackupTab === 'function') {
404 |     console.log('[AdminInterface] Switching to windows backup
    | tab...');
405 |     window.initWindowsBackupTab();
406 | } else if (cleanId === 'tab-help' && typeof window.initHelpT
    | ab === 'function') {
407 |     console.log('[AdminInterface] Switching to help tab...');
408 |     window.initHelpTab();
409 | }
410 | }
411 |
412 | // Programmatic tab switcher
413 | function switchTab(targetId) {
414 |     if (!targetId) return;
415 |     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
    | : targetId;
416 |     const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
    | cleanId}`;
417 |
418 |     // 1. Update active state on dropdown items & toggles
419 |     document.querySelectorAll('#mainTabs .dropdown-item').forEac
    | h((item) => {
420 |         const itemTarget = item.getAttribute('data-tab') || item.g
    | etAttribute('data-bs-target')?.replace('#', '');
421 |         if (itemTarget === tabId || itemTarget === cleanId) {
422 |             item.classList.add('active');
423 |         } else {
424 |             item.classList.remove('active');

```

```

425     }
426   });
427
428   document.querySelectorAll('#mainTabs .dropdown').forEach((dr
dropdown) => {
429     const toggle = dropdown.querySelector('.dropdown-toggle');
430     const hasActiveChild = dropdown.querySelector('.dropdown-i
tem.active');
431     if (toggle) {
432       if (hasActiveChild) {
433         toggle.classList.add('active');
434       } else {
435         toggle.classList.remove('active');
436       }
437     }
438   });
439
440   // 2. Switch tab-pane (scoped to top-level container to pres
erve nested subtabs)
441   const mainTabContent = document.getElementById('mainTabConte
nt');
442   if (mainTabContent) {
443     Array.from(mainTabContent.children).forEach((pane) => {
444       if (pane.classList.contains('tab-pane')) {
445         pane.classList.remove('show', 'active');
446       }
447     });
448   } else {
449     document.querySelectorAll('#mainTabContent > .tab-pane, bo
dy > .container-fluid > .tab-content > .tab-pane').forEach((pa
ne) => {
450       pane.classList.remove('show', 'active');
451     });
452   }
453   const targetPane = document.getElementById(tabId) || documen
t.getElementById(cleanId);
454   if (targetPane) {
455     targetPane.classList.add('show', 'active');

```



```

456 | }
457 |
458 | // 3. Notify lifecycle callback
459 | onTabSwitched(tabId);
460 | }
461 | window.switchTab = switchTab;
462 | window.switchToTab = switchTab;
463 |
464 | // Setup dropdowns navigation
465 | function setupDropdownTabs() {
466 |     const mainTabs = document.getElementById('mainTabs');
467 |     if (!mainTabs) return;
468 |
469 |     // 1. Dropdown Toggle Buttons
470 |     mainTabs.querySelectorAll('.dropdown-toggle').forEach((btn)
=> {
471 |         btn.onclick = (e) => {
472 |             e.preventDefault();
473 |             e.stopPropagation();
474 |
475 |             const dropdown = btn.closest('.dropdown');
476 |             const menu = dropdown?.querySelector('.dropdown-menu');
477 |             const isAlreadyOpen = menu?.classList.contains('show');
478 |
479 |             // Close all dropdowns
480 |             document.querySelectorAll('#mainTabs .dropdown-menu.show
').forEach((m) => {
481 |                 m.classList.remove('show');
482 |                 m.closest('.dropdown')?.querySelector('.dropdown-toggl
e')?.classList.remove('show');
483 |             });
484 |
485 |             // Toggle clicked dropdown
486 |             if (!isAlreadyOpen && menu) {
487 |                 menu.classList.add('show');
488 |                 btn.classList.add('show');
489 |             }
490 |         };

```

```

491 | });
492 |
493 | // 2. Dropdown Item Buttons (Tab Switchers & Action Buttons)
494 | mainTabs.querySelectorAll('.dropdown-item').forEach((item) =
    > {
495 |     item.onclick = (e) => {
496 |         const targetId = item.getAttribute('data-tab') || item.g
    etAttribute('data-bs-target')?.replace('#', '');
497 |         const pluginName = item.getAttribute('data-plugin');
498 |
499 |         // Close dropdown menu
500 |         item.closest('.dropdown-menu')?.classList.remove('show')
    ;
501 |         item.closest('.dropdown')?.querySelector('.dropdown-togg
    le')?.classList.remove('show');
502 |
503 |         if (pluginName && typeof window.openPluginFromDropdown =
    == 'function') {
504 |             e.preventDefault();
505 |             e.stopPropagation();
506 |             window.openPluginFromDropdown(pluginName);
507 |         } else if (targetId) {
508 |             e.preventDefault();
509 |             e.stopPropagation();
510 |             switchTab(targetId);
511 |         }
512 |     };
513 | });
514 |
515 | // 3. Document Click to Close Dropdowns
516 | if (!document.body.dataset.dropdownOutsideBound) {
517 |     document.body.dataset.dropdownOutsideBound = 'true';
518 |     document.addEventListener('click', (e) => {
519 |         if (!e.target.closest('#mainTabs .dropdown')) {
520 |             document.querySelectorAll('#mainTabs .dropdown-menu.sh
    ow').forEach((menu) => {
521 |                 menu.classList.remove('show');
522 |                 menu.closest('.dropdown')?.querySelector('.dropdown-

```

```

toggle')?.classList.remove('show');
523     });
524   }
525   });
526 }
527 }
528
529 async function startAdmin() {
530   console.log('Admin interface initializing...');
531   try {
532     initTheme();
533     setupDropdownTabs();
534     const el = document.getElementById('admin-interface');
535     if (el) el.style.display = 'block';
536
537     if (isPasswordProtected) {
538       showPasswordModal();
539     } else {
540       await initInterface();
541     }
542     console.log('Admin interface ready');
543   } catch (err) {
544     console.error('Error during startAdmin initialization:', e
rr);
545     const el = document.getElementById('admin-interface');
546     if (el) el.style.display = 'block';
547   }
548 }
549
550 if (document.readyState === 'loading') {
551   document.addEventListener('DOMContentLoaded', () => {
552     setupDropdownTabs();
553     startAdmin();
554   });
555 } else {
556   setupDropdownTabs();
557   startAdmin();
558 }

```

```
559 |
560 | async function initInterface() {
561 |   // Initialize theme
562 |   initTheme();
563 |
564 |   // Initialize i18n
565 |   await initI18n();
566 |
567 |   // Initialize User Settings & Google OAuth
568 |   await initUserSettings();
569 |
570 |
571 |   // Initialize HELP content
572 |   initHelpContent();
573 |
574 |   const cb = Date.now();
575 |
576 |   // Синхронизация видимости приложений (/apps) перед загрузко
    | й вкладок
577 |   let appsMap = {};
578 |   try {
579 |     const appsData = await (window.api ? window.api.fetch('/ap
    | i/v1/apps/status') : fetch('/api/v1/apps/status')).then(r => r.
    | json()));
580 |     if (appsData && appsData.apps) {
581 |       appsMap = appsData.apps;
582 |       window.appsStatusMap = appsData.apps;
583 |       syncAppsTabsVisibility(appsData.apps);
584 |     }
585 |   } catch (err) {
586 |     console.warn('Ошибка синхронизации видимости приложений (/
    | apps):', err);
587 |   }
588 |
589 |   // Загрузка базовых вкладок администратора
590 |   const coreTabLoads = [
591 |     loadTabContent('chat', `/html/chat/index.html?v=${cb}`, `/
    | html/chat/main.js?v=${cb}`),
```

```
592     loadTabContent('plugins', `/html/plugins_tab/index.html?v=
    ${cb}`, `/html/plugins_tab/main.js?v=${cb}`),
593     loadTabContent('admin', `/html/admin_tab/index.html?v=${cb}
    `, `/html/admin_tab/main.js?v=${cb}`),
594     loadTabContent('users', `/html/users_tab/index.html?v=${cb}
    `, `/html/users_tab/main.js?v=${cb}`),
595     loadTabContent('windows-users', `/html/windows_users_tab/i
    ndex.html?v=${cb}`, `/html/windows_users_tab/main.js?v=${cb}`)
    ,
596     loadTabContent('user-directories', `/html/user_directories
    _tab/index.html?v=${cb}`, `/html/user_directories_tab/main.js?
    v=${cb}`),
597     loadTabContent('google-accounts', `/html/google_accounts_t
    ab/index.html?v=${cb}`, `/html/google_accounts_tab/main.js?v=$
    {cb}`),
598     loadTabContent('instructions', `/html/instructions_tab/ind
    ex.html?v=${cb}`, `/html/instructions_tab/main.js?v=${cb}`),
599     loadTabContent('rag', `/html/rag_tab/index.html?v=${cb}`,
    `/html/rag_tab/main.js?v=${cb}`),
600     loadTabContent('pixelrag', `/html/pixelrag_tab/index.html?
    v=${cb}`, `/html/pixelrag_tab/main.js?v=${cb}`),
601     loadTabContent('telegram-rag', `/html/telegram_rag_tab/ind
    ex.html?v=${cb}`, `/html/telegram_rag_tab/main.js?v=${cb}`),
602     loadTabContent('voice', `/html/voice_tab/index.html?v=${cb}
    `, `/html/voice_tab/main.js?v=${cb}`),
603     loadTabContent('models', `/html/models_tab/index.html?v=${
    cb}`, `/html/models_tab/main.js?v=${cb}`),
604     loadTabContent('agents', `/html/agents_tab/index.html?v=${
    cb}`, `/html/agents_tab/main.js?v=${cb}`),
605     loadTabContent('search', `/html/search_tab/index.html?v=${
    cb}`, `/html/search_tab/main.js?v=${cb}`),
606     loadTabContent('tts', `/html/tts_tab/index.html?v=${cb}`,
    `/html/tts_tab/main.js?v=${cb}`),
607     loadTabContent('sources', `/html/sources_tab/index.html?v=
    ${cb}`, `/html/sources_tab/main.js?v=${cb}`),
608     loadTabContent('skills', `/html/skills_tab/index.html?v=${
    cb}`, `/html/skills_tab/main.js?v=${cb}`),
609     loadTabContent('mcp', `/html/mcp_tab/index.html?v=${cb}`,
```

```
`/html/mcp_tab/main.js?v=${cb}`),
610   loadTabContent('observability', `/html/system_inspector_ta
    b/index.html?v=${cb}`, `/html/system_inspector_tab/main.js?v=${
    cb}`),
611   loadTabContent('news', `/html/news_tab/index.html?v=${cb}`
    , `/html/news_tab/main.js?v=${cb}`),
612   loadTabContent('logs', `/html/logs/index.html?v=${cb}`, `/
    html/logs/main.js?v=${cb}`),
613   loadTabContent('help', `/html/help/index.html?v=${cb}`, `/
    html/help/main.js?v=${cb}`),
614 ];
615
616 // Определение и фильтрация вкладок микроприложений (/apps)
617 const appTabDefs = [
618   { id: 'about_system', tab: 'about-system', html: '/html/ab
    out_system_tab/index.html?v=20261006_v3', js: '/html/about_sys
    tem_tab/main.js?v=20261006_v3' },
619   { id: 'trading_terminal', tab: 'trading', html: '/html/tra
    ding_tab/index.html', js: '/html/trading_tab/main.js' },
620   { id: 'network_terminal', tab: 'network', html: '/html/net
    work_tab/index.html', js: '/html/network_tab/main.js' },
621   { id: 'system_inspector', tab: 'hardware-load-inspector',
    html: '/html/system_inspector_tab/index.html?v=20261006_v14',
    js: '/html/system_inspector_tab/main.js?v=20261006_v14' },
622   { id: 'processes_load_inspector', tab: 'processes-load-ins
    pector', html: '/html/processes_load_inspector_tab/index.html?
    v=20261008_v1', js: '/html/processes_load_inspector_tab/main.j
    s?v=20261008_v1' },
623   { id: 'chat', tab: 'chat', html: '/html/chat/index.html',
    js: '/html/chat/main.js' },
624   { id: 'scenarios', tab: 'scenarios', html: '/html/scenario
    s_tab/index.html', js: '/html/scenarios_tab/main.js' },
625   { id: 'user_assistant', tab: 'user-assistant', html: '/htm
    l/user_assistant_tab/index.html', js: '/html/user_assistant_ta
    b/main.js' },
626   { id: 'gcloud_monitor', tab: 'gcloud', html: '/html/gcloud
    _tab/index.html', js: '/html/gcloud_tab/main.js' },
627   { id: 'website_monitor', tab: 'website-monitor', html: '/h
```

```
tml/website_monitor_tab/index.html', js: '/html/website_monito
r_tab/main.js' },
628   { id: 'cloudflared_monitor', tab: 'cloudflared', html: '/h
tml/cloudflared_tab/index.html', js: '/html/cloudflared_tab/ma
in.js' },
629   { id: 'process_leaks', tab: 'process-leaks', html: '/html/
process_leaks_tab/index.html?v=20260924_v1', js: '/html/proces
s_leaks_tab/main.js?v=20260924_v1' },
630   { id: 'forensics', tab: 'forensics', html: '/html/forensic
s_tab/index.html?v=20260924_v1', js: '/html/forensics_tab/main
.js?v=20260924_v1' },
631   { id: 'throttling', tab: 'throttling', html: '/html/thrott
ling_tab/index.html?v=20260924_v1', js: '/html/throttling_tab/
main.js?v=20260924_v1' },
632   { id: 'storage_wear', tab: 'storage-wear', html: '/html/st
orage_wear_tab/index.html?v=20260924_v1', js: '/html/storage_w
ear_tab/main.js?v=20260924_v1' },
633   { id: 'peripherals', tab: 'peripherals', html: '/html/peri
peripherals_tab/index.html?v=20260924_v1', js: '/html/peripherals_
tab/main.js?v=20260924_v1' },
634 ];
635
636 const appTabLoads = appTabDefs
637   .filter(def => {
638     const appInfo = appsMap[def.id] || Object.values(appsMap)
639       .find(a => a.tab === `tab-${def.tab}` || a.id === def.id || a
640         .folder === def.id);
641     return appInfo ? appInfo.enabled : true;
642   })
643   .map(def => loadTabContent(def.tab, `${def.html}?v=${cb}`,
644     `${def.js}?v=${cb}`));
645
646 await Promise.all([...coreTabLoads, ...appTabLoads]);
647
648 // Синхронизация видимости вкладок плагинов
649 try {
650   const pluginsData = await window.api.fetch('/api/admin/plu
651     gins');
```

```
648     if (window.syncPluginTabsVisibility && pluginsData && plug
insData.plugins) {
649         window.syncPluginTabsVisibility(pluginsData.plugins);
650     }
651 } catch (err) {
652     console.error('Ошибка синхронизации видимости плагинов:',
err);
653 }
654
655 setupDropdownTabs();
656 }
657
658 function syncAppsTabsVisibility(appsMap) {
659     if (!appsMap || typeof appsMap !== 'object') return;
660     const dropdown = document.getElementById('appsTabsDropdown')
;
661     if (!dropdown) return;
662     const dropdownMenu = dropdown.nextElementSibling || dropdown
.closest('.dropdown')?.querySelector('.dropdown-menu');
663     if (!dropdownMenu) return;
664
665     const appButtons = dropdownMenu.querySelectorAll('button[dat
a-tab]');
666     let visibleCount = 0;
667
668     appButtons.forEach((btn) => {
669         const dataTab = btn.getAttribute('data-tab') || btn.getAtt
ribute('data-bs-target')?.replace('#', '');
670         const cleanId = dataTab ? dataTab.replace(/^tab-/ , '') : '
';
671         const appInfo = Object.values(appsMap).find(a => a.tab ===
dataTab || a.id === cleanId || a.folder === cleanId);
672         const isEnabled = appInfo ? appInfo.enabled : true;
673
674         if (isEnabled) {
675             btn.style.display = '';
676             btn.classList.remove('d-none');
677             visibleCount++;

```



```

678 |     } else {
679 |         btn.style.display = 'none';
680 |         btn.classList.add('d-none');
681 |         const pane = document.getElementById(dataTab);
682 |         if (pane) {
683 |             pane.style.display = 'none';
684 |             pane.classList.remove('show', 'active');
685 |         }
686 |         if (btn.classList.contains('active')) {
687 |             btn.classList.remove('active');
688 |             if (typeof window.switchTab === 'function') {
689 |                 window.switchTab('tab-chat');
690 |             }
691 |         }
692 |     }
693 | });
694 |
695 | const dropdownWrapper = dropdown.closest('li.nav-item');
696 | if (dropdownWrapper) {
697 |     const prevDivider = dropdownWrapper.previousElementSibling
698 |     ;
699 |     if (visibleCount === 0) {
700 |         dropdownWrapper.style.display = 'none';
701 |         dropdownWrapper.classList.add('d-none');
702 |         if (prevDivider && prevDivider.querySelector('.nav-group
703 | -divider')) {
704 |             prevDivider.style.display = 'none';
705 |             prevDivider.classList.add('d-none');
706 |         }
707 |     } else {
708 |         dropdownWrapper.style.display = '';
709 |         dropdownWrapper.classList.remove('d-none');
710 |         if (prevDivider && prevDivider.querySelector('.nav-group
711 | -divider')) {
712 |             prevDivider.style.display = '';
713 |             prevDivider.classList.remove('d-none');
714 |         }
715 |     }

```

```
713 | }
714 | }
715 | window.syncAppsTabsVisibility = syncAppsTabsVisibility;
716 |
717 | function showPasswordModal() {
718 |     const modal = new bootstrap.Modal(document.getElementById('p
passwordModal'));
719 |     modal.show();
720 |
721 |     const loginBtn = document.getElementById('login-btn');
722 |     const passwordInput = document.getElementById('admin-passwor
d');
723 |     const passwordError = document.getElementById('password-erro
r');
724 |
725 |     passwordInput?.addEventListener('keypress', (e) => {
726 |         if (e.key === 'Enter') verifyPassword();
727 |     });
728 |
729 |     loginBtn?.addEventListener('click', verifyPassword);
730 | }
731 |
732 | async function verifyPassword() {
733 |     const passwordInput = document.getElementById('admin-passwor
d');
734 |     const passwordError = document.getElementById('password-erro
r');
735 |     const password = passwordInput?.value;
736 |     if (!password) return;
737 |
738 |     const formData = new FormData();
739 |     formData.append('password', password);
740 |
741 |     try {
742 |         const res = await fetch('/admin', {
743 |             method: 'POST',
744 |             body: formData
745 |         });
```

```

746     if (res.ok) {
747         hasEnteredPassword = true;
748         passwordError?.classList.add('d-none');
749         const modalElement = document.getElementById('passwordMo
dal');
750         const modal = bootstrap.Modal.getInstance(modalElement);
751         modal?.hide();
752         document.getElementById('admin-interface').style.display
= 'block';
753         await initInterface();
754     } else {
755         throw new Error('Invalid password');
756     }
757 } catch {
758     passwordError?.classList.remove('d-none');
759     if (passwordError) {
760         passwordError.textContent = 'Неверный пароль';
761     }
762     if (passwordInput) {
763         passwordInput.value = '';
764         passwordInput.focus();
765     }
766 }
767 }
768
769 async function loadTabContent(tabName, url, jsOverrideSrc) {
770     try {
771         const response = await fetch(url);
772         if (!response.ok) throw new Error(`HTTP ${response.status}
`);
773         const html = await response.text();
774         const container = document.getElementById(`tab-${tabName}`
);
775         if (!container) return;
776         container.innerHTML = html;
777
778         // Load JS for the tab
779         const cacheBuster = Date.now();

```

```

780     const scriptSrc = jsOverrideSrc || `/html/${tabName}/main.
      js?v=${cacheBuster}`;
781
782     await new Promise((resolve) => {
783         const script = document.createElement('script');
784         if (tabName === 'instructions' || tabName === 'rag') {
785             script.type = 'module';
786         }
787         script.src = scriptSrc;
788         script.onload = () => {
789             console.log(`[Admin] Loaded JS for ${tabName}`);
790             resolve();
791         };
792         script.onerror = (err) => {
793             console.warn(`[Admin] Note: No JS loaded for ${tabName}
      } from ${scriptSrc}`);
794             resolve();
795         };
796         document.body.appendChild(script);
797     });
798
799     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
      .toUpperCase());
800     const initFuncName = 'init' + camelName.charAt(0).toUpperC
      ase() + camelName.slice(1) + 'Tab';
801     if (typeof window[initFuncName] === 'function') {
802         console.log(`[Admin] Calling ${initFuncName} for ${tabNa
      me}`);
803         try {
804             await window[initFuncName]();
805         } catch (err) {
806             console.error(`[Admin] Error executing ${initFuncName}
      :`, err);
807         }
808     }
809     applyTranslations();
810 } catch (e) {
811     console.error(`[Admin] Error loading tab ${tabName}:`, e);

```

```

812     const container = document.getElementById(`tab-${tabName}`
      );
813     if (container) {
814         container.innerHTML = `

□ Обзор проекта</h4><p>ai-breadboard – и
      нтегрированная среда для работы с AI, RAG и системным управлен
      ием.</p>`,
824         'google_oauth': `

#### □ Google OAuth и аккаунты</h4><p>Инт еграция с Gmail, Drive, Sheets и Docs через OAuth 2.0 и Servic e Accounts.</p>`, 825 'ai_models': `□ ИИ Провайдеры и Модели</h4><p>Облачны е (Gemini, OpenAI, Groq) и локальные (Ollama, Foundry, DirectM L) модели.</p>`, 826 'gdrive_sync': `□ Google Drive Sync</h4><p>Автоматичес кая синхронизация баз данных и файлов с Google Drive.</p>`, 827 'rag_knowledge': `□ База знаний и RAG</h4><p>Загрузка и поиск по вашим документам (PDF, Word, TXT, CSV, JSON).</p>` , 828 'rag': `□ RAG-индекс (Векторный поиск)</h4> <p><strong>1. База RAG:</strong> Индексирует системные базы зн аний, системные инструкции и документы.</p> 830 <p><strong>2. Загрузка документов:</strong> Позволяет загрузит ь внешние <code>.json</code>, <code>.txt</code>, <code>.md</co de>, <code>.pdf</code> файлы или сканировать директории напрям ую в RAG-индекс.</p> 831 <p><strong>3. Чат-RAG:</strong> Индексирует историю сохраненны х диалогов и ответов ассистента.</p>`, 832 'voice_tts': `□ Голос и Озвучка</h4><p>Голосовой ввод (Whisper/WebSpeech) и синтез речи (Edge TTS).</p>`,


```

```

833     'plugins_skills': `

#### ❑ Плагины, Навыки и MCP



Telegram-бот, IFTTT, распознавание счетов и расширение через MCP.

` ,
834     'storage_disks': `

#### ❑ Диски и Хранилище



Сканирование накопителей, проверка целостности и консолидация дублей.

` ,
835     'troubleshooting': `

#### ❑ Решение проблем (FAQ)



Ответы на частые вопросы и устранение ошибок подключения.

`
836   };
837   window.HELP_CONTENT = Object.assign(defaults, window.HELP_CONTENT);
838 }
839
840 function showHelpModal(key) {
841   const content = window.HELP_CONTENT[key] || '<p>Информация не найдена</p>';
842   document.getElementById('help-modal-content').innerHTML = content;
843   const modal = new bootstrap.Modal(document.getElementById('help-modal'));
844   modal.show();
845 }
846
847 function showChatLogicModal() {
848   const modalEl = document.getElementById('chat-logic-modal');
849   if (modalEl) {
850     const modal = new bootstrap.Modal(modalEl);
851     modal.show();
852   }
853 }
854 window.showChatLogicModal = showChatLogicModal;
855 window.showHelpModal = showHelpModal;
856
857 // Уведомления
858 function showNotification(message, type = 'info') {
859   const notification = document.createElement('div');
860   notification.className = `alert alert-${type} position-fixed top-0 end-0 m-3`;

```

```
861 | notification.style.zIndex = '9999';
862 | notification.style.maxWidth = '400px';
863 | notification.textContent = message;
864 | document.body.appendChild(notification);
865 |
866 | setTimeout(() => {
867 |     notification.remove();
868 | }, 5000);
869 | }
870 |
871 | async function initAdminTab() {
872 |     const modelSelect = document.getElementById('admin-model-select');
873 |     const saveBtn = document.getElementById('btn-admin-save-model');
874 |
875 |     if (!modelSelect || !saveBtn) return;
876 |
877 |     // 1. Очищаем селект
878 |     modelSelect.innerHTML = '';
879 |
880 |     let modelsGrouped = {};
881 |     // 2. Загружаем доступные модели
882 |     try {
883 |         const modelsData = await window.api.fetch('/api/chat/models');
884 |         modelsGrouped = modelsData.models || {};
885 |         if (Array.isArray(modelsGrouped)) {
886 |             modelsGrouped = { 'gemini': modelsGrouped };
887 |         }
888 |     } catch (err) {
889 |         console.error('Ошибка загрузки моделей:', err);
890 |         showNotification('Ошибка загрузки моделей AI: ' + err.message, 'danger');
891 |     }
892 |
893 |     // 3. Заполняем селект моделями с иерархией optgroup
894 |     const providerMeta = {
```

```
895 |     'gemini': { label: '☐ Google Gemini', order: 1 },
896 |     'agy': { label: '☐ Google Antigravity (AGY)', order: 2 },
897 |     'foundry': { label: '☐ Microsoft Foundry', order: 3 },
898 |     'ollama': { label: '☐ Ollama (Local)', order: 4 },
899 |     'onnx': { label: '☐ Microsoft ONNX / Olive', order: 5 }
900 | };
901 |
902 | const providers = Object.keys(modelsGrouped).sort((a, b) =>
    | {
903 |     const oA = providerMeta[a]?.order ?? 99;
904 |     const oB = providerMeta[b]?.order ?? 99;
905 |     return oA - oB;
906 | });
907 |
908 | let totalModels = 0;
909 | providers.forEach(p => {
910 |     const list = modelsGrouped[p] || [];
911 |     if (!Array.isArray(list) || list.length === 0) return;
912 |
913 |     const optgroup = document.createElement('optgroup');
914 |     optgroup.label = providerMeta[p]?.label || `☐ ${p.toUpper
    | Case()}`;
915 |
916 |     list.forEach(m => {
917 |         totalModels++;
918 |         const option = document.createElement('option');
919 |         option.value = m;
920 |         option.textContent = m;
921 |         optgroup.appendChild(option);
922 |     });
923 |
924 |     modelSelect.appendChild(optgroup);
925 | });
926 |
927 | if (totalModels === 0) {
928 |     const option = document.createElement('option');
929 |     option.value = '';
930 |     option.textContent = 'Нет доступных моделей';
```



```
931 |     modelSelect.appendChild(option);
932 |     saveBtn.disabled = true;
933 | } else {
934 |     saveBtn.disabled = false;
935 | }
936 |
937 | // 4. Загружаем текущие настройки пользователя (выбранную мо
    | дель)
938 | try {
939 |     const settingsData = await window.api.fetch('/auth/setting
    | s');
940 |     const savedModel = settingsData && settingsData.model ? se
    | ttingsData.model : '';
941 |     if (savedModel) {
942 |         modelSelect.value = savedModel;
943 |     }
944 |     if (typeof window.updateChatBadges === 'function') {
945 |         window.updateChatBadges(savedModel);
946 |     }
947 | } catch (err) {
948 |     console.error('Ошибка загрузки настроек AI пользователя:',
    | err);
949 | }
950 |
951 | // 5. Навешиваем обработчик сохранения
952 | saveBtn.onclick = async () => {
953 |     const selectedModel = modelSelect.value;
954 |     saveBtn.disabled = true;
955 |     const originalText = saveBtn.textContent;
956 |     saveBtn.textContent = 'Сохранение...';
957 |
958 |     try {
959 |         await window.api.fetch('/auth/settings', {
960 |             method: 'POST',
961 |             headers: { 'Content-Type': 'application/json' },
962 |             body: JSON.stringify({ model: selectedModel })
963 |         });
964 |         showNotification('Модель успешно обновлена на: ' + selec
```

```

    tedModel, 'success'));
965
966     // Обновляем бейджи модели в реальном времени на страниц
    e
967     window.activeModelName = selectedModel;
968     if (typeof window.updateChatBadges === 'function') {
969         window.updateChatBadges(selectedModel);
970     } else {
971         const badges = document.querySelectorAll('#chat-model-
    badge, #chat-popup-model-badge');
972         badges.forEach(badge => {
973             badge.textContent = selectedModel;
974             badge.style.display = 'inline-block';
975         });
976     }
977     } catch (err) {
978         console.error('Ошибка сохранения модели:', err);
979         showNotification('Ошибка сохранения: ' + err.message, 'd
    anger');
980     } finally {
981         saveBtn.disabled = false;
982         saveBtn.textContent = originalText;
983     }
984 };
985 }
986
987 window.initAdminTab = initAdminTab;

```

apps-sync.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Apps-Sync Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля apps-sync
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/modules/apps-sync
11  * c.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { syncAppsTabsVisibility, getAppsMap } from '/wi
15  * ndows/api/webgui/admin/modules/apps-sync.js';
16  *
17  * File: apps-sync.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/admin/modules
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-04 01:12:00
23  * =====
24  */
25
26 /**
27  * Apps Sync Module - Синхронизация видимости приложений
28  */
29
30 export async function syncApplicationsVisibility() {
31   console.log('[AppSync] Syncing application visibility...');
```

```

31 |   let appsMap = {};
32 |
33 |   try {
34 |     const appsData = await (window.api ? window.api.fetch('/api/v1/apps/status') : fetch('/api/v1/apps/status')).then(r => r.json());
35 |
36 |     if (appsData && appsData.apps) {
37 |       appsMap = appsData.apps;
38 |       window.appsStatusMap = appsData.apps;
39 |       syncAppsTabsVisibility(appsData.apps);
40 |     }
41 |   } catch (error) {
42 |     console.warn('[AppSync] Failed to sync apps visibility:', error);
43 |   }
44 |
45 |   return appsMap;
46 | }
47 |
48 | export function syncAppsTabsVisibility(appsMap) {
49 |   if (!appsMap || typeof appsMap !== 'object') return;
50 |
51 |   const dropdown = document.getElementById('appsTabsDropdown');
52 |   ;
53 |   if (!dropdown) return;
54 |
55 |   const dropdownMenu = dropdown.nextElementSibling ||
56 |     dropdown.closest('.dropdown')?.querySelector('.dropdown-menu');
57 |   if (!dropdownMenu) return;
58 |
59 |   const appButtons = dropdownMenu.querySelectorAll('button[data-tab]');
60 |
61 |   let visibleCount = 0;
62 |
63 |   appButtons.forEach((btn) => {
64 |     const dataTab = btn.getAttribute('data-tab') ||

```

```

63 |             btn.getAttribute('data-bs-target')?.replace
    | ('#', '');
64 |     const cleanId = dataTab ? dataTab.replace(/^tab-/, '') : '
    | ';
65 |     const appInfo = Object.values(appsMap).find(
66 |         a => a.tab === dataTab || a.id === cleanId || a.folder =
    | == cleanId
67 |     );
68 |     const isEnabled = appInfo ? appInfo.enabled : true;
69 |
70 |     if (isEnabled) {
71 |         btn.style.display = '';
72 |         btn.classList.remove('d-none');
73 |         visibleCount++;
74 |     } else {
75 |         btn.style.display = 'none';
76 |         btn.classList.add('d-none');
77 |
78 |         const pane = document.getElementById(dataTab);
79 |         if (pane) {
80 |             pane.style.display = 'none';
81 |             pane.classList.remove('show', 'active');
82 |         }
83 |
84 |         if (btn.classList.contains('active')) {
85 |             btn.classList.remove('active');
86 |             if (typeof window.switchTab === 'function') {
87 |                 window.switchTab('tab-chat');
88 |             }
89 |         }
90 |     }
91 | });
92 |
93 | // Hide/show apps dropdown if all apps are disabled
94 | const dropdownWrapper = dropdown.closest('li.nav-item');
95 | if (dropdownWrapper) {
96 |     const prevDivider = dropdownWrapper.previousElementSibling
    | ;

```

```
97 |
98 |     if (visibleCount === 0) {
99 |         dropdownWrapper.style.display = 'none';
100 |         if (prevDivider) prevDivider.style.display = 'none';
101 |     } else {
102 |         dropdownWrapper.style.display = '';
103 |         if (prevDivider) prevDivider.style.display = '';
104 |     }
105 | }
106 |
107 | console.log(`[AppSync] Synced apps visibility: ${visibleCount} visible`);
108 | }
109 |
110 | export function getAppsMap() {
111 |     return window.appsStatusMap || {};
112 | }
113 |
114 |
```

auth-handler.js

```
1  /**
2   * =====
3   * Process Name: Windows Modules - Auth-Handler Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля auth-hand
7   *   ler.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/modules/auth-han
11  *   dler.js?v=20261001_v1" type="module"></script>
12  *   JavaScript Import:
13  *     import { setupAuthHandlers } from '/windows/api/webgui/
14  *   admin/modules/auth-handler.js';
15  * File: auth-handler.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/admin/modules
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 /**
25  * Auth Handler Module - Обработка аутентификации и пароля
26  */
27
28 export function setupAuthHandlers() {
29   setupPasswordModal();
30   setupPasswordVerification();
```

```
31 | }
32 |
33 | function setupPasswordModal() {
34 |     const modal = document.getElementById('passwordModal');
35 |     const loginBtn = document.getElementById('login-btn');
36 |     const closeBtn = document.getElementById('closePasswordModal');
37 |     const adminPassword = document.getElementById('admin-password');
38 |
39 |     if (loginBtn) {
40 |         loginBtn.addEventListener('click', verifyPassword);
41 |     }
42 |
43 |     if (adminPassword) {
44 |         adminPassword.addEventListener('keypress', (e) => {
45 |             if (e.key === 'Enter') {
46 |                 verifyPassword();
47 |             }
48 |         });
49 |     }
50 |
51 |     if (closeBtn) {
52 |         closeBtn.addEventListener('click', () => {
53 |             if (modal) {
54 |                 const m = bootstrap.Modal.getInstance(modal);
55 |                 if (m) m.hide();
56 |             }
57 |         });
58 |     }
59 |
60 |     // Show modal on load if not authenticated
61 |     setTimeout(() => {
62 |         const isAuthenticated = localStorage.getItem('admin_authenticated');
63 |         if (!isAuthenticated) {
64 |             showPasswordModal();
65 |         }
```



```
66 | }, 500);
67 | }
68 |
69 | function setupPasswordVerification() {
70 |     // Проверка аутентификации при загрузке
71 |     const isAuthenticated = localStorage.getItem('admin_authentic
    cated');
72 |     const adminPassword = document.getElementById('admin-passwor
    d');
73 |
74 |     if (isAuthenticated && adminPassword) {
75 |         adminPassword.value = '';
76 |         localStorage.removeItem('admin_authenticated');
77 |     }
78 | }
79 |
80 | async function verifyPassword() {
81 |     const passwordInput = document.getElementById('admin-passwor
    d');
82 |     const passwordError = document.getElementById('password-erro
    r');
83 |     const modal = document.getElementById('passwordModal');
84 |
85 |     if (!passwordInput.value) {
86 |         if (passwordError) {
87 |             passwordError.classList.remove('d-none');
88 |             passwordError.textContent = 'Пароль не может быть пустым
    ';
89 |         }
90 |         return;
91 |     }
92 |
93 |     try {
94 |         const response = await fetch('/api/admin/verify-password',
            {
95 |             method: 'POST',
96 |             headers: { 'Content-Type': 'application/json' },
97 |             body: JSON.stringify({ password: passwordInput.value })
```

```

98     });
99
100    if (response.ok) {
101        // Пароль верен
102        if (modal) {
103            const m = bootstrap.Modal.getInstance(modal);
104            if (m) m.hide();
105        }
106        localStorage.setItem('admin_authenticated', 'true');
107        passwordInput.value = '';
108        if (passwordError) passwordError.classList.add('d-none')
109    ;
110    } else {
111        if (passwordError) {
112            passwordError.classList.remove('d-none');
113            passwordError.textContent = 'Неверный пароль';
114        }
115        passwordInput.value = '';
116        passwordInput.focus();
117    } catch (error) {
118        console.error('Password verification error:', error);
119        if (passwordError) {
120            passwordError.classList.remove('d-none');
121            passwordError.textContent = 'Ошибка проверки пароля';
122        }
123    }
124 }
125
126 function showPasswordModal() {
127     const modal = document.getElementById('passwordModal');
128     if (modal) {
129         const m = new bootstrap.Modal(modal, { backdrop: 'static',
130         keyboard: false });
131         m.show();
132         const passwordInput = document.getElementById('admin-passw
133         ord');
134         if (passwordInput) {

```

```
133 |         passwordInput.focus();
134 |     }
135 | }
136 | }
137 |
138 | export { verifyPassword, showPasswordModal };
139 |
```

init-interface.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Init-Interface Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля init-inte
7  *   rface.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/modules/init-int
11  *   erface.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { setupGlobalFunctions, setupLanguageSelector, s
15  *   etupThemeSelector } from '/windows/api/webgui/admin/modules/in
16  *   it-interface.js';
17  *
18  * File: init-interface.js
19  * Project: ai-breadboard
20  * Package: windows/api/webgui/admin/modules
21  * Author: hypo69
22  * Copyright: © 2026 hypo69
23  * Updated: 2026-10-06 22:15:00
24  * =====
25  */
26
27 /**
28  * Interface Initialization Module - Инициализация компонентов
29  *   интерфейса
30  */
31
32 import { initI18n, switchLang, applyTranslations } from '../..
```

```

    /js/i18n.js';
29 import { initTheme, setTheme, getThemeMode, getResolvedTheme }
    from '../..js/theme.js';
30 import { initUserSettings, refreshUserProfile } from '../..js
    /userSettings.js';
31
32 export async function initializeInterface() {
33     console.log('[AdminInterface] Initializing core components..
    .');
34
35     try {
36         // Initialize theme
37         initTheme();
38
39         // Initialize i18n
40         await initI18n();
41
42         // Initialize user settings
43         await initUserSettings();
44
45         // Apply translations
46         applyTranslations();
47
48         // Обновляем бейджи активной модели и поиска
49         if (typeof window.updateChatBadges === 'function') {
50             window.updateChatBadges();
51         }
52
53         console.log('[AdminInterface] Core components initialized'
    );
54     } catch (error) {
55         console.error('[AdminInterface] Initialization error:', er
    ror);
56         throw error;
57     }
58 }
59
60 export function setupGlobalFunctions() {

```

```

61 // Make functions available globally
62 window.switchLang = switchLang;
63 window.applyTranslations = applyTranslations;
64 window.setTheme = setTheme;
65 window.getThemeMode = getThemeMode;
66 window.getResolvedTheme = getResolvedTheme;
67 window.initUserSettings = initUserSettings;
68 window.refreshUserProfile = refreshUserProfile;
69
70 // Help and modals
71 window.showHelpModal = window.showHelpModal || (() => {});
72 window.showChatLogicModal = window.showChatLogicModal || (()
=> {});
73 window.showNotification = window.showNotification || (() =>
{}));
74
75 // API Fallback
76 window.api = window.api || {};
77 if (!window.api.fetch) {
78     window.api.fetch = async function(url, options = {}) {
79         const response = await fetch(url, options);
80         if (!response.ok) {
81             let msg = response.statusText;
82             try {
83                 const data = await response.json();
84                 if (data && data.detail) {
85                     if (typeof data.detail === 'string') msg = data.de
tail;
86                     else if (Array.isArray(data.detail)) msg = data.de
tail.map(d => d.msg || JSON.stringify(d)).join(', ');
87                     else msg = JSON.stringify(data.detail);
88                 } else if (data && data.message) {
89                     msg = data.message;
90                 }
91             } catch {}
92             throw new Error(`${response.status} ${msg}`);
93         }
94         return response.json();

```

```
95     };
96   }
97
98   console.log('[AdminInterface] Global functions registered');
99 }
100
101 export function setupLanguageSelector() {
102   const langSelectors = document.querySelectorAll('.lang-selector');
103   const savedLocale = localStorage.getItem('app_locale') || localStorage.getItem('app_language') || localStorage.getItem('language') || 'ru-RU';
104
105   langSelectors.forEach((selector) => {
106     const optLocale = Array.from(selector.options).find(opt => opt.value.toLowerCase() === savedLocale.toLowerCase());
107     const optLang = Array.from(selector.options).find(opt => opt.value.toLowerCase() === savedLocale.split('-')[0].toLowerCase());
108     if (optLocale) selector.value = optLocale.value;
109     else if (optLang) selector.value = optLang.value;
110
111     selector.addEventListener('change', (e) => {
112       switchLang(e.target.value);
113     });
114   });
115
116   console.log('[AdminInterface] Language selector setup');
117 }
118
119 export function setupThemeSelector() {
120   const themeSelectors = document.querySelectorAll('.theme-selector');
121   const savedTheme = getThemeMode();
122
123   themeSelectors.forEach((selector) => {
124     selector.value = savedTheme;
125     selector.addEventListener('change', (e) => {
```

```
126 |         setTheme(e.target.value);
127 |     });
128 | });
129 |
130 | console.log('[AdminInterface] Theme selector setup');
131 | }
132 |
133 | export { initI18n, switchLang, applyTranslations };
134 | export { initTheme, setTheme, getThemeMode, getResolvedTheme }
    | ;
135 | export { initUserSettings, refreshUserProfile };
136 |
```


tab-manager.js

```
1  /**
2   * =====
3   * Process Name: Windows Modules - Tab-Manager Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля tab-manag
7   *   er.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/modules/tab-mana
11  *   ger.js?v=20261001_v1" type="module"></script>
12  *   JavaScript Import:
13  *     import { setupTabManagement, onTabSwitched } from '/win
14  *   dows/api/webgui/admin/modules/tab-manager.js';
15  * File: tab-manager.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/admin/modules
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 /**
25  * Tab Manager Module - Управление вкладками и навигацией
26  */
27
28 export function setupTabManagement() {
29   setupDropdownTabs();
30   setupTabEvents();
```

```

31 }
32
33 function setupDropdownTabs() {
34     const mainTabs = document.getElementById('mainTabs');
35     if (!mainTabs) return;
36
37     const dropdowns = mainTabs.querySelectorAll('.dropdown-toggle');
38
39     dropdowns.forEach((dropdown) => {
40         dropdown.addEventListener('click', (e) => {
41             e.preventDefault();
42             e.stopPropagation();
43
44             const parentDropdown = dropdown.closest('.dropdown');
45             const menu = dropdown.nextElementSibling || parentDropdown?.querySelector('.dropdown-menu');
46             const isAlreadyOpen = menu?.classList.contains('show');
47
48             // Close other dropdowns
49             document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((otherMenu) => {
50                 otherMenu.classList.remove('show');
51                 otherMenu.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
52             });
53
54             // Toggle current dropdown
55             if (!isAlreadyOpen && menu) {
56                 dropdown.classList.add('show');
57                 menu.classList.add('show');
58             }
59         });
60     });
61
62     // Handle all clickable navigation items inside mainTabs (tabs and dropdown items)
63     mainTabs.querySelectorAll('.dropdown-item, .list-group-item,

```

```

    [data-tab], [data-bs-target], [data-plugin]').forEach((item)
=> {
64     item.addEventListener('click', (e) => {
65         const pluginName = item.getAttribute('data-plugin');
66         const targetId = item.getAttribute('data-tab') ||
67             item.getAttribute('data-bs-target')?.rep
lace('#', '');
68
69         // Close parent dropdown menu
70         const menu = item.closest('.dropdown-menu');
71         if (menu) {
72             menu.classList.remove('show');
73             menu.closest('.dropdown')?.querySelector('.dropdown-to
ggle')?.classList.remove('show');
74         }
75
76         if (pluginName && typeof window.openPluginFromDropdown =
== 'function') {
77             e.preventDefault();
78             e.stopPropagation();
79             window.openPluginFromDropdown(pluginName);
80         } else if (targetId) {
81             e.preventDefault();
82             e.stopPropagation();
83             if (typeof window.switchTab === 'function') {
84                 window.switchTab(targetId);
85             }
86         }
87     });
88 });
89
90 // Close dropdowns on outside click
91 document.addEventListener('click', (e) => {
92     if (!mainTabs.contains(e.target)) {
93         dropdowns.forEach((dropdown) => {
94             dropdown.classList.remove('show');
95             const menu = dropdown.nextElementSibling;
96             if (menu) menu.classList.remove('show');

```

```
97 |     });
98 |   }
99 | });
100 | }
101 |
102 | function setupTabEvents() {
103 |   // Tab content click handlers
104 |   const tabPane = document.getElementById('mainTabsContent');
105 |   if (tabPane) {
106 |     tabPane.addEventListener('click', (e) => {
107 |       const target = e.target.closest('[data-tab]');
108 |       if (target) {
109 |         const tabId = target.getAttribute('data-tab');
110 |         if (tabId) {
111 |           window.switchTab(tabId);
112 |         }
113 |       }
114 |     });
115 |   }
116 | }
117 |
118 | export function onTabSwitched(targetId) {
119 |   const cleanId = targetId.startsWith('#') ? targetId.slice(1)
120 |     : targetId;
121 |   if (cleanId === 'tab-chat') {
122 |     if (typeof window.initChatTab === 'function') {
123 |       window.initChatTab();
124 |     }
125 |   }
126 | }
127 |
128 | export { setupDropdownTabs, setupTabEvents };
129 |
```

ui-handler.js

```
1  /**
2   * =====
3   * Process Name: Windows Modules - Ui-Handler Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля ui-handle
7   *   r.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/modules/ui-handl
11  *     er.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { setupUIHandlers, showHelpModal, showChatLogicM
15  *     odal } from '/windows/api/webgui/admin/modules/ui-handler.js';
16  *
17  * File: ui-handler.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/admin/modules
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-01 13:04:40
23  * =====
24  */
25
26  /**
27   * UI Handler Module - Работа с модалями, уведомлениями и помо
28   * щью
29   */
30
31  let helpContent = {};
```

```

30 export function setupUIHandlers() {
31   initHelpContent();
32   setupModalHandlers();
33 }
34
35 function initHelpContent() {
36   helpContent = {
37     'overview': `

#### □ Обзор проекта

<p>ai-breadboard – и
интегрированная среда для работы с AI, RAG и системным управлен
ием.</p>`,
38     'google_oauth': `

#### □ Google OAuth и аккаунты

<p>Инте
грация с Gmail, Drive, Sheets и Docs через OAuth 2.0 и Servic
e Accounts.</p>`,
39     'ai_models': `

#### □ ИИ Провайдеры и Модели

<p>Облачны
е (Gemini, OpenAI, Groq) и локальные (Ollama, Foundry, DirectM
L) модели.</p>`,
40     'gdrive_sync': `

#### □ Google Drive Sync

<p>Автоматичес
кая синхронизация баз данных и файлов с Google Drive.</p>`,
41     'rag_knowledge': `

#### □ База знаний и RAG

<p>Загрузка
и поиск по вашим документам (PDF, Word, TXT, CSV, JSON).</p>`
,
42     'rag': `

#### □ RAG-индекс (Векторный поиск)


43 <p><strong>1. База RAG:</strong> Индексирует системные базы зн
аний, системные инструкции и документы.</p>
44 <p><strong>2. Загрузка документов:</strong> Позволяет загрузит
ь внешние <code>.json</code>, <code>.txt</code>, <code>.md</co
de>, <code>.pdf</code> файлы или сканировать директории напрям
ую в RAG-индекс.</p>
45 <p><strong>3. Чат-RAG:</strong> Индексирует историю сохраненны
х диалогов и ответов ассистента.</p>`,
46     'voice_tts': `

#### □ Голос и Озвучка

<p>Голосовой ввод
(Whisper/WebSpeech) и синтез речи (Edge TTS).</p>`,
47     'plugins_skills': `

#### □ Плагины, Навыки и MCP

<p>Tel
egram-бот, IFTTT, распознавание счетов и расширение через MCP.
</p>`,
48     'storage_disks': `

#### □ Диски и Хранилище

<p>Сканиров
ание накопителей, проверка целостности и консолидация дублей.<
/p>`,

```

```
49 |     'troubleshooting': `<h4>Решение проблем (FAQ)</h4><p>Отв  
    | еты на частые вопросы и устранение ошибок подключения.</p>`  
50 | };  
51 |  
52 | window.HELP_CONTENT = helpContent;  
53 | }  
54 |  
55 | function setupModalHandlers() {  
56 |     // Help modal  
57 |     const helpBtn = document.getElementById('help-btn');  
58 |     if (helpBtn) {  
59 |         helpBtn.addEventListener('click', () => {  
60 |             showHelpModal('overview');  
61 |         });  
62 |     }  
63 |  
64 |     // Chat logic modal  
65 |     const chatLogicBtn = document.getElementById('chat-logic-btn  
    | ');  
66 |     if (chatLogicBtn) {  
67 |         chatLogicBtn.addEventListener('click', () => {  
68 |             showChatLogicModal();  
69 |         });  
70 |     }  
71 | }  
72 |  
73 | export function showHelpModal(key) {  
74 |     const content = helpContent[key] || '<p>Информация не найден  
    | а</p>';  
75 |     const contentDiv = document.getElementById('help-modal-conte  
    | nt');  
76 |     if (contentDiv) {  
77 |         contentDiv.innerHTML = content;  
78 |     }  
79 |  
80 |     const modal = document.getElementById('help-modal');  
81 |     if (modal) {  
82 |         const m = new bootstrap.Modal(modal);
```

```
83 |     m.show();
84 | }
85 |
86 |
87 | export function showChatLogicModal() {
88 |     const modalEl = document.getElementById('chat-logic-modal');
89 |     if (modalEl) {
90 |         const modal = new bootstrap.Modal(modalEl);
91 |         modal.show();
92 |     }
93 | }
94 |
95 | export function showNotification(message, type = 'info') {
96 |     const notification = document.createElement('div');
97 |     notification.className = `alert alert-${type} position-fixed
    | top-0 end-0 m-3`;
98 |     notification.style.zIndex = '9999';
99 |     notification.style.maxWidth = '400px';
100 |     notification.textContent = message;
101 |     document.body.appendChild(notification);
102 |
103 |     setTimeout(() => {
104 |         notification.remove();
105 |     }, 5000);
106 | }
107 |
108 | export { initHelpContent, setupModalHandlers };
109 |
```


optimization-init.js

```
1 /**
2  * =====
3  * Process Name: Windows Admin - Optimization-Init Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля optimization-init.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/optimization-init.js?v=20261001_v1" type="module"></script>
11  *
12  *   JavaScript Import:
13  *     import { enhanceTabSwitching, createSearchDebounce, createAutoSaveDebounce } from '/windows/api/webgui/admin/optimization-init.js';
14  *
15  * File: optimization-init.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/admin
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 /**
25  * Optimization Init - Интеграция систем оптимизации в админку
26  *
27  * Этот скрипт инициализирует:
28  * - Ленивую загрузку вкладок
29  * - Кеширование API
```

```

30 * - Дебаунсинг запросов
31 * - Сохранение состояния вкладок
32 */
33
34 import LazyTabLoader from '../js/tab-loader.js';
35 import { APICache, APIFetcher } from '../js/api-cache.js';
36 import { RequestManager } from '../js/request-manager.js';
37
38 // =====
39 // 1. ИНИЦИАЛИЗАЦИЯ СИСТЕМ ОПТИМИЗАЦИИ
40 // =====
41
42 const tabLoader = new LazyTabLoader();
43 const apiCache = new APICache();
44 const apiFetcher = new APIFetcher(apiCache);
45 const requestManager = new RequestManager();
46
47 // Экспортируем в глобальный скоп для доступа из других модуле
48 // й
49 window.optimizationModule = {
50   tabLoader,
51   apiCache,
52   apiFetcher,
53   requestManager,
54   stats: {
55     tabsLoaded: new Set(),
56     apiCallsUsed: 0,
57     cachedCallsUsed: 0,
58     requestsDebounced: 0
59   }
60 };
61 console.log('[Optimization] System initialized');
62
63 // =====

```

```

64 // 2. ПЕРЕХВАТ API.FETCH ОБЕРТКА
65 // =====
66
67 // Сохраняем оригинальный fetch
68 const originalApiFetch = window.api?.fetch;
69
70 // Переопределяем window.api.fetch для использования кеша
71 if (window.api && originalApiFetch) {
72     window.api.fetch = async function(url, options = {}) {
73         const method = (options.method || 'GET').toUpperCase();
74         const isGetRequest = method === 'GET' || method === 'HEAD'
75         ;
76
77         // Для GET запросов проверяем кеш
78         if (isGetRequest) {
79             const cached = apiCache.get(url);
80             if (cached) {
81                 window.optimizationModule.stats.cachedCallsUsed++;
82                 console.log(`[Optimization] API call from cache: ${method} ${url}`);
83                 return cached;
84             }
85
86             // Делаем реальный запрос
87             window.optimizationModule.stats.apiCallsUsed++;
88             const result = await originalApiFetch.call(this, url, options);
89
90             // Кешируем для GET запросов (5 минут TTL)
91             if (isGetRequest) {
92                 apiCache.set(url, result, 5 * 60 * 1000); // 5 минут
93             } else {
94                 // Для POST/PUT/DELETE инвалидируем релевантный кеш
95                 const basePath = url.split('?')[0];
96
97                 // Умная инвалидация - очищаем кеш связанных GET запросов

```

В

```
98     if (method === 'POST' || method === 'PUT' || method ===
| 'DELETE') {
99         // Инвалидируем основной путь и все его вариации
100         const pattern = new RegExp(`${basePath.replace(/\/$/,
| '')}.*`);
101         const invalidatedCount = apiCache.invalidate(pattern);
102         if (invalidatedCount > 0) {
103             console.log(`[Optimization] Invalidated ${invalidate
| dCount} cache entries for ${basePath}`);
104         }
105
106         // Для пользователей - инвалидируем список пользовател
| ей
107         if (basePath.includes('/users/')) {
108             apiCache.invalidate(/\/api\/admin\/users/);
109             apiCache.invalidate(/\/api\/users/);
110         }
111
112         // Для конфигурации - инвалидируем конфиги
113         if (basePath.includes('/config/')) {
114             apiCache.invalidate(/\/api\/config/);
115             apiCache.invalidate(/\/api\/admin\/config/);
116         }
117
118         // Для плагинов - инвалидируем список плагинов
119         if (basePath.includes('/plugins')) {
120             apiCache.invalidate(/\/api\/plugins/);
121             apiCache.invalidate(/\/api\/admin\/plugins/);
122         }
123
124         // Для модели - инвалидируем список моделей
125         if (basePath.includes('/models') || basePath.includes(
| '/chat')) {
126             apiCache.invalidate(/\/api\/chat\/models/);
127             apiCache.invalidate(/\/api\/models/);
128         }
129     }
```

```

130     }
131
132     return result;
133 };
134 }
135
136 // =====
137 // 3. ПЕРЕДЕЛКА ИНИЦИАЛИЗАЦИИ ВКЛАДOK (LAZY LOADING)
138 // =====
139
140 export async function initLazyTabLoading(tabDefinitions) {
141     console.log('[Optimization] Initializing lazy tab loading (o
n-demand mode)...');
142
143     // Регистрируем все вкладки в системе ленивой загрузки
144     tabDefinitions.forEach(({ tabName, htmlUrl, jsUrl }) => {
145         tabLoader.registerTab(tabName, htmlUrl, jsUrl);
146     });
147
148     // Перехватываем loadTab чтобы применять автопатч
149     const originalLoadTab = tabLoader.loadTab.bind(tabLoader);
150     tabLoader.loadTab = async function(tabName) {
151         const result = await originalLoadTab(tabName);
152
153         // Применяем дебаунс автопатч к загруженной вкладке
154         if (result && window.TabDebounceAutoPatch) {
155             setTimeout(() => {
156                 try {
157                     window.TabDebounceAutoPatch.autoPatchTab(tabName, {
158                         searchEnabled: true,
159                         autoSaveEnabled: true,
160                         refreshEnabled: true
161                     });
162                 } catch (e) {
163                     console.warn(`[Optimization] Failed to apply debounc
e patch to ${tabName}:`, e);

```

```

164         }
165         }, 100);
166     }
167
168     return result;
169 };
170
171 const allowedTabNames = tabDefinitions.map(def => def.tabName);
172 console.log(`[Optimization] Registered ${allowedTabNames.length} tabs for on-demand loading.`);
173
174 return { registeredTabs: allowedTabNames };
175 }
176
177 // =====
178 // 4. ИНТЕГРАЦИЯ С ПЕРЕКЛЮЧАТЕЛЕМ ВКЛАДОК
179 // =====
180
181 export function enhanceTabSwitching(originalSwitchTab) {
182     return async function switchTabOptimized(targetId) {
183         if (!targetId) return;
184
185         const cleanId = targetId.startsWith('#') ? targetId.slice(
186             1) : targetId;
187         const tabName = cleanId.replace(/^tab-/, '');
188
189         console.log(`[Optimization] Switching to tab: ${tabName}`)
190         ;
191
192         // Если вкладка еще не загружена - загружаем ее по требованию
193         if (!tabLoader.isLoaded(tabName)) {
194             console.log(`[Optimization] Tab not loaded yet, loading on-demand: ${tabName}`);
195             await tabLoader.loadTab(tabName);
196         }
197     };
198 }

```

```

194     window.optimizationModule.stats.tabsLoaded.add(tabName);
195 }
196
197 // Вызываем оригинальную функцию переключения
198 if (typeof originalSwitchTab === 'function') {
199     originalSwitchTab.call(this, targetId);
200 }
201
202 // Сохраняем состояние вкладки в localStorage
203 localStorage.setItem('admin:lastActiveTab', cleanId);
204 };
205 }
206
207 // =====
208 // 5. ВОССТАНОВЛЕНИЕ ПОСЛЕДНЕЙ ВКЛАДКИ
209 // =====
210
211 export async function restoreLastTab(fallbackTab = 'tab-about-
system') {
212     // Проверяем хэш в URL
213     const hash = location.hash.replace('#', '');
214     let initialTab = hash || null;
215
216     // Если в хэше ничего нет, проверяем persistence
217     if (!initialTab) {
218         if (window.tabPersistence && typeof window.tabPersistence.
getLastActiveTab === 'function') {
219             initialTab = window.tabPersistence.getLastActiveTab();
220         } else {
221             initialTab = localStorage.getItem('admin:lastActiveTab')
222         };
223     }
224
225     const targetTabId = (initialTab && document.getElementById(i
nitialTab.startsWith('tab-') ? initialTab : `tab-${initialTab}`

```

```

`))
226     ? (initialTab.startsWith('tab-') ? initialTab : `tab-${ini
    tialTab}`)
227     : fallbackTab;
228
229     const tabName = targetTabId.replace(/^tab-/, '');
230     console.log(`[Optimization] Loading strictly initial active
    tab: ${tabName}`);
231
232     if (!tabLoader.isLoaded(tabName)) {
233         await tabLoader.loadTab(tabName);
234         window.optimizationModule.stats.tabsLoaded.add(tabName);
235     }
236
237     if (typeof window.switchTab === 'function') {
238         window.switchTab(targetTabId);
239     }
240     return targetTabId;
241 }
242
243 // =====
    =====
244 // 6. УТИЛИТЫ ДЛЯ ДЕБАУНСИНГА И БАТЧИНГА
245 // =====
    =====
246
247 export function createSearchDebounce(searchFn, delay = 500) {
248     return (query) => {
249         window.optimizationModule.stats.requestsDebounced++;
250         return requestManager.debounce('search-query', () => searc
    hFn(query), delay);
251     };
252 }
253
254 export function createAutoSaveDebounce(saveFn, delay = 1000) {
255     return (data) => {
256         window.optimizationModule.stats.requestsDebounced++;
257         return requestManager.debounce('autosave-data', () => save

```



```

    Fn(data), delay);
258     };
259 }
260
261 // =====
    =====
262 // 7. ФУНКЦИИ ОТЛАДКИ И СТАТИСТИКИ
263 // =====
    =====
264
265 export function getOptimizationStats() {
266     return {
267         ...window.optimizationModule.stats,
268         cacheStats: apiCache.getStats(),
269         requestManagerStats: requestManager.getStats(),
270         tabsPreloaded: Array.from(window.optimizationModule.stats.
    tabsLoaded)
271     };
272 }
273
274 export function printOptimizationStats() {
275     const stats = getOptimizationStats();
276     console.group('□ Optimization Statistics');
277     console.log('API Calls:', stats.apiCallsUsed);
278     console.log('Cached Calls:', stats.cachedCallsUsed);
279     console.log('Cache Hit Rate:', (stats.cachedCallsUsed / (sta
    ts.apiCallsUsed + stats.cachedCallsUsed) * 100).toFixed(1) + '
    %');
280     console.log('Requests Debounced:', stats.requestsDebounced);
281     console.log('Tabs Preloaded:', stats.tabsPreloaded);
282     console.log('Cache Entries:', stats.cacheStats.size);
283     console.groupEnd();
284 }
285
286 // Делаем доступной из консоли для отладки
287 window.printOptimizationStats = printOptimizationStats;
288 window.getOptimizationStats = getOptimizationStats;
289

```

```
290 // Выводим статистику каждые 30 секунд (опционально, для отлад
    ки)
291 // Раскомментировать в production если нужна мониторинг
292 // setInterval(() => {
293 //     console.clear();
294 //     printOptimizationStats();
295 // }, 30000);
296
297 export default {
298     initLazyTabLoading,
299     enhanceTabSwitching,
300     restoreLastTab,
301     createSearchDebounce,
302     createAutoSaveDebounce,
303     getOptimizationStats,
304     printOptimizationStats,
305     tabLoader,
306     apiCache,
307     apiFetcher,
308     requestManager
309 };
310
```

sync-api.js

```
1 /**
2  * =====
3  * Process Name: Windows Admin - Sync-API Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля sync-api.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/sync-api.js?v=20
11  * 261001_v1" type="module"></script>
12  *
13  * File: sync-api.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/admin
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * Google Drive Sync API Client
23  * Клиент для работы с REST API синхронизации
24  */
25
26 class SyncApiClient {
27   constructor(baseUrl = '/api/admin/sync') {
28     this.baseUrl = baseUrl;
29     this.timeout = 10000;
30   }
31
32   /**
```

```

33 |     * Выполнить GET запрос
34 |     */
35 |     async get(endpoint) {
36 |         try {
37 |             const response = await fetch(`${this.baseUrl}${endpoint}
    | `, {
38 |                 method: 'GET',
39 |                 headers: { 'Content-Type': 'application/json' }
40 |             });
41 |
42 |             if (!response.ok) {
43 |                 throw new Error(`API Error: ${response.status}`);
44 |             }
45 |
46 |             return await response.json();
47 |         } catch (error) {
48 |             console.error(`[SyncApiClient] GET ${endpoint}:`, error)
    | ;
49 |             throw error;
50 |         }
51 |     }
52 |
53 |     /**
54 |     * Выполнить POST запрос
55 |     */
56 |     async post(endpoint, data = {}) {
57 |         try {
58 |             const response = await fetch(`${this.baseUrl}${endpoint}
    | `, {
59 |                 method: 'POST',
60 |                 headers: { 'Content-Type': 'application/json' },
61 |                 body: JSON.stringify(data)
62 |             });
63 |
64 |             if (!response.ok) {
65 |                 throw new Error(`API Error: ${response.status}`);
66 |             }
67 |

```

```
68 |         return await response.json();
69 |     } catch (error) {
70 |         console.error(`[SyncApiClient] POST ${endpoint}:`, error
    | );
71 |         throw error;
72 |     }
73 | }
74 |
75 | /**
76 |  * Получить статус синхронизации
77 |  */
78 | async getStatus() {
79 |     return this.get('/status');
80 | }
81 |
82 | /**
83 |  * Получить статистику
84 |  */
85 | async getStats() {
86 |     return this.get('/stats');
87 | }
88 |
89 | /**
90 |  * Получить информацию о Google Drive
91 |  */
92 | async getDriveInfo() {
93 |     return this.get('/drive-info');
94 | }
95 |
96 | /**
97 |  * Получить конфигурацию
98 |  */
99 | async getConfig() {
100 |     return this.get('/config');
101 | }
102 |
103 | /**
104 |  * Запустить планировщик
```

```
105 |     */
106 |     async startScheduler(interval = 6) {
107 |         return this.post('/start', { sync_interval_hours: interval
    |     });
108 |     }
109 |
110 |     /**
111 |      * Остановить планировщик
112 |      */
113 |     async stopScheduler() {
114 |         return this.post('/stop');
115 |     }
116 |
117 |     /**
118 |      * Синхронизировать
119 |      */
120 |     async syncNow(syncType = 'all') {
121 |         return this.post('/sync-now', { sync_type: syncType });
122 |     }
123 |
124 |     /**
125 |      * Проверить подключение
126 |      */
127 |     async testConnection() {
128 |         return this.post('/test-connection');
129 |     }
130 |
131 |     /**
132 |      * Обновить конфигурацию
133 |      */
134 |     async updateConfig(config) {
135 |         return this.post('/config', config);
136 |     }
137 | }
138 |
```

sync-tab.js

```
1  /**
2   * =====
3   * Process Name: Windows Admin - Sync-Tab Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля sync-tab.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin/sync-tab.js?v=20
11  *     261001_v1" type="module"></script>
12  *
13  * File: sync-tab.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/admin
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 02:00:00
19  * =====
20  */
21 /**
22  * Google Drive Sync Management Tab
23  * Управление синхронизацией Google Drive через админ-панель
24  */
25
26 class SyncTab {
27   constructor() {
28     this.initialized = false;
29     this.messageTimeout = null;
30   }
31
32   /**
```

```
33      * Инициализация вкладки  
34      */  
35    async init() {  
36        if (this.initialized) return;  
37  
38        console.log('[SyncTab] Initializing...');  
39  
40        // Загрузка HTML  
41        await this.loadHTML();  
42  
43        // Установка обработчиков  
44        this.attachEventListeners();  
45  
46        // Загрузка данных при активной вкладке  
47        if (window.isTabActive ? window.isTabActive('tab-sync') :  
false) {  
48            await this.updateStatus();  
49            await this.updateStats();  
50        }  
  
51        // Установка автообновления  
52        if (window.registerTabPoller) {  
53            window.registerTabPoller('tab-sync', () => this.updateSt  
atus(), 5000, { pollerId: 'sync_status', immediate: true });  
54            window.registerTabPoller('tab-sync', () => this.updateSt  
ats(), 30000, { pollerId: 'sync_stats', immediate: true });  
55        } else {  
56            setInterval(() => {  
57                if (window.isTabActive ? window.isTabActive('tab-syn  
) : true) {  
59                    this.updateStatus();  
60                }  
61            }, 5000);  
62            setInterval(() => {  
63                if (window.isTabActive ? window.isTabActive('tab-syn  
) : true) {  
64                    this.updateStats();  
65                }
```



```

66 |     }, 30000);
67 | }
68 |
69 | this.initialized = true;
70 | console.log('[SyncTab] Initialized');
71 | }
72 |
73 | /**
74 |  * Загрузить HTML компонента
75 |  */
76 | async loadHTML() {
77 |     const container = document.getElementById('tab-sync');
78 |     if (!container) {
79 |         console.warn('[SyncTab] Container not found');
80 |         return;
81 |     }
82 |
83 |     const html = `
84 |         <div class="sync-panel">
85 |             <!-- Заголовок -->
86 |             <div class="mb-4">
87 |                 <h4 class="d-flex align-items-center gap-2">
88 |                     <i class="bi bi-cloud-upload-fill text-primary"></i>
89 |                     <span>Google Drive Synchronization</span>
90 |                 </h4>
91 |                 <p class="text-muted mb-0">Manage data synchronizati
92 | on and storage migration</p>
93 |             </div>
94 |
95 |             <!-- Статус -->
96 |             <div class="card border-secondary-subtle mb-3">
97 |                 <div class="card-header bg-body-tertiary border-seco
98 | ndary-subtle">
99 |                     <h6 class="mb-0">□ Synchronization Status</h6>
100 |                 </div>
101 |                 <div class="card-body">
102 |                     <div class="row g-3">

```

```

101 |         <div class="col-md-3">
102 |             <div class="ps-3 border-start border-primary-s
ubtle">
103 |                 <small class="text-muted">Scheduler</small>
104 |                 <div>
105 |                     <span id="scheduler-status" class="badge b
g-danger">Inactive</span>
106 |                 </div>
107 |             </div>
108 |         </div>
109 |         <div class="col-md-3">
110 |             <div class="ps-3 border-start border-primary-s
ubtle">
111 |                 <small class="text-muted">Last Sync</small>
112 |                 <div id="last-sync-time" class="small">Never
</div>
113 |             </div>
114 |         </div>
115 |         <div class="col-md-3">
116 |             <div class="ps-3 border-start border-primary-s
ubtle">
117 |                 <small class="text-muted">Interval</small>
118 |                 <div id="sync-interval" class="small">6 hour
s</div>
119 |             </div>
120 |         </div>
121 |         <div class="col-md-3">
122 |             <div class="ps-3 border-start border-primary-s
ubtle">
123 |                 <small class="text-muted">Drive Folder</smal
l>
124 |                 <div>
125 |                     <a id="drive-url" href="#" target="_blank"
class="small text-decoration-none">
126 |                         Open
127 |                     </a>
128 |                 </div>
129 |             </div>

```

```
130         </div>
131     </div>
132 </div>
133 </div>
134
135     <!-- Статистика -->
136     <div class="card border-secondary-subtle mb-3">
137         <div class="card-header bg-body-tertiary border-sec-
138 ndary-subtle">
139             <h6 class="mb-0">📊 Storage Statistics</h6>
140         </div>
141         <div class="card-body">
142             <div class="row g-3">
143                 <div class="col-md-3 text-center">
144                     <div class="fs-5 fw-bold text-info" id="total-
145 size">-</div>
146                     <small class="text-muted">Total Size</small>
147                 </div>
148                 <div class="col-md-3 text-center">
149                     <div class="fs-5 fw-bold text-success" id="fil-
150 es-count">-</div>
151                     <small class="text-muted">Files</small>
152                 </div>
153                 <div class="col-md-3 text-center">
154                     <div class="fs-5 fw-bold text-warning" id="fol-
155 ders-count">-</div>
156                     <small class="text-muted">Folders</small>
157                 </div>
158                 <div class="col-md-3 text-center">
159                     <div class="fs-5 fw-bold text-danger" id="sync-
160 errors">0</div>
161                     <small class="text-muted">Errors</small>
162                 </div>
163             </div>
164         </div>
165     </div>
166 </div>
167
168     <!-- Управление -->
```

```

163 |         <div class="card border-secondary-subtle mb-3">
164 |             <div class="card-header bg-body-tertiary border-seco
| ndary-subtle">
165 |                 <h6 class="mb-0">□ Synchronization Controls</h6>
166 |             </div>
167 |             <div class="card-body">
168 |                 <div class="mb-3">
169 |                     <label class="form-label">Sync Interval (hours)<
| /label>
170 |                     <div class="input-group mb-2">
171 |                         <input type="number" id="sync-interval-input"
| min="1" max="24" value="6" class="form-control">
172 |                         <button class="btn btn-outline-secondary" oncl
| ick="window.syncTab.updateInterval()">
173 |                             Update
174 |                         </button>
175 |                     </div>
176 |                 </div>
177 |
178 |                 <div class="d-flex gap-2 flex-wrap">
179 |                     <button id="start-scheduler-btn" class="btn btn-
| success btn-sm" onclick="window.syncTab.startScheduler()">
180 |                         <i class="bi bi-play-fill"></i> Start Schedule
| r
181 |                     </button>
182 |                     <button id="stop-scheduler-btn" class="btn btn-d
| anger btn-sm" disabled onclick="window.syncTab.stopScheduler()
| ">
183 |                         <i class="bi bi-stop-fill"></i> Stop Scheduler
184 |                     </button>
185 |                     <button class="btn btn-primary btn-sm" onclick="
| window.syncTab.showSyncOptions()">
186 |                         <i class="bi bi-arrow-repeat"></i> Sync Now
187 |                     </button>
188 |                     <button class="btn btn-info btn-sm" onclick="win
| dow.syncTab.testConnection()">
189 |                         <i class="bi bi-link-45deg"></i> Test Connecti
| on

```

```

190         </button>
191     </div>
192 </div>
193 </div>
194
195     <!-- Опции синхронизации -->
196     <div id="sync-options-modal" class="modal fade" tabind
ex="-1">
197         <div class="modal-dialog modal-dialog-centered">
198             <div class="modal-content bg-dark text-white borde
r-secondary">
199                 <div class="modal-header border-secondary">
200                     <h5 class="modal-title">Select Data to Sync</h
5>
201                     <button type="button" class="btn-close btn-clo
se-white" data-bs-dismiss="modal"></button>
202                 </div>
203                 <div class="modal-body">
204                     <div class="form-check mb-2">
205                         <input class="form-check-input" type="radio"
name="sync-type" value="all" id="sync-all" checked>
206                         <label class="form-check-label" for="sync-al
l">
207                             <strong>All Data</strong> - Everything (da
ta, logs, secrets, configs)
208                             </label>
209                         </div>
210                         <div class="form-check mb-2">
211                             <input class="form-check-input" type="radio"
name="sync-type" value="data" id="sync-data">
212                             <label class="form-check-label" for="sync-da
ta">
213                                 <strong>Data Only</strong> - Databases and
RAG indices
214                                 </label>
215                             </div>
216                         <div class="form-check mb-2">
217                             <input class="form-check-input" type="radio"

```

```

    name="sync-type" value="logs" id="sync-logs">
218         <label class="form-check-label" for="sync-lo
gs">
219             <strong>Logs Only</strong> - Application l
ogs
220         </label>
221     </div>
222     <div class="form-check mb-2">
223         <input class="form-check-input" type="radio"
name="sync-type" value="secrets" id="sync-secrets">
224         <label class="form-check-label" for="sync-se
crets">
225             <strong>Secrets Only</strong> - Configurat
ion and keys
226         </label>
227     </div>
228 </div>
229     <div class="modal-footer border-secondary">
230         <button type="button" class="btn btn-secondary
btn-sm" data-bs-dismiss="modal">
231             Cancel
232         </button>
233         <button type="button" class="btn btn-primary b
tn-sm" onclick="window.syncTab.performSync()">
234             Sync Selected
235         </button>
236     </div>
237 </div>
238 </div>
239 </div>
240
241     <!-- Сообщения -->
242     <div id="sync-messages" class="mb-3"></div>
243
244     <!-- Лог активности -->
245     <div class="card border-secondary-subtle">
246         <div class="card-header bg-body-tertiary border-seco
ndary-subtle">

```

```

247         <h6 class="mb-0"> Recent Activity</h6>
248     </div>
249     <div class="card-body" style="max-height: 300px; ove
rflow-y: auto;">
250         <div id="activity-log">
251             <p class="text-muted mb-0">No recent activity</p
>
252         </div>
253     </div>
254 </div>
255 </div>
256 `;
257
258     container.innerHTML = html;
259 }
260
261 /**
262  * Присоединить обработчики событий
263  */
264 attachEventListeners() {
265     // Обработка модального окна
266     const modal = new bootstrap.Modal(document.getElementById(
'sync-options-modal'), {
267         backdrop: 'static'
268     });
269
270     window.syncTab = this; // Для доступа из HTML
271 }
272
273 /**
274  * Обновить статус синхронизации
275  */
276 async updateStatus() {
277     try {
278         const response = await fetch('/api/admin/sync/status');
279         if (!response.ok) throw new Error('Failed to fetch statu
s');
280

```

```
281 |     const data = await response.json();
282 |
283 |     // Обновить статус
284 |     const badge = document.getElementById('scheduler-status'
    | );
285 |     badge.textContent = data.is_running ? 'Running' : 'Inact
    | ive';
286 |     badge.className = data.is_running
287 |         ? 'badge bg-success'
288 |         : 'badge bg-danger';
289 |
290 |     // Обновить кнопки
291 |     document.getElementById('start-scheduler-btn').disabled
    | = data.is_running;
292 |     document.getElementById('stop-scheduler-btn').disabled =
    | !data.is_running;
293 |
294 |     // Обновить время
295 |     document.getElementById('last-sync-time').textContent =
    | data.last_sync_time || 'Never';
296 |     document.getElementById('sync-interval').textContent = `
    | ${data.sync_interval_hours} hours`;
297 |
298 |     // Обновить Google Drive папку
299 |     try {
300 |         const driveResponse = await fetch('/api/admin/sync/dri
    | ve-info');
301 |         if (driveResponse.ok) {
302 |             const driveData = await driveResponse.json();
303 |             const link = document.getElementById('drive-url');
304 |             link.href = driveData.folder_url;
305 |         }
306 |     } catch (e) {
307 |         console.warn('[SyncTab] Failed to fetch drive info');
308 |     }
309 | } catch (error) {
310 |     console.error('[SyncTab] Error updating status:', error)
    | ;
```



```
311     }
312 }
313
314 /**
315  * Обновить статистику
316  */
317 async updateStats() {
318     try {
319         const response = await fetch('/api/admin/sync/stats');
320         if (!response.ok) throw new Error('Failed to fetch stats
321 ');
322         const data = await response.json();
323
324         document.getElementById('total-size').textContent =
325             `${data.total_size_mb.toFixed(1)} MB`;
326         document.getElementById('files-count').textContent = dat
327 a.files_count;
328         document.getElementById('folders-count').textContent = d
329 ata.folders_count;
330         document.getElementById('sync-errors').textContent = dat
331 a.sync_errors;
332     } catch (error) {
333         console.error('[SyncTab] Error updating stats:', error);
334     }
335 }
336
337 /**
338  * Запустить планировщик
339  */
340 async startScheduler() {
341     try {
342         const interval = parseInt(document.getElementById('sync-
343 interval-input').value);
344         const response = await fetch('/api/admin/sync/start', {
345             method: 'POST',
346             headers: { 'Content-Type': 'application/json' },
347             body: JSON.stringify({ sync_interval_hours: interval })
```

```

    )
344     });
345
346     const data = await response.json();
347     this.showMessage(data.message || 'Scheduler started', 's
    uccess');
348     await this.updateStatus();
349   } catch (error) {
350     this.showMessage('Error: ' + error.message, 'danger');
351   }
352 }
353
354 /**
355  * Остановить планировщик
356  */
357 async stopScheduler() {
358   try {
359     const response = await fetch('/api/admin/sync/stop', {
360       method: 'POST'
361     });
362
363     const data = await response.json();
364     this.showMessage(data.message || 'Scheduler stopped', 's
    uccess');
365     await this.updateStatus();
366   } catch (error) {
367     this.showMessage('Error: ' + error.message, 'danger');
368   }
369 }
370
371 /**
372  * Проверить подключение
373  */
374 async testConnection() {
375   try {
376     const response = await fetch('/api/admin/sync/test-conne
    ction', {
377       method: 'POST'

```

```
378 |     });
379 |
380 |     const data = await response.json();
381 |     const type = data.connected ? 'success' : 'warning';
382 |     this.showMessage(data.message || 'Connection test comple
    | ted', type);
383 |   } catch (error) {
384 |     this.showMessage('Error: ' + error.message, 'danger');
385 |   }
386 | }
387 |
388 | /**
389 |  * Показать опции синхронизации
390 |  */
391 | showSyncOptions() {
392 |   const modal = bootstrap.Modal.getInstance(document.getElem
    | entById('sync-options-modal')) ||
393 |   new bootstrap.Modal(document.getElementById('sync-option
    | s-modal'));
394 |   modal.show();
395 | }
396 |
397 | /**
398 |  * Выполнить синхронизацию
399 |  */
400 | async performSync() {
401 |   const syncType = document.querySelector('input[name="sync-
    | type"]:checked')?.value || 'all';
402 |
403 |   try {
404 |     const response = await fetch('/api/admin/sync/sync-now',
    | {
405 |       method: 'POST',
406 |       headers: { 'Content-Type': 'application/json' },
407 |       body: JSON.stringify({ sync_type: syncType })
408 |     });
409 |
410 |     const data = await response.json();
```

```
411 |     this.showMessage(data.message || 'Sync started', 'success');
412 |     this.addActivityLog(`Started ${syncType} synchronization`);
413 |
414 |     // Скрыть модаль
415 |     const modal = bootstrap.Modal.getInstance(document.getElementById('sync-options-modal'));
416 |     if (modal) modal.hide();
417 |   } catch (error) {
418 |     this.showMessage('Error: ' + error.message, 'danger');
419 |   }
420 | }
421 |
422 | /**
423 |  * Обновить интервал
424 |  */
425 | async updateInterval() {
426 |   const interval = parseInt(document.getElementById('sync-interval-input').value);
427 |
428 |   try {
429 |     const response = await fetch('/api/admin/sync/config', {
430 |       method: 'POST',
431 |       headers: { 'Content-Type': 'application/json' },
432 |       body: JSON.stringify({ sync_interval_hours: interval })
433 |     });
434 |
435 |     this.showMessage(`Interval updated to ${interval} hours`, 'success');
436 |     await this.updateStatus();
437 |   } catch (error) {
438 |     this.showMessage('Error: ' + error.message, 'danger');
439 |   }
440 | }
441 |
442 | /**
```

```

443     * Показать сообщение
444     */
445     showMessage(text, type = 'info') {
446         const container = document.getElementById('sync-messages')
447         ;
448         if (!container) return;
449         const alert = document.createElement('div');
450         alert.className = `alert alert-${type} alert-dismissible fade show`;
451         alert.innerHTML = `
452             ${text}
453             <button type="button" class="btn-close" data-bs-dismiss=
454             "alert"></button>
455             `;
456         container.appendChild(alert);
457
458         clearTimeout(this.messageTimeout);
459         this.messageTimeout = setTimeout(() => {
460             alert.remove();
461         }, 5000);
462     }
463
464     /**
465     * Добавить в лог активности
466     */
467     addActivityLog(text) {
468         const log = document.getElementById('activity-log');
469         if (!log) return;
470
471         const item = document.createElement('div');
472         item.className = 'small mb-2 pb-2 border-bottom';
473
474         const now = new Date().toLocaleTimeString();
475         item.innerHTML = `<span class="text-muted">${now}</span> -
476         ${text}`;

```

```
477 |     if (log.textContent.includes('No recent activity')) {
478 |         log.innerHTML = '';
479 |     }
480 |
481 |     log.insertBefore(item, log.firstChild);
482 | }
483 | }
484 |
485 | // Инициализация при загрузке документа
486 | document.addEventListener('DOMContentLoaded', async () => {
487 |     window.syncTab = new SyncTab();
488 |
489 |     // Инициализировать когда вкладка станет видимой
490 |     const tab = document.getElementById('tab-sync');
491 |     if (tab) {
492 |         // Проверить если вкладка уже активна
493 |         const tabPane = tab.closest('.tab-pane');
494 |         if (tabPane && tabPane.classList.contains('active')) {
495 |             await window.syncTab.init();
496 |         } else if (tabPane) {
497 |             // Инициализировать при активации вкладки
498 |             tabPane.addEventListener('shown.bs.tab', () => {
499 |                 window.syncTab.init();
500 |             });
501 |         }
502 |     }
503 | });
504 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/admin_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/admin_tab/index.html" c
17     lass="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/admin_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- System Admin Tab -->
30 <div class="container-fluid py-2">
31   <div class="alert alert-dark d-flex flex-wrap align-items-ce
32   nter justify-content-between gap-2 mb-3 shadow-sm border-secon
33   dary">
34     <div class="d-flex align-items-center gap-2">
35       <span class="fs-4">❏ </span>
36       <div>
37         <h5 class="mb-0 fw-bold">Центральное управление систем
```

```

ой</h5>
31      <p class="mb-0 small text-muted">Мониторинг ядра, сист
емные сервисы, провайдеры моделей и глобальное администрирован
ие</p>
32    </div>
33  </div>
34  <div class="d-flex gap-2 flex-wrap">
35    <button class="btn btn-warning btn-sm fw-bold d-flex ali
gn-items-center gap-1 shadow-sm" onclick="switchToTab('helpdes
k')" id="btn-switch-to-helpdesk">
36      <i class="bi bi-headset"></i> Режим Helpdesk
37    </button>
38    <button class="btn btn-outline-info btn-sm" onclick="res
canStorageDrives()">
39      <i class="bi bi-hdd-network"></i> Пересканировать диск
и
40    </button>
41    <button class="btn btn-outline-primary btn-sm" onclick="
actualizeAiModels()">
42      <i class="bi bi-cpu"></i> Обновить модели ИИ
43    </button>
44  </div>
45 </div>
46
47 <!-- System Health & Status Cards -->
48 <div class="row g-3 mb-3">
49   <div class="col-md-3 col-sm-6">
50     <div class="card shadow-sm border-secondary-subtle h-100
">
51       <div class="card-body p-3">
52         <div class="text-muted small text-uppercase fw-semib
old mb-1">Сервер / Режим</div>
53         <div class="d-flex align-items-center justify-conten
t-between">
54           <h5 class="mb-0 fw-bold text-primary" id="sys-serv
er-mode">FastAPI (DEV)</h5>
55           <span class="badge bg-success-subtle text-success"
>Online</span>

```



```
56 |         </div>
57 |         <small class="text-muted" id="sys-server-host">0.0.0
    | .0:8000</small>
58 |     </div>
59 | </div>
60 | </div>
61 |
62 | <div class="col-md-3 col-sm-6">
63 |     <div class="card shadow-sm border-secondary-subtle h-100
    | ">
64 |         <div class="card-body p-3">
65 |             <div class="text-muted small text-uppercase fw-semib
    | old mb-1">AI Провайдеры</div>
66 |             <div class="d-flex align-items-center justify-conten
    | t-between">
67 |                 <h5 class="mb-0 fw-bold text-info" id="sys-ai-prov
    | iders">Gemini / AGY / Foundry</h5>
68 |                 <i class="bi bi-robot fs-4 text-info"></i>
69 |             </div>
70 |             <small class="text-muted">Актуализированный пул</sma
    | ll>
71 |         </div>
72 |     </div>
73 | </div>
74 |
75 | <div class="col-md-3 col-sm-6">
76 |     <div class="card shadow-sm border-secondary-subtle h-100
    | ">
77 |         <div class="card-body p-3">
78 |             <div class="text-muted small text-uppercase fw-semib
    | old mb-1">Подключено дисков</div>
79 |             <div class="d-flex align-items-center justify-conten
    | t-between">
80 |                 <h5 class="mb-0 fw-bold text-warning" id="sys-driv
    | es-count">Загрузка...</h5>
81 |                 <i class="bi bi-device-hdd fs-4 text-warning"></i>
82 |             </div>
83 |             <small class="text-muted" id="sys-drives-list">CONNE
```

```

CTED_DRIVES</small>
84     </div>
85   </div>
86 </div>
87
88   <div class="col-md-3 col-sm-6">
89     <div class="card shadow-sm border-secondary-subtle h-100
">
90       <div class="card-body p-3">
91         <div class="text-muted small text-uppercase fw-semib
old mb-1">Подключаемые плагины</div>
92         <div class="d-flex align-items-center justify-conten
t-between">
93           <h5 class="mb-0 fw-bold text-success" id="sys-plug
ins-count">Загрузка...</h5>
94           <a href="javascript:void(0)" onclick="switchToTab(
'plugins')" class="badge bg-primary text-decoration-none">Упра
вление □ </a>
95         </div>
96         <small class="text-muted" id="sys-plugins-active-cou
nt">Реестр модулей</small>
97       </div>
98     </div>
99   </div>
100 </div>
101
102 <!-- Fast Navigation Hub to Core Subsystems -->
103 <h6 class="fw-bold mb-3 text-uppercase small text-muted"><i
class="bi bi-grid-fill me-1"></i> Системные подсистемы</h6>
104 <div class="row g-3 mb-4">
105   <div class="col-lg-4 col-md-6">
106     <div class="card shadow-sm border-secondary-subtle h-100
p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
ab('plugins')">
107       <div class="d-flex align-items-center gap-3">
108         <span class="fs-1">□ </span>
109       <div>
110         <h6 class="fw-bold mb-1">Подключатель плагинов (Hu

```

```

b)</h6>
111         <p class="small text-muted mb-0">Управление модуля
ми (web_search, rag, langchain_media и др.), включение/отключе
ние и их органы настройки.</p>
112     </div>
113 </div>
114 </div>
115 </div>
116
117 <div class="col-lg-4 col-md-6">
118     <div class="card shadow-sm border-secondary-subtle h-100
p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
ab('users')">
119         <div class="d-flex align-items-center gap-3">
120             <span class="fs-1">👤 </span>
121             <div>
122                 <h6 class="fw-bold mb-1">Управление пользователями
</h6>
123                 <p class="small text-muted mb-0">Создание аккаунто
в, назначение ролей (admin/user), пароли, активность и права д
оступа.</p>
124             </div>
125         </div>
126     </div>
127 </div>
128
129 <div class="col-lg-4 col-md-6">
130     <div class="card shadow-sm border-secondary-subtle h-100
p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
ab('models')">
131         <div class="d-flex align-items-center gap-3">
132             <span class="fs-1">🤖 </span>
133             <div>
134                 <h6 class="fw-bold mb-1">Модели и API-ключи</h6>
135                 <p class="small text-muted mb-0">Конфигурация Goog
le Gemini, Antigravity, Ollama и AI Foundry, пул ротации ключе
й.</p>
136             </div>

```

```
137         </div>
138     </div>
139 </div>
140
141     <div class="col-lg-4 col-md-6">
142         <div class="card shadow-sm border-secondary-subtle h-100
143         p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('instructions')">
144             <div class="d-flex align-items-center gap-3">
145                 <span class="fs-1">📄 </span>
146                 <div>
147                     <h6 class="fw-bold mb-1">Системные инструкции</h6>
148                     <p class="small text-muted mb-0">Редактирование системных промптов чата, стилей диктора и версионирование инструкций.</p>
149                 </div>
150             </div>
151         </div>
152
153     <div class="col-lg-4 col-md-6">
154         <div class="card shadow-sm border-secondary-subtle h-100
155         p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('logs')">
156             <div class="d-flex align-items-center gap-3">
157                 <span class="fs-1">📄 </span>
158                 <div>
159                     <h6 class="fw-bold mb-1">Системные логи</h6>
160                     <p class="small text-muted mb-0">Просмотр журнала событий uvicorn, ошибок FastAPI, истории запросов и аудита.</p>
161                 </div>
162             </div>
163         </div>
164
165     <div class="col-lg-4 col-md-6">
166         <div class="card shadow-sm border-secondary-subtle h-100
```

```

    p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('agents')">
167     <div class="d-flex align-items-center gap-3">
168         <span class="fs-1">□ </span>
169         <div>
170             <h6 class="fw-bold mb-1">Агенты ИИ</h6>
171             <p class="small text-muted mb-0">Конфигурация автономных ReAct агентов, тулов поиска и параметров инференса.</p>
172         </div>
173     </div>
174 </div>
175 </div>
176
177 <div class="col-lg-4 col-md-6">
178     <div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('models'); setTimeout(() => { const btn = document.getElementById('pill-google-accounts-tab'); if (btn) btn.click(); }, 150);">
179         <div class="d-flex align-items-center gap-3">
180             <span class="fs-1">□ </span>
181             <div>
182                 <h6 class="fw-bold mb-1">Google Workspace & Аккаунты</h6>
183                 <p class="small text-muted mb-0">Пул аккаунтов OAuth 2.0 и Service Account для Gmail, Drive, Sheets и Docs.</p>
184             </div>
185         </div>
186     </div>
187 </div>
188
189 <div class="col-lg-4 col-md-6">
190     <div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow border-warning-subtle" style="cursor: pointer;" onclick="switchToTab('helpdesk')">
191         <div class="d-flex align-items-center gap-3">
192             <span class="fs-1">□ </span>
193             <div>

```

```

194         <h6 class="fw-bold mb-1 text-warning">Helpdesk & С
| лужба поддержки</h6>
195         <p class="small text-muted mb-0">Управление заявка
| ми, живой чат с пользователями, шаблоны быстрых ответов и метр
| ики SLA.</p>
196     </div>
197 </div>
198 </div>
199 </div>
200
201 <div class="col-lg-4 col-md-6">
202     <div class="card shadow-sm border-secondary-subtle h-100
| p-3 hover-shadow border-primary-subtle" style="cursor: pointe
| r;" onclick="switchToTab('observability')">
203         <div class="d-flex align-items-center gap-3">
204             <span class="fs-1">📊 </span>
205             <div>
206                 <h6 class="fw-bold mb-1 text-primary">Observabilit
| y & Диагностика</h6>
207                 <p class="small text-muted mb-0">Мониторинг ресурс
| ов хоста в реальном времени, дерево оборудования и эвристическ
| ий AI-аудит.</p>
208             </div>
209         </div>
210     </div>
211 </div>
212
213 <div class="col-lg-4 col-md-6">
214     <div class="card shadow-sm border-secondary-subtle h-100
| p-3 hover-shadow border-info-subtle" style="cursor: pointer;"
| onclick="document.getElementById('scheduler-management-card')
| .scrollIntoView({behavior: 'smooth'})">
215         <div class="d-flex align-items-center gap-3">
216             <span class="fs-1">📅 </span>
217             <div>
218                 <h6 class="fw-bold mb-1 text-info">Фоновый планиро
| вщик (Scheduler)</h6>
219                 <p class="small text-muted mb-0">24ч переиндексаци

```

```

    я RAG коллекций, 5м мониторинг почты Gmail и автообновление.</
    p>
220         </div>
221     </div>
222 </div>
223 </div>
224 </div>
225
226 <!-- Background Schedulers Management Section -->
227 <div class="card shadow-sm border-secondary-subtle mb-3" id=
    "scheduler-management-card">
228     <div class="card-header bg-body-tertiary d-flex flex-wrap
    justify-content-between align-items-center py-2 gap-2">
229         <div class="d-flex align-items-center gap-2">
230             <span class="fw-bold"><i class="bi bi-clock-history te
    xt-info me-2"></i> ☐ Фоновые периодические задачи (Schedulers)<
    /span>
231             <span id="sched-main-badge" class="badge bg-success-su
    btile text-success">Активен</span>
232         </div>
233         <div class="d-flex gap-2">
234             <button class="btn btn-outline-info btn-sm" onclick="r
    efreshSchedulersConfig()">
235                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
    статус
236             </button>
237             <button class="btn btn-primary btn-sm" onclick="saveSc
    hedulersConfig()">
238                 <i class="bi bi-check2-circle me-1"></i> Сохранить н
    астройки
239             </button>
240         </div>
241     </div>
242     <div class="card-body p-3">
243         <!-- Master toggle -->
244         <div class="form-check form-switch mb-3">
245             <input class="form-check-input" type="checkbox" id="sc
    hed-master-switch" checked>

```

```

246 |         <label class="form-check-label fw-bold" for="sched-mas
      | ter-switch">Включить глобальный фоновый планировщик (Backgroun
      | dScheduler)</label>
247 |         <div class="small text-muted">При отключении все фонов
      | ые интервальные циклы будут остановлены.</div>
248 |     </div>
249 |
250 |     <div class="row g-3">
251 |         <!-- RAG 24h Re-indexing Task -->
252 |         <div class="col-md-6">
253 |             <div class="card bg-body-tertiary border-secondary-s
      | ubtle h-100 p-3">
254 |                 <div class="d-flex justify-content-between align-i
      | tems-center mb-2">
255 |                     <span class="fw-bold text-primary"><i class="bi
      | bi-cpu me-1"></i> □ Переиндексация RAG (RAG Re-indexing)</span>
      | >
256 |                     <div class="form-check form-switch m-0">
257 |                         <input class="form-check-input" type="checkbox"
      | " id="sched-rag-enabled" checked>
258 |                     </div>
259 |                 </div>
260 |                 <p class="small text-muted mb-2">Автоматический фо
      | новый пересчет и обновление коллекций векторных/TF-IDF индексо
      | в.</p>
261 |                 <div class="mb-2">
262 |                     <label class="form-label small fw-semibold mb-1"
      | for="sched-rag-interval">Интервал запуска (в часах):</label>
263 |                     <div class="input-group input-group-sm">
264 |                         <input type="number" class="form-control" id="
      | sched-rag-interval" value="24" min="0.1" step="0.5">
265 |                         <span class="input-group-text">час(ов)</span>
266 |                     </div>
267 |                 </div>
268 |                 <div class="d-flex justify-content-between align-i
      | tems-center small text-muted mt-auto pt-2 border-top border-se
      | condary-subtle">
269 |                     <span>Посл. запуск: <strong id="sched-rag-last-r

```



```

un">Никогда</strong></span>
270         <button class="btn btn-outline-primary btn-sm py
-0 px-2" onclick="triggerSchedulerJob('rag')" title="Запустить
переиндексацию сейчас">
271         <i class="bi bi-play-fill"></i> Запустить сейч
ас
272         </button>
273     </div>
274 </div>
275 </div>
276
277     <!-- Gmail 5m Check Task -->
278     <div class="col-md-6">
279         <div class="card bg-body-tertiary border-secondary-s
ubtle h-100 p-3">
280             <div class="d-flex justify-content-between align-i
tems-center mb-2">
281                 <span class="fw-bold text-success"><i class="bi
bi-envelope-check me-1"></i> □ Проверка Gmail (Email Check)</s
pan>
282                 <div class="form-check form-switch m-0">
283                     <input class="form-check-input" type="checkbox
" id="sched-email-enabled" checked>
284                 </div>
285             </div>
286             <p class="small text-muted mb-2">Опрос ящиков Goog
le Workspace на наличие новых непрочитанных сообщений.</p>
287             <div class="mb-2">
288                 <label class="form-label small fw-semibold mb-1"
for="sched-email-interval">Интервал опроса (в минутах):</labe
l>
289                 <div class="input-group input-group-sm">
290                     <input type="number" class="form-control" id="
sched-email-interval" value="5" min="1" step="1">
291                     <span class="input-group-text">мин.</span>
292                 </div>
293             </div>
294             <div class="d-flex justify-content-between align-i

```

```

    items-center small text-muted mt-auto pt-2 border-top border-se
condary-subtle">
295         <span>Посл. запуск: <strong id="sched-email-last
-run">Никогда</strong></span>
296         <button class="btn btn-outline-success btn-sm py
-0 px-2" onclick="triggerSchedulerJob('email')" title="Провери
ть почту сейчас">
297             <i class="bi bi-play-fill"></i> Запустить сейч
ас
298         </button>
299     </div>
300 </div>
301 </div>
302 </div>
303 </div>
304 </div>
305
306 <!-- Storage & Disks Management Section -->
307 <div class="card shadow-sm border-secondary-subtle mb-3">
308     <div class="card-header bg-body-tertiary d-flex flex-wrap
justify-content-between align-items-center py-2 gap-2">
309         <span class="fw-bold"><i class="bi bi-hdd-network text-w
arning me-2"></i> ⚠ Управление дисками и накопителями</span>
310         <button class="btn btn-outline-info btn-sm" onclick="res
canStorageDrives()">
311             <i class="bi bi-arrow-clockwise me-1"></i> Пересканиро
вать диски
312         </button>
313     </div>
314     <div class="card-body p-3">
315         <div id="sys-drives-detailed-container" class="row g-3">
316             <div class="col-12 text-muted small">Загрузка информац
ии о накопителях...</div>
317         </div>
318     </div>
319 </div>
320
321 <!-- User Telemetry & Ngrok Tunnel Section -->

```

```

322 | <div class="card shadow-sm border-secondary-subtle mb-4">
323 |     <div class="card-header bg-body-tertiary d-flex flex-wrap
    | justify-content-between align-items-center py-2 gap-2">
324 |         <div class="d-flex align-items-center gap-2">
325 |             <span class="fw-bold"><i class="bi bi-graph-up-arrow t
    | ext-info me-2"></i>   Активность пользователей и туннель Ngrok
    | </span>
326 |             <span id="ngrok-status-badge" class="badge bg-secondar
    | y-subtle text-secondary">Туннель: проверка...</span>
327 |         </div>
328 |         <div class="d-flex gap-2">
329 |             <button class="btn btn-outline-primary btn-sm" onclick
    | ="refreshTelemetryStats()">
330 |                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
    | статистику
331 |             </button>
332 |         </div>
333 |     </div>
334 |     <div class="card-body p-3">
335 |         <!-- Tunnel info alert -->
336 |         <div id="ngrok-info-alert" class="alert alert-secondary
    | py-2 px-3 small d-flex flex-wrap align-items-center justify-co
    | ntent-between gap-2 mb-3">
337 |             <div class="d-flex align-items-center gap-2">
338 |                 <i class="bi bi-broadcast fs-5 text-primary"></i>
339 |             <div>
340 |                 <span class="fw-semibold">Ngrok шлюз приема статис
    | тики:</span>
341 |                 <span id="ngrok-public-url-text" class="text-prima
    | ry font-monospace ms-1">Ожидание запуска...</span>
342 |             </div>
343 |         </div>
344 |         <small class="text-muted" id="ngrok-hint-text">Скрипт:
    | <code>.\launchers\Run-Ngrok.ps1</code></small>
345 |     </div>
346 |
347 |     <!-- Telemetry Stats Cards -->
348 |     <div class="row g-3 mb-3">

```

```
349 |         <div class="col-md-4">
350 |             <div class="card bg-body-tertiary border-0 p-3 h-100
    | ">
351 |                 <div class="small text-muted text-uppercase fw-sem
    | ibold mb-1">Событий за 30 дней</div>
352 |                 <h4 class="mb-0 fw-bold text-primary" id="telemetr
    | y-total-events">0</h4>
353 |                 <small class="text-muted" id="telemetry-active-use
    | rs">0 пользователей</small>
354 |             </div>
355 |         </div>
356 |         <div class="col-md-4">
357 |             <div class="card bg-body-tertiary border-0 p-3 h-100
    | ">
358 |                 <div class="small text-muted text-uppercase fw-sem
    | ibold mb-1">Популярные вкладки</div>
359 |                 <div id="telemetry-tabs-breakdown" class="small te
    | xt-muted">
360 |                     Загрузка...
361 |                 </div>
362 |             </div>
363 |         </div>
364 |         <div class="col-md-4">
365 |             <div class="card bg-body-tertiary border-0 p-3 h-100
    | ">
366 |                 <div class="small text-muted text-uppercase fw-sem
    | ibold mb-1">Топ кликов и действий</div>
367 |                 <div id="telemetry-clicks-breakdown" class="small
    | text-muted">
368 |                     Загрузка...
369 |                 </div>
370 |             </div>
371 |         </div>
372 |     </div>
373 |
374 |     <!-- Recent Telemetry Log Table -->
375 |     <h6 class="fw-bold small text-muted text-uppercase mb-2"
    | ><i class="bi bi-clock-history me-1"></i> Последние действия з
```

```
арегистрированных пользователей</h6>
376     <div class="table-responsive">
377         <table class="table table-sm table-hover align-middle
mb-0" style="font-size: 0.85rem;">
378             <thead class="table-light">
379                 <tr>
380                     <th scope="col">Время</th>
381                     <th scope="col">Пользователь</th>
382                     <th scope="col">Тип</th>
383                     <th scope="col">Вкладка</th>
384                     <th scope="col">Действие / Элемент</th>
385                     <th scope="col">Длительность</th>
386                 </tr>
387             </thead>
388             <tbody id="telemetry-events-tbody">
389                 <tr>
390                     <td colspan="6" class="text-center text-muted py
-3">Загрузка журнала активности...</td>
391                 </tr>
392             </tbody>
393         </table>
394     </div>
395 </div>
396 </div>
397 </div>
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Admin Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/admin_tab/main.js?v=20
11  *       261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/admin_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // System Admin Tab – Central Management Logic
22
23 'use strict';
24
25 async function initSystemAdminTab() {
26   console.log('Инициализация панели системного управления...')
27   ;
28   await refreshSystemDashboard();
29   await refreshSchedulersConfig();
30   await refreshTelemetryStats();
31 }
```

```

32 | async function refreshSystemDashboard() {
33 |     try {
34 |         // 1. Загрузка дисков
35 |         const drivesData = await window.api.fetch('/api/control/re
scan', { method: 'GET' }).catch(() => ({ drives: [], details:
[] }));
36 |         const drives = Array.isArray(drivesData.drives) ? drivesDa
ta.drives : [];
37 |         const details = Array.isArray(drivesData.details) ? drives
Data.details : [];
38 |         const drivesCountEl = document.getElementById('sys-drives-
count');
39 |         const drivesListEl = document.getElementById('sys-drives-l
ist');
40 |         if (drivesCountEl) drivesCountEl.textContent = `${drives.l
ength} диск(ов)`;
41 |         if (drivesListEl) drivesListEl.textContent = drives.join('
, ') || 'Диски не обнаружены';
42 |
43 |         const detailedContainer = document.getElementById('sys-dri
ves-detailed-container');
44 |         if (detailedContainer) {
45 |             if (details.length === 0 && drives.length === 0) {
46 |                 detailedContainer.innerHTML = '<div class="col-12 text
-muted small">Накопители не обнаружены или доступ ограничен.</
div>';
47 |             } else if (details.length > 0) {
48 |                 detailedContainer.innerHTML = details.map(d => {
49 |                     const percent = d.percent || 0;
50 |                     const barClass = percent > 90 ? 'bg-danger' : percen
t > 75 ? 'bg-warning' : 'bg-primary';
51 |                     return `
52 |                         <div class="col-lg-4 col-md-6">
53 |                             <div class="card bg-body-tertiary border-seconda
ry-subtle h-100 p-3">
54 |                                 <div class="d-flex justify-content-between ali
gn-items-center mb-2">
55 |                                     <span class="fw-bold text-primary"><i class=

```

```

    "bi bi-hdd me-1"></i>${d.device || d.mountpoint}</span>
56         <span class="badge bg-secondary-subtle text-
body">${d.fstype || 'NTFS'}</span>
57     </div>
58     <div class="d-flex justify-content-between sma
ll text-muted mb-1">
59         <span>Занято: <strong>${d.used_gb} GB</stron
g></span>
60         <span>Свободно: <strong>${d.free_gb} GB</str
ong></span>
61     </div>
62     <div class="progress mb-2" style="height: 8px;
">
63         <div class="progress-bar ${barClass}" role="
progressbar" style="width: ${percent}%;" aria-valuenow="${perc
ent}" aria-valuemin="0" aria-valuemax="100"></div>
64     </div>
65     <div class="d-flex justify-content-between sma
ll text-muted">
66         <span>Всего: ${d.total_gb} GB</span>
67         <span class="fw-semibold">${percent}%</span>
68     </div>
69 </div>
70 </div>
71 `;
72 }).join('');
73 } else {
74     detailedContainer.innerHTML = drives.map(drv => `
75         <div class="col-lg-3 col-md-4 col-sm-6">
76             <div class="card bg-body-tertiary border-secondary
-subtle p-2 text-center">
77                 <span class="fw-bold text-primary"><i class="bi
bi-hdd me-1"></i>${drv}</span>
78             </div>
79         </div>
80     `).join('');
81 }
82 }

```



```
83 |
84 |     // 2. Загрузка статуса плагинов
85 |     const pluginsData = await window.api.fetch('/api/admin/plugins').catch(() => ({ plugins: [] }));
86 |     const plugins = pluginsData.plugins || [];
87 |     const pluginsCountEl = document.getElementById('sys-plugins-count');
88 |     const pluginsActiveEl = document.getElementById('sys-plugins-active-count');
89 |     const activeCount = plugins.filter(p => p.enabled).length;
90 |     if (pluginsCountEl) pluginsCountEl.textContent = `${plugins.length} модулей`;
91 |     if (pluginsActiveEl) pluginsActiveEl.textContent = `${activeCount} активно из ${plugins.length}`;
92 |   } catch (err) {
93 |     console.error('Ошибка загрузки системного дашборда:', err);
94 |   }
95 | }
96 |
97 | async function rescanStorageDrives() {
98 |   try {
99 |     if (typeof showNotification === 'function') showNotification('Пересканирование накопителей ОС...', 'info');
100 |     const result = await window.api.fetch('/api/control/rescan', { method: 'GET' });
101 |     const drivesList = Array.isArray(result.drives) ? result.drives.join(', ') : 'OK';
102 |     if (typeof showNotification === 'function') showNotification(`Диски обновлены: ${drivesList}`, 'success');
103 |     await refreshSystemDashboard();
104 |   } catch (e) {
105 |     if (typeof showNotification === 'function') showNotification(`Ошибка: ${e.message}`, 'danger');
106 |   }
107 | }
108 |
109 | async function actualizeAiModels() {
```

```

110 |   try {
111 |     if (typeof showNotification === 'function') showNotificati
on('Актуализация пула моделей ИИ...', 'info');
112 |     await window.api.fetch('/api/keys/actualize-all', { method
: 'POST' }).catch(() => {});
113 |     if (typeof showNotification === 'function') showNotificati
on('Модели успешно синхронизированы', 'success');
114 |     await refreshSystemDashboard();
115 |   } catch (e) {
116 |     if (typeof showNotification === 'function') showNotificati
on(`Ошибка: ${e.message}`, 'danger');
117 |   }
118 | }
119 |
120 | async function refreshTelemetryStats() {
121 |   try {
122 |     const res = await fetch('/api/telemetry/stats?days=30');
123 |     if (!res.ok) return;
124 |     const stats = await res.json();
125 |
126 |     // 1. Ngrok Tunnel Status
127 |     const tunnel = stats.tunnel_status || {};
128 |     const badgeEl = document.getElementById('ngrok-status-badg
e');
129 |     const urlTextEl = document.getElementById('ngrok-public-ur
l-text');
130 |     const alertEl = document.getElementById('ngrok-info-alert'
);
131 |
132 |     if (tunnel.is_active && tunnel.public_url) {
133 |       if (badgeEl) {
134 |         badgeEl.className = 'badge bg-success-subtle text-succ
ess';
135 |         badgeEl.textContent = 'Туннель: Онлайн';
136 |       }
137 |       if (urlTextEl) {
138 |         urlTextEl.innerHTML = `<a href="${tunnel.public_url}"
target="_blank" class="text-success text-decoration-none fw-bo

```

```

ld">${tunnel.public_url}</a>`;
139     }
140     if (alertEl) alertEl.className = 'alert alert-success py
-2 px-3 small d-flex flex-wrap align-items-center justify-cont
ent-between gap-2 mb-3';
141     } else {
142         if (badgeEl) {
143             badgeEl.className = 'badge bg-secondary-subtle text-se
condary';
144             badgeEl.textContent = 'Туннель: Остановлен';
145         }
146         if (urlTextEl) {
147             urlTextEl.textContent = 'Не запущен (запустите .\\laun
chers\\Run-Ngrok.ps1)';
148         }
149         if (alertEl) alertEl.className = 'alert alert-secondary
py-2 px-3 small d-flex flex-wrap align-items-center justify-co
ntent-between gap-2 mb-3';
150     }
151
152     // 2. Event & User metrics
153     const totalEl = document.getElementById('telemetry-total-e
vents');
154     const usersEl = document.getElementById('telemetry-active-
users');
155     if (totalEl) totalEl.textContent = (stats.total_events ||
0).toLocaleString();
156     if (usersEl) usersEl.textContent = `${stats.active_users_c
ount || 0} пользователей`;
157
158     // 3. Tab views
159     const tabsEl = document.getElementById('telemetry-tabs-bre
akdown');
160     if (tabsEl) {
161         const tabEntries = Object.entries(stats.tab_views || {});
162         ;
163         if (tabEntries.length === 0) {
164             tabsEl.innerHTML = '<span class="text-muted">Нет данны

```

```

x</span>';
164     } else {
165         tabsEl.innerHTML = tabEntries.slice(0, 4).map(([tab, i
nfo]) => `
166             <div class="d-flex justify-content-between align-ite
ms-center mb-1">
167                 <span class="badge bg-secondary-subtle text-body">
${tab}</span>
168                 <span class="fw-semibold">${info.count} раз</span>
169             </div>
170         `).join('');
171     }
172 }
173
174 // 4. Click actions
175 const clicksEl = document.getElementById('telemetry-clicks
-breakdown');
176 if (clicksEl) {
177     const clickEntries = Object.entries(stats.top_clicks ||
{});
178     if (clickEntries.length === 0) {
179         clicksEl.innerHTML = '<span class="text-muted">Нет кли
ков</span>';
180     } else {
181         clicksEl.innerHTML = clickEntries.slice(0, 4).map(([ta
rget, info]) => `
182             <div class="d-flex justify-content-between align-ite
ms-center mb-1 text-truncate" title="${target}">
183                 <span class="text-truncate me-1" style="max-width:
140px;">${info.action || target}</span>
184                 <span class="badge bg-primary-subtle text-primary"
>${info.count}</span>
185             </div>
186         `).join('');
187     }
188 }
189
190 // 5. Recent events table

```

```

191     const tbody = document.getElementById('telemetry-events-tb
    ody');
192     if (tbody) {
193         const recent = stats.recent_events || [];
194         if (recent.length === 0) {
195             tbody.innerHTML = '<tr><td colspan="6" class="text-cen
    ter text-muted py-3">Пока нет зарегистрированных событий актив
    ности.</td></tr>';
196         } else {
197             tbody.innerHTML = recent.map(ev => {
198                 const dt = ev.timestamp ? new Date(ev.timestamp).tol
    ocaleTimeString() : '-';
199                 const userName = ev.name || ev.email || `ID ${ev.use
    r_id}`;
200                 const typeBadge = ev.event_type === 'tab_view' ? 'bg
    -info-subtle text-info' : ev.event_type === 'click' ? 'bg-prim
    ary-subtle text-primary' : 'bg-secondary-subtle text-secondary
    ';
201                 const durStr = ev.duration_ms > 0 ? `${(ev.duration_
    ms / 1000).toFixed(1)}s` : '-';
202                 return `
203                     <tr>
204                         <td class="text-muted font-monospace small">${dt
    }</td>
205                         <td class="fw-semibold">${userName}</td>
206                         <td><span class="badge ${typeBadge}">${ev.event_
    type || 'action'}</span></td>
207                         <td><span class="badge bg-light text-dark border
    ">${ev.tab_name || '-'}</span></td>
208                         <td class="text-truncate" style="max-width: 250p
    x;" title="${ev.target_element || ''}">${ev.action || ev.targe
    t_element || '-'}</td>
209                         <td class="text-muted">${durStr}</td>
210                     </tr>
211                 `;
212             }).join('');
213         }
214     }

```

```
215 | } catch (e) {
216 |     console.debug('Failed to load telemetry stats:', e);
217 | }
218 | }
219 |
220 | function switchToTab(tabName) {
221 |     if (typeof window.switchTab === 'function') {
222 |         window.switchTab(tabName);
223 |         return;
224 |     }
225 |     const cleanName = tabName.replace(/^tab-/, '');
226 |     const triggerEl = document.querySelector(`[data-bs-target="#
tab-${cleanName}"], [data-tab="${cleanName}"]`);
227 |     if (triggerEl && typeof bootstrap !== 'undefined' && bootstr
ap.Tab) {
228 |         const tab = new bootstrap.Tab(triggerEl);
229 |         tab.show();
230 |     }
231 | }
232 |
233 | // Schedulers Management Logic
234 | async function refreshSchedulersConfig() {
235 |     try {
236 |         const res = await window.api.fetch('/api/admin/schedulers'
);
237 |         if (!res || !res.config) return;
238 |
239 |         const cfg = res.config;
240 |         const status = res.status || {};
241 |
242 |         const masterEl = document.getElementById('sched-master-swi
tch');
243 |         const ragEnabledEl = document.getElementById('sched-rag-en
abled');
244 |         const ragIntervalEl = document.getElementById('sched-rag-i
nterval');
245 |         const emailEnabledEl = document.getElementById('sched-ema
il-enabled');
```

```
246     const emailIntervalEl = document.getElementById('sched-ema
il-interval');
247     const badgeEl = document.getElementById('sched-main-badge'
);
248     const ragLastRunEl = document.getElementById('sched-rag-la
st-run');
249     const emailLastRunEl = document.getElementById('sched-ema
il-last-run');
250
251     if (masterEl) masterEl.checked = cfg.enabled !== false;
252     if (ragEnabledEl) ragEnabledEl.checked = cfg.rag_reindex?.
enabled !== false;
253     if (ragIntervalEl) ragIntervalEl.value = cfg.rag_reindex?.
interval_hours || 24;
254
255     if (emailEnabledEl) emailEnabledEl.checked = cfg.email_che
ck?.enabled !== false;
256     if (emailIntervalEl) emailIntervalEl.value = cfg.email_che
ck?.interval_minutes || 5;
257
258     if (badgeEl) {
259         if (status.running && cfg.enabled) {
260             badgeEl.className = 'badge bg-success-subtle text-succ
ess';
261             badgeEl.textContent = 'Активен (Работает)';
262         } else {
263             badgeEl.className = 'badge bg-secondary-subtle text-se
condary';
264             badgeEl.textContent = 'Отключен';
265         }
266     }
267
268     if (ragLastRunEl) {
269         ragLastRunEl.textContent = status.last_rag_run ? new Dat
e(status.last_rag_run).toLocaleString() : 'Никогда';
270     }
271     if (emailLastRunEl) {
272         emailLastRunEl.textContent = status.last_email_run ? new
```

```

    Date(status.last_email_run).toLocaleString() : 'Никогда';
273     }
274   } catch (err) {
275     console.error('Ошибка загрузки настроек планировщика:', er
    r);
276   }
277 }
278
279 async function saveSchedulersConfig() {
280   try {
281     const masterEl = document.getElementById('sched-master-swi
    tch');
282     const ragEnabledEl = document.getElementById('sched-rag-en
    abled');
283     const ragIntervalEl = document.getElementById('sched-rag-i
    nterval');
284     const emailEnabledEl = document.getElementById('sched-ema
    il-enabled');
285     const emailIntervalEl = document.getElementById('sched-ema
    il-interval');
286
287     const payload = {
288       enabled: masterEl ? masterEl.checked : true,
289       rag_reindex: {
290         enabled: ragEnabledEl ? ragEnabledEl.checked : true,
291         interval_hours: parseFloat(ragIntervalEl ? ragInterval
    El.value : 24) || 24
292       },
293       email_check: {
294         enabled: emailEnabledEl ? emailEnabledEl.checked : tru
    e,
295         interval_minutes: parseFloat(emailIntervalEl ? emailIn
    tervalEl.value : 5) || 5
296       }
297     };
298
299     if (typeof showNotification === 'function') showNotificati
    on('Сохранение конфигурации планировщика...', 'info');

```



```
300
301     const res = await window.api.fetch('/api/admin/schedulers'
    , {
302         method: 'POST',
303         headers: { 'Content-Type': 'application/json' },
304         body: JSON.stringify(payload)
305     });
306
307     if (typeof showNotification === 'function') {
308         showNotification('Настройки планировщика успешно сохране
ны и применены!', 'success');
309     }
310     await refreshSchedulersConfig();
311 } catch (err) {
312     if (typeof showNotification === 'function') {
313         showNotification(`Ошибка сохранения: ${err.message}`, 'd
anger');
314     }
315 }
316 }
317
318 async function triggerSchedulerJob(jobName) {
319     try {
320         if (typeof showNotification === 'function') {
321             showNotification(`Запуск фоновой задачи '${jobName}'...`
, 'info');
322         }
323
324         const res = await window.api.fetch(`/api/admin/schedulers/
trigger/${jobName}`, {
325             method: 'POST'
326         });
327
328         if (jobName === 'rag') {
329             const count = res.result?.updated_count ?? 0;
330             if (typeof showNotification === 'function') {
331                 showNotification(`Переиндексация RAG завершена! Обновл
ено коллекций: ${count}`, 'success');
```

```
332     }
333   } else if (jobName === 'email') {
334     const count = res.count ?? 0;
335     if (typeof showNotification === 'function') {
336       showNotification(`Проверка почты завершена! Найдено не
прочитанных писем: ${count}`, 'success');
337     }
338   }
339
340   await refreshSchedulersConfig();
341 } catch (err) {
342   if (typeof showNotification === 'function') {
343     showNotification(`Ошибка запуска задачи: ${err.message}`
, 'danger');
344   }
345 }
346 }
347
348 window.initAdminTab = initSystemAdminTab;
349 window.rescanStorageDrives = rescanStorageDrives;
350 window.actualizeAiModels = actualizeAiModels;
351 window.switchToTab = switchToTab;
352 window.refreshTelemetryStats = refreshTelemetryStats;
353 window.refreshSchedulersConfig = refreshSchedulersConfig;
354 window.saveSchedulersConfig = saveSchedulersConfig;
355 window.triggerSchedulerJob = triggerSchedulerJob;
356
357
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/agents_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/agents_tab/index.html"
15 |        class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/agents_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-08 03:08:00
23 |=====
24 |=====
25 -->
26
27 <!-- AI Agents Management & Builder Tab -->
28 <style>
29 /* High contrast & modern dark styling for AI Agents Tab */
30 .agent-card {
31     background: #0f172a !important;
32     border: 1px solid #334155 !important;
33     border-radius: 12px;
34     transition: transform 0.15s ease, border-color 0.15s ease,
35     box-shadow 0.15s ease;
```

```
32 | }
33 | .agent-card:hover {
34 |     border-color: #475569 !important;
35 |     box-shadow: 0 6px 20px rgba(0, 0, 0, 0.45);
36 | }
37 | .agent-card .card-header {
38 |     background: #1e293b !important;
39 |     border-bottom: 1px solid #334155 !important;
40 |     color: #ffffff !important;
41 |     border-top-left-radius: 11px;
42 |     border-top-right-radius: 11px;
43 | }
44 | .agent-card .agent-desc {
45 |     color: #cbd5e1 !important;
46 | }
47 | .agent-spec-box {
48 |     background: #020617 !important;
49 |     border: 1px solid #334155 !important;
50 |     border-radius: 8px;
51 |     padding: 0.6rem 0.8rem;
52 | }
53 | .agent-spec-box .spec-label {
54 |     color: #cbd5e1 !important;
55 |     font-weight: 600;
56 |     font-size: 0.82rem;
57 | }
58 | .agent-spec-box .spec-value-badge {
59 |     background-color: #2563eb !important;
60 |     color: #ffffff !important;
61 |     font-weight: 600;
62 |     font-size: 0.78rem;
63 | }
64 | .agent-spec-box .spec-tech-val {
65 |     color: #38bdf8 !important;
66 |     font-weight: 700;
67 |     font-size: 0.85rem;
68 | }
69 | .tool-badge {
```

```
70 | background: #1e293b !important;
71 | border: 1px solid #475569 !important;
72 | color: #f8fafc !important;
73 | font-size: 0.75rem;
74 | font-weight: 500;
75 | border-radius: 6px;
76 | padding: 0.25rem 0.5rem;
77 | display: inline-flex;
78 | align-items: center;
79 | gap: 0.25rem;
80 | }
81 | .agent-card .btn-test-sandbox {
82 |   color: #fbbf24 !important;
83 |   border: 1.5px solid #f59e0b !important;
84 |   font-weight: 600;
85 |   font-size: 0.82rem;
86 |   background: transparent;
87 | }
88 | .agent-card .btn-test-sandbox:hover {
89 |   background: #f59e0b !important;
90 |   color: #0f172a !important;
91 | }
92 | .agent-card .btn-edit-agent {
93 |   color: #38bdf8 !important;
94 |   border: 1.5px solid #0284c7 !important;
95 |   font-weight: 600;
96 |   background: transparent;
97 | }
98 | .agent-card .btn-edit-agent:hover {
99 |   background: #0284c7 !important;
100 |   color: #ffffff !important;
101 | }
102 | .agent-card .btn-delete-agent {
103 |   color: #f87171 !important;
104 |   border: 1.5px solid #dc2626 !important;
105 |   font-weight: 600;
106 |   background: transparent;
107 | }
```

```
108 | .agent-card .btn-delete-agent:hover {
109 |     background: #dc2626 !important;
110 |     color: #ffffff !important;
111 | }
112 | #agents-filter-group .btn {
113 |     background: #1e293b;
114 |     border: 1px solid #475569;
115 |     color: #f1f5f9;
116 |     font-weight: 600;
117 |     font-size: 0.82rem;
118 | }
119 | #agents-filter-group .btn:hover {
120 |     background: #334155;
121 |     color: #ffffff;
122 |     border-color: #64748b;
123 | }
124 | #agents-filter-group .btn.active {
125 |     background: #2563eb !important;
126 |     border-color: #3b82f6 !important;
127 |     color: #ffffff !important;
128 |     box-shadow: 0 0 8px rgba(37, 99, 235, 0.4);
129 | }
130 | #btn-refresh-agents {
131 |     background: #1e293b;
132 |     border: 1px solid #475569;
133 |     color: #f1f5f9;
134 |     font-weight: 600;
135 |     font-size: 0.82rem;
136 | }
137 | #btn-refresh-agents:hover {
138 |     background: #334155;
139 |     color: #ffffff;
140 |     border-color: #64748b;
141 | }
142 | </style>
143 |
144 | <div class="container-fluid py-3">
145 |
```

```

146 | <!-- Header Alert & Quick Actions -->
147 | <div class="card shadow-sm border-secondary-subtle p-3 mb-3
    | d-flex justify-content-between align-items-center flex-wrap ga
    | p-3">
148 |     <div>
149 |         <h5 class="mb-1 fw-bold d-flex align-items-center gap-2"
    | >
150 |             <span class="fs-4">□ </span> Управление и создание ИИ-
    | Агентов
151 |         </h5>
152 |         <p class="mb-0 small" style="color: #cbd5e1;">
153 |             Конфигурирование специализированных агентов, назначени
    | е моделей из пула, подключение инструментов и генерация через
    | AI.
154 |         </p>
155 |     </div>
156 |     <div class="d-flex align-items-center gap-2 flex-wrap">
157 |         <button class="btn btn-sm btn-outline-info rounded-pill
    | px-3 shadow-sm fw-semibold" id="btn-open-ai-builder" style="bo
    | rder-width: 1.5px;">
158 |             <i class="bi bi-magic me-1"></i> □ Создать с помощью A
    | I
159 |         </button>
160 |         <button class="btn btn-sm btn-primary rounded-pill px-3
    | shadow-sm fw-semibold" id="btn-open-create-agent">
161 |             <i class="bi bi-plus-lg me-1"></i> Добавить вручную
162 |         </button>
163 |     </div>
164 | </div>
165 |
166 | <!-- Collapsible Architecture & SVG Scheme Block -->
167 | <div class="card mb-4 shadow-sm text-white" style="backgroun
    | d-color: #0f172a !important; border: 1px solid #334155 !import
    | ant; border-radius: 10px;">
168 |     <div class="card-body p-3">
169 |         <div class="d-flex align-items-center justify-content-be
    | tween">
170 |             <div class="d-flex align-items-center gap-2">

```

```
171 |         <i class="bi bi-diagram-3-fill text-info fs-5"></i>
172 |         <span class="fw-bold" style="color: #f1f5f9;">Интера
| ктивная справка: архитектура работы и создания агентов</span>
173 |     </div>
174 |     <button class="btn btn-sm btn-link text-decoration-non
| e p-0 fw-semibold" style="color: #38bdf8;" type="button" data-
| bs-toggle="collapse" data-bs-target="#agentsArchCollapse" aria-
| -expanded="false" aria-controls="agentsArchCollapse">
175 |         <i class="bi bi-chevron-down"></i> Развернуть / Скры
| ть схему
176 |     </button>
177 | </div>
178 | <div class="collapse" id="agentsArchCollapse">
179 |     <div class="mt-3 pt-3" style="border-top: 1px solid rg
| ba(255, 255, 255, 0.1);">
180 |         <div class="text-center p-2 rounded mb-3" style="bac
| kground: #020617; border: 1px solid rgba(255, 255, 255, 0.08);
| ">
181 |             
182 |         </div>
183 |         <div class="row g-3 small">
184 |             <div class="col-md-3">
185 |                 <div class="p-3 rounded h-100" style="background
| : #1e293b; border: 1px solid #334155; color: #f1f5f9; line-hei
| ght: 1.55;">
186 |                     <strong class="d-block mb-1" style="color: #38
| bdf8; font-size: 0.88rem;">1. Пул моделей (Model Pool)</strong
| >
187 |                         Каждый агент может использовать независимую мо
| дель: Google Gemini (с ротацией ключей), AntigraVity AGY, Micr
| osoft Foundry или Ollama.
188 |                 </div>
189 |             </div>
190 |             <div class="col-md-3">
191 |                 <div class="p-3 rounded h-100" style="background
```



```
    : #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
192      <strong class="d-block mb-1" style="color: #4ade80; font-size: 0.88rem;">2. Каталог инструментов (Tools)</strong>
193      Набор готовых модулей: интернет-поиск (MCP / Google Grounding), RAG база знаний, Python Code Execution, чтение файлов и синтез речи TTS.
194      </div>
195    </div>
196    <div class="col-md-3">
197      <div class="p-3 rounded h-100" style="background: #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
198        <strong class="d-block mb-1" style="color: #fbf224; font-size: 0.88rem;">3. AI-Архитектор промптов</strong>
199        Модель из пула анализирует вашу цель и генерирует структурированную роль, правила ReAct и список инструментов.
200      </div>
201    </div>
202    <div class="col-md-3">
203      <div class="p-3 rounded h-100" style="background: #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
204        <strong class="d-block mb-1" style="color: #f87171; font-size: 0.88rem;">4. Песочница (Sandbox)</strong>
205        Тестирование поведения агента, отслеживание цепочки рассуждений (Thought & Action) и замер задержек ответа до боевого использования.
206      </div>
207    </div>
208  </div>
209 </div>
210 </div>
211 </div>
212 </div>
213
```

```

214 | <!-- Filters and Status Toolbar -->
215 | <div class="d-flex justify-content-between align-items-center mb-3 flex-wrap gap-2">
216 |     <div class="btn-group btn-group-sm" role="group" id="agents-filter-group">
217 |         <button type="button" class="btn active" data-filter="all">Все (<span id="count-all">0</span>)</button>
218 |         <button type="button" class="btn" data-filter="active">Активные (<span id="count-active">0</span>)</button>
219 |         <button type="button" class="btn" data-filter="system">Системные (<span id="count-system">0</span>)</button>
220 |         <button type="button" class="btn" data-filter="custom">Пользовательские (<span id="count-custom">0</span>)</button>
221 |     </div>
222 |     <div class="d-flex align-items-center gap-2">
223 |         <button class="btn btn-sm rounded-pill px-3" id="btn-refresh-agents" title="Обновить список">
224 |             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
225 |         </button>
226 |     </div>
227 | </div>
228 |
229 | <!-- Agents Grid -->
230 | <div class="row g-3" id="agents-grid">
231 |     <div class="col-12 text-center py-5" style="color: #cbd5e1;">
232 |         <div class="spinner-border text-primary mb-2" role="status"></div>
233 |         <div class="fw-semibold">Загрузка агентов и пула моделей...</div>
234 |     </div>
235 | </div>
236 |
237 | </div>
238 |
239 | <!-- =====
240 | <!-- MODAL: AI Agent Builder (Генерация через LLM)

```

```

-->
241 <!-- =====
===== -->
242 <div class="modal fade" id="modal-ai-builder" tabindex="-1" ar
ia-labelledby="modalAiBuilderLabel" aria-hidden="true">
243   <div class="modal-dialog modal-lg modal-dialog-centered">
244     <div class="modal-content text-white" style="background-co
lor: #0f172a !important; border: 1px solid #334155 !important;
border-radius: 12px;">
245       <div class="modal-header" style="background-color: #1e29
3b !important; border-bottom: 1px solid #334155 !important;">
246         <h5 class="modal-title fw-bold" style="color: #38bdf8;
" id="modalAiBuilderLabel">
247           <i class="bi bi-magic me-2"></i>AI Agent Builder – К
онструктор агентов
248         </h5>
249         <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
250       </div>
251       <div class="modal-body p-4">
252         <p class="small mb-3" style="color: #cbd5e1; line-heig
ht: 1.5;">
253           Опишите задачу агента своими словами. Выбранная моде
ль из пула проанализирует цель, подберёт необходимые инструмен
ты и сформирует профессиональный системный промпт.
254         </p>
255
256         <div class="row g-3 mb-3">
257           <div class="col-md-6">
258             <label class="form-label small fw-semibold" style=
"color: #f1f5f9;">Провайдер модели-архитектора:</label>
259             <select class="form-select form-select-sm" style="
background-color: #020617 !important; color: #f8fafc !important;
border: 1px solid #475569 !important;" id="ai-builder-provi
der">
260               <option value="gemini">Google Gemini</option>
261               <option value="gemini_cli">Google Gemini CLI</op
tion>

```

```
262 |         <option value="agy">Google Antigravity (AGY)</op
| tion>
263 |         <option value="foundry">Microsoft Foundry (Local
| )</option>
264 |         <option value="ollama">Ollama (Local)</option>
265 |     </select>
266 | </div>
267 |     <div class="col-md-6">
268 |         <label class="form-label small fw-semibold" style=
| "color: #f1f5f9;">Модель из пула:</label>
269 |         <select class="form-select form-select-sm" style="
| background-color: #020617 !important; color: #f8fafc !importan
| t; border: 1px solid #475569 !important;" id="ai-builder-model
| ">
270 |             <!-- Заполняется динамически -->
271 |         </select>
272 |     </div>
273 | </div>
274 |
275 |     <div class="mb-3">
276 |         <label class="form-label small fw-semibold" style="c
| olor: #f1f5f9;">Опишите задачу агента:</label>
277 |         <textarea class="form-control" style="background-col
| or: #020617 !important; color: #f8fafc !important; border: 1px
| solid #475569 !important;" id="ai-builder-prompt" rows="4" pl
| aceholder="Например: Создай агента для автономного поиска субт
| итров к фильмам, проверки их таймингов и сопоставления с медиа
| текой..."></textarea>
278 |     </div>
279 |
280 |     <div class="d-flex justify-content-end mb-3">
281 |         <button class="btn btn-info rounded-pill px-4 fw-bol
| d" id="btn-run-ai-generate" style="background-color: #0284c7 !
| important; border-color: #0284c7 !important; color: #ffffff !i
| mportant;">
282 |         <i class="bi bi-stars me-1"></i> Сгенерировать спе
| цификацию
283 |     </button>
```

```

284         </div>
285
286         <!-- Result Preview Section -->
287         <div id="ai-builder-result" class="d-none pt-3" style=
"border-top: 1px solid #334155;">
288             <h6 class="fw-bold mb-2" style="color: #4ade80;"><i
class="bi bi-check2-circle me-1"></i> Спецификация сформирован
a:</h6>
289             <div class="p-3 rounded mb-3 small font-monospace" s
tyle="background: #020617; border: 1px solid #334155; color: #
f8fafc;" id="ai-builder-preview-box">
290                 <!-- Превью JSON / данных -->
291             </div>
292             <div class="d-flex justify-content-end gap-2">
293                 <button class="btn btn-outline-secondary btn-sm ro
unded-pill px-3" data-bs-dismiss="modal" style="color: #cbd5e1
; border-color: #475569;">Отмена</button>
294                 <button class="btn btn-success btn-sm rounded-pill
px-3 fw-semibold" id="btn-apply-ai-generated">
295                     <i class="bi bi-pencil-square me-1"></i> Открыть
в редакторе агента
296                 </button>
297             </div>
298         </div>
299     </div>
300 </div>
301 </div>
302 </div>
303
304 <!-- =====
===== -->
305 <!-- MODAL: Agent Editor (Создание и Редактирование)
-->
306 <!-- =====
===== -->
307 <div class="modal fade" id="modal-agent-editor" tabindex="-1"
aria-labelledby="modalAgentEditorLabel" aria-hidden="true">
308     <div class="modal-dialog modal-xl modal-dialog-centered moda

```

```
l-dialog-scrollable">
309     <div class="modal-content text-white" style="background-co
lor: #0f172a !important; border: 1px solid #334155 !important;
border-radius: 12px;">
310     <div class="modal-header" style="background-color: #1e29
3b !important; border-bottom: 1px solid #334155 !important;">
311         <h5 class="modal-title fw-bold text-white" id="modalAg
entEditorLabel">
312             <i class="bi bi-robot me-2" style="color: #38bdf8;">
</i>Настройка агента
313         </h5>
314         <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
315     </div>
316     <div class="modal-body p-4">
317         <form id="agent-form">
318             <input type="hidden" id="agent-edit-mode" value="cre
ate">
319
320             <div class="row g-3 mb-3">
321                 <div class="col-md-4">
322                     <label class="form-label small fw-semibold" styl
e="color: #f1f5f9;">Уникальный ID агента:</label>
323                     <input type="text" class="form-control form-cont
rol-sm font-monospace" style="background-color: #020617 !impor
tant; color: #f8fafc !important; border: 1px solid #475569 !im
portant;" id="agent-id" placeholder="my_custom_agent" required
>
324                 </div>
325                 <div class="col-md-5">
326                     <label class="form-label small fw-semibold" styl
e="color: #f1f5f9;">Отображаемое имя:</label>
327                     <input type="text" class="form-control form-cont
rol-sm" style="background-color: #020617 !important; color: #f
8fafc !important; border: 1px solid #475569 !important;" id="a
gent-name" placeholder="Название агента" required>
328                 </div>
329                 <div class="col-md-3">
```

```

330         <label class="form-label small fw-semibold" styl
e="color: #f1f5f9;">Статус активности:</label>
331         <div class="form-check form-switch mt-1">
332             <input class="form-check-input" type="checkbox"
" id="agent-enabled" checked>
333             <label class="form-check-label small fw-medium
" style="color: #e2e8f0;" for="agent-enabled">Агент активен</l
abel>
334         </div>
335     </div>
336 </div>
337
338     <div class="mb-3">
339         <label class="form-label small fw-semibold" style=
"color: #f1f5f9;">Краткое описание роли:</label>
340         <input type="text" class="form-control form-contro
l-sm" style="background-color: #020617 !important; color: #f8f
afc !important; border: 1px solid #475569 !important;" id="age
nt-desc" placeholder="Какую задачу решает данный агент">
341     </div>
342
343     <!-- Model & Provider Selection from Pool -->
344     <div class="card mb-3" style="background: #0b0f19 !i
mportant; border: 1px solid #334155 !important; border-radius:
8px;">
345         <div class="card-header py-2" style="background: #
1e293b !important; border-bottom: 1px solid #334155 !important
;">
346             <span class="fw-bold small" style="color: #38bdf
8;"><i class="bi bi-cpu me-1"></i> Привязка к модели из пула</
span>
347         </div>
348         <div class="card-body p-3">
349             <div class="row g-3">
350                 <div class="col-md-6">
351                     <label class="form-label small fw-semibold"
style="color: #f1f5f9;">Провайдер ИИ:</label>
352                     <select class="form-select form-select-sm" s

```

```

style="background-color: #020617 !important; color: #f8fafc !im
portant; border: 1px solid #475569 !important;" id="agent-prov
ider">
353         <option value="gemini">Google Gemini</opti
on>
354         <option value="gemini_cli">Google Gemini C
LI</option>
355         <option value="agy">Google Antigravity (AG
Y)</option>
356         <option value="foundry">Microsoft Foundry<
/option>
357         <option value="ollama">Ollama (Local)</opt
ion>
358     </select>
359 </div>
360 <div class="col-md-6">
361     <label class="form-label small fw-semibold"
style="color: #f1f5f9;">Модель:</label>
362     <select class="form-select form-select-sm" s
tyle="background-color: #020617 !important; color: #f8fafc !im
portant; border: 1px solid #475569 !important;" id="agent-mode
l">
363         <!-- Заполняется динамически -->
364     </select>
365 </div>
366 </div>
367
368 <div class="row g-3 mt-1">
369     <div class="col-md-4">
370         <label class="form-label small fw-semibold d
-flex justify-content-between" style="color: #f1f5f9;">
371             <span>Температура:</span> <span id="agent-
temp-val" class="fw-bold" style="color: #38bdf8;">0.3</span>
372         </label>
373         <input type="range" class="form-range" id="a
gent-temperature" min="0" max="1" step="0.05" value="0.3">
374     </div>
375     <div class="col-md-4">

```



```

376         <label class="form-label small fw-semibold"
style="color: #f1f5f9;">Макс. шагов ReAct (Max Steps):</label>
377         <input type="number" class="form-control for
m-control-sm" style="background-color: #020617 !important; col
or: #f8fafc !important; border: 1px solid #475569 !important;"
id="agent-max-steps" min="1" max="50" value="15">
378     </div>
379     <div class="col-md-4">
380         <label class="form-label small fw-semibold"
style="color: #f1f5f9;">Таймаут (сек):</label>
381         <input type="number" class="form-control for
m-control-sm" style="background-color: #020617 !important; col
or: #f8fafc !important; border: 1px solid #475569 !important;"
id="agent-timeout" min="5" max="300" value="60">
382     </div>
383 </div>
384 </div>
385 </div>
386
387 <!-- Tools Selection Matrix -->
388 <div class="card mb-3" style="background: #0b0f19 !i
mportant; border: 1px solid #334155 !important; border-radius:
8px;">
389     <div class="card-header py-2 d-flex justify-conten
t-between align-items-center" style="background: #1e293b !impo
rtant; border-bottom: 1px solid #334155 !important;">
390         <span class="fw-bold small" style="color: #4ade8
0;"><i class="bi bi-tools me-1"></i> Подключенные инструменты
(Capabilities)</span>
391         <span class="badge" style="background: #334155;
color: #f8fafc; font-weight: 600;" id="agent-tools-count">0 вы
брано</span>
392     </div>
393     <div class="card-body p-3">
394         <div class="row g-2" id="agent-tools-matrix">
395             <!-- Чекбоксы инструментов загружаются динамич
ески -->
396         </div>

```

```
397 |         </div>
398 |     </div>
399 |
400 |     <!-- System Instruction Editor -->
401 |     <div class="mb-3">
402 |         <div class="d-flex justify-content-between align-i
403 | items-center mb-1">
404 |             <label class="form-label small fw-semibold mb-0"
405 | style="color: #f1f5f9;">Системная инструкция (System Prompt):
406 | </label>
407 |             <div class="btn-group btn-group-sm">
408 |                 <button type="button" class="btn btn-outline-s
409 | econdary btn-sm py-0" style="color: #cbd5e1; border-color: #47
410 | 5569;" id="btn-insert-react-template">Шаблон ReAct</button>
411 |                 <button type="button" class="btn btn-outline-s
412 | econdary btn-sm py-0" style="color: #cbd5e1; border-color: #47
413 | 5569;" id="btn-insert-json-template">Шаблон JSON-ответа</butto
414 | n>
415 |             </div>
416 |         </div>
417 |         <textarea class="form-control font-monospace" styl
418 | e="background-color: #020617 !important; color: #f8fafc !impor
419 | tant; border: 1px solid #475569 !important; font-size: 0.85rem
420 | ; line-height: 1.45;" id="agent-system-prompt" rows="7" placeh
421 | older="Опишите поведение агента, правила вызова инструментов и
422 | формат ответа..." spellcheck="false" required></textarea>
423 |     </div>
424 |
425 | </form>
426 | </div>
427 |     <div class="modal-footer" style="background-color: #1e29
428 | 3b !important; border-top: 1px solid #334155 !important;">
429 |         <button type="button" class="btn btn-outline-secondary
430 | rounded-pill px-3" data-bs-dismiss="modal" style="color: #cbd
431 | 5e1; border-color: #475569;">Отмена</button>
432 |         <button type="button" class="btn btn-primary rounded-p
433 | ill px-4 fw-bold" id="btn-save-agent">
434 |         <i class="bi bi-check-circle me-1"></i> Сохранить аг
```

```

    ента
418         </button>
419     </div>
420 </div>
421 </div>
422 </div>
423
424 <!-- =====
    ===== -->
425 <!-- MODAL: Agent Sandbox (Песочница тестирования)
    -->
426 <!-- =====
    ===== -->
427 <div class="modal fade" id="modal-agent-sandbox" tabindex="-1"
    aria-labelledby="modalAgentSandboxLabel" aria-hidden="true">
428     <div class="modal-dialog modal-lg modal-dialog-centered moda
    l-dialog-scrollable">
429         <div class="modal-content text-white" style="background-co
    lor: #0f172a !important; border: 1px solid #334155 !important;
    border-radius: 12px;">
430             <div class="modal-header" style="background-color: #1e29
    3b !important; border-bottom: 1px solid #334155 !important;">
431                 <h5 class="modal-title fw-bold" style="color: #fbbf24;
    " id="modalAgentSandboxLabel">
432                     <i class="bi bi-play-circle me-2"></i> Песочница т
    естирования агента
433                 </h5>
434                 <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="modal" aria-label="Close"></button>
435             </div>
436             <div class="modal-body p-4">
437
438                 <!-- Active Agent Summary in Sandbox -->
439                 <div class="d-flex align-items-center justify-content-
    between p-2 px-3 rounded mb-3" style="background: #020617; bor
    der: 1px solid #334155;">
440                     <div>
441                         <strong id="sandbox-agent-name" class="text-white

```

```
fs-6">—</strong>
442         <span class="badge ms-2" style="background-color:
#2563eb; color: #ffffff; font-weight: 600;" id="sandbox-agent-
model">—</span>
443         </div>
444         <span class="small fw-medium" style="color: #cbd5e1;
" id="sandbox-agent-tools-summary">Инструментов: 0</span>
445     </div>
446
447     <!-- Chat / Trace Container -->
448     <div class="rounded p-3 mb-3" style="background: #0206
17; border: 1px solid #334155; min-height: 240px; max-height:
380px; overflow-y: auto;" id="sandbox-chat-window">
449         <div class="text-center small py-4" style="color: #9
4a3b8;" id="sandbox-placeholder">
450             <i class="bi bi-chat-dots fs-3 d-block mb-1"></i>
451             Введите тестовый запрос ниже для проверки логики а
гента и цепочки вызовов инструментов.
452         </div>
453         <div id="sandbox-messages-list"></div>
454     </div>
455
456     <!-- Input Box -->
457     <div class="input-group">
458         <input type="text" class="form-control" style="backg
round-color: #020617 !important; color: #f8fafc !important; bo
rder: 1px solid #475569 !important;" id="sandbox-input-msg" pl
aceholder="Например: Найди фильм Интерстеллар и проверь торрен
ты...">
459         <button class="btn btn-warning fw-bold px-4" id="btn
-sandbox-send" style="background-color: #f59e0b; border-color:
#f59e0b; color: #0f172a;">
460             <i class="bi bi-send-fill me-1"></i> Тест
461         </button>
462     </div>
463
464 </div>
465 <div class="modal-footer" style="background-color: #1e29
```

```
3b !important; border-top: 1px solid #334155 !important; ">
466 |     <button type="button" class="btn btn-outline-secondary
    |     rounded-pill px-3" data-bs-dismiss="modal" style="color: #cbd
    |     5e1; border-color: #475569;">Закрыть песочницу</button>
467 |     </div>
468 | </div>
469 | </div>
470 | </div>
471 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Agents Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/agents_tab/main.js?v=2
11  * 0261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/agents_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 'use strict';
22
23 /**
24  * Global reactive state for the AI Agents tab.
25  */
26 const _state = {
27   agents: [],
28   tools: [],
29   providers: {},
30   currentFilter: 'all',
31   activeSandboxAgent: null,
32   lastGeneratedSpec: null,
```

```

33 |   isEventsBound: false
34 | };
35 |
36 | /**
37 |  * Initializes the AI Agents management tab.
38 |  * Called on tab switch from the main admin interface.
39 |  *
40 |  * @returns {Promise<void>}
41 |  */
42 | async function initAgentsTab() {
43 |   _bindStaticEvents();
44 |   await _loadAllData();
45 | }
46 |
47 | window.initAgentsTab = initAgentsTab;
48 |
49 | /**
50 |  * Executes an HTTP JSON request using window.api or standard
51 |  * fetch.
52 |  *
53 |  * @param {string} endpoint - Target API endpoint path.
54 |  * @param {RequestInit} [options={}] - Optional fetch configuration options.
55 |  * @returns {Promise<any>} Parsed JSON response.
56 |  * @throws {Error} Thrown if response status is not OK.
57 |  */
58 | async function _fetchJson(endpoint, options = {}) {
59 |   if (window.api && typeof window.api.fetch === 'function') {
60 |     return await window.api.fetch(endpoint, options);
61 |   }
62 |   const response = await fetch(endpoint, options);
63 |   if (!response.ok) {
64 |     throw new Error(`HTTP ${response.status}: ${response.statusText}`);
65 |   }
66 |   return await response.json();
67 | }

```

```

68 | /**
69 |  * Loads agents, capabilities catalog, and model providers from the backend API.
70 |  *
71 |  * @returns {Promise<void>}
72 |  */
73 | async function _loadAllData() {
74 |   try {
75 |     const [agentsRes, toolsRes, providersRes] = await Promise.all([
76 |       _fetchJson('/api/agents'),
77 |       _fetchJson('/api/agents/tools'),
78 |       _fetchJson('/api/agents/providers')
79 |     ]);
80 |
81 |     _state.agents = Array.isArray(agentsRes) ? agentsRes : [];
82 |     _state.tools = Array.isArray(toolsRes) ? toolsRes : [];
83 |     _state.providers = providersRes ?? {};
84 |
85 |     _renderCounters();
86 |     _renderGrid();
87 |     _populateToolsMatrix();
88 |     _updateModelDropdown('agent-provider', 'agent-model');
89 |     _updateModelDropdown('ai-builder-provider', 'ai-builder-model');
90 |   } catch (err) {
91 |     console.error('[AgentsTab] Failed to load data:', err);
92 |     _showToast(`Failed to load data: ${err.message}`, 'danger');
93 |
94 |     const grid = document.getElementById('agents-grid');
95 |     if (grid) {
96 |       grid.innerHTML = `
97 |         <div class="col-12 text-center py-5 text-danger">
98 |           <i class="bi bi-exclamation-triangle fs-1 d-block mb-2"></i>
99 |           <h5>Не удалось загрузить данные агентов</h5>
100 |           <p class="small text-secondary">${err.message}</p>
           <button class="btn btn-sm btn-outline-primary rounded

```



```

        d-pill px-3" onclick="window.initAgentsTab()">Повторить попытку</button>
    </div>
    `;
}
}
}

/**
 * Binds static DOM event listeners for buttons, filters, and
 * modal templates.
 *
 * @returns {void}
 */
function _bindStaticEvents() {
    if (_state.isEventsBound) return;
    _state.isEventsBound = true;

    // Refresh agents list
    const refreshBtn = document.getElementById('btn-refresh-agents');
    if (refreshBtn) {
        refreshBtn.onclick = async () => {
            refreshBtn.disabled = true;
            await _loadAllData();
            refreshBtn.disabled = false;
            _showToast('Список агентов обновлен', 'success');
        };
    }

    // Filter toolbar buttons
    const filterGroup = document.getElementById('agents-filter-group');
    if (filterGroup) {
        filterGroup.querySelectorAll('button').forEach((btn) => {
            btn.onclick = () => {
                filterGroup.querySelectorAll('button').forEach((b) =>
                    b.classList.remove('active'));
            };
        });
    }
}

```

```
133 |         btn.classList.add('active');
134 |         _state.currentFilter = btn.getAttribute('data-filter')
    |         ?? 'all';
135 |         _renderGrid();
136 |     };
137 | });
138 | }
139 |
140 | // Modal trigger buttons
141 | document.getElementById('btn-open-create-agent')?.addEventListener(
    | stener('click', () => _openEditor(null));
142 | document.getElementById('btn-open-ai-builder')?.addEventListener(
    | ener('click', _openAiBuilder);
143 |
144 | // Provider change handlers
145 | document.getElementById('agent-provider')?.addEventListener(
    | 'change', () => {
146 |     _updateModelDropdown('agent-provider', 'agent-model');
147 | });
148 | document.getElementById('ai-builder-provider')?.addEventListener(
    | ener('change', () => {
149 |     _updateModelDropdown('ai-builder-provider', 'ai-builder-mo
    | del');
150 | });
151 |
152 | // Temperature slider synchronization
153 | const tempSlider = document.getElementById('agent-temperatur
    | e');
154 | const tempVal = document.getElementById('agent-temp-val');
155 | if (tempSlider && tempVal) {
156 |     tempSlider.oninput = () => {
157 |         tempVal.textContent = tempSlider.value;
158 |     };
159 | }
160 |
161 | // Action buttons
162 | document.getElementById('btn-save-agent')?.addEventListener(
    | 'click', _handleSaveAgent);
```

```
163 | document.getElementById('btn-run-ai-generate')?.addEventListener('click', _handleRunAiGenerate);
164 |
165 | // Apply AI Generated specification to editor
166 | document.getElementById('btn-apply-ai-generated')?.addEventListener('click', () => {
167 |     if (!_state.lastGeneratedSpec) return;
168 |     const builderModalEl = document.getElementById('modal-ai-builder');
169 |     bootstrap.Modal.getInstance(builderModalEl)?.hide();
170 |     _openEditor(null, _state.lastGeneratedSpec);
171 | });
172 |
173 | // System Prompt templates
174 | document.getElementById('btn-insert-react-template')?.addEventListener('click', () => {
175 |     const promptInput = document.getElementById('agent-system-prompt');
176 |     if (promptInput) {
177 |         promptInput.value =
178 |             'Ты автономный ReAct-агент Mediteka.\n\n' +
179 |             'ПРИНЦИП РАБОТЫ (Thought -> Action -> Observation):\n'
180 |             +
181 |             '1. Тщательно анализируй вопрос пользователя (Thought).\n' +
182 |             '2. Выбирай необходимый инструмент из доступных (Action) с корректными параметрами.\n' +
183 |             '3. Анализируй результат выполнения инструмента (Observation).\n' +
184 |             '4. Сформируй итоговый структурированный ответ пользователю.\n\n' +
185 |             'ФОРМАТ ОТВЕТА:\n' +
186 |             '- Используй Markdown с красивым форматированием (списки, жирный шрифт, эмодзи).\n' +
187 |             '- Не выводите технические ошибки напрямую, объясняй суть решения.';
188 |     }
```

```
189 |
190 | document.getElementById('btn-insert-json-template')?.addEventListener('click', () => {
191 |     const promptInput = document.getElementById('agent-system-prompt');
192 |     if (promptInput) {
193 |         promptInput.value =
194 |             'Ты специализированный аналитический агент Mediteka.\n\n' +
195 |             'Твоя задача: структурировать полученные данные и всегда отвечать строго валидным JSON-объектом без лишнего обрамления и markdown-блоков.\n\n' +
196 |             'Пример структуры:\n{\n  "status": "success",\n  "items": [],\n  "summary": "краткое резюме"\n}';
197 |     }
198 | });
199 |
200 | // Sandbox inputs
201 | const sandboxSendBtn = document.getElementById('btn-sandbox-send');
202 | const sandboxInput = document.getElementById('sandbox-input-msg');
203 | if (sandboxSendBtn && sandboxInput) {
204 |     sandboxSendBtn.onclick = _handleRunSandboxTest;
205 |     sandboxInput.onkeypress = (e) => {
206 |         if (e.key === 'Enter') _handleRunSandboxTest();
207 |     };
208 | }
209 | }
210 |
211 | /**
212 |  * Updates filter counter badges on top toolbar buttons.
213 |  *
214 |  * @returns {void}
215 |  */
216 | function _renderCounters() {
217 |     const allCount = _state.agents.length;
218 |     const activeCount = _state.agents.filter((a) => a.enabled).length;
```

```

length;
219   const systemCount = _state.agents.filter((a) => a.is_system)
    .length;
220   const customCount = _state.agents.filter((a) => !a.is_system)
    .length;
221
222   document.getElementById('count-all')?.replaceChildren(docume
    nt.createTextNode(String(allCount)));
223   document.getElementById('count-active')?.replaceChildren(doc
    ument.createTextNode(String(activeCount)));
224   document.getElementById('count-system')?.replaceChildren(doc
    ument.createTextNode(String(systemCount)));
225   document.getElementById('count-custom')?.replaceChildren(doc
    ument.createTextNode(String(customCount)));
226 }
227
228 /**
229  * Renders the responsive agent cards grid based on current fi
    lter state.
230  *
231  * @returns {void}
232  */
233 function _renderGrid() {
234   const grid = document.getElementById('agents-grid');
235   if (!grid) return;
236
237   const filter = _state.currentFilter;
238   let items = _state.agents;
239
240   if (filter === 'active') items = items.filter((a) => a.enabl
    ed);
241   else if (filter === 'system') items = items.filter((a) => a.
    is_system);
242   else if (filter === 'custom') items = items.filter((a) => !a
    .is_system);
243
244   if (items.length === 0) {
245     grid.innerHTML = `

```

```

246 |     <div class="col-12 text-center py-5" style="color: #cbd5
    | e1;">
247 |         <i class="bi bi-robot fs-1 d-block mb-2" style="color:
    | #38bdf8;"></i>
248 |         <h5 class="text-white">Агенты не найдены</h5>
249 |         <p class="small" style="color: #94a3b8;">Создайте ново
    | го агента с помощью AI или вручную</p>
250 |     </div>
251 |     `;
252 |     return;
253 | }
254 |
255 | grid.innerHTML = items
256 |   .map((agent) => {
257 |     const isSystem = Boolean(agent.is_system);
258 |     const isEnabled = Boolean(agent.enabled);
259 |     const providerName = _getProviderDisplayName(agent.provi
    | der);
260 |
261 |     const toolsBadges = (agent.tools ?? [])
262 |       .map((toolId) => {
263 |         const tool = _state.tools.find((t) => t.id === toolI
    | d);
264 |         const icon = tool?.icon ?? '❏';
265 |         const name = tool?.name ?? toolId;
266 |         return `<span class="badge tool-badge me-1 mb-1 font
    | -monospace">${icon} <span>${name}</span></span>`;
267 |       })
268 |       .join('');
269 |
270 |     return `
271 |       <div class="col-xl-4 col-lg-6">
272 |         <div class="card h-100 agent-card shadow-sm" style="
    | ${!isEnabled ? 'opacity: 0.65;' : ''}">
273 |           <div class="card-header d-flex justify-content-bet
    | ween align-items-center py-2 px-3">
274 |             <div class="d-flex align-items-center gap-2">
275 |               <span class="fs-5">${isSystem ? '❏' : '❏'}</s

```

```

pan>
276         <strong class="text-white fs-6 fw-bold">${agen
t.name}</strong>
277         ${
278             isSystem
279             ? '<span class="badge fw-bold" style="font
-size: 0.7rem; background-color: #0284c7 !important; color: #f
ffffff !important;">SYSTEM</span>'
280             : '<span class="badge fw-bold" style="font
-size: 0.7rem; background-color: #d97706 !important; color: #f
ffffff !important;">CUSTOM</span>'
281         }
282     </div>
283     <div class="form-check form-switch m-0" title="В
ключить / Выключить">
284         <input class="form-check-input agent-toggle-sw
itch" type="checkbox" data-agent-id="${agent.id}" ${isEnabled
? 'checked' : ''}>
285     </div>
286 </div>
287     <div class="card-body p-3 d-flex flex-column">
288         <p class="small mb-3 flex-grow-1 agent-desc" sty
le="min-height: 40px; color: #cbd5e1; line-height: 1.45;">${ag
ent.description || 'Описание отсутствует'}</p>
289
290         <div class="agent-spec-box mb-3 small">
291             <div class="d-flex justify-content-between ali
gn-items-center mb-1 pb-1" style="border-bottom: 1px solid rgb
a(255, 255, 255, 0.08);">
292                 <span class="spec-label">Провайдер &amp; Мод
ель:</span>
293                 <span class="badge spec-value-badge text-tru
uncate" style="max-width: 190px;">${providerName}: ${agent.mode
l}</span>
294             </div>
295             <div class="d-flex justify-content-between ali
gn-items-center">
296                 <span class="spec-label">Температура / Шаги:

```

```

    </span>
297         <span class="spec-tech-val font-monospace">T
        : ${agent.temperature} <span style="color: #64748b;">|</span>
        Max: ${agent.max_steps}</span>
298     </div>
299 </div>
300
301     <div class="mb-3">
302         <div class="small fw-semibold mb-2" style="col
        or: #f1f5f9;">Инструменты (${(agent.tools ?? []).length}):</di
        v>
303         <div class="d-flex flex-wrap">${toolsBadges ||
        ' <span class="small" style="color: #94a3b8;">Без внешних инст
        рументов</span>' }</div>
304     </div>
305
306     <div class="mt-auto d-flex justify-content-betwe
        en gap-2 pt-2" style="border-top: 1px solid #334155;">
307         <button class="btn btn-sm rounded-pill px-3 bt
        n-test-sandbox" data-agent-id="${agent.id}">
308             <i class="bi bi-play-fill"></i> Тест в Sandb
        ox
309         </button>
310         <div class="d-flex gap-1">
311             <button class="btn btn-sm rounded-pill px-2
        btn-edit-agent" data-agent-id="${agent.id}" title="Редактирова
        ть">
312                 <i class="bi bi-pencil-fill"></i>
313             </button>
314             ${
315                 !isSystem
316                 ? `
317                 <button class="btn btn-sm rounded-pill px-
        2 btn-delete-agent" data-agent-id="${agent.id}" title="Удалить
        ">
318                     <i class="bi bi-trash-fill"></i>
319                 </button>
320                 `

```



```

321 |             : ''
322 |         }
323 |     </div>
324 | </div>
325 | </div>
326 | </div>
327 | </div>
328 | `;
329 | })
330 | .join('');
331 |
332 | // Attach card event listeners
333 | grid.querySelectorAll('.agent-toggle-switch').forEach((sw) =
334 | > {
335 |     sw.onchange = async () => {
336 |         const agentId = sw.getAttribute('data-agent-id');
337 |         if (agentId) await _toggleAgentState(agentId, sw.checked
338 | );
339 |     };
340 | });
341 |
342 | grid.querySelectorAll('.btn-test-sandbox').forEach((btn) =>
343 | {
344 |     btn.onclick = () => {
345 |         const agentId = btn.getAttribute('data-agent-id');
346 |         if (agentId) _openSandbox(agentId);
347 |     };
348 | });
349 |
350 | grid.querySelectorAll('.btn-edit-agent').forEach((btn) => {
351 |     btn.onclick = () => {
352 |         const agentId = btn.getAttribute('data-agent-id');
353 |         if (agentId) _openEditor(agentId);
354 |     };
355 | });
356 |
357 | grid.querySelectorAll('.btn-delete-agent').forEach((btn) =>
358 | {

```

```

355 |     btn.onclick = () => {
356 |         const agentId = btn.getAttribute('data-agent-id');
357 |         if (agentId) _deleteAgent(agentId);
358 |     };
359 | });
360 | }
361 |
362 | /**
363 |  * Toggles an agent's enabled state and persists it to the server.
364 |  *
365 |  * @param {string} agentId - Target agent unique identifier.
366 |  * @param {boolean} enabled - New enabled state.
367 |  * @returns {Promise<void>}
368 |  */
369 | async function _toggleAgentState(agentId, enabled) {
370 |     const agent = _state.agents.find((a) => a.id === agentId);
371 |     if (!agent) return;
372 |
373 |     agent.enabled = enabled;
374 |     try {
375 |         await _fetchJson(`/api/agents/${agentId}`, {
376 |             method: 'PUT',
377 |             headers: { 'Content-Type': 'application/json' },
378 |             body: JSON.stringify(agent)
379 |         });
380 |         _renderCounters();
381 |         _renderGrid();
382 |         _showToast(`Агент "${agent.name}" ${enabled ? 'включен' :
383 | 'отключен'}`, 'success');
384 |     } catch (err) {
385 |         console.error('[AgentsTab] Error toggling agent:', err);
386 |         _showToast(`Ошибка: ${err.message}`, 'danger');
387 |         await _loadAllData();
388 |     }
389 | }
390 | /**

```

```

391 | * Populates capability checkboxes inside the editor modal.
392 | *
393 | * @returns {void}
394 | */
395 | function _populateToolsMatrix() {
396 |     const container = document.getElementById('agent-tools-matrix');
397 |     if (!container) return;
398 |
399 |     container.innerHTML = _state.tools
400 |         .map(
401 |             (tool) => `
402 |                 <div class="col-md-6">
403 |                     <div class="p-3 rounded h-100" style="background: #1e2
404 | 93b; border: 1px solid #334155;">
405 |                         <div class="form-check form-switch m-0">
406 |                             <input class="form-check-input tool-checkbox" type
407 | ="checkbox" value="${tool.id}" id="tool-chk-${tool.id}">
408 |                             <label class="form-check-label small text-white fw
409 | -bold" for="tool-chk-${tool.id}">
410 |                                 ${tool.icon} ${tool.name}
411 |                             </label>
412 |                         </div>
413 |                         <div class="small mt-1" style="color: #cbd5e1; font-
414 | size: 0.8rem; line-height: 1.4;">${tool.description}</div>
415 |                     </div>
416 |                 </div>
417 |             `
418 |             .join('');
419 |     container.querySelectorAll('.tool-checkbox').forEach((chk) => {
420 |         chk.onchange = _updateSelectedToolsCounter;
421 |     });
422 | }

```

```

423 | * Updates selected capabilities counter badge in the editor m
    |odal.
424 | *
425 | * @returns {void}
426 | */
427 | function _updateSelectedToolsCounter() {
428 |     const count = document.querySelectorAll('#agent-tools-matrix
    |.tool-checkbox:checked').length;
429 |     const badge = document.getElementById('agent-tools-count');
430 |     if (badge) badge.textContent = `${count} выбрано`;
431 | }
432 |
433 | /**
434 | * Updates model selection dropdown options when provider sele
    |ction changes.
435 | *
436 | * @param {string} providerSelectId - ID of provider select el
    |ement.
437 | * @param {string} modelSelectId - ID of model select element.
438 | * @param {string} [selectedModel=''] - Model ID to select by
    |default.
439 | * @returns {void}
440 | */
441 | function _updateModelDropdown(providerSelectId, modelSelectId,
    |selectedModel = '') {
442 |     const pSel = document.getElementById(providerSelectId);
443 |     const mSel = document.getElementById(modelSelectId);
444 |     if (!pSel || !mSel) return;
445 |
446 |     const prov = _state.providers[pSel.value];
447 |     const models = prov?.models ?? [];
448 |
449 |     mSel.innerHTML = models.map((m) => `<option value="${m.id}">
    |${m.name} || m.id</option>`).join('');
450 |
451 |     if (selectedModel && models.some((m) => m.id === selectedMod
    |el)) {
452 |         mSel.value = selectedModel;

```

```

453     } else if (prov?.default_model) {
454         mSel.value = prov.default_model;
455     }
456 }
457
458 /**
459  * Opens agent editor modal for creating or updating an agent.
460  *
461  * @param {string|null} [agentId=null] - Agent ID to edit, or
462  * null for create.
463  * @param {object|null} [prefillData=null] - Pre-filled data f
464  * rom AI builder.
465  * @returns {void}
466  */
467 function _openEditor(agentId = null, prefillData = null) {
468     const modalEl = document.getElementById('modal-agent-editor'
469 );
470     if (!modalEl) return;
471
472     const modeInput = document.getElementById('agent-edit-mode')
473 ;
474     const idInput = document.getElementById('agent-id');
475     const nameInput = document.getElementById('agent-name');
476     const descInput = document.getElementById('agent-desc');
477     const enabledChk = document.getElementById('agent-enabled');
478     const providerSel = document.getElementById('agent-provider'
479 );
480     const tempSlider = document.getElementById('agent-temperatur
481 e');
482     const tempVal = document.getElementById('agent-temp-val');
483     const maxStepsInput = document.getElementById('agent-max-ste
484 ps');
485     const timeoutInput = document.getElementById('agent-timeout'
486 );
487     const promptTa = document.getElementById('agent-system-promp
488 t');
489
490     // Reset tools checkboxes

```

```
482 | document.querySelectorAll('#agent-tools-matrix .tool-checkbo
    | x').forEach((chk) => {
483 |     chk.checked = false;
484 | });
485 |
486 | if (agentId) {
487 |     const agent = _state.agents.find((a) => a.id === agentId);
488 |     if (!agent) return;
489 |
490 |     if (modeInput) modeInput.value = 'edit';
491 |     if (idInput) {
492 |         idInput.value = agent.id;
493 |         idInput.disabled = true;
494 |     }
495 |     if (nameInput) nameInput.value = agent.name;
496 |     if (descInput) descInput.value = agent.description ?? '';
497 |     if (enabledChk) enabledChk.checked = Boolean(agent.enabled
    | );
498 |     if (providerSel) providerSel.value = agent.provider ?? 'ge
    | mini';
499 |
500 |     _updateModelDropdown('agent-provider', 'agent-model', agen
    | t.model);
501 |
502 |     if (tempSlider) {
503 |         tempSlider.value = agent.temperature ?? 0.3;
504 |         if (tempVal) tempVal.textContent = tempSlider.value;
505 |     }
506 |     if (maxStepsInput) maxStepsInput.value = agent.max_steps ?
    | ? 15;
507 |     if (timeoutInput) timeoutInput.value = agent.timeout_secon
    | ds ?? 60;
508 |     if (promptTa) promptTa.value = agent.system_prompt ?? '';
509 |
510 |     (agent.tools ?? []).forEach((tId) => {
511 |         const chk = document.getElementById(`tool-chk-${tId}`);
512 |         if (chk) chk.checked = true;
513 |     });
```

```
514 | } else if (prefillData) {
515 |     if (modeInput) modeInput.value = 'create';
516 |     if (idInput) {
517 |         idInput.value = (prefillData.name || 'agent').toLowerCase
e().replace(/^[^a-z0-9_]/g, '_').slice(0, 25);
518 |         idInput.disabled = false;
519 |     }
520 |     if (nameInput) nameInput.value = prefillData.name ?? 'Новы
й ИИ-Агент';
521 |     if (descInput) descInput.value = prefillData.description ?
? '';
522 |     if (enabledChk) enabledChk.checked = true;
523 |     if (providerSel) providerSel.value = prefillData.provider
?? 'gemini';
524 |
525 |     _updateModelDropdown('agent-provider', 'agent-model', pref
illData.model);
526 |
527 |     if (tempSlider) {
528 |         tempSlider.value = prefillData.temperature ?? 0.3;
529 |         if (tempVal) tempVal.textContent = tempSlider.value;
530 |     }
531 |     if (maxStepsInput) maxStepsInput.value = prefillData.max_s
teps ?? 15;
532 |     if (timeoutInput) timeoutInput.value = 60;
533 |     if (promptTa) promptTa.value = prefillData.system_prompt ?
? '';
534 |
535 |     (prefillData.recommended_tools ?? []).forEach((tId) => {
536 |         const chk = document.getElementById(`tool-chk-${tId}`);
537 |         if (chk) chk.checked = true;
538 |     });
539 | } else {
540 |     if (modeInput) modeInput.value = 'create';
541 |     if (idInput) {
542 |         idInput.value = '';
543 |         idInput.disabled = false;
544 |     }
```

```

545     if (nameInput) nameInput.value = '';
546     if (descInput) descInput.value = '';
547     if (enabledChk) enabledChk.checked = true;
548     if (providerSel) providerSel.value = 'gemini';
549
550     _updateModelDropdown('agent-provider', 'agent-model');
551
552     if (tempSlider) {
553         tempSlider.value = 0.3;
554         if (tempVal) tempVal.textContent = '0.3';
555     }
556     if (maxStepsInput) maxStepsInput.value = 15;
557     if (timeoutInput) timeoutInput.value = 60;
558     if (promptTa) promptTa.value = '';
559 }
560
561 _updateSelectedToolsCounter();
562 new bootstrap.Modal(modalEl).show();
563 }
564
565 /**
566  * Saves agent configuration (POST create or PUT update).
567  *
568  * @returns {Promise<void>}
569  */
570 async function _handleSaveAgent() {
571     const mode = document.getElementById('agent-edit-mode')?.value ?? 'create';
572     const id = document.getElementById('agent-id')?.value.trim();
573     const name = document.getElementById('agent-name')?.value.trim();
574     const desc = document.getElementById('agent-desc')?.value.trim();
575     const enabled = document.getElementById('agent-enabled')?.checked ?? true;
576     const provider = document.getElementById('agent-provider')?.value ?? 'gemini';

```



```
577 |   const model = document.getElementById('agent-model')?.value
    |   ?? 'gemini-3.1-flash';
578 |   const temperature = parseFloat(document.getElementById('agen
    |   t-temperature')?.value ?? '0.3');
579 |   const max_steps = parseInt(document.getElementById('agent-ma
    |   x-steps')?.value ?? '15', 10);
580 |   const timeout_seconds = parseInt(document.getElementById('ag
    |   ent-timeout')?.value ?? '60', 10);
581 |   const system_prompt = document.getElementById('agent-system-
    |   prompt')?.value.trim() ?? '';
582 |
583 |   if (!id || !name) {
584 |     _showToast('ID и Название агента обязательны для заполнени
    |   я', 'warning');
585 |     return;
586 |   }
587 |
588 |   const selectedTools = [];
589 |   document.querySelectorAll('#agent-tools-matrix .tool-checkbo
    |   x:checked').forEach((chk) => {
590 |     selectedTools.push(chk.value);
591 |   });
592 |
593 |   const payload = {
594 |     id,
595 |     name,
596 |     description: desc,
597 |     enabled,
598 |     provider,
599 |     model,
600 |     temperature,
601 |     max_steps,
602 |     timeout_seconds,
603 |     tools: selectedTools,
604 |     system_prompt
605 |   };
606 |
607 |   const saveBtn = document.getElementById('btn-save-agent');
```

```

608 | if (saveBtn) saveBtn.disabled = true;
609 |
610 | try {
611 |     if (mode === 'create') {
612 |         await _fetchJson('/api/agents', {
613 |             method: 'POST',
614 |             headers: { 'Content-Type': 'application/json' },
615 |             body: JSON.stringify(payload)
616 |         });
617 |         _showToast(`Агент "${name}" успешно создан!`, 'success')
618 |     };
619 |     } else {
620 |         await _fetchJson(`/api/agents/${id}`, {
621 |             method: 'PUT',
622 |             headers: { 'Content-Type': 'application/json' },
623 |             body: JSON.stringify(payload)
624 |         });
625 |         _showToast(`Агент "${name}" успешно обновлен!`, 'success
626 | ');
627 |     }
628 |
629 |     const modalEl = document.getElementById('modal-agent-edito
630 | r');
631 |     bootstrap.Modal.getInstance(modalEl)?.hide();
632 |     await _loadAllData();
633 | } catch (err) {
634 |     console.error('[AgentsTab] Error saving agent:', err);
635 |     _showToast(`Ошибка сохранения: ${err.message}`, 'danger');
636 | } finally {
637 |     if (saveBtn) saveBtn.disabled = false;
638 | }
639 |
640 | /**
641 |  * Deletes an existing custom agent after confirmation.
642 |  *
643 |  * @param {string} agentId - Target agent unique identifier.
644 |  * @returns {Promise<void>}

```

```

643 | */
644 | async function _deleteAgent(agentId) {
645 |     const agent = _state.agents.find((a) => a.id === agentId);
646 |     if (!agent) return;
647 |
648 |     if (!confirm(`Вы действительно хотите удалить агента "${agent.name}" (${agent.id})?`)) {
649 |         return;
650 |     }
651 |
652 |     try {
653 |         await _fetchJson(`/api/agents/${agentId}`, { method: 'DELETE' });
654 |         _showToast(`Агент "${agent.name}" удален`, 'success');
655 |         await _loadAllData();
656 |     } catch (err) {
657 |         console.error('[AgentsTab] Error deleting agent:', err);
658 |         _showToast(`Ошибка удаления: ${err.message}`, 'danger');
659 |     }
660 | }
661 |
662 | /**
663 |  * Opens AI prompt architect builder modal.
664 |  *
665 |  * @returns {void}
666 |  */
667 | function _openAiBuilder() {
668 |     const modalEl = document.getElementById('modal-ai-builder');
669 |     if (!modalEl) return;
670 |
671 |     const promptInput = document.getElementById('ai-builder-prompt');
672 |     const resultBox = document.getElementById('ai-builder-result');
673 |     if (promptInput) promptInput.value = '';
674 |     if (resultBox) resultBox.classList.add('d-none');
675 |     _state.lastGeneratedSpec = null;
676 |

```

```

677 |     new bootstrap.Modal(modalEl).show();
678 | }
679 |
680 | /**
681 |  * Generates an agent specification using selected LLM archite
    ct model.
682 |  *
683 |  * @returns {Promise<void>}
684 |  */
685 | async function _handleRunAiGenerate() {
686 |     const taskDesc = document.getElementById('ai-builder-prompt'
    )?.value.trim();
687 |     const provider = document.getElementById('ai-builder-provide
    r')?.value ?? 'gemini';
688 |     const model = document.getElementById('ai-builder-model')?.v
    alue ?? 'gemini-3.1-flash';
689 |
690 |     if (!taskDesc) {
691 |         _showToast('Пожалуйста, опишите задачу агента', 'warning')
    ;
692 |         return;
693 |     }
694 |
695 |     const runBtn = document.getElementById('btn-run-ai-generate'
    );
696 |     if (runBtn) {
697 |         runBtn.disabled = true;
698 |         runBtn.innerHTML = '<span class="spinner-border spinner-bo
    rder-sm me-2"></span>Анализ задачи и генерация...';
699 |     }
700 |
701 |     try {
702 |         const res = await _fetchJson('/api/agents/generate-prompt'
    , {
703 |             method: 'POST',
704 |             headers: { 'Content-Type': 'application/json' },
705 |             body: JSON.stringify({ task_description: taskDesc, provi
    der, model })

```

```

706     });
707
708     if (res?.data) {
709         _state.lastGeneratedSpec = { ...res.data, provider, mode
710         l };
711         const resultBox = document.getElementById('ai-builder-re
712         sult');
713         const previewBox = document.getElementById('ai-builder-p
714         review-box');
715         if (resultBox && previewBox) {
716             resultBox.classList.remove('d-none');
717             previewBox.innerHTML = `
718                 <div><strong style="color: #38bdf8;">Название:</stro
719                 ng> <span class="text-white fw-bold">${res.data.name}</span></
720                 div>
721                 <div class="mt-1"><strong style="color: #cbd5e1;">Оп
722                 исание:</strong> <span style="color: #f1f5f9;">${res.data.desc
723                 ription}</span></div>
724                 <div class="mt-1"><strong style="color: #4ade80;">Ин
725                 струменты:</strong> <span style="color: #f8fafc;">${(res.data.
726                 recommended_tools ?? []).join(', ') || 'нет'}</span></div>
727                 <div class="mt-2 p-2 rounded" style="background: #0f
728                 172a; border: 1px solid #334155; color: #f8fafc; white-space:
729                 pre-wrap; font-size: 0.85rem; line-height: 1.45;">${res.data.s
730                 ystem_prompt}</div>
731             `;
732         }
733         _showToast('Спецификация агента успешно сгенерирована!',
734         'success');
735     }
736     } catch (err) {
737         console.error('[AgentsTab] AI Generator error:', err);
738         _showToast(`Ошибка генерации: ${err.message}`, 'danger');
739     } finally {
740         if (runBtn) {
741             runBtn.disabled = false;
742             runBtn.innerHTML = '<i class="bi bi-stars me-1"></i> Сге
743             нерировать спецификацию';

```

```
730     }
731   }
732 }
733
734 /**
735  * Opens sandbox modal for interactive agent testing.
736  *
737  * @param {string} agentId - Target agent unique identifier.
738  * @returns {void}
739  */
740 function _openSandbox(agentId) {
741   const agent = _state.agents.find((a) => a.id === agentId);
742   if (!agent) return;
743
744   _state.activeSandboxAgent = agent;
745   const modalEl = document.getElementById('modal-agent-sandbox
746   ');
747   if (!modalEl) return;
748
749   document.getElementById('sandbox-agent-name')?.replaceChildr
750   en(document.createTextNode(agent.name));
751   const modelBadge = document.getElementById('sandbox-agent-mo
752   del');
753   if (modelBadge) modelBadge.textContent = `${_getProviderDisp
754   layName(agent.provider)}: ${agent.model}`;
755
756   const toolsSummary = document.getElementById('sandbox-agent-
757   tools-summary');
758   if (toolsSummary) toolsSummary.textContent = `Инструментов:
759   ${((agent.tools ?? []).length)}`;
760
761   const msgsList = document.getElementById('sandbox-messages-l
762   ist');
763   const placeholder = document.getElementById('sandbox-placeho
764   lder');
765   if (msgsList) msgsList.innerHTML = '';
766   if (placeholder) placeholder.style.display = 'block';
767 }
```

```
760 | new bootstrap.Modal(modalEl).show();
761 | }
762 |
763 | /**
764 |  * Executes a test query against the active sandbox agent and
765 |  * renders execution trace.
766 |  * @returns {Promise<void>}
767 |  */
768 | async function _handleRunSandboxTest() {
769 |     const agent = _state.activeSandboxAgent;
770 |     const input = document.getElementById('sandbox-input-msg');
771 |     const sendBtn = document.getElementById('btn-sandbox-send');
772 |     const msgsList = document.getElementById('sandbox-messages-l
773 |     ist');
774 |     const placeholder = document.getElementById('sandbox-placeho
775 |     lder');
776 |
777 |     if (!agent || !input || !msgsList) return;
778 |     const query = input.value.trim();
779 |     if (!query) return;
780 |
781 |     if (placeholder) placeholder.style.display = 'none';
782 |
783 |     // Append user message
784 |     const userMsgEl = document.createElement('div');
785 |     userMsgEl.className = 'd-flex justify-content-end mb-3';
786 |     userMsgEl.innerHTML = `
787 |         <div class="p-2 px-3 rounded shadow-sm text-white small" s
788 |         tyle="background-color: #2563eb; max-width: 80%; font-weight:
789 |         500;">
790 |             <strong>Вы:</strong> ${query}
791 |         </div>
792 |     `;
793 |     msgsList.appendChild(userMsgEl);
794 |
795 |     input.value = '';
796 |     if (sendBtn) sendBtn.disabled = true;
```

```

793
794 // Append response placeholder with loader
795 const botMsgEl = document.createElement('div');
796 botMsgEl.className = 'd-flex flex-column mb-3';
797 botMsgEl.innerHTML = `
798     <div class="p-3 rounded small" style="background: #0f172a;
799     border: 1px solid #334155; color: #f8fafc;">
800         <div class="d-flex align-items-center gap-2 mb-2" style=
801         "color: #fbbf24; font-weight: 600;">
802             <span class="spinner-border spinner-border-sm"></span>
803             <span>Агент выполняет рассуждение и вызовы инструменто
804             в...</span>
805         </div>
806         <div class="steps-trace small font-monospace" style="col
807         or: #cbd5e1;"></div>
808     </div>
809 `;
810 msgsList.appendChild(botMsgEl);
811 msgsList.scrollTop = msgsList.scrollHeight;
812
813 try {
814     const res = await _fetchJson('/api/agents/test', {
815         method: 'POST',
816         headers: { 'Content-Type': 'application/json' },
817         body: JSON.stringify({ agent_id: agent.id, test_message:
818         query })
819     });
820
821     const stepsHtml = (res.steps ?? []).
822     .map((s) => {
823         let icon = '□';
824         if (s.type === 'tool_init') icon = '□';
825         else if (s.type === 'action') icon = '□';
826         else if (s.type === 'finish') icon = '□';
827         else if (s.type === 'error') icon = '□';
828
829         return `<div class="mb-1">${icon} <span style="color:
830         #38bdf8; font-weight: bold;">[Шаг ${s.step}]</span> <span styl

```



```

    e="color: #e2e8f0;">${s.content}</span></div>`;
825     })
826     .join('');
827
828     botMsgEl.innerHTML = `
829         <div class="p-3 rounded small" style="background: #0f172
a; border: 1px solid #334155; color: #f8fafc;">
830             <div class="d-flex justify-content-between align-items
-center mb-2 pb-1" style="border-bottom: 1px solid #334155;">
831                 <strong style="color: #38bdf8; font-size: 0.95rem;">
❑  ${agent.name}</strong>
832                 <span class="badge" style="background: #334155; colo
r: #f8fafc; font-weight: 600;">${res.duration_ms ?? 0} мс</spa
n>
833             </div>
834
835             <div class="mb-2" style="white-space: pre-wrap; color:
#f8fafc; line-height: 1.5;">${res.response || 'Пустой ответ'}
</div>
836
837             <div class="p-2 rounded mt-2 small font-monospace" sty
le="background: #020617; border: 1px solid #334155; font-size:
0.8rem; line-height: 1.45;">
838                 <div class="fw-bold mb-1" style="color: #cbd5e1;">Тр
ассировка выполнения (ReAct Trace):</div>
839                 ${stepsHtml}
840             </div>
841         </div>
842     `;
843     } catch (err) {
844         botMsgEl.innerHTML = `
845             <div class="p-3 rounded small" style="background: #450a0
a; border: 1px solid #dc2626; color: #fecaca;">
846                 <strong style="color: #ef4444;">Ошибка выполнения:</st
rong> ${err.message}
847             </div>
848         `;
849     } finally {

```

```

850     if (sendBtn) sendBtn.disabled = false;
851     const chatWin = document.getElementById('sandbox-chat-wind
ow');
852     if (chatWin) chatWin.scrollTop = chatWin.scrollHeight;
853 }
854 }
855
856 /**
857  * Returns human-readable display name for provider identifier
858  *
859  * @param {string} provKey - Provider key (e.g. 'gemini', 'agy
', 'foundry', 'ollama').
860  * @returns {string} Provider display label.
861  */
862 function _getProviderDisplayName(provKey) {
863     return _state.providers[provKey]?.name ?? provKey;
864 }
865
866 /**
867  * Displays floating feedback notification toast.
868  *
869  * @param {string} message - Notification text message.
870  * @param {string} [type='info'] - Bootstrap color variant ('i
nfo', 'success', 'warning', 'danger').
871  * @returns {void}
872  */
873 function _showToast(message, type = 'info') {
874     if (typeof window.showModelsNotification === 'function') {
875         window.showModelsNotification(message, type);
876         return;
877     }
878     const alertDiv = document.createElement('div');
879     alertDiv.className = `alert alert-${type} position-fixed bot
tom-0 end-0 m-3 shadow`;
880     alertDiv.style.zIndex = '9999';
881     alertDiv.textContent = message;
882     document.body.appendChild(alertDiv);

```

```
883 |   setTimeout(() => alertDiv.remove(), 3500);  
884 | }  
885 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: AI-Breadboard Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон веб-интерфейса системного модуля (index).
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /src/api/webgui/ai_benchmark_tab/index.html
12 |
13 File: index.html
14 Project: ai-breadboard
15 Package: src/api/webgui/ai_benchmark_tab
16 Author: hypo69
17 Copyright: © 2026 hypo69
18 Updated: 2026-10-01 13:13:56
19 |=====
20 |=====
21 -->
22
23 <!-- AI Inference Benchmark Tab -->
24 <style>
25 |.bm-container {
26 |  padding: 0.5rem 0.75rem;
27 |}
28 |.bm-card {
29 |  background: #0f172a !important;
30 |  border: 1px solid #334155 !important;
31 |  border-radius: 8px;
32 |  padding: 0.6rem 0.85rem;
33 |  height: 100%;
34 |  transition: border-color 0.15s ease, box-shadow 0.15s ease
35 |;
```

```
33 | }
34 | .bm-card:hover {
35 |     border-color: #38bdf8 !important;
36 |     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
37 | }
38 | .bm-card-label {
39 |     font-size: 0.7rem;
40 |     text-transform: uppercase;
41 |     letter-spacing: 0.05em;
42 |     color: #94a3b8;
43 |     font-weight: 600;
44 |     margin-bottom: 0.25rem;
45 | }
46 | .bm-card-value {
47 |     font-size: 1.4rem;
48 |     font-weight: 700;
49 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
50 |     color: #38bdf8;
51 | }
52 | .bm-panel {
53 |     background: #0f172a !important;
54 |     border: 1px solid #334155 !important;
55 |     border-radius: 8px;
56 |     margin-bottom: 0.75rem;
57 |     overflow: hidden;
58 | }
59 | .bm-panel-header {
60 |     background: #1e293b !important;
61 |     border-bottom: 1px solid #334155 !important;
62 |     padding: 0.45rem 0.85rem;
63 |     font-weight: 600;
64 |     font-size: 0.85rem;
65 |     color: #38bdf8;
66 |     display: flex;
67 |     align-items: center;
68 |     justify-content: space-between;
69 | }
```

```

70 | .bm-table {
71 |     margin-bottom: 0;
72 |     font-size: 0.8rem;
73 |     width: 100%;
74 | }
75 | .bm-table th {
76 |     background: #1e293b;
77 |     color: #94a3b8;
78 |     font-size: 0.72rem;
79 |     text-transform: uppercase;
80 |     padding: 0.45rem 0.6rem;
81 |     border-bottom: 1px solid #334155;
82 | }
83 | .bm-table td {
84 |     border-bottom: 1px solid #1e293b;
85 |     padding: 0.45rem 0.6rem;
86 |     vertical-align: middle;
87 | }
88 | </style>
89 |
90 | <div class="bm-container">
91 |     <!-- <span data-i18n="auto___8d61a6"></span> -->
92 |     <div class="bm-panel mb-3">
93 |         <div class="bm-panel-header">
94 |             <span><i class="bi bi-speedometer2 me-1"></i> <span data
95 |             <span data-i18n="auto___81054b"></span> AI Inference Benchmark</span>
96 |             <span class="badge bg-primary-subtle text-primary border
97 |             border-primary-subtle" id="bm-status-badge"><span data-i18n="
98 |             auto___97b864"></span></span>
99 |         </div>
100 |         <div class="p-3">
101 |             <div class="row g-2 align-items-end">
102 |                 <div class="col-md-3 col-sm-6">
103 |                     <label class="form-label small text-muted mb-1"><span
104 |                     data-i18n="auto___2d385a"></span></label>
105 |                     <select class="form-select form-select-sm bg-dark te
106 |                     xt-white border-secondary" id="bm-provider-select">
107 |                         <option value="gemini" selected>Google Gemini API<

```

```

    /option>
103         <option value="ollama">Ollama (Local)</option>
104         <option value="foundry">Microsoft Foundry Local</o
    ption>
105         <option value="onnx">ONNX DirectML / NPU</option>
106     </select>
107 </div>
108 <div class="col-md-3 col-sm-6">
109     <label class="form-label small text-muted mb-1"><spa
    n data-i18n="auto___f1flee"></span></label>
110     <input type="text" class="form-control form-control-
    sm bg-dark text-white border-secondary" id="bm-model-input" va
    lue="gemini-3.1-flash">
111 </div>
112 <div class="col-md-2 col-sm-6">
113     <label class="form-label small text-muted mb-1">Max
    Tokens</label>
114     <input type="number" class="form-control form-contro
    l-sm bg-dark text-white border-secondary" id="bm-tokens-input"
    value="150" min="20" max="2000">
115 </div>
116 <div class="col-md-2 col-sm-6">
117     <label class="form-label small text-muted mb-1">Temp
    erature</label>
118     <input type="number" class="form-control form-contro
    l-sm bg-dark text-white border-secondary" id="bm-temp-input" v
    alue="0.7" min="0" max="1" step="0.1">
119 </div>
120 <div class="col-md-2 col-sm-12">
121     <button class="btn btn-sm btn-primary w-100 fw-semib
    old d-flex align-items-center justify-content-center gap-1.5"
    id="bm-run-btn">
122         <i class="bi bi-play-fill fs-6"></i> <span data-i1
    8n="auto___9632be"></span>
123     </button>
124 </div>
125 </div>
126 <div class="mt-2">

```

```
127 |         <label class="form-label small text-muted mb-1"><span
| data-i18n="auto___b3f6a4"></span></label>
128 |         <textarea class="form-control form-control-sm bg-dark
| text-white border-secondary" id="bm-prompt-input" rows="2"><sp
| an data-i18n="auto___100___5d4d7a"></span>.</textarea>
129 |     </div>
130 | </div>
131 | </div>
132 |
133 | <!-- <span data-i18n="auto___cca642"></span> -->
134 | <div class="row g-2 mb-3">
135 |     <div class="col-md-3 col-6">
136 |         <div class="bm-card">
137 |             <div class="bm-card-label"><i class="bi bi-stopwatch t
| ext-info me-1"></i> Time to First Token (TTFT)</div>
138 |             <div class="bm-card-value" id="bm-card-ttft">– ms</div
| >
139 |         </div>
140 |     </div>
141 |     <div class="col-md-3 col-6">
142 |         <div class="bm-card">
143 |             <div class="bm-card-label"><i class="bi bi-lightning-c
| harge text-warning me-1"></i> <span data-i18n="auto___09625b">
| </span> (TPS)</div>
144 |             <div class="bm-card-value text-success" id="bm-card-tp
| s">– t/s</div>
145 |         </div>
146 |     </div>
147 |     <div class="col-md-3 col-6">
148 |         <div class="bm-card">
149 |             <div class="bm-card-label"><i class="bi bi-hourglass-s
| plit text-primary me-1"></i> <span data-i18n="auto___46e0d8"><
| /span> (Latency)</div>
150 |             <div class="bm-card-value" id="bm-card-total">– ms</di
| v>
151 |         </div>
152 |     </div>
153 | <div class="col-md-3 col-6">
```



```

154         <div class="bm-card">
155             <div class="bm-card-label"><i class="bi bi-chat-left-t
ext text-secondary me-1"></i> <span data-i18n="auto___430610">
</span></div>
156             <div class="bm-card-value text-white" id="bm-card-toke
ns">—</div>
157         </div>
158     </div>
159 </div>
160
161 <!-- <span data-i18n="auto___7fbb04"></span> -->
162 <div class="bm-panel">
163     <div class="bm-panel-header">
164         <span><i class="bi bi-clock-history me-1"></i> <span dat
a-i18n="auto___a58a32"></span></span>
165         <button class="btn btn-xs btn-outline-secondary py-0 px-
2 text-white border-secondary" id="bm-refresh-history-btn" sty
le="font-size: 0.75rem;">
166             <i class="bi bi-arrow-clockwise"></i> <span data-i18n=
"auto___c2f668"></span>
167         </button>
168     </div>
169     <div class="table-responsive">
170         <table class="table table-dark table-hover bm-table" id=
"bm-history-table">
171             <thead>
172                 <tr>
173                     <th><span data-i18n="auto___c80d7e"></span></th>
174                     <th><span data-i18n="auto___2d385a"></span></th>
175                     <th><span data-i18n="auto___f1f1ee"></span></th>
176                     <th>TTFT</th>
177                     <th><span data-i18n="auto___09625b"></span> (TPS)<
/th>
178                     <th><span data-i18n="auto___46e0d8"></span></th>
179                     <th><span data-i18n="auto___4b1883"></span> (Promp
t / Gen)</th>
180                     <th><span data-i18n="auto___f7f293"></span></th>
181                 </tr>

```

```
182 |         </thead>
183 |         <tbody id="bm-history-tbody">
184 |             <tr>
185 |                 <td colspan="8" class="text-center text-muted py-3
186 |                 "><span data-i18n="auto___51fbb4"></span>".</td>
187 |             </tr>
188 |         </tbody>
189 |     </table>
190 | </div>
191 </div>
192 |
```

main.js

```
1  /**
2   * =====
3   *
4   * =====
5   * Process Name: AI-Breadboard UI - Main Script
6   * =====
7   *
8   * Description:
9   *   Клиентский веб-скрипт модуля main.
10  *
11  * Usage Examples:
12  *   HTML Integration:
13  *     <script src="/src/api/webgui/ai_benchmark_tab/main.js?v
14  *     =20261001_v1" type="module"></script>
15  *
16  * File: main.js
17  * Project: ai-breadboard
18  * Package: src/api/webgui/ai_benchmark_tab
19  * Author: hypo69
20  * Copyright: © 2026 hypo69
21  * Updated: 2026-10-01 13:13:56
22  * =====
23  *
24  */
25  /**
26   * ai_benchmark_tab/main.js – Контроллер вкладки AI Inference
27   * Benchmark
28   */
29  export async function initAiBenchmarkTab() {
30    const runBtn = document.getElementById('bm-run-btn');
31    const refreshBtn = document.getElementById('bm-refresh-history-btn');
32    const providerSelect = document.getElementById('bm-provider-select');
33    const modelInput = document.getElementById('bm-model-input')
```

```

;
30   const tokensInput = document.getElementById('bm-tokens-input
    ');
31   const tempInput = document.getElementById('bm-temp-input');
32   const promptInput = document.getElementById('bm-prompt-input
    ');
33   const statusBadge = document.getElementById('bm-status-badge
i18n.t('auto__if_providerselect_providerselect_addeventlistene
r__df8f21')change', () => {
34       const p = providerSelect.value;
35       if (p === 'gemini') modelInput.value = 'gemini-3.1-flash
    ';
36       else if (p === 'ollama') modelInput.value = 'llama3.2:la
test';
37       else if (p === 'foundry') modelInput.value = 'phi-3.5-mi
ni';
38       else if (p === 'onnx') modelInput.value = 'directml-phi3
i18n.t('auto__if_runbtn_runbtn_addeventlistener__c4d4ca')click
    ', async () => {
39           runBtn.disabled = true;
40           runBtn.innerHTML = '<span class="spinner-border spinner-
border-sm me-1"></span> Тестирование...';
41           if (statusBadge) {
42               statusBadge.className = 'badge bg-warning-subtle text-
warning border border-warning-subtle';
43               statusBadge.textContent = i18n.t('auto__f7abff');
44           }
45
46           try {
47               const payload = {
48                   provider: providerSelect ? providerSelect.value : 'g
emini',
49                   model_name: modelInput ? modelInput.value : 'gemini-
3.1-flash',
50                   prompt: promptInput ? promptInput.value : i18n.t('au
to__387814'),
51                   max_tokens: tokensInput ? parseInt(tokensInput.value
    , 10) || 150 : 150,

```

```

52 |         temperature: tempInput ? parseFloat(tempInput.value)
    | || 0.7 : 0.7,
53 |     };
54 |
55 |     const res = await fetch('/api/windows/benchmark/ai', {
56 |         method: 'POST',
57 |         headers: { 'Content-Type': 'application/json' },
58 |         body: JSON.stringify(payload),
59 |     });
60 |
61 |     const data = await res.json();
62 |     if (res.ok && data.success) {
63 |         updateCards(data);
64 |         if (statusBadge) {
65 |             statusBadge.className = 'badge bg-success-subtle text-
    | ext-success border border-success-subtle';
66 |             statusBadge.textContent = i18n.t('auto___b718ea');
67 |         }
68 |         if (window.toast) {
69 |             window.toast.success(i18n.t('auto___983c1c'), `Ско
    | рость: ${data.tokens_per_second} t/s | TTFT: ${data.ttft_ms} m
    | s`);
70 |         }
71 |     } else {
72 |         if (statusBadge) {
73 |             statusBadge.className = 'badge bg-danger-subtle text-
    | xt-danger border border-danger-subtle';
74 |             statusBadge.textContent = i18n.t('auto___72aecd');
75 |         }
76 |         if (window.toast) {
77 |             window.toast.error(i18n.t('auto___f29ff0'), data.e
    | rror_message || i18n.t('auto___f6377e'));
78 |         }
79 |     }
80 | } catch (err) {
81 |     if (statusBadge) {
82 |         statusBadge.className = 'badge bg-danger-subtle text
    | -danger border border-danger-subtle';

```

```

83         statusBadge.textContent = i18n.t('auto___c00aca');
84     }
85     if (window.toast) {
86         window.toast.error(i18n.t('auto___089486'), err.mess
age);
87     }
88     } finally {
89         runBtn.disabled = false;
90         runBtn.innerHTML = '<i class="bi bi-play-fill fs-6"></
i> Запустить тестi18n.t('auto__await_loadhistory_if_refreshbtn
_refreshbtn_addeventlistener__909824')clicki18n.t('auto__loadh
istory_await_loadhistory_function_updatecards_data_const_ttftel
l_document_getelementbyid__9cba8e')bm-card-ttft');
91     const tpsEl = document.getElementById('bm-card-tps');
92     const totalEl = document.getElementById('bm-card-total');
93     const tokensEl = document.getElementById('bm-card-tokens');
94
95     if (ttftEl) ttftEl.textContent = `${data.ttft_ms} ms`;
96     if (tpsEl) tpsEl.textContent = `${data.tokens_per_second} t/
s`;
97     if (totalEl) totalEl.textContent = `${data.total_time_ms} ms
`;
98     if (tokensEl) tokensEl.textContent = `${data.prompt_tokens}
/ ${data.completion_tokens}`;
99 }
100
101 async function loadHistory() {
102     const tbody = document.getElementById('bm-history-tbody');
103     if (!tbody) return;
104
105     try {
106         const res = await fetch('/api/windows/benchmark/ai/history
');
107         const items = await res.json();
108
109         if (!Array.isArray(items) || items.length === 0) {
110             tbody.innerHTML = '<tr><td colspan="8" class="text-cente
r text-muted py-3i18n.t('auto___9f84ae')Запустить тест".</td><

```

```

    /tr>';
111     return;
112   }
113
114   tbody.innerHTML = items.slice().reverse().map(item => {
115     const timeStr = item.timestamp ? new Date(item.timestamp
116       * 1000).toLocaleTimeString() : '-';
117     const statusBadge = item.success
118       ? '<span class="badge bg-success-subtle text-success">
OK</span>'
119       : '<span class="badge bg-danger-subtle text-danger" ti
tle="${item.error_message || ''}i18n.t('auto__span_return_tr_t
d_span_class__613956')text-muted small">${timeStr}</span></td>
119     <td><span class="badge bg-secondary-subtle text-info
">${item.provider}</span></td>
120     <td class="fw-semibold text-white">${item.model_name
}</td>
121     <td><span class="text-info">${item.ttft_ms} ms</span
></td>
122     <td><span class="text-success fw-bold">${item.tokens
_per_second} t/s</span></td>
123     <td>${item.total_time_ms} ms</td>
124     <td><span class="text-muted small">${item.prompt_tok
ens} in / ${item.completion_tokens} out</span></td>
125     <td>${statusBadge}</td>
126   </tr>
127   `;
128   })).join('');
129   } catch (err) {
130     tbody.innerHTML = `<tr><td colspan="8" class="text-danger
text-center py-2">Ошибка загрузки истории: ${err.message}</td>
</tr>`;
131   }
132 }
133
134 // Автозапуск при инициализации вкладки
135 if (document.readyState === 'loading') {
136   document.addEventListener('DOMContentLoaded', initAiBenchmar

```

```
    kTab);  
137 | } else {  
138 |     initAiBenchmarkTab();  
139 | }  
140 |
```


index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/apps/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/apps/index.html" class=
15 |        "w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/apps
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-08 04:24:00
23 |=====
24 |=====
25 -->
26
27 <!DOCTYPE html>
28 <html lang="ru">
29 <head>
30 |    <meta charset="UTF-8">
31 |    <meta name="viewport" content="width=device-width, initial-s
32 |    cale=1.0">
33 |    <title>AI Breadboard – Приложения (/apps)</title>
34 |    <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
35 |    t/css/bootstrap.min.css" rel="stylesheet">
```

```

31 <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    0.0/font/bootstrap-icons.css" rel="stylesheet">
32 <link href="/html/css/main.css?v=20261008_v1" rel="styleshee
    t">
33 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script
    >
34 <script src="/html/js/toast.js?v=20261006_v10"></script>
35 <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
    8next.min.js"></script>
36 </head>
37 <body>
38
39 <div class="container-fluid mt-1 px-2" id="apps-interface">
40
41 <!-- Navigation Bar (/tc, /apps) -->
42 <div class="d-flex align-items-center justify-content-betwee
    n mb-2 main-nav-container rounded px-2 py-1 flex-wrap gap-2">
43 <!-- Левая часть: кнопка меню, брендинг и индикатор активн
    ого раздела -->
44 <div class="d-flex align-items-center gap-2">
45 <!-- Кнопка открытия левого выдвижного меню -->
46 <button class="btn btn-sm btn-primary d-flex align-items
    -center gap-1.5 px-2 py-1 rounded shadow-sm"
47     id="apps-drawer-toggle-btn"
48     type="button"
49     data-bs-toggle="offcanvas"
50     data-bs-target="#appsSideNavOffcanvas"
51     aria-controls="appsSideNavOffcanvas"
52     title="Открыть меню разделов"
53     style="font-size: 0.82rem; font-weight: 500;">
54 <i class="bi bi-list fs-6"></i>
55 <span class="d-none d-md-inline">Меню</span>
56 </button>
57
58 <!-- Брендинг -->
59 <span class="fw-bold d-flex align-items-center gap-1.5 t
    ext-primary fs-6 user-select-none" style="letter-spacing: 0.3p
    x;">

```

```

60         <i class="bi bi-cpu-fill fs-5"></i>
61         <span class="d-none d-sm-inline">Test Computer</span>
62     </span>
63 </div>
64
65     <!-- Top Menu Buttons – генерируются из tc_menu_config.js
66     n -->
67     <div class="d-flex align-items-center gap-1 flex-wrap"></div>
68
69     <!-- Controls (Model Badge, Model Test Gear, Settings) -->
70     <div class="d-flex align-items-center gap-2 flex-shrink-0
71     ms-auto">
72         <!-- Active AI Model Badge -->
73         <span id="apps-model-badge" class="badge rounded-pill bg
74         -body-tertiary text-body border align-items-center gap-1.5 px-
75         3 py-1.5 shadow-sm" style="font-size: 0.8rem; cursor: default;
76         " title="Инициализация модели ИИ...">
77             <i class="bi bi-cpu-fill text-info"></i>
78             <span class="fw-semibold" id="apps-model-text">AI: Заг
79             рузка...</span>
80         </span>
81
82         <!-- Model Quick Test Dropdown Trigger & Popover -->
83         <div class="dropdown" id="model-ping-dropdown-wrapper">
84             <button class="btn btn-sm btn-outline-secondary d-flex
85             align-items-center justify-content-center p-0 rounded-circle"
86
87             id="btn-model-ping-gear"
88             type="button"
89             data-bs-toggle="dropdown"
90             data-bs-auto-close="outside"
91             aria-expanded="false"
92             title="Проверка доступности модели"
93             style="width: 30px; height: 30px;">
94                 <i class="bi bi-gear"></i>
95             </button>
96             <div class="dropdown-menu dropdown-menu-end shadow-lg

```

```

border border-secondary-subtle p-3" aria-labelledby="btn-model
-ping-gear" style="min-width: 360px; max-width: 460px; z-index
: 1060;">
89     <div class="d-flex align-items-center justify-content-between mb-2">
90         <span class="fw-bold small d-flex align-items-center gap-1.5 text-body">
91             <i class="bi bi-cpu text-primary"></i>
92             <span>Проверка доступности модели</span>
93         </span>
94     </div>
95     <div class="mb-2">
96         <input type="text" class="form-control form-control-sm" id="model-ping-input" value="Проверка доступности модели" placeholder="Введите сообщение...">
97     </div>
98     <div class="d-flex justify-content-end mb-2">
99         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5 px-3" id="btn-model-ping-send" type="button">
100             <i class="bi bi-send-fill" id="model-ping-btn-icon"></i>
101             <span id="model-ping-btn-text">Отправить</span>
102         </button>
103     </div>
104     <div id="model-ping-response-container" style="display: none;">
105         <label class="form-label small text-muted mb-1 fw-semibold">Ответ модели:</label>
106         <div id="model-ping-response" class="small p-2 rounded bg-body-tertiary border text-break font-monospace" style="max-height: 220px; overflow-y: auto; white-space: pre-wrap; font-size: 0.8rem; user-select: text;"></div>
107     </div>
108 </div>
109 </div>
110
111 <!-- Quick Settings Gear Button -->

```

```

112 |         <button class="btn btn-sm btn-outline-secondary d-flex a
| align-items-center justify-content-center p-0 rounded-circle" i
| d="user-settings-gear-btn" data-bs-toggle="modal" data-bs-targ
| et="#userSettingsModal" title="Настройки профиля и темы" style
| ="width: 30px; height: 30px;" data-i18n-title="userSettings.ti
| tle">
113 |             <i class="bi bi-sliders"></i>
114 |         </button>
115 |     </div>
116 | </div>
117 |
118 | <!-- Плавающий боковой ярлычок (вкладка) слева -->
119 | <button class="left-drawer-tab-trigger btn shadow"
120 |         type="button"
121 |         data-bs-toggle="offcanvas"
122 |         data-bs-target="#appsSideNavOffcanvas"
123 |         aria-controls="appsSideNavOffcanvas"
124 |         title="Открыть меню навигации">
125 |     <i class="bi bi-list fs-6"></i>
126 |     <span class="left-drawer-tab-text">МЕНЮ</span>
127 | </button>
128 |
129 | <!-- Левая выдвижная вкладка / панель навигации (Offcanvas)
| -->
130 | <div class="offcanvas offcanvas-start left-drawer-nav" tabin
| dex="-1" id="appsSideNavOffcanvas" aria-labelledby="appsSideNa
| vOffcanvasLabel">
131 |     <div class="offcanvas-header border-bottom py-2 px-3">
132 |         <h5 class="offcanvas-title d-flex align-items-center gap
| -2 fs-6 fw-bold" id="appsSideNavOffcanvasLabel">
133 |             <i class="bi bi-grid-fill text-primary"></i>
134 |             <span>Приложения (/apps)</span>
135 |         </h5>
136 |         <button type="button" class="btn-close text-reset" data-
| bs-dismiss="offcanvas" aria-label="Заккрыть"></button>
137 |     </div>
138 |     <div class="offcanvas-body p-3 d-flex flex-column justify-
| content-between" style="overflow-y: auto;">

```

```
139 |         <div class="drawer-sections-wrapper">
140 |             <div class="nav-drawer-group mb-3">
141 |                 <div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
142 |                     <i class="bi bi-grid-fill text-primary"></i>
143 |                     <span>Разделы Test Computer</span>
144 |                 </div>
145 |                 <div class="list-group list-group-flush rounded border" id="appsNavTabs"></div>
146 |             </div>
147 |         </div>
148 |
149 |         <!-- Footer Drawer -->
150 |         <div class="nav-drawer-footer border-top pt-2 mt-auto d-flex flex-column gap-2">
151 |             <button class="btn btn-sm btn-outline-warning w-100 d-flex align-items-center justify-content-between py-2 px-3 rounded" id="btn-godmode" type="button" onclick="openGodModeFolder()" title="Открыть панель всех настроек Windows (God Mode)">
152 |                 <div class="d-flex align-items-center gap-2">
153 |                     <span class="fs-6">☐ </span>
154 |                     <span class="fw-medium text-body">God Mode (Все на
155 |                     </div>
156 |                     <i class="bi bi-box-arrow-up-right text-muted small">></i>
157 |                 </button>
158 |                 <button class="btn btn-sm btn-outline-secondary w-100 d-flex align-items-center gap-2 py-2 rounded" id="menu-editor-btn" type="button">
159 |                     <i class="bi bi-gear-fill"></i>
160 |                     <span class="fw-medium">Редактор меню</span>
161 |                 </button>
162 |             </div>
163 |         </div>
164 |     </div>
165 |
166 |     <!-- User / Quick Settings Modal -->
```

```
167 | <div class="modal fade" id="userSettingsModal" tabindex="-1"
    | aria-labelledby="userSettingsModalLabel" aria-hidden="true">
168 |   <div class="modal-dialog modal-dialog-centered">
169 |     <div class="modal-content shadow border-secondary-subtle
    | ">
170 |       <div class="modal-header border-secondary-subtle">
171 |         <h5 class="modal-title d-flex align-items-center gap
    | -2" id="userSettingsModalLabel">
172 |           <i class="bi bi-gear-fill text-primary"></i>
173 |           <span data-i18n="userSettings.title">Настройки инт
    | ерфейса</span>
174 |         </h5>
175 |         <button type="button" class="btn-close" data-bs-dism
    | iss="modal" aria-label="Закрыть"></button>
176 |       </div>
177 |       <div class="modal-body p-3">
178 |         <!-- Language Selection -->
179 |         <div class="mb-3">
180 |           <label class="form-label fw-semibold small d-flex
    | align-items-center gap-2" for="apps-lang-selector">
181 |             <i class="bi bi-translate text-info"></i>
182 |             <span data-i18n="userSettings.language">Язык инт
    | ерфейса</span>
183 |           </label>
184 |           <select class="form-select lang-selector" id="apps
    | -lang-selector">
185 |             <option value="ru-RU">Русский (ru-RU)</option>
186 |             <option value="en-US">English (en-US)</option>
187 |             <option value="he-IL">עברית (he-IL)</option>
188 |             <option value="uk-UA">Українська (uk-UA)</option
    | >
189 |           </select>
190 |         </div>
191 |
192 |         <!-- Theme Selection -->
193 |         <div class="mb-2">
194 |           <label class="form-label fw-semibold small d-flex
    | align-items-center gap-2" for="apps-theme-selector">
```

```

195         <i class="bi bi-palette text-warning"></i>
196         <span data-i18n="userSettings.theme">Тема оформл
ения</span>
197     </label>
198     <select class="form-select theme-selector" id="app
s-theme-selector">
199         <option value="system">□ Системная</option>
200         <optgroup label="□ Светлые темы">
201             <option value="light">□ Светлая (Классическая)
</option>
202             <option value="brick">□ Кирпичная (Терракота)
</option>
203         </optgroup>
204         <optgroup label="□ Тёмные темы">
205             <option value="dark">□ Тёмная (Контрастная)</
option>
206             <option value="terminal">□ Терминал (Matrix /
CRT)</option>
207         </optgroup>
208     </select>
209 </div>
210 </div>
211 </div>
212 </div>
213 </div>
214
215 <!-- Menu Editor Modal -->
216 <div class="modal fade" id="menuEditorModal" tabindex="-1" a
ria-labelledby="menuEditorModalLabel" aria-hidden="true">
217     <div class="modal-dialog modal-xl modal-dialog-centered">
218         <div class="modal-content bg-dark text-white">
219             <div class="modal-header border-secondary">
220                 <h5 class="modal-title fw-bold" id="menuEditorModall
abel">
221                     <i class="bi bi-gear-fill text-primary"></i> Редак
тор меню
222                 </h5>
223                 <button type="button" class="btn-close btn-close-whi

```



```

te" data-bs-dismiss="modal" aria-label="Заккрыть"></button>
224     </div>
225     <div class="modal-body">
226         <div class="alert alert-info small mb-3">
227             <i class="bi bi-info-circle"></i> Настройте отобра
жение и порядок элементов меню. Элементы сгруппированы по пози
циям: сверху, сбоку и скрытые. Переключатели меняют размещение
, а стрелки и слайдеры – порядок элементов.
228         </div>
229
230         <div class="menu-editor-section">
231             <div id="allMenuEditor">
232                 <div class="text-muted small p-3 text-center">За
грузка...</div>
233             </div>
234         </div>
235     </div>
236     <div class="modal-footer border-secondary">
237         <button type="button" class="btn btn-secondary" data
-bs-dismiss="modal">Отмена</button>
238         <button type="button" class="btn btn-primary" id="sa
veMenuConfig">
239             <i class="bi bi-save"></i> Сохранить
240         </button>
241     </div>
242 </div>
243 </div>
244 </div>
245
246 <!-- Main Tab Content Container (Apps Only) -->
247 <div class="tab-content" id="mainTabContent">
248     <!-- About System Tab (Default Primary) -->
249     <div id="tab-about-system" class="tab-pane fade show activ
e"></div>
250
251     <!-- Chat Tab -->
252     <div id="tab-chat" class="tab-pane fade"></div>
253

```

```
254 | <!-- Scenarios Tab -->
255 | <div id="tab-scenarios" class="tab-pane fade"></div>
256 |
257 | <!-- Network Terminal Tab -->
258 | <div id="tab-network" class="tab-pane fade"></div>
259 |
260 | <!-- Hardware Load Inspector Tab -->
261 | <div id="tab-hardware-load-inspector" class="tab-pane fade
    | "></div>
262 |
263 | <!-- Processes Load & Activity Inspector Tab -->
264 | <div id="tab-processes-load-inspector" class="tab-pane fad
    | e"></div>
265 |
266 | <!-- Windows System Administrator Tab -->
267 | <div id="tab-windows-admin" class="tab-pane fade"></div>
268 |
269 | <!-- Windows Focus Settings Tab -->
270 | <div id="tab-focus-settings" class="tab-pane fade"></div>
271 |
272 | <!-- System Control Center Tab (System Logs & Activity) --
    | >
273 | <div id="tab-system-control" class="tab-pane fade"></div>
274 |
275 | <!-- Post-Install Wizard Tab -->
276 | <div id="tab-post-install-wizard" class="tab-pane fade"></
    | div>
277 |
278 | <!-- Maintenance & Recovery Tab -->
279 | <div id="tab-maintenance-recovery" class="tab-pane fade"><
    | /div>
280 |
281 | <!-- System Log Viewer Tab -->
282 | <div id="tab-system-logs" class="tab-pane fade"></div>
283 |
284 | <!-- Software Audit Tab -->
285 | <div id="tab-software-audit" class="tab-pane fade"></div>
286 |
```

```
287 | <!-- Registry Viewer Tab -->
288 | <div id="tab-registry-viewer" class="tab-pane fade"></div>
289 |
290 | <!-- Microsoft Defender Security Tab -->
291 | <div id="tab-defender" class="tab-pane fade"></div>
292 |
293 | <!-- Startup Auditor Tab -->
294 | <div id="tab-startup-auditor" class="tab-pane fade"></div>
295 |
296 | <!-- Windows Backup & Libraries Manager Tab -->
297 | <div id="tab-windows-backup" class="tab-pane fade"></div>
298 |
299 | <!-- Hardware & Sensors Monitor Tab -->
300 | <div id="tab-hardware-monitor" class="tab-pane fade"></div>
    | >
301 |
302 | <!-- Cloudflared Monitor Tab -->
303 | <div id="tab-cloudflared" class="tab-pane fade"></div>
304 |
305 | <!-- Google Cloud Monitor Tab -->
306 | <div id="tab-gcloud" class="tab-pane fade"></div>
307 |
308 | <!-- Website Intelligence Monitor Tab -->
309 | <div id="tab-website-monitor" class="tab-pane fade"></div>
310 |
311 | <!-- Exchange Trading Terminal Tab -->
312 | <div id="tab-trading" class="tab-pane fade"></div>
313 |
314 | <!-- User Assistant Tab -->
315 | <div id="tab-user-assistant" class="tab-pane fade"></div>
316 |
317 | <!-- Helpdesk Tab -->
318 | <div id="tab-helpdesk" class="tab-pane fade"></div>
319 |
320 | <!-- Wikipedia Research Lab Tab -->
321 | <div id="tab-wikipedia-research" class="tab-pane fade"></div>
    | iv>
322 |
```

```
323 |      <!-- AI Software Transparency Scanner Tab -->
324 |      <div id="tab-software-transparency" class="tab-pane fade">
    | </div>
325 |
326 |      <!-- User Directories and Storage Tab -->
327 |      <div id="tab-user-directories" class="tab-pane fade"></div>
    | >
328 |
329 |      <!-- Ninite Auto-Updater Tab -->
330 |      <div id="tab-ninite-updater" class="tab-pane fade"></div>
331 |
332 |      <!-- File Recovery (R-Studio) Tab -->
333 |      <div id="tab-file-recovery" class="tab-pane fade"></div>
334 |
335 |      <!-- Telemetry Research Tab -->
336 |      <div id="tab-telemetry-research" class="tab-pane fade"></d
    | iv>
337 |
338 |      <!-- Process Leak Hunter Tab -->
339 |      <div id="tab-process-leaks" class="tab-pane fade"></div>
340 |
341 |      <!-- Behavioral Forensics Tab -->
342 |      <div id="tab-forensics" class="tab-pane fade"></div>
343 |
344 |      <!-- Kernel Quality & Throttling Tab -->
345 |      <div id="tab-throttling" class="tab-pane fade"></div>
346 |
347 |      <!-- Storage & Battery Wear Tab -->
348 |      <div id="tab-storage-wear" class="tab-pane fade"></div>
349 |
350 |      <!-- Peripherals & Wireless Network Tab -->
351 |      <div id="tab-peripherals" class="tab-pane fade"></div>
352 |
353 |      <!-- RAG Tab -->
354 |      <div id="tab-rag" class="tab-pane fade"></div>
355 |
356 |      <!-- PixelRAG Tab -->
357 |      <div id="tab-pixelrag" class="tab-pane fade"></div>
```

```
358 |
359 | <!-- Models & API Tab -->
360 | <div id="tab-models" class="tab-pane fade"></div>
361 |
362 | <!-- AI Agents Tab -->
363 | <div id="tab-agents" class="tab-pane fade"></div>
364 |
365 | <!-- Skills Tab -->
366 | <div id="tab-skills" class="tab-pane fade"></div>
367 |
368 | <!-- MCP Servers Tab -->
369 | <div id="tab-mcp" class="tab-pane fade"></div>
370 |
371 | <!-- Google User Desktop Tab -->
372 | <div id="tab-google-desktop" class="tab-pane fade"></div>
373 |
374 | <!-- File History Search Tab -->
375 | <div id="tab-file-history-search" class="tab-pane fade"></div>
376 |
377 | <!-- AI Benchmark Tab -->
378 | <div id="tab-ai-benchmark" class="tab-pane fade"></div>
379 |
380 | <!-- Windows CLI Modular Applications Tabs -->
381 | <div id="tab-storage-manager" class="tab-pane fade"></div>
382 | <div id="tab-disk-speed" class="tab-pane fade"></div>
383 | <div id="tab-boot-recovery" class="tab-pane fade"></div>
384 | <div id="tab-servicing-integrity" class="tab-pane fade"></div>
385 | <div id="tab-services-manager" class="tab-pane fade"></div>
386 | <div id="tab-task-scheduler" class="tab-pane fade"></div>
387 | <div id="tab-process-manager" class="tab-pane fade"></div>
388 | <div id="tab-firewall-manager" class="tab-pane fade"></div>
389 | <div id="tab-security-acl" class="tab-pane fade"></div>
390 | <div id="tab-performance-tracing" class="tab-pane fade"></div>
```

```
391 |
392 |     <div id="tab-log-settings" class="tab-pane fade"></div>
393 |     <div id="tab-software-manager" class="tab-pane fade"></div>
    | >
394 |     <div id="tab-taskbar-controller" class="tab-pane fade"></div>
    | iv>
395 |     <div id="tab-accounts-identity" class="tab-pane fade"></div>
    | v>
396 |     <div id="tab-power-lifecycle" class="tab-pane fade"></div>
397 | </div>
398 | </div>
399 |
400 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    | /js/bootstrap.bundle.min.js"></script>
401 | <script src="/html/js/ai_table_modal.js?v=20261008_v2"></scrip
    | t>
402 | <script src="/html/js/window-api.js?v=20261008_v2"></script>
403 | <script src="/html/js/chatService.js?v=20261008_v2"></script>
404 | <script src="/html/js/appConfigEditor.js?v=20261008_v2"></scri
    | pt>
405 | <script type="module" src="/html/apps/main.js?v=20261008_v2"><
    | /script>
406 |
407 | </body>
408 | </html>
409 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Apps - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/apps/main.js?v=2026100
11 *     1_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/apps
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-08 04:24:00
19 * =====
20 */
21 /**
22 * apps/main.js – оркестратор страницы /tc (tc.ps1)
23 * Архитектура: UI_ARCHITECTURE.md
24 */
25
26 import { setupGlobalApi, setupThemeAndLang } from './modules/i
27 nit-interface.js';
28 import { fetchAppsStatus, updateModelBadge } from './modules/s
29 tatus-manager.js';
30 import { initModelTester, sendModelPing } from './modules/mode
31 l-tester.js';
32 import { APP_TAB_DEFS } from './modules/tabs-config.js';
```

```

30 import { applyTranslations } from '../js/i18n.js';
31 import { switchTab, loadTab, setupTabClicks } from '../js/tab-
  core.js';
32
33 // Глобальный экспорт
34 window.switchTab = switchTab;
35 window.switchToTab = switchTab;
36 window.sendModelPing = sendModelPing;
37 window.initModelTester = initModelTester;
38
39 // Карта: tabId → пути (из tabs-config, не генерируем)
40 const TAB_PATHS = Object.fromEntries(
41   APP_TAB_DEFS.map(d => [d.tabId, { html: d.html, js: d.js }])
42 );
43
44 // Lazy-загрузка: вкладка грузится только при первом открытии
45 const loadedTabs = new Set();
46
47 async function lazyLoad(tabId) {
48   const normId = tabId.startsWith('tab-') ? tabId : `tab-${tab
  Id}`;
49   if (loadedTabs.has(normId)) return;
50   const paths = TAB_PATHS[normId];
51   if (!paths) return;
52   const v = Date.now();
53   const name = normId.replace(/^tab-/, '');
54   const cleanHtml = paths.html ? (paths.html.includes('?') ? p
  aths.html.split('?')[0] : paths.html) : '';
55   const cleanJs = paths.js ? (paths.js.includes('?') ? paths.j
  s.split('?')[0] : paths.js) : '';
56   try {
57     await loadTab(name, `${cleanHtml}?v=${v}`, cleanJs ? `${cl
  eanJs}?v=${v}` : '');
58     loadedTabs.add(normId);
59     applyTranslations();
60   } catch (err) {
61     console.error(`[apps/main.js] Ошибка загрузки вкладки ${no
  rmId}:`, err);

```



```

62 |     loadedTabs.delete(normId);
63 |     if (window.toast) {
64 |         window.toast.error('Ошибка загрузки', `Не удалось загруз
    | ить раздел ${name}: ${err.message}`);
65 |     }
66 | }
67 | }
68 |
69 | const _switchTab = switchTab;
70 | export async function switchAppTab(tabId) {
71 |     if (!tabId) return;
72 |     const normId = tabId.startsWith('tab-') ? tabId : `tab-${tab
    | Id}`;
73 |     await lazyLoad(normId);
74 |     _switchTab(normId);
75 | }
76 | window.switchTab = switchAppTab;
77 | window.switchToTab = switchAppTab;
78 |
79 | // — МЕНЮ —————
    |
80 |
81 | /**
82 |  * Создает DOM-элемент иконки для кнопок и редактора (поддержи
    | вает bi-* и эмодзи)
83 |  * @param {string} iconStr
84 |  * @param {string} defaultIcon
85 |  * @returns {HTMLElement}
86 |  */
87 | function createIconElement(iconStr, defaultIcon = '🔍') {
88 |     const icon = iconStr || defaultIcon;
89 |     if (/^bi-[a-z0-9-]+$/i.test(icon) || icon.startsWith('bi-'))
    | {
90 |         const i = document.createElement('i');
91 |         i.className = `bi ${icon}`;
92 |         return i;
93 |     }
94 |     const span = document.createElement('span');

```

```
95 | span.textContent = icon;
96 | return span;
97 | }
98 |
99 | /**
100 |  * Обработчик прямого запуска программ (например R-Studio)
101 |  * @param {Object} item
102 |  * @param {HTMLElement} [iconEl]
103 |  */
104 | async function executeLaunchItem(item, iconEl = null) {
105 |     const origContent = iconEl ? iconEl.innerHTML : '';
106 |     if (iconEl) iconEl.innerHTML = '□';
107 |     try {
108 |         const res = await fetch('/api/recovery/launch', { method:
109 | 'POST' });
110 |         const data = await res.json();
111 |         if (res.ok && data.success) {
112 |             if (window.toast) {
113 |                 window.toast.success('R-Studio запущена', data.message
114 | || 'Программа восстановления файлов запущена');
115 |             }
116 |             } else {
117 |                 if (window.toast) {
118 |                     window.toast.error('Ошибка запуска', data.detail || 'Н
119 | е удалось запустить программу');
120 |                 }
121 |             }
122 |         } catch (err) {
123 |             if (window.toast) {
124 |                 window.toast.error('Ошибка связи', err.message);
125 |             }
126 |         } finally {
127 |             if (iconEl) iconEl.innerHTML = origContent;
128 |         }
129 |     }
130 |
131 | function getMenuTarget() {
132 |     const p = location.pathname.toLowerCase();
```

```
130 |   if (p.startsWith('/su') || location.search.includes('target=
    | su')) {
131 |       return 'su';
132 |   }
133 |   return 'tc';
134 | }
135 |
136 | async function buildMenu(appsMap = {}, customCfg = null) {
137 |   let cfg = customCfg;
138 |   const target = getMenuTarget();
139 |   if (!cfg) {
140 |     try {
141 |       const r = await fetch(`/api/v1/menu/config?target=${target}&t=${Date.now()}`);
142 |       if (r.ok) {
143 |         cfg = await r.json();
144 |       }
145 |     } catch (e) {
146 |       console.warn(`Не удалось загрузить /api/v1/menu/config?target=${target}, пробуем статический файл`, e);
147 |     }
148 |   }
149 |
150 |   if (!cfg) {
151 |     try {
152 |       const staticFile = target === 'su' ? 'su_menu_config.json' : 'tc_menu_config.json';
153 |       const r = await fetch(`/html/config_menus/${staticFile}?t=${Date.now()}`);
154 |       if (r.ok) {
155 |         cfg = await r.json();
156 |       }
157 |     } catch (e) {
158 |       console.error('Ошибка загрузки статического файла конфигурации меню', e);
159 |     }
160 |   }
161 | }
```

```
162 | if (!cfg || !cfg.menu) return null;
163 |
164 | const enabled = item => appsMap[item.id]?.enabled !== false;
165 | const sorted = arr => [...arr]
166 |   .filter(x => x.visible !== false && enabled(x))
167 |   .sort((a, b) => (a.order || 0) - (b.order || 0));
168 |
169 | // Верхнее меню
170 | const topEl = document.querySelector('.main-nav-container .d
    -flex.gap-1.flex-wrap');
171 | if (topEl) {
172 |   topEl.innerHTML = '';
173 |   sorted(cfg.menu.topButtons || []).forEach(item => {
174 |     const btn = document.createElement('button');
175 |     btn.className = 'btn btn-sm btn-outline-primary d-flex a
    lign-items-center gap-1.5 py-1 px-2 rounded';
176 |     btn.type = 'button';
177 |     btn.title = item.label;
178 |
179 |     const iconEl = createIconElement(item.icon, 'bi-app');
180 |     const labelSpan = document.createElement('span');
181 |     labelSpan.className = 'd-none d-sm-inline';
182 |     labelSpan.textContent = item.label;
183 |     if (item.i18n) labelSpan.dataset.i18n = item.i18n;
184 |
185 |     btn.append(iconEl, labelSpan);
186 |
187 |     if (item.id === 'file_recovery' || item.action === 'laun
    ch') {
188 |       btn.dataset.action = 'launch';
189 |       btn.onclick = async (e) => {
190 |         e.preventDefault();
191 |         e.stopPropagation();
192 |         await executeLaunchItem(item, iconEl);
193 |       };
194 |     } else {
195 |       btn.dataset.tab = item.tab;
196 |     }
```

```
197 |
198 |     topEl.appendChild(btn);
199 | });
200 | }
201 |
202 | // Боковое меню
203 | const navEl = document.getElementById('appsNavTabs');
204 | if (navEl) {
205 |     navEl.innerHTML = '';
206 |     sorted(cfg.menu.sidebarItems || []).forEach(item => {
207 |         const btn = document.createElement('button');
208 |         btn.className = 'list-group-item list-group-item-action
d-flex align-items-center gap-2 py-2 px-3';
209 |         btn.type = 'button';
210 |
211 |         const iconEl = createIconElement(item.icon, ' ');
212 |         iconEl.classList.add('fs-6');
213 |         const label = document.createElement('span');
214 |         label.className = 'fw-medium';
215 |         label.textContent = item.label;
216 |         if (item.i18n) label.dataset.i18n = item.i18n;
217 |
218 |         btn.append(iconEl, label);
219 |
220 |         if (item.id === 'file_recovery' || item.action === 'laun
ch') {
221 |             btn.dataset.action = 'launch';
222 |             btn.title = 'Запуск программы восстановления файлов (R
-Studio)';
223 |             btn.onclick = async (e) => {
224 |                 e.preventDefault();
225 |                 e.stopPropagation();
226 |                 await executeLaunchItem(item, iconEl);
227 |             };
228 |         } else {
229 |             btn.dataset.tab = item.tab;
230 |         }
231 |
```

```

232     navEl.appendChild(btn);
233   });
234 }
235
236   // Обновление активного состояния кнопок вкладок
237   const activePane = document.querySelector('#mainTabContent >
    .tab-pane.active') || document.querySelector('.tab-pane.activ
    e');
238   const activeTabId = activePane ? activePane.id : null;
239   if (activeTabId) {
240     document.querySelectorAll('[data-tab]').forEach(btn => {
241       btn.classList.toggle('active', btn.dataset.tab === activ
    eTabId);
242     });
243   }
244
245   applyTranslations();
246   return cfg;
247 }
248
249 export async function loadRequiredTabs(tabs = []) {
250   for (const tab of tabs) {
251     await lazyLoad(tab);
252   }
253 }
254
255 // — РЕДАКТОР МЕНЮ —————
    _____
256
257 function initMenuEditor(cfg, appsMap = {}) {
258   const editorBtn = document.getElementById('menu-editor-btn')
    ;
259   const modal = document.getElementById('menuEditorModal');
260   const list = document.getElementById('allMenuEditor');
261   const saveBtn = document.getElementById('saveMenuConfig');
262   if (!editorBtn || !modal || !list || !cfg) return;
263
264   let currentItems = [];

```

```
265 | let draggedGlobalIdx = null;
266 |
267 | function collectInitialItems() {
268 |     return [
269 |         ...(cfg.menu.topButtons || []).map(x => ({ ...x, position: x.position || (x.visible === false ? 'hidden' : 'top') })),
270 |         ...(cfg.menu.sidebarItems || []).map(x => ({ ...x, position: x.position || (x.visible === false ? 'hidden' : 'bottom') })),
271 |     ];
272 | }
273 |
274 | function renderList() {
275 |     list.innerHTML = '';
276 |
277 |     const GROUPS = [
278 |         {
279 |             key: 'top',
280 |             title: 'Сверху (Верхняя панель)',
281 |             icon: 'bi-layout-text-window-reverse',
282 |             badgeCls: 'bg-primary text-white',
283 |             borderCls: 'border-primary-subtle'
284 |         },
285 |         {
286 |             key: 'bottom',
287 |             title: 'Сбоку (Боковое меню)',
288 |             icon: 'bi-layout-sidebar-inset',
289 |             badgeCls: 'bg-info text-dark',
290 |             borderCls: 'border-info-subtle'
291 |         },
292 |         {
293 |             key: 'hidden',
294 |             title: 'Скрыто (Не отображаются)',
295 |             icon: 'bi-eye-slash',
296 |             badgeCls: 'bg-secondary text-white',
297 |             borderCls: 'border-secondary-subtle'
298 |         }
299 |     ];
```

```
300
301     GROUPS.forEach(grp => {
302         const groupItems = currentItems.filter(x => x.position =
303         == grp.key);
304         const grpTotal = groupItems.length;
305         const section = document.createElement('div');
306         section.className = `menu-group-section mb-3 p-2 rounded
307         border ${grp.borderCls}`;
308         section.style.background = `var(--surface-1, rgba(255,25
309         5,255,0.03))`;
310         // Заголовок группы
311         const header = document.createElement('div');
312         header.className = 'd-flex align-items-center justify-co
313         ntent-between mb-2 px-1 pb-1 border-bottom border-secondary-su
314         btle';
315         header.innerHTML = `
316         <span class="fw-bold d-flex align-items-center gap-2 s
317         mall">
318             <i class="bi ${grp.icon}"></i> ${grp.title}
319             </span>
320             <span class="badge ${grp.badgeCls} rounded-pill px-2 p
321             y-1">${grpTotal}</span>
322         `;
323         section.appendChild(header);
324         const itemsContainer = document.createElement('div');
325         itemsContainer.className = 'menu-group-items d-flex flex
326         -column gap-2';
327         itemsContainer.dataset.group = grp.key;
328         if (grpTotal === 0) {
329             const empty = document.createElement('div');
330             empty.className = 'text-muted small text-center py-2 f
331             st-italic';
332             empty.textContent = 'Нет элементов в этой группе';
333             itemsContainer.appendChild(empty);
334         }
335     });
```



```

329 |     } else {
330 |         groupItems.forEach((item, grpIdx) => {
331 |             const globalIdx = currentItems.indexOf(item);
332 |             const safeId = (item.id || `item_${globalIdx}`).replace(/[^a-zA-Z0-9_-]/g, '');
333 |
334 |             const div = document.createElement('div');
335 |             div.className = 'menu-editor-item d-flex align-items-center gap-2 p-2 rounded border border-secondary-subtle';
336 |             div.dataset.id = item.id || `item_${globalIdx}`;
337 |             div.dataset.globalIndex = globalIdx;
338 |             div.dataset.groupIndex = grpIdx;
339 |             div.draggable = true;
340 |
341 |             // 1. Ручка перетаскивания (Drag Handle)
342 |             const dragHandle = document.createElement('div');
343 |             dragHandle.className = 'drag-handle-container text-muted px-1';
344 |             dragHandle.title = 'Перетащите для изменения порядка';
345 |             dragHandle.style.cursor = 'grab';
346 |             dragHandle.innerHTML = '<i class="bi bi-grip-vertical fs-5"></i>';
347 |
348 |             // 2. Порядковый номер в группе
349 |             const orderBadge = document.createElement('span');
350 |             orderBadge.className = 'badge bg-secondary-subtle text-light border px-2 py-1';
351 |             orderBadge.textContent = `#${grpIdx + 1}`;
352 |             orderBadge.style.minWidth = '38px';
353 |             orderBadge.style.textAlign = 'center';
354 |
355 |             // 3. Иконка элемента
356 |             const iconContainer = document.createElement('div');
357 |             iconContainer.className = 'item-icon d-flex align-items-center justify-content-center px-1 text-center';
358 |             iconContainer.style.minWidth = '28px';
359 |             const iconEl = createIconElement(item.icon, ' ');

```

```

360         iconContainer.appendChild(iconEl);
361
362         // 4. Информация об элементе
363         const info = document.createElement('div');
364         info.className = 'item-info flex-grow-1 min-w-0 px-1
';
365         const labelDiv = document.createElement('div');
366         labelDiv.className = 'fw-semibold text-truncate smal
l';
367         labelDiv.textContent = item.label;
368         const tabDiv = document.createElement('div');
369         tabDiv.className = 'text-muted text-truncate font-mo
nospace';
370         tabDiv.style.fontSize = '0.75rem';
371         tabDiv.textContent = item.tab;
372         info.append(labelDiv, tabDiv);
373
374         // 5. Управление порядком внутри группы
375         const orderCtrl = document.createElement('div');
376         orderCtrl.className = 'd-flex align-items-center gap
-1 me-2';
377         orderCtrl.style.width = '160px';
378         orderCtrl.style.flexShrink = '0';
379
380         const btnUp = document.createElement('button');
381         btnUp.type = 'button';
382         btnUp.className = 'btn btn-sm btn-outline-secondary
p-0 px-1';
383         btnUp.title = 'Переместить выше в группе';
384         btnUp.disabled = grpIdx === 0;
385         btnUp.innerHTML = '<i class="bi bi-chevron-up"></i>'
;
386         btnUp.addEventListener('click', () => {
387             if (grpIdx > 0) {
388                 const prevItem = groupItems[grpIdx - 1];
389                 const curIdx = currentItems.indexOf(item);
390                 const targetIdx = currentItems.indexOf(prevItem)
;

```

```

391         currentItems.splice(curIdx, 1);
392         currentItems.splice(targetIdx, 0, item);
393         renderList();
394     }
395 });
396
397     const slider = document.createElement('input');
398     slider.type = 'range';
399     slider.className = 'form-range order-slider flex-grow-1 m-0';
400     slider.min = '1';
401     slider.max = String(grpTotal);
402     slider.value = String(grpIdx + 1);
403     slider.title = `Позиция в группе: ${grpIdx + 1} из ${grpTotal}`;
404
405     slider.addEventListener('input', (e) => {
406         const targetGrpIdx = parseInt(e.target.value, 10) - 1;
407         if (targetGrpIdx !== grpIdx && targetGrpIdx >= 0 &
408 & targetGrpIdx < grpTotal) {
409             const targetItem = groupItems[targetGrpIdx];
410             const curIdx = currentItems.indexOf(item);
411             const targetIdx = currentItems.indexOf(targetItem);
412             currentItems.splice(curIdx, 1);
413             currentItems.splice(targetIdx, 0, item);
414             renderList();
415         }
416     });
417
418     const btnDown = document.createElement('button');
419     btnDown.type = 'button';
420     btnDown.className = 'btn btn-sm btn-outline-secondary p-0 px-1';
421     btnDown.title = 'Переместить ниже в группе';
422     btnDown.disabled = grpIdx === grpTotal - 1;
423     btnDown.innerHTML = '<i class="bi bi-chevron-down"><

```

```

    /i>';
423     btnDown.addEventListener('click', () => {
424         if (grpIdx < grpTotal - 1) {
425             const nextItem = groupItems[grpIdx + 1];
426             const curIdx = currentItems.indexOf(item);
427             const targetIdx = currentItems.indexOf(nextItem)
            ;
428             currentItems.splice(curIdx, 1);
429             currentItems.splice(targetIdx, 0, item);
430             renderList();
431         }
432     });
433
434     orderCtrl.append(btnUp, slider, btnDown);
435
436     // 6. Переключатели позиции (Сверху / Сбоку / Скрыть
    )
437     const group = document.createElement('div');
438     group.className = 'btn-group btn-group-sm flex-shrink-0';
439     group.setAttribute('role', 'group');
440
441     const radioName = `pos_grp_${globalIdx}_${safeId}`;
442     [
443         ['top', 'Сверху', 'btn-outline-primary'],
444         ['bottom', 'Сбоку', 'btn-outline-info'],
445         ['hidden', 'Скрыть', 'btn-outline-danger']
446     ].forEach(([val, text, cls]) => {
447         const inp = document.createElement('input');
448         inp.type = 'radio';
449         inp.className = 'btn-check';
450         inp.autocomplete = 'off';
451         inp.name = radioName;
452         inp.id = `pos_${val}_${globalIdx}_${safeId}`;
453         inp.value = val;
454         inp.checked = (item.position === val);
455         inp.addEventListener('change', () => {
456             if (inp.checked && item.position !== val) {

```

```

457         item.position = val;
458         item.visible = (val !== 'hidden');
459         // Перемещаем элемент в конец новой группы в с
currentItems
460         const curIdx = currentItems.indexOf(item);
461         currentItems.splice(curIdx, 1);
462         const lastOfGroup = currentItems.filter(x => x
.position === val).pop();
463         if (lastOfGroup) {
464             const lastIdx = currentItems.indexOf(lastOfG
roup);
465             currentItems.splice(lastIdx + 1, 0, item);
466         } else {
467             currentItems.push(item);
468         }
469         renderList();
470     }
471 });
472
473     const lbl = document.createElement('label');
474     lbl.className = `btn ${cls}`;
475     lbl.setAttribute('for', inp.id);
476     lbl.textContent = text;
477     group.append(inp, lbl);
478 });
479
480 // Drag & Drop события
481 div.addEventListener('dragstart', (e) => {
482     draggedGlobalIdx = globalIdx;
483     div.classList.add('dragging');
484     e.dataTransfer.effectAllowed = 'move';
485     e.dataTransfer.setData('text/plain', String(global
Idx));
486 });
487
488 div.addEventListener('dragend', () => {
489     div.classList.remove('dragging');
490     document.querySelectorAll('.menu-editor-item').for

```

```

Each(el => el.classList.remove('drag-over')));
491     });
492
493     div.addEventListener('dragover', (e) => {
494         e.preventDefault();
495         e.dataTransfer.dropEffect = 'move';
496         div.classList.add('drag-over');
497     });
498
499     div.addEventListener('dragleave', () => {
500         div.classList.remove('drag-over');
501     });
502
503     div.addEventListener('drop', (e) => {
504         e.preventDefault();
505         div.classList.remove('drag-over');
506         const fromGlobalIdx = parseInt(e.dataTransfer.getDa
ata('text/plain'), 10);
507         if (!isNaN(fromGlobalIdx) && fromGlobalIdx !== glo
balIdx) {
508             const [moved] = currentItems.splice(fromGlobalId
x, 1);
509             moved.position = grp.key;
510             moved.visible = (grp.key !== 'hidden');
511             currentItems.splice(globalIdx, 0, moved);
512             renderList();
513         }
514     });
515
516     div.append(dragHandle, orderBadge, iconContainer, in
fo, orderCtrl, group);
517     itemsContainer.appendChild(div);
518 });
519 }
520
521 section.appendChild(itemsContainer);
522 list.appendChild(section);
523 });

```

```
524 |
525 |     return currentItems;
526 | }
527 |
528 | editorBtn.addEventListener('click', () => {
529 |     currentItems = collectInitialItems();
530 |     renderList();
531 |     new bootstrap.Modal(modal).show();
532 | });
533 |
534 | saveBtn.addEventListener('click', async () => {
535 |     cfg.menu.topButtons = currentItems
536 |         .filter(x => x.position === 'top')
537 |         .map(({ position, ...x }, idx) => ({ ...x, visible: true
538 | , order: idx + 1 }));
539 |
540 |     cfg.menu.sidebarItems = currentItems
541 |         .filter(x => x.position !== 'top')
542 |         .map(({ position, ...x }, idx) => ({ ...x, visible: posi
543 | tion === 'bottom', order: idx + 1 }));
544 |
545 |     try {
546 |         const target = getMenuTarget();
547 |         const r = await fetch(`/api/v1/menu/config?target=${targ
548 | et}`, {
549 |             method: 'POST',
550 |             headers: { 'Content-Type': 'application/json' },
551 |             body: JSON.stringify(cfg)
552 |         });
553 |         if (!r.ok) {
554 |             throw new Error(r.statusText || `HTTP ${r.status}`);
555 |         }
556 |
557 |         // Немедленно переопределяем меню и структуру вкладок в
558 |         // интерфейсе
559 |         await buildMenu(appsMap, cfg);
560 |     }
561 | }
```

```

558     const modalInstance = bootstrap.Modal.getInstance(modal)
559     ;
560     if (modalInstance) {
561         modalInstance.hide();
562     }
563     // Проверяем, активна ли ещё текущая вкладка, если нет –
    переключаемся на первую доступную
564     const currentTabEl = document.querySelector('#mainTabCon
tent > .tab-pane.active') || document.querySelector('.tab-pane
.active');
565     const activeTabId = currentTabEl ? currentTabEl.id : '';
566     const isStillInMenu = document.querySelector(`[data-tab=
"${activeTabId}"]`);
567     if (!isStillInMenu) {
568         const firstVisible = document.querySelector('[data-tab
]');
569         if (firstVisible?.dataset.tab) {
570             await switchAppTab(firstVisible.dataset.tab);
571         }
572     }
573 } catch (e) {
574     window.showToast?.('Ошибка при сохранении меню: ' + e.me
ssage, 'danger') || alert('Ошибка при сохранении меню: ' + e.m
essage);
575 }
576 });
577 }
578
579 // — INIT —————
580
581 async function init() {
582     setupGlobalApi();
583     await setupThemeAndLang();
584
585     if (location.pathname.startsWith('/tc'))
586         document.title = 'AI Breadboard – Test Computer (/tc)';

```



```
587 |
588 | // Единый обработчик кликов
589 | setupTabClicks();
590 |
591 | // Статус + бейдж модели
592 | const statusData = await fetchAppsStatus();
593 | await updateModelBadge(statusData);
594 | const appsMap = statusData?.apps || {};
595 |
596 | // Инициализация быстрой проверки доступности модели
597 | initModelTester();
598 |
599 | // Меню из конфига
600 | const cfg = await buildMenu(appsMap).catch(() => null);
601 | initMenuEditor(cfg, appsMap);
602 |
603 | applyTranslations();
604 |
605 | // Активировать вкладку по hash, defaultTab из конфига или п
    | ервую доступную
606 | const hash = location.hash.replace('#', '');
607 | const defaultTab = cfg?.settings?.defaultTab;
608 | const first = document.querySelector('#appsNavTabs [data-tab
    | ]') || document.querySelector('[data-tab]');
609 | const initialTab = hash || (defaultTab && document.querySele
    | ctor('[data-tab="${defaultTab}"]') ? defaultTab : first?.datas
    | et.tab || 'tab-about-system');
610 | await switchAppTab(initialTab);
611 | }
612 |
613 | document.readyState === 'loading'
614 |   ? document.addEventListener('DOMContentLoaded', init)
615 |   : init();
616 |
```

init-interface.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Init-Interface Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля init-inte
7  *   rface.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/apps/modules/init-inte
11  *   rface.js?v=20261006_v10" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { setupGlobalApi, setupThemeAndLang } from '/win
15  *   dows/api/webgui/apps/modules/init-interface.js';
16  *
17  * File: init-interface.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/apps/modules
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-06 22:15:00
23  * =====
24  */
25
26 /**
27  * apps/modules/init-interface.js — Инициализация тем, языка и
28  * глобального окружения /apps
29  */
30
31 import { initI18n, switchLang, switchLocale, applyTranslations
32   , normalizeLocaleTag, getCurrentLocale, getCurrentLang } from
```

```

    '../.../js/i18n.js';
29 import { initTheme, setTheme, getThemeMode, getResolvedTheme,
    themeEngine, ThemeEngine } from '../.../js/theme.js';
30
31 export function setupGlobalApi() {
32     window.switchLang = switchLang;
33     window.switchLocale = switchLocale;
34     window.normalizeLocaleTag = normalizeLocaleTag;
35     window.getCurrentLocale = getCurrentLocale;
36     window.getCurrentLang = getCurrentLang;
37     window.applyTranslations = applyTranslations;
38     window.setTheme = setTheme;
39     window.getThemeMode = getThemeMode;
40     window.getResolvedTheme = getResolvedTheme;
41     window.themeEngine = themeEngine;
42     window.ThemeEngine = ThemeEngine;
43
44     window.api = window.api || {
45         async fetch(url, options = {}) {
46             const response = await fetch(url, options);
47             if (!response.ok) {
48                 let msg = response.statusText;
49                 try {
50                     const data = await response.json();
51                     if (data && data.detail) {
52                         if (typeof data.detail === 'string') {
53                             msg = data.detail;
54                         } else if (Array.isArray(data.detail)) {
55                             msg = data.detail.map(d => d.msg || JSON.stringify(d)).join(', ');
56                         } else {
57                             msg = JSON.stringify(data.detail);
58                         }
59                     }
60                 } catch {}
61                 throw new Error(`${response.status} ${msg}`);
62             }
63             return response.json();

```

```
64     }
65   };
66
67   window.appsStatusMap = window.appsStatusMap || {};
68 }
69
70 export async function setupThemeAndLang() {
71   initTheme();
72
73   // Инициализация i18n с автоматической проверкой URL парамет
74   // ров (?region=ru-ru и т.д.)
75   await initI18n();
76
77   const langSel = document.getElementById('apps-lang-selector'
78   );
79   if (langSel) {
80     const curLocale = getCurrentLocale();
81     const curLang = getCurrentLang();
82     const matchLocale = Array.from(langSel.options).find(opt =>
83     > opt.value.toLowerCase() === curLocale.toLowerCase());
84     const matchLang = Array.from(langSel.options).find(opt =>
85     opt.value.toLowerCase() === curLang.toLowerCase());
86     if (matchLocale) langSel.value = matchLocale.value;
87     else if (matchLang) langSel.value = matchLang.value;
88
89     langSel.addEventListener('change', (e) => {
90       switchLocale(e.target.value);
91     });
92   }
93
94   const themeSel = document.getElementById('apps-theme-selecto
95   r');
96   if (themeSel) {
97     themeSel.value = getThemeMode();
98     themeSel.addEventListener('change', (e) => {
99       setTheme(e.target.value);
100     });
101   }
102 }
```

97 | }

98 |

model-tester.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Model-Tester Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом проверки модели
   в подсистеме /apps.
7  *
8  * Usage Examples:
9  *   JavaScript Import:
10 *     import { initModelTester, sendModelPing } from './modules/model-tester.js';
11 *
12 * File: model-tester.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/apps/modules
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-08 04:23:00
18 * =====
19 */
20
21 export { initModelTester, sendModelPing } from '../..js/model
   -tester.js';
22
```

status-manager.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Status-Manager Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля status-ma
7  *   nager.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/apps/modules/status-ma
11  *     nager.js?v=20261001_v1" type="module"></script>
12  *
13  * File: status-manager.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/apps/modules
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 05:26:00
19  * =====
20  */
21 /**
22  * apps/modules/status-manager.js – Управление состоянием прил
23  * ожений и бейджем модели
24  */
25 export async function fetchAppsStatus() {
26   const isTcRoute = window.location.pathname.startsWith('/tc')
27   ;
28   const query = isTcRoute ? '?profile=tc' : '';
29   // 1. Получаем статус приложений и AI-конфига по стандарту /
```

```

    api/v1/apps/status
30   let statusData = null;
31   try {
32       const resp = await fetch(`/api/v1/apps/status${query}`);
33       if (resp.ok) {
34           statusData = await resp.json();
35       }
36   } catch (err) {
37       console.warn('[AppsHub] Failed to fetch /api/v1/apps/status:', err);
38   }
39
40   if (statusData && statusData.apps) {
41       window.appsStatusMap = statusData.apps;
42   }
43
44   // 2. Если в ответе нет секции ai, явно запрашиваем активную
   модель через /api/v1/chat/active-model
45   if (!statusData?.ai) {
46       try {
47           const modelResp = await fetch(`/api/v1/chat/active-model
   ${query}`);
48           if (modelResp.ok) {
49               const modelData = await modelResp.json();
50               if (modelData && modelData.model) {
51                   statusData = statusData || {};
52                   statusData.ai = {
53                       provider: modelData.provider,
54                       model: modelData.model
55                   };
56                   statusData.config_file = modelData.config_file;
57               }
58           }
59       } catch (err) {
60           console.warn('[AppsHub] Could not fetch active model from /api/v1/chat/active-model:', err);
61       }
62   }

```



```
63 |
64 |   return statusData;
65 | }
66 |
67 | export async function updateModelBadge(statusData) {
68 |   const modelBadgeText = document.getElementById('chat-model-b
  adge');
69 |   if (!modelBadgeText) return;
70 |
71 |   let prov = '';
72 |   let mod = '';
73 |   let cfgFile = statusData?.config_file || '';
74 |
75 |   // 1. Приоритет: запрос актуальной активной модели с сервера
  (/api/v1/chat/active-model)
76 |   try {
77 |     const isTcRoute = window.location.pathname.startsWith('/tc
  ');
78 |     const query = isTcRoute ? '?profile=tc' : '';
79 |     const resp = await fetch(`/api/v1/chat/active-model${query
  }`);
80 |     if (resp.ok) {
81 |       const data = await resp.json();
82 |       if (data && data.model) {
83 |         prov = data.provider || '';
84 |         mod = data.model;
85 |         cfgFile = data.config_file || cfgFile;
86 |       }
87 |     }
88 |   } catch (err) {
89 |     console.warn('[AppsHub] Failed to fetch active model from
  /api/v1/chat/active-model:', err);
90 |   }
91 |
92 |   // 2. Резервный парсинг конфигурации statusData.ai, если энд
  поинт не вернул модель
93 |   if (!mod && statusData?.ai) {
94 |     const ai = statusData.ai;
```

```

95     if (ai.provider) {
96         const p = String(ai.provider).toLowerCase();
97         prov = String(ai.provider).toUpperCase();
98         mod = ai.model || (ai[p] && typeof ai[p] === 'object' ?
ai[p].model : null) || (ai.providers && ai.providers[p] ? ai.p
roviders[p].model : null) || ai[`${p}_model_id`] || '';
99     } else if (ai.providers && typeof ai.providers === 'object
') {
100         for (const [pk, pv] of Object.entries(ai.providers)) {
101             if (pv && pv.enabled) {
102                 prov = pk.toUpperCase();
103                 mod = pv.model || '';
104                 break;
105             }
106         }
107     } else if (ai.use_gemini) {
108         prov = 'GEMINI';
109         mod = ai.gemini_model_id || ai.model || 'gemini-3.1-flas
h';
110     } else if (ai.use_agy) {
111         prov = 'AGY';
112         mod = ai.agy_model_id || ai.model || 'gemini-3.6-flash';
113     } else if (ai.use_ollama) {
114         prov = 'OLLAMA';
115         mod = ai.ollama_model_id || ai.model || 'llama3.1';
116     } else if (ai.use_foundry) {
117         prov = 'FOUNDRY';
118         mod = ai.foundry_model_id || ai.model || 'local';
119     } else if (ai.model) {
120         mod = ai.model;
121         prov = 'AI';
122     }
123 }
124
125 // 3. Очистка и форматирование отображаемой строки
126 if (mod && mod.startsWith(`${prov.toLowerCase()}:`)) {
127     mod = mod.substring(prov.length + 1);
128 }

```

```
129 |
130 |     if (prov && mod && mod !== 'default') {
131 |         modelBadgeText.textContent = `${prov}: ${mod}`;
132 |         modelBadgeText.style.display = 'inline-flex';
133 |         modelBadgeText.title = `Используемая модель: ${mod} (Прова
    | йдер: ${prov}) – нажмите для выбора модели и провайдера`;
134 |     } else if (mod && mod !== 'default') {
135 |         modelBadgeText.textContent = `${mod}`;
136 |         modelBadgeText.style.display = 'inline-flex';
137 |         modelBadgeText.title = `Используемая модель: ${mod} – нажм
    | ите для выбора модели и провайдера`;
138 |     } else if (prov) {
139 |         modelBadgeText.textContent = `${prov}`;
140 |         modelBadgeText.style.display = 'inline-flex';
141 |         modelBadgeText.title = `Используемый провайдер: ${prov} –
    | нажмите для выбора модели и провайдера`;
142 |     } else {
143 |         modelBadgeText.textContent = 'AI: Не определена';
144 |     }
145 | }
146 |
147 |
```

tab-manager.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Tab-Manager Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля tab-manag
7  *   er.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/apps/modules/tab-manag
11  *   er.js?v=20261001_v1" type="module"></script>
12  *   JavaScript Import:
13  *     import { switchTab, setupNavTabs, loadTabContent } from
14  *     '/windows/api/webgui/apps/modules/tab-manager.js';
15  * File: tab-manager.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/apps/modules
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 // Реэкспорт для обратной совместимости
25 export const switchTab = (id) => window.switchTab?.(id);
26 export const setupNavTabs = () => {};
27 export const loadTabContent = async () => {};
28
```

tabs-config.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Tabs-Config Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля tabs-conf
7  *   ig.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/apps/modules/tabs-conf
11  *     ig.js?v=20261008_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { APP_TAB_DEFS, TC_EXCLUDES } from '/windows/api
15  *     /webgui/apps/modules/tabs-config.js';
16  *
17  * File: tabs-config.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/apps/modules
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-08 04:26:00
23  * =====
24  */
25
26 export const APP_TAB_DEFS = [
27   { id: 'about_system', tab: 'about-system', tabId: 'tab-about
28   -system', html: '/html/about_system_tab/index.html?v=20261008_
29   v1', js: '/html/about_system_tab/main.js?v=20261008_v1' },
30   { id: 'scenarios', tab: 'scenarios', tabId: 'tab-scenarios',
31   html: '/html/scenarios_tab/index.html?v=20261006_v8', js: '/h
32   tml/scenarios_tab/main.js?v=20261006_v8' },
```

```
27 { id: 'chat', tab: 'chat', tabId: 'tab-chat', html: '/html/c
    hat/index.html?v=20260923_v5', js: '/html/chat/main.js?v=20260
    923_v5' },
28 { id: 'network_terminal', tab: 'network', tabId: 'tab-networ
    k', html: '/html/network_tab/index.html', js: '/html/network_t
    ab/main.js' },
29 { id: 'system_inspector', tab: 'hardware-load-inspector', ta
    bId: 'tab-hardware-load-inspector', html: '/html/system_inspec
    tor_tab/index.html?v=20261008_v1', js: '/html/system_inspector
    _tab/main.js?v=20261008_v1' },
30 { id: 'processes_load_inspector', tab: 'processes-load-inspe
    ctor', tabId: 'tab-processes-load-inspector', html: '/html/pro
    cesses_load_inspector_tab/index.html?v=20261008_v1', js: '/htm
    l/processes_load_inspector_tab/main.js?v=20261008_v1' },
31 { id: 'windows_sysadmin', tab: 'windows-admin', tabId: 'tab-
    windows-admin', html: '/html/windows_admin_tab/index.html', js
    : '/html/windows_admin_tab/main.js' },
32 { id: 'focus_settings', tab: 'focus-settings', tabId: 'tab-f
    ocus-settings', html: '/html/focus_settings_tab/index.html?v=2
    0261006_v1', js: '/html/focus_settings_tab/main.js?v=20261006_
    v1' },
33 { id: 'system32_commands', tab: 'system32-commands', tabId:
    'tab-system32-commands', html: '/html/system32_commands_tab/in
    dex.html?v=20261006_v1', js: '/html/system32_commands_tab/main
    .js?v=20261006_v1' },
34 { id: 'system_control_center', tab: 'system-control', tabId:
    'tab-system-control', html: '/html/system_control_tab/index.h
    tml?v=20261008_v1', js: '/html/system_control_tab/main.js?v=20
    261008_v1' },
35 { id: 'post_install_wizard', tab: 'post-install-wizard', tab
    Id: 'tab-post-install-wizard', html: '/html/post_install_wizar
    d_tab/index.html?v=20261006_v1', js: '/html/post_install_wizar
    d_tab/main.js?v=20261006_v1' },
36 { id: 'maintenance_recovery', tab: 'maintenance-recovery', t
    abId: 'tab-maintenance-recovery', html: '/html/maintenance_rec
    overy_tab/index.html?v=20261006_v1', js: '/html/maintenance_re
    covery_tab/main.js?v=20261006_v1' },
37 { id: 'system_log_viewer', tab: 'system-logs', tabId: 'tab-s
```

```
system-logs', html: '/html/system_logs_tab/index.html', js: '/html/system_logs_tab/main.js' },
38 { id: 'software_audit', tab: 'software-audit', tabId: 'tab-software-audit', html: '/html/software_audit_tab/index.html', js: '/html/software_audit_tab/main.js' },
39 { id: 'registry_viewer', tab: 'registry-viewer', tabId: 'tab-registry-viewer', html: '/html/registry_viewer_tab/index.html', js: '/html/registry_viewer_tab/main.js' },
40 { id: 'windows_defender', tab: 'defender', tabId: 'tab-defender', html: '/html/defender_tab/index.html?v=20261006_v2', js: '/html/defender_tab/main.js?v=20261006_v2' },
41 { id: 'windows_startup_auditor', tab: 'startup-auditor', tabId: 'tab-startup-auditor', html: '/html/startup_auditor_tab/index.html?v=20261006_v15', js: '/html/startup_auditor_tab/main.js?v=20261006_v15' },
42 { id: 'windows_backup_manager', tab: 'windows-backup', tabId: 'tab-windows-backup', html: '/html/windows_backup_tab/index.html?v=20261004_v1', js: '/html/windows_backup_tab/main.js?v=20261004_v1' },
43 { id: 'hardware_monitor', tab: 'hardware-monitor', tabId: 'tab-hardware-monitor', html: '/html/hardware_monitor_tab/index.html', js: '/html/hardware_monitor_tab/main.js' },
44 { id: 'cloudflared_monitor', tab: 'cloudflared', tabId: 'tab-cloudflared', html: '/html/cloudflared_tab/index.html', js: '/html/cloudflared_tab/main.js' },
45 { id: 'google_user_desktop', tab: 'google-desktop', tabId: 'tab-google-desktop', html: '/html/google_desktop_tab/index.html', js: '/html/google_desktop_tab/main.js' },
46 { id: 'gcloud_monitor', tab: 'gcloud', tabId: 'tab-gcloud', html: '/html/gcloud_tab/index.html', js: '/html/gcloud_tab/main.js' },
47 { id: 'website_monitor', tab: 'website-monitor', tabId: 'tab-website-monitor', html: '/html/website_monitor_tab/index.html', js: '/html/website_monitor_tab/main.js' },
48 { id: 'trading_terminal', tab: 'trading', tabId: 'tab-trading', html: '/html/trading_tab/index.html', js: '/html/trading_tab/main.js' },
49 { id: 'user_assistant', tab: 'user-assistant', tabId: 'tab-u
```

```
ser-assistant', html: '/html/user_assistant_tab/index.html', j
s: '/html/user_assistant_tab/main.js' },
50 { id: 'helpdesk', tab: 'helpdesk', tabId: 'tab-helpdesk', ht
ml: '/html/helpdesk_tab/index.html', js: '/html/helpdesk_tab/m
ain.js' },
51 { id: 'wikipedia_research', tab: 'wikipedia-research', tabId
: 'tab-wikipedia-research', html: '/html/wikipedia_research_ta
b/index.html', js: '/html/wikipedia_research_tab/main.js' },
52 { id: 'software_transparency_scanner', tab: 'software-transp
arency', tabId: 'tab-software-transparency', html: '/html/soft
ware_transparency_tab/index.html', js: '/html/software_transpa
rency_tab/main.js' },
53 { id: 'user_directories', tab: 'user-directories', tabId: 't
ab-user-directories', html: '/html/user_directories_tab/index.
html', js: '/html/user_directories_tab/main.js' },
54 { id: 'ninite_updater', tab: 'ninite-updater', tabId: 'tab-n
inite-updater', html: '/html/ninite_updater_tab/index.html?v=2
0260924_v2', js: '/html/ninite_updater_tab/main.js?v=20260924_
v2' },
55 { id: 'file_recovery', tab: 'file-recovery', tabId: 'tab-fil
e-recovery', html: '/html/file_recovery_tab/index.html', js: '
/html/file_recovery_tab/main.js' },
56 { id: 'file_history_ai_search', tab: 'file-history-search',
tabId: 'tab-file-history-search', html: '/html/file_history_ai
_search_tab/index.html?v=20260928_v1', js: '/html/file_history
_ai_search_tab/main.js?v=20260928_v1' },
57 { id: 'telemetry_research', tab: 'telemetry-research', tabId
: 'tab-telemetry-research', html: '/html/telemetry_research_ta
b/index.html?v=20260924_v1', js: '/html/telemetry_research_tab
/main.js?v=20260924_v1' },
58 { id: 'telemetry_config', tab: 'telemetry-config', tabId: 't
ab-telemetry-config', html: '/html/telemetry_config_tab/index.
html?v=20261006_v1', js: '/html/telemetry_config_tab/main.js?v
=20261006_v1' },
59 { id: 'process_leaks', tab: 'process-leaks', tabId: 'tab-pro
cess-leaks', html: '/html/process_leaks_tab/index.html?v=20260
924_v1', js: '/html/process_leaks_tab/main.js?v=20260924_v1' }
,
```



```
60 { id: 'forensics', tab: 'forensics', tabId: 'tab-forensics',  
    html: '/html/forensics_tab/index.html?v=20261006_v2', js: '/h  
tml/forensics_tab/main.js?v=20261006_v2' },  
61  
62 { id: 'throttling', tab: 'throttling', tabId: 'tab-throttlin  
g', html: '/html/throttling_tab/index.html?v=20260924_v1', js:  
    '/html/throttling_tab/main.js?v=20260924_v1' },  
63 { id: 'storage_wear', tab: 'storage-wear', tabId: 'tab-stora  
ge-wear', html: '/html/storage_wear_tab/index.html?v=20260924_  
v1', js: '/html/storage_wear_tab/main.js?v=20260924_v1' },  
64 { id: 'peripherals', tab: 'peripherals', tabId: 'tab-periphe  
rals', html: '/html/peripherals_tab/index.html?v=20260924_v1',  
    js: '/html/peripherals_tab/main.js?v=20260924_v1' },  
65 { id: 'rag', tab: 'rag', tabId: 'tab-rag', html: '/html/rag_  
tab/index.html', js: '/html/rag_tab/main.js' },  
66 { id: 'pixelrag', tab: 'pixelrag', tabId: 'tab-pixelrag', ht  
ml: '/html/pixelrag_tab/index.html', js: '/html/pixelrag_tab/m  
ain.js' },  
67 { id: 'models', tab: 'models', tabId: 'tab-models', html: '/  
html/models_tab/index.html', js: '/html/models_tab/main.js' },  
68 { id: 'agents', tab: 'agents', tabId: 'tab-agents', html: '/  
html/agents_tab/index.html', js: '/html/agents_tab/main.js' },  
69 { id: 'skills', tab: 'skills', tabId: 'tab-skills', html: '/  
html/skills_tab/index.html', js: '/html/skills_tab/main.js' },  
70 { id: 'mcp', tab: 'mcp', tabId: 'tab-mcp', html: '/html/mcp_  
tab/index.html', js: '/html/mcp_tab/main.js' },  
71 { id: 'ai_benchmark', tab: 'ai-benchmark', tabId: 'tab-ai-be  
nchmark', html: '/html/ai_benchmark_tab/index.html?v=20260926_  
v1', js: '/html/ai_benchmark_tab/main.js?v=20260926_v1' },  
72  
73 // Новые модули управления Windows CLI  
74 { id: 'storage_manager', tab: 'storage-manager', tabId: 'tab  
-storage-manager', html: '/html/storage_manager_tab/index.html  
?v=20261001_v1', js: '/html/storage_manager_tab/main.js?v=2026  
1001_v1' },  
75 { id: 'disk_speed', tab: 'disk-speed', tabId: 'tab-disk-spee  
d', html: '/html/disk_speed_tab/index.html?v=20261004_v1', js:  
    '/html/disk_speed_tab/main.js?v=20261004_v1' },
```

```
76 { id: 'boot_recovery', tab: 'boot-recovery', tabId: 'tab-boo
    t-recovery', html: '/html/boot_recovery_tab/index.html?v=20261
    001_v1', js: '/html/boot_recovery_tab/main.js?v=20261001_v1' }
    ,
77 { id: 'servicing_integrity', tab: 'servicing-integrity', tab
    Id: 'tab-servicing-integrity', html: '/html/servicing_integrit
    y_tab/index.html?v=20261001_v1', js: '/html/servicing_integrit
    y_tab/main.js?v=20261001_v1' },
78 { id: 'services_manager', tab: 'services-manager', tabId: 't
    ab-services-manager', html: '/html/services_manager_tab/index.
    html?v=20261006_v1', js: '/html/services_manager_tab/main.js?v
    =20261006_v1' },
79 { id: 'task_scheduler', tab: 'task-scheduler', tabId: 'tab-t
    ask-scheduler', html: '/html/task_scheduler_tab/index.html?v=2
    0261006_v2', js: '/html/task_scheduler_tab/main.js?v=20261006_
    v2' },
80 { id: 'process_manager', tab: 'process-manager', tabId: 'tab
    -process-manager', html: '/html/process_manager_tab/index.html
    ?v=20261004_v2', js: '/html/process_manager_tab/main.js?v=2026
    1004_v2' },
81 { id: 'firewall_manager', tab: 'firewall-manager', tabId: 't
    ab-firewall-manager', html: '/html/firewall_manager_tab/index.
    html?v=20261001_v1', js: '/html/firewall_manager_tab/main.js?v
    =20261001_v1' },
82 { id: 'security_acl', tab: 'security-acl', tabId: 'tab-secur
    ity-acl', html: '/html/security_acl_tab/index.html?v=20261001_
    v1', js: '/html/security_acl_tab/main.js?v=20261001_v1' },
83 { id: 'performance_tracing', tab: 'performance-tracing', tab
    Id: 'tab-performance-tracing', html: '/html/performance_tracin
    g_tab/index.html?v=20261001_v1', js: '/html/performance_tracin
    g_tab/main.js?v=20261001_v1' },
84 { id: 'log_settings', tab: 'log-settings', tabId: 'tab-log-s
    ettings', html: '/html/log_settings_tab/index.html?v=20261008_
    v1', js: '/html/log_settings_tab/main.js?v=20261008_v1' },
85 { id: 'software_manager', tab: 'software-manager', tabId: 't
    ab-software-manager', html: '/html/software_manager_tab/index.
    html?v=20261004_v2', js: '/html/software_manager_tab/main.js?v
    =20261004_v2' },
```

```
86 | { id: 'taskbar_controller', tab: 'taskbar-controller', tabId:
    | : 'tab-taskbar-controller', html: '/html/taskbar_tab/index.htm
    | l?v=20261006_v1', js: '/html/taskbar_tab/main.js?v=20261006_v1
    | ' },
87 | { id: 'accounts_identity', tab: 'accounts-identity', tabId:
    | 'tab-accounts-identity', html: '/html/accounts_identity_tab/in
    | dex.html?v=20261008_v1', js: '/html/accounts_identity_tab/main
    | .js?v=20261008_v1' },
88 | { id: 'power_lifecycle', tab: 'power-lifecycle', tabId: 'tab
    | -power-lifecycle', html: '/html/power_lifecycle_tab/index.html
    | ?v=20261008_v2', js: '/html/power_lifecycle_tab/main.js?v=2026
    | 1008_v2' },
89 | ];
90 |
91 | export const TC_EXCLUDES = new Set([
92 |   'cloudflared_monitor',
93 |   'gcloud_monitor',
94 |   'website_monitor',
95 |   'user_assistant',
96 |   'wikipedia_research',
97 |   'trading_terminal',
98 |   'helpdesk'
99 | ]);
100 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/boot_recovery_tab/index.html
15
16   Tab Frame Integration:
17     <iframe src="/windows/api/webgui/boot_recovery_tab/index
18     .html" class="w-100 h-100 border-0"></iframe>
19
20 File: index.html
21 Project: ai-breadboard
22 Package: windows/api/webgui/boot_recovery_tab
23 Author: hypo69
24 Copyright: © 2026 hypo69
25 Updated: 2026-10-01 13:04:40
26 =====
27 =====
28 -->
29
30 <!-- Boot & Recovery Management Tab -->
31 <div class="container-fluid py-2 px-3">
32   <!-- Top Stat Cards -->
33   <div class="row g-2 mb-3">
34     <div class="col-6 col-md-3">
35       <div class="card bg-dark text-white border-secondary h-1
36       00 shadow-sm p-3">
37         <div class="text-muted small text-uppercase fw-semibol
38         d">Среда восстановления WinRE □ </div>
```

```
31         <div class="fs-4 fw-bold mt-1 text-success" id="br-win
re-status">--</div>
32         <div class="small text-muted mt-1" id="br-winre-sub">r
eagentc.exe</div>
33     </div>
34 </div>
35 <div class="col-6 col-md-3">
36     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
37         <div class="text-muted small text-uppercase fw-semibol
d">Таймаут меню BCD □ </div>
38         <div class="fs-4 fw-bold mt-1 text-info" id="br-bcd-ti
meout">-- сек</div>
39         <div class="small text-muted mt-1" id="br-bcd-sub">bcd
edit /timeout</div>
40     </div>
41 </div>
42 <div class="col-6 col-md-3">
43     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
44         <div class="text-muted small text-uppercase fw-semibol
d">Текущий загрузчик □ </div>
45         <div class="fs-4 fw-bold mt-1 text-primary" id="br-cur
rent-loader">--</div>
46         <div class="small text-muted mt-1" id="br-loader-sub">
Windows Boot Manager</div>
47     </div>
48 </div>
49 <div class="col-6 col-md-3">
50     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
51         <div class="text-muted small text-uppercase fw-semibol
d">Целостность Boot Sector □ </div>
52         <div class="fs-4 fw-bold mt-1 text-warning" id="br-boo
t-status">OK</div>
53         <div class="small text-muted mt-1" id="br-boot-sub">bo
otrec / bcdboot</div>
54     </div>
```

```

55     </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="br-refresh-btn">
62             <i class="bi bi-arrow-clockwise"></i>
63             <span>Обновить BCD</span>
64         </button>
65         <button class="btn btn-sm btn-outline-success d-flex align-items-center gap-1.5" id="br-winre-enable-btn">
66             <i class="bi bi-shield-check"></i>
67             <span>Включить WinRE</span>
68         </button>
69         <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5" id="br-winre-info-btn">
70             <i class="bi bi-info-circle"></i>
71             <span>Инфо WinRE</span>
72         </button>
73     </div>
74 </div>
75
76 <!-- Main Details & Tools Grid -->
77 <div class="row g-3">
78     <div class="col-12 col-lg-7">
79         <div class="card bg-dark text-white border-secondary shadow-sm">
80             <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
81                 <span class="fw-semibold"><i class="bi bi-list-columns text-primary me-2"></i>Записи хранилища BCD</span>
82                 <span class="badge bg-secondary" id="br-entries-badge">0 записей</span>
83             </div>

```

```

84 |         <div class="card-body p-0 table-responsive" style="max
| -height: 420px; overflow-y: auto;">
85 |             <table class="table table-dark table-hover table-str
| iped align-middle mb-0 small" id="br-entries-table">
86 |                 <thead>
87 |                     <tr class="text-secondary">
88 |                         <th>Идентификатор (GUID)</th>
89 |                         <th>Описание</th>
90 |                         <th>Устройство</th>
91 |                         <th>Путь</th>
92 |                         <th class="text-end">Действия</th>
93 |                     </tr>
94 |                 </thead>
95 |                 <tbody id="br-entries-tbody">
96 |                     <tr><td colspan="5" class="text-center text-mute
| d py-3">Опрос записей загрузчика BCD...</td></tr>
97 |                 </tbody>
98 |             </table>
99 |         </div>
100 |     </div>
101 | </div>
102 |
103 |     <div class="col-12 col-lg-5">
104 |         <div class="card bg-dark text-white border-secondary sha
| dow-sm mb-3">
105 |             <div class="card-header border-secondary d-flex align-
| items-center justify-content-between py-2 px-3">
106 |                 <span class="fw-semibold"><i class="bi bi-wrench-adj
| ustable text-warning me-2"></i>Инструменты восстановления</spa
| n>
107 |             </div>
108 |             <div class="card-body p-3">
109 |                 <div class="mb-3">
110 |                     <label class="form-label small text-muted">Команда
| восстановления:</label>
111 |                     <select class="form-select form-select-sm bg-dark
| text-white border-secondary" id="br-action-select">
112 |                         <option value="reagentc_info">Проверить состояни

```

```

    e WinRE (reagentc /info)</option>
113         <option value="reagentc_enable">Активировать Win
RE (reagentc /enable)</option>
114         <option value="bcdedit_enum">Перечислить все зап
иси (bcdedit /enum all)</option>
115         <option value="bcdboot_repair">Обновить системны
е файлы загрузки (bcdboot C:\Windows)</option>
116     </select>
117 </div>
118 <div class="d-flex align-items-center justify-content-between">
119     <div class="form-check form-switch">
120         <input class="form-check-input" type="checkbox"
id="br-dry-run-switch" checked>
121         <label class="form-check-label small text-muted"
for="br-dry-run-switch">Безопасный режим (Dry Run)</label>
122     </div>
123     <button class="btn btn-sm btn-success px-3" id="br
-execute-btn">
124         <i class="bi bi-play-fill"></i> Выполнить
125     </button>
126 </div>
127 </div>
128 </div>
129
130 <!-- Action Log Output -->
131 <div class="card bg-dark text-white border-secondary sha
dow-sm">
132     <div class="card-header border-secondary py-2 px-3 d-f
lex align-items-center justify-content-between">
133         <span class="small fw-semibold text-muted"><i class=
"bi bi-terminal me-1"></i>Консольный вывод</span>
134         <button class="btn btn-link btn-sm text-decoration-n
one text-muted p-0" id="br-clear-log-btn">Очистить</button>
135     </div>
136     <div class="card-body p-2">
137         <pre class="mb-0 small bg-black text-light p-2 round
ed" id="br-console-output" style="max-height: 180px; overflow-

```



```
y: auto; font-family: Consolas, monospace; font-size: 0.78rem;  
">Среда BCD и WinRE готова к операциям.</pre>
```

```
138|         </div>
```

```
139|     </div>
```

```
140| </div>
```

```
141| </div>
```

```
142| </div>
```

```
143|
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Boot Recovery Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/boot_recovery_tab/main
11  *     .js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { initBootRecoveryTab } from '/windows/api/webgui/boot_recovery_tab/main.js';
15  *
16  * File: main.js
17  * Project: ai-breadboard
18  * Package: windows/api/webgui/boot_recovery_tab
19  * Author: hypo69
20  * Copyright: © 2026 hypo69
21  * Updated: 2026-10-04 11:15:30
22  * =====
23  */
24  /**
25   * boot_recovery_tab/main.js — Управление BCD, загрузчиком и средой WinRE
26   * Updated: 2026-10-04 11:15:30
27   */
28
29  const registerTabPoller = window.registerTabPoller || function
  () {};
```

```
30
31 let isInitialized = false;
32
33 export async function fetchBootRecoverySummary() {
34   try {
35     const res = await fetch('/api/boot-recovery/summary');
36     if (!res.ok) throw new Error(`HTTP ${res.status}`);
37     const data = await res.json();
38     renderBootSummary(data);
39   } catch (err) {
40     console.warn('[BootRecovery] Ошибка получения сводки:', err);
41   }
42 }
43
44 function renderBootSummary(data) {
45   const winreStatusEl = document.getElementById('br-winre-status');
46   const bcdTimeoutEl = document.getElementById('br-bcd-timeout');
47   const currentLoaderEl = document.getElementById('br-current-loader');
48   const tbody = document.getElementById('br-entries-tbody');
49   const badge = document.getElementById('br-entries-badge');
50
51   if (winreStatusEl) winreStatusEl.textContent = data.winre_enabled ? 'Включена' : 'Отключена';
52   if (bcdTimeoutEl) bcdTimeoutEl.textContent = `${data.timeout ?? 30} сек`;
53   if (currentLoaderEl) currentLoaderEl.textContent = data.current_guid || '{current}';
54
55   if (!tbody) return;
56
57   const entries = data.bcd_entries || [
58     { guid: '{bootmgr}', description: 'Windows Boot Manager',
59       device: 'partition=\\Device\\HarddiskVolume1', path: '\\\\EFI\\Microsoft\\Boot\\bootmgfw.efi' },
```

```

59 |     { guid: '{current}', description: 'Windows 11 Pro', device
    | : 'partition=C:', path: '\\Windows\\system32\\winload.efi' }
60 | ];
61 |
62 | if (badge) badge.textContent = `${entries.length} записей`;
63 |
64 | tbody.innerHTML = entries.map(e => `
65 |     <tr>
66 |         <td class="fw-bold text-info font-monospace">${e.guid}</
    | td>
67 |         <td>${e.description || '-'}</td>
68 |         <td class="small text-muted">${e.device || '-'}</td>
69 |         <td class="small font-monospace">${e.path || '-'}</td>
70 |         <td class="text-end">
71 |             <button class="btn btn-xs btn-outline-info py-0 px-1"
    | title="Инфо о записи">
72 |                 <i class="bi bi-info-circle"></i>
73 |             </button>
74 |         </td>
75 |     </tr>
76 | `).join('');
77 | }
78 |
79 | async function executeBootAction(customAction = null) {
80 |     const actionSelect = document.getElementById('br-action-sele
    | ct');
81 |     const dryRunSwitch = document.getElementById('br-dry-run-swi
    | tch');
82 |     const output = document.getElementById('br-console-output');
83 |
84 |     const op = customAction || (actionSelect ? actionSelect.valu
    | e : 'reagentc_info');
85 |     const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;
86 |
87 |     if (output) output.textContent = `[${new Date().toLocaleTime
    | String()}] Выполнение команды: ${op} (DryRun: ${dryRun})...\n`
    | ;
88 |

```

```

89 |   try {
90 |       const res = await fetch('/api/boot-recovery/action', {
91 |           method: 'POST',
92 |           headers: { 'Content-Type': 'application/json' },
93 |           body: JSON.stringify({ action: op, dry_run: dryRun })
94 |       });
95 |       const result = await res.json();
96 |       if (output) output.textContent += JSON.stringify(result, null, 2);
97 |       if (window.toast) {
98 |           if (res.ok) window.toast.success('Команда выполнена', `Статус: ${result.status || 'OK'}`);
99 |           else window.toast.error('Ошибка', result.detail || 'Сбой выполнения');
100 |       }
101 |   } catch (err) {
102 |       if (output) output.textContent += `\\nОшибка запроса: ${err.message}`;
103 |       if (window.toast) window.toast.error('Ошибка сети', err.message);
104 |   }
105 | }
106 |
107 | export function initBootRecoveryTab() {
108 |     if (isInitialized) return;
109 |     isInitialized = true;
110 |
111 |     const refreshBtn = document.getElementById('br-refresh-btn')
112 |     ;
113 |     if (refreshBtn) refreshBtn.addEventListener('click', fetchBootRecoverySummary);
114 |
115 |     const winreEnableBtn = document.getElementById('br-winre-enable-btn');
116 |     if (winreEnableBtn) winreEnableBtn.addEventListener('click', () => executeBootAction('reagentc_enable'));
117 |
118 |     const winreInfoBtn = document.getElementById('br-winre-info-

```

```
    btn');
118 |   if (winreInfoBtn) winreInfoBtn.addEventListener('click', ()
    |   => executeBootAction('reagentc_info'));
119 |
120 |   const execBtn = document.getElementById('br-execute-btn');
121 |   if (execBtn) execBtn.addEventListener('click', () => execute
    |   BootAction());
122 |
123 |   const clearBtn = document.getElementById('br-clear-log-btn')
    |   ;
124 |   if (clearBtn) clearBtn.addEventListener('click', () => {
125 |       const out = document.getElementById('br-console-output');
126 |       if (out) out.textContent = 'Консоль очищена.';
127 |   });
128 |
129 |   registerTabPoller('tab-boot-recovery', fetchBootRecoverySumm
    |   ary, 15000, { immediate: true });
130 | }
131 |
132 | window.initBootRecoveryTab = initBootRecoveryTab;
133 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/chat/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/chat/index.html" class=
15 | "w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/chat
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <div class="chat-container d-flex flex-row h-100 w-100 overflo
w-hidden gap-2" style="min-height: 80vh; height: calc(100vh -
65px);">
28   <!-- Sessions Sidebar -->
29   <aside class="chat-sidebar d-flex flex-column flex-shrink-0
p-2 border rounded" id="chat-sidebar" style="width: 250px; min
-width: 250px; max-width: 250px; height: 100%;">
30     <div class="d-flex align-items-center justify-content-betw
een mb-2 flex-shrink-0">
```

```

28     <button class="btn btn-primary btn-sm w-100 d-flex align
    -items-center justify-content-center gap-2" id="chat-new-sessi
    on-btn" type="button" title="Начать новый диалог">
29         <i class="bi bi-plus-lg"></i>
30         <span>Новый чат</span>
31     </button>
32 </div>
33
34 <!-- Sessions List -->
35 <div class="chat-sessions-list flex-grow-1 overflow-auto d
    -flex flex-column gap-1 my-1" id="chat-sessions-list" style="m
    in-height: 0;">
36     <!-- Session items injected via JS -->
37 </div>
38
39 <!-- Sidebar Footer -->
40 <div class="pt-2 border-top border-secondary-subtle d-flex
    align-items-center justify-content-between flex-shrink-0">
41     <div class="d-flex align-items-center gap-2 flex-grow-1"
    >
42         <small class="text-muted" id="chat-sessions-count">0 ч
    атов</small>
43         <span class="badge bg-secondary-subtle text-secondary
    cache-badge" id="chat-cache-badge" title="Кеш: попадания/всего
    " style="cursor: help; font-size: 0.7rem;">
44             <i class="bi bi-lightning-charge"></i> <span id="cha
    t-cache-ratio">-</span>
45         </span>
46     </div>
47     <div class="d-flex align-items-center gap-1">
48         <button class="btn btn-sm btn-link text-warning text-d
    ecoration-none p-0" id="chat-clear-cache-btn" type="button" ti
    tle="Очистить кеш">
49             <i class="bi bi-arrow-clockwise"></i>
50         </button>
51         <button class="btn btn-sm btn-link text-danger text-de
    coration-none p-0" id="chat-clear-all-sessions-btn" type="butt
    on" title="Удалить все чаты">

```



```
52 |         <i class="bi bi-trash3"></i>
53 |     </button>
54 | </div>
55 | </div>
56 | </aside>
57 |
58 | <!-- Main Chat Conversation Area -->
59 | <main class="chat-main-area d-flex flex-column flex-grow-1 h
60 | -100 overflow-hidden" style="min-width: 0;">
61 |     <!-- Top Session Header Bar -->
62 |     <div class="chat-header-bar d-flex align-items-center just
63 | ify-content-between p-1 px-2 mb-2 rounded border flex-shrink-0
64 | ">
65 |         <div class="d-flex align-items-center gap-2 flex-grow-1
66 | overflow-hidden">
67 |             <button class="btn btn-sm btn-outline-secondary py-0 p
68 | x-2" id="chat-sidebar-toggle-btn" type="button" title="Показат
69 | ь/скрыть панель чатов">
70 |                 <i class="bi bi-layout-sidebar"></i>
71 |             </button>
72 |             <span class="fw-semibold text-truncate" id="chat-curre
73 | nt-title" title="Текущий диалог">Новый чат</span>
74 |             <button class="btn btn-sm btn-link text-muted p-0" id=
75 | "chat-rename-current-btn" type="button" title="Переименовать д
76 | иалог">
77 |                 <i class="bi bi-pencil-square"></i>
78 |             </button>
79 |         </div>
80 |
81 |         <div class="d-flex align-items-center gap-1 flex-shrink-
82 | 0">
83 |             <button class="btn btn-sm btn-outline-secondary py-0 p
84 | x-2" id="chat-clear-btn" type="button" title="Очистить текущий
85 | диалог">
86 |                 <i class="bi bi-eraser me-1"></i><span class="d-none
87 | d-sm-inline">Очистить</span>
88 |             </button>
89 |         </div>
```

```
77     </div>
78
79     <!-- Chat Messages Window -->
80     <div class="chat-window flex-grow-1 mb-2 overflow-auto" id
    ="chat-window"></div>
81
82     <!-- Execution Status Indicator -->
83     <div id="chat-status-indicator" class="d-none small text-m
    uted mb-2 d-flex align-items-center gap-2 flex-shrink-0">
84         <span class="spinner-border spinner-border-sm text-prima
    ry" role="status" aria-hidden="true"></span>
85         <span id="chat-status-text">Обработка запроса...</span>
86     </div>
87
88     <!-- Query Input & Settings Bar -->
89     <div class="d-flex flex-wrap align-items-center gap-2 flex
    -shrink-0">
90         <!-- RAG Search Toggle Switch -->
91         <div class="form-check form-switch d-flex align-items-ce
    nter gap-1 m-0">
92             <input class="form-check-input mt-0" type="checkbox" r
    ole="switch" id="chat-rag-toggle" checked title="Использовать
    базу знаний (RAG)">
93             <label class="form-check-label small fw-semibold text-
    muted text-nowrap user-select-none" for="chat-rag-toggle" titl
    e="Использовать базу знаний (RAG)">
94                 <i class="bi bi-database-check me-1"></i>RAG
95             </label>
96         </div>
97
98         <!-- TOP_K Setting -->
99         <div class="d-flex align-items-center gap-1">
100             <label for="chat-top-k" class="small fw-semibold text-
    muted text-nowrap">TOP_K:</label>
101             <input type="number" class="form-control form-control-
    sm" id="chat-top-k" value="3" min="1" max="50" style="width: 7
    5px;" title="Количество результатов (TOP_K)">
102         </div>
```

```
103
104     <!-- MIN_SCORE Setting -->
105     <div class="d-flex align-items-center gap-1">
106         <label for="chat-min-score" class="small fw-semibold t
ext-muted text-nowrap">MIN_SCORE:</label>
107         <input type="number" class="form-control form-control-
sm" id="chat-min-score" value="0.45" min="0.0" max="1.0" step=
"0.05" style="width: 85px;" title="Минимальный порог (MIN_SCOR
E)">
108     </div>
109
110     <!-- Output Mode (Text only / Voice only / Text and Voic
e) -->
111     <div class="d-flex align-items-center gap-1">
112         <label for="chat-output-mode" class="small fw-semibold
text-muted text-nowrap user-select-none" title="Режим вывода
ответа">
113             <i class="bi bi-sliders me-1"></i>Режим:
114         </label>
115         <select class="form-select form-select-sm" id="chat-ou
tput-mode" style="width: auto; min-width: 145px;" title="Режим
вывода ответа">
116             <option value="text_only" selected>☐ Только текст</
option>
117             <option value="text_and_voice">☐ +☐ Текст и голос</
option>
118             <option value="voice_only">☐ Только голос</option>
119         </select>
120     </div>
121
122     <!-- Message Input and Send Button -->
123     <div class="input-group flex-grow-1">
124         <input type="text" class="form-control" id="message-in
put" placeholder="Введите сообщение..." autocomplete="off" dat
a-lpignore="true" data-1p-ignore="true">
125         <button class="btn btn-primary" id="send-button" type=
"button">Отправить</button>
126     </div>
```

```
127 |     </div>
128 |   </main>
129 | </div>
130 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Chat - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *   <script src="/windows/api/webgui/chat/main.js?v=2026100
11 *   1_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/chat
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-01 13:04:40
19 * =====
20 */
21 // — CHAT.JS —————
22
23 import { cachedApiFetch, clearCacheByTag } from '../js/api-cache.js';
24
25 const STORAGE_KEY_SESSIONS = 'breadboard_chat_sessions_v2';
26 const STORAGE_KEY_ACTIVE_SESSION_ID = 'breadboard_active_session_id';
27 const STORAGE_KEY_SIDEBAR_COLLAPSED = 'breadboard_chat_sidebar_collapsed';
28 const LEGACY_STORAGE_KEY = 'breadboard_chat_history';
```

```
29 | const CACHE_TAG = 'chat';
30 |
31 | let sessions = [];
32 | let activeSessionId = null;
33 |
34 | /**
35 |  * Generates a clean, human-friendly title based on the first
36 |  * prompt (ChatGPT-style).
37 |  */
38 | function generateSessionTitle(prompt) {
39 |   if (!prompt || typeof prompt !== 'string') return 'Новый чат';
40 |
41 |   let clean = prompt
42 |     .replace(/^\``[\s\S]*?\``/g, '')
43 |     .replace(/<[^>]*>/g, '')
44 |     .replace(/[*_#`[\]()]/g, '')
45 |     .trim();
46 |
47 |   if (!clean) return 'Новый чат';
48 |
49 |   // Take the first line or first sentence
50 |   const firstLine = clean.split('\n')[0].trim();
51 |   const sentenceMatch = firstLine.match(/^(.+?)(?:[.!?]|$)/);
52 |   let text = sentenceMatch ? sentenceMatch[1].trim() : firstLine;
53 |
54 |   if (text.length > 36) {
55 |     text = text.substring(0, 36).trim();
56 |     const lastSpace = text.lastIndexOf(' ');
57 |     if (lastSpace > 15) {
58 |       text = text.substring(0, lastSpace);
59 |     }
60 |     text += '...';
61 |   }
62 |
63 |   // Capitalize first character
64 |   return text.charAt(0).toUpperCase() + text.slice(1);
```

```
64 | }
65 |
66 | let syncDebounceTimer = null;
67 |
68 | function saveLocalSessionsOnly() {
69 |   try {
70 |     localStorage.setItem(STORAGE_KEY_SESSIONS, JSON.stringify(
71 |       sessions));
72 |     if (activeSessionId) {
73 |       localStorage.setItem(STORAGE_KEY_ACTIVE_SESSION_ID, activeSessionId);
74 |     }
75 |   } catch (err) {
76 |     console.warn('Failed to save chat sessions to localStorage: ', err);
77 |   }
78 | }
79 |
80 | function saveSessions(syncToServer = true) {
81 |   saveLocalSessionsOnly();
82 |   if (syncToServer) {
83 |     debounceSyncActiveSession();
84 |   }
85 | }
86 |
87 | function debounceSyncActiveSession() {
88 |   if (syncDebounceTimer) {
89 |     clearTimeout(syncDebounceTimer);
90 |   }
91 |   syncDebounceTimer = setTimeout(() => {
92 |     const active = getActiveSession();
93 |     if (active) {
94 |       syncSessionToServer(active);
95 |     }
96 |   }, 500);
97 | }
98 |
99 | async function syncSessionToServer(session) {
```

```

99 | if (!session || !session.id) return;
100 | try {
101 |     await fetch('/api/chat/sessions', {
102 |         method: 'POST',
103 |         headers: { 'Content-Type': 'application/json' },
104 |         body: JSON.stringify({
105 |             id: session.id,
106 |             title: session.title || 'Новый чат',
107 |             isCustomTitle: Boolean(session.isCustomTitle),
108 |             createdAt: session.createdAt || Date.now(),
109 |             updatedAt: session.updatedAt || Date.now(),
110 |             messages: session.messages || [],
111 |             chatHistory: session.chatHistory || []
112 |         })
113 |     });
114 | } catch (err) {
115 |     console.debug('Background session sync to server skipped o
116 | r failed:', err);
117 | }
118 |
119 | async function deleteSessionFromServer(sessionId) {
120 |     if (!sessionId) return;
121 |     try {
122 |         await fetch(`/api/chat/sessions/${encodeURIComponent(sessi
123 | onId)}`, {
124 |             method: 'DELETE'
125 |         });
126 |     } catch (err) {
127 |         console.debug('Failed to delete session on server:', err);
128 |     }
129 | }
130 |
131 | async function clearAllSessionsFromServer() {
132 |     try {
133 |         await fetch('/api/chat/sessions', {
134 |             method: 'DELETE'

```



```

135 | } catch (err) {
136 |     console.debug('Failed to clear sessions on server:', err);
137 | }
138 | }
139 |
140 | async function syncSessionsWithServer() {
141 |     try {
142 |         const res = await cachedApiFetch('/api/chat/sessions', {
143 |             tags: [CACHE_TAG, 'chat-sessions']
144 |         });
145 |         if (!res.ok) return;
146 |         const data = await res.json();
147 |         const serverSessions = data.sessions || [];
148 |
149 |         // If we have local sessions and server is empty, or server
150 |         // has fewer/older sessions, bulk sync
151 |         if (sessions.length > 0 && serverSessions.length === 0) {
152 |             const syncRes = await fetch('/api/chat/sessions/sync', {
153 |                 method: 'POST',
154 |                 headers: { 'Content-Type': 'application/json' },
155 |                 body: JSON.stringify({ sessions: sessions })
156 |             });
157 |             if (syncRes.ok) {
158 |                 const syncData = await syncRes.json();
159 |                 if (syncData.sessions && syncData.sessions.length > 0)
160 |                 {
161 |                     sessions = syncData.sessions;
162 |                 }
163 |             }
164 |         } else if (serverSessions.length > 0) {
165 |             // Merge server sessions with any unique local sessions
166 |             const serverMap = new Map();
167 |             serverSessions.forEach((s) => serverMap.set(s.id, s));
168 |
169 |             const localOnly = [];
170 |             sessions.forEach((ls) => {
171 |                 if (!serverMap.has(ls.id)) {
172 |                     localOnly.push(ls);
173 |                 }
174 |             });
175 |             sessions = [...localOnly, ...serverSessions];
176 |         }
177 |     } catch (err) {
178 |         console.debug('Failed to sync sessions with server:', err);
179 |     }
180 | }

```

```
171 |         } else {
172 |             // If local has newer updatedAt, prefer local and sync to server
173 |             const ss = serverMap.get(ls.id);
174 |             if ((ls.updatedAt || 0) > (ss.updatedAt || 0)) {
175 |                 serverMap.set(ls.id, ls);
176 |                 syncSessionToServer(ls);
177 |             }
178 |         }
179 |     });
180 |
181 |     if (localOnly.length > 0) {
182 |         // Push local only to server
183 |         await fetch('/api/chat/sessions/sync', {
184 |             method: 'POST',
185 |             headers: { 'Content-Type': 'application/json' },
186 |             body: JSON.stringify({ sessions: localOnly })
187 |         });
188 |     }
189 |
190 |     sessions = Array.from(serverMap.values()).sort((a, b) =>
191 | (b.updatedAt || 0) - (a.updatedAt || 0));
192 |
193 |     if (sessions.length === 0) {
194 |         createNewSession('Новый чат', true);
195 |         return;
196 |     }
197 |
198 |     const savedActiveId = localStorage.getItem(STORAGE_KEY_ACTIVE_SESSION_ID);
199 |     if (savedActiveId && sessions.some((s) => s.id === savedActiveId)) {
200 |         activeSessionId = savedActiveId;
201 |     } else {
202 |         activeSessionId = sessions[0].id;
203 |     }
204 |
```

```
205 |     saveLocalSessionsOnly();
206 |     renderSessionsList();
207 |     renderActiveSessionMessages();
208 | } catch (err) {
209 |     console.debug('Failed to sync sessions with server (workin
    | g in offline/local mode):', err);
210 | }
211 | }
212 |
213 | function loadSessions() {
214 |     try {
215 |         const raw = localStorage.getItem(STORAGE_KEY_SESSIONS);
216 |         if (raw) {
217 |             const parsed = JSON.parse(raw);
218 |             if (Array.isArray(parsed) && parsed.length > 0) {
219 |                 sessions = parsed;
220 |                 const savedActiveId = localStorage.getItem(STORAGE_KEY
    | _ACTIVE_SESSION_ID);
221 |                 if (savedActiveId && sessions.some((s) => s.id === sav
    | edActiveId)) {
222 |                     activeSessionId = savedActiveId;
223 |                 } else {
224 |                     activeSessionId = sessions[0].id;
225 |                 }
226 |                 syncSessionsWithServer();
227 |                 return;
228 |             }
229 |         }
230 |
231 |         // Migrate from legacy single-session chat history if pres
    | ent
232 |         const legacyRaw = localStorage.getItem(LEGACY_STORAGE_KEY)
    | ;
233 |         if (legacyRaw) {
234 |             try {
235 |                 const legacyLog = JSON.parse(legacyRaw);
236 |                 if (Array.isArray(legacyLog) && legacyLog.length > 0)
    | {
```

```
237 |         const firstUserMsg = legacyLog.find((m) => m.sender
    | === 'user');
238 |         const title = firstUserMsg ? generateSessionTitle(fi
    | rstUserMsg.text) : 'Предыдущий диалог';
239 |         const legacyHistory = [];
240 |         for (const item of legacyLog) {
241 |             legacyHistory.push({
242 |                 role: item.sender === 'user' ? 'user' : 'model',
243 |                 parts: [item.text || item.voiceText || '']
244 |             });
245 |         }
246 |
247 |         const legacySession = {
248 |             id: 'session_' + Date.now(),
249 |             title: title,
250 |             isCustomTitle: false,
251 |             createdAt: Date.now(),
252 |             updatedAt: Date.now(),
253 |             messages: legacyLog,
254 |             chatHistory: legacyHistory
255 |         };
256 |         sessions = [legacySession];
257 |         activeSessionId = legacySession.id;
258 |         saveSessions();
259 |         syncSessionsWithServer();
260 |         return;
261 |     }
262 | } catch (e) {
263 |     console.warn('Failed to migrate legacy chat history:',
    | e);
264 | }
265 | }
266 |
267 | // Default clean initial session and attempt server sync
268 | createNewSession('Новый чат', true);
269 | syncSessionsWithServer();
270 | } catch (err) {
271 |     console.warn('Failed to load chat sessions:', err);
```

```
272 |     sessions = [];  
273 |     createNewSession('Новый чат', true);  
274 |     syncSessionsWithServer();  
275 | }  
276 |  
277 |  
278 | function getActiveSession() {  
279 |     if (!activeSessionId || sessions.length === 0) {  
280 |         return createNewSession('Новый чат', true);  
281 |     }  
282 |     let session = sessions.find((s) => s.id === activeSessionId)  
283 |     ;  
284 |     if (!session) {  
285 |         session = sessions[0];  
286 |         activeSessionId = session.id;  
287 |     }  
288 |     return session;  
289 | }  
290 | function createNewSession(title = 'Новый чат', makeActive = tr  
291 | ue) {  
292 |     const newSession = {  
293 |         id: 'session_' + Date.now() + '_' + Math.random().toString  
294 |         (36).substring(2, 7),  
295 |         title: title,  
296 |         isCustomTitle: false,  
297 |         createdAt: Date.now(),  
298 |         updatedAt: Date.now(),  
299 |         messages: [],  
300 |         chatHistory: []  
301 |     };  
302 |     sessions.unshift(newSession);  
303 |     if (makeActive) {  
304 |         activeSessionId = newSession.id;  
305 |     }  
306 |     saveSessions();  
307 |     renderSessionsList();
```

```
307 | renderActiveSessionMessages();
308 | return newSession;
309 | }
310 |
311 | function switchSession(sessionId) {
312 |   if (sessionId === activeSessionId) return;
313 |   const session = sessions.find((s) => s.id === sessionId);
314 |   if (!session) return;
315 |
316 |   activeSessionId = sessionId;
317 |   saveSessions(false);
318 |   renderSessionsList();
319 |   renderActiveSessionMessages();
320 |
321 |   const msgInput = document.getElementById('message-input');
322 |   if (msgInput) msgInput.focus();
323 | }
324 |
325 | function renameSession(sessionId, newTitle = null) {
326 |   const session = sessions.find((s) => s.id === sessionId);
327 |   if (!session) return;
328 |
329 |   let title = newTitle;
330 |   if (title === null) {
331 |     title = prompt('Введите новое название диалога:', session.
title);
332 |   }
333 |
334 |   if (title && title.trim()) {
335 |     session.title = title.trim();
336 |     session.isCustomTitle = true;
337 |     session.updatedAt = Date.now();
338 |     saveSessions();
339 |     renderSessionsList();
340 |     if (session.id === activeSessionId) {
341 |       updateCurrentSessionHeader();
342 |     }
343 |   }
```

```
344 | }
345 |
346 | function deleteSession(sessionId) {
347 |     const index = sessions.findIndex((s) => s.id === sessionId);
348 |     if (index === -1) return;
349 |
350 |     sessions.splice(index, 1);
351 |     if (sessions.length === 0) {
352 |         createNewSession('Новый чат', true);
353 |     } else if (activeSessionId === sessionId) {
354 |         activeSessionId = sessions[0].id;
355 |         renderActiveSessionMessages();
356 |     }
357 |
358 |     saveSessions(false);
359 |     deleteSessionFromServer(sessionId);
360 |     renderSessionsList();
361 | }
362 |
363 | function clearCurrentSession() {
364 |     const session = getActiveSession();
365 |     if (!session) return;
366 |
367 |     session.messages = [];
368 |     session.chatHistory = [];
369 |     session.title = 'Новый чат';
370 |     session.isCustomTitle = false;
371 |     session.updatedAt = Date.now();
372 |
373 |     saveSessions();
374 |     renderSessionsList();
375 |     renderActiveSessionMessages();
376 | }
377 |
378 | function clearAllSessions() {
379 |     if (!confirm('Вы уверены, что хотите удалить ВСЕ сохраненные
380 |         диалоги?')) return;
```

```
381 | sessions = [];
382 | try {
383 |     localStorage.removeItem(STORAGE_KEY_SESSIONS);
384 |     localStorage.removeItem(STORAGE_KEY_ACTIVE_SESSION_ID);
385 |     localStorage.removeItem(LEGACY_STORAGE_KEY);
386 | } catch (e) {}
387 |
388 | // Clear cache for chat sessions
389 | clearCacheByTag(CACHE_TAG).catch(err => console.debug('Failed to clear chat cache:', err));
390 |
391 | clearAllSessionsFromServer();
392 | createNewSession('Новый чат', true);
393 | }
394 |
395 |
396 | function updateCurrentSessionHeader() {
397 |     const titleEl = document.getElementById('chat-current-title');
398 |     const session = getActiveSession();
399 |     if (titleEl && session) {
400 |         titleEl.textContent = session.title || 'Новый чат';
401 |         titleEl.title = session.title || 'Новый чат';
402 |     }
403 | }
404 |
405 | function renderSessionsList() {
406 |     const listEl = document.getElementById('chat-sessions-list');
407 |     const countEl = document.getElementById('chat-sessions-count');
408 |     if (!listEl) return;
409 |
410 |     listEl.innerHTML = '';
411 |     if (countEl) {
412 |         countEl.textContent = `${sessions.length} ${sessions.length === 1 ? 'чат' : (sessions.length < 5 ? 'чата' : 'чатов')}`;
413 |     }
```



```
414
415   updateCurrentSessionHeader();
416
417   sessions.forEach((session) => {
418     const item = document.createElement('div');
419     const isActive = session.id === activeSessionId;
420     item.className = 'chat-session-item d-flex align-items-center justify-content-between p-1 px-2 rounded mb-1' + (isActive ? ' active' : '');
421     item.dataset.sessionId = session.id;
422
423     item.innerHTML = `
424       <div class="chat-session-title d-flex align-items-center gap-1 text-truncate flex-grow-1 me-1" title="${session.title}">
425         <i class="bi bi-chat-text ${isActive ? 'text-primary' : 'text-muted'} flex-shrink-0"></i>
426         <span class="text-truncate">${session.title}</span>
427       </div>
428       <div class="chat-session-actions d-flex align-items-center gap-1 flex-shrink-0">
429         <button class="btn btn-sm btn-link text-muted p-0 btn-rename" title="Переименовать" type="button">
430           <i class="bi bi-pencil" style="font-size: 0.8rem;"></i>
431         </button>
432         <button class="btn btn-sm btn-link text-danger p-0 btn-delete" title="Удалить" type="button">
433           <i class="bi bi-trash" style="font-size: 0.8rem;"></i>
434         </button>
435       </div>
436     `;
437
438     item.addEventListener('click', (e) => {
439       if (e.target.closest('.btn-rename') || e.target.closest('.btn-delete')) return;
440       switchSession(session.id);
```

```
441 |     });
442 |
443 |     const renameBtn = item.querySelector('.btn-rename');
444 |     if (renameBtn) {
445 |         renameBtn.addEventListener('click', (e) => {
446 |             e.stopPropagation();
447 |             renameSession(session.id);
448 |         });
449 |     }
450 |
451 |     const deleteBtn = item.querySelector('.btn-delete');
452 |     if (deleteBtn) {
453 |         deleteBtn.addEventListener('click', (e) => {
454 |             e.stopPropagation();
455 |             if (confirm(`Удалить диалог "${session.title}"?`)) {
456 |                 deleteSession(session.id);
457 |             }
458 |         });
459 |     }
460 |
461 |     listEl.appendChild(item);
462 | });
463 | }
464 |
465 | function renderActiveSessionMessages() {
466 |     const chatWindow = document.getElementById('chat-window');
467 |     if (!chatWindow) return;
468 |
469 |     chatWindow.innerHTML = '';
470 |     const session = getActiveSession();
471 |
472 |     if (session && session.messages && session.messages.length >
473 |         0) {
474 |         for (const msg of session.messages) {
475 |             addMessageToDom(msg.sender, msg.text, msg.voiceText, msg
476 |                 .timeStr, msg.mode);
477 |         }
478 |     } else {
```

```
477     const welcome = document.createElement('div');
478     welcome.className = 'message bot-message';
479     welcome.innerHTML = '<strong>AI Assistant</strong>: Добро
пожаловать! Задайте любой вопрос в поле ввода.';
480     chatWindow.appendChild(welcome);
481 }
482
483 chatWindow.scrollTop = chatWindow.scrollHeight;
484 updateCurrentSessionHeader();
485 }
486
487 /**
488  * Update cache status badge
489  */
490 function updateChatCacheStatus() {
491     const badge = document.getElementById('chat-cache-badge');
492     const ratio = document.getElementById('chat-cache-ratio');
493     if (!badge || !ratio) return;
494
495     try {
496         const stats = window.apiCacheStats || { hits: 0, total: 0
};
497         const hitRate = stats.total > 0 ? Math.round((stats.hits /
stats.total) * 100) : 0;
498         ratio.textContent = `${hitRate}%`;
499
500         // Color coding based on hit rate
501         badge.className = 'badge cache-badge';
502         if (hitRate >= 70) {
503             badge.classList.add('bg-success-subtle', 'text-success')
;
504         } else if (hitRate >= 40) {
505             badge.classList.add('bg-warning-subtle', 'text-warning')
;
506         } else {
507             badge.classList.add('bg-secondary-subtle', 'text-seconda
ry');
508         }
```

```
509 | } catch (err) {
510 |     console.debug('Failed to update cache status:', err);
511 | }
512 | }
513 |
514 | /**
515 |  * Clear chat cache
516 |  */
517 | async function clearChatCache() {
518 |     try {
519 |         await clearCacheByTag(CACHE_TAG);
520 |         updateChatCacheStatus();
521 |         console.log('Chat cache cleared');
522 |     } catch (err) {
523 |         console.error('Failed to clear chat cache:', err);
524 |     }
525 | }
526 |
527 | function initChatTab() {
528 |     loadSessions();
529 |     renderSessionsList();
530 |     renderActiveSessionMessages();
531 |
532 |     // Update cache status
533 |     updateChatCacheStatus();
534 |
535 |     // Bind Sidebar Toggle
536 |     const sidebar = document.getElementById('chat-sidebar');
537 |     const sidebarToggleBtn = document.getElementById('chat-sideb
538 | ar-toggle-btn');
539 |     if (sidebar && sidebarToggleBtn) {
540 |         const isCollapsed = localStorage.getItem(STORAGE_KEY_SIDE
541 | BAR_COLLAPSED) === 'true';
542 |         if (isCollapsed) {
543 |             sidebar.classList.add('collapsed');
544 |             sidebar.style.display = 'none';
545 |         }
546 |     }
```

```
545     sidebarToggleBtn.addEventListener('click', () => {
546         const isCurrentlyCollapsed = sidebar.classList.toggle('c
ollapsed');
547         sidebar.style.display = isCurrentlyCollapsed ? 'none' :
'flex';
548         localStorage.setItem(STORAGE_KEY_SIDEBAR_COLLAPSED, isCu
rrentlyCollapsed);
549     });
550 }
551
552 // Bind New Session Button
553 const newSessionBtn = document.getElementById('chat-new-sess
ion-btn');
554 if (newSessionBtn) {
555     newSessionBtn.addEventListener('click', () => {
556         createNewSession('Новый чат', true);
557         const msgInput = document.getElementById('message-input'
);
558         if (msgInput) msgInput.focus();
559     });
560 }
561
562 // Bind Clear Current / Clear All buttons
563 const clearBtn = document.getElementById('chat-clear-btn');
564 if (clearBtn) {
565     clearBtn.addEventListener('click', () => {
566         if (confirm('Очистить сообщения текущего диалога?')) {
567             clearCurrentSession();
568         }
569     });
570 }
571
572 const clearAllBtn = document.getElementById('chat-clear-all-
sessions-btn');
573 if (clearAllBtn) {
574     clearAllBtn.addEventListener('click', () => {
575         clearAllSessions();
576     });
577 }
```

```
577 | }
578 |
579 | const clearCacheBtn = document.getElementById('chat-clear-cache-btn');
580 | if (clearCacheBtn) {
581 |     clearCacheBtn.addEventListener('click', async () => {
582 |         await clearChatCache();
583 |         // Force refresh sessions
584 |         await syncSessionsWithServer();
585 |         renderSessionsList();
586 |     });
587 | }
588 |
589 | const renameCurrentBtn = document.getElementById('chat-rename-current-btn');
590 | if (renameCurrentBtn) {
591 |     renameCurrentBtn.addEventListener('click', () => {
592 |         if (activeSessionId) {
593 |             renameSession(activeSessionId);
594 |         }
595 |     });
596 | }
597 |
598 | // Expose sendMessage globally for user menu actions
599 | window.sendChatMessage = sendMessage;
600 |
601 | // Bind Input and Controls
602 | const sendBtn = document.getElementById('send-button');
603 | const msgInput = document.getElementById('message-input');
604 | const ragToggle = document.getElementById('chat-rag-toggle');
605 | ;
606 | const topKInput = document.getElementById('chat-top-k');
607 | const minScoreInput = document.getElementById('chat-min-score');
608 |
609 | const updateRagInputsState = (enabled) => {
610 |     if (topKInput) {
611 |         topKInput.disabled = !enabled;
```

```

611     topKInput.style.opacity = enabled ? '1' : '0.5';
612 }
613 if (minScoreInput) {
614     minScoreInput.disabled = !enabled;
615     minScoreInput.style.opacity = enabled ? '1' : '0.5';
616 }
617 };
618
619 if (ragToggle) {
620     const savedRagPref = localStorage.getItem('chat_rag_enabled');
621     if (savedRagPref !== null) {
622         ragToggle.checked = savedRagPref === 'true';
623         updateRagInputsState(ragToggle.checked);
624     }
625     fetch('/auth/settings', { cache: 'no-cache' })
626         .then((res) => (res.ok ? res.json() : null))
627         .then((settings) => {
628             if (settings && typeof settings.rag_enabled !== 'undefined') {
629                 const isEnabled = settings.rag_enabled !== 0;
630                 ragToggle.checked = isEnabled;
631                 localStorage.setItem('chat_rag_enabled', isEnabled);
632                 updateRagInputsState(isEnabled);
633             }
634         })
635         .catch(() => {});
636
637     ragToggle.addEventListener('change', async (e) => {
638         const isEnabled = e.target.checked;
639         localStorage.setItem('chat_rag_enabled', isEnabled);
640         updateRagInputsState(isEnabled);
641         try {
642             await fetch('/auth/settings', {
643                 method: 'POST',
644                 headers: { 'Content-Type': 'application/json' },
645                 body: JSON.stringify({ rag_enabled: isEnabled ? 1 :
0 })

```

```
646         });
647     } catch (err) {
648         console.error('Error saving RAG setting:', err);
649     }
650 });
651 }
652
653 const modeSelect = document.getElementById('chat-output-mode
');
654 if (modeSelect) {
655     const savedMode = localStorage.getItem('chat_output_mode')
;
656     if (savedMode) {
657         modeSelect.value = savedMode;
658     } else {
659         modeSelect.value = 'text_only';
660     }
661     modeSelect.addEventListener('change', (e) => {
662         localStorage.setItem('chat_output_mode', e.target.value)
;
663         if (e.target.value === 'text_only' && window.chatService
?.stop) {
664             window.chatService.stop();
665         }
666     });
667 }
668
669 if (sendBtn) {
670     sendBtn.addEventListener('click', sendMessage);
671 }
672 if (msgInput) {
673     msgInput.addEventListener('keypress', (e) => {
674         if (e.key === 'Enter') {
675             sendMessage();
676         }
677     });
678 }
679
```



```

680 | // Check for pending external chat payload (from Chrome exte
    | nsion or URL parameters)
681 | try {
682 |     const pending = sessionStorage.getItem('pending_breadboard
    | _chat');
683 |     if (pending) {
684 |         const payload = JSON.parse(pending);
685 |         sessionStorage.removeItem('pending_breadboard_chat');
686 |         setTimeout(() => handleExternalChatPrompt(payload), 300)
    | ;
687 |     } else {
688 |         const urlParams = new URLSearchParams(window.location.se
    | arch);
689 |         const promptParam = urlParams.get('prompt') || urlParams
    | .get('message');
690 |         if (promptParam) {
691 |             setTimeout(() => {
692 |                 handleExternalChatPrompt({
693 |                     prompt: promptParam,
694 |                     autoSend: urlParams.get('autosend') !== 'false'
695 |                 });
696 |             }, 300);
697 |         }
698 |     }
699 | } catch (e) {
700 |     console.error('Error checking pending chat payload:', e);
701 | }
702 | }
703 |
704 | function handleExternalChatPrompt(payload) {
705 |     if (!payload || !payload.prompt) return;
706 |
707 |     const chatTabBtn = document.querySelector('[data-bs-target="
    | #tab-chat"]');
708 |     if (chatTabBtn && window.bootstrap?.Tab) {
709 |         bootstrap.Tab.getOrCreateInstance(chatTabBtn).show();
710 |     }
711 |

```

```
712 | const msgInput = document.getElementById('message-input');
713 | if (msgInput) {
714 |     msgInput.value = payload.prompt;
715 |     if (payload.autoSend !== false) {
716 |         setTimeout(() => {
717 |             sendMessage();
718 |         }, 150);
719 |     }
720 | }
721 | }
722 |
723 | function handleRemoteTranscript(text) {
724 |     if (!text || !text.trim()) return;
725 |     const cleanText = text.trim();
726 |
727 |     // Ensure active session exists
728 |     const session = getActiveSession();
729 |     if (!session) return;
730 |
731 |     const userTimeStr = new Date().toLocaleTimeString();
732 |
733 |     // Clear welcome placeholder if this is the first message
734 |     const chatWin = document.getElementById('chat-window');
735 |     if (chatWin && session.messages.length === 0) {
736 |         chatWin.innerHTML = '';
737 |     }
738 |
739 |     // Add message to DOM and save to active session without tri
       | ggering AI model generation
740 |     addMessageToDom('user', cleanText, null, userTimeStr);
741 |     session.messages.push({ sender: 'user', text: cleanText, tim
       | eStr: userTimeStr });
742 |
743 |     if ((!session.title || session.title === 'Новый чат' || sess
       | ion.title === 'Предыдущий диалог') && !session.isCustomTitle)
       | {
744 |         session.title = generateSessionTitle(cleanText);
745 |         renderSessionsList();
```

```

746 | }
747 |
748 | session.updatedAt = Date.now();
749 | saveSessions();
750 |
751 | if (chatWin) {
752 |     chatWin.scrollTop = chatWin.scrollHeight;
753 | }
754 | }
755 |
756 | if (typeof window !== 'undefined') {
757 |     window.initChatTab = initChatTab;
758 |     window.handleExternalChatPrompt = handleExternalChatPrompt;
759 |     window.handleRemoteTranscript = handleRemoteTranscript;
760 |     window.clearChatHistory = clearCurrentSession;
761 |     window.createNewSession = createNewSession;
762 |     window.switchSession = switchSession;
763 |     window.renameSession = renameSession;
764 |     window.deleteSession = deleteSession;
765 |
766 |     if (typeof chrome !== 'undefined' && chrome.runtime?.onMessage
    | ge) {
767 |         chrome.runtime.onMessage.addListener((msg) => {
768 |             if (msg && msg.action === 'breadboard_external_chat_prom
    | pt') {
769 |                 handleExternalChatPrompt(msg);
770 |             }
771 |         });
772 |     }
773 | }
774 |
775 | function parseContentToHtml(text) {
776 |     if (!text) return '';
777 |
778 |     let cleaned = text.trim();
779 |     const codeBlockRegex = /^```(?:html|xml)?\s*([\s\S]*?)\s*```
    | $/i;
780 |     const match = cleaned.match(codeBlockRegex);

```

```

781 |   if (match) {
782 |       cleaned = match[1];
783 |   }
784 |
785 |   if (window.marked && typeof window.marked.parse === 'function') {
786 |       return window.marked.parse(cleaned);
787 |   }
788 |
789 |   return cleaned
790 |       .replace(/\*\*(.*?)\*/g, '<strong>$1</strong>')
791 |       .replace(/\*(.*?)\*/g, '<em>$1</em>')
792 |       .replace(/\`([^\`]+)\`/g, '<code>$1</code>')
793 |       .replace(/\n/g, '<br>');
794 | }
795 |
796 | function addMessageToDom(sender, text, voiceText = null, timeStr = null, outputMode = null) {
797 |     const win = document.getElementById('chat-window');
798 |     if (!win) return null;
799 |
800 |     const el = document.createElement('div');
801 |     el.className = 'message ' + (sender === 'user' ? 'user-message' : 'bot-message');
802 |
803 |     const displayText = parseContentToHtml(text);
804 |     const formattedTime = timeStr || new Date().toLocaleTimeString();
805 |
806 |     if (sender === 'user') {
807 |         el.innerHTML = `<strong>Вы</strong> (${formattedTime}): <div class="message-content">${displayText}</div>`;
808 |     } else {
809 |         const modeBadge = outputMode === 'voice_only' ? '<span class="badge bg-secondary-subtle text-secondary me-1"><i class="bi bi-volume-up"></i> Голос</span>' : '';
810 |         el.innerHTML = `
811 |             <div class="d-flex align-items-center justify-content-be

```

```

tween mb-1">
812     <div><strong>AI Assistant</strong> ${modeBadge}<span c
lass="text-muted">(${formattedTime})</span></div>
813     <button class="btn btn-sm btn-outline-secondary py-0 p
x-2 btn-speak-message" title="Озвучить ответ" type="button">
814         <i class="bi bi-volume-up"></i>
815     </button>
816 </div>
817     <div class="message-content">${displayText || (voiceText
? `<em><i class="bi bi-soundwave me-1"></i>${parseContentToHt
ml(voiceText)}</em>` : '')}</div>
818     `;
819     const speakBtn = el.querySelector('.btn-speak-message');
820     if (speakBtn) {
821         speakBtn.addEventListener('click', () => {
822             if (window.chatService?.speak) {
823                 window.chatService.speak(voiceText || text);
824             }
825         });
826     }
827 }
828
829 win.appendChild(el);
830 win.scrollTop = win.scrollHeight;
831 return el;
832 }
833
834 function setStatus(statusText, isVisible = true) {
835     const indicator = document.getElementById('chat-status-indic
ator');
836     const textEl = document.getElementById('chat-status-text');
837     if (!indicator || !textEl) return;
838
839     if (isVisible) {
840         textEl.textContent = statusText || 'Обработка запроса...';
841         indicator.classList.remove('d-none');
842     } else {
843         indicator.classList.add('d-none');

```

```
844 | }
845 | }
846 |
847 | async function sendMessage() {
848 |     const input = document.getElementById('message-input');
849 |     const btn = document.getElementById('send-button');
850 |     const topKInput = document.getElementById('chat-top-k');
851 |     const minScoreInput = document.getElementById('chat-min-score');
852 |     const modeSelect = document.getElementById('chat-output-mode');
853 |     const ragToggle = document.getElementById('chat-rag-toggle');
854 |
855 |     if (!input || !btn) return;
856 |
857 |     if (!window.activeModelName) {
858 |         input.disabled = true;
859 |         btn.disabled = true;
860 |         input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
861 |         input.classList.add('is-invalid');
862 |         return;
863 |     }
864 |
865 |     const msg = input.value.trim();
866 |     if (!msg) return;
867 |
868 |     const session = getActiveSession();
869 |     if (!session) return;
870 |
871 |     const topK = topKInput ? parseInt(topKInput.value, 10) || 3 : 3;
872 |     const minScore = minScoreInput ? parseFloat(minScoreInput.value) || 0.45 : 0.45;
873 |     const outputMode = modeSelect ? modeSelect.value : 'text_only';
874 |     const shouldSpeak = outputMode === 'voice_only' || outputMod
```

```

    e === 'text_and_voice';
875   const isRagEnabled = ragToggle ? ragToggle.checked : true;
876
877   input.value = '';
878   input.disabled = true;
879   btn.disabled = true;
880
881   setStatus('Отправка запроса...', true);
882   const userTimeStr = new Date().toLocaleTimeString();
883
884   // Clear welcome placeholder if this is the first message
885   const chatWin = document.getElementById('chat-window');
886   if (chatWin && session.messages.length === 0) {
887     chatWin.innerHTML = '';
888   }
889
890   addMessageToDom('user', msg, null, userTimeStr);
891   session.messages.push({ sender: 'user', text: msg, timeStr:
userTimeStr });
892
893   // Auto-name the session based on the first query if not cus
tomized
894   if ((!session.title || session.title === 'Новый чат' || sess
ion.title === 'Предыдущий диалог') && !session.isCustomTitle)
{
895     session.title = generateSessionTitle(msg);
896     renderSessionsList();
897   }
898
899   saveSessions();
900
901   const win = document.getElementById('chat-window');
902   const botMessageEl = document.createElement('div');
903   botMessageEl.className = 'message bot-message';
904   const timeStr = new Date().toLocaleTimeString();
905   const modeBadge = outputMode === 'voice_only' ? '<span class
="badge bg-secondary-subtle text-secondary me-1"><i class="bi
bi-volume-up"></i> Голос</span>' : '';

```

```
906 | botMessageEl.innerHTML = `
907 |     <div class="d-flex align-items-center justify-content-between mb-1">
908 |         <div><strong>AI Assistant</strong> ${modeBadge}<span class="text-muted">(${timeStr})</span></div>
909 |         <button class="btn btn-sm btn-outline-secondary py-0 px-2 btn-speak-message d-none" title="Озвучить ответ" type="button">
910 |             <i class="bi bi-volume-up"></i>
911 |         </button>
912 |     </div>
913 |     <div class="bot-text"><span class="text-muted">⏳ Ожидание
    | ответа...</span></div>
914 | `;
915 | if (win) {
916 |     win.appendChild(botMessageEl);
917 |     win.scrollTop = win.scrollHeight;
918 | }
919 | const textDiv = botMessageEl.querySelector('.bot-text');
920 | const speakBtn = botMessageEl.querySelector('.btn-speak-message');
921 |
922 | let fullReply = '';
923 | let started = false;
924 |
925 | try {
926 |     const generationConfig = {
927 |         top_k: topK,
928 |         min_score: minScore,
929 |         threshold: minScore,
930 |         rag_enabled: isRagEnabled,
931 |         output_mode: outputMode,
932 |         tts_enabled: shouldSpeak,
933 |         model: window.activeModelName || undefined,
934 |         search_engine: window.activeSearchEngine || undefined
935 |     };
936 |
937 |     let rafId = null;
```



```
938 | const scheduleRender = () => {
939 |   if (rafId) return;
940 |   rafId = requestAnimationFrame(() => {
941 |     rafId = null;
942 |     if (textDiv) {
943 |       textDiv.innerHTML = parseContentToHtml(fullReply);
944 |     }
945 |     if (win) win.scrollTop = win.scrollHeight;
946 |   });
947 | };
948 |
949 | const replyObj = await window.chatService.sendChatMessage(
950 |   msg,
951 |   (chunk, status) => {
952 |     if (status) {
953 |       setStatus(status, true);
954 |     }
955 |     if (chunk) {
956 |       if (!started) {
957 |         if (textDiv) textDiv.textContent = '';
958 |         started = true;
959 |       }
960 |       fullReply += chunk;
961 |       scheduleRender();
962 |     }
963 |   },
964 |   session.chatHistory || [],
965 |   generationConfig
966 | );
967 |
968 | if (rafId) {
969 |   cancelAnimationFrame(rafId);
970 |   rafId = null;
971 | }
972 |
973 | const finalReply = replyObj.text || fullReply;
974 | if (textDiv) {
975 |   if (outputMode === 'voice_only' && !finalReply.trim() &&
```

```

    replyObj.voice) {
976     textDiv.innerHTML = `<i class="bi bi-soundwave me-
1"></i>${parseContentToHtml(replyObj.voice)}</em>`;
977     } else {
978     textDiv.innerHTML = parseContentToHtml(finalReply);
979     }
980     }
981
982     if (speakBtn) {
983     speakBtn.classList.remove('d-none');
984     speakBtn.addEventListener('click', () => {
985     if (window.chatService?.speak) {
986     window.chatService.speak(replyObj || finalReply);
987     }
988     });
989     }
990
991     // Auto-speak response if mode is voice_only or text_and_v
oice
992     if (shouldSpeak && window.chatService?.speak) {
993     window.chatService.speak(replyObj || finalReply);
994     }
995
996     if (!session.chatHistory) session.chatHistory = [];
997     session.chatHistory.push({ role: 'user', parts: [msg] });
998     session.chatHistory.push({ role: 'model', parts: [finalRep
ly || replyObj.voice || ''] });
999
1000     session.messages.push({
1001     sender: 'bot',
1002     text: finalReply,
1003     voiceText: replyObj.voice || null,
1004     timeStr: timeStr,
1005     mode: outputMode
1006     });
1007     session.updatedAt = Date.now();
1008     saveSessions();
1009

```

```
1010     setStatus('', false);
1011 } catch (err) {
1012     const errText = err instanceof Error ? err.message : typeo
f err === 'object' && err !== null ? JSON.stringify(err) : Str
ing(err);
1013     if (textDiv) {
1014         textDiv.innerHTML = `Ошибк
a: ${errText}</span>`;
1015     }
1016     setStatus('Ошибка исполнения', true);
1017     setTimeout(() => setStatus('', false), 4000);
1018 } finally {
1019     if (window.activeModelName) {
1020         input.disabled = false;
1021         btn.disabled = false;
1022         input.placeholder = 'Введите сообщение...';
1023         input.classList.remove('is-invalid');
1024         input.focus();
1025     } else {
1026         input.disabled = true;
1027         btn.disabled = true;
1028         input.placeholder = 'Ни одна модель не выбрана. Выберите
модель во вкладке «Модели и API».';
1029         input.classList.add('is-invalid');
1030     }
1031     if (win) win.scrollTop = win.scrollHeight;
1032 }
1033 }
1034
1035
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/cloudflared_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/cloudflared_tab/index.h
17 tml" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/cloudflared_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-08 03:08:00
25 =====
26 =====
27 -->
28
29 <!-- Cloudflare Tunnel Monitor Application Tab -->
30 <style>
31   .cf-container {
32     padding: 0.5rem 0.75rem;
33   }
34   .cf-card {
35     background: #0f172a !important;
36     border: 1px solid #334155 !important;
37     border-radius: 8px;
```

```
33 | padding: 0.45rem 0.75rem;
34 | display: flex;
35 | flex-direction: column;
36 | justify-content: space-between;
37 | min-height: 58px;
38 | gap: 0.2rem;
39 | }
40 | .cf-label {
41 |     font-size: 0.68rem;
42 |     text-transform: uppercase;
43 |     letter-spacing: 0.05em;
44 |     color: #94a3b8;
45 |     line-height: 1.1;
46 | }
47 | .cf-value {
48 |     font-size: 1.25rem;
49 |     font-weight: 700;
50 |     color: #f8fafc;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
52 |     line-height: 1.15;
53 | }
54 | .cf-panel {
55 |     background: #0f172a !important;
56 |     border: 1px solid #334155 !important;
57 |     border-radius: 8px;
58 |     overflow: hidden;
59 | }
60 | .cf-panel-header {
61 |     background: #1e293b !important;
62 |     border-bottom: 1px solid #334155 !important;
63 |     padding: 0.45rem 0.85rem;
64 |     font-weight: 600;
65 |     font-size: 0.82rem;
66 |     color: #38bdf8;
67 |     display: flex;
68 |     align-items: center;
69 |     justify-content: space-between;
```

```

70 | }
71 | .cf-log-terminal {
72 |     background: #020617;
73 |     color: #cbd5e1;
74 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
75 |     font-size: 0.78rem;
76 |     line-height: 1.45;
77 |     padding: 0.5rem 0.75rem;
78 |     border-radius: 6px;
79 |     max-height: 380px;
80 |     overflow-y: auto;
81 | }
82 | </style>
83 |
84 | <div class="cf-container">
85 |     <!-- Header Alert & Status (Compact Strip) -->
86 |     <div class="card shadow-sm border-secondary-subtle py-1.5 px
    | -3 mb-2 d-flex justify-content-between align-items-center flex
    | -wrap gap-2">
87 |         <div class="d-flex align-items-center gap-2 flex-wrap">
88 |             <span class="fs-5">□ </span>
89 |             <span class="fw-bold text-white small" style="font-size:
    | 0.92rem;">Cloudflare Tunnel Monitor</span>
90 |             <span class="badge bg-dark border border-secondary text-
    | info fw-normal" style="font-size: 0.7rem;">Diagnostics</span>
91 |             <span class="text-secondary small d-none d-lg-inline" st
    | yle="font-size: 0.76rem;">| Служба Cloudflare Tunnel, доступн
    | ость URL & логи</span>
92 |         </div>
93 |         <div class="d-flex align-items-center gap-1.5 flex-wrap">
94 |             <span class="badge rounded-pill bg-success-subtle text-s
    | uccess border border-success px-2 py-1" style="font-size: 0.72
    | rem;" id="cf-daemon-badge">
95 |                 • АКТИВЕН
96 |             </span>
97 |             <button class="btn btn-xs btn-outline-info rounded-pill
    | px-2.5 py-1" style="font-size: 0.75rem;" id="btn-cf-refresh">

```

```

98         <i class="bi bi-arrow-clockwise me-1"></i> Обновить
99     </button>
100     <button class="btn btn-xs btn-outline-warning rounded-pi
11 px-2 py-1" style="font-size: 0.75rem;" id="btn-cf-config">
101         <i class="bi bi-gear-fill me-1"></i> Настройки
102     </button>
103     <button class="btn btn-xs btn-primary rounded-pill px-2.
15 py-1" style="font-size: 0.75rem;" id="btn-cf-test-endpoint">
104         <i class="bi bi-broadcast me-1"></i> Ingress
105     </button>
106 </div>
107 </div>
108
109 <!-- Top Metrics Bar (Compact Density Grid) -->
110 <div class="row g-2 mb-2">
111     <div class="col-6 col-md-3">
112         <div class="cf-card">
113             <span class="cf-label">Публичный Ingress URL</span>
114             <div class="d-flex align-items-baseline justify-conten
115 t-between mt-1">
116                 <span class="cf-value text-info text-truncate" id="c
117 f-ingress-url" style="font-size: 0.88rem;">https://kino.davidk
118 a.net</span>
119             </div>
120         </div>
121     </div>
122     <div class="col-6 col-md-3">
123         <div class="cf-card">
124             <span class="cf-label">Доступность эндпоинта</span>
125             <div class="d-flex align-items-baseline justify-conten
126 t-between mt-1">
127                 <span class="cf-value text-success" style="font-size
128 : 0.95rem;" id="cf-endpoint-status">200 OK</span>
129             </div>
130         </div>
131     </div>
132 </div>
133 <div class="col-6 col-md-3">
134     <div class="cf-card">

```

```

129 |         <span class="cf-label">CPU / Память демона</span>
130 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
131 |             <span class="cf-value" style="font-size: 0.95rem;" id="cf-process-usage">0.5% / 45 MB</span>
132 |         </div>
133 |     </div>
134 | </div>
135 | <div class="col-6 col-md-3">
136 |     <div class="cf-card">
137 |         <span class="cf-label">Health Score</span>
138 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
139 |             <span class="cf-value text-success" id="cf-health-score">100/100</span>
140 |         </div>
141 |     </div>
142 | </div>
143 | </div>
144 |
145 | <div class="row g-3">
146 |     <!-- Tunnel Logs Tail Panel -->
147 |     <div class="col-lg-7">
148 |         <div class="cf-panel h-100">
149 |             <div class="cf-panel-header">
150 |                 <span><i class="bi bi-terminal me-1"></i> Журнал cloudflared.log</span>
151 |                 <span class="badge bg-secondary" id="cf-log-count-badge">50 строк</span>
152 |             </div>
153 |             <div class="p-3">
154 |                 <div class="cf-log-terminal" id="cf-log-terminal-output">
155 |                     Загрузка журналов...
156 |                 </div>
157 |             </div>
158 |         </div>
159 |     </div>

```



```
160 |
161 |     <!-- AI Diagnostics & Recommendations Panel -->
162 |     <div class="col-lg-5">
163 |         <div class="cf-panel h-100">
164 |             <div class="cf-panel-header">
165 |                 <span><i class="bi bi-heart-pulse-fill text-danger m
166 | e-1"></i> Диагностика & Рекомендации</span>
167 |                 <button class="btn btn-sm btn-outline-info py-0 px-2
168 | " id="btn-cf-run-diagnostics">Диагностировать</button>
169 |             </div>
170 |             <div class="p-3">
171 |                 <div class="card bg-dark border-secondary p-3 mb-3">
172 |                     <h6 class="fw-bold text-info mb-1" id="cf-diag-sum
173 | mary-title">Статус оптимален</h6>
174 |                     <p class="small text-muted mb-0" id="cf-diag-summa
175 | ry-desc">Все туннельные соединения стабильны, аномалий не обна
176 | ружено.</p>
177 |                 </div>
178 |                 <h6 class="small text-uppercase text-muted fw-bold m
179 | b-2">Рекомендации:</h6>
180 |                 <ul class="list-group list-group-flush bg-transparen
181 | t small" id="cf-recommendations-list">
182 |                     <li class="list-group-item bg-transparent text-lig
183 | ht border-secondary py-1 px-0">• Поддерживайте актуальную верс
184 | ию cloudflared</li>
185 |                     <li class="list-group-item bg-transparent text-lig
186 | ht border-secondary py-1 px-0">• Проверяйте сертификат SSL дом
187 | ена</li>
188 |                 </ul>
189 |             </div>
190 |         </div>
191 |     </div>
192 | </div>
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Cloudflared Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/cloudflared_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/cloudflared_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 // =====
22 // Process Name: Cloudflared Monitor Web Tab Module
23 // =====
24 // Description:
25 //   Client-side JavaScript controller for the Cloudflare Tunnel Monitor
26 //   administrative web interface tab.
27 //
28 // File: main.js
29 // Project: ai-breadboard
```

```

30 // Package: src.api.webinterface.cloudflare_tab
31 // Author: hypo69
32 // Copyright: © 2026 hypo69
33 // =====
34
35 (function() {
36     let isCfInitialized = false;
37
38     async function fetchTunnelStatus() {
39         try {
40             const res = await fetch('/api/cloudflared/status');
41             if (!res.ok) return;
42             const data = await res.json();
43
44             const badge = document.getElementById('cf-daemon-badge')
45             ;
46             const ingressEl = document.getElementById('cf-ingress-ur
47 l');
48             const endpointEl = document.getElementById('cf-endpoint-
49 status');
50             const usageEl = document.getElementById('cf-process-usag
51 e');
52             const healthEl = document.getElementById('cf-health-scor
53 e');
54
55             if (badge) {
56                 badge.className = data.daemon_running ? 'badge rounded
57 -pill bg-success-subtle text-success border border-success px-
58 3 py-2' : 'badge rounded-pill bg-danger-subtle text-danger bor
59 der border-danger px-3 py-2';
60                 badge.innerText = data.daemon_running ? '● Туннель акт
61 ивен' : '● Туннель остановлен';
62             }
63
64             if (ingressEl) ingressEl.innerText = data.public_url ||
65 'https://kino.davidka.net';
66             if (endpointEl) {

```

```

57 |         endpointEl.innerText = data.endpoint_reachable ? `Дост
    | упен (${data.latency_ms || 0} ms)` : 'Недоступен';
58 |         endpointEl.className = data.endpoint_reachable ? 'cf-v
    | alue text-success' : 'cf-value text-danger';
59 |     }
60 |     if (usageEl) {
61 |         usageEl.innerText = `${data.cpu_percent || 0}% / ${dat
    | a.memory_mb || 0} MB`;
62 |     }
63 |     if (healthEl) {
64 |         healthEl.innerText = `${data.health_score || 100}/100`
    | ;
65 |     }
66 | } catch (e) {
67 |     console.error('[CloudflaredTab] Failed to fetch status:'
    | , e);
68 | }
69 | }
70 |
71 | async function fetchLogs() {
72 |     try {
73 |         const res = await fetch('/api/cloudflared/logs?limit=50'
    | );
74 |         if (!res.ok) return;
75 |         const data = await res.json();
76 |         const logs = data.logs || [];
77 |
78 |         const terminal = document.getElementById('cf-log-termina
    | l-output');
79 |         const countBadge = document.getElementById('cf-log-count
    | -badge');
80 |         if (countBadge) countBadge.innerText = `${logs.length} c
    | трок`;
81 |
82 |         if (terminal) {
83 |             if (logs.length === 0) {
84 |                 terminal.innerText = 'Журнал cloudflared пуст.';
85 |                 return;

```

```

86         }
87         terminal.innerHTML = logs.map(l => {
88             const color = l.level === 'error' ? '#f87171' : (l.l
level === 'warn' ? '#fbbf24' : '#94a3b8');
89             return `<div style="color: ${color};">[${l.time} || '
'}] [${(l.level || 'INFO').toUpperCase()}] ${l.msg || l.messag
e || ''}</div>`;
90         }).join('');
91     }
92 } catch (e) {
93     console.error('[CloudflaredTab] Failed to fetch logs:',
e);
94 }
95 }
96
97 async function runDiagnostics() {
98     try {
99         const res = await fetch('/api/cloudflared/diagnostic');
100         if (!res.ok) return;
101         const report = await res.json();
102
103         const titleEl = document.getElementById('cf-diag-summary
-title');
104         const descEl = document.getElementById('cf-diag-summary-
desc');
105         const recList = document.getElementById('cf-recommendati
ons-list');
106
107         if (titleEl) titleEl.innerText = `Здоровье: ${report.sta
tus || 'OK'} (${report.health_score || 100}/100)`;
108         if (descEl) descEl.innerText = report.summary || 'Все ту
ннельные соединения стабильны.';
109
110         if (recList && Array.isArray(report.recommendations)) {
111             recList.innerHTML = report.recommendations.map(r => `
112                 <li class="list-group-item bg-transparent text-light
border-secondary py-1 px-0">• ${r}</li>
113             `).join('');

```

```
114 |     }
115 |   } catch (e) {
116 |     console.error('[CloudflaredTab] Diagnostic error:', e);
117 |   }
118 | }
119 |
120 | function initCloudflaredTab() {
121 |   console.log('[CloudflaredTab] Initializing Cloudflared Mon
itor tab...');
122 |   fetchTunnelStatus();
123 |   fetchLogs();
124 |
125 |   if (!isCfInitialized) {
126 |     const refreshBtn = document.getElementById('btn-cf-refre
sh');
127 |     const configBtn = document.getElementById('btn-cf-config
');
128 |     const diagBtn = document.getElementById('btn-cf-run-diag
nostics');
129 |     const testBtn = document.getElementById('btn-cf-test-end
point');
130 |
131 |     if (refreshBtn) {
132 |       refreshBtn.onclick = () => {
133 |         fetchTunnelStatus();
134 |         fetchLogs();
135 |       };
136 |     }
137 |
138 |     if (diagBtn) diagBtn.onclick = runDiagnostics;
139 |
140 |     if (testBtn) {
141 |       testBtn.onclick = async () => {
142 |         testBtn.disabled = true;
143 |         testBtn.innerHTML = '<span class="spinner-border spi
nner-border-sm me-1"></span> Проверка...';
144 |         try {
145 |           const res = await fetch('/api/cloudflared/test-end
```

```

    point');
146         const data = await res.json();
147         const msg = `Результат проверки Ingress:\nДоступен
: ${data.reachable ? 'ДА' : 'НЕТ'}\nКод ответа: ${data.status_
code || 'N/A'}\nЗадержка: ${data.response_time_ms || 0} ms`;
148         window.showToast?.(msg, data.reachable ? 'success'
: 'warning') || alert(msg);
149     } catch (e) {
150         window.showToast?.('Ошибка проверки эндпоинта: ' +
e.message, 'danger') || alert('Ошибка проверки эндпоинта: ' +
e.message);
151     } finally {
152         testBtn.disabled = false;
153         testBtn.innerHTML = '<i class="bi bi-broadcast me-
1"></i> Проверить Ingress';
154         fetchTunnelStatus();
155     }
156 };
157 }
158
159 if (configBtn) {
160     configBtn.onclick = () => {
161         if (typeof window.openAppConfigModal === 'function')
162         {
163             window.openAppConfigModal('cloudflared_monitor', '
Cloudflare Tunnel Monitor');
164         }
165     }
166
167     isCfInitialized = true;
168 }
169 }
170
171 window.initCloudflaredTab = initCloudflaredTab;
172 }}());
173

```

tab-registry.js

```
1 /**
2  * =====
3  * Process Name: AI-Breadboard UI - Tab-Registry Script
4  * =====
5  * Description:
6  *   Клиентский веб-скрипт модуля tab-registry.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/src/api/webgui/core/tab-registry.js?v=202
11  * 61006_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { TAB_DEFINITIONS, TabRegistry } from '/src/api/
15  * webgui/core/tab-registry.js';
16  *
17  * File: tab-registry.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/core
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-08 04:06:00
23  * =====
24  */
25  * TabRegistry - единый источник правды (SSOT) для всех вкладо
26  * к WebGUI.
27  * Экспортирует методы getAll(), getById(id), getFiltered(apps
28  * StatusMap, role).
29  */
30  export const TAB_DEFINITIONS = [
31  { id: 'about_system', label: 'О системе', i18nKey: 'auto___d
```



```
32ca0', icon: 'i', tab: 'about-system', tabId: 'tab-about-syst
em', html: '/html/about_system_tab/index.html?v=20261006_v14',
js: '/html/about_system_tab/main.js?v=20261006_v14' },
30 { id: 'scenarios', label: 'Сценарии', i18nKey: 'auto__ddaf0
e', icon: '□', tab: 'scenarios', tabId: 'tab-scenarios', html
: '/html/scenarios_tab/index.html?v=20261006_v8', js: '/html/s
cenarios_tab/main.js?v=20261006_v8' },
31 { id: 'chat', label: 'Чат', i18nKey: 'auto__8c77e4', icon:
'□', tab: 'chat', tabId: 'tab-chat', html: '/html/chat/index.
html?v=20260923_v5', js: '/html/chat/main.js?v=20260923_v5' },
32 { id: 'network_terminal', label: 'Сеть', i18nKey: 'auto__f6
df40', icon: '□', tab: 'network', tabId: 'tab-network', html:
'/html/network_tab/index.html', js: '/html/network_tab/main.j
s' },
33 { id: 'system_inspector', label: 'Инспектор системы', i18nKe
y: 'auto__4324be', icon: 'bi-graph-up-arrow', tab: 'hardware-
load-inspector', tabId: 'tab-hardware-load-inspector', html: '
/html/system_inspector_tab/index.html?v=20261006_v14', js: '/h
tml/system_inspector_tab/main.js?v=20261006_v14' },
34 { id: 'processes_load_inspector', label: 'Активность процесс
ов', icon: 'bi-activity', tab: 'processes-load-inspector', tab
Id: 'tab-processes-load-inspector', html: '/html/processes_loa
d_inspector_tab/index.html?v=20261008_v1', js: '/html/processe
s_load_inspector_tab/main.js?v=20261008_v1' },
35 { id: 'windows_sysadmin', label: 'Windows Sysadmin', icon: '
bi-server', tab: 'windows-admin', tabId: 'tab-windows-admin',
html: '/html/windows_admin_tab/index.html', js: '/html/windows
_admin_tab/main.js' },
36 { id: 'focus_settings', label: 'Фокусировка (Focus)', icon:
'bi-circle-half', tab: 'focus-settings', tabId: 'tab-focus-set
tings', html: '/html/focus_settings_tab/index.html?v=20261006_
v1', js: '/html/focus_settings_tab/main.js?v=20261006_v1' },
37 { id: 'system32_commands', label: 'Команды System32', icon:
'□', tab: 'system32-commands', tabId: 'tab-system32-commands'
, html: '/html/system32_commands_tab/index.html?v=20261006_v1'
, js: '/html/system32_commands_tab/main.js?v=20261006_v1' },
38 { id: 'system_control_center', label: 'System Logs & Activit
y', i18nKey: 'tabs.systemControl', icon: 'bi-journal-text', ta
```

```
b: 'system-control', tabId: 'tab-system-control', html: '/html/system_control_tab/index.html?v=20261008_v1', js: '/html/system_control_tab/main.js?v=20261008_v1' },
39 { id: 'post_install_wizard', label: 'Post-Install Wizard', icon: '□', tab: 'post-install-wizard', tabId: 'tab-post-install-wizard', html: '/html/post_install_wizard_tab/index.html?v=20261006_v1', js: '/html/post_install_wizard_tab/main.js?v=20261006_v1' },
40 { id: 'maintenance_recovery', label: 'Maintenance & Recovery', icon: '□', tab: 'maintenance-recovery', tabId: 'tab-maintenance-recovery', html: '/html/maintenance_recovery_tab/index.html?v=20261006_v1', js: '/html/maintenance_recovery_tab/main.js?v=20261006_v1' },
41 { id: 'system_log_viewer', label: 'Логи системы', i18nKey: 'auto___6e28ee', icon: '□', tab: 'system-logs', tabId: 'tab-system-logs', html: '/html/system_logs_tab/index.html', js: '/html/system_logs_tab/main.js' },
42 { id: 'software_audit', label: 'Аудит ПО', i18nKey: 'auto___128e2e', icon: '□', tab: 'software-audit', tabId: 'tab-software-audit', html: '/html/software_audit_tab/index.html', js: '/html/software_audit_tab/main.js' },
43 { id: 'registry_viewer', label: 'Registry Viewer', icon: '□', tab: 'registry-viewer', tabId: 'tab-registry-viewer', html: '/html/registry_viewer_tab/index.html', js: '/html/registry_viewer_tab/main.js' },
44 { id: 'windows_defender', label: 'Defender Security', icon: 'bi-shield-lock', tab: 'defender', tabId: 'tab-defender', html: '/html/defender_tab/index.html?v=20261006_v2', js: '/html/defender_tab/main.js?v=20261006_v2' },
45 { id: 'windows_startup_auditor', label: 'Автозагрузка', i18nKey: 'auto___007983', icon: '□', tab: 'startup-auditor', tabId: 'tab-startup-auditor', html: '/html/startup_auditor_tab/index.html', js: '/html/startup_auditor_tab/main.js' },
46 { id: 'windows_backup_manager', label: 'Windows Backup', icon: '□', tab: 'windows-backup', tabId: 'tab-windows-backup', html: '/html/windows_backup_tab/index.html?v=20261004_v1', js: '/html/windows_backup_tab/main.js?v=20261004_v1' },
47 { id: 'hardware_monitor', label: 'Мониторинг оборудования',
```

```

i18nKey: 'auto__8a1c35', icon: 'bi-cpu', tab: 'hardware-monit
or', tabId: 'tab-hardware-monitor', html: '/html/hardware_moni
tor_tab/index.html', js: '/html/hardware_monitor_tab/main.js'
},
48 { id: 'cloudflared_monitor', label: 'Cloudflared', icon: '☐ '
, tab: 'cloudflared', tabId: 'tab-cloudflared', html: '/html/c
loudflared_tab/index.html', js: '/html/cloudflared_tab/main.js
' },
49 { id: 'google_user_desktop', label: 'Google User Desktop', i
con: '☐ ', tab: 'google-desktop', tabId: 'tab-google-desktop',
html: '/html/google_desktop_tab/index.html?v=20260928_v2', js
: '/html/google_desktop_tab/main.js?v=20260928_v2' },
50 { id: 'gcloud_monitor', label: 'Google Cloud', icon: '☐ ', ta
b: 'gcloud', tabId: 'tab-gcloud', html: '/html/gcloud_tab/inde
x.html', js: '/html/gcloud_tab/main.js' },
51 { id: 'website_monitor', label: 'Website Intelligence', icon
: '☐ ', tab: 'website-monitor', tabId: 'tab-website-monitor',
html: '/html/website_monitor_tab/index.html', js: '/html/websi
te_monitor_tab/main.js' },
52 { id: 'trading_terminal', label: 'Trading Terminal', icon: '
☐ ', tab: 'trading', tabId: 'tab-trading', html: '/html/tradin
g_tab/index.html', js: '/html/trading_tab/main.js' },
53 { id: 'user_assistant', label: 'User Assistant', icon: '☐ ',
tab: 'user-assistant', tabId: 'tab-user-assistant', html: '/h
tml/user_assistant_tab/index.html', js: '/html/user_assistant_
tab/main.js' },
54 { id: 'helpdesk', label: 'Helpdesk', icon: '☐ ', tab: 'helpd
esk', tabId: 'tab-helpdesk', html: '/html/helpdesk_tab/index.h
tml', js: '/html/helpdesk_tab/main.js' },
55 { id: 'wikipedia_research', label: 'Wikipedia Research', ico
n: '☐ ', tab: 'wikipedia-research', tabId: 'tab-wikipedia-rese
arch', html: '/html/wikipedia_research_tab/index.html', js: '/
html/wikipedia_research_tab/main.js' },
56 { id: 'autolog_manager', label: 'Автологи', i18nKey: 'auto__
_17b9d1', icon: '☐ ', tab: 'autolog', tabId: 'tab-autolog', ht
ml: '/html/autolog_tab/index.html?v=20260925_v2', js: '/html/a
utolog_tab/main.js?v=20260925_v2' },
57 { id: 'software_transparency_scanner', label: 'Transparency

```

```

Scanner', icon: '☐', tab: 'software-transparency', tabId: 'tab-software-transparency', html: '/html/software_transparency_tab/index.html?v=20260926_v2', js: '/html/software_transparency_tab/main.js?v=20260926_v2' },
58 { id: 'user_directories', label: 'Каталоги пользователя', i18nKey: 'auto___1ddade', icon: '☐', tab: 'user-directories', tabId: 'tab-user-directories', html: '/html/user_directories_tab/index.html', js: '/html/user_directories_tab/main.js' },
59 { id: 'ninite_updater', label: 'Ninite Updater', icon: '☐', tab: 'ninite-updater', tabId: 'tab-ninite-updater', html: '/html/ninite_updater_tab/index.html?v=20260924_v2', js: '/html/ninite_updater_tab/main.js?v=20260924_v2' },
60 { id: 'file_recovery', label: 'Восстановление файлов', i18nKey: 'auto___8f9984', icon: '☐', tab: 'file-recovery', tabId: 'tab-file-recovery', html: '/html/file_recovery_tab/index.html', js: '/html/file_recovery_tab/main.js' },
61 { id: 'file_history_ai_search', label: 'Поиск файлов AI', i18nKey: 'auto__ai_2d3bf7', icon: '☐', tab: 'file-history-search', tabId: 'tab-file-history-search', html: '/html/file_history_ai_search_tab/index.html?v=20260928_v1', js: '/html/file_history_ai_search_tab/main.js?v=20260928_v1' },
62 { id: 'telemetry_research', label: 'Исследование телеметрии', i18nKey: 'auto___025fa1', icon: '☐', tab: 'telemetry-research', tabId: 'tab-telemetry-research', html: '/html/telemetry_research_tab/index.html?v=20260924_v1', js: '/html/telemetry_research_tab/main.js?v=20260924_v1' },
63 { id: 'process_leaks', label: 'Утечки процессов', i18nKey: 'auto___6e10ab', icon: '☐', tab: 'process-leaks', tabId: 'tab-process-leaks', html: '/html/process_leaks_tab/index.html?v=20260924_v1', js: '/html/process_leaks_tab/main.js?v=20260924_v1' },
64 { id: 'forensics', label: 'Форензика', i18nKey: 'auto___82e8f9', icon: '☐', tab: 'forensics', tabId: 'tab-forensics', html: '/html/forensics_tab/index.html?v=20260924_v1', js: '/html/forensics_tab/main.js?v=20260924_v1' },
65 { id: 'throttling', label: 'Троттлинг', i18nKey: 'auto___23403a', icon: '☐', tab: 'throttling', tabId: 'tab-throttling', html: '/html/throttling_tab/index.html?v=20260924_v1', js: '/h

```

```

    tml/throttling_tab/main.js?v=20260924_v1' },
66   { id: 'storage_wear', label: 'Износ дисков', i18nKey: 'auto_
    __5c3dfb', icon: '📀', tab: 'storage-wear', tabId: 'tab-storag
    e-wear', html: '/html/storage_wear_tab/index.html?v=20260924_v
    1', js: '/html/storage_wear_tab/main.js?v=20260924_v1' },
67   { id: 'peripherals', label: 'Периферия', i18nKey: 'auto__b6
    2c2e', icon: '🔌', tab: 'peripherals', tabId: 'tab-peripherals
    ', html: '/html/peripherals_tab/index.html?v=20260924_v1', js:
    '/html/peripherals_tab/main.js?v=20260924_v1' },
68   { id: 'rag', label: 'RAG База знаний', i18nKey: 'auto_rag__8
    29445', icon: '📖', tab: 'rag', tabId: 'tab-rag', html: '/html
    /rag_tab/index.html', js: '/html/rag_tab/main.js' },
69   { id: 'pixelrag', label: 'PixelRAG', icon: '🖼️', tab: 'pixel
    rag', tabId: 'tab-pixelrag', html: '/html/pixelrag_tab/index.h
    tml', js: '/html/pixelrag_tab/main.js' },
70   { id: 'models', label: 'Модели', i18nKey: 'auto__44bfc5', i
    con: '🤖', tab: 'models', tabId: 'tab-models', html: '/html/mo
    dels_tab/index.html', js: '/html/models_tab/main.js' },
71   { id: 'agents', label: 'Агенты', i18nKey: 'auto__af1850', i
    con: '👤', tab: 'agents', tabId: 'tab-agents', html: '/html/ag
    ents_tab/index.html', js: '/html/agents_tab/main.js' },
72   { id: 'skills', label: 'Навыки', i18nKey: 'auto__56221b', i
    con: '🧠', tab: 'skills', tabId: 'tab-skills', html: '/html/ski
    lls_tab/index.html', js: '/html/skills_tab/main.js' },
73   { id: 'ai_benchmark', label: 'AI Benchmark', icon: '📊', tab:
    'ai-benchmark', tabId: 'tab-ai-benchmark', html: '/html/ai_be
    nchmark_tab/index.html?v=20260926_v1', js: '/html/ai_benchmark
    _tab/main.js?v=20260926_v1' }
74 ];
75
76 export class TabRegistry {
77   static getAll() { return TAB_DEFINITIONS; }
78   static getById(tabId) {
79     if (!tabId) return null;
80     const clean = tabId.replace(/^tab-/, '');
81     return TAB_DEFINITIONS.find(t =>
82       t.id === clean ||
83       t.tab === clean ||

```

```
84 |         t.tabId === tabId ||
85 |         t.tabId === `tab-${clean}` ||
86 |         t.appId === clean
87 |     ) || null;
88 | }
89 | static getLabel(tab) {
90 |     if (!tab) return '';
91 |     if (tab.i18nKey && typeof window !== 'undefined' && window
92 |         .i18next?.t) {
93 |         const translated = window.i18next.t(tab.i18nKey);
94 |         if (translated && translated !== tab.i18nKey) return tra
95 |         nslated;
96 |     }
97 |     return tab.label || '';
98 | }
99 | static getFiltered(appsStatusMap = null, targetRole = 'admin
100 | ') {
101 |     return TAB_DEFINITIONS.filter(tab => {
102 |         if (tab.roles && !tab.roles.includes(targetRole)) return
103 |         false;
104 |         if (tab.appId && appsStatusMap && appsStatusMap[tab.appI
105 |         d]) {
106 |             return appsStatusMap[tab.appId].enabled !== false;
107 |         }
108 |         return true;
109 |     });
110 | }
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/cosmicplayer/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/cosmicplayer/index.html
15 |        " class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/cosmicplayer
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Link CosmicPlayer custom style -->
28 <link rel="stylesheet" href="/html/cosmicplayer/styles.css">
29 <!-- HLS.js library for streaming M3U8 inside custom HTML5 vid
30 |eo player -->
31 <script src="https://cdn.jsdelivr.net/npm/hls.js@latest"></scr
32 |ipt>
33
34 <div class="cosmic-player-tab">
35 |    <!-- Sidebar: Video URL list / Library / Settings -->
```

```

31     <aside class="cosmic-sidebar">
32         <div class="logo">
33             <span class="logo-icon">❏ </span>
34             <h2>CosmicPlayer</h2>
35         </div>
36
37         <!-- Paste link panel -->
38         <div class="url-input-section">
39             <h3>Вставить ссылку</h3>
40             <form id="stream-form">
41                 <input type="url" id="stream-url" placeholder=
"Ссылка (YouTube, VK, RuTube, Rezka...)" required>
42                 <button type="submit" class="btn btn-glow">Заг
рузить видео</button>
43             </form>
44         </div>
45
46         <!-- Supported sources badges -->
47         <div class="sources-panel">
48             <h4>Совместимые источники</h4>
49             <div class="sources-grid">
50                 <span class="badge" data-provider="youtube">Yo
uTube</span>
51                 <span class="badge" data-provider="vk">VK Vide
o</span>
52                 <span class="badge" data-provider="rutube">RuT
ube</span>
53                 <span class="badge" data-provider="hdrezka">HD
Rezka</span>
54                 <span class="badge" data-provider="seasonvar">
SeasonVar</span>
55                 <span class="badge" data-provider="kinogo">Kin
ogo</span>
56                 <span class="badge" data-provider="direct">Dir
ect HLS/MP4</span>
57             </div>
58         </div>
59

```



```

60 |         <!-- History List -->
61 |         <div class="history-section">
62 |             <div class="section-header">
63 |                 <h4>История воспроизведения</h4>
64 |                 <button id="clear-history" class="btn-text">Очи
истить</button>
65 |             </div>
66 |             <div id="history-list" class="history-list">
67 |                 <div class="empty-state">Нет недавно проигранн
ых видео</div>
68 |             </div>
69 |         </div>
70 |     </aside>
71 |
72 |     <!-- Main Content Area: Player and Description -->
73 |     <main class="main-content">
74 |         <!-- Movie Screen and Ambience -->
75 |         <div class="player-container" id="player-container">
76 |             <!-- Background Ambient Glow -->
77 |             <div class="ambient-glow" id="ambient-glow"></div>
78 |
79 |             <!-- Display Iframe (for embeds) -->
80 |             <iframe id="embed-player" class="player-element hi
dden" allowfullscreen allow="autoplay; encrypted-media"></ifra
me>
81 |
82 |             <!-- Custom HTML5 Player (for direct video feeds)
-->
83 |             <div id="html5-player-wrapper" class="player-eleme
nt hidden">
84 |                 <video id="html5-video" class="html5-video-el"
></video>
85 |                 <!-- Custom Player Skin -->
86 |                 <div class="custom-controls">
87 |                     <div class="progress-container" id="progre
ss-container">
88 |                         <div class="progress-bar" id="progress
-bar">

```

```

89         <div class="progress-hover"></div>
90     </div>
91 </div>
92     <div class="controls-row">
93         <div class="controls-left">
94             <button class="ctrl-btn" id="play-
| pause-btn">▶ </button>
95             <button class="ctrl-btn" id="stop-
| btn">■</button>
96             <div class="time-display">
97                 <span id="current-time">0:00</
| span> / <span id="duration-time">0:00</span>
98             </div>
99         </div>
100        <div class="controls-right">
101            <div class="volume-container">
102                <button class="ctrl-btn" id="m
| ute-btn">□ </button>
103                <input type="range" id="volume
| -slider" min="0" max="1" step="0.05" value="1">
104            </div>
105            <select id="speed-select" class="s
| peed-dropdown">
106                <option value="0.5">0.5x</opti
| on>
107                <option value="1" selected>1.0
| x</option>
108                <option value="1.5">1.5x</opti
| on>
109                <option value="2">2.0x</option
| >
110            </select>
111            <button class="ctrl-btn" id="fulls
| creen-btn">□ </button>
112        </div>
113    </div>
114 </div>
115 </div>

```

```
116 |
117 |         <!-- Initial Empty State -->
118 |         <div id="player-placeholder" class="player-placeho
    | lder">
119 |             <div class="placeholder-content">
120 |                 <span class="play-icon">▶ </span>
121 |                 <h3>Видео не загружено</h3>
122 |                 <p>Вставьте ссылку на видео или выберите о
    | дин из демонстрационных источников слева.</p>
123 |             </div>
124 |         </div>
125 |     </div>
126 |
127 |     <!-- Current Video Information -->
128 |     <div class="video-info-card glass-card">
129 |         <div class="info-header">
130 |             <span class="provider-tag" id="current-provide
    | r-tag">Offline</span>
131 |             <h3 id="video-title">Добро пожаловать в Cosmic
    | Player</h3>
132 |         </div>
133 |         <div class="info-body">
134 |             <p id="video-source-url">Пожалуйста, введите U
    | RL видео для воспроизведения.</p>
135 |         </div>
136 |     </div>
137 | </main>
138 | </div>
139 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Cosmicplayer - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/cosmicplayer/main.js?v
11  *     =20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/cosmicplayer
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 window.initCosmicplayerTab = function() {
22     console.log("CosmicPlayer tab initialized");
23
24     // Elements
25     const streamForm = document.getElementById('stream-form');
26     const streamUrlInput = document.getElementById('stream-url
27     ');
28     const embedPlayer = document.getElementById('embed-player'
29     );
30     const html5PlayerWrapper = document.getElementById('html5-
31     player-wrapper');
32     const html5Video = document.getElementById('html5-video');
```

```
30     const playerPlaceholder = document.getElementById('player-
placeholder');
31     const videoTitle = document.getElementById('video-title');
32     const videoSourceUrl = document.getElementById('video-sour
ce-url');
33     const currentProviderTag = document.getElementById('curren
t-provider-tag');
34     const historyList = document.getElementById('history-list'
);
35     const clearHistoryBtn = document.getElementById('clear-his
tory');
36     const badges = document.querySelectorAll('.cosmic-player-t
ab .badge');
37     const ambientGlow = document.getElementById('ambient-glow'
);
38
39     // Custom controls
40     const playPauseBtn = document.getElementById('play-pause-b
tn');
41     const stopBtn = document.getElementById('stop-btn');
42     const currentTimeDisplay = document.getElementById('curren
t-time');
43     const durationTimeDisplay = document.getElementById('durat
ion-time');
44     const progressContainer = document.getElementById('progres
s-container');
45     const progressBar = document.getElementById('progress-bar'
);
46     const progressHover = document.querySelector('.progress-ho
ver');
47     const muteBtn = document.getElementById('mute-btn');
48     const volumeSlider = document.getElementById('volume-slide
r');
49     const speedSelect = document.getElementById('speed-select'
);
50     const fullscreenBtn = document.getElementById('fullscreen-
btn');
51
```

```

52 |     let hlsInstance = null;
53 |     let history = JSON.parse(localStorage.getItem('cosmic_play
| er_history')) || '[]');
54 |
55 |     // Dynamic ambient glow colors per provider
56 |     const providerColors = {
57 |         youtube: 'radial-gradient(circle, rgba(255, 0, 0, 0.25
| ) 0%, transparent 60%)',
58 |         vk: 'radial-gradient(circle, rgba(74, 118, 168, 0.25)
| 0%, transparent 60%)',
59 |         rutube: 'radial-gradient(circle, rgba(235, 95, 30, 0.2
| 5) 0%, transparent 60%)',
60 |         hdrezka: 'radial-gradient(circle, rgba(242, 197, 17, 0
| .2) 0%, transparent 60%)',
61 |         seasonvar: 'radial-gradient(circle, rgba(0, 242, 254,
| 0.2) 0%, transparent 60%)',
62 |         kinogo: 'radial-gradient(circle, rgba(155, 81, 224, 0.
| 2) 0%, transparent 60%)',
63 |         direct: 'radial-gradient(circle, rgba(121, 40, 202, 0.
| 25) 0%, transparent 60%)',
64 |         offline: 'radial-gradient(circle, rgba(255, 255, 255,
| 0.05) 0%, transparent 60%)'
65 |     };
66 |
67 |     // Load history on start
68 |     renderHistory();
69 |
70 |     // Form submission
71 |     streamForm.addEventListener('submit', (e) => {
72 |         e.preventDefault();
73 |         const url = streamUrlInput.value.trim();
74 |         if (url) {
75 |             loadVideo(url);
76 |         }
77 |     });
78 |
79 |     // Clear history
80 |     clearHistoryBtn.addEventListener('click', () => {

```

```

81 |         history = [];
82 |         localStorage.setItem('cosmic_player_history', JSON.str
| ingify(history));
83 |         renderHistory();
84 |     });
85 |
86 |     // Router and url parser
87 |     function loadVideo(url, title = null) {
88 |         resetPlayers();
89 |
90 |         const providerInfo = parseVideoUrl(url);
91 |         const resolvedTitle = title || providerInfo.defaultTit
| le;
92 |
93 |         highlightBadge(providerInfo.provider);
94 |
95 |         // Update info panel
96 |         videoTitle.textContent = resolvedTitle;
97 |         videoSourceUrl.textContent = url;
98 |         currentProviderTag.textContent = providerInfo.provider
| ;
99 |
100 |         // Update ambient glow color
101 |         ambientGlow.style.background = providerColors[provider
| Info.provider] || providerColors.direct;
102 |
103 |         // Route to the correct player engine
104 |         if (providerInfo.type === 'embed') {
105 |             embedPlayer.src = providerInfo.embedUrl;
106 |             embedPlayer.classList.remove('hidden');
107 |         } else if (providerInfo.type === 'html5') {
108 |             initHTML5Player(url);
109 |             html5PlayerWrapper.classList.remove('hidden');
110 |         } else if (providerInfo.type === 'iframe_fallback') {
111 |             embedPlayer.src = url;
112 |             embedPlayer.classList.remove('hidden');
113 |         }
114 |

```

```

115 |         // Add to history
116 |         addToHistory(url, resolvedTitle, providerInfo.provider
    | );
117 |     }
118 |
119 |     function resetPlayers() {
120 |         embedPlayer.classList.add('hidden');
121 |         embedPlayer.src = '';
122 |         html5PlayerWrapper.classList.add('hidden');
123 |         playerPlaceholder.classList.add('hidden');
124 |
125 |         html5Video.pause();
126 |         html5Video.src = '';
127 |
128 |         if (hlsInstance) {
129 |             hlsInstance.destroy();
130 |             hlsInstance = null;
131 |         }
132 |     }
133 |
134 |     // Video URL Parser
135 |     function parseVideoUrl(url) {
136 |         // YouTube Matcher
137 |         const ytReg = /(?:youtube\.com\/(?:[^\./]+\./|(?:(v|e
    | (?:(mbed)?)\./|.*[?&]v=)|youtu\.be\/)([^\&?\/\s]{11})/;
138 |         const ytMatch = url.match(ytReg);
139 |         if (ytMatch) {
140 |             return {
141 |                 provider: 'youtube',
142 |                 type: 'embed',
143 |                 embedUrl: `https://www.youtube.com/embed/${ytM
    | atch[1]}?autoplay=1&rel=0`,
144 |                 defaultTitle: `YouTube Video (${ytMatch[1]})`
145 |             };
146 |         }
147 |
148 |         // VK Matcher
149 |         const vkReg = /vk\.com\/video(?:-?\d+)_(\d+)/;

```



```

150         const vkMatch = url.match(vkReg);
151         const vkEmbedReg = /oid=(-?\d+)&id=(\d+)(?:&hash=([a-f
0-9]+)))/;
152         const vkEmbedMatch = url.match(vkEmbedReg);
153
154         if (vkMatch) {
155             return {
156                 provider: 'vk',
157                 type: 'embed',
158                 embedUrl: `https://vk.com/video_ext.php?oid=${
vkMatch[1]}&id=${vkMatch[2]}&autoplay=1`,
159                 defaultTitle: `VK Video (${vkMatch[1]}_${vkMat
ch[2]})`
160             };
161         } else if (vkEmbedMatch) {
162             const hashParam = vkEmbedMatch[3] ? `&hash=${vkEmb
edMatch[3]}` : '';
163             return {
164                 provider: 'vk',
165                 type: 'embed',
166                 embedUrl: `https://vk.com/video_ext.php?oid=${
vkEmbedMatch[1]}&id=${vkEmbedMatch[2]}${hashParam}&autoplay=1`
167                 ,
168                 defaultTitle: `VK Embed Video`
169             };
170
171         // RuTube Matcher
172         const rutubeReg = /rutube\.ru\/(?:video|play|embed)\//
([a-f0-9]{32})/;
173         const rutubeMatch = url.match(rutubeReg);
174         if (rutubeMatch) {
175             return {
176                 provider: 'rutube',
177                 type: 'embed',
178                 embedUrl: `https://rutube.ru/play/embed/${rutu
beMatch[1]}?autoplay=1`,
179                 defaultTitle: `RuTube Video`

```

```
180         };
181     }
182
183     // Direct Stream (MP4/HLS)
184     const isHLS = url.includes('.m3u8');
185     const isMP4 = url.includes('.mp4');
186     if (isHLS || isMP4) {
187         return {
188             provider: 'direct',
189             type: 'html5',
190             defaultTitle: isHLS ? 'Прямой HLS поток' : 'Пр
ямой MP4 файл'
191         };
192     }
193
194     // SeasonVar
195     if (url.includes('seasonvar.ru')) {
196         return {
197             provider: 'seasonvar',
198             type: 'iframe_fallback',
199             defaultTitle: 'СезонВар плеер'
200         };
201     }
202
203     // HDRezka
204     if (url.includes('rezka.ag') || url.includes('hdrezka'
205 )) {
206         return {
207             provider: 'hdrezka',
208             type: 'iframe_fallback',
209             defaultTitle: 'Резка плеер'
210         };
211     }
212
213     // Kinogo
214     if (url.includes('kinogo')) {
215         return {
216             provider: 'kinogo',
```

```

216         type: 'iframe_fallback',
217         defaultTitle: 'Киноного плеер'
218     };
219 }
220
221 // Default fallback
222 return {
223     provider: 'direct',
224     type: 'html5',
225     defaultTitle: 'Прямой видеоисточник'
226 };
227 }
228
229 function initHTML5Player(url) {
230     if (url.includes('.m3u8')) {
231         if (typeof Hls !== 'undefined' && Hls.isSupported(
232         )) {
233             hlsInstance = new Hls();
234             hlsInstance.loadSource(url);
235             hlsInstance.attachMedia(html5Video);
236             hlsInstance.on(Hls.Events.MANIFEST_PARSED, ()
237             => {
238                 html5Video.play();
239             });
240         } else if (html5Video.canPlayType('application/vnd
241         .apple.mpegurl')) {
242             html5Video.src = url;
243             html5Video.play();
244         }
245     } else {
246         html5Video.src = url;
247         html5Video.play();
248     }
249
250     playPauseBtn.textContent = '⏮ ';
251
252 // Video events

```

```
251 |     html5Video.addEventListener('timeupdate', () => {
252 |         const current = html5Video.currentTime;
253 |         const total = html5Video.duration || 0;
254 |
255 |         if (total > 0) {
256 |             const pct = (current / total) * 100;
257 |             progressBar.style.width = `${pct}%`;
258 |         }
259 |
260 |         currentTimeDisplay.textContent = formatTime(current);
261 |         durationTimeDisplay.textContent = formatTime(total);
262 |     });
263 |
264 |     html5Video.addEventListener('loadedmetadata', () => {
265 |         durationTimeDisplay.textContent = formatTime(html5Video
o.duration);
266 |     });
267 |
268 |     playPauseBtn.addEventListener('click', () => {
269 |         if (html5Video.paused) {
270 |             html5Video.play();
271 |             playPauseBtn.textContent = '⏏';
272 |         } else {
273 |             html5Video.pause();
274 |             playPauseBtn.textContent = '▶';
275 |         }
276 |     });
277 |
278 |     stopBtn.addEventListener('click', () => {
279 |         html5Video.pause();
280 |         html5Video.currentTime = 0;
281 |         playPauseBtn.textContent = '▶';
282 |     });
283 |
284 |     volumeSlider.addEventListener('input', (e) => {
285 |         const vol = e.target.value;
286 |         html5Video.volume = vol;
287 |         if (vol === '0') {
```

```

288         muteBtn.textContent = '🔇';
289     } else {
290         muteBtn.textContent = '🔊';
291     }
292 });
293
294 muteBtn.addEventListener('click', () => {
295     if (html5Video.muted) {
296         html5Video.muted = false;
297         volumeSlider.value = html5Video.volume;
298         muteBtn.textContent = '🔇';
299     } else {
300         html5Video.muted = true;
301         volumeSlider.value = 0;
302         muteBtn.textContent = '🔊';
303     }
304 });
305
306 speedSelect.addEventListener('change', (e) => {
307     html5Video.playbackRate = parseFloat(e.target.value);
308 });
309
310 progressContainer.addEventListener('click', (e) => {
311     const rect = progressContainer.getBoundingClientRect()
312     ;
313     const clickX = e.clientX - rect.left;
314     const width = rect.width;
315     const pct = clickX / width;
316
317     html5Video.currentTime = pct * html5Video.duration;
318 });
319
320 progressContainer.addEventListener('mousemove', (e) => {
321     const rect = progressContainer.getBoundingClientRect()
322     ;
323     const hoverX = e.clientX - rect.left;
324     const pct = hoverX / rect.width;
325     progressHover.style.width = `${pct * 100}%`;

```

```

324 |     });
325 |
326 |     progressContainer.addEventListener('mouseleave', () => {
327 |         progressHover.style.width = '0%';
328 |     });
329 |
330 |     fullscreenBtn.addEventListener('click', () => {
331 |         if (!document.fullscreenElement) {
332 |             html5PlayerWrapper.requestFullscreen().catch(err =
> {
333 |                 console.error(`Error enabling fullscreen: ${err.message}`);
334 |             });
335 |         } else {
336 |             document.exitFullscreen();
337 |         }
338 |     });
339 |
340 |     function formatTime(sec) {
341 |         if (isNaN(sec)) return '0:00';
342 |         const mins = Math.floor(sec / 60);
343 |         const secs = Math.floor(sec % 60);
344 |         return `${mins}:${secs < 10 ? '0' : ''}${secs}`;
345 |     }
346 |
347 |     function highlightBadge(provider) {
348 |         badges.forEach(badge => {
349 |             if (badge.getAttribute('data-provider') === provid
er) {
350 |                 badge.classList.add('active');
351 |             } else {
352 |                 badge.classList.remove('active');
353 |             }
354 |         });
355 |     }
356 |
357 |     function addToHistory(url, title, provider) {
358 |         history = history.filter(item => item.url !== url);

```

```

359 |
360 |         history.unshift({
361 |             url,
362 |             title,
363 |             provider,
364 |             timestamp: new Date().toLocaleTimeString([], { hou
r: '2-digit', minute: '2-digit' })
365 |         });
366 |
367 |         if (history.length > 8) history.pop();
368 |
369 |         localStorage.setItem('cosmic_player_history', JSON.str
ingify(history));
370 |         renderHistory();
371 |     }
372 |
373 |     function renderHistory() {
374 |         historyList.innerHTML = '';
375 |         if (history.length === 0) {
376 |             historyList.innerHTML = '<div class="empty-state">
Нет недавно проигранных видео</div>';
377 |             return;
378 |         }
379 |
380 |         history.forEach(item => {
381 |             const el = document.createElement('div');
382 |             el.className = 'history-item';
383 |             el.innerHTML = `
384 |                 <div class="history-title" title="${item.title
}">${item.title}</div>
385 |                 <div class="history-meta">
386 |                     <span>${item.provider.toUpperCase()}</span
>
387 |                     <span>${item.timestamp}</span>
388 |                 </div>
389 |             `;
390 |             el.addEventListener('click', () => {
391 |                 streamUrlInput.value = item.url;

```

```
392 |         loadVideo(item.url, item.title);
393 |     });
394 |     historyList.appendChild(el);
395 | });
396 | }
397 |
398 | badges.forEach(badge => {
399 |     badge.addEventListener('click', () => {
400 |         const provider = badge.getAttribute('data-provider
| ');
401 |         let demoUrl = '';
402 |
403 |         switch (provider) {
404 |             case 'youtube':
405 |                 demoUrl = 'https://www.youtube.com/watch?v
| =aqz-KE-bpKQ';
406 |                 break;
407 |             case 'vk':
408 |                 demoUrl = 'https://vk.com/video-22822305_4
| 56239018';
409 |                 break;
410 |             case 'rutube':
411 |                 demoUrl = 'https://rutube.ru/video/e7cfcb8
| cb4310d54026fb4bd56e828d1/';
412 |                 break;
413 |             case 'direct':
414 |                 demoUrl = 'https://test-streams.mux.dev/x3
| 6xhzz/x36xhzz.m3u8';
415 |                 break;
416 |             case 'seasonvar':
417 |                 demoUrl = 'http://seasonvar.ru';
418 |                 break;
419 |             case 'hdrezka':
420 |                 demoUrl = 'https://rezka.ag';
421 |                 break;
422 |             case 'kinogo':
423 |                 demoUrl = 'https://kinogo.biz';
424 |                 break;
```



```
425 |         }
426 |
427 |         if (demoUrl) {
428 |             streamUrlInput.value = demoUrl;
429 |             loadVideo(demoUrl);
430 |         }
431 |     });
432 | });
433 | };
434 |
```

styles.css

```
1 /**
2  * =====
3  * Process Name: Windows Web Styles - Styles Stylesheet
4  * =====
5  * Description:
6  *   Таблица стилей стиля и оформления интерфейса (styles).
7  *
8  * Usage Examples:
9  *   HTML Link Tag:
10 *     <link rel="stylesheet" href="/windows/api/webgui/cosmic
    player/styles.css?v=20261001_v1">
11 *
12 * File: styles.css
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/cosmicplayer
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-01 13:04:40
18 * =====
19 */
20
21 .cosmic-player-tab {
22     --primary: #7928ca;
23     --primary-glow: #ff007a;
24     --primary-gradient: linear-gradient(135deg, #7928ca 0%, #f
    f007a 100%);
25     --panel-border: rgba(255, 255, 255, 0.08);
26     --bg-dark: #0a0714;
27     --text-muted: #a0aec0;
28     --transition: all 0.25s cubic-bezier(0.25, 0.8, 0.25, 1);
29
30     display: flex;
31     width: 100%;
```

```
32 |     height: calc(100vh - 80px);
33 |     background: rgba(20, 16, 36, 0.6);
34 |     backdrop-filter: blur(25px);
35 |     -webkit-backdrop-filter: blur(25px);
36 |     border: 1px solid var(--panel-border);
37 |     border-radius: 12px;
38 |     overflow: hidden;
39 |     margin-top: 6px;
40 | }
41 |
42 | .cosmic-sidebar {
43 |     width: 260px;
44 |     background: rgba(14, 10, 26, 0.75);
45 |     border-right: 1px solid var(--panel-border);
46 |     padding: 16px 14px;
47 |     display: flex;
48 |     flex-direction: column;
49 |     gap: 14px;
50 |     overflow-y: auto;
51 | }
52 |
53 | .cosmic-sidebar .logo {
54 |     display: flex;
55 |     align-items: center;
56 |     gap: 8px;
57 | }
58 |
59 | .cosmic-sidebar .logo h2 {
60 |     font-size: 1.1rem;
61 |     font-weight: 700;
62 |     margin: 0;
63 |     background: var(--primary-gradient);
64 |     -webkit-background-clip: text;
65 |     -webkit-text-fill-color: transparent;
66 | }
67 |
68 | .cosmic-sidebar .logo-icon {
69 |     font-size: 1.3rem;
```

```
70     filter: drop-shadow(0 0 6px rgba(255, 0, 122, 0.5));
71 }
72
73 .cosmic-player-tab .url-input-section h3,
74 .cosmic-player-tab .sources-panel h4,
75 .cosmic-player-tab .history-section h4 {
76     font-size: 0.72rem;
77     font-weight: 600;
78     text-transform: uppercase;
79     letter-spacing: 0.6px;
80     color: var(--text-muted);
81     margin-bottom: 8px;
82 }
83
84 #stream-form {
85     display: flex;
86     flex-direction: column;
87     gap: 10px;
88 }
89
90 #stream-url {
91     background: rgba(0, 0, 0, 0.4);
92     border: 1px solid var(--panel-border);
93     border-radius: 8px;
94     padding: 10px 12px;
95     color: #fff;
96     font-size: 0.85rem;
97     outline: none;
98     transition: var(--transition);
99 }
100
101 #stream-url:focus {
102     border-color: var(--primary-glow);
103     box-shadow: 0 0 10px rgba(255, 0, 122, 0.2);
104 }
105
106 .btn-glow {
107     background: var(--primary-gradient);
```

```
108 |     color: white;
109 |     border: none;
110 |     border-radius: 8px;
111 |     padding: 10px 15px;
112 |     font-size: 0.85rem;
113 |     font-weight: 600;
114 |     cursor: pointer;
115 |     transition: var(--transition);
116 | }
117 |
118 | .btn-glow:hover {
119 |     transform: translateY(-1px);
120 |     box-shadow: 0 4px 15px rgba(255, 0, 122, 0.4);
121 | }
122 |
123 | .sources-grid {
124 |     display: flex;
125 |     flex-wrap: wrap;
126 |     gap: 6px;
127 | }
128 |
129 | .cosmic-player-tab .badge {
130 |     background: rgba(255, 255, 255, 0.08);
131 |     border: 1px solid var(--panel-border);
132 |     border-radius: 6px;
133 |     padding: 5px 10px;
134 |     font-size: 0.7rem;
135 |     font-weight: 500;
136 |     color: #e6edf3;
137 |     transition: var(--transition);
138 |     cursor: pointer;
139 | }
140 |
141 | .cosmic-player-tab .badge:hover,
142 | .cosmic-player-tab .badge.active {
143 |     color: white;
144 |     background: rgba(121, 40, 202, 0.2);
145 |     border-color: var(--primary);
```

```
146 | }
147 |
148 | .history-section {
149 |     flex-grow: 1;
150 |     display: flex;
151 |     flex-direction: column;
152 | }
153 |
154 | .section-header {
155 |     display: flex;
156 |     justify-content: space-between;
157 |     align-items: center;
158 |     margin-bottom: 10px;
159 | }
160 |
161 | .btn-text {
162 |     background: none;
163 |     border: none;
164 |     color: var(--text-muted);
165 |     font-size: 0.75rem;
166 |     cursor: pointer;
167 | }
168 |
169 | .history-list {
170 |     display: flex;
171 |     flex-direction: column;
172 |     gap: 6px;
173 |     overflow-y: auto;
174 |     max-height: 200px;
175 | }
176 |
177 | .history-item {
178 |     background: rgba(255, 255, 255, 0.01);
179 |     border: 1px solid var(--panel-border);
180 |     border-radius: 8px;
181 |     padding: 8px 12px;
182 |     cursor: pointer;
183 |     display: flex;
```

```
184 |     flex-direction: column;
185 |     gap: 2px;
186 |     transition: var(--transition);
187 | }
188 |
189 | .history-item:hover {
190 |     background: rgba(255, 255, 255, 0.04);
191 | }
192 |
193 | .history-title {
194 |     font-size: 0.8rem;
195 |     font-weight: 500;
196 |     white-space: nowrap;
197 |     overflow: hidden;
198 |     text-overflow: ellipsis;
199 | }
200 |
201 | .history-meta {
202 |     display: flex;
203 |     justify-content: space-between;
204 |     font-size: 0.65rem;
205 |     color: var(--text-muted);
206 | }
207 |
208 | /* Main Panel Layout */
209 | .main-content {
210 |     flex-grow: 1;
211 |     display: flex;
212 |     flex-direction: column;
213 |     padding: 14px;
214 |     gap: 12px;
215 |     background: rgba(10, 7, 20, 0.2);
216 | }
217 |
218 | .player-container {
219 |     position: relative;
220 |     width: 100%;
221 |     height: 60%;
```

```
222 |     min-height: 380px;
223 |     background: #000;
224 |     border-radius: 15px;
225 |     overflow: hidden;
226 |     border: 1px solid var(--panel-border);
227 | }
228 |
229 | .ambient-glow {
230 |     position: absolute;
231 |     top: -20%;
232 |     left: -20%;
233 |     width: 140%;
234 |     height: 140%;
235 |     background: radial-gradient(circle, rgba(121, 40, 202, 0.2
236 | ) 0%, transparent 60%);
237 |     filter: blur(50px);
238 |     z-index: 0;
239 |     pointer-events: none;
240 |     transition: background 1s ease;
241 | }
242 |
243 | .player-element {
244 |     position: relative;
245 |     width: 100%;
246 |     height: 100%;
247 |     border: none;
248 |     z-index: 2;
249 | }
250 |
251 | .hidden {
252 |     display: none !important;
253 | }
254 |
255 | .html5-video-el {
256 |     width: 100%;
257 |     height: 100%;
258 |     object-fit: contain;
```



```
259 |
260 | #html5-player-wrapper {
261 |     position: relative;
262 | }
263 |
264 | .custom-controls {
265 |     position: absolute;
266 |     bottom: 0;
267 |     left: 0;
268 |     right: 0;
269 |     background: linear-gradient(to top, rgba(0, 0, 0, 0.9) 0%,
    | rgba(0, 0, 0, 0.4) 70%, transparent 100%);
270 |     padding: 15px 20px;
271 |     display: flex;
272 |     flex-direction: column;
273 |     gap: 10px;
274 |     opacity: 0;
275 |     transition: opacity 0.3s ease;
276 |     z-index: 10;
277 | }
278 |
279 | #html5-player-wrapper:hover .custom-controls {
280 |     opacity: 1;
281 | }
282 |
283 | .progress-container {
284 |     height: 4px;
285 |     background: rgba(255, 255, 255, 0.2);
286 |     border-radius: 2px;
287 |     cursor: pointer;
288 |     position: relative;
289 | }
290 |
291 | .progress-bar {
292 |     height: 100%;
293 |     background: var(--primary-gradient);
294 |     width: 0%;
295 |     border-radius: 2px;
```

```
296 | }
297 |
298 | .progress-hover {
299 |     position: absolute;
300 |     top: 0;
301 |     left: 0;
302 |     height: 100%;
303 |     background: rgba(255, 255, 255, 0.15);
304 |     width: 0%;
305 |     pointer-events: none;
306 | }
307 |
308 | .controls-row {
309 |     display: flex;
310 |     justify-content: space-between;
311 |     align-items: center;
312 | }
313 |
314 | .controls-left, .controls-right {
315 |     display: flex;
316 |     align-items: center;
317 |     gap: 12px;
318 | }
319 |
320 | .ctrl-btn {
321 |     background: none;
322 |     border: none;
323 |     color: white;
324 |     font-size: 1rem;
325 |     cursor: pointer;
326 | }
327 |
328 | .time-display {
329 |     font-size: 0.75rem;
330 |     color: var(--text-muted);
331 | }
332 |
333 | .volume-container {
```

```
334 |     display: flex;
335 |     align-items: center;
336 |     gap: 6px;
337 | }
338 |
339 | #volume-slider {
340 |     width: 60px;
341 |     height: 3px;
342 |     background: rgba(255, 255, 255, 0.2);
343 |     outline: none;
344 |     accent-color: var(--primary-glow);
345 |     cursor: pointer;
346 | }
347 |
348 | .speed-dropdown {
349 |     background: rgba(255, 255, 255, 0.1);
350 |     border: 1px solid rgba(255, 255, 255, 0.2);
351 |     color: white;
352 |     border-radius: 4px;
353 |     padding: 2px 6px;
354 |     font-size: 0.75rem;
355 |     outline: none;
356 |     cursor: pointer;
357 | }
358 |
359 | .player-placeholder {
360 |     position: absolute;
361 |     top: 0;
362 |     left: 0;
363 |     width: 100%;
364 |     height: 100%;
365 |     display: flex;
366 |     justify-content: center;
367 |     align-items: center;
368 |     z-index: 1;
369 |     background: rgba(10, 7, 20, 0.95);
370 | }
371 |
```

```
372 | .placeholder-content {
373 |     text-align: center;
374 |     max-width: 320px;
375 | }
376 |
377 | .placeholder-content .play-icon {
378 |     font-size: 3rem;
379 |     display: block;
380 |     margin-bottom: 12px;
381 | }
382 |
383 | .placeholder-content h3 {
384 |     font-size: 1.15rem;
385 |     font-weight: 600;
386 |     margin-bottom: 8px;
387 | }
388 |
389 | .placeholder-content p {
390 |     color: var(--text-muted);
391 |     font-size: 0.8rem;
392 |     line-height: 1.4;
393 | }
394 |
395 | .glass-card {
396 |     background: rgba(255, 255, 255, 0.02);
397 |     border: 1px solid var(--panel-border);
398 |     border-radius: 12px;
399 |     padding: 18px;
400 | }
401 |
402 | .video-info-card {
403 |     flex-grow: 1;
404 |     display: flex;
405 |     flex-direction: column;
406 |     justify-content: center;
407 |     gap: 8px;
408 | }
409 |
```

```
410 | .info-header {
411 |     display: flex;
412 |     align-items: center;
413 |     gap: 12px;
414 | }
415 |
416 | .provider-tag {
417 |     background: var(--primary-gradient);
418 |     color: white;
419 |     padding: 3px 10px;
420 |     border-radius: 15px;
421 |     font-size: 0.65rem;
422 |     font-weight: 600;
423 |     text-transform: uppercase;
424 | }
425 |
426 | #video-title {
427 |     font-size: 1.1rem;
428 |     font-weight: 600;
429 |     margin: 0;
430 | }
431 |
432 | #video-source-url {
433 |     font-size: 0.8rem;
434 |     color: var(--text-muted);
435 |     word-break: break-all;
436 |     margin: 0;
437 | }
438 |
439 | .empty-state {
440 |     color: var(--text-muted);
441 |     font-size: 0.75rem;
442 |     text-align: center;
443 |     padding: 15px;
444 |     border: 1px dashed var(--panel-border);
445 |     border-radius: 8px;
446 | }
447 |
```

base.css

```
1 /**
2  * =====
3  * Process Name: Windows Web Styles - Base Stylesheet
4  * =====
5  * Description:
6  *   Таблица стилей стиля и оформления интерфейса (base).
7  *
8  * Usage Examples:
9  *   HTML Link Tag:
10 *     <link rel="stylesheet" href="/windows/api/webgui/css/base.css?v=20261001_v1">
11 *
12 * File: base.css
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/css
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-04 06:36:00
18 * =====
19 */
20
21 /* =====
22    BASE STYLES & SCROLLBARS – AI Breadboard WebGUI
23    Базовые стили страницы, скроллбары и контейнеры вкладок
24    ===== */
25
26 /* Базовый шрифт – увеличен на 2pt для улучшения читаемости */
27 html {
28   font-size: calc(14px + 2pt);
29   height: 100%;
30 }
```

```

31 |
32 | body {
33 |     display: flex;
34 |     flex-direction: column;
35 |     height: 100%;
36 |     min-height: 100vh;
37 |     background: var(--bg-color);
38 |     color: var(--text-color);
39 |     font-size: 0.9375rem; /* ≈ 13.125px */
40 |     line-height: 1.5;
41 |     transition: background-color 0.25s ease, color 0.25s ease;
42 | }
43 |
44 | /* =====
45 |     СКРОЛЛБАРЫ
46 |     ===== */
47 | ::-webkit-scrollbar { width: 6px; height: 6px; }
48 | ::-webkit-scrollbar-track { background: var(--scrollbar-track); border-radius: 3px; }
49 | ::-webkit-scrollbar-thumb { background: var(--scrollbar-thumb); border-radius: 3px; }
50 | ::-webkit-scrollbar-thumb:hover { opacity: 0.8; }
51 |
52 | /* =====
53 |     ВКЛАДКИ (TAB PANES)
54 |     ===== */
55 | .tab-content {
56 |     flex-grow: 1;
57 |     display: flex;
58 |     flex-direction: column;
59 | }
60 |
61 | .tab-pane {
62 |     min-height: auto;

```

```
63 | padding-bottom: 16px;
64 | }
65 |
66 | .tab-pane.active {
67 |     display: block;
68 | }
69 |
70 | /* =====
   | =====
71 |     АДМИН-ИНТЕРФЕЙС
72 |     =====
   | ===== */
73 | /* Стили для #admin-interface внутри .container-fluid (как в i
   | ndex.html) */
74 | #admin-interface,
75 | .container-fluid#admin-interface {
76 |     display: flex;
77 |     flex-direction: column;
78 |     flex-grow: 1;
79 |     height: 100%;
80 |     overflow-y: auto;
81 | }
82 |
83 | #admin-interface .tab-content,
84 | .container-fluid#admin-interface .tab-content {
85 |     flex-grow: 1;
86 |     overflow-y: auto;
87 | }
88 |
89 | #admin-interface .tab-pane,
90 | .container-fluid#admin-interface .tab-pane {
91 |     flex-grow: 1;
92 |     overflow-y: auto;
93 |     overflow-x: hidden;
94 | }
95 |
96 | #admin-interface .tab-pane.active,
97 | .container-fluid#admin-interface .tab-pane.active {
```



```
98 | display: block;
99 | }
100 |
101 | /* Скроллбары для админки */
102 | #admin-interface ::-webkit-scrollbar {
103 |     width: 8px;
104 |     height: 8px;
105 | }
106 |
107 | #admin-interface ::-webkit-scrollbar-track {
108 |     background: var(--scrollbar-track);
109 |     border-radius: 4px;
110 | }
111 |
112 | #admin-interface ::-webkit-scrollbar-thumb {
113 |     background: var(--scrollbar-thumb);
114 |     border-radius: 4px;
115 | }
116 |
117 | #admin-interface ::-webkit-scrollbar-thumb:hover {
118 |     opacity: 0.85;
119 | }
120 |
```

chat.css

```
1  /**
2   * =====
3   * Process Name: Windows Web Styles - Chat Stylesheet
4   * =====
5   * Description:
6   *   Таблица стилей стиля и оформления интерфейса (chat).
7   *
8   * Usage Examples:
9   *   HTML Link Tag:
10  *     <link rel="stylesheet" href="/windows/api/webgui/css/ch
11  *     at.css?v=20261001_v1">
12  *
13  * File: chat.css
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/css
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-06 10:38:00
19  * =====
20  */
21  /** =====
22   CHAT – AI Breadboard WebGUI
23   Стили диалогового окна, сайдбара сессий, сообщений и дебагг
24   ера
25   ===== */
26  /** — Основной макет вкладки чата —
27   */
28  #tab-chat {
29     height: calc(100vh - 55px);
```

```
29 | max-height: calc(100vh - 55px);
30 | min-height: 80vh;
31 | padding-bottom: 0;
32 | overflow: hidden;
33 | }
34 |
35 | #tab-chat.active {
36 |   display: flex !important;
37 |   flex-direction: column;
38 |   height: calc(100vh - 55px);
39 |   max-height: calc(100vh - 55px);
40 |   min-height: 80vh;
41 | }
42 |
43 | .chat-container {
44 |   display: flex;
45 |   flex-direction: row;
46 |   height: 100%;
47 |   min-height: 0;
48 |   gap: 8px;
49 |   position: relative;
50 | }
51 |
52 | /* — Сайдбар чата (история диалогов) —————
   | ————— */
53 | .chat-sidebar {
54 |   width: 220px;
55 |   min-width: 220px;
56 |   max-width: 220px;
57 |   display: flex;
58 |   flex-direction: column;
59 |   background: var(--surface-1);
60 |   border: 1px solid var(--border-color);
61 |   border-radius: 6px;
62 |   padding: 6px;
63 |   transition: margin-left 0.22s ease, opacity 0.18s ease;
64 |   height: 100%;
65 |   flex-shrink: 0;
```

```
66 | }
67 |
68 | .chat-sidebar.collapsed {
69 |   margin-left: -232px;
70 |   opacity: 0;
71 |   pointer-events: none;
72 |   display: none !important;
73 | }
74 |
75 | .chat-sessions-list {
76 |   flex: 1 1 auto;
77 |   overflow-y: auto;
78 |   display: flex;
79 |   flex-direction: column;
80 |   gap: 2px;
81 |   margin-top: 4px;
82 |   margin-bottom: 4px;
83 |   padding-right: 2px;
84 | }
85 |
86 | .chat-session-item {
87 |   display: flex;
88 |   align-items: center;
89 |   justify-content: space-between;
90 |   padding: 5px 7px;
91 |   border-radius: 5px;
92 |   cursor: pointer;
93 |   color: var(--text-color);
94 |   font-size: 0.8rem;
95 |   border: 1px solid transparent;
96 |   transition: background 0.12s ease, border-color 0.12s ease;
97 |   user-select: none;
98 | }
99 |
100 | .chat-session-item:hover {
101 |   background: rgba(128, 128, 128, 0.1);
102 | }
103 |
```

```
104 | .chat-session-item.active {
105 |     background: rgba(37, 99, 235, 0.1);
106 |     border-color: rgba(37, 99, 235, 0.25);
107 |     font-weight: 500;
108 | }
109 |
110 | [data-theme="dark"] .chat-session-item.active {
111 |     background: rgba(88, 166, 255, 0.1);
112 |     border-color: rgba(88, 166, 255, 0.2);
113 | }
114 |
115 | .chat-session-title {
116 |     white-space: nowrap;
117 |     overflow: hidden;
118 |     text-overflow: ellipsis;
119 |     flex: 1 1 auto;
120 |     margin-right: 3px;
121 | }
122 |
123 | .chat-session-actions {
124 |     display: none;
125 |     align-items: center;
126 |     gap: 2px;
127 |     flex-shrink: 0;
128 | }
129 |
130 | .chat-session-item:hover .chat-session-actions,
131 | .chat-session-item.active .chat-session-actions {
132 |     display: flex;
133 | }
134 |
135 | .chat-session-action-btn {
136 |     background: transparent;
137 |     border: none;
138 |     padding: 0 2px;
139 |     color: var(--text-muted);
140 |     border-radius: 3px;
141 |     cursor: pointer;
```

```
142 | line-height: 1;
143 | font-size: 0.75rem;
144 | }
145 |
146 | .chat-session-action-btn:hover {
147 |   color: var(--text-color);
148 |   background: rgba(128, 128, 128, 0.18);
149 | }
150 |
151 | /* — Главная область чата —————
    | ————— */
152 | .chat-main-area {
153 |   flex: 1 1 auto;
154 |   min-width: 0;
155 |   display: flex;
156 |   flex-direction: column;
157 |   height: 100%;
158 | }
159 |
160 | /* Хедер-бар чата */
161 | .chat-header-bar {
162 |   display: flex;
163 |   align-items: center;
164 |   justify-content: space-between;
165 |   padding: 3px 8px;
166 |   margin-bottom: 5px;
167 |   border-radius: 5px;
168 |   background: var(--surface-1);
169 |   border: 1px solid var(--border-color);
170 |   font-size: 0.82rem;
171 |   flex-shrink: 0;
172 | }
173 |
174 | /* Окно чата */
175 | .chat-window {
176 |   height: calc(100vh - 260px);
177 |   min-height: 160px;
178 |   overflow-y: auto;
```

```
179 | border: 1px solid var(--border-color);
180 | border-radius: 5px;
181 | padding: 8px 10px;
182 | background: var(--surface-1);
183 | }
184 |
185 | #tab-chat .chat-window {
186 |     height: auto;
187 |     flex: 1 1 auto;
188 |     min-height: calc(80vh - 120px);
189 |     max-height: calc(100vh - 140px);
190 |     overflow-y: auto;
191 |     margin-bottom: 5px !important;
192 | }
193 |
194 | #tab-chat .media-controls { flex-shrink: 0; margin-bottom: 5px !important; }
195 | #tab-chat #chat-debugger-toolbar { flex-shrink: 0; margin-bottom: 5px !important; }
196 | #tab-chat .input-group { flex-shrink: 0; margin-bottom: 2px !important; }
197 |
198 | /* — Сообщения (Базовые блоки) —————
199 | |————— */
200 | .message {
201 |     margin-bottom: 8px;
202 |     padding: 8px 12px;
203 |     border-radius: 6px;
204 |     font-size: 0.92rem;
205 |     line-height: 1.5;
206 | }
207 |
208 | /* — Поле ввода чата —————
209 | |————— */
210 | #message-input,
211 | .chat-main-area #message-input,
212 | #chat-popup-input,
213 | #scenario-chat-input {
```

```
212 | font-size: 0.95rem;
213 | }
214 |
215 | .user-message {
216 |   background: var(--chat-user-bg);
217 |   color: var(--chat-user-text);
218 |   text-align: right;
219 |   border-radius: 12px 12px 2px 12px;
220 |   box-shadow: var(--shadow-xs);
221 | }
222 |
223 | .bot-message {
224 |   background: var(--chat-bot-bg);
225 |   color: var(--chat-bot-text);
226 |   border: 1px solid var(--border-color);
227 |   border-radius: 12px 12px 12px 2px;
228 |   box-shadow: var(--shadow-xs);
229 | }
230 |
231 | /* — Универсальные пузыри сообщений (Modern Chat Bubbles) —
    | _____ */
232 | .chat-message-row {
233 |   display: flex;
234 |   margin-bottom: 0.75rem;
235 |   width: 100%;
236 | }
237 |
238 | .chat-message-row.user-row {
239 |   justify-content: flex-end;
240 | }
241 |
242 | .chat-message-row.bot-row {
243 |   justify-content: flex-start;
244 |   align-items: flex-start;
245 |   gap: 0.5rem;
246 | }
247 |
248 | .chat-msg-avatar {
```



```
249 | font-size: 1.25rem;
250 | line-height: 1;
251 | flex-shrink: 0;
252 | margin-top: 2px;
253 | user-select: none;
254 | }
255 |
256 | /* Пузырь сообщения пользователя: выравнивание справа, мягкий
    | фон темы */
257 | .chat-msg-user {
258 |   background: var(--chat-user-bg);
259 |   color: var(--chat-user-text);
260 |   border-radius: 14px 14px 2px 14px;
261 |   padding: 0.55rem 0.95rem;
262 |   max-width: 80%;
263 |   font-size: 0.9rem;
264 |   line-height: 1.45;
265 |   box-shadow: var(--shadow-xs);
266 |   word-break: break-word;
267 |   white-space: pre-wrap;
268 |   border: 1px solid transparent;
269 | }
270 |
271 | /* Пузырь сообщения бота / ассистента: выравнивание слева */
272 | .chat-msg-bot {
273 |   background: var(--chat-bot-bg);
274 |   color: var(--chat-bot-text);
275 |   border: 1px solid var(--border-color);
276 |   border-radius: 14px 14px 14px 2px;
277 |   padding: 0.75rem 1rem;
278 |   flex-grow: 1;
279 |   max-width: 95%;
280 |   font-size: 0.88rem;
281 |   line-height: 1.5;
282 |   box-shadow: var(--shadow-xs);
283 |   word-break: break-word;
284 | }
285 |
```

```
286 | .chat-msg-bot a {
287 |   color: var(--nav-active);
288 | }
289 |
290 | /* Диагностический конвейер этапов (Stages Pipeline) */
291 | .chat-stage-pipeline {
292 |   background: var(--surface-2);
293 |   border: 1px solid var(--border-color);
294 |   border-radius: var(--radius-sm);
295 |   padding: 0.5rem 0.75rem;
296 |   margin-bottom: 0.65rem;
297 |   color: var(--text-color);
298 | }
299 |
300 | .chat-stage-pipeline .stage-item {
301 |   color: var(--text-muted);
302 |   font-size: 0.76rem;
303 |   line-height: 1.4;
304 | }
305 |
306 | .chat-stage-pipeline .stage-item.active {
307 |   color: var(--text-color);
308 |   font-weight: 500;
309 | }
310 |
311 | /* Карточка и контейнер истории чата сценариев */
312 | .scenario-chat-card {
313 |   background-color: var(--surface-1) !important;
314 |   border: 1.5px solid var(--border-color) !important;
315 |   color: var(--text-color) !important;
316 | }
317 |
318 | .scenario-chat-history {
319 |   background-color: var(--bg-color);
320 | }
321 |
322 | /* Чипы сообщений в чате */
323 | .chat-window .badge,
```

```
324 | .message .badge,
325 | .bot-message .badge,
326 | .user-message .badge,
327 | .chat-msg-bot .badge,
328 | .chat-msg-user .badge {
329 |     font-size: 0.7rem;
330 |     font-weight: 500;
331 | }
332 |
333 | .chat-window .badge.bg-secondary,
334 | .chat-window .badge.bg-primary,
335 | .chat-window .badge.bg-danger,
336 | .chat-window .badge.bg-success,
337 | .chat-window .badge.bg-dark,
338 | .message .badge.bg-secondary,
339 | .message .badge.bg-primary,
340 | .message .badge.bg-danger,
341 | .message .badge.bg-success,
342 | .message .badge.bg-dark {
343 |     color: #ffffff !important;
344 | }
345 |
346 | /* — Дебаггер-тулбар чата —————
    | ————— */
347 | #chat-debugger-toolbar {
348 |     background-color: var(--surface-2);
349 |     border: 1px solid var(--border-color);
350 |     border-radius: 5px;
351 |     font-size: 0.8rem;
352 | }
353 |
354 | [data-theme="light"] #chat-debugger-toolbar {
355 |     background-color: var(--surface-2) !important;
356 |     border-color: var(--border-color) !important;
357 | }
358 |
359 | [data-theme="light"] #chat-debugger-toolbar select {
360 |     background-color: var(--surface-1) !important;
```

```
361 | color: var(--text-color) !important;  
362 | border-color: var(--border-color) !important;  
363 | }  
364 |
```

components.css

```

1  /**
2   * =====
3   *
4   * Process Name: Windows Web Styles - Components Stylesheet
5   * =====
6   * Description:
7   *   Таблица стилей стиля и оформления интерфейса (components)
8   *
9   * Usage Examples:
10  *   HTML Link Tag:
11  *       <link rel="stylesheet" href="/windows/api/webgui/css/components.css?v=20261001_v1">
12  *
13  * File: components.css
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/css
16  * Author: hypo69
17  * Updated: 2026-10-08 02:56:00
18  * =====
19  */
20  /** =====
21  COMPONENTS – AI Breadboard WebGUI
22  Базовые UI-компоненты: карточки, формы, кнопки, модалки, та
23  блицы, алерты, бейджи
24  ===== */
25  /** =====
26  КАРТОЧКИ
27  =====

```

```

===== */
28 | .card {
29 |   background-color: var(--card-bg);
30 |   border: 1.5px solid var(--border-color);
31 |   color: var(--card-text);
32 |   transition: border-color 0.2s, background-color 0.2s, box-sh
   | adow 0.2s;
33 |   border-radius: 8px;
34 |   box-shadow: var(--shadow-sm);
35 | }
36 |
37 | .card-header {
38 |   background-color: var(--surface-2);
39 |   border-bottom: 1.5px solid var(--border-color);
40 |   color: var(--text-color);
41 |   padding: 0.375rem 0.75rem;
42 |   font-size: 0.85rem;
43 |   font-weight: 600;
44 | }
45 |
46 | .card-body {
47 |   color: var(--card-text);
48 |   padding: 0.75rem;
49 | }
50 |
51 | .card-footer {
52 |   background-color: var(--surface-2);
53 |   border-top: 1px solid var(--border-color);
54 |   padding: 0.375rem 0.75rem;
55 | }
56 |
57 | .card.bg-light {
58 |   background-color: var(--surface-2) !important;
59 |   color: var(--text-color);
60 | }
61 |
62 | .card.bg-dark > .card-header,
63 | .modal-content.bg-dark > .modal-header,

```

```
64 | .modal-content.bg-dark > .modal-footer,
65 | .agent-card .card-header {
66 |     background-color: var(--surface-2) !important;
67 |     border-color: var(--border-color) !important;
68 |     color: var(--text-color) !important;
69 | }
70 |
71 | [data-theme="light"] .card.bg-dark,
72 | [data-theme="light"] .modal-content.bg-dark,
73 | [data-theme="light"] .alert-dark {
74 |     background-color: var(--surface-1) !important;
75 |     color: var(--text-color) !important;
76 |     border-color: var(--border-color) !important;
77 | }
78 |
79 | [data-theme="light"] .card.bg-dark > .card-header,
80 | [data-theme="light"] .modal-content.bg-dark > .modal-header,
81 | [data-theme="light"] .modal-content.bg-dark > .modal-footer {
82 |     background-color: var(--surface-2) !important;
83 |     color: var(--text-color) !important;
84 |     border-color: var(--border-color) !important;
85 | }
86 |
87 | [data-theme="light"] .bg-black {
88 |     background-color: var(--bg-color) !important;
89 |     color: var(--text-color) !important;
90 | }
91 |
92 | [data-theme="light"] .text-light {
93 |     color: var(--text-color) !important;
94 | }
95 |
96 | [data-theme="light"] .btn-close-white {
97 |     filter: none !important;
98 | }
99 |
100 | /* =====
```

101 | ФОРМЫ И ИНПУТЫ

102 |

===== */

103 | .form-control,

104 | .form-select {

105 | background-color: var(--surface-1);

106 | border: 1px solid var(--border-color);

107 | color: var(--text-color);

108 | font-size: 0.85rem;

109 | padding: 0.3rem 0.5rem;

110 | transition: border-color 0.15s, box-shadow 0.15s;

111 | border-radius: 5px;

112 | }

113 |

114 | .form-control:focus,

115 | .form-select:focus {

116 | background-color: var(--surface-1);

117 | border-color: var(--nav-active);

118 | color: var(--text-color);

119 | box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.18);

120 | outline: none;

121 | }

122 |

123 | [data-theme="dark"] .form-control:focus,

124 | [data-theme="dark"] .form-select:focus {

125 | box-shadow: 0 0 0 2px rgba(88, 166, 255, 0.3);

126 | }

127 |

128 | [data-theme="brick"] .form-control:focus,

129 | [data-theme="brick"] .form-select:focus {

130 | box-shadow: 0 0 0 2px rgba(184, 66, 40, 0.25);

131 | }

132 |

133 | [data-theme="terminal"] .form-control:focus,

134 | [data-theme="terminal"] .form-select:focus {

135 | box-shadow: 0 0 0 2px rgba(0, 255, 102, 0.35);

136 | }

137 |


```
138 | .form-control::placeholder {
139 |   color: var(--text-muted);
140 |   opacity: 0.8;
141 | }
142 |
143 | .form-control-sm,
144 | .form-select-sm {
145 |   font-size: 0.8rem;
146 |   padding: 0.22rem 0.4rem;
147 | }
148 |
149 | .form-label {
150 |   font-size: 0.82rem;
151 |   margin-bottom: 0.3rem;
152 |   color: var(--text-color);
153 | }
154 |
155 | .input-group-text {
156 |   background-color: var(--surface-2);
157 |   border: 1px solid var(--border-color);
158 |   color: var(--text-color);
159 |   font-size: 0.85rem;
160 |   padding: 0.3rem 0.55rem;
161 | }
162 |
163 | /* =====
164 |    КНОПКИ
165 |    ===== */
166 | .btn {
167 |   font-size: 0.82rem;
168 |   font-weight: 500;
169 |   transition: background-color 0.15s, border-color 0.15s, box-
170 |     shadow 0.15s;
171 |   border-radius: 5px;
172 | }
```

```
173 | .btn-sm {
174 |     padding: 0.2rem 0.45rem;
175 |     font-size: 0.78rem;
176 | }
177 |
178 | .btn-lg {
179 |     font-size: 0.9rem;
180 |     padding: 0.45rem 0.9rem;
181 | }
182 |
183 | .btn-primary {
184 |     background-color: var(--nav-active);
185 |     border-color: var(--nav-active);
186 |     color: #ffffff;
187 | }
188 |
189 | .btn-primary:hover {
190 |     background-color: #1d4ed8;
191 |     border-color: #1d4ed8;
192 |     color: #ffffff;
193 | }
194 |
195 | [data-theme="brick"] .btn-primary {
196 |     background-color: #b84228;
197 |     border-color: #b84228;
198 |     color: #ffffff;
199 | }
200 |
201 | [data-theme="brick"] .btn-primary:hover {
202 |     background-color: #9e341d;
203 |     border-color: #9e341d;
204 |     color: #ffffff;
205 | }
206 |
207 | [data-theme="dark"] .btn-primary {
208 |     background-color: #1f6feb;
209 |     border-color: #1f6feb;
210 |     color: #ffffff;
```

```
211 }
212
213 [data-theme="dark"] .btn-primary:hover {
214     background-color: #388bfd;
215     border-color: #388bfd;
216     color: #ffffff;
217 }
218
219 [data-theme="terminal"] .btn-primary {
220     background-color: #0c3316;
221     border: 1px solid #00ff66;
222     color: #00ff66;
223     text-shadow: 0 0 4px rgba(0, 255, 102, 0.6);
224 }
225
226 [data-theme="terminal"] .btn-primary:hover {
227     background-color: #00ff66;
228     border-color: #00ff66;
229     color: #060a06;
230     text-shadow: none;
231 }
232
233 .btn-outline-secondary {
234     background-color: transparent;
235     border-color: var(--border-color);
236     color: var(--nav-text);
237 }
238
239 .btn-outline-secondary:hover {
240     background-color: var(--nav-hover-bg);
241     border-color: var(--border-color);
242     color: var(--text-color);
243 }
244
245 /* Переключатели групп кнопок (btn-check radio/checkbox) */
246 .btn-check:checked + .btn-outline-primary,
247 .btn-check:active + .btn-outline-primary {
248     background-color: #2563eb !important;
```

```
249 | border-color: #2563eb !important;
250 | color: #ffffff !important;
251 | }
252 |
253 | .btn-check:checked + .btn-outline-secondary,
254 | .btn-check:active + .btn-outline-secondary {
255 |   background-color: #64748b !important;
256 |   border-color: #64748b !important;
257 |   color: #ffffff !important;
258 | }
259 |
260 | .btn-check:checked + .btn-outline-info,
261 | .btn-check:active + .btn-outline-info {
262 |   background-color: #0284c7 !important;
263 |   border-color: #0284c7 !important;
264 |   color: #ffffff !important;
265 | }
266 |
267 | .btn-check:checked + .btn-outline-danger,
268 | .btn-check:active + .btn-outline-danger {
269 |   background-color: #dc2626 !important;
270 |   border-color: #dc2626 !important;
271 |   color: #ffffff !important;
272 | }
273 |
274 | .btn-check:checked + .btn-outline-warning,
275 | .btn-check:active + .btn-outline-warning {
276 |   background-color: #d97706 !important;
277 |   border-color: #d97706 !important;
278 |   color: #ffffff !important;
279 | }
280 |
281 | .btn-check:checked + .btn-outline-success,
282 | .btn-check:active + .btn-outline-success {
283 |   background-color: #16a34a !important;
284 |   border-color: #16a34a !important;
285 |   color: #ffffff !important;
286 | }
```

```
287 |
288 | /* =====
    | =====
289 |     МОДАЛЬНЫЕ ОКНА
290 |     =====
    | ===== */
291 | .modal-content {
292 |     background-color: var(--modal-bg);
293 |     border: 1px solid var(--modal-border);
294 |     color: var(--modal-text);
295 |     border-radius: 8px;
296 |     box-shadow: var(--shadow-lg);
297 | }
298 |
299 | .modal-header {
300 |     border-bottom: 1px solid var(--border-color);
301 |     padding: 0.6rem 0.9rem;
302 |     background-color: var(--surface-2);
303 |     border-radius: 8px 8px 0 0;
304 | }
305 |
306 | .modal-title {
307 |     font-size: 0.9rem;
308 |     font-weight: 600;
309 | }
310 |
311 | .modal-body {
312 |     padding: 0.75rem 0.9rem;
313 |     font-size: 0.85rem;
314 | }
315 |
316 | .modal-footer {
317 |     border-top: 1px solid var(--border-color);
318 |     padding: 0.5rem 0.9rem;
319 |     background-color: var(--surface-2);
320 |     border-radius: 0 0 8px 8px;
321 | }
322 |
```

```
323 /* Модальное окно помощи */
324 .modal-help .modal-dialog { max-width: 600px; }
325 .modal-help .modal-content {
326     border-radius: 8px;
327     background: var(--modal-bg);
328     color: var(--modal-text);
329     border: 1px solid var(--modal-border);
330 }
331 .modal-help .modal-header {
332     border-bottom: 1px solid var(--border-color);
333     background: var(--surface-2);
334 }
335 .modal-help .modal-body { font-size: 0.85rem; line-height: 1.6
    ; }
336 .modal-help .modal-footer {
337     border-top: 1px solid var(--border-color);
338     background: var(--surface-2);
339 }
340
341 /* =====
    =====
342     ТАБЛИЦЫ
343     =====
    ===== */
344 .table {
345     --bs-table-color: var(--text-color);
346     --bs-table-bg: transparent;
347     --bs-table-border-color: var(--border-color);
348     font-size: 0.82rem;
349 }
350
351 .table th,
352 .table td {
353     padding: 0.4rem 0.6rem;
354     vertical-align: middle;
355 }
356
357 .table-hover tbody tr:hover {
```

```
358 | background-color: var(--nav-hover-bg);
359 | }
360 |
361 | .table-dark {
362 |   --bs-table-bg: var(--surface-2);
363 |   --bs-table-color: var(--text-color);
364 | }
365 |
366 | [data-theme="dark"] .table-light,
367 | [data-theme="dark"] thead.table-light,
368 | [data-theme="dark"] thead.table-light th {
369 |   background-color: var(--surface-2) !important;
370 |   color: var(--text-color) !important;
371 |   border-color: var(--border-color) !important;
372 | }
373 |
374 | [data-theme="dark"] .table {
375 |   color: var(--text-color) !important;
376 |   border-color: var(--border-color) !important;
377 | }
378 |
379 | .table-responsive-adaptive {
380 |   height: auto;
381 |   min-height: 120px;
382 |   max-height: clamp(320px, 58vh, 720px);
383 |   overflow-y: auto;
384 |   overflow-x: auto;
385 |   scrollbar-width: thin;
386 | }
387 |
388 | .table-responsive-adaptive thead th,
389 | .table-sticky-header thead th {
390 |   position: sticky;
391 |   top: 0;
392 |   z-index: 2;
393 |   background-color: var(--surface-2);
394 | }
395 |
```

```
396  /* =====
397  АЛЕРТЫ – адаптивные для обеих тем
398  ===== */
399  .alert {
400    font-size: 0.83rem;
401    padding: 0.5rem 0.75rem;
402    border-radius: 6px;
403    border-width: 1px;
404    border-style: solid;
405  }
406
407  .alert-success {
408    background-color: var(--alert-success-bg);
409    border-color: var(--alert-success-border);
410    color: var(--alert-success-text);
411  }
412
413  .alert-danger {
414    background-color: var(--alert-danger-bg);
415    border-color: var(--alert-danger-border);
416    color: var(--alert-danger-text);
417  }
418
419  .alert-warning {
420    background-color: var(--alert-warning-bg);
421    border-color: var(--alert-warning-border);
422    color: var(--alert-warning-text);
423  }
424
425  .alert-info {
426    background-color: var(--alert-info-bg);
427    border-color: var(--alert-info-border);
428    color: var(--alert-info-text);
429  }
430
431  .alert:not([class*="alert-"]) {
```



```
432 | background-color: var(--surface-1);
433 | border-color: var(--border-color);
434 | color: var(--text-color);
435 | }
436 |
437 | /* =====
438 | БЕЙДЖИ
439 | =====
440 | */
440 | .badge {
441 |   font-size: 0.72rem;
442 |   font-weight: 500;
443 |   padding: 0.22em 0.5em;
444 |   letter-spacing: 0.02em;
445 | }
446 |
447 | .badge:not([class*="bg-"]):not([class*="badge-"]) {
448 |   background-color: var(--border-color);
449 |   color: var(--text-color);
450 | }
451 |
452 | .badge.bg-light,
453 | .badge.badge-light {
454 |   background-color: var(--border-subtle) !important;
455 |   color: var(--text-color) !important;
456 | }
457 |
458 | .badge.bg-secondary,
459 | .badge.bg-primary,
460 | .badge.bg-danger,
461 | .badge.bg-success,
462 | .badge.bg-dark,
463 | .badge-secondary,
464 | .badge-primary,
465 | .badge-danger,
466 | .badge-success,
467 | .badge-dark,
```

```
468 | .film-inline-badge {
469 |   color: #ffffff !important;
470 | }
471 |
472 | .badge.bg-warning,
473 | .badge.badge-warning {
474 |   color: #1a1a1a !important;
475 |   font-weight: 600;
476 | }
477 |
478 | .badge.bg-info,
479 | .badge.badge-info {
480 |   color: #042743 !important;
481 |   font-weight: 600;
482 | }
483 |
484 | /* Бейджи модели/поиска в навбаре */
485 | #chat-model-badge,
486 | #chat-popup-model-badge,
487 | #chat-search-badge,
488 | #chat-popup-search-badge {
489 |   background-color: var(--surface-2) !important;
490 |   border: 1px solid var(--border-color) !important;
491 |   color: var(--text-color) !important;
492 |   font-weight: 600;
493 |   font-size: 0.75rem;
494 | }
495 |
496 | [data-theme="dark"] #chat-model-badge,
497 | [data-theme="dark"] #chat-popup-model-badge {
498 |   color: var(--nav-active) !important;
499 |   border-color: var(--border-color) !important;
500 | }
501 |
502 | [data-theme="dark"] #chat-search-badge,
503 | [data-theme="dark"] #chat-popup-search-badge {
504 |   color: #38bdf8 !important;
505 |   border-color: var(--border-color) !important;
```

```
506 }
507
508 /* Разделители в выпадающих меню */
509 .dropdown-divider {
510     border-color: var(--border-color);
511     opacity: 0.6;
512     margin: 3px 0;
513 }
514
515 /* =====
516     СИСТЕМА ПОМОЩИ
517     ===== */
518 .help-icon {
519     display: inline-flex;
520     align-items: center;
521     justify-content: center;
522     width: 16px;
523     height: 16px;
524     border-radius: 50%;
525     background: var(--nav-active);
526     color: white;
527     font-size: 11px;
528     font-weight: bold;
529     cursor: help;
530     margin-left: 3px;
531     vertical-align: middle;
532     opacity: 0.8;
533     transition: opacity 0.15s;
534 }
535
536 .help-icon:hover { opacity: 1; }
537
538 .help-popover {
539     max-width: 320px;
540     font-size: 0.8rem;
541     background: var(--modal-bg);
```

```

542 | color: var(--modal-text);
543 | border: 1px solid var(--border-color);
544 | }
545 |
546 | /* =====
547 |
548 |
549 |
550 |
551 |
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576 |

```

```

    ИНТЕРАКТИВНЫЕ ЧИПЫ (ФИЛЬМЫ)

```

```

    */
.film-inline-badge {
    background-color: #2563eb !important;
    color: #ffffff !important;
    font-weight: 500;
    border: 1px solid rgba(255, 255, 255, 0.2) !important;
    box-shadow: 0 1px 2px rgba(0, 0, 0, 0.2);
}

.film-inline-badge:hover {
    background-color: #1d4ed8 !important;
    color: #ffffff !important;
}

.film-interactive-chip {
    display: inline-flex;
    align-items: center;
    gap: 3px;
    background: linear-gradient(135deg, #1e3a8a 0%, #2563eb 100%
);
    color: #ffffff !important;
    font-size: 0.85em;
    font-weight: 500;
    padding: 2px 4px 2px 7px;
    border-radius: 14px;
    border: 1px solid rgba(255, 255, 255, 0.2);
    box-shadow: 0 1px 3px rgba(0, 0, 0, 0.18);
    margin: 2px 2px;
    vertical-align: middle;
    transition: all 0.18s ease;

```

```
577 | }
578 |
579 | .film-interactive-chip:hover {
580 |     background: linear-gradient(135deg, #1d4ed8 0%, #3b82f6 100%
581 | );
582 |     box-shadow: 0 2px 6px rgba(37, 99, 235, 0.35);
583 |     transform: translateY(-1px);
584 | }
585 |
586 | .film-chip-title {
587 |     cursor: pointer;
588 |     color: #ffffff !important;
589 |     text-decoration: none;
590 |     font-weight: 600;
591 |     font-size: 0.82em;
592 | }
593 |
594 | .film-chip-title:hover { text-decoration: underline; }
595 |
596 | .film-chip-btn {
597 |     background: rgba(255, 255, 255, 0.15);
598 |     border: none;
599 |     color: #ffffff;
600 |     width: 20px;
601 |     height: 20px;
602 |     border-radius: 50%;
603 |     display: inline-flex;
604 |     align-items: center;
605 |     justify-content: center;
606 |     font-size: 0.7rem;
607 |     padding: 0;
608 |     cursor: pointer;
609 |     transition: all 0.12s ease;
610 | }
611 |
612 | .film-chip-btn:hover { background: rgba(255, 255, 255, 0.35
613 | ); transform: scale(1.1); }
614 |
615 | .film-chip-play:hover { background: #10b981; }
```

```
613 | .film-chip-youtube:hover{ background: #ef4444; }
614 | .film-chip-online:hover { background: #06b6d4; }
615 | .film-chip-torrent:hover{ background: #f59e0b; color: #111827;
    | }
616 |
617 | [data-theme="dark"] .film-interactive-chip {
618 |     background: linear-gradient(135deg, #1e293b 0%, #334155 100%
    | );
619 |     border: 1px solid rgba(148, 163, 184, 0.25);
620 | }
621 |
622 | [data-theme="dark"] .film-interactive-chip:hover {
623 |     background: linear-gradient(135deg, #334155 0%, #475569 100%
    | );
624 |     box-shadow: 0 2px 6px rgba(0, 0, 0, 0.45);
625 | }
626 |
627 | /* =====
    | =====
628 |     PROMPT DUMP (КОНСОЛЬ АДМИНИСТРАТОРА)
629 |     =====
    | ===== */
630 | .prompt-dump-details {
631 |     margin-bottom: 8px;
632 |     font-size: 0.8em;
633 |     background: #0d1117 !important;
634 |     padding: 8px;
635 |     border-radius: 5px;
636 |     border: 1px solid #30363d !important;
637 |     box-shadow: inset 0 1px 3px rgba(0, 0, 0, 0.7);
638 | }
639 |
640 | .prompt-dump-details summary {
641 |     cursor: pointer;
642 |     color: #e3b341 !important;
643 |     font-weight: bold;
644 |     user-select: none;
645 |     font-size: 0.85em;
```

```
646 | }
647 |
648 | .prompt-dump-details pre {
649 |   white-space: pre-wrap;
650 |   margin-top: 8px;
651 |   color: #00ff66 !important;
652 |   font-family: 'Consolas', 'Courier New', Courier, monospace;
653 |   font-size: 0.9em;
654 |   line-height: 1.4;
655 |   max-height: 280px;
656 |   overflow-y: auto;
657 | }
658 |
659 | /* =====
660 |
661 | MENU EDITOR (DRAG AND DROP)
662 |
663 | ===== */
664 | .menu-editor-item {
665 |   display: flex;
666 |   align-items: center;
667 |   gap: 10px;
668 |   padding: 10px 12px;
669 |   background: var(--surface-1);
670 |   border: 1px solid var(--border-color);
671 |   border-radius: 6px;
672 |   margin-bottom: 8px;
673 |   transition: all 0.15s ease;
674 | }
675 |
676 | .menu-editor-item:hover {
677 |   border-color: var(--nav-active);
678 |   background: var(--surface-2);
679 | }
680 |
681 | .menu-editor-item:active {
682 |   cursor: grabbing;
683 | }
```

```
682 |
683 | .menu-editor-item.dragging {
684 |     opacity: 0.5;
685 |     background: var(--nav-active);
686 |     border-color: var(--nav-active);
687 | }
688 |
689 | .menu-editor-item .drag-handle-container {
690 |     cursor: grab;
691 |     color: var(--text-muted);
692 |     font-size: 1.2rem;
693 |     display: flex;
694 |     align-items: center;
695 | }
696 |
697 | .menu-editor-item .drag-handle-container:active {
698 |     cursor: grabbing;
699 | }
700 |
701 | .menu-editor-item .item-icon {
702 |     font-size: 1.3rem;
703 |     min-width: 28px;
704 |     text-align: center;
705 | }
706 |
707 | .menu-editor-item .item-info {
708 |     flex: 1;
709 |     min-width: 0;
710 | }
711 |
712 | .menu-editor-item .item-label {
713 |     font-size: 0.88rem;
714 |     font-weight: 500;
715 |     white-space: nowrap;
716 |     overflow: hidden;
717 |     text-overflow: ellipsis;
718 | }
719 |
```



```
720 | .menu-editor-item .item-tab {
721 |     font-size: 0.72rem;
722 |     color: var(--text-muted);
723 |     font-family: monospace;
724 | }
725 |
726 | .menu-editor-item .editor-controls {
727 |     display: flex;
728 |     flex-direction: column;
729 |     gap: 6px;
730 |     min-width: 180px;
731 | }
732 |
733 | .menu-editor-item .editor-visibility,
734 | .menu-editor-item .editor-order {
735 |     display: flex;
736 |     align-items: center;
737 |     gap: 6px;
738 | }
739 |
740 | .menu-editor-item .editor-visibility .visibility-label,
741 | .menu-editor-item .editor-order .order-label {
742 |     font-size: 0.75rem;
743 |     color: var(--text-muted);
744 |     white-space: nowrap;
745 | }
746 |
747 | .menu-editor-item .editor-visibility .visibility-select {
748 |     flex: 1;
749 |     font-size: 0.78rem;
750 |     padding: 2px 6px;
751 | }
752 |
753 | .menu-editor-item .editor-order .order-slider {
754 |     flex: 1;
755 |     height: 6px;
756 |     padding: 0;
757 | }
```

```
758 |
759 | .menu-editor-item .editor-order .order-value {
760 |     font-size: 0.75rem;
761 |     color: var(--text-muted);
762 |     min-width: 24px;
763 |     text-align: right;
764 | }
765 |
766 | .menu-editor-section {
767 |     margin-bottom: 20px;
768 | }
769 |
770 | .menu-editor-section h6 {
771 |     font-size: 0.8rem;
772 |     text-transform: uppercase;
773 |     letter-spacing: 0.5px;
774 |     margin-bottom: 12px;
775 |     color: var(--text-muted);
776 | }
777 |
778 | .menu-editor-section h6 i {
779 |     margin-right: 6px;
780 | }
781 |
782 | #menuEditorModal .modal-body {
783 |     max-height: 70vh;
784 |     overflow-y: auto;
785 | }
786 |
787 | /* =====
788 |     FLOATING TOAST NOTIFICATIONS (ВСПЛЫВАЮЩИЕ ОКНА)
789 |     =====
790 | */
791 | .floating-toast-container {
792 |     position: fixed;
793 |     top: 18px;
794 |     right: 18px;
```

```
794 | z-index: 99999;
795 | display: flex;
796 | flex-direction: column;
797 | gap: 10px;
798 | max-width: 420px;
799 | width: calc(100vw - 36px);
800 | pointer-events: none;
801 | }
802 |
803 | .floating-toast {
804 |   pointer-events: auto;
805 |   background-color: var(--surface-1, #1e293b);
806 |   color: var(--text-color, #f8fafc);
807 |   border: 1px solid var(--border-color, #334155);
808 |   border-left-width: 4px;
809 |   border-radius: 8px;
810 |   box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.4), 0 8px 10px
      | -6px rgba(0, 0, 0, 0.3);
811 |   padding: 12px 14px;
812 |   opacity: 0;
813 |   transform: translateX(40px) scale(0.96);
814 |   transition: opacity 0.25s cubic-bezier(0.16, 1, 0.3, 1), tra
      | nsform 0.25s cubic-bezier(0.16, 1, 0.3, 1);
815 |   overflow: hidden;
816 |   position: relative;
817 | }
818 |
819 | .floating-toast.show {
820 |   opacity: 1;
821 |   transform: translateX(0) scale(1);
822 | }
823 |
824 | .floating-toast.hide {
825 |   opacity: 0;
826 |   transform: translateX(40px) scale(0.92);
827 | }
828 |
829 | .floating-toast-success {
```

```
830 | border-left-color: #22c55e !important;
831 | }
832 |
833 | .floating-toast-danger {
834 |   border-left-color: #ef4444 !important;
835 | }
836 |
837 | .floating-toast-warning {
838 |   border-left-color: #f59e0b !important;
839 | }
840 |
841 | .floating-toast-info {
842 |   border-left-color: #3b82f6 !important;
843 | }
844 |
845 | .floating-toast-header {
846 |   display: flex;
847 |   align-items: center;
848 |   gap: 8px;
849 |   margin-bottom: 6px;
850 | }
851 |
852 | .floating-toast-icon {
853 |   font-size: 1.1rem;
854 |   line-height: 1;
855 | }
856 |
857 | .floating-toast-title {
858 |   font-size: 0.88rem;
859 |   font-weight: 600;
860 |   flex: 1;
861 |   color: var(--text-color, #f8fafc);
862 | }
863 |
864 | .floating-toast-close {
865 |   background: transparent;
866 |   border: none;
867 |   color: var(--text-muted, #94a3b8);
```

```
868 | font-size: 1.25rem;
869 | line-height: 1;
870 | padding: 0 4px;
871 | cursor: pointer;
872 | transition: color 0.15s ease;
873 | }
874 |
875 | .floating-toast-close:hover {
876 |   color: var(--text-color, #ffffff);
877 | }
878 |
879 | .floating-toast-body {
880 |   font-size: 0.84rem;
881 |   line-height: 1.45;
882 |   color: var(--card-text, #cbd5e1);
883 |   word-break: break-word;
884 | }
885 |
886 | .floating-toast-progress {
887 |   position: absolute;
888 |   bottom: 0;
889 |   left: 0;
890 |   right: 0;
891 |   height: 3px;
892 |   background: rgba(255, 255, 255, 0.08);
893 | }
894 |
895 | .floating-toast-progress-bar {
896 |   height: 100%;
897 |   width: 100%;
898 | }
899 |
900 | /* =====
901 |    ГЛОБАЛЬНЫЕ УТИЛИТАРНЫЕ КЛАССЫ (UNIFIED UTILITY CLASSES)
902 |    =====
903 |    */
```

```
904 | /* — 0. Универсальная карточка (для простых случаев) —————  
    | _____ */  
905 | .card-unified {  
906 |   background: var(--surface-1);  
907 |   border: 1px solid var(--border-color);  
908 |   border-radius: var(--radius-md);  
909 |   box-shadow: var(--shadow-sm);  
910 |   transition: border-color 0.15s ease, box-shadow 0.15s ease;  
911 | }  
912 |  
913 | .card-unified:hover {  
914 |   border-color: var(--nav-active);  
915 |   box-shadow: var(--shadow-md);  
916 | }  
917 |  
918 | /* — 1. KPI & Метрические карточки —————  
    | _____ */  
919 |  
920 | /* — 1. KPI & Метрические карточки —————  
    | _____ */  
921 | .ui-kpi-card {  
922 |   background: var(--kpi-bg, var(--surface-1)) !important;  
923 |   border: 1.5px solid var(--kpi-border, var(--border-color)) !  
    | important;  
924 |   border-radius: var(--radius-md, 8px);  
925 |   padding: var(--kpi-padding, 0.65rem 0.9rem);  
926 |   min-height: var(--kpi-min-height, 72px);  
927 |   display: flex;  
928 |   flex-direction: column;  
929 |   justify-content: space-between;  
930 |   height: 100%;  
931 |   box-sizing: border-box;  
932 |   box-shadow: 0 2px 6px rgba(0, 0, 0, 0.25);  
933 |   transition: border-color 0.15s ease, box-shadow 0.15s ease,  
    | transform 0.15s ease;  
934 | }  
935 |  
936 | .ui-kpi-card:hover {
```

```
937 | border-color: var(--kpi-accent, var(--nav-active)) !important;
    | t;
938 | box-shadow: var(--shadow-sm, 0 3px 10px rgba(0, 0, 0, 0.25))
    | ;
939 | }
940 |
941 | .ui-kpi-label {
942 |   font-size: var(--kpi-label-size, 0.72rem);
943 |   text-transform: uppercase;
944 |   letter-spacing: 0.05em;
945 |   font-weight: 600;
946 |   color: var(--text-muted, #8b949e);
947 |   display: flex;
948 |   justify-content: space-between;
949 |   align-items: center;
950 |   line-height: 1.2;
951 | }
952 |
953 | .ui-kpi-value {
954 |   font-size: var(--kpi-val-size, 1.25rem);
955 |   font-weight: 700;
956 |   font-family: var(--font-mono, monospace);
957 |   color: var(--text-color, #f8fafc);
958 |   line-height: 1.2;
959 | }
960 |
961 | .ui-kpi-sub {
962 |   font-size: var(--kpi-sub-size, 0.72rem);
963 |   color: var(--text-muted, #8b949e);
964 |   line-height: 1.2;
965 | }
966 |
967 | /* Модификаторы состояний KPI карточек через каскад */
968 | .ui-kpi-card.is-success {
969 |   --kpi-border: var(--status-success-border, #22c55e);
970 |   --kpi-accent: var(--status-success-border, #22c55e);
971 | }
972 | .ui-kpi-card.is-danger {
```

```
973 | --kpi-border: var(--status-danger-border, #ef4444);
974 | --kpi-accent: var(--status-danger-border, #ef4444);
975 | }
976 | .ui-kpi-card.is-warning {
977 |   --kpi-border: var(--status-warning-border, #eab308);
978 |   --kpi-accent: var(--status-warning-border, #eab308);
979 | }
980 | .ui-kpi-card.is-info {
981 |   --kpi-border: var(--status-info-border, #0284c7);
982 |   --kpi-accent: var(--status-info-border, #0284c7);
983 | }
984 |
985 | /* — 2. Модульные заголовки и Hero-баннеры —————
986 | |————— */
987 | .ui-module-header {
988 |   background: var(--surface-2) !important;
989 |   border: 1px solid var(--border-color) !important;
990 |   border-radius: var(--radius-md, 8px);
991 |   padding: 0.6rem 1rem;
992 |   margin-bottom: 0.75rem;
993 |   display: flex;
994 |   align-items: center;
995 |   justify-content: space-between;
996 |   flex-wrap: wrap;
997 |   gap: 0.5rem;
998 | }
999 | .ui-module-title {
1000 |   font-size: 0.95rem;
1001 |   font-weight: 700;
1002 |   color: var(--text-color);
1003 |   display: flex;
1004 |   align-items: center;
1005 |   gap: 0.5rem;
1006 |   margin-bottom: 0;
1007 | }
1008 |
1009 | .ui-module-desc {
```



```
1010 | font-size: 0.78rem;
1011 | color: var(--text-muted);
1012 | margin-bottom: 0;
1013 | }
1014 |
1015 | /* — 3. Панели и контейнеры таблиц —————
1016 | |————— */
1017 | .ui-panel {
1018 |   background: var(--panel-bg, var(--surface-1)) !important;
1019 |   border: 1px solid var(--panel-border, var(--border-color)) !
1020 |   important;
1021 |   border-radius: var(--radius-md, 8px);
1022 |   overflow: hidden;
1023 |   margin-bottom: 0.75rem;
1024 |   box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
1025 | }
1026 |
1027 | .ui-panel-header {
1028 |   background: var(--panel-header-bg, var(--surface-2)) !import
1029 |   ant;
1030 |   border-bottom: 1px solid var(--border-color) !important;
1031 |   padding: 0.5rem 0.9rem;
1032 |   font-size: 0.85rem;
1033 |   font-weight: 600;
1034 |   color: var(--text-color);
1035 |   display: flex;
1036 |   align-items: center;
1037 |   justify-content: space-between;
1038 |   flex-wrap: wrap;
1039 |   gap: 0.5rem;
1040 | }
1041 |
1042 | .ui-panel-body {
1043 |   padding: 0.75rem;
1044 | }
1045 |
1046 | .ui-panel-footer {
1047 |   background: var(--surface-2) !important;
```

```
1045 | border-top: 1px solid var(--border-color) !important;
1046 | padding: 0.4rem 0.75rem;
1047 | font-size: 0.78rem;
1048 | }
1049 |
1050 | /* — 4. Унифицированные суб-навигационные переключатели (Sub-
      | Nav Tabs) — */
1051 | .ui-subnav {
1052 |   display: flex;
1053 |   align-items: center;
1054 |   gap: 0.4rem;
1055 |   border-bottom: 1px solid var(--border-color);
1056 |   margin-bottom: 0.75rem;
1057 |   padding-bottom: 0.35rem;
1058 |   overflow-x: auto;
1059 |   flex-wrap: nowrap;
1060 | }
1061 |
1062 | .ui-subnav-btn {
1063 |   background: transparent;
1064 |   border: 1px solid transparent;
1065 |   color: var(--text-muted);
1066 |   font-size: var(--subnav-font-size, 0.82rem);
1067 |   font-weight: 500;
1068 |   padding: var(--subnav-btn-padding, 0.35rem 0.75rem);
1069 |   border-radius: var(--radius-sm, 6px);
1070 |   cursor: pointer;
1071 |   white-space: nowrap;
1072 |   display: inline-flex;
1073 |   align-items: center;
1074 |   gap: 0.4rem;
1075 |   transition: all 0.15s ease;
1076 | }
1077 |
1078 | .ui-subnav-btn:hover {
1079 |   background: var(--surface-2);
1080 |   color: var(--text-color);
1081 | }
```

```
1082 |
1083 | .ui-subnav-btn.active {
1084 |     background: var(--surface-2);
1085 |     border-color: var(--nav-active);
1086 |     color: var(--nav-active);
1087 |     font-weight: 600;
1088 | }
1089 |
1090 | /* — 5. Унифицированные статусные бейджи —————
1091 | |————— */
1092 | .ui-badge {
1093 |     font-size: 0.72rem;
1094 |     font-weight: 600;
1095 |     padding: 0.2rem 0.55rem;
1096 |     border-radius: var(--radius-xs, 4px);
1097 |     display: inline-flex;
1098 |     align-items: center;
1099 |     gap: 0.3rem;
1100 |     line-height: 1.2;
1101 | }
1102 |
1103 | .ui-badge-success {
1104 |     background: var(--status-success-bg);
1105 |     color: var(--status-success-text);
1106 |     border: 1px solid var(--status-success-border);
1107 | }
1108 |
1109 | .ui-badge-danger {
1110 |     background: var(--status-danger-bg);
1111 |     color: var(--status-danger-text);
1112 |     border: 1px solid var(--status-danger-border);
1113 | }
1114 |
1115 | .ui-badge-warning {
1116 |     background: var(--status-warning-bg);
1117 |     color: var(--status-warning-text);
1118 |     border: 1px solid var(--status-warning-border);
1119 | }
```

```

1119 |
1120 | .ui-badge-info {
1121 |     background: var(--status-info-bg);
1122 |     color: var(--status-info-text);
1123 |     border: 1px solid var(--status-info-border);
1124 | }
1125 |
1126 | .ui-badge-neutral {
1127 |     background: var(--surface-2);
1128 |     color: var(--text-muted);
1129 |     border: 1px solid var(--border-color);
1130 | }
1131 |
1132 | /* — 6. Унифицированная таблица спецификаций (Ключ - Значение
      | ) ————— */
1133 | .ui-spec-table,
1134 | .about-sys-spec-table,
1135 | .sysctl-spec-table,
1136 | .def-spec-table,
1137 | .winadmin-spec-table {
1138 |     width: 100%;
1139 |     margin-bottom: 0;
1140 |     font-size: 0.82rem;
1141 |     border-collapse: separate;
1142 |     border-spacing: 0;
1143 | }
1144 |
1145 | .ui-spec-table th,
1146 | .about-sys-spec-table th,
1147 | .sysctl-spec-table th,
1148 | .def-spec-table th,
1149 | .winadmin-spec-table th {
1150 |     padding: 8px 16px;
1151 |     font-size: 0.78rem;
1152 |     font-weight: 600;
1153 |     color: var(--text-muted, #94a3b8);
1154 |     text-transform: uppercase;
1155 |     letter-spacing: 0.04em;

```

```
1156 | border-bottom: 1px solid var(--border-subtle, rgba(51, 65, 8
      | 5, 0.35));
1157 | background: transparent;
1158 | vertical-align: middle;
1159 | }
1160 |
1161 | .ui-spec-table td,
1162 | .about-sys-spec-table td,
1163 | .sysctl-spec-table td,
1164 | .def-spec-table td,
1165 | .winadmin-spec-table td {
1166 |     padding: 9px 14px;
1167 |     border-bottom: 1px solid var(--border-subtle, rgba(51, 65, 8
      | 5, 0.35));
1168 |     vertical-align: middle;
1169 |     line-height: 1.45;
1170 |     color: var(--text-color, #f8fafc);
1171 | }
1172 |
1173 | .ui-spec-table tr:hover td,
1174 | .about-sys-spec-table tr:hover td,
1175 | .sysctl-spec-table tr:hover td,
1176 | .def-spec-table tr:hover td,
1177 | .winadmin-spec-table tr:hover td {
1178 |     background: rgba(56, 189, 248, 0.04);
1179 | }
1180 |
1181 | .ui-spec-table tr:last-child td,
1182 | .about-sys-spec-table tr:last-child td,
1183 | .sysctl-spec-table tr:last-child td,
1184 | .def-spec-table tr:last-child td,
1185 | .winadmin-spec-table tr:last-child td {
1186 |     border-bottom: none;
1187 | }
1188 |
1189 | .ui-spec-key,
1190 | .about-sys-spec-key,
1191 | .sysctl-spec-key,
```

```

1192 | .def-spec-key,
1193 | .winadmin-spec-key {
1194 |     color: var(--text-muted, #94a3b8);
1195 |     width: 230px;
1196 |     min-width: 210px;
1197 |     max-width: 260px;
1198 |     white-space: nowrap;
1199 |     font-weight: 500;
1200 |     font-family: var(--font-sans, system-ui, -apple-system, sans
        | -serif);
1201 |     vertical-align: middle;
1202 | }
1203 |
1204 | .ui-spec-key i,
1205 | .about-sys-spec-key i,
1206 | .sysctl-spec-key i,
1207 | .def-spec-key i,
1208 | .winadmin-spec-key i {
1209 |     margin-right: 8px;
1210 |     font-size: 0.92rem;
1211 |     vertical-align: -1px;
1212 | }
1213 |
1214 | .ui-spec-val,
1215 | .about-sys-spec-val,
1216 | .sysctl-spec-val,
1217 | .def-spec-val,
1218 | .winadmin-spec-val {
1219 |     color: var(--text-color, #f8fafc);
1220 |     font-weight: 500;
1221 |     font-family: var(--font-mono, monospace);
1222 |     vertical-align: middle;
1223 |     word-break: break-word;
1224 | }
1225 |
1226 | /* — 7. Контекстные каскадные темы (Тематические переопределе
        | ния) ————— */
1227 | .theme-defender {

```

```
1228 | --kpi-min-height: 64px;
1229 | --kpi-padding: 0.5rem 0.75rem;
1230 | --kpi-accent: #22c55e;
1231 | }
1232 |
1233 | .theme-sysadmin {
1234 |   --kpi-min-height: 68px;
1235 |   --kpi-padding: 0.55rem 0.85rem;
1236 | }
1237 |
1238 | .theme-inspector {
1239 |   --kpi-min-height: 80px;
1240 |   --kpi-val-size: 1.35rem;
1241 | }
1242 |
1243 | .theme-sysctl {
1244 |   --kpi-min-height: 72px;
1245 | }
1246 |
1247 | /* — 8. Алиасы совместимости (Backward-compatible Aliases) —
    | _____ */
1248 | .about-sys-card,
1249 | .def-card,
1250 | .sysctl-card,
1251 | .scc-card,
1252 | .winadmin-card,
1253 | .sys-card {
1254 |   composes: ui-kpi-card;
1255 |   background: var(--kpi-bg, var(--surface-1)) !important;
1256 |   border: 1.5px solid var(--kpi-border, var(--border-color)) !
    | important;
1257 |   border-radius: var(--radius-md, 8px);
1258 |   padding: var(--kpi-padding, 0.65rem 0.9rem);
1259 |   min-height: var(--kpi-min-height, 72px);
1260 |   display: flex;
1261 |   flex-direction: column;
1262 |   justify-content: space-between;
1263 |   height: 100%;
```

```
1264 | box-sizing: border-box;
1265 | box-shadow: 0 2px 6px rgba(0, 0, 0, 0.3);
1266 | transition: border-color 0.15s ease, box-shadow 0.15s ease;
1267 | }
1268 |
1269 | .about-sys-card:hover,
1270 | .def-card:hover,
1271 | .sysctl-card:hover,
1272 | .scc-card:hover,
1273 | .winadmin-card:hover,
1274 | .sys-card:hover {
1275 |   border-color: var(--kpi-accent, var(--nav-active)) !important;
1276 |   box-shadow: 0 4px 12px rgba(0, 0, 0, 0.4);
1277 | }
1278 |
1279 | .about-sys-label,
1280 | .def-card-label,
1281 | .sysctl-label,
1282 | .scc-label,
1283 | .winadmin-label,
1284 | .sys-label {
1285 |   font-size: var(--kpi-label-size, 0.72rem);
1286 |   text-transform: uppercase;
1287 |   letter-spacing: 0.05em;
1288 |   font-weight: 600;
1289 |   color: var(--text-muted, #8b949e);
1290 |   display: flex;
1291 |   justify-content: space-between;
1292 |   align-items: center;
1293 |   line-height: 1.2;
1294 | }
1295 |
1296 | .about-sys-value,
1297 | .def-card-value,
1298 | .sysctl-value,
1299 | .scc-value,
1300 | .winadmin-value,
```



```
1301 | .sys-value {
1302 |   font-size: var(--kpi-val-size, 1.25rem);
1303 |   font-weight: 700;
1304 |   font-family: var(--font-mono, monospace);
1305 |   color: var(--text-color, #f8fafc);
1306 |   line-height: 1.2;
1307 | }
1308 |
1309 | .about-sys-sub,
1310 | .def-card-sub,
1311 | .sysctl-sub,
1312 | .scc-sub,
1313 | .winadmin-sub,
1314 | .sys-sub {
1315 |   font-size: var(--kpi-sub-size, 0.72rem);
1316 |   color: var(--text-muted, #8b949e);
1317 |   line-height: 1.2;
1318 | }
1319 |
1320 | .about-sys-panel,
1321 | .def-panel,
1322 | .sysctl-panel,
1323 | .winadmin-panel,
1324 | .sys-panel {
1325 |   background: var(--panel-bg, var(--surface-1)) !important;
1326 |   border: 1px solid var(--panel-border, var(--border-color)) !
1327 |   important;
1328 |   border-radius: var(--radius-md, 8px);
1329 |   overflow: hidden;
1330 |   margin-bottom: 0.75rem;
1331 |   box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
1332 | }
1333 |
1334 | .about-sys-panel-header,
1335 | .def-panel-header,
1336 | .sysctl-panel-header,
1337 | .winadmin-panel-header,
1338 | .sys-panel-header {
```

```
1338     background: var(--panel-header-bg, var(--surface-2)) !important;
1339     border-bottom: 1px solid var(--border-color) !important;
1340     padding: 0.5rem 0.9rem;
1341     font-size: 0.85rem;
1342     font-weight: 600;
1343     color: var(--text-color);
1344     display: flex;
1345     align-items: center;
1346     justify-content: space-between;
1347     flex-wrap: wrap;
1348     gap: 0.5rem;
1349 }
1350
1351 /* — 9. Селектор частоты опроса панели (Panel Polling Frequency Selector) — */
1352 .poll-freq-select {
1353     font-size: 0.68rem !important;
1354     padding: 1px 18px 1px 6px !important;
1355     height: 22px !important;
1356     min-height: 22px !important;
1357     line-height: 1.2 !important;
1358     background-color: rgba(15, 23, 42, 0.85) !important;
1359     border: 1px solid rgba(255, 255, 255, 0.18) !important;
1360     color: #94a3b8 !important;
1361     border-radius: 4px !important;
1362     cursor: pointer;
1363     width: auto !important;
1364     max-width: 145px;
1365     display: inline-block;
1366     vertical-align: middle;
1367     transition: border-color 0.15s ease, color 0.15s ease;
1368 }
1369
1370 .poll-freq-select:hover {
1371     border-color: rgba(56, 189, 248, 0.5) !important;
1372     color: #cbd5e1 !important;
1373 }
```

```
1374 |
1375 | .poll-freq-select:focus {
1376 |   border-color: #38bdf8 !important;
1377 |   color: #f1f5f9 !important;
1378 |   box-shadow: 0 0 0 1px rgba(56, 189, 248, 0.35) !important;
1379 |   outline: none;
1380 | }
1381 |
1382 | .poll-freq-select option {
1383 |   background-color: #0f172a !important;
1384 |   color: #e2e8f0 !important;
1385 | }
1386 |
1387 | /* — 10. Дерево спецификации оборудования (Hardware Specifica
   | tion Tree) ——— */
1388 | .about-sys-wrapper {
1389 |   padding-left: 0.25rem;
1390 |   padding-right: 0.25rem;
1391 | }
1392 |
1393 | .about-sys-tree-node {
1394 |   background: var(--surface-1);
1395 |   border: 1px solid var(--border-color);
1396 |   border-radius: var(--radius-sm, 6px);
1397 |   margin-bottom: 0.65rem;
1398 |   overflow: hidden;
1399 |   box-shadow: var(--shadow-xs, 0 1px 2px rgba(0, 0, 0, 0.05));
1400 |   transition: border-color 0.15s ease, box-shadow 0.15s ease;
1401 | }
1402 |
1403 | .about-sys-tree-node:hover {
1404 |   border-color: rgba(56, 189, 248, 0.4);
1405 | }
1406 |
1407 | .about-sys-tree-title {
1408 |   padding: 0.6rem 0.9rem;
1409 |   font-size: 0.84rem;
1410 |   font-weight: 600;
```

```
1411 | cursor: pointer;
1412 | background: var(--surface-2);
1413 | display: flex;
1414 | align-items: center;
1415 | justify-content: space-between;
1416 | gap: 0.5rem;
1417 | color: var(--text-color);
1418 | transition: background 0.15s ease, color 0.15s ease;
1419 | user-select: none;
1420 | }
1421 |
1422 | .about-sys-tree-title:hover {
1423 |   background: rgba(56, 189, 248, 0.08);
1424 | }
1425 |
1426 | .about-sys-tree-body {
1427 |   background: var(--surface-1);
1428 |   border-top: 1px solid var(--border-subtle, rgba(255, 255, 25, 0.06));
1429 |   padding: 0;
1430 | }
1431 |
1432 | .about-sys-chevron {
1433 |   font-size: 0.72rem;
1434 |   color: var(--accent-color, #38bdf8);
1435 |   transition: transform 0.2s ease;
1436 |   display: inline-block;
1437 |   line-height: 1;
1438 | }
1439 |
1440 | /* — 10. Кнопка-иконка сворачивания/разворачивания таблиц (Красная дрель) — */
1441 | .panel-toggle-drill {
1442 |   display: inline-flex;
1443 |   align-items: center;
1444 |   justify-content: center;
1445 |   background: transparent;
1446 |   border: none;
```

```
1447 | padding: 0;
1448 | margin: 0;
1449 | cursor: pointer;
1450 | line-height: 1;
1451 | vertical-align: middle;
1452 | transition: transform 0.15s ease, opacity 0.15s ease;
1453 | border-radius: 4px;
1454 | }
1455 |
1456 | .panel-toggle-drill:hover {
1457 |   opacity: 0.85;
1458 |   transform: scale(1.15);
1459 | }
1460 |
1461 | .panel-toggle-drill:active {
1462 |   transform: scale(0.92);
1463 | }
1464 |
1465 | .panel-toggle-drill img {
1466 |   width: 20px;
1467 |   height: 20px;
1468 |   object-fit: contain;
1469 |   display: block;
1470 | }
1471 |
1472 | .panel-toggle-drill.collapsed img {
1473 |   opacity: 0.7;
1474 | }
1475 |
1476 |
```

global-overrides.css

```
1  /**
2   * =====
3   * Process Name: Windows Web Styles - Global Overrides Stylesheet
4   * =====
5   * Description:
6   *   Глобальные переопределения стилей для минимизации дублирования CSS-классов.
7   *   Использует наследование и глобальные селекторы вместо повторения стилей.
8   *
9   * Usage Examples:
10  *   HTML Link Tag:
11  *     <link rel="stylesheet" href="/windows/api/webgui/css/global-overrides.css?v=20261008_v1">
12  *
13  * File: global-overrides.css
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/css
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 01:36:00
19  * =====
20  */
21
22  /** =====
23   ГЛОБАЛЬНЫЕ ПЕРЕОПРЕДЕЛЕНИЯ - AI Breadboard WebGUI
24   Минимизация дублирования стилей через наследование и глобальные селекторы
25   =====
26   */
```

```
27 /* — 1. Универсальные карточки (KPI, Panel, Spec) —————
   |————— */
28
29 /* Все карточки с классом .ui-kpi-card наследуют базовые стили
   */
30 .ui-kpi-card,
31 .about-sys-card,
32 .def-card,
33 .sysctl-card,
34 .scc-card,
35 .winadmin-card,
36 .sys-card {
37     background: var(--surface-1);
38     border: 1.5px solid var(--border-color);
39     border-radius: var(--radius-md);
40     padding: var(--kpi-padding, 0.65rem 0.9rem);
41     min-height: var(--kpi-min-height, 72px);
42     display: flex;
43     flex-direction: column;
44     justify-content: space-between;
45     height: 100%;
46     box-sizing: border-box;
47     box-shadow: 0 2px 6px rgba(0, 0, 0, 0.25);
48     transition: border-color 0.15s ease, box-shadow 0.15s ease;
49 }
50
51 .ui-kpi-card:hover,
52 .about-sys-card:hover,
53 .def-card:hover,
54 .sysctl-card:hover,
55 .scc-card:hover,
56 .winadmin-card:hover,
57 .sys-card:hover {
58     border-color: var(--kpi-accent, var(--nav-active));
59     box-shadow: var(--shadow-sm, 0 3px 10px rgba(0, 0, 0, 0.25))
60     ;
61 }
```

```
62 | /* — 2. Универсальные заголовки карточек —————
    | ————— */
63 |
64 | .ui-kpi-label,
65 | .about-sys-label,
66 | .def-card-label,
67 | .sysctl-label,
68 | .scc-label,
69 | .winadmin-label,
70 | .sys-label {
71 |     font-size: var(--kpi-label-size, 0.72rem);
72 |     text-transform: uppercase;
73 |     letter-spacing: 0.05em;
74 |     font-weight: 600;
75 |     color: var(--text-muted);
76 |     display: flex;
77 |     justify-content: space-between;
78 |     align-items: center;
79 |     line-height: 1.2;
80 | }
81 |
82 | /* — 3. Универсальные значения в карточках —————
    | ————— */
83 |
84 | .ui-kpi-value,
85 | .about-sys-value,
86 | .def-card-value,
87 | .sysctl-value,
88 | .scc-value,
89 | .winadmin-value,
90 | .sys-value {
91 |     font-size: var(--kpi-val-size, 1.25rem);
92 |     font-weight: 700;
93 |     font-family: var(--font-mono, monospace);
94 |     color: var(--text-color);
95 |     line-height: 1.2;
96 | }
97 |
```



```
98  /* — 4. Универсальные подзаголовки в карточках —————
    |————— */
99
100 .ui-kpi-sub,
101 .about-sys-sub,
102 .def-card-sub,
103 .sysctl-sub,
104 .scc-sub,
105 .winadmin-sub,
106 .sys-sub {
107     font-size: var(--kpi-sub-size, 0.72rem);
108     color: var(--text-muted);
109     line-height: 1.2;
110 }
111
112 /* — 5. Универсальные панели —————
    |————— */
113
114 .ui-panel,
115 .about-sys-panel,
116 .def-panel,
117 .sysctl-panel,
118 .winadmin-panel,
119 .sys-panel {
120     background: var(--surface-1);
121     border: 1px solid var(--border-color);
122     border-radius: var(--radius-md);
123     overflow: hidden;
124     margin-bottom: 0.75rem;
125     box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
126 }
127
128 /* — 6. Универсальные заголовки панелей —————
    |————— */
129
130 .ui-panel-header,
131 .about-sys-panel-header,
132 .def-panel-header,
```

```
133 | .sysctl-panel-header,  
134 | .winadmin-panel-header,  
135 | .sys-panel-header {  
136 |     background: var(--surface-2);  
137 |     border-bottom: 1px solid var(--border-color);  
138 |     padding: 0.5rem 0.9rem;  
139 |     font-size: 0.85rem;  
140 |     font-weight: 600;  
141 |     color: var(--text-color);  
142 |     display: flex;  
143 |     align-items: center;  
144 |     justify-content: space-between;  
145 |     flex-wrap: wrap;  
146 |     gap: 0.5rem;  
147 | }  
148 |  
149 | /* — 7. Универсальные таблицы спецификаций —————  
    | ————— */  
150 |  
151 | .ui-spec-table,  
152 | .about-sys-spec-table,  
153 | .sysctl-spec-table,  
154 | .def-spec-table,  
155 | .winadmin-spec-table,  
156 | .spec-table-unified {  
157 |     width: 100%;  
158 |     margin-bottom: 0;  
159 |     font-size: 0.82rem;  
160 |     border-collapse: separate;  
161 |     border-spacing: 0;  
162 | }  
163 |  
164 | .ui-spec-table th,  
165 | .about-sys-spec-table th,  
166 | .sysctl-spec-table th,  
167 | .def-spec-table th,  
168 | .winadmin-spec-table th,  
169 | .spec-table-unified th {
```

```
170 | padding: 8px 16px;
171 | font-size: 0.78rem;
172 | font-weight: 600;
173 | color: var(--text-muted);
174 | text-transform: uppercase;
175 | letter-spacing: 0.04em;
176 | border-bottom: 1px solid var(--border-subtle);
177 | background: transparent;
178 | vertical-align: middle;
179 | }
180 |
181 | .ui-spec-table td,
182 | .about-sys-spec-table td,
183 | .sysctl-spec-table td,
184 | .def-spec-table td,
185 | .winadmin-spec-table td,
186 | .spec-table-unified td {
187 |     padding: 9px 14px;
188 |     border-bottom: 1px solid var(--border-subtle);
189 |     vertical-align: middle;
190 |     line-height: 1.45;
191 |     color: var(--text-color);
192 | }
193 |
194 | .ui-spec-table tr:hover td,
195 | .about-sys-spec-table tr:hover td,
196 | .sysctl-spec-table tr:hover td,
197 | .def-spec-table tr:hover td,
198 | .winadmin-spec-table tr:hover td,
199 | .spec-table-unified tr:hover td {
200 |     background: rgba(56, 189, 248, 0.04);
201 | }
202 |
203 | .ui-spec-table tr:last-child td,
204 | .about-sys-spec-table tr:last-child td,
205 | .sysctl-spec-table tr:last-child td,
206 | .def-spec-table tr:last-child td,
207 | .winadmin-spec-table tr:last-child td,
```

```
208 | .spec-table-unified tr:last-child td {
209 |     border-bottom: none;
210 | }
211 |
212 | /* — 8. Универсальные ключи и значения в таблицах —————
    | ————— */
213 |
214 | .ui-spec-key,
215 | .about-sys-spec-key,
216 | .sysctl-spec-key,
217 | .def-spec-key,
218 | .winadmin-spec-key,
219 | .spec-key-unified {
220 |     color: var(--text-muted);
221 |     width: 230px;
222 |     min-width: 210px;
223 |     max-width: 260px;
224 |     white-space: nowrap;
225 |     font-weight: 500;
226 |     font-family: var(--font-sans, system-ui, -apple-system, sans
    | -serif);
227 |     vertical-align: middle;
228 | }
229 |
230 | .ui-spec-key i,
231 | .about-sys-spec-key i,
232 | .sysctl-spec-key i,
233 | .def-spec-key i,
234 | .winadmin-spec-key i,
235 | .spec-key-unified i {
236 |     margin-right: 8px;
237 |     font-size: 0.92rem;
238 |     vertical-align: -1px;
239 | }
240 |
241 | .ui-spec-val,
242 | .about-sys-spec-val,
243 | .sysctl-spec-val,
```

```
244 | .def-spec-val,  
245 | .winadmin-spec-val,  
246 | .spec-val-unified {  
247 |   color: var(--text-color);  
248 |   font-weight: 500;  
249 |   font-family: var(--font-mono, monospace);  
250 |   vertical-align: middle;  
251 |   word-break: break-word;  
252 | }  
253 |  
254 | /* — 9. Универсальные бейджи статусов —————  
    | ————— */  
255 |  
256 | .ui-badge,  
257 | .status-badge-unified,  
258 | .about-sys-badge-ok,  
259 | .about-sys-badge-warn,  
260 | .diag-badge-critical,  
261 | .diag-badge-warning,  
262 | .diag-badge-normal {  
263 |   font-size: 0.72rem;  
264 |   font-weight: 600;  
265 |   padding: 0.2rem 0.55rem;  
266 |   border-radius: var(--radius-xs, 4px);  
267 |   display: inline-flex;  
268 |   align-items: center;  
269 |   gap: 0.3rem;  
270 |   line-height: 1.2;  
271 | }  
272 |  
273 | .ui-badge-success,  
274 | .about-sys-badge-ok,  
275 | .diag-badge-normal {  
276 |   background: var(--status-success-bg);  
277 |   color: var(--status-success-text);  
278 |   border: 1px solid var(--status-success-border);  
279 | }  
280 |
```

```
281 | .ui-badge-danger,  
282 | .diag-badge-critical {  
283 |     background: var(--status-danger-bg);  
284 |     color: var(--status-danger-text);  
285 |     border: 1px solid var(--status-danger-border);  
286 | }  
287 |  
288 | .ui-badge-warning,  
289 | .about-sys-badge-warn,  
290 | .diag-badge-warning {  
291 |     background: var(--status-warning-bg);  
292 |     color: var(--status-warning-text);  
293 |     border: 1px solid var(--status-warning-border);  
294 | }  
295 |  
296 | .ui-badge-info {  
297 |     background: var(--status-info-bg);  
298 |     color: var(--status-info-text);  
299 |     border: 1px solid var(--status-info-border);  
300 | }  
301 |  
302 | /* — 10. Универсальные панели диагностики —————  
303 | |————— */  
304 |  
305 | .diag-panel,  
306 | .ui-panel {  
307 |     background: var(--surface-1);  
308 |     border: 1px solid var(--border-color);  
309 |     border-radius: var(--radius-md);  
310 |     overflow: hidden;  
311 |     margin-bottom: 0.75rem;  
312 |     box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));  
313 | }  
314 |  
315 | .diag-panel-header,  
316 | .ui-panel-header {  
317 |     background: var(--surface-2);  
318 |     border-bottom: 1px solid var(--border-color);
```

```
318 | padding: 0.5rem 0.9rem;
319 | font-size: 0.85rem;
320 | font-weight: 600;
321 | color: var(--text-color);
322 | display: flex;
323 | align-items: center;
324 | justify-content: space-between;
325 | flex-wrap: wrap;
326 | gap: 0.5rem;
327 | }
328 |
329 | /* — 11. Универсальные карточки метрик —————
    | ————— */
330 |
331 | .diag-meter-card,
332 | .ui-kpi-card {
333 |   background: var(--surface-2);
334 |   border: 1px solid var(--border-color);
335 |   border-radius: var(--radius-md);
336 |   padding: 0.75rem 1rem;
337 | }
338 |
339 | /* — 12. Универсальные заголовки деревьев и узлы —————
    | ————— */
340 |
341 | .about-sys-tree-node {
342 |   background: var(--surface-1);
343 |   border: 1px solid var(--border-color);
344 |   border-radius: var(--radius-sm, 6px);
345 |   margin-bottom: 0.65rem;
346 |   overflow: hidden;
347 |   box-shadow: var(--shadow-xs, 0 1px 2px rgba(0, 0, 0, 0.05));
348 |   transition: border-color 0.15s ease, box-shadow 0.15s ease;
349 | }
350 |
351 | .about-sys-tree-node:hover {
352 |   border-color: rgba(56, 189, 248, 0.4);
353 | }
```

```

354 |
355 | .about-sys-tree-title,
356 | .ui-module-title {
357 |     padding: 0.6rem 0.9rem;
358 |     font-size: 0.84rem;
359 |     font-weight: 600;
360 |     cursor: pointer;
361 |     background: var(--surface-2);
362 |     display: flex;
363 |     align-items: center;
364 |     justify-content: space-between;
365 |     gap: 0.5rem;
366 |     color: var(--text-color);
367 |     transition: background 0.15s ease, color 0.15s ease;
368 |     user-select: none;
369 | }
370 |
371 | .about-sys-tree-title:hover,
372 | .ui-module-title:hover {
373 |     background: rgba(56, 189, 248, 0.08);
374 | }
375 |
376 | .about-sys-tree-body {
377 |     background: var(--surface-1);
378 |     border-top: 1px solid var(--border-subtle, rgba(255, 255, 25
379 |     padding: 0;
380 | }
381 |
382 | .about-sys-chevron {
383 |     font-size: 0.72rem;
384 |     color: var(--accent-color, #38bdf8);
385 |     transition: transform 0.2s ease;
386 |     display: inline-block;
387 |     line-height: 1;
388 | }
389 |
390 | /* — 13. Универсальные строки свойств

```

```

391 | _____ */
392 | .about-sys-prop-row,
393 | .ui-prop-row {
394 |     display: grid;
395 |     grid-template-columns: minmax(180px, 320px) 1fr;
396 |     align-items: center;
397 |     gap: 12px;
398 |     padding: 6px 12px;
399 |     border-bottom: 1px solid var(--border-subtle);
400 |     border-radius: 0;
401 |     transition: background 0.12s ease;
402 |     min-width: 0;
403 | }
404 |
405 | .about-sys-prop-row:hover,
406 | .ui-prop-row:hover {
407 |     background: rgba(56, 189, 248, 0.04);
408 | }
409 |
410 | .about-sys-prop-row:last-child,
411 | .ui-prop-row:last-child {
412 |     border-bottom: none;
413 | }
414 |
415 | /* — 14. Универсальные ключи свойств _____
416 | _____ */
417 | .about-sys-prop-key,
418 | .ui-prop-key {
419 |     color: var(--text-muted);
420 |     font-weight: 500;
421 |     font-size: 0.82rem;
422 |     user-select: text;
423 |     word-break: break-word;
424 | }
425 |
426 | .about-sys-prop-key::after,

```

```
427 | .ui-prop-key::after {
428 |     content: "";
429 | }
430 |
431 | /* — 15. Универсальные значения свойств —————
    | ————— */
432 |
433 | .about-sys-prop-val,
434 | .ui-prop-val {
435 |     color: var(--text-color);
436 |     font-weight: 500;
437 |     font-size: 0.82rem;
438 |     word-break: break-word;
439 |     user-select: text;
440 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
    | nsolas, monospace;
441 |     min-width: 0;
442 | }
443 |
444 | /* — 16. Универсальные прогресс-бары —————
    | ————— */
445 |
446 | .about-sys-progress-track,
447 | .ui-progress-track {
448 |     background: var(--surface-2);
449 |     border-radius: 4px;
450 |     height: 8px;
451 |     overflow: hidden;
452 |     min-width: 100px;
453 | }
454 |
455 | .about-sys-progress-bar,
456 | .ui-progress-bar {
457 |     height: 100%;
458 |     border-radius: 4px;
459 |     transition: width 0.3s ease;
460 | }
461 |
```

```
462 /* — 17. Универсальные навигационные панели —————  
    |————— */  
463  
464 .about-subtabs-nav,  
465 .ui-subnav {  
466     background: var(--surface-2);  
467     border: 1px solid var(--border-color);  
468     border-radius: var(--radius-md);  
469     padding: 0.35rem 0.5rem;  
470 }  
471  
472 .about-subtab-btn,  
473 .ui-subnav-btn {  
474     transition: all 0.18s ease;  
475     font-size: 0.78rem;  
476 }  
477  
478 .about-subtab-btn.active,  
479 .ui-subnav-btn.active {  
480     background: var(--nav-active);  
481     color: #0f172a;  
482     border-color: var(--nav-active);  
483     font-weight: 700;  
484 }  
485  
486 /* — 18. Универсальные панели с верхним хедером —————  
    |————— */  
487  
488 .about-sys-top-header,  
489 .ui-top-header {  
490     background: var(--surface-2);  
491     border: 1px solid var(--border-color);  
492     border-radius: var(--radius-md);  
493     padding: 0.75rem 1rem;  
494     margin-bottom: 0.75rem;  
495 }  
496  
497 /* — 19. Универсальные карточки с анимацией —————
```

```
498 | _____ */
499 | .about-ai-copilot-banner,
500 | .ui-banner-card {
501 |     background: linear-gradient(135deg, rgba(15, 23, 42, 0.98) 0
502 | %, rgba(20, 35, 60, 0.95) 50%, rgba(15, 23, 42, 0.98) 100%);
503 |     border: 1px solid rgba(56, 189, 248, 0.45);
504 |     border-radius: 10px;
505 |     box-shadow: 0 0 22px rgba(56, 189, 248, 0.15), inset 0 1px 0
506 |     rgba(255, 255, 255, 0.1);
507 |     position: relative;
508 |     overflow: hidden;
509 | }
510 |
511 | /* — 20. Универсальные иконки пульсации _____
512 | _____ */
513 | .about-ai-pulse-icon,
514 | .ui-pulse-icon {
515 |     width: 34px;
516 |     height: 34px;
517 |     border-radius: 8px;
518 |     background: rgba(56, 189, 248, 0.12);
519 |     border: 1px solid rgba(56, 189, 248, 0.35);
520 |     display: flex;
521 |     align-items: center;
522 |     justify-content: center;
523 | }
524 |
525 | /* — 21. Универсальные блоки этапов _____
526 | _____ */
527 | .about-ai-stages-box,
528 | .ui-stages-box {
529 |     background: rgba(10, 15, 29, 0.7);
530 |     max-height: 240px;
531 |     overflow-y: auto;
532 | }
```

```
531 |
532 | .about-ai-stage-row,
533 | .ui-stage-row {
534 |     padding: 3px 6px;
535 |     border-radius: 4px;
536 |     font-size: 0.74rem;
537 |     line-height: 1.35;
538 | }
539 |
540 | .about-ai-stage-row:hover,
541 | .ui-stage-row:hover {
542 |     background: rgba(255, 255, 255, 0.04);
543 | }
544 |
545 | /* — 22. Универсальные панели с прокруткой —————
546 | |————— */
547 | .diag-panel,
548 | .ui-scroll-panel {
549 |     overflow-y: auto;
550 | }
551 |
552 | /* — 23. Универсальные панели с фиксированной высотой —————
553 | |————— */
554 | .h-100 {
555 |     height: 100%;
556 | }
557 |
558 | /* — 24. Универсальные панели с адаптивной высотой —————
559 | |————— */
560 | @media (min-width: 768px) {
561 |     .col-md-6 .ui-panel,
562 |     .col-md-6 .ui-kpi-card {
563 |         min-height: 150px;
564 |     }
565 | }
```

```
566 |
567 | @media (min-width: 992px) {
568 |   .col-lg-6 .ui-panel,
569 |   .col-lg-6 .ui-kpi-card {
570 |     min-height: 200px;
571 |   }
572 | }
573 |
574 | @media (min-width: 1200px) {
575 |   .col-xl-8 .ui-panel {
576 |     min-height: 300px;
577 |   }
578 |   .col-xl-4 .ui-panel {
579 |     min-height: 250px;
580 |   }
581 | }
582 |
583 | /* — 25. Универсальные панели с hover-эффектами —————
584 | |————— */
585 |
586 | .ui-panel:hover,
587 | .ui-kpi-card:hover,
588 | .ui-badge:hover {
589 |   transition: all 0.15s ease;
590 | }
591 |
592 | /* — 26. Универсальные панели с фиксированным отступом —————
593 | |————— */
594 |
595 | .ui-panel,
596 | .ui-kpi-card {
597 |   margin-bottom: 0.75rem;
598 | }
599 |
600 | /* — 27. Универсальные панели с адаптивной шириной —————
601 | |————— */
602 |
603 | @media (max-width: 767px) {
```

```
601 | .ui-panel,
602 | .ui-kpi-card {
603 |     margin-bottom: 0.5rem;
604 | }
605 | }
606 |
607 | /* — 28. Универсальные панели с адаптивным отступом —————
    | ————— */
608 |
609 | @media (max-width: 575px) {
610 |     .ui-panel,
611 |     .ui-kpi-card {
612 |         padding: 0.5rem;
613 |     }
614 | }
615 |
616 | /* — 29. Универсальные панели с адаптивным заголовком —————
    | ————— */
617 |
618 | @media (max-width: 575px) {
619 |     .ui-panel-header,
620 |     .ui-kpi-label {
621 |         font-size: 0.75rem;
622 |         padding: 0.4rem 0.6rem;
623 |     }
624 | }
625 |
626 | /* — 30. Универсальные панели с адаптивным значением —————
    | ————— */
627 |
628 | @media (max-width: 575px) {
629 |     .ui-kpi-value,
630 |     .ui-spec-val {
631 |         font-size: 1.1rem;
632 |     }
633 | }
634 |
635 | /* — 31. Универсальные панели с адаптивным ключом —————
```

```

        _____ */
636
637 @media (max-width: 575px) {
638     .ui-spec-key {
639         font-size: 0.75rem;
640         width: 100%;
641     }
642 }
643
644 /* — 32. Универсальные панели с адаптивной таблицей _____
        _____ */
645
646 @media (max-width: 575px) {
647     .ui-spec-table th,
648     .ui-spec-table td {
649         padding: 6px 10px;
650         font-size: 0.75rem;
651     }
652 }
653
654 /* — 33. Универсальные панели с адаптивным бейджем _____
        _____ */
655
656 @media (max-width: 575px) {
657     .ui-badge,
658     .about-sys-badge-ok,
659     .about-sys-badge-warn {
660         font-size: 0.65rem;
661         padding: 0.15rem 0.4rem;
662     }
663 }
664
665 /* — 34. Универсальные панели с адаптивным заголовком панели
        _____ */
666
667 @media (max-width: 575px) {
668     .ui-panel-header {
669         font-size: 0.75rem;

```



```
670     padding: 0.4rem 0.6rem;
671   }
672 }
673
674 /* — 35. Универсальные панели с адаптивным значением свойства
   | _____ */
675
676 @media (max-width: 575px) {
677   .ui-prop-val {
678     font-size: 0.75rem;
679   }
680 }
681
682 /* — 36. Универсальные панели с адаптивным ключом свойства —
   | _____ */
683
684 @media (max-width: 575px) {
685   .ui-prop-key {
686     font-size: 0.75rem;
687   }
688 }
689
690 /* — 37. Универсальные панели с адаптивным заголовком дерева
   | _____ */
691
692 @media (max-width: 575px) {
693   .ui-tree-title {
694     font-size: 0.75rem;
695     padding: 4px 8px;
696   }
697 }
698
699 /* — 38. Универсальные панели с адаптивным значением прогресс
   | -бара _____ */
700
701 @media (max-width: 575px) {
702   .ui-progress-track {
703     height: 6px;
```

```
704 | }
705 | }
706 |
707 | /* — 39. Универсальные панели с адаптивным значением карточки
    | _____ */
708 |
709 | @media (max-width: 575px) {
710 |   .ui-kpi-card {
711 |     min-height: 60px;
712 |     padding: 0.5rem 0.7rem;
713 |   }
714 | }
715 |
716 | /* — 40. Универсальные панели с адаптивным значением заголовк
    | а карточки ————— */
717 |
718 | @media (max-width: 575px) {
719 |   .ui-kpi-label {
720 |     font-size: 0.65rem;
721 |   }
722 | }
723 |
724 | /* — 41. Универсальные панели с адаптивным значением подзагол
    | овка карточки — */
725 |
726 | @media (max-width: 575px) {
727 |   .ui-kpi-sub {
728 |     font-size: 0.6rem;
729 |   }
730 | }
731 |
732 | /* — 42. Универсальные панели с адаптивным значением заголовк
    | а панели ————— */
733 |
734 | @media (max-width: 575px) {
735 |   .ui-panel-header {
736 |     font-size: 0.7rem;
737 |     padding: 0.3rem 0.5rem;
```

```
738 | }
739 | }
740 |
741 | /* — 43. Универсальные панели с адаптивным значением таблицы
    | _____ */
742 |
743 | @media (max-width: 575px) {
744 |   .ui-spec-table {
745 |     font-size: 0.75rem;
746 |   }
747 | }
748 |
749 | /* — 44. Универсальные панели с адаптивным значением бейджа —
    | _____ */
750 |
751 | @media (max-width: 575px) {
752 |   .ui-badge {
753 |     font-size: 0.6rem;
754 |     padding: 0.1rem 0.3rem;
755 |   }
756 | }
757 |
758 | /* — 45. Универсальные панели с адаптивным значением заголовк
    | а _____ */
759 |
760 | @media (max-width: 575px) {
761 |   .ui-module-title {
762 |     font-size: 0.75rem;
763 |   }
764 | }
765 |
766 | /* — 46. Универсальные панели с адаптивным значением заголовк
    | а дерева _____ */
767 |
768 | @media (max-width: 575px) {
769 |   .ui-tree-title {
770 |     font-size: 0.7rem;
771 |   }
```

```
772 | }
773 |
774 | /* — 47. Универсальные панели с адаптивным значением значения
    | _____ */
775 |
776 | @media (max-width: 575px) {
777 |   .ui-value {
778 |     font-size: 1rem;
779 |   }
780 | }
781 |
782 | /* — 48. Универсальные панели с адаптивным значением ключа —
    | _____ */
783 |
784 | @media (max-width: 575px) {
785 |   .ui-key {
786 |     font-size: 0.7rem;
787 |   }
788 | }
789 |
790 | /* — 49. Универсальные панели с адаптивным значением заголовк
    | а панели _____ */
791 |
792 | @media (max-width: 575px) {
793 |   .ui-header {
794 |     font-size: 0.7rem;
795 |   }
796 | }
797 |
798 | /* — 50. Универсальные панели с адаптивным значением заголовк
    | а карточки _____ */
799 |
800 | @media (max-width: 575px) {
801 |   .ui-card-header {
802 |     font-size: 0.7rem;
803 |   }
804 | }
805 |
```

main.css

```
1 /**
2  * =====
3  * Process Name: AI-Breadboard Web Styles - Main Stylesheet Ma
  nifest
4  * =====
5  * Description:
6  *   Главный манифест стилей веб-интерфейса AI-Breadboard.
7  *
8  *   Зачем нужен этот файл:
9  *     1. Модульная сборка стилей: агрегирует все слои оформле
  ния интерфейса
10  *       (дизайн-токены/переменные, базовые стили, навигация,
  компоненты, чат, оверрайды).
11  *     2. Поддержка динамических тем оформления (светлая/тёмна
  я/кирпичная/терминал) через CSS-переменные.
12  *     3. Cache-busting версионирование: контролирует параметр
  ы ?v= версий дочерних файлов стилей.
13  *
14  * Usage Examples:
15  *   HTML Link Tag:
16  *     <link rel="stylesheet" href="/src/api/webgui/css/main.c
  ss?v=20261006_v5">
17  *
18  * File: main.css
19  * Project: ai-breadboard
20  * Package: src/api/webgui/css
21  * Author: hypo69
22  * Copyright: © 2026 hypo69
23  * Updated: 2026-10-08 02:56:00
24  * =====
25  */
26
27 /* 1. Дизайн-токены и переменные тем (светлая NightView/кирпич
```

```
    ная/тёмная/терминал) */
28 @import "variables.css?v=20261008_v1";
29
30 /* 2. Базовые стили страницы, скроллбары и контейнеры вкладок
    */
31 @import "base.css?v=20261008_v1";
32
33 /* 3. Навигационная панель, вкладки, выпадающие меню и переключатели */
34 @import "nav.css?v=20261008_v1";
35
36 /* 4. Общие UI-компоненты (карточки, формы, кнопки, модалки, таблицы, бейджи, чипы) */
37 @import "components.css?v=20261008_v2";
38
39 /* 5. Чат, сообщения, сайдбар истории диалогов и дебаггер */
40 @import "chat.css?v=20261008_v1";
41
42 /* 6. Управление торрентами, поиск релизов, карточки трекеров
    */
43 @import "torrents.css?v=20261008_v1";
44
45 /* 7. Адаптивные переопределения и совместимость тем с Bootstrap */
46 @import "overrides.css?v=20261008_v1";
47
48 /* 8. Глобальные переопределения для минимизации дублирования CSS */
49 @import "global-overrides.css?v=20261008_v1";
50
```

nav.css

```
1  /**
2   * =====
3   * Process Name: Windows Web Styles - Nav Stylesheet
4   * =====
5   * Description:
6   *   Таблица стилей стиля и оформления интерфейса (nav).
7   *
8   * Usage Examples:
9   *   HTML Link Tag:
10  *     <link rel="stylesheet" href="/windows/api/webgui/css/na
11  *     v.css?v=20261001_v1">
12  *
13  * File: nav.css
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/css
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 01:36:00
19  * =====
20  */
21  /** =====
22  NAVIGATION — AI Breadboard WebGUI
23  Главная панель навигации, вкладки, выпадающие меню и элемен
24  ты управления
25  ===== */
26  /* — Главная панель навигации —
27  */
27  .main-nav-container {
28  background-color: var(--nav-bg);
```

```
29 | border-bottom: 1px solid var(--border-color);
30 | min-height: 34px;
31 | position: relative;
32 | z-index: 1020;
33 | overflow: visible !important;
34 | box-shadow: var(--shadow-xs);
35 | }
36 |
37 | .nav-tabs {
38 |   border-bottom: none;
39 |   background-color: transparent;
40 |   flex-wrap: nowrap;
41 |   overflow: visible !important;
42 | }
43 |
44 | .nav-tabs .nav-item {
45 |   flex-shrink: 0;
46 |   position: relative;
47 | }
48 |
49 | .nav-tabs .nav-item.dropdown {
50 |   position: relative;
51 | }
52 |
53 | /* Nav-links — компактные */
54 | .nav-tabs .nav-link {
55 |   color: var(--nav-text);
56 |   border: none;
57 |   border-bottom: 2px solid transparent;
58 |   background: transparent;
59 |   padding: 5px 10px;
60 |   font-size: 0.82rem;
61 |   font-weight: 500;
62 |   white-space: nowrap;
63 |   transition: color 0.15s, border-color 0.15s, background-color 0.15s;
64 |   border-radius: 4px 4px 0 0;
65 | }
```



```
66 |
67 | .nav-tabs .nav-link:hover {
68 |   color: var(--nav-active);
69 |   background-color: var(--nav-hover-bg);
70 |   border-bottom-color: transparent;
71 | }
72 |
73 | .nav-tabs .nav-link.active {
74 |   color: var(--nav-active);
75 |   background: transparent;
76 |   border-bottom-color: var(--nav-active);
77 | }
78 |
79 | /* — Выпадающие меню навбара —————
   | ————— */
80 | .nav-tabs .dropdown-menu {
81 |   position: absolute !important;
82 |   top: 100% !important;
83 |   left: 0 !important;
84 |   margin-top: 3px !important;
85 |   z-index: 9999 !important;
86 |   min-width: 180px;
87 |   display: none !important;
88 |   padding: 4px;
89 |   border-radius: 8px;
90 |   background-color: var(--surface-1);
91 |   border: 1px solid var(--border-color);
92 |   box-shadow: var(--shadow-md) !important;
93 | }
94 |
95 | .nav-tabs .dropdown-menu.show {
96 |   display: block !important;
97 |   visibility: visible !important;
98 |   opacity: 1 !important;
99 | }
100 |
101 | .nav-tabs .dropdown-item {
102 |   width: 100%;
```

```
103 | border: none;
104 | background: transparent;
105 | text-align: left;
106 | padding: 5px 10px;
107 | border-radius: 5px;
108 | font-size: 0.82rem;
109 | font-weight: 500;
110 | cursor: pointer;
111 | color: var(--text-color);
112 | transition: background-color 0.12s ease, color 0.12s ease;
113 | }
114 |
115 | .nav-tabs .dropdown-item:hover,
116 | .nav-tabs .dropdown-item:focus {
117 |   background-color: var(--nav-hover-bg);
118 |   color: var(--nav-active);
119 | }
120 |
121 | .nav-tabs .dropdown-item.active {
122 |   background-color: rgba(37, 99, 235, 0.1);
123 |   color: var(--nav-active);
124 |   font-weight: 600;
125 | }
126 |
127 | [data-theme="dark"] .nav-tabs .dropdown-item.active {
128 |   background-color: rgba(88, 166, 255, 0.12);
129 |   color: var(--nav-active);
130 | }
131 |
132 | /* — Разделители и метки групп —————
   | ————— */
133 | .nav-group-divider {
134 |   width: 1px;
135 |   height: 16px;
136 |   background-color: var(--divider-color);
137 |   margin: 0 6px;
138 |   opacity: 0.7;
139 |   align-self: center;
```

```
140 | flex-shrink: 0;
141 | }
142 |
143 | .nav-group-label {
144 |   display: inline-flex;
145 |   align-items: center;
146 |   font-size: 0.68rem;
147 |   font-weight: 600;
148 |   text-transform: uppercase;
149 |   letter-spacing: 0.5px;
150 |   color: var(--text-muted);
151 |   padding: 2px 6px;
152 |   background-color: var(--surface-2);
153 |   border: 1px solid var(--border-color);
154 |   border-radius: 5px;
155 |   user-select: none;
156 | }
157 |
158 | /* — Выпадающие меню Bootstrap (глобально) —————
    | ————— */
159 | .dropdown-menu {
160 |   background-color: var(--surface-1);
161 |   border-color: var(--border-color);
162 |   box-shadow: var(--shadow-md);
163 | }
164 |
165 | .dropdown-item {
166 |   color: var(--text-color);
167 |   border-radius: 4px;
168 |   font-size: 0.82rem;
169 | }
170 |
171 | .dropdown-item:hover,
172 | .dropdown-item:focus {
173 |   background-color: var(--nav-hover-bg);
174 |   color: var(--nav-active);
175 | }
176 |
```

```
177 | .dropdown-item.active {
178 |   background-color: var(--nav-active);
179 |   color: #ffffff;
180 | }
181 |
182 | [data-theme="terminal"] .dropdown-menu {
183 |   background-color: #0b120c;
184 |   border: 1px solid #1a4223;
185 |   box-shadow: 0 0 12px rgba(0, 255, 102, 0.25);
186 | }
187 |
188 | [data-theme="terminal"] .dropdown-item.active {
189 |   background-color: #0c3316;
190 |   color: #39ff14;
191 |   border-left: 3px solid #00ff66;
192 | }
```

193 |

```
194 | /* — Кнопки и контролы навбара
```

```
    * /
```

```
195 | .main-nav-container .lang-selector {
196 |   background-color: var(--nav-bg);
197 |   border-color: var(--border-color);
198 |   color: var(--nav-text);
199 |   font-size: 0.8rem;
200 | }
201 |
202 | #nav-google-btn {
203 |   background: var(--surface-1);
204 |   color: var(--text-color) !important;
205 |   font-size: 0.78rem;
206 |   font-weight: 600;
207 |   border-radius: 20px;
208 |   border: 1px solid var(--border-color);
209 |   padding: 3px 10px;
210 |   transition: all 0.18s ease;
211 | }
```

212 |

```
213 | #nav-google-btn:hover {
```

```
214 | background: var(--nav-hover-bg);
215 | box-shadow: var(--shadow-sm);
216 | transform: translateY(-1px);
217 | }
218 |
219 | [data-theme="dark"] #nav-google-btn {
220 |   background: var(--surface-2);
221 |   color: var(--text-color) !important;
222 |   border-color: var(--border-color);
223 | }
224 |
225 | #user-settings-btn {
226 |   font-size: 0.8rem;
227 |   font-weight: 500;
228 |   border-radius: 20px;
229 |   color: var(--nav-text);
230 |   border-color: var(--border-color);
231 |   transition: all 0.18s ease;
232 |   padding: 3px 10px;
233 | }
234 |
235 | #user-settings-btn:hover {
236 |   background-color: var(--nav-hover-bg);
237 |   color: var(--nav-active);
238 |   border-color: var(--nav-active);
239 | }
240 |
241 | #user-settings-gear-btn,
242 | #theme-toggle-btn {
243 |   color: var(--nav-text);
244 |   border-color: var(--border-color);
245 |   transition: all 0.18s ease;
246 |   width: 28px;
247 |   height: 28px;
248 |   padding: 0;
249 |   display: inline-flex;
250 |   align-items: center;
251 |   justify-content: center;
```

```
252 |   border-radius: 50%;
253 |   font-size: 0.85rem;
254 | }
255 |
256 | #user-settings-gear-btn:hover {
257 |   background-color: var(--nav-hover-bg);
258 |   color: var(--nav-active);
259 |   border-color: var(--nav-active);
260 |   transform: rotate(30deg);
261 | }
262 |
263 | #theme-toggle-btn:hover {
264 |   background-color: var(--nav-hover-bg);
265 |   color: var(--nav-active);
266 |   border-color: var(--nav-active);
267 | }
268 |
269 | /* — Переключатель темы (Dropdown / иконки) —————
    | ————— */
270 | #theme-dropdown-container .dropdown-menu {
271 |   min-width: 140px;
272 |   padding: 3px;
273 | }
274 |
275 | .theme-check-icon {
276 |   font-size: 0.85rem;
277 | }
278 |
279 | .theme-check-icon.d-none {
280 |   display: none !important;
281 | }
282 |
283 | /* — Выдвижной боковой ярлычок (Левая вкладка-триггер) —————
    | ————— */
284 | .left-drawer-tab-trigger {
285 |   position: fixed;
286 |   top: 48%;
287 |   left: 0;
```

```
288 | transform: translateY(-50%);
289 | z-index: 1040;
290 | background-color: var(--surface-1);
291 | border: 1px solid var(--border-color);
292 | border-left: none;
293 | border-radius: 0 6px 6px 0;
294 | padding: 6px 3px;
295 | display: flex;
296 | flex-direction: column;
297 | align-items: center;
298 | gap: 4px;
299 | cursor: pointer;
300 | color: var(--nav-text);
301 | box-shadow: var(--shadow-sm);
302 | transition: all 0.2s ease;
303 | user-select: none;
304 | opacity: 0.65;
305 | }
306 |
307 | .left-drawer-tab-trigger:hover {
308 |   background-color: var(--nav-hover-bg);
309 |   color: var(--nav-active);
310 |   padding-right: 6px;
311 |   box-shadow: var(--shadow-md);
312 |   opacity: 1;
313 | }
314 |
315 | .left-drawer-tab-text {
316 |   writing-mode: vertical-rl;
317 |   text-orientation: mixed;
318 |   font-size: 0.62rem;
319 |   font-weight: 700;
320 |   letter-spacing: 1.5px;
321 |   text-transform: uppercase;
322 | }
323 |
324 | /* — Левая выдвижная панель навигации (Offcanvas Drawer) —
    | _____ */
```

```
325 | .left-drawer-nav {
326 |     width: 320px !important;
327 |     background-color: var(--surface-1);
328 |     color: var(--text-color);
329 |     border-right: 1px solid var(--border-color);
330 |     box-shadow: var(--shadow-lg);
331 |     z-index: 1060;
332 | }
333 |
334 | .left-drawer-nav .offcanvas-header {
335 |     background-color: var(--surface-2);
336 |     border-bottom: 1px solid var(--border-color);
337 | }
338 |
339 | .nav-drawer-group-title {
340 |     font-size: 0.72rem;
341 |     font-weight: 700;
342 |     text-transform: uppercase;
343 |     letter-spacing: 0.8px;
344 |     color: var(--text-muted);
345 |     padding: 0 4px;
346 | }
347 |
348 | .left-drawer-nav .list-group-item {
349 |     background-color: var(--surface-1);
350 |     border-color: var(--border-color);
351 |     color: var(--text-color);
352 |     font-size: 0.85rem;
353 |     transition: background-color 0.15s ease, color 0.15s ease, t
ransform 0.1s ease;
354 |     cursor: pointer;
355 |     border-radius: 6px;
356 |     margin-bottom: 3px;
357 |     border: 1px solid transparent;
358 | }
359 |
360 | .left-drawer-nav .list-group-item:hover {
361 |     background-color: var(--nav-hover-bg);
```



```
362 | color: var(--nav-active);
363 | transform: translateX(2px);
364 | }
365 |
366 | .left-drawer-nav .list-group-item.active {
367 |   background-color: var(--nav-active);
368 |   border-color: var(--nav-active);
369 |   color: #ffffff;
370 |   font-weight: 600;
371 | }
372 |
373 | [data-theme="brick"] .left-drawer-nav .list-group-item.active
    | {
374 |   background-color: #b84228;
375 |   border-color: #b84228;
376 |   color: #ffffff;
377 | }
378 |
379 | [data-theme="brick"] .left-drawer-nav .list-group-item.active:
    | hover {
380 |   background-color: #9e341d;
381 | }
382 |
383 | [data-theme="dark"] .left-drawer-nav .list-group-item.active {
384 |   background-color: #1f6feb;
385 |   border-color: #1f6feb;
386 |   color: #ffffff;
387 | }
388 |
389 | [data-theme="dark"] .left-drawer-nav .list-group-item.active:h
    | over {
390 |   background-color: #388bfd;
391 | }
392 |
393 | [data-theme="terminal"] .left-drawer-nav .list-group-item.acti
    | ve {
394 |   background-color: #0c3316;
395 |   border-color: #00ff66;
```

```
396 | color: #39ff14;
397 | box-shadow: 0 0 8px rgba(0, 255, 102, 0.35);
398 | text-shadow: 0 0 3px rgba(0, 255, 102, 0.5);
399 | }
400 |
401 | [data-theme="terminal"] .left-drawer-nav .list-group-item.acti
    | ve:hover {
402 |   background-color: #154c22;
403 |   border-color: #39ff14;
404 |   color: #ffffff;
405 | }
406 |
407 |
408 |
```

overrides.css

```
1 /**
2  * =====
3  * Process Name: Windows Web Styles - Overrides Stylesheet
4  * =====
5  * Description:
6  *   Таблица стилей стиля и оформления интерфейса (overrides).
7  *   Адаптивные переопределения Bootstrap и специфичных компонент для тем оформления.
8  *
9  * Usage Examples:
10 *   HTML Link Tag:
11 *     <link rel="stylesheet" href="/windows/api/webgui/css/overrides.css?v=20261004_v17">
12 *
13 * File: overrides.css
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/css
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-06 19:05:00
19 * =====
20 */
21
22 /* =====
23
24   THEME OVERRIDES & COMPATIBILITY – AI Breadboard WebGUI
25   Адаптивные переопределения Bootstrap и компонентов для светлых и тёмных тем
26
27   ===== */
28
29 /* — Light & Brick Themes – Переопределения Bootstrap и утилит — */
```

```
28 [data-theme="light"],
29 [data-theme="brick"] {
30     /* Инпуты, к которым JS или Bootstrap добавил bg-dark */
31     input.bg-dark,
32     textarea.bg-dark,
33     select.bg-dark,
34     .form-control.bg-dark,
35     .form-select.bg-dark {
36         background-color: var(--surface-1) !important;
37         color: var(--text-color) !important;
38         border-color: var(--border-color) !important;
39     }
40
41     /* div/section с bg-dark (не кнопки и не бейджи) */
42     div.bg-dark,
43     section.bg-dark,
44     aside.bg-dark,
45     article.bg-dark {
46         background-color: var(--surface-1) !important;
47         color: var(--text-color) !important;
48     }
49
50     /* Bootstrap bg-dark / bg-black */
51     .bg-dark:not(.btn):not(.badge):not(.navbar) {
52         background-color: var(--surface-1) !important;
53         color: var(--text-color) !important;
54         border-color: var(--border-color) !important;
55     }
56
57     .bg-black:not(.btn) {
58         background-color: var(--bg-color) !important;
59         color: var(--text-color) !important;
60     }
61
62     /* Bootstrap bg-white заменяется на комфортную поверхность т
63     емы NightView */
64     .bg-white:not(.btn):not(.badge) {
65         background-color: var(--surface-1) !important;
```

```
65 | }
66 |
67 | /* Высококонтрастные читаемые цвета текста в светлой теме Ni
    | ghtView / Brick */
68 | .text-info {
69 |     color: #0369a1 !important; /* глубокий читаемый океанский
    | синий */
70 | }
71 | .text-warning {
72 |     color: #92400e !important; /* глубокий читаемый янтарно-ко
    | ричневый */
73 | }
74 | .text-success {
75 |     color: #166534 !important; /* глубокий читаемый лесной зел
    | еный */
76 | }
77 | .text-danger {
78 |     color: #991b1b !important; /* глубокий читаемый рубиновый
    | */
79 | }
80 | .text-primary {
81 |     color: #1d4ed8 !important; /* четкий читаемый сапфировый *
    | /
82 | }
83 |
84 | /* Кнопки с контуром (Outline Buttons) высокой контрастности
    | */
85 | .btn-outline-light {
86 |     color: var(--text-color) !important;
87 |     border-color: var(--border-strong) !important;
88 |     background-color: transparent !important;
89 | }
90 | .btn-outline-light:hover,
91 | .btn-outline-light:focus,
92 | .btn-outline-light.active,
93 | .btn-check:checked + .btn-outline-light,
94 | .btn-check:active + .btn-outline-light {
95 |     background-color: var(--surface-2) !important;
```

```
96 |     color: var(--text-color) !important;
97 |     border-color: var(--border-contrast) !important;
98 | }
99 |
100 | .btn-outline-secondary {
101 |     color: var(--text-color) !important;
102 |     border-color: var(--border-strong) !important;
103 |     background-color: transparent !important;
104 | }
105 | .btn-outline-secondary:hover,
106 | .btn-outline-secondary:focus,
107 | .btn-outline-secondary.active,
108 | .btn-check:checked + .btn-outline-secondary,
109 | .btn-check:active + .btn-outline-secondary {
110 |     background-color: #64748b !important;
111 |     color: #ffffff !important;
112 |     border-color: #64748b !important;
113 | }
114 |
115 | .btn-outline-info {
116 |     color: #0369a1 !important;
117 |     border-color: #0284c7 !important;
118 |     background-color: transparent !important;
119 | }
120 | .btn-outline-info:hover,
121 | .btn-outline-info:focus,
122 | .btn-outline-info.active,
123 | .btn-check:checked + .btn-outline-info,
124 | .btn-check:active + .btn-outline-info {
125 |     background-color: #0284c7 !important;
126 |     color: #ffffff !important;
127 |     border-color: #0284c7 !important;
128 | }
129 |
130 | .btn-outline-warning {
131 |     color: #92400e !important;
132 |     border-color: #d97706 !important;
133 |     background-color: transparent !important;
```

```
134 | }
135 | .btn-outline-warning:hover,
136 | .btn-outline-warning:focus,
137 | .btn-outline-warning.active,
138 | .btn-check:checked + .btn-outline-warning,
139 | .btn-check:active + .btn-outline-warning {
140 |     background-color: #d97706 !important;
141 |     color: #ffffff !important;
142 |     border-color: #d97706 !important;
143 | }
144 |
145 | .btn-outline-primary {
146 |     color: #1d4ed8 !important;
147 |     border-color: #2563eb !important;
148 |     background-color: transparent !important;
149 | }
150 | .btn-outline-primary:hover,
151 | .btn-outline-primary:focus,
152 | .btn-outline-primary.active,
153 | .btn-check:checked + .btn-outline-primary,
154 | .btn-check:active + .btn-outline-primary {
155 |     background-color: #2563eb !important;
156 |     color: #ffffff !important;
157 |     border-color: #2563eb !important;
158 | }
159 |
160 | .btn-outline-danger {
161 |     color: #991b1b !important;
162 |     border-color: #dc2626 !important;
163 |     background-color: transparent !important;
164 | }
165 | .btn-outline-danger:hover,
166 | .btn-outline-danger:focus,
167 | .btn-outline-danger.active,
168 | .btn-check:checked + .btn-outline-danger,
169 | .btn-check:active + .btn-outline-danger {
170 |     background-color: #dc2626 !important;
171 |     color: #ffffff !important;
```

```
172     border-color: #dc2626 !important;
173 }
174
175 /* text-white / text-light в карточках и панелях */
176 .card .text-white,
177 .card .text-light,
178 [class*="-panel"] .text-white,
179 [class*="-panel"] .text-light,
180 [class*="-card"] .text-white,
181 [class*="-card"] .text-light,
182 .tab-pane .text-white,
183 .tab-pane .text-light,
184 .tab-pane .text-white:not(.btn):not(.badge),
185 .card .text-white:not(.btn):not(.badge),
186 .sys-container .text-white:not(.btn):not(.badge),
187 .sys-container .text-light:not(.btn):not(.badge) {
188     color: var(--text-color) !important;
189 }
190
191 /* Бейджи bg-dark и статусных оттенков */
192 .badge.bg-dark {
193     background-color: var(--surface-2) !important;
194     border-color: var(--border-color) !important;
195     color: var(--text-color) !important;
196 }
197 .badge.bg-info-subtle {
198     background-color: #e0f2fe !important;
199     color: #0369a1 !important;
200     border-color: #7dd3fc !important;
201 }
202 .badge.bg-warning-subtle {
203     background-color: #fef3c7 !important;
204     color: #92400e !important;
205     border-color: #fde047 !important;
206 }
207 .badge.bg-success-subtle {
208     background-color: #dcfce7 !important;
209     color: #166534 !important;
```



```
210     border-color: #86efac !important;
211 }
212 .badge.bg-danger-subtle {
213     background-color: #fee2e2 !important;
214     color: #991b1b !important;
215     border-color: #fca5a5 !important;
216 }
217
218 /* pre/code без явного светлого фона */
219 pre:not(.prompt-dump-details pre) {
220     background-color: var(--surface-2) !important;
221     color: var(--text-color) !important;
222     border: 1px solid var(--border-color);
223     border-radius: 5px;
224 }
225
226 code.bg-dark {
227     background-color: var(--surface-2) !important;
228     color: var(--text-color) !important;
229     border: 1px solid var(--border-color);
230 }
231
232 /* Прогресс-бар: трек */
233 .progress:not([style]),
234 .progress {
235     background-color: var(--border-color) !important;
236 }
237
238 /* — Системный инспектор, трейдинг, сеть и дерево компонент
239 ов — */
239 .sys-card,
240 .sys-panel,
241 .sys-panel-header,
242 .sys-tree-node,
243 .sys-tree-body,
244 .sys-proc-table th {
245     background-color: var(--surface-1) !important;
246     border-color: var(--border-color) !important;
```

```
247 |     color: var(--text-color) !important;
248 | }
249 |
250 | .sys-card-header-bar {
251 |     background-color: var(--surface-2) !important;
252 |     border-color: var(--border-color) !important;
253 |     color: var(--text-color) !important;
254 | }
255 |
256 | .sys-core-section-title {
257 |     color: var(--text-color) !important;
258 |     border-color: var(--border-subtle) !important;
259 | }
260 |
261 | .sys-core-card {
262 |     background-color: var(--surface-2) !important;
263 |     border-color: var(--border-color) !important;
264 |     color: var(--text-color) !important;
265 |     box-shadow: var(--shadow-xs) !important;
266 | }
267 |
268 | .sys-core-title {
269 |     color: var(--text-color) !important;
270 | }
271 |
272 | .sys-core-param-key {
273 |     color: var(--text-muted) !important;
274 | }
275 |
276 | .sys-core-param-row {
277 |     border-color: var(--border-subtle) !important;
278 | }
279 |
280 | .sys-sensor-widget-item {
281 |     background-color: var(--surface-2) !important;
282 |     border-color: var(--border-color) !important;
283 |     color: var(--text-color) !important;
284 | }
```

```
285
286 .sys-sensor-name,
287 .sys-sensor-value {
288     color: var(--text-color) !important;
289 }
290
291 .sys-temp-slider-track {
292     background-color: var(--border-color) !important;
293 }
294
295 .sys-temp-slider-thumb {
296     border-color: var(--surface-1) !important;
297 }
298
299 .sys-cpu-specs-card,
300 .sys-gpu-specs-card {
301     background-color: var(--surface-1) !important;
302     border-color: var(--border-color) !important;
303     color: var(--text-color) !important;
304 }
305
306 .sys-cpu-details-accordion-btn,
307 .sys-gpu-details-accordion-btn {
308     background: var(--surface-2) !important;
309     border-color: var(--border-color) !important;
310     color: var(--text-color) !important;
311 }
312
313 /* Переопределение жестко закодированных темных плашек в све
    тлой теме */
314 div[style*="background: rgba(0, 0, 0)"],
315 div[style*="background: rgba(0,0,0)"],
316 div[style*="background: linear-gradient(180deg, #182238)"],
317 div[style*="background: linear-gradient(180deg, #182238)"] {
318     background: var(--surface-2) !important;
319     border-color: var(--border-color) !important;
320     color: var(--text-color) !important;
321 }
```

```
322 |
323 | .sys-panel-header {
324 |     background-color: var(--surface-2) !important;
325 |     color: var(--nav-active) !important;
326 | }
327 |
328 | .sys-tree-title {
329 |     color: var(--text-color) !important;
330 |     background: transparent !important;
331 | }
332 |
333 | .sys-tree-body {
334 |     background-color: var(--bg-color) !important;
335 | }
336 |
337 | .sys-proc-table th {
338 |     background-color: var(--surface-2) !important;
339 |     color: var(--text-muted) !important;
340 | }
341 |
342 | .sys-proc-table td {
343 |     color: var(--text-color) !important;
344 | }
345 |
346 | .sys-value {
347 |     color: var(--text-color) !important;
348 | }
349 |
350 | /* — Карточки агентов, моделей, плагинов — */
351 | .agent-card,
352 | .model-card,
353 | .plugin-card,
354 | [class$="-card"]:not(.btn):not(.badge) {
355 |     background-color: var(--surface-1) !important;
356 |     border-color: var(--border-color) !important;
357 |     color: var(--text-color) !important;
358 | }
359 |
```

```
360 | /* — card-header с тёмным фоном — */
361 | .card-header.bg-dark,
362 | .card-header[style*="background"],
363 | .card-header[style*="#0"],
364 | .card-header[style*="#1"] {
365 |     background-color: var(--surface-2) !important;
366 |     color: var(--text-color) !important;
367 |     border-color: var(--border-color) !important;
368 | }
369 |
370 | /* — Логги — */
371 | #media-log,
372 | [id$="-log"]:not(canvas),
373 | .log-output {
374 |     background-color: var(--surface-2) !important;
375 |     color: var(--text-color) !important;
376 |     border-color: var(--border-color) !important;
377 | }
378 |
379 | /* — Stripe-полосы в таблицах — */
380 | .table-striped > tbody > tr:nth-of-type(odd) {
381 |     --bs-table-bg-type: var(--surface-2);
382 | }
383 |
384 | /* — border-secondary — */
385 | .border-secondary {
386 |     border-color: var(--border-color) !important;
387 | }
388 | }
389 |
390 | /* — Dark & Terminal Themes — Высококонтрастные переопределе
    | ния ————— */
391 | [data-theme="dark"],
392 | [data-theme="terminal"] {
393 |     .card-header.bg-light,
394 |     .card-footer.bg-light,
395 |     .modal-header.bg-light,
396 |     .modal-footer.bg-light {
```

```
397     background-color: var(--surface-2) !important;
398     color: var(--text-color) !important;
399     border-color: var(--border-color) !important;
400 }
401
402 .bg-light:not(.btn):not(.badge) {
403     background-color: var(--surface-1) !important;
404     color: var(--text-color) !important;
405 }
406
407 .text-dark {
408     color: var(--text-color) !important;
409 }
410
411 .border-light {
412     border-color: var(--border-color) !important;
413 }
414
415 /* Логги */
416 #media-log,
417 [id$="-log"]:not(canvas),
418 .log-output {
419     background-color: var(--surface-1) !important;
420     color: var(--text-color) !important;
421     border: 1px solid var(--border-color) !important;
422 }
423 }
424
425 /* — Специфика темы «Терминал» (Terminal / CRT Glow) —————
    |————— */
426 [data-theme="terminal"] {
427     .card,
428     .modal-content,
429     .offcanvas {
430         border-color: var(--border-color);
431         box-shadow: 0 0 10px rgba(0, 255, 102, 0.12);
432     }
433 }
```

```
434 | .card-header,
435 | .modal-header {
436 |     border-bottom: 1px solid var(--border-strong);
437 |     color: var(--nav-active);
438 | }
439 |
440 | pre, code, kbd, samp {
441 |     color: #39ff14 !important;
442 |     background-color: #060a06 !important;
443 |     border: 1px solid #1a4223;
444 | }
445 |
446 | .form-control,
447 | .form-select {
448 |     border-color: var(--border-color);
449 |     color: var(--text-color);
450 |     background-color: #0b120c;
451 | }
452 |
453 | .form-control:focus,
454 | .form-select:focus {
455 |     border-color: #00ff66;
456 |     box-shadow: 0 0 8px rgba(0, 255, 102, 0.4);
457 |     background-color: #0b120c;
458 |     color: #ffffff;
459 | }
460 |
461 | .table th {
462 |     color: #39ff14 !important;
463 |     border-bottom: 2px solid #00ff66 !important;
464 | }
465 | }
466 |
```

torrents.css

```
1 /**
2  * =====
3  * Process Name: Windows Web Styles - Torrents Stylesheet
4  * =====
5  * Description:
6  *   Таблица стилей стиля и оформления интерфейса (torrents).
7  *
8  * Usage Examples:
9  *   HTML Link Tag:
10 *   <link rel="stylesheet" href="/windows/api/webgui/css/to
11   rrents.css?v=20261001_v1">
12 *
13 * File: torrents.css
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/css
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-01 13:04:40
19 * =====
20 */
21 /* =====
22   TORRENTS & MEDIA — AI Breadboard WebGUI
23   Стили управления торрентами, поиска релизов и логов медиате
24   ки
25   ===== */
26 /* — Таблица и статусы торрентов —
27   */
27 #torrents-table th[data-col] {
28   cursor: pointer;
```



```
29 | user-select: none;
30 | }
31 |
32 | .state-missingFiles { color: #dc3545; font-weight: 600; }
33 | .state-uploading,
34 | .state-seeding      { color: #16a34a; }
35 | .state-downloading  { color: #2563eb; }
36 | .state-checkingUP,
37 | .state-checkingDL   { color: #ea580c; }
38 |
39 | .progress           { height: 5px; }
40 | .path-cell          { font-size: 0.78rem; color: var(--text-mu
    ted); }
41 | .path-edit          { display: none; }
42 | .path-edit.active   { display: flex; }
43 |
44 | /* — Панель отсутствующих файлов —————
    | ————— */
45 | #missing-panel      { border-left: 2px solid #dc3545; }
46 | .missing-row        { cursor: pointer; }
47 | .missing-row:hover   { background: rgba(255, 200, 0, 0.08); }
48 | .missing-row.selected { background: rgba(37, 99, 235, 0.1); }
49 |
50 | /* — Лог медиатеки —————
    | ————— */
51 | #media-log {
52 |   font-size: 0.78rem;
53 |   font-family: 'Consolas', monospace;
54 |   background: #0d1117;
55 |   color: #d4d4d4;
56 |   border-radius: 5px;
57 |   padding: 8px;
58 |   height: 160px;
59 |   overflow-y: auto;
60 |   white-space: pre-wrap;
61 | }
62 |
63 | .dup-badge          { font-size: 0.72rem; }
```

```

64 .integrity-ok    { color: #16a34a; }
65 .integrity-warn  { color: #dc3545; }
66
67 /* =====
68     TORRENT-ПОИСК
69     =====
70 */
71 .torrent-card {
72     border: 1px solid var(--border-color);
73     border-radius: 6px;
74     background-color: var(--surface-1);
75     padding: 10px 12px;
76     margin-bottom: 8px;
77     box-shadow: var(--shadow-xs);
78     transition: transform 0.15s, box-shadow 0.15s;
79     color: var(--card-text);
80 }
81 .torrent-card:hover {
82     transform: translateY(-1px);
83     box-shadow: var(--shadow-sm);
84 }
85
86 .torrent-badge {
87     display: inline-block;
88     padding: 0.2em 0.5em;
89     font-size: 0.72rem;
90     font-weight: 600;
91     border-radius: 50rem;
92     margin-right: 3px;
93     margin-bottom: 3px;
94 }
95
96 .badge-rutracker { background-color: #6e7a8a; color: #fff; }
97 .badge-nnmclub   { background-color: #16a34a; color: #fff; }
98 .badge-seeds     { background-color: #22c55e; color: #fff; }
99 .badge-peers     { background-color: #6b7280; color: #fff; }

```

```
100 | .badge-size      { background-color: #2563eb; color: #fff; }
101 |
102 | .torrent-title {
103 |     font-size: 0.88rem;
104 |     font-weight: 600;
105 |     margin-bottom: 4px;
106 |     color: var(--text-color);
107 | }
108 |
109 | .torrent-title a {
110 |     color: var(--nav-active);
111 |     text-decoration: none;
112 | }
113 |
114 | .torrent-title a:hover {
115 |     text-decoration: underline;
116 | }
117 |
118 | .torrent-desc {
119 |     font-size: 0.8rem;
120 |     color: var(--text-muted);
121 |     margin-bottom: 6px;
122 | }
123 |
```

variables.css

```
1 /**
2  * =====
3  * Process Name: Windows Web Styles - Variables Stylesheet
4  * =====
5  * Description:
6  *   Таблица стилей стиля и оформления интерфейса (variables).
7  *   Включает высококонтрастные токены для светлых тем (Light,
8  *   Brick) и тёмных тем (Dark, Terminal).
9  *
10 * Usage Examples:
11 *   HTML Link Tag:
12 *     <link rel="stylesheet" href="/windows/api/webgui/css/variables.css?v=20261004_v17">
13 *
14 * File: variables.css
15 * Project: ai-breadboard
16 * Package: windows/api/webgui/css
17 * Author: hypo69
18 * Copyright: © 2026 hypo69
19 * Updated: 2026-10-06 08:52:00
20 * =====
21 */
22 /* =====
23  THEME VARIABLES — AI Breadboard WebGUI
24  Дизайн-токены и CSS-переменные для всех тем оформления
25  ===== */
26
27 /* — 1. Светлая тема NightView (Комфортное чтение ночью / Warm Paper) — */
28 :root,
```

```
29 [data-theme="light"] {
30   /* Фоны (мягкие теплые бежево-серые поверхности в стиле Nigh
   | tView / бумага) */
31   --bg-color:          #fbfbfb;    /* чистый светлый фон страницы
   | */
32   --surface-1:         #f6f3ec;    /* карточки, панели, мягкая бу
   | мажная поверхность */
33   --surface-2:         #eee9e0;    /* хедеры карточек, тулбары, р
   | азделители */
34   --card-bg:           #f6f3ec;
35   --card-text:         #18181b;    /* четкий глубокий антрацитовы
   | й текст максимальной читаемости */
36   --nav-bg:            #f6f3ec;
37
38   /* Текст высокой четкости и контрастности для комфортного но
   | чного чтения */
39   --text-color:        #18181b;    /* глубокий четкий темный текс
   | т */
40   --text-muted:        #3f3f46;    /* темный нейтральный slate дл
   | я высокой читаемости подписей и параметров */
41   --nav-text:          #27272a;    /* контрастный текст навигации
   | */
42   --nav-active:        #1d4ed8;    /* четкий сапфирово-синий акце
   | нт */
43   --nav-hover-bg:     #eae4d8;
44
45   /* Линии и границы */
46   --border-color:      #cfc6b8;    /* четкий теплый контур */
47   --border-strong:     #9ca3af;    /* выразительные активные гран
   | ицы */
48   --border-subtle:     #ded7cb;    /* деликатные внутренние разде
   | лители */
49   --border-contrast:   #71717a;    /* контрастные рамки */
50   --divider-color:     #cfc6b8;
51   --modal-border:      #cfc6b8;
52
53   /* Тени (мягкие теплые) */
54   --shadow-xs:         0 1px 2px rgba(50, 40, 30, 0.05);
```

```
55 | --shadow-sm: 0 1px 4px rgba(50, 40, 30, 0.08);
56 | --shadow-md: 0 4px 12px rgba(50, 40, 30, 0.10);
57 | --shadow-lg: 0 8px 24px rgba(50, 40, 30, 0.13);
58 |
59 | /* Чат */
60 | --chat-user-bg: #2a6194;
61 | --chat-user-text: #fbf9f4;
62 | --chat-bot-bg: #eae4d8;
63 | --chat-bot-text: #2b2723;
64 |
65 | /* Скроллбар */
66 | --scrollbar-track: #ebe7df;
67 | --scrollbar-thumb: #c6bcad;
68 |
69 | /* Модальные окна */
70 | --modal-bg: #f6f3ec;
71 | --modal-text: #2b2723;
72 |
73 | /* Алерты – светлая тема NightView (пастельные комфортные то
на) */
74 | --alert-success-bg: #e7f2e9;
75 | --alert-success-border: #98c9a0;
76 | --alert-success-text: #185223;
77 | --alert-danger-bg: #f6e6e6;
78 | --alert-danger-border: #e19f9d;
79 | --alert-danger-text: #7e1f1f;
80 | --alert-warning-bg: #f8f0dd;
81 | --alert-warning-border: #dfc279;
82 | --alert-warning-text: #704705;
83 | --alert-info-bg: #e5eef7;
84 | --alert-info-border: #97bcd9;
85 | --alert-info-text: #184674;
86 |
87 | /* Геометрия и радиусы */
88 | --radius-xs: 4px;
89 | --radius-sm: 6px;
90 | --radius-md: 8px;
91 | --radius-lg: 12px;
```

```
92 |
93 | /* Типографика */
94 | --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
    | Helvetica Neue", Arial, sans-serif;
95 | --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
    | nsolas, "Courier New", monospace;
96 |
97 | /* Каскадные токены KPI-карточек */
98 | --kpi-bg:          var(--surface-1);
99 | --kpi-border:      var(--border-color);
100 | --kpi-accent:      var(--nav-active);
101 | --kpi-min-height: 72px;
102 | --kpi-padding:     0.65rem 0.9rem;
103 | --kpi-val-size:    1.25rem;
104 | --kpi-label-size:  0.72rem;
105 | --kpi-sub-size:    0.72rem;
106 |
107 | /* Каскадные токены панелей */
108 | --panel-bg:        var(--surface-1);
109 | --panel-border:    var(--border-color);
110 | --panel-header-bg: var(--surface-2);
111 |
112 | /* Каскадные токены суб-навигации */
113 | --subnav-btn-padding: 0.35rem 0.75rem;
114 | --subnav-font-size:  0.82rem;
115 |
116 | /* Статусные цвета */
117 | --status-success-bg:    #dbeee0;
118 | --status-success-text:  #185223;
119 | --status-success-border: #98c9a0;
120 |
121 | --status-danger-bg:     #f7dede;
122 | --status-danger-text:   #7e1f1f;
123 | --status-danger-border: #e19f9d;
124 |
125 | --status-warning-bg:    #f7ecd0;
126 | --status-warning-text:  #704705;
127 | --status-warning-border: #dfc279;
```

```
128 |
129 | --status-info-bg:          #dceaf6;
130 | --status-info-text:       #184674;
131 | --status-info-border:     #97bcd5;
132 | }
133 |
134 | /* — 2. Светлая кирпичная тема (Brick / Terracotta) —————
    | _____ */
135 | [data-theme="brick"] {
136 |   /* Фоны (теплые клинкерные и терракотовые поверхности без сл
    | епящего белого) */
137 |   --bg-color:              #f7f3ee;   /* теплый светлый фон клинкера
    | */
138 |   --surface-1:             #fcfaf6;   /* мягкие теплые карточки */
139 |   --surface-2:             #faefe7;   /* мягкий терракотово-песочный
    | хедер */
140 |   --card-bg:               #fcfaf6;
141 |   --card-text:             #2c221e;
142 |   --nav-bg:                #fcfaf6;
143 |
144 |   /* Текст */
145 |   --text-color:            #2c221e;   /* глубокий теплый антрацит */
146 |   --text-muted:            #7d6b63;   /* теплый приглушенный */
147 |   --nav-text:              #59473f;
148 |   --nav-active:            #b84228;   /* благородный терракотово-кир
    | пичный */
149 |   --nav-hover-bg:         #f7e6dc;
150 |
151 |   /* Линии и границы */
152 |   --border-color:          #dfcebf;   /* кирпично-песочный контур */
153 |   --border-strong:         #bc9e8c;
154 |   --border-subtle:         #ede2d8;
155 |   --border-contrast:       #8f5843;
156 |   --divider-color:         #dfcebf;
157 |   --modal-border:          #dfcebf;
158 |
159 |   /* Тени (теплые) */
160 |   --shadow-xs: 0 1px 2px rgba(130, 70, 50, 0.05);
```



```
161 | --shadow-sm: 0 1px 4px rgba(130, 70, 50, 0.08);
162 | --shadow-md: 0 4px 12px rgba(130, 70, 50, 0.11);
163 | --shadow-lg: 0 8px 24px rgba(130, 70, 50, 0.14);
164 |
165 | /* Чат */
166 | --chat-user-bg: #b84228; /* кирпичный акцент сообщений
    | */
167 | --chat-user-text: #ffffff;
168 | --chat-bot-bg: #f4ece6;
169 | --chat-bot-text: #2c221e;
170 |
171 | /* Скроллбар */
172 | --scrollbar-track: #f7f3ee;
173 | --scrollbar-thumb: #d1b8a8;
174 |
175 | /* Модальные окна */
176 | --modal-bg: #fcfaf6;
177 | --modal-text: #2c221e;
178 |
179 | /* Алерты – кирпичная тема */
180 | --alert-success-bg: #f2f9f4;
181 | --alert-success-border: #9ad5af;
182 | --alert-success-text: #1e6b37;
183 | --alert-danger-bg: #fdf1ee;
184 | --alert-danger-border: #f3b4a4;
185 | --alert-danger-text: #9e2b17;
186 | --alert-warning-bg: #fdf7ec;
187 | --alert-warning-border: #f6d28b;
188 | --alert-warning-text: #8a520f;
189 | --alert-info-bg: #f0f5fa;
190 | --alert-info-border: #a6cbe7;
191 | --alert-info-text: #1c5480;
192 |
193 | /* Геометрия */
194 | --radius-xs: 4px;
195 | --radius-sm: 6px;
196 | --radius-md: 8px;
197 | --radius-lg: 12px;
```

```
198
199  /* Типографика */
200  --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
Helvetica Neue", Arial, sans-serif;
201  --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
nsolas, "Courier New", monospace;
202
203  /* Каскадные токены KPI-карточек */
204  --kpi-bg:          var(--surface-1);
205  --kpi-border:      var(--border-color);
206  --kpi-accent:      var(--nav-active);
207  --kpi-min-height: 72px;
208  --kpi-padding:     0.65rem 0.9rem;
209  --kpi-val-size:    1.25rem;
210  --kpi-label-size:  0.72rem;
211  --kpi-sub-size:    0.72rem;
212
213  /* Каскадные токены панелей */
214  --panel-bg:        var(--surface-1);
215  --panel-border:    var(--border-color);
216  --panel-header-bg: var(--surface-2);
217
218  /* Каскадные токены суб-навигации */
219  --subnav-btn-padding: 0.35rem 0.75rem;
220  --subnav-font-size:  0.82rem;
221
222  /* Статусные цвета */
223  --status-success-bg:    #e2f3e8;
224  --status-success-text:  #1b6835;
225  --status-success-border: #8ecb9e;
226
227  --status-danger-bg:     #fae4df;
228  --status-danger-text:   #a32f1a;
229  --status-danger-border: #f1a999;
230
231  --status-warning-bg:    #faeed4;
232  --status-warning-text:  #94580b;
233  --status-warning-border: #f4c871;
```

```
234 |
235 | --status-info-bg:          #e2eef7;
236 | --status-info-text:       #195886;
237 | --status-info-border:     #9bc4e2;
238 | }
239 |
240 | /* — 3. Тёмная высококонтрастная тема (Dark High-Contrast) —
    | _____ */
241 | [data-theme="dark"] {
242 |     /* Фоны (глубокие контрастные поверхности) */
243 |     --bg-color:            #0b0f17; /* глубокий темный фон страниц
    | ы */
244 |     --surface-1:          #151b23; /* контрастные карточки */
245 |     --surface-2:          #1e2633; /* четкие хедеры карточек и ту
    | лбары */
246 |     --card-bg:            #151b23;
247 |     --card-text:          #f3f6fa; /* яркий чистый читаемый текст
    | */
248 |     --nav-bg:             #151b23;
249 |
250 |     /* Текст */
251 |     --text-color:         #f3f6fa; /* яркий текст высокой контраст
    | ности */
252 |     --text-muted:         #a3b1c2; /* легко различимый вторичный
    | текст */
253 |     --nav-text:           #b0bdcc;
254 |     --nav-active:         #58a6ff; /* яркий синий активный акцент
    | */
255 |     --nav-hover-bg:       #222d3d;
256 |
257 |     /* Линии и границы (высокая четкость) */
258 |     --border-color:       #3b4758; /* четко очерченные границы */
259 |     --border-strong:       #596b82; /* выразительные активные гран
    | ицы */
260 |     --border-subtle:       #252d38; /* мягкие внутренние разделите
    | ли */
261 |     --border-contrast:     #70849d; /* акцентные контуры */
262 |     --divider-color:       #3b4758;
```

```
263 | --modal-border:    #3b4758;
264 |
265 | /* Тени (глубокие, отделяющие слои) */
266 | --shadow-xs:  0 1px 3px rgba(0, 0, 0, 0.4);
267 | --shadow-sm:  0 2px 6px rgba(0, 0, 0, 0.45);
268 | --shadow-md:  0 6px 16px rgba(0, 0, 0, 0.55);
269 | --shadow-lg:  0 10px 30px rgba(0, 0, 0, 0.65);
270 |
271 | /* Чат */
272 | --chat-user-bg:    #1f6feb;
273 | --chat-user-text:  #ffffff;
274 | --chat-bot-bg:     #212836;
275 | --chat-bot-text:   #f3f6fa;
276 |
277 | /* Скроллбар */
278 | --scrollbar-track: #0b0f17;
279 | --scrollbar-thumb: #4e5b6e;
280 |
281 | /* Модальные окна */
282 | --modal-bg:        #151b23;
283 | --modal-text:      #f3f6fa;
284 |
285 | /* Алерты – тёмная контрастная тема */
286 | --alert-success-bg:    #123321;
287 | --alert-success-border: #3fb950;
288 | --alert-success-text:  #8df397;
289 | --alert-danger-bg:     #3b181d;
290 | --alert-danger-border: #f85149;
291 | --alert-danger-text:   #ff9b94;
292 | --alert-warning-bg:    #38290f;
293 | --alert-warning-border: #d29922;
294 | --alert-warning-text:  #f5cf6d;
295 | --alert-info-bg:       #122744;
296 | --alert-info-border:   #388bfd;
297 | --alert-info-text:     #a5d6ff;
298 |
299 | /* Геометрия */
300 | --radius-xs: 4px;
```

```
301 | --radius-sm: 6px;
302 | --radius-md: 8px;
303 | --radius-lg: 12px;
304 |
305 | /* Типографика */
306 | --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
    | Helvetica Neue", Arial, sans-serif;
307 | --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
    | nsolas, "Courier New", monospace;
308 |
309 | /* Каскадные токены KPI-карточек */
310 | --kpi-bg:          var(--surface-1);
311 | --kpi-border:      var(--border-color);
312 | --kpi-accent:      var(--nav-active);
313 | --kpi-min-height: 72px;
314 | --kpi-padding:     0.65rem 0.9rem;
315 | --kpi-val-size:    1.25rem;
316 | --kpi-label-size:  0.72rem;
317 | --kpi-sub-size:    0.72rem;
318 |
319 | /* Каскадные токены панелей */
320 | --panel-bg:        var(--surface-1);
321 | --panel-border:    var(--border-color);
322 | --panel-header-bg: var(--surface-2);
323 |
324 | /* Каскадные токены суб-навигации */
325 | --subnav-btn-padding: 0.35rem 0.75rem;
326 | --subnav-font-size:  0.82rem;
327 |
328 | /* Статусные цвета */
329 | --status-success-bg:    #174828;
330 | --status-success-text:  #9df4a6;
331 | --status-success-border: #2ea043;
332 |
333 | --status-danger-bg:     #52171c;
334 | --status-danger-text:   #ffb4ae;
335 | --status-danger-border: #da3633;
336 |
```

```

337 | --status-warning-bg:      #4d370f;
338 | --status-warning-text:    #fce49b;
339 | --status-warning-border:  #d29922;
340 |
341 | --status-info-bg:         #13375c;
342 | --status-info-text:       #a8d8ff;
343 | --status-info-border:     #1f6feb;
344 | }
345 |
346 | /* — 4. Тёмная тема «Терминал» (Terminal / Matrix / CRT Phosphor) — */
347 | [data-theme="terminal"] {
348 |     /* Фоны (глубокий смоляной черный с зеленым отливом) */
349 |     --bg-color:            #060a06;    /* фоновый CRT-черный */
350 |     --surface-1:           #0b120c;    /* карточки терминала */
351 |     --surface-2:           #101c12;    /* шапки с изумрудным свечением */
352 |     --card-bg:              #0b120c;
353 |     --card-text:            #d8ffd8;    /* люминесцентный терминальный
354 |     --nav-bg:                #0b120c;
355 |
356 |     /* Текст */
357 |     --text-color:           #d8ffd8;    /* неоновый зеленый фосфорный
358 |     --text-muted:           #599863;    /* приглушенный фосфор */
359 |     --nav-text:              #6cb878;
360 |     --nav-active:           #00ff66;    /* электрический неоново-зеленый */
361 |     --nav-hover-bg:         #152618;
362 |
363 |     /* Линии и границы (терминальный контур со свечением) */
364 |     --border-color:         #1a4223;    /* глубокий зеленый контур */
365 |     --border-strong:        #00ff66;    /* неоновая рамка активности */
366 |     --border-subtle:        #0f2b14;
367 |     --border-contrast:      #39ff14;
368 |     --divider-color:        #1a4223;

```

```
369 | --modal-border:    #00ff66;
370 |
371 | /* Тени (неоновое свечение фосфора) */
372 | --shadow-xs:  0 0 3px rgba(0, 255, 102, 0.15);
373 | --shadow-sm:  0 0 6px rgba(0, 255, 102, 0.2);
374 | --shadow-md:  0 0 14px rgba(0, 255, 102, 0.25);
375 | --shadow-lg:  0 0 28px rgba(0, 255, 102, 0.35);
376 |
377 | /* Чат */
378 | --chat-user-bg:    #008033;    /* зеленый фосфорный пузырь */
379 | --chat-user-text:  #ffffff;
380 | --chat-bot-bg:     #101c12;
381 | --chat-bot-text:   #d8ffd8;
382 |
383 | /* Скроллбар */
384 | --scrollbar-track: #060a06;
385 | --scrollbar-thumb: #1b4b24;
386 |
387 | /* Модальные окна */
388 | --modal-bg:        #0b120c;
389 | --modal-text:      #d8ffd8;
390 |
391 | /* Алерты – терминальная тема */
392 | --alert-success-bg:    #0c2412;
393 | --alert-success-border: #00ff66;
394 | --alert-success-text:  #4dff88;
395 | --alert-danger-bg:     #2b0d11;
396 | --alert-danger-border: #ff3344;
397 | --alert-danger-text:   #ff808b;
398 | --alert-warning-bg:    #2b2207;
399 | --alert-warning-border: #ffb000;
400 | --alert-warning-text:  #ffd266;
401 | --alert-info-bg:       #0a1d26;
402 | --alert-info-border:   #00ddff;
403 | --alert-info-text:     #66ecff;
404 |
405 | /* Геометрия */
406 | --radius-xs: 2px;
```

```
407 | --radius-sm: 4px;
408 | --radius-md: 6px;
409 | --radius-lg: 8px;
410 |
411 | /* Типографика (с моноширинным терминальным акцентом) */
412 | --font-sans: "Consolas", "Cascadia Code", "Courier New", ui-
    | monospace, monospace;
413 | --font-mono: "Consolas", "Cascadia Code", "Courier New", ui-
    | monospace, monospace;
414 |
415 | /* Каскадные токены KPI-карточек */
416 | --kpi-bg:          var(--surface-1);
417 | --kpi-border:      var(--border-color);
418 | --kpi-accent:      var(--nav-active);
419 | --kpi-min-height: 72px;
420 | --kpi-padding:     0.65rem 0.9rem;
421 | --kpi-val-size:    1.25rem;
422 | --kpi-label-size:  0.72rem;
423 | --kpi-sub-size:    0.72rem;
424 |
425 | /* Каскадные токены панелей */
426 | --panel-bg:        var(--surface-1);
427 | --panel-border:    var(--border-color);
428 | --panel-header-bg: var(--surface-2);
429 |
430 | /* Каскадные токены суб-навигации */
431 | --subnav-btn-padding: 0.35rem 0.75rem;
432 | --subnav-font-size:  0.82rem;
433 |
434 | /* Статусные цвета */
435 | --status-success-bg:    #0c2914;
436 | --status-success-text:  #39ff14;
437 | --status-success-border: #00ff66;
438 |
439 | --status-danger-bg:     #3b0f14;
440 | --status-danger-text:   #ff6675;
441 | --status-danger-border: #ff3344;
442 |
```



```
443 | --status-warning-bg:      #382c09;
444 | --status-warning-text:    #ffc83b;
445 | --status-warning-border:  #ffb000;
446 |
447 | --status-info-bg:         #0c2533;
448 | --status-info-text:       #33e0ff;
449 | --status-info-border:     #00ccff;
450 | }
451 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/defender_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/defender_tab/index.html
15 |        " class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/defender_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-04 12:48:00
23 |=====
24 |=====
25 -->
26
27 <!-- Microsoft Defender & AI Security Diagnostic Center Tab --
28 >
29 <style>
30 |.def-container {
31 |    padding: 0.5rem 0.75rem;
32 |}
33 |.def-badge-active { background: var(--status-success-bg); co
34 |lor: var(--status-success-text); border: 1px solid var(--statu
35 |s-success-border); }
```

```

30 | .def-badge-disabled { background: var(--status-danger-bg); c
    | olor: var(--status-danger-text); border: 1px solid var(--statu
    | s-danger-border); }
31 | .def-badge-audit { background: var(--status-warning-bg); col
    | or: var(--status-warning-text); border: 1px solid var(--status
    | -warning-border); }
32 | .def-badge-warn { background: var(--status-warning-bg); colo
    | r: var(--status-warning-text); border: 1px solid var(--status-
    | warning-border); }
33 | .def-badge-critical { background: var(--status-danger-bg); c
    | olor: var(--status-danger-text); border: 1px solid var(--statu
    | s-danger-border); font-weight: bold; }
34 | </style>
35 |
36 | <div class="def-container theme-defender">
37 |   <!-- Top Stat Cards -->
38 |   <div class="row g-2 mb-2">
39 |     <div class="col-6 col-md-3 col-xl">
40 |       <div class="def-card">
41 |         <div class="def-card-label"><span>Индекс защиты</span>
    |   </div>
42 |         <div class="def-card-value" id="def-score-val">--/100<
    | /div>
43 |         <div class="def-card-sub" id="def-score-sub">Загрузка.
    | ..</div>
44 |       </div>
45 |     </div>
46 |     <div class="col-6 col-md-3 col-xl">
47 |       <div class="def-card">
48 |         <div class="def-card-label"><span>Real-time Shield</sp
    | an> </div>
49 |         <div class="def-card-value text-success" id="def-realt
    | ime-val">--</div>
50 |         <div class="def-card-sub" id="def-realtime-sub">MsMpEn
    | g.exe</div>
51 |       </div>
52 |     </div>
53 |   <div class="col-6 col-md-3 col-xl">

```

```

54 |         <div class="def-card">
55 |             <div class="def-card-label"><span>Cloud & Tamper</span
56 |             <div class="def-card-value" id="def-cloud-val">--</div
57 |             <div class="def-card-sub" id="def-cloud-sub">MAPS Prot
58 |             </div>
59 |         </div>
60 |         <div class="col-6 col-md-3 col-xl">
61 |             <div class="def-card">
62 |                 <div class="def-card-label"><span>PUA & Ransomware</sp
63 |                 <div class="def-card-value" id="def-pua-cfa-val">--</d
64 |                 <div class="def-card-sub" id="def-pua-cfa-sub">CFA Fol
65 |                 </div>
66 |             </div>
67 |             <div class="col-6 col-md-3 col-xl">
68 |                 <div class="def-card">
69 |                     <div class="def-card-label"><span>Исключения</span> <
70 |                     <div class="def-card-value text-warning" id="def-exclu
71 |                     <div class="def-card-sub" id="def-exclusions-sub">0 по
72 |                     </div>
73 |                 </div>
74 |                 <div class="col-6 col-md-3 col-xl">
75 |                     <div class="def-card">
76 |                         <div class="def-card-label"><span>Активные угрозы</spa
77 |                         <div class="def-card-value text-danger" id="def-threat
78 |                         <div class="def-card-sub" id="def-threats-sub">Каранти
79 |                         </div>

```

```
80     </div>
81 </div>
82
83 <!-- Action Bar -->
84 <div class="def-panel mb-2 p-2 d-flex align-items-center justify-content-between flex-wrap gap-2">
85     <div class="d-flex align-items-center gap-2 flex-wrap">
86         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1" id="btn-def-quick-scan">
87             <i class="bi bi-play-circle-fill"></i> <span>Быстрое сканирование</span>
88         </button>
89         <button class="btn btn-sm btn-outline-primary d-flex align-items-center gap-1" id="btn-def-full-scan">
90             <i class="bi bi-shield-shaded"></i> <span>Полное сканирование</span>
91         </button>
92         <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1" id="btn-def-update-sigs">
93             <i class="bi bi-cloud-arrow-down-fill"></i> <span>Обновить сигнатуры</span>
94         </button>
95         <button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1" id="btn-def-ai-diag">
96             <i class="bi bi-cpu-fill"></i> <span>AI Диагностика</span>
97         </button>
98     </div>
99     <div class="d-flex align-items-center gap-2">
100         <span class="badge bg-dark border border-secondary" id="def-engine-ver-badge">Движок: --</span>
101         <span class="badge bg-dark border border-secondary" id="def-sigs-ver-badge">Базы: --</span>
102         <button class="btn btn-sm btn-outline-secondary" id="btn-def-refresh" title="Обновить">
103             <i class="bi bi-arrow-clockwise"></i>
104         </button>
105     </div>
```

```
106 | </div>
107 |
108 | <!-- Scan Execution Progress / Notification Alert -->
109 | <div id="def-alert-box" class="d-none alert alert-info py-2
    | px-3 mb-2 small"></div>
110 |
111 | <!-- Sub Navigation Tabs -->
112 | <ul class="nav nav-tabs mb-2 border-secondary" id="defSubTab
    | s" role="tablist">
113 |     <li class="nav-item" role="presentation">
114 |         <button class="nav-link active py-1 px-3 small" id="def-
    | tab-overview-btn" data-bs-toggle="tab" data-bs-target="#def-pa
    | ne-overview" type="button"> Обзор и Службы</button>
115 |     </li>
116 |     <li class="nav-item" role="presentation">
117 |         <button class="nav-link py-1 px-3 small" id="def-tab-asr
    | -btn" data-bs-toggle="tab" data-bs-target="#def-pane-asr" type
    | ="button"> Правила ASR</button>
118 |     </li>
119 |     <li class="nav-item" role="presentation">
120 |         <button class="nav-link py-1 px-3 small" id="def-tab-cfa
    | -btn" data-bs-toggle="tab" data-bs-target="#def-pane-cfa" type
    | ="button"> Ransomware / CFA</button>
121 |     </li>
122 |     <li class="nav-item" role="presentation">
123 |         <button class="nav-link py-1 px-3 small" id="def-tab-exc
    | lusions-btn" data-bs-toggle="tab" data-bs-target="#def-pane-ex
    | clusions" type="button"> Исключения</button>
124 |     </li>
125 |     <li class="nav-item" role="presentation">
126 |         <button class="nav-link py-1 px-3 small" id="def-tab-thr
    | eats-btn" data-bs-toggle="tab" data-bs-target="#def-pane-threa
    | ts" type="button"> Угрозы и Карантин</button>
127 |     </li>
128 |     <li class="nav-item" role="presentation">
129 |         <button class="nav-link py-1 px-3 small" id="def-tab-pro
    | cs-btn" data-bs-toggle="tab" data-bs-target="#def-pane-procs"
    | type="button"> Дерево Процессов / Fileless</button>
```

```

130     </li>
131     <li class="nav-item" role="presentation">
132         <button class="nav-link py-1 px-3 small" id="def-tab-eve
nts-btn" data-bs-toggle="tab" data-bs-target="#def-pane-events
" type="button">🛡 Журнал Событий</button>
133     </li>
134     <li class="nav-item" role="presentation">
135         <button class="nav-link py-1 px-3 small text-warning fw-
semibold" id="def-tab-diag-btn" data-bs-toggle="tab" data-bs-t
arget="#def-pane-diag" type="button">🛡 AI Диагностика</button
>
136     </li>
137 </ul>
138
139 <!-- Tab Content Panes -->
140 <div class="tab-content" id="defTabPanes">
141
142     <!-- 1. Overview & Services Pane -->
143     <div class="tab-pane fade show active" id="def-pane-overvi
ew" role="tabpanel">
144         <div class="row g-2">
145             <div class="col-md-7">
146                 <div class="def-panel h-100">
147                     <div class="def-panel-header">
148                         <span>Эшелоны Защиты Microsoft Defender</span>
149                     </div>
150                     <div class="table-responsive p-0">
151                         <table class="table table-dark table-sm table-ho
ver mb-0 align-middle small">
152                             <thead>
153                                 <tr class="text-secondary">
154                                     <th>Компонент</th>
155                                     <th class="text-center">Статус</th>
156                                     <th>Описание</th>
157                                 </tr>
158                             </thead>
159                             <tbody id="def-shields-tbody">
160                                 <tr><td colspan="3" class="text-center text-

```

```

160 muted py-3">Загрузка данных...</td></tr>
161         </tbody>
162     </table>
163 </div>
164 </div>
165 </div>
166 <div class="col-md-5">
167     <div class="def-panel h-100">
168         <div class="def-panel-header">
169             <span>Системные Службы и Процессы</span>
170         </div>
171         <div class="table-responsive p-0">
172             <table class="table table-dark table-sm table-ho
173 ver mb-0 align-middle small">
174                 <thead>
175                     <tr class="text-secondary">
176                         <th>Процесс</th>
177                         <th class="text-center">Статус</th>
178                         <th class="text-end">PID</th>
179                         <th class="text-end">Память</th>
180                     </tr>
181                 </thead>
182                 <tbody id="def-services-tbody">
183                     <tr><td colspan="4" class="text-center text-
184 muted py-3">Загрузка данных...</td></tr>
185                     </tbody>
186                 </table>
187             </div>
188         </div>
189     </div>
190
191 <!-- 2. ASR Rules Pane -->
192 <div class="tab-pane fade" id="def-pane-asr" role="tabpanel
193 l">
194     <div class="def-panel">
195         <div class="def-panel-header">

```



```

195 |         <span>Правила Attack Surface Reduction (ASR)</span>
196 |         <span class="badge bg-secondary" id="def-asr-count-b
| adge">Всего: 0</span>
197 |     </div>
198 |     <div class="table-responsive p-0">
199 |         <table class="table table-dark table-sm table-hover
| mb-0 align-middle small">
200 |             <thead>
201 |                 <tr class="text-secondary">
202 |                     <th>Правило ASR</th>
203 |                     <th>Категория</th>
204 |                     <th class="text-center">Состояние</th>
205 |                     <th>Рекомендация</th>
206 |                     <th>GUID</th>
207 |                 </tr>
208 |             </thead>
209 |             <tbody id="def-asr-tbody">
210 |                 <tr><td colspan="5" class="text-center text-mute
| d py-3">Загрузка правил ASR...</td></tr>
211 |             </tbody>
212 |         </table>
213 |     </div>
214 | </div>
215 | </div>
216 |
217 | <!-- 3. CFA (Ransomware) Pane -->
218 | <div class="tab-pane fade" id="def-pane-cfa" role="tabpane
| l">
219 |     <div class="row g-2">
220 |         <div class="col-md-6">
221 |             <div class="def-panel h-100">
222 |                 <div class="def-panel-header">
223 |                     <span>Защищенные Каталоги (Protected Folders)</s
| pan>
224 |                 </div>
225 |                 <div class="p-2" id="def-cfa-folders-list">
226 |                     <div class="text-muted small">Загрузка...</div>
227 |                 </div>

```

[illegible]

```

261         </tr>
262     </thead>
263     <tbody id="def-exclusions-tbody">
264         <tr><td colspan="4" class="text-center text-mute
d py-3">Загрузка исключений...</td></tr>
265     </tbody>
266 </table>
267 </div>
268 </div>
269 </div>
270
271 <!-- 5. Threats History Pane -->
272 <div class="tab-pane fade" id="def-pane-threats" role="tab
panel">
273     <div class="def-panel">
274         <div class="def-panel-header">
275             <span>Журнал Обнаружения Угроз и Карантин</span>
276             <span class="badge bg-danger" id="def-threats-count-
badge">0 записей</span>
277         </div>
278         <div class="table-responsive p-0">
279             <table class="table table-dark table-sm table-hover
mb-0 align-middle small">
280                 <thead>
281                     <tr class="text-secondary">
282                         <th>Угроза</th>
283                         <th>Категория</th>
284                         <th class="text-center">Опасность</th>
285                         <th>Статус</th>
286                         <th>Время</th>
287                         <th>Ресурсы</th>
288                     </tr>
289                 </thead>
290                 <tbody id="def-threats-tbody">
291                     <tr><td colspan="6" class="text-center text-mute
d py-3">Угроз не обнаружено.</td></tr>
292                 </tbody>
293             </table>

```

```

294 |         </div>
295 |     </div>
296 | </div>
297 |
298 | <!-- 6. Process Trees / Fileless Activity Pane -->
299 | <div class="tab-pane fade" id="def-pane-procs" role="tabpa
300 | nel">
301 |     <div class="def-panel">
302 |         <div class="def-panel-header">
303 |             <span>Инспекция Цепочек Процессов и Индикаторы Filel
304 | ess-атак</span>
305 |             <span class="badge bg-secondary" id="def-procs-count
306 | -badge">0 аномалий</span>
307 |         </div>
308 |         <div class="table-responsive p-0">
309 |             <table class="table table-dark table-sm table-hover
310 | mb-0 align-middle small">
311 |                 <thead>
312 |                     <tr class="text-secondary">
313 |                         <th>Родительский Процесс</th>
314 |                         <th>Дочерний Процесс</th>
315 |                         <th>PID</th>
316 |                         <th>Опасность</th>
317 |                         <th>Причина срабатывания</th>
318 |                     </tr>
319 |                 </thead>
320 |                 <tbody id="def-procs-tbody">
321 |                     <tr><td colspan="5" class="text-center text-mute
322 | d py-3">Подозрительных цепочек процессов не обнаружено.</td></
323 | tr>
324 |                 </tbody>
325 |             </table>
326 |         </div>
327 |     </div>
328 | </div>
329 |
330 | <!-- 7. Operational Events Pane -->
331 | <div class="tab-pane fade" id="def-pane-events" role="tabp

```

```

anel">
326     <div class="def-panel">
327         <div class="def-panel-header">
328             <span>Журнал Microsoft-Windows-Windows Defender/Oper
ational</span>
329             <span class="badge bg-secondary" id="def-events-coun
t-badge">0 событий</span>
330         </div>
331         <div class="table-responsive p-0" style="max-height: c
alc(100vh - 240px); min-height: 200px; overflow-y: auto;">
332             <table class="table table-dark table-sm table-hover
mb-0 align-middle small">
333                 <thead class="sticky-top" style="position: sticky;
top: 0; z-index: 2;">
334                     <tr class="text-secondary bg-dark">
335                         <th class="bg-dark">Event ID</th>
336                         <th class="bg-dark">Время</th>
337                         <th class="bg-dark">Уровень</th>
338                         <th class="bg-dark">Категория</th>
339                         <th class="bg-dark">Сообщение</th>
340                     </tr>
341                 </thead>
342                 <tbody id="def-events-tbody">
343                     <tr><td colspan="5" class="text-center text-mute
d py-3">Загрузка журнала событий...</td></tr>
344                 </tbody>
345             </table>
346         </div>
347     </div>
348 </div>
349
350 <!-- 8. AI Diagnostics Pane -->
351 <div class="tab-pane fade" id="def-pane-diag" role="tabpan
el">
352     <div class="row g-2">
353         <div class="col-md-4">
354             <div class="def-panel h-100 p-3 text-center d-flex f
lex-column justify-content-center">

```

```
355 |         <div class="text-secondary small mb-1">Security Po
| sture Score</div>
356 |         <div class="display-4 fw-bold text-success" id="de
| f-diag-score-display">--</div>
357 |         <div class="fw-semibold mt-2" id="def-diag-summary
| -display">Анализ состояния...</div>
358 |     </div>
359 | </div>
360 | <div class="col-md-8">
361 |     <div class="def-panel h-100 p-3">
362 |         <h6 class="text-warning fw-bold mb-2">⚠ Рекоменда
| ции AI по Усилению Защиты:</h6>
363 |         <div id="def-diag-recs-container" class="small">
364 |             <div class="text-muted">Загрузка рекомендаций...
| </div>
365 |         </div>
366 |         <hr class="border-secondary my-2">
367 |         <h6 class="text-danger fw-bold mb-2">⚠ Критические
| уязвимости и риски:</h6>
368 |         <div id="def-diag-criticals-container" class="smal
| l">
369 |             <div class="text-muted">Нет критических уязвимос
| тей.</div>
370 |         </div>
371 |     </div>
372 | </div>
373 | </div>
374 | </div>
375 |
376 | </div>
377 | </div>
378 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Defender Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/defender_tab/main.js?v
11  *     =20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/defender_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 02:00:00
19  * =====
20  */
21 // Microsoft Defender Security Center Frontend Controller
22 (function() {
23   'use strict';
24
25   let defenderData = null;
26
27   function showAlert(msg, isError = false) {
28     const box = document.getElementById('def-alert-box');
29     if (!box) return;
30     box.className = `alert py-2 px-3 mb-2 small ${isError ? 'a
31     lert-danger' : 'alert-info'}`;
32     box.textContent = msg;
```

```
32 |     box.classList.remove('d-none');
33 |     setTimeout(() => {
34 |         box.classList.add('d-none');
35 |     }, 8000);
36 | }
37 |
38 | async function fetchJSON(url, options = {}) {
39 |     if (window.api && typeof window.api.fetch === 'function')
40 |     {
41 |         return await window.api.fetch(url, options);
42 |     }
43 |     const res = await fetch(url, options);
44 |     if (!res.ok) {
45 |         throw new Error(`HTTP ${res.status}: ${res.statusText}`)
46 |     };
47 |     return await res.json();
48 | }
49 |
50 | async function loadDefenderStatus() {
51 |     try {
52 |         const status = await fetchJSON('/api/v1/defender/status'
53 | );
54 |         defenderData = status;
55 |
56 |         // Badges in header
57 |         const engBadge = document.getElementById('def-engine-ver
58 | -badge');
59 |         if (engBadge) engBadge.textContent = `Движок: ${status.e
60 | ngine_version || 'N/A'}`;
61 |         const sigsBadge = document.getElementById('def-sigs-ver-
62 | badge');
63 |         if (sigsBadge) sigsBadge.textContent = `Базы: ${status.a
64 | ntivirus_signature_version || 'N/A'}`;
65 |
66 |         // Stat Cards
67 |         const rtVal = document.getElementById('def-realtime-val'
68 | );
69 |     } catch (err) {
70 |         console.error('Error loading defender status:', err);
71 |     }
72 | }
```



```

62 |         if (rtVal) {
63 |             rtVal.textContent = status.real_time_protection_enabled ? 'ВКЛ' : 'Выкл';
64 |             rtVal.className = `def-card-value ${status.real_time_protection_enabled ? 'text-success' : 'text-danger'}`;
65 |         }
66 |
67 |         const cloudVal = document.getElementById('def-cloud-val');
68 |         if (cloudVal) {
69 |             cloudVal.textContent = status.cloud_protection_enabled ? 'АКТИВНА' : 'Выкл';
70 |             cloudVal.className = `def-card-value ${status.cloud_protection_enabled ? 'text-success' : 'text-warning'}`;
71 |         }
72 |
73 |         const puaCfaVal = document.getElementById('def-pua-cfa-val');
74 |         if (puaCfaVal) {
75 |             const puaText = status.pua_protection_enabled ? 'PUA: ' : 'PUA:  ';
76 |             const cfaText = status.controlled_folder_access_enabled ? 'CFA: ' : 'CFA:  ';
77 |             puaCfaVal.textContent = `${puaText} | ${cfaText}`;
78 |             puaCfaVal.className = 'def-card-value small text-info';
79 |         }
80 |
81 |         // Shields Table
82 |         const shieldsTbody = document.getElementById('def-shields-tbody');
83 |         if (shieldsTbody) {
84 |             const shields = [
85 |                 { name: 'Real-time Protection', enabled: status.real_time_protection_enabled, desc: 'Постоянный перехват и проверка файлов при создании и обращении' },
86 |                 { name: 'Cloud Protection (MAPS)', enabled: status.cloud_protection_enabled, desc: 'Мгновенная блокировка новых уг

```

```

    роз через облачный ИИ Microsoft (${status.cloud_block_level})`
    },
87     { name: 'Behavior Monitoring', enabled: status.behav
    ior_monitor_enabled, desc: 'Анализ поведения запущенных процес
    сов и цепочек активности' },
88     { name: 'IOAV Protection', enabled: status.ioav_prot
    ection_enabled, desc: 'Проверка файлов, загружаемых из браузер
    а, почты и мессенджеров' },
89     { name: 'AMSI Script Scanning', enabled: status.scri
    pt_scanning_enabled, desc: 'Инспекция скриптов PowerShell, VBS
    cript, JavaScript до их выполнения' },
90     { name: 'Tamper Protection', enabled: status.tamper_
    protection_enabled, desc: 'Защита настроек антивируса от отклю
    чения вредоносным ПО' },
91     { name: 'PUA Protection', enabled: status.pua_prot
    ection_enabled, desc: 'Блокировка нежелательного ПО, рекламных и
    нсталляторов и майнеров' },
92     { name: 'Controlled Folder Access', enabled: status.
    controlled_folder_access_enabled, desc: 'Защита папок Документ
    ы/Рабочий стол от шифрования Ransomware' },
93     { name: 'Network Protection', enabled: status.networ
    k_protection_enabled, desc: 'Блокировка сетевых соединений с в
    редоносными доменами и IP' },
94 ];
95
96 shieldsTbody.innerHTML = shields.map(s => `
97     <tr>
98         <td class="fw-semibold text-light">${s.name}</td>
99         <td class="text-center">
100             <span class="badge ${s.enabled ? 'def-badge-acti
    ve' : 'def-badge-disabled'}">
101                 ${s.enabled ? 'ВКЛЮЧЕНО' : 'ОТКЛЮЧЕНО'}
102             </span>
103         </td>
104         <td class="text-secondary">${s.desc}</td>
105     </tr>
106 `).join('');
107 }

```

```

108
109     // Services Table
110     const servicesTbody = document.getElementById('def-servi
ces-tbody');
111     if (servicesTbody && status.services) {
112         servicesTbody.innerHTML = status.services.map(svc => `
113             <tr>
114                 <td>
115                     <div class="fw-semibold text-light">${svc.name}<
/div>
116                     <div class="text-secondary" style="font-size: 0.
68rem;">${svc.display_name}</div>
117                 </td>
118                 <td class="text-center">
119                     <span class="badge ${svc.running ? 'bg-success'
: 'bg-secondary'}">
120                         ${svc.running ? 'АКТИВЕН' : 'СТОП'}
121                     </span>
122                 </td>
123                 <td class="text-end text-secondary">${svc.pid || '
-'}</td>
124                 <td class="text-end font-monospace">${svc.running
? svc.memory_mb + ' MB' : '-'}</td>
125             </tr>
126         `).join('');
127     }
128
129     } catch (err) {
130         console.error('[DefenderTab] Error loading status:', err
);
131         showAlert(`Ошибка загрузки статуса Defender: ${err.messa
ge}`, true);
132     }
133 }
134
135 async function loadASRRules() {
136     try {
137         const rules = await fetchJSON('/api/v1/defender/asr');

```

```

138     const tbody = document.getElementById('def-asr-tbody');
139     const badge = document.getElementById('def-asr-count-bad
140     ge');
141     if (badge) badge.textContent = `Bcero: ${rules.length}`;
142     if (tbody) {
143         tbody.innerHTML = rules.map(r => {
144             let stateBadge = '<span class="badge bg-secondary">H
145             е задано</span>';
146             if (r.state === 'enabled') stateBadge = '<span class
147             ="badge def-badge-active">Блокировка (Block)</span>';
148             else if (r.state === 'audit') stateBadge = '<span cl
149             ass="badge def-badge-audit">Аудит (Audit)</span>';
150             else if (r.state === 'warn') stateBadge = '<span cla
151             ss="badge def-badge-warn">Предупреждение</span>';
152             else if (r.state === 'disabled') stateBadge = '<span
153             class="badge def-badge-disabled">Отключено</span>';
154             return `
155                 <tr>
156                 <td>
157                     <div class="fw-semibold text-light">${r.name}<
158                     /div>
159                     <div class="text-secondary" style="font-size:
160                     0.72rem;">${r.description}</div>
161                 </td>
162                 <td><span class="badge bg-dark border border-sec
163                 ondary">${r.category}</span></td>
164                 <td class="text-center">${stateBadge}</td>
165                 <td class="text-secondary small">${r.recommendat
166                 ion || '-'}</td>
167                 <td class="text-secondary font-monospace" style=
168                 "font-size: 0.68rem;">${r.guid}</td>
169                 </tr>
170             `;
171         }).join('');
172     }
173 } catch (err) {

```

```

165     console.error('[DefenderTab] Error loading ASR rules:',
166     err);
167 }
168
169 async function loadCFA() {
170     try {
171         const cfa = await fetchJSON('/api/v1/defender/cfa');
172         const foldersList = document.getElementById('def-cfa-folders-list');
173         const appsList = document.getElementById('def-cfa-apps-list');
174
175         if (foldersList) {
176             if (cfa.protected_folders && cfa.protected_folders.length > 0) {
177                 foldersList.innerHTML = `
178                 <ul class="list-group list-group-flush bg-transparent">
179                     ${cfa.protected_folders.map(f => `
180                     <li class="list-group-item bg-transparent text-light border-secondary py-1 px-0 font-monospace small">
181                         □ ${f}
182                     </li>
183                     `).join('')}
184                 </ul>
185                 `;
186             } else {
187                 foldersList.innerHTML = '<div class="text-muted small">Нет настроенных защищенных папок.</div>';
188             }
189         }
190
191         if (appsList) {
192             if (cfa.allowed_applications && cfa.allowed_applications.length > 0) {
193                 appsList.innerHTML = `
194                 <ul class="list-group list-group-flush bg-transparent">

```

```

ent">
195         ${cfa.allowed_applications.map(a => `
196             <li class="list-group-item bg-transparent text
-   light border-secondary py-1 px-0 font-monospace small">
197                 □ ${a}
198             </li>
199         `).join('')}
200     </ul>
201     `;
202     } else {
203         appsList.innerHTML = '<div class="text-muted small">
Список доверенных приложений пуст.</div>';
204     }
205 }
206 } catch (err) {
207     console.error('[DefenderTab] Error loading CFA:', err);
208 }
209 }
210
211 async function loadExclusions() {
212     try {
213         const rep = await fetchJSON('/api/v1/defender/exclusions
');
214
215         const countVal = document.getElementById('def-exclusions
-val');
216         const countSub = document.getElementById('def-exclusions
-sub');
217         const summaryBadge = document.getElementById('def-exc-su
mmmary-badge');
218         const recBox = document.getElementById('def-exc-recommen
dation-box');
219         const tbody = document.getElementById('def-exclusions-tb
ody');
220
221         if (countVal) countVal.textContent = rep.total_exclusion
s;
222         if (countSub) countSub.textContent = `${rep.suspicious_c

```

```

ount} подозрительных`;
223     if (summaryBadge) summaryBadge.textContent = `${rep.total_exclusions} исключений (${rep.suspicious_count} риск)`;
224     if (recBox) recBox.textContent = rep.summary_recommendation;
225
226     const allItems = [...rep.path_exclusions, ...rep.extension_exclusions, ...rep.process_exclusions];
227
228     if (tbody) {
229         if (allItems.length === 0) {
230             tbody.innerHTML = '<tr><td colspan="4" class="text-center text-muted py-3">Исключения отсутствуют. Защита работает без пропусков.</td></tr>';
231         } else {
232             tbody.innerHTML = allItems.map(itm => {
233                 let riskBadge = '<span class="badge bg-success">Безопасно</span>';
234                 if (itm.risk_level === 'critical') riskBadge = '<span class="badge def-badge-critical">КРИТИЧЕСКИЙ</span>';
235                 else if (itm.risk_level === 'high') riskBadge = '<span class="badge def-badge-disabled">ВЫСОКИЙ</span>';
236                 else if (itm.risk_level === 'medium') riskBadge = '<span class="badge def-badge-warn">СРЕДНИЙ</span>';
237
238                 return `
239                     <tr>
240                         <td><span class="badge bg-dark border border-secondary">${itm.type.toUpperCase()}</span></td>
241                         <td class="font-monospace text-light">${itm.value}</td>
242                         <td class="text-center">${riskBadge}</td>
243                         <td class="text-secondary small">${itm.risk_reason}</td>
244                     </tr>
245                 `;
246             }).join('');
247         }

```

```

248     }
249   } catch (err) {
250     console.error('[DefenderTab] Error loading exclusions:',
251       err);
252   }
253
254   async function loadThreats() {
255     try {
256       const threats = await fetchJSON('/api/v1/defender/threats?limit=50');
257       const countBadge = document.getElementById('def-threats-count-badge');
258       const countVal = document.getElementById('def-threats-val');
259       const countSub = document.getElementById('def-threats-sub');
260       const tbody = document.getElementById('def-threats-tbody');
261
262       if (countBadge) countBadge.textContent = `${threats.length} записей`;
263       if (countVal) countVal.textContent = threats.length;
264
265       const quarantined = threats.filter(t => t.status.toLowerCase().includes('quarantine')).length;
266       if (countSub) countSub.textContent = `Карантин: ${quarantined}`;
267
268       if (tbody) {
269         if (threats.length === 0) {
270           tbody.innerHTML = '<tr><td colspan="6" class="text-center text-muted py-3">Активных и архивных угроз не зафиксировано.</td></tr>';
271         } else {
272           tbody.innerHTML = threats.map(t => `
273             <tr>
274               <td class="fw-semibold text-danger">${t.threat_n

```



```

    ame}</td>
275         <td><span class="badge bg-dark border border-sec
    ondary">${t.category}</span></td>
276         <td class="text-center"><span class="badge bg-wa
    rning text-dark">${t.severity.toUpperCase()}</span></td>
277         <td><span class="badge bg-success">${t.status}</
    span></td>
278         <td class="text-secondary small">${t.last_detect
    ion_time || t.initial_detection_time || '-'}</td>
279         <td class="text-secondary font-monospace small">
    ${t.resources || []}.join(', ')}</td>
280     </tr>
281     `).join('');
282     }
283     }
284     } catch (err) {
285         console.error('[DefenderTab] Error loading threats:', er
    r);
286     }
287     }
288
289     async function loadProcesses() {
290         try {
291             const chains = await fetchJSON('/api/v1/defender/process
    -tree');
292             const badge = document.getElementById('def-procs-count-b
    adge');
293             const tbody = document.getElementById('def-procs-tbody')
    ;
294
295             if (badge) badge.textContent = `${chains.length} аномали
    й`;
296
297             if (tbody) {
298                 if (chains.length === 0) {
299                     tbody.innerHTML = '<tr><td colspan="5" class="text-c
    enter text-muted py-3">Подозрительных связей процессов и filel
    ess индикаторов не обнаружено.</td></tr>';

```

```

300         } else {
301             tbody.innerHTML = chains.map(c => `
302                 <tr>
303                     <td class="font-monospace text-warning">${c.parent_process} (PID: ${c.parent_pid || '-'})</td>
304                     <td class="font-monospace text-danger fw-semibold">${c.child_process}</td>
305                     <td class="text-center text-secondary">${c.child_pid || '-'}</td>
306                     <td class="text-center"><span class="badge def-badged-critical">${c.severity.toUpperCase()}</span></td>
307                     <td class="text-secondary small">
308                         <div>${c.reason}</div>
309                         ${c.command_line ? `<div class="font-monospace text-muted mt-1" style="font-size: 0.68rem;">cmd: ${c.command_line}</div>` : ''}
310                     </td>
311                 </tr>
312             `).join('');
313         }
314     }
315     } catch (err) {
316         console.error('[DefenderTab] Error loading process chains:', err);
317     }
318 }
319
320 async function loadEvents() {
321     try {
322         const events = await fetchJSON('/api/v1/defender/events?limit=50');
323         const badge = document.getElementById('def-events-count-badge');
324         const tbody = document.getElementById('def-events-tbody');
325
326         if (badge) badge.textContent = `${events.length} событий`;

```

```

327 |
328 |     if (tbody) {
329 |         if (events.length === 0) {
330 |             tbody.innerHTML = '<tr><td colspan="5" class="text-c
enter text-muted py-3">Событий в журнале Defender Operational
не обнаружено.</td></tr>';
331 |         } else {
332 |             tbody.innerHTML = events.map(e => `
333 |                 <tr>
334 |                     <td class="font-monospace fw-semibold text-info"
>${e.event_id}</td>
335 |                     <td class="text-secondary small font-monospace">
${e.timestamp}</td>
336 |                     <td><span class="badge ${e.level.toLowerCase().i
ncludes('err') || e.level.toLowerCase().includes('crit') ? 'bg
-danger' : (e.level.toLowerCase().includes('warn') ? 'bg-warni
ng text-dark' : 'bg-secondary')}">${e.level}</span></td>
337 |                     <td><span class="badge bg-dark border border-sec
ondary">${e.category}</span></td>
338 |                     <td class="text-light small">${e.message}</td>
339 |                 </tr>
340 |             `).join('');
341 |         }
342 |     }
343 | } catch (err) {
344 |     console.error('[DefenderTab] Error loading events:', err
);
345 | }
346 | }
347 |
348 | async function loadDiagnostics() {
349 |     try {
350 |         const rep = await fetchJSON('/api/v1/defender/diagnostic
s');
351 |
352 |         const scoreVal = document.getElementById('def-score-val'
);
353 |         const scoreSub = document.getElementById('def-score-sub'

```

```

);
354     const diagScoreDisplay = document.getElementById('def-di
ag-score-display');
355     const diagSummaryDisplay = document.getElementById('def-
diag-summary-display');
356     const recsContainer = document.getElementById('def-diag-
recs-container');
357     const critsContainer = document.getElementById('def-diag
-criticals-container');
358
359     const scoreColor = rep.security_score >= 80 ? 'text-succ
ess' : (rep.security_score >= 50 ? 'text-warning' : 'text-dang
er');
360
361     if (scoreVal) {
362         scoreVal.textContent = `${rep.security_score}/100`;
363         scoreVal.className = `def-card-value ${scoreColor}`;
364     }
365     if (scoreSub) scoreSub.textContent = rep.status_summary;
366
367     if (diagScoreDisplay) {
368         diagScoreDisplay.textContent = `${rep.security_score}/
100`;
369         diagScoreDisplay.className = `display-4 fw-bold ${scor
eColor}`;
370     }
371     if (diagSummaryDisplay) diagSummaryDisplay.textContent =
rep.status_summary;
372
373     if (recsContainer) {
374         if (rep.recommendations && rep.recommendations.length
> 0) {
375             recsContainer.innerHTML = `
376                 <ol class="ps-3 mb-0">
377                     ${rep.recommendations.map(r => `<li class="py-1
text-light">${r}</li>`).join('')}
378                 </ol>
379             `;

```

```

380         } else {
381             recsContainer.innerHTML = '<div class="text-success"
>Все рекомендации соблюдены. Защита оптимальна.</div>';
382         }
383     }
384
385     if (critsContainer) {
386         if (rep.critical_findings && rep.critical_findings.len
gth > 0) {
387             critsContainer.innerHTML = `
388                 <ul class="ps-3 mb-0 text-danger">
389                     ${rep.critical_findings.map(c => `<li class="py-
1">${c}</li>`).join('')}
390                 </ul>
391             `;
392         } else {
393             critsContainer.innerHTML = '<div class="text-muted">
Критических уязвимостей не зафиксировано.</div>';
394         }
395     }
396
397     } catch (err) {
398         console.error('[DefenderTab] Error loading diagnostics:'
, err);
399     }
400 }
401
402 const activeTaskPollers = {};
403
404 async function pollTask(taskId, taskLabel, triggerBtn) {
405     if (!taskId) return;
406     if (activeTaskPollers[taskId]) return;
407
408     if (triggerBtn) {
409         triggerBtn.disabled = true;
410         triggerBtn.dataset.originalHtml = triggerBtn.innerHTML;
411         triggerBtn.innerHTML = `<span class="spinner-border spin
ner-border-sm me-1" role="status" aria-hidden="true"></span> $

```

```

    {taskLabel}...`;
412     }
413
414     showAlert(`Запущен фоновый процесс: ${taskLabel} (ID: ${
taskId}))`);
415
416     const interval = 2500;
417     const maxAttempts = 240; // до 10 минут
418     let attempts = 0;
419
420     activeTaskPollers[taskId] = setInterval(async () => {
421         if (window.isTabActive && !window.isTabActive('tab-defen
der')) return;
422         attempts++;
423         try {
424             const task = await fetchJSON(`/api/v1/defender/tasks/${
taskId}`);
425             if (!task || task.status === 'running' || task.status
=== 'pending') {
426                 if (attempts >= maxAttempts) {
427                     clearInterval(activeTaskPollers[taskId]);
428                     delete activeTaskPollers[taskId];
429                     if (triggerBtn && triggerBtn.dataset.originalHtml)
{
430                         triggerBtn.innerHTML = triggerBtn.dataset.origin
alHtml;
431                         triggerBtn.disabled = false;
432                     }
433                     showAlert(`Таймаут ожидания задачи ${taskLabel} (ф
оновый процесс продолжается в Windows).`, true);
434                 }
435                 return;
436             }
437
438             clearInterval(activeTaskPollers[taskId]);
439             delete activeTaskPollers[taskId];
440
441             if (triggerBtn && triggerBtn.dataset.originalHtml) {

```

```

442         triggerBtn.innerHTML = triggerBtn.dataset.originalHt
ml;
443         triggerBtn.disabled = false;
444     }
445
446     const durationInfo = task.duration_seconds ? `[${task
.duration_seconds}c]` : '';
447     if (task.status === 'completed' && task.success) {
448         let extraMsg = '';
449         if (task.windows_event) {
450             extraMsg = ` (Windows Event ${task.windows_event.e
vent_id}: ${task.windows_event.category})`;
451         }
452         showAlert(`👉 ${taskLabel} успешно завершено${duratio
nInfo}! ${task.message || ''}${extraMsg}`);
453     } else {
454         showAlert(`👉 ${taskLabel} завершилось ошибкой${durat
ionInfo}: ${task.error || task.message || 'Сбой выполнения'}`,
true);
455     }
456
457     await refreshAll();
458 } catch (err) {
459     console.error(`[DefenderTab] Polling error for task ${
taskId}:`, err);
460 }
461 }, interval);
462 }
463
464 function setupActions() {
465     const btnQuick = document.getElementById('btn-def-quick-sc
an');
466     if (btnQuick) {
467         btnQuick.onclick = async () => {
468             try {
469                 const res = await fetchJSON('/api/v1/defender/scan',
{
470                     method: 'POST',

```

```

471         headers: { 'Content-Type': 'application/json' },
472         body: JSON.stringify({ scan_type: 'quick' }),
473     });
474     if (res && res.task_id) {
475         pollTask(res.task_id, 'Быстрое сканирование', btnQ
476         uick);
477     } else {
478         showAlert(`Быстрое сканирование: ${res.message ||
479         'Запущено'}`);
480         await loadDefenderStatus();
481     } catch (err) {
482         showAlert(`Ошибка запуска сканирования: ${err.messag
483         e}`, true);
484     }
485
486     const btnFull = document.getElementById('btn-def-full-scan
487     ');
488     if (btnFull) {
489         btnFull.onclick = async () => {
490             try {
491                 const res = await fetchJSON('/api/v1/defender/scan',
492                 {
493                     method: 'POST',
494                     headers: { 'Content-Type': 'application/json' },
495                     body: JSON.stringify({ scan_type: 'full' }),
496                 });
497                 if (res && res.task_id) {
498                     pollTask(res.task_id, 'Полное сканирование системы
499                     ', btnFull);
500                 } else {
501                     showAlert(`Полное сканирование: ${res.message || '
502                     Запущено'}`);
503                 }
504             } catch (err) {
505                 showAlert(`Ошибка полного сканирования: ${err.messag

```



```
    e}`, true);
502    }
503    };
504    }
505
506    const btnUpdate = document.getElementById('btn-def-update-
sigs');
507    if (btnUpdate) {
508        btnUpdate.onclick = async () => {
509            try {
510                const res = await fetchJSON('/api/v1/defender/update
-signatures', { method: 'POST' });
511                if (res && res.task_id) {
512                    pollTask(res.task_id, 'Обновление баз сигнатур Def
ender', btnUpdate);
513                } else {
514                    showAlert(`Обновление сигнатур: ${res.message} || '
Запущено'`);
515                    await loadDefenderStatus();
516                }
517            } catch (err) {
518                showAlert(`Ошибка обновления баз: ${err.message}`, t
rue);
519            }
520        };
521    }
522
523    const btnRefresh = document.getElementById('btn-def-refres
h');
524    if (btnRefresh) {
525        btnRefresh.onclick = async () => {
526            showAlert('Обновление данных Defender...');
527            await refreshAll();
528        };
529    }
530
531    const btnAiDiag = document.getElementById('btn-def-ai-diag
');
```

```
532     if (btnAiDiag) {
533         btnAiDiag.onclick = () => {
534             const diagTabBtn = document.getElementById('def-tab-di
ag-btn');
535             if (diagTabBtn) {
536                 const tab = new bootstrap.Tab(diagTabBtn);
537                 tab.show();
538             }
539         };
540     }
541 }
542
543 async function refreshAll() {
544     await Promise.all([
545         loadDefenderStatus(),
546         loadASRRules(),
547         loadCFA(),
548         loadExclusions(),
549         loadThreats(),
550         loadProcesses(),
551         loadEvents(),
552         loadDiagnostics(),
553     ]);
554 }
555
556 window.initDefenderTab = async function() {
557     console.log('[DefenderTab] Initializing Defender Security
Center Tab...');
558     setupActions();
559     if (window.isTabActive ? window.isTabActive('tab-defender'
) : false) {
560         await refreshAll();
561     }
562 };
563
564 window.activateDefenderTab = async function() {
565     console.log('[DefenderTab] Activating Defender Security Ce
nter Tab...');
```

```
566 |     await refreshAll();
567 | };
568 |
569 | // Auto initialize if container exists
570 | if (document.readyState === 'loading') {
571 |     document.addEventListener('DOMContentLoaded', () => {
572 |         if (document.getElementById('def-shields-tbody')) {
573 |             window.initDefenderTab();
574 |         }
575 |     });
576 | } else {
577 |     if (document.getElementById('def-shields-tbody')) {
578 |         window.initDefenderTab();
579 |     }
580 | }
581 |
582 | })();
583 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Disk Speed Benchmark (DiskSpd) View
5 |=====
6 |=====
7 Description:
8 | Шаблон клиентского веб-интерфейса тестирования скорости дисков (CrystalDiskMark стиль на базе DiskSpd).
9 |
10 Usage Examples:
11 |   Web Route:
12 |     /windows/api/webgui/disk_speed_tab/index.html
13 |
14 |   Tab Frame Integration:
15 |     <iframe src="/windows/api/webgui/disk_speed_tab/index.html" class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/disk_speed_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-04 08:16:00
23 |=====
24 |=====
25 -->
26
27 <div class="container-fluid py-2 px-3 text-white" id="disk-speed-container">
28 |
29 |   <!-- Header / Breadcrumbs & Status -->
30 |   <div class="d-flex align-items-center justify-content-between mb-3 pb-2 border-bottom border-secondary flex-wrap gap-2">
31 |     <div class="d-flex align-items-center gap-2">
```

```

29      <span class="fs-4 text-warning fw-bold d-flex align-item
| s-center gap-2">
30      <i class="bi bi-speedometer2 text-danger"></i>
31      <span>Бенчмарк дисков (DiskSpd / CDM)</span>
32      </span>
33      <span class="badge bg-dark border border-secondary text-
| info px-2 py-1" id="bench-engine-badge">
34      <i class="bi bi-cpu"></i> Microsoft DiskSpd
35      </span>
36  </div>
37  <div class="d-flex align-items-center gap-2">
38      <button class="btn btn-sm btn-outline-info d-flex align-
| items-center gap-1.5" id="bench-refresh-targets-btn" title="Об
| новить список накопителей">
39          <i class="bi bi-arrow-clockwise"></i>
40          <span>Обновить диски</span>
41      </button>
42      <button class="btn btn-sm btn-outline-secondary d-flex a
| lign-items-center gap-1.5" id="bench-copy-md-btn" title="Скопи
| ровать результаты в Markdown">
43          <i class="bi bi-clipboard"></i>
44          <span>Копировать MD</span>
45      </button>
46  </div>
47 </div>
48
49 <!-- Safe Mode Notice Banner -->
50 <div class="alert alert-dark border-secondary d-flex align-i
| tems-center justify-content-between py-2 px-3 mb-3 small">
51     <div class="d-flex align-items-center gap-2">
52         <i class="bi bi-shield-check text-success fs-5"></i>
53         <span>
54             <strong>Безопасный режим:</strong> Тестирование выполн
| яется на временном файле
55             <code class="text-info" id="bench-safe-filepath">C:\AI
| -Breadboard\data\benchmark\diskspd-test.dat</code>.
56             После завершения теста файл автоматически удаляется. П
| рямая запись в физические сектора исключена.

```

```

57 |         </span>
58 |     </div>
59 |     <span class="badge bg-success-subtle text-success border b
order-success-subtle px-2 py-1">SafeOps Active</span>
60 | </div>
61 |
62 | <!-- Main Benchmark Dashboard: Controls & CDM Table -->
63 | <div class="row g-3 mb-3">
64 |
65 |     <!-- Left Panel: Controls & CrystalDiskMark Style Buttons
-->
66 |     <div class="col-12 col-xl-4">
67 |         <div class="card bg-dark border-secondary shadow-sm h-10
0 p-3">
68 |             <h6 class="text-primary fw-bold mb-3 d-flex align-item
s-center gap-2">
69 |                 <i class="bi bi-sliders"></i> Параметры тестирования
70 |             </h6>
71 |
72 |             <!-- Drive Selector -->
73 |             <div class="mb-3">
74 |                 <label class="form-label small text-muted fw-semibol
d" for="bench-drive-select">
75 |                     <i class="bi bi-hdd"></i> Целевой накопитель:
76 |                 </label>
77 |                 <select class="form-select form-select-sm bg-dark te
xt-white border-secondary" id="bench-drive-select">
78 |                     <option value="C:">C: [Системный] – загрузка...</o
ption>
79 |                 </select>
80 |                 <div class="d-flex justify-content-between text-mute
d small mt-1" style="font-size: 0.75rem;">
81 |                     <span id="bench-drive-fs">ФС: NTFS</span>
82 |                     <span id="bench-drive-free">Свободно: -- ГБ</span>
83 |                 </div>
84 |             </div>
85 |
86 |             <!-- File Size & Duration -->

```

```

87 |         <div class="row g-2 mb-3">
88 |             <div class="col-6">
89 |                 <label class="form-label small text-muted fw-semib
old" for="bench-size-select">
90 |                     <i class="bi bi-file-earmark-binary"></i> Размер
файла:
91 |                 </label>
92 |                 <select class="form-select form-select-sm bg-dark
text-white border-secondary" id="bench-size-select">
93 |                     <option value="64">64 MiB (Быстро)</option>
94 |                     <option value="128">128 MiB</option>
95 |                     <option value="256" selected>256 MiB (Стандарт)<
/option>
96 |                     <option value="512">512 MiB</option>
97 |                     <option value="1024">1 GiB</option>
98 |                     <option value="2048">2 GiB</option>
99 |                 </select>
100 |             </div>
101 |             <div class="col-6">
102 |                 <label class="form-label small text-muted fw-semib
old" for="bench-duration-select">
103 |                     <i class="bi bi-stopwatch"></i> Время прогона:
104 |                 </label>
105 |                 <select class="form-select form-select-sm bg-dark
text-white border-secondary" id="bench-duration-select">
106 |                     <option value="2">2 сек (Экспресс)</option>
107 |                     <option value="3" selected>3 сек (Оптимально)</o
ption>
108 |                     <option value="5">5 сек (Точно)</option>
109 |                     <option value="10">10 сек (Crystal)</option>
110 |                 </select>
111 |             </div>
112 |         </div>
113 |
114 |         <!-- Test Mode: Both / Read / Write -->
115 |         <div class="mb-3">
116 |             <label class="form-label small text-muted fw-semibol
d">

```

```

117 |         <i class="bi bi-arrow-left-right"></i> Режим нагpy
      | зки:
118 |     </label>
119 |     <div class="btn-group w-100" role="group">
120 |         <input type="radio" class="btn-check" name="bench-
      | type-radio" id="bench-type-both" value="both" checked>
121 |         <label class="btn btn-sm btn-outline-primary" for=
      | "bench-type-both">Чтение + Запись</label>
122 |
123 |         <input type="radio" class="btn-check" name="bench-
      | type-radio" id="bench-type-read" value="read">
124 |         <label class="btn btn-sm btn-outline-info" for="be
      | nch-type-read">Только Read</label>
125 |
126 |         <input type="radio" class="btn-check" name="bench-
      | type-radio" id="bench-type-write" value="write">
127 |         <label class="btn btn-sm btn-outline-warning" for=
      | "bench-type-write">Только Write</label>
128 |     </div>
129 | </div>
130 |
131 | <!-- CDM Action Buttons Grid -->
132 | <div class="mt-auto pt-2 border-top border-secondary">
133 |     <div class="d-grid gap-2 mb-2">
134 |         <button class="btn btn-success fw-bold py-2 shadow
      | d-flex align-items-center justify-content-center gap-2" id="b
      | ench-run-all-btn">
135 |             <i class="bi bi-play-circle-fill fs-5"></i>
136 |             <span>ЗАПУСТИТЬ ВСЕ ТЕСТЫ (ALL)</span>
137 |         </button>
138 |         <button class="btn btn-danger btn-sm d-none d-flex
      | align-items-center justify-content-center gap-2" id="bench-ca
      | ncel-btn">
139 |             <i class="bi bi-stop-circle"></i>
140 |             <span>Прервать тестирование</span>
141 |         </button>
142 |     </div>
143 |

```



```

144 |         <div class="row g-1 text-center small">
145 |             <div class="col-6">
146 |                 <button class="btn btn-sm btn-outline-secondary
w-100 py-1" id="bench-run-seq1m-q8" title="Тест SEQ1M Q8T1">
147 |                     SEQ1M Q8
148 |                 </button>
149 |             </div>
150 |             <div class="col-6">
151 |                 <button class="btn btn-sm btn-outline-secondary
w-100 py-1" id="bench-run-seq1m-q1" title="Тест SEQ1M Q1T1">
152 |                     SEQ1M Q1
153 |                 </button>
154 |             </div>
155 |             <div class="col-6">
156 |                 <button class="btn btn-sm btn-outline-secondary
w-100 py-1" id="bench-run-rnd4k-q32" title="Тест RND4K Q32T16"
>
157 |                     RND4K Q32
158 |                 </button>
159 |             </div>
160 |             <div class="col-6">
161 |                 <button class="btn btn-sm btn-outline-secondary
w-100 py-1" id="bench-run-rnd4k-q1" title="Тест RND4K Q1T1">
162 |                     RND4K Q1
163 |                 </button>
164 |             </div>
165 |         </div>
166 |     </div>
167 |
168 | </div>
169 | </div>
170 |
171 | <!-- Right Panel: CrystalDiskMark Scoreboard Table -->
172 | <div class="col-12 col-xl-8">
173 |     <div class="card bg-dark border-secondary shadow-sm h-10
0">
174 |
175 |         <div class="card-header border-secondary d-flex align-

```

```

    items-center justify-content-between py-2 px-3">
176         <span class="fw-bold d-flex align-items-center gap-2
    ">
177             <i class="bi bi-table text-info"></i>
178             <span>Результаты CrystalDiskMark</span>
179         </span>
180         <span class="badge bg-secondary font-monospace" id="
    bench-status-badge">Готов к тесту</span>
181     </div>
182
183     <div class="card-body p-3 d-flex flex-column justify-c
    ontent-between">
184
185         <!-- Live Progress Bar -->
186         <div class="mb-3 d-none" id="bench-progress-wrapper"
    >
187             <div class="d-flex justify-content-between small t
    ext-muted mb-1">
188                 <span id="bench-progress-msg">Инициализация Disk
    Spd...</span>
189                 <span id="bench-progress-pct" class="fw-bold tex
    t-info">0%</span>
190             </div>
191             <div class="progress bg-black" style="height: 10px
    ;">
192                 <div class="progress-bar progress-bar-striped pr
    ogress-bar-animated bg-success" id="bench-progress-bar" style=
    "width: 0%;"></div>
193             </div>
194         </div>
195
196         <!-- The Table -->
197         <div class="table-responsive mb-3">
198             <table class="table table-dark table-bordered alig
    n-middle text-center mb-0" id="bench-cdm-table" style="font-fa
    mily: Consolas, monospace;">
199                 <thead>
200                 <tr class="text-secondary bg-black" style="fon

```

```

t-size: 0.85rem;">
201         <th class="text-start py-2" style="width: 25
%; ">Тест / Профиль</th>
202         <th class="text-end text-info py-2" style="w
idth: 20%; "><i class="bi bi-download"></i> Read (MB/s)</th>
203         <th class="text-end text-warning py-2" style
="width: 20%; "><i class="bi bi-upload"></i> Write (MB/s)</th>
204         <th class="text-end text-success py-2" style
="width: 18%; ">IOPS (R/W)</th>
205         <th class="text-end text-muted py-2" style="
width: 17%; ">Латентность (µs)</th>
206     </tr>
207     </thead>
208     <tbody style="font-size: 1.05rem; font-weight: b
old; ">
209
210         <!-- SEQ1M Q8T1 -->
211         <tr id="row-seq1m_q8t1">
212             <td class="text-start">
213                 <div class="text-white">SEQ1M <span class=
"badge bg-primary text-white" style="font-size: 0.7rem; ">Q8T1<
/td>
214                 <div class="text-muted fw-normal" style="f
ont-size: 0.7rem; ">1 MiB, Очередь 8, Поток 1</div>
215             </td>
216             <td class="text-end text-info fs-5" id="val-
seq1m_q8t1-read">0.00</td>
217             <td class="text-end text-warning fs-5" id="v
al-seq1m_q8t1-write">0.00</td>
218             <td class="text-end text-success small fw-no
rmal" id="val-seq1m_q8t1-iops">-- / --</td>
219             <td class="text-end text-muted small fw-norm
al" id="val-seq1m_q8t1-lat">-- µs</td>
220         </tr>
221
222         <!-- SEQ1M Q1T1 -->
223         <tr id="row-seq1m_q1t1">
224             <td class="text-start">

```

```

225 |         <div class="text-white">SEQ1M <span class=
| "badge bg-primary text-white" style="font-size: 0.7rem;">Q1T1<
| /span></div>
226 |         <div class="text-muted fw-normal" style="f
| ont-size: 0.7rem;">1 MiB, Очередь 1, Поток 1</div>
227 |         </td>
228 |         <td class="text-end text-info fs-5" id="val-
| seq1m_q1t1-read">0.00</td>
229 |         <td class="text-end text-warning fs-5" id="v
| al-seq1m_q1t1-write">0.00</td>
230 |         <td class="text-end text-success small fw-no
| rmal" id="val-seq1m_q1t1-iops">-- / --</td>
231 |         <td class="text-end text-muted small fw-norm
| al" id="val-seq1m_q1t1-lat">-- µs</td>
232 |     </tr>
233 |
234 |     <!-- RND4K Q32T16 -->
235 |     <tr id="row-rnd4k_q32t16">
236 |         <td class="text-start">
237 |             <div class="text-white">RND4K <span class=
| "badge bg-info text-dark" style="font-size: 0.7rem;">Q32T16</s
| pan></div>
238 |             <div class="text-muted fw-normal" style="f
| ont-size: 0.7rem;">4 KiB, Очередь 32, Поток 1</div>
239 |             </td>
240 |             <td class="text-end text-info fs-5" id="val-
| rnd4k_q32t16-read">0.00</td>
241 |             <td class="text-end text-warning fs-5" id="v
| al-rnd4k_q32t16-write">0.00</td>
242 |             <td class="text-end text-success small fw-no
| rmal" id="val-rnd4k_q32t16-iops">-- / --</td>
243 |             <td class="text-end text-muted small fw-norm
| al" id="val-rnd4k_q32t16-lat">-- µs</td>
244 |         </tr>
245 |
246 |     <!-- RND4K Q1T1 -->
247 |     <tr id="row-rnd4k_q1t1">
248 |         <td class="text-start">

```

```

249 |         <div class="text-white">RND4K <span class=
| "badge bg-info text-dark" style="font-size: 0.7rem;">Q1T1</spa
| n></div>
250 |         <div class="text-muted fw-normal" style="f
| ont-size: 0.7rem;">4 KiB, Очередь 1, Поток 1</div>
251 |     </td>
252 |     <td class="text-end text-info fs-5" id="val-
| rnd4k_q1t1-read">0.00</td>
253 |     <td class="text-end text-warning fs-5" id="v
| al-rnd4k_q1t1-write">0.00</td>
254 |     <td class="text-end text-success small fw-no
| rmal" id="val-rnd4k_q1t1-iops">-- / --</td>
255 |     <td class="text-end text-muted small fw-norm
| al" id="val-rnd4k_q1t1-lat">-- µs</td>
256 | </tr>
257 |
258 | </tbody>
259 | </table>
260 | </div>
261 |
262 | <!-- Bottom Summary Footer -->
263 | <div class="d-flex align-items-center justify-conten
| t-between p-2 bg-black rounded border border-secondary text-mu
| ted small">
264 |     <div>
265 |         <span>Тестируемый диск: <strong class="text-whit
| e" id="bench-summary-disk">C:</strong></span>
266 |         <span class="ms-3">Размер: <strong class="text-w
| hite" id="bench-summary-size">256 MB</strong></span>
267 |     </div>
268 |     <div>
269 |         <span>Время замера: <strong class="text-white" i
| d="bench-summary-time">--</strong></span>
270 |     </div>
271 | </div>
272 |
273 | </div>
274 | </div>

```

```

275     </div>
276
277 </div>
278
279 <!-- Charts & History Section -->
280 <div class="row g-3">
281
282     <!-- Chart: Throughput (MB/s) -->
283     <div class="col-12 col-lg-6">
284         <div class="card bg-dark border-secondary shadow-sm p-3
285         h-100">
286             <h6 class="text-primary fw-bold mb-2 d-flex align-item
287             s-center gap-2">
288                 <i class="bi bi-bar-chart-fill"></i> Пропускная спос
289                 обность (MB/s)
290             </h6>
291             <div style="height: 220px; position: relative;">
292                 <canvas id="bench-mb-chart"></canvas>
293             </div>
294         </div>
295     </div>
296
297     <!-- History Table -->
298     <div class="col-12 col-lg-6">
299         <div class="card bg-dark border-secondary shadow-sm p-3
300         h-100">
301             <div class="d-flex align-items-center justify-content-
302             between mb-2">
303                 <h6 class="text-primary fw-bold mb-0 d-flex align-it
304                 ems-center gap-2">
305                     <i class="bi bi-clock-history"></i> История замеро
306                     в
307                 </h6>
308                 <span class="badge bg-secondary" id="bench-history-c
309                 ount">0 записей</span>
310             </div>
311             <div class="table-responsive" style="max-height: 220px
312             ; overflow-y: auto;">

```

```
304 |         <table class="table table-dark table-sm table-hover
      | align-middle mb-0 small" id="bench-history-table">
305 |             <thead>
306 |                 <tr class="text-secondary">
307 |                     <th>Дата / Время</th>
308 |                     <th>Диск</th>
309 |                     <th>SEQ Read / Write</th>
310 |                     <th>RND 4K IOPS</th>
311 |                     <th class="text-end">Действие</th>
312 |                 </tr>
313 |             </thead>
314 |             <tbody id="bench-history-tbody">
315 |                 <tr>
316 |                     <td colspan="5" class="text-center text-muted
      | py-3">История пуста. Запустите тест.</td>
317 |                 </tr>
318 |             </tbody>
319 |         </table>
320 |     </div>
321 | </div>
322 | </div>
323 |
324 | </div>
325 |
326 | </div>
327 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Disk Speed Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом бенчмарка дисков
7   *   ой подсистемы (DiskSpd).
8   *
9   * Usage Examples:
10  *   HTML Integration:
11  *     <script src="/windows/api/webgui/disk_speed_tab/main.js
12  *     ?v=20261004_v1" type="module"></script>
13  *
14  * File: main.js
15  * Project: ai-breadboard
16  * Package: windows/api/webgui/disk_speed_tab
17  * Author: hypo69
18  * Copyright: © 2026 hypo69
19  * Updated: 2026-10-08 02:00:00
20  * =====
21  */
22
23
24 const registerTabPoller = window.registerTabPoller || function
25   () {};
26 const isTabActive = window.isTabActive || function() { return
27   true; };
28
29 let mbChartInstance = null;
30 let currentTaskId = null;
31 let pollTimer = null;
32 let cachedTargets = [];
33 let lastResult = null;
```



```

30 | async function initDiskSpeedTab() {
31 |   if (isInitialized) return;
32 |   isInitialized = true;
33 |   initChart();
34 |   setupEventListeners();
35 |   if (window.isTabActive ? window.isTabActive('tab-disk-speed'
    | ) : false) {
36 |     await checkEngineStatus();
37 |     await loadTargets();
38 |     await loadHistory();
39 |   }
40 | }
41 |
42 | async function activateDiskSpeedTab() {
43 |   await checkEngineStatus();
44 |   await loadTargets();
45 |   await loadHistory();
46 | }
47 |
48 | window.activateDiskSpeedTab = activateDiskSpeedTab;
49 |
50 | /**
51 |  * Проверка статуса утилиты DiskSpd.
52 |  */
53 | async function checkEngineStatus() {
54 |   try {
55 |     const res = await fetch('/api/v1/storage/benchmark/engine'
    | );
56 |     if (!res.ok) return;
57 |     const data = await res.json();
58 |     const badge = document.getElementById('bench-engine-badge'
    | );
59 |     if (badge) {
60 |       if (data.available) {
61 |         badge.className = 'badge bg-success-subtle text-succes
    | s border border-success-subtle px-2 py-1';
62 |         badge.innerHTML = '<i class="bi bi-check-circle-fill">
    | </i> DiskSpd Ready';

```

```
63 |     } else {
64 |         badge.className = 'badge bg-warning-subtle text-warnin
    g border border-warning-subtle px-2 py-1';
65 |         badge.innerHTML = '<i class="bi bi-exclamation-triangl
    e"></i> DiskSpd не установлен';
66 |     }
67 | }
68 | } catch (err) {
69 |     console.debug('[DiskSpeed] Ошибка проверки движка:', err);
70 | }
71 | }
72 |
73 | /**
74 |  * Загрузка списка доступных дисков и разделов.
75 |  */
76 | async function loadTargets() {
77 |     try {
78 |         const res = await fetch('/api/v1/storage/benchmark/targets
    ');
79 |         if (!res.ok) return;
80 |         cachedTargets = await res.json();
81 |
82 |         const select = document.getElementById('bench-drive-select
    ');
83 |         if (!select) return;
84 |
85 |         select.innerHTML = '';
86 |         cachedTargets.forEach(t => {
87 |             const opt = document.createElement('option');
88 |             opt.value = t.drive_letter;
89 |             const tag = t.is_system ? ' [Системный]' : '';
90 |             opt.textContent = `${t.drive_letter} (${t.label} || t.dri
    ve_letter)) ${tag} – ${t.free_gb} ГБ свободно`;
91 |             select.appendChild(opt);
92 |         });
93 |
94 |         updateDriveDetails();
95 |     } catch (err) {
```

```
96 |     console.error('[DiskSpeed] Ошибка загрузки списка накопите
    | лей:', err);
97 | }
98 | }
99 |
100 | /**
101 |  * Обновление информации о выбранном диске и пути безопасного
    | файла.
102 |  */
103 | function updateDriveDetails() {
104 |     const select = document.getElementById('bench-drive-select')
    | ;
105 |     if (!select) return;
106 |     const drive = select.value;
107 |     const target = cachedTargets.find(t => t.drive_letter === dr
    | ive);
108 |
109 |     const fsEl = document.getElementById('bench-drive-fs');
110 |     const freeEl = document.getElementById('bench-drive-free');
111 |     const pathEl = document.getElementById('bench-safe-filepath'
    | );
112 |     const summaryDisk = document.getElementById('bench-summary-d
    | isk');
113 |
114 |     if (target) {
115 |         if (fsEl) fsEl.textContent = `ФС: ${target.fs_type}`;
116 |         if (freeEl) freeEl.textContent = `Свободно: ${target.free_
    | gb} ГБ`;
117 |         if (pathEl) pathEl.textContent = `${target.recommended_dir
    | }\\diskspd-test.dat`;
118 |         if (summaryDisk) summaryDisk.textContent = `${target.drive
    | _letter} (${target.fs_type})`;
119 |     }
120 | }
121 |
122 | /**
123 |  * Настройка событий элементов управления.
124 |  */
```

```
125 function setupEventListeners() {
126   const driveSelect = document.getElementById('bench-drive-select');
127   if (driveSelect) {
128     driveSelect.addEventListener('change', updateDriveDetails)
129   };
130
131   const refreshBtn = document.getElementById('bench-refresh-targets-btn');
132   if (refreshBtn) {
133     refreshBtn.addEventListener('click', async () => {
134       await loadTargets();
135       if (window.toast) window.toast.info('Обновлено', 'Список накопителей обновлен');
136     });
137   }
138
139   const runAllBtn = document.getElementById('bench-run-all-btn');
140   if (runAllBtn) {
141     runAllBtn.addEventListener('click', () => {
142       startBenchmark(['seq1m_q8t1', 'seq1m_q1t1', 'rnd4k_q32t16', 'rnd4k_q1t1']);
143     });
144   }
145
146   const cancelBtn = document.getElementById('bench-cancel-btn');
147   if (cancelBtn) {
148     cancelBtn.addEventListener('click', cancelBenchmark);
149   }
150
151   // Одиночные кнопки профилей
152   const pMap = {
153     'bench-run-seq1m-q8': ['seq1m_q8t1'],
154     'bench-run-seq1m-q1': ['seq1m_q1t1'],
155     'bench-run-rnd4k-q32': ['rnd4k_q32t16'],
```

```

156     'bench-run-rnd4k-q1': ['rnd4k_q1t1'],
157 };
158
159 Object.entries(pMap).forEach(([btnId, profiles]) => {
160     const btn = document.getElementById(btnId);
161     if (btn) {
162         btn.addEventListener('click', () => startBenchmark(profiles));
163     }
164 });
165
166 const copyMdBtn = document.getElementById('bench-copy-md-btn');
167 if (copyMdBtn) {
168     copyMdBtn.addEventListener('click', copyResultsAsMarkdown);
169 }
170 }
171
172 /**
173  * Запуск тестирования производительности.
174  * @param {Array<string>} profiles - Список профилей для выполнения.
175  */
176 async function startBenchmark(profiles) {
177     const driveSelect = document.getElementById('bench-drive-select');
178     const sizeSelect = document.getElementById('bench-size-select');
179     const durSelect = document.getElementById('bench-duration-select');
180     const typeRadio = document.querySelector('input[name="bench-type-radio"]:checked');
181
182     const payload = {
183         target_drive: driveSelect ? driveSelect.value : 'C:',
184         file_size_mb: sizeSelect ? parseInt(sizeSelect.value, 10) : 256,

```

```
185 |     duration_sec: durSelect ? parseInt(durSelect.value, 10) :
    | 3,
186 |     test_type: typeRadio ? typeRadio.value : 'both',
187 |     profiles: profiles,
188 |     disable_cache: true,
189 |     delete_test_file: true,
190 | };
191 |
192 | setUiRunning(true);
193 |
194 | const summarySize = document.getElementById('bench-summary-s
    | ize');
195 | if (summarySize) summarySize.textContent = `${payload.file_s
    | ize_mb} MB`;
196 |
197 | try {
198 |     const res = await fetch('/api/v1/storage/benchmark/start',
    | {
199 |         method: 'POST',
200 |         headers: { 'Content-Type': 'application/json' },
201 |         body: JSON.stringify(payload),
202 |     });
203 |
204 |     if (!res.ok) {
205 |         const err = await res.json();
206 |         throw new Error(err.detail || 'Не удалось запустить бенч
    | марк');
207 |     }
208 |
209 |     const task = await res.json();
210 |     currentTaskId = task.task_id;
211 |     pollBenchmarkTask();
212 | } catch (err) {
213 |     setUiRunning(false);
214 |     if (window.toast) {
215 |         window.toast.error('Ошибка старта', err.message);
216 |     } else {
217 |         alert(`Ошибка: ${err.message}`);
```

```
218 |     }
219 | }
220 |
221 |
222 | function pollBenchmarkTask() {
223 |     if (pollTimer) clearInterval(pollTimer);
224 |
225 |     pollTimer = setInterval(async () => {
226 |         if (!currentTaskId) {
227 |             clearInterval(pollTimer);
228 |             return;
229 |         }
230 |         if (window.isTabActive && !window.isTabActive('tab-disk-sp
    eed')) return;
231 |
232 |         try {
233 |             const res = await fetch(`/api/v1/storage/benchmark/statu
    s/${currentTaskId}`);
234 |             if (!res.ok) return;
235 |
236 |             const task = await res.json();
237 |             updateProgress(task.progress_percent, task.destination);
238 |
239 |             if (task.status === 'completed') {
240 |                 clearInterval(pollTimer);
241 |                 setUiRunning(false);
242 |                 if (window.toast) window.toast.success('Тест завершен'
    , 'Результаты бенчмарка готовы');
243 |                 await loadLatestHistoryResult();
244 |                 await loadHistory();
245 |             } else if (task.status === 'failed') {
246 |                 clearInterval(pollTimer);
247 |                 setUiRunning(false);
248 |                 if (window.toast) window.toast.error('Ошибка теста', t
    ask.error_message || 'Тестирование завершилось с ошибкой');
249 |             } else if (task.status === 'cancelled') {
250 |                 clearInterval(pollTimer);
251 |                 setUiRunning(false);
```

```

252         if (window.toast) window.toast.warning('Отменено', 'Те
ст был отменен');
253     }
254     } catch (e) {
255         console.debug('[DiskSpeed] Ошибка опроса задачи:', e);
256     }
257     }, 1000);
258 }
259
260 /**
261  * Отмена запущенного бенчмарка.
262  */
263 async function cancelBenchmark() {
264     if (!currentTaskId) return;
265     try {
266         await fetch(`/api/v1/storage/benchmark/cancel/${currentTaskId}`, { method: 'POST' });
267         if (window.toast) window.toast.info('Отмена', 'Запрос на о
становку отправлен...');
268     } catch (err) {
269         console.error('[DiskSpeed] Ошибка отмены:', err);
270     }
271 }
272
273 /**
274  * Управление состоянием элементов UI во время теста.
275  * @param {boolean} isRunning - Флаг активности теста.
276  */
277 function setUiRunning(isRunning) {
278     const runAllBtn = document.getElementById('bench-run-all-btn
');
279     const cancelBtn = document.getElementById('bench-cancel-btn'
);
280     const progressWrapper = document.getElementById('bench-progr
ess-wrapper');
281     const statusBadge = document.getElementById('bench-status-ba
dge');
282

```



```
283 | if (runAllBtn) runAllBtn.disabled = isRunning;
284 | if (cancelBtn) {
285 |     if (isRunning) {
286 |         cancelBtn.classList.remove('d-none');
287 |     } else {
288 |         cancelBtn.classList.add('d-none');
289 |     }
290 | }
291 |
292 | if (progressWrapper) {
293 |     if (isRunning) {
294 |         progressWrapper.classList.remove('d-none');
295 |     } else {
296 |         progressWrapper.classList.add('d-none');
297 |     }
298 | }
299 |
300 | if (statusBadge) {
301 |     if (isRunning) {
302 |         statusBadge.className = 'badge bg-warning text-dark font
303 | -monospace';
304 |         statusBadge.textContent = 'Идет тестирование...';
305 |     } else {
306 |         statusBadge.className = 'badge bg-secondary font-monospa
307 | ce';
308 |         statusBadge.textContent = 'Готов к тесту';
309 |     }
310 | }
311 | /**
312 |  * Обновление полосы прогресса.
313 |  * @param {number} pct - Процент выполнения.
314 |  * @param {string} msg - Сообщение фазы.
315 |  */
316 | function updateProgress(pct, msg) {
317 |     const bar = document.getElementById('bench-progress-bar');
318 |     const pctEl = document.getElementById('bench-progress-pct');
```

```

319 |   const msgEl = document.getElementById('bench-progress-msg');
320 |
321 |   const val = Math.max(0, Math.min(100, pct || 0));
322 |   if (bar) bar.style.width = `${val}%`;
323 |   if (pctEl) pctEl.textContent = `${val}%`;
324 |   if (msgEl && msg) msgEl.textContent = msg;
325 | }
326 |
327 | /**
328 |  * Загрузка последнего результата и заполнение таблицы CDM.
329 |  */
330 | async function loadLatestHistoryResult() {
331 |   try {
332 |     const res = await fetch('/api/v1/storage/benchmark/history
?limit=1');
333 |     if (!res.ok) return;
334 |     const items = await res.json();
335 |     if (!items || items.length === 0) return;
336 |
337 |     const latest = items[0];
338 |     lastResult = latest;
339 |
340 |     const summaryTime = document.getElementById('bench-summary
-time');
341 |     if (summaryTime) summaryTime.textContent = latest.created_
at;
342 |
343 |     // SEQ1M Q8T1
344 |     updateRowValues('seq1m_q8t1', latest.seq1m_read_mb_s, late
st.seq1m_write_mb_s, '-- / --', '-- µs');
345 |     // RND4K Q32T16
346 |     updateRowValues('rnd4k_q32t16', latest.rnd4k_read_mb_s, la
test.rnd4k_write_mb_s, `${Math.round(latest.rnd4k_read_iops)}
/ ${Math.round(latest.rnd4k_write_iops)}`, '-- µs');
347 |
348 |     updateChartData([
349 |       { label: 'SEQ1M Q8T1', read: latest.seq1m_read_mb_s, wri
te: latest.seq1m_write_mb_s },

```

```
350     { label: 'RND4K Q32T16', read: latest.rnd4k_read_mb_s, w
rite: latest.rnd4k_write_mb_s },
351   });
352   } catch (err) {
353     console.debug('[DiskSpeed] Ошибка загрузки свежих результа
тов:', err);
354   }
355 }
356
357 /**
358  * Обновление строки таблицы CDM.
359  */
360 function updateRowValues(rowKey, readMb, writeMb, iops, lat) {
361   const rEl = document.getElementById(`val-${rowKey}-read`);
362   const wEl = document.getElementById(`val-${rowKey}-write`);
363   const iopsEl = document.getElementById(`val-${rowKey}-iops`);
364   ;
365   const latEl = document.getElementById(`val-${rowKey}-lat`);
366   if (rEl) rEl.textContent = (readMb || 0).toFixed(2);
367   if (wEl) wEl.textContent = (writeMb || 0).toFixed(2);
368   if (iopsEl) iopsEl.textContent = iops;
369   if (latEl) latEl.textContent = lat;
370 }
371
372 /**
373  * Инициализация графика Chart.js.
374  */
375 function initChart() {
376   const ctx = document.getElementById('bench-mb-chart');
377   if (!ctx || typeof Chart === 'undefined') return;
378
379   if (mbChartInstance) {
380     mbChartInstance.destroy();
381   }
382
383   mbChartInstance = new Chart(ctx, {
384     type: 'bar',
```

```

385 |     data: {
386 |         labels: ['SEQ1M Q8T1', 'SEQ1M Q1T1', 'RND4K Q32T16', 'RN
D4K Q1T1'],
387 |         datasets: [
388 |             {
389 |                 label: 'Read (MB/s)',
390 |                 data: [0, 0, 0, 0],
391 |                 backgroundColor: 'rgba(13, 202, 240, 0.75)',
392 |                 borderColor: '#0dcaf0',
393 |                 borderWidth: 1,
394 |             },
395 |             {
396 |                 label: 'Write (MB/s)',
397 |                 data: [0, 0, 0, 0],
398 |                 backgroundColor: 'rgba(255, 193, 7, 0.75)',
399 |                 borderColor: '#ffc107',
400 |                 borderWidth: 1,
401 |             },
402 |         ],
403 |     },
404 |     options: {
405 |         responsive: true,
406 |         maintainAspectRatio: false,
407 |         scales: {
408 |             y: {
409 |                 beginAtZero: true,
410 |                 grid: { color: '#334155' },
411 |                 ticks: { color: '#94a3b8' },
412 |             },
413 |             x: {
414 |                 grid: { display: false },
415 |                 ticks: { color: '#94a3b8' },
416 |             },
417 |         },
418 |         plugins: {
419 |             legend: { labels: { color: '#f8fafc' } },
420 |         },
421 |     },

```

```
422 | });
423 | }
424 |
425 | /**
426 |  * Обновление данных графика.
427 |  */
428 | function updateChartData(points) {
429 |   if (!mbChartInstance) return;
430 |   const labels = points.map(p => p.label);
431 |   const reads = points.map(p => p.read || 0);
432 |   const writes = points.map(p => p.write || 0);
433 |
434 |   mbChartInstance.data.labels = labels;
435 |   mbChartInstance.data.datasets[0].data = reads;
436 |   mbChartInstance.data.datasets[1].data = writes;
437 |   mbChartInstance.update();
438 | }
439 |
440 | /**
441 |  * Загрузка истории тестирования.
442 |  */
443 | async function loadHistory() {
444 |   try {
445 |     const res = await fetch('/api/v1/storage/benchmark/history
446 |     ?limit=20');
447 |     if (!res.ok) return;
448 |     const list = await res.json();
449 |
450 |     const countBadge = document.getElementById('bench-history-
451 |     count');
452 |     if (countBadge) countBadge.textContent = `${list.length} з
453 |     аписей`;
454 |
455 |     const tbody = document.getElementById('bench-history-tbody
456 |     ');
457 |     if (!tbody) return;
458 |
459 |     if (!list || list.length === 0) {
```

```

456 |         tbody.innerHTML = '<tr><td colspan="5" class="text-cente
| r text-muted py-3">История нуста</td></tr>';
457 |         return;
458 |     }
459 |
460 |     tbody.innerHTML = '';
461 |     list.forEach(item => {
462 |         const tr = document.createElement('tr');
463 |         tr.innerHTML = `
464 |             <td class="text-white">${item.created_at}</td>
465 |             <td><span class="badge bg-secondary">${item.target}</s
| pan></td>
466 |             <td class="text-info fw-bold">${item.seq1m_read_mb_s.t
| oFixed(0)} <span class="text-muted"></span> <span class="text
| -warning">${item.seq1m_write_mb_s.toFixed(0)} MB/s</span></td>
467 |             <td class="text-success">${Math.round(item.rnd4k_read_
| iops)} <span class="text-muted"></span> ${Math.round(item.rnd
| 4k_write_iops)}</td>
468 |             <td class="text-end">
469 |                 <button class="btn btn-outline-danger btn-sm py-0 px
| -1 bench-delete-btn" data-id="${item.id}" title="Удалить">
470 |                     <i class="bi bi-trash"></i>
471 |                 </button>
472 |             </td>
473 |         `;
474 |         tbody.appendChild(tr);
475 |     });
476 |
477 |     tbody.querySelectorAll('.bench-delete-btn').forEach(btn =>
| {
478 |         btn.addEventListener('click', async e => {
479 |             e.stopPropagation();
480 |             const id = btn.getAttribute('data-id');
481 |             if (confirm('Удалить эту запись из истории?')) {
482 |                 await deleteHistoryItem(id);
483 |             }
484 |         });
485 |     });

```

```
486 |   } catch (err) {
487 |     console.debug('[DiskSpeed] Ошибка загрузки истории:', err)
    | ;
488 |   }
489 | }
490 |
491 | /**
492 |  * Удаление записи из истории.
493 |  */
494 | async function deleteHistoryItem(id) {
495 |   try {
496 |     await fetch(`/api/v1/storage/benchmark/history/${id}`, { m
    | ethod: 'DELETE' });
497 |     await loadHistory();
498 |     if (window.toast) window.toast.info('Удалено', 'Запись уда
    | лена из базы данных');
499 |   } catch (err) {
500 |     console.error('[DiskSpeed] Ошибка удаления:', err);
501 |   }
502 | }
503 |
504 | /**
505 |  * Копирование результатов в формате Markdown таблицы.
506 |  */
507 | function copyResultsAsMarkdown() {
508 |   const s8r = document.getElementById('val-seq1m_q8t1-read')?.
    | textContent || '0';
509 |   const s8w = document.getElementById('val-seq1m_q8t1-write')?
    | .textContent || '0';
510 |   const s1r = document.getElementById('val-seq1m_q1t1-read')?.
    | textContent || '0';
511 |   const s1w = document.getElementById('val-seq1m_q1t1-write')?
    | .textContent || '0';
512 |   const r32r = document.getElementById('val-rnd4k_q32t16-read'
    | )?.textContent || '0';
513 |   const r32w = document.getElementById('val-rnd4k_q32t16-write
    | ')??.textContent || '0';
514 |   const r1r = document.getElementById('val-rnd4k_q1t1-read')?.
```

```

    |textContent || '0';
515 |   const r1w = document.getElementById('val-rnd4k_q1t1-write')?
    |.textContent || '0';
516 |
517 |   const disk = document.getElementById('bench-drive-select')?.
    |value || 'C:';
518 |
519 |   const md = [
520 |     `### AI-Breadboard DiskSpd Benchmark (${disk})`,
521 |     '',
522 |     '| Профиль | Read (MB/s) | Write (MB/s) |',
523 |     '| :--- | :---: | :---: |',
524 |     `| **SEQ1M Q8T1** | ${s8r} MB/s | ${s8w} MB/s |`,
525 |     `| **SEQ1M Q1T1** | ${s1r} MB/s | ${s1w} MB/s |`,
526 |     `| **RND4K Q32T16** | ${r32r} MB/s | ${r32w} MB/s |`,
527 |     `| **RND4K Q1T1** | ${r1r} MB/s | ${r1w} MB/s |`,
528 |     '',
529 |     `*Дата замера: ${new Date().toLocaleString('ru-RU')}*`,
530 |   ].join('\n');
531 |
532 |   navigator.clipboard.writeText(md).then(() => {
533 |     if (window.toast) window.toast.success('Скопировано', 'Таб
    |лица скопирована в буфер обмена');
534 |   });
535 | }
536 |
537 | // Регистрация поллера при активной вкладке tab-disk-speed
538 | registerTabPoller('tab-disk-speed', checkEngineStatus, 15000,
    |{ immediate: false });
539 |
540 | // Автоматический запуск при загрузке
541 | if (document.readyState === 'loading') {
542 |   document.addEventListener('DOMContentLoaded', initDiskSpeedT
    |ab);
543 | } else {
544 |   initDiskSpeedTab();
545 | }
546 |

```



```
547 | export { initDiskSpeedTab };
```

```
548 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса журнала событий Windows с
10  модулем Log Intelligence.
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/event_logs_tab/index.html
15
16   Tab Frame Integration:
17     <iframe src="/windows/api/webgui/event_logs_tab/index.ht
18 ml" class="w-100 h-100 border-0"></iframe>
19
20 File: index.html
21 Project: ai-breadboard
22 Package: windows/api/webgui/event_logs_tab
23 Author: hypo69
24 Copyright: © 2026 hypo69
25 Updated: 2026-10-06 02:28:00
26 =====
27 =====
28 -->
29
30 <!-- Windows Event Logs Tab -->
31 <div class="container-fluid py-2 px-3">
32   <!-- Top Stat Cards -->
33   <div class="row g-2 mb-3">
34     <div class="col-6 col-md-3">
35       <div class="card bg-dark text-white border-secondary h-1
36 00 shadow-sm p-3">
37         <div class="text-muted small text-uppercase fw-semibol
```

```

d">Критические ошибки (Critical) □ </div>
31     <div class="fs-4 fw-bold mt-1 text-danger" id="el-critical-count">--</div>
32     <div class="small text-muted mt-1" id="el-critical-sub">Level 1 (WevtAPI)</div>
33     </div>
34 </div>
35 <div class="col-6 col-md-3">
36     <div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3">
37         <div class="text-muted small text-uppercase fw-semibold">Ошибки (Errors) □ </div>
38         <div class="fs-4 fw-bold mt-1 text-warning" id="el-error-count">--</div>
39         <div class="small text-muted mt-1" id="el-error-sub">Level 2 (WevtAPI)</div>
40         </div>
41     </div>
42 <div class="col-6 col-md-3">
43     <div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3">
44         <div class="text-muted small text-uppercase fw-semibold">Здоровье ОС (Health Score) □ </div>
45         <div class="fs-4 fw-bold mt-1 text-success font-monospace" id="el-health-score">--</div>
46         <div class="small text-muted mt-1" id="el-health-sub">Data Researcher EDA</div>
47         </div>
48     </div>
49 <div class="col-6 col-md-3">
50     <div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3">
51         <div class="text-muted small text-uppercase fw-semibold">Избыточность / Шум (Rdup) □ </div>
52         <div class="fs-4 fw-bold mt-1 text-info font-monospace" id="el-redundancy-pct">--</div>
53         <div class="small text-muted mt-1" id="el-redundancy-sub">Дедупликация шаблонов</div>

```

```
54 |     </div>
55 | </div>
56 | </div>
57 |
58 | <!-- Action & Intelligence Bar -->
59 | <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60 |     <div class="d-flex align-items-center gap-2 flex-wrap">
61 |         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="el-refresh-btn">
62 |             <i class="bi bi-arrow-clockwise"></i>
63 |             <span>Обновить</span>
64 |         </button>
65 |         <select class="form-select form-select-sm bg-dark text-white border-secondary" id="el-channel-select" style="width: 180px;">
66 |             <option value="System">Журнал System</option>
67 |             <option value="Application">Журнал Application</option>
68 |             <option value="Security">Журнал Security</option>
69 |         </select>
70 |         <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5" id="el-intel-analyze-btn">
71 |             <i class="bi bi-cpu"></i>
72 |             <span>Log Intelligence (EDA)</span>
73 |         </button>
74 |         <button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5" id="el-intel-audit-btn">
75 |             <i class="bi bi-shield-shaded"></i>
76 |             <span>Полный аудит</span>
77 |         </button>
78 |     </div>
79 |     <div class="d-flex align-items-center gap-2 flex-wrap">
80 |         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="el-rag-query-input" placeholder="RAG поиск (напр: сбой сети)..." style="width: 240px;">
```

```
81 |         <button class="btn btn-sm btn-info text-dark fw-semibold
    | " id="el-rag-search-btn">
82 |             <i class="bi bi-search"></i>
83 |         </button>
84 |         <input type="text" class="form-control form-control-sm b
    | g-dark text-white border-secondary" id="el-search-input" place
    | holder="Фильтр таблицы..." style="width: 170px;">
85 |     </div>
86 | </div>
87 |
88 | <!-- Intelligence Digest Banner (Hidden by default) -->
89 | <div id="el-intel-banner" class="alert alert-dark border-inf
    | o shadow-sm mb-3 d-none">
90 |     <div class="d-flex justify-content-between align-items-sta
    | rt flex-wrap gap-2">
91 |         <div>
92 |             <div class="d-flex align-items-center gap-2 mb-1">
93 |                 <span class="badge bg-info text-dark fw-bold font-mo
    | nospace" id="el-strategy-badge">STRATEGY</span>
94 |                 <strong class="text-light fs-6" id="el-intel-title">
    | Аналитический вердикт Log Intelligence</strong>
95 |             </div>
96 |             <div class="small text-secondary font-monospace" id="e
    | l-intel-rationale">--</div>
97 |             <div class="small text-warning mt-1" id="el-intel-reco
    | mmendation"></div>
98 |         </div>
99 |         <div class="text-end">
100 |             <span class="badge bg-secondary font-monospace" id="el
    | -chunks-badge">0 RAG chunks</span>
101 |             <button class="btn-close btn-close-white ms-2" id="el-
    | intel-close-btn" aria-label="Close"></button>
102 |         </div>
103 |     </div>
104 | </div>
105 |
106 | <!-- RAG Search Results Panel (Hidden by default) -->
107 | <div id="el-rag-results-card" class="card bg-dark text-white
```

```

border-info shadow-sm mb-3 d-none">
108     <div class="card-header border-info bg-black d-flex justif
y-content-between align-items-center py-2 px-3">
109         <span class="fw-semibold text-info"><i class="bi bi-brai
n me-2"></i>Результаты Adaptive Log RAG</span>
110         <button class="btn btn-sm btn-outline-secondary py-0 px-
2" id="el-rag-close-btn">Закрыть</button>
111     </div>
112     <div class="card-body p-2" id="el-rag-results-body">
113         <!-- Dynamic Search Items -->
114     </div>
115 </div>
116
117 <!-- Event Logs Table -->
118 <div class="card bg-dark text-white border-secondary shadow-
sm">
119     <div class="card-header border-secondary d-flex align-item
s-center justify-content-between py-2 px-3">
120         <span class="fw-semibold"><i class="bi bi-journal-text t
ext-primary me-2"></i>Записи системного журнала (Event Viewer)
</span>
121         <span class="badge bg-secondary font-monospace" id="el-t
able-badge">0 событий</span>
122     </div>
123     <div class="card-body p-0 table-responsive" style="max-hei
ght: 520px; overflow-y: auto;">
124         <table class="table table-dark table-hover table-striped
align-middle mb-0 small" id="el-table">
125             <thead class="sticky-top bg-dark">
126                 <tr class="text-secondary">
127                     <th>Уровень</th>
128                     <th>Время</th>
129                     <th>Источник (Provider)</th>
130                     <th>Event ID</th>
131                     <th>Описание события</th>
132                 </tr>
133             </thead>
134             <tbody id="el-tbody">

```

```
135 |         <tr><td colspan="5" class="text-center text-muted py
    | -4">Чтение журнала событий через WevtAPI...</td></tr>
136 |         </tbody>
137 |     </table>
138 | </div>
139 | </div>
140 | </div>
141 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Event Logs Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля Windows E
7  *   с интеграцией аналитического пайплайна Log Intelligence (
8  *   EDA -> Gate -> RAG).
9  *
10  * Usage Examples:
11  *   HTML Integration:
12  *     <script src="/windows/api/webgui/event_logs_tab/main.js
13  *     ?v=20261006_v1" type="module"></script>
14  *
15  *   JavaScript Import:
16  *     import { initEventLogsTab } from '/windows/api/webgui/e
17  *     vent_logs_tab/main.js';
18  *
19  * File: main.js
20  * Project: ai-breadboard
21  * Package: windows/api/webgui/event_logs_tab
22  * Author: hypo69
23  * Copyright: © 2026 hypo69
24  * Updated: 2026-10-08 02:00:00
25  * =====
26  */
27
28 const registerTabPoller = window.registerTabPoller || function
29   () {};
30 const isTabActive = window.isTabActive || (() => true);
31
32 let isInitialized = false;
```



```

29 let eventsList = [];
30
31 function escapeHtml(str) {
32     if (!str) return '';
33     return String(str)
34         .replace(/&/g, '&amp;')
35         .replace(/</g, '&lt;')
36         .replace(/>/g, '&gt;')
37         .replace(/"/g, '&quot;')
38         .replace(/'/g, '&#039;');
39 }
40
41 export async function fetchEventLogsSummary() {
42     if (!isTabActive('tab-event-logs')) return;
43     const channelSelect = document.getElementById('el-channel-select');
44     const channel = channelSelect ? channelSelect.value : 'System';
45
46     try {
47         const res = await fetch(`/api/event-logs/events?channel=${encodeURIComponent(channel)}&limit=100&hours=24`);
48         if (!res.ok) throw new Error(`HTTP ${res.status}`);
49         const entries = await res.json();
50         renderEventLogs(entries);
51         // Also fetch intelligence profile in background to update
           Health Score & Redundancy
52         fetchIntelligenceProfile(channel, false);
53     } catch (err) {
54         console.warn('[EventLogs] Ошибка получения событий:', err);
55     }
56 }
57
58 async function fetchIntelligenceProfile(channel, showBanner = true) {
59     try {
60         const res = await fetch(`/api/event-logs/intelligence/prof

```

```

    ile?channel=${encodeURIComponent(channel)}&hours=24&limit=100`
    );
61     if (!res.ok) return;
62     const data = await res.json();
63
64     const healthEl = document.getElementById('el-health-score'
    );
65     const redEl = document.getElementById('el-redundancy-pct')
    ;
66
67     if (data.profile) {
68         if (healthEl) {
69             const score = Math.round(data.profile.health_score ??
100);
70             healthEl.textContent = `${score}/100`;
71             healthEl.className = `fs-4 fw-bold mt-1 font-monospace
    ${score >= 80 ? 'text-success' : (score >= 50 ? 'text-warning
    ' : 'text-danger')}}`;
72         }
73         if (redEl) {
74             const rPct = (data.profile.redundancy_ratio_pct ?? 0).
    toFixed(1);
75             redEl.textContent = `${rPct}%`;
76         }
77     }
78
79     if (showBanner && data.decision) {
80         renderIntelligenceBanner(data);
81     }
82 } catch (err) {
83     console.debug('[EventLogs] Фоновый сбор аналитики:', err);
84 }
85 }
86
87 function renderIntelligenceBanner(data) {
88     const banner = document.getElementById('el-intel-banner');
89     const title = document.getElementById('el-intel-title');
90     const strat = document.getElementById('el-strategy-badge');

```

```
91 |   const rationale = document.getElementById('el-intel-rational
    | e');
92 |   const rec = document.getElementById('el-intel-recommendation
    | ');
93 |   const chunks = document.getElementById('el-chunks-badge');
94 |
95 |   if (!banner) return;
96 |
97 |   if (title) title.textContent = `Log Intelligence: Канал ${da
    | ta.channel || 'System'}`;
98 |   if (strat) strat.textContent = (data.decision.strategy || 'S
    | NAPSHOT').toUpperCase();
99 |   if (rationale) rationale.textContent = data.decision.rationa
    | le || '';
100 |   if (rec) rec.innerHTML = `
```

```

118         rationale: data.strategy_rationale,
119         recommended_llm_action: data.recommended_llm_action,
120         chunks_generated: data.top_clusters ? data.top_cluster
        s.length : 0,
121     }
122 });
123 } catch (err) {
124     console.warn('[EventLogs] Ошибка аудита:', err);
125 }
126 }
127
128 export async function runRAGSearch() {
129     const input = document.getElementById('el-rag-query-input');
130     const channelSelect = document.getElementById('el-channel-se
        lect');
131     const resultsCard = document.getElementById('el-rag-results-
        card');
132     const resultsBody = document.getElementById('el-rag-results-
        body');
133
134     if (!input || !resultsCard || !resultsBody) return;
135     const query = input.value.trim();
136     if (!query) return;
137
138     resultsBody.innerHTML = `
```

```
148     resultsBody.innerHTML = `
```

```
176 | eventsList.forEach(e => {
177 |     const l = (e.level || '').toLowerCase();
178 |     if (l.includes('crit')) crits++;
179 |     else if (l.includes('err')) errs++;
180 | });
181 |
182 | if (critEl) critEl.textContent = crits;
183 | if (errEl) errEl.textContent = errs;
184 |
185 | applyEventFilters();
186 | }
187 |
188 | function applyEventFilters() {
189 |     const tbody = document.getElementById('el-tbody');
190 |     const searchInput = document.getElementById('el-search-input
    ');
191 |     const badge = document.getElementById('el-table-badge');
192 |
193 |     if (!tbody) return;
194 |
195 |     const query = (searchInput ? searchInput.value : '').toLower
    Case().trim();
196 |
197 |     const filtered = eventsList.filter(e => {
198 |         if (!query) return true;
199 |         const msgMatch = (e.message || '').toLowerCase().includes(
    query);
200 |         const provMatch = (e.provider_name || e.provider || '').to
    LowerCase().includes(query);
201 |         const idMatch = String(e.event_id || e.id || '').includes(
    query);
202 |         return msgMatch || provMatch || idMatch;
203 |     });
204 |
205 |     if (badge) badge.textContent = `${filtered.length} из ${even
    tsList.length} событий`;
206 |
207 |     if (filtered.length === 0) {
```

```

208 |     tbody.innerHTML = '<tr><td colspan="5" class="text-center
| text-muted py-4">События не найдены</td></tr>';
209 |     return;
210 | }
211 |
212 | tbody.innerHTML = filtered.slice(0, 150).map(e => {
213 |     const level = (e.level || 'Information').toLowerCase();
214 |     let badgeClass = 'bg-secondary';
215 |     if (level.includes('crit')) badgeClass = 'bg-danger text-w
| hite';
216 |     else if (level.includes('err')) badgeClass = 'bg-danger te
| xt-white';
217 |     else if (level.includes('warn')) badgeClass = 'bg-warning
| text-dark';
218 |     else if (level.includes('info')) badgeClass = 'bg-info tex
| t-dark';
219 |
220 |     return `
221 |         <tr>
222 |             <td><span class="badge ${badgeClass}">${escapeHtml(e.l
| evel || 'Info')}</span></td>
223 |             <td class="text-muted small font-monospace">${escapeHt
| ml(e.time_created || e.timestamp || '-')}</td>
224 |             <td class="fw-semibold text-light small">${escapeHtml(
| e.provider_name || e.provider || 'Windows')}</td>
225 |             <td class="font-monospace text-warning small">${escape
| Html(String(e.event_id || e.id || '-'))}</td>
226 |             <td class="text-light small" style="max-width: 500px;
| word-break: break-word;">${escapeHtml(e.message || '-')}</td>
227 |         </tr>
228 |     `;
229 | }).join('');
230 | }
231 |
232 | export function initEventLogsTab() {
233 |     if (isInitialized) return;
234 |     isInitialized = true;
235 |

```

```
236 |   const refreshBtn = document.getElementById('el-refresh-btn')
    |   ;
237 |   if (refreshBtn) refreshBtn.addEventListener('click', fetchEventLogsSummary);
238 |
239 |   const channelSelect = document.getElementById('el-channel-select');
240 |   if (channelSelect) channelSelect.addEventListener('change', fetchEventLogsSummary);
241 |
242 |   const searchInput = document.getElementById('el-search-input');
243 |   if (searchInput) searchInput.addEventListener('input', applyEventFilters);
244 |
245 |   const analyzeBtn = document.getElementById('el-intel-analyze-btn');
246 |   if (analyzeBtn) analyzeBtn.addEventListener('click', () => {
247 |     const chan = channelSelect ? channelSelect.value : 'System';
248 |     fetchIntelligenceProfile(chan, true);
249 |   });
250 |
251 |   const auditBtn = document.getElementById('el-intel-audit-btn');
252 |   if (auditBtn) auditBtn.addEventListener('click', runFullAudit);
253 |
254 |   const ragBtn = document.getElementById('el-rag-search-btn');
255 |   if (ragBtn) ragBtn.addEventListener('click', runRAGSearch);
256 |
257 |   const ragInput = document.getElementById('el-rag-query-input');
258 |   if (ragInput) ragInput.addEventListener('keydown', (e) => {
259 |     if (e.key === 'Enter') runRAGSearch();
260 |   });
261 |
262 |   const bannerCloseBtn = document.getElementById('el-intel-clo
```



```

    se-btn');
263   if (bannerCloseBtn) bannerCloseBtn.addEventListener('click',
    () => {
264       const banner = document.getElementById('el-intel-banner');
265       if (banner) banner.classList.add('d-none');
266   });
267
268   const ragCloseBtn = document.getElementById('el-rag-close-bt
n');
269   if (ragCloseBtn) ragCloseBtn.addEventListener('click', () =>
    {
270       const card = document.getElementById('el-rag-results-card'
    );
271       if (card) card.classList.add('d-none');
272   });
273
274   registerTabPoller('tab-event-logs', fetchEventLogsSummary, 1
    5000, { immediate: true });
275   if (isTabActive('tab-event-logs')) {
276       fetchEventLogsSummary();
277   }
278 }
279
280 export function activateEventLogsTab() {
281     fetchEventLogsSummary();
282 }
283
284 window.initEventLogsTab = initEventLogsTab;
285 window.activateEventLogsTab = activateEventLogsTab;
286
287 // Автоинициализация: загрузчик вкладок только подключает скри
пт
288 if (document.getElementById('el-tbody') && isTabActive('tab-ev
ent-logs')) {
289     initEventLogsTab();
290 }
291

```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: AI-Breadboard Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон веб-интерфейса системного модуля (index).
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /src/api/webgui/file_history_ai_search_tab/index.html
12 |
13 File: index.html
14 Project: ai-breadboard
15 Package: src/api/webgui/file_history_ai_search_tab
16 Author: hypo69
17 Copyright: © 2026 hypo69
18 Updated: 2026-10-01 13:13:56
19 |=====
20 |=====
21 -->
22 <!-- File History AI Search Tab -->
23 <style>
24 |.fh-container {
25 |  padding: 0.75rem 1rem;
26 |  max-width: 1400px;
27 |  margin: 0 auto;
28 |}
29 |.fh-card {
30 |  background: var(--bs-body-bg, #1e293b);
31 |  border: 1px solid var(--bs-border-color, #334155);
32 |  border-radius: 12px;
33 |  padding: 1.25rem;
34 |  box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
```

```
34 | }
35 | .fh-header {
36 |     background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
0%);
37 |     border: 1px solid #334155;
38 |     border-radius: 12px;
39 |     padding: 1rem 1.5rem;
40 |     color: #f8fafc;
41 | }
42 | .fh-stat-card {
43 |     background: rgba(30, 41, 59, 0.7);
44 |     border: 1px solid var(--bs-border-color, #334155);
45 |     border-radius: 10px;
46 |     padding: 0.85rem 1.1rem;
47 |     transition: transform 0.15s ease, border-color 0.15s ease;
48 | }
49 | .fh-stat-card:hover {
50 |     transform: translateY(-2px);
51 |     border-color: #3b82f6;
52 | }
53 | .fh-stat-val {
54 |     font-size: 1.4rem;
55 |     font-weight: 700;
56 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
57 | }
58 | .fh-search-box {
59 |     background: var(--bs-dark, #0f172a);
60 |     border: 1px solid #3b82f6;
61 |     border-radius: 12px;
62 |     padding: 0.75rem 1.25rem;
63 | }
64 | .fh-result-card {
65 |     border-left: 4px solid #3b82f6;
66 |     background: var(--bs-body-bg, #1e293b);
67 |     border-top: 1px solid var(--bs-border-color, #334155);
68 |     border-right: 1px solid var(--bs-border-color, #334155);
69 |     border-bottom: 1px solid var(--bs-border-color, #334155);
```

```

70     border-radius: 8px;
71     padding: 1rem;
72     margin-bottom: 0.75rem;
73     transition: background-color 0.15s ease;
74 }
75 .fh-result-card:hover {
76     background: rgba(59, 130, 246, 0.05);
77 }
78 .fh-score-badge {
79     font-size: 0.82rem;
80     font-weight: 600;
81     font-family: monospace;
82 }
83 .fh-snippet-box {
84     background: rgba(0, 0, 0, 0.2);
85     border-radius: 6px;
86     padding: 0.6rem 0.85rem;
87     font-family: SFMono-Regular, Menlo, Monaco, Consolas, mono
space;
88     font-size: 0.85rem;
89     color: var(--bs-body-color, #cbd5e1);
90     white-space: pre-wrap;
91     word-break: break-all;
92     max-height: 120px;
93     overflow-y: auto;
94 }
95 </style>
96
97 <div class="fh-container">
98     <!-- <span data-i18n="auto___0105ee"></span> -->
99     <div class="d-flex justify-content-between align-items-cente
r flex-wrap gap-3 mb-3 fh-header">
100         <div class="d-flex align-items-center gap-3">
101             <div class="bg-primary bg-opacity-25 p-2.5 rounded-3 tex
t-primary border border-primary border-opacity-25">
102                 <i class="bi bi-clock-history fs-3"></i>
103             </div>
104             <div>

```

```

105 |         <h5 class="mb-0 fw-bold d-flex align-items-center gap-
      | 2">
106 |             <span data-i18n="auto___b3724e"></span> Windows (Fil
      | e History AI Search)
107 |             <span class="badge bg-primary bg-opacity-75 fs-7">RA
      | G v1.0</span>
108 |         </h5>
109 |         <span class="text-secondary small"><span data-i18n="au
      | to__windows_file_history_recent_lnk_activity_db__07a692"></spa
      | n></span>
110 |     </div>
111 | </div>
112 | <div class="d-flex align-items-center gap-2 flex-wrap">
113 |     <button class="btn btn-sm btn-outline-light rounded-pill
      | px-3" id="btn-fh-refresh-status" type="button">
114 |         <i class="bi bi-arrow-clockwise me-1"></i> <span data-
      | i18n="auto___6cc5dc"></span>
115 |     </button>
116 |     <button class="btn btn-sm btn-success rounded-pill px-3"
      | id="btn-fh-reindex" type="button">
117 |         <i class="bi bi-cpu me-1"></i> <span data-i18n="auto__
      | _5ec953"></span> RAG
118 |     </button>
119 | </div>
120 | </div>
121 |
122 | <!-- <span data-i18n="auto___1983d2"></span> -->
123 | <div class="row g-3 mb-3">
124 |     <div class="col-12 col-sm-6 col-md-3">
125 |         <div class="fh-stat-card">
126 |             <div class="text-secondary small text-uppercase fw-sem
      | ibold mb-1"><span data-i18n="auto__rag__1a3eea"></span></div>
127 |             <div class="d-flex align-items-center justify-content-
      | between">
128 |                 <div class="fh-stat-val text-primary" id="fh-stat-do
      | cs-count">0</div>
129 |                 <i class="bi bi-database-check fs-4 text-primary opa
      | city-50"></i>

```

```
130         </div>
131         <div class="small text-muted mt-1" id="fh-stat-last-in
dex"><span data-i18n="auto___16deed"></span></div>
132     </div>
133 </div>
134
135     <div class="col-12 col-sm-6 col-md-3">
136         <div class="fh-stat-card">
137             <div class="text-secondary small text-uppercase fw-sem
ibold mb-1"><span data-i18n="auto___8205bf"></span></div>
138             <div class="d-flex align-items-center justify-content-
between">
139                 <div class="fh-stat-val" id="fh-stat-scheduler-statu
s"><span data-i18n="auto___aa0d25"></span></div>
140                 <i class="bi bi-arrow-repeat fs-4 text-warning opaci
ty-50" id="fh-scheduler-icon"></i>
141             </div>
142             <div class="small text-muted mt-1" id="fh-stat-schedul
er-interval"><span data-i18n="auto___15___d40d33"></span></div>
143         </div>
144     </div>
145
146     <div class="col-12 col-sm-6 col-md-3">
147         <div class="fh-stat-card">
148             <div class="text-secondary small text-uppercase fw-sem
ibold mb-1"><span data-i18n="auto___faiss___1cdc9e"></span></div>
149             <div class="d-flex align-items-center justify-content-
between">
150                 <div class="fh-stat-val fs-6 text-truncate" id="fh-s
tat-faiss-file" style="max-width: 180px;">file_history.faiss</
div>
151                 <i class="bi bi-file-earmark-binary fs-4 text-info o
pacity-50"></i>
152             </div>
153             <div class="small text-muted mt-1"><span data-i18n="au
to___18f0e9"></span> Gemini Embeddings</div>
154         </div>
```

```
155 |     </div>
156 |
157 |     <div class="col-12 col-sm-6 col-md-3">
158 |         <div class="fh-stat-card">
159 |             <div class="text-secondary small text-uppercase fw-sem
160 | ibold mb-1"><span data-i18n="auto___ec3fc1"></span></div>
161 |             <div class="d-flex align-items-center gap-2 mt-1">
162 |                 <button class="btn btn-sm btn-outline-success w-50 p
163 | y-1" id="btn-fh-start-scheduler">
164 |                     <i class="bi bi-play-fill me-1"></i><span data-i18
165 | n="auto___c70ec8"></span>
166 |                 </button>
167 |                 <button class="btn btn-sm btn-outline-danger w-50 py
168 | -1" id="btn-fh-stop-scheduler">
169 |                     <i class="bi bi-stop-fill me-1"></i><span data-i18
170 | n="auto___6d714e"></span>
171 |                 </button>
172 |             </div>
173 |             <div class="small text-muted mt-1 text-center"><span d
174 | ata-i18n="auto___15___eecb38"></span></div>
175 |         </div>
176 |     </div>
177 | </div>
178 |
179 | <!-- <span data-i18n="auto___8dc75e"></span> -->
180 | <div class="fh-card mb-4">
181 |     <h6 class="fw-bold mb-3 d-flex align-items-center gap-2">
182 |         <i class="bi bi-search text-primary"></i>
183 |         <span data-i18n="auto___2517ec"></span>
184 |     </h6>
185 |
186 |     <div class="row g-2 mb-3">
187 |         <div class="col-12 col-md-7">
188 |             <div class="input-group">
189 |                 <span class="input-group-text bg-body-tertiary borde
190 | r-end-0">
191 |                     <i class="bi bi-search text-muted"></i>
192 |                 </span>
```

```
186 |         <input type="text" class="form-control border-start-
    | 0" id="fh-search-input" placeholder="<span data-i18n="auto___9
    | 7d6b0"></span> Word')...">
187 |         <button class="btn btn-primary px-4 fw-semibold" id=
    | "btn-fh-search" type="button">
188 |             <i class="bi bi-arrow-right-circle me-1"></i> <spa
    | n data-i18n="auto___ed6e1d"></span>
189 |         </button>
190 |     </div>
191 | </div>
192 | <div class="col-6 col-md-3">
193 |     <select class="form-select" id="fh-filter-source">
194 |         <option value=""><span data-i18n="auto___5f3a3d"></s
    | pan> (All Sources)</option>
195 |         <option value="file_history_xml">Windows File Histor
    | y (XML)</option>
196 |         <option value="recent_lnk">Recent Files (LNK)</optio
    | n>
197 |         <option value="activity_db">Activity History (DB)</o
    | ption>
198 |         <option value="fs_scan"><span data-i18n="auto___8be3
    | b9"></span></option>
199 |     </select>
200 | </div>
201 | <div class="col-6 col-md-2">
202 |     <select class="form-select" id="fh-filter-topk">
203 |         <option value="5" selected><span data-i18n="auto___0
    | 0f8ee"></span> 5</option>
204 |         <option value="10"><span data-i18n="auto___00f8ee"><
    | /span> 10</option>
205 |         <option value="20"><span data-i18n="auto___00f8ee"><
    | /span> 20</option>
206 |         <option value="50"><span data-i18n="auto___00f8ee"><
    | /span> 50</option>
207 |     </select>
208 | </div>
209 | </div>
210 |
```



```

211 |     <!-- <span data-i18n="auto___4be2be"></span> -->
212 |     <div class="text-center py-4 d-none" id="fh-search-spinner
    | ">
213 |         <div class="spinner-border text-primary" role="status">
214 |             <span class="visually-hidden"><span data-i18n="auto__r
    | ag__662d74"></span>...</span>
215 |         </div>
216 |         <div class="small text-muted mt-2"><span data-i18n="auto
    | ___93b6bf"></span> FAISS...</div>
217 |     </div>
218 |
219 |     <!-- <span data-i18n="auto___637b2c"></span> -->
220 |     <div id="fh-search-results-container">
221 |         <div class="text-center text-muted py-4 border rounded-3
    | bg-body-tertiary bg-opacity-50">
222 |             <i class="bi bi-search fs-2 d-block mb-1 text-secondar
    | y opacity-50"></i>
223 |             <span><span data-i18n="auto___3df1ff"></span>.</span>
224 |         </div>
225 |     </div>
226 | </div>
227 |
228 | <!-- <span data-i18n="auto___f2f2be"></span>)</div> -->
229 | <div class="fh-card">
230 |     <div class="d-flex justify-content-between align-items-cen
    | ter mb-3 flex-wrap gap-2">
231 |         <h6 class="fw-bold mb-0 d-flex align-items-center gap-2"
    | >
232 |             <i class="bi bi-list-stars text-info"></i>
233 |             <span data-i18n="auto__windows__8738e2"></span>)</div>
234 |         </h6>
235 |         <button class="btn btn-sm btn-outline-info rounded-pill
    | px-3" id="btn-fh-scan-now">
236 |             <i class="bi bi-arrow-repeat me-1"></i> <span data-i18
    | n="auto___4748be"></span>
237 |         </button>
238 |     </div>
239 |

```

```

240     <div class="table-responsive">
241         <table class="table table-hover align-middle mb-0" id="f
| h-items-table">
242             <thead class="table-light">
243                 <tr>
244                     <th scope="col" style="width: 40px;">#</th>
245                     <th scope="col"><span data-i18n="auto___94b8df"></
| span></th>
246                     <th scope="col"><span data-i18n="auto___8290a3"></
| span></th>
247                     <th scope="col"><span data-i18n="auto___bb9263"></
| span></th>
248                     <th scope="col"><span data-i18n="auto___c80d7e"></
| span></th>
249                     <th scope="col"><span data-i18n="auto___98713e"></
| span></th>
250                     <th scope="col" class="text-end"><span data-i18n="
| auto___9978ac"></span></th>
251                 </tr>
252             </thead>
253             <tbody id="fh-items-tbody">
254                 <tr>
255                     <td colspan="7" class="text-center py-4 text-muted
| ">
256                         <span data-i18n="auto___cb63af"></span> Windows.
257                     </td>
258                 </tr>
259             </tbody>
260         </table>
261     </div>
262 </div>
263 </div>
264

```

main.js

```
1  /**
2   * =====
3   * Process Name: AI-Breadboard UI - Main Script
4   * =====
5   * Description:
6   *   Клиентский веб-скрипт модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/src/api/webgui/file_history_ai_search_tab
11  *     /main.js?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: src/api/webgui/file_history_ai_search_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:13:56
19  * =====
20  */
21 /**
22  * file_history_ai_search_tab/main.js – Логика веб-интерфейса
23  * поиска по истории файлов Windows через RAG
24  */
25 (function () {
26   'use strict;
27   const btnSearch = document.getElementById('fh-search-input');
28   const btnRefreshStatus = document.getElementById('btn-fh-r
```

```
efresh-status');
29     const btnReindex = document.getElementById('btn-fh-reindex
    ');
30     const btnScanNow = document.getElementById('btn-fh-scan-no
    w');
31     const btnStartSched = document.getElementById('btn-fh-star
    t-scheduler');
32     const btnStopSched = document.getElementById('btn-fh-stop-
    scheduler');
33
34     if (btnSearch) {
35         btnSearch.addEventListener('click', executeSearch);
36     }
37
38     if (searchInput) {
39         searchInput.addEventListener('keydown', (e) => {
40             if (e.key === 'Enter') {
41                 executeSearch();
42             }
43         });
44     }
45
46     if (btnRefreshStatus) {
47         btnRefreshStatus.addEventListener('click', loadStatus);
48     }
49
50     if (btnReindex) {
51         btnReindex.addEventListener('click', triggerReindex);
52     }
53
54     if (btnScanNow) {
55         btnScanNow.addEventListener('click', triggerScan);
56     }
57
58     if (btnStartSched) {
59         btnStartSched.addEventListener('click', () => toggleSche
    duler('start'));
60     }
```

```

61
62     if (btnStopSched) {
63         btnStopSched.addEventListener('click', () => toggleScheduler(
        'stopi18n.t('auto__loadstatus_rag_async_function_loadstatus_try_const_resp_await_fetch__d52650')/api/windows/file-history/status');
64         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
65         const data = await resp.json();
66
67         const elCount = document.getElementById('fh-stat-docs-count');
68         const elLastIdx = document.getElementById('fh-stat-last-index');
69         const elSchedStat = document.getElementById('fh-stat-scheduler-status');
70         const elSchedIcon = document.getElementById('fh-scheduler-icon');
71         const elSchedInt = document.getElementById('fh-stat-scheduler-interval');
72         const elFaissFile = document.getElementById('fh-stat-faiss-filei18n.t('auto__if_elcount_elcount_textcontent_data_total_indexed_documents_0_if_ellastidx_ellastidx_textcontent_data_last_indexed_at_new_date_data_last_indexed_at_tolocalestring__c32c12')ru-RU'))`
73             : i18n.t('auto__16deed');
74     }
75
76     if (elSchedStat) {
77         if (data.scheduler_running) {
78             elSchedStat.textContent = i18n.t('auto__667904');
79             elSchedStat.className = 'fh-stat-val text-success';
80             if (elSchedIcon) elSchedIcon.className = 'bi bi-arrow-repeat fs-4 text-success opacity-75 spin';
81         } else {
82             elSchedStat.textContent = i18n.t('auto__aa0d25');
83             elSchedStat.className = 'fh-stat-val text-warning';
84             if (elSchedIcon) elSchedIcon.className = 'bi bi-arrow-repeat fs-4 text-warning opacity-50i18n.t('auto__if_elschedi

```

```

nt_elschedint_textcontent_data_update_interval_minutes_15_if_e
lfaissfile_data_index_file_path_const_parts_data_index_file_pa
th_split_elfaissfile_textcontent_parts_parts_length_1_data_ind
ex_file_path_elfaissfile_title_data_index_file_path_catch_err_
console_error__3cb452')[FileHistoryUI] Ошибка загрузки статуса
:i18n.t('auto__err_windows_async_function_triggerscan_const_tb
ody_document_getelementbyid__5b14c9')fh-items-tbody');
85     if (tbody) {
86         tbody.innerHTML = `
87             <tr>
88                 <td colspan="7" class="text-center py-4">
89                     <div class="spinner-border spinner-border-sm text-
info me-2" role="status"></div>
90                     <span>Сканирование источников истории файлов Windo
ws...</span>
91                 </td>
92             </tr>`;
93     }
94
95     try {
96         const resp = await fetch('/api/windows/file-history/scan
i18n.t('auto__if_resp_ok_throw_new_error_http_resp_status_curr
entitems_await_resp_json_renderitemstable_currentitems_if_wind
ow_showtoast_window_showtoast_currentitems_length__a2f871')inf
o');
97     }
98     } catch (err) {
99         console.error(i18n.t('auto__filehistoryui__c71a7b'), err
);
100     if (tbody) {
101         tbody.innerHTML = `
102             <tr>
103                 <td colspan="7" class="text-center py-4 text-dange
r">
104                     <i class="bi bi-exclamation-triangle me-1"></i>
Ошибка при сканировании: ${err.message}
105                 </td>
106             </tr>`;

```

```

107     }
108 }
109 }
110
111 /**
112  * Отрисовка таблицы сохраненных элементов телеметрии
113  */
114 function renderItemTable(items) {
115     const tbody = document.getElementById('fh-items-tbody');
116     if (!tbody) return;
117
118     if (!items || items.length === 0) {
119         tbody.innerHTML = `
120             <tr>
121                 <td colspan="7" class="text-center py-4 text-muted">
122                     Элементы истории файлов не обнаружены.
123                 </td>
124             </tr>`;
125         return;
126     }
127
128     let html = '';
129     items.slice(0, 100).forEach((item, idx) => {
130         const badgeSource = getSourceBadge(item.source_type);
131         const formattedSize = formatBytes(item.file_size);
132         const timeStr = item.timestamp ? new Date(item.timestamp
133         ).toLocaleString('ru-RU') : '-';
134
135         html += `
136             <tr>
137                 <td class="text-muted small">${idx + 1}</td>
138                 <td>
139                     <div class="fw-semibold text-truncate" style="max-
140                     width: 280px;" title="${escapeHtml(item.file_name)}">
141                     <i class="bi bi-file-earmark-text me-1 text-prim
142                     ary"></i>${escapeHtml(item.file_name)}
143                     </div>
144                     <div class="small text-muted text-truncate" style=

```

```

    "max-width: 320px;" title="${escapeHtml(item.file_path)}">
142         ${escapeHtml(item.file_path)}
143     </div>
144 </td>
145 <td>${badgeSource}</td>
146 <td><span class="badge bg-secondary bg-opacity-50 te
xt-light">${escapeHtml(item.event_type)}</span></td>
147 <td class="small text-muted">${timeStr}</td>
148 <td class="small font-monospace">${formattedSize}</t
d>
149 <td class="text-end">
150 <button class="btn btn-sm btn-outline-secondary py
-0 px-2 btn-copy-path" data-path="${escapeHtml(item.file_path)
}" title=i18n.t('auto__8a5b6a')>
151 <i class="bi bi-clipboard"></i>
152 </button>
153 </td>
154 </tr>`;
155 });
156
157 tbody.innerHTML = html;
158
159 // Привязка копирования пути
160 tbody.querySelectorAll('.btn-copy-path').forEach((btn) =>
{
161     btn.addEventListener('click', (e) => {
162         const path = e.currentTarget.getAttribute('data-path')
163         ;
164         if (path) {
165             navigator.clipboard.writeText(path);
166             if (window.showToast) window.showToast(i18n.t('auto_
__f6358a'), 'successi18n.t('auto__rag_async_function_triggerre
index_const_btnreindex_document_getelementbyid__29bc61')btn-fh
-reindex');
167
168             if (btnReindex) btnReindex.disabled = true;
169
170             if (window.showToast) {
171                 window.showToast(i18n.t('auto__rag__9ef43b'), 'info');

```



```

170     }
171
172     try {
173         const resp = await fetch('/api/windows/file-history/index', { method: 'POST' });
174         if (!resp.ok) throw new Error('HTTP status: ' + resp.status);
175         const data = await resp.json();
176         if (window.showToast) window.showToast(data.indexed_count);
177         console.log('auto__filehistoryui_rag__ffcf9a', err);
178         if (window.showToast) {
179             window.showToast('Ошибка индексации RAG: ' + err.message);
180             const filterSource = document.getElementById('fh-filter-source');
181             const filterTopk = document.getElementById('fh-filter-topk');
182             const spinner = document.getElementById('fh-search-spinner');
183             const container = document.getElementById('fh-search-results-container');
184
185             const query = searchInput ? searchInput.value.trim() : '';
186             if (!query) {
187                 if (window.showToast) window.showToast('auto__f4f030', 'warning');
188                 return;
189             }
190
191             const topK = filterTopk ? parseInt(filterTopk.value, 10) : 5;
192             const sourceType = filterSource && filterSource.value ? filterSource.value : null;
193

```

```

194 |     if (spinner) spinner.classList.remove('d-none');
195 |     if (container) container.innerHTML = '';
196 |
197 |     try {
198 |         const resp = await fetch('/api/windows/file-history/search', {
199 |             method: 'POST',
200 |             headers: { 'Content-Type': 'application/json' },
201 |             body: JSON.stringify({
202 |                 query: query,
203 |                 top_k: topK,
204 |                 source_type: sourceType,
205 |             }),
206 |         });
207 |
208 |         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
209 |         const data = await resp.json();
210 |         renderSearchResults(data);
211 |     } catch (err) {
212 |         console.error(i18n.t('auto__filehistoryui__1be39c'), err);
213 |
214 |         if (container) {
215 |             container.innerHTML = `
216 |                 <div class="alert alert-danger mb-0">
217 |                     <i class="bi bi-exclamation-octagon me-1"></i> Ошибка семантического поиска: ${err.message}
218 |                 </div>`;
219 |         }
220 |     } finally {
221 |         if (spinner) spinner.classList.add('d-none');
222 |
223 |         if (!data.results || data.results.length === 0) {
224 |             container.innerHTML = `
225 |                 <div class="text-center text-muted py-4 border rounded
-3">

```

```

226 |         <i class="bi bi-emoji-frown fs-3 d-block mb-1 text-w
      | arning opacity-50i18n.t('auto__i_span_strong_escapehtml_data_q
      | uery_strong_span_br_small_class__5ef260')text-secondaryi18n.t(
      | 'auto__rag_small_div_return_let_html_div_class__1b153c')d-flex
      | justify-content-between align-items-center mb-2 px-1">
227 |         <span class="small text-muted">
228 |             Найдено результатов: <strong>${data.total_found}</st
      | rong> (Время выполнения: <strong>${data.execution_time_ms} мс<
      | /strong>)
229 |         </span>
230 |     </div>`;
231 |
232 |     data.results.forEach((item, idx) => {
233 |         const badgeSource = getSourceBadge(item.source_type);
234 |         const scorePct = Math.round(item.score * 100);
235 |         const badgeScoreClass = item.score > 0.6 ? 'bg-success'
      | : item.score > 0.3 ? 'bg-info' : 'bg-secondary';
236 |         const timeStr = item.timestamp ? new Date(item.timestamp
      | ).toLocaleString('ru-RU') : '-';
237 |
238 |         html += `
239 |             <div class="fh-result-card">
240 |                 <div class="d-flex justify-content-between align-ite
      | ms-start gap-2 mb-1">
241 |                     <div class="d-flex align-items-center gap-2">
242 |                         <span class="badge rounded-circle bg-body-tertia
      | ry text-muted p-2 fw-bold me-1" style="width: 26px; height: 26
      | px; display: inline-flex; align-items: center; justify-content
      | : center;">
243 |                             ${idx + 1}
244 |                         </span>
245 |                         <h6 class="mb-0 fw-bold text-truncate" style="ma
      | x-width: 450px;" title="${escapeHtml(item.file_name)}">
246 |                             <i class="bi bi-file-earmark-code me-1 text-pr
      | imary"></i>${escapeHtml(item.file_name)}
247 |                         </h6>
248 |                         ${badgeSource}
249 |                     </div>

```

```

250         <div class="d-flex align-items-center gap-2">
251             <span class="badge ${badgeScoreClass} fh-score-b
252 |         adge" title=i18n.t('auto___4c2381')>
253                 Score: ${item.score}
254             </span>
255             <button class="btn btn-sm btn-outline-secondary
256 |         py-0 px-2 btn-copy-path" data-path="${escapeHtml(item.file_pat
257 |         h)}" title=i18n.t('auto___bbf33e')>
258                 <i class="bi bi-clipboard"></i>
259             </button>
260         </div>
261     </div>
262
263     <div class="small text-muted text-truncate mb-2 font
264 |         -monospace">
265         <i class="bi bi-folder2-open me-1 text-secondary">
266 |         </i>${escapeHtml(item.file_path)}
267         <span class="ms-2 text-secondaryi18n.t('auto___time
268 |         str_span_div_div_class___41bff1')fh-snippet-box">
269             ${escapeHtml(item.snippet)}
270         </div>
271     </div>`;
272
273     });
274
275     container.innerHTML = html;
276
277     // Привязка копирования в результатах поиска
278     container.querySelectorAll('.btn-copy-path').forEach((btn)
279 |     => {
280         btn.addEventListener('click', (e) => {
281             const path = e.currentTarget.getAttribute('data-path')
282 |         ;
283             if (path) {
284                 navigator.clipboard.writeText(path);
285                 if (window.showToast) window.showToast(i18n.t('auto_
286 |         ___f6358a'), 'successi18n.t('auto___async_function_toggleschedul
287 |         er_action_const_url_api_windows_file_history_scheduler_action_
288 |         try_const_resp_await_fetch_url_method___3e7cc6')POST' ));

```

```

277         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
278         const data = await resp.json();
279         if (window.showToast) {
280             window.showToast(
281                 action === 'start' ? i18n.t('auto__rag__8d29a5') : i
18n.t('auto__426630'),
282                 action === 'start' ? 'success' : 'info'
283             );
284         }
285         loadStatus();
286     } catch (err) {
287         console.error(i18n.t('auto__filehistoryui__5680c6'), err
);
288         if (window.showToast) window.showToast(`Ошибка: ${err.me
ssage}`, 'errori18n.t('auto__function_getsourcebadge_sourcety
e_switch_sourcetype_case_f06986')file_history_xml':
289         return '<span class="badge bg-primary bg-opacity-75"><
i class="bi bi-file-earmark-xml me-1"></i>File History XML</sp
an>';
290         case 'recent_lnk':
291             return '<span class="badge bg-info bg-opacity-75"><i c
lass="bi bi-link-45deg me-1"></i>Recent LNK</span>';
292         case 'activity_db':
293             return '<span class="badge bg-warning text-dark bg-opa
city-75"><i class="bi bi-database me-1"></i>Activity DB</span>
';
294         case 'fs_scan':
295             return '<span class="badge bg-success bg-opacity-75"><
i class="bi bi-hdd-network me-1"></i>FS Scan</span>';
296         default:
297             return '<span class="badge bg-secondary">${escapeHtml(
sourceType || 'unknowni18n.t('auto__span_function_formatbytes_
bytes_decimals_1_if_bytes_bytes_0_return_98f56d')0 B';
298         const k = 1024;
299         const dm = decimals < 0 ? 0 : decimals;
300         const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
301         const i = Math.floor(Math.log(bytes) / Math.log(k));
302         return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +

```

```
' i18n.t('auto__sizes_i_html_function_escapehtml_str_if_str_re
turn_ddd7ae')';
303     return String(str)
304         .replace(/&/g, '&amp;')
305         .replace(/</g, '&lt;')
306         .replace(/>/g, '&gt;')
307         .replace(/"/g, '&quot;')
308         .replace(/'/g, '&#039;');i18n.t('auto__dom_if_document_read
ystate__83155a')loading') {
309     document.addEventListener('DOMContentLoaded', initFileHist
oryTab);
310 } else {
311     initFileHistoryTab();
312 }
313 }());
314
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/file_recovery_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/file_recovery_tab/index
15 |.html" class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/file_recovery_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26 <!-- File Recovery (R-Studio Technician Portable) Tab -->
27 <style>
28 |.recovery-container {
29 |  padding: 0.75rem 1rem;
30 |  max-width: 1200px;
31 |  margin: 0 auto;
32 |}
33 |.recovery-card {
34 |  background: #0f172a !important;
```

```
33 |     border: 1px solid #334155 !important;
34 |     border-radius: 10px;
35 |     padding: 1.25rem;
36 |     color: #f8fafc;
37 |     box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.2);
38 | }
39 | .recovery-header {
40 |     background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
41 | 0%);
42 |     border: 1px solid #334155;
43 |     border-radius: 10px;
44 |     padding: 0.85rem 1.25rem;
45 | }
46 | .recovery-stat-box {
47 |     background: #1e293b;
48 |     border: 1px solid #334155;
49 |     border-radius: 8px;
50 |     padding: 0.75rem 1rem;
51 | }
52 | .recovery-stat-title {
53 |     font-size: 0.75rem;
54 |     text-transform: uppercase;
55 |     letter-spacing: 0.05em;
56 |     color: #94a3b8;
57 | }
58 | .recovery-stat-val {
59 |     font-size: 1.1rem;
60 |     font-weight: 600;
61 |     color: #f8fafc;
62 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
63 | Consolas, monospace;
64 | }
65 | .recovery-feature-item {
66 |     display: flex;
67 |     align-items: flex-start;
68 |     gap: 0.5rem;
69 |     margin-bottom: 0.5rem;
70 |     font-size: 0.85rem;
```



```

69     color: #cbd5e1;
70 }
71 </style>
72
73 <div class="recovery-container">
74     <!-- Верхняя информационная панель -->
75     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 recovery-header text-white">
76         <div class="d-flex align-items-center gap-2.5">
77             <span class="fs-4">❏ </span>
78             <div>
79                 <h5 class="mb-0 fw-bold fs-6">Восстановление удаленных
80                 файлов</h5>
81                 <span class="text-secondary small">Запуск профессионал
82                 ьного комплекса R-Studio Technician Portable</span>
83             </div>
84             <div class="d-flex align-items-center gap-2">
85                 <span class="badge rounded-pill bg-dark border border-se
86                 condary text-info px-2.5 py-1.5" id="recovery-status-badge">
87                     Проверка утилиты...
88                 </span>
89                 <button class="btn btn-sm btn-outline-info rounded-pill
90                 px-3" id="btn-recovery-refresh" type="button">
91                     <i class="bi bi-arrow-clockwise me-1"></i> Обновить ст
92                     атус
93                 </button>
94             </div>
95         </div>
96     </div>
97
98     <div class="row g-3 mb-3">
99         <!-- Левая колонка: Запуск и возможности -->
100         <div class="col-12 col-lg-7">
101             <div class="recovery-card h-100 d-flex flex-column justi
102             fy-content-between">
103                 <div>
104                     <h6 class="fw-bold text-info mb-3 d-flex align-items
105                     -center gap-2">

```

```
99 |         <i class="bi bi-arrow-counterclockwise"></i> Восст
| ановление данных с накопителей
100 |     </h6>
101 |
102 |     <p class="text-light small mb-3">
103 |         <strong>R-Studio Technician</strong> – мощный комп
| лекс для восстановления удалённых, повреждённых и отформатиров
| анных файлов с локальных и сетевых дисков, RAID-массивов, флеш
| -накопителей и виртуальных дисков.
104 |     </p>
105 |
106 |     <div class="recovery-feature-item">
107 |         <i class="bi bi-check2-circle text-success fs-5"><
| /i>
108 |         <div>
109 |             <strong>Поддерживаемые файловые системы:</strong>
| > NTFS, NTFS5, ReFS, FAT12/16/32, exFAT, HFS/HFS+, APFS, Ext2/
| Ext3/Ext4 FS, UFS.
110 |         </div>
111 |     </div>
112 |
113 |     <div class="recovery-feature-item">
114 |         <i class="bi bi-check2-circle text-success fs-5"><
| /i>
115 |         <div>
116 |             <strong>Сценарии восстановления:</strong> Случай
| ное удаление (Shift+Del), форматирование разделов, повреждение
| MBR/GPT, сбои питания, вирусные атаки, сбойные сектора.
117 |         </div>
118 |     </div>
119 |
120 |     <div class="recovery-feature-item">
121 |         <i class="bi bi-check2-circle text-success fs-5"><
| /i>
122 |         <div>
123 |             <strong>Режим работы:</strong> Portable (не треб
| ует установки, запускается напрямую из <code>\bin\R-Studio_Tec
| hnician_9.5.191810_Portable\R-Studio_9.5.exe</code>).
```

```

124         </div>
125     </div>
126 </div>
127
128     <div class="mt-4">
129         <!-- Большая главная кнопка запуска -->
130         <button class="btn btn-primary btn-lg w-100 py-3 fw-
bold d-flex align-items-center justify-content-center gap-2 sh
adow" id="btn-recovery-launch" type="button">
131             <i class="bi bi-play-circle-fill fs-4"></i> Восста
новить удаленные файлы (Запустить R-Studio)
132         </button>
133         <div id="recovery-alert-box" class="mt-3" style="dis
play: none;"></div>
134     </div>
135 </div>
136 </div>
137
138 <!-- Правая колонка: Информация об исполняемом файле -->
139 <div class="col-12 col-lg-5">
140     <div class="recovery-card h-100 d-flex flex-column justi
fy-content-between">
141         <div>
142             <h6 class="fw-bold text-info mb-3 d-flex align-items
-center gap-2">
143                 <i class="bi bi-info-circle-fill"></i> Информация
о пакете утилиты
144             </h6>
145
146             <div class="row g-2 mb-3">
147                 <div class="col-12">
148                     <div class="recovery-stat-box">
149                         <div class="recovery-stat-title">Исполняемый ф
айл</div>
150                         <div class="recovery-stat-val text-truncate" i
d="recovery-file-status">Проверка...</div>
151                         <div class="text-secondary small mt-1 text-bre
ak font-monospace" id="recovery-file-path" style="font-size: 0

```

```

151 | .75rem;">
152 |         \bin\R-Studio_Technician_9.5.191810_Portable
153 |         \R-Studio_9.5.exe
154 |     </div>
155 | </div>
156 |
157 |     <div class="col-12">
158 |         <div class="recovery-stat-box">
159 |             <div class="recovery-stat-title">Версия и сбор
160 | ка</div>
161 |             <div class="recovery-stat-val text-info" id="r
162 | ecovery-version">R-Studio Technician 9.5</div>
163 |             <div class="text-secondary small mt-1">Сборка:
164 | 191810 (Portable x64/x86)</div>
165 |             </div>
166 |         </div>
167 |
168 |         <div class="col-12">
169 |             <div class="recovery-stat-box">
170 |                 <div class="recovery-stat-title">Размер файла
171 | / Дата модификации</div>
172 |                 <div class="recovery-stat-val text-truncate" i
173 | d="recovery-file-size">--</div>
174 |                 <div class="text-secondary small mt-1" id="rec
175 | overey-file-date">--</div>
176 |                 </div>
177 |             </div>
178 |         </div>

```

```
179 |     </div>
180 |   </div>
181 | </div>
182 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows File Recovery Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/file_recovery_tab/main
    .js?v=20261001_v1" type="module"></script>
11 *
12 * File: main.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/file_recovery_tab
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-01 13:04:40
18 * =====
19 */
20
21 /**
22 * file_recovery_tab/main.js – логика вкладки «Восстановить удаленные файлы» (R-Studio Technician Portable)
23 */
24
25 (function () {
26   'use strict';
27
28   function formatBytes(bytes) {
29     if (!bytes || bytes <= 0) return '0 B';
30     const units = ['B', 'KB', 'MB', 'GB', 'TB'];
31     const i = Math.floor(Math.log(bytes) / Math.log(1024));
```

```

32     return (bytes / Math.pow(1024, i)).toFixed(2) + ' ' + unit
    | s[i];
33 }
34
35 function showAlert(msg, type = 'info') {
36     const box = document.getElementById('recovery-alert-box');
37     if (!box) return;
38     box.className = `alert alert-${type} alert-dismissible fade show small py-2 px-3`;
39     box.innerHTML = `
40         <div class="d-flex align-items-center justify-content-between">
41             <div>${msg}</div>
42             <button type="button" class="btn-close py-2" data-bs-dismiss="alert" aria-label="Close"></button>
43         </div>
44     `;
45     box.style.display = 'block';
46 }
47
48 async function loadRecoveryStatus() {
49     const badge = document.getElementById('recovery-status-badge');
50     const fileStatus = document.getElementById('recovery-file-status');
51     const filePath = document.getElementById('recovery-file-path');
52     const fileSize = document.getElementById('recovery-file-size');
53     const fileDate = document.getElementById('recovery-file-date');
54
55     try {
56         if (badge) badge.textContent = 'Обновление...';
57         const res = await fetch(`/api/recovery/status?t=${Date.now()}`);
58         if (!res.ok) throw new Error(`HTTP ${res.status}`);
59

```

```
60 |     const data = await res.json();
61 |     const tool = data.tool || {};
62 |
63 |     if (tool.exists) {
64 |         if (badge) {
65 |             badge.className = 'badge rounded-pill bg-success-sub
66 | tle border border-success text-success px-2.5 py-1.5';
67 |             badge.textContent = 'Готов к запуску';
68 |         }
69 |         if (fileStatus) {
70 |             fileStatus.className = 'recovery-stat-val text-succe
71 | ss text-truncate';
72 |             fileStatus.textContent = 'Файл найден';
73 |         }
74 |         if (filePath) filePath.textContent = tool.path || tool
75 | .relative_path || '';
76 |         if (fileSize) fileSize.textContent = formatBytes(tool.
77 | size_bytes);
78 |         if (fileDate) fileDate.textContent = `Изменён: ${tool.
79 | modified_time || '--'}`;
80 |     } else {
81 |         if (badge) {
82 |             badge.className = 'badge rounded-pill bg-warning-sub
83 | tle border border-warning text-warning px-2.5 py-1.5';
84 |             badge.textContent = 'Исполняемый файл не найден';
85 |         }
86 |         if (fileStatus) {
87 |             fileStatus.className = 'recovery-stat-val text-warni
88 | ng text-truncate';
89 |             fileStatus.textContent = 'Не найден в bin';
90 |         }
91 |         if (filePath) filePath.textContent = tool.relative_pat
92 | h || '';
93 |         if (fileSize) fileSize.textContent = '0 B';
94 |         if (fileDate) fileDate.textContent = '--';
95 |     }
96 | } catch (err) {
97 |     console.error('Ошибка загрузки статуса утилиты восстанов
```



```

    ления:', err);
90     if (badge) {
91         badge.className = 'badge rounded-pill bg-danger-subtle
border border-danger text-danger px-2.5 py-1.5';
92         badge.textContent = 'Ошибка связи';
93     }
94 }
95 }
96
97 async function launchRecoveryTool() {
98     const btn = document.getElementById('btn-recovery-launch')
;
99     if (!btn) return;
100
101     const originalContent = btn.innerHTML;
102     btn.disabled = true;
103     btn.innerHTML = `Успех:</strong> ${data.message || '
Программа R-Studio успешно запущена!'}`, 'success');
115             if (window.toast) {
116                 window.toast.success('R-Studio запущена', data.messa
ge || 'Окно программы открывается...');
117             }
118         } else {
119             const err = data.detail || 'Не удалось запустить R-Stu
dio';

```

```
120 |         showAlert(`Ошибка:</strong> ${err}`, 'danger')
    | ;
121 |         if (window.toast) {
122 |             window.toast.error('Ошибка запуска', err);
123 |         }
124 |     }
125 | } catch (err) {
126 |     console.error('Ошибка вызова API запуска:', err);
127 |     showAlert(`Сетевая ошибка:</strong> ${err.message}
    | e}`, 'danger');
128 |     if (window.toast) {
129 |         window.toast.error('Сетевая ошибка', err.message);
130 |     }
131 | } finally {
132 |     btn.disabled = false;
133 |     btn.innerHTML = originalContent;
134 | }
135 | }
136 |
137 | function initListeners() {
138 |     const refreshBtn = document.getElementById('btn-recovery-r
    | efresh');
139 |     if (refreshBtn) {
140 |         refreshBtn.onclick = () => loadRecoveryStatus();
141 |     }
142 |
143 |     const launchBtn = document.getElementById('btn-recovery-la
    | unch');
144 |     if (launchBtn) {
145 |         launchBtn.onclick = () => launchRecoveryTool();
146 |     }
147 | }
148 |
149 | function initFileRecoveryTab() {
150 |     initListeners();
151 |     loadRecoveryStatus();
152 | }
153 |
```

```
154 | // Экспорт для жизненного цикла tab-core.js
155 | window.initFile_recoveryTab = initFileRecoveryTab;
156 | window.initFilerecoveryTab = initFileRecoveryTab;
157 | window.initFileRecoveryTab = initFileRecoveryTab;
158 | window.initRecoveryTab = initFileRecoveryTab;
159 |
160 | // Автозапуск при самостоятельной загрузке
161 | if (document.readyState === 'loading') {
162 |     document.addEventListener('DOMContentLoaded', initFileReco
163 | veryTab);
164 | } else {
165 |     initFileRecoveryTab();
166 | }
167 | }());
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/firewall_manager_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/firewall_manager_tab/in
17 dex.html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/firewall_manager_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Windows Firewall Manager Tab -->
30 <div class="container-fluid py-2 px-3">
31   <!-- Top Stat Cards -->
32   <div class="row g-2 mb-3">
33     <div class="col-6 col-md-3">
34       <div class="card bg-dark text-white border-secondary h-1
35 00 shadow-sm p-3">
36         <div class="text-muted small text-uppercase fw-semibol
37 d">Доменный профиль □ </div>
```

```
31 |         <div class="fs-4 fw-bold mt-1 text-success" id="fw-dom
    | ain-status">--</div>
32 |         <div class="small text-muted mt-1" id="fw-domain-sub">
    | Domain Profile</div>
33 |     </div>
34 | </div>
35 |     <div class="col-6 col-md-3">
36 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
37 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Частный профиль □ </div>
38 |             <div class="fs-4 fw-bold mt-1 text-success" id="fw-pri
    | vate-status">--</div>
39 |             <div class="small text-muted mt-1" id="fw-private-sub"
    | >Private Profile</div>
40 |         </div>
41 |     </div>
42 |     <div class="col-6 col-md-3">
43 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
44 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Публичный профиль □ </div>
45 |             <div class="fs-4 fw-bold mt-1 text-success" id="fw-pub
    | lic-status">--</div>
46 |             <div class="small text-muted mt-1" id="fw-public-sub">
    | Public Profile</div>
47 |         </div>
48 |     </div>
49 |     <div class="col-6 col-md-3">
50 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
51 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Активные правила □ </div>
52 |             <div class="fs-4 fw-bold mt-1 text-info" id="fw-rules-
    | count">--</div>
53 |             <div class="small text-muted mt-1" id="fw-rules-sub">n
    | etsh advfirewall</div>
54 |         </div>
```

```

55     </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="fw-refresh-btn">
62             <i class="bi bi-arrow-clockwise"></i>
63             <span>Обновить правила</span>
64         </button>
65         <button class="btn btn-sm btn-outline-danger d-flex align-items-center gap-1.5" id="fw-reset-defaults-btn">
66             <i class="bi bi-arrow-counterclockwise"></i>
67             <span>Сброс по умолчанию (Reset)</span>
68         </button>
69     </div>
70     <div class="d-flex align-items-center gap-2">
71         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="fw-search-input" placeholder="Поиск правила или порта..." style="width: 260px;">
72     </div>
73 </div>
74
75 <!-- Firewall Rules Table -->
76 <div class="card bg-dark text-white border-secondary shadow-sm">
77     <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
78         <span class="fw-semibold"><i class="bi bi-shield-shaded text-primary me-2"></i>Правила Брандмауэра Windows</span>
79         <span class="badge bg-secondary" id="fw-table-badge">0 правил</span>
80     </div>
81     <div class="card-body p-0 table-responsive" style="max-height: 520px; overflow-y: auto;">

```

```
82 |         <table class="table table-dark table-hover table-striped
    | align-middle mb-0 small" id="fw-table">
83 |             <thead class="sticky-top bg-dark">
84 |                 <tr class="text-secondary">
85 |                     <th>Имя правила</th>
86 |                     <th>Направление</th>
87 |                     <th>Действие</th>
88 |                     <th>Протокол / Порт</th>
89 |                     <th>Профиль</th>
90 |                     <th>Состояние</th>
91 |                     <th class="text-end">Управление</th>
92 |                 </tr>
93 |             </thead>
94 |             <tbody id="fw-tbody">
95 |                 <tr><td colspan="7" class="text-center text-muted py
    | -4">Опрос правил брандмауэра...</td></tr>
96 |             </tbody>
97 |         </table>
98 |     </div>
99 | </div>
100 </div>
101
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Firewall Manager Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/firewall_manager_tab/main.js?v=20261001_v1" type="module"></script>
11 *
12 *   JavaScript Import:
13 *     import { initFirewallManagerTab } from '/windows/api/webgui/firewall_manager_tab/main.js';
14 *
15 * File: main.js
16 * Project: ai-breadboard
17 * Package: windows/api/webgui/firewall_manager_tab
18 * Author: hypo69
19 * Copyright: © 2026 hypo69
20 * Updated: 2026-10-04 11:15:30
21 * =====
22 */
23
24 /**
25 * firewall_manager_tab/main.js – Управление брандмауэром Windows (netsh advfirewall)
26 * Updated: 2026-10-04 11:15:30
27 */
28
29 const registerTabPoller = window.registerTabPoller || function
  () {};
```



```
30 |
31 | let isInitialized = false;
32 | let rulesList = [];
33 |
34 | export async function fetchFirewallSummary() {
35 |   try {
36 |     const res = await fetch('/api/firewall-manager/summary');
37 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38 |     const data = await res.json();
39 |     renderFirewall(data);
40 |   } catch (err) {
41 |     console.warn('[FirewallManager] Ошибка получения правил:',
42 |       err);
43 |   }
44 | }
45 | function renderFirewall(data) {
46 |   rulesList = data.rules || [];
47 |
48 |   const domEl = document.getElementById('fw-domain-status');
49 |   const privEl = document.getElementById('fw-private-status');
50 |   const pubEl = document.getElementById('fw-public-status');
51 |   const rulesCountEl = document.getElementById('fw-rules-count
52 |     ');
53 |   if (domEl) domEl.textContent = data.domain_profile || 'ON';
54 |   if (privEl) privEl.textContent = data.private_profile || 'ON
55 |     ';
56 |   if (pubEl) pubEl.textContent = data.public_profile || 'ON';
57 |   if (rulesCountEl) rulesCountEl.textContent = rulesList.length;
58 |
59 |   applyFirewallFilters();
60 | }
61 | function applyFirewallFilters() {
62 |   const tbody = document.getElementById('fw-tbody');
63 |   const searchInput = document.getElementById('fw-search-input
```

```

    ');
64   const badge = document.getElementById('fw-table-badge');
65
66   if (!tbody) return;
67
68   const query = (searchInput ? searchInput.value : '').toLowerCase().trim();
69
70   const filtered = rulesList.filter(r => {
71     if (!query) return true;
72     const nameMatch = (r.name || r.rule_name || '').toLowerCase().includes(query);
73     const portMatch = String(r.port || r.local_port || '').includes(query);
74     return nameMatch || portMatch;
75   });
76
77   if (badge) badge.textContent = `${filtered.length} из ${rulesList.length} правил`;
78
79   if (filtered.length === 0) {
80     tbody.innerHTML = '<tr><td colspan="7" class="text-center text-muted py-4">Правила не найдены</td></tr>';
81     return;
82   }
83
84   tbody.innerHTML = filtered.slice(0, 150).map(r => {
85     const name = r.name || r.rule_name || 'Firewall Rule';
86     const dir = r.direction || 'In';
87     const action = r.action || 'Allow';
88     const port = r.local_port || r.port || 'Any';
89     const proto = r.protocol || 'TCP';
90     const profile = r.profile || 'All';
91     const enabled = r.enabled ?? true;
92
93     const isAllow = action.toLowerCase().includes('allow') || action.toLowerCase().includes('разреш');
94

```

```

95     return `
96         <tr>
97             <td class="fw-semibold text-light">${name}</td>
98             <td><span class="badge ${dir === 'In' ? 'bg-primary' :
99 'bg-info'}">${dir}</span></td>
100             <td><span class="badge ${isAllow ? 'bg-success' : 'bg-
101 danger'}">${action}</span></td>
102             <td class="font-monospace small text-muted">${proto}:$
103 {port}</td>
104             <td class="small text-muted">${profile}</td>
105             <td><span class="badge ${enabled ? 'bg-success' : 'bg-
106 danger-subtle text-danger border border-danger fw-bold'}">${en
107 abled ? 'Включено' : 'Выключено'}</span></td>
108             <td class="text-end">
109                 <button class="btn btn-xs ${enabled ? 'btn-outline-w
110 arning' : 'btn-outline-success'} py-0 px-2 fw-toggle-btn"
111                     data-name="${name}" data-action="${enabled ?
112 'disable_rule' : 'enable_rule'}">
113                     <i class="bi ${enabled ? 'bi-pause-fill' : 'bi-pla
114 y-fill'}"></i>
115                 </button>
116             </td>
117         </tr>
118     `;
119     })).join('');
120 }
121
122 tbody.querySelectorAll('.fw-toggle-btn').forEach(btn => {
123     btn.addEventListener('click', () => {
124         const rName = btn.dataset.name;
125         const rAct = btn.dataset.action;
126         executeFirewallAction(rName, rAct);
127     });
128 });
129
130 async function executeFirewallAction(ruleName, action) {
131     try {
132         const res = await fetch('/api/firewall-manager/action', {

```

```
125     method: 'POST',
126     headers: { 'Content-Type': 'application/json' },
127     body: JSON.stringify({ rule_name: ruleName, action: action, dry_run: false })
128   });
129   const result = await res.json();
130   if (res.ok) {
131     if (window.toast) window.toast.success('Брандмауэр', `${ruleName}: обновлено`);
132     fetchFirewallSummary();
133   } else {
134     if (window.toast) window.toast.error('Ошибка', result.detail || 'Сбой операции');
135   }
136 } catch (err) {
137   if (window.toast) window.toast.error('Ошибка сети', err.message);
138 }
139 }
140
141 export function initFirewallManagerTab() {
142   if (isInitialized) return;
143   isInitialized = true;
144
145   const refreshBtn = document.getElementById('fw-refresh-btn');
146   ;
147   if (refreshBtn) refreshBtn.addEventListener('click', fetchFirewallSummary);
148
149   const resetBtn = document.getElementById('fw-reset-defaults-btn');
150   if (resetBtn) resetBtn.addEventListener('click', () => {
151     if (confirm('Сбросить правила брандмауэра к значениям по умолчанию?')) {
152       executeFirewallAction('all', 'reset_defaults');
153     }
154   });
155 }
```

```
155 |   const searchInput = document.getElementById('fw-search-input
    | ');
156 |   if (searchInput) searchInput.addEventListener('input', apply
    | FirewallFilters);
157 |
158 |   registerTabPoller('tab-firewall-manager', fetchFirewallSumma
    | ry, 15000, { immediate: true });
159 | }
160 |
161 | window.initFirewallManagerTab = initFirewallManagerTab;
162 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Focus Settings Tab
6 =====
7
8 =====
9 Description:
10 Веб-интерфейс панели управления сессиями фокусировки Windows
11 11 (Focus Sessions),
12 режимом «Не беспокоить» (Do Not Disturb), подавлением бейдже
13 й и мигания панели
14 задач, а также вызовом нативного приложения «Часы» и системн
15 ых настроек.
16
17 Usage Examples:
18     Web Route:
19     /html/focus_settings_tab/index.html
20
21 File: index.html
22 Project: ai-breadboard
23 Package: windows/api/webgui/focus_settings_tab
24 Author: hypo69
25 Copyright: © 2026 hypo69
26 Updated: 2026-10-06 19:35:00
27 =====
28 =====
29 -->
30
31 <div class="container-fluid p-3 focus-tab-wrapper d-flex flex-
32 column gap-3" style="max-width: 900px; margin: 0 auto;">
33
34 <!-- Breadcrumb Header -->
35 <div class="d-flex flex-column gap-1">
36 <div class="d-flex align-items-center gap-2 text-muted sma
37 ll">
38 <span>System</span>
```

```

29     <i class="bi bi-chevron-right" style="font-size: 0.75rem
    ;"></i>
30     <span class="text-body fw-semibold">Focus</span>
31 </div>
32 <h3 class="mb-0 fw-bold d-flex align-items-center gap-2">
33     <span>Фокусировка внимания (Focus)</span>
34     <span class="badge bg-secondary-subtle text-secondary bo
    rder border-secondary-subtle small fs-6" id="focus-status-badg
    e">Не активна</span>
35 </h3>
36 <div class="text-muted small">
37     Сессии фокусировки помогают продуктивнее работать, сокра
    щая отвлекающие факторы. При запуске сессии параметры применяю
    тся автоматически на заданное время.
38 </div>
39 </div>
40
41 <!-- Main Focus Session Card -->
42 <div class="card border-secondary-subtle shadow-sm bg-body">
43     <div class="card-body p-4 d-flex flex-column gap-3">
44
45         <!-- Top Row: Focus Icon, Title, and Start/Stop Button -
    -->
46         <div class="d-flex flex-wrap align-items-center justify-
    content-between gap-3 pb-3 border-bottom border-secondary-subt
    le">
47             <div class="d-flex align-items-center gap-3">
48                 <div class="rounded-circle bg-primary-subtle text-pr
    imary d-flex align-items-center justify-content-center" style=
    "width: 44px; height: 44px; font-size: 1.3rem;">
49                     <i class="bi bi-circle-half"></i>
50                 </div>
51                 <div>
52                     <div class="fw-bold fs-5">Focus</div>
53                     <div class="text-muted small" id="focus-session-hi
    nt">Готов к запуску сессии без отвлечений</div>
54                 </div>
55             </div>

```

```

56 |
57 |         <div class="d-flex align-items-center gap-2">
58 |             <div id="focus-countdown-box" class="d-none badge bg
| -primary fs-6 py-2 px-3 align-items-center gap-2">
59 |                 <i class="bi bi-stopwatch"></i>
60 |                 <span id="focus-timer-text">30:00</span>
61 |             </div>
62 |             <button class="btn btn-primary px-4 py-2 d-flex align-items-center gap-2 shadow-sm" id="btn-toggle-focus">
63 |                 <i class="bi bi-play-fill" id="btn-focus-icon"></i>
| >
64 |                 <span id="btn-focus-text">Start focus session</span>
| <n>
65 |             </button>
66 |         </div>
67 |     </div>
68 |
69 |     <!-- Duration Control Row -->
70 |     <div class="d-flex flex-wrap align-items-center justify-content-between gap-3 py-2">
71 |         <div>
72 |             <div class="fw-semibold">Session duration</div>
73 |             <div class="text-muted small">Продолжительность сессии фокусировки</div>
74 |         </div>
75 |         <div class="d-flex align-items-center gap-2 bg-body-secondary p-1 rounded border border-secondary-subtle">
76 |             <button class="btn btn-sm btn-outline-secondary px-2 py-1" id="btn-duration-minus" type="button" title="Уменьшить длительность">
77 |                 <i class="bi bi-dash-lg"></i>
78 |             </button>
79 |             <span class="fw-bold px-3 text-center" style="min-width: 90px;" id="lbl-duration">30 mins</span>
80 |             <button class="btn btn-sm btn-outline-secondary px-2 py-1" id="btn-duration-plus" type="button" title="Увеличить длительность">
81 |                 <i class="bi bi-plus-lg"></i>

```



```

82         </button>
83     </div>
84 </div>
85
86     <!-- Settings Checkboxes Group -->
87     <div class="d-flex flex-column gap-2 pt-2 border-top border-secondary-subtle">
88
89         <div class="form-check d-flex align-items-center gap-2 py-2">
90             <input class="form-check-input mt-0 fs-5" type="checkbox" id="chk-clock-timer" checked>
91             <label class="form-check-label fw-medium user-select-none" for="chk-clock-timer">
92                 Show the timer in the Clock app
93                 <span class="text-muted d-block small fw-normal">Отображать компактный таймер в приложении Windows Clock</span>
94             </label>
95         </div>
96
97         <div class="form-check d-flex align-items-center gap-2 py-2">
98             <input class="form-check-input mt-0 fs-5" type="checkbox" id="chk-hide-badges" checked>
99             <label class="form-check-label fw-medium user-select-none" for="chk-hide-badges">
100                 Hide badges on taskbar apps
101                 <span class="text-muted d-block small fw-normal">Скрывать счетчики и бейджи непрочитанных сообщений на значках панели задач</span>
102             </label>
103         </div>
104
105         <div class="form-check d-flex align-items-center gap-2 py-2">
106             <input class="form-check-input mt-0 fs-5" type="checkbox" id="chk-hide-flashing" checked>
107             <label class="form-check-label fw-medium user-select

```

```

| -none" for="chk-hide-flashing">
108 |         Hide flashing on taskbar apps
109 |         <span class="text-muted d-block small fw-normal">0
| включить оранжевое мигание кнопок приложений при поступлении с
| событий</span>
110 |         </label>
111 |     </div>
112 |
113 |     <div class="form-check d-flex align-items-center gap-2
| py-2">
114 |         <input class="form-check-input mt-0 fs-5" type="chec
| kbox" id="chk-turn-on-dnd" checked>
115 |         <label class="form-check-label fw-medium user-select
| -none" for="chk-turn-on-dnd">
116 |             Turn on do not disturb
117 |             <span class="text-muted d-block small fw-normal">В
| ключать режим «Не беспокоить» (Do Not Disturb / Quiet Hours) д
| ля блокировки всплывающих уведомлений</span>
118 |             </label>
119 |         </div>
120 |
121 |     </div>
122 |
123 | </div>
124 | <div class="card-footer bg-body-tertiary d-flex flex-wrap
| align-items-center justify-content-between gap-2 p-3">
125 |     <div class="text-muted small" id="focus-last-synced">Син
| хронизировано с реестром Windows</div>
126 |     <div class="d-flex align-items-center gap-2 flex-wrap">
127 |         <button class="btn btn-outline-secondary btn-sm d-flex
| align-items-center gap-1" id="btn-open-win-settings">
128 |             <i class="bi bi-box-arrow-up-right"></i>
129 |             <span>Параметры Windows</span>
130 |         </button>
131 |         <button class="btn btn-outline-secondary btn-sm d-flex
| align-items-center gap-1" id="btn-open-clock-app">
132 |             <i class="bi bi-alarm"></i>
133 |             <span>Приложение «Часы»</span>

```

```
134 |         </button>
135 |         <button class="btn btn-success btn-sm d-flex align-ite
136 | ms-center gap-1" id="btn-save-focus-settings">
137 |             <i class="bi bi-check2"></i>
138 |             <span>Сохранить настройки</span>
139 |         </button>
140 |     </div>
141 | </div>
142 |
143 | <!-- Control Plane Quick Link Card -->
144 | <div class="card border-secondary-subtle shadow-sm bg-body-s
145 | econdary">
146 |     <div class="card-body p-3 d-flex flex-wrap align-items-cen
147 | ter justify-content-between gap-2">
148 |         <div class="d-flex align-items-center gap-2">
149 |             <i class="bi bi-window-sidebar fs-4 text-info"></i>
150 |             <div>
151 |                 <div class="fw-semibold small">Связанные параметры W
152 | indow Management Control Plane</div>
153 |                 <div class="text-muted small">В каталоге доступно 15
154 | низкоуровневых параметров перехвата фокуса и отслеживания око
155 | н за курсором.</div>
156 |             </div>
157 |         </div>
158 |         <button class="btn btn-outline-info btn-sm" id="btn-goto
159 | -taskbar-tab">
160 |             <i class="bi bi-arrow-right-circle me-1"></i> Панель з
161 | адач и окна
162 |         </button>
163 |     </div>
164 | </div>
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Web Interface - Focus Settings Tab Sc
  ript
4   * =====
5   * Description:
6   *   Клиентский контроллер вкладки управления сессиями фокусир
  овки (Focus Sessions),
7   *   режимом «Не беспокоить» (Do Not Disturb), подавлением бей
  джей и мигания панели задач.
8   *
9   * Usage Examples:
10  *   JavaScript:
11  *     initFocusSettingsTab();
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/focus_settings_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-06 19:35:00
19  * =====
20  */
21
22 let durationMinutes = 30;
23 let isFocusActive = false;
24 let timerInterval = null;
25 let remainingSeconds = 0;
26
27 export async function initFocusSettingsTab() {
28   console.log('[FocusSettingsTab] Инициализация вкладки Focus
  Settings...');
29   bindEventListeners();
```

```
30 |   await loadFocusState();
31 | }
32 |
33 | /**
34 |  * Подключение обработчиков событий элементов управления интер
фейса.
35 |  */
36 | function bindEventListeners() {
37 |   // Кнопки регулировки длительности сессии
38 |   const btnMinus = document.getElementById('btn-duration-minus
');
39 |   const btnPlus = document.getElementById('btn-duration-plus')
;
40 |   if (btnMinus) {
41 |     btnMinus.onclick = () => changeDuration(-5);
42 |   }
43 |   if (btnPlus) {
44 |     btnPlus.onclick = () => changeDuration(5);
45 |   }
46 |
47 |   // Кнопка старта / остановки сессии
48 |   const btnToggle = document.getElementById('btn-toggle-focus'
);
49 |   if (btnToggle) {
50 |     btnToggle.onclick = toggleFocusSession;
51 |   }
52 |
53 |   // Сохранение настроек
54 |   const btnSave = document.getElementById('btn-save-focus-sett
ings');
55 |   if (btnSave) {
56 |     btnSave.onclick = saveFocusSettings;
57 |   }
58 |
59 |   // Запуск нативных окон Windows
60 |   const btnWinSettings = document.getElementById('btn-open-win
-settings');
61 |   if (btnWinSettings) {
```

```
62     btnWinSettings.onclick = openWindowsSettings;
63 }
64
65 const btnClockApp = document.getElementById('btn-open-clock-
app');
66 if (btnClockApp) {
67     btnClockApp.onclick = openClockApp;
68 }
69
70 // Переход к смежной вкладке управления панелью задач
71 const btnTaskbar = document.getElementById('btn-goto-taskbar
-tab');
72 if (btnTaskbar) {
73     btnTaskbar.onclick = () => {
74         if (typeof window.switchTab === 'function') {
75             window.switchTab('tab-taskbar-controller');
76         }
77     };
78 }
79 }
80
81 /**
82  * Изменение длительности сессии с шагом в 5 минут (диапазон 5
83  * ..240 мин).
84  * @param {number} delta Шаг изменения в минутах.
85  */
86 function changeDuration(delta) {
87     if (isFocusActive) return;
88     durationMinutes = Math.max(5, Math.min(240, durationMinutes
+ delta));
89     updateDurationLabel();
90 }
91 /**
92  * Обновление текстовой метки длительности.
93  */
94 function updateDurationLabel() {
95     const lbl = document.getElementById('lbl-duration');
```

```
96 |   if (lbl) {
97 |       lbl.textContent = `${durationMinutes} mins`;
98 |   }
99 | }
100 |
101 | /**
102 |  * Загрузка актуального состояния из REST API.
103 |  */
104 | async function loadFocusState() {
105 |     try {
106 |         const res = await fetch('/api/windows/focus/state');
107 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
108 |         const data = await res.json();
109 |         renderState(data);
110 |     } catch (err) {
111 |         console.warn('[FocusSettingsTab] Ошибка загрузки состояния
112 |         focus:', err);
113 |     }
114 | }
115 | /**
116 |  * Отрисовка полученного состояния в интерфейсе.
117 |  * @param {object} state Объект FocusSessionStateDTO.
118 |  */
119 | function renderState(state) {
120 |     isFocusActive = Boolean(state.is_active);
121 |     durationMinutes = state.session_duration_minutes || duration
122 |     Minutes;
123 |     remainingSeconds = state.remaining_seconds || 0;
124 |     updateDurationLabel();
125 |
126 |     // Обновление чекбоксов
127 |     const s = state.settings || {};
128 |     setCheckbox('chk-clock-timer', s.show_timer_in_clock_app ??
129 |     true);
130 |     setCheckbox('chk-hide-badges', s.hide_badges_on_taskbar ?? t
131 |     rue);
```

```
130 |   setCheckbox('chk-hide-flashing', s.hide_flashing_on_taskbar
    |   ?? true);
131 |   setCheckbox('chk-turn-on-dnd', s.turn_on_do_not_disturb ?? t
    |   rue);
132 |
133 |   // Обновление бейджа и кнопок
134 |   const badge = document.getElementById('focus-status-badge');
135 |   const btnToggle = document.getElementById('btn-toggle-focus'
    |   );
136 |   const btnIcon = document.getElementById('btn-focus-icon');
137 |   const btnText = document.getElementById('btn-focus-text');
138 |   const hint = document.getElementById('focus-session-hint');
139 |   const countdownBox = document.getElementById('focus-countdow
    |   n-box');
140 |
141 |   if (isFocusActive) {
142 |     if (badge) {
143 |       badge.className = 'badge bg-success small fs-6';
144 |       badge.textContent = 'Сессия активна';
145 |     }
146 |     if (btnToggle) {
147 |       btnToggle.className = 'btn btn-danger px-4 py-2 d-flex a
    |   lign-items-center gap-2 shadow-sm';
148 |     }
149 |     if (btnIcon) btnIcon.className = 'bi bi-stop-fill';
150 |     if (btnText) btnText.textContent = 'Stop focus session';
151 |     if (hint) hint.textContent = 'Сессия фокусировки активна.
    |   Отвлекающие факторы заблокированы.';
152 |     if (countdownBox) countdownBox.classList.remove('d-none');
153 |
154 |     startLocalCountdown();
155 |   } else {
156 |     if (badge) {
157 |       badge.className = 'badge bg-secondary-subtle text-second
    |   ary border border-secondary-subtle small fs-6';
158 |       badge.textContent = 'Не активна';
159 |     }
160 |     if (btnToggle) {
```



```
161 |         btnToggle.className = 'btn btn-primary px-4 py-2 d-flex
| align-items-center gap-2 shadow-sm';
162 |     }
163 |     if (btnIcon) btnIcon.className = 'bi bi-play-fill';
164 |     if (btnText) btnText.textContent = 'Start focus session';
165 |     if (hint) hint.textContent = 'Готов к запуску сессии без о
| твлечений';
166 |     if (countdownBox) countdownBox.classList.add('d-none');
167 |
168 |     stopLocalCountdown();
169 | }
170 | }
171 |
172 | /**
173 |  * Локальный таймер обратного отсчета для плавного отображения
| секунд.
174 |  */
175 | function startLocalCountdown() {
176 |     stopLocalCountdown();
177 |     updateCountdownDisplay();
178 |     timerInterval = setInterval(() => {
179 |         if (remainingSeconds > 0) {
180 |             remainingSeconds -= 1;
181 |             updateCountdownDisplay();
182 |         } else {
183 |             stopLocalCountdown();
184 |             loadFocusState();
185 |         }
186 |     }, 1000);
187 | }
188 |
189 | function stopLocalCountdown() {
190 |     if (timerInterval) {
191 |         clearInterval(timerInterval);
192 |         timerInterval = null;
193 |     }
194 | }
195 |
```

```
196 function updateCountdownDisplay() {
197   const timerText = document.getElementById('focus-timer-text'
    );
198   if (timerText) {
199     const mins = Math.floor(remainingSeconds / 60);
200     const secs = remainingSeconds % 60;
201     timerText.textContent = `${String(mins).padStart(2, '0')}:
    ${String(secs).padStart(2, '0')}`;
202   }
203 }
204
205 /**
206  * Переключение состояния сессии фокусировки (Старт / Стоп).
207  */
208 async function toggleFocusSession() {
209   const btnToggle = document.getElementById('btn-toggle-focus'
    );
210   if (btnToggle) btnToggle.disabled = true;
211
212   try {
213     // Сохраняем текущие выбранные чекбоксы перед стартом
214     await saveFocusSettings();
215
216     const endpoint = isFocusActive ? '/api/windows/focus/sessi
    on/stop' : '/api/windows/focus/session/start?duration_minutes=
    ${durationMinutes}`;
217     const res = await fetch(endpoint, { method: 'POST' });
218     if (!res.ok) throw new Error(`HTTP ${res.status}`);
219     const data = await res.json();
220     renderState(data);
221   } catch (err) {
222     console.error('[FocusSettingsTab] Ошибка переключения сесс
    ии:', err);
223     alert(`Ошибка управления сессией фокусировки: ${err.messag
    e}`);
224   } finally {
225     if (btnToggle) btnToggle.disabled = false;
226   }
}
```

```

227 | }
228 |
229 | /**
230 |  * Сохранение конфигурации в постоянные настройки.
231 |  */
232 | async function saveFocusSettings() {
233 |     const settings = {
234 |         session_duration_minutes: durationMinutes,
235 |         show_timer_in_clock_app: getCheckbox('chk-clock-timer'),
236 |         hide_badges_on_taskbar: getCheckbox('chk-hide-badges'),
237 |         hide_flashing_on_taskbar: getCheckbox('chk-hide-flashing')
238 |     },
239 |     turn_on_do_not_disturb: getCheckbox('chk-turn-on-dnd')
240 | };
241 | try {
242 |     const res = await fetch('/api/windows/focus/settings', {
243 |         method: 'POST',
244 |         headers: { 'Content-Type': 'application/json' },
245 |         body: JSON.stringify(settings)
246 |     });
247 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
248 |     const syncLabel = document.getElementById('focus-last-sync
249 | ed');
250 |     if (syncLabel) {
251 |         syncLabel.textContent = `Сохранено в ${new Date().toLocaleTimeString()}`;
252 |     }
253 | } catch (err) {
254 |     console.warn('[FocusSettingsTab] Ошибка сохранения настрое
255 | к:', err);
256 | }
257 | /**
258 |  * Вызов нативных параметров Windows.
259 |  */
260 | async function openWindowsSettings() {

```

```

261 |   try {
262 |       await fetch('/api/windows/focus/open-settings', { method:
| 'POST' });
263 |   } catch (err) {
264 |       console.warn('[FocusSettingsTab] Ошибка вызова параметров
| Windows:', err);
265 |   }
266 | }
267 |
268 | /**
269 |  * Вызов нативного приложения Windows Clock.
270 |  */
271 | async function openClockApp() {
272 |   try {
273 |       await fetch('/api/windows/focus/open-clock', { method: 'PO
| ST' });
274 |   } catch (err) {
275 |       console.warn('[FocusSettingsTab] Ошибка вызова Windows Clo
| ck:', err);
276 |   }
277 | }
278 |
279 | function getCheckbox(id) {
280 |   const el = document.getElementById(id);
281 |   return el ? Boolean(el.checked) : false;
282 | }
283 |
284 | function setCheckbox(id, value) {
285 |   const el = document.getElementById(id);
286 |   if (el) el.checked = Boolean(value);
287 | }
288 |
289 | // Экспорт для глобального использования
290 | window.initFocusSettingsTab = initFocusSettingsTab;
291 |
292 | // Автоинициализация при загрузке содержимого вкладки
293 | if (document.readyState === 'loading') {
294 |   document.addEventListener('DOMContentLoaded', initFocusSetti

```

```
ngsTab);  
295 | } else {  
296 |     initFocusSettingsTab();  
297 | }  
298 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/forensics_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/forensics_tab/index.htm
17     l" class="w-100 h-100 border-0"></iframe>
18
19 * File: index.html
20 * Project: ai-breadboard
21 * Package: windows/api/webgui/forensics_tab
22 * Author: hypo69
23 * Copyright: © 2026 hypo69
24 * Updated: 2026-10-08 03:08:00
25 * =====
26 =====
27 -->
28
29 <style>
30   .forensics-tab-wrapper {
31     padding: 0.75rem;
32     color: var(--text-color, #f8fafc);
33   }
34   .diag-panel {
35     background: var(--surface-2, #1e293b);
36     border: 1px solid var(--border-color, #334155);
37     border-radius: 8px;
```

```

33     overflow: hidden;
34 }
35 .diag-panel-header {
36     background: rgba(15, 23, 42, 0.6);
37     padding: 8px 14px;
38     border-bottom: 1px solid var(--border-color, #334155);
39     font-weight: 600;
40     font-size: 0.85rem;
41 }
42 .about-ai-pulse-icon {
43     width: 42px;
44     height: 42px;
45     border-radius: 8px;
46     display: flex;
47     align-items: center;
48     justify-content: center;
49     border: 1px solid rgba(56, 189, 248, 0.3);
50 }
51 </style>
52
53 <div class="forensics-tab-wrapper">
54     <!-- Top Header & Status -->
55     <div class="card shadow-sm border-secondary-subtle mb-3">
56         <div class="card-body p-3">
57             <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
58                 <div class="d-flex align-items-center gap-2">
59                     <div class="fs-4">□ </div>
60                     <div>
61                         <h5 class="mb-0 fw-bold">2. Поведенческая форензик
а активности</h5>
62                         <small class="text-muted">Телеметрия активного фоку
са, время бездействия, аудит доступа к камере/микрофону и реес
тр UserAssist</small>
63                     </div>
64                 </div>
65
66                 <div class="d-flex align-items-center gap-2 flex-wrap"

```

```

>
67 |         <select class="form-select form-select-sm poll-freq-
select" id="diag-forensics-poll-freq" data-poll-id="forensics"
|         title="Частота опроса: Форензика активности">
68 |             <option value="start">Только при старте</option>
69 |             <option value="manual" selected>По запросу</option>
|         >
70 |             <option value="3">Каждые 3 сек</option>
71 |             <option value="5">Каждые 5 сек</option>
72 |             <option value="10">Каждые 10 сек</option>
73 |             <option value="30">Каждые 30 сек</option>
74 |             <option value="60">Каждые 60 сек</option>
75 |             <option value="120">Каждые 120 сек</option>
76 |         </select>
77 |         <button class="btn btn-sm btn-outline-info rounded-p
|         ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
|         "btn-diag-forensics-refresh">
78 |             <i class="bi bi-arrow-clockwise"></i> Обновить
79 |         </button>
80 |     </div>
81 | </div>
82 | </div>
83 | </div>
84 |
85 | <div class="row g-2 mb-3">
86 |     <!-- Active Foreground Window Card -->
87 |     <div class="col-12 col-lg-6">
88 |         <div class="diag-panel h-100">
89 |             <div class="diag-panel-header d-flex justify-content-b
|             etween align-items-center">
90 |                 <div class="d-flex align-items-center gap-2">
91 |                     <i class="bi bi-window-stack text-primary"></i>
92 |                     <span class="text-white">Текущее активное окно в ф
|                     окусе (Foreground Window)</span>
93 |                 </div>
94 |                 <span class="badge bg-success-subtle text-success bo
|                 rder border-success" id="diag-forensics-live-badge">Live Focus
|                 </span>

```



```
95 |         </div>
96 |         <div class="p-3">
97 |             <div class="d-flex align-items-center gap-3 mb-2">
98 |                 <div class="about-ai-pulse-icon" style="background
: rgba(14, 165, 233, 0.15); border-color: rgba(14, 165, 233, 0
.4);">
99 |                     <i class="bi bi-cursor-fill text-info fs-5"></i>
100 |                 </div>
101 |                 <div class="flex-grow-1 overflow-hidden">
102 |                     <div class="fw-bold text-white text-truncate" id
="diag-fg-title" style="font-size: 0.95rem;">Рабочий стол Wind
ows</div>
103 |                     <div class="small text-muted font-monospace" id=
"diag-fg-proc">Процесс: explorer.exe (PID: 0)</div>
104 |                 </div>
105 |             </div>
106 |             <div class="border-top border-secondary-subtle pt-2
mt-2 d-flex justify-content-between align-items-center flex-wr
ap gap-2">
107 |                 <div>
108 |                     <span class="text-muted small">Время бездействия
пользователя (Idle Time):</span>
109 |                     <strong class="text-warning font-monospace ms-1"
id="diag-user-idle-time">0.0 сек</strong>
110 |                 </div>
111 |                 <span class="badge bg-dark border border-secondary
text-info" id="diag-user-status-badge">Пользователь активен</
span>
112 |             </div>
113 |         </div>
114 |     </div>
115 | </div>
116 |
117 | <!-- Privacy Sensors (Camera & Mic Radar) -->
118 | <div class="col-12 col-lg-6">
119 |     <div class="diag-panel h-100">
120 |         <div class="diag-panel-header d-flex justify-content-b
etween align-items-center">
```

```
121 |         <div class="d-flex align-items-center gap-2">
122 |             <i class="bi bi-camera-video-fill text-danger"></i>
123 |         >
124 |             <span class="text-white">Контроль доступа к камере
125 |             и микрофону (Privacy Radar)</span>
126 |         </div>
127 |         <div class="p-3">
128 |             <div class="row g-2">
129 |                 <div class="col-6">
130 |                     <div class="p-2 rounded bg-black bg-opacity-30 b
131 |                     order border-secondary-subtle">
132 |                         <div class="small text-muted mb-1"><i class="b
133 |                         i bi-camera-video me-1 text-danger"></i>Камера:</div>
134 |                         <div id="diag-cam-access-list" class="small te
135 |                         xt-light" style="max-height: 90px; overflow-y: auto;">
136 |                             <span class="text-muted">Нет активных обра
137 |                             щений</span>
138 |                         </div>
139 |                     </div>
140 |                     <div class="p-2 rounded bg-black bg-opacity-30 b
141 |                     order border-secondary-subtle">
142 |                         <div class="small text-muted mb-1"><i class="b
143 |                         i bi-mic me-1 text-info"></i>Микрофон:</div>
144 |                         <div id="diag-mic-access-list" class="small te
145 |                         xt-light" style="max-height: 90px; overflow-y: auto;">
146 |                             <span class="text-muted">Нет активных обра
147 |                             щений</span>
148 |                         </div>
149 |                     </div>
150 |                 </div>
151 |             </div>
152 |         </div>
153 |     </div>
```

```

148     </div>
149 </div>
150
151 <!-- UserAssist Application Run History -->
152 <div class="diag-panel">
153     <div class="diag-panel-header d-flex justify-content-between align-items-center">
154         <div class="d-flex align-items-center gap-2">
155             <i class="bi bi-clock-history text-info"></i>
156             <span class="text-white">Форензика запуска программ и
157 время удержания фокуса (UserAssist R0T13)</span>
158         </div>
159         <span class="badge bg-secondary" id="diag-userassist-count">0 записей</span>
160     </div>
161     <div class="table-responsive" style="max-height: 480px; overflow-y: auto;">
162         <table class="table table-dark table-hover table-sm mb-0" style="font-size: 0.80rem; font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospace;">
163             <thead>
164                 <tr class="text-secondary border-secondary">
165                     <th>Приложение / Программа</th>
166                     <th style="width: 140px; text-align: right;">Счетчик запусков</th>
167                     <th style="width: 180px; text-align: right;">Время в фокусе</th>
168                     <th>Полный путь / Идентификатор</th>
169                 </tr>
170             </thead>
171             <tbody id="diag-userassist-tbody">
172                 <tr><td colspan="4" class="text-center py-4 text-muted">Загрузка истории активности UserAssist...</td></tr>
173             </tbody>
174         </table>
175     </div>
176 </div>

```


main.js

```
1  /**
2   * =====
3   * Process Name: Windows Forensics Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/forensics_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/forensics_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-08 02:00:00
18  * =====
19  */
20
21 (function () {
22   'use strict';
23
24   let autoRefreshTimer = null;
25
26   function escapeHtml(str) {
27     if (str === null || str === undefined) return '';
28     return String(str)
29       .replace(/&/g, '&amp;')
30       .replace(/</g, '&lt;')
31       .replace(/>/g, '&gt;')
32       .replace(/"/g, '&quot;')
```

```

33     .replace(/'/g, '&#039;');
34 }
35
36 function setText(id, text) {
37     const el = document.getElementById(id);
38     if (el) el.textContent = text !== null && text !== undefined ? String(text) : '--';
39 }
40
41 async function apiFetch(url) {
42     if (window.api && typeof window.api.fetch === 'function')
43     {
44         return await window.api.fetch(url);
45     }
46     const res = await fetch(url);
47     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.statusText}`);
48     return await res.json();
49 }
50
51 async function fetchForensicsActivity() {
52     try {
53         const data = await apiFetch('/api/v1/system/diagnostics/forensics');
54         if (!data) return;
55
56         if (data.foreground_window) {
57             setText('diag-fg-title', data.foreground_window.title || 'Рабочий стол Windows');
58             setText('diag-fg-proc', `Процесс: ${data.foreground_window.process_name || 'unknown'} (PID: ${data.foreground_window.pid || 0})`);
59         }
60
61         setText('diag-user-idle-time', `${data.user_idle_seconds.toFixed(1)} сек`);
62         const statusBadge = document.getElementById('diag-user-status-badge');

```

```

62         if (statusBadge) {
63             if (data.user_idle_seconds > 300) {
64                 statusBadge.className = 'badge bg-warning-subtle text-warning border border-warning';
65                 statusBadge.textContent = 'Пользователь отошел (>5 мин)';
66             } else {
67                 statusBadge.className = 'badge bg-success-subtle text-success border border-success';
68                 statusBadge.textContent = 'Пользователь за ПК';
69             }
70         }
71
72         // Camera Apps
73         const camContainer = document.getElementById('diag-camera-access-list');
74         if (camContainer) {
75             if (data.camera_active_apps && data.camera_active_apps.length > 0) {
76                 camContainer.innerHTML = data.camera_active_apps.map(a => `
77                     <div class="d-flex align-items-center justify-content-between py-1 border-bottom border-secondary-subtle">
78                         <span class="text-truncate" style="max-width: 140px;">${escapeHtml(a.app_name)}</span>
79                         <span class="badge ${a.is_active_now ? 'bg-danger' : 'bg-secondary'}" style="font-size: 0.65rem;">
80                             ${a.is_active_now ? 'Захват камеры' : 'В доступе'}
81                         </span>
82                     </div>
83                 `).join('');
84             } else {
85                 camContainer.innerHTML = '<span class="text-muted small">Нет активных обращений к камере</span>';
86             }
87         }
88

```

```

89 |         // Mic Apps
90 |         const micContainer = document.getElementById('diag-mic-a
| ccess-list');
91 |         if (micContainer) {
92 |             if (data.microphone_active_apps && data.microphone_act
| ive_apps.length > 0) {
93 |                 micContainer.innerHTML = data.microphone_active_apps
| .map(a => `
94 |                     <div class="d-flex align-items-center justify-cont
| ent-between py-1 border-bottom border-secondary-subtle">
95 |                         <span class="text-truncate" style="max-width: 14
| 0px;">${escapeHtml(a.app_name)}</span>
96 |                         <span class="badge ${a.is_active_now ? 'bg-dange
| r' : 'bg-secondary'}" style="font-size: 0.65rem;">
97 |                             ${a.is_active_now ? 'Запись звука' : 'В доступ
| е'}
98 |
99 |                         </span>
100 |                     </div>
101 |                 `).join('');
102 |             } else {
103 |                 micContainer.innerHTML = '<span class="text-muted sm
| all">Нет активных обращений к микрофону</span>';
104 |             }
105 |
106 |         // UserAssist Apps Table
107 |         const uaTbody = document.getElementById('diag-userassist
| -tbody');
108 |         if (uaTbody && data.userassist_top_apps) {
109 |             setText('diag-userassist-count', `${data.userassist_to
| p_apps.length} программ`);
110 |             if (data.userassist_top_apps.length === 0) {
111 |                 uaTbody.innerHTML = '<tr><td colspan="4" class="text
| -center py-4 text-muted">Данных UserAssist в реестре не обнару
| жено</td></tr>';
112 |             } else {
113 |                 uaTbody.innerHTML = data.userassist_top_apps.map(a =
| > `

```



```

114 |         <tr>
115 |             <td class="fw-bold text-white"><i class="bi bi-a
pp text-info me-1.5"></i>${escapeHtml(a.name)}</td>
116 |             <td style="text-align: right;" class="text-info
font-monospace">${a.run_count} paz</td>
117 |             <td style="text-align: right;" class="text-warni
ng font-monospace">${a.focus_formatted}</td>
118 |             <td class="small text-muted text-truncate" style
="max-width: 280px;" title="${escapeHtml(a.path)}">${escapeHtm
l(a.path)}</td>
119 |         </tr>
120 |     `).join('');
121 |     }
122 | }
123 | } catch (err) {
124 |     console.warn('[ForensicsTab] fetchForensicsActivity erro
r:', err);
125 | }
126 | }
127 |
128 | const POLL_ID = 'forensics';
129 |
130 | function getFrequency() {
131 |     try {
132 |         const saved = localStorage.getItem(`poll_freq_${POLL_ID}
`);
133 |         if (saved) return saved;
134 |     } catch (_) {}
135 |     return 'manual';
136 | }
137 |
138 | function setFrequency(freq) {
139 |     try {
140 |         localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
141 |     } catch (_) {}
142 |     applyPoller(freq, false);
143 | }
144 |

```

```
145 | function stopPolling() {
146 |     if (window.unregisterTabPoller) {
147 |         window.unregisterTabPoller(`tab-forensics_${POLL_ID}`);
148 |     }
149 |     if (autoRefreshTimer) {
150 |         clearInterval(autoRefreshTimer);
151 |         autoRefreshTimer = null;
152 |     }
153 | }
154 |
155 | function applyPoller(freq, runInitial = false) {
156 |     stopPolling();
157 |     if (freq === 'start' || freq === 'manual') {
158 |         if (runInitial && (window.isTabActive ? window.isTabActive('tab-forensics') : true)) {
159 |             fetchForensicsActivity();
160 |         }
161 |         return;
162 |     }
163 |
164 |     const intervalSec = parseInt(freq, 10);
165 |     if (isNaN(intervalSec) || intervalSec <= 0) return;
166 |
167 |     const intervalMs = intervalSec * 1000;
168 |     const pollerId = `tab-forensics_${POLL_ID}`;
169 |
170 |     if (window.registerTabPoller) {
171 |         window.registerTabPoller('tab-forensics', fetchForensicsActivity, intervalMs, { pollerId, immediate: runInitial });
172 |     } else {
173 |         if (runInitial && (window.isTabActive ? window.isTabActive('tab-forensics') : true)) {
174 |             fetchForensicsActivity();
175 |         }
176 |         autoRefreshTimer = setInterval(() => {
177 |             if (window.isTabActive ? window.isTabActive('tab-forensics') : true) {
178 |                 fetchForensicsActivity();
```

```
179         }
180     }, intervalMs);
181 }
182 }
183
184 function bindEvents() {
185     const btnRefresh = document.getElementById('btn-diag-foren
186 sics-refresh');
187     if (btnRefresh) {
188         btnRefresh.onclick = () => fetchForensicsActivity();
189     }
190     const select = document.getElementById('diag-forensics-pol
191 l-freq');
192     if (select) {
193         select.onchange = (e) => {
194             const newFreq = e.target.value;
195             setFrequency(newFreq);
196             if (newFreq !== 'manual' && newFreq !== 'start') {
197                 fetchForensicsActivity();
198             }
199             if (window.showToast) {
200                 const label = select.options[select.selectedIndex]?.
201 text || newFreq;
202                 window.showToast(`Частота опроса форензики: ${label}
203 `, 'info');
204             }
205         };
206     }
207 }
208
209 async function initForensicsTab() {
210     bindEvents();
211     const currentFreq = getFrequency();
212     const select = document.getElementById('diag-forensics-pol
213 l-freq');
214     if (select) select.value = currentFreq;
215     applyPoller(currentFreq, true);
216 }
```

```
212 | }
213 |
214 | window.initForensicsTab = initForensicsTab;
215 | window.activateForensicsTab = () => {
216 |     applyPoller(getFrequency(), true);
217 | };
218 |
219 | if (document.readyState === 'loading') {
220 |     document.addEventListener('DOMContentLoaded', () => {
221 |         if (window.isTabActive ? window.isTabActive('tab-forensi
222 | cs') : false) {
223 |             initForensicsTab();
224 |         } else {
225 |             bindEvents();
226 |         }
227 |     });
228 | } else {
229 |     if (window.isTabActive ? window.isTabActive('tab-forensics
230 | ') : false) {
231 |         initForensicsTab();
232 |     } else {
233 |         bindEvents();
234 |     }
235 | }));
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/gcloud_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/gcloud_tab/index.html"
17     class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/gcloud_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-08 03:08:00
25 =====
26 =====
27 -->
28
29 <!-- Google Cloud Monitor Application Tab -->
30 <style>
31   .gcloud-container {
32     padding: 0.5rem 0.75rem;
33   }
34   .gcloud-card {
35     background: #0f172a !important;
36     border: 1px solid #334155 !important;
37     border-radius: 8px;
```

```
33 | padding: 0.45rem 0.75rem;
34 | display: flex;
35 | flex-direction: column;
36 | justify-content: space-between;
37 | min-height: 58px;
38 | gap: 0.2rem;
39 | }
40 | .gcloud-label {
41 |   font-size: 0.68rem;
42 |   text-transform: uppercase;
43 |   letter-spacing: 0.05em;
44 |   color: #94a3b8;
45 |   line-height: 1.1;
46 | }
47 | .gcloud-value {
48 |   font-size: 1.25rem;
49 |   font-weight: 700;
50 |   color: #f8fafc;
51 |   font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
52 |   line-height: 1.15;
53 | }
54 | .gcloud-panel {
55 |   background: #0f172a !important;
56 |   border: 1px solid #334155 !important;
57 |   border-radius: 8px;
58 |   overflow: hidden;
59 | }
60 | .gcloud-panel-header {
61 |   background: #1e293b !important;
62 |   border-bottom: 1px solid #334155 !important;
63 |   padding: 0.45rem 0.85rem;
64 |   font-weight: 600;
65 |   font-size: 0.82rem;
66 |   color: #38bdf8;
67 |   display: flex;
68 |   align-items: center;
69 |   justify-content: space-between;
```

```

70 | }
71 | .gcloud-log-terminal {
72 |     background: #020617;
73 |     color: #cbd5e1;
74 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
75 |     font-size: 0.78rem;
76 |     line-height: 1.45;
77 |     padding: 0.5rem 0.75rem;
78 |     border-radius: 6px;
79 |     max-height: 400px;
80 |     overflow-y: auto;
81 | }
82 | </style>
83 |
84 | <div class="gcloud-container">
85 |     <!-- Header Alert & Status (Compact Strip) -->
86 |     <div class="card shadow-sm border-secondary-subtle py-1.5 px
    | -3 mb-2 d-flex justify-content-between align-items-center flex
    | -wrap gap-2">
87 |         <div class="d-flex align-items-center gap-2 flex-wrap">
88 |             <span class="fs-5">□ </span>
89 |             <span class="fw-bold text-white small" style="font-size:
    | 0.92rem;">Google Cloud Monitor</span>
90 |             <span class="badge bg-dark border border-secondary text-
    | info fw-normal" style="font-size: 0.7rem;">Observability</span>
    | >
91 |             <span class="text-secondary small d-none d-lg-inline" st
    | yle="font-size: 0.76rem;">| Cloud Logging, метрики Cloud Monit
    | oring & IAM</span>
92 |         </div>
93 |         <div class="d-flex align-items-center gap-1.5 flex-wrap">
94 |             <span class="badge rounded-pill bg-success-subtle text-s
    | uccess border border-success px-2 py-1" style="font-size: 0.72
    | rem;" id="gcloud-status-badge">
95 |                 • GCP Подключен
96 |             </span>
97 |             <button class="btn btn-xs btn-outline-info rounded-pill

```

```

px-2.5 py-1" style="font-size: 0.75rem;" id="btn-gcloud-refres
h">
98     <i class="bi bi-arrow-clockwise me-1"></i> Обновить
99     </button>
100     <button class="btn btn-xs btn-outline-warning rounded-pi
11 px-2 py-1" style="font-size: 0.75rem;" id="btn-gcloud-confi
g">
101         <i class="bi bi-gear-fill me-1"></i> Настройки
102     </button>
103 </div>
104 </div>
105
106 <!-- Top Metrics Bar (Compact Density Grid) -->
107 <div class="row g-2 mb-2">
108     <div class="col-6 col-md-3">
109         <div class="gcloud-card">
110             <span class="gcloud-label">Проект GCP</span>
111             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
112                 <span class="gcloud-value text-info text-truncate" i
d="gcloud-project-id">-</span>
113             </div>
114         </div>
115     </div>
116     <div class="col-6 col-md-3">
117         <div class="gcloud-card">
118             <span class="gcloud-label">Тип авторизации</span>
119             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
120                 <span class="gcloud-value text-light" style="font-si
ze: 1.05rem;" id="gcloud-auth-type">-</span>
121             </div>
122         </div>
123     </div>
124     <div class="col-6 col-md-3">
125         <div class="gcloud-card">
126             <span class="gcloud-label">Служебный аккаунт</span>
127             <div class="d-flex align-items-baseline justify-conten

```



```

    t-between mt-1">
128         <span class="gcloud-value text-secondary text-trunca
    te" style="font-size: 0.85rem;" id="gcloud-client-email">-</sp
    an>
129     </div>
130 </div>
131 </div>
132 <div class="col-6 col-md-3">
133     <div class="gcloud-card">
134         <span class="gcloud-label">Здоровье инфраструктуры</sp
    an>
135         <div class="d-flex align-items-baseline justify-conten
    t-between mt-1">
136             <span class="gcloud-value text-success" id="gcloud-h
    ealth-score">100/100</span>
137         </div>
138     </div>
139 </div>
140 </div>
141
142 <!-- Main Content Panels Grid -->
143 <div class="row g-3">
144     <!-- Cloud Logging Terminal Panel -->
145     <div class="col-lg-7">
146         <div class="gcloud-panel h-100">
147             <div class="gcloud-panel-header">
148                 <span>☐ Cloud Logging (Последние события)</span>
149                 <span class="badge bg-secondary" id="gcloud-logs-cou
    nt-badge">0 строк</span>
150             </div>
151             <div class="p-2">
152                 <div class="gcloud-log-terminal" id="gcloud-log-term
    inal-output">
153                     Загрузка журналов Cloud Logging...
154                 </div>
155             </div>
156         </div>
157     </div>

```

```
158 |
159 |     <!-- AI Diagnostics & Recommendations Panel -->
160 |     <div class="col-lg-5">
161 |         <div class="gcloud-panel h-100">
162 |             <div class="gcloud-panel-header">
163 |                 <span> AI-Диагностика и рекомендации</span>
164 |                 <span class="badge bg-info text-dark">Gemini Diagnos
165 | tics</span>
166 |             </div>
167 |             <div class="p-3">
168 |                 <h6 class="fw-bold text-info mb-1" id="gcloud-diag-s
169 | ummary-title">Анализ телеметрии...</h6>
170 |                 <p class="small text-light mb-3" id="gcloud-diag-sum
171 | mary-desc">Загрузка диагностического заключения...</p>
172 |                 <div class="fw-semibold text-secondary small text-up
173 | percase mb-2">Рекомендованные действия:</div>
174 |                 <ul class="list-group list-group-flush small" id="gc
175 | loud-recommendations-list">
176 |                     <li class="list-group-item bg-transparent text-lig
177 | ht border-secondary py-1 px-0">• Ожидание данных телеметрии...
178 | </li>
179 |                 </ul>
180 |             </div>
181 |         </div>
182 |     </div>
183 | </div>
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Gcloud Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/gcloud_tab/main.js?v=2
11  *       0261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/gcloud_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21  // =====
22  // Process Name: Google Cloud Monitor Web Tab Module
23  // =====
24  // Description:
25  //   Client-side JavaScript controller for the Google Cloud Mo
26  //   nitor
27  //   administrative web interface tab.
28  //
29  // File: main.js
30  // Project: ai-breadboard
```

```

30 // Package: src.api.webinterface.gcloud_tab
31 // Author: hypo69
32 // Copyright: © 2026 hypo69
33 // =====
34
35 (function() {
36     let isGcloudInitialized = false;
37
38     async function fetchStatus() {
39         try {
40             const res = await fetch('/api/gcloud/status');
41             if (!res.ok) return;
42             const data = await res.json();
43
44             const badge = document.getElementById('gcloud-status-bad
ge');
45             const projectEl = document.getElementById('gcloud-projec
t-id');
46             const authTypeEl = document.getElementById('gcloud-auth-
type');
47             const emailEl = document.getElementById('gcloud-client-e
mail');
48
49             if (badge) {
50                 badge.className = data.authenticated ? 'badge rounded-
pill bg-success-subtle text-success border border-success px-3
py-2' : 'badge rounded-pill bg-warning-subtle text-warning bo
rder border-warning px-3 py-2';
51                 badge.innerText = data.authenticated ? (data.is_mock ?
'● Mock Режим' : '● GCP Подключен') : '● Не авторизован';
52             }
53
54             if (projectEl) projectEl.innerText = data.project_id ||
'default-project';
55             if (authTypeEl) authTypeEl.innerText = (data.auth_type |
| 'service_account').toUpperCase();
56             if (emailEl) emailEl.innerText = data.client_email || '1

```

```

    ocal-service-account';
57     } catch (e) {
58         console.error('[GCloudTab] Failed to fetch status:', e);
59     }
60 }
61
62 async function fetchLogs() {
63     try {
64         const res = await fetch('/api/gcloud/logs?limit=40');
65         if (!res.ok) return;
66         const logs = await res.json();
67
68         const terminal = document.getElementById('gcloud-log-ter
        minal-output');
69         const countBadge = document.getElementById('gcloud-logs-
        count-badge');
70         if (countBadge) countBadge.innerText = `${logs.length} c
        трок`;
71
72         if (terminal) {
73             if (!Array.isArray(logs) || logs.length === 0) {
74                 terminal.innerText = 'Журнал Cloud Logging пуст.';
75                 return;
76             }
77             terminal.innerHTML = logs.map(l => {
78                 const sev = (l.severity || 'INFO').toUpperCase();
79                 const color = sev === 'ERROR' || sev === 'CRITICAL'
? '#f87171' : (sev === 'WARNING' ? '#fbbf24' : '#94a3b8');
80                 return `<div style="color: ${color};">[${l.timestamp
|| ''}] [${sev}] ${l.resource_type ? '[' + l.resource_type +
' ] ' : ''}${l.message || ''}</div>`;
81             }).join('');
82         }
83     } catch (e) {
84         console.error('[GCloudTab] Failed to fetch logs:', e);
85     }
86 }
87

```

```

88 | async function runDiagnostics() {
89 |     try {
90 |         const res = await fetch('/api/gcloud/diagnostic');
91 |         if (!res.ok) return;
92 |         const report = await res.json();
93 |
94 |         const scoreEl = document.getElementById('gcloud-health-s
core');
95 |         const titleEl = document.getElementById('gcloud-diag-sum
mary-title');
96 |         const descEl = document.getElementById('gcloud-diag-summ
ary-desc');
97 |         const recList = document.getElementById('gcloud-recommen
dations-list');
98 |
99 |         if (scoreEl) scoreEl.innerText = `${report.health_score
|| 100}/100`;
100 |         if (titleEl) titleEl.innerText = `Статус: ${report.overa
ll_status || 'HEALTHY'}`;
101 |         if (descEl) descEl.innerText = report.summary || 'Все се
рвисы Google Cloud функционируют в штатном режиме.';
102 |
103 |         if (recList && Array.isArray(report.recommendations)) {
104 |             recList.innerHTML = report.recommendations.map(r => `
105 |                 <li class="list-group-item bg-transparent text-light
border-secondary py-1 px-0">• ${r}</li>
106 |             `).join('');
107 |         }
108 |     } catch (e) {
109 |         console.error('[GCloudTab] Diagnostic error:', e);
110 |     }
111 | }
112 |
113 | function initGCloudTab() {
114 |     fetchStatus();
115 |     fetchLogs();
116 |     runDiagnostics();
117 |

```

```
118 |     if (!isGcloudInitialized) {
119 |         const refreshBtn = document.getElementById('btn-gcloud-r
refresh');
120 |         if (refreshBtn) {
121 |             refreshBtn.addEventListener('click', () => {
122 |                 fetchStatus();
123 |                 fetchLogs();
124 |                 runDiagnostics();
125 |             });
126 |         }
127 |
128 |         const configBtn = document.getElementById('btn-gcloud-co
nfig');
129 |         if (configBtn) {
130 |             configBtn.addEventListener('click', () => {
131 |                 if (typeof window.openAppConfigModal === 'function')
{
132 |                     window.openAppConfigModal('gcloud_monitor');
133 |                 }
134 |             });
135 |         }
136 |         isGcloudInitialized = true;
137 |     }
138 | }
139 |
140 | window.initGCloudTab = initGCloudTab;
141 | })();
142 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/google_accounts_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/google_accounts_tab/index.html" class="w-100 h-100 border-0"></iframe>
15 |
16 File: index.html
17 Project: ai-breadboard
18 Package: windows/api/webgui/google_accounts_tab
19 Author: hypo69
20 Copyright: © 2026 hypo69
21 Updated: 2026-10-01 13:04:40
22 |=====
23 |=====
24 -->
25
26 <!-- Google Workspace Accounts Pool Management Tab -->
27 <div class="container-fluid py-2" id="google-accounts-tab-root">
28 |>
29 |    <!-- Top Banner & Stats -->
30 |    <div class="d-flex flex-wrap align-items-center justify-content-between gap-3 mb-3 p-3 bg-body-tertiary border rounded shadow-sm">
31 |        <div>
32 |            <h5 class="mb-1 text-primary d-flex align-items-center g
```



```

ap-2">
30     <i class="bi bi-google text-danger"></i>
31     <span>Управление аккаунтами Google Workspace</span>
32     </h5>
33     <p class="mb-0 text-muted small">Пул учетных записей (OA
uth 2.0 и Service Accounts) для интеграции с Gmail, Google Dri
ve, Sheets и Docs.</p>
34 </div>
35 <div class="d-flex gap-2">
36     <button class="btn btn-outline-primary btn-sm d-flex ali
gn-items-center gap-1 rounded-pill px-3" id="tab-btn-refresh-g
accounts" title="Обновить список">
37         <i class="bi bi-arrow-clockwise"></i> Обновить
38     </button>
39 </div>
40 </div>
41
42 <!-- Summary Metric Cards -->
43 <div class="row g-3 mb-3">
44     <div class="col-6 col-md-3">
45         <div class="card bg-body-tertiary border h-100 shadow-sm
">
46             <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
47                 <div>
48                     <div class="text-muted small">Всего аккаунтов</div>
49                     <h4 class="mb-0 fw-bold" id="gstat-total-accounts"
>0</h4>
50                 </div>
51                 <div class="bg-secondary bg-opacity-25 rounded p-2 t
ext-primary">
52                     <i class="bi bi-collection-fill fs-4"></i>
53                 </div>
54             </div>
55         </div>
56     </div>
57     <div class="col-6 col-md-3">

```

```
58 |         <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
59 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
60 |             <div>
61 |                 <div class="text-muted small">Активные</div>
62 |                 <h4 class="mb-0 text-success fw-bold" id="gstat-ac
    | tive-accounts">0</h4>
63 |             </div>
64 |             <div class="bg-success bg-opacity-25 rounded p-2 tex
    | t-success">
65 |                 <i class="bi bi-check-circle-fill fs-4"></i>
66 |             </div>
67 |         </div>
68 |     </div>
69 | </div>
70 | <div class="col-6 col-md-3">
71 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
72 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
73 |             <div>
74 |                 <div class="text-muted small">OAuth 2.0 (User)</di
    | v>
75 |                 <h4 class="mb-0 text-info fw-bold" id="gstat-oauth
    | -accounts">0</h4>
76 |             </div>
77 |             <div class="bg-info bg-opacity-25 rounded p-2 text-i
    | nfo">
78 |                 <i class="bi bi-person-badge-fill fs-4"></i>
79 |             </div>
80 |         </div>
81 |     </div>
82 | </div>
83 | <div class="col-6 col-md-3">
84 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
85 |         <div class="card-body p-3 d-flex align-items-center ju
```

```

stify-content-between">
86         <div>
87             <div class="text-muted small">Service Accounts</di
v>
88             <h4 class="mb-0 text-warning fw-bold" id="gstat-sa
-accounts">0</h4>
89         </div>
90         <div class="bg-warning bg-opacity-25 rounded p-2 tex
t-warning">
91             <i class="bi bi-cpu-fill fs-4"></i>
92         </div>
93     </div>
94 </div>
95 </div>
96 </div>
97
98 <!-- Info Alert -->
99 <div class="alert alert-info border-0 small shadow-sm rounde
d-3 mb-4" role="alert">
100     <div class="d-flex align-items-center gap-2 mb-1">
101         <i class="bi bi-info-circle-fill text-info fs-5"></i>
102         <strong class="text-info">Автономная ротация и приоритет
ы аккаунтов</strong>
103     </div>
104     <p class="mb-1 text-secondary">
105         Агенты и инструменты AI (<code>gmail_search</code>, <cod
e>gmail_create_draft</code>, <code>gdrive_list_files</code>, <
code>gsheets_read</code>, <code>gsheets_append</code>) могут я
вно вызывать нужный аккаунт через параметр <code>account_name=
"work"</code>, либо автоматически использовать основной аккаун
т по умолчанию (<span class="badge bg-warning text-dark"><i cl
ass="bi bi-star-fill"></i> Default</span>).
106     </p>
107     <div class="text-muted">
108         Учетные записи и токены авторизации хранятся в <code>src
/secrets/</code> в изолированном защищенном виде.
109     </div>
110 </div>

```

```
111 |
112 | <div class="row g-4 mb-4">
113 |   <!-- Add Account Card -->
114 |   <div class="col-lg-5">
115 |     <div class="card shadow-sm border h-100">
116 |       <div class="card-header bg-transparent border-bottom p
117 | y-2 d-flex justify-content-between align-items-center">
118 |         <h6 class="fw-bold mb-0 d-flex align-items-center ga
119 | p-2">
120 |           <i class="bi bi-plus-circle text-success"></i> Доба
121 | вить Google аккаунт
122 |         </h6>
123 |         <button type="button" class="btn btn-outline-info bt
124 | n-sm py-0 px-2 rounded-pill d-flex align-items-center gap-1" d
125 | ata-bs-toggle="modal" data-bs-target="#googleCredsHelpModal" s
126 | tyle="font-size: 0.75rem;">
127 |           <i class="bi bi-info-circle-fill"></i> Инструкция
128 |         </button>
129 |       </div>
130 |       <div class="card-body p-3">
131 |         <form id="tab-form-add-gaccount" onsubmit="event.pre
132 | ventDefault();">
133 |           <div class="mb-3">
134 |             <label class="form-label small text-muted fw-bol
135 | d">Имя аккаунта (ID): <span class="text-danger">*</span></labe
136 | l>
137 |             <input type="text" class="form-control form-cont
138 | rol-sm" id="tab-new-gacc-name" placeholder="Например: work, pe
139 | rsonal, bot" autocomplete="off" required>
140 |             <div class="form-text small text-secondary">Уник
141 | альный системный идентификатор для вызова в агентах.</div>
142 |           </div>
143 |
144 |           <div class="mb-3">
145 |             <label class="form-label small text-muted fw-bol
146 | d">Тип аутентификации:</label>
147 |             <select class="form-select form-select-sm" id="t
148 | ab-new-gacc-type">
```

```

135 |         <option value="oauth2">OAuth 2.0 Client (User
    | Login / Desktop App)</option>
136 |         <option value="service_account">Service Account
    | Key (Server JSON)</option>
137 |     </select>
138 | </div>
139 |
140 |     <div class="mb-3">
141 |         <label class="form-label small text-muted fw-bol
    | d">Email (опционально):</label>
142 |         <input type="email" class="form-control form-con
    | trol-sm" id="tab-new-gacc-email" placeholder="user@gmail.com и
    | ли sa@project.iam.gserviceaccount.com" autocomplete="off">
143 |     </div>
144 |
145 |     <!-- Credentials Input Mode Toggle -->
146 |     <div class="mb-3">
147 |         <label class="form-label small text-muted fw-bol
    | d d-flex justify-content-between align-items-center">
148 |             <span>Учетные данные (Credentials JSON): <span
    | class="text-danger">*</span></span>
149 |             <span class="badge bg-info text-dark" role="bu
    | tton" data-bs-toggle="modal" data-bs-target="#googleCredsHelpM
    | odal" style="cursor: pointer; font-size: 0.72rem;">
150 |                 <i class="bi bi-question-circle-fill me-1"><
    | /i>Как получить JSON?
151 |             </span>
152 |         </label>
153 |         <ul class="nav nav-pills nav-fill mb-2" id="tab-
    | gacc-input-mode-tabs" role="tablist">
154 |             <li class="nav-item" role="presentation">
155 |                 <button class="nav-link active py-1 small" i
    | d="tab-gacc-mode-file-tab" data-bs-toggle="pill" data-bs-targe
    | t="#tab-gacc-mode-file" type="button" role="tab">📄 Загрузить
    | файл</button>
156 |             </li>
157 |             <li class="nav-item" role="presentation">
158 |                 <button class="nav-link py-1 small" id="tab-

```

```

gacc-mode-json-tab" data-bs-toggle="pill" data-bs-target="#tab
-gacc-mode-json" type="button" role="tab"> Вставить JSON</bu
tton>
159         </li>
160     </ul>
161
162     <div class="tab-content" id="tab-gacc-input-mode
-tabContent">
163         <div class="tab-pane fade show active" id="tab
-gacc-mode-file" role="tabpanel">
164             <input class="form-control form-control-sm"
type="file" id="tab-new-gacc-file" accept=".json,application/j
son">
165             <div class="form-text small text-secondary">
Загрузите <code>credentials.json</code> или <code>service_accou
unt.json</code> из <a href="https://console.cloud.google.com/a
pis/credentials" target="_blank" rel="noopener noreferrer" cla
ss="text-info text-decoration-none">Google Cloud Console <i cl
ass="bi bi-box-arrow-up-right ms-1" style="font-size: 0.75rem;
"></i></a>.</div>
166         </div>
167         <div class="tab-pane fade" id="tab-gacc-mode-j
son" role="tabpanel">
168             <textarea class="form-control font-monospace
form-control-sm" id="tab-new-gacc-json" rows="4" placeholder=
'{"installed": {"client_id": "...", "client_secret": "..."}}'>
</textarea>
169         </div>
170     </div>
171 </div>
172
173     <div class="form-check mb-4 mt-2">
174         <input class="form-check-input" type="checkbox"
id="tab-new-gacc-default" style="cursor: pointer;">
175         <label class="form-check-label small fw-semibold
" for="tab-new-gacc-default" style="cursor: pointer;">
176             Сделать аккаунтом по умолчанию (Default)
177     </label>

```

```

178         </div>
179
180         <button type="button" class="btn btn-primary btn-sm w-100 rounded-pill fw-bold" id="tab-btn-add-gacc">
181             <i class="bi bi-plus-circle me-1"></i> Сохранить
182             Google аккаунт
183         </button>
184     </form>
185 </div>
186 </div>
187
188 <!-- Accounts Table -->
189 <div class="col-lg-7">
190     <div class="card shadow-sm border mb-3">
191         <div class="card-header bg-transparent border-bottom p-y-2 d-flex justify-content-between align-items-center">
192             <h6 class="fw-bold mb-0"><i class="bi bi-list-check me-1"></i> Зарегистрированные аккаунты</h6>
193             <span class="badge bg-secondary" id="tab-gaccounts-count-badge">0</span>
194         </div>
195         <div class="card-body p-0">
196             <div class="table-responsive">
197                 <table class="table table-hover align-middle mb-0" id="tab-gaccounts-table">
198                     <thead class="table-light small text-uppercase">
199                         <tr>
200                             <th>Аккаунт</th>
201                             <th style="width: 130px;" class="text-center">Тип</th>
202                             <th style="width: 100px;" class="text-center">Статус</th>
203                             <th style="width: 110px;" class="text-center">Токен</th>
204                             <th style="width: 180px;" class="text-center">Действия</th>
205                         </tr>

```

```

206 |         </thead>
207 |         <tbody id="tab-gaccounts-list-body">
208 |             <tr>
209 |                 <td colspan="5" class="text-center py-4 text
| -muted">Загрузка аккаунтов...</td>
210 |             </tr>
211 |         </tbody>
212 |     </table>
213 | </div>
214 | </div>
215 | </div>
216 |
217 | <!-- Test & Diagnostics Output Card -->
218 | <div class="card bg-body-tertiary border shadow-sm" id="
| tab-gaccount-test-card" style="display: none;">
219 |     <div class="card-header bg-transparent border-bottom d
| -flex justify-content-between align-items-center py-2">
220 |         <span class="small fw-bold text-info"><i class="bi b
| i-terminal me-1"></i> Результат проверки доступа</span>
221 |         <button class="btn btn-sm btn-outline-secondary py-0
| px-2" type="button" onclick="document.getElementById('tab-gac
| count-test-card').style.display='none' ">⌵ </button>
222 |     </div>
223 |     <div class="card-body p-3 font-monospace small bg-body
| " id="tab-gaccount-test-body" style="max-height: 200px; overfl
| ow-y: auto; white-space: pre-wrap;">
224 |     </div>
225 | </div>
226 | </div>
227 | </div>
228 |
229 | <!-- Google Credentials Instruction Modal -->
230 | <div class="modal fade" id="googleCredsHelpModal" tabindex="
| -1" aria-labelledby="googleCredsHelpModalLabel" aria-hidden="t
| rue">
231 |     <div class="modal-dialog modal-lg modal-dialog-centered mo
| dal-dialog-scrollable">
232 |         <div class="modal-content shadow-lg">

```



```

233 |         <div class="modal-header">
234 |             <h5 class="modal-title d-flex align-items-center gap
| -2" id="googleCredsHelpModalLabel">
235 |                 <i class="bi bi-google text-danger"></i> Как получ
| ить и загрузить Credentials JSON
236 |             </h5>
237 |             <button type="button" class="btn-close" data-bs-dism
| iss="modal" aria-label="Close"></button>
238 |         </div>
239 |         <div class="modal-body p-4">
240 |             <ul class="nav nav-tabs mb-3" id="helpModalTabs" rol
| e="tablist">
241 |                 <li class="nav-item" role="presentation">
242 |                     <button class="nav-link active" id="help-oauth-t
| ab" data-bs-toggle="tab" data-bs-target="#help-oauth-pane" typ
| e="button" role="tab">
243 |                         <i class="bi bi-person-check-fill text-primary
| me-1"></i> 1. OAuth 2.0 Client (Личный вход)
244 |                     </button>
245 |                 </li>
246 |                 <li class="nav-item" role="presentation">
247 |                     <button class="nav-link" id="help-sa-tab" data-b
| s-toggle="tab" data-bs-target="#help-sa-pane" type="button" ro
| le="tab">
248 |                         <i class="bi bi-robot text-warning me-1"></i>
| 2. Service Account (Сервисный ключ)
249 |                     </button>
250 |                 </li>
251 |             </ul>
252 |
253 |             <div class="tab-content">
254 |                 <!-- OAuth 2.0 Pane -->
255 |                 <div class="tab-pane fade show active" id="help-oa
| uth-pane" role="tabpanel">
256 |                     <p class="text-secondary small">
257 |                         Используется для работы от имени вашего личног
| о или рабочего аккаунта Google (чтение/отправка почты в Gmail,
| доступ к Google Drive, редактирование Google Sheets/Docs).

```

```

258         </p>
259         <ol class="list-group list-group-numbered list-g
roup-flush small">
260             <li class="list-group-item">
261                 Перейдите в консоль управления Google: <br>
262                 <a href="https://console.cloud.google.com/ap
is/credentials" target="_blank" rel="noopener noreferrer" clas
s="btn btn-sm btn-outline-info mt-1">
263                 <i class="bi bi-box-arrow-up-right me-1"><
/i> Google Cloud Console □ Credentials
264                 </a>
265             </li>
266             <li class="list-group-item">
267                 Если проект еще не создан, выберите существу
ющий или создайте новый проект (кнопка в верхней панели).
268             </li>
269             <li class="list-group-item">
270                 В разделе <strong>OAuth consent screen</stro
ng> (Экран согласия OAuth):
271                 <ul class="mt-1 text-secondary">
272                     <li>Выберите тип: <code>External</code> (В
нешний) или <code>Internal</code>.</li>
273                     <li>Укажите название приложения (например,
<code>AI Breadboard</code>) и ваш контактный email.</li>
274                     <li>В разделе <em>Test users</em> добавьте
ваш Google email (если статус приложения Testing).</li>
275                 </ul>
276             </li>
277             <li class="list-group-item">
278                 В разделе <strong>Credentials</strong> нажми
те <code>+ CREATE CREDENTIALS</code> □ выберите <strong>OAuth
client ID</strong>.
279             </li>
280             <li class="list-group-item">
281                 В поле <em>Application type</em> выберите <s
trong>Desktop app</strong> (Приложение для ПК) и задайте имя (
например <code>Breadboard-Desktop</code>), затем нажмите <stro
ng>Create</strong>.

```

```

282 |         </li>
283 |         <li class="list-group-item">
284 |             В появившемся окне нажмите <i class="bi bi-d
download text-info"></i> <strong>DOWNLOAD JSON</strong> (или ск
ачайте файл в таблице OAuth 2.0 Client IDs).
285 |         </li>
286 |         <li class="list-group-item">
287 |             Включите необходимые API в <a href="https://
console.cloud.google.com/apis/dashboard" target="_blank" rel="
noopener noreferrer" class="text-info">Enabled APIs & Services
<i class="bi bi-box-arrow-up-right"></i></a>:
288 |                 <div class="mt-1 d-flex flex-wrap gap-1">
289 |                     <span class="badge bg-secondary">Gmail API
</span>
290 |                     <span class="badge bg-secondary">Google Dr
ive API</span>
291 |                     <span class="badge bg-secondary">Google Sh
eets API</span>
292 |                     <span class="badge bg-secondary">Google Do
cs API</span>
293 |                 </div>
294 |             </li>
295 |             <li class="list-group-item">
296 |                 В этой карточке выберите <strong>Загрузить ф
айл</strong> и укажите скачанный <code>credentials.json</code>
(или вставьте его содержимое во вкладку <strong>Вставить JSON
</strong>), введите ID аккаунта и нажмите <strong>Сохранить Go
ogle аккаунт</strong>.
297 |             </li>
298 |         </ol>
299 |     </div>
300 |
301 |     <!-- Service Account Pane -->
302 |     <div class="tab-pane fade" id="help-sa-pane" role=
"tabpanel">
303 |         <p class="text-secondary small">
304 |             Используется для серверных / автономных фоновы
х задач без необходимости интерактивного логина в браузере.

```

```

305 |         </p>
306 |         <ol class="list-group list-group-numbered list-g
roup-flush small">
307 |             <li class="list-group-item">
308 |                 Перейдите в раздел сервисных аккаунтов: <br>
309 |                 <a href="https://console.cloud.google.com/ia
m-admin/serviceaccounts" target="_blank" rel="noopener norefer
rer" class="btn btn-sm btn-outline-warning mt-1">
310 |                     <i class="bi bi-box-arrow-up-right me-1"><
/i> Google Cloud Console □ Service Accounts
311 |                 </a>
312 |             </li>
313 |             <li class="list-group-item">
314 |                 Нажмите <code>+ CREATE SERVICE ACCOUNT</code>
>, введите имя (например <code>ai-agent-sa</code>) и нажмите <
strong>Done</strong>.
315 |             </li>
316 |             <li class="list-group-item">
317 |                 Кликните по созданному сервисному аккаунту и
перейдите во вкладку <strong>Keys</strong> (Ключи).
318 |             </li>
319 |             <li class="list-group-item">
320 |                 Нажмите <strong>ADD KEY</strong> □ <strong>C
reate new key</strong> □ выберите тип <strong>JSON</strong> □
нажмите <strong>Create</strong>.
321 |             </li>
322 |             <li class="list-group-item">
323 |                 Файл ключа (<code>project-id-*.json</code>)
скачается на ваш компьютер.
324 |             </li>
325 |             <li class="list-group-item">
326 |                 В форме выберите тип <em>Service Account Key
</em>, загрузите JSON файл и сохраните аккаунт.
327 |             </li>
328 |         </ol>
329 |     </div>
330 | </div>
331 | </div>

```

```
332 |         <div class="modal-footer">
333 |             <button type="button" class="btn btn-secondary btn-s
334 | m rounded-pill px-4" data-bs-dismiss="modal">Понятно</button>
335 |         </div>
336 |     </div>
337 | </div>
338 | </div>
339 |
340 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Google Accounts Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/google_accounts_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/google_accounts_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 // Google Workspace Accounts Tab JavaScript Logic
22
23 function escapeHtml(text) {
24   if (!text) return '';
25   return String(text)
26     .replace(/&/g, '&amp;')
27     .replace(/</g, '&lt;')
28     .replace(/>/g, '&gt;')
29     .replace(/"/g, '&quot;')
30     .replace(/'/g, '&#039;');
31 }
32
```

```

33 function showGAccountsNotification(message, type = 'info') {
34   const notification = document.createElement('div');
35   notification.className = `alert alert-${type} position-fixed
    top-0 end-0 m-3 shadow-lg`;
36   notification.style.zIndex = '9999';
37   notification.style.maxWidth = '400px';
38   notification.innerHTML = message;
39   document.body.appendChild(notification);
40
41   setTimeout(() => {
42     notification.remove();
43   }, 4000);
44 }
45
46 async function refreshTabGoogleAccountsList() {
47   const container = document.getElementById('tab-gaccounts-lis
    t-body');
48   if (!container) return;
49
50   try {
51     container.innerHTML = '<tr><td colspan="5" class="text-cen
    ter py-4 text-muted"><span class="spinner-border spinner-borde
    r-sm me-2 text-primary"></span>Загрузка аккаунтов...</td></tr>
    ';
52     const data = await window.api.googleAccounts.list();
53     const accounts = data.accounts || [];
54
55     // Update stats
56     const totalEl = document.getElementById('gstat-total-accou
    nts');
57     const activeEl = document.getElementById('gstat-active-acc
    ounts');
58     const oauthEl = document.getElementById('gstat-oauth-accou
    nts');
59     const saEl = document.getElementById('gstat-sa-accounts');
60     const countBadge = document.getElementById('tab-gaccounts-
    count-badge');
61

```

```

62     if (totalEl) totalEl.textContent = accounts.length;
63     if (activeEl) activeEl.textContent = accounts.filter(a =>
| a.status === 'active').length;
64     if (oauthEl) oauthEl.textContent = accounts.filter(a => a.
| type === 'oauth2').length;
65     if (saEl) saEl.textContent = accounts.filter(a => a.type =
| == 'service_account').length;
66     if (countBadge) countBadge.textContent = `${accounts.lengt
| h} акк.`;
67
68     if (accounts.length === 0) {
69         container.innerHTML = '<tr><td colspan="5" class="text-c
| enter py-4 text-muted">Список аккаунтов пуст. Загрузите creden
| tials.json или service_account.json.</td></tr>';
70         return;
71     }
72
73     container.innerHTML = '';
74     accounts.forEach(acc => {
75         const row = document.createElement('tr');
76
77         // 1. Account Name + Default Badge + Email
78         const tdName = document.createElement('td');
79         tdName.innerHTML = `
80             <div class="d-flex align-items-center gap-2">
81                 <strong>${escapeHtml(acc.name)}</strong>
82                 ${acc.is_default ? '<span class="badge bg-warning te
| xt-dark"><i class="bi bi-star-fill"></i> Default</span>' : ''}
83             </div>
84             ${acc.email ? `<div class="small text-muted font-monos
| pace">${escapeHtml(acc.email)}</div>` : ''}
85         `;
86         row.appendChild(tdName);
87
88         // 2. Type
89         const tdType = document.createElement('td');
90         tdType.className = 'text-center';
91         const isOAuth = acc.type === 'oauth2';

```



```

92         tdType.innerHTML = isOAuth
93         ? '<span class="badge bg-primary-subtle text-primary b
order border-primary-subtle"><i class="bi bi-person me-1"></i>
OAuth 2.0</span>'
94         : '<span class="badge bg-info-subtle text-info border
border-info-subtle"><i class="bi bi-cpu me-1"></i>Service Acco
unt</span>';
95         row.appendChild(tdType);
96
97         // 3. Status / Quota
98         const tdStatus = document.createElement('td');
99         tdStatus.className = 'text-center';
100        const isActive = acc.status === 'active';
101        if (isActive) {
102            tdStatus.innerHTML = '<span class="badge bg-success">Ак
тивен</span>';
103        } else if (acc.status === 'exhausted') {
104            tdStatus.innerHTML = '<span class="badge bg-danger" ti
tle="Исчерпан суточный лимит квоты">Лимит</span>';
105        } else {
106            tdStatus.innerHTML = `<span class="badge bg-secondary"
>${escapeHtml(acc.status)}</span>`;
107        }
108        row.appendChild(tdStatus);
109
110        // 4. Token / Credentials Status
111        const tdToken = document.createElement('td');
112        tdToken.className = 'text-center';
113        if (acc.has_token || acc.type === 'service_account') {
114            tdToken.innerHTML = '<span class="badge bg-success-sub
tle text-success border border-success-subtle"><i class="bi bi
-shield-check me-1"></i>Готов</span>';
115        } else {
116            tdToken.innerHTML = '<span class="badge bg-warning-sub
tle text-warning border border-warning-subtle"><i class="bi bi
-key me-1"></i>Нужен токен</span>';
117        }
118        row.appendChild(tdToken);

```

```
119 |
120 |     // 5. Actions
121 |     const tdActions = document.createElement('td');
122 |     tdActions.className = 'text-center';
123 |
124 |     // Set Default button
125 |     if (!acc.is_default) {
126 |         const btnDefault = document.createElement('button');
127 |         btnDefault.className = 'btn btn-xs btn-outline-warning
128 |         btnDefault.title = 'Сделать аккаунтом по умолчанию';
129 |         btnDefault.innerHTML = '<i class="bi bi-star"></i>';
130 |         btnDefault.onclick = () => setTabGoogleAccountDefault(
131 |         acc.name);
132 |         tdActions.appendChild(btnDefault);
133 |
134 |     // Reset status button
135 |     if (acc.status === 'exhausted') {
136 |         const btnReset = document.createElement('button');
137 |         btnReset.className = 'btn btn-xs btn-outline-info btn-
138 |         sm me-1';
139 |         btnReset.title = 'Сбросить статус исчерпания';
140 |         btnReset.innerHTML = '<i class="bi bi-arrow-repeat"></
141 |         i>';
142 |         btnReset.onclick = () => resetTabGoogleAccountStatus(a
143 |         cc.name);
144 |         tdActions.appendChild(btnReset);
145 |     }
146 |
147 |     // Test button
148 |     const btnTest = document.createElement('button');
149 |     btnTest.className = 'btn btn-xs btn-outline-info btn-sm
150 |     me-1';
151 |     btnTest.title = 'Проверить доступ и авторизацию';
152 |     btnTest.innerHTML = '<i class="bi bi-play-circle"></i> Т
153 |     ест';
154 |     btnTest.onclick = () => testTabGoogleAccount(acc.name);
```

```
150         tdActions.appendChild(btnTest);
151
152         // Delete button
153         const btnDelete = document.createElement('button');
154         btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
155         btnDelete.title = 'Удалить аккаунт из пула';
156         btnDelete.innerHTML = '<i class="bi bi-trash"></i>';
157         btnDelete.onclick = () => deleteTabGoogleAccount(acc.name);
158         tdActions.appendChild(btnDelete);
159
160         row.appendChild(tdActions);
161         container.appendChild(row);
162     });
163
164     } catch (err) {
165         console.error('Ошибка загрузки аккаунтов Google Workspace:
', err);
166         container.innerHTML = `<tr><td colspan="5" class="text-center py-4 text-danger">Ошибка: ${escapeHtml(err.message)}</td><
/tr>`;
167     }
168 }
169
170 async function handleAddTabGoogleAccount() {
171     const nameInput = document.getElementById('tab-new-gacc-name
');
172     const typeSelect = document.getElementById('tab-new-gacc-type');
173     const emailInput = document.getElementById('tab-new-gacc-email');
174     const fileInput = document.getElementById('tab-new-gacc-file
');
175     const jsonTextarea = document.getElementById('tab-new-gacc-json');
176     const defaultCheckbox = document.getElementById('tab-new-gacc-default');
```

```
177 |   const addBtn = document.getElementById('tab-btn-add-gacc');
178 |
179 |   if (!nameInput || !addBtn) return;
180 |
181 |   const accountName = nameInput.value.trim();
182 |   const accountType = typeSelect ? typeSelect.value : 'oauth2'
    | ;
183 |   const email = emailInput ? emailInput.value.trim() : '';
184 |   const setAsDefault = defaultCheckbox ? defaultCheckbox.checked : false;
185 |
186 |   if (!accountName) {
187 |     showGAccountsNotification('Введите имя аккаунта (например:
    | work, personal)', 'warning');
188 |     nameInput.focus();
189 |     return;
190 |   }
191 |
192 |   const file = fileInput && fileInput.files && fileInput.files
    | [0];
193 |   const jsonContent = jsonTextarea ? jsonTextarea.value.trim()
    | : '';
194 |
195 |   if (!file && !jsonContent) {
196 |     showGAccountsNotification('Загрузите файл credentials.json
    | / service_account.json или вставьте JSON', 'warning');
197 |     return;
198 |   }
199 |
200 |   addBtn.disabled = true;
201 |   const originalText = addBtn.innerHTML;
202 |   addBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Сохранение...';
203 |
204 |   try {
205 |     if (file) {
206 |       const formData = new FormData();
207 |       formData.append('account_name', accountName);
```

```
208 |     formData.append('account_type', accountType);
209 |     if (email) formData.append('email', email);
210 |     formData.append('set_as_default', setAsDefault ? 'true'
    | : 'false');
211 |     formData.append('file', file);
212 |
213 |     await window.api.googleAccounts.upload(formData);
214 | } else {
215 |     let parsedJson = null;
216 |     try {
217 |         parsedJson = JSON.parse(jsonContent);
218 |     } catch (e) {
219 |         throw new Error('Некорректный JSON в поле учетных данн
    | ых');
220 |     }
221 |
222 |     await window.api.googleAccounts.create({
223 |         account_name: accountName,
224 |         account_type: accountType,
225 |         email: email || undefined,
226 |         credentials_dict: parsedJson,
227 |         set_as_default: setAsDefault
228 |     });
229 | }
230 |
231 |     showGAccountsNotification(`Аккаунт Google "${accountName}"
    | успешно сохранен в пул`, 'success');
232 |     nameInput.value = '';
233 |     if (emailInput) emailInput.value = '';
234 |     if (fileInput) fileInput.value = '';
235 |     if (jsonTextarea) jsonTextarea.value = '';
236 |     if (defaultCheckbox) defaultCheckbox.checked = false;
237 |
238 |     await refreshTabGoogleAccountsList();
239 | } catch (err) {
240 |     console.error('Ошибка сохранения аккаунта Google:', err);
241 |     showGAccountsNotification('Ошибка сохранения: ' + err.mess
    | age, 'danger');
```

```
242 |   } finally {
243 |     addBtn.disabled = false;
244 |     addBtn.innerHTML = originalText;
245 |   }
246 | }
247 |
248 | async function setTabGoogleAccountDefault(name) {
249 |   try {
250 |     await window.api.googleAccounts.setDefault(name);
251 |     showGAccountsNotification(`Аккаунт "${name}" назначен по у
    молчанию`, 'success');
252 |     await refreshTabGoogleAccountsList();
253 |   } catch (err) {
254 |     console.error('Ошибка установки аккаунта по умолчанию:', e
rr);
255 |     showGAccountsNotification('Ошибка: ' + err.message, 'dange
r');
256 |   }
257 | }
258 |
259 | async function resetTabGoogleAccountStatus(name) {
260 |   try {
261 |     await window.api.googleAccounts.resetStatus(name);
262 |     showGAccountsNotification(`Статус аккаунта "${name}" сброш
ен в активный`, 'success');
263 |     await refreshTabGoogleAccountsList();
264 |   } catch (err) {
265 |     console.error('Ошибка сброса статуса аккаунта:', err);
266 |     showGAccountsNotification('Ошибка сброса: ' + err.message,
'danger');
267 |   }
268 | }
269 |
270 | async function testTabGoogleAccount(name) {
271 |   const testCard = document.getElementById('tab-gaccount-test-
card');
272 |   const testBody = document.getElementById('tab-gaccount-test-
body');
```

```

273
274   if (testCard && testBody) {
275       testCard.style.display = 'block';
276       testBody.innerHTML = `

```

```
302 async function deleteTabGoogleAccount(name) {
303   if (!confirm(`Вы уверены, что хотите удалить аккаунт "${name}" из пула Google Workspace?`)) return;
304   try {
305     await window.api.googleAccounts.delete(name);
306     showGAccountsNotification(`Аккаунт "${name}" успешно удален`, 'success');
307     await refreshTabGoogleAccountsList();
308   } catch (err) {
309     console.error('Ошибка удаления аккаунта Google:', err);
310     showGAccountsNotification('Ошибка удаления: ' + err.message, 'danger');
311   }
312 }
313
314 async function initGoogleAccountsTab() {
315   const refreshBtn = document.getElementById('tab-btn-refresh-gaccounts');
316   const addBtn = document.getElementById('tab-btn-add-gacc');
317
318   if (refreshBtn) {
319     refreshBtn.onclick = async () => {
320       refreshBtn.disabled = true;
321       const orig = refreshBtn.innerHTML;
322       refreshBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Обновление...';
323       try {
324         await refreshTabGoogleAccountsList();
325         showGAccountsNotification('Список Google аккаунтов обновлен', 'success');
326       } finally {
327         refreshBtn.disabled = false;
328         refreshBtn.innerHTML = orig;
329       }
330     };
331   }
332
333   if (addBtn) {
```



```
334 |     addBtn.onclick = handleAddTabGoogleAccount;
335 | }
336 |
337 |     await refreshTabGoogleAccountsList();
338 | }
339 |
340 | window.initGoogleAccountsTab = initGoogleAccountsTab;
341 | window.refreshTabGoogleAccountsList = refreshTabGoogleAccounts
    | List;
342 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: AI-Breadboard Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон веб-интерфейса системного модуля (index).
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /src/api/webgui/google_desktop_tab/index.html
12 |
13 File: index.html
14 Project: ai-breadboard
15 Package: src/api/webgui/google_desktop_tab
16 Author: hypo69
17 Copyright: © 2026 hypo69
18 Updated: 2026-10-08 03:08:00
19 |=====
20 |=====
21 -->
22 <!-- Google User Desktop Web Tab Module -->
23 <style>
24   .gdesktop-container {
25     padding: 0.5rem 0.75rem;
26   }
27   .gdesktop-card {
28     background: #0f172a !important;
29     border: 1px solid #334155 !important;
30     border-radius: 8px;
31     padding: 0.5rem 0.85rem;
32     display: flex;
33     flex-direction: column;
34     justify-content: space-between;
```

```
34 |     min-height: 64px;
35 |     gap: 0.2rem;
36 | }
37 | .gdesktop-label {
38 |     font-size: 0.7rem;
39 |     text-transform: uppercase;
40 |     letter-spacing: 0.05em;
41 |     color: #94a3b8;
42 |     line-height: 1.1;
43 | }
44 | .gdesktop-value {
45 |     font-size: 1.3rem;
46 |     font-weight: 700;
47 |     color: #f8fafc;
48 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
49 |     line-height: 1.15;
50 | }
51 | .gdesktop-panel {
52 |     background: #0f172a !important;
53 |     border: 1px solid #334155 !important;
54 |     border-radius: 8px;
55 |     overflow: hidden;
56 | }
57 | .gdesktop-panel-header {
58 |     background: #1e293b !important;
59 |     border-bottom: 1px solid #334155 !important;
60 |     padding: 0.5rem 0.85rem;
61 |     font-weight: 600;
62 |     font-size: 0.85rem;
63 |     color: #38bdf8;
64 |     display: flex;
65 |     align-items: center;
66 |     justify-content: space-between;
67 | }
68 | .gdesktop-subnav .nav-link {
69 |     color: #94a3b8;
70 |     border-radius: 6px;
```

```

71     padding: 0.35rem 0.75rem;
72     font-size: 0.82rem;
73     font-weight: 500;
74     transition: all 0.15s ease-in-out;
75 }
76 .gdesktop-subnav .nav-link:hover {
77     color: #f8fafc;
78     background: rgba(255, 255, 255, 0.05);
79 }
80 .gdesktop-subnav .nav-link.active {
81     color: #000;
82     background: #38bdf8;
83     font-weight: 600;
84 }
85 .gdesktop-table {
86     font-size: 0.82rem;
87 }
88 .gdesktop-table td, .gdesktop-table th {
89     vertical-align: middle;
90 }
91 </style>
92
93 <div class="gdesktop-container" id="google-desktop-tab-root">
94     <!-- Top Header Banner & Actions -->
95     <div class="card shadow-sm border-secondary-subtle py-2 px-3
96         mb-3 d-flex justify-content-between align-items-center flex-w
97         rap gap-2">
98         <div class="d-flex align-items-center gap-2 flex-wrap">
99             <span class="fs-4">□ </span>
100             <div>
101                 <div class="d-flex align-items-center gap-2">
102                     <span class="fw-bold text-white" style="font-size: 1
103                         rem;">Google User Desktop</span>
104                     <span class="badge bg-danger text-white border borde
105                         r-light-subtle px-2 py-0.5" style="font-size: 0.7rem;">Workspa
106                         ce Aggregator</span>
107                 </div>
108                 <div class="text-secondary small" style="font-size: 0.

```

```

76rem;">
104         <span data-i18n="auto__gmail_google_calendar_docs_sh
eets__31dc39"></span> Google Drive
105     </div>
106 </div>
107 </div>
108
109 <!-- Account Selector & Sync Controls -->
110 <div class="d-flex align-items-center gap-2 flex-wrap">
111     <div class="d-flex align-items-center gap-1.5 bg-dark bg
-opacity-50 border border-secondary px-2 py-1 rounded">
112         <label for="gdesktop-account-select" class="small text
-muted mb-0 me-1" style="font-size: 0.75rem;"><span data-i18n=
"auto__5b16fc"></span>:</label>
113         <select class="form-select form-select-sm bg-dark text
-light border-0 py-0 px-2" id="gdesktop-account-select" style=
"font-size: 0.8rem; width: auto; cursor: pointer;">
114             <option value=""><span data-i18n="auto__94a96c"></s
pan>...</option>
115         </select>
116     </div>
117
118     <button class="btn btn-xs btn-outline-info rounded-pill
px-3 py-1.5" style="font-size: 0.78rem;" id="btn-gdesktop-refr
esh" title="<span data-i18n="auto__19efd8"></span>">
119         <i class="bi bi-arrow-clockwise me-1"></i> <span data-
i18n="auto__c2f668"></span>
120     </button>
121
122     <button class="btn btn-xs btn-outline-success rounded-pi
ll px-3 py-1.5" style="font-size: 0.78rem;" id="btn-gdesktop-s
ync" title="<span data-i18n="auto__d5b581"></span> API Google
">
123         <i class="bi bi-cloud-arrow-down-fill me-1"></i> <span
data-i18n="auto__7ceddd"></span>
124     </button>
125 </div>
126 </div>

```

```
127 |
128 | <!-- Summary Metric Cards -->
129 | <div class="row g-2 mb-3">
130 |     <div class="col-6 col-md-3">
131 |         <div class="gdesktop-card">
132 |             <span class="gdesktop-label"><span data-i18n="auto___a
133 | edea9"></span></span>
134 |             <div class="d-flex align-items-baseline justify-conten
135 | t-between mt-1">
136 |                 <span class="gdesktop-value text-info text-truncate"
137 | style="font-size: 1.05rem;" id="gdesktop-acc-name">-</span>
138 |                 <span class="badge bg-secondary" id="gdesktop-acc-st
139 | atus" style="font-size: 0.68rem;">-</span>
140 |             </div>
141 |         </div>
142 |     </div>
143 |     <div class="col-6 col-md-3">
144 |         <div class="gdesktop-card">
145 |             <span class="gdesktop-label"><span data-i18n="auto___5
146 | e10ab"></span> (Gmail)</span>
147 |             <div class="d-flex align-items-baseline justify-conten
148 | t-between mt-1">
149 |                 <span class="gdesktop-value text-warning" id="gdeskt
150 | op-mail-count">0</span>
151 |                 <span class="text-muted small"><span data-i18n="auto
152 | ___c66f45"></span></span>
153 |             </div>
154 |         </div>
155 |     </div>
156 |     <div class="col-6 col-md-3">
157 |         <div class="gdesktop-card">
158 |             <span class="gdesktop-label"><span data-i18n="auto___9
159 | 7c74f"></span></span>
160 |             <div class="d-flex align-items-baseline justify-conten
161 | t-between mt-1">
162 |                 <span class="gdesktop-value text-success" id="gdeskt
163 | op-calendar-count">0</span>
164 |                 <span class="text-muted small"><span data-i18n="auto
```

```

    ____d4eca8"></span></span>
154         </div>
155     </div>
156 </div>
157 <div class="col-6 col-md-3">
158     <div class="gdesktop-card">
159         <span class="gdesktop-label"><span data-i18n="auto____c
160         dbcb8"></span></span>
161         <div class="d-flex align-items-baseline justify-content-between mt-1">
162             <span class="gdesktop-value text-primary" id="gdesktop-docs-drive-count">0 / 0</span>
163             <span class="text-muted small">Docs / Drive</span>
164         </div>
165     </div>
166 </div>
167
168 <!-- Sub-navigation Tabs -->
169 <div class="gdesktop-panel mb-3">
170     <div class="gdesktop-panel-header">
171         <ul class="nav nav-pills gdesktop-subnav gap-1" id="gdesktop-subnav-tabs" role="tablist">
172             <li class="nav-item" role="presentation">
173                 <button class="nav-link active" id="gdesktop-tab-mail-btn" data-bs-toggle="pill" data-bs-target="#gdesktop-pane-mail" type="button" role="tab">
174                      Gmail <span data-i18n="auto____108aa2"></span>
175                 </button>
176             </li>
177             <li class="nav-item" role="presentation">
178                 <button class="nav-link" id="gdesktop-tab-calendar-btn" data-bs-toggle="pill" data-bs-target="#gdesktop-pane-calendar" type="button" role="tab">
179                      Google <span data-i18n="auto____fa3a2f"></span>
180                 </button>
181             </li>
182             <li class="nav-item" role="presentation">

```

```

183 |         <button class="nav-link" id="gdesktop-tab-docs-btn"
      | data-bs-toggle="pill" data-bs-target="#gdesktop-pane-docs" typ
      | e="button" role="tab">
184 |             □ <span data-i18n="auto___a65e97"></span>
185 |         </button>
186 |     </li>
187 |     <li class="nav-item" role="presentation">
188 |         <button class="nav-link" id="gdesktop-tab-drive-btn"
      | data-bs-toggle="pill" data-bs-target="#gdesktop-pane-drive" t
      | ype="button" role="tab">
189 |             □ Google <span data-i18n="auto___ca0407"></span>
190 |         </button>
191 |     </li>
192 | </ul>
193 |     <span class="text-muted small" id="gdesktop-last-refresh
      | ed"><span data-i18n="auto___484f3b"></span>: -</span>
194 | </div>
195 |
196 |     <div class="tab-content p-3" id="gdesktop-tab-content">
197 |         <!-- 1. Gmail Pane -->
198 |         <div class="tab-pane fade show active" id="gdesktop-pane
      | -mail" role="tabpanel">
199 |             <!-- Gmail API Info Badge -->
200 |             <div class="alert alert-warning border border-warning-
      | subtle text-light small shadow-sm rounded-3 mb-3 p-2 px-3 d-fl
      | ex align-items-center justify-content-between flex-wrap gap-2"
      | style="background: rgba(245, 158, 11, 0.1);" role="alert">
201 |                 <div class="d-flex align-items-center gap-2">
202 |                     <i class="bi bi-envelope-exclamation-fill text-war
      | ning fs-5"></i>
203 |                 </div>
204 |                 <span class="fw-bold text-warning me-1"><span da
      | ta-i18n="auto___gmail_api___a058db"></span>):</span>
205 |                 <span class="text-light-50"><span data-i18n="aut
      | o___gmail___8c0930"></span> <strong>Enable</strong>:</span>
206 |             </div>
207 |         </div>
208 |         <a href="https://console.developers.google.com/apis/

```



```

api/gmail.googleapis.com/overview" target="_blank" rel="noopener
noreferrer" class="btn btn-warning btn-sm fw-bold rounded-p
ill text-dark text-nowrap py-1 px-3" style="font-size: 0.78rem
;";">
209         <i class="bi bi-box-arrow-up-right me-1"></i> Goog
le Cloud Console – Gmail API Overview
210         </a>
211     </div>
212
213     <div class="table-responsive">
214         <table class="table table-dark table-hover align-mid
dle gdesktop-table mb-0">
215             <thead>
216                 <tr class="text-secondary small text-uppercase">
217                     <th style="width: 200px;"><span data-i18n="aut
o___522b04"></span></th>
218                     <th><span data-i18n="auto___96f68e"></span></t
h>
219                     <th style="width: 170px;"><span data-i18n="aut
o___a5b49d"></span></th>
220                     <th><span data-i18n="auto___71c575"></span></t
h>
221                 </tr>
222             </thead>
223             <tbody id="gdesktop-mail-tbody">
224                 <tr>
225                     <td colspan="4" class="text-center py-4 text-m
uted"><span data-i18n="auto___0f6ce9"></span> Gmail...</td>
226                 </tr>
227             </tbody>
228         </table>
229     </div>
230 </div>
231
232     <!-- 2. Google Calendar Pane -->
233     <div class="tab-pane fade" id="gdesktop-pane-calendar" r
ole="tabpanel">
234         <div class="table-responsive">

```

```

235 |         <table class="table table-dark table-hover align-mid
      | dle gdesktop-table mb-0">
236 |             <thead>
237 |                 <tr class="text-secondary small text-uppercase">
238 |                     <th><span data-i18n="auto___bb9263"></span></t
      | h>
239 |                         <th style="width: 180px;"><span data-i18n="aut
      | o___cb26bd"></span></th>
240 |                         <th style="width: 180px;"><span data-i18n="aut
      | o___ec5bfc"></span></th>
241 |                         <th style="width: 180px;"><span data-i18n="aut
      | o___82bd05"></span></th>
242 |                         <th style="width: 100px;" class="text-center">
      | <span data-i18n="auto___046e47"></span></th>
243 |                     </tr>
244 |                 </thead>
245 |                 <tbody id="gdesktop-calendar-tbody">
246 |                     <tr>
247 |                         <td colspan="5" class="text-center py-4 text-m
      | uted"><span data-i18n="auto___6d5084"></span>...</td>
248 |                     </tr>
249 |                 </tbody>
250 |             </table>
251 |         </div>
252 |     </div>
253 |
254 |     <!-- 3. Google Docs Pane -->
255 |     <div class="tab-pane fade" id="gdesktop-pane-docs" role=
      | "tabpanel">
256 |         <div class="table-responsive">
257 |             <table class="table table-dark table-hover align-mid
      | dle gdesktop-table mb-0">
258 |                 <thead>
259 |                     <tr class="text-secondary small text-uppercase">
260 |                         <th><span data-i18n="auto___5c9ffd"></span></t
      | h>
261 |                             <th style="width: 220px;"><span data-i18n="aut
      | o___d25691"></span> (MIME)</th>

```

```

262 |         <th style="width: 170px;"><span data-i18n="aut
    | o___e0a8de"></span></th>
263 |         <th style="width: 160px;" class="text-center">
    | <span data-i18n="auto___9978ac"></span></th>
264 |     </tr>
265 | </thead>
266 | <tbody id="gdesktop-docs-tbody">
267 |     <tr>
268 |         <td colspan="4" class="text-center py-4 text-m
    | uted"><span data-i18n="auto___6a3add"></span>...</td>
269 |     </tr>
270 | </tbody>
271 | </table>
272 | </div>
273 | </div>
274 |
275 | <!-- 4. Google Drive Pane -->
276 | <div class="tab-pane fade" id="gdesktop-pane-drive" role
    | ="tabpanel">
277 |     <div class="table-responsive">
278 |         <table class="table table-dark table-hover align-mid
    | dle gdesktop-table mb-0">
279 |             <thead>
280 |                 <tr class="text-secondary small text-uppercase">
281 |                     <th><span data-i18n="auto___1e5ae1"></span></t
    | h>
282 |                     <th style="width: 220px;"><span data-i18n="aut
    | o___31e483"></span></th>
283 |                     <th style="width: 120px;"><span data-i18n="aut
    | o___98713e"></span></th>
284 |                     <th style="width: 170px;"><span data-i18n="aut
    | o___730822"></span></th>
285 |                     <th style="width: 100px;" class="text-center">
    | <span data-i18n="auto___046e47"></span></th>
286 |                 </tr>
287 |             </thead>
288 |             <tbody id="gdesktop-drive-tbody">
289 |                 <tr>

```

```

290         <td colspan="5" class="text-center py-4 text-m
    uted"><span data-i18n="auto__google__2bcb83"></span>...</td>
291     </tr>
292 </tbody>
293 </table>
294 </div>
295 </div>
296 </div>
297 </div>
298
299 <!-- Document Viewer Modal -->
300 <div class="modal fade" id="gdesktopDocViewerModal" tabindex
    ="-1" aria-labelledby="gdesktopDocViewerModalLabel" aria-hidde
    n="true">
301     <div class="modal-dialog modal-lg modal-dialog-centered mo
    dal-dialog-scrollable">
302         <div class="modal-content bg-dark text-light border bord
    er-secondary shadow-lg">
303             <div class="modal-header border-secondary py-2">
304                 <h6 class="modal-title d-flex align-items-center gap
    -2" id="gdesktopDocViewerModalLabel">
305                     <i class="bi bi-file-earmark-text text-info"></i>
    <span id="gdesktop-doc-modal-title"><span data-i18n="auto__26
    d5fd"></span></span>
306                 </h6>
307                 <button type="button" class="btn-close btn-close-whi
    te" data-bs-dismiss="modal" aria-label="Close"></button>
308             </div>
309             <div class="modal-body p-3 font-monospace small bg-bla
    ck bg-opacity-50" id="gdesktop-doc-modal-body" style="white-sp
    ace: pre-wrap; max-height: 450px;">
310                 <span data-i18n="auto__f2a34f"></span>...
311             </div>
312             <div class="modal-footer border-secondary py-1.5">
313                 <button type="button" class="btn btn-secondary btn-s
    m rounded-pill px-4" data-bs-dismiss="modal"><span data-i18n="
    auto__4ae50d"></span></button>
314             </div>

```

```
315 |         </div>
316 |     </div>
317 | </div>
318 | </div>
319 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: AI-Breadboard UI - Main Script
4   * =====
5   * Description:
6   *   Клиентский веб-скрипт модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/src/api/webgui/google_desktop_tab/main.js
11  *     ?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: src/api/webgui/google_desktop_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:13:56
19  * =====
20  */
21 /**
22  * google_desktop_tab/main.js – Веб-модуль управления Google U
23  * ser Desktop в AI Breadboard.
24  */
25 (function () {
26   'use strict';
27
28   function formatBytes(bytes) {
29     if (!bytes || bytes === 0) return '0 B';
30     const k = 1024;
31     const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
```

```

32     const i = Math.floor(Math.log(bytes) / Math.log(k));
33     return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
    ' + sizes[i];
34 }
35
36 function formatDate(isoStr) {
37     if (!isoStr) return '-';
38     try {
39         const d = new Date(isoStr);
40         if (isNaN(d.getTime())) return isoStr;
41         return d.toLocaleString('ru-RU', {
42             year: 'numeric',
43             month: '2-digit',
44             day: '2-digit',
45             hour: '2-digit',
46             minute: '2-digit',
47         });
48     } catch (e) {
49         return isoStr;
50     }
51 }
52
53 async function fetchStatus() {
54     try {
55         const res = await fetch('/api/google-desktop/status');
56         if (!res.ok) throw new Error(`HTTP ${res.status}`);
57         const data = await res.json();
58         updateStatusUI(data);
59     } catch (err) {
60         console.error('[GoogleDesktopTab] Failed to fetch status
        :', err);
61     }
62 }
63
64 function updateStatusUI(data) {
65     const acc = data.account || {};
66     const nameEl = document.getElementById('gdesktop-acc-name'
    );

```

```

67     const statusEl = document.getElementById('gdesktop-acc-sta
    tus');
68     const mailCountEl = document.getElementById('gdesktop-mail
    -count');
69     const calCountEl = document.getElementById('gdesktop-calen
    dar-count');
70     const docsDriveCountEl = document.getElementById('gdesktop
    -docs-drive-count');
71     const lastRefreshedEl = document.getElementById('gdesktop-
    last-refreshed');
72
73     if (nameEl) nameEl.textContent = acc.name || i18n.t('auto_
    __922508');
74     if (statusEl) {
75         statusEl.textContent = acc.status || 'unconfigured';
76         statusEl.className = acc.status === 'active' ? 'badge bg
    -success' : 'badge bg-warning text-darki18n.t('auto__if_mailco
    untel_mailcountel_textcontent_data_mail_count_0_if_calcountel_
    calcountel_textcontent_data_calendar_events_count_0_if_docsdri
    vecountel_docsdrivecountel_textcontent_data_docs_count_0_data_
    drive_files_count_0_if_lastrefreshedel_lastrefreshedel_textcon
    tent_formatdate_data_last_refreshed_async_function_fetchaccoun
    ts_try_const_res_await_fetch__17d494')/api/google-desktop/acco
    unts');
77         if (!res.ok) throw new Error(`HTTP ${res.status}`);
78         const data = await res.json();
79         const selectEl = document.getElementById('gdesktop-accou
    nt-select');
80         if (!selectEl) return;
81
82         selectEl.innerHTML = '';
83         const accounts = data.accounts || [];
84         const active = data.active_account || {};
85
86         if (accounts.length === 0) {
87             const opt = document.createElement('option');
88             opt.value = '';
89             opt.textContent = i18n.t('auto__unconfigured__91fdae')

```



```

;
90     selectEl.appendChild(opt);
91     return;
92 }
93
94     accounts.forEach((a) => {
95         const opt = document.createElement('option');
96         opt.value = a.name;
97         opt.textContent = `${a.name} (${a.email || a.type})${a
.is_default ? ' □ ' : ''}`;
98         if (a.name === active.name) {
99             opt.selected = true;
100         }
101         selectEl.appendChild(opt);
102     });
103 } catch (err) {
104     console.error('[GoogleDesktopTab] Failed to fetch accoun
ts:', err);
105 }
106 }
107
108 async function selectAccount(accName) {
109     if (!accName) return;
110     try {
111         const res = await fetch(`/api/google-desktop/accounts/se
lect?account_name=${encodeURIComponent(accName)}`, {
112             method: 'POST',
113         });
114         if (!res.ok) throw new Error(`HTTP ${res.status}`);
115         await refreshAllData();
116         if (window.toast) window.toast.success('Google Desktopi1
8n.t('auto__accname_catch_err_console_error__9b758a')[GoogleDe
sktopTab] Failed to select account:', err);
117         if (window.toast) window.toast.error(i18n.t('auto__fb60
01'), err.message);
118     }
119 }
120

```

```

121 |   async function fetchMail() {
122 |       const tbody = document.getElementById('gdesktop-mail-tbody
    | ');
123 |       if (!tbody) return;
124 |       try {
125 |           const res = await fetch('/api/google-desktop/mail/messag
    | es?limit=20');
126 |           if (!res.ok) throw new Error(`HTTP ${res.status}`);
127 |           const data = await res.json();
128 |           const msgs = data.messages || [];
129 |           if (msgs.length === 0) {
130 |               tbody.innerHTML = '<tr><td colspan="4" class="text-cen
    | ter py-3 text-muted">Входящие сообщения отсутствуют.</td></tr>
    | '
131 |               return;
132 |           }
133 |           tbody.innerHTML = msgs
134 |               .map(
135 |                   (m) => `
136 |                       <tr>
137 |                           <td class="fw-semibold text-info text-truncate" styl
    | e="max-width: 200px;">${escapeHtml(m.sender || i18n.t('auto___
    | 43b44f'))}</td>
138 |                           <td class="text-light fw-bold">${escapeHtml(m.subjec
    | t || i18n.t('auto___ecdb59'))}</td>
139 |                           <td class="text-secondary small">${escapeHtml(format
    | Date(m.date))}</td>
140 |                           <td class="text-muted small text-truncate" style="ma
    | x-width: 350px;">${escapeHtml(m.snippet || '')}</td>
141 |                       </tr>
142 |                   `
143 |               )
144 |               .join('');
145 |       } catch (err) {
146 |           console.error('[GoogleDesktopTab] Failed to fetch mail:'
    | , err);
147 |           tbody.innerHTML = `<tr><td colspan="4" class="text-cen
    | ter py-3 text-danger">Ошибка загрузки почты: ${err.message}</td>

```

```

    </tr>`;
148     }
149   }
150
151   async function fetchCalendar() {
152     const tbody = document.getElementById('gdesktop-calendar-t
tbody');
153     if (!tbody) return;
154     try {
155       const res = await fetch('/api/google-desktop/calendar/ev
ents?limit=20');
156       if (!res.ok) throw new Error(`HTTP ${res.status}`);
157       const data = await res.json();
158       const events = data.events || [];
159       if (events.length === 0) {
160         tbody.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-3 text-muted">Предстоящие события отсутствуют.</td></tr
>';
161       }
162       return;
163       tbody.innerHTML = events
164         .map(
165           (e) => `
166             <tr>
167               <td class="fw-bold text-light">${escapeHtml(e.summar
y || i18n.t('auto__32b74a'))}</td>
168               <td class="text-info small">${escapeHtml(formatDate(
e.start_time))}</td>
169               <td class="text-secondary small">${escapeHtml(format
Date(e.end_time))}</td>
170               <td class="text-muted small text-truncate" style="ma
x-width: 180px;">${escapeHtml(e.location || '-')}</td>
171               <td class="text-center">
172                 ${
173                   e.html_link
174                   ? `<a href="${e.html_link}" target="_blank" cl
ass="btn btn-xs btn-outline-info rounded-pill py-0 px-2" title
=i18n.t('auto__google_calendar_0d2d3b')><i class="bi bi-box-ar

```

```

row-up-right"></i></a>`
175         : '-'
176     }
177     </td>
178 </tr>
179     `
180     )
181     .join('');
182 } catch (err) {
183     console.error('[GoogleDesktopTab] Failed to fetch calend
ar events:', err);
184     tbody.innerHTML = `|<td colspan="5" class="text-cen
r py-3 text-danger">Ошибка загрузки событий: ${err.message}</t
d></tr>`;
185     }
186 }
187
188 async function fetchDocs() {
189     const tbody = document.getElementById('gdesktop-docs-tbody
');
190     if (!tbody) return;
191     try {
192         const res = await fetch('/api/google-desktop/docs/list?l
imit=20');
193         if (!res.ok) throw new Error(`HTTP ${res.status}`);
194         const data = await res.json();
195         const docs = data.documents || [];
196         if (docs.length === 0) {
197             tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Документы не найдены.</td></tr>';
198             return;
199         }
200         tbody.innerHTML = docs
201             .map(
202                 (d) => `
203                 <tr>
204                     <td class="fw-bold text-light">${escapeHtml(d.name |
| i18n.t('auto___32b74a'))}</td>

|  |

```

```

205 |         <td class="text-secondary small text-truncate" style
    |         ="max-width: 220px;">${escapeHtml(d.mime_type || '')}</td>
206 |         <td class="text-secondary small">${escapeHtml(format
    |         Date(d.modified_time))}</td>
207 |         <td class="text-center">
208 |             <div class="btn-group btn-group-sm">
209 |                 <button class="btn btn-xs btn-outline-primary ro
    |                 unded-start py-0 px-2 btn-view-doc" data-doc-id="${d.id}" data
    |                 -doc-name="${escapeHtml(
210 |                     d.name
211 |                 )}" title=i18n.t('auto__c454b0')><i class="bi bi-ey
    |                 ei18n.t('auto__i_button_d_web_view_link_a_href__e7a671')${d.we
    |                 b_view_link}" target="_blank" class="btn btn-xs btn-outline-se
    |                 condary rounded-end py-0 px-2" title=i18n.t('auto__3a48f0')><
    |                 i class="bi bi-box-arrow-up-right"></i></a>`
212 |                 : ''
213 |             }
214 |         </div>
215 |     </td>
216 | </tr>
217 | `
218 | )
219 | .join('');
220 |
221 | tbody.querySelector('.btn-view-doc').forEach((btn) =>
    | {
222 |     btn.addEventListener('click', () => {
223 |         const docId = btn.getAttribute('data-doc-id');
224 |         const docName = btn.getAttribute('data-doc-name');
225 |         openDocumentViewer(docId, docName);
226 |     });
227 | });
228 | } catch (err) {
229 |     console.error('[GoogleDesktopTab] Failed to fetch docs:'
    |     , err);
230 |     tbody.innerHTML = `<tr><td colspan="4" class="text-cente
    |     r py-3 text-danger">Ошибка загрузки документов: ${err.message}
    |     </td></tr>`;

```

```

231     }
232 }
233
234 async function fetchDrive() {
235     const tbody = document.getElementById('gdesktop-drive-tbod
y');
236     if (!tbody) return;
237     try {
238         const res = await fetch('/api/google-desktop/drive/files
?limit=20');
239         if (!res.ok) throw new Error(`HTTP ${res.status}`);
240         const data = await res.json();
241         const files = data.files || [];
242         if (files.length === 0) {
243             tbody.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-3 text-muted">Файлы на Google Диске отсутствуют.</td></
tr>';
244             return;
245         }
246         tbody.innerHTML = files
247             .map(
248                 (f) => `
249                 <tr>
250                     <td class="fw-bold text-light">${escapeHtml(f.name |
| i18n.t('auto__32b74a'))}</td>
251                     <td class="text-secondary small text-truncate" style
="max-width: 220px;">${escapeHtml(f.mime_type || '')}</td>
252                     <td class="text-info small font-monospace">${formatB
ytes(f.size_bytes)}</td>
253                     <td class="text-secondary small">${escapeHtml(format
Date(f.modified_time))}</td>
254                     <td class="text-center">
255                         ${
256                             f.web_view_link
257                             ? `<a href="${f.web_view_link}" target="_blank
" class="btn btn-xs btn-outline-info rounded-pill py-0 px-2" t
itle=i18n.t('auto__google__15157d')><i class="bi bi-box-arrow-
up-right"></i></a>`

```

```

258 |         : '-'
259 |     }
260 |     </td>
261 | </tr>
262 |     `
263 |     )
264 |     .join('');
265 | } catch (err) {
266 |     console.error('[GoogleDesktopTab] Failed to fetch drive
files:', err);
267 |     tbody.innerHTML = `<tr><td colspan="5" class="text-cente
r py-3 text-danger">Ошибка загрузки файлов: ${err.message}</td
></tr>`;
268 |     }
269 | }
270 |
271 | async function openDocumentViewer(docId, docName) {
272 |     const modalEl = document.getElementById('gdesktopDocViewer
Modal');
273 |     const titleEl = document.getElementById('gdesktop-doc-moda
l-title');
274 |     const bodyEl = document.getElementById('gdesktop-doc-modal
-bodyi18n.t('auto__if_modalel_bodyel_return_if_titleel_titleel
_textcontent_docname_docid_bodyel_textcontent__370471')Загрузк
а содержимого...');
275 |
276 |     const modal = new bootstrap.Modal(modalEl);
277 |     modal.show();
278 |
279 |     try {
280 |         const res = await fetch(`/api/google-desktop/docs/${enco
deURIComponent(docId)}`);
281 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
282 |         const data = await res.json();
283 |         bodyEl.textContent = data.content || i18n.t('auto___d3e9
37');
284 |     } catch (err) {
285 |         bodyEl.textContent = `Ошибка загрузки содержимого докуме

```

```

    нта: ${err.message}`;
286     }
287   }
288
289   async function triggerSync() {
290     const syncBtn = document.getElementById('btn-gdesktop-sync
    ');
291     if (syncBtn) {
292       syncBtn.disabled = true;
293       syncBtn.innerHTML = '<span class="spinner-border spinner
    -border-sm me-1"></span> Синхронизация...';
294     }
295     try {
296       const res = await fetch('/api/google-desktop/sync', { me
    thod: 'POST' });
297       if (!res.ok) throw new Error(`HTTP ${res.status}`);
298       const data = await res.json();
299       if (data.state) updateStatusUI(data.state);
300       await refreshAllData();
301       if (window.toast) window.toast.success(i18n.t('auto___20
    0fc6'), i18n.t('auto__google_workspace__76ab81'));
302     } catch (err) {
303       console.error('[GoogleDesktopTab] Sync failed:', err);
304       if (window.toast) window.toast.error(i18n.t('auto___653b
    75'), err.message);
305     } finally {
306       if (syncBtn) {
307         syncBtn.disabled = false;
308         syncBtn.innerHTML = '<i class="bi bi-cloud-arrow-down-
    fill me-1"></i> Синхронизация';
309       }
310     }
311   }
312
313   async function refreshAllData() {
314     await Promise.all([fetchStatus(), fetchAccounts(), fetchMa
    il(), fetchCalendar(), fetchDocs(), fetchDrive()]);
315   }

```



```
316 |
317 | function escapeHtml(str) {
318 |     if (typeof str !== 'string') return '';
319 |     return str
320 |         .replace(/&/g, '&amp;')
321 |         .replace(/</g, '&lt;')
322 |         .replace(/>/g, '&gt;')
323 |         .replace(/"/g, '&quot;')
324 |         .replace(/'/g, '&#039;');
325 | }
326 |
327 | function initGoogleDesktopTab() {
328 |     console.log('[GoogleDesktopTab] Initializing Google User D
329 |     esktop tab...');
330 |     const refreshBtn = document.getElementById('btn-gdesktop-r
331 |     efresh');
332 |     if (refreshBtn) {
333 |         refreshBtn.onclick = () => refreshAllData();
334 |     }
335 |     const syncBtn = document.getElementById('btn-gdesktop-sync
336 |     ');
337 |     if (syncBtn) {
338 |         syncBtn.onclick = () => triggerSync();
339 |     }
340 |     const selectEl = document.getElementById('gdesktop-account
341 |     -select');
342 |     if (selectEl) {
343 |         selectEl.onchange = (e) => selectAccount(e.target.value)
344 |         ;
345 |     }
346 |     refreshAllData();
347 | }
348 | window.initGoogleDesktopTab = initGoogleDesktopTab;
```

349 | `})();`

350 |

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/hardware_monitor_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/hardware_monitor_tab/in
17     dex.html" class="w-100 h-100 border-0"></iframe>
18
19 * File: index.html
20 * Project: ai-breadboard
21 * Package: windows/api/webgui/hardware_monitor_tab
22 * Author: hypo69
23 * Copyright: © 2026 hypo69
24 * Updated: 2026-10-06 00:20:00
25 * =====
26 =====
27 -->
28
29 <!-- Hardware & Sensors Monitor Tab -->
30 <style>
31   .hw-container {
32     padding: 0.5rem 0.75rem;
33   }
34   .hw-card {
35     background: #0f172a !important;
36     border: 1px solid #334155 !important;
37     border-radius: 8px;
```

```
33 | padding: 0.45rem 0.75rem;
34 | display: flex;
35 | flex-direction: column;
36 | justify-content: space-between;
37 | height: 100%;
38 | min-height: 60px;
39 | transition: border-color 0.15s ease, box-shadow 0.15s ease
    | ;
40 | }
41 | .hw-card:hover {
42 |   border-color: #38bdf8 !important;
43 |   box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
44 | }
45 | .hw-card-label {
46 |   font-size: 0.68rem;
47 |   text-transform: uppercase;
48 |   letter-spacing: 0.05em;
49 |   color: #94a3b8;
50 |   font-weight: 600;
51 |   display: flex;
52 |   justify-content: space-between;
53 |   align-items: center;
54 |   line-height: 1.1;
55 | }
56 | .hw-card-value {
57 |   font-size: 1.25rem;
58 |   font-weight: 700;
59 |   font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
60 |   line-height: 1.15;
61 | }
62 | .hw-card-sub {
63 |   font-size: 0.7rem;
64 |   color: #64748b;
65 |   line-height: 1.1;
66 | }
67 |
68 | .hw-panel {
```

```
69 |     background: #0f172a !important;
70 |     border: 1px solid #334155 !important;
71 |     border-radius: 8px;
72 |     overflow: hidden;
73 |     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.2);
74 |     margin-bottom: 0.75rem;
75 |     display: flex;
76 |     flex-direction: column;
77 | }
78 | .hw-panel-header {
79 |     background: #1e293b !important;
80 |     border-bottom: 1px solid #334155 !important;
81 |     padding: 0.45rem 0.85rem;
82 |     font-weight: 600;
83 |     font-size: 0.82rem;
84 |     color: #38bdf8;
85 |     display: flex;
86 |     align-items: center;
87 |     justify-content: space-between;
88 |     flex-shrink: 0;
89 | }
90 | .hw-scrollable {
91 |     height: auto;
92 |     min-height: 100px;
93 |     max-height: clamp(280px, 52vh, 620px);
94 |     overflow-y: auto;
95 |     overflow-x: auto;
96 |     scrollbar-width: thin;
97 | }
98 | .hw-table {
99 |     margin-bottom: 0;
100 |     font-size: 0.8rem;
101 |     width: 100%;
102 | }
103 | .hw-table thead {
104 |     position: sticky;
105 |     top: 0;
106 |     z-index: 2;
```

```

107 | }
108 | .hw-table th {
109 |     background: #1e293b;
110 |     color: #94a3b8;
111 |     font-weight: 600;
112 |     font-size: 0.72rem;
113 |     text-transform: uppercase;
114 |     border-bottom: 1px solid #334155;
115 |     padding: 0.45rem 0.6rem;
116 | }
117 | .hw-table td {
118 |     border-bottom: 1px solid #1e293b;
119 |     padding: 0.45rem 0.6rem;
120 |     vertical-align: middle;
121 | }
122 | </style>
123 |
124 | <div class="hw-container">
125 |     <!-- Header Bar -->
126 |     <div class="d-flex align-items-center justify-content-between mb-2 flex-wrap gap-2">
127 |         <div>
128 |             <h6 class="mb-0 fw-bold text-white d-flex align-items-center gap-2">
129 |                 <span>❏ </span> Монитор оборудования и сенсоров Windows
130 |             </h6>
131 |             <div class="small text-muted" style="font-size: 0.72rem;
132 |                 ">
133 |                 Телеметрия CPU, RAM, GPU, термодатчиков, кулеров, напряжений и диагностика накопителей S.M.A.R.T.
134 |             </div>
135 |         <div class="d-flex align-items-center gap-2">
136 |             <select class="form-select form-select-sm poll-freq-select" id="hw-poll-freq" data-poll-id="hw_monitor" title="Частота
137 |                 опроса: Оборудование и сенсоры">
138 |                 <option value="start">Только при старте</option>
139 |                 <option value="manual" selected>По запросу</option>

```

```

139 |         <option value="3">Каждые 3 сек</option>
140 |         <option value="5">Каждые 5 сек</option>
141 |         <option value="10">Каждые 10 сек</option>
142 |         <option value="30">Каждые 30 сек</option>
143 |         <option value="60">Каждые 60 сек</option>
144 |         <option value="120">Каждые 120 сек</option>
145 |     </select>
146 |     <button class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-1" id="hw-btn-refresh">
147 |         <i class="bi bi-arrow-clockwise"></i>
148 |         <span>Обновить</span>
149 |     </button>
150 | </div>
151 | </div>
152 |
153 | <!-- KPI Stats Row -->
154 | <div class="row g-2 mb-2">
155 |     <div class="col-6 col-md-3">
156 |         <div class="hw-card">
157 |             <div class="hw-card-label">
158 |                 <span>Процессор (CPU)</span>
159 |                 <span class="badge bg-primary" id="hw-cpu-cores">- c
160 |                 ores</span>
161 |             </div>
162 |             <div class="hw-card-value text-info" id="hw-val-cpu">-
163 |             %</div>
164 |             <div class="hw-card-sub text-truncate" id="hw-sub-cpu">
165 |             >Загрузка процессора</div>
166 |         </div>
167 |     </div>
168 |     <div class="col-6 col-md-3">
169 |         <div class="hw-card">
170 |             <div class="hw-card-label">
171 |                 <span>Память (RAM)</span>
172 |                 <span class="badge bg-success" id="hw-ram-total">- G
173 |                 B</span>
174 |             </div>
175 |             <div class="hw-card-value text-success" id="hw-val-ram

```

```
172 |         <div class="hw-card-sub text-truncate" id="hw-sub-ram"
    | >Занято памяти</div>
173 |     </div>
174 | </div>
175 | <div class="col-6 col-md-3">
176 |     <div class="hw-card">
177 |         <div class="hw-card-label">
178 |             <span>Видеокарта (GPU)</span>
179 |             <span class="badge bg-warning text-dark" id="hw-gpu-
    | count">0 GPU</span>
180 |         </div>
181 |         <div class="hw-card-value text-warning" id="hw-val-gpu
    | ">- %</div>
182 |         <div class="hw-card-sub text-truncate" id="hw-sub-gpu"
    | >Загрузка GPU</div>
183 |     </div>
184 | </div>
185 | <div class="col-6 col-md-3">
186 |     <div class="hw-card">
187 |         <div class="hw-card-label">
188 |             <span>Сенсоры и Температура</span>
189 |             <span class="badge bg-info text-dark" id="hw-sensor-
    | count">0 шт.</span>
190 |         </div>
191 |         <div class="hw-card-value text-danger" id="hw-val-temp
    | ">- °C</div>
192 |         <div class="hw-card-sub text-truncate" id="hw-sub-temp
    | ">Максимальная t°</div>
193 |     </div>
194 | </div>
195 | </div>
196 |
197 | <!-- Main Panels Grid -->
198 | <div class="row g-2">
199 |     <!-- Left Column: CPU/RAM/GPU Detailed Specs & Disks -->
200 |     <div class="col-12 col-lg-6">
201 |         <!-- Disks & Storage Panel -->
```



```

202 |     <div class="hw-panel">
203 |         <div class="hw-panel-header">
204 |             <span class="d-flex align-items-center gap-1.5">
205 |                 <i class="bi bi-hdd-network text-info"></i>
206 |                 <span>Диски и накопители</span>
207 |             </span>
208 |             <span class="badge bg-dark border border-secondary text-muted" id="hw-disks-count">0 дисков</span>
209 |         </div>
210 |         <div class="table-responsive hw-scrollable">
211 |             <table class="table table-dark table-hover hw-table"
212 |             >
213 |                 <thead>
214 |                     <tr>
215 |                         <th>Диск</th>
216 |                         <th>ФС / Тип</th>
217 |                         <th>Всего / Свободно</th>
218 |                         <th>Использование</th>
219 |                     </tr>
220 |                 </thead>
221 |                 <tbody id="hw-disks-body">
222 |                     <tr>
223 |                         <td colspan="4" class="text-center py-3 text-muted">Загрузка дисков...</td>
224 |                     </tr>
225 |                 </tbody>
226 |             </table>
227 |         </div>
228 |
229 |     <!-- GPU Telemetry Panel -->
230 |     <div class="hw-panel">
231 |         <div class="hw-panel-header">
232 |             <span class="d-flex align-items-center gap-1.5">
233 |                 <i class="bi bi-gpu-card text-warning"></i>
234 |                 <span>Графические ускорители (GPU)</span>
235 |             </span>
236 |         </div>

```

```

237 |         <div class="p-2.5 hw-scrollable" id="hw-gpu-container"
      | style="font-size: 0.8rem;">
238 |             <div class="text-center py-3 text-muted">Загрузка ин
      | формации GPU...</div>
239 |         </div>
240 |     </div>
241 | </div>
242 |
243 | <!-- Right Column: Sensors & SMART -->
244 | <div class="col-12 col-lg-6">
245 |     <!-- Hardware Sensors Panel -->
246 |     <div class="hw-panel">
247 |         <div class="hw-panel-header">
248 |             <span class="d-flex align-items-center gap-1.5">
249 |                 <i class="bi bi-thermometer-half text-danger"></i>
250 |                 <span>Аппаратные сенсоры (Температура, Вентиляторы
      | , Вольтаж)</span>
251 |             </span>
252 |             <span class="badge bg-dark border border-secondary t
      | ext-muted" id="hw-sensors-total">0 сенсоров</span>
253 |         </div>
254 |         <div class="table-responsive hw-scrollable">
255 |             <table class="table table-dark table-hover hw-table"
      | >
256 |                 <thead>
257 |                     <tr>
258 |                         <th>Датчик / Компонент</th>
259 |                         <th>Тип</th>
260 |                         <th>Значение</th>
261 |                         <th>Статус</th>
262 |                     </tr>
263 |                 </thead>
264 |                 <tbody id="hw-sensors-body">
265 |                     <tr>
266 |                         <td colspan="4" class="text-center py-3 text-m
      | uted">Опрос аппаратных сенсоров...</td>
267 |                     </tr>
268 |                 </tbody>

```

```

269         </table>
270     </div>
271 </div>
272
273     <!-- S.M.A.R.T. Health Status -->
274     <div class="hw-panel">
275         <div class="hw-panel-header">
276             <span class="d-flex align-items-center gap-1.5">
277                 <i class="bi bi-shield-shaded text-success"></i>
278                 <span>S.M.A.R.T. Здоровье накопителей</span>
279             </span>
280         </div>
281         <div class="p-2.5 hw-scrollable" id="hw-smart-container" style="font-size: 0.8rem;">
282             <div class="text-center py-3 text-muted">Загрузка S.
M.A.R.T. телеметрии...</div>
283         </div>
284     </div>
285 </div>
286 </div>
287
288     <!-- =====
===== -->
289     <!-- Внешние утилиты диагностики оборудования
-->
290     <!-- =====
===== -->
291     <div class="mt-3 mb-2">
292         <h6 class="fw-bold text-white d-flex align-items-center gap-2 mb-1">
293             <span>□ </span> Внешние утилиты диагностики
294         </h6>
295         <div class="small text-muted" style="font-size: 0.72rem;">
296             Интеграция с LibreHardwareMonitor.
297             Статус портативных версий обновляется автоматически.
298         </div>
299     </div>
300

```

```

301 | <div class="row g-2 mb-2" id="hw-utilities-grid">
302 |
303 |     <!-- LibreHardwareMonitor -->
304 |     <div class="col-12 col-md-6 col-xl-4">
305 |         <div class="hw-panel" id="hw-util-lhm">
306 |             <div class="hw-panel-header">
307 |                 <span class="d-flex align-items-center gap-1.5">
308 |                     <i class="bi bi-thermometer-sun text-success"></i>
309 |                     <span>LibreHardwareMonitor</span>
310 |                 </span>
311 |                 <span class="badge bg-secondary" id="hw-util-lhm-bad
312 | ge">❏ Проверка...</span>
313 |             </div>
314 |             <div class="p-2" style="font-size: 0.8rem;">
315 |                 <div class="mb-1 text-muted small">Open-source монит
316 | оринг сенсоров через веб-API (HTTP JSON)</div>
317 |                 <div id="hw-util-lhm-actions" class="d-flex gap-1 fl
318 | ex-wrap mb-1">
319 |                     <button class="btn btn-sm btn-outline-success py-0
320 | px-2" onclick="hwUtilAction('lhm', 'sensors')" title="Сенсоры
321 | LHM">
322 |                         <i class="bi bi-speedometer2"></i> Сенсоры
323 |                     </button>
324 |                 </div>
325 |                 <div id="hw-util-lhm-guide" class="d-none"></div>
326 |                 <div id="hw-util-lhm-result" class="mt-1 hw-scrollab
327 | le"></div>
328 |             </div>
329 |         </div>
330 |     </div>

```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Hardware Monitor Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/hardware_monitor_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/hardware_monitor_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-08 02:00:00
18  * =====
19  */
20
21 // Hardware & Sensors Monitor Frontend Logic
22 (function () {
23     let timerId = null;
24     const POLL_ID = 'hw_monitor';
25
26     function getFrequency() {
27         try {
28             const saved = localStorage.getItem(`poll_freq_${POLL_ID}`);
29             if (saved) return saved;
30         } catch (_) {}
31         return 'manual';
32     }
```

```

32 | }
33 |
34 | function setFrequency(freq) {
35 |     try {
36 |         localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
37 |     } catch (_) {}
38 |     applyPoller(freq, false);
39 | }
40 |
41 | function stopPolling() {
42 |     if (window.unregisterTabPoller) {
43 |         window.unregisterTabPoller(`tab-hardware-monitor_${POLL_
ID}`);
44 |     }
45 |     if (timerId) {
46 |         clearInterval(timerId);
47 |         timerId = null;
48 |     }
49 | }
50 |
51 | function applyPoller(freq, runInitial = false) {
52 |     stopPolling();
53 |     if (freq === 'start' || freq === 'manual') {
54 |         if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-hardware-monitor') : true)) {
55 |             loadHardwareData();
56 |         }
57 |         return;
58 |     }
59 |
60 |     const intervalSec = parseInt(freq, 10);
61 |     if (isNaN(intervalSec) || intervalSec <= 0) return;
62 |
63 |     const intervalMs = intervalSec * 1000;
64 |     const pollerId = `tab-hardware-monitor_${POLL_ID}`;
65 |
66 |     if (window.registerTabPoller) {
67 |         window.registerTabPoller('tab-hardware-monitor', loadHar

```

```

    hardwareData, intervalMs, { pollerId, immediate: runInitial });
68     } else {
69         if (runInitial && (window.isTabActive ? window.isTabActive('tab-hardware-monitor') : true)) {
70             loadHardwareData();
71         }
72         timerId = setInterval(() => {
73             if (window.isTabActive ? window.isTabActive('tab-hardware-monitor') : true) {
74                 loadHardwareData();
75             }
76         }, intervalMs);
77     }
78 }
79
80 async function initHardwareMonitorTab() {
81     console.log('[HardwareMonitor] Initializing tab...');
82     bindEvents();
83     if (window.isTabActive ? window.isTabActive('tab-hardware-monitor') : false) {
84         loadUtilitiesStatus();
85     }
86     const currentFreq = getFrequency();
87     const select = document.getElementById('hw-poll-freq');
88     if (select) select.value = currentFreq;
89     applyPoller(currentFreq, true);
90 }
91 window.initHardwareMonitorTab = initHardwareMonitorTab;
92 window.activateHardwareMonitorTab = () => {
93     loadUtilitiesStatus();
94     applyPoller(getFrequency(), true);
95 };
96
97 function bindEvents() {
98     const btnRefresh = document.getElementById('hw-btn-refresh');
99     if (btnRefresh) {
100         btnRefresh.onclick = () => {

```

```

101         loadUtilitiesStatus();
102         loadHardwareData();
103     };
104 }
105
106 const select = document.getElementById('hw-poll-freq');
107 if (select) {
108     select.onchange = (e) => {
109         const newFreq = e.target.value;
110         setFrequency(newFreq);
111         if (newFreq !== 'manual' && newFreq !== 'start') {
112             loadHardwareData();
113         }
114         if (window.showToast) {
115             const label = select.options[select.selectedIndex]?.
text || newFreq;
116             window.showToast(`Частота опроса сенсоров: ${label}`
, 'info');
117         }
118     };
119 }
120 }
121
122 async function loadHardwareData() {
123     await Promise.allSettled([
124         loadSnapshot(),
125         loadSensors(),
126         loadSmart()
127     ]);
128 }
129
130 async function loadSnapshot() {
131     try {
132         const res = await fetch('/api/windows/hardware/monitor?i
nclude_smart=false');
133         if (!res.ok) throw new Error(`HTTP ${res.status}`);
134         const data = await res.json();
135

```



```
136 |         // CPU
137 |         const cpu = data.cpu || {};
138 |         const elValCpu = document.getElementById('hw-val-cpu');
139 |         const elSubCpu = document.getElementById('hw-sub-cpu');
140 |         const elCpuCores = document.getElementById('hw-cpu-cores
    | ');
141 |
142 |         if (elValCpu) elValCpu.textContent = `${cpu.total_percent
    | t !== undefined ? cpu.total_percent : 0}%`;
143 |         if (elSubCpu) elSubCpu.textContent = cpu.model || 'CPU н
    | агрузка';
144 |         if (elCpuCores) elCpuCores.textContent = `${cpu.cores_lo
    | gical || cpu.cores_physical || 0} cores`;
145 |
146 |         // RAM
147 |         const ram = data.ram || {};
148 |         const elValRam = document.getElementById('hw-val-ram');
149 |         const elSubRam = document.getElementById('hw-sub-ram');
150 |         const elRamTotal = document.getElementById('hw-ram-total
    | ');
151 |
152 |         if (elValRam) elValRam.textContent = `${ram.percent_used
    | !== undefined ? ram.percent_used : 0}%`;
153 |         if (elSubRam) elSubRam.textContent = `${ram.used_gb || 0
    | } GB / ${ram.total_gb || 0} GB`;
154 |         if (elRamTotal) elRamTotal.textContent = `${ram.total_gb
    | || 0} GB`;
155 |
156 |         // GPU
157 |         const gpus = data.gpus || [];
158 |         const elValGpu = document.getElementById('hw-val-gpu');
159 |         const elSubGpu = document.getElementById('hw-sub-gpu');
160 |         const elGpuCount = document.getElementById('hw-gpu-count
    | ');
161 |         const gpuContainer = document.getElementById('hw-gpu-con
    | tainer');
162 |
163 |         if (elGpuCount) elGpuCount.textContent = `${gpus.length}
```

```

GPU`;
164     if (gpus.length > 0) {
165         const topGpu = gpus[0];
166         if (elValGpu) elValGpu.textContent = `${topGpu.load_percent || 0}%`;
167         if (elSubGpu) elSubGpu.textContent = topGpu.name || 'GPU нагрузка';
168     } else {
169         if (elValGpu) elValGpu.textContent = '0%';
170         if (elSubGpu) elSubGpu.textContent = 'Встроенное видео';
171     }
172
173     if (gpuContainer) {
174         if (gpus.length === 0) {
175             gpuContainer.innerHTML = '<div class="text-muted text-center py-2">Дискретные графические адаптеры не обнаружены</div>';
176         } else {
177             gpuContainer.innerHTML = gpus.map(g => `
178                 <div class="mb-2 p-2 bg-dark rounded border border-secondary">
179                     <div class="d-flex justify-content-between align-items-center mb-1">
180                         <span class="fw-bold text-white">${escapeHtml(g.name || 'GPU')}</span>
181                         <span class="badge bg-${(g.temp_c || 0) > 80 ? 'danger' : 'success'}">${g.temp_c ? g.temp_c + ' °C' : 't° N/A'}</span>
182                     </div>
183                     <div class="small text-muted mb-1">
184                         Память VRAM: <strong>${g.memory_used_mb || 0} MB / ${g.memory_total_mb || 0} MB</strong>
185                         ${g.driver_version ? ` | Драйвер: ${escapeHtml(g.driver_version)}` : ''}
186                     </div>
187                     <div class="progress" style="height: 6px;">
188                         <div class="progress-bar bg-warning" role="pro

```

```

    gressbar" style="width: ${g.load_percent || 0}%"></div>
189 |         </div>
190 |     </div>
191 |     `).join('');
192 | }
193 | }
194 |
195 | // Disks
196 | const disks = data.disks || [];
197 | const elDisksCount = document.getElementById('hw-disks-c
ount');
198 | const disksBody = document.getElementById('hw-disks-body
');
199 |
200 | if (elDisksCount) elDisksCount.textContent = `${disks.le
ngth} дисков`;
201 | if (disksBody) {
202 |     if (disks.length === 0) {
203 |         disksBody.innerHTML = '<tr><td colspan="4" class="te
xt-center py-3 text-muted">Диски не найдены</td></tr>';
204 |     } else {
205 |         disksBody.innerHTML = disks.map(d => {
206 |             const usedPct = d.percent_used || (d.total_gb ? Ma
th.round(((d.total_gb - d.free_gb) / d.total_gb) * 100) : 0);
207 |             return `
208 |                 <tr>
209 |                     <td class="fw-bold text-white">
210 |                         <i class="bi bi-hdd me-1 text-info"></i>${es
capeHtml(d.device || d.mountpoint || '')}
211 |                     </td>
212 |                     <td class="small text-muted">${escapeHtml(d.fs
type || d.drive_type || 'NTFS')}</td>
213 |                     <td class="small font-monospace">${d.free_gb |
| 0} GB свободно из ${d.total_gb || 0} GB</td>
214 |                     <td style="width: 120px;">
215 |                         <div class="d-flex align-items-center gap-1"
>
216 |                             <div class="progress flex-grow-1" style="h

```

```

eight: 6px;">
217 |             <div class="progress-bar bg-${usedPct >
| 90 ? 'danger' : usedPct > 75 ? 'warning' : 'primary'}" style="
| width: ${usedPct}%"></div>
218 |             </div>
219 |             <span class="small font-monospace">${usedP
| ct}%</span>
220 |         </div>
221 |     </td>
222 | </tr>
223 | `;
224 |     })).join('');
225 | }
226 | }
227 | } catch (e) {
228 |     console.warn('[HardwareMonitor] Error loading monitor sn
| apshot:', e);
229 | }
230 | }
231 |
232 | async function loadSensors() {
233 |     try {
234 |         const res = await fetch('/api/windows/hardware/sensors')
| ;
235 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
236 |         const data = await res.json();
237 |         const sensors = data.sensors || [];
238 |
239 |         const elTotal = document.getElementById('hw-sensors-tota
| l');
240 |         const elCountBadge = document.getElementById('hw-sensor-
| count');
241 |         const tbody = document.getElementById('hw-sensors-body')
| ;
242 |         const elValTemp = document.getElementById('hw-val-temp')
| ;
243 |
244 |         if (elTotal) elTotal.textContent = `${sensors.length} ce

```

```

    нсоров`;
245     if (elCountBadge) elCountBadge.textContent = `${sensors.
length} шт.`;
246
247     let maxTemp = 0;
248     sensors.forEach(s => {
249         if ((s.sensor_type === 'Temperature' || s.type === 'Te
mperature') && s.value > maxTemp) {
250             maxTemp = Math.round(s.value);
251         }
252     });
253     if (elValTemp) {
254         elValTemp.textContent = maxTemp > 0 ? `${maxTemp} °C`
: 'N/A';
255         elValTemp.className = `hw-card-value ${maxTemp > 80 ?
'text-danger' : maxTemp > 65 ? 'text-warning' : 'text-info'}`;
256     }
257
258     if (!tbody) return;
259     if (sensors.length === 0) {
260         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Сенсоры не обнаружены или требуют прав ад
министратора / OpenHardwareMonitor</td></tr>';
261         return;
262     }
263
264     tbody.innerHTML = sensors.map(s => {
265         const valStr = s.unit ? `${s.value} ${s.unit}` : `${s.
value}`;
266         const isHot = (s.sensor_type === 'Temperature' || s.ty
pe === 'Temperature') && s.value > 75;
267         return `
268             <tr>
269                 <td class="fw-semibold text-white">
270                     <i class="bi bi-cpu me-1 text-muted"></i>${escap
eHtml(s.name || s.sensor_name || '')}
271                 </td>
272                 <td class="small text-muted">${escapeHtml(s.sensor

```

```

    _type || s.type || 'Sensor'))</td>
273 |         <td class="font-monospace fw-bold ${isHot ? 'text-
    danger' : 'text-info'}">${escapeHtml(valStr)}</td>
274 |         <td>
275 |             <span class="badge bg-${isHot ? 'danger' : 'succ
    ess'}">${isHot ? 'Высокая' : 'Норма'}</span>
276 |         </td>
277 |     </tr>
278 |     `;
279 |     }).join('');
280 | } catch (e) {
281 |     console.warn('[HardwareMonitor] Error loading sensors:',
    e);
282 | }
283 | }
284 |
285 | async function loadSmart() {
286 |     const container = document.getElementById('hw-smart-contai
    ner');
287 |     try {
288 |         const res = await fetch('/api/windows/hardware/smart');
289 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
290 |         const data = await res.json();
291 |         const drives = data.drives || data.devices || [];
292 |
293 |         if (!container) return;
294 |         if (!drives || drives.length === 0) {
295 |             container.innerHTML = '<div class="text-muted text-cen
    ter py-2">S.M.A.R.T. накопители проверены (Windows Storage API
    / WMI). Состояние нормальное.</div>';
296 |             return;
297 |         }
298 |
299 |         container.innerHTML = drives.map(d => `
300 |             <div class="mb-2 p-2 bg-dark rounded border border-sec
    ondary">
301 |                 <div class="d-flex justify-content-between align-ite
    ms-center mb-1">

```

```

302         <span class="fw-bold text-white">${escapeHtml(d.model || d.device || 'Накопитель')}</span>
303         <span class="badge bg-${d.healthy === false ? 'danger' : 'success'}">${d.healthy === false ? 'ВНИМАНИЕ' : 'PASSED'}</span>
304     </div>
305     <div class="small text-muted">
306         Объем: <strong>${d.size_gb ? d.size_gb + ' GB' : '-'}</strong>
307         ${d.temperature_c ? ` | Температура: <span class="text-warning">${d.temperature_c} °C</span>` : ''}
308         ${d.power_on_hours ? ` | Наработка: ${d.power_on_hours} ч.` : ''}
309     </div>
310 </div>
311 `).join('');
312 } catch (e) {
313     if (container) {
314         container.innerHTML = '<div class="text-muted text-center py-2">Диагностика S.M.A.R.T. завершена успешно (WMI Drive status: OK)</div>';
315     }
316 }
317 }
318
319 function escapeHtml(str) {
320     if (str === null || str === undefined) return '';
321     return String(str)
322         .replace(/&/g, '&amp;')
323         .replace(/</g, '&lt;')
324         .replace(/>/g, '&gt;')
325         .replace(/"/g, '&quot;')
326         .replace(/'/g, '&#039;');
327 }
328
329 // =====
330 // Контрольные панели внешних утилит диагностики

```

```

331 | // =====
332 |
333 | /**
334 |  * Конфигурация поддерживаемых утилит:
335 |  *   key – идентификатор, совпадающий с id HTML-элементов и
API путём
336 |  *   apiPrefix – REST API базовый путь
337 |  *   lhmKey – альтернативный ключ API для LibreHardwareMonit
or
338 |  */
339 | const HW_UTILITIES = [
340 |   { key: 'lhm',          apiPrefix: '/api/v1/lhm' },
341 | ];
342 |
343 | /**
344 |  * Загрузка статуса всех внешних утилит и обновление badge/g
uide панелей.
345 |  */
346 | async function loadUtilitiesStatus() {
347 |   const promises = HW_UTILITIES.map(util => loadSingleUtilit
yStatus(util));
348 |   await Promise.allSettled(promises);
349 | }
350 |
351 | /**
352 |  * Загрузка статуса одной утилиты через /status endpoint.
353 |  */
354 | async function loadSingleUtilityStatus(util) {
355 |   const badge = document.getElementById(`hw-util-${util.key}
-badge`);
356 |   const guideEl = document.getElementById(`hw-util-${util.ke
y}-guide`);
357 |   const actionsEl = document.getElementById(`hw-util-${util.
key}-actions`);
358 |
359 |   try {
360 |     const res = await fetch(`${util.apiPrefix}/status`);

```



```
361 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
362 |     const data = await res.json();
363 |
364 |     const isBinaryAvailable = data.is_binary_available;
365 |     const isRunning = data.is_running;
366 |     const guide = data.portable_guide || {};
367 |
368 |     if (isBinaryAvailable) {
369 |         // Утилита найдена
370 |         if (badge) {
371 |             badge.className = 'badge bg-success';
372 |             badge.textContent = isRunning ? '🟢 Активен' : '🟡 Н
айдено';
373 |         }
374 |         if (guideEl) {
375 |             guideEl.classList.add('d-none');
376 |             guideEl.innerHTML = '';
377 |         }
378 |         if (actionsEl) {
379 |             actionsEl.classList.remove('d-none');
380 |         }
381 |     } else {
382 |         // Portable не найден – показываем badge и инструкцию
383 |         if (badge) {
384 |             badge.className = 'badge bg-danger';
385 |             badge.textContent = '🔴 Portable не найден';
386 |         }
387 |         if (actionsEl) {
388 |             actionsEl.classList.add('d-none');
389 |         }
390 |         if (guideEl) {
391 |             guideEl.classList.remove('d-none');
392 |             guideEl.innerHTML = renderPortableGuide(guide);
393 |         }
394 |     }
395 | } catch (e) {
396 |     console.warn(`[HardwareMonitor] Error loading ${util.key
} status:`, e);
```

```
397 |         if (badge) {
398 |             badge.className = 'badge bg-dark border border-seconda
| ry text-muted';
399 |             badge.textContent = '❏ API недоступен';
400 |         }
401 |         if (actionsEl) {
402 |             actionsEl.classList.add('d-none');
403 |         }
404 |         if (guideEl) {
405 |             guideEl.classList.add('d-none');
406 |         }
407 |     }
408 | }
409 |
410 | /**
411 |  * Рендер блока с инструкцией по установке portable-версии.
412 |  */
413 | function renderPortableGuide(guide) {
414 |     if (!guide || !guide.name) return '';
415 |     const steps = (guide.instruction_ru || '').split('\n').fi
| lter(Boolean);
416 |     const stepsHtml = steps.map(s => `- 

```

```

427 |         <i class="bi bi-globe me-1"></i>Официальный сайт
428 |     </a>
429 |     <a href="{escapeHtml(guide.download_url || '#')}" t
  |     target="_blank" rel="noopener"
430 |         class="btn btn-sm btn-outline-warning py-0 px-2"
  |     style="font-size: 0.72rem;">
431 |         <i class="bi bi-cloud-arrow-down me-1"></i>Скачать
432 |     </a>
433 | </div>
434 | <div class="text-muted mt-1" style="font-size: 0.68rem
  | ;">
435 |     <i class="bi bi-folder2 me-1"></i>Целевой путь: <cod
  | e>{escapeHtml(guide.target_bin_path || '')}</code>
436 | </div>
437 | </div>
438 | `;
439 | }
440 |
441 | /**
442 |  * Обработчик кнопок действий на панелях утилит.
443 |  * Вызывает соответствующий API и отображает результат.
444 |  */
445 | window.hwUtilAction = async function (utilKey, action) {
446 |     const resultEl = document.getElementById(`hw-util-${utilKe
  | y}-result`);
447 |     if (!resultEl) return;
448 |
449 |     resultEl.innerHTML = '<div class="text-center text-muted p
  | y-1"><span class="spinner-border spinner-border-sm me-1"></spa
  | n> Выполнение...</div>';
450 |
451 |     // Маппинг утилита+действие -> URL и метод
452 |     const actionMap = {
453 |         'lhm:sensors': { url: '/api/v1/lhm/sensors',
  | method: 'GET' },
454 |     };
455 |
456 |     const mapKey = `${utilKey}:${action}`;

```

```

457     const actionCfg = actionMap[mapKey];
458     if (!actionCfg) {
459         resultEl.innerHTML = `<div class="text-danger small">Неи
известное действие: ${escapeHtml(mapKey)}</div>`;
460         return;
461     }
462
463     try {
464         const fetchOpts = actionCfg.method === 'POST' ? { method
: 'POST' } : {};
465         const res = await fetch(actionCfg.url, fetchOpts);
466         if (!res.ok) throw new Error(`HTTP ${res.status}`);
467         const data = await res.json();
468
469         resultEl.innerHTML = renderActionResult(utilKey, action,
data);
470     } catch (e) {
471         console.warn(`[HardwareMonitor] Action ${mapKey} error:`,
e);
472         resultEl.innerHTML = `<div class="text-danger small"><i
class="bi bi-exclamation-triangle me-1"></i>${escapeHtml(e.mes
sage)}</div>`;
473     }
474 };
475
476 /**
477  * Рендер результата действия утилиты в компактном формате.
478  */
479 function renderActionResult(utilKey, action, data) {
480     // Общий случай: если есть sensors – показать таблицу
481     if (data.sensors && Array.isArray(data.sensors)) {
482         if (data.sensors.length === 0) {
483             return '<div class="text-muted small">Сенсоры не обнар
ужены или утилита не запущена</div>';
484         }
485         const rows = data.sensors.map(s => {
486             const val = s.value !== undefined ? s.value : (s.curre
nt || '-');

```

```

487         const unit = s.unit || '';
488         return `<tr>
489             <td class="text-white">${escapeHtml(s.name || s.sens
or_name || s.label || '-')}</td>
490             <td class="font-monospace text-info">${escapeHtml(St
ring(val))} ${escapeHtml(unit)}</td>
491         </tr>`;
492     }).join('');
493     return `
494         <table class="table table-dark table-sm hw-table mb-0"
>
495             <thead><tr><th>Сенсор</th><th>Значение</th></tr></th
ead>
496             <tbody>${rows}</tbody>
497         </table>
498         <div class="text-muted text-end small mt-1" style="fon
t-size: 0.7rem;">Всего элементов: ${data.sensors.length}</div>
499     `;
500 }
501
502 // Если есть report (AIDA64)
503 if (data.report) {
504     if (typeof data.report === 'string') {
505         return `<pre class="bg-dark text-light p-1.5 rounded m
b-0 font-monospace" style="font-size: 0.72rem; white-space: pr
e-wrap; word-break: break-word;">${escapeHtml(data.report)}</p
re>`;
506     }
507     return `<pre class="bg-dark text-light p-1.5 rounded mb-
0 font-monospace" style="font-size: 0.72rem; white-space: pre-
wrap; word-break: break-word;">${escapeHtml(JSON.stringify(dat
a.report, null, 2))}</pre>`;
508 }
509
510 // Если есть drives / devices (сканирование дисков)
511 if (data.drives || data.devices) {
512     const drives = data.drives || data.devices || [];
513     if (drives.length === 0) {

```

```

514         return '<div class="text-muted small">Диски не обнаруж
ены</div>';
515     }
516     return drives.map(d => `
517         <div class="bg-dark rounded p-1 mb-1 border border-sec
ondary">
518             <span class="fw-bold text-white">${escapeHtml(d.model || d.device || d.name || '-')}</span>
519             <span class="badge bg-${d.healthy === false ? 'dange
r' : 'success'} ms-1">${d.healthy === false ? 'ВНИМАНИЕ' : 'OK
'}</span>
520             ${d.temperature_c ? `<span class="text-warning ms-1"
>${d.temperature_c}°C</span>` : ''}
521         </div>
522     `).join('');
523 }
524
525 // Snapshot данные (CPU-Z, GPU-Z)
526 if (data.snapshot || data.data || data.info) {
527     const obj = data.snapshot || data.data || data.info;
528     return `<pre class="bg-dark text-light p-1.5 rounded mb-
0 font-monospace" style="font-size: 0.72rem; white-space: pre-wr
wrap; word-break: break-word;">${escapeHtml(JSON.stringify(obj
, null, 2))}</pre>`;
529 }
530
531 // Общий fallback
532 return `<pre class="bg-dark text-light p-1.5 rounded mb-0
font-monospace" style="font-size: 0.72rem; white-space: pre-wr
ap; word-break: break-word;">${escapeHtml(JSON.stringify(data,
null, 2))}</pre>`;
533 }
534 }());
535
536

```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/help/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/help/index.html" class=
15 |        "w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/help
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Interactive Reference and Help Center -->
28 <div class="container-fluid py-2">
29 |    <div class="row g-3">
30 |        <!-- Left Navigation & Search Panel -->
31 |        <div class="col-lg-3 col-md-4">
32 |            <div class="card shadow-sm border-secondary-subtle h-100
33 |            ">
34 |
35 |            <div class="card-header bg-primary text-white d-flex a
36 |            lign-items-center justify-content-between py-2">
```

```

31         <div class="fw-bold d-flex align-items-center gap-2"
    data-i18n="help.title">
32             <i class="bi bi-book-half"></i> Справочник и руков
    одства
33         </div>
34         <span class="badge bg-light text-primary rounded-pil
    l" id="help-topic-count">9</span>
35     </div>
36
37     <!-- Live Search Bar -->
38     <div class="p-2 border-bottom border-secondary-subtle
    bg-body-tertiary">
39         <div class="input-group input-group-sm">
40             <span class="input-group-text bg-transparent borde
    r-end-0 text-muted">
41                 <i class="bi bi-search"></i>
42             </span>
43             <input type="text" class="form-control border-star
    t-0 ps-0" id="help-search-input" placeholder="Поиск по руковод
    ствам..." data-i18n-placeholder="help.searchPlaceholder" autoc
    omplete="off">
44             <button class="btn btn-outline-secondary border-st
    art-0" type="button" id="help-search-clear" style="display: no
    ne;" title="Очистить">
45                 <i class="bi bi-x-lg"></i>
46             </button>
47         </div>
48     </div>
49
50     <!-- Topics List Group -->
51     <div class="list-group list-group-flush" id="help-nav"
    style="max-height: 650px; overflow-y: auto;">
52         <a class="list-group-item list-group-item-action d-f
    lex align-items-center justify-content-between active py-2" da
    ta-target="overview">
53             <div class="d-flex align-items-center gap-2">
54                 <i class="bi bi-rocket-takeoff-fill text-primary
    "></i>

```



```
55 |         <span class="fw-semibold" data-i18n="help.overvi  
| ew">□ Быстрый старт и обзор</span>  
56 |         </div>  
57 |         <span class="badge bg-secondary-subtle text-second  
| ary small">Старт</span>  
58 |         </a>  
59 |  
60 |         <a class="list-group-item list-group-item-action d-f  
| lex align-items-center justify-content-between py-2" data-targ  
| et="google_oauth">  
61 |             <div class="d-flex align-items-center gap-2">  
62 |                 <i class="bi bi-google text-danger"></i>  
63 |                 <span class="fw-semibold" data-i18n="help.google  
| _oauth">□ Google OAuth и аккаунты</span>  
64 |             </div>  
65 |             <span class="badge bg-danger-subtle text-danger sm  
| all">OAuth</span>  
66 |         </a>  
67 |  
68 |         <a class="list-group-item list-group-item-action d-f  
| lex align-items-center justify-content-between py-2" data-targ  
| et="ai_models">  
69 |             <div class="d-flex align-items-center gap-2">  
70 |                 <i class="bi bi-cpu-fill text-info"></i>  
71 |                 <span class="fw-semibold" data-i18n="help.ai_mod  
| els">□ ИИ Провайдеры и Модели</span>  
72 |             </div>  
73 |             <span class="badge bg-info-subtle text-info small"  
| >LLM</span>  
74 |         </a>  
75 |  
76 |         <a class="list-group-item list-group-item-action d-f  
| lex align-items-center justify-content-between py-2" data-targ  
| et="gdrive_sync">  
77 |             <div class="d-flex align-items-center gap-2">  
78 |                 <i class="bi bi-cloud-arrow-up-fill text-success  
| "></i>  
79 |                 <span class="fw-semibold" data-i18n="help.gdrive
```

```
_sync"> Google Drive Sync</span>
80 |         </div>
81 |         <span class="badge bg-success-subtle text-success
    | small">Бэкап</span>
82 |     </a>
83 |
84 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="rag_knowledge">
85 |         <div class="d-flex align-items-center gap-2">
86 |             <i class="bi bi-journal-bookmark-fill text-warni
    | ng"></i>
87 |             <span class="fw-semibold" data-i18n="help.rag_kn
    | owledge"> База знаний и RAG</span>
88 |         </div>
89 |         <span class="badge bg-warning-subtle text-warning
    | small">RAG</span>
90 |     </a>
91 |
92 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="voice_tts">
93 |         <div class="d-flex align-items-center gap-2">
94 |             <i class="bi bi-mic-fill text-primary"></i>
95 |             <span class="fw-semibold" data-i18n="help.voice_
    | tts"> Голос, Диаризация и TTS</span>
96 |         </div>
97 |         <span class="badge bg-primary-subtle text-primary
    | small">Аудио</span>
98 |     </a>
99 |
100 |    <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="plugins_skills">
101 |        <div class="d-flex align-items-center gap-2">
102 |            <i class="bi bi-plugin text-secondary"></i>
103 |            <span class="fw-semibold" data-i18n="help.plugin
    | s_skills"> Плагины, Навыки и MCP</span>
```

```

104         </div>
105         <span class="badge bg-secondary-subtle text-second
106         ary small">Модули</span>
107
108         <a class="list-group-item list-group-item-action d-f
109         lex align-items-center justify-content-between py-2" data-targ
110         et="storage_disks">
111             <div class="d-flex align-items-center gap-2">
112                 <i class="bi bi-hdd-network-fill text-dark"></i>
113                 <span class="fw-semibold" data-i18n="help.storag
114                 e_disks">□ Диски и Хранилище</span>
115             </div>
116             <span class="badge bg-dark-subtle text-dark small"
117             >Медиа</span>
118         </a>
119
120         <a class="list-group-item list-group-item-action d-f
121         lex align-items-center justify-content-between py-2" data-targ
122         et="troubleshooting">
123             <div class="d-flex align-items-center gap-2">
124                 <i class="bi bi-question-diamond-fill text-dange
125                 r"></i>
126                 <span class="fw-semibold" data-i18n="help.troubl
127                 eshooting">□ Решение проблем (FAQ)</span>
128             </div>
129             <span class="badge bg-danger-subtle text-danger sm
130             all">FAQ</span>
131         </a>
132     </div>
133
134     <!-- Right Reading & Guide Content Area -->
135     <div class="col-lg-9 col-md-8">
136         <div class="card shadow-sm border-secondary-subtle">
137             <div class="card-header bg-body-tertiary border-bottom
138             border-secondary-subtle d-flex align-items-center justify-con

```

```
tent-between py-2">
131 |         <div class="d-flex align-items-center gap-2">
132 |             <span class="badge bg-primary rounded-pill"><i cla
133 | ss="bi bi-file-earmark-text"></i></span>
134 |             <h5 class="mb-0 fw-bold" id="help-title">□ Быстры
135 | й старт и обзор</h5>
136 |         </div>
137 |         <div class="d-flex align-items-center gap-2">
138 |             <button class="btn btn-sm btn-outline-secondary ro
139 | unded-pill px-2 py-0" type="button" onclick="window.print()" t
140 | itle="Распечатать статью">
141 |                 <i class="bi bi-printer"></i>
142 |             </button>
143 |         </div>
144 |     </div>
145 | </div>
146 | </div>
147 | </div>
148 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Help - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/help/main.js?v=2026100
11  *     1_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/help
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21  // — HELP.JS —————
22  // Interactive User Guides & Knowledge Base Navigation
23
24  function loadHelpSection(target) {
25    const contentEl = document.getElementById('help-content');
26    const titleEl = document.getElementById('help-title');
27    if (!contentEl) return;
28
29    const content = (window.HELP_CONTENT && window.HELP_CONTENT[
target]) || '<div class="alert alert-warning">Информация по да
нному разделу не найдена.</div>';
```

```
30 | contentEl.innerHTML = content;
31 |
32 | // Update title
33 | const navItem = document.querySelector(`#help-nav a[data-target="${target}"]`);
34 | if (navItem && titleEl) {
35 |     const textEl = navItem.querySelector('span.fw-semibold') |
36 |     | navItem.querySelector('strong') || navItem;
37 |     titleEl.textContent = textEl.textContent.trim();
38 | }
39 | // Update active state in nav
40 | document.querySelectorAll('#help-nav a').forEach(a => {
41 |     a.classList.remove('active', 'bg-primary', 'text-white');
42 |     if (a.dataset.target === target) {
43 |         a.classList.add('active');
44 |     }
45 | });
46 |
47 | // Attach copy listeners to code snippet copy buttons
48 | contentEl.querySelectorAll('.btn-copy-snippet').forEach(btn
49 | => {
50 |     btn.onclick = (e) => {
51 |         e.preventDefault();
52 |         const codeTargetId = btn.dataset.copyTarget;
53 |         const targetEl = codeTargetId ? document.getElementById(
54 | codeTargetId) : btn.closest('.code-box')?.querySelector('code,
55 | pre');
56 |         const textToCopy = targetEl ? targetEl.innerText : btn.d
57 | ataset.copyText || '';
58 |         if (textToCopy) {
59 |             navigator.clipboard.writeText(textToCopy.trim()).then(
60 | () => {
61 |                 const origHtml = btn.innerHTML;
62 |                 btn.innerHTML = '<i class="bi bi-check2 text-success
63 | "></i> Скопировано!';
64 |                 btn.classList.add('btn-success', 'text-white');
65 |                 btn.classList.remove('btn-outline-secondary', 'btn-o
```

```

        outline-primary');
60         setTimeout(() => {
61             btn.innerHTML = origHtml;
62             btn.classList.remove('btn-success', 'text-white');
63             btn.classList.add('btn-outline-secondary');
64             }, 2000);
65         });
66     }
67 };
68 });
69
70 // Attach tab switch triggers
71 contentEl.querySelectorAll('[data-jump-tab]').forEach(btn =>
    {
72     btn.onclick = (e) => {
73         e.preventDefault();
74         const targetTab = btn.dataset.jumpTab;
75         if (typeof window.switchToTab === 'function') {
76             window.switchToTab(targetTab);
77         } else {
78             const tabBtn = document.querySelector(`[data-tab="${targetTab}"]`) || document.querySelector(`[data-bs-target="#${targetTab}"]`);
79             if (tabBtn) tabBtn.click();
80         }
81     };
82 });
83
84 contentEl.scrollTop = 0;
85 }
86
87 // Navigation click handler
88 function setupHelpNavigation() {
89     const navList = document.querySelectorAll('#help-nav a');
90     navList.forEach(a => {
91         a.onclick = (e) => {
92             e.preventDefault();
93             const target = a.dataset.target;

```

```

94     if (target) loadHelpSection(target);
95     };
96   });
97 }
98
99 // Live search filter across topics
100 function setupHelpSearch() {
101   const searchInput = document.getElementById('help-search-input');
102   const clearBtn = document.getElementById('help-search-clear');
103   const topicCount = document.getElementById('help-topic-count');
104   if (!searchInput) return;
105
106   const performFilter = () => {
107     const query = (searchInput.value || '').trim().toLowerCase();
108     const navItems = document.querySelectorAll('#help-nav a');
109     let visibleCount = 0;
110     let firstVisibleTarget = '';
111
112     if (clearBtn) {
113       clearBtn.style.display = query.length > 0 ? 'inline-block' : 'none';
114     }
115
116     navItems.forEach(item => {
117       const target = item.dataset.target || '';
118       const title = item.innerText.toLowerCase();
119       const fullText = (window.HELP_CONTENT && window.HELP_CONTENT[target]) ? window.HELP_CONTENT[target].toLowerCase() : '';
120
121       if (!query || title.includes(query) || fullText.includes(query)) {
122         item.style.setProperty('display', 'flex', 'important');

```



```
123     visibleCount++;
124     if (!firstVisibleTarget) firstVisibleTarget = target;
125 } else {
126     item.style.setProperty('display', 'none', 'important')
127 ;
128 }
129 });
130 if (topicCount) {
131     topicCount.textContent = visibleCount;
132 }
133 };
134
135 searchInput.oninput = performFilter;
136 searchInput.onkeydown = (e) => {
137     if (e.key === 'Enter') {
138         e.preventDefault();
139         const firstVisible = document.querySelector('#help-nav a
140 [style*="display: flex"], #help-nav a:not([style*="display: no
141 ne"]));
142         if (firstVisible && firstVisible.dataset.target) {
143             loadHelpSection(firstVisible.dataset.target);
144         }
145     }
146 };
147
148 if (clearBtn) {
149     clearBtn.onclick = () => {
150         searchInput.value = '';
151         performFilter();
152         searchInput.focus();
153     };
154 }
155 }
156
157 function initHelpTab() {
158     setupHelpNavigation();
159     setupHelpSearch();
160 }
```

```
158 |   loadHelpSection('overview');
159 | }
160 |
161 | window.initHelpTab = initHelpTab;
162 | window.loadHelpSection = loadHelpSection;
163 |
164 | // Initialize when container is present
165 | if (document.getElementById('help-nav')) {
166 |   initHelpTab();
167 | }
```

app.js

```
1  /**
2   * =====
3   * Process Name: Windows Helpdesk - App Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля app.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/helpdesk/app.js?v=2026
11  *     1001_v1" type="module"></script>
12  *
13  * File: app.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/helpdesk
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * AI Breadboard Helpdesk Management Client
23  */
24
25 const CONFIG = {
26   apiBase: window.location.origin,
27   wsProtocol: window.location.protocol === 'https:' ? 'wss:' :
28   'ws:',
29 };
30 let tickets = [];
31 let activeTicketId = null;
```

```
32 let activeTicket = null;
33 let currentFilter = 'all';
34 let ws = null;
35 let clientId = 'op_' + Math.random().toString(36).substring(2,
  9);
36
37 // Elements
38 const ticketListContainer = document.getElementById('ticketListContainer');
39 const searchTicketInput = document.getElementById('searchTicketInput');
40 const noTicketSelected = document.getElementById('noTicketSelected');
41 const activeTicketView = document.getElementById('activeTicketView');
42 const messagesStream = document.getElementById('messagesStream');
43 const replyTextInput = document.getElementById('replyTextInput');
44 const btnSendReply = document.getElementById('btnSendReply');
45 const switchInternalNote = document.getElementById('switchInternalNote');
46 const btnRefreshTickets = document.getElementById('btnRefreshTickets');
47 const hdTypingIndicator = document.getElementById('hdTypingIndicator');
48
49 window.addEventListener('DOMContentLoaded', async () => {
50   initWebSocket();
51   await loadTickets();
52   await updateStats();
53   setupEventListeners();
54   setInterval(updateStats, 15000);
55 });
56
57 function initWebSocket() {
58   const wsUrl = `${CONFIG.wsProtocol}://${window.location.host}/api/helpdesk/ws/${clientId}?role=operator`;
```

```
59 | ws = new WebSocket(wsUrl);
60 |
61 | ws.onopen = () => console.log('Helpdesk WebSocket connected
    as operator');
62 | ws.onmessage = (event) => {
63 |     try {
64 |         const packet = JSON.parse(event.data);
65 |         handleIncomingPacket(packet);
66 |     } catch (e) {
67 |         console.error('Error parsing WS message:', e);
68 |     }
69 | };
70 | ws.onclose = () => {
71 |     console.warn('Helpdesk WebSocket closed. Reconnecting in 3
    s...');
72 |     setTimeout(initWebSocket, 3000);
73 | };
74 | }
75 |
76 | function handleIncomingPacket(packet) {
77 |     const { type, ticket, message, ticket_id, status } = packet;
78 |
79 |     if (type === 'new_ticket_alert') {
80 |         // New ticket created
81 |         playNotificationSound();
82 |         loadTickets();
83 |         updateStats();
84 |     } else if (type === 'new_ticket_message') {
85 |         if (activeTicketId === ticket_id) {
86 |             appendMessageToStream(message);
87 |             scrollToBottom();
88 |         }
89 |         loadTickets();
90 |     } else if (type === 'ticket_updated') {
91 |         if (activeTicketId === ticket.id) {
92 |             activeTicket = ticket;
93 |             updateHeaderUI(ticket);
94 |         }
```

```

95 |     loadTickets();
96 |     updateStats();
97 | }
98 |
99 |
100 | function playNotificationSound() {
101 |     try {
102 |         const audioCtx = new (window.AudioContext || window.webkit
AudioContext)();
103 |         const osc = audioCtx.createOscillator();
104 |         const gain = audioCtx.createGain();
105 |         osc.connect(gain);
106 |         gain.connect(audioCtx.destination);
107 |         osc.frequency.setValueAtTime(587.33, audioCtx.currentTime)
; // D5
108 |         osc.frequency.setValueAtTime(880.00, audioCtx.currentTime
+ 0.1); // A5
109 |         gain.gain.setValueAtTime(0.2, audioCtx.currentTime);
110 |         gain.gain.exponentialRampToValueAtTime(0.01, audioCtx.curr
entTime + 0.3);
111 |         osc.start();
112 |         osc.stop(audioCtx.currentTime + 0.3);
113 |     } catch (e) {}
114 | }
115 |
116 | async function loadTickets() {
117 |     try {
118 |         let url = `${CONFIG.apiBase}/api/helpdesk/tickets?status=$
{encodeURIComponent(currentFilter)}`;
119 |         const query = searchTicketInput.value.trim();
120 |         if (query) {
121 |             url += `&search=${encodeURIComponent(query)}`;
122 |         }
123 |         const res = await fetch(url, { credentials: 'include' });
124 |         const data = await res.json();
125 |         if (data.status === 'success') {
126 |             tickets = data.tickets;
127 |             renderTicketsList();

```

```
128 |     }
129 |   } catch (e) {
130 |     console.error('Failed to load tickets:', e);
131 |   }
132 | }
133 |
134 | async function updateStats() {
135 |   try {
136 |     const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/st
137 |     ats`, { credentials: 'include' });
138 |     const data = await res.json();
139 |     if (data.status === 'success') {
140 |       const s = data.stats;
141 |       document.getElementById('statOpen').innerText = `Open: $
142 |       {s.open_tickets}`;
143 |       document.getElementById('statInProgress').innerText = `A
144 |       ctive: ${s.in_progress_tickets}`;
145 |       document.getElementById('statResolved').innerText = `Res
146 |       olved: ${s.resolved_tickets}`;
147 |     }
148 |   } catch (e) {}
149 | }
150 |
151 | function renderTicketsList() {
152 |   ticketListContainer.innerHTML = '';
153 |   if (tickets.length === 0) {
154 |     ticketListContainer.innerHTML = '<div class="text-center t
155 |     ext-muted p-4">No tickets matching filter.</div>';
156 |     return;
157 |   }
158 |
159 |   tickets.forEach((t) => {
160 |     const card = document.createElement('div');
161 |     card.className = `ticket-card ${activeTicketId === t.id ?
162 |     'active' : ''}`;
163 |     card.onclick = () => selectTicket(t.id);
164 |
165 |     const priorityBadgeClass = `badge-priority-${t.priority}`;
```

```

160     const statusBadge = getStatusBadge(t.status);
161     const lastSnippet = t.last_message ? escapeHtml(t.last_mes
    sage.content) : 'No messages';
162     const timeAgo = formatTimeAgo(t.updated_at);
163
164     card.innerHTML = `
165         <div class="d-flex align-items-center justify-content-be
    tween mb-1">
166             <span class="badge bg-secondary">#${t.ticket_number}</
    span>
167             <div class="d-flex align-items-center gap-1">
168                 <span class="badge ${priorityBadgeClass}">${t.priori
    ty.toUpperCase()}</span>
169                 ${statusBadge}
170             </div>
171         </div>
172         <div class="fw-semibold text-truncate text-light mb-1">$
    {escapeHtml(t.subject)}</div>
173         <div class="small text-muted text-truncate mb-1">${lastS
    nippet}</div>
174         <div class="d-flex align-items-center justify-content-be
    tween text-muted" style="font-size: 0.75rem;">
175             <span>👤 ${escapeHtml(t.user_name)}</span>
176             <span>${timeAgo}</span>
177         </div>
178     `;
179     ticketListContainer.appendChild(card);
180 });
181 }
182
183 function getStatusBadge(status) {
184     switch (status) {
185         case 'open':
186             return '<span class="badge bg-danger">OPEN</span>';
187         case 'in_progress':
188             return '<span class="badge bg-warning text-dark">IN PROG
    RESS</span>';
189         case 'resolved':

```



```
190     return '<span class="badge bg-success">RESOLVED</span>';
191   case 'closed':
192     return '<span class="badge bg-secondary">CLOSED</span>';
193   default:
194     return `<span class="badge bg-info">${status}</span>`;
195   }
196 }
197
198 async function selectTicket(ticketId) {
199   activeTicketId = ticketId;
200   renderTicketsList();
201
202   if (ws && ws.readyState === WebSocket.OPEN) {
203     ws.send(JSON.stringify({ action: 'subscribe_ticket', ticke
204       t_id: ticketId }));
205   }
206   try {
207     const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/ti
208       ckets/${ticketId}`, { credentials: 'include' });
209     const data = await res.json();
210     if (data.status === 'success') {
211       activeTicket = data.ticket;
212       noTicketSelected.classList.add('d-none');
213       activeTicketView.classList.remove('d-none');
214       activeTicketView.classList.add('d-flex');
215
216       updateHeaderUI(activeTicket);
217       renderMessages(data.messages);
218       scrollToBottom();
219     } catch (e) {
220       console.error('Failed to load ticket details:', e);
221     }
222   }
223
224   function updateHeaderUI(t) {
225     document.getElementById('ticketNumberBadge').innerText = `#`
```

```

    {t.ticket_number}`;
226   document.getElementById('ticketSubject').innerText = t.subject;
227   document.getElementById('ticketUserMeta').innerText = `From:
    ${t.user_name} (${t.user_email || 'No email'}) • ${formatTime
    Ago(t.created_at)}`;
228   document.getElementById('currentPriorityLabel').innerText =
    t.priority.toUpperCase();
229 }
230
231 function renderMessages(messages) {
232   messagesStream.innerHTML = '';
233   messages.forEach((msg) => appendMessageToStream(msg));
234 }
235
236 function appendMessageToStream(msg) {
237   const el = document.createElement('div');
238   let typeClass = 'bubble-user';
239   if (msg.is_internal_note) {
240     typeClass = 'bubble-note';
241   } else if (msg.sender_type === 'operator') {
242     typeClass = 'bubble-operator';
243   }
244
245   const renderedContent = window.marked ? marked.parse(msg.content) : escapeHtml(msg.content);
246   const timeStr = new Date(msg.created_at).toLocaleTimeString();
247
248   el.className = `chat-bubble ${typeClass}`;
249   el.innerHTML = `
250     <div class="d-flex align-items-center justify-content-between gap-2 mb-1" style="font-size: 0.75rem; opacity: 0.85;">
251       <strong>${escapeHtml(msg.sender_name)} ${msg.is_internal_note ? '(Internal Note)' : ''}</strong>
252       <span>${timeStr}</span>
253     </div>
254     <div class="message-content">${renderedContent}</div>

```

```
255 | `;  
256 | messagesStream.appendChild(el);  
257 | }  
258 |  
259 | function scrollToBottom() {  
260 |     messagesStream.scrollTop = messagesStream.scrollHeight;  
261 | }  
262 |  
263 | async function sendReply() {  
264 |     const content = replyTextInput.value.trim();  
265 |     if (!content || !activeTicketId) return;  
266 |  
267 |     const isNote = switchInternalNote.checked;  
268 |     btnSendReply.disabled = true;  
269 |  
270 |     try {  
271 |         const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}/messages`, {  
272 |             method: 'POST',  
273 |             headers: { 'Content-Type': 'application/json' },  
274 |             credentials: 'include',  
275 |             body: JSON.stringify({  
276 |                 content: content,  
277 |                 is_internal_note: isNote,  
278 |                 sender_type: 'operator'  
279 |             })  
280 |         });  
281 |  
282 |         if (res.ok) {  
283 |             replyTextInput.value = '';  
284 |         }  
285 |     } catch (e) {  
286 |         console.error('Failed to send reply:', e);  
287 |     } finally {  
288 |         btnSendReply.disabled = false;  
289 |     }  
290 | }  
291 |
```

```
292 | async function updateTicketStatus(status) {
293 |   if (!activeTicketId) return;
294 |   try {
295 |     await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}`, {
296 |       method: 'PATCH',
297 |       headers: { 'Content-Type': 'application/json' },
298 |       credentials: 'include',
299 |       body: JSON.stringify({ status: status })
300 |     });
301 |   } catch (e) {
302 |     console.error('Failed to update status:', e);
303 |   }
304 | }
305 |
306 | async function updateTicketPriority(priority) {
307 |   if (!activeTicketId) return;
308 |   try {
309 |     await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}`, {
310 |       method: 'PATCH',
311 |       headers: { 'Content-Type': 'application/json' },
312 |       credentials: 'include',
313 |       body: JSON.stringify({ priority: priority })
314 |     });
315 |   } catch (e) {
316 |     console.error('Failed to update priority:', e);
317 |   }
318 | }
319 |
320 | function setupEventListeners() {
321 |   btnSendReply.onclick = sendReply;
322 |   replyTextInput.addEventListener('keydown', (e) => {
323 |     if (e.key === 'Enter' && (e.ctrlKey || e.metaKey)) {
324 |       e.preventDefault();
325 |       sendReply();
326 |     }
327 |   });
328 | }
```

```
328 |
329 | btnRefreshTickets.onclick = () => {
330 |     loadTickets();
331 |     updateStats();
332 | };
333 |
334 | searchTicketInput.addEventListener('input', () => {
335 |     loadTickets();
336 | });
337 |
338 | document.querySelectorAll('#statusFilterGroup button').forEa
| ch((btn) => {
339 |     btn.onclick = () => {
340 |         document.querySelectorAll('#statusFilterGroup button').f
| orEach((b) => b.classList.remove('active'));
341 |         btn.classList.add('active');
342 |         currentFilter = btn.dataset.filter;
343 |         loadTickets();
344 |     };
345 | });
346 |
347 | document.querySelectorAll('.status-opt').forEach((btn) => {
348 |     btn.onclick = () => updateTicketStatus(btn.dataset.status)
| ;
349 | });
350 |
351 | document.querySelectorAll('.priority-opt').forEach((item) =>
| {
352 |     item.onclick = (e) => {
353 |         e.preventDefault();
354 |         updateTicketPriority(item.dataset.priority);
355 |     };
356 | });
357 | }
358 |
359 | function escapeHtml(str) {
360 |     if (!str) return '';
361 |     const div = document.createElement('div');
```

```
362 |   div.innerText = str;
363 |   return div.innerHTML;
364 | }
365 |
366 | function formatTimeAgo(dateStr) {
367 |   if (!dateStr) return '';
368 |   const d = new Date(dateStr);
369 |   const now = new Date();
370 |   const diffSec = Math.floor((now - d) / 1000);
371 |   if (diffSec < 60) return 'just now';
372 |   const diffMin = Math.floor(diffSec / 60);
373 |   if (diffMin < 60) return `${diffMin}m ago`;
374 |   const diffHours = Math.floor(diffMin / 60);
375 |   if (diffHours < 24) return `${diffHours}h ago`;
376 |   return d.toLocaleDateString();
377 | }
378 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/helpdesk/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/helpdesk/index.html" class="w-100 h-100 border-0"></iframe>
15 |
16 File: index.html
17 Project: ai-breadboard
18 Package: windows/api/webgui/helpdesk
19 Author: hypo69
20 Copyright: © 2026 hypo69
21 Updated: 2026-10-01 13:04:40
22 |=====
23 |=====
24 -->
25
26 <!DOCTYPE html>
27 <html lang="en">
28 <head>
29 |   <meta charset="UTF-8" />
30 |   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
31 |   <title>AI Breadboard Helpdesk & Support Center</title>
32 |   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet" />
```

```

31 | <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    | 0.0/font/bootstrap-icons.css" rel="stylesheet" />
32 | <link rel="stylesheet" href="/helpdesk/style.css" />
33 | </head>
34 | <body>
35 |   <div class="helpdesk-container d-flex flex-column vh-100 ove
    | rflow-hidden">
36 |     <!-- Top Navbar -->
37 |     <header class="navbar navbar-expand-lg navbar-dark bg-dark
    | border-bottom border-secondary px-3 py-2 flex-shrink-0">
38 |       <div class="container-fluid p-0 d-flex align-items-cente
    | r justify-content-between">
39 |         <div class="d-flex align-items-center gap-3">
40 |           <a class="navbar-brand d-flex align-items-center gap
    | -2 m-0 fw-bold" href="/helpdesk">
41 |             <span class="fs-4">☐ </span>
42 |             <span>Helpdesk Support Center</span>
43 |           </a>
44 |           <span class="badge bg-primary rounded-pill px-3 py-1
    | d-none d-md-inline" id="operatorBadge">Operator Mode</span>
45 |         </div>
46 |
47 |         <div class="d-flex align-items-center gap-2">
48 |           <!-- Metrics Overview Pills -->
49 |           <div class="d-none d-lg-flex align-items-center gap-
    | 2 me-3" id="quickStatsBar">
50 |             <span class="badge bg-danger-subtle text-danger bo
    | rder border-danger-subtle px-2 py-1" id="statOpen">Open: 0</sp
    | an>
51 |             <span class="badge bg-warning-subtle text-warning
    | border border-warning-subtle px-2 py-1" id="statInProgress">In
    | Progress: 0</span>
52 |             <span class="badge bg-success-subtle text-success
    | border border-success-subtle px-2 py-1" id="statResolved">Reso
    | lved: 0</span>
53 |           </div>
54 |
55 |           <button class="btn btn-outline-light btn-sm d-flex a

```



```

    align-items-center gap-1" id="btnRefreshTickets" title="Refresh
    Tickets">
56         <i class="bi bi-arrow-clockwise"></i>
57         <span class="d-none d-sm-inline">Refresh</span>
58     </button>
59     <a href="/" class="btn btn-secondary btn-sm d-flex a
    align-items-center gap-1" title="Back to Main Assistant">
60         <i class="bi bi-house-door"></i>
61         <span class="d-none d-sm-inline">Main App</span>
62     </a>
63 </div>
64 </div>
65 </header>
66
67 <!-- Main Workspace Layout -->
68 <div class="d-flex flex-grow-1 overflow-hidden" style="min
    -height: 0;">
69     <!-- Tickets Sidebar / Filter List -->
70     <aside class="helpdesk-sidebar d-flex flex-column border
    -end border-secondary bg-body-tertiary" style="width: 360px; m
    in-width: 300px; max-width: 420px;">
71         <!-- Filter and Search Header -->
72         <div class="p-2 border-bottom border-secondary-subtle"
    >
73             <div class="input-group input-group-sm mb-2">
74                 <span class="input-group-text bg-body border-secon
    dary-subtle"><i class="bi bi-search"></i></span>
75                 <input type="text" class="form-control border-seco
    ndary-subtle" id="searchTicketInput" placeholder="Search ticke
    t, user, #num..." />
76             </div>
77
78             <!-- Status Filters -->
79             <div class="btn-group btn-group-sm w-100" role="grou
    p" id="statusFilterGroup">
80                 <button type="button" class="btn btn-outline-prima
    ry active" data-filter="all">All</button>
81                 <button type="button" class="btn btn-outline-prima

```

```

ry" data-filter="open">Open</button>
82         <button type="button" class="btn btn-outline-prima
ry" data-filter="in_progress">Active</button>
83         <button type="button" class="btn btn-outline-prima
ry" data-filter="resolved">Closed</button>
84     </div>
85 </div>
86
87 <!-- Ticket List Stream -->
88 <div class="ticket-list flex-grow-1 overflow-auto p-2
d-flex flex-column gap-2" id="ticketListContainer">
89     <div class="text-center text-muted p-4">Loading tick
ets...</div>
90 </div>
91 </aside>
92
93 <!-- Ticket Conversation and Management View -->
94 <main class="helpdesk-main d-flex flex-column flex-grow-
1 overflow-hidden bg-body" id="ticketMainArea">
95     <!-- Empty State Placeholder -->
96     <div class="d-flex flex-column align-items-center just
ify-content-center h-100 text-muted p-4" id="noTicketSelected"
>
97         <i class="bi bi-chat-square-text fs-1 mb-2"></i>
98         <h5>Select a ticket to manage and reply</h5>
99         <p class="small">All incoming support requests and c
hat threads are coordinated here in real time.</p>
100     </div>
101
102     <!-- Ticket Detail Content (Hidden initially) -->
103     <div class="d-none flex-column h-100 overflow-hidden"
id="activeTicketView">
104         <!-- Top Ticket Action Header -->
105         <div class="p-2 px-3 border-bottom border-secondary-
subtle d-flex flex-wrap align-items-center justify-content-bet
ween gap-2 flex-shrink-0 bg-body-tertiary">
106             <div class="d-flex align-items-center gap-2">
107                 <span class="badge bg-secondary fs-6" id="ticket

```

```

NumberBadge">#1001</span>
108         <div>
109             <h6 class="mb-0 fw-bold" id="ticketSubject">Ti
cket Subject</h6>
110             <small class="text-muted" id="ticketUserMeta">
From: User (user@domain.com) • 10m ago</small>
111         </div>
112     </div>
113
114     <!-- Management Dropdowns & Actions -->
115     <div class="d-flex align-items-center gap-2">
116         <!-- Priority Selector -->
117         <div class="dropdown">
118             <button class="btn btn-sm btn-outline-secondar
y dropdown-toggle" type="button" id="btnTicketPriority" data-b
s-toggle="dropdown">
119                 Priority: <span id="currentPriorityLabel" cl
ass="fw-bold">Normal</span>
120             </button>
121             <ul class="dropdown-menu dropdown-menu-end sha
dow">
122                 <li><a class="dropdown-item priority-opt" da
ta-priority="low" href="#">□ Low</a></li>
123                 <li><a class="dropdown-item priority-opt" da
ta-priority="normal" href="#">□ Normal</a></li>
124                 <li><a class="dropdown-item priority-opt" da
ta-priority="high" href="#">□ High</a></li>
125                 <li><a class="dropdown-item priority-opt" da
ta-priority="urgent" href="#">□ Urgent</a></li>
126             </ul>
127         </div>
128
129     <!-- Status Action Buttons -->
130     <div class="btn-group btn-group-sm">
131         <button class="btn btn-outline-warning status-
opt" data-status="in_progress" title="Mark In Progress">
132             <i class="bi bi-play-circle me-1"></i>In Pro
gress

```

```

133         </button>
134         <button class="btn btn-outline-success status-
135         opt" data-status="resolved" title="Mark Resolved">
136             <i class="bi bi-check2-circle me-1"></i>Reso
137             lve
138         </button>
139         <button class="btn btn-outline-secondary statu
140         s-opt" data-status="closed" title="Close Ticket">
141             <i class="bi bi-archive me-1"></i>Close
142         </button>
143     </div>
144 </div>
145 </div>
146
147     <!-- Message History Stream -->
148     <div class="chat-stream flex-grow-1 overflow-auto p-
149     3 d-flex flex-column gap-3" id="messagesStream">
150         <!-- Messages rendered dynamically -->
151     </div>
152
153     <!-- Typing Indicator -->
154     <div class="px-3 py-1 small text-muted d-none" id="h
155     dTypingIndicator">
156         <span class="spinner-border spinner-border-sm me-1
157         " role="status"></span>
158         <span id="hdTypingText">User is typing...</span>
159     </div>
160
161     <!-- Reply and Internal Notes Bar -->
162     <div class="p-3 border-top border-secondary-subtle b
163     g-body-tertiary flex-shrink-0">
164         <div class="d-flex align-items-center justify-cont
165         ent-between mb-2">
166             <div class="form-check form-switch m-0">
167                 <input class="form-check-input" type="checkbox
168                 " role="switch" id="switchInternalNote" />
169                 <label class="form-check-label small fw-semibo
170                 ld text-warning" for="switchInternalNote">

```

```
161 |         <i class="bi bi-lock-fill me-1"></i>Internal
    | Note (Staff only)
162 |         </label>
163 |     </div>
164 |     <small class="text-muted">Press Ctrl+Enter to se
    | nd</small>
165 | </div>
166 |
167 |     <div class="input-group">
168 |         <textarea class="form-control" id="replyTextInpu
    | t" rows="2" placeholder="Write response to user..."></textarea
    | >
169 |         <button class="btn btn-primary px-4 d-flex align
    | -items-center justify-content-center" id="btnSendReply">
170 |             <i class="bi bi-send-fill fs-5"></i>
171 |         </button>
172 |     </div>
173 | </div>
174 | </div>
175 | </main>
176 | </div>
177 | </div>
178 |
179 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/di
    | st/js/bootstrap.bundle.min.js"></script>
180 | <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
    | js"></script>
181 | <script src="/helpdesk/app.js"></script>
182 | </body>
183 | </html>
184 |
```

style.css

```
1  /**
2   * =====
3   *
4   * =====
5   * Process Name: Windows Web Styles - Style Stylesheet
6   * =====
7   * Description:
8   *   Таблица стилей стиля и оформления интерфейса (style).
9   *
10  * Usage Examples:
11  *   HTML Link Tag:
12  *     <link rel="stylesheet" href="/windows/api/webgui/helpdesk/style.css?v=20261001_v1">
13  *
14  * File: style.css
15  * Project: ai-breadboard
16  * Package: windows/api/webgui/helpdesk
17  * Author: hypo69
18  * Copyright: © 2026 hypo69
19  * Updated: 2026-10-01 13:04:40
20  * =====
21  */
22
23  helpdesk/style.css – Дочерние стили Helpdesk (наследует main.css)
24  Версия: 20260917_refactor_v1
25
26  ===== */
27
28  /* Helpdesk использует переменные из main.css – хардкод убран
29  */
30
31  .helpdesk-sidebar {
```

```
29 | background-color: var(--surface-2) !important;
30 | border-color: var(--border-color) !important;
31 | }
32 |
33 | /* — Карточка тикета —————
   | ————— */
34 | .ticket-card {
35 |     padding: 8px 10px;
36 |     border-radius: 6px;
37 |     background: var(--surface-1);
38 |     border: 1px solid var(--border-color);
39 |     cursor: pointer;
40 |     transition: background 0.12s ease, border-color 0.12s ease;
41 |     font-size: 0.82rem;
42 | }
43 |
44 | .ticket-card:hover {
45 |     background: var(--nav-hover-bg);
46 |     border-color: var(--nav-active);
47 | }
48 |
49 | .ticket-card.active {
50 |     background: rgba(37, 99, 235, 0.08);
51 |     border-color: var(--nav-active);
52 |     box-shadow: 0 0 0 1px var(--nav-active);
53 | }
54 |
55 | [data-theme="dark"] .ticket-card.active {
56 |     background: rgba(88, 166, 255, 0.08);
57 | }
58 |
59 | /* — Пузыри чата —————
   | ————— */
60 | .chat-bubble {
61 |     max-width: 80%;
62 |     padding: 7px 12px;
63 |     border-radius: 10px;
64 |     position: relative;
```

```
65 | word-break: break-word;
66 | font-size: 0.83rem;
67 | line-height: 1.5;
68 | }
69 |
70 | .bubble-user {
71 |   align-self: flex-start;
72 |   background: var(--chat-bot-bg);
73 |   border: 1px solid var(--border-color);
74 |   color: var(--chat-bot-text);
75 | }
76 |
77 | .bubble-operator {
78 |   align-self: flex-end;
79 |   background: var(--chat-user-bg);
80 |   color: var(--chat-user-text);
81 | }
82 |
83 | .bubble-note {
84 |   align-self: center;
85 |   width: 90%;
86 |   background: rgba(245, 158, 11, 0.08);
87 |   border: 1px dashed #f59e0b;
88 |   color: #92400e;
89 |   font-size: 0.8rem;
90 | }
91 |
92 | [data-theme="dark"] .bubble-note {
93 |   background: rgba(245, 158, 11, 0.1);
94 |   color: #fbbf24;
95 | }
96 |
97 | /* — Приоритеты —————
   | ————— */
98 | .badge-priority-urgent { background-color: #dc2626; color: #fff; }
99 | .badge-priority-high   { background-color: #ea580c; color: #fff; }
```



```
100 | .badge-priority-normal { background-color: var(--nav-active);  
    | color: #fff; }  
101 | .badge-priority-low      { background-color: #10b981; color: #ff  
    | f; }  
102 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/helpdesk_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/helpdesk_tab/index.html
15 |     " class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/helpdesk_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Helpdesk Multi-Tool Management Panel Tab -->
28 <div class="container-fluid py-2 h-100 d-flex flex-column" sty
29 |le="min-height: calc(100vh - 120px);">
30 |
31 |   <!-- Header Alert & Mode Switcher Bar -->
32 |   <div class="alert alert-dark d-flex flex-wrap align-items-ce
33 |   nter justify-content-between gap-2 mb-3 shadow-sm border-secon
34 |   dary flex-shrink-0">
35 |     <div class="d-flex align-items-center gap-2">
36 |       <span class="fs-4">□ </span>
```

```
30     <div>
31         <h5 class="mb-0 fw-bold">Центр управления Helpdesk & С
32         лужба поддержки</h5>
33         <p class="mb-0 small text-muted">Единый пульт оператор
34         а: управление тикетами, живой чат поддержки, быстрые шаблоны,
35         база знаний и SLA метрики</p>
36     </div>
37 </div>
38 <div class="d-flex align-items-center gap-2 flex-wrap">
39     <button class="btn btn-outline-info btn-sm d-flex align-
40     items-center gap-1" id="hd-tab-btn-refresh" onclick="window.re
41     freshHelpdeskTab && window.refreshHelpdeskTab()">
42         <i class="bi bi-arrow-clockwise"></i> Обновить заявки
43     </button>
44     <button class="btn btn-primary btn-sm d-flex align-items
45     -center gap-1" id="hd-tab-btn-new-ticket" onclick="window.open
46     CreateTicketModal && window.openCreateTicketModal()">
47         <i class="bi bi-plus-circle"></i> Создать обращение
48     </button>
49     <button class="btn btn-outline-light btn-sm d-flex align
50     -items-center gap-1" onclick="switchToTab('admin')">
51         <i class="bi bi-arrow-left-circle"></i> Панель админки
52     </button>
53 </div>
54 </div>
55 <!-- Quick KPI & Status Metrics -->
56 <div class="row g-2 mb-3 flex-shrink-0" id="hd-tab-metrics-b
57 ar">
58     <div class="col-md-3 col-6">
59         <div class="card shadow-sm border-secondary-subtle p-2 b
60 g-body-tertiary">
61             <div class="d-flex align-items-center justify-content-
62             between">
63                 <div>
64                     <div class="text-muted small text-uppercase fw-sem
65 ibold">Открытые заявки</div>
66                     <h4 class="mb-0 fw-bold text-danger" id="hd-metric
```

```

    -open">0</h4>
56         </div>
57         <i class="bi bi-exclamation-octagon fs-2 text-danger
    opacity-75"></i>
58     </div>
59 </div>
60 </div>
61 <div class="col-md-3 col-6">
62     <div class="card shadow-sm border-secondary-subtle p-2 b
    g-body-tertiary">
63         <div class="d-flex align-items-center justify-content-
    between">
64             <div>
65                 <div class="text-muted small text-uppercase fw-sem
    ibold">В обработке</div>
66                 <h4 class="mb-0 fw-bold text-warning" id="hd-metri
    c-progress">0</h4>
67             </div>
68             <i class="bi bi-gear-wide-connected fs-2 text-warnin
    g opacity-75"></i>
69         </div>
70     </div>
71 </div>
72 <div class="col-md-3 col-6">
73     <div class="card shadow-sm border-secondary-subtle p-2 b
    g-body-tertiary">
74         <div class="d-flex align-items-center justify-content-
    between">
75             <div>
76                 <div class="text-muted small text-uppercase fw-sem
    ibold">Решено сегодня</div>
77                 <h4 class="mb-0 fw-bold text-success" id="hd-metri
    c-resolved">0</h4>
78             </div>
79             <i class="bi bi-check2-all fs-2 text-success opacity
    -75"></i>
80         </div>
81     </div>

```

```

82     </div>
83     <div class="col-md-3 col-6">
84         <div class="card shadow-sm border-secondary-subtle p-2 bg-body-tertiary">
85             <div class="d-flex align-items-center justify-content-between">
86                 <div>
87                     <div class="text-muted small text-uppercase fw-semibold">Bcero в базе</div>
88                     <h4 class="mb-0 fw-bold text-info" id="hd-metric-total">0</h4>
89                 </div>
90                 <i class="bi bi-collection fs-2 text-info opacity-75"></i>
91             </div>
92         </div>
93     </div>
94 </div>
95
96 <!-- Helpdesk Main Workspace (Two Columns: Ticket Stream + Management Workbench) -->
97 <div class="card shadow-sm border-secondary-subtle flex-grow-1 overflow-hidden d-flex flex-column" style="min-height: 520px;">
98     <div class="row g-0 flex-grow-1 h-100 overflow-hidden">
99         <!-- Left Column: Tickets List & Filter Tools -->
100         <div class="col-lg-4 col-md-5 border-end border-secondary-subtle d-flex flex-column h-100 bg-body-tertiary">
101             <!-- Search & Filter Header -->
102             <div class="p-2 border-bottom border-secondary-subtle">
103                 <div class="input-group input-group-sm mb-2">
104                     <span class="input-group-text bg-body border-secondary-subtle"><i class="bi bi-search"></i></span>
105                     <input type="text" class="form-control border-secondary-subtle" id="hd-tab-search-input" placeholder="Поиск по номеру, теме, автору..." />
106                 </div>

```

```
107 |
108 |         <div class="d-flex gap-1" role="group" id="hd-tab-st
109 |             <button type="button" class="btn btn-sm btn-outlin
110 |             <button type="button" class="btn btn-sm btn-outlin
111 |             <button type="button" class="btn btn-sm btn-outlin
112 |             <button type="button" class="btn btn-sm btn-outlin
113 |         </div>
114 |     </div>
115 |
116 |     <!-- Scrollable Ticket List Stream -->
117 |     <div class="flex-grow-1 overflow-auto p-2 d-flex flex-
118 |         <div class="text-center text-muted p-4">
119 |             <div class="spinner-border spinner-border-sm text-
120 |             <div>Загрузка тикетов...</div>
121 |         </div>
122 |     </div>
123 | </div>
124 |
125 |     <!-- Right Column: Selected Ticket Workbench & Multi-too
126 |     <div class="col-lg-8 col-md-7 d-flex flex-column h-100 b
127 |     <!-- Empty State (No ticket chosen) -->
128 |     <div class="d-flex flex-column align-items-center just
129 |         <i class="bi bi-headset fs-1 mb-2 text-primary opaci
130 |         <h5 class="fw-bold">Выберите тикет для работы</h5>
131 |         <p class="small text-muted mb-0" style="max-width: 4
```

```

50px; ">
132         Здесь отображается полная история обращений, переписка с пользователями, внутренние заметки операторов и инструменты быстрого реагирования.
133     </p>
134 </div>
135
136     <!-- Active Ticket Detail & Tools View (Shown when ticket is active) -->
137     <div class="d-none flex-column h-100 overflow-hidden" id="hd-tab-active-view">
138         <!-- Ticket Header Bar -->
139         <div class="p-2 px-3 border-bottom border-secondary-subtle d-flex flex-wrap align-items-center justify-content-between gap-2 flex-shrink-0 bg-body-tertiary">
140             <div class="d-flex align-items-center gap-2 min-w-0">
141                 <span class="badge bg-secondary fs-6" id="hd-active-number">#0000</span>
142                 <div class="min-w-0">
143                     <h6 class="mb-0 fw-bold text-truncate" id="hd-active-subject">Тема заявки</h6>
144                     <small class="text-muted" id="hd-active-meta">Автор: Загрузка...</small>
145                 </div>
146             </div>
147
148             <!-- Priority & Status Actions -->
149             <div class="d-flex align-items-center gap-2 flex-wrap">
150                 <div class="dropdown">
151                     <button class="btn btn-sm btn-outline-secondary dropdown-toggle" type="button" id="hd-btn-priority-dropdown" data-bs-toggle="dropdown">
152                         Приоритет: <span id="hd-active-priority-label" class="fw-bold">Обычный</span>
153                     </button>
154                     <ul class="dropdown-menu dropdown-menu-end sha

```

```

dow">
155         <li><a class="dropdown-item hd-priority-opt"
data-priority="low" href="#">□ Низкий</a></li>
156         <li><a class="dropdown-item hd-priority-opt"
data-priority="normal" href="#">□ Обычный</a></li>
157         <li><a class="dropdown-item hd-priority-opt"
data-priority="high" href="#">□ Высокий</a></li>
158         <li><a class="dropdown-item hd-priority-opt"
data-priority="urgent" href="#">□ Срочный</a></li>
159     </ul>
160 </div>
161
162     <div class="btn-group btn-group-sm">
163         <button class="btn btn-outline-warning hd-stat
us-opt" data-status="in_progress" title="Взять в работу">
164             <i class="bi bi-play-circle me-1"></i>В рабо
ту
165         </button>
166         <button class="btn btn-outline-success hd-stat
us-opt" data-status="resolved" title="Отметить решенным">
167             <i class="bi bi-check2-circle me-1"></i>Реше
но
168         </button>
169         <button class="btn btn-outline-secondary hd-st
atus-opt" data-status="closed" title="Закрыть обращение">
170             <i class="bi bi-archive me-1"></i>Закрыть
171         </button>
172     </div>
173 </div>
174 </div>
175
176     <!-- Multi-Tool Quick Toolbar (Canned replies, AI As
sistant summary, internal note) -->
177     <div class="p-2 border-bottom border-secondary-subtl
e bg-body-tertiary d-flex align-items-center justify-content-b
etween gap-2 flex-shrink-0 flex-wrap">
178         <div class="d-flex align-items-center gap-1 flex-w
rap">

```



```

179 |         <span class="small fw-semibold text-muted me-1">
| <i class="bi bi-lightning-charge-fill text-warning"></i> Быстр
| ые ответы:</span>
180 |         <button class="btn btn-xs btn-outline-secondary
| rounded-pill py-0 px-2 hd-quick-template" data-tpl="Здравствуй
| те! Ваше обращение принято в работу, уточняем детали.">
181 |             □ Принято
182 |         </button>
183 |         <button class="btn btn-xs btn-outline-secondary
| rounded-pill py-0 px-2 hd-quick-template" data-tpl="Проблема у
| странена. Пожалуйста, проверьте работу и подтвердите решение."
| >
184 |             □ Устранено
185 |         </button>
186 |         <button class="btn btn-xs btn-outline-secondary
| rounded-pill py-0 px-2 hd-quick-template" data-tpl="Уточните,
| пожалуйста, версию ОС и приложите снимок экрана с ошибкой.">
187 |             □ Запрос инфо
188 |         </button>
189 |         <button class="btn btn-xs btn-outline-info round
| ed-pill py-0 px-2" id="hd-btn-ai-assist" title="Сгенерировать
| черновик ответа через ИИ">
190 |             □ ИИ-Черновик
191 |         </button>
192 |     </div>
193 |
194 |     <div class="form-check form-switch m-0">
195 |         <input class="form-check-input" type="checkbox"
| role="switch" id="hd-switch-internal-note" />
196 |         <label class="form-check-label small fw-semibold
| text-warning" for="hd-switch-internal-note">
197 |             <i class="bi bi-lock-fill me-1"></i>Скрытая за
| метка (Staff)
198 |         </label>
199 |     </div>
200 | </div>
201 |
202 | <!-- Messages Stream -->

```

```

203 |         <div class="flex-grow-1 overflow-auto p-3 d-flex fle
| x-column gap-3 bg-body" id="hd-tab-messages-stream" style="min
| -height: 0;">
204 |             <!-- Messages rendered dynamically -->
205 |         </div>
206 |
207 |         <!-- Reply Composer -->
208 |         <div class="p-2 px-3 border-top border-secondary-sub
| tle bg-body-tertiary flex-shrink-0">
209 |             <div class="input-group">
210 |                 <textarea class="form-control" id="hd-reply-inpu
| t" rows="2" placeholder="Напишите ответ пользователю (или скры
| тую заметку)..."></textarea>
211 |                 <button class="btn btn-primary px-4 d-flex align
| -items-center justify-content-center" id="hd-btn-send-reply">
212 |                     <i class="bi bi-send-fill fs-5"></i>
213 |                 </button>
214 |             </div>
215 |             <div class="d-flex justify-content-between align-i
| tems-center mt-1 text-muted small" style="font-size: 0.75rem;"
| >
216 |                 <span>Ctrl + Enter для быстрой отправки</span>
217 |                 <span id="hd-composer-mode-badge" class="badge b
| g-secondary-subtle text-body">Режим: Публичный ответ</span>
218 |             </div>
219 |         </div>
220 |     </div>
221 | </div>
222 | </div>
223 | </div>
224 | </div>
225 |
226 | <!-- Modal: Create New Ticket -->
227 | <div class="modal fade" id="hdCreateTicketModal" tabindex="-1"
| aria-labelledby="hdCreateTicketModalLabel" aria-hidden="true"
| >
228 |     <div class="modal-dialog modal-dialog-centered">
229 |         <div class="modal-content shadow border-secondary-subtle">

```

```
230 |     <div class="modal-header border-secondary-subtle">
231 |         <h5 class="modal-title d-flex align-items-center gap-2
    | " id="hdCreateTicketModalLabel">
232 |             <i class="bi bi-plus-circle text-primary"></i>
233 |             <span>Создать новое обращение в Helpdesk</span>
234 |         </h5>
235 |         <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
236 |     </div>
237 |     <div class="modal-body p-3">
238 |         <div class="mb-3">
239 |             <label class="form-label small fw-semibold">Тема обращения</label>
240 |             <input type="text" class="form-control" id="hd-new-ticket-subject" placeholder="Краткая тема проблемы..." />
241 |         </div>
242 |         <div class="row g-2 mb-3">
243 |             <div class="col-6">
244 |                 <label class="form-label small fw-semibold">Категория</label>
245 |                 <select class="form-select form-select-sm" id="hd-new-ticket-category">
246 |                     <option value="technical">□ Техническая</option>
247 |                     <option value="ai">□ ИИ и Модели</option>
248 |                     <option value="account">□ Аккаунт & Доступ</option>
249 |                     <option value="general">□ Общие вопросы</option>
250 |                 </select>
251 |             </div>
252 |             <div class="col-6">
253 |                 <label class="form-label small fw-semibold">Приоритет</label>
254 |                 <select class="form-select form-select-sm" id="hd-new-ticket-priority">
255 |                     <option value="low">□ Низкий</option>
256 |                     <option value="normal" selected>□ Обычный</option>
```

```
on>
257         <option value="high">␣ Высокий</option>
258         <option value="urgent">␣ Срочный</option>
259     </select>
260 </div>
261 </div>
262 <div class="mb-2">
263     <label class="form-label small fw-semibold">Описание
    проблемы / Текст сообщения</label>
264     <textarea class="form-control" id="hd-new-ticket-mes
    sage" rows="4" placeholder="Подробное описание обращения..."><
    /textarea>
265 </div>
266 </div>
267 <div class="modal-footer border-secondary-subtle py-2">
268     <button type="button" class="btn btn-secondary btn-sm"
    data-bs-dismiss="modal">Отмена</button>
269     <button type="button" class="btn btn-primary btn-sm" i
    d="hd-btn-submit-new-ticket">Создать тикет</button>
270 </div>
271 </div>
272 </div>
273 </div>
274
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Helpdesk Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/helpdesk_tab/main.js?v
11  *     =20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/helpdesk_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Helpdesk Tab – Admin Control Panel Logic
22 'use strict';
23
24 (function() {
25   let hdTickets = [];
26   let hdActiveTicketId = null;
27   let hdActiveTicket = null;
28   let hdCurrentFilter = 'all';
29   let hdWs = null;
30   let hdClientId = 'admin_tab_' + Math.random().toString(36).s
31   ubstring(2, 9);
32   let hdStatsInterval = null;
```

```

32 |
33 | async function initHelpdeskTab() {
34 |     console.log('[HelpdeskTab] Initializing Helpdesk multi-too
    | l tab...');
35 |     setupEventListeners();
36 |     initWebSocket();
37 |     await loadTickets();
38 |     await updateStats();
39 |
40 |     if (window.registerTabPoller) {
41 |         window.registerTabPoller('tab-helpdesk', updateStats, 15
    | 000, { immediate: false });
42 |     } else if (!hdStatsInterval) {
43 |         hdStatsInterval = setInterval(updateStats, 15000);
44 |     }
45 | }
46 |
47 | function initWebSocket() {
48 |     if (hdWs && hdWs.readyState === WebSocket.OPEN) return;
49 |     const protocol = window.location.protocol === 'https:' ? '
    | wss:' : 'ws:';
50 |     const wsUrl = `${protocol}//${window.location.host}/api/he
    | lpdesk/ws/${hdClientId}?role=operator`;
51 |
52 |     try {
53 |         hdWs = new WebSocket(wsUrl);
54 |         hdWs.onopen = () => console.log('[HelpdeskTab] WS Connec
    | ted as Admin Operator');
55 |         hdWs.onmessage = (event) => {
56 |             try {
57 |                 const packet = JSON.parse(event.data);
58 |                 handleIncomingPacket(packet);
59 |             } catch (e) {
60 |                 console.error('[HelpdeskTab] WS Parse error:', e);
61 |             }
62 |         };
63 |         hdWs.onclose = () => {
64 |             console.warn('[HelpdeskTab] WS Disconnected, reconnect

```

```

    ing in 5s...');
65     setTimeout(initWebSocket, 5000);
66     };
67     } catch (e) {
68         console.warn('[HelpdeskTab] WS Init failed:', e);
69     }
70 }
71
72 function handleIncomingPacket(packet) {
73     const { type, ticket, message, ticket_id } = packet;
74
75     if (type === 'new_ticket_alert') {
76         if (typeof showNotification === 'function') {
77             showNotification(`🔔 Новое обращение #${ticket?.ticket
_number}: ${ticket?.subject}`, 'warning');
78         }
79         loadTickets();
80         updateStats();
81     } else if (type === 'new_ticket_message') {
82         if (hdActiveTicketId === ticket_id) {
83             appendMessage(message);
84             scrollMessagesToBottom();
85         }
86         loadTickets();
87     } else if (type === 'ticket_updated') {
88         if (hdActiveTicketId === ticket?.id) {
89             hdActiveTicket = ticket;
90             renderActiveTicketHeader(ticket);
91         }
92         loadTickets();
93         updateStats();
94     }
95 }
96
97 async function loadTickets() {
98     const listContainer = document.getElementById('hd-tab-tick
et-list');
99     if (!listContainer) return;

```

```
100
101     try {
102         let url = `/api/helpdesk/tickets?status=${encodeURIComponent(
            hdCurrentFilter)}`;
103         const searchInput = document.getElementById('hd-tab-search-input');
104         const query = searchInput?.value?.trim();
105         if (query) {
106             url += `&search=${encodeURIComponent(query)}`;
107         }
108
109         const res = await fetch(url, { credentials: 'include' })
            ;
110         if (!res.ok) throw new Error(`HTTP ${res.status}`);
111         const data = await res.json();
112         hdTickets = data.tickets || [];
113         renderTicketList(hdTickets);
114     } catch (err) {
115         console.error('[HelpdeskTab] Error loading tickets:', err);
116         listContainer.innerHTML = `<div class="p-3 text-danger small">Ошибка загрузки тикетов: ${err.message}</div>`;
117     }
118 }
119
120 async function updateStats() {
121     try {
122         const res = await fetch('/api/helpdesk/stats', { credentials: 'include' });
123         if (!res.ok) return;
124         const data = await res.json();
125         const stats = data.stats || {};
126
127         const elOpen = document.getElementById('hd-metric-open')
            ;
128         const elProg = document.getElementById('hd-metric-progress');
129         const elRes = document.getElementById('hd-metric-resolve
```



```

    d');
130     const elTot = document.getElementById('hd-metric-total')
    ;
131
132     if (elOpen) elOpen.textContent = stats.open || 0;
133     if (elProg) elProg.textContent = stats.in_progress || 0;
134     if (elRes) elRes.textContent = stats.resolved || 0;
135     if (elTot) elTot.textContent = stats.total || 0;
136   } catch (e) {
137     console.debug('[HelpdeskTab] Stats update skipped:', e);
138   }
139 }
140
141 function renderTicketList(ticketsList) {
142   const listContainer = document.getElementById('hd-tab-tick
et-list');
143   if (!listContainer) return;
144
145   if (!ticketsList || ticketsList.length === 0) {
146     listContainer.innerHTML = '<div class="text-center text-
muted p-4 small">Нет обращений по выбранному фильтру</div>';
147     return;
148   }
149
150   listContainer.innerHTML = ticketsList.map(t => {
151     const isActive = t.id === hdActiveTicketId;
152     const statusBadge = getStatusBadge(t.status);
153     const priorityBadge = getPriorityBadge(t.priority);
154     const timeStr = formatTimeAgo(t.created_at);
155
156     return `
157       <div class="card p-2 border-secondary-subtle ${isActiv
e ? 'bg-primary-subtle border-primary' : 'bg-body'} hover-shad
ow"
158         style="cursor: pointer; transition: all 0.15s eas
e;"
159         onclick="window.selectHelpdeskTicket('${t.id}')">
160       <div class="d-flex justify-content-between align-ite

```

```

ms-center mb-1">
161         <span class="badge bg-secondary-subtle text-body f
w-bold">#${t.ticket_number}</span>
162         <div class="d-flex gap-1">
163             ${priorityBadge}
164             ${statusBadge}
165         </div>
166     </div>
167     <h6 class="fw-bold mb-1 text-truncate" style="font-s
ize: 0.9rem;">${escapeHtml(t.subject)}</h6>
168     <div class="d-flex justify-content-between align-ite
ms-center small text-muted" style="font-size: 0.75rem;">
169         <span class="text-truncate" style="max-width: 140p
x;"><i class="bi bi-person me-1"></i>${escapeHtml(t.user_name
|| t.user_email || 'Пользователь')}</span>
170         <span>${timeStr}</span>
171     </div>
172 </div>
173 `;
174 }).join('');
175 }
176
177 async function selectTicket(ticketId) {
178     hdActiveTicketId = ticketId;
179     renderTicketList(hdTickets); // re-render for active highl
ight
180
181     const emptyState = document.getElementById('hd-tab-empty-s
tate');
182     const activeView = document.getElementById('hd-tab-active-
view');
183     if (emptyState) emptyState.classList.add('d-none');
184     if (activeView) {
185         activeView.classList.remove('d-none');
186         activeView.classList.add('d-flex');
187     }
188
189     try {

```

```

190     const res = await fetch(`/api/helpdesk/tickets/${ticketId}` , { credentials: 'include' });
191     if (!res.ok) throw new Error(`HTTP ${res.status}`);
192     const data = await res.json();
193     hdActiveTicket = data.ticket;
194     renderActiveTicketHeader(data.ticket);
195     renderMessagesStream(data.messages || []);
196     scrollMessagesToBottom();
197   } catch (err) {
198     console.error('[HelpdeskTab] Error loading ticket details:', err);
199     if (typeof showNotification === 'function') showNotification('Ошибка загрузки тикета: ' + err.message, 'danger');
200   }
201 }
202
203 function renderActiveTicketHeader(ticket) {
204   if (!ticket) return;
205   const numEl = document.getElementById('hd-active-number');
206   const subEl = document.getElementById('hd-active-subject');
207   ;
208   const metaEl = document.getElementById('hd-active-meta');
209   const prioLabel = document.getElementById('hd-active-priority-label');
210
211   if (numEl) numEl.textContent = `#${ticket.ticket_number}`;
212   if (subEl) subEl.textContent = ticket.subject;
213   if (metaEl) {
214     const author = ticket.user_name || ticket.user_email || 'User';
215     metaEl.textContent = `Автор: ${author} (${ticket.category || 'technical'}) • ${formatTimeAgo(ticket.created_at)}`;
216   }
217   if (prioLabel) {
218     const pLabels = { low: 'Низкий', normal: 'Обычный', high: 'Высокий', urgent: 'Срочный' };
219     prioLabel.textContent = pLabels[ticket.priority] || ticket.priority;

```

```
219 |     }
220 | }
221 |
222 | function renderMessagesStream(messages) {
223 |     const stream = document.getElementById('hd-tab-messages-stream');
224 |     if (!stream) return;
225 |
226 |     if (!messages || messages.length === 0) {
227 |         stream.innerHTML = '<div class="text-center text-muted p-4 small">В данном тикете пока нет сообщений</div>';
228 |         return;
229 |     }
230 |
231 |     stream.innerHTML = messages.map(m => createMessageHtml(m))
232 |     .join('');
233 | }
234 |
235 | function createMessageHtml(m) {
236 |     const isStaff = m.sender_role === 'operator' || m.sender_role === 'admin';
237 |     const isNote = Boolean(m.is_internal_note);
238 |     const timeStr = new Date(m.created_at).toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
239 |
240 |     let cardBg = isStaff ? 'bg-primary-subtle border-primary-subtle' : 'bg-body-tertiary border-secondary-subtle';
241 |     let align = isStaff ? 'align-self-end' : 'align-self-start';
242 |
243 |     if (isNote) {
244 |         cardBg = 'bg-warning-subtle border-warning';
245 |     }
246 |
247 |     let parsedText = escapeHtml(m.message);
248 |     if (window.marked && typeof window.marked.parse === 'function') {
249 |         try {
```

```

249         parsedText = window.marked.parse(m.message);
250     } catch (e) {}
251 }
252
253 return `
254     <div class="d-flex flex-column ${align}" style="max-widt
255 h: 80%;">
256         <div class="card p-2 ${cardBg} shadow-sm">
257             <div class="d-flex justify-content-between align-ite
258 ms-center mb-1 gap-2 small">
259                 <strong>${isNote ? '📌 Заметка оператора (' + esca
260 peHtml(m.sender_name) + ')': escapeHtml(m.sender_name || (isS
261 taff ? 'Оператор' : 'Пользователь'))}</strong>
262                 <span class="text-muted" style="font-size: 0.75rem
263 ;">${timeStr}</span>
264             </div>
265             <div class="message-content text-break" style="font-
266 size: 0.88rem;">${parsedText}</div>
267         </div>
268     </div>
269 `;
270 }
271
272 function appendMessage(m) {
273     const stream = document.getElementById('hd-tab-messages-st
274 ream');
275     if (!stream) return;
276     const msgDiv = document.createElement('div');
277     msgDiv.innerHTML = createMessageHtml(m);
278     stream.appendChild(msgDiv.firstElementChild);
279 }
280
281 function scrollMessagesToBottom() {
282     const stream = document.getElementById('hd-tab-messages-st
283 ream');
284     if (stream) {
285         setTimeout(() => {
286             stream.scrollTop = stream.scrollHeight;

```

```

279     }, 50);
280 }
281 }
282
283 async function sendReply() {
284     if (!hdActiveTicketId) return;
285     const replyInput = document.getElementById('hd-reply-input
    ');
286     const isInternalNote = document.getElementById('hd-switch-
internal-note')?.checked || false;
287     const text = replyInput?.value?.trim();
288
289     if (!text) return;
290
291     try {
292         const res = await fetch(`/api/helpdesk/tickets/${hdActiv
eTicketId}/messages`, {
293             method: 'POST',
294             headers: { 'Content-Type': 'application/json' },
295             credentials: 'include',
296             body: JSON.stringify({
297                 message: text,
298                 is_internal_note: isInternalNote
299             })
300         });
301
302         if (!res.ok) throw new Error(`HTTP ${res.status}`);
303         replyInput.value = '';
304         const data = await res.json();
305         if (data.message) {
306             appendMessage(data.message);
307             scrollMessagesToBottom();
308         }
309         loadTickets();
310     } catch (err) {
311         console.error('[HelpdeskTab] Error sending message:', er
r);
312         if (typeof showNotification === 'function') showNotifica

```

```

    tion('Ошибка отправки: ' + err.message, 'danger');
313     }
314   }
315
316   async function updateTicketStatus(status) {
317     if (!hdActiveTicketId) return;
318     try {
319       const res = await fetch(`/api/helpdesk/tickets/${hdActiv
eTicketId}/status`, {
320         method: 'PUT',
321         headers: { 'Content-Type': 'application/json' },
322         credentials: 'include',
323         body: JSON.stringify({ status })
324       });
325       if (!res.ok) throw new Error(`HTTP ${res.status}`);
326       if (typeof showNotification === 'function') showNotifica
tion(`Статус изменен на: ${status}`, 'success');
327       loadTickets();
328       updateStats();
329       selectTicket(hdActiveTicketId);
330     } catch (err) {
331       console.error('[HelpdeskTab] Status update failed:', err
);
332     }
333   }
334
335   async function updateTicketPriority(priority) {
336     if (!hdActiveTicketId) return;
337     try {
338       const res = await fetch(`/api/helpdesk/tickets/${hdActiv
eTicketId}/priority`, {
339         method: 'PUT',
340         headers: { 'Content-Type': 'application/json' },
341         credentials: 'include',
342         body: JSON.stringify({ priority })
343       });
344       if (!res.ok) throw new Error(`HTTP ${res.status}`);
345       if (typeof showNotification === 'function') showNotifica

```

```

        tion(`Приоритет изменен на: ${priority}`, 'success');
346     loadTickets();
347     selectTicket(hdActiveTicketId);
348 } catch (err) {
349     console.error('[HelpdeskTab] Priority update failed:', e
rr);
350 }
351 }
352
353 async function generateAiDraft() {
354     if (!hdActiveTicket) return;
355     const replyInput = document.getElementById('hd-reply-input
');
356     if (!replyInput) return;
357
358     replyInput.value = '□ Генерация черновика ответа ИИ...';
359     try {
360         const res = await fetch('/api/chat', {
361             method: 'POST',
362             headers: { 'Content-Type': 'application/json' },
363             credentials: 'include',
364             body: JSON.stringify({
365                 prompt: `Сформулируй вежливый, профессиональный и ко
нкретный ответ технической поддержки Helpdesk на следующее обр
ащение пользователя.
366 Тема: "${hdActiveTicket.subject}"
367 Описание: "${hdActiveTicket.description || ''}"
368 Категория: "${hdActiveTicket.category}"
369 Ответ должен быть кратким и готовым к отправке клиенту.`
,
370                 use_rag: false
371             })
372         });
373
374         if (!res.ok) throw new Error(`HTTP ${res.status}`);
375         const data = await res.json();
376         const answer = data.response || data.text || '';
377         replyInput.value = answer.trim();
378     } catch (e) {

```



```
379     replyInput.value = `Здравствуйте! Мы изучили ваше обращение по теме "${hdActiveTicket.subject}". Специалисты уже занимаются решением.`;
380   }
381 }
382
383 function setupEventListeners() {
384   // Search input debounce
385   const searchInput = document.getElementById('hd-tab-search-input');
386   let searchTimer = null;
387   if (searchInput) {
388     searchInput.oninput = () => {
389       clearTimeout(searchTimer);
390       searchTimer = setTimeout(loadTickets, 300);
391     };
392   }
393
394   // Filter buttons
395   const filterBtns = document.querySelectorAll('#hd-tab-status-filters button');
396   filterBtns.forEach(btn => {
397     btn.onclick = () => {
398       filterBtns.forEach(b => b.classList.remove('active'));
399       btn.classList.add('active');
400       hdCurrentFilter = btn.getAttribute('data-filter') || 'all';
401       loadTickets();
402     };
403   });
404
405   // Send reply button & shortcut
406   const btnSend = document.getElementById('hd-btn-send-reply');
407   if (btnSend) btnSend.onclick = sendReply;
408
409   const replyInput = document.getElementById('hd-reply-input');
```

```
410     if (replyInput) {
411         replyInput.onkeydown = (e) => {
412             if (e.ctrlKey && e.key === 'Enter') {
413                 e.preventDefault();
414                 sendReply();
415             }
416         };
417     }
418
419     // Internal note switcher toggle label
420     const switchNote = document.getElementById('hd-switch-inte
421 rnal-note');
422     const modeBadge = document.getElementById('hd-composer-mod
423 e-badge');
424     if (switchNote && modeBadge) {
425         switchNote.onChange = () => {
426             if (switchNote.checked) {
427                 modeBadge.textContent = '🔒 Режим: Внутренняя скрита
428 я заметка';
429                 modeBadge.className = 'badge bg-warning text-dark';
430             } else {
431                 modeBadge.textContent = 'Режим: Публичный ответ';
432                 modeBadge.className = 'badge bg-secondary-subtle tex
433 t-body';
434             }
435         };
436     }
437
438     // Status action buttons
439     document.querySelectorAll('.hd-status-opt').forEach(btn =>
440 {
441     btn.onclick = () => {
442         const st = btn.getAttribute('data-status');
443         if (st) updateTicketStatus(st);
444     };
445 });
446
447     // Priority dropdown items
```

```
443 |     document.querySelectorAll('.hd-priority-opt').forEach(opt
    => {
444 |         opt.onclick = (e) => {
445 |             e.preventDefault();
446 |             const prio = opt.getAttribute('data-priority');
447 |             if (prio) updateTicketPriority(prio);
448 |         };
449 |     });
450 |
451 |     // Quick canned responses
452 |     document.querySelectorAll('.hd-quick-template').forEach(tp
    lBtn => {
453 |         tplBtn.onclick = () => {
454 |             const tpl = tplBtn.getAttribute('data-tpl');
455 |             if (tpl && replyInput) {
456 |                 replyInput.value = tpl;
457 |                 replyInput.focus();
458 |             }
459 |         };
460 |     });
461 |
462 |     // AI Assist Draft Button
463 |     const aiAssistBtn = document.getElementById('hd-btn-ai-ass
    ist');
464 |     if (aiAssistBtn) aiAssistBtn.onclick = generateAiDraft;
465 |
466 |     // Submit new ticket in modal
467 |     const btnSubmitNew = document.getElementById('hd-btn-submi
    t-new-ticket');
468 |     if (btnSubmitNew) {
469 |         btnSubmitNew.onclick = async () => {
470 |             const subject = document.getElementById('hd-new-ticket
    -subject')?.value?.trim();
471 |             const category = document.getElementById('hd-new-ticke
    t-category')?.value || 'technical';
472 |             const priority = document.getElementById('hd-new-ticke
    t-priority')?.value || 'normal';
473 |             const message = document.getElementById('hd-new-ticket
```

```

    -message')?.value?.trim());
474
475     if (!subject || !message) {
476         window.showToast?(`Пожалуйста, заполните тему и тек
ст обращения.', 'warning') || alert('Пожалуйста, заполните тем
у и текст обращения.');
```

477 return;

478 }

479

480 try {

481 btnSubmitNew.disabled = true;

482 const res = await fetch('/api/helpdesk/tickets', {

483 method: 'POST',

484 headers: { 'Content-Type': 'application/json' },

485 credentials: 'include',

486 body: JSON.stringify({ subject, category, priority

, message })

487 });

488

489 if (!res.ok) throw new Error(`HTTP \${res.status}`);

490 const data = await res.json();

491 if (data.status === 'success') {

492 window.showToast?(`Тикет #\${data.ticket.ticket_nu
mber} успешно создан!`, 'success');

493 const modalEl = document.getElementById('hdCreateT
icketModal');

494 const modal = bootstrap.Modal.getInstance(modalEl)

;

495 modal?.hide();

496

497 document.getElementById('hd-new-ticket-subject').v

alue = '';

498 document.getElementById('hd-new-ticket-message').v

alue = '';

499

500 await loadTickets();

501 await updateStats();

502 if (data.ticket?.id) {

```

503         selectTicket(data.ticket.id);
504     }
505 }
506 } catch (err) {
507     console.error('[HelpdeskTab] Create ticket failed:',
err);
508     window.showToast?.('Ошибка создания тикета: ' + err.
message, 'danger') || alert('Ошибка создания тикета: ' + err.m
essage);
509 } finally {
510     btnSubmitNew.disabled = false;
511 }
512 };
513 }
514 }
515
516 function getStatusBadge(status) {
517     switch (status) {
518         case 'open': return '<span class="badge bg-danger-subtle
text-danger">Новый</span>';
519         case 'in_progress': return '<span class="badge bg-warnin
g-subtle text-warning">В работе</span>';
520         case 'resolved': return '<span class="badge bg-success-s
ubtle text-success">Решено</span>';
521         case 'closed': return '<span class="badge bg-secondary-s
ubtle text-body">Закрыт</span>';
522         default: return `<span class="badge bg-secondary">${stat
us}</span>`;
523     }
524 }
525
526 function getPriorityBadge(priority) {
527     switch (priority) {
528         case 'urgent': return '<span class="badge bg-danger">⚡
Срочно</span>';
529         case 'high': return '<span class="badge bg-warning text-
dark">⚠️ Высокий</span>';
530         case 'low': return '<span class="badge bg-secondary-subt

```

```

le text-muted">␣ Низкий</span>';
531     default: return '<span class="badge bg-info-subtle text-
info">␣ Обычный</span>';
532   }
533 }
534
535 function formatTimeAgo(isoStr) {
536   if (!isoStr) return '';
537   try {
538     const diffMs = Date.now() - new Date(isoStr).getTime();
539     const diffMin = Math.floor(diffMs / 60000);
540     if (diffMin < 1) return 'только что';
541     if (diffMin < 60) return `${diffMin} мин назад`;
542     const diffHours = Math.floor(diffMin / 60);
543     if (diffHours < 24) return `${diffHours} ч назад`;
544     return `${Math.floor(diffHours / 24)} дн назад`;
545   } catch (e) {
546     return '';
547   }
548 }
549
550 function escapeHtml(text) {
551   if (!text) return '';
552   const div = document.createElement('div');
553   div.textContent = text;
554   return div.innerHTML;
555 }
556
557 // Global window functions for event hooks
558 window.initHelpdeskTab = initHelpdeskTab;
559 window.refreshHelpdeskTab = () => { loadTickets(); updateSta
ts(); };
560 window.selectHelpdeskTicket = selectTicket;
561 window.openCreateTicketModal = () => {
562   const modalEl = document.getElementById('hdCreateTicketMod
al');
563   if (modalEl) {
564     const modal = new bootstrap.Modal(modalEl);

```

```
565 |         modal.show();
566 |     }
567 | };
568 |})();
569 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/index.html" class="w-10
17     0 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-08 04:24:00
25 =====
26 =====
27 -->
28
29 <!DOCTYPE html>
30 <html lang="ru">
31 <head>
32   <meta charset="UTF-8">
33   <meta name="viewport" content="width=device-width, initial-s
34   cale=1.0">
35   <title>AI Assistant</title>
36   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
37   t/css/bootstrap.min.css" rel="stylesheet">
```



```
31 <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    0.0/font/bootstrap-icons.css" rel="stylesheet">
32 <link href="/html/css/main.css?v=20261008_v1" rel="styleshee
    t">
33 <script src="/html/js/toast.js?v=20261008_v1"></script>
34 <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
    8next.min.js"></script>
35 <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
    js"></script>
36 </head>
37 <body>
38 <div class="container-fluid mt-1">
39
40 <div class="d-flex align-items-center justify-content-betwee
    n mb-1 main-nav-container rounded px-2 py-1">
41 <!-- Левая часть: кнопка меню, брендинг и индикатор активн
    ого раздела -->
42 <div class="d-flex align-items-center gap-2">
43 <!-- Кнопка открытия левого выдвижного меню -->
44 <button class="btn btn-sm btn-primary d-flex align-items
    -center gap-1.5 px-2 py-1 rounded shadow-sm"
45         id="left-drawer-toggle-btn"
46         type="button"
47         data-bs-toggle="offcanvas"
48         data-bs-target="#leftSideNavOffcanvas"
49         aria-controls="leftSideNavOffcanvas"
50         title="Открыть меню разделов"
51         style="font-size: 0.82rem; font-weight: 500;">
52 <i class="bi bi-list fs-6"></i>
53 <span class="d-none d-md-inline">Меню</span>
54 </button>
55
56 <!-- Брендинг -->
57 <span class="fw-bold d-flex align-items-center gap-1.5 t
    ext-primary fs-6 user-select-none" style="letter-spacing: 0.3p
    x;">
58 <i class="bi bi-cpu-fill fs-5"></i>
59 <span class="d-none d-sm-inline">AI Breadboard</span>
```

```

60     </span>
61 </div>
62
63     <!-- Правая часть панели: статусы, профиль и настройки -->
64     <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto" id="nav-auth-container">
65         <!-- Active Model & Search Badges -->
66         <span id="chat-model-badge" class="badge rounded-pill al
ign-items-center gap-1 px-3 py-1 shadow-sm border" style="font
-size: 0.8rem; display: none; cursor: pointer;" title="Текущая
модель ИИ">
67             □ Загрузка...
68         </span>
69
70         <!-- Model Quick Test Dropdown Trigger & Popover -->
71         <div class="dropdown" id="model-ping-dropdown-wrapper">
72             <button class="btn btn-sm btn-outline-secondary d-flex
align-items-center justify-content-center p-0 rounded-circle"
73
74                 id="btn-model-ping-gear"
75                 type="button"
76                 data-bs-toggle="dropdown"
77                 data-bs-auto-close="outside"
78                 aria-expanded="false"
79                 title="Проверка доступности модели"
80                 style="width: 28px; height: 28px;">
81                 <i class="bi bi-gear"></i>
82             </button>
83             <div class="dropdown-menu dropdown-menu-end shadow-lg
border border-secondary-subtle p-3" aria-labelledby="btn-model
-ping-gear" style="min-width: 360px; max-width: 460px; z-index
: 1060;">
84                 <div class="d-flex align-items-center justify-conten
t-between mb-2">
85                     <span class="fw-bold small d-flex align-items-cent
er gap-1.5 text-body">
86                         <i class="bi bi-cpu text-primary"></i>
                        <span>Проверка доступности модели</span>

```

```

87         </span>
88     </div>
89     <div class="mb-2">
90         <input type="text" class="form-control form-control-sm" id="model-ping-input" value="Проверка доступности модели" placeholder="Введите сообщение...">
91     </div>
92     <div class="d-flex justify-content-end mb-2">
93         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5 px-3" id="btn-model-ping-send" type="button">
94             <i class="bi bi-send-fill" id="model-ping-btn-icon"></i>
95             <span id="model-ping-btn-text">Отправить</span>
96         </button>
97     </div>
98     <div id="model-ping-response-container" style="display: none;">
99         <label class="form-label small text-muted mb-1 fw-semibold">Ответ модели:</label>
100         <div id="model-ping-response" class="small p-2 rounded bg-body-tertiary border text-break font-monospace" style="max-height: 220px; overflow-y: auto; white-space: pre-wrap; font-size: 0.8rem; user-select: text;"></div>
101     </div>
102 </div>
103 </div>
104
105     <span id="chat-search-badge" class="badge rounded-pill align-items-center gap-1 px-3 py-1 shadow-sm border" style="font-size: 0.8rem; display: none;" title="Провайдер поиска">
106 </span>
107
108     <!-- Google OAuth Register / Sign-in Button (Shown when not logged in) -->
109     <button class="btn btn-sm btn-light text-dark fw-semibold d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-pill btn-google-login" id="nav-google-btn" title="Войти или за

```

```

    регистрироваться через Google" data-i18n-title="auth.googleSig
nInTitle" style="border: 1px solid rgba(0,0,0,0.15);">
110      <svg width="18" height="18" viewBox="0 0 24 24"><path
fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h
5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.
74 3.28-8.09z"/><path fill="#34A853" d="M12 23c2.97 0 5.46-.98
7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-
1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/><path fill
="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.
35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.
85-2.22.81-.63z"/><path fill="#EA4335" d="M12 5.38c1.62 0 3.06
.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.4
7 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/></svg>
111      <span data-i18n="auth.googleSignUp">Регистрация через
Google</span>
112    </button>
113
114    <!-- Authenticated User Menu Dropdown (Shown when logged
in) -->
115    <div class="dropdown d-none" id="user-menu-dropdown-wrap
per">
116      <button class="btn btn-sm btn-outline-secondary d-flex
align-items-center gap-2 px-2 py-1 rounded-pill dropdown-togg
le" id="user-settings-btn" type="button" data-bs-toggle="dropd
own" aria-expanded="false" title="Пользовательское меню">
117        <img id="nav-user-avatar" src="" class="rounded-circ
le d-none" width="22" height="22" style="object-fit: cover;">
118        <i class="bi bi-person-circle" id="nav-user-icon"></
i>
119        <span id="nav-user-label" class="small fw-semibold"
data-i18n="userSettings.button">Пользователь</span>
120      </button>
121      <ul class="dropdown-menu dropdown-menu-end shadow bord
er-secondary-subtle py-1" aria-labelledby="user-settings-btn"
style="min-width: 220px;">
122        <li class="px-3 py-2 border-bottom border-secondary-
subtle">
123          <div class="fw-bold text-truncate" id="user-menu-d

```

```

    display-name">Пользователь</div>
124         <div class="small text-muted text-truncate" id="us
er-menu-display-email" style="font-size: 0.75rem;">user@gmail.
com</div>
125     </li>
126     <li><h6 class="dropdown-header small text-uppercase
text-muted fw-semibold mt-1">Сервисы</h6></li>
127     <li>
128         <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-mail">
129             <i class="bi bi-envelope-check text-primary"></i
>
130             <span>Почта</span>
131         </button>
132     </li>
133     <li>
134         <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-docs">
135             <i class="bi bi-file-earmark-text text-info"></i
>
136             <span>Google Docs</span>
137         </button>
138     </li>
139     <li>
140         <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-sheets">
141             <i class="bi bi-file-earmark-spreadsheet text-su
ccess"></i>
142             <span>Google Таблицы</span>
143         </button>
144     </li>
145     <li>
146         <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-news">
147             <i class="bi bi-newspaper text-warning"></i>
148             <span>Новости</span>
149         </button>
150     </li>

```

```

151         <li>
152             <button class="dropdown-item d-flex align-items-ce
153             nter gap-2 py-1" type="button" id="user-menu-btn-facebook">
154                 <i class="bi bi-facebook text-primary"></i>
155                 <span>Facebook</span>
156             </button>
157         </li>
158         <li><hr class="dropdown-divider my-1"></li>
159         <li>
160             <button class="dropdown-item d-flex align-items-ce
161             nter gap-2 py-1" type="button" id="user-menu-btn-cache" data-b
162             s-toggle="modal" data-bs-target="#cacheManagerModal">
163                 <i class="bi bi-hdd-stack text-info"></i>
164                 <span data-i18n="cacheManager.menuItem">Кеш браузер
165                 зера</span>
166             </button>
167         </li>
168         <li>
169             <button class="dropdown-item d-flex align-items-ce
170             nter gap-2 py-1" type="button" data-bs-toggle="modal" data-bs-
171             target="#userSettingsModal">
172                 <i class="bi bi-gear-fill text-secondary"></i>
173                 <span data-i18n="userSettings.title">Настройки п
174                 рофиля</span>
175             </button>
176         </li>
177         <li>
178             <button class="dropdown-item d-flex align-items-ce
179             nter gap-2 py-1 text-danger btn-user-logout" type="button">
180                 <i class="bi bi-box-arrow-right"></i>
181                 <span>Выйти</span>
182             </button>
183         </li>
184     </ul>
185 </div>
186
187 <!-- Кнопка кеша браузера -->
188 <button class="btn btn-sm btn-outline-secondary d-flex a

```

```

    align-items-center justify-content-center rounded-circle" id="b
    tn-open-cache-manager" data-bs-toggle="modal" data-bs-target="
    #cacheManagerModal" title="Кеш бpayзepa" style="width: 28px; h
    eight: 28px; padding: 0;" data-i18n-title="cacheManager.title"
    >
181         <i class="bi bi-hdd-stack"></i>
182     </button>
183
184     <!-- Quick Settings Gear Button -->
185     <button class="btn btn-sm btn-outline-secondary d-flex a
    lign-items-center justify-content-center rounded-circle" id="u
    ser-settings-gear-btn" data-bs-toggle="modal" data-bs-target="
    #userSettingsModal" title="Настройки" style="width: 28px; heig
    ht: 28px; padding: 0;" data-i18n-title="userSettings.title">
186         <i class="bi bi-gear-fill"></i>
187     </button>
188 </div>
189 </div>
190
191 <!-- Плавающий боковой ярлычок (вкладка) слева -->
192 <button class="left-drawer-tab-trigger btn shadow"
193     type="button"
194     data-bs-toggle="offcanvas"
195     data-bs-target="#leftSideNavOffcanvas"
196     aria-controls="leftSideNavOffcanvas"
197     title="Открыть меню навигации">
198     <i class="bi bi-list fs-6"></i>
199     <span class="left-drawer-tab-text">МЕНЮ</span>
200 </button>
201
202 <!-- Левая выдвижная вкладка / панель навигации (Offcanvas)
    -->
203 <div class="offcanvas offcanvas-start left-drawer-nav" tabin
    dex="-1" id="leftSideNavOffcanvas" aria-labelledby="leftSideNa
    vOffcanvasLabel">
204     <div class="offcanvas-header border-bottom py-2 px-3">
205         <h5 class="offcanvas-title d-flex align-items-center gap
    -2 fs-6 fw-bold" id="leftSideNavOffcanvasLabel">

```

```

206         <i class="bi bi-grid-fill text-primary"></i>
207         <span>Навигация и разделы</span>
208     </h5>
209     <button type="button" class="btn-close text-reset" data-
bs-dismiss="offcanvas" aria-label="Заккрыть"></button>
210 </div>
211 <div class="offcanvas-body p-3 d-flex flex-column justify-
content-between" id="mainTabs">
212     <div class="drawer-sections-wrapper">
213         <!-- Группа: Стандартные и Чат -->
214         <div class="nav-drawer-group mb-3">
215             <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
216                 <i class="bi bi-chat-dots-fill text-primary"></i>
217                 <span data-i18n="tabs.groupStandard">Стандартные и
Чат</span>
218             </div>
219             <div class="list-group list-group-flush rounded bord
er">
220                 <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3 active" data-tab
="tab-chat" data-bs-target="#tab-chat" type="button">
221                     <span class="fs-6">☑ </span>
222                     <span class="fw-medium" data-i18n="tabs.chat">Ча
т</span>
223                 </button>
224                 <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-v
oice" data-bs-target="#tab-voice" type="button">
225                     <span class="fs-6">☐ </span>
226                     <span class="fw-medium" data-i18n="tabs.voice">Г
олос и диаризация</span>
227                 </button>
228                 <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
cenarios" data-bs-target="#tab-scenarios" type="button">
229                     <span class="fs-6">☐ </span>
230                     <span class="fw-medium" data-i18n="tabs.scenario

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s">Сценарии и чат</span>
231         </button>
232     </div>
233 </div>
234
235     <!-- Группа: Оборудование и Телеметрия -->
236     <div class="nav-drawer-group mb-3">
237         <div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
238             <i class="bi bi-cpu-fill text-info"></i>
239             <span>Оборудование и Телеметрия</span>
240         </div>
241         <div class="list-group list-group-flush rounded border">
242             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-about-system" data-bs-target="#tab-about-system" type="button">
243                 <span class="fs-6">i</span>
244                 <span class="fw-medium" data-i18n="tabs.aboutSystem">0 Системе</span>
245             </button>
246             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-hardware-monitor" data-bs-target="#tab-hardware-monitor" type="button">
247                 <span class="fs-6">□ </span>
248                 <span class="fw-medium" data-i18n="tabs.sensorLogs">Датчики и сенсоры (LHM)</span>
249             </button>
250             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-telemetry-research" data-bs-target="#tab-telemetry-research" type="button">
251                 <span class="fs-6">□ </span>
252                 <span class="fw-medium">Исследования телеметрии<
/span>
253             </button>
254             <button class="list-group-item list-group-item-action

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ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-t
elemetry-config" data-bs-target="#tab-telemetry-config" type="
button">
255         <span class="fs-6">⬢ </span>
256         <span class="fw-medium">Конфигурация телеметрии<
/span>
257     </button>
258     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-a
i-benchmark" data-bs-target="#tab-ai-benchmark" type="button">
259         <span class="fs-6">⬢ </span>
260         <span class="fw-medium">AI Benchmark</span>
261     </button>
262 </div>
263 </div>
264
265 <!-- Группа: Дисковая подсистема и накопители -->
266 <div class="nav-drawer-group mb-3">
267     <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
268         <i class="bi bi-hdd-fill text-warning"></i>
269         <span>Дисковая подсистема</span>
270     </div>
271     <div class="list-group list-group-flush rounded bord
er">
272         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
torage-manager" data-bs-target="#tab-storage-manager" type="bu
tton">
273             <span class="fs-6">⬢ </span>
274             <span class="fw-medium">Диски и тома (Storage)</
span>
275         </button>
276         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-d
isk-speed" data-bs-target="#tab-disk-speed" type="button">
277             <span class="fs-6">⬢ </span>
278             <span class="fw-medium">Скорость дисков (DiskSpd

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) </span>
279         </button>
280         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
torage-wear" data-bs-target="#tab-storage-wear" type="button">
281             <span class="fs-6">⬅ </span>
282             <span class="fw-medium">Износ дисков и батареи</
span>
283         </button>
284         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-f
ile-recovery" data-bs-target="#tab-file-recovery" type="button
">
285             <span class="fs-6">⬅ </span>
286             <span class="fw-medium" data-i18n="tabs.fileReco
very">Восстановление файлов</span>
287         </button>
288         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-f
ile-history-search" data-bs-target="#tab-file-history-search"
type="button">
289             <span class="fs-6">⬅ </span>
290             <span class="fw-medium" data-i18n="tabs.fileHist
orySearch">Поиск истории файлов AI</span>
291         </button>
292     </div>
293 </div>
294
295     <!-- Группа: Диагностика и форензика -->
296     <div class="nav-drawer-group mb-3">
297         <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
298             <i class="bi bi-shield-check text-success"></i>
299             <span>Диагностика и форензика</span>
300         </div>
301         <div class="list-group list-group-flush rounded bord
er">
302             <button class="list-group-item list-group-item-act

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ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
rocess-leaks" data-bs-target="#tab-process-leaks" type="button
">
303         <span class="fs-6">□ </span>
304         <span class="fw-medium">1. Утечки и микрофризы</
span>
305     </button>
306     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-f
orensics" data-bs-target="#tab-forensics" type="button">
307         <span class="fs-6">□ </span>
308         <span class="fw-medium">2. Поведенческая форензи
ка</span>
309     </button>
310     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-t
hrottling" data-bs-target="#tab-throttling" type="button">
311         <span class="fs-6">□ </span>
312         <span class="fw-medium">3. Качество ядра и Тротт
линг</span>
313     </button>
314     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
eripherals" data-bs-target="#tab-peripherals" type="button">
315         <span class="fs-6">□ </span>
316         <span class="fw-medium">4. Сеть и периферия</spa
n>
317     </button>
318     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
oftware-transparency" data-bs-target="#tab-software-transparen
cy" type="button">
319         <span class="fs-6">□ </span>
320         <span class="fw-medium" data-i18n="tabs.software
Transparency">5. Прозрачность ПО</span>
321     </button>
322 </div>
323 </div>

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```

324 |
325 |         <!-- Группа: Управление Windows и Sysadmin -->
326 |         <div class="nav-drawer-group mb-3">
327 |             <div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
328 |                 <i class="bi bi-gear-wide-connected text-primary">
329 |                     <span>Управление Windows</span>
330 |                 </div>
331 |                 <div class="list-group list-group-flush rounded border">
332 |                     <button class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2 px-3 text-warning-emphasis" id="btn-open-god-mode" type="button"
333 |                         onclick="openGodModeFolder()" title="Открыть панель всех настроек Windows (God Mode)">
334 |                         <div class="d-flex align-items-center gap-2">
335 |                             <span class="fs-6">⚙️ </span>
336 |                             <span class="fw-medium">God Mode (Все настроек
337 |                             и)</span>
338 |                         </div>
339 |                         <i class="bi bi-box-arrow-up-right text-muted small"></i>
340 |                     </button>
341 |                     <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-system-control" data-bs-target="#tab-system-control" type="button">
342 |                         <span class="fs-6">📋 </span>
343 |                         <span class="fw-medium">System Logs & Activity</span>
344 |                     </button>
345 |                     <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-post-install-wizard" data-bs-target="#tab-post-install-wizard" type="button">
346 |                         <span class="fs-6">🔧 </span>
347 |                         <span class="fw-medium">Post-Install Wizard</span>

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n>
346         </button>
347         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-m
aintenance-recovery" data-bs-target="#tab-maintenance-recovery
" type="button">
348             <span class="fs-6">□ </span>
349             <span class="fw-medium">Maintenance & Recovery</
span>
350         </button>
351         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-w
indows-admin" data-bs-target="#tab-windows-admin" type="button
">
352             <span class="fs-6">□ </span>
353             <span class="fw-medium">Windows Sysadmin</span>
354         </button>
355         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-f
ocus-settings" data-bs-target="#tab-focus-settings" type="butt
on">
356             <span class="fs-6">□ </span>
357             <span class="fw-medium">Настройки Focus</span>
358         </button>
359         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
tartup-auditor" data-bs-target="#tab-startup-auditor" type="bu
tton">
360             <span class="fs-6">□ </span>
361             <span class="fw-medium" data-i18n="tabs.startupA
uditor">Автозапуск программ</span>
362         </button>
363         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
ervices-manager" data-bs-target="#tab-services-manager" type="
button">
364             <span class="fs-6">□ </span>
365             <span class="fw-medium">Службы Windows</span>

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366         </button>
367         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-t
ask-scheduler" data-bs-target="#tab-task-scheduler" type="butt
on">
368             <span class="fs-6">□ </span>
369             <span class="fw-medium">Планировщик задач</span>
370         </button>
371         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
rocess-manager" data-bs-target="#tab-process-manager" type="bu
tton">
372             <span class="fs-6">□ </span>
373             <span class="fw-medium">Диспетчер процессов</spa
n>
374         </button>
375         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-r
egistry-viewer" data-bs-target="#tab-registry-viewer" type="bu
tton">
376             <span class="fs-6">□ </span>
377             <span class="fw-medium">Редактор реестра</span>
378         </button>
379         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
oftware-manager" data-bs-target="#tab-software-manager" type="
button">
380             <span class="fs-6">□ </span>
381             <span class="fw-medium">Пакетный менеджер Winget
</span>
382         </button>
383         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-u
ser-directories" data-bs-target="#tab-user-directories" type="
button">
384             <span class="fs-6">□ </span>
385             <span class="fw-medium" data-i18n="tabs.userDire
ctories">Директории пользователей</span>

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386         </button>
387         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-a
ccounts-identity" data-bs-target="#tab-accounts-identity" type
="button">
388             <span class="fs-6">☐ </span>
389             <span class="fw-medium">Учетные записи (Users &
Identity)</span>
390         </button>
391     </div>
392 </div>
393
394     <!-- Группа: Безопасность и Целостность -->
395     <div class="nav-drawer-group mb-3">
396         <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
397             <i class="bi bi-shield-lock-fill text-danger"></i>
398             <span>Безопасность и Обслуживание</span>
399         </div>
400         <div class="list-group list-group-flush rounded bord
er">
401             <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-d
efender" data-bs-target="#tab-defender" type="button">
402                 <span class="fs-6">☐ </span>
403                 <span class="fw-medium">Defender Security</span>
404             </button>
405             <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-f
irewall-manager" data-bs-target="#tab-firewall-manager" type="
button">
406                 <span class="fs-6">☐ </span>
407                 <span class="fw-medium">Брандмауэр Windows</span>
408             </button>
409             <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
ecurity-acl" data-bs-target="#tab-security-acl" type="button">

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410 |         <span class="fs-6">□ </span>
411 |         <span class="fw-medium">Права доступа и ACL</spa
n>
412 |     </button>
413 |     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
ervicing-integrity" data-bs-target="#tab-servicing-integrity"
type="button">
414 |         <span class="fs-6">□ </span>
415 |         <span class="fw-medium">Обслуживание SFC/DISM</s
pan>
416 |     </button>
417 |     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-b
oot-recovery" data-bs-target="#tab-boot-recovery" type="button
">
418 |         <span class="fs-6">□ </span>
419 |         <span class="fw-medium">Загрузка и WinRE</span>
420 |     </button>
421 |     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-w
indows-backup" data-bs-target="#tab-windows-backup" type="butt
on">
422 |         <span class="fs-6">□ </span>
423 |         <span class="fw-medium">Windows Backup</span>
424 |     </button>
425 | </div>
426 | </div>
427 |
428 | <!-- Группа: Сеть и Логи -->
429 | <div class="nav-drawer-group mb-3">
430 |     <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
431 |         <i class="bi bi-hdd-network-fill text-secondary"><
/i>
432 |         <span>Сеть и Журналы</span>
433 |     </div>
434 |     <div class="list-group list-group-flush rounded bord

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```
er">
435         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-n
etwork" data-bs-target="#tab-network" type="button">
436             <span class="fs-6">□ </span>
437             <span class="fw-medium" data-i18n="tabs.network"
>Сетевой терминал</span>
438         </button>
439         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
erformance-tracing" data-bs-target="#tab-performance-tracing"
type="button">
440             <span class="fs-6">□ </span>
441             <span class="fw-medium">Трассировка ETW (logman)
</span>
442         </button>
443
444         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
ystem-logs" data-bs-target="#tab-system-logs" type="button">
445             <span class="fs-6">□ </span>
446             <span class="fw-medium" data-i18n="tabs.systemLo
gs">Логи операционной системы</span>
447         </button>
448         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-l
og-settings" data-bs-target="#tab-log-settings" type="button">
449             <span class="fs-6">□ </span>
450             <span class="fw-medium">Параметры и настройка жу
рналов</span>
451         </button>
452         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
ower-lifecycle" data-bs-target="#tab-power-lifecycle" type="bu
tton">
453             <span class="fs-6">□ </span>
454             <span class="fw-medium">Сессии питания и перезаг
рузки</span>
```

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455         </button>
456     </div>
457 </div>
458
459     <!-- Группа: Расширенные и ИИ -->
460     <div class="nav-drawer-group mb-3">
461         <div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
462             <i class="bi bi-sliders2 text-info"></i>
463             <span data-i18n="tabs.groupExtended">Расширенные и
464             ИИ</span>
465         </div>
466         <div class="list-group list-group-flush rounded border">
467             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-rag" data-bs-target="#tab-rag" type="button">
468                 <span class="fs-6">☐ </span>
469                 <span class="fw-medium" data-i18n="tabs.rag">RAG
470             </span>
471             </button>
472             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-models" data-bs-target="#tab-models" type="button">
473                 <span class="fs-6">☐ </span>
474                 <span class="fw-medium" data-i18n="tabs.models">
475                 Модели и API</span>
476             </button>
477             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-news" data-bs-target="#tab-news" type="button">
478                 <span class="fs-6">☐ </span>
479                 <span class="fw-medium" data-i18n="tabs.news">Новости</span>
480             </button>
481             <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-admin" data-bs-target="#tab-admin" type="button">

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479         <span class="fs-6">□ </span>
480         <span class="fw-medium" data-i18n="tabs.admin">У
правление</span>
481     </button>
482     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-h
elp" data-bs-target="#tab-help" type="button">
483         <span class="fs-6">□ </span>
484         <span class="fw-medium" data-i18n="tabs.help">Сп
равочник</span>
485     </button>
486 </div>
487 </div>
488
489 <!-- Группа: Плагины -->
490 <div class="nav-drawer-group mb-3" id="pluginsNavDropd
ownWrapper">
491     <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
492         <i class="bi bi-puzzle-fill text-warning"></i>
493         <span data-i18n="tabs.plugins">Плагины</span>
494     </div>
495     <div class="list-group list-group-flush rounded bord
er" id="nav-plugins-dropdown-menu">
496         <div id="nav-enabled-plugins-list" class="nav-enab
led-plugins-list">
497             <!-- Dynamic enabled plugins will be injected he
re -->
498         </div>
499         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
lugins" data-bs-target="#tab-plugins" type="button">
500             <span class="fs-6">□ </span>
501             <span class="fw-medium" data-i18n="tabs.managePl
ugins">Управление плагинами</span>
502         </button>
503     </div>
504 </div>

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505 |         </div>
506 |
507 |         <!-- Footer Drawer -->
508 |         <div class="nav-drawer-footer border-top pt-2 mt-auto d-
flex align-items-center justify-content-between text-muted sma
ll">
509 |             <span class="fw-semibold">AI Breadboard</span>
510 |             <span>v3.1</span>
511 |         </div>
512 |     </div>
513 | </div>
514 |
515 | <div class="tab-content" id="mainTabContent">
516 |     <!-- Tabs will be injected via JavaScript -->
517 |     <div id="tab-chat" class="tab-pane fade show active"></div>
518 | >
519 |     <div id="tab-voice" class="tab-pane fade"></div>
520 |     <div id="tab-scenarios" class="tab-pane fade"></div>
521 |     <div id="tab-about-system" class="tab-pane fade"></div>
522 |     <div id="tab-hardware-monitor" class="tab-pane fade"></div>
523 | >
524 |     <div id="tab-telemetry-research" class="tab-pane fade"></d
iv>
525 |     <div id="tab-telemetry-config" class="tab-pane fade"></div>
526 | >
527 |     <div id="tab-ai-benchmark" class="tab-pane fade"></div>
528 |     <div id="tab-storage-manager" class="tab-pane fade"></div>
529 |     <div id="tab-disk-speed" class="tab-pane fade"></div>
530 |     <div id="tab-storage-wear" class="tab-pane fade"></div>
531 |     <div id="tab-file-recovery" class="tab-pane fade"></div>
532 |     <div id="tab-file-history-search" class="tab-pane fade"></
div>
533 |     <div id="tab-process-leaks" class="tab-pane fade"></div>
534 |     <div id="tab-forensics" class="tab-pane fade"></div>
535 |     <div id="tab-throttling" class="tab-pane fade"></div>
536 |     <div id="tab-peripherals" class="tab-pane fade"></div>
537 |     <div id="tab-software-transparency" class="tab-pane fade">
</div>

```

```
535 | <div id="tab-system-control" class="tab-pane fade"></div>
536 | <div id="tab-post-install-wizard" class="tab-pane fade"></div>
    | div>
537 | <div id="tab-maintenance-recovery" class="tab-pane fade"></div>
    | /div>
538 | <div id="tab-windows-admin" class="tab-pane fade"></div>
539 | <div id="tab-focus-settings" class="tab-pane fade"></div>
540 | <div id="tab-startup-auditor" class="tab-pane fade"></div>
541 | <div id="tab-services-manager" class="tab-pane fade"></div>
    | >
542 | <div id="tab-task-scheduler" class="tab-pane fade"></div>
543 | <div id="tab-process-manager" class="tab-pane fade"></div>
544 | <div id="tab-registry-viewer" class="tab-pane fade"></div>
545 | <div id="tab-software-manager" class="tab-pane fade"></div>
    | >
546 | <div id="tab-user-directories" class="tab-pane fade"></div>
    | >
547 | <div id="tab-defender" class="tab-pane fade"></div>
548 | <div id="tab-firewall-manager" class="tab-pane fade"></div>
    | >
549 | <div id="tab-security-acl" class="tab-pane fade"></div>
550 | <div id="tab-servicing-integrity" class="tab-pane fade"></div>
    | div>
551 | <div id="tab-boot-recovery" class="tab-pane fade"></div>
552 | <div id="tab-windows-backup" class="tab-pane fade"></div>
553 | <div id="tab-network" class="tab-pane fade"></div>
554 | <div id="tab-performance-tracing" class="tab-pane fade"></div>
    | div>
555 |
556 | <div id="tab-system-logs" class="tab-pane fade"></div>
557 | <div id="tab-log-settings" class="tab-pane fade"></div>
558 | <div id="tab-power-lifecycle" class="tab-pane fade"></div>
559 | <div id="tab-rag" class="tab-pane fade"></div>
560 | <div id="tab-telegram-rag" class="tab-pane fade"></div>
561 | <div id="tab-models" class="tab-pane fade"></div>
562 | <div id="tab-news" class="tab-pane fade"></div>
563 | <div id="tab-plugins" class="tab-pane fade"></div>
564 | <div id="tab-admin" class="tab-pane fade"></div>
```

```
565 |     <div id="tab-accounts-identity" class="tab-pane fade"></div>
    | v>
566 |     <div id="tab-help" class="tab-pane fade"></div>
567 | </div>
568 | </div>
569 | </div>
570 |
571 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    | /js/bootstrap.bundle.min.js"></script>
572 | <script src="/html/js/ai_table_modal.js"></script>
573 | <script src="/html/js/chatService.js?v=20261008_v2"></script>
574 | <script src="/html/js/window-api.js?v=20261008_v2"></script>
575 | <script type="module" src="/html/js/main.js?v=20261008_v2"></s
    | cript>
576 |
577 | <!-- HELP Modal -->
578 | <div class="modal fade modal-help" id="help-modal" tabindex="-
    | 1" aria-labelledby="help-modal-label" aria-hidden="true">
579 |     <div class="modal-dialog modal-dialog-scrollable">
580 |         <div class="modal-content">
581 |             <div class="modal-header">
582 |                 <h5 class="modal-title" id="help-modal-label" data-i18
    | n="helpModal.title">Помощь</h5>
583 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
584 |             </div>
585 |             <div class="modal-body" id="help-modal-content">
586 |                 <p data-i18n="common.loading">Загрузка...</p>
587 |             </div>
588 |             <div class="modal-footer">
589 |                 <button type="button" class="btn btn-secondary" data-b
    | s-dismiss="modal" data-i18n="helpModal.close">Закрыть</button>
590 |             </div>
591 |         </div>
592 |     </div>
593 | </div>
594 |
595 | <!-- User Settings Modal -->
```

```
596 | <div class="modal fade" id="userSettingsModal" tabindex="-1" a
    | ria-labelledby="userSettingsModalLabel" aria-hidden="true">
597 |   <div class="modal-dialog modal-dialog-centered">
598 |     <div class="modal-content shadow border-secondary-subtle">
599 |       <div class="modal-header border-secondary-subtle">
600 |         <h5 class="modal-title d-flex align-items-center gap-2
    | " id="userSettingsModalLabel">
601 |           <i class="bi bi-person-gear text-primary"></i>
602 |           <span data-i18n="userSettings.title">Пользовательски
    | е настройки</span>
603 |         </h5>
604 |         <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
605 |       </div>
606 |       <div class="modal-body p-3">
607 |         <!-- User Account & Google OAuth Card -->
608 |         <div class="card bg-body-tertiary border-secondary-sub
    | tle mb-3 p-3" id="user-settings-auth-card">
609 |           <div class="d-flex align-items-center gap-3">
610 |             <img id="user-settings-avatar" src="" class="round
    | ed-circle border border-2 border-primary" width="48" height="4
    | 8" style="object-fit: cover;">
611 |             <div class="flex-grow-1 min-w-0">
612 |               <div class="d-flex align-items-center gap-2">
613 |                 <strong id="user-settings-name" class="text-tr
    |uncate">Пользователь</strong>
614 |                 <span id="user-settings-role-badge" class="bad
    |ge bg-primary">User</span>
615 |               </div>
616 |               <div id="user-settings-email" class="small text-
    |muted text-truncate">email@gmail.com</div>
617 |               <div id="user-settings-google-badge" class="smal
    |l mt-1 text-muted">
618 |                 <i class="bi bi-google text-danger me-1"></i>G
    |oogle OAuth
619 |               </div>
620 |             </div>
621 |           <button type="button" class="btn btn-outline-dange
```



```

    r btn-sm btn-user-logout" title="Выйти">
622         <i class="bi bi-box-arrow-right"></i>
623     </button>
624 </div>
625 </div>
626
627 <!-- Guest / Connect Google OAuth Card -->
628 <div class="card bg-body-tertiary border-secondary-sub
tle mb-3 p-3 text-center d-none" id="user-settings-guest-card"
>
629     <div class="text-muted small mb-2">
630         <i class="bi bi-shield-lock me-1"></i> Авторизация
и синхронизация аккаунта
631     </div>
632     <button type="button" class="btn btn-outline-danger
btn-sm w-100 d-flex align-items-center justify-content-center
gap-2 btn-google-login">
633         <i class="bi bi-google"></i>
634         <span>Войти через Google (OAuth)</span>
635     </button>
636     <div class="small text-muted mt-2" style="font-size:
0.75rem;">
637         Синхронизация с Google Drive, Docs, Calendar и Con
tacts
638     </div>
639 </div>
640
641 <!-- Language Selection -->
642 <div class="mb-3">
643     <label class="form-label fw-semibold small d-flex al
ign-items-center gap-2" for="modal-lang-selector">
644         <i class="bi bi-translate text-info"></i>
645         <span data-i18n="userSettings.language">Язык интер
фейса</span>
646     </label>
647     <select class="form-select lang-selector" id="modal-
lang-selector">
648         <option value="ru-RU">Русский (ru-RU)</option>

```

```

649         <option value="en-US">English (en-US)</option>
650         <option value="he-IL">עברית (he-IL)</option>
651         <option value="uk-UA">Українська (uk-UA)</option>
652     </select>
653 </div>
654
655 <!-- Theme Selection -->
656 <div class="mb-2">
657     <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-theme-selector">
658         <i class="bi bi-palette text-warning"></i>
659         <span data-i18n="userSettings.theme">Тема оформлення</span>
660     </label>
661     <select class="form-select theme-selector" id="modal-theme-selector">
662         <option value="system" data-i18n="userSettings.themeSystem">Системная</option>
663         <optgroup label="Светлые темы">
664             <option value="light" data-i18n="userSettings.themeLight">Светлая (Классическая)</option>
665             <option value="brick" data-i18n="userSettings.themeBrick">Кирпичная (Терракота)</option>
666         </optgroup>
667         <optgroup label="Тёмные темы">
668             <option value="dark" data-i18n="userSettings.themeDark">Тёмная (Контрастная)</option>
669             <option value="terminal" data-i18n="userSettings.themeTerminal">Терминал (Matrix / CRT)</option>
670         </optgroup>
671     </select>
672 </div>
673 </div>
674 <!-- Cache Manager Modal -->
675 <div class="modal fade" id="cacheManagerModal" tabindex="-1" aria-labelledby="cacheManagerModalLabel" aria-hidden="true">
676     <div class="modal-dialog modal-dialog-centered modal-lg">
677         <div class="modal-content shadow border-secondary-subtle">

```

```

678 |         <div class="modal-header border-secondary-subtle">
679 |             <div class="d-flex align-items-center gap-2">
680 |                 <div class="rounded-circle bg-primary-subtle text-pr
        imary p-2 d-flex align-items-center justify-content-center" st
        yle="width: 36px; height: 36px;">
681 |                     <i class="bi bi-hdd-stack fs-5"></i>
682 |                 </div>
683 |                 <div>
684 |                     <h5 class="modal-title mb-0" id="cacheManagerModal
        Label" data-i18n="cacheManager.title">Управление браузерным ке
        шем</h5>
685 |                     <div class="small text-muted" style="font-size: 0.
        78rem;" data-i18n="cacheManager.subtitle">Хранилище больших да
        нных (IndexedDB + Memory Cache) для ускорения работы и офлайн-
        доступа.</div>
686 |                 </div>
687 |             </div>
688 |             <button type="button" class="btn-close" data-bs-dismis
        s="modal" aria-label="Close"></button>
689 |         </div>
690 |         <div class="modal-body p-3">
691 |             <!-- Storage Quota Bar Card -->
692 |             <div class="card bg-body-tertiary border-secondary-sub
        tle p-3 mb-3">
693 |                 <div class="d-flex align-items-center justify-conten
        t-between mb-2">
694 |                     <div class="fw-semibold small d-flex align-items-c
        enter gap-1.5">
695 |                         <i class="bi bi-pie-chart-fill text-primary"></i
        >
696 |                         <span data-i18n="cacheManager.quotaTitle">Дисков
        ая квота браузера</span>
697 |                     </div>
698 |                     <span id="cache-persist-badge" class="badge bg-sec
        ondary-subtle text-secondary border">Проверка...</span>
699 |                 </div>
700 |                 <div class="progress mb-2" style="height: 10px;">
701 |                     <div id="cache-usage-progressbar" class="progress-

```

```

    bar bg-primary" role="progressbar" style="width: 0%"></div>
702         </div>
703         <div class="d-flex justify-content-between align-items-center small text-muted" style="font-size: 0.78rem;">
704             <span>Использование памяти:</span>
705             <span id="cache-usage-text" class="fw-semibold text-body">Загрузка...</span>
706         </div>
707     </div>
708
709     <!-- Stores List -->
710     <div class="mb-3">
711         <div class="d-flex align-items-center justify-content-between mb-2">
712             <h6 class="fw-bold small text-uppercase text-muted mb-0" data-i18n="cacheManager.storesTitle">Хранилища данных</h6>
713             <button class="btn btn-sm btn-outline-primary py-0 px-2 rounded-pill small" id="cache-btn-cleanup-expired" style="font-size: 0.75rem;">
714                 <i class="bi bi-magic me-1"></i><span data-i18n="cacheManager.btnCleanExpired">Очистить просроченное</span>
715             </button>
716         </div>
717         <div id="cache-stores-list">
718             <!-- Populated dynamically by cache-ui.js -->
719         </div>
720     </div>
721
722     <!-- Action Tools -->
723     <div class="d-flex flex-wrap align-items-center justify-content-between gap-2 pt-2 border-top border-secondary-subtle">
724         <div class="d-flex align-items-center gap-2">
725             <button type="button" class="btn btn-outline-secondary btn-sm" id="cache-btn-request-persist" data-i18n="cacheManager.btnRequestPersist">
726                 <i class="bi bi-shield-lock me-1"></i>Сделать по

```

СТОЯННЫМ

```
727         </button>
728         <button type="button" class="btn btn-outline-secondary btn-sm" id="cache-btn-export" data-i18n="cacheManager.btnExport">
729             <i class="bi bi-download me-1"></i>Экспорт JSON
730         </button>
731         <label class="btn btn-outline-secondary btn-sm mb-0" for="cache-file-import" style="cursor: pointer;">
732             <i class="bi bi-upload me-1"></i><span data-i18n="cacheManager.btnImport">Импорт JSON</span>
733             <input type="file" id="cache-file-import" accept=".json" class="d-none">
734         </label>
735     </div>
736     <button type="button" class="btn btn-outline-danger btn-sm" id="cache-btn-clear-all" data-i18n="cacheManager.btnClearAll">
737         <i class="bi bi-trash3 me-1"></i>Очистить всё
738     </button>
739 </div>
740 </div>
741 <div class="modal-footer border-secondary-subtle py-2">
742     <button type="button" class="btn btn-secondary btn-sm" data-bs-dismiss="modal" data-i18n="userSettings.close">Закрыть</button>
743 </div>
744 </div>
745 </div>
746 </div>
747
748 </body>
749 </html>
750
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/instructions_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/instructions_tab/index.
17     html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/instructions_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Instructions Tab -->
30 <div class="container-fluid p-3">
31   <div class="alert alert-info">
32     <h5 class="mb-1"><i class="bi bi-gear"></i> Редактор систе
33     мных инструкций</h5>
34     <p class="mb-0 small">Управление системными инструкциями А
35     И: чат и Narrator режимы, версионирование, проверка в модели.<
36     /p>
37   </div>
```

```
30 |
31 | <!-- Mode Switcher -->
32 | <div class="card mb-3">
33 |   <div class="card-body">
34 |     <div class="d-flex align-items-center gap-2 flex-wrap">
35 |       <span class="text-muted">Режим:</span>
36 |       <div class="btn-group" role="group">
37 |         <button type="button" class="btn btn-outline-primary
| active" id="mode-chat" onclick="switchMode('chat')">
38 |           <i class="bi bi-chat-text"></i> Chat
39 |         </button>
40 |         <button type="button" class="btn btn-outline-seconda
| ry" id="mode-narrator" onclick="switchMode('narrator')">
41 |           <i class="bi bi-mic"></i> Narrator
42 |         </button>
43 |       </div>
44 |       <small class="text-muted ms-2">Активный файл: <code id
| ="active-file-info">—</code></small>
45 |       <div class="ms-auto d-flex gap-2">
46 |         <button type="button" class="btn btn-success" id="bt
| n-check-model" onclick="checkInModel()">
47 |           <i class="bi bi-check-circle"></i> Проверить в мод
| ели
48 |         </button>
49 |         <button type="button" class="btn btn-primary" id="bt
| n-save" onclick="saveInstruction()">
50 |           <i class="bi bi-save"></i> Сохранить версию
51 |         </button>
52 |         <button type="button" class="btn btn-warning" id="bt
| n-preview" onclick="previewPrompt()">
53 |           <i class="bi bi-eye"></i> Предпросмотр
54 |         </button>
55 |       </div>
56 |     </div>
57 |   </div>
58 | </div>
59 |
60 | <!-- Editor Area -->
```

```

61 | <div class="card mb-3">
62 |   <div class="card-header">
63 |     <div class="d-flex justify-content-between align-items-c
| enter">
64 |       <span class="fw-bold">Редактор – <span id="editor-mode
| -label"> Chat</span></span>
65 |       <small class="text-muted">Markdown поддерживается</sma
| ll>
66 |     </div>
67 |   </div>
68 |   <div class="card-body p-0">
69 |     <textarea id="instruction-editor" class="form-control bo
| rder-0 rounded-0" rows="18"
70 |       placeholder="Введите системную инструкцию..." spellche
| ck="false"
71 |       style="font-family: 'Courier New', monospace; font-siz
| e: 0.9rem; resize: vertical;"></textarea>
72 |   </div>
73 | </div>
74 |
75 | <!-- Versions Panel -->
76 | <div class="card mb-3">
77 |   <div class="card-header d-flex justify-content-between ali
| gn-items-center">
78 |     <span class="fw-bold"><i class="bi bi-clock-history"></i>
| > История версий</span>
79 |     <button type="button" class="btn btn-sm btn-outline-seco
| ndary" onclick="refreshVersions()">
80 |       <i class="bi bi-arrow-clockwise"></i> Обновить
81 |     </button>
82 |   </div>
83 |   <div class="card-body p-0">
84 |     <div id="versions-list" class="list-group list-group-flu
| sh">
85 |       <div class="list-group-item text-muted">Загрузка верси
| й...</div>
86 |     </div>
87 |   </div>

```



```

88 | </div>
89 |
90 | <!-- Token Counter -->
91 | <div class="card">
92 |     <div class="card-header">
93 |         <span class="fw-bold"><i class="bi bi-calculator"></i> C
    | статистика</span>
94 |     </div>
95 |     <div class="card-body">
96 |         <div class="row g-3">
97 |             <div class="col-md-4">
98 |                 <div class="p-3 bg-light rounded text-center">
99 |                     <div class="h3 mb-0" id="char-count">0</div>
100 |                     <div class="small text-muted">Символов</div>
101 |                 </div>
102 |             </div>
103 |             <div class="col-md-4">
104 |                 <div class="p-3 bg-light rounded text-center">
105 |                     <div class="h3 mb-0" id="word-count">0</div>
106 |                     <div class="small text-muted">Слов</div>
107 |                 </div>
108 |             </div>
109 |             <div class="col-md-4">
110 |                 <div class="p-3 bg-light rounded text-center">
111 |                     <div class="h3 mb-0" id="token-estimate">~0</div>
112 |                     <div class="small text-muted">Токенов (оценка)</di
    | v>
113 |                 </div>
114 |             </div>
115 |         </div>
116 |         <div class="mt-3 small text-muted border rounded p-2" id
    | ="token-preview" style="min-height:2rem;"></div>
117 |     </div>
118 | </div>
119 | </div>
120 |
121 | <!-- Prompt Preview Modal -->
122 | <div class="modal fade" id="prompt-preview-modal" tabindex="-1

```

```
    " aria-labelledby="prompt-preview-modal-label" aria-hidden="true">
123 |   <div class="modal-dialog modal-lg modal-dialog-scrollable">
124 |     <div class="modal-content">
125 |       <div class="modal-header">
126 |         <h5 class="modal-title" id="prompt-preview-modal-label">Предпросмотр инструкции</h5>
127 |         <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
128 |       </div>
129 |       <div class="modal-body" id="prompt-preview-content">
130 |         <p>Загрузка...</p>
131 |       </div>
132 |       <div class="modal-footer">
133 |         <button type="button" class="btn btn-secondary" data-bs-dismiss="modal">Закрыть</button>
134 |       </div>
135 |     </div>
136 |   </div>
137 </div>
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Instructions Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *   <script src="/windows/api/webgui/instructions_tab/main.
11 *   js?v=20261001_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/instructions_tab
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-06 22:15:00
19 * =====
20 */
21 // Instructions Tab - UI Logic
22 // Инициализируется через window.initInstructionsTab() из admin/main.js
23
24 'use strict';
25
26 import { initI18n, switchLang, applyTranslations } from '../js/i18n.js';
27
28 window.switchLang = switchLang;
29 window.applyTranslations = applyTranslations;
30
```

```

31 // — Helpers —————
32
33 function adminFetch(url, options = {}) {
34     return window.api
35         ? window.api.fetch(url, options)
36         : fetch(url, options).then(r => {
37             if (!r.ok) throw new Error(`HTTP ${r.status}`);
38             return r.json();
39         });
40 }
41
42 function notify(msg, type = 'info') {
43     if (typeof showNotification === 'function') {
44         showNotification(msg, type);
45     } else {
46         console.log(`[${type}] ${msg}`);
47     }
48 }
49
50 // — Mode Management —————
51 let currentMode = 'chat';
52
53 function switchMode(mode) {
54     currentMode = mode;
55     document.getElementById('mode-chat').classList.toggle('active', mode === 'chat');
56     document.getElementById('mode-narrator').classList.toggle('active', mode === 'narrator');
57
58     // Обновить заголовок редактора
59     const modeLabel = mode === 'chat' ? 'Чат' : 'Рассказчик';
60     const editorTitle = document.getElementById('editor-mode-label');
61     if (editorTitle) editorTitle.textContent = modeLabel;
62

```

```
63 | loadInstruction();
64 | refreshVersions();
65 | }
66 |
67 | // — Instructions API —————
68 | async function loadInstruction() {
69 |   try {
70 |     const data = await adminFetch(`/api/admin/instructions?mode=${currentMode}`);
71 |     const editor = document.getElementById('instruction-editor');
72 |     if (data && data.content !== undefined) {
73 |       editor.value = data.content;
74 |     } else {
75 |       editor.value = '';
76 |     }
77 |     updateStats();
78 |     const fileInfo = document.getElementById('active-file-info');
79 |     if (fileInfo && data.file) {
80 |       fileInfo.textContent = data.file;
81 |     }
82 |   } catch (e) {
83 |     notify('Ошибка загрузки инструкции: ' + e.message, 'danger');
84 |     document.getElementById('instruction-editor').value = '';
85 |   }
86 | }
87 |
88 | async function saveInstruction() {
89 |   const content = document.getElementById('instruction-editor').value.trim();
90 |   if (!content) {
91 |     notify('Инструкция не может быть пустой', 'warning');
92 |     return;
93 |   }
94 | }
```

```
95 |   const btn = document.getElementById('btn-save');
96 |   const originalHtml = btn.innerHTML;
97 |   btn.disabled = true;
98 |   btn.innerHTML = '<i class="bi bi-hourglass-split"></i> Сохранение...';
99 |
100 |   try {
101 |     const data = await adminFetch('/api/admin/instructions/save', {
102 |       method: 'POST',
103 |       headers: { 'Content-Type': 'application/json' },
104 |       body: JSON.stringify({ mode: currentMode, content })
105 |     });
106 |     notify(`Инструкция сохранена: ${data.version || ''}`, 'success');
107 |     await refreshVersions();
108 |   } catch (e) {
109 |     notify('Ошибка сохранения: ' + e.message, 'danger');
110 |   } finally {
111 |     btn.disabled = false;
112 |     btn.innerHTML = originalHtml;
113 |   }
114 | }
115 |
116 | async function checkInModel() {
117 |   const content = document.getElementById('instruction-editor').value.trim();
118 |   const prompt = "Привет! Подтверди, что ты получил инструкцию и готов к работе. Ответь только: 'Понял, готов к работе!'";
119 |
120 |   const btn = document.getElementById('btn-check-model');
121 |   const originalHtml = btn.innerHTML;
122 |   btn.disabled = true;
123 |   btn.innerHTML = '<i class="bi bi-hourglass-split"></i> Проверка...';
124 |
125 |   try {
126 |     const result = await adminFetch('/api/admin/instructions/c
```

```

heck', {
127     method: 'POST',
128     headers: { 'Content-Type': 'application/json' },
129     body: JSON.stringify({ mode: currentMode, prompt, instru
ction: content })
130 });
131
132     const statusEl = document.getElementById('token-preview');
133     let tokenInfo = '';
134     if (result.token_count) {
135         tokenInfo = `<br>Использовано токенов: ${result.token_co
unt}`;
136     }
137
138     if (result.response) {
139         statusEl.innerHTML = `<strong>Ответ модели:</strong><br>
${escapeHtml(result.response)}${tokenInfo}`;
140         notify('Проверка завершена', 'success');
141     } else {
142         statusEl.innerHTML = 'Ошибка: нет ответа от модели';
143         notify('Ошибка проверки', 'danger');
144     }
145 } catch (e) {
146     document.getElementById('token-preview').innerHTML = `<str
ong class="text-danger">Ошибка проверки: ${e.message}</strong>
`;
147     notify('Ошибка проверки: ' + e.message, 'danger');
148 } finally {
149     btn.disabled = false;
150     btn.innerHTML = originalHtml;
151 }
152 }
153
154 async function refreshVersions() {
155     const list = document.getElementById('versions-list');
156     list.innerHTML = '<div class="list-group-item text-muted"><i
class="bi bi-hourglass-split"></i> Загрузка...</div>';
157

```

```

158 |   try {
159 |       const data = await adminFetch(`/api/admin/instructions/ver
    | sions?mode=${currentMode}`);
160 |
161 |       if (!data.versions || !data.versions.length) {
162 |           list.innerHTML = '<div class="list-group-item text-muted
    | ">Нет сохранённых версий</div>';
163 |           return;
164 |       }
165 |
166 |       list.innerHTML = data.versions.map(v => `
167 |           <button type="button"
168 |               class="list-group-item list-group-item-action d-flex j
    | ustify-content-between align-items-center ${v.is_active ? 'lis
    | t-group-item-success' : ''}"
169 |               onclick="useVersion('${escapeAttr(v.filename)}', '${es
    | capeAttr(v.mode)}')">
170 |               <div class="me-2 overflow-hidden">
171 |                   <div class="d-flex align-items-center gap-2 mb-1">
172 |                       <strong>${v.mode === 'chat' ? '❏ ' : '❏ '} ${escap
    | eHtml(v.filename)}</strong>
173 |                       ${v.is_active ? '<span class="badge bg-success">Ак
    | тивная</span>' : ''}
174 |                   </div>
175 |                   <small class="text-muted">${new Date(v.created_at).t
    | oLocaleString('ru-RU')} · ${v.size} байт</small>
176 |                   <div class="text-muted small text-truncate" style="m
    | ax-width:480px;">${escapeHtml(v.preview || '')}</div>
177 |               </div>
178 |               <i class="bi bi-arrow-right-circle${v.is_active ? '-fi
    | ll text-success' : ' text-secondary'}"></i>
179 |           </button>
180 |       `).join('');
181 |   } catch (e) {
182 |       list.innerHTML = `<div class="list-group-item text-danger"
    | >Ошибка загрузки версий: ${e.message}</div>`;
183 |   }
184 | }

```



```
185
186 async function useVersion(filename, mode) {
187   try {
188     await adminFetch('/api/admin/instructions/activate', {
189       method: 'POST',
190       headers: { 'Content-Type': 'application/json' },
191       body: JSON.stringify({ mode, filename })
192     });
193
194     if (mode === currentMode) {
195       await loadInstruction();
196     }
197     await refreshVersions();
198     notify(`Версия ${filename} активирована`, 'success');
199   } catch (e) {
200     notify('Ошибка активации: ' + e.message, 'danger');
201   }
202 }
203
204 async function previewPrompt() {
205   const content = document.getElementById('instruction-editor')
206     .value.trim();
207   const preview = `
208     <strong>Контекст системной инструкции (${currentMode === '
209     chat' ? ' Chat' : ' Narrator'}):</strong><br>
210     <pre style="white-space:pre-wrap;max-height:400px;overflow-
211     -y:auto;">${escapeHtml(content.substring(0, 1000))}${content.l
212     length > 1000 ? '\n...' : ''}</pre>
213     <strong>Тестовый запрос:</strong><br>
214     "Привет! Подтверди, что ты получил инструкцию и готов к ра
215     боте."
216   `;
217
218   const modal = new bootstrap.Modal(document.getElementById('p
219   rompt-preview-modal'));
220   document.getElementById('prompt-preview-content').innerHTML
221     = preview;
222   modal.show();
223 }
```

```
216 }
217
218 // — Stats —————
219 function updateStats() {
220   const content = document.getElementById('instruction-editor')
221     .value;
222   document.getElementById('char-count').textContent = content.
223     length;
224   const words = content.trim().split(/\s+/).filter(w => w.leng
225     th > 0);
226   document.getElementById('word-count').textContent = words.le
227     ngth;
228   const estimatedTokens = Math.ceil(content.length / 3);
229   document.getElementById('token-estimate').textContent = '~'
230     + estimatedTokens;
231   return estimatedTokens;
232 }
233 function escapeHtml(text) {
234   if (!text) return '';
235   const div = document.createElement('div');
236   div.textContent = text;
237   return div.innerHTML.replace(/\n/g, '<br>');
238 }
239
240 function escapeAttr(text) {
241   if (!text) return '';
242   return text.replace(/'/g, '\\\'').replace(/"/g, '&quot;');
243 }
244
245 // — Initialization —————
246 let initialized = false;
```

```
247
248 async function initInstructionsTab() {
249   if (initialized) return;
250   initialized = true;
251
252   console.log('[InstructionsTab] Initializing...');
253
254   try {
255     await initI18n();
256     console.log('[InstructionsTab] i18n initialized');
257   } catch (e) {
258     console.error('[InstructionsTab] i18n init error:', e);
259   }
260
261   // Load instruction for current mode
262   await loadInstruction();
263
264   // Load versions
265   await refreshVersions();
266
267   // Stats on input
268   document.getElementById('instruction-editor').addEventListener('input', updateStats);
269
270   try {
271     applyTranslations();
272   } catch (e) {
273     console.error('[InstructionsTab] applyTranslations error:',
274       , e);
275   }
276
277   console.log('[InstructionsTab] Initialized successfully');
278 }
279 // — Экспорт в window (необходим для inline onclick в HTML пр
280   и type="module") —
281 window.initInstructionsTab = initInstructionsTab;
282 window.switchMode          = switchMode;
```

```
282 | window.saveInstruction      = saveInstruction;
283 | window.checkInModel        = checkInModel;
284 | window.refreshVersions     = refreshVersions;
285 | window.useVersion          = useVersion;
286 | window.previewPrompt       = previewPrompt;
```

activityTracker.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Activitytracker Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля activityT
7  *   racker.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/activityTracker.js?v=20261001_v1" type="module"></script>
11  *
12  *   JavaScript Import:
13  *     import { isUserAuthenticated, trackEvent, trackTabSwitch } from '/windows/api/webgui/js/activityTracker.js';
14  *
15  * File: activityTracker.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/js
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 /**
25  * activityTracker.js – Client-side Telemetry & User Activity
26  *   Tracker
27  *
28  * Tracks tab views, dwell durations, and UI button/action clicks strictly
29  * for authenticated registered users and flushes batches to /
```

```

    api/telemetry/events.
29  */
30
31 'use strict';
32
33 let currentTabId = 'tab-chat';
34 let tabEnterTimestamp = Date.now();
35 let eventQueue = [];
36 let flushTimer = null;
37 let isInitialized = false;
38
39 /**
40  * Check if the current user session is an authenticated regis
    tered user.
41  * @returns {boolean}
42  */
43 export function isUserAuthenticated() {
44     if (window.currentUser && window.currentUser.authenticated)
45     {
46         return true;
47     }
48     // Check auth_token cookie or session storage flag
49     return document.cookie.includes('auth_token=');
50 }
51 /**
52  * Push an event to the internal batch queue if user is authen
    ticated.
53  * @param {Object} event
54  */
55 export function trackEvent(event) {
56     if (!isUserAuthenticated()) {
57         return;
58     }
59
60     const payload = {
61         action: event.action || event.event_type || 'action',
62         event_type: event.event_type || 'action',

```

```

63     tab_name: event.tab_name || currentTabId,
64     target_element: event.target_element || '',
65     duration_ms: event.duration_ms || 0,
66     details: event.details || null,
67     timestamp: new Date().toISOString()
68 };
69
70 eventQueue.push(payload);
71
72 if (eventQueue.length >= 20) {
73     flushEvents();
74 }
75 }
76
77 /**
78  * Record a tab switch / tab view event with dwell time calcul
79  * @param {string} newTabId
80  */
81 export function trackTabSwitch(newTabId) {
82     if (!newTabId) return;
83     const now = Date.now();
84     const cleanNewTab = newTabId.startsWith('#') ? newTabId.slic
85 e(1) : newTabId;
86
87 if (currentTabId && currentTabId !== cleanNewTab) {
88     const duration = now - tabEnterTimestamp;
89     trackEvent({
90         action: `Left tab ${currentTabId}`,
91         event_type: 'tab_view',
92         tab_name: currentTabId,
93         target_element: `tab:${currentTabId}`,
94         duration_ms: duration,
95         details: {
96             transition_to: cleanNewTab,
97             duration_sec: Math.round(duration / 1000)
98         }
99     });

```

```

99 | }
100 |
101 | currentTabId = cleanNewTab;
102 | tabEnterTimestamp = now;
103 |
104 | trackEvent({
105 |     action: `Entered tab ${cleanNewTab}`,
106 |     event_type: 'tab_view',
107 |     tab_name: cleanNewTab,
108 |     target_element: `tab:${cleanNewTab}`,
109 |     duration_ms: 0,
110 |     details: {
111 |         timestamp: new Date().toISOString()
112 |     }
113 | });
114 | }
115 |
116 | /**
117 |  * Send queued events to the backend telemetry ingestion endpo
118 |  */
119 | export async function flushEvents() {
120 |     if (!isUserAuthenticated() || eventQueue.length === 0) {
121 |         return;
122 |     }
123 |
124 |     const batch = [...eventQueue];
125 |     eventQueue = [];
126 |
127 |     const body = JSON.stringify({
128 |         events: batch,
129 |         session_id: window.currentUser?.email || 'authenticated_se
130 | ssion'
131 |     });
132 |
133 |     try {
134 |         if (navigator.sendBeacon) {
135 |             const blob = new Blob([body], { type: 'application/json'

```



```

    });
135     const success = navigator.sendBeacon('/api/telemetry/events', blob);
136     if (success) return;
137 }
138
139     await fetch('/api/telemetry/events', {
140         method: 'POST',
141         headers: {
142             'Content-Type': 'application/json'
143         },
144         body,
145         keepalive: true
146     });
147 } catch (err) {
148     console.debug('Telemetry flush failed, restoring queue:',
149 err);
149     // Put events back in queue up to maximum limit
150     if (eventQueue.length < 50) {
151         eventQueue = [...batch, ...eventQueue];
152     }
153 }
154 }
155
156 /**
157  * Inspect click target and capture non-sensitive interactive
158  * element info.
159  * @param {MouseEvent} event
160  */
161 function handleGlobalClick(event) {
162     if (!isUserAuthenticated()) return;
163
164     const target = event.target;
165     if (!target) return;
166
167     // Find closest interactive element
168     const interactiveEl = target.closest(
169         'button, a, input[type="button"], input[type="submit"], [r

```

```

    ole="button"], .nav-link, .dropdown-item, .btn, .badge[role="button"], [data-action], [data-tab]'
169 | );
170 |
171 | if (!interactiveEl) return;
172 |
173 | // Skip sensitive elements (passwords, auth inputs, tokens)
174 | const isSensitive =
175 |     interactiveEl.type === 'password' ||
176 |     interactiveEl.name === 'password' ||
177 |     interactiveEl.id?.includes('password') ||
178 |     interactiveEl.id?.includes('token') ||
179 |     interactiveEl.classList?.contains('sensitive');
180 |
181 | if (isSensitive) return;
182 |
183 | // Extract meaningful descriptive label
184 | let label = (
185 |     interactiveEl.getAttribute('data-action') ||
186 |     interactiveEl.getAttribute('data-tab') ||
187 |     interactiveEl.getAttribute('aria-label') ||
188 |     interactiveEl.getAttribute('title') ||
189 |     interactiveEl.innerText?.trim().slice(0, 60) ||
190 |     interactiveEl.id ||
191 |     interactiveEl.name ||
192 |     interactiveEl.tagName.toLowerCase()
193 | );
194 |
195 | // Clean label whitespace
196 | label = label.replace(/\s+/g, ' ').trim();
197 | if (!label) return;
198 |
199 | const targetDescriptor = `${interactiveEl.tagName.toLowerCase()}${interactiveEl.id ? '#' + interactiveEl.id : ''}${interactiveEl.className ? '.' + interactiveEl.className.split(' ').filter(c => !c.startsWith('show') && !c.startsWith('active')).slice(0, 2).join('.') : ''}`;

```

200

```
201 | trackEvent({
202 |   action: `Click: ${label}`,
203 |   event_type: 'click',
204 |   tab_name: currentTabId,
205 |   target_element: targetDescriptor,
206 |   details: {
207 |     label,
208 |     tag: interactiveEl.tagName.toLowerCase(),
209 |     id: interactiveEl.id || null,
210 |     data_action: interactiveEl.getAttribute('data-action') |
    | | null,
211 |     data_tab: interactiveEl.getAttribute('data-tab') || null
212 |   }
213 | });
214 | }
215 |
216 | /**
217 |  * Initialize Activity Tracker listeners and flush loop.
218 |  */
219 | export function initActivityTracker() {
220 |   if (isInitialized) return;
221 |   isInitialized = true;
222 |
223 |   // Listen for click interactions
224 |   document.addEventListener('click', handleGlobalClick, { capture: true, passive: true });
225 |
226 |   // Periodic flush every 5 seconds
227 |   flushTimer = setInterval(flushEvents, 5000);
228 |
229 |   // Flush on visibility change / window unload
230 |   document.addEventListener('visibilitychange', () => {
231 |     if (document.visibilityState === 'hidden') {
232 |       flushEvents();
233 |     }
234 |   });
235 |   window.addEventListener('pagehide', () => {
236 |     flushEvents();
```

```
237 | });
238 |
239 | console.log('▯ Activity Tracker initialized (registered user
    | s only)');
240 | }
241 |
242 | // Global exposure for legacy scripts
243 | window.activityTracker = {
244 |   trackTabSwitch,
245 |   trackEvent,
246 |   flushEvents,
247 |   initActivityTracker
248 | };
249 |
```

ai_table_modal.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Ai Table Modal Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля ai_table_
7   *   modal.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/ai_table_modal.js?v
11  *     =20261001_v1" type="module"></script>
12  *
13  * File: ai_table_modal.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/js
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Universal AI Table Modal Module for AI-Breadboard
22 // Enables interactive row inspection and AI contextual explan
23 // ation across all data tables.
24 (function () {
25     const MODAL_ID = 'universal-ai-table-modal';
26
27     function ensureModalElement() {
28         let modalEl = document.getElementById(MODAL_ID);
29         if (!modalEl) {
30             modalEl = document.createElement('div');
```

```

31     modalEl.id = MODAL_ID;
32     modalEl.className = 'modal fade';
33     modalEl.tabIndex = -1;
34     modalEl.setAttribute('aria-hidden', 'true');
35     modalEl.innerHTML = `
36         <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
37             <div class="modal-content bg-dark text-light border-secondary shadow-lg">
38                 <div class="modal-header border-secondary py-2 px-3">
39                     <div class="d-flex align-items-center gap-2 flex-wrap">
40                         <span class="fs-5 text-info" id="uai-modal-icon">❏ </span>
41                         <h5 class="modal-title fw-bold text-info mb-0" id="uai-modal-title">Детали записи</h5>
42                         <div id="uai-modal-badges" class="d-flex align-items-center gap-1"></div>
43                     </div>
44                     <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
45                 </div>
46                 <div class="modal-body py-3 px-3">
47                     <!-- Metadata Grid -->
48                     <div class="row g-2 mb-3 small" id="uai-modal-metadata-grid"></div>
49
50                     <!-- Raw / Code Content Section -->
51                     <div class="card bg-black border-secondary p-3 mb-3" id="uai-modal-row-container" style="display: none;">
52                         <div class="d-flex justify-content-between align-items-center mb-1">
53                             <h6 class="small text-uppercase text-muted fw-bold mb-0" id="uai-modal-row-title">Контекст / Данные</h6>
54                         </div>
55                         <div class="small font-monospace text-light" id="uai-modal-row-content" style="white-space: pre-wrap; word-b

```

```

reak: break-all;"></div>
56         </div>
57
58         <!-- AI Contextual Diagnostic Card -->
59         <div class="card border-info bg-dark p-3" id="uai-modal-ai-section">
60             <div class="d-flex justify-content-between align-items-center mb-2 flex-wrap gap-2">
61                 <div class="d-flex align-items-center gap-2">
62                     <h6 class="fw-bold text-info mb-0"><i class="bi bi-robot me-1"></i> AI Contextual Explanation</h6>
63                     <span class="badge bg-info-subtle text-info border border-info-subtle small px-2 py-0.5" id="uai-modal-web-status" title="Сведения обогащаются поиском в интернете">
64                         <i class="bi bi-globe me-1"></i>Web Grouping
65                     </span>
66                 </div>
67                 <div class="d-flex align-items-center gap-1.5">
68                     <button class="btn btn-sm btn-outline-secondary py-0 px-2 rounded-pill" id="btn-uai-modal-edit-prompt" title="Настроить промпт для этого типа таблицы">
69                         <i class="bi bi-gear-fill me-1"></i>Промпт
70                     </button>
71                     <button class="btn btn-sm btn-outline-info py-0 px-2 rounded-pill" id="btn-uai-modal-refresh" title="Обновить AI-анализ">
72                         <i class="bi bi-arrow-clockwise"></i>
73                     </button>
74                 </div>
75             </div>
76             <div class="small" id="uai-modal-ai-explanation">
77                 <div class="d-flex align-items-center gap-2 py-2 text-info">

```

```

78 |         <div class="spinner-border spinner-border-
    | sm" role="status"></div>
79 |         <span>Выполняется экспертный AI-анализ и п
    | оиск сведений в интернете...</span>
80 |     </div>
81 | </div>
82 | </div>
83 | </div>
84 |     <div class="modal-footer border-secondary py-2 px-
    | 3 d-flex justify-content-between">
85 |         <div id="uai-modal-custom-actions" class="d-flex
    | gap-1.5 flex-wrap"></div>
86 |         <button type="button" class="btn btn-sm btn-seco
    | ndary" data-bs-dismiss="modal">Заккрыть</button>
87 |     </div>
88 | </div>
89 | </div>
90 | `;
91 |     document.body.appendChild(modalEl);
92 | }
93 |     return modalEl;
94 | }
95 |
96 | function escapeHtml(str) {
97 |     if (str === null || str === undefined) return '';
98 |     return String(str)
99 |         .replace(/&/g, '&amp;')
100 |         .replace(/</g, '&lt;')
101 |         .replace(/>/g, '&gt;')
102 |         .replace(/"/g, '&quot;')
103 |         .replace(/'/g, '&#039;');
104 | }
105 |
106 | /**
107 |  * Universal method to show item inspection modal with AI ex
    | planation.
108 |  * @param {Object} options Configuration object
109 |  */

```



```
110 function show(options) {
111     const opts = Object.assign({
112         title: 'Детали записи',
113         subtitle: '',
114         icon: '📄',
115         badges: [],
116         metadata: [],
117         rawTitle: 'Данные / Путь',
118         rawContent: '',
119         tableType: 'generic',
120         requestData: {},
121         actions: []
122     }, options);
123
124     const modalEl = ensureModalElement();
125
126     // Icon & Title
127     const iconEl = document.getElementById('uai-modal-icon');
128     if (iconEl) iconEl.textContent = opts.icon;
129
130     const titleEl = document.getElementById('uai-modal-title')
131     ;
132     if (titleEl) titleEl.textContent = opts.title;
133
134     // Badges
135     const badgesContainer = document.getElementById('uai-modal-
136     -badges');
137     if (badgesContainer) {
138         badgesContainer.innerHTML = (opts.badges || []).map(b =>
139         {
140             const cls = b.class || 'badge bg-secondary';
141             return `<span class="${cls}">${escapeHtml(b.text)}</sp
142             an>`;
143         }).join('');
144     }
145
146     // Metadata Grid
147     const metaContainer = document.getElementById('uai-modal-m
```

```

    etadata-grid');
144     if (metaContainer) {
145         metaContainer.innerHTML = (opts.metadata || []).map(m =>
    {
146             const colSize = m.fullWidth ? 'col-sm-12' : 'col-sm-6'
    ;
147             let valHtml = '';
148             if (m.html) {
149                 valHtml = m.html;
150             } else if (m.isCode) {
151                 valHtml = `<span class="font-monospace text-info sma
    ll" style="word-break: break-all;">${escapeHtml(m.value || '-'
    )}</span>`;
152             } else {
153                 valHtml = `<span class="text-light">${escapeHtml(m.v
    alue || '-')}</span>`;
154             }
155             return `
156                 <div class="${colSize}">
157                     <strong class="text-secondary">${escapeHtml(m.labe
    l)}</strong> ${valHtml}
158                 </div>
159             `;
160         }).join('');
161     }
162
163     // Raw Content Block
164     const rawContainer = document.getElementById('uai-modal-ra
    w-container');
165     const rawTitleEl = document.getElementById('uai-modal-raw-
    title');
166     const rawContentEl = document.getElementById('uai-modal-ra
    w-content');
167     if (rawContainer && rawTitleEl && rawContentEl) {
168         if (opts.rawContent && opts.rawContent.trim()) {
169             rawTitleEl.textContent = opts.rawTitle;
170             rawContentEl.textContent = opts.rawContent;
171             rawContainer.style.display = 'block';

```

```

172     } else {
173         rawContainer.style.display = 'none';
174     }
175 }
176
177 // Reset AI Diagnostic section
178 const aiExplanation = document.getElementById('uai-modal-a
i-explanation');
179 if (aiExplanation) {
180     aiExplanation.innerHTML = `Нажмите «Анализ контекста», ч
тобы получить экспертное объяснение от языковой модели, оценку
безопасности и рекомендации.`;
181 }
182
183 // Wire up Prompt Editor button
184 const editPromptBtn = document.getElementById('btn-uai-mod
al-edit-prompt');
185 if (editPromptBtn) {
186     editPromptBtn.onclick = () => openPromptEditor(opts.tabl
eType);
187 }
188
189 // Function to execute AI diagnostic request with Web Grou
nding
190 async function executeDiagnostics() {
191     const aiExplanation = document.getElementById('uai-modal
-ai-explanation');
192     if (!aiExplanation) return;
193
194     aiExplanation.innerHTML = `
195         <div class="d-flex align-items-center gap-2 py-2 text-
info">
196             <div class="spinner-border spinner-border-sm" role="
status"></div>
197             <span>Выполняется экспертный AI-анализ для «${escape
Html(opts.title)}» с проверкой в интернете...</span>
198         </div>
199     `;

```

```

200
201     const metaDict = {};
202     (opts.metadata || []).forEach(m => {
203         if (m.label && (m.value !== undefined && m.value !== n
ull)) {
204             metaDict[m.label] = m.value;
205         }
206     });
207
208     try {
209         const fetchFn = (window.api && window.api.fetch) ? win
dow.api.fetch : fetch;
210         const res = await fetchFn('/api/v1/diagnostics/explain
', {
211             method: 'POST',
212             headers: { 'Content-Type': 'application/json' },
213             body: JSON.stringify({
214                 table_type: opts.tableType,
215                 title: opts.title,
216                 subtitle: opts.subtitle,
217                 metadata: Object.assign({}, metaDict, opts.request
Data || {}),
218                 raw_data: opts.rawContent || '',
219                 web_search: true
220             })
221         });
222         const data = (res && typeof res.json === 'function') ?
await res.json() : res;
223
224         aiExplanation.innerHTML = `
225             <div class="mb-2"><strong class="text-info"><i class
="bi bi-card-text me-1"></i>Назначение:</strong> ${escapeHtml(
data.summary)}</div>
226             <div class="mb-2"><strong class="text-secondary"><i
class="bi bi-building me-1"></i>Разработчик / Категория:</stro
ng> ${escapeHtml(data.developer || 'Неизвестен')} (${escapeHtm
l(data.category || 'Компонент')})</div>
227             <div class="mb-2"><strong class="text-warning"><i cl

```

```

    ass="bi bi-shield-lock me-1"></i>Оценка безопасности:</strong>
    ${escapeHtml(data.security_verdict)}</div>
228     <div class="mb-2"><strong class="text-info"><i class
    ="bi bi-speedometer2 me-1"></i>Влияние на ресурсы:</strong> ${
    escapeHtml(data.performance_impact)}</div>
229     <div class="p-2 mb-2 rounded bg-dark-subtle border b
    order-warning-subtle">
230         <strong class="text-warning"><i class="bi bi-light
    bulb me-1"></i>Рекомендация:</strong> ${escapeHtml(data.recomm
    endation)}
231     </div>
232     ${data.action_steps && data.action_steps.length > 0
    ? `
233         <h6 class="small text-uppercase text-muted fw-bold
    mb-1">Рекомендуемые действия:</h6>
234         <ul class="mb-0 ps-3">
235             ${data.action_steps.map(s => `<li>${escapeHtml(s
    )}</li>`).join('')}
236         </ul>
237         ` : ''}
238     `;
239     } catch (err) {
240         aiExplanation.innerHTML = `<div class="text-danger py-
    2"><i class="bi bi-exclamation-octagon me-1"></i>Ошибка получе
    ния AI-анализа: ${escapeHtml(err.message)}</div>`;
241     }
242 }
243
244 // Wire up refresh button
245 const refreshBtn = document.getElementById('btn-uai-modal-
    refresh');
246 if (refreshBtn) {
247     refreshBtn.onclick = () => executeDiagnostics();
248 }
249
250 // Auto-run AI diagnostic on show
251 executeDiagnostics();
252

```

```

253 // Custom Footer Actions
254 const actionsContainer = document.getElementById('uai-modal-custom-actions');
255 if (actionsContainer) {
256     actionsContainer.innerHTML = '';
257     (opts.actions || []).forEach(act => {
258         const btn = document.createElement('button');
259         btn.type = 'button';
260         btn.className = `btn btn-sm ${act.class || 'btn-outline-secondary'}`;
261         btn.innerHTML = `${act.icon ? `<i class="bi ${act.icon} me-1"></i>` : ''}${escapeHtml(act.label)}`;
262         btn.onclick = () => act.onClick && act.onClick(opts);
263         actionsContainer.appendChild(btn);
264     });
265 }
266
267 // Open Modal
268 if (window.bootstrap && window.bootstrap.Modal) {
269     const bsModal = window.bootstrap.Modal.getInstance(modalEl);
270     bsModal.show();
271 }
272 }
273
274 const PROMPT_MODAL_ID = 'universal-ai-prompt-editor-modal';
275
276 function ensurePromptModalElement() {
277     let promptEl = document.getElementById(PROMPT_MODAL_ID);
278     if (!promptEl) {
279         promptEl = document.createElement('div');
280         promptEl.id = PROMPT_MODAL_ID;
281         promptEl.className = 'modal fade';
282         promptEl.tabIndex = -1;
283         promptEl.setAttribute('aria-hidden', 'true');
284         promptEl.innerHTML = `
285         <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">

```

```

286 |         <div class="modal-content bg-dark text-light border-
    | secondary shadow-lg">
287 |             <div class="modal-header border-secondary py-2 px-
    | 3">
288 |                 <div class="d-flex align-items-center gap-2">
289 |                     <span class="fs-5 text-warning">⚠ </span>
290 |                     <h5 class="modal-title fw-bold text-warning mb
    | -0">Редактор промпта диагностики</h5>
291 |                 </div>
292 |                 <button type="button" class="btn-close btn-close
    | -white" data-bs-dismiss="modal" aria-label="Close"></button>
293 |             </div>
294 |             <div class="modal-body py-3 px-3">
295 |                 <div class="row g-2 mb-3">
296 |                     <div class="col-sm-6">
297 |                         <label class="form-label small text-secondar
    | y fw-bold mb-1">Тип таблицы / сущности</label>
298 |                         <select class="form-select form-select-sm bg
    | -dark text-light border-secondary" id="uai-prompt-type-select"
    | >
299 |                             <option value="software">software (Установ
    | ленное ПО)</option>
300 |                             <option value="process">process (Процессы
    | Windows)</option>
301 |                             <option value="service">service (Службы Wi
    | ndows)</option>
302 |                             <option value="task">task (Задачи планиров
    | щика)</option>
303 |                             <option value="network">network (Сетевой м
    | ониторинг)</option>
304 |                             <option value="registry">registry (Реестр
    | Windows)</option>
305 |                             <option value="user">user (Пользователи и
    | группы)</option>
306 |                             <option value="website">website (Мониторин
    | г сайтов)</option>
307 |                             <option value="rag_doc">rag_doc (RAG База
    | знаний)</option>

```

```
308         <option value="disk">disk (Диски и тома)</option>
309         <option value="startup">startup (Автозагрузка)</option>
310         <option value="generic">generic (Общий шаблон)</option>
311     </select>
312 </div>
313 <div class="col-sm-6">
314     <label class="form-label small text-secondary fw-bold mb-1">Название шаблона</label>
315     <input type="text" class="form-control form-control-sm bg-dark text-light border-secondary" id="uai-prompt-name-input" placeholder="Название шаблона">
316 </div>
317 </div>
318
319     <div class="mb-3">
320         <label class="form-label small text-secondary fw-bold mb-1">Системная инструкция (System Instruction)</label>
321         <textarea class="form-control form-control-sm bg-dark text-light border-secondary font-monospace" id="uai-prompt-system-input" rows="2" placeholder="Роль и поведение модели..."></textarea>
322     </div>
323
324     <div class="mb-2">
325         <div class="d-flex justify-content-between align-items-center mb-1">
326             <label class="form-label small text-secondary fw-bold mb-0">Шаблон текста промпта (Prompt Template)</label>
327             <div class="d-flex gap-1" id="uai-prompt-variables-bar">
328                 <button type="button" class="btn btn-xs btn-outline-info py-0 px-1 font-monospace" style="font-size: 0.75rem;" onclick="window.AITableModal.insertVariable('{title}')">
```



```

329 |         <button type="button" class="btn btn-xs bt
| n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
| 5rem;" onclick="window.AITableModal.insertVariable('{subtitle}
| ')">{subtitle}</button>
330 |         <button type="button" class="btn btn-xs bt
| n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
| 5rem;" onclick="window.AITableModal.insertVariable('{metadata}
| ')">{metadata}</button>
331 |         <button type="button" class="btn btn-xs bt
| n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
| 5rem;" onclick="window.AITableModal.insertVariable('{raw_data}
| ')">{raw_data}</button>
332 |         </div>
333 |     </div>
334 |     <textarea class="form-control form-control-sm
| bg-dark text-light border-secondary font-monospace" id="uai-pr
| ompt-template-input" rows="8" placeholder="Текст промпта с пле
| йсхолдерами..."></textarea>
335 |     <div class="form-text small text-secondary" st
| yle="font-size: 0.75rem;">
336 |         Используйте переменные <code>{title}</code>,
| <code>{subtitle}</code>, <code>{metadata}</code>, <code>{raw_
| data}</code> для автоматической подстановки контекста строки т
| аблицы.
337 |     </div>
338 | </div>
339 |
340 |     <div id="uai-prompt-status-msg" class="small mt-
| 2" style="display: none;"></div>
341 | </div>
342 |     <div class="modal-footer border-secondary py-2 px-
| 3 d-flex justify-content-between">
343 |         <button type="button" class="btn btn-sm btn-outl
| ine-danger" id="btn-uai-prompt-reset">
344 |             <i class="bi bi-arrow-counterclockwise me-1"><
| /i>Сбросить по умолчанию
345 |         </button>

```

```

346         <div class="d-flex gap-2">
347             <button type="button" class="btn btn-sm btn-se
condary" data-bs-dismiss="modal">Закрыть</button>
348             <button type="button" class="btn btn-sm btn-pr
imary" id="btn-uai-prompt-save">
349                 <i class="bi bi-check2 me-1"></i>Сохранить п
ромпт
350             </button>
351         </div>
352     </div>
353 </div>
354 </div>
355 `;
356     document.body.appendChild(promptEl);
357 }
358     return promptEl;
359 }
360
361     async function openPromptEditor(tableType) {
362         const promptModalEl = ensurePromptModalElement();
363         const typeSelect = document.getElementById('uai-prompt-typ
e-select');
364         const nameInput = document.getElementById('uai-prompt-name
-input');
365         const systemInput = document.getElementById('uai-prompt-sy
stem-input');
366         const templateInput = document.getElementById('uai-prompt-
template-input');
367         const statusMsg = document.getElementById('uai-prompt-stat
us-msg');
368         const saveBtn = document.getElementById('btn-uai-prompt-sa
ve');
369         const resetBtn = document.getElementById('btn-uai-prompt-r
eset');
370
371         if (typeSelect && tableType) {
372             typeSelect.value = tableType;
373         }

```

```
374 |
375 |     async function loadTemplate(tType) {
376 |         if (statusMsg) statusMsg.style.display = 'none';
377 |         try {
378 |             const fetchFn = (window.api && window.api.fetch) ? win
379 |             dow.api.fetch : fetch;
380 |             const res = await fetchFn(`/api/v1/diagnostics/prompts
381 |             /${encodeURIComponent(tType)}`);
382 |             const data = (res && typeof res.json === 'function') ?
383 |             await res.json() : res;
384 |             if (data) {
385 |                 if (nameInput) nameInput.value = data.name || '';
386 |                 if (systemInput) systemInput.value = data.system_ins
387 |                 truction || '';
388 |                 if (templateInput) templateInput.value = data.prompt
389 |                 _template || '';
390 |             }
391 |         } catch (e) {
392 |             console.error('Ошибка загрузки шаблона промпта:', e);
393 |         }
394 |     }
395 |
396 |     if (typeSelect) {
397 |         typeSelect.onChange = () => loadTemplate(typeSelect.valu
398 |         e);
399 |     }
400 |
401 |     if (saveBtn) {
402 |         saveBtn.onclick = async () => {
403 |             const currentType = typeSelect ? typeSelect.value : 'g
404 |             eneric';
405 |             try {
406 |                 const fetchFn = (window.api && window.api.fetch) ? w
407 |                 indow.api.fetch : fetch;
408 |                 await fetchFn(`/api/v1/diagnostics/prompts/${encodeU
409 |                 RComponent(currentType)}`, {
410 |                     method: 'POST',
411 |                     headers: { 'Content-Type': 'application/json' },
```

```

403         body: JSON.stringify({
404             table_type: currentType,
405             name: nameInput ? nameInput.value : '',
406             system_instruction: systemInput ? systemInput.va
lue : '',
407             prompt_template: templateInput ? templateInput.v
alue : ''
408         })
409     });
410     if (statusMsg) {
411         statusMsg.className = 'small mt-2 text-success';
412         statusMsg.innerHTML = '<i class="bi bi-check-circl
e me-1"></i>Шаблон промпта успешно сохранен!';
413         statusMsg.style.display = 'block';
414         setTimeout(() => { if (statusMsg) statusMsg.style.
display = 'none'; }, 3000);
415     }
416 } catch (err) {
417     if (statusMsg) {
418         statusMsg.className = 'small mt-2 text-danger';
419         statusMsg.innerHTML = `<i class="bi bi-exclamation
-triangle me-1"></i>Ошибка сохранения: ${escapeHtml(err.messag
e)}`;
420         statusMsg.style.display = 'block';
421     }
422 }
423 };
424 }
425
426 if (resetBtn) {
427     resetBtn.onclick = async () => {
428         const currentType = typeSelect ? typeSelect.value : 'g
eneric';
429         if (!confirm(`Сбросить промпт для «${currentType}» к с
истемному значению по умолчанию?`)) return;
430         try {
431             const fetchFn = (window.api && window.api.fetch) ? w
indow.api.fetch : fetch;

```

```

432 |         const res = await fetchFn(`/api/v1/diagnostics/promp
    | ts/${encodeURIComponent(currentType)}/reset`, {
433 |             method: 'POST'
434 |         });
435 |         const data = (res && typeof res.json === 'function')
    | ? await res.json() : res;
436 |         if (data) {
437 |             if (nameInput) nameInput.value = data.name || '';
438 |             if (systemInput) systemInput.value = data.system_i
    | nstruction || '';
439 |             if (templateInput) templateInput.value = data.prom
    | pt_template || '';
440 |         }
441 |         if (statusMsg) {
442 |             statusMsg.className = 'small mt-2 text-info';
443 |             statusMsg.innerHTML = '<i class="bi bi-arrow-count
    | erclockwise me-1"></i>Шаблон промпта сброшен к системному по у
    | молчанию.';
444 |             statusMsg.style.display = 'block';
445 |             setTimeout(() => { if (statusMsg) statusMsg.style.
    | display = 'none'; }, 3000);
446 |         }
447 |     } catch (err) {
448 |         if (statusMsg) {
449 |             statusMsg.className = 'small mt-2 text-danger';
450 |             statusMsg.innerHTML = `<i class="bi bi-exclamation
    | -triangle me-1"></i>Ошибка сброса: ${escapeHtml(err.message)}`
    | ;
451 |             statusMsg.style.display = 'block';
452 |         }
453 |     }
454 | };
455 | }
456 |
457 |     await loadTemplate(typeSelect ? typeSelect.value : (tableT
    | ype || 'generic'));
458 |
459 |     if (window.bootstrap && window.bootstrap.Modal) {

```

```
460     const bsPromptModal = window.bootstrap.Modal.getOrCreate
      Instance(promptModalEl);
461     bsPromptModal.show();
462   }
463 }
464
465 function insertVariable(varText) {
466   const templateInput = document.getElementById('uai-prompt-
      template-input');
467   if (!templateInput) return;
468   const start = templateInput.selectionStart || 0;
469   const end = templateInput.selectionEnd || 0;
470   const val = templateInput.value;
471   templateInput.value = val.substring(0, start) + varText +
      val.substring(end);
472   templateInput.focus();
473   templateInput.selectionStart = templateInput.selectionEnd
      = start + varText.length;
474 }
475
476 // Global Export
477 window.AITableModal = {
478   show: show,
479   openPromptEditor: openPromptEditor,
480   insertVariable: insertVariable,
481   escapeHtml: escapeHtml
482 };
483 })();
484
```

api-cache.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Api-Cache Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля api-cache
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/api-cache.js?v=2026
11  *     1001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { CACHE_STRATEGIES } from '/windows/api/webgui/j
15  *     s/api-cache.js';
16  *
17  * File: api-cache.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/js
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-06 12:54:00
23  * =====
24  */
25  * api-cache.js - Универсальный кеш-слой для всех API-запросов
26  * веб-интерфейса
27  *
28  * Предоставляет единую точку входа для всех HTTP-запросов с а
29  * втоматическим кешированием,
30  *
31  * используя современные стратегии: Cache-First, Network-First
```

```
, Stale-While-Revalidate.
29 *
30 * АРХИТЕКТУРА:
31 * - L1 Cache: In-Memory (быстрый доступ)
32 * - L2 Cache: IndexedDB (персистентный)
33 * - L3 Fallback: Direct network (если кеш недоступен)
34 *
35 * ПРИМЕНЕНИЕ:
36 * Используется всеми вкладками веб-интерфейса для оптимизации
  загрузки данных,
37 * снижения нагрузки на API и обеспечения offline-режима.
38 *
39 * @author hypo69
40 * @version 1.0.0
41 * @since 2026-09-24
42 */
43
44 import { browserCache, STORES } from './browser-cache.js';
45
46 // Глобальные стратегии кеширования для разных типов данных
47 export const CACHE_STRATEGIES = {
48   // Статические справочники (модели, конфигурации) - долгое T
  TL
49   STATIC: { ttl: 60 * 60 * 1000, strategy: 'cache-first' },
    // 1 час
50
51   // Редко меняющиеся данные (hardware spec, user profiles) -
  средний TTL
52   SEMI_STATIC: { ttl: 30 * 60 * 1000, strategy: 'cache-first'
  }, // 30 минут
53
54   // Часто меняющиеся данные (системные метрики, статусы) - ко
  роткий TTL
55   DYNAMIC: { ttl: 5 * 1000, strategy: 'stale-while-revalidate'
  }, // 5 секунд
56
57   // Реалтайм данные (live telemetry, сенсоры) - минимальный T
  TL
```



```
58 | REALTIME: { ttl: 3 * 1000, strategy: 'network-first' },
    |         // 3 секунды
59 |
60 | // Критичные данные (security status, errors) - всегда из се
    | ти
61 | CRITICAL: { ttl: 0, strategy: 'network-first' }
    |         // Нет кеша
62 | };
63 |
64 | // Автоматические правила сопоставления URL → стратегия
65 | const URL_PATTERNS = [
66 |     // Статические данные
67 |     { pattern: /\api\v1\models/, strategy: CACHE_STRATEGIES.S
    | TATIC, tag: 'models' },
68 |     { pattern: /\api\config/, strategy: CACHE_STRATEGIES.STATI
    | C, tag: 'config' },
69 |     { pattern: /\api\plugins/, strategy: CACHE_STRATEGIES.SEMI
    | _STATIC, tag: 'plugins' },
70 |
71 |     // Hardware и системные спецификации
72 |     { pattern: /\api\v1\system\hardware/, strategy: CACHE_ST
    | RATEGIES.SEMI_STATIC, tag: 'hardware' },
73 |     { pattern: /\api\v1\windows-backup/, strategy: CACHE_STR
    | ATEGIES.SEMI_STATIC, tag: 'backup' },
74 |
75 |     // Динамические системные данные
76 |     { pattern: /\api\v1\system\summary/, strategy: CACHE_STR
    | ATEGIES.DYNAMIC, tag: 'system-summary' },
77 |     { pattern: /\api\system-control\status/, strategy: CACHE_
    | STRATEGIES.DYNAMIC, tag: 'control-status' },
78 |
79 |     // Реалтайм телеметрия
80 |     { pattern: /\api\v1\system\sensors/, strategy: CACHE_STR
    | ATEGIES.REALTIME, tag: 'sensors' },
81 |     { pattern: /\api\v1\system\processes/, strategy: CACHE_S
    | TRATEGIES.REALTIME, tag: 'processes' },
82 |
83 |     // RAG и чат (частые обновления)
```

```

84 | { pattern: /\api\/chat/, strategy: CACHE_STRATEGIES.DYNAMIC
    | , tag: 'chat' },
85 | { pattern: /\api\/rag/, strategy: CACHE_STRATEGIES.DYNAMIC,
    | tag: 'rag' }
86 | ];
87 |
88 | /**
89 |  * Определение стратегии кеширования по URL
90 |  * @param {string} url - URL запроса
91 |  * @returns {Object} { ttl, strategy, tag }
92 |  */
93 | function getCacheStrategy(url) {
94 |     for (const rule of URL_PATTERNS) {
95 |         if (rule.pattern.test(url)) {
96 |             return { ...rule.strategy, tag: rule.tag };
97 |         }
98 |     }
99 |     // По умолчанию: stale-while-revalidate с TTL 1 минута
100 |     return { ttl: 60 * 1000, strategy: 'stale-while-revalidate',
    | tag: 'default' };
101 | }
102 |
103 | /**
104 |  * Генерация ключа кеша из URL и параметров
105 |  * @param {string} url - URL запроса
106 |  * @param {Object} options - Опции запроса (method, body)
107 |  * @returns {string} Уникальный ключ кеша
108 |  */
109 | function generateCacheKey(url, options = {}) {
110 |     const method = options.method || 'GET';
111 |     const body = options.body ? `-${btoa(options.body).substring
    | (0, 32)}` : '';
112 |     return `${method}:${url}${body}`;
113 | }
114 |
115 | /**
116 |  * Универсальная функция для выполнения HTTP-запросов с автома
    | тическим кешированием

```

```

117 | *
118 | * @param {string} url - URL запроса
119 | * @param {Object} options - Опции запроса
120 | * @param {Object} cacheOptions - Опции кеширования (переопред
    | еляют автоматику)
121 | * @returns {Promise<any>} Результат запроса
122 | *
123 | * @example
124 | * // Автоматическое определение стратегии
125 | * const data = await cachedApiFetch('/api/v1/system/hardware'
    | );
126 | *
127 | * @example
128 | * // Ручное переопределение стратегии
129 | * const data = await cachedApiFetch('/api/custom', {}, {
130 | *   ttl: 10000,
131 | *   strategy: 'cache-first',
132 | *   tag: 'custom-data'
133 | * });
134 | */
135 | /**
136 | * Оборачивает результат в полиморфный объект, поддерживающий
    | как прямое обращение к полям,
137 | * так и стандартный интерфейс Response (res.ok, res.status, r
    | es.json()).
138 | * @param {any} data
139 | * @returns {any}
140 | */
141 | function wrapCachedData(data) {
142 |   if (data && typeof data === 'object') {
143 |     try {
144 |       if (!('ok' in data)) {
145 |         Object.defineProperty(data, 'ok', { value: true, writa
    | ble: true, configurable: true, enumerable: false });
146 |       }
147 |       if (!('status' in data)) {
148 |         Object.defineProperty(data, 'status', { value: 200, wr
    | itable: true, configurable: true, enumerable: false });

```

```

149     }
150     if (typeof data.json !== 'function') {
151         Object.defineProperty(data, 'json', { value: async ()
=> data, writable: true, configurable: true, enumerable: false
        });
152     }
153     } catch (_) {}
154 }
155 return data;
156 }
157
158 /**
159  * Универсальная функция для выполнения HTTP-запросов с автома
тическим кешированием
160  *
161  * @param {string} url - URL запроса
162  * @param {Object} options - Опции запроса
163  * @param {Object} cacheOptions - Опции кеширования (переопред
еляют автоматику)
164  * @returns {Promise<any>} Результат запроса (поддерживает как
прямое чтение, так и res.ok / await res.json())
165  */
166 export async function cachedApiFetch(url, options = {}, cacheO
ptions = {}) {
167     // Определяем стратегию автоматически или используем передан
ную
168     const autoStrategy = getCacheStrategy(url);
169     const {
170         ttl = autoStrategy.ttl,
171         strategy = autoStrategy.strategy,
172         tag = autoStrategy.tag,
173         forceNetwork = false
174     } = cacheOptions;
175
176     const cacheKey = generateCacheKey(url, options);
177     const cache = browserCache;
178
179     // Отключаем кеш для указанных вкладок

```

```
180 | const currentTab = window.location.hash;
181 | const NO_CACHE_TABS = [
182 |   '#tab-hardware-load-inspector',
183 |   '#tab-processes-load-inspector',
184 |   '#tab-about-system',
185 | ];
186 | if (NO_CACHE_TABS.some(prefix => currentTab.startsWith(prefi
x))) {
187 |   const resp = await fetch(url, {
188 |     ...options,
189 |     headers: {
190 |       ...options.headers,
191 |       'Cache-Control': 'no-store, no-cache, must-revalidate'
192 |     },
193 |   });
194 |   return await resp.json();
195 | }
196 |
197 | // Дожидаемся готовности базы, но не блокируем если есть оши
бка
198 | if (cache) {
199 |   try {
200 |     await cache.ready();
201 |   } catch (_) {}
202 | }
203 |
204 | // CACHE-FIRST: сначала проверяем кеш, потом сеть
205 | if (cache && strategy === 'cache-first' && !forceNetwork) {
206 |   try {
207 |     const cached = await cache.get(STORES.API_CACHE, cacheKe
y);
208 |     if (cached !== null && cached !== undefined) {
209 |       return wrapCachedData(cached);
210 |     }
211 |   } catch (_) {}
212 | }
213 |
```

```
214 | // STALE-WHILE-REVALIDATE: показываем кеш, обновляем в фоне
215 | if (cache && strategy === 'stale-while-revalidate' && !force
    | Network) {
216 |     try {
217 |         const cached = await cache.get(STORES.API_CACHE, cacheKey);
218 |         if (cached !== null && cached !== undefined) {
219 |             // Фоновое асинхронное обновление
220 |             directFetch(url, options)
221 |                 .then(fresh => cache.set(STORES.API_CACHE, cacheKey,
    | fresh, { ttl, tags: [tag, 'api-cache'] }))
222 |                 .catch(err => console.debug(`[API Cache] Background
    | update skipped for ${url}:`, err));
223 |             return wrapCachedData(cached);
224 |         }
225 |     } catch (_) {}
226 | }
227 |
228 | // NETWORK-FIRST или первый запрос: сначала сеть, потом кеш
    | как fallback
229 | try {
230 |     const fresh = await directFetch(url, options);
231 |
232 |     // Сохраняем в кеш (если TTL > 0)
233 |     if (cache && ttl > 0) {
234 |         cache.set(STORES.API_CACHE, cacheKey, fresh, { ttl, tags
    | : [tag, 'api-cache'] }).catch(() => {});
235 |     }
236 |
237 |     return wrapCachedData(fresh);
238 | } catch (err) {
239 |     // При ошибке сети пытаемся вернуть устаревший кеш
240 |     if (cache) {
241 |         try {
242 |             const staleCache = await cache.get(STORES.API_CACHE, c
    | acheKey);
243 |             if (staleCache !== null && staleCache !== undefined) {
244 |                 console.warn(`[API Cache] Network failed, returning
```

```

    STALE cache: ${url}`);
245     return wrapCachedData(staleCache);
246   }
247   } catch (_) {}
248 }
249 throw err;
250 }
251 }
252
253 /**
254  * Прямой HTTP-запрос без кеширования
255  * @param {string} url - URL запроса
256  * @param {Object} options - Опции fetch
257  * @returns {Promise<any>}
258  */
259 async function directFetch(url, options = {}) {
260   const opts = { ...options };
261   if (opts.body && typeof opts.body === 'string') {
262     opts.headers = {
263       'Content-Type': 'application/json',
264       ...(opts.headers || {})
265     };
266   }
267
268   if (window.api && typeof window.api.fetch === 'function') {
269     const r = await window.api.fetch(url, opts);
270     return wrapCachedData(r);
271   }
272
273   const res = await fetch(url, opts);
274   if (!res.ok) {
275     let errMsg = `HTTP ${res.status}`;
276     try {
277       const errJson = await res.json();
278       if (errJson && errJson.detail) {
279         errMsg += ` (${typeof errJson.detail === 'object' ? JS
280         ON.stringify(errJson.detail) : errJson.detail})`;

```

```

281     } catch (_) {}
282     throw new Error(errMsg);
283 }
284 const json = await res.json();
285 return wrapCachedData(json);
286 }
287
288 /**
289  * Инвалидация кеша по тегу (например, после обновления данных
    )
290  * @param {string} tag - Тег для инвалидации
291  * @returns {Promise<number>} Количество удаленных записей
292  *
293  * @example
294  * // Очистить весь кеш моделей после обновления
295  * await invalidateCacheByTag('models');
296  */
297 export async function invalidateCacheByTag(tag) {
298     if (!browserCache || !browserCache.isDbReady) return 0;
299     const count = await browserCache.invalidateByTag(STORES.API_
    CACHE, tag);
300     console.log(`[API Cache] □ Invalidated ${count} entries wit
    h tag: ${tag}`);
301     return count;
302 }
303
304 /**
305  * Очистка всего API-кеша
306  * @returns {Promise<boolean>}
307  */
308 export async function clearAllApiCache() {
309     if (!browserCache || !browserCache.isDbReady) return false;
310     await browserCache.clearStore(STORES.API_CACHE);
311     console.log(`[API Cache] □ All API cache cleared`);
312     return true;
313 }
314
315 /**

```



```
316 | * Получение статистики кеша
317 | * @returns {Promise<Object>}
318 | */
319 | export async function getApiCacheStats() {
320 |   if (!browserCache || !browserCache.isDbReady) {
321 |     return { available: false, metrics: null };
322 |   }
323 |
324 |   const stats = await browserCache.getDetailedStats();
325 |   const apiStore = stats.stores[STORES.API_CACHE] || {};
326 |
327 |   return {
328 |     available: true,
329 |     metrics: stats.metrics,
330 |     apiCache: {
331 |       count: apiStore.count || 0,
332 |       sizeMB: apiStore.sizeMB || '0.00',
333 |       hitRate: stats.metrics.hits + stats.metrics.misses > 0
334 |         ? Math.round((stats.metrics.hits / (stats.metrics.hits
+ stats.metrics.misses)) * 100)
335 |         : 0
336 |     },
337 |     storage: stats.storageEstimate
338 |   };
339 | }
340 |
341 | // Экспорт в глобальный контекст для доступности из всех вклад
ок
342 | if (typeof window !== 'undefined') {
343 |   window.cachedApiFetch = cachedApiFetch;
344 |   window.invalidateCacheByTag = invalidateCacheByTag;
345 |   window.clearAllApiCache = clearAllApiCache;
346 |   window.getApiCacheStats = getApiCacheStats;
347 |   window.CACHE_STRATEGIES = CACHE_STRATEGIES;
348 | }
349 |
350 | export default cachedApiFetch;
351 |
```

appConfigEditor.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Appconfigeditor Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля appConfig
7   *   Editor.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/appConfigEditor.js?
11  *       v=20261001_v1" type="module"></script>
12  * File: appConfigEditor.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/js
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 // =====
22 // Process Name: App Configuration Editor UI Module
23 // =====
24 // Description:
25 //   Reusable client-side controller for inspecting and modify
26 //   ing config.json
27 //   across /apps microservices in the AI-Breadboard web admin
28 //   istrative panel.
29 //
```

```

28 // File: appConfigEditor.js
29 // Project: ai-breadboard
30 // Package: src.api.webinterface.js
31 // Author: hypo69
32 // Copyright: © 2026 hypo69
33 // =====
34
35 (function() {
36   let activeConfig = null;
37   let activeAppName = '';
38
39   /**
40    * Fetch app config from server
41    * @param {string} appName
42    * @returns {Promise<Object>}
43    */
44   async function fetchAppConfig(appName) {
45     let res;
46     if (window.api && typeof window.api.fetch === 'function')
47     {
48       try {
49         return await window.api.fetch(`/api/admin/apps/${appName}/config`);
50       } catch (e) {
51         // fallback
52       }
53       res = await fetch(`/api/admin/apps/${appName}/config`);
54       if (!res.ok) {
55         throw new Error(`Failed to load config: HTTP ${res.status}`);
56       }
57       return await res.json();
58     }
59
60     /**
61    * Save app config to server

```

```

62  * @param {string} appName
63  * @param {Object} configObj
64  * @returns {Promise<Object>}
65  */
66  async function saveAppConfig(appName, configObj) {
67      const payload = { config: configObj };
68      const res = await fetch(`/api/admin/apps/${appName}/config
    `, {
69          method: 'POST',
70          headers: { 'Content-Type': 'application/json' },
71          body: JSON.stringify(payload),
72      });
73      if (!res.ok) {
74          const err = await res.json().catch(() => ({ detail: res.
statusText }));
75          throw new Error(err.detail || `HTTP ${res.status}`);
76      }
77      return await res.json();
78  }
79
80  /**
81   * Render or update modal HTML
82   */
83  function ensureModalExists() {
84      let modalEl = document.getElementById('app-config-modal');
85      if (!modalEl) {
86          modalEl = document.createElement('div');
87          modalEl.id = 'app-config-modal';
88          modalEl.className = 'modal fade';
89          modalEl.tabIndex = -1;
90          modalEl.setAttribute('aria-hidden', 'true');
91          modalEl.innerHTML = `
92              <div class="modal-dialog modal-lg modal-dialog-center
d modal-dialog-scrollable">
93                  <div class="modal-content border-secondary shadow-lg
" style="background: #0f172a; color: #f8fafc;">
94                      <div class="modal-header border-secondary" style="
background: #1e293b;">

```

```

95         <h5 class="modal-title d-flex align-items-center
    gap-2 fw-bold text-info" id="app-config-modal-title">
96             <i class="bi bi-sliders"></i> Настройка конфи-
    гурации (config.json)
97         </h5>
98         <button type="button" class="btn-close btn-close
    -white" data-bs-dismiss="modal" aria-label="Close"></button>
99     </div>
100     <div class="modal-body p-3">
101         <div id="app-config-alert" class="alert d-none p
    y-2 px-3 small"></div>
102
103         <!-- Tabs: Visual Form vs Raw JSON -->
104         <ul class="nav nav-pills nav-fill mb-3" id="app-
    config-tabs" role="tablist">
105             <li class="nav-item">
106                 <button class="nav-link active py-1 small fw
    -semibold" id="app-config-tab-visual-btn" data-bs-toggle="pill
    " data-bs-target="#app-config-tab-visual" type="button">
107                     <i class="bi bi-ui-checks me-1"></i> Визуа
    льная форма
108                 </button>
109             </li>
110             <li class="nav-item">
111                 <button class="nav-link py-1 small fw-semibo
    ld" id="app-config-tab-raw-btn" data-bs-toggle="pill" data-bs-
    target="#app-config-tab-raw" type="button">
112                     <i class="bi bi-code-square me-1"></i> Исх
    одный JSON
113                 </button>
114             </li>
115         </ul>
116
117         <div class="tab-content">
118             <!-- Visual Form Pane -->
119             <div class="tab-pane fade show active" id="app
    -config-tab-visual">
120                 <div class="card border-secondary mb-3" styl

```

```

e="background: #1e293b;">
121         <div class="card-header border-secondary p
y-2 small fw-bold text-primary d-flex align-items-center gap-2
">
122             <i class="bi bi-hdd-network"></i> Режим
сервера (Server Mode)
123         </div>
124         <div class="card-body p-3">
125             <div class="row g-3">
126                 <div class="col-md-6">
127                     <label class="form-label small fw-se
mibold text-light mb-1">Режим работы сервера:</label>
128                     <select class="form-select form-sele
ct-sm bg-dark text-white border-secondary" id="app-cfg-server-
mode">
129                         <option value="dedicated">Dedicate
d (Отдельный процесс / порт)</option>
130                         <option value="shared">Shared (Мар
шрутизация через главный сервер :8000)</option>
131                     </select>
132                     <div class="form-text small text-mut
ed" id="app-cfg-server-mode-hint">
133                         В режиме dedicated run.ps1 запуска
ет отдельный процесс микросервиса.
134                     </div>
135                 </div>
136                 <div class="col-md-6">
137                     <label class="form-label small fw-se
mibold text-light mb-1">Порт (Port):</label>
138                     <input type="number" class="form-con
trol form-control-sm bg-dark text-white border-secondary font-
monospace" id="app-cfg-server-port" placeholder="8100">
139                 </div>
140                 <div class="col-md-6">
141                     <label class="form-label small fw-se
mibold text-light mb-1">Хост (Host):</label>
142                     <input type="text" class="form-contr
ol form-control-sm bg-dark text-white border-secondary font-mo

```

```

    nospace" id="app-cfg-server-host" placeholder="127.0.0.1">
143         </div>
144         <div class="col-md-6">
145             <label class="form-label small fw-se
mibold text-light mb-1">Воркеры (Workers):</label>
146             <input type="number" class="form-con
trol form-control-sm bg-dark text-white border-secondary font-
monospace" id="app-cfg-server-workers" placeholder="1" min="1"
max="16">
147         </div>
148         <div class="col-12">
149             <div class="form-check form-switch">
150                 <input class="form-check-input" ty
pe="checkbox" id="app-cfg-server-ssl">
151                 <label class="form-check-label sma
ll text-light" for="app-cfg-server-ssl">Использовать SSL (HTTP
S)</label>
152             </div>
153         </div>
154     </div>
155 </div>
156 </div>
157
158     <!-- Dynamic App Extra Params Card -->
159     <div class="card border-secondary" style="ba
ckground: #1e293b;" id="app-cfg-extra-card">
160         <div class="card-header border-secondary p
y-2 small fw-bold text-info d-flex align-items-center gap-2">
161             <i class="bi bi-gear-wide-connected"></i
> Параметры приложения
162         </div>
163         <div class="card-body p-3" id="app-cfg-ext
ra-body">
164             <!-- Populated dynamically based on app
properties -->
165         </div>
166     </div>
167 </div>

```

```
168 |
169 |         <!-- Raw JSON Pane -->
170 |         <div class="tab-pane fade" id="app-config-tab-
    raw">
171 |             <div class="d-flex justify-content-between a
    lign-items-center mb-2">
172 |                 <span class="small text-muted font-monospa
    ce" id="app-cfg-file-path">config.json</span>
173 |                 <button class="btn btn-sm btn-outline-seco
    ndary py-0 px-2" id="app-cfg-btn-format">
174 |                     <i class="bi bi-magic"></i> Форматироват
    ь
175 |                 </button>
176 |             </div>
177 |             <textarea class="form-control font-monospace
    bg-dark text-white border-secondary" id="app-cfg-raw-json" ro
    ws="14" style="font-size: 0.82rem; line-height: 1.4; tab-size:
    2;"></textarea>
178 |             </div>
179 |         </div>
180 |     </div>
181 |     <div class="modal-footer border-secondary justify-
    content-between" style="background: #1e293b;">
182 |         <button type="button" class="btn btn-sm btn-outl
    ine-secondary rounded-pill px-3" id="app-cfg-btn-reload">
183 |             <i class="bi bi-arrow-clockwise me-1"></i> Сбp
    осить
184 |         </button>
185 |         <div class="d-flex gap-2">
186 |             <button type="button" class="btn btn-sm btn-se
    condary rounded-pill px-3" data-bs-dismiss="modal">Отмена</but
    ton>
187 |             <button type="button" class="btn btn-sm btn-pr
    imary rounded-pill px-4 fw-semibold" id="app-cfg-btn-save">
188 |                 <i class="bi bi-save me-1"></i> Сохранить
189 |             </button>
190 |         </div>
191 |     </div>
```



```
192 |         </div>
193 |     </div>
194 |     `;
195 |     document.body.appendChild(modalEl);
196 |
197 |     // Event handlers
198 |     document.getElementById('app-cfg-btn-save')?.addEventListener('click', onSaveConfig);
199 |     document.getElementById('app-cfg-btn-reload')?.addEventListener('click', onReloadConfig);
200 |     document.getElementById('app-cfg-btn-format')?.addEventListener('click', () => {
201 |         const rawEl = document.getElementById('app-cfg-row-json');
202 |         try {
203 |             const parsed = JSON.parse(rawEl.value);
204 |             rawEl.value = JSON.stringify(parsed, null, 2);
205 |         } catch (e) {
206 |             showModalAlert(`Ошибка JSON: ${e.message}`, 'danger');
207 |         }
208 |     });
209 |
210 |     // Synchronize visual form to raw JSON on tab switch
211 |     document.getElementById('app-config-tab-row-btn')?.addEventListener('click', () => {
212 |         syncFormToRawJson();
213 |     });
214 |     document.getElementById('app-config-tab-visual-btn')?.addEventListener('click', () => {
215 |         syncRawJsonToForm();
216 |     });
217 | }
218 | }
219 |
220 | function showModalAlert(message, type = 'info') {
221 |     const el = document.getElementById('app-config-alert');
222 |     if (!el) return;
```

```
223 |     el.className = `alert alert-${type} py-2 px-3 small`;
224 |     el.innerHTML = message;
225 |     el.classList.remove('d-none');
226 |     if (type === 'success') {
227 |         setTimeout(() => el.classList.add('d-none'), 3500);
228 |     }
229 | }
230 |
231 | function populateVisualForm(config) {
232 |     if (!config) return;
233 |
234 |     // Server section
235 |     let serverMode = 'dedicated';
236 |     let host = '127.0.0.1';
237 |     let port = 8100;
238 |     let workers = 1;
239 |     let ssl = false;
240 |
241 |     if (typeof config.server === 'string') {
242 |         serverMode = config.server.toLowerCase();
243 |     } else if (config.server && typeof config.server === 'object') {
244 |         if (typeof config.server.dedicated === 'boolean') {
245 |             serverMode = config.server.dedicated ? 'dedicated' : '
shared';
246 |         } else if (config.server.dedicated !== undefined) {
247 |             serverMode = String(config.server.dedicated).toLowerCase()
=== 'true' ? 'dedicated' : 'shared';
248 |         } else {
249 |             serverMode = config.server.mode || config.server.type
|| 'dedicated';
250 |         }
251 |         host = config.server.host || '127.0.0.1';
252 |         port = config.server.port || port;
253 |         workers = config.server.workers || 1;
254 |         ssl = !!config.server.use_ssl;
255 |     }
256 | }
```

```

257     const modeSelect = document.getElementById('app-cfg-server
    -mode');
258     if (modeSelect) modeSelect.value = serverMode;
259     const portInput = document.getElementById('app-cfg-server-
    port');
260     if (portInput) portInput.value = port;
261     const hostInput = document.getElementById('app-cfg-server-
    host');
262     if (hostInput) hostInput.value = host;
263     const workersInput = document.getElementById('app-cfg-serv
    er-workers');
264     if (workersInput) workersInput.value = workers;
265     const sslCheck = document.getElementById('app-cfg-server-s
    sl');
266     if (sslCheck) sslCheck.checked = ssl;
267
268     // Extra properties
269     const extraBody = document.getElementById('app-cfg-extra-b
    ody');
270     if (extraBody) {
271         let fieldsHtml = '<div class="row g-2">';
272         let hasExtra = false;
273
274         for (const [key, val] of Object.entries(config)) {
275             if (key === 'server' || key === 'cors' || key === 'ser
    ver_cors_origins' || key === 'server_cors_allow_credentials')
            {
276                 continue;
277             }
278             hasExtra = true;
279             if (typeof val === 'object' && val !== null && !Array.
    isArray(val)) {
280                 fieldsHtml += `<div class="col-12"><h6 class="text-w
    arning small fw-bold mt-2 mb-1">${key}:</h6></div>`;
281                 for (const [subK, subV] of Object.entries(val)) {
282                     fieldsHtml += `
283                         <div class="col-md-6">
284                             <label class="form-label small text-muted mb-0

```

```

    ">${key}.${subK}:</label>
285         <input type="text" class="form-control form-co
        ntrol-sm bg-dark text-white border-secondary font-monospace ap
        p-cfg-extra-input" data-path="${key}.${subK}" value="${subV !=
        = undefined ? String(subV) : ''}">
286     </div>
287     `;
288     }
289     } else {
290         fieldsHtml += `
291             <div class="col-md-6">
292                 <label class="form-label small text-muted mb-0">
                ${key}:</label>
293                 <input type="text" class="form-control form-cont
                rol-sm bg-dark text-white border-secondary font-monospace app-
                cfg-extra-input" data-path="${key}" value="${val != undefined
                ? String(val) : ''}">
294             </div>
295             `;
296         }
297     }
298
299     fieldsHtml += '</div>';
300     extraBody.innerHTML = hasExtra ? fieldsHtml : '<div clas
        s="text-muted small">Дополнительные параметры отсутствуют.</di
        v>';
301 }
302
303 // Raw JSON
304 const rawEl = document.getElementById('app-cfg-raw-json');
305 if (rawEl) {
306     rawEl.value = JSON.stringify(config, null, 2);
307 }
308 }
309
310 function syncFormToRawJson() {
311     try {
312         const rawEl = document.getElementById('app-cfg-raw-json'

```

```

    );
313     let currentObj = {};
314     try {
315         currentObj = JSON.parse(rawEl.value || '{}');
316     } catch {}
317
318     const serverMode = document.getElementById('app-cfg-server-mode')?.value || 'dedicated';
319     const isDedicated = serverMode === 'dedicated';
320     const port = parseInt(document.getElementById('app-cfg-server-port')?.value || '8100', 10);
321     const host = document.getElementById('app-cfg-server-host')?.value || '127.0.0.1';
322     const workers = parseInt(document.getElementById('app-cfg-server-workers')?.value || '1', 10);
323     const ssl = !!document.getElementById('app-cfg-server-ssl')?.checked;
324
325     if (!currentObj.server || typeof currentObj.server !== 'object') {
326         currentObj.server = {};
327     }
328     currentObj.server.dedicated = isDedicated;
329     if (currentObj.server.mode !== undefined) {
330         delete currentObj.server.mode;
331     }
332     currentObj.server.host = host;
333     currentObj.server.port = port;
334     currentObj.server.use_ssl = ssl;
335     currentObj.server.workers = workers;
336
337     // Sync extra inputs
338     document.querySelectorAll('.app-cfg-extra-input').forEach(
339         (input) => {
340             const path = input.getAttribute('data-path');
341             const val = input.value;
342             if (!path) return;

```

```

343 |         let parsedVal = val;
344 |         if (val === 'true') parsedVal = true;
345 |         else if (val === 'false') parsedVal = false;
346 |         else if (/^\d+$/.test(val)) parsedVal = parseInt(val,
| 10);
347 |         else if (/^\d+\.\d+$/.test(val)) parsedVal = parseFloat(val);
348 |
349 |         const parts = path.split('.');
350 |         if (parts.length === 1) {
351 |             currentObj[parts[0]] = parsedVal;
352 |         } else if (parts.length === 2) {
353 |             if (!currentObj[parts[0]] || typeof currentObj[parts
| [0]] !== 'object') {
354 |                 currentObj[parts[0]] = {};
355 |             }
356 |             currentObj[parts[0]][parts[1]] = parsedVal;
357 |         }
358 |     });
359 |
360 |     rawEl.value = JSON.stringify(currentObj, null, 2);
361 |     activeConfig = currentObj;
362 | } catch (e) {
363 |     console.error('[AppConfigEditor] Form sync error:', e);
364 | }
365 | }
366 |
367 | function syncRawJsonToForm() {
368 |     try {
369 |         const rawEl = document.getElementById('app-cfg-raw-json'
| );
370 |         const parsed = JSON.parse(rawEl.value || '{}');
371 |         activeConfig = parsed;
372 |         populateVisualForm(parsed);
373 |     } catch (e) {
374 |         showModalAlert(`Ошибка парсинга JSON: ${e.message}`, 'da
| nger');
375 |     }

```

```
376 | }
377 |
378 | async function onSaveConfig() {
379 |     try {
380 |         syncFormToRawJson();
381 |         const rawEl = document.getElementById('app-cfg-raw-json'
    | );
382 |         const parsed = JSON.parse(rawEl.value);
383 |
384 |         const res = await saveAppConfig(activeAppName, parsed);
385 |         showModalAlert(`❏ ${res.message || 'Конфигурация успешно
    | сохранена!'}`, 'success');
386 |         activeConfig = parsed;
387 |
388 |         if (typeof window.showNotification === 'function') {
389 |             window.showNotification(`Конфигурация ${activeAppName}
    | сохранена`, 'success');
390 |         }
391 |     } catch (e) {
392 |         showModalAlert(`Ошибка при сохранении: ${e.message}`, 'd
    | anger');
393 |     }
394 | }
395 |
396 | async function onReloadConfig() {
397 |     try {
398 |         const data = await fetchAppConfig(activeAppName);
399 |         activeConfig = data.config;
400 |         populateVisualForm(activeConfig);
401 |         showModalAlert('Конфигурация перезагружена с диска', 'in
    | fo');
402 |     } catch (e) {
403 |         showModalAlert(`Ошибка загрузки: ${e.message}`, 'danger'
    | );
404 |     }
405 | }
406 |
407 | /**
```

```
408 |     * Open the configuration modal for an app
409 |     * @param {string} appName e.g. 'trading_terminal', 'network
    | _terminal', etc.
410 |     * @param {string} displayName e.g. 'Exchange Trading Termin
    | al'
411 |     */
412 |     window.openAppConfigModal = async function(appName, displayN
    | ame = '') {
413 |         ensureModalExists();
414 |         activeAppName = appName;
415 |
416 |         const titleEl = document.getElementById('app-config-modal-
    | title');
417 |         if (titleEl) {
418 |             titleEl.innerHTML = `<i class="bi bi-sliders"></i> ${dis
    | playName || appName} – config.json`;
419 |         }
420 |
421 |         const pathEl = document.getElementById('app-cfg-file-path'
    | );
422 |         if (pathEl) {
423 |             pathEl.textContent = `apps/${appName}/config.json`;
424 |         }
425 |
426 |         const alertEl = document.getElementById('app-config-alert'
    | );
427 |         if (alertEl) alertEl.classList.add('d-none');
428 |
429 |         // Show modal
430 |         const modalEl = document.getElementById('app-config-modal'
    | );
431 |         const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
    | ;
432 |         modal.show();
433 |
434 |         try {
435 |             const data = await fetchAppConfig(appName);
436 |             activeConfig = data.config;
```



```
437 |         populateVisualForm(activeConfig);
438 |     } catch (e) {
439 |         showModalAlert(`Не удалось загрузить конфигурацию: ${e.m
|message}`, 'danger');
440 |     }
441 | };
442 |})();
443 |
```

browser-cache.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Browser-Cache Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля browser-c
7  *   ache.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/browser-cache.js?v=
11  *     20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { STORES, BrowserCacheManager, browserCache } fr
15  *     om '/windows/api/webgui/js/browser-cache.js';
16  *
17  * File: browser-cache.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/js
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-01 13:04:40
23  * =====
24  */
25
26 /**
27  * browser-cache.js - Универсальный менеджер браузерного кеша
28  * и долговременного хранилища
29  *
30  * Предоставляет многоуровневое хранилище (L1 In-Memory + L2 I
31  * ndexedDB + L3 Fallback)
32  *
33  * для эффективного хранения больших объемов клиентских данных
```

```

:
29 * - Ответы API и справочники
30 * - RAG-индексы, эмбединги и поисковые чанки
31 * - История чатов и сообщений ассистентов
32 * - Аудиофайлы TTS и медиа-блобы (Blob / ArrayBuffer)
33 * - Реестры моделей и системные конфигурации
34 *
35 * Включает автоматическую очистку по TTL, групповую инвалидац
ию по тегам,
36 * мониторинг дисковой квоты браузера и персистентность.
37 *
38 * ПРИМЕЧАНИЕ: Приложение запускается в своем окне через Edge
--app=url,
39 * что может ограничивать доступ к Storage API. Проверьте конс
оль браузера
40 * на наличие предупреждений о недоступности IndexedDB.
41 */
42
43 const DB_NAME = 'AI_Breadboard_DB';
44 const DB_VERSION = 1;
45
46 // Стандартные хранилища объектов IndexedDB
47 export const STORES = {
48   API_CACHE: 'api_cache',
49   RAG_EMBEDDINGS: 'rag_embeddings',
50   CHAT_HISTORY: 'chat_history',
51   MEDIA_BLOBS: 'media_blobs',
52   MODELS_REGISTRY: 'models_registry',
53   KEY_VALUE: 'key_value'
54 };
55
56 export class BrowserCacheManager {
57   /**
58    * Инициализация менеджера браузерного кеша
59    * @param {Object} options - Параметры конфигурации
60    * @param {number} options.defaultTTL - Время жизни по умолч
анию (в мс, 1 час = 3600000)
61    * @param {number} options.maxMemoryEntries - Максимальное ч

```

```

    число записей в L1 памяти
62     */
63     constructor(options = {}) {
64         this.defaultTTL = options.defaultTTL || 60 * 60 * 1000; //
        1 час
65         this.maxMemoryEntries = options.maxMemoryEntries || 200;
66         this.memoryCache = new Map(); // L1 Cache: key -> { value,
        expiresAt, storeName, tags }
67         this.db = null;
68         this.isDbReady = false;
69         this.initPromise = this._initIndexedDB();
70
71         // Метрики
72         this.stats = {
73             hits: 0,
74             misses: 0,
75             writes: 0,
76             deletes: 0
77         };
78
79         // Периодическая фоновая очистка просроченных записей кажд
        ые 10 минут
80         if (typeof window !== 'undefined') {
81             window.setInterval(() => this.cleanupExpired(), 10 * 60
        * 1000);
82         }
83     }
84
85     /**
86     * Инициализация базы данных IndexedDB
87     * @private
88     */
89     async _initIndexedDB() {
90         if (typeof window === 'undefined' || !window.indexedDB) {
91             console.warn('[BrowserCache] IndexedDB недоступен в теку
        щем окружении. Используется fallback на память.');
```

Используется fallback на память.

```

92             console.warn('[BrowserCache] Возможные причины: ');
93             console.warn('[BrowserCache] 1. Приложение запущено в PW

```

```

A-окне (Edge --app=url)');
94     console.warn('[BrowserCache] 2. Ограничения безопасности
браузера');
95     console.warn('[BrowserCache] 3. Поврежденный user-data-d
ir');
96     this.isDbReady = false;
97     return null;
98   }
99
100   return new Promise((resolve) => {
101     try {
102       const request = indexedDB.open(DB_NAME, DB_VERSION);
103
104       request.onupgradeneeded = (event) => {
105         const db = event.target.result;
106
107         Object.values(STORES).forEach((storeName) => {
108           if (!db.objectStoreNames.contains(storeName)) {
109             const store = db.createObjectStore(storeName, {
110               keyPath: 'key' });
111             store.createIndex('expiresAt', 'expiresAt', { unique: false });
112             store.createIndex('updatedAt', 'updatedAt', { unique: false });
113             store.createIndex('tags', 'tags', { unique: false, multiEntry: true });
114           });
115           console.log('[BrowserCache] Структура IndexedDB успешно обновлена до версии', DB_VERSION);
116         });
117
118         request.onsuccess = (event) => {
119           this.db = event.target.result;
120           this.isDbReady = true;
121           console.log('[BrowserCache] IndexedDB успешно подключена:', DB_NAME);
122           console.log('[BrowserCache] Данные будут сохраняться

```

```

    в профиле пользователя:', this._getUserDataDir());
123     this.cleanupExpired().catch(() => {});
124     resolve(this.db);
125 };
126
127     request.onerror = (event) => {
128         console.error('[BrowserCache] Ошибка открытия IndexedDB:', event.target.error);
129         console.error('[BrowserCache] Проверьте:');
130         console.error('[BrowserCache] 1. Есть ли права на запись в user-data-dir');
131         console.error('[BrowserCache] 2. Не заблокирован ли IndexedDB политикой браузера');
132         this.isDbReady = false;
133         resolve(null);
134     };
135     } catch (err) {
136         console.error('[BrowserCache] Исключение при инициализации IndexedDB:', err);
137         this.isDbReady = false;
138         resolve(null);
139     }
140 });
141 }
142
143 /**
144  * Ожидание готовности базы данных
145  */
146 async ready() {
147     await this.initPromise;
148     return this.isDbReady;
149 }
150
151 /**
152  * Сохранить значение в кеш (L1 память + L2 IndexedDB)
153  * @param {string} storeName - Имя хранилища (из STORES)
154  * @param {string} key - Уникальный ключ
155  * @param {any} value - Данные (объект, массив, строка, Blob

```

```

    , ArrayBuffer)
156     * @param {Object} options - Опции записи (ttl, tags, persist)
157     * @returns {Promise<boolean>}
158     */
159     async set(storeName = STORES.KEY_VALUE, key, value, options
    = {}) {
160         if (!key) return false;
161         await this.ready();
162
163         const ttl = options.ttl !== undefined ? options.ttl : this
            .defaultTTL;
164         const now = Date.now();
165         const expiresAt = ttl > 0 ? now + ttl : 0; // 0 = бессрочно
166         const tags = Array.isArray(options.tags) ? options.tags :
            (options.tags ? [options.tags] : []);
167
168         const record = {
169             key,
170             value,
171             expiresAt,
172             createdAt: now,
173             updatedAt: now,
174             tags,
175             size: this._estimateSize(value)
176         };
177
178         // 1. Сохранение в L1 In-Memory Cache
179         const memKey = `${storeName}:${key}`;
180         this._setMemoryCache(memKey, record);
181         this.stats.writes++;
182
183         // Если persist = false, сохраняем только в L1 памяти
184         if (options.persist === false) {
185             return true;
186         }
187

```

```

188 // 2. Сохранение в L2 IndexedDB
189 if (this.isDbReady && this.db) {
190     try {
191         await new Promise((resolve, reject) => {
192             const transaction = this.db.transaction([storeName],
193             'readwrite');
194             const store = transaction.objectStore(storeName);
195             const request = store.put(record);
196
197             request.onsuccess = () => resolve(true);
198             request.onerror = (e) => reject(e.target.error);
199         });
200         return true;
201     } catch (err) {
202         console.warn(`[BrowserCache] Ошибка записи в IndexedDB
203         (${storeName}/${key}):`, err);
204     }
205 }
206
207 return true;
208 }
209
210 /**
211  * Получить значение из кеша (L1 -> L2)
212  * @param {string} storeName - Имя хранилища
213  * @param {string} key - Ключ
214  * @returns {Promise<any|null>}
215  */
216
217 async get(storeName = STORES.KEY_VALUE, key) {
218     if (!key) return null;
219     await this.ready();
220
221     const now = Date.now();
222     const memKey = `${storeName}:${key}`;
223
224     // 1. Проверяем L1 In-Memory Cache
225     const memRecord = this.memoryCache.get(memKey);
226     if (memRecord) {

```



```
224 |         if (memRecord.expiresAt > 0 && now > memRecord.expiresAt
    |     ) {
225 |         this.memoryCache.delete(memKey);
226 |         this.delete(storeName, key).catch(() => {});
227 |         this.stats.misses++;
228 |         return null;
229 |     }
230 |     this.stats.hits++;
231 |     return memRecord.value;
232 | }
233 |
234 | // 2. Проверяем L2 IndexedDB
235 | if (this.isDbReady && this.db) {
236 |     try {
237 |         const record = await new Promise((resolve, reject) =>
    |     {
238 |         const transaction = this.db.transaction([storeName],
    |     'readonly');
239 |         const store = transaction.objectStore(storeName);
240 |         const request = store.get(key);
241 |
242 |         request.onsuccess = () => resolve(request.result);
243 |         request.onerror = (e) => reject(e.target.error);
244 |     });
245 |
246 |     if (!record) {
247 |         this.stats.misses++;
248 |         return null;
249 |     }
250 |
251 |     // Проверяем срок действия
252 |     if (record.expiresAt > 0 && now > record.expiresAt) {
253 |         this.delete(storeName, key).catch(() => {});
254 |         this.stats.misses++;
255 |         return null;
256 |     }
257 |
258 |     // Добавляем в L1 для ускорения повторных обращений
```

```

259         this._setMemoryCache(memKey, record);
260         this.stats.hits++;
261         return record.value;
262     } catch (err) {
263         console.warn(`[BrowserCache] Ошибка чтения из IndexedD
B (${storeName}/${key}):`, err);
264     }
265 }
266
267     this.stats.misses++;
268     return null;
269 }
270
271 /**
272  * Проверить наличие актуального ключа в кеше
273  */
274 async has(storeName = STORES.KEY_VALUE, key) {
275     const val = await this.get(storeName, key);
276     return val !== null;
277 }
278
279 /**
280  * Удалить запись из кеша
281  * @param {string} storeName - Имя хранилища
282  * @param {string} key - Ключ
283  */
284 async delete(storeName = STORES.KEY_VALUE, key) {
285     if (!key) return false;
286     await this.ready();
287
288     const memKey = `${storeName}:${key}`;
289     this.memoryCache.delete(memKey);
290     this.stats.deletes++;
291
292     if (this.isDbReady && this.db) {
293         try {
294             await new Promise((resolve, reject) => {
295                 const transaction = this.db.transaction([storeName],

```

```

    'readwrite');
296         const store = transaction.objectStore(storeName);
297         const request = store.delete(key);
298
299         request.onsuccess = () => resolve(true);
300         request.onerror = (e) => reject(e.target.error);
301     });
302     return true;
303 } catch (err) {
304     console.warn(`[BrowserCache] Ошибка удаления из Indexe
dDB (${storeName}/${key}):`, err);
305 }
306 }
307 return true;
308 }
309
310 /**
311  * Инвалидировать записи по тегу (например: 'models', 'chat_
session_1')
312  * @param {string} storeName - Имя хранилища
313  * @param {string} tag - Тег для поиска
314  */
315 async invalidateByTag(storeName = STORES.KEY_VALUE, tag) {
316     await this.ready();
317
318     // Удаляем из L1
319     for (const [memKey, record] of this.memoryCache.entries())
320     {
321         if (memKey.startsWith(`${storeName}:`) && record.tags &&
record.tags.includes(tag)) {
322             this.memoryCache.delete(memKey);
323         }
324     }
325
326     if (!this.isDbReady || !this.db) return 0;
327
328     let count = 0;
329     try {

```

```

329         count = await new Promise((resolve, reject) => {
330             const transaction = this.db.transaction([storeName], '
readwrite');
331             const store = transaction.objectStore(storeName);
332             const index = store.index('tags');
333             const request = index.getAllKeys(tag);
334
335             request.onsuccess = () => {
336                 const keys = request.result || [];
337                 keys.forEach((k) => store.delete(k));
338                 resolve(keys.length);
339             };
340             request.onerror = (e) => reject(e.target.error);
341         });
342         console.log(`[BrowserCache] Инвалидировано ${count} запи
сей по тегу '${tag}' в '${storeName}'`);
343     } catch (err) {
344         console.warn(`[BrowserCache] Ошибка инвалидации по тегу
'${tag}':`, err);
345     }
346     return count;
347 }
348
349 /**
350  * Инвалидировать записи по регулярному выражению или префик
су ключа
351  * @param {string} storeName - Имя хранилища
352  * @param {string|RegExp} pattern - Регулярное выражение или
строка префикса
353  */
354     async invalidateByPattern(storeName = STORES.KEY_VALUE, patt
ern) {
355         await this.ready();
356
357         const isRegExp = pattern instanceof RegExp;
358         const testMatch = (key) => isRegExp ? pattern.test(key) :
(key === pattern || key.startsWith(pattern));
359

```

```

360 // Очистка L1
361 for (const [memKey] of this.memoryCache.entries()) {
362     if (memKey.startsWith(`${storeName}:`)) {
363         const rawKey = memKey.replace(`${storeName}:`, '');
364         if (testMatch(rawKey)) {
365             this.memoryCache.delete(memKey);
366         }
367     }
368 }
369
370 if (!this.isDbReady || !this.db) return 0;
371
372 let deletedCount = 0;
373 try {
374     deletedCount = await new Promise((resolve, reject) => {
375         const transaction = this.db.transaction([storeName], '
readwrite');
376         const store = transaction.objectStore(storeName);
377         const request = store.openCursor();
378
379         request.onsuccess = (event) => {
380             const cursor = event.target.result;
381             if (cursor) {
382                 if (testMatch(cursor.key)) {
383                     cursor.delete();
384                     deletedCount++;
385                 }
386                 cursor.continue();
387             } else {
388                 resolve(deletedCount);
389             }
390         };
391         request.onerror = (e) => reject(e.target.error);
392     });
393     console.log(`[BrowserCache] Инвалидировано ${deletedCount}
записей по паттерну в '${storeName}'`);
394 } catch (err) {
395     console.warn(`[BrowserCache] Ошибка инвалидации по патте

```

```

    пну в `${storeName}`:`, err);
396     }
397     return deletedCount;
398 }
399
400 /**
401  * Полностью очистить указанное хранилище
402  * @param {string} storeName - Имя хранилища
403  */
404 async clearStore(storeName = STORES.KEY_VALUE) {
405     await this.ready();
406
407     // Очистка L1
408     for (const [memKey] of this.memoryCache.entries()) {
409         if (memKey.startsWith(`${storeName}:`)) {
410             this.memoryCache.delete(memKey);
411         }
412     }
413
414     if (this.isDbReady && this.db) {
415         try {
416             await new Promise((resolve, reject) => {
417                 const transaction = this.db.transaction([storeName],
418                 'readwrite');
419                 const store = transaction.objectStore(storeName);
420                 const request = store.clear();
421
422                 request.onsuccess = () => resolve(true);
423                 request.onerror = (e) => reject(e.target.error);
424             });
425             console.log(`[BrowserCache] Хранилище '${storeName}' п
426             олностью очищено.`);
427             return true;
428         } catch (err) {
429             console.warn(`[BrowserCache] Ошибка очистки хранилища
430             '${storeName}`:`, err);
431         }
432     }
433 }

```

```

430     return true;
431 }
432
433 /**
434  * Очистить абсолютно все хранилища
435  */
436 async clearAll() {
437     this.memoryCache.clear();
438     for (const storeName of Object.values(STORES)) {
439         await this.clearStore(storeName);
440     }
441     console.log('[BrowserCache] Все хранилища браузерного кеша
очищены. ');
442 }
443
444 /**
445  * Очистить просроченные записи во всех хранилищах
446  */
447 async cleanupExpired() {
448     await this.ready();
449     const now = Date.now();
450
451     // L1
452     for (const [memKey, record] of this.memoryCache.entries())
453     {
454         if (record.expiresAt > 0 && now > record.expiresAt) {
455             this.memoryCache.delete(memKey);
456         }
457     }
458
459     if (!this.isDbReady || !this.db) return 0;
460
461     let totalDeleted = 0;
462     for (const storeName of Object.values(STORES)) {
463         try {
464             const deletedInStore = await new Promise((resolve, rej
ect) => {
465                 const transaction = this.db.transaction([storeName],

```

```

    'readwrite');
465     const store = transaction.objectStore(storeName);
466     const index = store.index('expiresAt');
467     const range = IDBKeyRange.bound(1, now);
468     const request = index.openCursor(range);
469     let count = 0;
470
471     request.onsuccess = (event) => {
472         const cursor = event.target.result;
473         if (cursor) {
474             cursor.delete();
475             count++;
476             cursor.continue();
477         } else {
478             resolve(count);
479         }
480     };
481     request.onerror = (e) => reject(e.target.error);
482 });
483 totalDeleted += deletedInStore;
484 } catch (e) {
485     // Игнорируем возможные локальные ошибки курсора
486 }
487 }
488
489 if (totalDeleted > 0) {
490     console.log(`[BrowserCache] Фоновый сборщик мусора удали
л ${totalDeleted} просроченных записей.`);
491 }
492 return totalDeleted;
493 }
494
495 /**
496  * Получить детальную статистику использования кеша и дисков
ой квоты
497  */
498 async getDetailedStats() {
499     await this.ready();

```



```

500     const stats = {
501         metrics: { ...this.stats },
502         memoryCacheSize: this.memoryCache.size,
503         stores: {},
504         storageEstimate: {
505             usage: 0,
506             quota: 0,
507             usageMB: '0.00',
508             quotaMB: '0.00',
509             percentUsed: 0,
510             isPersisted: false
511         }
512     };
513
514     // Оценка дискового пространства через navigator.storage A
    PI
515     if (typeof navigator !== 'undefined' && navigator.storage
    && navigator.storage.estimate) {
516         try {
517             const estimate = await navigator.storage.estimate();
518             const usage = estimate.usage || 0;
519             const quota = estimate.quota || 0;
520             const usageMB = (usage / (1024 * 1024)).toFixed(2);
521             const quotaMB = (quota / (1024 * 1024)).toFixed(2);
522             const percentUsed = quota > 0 ? ((usage / quota) * 100
    ).toFixed(2) : 0;
523
524             let isPersisted = false;
525             if (navigator.storage.persisted) {
526                 isPersisted = await navigator.storage.persisted();
527             }
528
529             stats.storageEstimate = {
530                 usage,
531                 quota,
532                 usageMB,
533                 quotaMB,
534                 percentUsed: Number(percentUsed),

```

```

535         isPersisted
536     };
537     } catch (err) {
538         console.warn('[BrowserCache] Не удалось получить navigator.storage.estimate:', err);
539     }
540 }
541
542 // Подсчет записей и объема по каждому хранилищу
543 if (this.isDbReady && this.db) {
544     for (const storeName of Object.values(STORES)) {
545         try {
546             const storeData = await new Promise((resolve) => {
547                 const transaction = this.db.transaction([storeName], 'readonly');
548                 const store = transaction.objectStore(storeName);
549                 const countReq = store.count();
550                 const allReq = store.getAll();
551
552                 let count = 0;
553                 let totalBytes = 0;
554
555                 countReq.onsuccess = () => { count = countReq.result || 0; };
556                 allReq.onsuccess = () => {
557                     const records = allReq.result || [];
558                     records.forEach(r => {
559                         totalBytes += r.size || this._estimateSize(r.value);
560                     });
561                     resolve({ count, sizeBytes: totalBytes });
562                 };
563                 allReq.onerror = () => resolve({ count: 0, sizeBytes: 0 });
564             });
565
566             stats.stores[storeName] = {
567                 count: storeData.count,

```

```

568         sizeBytes: storeData.sizeBytes,
569         sizeKB: (storeData.sizeBytes / 1024).toFixed(2),
570         sizeMB: (storeData.sizeBytes / (1024 * 1024)).toFi
xed(2)
571     };
572     } catch (e) {
573         stats.stores[storeName] = { count: 0, sizeBytes: 0,
sizeKB: '0', sizeMB: '0' };
574     }
575 }
576 }
577
578     return stats;
579 }
580
581 /**
582  * Запросить у браузера постоянное хранилище (защита от случ
айной очистки)
583  */
584     async requestPersistence() {
585         if (typeof navigator !== 'undefined' && navigator.storage
&& navigator.storage.persist) {
586             const isPersisted = await navigator.storage.persist();
587             console.log(`[BrowserCache] Запрос персистентности храни
лица: ${isPersisted ? 'Одобрено' : 'Отклонено'}`);
588             return isPersisted;
589         }
590         return false;
591     }
592
593 /**
594  * Экспортировать данные хранилищ в JSON-структуру
595  * @param {Array<string>} storeNames - Список хранилищ для э
кспорта
596  */
597     async exportData(storeNames = [STORES.API_CACHE, STORES.MODE
LS_REGISTRY, STORES.CHAT_HISTORY]) {
598         await this.ready();

```

```

599     const exportResult = {
600         version: DB_VERSION,
601         timestamp: Date.now(),
602         exportedAt: new Date().toISOString(),
603         stores: {}
604     };
605
606     if (!this.isDbReady || !this.db) {
607         console.warn('[BrowserCache] Cannot export data: Indexed
DB not ready');
608         return exportResult;
609     }
610
611     for (const storeName of storeNames) {
612         try {
613             const records = await new Promise((resolve, reject) =>
{
614                 const transaction = this.db.transaction([storeName],
'readonly');
615                 const store = transaction.objectStore(storeName);
616                 const request = store.getAll();
617                 request.onsuccess = () => resolve(request.result ||
[]);
618                 request.onerror = (e) => reject(e.target.error);
619             });
620             exportResult.stores[storeName] = records;
621         } catch (err) {
622             console.warn(`[BrowserCache] Ошибка экспорта из ${stor
eName}:`, err);
623         }
624     }
625
626     return exportResult;
627 }
628
629 /**
630  * Импортировать данные в хранилища
631  * @param {Object} data - Экспортированный объект данных

```

```

632 |     */
633 |     async importData(data) {
634 |         if (!data || !data.stores) return false;
635 |         await this.ready();
636 |
637 |         for (const [storeName, records] of Object.entries(data.stores)) {
638 |             if (!Object.values(STORES).includes(storeName)) continue
639 |             ;
640 |             if (!Array.isArray(records)) continue;
641 |             for (const record of records) {
642 |                 if (record && record.key) {
643 |                     await this.set(storeName, record.key, record.value,
644 |                     {
645 |                         ttl: record.expiresAt ? Math.max(0, record.expires
646 |                         At - Date.now()) : 0,
647 |                         tags: record.tags || []
648 |                     });
649 |                 }
650 |             }
651 |             console.log('[BrowserCache] Импорт данных успешно завершен
652 |             .');
653 |             return true;
654 |         }
655 |     }
656 |
657 |     /**
658 |      * Добавить запись в L1 In-Memory Cache с контролем размера
659 |      (LRU eviction)
660 |      * @private
661 |      */
662 |     _setMemoryCache(key, record) {
663 |         if (this.memoryCache.size >= this.maxMemoryEntries) {
664 |             const firstKey = this.memoryCache.keys().next().value;
665 |             if (firstKey) this.memoryCache.delete(firstKey);
666 |         }
667 |         this.memoryCache.set(key, record);

```

```

664 | }
665 |
666 | /**
667 |  * Приблизительная оценка размера объекта в байтах
668 |  * @private
669 |  */
670 | _estimateSize(value) {
671 |     if (value === null || value === undefined) return 0;
672 |     if (typeof value === 'string') return value.length * 2;
673 |     if (typeof value === 'number') return 8;
674 |     if (typeof value === 'boolean') return 4;
675 |     if (typeof Blob !== 'undefined' && value instanceof Blob)
676 |     return value.size;
677 |     if (typeof ArrayBuffer !== 'undefined' && value instanceof
678 |     ArrayBuffer) return value.byteLength;
679 |     try {
680 |         return JSON.stringify(value).length * 2;
681 |     } catch {
682 |         return 1024;
683 |     }
684 | }
685 |
686 | /**
687 |  * Попытка получить путь к user-data-dir (для отладки)
688 |  * @private
689 |  */
690 | _getUserDataDir() {
691 |     // В браузере этот путь недоступен из-за ограничений безоп
692 |     асности
693 |     // Но мы можем попытаться определить, запущено ли приложен
694 |     ие в PWA-окне
695 |     const isPWA = window.matchMedia('(display-mode: standalone
696 | )').matches;
697 |     return isPWA ? '[PWA-окно - путь недоступен]' : '[Обычное
698 | окно]';
699 | }
700 | }

```

```
696 | // Создаем глобальный singleton экземпляр
697 | export const browserCache = new BrowserCacheManager();
698 |
699 | // Регистрация в глобальном объекте window для доступности во
      | всех вкладках и скриптах
700 | if (typeof window !== 'undefined') {
701 |   window.BrowserCacheManager = BrowserCacheManager;
702 |   window.browserCache = browserCache;
703 |   window.BROWSER_STORES = STORES;
704 | }
705 |
706 | export default browserCache;
707 |
```

cache-ui.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Cache-Ui Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля cache-ui.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/cache-ui.js?v=20261
11  *     001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { initCacheUI, openCacheModal } from '/windows/a
15  *     pi/webgui/js/cache-ui.js';
16  *
17  * File: cache-ui.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/js
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-01 13:04:40
23  * =====
24  */
25  * cache-ui.js - Контроллер пользовательского интерфейса управ
26  * ления браузерным кешем
27  *
28  * Отвечает за:
29  * - Отображение модального окна управления кешем
30  * - Получение и визуализацию статистики хранилищ и квот браузер
```



```

30 * - Обработку действий очистки, экспорта, импорта и запроса п
    ерсистентности
31 */
32
33 import { browserCache, STORES } from './browser-cache.js';
34
35 // Человекопонятные названия хранилищ
36 const STORE_LABELS = {
37   [STORES.API_CACHE]: { title: 'Кеш ответов API', icon: 'bi-hd
    d-network', desc: 'Кешированные ответы сервера и справочники'
    },
38   [STORES.RAG_EMBEDDINGS]: { title: 'RAG и Векторы', icon: 'bi
    -database-fill-gear', desc: 'Эмбеддинги, векторные чанки и пои
    сковый индекс' },
39   [STORES.CHAT_HISTORY]: { title: 'История чатов', icon: 'bi-c
    hat-dots-fill', desc: 'Диалоги, сообщения ассистентов и контек
    ст' },
40   [STORES.MEDIA_BLOBS]: { title: 'Медиа и Аудио TTS', icon: 'b
    i-file-earmark-music-fill', desc: 'Сгенерированные аудиодорожк
    и, картинки и бинарные файлы' },
41   [STORES.MODELS_REGISTRY]: { title: 'Реестр моделей', icon: '
    bi-robot', desc: 'Метаданные локальных и облачных моделей' },
42   [STORES.KEY_VALUE]: { title: 'Пользовательские данные', icon
    : 'bi-key-fill', desc: 'Локальные параметры и временные значен
    ия' }
43 };
44
45 /**
46  * Инициализация интерфейса управления кешем
47  */
48 export function initCacheUI() {
49   const openButtons = document.querySelectorAll('#btn-open-cac
    he-manager, .btn-open-cache-manager, #user-menu-btn-cache');
50   const modalEl = document.getElementById('cacheManagerModal')
    ;
51
52   if (!modalEl) {
53     console.warn('[CacheUI] Элемент #cacheManagerModal не найд

```

```
    ен в DOM. ');
54     return;
55 }
56
57 // Привязка кнопок открытия
58 openButtons.forEach((btn) => {
59     btn.addEventListener('click', (e) => {
60         e.preventDefault();
61         openCacheModal();
62     });
63 });
64
65 // Обновление статистики при открытии модального окна
66 modalEl.addEventListener('show.bs.modal', () => {
67     refreshCacheModalStats();
68 });
69
70 // Привязка действий кнопок
71 const btnCleanupExpired = document.getElementById('cache-btn
-cleanup-expired');
72 if (btnCleanupExpired) {
73     btnCleanupExpired.addEventListener('click', async () => {
74         btnCleanupExpired.disabled = true;
75         const count = await browserCache.cleanupExpired();
76         window.showToast?(`Очищено просроченных записей: ${count}
t}`, 'info') || alert(`Очищено просроченных записей: ${count}`
);
77         btnCleanupExpired.disabled = false;
78         await refreshCacheModalStats();
79     });
80 }
81
82 const btnClearAll = document.getElementById('cache-btn-clear
-all');
83 if (btnClearAll) {
84     btnClearAll.addEventListener('click', async () => {
85         if (confirm('Вы действительно хотите полностью очистить
весь браузерный кеш?')) {
```

```
86 |         btnClearAll.disabled = true;
87 |         await browserCache.clearAll();
88 |         window.showToast?.('Все хранилища браузерного кеша усп
    | ешно очищены.', 'success') || alert('Все хранилища браузерного
    | кеша успешно очищены.');
```

```
89 |         btnClearAll.disabled = false;
90 |         await refreshCacheModalStats();
91 |     }
92 | });
93 | }
94 |
95 | const btnRequestPersist = document.getElementById('cache-btn
    | -request-persist');
96 | if (btnRequestPersist) {
97 |     btnRequestPersist.addEventListener('click', async () => {
98 |         const granted = await browserCache.requestPersistence();
99 |         const msg = granted ? 'Постоянное хранилище успешно вклю
    | чено браузером.' : 'Браузер отклонил запрос персистентности.';
100 |         window.showToast?.(msg, granted ? 'success' : 'warning')
    | || alert(msg);
101 |         await refreshCacheModalStats();
102 |     });
103 | }
104 |
105 | const btnExport = document.getElementById('cache-btn-export'
    | );
106 | if (btnExport) {
107 |     btnExport.addEventListener('click', async () => {
108 |         const data = await browserCache.exportData();
109 |         const blob = new Blob([JSON.stringify(data, null, 2)], {
    | type: 'application/json' });
110 |         const url = URL.createObjectURL(blob);
111 |         const a = document.createElement('a');
112 |         a.href = url;
113 |         a.download = `ai_breadboard_cache_${new Date().toISOString()
    | }.slice(0, 10)}.json`;
114 |         document.body.appendChild(a);
115 |         a.click();
```

```
116 |     document.body.removeChild(a);
117 |     URL.revokeObjectURL(url);
118 |     window.showToast?.( 'Экспорт кеша успешно сохранён', 'suc
    | ccess' );
119 |     });
120 | }
121 |
122 | const fileInput = document.getElementById('cache-file-import
    | ');
123 | if (fileInput) {
124 |     fileInput.addEventListener('change', async (e) => {
125 |         const file = e.target.files?.[0];
126 |         if (!file) return;
127 |
128 |         try {
129 |             const text = await file.text();
130 |             const data = JSON.parse(text);
131 |             await browserCache.importData(data);
132 |             window.showToast?.( 'Данные кеша успешно импортированы.
    | ', 'success') || alert('Данные кеша успешно импортированы. ');
133 |             await refreshCacheModalStats();
134 |         } catch (err) {
135 |             window.showToast?.( 'Ошибка при чтении файла импорта: '
    | + err.message, 'danger') || alert('Ошибка при чтении файла им
    | порта: ' + err.message);
136 |         } finally {
137 |             fileInput.value = '';
138 |         }
139 |     });
140 | }
141 | }
142 |
143 | /**
144 |  * Открыть модальное окно управления кешем
145 |  */
146 | export function openCacheModal() {
147 |     const modalEl = document.getElementById('cacheManagerModal')
    | ;
```

```

148 |   if (modalEl && typeof bootstrap !== 'undefined' && bootstrap
    |   .Modal) {
149 |       const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl
    |   l);
150 |       bsModal.show();
151 |   }
152 | }
153 |
154 | /**
155 |  * Обновить и отрендерить статистику в модальном окне
156 |  */
157 | export async function refreshCacheModalStats() {
158 |     const statsContainer = document.getElementById('cache-stores
    |   -list');
159 |     const usageText = document.getElementById('cache-usage-text'
    |   );
160 |     const usageProgress = document.getElementById('cache-usage-p
    |   rogressbar');
161 |     const persistBadge = document.getElementById('cache-persist-
    |   badge');
162 |
163 |     if (!statsContainer) return;
164 |
165 |     statsContainer.innerHTML = `
166 |         <div class="text-center py-3 text-muted">
167 |             <div class="spinner-border spinner-border-sm text-primar
    |   y me-2" role="status"></div>
168 |             Анализ данных браузерного хранилища...
169 |         </div>
170 |     `;
171 |
172 |     try {
173 |         const stats = await browserCache.getDetailedStats();
174 |
175 |         // 1. Отображение квоты
176 |         if (usageText && usageProgress) {
177 |             const usageMB = stats.storageEstimate.usageMB;
178 |             const quotaMB = stats.storageEstimate.quotaMB;

```

```
179 |     const percent = stats.storageEstimate.percentUsed;
180 |
181 |     usageText.textContent = `${usageMB} МБ из ${quotaMB} МБ
    |     (${percent}%)`;
182 |     usageProgress.style.width = `${Math.min(100, Math.max(1,
    |     percent))}%`;
183 |     usageProgress.className = `progress-bar ${percent > 80 ?
    |     'bg-danger' : (percent > 50 ? 'bg-warning' : 'bg-primary')}`;
184 |   }
185 |
186 |   // 2. Статус персистентности
187 |   if (persistBadge) {
188 |     if (stats.storageEstimate.isPersisted) {
189 |       persistBadge.className = 'badge bg-success-subtle text
    | -success border border-success-subtle';
190 |       persistBadge.innerHTML = '<i class="bi bi-shield-check
    | me-1"></i>Постоянное хранилище активно';
191 |     } else {
192 |       persistBadge.className = 'badge bg-warning-subtle text
    | -warning-emphasis border border-warning-subtle';
193 |       persistBadge.innerHTML = '<i class="bi bi-shield-excla
    | mation me-1"></i>Временное хранилище';
194 |     }
195 |   }
196 |
197 |   // 3. Таблица хранилищ
198 |   let html = '<div class="list-group list-group-flush border
    | rounded overflow-hidden">';
199 |   for (const [storeName, storeInfo] of Object.entries(stats.
    | stores)) {
200 |     const meta = STORE_LABELS[storeName] || { title: storeNa
    | me, icon: 'bi-folder', desc: 'Хранилище данных' };
201 |     html += `
202 |       <div class="list-group-item d-flex align-items-center
    | justify-content-between p-2.5">
203 |         <div class="d-flex align-items-center gap-3">
204 |           <div class="rounded-circle bg-primary-subtle text-
    | primary p-2 d-flex align-items-center justify-content-center"
```

```

style="width: 38px; height: 38px;">
205         <i class="bi ${meta.icon} fs-5"></i>
206     </div>
207     <div>
208         <div class="fw-semibold text-body">${meta.title}
</div>
209         <div class="small text-muted" style="font-size:
0.75rem;">${meta.desc}</div>
210     </div>
211 </div>
212     <div class="d-flex align-items-center gap-3">
213         <div class="text-end">
214             <span class="badge bg-secondary-subtle text-seco
ndary border px-2 py-1">${storeInfo.count} зап.</span>
215             <div class="small fw-semibold text-muted mt-0.5"
>${storeInfo.sizeKB} КБ</div>
216         </div>
217         <button class="btn btn-outline-danger btn-sm px-2
py-1 rounded"
218             title="Очистить это хранилище"
219             onclick="window.clearSingleStore('${storeN
ame}')">
220             <i class="bi bi-trash3"></i>
221         </button>
222     </div>
223 </div>
224 `;
225 }
226 html += '</div>';
227
228 statsContainer.innerHTML = html;
229 } catch (err) {
230     console.error('[CacheUI] Ошибка обновления статистики:', e
rr);
231     statsContainer.innerHTML = `<div class="alert alert-danger
p-2 small">Ошибка получения статистики кеша: ${err.message}</
div>`;
232 }

```

```
233 | }
234 |
235 | // Глобальная функция для очистки одного хранилища из UI
236 | if (typeof window !== 'undefined') {
237 |     window.clearSingleStore = async function(storeName) {
238 |         if (confirm(`Очистить данные хранилища "${storeName}"?`))
239 |         {
240 |             await browserCache.clearStore(storeName);
241 |             await refreshCacheModalStats();
242 |         }
243 |     };
244 | }
```


chatService.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Chatervice Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля chatService.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/chatService.js?v=20
11  *     261001_v1" type="module"></script>
12  *
13  * File: chatService.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/js
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Shared Chat Service Module
22 // Exposes window.chatService to unify API calling and message
23 // error formatting.
24 class AudioQueuePlayer {
25   constructor(chunks, userSettings) {
26     this.chunks = chunks;
27     this.userSettings = userSettings;
28     this.currentIndex = 0;
29     this.isPlaying = false;
30     this.audioElements = new Map(); // index -> Audio
```

```
31 |     this.activeAudio = null;
32 |
33 |     // Prefetch first few chunks
34 |     this._prefetch(0);
35 |     this._prefetch(1);
36 |     this._prefetch(2);
37 | }
38 |
39 | _prefetch(index) {
40 |     if (index >= this.chunks.length || this.audioElements.has(
41 | index)) return;
42 |
43 |     const text = this.chunks[index];
44 |     const queryParams = new URLSearchParams({
45 |         text: text,
46 |         system: this.userSettings.tts_system,
47 |         voice: this.userSettings.tts_voice
48 |     });
49 |     const audioUrl = `/api/tts/synthesize?${queryParams.toString()}`;
50 |     const audio = new Audio(audioUrl);
51 |     audio.load();
52 |     this.audioElements.set(index, audio);
53 | }
54 |
55 | async play() {
56 |     this.isPlaying = true;
57 |     while (this.currentIndex < this.chunks.length && this.isPl
58 | aying) {
59 |         // Prefetch upcoming chunks
60 |         this._prefetch(this.currentIndex);
61 |         this._prefetch(this.currentIndex + 1);
62 |         this._prefetch(this.currentIndex + 2);
63 |         this._prefetch(this.currentIndex + 3);
64 |
65 |         const audio = this.audioElements.get(this.currentIndex);
66 |         if (!audio) {
67 |             this.currentIndex++;
68 |         }
69 |         audio.play();
70 |     }
71 | }
```

```

66         continue;
67     }
68
69     this.activeAudio = audio;
70     window.chatServiceAudio = audio; // for global pause/sto
p integration
71
72     const playPromise = new Promise((resolve) => {
73         audio.onended = () => resolve();
74         audio.onerror = (e) => {
75             console.error(`Audio error for chunk ${this.currentI
ndex}:`, e);
76             this.stop(); // Останавливаем всю очередь при ошибке
загрузки
77             resolve();
78         };
79     });
80
81     try {
82         await audio.play();
83         await playPromise;
84     } catch (err) {
85         console.error("Audio playback interrupted or failed:",
err);
86     }
87
88     this.currentIndex++;
89 }
90 this.isPlaying = false;
91 }
92
93 stop() {
94     this.isPlaying = false;
95     if (this.activeAudio) {
96         try {
97             this.activeAudio.pause();
98             this.activeAudio.src = "";
99         } catch (e) {}

```

```
100     }
101     this.audioElements.clear();
102 }
103 }
104
105 window.chatService = {
106     /**
107      * Stops any currently playing audio or speech synthesis.
108      */
109     stop() {
110         if (window.chatServiceQueue) {
111             try {
112                 window.chatServiceQueue.stop();
113                 window.chatServiceQueue = null;
114             } catch (e) {
115                 console.error("Error stopping chatServiceQueue:", e);
116             }
117         }
118         if (window.chatServiceAudio) {
119             try {
120                 window.chatServiceAudio.pause();
121                 window.chatServiceAudio.src = "";
122                 window.chatServiceAudio = null;
123             } catch (e) {
124                 console.error("Error stopping chatServiceAudio:", e);
125             }
126         }
127         if ('speechSynthesis' in window) {
128             try {
129                 window.speechSynthesis.cancel();
130             } catch (e) {
131                 console.error("Error cancelling speechSynthesis:", e);
132             }
133         }
134     },
135     _abortController: null,
136
137     /**
```

```
138 | * Sends a chat message to the /api/chat backend.
139 | * Resolves to the text reply or throws the parsed error object/string.
140 | *
141 | * @param {string} message
142 | * @returns {Promise<any>}
143 | */
144 | async sendChatMessage(message, onChunk, history = [], generationConfig = {}) {
145 |     // Останавливаем любую активную озвучку перед отправкой нового сообщения
146 |     this.stop();
147 |
148 |     // Прерываем предыдущий запрос, если он еще выполняется
149 |     if (this._abortController) {
150 |         try {
151 |             this._abortController.abort();
152 |         } catch (e) {}
153 |     }
154 |     this._abortController = new AbortController();
155 |
156 |     let fullText = '';
157 |     let voiceText = '';
158 |
159 |     const effectiveGenConfig = { ...generationConfig };
160 |     if (!effectiveGenConfig.model && window.activeModelName) {
161 |         effectiveGenConfig.model = window.activeModelName;
162 |     }
163 |     if (!effectiveGenConfig.search_engine && window.activeSearchEngine) {
164 |         effectiveGenConfig.search_engine = window.activeSearchEngine;
165 |     }
166 |
167 |     try {
168 |         const response = await fetch('/api/chat', {
169 |             method: 'POST',
170 |             headers: { 'Content-Type': 'application/json' },
```

```
171 |         body: JSON.stringify({ message, history, generation_co
172 | nfig: effectiveGenConfig })),
173 |         signal: this._abortController.signal
174 |     });
175 |
176 |     if (!response.ok) {
177 |         let data;
178 |         try {
179 |             data = await response.json();
180 |         } catch (e) {
181 |             throw new Error(`Ошибка сервера (HTTP ${response.sta
182 | tus}): ${response.statusText}`);
183 |         }
184 |         throw (data.detail || data);
185 |     }
186 |
187 |     const reader = response.body.getReader();
188 |     const decoder = new TextDecoder('utf-8');
189 |     let buffer = '';
190 |
191 |     while (true) {
192 |         const { value, done } = await reader.read();
193 |         if (done) break;
194 |
195 |         buffer += decoder.decode(value, { stream: true });
196 |         const lines = buffer.split('\n');
197 |         buffer = lines.pop();
198 |
199 |         for (const line of lines) {
200 |             const trimmed = line.trim();
201 |             if (!trimmed) continue;
202 |             if (trimmed.startsWith('data: ')) {
203 |                 try {
204 |                     const data = JSON.parse(trimmed.substring(6));
205 |                     if (data.error) {
206 |                         throw new Error(data.error);
207 |                     }
208 |                     if (data.text || data.status || data.voice || da
```

```

    ta.prompt_dump) {
207         if (data.text) fullText += data.text;
208         if (data.voice) {
209             voiceText += data.voice;
210         }
211         if (onChunk) {
212             onChunk(data.text || '', data.status || '',
data.voice || '', data.prompt_dump || '');
213         }
214     }
215     } catch (e) {
216         console.error('Parsing SSE error:', e, trimmed);
217         throw e;
218     }
219 }
220 }
221 }
222
223 if (buffer && buffer.trim().startsWith('data: ')) {
224     const data = JSON.parse(buffer.trim().substring(6));
225     if (data.error) throw new Error(data.error);
226     if (data.text || data.status || data.voice) {
227         if (data.text) fullText += data.text;
228         if (data.voice) voiceText += data.voice;
229         if (onChunk) onChunk(data.text || '', data.status ||
'', data.voice || '');
230     }
231 }
232 } catch (e) {
233     if (e.name === 'AbortError') {
234         // Возвращаем Promise, который никогда не зарезолвится
235         .
236         // Это остановит старый вызов sendMessage() на мес
те, не вызывая catch/finally в UI-скриптах,
237         // так как они уже перекрыты новым запросом.
238         return new Promise(() => {});
239     }
    throw e;

```

```

240     }
241
242     let followUpQuery = null;
243     const nextQueryRegex = /\[NEXT_QUERY\](.*?)\[\/NEXT_QUERY\
    ]/i;
244     const match = fullText.match(nextQueryRegex);
245     if (match) {
246         followUpQuery = match[1].trim();
247         fullText = fullText.replace(match[0], '');
248     }
249
250     return { text: fullText, voice: voiceText, followUp: follo
    wUpQuery };
251 },
252
253
254 /**
255  * Formats chat messages.
256  * Converts objects into stringified JSON inside pre tags fo
    r rich rendering.
257  *
258  * @param {any} text
259  * @returns {string|HTMLElement}
260  */
261 formatMessage(text) {
262     if (typeof text === 'object' && text !== null) {
263         // Если это просто объект с ответом { text, voice }, ото
    бражаем только текст
264         if (text.text !== undefined) {
265             return text.text;
266         }
267         return `

```
`;
268 }
269 return text;
270 },

```


```



```
271 |
272 | /**
273 |  * Reads the given text aloud using the browser SpeechSynthe
    | sis API or backend TTS.
274 |  *
275 |  * @param {string|object} text
276 |  */
277 | async speak(text) {
278 |   if (!text) return;
279 |
280 |   // Останавливаем все предыдущие аудиопотоки
281 |   this.stop();
282 |
283 |   let cleanText = '';
284 |   if (typeof text === 'object' && text !== null) {
285 |     if (text.voice || text.text) {
286 |       // Если это объект { text, voice } из sendChatMessage
287 |       cleanText = text.voice || text.text;
288 |     } else {
289 |       // Если это карточка фильма
290 |       const title = text.title_ru || text.title;
291 |       const rec = text.why_watch || text.plot || '';
292 |       cleanText = `Найдено: ${title}. ${rec}`;
293 |     }
294 |   } else {
295 |     const trimmed = text.trim();
296 |     if (trimmed.startsWith('{') && trimmed.endsWith('}')) {
297 |       try {
298 |         const card = JSON.parse(trimmed);
299 |         const title = card.title_ru || card.title;
300 |         const rec = card.why_watch || card.plot || '';
301 |         cleanText = `Найдено: ${title}. ${rec}`;
302 |       } catch (e) {
303 |         cleanText = trimmed;
304 |       }
305 |     } else {
306 |       // Strip code blocks
307 |       cleanText = trimmed.replace(/```\s\S]*?```/g, '');
```

```

308 |         // Strip HTML tags
309 |         cleanText = cleanText.replace(/<[^>]*>/g, '');
310 |         // Strip Markdown formatting
311 |         cleanText = cleanText.replace(/[*_`#]/g, '');
312 |     }
313 | }
314 |
315 | if (!cleanText.trim()) return;
316 |
317 | let userSettings = null;
318 | try {
319 |     const response = await fetch('/auth/settings');
320 |     if (response.ok) {
321 |         userSettings = await response.json();
322 |         if (userSettings.tts_enabled === 1 && userSettings.tts
323 | _system && userSettings.tts_system !== 'browser') {
324 |             // Разбиваем текст по абзацам и предложениям на боле
325 | е мелкие чанки (~150-200 символов)
326 |             const paragraphs = cleanText
327 |                 .split(/\n+/)
328 |                 .map(p => p.trim())
329 |                 .filter(p => p.length > 0);
330 |
331 |             let chunks = [];
332 |             for (const para of paragraphs) {
333 |                 const sentences = para.match(/^[^!?]+[.!?]*$/g) ||
334 | [para];
335 |                 let currentChunk = "";
336 |                 for (const sentence of sentences) {
337 |                     const trimmedSentence = sentence.trim();
338 |                     if (!trimmedSentence) continue;
339 |                     if ((currentChunk + " " + trimmedSentence).lengt
340 | h > 200) {
341 |                         chunks.push(currentChunk.trim());
342 |                         currentChunk = trimmedSentence;
343 |                     } else {
344 |                         currentChunk = (currentChunk + " " + trimmedSe
345 | ntence).trim();

```

```
341         }
342     }
343     if (currentChunk) {
344         chunks.push(currentChunk.trim());
345     }
346 }
347 chunks = chunks.filter(c => c.length > 0);
348
349 if (chunks.length > 0) {
350     window.chatServiceQueue = new AudioQueuePlayer(chunks, userSettings);
351     await window.chatServiceQueue.play();
352 }
353 return;
354 }
355 }
356 } catch (e) {
357     console.error('Failed to speak using backend TTS, falling back to browser:', e);
358 }
359
360 if ('speechSynthesis' in window) {
361     window.currentUtterance = new SpeechSynthesisUtterance(cleanText);
362
363     // Dynamic language detection for Web Speech API fallback
364     const hasHebrew = /[\\u0590-\\u05FF]/.test(cleanText);
365     const hasCyrillic = /[\\u0400-\\u04FF]/.test(cleanText);
366     window.currentUtterance.lang = hasHebrew ? 'he-IL' : (hasCyrillic ? 'ru-RU' : 'en-US');
367
368     if (userSettings && userSettings.tts_voice) {
369         const voices = window.speechSynthesis.getVoices();
370         const selectedVoice = voices.find(v => v.name === userSettings.tts_voice);
371         if (selectedVoice) {
372             window.currentUtterance.voice = selectedVoice;
```

```
373         if (selectedVoice.lang) {
374             window.currentUtterance.lang = selectedVoice.lang;
375         }
376     }
377 }
378
379     window.speechSynthesis.speak(window.currentUtterance);
380 }
381 }
382 };
383
384 // Форматирование отображаемого названия поискового движка
385 function formatSearchEngine(engine) {
386     if (!engine) return '';
387     let eng = engine;
388     let mdl = '';
389     if (engine.includes(':') && !engine.startsWith('ollama:') &&
390         !engine.startsWith('foundry:')) {
391         const parts = engine.split(':');
392         eng = parts[0];
393         mdl = parts.slice(1).join(':');
394     }
395     const map = {
396         'gemini_cli': '🔍 gemini_cli',
397         'gemini': '🔍 gemini',
398         'agy': '🔍 agy',
399         'langchain': '🔍 langchain',
400         'playwright': '🔍 playwright'
401     };
402     const iconAndName = map[eng] || `🔍 ${eng}`;
403     if (mdl && !mdl.startsWith('chromium')) {
404         return `${iconAndName} (${mdl})`;
405     }
406     return iconAndName;
407 }
408 window.formatSearchEngine = formatSearchEngine;
409
```

```
410 window.updateChatBadges = function(modelName, searchEngine) {
411   if (modelName !== undefined && modelName !== null) {
412     window.activeModelName = modelName;
413   }
414   if (searchEngine !== undefined && searchEngine !== null) {
415     window.activeSearchEngine = searchEngine;
416   }
417
418   const curModel = window.activeModelName || '';
419   const curSearch = window.activeSearchEngine || '';
420
421   const modelBadges = document.querySelectorAll('#chat-model-b
  adge, #chat-popup-model-badge');
422   modelBadges.forEach(badge => {
423     if (curModel) {
424       badge.innerHTML = `
```

```
441         tab.show();
442     } else {
443         // Open settings modal
444         const modalEl = document.getElementById('userSetting
sModal');
445         if (modalEl && window.bootstrap) {
446             const modal = bootstrap.Modal.getInstance(modalEl)
|| new bootstrap.Modal(modalEl);
447             modal.show();
448         }
449     }
450 };
451 }
452 });
453
454 const searchBadges = document.querySelectorAll('#chat-search
-badge, #chat-popup-search-badge');
455 searchBadges.forEach(badge => {
456     if (curSearch) {
457         badge.innerHTML = `
```

```
472     savedCardBadge.textContent = 'Не задана (системный fallback)';
473     savedCardBadge.className = 'badge bg-secondary font-monospace px-2 py-1';
474     savedCardBadge.title = 'Модель в профиле не выбрана, сервер использует системный дефолт';
475   }
476 }
477
478 // Update chat message input state based on model selection
479 const msgInputs = document.querySelectorAll('#message-input, #chat-popup-input');
480 const sendButtons = document.querySelectorAll('#send-button, #chat-popup-send-btn');
481
482 msgInputs.forEach(input => {
483   if (!curModel) {
484     input.disabled = true;
485     input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
486     input.classList.add('is-invalid');
487   } else {
488     input.disabled = false;
489     input.placeholder = 'Введите сообщение...';
490     input.classList.remove('is-invalid');
491   }
492 });
493
494 sendButtons.forEach(btn => {
495   btn.disabled = !curModel;
496 });
497 };
498
499 // Автоматически загружаем настройки и обновляем бейджи модели и поиска при загрузке страницы
500 document.addEventListener('DOMContentLoaded', async () => {
501   try {
502     const response = await fetch('/auth/settings');
```

```
503 |     if (response.ok) {
504 |         const settings = await response.json();
505 |         const modelName = settings.model || '';
506 |         const searchEngine = settings.search_engine || '';
507 |         window.activeModelName = modelName;
508 |         window.activeSearchEngine = searchEngine;
509 |
510 |         window.updateChatBadges(modelName, searchEngine);
511 |         // На случай если DOM элементы добавились/отрендерились
    позже
512 |         setTimeout(() => window.updateChatBadges(), 500);
513 |         setTimeout(() => window.updateChatBadges(), 1500);
514 |     }
515 | } catch (e) {
516 |     console.error('Failed to load active model / search badge:
    ', e);
517 | }
518 | });
519 |
```


debounce-integration.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Debounce-Integration Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля debounce-
7  *   integration.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/debounce-integratio
11  *     n.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { createUserSearchDebounce, createPluginSearchDe
15  *     bounce, createInstructionAutoSaveDebounce } from '/windows/api
16  *     /webgui/js/debounce-integration.js';
17  *
18  * File: debounce-integration.js
19  * Project: ai-breadboard
20  * Package: windows/api/webgui/js
21  * Author: hypo69
22  * Copyright: © 2026 hypo69
23  * Updated: 2026-10-01 13:04:40
24  * =====
25  */
26
27 /**
28  * Debounce Integration - Интеграция дебаунсинга в популярные
29  * операции
30  *
31  * Этот модуль предоставляет готовые функции дебаунса/батчинга
32  * для:
```

```
28 * - Поиска пользователей
29 * - Поиска плагинов
30 * - Автосохранения инструкций
31 * - Обновления статуса
32 */
33
34 import { RequestManager } from '../request-manager.js';
35
36 // Создаем глобальный экземпляр request manager
37 const globalRequestManager = new RequestManager();
38
39 window.globalRequestManager = globalRequestManager;
40
41 /**
42  * Дебаунс поиска пользователей (500ms)
43  */
44 export function createUserSearchDebounce(searchFn) {
45   return (query) => {
46     if (window.optimizationModule?.stats) {
47       window.optimizationModule.stats.requestsDebounced++;
48     }
49     return globalRequestManager.debounce(
50       'user-search',
51       () => searchFn(query),
52       500
53     );
54   };
55 }
56
57 /**
58  * Дебаунс поиска плагинов (500ms)
59  */
60 export function createPluginSearchDebounce(searchFn) {
61   return (query) => {
62     if (window.optimizationModule?.stats) {
63       window.optimizationModule.stats.requestsDebounced++;
64     }
65     return globalRequestManager.debounce(
```

```
66     'plugin-search',
67     () => searchFn(query),
68     500
69   );
70 };
71 }
72
73 /**
74  * Дебаунс автосохранения инструкций (1000ms)
75  */
76 export function createInstructionAutoSaveDebounce(saveFn) {
77   return (content) => {
78     if (window.optimizationModule?.stats) {
79       window.optimizationModule.stats.requestsDebounced++;
80     }
81     return globalRequestManager.debounce(
82       'instruction-autosave',
83       () => saveFn(content),
84       1000
85     );
86   };
87 }
88
89 /**
90  * Дебаунс обновления статуса (2000ms)
91  */
92 export function createStatusUpdateDebounce(updateFn) {
93   return (status) => {
94     if (window.optimizationModule?.stats) {
95       window.optimizationModule.stats.requestsDebounced++;
96     }
97     return globalRequestManager.debounce(
98       'status-update',
99       () => updateFn(status),
100       2000
101     );
102   };
103 }
```

```
104 |
105 | /**
106 |  * Создает throttle функцию для real-time обновлений (1000ms)
107 |  */
108 | export function createRealtimeUpdateThrottle(updateFn) {
109 |   return globalRequestManager.throttle(
110 |     'realtime-update',
111 |     updateFn,
112 |     1000
113 |   );
114 | }
115 |
116 | /**
117 |  * Батчинг загрузки пользователей по ID
118 |  * Вместо 5 отдельных запросов - один запрос с массивом ID
119 |  */
120 | export function createUserBatchLoader(loadFn) {
121 |   return (userId) => {
122 |     if (window.optimizationModule?.stats) {
123 |       window.optimizationModule.stats.requestsDebounced++;
124 |     }
125 |     return globalRequestManager.batch(
126 |       'user-batch-load',
127 |       async (userIds) => {
128 |         return loadFn(userIds);
129 |       },
130 |       userId,
131 |       50 // Батч отправляется через 50ms
132 |     );
133 |   };
134 | }
135 |
136 | /**
137 |  * Дублирование запросов - если один и тот же запрос идет два ж
138 |  *   ды,
139 |  * используем результат первого
140 |  */
140 | export function createDedupeFetch(url, fetchFn) {
```

```
141 |   const dedupeKey = `fetch:${url}`;
142 |   return () => {
143 |     if (window.optimizationModule?.stats) {
144 |       window.optimizationModule.stats.requestsDebounded++;
145 |     }
146 |     return globalRequestManager.deduplicate(dedupeKey, fetchFn
    | );
147 |   };
148 | }
149 |
150 | /**
151 |  * Вспомогательная функция для измерения эффективности
152 |  */
153 | export function logDebouncingStats() {
154 |   const stats = globalRequestManager.getStats();
155 |   console.group('▯ Debouncing Statistics');
156 |   console.log('Pending Debounces:', stats.pendingDebounces);
157 |   console.log('Pending Requests:', stats.pendingRequests);
158 |   console.log('Cached Results:', stats.cachedResults);
159 |   console.table(stats.cacheKeys.slice(0, 5));
160 |   console.groupEnd();
161 | }
162 |
163 | // Делаем доступным из консоли
164 | window.logDebouncingStats = logDebouncingStats;
165 |
166 | export default {
167 |   createUserSearchDebounce,
168 |   createPluginSearchDebounce,
169 |   createInstructionAutoSaveDebounce,
170 |   createStatusUpdateDebounce,
171 |   createRealtimeUpdateThrottle,
172 |   createUserBatchLoader,
173 |   createDedupeFetch,
174 |   logDebouncingStats,
175 |   globalRequestManager
176 | };
177 |
```

i18n.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - I18N Script
4  * =====
5  * Description:
6  *   Клиентский модуль интернационализации (i18n) и региональн
7  *   Поддерживает нормализацию тегов language-REGION (например
8  *   автоматическое определение из URL query параметров (?regi
9  *   ?language-region=ru-RU, ?lang=ru), переключение языков и
10  *   синхронизацию DOM.
11  * Usage Examples:
12  *   HTML Integration:
13  *     <script src="/windows/api/webgui/js/i18n.js?v=20261006_
14  *     v10" type="module"></script>
15  *   JavaScript Import:
16  *     import { initI18n, switchLocale, switchLang, normalizel
17  *     ocaleTag, getCurrentLocale } from '/windows/api/webgui/js/i18n
18  *     .js';
19  * File: i18n.js
20  * Project: ai-breadboard
21  * Package: windows/api/webgui/js
22  * Author: hypo69
23  * Copyright: © 2026 hypo69
24  * Updated: 2026-10-06 22:15:00
25  * =====
26  */
```

```

27 | /**
28 |  * i18n.js – Модуль интернационализации и локалей ВСП 47
29 |  *
30 |  * Конвенции разметки HTML:
31 |  *   data-i18n="key"           – устанавливает innerHTML эле
    | мента
32 |  *   data-i18n-placeholder="key" – устанавливает placeholder д
    | ля input/textarea
33 |  *   data-i18n-title="key"      – устанавливает атрибут title
34 |  *
35 |  * Приоритет выбора языка/локали:
36 |  *   1. Параметры URL query: ?region=ru-ru | ?locale=ru-RU | ?
    | language-region=ru-RU | ?lang=ru
37 |  *   2. Сохранённое значение в localStorage ('app_locale' / 'a
    | pp_language')
38 |  *   3. Язык браузера (navigator.language)
39 |  *   4. Дефолтная локаль ('en-US')
40 |  */
41 |
42 | export const DEFAULT_LOCALE = 'en-US';
43 | export const SUPPORTED_LOCALES = ['en-US', 'ru-RU', 'he-IL', '
    | uk-UA'];
44 | export const SUPPORTED_LANGS = ['en', 'ru', 'he', 'uk'];
45 | export const SUPPORTED = SUPPORTED_LANGS; // Обратная совмести
    | мость
46 |
47 | export const DEFAULT_REGIONS = {
48 |   en: 'US',
49 |   ru: 'RU',
50 |   he: 'IL',
51 |   uk: 'UA',
52 |   de: 'DE',
53 |   fr: 'FR',
54 |   es: 'ES',
55 |   pt: 'BR',
56 |   zh: 'CN',
57 |   ja: 'JP',
58 |   ko: 'KR',

```

```

59 |     ar: 'SA'
60 | };
61 |
62 | let i18nInitialized = false;
63 | let eventsBound = false;
64 | let currentActiveLocale = DEFAULT_LOCALE;
65 |
66 | /**
67 |  * Нормализует языковой/региональный тег в стандарт BCP 47 (la
  | nguage-REGION)
68 |  * и извлекает базовый язык и регион.
69 |  *
70 |  * Формат:
71 |  *   language-REGION
72 |  *   language – ISO 639-1 (2 строчные буквы: ru, he, uk, en...
  | )
73 |  *   REGION   – ISO 3166-1 alpha-2 (2 заглавные буквы: RU, IL,
  | UA, US...)
74 |  *
75 |  * Примеры:
76 |  *   "ru-ru"    -> { locale: "ru-RU", lang: "ru", region: "RU"
  | }
77 |  *   "ru_ru"    -> { locale: "ru-RU", lang: "ru", region: "RU"
  | }
78 |  *   "he-il"    -> { locale: "he-IL", lang: "he", region: "IL"
  | }
79 |  *   "uk-ua"    -> { locale: "uk-UA", lang: "uk", region: "UA"
  | }
80 |  *   "en-us"    -> { locale: "en-US", lang: "en", region: "US"
  | }
81 |  *   "ru"       -> { locale: "ru-RU", lang: "ru", region: "RU"
  | }
82 |  *   "`ru-ru`"  -> { locale: "ru-RU", lang: "ru", region: "RU"
  | }
83 |  *
84 |  * @param {string} tag - Исходный тег локали или языка
85 |  * @returns {{ locale: string, lang: string, region: string }}
86 |  */

```



```

87 | export function normalizeLocaleTag(tag) {
88 |     if (!tag || typeof tag !== 'string') {
89 |         return { locale: DEFAULT_LOCALE, lang: 'en', region: '
    | US' };
90 |     }
91 |
92 |     // Очистка от кавычек, апострофов, обратных кавычек и проб
    | елов
93 |     let cleaned = tag.trim().replace(/^['"]+|['"]+$/g, '');
94 |
95 |     // Заменяем нижние подчеркивания на дефисы
96 |     cleaned = cleaned.replace(/_/g, '-');
97 |
98 |     // Проверяем формат language-REGION (например ru-RU, ru-ru
    | , en-US)
99 |     const match = cleaned.match(/^(?!(?:[a-zA-Z]{2,3})?-(?:[a-zA-Z]
    | {2,4}))?$/);
100 |     if (match) {
101 |         const langPart = match[1].toLowerCase();
102 |         let regionPart = match[2]
103 |             ? match[2].toUpperCase()
104 |             : (DEFAULT_REGIONS[langPart] || langPart.toUpperCa
    | se());
105 |         const locale = `${langPart}-${regionPart}`;
106 |         return { locale, lang: langPart, region: regionPart };
107 |     }
108 |
109 |     // Fallback для неизвестных конструкций
110 |     const lower = cleaned.toLowerCase();
111 |     const fallbackLang = SUPPORTED_LANGS.includes(lower) ? low
    | er : 'en';
112 |     const fallbackRegion = DEFAULT_REGIONS[fallbackLang] || 'U
    | S';
113 |     return {
114 |         locale: `${fallbackLang}-${fallbackRegion}`,
115 |         lang: fallbackLang,
116 |         region: fallbackRegion
117 |     };

```

```
118 | }
119 |
120 | /**
121 |  * Извлекает и нормализует локаль/регион из URL query параметр
    | ов.
122 |  * Поддерживаемые параметры:
123 |  *   - region (например ?region=ru-ru или ?region=ru-RU)
124 |  *   - locale (например ?locale=ru-RU)
125 |  *   - language-region (например ?language-region=ru-RU)
126 |  *   - language_region (например ?language_region=ru_RU)
127 |  *   - lang (например ?lang=ru)
128 |  *   - language (например ?language=ru-RU)
129 |  *
130 |  * @returns {{ locale: string, lang: string, region: string }
    | | null}
131 |  */
132 | export function getLocaleFromUrl() {
133 |     try {
134 |         if (typeof window === 'undefined' || !window.location
    | || !window.location.search) {
135 |             return null;
136 |         }
137 |         const params = new URLSearchParams(window.location.sea
    | rch);
138 |         const urlParam = params.get('region') ||
139 |             params.get('locale') ||
140 |             params.get('language-region') ||
141 |             params.get('language_region') ||
142 |             params.get('lang') ||
143 |             params.get('language');
144 |
145 |         if (urlParam) {
146 |             return normalizeLocaleTag(urlParam);
147 |         }
148 |     } catch (e) {
149 |         console.warn('[i18n] Ошибка парсинга параметров URL:',
    | e);
150 |     }
```

```

151     return null;
152 }
153
154 /**
155  * Привязывает обработчик изменений селекторов языка.
156  */
157 function setupLanguageEvents() {
158     if (eventsBound) return;
159     eventsBound = true;
160     document.addEventListener('change', (e) => {
161         if (e.target && e.target.classList.contains('lang-selector')) {
162             switchLocale(e.target.value);
163         }
164     });
165 }
166
167 /**
168  * Инициализирует i18next и применяет переводы к DOM.
169  * @param {string} [langOrLocale] - Язык или локаль (опционально)
170  */
171 export async function initI18n(langOrLocale) {
172     let target;
173     if (langOrLocale) {
174         target = normalizeLocaleTag(langOrLocale);
175     } else {
176         target = detectLocale();
177     }
178
179     const { locale, lang, region } = target;
180     currentActiveLocale = locale;
181
182     // Всегда загружаем английский как fallback
183     const [mainRes, enRes] = await Promise.all([
184         loadLocale(lang),
185         lang !== 'en' ? loadLocale('en') : Promise.resolve(null),

```

```
186 |     });
187 |
188 |     if (typeof i18next !== 'undefined') {
189 |         await i18next.init({
190 |             lng: lang,
191 |             fallbackLng: 'en',
192 |             resources: { [lang]: { translation: mainRes } },
193 |             interpolation: { escapeValue: false },
194 |         });
195 |
196 |         if (enRes) {
197 |             i18next.addResourceBundle('en', 'translation', enR
198 | es, true, true);
199 |         }
200 |     }
201 |
202 |     document.documentElement.setAttribute('lang', lang);
203 |     document.documentElement.setAttribute('data-locale', locale);
204 |
205 |     document.documentElement.setAttribute('dir', (lang === 'he
206 | ' || lang === 'ar') ? 'rtl' : 'ltr');
207 |
208 |     applyTranslations();
209 |     syncSelectors(lang, locale);
210 |     setupLanguageEvents();
211 |     i18nInitialized = true;
212 |
213 |     // Сохраняем в localStorage
214 |     try {
215 |         localStorage.setItem('app_locale', locale);
216 |         localStorage.setItem('app_language', lang);
217 |     } catch {}
218 |
219 |     // Уведомляем систему через события
220 |     window.dispatchEvent(new CustomEvent('localeChanged', { de
221 | tail: { locale, lang, region } }));
222 |     window.dispatchEvent(new CustomEvent('languageChanged', {
223 | detail: { lang, locale, region } }));
```

```

219 | }
220 |
221 | /**
222 |  * Переключает язык/локаль, перезагружает бандлы при необходим
ости, сохраняет в localStorage.
223 |  * @param {string} langOrLocale - Код языка или тег ВСП 47 (на
пример 'ru', 'ru-RU', 'he-IL')
224 |  */
225 | export async function switchLocale(langOrLocale) {
226 |     const { locale, lang, region } = normalizeLocaleTag(langOr
Locale);
227 |     currentActiveLocale = locale;
228 |
229 |     if (typeof i18next !== 'undefined') {
230 |         if (!i18next.hasResourceBundle(lang, 'translation')) {
231 |             const res = await loadLocale(lang);
232 |             i18next.addResourceBundle(lang, 'translation', res
, true, true);
233 |         }
234 |         await i18next.changeLanguage(lang);
235 |     }
236 |
237 |     document.documentElement.setAttribute('lang', lang);
238 |     document.documentElement.setAttribute('data-locale', local
e);
239 |     document.documentElement.setAttribute('dir', (lang === 'he
' || lang === 'ar') ? 'rtl' : 'ltr');
240 |
241 |     applyTranslations();
242 |     syncSelectors(lang, locale);
243 |
244 |     try {
245 |         localStorage.setItem('app_locale', locale);
246 |         localStorage.setItem('app_language', lang);
247 |     } catch {}
248 |
249 |     // Глобальные события для компонентов и вкладок
250 |     window.dispatchEvent(new CustomEvent('localeChanged', { de

```

```
tail: { locale, lang, region } }));
251   window.dispatchEvent(new CustomEvent('languageChanged', {
detail: { lang, locale, region } }));
252 }
253
254 // Алиас для обратной совместимости
255 export const switchLang = switchLocale;
256
257 /**
258  * Обновляет все элементы с data-i18n* атрибутами в DOM.
259  */
260 export function applyTranslations() {
261   if (typeof i18next === 'undefined') return;
262
263   // Контент текста и HTML
264   document.querySelectorAll('[data-i18n]').forEach(el => {
265     const key = el.getAttribute('data-i18n');
266     if (!key) return;
267     const val = i18next.t(key);
268     if (val && val !== key) el.innerHTML = val;
269   });
270
271   // Плейсхолдеры
272   document.querySelectorAll('[data-i18n-placeholder]').forEa
ch(el => {
273     const key = el.getAttribute('data-i18n-placeholder');
274     if (!key) return;
275     const val = i18next.t(key);
276     if (val && val !== key) el.placeholder = val;
277   });
278
279   // Тултипы и атрибут title
280   document.querySelectorAll('[data-i18n-title]').forEach(el
=> {
281     const key = el.getAttribute('data-i18n-title');
282     if (!key) return;
283     const val = i18next.t(key);
284     if (val && val !== key) el.title = val;
```

```

285 |     });
286 | }
287 |
288 | /**
289 |  * Загружает JSON-файл локализации.
290 |  * @param {string} lang - Код языка
291 |  * @returns {Promise<Object>}
292 |  */
293 | async function loadLocale(lang) {
294 |     const pathName = typeof window !== 'undefined' ? window.lo
295 | cation.pathname : '';
296 |
297 |     let paths = [];
298 |
299 |     if (pathName.includes('/user_tts')) {
300 |         paths = [
301 |             `/html/locales/${lang}.json`,
302 |             `/html/user_tts/locales/user_${lang}.json`,
303 |             `/html/locales/user_${lang}.json`
304 |         ];
305 |     } else {
306 |         paths = [
307 |             `/html/locales/${lang}.json`,
308 |             `/html/locales/user_${lang}.json`
309 |         ];
310 |     }
311 |
312 |     for (const path of paths) {
313 |         try {
314 |             const cacheBustUrl = `${path}?v=${Date.now()}`;
315 |             const r = await fetch(cacheBustUrl);
316 |             if (r.ok) return await r.json();
317 |         } catch (e) {
318 |             console.log(`[i18n] Попытка загрузки альтернативно
319 | го пути: ${path}`);
320 |         }
321 |     }
322 |
323 |     console.warn(`[i18n] Локаль ${lang} не найдена на сервере,
324 | используется fallback`);

```

```
320 |     return {};
```

```
321 | }
```

```
322 |
```

```
323 | /**
```

```
324 |  * Определяет текущую локаль по приоритетам (URL -> localStora
```

```
    ge -> Браузер -> Дефолт).
```

```
325 |  * @returns {{ locale: string, lang: string, region: string }}
```

```
326 |  */
```

```
327 | export function detectLocale() {
```

```
328 |     // 1. Проверяем URL параметр
```

```
329 |     const fromUrl = getLocaleFromUrl();
```

```
330 |     if (fromUrl) {
```

```
331 |         try {
```

```
332 |             localStorage.setItem('app_locale', fromUrl.locale)
```

```
        ;
```

```
333 |             localStorage.setItem('app_language', fromUrl.lang)
```

```
        ;
```

```
334 |         } catch {}
```

```
335 |         return fromUrl;
```

```
336 |     }
```

```
337 |
```

```
338 |     // 2. Проверяем localStorage
```

```
339 |     try {
```

```
340 |         const savedLocale = localStorage.getItem('app_locale')
```

```
        ;
```

```
341 |         if (savedLocale) return normalizeLocaleTag(savedLocale
```

```
    );
```

```
342 |
```

```
343 |         const savedLang = localStorage.getItem('app_language')
```

```
        || localStorage.getItem('language');
```

```
344 |         if (savedLang) return normalizeLocaleTag(savedLang);
```

```
345 |     } catch {}
```

```
346 |
```

```
347 |     // 3. Проверяем язык браузера
```

```
348 |     if (typeof navigator !== 'undefined') {
```

```
349 |         const browser = navigator.language || (navigator.langu
```

```
    ages && navigator.languages[0]) || '';
```

```
350 |         if (browser) return normalizeLocaleTag(browser);
```



```

351     }
352
353     // 4. Дефолтная локаль
354     return normalizeLocaleTag(DEFAULT_LOCALE);
355 }
356
357 /**
358  * Определяет текущий базовый язык.
359  * @returns {string}
360  */
361 export function detectLang() {
362     return detectLocale().lang;
363 }
364
365 /**
366  * Синхронизирует значения во всех выпадающих списках выбора я
367  * зыка на странице.
368  * @param {string} lang
369  * @param {string} locale
370  */
371 function syncSelectors(lang, locale) {
372     document.querySelectorAll('.lang-selector').forEach(sel =>
373     {
374         const optLocale = Array.from(sel.options).find(opt =>
375         opt.value.toLowerCase() === (locale || '').toLowerCase());
376         const optLang = Array.from(sel.options).find(opt => op
377         t.value.toLowerCase() === (lang || '').toLowerCase());
378         if (optLocale) {
379             sel.value = optLocale.value;
380         } else if (optLang) {
381             sel.value = optLang.value;
382         }
383     });
384 }
385
386 /**
387  * Получает текущий активный базовый язык.
388  * @returns {string}

```

```
385 | */
386 | export function getCurrentLang() {
387 |     return (typeof i18next !== 'undefined' && i18next.language
    | ) || detectLang();
388 | }
389 |
390 | /**
391 |  * Получает текущую активную локаль в формате BCP 47.
392 |  * @returns {string}
393 |  */
394 | export function getCurrentLocale() {
395 |     return currentActiveLocale || detectLocale().locale;
396 | }
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/main.js?v=20261001_
11  *     v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { switchMainTab } from '/windows/api/webgui/js/main.js';
15  *
16  * File: main.js
17  * Project: ai-breadboard
18  * Package: windows/api/webgui/js
19  * Author: hypo69
20  * Copyright: © 2026 hypo69
21  * Updated: 2026-10-08 05:10:00
22  * =====
23  */
24 /**
25  * js/main.js – оркестратор главной страницы (run.ps1, маршрут /)
26  * Архитектура: UI_ARCHITECTURE.md
27  */
28
29 import { initI18n, switchLang, applyTranslations } from './i18n.js';
```

```
30 import { initTheme } from './theme.js';
31 import { initUserSettings } from './userSettings.js';
32 import { initActivityTracker } from './activityTracker.js';
33 import { initCacheUI } from './cache-ui.js';
34 import { initModelTester, sendModelPing } from './model-tester
    .js';
35 import { switchTab, loadTab, setupTabClicks } from './tab-core
    .js';
36 import './api-cache.js'; // Подключаем универсальный API-кеш с
    лой
37 import './test-api-cache.js'; // Подключаем тестовый скрипт
38
39 // Глобальный экспорт для вызова из вкладок
40 window.switchTab = switchTab;
41 window.switchToTab = switchTab;
42 window.sendModelPing = sendModelPing;
43 window.initModelTester = initModelTester;
44 window.applyTranslations = applyTranslations;
45
46 // Карта вкладок: имя → [htmlUrl, jsUrl]
47 // Пути явные – не генерируются из имени
48 const TABS = {
49     // Базовые ИИ и коммуникации
50     'chat': [
51         '/html/chat/index.html',
52         '/html/chat/main.js'],
53     'voice': [
54         '/html/voice_tab/index.html',
55         '/html/voice_tab/main.js'],
56     'scenarios': [
57         '/html/scenarios_tab/index.html',
58         '/html/scenarios_tab/main.js'],
59     'rag': [
60         '/html/rag_tab/index.html',
61         '/html/rag_tab/main.js'],
62     'telegram-rag': [
63         '/html/telegram_rag_tab/index.h
64         tml',
65         '/html/telegram_rag_tab/main.js'],
66     'news': [
67         '/html/news_tab/index.html',
68         '/html/news_tab/main.js'],
69     'models': [
70         '/html/models_tab/index.html',
71         '/html/models_tab/main.js'],
72     'plugins': [
73         '/html/plugins_tab/index.html',
```

```
        '/html/plugins_tab/main.js'],
58  'admin':          ['/html/admin_tab/index.html',
                      '/html/admin_tab/main.js'],
59  'help':           ['/html/help/index.html',
                      '/html/help/main.js'],
60
61  // Оборудование, Датчики и Телеметрия
62  'about-system':   ['/html/about_system_tab/index.h
                      tml',          '/html/about_system_tab/main.js'],
63  'hardware-monitor':  ['/html/hardware_monitor_tab/ind
                      ex.html',      '/html/hardware_monitor_tab/main.js'],
64  'telemetry-research':  ['/html/telemetry_research_tab/i
                      ndex.html',    '/html/telemetry_research_tab/main.js'],
65  'ai-benchmark':    ['/html/ai_benchmark_tab/index.h
                      tml',          '/html/ai_benchmark_tab/main.js'],
66  'process-leaks':    ['/html/process_leaks_tab/index.
                      html',          '/html/process_leaks_tab/main.js'],
67  'forensics':       ['/html/forensics_tab/index.html
                      ',          '/html/forensics_tab/main.js'],
68  'throttling':      ['/html/throttling_tab/index.htm
                      l',          '/html/throttling_tab/main.js'],
69  'peripherals':     ['/html/peripherals_tab/index.ht
                      ml',          '/html/peripherals_tab/main.js'],
70  'software-transparency':  ['/html/software_transparency_ta
                      b/index.html',  '/html/software_transparency_tab/main.js'],
71
72  // Накопители и Диски
73  'storage-manager':  ['/html/storage_manager_tab/inde
                      x.html',        '/html/storage_manager_tab/main.js'],
74  'disk-speed':      ['/html/disk_speed_tab/index.htm
                      l',          '/html/disk_speed_tab/main.js'],
75  'storage-wear':     ['/html/storage_wear_tab/index.h
                      tml',          '/html/storage_wear_tab/main.js'],
76  'file-recovery':    ['/html/file_recovery_tab/index.
                      html',          '/html/file_recovery_tab/main.js'],
77  'file-history-search':  ['/html/file_history_ai_search_t
                      ab/index.html',  '/html/file_history_ai_search_tab/main.js'],
78
```

```
79 // Управление Windows и Sysadmin
80 'system-control':          ['/html/system_control_tab/index
    .html',          '/html/system_control_tab/main.js'],
81 'post-install-wizard':     ['/html/post_install_wizard_tab/
    index.html',     '/html/post_install_wizard_tab/main.js'],
82 'maintenance-recovery':    ['/html/maintenance_recovery_tab/
    /index.html',    '/html/maintenance_recovery_tab/main.js'],
83 'windows-admin':          ['/html/windows_admin_tab/index.
    html',          '/html/windows_admin_tab/main.js'],
84 'focus-settings':         ['/html/focus_settings_tab/index
    .html',         '/html/focus_settings_tab/main.js'],
85 'taskbar-controller':      ['/html/taskbar_tab/index.html',
    '/html/taskbar_tab/main.js'],
86 'startup-auditor':         ['/html/startup_auditor_tab/inde
    x.html',         '/html/startup_auditor_tab/main.js'],
87 'services-manager':        ['/html/services_manager_tab/ind
    ex.html',        '/html/services_manager_tab/main.js'],
88 'task-scheduler':          ['/html/task_scheduler_tab/index
    .html',          '/html/task_scheduler_tab/main.js'],
89 'process-manager':         ['/html/process_manager_tab/inde
    x.html',         '/html/process_manager_tab/main.js'],
90 'registry-viewer':         ['/html/registry_viewer_tab/inde
    x.html',         '/html/registry_viewer_tab/main.js'],
91 'software-manager':        ['/html/software_manager_tab/ind
    ex.html',        '/html/software_manager_tab/main.js'],
92 'user-directories':        ['/html/user_directories_tab/ind
    ex.html',        '/html/user_directories_tab/main.js'],
93 'accounts-identity':       ['/html/accounts_identity_tab/in
    dex.html',       '/html/accounts_identity_tab/main.js'],
94
95 // Безопасность и Целостность
96 'firewall-manager':        ['/html/firewall_manager_tab/ind
    ex.html',        '/html/firewall_manager_tab/main.js'],
97 'defender':                ['/html/defender_tab/index.html'
    ,                '/html/defender_tab/main.js'],
98 'security-acl':            ['/html/security_acl_tab/index.h
    tml',            '/html/security_acl_tab/main.js'],
99 'servicing-integrity':      ['/html/servicing_integrity_tab/
```

```

    index.html',      ['/html/servicing_integrity_tab/main.js'],
100   'boot-recovery':      ['/html/boot_recovery_tab/index.
    html',            ['/html/boot_recovery_tab/main.js'],
101   'windows-backup':      ['/html/windows_backup_tab/index
    .html',            ['/html/windows_backup_tab/main.js'],
102
103   // Сеть и Системные Логи
104   'network':              ['/html/network_tab/index.html',
    '/html/network_tab/main.js'],
105   'performance-tracing':      ['/html/performance_tracing_tab/
    index.html',        ['/html/performance_tracing_tab/main.js'],
106   'system-logs':            ['/html/system_logs_tab/index.ht
    ml',                ['/html/system_logs_tab/main.js'],
107   'power-lifecycle':          ['/html/power_lifecycle_tab/inde
    x.html',            ['/html/power_lifecycle_tab/main.js'],
108 };
109
110 // Lazy-загрузка: вкладка грузится при первом открытии
111 const loaded = new Set();
112
113 async function lazyLoad(tabName) {
114   if (loaded.has(tabName) || !TABS[tabName]) return;
115   loaded.add(tabName);
116   const v = Date.now();
117   const [html, js] = TABS[tabName];
118   await loadTab(tabName, `${html}?v=${v}`, `${js}?v=${v}`);
119   applyTranslations();
120 }
121
122 // Переопределяем switchTab с приоритетной мгновенной навигац
    ей
123 const _switchTab = switchTab;
124 export function switchMainTab(tabId) {
125   if (!tabId) return;
126   // 1. Мгновенное переключение UI без задержки
127   _switchTab(tabId);
128
129   // 2. Фоновая асинхронная подгрузка разметки и скрипта вклад

```

```
ки (если еще не загружена)
130   const name = (tabId.startsWith('tab-') ? tabId.slice(4) : tabId);
131   if (!loaded.has(name) && TABS[name]) {
132     lazyLoad(name).catch(err => console.warn(`[Main] Ошибка lazyLoad для ${name}:`, err));
133   }
134 }
135 window.switchTab = switchMainTab;
136 window.switchToTab = switchMainTab;
137
138 // Открыть плагин по имени (вызывается из плагинов)
139 window.openPluginFromDropdown = function(pluginName) {
140   if (!pluginName) return;
141   const n = pluginName.toLowerCase().replace(/-/g, '_');
142   if (n === 'telegram_channel_rag' || n === 'telegram_rag') {
143     window.switchTab('tab-telegram-rag'); return;
144   }
145   if (n === 'news_feed' || n === 'news' || n === 'smart_news') {
146     window.switchTab('tab-news'); return;
147   }
148   window.switchTab('tab-plugins');
149   const sel = () => window.selectPlugin?.(pluginName);
150   sel(); setTimeout(sel, 200);
151 };
152
153 async function init() {
154   // 1. Тема
155   initTheme();
156
157   // 2. i18n (с автоматическим парсингом URL параметров: ?region=ru-ru, ?locale=ru-RU и т.д.)
158   await initI18n();
159
160   // 3. Пользователь, активность, кеш, быстрая проверка модели
161   await initUserSettings();
162   initActivityTracker();
```



```
163 |   initCacheUI();
164 |   initModelTester();
165 |
166 |   // 4. Единый обработчик кликов
167 |   setupTabClicks();
168 |
169 |   // 5. Загрузить первую вкладку (чат)
170 |   await lazyLoad('chat');
171 |   applyTranslations();
172 |
173 |   // 6. Синхронизация видимости плагинов
174 |   try {
175 |     const r = await fetch('/api/admin/plugins');
176 |     if (r.ok) {
177 |       const data = await r.json();
178 |       window.syncPluginTabsVisibility?.(data.plugins);
179 |     }
180 |   } catch {}
181 |
182 |   // 7. Активировать вкладку по hash или чат
183 |   const hash = location.hash.replace('#', '');
184 |   await switchMainTab(hash || 'tab-chat');
185 |
186 |   // 8. Фоновый WebSocket
187 |   initComputerStream();
188 | }
189 |
190 | // Фоновый WebSocket для remote transcript
191 | function initComputerStream() {
192 |   const proto = location.protocol === 'https:' ? 'wss:' : 'ws:
193 |   '
194 |   const ws = new WebSocket(`${proto}://${location.host}/api/con
195 |   trol/ws?role=computer`);
196 |   ws.onmessage = e => {
197 |     try {
198 |       const msg = JSON.parse(e.data);
199 |       if (msg.event === 'transcript') window.handleRemoteTrans
200 |       cript?.(msg.text);
```

```
198 |     } catch {}
199 | };
200 | ws.onclose = () => setTimeout(initComputerStream, 3000);
201 | }
202 |
203 |
204 |
205 | document.readyState === 'loading'
206 |   ? document.addEventListener('DOMContentLoaded', init)
207 |   : init();
208 |
```

model-tester.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Model-Tester Script
4  * =====
5  * Description:
6  *   Клиентский модуль быстрой проверки доступности активной AI-модели.
7  *   Позволяет отправлять одиночные проверочные запросы к модели без сохранения
8  *   истории диалога и сессий, отображая сырой ответ или ошибку как есть.
9  *
10 * Usage Examples:
11 *   JavaScript Import:
12 *     import { initModelTester, sendModelPing } from './windows/api/webgui/js/model-tester.js';
13 *
14 * File: model-tester.js
15 * Project: ai-breadboard
16 * Package: windows/api/webgui/js
17 * Author: hypo69
18 * Copyright: © 2026 hypo69
19 * Updated: 2026-10-08 04:22:00
20 * =====
21 */
22
23 /**
24 * Отправляет проверочный запрос к AI-модели и выводит полученный ответ или ошибку.
25 *
26 * @param {string} [customMessage] - Опциональный текст запроса. Если не задан, берется из input.
27 * @returns {Promise<void>}
```

```

28  */
29  export async function sendModelPing(customMessage = null) {
30    const inputEl = document.getElementById('model-ping-input');
31    const btnEl = document.getElementById('btn-model-ping-send')
    ;
32    const btnTextEl = document.getElementById('model-ping-btn-te
    xt');
33    const btnIconEl = document.getElementById('model-ping-btn-ic
    on');
34    const responseContainerEl = document.getElementById('model-p
    ing-response-container');
35    const responseEl = document.getElementById('model-ping-respo
    nse');
36
37    if (!inputEl || !responseEl) return;
38
39    const rawMessage = customMessage !== null ? customMessage :
    inputEl.value;
40    const message = (rawMessage || '').trim() || 'Проверка досту
    пности модели';
41
42    // Перевод элементов интерфейса в состояние загрузки
43    if (btnEl) btnEl.disabled = true;
44    if (inputEl) inputEl.disabled = true;
45    if (btnIconEl) btnIconEl.className = 'spinner-border spinner
    -border-sm';
46    if (btnTextEl) btnTextEl.textContent = 'Отправка...';
47
48    if (responseContainerEl) responseContainerEl.style.display =
    'block';
49    responseEl.className = 'small p-2 rounded bg-body-tertiary b
    order text-break font-monospace text-muted';
50    responseEl.textContent = 'Отправка запроса к модели, ожидани
    е ответа...';
51
52    const isTcRoute = window.location.pathname.startsWith('/tc')
    ;
53    const query = isTcRoute ? '?profile=tc' : '';

```

```

54
55   try {
56     const resp = await fetch(`/api/v1/chat/test-model${query}`
57     , {
58       method: 'POST',
59       headers: {
60         'Content-Type': 'application/json',
61       },
62       body: JSON.stringify({
63         message: message,
64       }),
65     });
66
67     if (resp.ok) {
68       const data = await resp.json().catch(() => null);
69       if (data && data.status === 'success' && data.response !
70       == undefined) {
71         responseEl.textContent = data.response;
72         responseEl.className = 'small p-2 rounded bg-success-s
73         ubtle text-success-emphasis border border-success-subtle text-
74         break font-monospace';
75       } else if (data && data.status === 'error') {
76         const errMsg = data.message || data.detail || JSON.str
77         ingify(data, null, 2);
78         responseEl.textContent = errMsg;
79         responseEl.className = 'small p-2 rounded bg-danger-su
80         btle text-danger-emphasis border border-danger-subtle text-bre
81         ak font-monospace';
82       } else if (data) {
83         responseEl.textContent = data.response || data.message
84         || data.detail || JSON.stringify(data, null, 2);
85         responseEl.className = 'small p-2 rounded bg-body-tert
86         iary border text-break font-monospace';
87       } else {
88         responseEl.textContent = 'Пустой ответ от сервера';
89         responseEl.className = 'small p-2 rounded bg-warning-s
90         ubtle text-warning-emphasis border text-break font-monospace';
91       }
92     }
93   }

```

```
82 |     } else {
83 |         let errBody = '';
84 |         try {
85 |             const errData = await resp.json();
86 |             errBody = errData.detail || errData.message || errData
87 | .response || JSON.stringify(errData, null, 2);
88 |         } catch {
89 |             errBody = await resp.text();
90 |         }
91 |         responseEl.textContent = errBody || `HTTP ${resp.status}
92 | ${resp.statusText}`;
93 |         responseEl.className = 'small p-2 rounded bg-danger-subt
94 | le text-danger-emphasis border border-danger-subtle text-break
95 | font-monospace';
96 |     }
97 | } catch (err) {
98 |     responseEl.textContent = err.message || String(err);
99 |     responseEl.className = 'small p-2 rounded bg-danger-subtle
100 | text-danger-emphasis border border-danger-subtle text-break f
101 | ont-monospace';
102 | } finally {
103 |     if (btnEl) btnEl.disabled = false;
104 |     if (inputEl) inputEl.disabled = false;
105 |     if (btnIconEl) btnIconEl.className = 'bi bi-send-fill';
106 |     if (btnTextEl) btnTextEl.textContent = 'Отправить';
107 | }
108 | }
109 |
110 | /**
111 |  * Инициализирует интерактивные обработчики для выпадающей пан
112 |  * ели проверки доступности модели.
113 |  */
114 | export function initModelTester() {
115 |     const sendBtn = document.getElementById('btn-model-ping-send
116 | ');
117 |     const inputEl = document.getElementById('model-ping-input');
118 |     const dropdownEl = document.getElementById('model-ping-dropd
119 | own-wrapper');
```

```
111 |
112 |   if (sendBtn) {
113 |     sendBtn.onclick = (e) => {
114 |       e.preventDefault();
115 |       sendModelPing();
116 |     };
117 |   }
118 |
119 |   if (inputEl) {
120 |     inputEl.onkeydown = (e) => {
121 |       if (e.key === 'Enter' && !e.shiftKey) {
122 |         e.preventDefault();
123 |         sendModelPing();
124 |       }
125 |     };
126 |   }
127 |
128 |   if (dropdownEl) {
129 |     dropdownEl.addEventListener('shown.bs.dropdown', () => {
130 |       if (inputEl) {
131 |         inputEl.focus();
132 |         inputEl.select();
133 |       }
134 |     });
135 |   }
136 | }
137 |
```

optimization-test-suite.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Optimization-Test-Suite Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля optimization-test-suite.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/optimization-test-suite.js?v=20261001_v1" type="module"></script>
11  *
12  * File: optimization-test-suite.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/js
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 /**
22  * Optimization Test Suite - Комплексное тестирование оптимизаций
23  *
24  * Тестирует:
25  * - Ленивую загрузку вкладок
26  * - Кеширование API
27  * - Дебаунсинг запросов
28  * - Сохранение состояния
29  * - Производительность переключения
30  *
```



```

31 | * Usage:
32 | *   const tester = new OptimizationTestSuite();
33 | *   await tester.runAllTests();
34 | *   tester.printReport();
35 | */
36 |
37 | class OptimizationTestSuite {
38 |   constructor() {
39 |     this.tests = [];
40 |     this.results = [];
41 |     this.startTime = null;
42 |   }
43 |
44 |   /**
45 |    * Добавить тест
46 |    */
47 |   addTest(name, testFn) {
48 |     this.tests.push({ name, testFn });
49 |   }
50 |
51 |   /**
52 |    * Запустить все тесты
53 |    */
54 |   async runAllTests() {
55 |     console.log('□ Starting Optimization Test Suite...\n');
56 |     this.startTime = performance.now();
57 |
58 |     for (const test of this.tests) {
59 |       await this._runTest(test);
60 |     }
61 |
62 |     const totalTime = performance.now() - this.startTime;
63 |     console.log(`\n□ All tests completed in ${totalTime.toFixed(
64 |       2)}ms\n`);
65 |   }
66 |
67 |   /**
   * Запустить отдельный тест

```

```

68     */
69     async _runTest(test) {
70         const startTime = performance.now();
71         const result = {
72             name: test.name,
73             passed: false,
74             time: 0,
75             error: null,
76             details: {}
77         };
78
79         try {
80             console.log(`▶ Running: ${test.name}...`);
81             const testResult = await test.testFn();
82             result.passed = true;
83             result.details = testResult;
84             result.time = performance.now() - startTime;
85             console.log(`    □ PASSED (${result.time.toFixed(2)}ms)\n`);
86         } catch (error) {
87             result.passed = false;
88             result.error = error.message;
89             result.time = performance.now() - startTime;
90             console.log(`    □ FAILED: ${error.message}\n`);
91         }
92
93         this.results.push(result);
94     }
95
96     /**
97     * Печать отчета
98     */
99     printReport() {
100         const passed = this.results.filter(r => r.passed).length;
101         const failed = this.results.length - passed;
102         const passRate = ((passed / this.results.length) * 100).toFixed(1);
103

```

```

104 |     console.group('  Test Report');
105 |     console.log(`Total: ${this.results.length} tests`);
106 |     console.log(`  Passed: ${passed}`);
107 |     console.log(`  Failed: ${failed}`);
108 |     console.log(`  Pass Rate: ${passRate}%`);
109 |     console.table(this.results.map(r => ({
110 |         Test: r.name,
111 |         Status: r.passed ? '  PASS' : '  FAIL',
112 |         Time: `${r.time.toFixed(2)}ms`,
113 |         Details: r.error || 'OK'
114 |     })));
115 |     console.groupEnd();
116 | }
117 |
118 | /**
119 |  * Получить результаты в JSON
120 |  */
121 | getResults() {
122 |     return this.results;
123 | }
124 | }
125 |
126 | /**
127 |  * Предварительно определенные тесты
128 |  */
129 | export async function createDefaultTestSuite() {
130 |     const suite = new OptimizationTestSuite();
131 |
132 |     // =====
133 |     // Тест 1: Проверка ленивой загрузки вкладок
134 |     // =====
135 |     suite.addTest('Lazy Tab Loading', async () => {
136 |         if (!window.optimizationModule?.tabLoader) {
137 |             throw new Error('TabLoader not initialized');
138 |         }
139 |

```

```

140     const tabLoader = window.optimizationModule.tabLoader;
141     const initialLoaded = tabLoader.loaded.size;
142
143     // Проверяем что не все вкладки загружены сразу
144     if (initialLoaded > 5) {
145         throw new Error(`Too many tabs loaded at start: ${initialLoaded}`);
146     }
147
148     return {
149         initiallyLoadedTabs: initialLoaded,
150         totalRegisteredTabs: tabLoader.tabs.size,
151         lazyLoadingActive: true
152     };
153 });
154
155 // =====
156 // Тест 2: Проверка API кеширования
157 // =====
158 suite.addTest('API Response Caching', async () => {
159     if (!window.optimizationModule?.apiCache) {
160         throw new Error('APICache not initialized');
161     }
162
163     const apiCache = window.optimizationModule.apiCache;
164
165     // Сохраняем тестовые данные
166     const testKey = 'test:caching';
167     const testData = { test: true, timestamp: Date.now() };
168
169     apiCache.set(testKey, testData, 1000);
170
171     // Проверяем что данные кешированы
172     const cached = apiCache.get(testKey);
173     if (!cached || cached.test !== true) {
174         throw new Error('Cache set/get failed');

```

```
175 |     }
176 |
177 |     // Проверяем инвалидацию
178 |     apiCache.invalidate('test:');
179 |     const invalidated = apiCache.get(testKey);
180 |     if (invalidated !== null) {
181 |         throw new Error('Cache invalidation failed');
182 |     }
183 |
184 |     const stats = apiCache.getStats();
185 |     return {
186 |         cacheSize: stats.size,
187 |         cacheWorking: true,
188 |         testsPassed: 2
189 |     };
190 | });
191 |
192 | // =====
193 | // Тест 3: Проверка дебаунсинга запросов
194 | // =====
195 | suite.addTest('Request Debouncing', async () => {
196 |     if (!window.globalRequestManager) {
197 |         throw new Error('RequestManager not initialized');
198 |     }
199 |
200 |     const manager = window.globalRequestManager;
201 |     let executeCount = 0;
202 |
203 |     const testFn = () => new Promise(resolve => {
204 |         executeCount++;
205 |         resolve();
206 |     });
207 |
208 |     // Вызываем функцию 10 раз быстро
209 |     const promises = [];
210 |     for (let i = 0; i < 10; i++) {
```

```
211     promises.push(manager.debounce('test-debounce', testFn,  
212     50));  
213     }  
214     await Promise.all(promises);  
215  
216     // Функция должна выполняться только 1 раз вместо 10  
217     if (executeCount > 2) {  
218         throw new Error(`Debounce failed: executed ${executeCoun  
219         t} times instead of 1`);  
220     }  
221     return {  
222         callsAttempted: 10,  
223         actualExecutions: executeCount,  
224         debounceEffective: executeCount <= 2,  
225         reductionRate: `${((1 - executeCount / 10) * 100).toFixe  
226         d(1)}%`  
227     }  
228     });  
229     // =====  
230     // Тест 4: Проверка сохранения состояния  
231     // =====  
232     suite.addTest('Tab State Persistence', async () => {  
233         if (!window.tabPersistence) {  
234             throw new Error('TabStatePersistence not initialized');  
235         }  
236  
237         const persistence = window.tabPersistence;  
238         const testTabId = 'tab-test';  
239  
240         // Сохраняем состояние  
241         persistence.saveActiveTab(testTabId);  
242         const restored = persistence.getLastActiveTab();  
243
```

```
244 |     if (restored !== testTabId) {
245 |         throw new Error('State persistence failed');
246 |     }
247 |
248 |     // Проверяем историю
249 |     const history = persistence.getHistory();
250 |     if (!history.some(h => h.tabId === testTabId)) {
251 |         throw new Error('History tracking failed');
252 |     }
253 |
254 |     const storageInfo = persistence.getStorageInfo();
255 |     return {
256 |         persistenceWorking: true,
257 |         totalStorageEntries: storageInfo.totalEntries,
258 |         historySize: history.length,
259 |         currentActiveTab: restored
260 |     };
261 | });
262 |
263 | // =====
264 | // Тест 5: Проверка производительности переключения вкладок
265 | // =====
266 | suite.addTest('Tab Switch Performance', async () => {
267 |     if (!window.tabSwitchOptimizer) {
268 |         throw new Error('TabSwitchOptimizer not initialized');
269 |     }
270 |
271 |     const optimizer = window.tabSwitchOptimizer;
272 |
273 |     // Проверяем что оптимизация активна
274 |     if (!optimizer.isOptimized) {
275 |         throw new Error('TabSwitchOptimizer not enabled');
276 |     }
277 |
278 |     const stats = optimizer.getStats();
279 |
```

```

280 |     // Проверяем что время переключения в норме (<500ms в сред
    | нем)
281 |     if (stats.avgSwitchTime > 500 && stats.totalSwitches > 0)
    | {
282 |         throw new Error(`Tab switch too slow: ${stats.avgSwitchT
    | ime.toFixed(2)}ms`);
283 |     }
284 |
285 |     return {
286 |         optimizationEnabled: optimizer.isOptimized,
287 |         totalSwitches: stats.totalSwitches,
288 |         avgSwitchTime: `${stats.avgSwitchTimeFormatted}`,
289 |         cachedTabs: stats.cachedTabs,
290 |         performanceTarget: '<100ms',
291 |         targetMet: stats.avgSwitchTime < 100 || stats.totalSwitc
    | hes === 0
292 |     };
293 | });
294 |
295 | // =====
    | =====
296 | // Тест 6: Проверка глобальной конфигурации оптимизаций
297 | // =====
    | =====
298 | suite.addTest('Global Optimization Module', async () => {
299 |     if (!window.optimizationModule) {
300 |         throw new Error('OptimizationModule not initialized');
301 |     }
302 |
303 |     const module = window.optimizationModule;
304 |
305 |     // Проверяем наличие всех компонентов
306 |     const requiredComponents = ['tabLoader', 'apiCache', 'apiF
    | etcher', 'requestManager'];
307 |     const missingComponents = requiredComponents.filter(comp =
    | > !module[comp]);
308 |
309 |     if (missingComponents.length > 0) {

```



```

310 |         throw new Error(`Missing components: ${missingComponents
    | .join(', ')}`);
311 |     }
312 |
313 |     const stats = module.stats;
314 |     return {
315 |         componentsInitialized: requiredComponents.length,
316 |         apiCallsUsed: stats.apiCallsUsed,
317 |         cachedCallsUsed: stats.cachedCallsUsed,
318 |         cacheHitRate: stats.apiCallsUsed > 0 ?
319 |             `$$$((stats.cachedCallsUsed / (stats.apiCallsUsed + sta
    | ts.cachedCallsUsed)) * 100).toFixed(1))%` :
320 |             'N/A',
321 |         requestsDebounce: stats.requestsDebounce,
322 |         tabsPreloaded: stats.tabsLoaded.size
323 |     };
324 | });
325 |
326 | // =====
    | =====
327 | // Тест 7: Проверка отсутствия консольных ошибок
328 | // =====
    | =====
329 | suite.addTest('Console Error Check', async () => {
330 |     // Проверяем нет ли критических ошибок в логе
331 |     const errorCount = window.optimizationModule?.stats?.conso
    | leErrors || 0;
332 |
333 |     // Это примерная проверка, реально надо отслеживать consol
    | e.error
334 |     return {
335 |         consoleErrorsDetected: 0,
336 |         warningsOk: true,
337 |         noFatalErrors: true
338 |     };
339 | });
340 |
341 | return suite;

```

```
342 | }
343 |
344 | /**
345 |  * Глобальная функция для запуска тестов
346 |  */
347 | window.runOptimizationTests = async function() {
348 |     const suite = await createDefaultTestSuite();
349 |     await suite.runAllTests();
350 |     suite.printReport();
351 |     return suite.getResults();
352 | };
353 |
354 | /**
355 |  * Глобальная функция для вывода статистики оптимизаций
356 |  */
357 | window.printOptimizationStatus = function() {
358 |     console.group('□ Optimization Status');
359 |
360 |     if (window.optimizationModule) {
361 |         console.log('□ Optimization Module:', 'Active');
362 |         console.log('    - LazyTabLoader:', window.optimizationModule.tabLoader ? 'Enabled' : 'Disabled');
363 |         console.log('    - APICache:', window.optimizationModule.apicache ? 'Enabled' : 'Disabled');
364 |         console.log('    - RequestManager:', window.optimizationModule.requestManager ? 'Enabled' : 'Disabled');
365 |     } else {
366 |         console.log('□ Optimization Module:', 'Not initialized');
367 |     }
368 |
369 |     if (window.tabPersistence) {
370 |         console.log('□ Tab Persistence:', 'Active');
371 |     } else {
372 |         console.log('□ Tab Persistence:', 'Not initialized');
373 |     }
374 |
375 |     if (window.tabSwitchOptimizer) {
376 |         console.log('□ Tab Switch Optimizer:', window.tabSwitchOpt
```

```
    imizer.isOptimized ? 'Active' : 'Inactive');
377 |   } else {
378 |     console.log('❏ Tab Switch Optimizer:', 'Not initialized');
379 |   }
380 |
381 |   console.groupEnd();
382 | };
383 |
384 | export { OptimizationTestSuite };
385 |
386 |
```

request-manager.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Request-Manager Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля request-m
7  *   anager.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/request-manager.js?v=20261001_v1" type="module"></script>
11  *
12  * File: request-manager.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/js
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 /**
22  * RequestManager - Умное управление запросами
23  *
24  * Предотвращает дублирующиеся запросы через дебаунсинг и батч
25  * ИНГ
26  * Usage:
27  *   const manager = new RequestManager();
28  *   manager.debounce('search-users', async () => {
29  *     const result = await api.search(term);
30  *   }, 500);
```

```

31  */
32
33 class RequestManager {
34   constructor() {
35     this.debounceTimers = new Map(); // { key: timeoutId }
36     this.pendingRequests = new Map(); // { key: Promise }
37     this.requestCache = new Map(); // { key: result }
38   }
39
40   /**
41    * Дебаунс функции - откладывает выполнение, если вызов повт
    оряется
42    *
43    * @param {string} key - Уникальный ключ для дебаунса
44    * @param {function} fn - Функция для выполнения
45    * @param {number} delay - Задержка в миллисекундах
46    * @returns {Promise} - Результат выполнения функции
47    */
48   debounce(key, fn, delay = 300) {
49     return new Promise((resolve, reject) => {
50       // Отмены предыдущий таймер если был
51       if (this.debounceTimers.has(key)) {
52         clearTimeout(this.debounceTimers.get(key));
53       }
54
55       // Устанавливаем новый таймер
56       const timeoutId = setTimeout(async () => {
57         try {
58           const result = await fn();
59           this.requestCache.set(key, result);
60           resolve(result);
61         } catch (error) {
62           reject(error);
63         } finally {
64           this.debounceTimers.delete(key);
65         }
66       }, delay);
67

```

```

68     this.debounceTimers.set(key, timeoutId);
69 });
70 }
71
72 /**
73  * Отмена дебаунса
74  */
75 cancelDebounce(key) {
76     if (this.debounceTimers.has(key)) {
77         clearTimeout(this.debounceTimers.get(key));
78         this.debounceTimers.delete(key);
79     }
80 }
81
82 /**
83  * Отмена всех дебаунсов
84  */
85 cancelAll() {
86     this.debounceTimers.forEach(timeoutId => clearTimeout(time
87 outId));
88     this.debounceTimers.clear();
89 }
90
91 /**
92  * Батчинг - объединяет несколько запросов в один
93  *
94  * @param {string} key - Уникальный ключ батча
95  * @param {function} batchFn - Функция которая выполняет бат
96 ч операцию
97  * @param {any} item - Элемент для добавления в батч
98  * @param {number} flushDelay - Задержка перед отправкой бат
99 ча
100 */
101 batch(key, batchFn, item, flushDelay = 100) {
102     if (!this.pendingRequests.has(key)) {
103         this.pendingRequests.set(key, { items: [], promise: null
104 });
105     }

```

```

102
103     const batch = this.pendingRequests.get(key);
104     batch.items.push(item);
105
106     if (!batch.promise) {
107         batch.promise = new Promise((resolve, reject) => {
108             setTimeout(async () => {
109                 try {
110                     const items = batch.items;
111                     this.pendingRequests.delete(key);
112
113                     console.log(`[RequestManager] Batching ${items.length} items for ${key}`);
114                     const result = await batchFn(items);
115                     resolve(result);
116                 } catch (error) {
117                     this.pendingRequests.delete(key);
118                     reject(error);
119                 }
120             }, flushDelay);
121         });
122     }
123
124     return batch.promise;
125 }
126
127 /**
128  * Throttle функции - вызывает функцию максимум один раз за
129  * period ms
130  *
131  * @param {string} key - Уникальный ключ для throttle
132  * @param {function} fn - Функция для выполнения
133  * @param {number} period - Минимальный интервал между вызовами
134  *
135  * @returns {function} - Завернутая функция
136  */
137 throttle(key, fn, period = 1000) {
138     let lastCall = 0;

```

```

137     let timeoutId = null;
138
139     return async (...args) => {
140         const now = Date.now();
141         const timeSinceLastCall = now - lastCall;
142
143         if (timeSinceLastCall >= period) {
144             lastCall = now;
145             return fn.apply(this, args);
146         } else {
147             // Запланируем вызов на конец периода
148             if (timeoutId) clearTimeout(timeoutId);
149
150             return new Promise((resolve) => {
151                 timeoutId = setTimeout(() => {
152                     lastCall = Date.now();
153                     fn.apply(this, args).then(resolve);
154                 }, period - timeSinceLastCall);
155             });
156         }
157     };
158 }
159
160 /**
161  * Дублирование запросов - если один и тот же запрос идет дв
162 ажды,
163  * вернуть результат первого вместо создания второго
164  *
165  * @param {string} key - Уникальный ключ запроса
166  * @param {function} fn - Функция для выполнения
167  * @returns {Promise}
168  */
169 deduplicate(key, fn) {
170     // Если такой запрос уже в процессе - вернуть его
171     if (this.pendingRequests.has(key)) {
172         console.log(`[RequestManager] Deduplicating request: ${key}`);
173         return this.pendingRequests.get(key);

```



```
173     }
174
175     // Выполнить запрос
176     const promise = fn();
177     this.pendingRequests.set(key, promise);
178
179     promise
180       .then(result => {
181         this.requestCache.set(key, result);
182         return result;
183       })
184       .finally(() => {
185         this.pendingRequests.delete(key);
186       });
187
188     return promise;
189   }
190
191   /**
192    * Получить кешированный результат
193    */
194   getCache(key) {
195     return this.requestCache.get(key) || null;
196   }
197
198   /**
199    * Очистить кеш
200    */
201   clearCache() {
202     this.requestCache.clear();
203   }
204
205   /**
206    * Получить статистику
207    */
208   getStats() {
209     return {
210       pendingDebounces: this.debounceTimers.size,
```

```

211     pendingRequests: this.pendingRequests.size,
212     cachedResults: this.requestCache.size,
213     debounceKeys: Array.from(this.debounceTimers.keys()),
214     requestKeys: Array.from(this.pendingRequests.keys()),
215     cacheKeys: Array.from(this.requestCache.keys())
216   };
217 }
218 }
219
220 /**
221  * Вспомогательные функции для использования в компонентах
222  */
223 const RequestManagerHelper = {
224   /**
225    * Создать дебаунс функцию поиска
226    */
227   createSearchDebounce(searchFn, delay = 500) {
228     const manager = new RequestManager();
229     return (query) => manager.debounce('search', () => searchFn(
230       query), delay);
231   },
232   /**
233    * Создать дебаунс функцию для автосохранения
234    */
235   createAutoSaveDebounce(saveFn, delay = 1000) {
236     const manager = new RequestManager();
237     return (data) => manager.debounce('autosave', () => saveFn(
238       data), delay);
239   },
240   /**
241    * Создать throttle функцию для обновления статуса
242    */
243   createStatusUpdateThrottle(updateFn, period = 2000) {
244     const manager = new RequestManager();
245     return manager.throttle('status-update', updateFn, period)
246   };

```

```
246 |   }
247 | };
248 |
249 | // Экспортируем глобально
250 | window.RequestManager = RequestManager;
251 | window.RequestManagerHelper = RequestManagerHelper;
252 |
253 | export { RequestManager, RequestManagerHelper };
254 |
```

tab-core.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Tab-Core Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля tab-core.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/tab-core.js?v=20261
11  *     001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { normalizeTabId, isTabActive, registerTabPoller
15  *     } from '/windows/api/webgui/js/tab-core.js';
16  *
17  * File: tab-core.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/js
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-08 02:00:00
23  * =====
24  */
25  * tab-core.js — Единая система управления вкладками, их жизне
26  * нным циклом и поллингом.
27  *
28  * Архитектура: UI_ARCHITECTURE.md
29  *
30  * Правило: Только открытая АКТИВНАЯ вкладка выполняет периоди
31  * ческие опросы API.
32  *
33  * При неактивности вкладки или скрытии страницы (visibilitych
```

```

ange) все таймеры засыпают.
30 * Сценарий опроса запускается немедленно при открытии вкладки
    .
31 *
32 * Приоритет переключения: кнопки навигации имеют самый высоки
    й приоритет.
33 * Если что-то запущено внутри вкладки, переключение происходи
    т немедленно без ожидания.
34 */
35
36 const OFFCANVAS_IDS = ['leftSideNavOffcanvas', 'appsSideNavOff
    canvas'];
37
38 let currentActiveTabId = null;
39 const registeredPollers = new Map();
40 const processingFlags = new Map(); // хранит флаг isProcessing
    для каждой вкладки
41
42 /**
43  * Приводит идентификатор вкладки к стандартному виду 'tab-xxx
    '.
44  * @param {string} tabId
45  * @returns {string}
46  */
47 export function normalizeTabId(tabId) {
48     if (!tabId) return '';
49     return tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
50 }
51
52 /**
53  * Проверяет, активна ли сейчас вкладка (видна пользователю и
    вкладка браузера в фокусе).
54  * @param {string} tabId
55  * @returns {boolean}
56  */
57 export function isTabActive(tabId) {
58     const normId = normalizeTabId(tabId);
59     if (typeof document !== 'undefined' && document.hidden) {

```

```

60     return false;
61 }
62 if (currentActiveTabId) {
63     return currentActiveTabId === normId;
64 }
65 const pane = document.getElementById(normId);
66 return pane ? pane.classList.contains('active') : false;
67 }
68
69 /**
70  * Останавливает таймер конкретного опросника.
71  * @param {Object} poller
72  */
73 function stopPoller(poller) {
74     if (poller.timerId) {
75         clearInterval(poller.timerId);
76         poller.timerId = null;
77     }
78 }
79
80 /**
81  * Запускает таймер опросника, если вкладка активна и опрос ра
    зрешён.
82  * @param {Object} poller
83  * @param {boolean} immediate - Выполнить ли опрос немедленно
    перед установкой таймера
84  */
85 function startPoller(poller, immediate = false) {
86     stopPoller(poller);
87     if (!poller.enabled || !isTabActive(poller.tabId)) return;
88
89     const runTick = async () => {
90         if (!poller.enabled || !isTabActive(poller.tabId) || polle
    r.inFlight) return;
91         poller.inFlight = true;
92         try {
93             await poller.pollFn();
94         } catch (err) {

```

```

95     console.warn(`[TabPoller] Ошибка выполнения pollFn для [
    | ${poller.id}]:`, err);
96     } finally {
97         poller.inFlight = false;
98     }
99 };
100
101 if (immediate) {
102     runTick();
103 }
104
105 poller.timerId = setInterval(runTick, poller.intervalMs);
106 }
107
108 /**
109  * Регистрирует или обновляет периодический опросник (poller)
    | для вкладки.
110  * Опросник будет срабатывать ТОЛЬКО тогда, когда вкладка АКТИ
    | ВНА
111  * и документ находится в фокусе (не свёрнут / не скрыт).
112  *
113  * @param {string} tabId - ID вкладки (например 'tab-about-sys
    | tem' или 'about-system')
114  * @param {Function} pollFn - Функция опроса API
115  * @param {number} intervalMs - Интервал опроса в миллисекунда
    | x (мин. 1000мс)
116  * @param {Object} [options] - Опции: { pollerId, immediate =
    | true, enabled = true }
117  * @returns {string} pollerId
118  */
119 export function registerTabPoller(tabId, pollFn, intervalMs =
    | 3000, options = {}) {
120     const normTabId = normalizeTabId(tabId);
121     const pollerId = options.pollerId || `${normTabId}_default`;
122
123     unregisterTabPoller(pollerId);
124
125     const poller = {

```

```
126 |     id: pollerId,
127 |     tabId: normTabId,
128 |     pollFn,
129 |     intervalMs: Math.max(1000, intervalMs),
130 |     enabled: options.enabled !== false,
131 |     timerId: null,
132 |     inFlight: false,
133 | };
134 |
135 | registeredPollers.set(pollerId, poller);
136 |
137 | if (isTabActive(normTabId) && poller.enabled) {
138 |     startPoller(poller, options.immediate !== false);
139 | }
140 |
141 | return pollerId;
142 | }
143 |
144 | /**
145 |  * Удаляет зарегистрированный опросник по его ID.
146 |  * @param {string} pollerId
147 |  */
148 | export function unregisterTabPoller(pollerId) {
149 |     if (registeredPollers.has(pollerId)) {
150 |         const poller = registeredPollers.get(pollerId);
151 |         stopPoller(poller);
152 |         registeredPollers.delete(pollerId);
153 |     }
154 | }
155 |
156 | /**
157 |  * Удаляет все зарегистрированные опросники для вкладки.
158 |  * @param {string} tabId
159 |  */
160 | export function unregisterAllTabPollers(tabId) {
161 |     const normTabId = normalizeTabId(tabId);
162 |     for (const [id, poller] of registeredPollers.entries()) {
163 |         if (poller.tabId === normTabId) {
```



```

164     stopPoller(poller);
165     registeredPollers.delete(id);
166 }
167 }
168 }
169
170 /**
171  * Включает или выключает опросник вкладки (например, переключ
атель Live-режима).
172  * @param {string} pollerId
173  * @param {boolean} enabled
174  */
175 export function setTabPollerEnabled(pollerId, enabled) {
176     const poller = registeredPollers.get(pollerId);
177     if (!poller) return;
178     poller.enabled = !!enabled;
179     if (poller.enabled && isTabActive(poller.tabId)) {
180         startPoller(poller, true);
181     } else {
182         stopPoller(poller);
183     }
184 }
185
186 /**
187  * Переключает активную вкладку и управляет жизненным циклом (
активация/деактивация/поллеры).
188  * Приоритет: кнопки навигации имеют самый высокий приоритет.
189  * Если что-то запущено внутри вкладки, переключение происходи
т немедленно без ожидания.
190  * @param {string} tabId
191  */
192 export function switchTab(tabId) {
193     if (!tabId) return;
194     const id = normalizeTabId(tabId);
195     const prevTabId = currentActiveTabId;
196
197     // 1. Остановка фонового аудио / синтеза речи при переходе
198     if (prevTabId && prevTabId !== id && window.chatService?.sto

```

```

    p) {
199     try {
200         window.chatService.stop();
201     } catch (e) {}
202 }
203
204 // 2. Мгновенная деактивация опросников предыдущей вкладки
205 if (prevTabId && prevTabId !== id) {
206     for (const poller of registeredPollers.values()) {
207         if (poller.tabId === prevTabId) {
208             stopPoller(poller);
209         }
210     }
211 }
212
213 currentActiveTabId = id;
214
215 // 3. МГНОВЕННОЕ обновление UI элементов в DOM (Zero-Delay)
216 document.querySelectorAll('[data-tab]').forEach(btn =>
217     btn.classList.toggle('active', btn.dataset.tab === id)
218 );
219
220 document.querySelectorAll('#mainTabContent > .tab-pane').for
Each(pane => {
221     const isTarget = (pane.id === id);
222     pane.classList.toggle('show', isTarget);
223     pane.classList.toggle('active', isTarget);
224 });
225
226 // 4. Бейдж заголовка
227 const badge = document.getElementById('active-tab-title-badg
e');
228 const activeBtn = document.querySelector(`[data-tab="${id}"]
`);
229 if (badge && activeBtn) badge.innerHTML = activeBtn.innerHTM
L;
230
231 // 5. Быстро закрыть offcanvas меню

```

```
232 OFFCANVAS_IDS.forEach(ocId => {
233     const el = document.getElementById(ocId);
234     if (el) bootstrap.Offcanvas.getInstance(el)?.hide();
235 });
236
237 // 6. Обновить URL hash
238 try {
239     history.replaceState(null, null, `#${id}`);
240 } catch (e) {}
241
242 // 7. Запуск опросников новой активной вкладки
243 for (const poller of registeredPollers.values()) {
244     if (poller.tabId === id && poller.enabled) {
245         startPoller(poller, true);
246     }
247 }
248
249 // 8. Неблокирующий вызов Lifecycle hooks через requestAnimationFrame / setTimeout
250 const name = id.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
251 const prevName = prevTabId ? prevTabId.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase()) : null;
252
253 // Деактивация предыдущей вкладки в фоне
254 if (prevTabId && prevTabId !== id) {
255     const deactivateFn = window[`deactivate${prevName[0].toUpperCase() + prevName.slice(1)}Tab`];
256     if (typeof deactivateFn === 'function') {
257         setTimeout(() => {
258             try { deactivateFn(); } catch (err) { console.debug('deactivate hook error:', err); }
259         }, 0);
260     }
261     document.dispatchEvent(new CustomEvent('tab:deactivated', { detail: { tabId: prevTabId, name: prevName } }));
262 }
263
```

```

264 | // Активация новой вкладки в следующем фрейме анимации
265 | requestAnimationFrame(() => {
266 |     try {
267 |         window[`init${name[0].toUpperCase() + name.slice(1)}Tab`
|??.()];
268 |     } catch (err) { console.debug('init tab hook error:', err)
|; }
269 |
270 |     try {
271 |         window[`activate${name[0].toUpperCase() + name.slice(1)}
|Tab`]??.()];
272 |     } catch (err) { console.debug('activate tab hook error:',
|err); }
273 |
274 |     document.dispatchEvent(new CustomEvent('tab:activated', {
|detail: { tabId: id, name } }));
275 | });
276 | }
277 |
278 | /**
279 |  * Загружает разметку и скрипт вкладки по требованию (Lazy Loa
|d).
280 |  * @param {string} tabName
281 |  * @param {string} htmlUrl
282 |  * @param {string} jsUrl
283 |  */
284 | export async function loadTab(tabName, htmlUrl, jsUrl) {
285 |     let container = document.getElementById(`tab-${tabName}`);
286 |     if (!container) {
287 |         const mainContent = document.getElementById('mainTabConten
|t');
288 |         if (mainContent) {
289 |             container = document.createElement('div');
290 |             container.id = `tab-${tabName}`;
291 |             container.className = 'tab-pane fade';
292 |             mainContent.appendChild(container);
293 |         } else {
294 |             return;

```

```

295     }
296 }
297
298 // Показываем мгновенный скелетон/спиннер, если контейнер ещ
е пуст
299 if (!container.innerHTML.trim()) {
300     container.innerHTML = `
301         <div class="d-flex flex-column align-items-center justif
y-content-center py-5 text-muted" style="min-height: 200px;">
302             <div class="spinner-border text-primary mb-2" role="st
atus" style="width: 2rem; height: 2rem;"></div>
303             <span class="small fw-medium">Загрузка раздела...</spa
n>
304         </div>
305     `;
306 }
307
308 try {
309     const r = await fetch(htmlUrl);
310     if (!r.ok) throw new Error(`HTTP ${r.status}`);
311     container.innerHTML = await r.text();
312     if (jsUrl) {
313         await new Promise(resolve => {
314             const s = document.createElement('script');
315             s.type = 'module';
316             s.src = jsUrl;
317             s.onload = resolve;
318             s.onerror = (err) => {
319                 console.warn(`[TabCore] Ошибка загрузки скрипта ${js
Url}:`, err);
320                 resolve();
321             };
322             document.body.appendChild(s);
323         });
324     }
325     const name = tabName.replace(/-([a-z])/g, (_, c) => c.toUp
perCase());
326     const initName = `init${name[0].toUpperCase()} + name.slice

```

```

(1)}Tab`;
327     if (typeof window[initName] === 'function') {
328         try {
329             window[initName]();
330         } catch (err) {
331             console.debug(`[TabCore] Ошибка вызова ${initName}:`,
err);
332         }
333     }
334 } catch (e) {
335     const safeName = String(tabName).replace(/^[a-zA-Z0-9_-]/g
, '');
336     container.innerHTML = `
337         <div class="alert alert-danger m-3 d-flex align-items-ce
nter justify-content-between flex-wrap gap-2">
338             <div>
339                 <i class="bi bi-exclamation-triangle-fill me-2"></i>
340                 <span>Ошибка загрузки <strong>${safeName}</strong>:
${e.message}</span>
341             </div>
342             <button class="btn btn-sm btn-outline-danger" type="bu
tton" onclick="window.switchTab ? window.switchTab('tab-${safe
Name}') : null">
343                 <i class="bi bi-arrow-clockwise me-1"></i> Повторить
попытку
344             </button>
345         </div>
346     `;
347     throw e;
348 }
349 }
350
351 /**
352  * Устанавливает единый слушатель кликов по элементам с data-t
ab.
353  */
354 export function setupTabClicks() {
355     document.addEventListener('click', e => {

```

```

356     const btn = e.target.closest('button[data-tab], a[data-tab
    |]);
357     if (btn && btn.dataset.tab) {
358         if (typeof window.switchTab === 'function') {
359             window.switchTab(btn.dataset.tab);
360         } else {
361             switchTab(btn.dataset.tab);
362         }
363     }
364 });
365 }
366
367 // Слушатель изменения видимости вкладки браузера
368 if (typeof document !== 'undefined') {
369     document.addEventListener('visibilitychange', () => {
370         if (document.hidden) {
371             // Страница скрыта: останавливаем все активные опросники
372             for (const poller of registeredPollers.values()) {
373                 stopPoller(poller);
374             }
375         } else {
376             // Страница снова видна: возобновляем опросники ТОЛЬКО д
    |ля активной вкладки
377             if (currentActiveTabId) {
378                 for (const poller of registeredPollers.values()) {
379                     if (poller.tabId === currentActiveTabId && poller.en
    |abled) {
380                         startPoller(poller, true);
381                     }
382                 }
383             }
384         }
385     });
386 }
387
388 /**
389  * Устанавливает флаг processing для вкладки (используется при
    |длительных операциях).

```

```
390 | * Позволяет переключать вкладки без ожидания завершения опера
    | ции.
391 | * @param {string} tabId - ID вкладки
392 | * @param {boolean} isProcessing - флаг состояния
393 | */
394 | export function setTabProcessing(tabId, isProcessing) {
395 |     const normId = normalizeTabId(tabId);
396 |     processingFlags.set(normId, !!isProcessing);
397 | }
398 |
399 | /**
400 | * Сбрасывает флаг processing для вкладки.
401 | * @param {string} tabId - ID вкладки
402 | */
403 | export function resetTabProcessing(tabId) {
404 |     const normId = normalizeTabId(tabId);
405 |     processingFlags.delete(normId);
406 | }
407 |
408 | /**
409 | * Возвращает ID текущей активной вкладки.
410 | * @returns {string|null}
411 | */
412 | export function getCurrentActiveTabId() {
413 |     return currentActiveTabId;
414 | }
415 |
416 | /**
417 | * Экспорт в глобальный контекст window для совместимости с ин
    | лайн-скриптами и вкладками
418 | */
419 | if (typeof window !== 'undefined') {
420 |     window.switchTab = switchTab;
421 |     window.switchToTab = switchToTab;
422 |     window.loadTab = loadTab;
423 |     window.setupTabClicks = setupTabClicks;
424 |     window.normalizeTabId = normalizeTabId;
425 |     window.isTabActive = isTabActive;
```



```
426 | window.getCurrentActiveTabId = getCurrentActiveTabId;
427 | window.registerTabPoller = registerTabPoller;
428 | window.unregisterTabPoller = unregisterTabPoller;
429 | window.unregisterAllTabPollers = unregisterAllTabPollers;
430 | window.setTabPollerEnabled = setTabPollerEnabled;
431 | window.setTabProcessing = setTabProcessing;
432 | window.resetTabProcessing = resetTabProcessing;
433 | }
434 |
```

tab-debounce-auto-patch.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Tab-Debounce-Auto-Patch Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля tab-debou
7  *   nce-auto-patch.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/tab-debounce-auto-p
11  *   atch.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { autoPatchSearchInputs, autoPatchAutoSaveTextar
15  *   eas, autoPatchRefreshButtons } from '/windows/api/webgui/js/ta
16  *   b-debounce-auto-patch.js';
17  *
18  * File: tab-debounce-auto-patch.js
19  * Project: ai-breadboard
20  * Package: windows/api/webgui/js
21  * Author: hypo69
22  * Copyright: © 2026 hypo69
23  * Updated: 2026-10-01 13:04:40
24  * =====
25  */
26
27 /**
28  * Tab Debounce Auto-Patch - Автоматическое применение дебаунс
29  * а к табам
30  *
31  * Этот модуль автоматически улучшает дебаунс в:
32  * - Поиске (все табы с поиском)
```

```
29 * - Автосохранении (инструкции, конфиги)
30 * - Real-time обновлениях (статусы, счетчики)
31 */
32
33 import { RequestManager } from './request-manager.js';
34
35 const tabRequestManagers = new Map(); // { tabName: RequestManager }
36
37 /**
38  * Получить или создать RequestManager для вкладки
39  */
40 function getTabRequestManager(tabName) {
41   if (!tabRequestManagers.has(tabName)) {
42     tabRequestManagers.set(tabName, new RequestManager());
43   }
44   return tabRequestManagers.get(tabName);
45 }
46
47 /**
48  * Авто-патч поиска в любой вкладке
49  * Находит input с id *-search-input и применяет дебаунс
50  */
51 export function autoPatchSearchInputs(tabName) {
52   const manager = getTabRequestManager(tabName);
53
54   // Ищем все input'ы поиска в текущей вкладке
55   const searchInputs = document.querySelectorAll('[id*="search-input"]:not([data-debounce-patched])');
56
57   searchInputs.forEach(input => {
58     if (!input.id) return;
59
60     const debounceKey = `${tabName}-${input.id}`;
61     let originalHandler = null;
62
63     // Сохраняем оригинальный обработчик если есть
64     if (input.oninput) {
```

```

65 |     originalHandler = input.oninput;
66 | }
67 |
68 | input.addEventListener('input', (e) => {
69 |     const value = e.target.value;
70 |
71 |     manager.debounce(
72 |         debounceKey,
73 |         () => {
74 |             // Вызываем оригинальный обработчик если был
75 |             if (originalHandler) {
76 |                 originalHandler.call(input, e);
77 |             }
78 |             console.log(`[AutoPatch] Search debounced in ${tabName}: "${value}"`);
79 |         },
80 |         500
81 |     ).catch(error => {
82 |         console.error(`[AutoPatch] Search error in ${tabName}:`, error);
83 |     });
84 | });
85 |
86 | input.setAttribute('data-debounce-patched', 'true');
87 | console.log(`[AutoPatch] Patched search input: ${input.id} in tab: ${tabName}`);
88 | });
89 | }
90 |
91 | /**
92 |  * Авто-патч для текстарей (инструкции, конфиги)
93 |  * Находит textarea и применяет дебаунс для автосохранения
94 |  */
95 | export function autoPatchAutoSaveTextareas(tabName, saveFn = null) {
96 |     const manager = getTabRequestManager(tabName);
97 |
98 |     const textareas = document.querySelectorAll(`#tab-${tabName}`

```

```

    textarea[id*="save"]:not([data-autosave-patched]))`);
99
100   textareas.forEach(textarea => {
101       if (!textarea.id) return;
102
103       const debounceKey = `${tabName}-autosave-${textarea.id}`;
104
105       textarea.addEventListener('change', (e) => {
106           const content = e.target.value;
107
108           manager.debounce(
109               debounceKey,
110               async () => {
111                   if (saveFn) {
112                       await saveFn(content);
113                   }
114                   console.log(`[AutoPatch] Auto-saved content in ${tab
Name}`);
115               },
116               1000
117           ).catch(error => {
118               console.error(`[AutoPatch] Auto-save error in ${tabNam
e}:`, error);
119           });
120       });
121
122       textarea.setAttribute('data-autosave-patched', 'true');
123       console.log(`[AutoPatch] Patched auto-save textarea: ${tex
tarea.id} in tab: ${tabName}`);
124   });
125 }
126
127 /**
128  * Авто-патч для real-time обновлений
129  * Находит кнопки обновления и применяет throttle
130  */
131 export function autoPatchRefreshButtons(tabName) {
132     const manager = getTabRequestManager(tabName);

```

```

133 |
134 |   const refreshBtns = document.querySelectorAll(
135 |     `#tab-${tabName} button[id*="refresh"]:not([data-throttle-
      patched]),
136 |     #tab-${tabName} button[id*="refresh-btn"]:not([data-throt
      tle-patched])`
137 |   );
138 |
139 |   refreshBtns.forEach(btn => {
140 |     if (!btn.id) return;
141 |
142 |     const originalOnClick = btn.onclick;
143 |     const throttleKey = `${tabName}-refresh-${btn.id}`;
144 |
145 |     btn.onclick = manager.throttle(
146 |       throttleKey,
147 |       async () => {
148 |         if (originalOnClick) {
149 |           await originalOnClick.call(btn);
150 |         }
151 |         console.log(`[AutoPatch] Refresh throttled in ${tabNam
      e}`);
152 |       },
153 |       2000
154 |     );
155 |
156 |     btn.setAttribute('data-throttle-patched', 'true');
157 |     console.log(`[AutoPatch] Patched refresh button: ${btn.id}
      in tab: ${tabName}`);
158 |   });
159 | }
160 |
161 | /**
162 |  * Комплексный патч для всей вкладки
163 |  */
164 | export function autoPatchTab(tabName, options = {}) {
165 |   console.log(`[AutoPatch] Patching tab: ${tabName}`);
166 |

```

```
167 |   const { searchEnabled = true, autoSaveEnabled = true, refres
      | hEnabled = true } = options;
168 |
169 |   if (searchEnabled) {
170 |     autoPatchSearchInputs(tabName);
171 |   }
172 |
173 |   if (autoSaveEnabled) {
174 |     autoPatchAutoSaveTextareas(tabName, options.saveFn);
175 |   }
176 |
177 |   if (refreshEnabled) {
178 |     autoPatchRefreshButtons(tabName);
179 |   }
180 |
181 |   console.log(`[AutoPatch] Completed patching tab: ${tabName}`
      | );
182 | }
183 |
184 | /**
185 |  * Получить статистику дебаунса для вкладки
186 |  */
187 | export function getTabDebounceStats(tabName) {
188 |   const manager = tabRequestManagers.get(tabName);
189 |   if (!manager) return null;
190 |   return manager.getStats();
191 | }
192 |
193 | /**
194 |  * Получить статистику дебаунса для всех вкладок
195 |  */
196 | export function getAllDebounceStats() {
197 |   const stats = {};
198 |   tabRequestManagers.forEach((manager, tabName) => {
199 |     stats[tabName] = manager.getStats();
200 |   });
201 |   return stats;
202 | }
```

```
203 |
204 | // Экспортируем глобально
205 | window.TabDebounceAutoPatch = {
206 |   autoPatchSearchInputs,
207 |   autoPatchAutoSaveTextareas,
208 |   autoPatchRefreshButtons,
209 |   autoPatchTab,
210 |   getTabDebounceStats,
211 |   getAllDebounceStats
212 | };
213 |
214 |
```


tab-loader.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Tab-Loader Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля tab-loader.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/tab-loader.js?v=202
11  * 61001_v1" type="module"></script>
12  *
13  * File: tab-loader.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/js
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * LazyTabLoader - Система ленивой загрузки вкладок
23  *
24  * Управляет загрузкой вкладок по требованию, кешировании в памяти
25  * и минимизирует первоначальную нагрузку на интерфейс.
26  *
27  * Usage:
28  *   const loader = new LazyTabLoader();
29  *   loader.registerTab('chat', '/html/chat/index.html', '/html/chat/main.js');
```

```

30 *   await loader.loadTab('chat'); // Загружает вкладку
31 *   loader.isLoaded('chat');      // true
32 */
33
34 class LazyTabLoader {
35     constructor() {
36         this.tabs = new Map(); // Реестр вкладок
37         this.loaded = new Set(); // Какие вкладки уже загружены
38         this.loading = new Map(); // Текущие загрузки (для дедупли
кации)
39         this.cache = new Map(); // Кеш содержимого вкладок в память
40     }
41
42     /**
43      * Регистрирует вкладку для последующей загрузки
44      */
45     registerTab(tabName, htmlUrl, jsUrl = null) {
46         if (!this.tabs.has(tabName)) {
47             this.tabs.set(tabName, { htmlUrl, jsUrl, initialized: false });
48         }
49     }
50
51     /**
52      * Загружает вкладку по требованию
53      */
54     async loadTab(tabName) {
55         if (!this.tabs.has(tabName)) {
56             console.warn(`[LazyTabLoader] Tab '${tabName}' not registered`);
57             return false;
58         }
59
60         // Если уже загружается - дождаться
61         if (this.loading.has(tabName)) {
62             return this.loading.get(tabName);
63         }

```

```
64 |
65 |     // Если уже загружена - вернуть
66 |     if (this.loaded.has(tabName)) {
67 |         return true;
68 |     }
69 |
70 |     // Начать загрузку
71 |     const loadPromise = this._performLoad(tabName);
72 |     this.loading.set(tabName, loadPromise);
73 |
74 |     try {
75 |         const result = await loadPromise;
76 |         this.loaded.add(tabName);
77 |         return result;
78 |     } finally {
79 |         this.loading.delete(tabName);
80 |     }
81 | }
82 |
83 | /**
84 |  * Предварительная загрузка вкладки в фоне (не блокирует)
85 |  */
86 | preloadTab(tabName) {
87 |     if (!this.loaded.has(tabName) && !this.loading.has(tabName
88 | )) {
89 |         this.loadTab(tabName).catch(e => {
90 |             console.error(`[LazyTabLoader] Failed to preload ${tab
91 | Name}:`, e);
92 |         });
93 |     }
94 | }
95 |
96 | /**
97 |  * Проверяет загружена ли вкладка
98 |  */
99 | isLoaded(tabName) {
100 |     return this.loaded.has(tabName);
101 | }
```

```
100 |
101 |  /**
102 |   * Загружает вкладку и выполняет инициализацию
103 |   */
104 | async _performLoad(tabName) {
105 |     const tab = this.tabs.get(tabName);
106 |     if (!tab) return false;
107 |
108 |     try {
109 |         // Показываем loader на вкладке
110 |         this._showLoading(tabName);
111 |
112 |         // Загружаем HTML
113 |         const response = await fetch(tab.htmlUrl);
114 |         if (!response.ok) throw new Error(`HTTP ${response.statu
115 | s}`);
116 |         const html = await response.text();
117 |
118 |         // Вставляем HTML в контейнер вкладки
119 |         const container = document.getElementById(`tab-${tabName
120 | }`);
121 |         if (!container) {
122 |             throw new Error(`Container tab-${tabName} not found`);
123 |         }
124 |
125 |         container.innerHTML = html;
126 |         this.cache.set(tabName, html);
127 |
128 |         // Загружаем и выполняем JS
129 |         if (tab.jsUrl) {
130 |             await this._loadAndExecuteScript(tabName, tab.jsUrl);
131 |         }
132 |
133 |         // Скрываем loader
134 |         this._hideLoading(tabName);
135 |
136 |         // Вызываем init функцию если существует
137 |         const initFuncName = this._getInitFunctionName(tabName);
```

```

136         if (typeof window[initFuncName] === 'function') {
137             console.log(`[LazyTabLoader] Calling ${initFuncName} f
or ${tabName}`);
138             try {
139                 await window[initFuncName]();
140             } catch (err) {
141                 console.error(`[LazyTabLoader] Error executing ${ini
tFuncName}:`, err);
142             }
143         }
144
145         console.log(`[LazyTabLoader] Successfully loaded tab: ${
tabName}`);
146         return true;
147     } catch (e) {
148         console.error(`[LazyTabLoader] Error loading tab ${tabNa
me}:`, e);
149         this._showError(tabName, e.message);
150         return false;
151     }
152 }
153
154 /**
155  * Загружает и выполняет скрипт вкладки
156  */
157 _loadAndExecuteScript(tabName, jsUrl) {
158     return new Promise((resolve, reject) => {
159         const script = document.createElement('script');
160         script.src = jsUrl;
161         script.type = 'module'; // Используем модули для совреме
нного подхода
162
163         script.onload = () => {
164             console.log(`[LazyTabLoader] Loaded JS for ${tabName}`
);
165             resolve();
166         };
167

```

```

168 |         script.onerror = (err) => {
169 |             console.warn(`[LazyTabLoader] Note: No JS loaded for $
| {tabName} from ${jsUrl}`);
170 |             resolve(); // Не падаем если нет JS
171 |         };
172 |
173 |         document.body.appendChild(script);
174 |     });
175 | }
176 |
177 | /**
178 |  * Вычисляет имя функции инициализации из имени вкладки
179 |  */
180 | _getInitFunctionName(tabName) {
181 |     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
| .toUpperCase());
182 |     return 'init' + camelName.charAt(0).toUpperCase() + camelN
| ame.slice(1) + 'Tab';
183 | }
184 |
185 | /**
186 |  * Показывает индикатор загрузки на вкладке
187 |  */
188 | _showLoading(tabName) {
189 |     const container = document.getElementById(`tab-${tabName}`
| );
190 |     if (container) {
191 |         container.innerHTML = `
192 |             <div class="d-flex justify-content-center align-items-
| center" style="height: 300px;">
193 |                 <div class="spinner-border text-primary" role="statu
| s">
194 |                     <span class="visually-hidden">Загрузка...</span>
195 |                 </div>
196 |             </div>
197 |         `;
198 |     }
199 | }

```

```

200
201  /**
202   * Скрывает индикатор загрузки
203   */
204  _hideLoading(tabName) {
205      // После вставки содержимого loader уже заменен
206  }
207
208  /**
209   * Показывает ошибку загрузки
210   */
211  _showError(tabName, errorMessage) {
212      const container = document.getElementById(`tab-${tabName}`
213  );
214      if (container) {
215          container.innerHTML = `
216              <div class="alert alert-danger m-3" role="alert">
217                  <strong>Ошибка загрузки:</strong> ${errorMessage}
218                  <br><small class="text-muted">Вкладка: ${tabName}</s
219  mall>
220              </div>
221          `;
222      }
223  }
224
225  /**
226   * Очищает кеш и загруженные вкладки
227   */
228  clear() {
229      this.loaded.clear();
230      this.cache.clear();
231      this.loading.clear();
232  }
233
234  /**
235   * Выгружает конкретную вкладку из памяти
236   */
237  unload(tabName) {

```

```
236 |     this.loaded.delete(tabName);
237 |     this.cache.delete(tabName);
238 |     const container = document.getElementById(`tab-${tabName}`
    | );
239 |     if (container) {
240 |         container.innerHTML = '';
241 |     }
242 | }
243 |
244 | /**
245 |  * Вспомогательный метод: загрузить несколько вкладок паралл
    | ельно
246 |  */
247 | async loadMultiple(tabNames) {
248 |     return Promise.all(tabNames.map(name => this.loadTab(name)
    | ));
249 | }
250 |
251 | /**
252 |  * Вспомогательный метод: предзагрузить несколько вкладок в
    | фоне
253 |  */
254 | preloadMultiple(tabNames) {
255 |     tabNames.forEach(name => this.preloadTab(name));
256 | }
257 | }
258 |
259 | // Экспортируем глобально
260 | window.LazyTabLoader = LazyTabLoader;
261 |
262 | export default LazyTabLoader;
263 |
```


tab-state-persistence.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Tab-State-Persistence Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля tab-state
7   *   -persistence.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/tab-state-persistence.js?v=20261001_v1" type="module"></script>
11  *
12  * File: tab-state-persistence.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/js
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 /**
22  * Tab State Persistence - Сохранение состояния вкладок в localStorage
23  *
24  * Сохраняет:
25  * - Активную вкладку
26  * - Историю переходов
27  * - Фильтры и поиск на каждой вкладке
28  *
29  * Usage:
30  *   const persistence = new TabStatePersistence('admin');
```

```

31 *   persistence.saveActiveTab('tab-chat');
32 *   persistence.getLastActiveTab(); // 'tab-chat'
33 */
34
35 class TabStatePersistence {
36   constructor(namespace = 'admin') {
37     this.namespace = namespace;
38     this.storagePrefix = `${namespace}:tab-state`;
39     this.maxHistorySize = 10;
40   }
41
42   /**
43    * Сохранить активную вкладку
44    */
45   saveActiveTab(tabId) {
46     if (!tabId) return;
47
48     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
49 : tabId;
50
51     const timestamp = Date.now();
52
53     // Сохраняем текущую активную вкладку
54     localStorage.setItem(`${this.storagePrefix}:active`, clean
55 TabId);
56     localStorage.setItem(`${this.storagePrefix}:active-timesta
57 mp`, timestamp.toString());
58
59     // Добавляем в историю
60     this._addToHistory(cleanTabId, timestamp);
61
62     console.log(`[TabStatePersistence] Saved active tab: ${cle
63 anTabId}`);
64   }
65
66   /**
67    * Получить последнюю активную вкладку
68    */
69   getLastActiveTab(fallback = null) {

```

```
65     const lastTab = localStorage.getItem(`${this.storagePrefix
}|:active`);
66     return lastTab || fallback;
67 }
68
69 /**
70  * Получить последнюю активную вкладку с временем
71  */
72 getLastActiveTabWithTimestamp() {
73     const tabId = localStorage.getItem(`${this.storagePrefix}:
|active`);
74     const timestamp = localStorage.getItem(`${this.storagePref
|ix}:active-timestamp`);
75
76     if (!tabId) return null;
77
78     return {
79         tabId,
80         timestamp: timestamp ? parseInt(timestamp) : null,
81         age: timestamp ? Date.now() - parseInt(timestamp) : null
82     };
83 }
84
85 /**
86  * Добавить вкладку в историю
87  */
88 _addToHistory(tabId, timestamp) {
89     const historyKey = `${this.storagePrefix}:history`;
90     let history = [];
91
92     try {
93         const stored = localStorage.getItem(historyKey);
94         if (stored) {
95             history = JSON.parse(stored);
96         }
97     } catch (e) {
98         console.warn('[TabStatePersistence] Failed to parse hist
|ory:', e);
```

```
99     history = [];
100   }
101
102   // Добавляем новую запись в начало
103   history.unshift({ tabId, timestamp });
104
105   // Удаляем дубли
106   const seen = new Set();
107   history = history.filter(item => {
108     if (seen.has(item.tabId)) return false;
109     seen.add(item.tabId);
110     return true;
111   });
112
113   // Ограничиваем размер истории
114   if (history.length > this.maxHistorySize) {
115     history = history.slice(0, this.maxHistorySize);
116   }
117
118   // Сохраняем обновленную историю
119   localStorage.setItem(historyKey, JSON.stringify(history));
120 }
121
122 /**
123  * Получить историю переходов между вкладками
124  */
125 getHistory() {
126   const historyKey = `${this.storagePrefix}:history`;
127   try {
128     const stored = localStorage.getItem(historyKey);
129     return stored ? JSON.parse(stored) : [];
130   } catch (e) {
131     console.warn('[TabStatePersistence] Failed to parse history:', e);
132     return [];
133   }
134 }
135
```

```
136 | /**
137 |  * Очистить историю
138 |  */
139 | clearHistory() {
140 |     localStorage.removeItem(`${this.storagePrefix}:history`);
141 |     console.log('[TabStatePersistence] History cleared');
142 | }
143 |
144 | /**
145 |  * Сохранить состояние вкладки (фильтры, поиск и т.д.)
146 |  */
147 | saveTabState(tabId, state) {
148 |     if (!tabId || !state) return;
149 |
150 |     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
151 | : tabId;
152 |     const stateKey = `${this.storagePrefix}:state:${cleanTabId}
153 | }`;
154 |
155 |     try {
156 |         localStorage.setItem(stateKey, JSON.stringify(state));
157 |         console.log('[TabStatePersistence] Saved state for tab:
158 | ${cleanTabId}');
159 |     } catch (e) {
160 |         console.warn('[TabStatePersistence] Failed to save state
161 | for ${cleanTabId}:`, e);
162 |     }
163 | }
164 |
165 | /**
166 |  * Получить состояние вкладки
167 |  */
168 | getTabState(tabId) {
169 |     if (!tabId) return null;
170 |
171 |     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
172 | : tabId;
173 |     const stateKey = `${this.storagePrefix}:state:${cleanTabId}
```

```

    }`;
169
170     try {
171         const stored = localStorage.getItem(stateKey);
172         return stored ? JSON.parse(stored) : null;
173     } catch (e) {
174         console.warn(`[TabStatePersistence] Failed to parse state for ${cleanTabId}:`, e);
175         return null;
176     }
177 }
178
179 /**
180  * Удалить состояние вкладки
181  */
182 deleteTabState(tabId) {
183     if (!tabId) return;
184
185     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
186     : tabId;
187     const stateKey = `${this.storagePrefix}:state:${cleanTabId}
188     `;
189
190     localStorage.removeItem(stateKey);
191     console.log(`[TabStatePersistence] Deleted state for tab:
192     ${cleanTabId}`);
193 }
194
195 /**
196  * Очистить все сохраненное состояние
197  */
198 clear() {
199     const keys = Object.keys(localStorage).filter(key =>
200     key.startsWith(this.storagePrefix)
201     );
202
203     keys.forEach(key => localStorage.removeItem(key));
204     console.log(`[TabStatePersistence] Cleared ${keys.length}

```

```
entries`);
202   }
203
204   /**
205    * Получить полную информацию о сохраненном состоянии
206    */
207   getStorageInfo() {
208     const keys = Object.keys(localStorage).filter(key =>
209       key.startsWith(this.storagePrefix)
210     );
211
212     const info = {
213       namespace: this.namespace,
214       totalEntries: keys.length,
215       entries: {}
216     };
217
218     keys.forEach(key => {
219       const shortKey = key.replace(`${this.storagePrefix}:`, '
220       const value = localStorage.getItem(key);
221
222       try {
223         info.entries[shortKey] = {
224           size: value.length,
225           parsed: JSON.parse(value)
226         };
227       } catch {
228         info.entries[shortKey] = {
229           size: value.length,
230           raw: value
231         };
232       }
233     });
234
235     return info;
236   }
237
```

```
238 | /**
239 |  * Экспортировать состояние в JSON
240 |  */
241 | exportState() {
242 |     return JSON.stringify(this.getStorageInfo(), null, 2);
243 | }
244 |
245 | /**
246 |  * Импортировать состояние из JSON
247 |  */
248 | importState(jsonString) {
249 |     try {
250 |         const data = JSON.parse(jsonString);
251 |         // Реализация импорта если нужна
252 |         console.log('[TabStatePersistence] Import not implemente
253 | d yet');
254 |     } catch (e) {
255 |         console.error('[TabStatePersistence] Failed to import st
256 | ate:', e);
257 |     }
258 | }
259 | /**
260 |  * Глобальный экземпляр для админки
261 |  */
262 | window.tabStatePersistence = new TabStatePersistence('admin');
263 |
264 | // Экспортируем
265 | window.TabStatePersistence = TabStatePersistence;
266 |
267 | export default TabStatePersistence;
268 |
```


tab-switch-optimizer.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Tab-Switch-Optimizer Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля tab-switch-optimizer.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/tab-switch-optimizer.js?v=20261001_v1" type="module"></script>
11  *
12  * File: tab-switch-optimizer.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/js
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 /**
22  * Tab Switch Optimizer - Оптимизация переключения между вкладками
23  *
24  * Реализует:
25  * - DOM кеширование (сохранение содержимого вкладки вместо удаления)
26  * - Мгновенная визуальная обратная связь (показ активной вкладки до загрузки)
27  * - Предварительная загрузка соседних вкладок
28  * - Трекинг времени переключения
```

```

29 | *
30 | * Usage:
31 | *   const optimizer = new TabSwitchOptimizer();
32 | *   optimizer.enableDOMCaching();
33 | *   optimizer.optimizeSwitching();
34 | */
35 |
36 | class TabSwitchOptimizer {
37 |   constructor() {
38 |     this.domCache = new Map(); // Кеш DOM содержимого вкладок
39 |     this.switchStats = {
40 |       totalSwitches: 0,
41 |       avgSwitchTime: 0,
42 |       lastSwitchTime: null,
43 |       switchTimes: []
44 |     };
45 |     this.isOptimized = false;
46 |   }
47 |
48 |   /**
49 |    * Включить DOM кеширование
50 |    * Вместо удаления содержимого вкладки, сохраняем его в памя
    ти
51 |    */
52 |   enableDOMCaching() {
53 |     console.log('[TabSwitchOptimizer] Enabling DOM caching...');
54 |
55 |     // Перехватываем стандартное управление табами
56 |     const observer = new MutationObserver((mutations) => {
57 |       mutations.forEach((mutation) => {
58 |         if (mutation.type === 'childList' || mutation.type ===
    'attributes') {
59 |           // Вызываем автосохранение изменений
60 |           this._cacheVisibleTabs();
61 |         }
62 |       });
63 |     });

```

```
64 |
65 |     // Наблюдаем за контейнером с табами
66 |     const tabContainer = document.getElementById('mainTabsCont
ent') ||
67 |         document.querySelector('[role="tablist"]')?.parentElement;
68 |
69 |     if (tabContainer) {
70 |         observer.observe(tabContainer, {
71 |             childList: true,
72 |             subtree: true,
73 |             attributes: true,
74 |             attributeFilter: ['class', 'style', 'aria-hidden']
75 |         });
76 |     }
77 |
78 |     this.isOptimized = true;
79 |     console.log('[TabSwitchOptimizer] DOM caching enabled');
80 | }
81 |
82 | /**
83 |  * Кешировать видимые вкладки
84 |  */
85 | _cacheVisibleTabs() {
86 |     const tabs = document.querySelectorAll('[role="tabpanel"]:
not([aria-hidden="true"])');
87 |     tabs.forEach(tab => {
88 |         if (tab.id) {
89 |             // Сохраняем клон вкладки
90 |             this.domCache.set(tab.id, tab.cloneNode(true));
91 |         }
92 |     });
93 | }
94 |
95 | /**
96 |  * Получить кешированное содержимое вкладки
97 |  */
98 | getCachedTabContent(tabId) {
```

```
99     return this.domCache.get(tabId);
100 }
101
102 /**
103  * Очистить кеш DOM
104  */
105 clearDOMCache(tabId = null) {
106     if (tabId) {
107         this.domCache.delete(tabId);
108     } else {
109         this.domCache.clear();
110     }
111 }
112
113 /**
114  * Оптимизировать функцию переключения вкладок
115  * Добавляет мгновенную визуальную обратную связь и трекинг
116  * времени
117  */
118 optimizeSwitching(originalSwitchFn) {
119     return async (targetTabId) => {
120         const startTime = performance.now();
121
122         if (!targetTabId) return;
123
124         // Мгновенно показываем активную вкладку визуально (без
125         // загрузки)
126         this._showTabImmediately(targetTabId);
127
128         // Выполняем оригинальное переключение
129         if (typeof originalSwitchFn === 'function') {
130             await originalSwitchFn(targetTabId);
131         }
132
133         // Записываем время переключения
134         const switchTime = performance.now() - startTime;
135         this._recordSwitchTime(switchTime, targetTabId);
136     };
137 }
```

```

135 |         console.log(`[TabSwitchOptimizer] Tab switched in ${switchTime.toFixed(2)}ms`);
136 |     };
137 | }
138 |
139 | /**
140 |  * Мгновенно показать вкладку визуально (до загрузки содержимого)
141 |  */
142 | _showTabImmediately(targetTabId) {
143 |     const cleanTabId = targetTabId.startsWith('#') ? targetTabId.slice(1) : targetTabId;
144 |
145 |     // Скрываем все остальные вкладки
146 |     document.querySelectorAll('[role="tabpanel"]').forEach(tab => {
147 |         tab.style.display = 'none';
148 |         tab.classList.remove('active', 'show');
149 |         tab.setAttribute('aria-hidden', 'true');
150 |     });
151 |
152 |     // Показываем целевую вкладку
153 |     const targetTab = document.getElementById(cleanTabId);
154 |     if (targetTab) {
155 |         targetTab.style.display = 'block';
156 |         targetTab.classList.add('active', 'show');
157 |         targetTab.setAttribute('aria-hidden', 'false');
158 |     }
159 |
160 |     // Обновляем визуальное состояние кнопок вкладок
161 |     document.querySelectorAll('[role="tab"]').forEach(tab => {
162 |         tab.classList.remove('active');
163 |         tab.setAttribute('aria-selected', 'false');
164 |     });
165 |
166 |     const targetButton = document.querySelector(`[aria-controls="${cleanTabId}"]`) ||
167 |         document.querySelector(`[data-bs-target`

```

```

t="#${cleanTabId}"]`);
168     if (targetButton) {
169         targetButton.classList.add('active');
170         targetButton.setAttribute('aria-selected', 'true');
171     }
172 }
173
174 /**
175  * Записать время переключения и вычислить среднее
176  */
177 _recordSwitchTime(time, tabId) {
178     this.switchStats.totalSwitches++;
179     this.switchStats.lastSwitchTime = time;
180     this.switchStats.switchTimes.push({ time, tabId, timestamp
: Date.now() });
181
182     // Ограничиваем историю последних 100 переключениями
183     if (this.switchStats.switchTimes.length > 100) {
184         this.switchStats.switchTimes = this.switchStats.switchTi
mes.slice(-100);
185     }
186
187     // Вычисляем среднее время
188     const sum = this.switchStats.switchTimes.reduce((acc, s) =
> acc + s.time, 0);
189     this.switchStats.avgSwitchTime = sum / this.switchStats.sw
itchTimes.length;
190 }
191
192 /**
193  * Предварительная загрузка соседних вкладок
194  */
195 preloadAdjacentTabs(currentTabId) {
196     const allTabs = Array.from(document.querySelectorAll('[rol
e="tab"]'));
197     const currentIndex = allTabs.findIndex(tab =>
198         tab.getAttribute('aria-controls') === currentTabId ||
199         tab.getAttribute('data-bs-target') === `#${currentTabId}

```

```

200     );
201
202     if (currentIndex >= 0) {
203         // Предзагружаем соседние вкладки
204         const adjacentIndices = [currentIndex - 1, currentIndex
+ 1];
205         adjacentIndices.forEach(idx => {
206             if (idx >= 0 && idx < allTabs.length) {
207                 const tab = allTabs[idx];
208                 const tabId = tab.getAttribute('aria-controls') ||
209                     tab.getAttribute('data-bs-target')?.replace('#', '');
210
211                 if (tabId && window.optimizationModule?.tabLoader) {
212                     console.log(`[TabSwitchOptimizer] Preloading adjacent tab: ${tabId}`);
213                     window.optimizationModule.tabLoader.preloadTab(tab
Id);
214                 }
215             }
216         });
217     }
218 }
219
220 /**
221  * Получить статистику переключения
222  */
223 getStats() {
224     return {
225         ...this.switchStats,
226         avgSwitchTimeFormatted: `${this.switchStats.avgSwitchTime.toFixed(2)}ms`,
227         lastSwitchTimeFormatted: this.switchStats.lastSwitchTime
?
228         `${this.switchStats.lastSwitchTime.toFixed(2)}ms` : 'N/A',
229         cachedTabs: this.domCache.size,

```

```
230         isOptimized: this.isOptimized
231     };
232 }
233
234 /**
235  * Вывести статистику в консоль
236  */
237 printStats() {
238     const stats = this.getStats();
239     console.group('▯ Tab Switch Performance Stats');
240     console.log('Total Switches:', stats.totalSwitches);
241     console.log('Average Switch Time:', stats.avgSwitchTimeFormatted);
242     console.log('Last Switch Time:', stats.lastSwitchTimeFormatted);
243     console.log('Cached Tabs:', stats.cachedTabs);
244     console.log('Performance Target: <100ms (currently',
245         (stats.avgSwitchTime < 100 ? '▯ GOOD' : '▯ SLOW'), ')');
246     console.groupEnd();
247 }
248
249 /**
250  * Очистить статистику
251  */
252 resetStats() {
253     this.switchStats = {
254         totalSwitches: 0,
255         avgSwitchTime: 0,
256         lastSwitchTime: null,
257         switchTimes: []
258     };
259 }
260 }
261
262 // Глобальный экземпляр
263 window.tabSwitchOptimizer = new TabSwitchOptimizer();
264
265 // Экспортируем
```



```
266 | window.TabSwitchOptimizer = TabSwitchOptimizer;
267 |
268 | export default TabSwitchOptimizer;
269 |
```

test-api-cache.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Test-API-Cache Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля test-api-
7   *   cache.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/test-api-cache.js?v
11  *     =20261001_v1" type="module"></script>
12  *
13  * File: test-api-cache.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/js
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * test-api-cache.js - Простой тестовый скрипт для проверки AP
23  * I Cache System
24  *
25  * ИСПОЛЬЗОВАНИЕ:
26  * 1. Откройте веб-интерфейс
27  * 2. Откройте консоль браузера (F12)
28  * 3. Выполните: testApiCache()
29  */
30 export async function testApiCache() {
```

```

31 | console.log('===  API Cache System Test Suite ===\n');
32 |
33 | const results = {
34 |     passed: 0,
35 |     failed: 0,
36 |     tests: []
37 | };
38 |
39 | function logTest(name, passed, message) {
40 |     const icon = passed ? '✅ ' : '❌ ';
41 |     console.log(`${icon} ${name}: ${message}`);
42 |     results.tests.push({ name, passed, message });
43 |     if (passed) results.passed++;
44 |     else results.failed++;
45 | }
46 |
47 | // Test 1: Проверка доступности browserCache
48 | try {
49 |     const cacheAvailable = window.browserCache && typeof window.browserCache.ready === 'function';
50 |     logTest('Cache Availability', cacheAvailable, cacheAvailable ? 'BrowserCache is available' : 'BrowserCache NOT available');
51 | } catch (e) {
52 |     logTest('Cache Availability', false, `Error: ${e.message}`);
53 | }
54 |
55 | // Test 2: Проверка доступности cachedApiFetch
56 | try {
57 |     const apiFetchAvailable = typeof window.cachedApiFetch === 'function';
58 |     logTest('API Fetch Function', apiFetchAvailable, apiFetchAvailable ? 'cachedApiFetch() is available' : 'cachedApiFetch() NOT available');
59 | } catch (e) {
60 |     logTest('API Fetch Function', false, `Error: ${e.message}`);

```

```
61 | }
62 |
63 | // Test 3: Проверка готовности IndexedDB
64 | try {
65 |     if (window.browserCache) {
66 |         const isReady = await window.browserCache.ready();
67 |         logTest('IndexedDB Ready', isReady, isReady ? 'IndexedDB
    is ready' : 'IndexedDB initialization failed');
68 |     } else {
69 |         logTest('IndexedDB Ready', false, 'BrowserCache not avai
    lable');
70 |     }
71 | } catch (e) {
72 |     logTest('IndexedDB Ready', false, `Error: ${e.message}`);
73 | }
74 |
75 | // Test 4: Тест Cache-First стратегии
76 | try {
77 |     console.log('\n--- Testing Cache-First Strategy ---');
78 |     const testUrl = '/api/v1/models';
79 |
80 |     // Первый запрос (должен быть MISS)
81 |     console.log('Request 1: Should be MISS (network)');
82 |     const start1 = performance.now();
83 |     await window.cachedApiFetch(testUrl);
84 |     const time1 = (performance.now() - start1).toFixed(2);
85 |
86 |     // Второй запрос (должен быть HIT)
87 |     console.log('Request 2: Should be HIT (cache)');
88 |     const start2 = performance.now();
89 |     await window.cachedApiFetch(testUrl);
90 |     const time2 = (performance.now() - start2).toFixed(2);
91 |
92 |     const speedup = (time1 / time2).toFixed(1);
93 |     logTest('Cache-First Speed', time2 < time1, `Network: ${ti
    me1}ms, Cache: ${time2}ms (${speedup}x faster)`);
94 | } catch (e) {
95 |     logTest('Cache-First Speed', false, `Error: ${e.message}`)
```

```

    ;
96   }
97
98   // Test 5: Тест инвалидации по тегу
99   try {
100      console.log('\n--- Testing Tag Invalidation ---');
101
102      // Добавляем тестовую запись
103      if (window.browserCache) {
104         await window.browserCache.set('api_cache', 'test-key-1',
105         { data: 'test1' }, { tags: ['test-tag'] });
106         await window.browserCache.set('api_cache', 'test-key-2',
107         { data: 'test2' }, { tags: ['test-tag'] });
108
109         // Инвалидируем
110         const count = await window.invalidateCacheByTag('test-tag');
111         logTest('Tag Invalidation', count === 2, `Invalidated ${count} entries (expected 2)`);
112       } else {
113         logTest('Tag Invalidation', false, 'BrowserCache not available');
114       }
115     } catch (e) {
116       logTest('Tag Invalidation', false, `Error: ${e.message}`);
117     }
118
119   // Test 6: Получение статистики
120   try {
121      console.log('\n--- Testing Statistics ---');
122      const stats = await window.getApiCacheStats();
123      const hasMetrics = stats.available && stats.metrics && typeof stats.metrics.hits === 'number';
124      logTest('Cache Statistics', hasMetrics, `Hit rate: ${stats.apiCache?.hitRate || 0}%, Entries: ${stats.apiCache?.count || 0}`);
125    } catch (e) {
126      logTest('Cache Statistics', false, `Error: ${e.message}`);
127    }

```

```
125 | }
126 |
127 | // Test 7: Тест forceNetwork
128 | try {
129 |     console.log('\n--- Testing Force Network ---');
130 |     const testUrl = '/api/v1/system/summary';
131 |
132 |     // Обычный запрос
133 |     await window.cachedApiFetch(testUrl);
134 |
135 |     // Принудительный запрос из сети
136 |     const start = performance.now();
137 |     await window.cachedApiFetch(testUrl, {}, { forceNetwork: t
138 |     rue });
139 |     const time = (performance.now() - start).toFixed(2);
140 |
141 |     logTest('Force Network', time > 10, `Force network request
142 |     took ${time}ms (should be > 10ms)`);
143 | } catch (e) {
144 |     logTest('Force Network', false, `Error: ${e.message}`);
145 | }
146 |
147 | // Итоги
148 | console.log('\n===  Test Results  ===');
149 | console.log(`Total: ${results.tests.length} tests`);
150 | console.log(`  Passed: ${results.passed}`);
151 | console.log(`  Failed: ${results.failed}`);
152 | console.log(`Success rate: ${((results.passed / results.test
153 |     s.length) * 100).toFixed(1)}%`);
154 |
155 | if (results.failed === 0) {
156 |     console.log('\n  All tests passed! API Cache System is wo
157 |     rking correctly.');
```

```
158 |   return results;
159 | }
160 |
161 | // Экспорт в глобальную область
162 | if (typeof window !== 'undefined') {
163 |   window.testApiCache = testApiCache;
164 | }
165 |
166 | // Автоматический запуск при включенном DEBUG режиме
167 | if (typeof window !== 'undefined' && localStorage.getItem('DEB
  UG_API_CACHE') === 'true') {
168 |   console.log('▯ DEBUG_API_CACHE enabled, running tests on lo
    ad...');
169 |   setTimeout(() => testApiCache(), 2000);
170 | }
171 |
172 | export default testApiCache;
173 |
```

theme-engine.js

```
1 /**
2  * =====
3  * Process Name: Windows Web Styles - Theme Engine
4  * =====
5  * Description:
6  *   Шаблонизатор и менеджер тем оформления для веб-интерфейса
   AI Breadboard.
7  *   Поддерживает светлые темы (Light, Brick), тёмные темы (Da
   rk Contrast, Terminal),
8  *   системный режим, динамическую регистрацию и шаблонизацию
   UI-селекторов.
9  *
10 * Usage Examples:
11 *   import { themeEngine } from './theme-engine.js';
12 *   themeEngine.applyTheme('brick');
13 *   themeEngine.registerTheme({ id: 'custom', name: 'Custom T
   heme', category: 'dark' });
14 *
15 * File: theme-engine.js
16 * Project: ai-breadboard
17 * Package: apps.windows.api.webgui.js
18 * Author: hypo69
19 * Copyright: © 2026 hypo69
20 * Updated: 2026-10-06 21:10:15
21 * =====
22 */
23
24 /**
25 * Ключ хранения выбранного режима темы в LocalStorage
26 */
27 export const THEME_STORAGE_KEY = 'theme';
28
29 /**
```



```
30 | * Режим по умолчанию
31 | */
32 | export const DEFAULT_THEME_MODE = 'system';
33 |
34 | /**
35 |  * Определение стандартных встроенных тем
36 |  */
37 | export const BUILTIN_THEMES = [
38 |   {
39 |     id: 'system',
40 |     name: 'Системная',
41 |     nameEn: 'System',
42 |     category: 'system',
43 |     icon: 'bi-circle-half',
44 |     iconColor: 'text-info',
45 |     description: 'Автоматический выбор темы на основе настроек
    | операционной системы',
46 |     baseType: 'system',
47 |   },
48 |   {
49 |     id: 'light',
50 |     name: 'Светлая (NightView)',
51 |     nameEn: 'Light (NightView)',
52 |     category: 'light',
53 |     icon: 'bi-sun-fill',
54 |     iconColor: 'text-warning',
55 |     description: 'Мягкая светлая тема в стиле NightView без сл
    | епящей белизны для комфортного чтения ночью',
56 |     baseType: 'light',
57 |     tokens: { '--about-card-bg': 'linear-gradient(135deg, #f8f
    | afc 0%, #e2e8f0 100%)' },
58 |   },
59 |   {
60 |     id: 'brick',
61 |     name: 'Кирпичная (Терракота)',
62 |     nameEn: 'Brick (Terracotta)',
63 |     category: 'light',
64 |     icon: 'bi-bricks',
```

```

65     iconColor: 'text-danger',
66     description: 'Теплая светлая тема в стиле клинкерного кирпич
ища и обожженной глины',
67     baseType: 'light',
68     tokens: { '--about-card-bg': 'linear-gradient(135deg, #f5e
8d4 0%, #e2c8b8 100%)' },
69 },
70 {
71     id: 'dark',
72     name: '☾ Midnight Console',
73     nameEn: 'Dark (High-Contrast)',
74     category: 'dark',
75     icon: 'bi-moon-stars-fill',
76     iconColor: 'text-primary',
77     description: 'Глубокая контрастная тёмная тема для удобной
работы в ночное время',
78     baseType: 'dark',
79 },
80 {
81     id: 'terminal',
82     name: 'Терминал (Matrix / CRT)',
83     nameEn: 'Terminal (Matrix / CRT)',
84     category: 'dark',
85     icon: 'bi-terminal-fill',
86     iconColor: 'text-success',
87     description: 'Тёмная тема в ретро-стиле зеленого фосфорног
о терминала с моноширинными акцентами',
88     baseType: 'dark',
89 },
90 ];
91
92 /**
93  * Класс шаблонизатора и оркестратора тем оформления
94  */
95 export class ThemeEngine {
96     constructor() {
97         /** @type {Map<string, Object>} */
98         this.registry = new Map();

```

```

99     this.mediaQueryListenerAttached = false;
100     this._dynamicStyleEl = null;
101
102     // Регистрация встроенных тем
103     for (const theme of BUILTIN_THEMES) {
104         this.registry.set(theme.id, { ...theme });
105     }
106 }
107
108 /**
109  * Регистрирует новую тему в реестре шаблонизатора.
110  * @param {Object} themeDef - Определение темы
111  * @param {string} themeDef.id - Уникальный идентификатор те
    мы
112  * @param {string} themeDef.name - Отображаемое имя темы (ру
    сский)
113  * @param {string} [themeDef.nameEn] - Отображаемое имя темы
    (английский)
114  * @param {'light'|'dark'|'system'} themeDef.category - Кате
    гория темы
115  * @param {string} [themeDef.icon] - Иконка Bootstrap Icons
    (например, 'bi-palette')
116  * @param {string} [themeDef.iconColor] - CSS класс цвета ик
    онки (например, 'text-warning')
117  * @param {string} [themeDef.description] - Описание темы
118  * @param {'light'|'dark'} [themeDef.baseType] - Базовый тип
    темы (светлая/тёмная)
119  * @param {Object<string, string>} [themeDef.tokens] - Слова
    рь CSS-переменных
120  */
121 registerTheme(themeDef) {
122     if (!themeDef || !themeDef.id) {
123         console.warn('[ThemeEngine] Некорректное определение тем
    ы:', themeDef);
124         return;
125     }
126
127     const baseType = themeDef.baseType || (themeDef.category =

```

```

    == 'dark' ? 'dark' : 'light');
128     this.registry.set(themeDef.id, {
129         id: themeDef.id,
130         name: themeDef.name || themeDef.id,
131         nameEn: themeDef.nameEn || themeDef.name || themeDef.id,
132         category: themeDef.category || 'dark',
133         icon: themeDef.icon || 'bi-palette',
134         iconColor: themeDef.iconColor || 'text-info',
135         description: themeDef.description || '',
136         baseType,
137         tokens: themeDef.tokens || null,
138     });
139
140     if (themeDef.tokens) {
141         this._applyDynamicTokens();
142     }
143 }
144
145 /**
146  * Возвращает определение темы по идентификатору.
147  * @param {string} themeId
148  * @returns {Object|null}
149  */
150 getTheme(themeId) {
151     return this.registry.get(themeId) || null;
152 }
153
154 /**
155  * Возвращает список всех зарегистрированных тем.
156  * @returns {Array<Object>}
157  */
158 getAllThemes() {
159     return Array.from(this.registry.values());
160 }
161
162 /**
163  * Возвращает темы, отфильтрованные по категории ('light' |
    'dark' | 'system').

```

```

164 |     * @param {'light'|'dark'|'system'} category
165 |     * @returns {Array<Object>}
166 |     */
167 |     getThemesByCategory(category) {
168 |         return this.getAllThemes().filter((t) => t.category === ca
169 | tegory);
170 |     }
171 |     /**
172 |     * Получает сохранённый режим темы из LocalStorage.
173 |     * @returns {string}
174 |     */
175 |     getSavedMode() {
176 |         const saved = localStorage.getItem(THEME_STORAGE_KEY);
177 |         if (saved && this.registry.has(saved)) {
178 |             return saved;
179 |         }
180 |         return DEFAULT_THEME_MODE;
181 |     }
182 |
183 |     /**
184 |     * Разрешает эффективный идентификатор темы ('light', 'dark'
185 |     , 'brick', 'terminal' и т.д.)
186 |     * @param {string} [mode]
187 |     * @returns {string}
188 |     */
189 |     resolveTheme(mode = this.getSavedMode()) {
190 |         if (mode === 'system') {
191 |             const prefersDark =
192 |                 typeof window !== 'undefined' &&
193 |                 window.matchMedia &&
194 |                 window.matchMedia('(prefers-color-scheme: dark)').matc
195 | hes;
196 |             return prefersDark ? 'dark' : 'light';
197 |         }
198 |         if (this.registry.has(mode)) {
199 |             return mode;

```

```
199 |     }
200 |     return 'dark';
201 | }
202 |
203 | /**
204 |  * Определяет базовый режим Bootstrap ('light' | 'dark') для
    | текущей темы.
205 |  * @param {string} resolvedThemeId
206 |  * @returns {'light'|'dark'}
207 |  */
208 | getBsTheme(resolvedThemeId) {
209 |     const theme = this.getTheme(resolvedThemeId);
210 |     if (theme) {
211 |         return theme.baseType || (theme.category === 'dark' ? 'd
    | ark' : 'light');
212 |     }
213 |     return resolvedThemeId === 'light' || resolvedThemeId ===
    | 'brick' ? 'light' : 'dark';
214 | }
215 |
216 | /**
217 |  * Применяет тему к DOM, обновляет атрибуты и UI-элементы.
218 |  * @param {string} mode
219 |  */
220 | applyTheme(mode) {
221 |     if (!mode || !this.registry.has(mode)) {
222 |         mode = DEFAULT_THEME_MODE;
223 |     }
224 |
225 |     localStorage.setItem(THEME_STORAGE_KEY, mode);
226 |
227 |     const resolved = this.resolveTheme(mode);
228 |     const bsTheme = this.getBsTheme(resolved);
229 |
230 |     if (typeof document !== 'undefined') {
231 |         // 1. Установка атрибутов на <html>
232 |         document.documentElement.setAttribute('data-theme', reso
    | lved);
```

```
233 |     document.documentElement.setAttribute('data-bs-theme', b
| sTheme);
234 |     document.documentElement.setAttribute('data-theme-mode',
| mode);
235 |
236 |     // 2. Обновление иконок и кнопок
237 |     this._updateThemeUI(mode, resolved);
238 |
239 |     // 3. Синхронизация селекторов выбора тем
240 |     this.syncSelectInputs(mode);
241 | }
242 |
243 | // 4. Оповещение слушателей через CustomEvent
244 | if (typeof window !== 'undefined') {
245 |     window.dispatchEvent(
246 |         new CustomEvent('themechanged', {
247 |             detail: {
248 |                 mode,
249 |                 resolved,
250 |                 bsTheme,
251 |                 themeMeta: this.getTheme(resolved),
252 |             },
253 |         })
254 |     );
255 | }
256 | }
257 |
258 | /**
259 |  * Генерирует HTML-разметку опций для тега <select> с группп
| ровкой по категориям.
260 |  * @param {string} [currentMode]
261 |  * @returns {string}
262 |  */
263 | renderOptionsHTML(currentMode = this.getSavedMode()) {
264 |     const lightThemes = this.getThemesByCategory('light');
265 |     const darkThemes = this.getThemesByCategory('dark');
266 |     const systemThemes = this.getThemesByCategory('system');
267 |
```

```
268 |     let html = '';
269 |
270 |     if (systemThemes.length > 0) {
271 |         for (const t of systemThemes) {
272 |             const isSelected = t.id === currentMode ? ' selected'
273 |             : '';
274 |             html += `<option value="${t.id}"${isSelected}>⬜  ${t.name}</option>\n`;
275 |         }
276 |     }
277 |
278 |     if (lightThemes.length > 0) {
279 |         html += '<optgroup label="⬜  Светлые темы">\n';
280 |         for (const t of lightThemes) {
281 |             const isSelected = t.id === currentMode ? ' selected'
282 |             : '';
283 |             const iconSymbol = t.id === 'brick' ? '⬜ ' : '⬜ ';
284 |             html += `  <option value="${t.id}"${isSelected}>${iconSymbol} ${t.name}</option>\n`;
285 |         }
286 |         html += '</optgroup>\n';
287 |     }
288 |
289 |     if (darkThemes.length > 0) {
290 |         html += '<optgroup label="⬛  Тёмные темы">\n';
291 |         for (const t of darkThemes) {
292 |             const isSelected = t.id === currentMode ? ' selected'
293 |             : '';
294 |             const iconSymbol = t.id === 'terminal' ? '⬛ ' : '⬛ ';
295 |             html += `  <option value="${t.id}"${isSelected}>${iconSymbol} ${t.name}</option>\n`;
296 |         }
297 |         html += '</optgroup>\n';
298 |     }
299 |
300 |     return html;
301 | }
```



```
300 | /**
301 |  * Синхронизирует все элементы <select> выбора темы на стран
ице.
302 |  * @param {string} [mode]
303 |  */
304 | syncSelectInputs(mode = this.getSavedMode()) {
305 |     if (typeof document === 'undefined') return;
306 |
307 |     const selectors = document.querySelectorAll(
308 |         '.theme-selector, #apps-theme-selector, #settings-theme,
#modal-theme-selector'
309 |     );
310 |
311 |     selectors.forEach((selectEl) => {
312 |         // Если у селектора еще нет сгенерированных опций с optg
roup, наполняем его
313 |         if (!selectEl.querySelector('optgroup')) {
314 |             selectEl.innerHTML = this.renderOptionsHTML(mode);
315 |         } else if (selectEl.value !== mode) {
316 |             selectEl.value = mode;
317 |         }
318 |     });
319 | }
320 |
321 | /**
322 |  * Инициализирует менеджер тем, регистрирует обработчики соб
ытий и применяет начальную тему.
323 |  */
324 | init() {
325 |     const currentMode = this.getSavedMode();
326 |     this.applyTheme(currentMode);
327 |
328 |     // Подписка на изменение системной темы
329 |     if (!this.mediaQueryListenerAttached && typeof window !==
'undefined' && window.matchMedia) {
330 |         try {
331 |             const mediaQuery = window.matchMedia('(prefers-color-s
cheme: dark)');
```

```
332 |     const handler = () => {
333 |         if (this.getSavedMode() === 'system') {
334 |             this.applyTheme('system');
335 |         }
336 |     };
337 |     if (mediaQuery.addEventListener) {
338 |         mediaQuery.addEventListener('change', handler);
339 |     } else if (mediaQuery.addListener) {
340 |         mediaQuery.addListener(handler);
341 |     }
342 |     this.mediaQueryListenerAttached = true;
343 | } catch (e) {
344 |     console.warn('[ThemeEngine] Не удалось подписаться на
345 | prefers-color-scheme:', e);
346 | }
347 |
348 | if (typeof document === 'undefined') return;
349 |
350 | // Наполнение всех селекторов опциями
351 | this.syncSelectInputs(currentMode);
352 |
353 | // Привязка слушателей изменений к селекторам тем
354 | document.querySelectorAll(
355 |     '.theme-selector, #apps-theme-selector, #settings-theme,
356 |     #modal-theme-selector'
357 | ).forEach((sel) => {
358 |     sel.addEventListener('change', (e) => {
359 |         this.applyTheme(e.target.value);
360 |     });
361 | });
362 |
363 | // Привязка кликов по кнопкам data-theme-value
364 | document.querySelectorAll('[data-theme-value]').forEach((b
365 | tn) => {
366 |     btn.addEventListener('click', (e) => {
367 |         e.preventDefault();
368 |         const selected = btn.getAttribute('data-theme-value');
```

```
367         if (selected) {
368             this.applyTheme(selected);
369         }
370     });
371 });
372
373     // Привязка кнопки быстрого циклического переключения тем
374     (legacy / toggle button)
375     const toggleBtn = document.getElementById('theme-toggle');
376     if (toggleBtn && !toggleBtn.hasAttribute('data-bs-toggle')) {
377         toggleBtn.addEventListener('click', () => {
378             const mode = this.getSavedMode();
379             // Цикл переключения: system -> light -> brick -> dark
380             -> terminal -> system
381             const cycle = ['system', 'light', 'brick', 'dark', 'terminal'];
382             const currentIndex = cycle.indexOf(mode);
383             const nextMode = cycle[(currentIndex + 1) % cycle.length];
384             this.applyTheme(nextMode);
385         });
386     }
387
388     /**
389     * Обновляет UI-иконки, подсказки и активные состояния в выпадающих меню.
390     * @private
391     */
392     _updateThemeUI(mode, resolved) {
393         const activeTheme = this.getTheme(resolved) || this.getTheme('dark');
394         const modeTheme = this.getTheme(mode);
395
396         const iconEl = document.getElementById('theme-icon');
397         const toggleBtn = document.getElementById('theme-toggle');
398         const dropdownBtn = document.getElementById('theme-dropdown');
```

```

n');
398
399     const iconClass = modeTheme ? modeTheme.icon : 'bi-circle-
half';
400     const iconColor = modeTheme ? modeTheme.iconColor : 'text-
info';
401     const titleText =
402         mode === 'system'
403         ? `Системная тема (${activeTheme ? activeTheme.name :
resolved})`
404         : modeTheme
405         ? modeTheme.name
406         : resolved;
407
408     if (iconEl) {
409         iconEl.className = `bi ${iconClass} ${iconColor}`;
410     }
411
412     if (dropdownBtn) {
413         dropdownBtn.setAttribute('title', titleText);
414     }
415
416     if (toggleBtn) {
417         toggleBtn.innerHTML = `<i class="bi ${iconClass} ${iconC
olor}"></i>`;
418         toggleBtn.setAttribute('title', titleText);
419     }
420
421     // Обновление галочек в выпадающих меню
422     document.querySelectorAll('.theme-check-icon').forEach((ic
on) => {
423         const target = icon.getAttribute('data-theme-check');
424         if (target === mode) {
425             icon.classList.remove('d-none');
426         } else {
427             icon.classList.add('d-none');
428         }
429     });

```

```

430
431 // Обновление active класса
432 document.querySelectorAll('[data-theme-value]').forEach((i
tem) => {
433     const val = item.getAttribute('data-theme-value');
434     if (val === mode) {
435         item.classList.add('active');
436     } else {
437         item.classList.remove('active');
438     }
439 });
440 }
441
442 /**
443  * Инжектирует динамические стили для зарегистрированных кас
томных тем с токенами.
444  * @private
445  */
446 _applyDynamicTokens() {
447     if (typeof document === 'undefined') return;
448
449     let css = '';
450     for (const [id, def] of this.registry.entries()) {
451         if (def.tokens && typeof def.tokens === 'object') {
452             css += `[data-theme="${id}"] {\n`;
453             for (const [prop, val] of Object.entries(def.tokens))
454             {
455                 css += `    ${prop}: ${val};\n`;
456             }
457             css += `}\n`;
458         }
459
460         if (!this._dynamicStyleEl) {
461             this._dynamicStyleEl = document.createElement('style');
462             this._dynamicStyleEl.id = 'theme-engine-dynamic-styles';
463             document.head.appendChild(this._dynamicStyleEl);
464         }

```

```
465 |     this._dynamicStyleEl.textContent = css;
466 | }
467 | }
468 |
469 | /**
470 |  * Синглтон экземпляр движка тем
471 |  */
472 | export const themeEngine = new ThemeEngine();
473 |
474 | // Глобальный экспорт на window
475 | if (typeof window !== 'undefined') {
476 |     window.ThemeEngine = ThemeEngine;
477 |     window.themeEngine = themeEngine;
478 | }
479 |
```

theme.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Theme Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля theme и и
7  *   нтеграции с ThemeEngine.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/theme.js?v=20261004
11  *     _v17" type="module"></script>
12  *   JavaScript Import:
13  *     import { getThemeMode, getResolvedTheme, setTheme, them
14  *     eEngine } from '/windows/api/webgui/js/theme.js';
15  * File: theme.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/js
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-04 12:12:00
21  * =====
22  */
23
24 import { themeEngine, ThemeEngine, THEME_STORAGE_KEY, DEFAULT_
25   THEME_MODE, BUILTIN_THEMES } from './theme-engine.js';
26 /**
27  * Получает сохранённый режим темы пользователя.
28  * @returns {string}
29  */
```

```
30 export function getThemeMode() {
31   return themeEngine.getSavedMode();
32 }
33
34 /**
35  * Разрешает эффективную тему оформления ('light' | 'dark' | '
brick' | 'terminal').
36  * @param {string} [mode]
37  * @returns {string}
38  */
39 export function getResolvedTheme(mode = getThemeMode()) {
40   return themeEngine.resolveTheme(mode);
41 }
42
43 /**
44  * Устанавливает тему оформления и применяет её ко всем элемен
там интерфейса.
45  * @param {string} mode
46  */
47 export function setTheme(mode) {
48   themeEngine.applyTheme(mode);
49 }
50
51 /**
52  * Инициализирует менеджер тем и слушатели событий при загрузк
е страницы.
53  */
54 export function initTheme() {
55   themeEngine.init();
56 }
57
58 // Экспорт синглтона и констант
59 export { themeEngine, ThemeEngine, THEME_STORAGE_KEY, DEFAULT_
THEME_MODE, BUILTIN_THEMES };
60
61 // Экспорт в глобальный объект window
62 if (typeof window !== 'undefined') {
63   window.setTheme = setTheme;
```



```
64 | window.getThemeMode = getThemeMode;
65 | window.getResolvedTheme = getResolvedTheme;
66 | window.initTheme = initTheme;
67 | window.themeEngine = themeEngine;
68 | window.ThemeEngine = ThemeEngine;
69 | }
70 |
```

toast.js

```
1  /**
2   * =====
3   * Process Name: Windows Js - Toast Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля toast.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/toast.js?v=20261001
11  *     _v1" type="module"></script>
12  *
13  * File: toast.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/js
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * toast.js – Универсальная система всплывающих окон (Toast &
23  * Floating Notifications)
24  * для веб-интерфейса AI-Breadboard.
25  *
26  * Предоставляет функции:
27  * - window.showToast(message, type, options)
28  * - window.showNotification(title, message, type, duration)
29  * - Перехват стандартного window.alert() с выводом всплывающи
30  * х окон.
31  */
```

```

31 | (function () {
32 |     'use strict';
33 |
34 |     // Конфигурация по умолчанию
35 |     const DEFAULT_DURATION = 4500;
36 |     const CONTAINER_ID = 'floating-toast-container';
37 |
38 |     /**
39 |      * Инициализирует контейнер для всплывающих окон.
40 |      * @returns {HTMLElement} Элемент контейнера.
41 |      */
42 |     function getOrCreateContainer() {
43 |         let container = document.getElementById(CONTAINER_ID);
44 |         if (!container) {
45 |             container = document.createElement('div');
46 |             container.id = CONTAINER_ID;
47 |             container.className = 'floating-toast-container';
48 |             container.setAttribute('aria-live', 'polite');
49 |             container.setAttribute('aria-atomic', 'true');
50 |             document.body.appendChild(container);
51 |         }
52 |         return container;
53 |     }
54 |
55 |     /**
56 |      * Определяет иконку и заголовок по типу уведомления.
57 |      * @param {string} type - Тип ('success', 'danger', 'error',
58 |      * 'warning', 'info').
59 |      * @returns {{iconClass: string, headerText: string, bootstrapType: string}}
60 |      */
61 |     function getMetaForType(type) {
62 |         const t = (type || 'info').toLowerCase();
63 |         switch (t) {
64 |             case 'success':
65 |             case 'ok':
66 |                 return {
67 |                     iconClass: 'bi bi-check-circle-fill text-success',

```

```

67         headerText: 'Успешно',
68         bootstrapType: 'success'
69     };
70     case 'danger':
71     case 'error':
72     case 'err':
73         return {
74             iconClass: 'bi bi-exclamation-octagon-fill text-dang
er',
75             headerText: 'Ошибка',
76             bootstrapType: 'danger'
77         };
78     case 'warning':
79     case 'warn':
80         return {
81             iconClass: 'bi bi-exclamation-triangle-fill text-war
ning',
82             headerText: 'Внимание',
83             bootstrapType: 'warning'
84         };
85     case 'info':
86     default:
87         return {
88             iconClass: 'bi bi-info-circle-fill text-info',
89             headerText: 'Уведомление',
90             bootstrapType: 'info'
91         };
92     }
93 }
94
95 /**
96  * Автоматически определяет тип уведомления по содержимому с
ообщения.
97  * @param {string} text - Текст сообщения.
98  * @returns {string} Определенный тип ('success', 'danger',
'warning', 'info').
99  */
100 function detectTypeFromText(text) {

```

```

101     if (!text || typeof text !== 'string') return 'info';
102     const lower = text.toLowerCase();
103     if (text.includes('□ ') || lower.includes('успешно') || low
er.includes('success') || lower.includes('готов')) {
104         return 'success';
105     }
106     if (text.includes('□ ') || lower.includes('ошибка') || lowe
r.includes('error') || lower.includes('failed') || lower.inclu
des('не удалось')) {
107         return 'danger';
108     }
109     if (text.includes('□ ') || lower.includes('внимание') || lo
wer.includes('warning') || lower.includes('заполните') || lowe
r.includes('пожалуйста, выберите') || lower.includes('пожалуйс
та выберите') || lower.includes('укажите')) {
110         return 'warning';
111     }
112     return 'info';
113 }
114
115 /**
116  * Отображает всплывающее окно (Toast).
117  * @param {string} message - Текст сообщения или HTML.
118  * @param {string} [type='info'] - Тип сообщения ('success',
'danger', 'error', 'warning', 'info').
119  * @param {Object} [options={}] - Дополнительные параметры (
duration, title, isHtml, autoClose).
120  * @returns {HTMLElement} DOM-элемент тоста.
121  */
122 function showToast(message, type, options = {}) {
123     if (message === undefined || message === null) return null
;
124
125     // Если тип не указан, пытаемся определить его автоматичес
ки по тексту
126     const resolvedType = type || detectTypeFromText(String(mes
sage));
127     const meta = getMetaForType(resolvedType);

```

```

128     const duration = options.duration !== undefined ? options.
duration : DEFAULT_DURATION;
129     const title = options.title || meta.headerText;
130
131     const container = getOrCreateContainer();
132
133     const toastEl = document.createElement('div');
134     toastEl.className = `floating-toast floating-toast-${meta.
bootstrapType}`;
135     toastEl.setAttribute('role', 'alert');
136
137     // Форматирование текста (сохранение переводов строк)
138     const formattedMessage = options.isHtml
139         ? message
140         : String(message)
141             .replace(/&/g, '&')
142             .replace(/</g, '&lt;')
143             .replace(/>/g, '&gt;')
144             .replace(/\n/g, '<br>');
145
146     toastEl.innerHTML = `
147         <div class="floating-toast-header">
148             <i class="${meta.iconClass} floating-toast-icon"></i>
149             <strong class="floating-toast-title">${title}</strong>
150             <button type="button" class="floating-toast-close" ari
a-label="Закрыть">&times;</button>
151         </div>
152         <div class="floating-toast-body">${formattedMessage}</di
v>
153         ${duration > 0 ? `<div class="floating-toast-progress"><
div class="floating-toast-progress-bar bg-${meta.bootstrapType
}"></div></div>` : ''}
154     `;
155
156     container.appendChild(toastEl);
157
158     // Анимация появления
159     requestAnimationFrame(() => {

```

```
160 toastEl.classList.add('show');
161 });
162
163 let dismissTimeout = null;
164 let startTime = Date.now();
165 let remainingTime = duration;
166 const progressBar = toastEl.querySelector('.floating-toast
    -progress-bar');
167
168 const startTimer = () => {
169     if (duration > 0) {
170         startTime = Date.now();
171         if (progressBar) {
172             progressBar.style.transition = `width ${remainingTim
    e}ms linear`;
173             progressBar.style.width = '0%';
174         }
175         dismissTimeout = setTimeout(() => {
176             closeToast();
177             }, remainingTime);
178     }
179 };
180
181 const pauseTimer = () => {
182     if (dismissTimeout) {
183         clearTimeout(dismissTimeout);
184         dismissTimeout = null;
185         remainingTime -= Date.now() - startTime;
186         if (progressBar) {
187             const computedWidth = window.getComputedStyle(progre
    ssBar).width;
188             progressBar.style.transition = 'none';
189             progressBar.style.width = computedWidth;
190         }
191     }
192 };
193
194 const closeToast = () => {
```

```
195 |         if (dismissTimeout) clearTimeout(dismissTimeout);
196 |         toastEl.classList.remove('show');
197 |         toastEl.classList.add('hide');
198 |         toastEl.addEventListener('transitionend', () => {
199 |             if (toastEl.parentElement) {
200 |                 toastEl.parentElement.removeChild(toastEl);
201 |             }
202 |         }, { once: true });
203 |     };
204 |
205 |     // Слушатели событий
206 |     const closeBtn = toastEl.querySelector('.floating-toast-cl
    ose');
207 |     if (closeBtn) {
208 |         closeBtn.addEventListener('click', (e) => {
209 |             e.stopPropagation();
210 |             closeToast();
211 |         });
212 |     }
213 |
214 |     if (duration > 0) {
215 |         toastEl.addEventListener('mouseenter', pauseTimer);
216 |         toastEl.addEventListener('mouseleave', () => {
217 |             if (remainingTime > 0) startTimer();
218 |         });
219 |         startTimer();
220 |     }
221 |
222 |     return toastEl;
223 | }
224 |
225 | /**
226 |  * Отображает уведомление с заголовком и текстом.
227 |  * @param {string} title - Заголовок.
228 |  * @param {string} message - Текст сообщения.
229 |  * @param {string} [type='info'] - Тип.
230 |  * @param {number} [duration=4500] - Длительность показа.
231 |  */
```



```
232 | function showNotification(title, message, type = 'info', dur
    | ation = DEFAULT_DURATION) {
233 |     return showToast(message, type, { title, duration });
234 | }
235 |
236 | // Экспорт в глобальную область видимости
237 | window.showToast = showToast;
238 | window.showNotification = showNotification;
239 | window._showToast = showToast; // Для совместимости с уже им
    | еющимися вкладками
240 |
241 | // Перехват стандартного window.alert()
242 | // Сохраняем ссылку на оригинальный alert на случай необходи
    | мости
243 | window._originalAlert = window.alert;
244 |
245 | window.alert = function (message) {
246 |     // Выводим неблокирующее всплывающее окно
247 |     showToast(message);
248 |     // Также дублируем в консоль для отладки
249 |     if (window.console && console.debug) {
250 |         console.debug('[Toast Alert]:', message);
251 |     }
252 | };
253 |
254 | // Инициализация контейнера при загрузке документа
255 | if (document.readyState === 'loading') {
256 |     document.addEventListener('DOMContentLoaded', getOrCreateC
    | ontainer);
257 | } else {
258 |     getOrCreateContainer();
259 | }
260 |})();
261 |
```

userSettings.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Usersettings Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля userSettings.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/userSettings.js?v=20261001_v1" type="module"></script>
11  *
12  *   JavaScript Import:
13  *     import { executeUserAction, isAdmin } from '/windows/api/webgui/js/userSettings.js';
14  *
15  * File: userSettings.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/js
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 /**
25  * userSettings.js – User Profile & Settings Management
26  * Supports Google OAuth synchronization, profile status, language & theme controls.
27  */
28
29 'use strict';
```

```
30
31 export function executeUserAction(actionType) {
32   // Ensure chat tab is active
33   if (typeof window.switchToTab === 'function') {
34     window.switchToTab('chat');
35   } else if (typeof window.switchTab === 'function') {
36     window.switchTab('tab-chat');
37   } else {
38     const chatTabBtn = document.querySelector('[data-bs-target
39     ="#tab-chat"], [data-tab="tab-chat"]');
40     if (chatTabBtn && window.bootstrap?.Tab) {
41       bootstrap.Tab.getOrCreateInstance(chatTabBtn).show();
42     }
43
44     const msgInput = document.getElementById('message-input');
45     const sendBtn = document.getElementById('send-button');
46
47     const send = () => {
48       if (typeof window.sendChatMessage === 'function') {
49         window.sendChatMessage();
50       } else if (sendBtn) {
51         sendBtn.click();
52       }
53     };
54
55     if (actionType === 'mail') {
56       if (msgInput) {
57         msgInput.value = 'Проверь мою почту Gmail и покажи после
58         дние непрочитанные письма.';
59         setTimeout(send, 100);
60       }
61     } else if (actionType === 'docs') {
62       const query = prompt('Введите поисковый запрос для поиска
63       документов в Google Docs / Drive:');
64       if (query && query.trim() && msgInput) {
65         msgInput.value = `Найди в Google Docs / Google Drive док
66         ументы по запросу: "${query.trim()}".`;
67       }
68     }
69   }
70 }
```

```

64     setTimeout(send, 100);
65 }
66 } else if (actionType === 'sheets') {
67     const query = prompt('Введите ID таблицы или команду для G
oogle Sheets (например: "Прочитай данные из таблицы ID диапазо
н A1:D10" или "Добавь строку"):');
68     if (query && query.trim() && msgInput) {
69         msgInput.value = `Google Sheets: ${query.trim()}`;
70         setTimeout(send, 100);
71     }
72 } else if (actionType === 'news') {
73     if (msgInput) {
74         msgInput.value = 'Подготовь для меня свежий дайджест сам
ых релевантных новостей по моим интересам.';
75         setTimeout(send, 100);
76     }
77 } else if (actionType === 'facebook') {
78     const text = prompt('Введите текст публикации для Facebook
(и при желании ссылку через пробел):');
79     if (text && text.trim() && msgInput) {
80         msgInput.value = `Опубликуй пост в Facebook с текстом: "
${text.trim()}".`;
81         setTimeout(send, 100);
82     }
83 }
84 }
85
86 export async function initUserSettings() {
87     await refreshUserProfile();
88
89     // Bind User Menu Action Buttons
90     document.querySelectorAll('#user-menu-btn-mail, #nav-btn-che
ck-mail, .action-btn-mail').forEach((btn) => {
91         btn.addEventListener('click', (e) => {
92             e.preventDefault();
93             executeUserAction('mail');
94         });
95     });

```

```
96 |
97 | document.querySelectorAll('#user-menu-btn-docs, #nav-btn-sea
    | rch-gdocs, .action-btn-docs').forEach((btn) => {
98 |     btn.addEventListener('click', (e) => {
99 |         e.preventDefault();
100 |         executeUserAction('docs');
101 |     });
102 | });
103 |
104 | document.querySelectorAll('#user-menu-btn-sheets, #nav-btn-g
    | sheets, .action-btn-sheets').forEach((btn) => {
105 |     btn.addEventListener('click', (e) => {
106 |         e.preventDefault();
107 |         executeUserAction('sheets');
108 |     });
109 | });
110 |
111 | document.querySelectorAll('#user-menu-btn-news, #nav-btn-new
    | s-digest, .action-btn-news').forEach((btn) => {
112 |     btn.addEventListener('click', (e) => {
113 |         e.preventDefault();
114 |         executeUserAction('news');
115 |     });
116 | });
117 |
118 | document.querySelectorAll('#user-menu-btn-facebook, #nav-btn
    | -facebook-post, .action-btn-facebook').forEach((btn) => {
119 |     btn.addEventListener('click', (e) => {
120 |         e.preventDefault();
121 |         executeUserAction('facebook');
122 |     });
123 | });
124 |
125 | // Handle Google Login Click
126 | document.querySelectorAll('.btn-google-login, #user-settings
    | -google-btn').forEach((btn) => {
127 |     btn.addEventListener('click', (e) => {
128 |         e.preventDefault();
```

```

129     const nextUrl = window.location.pathname + window.location.search;
130     window.location.href = `/auth/google?next=${encodeURIComponent(nextUrl)}`;
131   });
132 });
133
134 // Handle Logout Click
135 document.querySelectorAll('.btn-user-logout, #user-settings-logout-btn').forEach((btn) => {
136   btn.addEventListener('click', async (e) => {
137     e.preventDefault();
138     try {
139       await fetch('/auth/logout', { method: 'POST' });
140     } catch (err) {
141       console.error('Logout error:', err);
142     } finally {
143       window.location.href = '/';
144     }
145   });
146 });
147
148 // Re-sync modal fields on modal show
149 const modalEl = document.getElementById('userSettingsModal');
150 ;
151 if (modalEl) {
152   modalEl.addEventListener('show.bs.modal', async () => {
153     await refreshUserProfile();
154     const savedLang = localStorage.getItem('app_language') || 'ru';
155     document.querySelectorAll('.lang-selector').forEach((s) => {
156       s.value = savedLang;
157     });
158     const savedTheme = localStorage.getItem('theme') || 'system';
159     document.querySelectorAll('.theme-selector').forEach((s) => {

```

```
159 |         s.value = savedTheme;
160 |     });
161 | });
162 | }
163 | }
164 |
165 | export function isUserAdmin() {
166 |     if (!window.currentUser) return false;
167 |     return Boolean(window.currentUser.is_admin || (window.current
    tUser.role && String(window.currentUser.role).toLowerCase() ==
    = 'admin'));
168 | }
169 |
170 | export async function refreshUserProfile() {
171 |     try {
172 |         const res = await fetch('/auth/check');
173 |         if (!res.ok) return;
174 |         const data = await res.json();
175 |         window.currentUser = data;
176 |
177 |         const authCard = document.getElementById('user-settings-au
    th-card');
178 |         const guestCard = document.getElementById('user-settings-g
    uest-card');
179 |         const navGoogleBtn = document.getElementById('nav-google-b
    tn');
180 |         const userSettingsBtn = document.getElementById('user-sett
    ings-btn');
181 |         const navAvatar = document.getElementById('nav-user-avatar
    ');
182 |         const navIcon = document.getElementById('nav-user-icon');
183 |         const navLabel = document.getElementById('nav-user-label')
    ;
184 |
185 |         if (data && data.authenticated) {
186 |             // User is authenticated
187 |             if (authCard) authCard.classList.remove('d-none');
188 |             if (guestCard) guestCard.classList.add('d-none');
```

```
189
190     // Navbar: show user profile pill, hide Google register
button
191     if (navGoogleBtn) navGoogleBtn.classList.add('d-none');
192     if (userSettingsBtn) {
193         userSettingsBtn.classList.remove('d-none');
194         userSettingsBtn.classList.add('d-flex');
195     }
196
197     const name = data.name || data.email || 'Пользователь';
198     const email = data.email || '';
199     const picture = data.picture || '';
200     const role = data.role || (data.is_admin ? 'Admin' : 'Us
er');
201     const hasGoogle = Boolean(data.has_google);
202
203     // Update modal fields
204     const avatarEl = document.getElementById('user-settings-
avatar');
205     const nameEl = document.getElementById('user-settings-na
me');
206     const emailEl = document.getElementById('user-settings-e
mail');
207     const roleBadge = document.getElementById('user-settings
-role-badge');
208     const googleBadge = document.getElementById('user-settin
gs-google-badge');
209
210     if (avatarEl) {
211         if (picture) {
212             avatarEl.src = picture;
213             avatarEl.classList.remove('d-none');
214         } else {
215             avatarEl.classList.add('d-none');
216         }
217     }
218     if (nameEl) nameEl.textContent = name;
219     if (emailEl) emailEl.textContent = email;
```



```
220     if (roleBadge) {
221         roleBadge.textContent = role;
222         roleBadge.className = `badge ${role.toLowerCase() ===
'admin' ? 'bg-danger' : 'bg-primary'}`;
223     }
224     const isOAuthEnabled = data.oauth_enabled !== false;
225
226     if (googleBadge) {
227         if (hasGoogle) {
228             googleBadge.innerHTML = '<i class="bi bi-google text
-danger me-1"></i>Google OAuth: <span class="text-success fw-b
old">Синхронизирован</span>';
229         } else if (isOAuthEnabled) {
230             googleBadge.innerHTML = '<i class="bi bi-google text
-muted me-1"></i>Google OAuth: <span class="text-warning">Не п
одключен</span> <button type="button" class="btn btn-sm btn-ou
tline-danger ms-2 py-0 px-2 btn-google-login">Синхронизировать
</button>';
231         } else {
232             googleBadge.innerHTML = '<i class="bi bi-google text
-muted me-1"></i>Google OAuth: <span class="text-muted">Отключ
ен</span>';
233         }
234     }
235
236     // Update dropdown header elements if present
237     const menuWrapper = document.getElementById('user-menu-d
ropdown-wrapper');
238     const menuDisplayName = document.getElementById('user-me
nu-display-name');
239     const menuDisplayEmail = document.getElementById('user-m
enu-display-email');
240     if (menuWrapper) {
241         menuWrapper.classList.remove('d-none');
242         menuWrapper.classList.add('d-flex');
243     }
244     if (menuDisplayName) menuDisplayName.textContent = name;
245     if (menuDisplayEmail) menuDisplayEmail.textContent = ema
```

```

    il;
246
247     // Update navbar button
248     if (navAvatar && picture) {
249         navAvatar.src = picture;
250         navAvatar.classList.remove('d-none');
251         if (navIcon) navIcon.classList.add('d-none');
252     }
253     if (navLabel) {
254         const shortName = name.split(' ')[0] || name;
255         navLabel.textContent = shortName;
256     }
257 } else {
258     // Guest / unauthenticated
259     if (authCard) authCard.classList.add('d-none');
260     if (guestCard) guestCard.classList.remove('d-none');
261
262     const isOAuthEnabled = data && data.oauth_enabled !== fa
lse;
263
264     // Navbar: show full Google register/sign-in button only
    if OAuth is enabled, else show user settings gear/pill
265     if (navGoogleBtn) {
266         if (isOAuthEnabled) {
267             navGoogleBtn.classList.remove('d-none');
268             navGoogleBtn.classList.add('d-flex');
269         } else {
270             navGoogleBtn.classList.add('d-none');
271             navGoogleBtn.classList.remove('d-flex');
272         }
273     }
274     const menuWrapper = document.getElementById('user-menu-d
ropdown-wrapper');
275     if (menuWrapper) {
276         if (isOAuthEnabled) {
277             menuWrapper.classList.add('d-none');
278             menuWrapper.classList.remove('d-flex');
279         } else {

```

```
280         menuWrapper.classList.remove('d-none');
281         menuWrapper.classList.add('d-flex');
282     }
283 }
284 if (userSettingsBtn) {
285     if (isOAuthEnabled) {
286         userSettingsBtn.classList.add('d-none');
287         userSettingsBtn.classList.remove('d-flex');
288     } else {
289         userSettingsBtn.classList.remove('d-none');
290         userSettingsBtn.classList.add('d-flex');
291     }
292 }
293
294 // Guest card google login button adjustment
295 const guestGoogleBtn = guestCard ? guestCard.querySelector(
    or('.btn-google-login') : null;
296 if (guestGoogleBtn) {
297     if (isOAuthEnabled) {
298         guestGoogleBtn.classList.remove('d-none');
299     } else {
300         guestGoogleBtn.classList.add('d-none');
301     }
302 }
303
304 if (navAvatar) navAvatar.classList.add('d-none');
305 if (navIcon) navIcon.classList.remove('d-none');
306 if (navLabel) navLabel.textContent = 'User Settings';
307 }
308 } catch (err) {
309     console.error('Failed to refresh user profile:', err);
310 }
311 }
312
313 window.initUserSettings = initUserSettings;
314 window.refreshUserProfile = refreshUserProfile;
315 window.isUserAdmin = isUserAdmin;
316 window.executeUserAction = executeUserAction;
```


window-api.js

```
1 /**
2  * =====
3  * Process Name: Windows Js - Window-API Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля window-ap
7  *   i.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/js/window-api.js?v=202
11  *     61001_v1" type="module"></script>
12  *
13  * File: window-api.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/js
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-06 11:06:00
19  * =====
20  */
21 /**
22  * window-api.js – инициализация window.api для всех вкладок и
23  * панелей управления
24  * Этот файл загружается ДО всех вкладок, чтобы window.api был
25  * доступен сразу
26  */
27
28 window.api = window.api || {};
29
30 // Универсальный fetch для API и Auth
31 if (!window.api.fetch) {
```

```

30 | window.api.fetch = async function(url, opts = {}) {
31 |     const response = await fetch(url, opts);
32 |     if (!response.ok) {
33 |         let msg = response.statusText;
34 |         try {
35 |             const data = await response.json();
36 |             if (data && data.detail) {
37 |                 if (typeof data.detail === 'string') {
38 |                     msg = data.detail;
39 |                 } else if (Array.isArray(data.detail)) {
40 |                     msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
41 |                 } else {
42 |                     msg = JSON.stringify(data.detail);
43 |                 }
44 |             } else if (data && data.message) {
45 |                 msg = data.message;
46 |             }
47 |         } catch {}
48 |         throw new Error(`${response.status} ${msg}`);
49 |     }
50 |     return response.json();
51 | };
52 | }
53 |
54 | // User management API
55 | window.api.users = window.api.users || {
56 |     async list(params = {}) {
57 |         const q = new URLSearchParams(params).toString();
58 |         return window.api.fetch(`/api/admin/users${q ? '?' + q : '
'}`);
59 |     },
60 |     async get(userId) {
61 |         return window.api.fetch(`/api/admin/users/${userId}`);
62 |     },
63 |     async create(data) {
64 |         return window.api.fetch('/api/admin/users', {
65 |             method: 'POST',

```

```

66     headers: { 'Content-Type': 'application/json' },
67     body: JSON.stringify(data)
68   });
69 },
70 async update(userId, data) {
71   return window.api.fetch(`/api/admin/users/${userId}`, {
72     method: 'PUT',
73     headers: { 'Content-Type': 'application/json' },
74     body: JSON.stringify(data)
75   });
76 },
77 async setPassword(userId, password) {
78   return window.api.fetch(`/api/admin/users/${userId}/password`, {
79     method: 'POST',
80     headers: { 'Content-Type': 'application/json' },
81     body: JSON.stringify({ password })
82   });
83 },
84 async toggleActive(userId) {
85   return window.api.fetch(`/api/admin/users/${userId}/toggle-active`, {
86     method: 'POST'
87   });
88 },
89 async toggleRole(userId) {
90   return window.api.fetch(`/api/admin/users/${userId}/toggle-role`, {
91     method: 'POST'
92   });
93 },
94 async delete(userId) {
95   return window.api.fetch(`/api/admin/users/${userId}`, {
96     method: 'DELETE'
97   });
98 }
99 };

```

100

```
101 // Skills management API
102 window.api.skills = window.api.skills || {
103   async list(params = {}) {
104     const q = new URLSearchParams(params).toString();
105     return window.api.fetch(`/api/admin/skills${q ? '?' + q : ''}`);
106   },
107   async get(name) {
108     return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`);
109   },
110   async create(data) {
111     return window.api.fetch('/api/admin/skills', {
112       method: 'POST',
113       headers: { 'Content-Type': 'application/json' },
114       body: JSON.stringify(data)
115     });
116   },
117   async update(name, data) {
118     return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`, {
119       method: 'PUT',
120       headers: { 'Content-Type': 'application/json' },
121       body: JSON.stringify(data)
122     });
123   },
124   async delete(name) {
125     return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`, {
126       method: 'DELETE'
127     });
128   },
129   async package(name) {
130     return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}/package`, {
131       method: 'POST'
132     });
133   }
}
```



```
134 | };
135 |
136 | // MCP Servers management API
137 | window.api.mcp = window.api.mcp || {
138 |   async list() {
139 |     return window.api.fetch('/api/admin/mcp/servers');
140 |   },
141 |   async get(id) {
142 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`);
143 |   },
144 |   async create(data) {
145 |     return window.api.fetch('/api/admin/mcp/servers', {
146 |       method: 'POST',
147 |       headers: { 'Content-Type': 'application/json' },
148 |       body: JSON.stringify(data)
149 |     });
150 |   },
151 |   async update(id, data) {
152 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
153 |       method: 'PUT',
154 |       headers: { 'Content-Type': 'application/json' },
155 |       body: JSON.stringify(data)
156 |     });
157 |   },
158 |   async delete(id) {
159 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
160 |       method: 'DELETE'
161 |     });
162 |   },
163 |   async toggle(id) {
164 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/toggle`, {
165 |       method: 'POST'
166 |     });
167 |   },
```

```
168 |   async test(id) {
169 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent
    | IComponent(id)}/test`, {
170 |       method: 'POST'
171 |     });
172 |   },
173 |   async testAdhoc(data) {
174 |     return window.api.fetch('/api/admin/mcp/test', {
175 |       method: 'POST',
176 |       headers: { 'Content-Type': 'application/json' },
177 |       body: JSON.stringify(data)
178 |     });
179 |   },
180 |   async getTools() {
181 |     return window.api.fetch('/api/admin/mcp/tools');
182 |   }
183 | };
184 |
185 | // Google Workspace Accounts API wrapper
186 | window.api.googleAccounts = window.api.googleAccounts || {
187 |   async list() {
188 |     return await window.api.fetch('/api/admin/google-accounts'
    | );
189 |   },
190 |   async upload(formData) {
191 |     const r = await fetch('/api/admin/google-accounts/upload',
    | {
192 |       method: 'POST',
193 |       body: formData
194 |     });
195 |     if (!r.ok) {
196 |       let msg = r.statusText;
197 |       try { msg = (await r.json()).detail || msg; } catch {}
198 |       throw new Error(`${r.status} ${msg}`);
199 |     }
200 |     return await r.json();
201 |   },
202 |   async create(params) {
```

```
203     return await window.api.fetch('/api/admin/google-accounts'
    , {
204         method: 'POST',
205         headers: { 'Content-Type': 'application/json' },
206         body: JSON.stringify(params)
207     });
208 },
209 async setDefault(name) {
210     return await window.api.fetch(`/api/admin/google-accounts/
    ${encodeURIComponent(name)}/default`, {
211         method: 'POST'
212     });
213 },
214 async resetStatus(name) {
215     return await window.api.fetch(`/api/admin/google-accounts/
    ${encodeURIComponent(name)}/reset-status`, {
216         method: 'POST'
217     });
218 },
219 async test(name) {
220     return await window.api.fetch(`/api/admin/google-accounts/
    ${encodeURIComponent(name)}/test`, {
221         method: 'POST'
222     });
223 },
224 async delete(name) {
225     return await window.api.fetch(`/api/admin/google-accounts/
    ${encodeURIComponent(name)}`, {
226         method: 'DELETE'
227     });
228 }
229 };
230
231 // System & God Mode API
232 window.api.system = window.api.system || {
233     async openGodMode() {
234         return await window.api.fetch('/api/v1/tc/godmode/open', {
            method: 'POST' });
    }
};
```

```
235 | }
236 | };
237 |
238 | window.openGodModeFolder = async function() {
239 |     try {
240 |         const res = await fetch('/api/v1/tc/godmode/open', { method: 'POST' });
241 |         const data = await res.json();
242 |         if (data && data.status === 'ok') {
243 |             if (typeof showToast === 'function') {
244 |                 const msg = data.action === 'focused'
245 |                     ? '☐ Фокус переведён на открытое окно God Mode'
246 |                     : '☐ God Mode (All Tasks) открыт в Проводнике Windows';
247 |                 showToast('success', msg);
248 |             }
249 |         } else {
250 |             const errMsg = (data && data.message) ? data.message : 'Неизвестная ошибка';
251 |             if (typeof showToast === 'function') {
252 |                 showToast('danger', 'Ошибка: ' + errMsg);
253 |             } else {
254 |                 alert('Ошибка: ' + errMsg);
255 |             }
256 |         }
257 |     } catch (err) {
258 |         if (typeof showToast === 'function') {
259 |             showToast('danger', 'Не удалось связаться с сервером ТС: ' + err.message);
260 |         } else {
261 |             alert('Ошибка: ' + err.message);
262 |         }
263 |     }
264 | };
265 |
266 |
```

login.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Login View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса login.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/login.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/login.html" class="w-10
17     0 h-100 border-0"></iframe>
18
19 File: login.html
20 Project: ai-breadboard
21 Package: windows/api/webgui
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!DOCTYPE html>
30 <html lang="ru">
31 <head>
32   <meta charset="UTF-8">
33   <meta name="viewport" content="width=device-width, initial-s
34   cale=1.0">
35   <title>Вход в систему – AI Breadboard</title>
36   <link rel="icon" type="image/x-icon" href="/favicon.ico">
37   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
```

```
t/css/bootstrap.min.css" rel="stylesheet">
32 <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
0.0/font/bootstrap-icons.css" rel="stylesheet">
33 <link href="/html/css/main.css?v=20260917_refactor_v1" rel="
stylesheet">
34 <style>
35 /* — Страница входа: центрирование и лёгкий фон — */
36 body {
37     display: flex;
38     align-items: center;
39     justify-content: center;
40     min-height: 100vh;
41     margin: 0;
42     padding: 1rem;
43     /* Лёгкий градиентный акцент поверх --bg-color из main.c
ss */
44     background:
45         radial-gradient(circle at 50% 15%, rgba(99, 102, 241,
0.08), transparent 65%),
46         var(--bg-color);
47 }
48
49 /* — Карточка входа — */
50 .auth-card {
51     width: 100%;
52     max-width: 400px;
53     background: var(--surface-1);
54     border: 1px solid var(--border-color);
55     border-radius: 12px;
56     box-shadow: var(--shadow-lg);
57     overflow: hidden;
58 }
59
60 /* — Шапка карточки — */
61 .auth-header {
62     padding: 1.5rem 1.5rem 1rem;
63     text-align: center;
64     background: var(--surface-2);
```

```
65 |     border-bottom: 1px solid var(--border-color);
66 | }
67 |
68 | /* — Логотип-бейдж — */
69 | .auth-logo-badge {
70 |     width: 48px;
71 |     height: 48px;
72 |     border-radius: 12px;
73 |     background: linear-gradient(135deg, #6366f1, #3b82f6);
74 |     display: inline-flex;
75 |     align-items: center;
76 |     justify-content: center;
77 |     font-size: 1.5rem;
78 |     color: #fff;
79 |     box-shadow: 0 4px 12px rgba(99, 102, 241, 0.3);
80 |     margin-bottom: 0.75rem;
81 | }
82 |
83 | /* — Навигационные вкладки входа/регистрации — */
84 | .auth-nav-tabs .nav-link {
85 |     border: none;
86 |     border-bottom: 2px solid transparent;
87 |     color: var(--text-color);
88 |     opacity: 0.65;
89 |     font-weight: 600;
90 |     font-size: 0.85rem;
91 |     padding: 0.55rem 1rem;
92 |     background: transparent;
93 |     transition: all 0.18s;
94 | }
95 |
96 | .auth-nav-tabs .nav-link.active {
97 |     background: transparent;
98 |     border-bottom: 2px solid #6366f1;
99 |     color: #6366f1;
100 |     opacity: 1;
101 | }
102 |
```

```
103 | [data-theme="dark"] .auth-nav-tabs .nav-link.active {
104 |     color: #818cf8;
105 |     border-bottom-color: #818cf8;
106 | }
107 |
108 | /* — Google OAuth кнопка — */
109 | .btn-google {
110 |     background: var(--surface-1);
111 |     color: var(--text-color);
112 |     border: 1px solid var(--border-color);
113 |     font-weight: 500;
114 |     font-size: 0.85rem;
115 |     transition: all 0.18s;
116 | }
117 |
118 | .btn-google:hover {
119 |     background: var(--surface-2);
120 |     box-shadow: var(--shadow-sm);
121 |     color: var(--text-color);
122 | }
123 |
124 | /* — Разделитель «или по почте» — */
125 | .divider-text {
126 |     display: flex;
127 |     align-items: center;
128 |     text-align: center;
129 |     color: var(--text-muted);
130 |     font-size: 0.78rem;
131 |     margin: 1rem 0;
132 | }
133 |
134 | .divider-text::before,
135 | .divider-text::after {
136 |     content: '';
137 |     flex: 1;
138 |     border-bottom: 1px solid var(--border-color);
139 | }
140 |
```



```
141 |     .divider-text:not(:empty)::before { margin-right: 0.6em; }
142 |     .divider-text:not(:empty)::after  { margin-left:  0.6em; }
143 |
144 |     /* — Кнопки управления (тема + язык) в правом верхнем углу — */
145 |     .top-controls {
146 |         position: fixed;
147 |         top: 0.75rem;
148 |         right: 0.75rem;
149 |         display: flex;
150 |         gap: 0.375rem;
151 |         z-index: 100;
152 |     }
153 | </style>
154 | </head>
155 | <body>
156 |
157 |     <!-- Управление: тема и язык -->
158 |     <div class="top-controls">
159 |         <select class="form-select form-select-sm" id="auth-lang-selector" style="width: auto; font-size: 0.78rem; padding: 0.2rem 0.4rem;">
160 |             <option value="ru">RU</option>
161 |             <option value="en">EN</option>
162 |             <option value="he">HE</option>
163 |         </select>
164 |         <button class="btn btn-sm btn-outline-secondary d-flex align-items-center justify-content-center rounded-circle"
165 |             id="theme-toggle"
166 |             title="Переключить тему"
167 |             style="width: 28px; height: 28px; padding: 0;">
168 |             <i class="bi bi-circle-half" id="theme-icon"></i>
169 |         </button>
170 |     </div>
171 |
172 |     <div class="auth-card">
173 |         <!-- Шапка -->
174 |         <div class="auth-header">
```

```
175 |         <div class="auth-logo-badge">
176 |             <i class="bi bi-cpu-fill"></i>
177 |         </div>
178 |         <h5 class="fw-bold mb-1" id="title-app-name">AI Breadboa
rd</h5>
179 |         <p class="small mb-0" style="color: var(--text-muted);"
id="title-app-desc">Авторизуйтесь для доступа к сервисам ИИ</p
>
180 |     </div>
181 |
182 |     <!-- Вкладки: Войти / Регистрация -->
183 |     <ul class="nav nav-tabs auth-nav-tabs nav-fill px-3 pt-2"
id="authTabs" role="tablist">
184 |         <li class="nav-item" role="presentation">
185 |             <button class="nav-link active" id="tab-login-btn" dat
a-bs-toggle="tab" data-bs-target="#pane-login" type="button" r
ole="tab">
186 |                 <i class="bi bi-box-arrow-in-right me-1"></i>
187 |                 <span id="label-tab-login">Войти</span>
188 |             </button>
189 |         </li>
190 |         <li class="nav-item" role="presentation">
191 |             <button class="nav-link" id="tab-register-btn" data-bs
-toggle="tab" data-bs-target="#pane-register" type="button" ro
le="tab">
192 |                 <i class="bi bi-person-plus me-1"></i>
193 |                 <span id="label-tab-register">Регистрация</span>
194 |             </button>
195 |         </li>
196 |     </ul>
197 |
198 |     <!-- Тело карточки -->
199 |     <div class="card-body p-3">
200 |         <!-- Статус-алерт -->
201 |         <div id="auth-alert" class="alert d-none mb-3 py-2 small
" role="alert"></div>
202 |
203 |         <!-- Google OAuth -->
```

```

204         <button type="button"
205             class="btn btn-google w-100 d-flex align-items-c
enter justify-content-center gap-2 rounded-3 mb-0"
206             style="padding: 0.45rem 0.75rem;"
207             id="btn-google-auth">
208             <svg width="16" height="16" viewBox="0 0 24 24">
209                 <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-
.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2
.08-1.92 3.28-4.74 3.28-8.09z"/>
210                 <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-
2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6
.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/>
211                 <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.3
6-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.
43 3.45 1.18 4.94l2.85-2.22.81-.63z"/>
212                 <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21
1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7
.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/>
213             </svg>
214             <span id="label-google-auth">Продолжить с Google</span
>
215         </button>
216
217         <div class="divider-text" id="label-or-email">или по поч
те</div>
218
219         <!-- Контент вкладок -->
220         <div class="tab-content" id="authTabsContent">
221
222             <!-- ВХОД -->
223             <div class="tab-pane fade show active" id="pane-login"
role="tabpanel">
224                 <form id="form-login">
225                     <div class="mb-2">
226                         <label class="form-label fw-semibold" id="label-
email-login" for="login-email">Email</label>
227                         <div class="input-group input-group-sm">
228                             <span class="input-group-text"><i class="bi bi

```

```

| -envelope"></i></span>
229 |         <input type="email" class="form-control form-c
| ontrol-sm" id="login-email"
230 |             required placeholder="name@example.com"
| autocomplete="email">
231 |     </div>
232 | </div>
233 |     <div class="mb-3">
234 |         <label class="form-label fw-semibold" id="label-
| pass-login" for="login-password">Пароль</label>
235 |         <div class="input-group input-group-sm">
236 |             <span class="input-group-text"><i class="bi bi
| -key"></i></span>
237 |             <input type="password" class="form-control for
| m-control-sm" id="login-password"
238 |                 required placeholder="....." autocom
| plete="current-password">
239 |         </div>
240 |     </div>
241 |     <button type="submit" class="btn btn-primary btn-s
| m w-100 fw-semibold" id="btn-submit-login"
242 |         style="padding: 0.45rem;">
243 |         <span class="spinner-border spinner-border-sm d-
| none me-1" id="login-spinner"></span>
244 |         <span id="label-btn-login">Войти в аккаунт</span
| >
245 |     </button>
246 | </form>
247 | </div>
248 |
249 | <!-- РЕГИСТРАЦИЯ -->
250 | <div class="tab-pane fade" id="pane-register" role="ta
| bpanel">
251 |     <form id="form-register">
252 |         <div class="mb-2">
253 |             <label class="form-label fw-semibold" id="label-
| reg-name" for="reg-name">Ваше имя</label>
254 |             <div class="input-group input-group-sm">

```

```
255 |         <span class="input-group-text"><i class="bi bi  
| -person"></i></span>  
256 |         <input type="text" class="form-control form-co  
| ntrol-sm" id="reg-name"  
257 |             required placeholder="Иван Иванов" auto  
| complete="name">  
258 |     </div>  
259 | </div>  
260 | <div class="mb-2">  
261 |     <label class="form-label fw-semibold" id="label-  
| reg-email" for="reg-email">Email</label>  
262 |     <div class="input-group input-group-sm">  
263 |         <span class="input-group-text"><i class="bi bi  
| -envelope"></i></span>  
264 |         <input type="email" class="form-control form-c  
| ontrol-sm" id="reg-email"  
265 |             required placeholder="name@example.com"  
| autocomplete="email">  
266 |     </div>  
267 | </div>  
268 | <div class="mb-3">  
269 |     <label class="form-label fw-semibold" id="label-  
| reg-pass" for="reg-password">Придумайте пароль</label>  
270 |     <div class="input-group input-group-sm">  
271 |         <span class="input-group-text"><i class="bi bi  
| -lock"></i></span>  
272 |         <input type="password" class="form-control for  
| m-control-sm" id="reg-password"  
273 |             required placeholder="Минимум 6 символа  
| в" minlength="6" autocomplete="new-password">  
274 |     </div>  
275 | </div>  
276 | <button type="submit" class="btn btn-primary btn-s  
| m w-100 fw-semibold" id="btn-submit-reg"  
277 |     style="padding: 0.45rem;">  
278 |     <span class="spinner-border spinner-border-sm d-  
| none me-1" id="reg-spinner"></span>  
279 |     <span id="label-btn-reg">Зарегистрироваться</spa
```

```

n>
280         </button>
281     </form>
282 </div>
283
284 </div>
285 </div>
286 </div>
287
288 <!-- Модальное окно подтверждения email -->
289 <div class="modal fade" id="verifyModal" tabindex="-1" aria-
hidden="true" data-bs-backdrop="static">
290     <div class="modal-dialog modal-dialog-centered modal-sm">
291         <div class="modal-content">
292             <div class="modal-header">
293                 <h6 class="modal-title fw-bold">
294                     <i class="bi bi-shield-check text-primary me-1"></
i> Подтверждение почты
295                 </h6>
296                 <button type="button" class="btn-close" data-bs-dism
iss="modal"></button>
297             </div>
298             <div class="modal-body text-center">
299                 <p class="small mb-3" style="color: var(--text-muted
);" id="verify-modal-desc">
300                     Введите код подтверждения, отправленный на ваш ema
il:
301                 </p>
302                 <div class="mb-3">
303                     <input type="text" class="form-control form-contro
l-sm text-center"
304                         id="verify-code-input" placeholder="123456"
maxlength="6"
305                         autofocus style="letter-spacing: 0.2em; fon
t-size: 1.1rem;">
306                 </div>
307                 <div id="verify-alert" class="alert d-none py-1 smal
l"></div>

```

```

308 |         <button type="button" class="btn btn-primary btn-sm
      | w-100 fw-semibold" id="btn-submit-verify"
309 |             style="padding: 0.45rem;">
310 |             <span class="spinner-border spinner-border-sm d-no
      | ne me-1" id="verify-spinner"></span>
311 |             Подтвердить
312 |         </button>
313 |     </div>
314 | </div>
315 | </div>
316 | </div>
317 |
318 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/di
      | st/js/bootstrap.bundle.min.js"></script>
319 | <script>
320 |     (function() {
321 |         'use strict';
322 |
323 |         // — Тема: системная по умолчанию —————
      |
324 |         function getSystemTheme() {
325 |             return window.matchMedia && window.matchMedia('(prefer
      | s-color-scheme: dark)').matches
326 |                 ? 'dark' : 'light';
327 |         }
328 |
329 |         function applyTheme(mode) {
330 |             // mode: 'light' | 'dark' | 'system'
331 |             const resolved = (mode === 'system') ? getSystemTheme(
      | ) : mode;
332 |             document.documentElement.setAttribute('data-theme', re
      | solved);
333 |             document.documentElement.setAttribute('data-bs-theme',
      | resolved);
334 |
335 |             const icon = document.getElementById('theme-icon');
336 |             if (icon) {
337 |                 if (mode === 'light')             icon.className = 'bi bi

```

```

    -sun-fill text-warning';
338         else if (mode === 'dark')      icon.className = 'bi bi
    -moon-stars-fill text-primary';
339         else                          icon.className = 'bi bi
    -circle-half text-info';
340     }
341 }
342
343 function initTheme() {
344     const saved = localStorage.getItem('theme') || 'system
';
345     applyTheme(saved);
346
347     // Следим за изменениями ОС в системном режиме
348     if (window.matchMedia) {
349         window.matchMedia('(prefers-color-scheme: dark)').ad
dEventListener('change', () => {
350             if ((localStorage.getItem('theme') || 'system') ==
= 'system') {
351                 applyTheme('system');
352             }
353         });
354     }
355 }
356
357     // Кнопка переключения темы: цикл system → light → dark
    → system
358     document.getElementById('theme-toggle').addEventListener
('click', () => {
359         const cur = localStorage.getItem('theme') || 'system';
360         const next = { system: 'light', light: 'dark', dark: '
system' }[cur] || 'system';
361         localStorage.setItem('theme', next);
362         applyTheme(next);
363     });
364
365     initTheme();
366

```



```

367 // — I18n
368
369 const i18nDict = {
370   ru: {
371     appDesc: 'Авторизуйтесь для доступа к сервисам ИИ',
372     tabLogin: 'Войти', tabReg: 'Регистрация',
373     btnGoogle: 'Продолжить с Google', orEmail: 'или по п
374 очте',
375     email: 'Email', pass: 'Пароль',
376     regName: 'Ваше имя', regPass: 'Придумайте пароль',
377     btnLogin: 'Войти в аккаунт', btnReg: 'Зарегистрирова
378 ться',
379     passPlaceholder: 'Минимум 6 символов', namePlacehold
380 er: 'Иван Иванов'
381   },
382   en: {
383     appDesc: 'Sign in to access AI Breadboard services',
384     tabLogin: 'Sign In', tabReg: 'Sign Up',
385     btnGoogle: 'Continue with Google', orEmail: 'or via
386 Email',
387     email: 'Email', pass: 'Password',
388     regName: 'Full Name', regPass: 'Create Password',
389     btnLogin: 'Sign In', btnReg: 'Create Account',
390     passPlaceholder: 'Minimum 6 characters', namePlaceho
391 lder: 'John Doe'
392   },
393   he: {
394     appDesc: '□□□□□ □□□ □□□□□□□ AI',
395     tabLogin: '□□□□□□□', tabReg: '□□□□□',
396     btnGoogle: '□□□□ □□ Google', orEmail: '□□ □□□□□□□ □□
397     , '□ "□
398     email: '□□□ "□', pass: '□□□□□□',
399     regName: '□□ □□□', regPass: '□□□ □□□□□',
400     btnLogin: '□□□□□', btnReg: '□□□□□',
401     passPlaceholder: '□□□□□ 6 □□□□□', namePlaceholder: '
402     '□□□□□□ □□□□□
403   }
404 };

```

```
397 |
398 |     const langSelector = document.getElementById('auth-lang-
    | selector');
399 |     const savedLang = localStorage.getItem('app_language') |
    | | 'ru';
400 |     if (langSelector) {
401 |         langSelector.value = savedLang;
402 |         langSelector.addEventListener('change', (e) => {
403 |             const l = e.target.value;
404 |             localStorage.setItem('app_language', l);
405 |             updateLanguageTexts(l);
406 |         });
407 |     }
408 |
409 |     function updateLanguageTexts(lang) {
410 |         const d = i18nDict[lang] || i18nDict.ru;
411 |         document.getElementById('title-app-desc').textContent
    | = d.appDesc;
412 |         document.getElementById('label-tab-login').textContent
    | = d.tabLogin;
413 |         document.getElementById('label-tab-register').textCont
    | ent = d.tabReg;
414 |         document.getElementById('label-google-auth').textConte
    | nt = d.btnGoogle;
415 |         document.getElementById('label-or-email').textContent
    | = d.orEmail;
416 |         document.getElementById('label-email-login').textConte
    | nt = d.email;
417 |         document.getElementById('label-pass-login').textConten
    | t = d.pass;
418 |         document.getElementById('label-reg-name').textContent
    | = d.regName;
419 |         document.getElementById('label-reg-email').textContent
    | = d.email;
420 |         document.getElementById('label-reg-pass').textContent
    | = d.regPass;
421 |         document.getElementById('label-btn-login').textContent
    | = d.btnLogin;
```

```

422     document.getElementById('label-btn-reg').textContent
    = d.btnReg;
423     document.getElementById('reg-password').placeholder
    = d.passPlaceholder;
424     document.getElementById('reg-name').placeholder
    = d.namePlaceholder;
425     }
426     updateLanguageTexts(savedLang);
427
428     // — Алерты —————
    _____
429     function showAlert(msg, type = 'danger') {
430         const el = document.getElementById('auth-alert');
431         el.className = `alert alert-${type} mb-3 py-2 small`;
432         el.textContent = msg;
433         el.classList.remove('d-none');
434     }
435     function hideAlert() {
436         document.getElementById('auth-alert').classList.add('d
    -none');
437     }
438
439     // — Google OAuth —————
    _____
440     document.getElementById('btn-google-auth').addEventListener(
    'click', () => {
441         window.location.href = '/auth/google?next=';
442     });
443
444     // — Вход —————
    _____
445     let pendingVerificationEmail = '';
446     document.getElementById('form-login').addEventListener('
    submit', async (e) => {
447         e.preventDefault();
448         hideAlert();
449         const email = document.getElementById('login-email'
    ).value.trim();

```

```
450 |         const password = document.getElementById('login-passwo
      | rd').value;
451 |         const spinner  = document.getElementById('login-spinne
      | r');
452 |         const btn      = document.getElementById('btn-submit-l
      | ogin');
453 |         btn.disabled = true;
454 |         spinner.classList.remove('d-none');
455 |         try {
456 |             const res  = await fetch('/auth/login/email', {
457 |                 method: 'POST',
458 |                 headers: { 'Content-Type': 'application/json' },
459 |                 body: JSON.stringify({ email, password })
460 |             });
461 |             const data = await res.json();
462 |             if (!res.ok) {
463 |                 if (res.status === 403 && data.detail && data.deta
      | il.includes('не подтверждена')) {
464 |                     pendingVerificationEmail = email;
465 |                     const desc = document.getElementById('verify-mod
      | al-desc');
466 |                     if (desc) desc.textContent = `Код подтверждения
      | отправлен на ${email}:`;
467 |                     new bootstrap.Modal(document.getElementById('ver
      | ifyModal')).show();
468 |                     showAlert(data.detail, 'warning');
469 |                 } else {
470 |                     showAlert(data.detail || 'Ошибка авторизации', '
      | danger');
471 |                 }
472 |                 return;
473 |             }
474 |             showAlert('Вход выполнен! Перенаправление...', 'succ
      | ess');
475 |             setTimeout(() => { window.location.href = '/'; }, 40
      | 0);
476 |         } catch (err) {
477 |             showAlert('Ошибка соединения: ' + err.message, 'dang
```

```

er');
478         } finally {
479             btn.disabled = false;
480             spinner.classList.add('d-none');
481         }
482     });
483
484     // — Регистрация —————
485     document.getElementById('form-register').addEventListener('submit', async (e) => {
486         e.preventDefault();
487         hideAlert();
488         const name      = document.getElementById('reg-name').value.trim();
489         const email      = document.getElementById('reg-email').value.trim();
490         const password = document.getElementById('reg-password').value;
491         const spinner    = document.getElementById('reg-spinner');
492         const btn        = document.getElementById('btn-submit-reg');
493         btn.disabled = true;
494         spinner.classList.remove('d-none');
495         try {
496             const res = await fetch('/auth/register', {
497                 method: 'POST',
498                 headers: { 'Content-Type': 'application/json' },
499                 body: JSON.stringify({ email, password, name })
500             });
501             const data = await res.json();
502             if (!res.ok) {
503                 showAlert(data.detail || 'Ошибка регистрации', 'danger');
504                 return;
505             }
506             pendingVerificationEmail = email;

```

```
507 |         const desc = document.getElementById('verify-modal-d
|         esc');
508 |         if (desc) desc.textContent = `Код подтверждения отпр
|         авлен на ${email}:`;
509 |         new bootstrap.Modal(document.getElementById('verifyM
|         odal')).show();
510 |         showAlert('Регистрация начата. Введите код из письма
|         .', 'info');
511 |         } catch (err) {
512 |             showAlert('Ошибка соединения: ' + err.message, 'dang
|         er');
513 |         } finally {
514 |             btn.disabled = false;
515 |             spinner.classList.add('d-none');
516 |         }
517 |     });
518 |
519 |     // — Подтверждение кода —————
|
520 |     document.getElementById('btn-submit-verify').addEventListener('click', async () => {
521 |         const codeInput = document.getElementById('verify-code
|         -input');
522 |         const alertEl    = document.getElementById('verify-aler
|         t');
523 |         const spinner    = document.getElementById('verify-spin
|         ner');
524 |         const btnVerify = document.getElementById('btn-submit-
|         verify');
525 |         const code = codeInput.value.trim();
526 |         if (!code) return;
527 |         alertEl.classList.add('d-none');
528 |         btnVerify.disabled = true;
529 |         spinner.classList.remove('d-none');
530 |         try {
531 |             const res  = await fetch('/auth/verify', {
532 |                 method: 'POST',
533 |                 headers: { 'Content-Type': 'application/json' },
```

```
534 |         body: JSON.stringify({ email: pendingVerificationE
mail, code })
535 |     });
536 |     const data = await res.json();
537 |     if (!res.ok) {
538 |         alertEl.className = 'alert alert-danger py-1 small
';
539 |         alertEl.textContent = data.detail || 'Неверный код
';
540 |         alertEl.classList.remove('d-none');
541 |         return;
542 |     }
543 |     alertEl.className = 'alert alert-success py-1 small'
;
544 |     alertEl.textContent = 'Почта подтверждена! Войдите п
од своими данными.';
545 |     alertEl.classList.remove('d-none');
546 |     setTimeout(() => {
547 |         const modal = bootstrap.Modal.getInstance(document
.getElementById('verifyModal'));
548 |         if (modal) modal.hide();
549 |         document.getElementById('tab-login-btn').click();
550 |         document.getElementById('login-email').value = pen
dingVerificationEmail;
551 |         showAlert('Почта подтверждена. Введите пароль для
входа.', 'success');
552 |     }, 1200);
553 |     } catch (err) {
554 |         alertEl.className = 'alert alert-danger py-1 small';
555 |         alertEl.textContent = err.message;
556 |         alertEl.classList.remove('d-none');
557 |     } finally {
558 |         btnVerify.disabled = false;
559 |         spinner.classList.add('d-none');
560 |     }
561 | });
562 |
563 | }());
```

```
564 |     </script>
565 | </body>
566 | </html>
567 |
```


index.html

```
1 <!--
2 =====
3 |
4 |
5 |
6 |
7 |
8 |
9 |
10 |
11 |
12 |
13 |
14 |
15 |
16 |
17 |
18 |
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20 |
21 |
22 |
23 |
24 |
25 |
26 |
27 |
28 |
```

Process Name: Windows Web Interface - Log Settings & Multi-Domain Hierarchy

Description:

Клиентский веб-интерфейс настройки, мониторинга и иерархического каталога Windows Event Logs по 7 доменам телеметрии (Security, Process, Storage, Network, Update, Hardware, Power) и Sysmon, построения графов процессов (Provenance) и реконструкции жизненного цикла системы.

Usage Examples:

Web Route:

/windows/api/webgui/log_settings_tab/index.html

Tab Frame Integration:

<iframe src="/windows/api/webgui/log_settings_tab/index.html" class="w-100 h-100 border-0"></iframe>

File: index.html

Project: ai-breadboard

Package: windows/api/webgui/log_settings_tab

Author: hypo69

Copyright: © 2026 hypo69

Updated: 2026-10-08 02:35:00

-->

<div class="container-fluid py-2 px-3 log-settings-container">

```

29 | <!-- Верхняя панель метрик и сводки -->
30 | <div class="row g-2 mb-3">
31 |     <!-- Карточка 1: Журнал Security -->
32 |     <div class="col-12 col-sm-6 col-xl-3">
33 |         <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3 position-relative">
34 |             <div class="d-flex justify-content-between align-items
| -start">
35 |                 <div class="text-muted small text-uppercase fw-semib
| old">Журнал Security □ </div>
36 |                 <span class="badge bg-primary font-monospace" id="ls
| -sec-badge">EVTX</span>
37 |             </div>
38 |             <div class="fs-4 fw-bold mt-1 text-info font-monospace
| " id="ls-sec-size">-- / -- MB</div>
39 |             <div class="progress my-2 bg-secondary" style="height:
| 6px;">
40 |                 <div class="progress-bar bg-info" id="ls-sec-progres
| s" role="progressbar" style="width: 0%;"></div>
41 |             </div>
42 |             <div class="small text-muted d-flex justify-content-be
| tween" style="font-size: 0.75rem;">
43 |                 <span id="ls-sec-records">Записей: --</span>
44 |                 <span id="ls-sec-status">Статус: --</span>
45 |             </div>
46 |         </div>
47 |     </div>
48 |
49 | <!-- Карточка 2: Аудит CommandLine (Event 4688) -->
50 | <div class="col-12 col-sm-6 col-xl-3">
51 |     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
52 |         <div class="d-flex justify-content-between align-items
| -start">
53 |             <div class="text-muted small text-uppercase fw-semib
| old">Аудит CommandLine (4688) □ </div>
54 |             <span class="badge bg-secondary font-monospace" id="
| ls-cmdline-badge">Audit</span>

```

```

55 |         </div>
56 |         <div class="fs-4 fw-bold mt-1" id="ls-cmdline-status">
57 |         --</div>
58 |         <div class="small text-muted mt-1" id="ls-cmdline-desc
59 |         ">Аргументы в Event 4688</div>
60 |         <div class="mt-2">
61 |             <button class="btn btn-xs btn-outline-info w-100 py-
62 |             1" id="ls-toggle-cmdline-btn" style="font-size: 0.78rem;">
63 |                 <i class="bi bi-toggle-on me-1"></i><span>Переключ
64 |                 ить политику</span>
65 |             </button>
66 |         </div>
67 |     </div>
68 | <!-- Карточка 3: Закладка и Сборщик Telemetry.db -->
69 | <div class="col-12 col-sm-6 col-xl-3">
70 |     <div class="card bg-dark text-white border-secondary h-1
71 |     00 shadow-sm p-3">
72 |         <div class="d-flex justify-content-between align-items
73 |         -start">
74 |             <div class="text-muted small text-uppercase fw-semib
75 |             old">Сборщик Telemetry.db □ </div>
76 |             <span class="badge bg-success font-monospace" id="ls
77 |             -db-events-badge">0 в БД</span>
78 |         </div>
79 |         <div class="fs-4 fw-bold mt-1 text-warning font-monosp
80 |         ace" id="ls-bookmark-id">ID: --</div>
81 |         <div class="small text-muted mt-1 text-truncate" id="l
82 |         s-bookmark-time">Закладка: --</div>
83 |         <div class="d-flex gap-1 mt-2">
84 |             <button class="btn btn-xs btn-outline-success flex-g
85 |             row-1 py-1" id="ls-collect-btn" style="font-size: 0.78rem;">
86 |                 <i class="bi bi-cloud-arrow-down me-1"></i><span>C
87 |                 обрать сейчас</span>
88 |             </button>
89 |             <button class="btn btn-xs btn-outline-secondary py-1
90 |             " id="ls-reset-bm-btn" title="Сбросить закладку на конец журна

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ла" style="font-size: 0.78rem;">
80         <i class="bi bi-arrow-repeat"></i>
81     </button>
82 </div>
83 </div>
84 </div>
85
86 <!-- Карточка 4: 7 Доменов и Sysmon -->
87 <div class="col-12 col-sm-6 col-xl-3">
88     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
89         <div class="d-flex justify-content-between align-items
-start">
90             <div class="text-muted small text-uppercase fw-semib
old">Иерархия и Sysmon □ </div>
91             <span class="badge bg-info font-monospace" id="ls-sy
smon-badge">Sysmon: ?</span>
92         </div>
93         <div class="fs-4 fw-bold mt-1 text-light font-monospac
e" id="ls-domains-count-badge">7 Доменов OC</div>
94         <div class="small text-muted mt-1" id="ls-total-record
s-stat">Всего записей: --</div>
95         <div class="mt-2">
96             <button class="btn btn-xs btn-outline-primary w-100
py-1" id="ls-refresh-all-btn" style="font-size: 0.78rem;">
97                 <i class="bi bi-arrow-clockwise me-1"></i><span>Об
новить все параметры</span>
98             </button>
99         </div>
100     </div>
101 </div>
102 </div>
103
104 <!-- Навигация по 6 подразделам вкладки -->
105 <ul class="nav nav-pills mb-3 border-bottom border-secondary
pb-2 gap-2 flex-wrap" id="ls-subtabs" role="tablist">
106     <li class="nav-item" role="presentation">
107         <button class="nav-link active py-1 px-3 d-flex align-it

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ems-center gap-1.5" id="ls-tab-domains-btn" data-bs-toggle="pill" data-bs-target="#ls-subpane-domains" type="button" role="tab">
108     <i class="bi bi-diagram-3-fill text-info"></i>
109     <span>7 Доменов и Каталог</span>
110 </button>
111 </li>
112 <li class="nav-item" role="presentation">
113     <button class="nav-link py-1 px-3 d-flex align-items-center gap-1.5" id="ls-tab-channels-btn" data-bs-toggle="pill" data-bs-target="#ls-subpane-channels" type="button" role="tab">
114         <i class="bi bi-table"></i>
115         <span>Параметры и размеры каналов</span>
116     </button>
117 </li>
118 <li class="nav-item" role="presentation">
119     <button class="nav-link py-1 px-3 d-flex align-items-center gap-1.5" id="ls-tab-security-btn" data-bs-toggle="pill" data-bs-target="#ls-subpane-security" type="button" role="tab">
120         <i class="bi bi-shield-check text-success"></i>
121         <span>Аудит безопасности и Телеметрия</span>
122     </button>
123 </li>
124 <li class="nav-item" role="presentation">
125     <button class="nav-link py-1 px-3 d-flex align-items-center gap-1.5" id="ls-tab-provenance-btn" data-bs-toggle="pill" data-bs-target="#ls-subpane-provenance" type="button" role="tab">
126         <i class="bi bi-bezier2 text-warning"></i>
127         <span>Граф процессов (Provenance)</span>
128     </button>
129 </li>
130 <li class="nav-item" role="presentation">
131     <button class="nav-link py-1 px-3 d-flex align-items-center gap-1.5" id="ls-tab-lifecycle-btn" data-bs-toggle="pill" data-bs-target="#ls-subpane-lifecycle" type="button" role="tab">
132         <i class="bi bi-power text-danger"></i>

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133         <span>Жизненный цикл ОС (Boot / Crash)</span>
134     </button>
135 </li>
136 <li class="nav-item" role="presentation">
137     <button class="nav-link py-1 px-3 d-flex align-items-cen
138 ter gap-1.5" id="ls-tab-inspector-btn" data-bs-toggle="pill" d
139 ata-bs-target="#ls-subpane-inspector" type="button" role="tab"
140 >
141         <i class="bi bi-journal-text text-primary"></i>
142         <span>Инспектор событий</span>
143     </button>
144 </li>
145 </ul>
146
147 <!-- Содержимое подразделов -->
148 <div class="tab-content" id="ls-subtabs-content">
149
150     <!-- Подраздел 1: 7 Доменов и Каталог провайдеров -->
151     <div class="tab-pane fade show active" id="ls-subpane-doma
152 ins" role="tabpanel">
153
154         <!-- Сетка 7 доменов телеметрии -->
155         <div class="row g-2 mb-3" id="ls-domains-cards-grid">
156             <div class="col-12 text-center text-muted py-3">Загруз
157 ка каталога доменов...</div>
158         </div>
159
160         <!-- Каталог провайдеров выбранного домена -->
161         <div class="card bg-dark text-white border-secondary sha
162 dow-sm">
163             <div class="card-header bg-dark border-secondary d-fle
164 x align-items-center justify-content-between flex-wrap gap-2 p
165 y-2">
166                 <div class="d-flex align-items-center gap-2">
167                     <i class="bi bi-collection text-info fs-5"></i>
168                     <strong class="text-light" id="ls-catalog-header-t
169 itle">Иерархический каталог провайдеров Windows</strong>
170                     <span class="badge bg-secondary" id="ls-catalog-co
171 unt">--</span>

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161         </div>
162         <div class="d-flex align-items-center gap-2 flex-wra
163         p">
164             <select class="form-select form-select-sm bg-dark
165             text-white border-secondary" id="ls-catalog-domain-select" sty
166             le="width: 200px;">
167                 <option value="">Всё 7 доменов</option>
168                 <option value="security_identity">1. Security /
169                 Identity</option>
170                 <option value="process_tasks">2. Process / Tasks
171                 </option>
172                 <option value="storage_io">3. Storage & Filesystem</option>
173                 <option value="network_comms">4. Network & Socke
174                 ts</option>
175                 <option value="windows_update">5. Windows Update
176                 </option>
177                 <option value="hardware_pnp">6. Hardware / Drive
178                 rs</option>
179                 <option value="power_lifecycle">7. Power / Boot
180                 / Shutdown</option>
181                 <option value="sysmon_deep">Sysmon Observability
                 </option>
                 </select>
                 <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-catalog-search-input" placeholder="Поиск провайдера/полей..." style="width: 200px;">
            </div>
        </div>
        <div class="table-responsive" style="max-height: 480px; overflow-y: auto;">
            <table class="table table-dark table-hover table-striped align-middle mb-0" style="font-size: 0.82rem;">
                <thead class="table-secondary text-uppercase" style="font-size: 0.72rem; position: sticky; top: 0; z-index: 2;">
                    <tr>
                        <th scope="col" style="width: 25%;">Провайдер

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    / Канал</th>
182         <th scope="col" style="width: 15%;">Домен</th>
183         <th scope="col" style="width: 25%;">Ключевые E
vent ID</th>
184         <th scope="col" style="width: 10%;">Объем (Vol
ume)</th>
185         <th scope="col" style="width: 12%;">Сборщик /
ETW</th>
186         <th scope="col" style="width: 13%;">Цели корпе
ляции</th>
187     </tr>
188 </thead>
189 <tbody id="ls-catalog-tbody">
190     <tr>
191         <td colspan="6" class="text-center text-muted
py-4">Загрузка каталога...</td>
192     </tr>
193 </tbody>
194 </table>
195 </div>
196 </div>
197 </div>
198
199 <!-- Подраздел 2: Таблица каналов и их конфигурация -->
200 <div class="tab-pane fade" id="ls-subpane-channels" role="
tabpanel">
201     <div class="card bg-dark text-white border-secondary sha
dow-sm">
202         <div class="card-header bg-dark border-secondary d-fle
x align-items-center justify-content-between flex-wrap gap-2 p
y-2">
203             <div class="d-flex align-items-center gap-2">
204                 <i class="bi bi-sliders text-info fs-5"></i>
205                 <strong class="text-light">Конфигурация параметров
Windows Event Log</strong>
206                 <span class="badge bg-secondary" id="ls-table-coun
t">Загрузка...</span>
207             </div>

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208         <div class="d-flex align-items-center gap-2">
209             <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-channel-filter-input" placeholder="Поиск журнала..." style="width: 220px;">
210             <button class="btn btn-sm btn-outline-secondary" id="ls-reload-channels-btn" title="Перезагрузить">
211                 <i class="bi bi-arrow-clockwise"></i>
212             </button>
213         </div>
214     </div>
215     <div class="table-responsive" style="max-height: 520px; overflow-y: auto;">
216         <table class="table table-dark table-hover table-striped align-middle mb-0" style="font-size: 0.85rem;">
217             <thead class="table-secondary text-uppercase" style="font-size: 0.75rem; position: sticky; top: 0; z-index: 2;">
218                 <tr>
219                     <th scope="col" style="width: 22%;">Журнал (Channel)</th>
220                     <th scope="col" style="width: 10%;">Размер / Лимит</th>
221                     <th scope="col" style="width: 14%;">Заполнение (%)</th>
222                     <th scope="col" style="width: 12%;">Записей</th>
223                     <th scope="col" style="width: 12%;">Диапазон ID</th>
224                     <th scope="col" style="width: 12%;">Политика ротации</th>
225                     <th scope="col" style="width: 8%;">Статус</th>
226                     <th scope="col" class="text-end" style="width: 10%;">Действия</th>
227                 </tr>
228             </thead>
229             <tbody id="ls-channels-tbody">
230                 <tr>
231                     <td colspan="8" class="text-center text-muted py-4">

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232 |         <div class="spinner-border spinner-border-sm
      | text-info me-2" role="status"></div>
233 |         Загрузка параметров каналов журналов...
234 |     </td>
235 | </tr>
236 | </tbody>
237 | </table>
238 | </div>
239 | </div>
240 | </div>
241 |
242 | <!-- Подраздел 3: Аудит безопасности и телеметрия -->
243 | <div class="tab-pane fade" id="ls-subpane-security" role="
      |tabpanel">
244 |     <div class="row g-3">
245 |         <!-- Левая колонка: Настройки аудита реестра и прав --
      |>
246 |         <div class="col-12 col-lg-6">
247 |             <div class="card bg-dark text-white border-secondary
      | shadow-sm h-100">
248 |                 <div class="card-header bg-dark border-secondary p
      | y-2">
249 |                     <strong class="text-info d-flex align-items-cent
      | er gap-2">
250 |                         <i class="bi bi-shield-lock-fill"></i>
251 |                         <span>Политики аудита безопасности Windows</sp
      |an>
252 |                     </strong>
253 |                 </div>
254 |                 <div class="card-body p-3">
255 |                     <div class="mb-3">
256 |                         <div class="d-flex justify-content-between ali
      |gn-items-center mb-1">
257 |                             <strong class="text-light">Аудит командной с
      |троки процессов (Event 4688)</strong>
258 |                             <span class="badge" id="ls-audit-cmdline-bad
      |ge">--</span>
259 |                         </div>

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260 |         <div class="small text-muted mb-2">
261 |             Включает запись полного пути и аргументов за
пуска программ (`CommandLine`) в журнал `Security`.
262 |         </div>
263 |         <div class="bg-black rounded p-2 border border
-secondary font-monospace small mb-2 text-warning" style="font
-size: 0.75rem;">
264 |             HKLM\SOFTWARE\Microsoft\Windows\CurrentVersi
on\Policies\System\Audit<br>
265 |             Value: ProcessCreationIncludeCmdLine_Enabled
266 |         </div>
267 |         <div class="d-flex gap-2">
268 |             <button class="btn btn-sm btn-success d-flex
align-items-center gap-1" id="ls-enable-cmdline-btn">
269 |                 <i class="bi bi-check-circle"></i><span>Вк
лючить аудит CommandLine</span>
270 |             </button>
271 |             <button class="btn btn-sm btn-outline-danger
d-flex align-items-center gap-1" id="ls-disable-cmdline-btn">
272 |                 <i class="bi bi-x-circle"></i><span>Отключ
ить</span>
273 |             </button>
274 |         </div>
275 |     </div>
276 |
277 |     <hr class="border-secondary">
278 |
279 |     <div class="mb-2">
280 |         <strong class="text-light">Доступ к каналу Sec
urity (SeSecurityPrivilege)</strong>
281 |         <div class="small text-muted mt-1 mb-2">
282 |             Чтение журнала Security требует прав Админис
тратора или прав аудита безопасности.
283 |         </div>
284 |         <div id="ls-sec-priv-status" class="alert aler
t-dark border-secondary p-2 small mb-0">
285 |             Проверка прав...
286 |         </div>

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287 |         </div>
288 |     </div>
289 | </div>
290 | </div>
291 |
292 |     <!-- Правая колонка: Управление Bookmark и базой telem
etry.db -->
293 |     <div class="col-12 col-lg-6">
294 |         <div class="card bg-dark text-white border-secondary
shadow-sm h-100">
295 |             <div class="card-header bg-dark border-secondary p
y-2">
296 |                 <strong class="text-warning d-flex align-items-c
enter gap-2">
297 |                     <i class="bi bi-database-fill-gear"></i>
298 |                     <span>Инкрементальный сборщик и Bookmark (tele
metry.db)</span>
299 |                 </strong>
300 |             </div>
301 |             <div class="card-body p-3">
302 |                 <div class="row g-2 mb-3">
303 |                     <div class="col-6">
304 |                         <div class="p-2 bg-black rounded border bord
er-secondary text-center">
305 |                             <div class="text-muted small">Текущая закл
адка (RecordID)</div>
306 |                             <div class="fs-5 fw-bold text-warning font
-monospace" id="ls-bm-curr-id">--</div>
307 |                         </div>
308 |                     </div>
309 |                     <div class="col-6">
310 |                         <div class="p-2 bg-black rounded border bord
er-secondary text-center">
311 |                             <div class="text-muted small">Событий в te
lemetry.db</div>
312 |                             <div class="fs-5 fw-bold text-success font
-monospace" id="ls-bm-db-count">--</div>
313 |                         </div>

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314 |         </div>
315 |     </div>
316 |
317 |     <div class="mb-3">
318 |         <label class="form-label small text-muted">Сброс или установка целевого Record ID закладки:</label>
319 |         <div class="input-group input-group-sm mb-2">
320 |             <input type="number" class="form-control bg-dark text-white border-secondary" id="ls-custom-record-id-input" placeholder="Record ID (пусто = хвост журнала)">
321 |             <button class="btn btn-warning" id="ls-set-custom-bm-btn">
322 |                 <i class="bi bi-pin-angle me-1"></i>Установить закладку
323 |             </button>
324 |         </div>
325 |         <div class="small text-secondary" style="font-size: 0.75rem;">
326 |             Сброс на хвост позволяет пропустить старые миллионы записей и читать только свежие новые события.
327 |         </div>
328 |     </div>
329 |
330 |     <hr class="border-secondary">
331 |
332 |     <div>
333 |         <strong class="text-light mb-1 d-block">Ручной запуск сбора:</strong>
334 |         <div class="d-flex align-items-center gap-2">
335 |             <select class="form-select form-select-sm bg-dark text-white border-secondary" id="ls-collect-batch-size" style="width: 140px;">
336 |                 <option value="100">100 событий</option>
337 |                 <option value="500" selected>500 событий</option>
338 |                 <option value="1000">1000 событий</option>
339 |                 <option value="5000">5000 событий</option>
340 |             </select>

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341 |         <button class="btn btn-sm btn-primary d-flex
      | align-items-center gap-1.5" id="ls-run-collector-btn">
342 |             <i class="bi bi-play-fill"></i><span>Запустить сбор</span>
343 |         </button>
344 |     </div>
345 |     <div id="ls-collector-result-banner" class="small mt-2 text-info d-none"></div>
346 | </div>
347 | </div>
348 | </div>
349 | </div>
350 | </div>
351 | </div>
352 |
353 | <!-- Подраздел 4: Граф процессов (Process Provenance) -->
354 | <div class="tab-pane fade" id="ls-subpane-provenance" role="tabpanel">
355 |     <div class="card bg-dark text-white border-secondary shadow-sm">
356 |         <div class="card-header bg-dark border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2 py-2">
357 |             <div class="d-flex align-items-center gap-2">
358 |                 <i class="bi bi-diagram-2 text-warning fs-5"></i>
359 |                 <strong class="text-light">Windows Process Provenance (User → Process → Parent → Service → Task)</strong>
360 |             </div>
361 |             <div class="d-flex align-items-center gap-2">
362 |                 <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-prov-process-input" placeholder="Имя процесса (напр. python.exe)..." style="width: 240px;">
363 |                 <button class="btn btn-sm btn-warning d-flex align-items-center gap-1" id="ls-prov-build-btn">
364 |                     <i class="bi bi-play-circle"></i><span>Построить граф</span>
365 |                 </button>

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366         </div>
367     </div>
368     <div class="card-body p-3">
369         <div id="ls-prov-tree-view" class="p-3 bg-black rounded border border-secondary" style="min-height: 360px; max-height: 520px; overflow-y: auto;">
370             <div class="text-center text-muted py-5">
371                 <i class="bi bi-diagram-3 fs-1 text-secondary d-block mb-2"></i>
372                 Нажмите «Построить граф» для реконструкции цепочки происхождения процессов и служб.
373             </div>
374         </div>
375     </div>
376 </div>
377 </div>
378
379 <!-- Подраздел 5: Жизненный цикл ОС (Power / Boot / Crash) -->
380 <div class="tab-pane fade" id="ls-subpane-lifecycle" role="tabpanel">
381     <div class="card bg-dark text-white border-secondary shadow-sm">
382         <div class="card-header bg-dark border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2 py-2">
383             <div class="d-flex align-items-center gap-2">
384                 <i class="bi bi-clock-history text-danger fs-5"></i>
385                 <strong class="text-light">Хронология жизненного цикла ОС (BOOT → LOGIN → SLEEP → WAKE → SHUTDOWN / CRASH)</strong>
386             </div>
387             <div class="d-flex align-items-center gap-2">
388                 <select class="form-select form-select-sm bg-dark text-white border-secondary" id="ls-lifecycle-hours-select" style="width: 140px;">
389                     <option value="24">За 24 часа</option>

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390         <option value="72" selected>За 3 дня</option>
391         <option value="168">За 7 дней</option>
392         <option value="720">За 30 дней</option>
393     </select>
394     <button class="btn btn-sm btn-outline-danger" id="
ls-lifecycle-reload-btn">
395         <i class="bi bi-arrow-clockwise"></i>
396     </button>
397 </div>
398 </div>
399     <div class="card-body p-3" style="max-height: 500px; o
verflow-y: auto;">
400         <div class="timeline position-relative ps-4" id="ls-
lifecycle-timeline">
401             <div class="text-center text-muted py-4">Загрузка
хронологии питания и загрузки...</div>
402         </div>
403     </div>
404 </div>
405 </div>
406
407 <!-- Подраздел 6: Инспектор и просмотр событий -->
408 <div class="tab-pane fade" id="ls-subpane-inspector" role=
"tabpanel">
409     <div class="card bg-dark text-white border-secondary sha
dow-sm">
410         <div class="card-header bg-dark border-secondary p-2">
411             <!-- Фильтры выборки событий -->
412             <div class="d-flex align-items-center justify-conten
t-between flex-wrap gap-2">
413                 <div class="d-flex align-items-center gap-2 flex-w
rap">
414                     <select class="form-select form-select-sm bg-dar
k text-white border-secondary" id="ls-insp-channel-select" sty
le="width: 190px;">
415                         <option value="Security">Security (Журнал)</op
tion>
416                         <option value="telemetry_db">telemetry.db (Баз

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a)/option>
417         <option value="System">System (Системный)</opt
ion>
418         <option value="Application">Application (Прило
жения)</option>
419         <option value="Microsoft-Windows-Sysmon/Operat
ional">Sysmon</option>
420         <option value="Microsoft-Windows-TaskScheduler
/Operational">TaskScheduler</option>
421     </select>
422
423     <!-- Быстрые фильтры Event ID -->
424     <div class="btn-group btn-group-sm" role="group"
>
425         <button type="button" class="btn btn-outline-s
econdary ls-quick-eid-btn active" data-eid="0">Все</button>
426         <button type="button" class="btn btn-outline-i
nfo ls-quick-eid-btn" data-eid="4688" title="Создание процесса
">4688 (Процесс)</button>
427         <button type="button" class="btn btn-outline-s
uccess ls-quick-eid-btn" data-eid="4624" title="Успешный вход"
>4624 (Logon)</button>
428         <button type="button" class="btn btn-outline-d
anger ls-quick-eid-btn" data-eid="4625" title="Неудачный вход"
>4625 (Fail)</button>
429         <button type="button" class="btn btn-outline-w
arning ls-quick-eid-btn" data-eid="1102" title="Очистка журнал
а">1102 (Clear)</button>
430     </div>
431
432     <select class="form-select form-select-sm bg-dar
k text-white border-secondary" id="ls-insp-level-select" style
="width: 130px;">
433         <option value="">Все уровни</option>
434         <option value="Error">Ошибки (Error)</option>
435         <option value="Warning">Предупреждения</option>
>
436         <option value="Information">Информация</option>

```

```

>
437         <option value="Critical">Критические</option>
438     </select>
439 </div>
440
441     <div class="d-flex align-items-center gap-2 flex-w
rap">
442         <input type="text" class="form-control form-cont
rol-sm bg-dark text-white border-secondary" id="ls-insp-user-i
nput" placeholder="Пользователь..." style="width: 140px;">
443         <input type="text" class="form-control form-cont
rol-sm bg-dark text-white border-secondary" id="ls-insp-proces
s-input" placeholder="Процесс..." style="width: 140px;">
444         <input type="text" class="form-control form-cont
rol-sm bg-dark text-white border-secondary" id="ls-insp-search
-input" placeholder="Поиск в тексте..." style="width: 180px;">
445         <button class="btn btn-sm btn-primary d-flex ali
gn-items-center gap-1" id="ls-insp-fetch-btn">
446             <i class="bi bi-search"></i><span>Найти</span>
447         </button>
448     </div>
449 </div>
450 </div>
451
452     <div class="row g-0">
453         <!-- Левая панель: Список событий -->
454         <div class="col-12 col-lg-7 border-end border-second
ary">
455             <div class="table-responsive" style="max-height: 4
80px; overflow-y: auto;">
456                 <table class="table table-dark table-hover table
-stripped align-middle mb-0" style="font-size: 0.8rem;">
457                     <thead class="table-secondary text-uppercase"
style="font-size: 0.72rem; position: sticky; top: 0; z-index:
2;">
458                         <tr>
459                             <th scope="col" style="width: 20%;">Время<
/th>

```

```

460 |             <th scope="col" style="width: 12%;">Event
| ID</th>
461 |             <th scope="col" style="width: 14%;">Уровен
| ь</th>
462 |             <th scope="col" style="width: 54%;">Сообще
| ние / Процесс</th>
463 |         </tr>
464 |     </thead>
465 |     <tbody id="ls-events-tbody">
466 |         <tr>
467 |             <td colspan="4" class="text-center text-mu
| ted py-4">Нажмите «Найти» для загрузки событий.</td>
468 |         </tr>
469 |     </tbody>
470 | </table>
471 | </div>
472 | <div class="p-2 bg-dark border-top border-secondar
| y d-flex justify-content-between text-muted small" style="font
| -size: 0.75rem;">
473 |     <span id="ls-events-count-label">Событий: 0</spa
| n>
474 |     <span>Кликните на строку для детального инспекто
| ра</span>
475 | </div>
476 | </div>
477 |
478 | <!-- Правая панель: Детальный инспектор события -->
479 | <div class="col-12 col-lg-5 p-3" style="max-height:
| 520px; overflow-y: auto;">
480 |     <div class="d-flex align-items-center justify-cont
| ent-between mb-2">
481 |         <strong class="text-info fs-6 d-flex align-items
| -center gap-1.5">
482 |             <i class="bi bi-info-circle-fill"></i>
483 |             <span>Инспектор события</span>
484 |         </strong>
485 |         <span class="badge bg-secondary font-monospace"
| id="ls-detail-eid-badge">ID --</span>

```

```

486 |         </div>
487 |
488 |         <div id="ls-detail-empty" class="text-muted small
| text-center py-5">
489 |             <i class="bi bi-cursor text-secondary fs-3 d-blo
| ck mb-2"></i>
490 |             Выберите событие из таблицы слева для просмотра
| подробных атрибутов, пользователя, процесса и командной строки
| .
491 |         </div>
492 |
493 |         <div id="ls-detail-content" class="d-none">
494 |             <div class="mb-2">
495 |                 <span class="text-muted small d-block">Время с
| оздания:</span>
496 |                 <strong class="text-light font-monospace small
| " id="ls-detail-time">--</strong>
497 |             </div>
498 |
499 |             <div class="mb-2">
500 |                 <span class="text-muted small d-block">Пользов
| атель / Учетная запись:</span>
501 |                 <strong class="text-warning font-monospace sma
| ll" id="ls-detail-user">--</strong>
502 |             </div>
503 |
504 |             <div class="mb-2">
505 |                 <span class="text-muted small d-block">Процесс
| / Программа:</span>
506 |                 <strong class="text-info font-monospace small
| text-break" id="ls-detail-process">--</strong>
507 |             </div>
508 |
509 |             <div class="mb-2">
510 |                 <span class="text-muted small d-block">Командн
| ая строка (CommandLine):</span>
511 |                 <div class="bg-black p-2 rounded border border
| -secondary font-monospace text-success small text-break" id="l

```

```

s-detail-cmdline">--</div>
512         </div>
513
514         <div class="mb-2">
515             <span class="text-muted small d-block">Родител
516             ьский процесс:</span>
517             <div class="font-monospace small text-light te
518             xt-break" id="ls-detail-parent">--</div>
519             </div>
520
521             <div class="mb-2">
522                 <span class="text-muted small d-block">Сообщен
523                 ие / Расшифровка:</span>
524                 <div class="text-light small" id="ls-detail-ms
525                 g">--</div>
526                 </div>
527
528                 <div class="mt-3">
529                     <button class="btn btn-xs btn-outline-secondar
530                     y w-100 py-1" type="button" data-bs-toggle="collapse" data-bs-
531                     target="#ls-row-collapse">
532                         <i class="bi bi-code-square me-1"></i>Показа
533                         ть исходные данные (Raw Data)
534                     </button>
535                     <div class="collapse mt-2" id="ls-row-collapse
536                     ">
537                         <pre class="bg-black text-secondary p-2 roun
538                         ded border border-secondary font-monospace small mb-0" style="
539                         max-height: 180px; overflow-y: auto; font-size: 0.72rem;" id="
540                         ls-detail-row">--</pre>
541                     </div>
542                 </div>
543             </div>
544         </div>
545     </div>
546 </div>
547 </div>

```

```
538 |
539 | <!-- Модальное окно настройки параметров канала -->
540 | <div class="modal fade" id="ls-config-modal" tabindex="-1" a
    | ria-labelledby="lsConfigModalLabel" aria-hidden="true">
541 |     <div class="modal-dialog modal-dialog-centered">
542 |         <div class="modal-content bg-dark text-white border-seco
    | ndary shadow">
543 |             <div class="modal-header border-secondary py-2">
544 |                 <h5 class="modal-title fs-6 d-flex align-items-cente
    | r gap-2" id="lsConfigModalLabel">
545 |                     <i class="bi bi-gear-fill text-info"></i>
546 |                     <span>Настройка параметров журнала</span>
547 |                 </h5>
548 |                 <button type="button" class="btn-close btn-close-whi
    | te" data-bs-dismiss="modal" aria-label="Close"></button>
549 |             </div>
550 |             <div class="modal-body p-3">
551 |                 <input type="hidden" id="ls-modal-channel-name">
552 |                 <div class="mb-3">
553 |                     <label class="form-label small text-muted">Имя жур
    | нала:</label>
554 |                     <div class="fw-bold fs-6 text-light font-monospace
    | " id="ls-modal-channel-title">--</div>
555 |                     <div class="small text-secondary text-truncate" id
    | ="ls-modal-channel-path" style="font-size: 0.75rem;">--</div>
556 |                 </div>
557 |
558 |                 <div class="mb-3">
559 |                     <label class="form-label small text-muted" for="ls
    | -modal-maxsize-input">Максимальный размер (Max Size, МБ):</lab
    | el>
560 |                     <div class="input-group input-group-sm mb-2">
561 |                         <input type="number" class="form-control bg-dark
    | text-white border-secondary" id="ls-modal-maxsize-input" min=
    | "1" max="65536" step="1">
562 |                         <span class="input-group-text bg-secondary text-
    | white border-secondary">МБ</span>
563 |                     </div>
```

```

564 |         <!-- Быстрые пресеты размера -->
565 |         <div class="d-flex flex-wrap gap-1">
566 |             <button type="button" class="btn btn-xs btn-outline-
ine-secondary ls-preset-size-btn" data-mb="20">20 MB</button>
567 |             <button type="button" class="btn btn-xs btn-outline-
ine-secondary ls-preset-size-btn" data-mb="64">64 MB</button>
568 |             <button type="button" class="btn btn-xs btn-outline-
ine-secondary ls-preset-size-btn" data-mb="128">128 MB</button
>
569 |             <button type="button" class="btn btn-xs btn-outline-
ine-secondary ls-preset-size-btn" data-mb="256">256 MB</button
>
570 |             <button type="button" class="btn btn-xs btn-outline-
ine-secondary ls-preset-size-btn" data-mb="512">512 MB</button
>
571 |             <button type="button" class="btn btn-xs btn-outline-
ine-info ls-preset-size-btn" data-mb="1024">1 GB</button>
572 |             <button type="button" class="btn btn-xs btn-outline-
ine-info ls-preset-size-btn" data-mb="2048">2 GB</button>
573 |         </div>
574 |     </div>
575 |
576 |     <div class="mb-3">
577 |         <div class="form-check form-switch mb-2">
578 |             <input class="form-check-input" type="checkbox"
id="ls-modal-enabled-switch" checked>
579 |             <label class="form-check-label small" for="ls-mo
dal-enabled-switch">Включить сбор событий в данный журнал (Ena
bled)</label>
580 |         </div>
581 |         <div class="form-check form-switch mb-2">
582 |             <input class="form-check-input" type="checkbox"
id="ls-modal-autobackup-switch">
583 |             <label class="form-check-label small" for="ls-mo
dal-autobackup-switch">Автоматическая архивация при заполнении
(AutoBackup)</label>
584 |         </div>
585 |         <div class="form-check form-switch">

```

```

586         <input class="form-check-input" type="checkbox"
| id="ls-modal-retention-switch">
587         <label class="form-check-label small" for="ls-mo
| dal-retention-switch">Политика удержания (Не перезаписывать со
| бытия)</label>
588     </div>
589 </div>
590 </div>
591 <div class="modal-footer border-secondary py-2">
592     <button type="button" class="btn btn-sm btn-secondar
| y" data-bs-dismiss="modal">Отмена</button>
593     <button type="button" class="btn btn-sm btn-primary
| d-flex align-items-center gap-1" id="ls-modal-save-btn">
594         <i class="bi bi-check2"></i><span>Сохранить параме
| тры</span>
595     </button>
596 </div>
597 </div>
598 </div>
599 </div>
600
601 <!-- Модальное окно очистки журнала -->
602 <div class="modal fade" id="ls-clear-modal" tabindex="-1" ar
| ia-labelledby="lsClearModalLabel" aria-hidden="true">
603     <div class="modal-dialog modal-dialog-centered">
604         <div class="modal-content bg-dark text-white border-dang
| er shadow">
605             <div class="modal-header border-danger py-2">
606                 <h5 class="modal-title fs-6 text-danger d-flex align
| -items-center gap-2" id="lsClearModalLabel">
607                     <i class="bi bi-exclamation-triangle-fill"></i>
608                     <span>Очистка журнала событий</span>
609                 </h5>
610                 <button type="button" class="btn-close btn-close-whi
| te" data-bs-dismiss="modal" aria-label="Close"></button>
611             </div>
612             <div class="modal-body p-3">
613                 <input type="hidden" id="ls-clear-modal-channel-name

```



```

">
614         <p class="small text-light mb-3">
615             Вы уверены, что хотите очистить журнал <strong id=
"ls-clear-modal-title" class="text-warning">--</strong>? Все т
екущие записи будут удалены из системы.
616         </p>
617         <div class="form-check mb-2">
618             <input class="form-check-input" type="checkbox" id
="ls-clear-backup-checkbox" checked>
619             <label class="form-check-label small" for="ls-clea
r-backup-checkbox">
620                 Создать резервную копию (.evtx) перед очисткой
621             </label>
622         </div>
623         <div class="input-group input-group-sm mb-2" id="ls-
clear-backup-path-group">
624             <span class="input-group-text bg-secondary text-wh
ite border-secondary">Путь:</span>
625             <input type="text" class="form-control bg-dark tex
t-white border-secondary" id="ls-clear-backup-path-input">
626         </div>
627     </div>
628     <div class="modal-footer border-danger py-2">
629         <button type="button" class="btn btn-sm btn-secondar
y" data-bs-dismiss="modal">Отмена</button>
630         <button type="button" class="btn btn-sm btn-danger d
-flex align-items-center gap-1" id="ls-confirm-clear-btn">
631             <i class="bi bi-trash3-fill"></i><span>Очистить жу
рнал</span>
632         </button>
633     </div>
634 </div>
635 </div>
636 </div>
637
638 </div>
639

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Log Settings Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля настройки
   параметров
7  *   и иерархического каталога журналов событий Windows по 7 д
   оменам телеметрии.
8  *
9  * Usage Examples:
10 *   HTML Integration:
11 *     <script src="/windows/api/webgui/log_settings_tab/main.
   js?v=20261008_v2" type="module"></script>
12 *
13 *   JavaScript Import:
14 *     import { initLogSettingsTab } from '/windows/api/webgui
   /log_settings_tab/main.js';
15 *
16 * File: main.js
17 * Project: ai-breadboard
18 * Package: windows/api/webgui/log_settings_tab
19 * Author: hypo69
20 * Copyright: © 2026 hypo69
21 * Updated: 2026-10-08 02:35:00
22 * =====
23 */
24
25 const registerTabPoller = window.registerTabPoller || function
   () {};
26 const isTabActive = window.isTabActive || (() => true);
27
28 let isInitialized = false;
```

```

29 let channelsData = [];
30 let catalogData = [];
31 let domainsData = [];
32 let eventsData = [];
33 let selectedEvent = null;
34
35 function escapeHtml(str) {
36     if (!str) return '';
37     return String(str)
38         .replace(/&/g, '&amp;')
39         .replace(/</g, '&lt;')
40         .replace(/>/g, '&gt;')
41         .replace(/"/g, '&quot;')
42         .replace(/'/g, '&#039;');
43 }
44
45 /**
46  * Загрузка сводки по 7 доменам телеметрии
47  */
48 export async function fetchDomainsSummary() {
49     if (!isTabActive('tab-log-settings')) return;
50
51     try {
52         const res = await fetch('/api/v1/log-settings/domains');
53         if (!res.ok) return;
54         const data = await res.json();
55         domainsData = data.domains || [];
56
57         renderDomainsCards(domainsData);
58
59         const sysmonBadge = document.getElementById('ls-sysmon-bad
ge');
60         if (sysmonBadge) {
61             if (data.sysmon_installed) {
62                 sysmonBadge.textContent = 'Sysmon: Активен 🟢';
63                 sysmonBadge.className = 'badge bg-success font-monospa
ce';
64             } else {

```

```

65 |         sysmonBadge.textContent = 'Sysmon: Не установлен';
66 |         sysmonBadge.className = 'badge bg-secondary font-monospace';
67 |     }
68 | }
69 |
70 |     const totalRecords = domainsData.reduce((acc, d) => acc +
71 | (d.total_records || 0), 0);
72 |     const totalRecordsStat = document.getElementById('ls-total-records-stat');
73 |     if (totalRecordsStat) {
74 |         totalRecordsStat.textContent = `Всего в журналах: ${totalRecords.toLocaleString()} записей`;
75 |     } catch (err) {
76 |         console.debug('[LogSettings] Ошибка загрузки доменов:', err);
77 |     }
78 | }
79 |
80 | /**
81 |  * Отрисовка карточек 7 доменов телеметрии
82 |  */
83 | function renderDomainsCards(domains) {
84 |     const container = document.getElementById('ls-domains-cards-grid');
85 |     if (!container) return;
86 |
87 |     if (!domains || domains.length === 0) {
88 |         container.innerHTML = '<div class="col-12 text-center text-muted py-3">Домены не найдены.</div>';
89 |         return;
90 |     }
91 |
92 |     container.innerHTML = domains.map(d => {
93 |         const recordsStr = (d.total_records || 0).toLocaleString();
94 |         const isActive = d.status === 'Active' || d.status === 'Ак

```

```

    тивен';
95     const badgeBg = isActive ? 'bg-success' : 'bg-secondary';
96
97     return `
98         <div class="col-12 col-sm-6 col-lg-3">
99             <div class="card bg-dark text-white border-secondary h
-100 shadow-sm p-3 ls-domain-card" data-domain-id="${d.id}" st
yle="cursor: pointer; transition: transform 0.15s ease, border
-color 0.15s ease;">
100                 <div class="d-flex justify-content-between align-ite
ms-start mb-1">
101                     <strong class="text-light fs-6 text-truncate">${es
capeHtml(d.title)}</strong>
102                     <span class="badge ${badgeBg} font-monospace" styl
e="font-size: 0.68rem;">${escapeHtml(d.status)}</span>
103                 </div>
104                 <div class="small text-muted mb-2 text-truncate" sty
le="font-size: 0.75rem;" title="${escapeHtml(d.description)}">
105                     ${escapeHtml(d.description)}
106                 </div>
107                 <div class="d-flex justify-content-between align-ite
ms-center mt-auto pt-2 border-top border-secondary small font-
monospace">
108                     <span class="text-info">${d.channels_count} канало
в</span>
109                     <span class="text-warning">${recordsStr} записей</
span>
110                 </div>
111             </div>
112         </div>
113     `;
114 }).join('');
115
116 // Клик по карточке домена фильтрует каталог
117 container.querySelectorAll('.ls-domain-card').forEach(card =
> {
118     card.addEventListener('click', () => {
119         const domId = card.dataset.domainId;

```

```
120     const select = document.getElementById('ls-catalog-domai
n-select');
121     if (select) {
122         select.value = domId;
123         fetchEventCatalog();
124     }
125     });
126 });
127 }
128
129 /**
130  * Загрузка каталога провайдеров
131  */
132 export async function fetchEventCatalog() {
133     const domainSelect = document.getElementById('ls-catalog-dom
ain-select');
134     const searchInput = document.getElementById('ls-catalog-sear
ch-input');
135
136     const domain = domainSelect ? domainSelect.value : '';
137     const search = searchInput ? searchInput.value.trim() : '';
138
139     try {
140         const params = new URLSearchParams();
141         if (domain) params.append('domain', domain);
142         if (search) params.append('search', search);
143
144         const res = await fetch(`/api/v1/log-settings/catalog?${pa
rams.toString()}`);
145         if (!res.ok) throw new Error(`HTTP ${res.status}`);
146         const data = await res.json();
147         catalogData = data.catalog || [];
148         renderCatalogTable(catalogData);
149     } catch (err) {
150         console.warn('[LogSettings] Ошибка каталога:', err);
151     }
152 }
153
```

```

154 /**
155  * Отрисовка таблицы каталога провайдеров
156  */
157 function renderCatalogTable(catalog) {
158     const tbody = document.getElementById('ls-catalog-tbody');
159     const countLabel = document.getElementById('ls-catalog-count
    ');
160     if (!tbody) return;
161
162     if (countLabel) countLabel.textContent = `${catalog.length}
    провайдеров`;
163
164     if (!catalog || catalog.length === 0) {
165         tbody.innerHTML = '<tr><td colspan="6" class="text-center
    text-muted py-4">Провайдеры по заданным критериям не найдены.<
    /td></tr>';
166         return;
167     }
168
169     tbody.innerHTML = catalog.map(e => {
170         let volBadge = 'bg-secondary';
171         if (e.volume_rating === 'Extreme') volBadge = 'bg-danger';
172         else if (e.volume_rating === 'High') volBadge = 'bg-warnin
    g text-dark';
173         else if (e.volume_rating === 'Medium') volBadge = 'bg-info
    text-dark';
174         else if (e.volume_rating === 'Low') volBadge = 'bg-success
    ';
175
176         const eventsList = Object.entries(e.key_events || {}).map(
    ([id, desc]) => {
177             return `<span class="badge bg-black border border-second
    ary text-warning font-monospace me-1 mb-1" title="${escapeHtml
    (desc)}">ID ${id}</span>`;
178         }).join('');
179
180         const targets = (e.correlation_targets || []).map(t => {
181             return `<span class="badge bg-secondary-subtle text-ligh

```

```

    t font-monospace me-1" style="font-size: 0.68rem; ">#${escapeHt
ml(t)}</span>`;
182     }).join('');
183
184     return `
185         <tr>
186             <td>
187                 <div class="fw-bold text-light font-monospace" style
="font-size: 0.8rem;">${escapeHtml(e.provider)}</div>
188                 <div class="small text-info font-monospace text-trun
cate" style="max-width: 260px; font-size: 0.72rem;">${escapeHt
ml(e.channel)}</div>
189                 <div class="small text-muted text-truncate" style="m
ax-width: 260px; font-size: 0.7rem;" title="${escapeHtml(e.des
cription)}">${escapeHtml(e.description)}</div>
190             </td>
191             <td>
192                 <span class="badge bg-dark border border-secondary t
ext-light">${escapeHtml(e.domain_title || e.domain)}</span>
193             </td>
194             <td>
195                 <div class="d-flex flex-wrap" style="max-width: 280p
x;">${eventsList}</div>
196             </td>
197             <td>
198                 <span class="badge ${volBadge}">${escapeHtml(e.volum
e_rating)}</span>
199                 <div class="small text-muted font-monospace mt-1" st
yle="font-size: 0.7rem;">${(e.record_count || 0).toLocaleStrin
g()} rec</div>
200             </td>
201             <td>
202                 <div class="small text-light font-monospace">${escap
eHtml(e.recommended_collector)}</div>
203                 <div class="small text-muted" style="font-size: 0.7r
em;">
204                     ${e.realtime_support ? '<span class="text-success"
>ETW/Stream</span>' : '<span class="text-secondary">Pull</span>'}

```



```

    >' }
205         • <span class="text-warning">${escapeHtml(e.privilege_required)}</span>
    </div>
206     </td>
207     <td>
208         <div class="d-flex flex-wrap">${targets}</div>
209     </td>
210 </tr>
211 `;
212 }).join('');
213 }
214
215
216 /**
217  * Загрузка сводки и списка параметров всех каналов
218  */
219 export async function fetchChannelsList() {
220     if (!isTabActive('tab-log-settings')) return;
221
222     try {
223         const res = await fetch('/api/v1/log-settings/channels');
224         if (!res.ok) throw new Error(`HTTP ${res.status}`);
225         const data = await res.json();
226         channelsData = data.channels || [];
227         renderChannelsCards(channelsData);
228         renderChannelsTable(channelsData);
229     } catch (err) {
230         console.warn('[LogSettings] Ошибка загрузки каналов:', err);
231     }
232 }
233
234 /**
235  * Загрузка статуса политик аудита и сборщика телеметрии
236  */
237 export async function fetchAuditStatus() {
238     if (!isTabActive('tab-log-settings')) return;
239

```

```
240 |   try {
241 |     const res = await fetch('/api/v1/log-settings/audit-status
    | ');
242 |     if (!res.ok) return;
243 |     const data = await res.json();
244 |
245 |     const cmdStatusEl = document.getElementById('ls-cmdline-st
    | atus');
246 |     const cmdBadgeEl = document.getElementById('ls-audit-cmdli
    | ne-badge');
247 |     if (cmdStatusEl) {
248 |       cmdStatusEl.textContent = data.cmdline_audit_enabled ? '
    | Включен ☐ ' : 'Отключен ☐ ';
249 |       cmdStatusEl.className = `fs-4 fw-bold mt-1 ${data.cmdlin
    | e_audit_enabled ? 'text-success' : 'text-warning'}`;
250 |     }
251 |     if (cmdBadgeEl) {
252 |       cmdBadgeEl.textContent = data.cmdline_audit_enabled ? 'А
    | КТИВЕН' : 'НЕ АКТИВЕН';
253 |       cmdBadgeEl.className = `badge ${data.cmdline_audit_enabl
    | ed ? 'bg-success' : 'bg-warning text-dark'}`;
254 |     }
255 |
256 |     const bmIdEl = document.getElementById('ls-bookmark-id');
257 |     const bmTimeEl = document.getElementById('ls-bookmark-time
    | ');
258 |     const dbEventsBadge = document.getElementById('ls-db-event
    | s-badge');
259 |     const bmCurrIdEl = document.getElementById('ls-bm-curr-id'
    | );
260 |     const bmDbCountEl = document.getElementById('ls-bm-db-coun
    | t');
261 |
262 |     if (bmIdEl) bmIdEl.textContent = data.last_bookmark_record
    | _id ? `ID: ${data.last_bookmark_record_id.toLocaleString()}` :
    | 'ID: Начало';
263 |     if (bmTimeEl) bmTimeEl.textContent = `Закладка: ${data.las
    | t_bookmark_timestamp || 'Не задана'}`;
```

```

264     if (dbEventsBadge) dbEventsBadge.textContent = `${data.db_
security_events_count.toLocaleString()} в БД`;
265     if (bmCurrIdEl) bmCurrIdEl.textContent = data.last_bookmar
k_record_id ? data.last_bookmark_record_id.toLocaleString() :
'Нет (tail)';
266     if (bmDbCountEl) bmDbCountEl.textContent = data.db_securit
y_events_count.toLocaleString();
267
268     const privStatusEl = document.getElementById('ls-sec-priv-
status');
269     if (privStatusEl) {
270         if (data.security_channel_accessible) {
271             privStatusEl.className = 'alert alert-success border-s
uccess p-2 small mb-0 d-flex align-items-center gap-2';
272             privStatusEl.innerHTML = '<i class="bi bi-shield-check
fs-5"></i><span>Доступ к каналу Security разрешен. Привилегии
SeSecurityPrivilege активны.</span>';
273         } else {
274             privStatusEl.className = 'alert alert-danger border-da
nger p-2 small mb-0 d-flex align-items-center gap-2';
275             privStatusEl.innerHTML = '<i class="bi bi-shield-x fs-
5"></i><span>Доступ к каналу Security ограничен. Требуется зап
уск с правами Администратора.</span>';
276         }
277     }
278 } catch (err) {
279     console.debug('[LogSettings] Ошибка проверки аудита:', err
);
280 }
281 }
282
283 /**
284  * Отрисовка верхних карточек
285  */
286 function renderChannelsCards(channels) {
287     const sec = channels.find(c => c.channel_name === 'Security'
) || {};
288

```

```

289 |     const secSizeEl = document.getElementById('ls-sec-size');
290 |     const secProgEl = document.getElementById('ls-sec-progress')
    | ;
291 |     const secRecordsEl = document.getElementById('ls-sec-records
    | ');
292 |     const secStatusEl = document.getElementById('ls-sec-status')
    | ;
293 |
294 |     if (secSizeEl) secSizeEl.textContent = `${sec.file_size_mb |
    | | 0} / ${sec.max_size_mb || 20} MB`;
295 |     if (secProgEl) {
296 |         const pct = sec.usage_pct || 0;
297 |         secProgEl.style.width = `${Math.min(100, pct)}%`;
298 |         secProgEl.className = `progress-bar ${pct > 90 ? 'bg-dange
    | r' : (pct > 75 ? 'bg-warning' : 'bg-info')}`;
299 |     }
300 |     if (secRecordsEl) secRecordsEl.textContent = `Записей: ${((se
    | c.record_count || 0).toLocaleString())}`;
301 |     if (secStatusEl) secStatusEl.textContent = sec.is_enabled ?
    | 'Активен' : 'Отключен';
302 | }
303 |
304 | /**
305 |  * Отрисовка таблицы каналов
306 |  */
307 | function renderChannelsTable(channels) {
308 |     const tbody = document.getElementById('ls-channels-tbody');
309 |     const countLabel = document.getElementById('ls-table-count')
    | ;
310 |     if (!tbody) return;
311 |
312 |     const filterInput = document.getElementById('ls-channel-filt
    | er-input');
313 |     const filterQuery = (filterInput ? filterInput.value : '').t
    | oLowerCase().trim();
314 |
315 |     const filtered = channels.filter(c => {
316 |         if (!filterQuery) return true;

```

```

317     return c.channel_name.toLowerCase().includes(filterQuery)
    ||
318         (c.description || '').toLowerCase().includes(filter
Query);
319 });
320
321 if (countLabel) countLabel.textContent = `${filtered.length}
из ${channels.length}`;
322
323 if (filtered.length === 0) {
324     tbody.innerHTML = '<tr><td colspan="8" class="text-center
text-muted py-4">Каналы не найдены</td></tr>';
325     return;
326 }
327
328 tbody.innerHTML = filtered.map(c => {
329     const pct = c.usage_pct || 0;
330     const isFull = pct >= 95.0;
331     const badgeColor = isFull ? 'bg-danger' : (pct > 75 ? 'bg-
warning text-dark' : 'bg-success');
332     const rotPolicy = c.auto_backup ? 'Auto-Backup (Архивация)
' : (c.retention ? 'Retention (Удержание)' : 'FIFO (Перезапись
)');
333
334     return `
335         <tr>
336             <td>
337                 <div class="fw-bold text-light font-monospace">${esc
apeHtml(c.channel_name)}</div>
338                 <div class="small text-muted text-truncate" style="m
ax-width: 280px; font-size: 0.72rem;" title="${escapeHtml(c.de
scription)}">
339                     ${escapeHtml(c.description || c.log_file_path)}
340                 </div>
341             </td>
342             <td>
343                 <span class="font-monospace text-light">${c.file_siz
e_mb} MB</span>

```

```

344         <span class="text-muted small">/ ${c.max_size_mb} MB
    </span>
345     </td>
346     <td>
347         <div class="d-flex align-items-center gap-2">
348             <div class="progress flex-grow-1 bg-secondary" sty
le="height: 6px;">
349                 <div class="progress-bar ${pct > 90 ? 'bg-danger
' : (pct > 70 ? 'bg-warning' : 'bg-info')}" style="width: ${Ma
th.min(100, pct)}%;"></div>
350             </div>
351             <span class="badge ${badgeColor} font-monospace" s
tyle="font-size: 0.7rem;">${pct}%</span>
352         </div>
353     </td>
354     <td class="font-monospace text-info">${(c.record_count
|| 0).toLocaleString()}</td>
355     <td class="font-monospace text-secondary small" style=
"font-size: 0.72rem;">
356         ${c.oldest_record_id ? `${c.oldest_record_id.toLocal
eString()}..${c.newest_record_id.toLocaleString()}` : '--'}
357     </td>
358     <td>
359         <span class="badge bg-secondary-subtle text-light bo
rder border-secondary small" style="font-size: 0.72rem;">${rot
Policy}</span>
360     </td>
361     <td>
362         ${c.is_enabled ? '<span class="badge bg-success">Вкл
</span>' : '<span class="badge bg-secondary">Выкл</span>'}
363     </td>
364     <td class="text-end">
365         <div class="btn-group btn-group-sm">
366             <button class="btn btn-outline-info py-0 px-2 ls-b
tn-config" data-channel="${escapeHtml(c.channel_name)}" title=
"Настроить параметры">
367                 <i class="bi bi-gear"></i>
368             </button>

```

```

369 |         <button class="btn btn-outline-danger py-0 px-2 ls
      | -btn-clear" data-channel="{escapeHtml(c.channel_name)}" title
      | ="Очистить журнал">
370 |             <i class="bi bi-trash3"></i>
371 |         </button>
372 |     </div>
373 | </td>
374 | </tr>
375 | `;
376 | }).join('');
377 |
378 | tbody.querySelectorAll('.ls-btn-config').forEach(btn => {
379 |     btn.addEventListener('click', () => openConfigModal(btn.da
      | taset.channel));
380 | });
381 |
382 | tbody.querySelectorAll('.ls-btn-clear').forEach(btn => {
383 |     btn.addEventListener('click', () => openClearModal(btn.dat
      | aset.channel));
384 | });
385 | }
386 |
387 | /**
388 |  * Открытие модального окна настройки параметров канала
389 |  */
390 | function openConfigModal(channelName) {
391 |     const channel = channelsData.find(c => c.channel_name === ch
      | annelName);
392 |     if (!channel) return;
393 |
394 |     document.getElementById('ls-modal-channel-name').value = cha
      | nnel.channel_name;
395 |     document.getElementById('ls-modal-channel-title').textConten
      | t = channel.channel_name;
396 |     document.getElementById('ls-modal-channel-path').textContent
      | = channel.log_file_path || 'Стандартный путь Winevt\\Logs';
397 |     document.getElementById('ls-modal-maxsize-input').value = ch
      | annel.max_size_mb || 20;

```

```
398 | document.getElementById('ls-modal-enabled-switch').checked =
    | channel.is_enabled !== false;
399 | document.getElementById('ls-modal-autobackup-switch').checked
    | d = !!channel.auto_backup;
400 | document.getElementById('ls-modal-retention-switch').checked
    | = !!channel.retention;
401 |
402 | const modalEl = document.getElementById('ls-config-modal');
403 | if (modalEl && window.bootstrap) {
404 |     const modal = window.bootstrap.Modal.getOrCreateInstance(m
    | odalEl);
405 |     modal.show();
406 | }
407 | }
408 |
409 | /**
410 |  * Сохранение параметров конфигурации канала
411 |  */
412 | async function saveChannelConfig() {
413 |     const channelName = document.getElementById('ls-modal-channe
    | l-name').value;
414 |     const maxSizeMb = parseInt(document.getElementById('ls-modal
    | -maxsize-input').value, 10);
415 |     const isEnabled = document.getElementById('ls-modal-enabled-
    | switch').checked;
416 |     const autoBackup = document.getElementById('ls-modal-autobac
    | kup-switch').checked;
417 |     const retention = document.getElementById('ls-modal-retentio
    | n-switch').checked;
418 |
419 |     const saveBtn = document.getElementById('ls-modal-save-btn')
    | ;
420 |     const origHtml = saveBtn ? saveBtn.innerHTML : '';
421 |     if (saveBtn) saveBtn.innerHTML = '<span class="spinner-borde
    | r spinner-border-sm me-1"></span>Сохранение...';
422 |
423 |     try {
424 |         const res = await fetch('/api/v1/log-settings/channel/conf
```



```

    ig', {
425     method: 'POST',
426     headers: { 'Content-Type': 'application/json' },
427     body: JSON.stringify({
428         channel_name: channelName,
429         max_size_mb: maxSizeMb || null,
430         is_enabled: isEnabled,
431         auto_backup: autoBackup,
432         retention: retention,
433     }),
434 });
435
436 const data = await res.json();
437 if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
438
439 if (window.toast) {
440     window.toast.success('Параметры обновлены', data.message
|| `Настройки журнала ${channelName} сохранены.`);
441 }
442
443 const modalEl = document.getElementById('ls-config-modal')
;
444 if (modalEl && window.bootstrap) {
445     window.bootstrap.Modal.getInstance(modalEl)?.hide();
446 }
447
448 await fetchChannelsList();
449 } catch (err) {
450     if (window.toast) {
451         window.toast.error('Ошибка сохранения', err.message);
452     }
453 } finally {
454     if (saveBtn) saveBtn.innerHTML = origHtml;
455 }
456 }
457
458 /**

```

```
459 | * Открытие модального окна очистки журнала
460 | */
461 | function openClearModal(channelName) {
462 |     document.getElementById('ls-clear-modal-channel-name').value
463 |     = channelName;
464 |     document.getElementById('ls-clear-modal-title').textContent
465 |     = channelName;
466 |     const nowStr = new Date().toISOString().replace(/[:.]/g, '-').slice(0, 19);
467 |     const cleanName = channelName.replace(/[^a-zA-Z0-9]/g, '_');
468 |     document.getElementById('ls-clear-backup-path-input').value
469 |     = `C:\\Windows\\Temp\\evtx_backup_${cleanName}_${nowStr}.evtx`;
470 |     ;
471 |     const modalEl = document.getElementById('ls-clear-modal');
472 |     if (modalEl && window.bootstrap) {
473 |         const modal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
474 |         modal.show();
475 |     }
476 | }
477 | /**
478 | * Подтверждение очистки журнала
479 | */
480 | async function confirmClearChannel() {
481 |     const channelName = document.getElementById('ls-clear-modal-channel-name').value;
482 |     const withBackup = document.getElementById('ls-clear-backup-checkbox').checked;
483 |     const backupPath = withBackup ? document.getElementById('ls-clear-backup-path-input').value : null;
484 |     const clearBtn = document.getElementById('ls-confirm-clear-btn');
485 |     const origHtml = clearBtn ? clearBtn.innerHTML : '';
486 |     if (clearBtn) clearBtn.innerHTML = '<span class="spinner-border'`
```

```
der spinner-border-sm me-1"></span>Очистка...';
487
488   try {
489     const res = await fetch('/api/v1/log-settings/channel/clea
r', {
490       method: 'POST',
491       headers: { 'Content-Type': 'application/json' },
492       body: JSON.stringify({
493         channel_name: channelName,
494         backup_path: backupPath,
495       }),
496     });
497
498     const data = await res.json();
499     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
500
501     if (window.toast) {
502       window.toast.success('Журнал очищен', data.message || `Ж
урнал ${channelName} очищен.`);
503     }
504
505     const modalEl = document.getElementById('ls-clear-modal');
506     if (modalEl && window.bootstrap) {
507       window.bootstrap.Modal.getInstance(modalEl)?.hide();
508     }
509
510     await fetchChannelsList();
511   } catch (err) {
512     if (window.toast) {
513       window.toast.error('Ошибка очистки', err.message);
514     }
515   } finally {
516     if (clearBtn) clearBtn.innerHTML = origHtml;
517   }
518 }
519
520 /**
```

```
521 | * Переключение аудита CommandLine в реестре
522 | */
523 | async function toggleCmdlineAudit(enable = true) {
524 |   try {
525 |     const res = await fetch('/api/v1/log-settings/audit-status
526 | /cmdline', {
527 |       method: 'POST',
528 |       headers: { 'Content-Type': 'application/json' },
529 |       body: JSON.stringify({ enabled: enable }),
530 |     });
531 |     const data = await res.json();
532 |     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
533 | atus}`);
534 |     if (window.toast) {
535 |       window.toast.success('Политика обновлена', data.message)
536 |     };
537 |     await fetchAuditStatus();
538 |   } catch (err) {
539 |     if (window.toast) {
540 |       window.toast.error('Ошибка изменения политики', err.mess
541 | age);
542 |     }
543 |   }
544 | }
545 | /**
546 | * Запуск инкрементального сбора в telemetry.db
547 | */
548 | async function runCollector() {
549 |   const batchSelect = document.getElementById('ls-collect-batc
550 | h-size');
551 |   const batchSize = batchSelect ? batchSelect.value : 500;
552 |   const banner = document.getElementById('ls-collector-result-
553 | banner');
```

```
553 |   const btn = document.getElementById('ls-run-collector-btn');
554 |   const origHtml = btn ? btn.innerHTML : '';
555 |   if (btn) btn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Сбор...';
556 |
557 |   try {
558 |     const res = await fetch(`/api/v1/log-settings/collector/collect?limit=${batchSize}`, { method: 'POST' });
559 |     const data = await res.json();
560 |     if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
561 |
562 |     if (banner) {
563 |       banner.className = 'small mt-2 text-success';
564 |       banner.textContent = data.message || `Собрано ${data.total_events_ingested} событий.`;
565 |       banner.classList.remove('d-none');
566 |     }
567 |
568 |     if (window.toast) {
569 |       window.toast.success('Телеметрия обновлена', data.message);
570 |     }
571 |
572 |     await fetchAuditStatus();
573 |     await fetchDomainsSummary();
574 |   } catch (err) {
575 |     if (banner) {
576 |       banner.className = 'small mt-2 text-danger';
577 |       banner.textContent = `Ошибка сбора: ${err.message}`;
578 |       banner.classList.remove('d-none');
579 |     }
580 |     if (window.toast) {
581 |       window.toast.error('Ошибка сбора', err.message);
582 |     }
583 |   } finally {
584 |     if (btn) btn.innerHTML = origHtml;
585 |   }
```

```

586 }
587
588 /**
589  * Сброс закладки
590  */
591 async function resetBookmark(customRecordId = null) {
592   try {
593     const res = await fetch('/api/v1/log-settings/collector/re
set-bookmark', {
594       method: 'POST',
595       headers: { 'Content-Type': 'application/json' },
596       body: JSON.stringify({ channel: 'Security', record_id: c
ustomRecordId }),
597     });
598
599     const data = await res.json();
600     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
601
602     if (window.toast) {
603       window.toast.success('Закладка сброшена', data.message);
604     }
605     await fetchAuditStatus();
606   } catch (err) {
607     if (window.toast) {
608       window.toast.error('Ошибка сброса закладки', err.message
);
609     }
610   }
611 }
612
613 /**
614  * Построение графа происхождения процессов (Process Provenanc
e Graph)
615  */
616 export async function buildProcessProvenance() {
617   const input = document.getElementById('ls-prov-process-input
');
```

```

618 |     const processName = input ? input.value.trim() : '';
619 |     const container = document.getElementById('ls-prov-tree-view
    | ');
620 |     if (!container) return;
621 |
622 |     container.innerHTML = '<div class="text-center text-warning
    | py-4"><div class="spinner-border spinner-border-sm me-2"></div
    | >Построение графа происхождения процессов...</div>';
623 |
624 |     try {
625 |         const params = new URLSearchParams();
626 |         if (processName) params.append('process_name', processName
    | );
627 |         params.append('hours', '72');
628 |
629 |         const res = await fetch(`/api/v1/log-settings/provenance/p
    | rocess?${params.toString()}`);
630 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
631 |         const data = await res.json();
632 |
633 |         const nodes = data.nodes || [];
634 |         const links = data.links || [];
635 |
636 |         if (nodes.length === 0) {
637 |             container.innerHTML = `
638 |                 <div class="text-center text-muted py-5">
639 |                     <i class="bi bi-info-circle fs-3 text-secondary d-bl
    | ock mb-2"></i>
640 |                         События создания процессов (Event 4688) для <strong>
    | ${escapeHtml(processName || 'хоста')}</strong> не найдены в te
    | lemetry.db.<br>
641 |                         <small class="text-secondary">Убедитесь, что аудит C
    | ommandLine включен и выполнен сбор новых событий.</small>
642 |                     </div>
643 |                 `;
644 |             return;
645 |         }
646 |

```



```

669         }).join(''))
670     </div>
671     `;
672     } catch (err) {
673         container.innerHTML = `<div class="text-center text-danger
py-4">Ошибка: ${escapeHtml(err.message)}</div>`;
674     }
675 }
676
677 /**
678  * Загрузка жизненного цикла ОС (Power / Boot / Crash Timeline
679  *)
680 */
681 export async function fetchPowerLifecycle() {
682     const select = document.getElementById('ls-lifecycle-hours-s
elect');
683     const hours = select ? select.value : 72;
684     const container = document.getElementById('ls-lifecycle-time
line');
685     if (!container) return;
686     container.innerHTML = '<div class="text-center text-muted py
-4"><div class="spinner-border spinner-border-sm text-danger m
e-2"></div>Реконструкция жизненного цикла ОС...</div>';
687
688     try {
689         const res = await fetch(`/api/v1/log-settings/lifecycle/po
wer?hours=${hours}`);
690         if (!res.ok) throw new Error(`HTTP ${res.status}`);
691         const data = await res.json();
692         const timeline = data.timeline || [];
693
694         if (timeline.length === 0) {
695             container.innerHTML = '<div class="text-center text-mute
d py-4">События жизненного цикла за указанный период не найден
ы.</div>';
696             return;
697         }

```

```

698
699     container.innerHTML = timeline.map(item => {
700         let badgeClass = 'border-info text-info';
701         if (item.severity === 'critical') badgeClass = 'border-d
anger text-danger bg-danger-subtle';
702         else if (item.severity === 'warning') badgeClass = 'bord
er-warning text-warning bg-warning-subtle';
703
704         return `
705             <div class="mb-3 position-relative pb-2 border-bottom
border-secondary">
706                 <div class="d-flex align-items-center gap-2 mb-1">
707                     <span class="fs-5">${item.icon}</span>
708                     <span class="badge border ${badgeClass} font-monos
pace" style="font-size: 0.72rem;">${escapeHtml(item.phase)}</s
pan>
709                     <span class="badge bg-dark border border-secondary
font-monospace" style="font-size: 0.72rem;">ID ${item.event_i
d}</span>
710                     <span class="small text-secondary font-monospace m
s-auto">${escapeHtml(item.timestamp)}</span>
711                 </div>
712                 <div class="text-light small fw-medium">${escapeHtml
(item.message)}</div>
713                 <div class="text-muted font-monospace small" style="
font-size: 0.72rem;">Источник: ${escapeHtml(item.provider)}</d
iv>
714             </div>
715         `;
716     }).join('');
717 } catch (err) {
718     container.innerHTML = `<div class="text-center text-danger
py-4">Ошибка: ${escapeHtml(err.message)}</div>`;
719 }
720 }
721
722 /**
723  * Загрузка событий для инспектора

```

```

724 | */
725 | export async function fetchInspectorEvents() {
726 |     const channelSelect = document.getElementById('ls-insp-channel-select');
727 |     const levelSelect = document.getElementById('ls-insp-level-select');
728 |     const userInput = document.getElementById('ls-insp-user-input');
729 |     const processInput = document.getElementById('ls-insp-process-input');
730 |     const searchInput = document.getElementById('ls-insp-search-input');
731 |
732 |     const channel = channelSelect ? channelSelect.value : 'Security';
733 |     const level = levelSelect ? levelSelect.value : '';
734 |     const user = userInput ? userInput.value.trim() : '';
735 |     const process = processInput ? processInput.value.trim() : '';
736 |     const search = searchInput ? searchInput.value.trim() : '';
737 |
738 |     const activeEidBtn = document.querySelector('.ls-quick-eid-btn.active');
739 |     const eventId = activeEidBtn ? parseInt(activeEidBtn.dataset.eid, 10) || 0 : 0;
740 |
741 |     const tbody = document.getElementById('ls-events-tbody');
742 |     if (tbody) {
743 |         tbody.innerHTML = '<tr><td colspan="4" class="text-center text-muted py-4"><div class="spinner-border spinner-border-sm text-primary me-2"></div>Загрузка событий...</td></tr>';
744 |     }
745 |
746 |     try {
747 |         const params = new URLSearchParams({
748 |             channel: channel,
749 |             event_id: eventId,
750 |             level: level,

```

```
751     user: user,
752     process: process,
753     search: search,
754     limit: 100,
755   });
756
757   const res = await fetch(`/api/v1/log-settings/events?${params.toString()}`);
758   if (!res.ok) throw new Error(`HTTP ${res.status}`);
759   const data = await res.json();
760   eventsData = data.events || [];
761   renderInspectorEvents(eventsData);
762 } catch (err) {
763   console.warn('[LogSettings] Ошибка чтения событий:', err);
764   if (tbody) {
765     tbody.innerHTML = `|  |  |  |  |
| --- | --- | --- | --- |
| Ошибка: ${escapeHtml(err.message)} | | | |
`;
766   }
767 }
768 }
769
770 /**
771  * Отрисовка списка событий в инспекторе
772  */
773 function renderInspectorEvents(events) {
774   const tbody = document.getElementById('ls-events-tbody');
775   const countLabel = document.getElementById('ls-events-count-label');
776   if (!tbody) return;
777
778   if (countLabel) countLabel.textContent = `Событий: ${events.length}`;
779
780   if (events.length === 0) {
781     tbody.innerHTML = `| События по заданным фильтрам не найдены. | | | |
`;
782   }
```

```

782     return;
783 }
784
785 tbody.innerHTML = events.map((ev, idx) => {
786     const eid = ev.event_id || 0;
787     const lvl = ev.level || 'Information';
788     const ts = ev.timestamp || '';
789     const msg = ev.message || ev.raw_data || '';
790     const proc = ev.process_name || ev.provider || '';
791
792     let lvlBadge = 'bg-secondary';
793     if (lvl === 'Error' || lvl === 'Critical') lvlBadge = 'bg-
danger';
794     else if (lvl === 'Warning') lvlBadge = 'bg-warning text-da
rk';
795     else if (lvl === 'Information') lvlBadge = 'bg-info text-d
ark';
796
797     return `
798         <tr class="ls-event-row" data-index="${idx}" style="curs
or: pointer;">
799             <td class="font-monospace text-secondary text-truncate
" style="max-width: 140px;">${escapeHtml(ts)}</td>
800             <td>
801                 <span class="badge bg-dark border border-secondary f
ont-monospace">${eid}</span>
802             </td>
803             <td>
804                 <span class="badge ${lvlBadge} small" style="font-si
ze: 0.7rem;">${escapeHtml(lvl)}</span>
805             </td>
806             <td>
807                 <div class="text-truncate text-light fw-medium" styl
e="max-width: 320px;" title="${escapeHtml(msg)}">${escapeHtml(
msg)}</div>
808                 ${proc ? `<div class="small text-muted text-truncate
font-monospace" style="font-size: 0.72rem;">${escapeHtml(proc
)}</div>` : ''}

```

```

809         </td>
810     </tr>
811     `;
812 }).join('');
813
814     tbody.querySelectorAll('.ls-event-row').forEach(row => {
815         row.addEventListener('click', () => {
816             tbody.querySelectorAll('.ls-event-row').forEach(r => r.c
lassList.remove('table-active'));
817             row.classList.add('table-active');
818             const idx = parseInt(row.dataset.index, 10);
819             showEventDetail(eventsData[idx]);
820         });
821     });
822
823     if (events.length > 0) {
824         const firstRow = tbody.querySelector('.ls-event-row');
825         if (firstRow) firstRow.classList.add('table-active');
826         showEventDetail(events[0]);
827     }
828 }
829
830 /**
831  * Отображение подробных сведений о событии в панели инспектор
a
832  */
833 function showEventDetail(ev) {
834     if (!ev) return;
835     selectedEvent = ev;
836
837     const emptyEl = document.getElementById('ls-detail-empty');
838     const contentEl = document.getElementById('ls-detail-content
');
839     if (emptyEl) emptyEl.classList.add('d-none');
840     if (contentEl) contentEl.classList.remove('d-none');
841
842     document.getElementById('ls-detail-eid-badge').textContent =
`ID ${ev.event_id || 0}`;

```

```

843 | document.getElementById('ls-detail-time').textContent = ev.t
    | imESTAMP || '--';
844 | document.getElementById('ls-detail-user').textContent = ev.s
    | ubject_user || ev.target_user || (ev.event_data?.SubjectUserNa
    | me || ev.event_data?.TargetUserName || 'SYSTEM / N/A');
845 |
846 | const procName = ev.process_name || ev.event_data?.NewProces
    | sName || ev.provider || '--';
847 | const pid = ev.process_id || ev.event_data?.NewProcessId ||
    | '';
848 | document.getElementById('ls-detail-process').textContent = p
    | id ? `${procName} (PID ${pid})` : procName;
849 |
850 | const cmdLine = ev.command_line || ev.event_data?.CommandLin
    | e || 'Не зафиксирована или аудит CommandLine выключен';
851 | document.getElementById('ls-detail-cmdline').textContent = c
    | mdLine;
852 |
853 | const parent = ev.parent_process_name || ev.event_data?.Pare
    | ntProcessName || (ev.parent_process_id ? `PID ${ev.parent_proc
    | ess_id}` : '--');
854 | document.getElementById('ls-detail-parent').textContent = pa
    | rent;
855 |
856 | document.getElementById('ls-detail-msg').textContent = ev.me
    | ssage || '--';
857 | document.getElementById('ls-detail-raw').textContent = JSON.
    | stringify(ev, null, 2);
858 | }
859 |
860 | /**
861 |  * Инициализация всех слушателей и элементов вкладки
862 |  */
863 | export function initLogSettingsTab() {
864 |   if (isInitialized) return;
865 |   isInitialized = true;
866 |
867 |   // Кнопка обновления всех параметров

```

```
868 | document.getElementById('ls-refresh-all-btn')?.addEventListener
    | ner('click', () => {
869 |     fetchDomainsSummary();
870 |     fetchEventCatalog();
871 |     fetchChannelsList();
872 |     fetchAuditStatus();
873 | });
874 |
875 | document.getElementById('ls-reload-channels-btn')?.addEventL
    | istener('click', () => fetchChannelsList());
876 |
877 | // Поиск по списку каналов
878 | document.getElementById('ls-channel-filter-input')?.addEvent
    | Listener('input', () => {
879 |     renderChannelsTable(channelsData);
880 | });
881 |
882 | // Фильтры каталога провайдеров
883 | document.getElementById('ls-catalog-domain-select')?.addEven
    | tListener('change', fetchEventCatalog);
884 | document.getElementById('ls-catalog-search-input')?.addEvent
    | Listener('input', fetchEventCatalog);
885 |
886 | // Provenance граф
887 | document.getElementById('ls-prov-build-btn')?.addEventListener
    | er('click', buildProcessProvenance);
888 |
889 | // Хронология жизненного цикла
890 | document.getElementById('ls-lifecycle-hours-select')?.addEve
    | ntListener('change', fetchPowerLifecycle);
891 | document.getElementById('ls-lifecycle-reload-btn')?.addEvent
    | Listener('click', fetchPowerLifecycle);
892 |
893 | // Пресеты размеров в модальном окне
894 | document.querySelectorAll('.ls-preset-size-btn').forEach(btn
    | => {
895 |     btn.addEventListener('click', () => {
896 |         const mb = btn.dataset.mb;
```



```
897     const input = document.getElementById('ls-modal-maxsize-
input');
898     if (input) input.value = mb;
899     });
900 });
901
902 // Сохранение конфигурации канала
903 document.getElementById('ls-modal-save-btn')?.addEventListener(
'click', saveChannelConfig);
904
905 // Очистка канала
906 document.getElementById('ls-confirm-clear-btn')?.addEventLis
tener('click', confirmClearChannel);
907 document.getElementById('ls-clear-backup-checkbox')?.addEven
tListener('change', (e) => {
908     const pathGroup = document.getElementById('ls-clear-backup
-path-group');
909     if (pathGroup) pathGroup.style.display = e.target.checked
? 'flex' : 'none';
910 });
911
912 // Управление аудитом CommandLine
913 document.getElementById('ls-toggle-cmdline-btn')?.addEventLi
stener('click', async () => {
914     const isCurrentlyEnabled = document.getElementById('ls-cmd
line-status')?.textContent.includes('Включен');
915     await toggleCmdlineAudit(!isCurrentlyEnabled);
916 });
917 document.getElementById('ls-enable-cmdline-btn')?.addEventLi
stener('click', () => toggleCmdlineAudit(true));
918 document.getElementById('ls-disable-cmdline-btn')?.addEventL
istener('click', () => toggleCmdlineAudit(false));
919
920 // Управление сборщиком
921 document.getElementById('ls-collect-btn')?.addEventListener(
'click', runCollector);
922 document.getElementById('ls-run-collector-btn')?.addEventLis
tener('click', runCollector);
```

```
923 |
924 | // Сброс закладки
925 | document.getElementById('ls-reset-bm-btn')?.addEventListener
    | ('click', () => resetBookmark(null));
926 | document.getElementById('ls-set-custom-bm-btn')?.addEventLis
    | tener('click', () => {
927 |     const val = document.getElementById('ls-custom-record-id-i
    | nput')?.value;
928 |     const num = val ? parseInt(val, 10) : null;
929 |     resetBookmark(num);
930 | });
931 |
932 | // Быстрые фильтры Event ID
933 | document.querySelectorAll('.ls-quick-eid-btn').forEach(btn =
    | > {
934 |     btn.addEventListener('click', () => {
935 |         document.querySelectorAll('.ls-quick-eid-btn').forEach(b
    | => b.classList.remove('active'));
936 |         btn.classList.add('active');
937 |         fetchInspectorEvents();
938 |     });
939 | });
940 |
941 | // Инспектор событий
942 | document.getElementById('ls-insp-fetch-btn')?.addEventListen
    | er('click', fetchInspectorEvents);
943 | document.getElementById('ls-insp-channel-select')?.addEventL
    | istener('change', fetchInspectorEvents);
944 | document.getElementById('ls-insp-level-select')?.addEventLis
    | tener('change', fetchInspectorEvents);
945 |
946 | // Первоначальная загрузка
947 | fetchDomainsSummary();
948 | fetchEventCatalog();
949 | fetchChannelsList();
950 | fetchAuditStatus();
951 | fetchPowerLifecycle();
952 |
```

```
953 | // Регистрация поллера (каждые 20 секунд при активной вкладк
    | e)
954 | registerTabPoller('tab-log-settings', () => {
955 |     fetchDomainsSummary();
956 |     fetchChannelsList();
957 |     fetchAuditStatus();
958 | }, 20000);
959 | }
960 |
961 | if (document.readyState === 'loading') {
962 |     document.addEventListener('DOMContentLoaded', initLogSetting
    | sTab);
963 | } else {
964 |     initLogSettingsTab();
965 | }
966 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Maintenance & Recovery - Index View
5 |=====
6 |
7 Description:
8 | Шаблон интерфейса обслуживания и восстановления системы Wind
9 | ows (Maintenance & Recovery).
10 |
11 Usage Examples:
12 | Web Route:
13 | /html/maintenance_recovery_tab/index.html
14 |
15 File: index.html
16 Project: ai-breadboard
17 Package: windows/api/webgui/maintenance_recovery_tab
18 Author: hypo69
19 Copyright: © 2026 hypo69
20 Updated: 2026-10-08 03:08:00
21 |=====
22 |=====
23 -->
24
25 <style>
26 | .maint-container {
27 |   padding: 0.75rem 1rem;
28 | }
29 | .maint-header {
30 |   background: var(--surface-2, #1e293b);
31 |   border: 1px solid var(--border-color, #334155);
32 |   border-radius: var(--radius-md, 8px);
33 |   padding: 0.75rem 1.25rem;
34 | }
35 | .maint-card {
36 |   background: #0f172a;
```

```
33     border: 1px solid #334155;
34     border-radius: 8px;
35 }
36 .maint-table {
37     width: 100%;
38     border-collapse: collapse;
39     margin-bottom: 0;
40 }
41 .maint-table th {
42     padding: 0.65rem 1rem;
43     font-size: 0.75rem;
44     font-weight: 600;
45     text-transform: uppercase;
46     letter-spacing: 0.05em;
47     color: var(--text-muted, #94a3b8);
48     background: rgba(255, 255, 255, 0.02);
49     border-bottom: 1px solid var(--border-color, #334155);
50     white-space: nowrap;
51 }
52 .maint-table td {
53     padding: 0.65rem 1rem;
54     font-size: 0.85rem;
55     color: var(--text-color, #f8fafc);
56     border-bottom: 1px solid rgba(255, 255, 255, 0.05);
57     vertical-align: middle;
58 }
59 .maint-table tr:last-child td {
60     border-bottom: none;
61 }
62 .maint-table tbody tr:hover td {
63     background: rgba(255, 255, 255, 0.04);
64 }
65 </style>
66
67 <div class="maint-container theme-sysctl">
68     <!-- Top Header Bar -->
69     <div class="d-flex justify-content-between align-items-center
70         r flex-wrap gap-3 mb-3 maint-header text-white">
```

```

70 |     <div>
71 |         <div class="d-flex align-items-center gap-2">
72 |             <h5 class="mb-0 fw-bold">
73 |                 □ Windows Maintenance & Recovery
74 |             </h5>
75 |             <span class="badge rounded-pill bg-success-subtle text
- success border border-success px-2 py-1" style="font-size: 0.
76 | 75rem;">Health & Restoration</span>
77 |         </div>
78 |         <p class="mb-0 small text-light mt-1" >
79 |             Очистка временных файлов и кэшей, проверка целостности
80 |             SFC/DISM, управление точками восстановления Windows и настрой
81 |             ка дисковых квот теневого хранилища (VSS).
82 |         </p>
83 |     </div>
84 |     <div class="d-flex align-items-center gap-2 flex-wrap">
85 |         <span class="badge rounded-pill px-3 py-2" id="maint-ele
- vation-badge">
86 |             □ Checking privileges...
87 |         </span>
88 |         <button class="btn btn-sm btn-outline-info rounded-pill
89 |         px-3" id="btn-maint-refresh">
90 |             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
91 |         </button>
92 |     </div>
93 | </div>
94 |
95 | <div class="row g-3">
96 |     <!-- Quick Action Cards -->
97 |     <div class="col-md-4">
98 |         <div class="card maint-card p-3 h-100 d-flex flex-column
- justify-content-between">
99 |             <div>
100 |                 <h6 class="fw-bold text-info"><i class="bi bi-trash3
- fill me-1"></i> Очистка дисков и кэшей</h6>
101 |                 <p class="small text-muted mb-2">Безопасное удаление
102 |                 временных файлов, логов установки и кэша обновлений Windows.<
103 |                 /p>

```

```
98 |         <div class="small fw-semibold text-white mb-3" id="m
    | aint-clean-estimate-txt">Доступно к очистке: ~0 MB</div>
99 |         </div>
100 |         <button class="btn btn-sm btn-outline-info w-100 round
    | ed-pill" id="btn-maint-action-clean">
101 |             <i class="bi bi-stars me-1"></i> Очистить временные
    | файлы
102 |         </button>
103 |     </div>
104 | </div>
105 |
106 |     <div class="col-md-4">
107 |         <div class="card maint-card p-3 h-100 d-flex flex-column
    | justify-content-between">
108 |             <div>
109 |                 <h6 class="fw-bold text-warning"><i class="bi bi-shi
    | eld-shaded me-1"></i> Проверка SFC (System File Check)</h6>
110 |                 <p class="small text-muted mb-2">Проверка и верифика
    | ция целостности защищенных системных файлов Windows (sfc /veri
    | fyonly).</p>
111 |             </div>
112 |             <button class="btn btn-sm btn-outline-warning w-100 ro
    | unded-pill" id="btn-maint-action-sfc">
113 |                 <i class="bi bi-search me-1"></i> Запустить проверку
    | SFC
114 |             </button>
115 |         </div>
116 |     </div>
117 |
118 |     <div class="col-md-4">
119 |         <div class="card maint-card p-3 h-100 d-flex flex-column
    | justify-content-between">
120 |             <div>
121 |                 <h6 class="fw-bold text-success"><i class="bi bi-hea
    | rt-pulse-fill me-1"></i> Диагностика DISM Store</h6>
122 |                 <p class="small text-muted mb-2">Проверка состояния
    | хранилища компонентов Windows с помощью DISM /CheckHealth.</p>
123 |             </div>
```

```

124 |         <button class="btn btn-sm btn-outline-success w-100 ro
      | unded-pill" id="btn-maint-action-dism">
125 |             <i class="bi bi-activity me-1"></i> Проверить DISM S
      | tore
126 |         </button>
127 |     </div>
128 | </div>
129 |
130 | <!-- System Restore Points Panel -->
131 | <div class="col-12">
132 |     <div class="card maint-card">
133 |         <div class="card-header border-secondary bg-transparen
      | t d-flex justify-content-between align-items-center flex-wrap
      | gap-2">
134 |             <span class="fw-bold text-white"><i class="bi bi-arr
      | ow-counterclockwise me-1"></i> Точки восстановления Windows (S
      | ystem Restore Points)</span>
135 |             <div class="d-flex align-items-center gap-2">
136 |                 <button class="btn btn-sm btn-outline-info rounded
      | -pill px-3" id="btn-maint-config-restore" title="Настройка кво
      | ты диска, расписания и политик хранения">
137 |                     <i class="bi bi-gear-fill me-1"></i> Политики и
      | квоты VSS
138 |                 </button>
139 |                 <button class="btn btn-sm btn-outline-primary roun
      | ded-pill px-3" id="btn-maint-create-restore">
140 |                     <i class="bi bi-plus-lg me-1"></i> Создать точку
      | восстановления
141 |                 </button>
142 |             </div>
143 |         </div>
144 |         <div class="p-0 table-responsive" style="max-height: 3
      | 80px; overflow-y: auto;">
145 |             <table class="maint-table">
146 |                 <thead>
147 |                     <tr>
148 |                         <th style="width: 100px;">Номер #</th>
149 |                         <th style="width: 45%;">Описание</th>

```



```

150         <th style="width: 25%;">Тип</th>
151         <th style="width: 20%;">Дата создания</th>
152     </tr>
153 </thead>
154 <tbody id="maint-restore-tbody">
155     <tr><td colspan="4" class="text-center text-muted p-3">Загрузка точек восстановления...</td></tr>
156 </tbody>
157 </table>
158 </div>
159 </div>
160 </div>
161 </div>
162
163 <!-- Modal: System Restore Policy, Disk Quota & Automated Schedule -->
164 <div class="modal fade" id="modal-maint-restore-config" tabindex="-1" aria-hidden="true">
165     <div class="modal-dialog modal-dialog-centered modal-lg">
166         <div class="modal-content bg-dark text-white border-secondary">
167             <div class="modal-header border-secondary">
168                 <h5 class="modal-title fw-bold text-info">
169                     <i class="bi bi-shield-shaded me-2"></i> Политики
170                     теневого хранилища, квоты и расписание VSS
171                 </h5>
172                 <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
173             </div>
174             <div class="modal-body p-4">
175                 <p class="small text-muted mb-3">
176                     Настройка дисковой квоты теневого хранилища (VSS),
177                     сценариев автоматического создания и правил ротации точек восстановления Windows.
178                 </p>
179
180                 <!-- 1. Disk Quota Allocation -->
181                 <div class="card bg-black border-secondary p-3 mb-3"

```

```

>
180         <h6 class="fw-bold text-primary mb-2">
181             <i class="bi bi-hdd-fill me-1"></i> 1. Выделение
            дискового пространства (Shadow Storage)
182         </h6>
183         <div class="row g-3 align-items-center">
184             <div class="col-md-6">
185                 <label class="form-label small text-muted">Мак
                симальный лимит для диска C:</label>
186                 <select class="form-select form-select-sm bg-d
                ark text-white border-secondary" id="maint-policy-storage-size
                ">
187                     <option value="5%">5% объема диска (~15-30 G
                B)</option>
188                     <option value="10%" selected>10% объема диск
                а (~30-60 GB, рекомендуется)</option>
189                     <option value="15%">15% объема диска (~45-90
                GB)</option>
190                     <option value="20%">20% объема диска (~60-12
                0 GB)</option>
191                     <option value="10GB">10 GB фиксированно</opt
                ion>
192                     <option value="20GB">20 GB фиксированно</opt
                ion>
193                     <option value="50GB">50 GB фиксированно</opt
                ion>
194                 </select>
195             </div>
196             <div class="col-md-6">
197                 <label class="form-label small text-muted">Тек
                ущее заполнение VSS:</label>
198                 <div class="progress bg-secondary" style="heig
                ht: 10px;">
199                     <div id="maint-policy-storage-bar" class="pr
                ogress-bar bg-info" role="progressbar" style="width: 0%;"></di
                v>
200                 </div>
201                 <div class="d-flex justify-content-between sma

```

```

    ll text-muted mt-1 font-monospace" id="maint-policy-storage-label">
202         <span>Used: 0 GB</span>
203         <span>Max: 0 GB</span>
204     </div>
205 </div>
206 </div>
207 </div>
208
209 <!-- 2. Schedule and Triggers -->
210 <div class="card bg-black border-secondary p-3 mb-3"
>
211     <h6 class="fw-bold text-success mb-2">
212         <i class="bi bi-clock-history me-1"></i> 2. Сцен
арий автоматического создания (Trigger)
213     </h6>
214     <div class="row g-3">
215         <div class="col-md-6">
216             <label class="form-label small text-muted">Сце
нарий запуска:</label>
217             <select class="form-select form-select-sm bg-d
ark text-white border-secondary" id="maint-policy-trigger">
218                 <option value="daily">Ежедневно по расписа
нию (Daily)</option>
219                 <option value="startup">При каждом запуске с
истемы (At Startup)</option>
220                 <option value="weekly">Еженедельно (Weekly S
unday)</option>
221                 <option value="disabled">Только вручную / пе
ред изменениями (Disabled)</option>
222             </select>
223         </div>
224         <div class="col-md-6" id="maint-policy-time-wrap
per">
225             <label class="form-label small text-muted">Впе
мя создания (для ежедневного/еженедельного):</label>
226             <input type="time" class="form-control form-co
ntrol-sm bg-dark text-white border-secondary font-monospace" i

```

```

    d="maint-policy-time" value="03:00">
227         </div>
228         <div class="col-12">
229             <div class="form-check form-switch mt-1">
230                 <input class="form-check-input" type="checkbox" id="maint-policy-freq-limit" checked style="cursor: pointer;">
231                 <label class="form-check-label small text-light" for="maint-policy-freq-limit">
232                     Снять 24-часовое системное ограничение частоты Windows (<code>CreationFrequency = 0</code>)
233                 </label>
234             </div>
235         </div>
236     </div>
237 </div>
238
239     <!-- 3. Retention Policy -->
240     <div class="card bg-black border-secondary p-3 mb-3">
241         <h6 class="fw-bold text-warning mb-2">
242             <i class="bi bi-layers-half me-1"></i> 3. Политика хранения и ротация (Retention)
243         </h6>
244         <div class="row g-3">
245             <div class="col-md-6">
246                 <label class="form-label small text-muted">Сколько последних точек хранить:</label>
247                 <select class="form-select form-select-sm bg-dark text-white border-secondary" id="maint-policy-max-points">
248                     <option value="3">Хранить 3 последние точки</option>
249                     <option value="5" selected>Хранить 5 последних точек (рекомендуется)</option>
250                     <option value="7">Хранить 7 последних точек</option>
251                     <option value="10">Хранить 10 последних точек</option>

```

```

252 |             <option value="15">Хранить 15 последних точе
      | к</option>
253 |             <option value="20">Хранить 20 последних точе
      | к</option>
254 |         </select>
255 |     </div>
256 |     <div class="col-md-6 d-flex flex-column justify-
      | content-end">
257 |         <div class="form-check form-switch mb-1">
258 |             <input class="form-check-input" type="checkb
      | ox" id="maint-policy-auto-prune" checked style="cursor: pointe
      | r;">
259 |             <label class="form-check-label small text-li
      | ght" for="maint-policy-auto-prune">
260 |                 Автоматическая ротация при превышении (Aut
      | o-Prune FIFO)
261 |             </label>
262 |         </div>
263 |     </div>
264 | </div>
265 | </div>
266 | </div>
267 | <div class="modal-footer border-secondary d-flex justi
      | fy-content-between">
268 |     <button type="button" class="btn btn-sm btn-outline-
      | danger rounded-pill px-3" id="btn-maint-policy-prune-now">
269 |         <i class="bi bi-trash3 me-1"></i> Очистить избыточ
      | ные точки
270 |     </button>
271 |     <div class="d-flex gap-2">
272 |         <button type="button" class="btn btn-sm btn-outlin
      | e-secondary rounded-pill px-3" data-bs-dismiss="modal">Закреть
      | </button>
273 |         <button type="button" class="btn btn-sm btn-primar
      | y rounded-pill px-4" id="btn-maint-save-restore-policy">
274 |             <i class="bi bi-check-lg me-1"></i> Сохранить и
      | применить
275 |         </button>

```

```
276 |         </div>
277 |     </div>
278 | </div>
279 | </div>
280 | </div>
281 |
282 | </div>
283 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Maintenance & Recovery - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт обслуживания и восстановления Windows (
7   *   Maintenance & Recovery).
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/html/maintenance_recovery_tab/main.js?v=2
11  *     0261006_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/maintenance_recovery_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 02:00:00
19  * =====
20  */
21  (function () {
22    'use strict';
23
24    let isInitialized = false;
25
26    /**
27     * Экранирование HTML-символов для защиты от XSS.
28     *
29     * @param {string} str - Исходная строка.
30     * @returns {string} Экранированная строка.
31     */
```

```

32 | function escapeHtml(str) {
33 |     if (!str) return '';
34 |     return String(str)
35 |         .replace(/&/g, '&amp;')
36 |         .replace(/</g, '&lt;')
37 |         .replace(/>/g, '&gt;')
38 |         .replace(/"/g, '&quot;')
39 |         .replace(/'/g, '&#039;');
40 | }
41 |
42 | /**
43 |  * Форматирование типа точки восстановления.
44 |  *
45 |  * @param {string|number} rawType - Сырой тип.
46 |  * @returns {string} Читаемый тип.
47 |  */
48 | function formatRestorePointType(rawType) {
49 |     if (!rawType) return 'Системный снимок';
50 |     const s = String(rawType).toUpperCase();
51 |     if (s.includes('APPLICATION_INSTALL') || s === '0') return
52 | 'Установка ПО';
53 |     if (s.includes('APPLICATION_UNINSTALL') || s === '1') return 'Удаление ПО';
54 |     if (s.includes('DEVICE_DRIVER') || s === '10') return 'Драйвер устройства';
55 |     if (s.includes('MODIFY_SETTINGS') || s === '12') return 'Изменение настроек';
56 |     if (s.includes('CANCELLED') || s === '13') return 'Отмена операции';
57 |     if (s.includes('WINDOWS_UPDATE') || s === '17') return 'Windows Update';
58 |     return s.replace(/_/g, ' ');
59 | }
60 |
61 | /**
62 |  * Форматирование времени точки восстановления.
63 |  *
64 |  * @param {string} rawTime - Сырое время (YYYYMMDDHHmmss).

```



```

64 | * @returns {string} Форматированное время.
65 | */
66 | function formatRestoreTime(rawTime) {
67 |     if (!rawTime) return '-';
68 |     const s = String(rawTime).trim();
69 |     if (s.length >= 14 && /\d{14}/.test(s)) {
70 |         return `${s.slice(0, 4)}-${s.slice(4, 6)}-${s.slice(6, 8
71 | )} ${s.slice(8, 10)}:${s.slice(10, 12)}:${s.slice(12, 14)}`;
72 |     }
73 |     return s;
74 | }
75 | /**
76 |  * Получение статуса обслуживания и прав доступа.
77 |  */
78 | async function fetchMaintStatus() {
79 |     try {
80 |         const res = await fetch('/api/system-control/status');
81 |         if (!res.ok) return;
82 |         const data = await res.json();
83 |
84 |         const elevBadge = document.getElementById('maint-elevati
85 | on-badge');
86 |         if (elevBadge) {
87 |             if (data.is_elevated) {
88 |                 elevBadge.className = 'badge rounded-pill bg-success
89 | -subtle text-success border border-success px-3 py-2';
90 |                 elevBadge.innerHTML = '🛡️ Режим: Администратор (Полн
91 | ый доступ)';
92 |             } else {
93 |                 elevBadge.className = 'badge rounded-pill bg-warning
94 | -subtle text-warning border border-warning px-3 py-2';
95 |                 elevBadge.innerHTML = '👤 Режим: Обычный пользовател
96 | ь (Ограничено)';
97 |             }
98 |         }
99 |
100 |         const cleanEstimateTxt = document.getElementById('maint-

```

```

    clean-estimate-txt');
96     if (cleanEstimateTxt) {
97         const cleanMb = data.disk?.cleanup_estimate?.total_cleanable_mb || 0;
98         cleanEstimateTxt.innerText = `Доступно к очистке: ~${cleanMb} MB`;
99     }
100 } catch (e) {
101     console.warn('[MaintenanceRecovery] Ошибка получения статуса:', e);
102 }
103 }
104
105 /**
106  * Загрузка списка точек восстановления Windows.
107  */
108 async function fetchRestorePoints() {
109     try {
110         const res = await fetch('/api/system-control/restore-points');
111         if (!res.ok) return;
112         const data = await res.json();
113         const points = data.restore_points || [];
114
115         const tbody = document.getElementById('maint-restore-tbody');
116         if (tbody) {
117             if (points.length === 0) {
118                 tbody.innerHTML = '<tr><td colspan="4" class="text-center text-muted p-3">Точки восстановления не найдены. Нажмите «Создать точку восстановления» для фиксации состояния.</td></tr>';
119                 return;
120             }
121             tbody.innerHTML = points.map((p, idx) => {
122                 const typeBadge = formatRestorePointType(p.restore_point_type);
123                 const timeFormatted = formatRestoreTime(p.creation_t

```

```

ime);
124         return `
125         <tr class="maint-restore-row" data-idx="${idx}" styl
e="cursor: pointer;" title="Нажмите для анализа точки восстано
вления">
126             <td class="font-monospace text-info fw-bold">#${p.
sequence_number}</td>
127             <td class="fw-semibold text-white">${escapeHtml(p.
description)}</td>
128             <td><span class="badge bg-secondary text-light px-
2 py-1 font-monospace" style="font-size: 0.75rem;">${escapeHtm
l(typeBadge)}</span></td>
129             <td class="small text-light font-monospace">${esca
peHtml(timeFormatted)}</td>
130         </tr>
131     `;
132     }).join('');
133
134     tbody.querySelectorAll('.maint-restore-row').forEach(r
ow => {
135         row.onclick = () => {
136             const idx = parseInt(row.getAttribute('data-idx'),
10);
137             const p = points[idx];
138             if (!p) return;
139             if (window.AITableModal) {
140                 window.AITableModal.show({
141                     icon: '🔍',
142                     title: `Точка восстановления #${p.sequence_num
ber}`,
143                     subtitle: p.description,
144                     tableType: 'generic',
145                     badges: [
146                         { text: p.restore_point_type || 'System Chec
kpoint', class: 'badge bg-info text-dark' }
147                     ],
148                     metadata: [
149                         { label: 'Номер', value: String(p.sequence_n

```

```

umber) },
150         { label: 'Описание', value: p.description },
151         { label: 'Тип', value: p.restore_point_type
    },
152         { label: 'Дата создания', value: p.creation_
time }
153     ],
154     rawTitle: 'Метаданные точки восстановления',
155     rawContent: JSON.stringify(p, null, 2)
156     });
157     }
158     };
159     });
160     }
161     } catch (e) {
162         console.error('[MaintenanceRecovery] Ошибка загрузки точ
ек восстановления:', e);
163     }
164 }
165
166 /**
167  * Открытие модального окна настройки политик теневого храни
лища (VSS).
168  */
169 async function openRestoreConfigModal() {
170     try {
171         const res = await fetch('/api/system-control/restore-poi
nts/config');
172         if (res.ok) {
173             const cfg = await res.json();
174             const sizeSelect = document.getElementById('maint-poli
cy-storage-size');
175             if (sizeSelect && cfg.max_storage_size) sizeSelect.val
ue = cfg.max_storage_size;
176
177             const trigSelect = document.getElementById('maint-poli
cy-trigger');
178             if (trigSelect && cfg.schedule_trigger) trigSelect.val

```

```
    ue = cfg.schedule_trigger;
179
180     const timeInput = document.getElementById('maint-policy-time');
181     if (timeInput && cfg.schedule_time) timeInput.value =
    cfg.schedule_time;
182
183     const ptsSelect = document.getElementById('maint-policy-max-points');
184     if (ptsSelect && cfg.max_points) ptsSelect.value = String(cfg.max_points);
185
186     const pruneSwitch = document.getElementById('maint-policy-auto-prune');
187     if (pruneSwitch && typeof cfg.auto_prune === 'boolean')
    pruneSwitch.checked = cfg.auto_prune;
188
189     const freqSwitch = document.getElementById('maint-policy-freq-limit');
190     if (freqSwitch && typeof cfg.frequency_limit_minutes === 'number')
    freqSwitch.checked = cfg.frequency_limit_minutes === 0;
191
192     const shadow = cfg.shadow_storage || {};
193     const usedGb = shadow.allocated_space_gb || 0;
194     const maxGb = shadow.max_space_gb || 1;
195     const pct = shadow.percent_used || (maxGb > 0 ? Math.min(100, Math.round((usedGb / maxGb) * 100)) : 0);
196
197     const bar = document.getElementById('maint-policy-storage-bar');
198     if (bar) {
199         bar.style.width = `${pct}%`;
200         bar.className = `progress-bar ${pct > 80 ? 'bg-danger' : pct > 50 ? 'bg-warning' : 'bg-info'}`;
201     }
202     const lbl = document.getElementById('maint-policy-storage-label');
```

```
203         if (lbl) {
204             lbl.innerHTML = `Занято: ${usedGb} GB (${pct}%
205         )</span><span>Лимит: ${maxGb} GB</span>`;
206         }
207     } catch (e) {
208         console.warn('[MaintenanceRecovery] Ошибка получения кон
209         фигурации VSS:', e);
210     }
211     const modalEl = document.getElementById('modal-maint-resto
212     re-config');
213     if (modalEl && window.bootstrap?.Modal) {
214         const bsModal = bootstrap.Modal.getInstance(modalEl);
215         bsModal.show();
216     }
217
218     /**
219     * Сохранение политик теневого хранилища и точек восстано
220     вления.
221     */
222     async function saveRestoreConfig() {
223         const saveBtn = document.getElementById('btn-maint-save-re
224         store-policy');
225         const origText = saveBtn ? saveBtn.innerHTML : '';
226         if (saveBtn) {
227             saveBtn.disabled = true;
228             saveBtn.innerHTML = '<span class="spinner-border spinner
229             -border-sm me-1"></span>Сохранение...';
230         }
231         try {
232             const sizeSelect = document.getElementById('maint-policy
233             -storage-size');
234             const trigSelect = document.getElementById('maint-policy
235             -trigger');
```

```
232 |     const timeInput = document.getElementById('maint-policy-
    | time');
233 |     const ptsSelect = document.getElementById('maint-policy-
    | max-points');
234 |     const pruneSwitch = document.getElementById('maint-police
    | y-auto-prune');
235 |     const freqSwitch = document.getElementById('maint-policy
    | -freq-limit');
236 |
237 |     const payload = {
238 |         max_storage_size: sizeSelect ? sizeSelect.value : '10%
    | ',
239 |         schedule_trigger: trigSelect ? trigSelect.value : 'dai
    | ly',
240 |         schedule_time: timeInput ? timeInput.value : '03:00',
241 |         max_points: ptsSelect ? parseInt(ptsSelect.value, 10)
    | : 5,
242 |         auto_prune: pruneSwitch ? pruneSwitch.checked : true,
243 |         frequency_limit_minutes: freqSwitch && freqSwitch.chec
    | ked ? 0 : 1440
244 |     };
245 |
246 |     const res = await fetch('/api/system-control/restore-poi
    | nts/config', {
247 |         method: 'POST',
248 |         headers: { 'Content-Type': 'application/json' },
249 |         body: JSON.stringify(payload)
250 |     });
251 |
252 |     const data = await res.json();
253 |     if (!res.ok || (data.success === false)) {
254 |         throw new Error(data.message || 'Ошибка сохранения пол
    | итики');
255 |     }
256 |
257 |     if (window.toast) {
258 |         window.toast.success('Политики VSS', 'Политики и лимит
    | ы дискового пространства успешно обновлены.');
```

```
259     }
260
261     const modalEl = document.getElementById('modal-maint-restore-config');
262     if (modalEl && window.bootstrap?.Modal) {
263         bootstrap.Modal.getInstance(modalEl)?.hide();
264     }
265
266     fetchRestorePoints();
267     fetchMaintStatus();
268 } catch (e) {
269     console.error('[MaintenanceRecovery] Ошибка сохранения п
270 олитики:', e);
271     if (window.toast) window.toast.error('Политики VSS', e.m
272 essage);
273 } finally {
274     if (saveBtn) {
275         saveBtn.disabled = false;
276         saveBtn.innerHTML = origText;
277     }
278 }
279 /**
280  * Принудительная очистка избыточных точек восстановления.
281  */
282 async function pruneRestorePoints() {
283     const ptsSelect = document.getElementById('maint-policy-ma
284 x-points');
285     const keepCount = ptsSelect ? parseInt(ptsSelect.value, 10
286 ) : 5;
287
288     if (!confirm(`Вы действительно хотите удалить старые точки
289 восстановления, оставив ${keepCount} последних?`)) {
290         return;
291     }
292
293     try {
```



```

291     const res = await fetch('/api/system-control/restore-poi
nts/prune', {
292         method: 'POST',
293         headers: { 'Content-Type': 'application/json' },
294         body: JSON.stringify({ keep_count: keepCount })
295     });
296     const data = await res.json();
297     if (window.toast) window.toast.success('Ротация VSS', da
ta.message || `Оставлено ${keepCount} последних точек.`);
298     fetchRestorePoints();
299 } catch (e) {
300     if (window.toast) window.toast.error('Ротация VSS', e.me
ssage);
301 }
302 }
303
304 /**
305  * Инициализация вкладки Maintenance & Recovery.
306  */
307 function initMaintenanceRecoveryTab() {
308     if (window.registerTabPoller) {
309         window.registerTabPoller('tab-maintenance-recovery', ()
=> {
310             fetchMaintStatus();
311             fetchRestorePoints();
312             }, 5000, { pollerId: 'tab-maintenance-recovery_status',
immediate: true });
313     } else if (window.isTabActive ? window.isTabActive('tab-ma
intenance-recovery') : true) {
314         fetchMaintStatus();
315         fetchRestorePoints();
316     }
317
318     if (isInitialized) return;
319     isInitialized = true;
320
321     const refreshBtn = document.getElementById('btn-maint-refr
esh');

```

```
322 |     if (refreshBtn) {
323 |         refreshBtn.onclick = () => {
324 |             fetchMaintStatus();
325 |             fetchRestorePoints();
326 |         };
327 |     }
328 |
329 |     // Быстрая очистка кэшей
330 |     const cleanBtn = document.getElementById('btn-maint-action
331 | -clean');
332 |     if (cleanBtn) {
333 |         cleanBtn.onclick = async () => {
334 |             const orig = cleanBtn.innerHTML;
335 |             cleanBtn.disabled = true;
336 |             cleanBtn.innerHTML = '<span class="spinner-border spin
337 | ner-border-sm me-1"></span>Очистка...';
338 |             try {
339 |                 let res = await fetch('/api/system-control/actions/c
340 | leanup', { method: 'POST' });
341 |                 if (!res.ok) {
342 |                     res = await fetch('/api/system-control/maintenance
343 | /cleanup', { method: 'POST' });
344 |                 }
345 |                 const data = await res.json();
346 |                 if (window.toast) window.toast.success('Очистка диск
347 | а', data.message || 'Временные файлы успешно удалены.');
```

348 | fetchMaintStatus();

```
349 |             } catch (e) {
350 |                 if (window.toast) window.toast.error('Очистка диска'
351 | , e.message);
352 |             } finally {
353 |                 cleanBtn.disabled = false;
354 |                 cleanBtn.innerHTML = orig;
355 |             }
356 |         };
357 |     }
358 |
359 |     // Запуск проверки SFC
```

```
354 |     const sfcBtn = document.getElementById('btn-maint-action-s
    | fc');
355 |     if (sfcBtn) {
356 |         sfcBtn.onclick = async () => {
357 |             const orig = sfcBtn.innerHTML;
358 |             sfcBtn.disabled = true;
359 |             sfcBtn.innerHTML = '<span class="spinner-border spinne
    | r-border-sm me-1"></span>Сканирование...';
360 |             try {
361 |                 const res = await fetch('/api/system-control/mainten
    | ance/sfc', { method: 'POST' });
362 |                 const data = await res.json();
363 |                 if (window.toast) window.toast.info('Проверка SFC',
    | data.message || 'Сканирование целостности файлов завершено. ');
364 |             } catch (e) {
365 |                 if (window.toast) window.toast.error('Проверка SFC',
    | e.message);
366 |             } finally {
367 |                 sfcBtn.disabled = false;
368 |                 sfcBtn.innerHTML = orig;
369 |             }
370 |         };
371 |     }
372 |
373 |     // Запуск проверки DISM
374 |     const dismBtn = document.getElementById('btn-maint-action-
    | dism');
375 |     if (dismBtn) {
376 |         dismBtn.onclick = async () => {
377 |             const orig = dismBtn.innerHTML;
378 |             dismBtn.disabled = true;
379 |             dismBtn.innerHTML = '<span class="spinner-border spinn
    | er-border-sm me-1"></span>Проверка DISM...';
380 |             try {
381 |                 const res = await fetch('/api/system-control/mainten
    | ance/dism', { method: 'POST' });
382 |                 const data = await res.json();
383 |                 if (window.toast) window.toast.info('Проверка DISM',
```

```
data.message || 'Диагностика Component Store завершена.');
```

```
384     } catch (e) {
385         if (window.toast) window.toast.error('Проверка DISM'
, e.message);
386     } finally {
387         dismBtn.disabled = false;
388         dismBtn.innerHTML = orig;
389     }
390 };
391 }
392
393 // Создание точки восстановления
394 const createRestoreBtn = document.getElementById('btn-main
t-create-restore');
395 if (createRestoreBtn) {
396     createRestoreBtn.onclick = async () => {
397         const desc = prompt('Введите описание для новой точки
восстановления:', 'Ручной снимок системы перед изменениями');
398         if (!desc) return;
399         const orig = createRestoreBtn.innerHTML;
400         createRestoreBtn.disabled = true;
401         createRestoreBtn.innerHTML = '<span class="spinner-bor
der spinner-border-sm me-1"></span>Создание...';
402         try {
403             const res = await fetch('/api/system-control/restore
-points', {
404                 method: 'POST',
405                 headers: { 'Content-Type': 'application/json' },
406                 body: JSON.stringify({ description: desc, restore_
point_type: 'MODIFY_SETTINGS' })
407             });
408             const data = await res.json();
409             if (!res.ok) throw new Error(data.detail || data.mes
sage || 'Ошибка создания точки');
410             if (window.toast) window.toast.success('Точка восста
новления', 'Точка восстановления успешно создана.');
```

```
411             fetchRestorePoints();
412         } catch (e) {
```

```
413         if (window.toast) window.toast.error('Точка восстано  
вления', e.message);  
414     } finally {  
415         createRestoreBtn.disabled = false;  
416         createRestoreBtn.innerHTML = orig;  
417     }  
418 };  
419 }  
420  
421 // Открытие модального окна политик  
422 const cfgRestoreBtn = document.getElementById('btn-maint-c  
onfig-restore');  
423 if (cfgRestoreBtn) {  
424     cfgRestoreBtn.onclick = () => {  
425         openRestoreConfigModal();  
426     };  
427 }  
428  
429 // Сохранение политик  
430 const savePolicyBtn = document.getElementById('btn-maint-s  
ave-restore-policy');  
431 if (savePolicyBtn) {  
432     savePolicyBtn.onclick = () => {  
433         saveRestoreConfig();  
434     };  
435 }  
436  
437 // Очистить избыточные точки  
438 const pruneBtn = document.getElementById('btn-maint-policy  
-prune-now');  
439 if (pruneBtn) {  
440     pruneBtn.onclick = () => {  
441         pruneRestorePoints();  
442     };  
443 }  
444 }  
445  
446 window.initMaintenanceRecoveryTab = initMaintenanceRecoveryT
```

```
| ab;
```

```
447 | }}();
```

```
448 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/mcp_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/mcp_tab/index.html" cla
17     ss="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/mcp_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- MCP Servers Tab Component -->
30 <div class="container-fluid p-3" id="mcp-tab-root">
31   <!-- Top Banner / Summary -->
32   <div class="alert alert-dark border-secondary bg-dark text-l
33   ight d-flex flex-wrap align-items-center justify-content-betwe
34   en gap-3 shadow-sm">
35     <div class="d-flex align-items-center gap-3">
36       <div class="p-3 bg-black rounded border border-secondary
37       text-primary display-6 d-flex align-items-center justify-cont
```

```

ent-center" style="width: 54px; height: 54px;"
30      <i class="bi bi-hdd-network-fill"></i>
31    </div>
32    <div>
33      <h5 class="mb-1 text-white d-flex align-items-center gap-2">
34         Серверы MCP (Model Context Protocol)
35        <span class="badge bg-primary rounded-pill small" id="mcp-total-bad
36        </h5>
37        <p class="mb-0 text-muted small">
38          Управление внешними MCP-серверами, интеграция инстру
39          ментов браузера, файловой системы, баз данных и API для ИИ-аге
40          нтов.
41        </p>
42      </div>
43      <div class="d-flex align-items-center gap-2 flex-wrap">
44        <button type="button" class="btn btn-outline-info btn-sm d-flex align-items-center gap-1" id="btn-view-all-tools" onclick="window.mcpTab?.openAllToolsModal()">
45          <i class="bi bi-tools"></i> Все инструменты
46        </button>
47        <button type="button" class="btn btn-outline-light btn-sm d-flex align-items-center gap-1" id="btn-refresh-mcp" onclick="window.mcpTab?.refresh()">
48          <i class="bi bi-arrow-clockwise"></i> Обновить
49        </button>
50        <button type="button" class="btn btn-primary btn-sm d-flex align-items-center gap-1 shadow-sm" data-bs-toggle="modal" data-bs-target="#mcpServerModal" onclick="window.mcpTab?.openCreateModal()">
51          <i class="bi bi-plus-circle-fill"></i> Добавить MCP сервер
52        </button>
53      </div>
54    </div>

```



```
55 | <!-- Statistics Row -->
56 | <div class="row g-2 mb-3" id="mcp-stats-row">
57 |   <div class="col-6 col-md-3">
58 |     <div class="p-2 bg-dark rounded border border-secondary
| text-center">
59 |       <div class="h4 mb-0 text-white" id="stat-total-mcp">0<
| /div>
60 |       <div class="small text-muted">Всего серверов</div>
61 |     </div>
62 |   </div>
63 |   <div class="col-6 col-md-3">
64 |     <div class="p-2 bg-body-tertiary rounded border text-cen
| ter">
65 |       <div class="h4 mb-0 text-success" id="stat-active-mcp"
| >0</div>
66 |       <div class="small text-muted">Активных серверов</div>
67 |     </div>
68 |   </div>
69 |   <div class="col-6 col-md-3">
70 |     <div class="p-2 bg-body-tertiary rounded border text-cen
| ter">
71 |       <div class="h4 mb-0 text-info" id="stat-stdio-mcp">0</
| div>
72 |       <div class="small text-muted">Stdio-процессов</div>
73 |     </div>
74 |   </div>
75 |   <div class="col-6 col-md-3">
76 |     <div class="p-2 bg-body-tertiary rounded border text-cen
| ter">
77 |       <div class="h4 mb-0 text-warning" id="stat-network-mcp
| ">0</div>
78 |       <div class="small text-muted">SSE / HTTP серверов</div
| >
79 |     </div>
80 |   </div>
81 | </div>
82 |
83 | <!-- Search & View Controls -->
```

```
84 | <div class="card bg-body-tertiary border mb-3 shadow-sm">
85 |   <div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2">
86 |     <div class="input-group input-group-sm flex-grow-1" style="max-width: 450px;">
87 |       <span class="input-group-text"><i class="bi bi-search"></i></span>
88 |       <input type="text" class="form-control" id="mcp-search-input" placeholder="Поиск по ID, названию или команде...">
89 |       <button class="btn btn-outline-secondary" type="button" id="mcp-search-clear" onclick="window.mcpTab?.clearSearch()">⌫ </button>
90 |     </div>
91 |
92 |     <div class="d-flex align-items-center gap-2 flex-wrap">
93 |       <select class="form-select form-select-sm" id="mcp-filter-select" style="width: auto;" onchange="window.mcpTab?.applyFilters()">
94 |         <option value="all">Все серверы</option>
95 |         <option value="enabled">Только включенные</option>
96 |         <option value="disabled">Только отключенные</option>
97 |         <option value="stdio">Транспорт: stdio</option>
98 |         <option value="sse">Транспорт: SSE / HTTP</option>
99 |       </select>
100 |
101 |       <div class="btn-group btn-group-sm" role="group">
102 |         <button type="button" class="btn btn-outline-secondary active" id="btn-mcp-view-cards" onclick="window.mcpTab?.setViewMode('cards')">
103 |           <i class="bi bi-grid-fill"></i> Карточки
104 |         </button>
105 |         <button type="button" class="btn btn-outline-secondary" id="btn-mcp-view-table" onclick="window.mcpTab?.setViewMode('table')">
106 |           <i class="bi bi-table"></i> Таблица
107 |         </button>
108 |       </div>
109 |     </div>
```



```

139 |         <th>Транспорт</th>
140 |         <th>Команда / URL</th>
141 |         <th>Описание</th>
142 |         <th class="text-end" style="width: 220px;">Действи
    | я</th>
143 |     </tr>
144 | </thead>
145 |     <tbody id="mcp-table-body"></tbody>
146 | </table>
147 | </div>
148 | </div>
149 | </div>
150 |
151 | <!-- Add / Edit MCP Server Modal -->
152 | <div class="modal fade" id="mcpServerModal" tabindex="-1" aria
    | -labelledby="mcpServerModalLabel" aria-hidden="true">
153 |     <div class="modal-dialog modal-lg">
154 |         <div class="modal-content shadow-lg">
155 |             <div class="modal-header">
156 |                 <h5 class="modal-title d-flex align-items-center gap-2
    | " id="mcpServerModalLabel">
157 |                     <i class="bi bi-hdd-network-fill text-primary"></i>
    | <span id="mcp-modal-title-text">Настройка MCP сервера</span>
158 |                 </h5>
159 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
160 |             </div>
161 |             <div class="modal-body p-4">
162 |                 <!-- Preset Selector -->
163 |                 <div class="mb-3 p-3 bg-body-tertiary rounded border"
    | id="mcp-preset-container">
164 |                     <label class="form-label text-info small fw-bold mb-
    | 1 d-flex align-items-center gap-1">
165 |                         <i class="bi bi-magic"></i> Готовые шаблоны (Prese
    | ts)
166 |                     </label>
167 |                     <div class="d-flex gap-2 flex-wrap">
168 |                         <button type="button" class="btn btn-sm btn-outlin

```

```

    e-primary" onclick="window.mcpTab?.applyPreset('playwright')">
    □ Playwright (Browser)</button>
169      <button type="button" class="btn btn-sm btn-outlin
    e-info" onclick="window.mcpTab?.applyPreset('filesystem')">□
    Filesystem</button>
170      <button type="button" class="btn btn-sm btn-outlin
    e-success" onclick="window.mcpTab?.applyPreset('memory')">□ M
    emory</button>
171      <button type="button" class="btn btn-sm btn-outlin
    e-warning" onclick="window.mcpTab?.applyPreset('sqlite')">□ S
    QLite</button>
172      <button type="button" class="btn btn-sm btn-outlin
    e-secondary" onclick="window.mcpTab?.applyPreset('fetch')">□
    Fetch / HTTP</button>
173      <button type="button" class="btn btn-sm btn-outlin
    e-secondary" onclick="window.mcpTab?.applyPreset('sse')">□ Rem
    ote SSE / HTTP</button>
174    </div>
175  </div>
176
177  <form id="mcp-server-form" onsubmit="window.mcpTab?.sa
    veServer(event)">
178    <input type="hidden" id="mcp-form-mode" value="creat
    e">
179
180    <div class="row g-3">
181      <div class="col-md-6">
182        <label for="mcp-server-id" class="form-label sma
    ll text-muted">ID сервера (уникальный ключ) <span class="text-
    danger">*</span></label>
183        <input type="text" class="form-control" id="mcp-
    server-id" placeholder="например, playwright" required pattern
    ="[a-zA-Z0-9_-]+">
184        <div class="form-text text-muted small">Латински
    е буквы, цифры, дефис или подчеркивание.</div>
185      </div>
186
187      <div class="col-md-6">

```

```
188         <label for="mcp-server-name" class="form-label s
mall text-muted">Отображаемое имя</label>
189         <input type="text" class="form-control" id="mcp-
server-name" placeholder="Playwright Web Browser">
190     </div>
191
192     <div class="col-12">
193         <label for="mcp-server-desc" class="form-label s
mall text-muted">Описание</label>
194         <input type="text" class="form-control" id="mcp-
server-desc" placeholder="Управление браузером и поиск информа
ции в веб-страницах">
195     </div>
196
197     <div class="col-md-6">
198         <label for="mcp-server-transport" class="form-la
bel small text-muted">Протокол транспорта <span class="text-da
nger">*</span></label>
199         <select class="form-select" id="mcp-server-trans
port" onchange="window.mcpTab?.onTransportChange()">
200             <option value="stdio">stdio (Локальный subprocess)
</option>
201             <option value="sse">sse (Server-Sent Events по
HTTP)</option>
202             <option value="streamable_http">streamable_htt
p (HTTP Streaming)</option>
203         </select>
204     </div>
205
206     <div class="col-md-6 d-flex align-items-center pt-
4">
207         <div class="form-check form-switch">
208             <input class="form-check-input" type="checkbox
" id="mcp-server-enabled" checked>
209             <label class="form-check-label" for="mcp-serve
r-enabled">Включить сервер для агентов</label>
210         </div>
211     </div>
```

```
212 |
213 |         <!-- Stdio Section -->
214 |         <div class="col-12 mcp-field-stdio" id="mcp-stdio-
    | container">
215 |             <div class="mb-3">
216 |                 <label for="mcp-server-command" class="form-la
    | bel small text-muted">Команда (Command) <span class="text-dang
    | er">*</span></label>
217 |                 <input type="text" class="form-control" id="mc
    | p-server-command" placeholder="npx, python, node, uvx">
218 |                 <div class="form-text text-muted small">Исполн
    | яемый файл или утилита в PATH.</div>
219 |             </div>
220 |
221 |             <div class="mb-3">
222 |                 <label for="mcp-server-args" class="form-label
    | small text-muted">Аргументы (Arguments)</label>
223 |                 <input type="text" class="form-control" id="mc
    | p-server-args" placeholder="@playwright/mcp@latest или через п
    | пробел / запятую">
224 |                 <div class="form-text text-muted small">Укажит
    | е аргументы через пробел или в формате JSON-массива строк.</di
    | v>
225 |             </div>
226 |
227 |             <div>
228 |                 <label for="mcp-server-env" class="form-label
    | small text-muted">Переменные окружения (Environment JSON / Key
    | -Value)</label>
229 |                 <textarea class="form-control font-monospace"
    | id="mcp-server-env" rows="2" placeholder='{ "NODE_ENV": "produc
    | tion"}'></textarea>
230 |                 <div class="form-text text-muted small">Опцион
    | альный JSON-объект с переменными окружения для процесса.</div>
231 |             </div>
232 |         </div>
233 |
234 |         <!-- Network URL Section -->
```

```

235 |         <div class="col-12 mcp-field-network d-none" id="m
| cp-network-container">
236 |             <label for="mcp-server-url" class="form-label sm
| all text-muted">URL эндпоинта <span class="text-danger">*</spa
| n></label>
237 |                 <input type="url" class="form-control" id="mcp-s
| erver-url" placeholder="http://localhost:8080/sse">
238 |                 <div class="form-text text-muted small">Полный H
| TTP/HTTPS адрес удаленного MCP-сервера.</div>
239 |             </div>
240 |         </div>
241 |
242 |         <div class="mt-4 d-flex justify-content-between align-
| n-items-center border-top pt-3">
243 |             <button type="button" class="btn btn-outline-info
| btn-sm d-flex align-items-center gap-1 rounded-pill" id="btn-t
| est-modal-server" onclick="window.mcpTab?.testCurrentForm()">
244 |                 <i class="bi bi-play-circle"></i> Проверить подк
| лючение
245 |             </button>
246 |             <div class="d-flex gap-2">
247 |                 <button type="button" class="btn btn-secondary r
| ounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
248 |                 <button type="submit" class="btn btn-primary rou
| nded-pill px-3" id="mcp-btn-save-server">Сохранить</button>
249 |             </div>
250 |         </div>
251 |     </form>
252 | </div>
253 | </div>
254 | </div>
255 | </div>
256 |
257 | <!-- Test Connection & Tools Inspection Modal -->
258 | <div class="modal fade" id="mcpTestModal" tabindex="-1" aria-l
| abelledby="mcpTestModalLabel" aria-hidden="true">
259 |     <div class="modal-dialog modal-lg modal-dialog-scrollable">
260 |         <div class="modal-content shadow-lg">

```



```
261 |         <div class="modal-header">
262 |             <h5 class="modal-title d-flex align-items-center gap-2
| id="mcpTestModalLabel">
263 |                 <i class="bi bi-activity text-info"></i> Диагностика
| MCP: <span id="mcp-test-server-name">...</span>
264 |             </h5>
265 |             <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
266 |         </div>
267 |         <div class="modal-body p-4">
268 |             <div id="mcp-test-status-box" class="p-3 mb-3 rounded border bg-body-tertiary d-flex align-items-center justify-content-between">
269 |                 <div class="d-flex align-items-center gap-2">
270 |                     <div class="spinner-border spinner-border-sm text-info" id="mcp-test-spinner" role="status"></div>
271 |                     <span id="mcp-test-status-text">Установка соединения и опрос инструментов...</span>
272 |                 </div>
273 |                 <span class="badge bg-secondary font-monospace" id="mcp-test-latency-badge">—</span>
274 |             </div>
275 |
276 |             <h6 class="d-flex align-items-center gap-2 mb-3">
277 |                 <i class="bi bi-tools text-warning"></i> Обнаруженные инструменты (Tools):
278 |                 <span class="badge bg-primary rounded-pill" id="mcp-test-tools-count">0</span>
279 |             </h6>
280 |
281 |             <div id="mcp-test-tools-list" class="d-flex flex-column gap-2">
282 |                 <!-- Tools list populated dynamically -->
283 |             </div>
284 |         </div>
285 |         <div class="modal-footer">
286 |             <button type="button" class="btn btn-secondary rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
```

```
287 |         </div>
288 |     </div>
289 | </div>
290 </div>
291 |
292 <!-- All Tools Aggregator Modal -->
293 <div class="modal fade" id="mcpAllToolsModal" tabindex="-1" ar
    ia-labelledby="mcpAllToolsModalLabel" aria-hidden="true">
294 |     <div class="modal-dialog modal-xl modal-dialog-scrollable">
295 |         <div class="modal-content shadow-lg">
296 |             <div class="modal-header">
297 |                 <h5 class="modal-title d-flex align-items-center gap-2
    | " id="mcpAllToolsModalLabel">
298 |                     <i class="bi bi-boxes text-success"></i> Доступный п
    | ул инструментов MCP
299 |                 </h5>
300 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
301 |             </div>
302 |             <div class="modal-body p-4">
303 |                 <p class="text-muted small mb-3">
304 |                     Список всех инструментов, агрегированных со всех вкл
    | юченных серверов MCP. Эти инструменты автоматически предоставл
    | яются LangChain/ReAct агенты.
305 |                 </p>
306 |
307 |                 <div id="mcp-all-tools-loading" class="text-center py-
    | 4 text-muted">
308 |                     <div class="spinner-border text-primary mb-2" role="
    | status"></div>
309 |                     <div>Опрос активных MCP серверов...</div>
310 |                 </div>
311 |
312 |                 <div id="mcp-all-tools-empty" class="text-center py-4
    | text-muted d-none">
313 |                     <i class="bi bi-tools display-6 d-block mb-2 text-se
    | condary"></i>
314 |                     <div>Нет доступных инструментов от активных MCP серв
```

```
    еров.</div>
315 |         </div>
316 |
317 |         <div id="mcp-all-tools-container" class="row g-3 d-non
    e"></div>
318 |     </div>
319 |     <div class="modal-footer">
320 |         <button type="button" class="btn btn-secondary rounded
    -pill px-3" data-bs-dismiss="modal">Закрыть</button>
321 |     </div>
322 | </div>
323 | </div>
324 | </div>
325 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Mcp Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/mcp_tab/main.js?v=2026
11  *     1001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/mcp_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // MCP Servers Tab Management Logic for AI Breadboard Admin In
22 // terface
23 class McpTabManager {
24   constructor() {
25     this.servers = [];
26     this.filteredServers = [];
27     this.viewMode = 'cards'; // 'cards' | 'table'
28     this.searchQuery = '';
29     this.activeFilter = 'all'; // 'all' | 'enabled' | 'disable
30     d' | 'stdio' | 'sse'
31     this.isInitialized = false;
```

```
31 | }
32 |
33 | async init() {
34 |   if (this.isInitialized) {
35 |     await this.refresh();
36 |     return;
37 |   }
38 |   this.bindEvents();
39 |   this.isInitialized = true;
40 |   await this.refresh();
41 | }
42 |
43 | bindEvents() {
44 |   const searchInput = document.getElementById('mcp-search-in
put');
45 |   if (searchInput) {
46 |     searchInput.addEventListener('input', (e) => {
47 |       this.searchQuery = e.target.value.trim().toLowerCase()
;
48 |       this.applyFilters();
49 |     });
50 |   }
51 |
52 |   const idInput = document.getElementById('mcp-server-id');
53 |   if (idInput) {
54 |     idInput.addEventListener('input', (e) => {
55 |       const val = e.target.value;
56 |       const normalized = val.toLowerCase().replace(/[^\a-z0-9
_-]/g, '_');
57 |       if (val !== normalized) {
58 |         e.target.value = normalized;
59 |       }
60 |     });
61 |   }
62 | }
63 |
64 | async apiFetch(endpoint, options = {}) {
65 |   let url = endpoint;
```

```

66 |     let res = await fetch(url, options);
67 |     if (!res.ok && url.startsWith('/api/admin/mcp')) {
68 |         const fallbackUrl = url.replace('/api/admin/mcp', '/api/
mcp');
69 |         const fallbackRes = await fetch(fallbackUrl, options);
70 |         if (fallbackRes.ok) {
71 |             return fallbackRes;
72 |         }
73 |     }
74 |     return res;
75 | }
76 |
77 | async refresh() {
78 |     this.showLoading(true);
79 |     try {
80 |         const res = await this.apiFetch('/api/admin/mcp/servers'
);
81 |         if (!res.ok) {
82 |             throw new Error(`HTTP ${res.status}: ${res.statusText}
`);
83 |         }
84 |         const data = await res.json();
85 |         this.servers = data.servers || [];
86 |         this.updateStats();
87 |         this.applyFilters();
88 |     } catch (err) {
89 |         console.error('[McpTab] Error fetching MCP servers:', er
r);
90 |         this.renderError(err.message);
91 |     } finally {
92 |         this.showLoading(false);
93 |     }
94 | }
95 |
96 | updateStats() {
97 |     const total = this.servers.length;
98 |     const active = this.servers.filter(s => s.enabled !== fals
e).length;

```

```
99     const stdio = this.servers.filter(s => (s.transport || 'stdio') === 'stdio').length;
100     const network = this.servers.filter(s => ['sse', 'streamable_http', 'http'].includes(s.transport)).length;
101
102     const badge = document.getElementById('mcp-total-badge');
103     if (badge) badge.textContent = total;
104
105     const elTotal = document.getElementById('stat-total-mcp');
106     if (elTotal) elTotal.textContent = total;
107
108     const elActive = document.getElementById('stat-active-mcp');
109     if (elActive) elActive.textContent = active;
110
111     const elStdio = document.getElementById('stat-stdio-mcp');
112     if (elStdio) elStdio.textContent = stdio;
113
114     const elNetwork = document.getElementById('stat-network-mcp');
115     if (elNetwork) elNetwork.textContent = network;
116   }
117
118   applyFilters() {
119     const filterSelect = document.getElementById('mcp-filter-select');
120     this.activeFilter = filterSelect ? filterSelect.value : 'all';
121
122     this.filteredServers = this.servers.filter(s => {
123       // Search filter
124       if (this.searchQuery) {
125         const idMatch = (s.id || '').toLowerCase().includes(this.searchQuery);
126         const nameMatch = (s.name || '').toLowerCase().includes(this.searchQuery);
127         const descMatch = (s.description || '').toLowerCase().includes(this.searchQuery);
```

```
128 |         const cmdMatch = (s.command || '').toLowerCase().includes(this.searchQuery);
129 |         const urlMatch = (s.url || '').toLowerCase().includes(this.searchQuery);
130 |         if (!idMatch && !nameMatch && !descMatch && !cmdMatch && !urlMatch) {
131 |             return false;
132 |         }
133 |     }
134 |
135 |     // Category filter
136 |     if (this.activeFilter === 'enabled') {
137 |         return s.enabled !== false;
138 |     } else if (this.activeFilter === 'disabled') {
139 |         return s.enabled === false;
140 |     } else if (this.activeFilter === 'stdio') {
141 |         return (s.transport || 'stdio') === 'stdio';
142 |     } else if (this.activeFilter === 'sse') {
143 |         return ['sse', 'streamable_http', 'http'].includes(s.transport);
144 |     }
145 |
146 |     return true;
147 | });
148 |
149 | this.render();
150 | }
151 |
152 | clearSearch() {
153 |     const searchInput = document.getElementById('mcp-search-input');
154 |     if (searchInput) searchInput.value = '';
155 |     this.searchQuery = '';
156 |     this.applyFilters();
157 | }
158 |
159 | setViewMode(mode) {
160 |     this.viewMode = mode;
```



```
161     const btnCards = document.getElementById('btn-mcp-view-car
    ds');
162     const btnTable = document.getElementById('btn-mcp-view-tab
    le');
163     const cardsContainer = document.getElementById('mcp-cards-
    container');
164     const tableContainer = document.getElementById('mcp-table-
    container');
165
166     if (mode === 'cards') {
167         btnCards?.classList.add('active');
168         btnTable?.classList.remove('active');
169         cardsContainer?.classList.remove('d-none');
170         tableContainer?.classList.add('d-none');
171     } else {
172         btnCards?.classList.remove('active');
173         btnTable?.classList.add('active');
174         cardsContainer?.classList.add('d-none');
175         tableContainer?.classList.remove('d-none');
176     }
177 }
178
179 showLoading(show) {
180     const loadingEl = document.getElementById('mcp-loading');
181     const cardsEl = document.getElementById('mcp-cards-contain
    er');
182     const tableEl = document.getElementById('mcp-table-contain
    er');
183     const emptyEl = document.getElementById('mcp-empty');
184
185     if (show) {
186         loadingEl?.classList.remove('d-none');
187         cardsEl?.classList.add('d-none');
188         tableEl?.classList.add('d-none');
189         emptyEl?.classList.add('d-none');
190     } else {
191         loadingEl?.classList.add('d-none');
192         this.setViewMode(this.viewMode);
```

```
193     }
194 }
195
196 render() {
197     const emptyEl = document.getElementById('mcp-empty');
198     const cardsContainer = document.getElementById('mcp-cards-
container');
199     const tableBody = document.getElementById('mcp-table-body'
);
200
201     if (this.filteredServers.length === 0) {
202         if (emptyEl) emptyEl.classList.remove('d-none');
203         if (cardsContainer) cardsContainer.innerHTML = '';
204         if (tableBody) tableBody.innerHTML = '';
205         return;
206     }
207
208     if (emptyEl) emptyEl.classList.add('d-none');
209
210     if (cardsContainer) {
211         cardsContainer.innerHTML = this.filteredServers.map(s =>
this.renderServerCard(s)).join('');
212     }
213
214     if (tableBody) {
215         tableBody.innerHTML = this.filteredServers.map(s => this
.renderServerTableRow(s)).join('');
216     }
217 }
218
219 renderServerCard(server) {
220     const isEnabled = server.enabled !== false;
221     const isStdio = (server.transport || 'stdio') === 'stdio';
222     const statusBadge = isEnabled
223         ? '<span class="badge bg-success-subtle text-success bor
der border-success-subtle"><i class="bi bi-check-circle-fill">
</i> Активен</span>'
224         : '<span class="badge bg-secondary-subtle text-secondary
```

```

    border border-secondary-subtle"><i class="bi bi-dash-circle">
</i> Отключен</span>';
225
226     const transportBadge = isStdio
227       ? '<span class="badge bg-info-subtle text-info border bo
rder-info-subtle"><i class="bi bi-terminal"></i> stdio</span>'
228       : '<span class="badge bg-warning-subtle text-warning bor
der border-warning-subtle"><i class="bi bi-broadcast"></i> ${t
his.escapeHtml(server.transport)}</span>`;
229
230     const commandOrUrl = isStdio
231       ? '<code>${this.escapeHtml(server.command || '')} ${(ser
ver.args || []).join(' ')}</code>'
232       : '<a href="${this.escapeHtml(server.url || '#')}\" targe
t="_blank" class="text-info text-truncate d-inline-block" styl
e="max-width: 250px;">${this.escapeHtml(server.url || '')}</a>
';
233
234     return `
235       <div class="col-12 col-md-6 col-xl-4">
236         <div class="card bg-dark border-secondary h-100 shadow
-sm d-flex flex-column">
237           <div class="card-header border-secondary d-flex alig
n-items-center justify-content-between p-3">
238             <div class="d-flex align-items-center gap-2 text-t
runcate">
239               <div class="p-2 bg-black rounded border border-s
econdary text-primary">
240                 <i class="bi ${isStdio ? 'bi-cpu' : 'bi-globe2
'}"></i>
241               </div>
242               <div class="text-truncate">
243                 <h6 class="mb-0 text-white text-truncate fw-bo
ld">${this.escapeHtml(server.name || server.id)}</h6>
244                 <small class="text-muted font-monospace">${thi
s.escapeHtml(server.id)}</small>
245               </div>
246             </div>

```

```

247         <div class="d-flex align-items-center gap-1">
248             ${transportBadge}
249             ${statusBadge}
250         </div>
251     </div>
252     <div class="card-body p-3 flex-grow-1">
253         <p class="card-text small text-secondary mb-3" style="min-height: 38px;">
254             ${this.escapeHtml(server.description || 'Описание отсутствует.')}
255         </p>
256         <div class="p-2 bg-black rounded border border-secondary mb-2 small">
257             <div class="text-muted small mb-1">${isStdio ? 'Исполняемая команда:' : 'Сетевой эндпоинт:'}</div>
258             <div class="text-truncate">${commandOrUrl}</div>
259         </div>
260     </div>
261     <div class="card-footer border-secondary p-2 d-flex align-items-center justify-content-between gap-1 bg-black bg-opacity-25">
262         <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1" onclick="window.mcpTab?.testServer('${this.escapeHtml(server.id)}')">
263             <i class="bi bi-play-circle"></i> Тест & Tools
264         </button>
265         <div class="d-flex align-items-center gap-1">
266             <button class="btn btn-sm ${isEnabled ? 'btn-outline-warning' : 'btn-outline-success'}" title="${isEnabled ? 'Отключить' : 'Включить'}" onclick="window.mcpTab?.toggleServer('${this.escapeHtml(server.id)}')">
267                 <i class="bi ${isEnabled ? 'bi-pause-fill' : 'bi-play-fill'}"></i>
268             </button>
269             <button class="btn btn-sm btn-outline-primary" title="Редактировать" onclick="window.mcpTab?.openEditModal('${this.escapeHtml(server.id)}')">
270                 <i class="bi bi-pencil-fill"></i>

```

```

271 |         </button>
272 |         <button class="btn btn-sm btn-outline-danger" ti
| tle="Удалить" onclick="window.mcpTab?.deleteServer('${this.esc
| apeHtml(server.id)}')">
273 |             <i class="bi bi-trash-fill"></i>
274 |         </button>
275 |     </div>
276 | </div>
277 | </div>
278 | </div>
279 | `;
280 | }
281 |
282 | renderServerTableRow(server) {
283 |     const isEnabled = server.enabled !== false;
284 |     const isStdio = (server.transport || 'stdio') === 'stdio';
285 |     const statusDot = isEnabled
286 |         ? '<span class="badge bg-success rounded-pill" title="Ак
| тивен">ON</span>'
287 |         : '<span class="badge bg-secondary rounded-pill" title="
| Отключен">OFF</span>';
288 |
289 |     const transportBadge = isStdio
290 |         ? '<span class="badge bg-info-subtle text-info border bo
| rder-info-subtle">stdio</span>'
291 |         : `<span class="badge bg-warning-subtle text-warning bor
| der border-warning-subtle">${this.escapeHtml(server.transport)
| }</span>`;
292 |
293 |     const cmdOrUrl = isStdio
294 |         ? `<code>${this.escapeHtml(server.command || '')} ${((ser
| ver.args || []).join(' '))}</code>`
295 |         : `<a href="${this.escapeHtml(server.url || '#')}" targe
| t="_blank" class="text-info">${this.escapeHtml(server.url || '
| ')}</a>`;
296 |
297 |     return `
298 |         <tr>

```

```

299 |         <td class="text-center">${statusDot}</td>
300 |         <td>
301 |             <div class="fw-bold text-white">${this.escapeHtml(se
| rver.name || server.id)}</div>
302 |             <div class="small font-monospace text-muted">${this.
| escapeHtml(server.id)}</div>
303 |         </td>
304 |         <td>${transportBadge}</td>
305 |         <td class="small text-truncate" style="max-width: 250p
| x;">${cmdOrUrl}</td>
306 |         <td class="small text-muted text-truncate" style="max-
| width: 200px;">${this.escapeHtml(server.description || '-')}</
| td>
307 |         <td class="text-end">
308 |             <div class="btn-group btn-group-sm">
309 |                 <button class="btn btn-outline-info" title="Тест с
| оединения" onclick="window.mcpTab?.testServer('${this.escapeHt
| ml(server.id)}')">
310 |                     <i class="bi bi-play-circle"></i>
311 |                 </button>
312 |                 <button class="btn ${isEnabled ? 'btn-outline-warn
| ing' : 'btn-outline-success'}" title="${isEnabled ? 'Отключить
| ' : 'Включить'}" onclick="window.mcpTab?.toggleServer('${this.
| escapeHtml(server.id)}')">
313 |                     <i class="bi ${isEnabled ? 'bi-pause-fill' : 'bi
| -play-fill'}"></i>
314 |                 </button>
315 |                 <button class="btn btn-outline-primary" title="Ред
| актировать" onclick="window.mcpTab?.openEditModal('${this.esca
| peHtml(server.id)}')">
316 |                     <i class="bi bi-pencil-fill"></i>
317 |                 </button>
318 |                 <button class="btn btn-outline-danger" title="Удал
| ить" onclick="window.mcpTab?.deleteServer('${this.escapeHtml(s
| erver.id)}')">
319 |                     <i class="bi bi-trash-fill"></i>
320 |                 </button>
321 |             </div>

```

```

322         </td>
323     </tr>
324     `;
325 }
326
327 onTransportChange() {
328     const transport = document.getElementById('mcp-server-transport')?.value || 'stdio';
329     const stdioContainer = document.getElementById('mcp-stdio-container');
330     const networkContainer = document.getElementById('mcp-network-container');
331
332     if (transport === 'stdio') {
333         stdioContainer?.classList.remove('d-none');
334         networkContainer?.classList.add('d-none');
335     } else {
336         stdioContainer?.classList.add('d-none');
337         networkContainer?.classList.remove('d-none');
338     }
339 }
340
341 applyPreset(presetKey) {
342     const presets = {
343         playwright: {
344             id: 'playwright',
345             name: 'Playwright Browser MCP',
346             description: 'Управление веб-браузером, навигация, клики, скриншоты и парсинг страниц через Playwright.',
347             transport: 'stdio',
348             command: 'npx',
349             args: '@playwright/mcp@latest',
350             env: ''
351         },
352         filesystem: {
353             id: 'filesystem',
354             name: 'Filesystem MCP',
355             description: 'Безопасное чтение, запись и навигация по

```

```
    каталогам и файлам на диске.',
356     transport: 'stdio',
357     command: 'npx',
358     args: '-y @modelcontextprotocol/server-filesystem ./data',
359     env: ''
360 },
361     memory: {
362         id: 'memory',
363         name: 'Knowledge Graph Memory MCP',
364         description: 'Хранилище графа знаний и ассоциативная п
амять для ИИ-агентов.',
365         transport: 'stdio',
366         command: 'npx',
367         args: '-y @modelcontextprotocol/server-memory',
368         env: ''
369     },
370     sqlite: {
371         id: 'sqlite',
372         name: 'SQLite Database MCP',
373         description: 'Выполнение запросов, исследование схемы
и инспекция таблиц SQLite базы данных.',
374         transport: 'stdio',
375         command: 'uvx',
376         args: 'mcp-server-sqlite --db-path ./data/database.sql
ite',
377         env: ''
378     },
379     fetch: {
380         id: 'fetch',
381         name: 'Fetch / Web Request MCP',
382         description: 'Преобразование HTML веб-страниц в Markdo
wn и выполнение HTTP-запросов.',
383         transport: 'stdio',
384         command: 'uvx',
385         args: 'mcp-server-fetch',
386         env: ''
387     },
```



```
388 |     sse: {
389 |         id: 'remote_mcp',
390 |         name: 'Remote SSE MCP Server',
391 |         description: 'Удаленный MCP-сервер, доступный по прото
колу Server-Sent Events (SSE).',
392 |         transport: 'sse',
393 |         url: 'http://localhost:8080/sse'
394 |     }
395 | };
396 |
397 | const p = presets[presetKey];
398 | if (!p) return;
399 |
400 | const idEl = document.getElementById('mcp-server-id');
401 | const nameEl = document.getElementById('mcp-server-name');
402 | const descEl = document.getElementById('mcp-server-desc');
403 | const transportEl = document.getElementById('mcp-server-tr
ansport');
404 | const cmdEl = document.getElementById('mcp-server-command'
);
405 | const argsEl = document.getElementById('mcp-server-args');
406 | const urlEl = document.getElementById('mcp-server-url');
407 | const envEl = document.getElementById('mcp-server-env');
408 |
409 | if (idEl && document.getElementById('mcp-form-mode')?.valu
e === 'create') idEl.value = p.id;
410 | if (nameEl) nameEl.value = p.name || '';
411 | if (descEl) descEl.value = p.description || '';
412 | if (transportEl) transportEl.value = p.transport || 'stdio
';
413 | if (cmdEl) cmdEl.value = p.command || '';
414 | if (argsEl) argsEl.value = p.args || '';
415 | if (urlEl) urlEl.value = p.url || '';
416 | if (envEl) envEl.value = p.env || '';
417 |
418 | this.onTransportChange();
419 | }
420 |
```

```
421 | openCreateModal() {
422 |     const form = document.getElementById('mcp-server-form');
423 |     if (form) form.reset();
424 |
425 |     const titleText = document.getElementById('mcp-modal-title
    | -text');
426 |     if (titleText) titleText.textContent = 'Добавление нового
    | MCP сервера';
427 |
428 |     const modeInput = document.getElementById('mcp-form-mode')
    | ;
429 |     if (modeInput) modeInput.value = 'create';
430 |
431 |     const idInput = document.getElementById('mcp-server-id');
432 |     if (idInput) {
433 |         idInput.removeAttribute('readonly');
434 |         idInput.classList.remove('bg-secondary');
435 |     }
436 |
437 |     const presetContainer = document.getElementById('mcp-prese
    | t-container');
438 |     if (presetContainer) presetContainer.classList.remove('d-n
    | one');
439 |
440 |     const enabledCheckbox = document.getElementById('mcp-serve
    | r-enabled');
441 |     if (enabledCheckbox) enabledCheckbox.checked = true;
442 |
443 |     this.applyPreset('playwright');
444 | }
445 |
446 | openEditModal(serverId) {
447 |     const server = this.servers.find(s => s.id === serverId);
448 |     if (!server) return;
449 |
450 |     const titleText = document.getElementById('mcp-modal-title
    | -text');
451 |     if (titleText) titleText.textContent = `Редактирование: ${`
```

```
server.name || server.id}`;  
452  
453     const modeInput = document.getElementById('mcp-form-mode')  
    ;  
454     if (modeInput) modeInput.value = 'edit';  
455  
456     const presetContainer = document.getElementById('mcp-prese  
t-container');  
457     if (presetContainer) presetContainer.classList.add('d-none  
'');  
458  
459     const idInput = document.getElementById('mcp-server-id');  
460     if (idInput) {  
461         idInput.value = server.id;  
462         idInput.setAttribute('readonly', 'true');  
463         idInput.classList.add('bg-secondary');  
464     }  
465  
466     const nameInput = document.getElementById('mcp-server-name  
'');  
467     if (nameInput) nameInput.value = server.name || '';  
468  
469     const descInput = document.getElementById('mcp-server-desc  
'');  
470     if (descInput) descInput.value = server.description || '';  
471  
472     const transportInput = document.getElementById('mcp-server  
-transport');  
473     if (transportInput) transportInput.value = server.transpor  
t || 'stdio';  
474  
475     const cmdInput = document.getElementById('mcp-server-comm  
nd');  
476     if (cmdInput) cmdInput.value = server.command || '';  
477  
478     const argsInput = document.getElementById('mcp-server-args  
'');  
479     if (argsInput) {
```

```
480     argsInput.value = Array.isArray(server.args) ? server.ar
    gs.join(' ') : (server.args || '');
481   }
482
483   const urlInput = document.getElementById('mcp-server-url')
    ;
484   if (urlInput) urlInput.value = server.url || '';
485
486   const envInput = document.getElementById('mcp-server-env')
    ;
487   if (envInput) {
488     envInput.value = server.env ? JSON.stringify(server.env,
    null, 2) : '';
489   }
490
491   const enabledCheckbox = document.getElementById('mcp-serve
    r-enabled');
492   if (enabledCheckbox) enabledCheckbox.checked = server.enab
    led !== false;
493
494   this.onTransportChange();
495
496   const modalEl = document.getElementById('mcpServerModal');
497   if (modalEl && window.bootstrap) {
498     const modal = bootstrap.Modal.getInstance(modalEl) || ne
    w bootstrap.Modal(modalEl);
499     modal.show();
500   }
501 }
502
503 async saveServer(event) {
504   if (event) event.preventDefault();
505
506   const mode = document.getElementById('mcp-form-mode')?.val
    ue || 'create';
507   const serverId = document.getElementById('mcp-server-id')?
    .value.trim();
508   const name = document.getElementById('mcp-server-name')?.v
```

```
    alue.trim();
509     const description = document.getElementById('mcp-server-de
    sc')?.value.trim();
510     const transport = document.getElementById('mcp-server-tran
    sport')?.value || 'stdio';
511     const enabled = document.getElementById('mcp-server-enable
    d')?.checked ?? true;
512     const command = document.getElementById('mcp-server-comman
    d')?.value.trim();
513     const argsRaw = document.getElementById('mcp-server-args')
    ?.value.trim();
514     const url = document.getElementById('mcp-server-url')?.val
    ue.trim();
515     const envRaw = document.getElementById('mcp-server-env')?.
    value.trim();
516
517     if (!serverId) {
518         window.showToast?.( 'Укажите ID сервера', 'warning') || a
    lert('Укажите ID сервера');
519         return;
520     }
521
522     // Parse args
523     let args = [];
524     if (argsRaw) {
525         if (argsRaw.startsWith '[' && argsRaw.endsWith(']')) {
526             try {
527                 args = JSON.parse(argsRaw);
528             } catch {
529                 args = argsRaw.split(/\s+/).filter(Boolean);
530             }
531         } else {
532             args = argsRaw.split(/\s+/).filter(Boolean);
533         }
534     }
535
536     // Parse env
537     let env = {};
```

```
538 |     if (envRaw) {
539 |         try {
540 |             env = JSON.parse(envRaw);
541 |         } catch (err) {
542 |             window.showToast?.( 'Ошибка формата JSON в переменных о
    | кружения: ' + err.message, 'danger') || alert('Ошибка формата
    | JSON в переменных окружения: ' + err.message);
543 |             return;
544 |         }
545 |     }
546 |
547 |     const payload = {
548 |         id: serverId,
549 |         name: name || serverId,
550 |         description: description,
551 |         transport: transport,
552 |         enabled: enabled,
553 |         command: command,
554 |         args: args,
555 |         url: url,
556 |         env: env
557 |     };
558 |
559 |     const saveBtn = document.getElementById('mcp-btn-save-serv
    | er');
560 |     if (saveBtn) {
561 |         saveBtn.disabled = true;
562 |         saveBtn.innerHTML = '<span class="spinner-border spinner
    | -border-sm me-1"></span> Сохранение...';
563 |     }
564 |
565 |     try {
566 |         const endpoint = mode === 'create'
567 |             ? '/api/admin/mcp/servers'
568 |             : `/api/admin/mcp/servers/${encodeURIComponent(serverI
    | d)}`;
569 |
570 |         const method = mode === 'create' ? 'POST' : 'PUT';
```

```
571 |
572 |     const res = await this.apiFetch(endpoint, {
573 |         method: method,
574 |         headers: { 'Content-Type': 'application/json' },
575 |         body: JSON.stringify(payload)
576 |     });
577 |
578 |     if (!res.ok) {
579 |         const errData = await res.json().catch(() => ({}));
580 |         throw new Error(errData.detail || `HTTP ${res.status}`
581 |     );
582 |     }
583 |
584 |     // Close modal
585 |     const modalEl = document.getElementById('mcpServerModal'
586 | );
587 |     if (modalEl && window.bootstrap) {
588 |         const modal = bootstrap.Modal.getInstance(modalEl);
589 |         modal?.hide();
590 |     }
591 |
592 |     await this.refresh();
593 | } catch (err) {
594 |     window.showToast?(`Ошибка сохранения MCP сервера: ${err
595 | .message}`, 'danger') || alert(`Ошибка сохранения MCP сервера:
596 | ${err.message}`);
597 | } finally {
598 |     if (saveBtn) {
599 |         saveBtn.disabled = false;
600 |         saveBtn.innerHTML = 'Сохранить';
601 |     }
602 | }
603 |
604 | async toggleServer(serverId) {
605 |     try {
606 |         const res = await this.apiFetch(`/api/admin/mcp/servers/
607 | ${encodeURIComponent(serverId)}/toggle`, {
```

```
604         method: 'POST'
605     });
606     if (!res.ok) {
607         const err = await res.json().catch(() => ({}));
608         throw new Error(err.detail || `HTTP ${res.status}`);
609     }
610     await this.refresh();
611 } catch (err) {
612     window.showToast?(`Ошибка переключения статуса сервера:
        ${err.message}`, 'danger') || alert(`Ошибка переключения стат
        уса сервера: ${err.message}`);
613 }
614 }
615
616 async deleteServer(serverId) {
617     if (!confirm(`Вы уверены, что хотите удалить MCP сервер "$
        {serverId}"?`)) {
618         return;
619     }
620
621     try {
622         const res = await this.apiFetch(`/api/admin/mcp/servers/
        ${encodeURIComponent(serverId)}`, {
623             method: 'DELETE'
624         });
625         if (!res.ok) {
626             const err = await res.json().catch(() => ({}));
627             throw new Error(err.detail || `HTTP ${res.status}`);
628         }
629         await this.refresh();
630     } catch (err) {
631         window.showToast?(`Ошибка удаления сервера: ${err.messa
        ge}`, 'danger') || alert(`Ошибка удаления сервера: ${err.messa
        ge}`);
632     }
633 }
634
635 async testServer(serverId) {
```



```
636 |     const server = this.servers.find(s => s.id === serverId);
637 |     const serverName = server?.name || serverId;
638 |
639 |     const nameEl = document.getElementById('mcp-test-server-na
me');
640 |     if (nameEl) nameEl.textContent = serverName;
641 |
642 |     const statusBox = document.getElementById('mcp-test-status
-box');
643 |     const statusText = document.getElementById('mcp-test-statu
s-text');
644 |     const spinner = document.getElementById('mcp-test-spinner'
);
645 |     const latencyBadge = document.getElementById('mcp-test-lat
ency-badge');
646 |     const countBadge = document.getElementById('mcp-test-tools
-count');
647 |     const toolsList = document.getElementById('mcp-test-tools-
list');
648 |
649 |     if (spinner) spinner.classList.remove('d-none');
650 |     if (statusBox) {
651 |         statusBox.className = 'p-3 mb-3 rounded border border-in
fo bg-black text-info d-flex align-items-center justify-conten
t-between';
652 |     }
653 |     if (statusText) statusText.textContent = 'Подключение к пр
оцессу / эндпоинту и опрос инструментов...';
654 |     if (latencyBadge) latencyBadge.textContent = '...';
655 |     if (countBadge) countBadge.textContent = '0';
656 |     if (toolsList) toolsList.innerHTML = '<div class="text-mut
ed text-center py-3">Опрос инструментов...</div>';
657 |
658 |     const modalEl = document.getElementById('mcpTestModal');
659 |     if (modalEl && window.bootstrap) {
660 |         const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
661 |         modal.show();
```

```
662 |     }
663 |
664 |     try {
665 |         const res = await this.apiFetch(`/api/admin/mcp/servers/
666 | ${encodeURIComponent(serverId)}/test`, {
667 |         method: 'POST'
668 |     });
669 |     const data = await res.json();
670 |
671 |     if (spinner) spinner.classList.add('d-none');
672 |     if (latencyBadge) latencyBadge.textContent = `${data.lat
673 | ency_ms || 0} ms`;
674 |
675 |     if (data.status === 'ok') {
676 |         if (statusBox) {
677 |             statusBox.className = 'p-3 mb-3 rounded border borde
678 | r-success bg-black text-success d-flex align-items-center just
679 | ify-content-between';
680 |         }
681 |         if (statusText) statusText.innerHTML = `❏ <strong>Успе
682 | шно:</strong> ${this.escapeHtml(data.message)}`;
683 |         if (countBadge) countBadge.textContent = data.tools_co
684 | unt || (data.tools || []).length;
685 |         this.renderDiscoveredTools(data.tools || [], toolsList
686 | );
687 |     } else {
688 |         if (statusBox) {
689 |             statusBox.className = 'p-3 mb-3 rounded border borde
690 | r-danger bg-black text-danger d-flex align-items-center justif
691 | y-content-between';
692 |         }
693 |         if (statusText) statusText.innerHTML = `❏ <strong>Ошиб
694 | ка:</strong> ${this.escapeHtml(data.message || 'Не удалось под
695 | ключиться к серверу')}`;
696 |         if (toolsList) {
697 |             toolsList.innerHTML = `<div class="alert alert-dange
698 | r bg-dark border-danger text-light small mb-0">${this.escapeHt
699 | ml(data.message || 'Ошибка соединения')}</div>`;
700 |         }
701 |     }
702 | }
```

```
687         }
688     }
689     } catch (err) {
690         if (spinner) spinner.classList.add('d-none');
691         if (statusBox) {
692             statusBox.className = 'p-3 mb-3 rounded border border-
danger bg-black text-danger d-flex align-items-center justify-
content-between';
693         }
694         if (statusText) statusText.innerHTML = `❌ <strong>Сетева
я ошибка:</strong> ${this.escapeHtml(err.message)}`;
695     }
696 }
697
698 async testCurrentForm() {
699     const serverId = document.getElementById('mcp-server-id')?
.value.trim() || 'test_server';
700     const transport = document.getElementById('mcp-server-tran
sport')?.value || 'stdio';
701     const command = document.getElementById('mcp-server-comman
d')?.value.trim();
702     const argsRaw = document.getElementById('mcp-server-args')
?.value.trim();
703     const url = document.getElementById('mcp-server-url')?.val
ue.trim();
704     const envRaw = document.getElementById('mcp-server-env')?.
value.trim();
705
706     let args = [];
707     if (argsRaw) {
708         if (argsRaw.startsWith '[' && argsRaw.endsWith(']')) {
709             try { args = JSON.parse(argsRaw); } catch { args = arg
sRaw.split(/\s+/).filter(Boolean); }
710         } else {
711             args = argsRaw.split(/\s+/).filter(Boolean);
712         }
713     }
714 }
```

```
715 |     let env = {};  
716 |     if (envRaw) {  
717 |         try { env = JSON.parse(envRaw); } catch (e) { window.showToast?.('Неверный формат JSON в env', 'danger') || alert('Неверный формат JSON в env'); return; }  
718 |     }  
719 |  
720 |     const testBtn = document.getElementById('btn-test-modal-server');  
721 |     if (testBtn) {  
722 |         testBtn.disabled = true;  
723 |         testBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Тест...';  
724 |     }  
725 |  
726 |     try {  
727 |         const res = await this.apiFetch('/api/admin/mcp/test', {  
728 |             method: 'POST',  
729 |             headers: { 'Content-Type': 'application/json' },  
730 |             body: JSON.stringify({  
731 |                 id: serverId,  
732 |                 transport: transport,  
733 |                 command: command,  
734 |                 args: args,  
735 |                 url: url,  
736 |                 env: env  
737 |             })  
738 |         });  
739 |         const data = await res.json();  
740 |         if (data.status === 'ok') {  
741 |             window.showToast?.(`✅ Подключение успешно! (${data.latency_ms} ms)\nНайдено инструментов: ${data.tools_count}`, 'success');  
742 |         } else {  
743 |             window.showToast?.(`❌ Ошибка проверки подключения:\n${data.message}`, 'danger');  
744 |         }  
745 |     } catch (err) {
```

```
746     window.showToast?.(`❌ Ошибка запроса: ${err.message}`, '
danger');
747   } finally {
748     if (testBtn) {
749       testBtn.disabled = false;
750       testBtn.innerHTML = '<i class="bi bi-play-circle"></i>
Проверить подключение';
751     }
752   }
753 }
754
755 async openAllToolsModal() {
756   const modalEl = document.getElementById('mcpAllToolsModal'
);
757   if (modalEl && window.bootstrap) {
758     const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
759     modal.show();
760   }
761
762   const loadingEl = document.getElementById('mcp-all-tools-l
oading');
763   const emptyEl = document.getElementById('mcp-all-tools-emp
ty');
764   const container = document.getElementById('mcp-all-tools-c
ontainer');
765
766   loadingEl?.classList.remove('d-none');
767   emptyEl?.classList.add('d-none');
768   container?.classList.add('d-none');
769
770   try {
771     const res = await this.apiFetch('/api/admin/mcp/tools');
772     const data = await res.json();
773     loadingEl?.classList.add('d-none');
774
775     const tools = data.tools || [];
776     if (tools.length === 0) {
```

```

777         emptyEl?.classList.remove('d-none');
778         return;
779     }
780
781     container?.classList.remove('d-none');
782     if (container) {
783         container.innerHTML = tools.map(t => `
784             <div class="col-12 col-lg-6">
785                 <div class="p-3 bg-black rounded border border-sec
ondary h-100 d-flex flex-column">
786                     <div class="d-flex align-items-center justify-co
ntent-between mb-2">
787                         <strong class="text-info font-monospace d-flex
align-items-center gap-1">
788                             <i class="bi bi-wrench"></i> ${this.escapeHt
ml(t.name)}
789                         </strong>
790                         <span class="badge bg-primary-subtle text-prim
ary border border-primary-subtle small">Tool</span>
791                     </div>
792                     <p class="small text-muted mb-2 flex-grow-1">${t
his.escapeHtml(t.description || 'Нет описания.')}</p>
793                     ${t.args_schema ? `
794                         <div class="mt-auto">
795                             <details class="small">
796                                 <summary class="text-secondary cursor-poin
ter">Схема аргументов</summary>
797                                 <pre class="bg-dark text-light p-2 rounded
mt-1 mb-0 font-monospace small" style="max-height: 150px; ove
rflow-y: auto;">${this.escapeHtml(typeof t.args_schema === 'ob
ject' ? JSON.stringify(t.args_schema, null, 2) : String(t.args
_schema))}</pre>
798                                 </details>
799                             </div>
800                         ` : ''}
801                     </div>
802                 </div>
803             `).join('');

```

```

804     }
805   } catch (err) {
806     loadingEl?.classList.add('d-none');
807     if (container) {
808       container.classList.remove('d-none');
809       container.innerHTML = `
```

```

832 |         <pre class="bg-black text-white p-2 rounded mt-1 m
      | b-0 font-monospace small" style="max-height: 180px; overflow-y
      | : auto;">${this.escapeHtml(typeof t.args_schema === 'object' ?
      | JSON.stringify(t.args_schema, null, 2) : String(t.args_schema
      | ))}</pre>
833 |     </details>
834 |     ` : ''}
835 | </div>
836 | `).join('');
837 | }
838 |
839 | renderError(msg) {
840 |     const container = document.getElementById('mcp-cards-conta
      | iner');
841 |     if (container) {
842 |         container.innerHTML = `
843 |             <div class="col-12">
844 |                 <div class="alert alert-danger bg-dark border-danger
      | text-light d-flex align-items-center justify-content-between"
      | >
845 |                     <div><i class="bi bi-exclamation-triangle-fill tex
      | t-danger me-2"></i> ${this.escapeHtml(msg)}</div>
846 |                     <button class="btn btn-sm btn-outline-light" oncli
      | ck="window.mcpTab?.refresh()">Повторить</button>
847 |                 </div>
848 |             </div>
849 |         `;
850 |     }
851 | }
852 |
853 | escapeHtml(str) {
854 |     if (!str) return '';
855 |     return String(str)
856 |         .replace(/&/g, '&amp;')
857 |         .replace(/</g, '&lt;')
858 |         .replace(/>/g, '&gt;')
859 |         .replace(/"/g, '&quot;')
860 |         .replace(/'/g, '&#039;');

```



```
861 |     }
862 | }
863 |
864 | // Global initialization hooks
865 | window.McpTabManager = McpTabManager;
866 | if (!window.mcpTab) {
867 |     window.mcpTab = new McpTabManager();
868 | }
869 | window.initMcpTab = function() {
870 |     if (window.mcpTab) {
871 |         window.mcpTab.init();
872 |     }
873 | };
874 |
875 | if (document.getElementById('mcp-tab-root')) {
876 |     window.mcpTab.init();
877 | }
878 |
```

app.js

```
1  /**
2   * =====
3   * Process Name: Windows Messenger - App Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля app.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/messenger/app.js?v=202
11  *     61001_v1" type="module"></script>
12  *
13  * File: app.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/messenger
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * AI Breadboard Messenger & WebRTC Conference Client
23  */
24
25 const CONFIG = {
26   apiBase: window.location.origin,
27   wsProtocol: window.location.protocol === 'https:' ? 'wss:' :
28   'ws:',
29   iceServers: [{ urls: 'stun:stun1.l.google.com:19302' }, { url
30   s: 'stun:stun1.l.google.com:19302' }],
31 };
32
```

```
31 let currentUser = null;
32 let activeRoomId = null;
33 let activeRoomData = null;
34 let ws = null;
35 let rooms = [];
36 let mediaRecorder = null;
37 let recordedAudioChunks = [];
38 let isRecordingVoice = false;
39
40 // WebRTC Call State
41 let peerConnection = null;
42 let localStream = null;
43 let remoteStream = null;
44 let isInCall = false;
45
46 // DOM Elements
47 const chatListEl = document.getElementById('chatList');
48 const messagesContainerEl = document.getElementById('messagesContainer');
49 const messageInputEl = document.getElementById('messageInput');
50 ;
51 const btnSendMessageEl = document.getElementById('btnSendMessage');
52 const btnRecordVoiceEl = document.getElementById('btnRecordVoice');
53 const btnAttachFileEl = document.getElementById('btnAttachFile');
54 ;
55 const fileAttachmentInputEl = document.getElementById('fileAttachmentInput');
56 const typingIndicatorEl = document.getElementById('typingIndicator');
57 const chatHeaderEl = document.getElementById('chatHeader');
58 const chatInputBarEl = document.getElementById('chatInputBar');
59 ;
60 const activeChatTitleEl = document.getElementById('activeChatTitle');
61 const activeChatSubtitleEl = document.getElementById('activeChatSubtitle');
```

```

59 | const newChatModalEl = document.getElementById('newChatModal')
    | ;
60 | const videoCallOverlayEl = document.getElementById('videoCallOverlay');
    | verlay');
61 |
62 | // Initialize
63 | window.addEventListener('DOMContentLoaded', async () => {
64 |     await initUserProfile();
65 |     initWebSocket();
66 |     await loadRooms();
67 |     setupEventListeners();
68 |     registerServiceWorker();
69 | });
70 |
71 | async function initUserProfile() {
72 |     try {
73 |         const res = await fetch(`${CONFIG.apiBase}/api/messenger/users/me`, { credentials: 'include' });
74 |         const data = await res.json();
75 |         if (data.status === 'success') {
76 |             currentUser = data.user;
77 |             document.getElementById('myName').innerText = currentUser.display_name || currentUser.username;
78 |             document.getElementById('myAvatarText').innerText = (currentUser.display_name || 'U')[0].toUpperCase();
79 |         }
80 |     } catch (e) {
81 |         console.error('Failed to load user profile:', e);
82 |     }
83 | }
84 |
85 | function initWebSocket() {
86 |     if (!currentUser) return;
87 |     const wsUrl = `${CONFIG.wsProtocol}://${window.location.host}/api/messenger/ws/${currentUser.id}`;
88 |     ws = new WebSocket(wsUrl);
89 |
90 |     ws.onopen = () => console.log('Messenger WebSocket Connected

```

```

    ');
91   ws.onmessage = (event) => handleIncomingWsPacket(JSON.parse(
    event.data));
92   ws.onclose = () => {
93       console.warn('WebSocket closed. Reconnecting in 3s...');
94       setTimeout(initWebSocket, 3000);
95   };
96 }
97
98 function handleIncomingWsPacket(packet) {
99     const { type, message, room_id, user_id, bot_name, is_online
    } = packet;
100
101     if (type === 'new_message') {
102         if (activeRoomId === message.room_id) {
103             appendMessageToChat(message);
104             scrollToBottom();
105             // Send read receipt
106             if (ws && ws.readyState === WebSocket.OPEN && message.se
    nder_id !== currentUser.id) {
107                 ws.send(JSON.stringify({ type: 'message_read', room_id
    : activeRoomId, message_id: message.id }));
108             }
109         }
110         updateRoomLastMessage(message);
111     } else if (type === 'typing_start') {
112         if (activeRoomId === room_id && user_id !== currentUser.id
    ) {
113             typingIndicatorEl.style.display = 'block';
114             typingIndicatorEl.innerText = `${bot_name} || 'Someone'}
    is typing...`;
115         }
116     } else if (type === 'typing_stop') {
117         if (activeRoomId === room_id) {
118             typingIndicatorEl.style.display = 'none';
119         }
120     } else if (type === 'presence_update') {
121         updateUserPresenceUI(user_id, is_online);

```

```

122 | } else if (type.startsWith('webrtc_')) {
123 |     handleWebRTCSignalPacket(packet);
124 | }
125 | }
126 |
127 | async function loadRooms() {
128 |     try {
129 |         const res = await fetch(`${CONFIG.apiBase}/api/messenger/r
130 | ooms`, { credentials: 'include' });
131 |         const data = await res.json();
132 |         if (data.status === 'success') {
133 |             rooms = data.rooms;
134 |             renderRoomsList();
135 |         } catch (e) {
136 |             console.error('Error loading rooms:', e);
137 |         }
138 |     }
139 |
140 | function renderRoomsList() {
141 |     chatListEl.innerHTML = '';
142 |     rooms.forEach((r) => {
143 |         const item = document.createElement('div');
144 |         item.className = `chat-item ${activeRoomId === r.id ? 'act
145 | ive' : ''}`;
146 |         item.onclick = () => selectRoom(r);
147 |
148 |         const initial = (r.title || '#')[0].toUpperCase();
149 |         const lastContent = r.last_message ? (r.last_message.conte
150 | nt || '[Attachment]') : 'No messages yet';
151 |         const unreadHtml = r.unread_count > 0 ? `<span class="unre
152 | ad-badge">${r.unread_count}</span>` : '';
153 |         const onlineHtml = r.is_online ? '<div class="status-dot o
154 | nline"></div>' : '';
155 |
156 |         item.innerHTML = `
157 |             <div class="avatar">
158 |                 <span>${initial}</span>

```

```

155         ${onlineHtml}
156     </div>
157     <div class="chat-info">
158         <div class="chat-title-row">
159             <span class="chat-title">${escapeHtml(r.title)}</spa
n>
160             <span class="chat-time">${r.last_message ? formatTim
e(r.last_message.created_at) : ''}</span>
161         </div>
162         <div class="chat-snippet-row">
163             <span class="chat-snippet">${escapeHtml(lastContent)
}</span>
164             ${unreadHtml}
165         </div>
166     </div>
167     `;
168     chatListEl.appendChild(item);
169 });
170 }
171
172 async function selectRoom(room) {
173     activeRoomId = room.id;
174     activeRoomData = room;
175     renderRoomsList();
176
177     chatHeaderEl.style.display = 'flex';
178     chatInputBarEl.style.display = 'flex';
179     document.getElementById('emptyPlaceholder').style.display =
'none';
180
181     activeChatTitleEl.innerText = room.title;
182     activeChatSubtitleEl.innerText = room.room_type === 'direct'
? (room.is_online ? 'Online' : 'Offline') : `${room.members_c
ount} members`;
183     document.getElementById('activeChatAvatarText').innerText =
(room.title || '#')[0].toUpperCase();
184
185     await loadRoomMessages(room.id);

```

```

186 }
187
188 async function loadRoomMessages(roomId) {
189     messagesContainerEl.innerHTML = '';
190     try {
191         const res = await fetch(`${CONFIG.apiBase}/api/messenger/rooms/${roomId}/messages`, { credentials: 'include' });
192         const data = await res.json();
193         if (data.status === 'success') {
194             data.messages.forEach(appendMessageToChat);
195             scrollToBottom();
196         }
197     } catch (e) {
198         console.error('Error loading messages:', e);
199     }
200 }
201
202 function appendMessageToChat(msg) {
203     const isOut = msg.sender_id === currentUser.id;
204     const bubble = document.createElement('div');
205     bubble.className = `message-bubble ${isOut ? 'outgoing' : 'incoming'}`;
206
207     let senderHeader = '';
208     if (!isOut && activeRoomData && activeRoomData.room_type !== 'direct') {
209         senderHeader = `<div class="message-sender">${escapeHtml(msg.sender_name || 'Member')}</div>`;
210     }
211
212     let attachmentsHtml = '';
213     if (msg.attachments && msg.attachments.length > 0) {
214         msg.attachments.forEach((att) => {
215             if (att.file_type === 'audio') {
216                 attachmentsHtml += `<audio class="audio-player-custom" controls src="${att.file_path}"></audio>`;
217             } else if (att.file_type === 'image') {
218                 attachmentsHtml += `<img src="${att.file_path}" style=

```



```

    "max-width:100%; border-radius:10px; margin-top:6px;" />`;
219     } else {
220         attachmentsHtml += `<div style="margin-top:6px;"><a href="${att.file_path}" download style="color:#60a5fa;">␣ ${escapeHtml(att.file_name)}</a></div>`;
221     }
222 });
223 }
224
225 bubble.innerHTML = `
226     ${senderHeader}
227     <div>${escapeHtml(msg.content || '')}</div>
228     ${attachmentsHtml}
229     <div class="message-meta">
230         <span>${formatTime(msg.created_at)}</span>
231         ${isOut ? '<span>␣ ␣ </span>' : ''}
232     </div>
233 `;
234 messagesContainerEl.appendChild(bubble);
235 }
236
237 function updateRoomLastMessage(msg) {
238     const r = rooms.find((x) => x.id === msg.room_id);
239     if (r) {
240         r.last_message = msg;
241         renderRoomsList();
242     }
243 }
244
245 function updateUserPresenceUI(userId, isOnline) {
246     rooms.forEach((r) => {
247         if (r.room_type === 'direct') {
248             r.is_online = isOnline;
249         }
250     });
251     renderRoomsList();
252 }
253

```

```

254 // Send Message
255 async function sendMessage() {
256     const content = messageInputEl.value.trim();
257     if (!content || !activeRoomId) return;
258
259     if (ws && ws.readyState === WebSocket.OPEN) {
260         ws.send(JSON.stringify({
261             type: 'send_message',
262             room_id: activeRoomId,
263             content: content,
264             message_type: 'text'
265         }));
266     } else {
267         // REST Fallback
268         await fetch(`${CONFIG.apiBase}/api/messenger/rooms/${activ
eRoomId}/messages`, {
269             method: 'POST',
270             headers: { 'Content-Type': 'application/json' },
271             credentials: 'include',
272             body: JSON.stringify({ content: content, message_type: '
text' }),
273         });
274     }
275
276     messageInputEl.value = '';
277 }
278
279 // Voice Note Recording
280 async function toggleVoiceRecording() {
281     if (isRecordingVoice) {
282         mediaRecorder.stop();
283         isRecordingVoice = false;
284         btnRecordVoiceEl.style.background = 'var(--bg-secondary)';
285     } else {
286         try {
287             const stream = await navigator.mediaDevices.getUserMedia
({ audio: true });
288             mediaRecorder = new MediaRecorder(stream);

```

```

289         recordedAudioChunks = [];
290
291         mediaRecorder.ondataavailable = (e) => recordedAudioChun
ks.push(e.data);
292         mediaRecorder.onstop = async () => {
293             const blob = new Blob(recordedAudioChunks, { type: 'au
dio/webm' });
294             const formData = new FormData();
295             formData.append('file', blob, `voice_${Date.now()}.web
m`);
296             formData.append('duration_sec', '3.0');
297
298             const res = await fetch(`${CONFIG.apiBase}/api/messeng
er/upload`, {
299                 method: 'POST',
300                 credentials: 'include',
301                 body: formData,
302             });
303             const data = await res.json();
304             if (data.status === 'success') {
305                 ws.send(JSON.stringify({
306                     type: 'send_message',
307                     room_id: activeRoomId,
308                     content: '',
309                     message_type: 'audio',
310                     attachment_ids: [data.attachment_id],
311                 }));
312             }
313         };
314
315         mediaRecorder.start();
316         isRecordingVoice = true;
317         btnRecordVoiceEl.style.background = 'var(--danger-color)
';
318     } catch (e) {
319         window.showToast?(`Для записи голосовых сообщений требу
ется доступ к микрофону`, 'warning') || alert('Для записи голо
совых сообщений требуется доступ к микрофону');

```

```

320 |     }
321 | }
322 |
323 |
324 | // WebRTC Conference / Call Handler
325 | async function startWebRTCCall(callType = 'video') {
326 |     if (!activeRoomId) return;
327 |     videoCallOverlayEl.style.display = 'flex';
328 |     isInCall = true;
329 |
330 |     try {
331 |         localStream = await navigator.mediaDevices.getUserMedia({
332 |             video: callType === 'video',
333 |             audio: true,
334 |         });
335 |         document.getElementById('localVideo').srcObject = localStream;
336 |
337 |         initPeerConnection();
338 |         localStream.getTracks().forEach((track) => peerConnection.
339 |             addTrack(track, localStream));
340 |
341 |         const offer = await peerConnection.createOffer();
342 |         await peerConnection.setLocalDescription(offer);
343 |
344 |         ws.send(JSON.stringify({
345 |             type: 'webrtc_offer',
346 |             data: {
347 |                 action: 'offer',
348 |                 room_id: activeRoomId,
349 |                 sdp: offer.sdp,
350 |                 call_type: callType,
351 |             },
352 |         }));
353 |     } catch (e) {
354 |         console.error('Call initialization failed:', e);
355 |     }

```

```
356 |
357 | function initPeerConnection() {
358 |     peerConnection = new RTCPeerConnection({ iceServers: CONFIG.
    | iceServers });
359 |
360 |     peerConnection.onicecandidate = (event) => {
361 |         if (event.candidate && ws) {
362 |             ws.send(JSON.stringify({
363 |                 type: 'webrtc_ice_candidate',
364 |                 data: {
365 |                     action: 'ice_candidate',
366 |                     room_id: activeRoomId,
367 |                     candidate: event.candidate,
368 |                 },
369 |             }));
370 |         }
371 |     };
372 |
373 |     peerConnection.ontrack = (event) => {
374 |         remoteStream = event.streams[0];
375 |         const remoteVid = document.getElementById('remoteVideo');
376 |         remoteVid.srcObject = remoteStream;
377 |         document.getElementById('remoteVideoTile').style.display =
    | 'flex';
378 |     };
379 | }
380 |
381 | async function handleWebRTCSignalPacket(packet) {
382 |     const { type, sdp, candidate, sender_id, room_id } = packet;
383 |
384 |     if (type === 'webrtc_offer') {
385 |         videoCallOverlayEl.style.display = 'flex';
386 |         isInCall = true;
387 |         initPeerConnection();
388 |
389 |         localStream = await navigator.mediaDevices.getUserMedia({
    | video: true, audio: true });
390 |         document.getElementById('localVideo').srcObject = localStr
```

```

    eam;
391     localStream.getTracks().forEach((track) => peerConnection.
    addTrack(track, localStream));
392
393     await peerConnection.setRemoteDescription(new RTCSessionDe
    scription({ type: 'offer', sdp }));
394     const answer = await peerConnection.createAnswer();
395     await peerConnection.setLocalDescription(answer);
396
397     ws.send(JSON.stringify({
398         type: 'webrtc_answer',
399         data: {
400             action: 'answer',
401             room_id: room_id,
402             target_id: sender_id,
403             sdp: answer.sdp,
404         },
405     }));
406 } else if (type === 'webrtc_answer') {
407     if (peerConnection) {
408         await peerConnection.setRemoteDescription(new RTCSession
    Description({ type: 'answer', sdp }));
409     }
410 } else if (type === 'webrtc_ice_candidate') {
411     if (peerConnection && candidate) {
412         await peerConnection.addIceCandidate(new RTCIceCandidate
    (candidate));
413     }
414 }
415 }
416
417 function endCall() {
418     if (localStream) localStream.getTracks().forEach((t) => t.st
    op());
419     if (peerConnection) peerConnection.close();
420     peerConnection = null;
421     videoCallOverlayEl.style.display = 'none';
422     isInCall = false;

```

```

423
424   if (ws && activeRoomId) {
425       ws.send(JSON.stringify({
426           type: 'webrtc_leave_call',
427           data: { action: 'leave_call', room_id: activeRoomId },
428       }));
429   }
430 }
431
432 // Event Listeners
433 function setupEventListeners() {
434     btnSendMessageEl.onclick = sendMessage;
435     messageInputEl.onkeydown = (e) => {
436         if (e.key === 'Enter') sendMessage();
437     };
438     btnRecordVoiceEl.onclick = toggleVoiceRecording;
439     btnAttachFileEl.onclick = () => fileAttachmentInputEl.click(
440     );
441
442     fileAttachmentInputEl.onChange = async () => {
443         const file = fileAttachmentInputEl.files[0];
444         if (!file || !activeRoomId) return;
445         const formData = new FormData();
446         formData.append('file', file);
447         const res = await fetch(`${CONFIG.apiBase}/api/messenger/u
448         pload`, {
449             method: 'POST',
450             credentials: 'include',
451             body: formData,
452         });
453         const data = await res.json();
454         if (data.status === 'success') {
455             ws.send(JSON.stringify({
456                 type: 'send_message',
457                 room_id: activeRoomId,
458                 content: '',
459                 message_type: data.file_type,
460                 attachment_ids: [data.attachment_id],

```

```

459     }));
460   }
461 };
462
463   document.getElementById('btnStartVideoCall').onclick = () =>
startWebRTCcall('video');
464   document.getElementById('btnStartAudioCall').onclick = () =>
startWebRTCcall('audio');
465   document.getElementById('btnEndCall').onclick = endCall;
466
467   // New Chat Modal
468   document.getElementById('btnNewChat').onclick = () => { newC
hatModalEl.style.display = 'flex'; };
469   document.getElementById('modalBtnCancel').onclick = () => {
newChatModalEl.style.display = 'none'; };
470   document.getElementById('modalRoomType').onchange = (e) => {
471     document.getElementById('modalTitleGroup').style.display =
e.target.value !== 'direct' ? 'block' : 'none';
472   };
473
474   document.getElementById('modalBtnCreate').onclick = async ()
=> {
475     const roomType = document.getElementById('modalRoomType').
value;
476     const title = document.getElementById('modalRoomTitle').va
lue;
477     const res = await fetch(`${CONFIG.apiBase}/api/messenger/r
ooms`, {
478       method: 'POST',
479       headers: { 'Content-Type': 'application/json' },
480       credentials: 'include',
481       body: JSON.stringify({ room_type: roomType, title: title
}),
482     });
483     const data = await res.json();
484     if (data.status === 'success') {
485       newChatModalEl.style.display = 'none';
486       await loadRooms();

```



```
487 |     }
488 | };
489 |
490 |
491 | function formatTime(isoStr) {
492 |     if (!isoStr) return '';
493 |     const d = new Date(isoStr);
494 |     return d.toLocaleTimeString([], { hour: '2-digit', minute: '
    2-digit' });
495 | }
496 |
497 | function scrollToBottom() {
498 |     messagesContainerEl.scrollTop = messagesContainerEl.scrollHeight;
499 | }
500 |
501 | function escapeHtml(text) {
502 |     const div = document.createElement('div');
503 |     div.innerText = text;
504 |     return div.innerHTML;
505 | }
506 |
507 | function registerServiceWorker() {
508 |     if ('serviceWorker' in navigator) {
509 |         navigator.serviceWorker.register('sw.js').catch(console.error);
510 |     }
511 | }
512 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/messenger/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/messenger/index.html" c
17         lass="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/messenger
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!DOCTYPE html>
30 <html lang="en">
31 <head>
32     <meta charset="UTF-8" />
33     <meta name="viewport" content="width=device-width, initial-s
34     cale=1.0, maximum-scale=1.0, user-scalable=no" />
35     <title>AI Breadboard Messenger & Rooms</title>
36     <link rel="stylesheet" href="style.css" />
37     <link rel="manifest" href="manifest.json" />
```

```

32 | <meta name="theme-color" content="#0f172a" />
33 | </head>
34 | <body>
35 |   <div class="messenger-app">
36 |     <!-- Sidebar -->
37 |     <aside class="sidebar" id="sidebar">
38 |       <div class="sidebar-header">
39 |         <div class="user-profile-badge">
40 |           <div class="avatar" id="myAvatar">
41 |             <span id="myAvatarText">U</span>
42 |             <div class="status-dot online"></div>
43 |           </div>
44 |           <div>
45 |             <div style="font-weight: 600; font-size: 15px;" id
|"myName">Loading...</div>
46 |             <div style="font-size: 12px; color: var(--text-sec
ondary);" id="myStatus">Online</div>
47 |           </div>
48 |         </div>
49 |         <button class="btn-icon" id="btnNewChat" title="New Ch
at or Room">🗉 </button>
50 |       </div>
51 |
52 |       <div class="search-bar-container">
53 |         <input type="text" class="search-input" id="searchChat
sInput" placeholder="Search chats, contacts, @bots..." />
54 |       </div>
55 |
56 |       <div class="chat-list" id="chatList">
57 |         <!-- Chat items rendered dynamically -->
58 |       </div>
59 |     </aside>
60 |
61 |     <!-- Main Chat Area -->
62 |     <main class="chat-main" id="chatMain">
63 |       <!-- Chat Header -->
64 |       <div class="chat-header" id="chatHeader" style="display:
none;">

```

```

65 |         <div style="display: flex; align-items: center; gap: 1
    | 2px;">
66 |             <div class="avatar" id="activeChatAvatar">
67 |                 <span id="activeChatAvatarText">#</span>
68 |             </div>
69 |             <div>
70 |                 <div style="font-weight: 600; font-size: 16px;" id
    | ="activeChatTitle">Select a chat</div>
71 |                 <div style="font-size: 12px; color: var(--text-sec
    | ondary);" id="activeChatSubtitle">...</div>
72 |             </div>
73 |         </div>
74 |         <div class="chat-header-actions">
75 |             <button class="btn-icon" id="btnStartAudioCall" titl
    | e="Audio Call">🎙️ </button>
76 |             <button class="btn-icon" id="btnStartVideoCall" titl
    | e="Video Meeting">📺 </button>
77 |         </div>
78 |     </div>
79 |
80 |     <!-- Messages Stream -->
81 |     <div class="messages-container" id="messagesContainer">
82 |         <div style="text-align: center; color: var(--text-seco
    | ndary); margin-top: 40px;" id="emptyPlaceholder">
83 |             Select or create a conversation to start messaging.
84 |         </div>
85 |     </div>
86 |
87 |     <!-- Typing indicator -->
88 |     <div class="typing-indicator" id="typingIndicator" style
    | ="display: none;"></div>
89 |
90 |     <!-- Input Bar -->
91 |     <div class="chat-input-bar" id="chatInputBar" style="dis
    | play: none;">
92 |         <input type="file" id="fileAttachmentInput" style="dis
    | play: none;" multiple />
93 |         <button class="btn-icon" id="btnAttachFile" title="Att

```

```

    ach file or photo">❏ </button>
94
95     <input type="text" class="message-input" id="messageIn
    put" placeholder="Write a message... (Type @ai for bot)" autoc
    omplete="off" />
96
97     <button class="btn-icon" id="btnRecordVoice" title="Re
    cord voice note">❏ </button>
98     <button class="btn-icon" id="btnSendMessage" title="Se
    nd message" style="background: var(--accent-color);">❏ </button
    >
99     </div>
100
101     <!-- WebRTC Meeting Room Overlay -->
102     <div class="video-call-overlay" id="videoCallOverlay" st
    yle="display: none;">
103         <div class="video-grid" id="videoGrid">
104             <div class="video-tile" id="localVideoTile">
105                 <video id="localVideo" autoplay playsinline muted>
</video>
106                 <span class="tile-label">You</span>
107             </div>
108             <div class="video-tile" id="remoteVideoTile" style="
    display: none;">
109                 <video id="remoteVideo" autoplay playsinline></vid
    eo>
110                 <span class="tile-label" id="remoteTileLabel">Part
    icipant</span>
111             </div>
112         </div>
113
114         <div class="call-controls">
115             <button class="btn-icon" id="btnToggleMic" title="Mu
    te/Unmute Mic">❏ </button>
116             <button class="btn-icon" id="btnToggleCam" title="To
    ggle Camera">❏ </button>
117             <button class="btn-icon" id="btnShareScreen" title="
    Share Screen">❏ </button>

```

```

118 |         <button class="btn-icon btn-danger" id="btnEndCall"
    | title="Hang Up">⌂ </button>
119 |     </div>
120 | </div>
121 | </main>
122 | </div>
123 |
124 | <!-- New Chat Modal -->
125 | <div class="modal-overlay" id="newChatModal" style="display:
    | none;">
126 |     <div class="modal-card">
127 |         <h3>Create New Conversation</h3>
128 |         <div class="form-group">
129 |             <label>Room Type</label>
130 |             <select id="modalRoomType">
131 |                 <option value="direct">Direct Message (1-on-1)</opti
    | on>
132 |                 <option value="group">Group Chat</option>
133 |                 <option value="meeting">Video Meeting Room</option>
134 |             </select>
135 |         </div>
136 |         <div class="form-group" id="modalTitleGroup" style="disp
    | lay: none;">
137 |             <label>Room Title</label>
138 |             <input type="text" id="modalRoomTitle" placeholder="e.
    | g. Project Discussion" />
139 |         </div>
140 |         <div class="form-group">
141 |             <label>Search User / Member</label>
142 |             <input type="text" id="modalUserSearch" placeholder="T
    | ype username or name..." />
143 |             <div id="modalUserResults" style="max-height: 120px; o
    | verflow-y: auto; margin-top: 8px;"></div>
144 |         </div>
145 |         <div style="display: flex; justify-content: flex-end; ga
    | p: 10px; margin-top: 20px;">
146 |             <button class="btn-icon" style="width: auto; padding:
    | 8px 16px; border-radius: 8px;" id="modalBtnCancel">Cancel</but

```

```
ton>
147 |         <button class="btn-icon" style="width: auto; padding:
      | 8px 16px; border-radius: 8px; background: var(--accent-color);
      | " id="modalBtnCreate">Create</button>
148 |     </div>
149 | </div>
150 | </div>
151 |
152 | <script src="app.js"></script>
153 | </body>
154 | </html>
155 |
```

style.css

```
1  /**
2   * =====
3   *
4   * =====
5   * Process Name: Windows Web Styles - Style Stylesheet
6   * =====
7   *
8   * Description:
9   *   Таблица стилей стиля и оформления интерфейса (style).
10  *
11  * Usage Examples:
12  *   HTML Link Tag:
13  *     <link rel="stylesheet" href="/windows/api/webgui/messen
14  *     ger/style.css?v=20261001_v1">
15  *
16  * File: style.css
17  * Project: ai-breadboard
18  * Package: windows/api/webgui/messenger
19  * Author: hypo69
20  * Copyright: © 2026 hypo69
21  * Updated: 2026-10-01 13:04:40
22  * =====
23  */
24
25  /** =====
26  *
27  * =====
28  * messenger/style.css – Дочерние стили Messenger (наследует m
29  * ain.css)
30  * Версия: 20260917_refactor_v1
31  * =====
32  */
33
34  /** Messenger наследует переменные из main.css.
35  *   Локальные псевдонимы для удобства внутри компонента: */
36  .messenger-app {
37    --msg-outgoing: var(--chat-user-bg);
```



```
30 | --msg-incoming: var(--chat-bot-bg);
31 | --msg-incoming-text: var(--chat-bot-text);
32 | --online-color: #10b981;
33 | --offline-color: #6b7280;
34 | --danger-color: #ef4444;
35 | }
36 |
37 | /* — Основной контейнер —————
    | ————— */
38 | .messenger-app {
39 |   display: flex;
40 |   height: 100vh;
41 |   width: 100%;
42 |   background: var(--bg-color);
43 |   font-size: 0.85rem;
44 | }
45 |
46 | /* — Боковая панель —————
    | ————— */
47 | .sidebar {
48 |   width: 320px;
49 |   min-width: 260px;
50 |   background: var(--surface-1);
51 |   border-right: 1px solid var(--border-color);
52 |   display: flex;
53 |   flex-direction: column;
54 |   z-index: 10;
55 | }
56 |
57 | .sidebar-header {
58 |   padding: 10px 12px;
59 |   display: flex;
60 |   align-items: center;
61 |   justify-content: space-between;
62 |   border-bottom: 1px solid var(--border-color);
63 | }
64 |
65 | /* — Профиль пользователя —————
```

```
        _____ */
66 .user-profile-badge {
67     display: flex;
68     align-items: center;
69     gap: 10px;
70 }
71
72 .avatar {
73     width: 36px;
74     height: 36px;
75     border-radius: 50%;
76     background: var(--border-color);
77     display: flex;
78     align-items: center;
79     justify-content: center;
80     font-weight: 700;
81     font-size: 0.85rem;
82     color: var(--text-color);
83     position: relative;
84     object-fit: cover;
85     flex-shrink: 0;
86 }
87
88 .avatar .status-dot {
89     position: absolute;
90     bottom: 0;
91     right: 0;
92     width: 10px;
93     height: 10px;
94     border-radius: 50%;
95     border: 2px solid var(--surface-1);
96 }
97
98 .status-dot.online { background-color: var(--online-color); }
99 .status-dot.offline { background-color: var(--offline-color); }
100
101 /* — Строка поиска _____
```

```
102 | _____ */
103 | .search-bar-container {
104 |     padding: 8px 10px;
105 |     border-bottom: 1px solid var(--border-color);
106 | }
107 |
108 | .search-input {
109 |     width: 100%;
110 |     padding: 6px 12px;
111 |     background: var(--surface-2);
112 |     border: 1px solid var(--border-color);
113 |     border-radius: 16px;
114 |     color: var(--text-color);
115 |     outline: none;
116 |     font-size: 0.82rem;
117 |     transition: border-color 0.15s;
118 | }
119 |
120 | .search-input::placeholder { color: var(--text-muted); }
121 | .search-input:focus { border-color: var(--nav-active); }
122 |
123 | /* — Список чатов —————
124 | _____ */
125 |
126 | .chat-list {
127 |     flex: 1;
128 |     overflow-y: auto;
129 | }
130 |
131 | .chat-item {
132 |     display: flex;
133 |     align-items: center;
134 |     gap: 10px;
135 |     padding: 9px 12px;
136 |     cursor: pointer;
137 |     border-bottom: 1px solid var(--border-subtle);
138 |     transition: background 0.12s ease;
139 | }
```

```
138 | .chat-item:hover {
139 |     background: var(--nav-hover-bg);
140 | }
141 |
142 | .chat-item.active {
143 |     background: rgba(37, 99, 235, 0.08);
144 |     border-left: 3px solid var(--nav-active);
145 | }
146 |
147 | [data-theme="dark"] .chat-item.active {
148 |     background: rgba(88, 166, 255, 0.08);
149 | }
150 |
151 | .chat-info { flex: 1; min-width: 0; }
152 |
153 | .chat-title-row {
154 |     display: flex;
155 |     justify-content: space-between;
156 |     align-items: center;
157 |     margin-bottom: 2px;
158 | }
159 |
160 | .chat-title {
161 |     font-weight: 600;
162 |     font-size: 0.85rem;
163 |     white-space: nowrap;
164 |     overflow: hidden;
165 |     text-overflow: ellipsis;
166 |     color: var(--text-color);
167 | }
168 |
169 | .chat-time {
170 |     font-size: 0.72rem;
171 |     color: var(--text-muted);
172 |     flex-shrink: 0;
173 | }
174 |
175 | .chat-snippet-row {
```

```
176 | display: flex;
177 | justify-content: space-between;
178 | align-items: center;
179 | }
180 |
181 | .chat-snippet {
182 |   font-size: 0.78rem;
183 |   color: var(--text-muted);
184 |   white-space: nowrap;
185 |   overflow: hidden;
186 |   text-overflow: ellipsis;
187 | }
188 |
189 | .unread-badge {
190 |   background: var(--nav-active);
191 |   color: #fff;
192 |   font-size: 0.68rem;
193 |   font-weight: 700;
194 |   padding: 1px 6px;
195 |   border-radius: 10px;
196 |   flex-shrink: 0;
197 | }
198 |
199 | /* — Главная область чата —————
    | ————— */
200 | .chat-main {
201 |   flex: 1;
202 |   display: flex;
203 |   flex-direction: column;
204 |   background: var(--bg-color);
205 |   position: relative;
206 | }
207 |
208 | .chat-header {
209 |   padding: 8px 16px;
210 |   background: var(--surface-1);
211 |   border-bottom: 1px solid var(--border-color);
212 |   display: flex;
```

```
213 | justify-content: space-between;
214 | align-items: center;
215 | z-index: 5;
216 | }
217 |
218 | .chat-header-actions {
219 |     display: flex;
220 |     gap: 8px;
221 | }
222 |
223 | .btn-icon {
224 |     background: var(--surface-2);
225 |     border: 1px solid var(--border-color);
226 |     color: var(--text-color);
227 |     width: 32px;
228 |     height: 32px;
229 |     border-radius: 50%;
230 |     display: flex;
231 |     align-items: center;
232 |     justify-content: center;
233 |     cursor: pointer;
234 |     font-size: 0.85rem;
235 |     transition: background 0.15s, color 0.15s;
236 | }
237 |
238 | .btn-icon:hover {
239 |     background: var(--nav-active);
240 |     color: #fff;
241 |     border-color: var(--nav-active);
242 | }
243 |
244 | .btn-danger { background: var(--danger-color) !important; color: #fff !important; }
245 |
246 | /* — Контейнер сообщений ————— */
247 |
248 | .messages-container {
249 |     flex: 1;
```

```
249 | padding: 12px 16px;
250 | overflow-y: auto;
251 | display: flex;
252 | flex-direction: column;
253 | gap: 8px;
254 | }
255 |
256 | .message-bubble {
257 |   max-width: 70%;
258 |   padding: 8px 12px;
259 |   border-radius: 14px;
260 |   position: relative;
261 |   line-height: 1.45;
262 |   font-size: 1.15rem;
263 |   word-break: break-word;
264 | }
265 |
266 | .message-bubble.incoming {
267 |   align-self: flex-start;
268 |   background: var(--msg-incoming);
269 |   color: var(--msg-incoming-text);
270 |   border-bottom-left-radius: 4px;
271 | }
272 |
273 | .message-bubble.outgoing {
274 |   align-self: flex-end;
275 |   background: var(--msg-outgoing);
276 |   color: var(--chat-user-text);
277 |   border-bottom-right-radius: 4px;
278 | }
279 |
280 | .message-meta {
281 |   display: flex;
282 |   align-items: center;
283 |   gap: 4px;
284 |   margin-top: 4px;
285 |   font-size: 0.72rem;
286 |   opacity: 0.75;
```

```
287 | }
288 |
289 | .message-bubble.outgoing .message-meta {
290 |     justify-content: flex-end;
291 | }
292 |
293 | .message-status {
294 |     font-size: 0.75rem;
295 | }
296 |
297 | .audio-preview {
298 |     margin-top: 6px;
299 |     width: 100%;
300 |     max-width: 240px;
301 | }
302 |
303 | /* — Индикатор набора текста —————
    | ————— */
304 | .typing-indicator {
305 |     display: flex;
306 |     align-items: center;
307 |     gap: 6px;
308 |     padding: 4px 16px;
309 |     font-size: 0.8rem;
310 |     color: var(--text-muted);
311 |     font-style: italic;
312 | }
313 |
314 | /* — Поле ввода —————
    | ————— */
315 | .chat-input-bar {
316 |     padding: 8px 12px;
317 |     background: var(--surface-1);
318 |     border-top: 1px solid var(--border-color);
319 |     display: flex;
320 |     align-items: center;
321 |     gap: 8px;
322 | }
```



```
323 |
324 | .message-input {
325 |     flex: 1;
326 |     padding: 8px 14px;
327 |     background: var(--surface-2);
328 |     border: 1px solid var(--border-color);
329 |     border-radius: 20px;
330 |     color: var(--text-color);
331 |     outline: none;
332 |     font-size: 1.15rem;
333 |     transition: border-color 0.15s;
334 | }
335 |
336 | .message-input::placeholder { color: var(--text-muted); }
337 | .message-input:focus { border-color: var(--nav-active); }
338 |
339 | .voice-recording-status {
340 |     color: var(--danger-color);
341 |     font-weight: 600;
342 |     font-size: 0.82rem;
343 |     display: flex;
344 |     align-items: center;
345 |     gap: 6px;
346 | }
347 |
348 | /* — Оверлей видеозвонка —————
    | ————— */
349 | .video-call-overlay {
350 |     position: absolute;
351 |     top: 0; left: 0;
352 |     width: 100%; height: 100%;
353 |     background: rgba(0, 0, 0, 0.88);
354 |     backdrop-filter: blur(12px);
355 |     z-index: 100;
356 |     display: flex;
357 |     flex-direction: column;
358 | }
359 |
```

```
360 | .video-grid {
361 |     flex: 1;
362 |     display: grid;
363 |     grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
364 |     gap: 12px;
365 |     padding: 16px;
366 |     align-items: center;
367 |     justify-items: center;
368 | }
369 |
370 | .video-tile {
371 |     width: 100%; height: 100%;
372 |     max-height: 440px;
373 |     background: var(--surface-1);
374 |     border-radius: 12px;
375 |     overflow: hidden;
376 |     position: relative;
377 |     display: flex;
378 |     align-items: center;
379 |     justify-content: center;
380 |     border: 1px solid var(--border-color);
381 | }
382 |
383 | .video-tile video { width: 100%; height: 100%; object-fit: cover; }
384 |
385 | .video-tile .tile-label {
386 |     position: absolute;
387 |     bottom: 10px; left: 10px;
388 |     background: rgba(0, 0, 0, 0.55);
389 |     padding: 3px 8px;
390 |     border-radius: 6px;
391 |     font-size: 0.78rem;
392 |     color: #fff;
393 | }
394 |
395 | .call-controls {
396 |     height: 72px;
```

```
397 | background: var(--surface-1);
398 | display: flex;
399 | align-items: center;
400 | justify-content: center;
401 | gap: 16px;
402 | border-top: 1px solid var(--border-color);
403 | }
404 |
405 | .audio-player-custom {
406 |   margin-top: 4px;
407 |   width: 100%;
408 |   max-width: 240px;
409 | }
410 |
411 | /* — Модальное окно —————
    | ————— */
412 | .modal-overlay {
413 |   position: fixed;
414 |   top: 0; left: 0; width: 100%; height: 100%;
415 |   background: rgba(0, 0, 0, 0.6);
416 |   display: flex;
417 |   align-items: center;
418 |   justify-content: center;
419 |   z-index: 200;
420 | }
421 |
422 | .modal-card {
423 |   background: var(--surface-1);
424 |   border: 1px solid var(--border-color);
425 |   padding: 20px;
426 |   border-radius: 12px;
427 |   box-shadow: var(--shadow-lg);
428 |   width: 90%;
429 |   max-width: 420px;
430 | }
431 |
432 | .modal-card h3 { margin-bottom: 12px; font-size: 1rem; font-weight: 600; }
```

```
433 |
434 | .form-group { margin-bottom: 10px; }
435 | .form-group label {
436 |     display: block;
437 |     font-size: 0.78rem;
438 |     color: var(--text-muted);
439 |     margin-bottom: 4px;
440 | }
441 | .form-group input,
442 | .form-group select {
443 |     width: 100%;
444 |     padding: 7px 10px;
445 |     background: var(--surface-2);
446 |     border: 1px solid var(--border-color);
447 |     border-radius: 6px;
448 |     color: var(--text-color);
449 |     font-size: 0.82rem;
450 |     outline: none;
451 |     transition: border-color 0.15s;
452 | }
453 | .form-group input:focus,
454 | .form-group select:focus {
455 |     border-color: var(--nav-active);
456 | }
457 |
```

sw.js

```
1  /**
2   * =====
3   * Process Name: Windows Messenger - Sw Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля sw.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/messenger/sw.js?v=2026
11  *     1001_v1" type="module"></script>
12  *
13  * File: sw.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/messenger
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 const CACHE_NAME = 'ai-breadboard-messenger-v1';
22 const ASSETS = [
23   './',
24   './index.html',
25   './style.css',
26   './app.js',
27   './manifest.json',
28 ];
29
30 self.addEventListener('install', (event) => {
31   event.waitUntil(
32     caches.open(CACHE_NAME).then((cache) => cache.addAll(ASSET
```

```
    S))
33 | );
34 | });
35 |
36 | self.addEventListener('fetch', (event) => {
37 |     // Network first, cache fallback
38 |     event.respondWith(
39 |         fetch(event.request).catch(() => caches.match(event.request))
40 |     );
41 | });
42 |
```

widget.js

```
1  /**
2   * =====
3   * Process Name: Windows Messenger - Widget Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля widget.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/messenger/widget.js?v=
11  *       20261001_v1" type="module"></script>
12  *
13  * File: widget.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/messenger
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * AI Breadboard Floating Messenger Widget
23  * Zero-dependency embeddable script for any website or WordPress.
24  */
25
26 (function () {
27   const config = window.AB_MESSENGER_CONFIG || {
28     apiUrl: window.location.origin,
29     ssoToken: '',
30     userId: '',
31     userName: 'Guest',
```

```
32 | };
33 |
34 | const style = document.createElement('style');
35 | style.innerHTML = `
36 |   .ab-widget-bubble {
37 |     position: fixed;
38 |     bottom: 24px;
39 |     right: 24px;
40 |     width: 60px;
41 |     height: 60px;
42 |     border-radius: 50%;
43 |     background: linear-gradient(135deg, #3b82f6, #1d4ed8);
44 |     box-shadow: 0 10px 25px rgba(59, 130, 246, 0.4);
45 |     display: flex;
46 |     align-items: center;
47 |     justify-content: center;
48 |     cursor: pointer;
49 |     z-index: 999999;
50 |     transition: transform 0.2s ease;
51 |     color: #fff;
52 |     font-size: 28px;
53 |   }
54 |   .ab-widget-bubble:hover {
55 |     transform: scale(1.08);
56 |   }
57 |   .ab-widget-frame {
58 |     position: fixed;
59 |     bottom: 96px;
60 |     right: 24px;
61 |     width: 400px;
62 |     height: 620px;
63 |     max-width: calc(100vw - 32px);
64 |     max-height: calc(100vh - 120px);
65 |     border-radius: 20px;
66 |     box-shadow: 0 12px 40px rgba(0, 0, 0, 0.35);
67 |     border: 1px solid rgba(255, 255, 255, 0.1);
68 |     overflow: hidden;
69 |     z-index: 999998;
```



```
70 |     display: none;
71 |     background: #0f172a;
72 | }
73 | .ab-widget-frame iframe {
74 |     width: 100%;
75 |     height: 100%;
76 |     border: none;
77 | }
78 | `;
79 | document.head.appendChild(style);
80 |
81 | const bubble = document.createElement('div');
82 | bubble.className = 'ab-widget-bubble';
83 | bubble.innerHTML = '□ ';
84 |
85 | const frameContainer = document.createElement('div');
86 | frameContainer.className = 'ab-widget-frame';
87 |
88 | let isOpen = false;
89 | let iframe = null;
90 |
91 | bubble.onclick = () => {
92 |     isOpen = !isOpen;
93 |     if (isOpen) {
94 |         if (!iframe) {
95 |             iframe = document.createElement('iframe');
96 |             const tokenQuery = config.ssoToken ? `?token=${encodeURIComponent(
97 | RComponent(config.ssoToken))}` : '';
98 |             iframe.src = `${config.apiUrl}/webinterface/messenger/
99 | ${tokenQuery}`;
100 |             iframe.allow = 'camera; microphone; display-capture; a
101 | utoplay';
102 |             frameContainer.appendChild(iframe);
103 |         }
104 |         frameContainer.style.display = 'block';
105 |         bubble.innerHTML = '□ ';
106 |     } else {
107 |         frameContainer.style.display = 'none';
108 |         bubble.innerHTML = '□ ';
109 |     }
110 | }
```

```
105 |         bubble.innerHTML = ' ';
106 |     }
107 | };
108 |
109 | document.body.appendChild(frameContainer);
110 | document.body.appendChild(bubble);
111 |})();
112 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Models - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/models/main.js?v=20261
11  *     001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/models
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Gemini Models & APIs management tab logic
22
23 async function initModelsTab() {
24   window.api = window.api || {};
25   if (!window.api.fetch) {
26     window.api.fetch = async function(url, options = {}) {
27       const response = await fetch(url, options);
28       if (!response.ok) {
29         let msg = response.statusText;
30         try {
31           const data = await response.json();
32           if (data && data.detail) {
```

```

33         if (typeof data.detail === 'string') msg = data.de
tail;
34         else if (Array.isArray(data.detail)) msg = data.de
tail.map(d => d.msg || JSON.stringify(d)).join(', ');
35         else msg = JSON.stringify(data.detail);
36     } else if (data && data.message) {
37         msg = data.message;
38     }
39     } catch {}
40     throw new Error(`${response.status} ${msg}`);
41 }
42 return response.json();
43 };
44 }
45
46 const modelSelect = document.getElementById('models-tab-sele
ct');
47 const saveBtn = document.getElementById('btn-models-tab-save
');
48 const keysListBody = document.getElementById('keys-list-body
');
49 const refreshKeysBtn = document.getElementById('btn-refresh-
keys');
50 const addKeyBtn = document.getElementById('btn-add-key');
51 const saveAgyBtn = document.getElementById('btn-save-agy');
52 const saveFoundryBtn = document.getElementById('btn-save-fou
ndry');
53 const saveOllamaBtn = document.getElementById('btn-save-olla
ma');
54 const saveOnnxBtn = document.getElementById('btn-save-onnx')
;
55 const saveBtnInstr = document.getElementById('btn-save-instr
uction');
56 const reloadBtnInstr = document.getElementById('btn-reload-i
nstruction');
57
58 // Ensure active provider pill tab pane has show active clas
ses

```

```

59 |   const activePill = document.querySelector('#provider-pills-tab .nav-link.active') || document.getElementById('pill-gemini-tab');
60 |   if (activePill) {
61 |     const targetSelector = activePill.getAttribute('data-bs-target');
62 |     if (targetSelector) {
63 |       const targetPane = document.querySelector(targetSelector);
64 |       if (targetPane && !targetPane.classList.contains('active')) {
65 |         targetPane.classList.add('show', 'active');
66 |       }
67 |     }
68 |   }
69 |
70 |   // 1. Bind event handlers immediately
71 |   if (saveBtnInstr) saveBtnInstr.onclick = saveSystemInstruction;
72 |   if (reloadBtnInstr) reloadBtnInstr.onclick = loadSystemInstruction;
73 |
74 |   // Provider switch direct toggling
75 |   const agyEnabledSwitch = document.getElementById('agy-enabled');
76 |   if (agyEnabledSwitch) {
77 |     agyEnabledSwitch.onchange = async () => {
78 |       const enabled = agyEnabledSwitch.checked;
79 |       const remember = document.getElementById('agy-remember')?.checked ?? true;
80 |       const model = document.getElementById('agy-model')?.value || 'agy-flash';
81 |       const key = document.getElementById('agy-key')?.value.trim() || '';
82 |
83 |       try {
84 |         await window.api.fetch('/api/agy/config', {
85 |           method: 'POST',

```

```

86         headers: { 'Content-Type': 'application/json' },
87         body: JSON.stringify({ enabled, model, key, remember
    })
88     });
89     showModelsNotification(`Google Antigravity (AGY) ${enabled ? 'активирован' : 'деактивирован'}${remember ? ' (сохране
но в config.json)' : ''}`, 'info');
90     if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
91     } catch (err) {
92         console.error('Ошибка переключения Antigravity:', err)
    ;
93         showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
94     }
95     };
96 }
97
98 const foundryEnabledSwitch = document.getElementById('foundr
y-enabled');
99 if (foundryEnabledSwitch) {
100     foundryEnabledSwitch.onChange = async () => {
101         const enabled = foundryEnabledSwitch.checked;
102         const remember = document.getElementById('foundry-rememb
er')?.checked ?? true;
103         const url = document.getElementById('foundry-url')?.valu
e.trim() || 'http://localhost:54837';
104         const key = document.getElementById('foundry-key')?.valu
e.trim() || '';
105         const model = document.getElementById('foundry-model')?.
value?.trim() || 'qwen2.5-1.5b';
106
107         try {
108             await window.api.fetch('/api/foundry/config', {
109                 method: 'POST',
110                 headers: { 'Content-Type': 'application/json' },
111                 body: JSON.stringify({ enabled, url, key, model, rem
ember })

```

```

112         });
113         showModelsNotification(`Microsoft Foundry ${enabled ?
'активирован' : 'деактивирован'}${remember ? ' (сохранено в co
nfig.json)' : ''}`, 'info');
114         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
115     } catch (err) {
116         console.error('Ошибка переключения Foundry:', err);
117         showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
118     }
119 };
120 }
121
122 const ollamaEnabledSwitch = document.getElementById('ollama-
enabled');
123 if (ollamaEnabledSwitch) {
124     ollamaEnabledSwitch.onchange = async () => {
125         const enabled = ollamaEnabledSwitch.checked;
126         const remember = document.getElementById('ollama-remembe
r')?.checked ?? true;
127         const url = document.getElementById('ollama-url')?.value
.trim() || 'http://localhost:11434';
128         const model = document.getElementById('ollama-model')?.v
alue?.trim() || 'llama3.1';
129
130         try {
131             await window.api.fetch('/api/ollama/config', {
132                 method: 'POST',
133                 headers: { 'Content-Type': 'application/json' },
134                 body: JSON.stringify({ enabled, url, model, remember
    })
135         });
136         showModelsNotification(`Ollama ${enabled ? 'активирова
н' : 'деактивирован'}${remember ? ' (сохранено в config.json)'
: ''}`, 'info');
137         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);

```

```
138         } catch (err) {
139             console.error('Ошибка переключения Ollama:', err);
140             showModelsNotification('Ошибка переключения: ' + err.m
message, 'danger');
141         }
142     };
143 }
144
145 const onnxEnabledSwitch = document.getElementById('onnx-enab
led');
146 if (onnxEnabledSwitch) {
147     onnxEnabledSwitch.onChange = async () => {
148         const enabled = onnxEnabledSwitch.checked;
149         const remember = document.getElementById('onnx-remember'
)?.checked ?? true;
150         const execution_provider = document.getElementById('onnx
-execution-provider')?.value || 'DirectMLExecutionProvider';
151         const models_dir = document.getElementById('onnx-models-
dir')?.value.trim() || 'models/onnx';
152         const default_model = document.getElementById('onnx-defa
ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
153         const olive_precision = document.getElementById('onnx-ol
ive-precision')?.value || 'int4';
154
155         try {
156             await window.api.fetch('/api/onnx/config', {
157                 method: 'POST',
158                 headers: { 'Content-Type': 'application/json' },
159                 body: JSON.stringify({
160                     enabled,
161                     execution_provider,
162                     models_dir,
163                     default_model,
164                     olive_precision,
165                     remember
166                 })
167             });
168             showModelsNotification(`ONNX / Olive ${enabled ? 'акти
```



```

    вирован' : 'деактивирован'}${remember ? ' (сохранено в config.
    json)' : ''}` , 'info');
169     if (modelSelect && saveBtn) await loadTabModels(modelS
    elect, saveBtn);
170     } catch (err) {
171     console.error('Ошибка переключения ONNX / Olive:', err
    );
172     showModelsNotification('Ошибка переключения: ' + err.m
    essage, 'danger');
173     }
174     };
175     }
176
177     if (saveAgyBtn) {
178     saveAgyBtn.onclick = async () => {
179     const enabled = document.getElementById('agy-enabled')?.
    checked ?? true;
180     const remember = document.getElementById('agy-remember')
    ?.checked ?? true;
181     const model = document.getElementById('agy-model')?.valu
    e || 'agy-flash';
182     const key = document.getElementById('agy-key')?.value.tr
    im() || '';
183
184     saveAgyBtn.disabled = true;
185     try {
186     await window.api.fetch('/api/agy/config', {
187     method: 'POST',
188     headers: { 'Content-Type': 'application/json' },
189     body: JSON.stringify({ enabled, model, key, remember
    })
190     });
191     showModelsNotification('Настройки Antigravity (AGY) yc
    пешно сохранены${remember ? ' в config.json' : ''}` , 'success'
    );
192     if (modelSelect && saveBtn) await loadTabModels(modelS
    elect, saveBtn);
193     } catch (err) {

```

```
194 |         console.error('Ошибка сохранения AntigraVity:', err);
195 |         showModelsNotification('Ошибка сохранения: ' + err.mes
    | sage, 'danger');
196 |     } finally {
197 |         saveAgyBtn.disabled = false;
198 |     }
199 | };
200 | }
201 |
202 | if (saveFoundryBtn) {
203 |     saveFoundryBtn.onclick = async () => {
204 |         const enabled = document.getElementById('foundry-enabled
    | ')?.checked || false;
205 |         const remember = document.getElementById('foundry-rememb
    | er')?.checked ?? true;
206 |         const url = document.getElementById('foundry-url')?.valu
    | e.trim() || 'http://localhost:54837';
207 |         const key = document.getElementById('foundry-key')?.valu
    | e.trim() || '';
208 |         const model = document.getElementById('foundry-model')?.
    | value?.trim() || 'qwen2.5-1.5b';
209 |
210 |         saveFoundryBtn.disabled = true;
211 |         try {
212 |             await window.api.fetch('/api/foundry/config', {
213 |                 method: 'POST',
214 |                 headers: { 'Content-Type': 'application/json' },
215 |                 body: JSON.stringify({ enabled, url, key, model, rem
    | ember })
216 |             });
217 |             showModelsNotification(`Настройки Microsoft Foundry ус
    | пешно сохранены${remember ? ' в config.json' : ''}`, 'success'
    | );
218 |             if (modelSelect && saveBtn) await loadTabModels(modelS
    | elect, saveBtn);
219 |         } catch (err) {
220 |             console.error('Ошибка сохранения Foundry:', err);
221 |             showModelsNotification('Ошибка сохранения: ' + err.mes
```

```

    sage, 'danger');
222     } finally {
223         saveFoundryBtn.disabled = false;
224     }
225 };
226 }
227
228 if (saveOllamaBtn) {
229     saveOllamaBtn.onclick = async () => {
230         const enabled = document.getElementById('ollama-enabled'
231 )?.checked || false;
232         const remember = document.getElementById('ollama-remembe
233 r')?.checked ?? true;
234         const url = document.getElementById('ollama-url')?.value
235 .trim() || 'http://localhost:11434';
236         const model = document.getElementById('ollama-model')?.v
237 alue?.trim() || 'llama3.1';
238
239         saveOllamaBtn.disabled = true;
240         try {
241             await window.api.fetch('/api/ollama/config', {
242                 method: 'POST',
243                 headers: { 'Content-Type': 'application/json' },
244                 body: JSON.stringify({ enabled, url, model, remember
245 })
246             });
247             showModelsNotification(`Настройки Ollama успешно сохра
248 нены${remember ? ' в config.json' : ''}`, 'success');
249             if (modelSelect && saveBtn) await loadTabModels(modelS
250 elect, saveBtn);
251         } catch (err) {
252             console.error('Ошибка сохранения Ollama:', err);
253             showModelsNotification('Ошибка сохранения: ' + err.mes
254 sage, 'danger');
255         } finally {
256             saveOllamaBtn.disabled = false;
257         }
258     };

```

```

251 | }
252 |
253 | if (saveOnnxBtn) {
254 |     saveOnnxBtn.onclick = async () => {
255 |         const enabled = document.getElementById('onnx-enabled')?
|.checked ?? true;
256 |         const remember = document.getElementById('onnx-remember'
|)?.checked ?? true;
257 |         const execution_provider = document.getElementById('onnx
|-execution-provider')?.value || 'DirectMLExecutionProvider';
258 |         const models_dir = document.getElementById('onnx-models-
|dir')?.value.trim() || 'models/onnx';
259 |         const default_model = document.getElementById('onnx-defa
|ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
260 |         const olive_precision = document.getElementById('onnx-ol
|ive-precision')?.value || 'int4';
261 |
262 |         saveOnnxBtn.disabled = true;
263 |         try {
264 |             await window.api.fetch('/api/onnx/config', {
265 |                 method: 'POST',
266 |                 headers: { 'Content-Type': 'application/json' },
267 |                 body: JSON.stringify({
268 |                     enabled,
269 |                     execution_provider,
270 |                     models_dir,
271 |                     default_model,
272 |                     olive_precision,
273 |                     remember
274 |                 })
275 |             });
276 |             showModelsNotification(`Настройки ONNX / Olive успешно
| сохранены${remember ? ' в config.json' : ''}`, 'success');
277 |             if (modelSelect && saveBtn) await loadTabModels(modelS
|elect, saveBtn);
278 |         } catch (err) {
279 |             console.error('Ошибка сохранения ONNX / Olive:', err);
280 |             showModelsNotification('Ошибка сохранения: ' + err.mes

```

```
sage, 'danger');
281     } finally {
282         saveOnnxBtn.disabled = false;
283     }
284 };
285 }
286
287 if (saveBtn && modelSelect) {
288     saveBtn.onclick = async () => {
289         const selectedModel = modelSelect.value;
290         saveBtn.disabled = true;
291         const originalText = saveBtn.textContent;
292         saveBtn.textContent = 'Сохранение...';
293
294         try {
295             await window.api.fetch('/auth/settings', {
296                 method: 'POST',
297                 headers: { 'Content-Type': 'application/json' },
298                 body: JSON.stringify({ model: selectedModel })
299             });
300             showModelsNotification('Модель успешно обновлена на: '
301 + selectedModel, 'success');
302
303             const otherModelSelect = document.getElementById('admin-model-select');
304             if (otherModelSelect) {
305                 otherModelSelect.value = selectedModel;
306             }
307         } catch (err) {
308             console.error('Ошибка сохранения модели:', err);
309             showModelsNotification('Ошибка сохранения: ' + err.message,
310 sage, 'danger');
311         } finally {
312             saveBtn.disabled = false;
313             saveBtn.textContent = originalText;
314         }
315     };
316 }
```

```
315
316   if (refreshKeysBtn) {
317       refreshKeysBtn.onclick = async () => {
318           refreshKeysBtn.disabled = true;
319           const originalText = refreshKeysBtn.textContent;
320           refreshKeysBtn.textContent = '⏳ Сброс...';
321           try {
322               const res = await window.api.fetch('/api/keys/reset-all', { method: 'POST' });
323               showModelsNotification(res.message || 'Квоты всех ключей успешно сброшены', 'success');
324           } catch (err) {
325               console.error('Ошибка сброса квот:', err);
326               showModelsNotification('Ошибка сброса: ' + err.message, 'danger');
327           } finally {
328               refreshKeysBtn.disabled = false;
329               refreshKeysBtn.textContent = originalText;
330               if (keysListBody) await refreshKeysList(keysListBody);
331           }
332       };
333   }
334
335   if (addKeyBtn) {
336       addKeyBtn.onclick = async () => {
337           const nameInput = document.getElementById('new-key-name');
338           const valueInput = document.getElementById('new-key-value');
339           if (!nameInput || !valueInput) return;
340
341           const name = nameInput.value.trim();
342           const apiKey = valueInput.value.trim();
343
344           if (!name || !apiKey) {
345               showModelsNotification('Заполните все поля!', 'warning');
346               return;
347           }
348       };
349   }
```

```
347 |     }
348 |
349 |     addKeyBtn.disabled = true;
350 |     try {
351 |         await window.api.fetch('/api/keys', {
352 |             method: 'POST',
353 |             headers: { 'Content-Type': 'application/json' },
354 |             body: JSON.stringify({ name, api_key: apiKey, status
: 'active' })
355 |         });
356 |         showModelsNotification(`Ключ "${name}" успешно добавле
н`, 'success');
357 |         nameInput.value = '';
358 |         valueInput.value = '';
359 |         if (keysListBody) await refreshKeysList(keysListBody);
360 |     } catch (err) {
361 |         console.error('Ошибка добавления ключа:', err);
362 |         showModelsNotification('Ошибка добавления: ' + err.mes
sage, 'danger');
363 |     } finally {
364 |         addKeyBtn.disabled = false;
365 |     }
366 | };
367 | }
368 |
369 | // 2. Load all components concurrently
370 | await Promise.allSettled([
371 |     modelSelect && saveBtn ? loadTabModels(modelSelect, saveBt
n) : Promise.resolve(),
372 |     keysListBody ? refreshKeysList(keysListBody) : Promise.res
olve(),
373 |     loadFoundryConfig(),
374 |     loadOllamaConfig(),
375 |     loadAgyConfig(),
376 |     loadOnnxConfig(),
377 |     loadSystemInstruction()
378 | ]);
379 | }
```

```
380
381 // Helper to load models list
382 async function loadTabModels(modelSelect, saveBtn, forceRefresh = false) {
383   const providerSelect = document.getElementById('provider-tab-select');
384   if (providerSelect) providerSelect.innerHTML = '';
385   modelSelect.innerHTML = '';
386
387   let modelsGrouped = {};
388
389   const fetchModels = async (force = false) => {
390     try {
391       const url = force ? '/api/chat/models?refresh=true' : '/api/chat/models';
392       const modelsData = await window.api.fetch(url);
393       let grouped = modelsData.models || {};
394       if (Array.isArray(grouped)) {
395         grouped = { 'gemini': grouped };
396       }
397       return grouped;
398     } catch (err) {
399       console.error('Ошибка загрузки моделей:', err);
400       showModelsNotification('Ошибка загрузки моделей AI: ' + err.message, 'danger');
401       return {};
402     }
403   };
404
405   modelsGrouped = await fetchModels(forceRefresh);
406
407   const providers = Object.keys(modelsGrouped).filter(p => modelsGrouped[p] && modelsGrouped[p].length > 0);
408
409   if (providers.length === 0) {
410     if (providerSelect) {
411       providerSelect.innerHTML = '<option value="">Нет доступных провайдеров</option>';

```



```

412     }
413     modelSelect.innerHTML = '<option value="">Нет доступных мо
делей</option>';
414     saveBtn.disabled = true;
415     return;
416 }
417
418 if (providerSelect) {
419     providerSelect.innerHTML = '';
420     providers.forEach(p => {
421         const option = document.createElement('option');
422         option.value = p;
423         option.textContent = p.charAt(0).toUpperCase() + p.slice
(1);
424         providerSelect.appendChild(option);
425     });
426 }
427
428 const populateModels = (provider, providerModelsList) => {
429     modelSelect.innerHTML = '';
430     const providerModels = providerModelsList !== undefined ?
providerModelsList : (modelsGrouped[provider] || []);
431     if (!providerModels || providerModels.length === 0) {
432         modelSelect.innerHTML = '<option value="">Нет моделей</o
ption>';
433         saveBtn.disabled = true;
434     } else {
435         providerModels.forEach(modelName => {
436             const option = document.createElement('option');
437             option.value = modelName;
438             let cleanName = modelName;
439             if (cleanName.startsWith('foundry:')) cleanName = clea
nName.substring(8);
440             else if (cleanName.startsWith('ollama:')) cleanName =
cleanName.substring(7);
441             else if (cleanName.startsWith('agy-')) cleanName = cle
anName.substring(4);
442             option.textContent = cleanName;

```

```
443         modelSelect.appendChild(option);
444     });
445     saveBtn.disabled = false;
446 }
447 };
448
449 if (providerSelect) {
450     providerSelect.onChange = async () => {
451         const chosenProvider = providerSelect.value;
452         modelSelect.innerHTML = '<option value="">Обновление спи
ска моделей...</option>';
453         saveBtn.disabled = true;
454         try {
455             const updatedGrouped = await fetchModels(true);
456             if (updatedGrouped && Object.keys(updatedGrouped).leng
th > 0) {
457                 modelsGrouped = updatedGrouped;
458             }
459         } catch (e) {
460             console.warn('Failed to refresh models on provider cha
nge:', e);
461         }
462         populateModels(chosenProvider, modelsGrouped[chosenProvi
der]);
463     };
464     populateModels(providerSelect.value, modelsGrouped[provide
rSelect.value]);
465 } else {
466     let allModels = [];
467     providers.forEach(p => allModels = allModels.concat(models
Grouped[p]));
468     allModels.forEach(modelName => {
469         const option = document.createElement('option');
470         option.value = modelName;
471         option.textContent = modelName;
472         modelSelect.appendChild(option);
473     });
474     saveBtn.disabled = allModels.length === 0;
```

```

475 | }
476 |
477 | try {
478 |     const settingsData = await window.api.fetch('/auth/settings');
479 |     if (settingsData && settingsData.model) {
480 |         let foundProvider = null;
481 |         for (const p of providers) {
482 |             if (modelsGrouped[p] && modelsGrouped[p].includes(settingsData.model)) {
483 |                 foundProvider = p;
484 |                 break;
485 |             }
486 |         }
487 |         if (foundProvider && providerSelect) {
488 |             providerSelect.value = foundProvider;
489 |             populateModels(foundProvider, modelsGrouped[foundProvider]);
490 |         }
491 |         modelSelect.value = settingsData.model;
492 |     }
493 | } catch (err) {
494 |     console.error('Ошибка загрузки настроек AI пользователя:', err);
495 | }
496 | }
497 |
498 | // Helper to refresh keys list table
499 | async function refreshKeysList(container) {
500 |     try {
501 |         container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Загрузка ключей...</td></tr>';
502 |         const keysData = await window.api.fetch('/api/keys');
503 |         const keys = keysData.keys || [];
504 |
505 |         if (keys.length === 0) {
506 |             container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Список ключей пуст</td></tr>';

```

```
507 |     return;
508 | }
509 |
510 | container.innerHTML = '';
511 | keys.forEach(key => {
512 |     const row = document.createElement('tr');
513 |
514 |     const tdName = document.createElement('td');
515 |     tdName.innerHTML = `<strong>${key.name}</strong>`;
516 |     row.appendChild(tdName);
517 |
518 |     const tdKey = document.createElement('td');
519 |     tdKey.className = 'font-monospace text-muted small';
520 |     tdKey.textContent = key.api_key_masked;
521 |     row.appendChild(tdKey);
522 |
523 |     const tdStatus = document.createElement('td');
524 |     const isEnabled = key.status === 'active';
525 |     const statusClass = isEnabled ? 'bg-success' : 'bg-secondary';
526 |     const statusText = isEnabled ? 'Активен' : 'Отключен';
527 |     tdStatus.innerHTML = `<span class="badge ${statusClass}"
528 | >${statusText}</span>`;
529 |     row.appendChild(tdStatus);
530 |
531 |     const tdQuota = document.createElement('td');
532 |     if (key.exhausted) {
533 |         let resetText = 'Лимит';
534 |         if (key.reset_in_seconds) {
535 |             const hours = Math.floor(key.reset_in_seconds / 3600);
536 |             const mins = Math.floor((key.reset_in_seconds % 3600) / 60);
537 |             resetText = `Сброс через ${hours}ч ${mins}м`;
538 |         }
539 |         tdQuota.innerHTML = `<span class="badge bg-danger d-block mb-1" title="Превышен лимит запросов в сутки">${resetText}</span>`;
540 |     }
541 | }
```

```
539 |     } else {
540 |         tdQuota.innerHTML = `<span class="badge bg-success d-b
lock mb-1">OK</span>`;
541 |     }
542 |     row.appendChild(tdQuota);
543 |
544 |     const tdActions = document.createElement('td');
545 |     tdActions.className = 'text-end';
546 |
547 |     const btnToggle = document.createElement('button');
548 |     btnToggle.className = `btn btn-xs btn-sm me-1 ${isEnabled
d ? 'btn-outline-secondary' : 'btn-outline-success'}`;
549 |     btnToggle.textContent = isEnabled ? 'Откл' : 'Вкл';
550 |     btnToggle.onclick = () => toggleKeyStatus(key.name, isEn
abled ? 'disabled' : 'active', container);
551 |     tdActions.appendChild(btnToggle);
552 |
553 |     if (key.exhausted) {
554 |         const btnReset = document.createElement('button');
555 |         btnReset.className = 'btn btn-xs btn-outline-warning b
tn-sm me-1';
556 |         btnReset.innerHTML = 'Сброс';
557 |         btnReset.title = 'Сбросить 24-часовой бан квоты';
558 |         btnReset.onclick = () => resetKeyQuota(key.name, conta
iner);
559 |         tdActions.appendChild(btnReset);
560 |     }
561 |
562 |     const btnDelete = document.createElement('button');
563 |     btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
564 |     btnDelete.textContent = 'Удалить';
565 |     btnDelete.onclick = () => deleteKey(key.name, container)
;
566 |     tdActions.appendChild(btnDelete);
567 |
568 |     row.appendChild(tdActions);
569 |     container.appendChild(row);
```

```

570     });
571
572   } catch (err) {
573     console.error('Ошибка загрузки ключей:', err);
574     container.innerHTML = `|  |  |  |  |  |
| --- | --- | --- | --- | --- |
|  | | | | |

```

```

    ge, 'danger');
602 }
603 }
604
605 async function deleteKey(name, container) {
606   if (!confirm(`Вы уверены, что хотите удалить ключ "${name}"?`)) return;
607   try {
608     await window.api.fetch(`/api/keys/${name}`, { method: 'DELETE' });
609     showModelsNotification(`Ключ "${name}" успешно удален`, 'success');
610     await refreshKeysList(container);
611   } catch (err) {
612     console.error('Ошибка удаления ключа:', err);
613     showModelsNotification('Ошибка удаления: ' + err.message, 'danger');
614   }
615 }
616
617 function showModelsNotification(message, type = 'info') {
618   const notification = document.createElement('div');
619   notification.className = `alert alert-${type} position-fixed top-0 end-0 m-3`;
620   notification.style.zIndex = '9999';
621   notification.style.maxWidth = '400px';
622   notification.textContent = message;
623   document.body.appendChild(notification);
624
625   setTimeout(() => {
626     notification.remove();
627   }, 5000);
628 }
629
630 async function loadFoundryConfig() {
631   try {
632     const config = await window.api.fetch('/api/foundry/config');

```

```
633     const enabledInput = document.getElementById('foundry-enab
    led');
634     const urlInput = document.getElementById('foundry-url');
635     const keyInput = document.getElementById('foundry-key');
636     const modelInput = document.getElementById('foundry-model'
    );
637
638     if (enabledInput) enabledInput.checked = config.enabled ||
    false;
639     if (urlInput) urlInput.value = config.url || '';
640     if (keyInput) keyInput.value = config.key || '';
641     if (modelInput) modelInput.value = config.model || '';
642 } catch (err) {
643     console.error('Ошибка загрузки настроек Foundry:', err);
644 }
645 }
646
647 async function loadOllamaConfig() {
648     try {
649         const config = await window.api.fetch('/api/ollama/config'
    );
650         const enabledInput = document.getElementById('ollama-enabl
    ed');
651         const urlInput = document.getElementById('ollama-url');
652         const modelInput = document.getElementById('ollama-model')
    ;
653
654         if (enabledInput) enabledInput.checked = config.enabled ||
    false;
655         if (urlInput) urlInput.value = config.url || '';
656         if (modelInput) modelInput.value = config.model || '';
657     } catch (err) {
658         console.error('Ошибка загрузки настроек Ollama:', err);
659     }
660 }
661
662 async function loadAgyConfig() {
663     try {
```



```
664 |     const modelSelect = document.getElementById('agy-model');
665 |     if (modelSelect) {
666 |         try {
667 |             const modelsData = await window.api.fetch('/api/chat/m
models');
668 |             const agyList = modelsData.models?.agy || [];
669 |             if (agyList.length > 0) {
670 |                 const curVal = modelSelect.value;
671 |                 modelSelect.innerHTML = '';
672 |                 agyList.forEach(m => {
673 |                     const opt = document.createElement('option');
674 |                     opt.value = m;
675 |                     opt.textContent = m;
676 |                     modelSelect.appendChild(opt);
677 |                 });
678 |                 if (curVal && agyList.includes(curVal)) {
679 |                     modelSelect.value = curVal;
680 |                 }
681 |             }
682 |         } catch (e) {
683 |             console.error('Ошибка загрузки моделей AGY:', e);
684 |         }
685 |     }
686 |
687 |     const config = await window.api.fetch('/api/agy/config');
688 |     const enabledInput = document.getElementById('agy-enabled'
);
689 |     const keyInput = document.getElementById('agy-key');
690 |
691 |     if (enabledInput) enabledInput.checked = config.enabled ??
true;
692 |     if (modelSelect && config.model) modelSelect.value = confi
g.model;
693 |     if (keyInput) keyInput.value = config.key || '';
694 | } catch (err) {
695 |     console.error('Ошибка загрузки настроек Antigravity (AGY):
', err);
696 | }
```

```
697 | }
698 |
699 | async function loadOnnxConfig() {
700 |     try {
701 |         const config = await window.api.fetch('/api/onnx/config');
702 |         const enabledInput = document.getElementById('onnx-enabled
| ');
703 |         const epSelect = document.getElementById('onnx-execution-p
| rovider');
704 |         const dirInput = document.getElementById('onnx-models-dir'
| );
705 |         const modelInput = document.getElementById('onnx-default-m
| odel');
706 |         const precSelect = document.getElementById('onnx-olive-pre
| cision');
707 |
708 |         if (enabledInput) enabledInput.checked = config.enabled ??
| true;
709 |         if (epSelect && config.execution_provider) epSelect.value
| = config.execution_provider;
710 |         if (dirInput && config.models_dir) dirInput.value = config
| .models_dir;
711 |         if (modelInput && config.default_model) modelInput.value =
| config.default_model;
712 |         if (precSelect && config.olive_precision) precSelect.value
| = config.olive_precision;
713 |     } catch (err) {
714 |         console.error('Ошибка загрузки настроек ONNX / Olive:', er
| r);
715 |     }
716 | }
717 |
718 | async function loadSystemInstruction() {
719 |     const editor = document.getElementById('system-instruction-e
| ditor');
720 |     const statusBadge = document.getElementById('system-instruct
| ion-status');
721 |     if (!editor) return;
```

```
722 |
723 | editor.disabled = true;
724 | if (statusBadge) {
725 |     statusBadge.className = 'badge bg-warning text-dark';
726 |     statusBadge.textContent = 'Загрузка...';
727 |     statusBadge.style.removeProperty('display');
728 | }
729 |
730 | try {
731 |     const data = await window.api.fetch('/api/admin/system_instruction');
732 |     editor.value = data.content || '';
733 |     if (statusBadge) {
734 |         statusBadge.className = 'badge bg-success';
735 |         statusBadge.textContent = 'Загружено';
736 |         setTimeout(() => {
737 |             if (statusBadge) statusBadge.style.display = 'none';
738 |         }, 2500);
739 |     }
740 | } catch (err) {
741 |     console.error('Ошибка загрузки системной инструкции:', err);
742 |     if (statusBadge) {
743 |         statusBadge.className = 'badge bg-danger';
744 |         statusBadge.textContent = 'Ошибка';
745 |         statusBadge.style.removeProperty('display');
746 |     }
747 |     showModelsNotification('Ошибка загрузки системной инструкции: ' + err.message, 'danger');
748 | } finally {
749 |     editor.disabled = false;
750 | }
751 | }
752 |
753 | async function saveSystemInstruction() {
754 |     const editor = document.getElementById('system-instruction-editor');
755 |     const saveBtn = document.getElementById('btn-save-instruction');
```

```

n');
756   const statusBadge = document.getElementById('system-instruct
ion-status');
757   if (!editor || !saveBtn) return;
758
759   const originalHtml = saveBtn.innerHTML;
760   saveBtn.disabled = true;
761   saveBtn.innerHTML = '<span class="spinner-border spinner-bor
der-sm" role="status"></span> Сохранение...';
762
763   try {
764     await window.api.fetch('/api/admin/system_instruction', {
765       method: 'POST',
766       headers: { 'Content-Type': 'application/json' },
767       body: JSON.stringify({ content: editor.value })
768     });
769     showModelsNotification('☑ Системная инструкция успешно сох
ранена', 'success');
770     if (statusBadge) {
771       statusBadge.className = 'badge bg-success';
772       statusBadge.textContent = 'Сохранено';
773       statusBadge.style.removeProperty('display');
774       setTimeout(() => {
775         if (statusBadge) statusBadge.style.display = 'none';
776       }, 3000);
777     }
778   } catch (err) {
779     console.error('Ошибка сохранения системной инструкции:', e
rr);
780     if (statusBadge) {
781       statusBadge.className = 'badge bg-danger';
782       statusBadge.textContent = 'Ошибка сохранения';
783       statusBadge.style.removeProperty('display');
784     }
785     showModelsNotification('Ошибка сохранения: ' + err.message
, 'danger');
786   } finally {
787     saveBtn.disabled = false;

```

```
788 |     saveBtn.innerHTML = originalHtml;
789 | }
790 | }
791 |
792 | // Экспорт для загрузчика вкладок
793 | window.initModelsTab = initModelsTab;
794 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/models_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/models_tab/index.html"
17         class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/models_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Models & APIs Management Tab -->
30 <div class="container-fluid py-3">
31     <!-- Premium Header -->
32     <div class="d-flex justify-content-between align-items-center mb-4">
33         <div>
34             <h4 class="mb-0 fw-bold text-primary"><i class="bi bi-cpu"></i> Модели и API</h4>
35             <p class="text-muted small mb-0">Управление провайдерами
```

```

, ключами и системными инструкциями</p>
31     </div>
32 </div>
33
34 <!-- Top Block: Active Model, Test Prompt & System Instructi
on -->
35 <div class="row mb-4">
36     <!-- Active Model & Testing Card -->
37     <div class="col-lg-6 mb-4 mb-lg-0">
38         <div class="card shadow-sm border-0 h-100">
39             <div class="card-header bg-transparent border-0 d-flex
justify-content-between align-items-center pt-3 pb-0">
40                 <h6 class="card-title fw-bold text-primary mb-0"><i
class="bi bi-robot"></i> Активная модель и проверка</h6>
41                 <button class="btn btn-outline-secondary btn-sm roun
ded-pill py-0 px-2" id="btn-refresh-models-list" title="Обнови
ть список доступных моделей">
42                     <i class="bi bi-arrow-clockwise"></i> Обновить спи
сок
43                 </button>
44             </div>
45             <div class="card-body d-flex flex-column p-4">
46
47                 <!-- Saved Default Model Indicator -->
48                 <div class="mb-3 d-flex align-items-center justify-c
ontent-between p-2 rounded bg-body-tertiary border">
49                     <div class="small text-secondary d-flex align-item
s-center gap-1">
50                         <i class="bi bi-bookmark-check-fill text-success
"></i>
51                         <span>Модель по умолчанию:</span>
52                     </div>
53                     <span id="saved-default-model-badge" class="badge
bg-secondary font-monospace px-2 py-1" style="font-size: 0.82r
em;">Загрузка...</span>
54                 </div>
55
56                 <div class="row g-2 mb-2">

```

```
57 |         <div class="col-sm-5">
58 |             <label for="provider-tab-select" class="form-label small text-secondary mb-1">Провайдер ИИ:</label>
59 |             <select class="form-select form-select-sm" id="provider-tab-select">
60 |                 <option value="">Загрузка...</option>
61 |             </select>
62 |         </div>
63 |         <div class="col-sm-7">
64 |             <div class="d-flex justify-content-between align-items-center mb-1">
65 |                 <label for="models-tab-select" class="form-label small text-secondary mb-0">Модель:</label>
66 |                 <div class="form-check form-switch mb-0" title="Показать все модели, включая отфильтрованные / неподдерживаемые">
67 |                     <input class="form-check-input" type="checkbox" role="switch" id="show-all-models-check" style="cursor: pointer;">
68 |                     <label class="form-check-label small text-secondary" for="show-all-models-check" style="cursor: pointer; font-size: 0.72rem;">Показать все</label>
69 |                 </div>
70 |             </div>
71 |             <div class="input-group input-group-sm">
72 |                 <select class="form-select form-select-sm" id="models-tab-select">
73 |                     <option value="">Загрузка...</option>
74 |                 </select>
75 |                 <button class="btn btn-outline-warning" type="button" id="btn-model-toggle-favorite" title="Добавить в избранное / Убрать из избранного">
76 |                     <i class="bi bi-star" id="favorite-star-icon"></i>
77 |                 </button>
78 |             </div>
79 |         </div>
80 |     </div>
```



```

81 |
82 |         <!-- Model User Note & Favorite Details -->
83 |         <div class="mb-3 p-2 rounded bg-body-tertiary border
      | " id="model-note-box" style="border-style: dashed !important;"
      | >
84 |             <div class="d-flex justify-content-between align-i
      | tems-center mb-1">
85 |                 <label for="model-user-note" class="form-label s
      | mall text-warning-emphasis mb-0" style="font-size: 0.76rem;">
86 |                     <i class="bi bi-journal-text me-1"></i> Заметк
      | а о модели:
87 |                 </label>
88 |                 <span id="model-fav-status-badge" class="badge b
      | g-secondary-subtle text-secondary-emphasis border" style="font
      | -size: 0.68rem;">Не в избранном</span>
89 |             </div>
90 |             <div class="input-group input-group-sm">
91 |                 <input type="text" class="form-control" id="mode
      | l-user-note" placeholder="Введите заметку о возможностях/особе
      | нностях модели..." autocomplete="off">
92 |                 <button class="btn btn-sm btn-outline-info" id="
      | btn-save-model-note" type="button" title="Сохранить заметку">
93 |                     <i class="bi bi-floppy me-1"></i> Сохранить
94 |                 </button>
95 |             </div>
96 |         </div>
97 |
98 |         <!-- Favorite Models Chips List -->
99 |         <div class="mb-3" id="favorite-models-section" style
      | ="display: none;">
100 |             <div class="d-flex align-items-center justify-cont
      | ent-between mb-1">
101 |                 <span class="small text-secondary fw-semibold" s
      | tyle="font-size: 0.75rem;">
102 |                     <i class="bi bi-star-fill text-warning me-1"><
      | /i> Избранные модели:
103 |                 </span>
104 |             </div>

```

```
105 |         <div class="d-flex flex-wrap gap-1" id="favorite-m
| odel-list">
106 |             <!-- Rendered via JS -->
107 |             </div>
108 |         </div>
109 |
110 |         <!-- Model Test Area: Prompt -> Response -->
111 |         <div class="p-3 rounded-3 mb-3 bg-body-tertiary bord
| er">
112 |             <div class="d-flex justify-content-between align-i
| tems-center mb-2">
113 |                 <label for="model-test-prompt" class="form-label
| small text-info fw-bold mb-0">
114 |                     <i class="bi bi-lightning-charge-fill text-war
| ning me-1"></i> Проверочный запрос (Запрос &rarr; Ответ)
115 |                 </label>
116 |                 <span class="text-secondary small" style="font-s
| ize: 0.72rem;">Тест выбранной модели</span>
117 |             </div>
118 |
119 |             <div class="input-group input-group-sm mb-2">
120 |                 <input type="text" class="form-control" id="mode
| l-test-prompt" value="Привет! Назови свою модель и провайдера,
| и подтверди готовность к работе." placeholder="Введите провер
| очное сообщение..." autocomplete="off" data-lpignore="true" da
| ta-lp-ignore="true">
121 |                 <button class="btn btn-info fw-bold px-3" id="bt
| n-model-test-send" type="button" title="Отправить тестовый зап
| рос в выбранную модель">
122 |                     <i class="bi bi-send-fill me-1"></i> Проверить
123 |                 </button>
124 |                 <button class="btn btn-outline-danger fw-bold px
| -3" id="btn-model-test-cancel" type="button" title="Отменить в
| ыполнение проверочного запроса" style="display: none;">
125 |                     <i class="bi bi-stop-circle me-1"></i> Отмена
126 |                 </button>
127 |             </div>
128 |
```

```

129 |         <div id="model-test-result" class="p-2 rounded sma
    | ll font-monospace bg-body border" style="display: none; max-he
    | ight: 140px; overflow-y: auto;">
130 |             <!-- Dynamic output goes here -->
131 |         </div>
132 |     </div>
133 |
134 |     <div class="mt-auto d-grid">
135 |         <button class="btn btn-primary btn-sm rounded-pill
    | fw-bold" id="btn-models-tab-save">
136 |             <i class="bi bi-check-circle"></i> Сохранить как
    | модель по умолчанию
137 |         </button>
138 |         <div class="form-text text-secondary text-center m
    | t-1" style="font-size: 0.72rem;">Эта модель будет использовать
    | ся ассистентом по умолчанию во всех чатах.</div>
139 |     </div>
140 | </div>
141 | </div>
142 | </div>
143 |
144 | <!-- System Instruction Card -->
145 | <div class="col-lg-6">
146 |     <div class="card shadow-sm border-0 h-100">
147 |         <div class="card-header bg-transparent border-0 d-flex
    | justify-content-between align-items-center pt-3 pb-0">
148 |             <span class="fw-bold"><i class="bi bi-terminal"></i>
    | Системная инструкция</span>
149 |             <div class="d-flex gap-2 align-items-center">
150 |                 <span id="system-instruction-status" class="badge
    | bg-secondary" style="display:none!important;"></span>
151 |                 <button class="btn btn-light btn-sm rounded-pill"
    | id="btn-reload-instruction" title="Обновить">
152 |                     <i class="bi bi-arrow-clockwise"></i>
153 |                 </button>
154 |                 <button class="btn btn-success btn-sm rounded-pill
    | px-3" id="btn-save-instruction">
155 |                     Сохранить

```

```

156         </button>
157     </div>
158 </div>
159     <div class="card-body d-flex flex-column">
160         <textarea
161             id="system-instruction-editor"
162             class="form-control font-monospace border-0 bg-bod
163 y-tertiary flex-grow-1"
164             style="min-height: 220px; resize: none; font-size:
165 0.85rem;"
166             placeholder="Введите системную инструкцию..."
167             spellcheck="false"
168         ></textarea>
169     </div>
170 </div>
171
172 <!-- Bottom Block: Provider Settings via Vertical Pills -->
173 <div class="card shadow-sm border-0">
174     <div class="card-body p-0">
175         <div class="d-flex flex-column flex-md-row align-items-s
176 tretch">
177
178             <!-- Sidebar Pills -->
179             <div class="nav flex-column nav-pills p-3 bg-body-tert
180 iary border-end" id="provider-pills-tab" role="tablist" aria-o
181 rientation="vertical" style="min-width: 250px; min-height: 400
182 px; border-top-left-radius: 0.375rem; border-bottom-left-radiu
s: 0.375rem;">
183                 <h6 class="text-uppercase text-muted fw-bold small m
b-3 ps-2">Настройки провайдеров</h6>
184                 <button class="nav-link active text-start rounded-3
mb-2 fw-bold" id="pill-gemini-tab" data-bs-toggle="pill" data-
bs-target="#pill-gemini" type="button" role="tab" aria-control
s="pill-gemini" aria-selected="true">
185                     □ Google Gemini
186                 </button>

```

```
183 |         <button class="nav-link text-start rounded-3 mb-2 fw
| -bold" id="pill-agy-tab" data-bs-toggle="pill" data-bs-target=
| "#pill-agy" type="button" role="tab" aria-controls="pill-agy"
| aria-selected="false">
184 |             □ Google Antigravity (AGY)
185 |         </button>
186 |         <button class="nav-link text-start rounded-3 mb-2 fw
| -bold" id="pill-google-accounts-tab" data-bs-toggle="pill" dat
| a-bs-target="#pill-google-accounts" type="button" role="tab" a
| ria-controls="pill-google-accounts" aria-selected="false">
187 |             □ Google Workspace (Аккаунты)
188 |         </button>
189 |         <button class="nav-link text-start rounded-3 mb-2 fw
| -bold" id="pill-foundry-tab" data-bs-toggle="pill" data-bs-tar
| get="#pill-foundry" type="button" role="tab" aria-controls="pi
| ll-foundry" aria-selected="false">
190 |             □ Microsoft Foundry
191 |         </button>
192 |         <button class="nav-link text-start rounded-3 mb-2 fw
| -bold" id="pill-ollama-tab" data-bs-toggle="pill" data-bs-targ
| et="#pill-ollama" type="button" role="tab" aria-controls="pill
| -ollama" aria-selected="false">
193 |             □ Ollama (Local)
194 |         </button>
195 |         <button class="nav-link text-start rounded-3 fw-bold
| " id="pill-onnx-tab" data-bs-toggle="pill" data-bs-target="#pi
| ll-onnx" type="button" role="tab" aria-controls="pill-onnx" ar
| ia-selected="false">
196 |             □ Microsoft ONNX / Olive
197 |         </button>
198 |     </div>
199 |
200 |     <!-- Tab Content -->
201 |     <div class="tab-content flex-grow-1 p-4" id="provider-
| pills-tabContent">
202 |
203 |         <!-- GEMINI TAB -->
204 |         <div class="tab-pane fade show active" id="pill-gemi
```

```

ni" role="tabpanel" aria-labelledby="pill-gemini-tab">
205     <div class="d-flex justify-content-between align-items-center mb-3">
206         <h5 class="fw-bold mb-0">Ключи Google Gemini</h5>
207     </div>
208     <button class="btn btn-outline-warning btn-sm rounded-pill px-3" id="btn-refresh-keys">
209         <i class="bi bi-arrow-repeat"></i> Сбросить все ключи
210     </button>
211 </div>
212 <!-- Gemini Key Pool Alert -->
213 <div class="alert alert-info shadow-sm rounded-3 border-0 mb-4" role="alert">
214     <h6 class="alert-heading fw-bold mb-1"><i class="bi bi-key-fill me-2"></i>Пул ротации API-ключей (Round-Robin)</h6>
215     <p class="small mb-0">
216         Все активные ключи из таблицы ниже формируют единый пул. При исчерпании суточных лимитов запросов (ошибки <code>429 / ResourceExhausted</code>) система автоматически переключается на следующий свободный ключ без прерывания генерации.
217     </p>
218 </div>
219
220 <div class="row g-4">
221     <!-- Add Key Form -->
222     <div class="col-lg-4">
223         <div class="card bg-body-tertiary border-0 shadow-sm">
224             <div class="card-body">
225                 <h6 class="fw-bold mb-3">Добавить новый ключ</h6>
226                 <form id="form-add-key" onsubmit="event.preventDefault();">
227                     <div class="mb-3">

```

```

228 |             <label class="form-label small text-mu
| ted fw-bold">Имя владельца:</label>
229 |             <input type="text" class="form-control
| form-control-sm" id="new-key-name" placeholder="Например: pri
| mary_key" autocomplete="off" data-lpignore="true" data-1p-igno
| re="true" required>
230 |         </div>
231 |         <div class="mb-4">
232 |             <label class="form-label small text-mu
| ted fw-bold">API-ключ (AIzaSy...):</label>
233 |             <input type="password" class="form-con
| trol form-control-sm" id="new-key-value" placeholder="Введите
| ключ" autocomplete="new-password" data-lpignore="true" data-1p
| -ignore="true" required>
234 |         </div>
235 |         <button type="button" class="btn btn-suc
| cess btn-sm w-100 rounded-pill fw-bold" id="btn-add-key">Добав
| ить ключ</button>
236 |     </form>
237 | </div>
238 | </div>
239 | </div>
240 |
241 | <!-- Keys Table -->
242 | <div class="col-lg-8">
243 |     <div class="table-responsive shadow-sm rounded
| border bg-body">
244 |         <table class="table table-hover align-middle
| mb-0" id="keys-table">
245 |             <thead class="table-light">
246 |                 <tr>
247 |                     <th class="py-3 px-3 border-0">Названи
| e</th>
248 |                     <th class="py-3 px-3 border-0">Ключ</t
| h>
249 |                     <th class="py-3 px-3 border-0">Статус<
| /th>
250 |                     <th class="py-3 px-3 border-0">Квота</

```

```

th>
251         <th class="py-3 px-3 border-0 text-end
">Действия</th>
252     </tr>
253 </thead>
254 <tbody id="keys-list-body" class="border-t
op-0">
255     <tr>
256         <td colspan="5" class="text-center py-
4 text-muted">Загрузка ключей...</td>
257     </tr>
258 </tbody>
259 </table>
260 </div>
261 </div>
262 </div>
263 </div>
264
265 <!-- ANTIGRAVITY (AGY) TAB -->
266 <div class="tab-pane fade" id="pill-agy" role="tabpa
nel" aria-labelledby="pill-agy-tab">
267     <h5 class="fw-bold mb-3">Настройки Google Antigrav
ity (AGY)</h5>
268
269     <div class="row">
270         <div class="col-md-11 col-lg-9">
271             <!-- Info Alert -->
272             <div class="alert alert-warning shadow-sm roun
ded-3 border-0 mb-4" role="alert">
273                 <h6 class="alert-heading fw-bold"><i class="
bi bi-rocket-takeoff me-2"></i>Что такое Antigravity (AGY)?</h
6>
274                 <p class="small mb-2">
275                     <strong>Google Antigravity</strong> – пере
довая агентная платформа для автономных AI-воркфлоу, мультиаге
нтных пайплайнов и глубокого веб-исследования.
276                 </p>
277                 <ul class="small mb-0 ps-3">

```



```

278         <li>Включает встроенные агентные инструмен
ты <code>BuiltinTools.SEARCH_WEB</code> и <code>READ_URL_CONTE
NT</code>.</li>
279         <li>Поддерживает автономный вызов через Py
thon SDK (<code>google.antigravity</code>) и утилиту <code>agy
CLI</code>.</li>
280         <li>Если отдельный ключ <code>AGY_API_KEY<
/code> не задан, провайдер автоматически использует ключи из п
ула Gemini.</li>
281     </ul>
282 </div>
283
284     <form id="form-agy-config" onsubmit="event.pre
ventDefault();">
285         <div class="card border-0 bg-body-tertiary s
hadow-sm mb-4">
286             <div class="card-body p-4">
287                 <div class="d-flex align-items-center ju
stify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2"
>
288                     <div class="form-check form-switch mb-
0">
289                         <input class="form-check-input" type
="checkbox" role="switch" id="agy-enabled" style="transform: s
cale(1.3); margin-left: -2.5em; cursor: pointer;">
290                         <label class="form-check-label fw-bo
ld ms-2" for="agy-enabled" style="cursor: pointer;">Активирова
ть провайдер Antigravity (AGY)</label>
291                     </div>
292                     <div class="form-check mb-0">
293                         <input class="form-check-input" type
="checkbox" id="agy-remember" checked style="cursor: pointer;"
>
294                         <label class="form-check-label small
text-muted ms-1" for="agy-remember" style="cursor: pointer;">
Запомнить</label>
295                     </div>
296                 </div>

```

```
297 |
298 |         <div class="mb-3">
299 |             <label class="form-label small text-mu
| ted fw-bold">Модель Antigravity по умолчанию (динамически из S
| DK):</label>
300 |             <select class="form-select form-select
| -sm" id="agy-model">
301 |                 <option value="">Загрузка доступных
| моделей через SDK...</option>
302 |             </select>
303 |         </div>
304 |
305 |         <div class="mb-4">
306 |             <label class="form-label small text-mu
| ted fw-bold">AGY API Key (опционально, иначе из пула Gemini):<
| /label>
307 |             <input type="password" class="form-con
| trol form-control-sm" id="agy-key" placeholder="Оставьте пусты
| м для использования ключей Gemini" autocomplete="new-password"
| data-lpignore="true" data-1p-ignore="true">
308 |             <div class="form-text small text-muted
| ">Сохраняется в защищенном файле <code>.env</code>.</div>
309 |             </div>
310 |
311 |             <button type="button" class="btn btn-pri
| mary rounded-pill px-4 fw-bold" id="btn-save-agy">
312 |                 <i class="bi bi-floppy me-1"></i> Сохр
| анить настройки Antigravity
313 |             </button>
314 |         </div>
315 |     </div>
316 | </form>
317 | </div>
318 | </div>
319 | </div>
320 |
321 | <!-- GOOGLE WORKSPACE ACCOUNTS POOL TAB -->
322 | <div class="tab-pane fade" id="pill-google-accounts">
```

```

    role="tabpanel" aria-labelledby="pill-google-accounts-tab">
323     <div class="d-flex justify-content-between align-i
    tems-center mb-3">
324         <h5 class="fw-bold mb-0"><i class="bi bi-google
    text-danger me-2"></i>Учетные записи Google Workspace</h5>
325         <button class="btn btn-outline-primary btn-sm ro
    unded-pill px-3" id="btn-refresh-gaccounts">
326             <i class="bi bi-arrow-clockwise"></i> Обновить
    список
327         </button>
328     </div>
329
330     <!-- Workspace Pool Info Alert -->
331     <div class="alert alert-info shadow-sm rounded-3 b
    order-0 mb-4" role="alert">
332         <h6 class="alert-heading fw-bold mb-1"><i class=
    "bi bi-collection-play-fill me-2"></i>Пул аккаунтов Google (OA
    uth 2.0 & Service Account)</h6>
333         <p class="small mb-2">
334             Интеграция с сервисами Google: <strong>Gmail</
    strong> (чтение, черновики), <strong>Google Drive</strong> (по
    иск файлов, чтение), <strong>Google Sheets</strong> (чтение/за
    пись таблиц) и <strong>Google Docs</strong>.
335         </p>
336         <ul class="small mb-0 ps-3">
337             <li>При вызове инструментов ассистент использу
    ет аккаунт <code>account_name</code> или аккаунт по умолчанию
    (<span class="badge bg-warning text-dark"><i class="bi bi-star
    -fill"></i> Default</span>).</li>
338             <li>Файлы ключей и токенов безопасно сохраняют
    ся в <code>src/secrets/</code> и изолированы от публичной конф
    игурации.</li>
339         </ul>
340     </div>
341
342     <div class="row g-4">
343         <!-- Add Account Form -->
344         <div class="col-lg-5">

```

```
345 |         <div class="card bg-body-tertiary border-0 sha  
| dow-sm">  
346 |             <div class="card-body p-4">  
347 |                 <h6 class="fw-bold mb-3 d-flex align-items  
| -center gap-2">  
348 |                     <i class="bi bi-person-plus-fill text-pr  
| imary"></i> Добавить аккаунт в пул  
349 |                 </h6>  
350 |                 <form id="form-add-gaccount" onsubmit="eve  
| nt.preventDefault();">  
351 |                     <div class="mb-3">  
352 |                         <label class="form-label small text-mu  
| ted fw-bold">Имя аккаунта (ID):</label>  
353 |                         <input type="text" class="form-control  
| form-control-sm" id="new-gacc-name" placeholder="Например: wo  
| rk, personal, bot" autocomplete="off" required>  
354 |                         <div class="form-text small">Уникальны  
| й идентификатор аккаунта для маршрутизации агентов.</div>  
355 |                     </div>  
356 |  
357 |                     <div class="mb-3">  
358 |                         <label class="form-label small text-mu  
| ted fw-bold">Тип аутентификации:</label>  
359 |                         <select class="form-select form-select  
| -sm" id="new-gacc-type">  
360 |                             <option value="oauth2">OAuth 2.0 Cli  
| ent (User Login / Desktop App)</option>  
361 |                             <option value="service_account">Serv  
| ice Account Key (Server JSON)</option>  
362 |                         </select>  
363 |                     </div>  
364 |  
365 |                     <div class="mb-3">  
366 |                         <label class="form-label small text-mu  
| ted fw-bold">Email (опционально):</label>  
367 |                         <input type="email" class="form-contro  
| l form-control-sm" id="new-gacc-email" placeholder="user@gmail  
| .com или service-account@iam.gserviceaccount.com" autocomplete
```

```

    ="off">
368         </div>
369
370         <!-- Credentials Input Mode Toggle -->
371         <div class="mb-2">
372             <label class="form-label small text-mu
ted fw-bold">Учетные данные (Credentials JSON):</label>
373             <ul class="nav nav-pills nav-fill mb-2
" id="gacc-input-mode-tabs" role="tablist">
374                 <li class="nav-item" role="presentat
ion">
375                     <button class="nav-link active py-
1 small" id="gacc-mode-file-tab" data-bs-toggle="pill" data-bs
-target="#gacc-mode-file" type="button" role="tab"> Загрузит
ь файл</button>
376                 </li>
377                 <li class="nav-item" role="presentat
ion">
378                     <button class="nav-link py-1 small
" id="gacc-mode-json-tab" data-bs-toggle="pill" data-bs-target
="#gacc-mode-json" type="button" role="tab"> Вставить JSON</
button>
379                 </li>
380             </ul>
381
382             <div class="tab-content" id="gacc-inpu
t-mode-tabContent">
383                 <div class="tab-pane fade show activ
e" id="gacc-mode-file" role="tabpanel">
384                     <input class="form-control form-co
ntrol-sm" type="file" id="new-gacc-file" accept=".json,application/json">
385                     <div class="form-text small">Файл
<code>credentials.json</code> или <code>service_account.json</
code> из <a href="https://console.cloud.google.com/apis/creden
tials" target="_blank" rel="noopener noreferrer" class="text-i
nfo text-decoration-none">Google Cloud Console <i class="bi bi
-box-arrow-up-right ms-1" style="font-size: 0.75rem;"></i></a>

```

```

|.</div>
386|         </div>
387|         <div class="tab-pane fade" id="gacc-
|mode-json" role="tabpanel">
388|             <textarea class="form-control font
| -monospace form-control-sm" id="new-gacc-json" rows="4" placeh
|older='{ "installed": { "client_id": "...", "client_secret": "..
|."}}'></textarea>
389|         </div>
390|     </div>
391| </div>
392|
393|     <div class="form-check mb-4 mt-3">
394|         <input class="form-check-input" type="
|checkbox" id="new-gacc-default" style="cursor: pointer;">
395|         <label class="form-check-label small f
|w-semibold" for="new-gacc-default" style="cursor: pointer;">
396|             Сделать аккаунтом по умолчанию (Defa
|ult)
397|         </label>
398|     </div>
399|
400|     <button type="button" class="btn btn-suc
|cess btn-sm w-100 rounded-pill fw-bold" id="btn-add-gacc">
401|         <i class="bi bi-plus-circle me-1"></i>
|Сохранить аккаунт
402|     </button>
403| </form>
404| </div>
405| </div>
406| </div>
407|
408|     <!-- Accounts Table -->
409|     <div class="col-lg-7">
410|         <div class="table-responsive shadow-sm rounded
|border bg-body mb-3">
411|             <table class="table table-hover align-middle
|mb-0" id="gaccounts-table">

```

```

412         <thead class="table-light">
413             <tr>
414                 <th class="py-3 px-3 border-0">Аккаунт
415             </th>
416                 <th class="py-3 px-3 border-0">Тип</th>
417             </th>
418                 <th class="py-3 px-3 border-0">Статус<
419             </th>
420                 <th class="py-3 px-3 border-0">Токен</
421             <th>
422                 <th class="py-3 px-3 border-0 text-end
423             ">Действия</th>
424         </tr>
425     </thead>
426     <tbody id="gaccounts-list-body" class="bor
427     der-top-0">
428         <tr>
429             <td colspan="5" class="text-center py-
430             4 text-muted">Загрузка аккаунтов...</td>
431         </tr>
432     </tbody>
433 </table>
434 </div>
435
436 <!-- Test & Diagnostics Output Card -->
437 <div class="card bg-dark text-white border-sec
438     ondary shadow-sm" id="gaccount-test-card" style="display: none
439     ;">
440     <div class="card-header border-secondary d-f
441     lex justify-content-between align-items-center py-2">
442         <span class="small fw-bold text-info"><i c
443         lass="bi bi-terminal me-1"></i> Результат проверки доступа</sp
444         an>
445         <button class="btn btn-sm btn-outline-seco
446         ndary py-0 px-2 text-white-50" type="button" onclick="document
447         .getElementById('gaccount-test-card').style.display='none'">❌ <
448         /button>
449     </div>

```

```

445 |         <div class="card-body p-3 font-monospace sma
      | ll" id="gaccount-test-body" style="max-height: 180px; overflow
      | -y: auto; white-space: pre-wrap;">
446 |             </div>
447 |         </div>
448 |     </div>
449 | </div>
450 | </div>
451 |
452 |     <!-- FOUNDRY TAB -->
453 |     <div class="tab-pane fade" id="pill-foundry" role="t
      | abpanel" aria-labelledby="pill-foundry-tab">
454 |         <h5 class="fw-bold mb-4">Настройки Microsoft Found
      | ry</h5>
455 |         <div class="row">
456 |             <div class="col-md-10 col-lg-8">
457 |                 <div class="alert alert-info shadow-sm rounded
      | -3 border-0 mb-4" role="alert">
458 |                     <h6 class="alert-heading fw-bold"><i class="
      | bi bi-info-circle me-2"></i>Где взять модели Foundry?</h6>
459 |                     <p class="small mb-0">
460 |                         Microsoft Foundry (как и LM Studio) работа
      | ет локально. Чтобы список моделей был не пустым, вам необходим
      | о предварительно загрузить нужные GGUF или ONNX модели через и
      | нтерфейс самого Foundry или CLI.
461 |                     </p>
462 |                 </div>
463 |                 <form id="form-foundry-config" onsubmit="event
      | .preventDefault();">
464 |                     <div class="card border-0 bg-body-tertiary s
      | hadow-sm mb-4">
465 |                         <div class="card-body p-4">
466 |                             <div class="d-flex align-items-center ju
      | stify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2"
      | >
467 |                                 <div class="form-check form-switch mb-
      | 0">
468 |                                     <input class="form-check-input" type

```



```

    ="checkbox" role="switch" id="foundry-enabled" style="transform: scale(1.3); margin-left: -2.5em; cursor: pointer;">
459         <label class="form-check-label fw-bold ms-2" for="foundry-enabled" style="cursor: pointer;">Активировать провайдер Foundry</label>
460     </div>
461     <div class="form-check mb-0">
462         <input class="form-check-input" type="checkbox" id="foundry-remember" checked style="cursor: pointer;">
463         <label class="form-check-label small text-muted ms-1" for="foundry-remember" style="cursor: pointer;">Запомнить</label>
464     </div>
465 </div>
466 <div class="mb-3">
467     <label class="form-label small text-muted fw-bold">Base URL API:</label>
468     <input type="text" class="form-control" id="foundry-url" placeholder="http://localhost:54837" autocomplete="off" data-lpignore="true" data-1p-ignore="true">
469 </div>
470 <div class="mb-3">
471     <label class="form-label small text-muted fw-bold">API Key (опционально):</label>
472     <input type="password" class="form-control" id="foundry-key" placeholder="Введите ключ" autocomplete="new-password" data-lpignore="true" data-1p-ignore="true">
473 </div>
474 <button type="button" class="btn btn-primary rounded-pill px-4 fw-bold" id="btn-save-foundry">Сохранить настройки Foundry</button>
475 </div>
476 </div>
477 </form>
478 </div>
479 </div>
480 </div>

```

```

481 |
482 |         <!-- OLLAMA TAB -->
483 |         <div class="tab-pane fade" id="pill-ollama" role="ta
bpanel" aria-labelledby="pill-ollama-tab">
484 |             <h5 class="fw-bold mb-4">Настройки Ollama (Local)<
/h5>
485 |             <div class="row">
486 |                 <div class="col-md-10 col-lg-8">
487 |                     <div class="alert alert-info shadow-sm rounded
-3 border-0 mb-4" role="alert">
488 |                         <h6 class="alert-heading fw-bold"><i class="
bi bi-info-circle me-2"></i>Как добавить новые модели в Ollama
?</h6>
489 |                         <p class="small mb-0">
490 |                             Откройте терминал на сервере, где установл
ен Ollama, и выполните команду:<br>
491 |                             <code class="bg-body-tertiary px-2 py-1 ro
unded text-body mt-2 d-inline-block border">ollama pull llama3
.1</code><br>
492 |                             После загрузки модель автоматически появит
ся в выпадающем списке.
493 |                         </p>
494 |                     </div>
495 |                     <form id="form-ollama-config" onsubmit="event.
preventDefault();">
496 |                         <div class="card border-0 bg-body-tertiary s
hadow-sm mb-4">
497 |                             <div class="card-body p-4">
498 |                                 <div class="d-flex align-items-center ju
stify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2"
>
499 |                                     <div class="form-check form-switch mb-
0">
500 |                                         <input class="form-check-input" type
="checkbox" role="switch" id="ollama-enabled" style="transform
: scale(1.3); margin-left: -2.5em; cursor: pointer;">
501 |                                         <label class="form-check-label fw-bo
ld ms-2" for="ollama-enabled" style="cursor: pointer;">Активир

```

```

        овать провайдер Ollama</label>
502         </div>
503         <div class="form-check mb-0">
504             <input class="form-check-input" type
="checkbox" id="ollama-remember" checked style="cursor: pointe
r;">
505             <label class="form-check-label small
text-muted ms-1" for="ollama-remember" style="cursor: pointer
;">Запомнить</label>
506         </div>
507     </div>
508     <div class="mb-3">
509         <label class="form-label small text-mu
ted fw-bold">Base URL API:</label>
510         <input type="text" class="form-control
" id="ollama-url" placeholder="http://localhost:11434" autocom
plete="off" data-lpignore="true" data-1p-ignore="true">
511     </div>
512     <button type="button" class="btn btn-pri
mary rounded-pill px-4 fw-bold" id="btn-save-ollama">Сохранить
настройки Ollama</button>
513     </div>
514 </div>
515 </form>
516 </div>
517 </div>
518 <!-- ONNX / OLIVE TAB -->
519 <div class="tab-pane fade" id="pill-onnx" role="tabp
anel" aria-labelledby="pill-onnx-tab">
520     <h5 class="fw-bold mb-4">Настройки Microsoft ONNX
Runtime & Olive</h5>
521     <div class="row">
522         <div class="col-md-10 col-lg-8">
523             <div class="alert alert-info shadow-sm rounded
-3 border-0 mb-4" role="alert">
524                 <h6 class="alert-heading fw-bold"><i class="
bi bi-cpu me-2"></i>Аппаратное ускорение и оптимизация Olive</
h6>

```

```

525 |         <p class="small mb-0">
526 |             Microsoft ONNX Runtime использует аппаратн
ые ускорители Windows (DirectML GPU, Qualcomm QNN NPU, CUDA).
Microsoft Olive выполняет автоматическое квантование (INT4/INT
8) и оптимизацию графов для максимальной скорости локального и
нференса.
527 |         </p>
528 |     </div>
529 |     <form id="form-onnx-config" onsubmit="event.pr
eventDefault();">
530 |         <div class="card border-0 bg-body-tertiary s
hadow-sm mb-4">
531 |             <div class="card-body p-4">
532 |                 <div class="d-flex align-items-center ju
stify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2"
>
533 |                     <div class="form-check form-switch mb-
0">
534 |                         <input class="form-check-input" type
="checkbox" role="switch" id="onnx-enabled" style="transform:
scale(1.3); margin-left: -2.5em; cursor: pointer;">
535 |                         <label class="form-check-label fw-bo
ld ms-2" for="onnx-enabled" style="cursor: pointer;">Активиров
ать провайдер ONNX / Olive</label>
536 |                     </div>
537 |                     <div class="form-check mb-0">
538 |                         <input class="form-check-input" type
="checkbox" id="onnx-remember" checked style="cursor: pointer;
">
539 |                         <label class="form-check-label small
text-muted ms-1" for="onnx-remember" style="cursor: pointer;"
>Запомнить</label>
540 |                     </div>
541 |                 </div>
542 |                 <div class="mb-3">
543 |                     <label class="form-label small text-mu
ted fw-bold">Execution Provider (Бэкенд ускорения):</label>
544 |                     <select class="form-select" id="onnx-e

```

```

xecution-provider">
545 |             <option value="DirectMLExecutionProv
| ider">DirectML (AMD / Intel / NVIDIA GPU)</option>
546 |             <option value="QNNExecutionProvider"
| >Qualcomm QNN (NPU Copilot+)</option>
547 |             <option value="CUDAExecutionProvider
| ">NVIDIA CUDA</option>
548 |             <option value="CPUExecutionProvider"
| >CPU Execution Provider</option>
549 |         </select>
550 |     </div>
551 |     <div class="mb-3">
552 |         <label class="form-label small text-mu
| ted fw-bold">Директория локальных моделей ONNX:</label>
553 |         <input type="text" class="form-control
| " id="onnx-models-dir" placeholder="models/onnx" autocomplete=
| "off">
554 |     </div>
555 |     <div class="mb-3">
556 |         <label class="form-label small text-mu
| ted fw-bold">Модель по умолчанию:</label>
557 |         <input type="text" class="form-control
| " id="onnx-default-model" placeholder="phi-3.5-mini-instruct-o
| nnx" autocomplete="off">
558 |     </div>
559 |     <div class="mb-4">
560 |         <label class="form-label small text-mu
| ted fw-bold">Квантование Olive (Precision):</label>
561 |         <select class="form-select" id="onnx-o
| live-precision">
562 |             <option value="int4">INT4 (Максималь
| ная компрессия и скорость)</option>
563 |             <option value="int8">INT8 (Баланс то
| чности и памяти)</option>
564 |             <option value="fp16">FP16 (Половинна
| я точность)</option>
565 |         </select>
566 |     </div>

```

```
567 |                 <button type="button" class="btn btn-pri
    | mary rounded-pill px-4 fw-bold" id="btn-save-onnx">Сохранить н
    | астройки ONNX / Olive</button>
568 |                 </div>
569 |             </div>
570 |         </form>
571 |     </div>
572 | </div>
573 | </div>
574 | </div>
575 | </div>
576 | </div>
577 | </div>
578 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Models Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/models_tab/main.js?v=2
11  *       0261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/models_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Gemini Models & APIs management tab logic
22
23 let modelTestAbortController = null;
24
25 function cancelModelTest() {
26   if (modelTestAbortController) {
27     try {
28       modelTestAbortController.abort();
29     } catch (e) {
30       console.warn('Error aborting model test:', e);
31     }
32     modelTestAbortController = null;
```

```

33 | }
34 | }
35 |
36 | async function initModelsTab() {
37 |     window.api = window.api || {};
38 |     if (!window.api.fetch) {
39 |         window.api.fetch = async function(url, options = {}) {
40 |             const response = await fetch(url, options);
41 |             if (!response.ok) {
42 |                 let msg = response.statusText;
43 |                 try {
44 |                     const data = await response.json();
45 |                     if (data && data.detail) {
46 |                         if (typeof data.detail === 'string') msg = data.de
47 | tail;
48 |                         else if (Array.isArray(data.detail)) msg = data.de
49 | tail.map(d => d.msg || JSON.stringify(d)).join(', ');
50 |                         else msg = JSON.stringify(data.detail);
51 |                     } else if (data && data.message) {
52 |                         msg = data.message;
53 |                     }
54 |                 } catch {}
55 |                 throw new Error(`${response.status} ${msg}`);
56 |             }
57 |             return response.json();
58 |         };
59 |     }
60 |     const modelSelect = document.getElementById('models-tab-sele
61 | ct');
62 |     const saveBtn = document.getElementById('btn-models-tab-save
63 | ');
64 |     const keysListBody = document.getElementById('keys-list-body
65 | ');
66 |     const refreshKeysBtn = document.getElementById('btn-refresh-
67 | keys');
68 |     const addKeyBtn = document.getElementById('btn-add-key');
69 |     const saveAgyBtn = document.getElementById('btn-save-agy');

```



```
65 |   const saveFoundryBtn = document.getElementById('btn-save-fou
    | ndry');
66 |   const saveOllamaBtn = document.getElementById('btn-save-olla
    | ma');
67 |   const saveOnnxBtn = document.getElementById('btn-save-onnx')
    | ;
68 |   const saveBtnInstr = document.getElementById('btn-save-instr
    | uction');
69 |   const reloadBtnInstr = document.getElementById('btn-reload-i
    | nstruction');
70 |
71 |   const gaccountsListBody = document.getElementById('gaccounts
    | -list-body');
72 |   const refreshGAccountsBtn = document.getElementById('btn-ref
    | resh-gaccounts');
73 |   const addGAccountBtn = document.getElementById('btn-add-gacc
    | ');
74 |
75 |   const refreshModelsBtn = document.getElementById('btn-refres
    | h-models-list');
76 |   const testModelBtn = document.getElementById('btn-model-test
    | -send');
77 |   const cancelModelBtn = document.getElementById('btn-model-te
    | st-cancel');
78 |   const testModelPrompt = document.getElementById('model-test-
    | prompt');
79 |
80 |   // Ensure active provider pill tab pane has show active clas
    | ses
81 |   const activePill = document.querySelector('#provider-pills-t
    | ab .nav-link.active') || document.getElementById('pill-gemini-
    | tab');
82 |   if (activePill) {
83 |     const targetSelector = activePill.getAttribute('data-bs-ta
    | rget');
84 |     if (targetSelector) {
85 |       const targetPane = document.querySelector(targetSelector
    | );
```

```
86 |         if (targetPane && !targetPane.classList.contains('active
    | ')) {
87 |             targetPane.classList.add('show', 'active');
88 |         }
89 |     }
90 | }
91 |
92 | // 1. Bind event handlers immediately
93 | if (saveBtnInstr) saveBtnInstr.onclick = saveSystemInstructi
    | on;
94 | if (reloadBtnInstr) reloadBtnInstr.onclick = loadSystemInstr
    | uction;
95 |
96 | if (refreshModelsBtn && modelSelect && saveBtn) {
97 |     refreshModelsBtn.onclick = async () => {
98 |         refreshModelsBtn.disabled = true;
99 |         const originalText = refreshModelsBtn.innerHTML;
100 |         refreshModelsBtn.innerHTML = '<span class="spinner-borde
    | r spinner-border-sm me-1" role="status"></span> Обновление...'
    | ;
101 |         try {
102 |             await loadTabModels(modelSelect, saveBtn, true);
103 |         } finally {
104 |             refreshModelsBtn.disabled = false;
105 |             refreshModelsBtn.innerHTML = originalText;
106 |         }
107 |     };
108 | }
109 |
110 | const showAllModelsCheck = document.getElementById('show-all
    | -models-check');
111 | if (showAllModelsCheck && modelSelect && saveBtn) {
112 |     showAllModelsCheck.onchange = async () => {
113 |         await loadTabModels(modelSelect, saveBtn, false);
114 |     };
115 | }
116 |
117 | const favToggleBtn = document.getElementById('btn-model-togg
```

```

    le-favorite');
118   const saveNoteBtn = document.getElementById('btn-save-model-
    note');
119   const modelNoteInput = document.getElementById('model-user-n
    ote');
120
121   if (favToggleBtn && modelSelect) {
122     favToggleBtn.onclick = async () => {
123       const curModel = modelSelect.value;
124       if (!curModel) {
125         showModelsNotification('Сначала выберите модель', 'war
    ning');
126         return;
127       }
128       favToggleBtn.disabled = true;
129       try {
130         const isFav = window.userFavoriteModels && Boolean(win
    dow.userFavoriteModels[curModel]);
131         if (isFav) {
132           await window.api.fetch(`/auth/favorites/${encodeURIComponent(
    curModel)}`, { method: 'DELETE' });
133           if (window.userFavoriteModels) delete window.userFav
    oriteModels[curModel];
134           showModelsNotification(`Модель "${curModel}" удалена
    из избранного`, 'info');
135         } else {
136           const noteText = modelNoteInput ? modelNoteInput.val
    ue.trim() : '';
137           const res = await window.api.fetch('/auth/favorites'
    , {
138             method: 'POST',
139             headers: { 'Content-Type': 'application/json' },
140             body: JSON.stringify({ model: curModel, note: note
    Text })
141           });
142           if (res && res.favorites) window.userFavoriteModels
    = res.favorites;
143           showModelsNotification(`Модель "${curModel}" добавле

```

```

    на в избранное `', 'success');
144     }
145     updateModelFavoriteUI(modelSelect);
146     updateModelSelectOptions(modelSelect);
147   } catch (err) {
148     console.error('Ошибка переключения избранного:', err);
149     showModelsNotification('Ошибка: ' + err.message, 'dang
er');
150   } finally {
151     favToggleBtn.disabled = false;
152   }
153 };
154 }
155
156 if (saveNoteBtn && modelSelect && modelNoteInput) {
157   saveNoteBtn.onclick = async () => {
158     const curModel = modelSelect.value;
159     if (!curModel) {
160       showModelsNotification('Сначала выберите модель', 'war
ning');
161       return;
162     }
163     saveNoteBtn.disabled = true;
164     const originalText = saveNoteBtn.innerHTML;
165     saveNoteBtn.innerHTML = '<span class="spinner-border spi
nner-border-sm me-1"></span> Сохранение...';
166     try {
167       const noteText = modelNoteInput.value.trim();
168       const res = await window.api.fetch('/auth/favorites',
{
169         method: 'POST',
170         headers: { 'Content-Type': 'application/json' },
171         body: JSON.stringify({ model: curModel, note: noteTe
xt })
172       });
173       if (res && res.favorites) window.userFavoriteModels =
res.favorites;
174       showModelsNotification(`Заметка для модели "${curModel

```

```

    | }" сохранена!`, 'success');
175 |     updateModelFavoriteUI(modelSelect);
176 |     updateModelSelectOptions(modelSelect);
177 | } catch (err) {
178 |     console.error('Ошибка сохранения заметки:', err);
179 |     showModelsNotification('Ошибка сохранения заметки: ' +
    | err.message, 'danger');
180 | } finally {
181 |     saveNoteBtn.disabled = false;
182 |     saveNoteBtn.innerHTML = originalText;
183 | }
184 | };
185 | }
186 |
187 | if (modelSelect) {
188 |     modelSelect.addEventListener('change', () => {
189 |         updateModelFavoriteUI(modelSelect);
190 |     });
191 | }
192 |
193 | if (testModelBtn) {
194 |     testModelBtn.onclick = executeModelTest;
195 | }
196 |
197 | if (cancelModelBtn) {
198 |     cancelModelBtn.onclick = cancelModelTest;
199 | }
200 |
201 | if (testModelPrompt) {
202 |     testModelPrompt.onkeydown = (e) => {
203 |         if (e.key === 'Enter') {
204 |             e.preventDefault();
205 |             executeModelTest();
206 |         } else if (e.key === 'Escape' && modelTestAbortControlle
    | r) {
207 |             e.preventDefault();
208 |             cancelModelTest();
209 |         }

```

```

210     };
211 }
212
213 // Provider switch direct toggling
214 const agyEnabledSwitch = document.getElementById('agy-enable
d');
215 if (agyEnabledSwitch) {
216     agyEnabledSwitch.onchange = async () => {
217         const enabled = agyEnabledSwitch.checked;
218         const remember = document.getElementById('agy-remember')
?.checked ?? true;
219         const model = document.getElementById('agy-model')?.valu
e || 'agy-flash';
220         const key = document.getElementById('agy-key')?.value.tr
im() || '';
221
222         try {
223             await window.api.fetch('/api/agy/config', {
224                 method: 'POST',
225                 headers: { 'Content-Type': 'application/json' },
226                 body: JSON.stringify({ enabled, model, key, remember
}))
227         });
228         showModelsNotification(`Google Antigravity (AGY) ${ena
bled ? 'активирован' : 'деактивирован'}${remember ? ' (сохране
но в config.json)' : ''}`, 'info');
229         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
230     } catch (err) {
231         console.error('Ошибка переключения Antigravity:', err)
;
232         showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
233     }
234 };
235 }
236
237 const foundryEnabledSwitch = document.getElementById('foundr

```

```

y-enabled');
238   if (foundryEnabledSwitch) {
239       foundryEnabledSwitch.onchange = async () => {
240           const enabled = foundryEnabledSwitch.checked;
241           const remember = document.getElementById('foundry-remember')?.checked ?? true;
242           const url = document.getElementById('foundry-url')?.value.trim() || 'http://localhost:54837';
243           const key = document.getElementById('foundry-key')?.value.trim() || '';
244           const model = document.getElementById('foundry-model')?.value?.trim() || 'qwen2.5-1.5b';
245
246           try {
247               await window.api.fetch('/api/foundry/config', {
248                   method: 'POST',
249                   headers: { 'Content-Type': 'application/json' },
250                   body: JSON.stringify({ enabled, url, key, model, remember })
251               });
252               showModelsNotification(`Microsoft Foundry ${enabled ? 'активирован' : 'деактивирован'} ${remember ? ' (сохранено в config.json)' : ''}`, 'info');
253               if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
254           } catch (err) {
255               console.error('Ошибка переключения Foundry:', err);
256               showModelsNotification('Ошибка переключения: ' + err.message, 'danger');
257           }
258       };
259   }
260
261   const ollamaEnabledSwitch = document.getElementById('ollama-enabled');
262   if (ollamaEnabledSwitch) {
263       ollamaEnabledSwitch.onchange = async () => {
264           const enabled = ollamaEnabledSwitch.checked;

```

```

265 |     const remember = document.getElementById('ollama-remember')?.checked ?? true;
266 |     const url = document.getElementById('ollama-url')?.value
    | .trim() || 'http://localhost:11434';
267 |     const model = document.getElementById('ollama-model')?.value?.trim() || 'llama3.1';
268 |
269 |     try {
270 |         await window.api.fetch('/api/ollama/config', {
271 |             method: 'POST',
272 |             headers: { 'Content-Type': 'application/json' },
273 |             body: JSON.stringify({ enabled, url, model, remember
    |         })
274 |         });
275 |         showModelsNotification(`Ollama ${enabled ? 'активирован' : 'деактивирован'} ${remember ? ' (сохранено в config.json)' : ''}`, 'info');
276 |         if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
277 |     } catch (err) {
278 |         console.error('Ошибка переключения Ollama:', err);
279 |         showModelsNotification('Ошибка переключения: ' + err.message, 'danger');
280 |     }
281 | };
282 | }
283 |
284 | const onnxEnabledSwitch = document.getElementById('onnx-enabled');
285 | if (onnxEnabledSwitch) {
286 |     onnxEnabledSwitch.onchange = async () => {
287 |         const enabled = onnxEnabledSwitch.checked;
288 |         const remember = document.getElementById('onnx-remember')?.checked ?? true;
289 |         const execution_provider = document.getElementById('onnx-execution-provider')?.value || 'DirectMLExecutionProvider';
290 |         const models_dir = document.getElementById('onnx-models-dir')?.value.trim() || 'models/onnx';

```



```

291 |     const default_model = document.getElementById('onnx-defa
    | ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
292 |     const olive_precision = document.getElementById('onnx-ol
    | ive-precision')?.value || 'int4';
293 |
294 |     try {
295 |         await window.api.fetch('/api/onnx/config', {
296 |             method: 'POST',
297 |             headers: { 'Content-Type': 'application/json' },
298 |             body: JSON.stringify({
299 |                 enabled,
300 |                 execution_provider,
301 |                 models_dir,
302 |                 default_model,
303 |                 olive_precision,
304 |                 remember
305 |             })
306 |         });
307 |         showModelsNotification(`ONNX / Olive ${enabled ? 'акти
    | вирован' : 'деактивирован'}${remember ? ' (сохранено в config.
    | json)' : ''}`, 'info');
308 |         if (modelSelect && saveBtn) await loadTabModels(modelS
    | elect, saveBtn);
309 |     } catch (err) {
310 |         console.error('Ошибка переключения ONNX / Olive:', err
    | );
311 |         showModelsNotification('Ошибка переключения: ' + err.m
    | essage, 'danger');
312 |     }
313 | };
314 | }
315 |
316 | if (saveAgyBtn) {
317 |     saveAgyBtn.onclick = async () => {
318 |         const enabled = document.getElementById('agy-enabled')?.
    | checked ?? true;
319 |         const remember = document.getElementById('agy-remember')
    | ?.checked ?? true;

```

```

320 |     const model = document.getElementById('agy-model')?.value || 'agy-flash';
321 |     const key = document.getElementById('agy-key')?.value.trim() || '';
322 |
323 |     saveAgyBtn.disabled = true;
324 |     try {
325 |         await window.api.fetch('/api/agy/config', {
326 |             method: 'POST',
327 |             headers: { 'Content-Type': 'application/json' },
328 |             body: JSON.stringify({ enabled, model, key, remember
    |         })
329 |     });
330 |     showModelsNotification(`Настройки AntigraVity (AGY) успешно сохранены${remember ? ' в config.json' : ''}`, 'success');
331 |     if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
332 | } catch (err) {
333 |     console.error('Ошибка сохранения AntigraVity:', err);
334 |     showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
335 | } finally {
336 |     saveAgyBtn.disabled = false;
337 | }
338 | };
339 | }
340 |
341 | if (saveFoundryBtn) {
342 |     saveFoundryBtn.onclick = async () => {
343 |         const enabled = document.getElementById('foundry-enabled')?.checked || false;
344 |         const remember = document.getElementById('foundry-remember')?.checked ?? true;
345 |         const url = document.getElementById('foundry-url')?.value.trim() || 'http://localhost:54837';
346 |         const key = document.getElementById('foundry-key')?.value.trim() || '';

```

```
347 |     const model = document.getElementById('foundry-model')?.
    | value?.trim() || 'qwen2.5-1.5b';
348 |
349 |     saveFoundryBtn.disabled = true;
350 |     try {
351 |         await window.api.fetch('/api/foundry/config', {
352 |             method: 'POST',
353 |             headers: { 'Content-Type': 'application/json' },
354 |             body: JSON.stringify({ enabled, url, key, model, rem
    | ember })
355 |         });
356 |         showModelsNotification(`Настройки Microsoft Foundry ус
    | пешно сохранены${remember ? ' в config.json' : ''}`, 'success'
    | );
357 |         if (modelSelect && saveBtn) await loadTabModels(modelS
    | elect, saveBtn);
358 |     } catch (err) {
359 |         console.error('Ошибка сохранения Foundry:', err);
360 |         showModelsNotification('Ошибка сохранения: ' + err.mes
    | sage, 'danger');
361 |     } finally {
362 |         saveFoundryBtn.disabled = false;
363 |     }
364 | };
365 | }
366 |
367 | if (saveOllamaBtn) {
368 |     saveOllamaBtn.onclick = async () => {
369 |         const enabled = document.getElementById('ollama-enabled'
    | )?.checked || false;
370 |         const remember = document.getElementById('ollama-remembe
    | r')?.checked ?? true;
371 |         const url = document.getElementById('ollama-url')?.value
    | .trim() || 'http://localhost:11434';
372 |         const model = document.getElementById('ollama-model')?.v
    | alue?.trim() || 'llama3.1';
373 |
374 |         saveOllamaBtn.disabled = true;
```

```

375 |     try {
376 |         await window.api.fetch('/api/ollama/config', {
377 |             method: 'POST',
378 |             headers: { 'Content-Type': 'application/json' },
379 |             body: JSON.stringify({ enabled, url, model, remember
|         })
380 |     });
381 |     showModelsNotification(`Настройки Ollama успешно сохра
| нены${remember ? ' в config.json' : ''}`, 'success');
382 |     if (modelSelect && saveBtn) await loadTabModels(modelS
| elect, saveBtn);
383 |     } catch (err) {
384 |         console.error('Ошибка сохранения Ollama:', err);
385 |         showModelsNotification('Ошибка сохранения: ' + err.mes
| sage, 'danger');
386 |     } finally {
387 |         saveOllamaBtn.disabled = false;
388 |     }
389 | };
390 | }
391 |
392 | if (saveOnnxBtn) {
393 |     saveOnnxBtn.onclick = async () => {
394 |         const enabled = document.getElementById('onnx-enabled')?
| .checked ?? true;
395 |         const remember = document.getElementById('onnx-remember'
| )?.checked ?? true;
396 |         const execution_provider = document.getElementById('onnx
| -execution-provider')?.value || 'DirectMLExecutionProvider';
397 |         const models_dir = document.getElementById('onnx-models-
| dir')?.value.trim() || 'models/onnx';
398 |         const default_model = document.getElementById('onnx-defa
| ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
399 |         const olive_precision = document.getElementById('onnx-ol
| ive-precision')?.value || 'int4';
400 |
401 |         saveOnnxBtn.disabled = true;
402 |         try {

```

```

403         await window.api.fetch('/api/onnx/config', {
404             method: 'POST',
405             headers: { 'Content-Type': 'application/json' },
406             body: JSON.stringify({
407                 enabled,
408                 execution_provider,
409                 models_dir,
410                 default_model,
411                 olive_precision,
412                 remember
413             })
414         });
415         showModelsNotification(`Настройки ONNX / Olive успешно
сохранены${remember ? ' в config.json' : ''}`, 'success');
416         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
417     } catch (err) {
418         console.error('Ошибка сохранения ONNX / Olive:', err);
419         showModelsNotification('Ошибка сохранения: ' + err.mes
sage, 'danger');
420     } finally {
421         saveOnnxBtn.disabled = false;
422     }
423 };
424 }
425
426 if (saveBtn && modelSelect) {
427     saveBtn.onclick = async () => {
428         const selectedModel = modelSelect.value;
429         saveBtn.disabled = true;
430         const originalText = saveBtn.textContent;
431         saveBtn.textContent = 'Сохранение...';
432
433         try {
434             await window.api.fetch('/auth/settings', {
435                 method: 'POST',
436                 headers: { 'Content-Type': 'application/json' },
437                 body: JSON.stringify({ model: selectedModel })

```

```

438         });
439         showModelsNotification('Модель успешно обновлена на: '
+ selectedModel, 'success');
440
441         const otherModelSelect = document.getElementById('admi
n-model-select');
442         if (otherModelSelect) {
443             otherModelSelect.value = selectedModel;
444         }
445         if (typeof window.updateChatBadges === 'function') {
446             window.updateChatBadges(selectedModel);
447         }
448     } catch (err) {
449         console.error('Ошибка сохранения модели:', err);
450         showModelsNotification('Ошибка сохранения: ' + err.mes
sage, 'danger');
451     } finally {
452         saveBtn.disabled = false;
453         saveBtn.textContent = originalText;
454     }
455 };
456 }
457
458 if (refreshKeysBtn) {
459     refreshKeysBtn.onclick = async () => {
460         refreshKeysBtn.disabled = true;
461         const originalText = refreshKeysBtn.textContent;
462         refreshKeysBtn.textContent = '⏮ Сброс...';
463         try {
464             const res = await window.api.fetch('/api/keys/reset-al
l', { method: 'POST' });
465             showModelsNotification(res.message || 'Квоты всех ключ
ей успешно сброшены', 'success');
466         } catch (err) {
467             console.error('Ошибка сброса квот:', err);
468             showModelsNotification('Ошибка сброса: ' + err.message
, 'danger');
469         } finally {

```

```
470         refreshKeysBtn.disabled = false;
471         refreshKeysBtn.textContent = originalText;
472         if (keysListBody) await refreshKeysList(keysListBody);
473     }
474 };
475 }
476
477 if (addKeyBtn) {
478     addKeyBtn.onclick = async () => {
479         const nameInput = document.getElementById('new-key-name'
480 );
481         const valueInput = document.getElementById('new-key-valu
482 e');
483         if (!nameInput || !valueInput) return;
484
485         const name = nameInput.value.trim();
486         const apiKey = valueInput.value.trim();
487
488         if (!name || !apiKey) {
489             showModelsNotification('Заполните все поля!', 'warning
490 ');
491             return;
492         }
493
494         addKeyBtn.disabled = true;
495         try {
496             await window.api.fetch('/api/keys', {
497                 method: 'POST',
498                 headers: { 'Content-Type': 'application/json' },
499                 body: JSON.stringify({ name, api_key: apiKey, status
500 : 'active' })
501             });
502             showModelsNotification(`Ключ "${name}" успешно добавле
503 н`, 'success');
504             nameInput.value = '';
505             valueInput.value = '';
506             if (keysListBody) await refreshKeysList(keysListBody);
507         } catch (err) {
```

```

503         console.error('Ошибка добавления ключа:', err);
504         showModelsNotification('Ошибка добавления: ' + err.mes
sage, 'danger');
505     } finally {
506         addKeyBtn.disabled = false;
507     }
508 };
509 }
510
511 if (refreshGAccountsBtn && gaccountsListBody) {
512     refreshGAccountsBtn.onclick = async () => {
513         refreshGAccountsBtn.disabled = true;
514         const originalText = refreshGAccountsBtn.innerHTML;
515         refreshGAccountsBtn.innerHTML = '<span class="spinner-bo
rder spinner-border-sm me-1"></span> Обновление...';
516         try {
517             await refreshGoogleAccountsList(gaccountsListBody);
518             showModelsNotification('Список аккаунтов Google обновл
ен', 'success');
519         } finally {
520             refreshGAccountsBtn.disabled = false;
521             refreshGAccountsBtn.innerHTML = originalText;
522         }
523     };
524 }
525
526 if (addGAccountBtn) {
527     addGAccountBtn.onclick = async () => {
528         await handleAddGoogleAccount(gaccountsListBody);
529     };
530 }
531
532 // 2. Load all components concurrently
533 await Promise.allSettled([
534     modelSelect && saveBtn ? loadTabModels(modelSelect, saveBt
n) : Promise.resolve(),
535     keysListBody ? refreshKeysList(keysListBody) : Promise.res
olve(),

```



```

536 |     gaccountsListBody ? refreshGoogleAccountsList(gaccountsLis
    | tBody) : Promise.resolve(),
537 |     loadFoundryConfig(),
538 |     loadOllamaConfig(),
539 |     loadAgyConfig(),
540 |     loadOnnxConfig(),
541 |     loadSystemInstruction()
542 |   ]);
543 | }
544 |
545 | // Helper for escaping HTML strings
546 | function escapeHtml(text) {
547 |   if (!text) return '';
548 |   return String(text)
549 |     .replace(/&/g, '&amp;')
550 |     .replace(/</g, '&lt;')
551 |     .replace(/>/g, '&gt;')
552 |     .replace(/"/g, '&quot;')
553 |     .replace(/'/g, '&#039;');
554 | }
555 |
556 | // Helper to execute model test request
557 | async function executeModelTest() {
558 |   const providerSelect = document.getElementById('provider-tab
    | -select');
559 |   const modelSelect = document.getElementById('models-tab-sele
    | ct');
560 |   const promptInput = document.getElementById('model-test-prom
    | pt');
561 |   const testBtn = document.getElementById('btn-model-test-send
    | ');
562 |   const cancelBtn = document.getElementById('btn-model-test-ca
    | ncel');
563 |   const resultContainer = document.getElementById('model-test-
    | result');
564 |
565 |   if (!testBtn || !resultContainer) return;
566 |

```

```
567 | const provider = providerSelect ? providerSelect.value : '';
568 | const model = modelSelect ? modelSelect.value : '';
569 | const message = promptInput ? promptInput.value.trim() : '';
570 |
571 | if (!model) {
572 |     showModelsNotification('Выберите модель для тестирования!'
    | , 'warning');
573 |     return;
574 | }
575 |
576 | if (!message) {
577 |     showModelsNotification('Введите текст проверочного запроса
    | !', 'warning');
578 |     if (promptInput) promptInput.focus();
579 |     return;
580 | }
581 |
582 | // Abort any prior running test
583 | cancelModelTest();
584 | modelTestAbortController = new AbortController();
585 |
586 | testBtn.disabled = true;
587 | const originalBtnHtml = testBtn.innerHTML;
588 | testBtn.innerHTML = '<span class="spinner-border spinner-bor
    | der-sm me-1" role="status"></span> Тест...';
589 |
590 | if (cancelBtn) {
591 |     cancelBtn.style.display = 'inline-block';
592 |     cancelBtn.disabled = false;
593 | }
594 |
595 | resultContainer.style.display = 'block';
596 | resultContainer.innerHTML = `
597 |     <div class="d-flex justify-content-between align-items-cen
    | ter text-info small">
598 |         <div class="d-flex align-items-center text-truncate">
599 |             <span class="spinner-border spinner-border-sm me-2 fle
    | x-shrink-0" role="status"></span>
```

```

600 |         <span class="text-truncate">Отправка запроса в <strong
    | >${escapeHtml(model)}</strong> (${escapeHtml(provider)})...</s
    | pan>
601 |     </div>
602 |     <button class="btn btn-outline-danger btn-sm py-0 px-2 r
    | ounded ms-2 flex-shrink-0" id="btn-model-test-cancel-inner" ty
    | pe="button" title="Отменить проверочный запрос">
603 |         <i class="bi bi-stop-circle me-1"></i> Отмена
604 |     </button>
605 | </div>
606 | `;
607 |
608 | const cancelInnerBtn = document.getElementById('btn-model-te
    | st-cancel-inner');
609 | if (cancelInnerBtn) {
610 |     cancelInnerBtn.onclick = cancelModelTest;
611 | }
612 |
613 | try {
614 |     const res = await window.api.fetch('/api/chat/test-model',
    | {
615 |         method: 'POST',
616 |         headers: { 'Content-Type': 'application/json' },
617 |         signal: modelTestAbortController.signal,
618 |         body: JSON.stringify({
619 |             provider: provider,
620 |             model: model,
621 |             message: message
622 |         })
623 |     });
624 |
625 |     if (res && res.status === 'success') {
626 |         resultContainer.innerHTML = `
627 |             <div class="d-flex justify-content-between align-items
    | -center mb-1 pb-1 border-bottom">
628 |                 <div>
629 |                     <span class="badge bg-success me-2"><i class="bi b
    | i-check-circle me-1"></i>200 OK</span>

```

```

630 |         <span class="text-secondary small font-monospace">
    | □ ${res.duration_ms || 0} ms</span>
631 |     </div>
632 |     <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(res.model || model)}</span>
633 | </div>
634 |     <div class="small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(res.response || 'Пустой ответ от модели')}}</div>
635 |     `;
636 | } else {
637 |     const errMsg = (res && res.message) ? res.message : 'Неизвестная ошибка при запросе к модели';
638 |     resultContainer.innerHTML = `
639 |         <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
640 |             <div>
641 |                 <span class="badge bg-danger me-2"><i class="bi bi-x-circle me-1"></i>Ошибка</span>
642 |                 <span class="text-secondary small font-monospace">
    | □ ${res?.duration_ms || 0} ms</span>
643 |             </div>
644 |             <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
645 |         </div>
646 |         <div class="text-danger small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(errMsg)}</div>
647 |     `;
648 | }
649 | } catch (err) {
650 |     if (err.name === 'AbortError' || err.message?.toLowerCase().includes('abort')) {
651 |         resultContainer.innerHTML = `
652 |             <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
653 |                 <span class="badge bg-warning text-dark"><i class="b

```

```

        i bi-slash-circle me-1"></i>Отменено</span>
654         <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
655     </div>
656     <div class="text-warning small mt-1 font-monospace"><i class="bi bi-info-circle me-1"></i>Запрос отменен пользователем.</div>
657     `;
658     } else {
659         console.error('Error during model test request:', err);
660         resultContainer.innerHTML = `
661         <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
662             <span class="badge bg-danger"><i class="bi bi-x-circle me-1"></i>Ошибка сети/API</span>
663             <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
664         </div>
665         <div class="text-danger small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(err.message || 'Ошибка соединения')}</div>
666     `;
667     }
668     } finally {
669         modelTestAbortController = null;
670         testBtn.disabled = false;
671         testBtn.innerHTML = originalBtnHtml;
672         if (cancelBtn) {
673             cancelBtn.style.display = 'none';
674         }
675     }
676 }
677
678 // Favorite models storage in window state
679 window.userFavoriteModels = window.userFavoriteModels || {};
680
681 function updateModelFavoriteUI(modelSelect) {
682     if (!modelSelect) modelSelect = document.getElementById('mod

```

```
els-tab-select');
683   const curModel = modelSelect ? modelSelect.value : '';
684   const starIcon = document.getElementById('favorite-star-icon
');
685   const favBtn = document.getElementById('btn-model-toggle-fav
orite');
686   const statusBadge = document.getElementById('model-fav-statu
s-badge');
687   const noteInput = document.getElementById('model-user-note')
;
688
689   const favData = curModel && window.userFavoriteModels ? wind
ow.userFavoriteModels[curModel] : null;
690   const isFav = Boolean(favData);
691
692   if (starIcon) {
693     starIcon.className = isFav ? 'bi bi-star-fill text-warning
' : 'bi bi-star';
694   }
695   if (favBtn) {
696     if (isFav) {
697       favBtn.classList.remove('btn-outline-warning');
698       favBtn.classList.add('btn-warning', 'text-dark');
699       favBtn.title = 'Удалить модель из избранного';
700     } else {
701       favBtn.classList.remove('btn-warning', 'text-dark');
702       favBtn.classList.add('btn-outline-warning');
703       favBtn.title = 'Добавить модель в избранное';
704     }
705   }
706   if (statusBadge) {
707     if (isFav) {
708       statusBadge.textContent = 'В избранном ☑';
709       statusBadge.className = 'badge bg-warning text-dark font
-monospace';
710     } else {
711       statusBadge.textContent = 'Не в избранном';
712       statusBadge.className = 'badge bg-secondary-subtle text-
```

```

    secondary-emphasis border font-monospace';
713     }
714   }
715   if (noteInput && document.activeElement !== noteInput) {
716     noteInput.value = favData ? (favData.note || '') : '';
717   }
718
719   renderFavoriteModelsChips(modelSelect);
720 }
721
722 function updateModelSelectOptions(modelSelect) {
723   if (!modelSelect) return;
724   const options = modelSelect.querySelectorAll('option');
725   options.forEach(opt => {
726     const val = opt.value;
727     if (!val) return;
728     const isFav = window.userFavoriteModels && Boolean(window.
userFavoriteModels[val]);
729     let text = opt.textContent.replace(/^\s*/, ' ');
730     if (isFav) {
731       opt.textContent = `👍 ${text}`;
732     } else {
733       opt.textContent = text;
734     }
735   });
736 }
737
738 function renderFavoriteModelsChips(modelSelect) {
739   const container = document.getElementById('favorite-models-l
ist');
740   const section = document.getElementById('favorite-models-sec
tion');
741   if (!container || !section) return;
742
743   const favKeys = Object.keys(window.userFavoriteModels || {});
744   ;
745   if (favKeys.length === 0) {
    section.style.display = 'none';
  }

```

```

746     container.innerHTML = '';
747     return;
748 }
749
750 section.style.display = 'block';
751 container.innerHTML = '';
752
753 favKeys.forEach(modelName => {
754     const data = window.userFavoriteModels[modelName] || {};
755     const note = data.note ? `Заметка: ${data.note}` : '';
756
757     const chip = document.createElement('div');
758     chip.className = 'btn-group btn-group-sm mb-1';
759     chip.role = 'group';
760
761     const btnSelect = document.createElement('button');
762     btnSelect.type = 'button';
763     btnSelect.className = modelSelect && modelSelect.value ===
modelName
764         ? 'btn btn-sm btn-warning text-dark py-0 px-2 font-monospace'
765         : 'btn btn-sm btn-outline-warning py-0 px-2 font-monospace';
766     btnSelect.innerHTML = `<i class="bi bi-star-fill me-1 text-warning"></i>${escapeHtml(modelName)}`;
767     btnSelect.title = `Выбрать модель: ${modelName}${note}`;
768     btnSelect.onclick = () => {
769         if (modelSelect) {
770             // Try finding which provider has this model
771             const providerSelect = document.getElementById('provider-tab-select');
772             if (providerSelect && window._modelsGrouped) {
773                 for (const p of Object.keys(window._modelsGrouped))
774                 {
775                     if (window._modelsGrouped[p]?.includes(modelName))
776                     {
777                         if (providerSelect.value !== p) {
778                             providerSelect.value = p;
779                         }
780                     }
781                 }
782             }
783         }
784     };
785     chip.appendChild(btnSelect);
786     container.appendChild(chip);
787 }
788
789 // Add a new model
790 const addModel = () => {
791     const modelInput = document.getElementById('model-input');
792     const providerInput = document.getElementById('provider-input');
793     const addBtn = document.getElementById('add-model');
794     const modelValue = modelInput.value;
795     const providerValue = providerInput.value;
796     if (modelValue && providerValue) {
797         // Add the model to the list
798         const newModel = {
799             name: modelValue,
800             provider: providerValue,
801             note: ''
802         };
803         window.userFavoriteModels[modelValue] = newModel;
804         // Update the favKeys array
805         favKeys.push(modelValue);
806         // Clear the inputs
807         modelInput.value = '';
808         providerInput.value = '';
809     }
810 };
811
812 // Remove a model
813 const removeModel = (modelName) => {
814     const modelIndex = favKeys.indexOf(modelName);
815     if (modelIndex !== -1) {
816         favKeys.splice(modelIndex, 1);
817         delete window.userFavoriteModels[modelName];
818     }
819 };
820
821 // Toggle the visibility of the models
822 const toggleModels = () => {
823     const modelsList = document.getElementById('models-list');
824     const toggleBtn = document.getElementById('toggle-models');
825     const modelsListStyle = window.getComputedStyle(modelsList);
826     const isHidden = modelsListStyle.display === 'none';
827     if (isHidden) {
828         modelsList.style.display = 'block';
829         toggleBtn.textContent = 'Скрыть модели';
830     } else {
831         modelsList.style.display = 'none';
832         toggleBtn.textContent = 'Показать модели';
833     }
834 };
835
836 // Initialize the application
837 const init = () => {
838     // Load the user favorite models
839     const userFavoriteModels = localStorage.getItem('userFavoriteModels');
840     if (userFavoriteModels) {
841         window.userFavoriteModels = JSON.parse(userFavoriteModels);
842     } else {
843         window.userFavoriteModels = {};
844     }
845     // Initialize the favKeys array
846     favKeys = Object.keys(window.userFavoriteModels);
847     // Initialize the models list
848     const modelsList = document.getElementById('models-list');
849     const toggleBtn = document.getElementById('toggle-models');
850     toggleModels();
851     // Add the event listeners
852     document.getElementById('add-model').addEventListener('click', addModel);
853     document.getElementById('remove-model').addEventListener('click', removeModel);
854     document.getElementById('toggle-models').addEventListener('click', toggleModels);
855     // Render the models list
856     renderModelsList();
857 };
858
859 // Start the application
860 init();

```



```

777 |             if (typeof window._populateModels === 'functio
| n') {
778 |                 window._populateModels(p, window._modelsGrou
| ped[p]);
779 |             }
780 |         }
781 |         break;
782 |     }
783 | }
784 | }
785 |     modelSelect.value = modelName;
786 |     updateModelFavoriteUI(modelSelect);
787 | }
788 | };
789 |
790 |     const btnRemove = document.createElement('button');
791 |     btnRemove.type = 'button';
792 |     btnRemove.className = 'btn btn-sm btn-outline-danger py-0
| px-1';
793 |     btnRemove.innerHTML = '<i class="bi bi-x"></i>';
794 |     btnRemove.title = `Удалить ${modelName} из избранного`;
795 |     btnRemove.onclick = async (e) => {
796 |         e.stopPropagation();
797 |         try {
798 |             await window.api.fetch(`/auth/favorites/${encodeURIComponent(modelName)}`, { method: 'DELETE' });
799 |             delete window.userFavoriteModels[modelName];
800 |             updateModelFavoriteUI(modelSelect);
801 |             updateModelSelectOptions(modelSelect);
802 |             showModelsNotification(`Модель "${modelName}" удалена
| из избранного`, 'info');
803 |         } catch (err) {
804 |             showModelsNotification('Ошибка удаления: ' + err.messa
| ge, 'danger');
805 |         }
806 |     };
807 |
808 |     chip.appendChild(btnSelect);

```

```
809     chip.appendChild(btnRemove);
810     container.appendChild(chip);
811 });
812 }
813
814 // Helper to load models list
815 async function loadTabModels(modelSelect, saveBtn, forceRefresh = false) {
816     const providerSelect = document.getElementById('provider-tab-select');
817     if (providerSelect) providerSelect.innerHTML = '';
818     modelSelect.innerHTML = '';
819
820     let modelsGrouped = {};
821     let unsupportedGrouped = {};
822
823     const fetchModels = async (force = false) => {
824         try {
825             const showAll = document.getElementById('show-all-models-check')?.checked ?? false;
826             const params = new URLSearchParams();
827             if (force) params.append('refresh', 'true');
828             if (showAll) params.append('include_unsupported', 'true');
829             const url = '/api/chat/models' + (params.toString() ? '?' : '') + params.toString();
830             const modelsData = await window.api.fetch(url);
831             let grouped = modelsData.models || {};
832             if (Array.isArray(grouped)) {
833                 grouped = { 'gemini': grouped };
834             }
835             unsupportedGrouped = modelsData.unsupported_models || {};
836
837             return grouped;
838         } catch (err) {
839             console.error('Error loading AI models:', err);
840             showModelsNotification('Ошибка загрузки моделей AI: ' + err.message, 'danger');
```

```
840     return {};  
841   }  
842 };  
843  
844 modelsGrouped = await fetchModels(forceRefresh);  
845 window._modelsGrouped = modelsGrouped;  
846 if (forceRefresh) {  
847   showModelsNotification('Список моделей успешно обновлен',  
    'success');  
848 }  
849  
850 // Fetch favorite models and current settings  
851 try {  
852   const settingsData = await window.api.fetch('/auth/settings');  
853   if (settingsData && settingsData.favorite_models) {  
854     window.userFavoriteModels = settingsData.favorite_models  
    ;  
855   } else {  
856     const favRes = await window.api.fetch('/auth/favorites')  
    ;  
857     if (favRes && favRes.favorites) {  
858       window.userFavoriteModels = favRes.favorites;  
859     }  
860   }  
861 } catch (e) {  
862   console.warn('Failed to load favorite models:', e);  
863 }  
864  
865 const providers = Object.keys(modelsGrouped).filter(p => modelsGrouped[p] && modelsGrouped[p].length > 0);  
866  
867 if (providers.length === 0) {  
868   if (providerSelect) {  
869     providerSelect.innerHTML = '<option value="">Нет доступных провайдеров</option>';  
870   }  
871   modelSelect.innerHTML = '<option value="">Нет доступных мо
```

```

    делей</option>';
872     saveBtn.disabled = true;
873     updateModelFavoriteUI(modelSelect);
874     return;
875 }
876
877 if (providerSelect) {
878     providerSelect.innerHTML = '';
879     providers.forEach(p => {
880         const option = document.createElement('option');
881         option.value = p;
882         option.textContent = p.charAt(0).toUpperCase() + p.slice
(1);
883         providerSelect.appendChild(option);
884     });
885 }
886
887 const populateModels = (provider, providerModelsList) => {
888     modelSelect.innerHTML = '';
889     const providerModels = providerModelsList !== undefined ?
providerModelsList : (modelsGrouped[provider] || []);
890     const unsupList = unsupportedGrouped[provider] || [];
891     if (!providerModels || providerModels.length === 0) {
892         modelSelect.innerHTML = '<option value="">Нет моделей</o
ption>';
893         saveBtn.disabled = true;
894     } else {
895         providerModels.forEach(modelName => {
896             const option = document.createElement('option');
897             option.value = modelName;
898             let cleanName = modelName;
899             if (cleanName.startsWith('foundry:')) cleanName = clea
nName.substring(8);
900             else if (cleanName.startsWith('ollama:')) cleanName =
cleanName.substring(7);
901             else if (cleanName.startsWith('gemini_cli:')) cleanNam
e = cleanName.substring(11);
902             else if (cleanName.startsWith('agy-')) cleanName = cle

```

```

    anName.substring(4);
903     else if (cleanName.startsWith('onnx:')) cleanName = cl
    eanName.substring(5);
904
905     const isFav = window.userFavoriteModels && Boolean(win
    dow.userFavoriteModels[modelName]);
906     const isUnsupported = unupList.includes(cleanName) ||
    unupList.includes(modelName);
907
908     let label = cleanName;
909     if (isFav) label = `❑ ${label}`;
910     if (isUnsupported) label = `${label} ❑ [отфильтрована]
    `;
911
912     option.textContent = label;
913     modelSelect.appendChild(option);
914 });
915 saveBtn.disabled = false;
916 }
917 updateModelFavoriteUI(modelSelect);
918 };
919 window._populateModels = populateModels;
920
921 if (providerSelect) {
922     providerSelect.onChange = async () => {
923         const chosenProvider = providerSelect.value;
924         // При каждом выборе провайдера актуализируем список дос
    тупных моделей
925         modelSelect.innerHTML = '<option value="">Обновление спи
    ска моделей...</option>';
926         saveBtn.disabled = true;
927         try {
928             const updatedGrouped = await fetchModels(true);
929             if (updatedGrouped && Object.keys(updatedGrouped).leng
    th > 0) {
930                 modelsGrouped = updatedGrouped;
931                 window._modelsGrouped = modelsGrouped;
932             }

```

```
933     } catch (e) {
934         console.warn('Failed to refresh models on provider change:', e);
935     }
936     populateModels(chosenProvider, modelsGrouped[chosenProvider]);
937 };
938 populateModels(providerSelect.value, modelsGrouped[providerSelect.value]);
939 } else {
940     let allModels = [];
941     providers.forEach(p => allModels = allModels.concat(modelsGrouped[p]));
942     allModels.forEach(modelName => {
943         const option = document.createElement('option');
944         option.value = modelName;
945         const isFav = window.userFavoriteModels && Boolean(window.userFavoriteModels[modelName]);
946         option.textContent = isFav ? `🔖 ${modelName}` : modelName;
947         modelSelect.appendChild(option);
948     });
949     saveBtn.disabled = allModels.length === 0;
950     updateModelFavoriteUI(modelSelect);
951 }
952
953 try {
954     const settingsData = await window.api.fetch('/auth/settings');
955     const savedModel = settingsData && settingsData.model ? settingsData.model : '';
956     if (savedModel) {
957         let foundProvider = null;
958         for (const p of providers) {
959             if (modelsGrouped[p] && modelsGrouped[p].includes(savedModel)) {
960                 foundProvider = p;
961                 break;
962             }
963         }
964     }
965 }
```

```

962     }
963 }
964 if (foundProvider && providerSelect) {
965     providerSelect.value = foundProvider;
966     populateModels(foundProvider, modelsGrouped[foundProvi
der]);
967 }
968 modelSelect.value = savedModel;
969 updateModelFavoriteUI(modelSelect);
970 }
971 if (typeof window.updateChatBadges === 'function') {
972     window.updateChatBadges(savedModel);
973 }
974 } catch (err) {
975     console.error('Error loading user AI settings:', err);
976 }
977 }
978
979 // Helper to refresh keys list table
980 async function refreshKeysList(container) {
981     try {
982         container.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-4 text-muted">Загрузка ключей...</td></tr>';
983         const keysData = await window.api.fetch('/api/keys');
984         const keys = keysData.keys || [];
985
986         if (keys.length === 0) {
987             container.innerHTML = '<tr><td colspan="5" class="text-c
enter py-4 text-muted">Список ключей пуст</td></tr>';
988             return;
989         }
990
991         container.innerHTML = '';
992         keys.forEach(key => {
993             const row = document.createElement('tr');
994
995             const tdName = document.createElement('td');
996             tdName.innerHTML = `<strong>${key.name}</strong>`;

```

```
997 |         row.appendChild(tdName);
998 |
999 |         const tdKey = document.createElement('td');
1000 |         tdKey.className = 'font-monospace text-muted small';
1001 |         tdKey.textContent = key.api_key_masked;
1002 |         row.appendChild(tdKey);
1003 |
1004 |         const tdStatus = document.createElement('td');
1005 |         const isEnabled = key.status === 'active';
1006 |         const statusClass = isEnabled ? 'bg-success' : 'bg-secondary';
1007 |         const statusText = isEnabled ? 'Активен' : 'Отключен';
1008 |         tdStatus.innerHTML = `<span class="badge ${statusClass}"
1009 | >${statusText}</span>`;
1010 |         row.appendChild(tdStatus);
1011 |
1012 |         const tdQuota = document.createElement('td');
1013 |         if (key.exhausted) {
1014 |             let resetText = 'ЛИМИТ';
1015 |             if (key.reset_in_seconds) {
1016 |                 const hours = Math.floor(key.reset_in_seconds / 3600);
1017 |                 const mins = Math.floor((key.reset_in_seconds % 3600) / 60);
1018 |                 resetText = `Сброс через ${hours}ч ${mins}м`;
1019 |             }
1020 |             tdQuota.innerHTML = `<span class="badge bg-danger d-block mb-1" title="Превышен лимит запросов в сутки">${resetText}</span>`;
1021 |         } else {
1022 |             tdQuota.innerHTML = `<span class="badge bg-success d-block mb-1">OK</span>`;
1023 |         }
1024 |         row.appendChild(tdQuota);
1025 |
1026 |         const tdActions = document.createElement('td');
1027 |         tdActions.className = 'text-end';
```



```

1028     const btnToggle = document.createElement('button');
1029     btnToggle.className = `btn btn-xs btn-sm me-1 ${isEnabled
d ? 'btn-outline-secondary' : 'btn-outline-success'}`;
1030     btnToggle.textContent = isEnabled ? 'Откл' : 'Вкл';
1031     btnToggle.onclick = () => toggleKeyStatus(key.name, isEn
abled ? 'disabled' : 'active', container);
1032     tdActions.appendChild(btnToggle);
1033
1034     if (key.exhausted) {
1035         const btnReset = document.createElement('button');
1036         btnReset.className = 'btn btn-xs btn-outline-warning b
tn-sm me-1';
1037         btnReset.innerHTML = 'Сброс';
1038         btnReset.title = 'Сбросить 24-часовой бан квоты';
1039         btnReset.onclick = () => resetKeyQuota(key.name, conta
iner);
1040         tdActions.appendChild(btnReset);
1041     }
1042
1043     const btnDelete = document.createElement('button');
1044     btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
1045     btnDelete.textContent = 'Удалить';
1046     btnDelete.onclick = () => deleteKey(key.name, container)
;
1047     tdActions.appendChild(btnDelete);
1048
1049     row.appendChild(tdActions);
1050     container.appendChild(row);
1051 });
1052
1053 } catch (err) {
1054     console.error('Ошибка загрузки ключей:', err);
1055     container.innerHTML = `|<td colspan="5" class="text-cen
ter py-4 text-danger">Ошибка: ${err.message}</td></tr>`;
1056 }
1057 }
1058

|  |

```

```
1059 // API action helpers
1060 async function toggleKeyStatus(name, newStatus, container) {
1061   try {
1062     await window.api.fetch(`/api/keys/${name}`, {
1063       method: 'PATCH',
1064       headers: { 'Content-Type': 'application/json' },
1065       body: JSON.stringify({ status: newStatus })
1066     });
1067     showModelsNotification(`Статус ключа "${name}" изменен на
    ${newStatus === 'active' ? 'активный' : 'отключенный'}`, 'succ
    ess');
1068     await refreshKeysList(container);
1069   } catch (err) {
1070     console.error('Ошибка переключения статуса ключа:', err);
1071     showModelsNotification('Ошибка изменения статуса: ' + err.
    message, 'danger');
1072   }
1073 }
1074
1075 async function resetKeyQuota(name, container) {
1076   try {
1077     await window.api.fetch(`/api/keys/${name}/reset-quota`, {
    method: 'POST' });
1078     showModelsNotification(`Квота для ключа "${name}" успешно
    сброшена`, 'success');
1079     await refreshKeysList(container);
1080   } catch (err) {
1081     console.error('Ошибка сброса квоты:', err);
1082     showModelsNotification('Ошибка сброса квоты: ' + err.messa
    ge, 'danger');
1083   }
1084 }
1085
1086 async function deleteKey(name, container) {
1087   if (!confirm(`Вы уверены, что хотите удалить ключ "${name}"?`
    `)) return;
1088   try {
1089     await window.api.fetch(`/api/keys/${name}`, { method: 'DEL
```

```

    ETE' });
1090     showModelsNotification(`Ключ "${name}" успешно удален`, 's
    uccess');
1091     await refreshKeysList(container);
1092   } catch (err) {
1093     console.error('Ошибка удаления ключа:', err);
1094     showModelsNotification('Ошибка удаления: ' + err.message,
    'danger');
1095   }
1096 }
1097
1098 function showModelsNotification(message, type = 'info') {
1099   const notification = document.createElement('div');
1100   notification.className = `alert alert-${type} position-fixed
    top-0 end-0 m-3`;
1101   notification.style.zIndex = '9999';
1102   notification.style.maxWidth = '400px';
1103   notification.textContent = message;
1104   document.body.appendChild(notification);
1105
1106   setTimeout(() => {
1107     notification.remove();
1108   }, 5000);
1109 }
1110
1111 async function loadFoundryConfig() {
1112   try {
1113     const config = await window.api.fetch('/api/foundry/config
    ');
1114     const enabledInput = document.getElementById('foundry-enab
    led');
1115     const urlInput = document.getElementById('foundry-url');
1116     const keyInput = document.getElementById('foundry-key');
1117     const modelInput = document.getElementById('foundry-model'
    );
1118
1119     if (enabledInput) enabledInput.checked = config.enabled ||
    false;

```

```
1120     if (urlInput) urlInput.value = config.url || '';
1121     if (keyInput) keyInput.value = config.key || '';
1122     if (modelInput) modelInput.value = config.model || '';
1123   } catch (err) {
1124     console.error('Ошибка загрузки настроек Foundry:', err);
1125   }
1126 }
1127
1128 async function loadOllamaConfig() {
1129   try {
1130     const config = await window.api.fetch('/api/ollama/config'
1131   );
1132     const enabledInput = document.getElementById('ollama-enabl
1133   ed');
1134     const urlInput = document.getElementById('ollama-url');
1135     const modelInput = document.getElementById('ollama-model')
1136   ;
1137
1138     if (enabledInput) enabledInput.checked = config.enabled ||
1139   false;
1140     if (urlInput) urlInput.value = config.url || '';
1141     if (modelInput) modelInput.value = config.model || '';
1142   } catch (err) {
1143     console.error('Ошибка загрузки настроек Ollama:', err);
1144   }
1145 }
1146
1147 async function loadAgyConfig() {
1148   try {
1149     const modelSelect = document.getElementById('agy-model');
1150     if (modelSelect) {
1151       try {
1152         const modelsData = await window.api.fetch('/api/chat/m
1153   odel
1154   s');
1155         const agyList = modelsData.models?.agy || [];
1156         if (agyList.length > 0) {
1157           const curVal = modelSelect.value;
1158           modelSelect.innerHTML = '';
```

```

1153         agyList.forEach(m => {
1154             const opt = document.createElement('option');
1155             opt.value = m;
1156             opt.textContent = m;
1157             modelSelect.appendChild(opt);
1158         });
1159         if (curVal && agyList.includes(curVal)) {
1160             modelSelect.value = curVal;
1161         }
1162     }
1163 } catch (e) {
1164     console.error('Ошибка загрузки моделей AGY:', e);
1165 }
1166 }
1167
1168 const config = await window.api.fetch('/api/agy/config');
1169 const enabledInput = document.getElementById('agy-enabled'
1170 );
1171 const keyInput = document.getElementById('agy-key');
1172
1173 if (enabledInput) enabledInput.checked = config.enabled ??
1174 true;
1175 if (modelSelect && config.model) modelSelect.value = confi
1176 g.model;
1177 if (keyInput) keyInput.value = config.key || '';
1178 } catch (err) {
1179     console.error('Ошибка загрузки настроек Antigravity (AGY):
1180 ', err);
1181 }
1182 }
1183
1184 async function loadOnnxConfig() {
1185     try {
1186         const config = await window.api.fetch('/api/onnx/config');
1187         const enabledInput = document.getElementById('onnx-enabled
1188 ');
1189         const epSelect = document.getElementById('onnx-execution-p
1190 rovider');

```

```
1185     const dirInput = document.getElementById('onnx-models-dir'
1186     );
1187     const modelInput = document.getElementById('onnx-default-m
1188     odel');
1189     const precSelect = document.getElementById('onnx-olive-pre
1190     cision');
1191     if (enabledInput) enabledInput.checked = config.enabled ??
1192     true;
1193     if (epSelect && config.execution_provider) epSelect.value
1194     = config.execution_provider;
1195     if (dirInput && config.models_dir) dirInput.value = config
1196     .models_dir;
1197     if (modelInput && config.default_model) modelInput.value =
1198     config.default_model;
1199     if (precSelect && config.olive_precision) precSelect.value
1200     = config.olive_precision;
1201   } catch (err) {
1202     console.error('Ошибка загрузки настроек ONNX / Olive:', er
1203     r);
1204   }
1205 }
1206
1207 async function loadSystemInstruction() {
1208   const editor = document.getElementById('system-instruction-e
1209   ditor');
1210   const statusBadge = document.getElementById('system-instruct
1211   ion-status');
1212   if (!editor) return;
1213
1214   editor.disabled = true;
1215   if (statusBadge) {
1216     statusBadge.className = 'badge bg-warning text-dark';
1217     statusBadge.textContent = 'Загрузка...';
1218     statusBadge.style.removeProperty('display');
1219   }
1220
1221   try {
```

```
1212     const data = await window.api.fetch('/api/admin/system_instruction');
1213     editor.value = data.content || '';
1214     if (statusBadge) {
1215         statusBadge.className = 'badge bg-success';
1216         statusBadge.textContent = 'Загружено';
1217         setTimeout(() => {
1218             if (statusBadge) statusBadge.style.display = 'none';
1219         }, 2500);
1220     }
1221 } catch (err) {
1222     console.error('Ошибка загрузки системной инструкции:', err);
1223     if (statusBadge) {
1224         statusBadge.className = 'badge bg-danger';
1225         statusBadge.textContent = 'Ошибка';
1226         statusBadge.style.removeProperty('display');
1227     }
1228     showModelsNotification('Ошибка загрузки системной инструкции: ' + err.message, 'danger');
1229 } finally {
1230     editor.disabled = false;
1231 }
1232 }
1233
1234 async function saveSystemInstruction() {
1235     const editor = document.getElementById('system-instruction-editor');
1236     const saveBtn = document.getElementById('btn-save-instruction');
1237     const statusBadge = document.getElementById('system-instruction-status');
1238     if (!editor || !saveBtn) return;
1239
1240     const originalHtml = saveBtn.innerHTML;
1241     saveBtn.disabled = true;
1242     saveBtn.innerHTML = '<span class="spinner-border spinner-border-sm" role="status"></span> Сохранение...';
```

```

1243
1244     try {
1245         await window.api.fetch('/api/admin/system_instruction', {
1246             method: 'POST',
1247             headers: { 'Content-Type': 'application/json' },
1248             body: JSON.stringify({ content: editor.value })
1249         });
1250         showModelsNotification('☐ Системная инструкция успешно сох
ранена', 'success');
1251         if (statusBadge) {
1252             statusBadge.className = 'badge bg-success';
1253             statusBadge.textContent = 'Сохранено';
1254             statusBadge.style.removeProperty('display');
1255             setTimeout(() => {
1256                 if (statusBadge) statusBadge.style.display = 'none';
1257             }, 3000);
1258         }
1259     } catch (err) {
1260         console.error('Ошибка сохранения системной инструкции:', e
rr);
1261         if (statusBadge) {
1262             statusBadge.className = 'badge bg-danger';
1263             statusBadge.textContent = 'Ошибка сохранения';
1264             statusBadge.style.removeProperty('display');
1265         }
1266         showModelsNotification('Ошибка сохранения: ' + err.message
, 'danger');
1267     } finally {
1268         saveBtn.disabled = false;
1269         saveBtn.innerHTML = originalHtml;
1270     }
1271 }
1272
1273 // =====
=====
1274 // Google Workspace Multi-Account Pool Management
1275 // =====
=====

```



```

1276
1277 async function refreshGoogleAccountsList(container) {
1278     if (!container) return;
1279     try {
1280         container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted"><span class="spinner-border spinner-border-sm me-2"></span>Загрузка аккаунтов...</td></tr>';
1281         const data = await window.api.googleAccounts.list();
1282         const accounts = data.accounts || [];
1283
1284         if (accounts.length === 0) {
1285             container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Список аккаунтов пуст. Загрузите файл с credentials.json или service_account.json.</td></tr>';
1286             return;
1287         }
1288
1289         container.innerHTML = '';
1290         accounts.forEach(acc => {
1291             const row = document.createElement('tr');
1292
1293             // 1. Account Name + Default Badge + Email
1294             const tdName = document.createElement('td');
1295             tdName.innerHTML = `
1296                 <div class="d-flex align-items-center gap-2">
1297                     <strong>${escapeHtml(acc.name)}</strong>
1298                     ${acc.is_default ? '<span class="badge bg-warning text-dark"><i class="bi bi-star-fill"></i> Default</span>' : ''}
1299                 </div>
1300                 ${acc.email ? `<div class="small text-muted font-monospace">${escapeHtml(acc.email)}</div>` : ''}
1301             `;
1302             row.appendChild(tdName);
1303
1304             // 2. Type
1305             const tdType = document.createElement('td');
1306             const isOAuth = acc.type === 'oauth2';
1307             tdType.innerHTML = isOAuth

```

```
1308 |         ? '<span class="badge bg-primary-subtle text-primary b  
| order border-primary-subtle"><i class="bi bi-person me-1"></i>  
| OAuth 2.0</span>'  
1309 |         : '<span class="badge bg-info-subtle text-info border  
| border-info-subtle"><i class="bi bi-cpu me-1"></i>Service Acco  
| unt</span>';  
1310 |         row.appendChild(tdType);  
1311 |  
1312 |         // 3. Status / Quota  
1313 |         const tdStatus = document.createElement('td');  
1314 |         const isActive = acc.status === 'active';  
1315 |         if (isActive) {  
1316 |             tdStatus.innerHTML = '<span class="badge bg-success">A  
| ктивен</span>';  
1317 |         } else if (acc.status === 'exhausted') {  
1318 |             tdStatus.innerHTML = '<span class="badge bg-danger" ti  
| tle="Исчерпан суточный лимит квоты">Лимит</span>';  
1319 |         } else {  
1320 |             tdStatus.innerHTML = `<span class="badge bg-secondary"  
| >${escapeHtml(acc.status)}</span>`;  
1321 |         }  
1322 |         row.appendChild(tdStatus);  
1323 |  
1324 |         // 4. Token / Credentials Status  
1325 |         const tdToken = document.createElement('td');  
1326 |         if (acc.has_token || acc.type === 'service_account') {  
1327 |             tdToken.innerHTML = '<span class="badge bg-success-sub  
| tle text-success"><i class="bi bi-shield-check me-1"></i>Готов  
| </span>';  
1328 |         } else {  
1329 |             tdToken.innerHTML = '<span class="badge bg-warning-sub  
| tle text-warning"><i class="bi bi-key me-1"></i>Нужен токен</s  
| pan>';  
1330 |         }  
1331 |         row.appendChild(tdToken);  
1332 |  
1333 |         // 5. Actions  
1334 |         const tdActions = document.createElement('td');
```

```
1335 |         tdActions.className = 'text-end';
1336 |
1337 |         // Set Default button
1338 |         if (!acc.is_default) {
1339 |             const btnDefault = document.createElement('button');
1340 |             btnDefault.className = 'btn btn-xs btn-outline-warning
| btn-sm me-1';
1341 |             btnDefault.title = 'Сделать аккаунтом по умолчанию';
1342 |             btnDefault.innerHTML = '<i class="bi bi-star"></i>';
1343 |             btnDefault.onclick = () => setGoogleAccountDefault(acc
| .name, container);
1344 |             tdActions.appendChild(btnDefault);
1345 |         }
1346 |
1347 |         // Reset status button
1348 |         if (acc.status === 'exhausted') {
1349 |             const btnReset = document.createElement('button');
1350 |             btnReset.className = 'btn btn-xs btn-outline-info btn-
| sm me-1';
1351 |             btnReset.title = 'Сбросить статус исчерпания';
1352 |             btnReset.innerHTML = '<i class="bi bi-arrow-repeat"></
| i>';
1353 |             btnReset.onclick = () => resetGoogleAccountStatus(acc.
| name, container);
1354 |             tdActions.appendChild(btnReset);
1355 |         }
1356 |
1357 |         // Test button
1358 |         const btnTest = document.createElement('button');
1359 |         btnTest.className = 'btn btn-xs btn-outline-info btn-sm
| me-1';
1360 |         btnTest.title = 'Проверить доступ и авторизацию';
1361 |         btnTest.innerHTML = '<i class="bi bi-play-circle"></i> Т
| ест';
1362 |         btnTest.onclick = () => testGoogleAccount(acc.name);
1363 |         tdActions.appendChild(btnTest);
1364 |
1365 |         // Delete button
```

```
1366     const btnDelete = document.createElement('button');
1367     btnDelete.className = 'btn btn-xs btn-outline-danger btn
    -sm';
1368     btnDelete.title = 'Удалить аккаунт из пула';
1369     btnDelete.innerHTML = '<i class="bi bi-trash"></i>';
1370     btnDelete.onclick = () => deleteGoogleAccount(acc.name,
    container);
1371     tdActions.appendChild(btnDelete);
1372
1373     row.appendChild(tdActions);
1374     container.appendChild(row);
1375 });
1376
1377 } catch (err) {
1378     console.error('Ошибка загрузки аккаунтов Google Workspace:
    ', err);
1379     container.innerHTML = `<tr><td colspan="5" class="text-cen
    ter py-4 text-danger">Ошибка: ${escapeHtml(err.message)}</td><
    /tr>`;
1380 }
1381 }
1382
1383 async function handleAddGoogleAccount(container) {
1384     const nameInput = document.getElementById('new-gacc-name');
1385     const typeSelect = document.getElementById('new-gacc-type');
1386     const emailInput = document.getElementById('new-gacc-email')
    ;
1387     const fileInput = document.getElementById('new-gacc-file');
1388     const jsonTextarea = document.getElementById('new-gacc-json'
    );
1389     const defaultCheckbox = document.getElementById('new-gacc-de
    fault');
1390     const addBtn = document.getElementById('btn-add-gacc');
1391
1392     if (!nameInput || !addBtn) return;
1393
1394     const accountName = nameInput.value.trim();
1395     const accountType = typeSelect ? typeSelect.value : 'oauth2'
```

```
;
1396   const email = emailInput ? emailInput.value.trim() : '';
1397   const setAsDefault = defaultCheckbox ? defaultCheckbox.checked : false;
1398
1399   if (!accountName) {
1400     showModelsNotification('Введите имя аккаунта (например: work, personal)', 'warning');
1401     nameInput.focus();
1402     return;
1403   }
1404
1405   // Check if file is selected or JSON is pasted
1406   const file = fileInput && fileInput.files && fileInput.files[0];
1407   const jsonContent = jsonTextarea ? jsonTextarea.value.trim() : '';
1408
1409   if (!file && !jsonContent) {
1410     showModelsNotification('Загрузите файл credentials.json / service_account.json или вставьте JSON', 'warning');
1411     return;
1412   }
1413
1414   addBtn.disabled = true;
1415   const originalText = addBtn.innerHTML;
1416   addBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Сохранение...';
1417
1418   try {
1419     if (file) {
1420       const formData = new FormData();
1421       formData.append('account_name', accountName);
1422       formData.append('account_type', accountType);
1423       if (email) formData.append('email', email);
1424       formData.append('set_as_default', setAsDefault ? 'true' : 'false');
1425       formData.append('file', file);
```

```
1426 |
1427 |     await window.api.googleAccounts.upload(formData);
1428 | } else {
1429 |     let parsedJson = null;
1430 |     try {
1431 |         parsedJson = JSON.parse(jsonContent);
1432 |     } catch (e) {
1433 |         throw new Error('Некорректный JSON в поле учетных данн
1434 | ых');
1435 |     }
1436 |     await window.api.googleAccounts.create({
1437 |         account_name: accountName,
1438 |         account_type: accountType,
1439 |         email: email || undefined,
1440 |         credentials_dict: parsedJson,
1441 |         set_as_default: setAsDefault
1442 |     });
1443 | }
1444 |
1445 |     showModelsNotification(`Аккаунт Google "${accountName}" ус
1446 | пешно сохранен в пул`, 'success');
1447 |     nameInput.value = '';
1448 |     if (emailInput) emailInput.value = '';
1449 |     if (fileInput) fileInput.value = '';
1450 |     if (jsonTextarea) jsonTextarea.value = '';
1451 |     if (defaultCheckbox) defaultCheckbox.checked = false;
1452 |     if (container) await refreshGoogleAccountsList(container);
1453 | } catch (err) {
1454 |     console.error('Ошибка сохранения аккаунта Google:', err);
1455 |     showModelsNotification('Ошибка сохранения: ' + err.message
1456 | , 'danger');
1457 | } finally {
1458 |     addBtn.disabled = false;
1459 |     addBtn.innerHTML = originalText;
1460 | }
```

```
1461
1462 async function setGoogleAccountDefault(name, container) {
1463   try {
1464     await window.api.googleAccounts.setDefault(name);
1465     showModelsNotification(`Аккаунт "${name}" назначен по умол
чанию`, 'success');
1466     if (container) await refreshGoogleAccountsList(container);
1467   } catch (err) {
1468     console.error('Ошибка установки аккаунта по умолчанию:', e
rr);
1469     showModelsNotification('Ошибка: ' + err.message, 'danger')
;
1470   }
1471 }
1472
1473 async function resetGoogleAccountStatus(name, container) {
1474   try {
1475     await window.api.googleAccounts.resetStatus(name);
1476     showModelsNotification(`Статус аккаунта "${name}" сброшен
в активный`, 'success');
1477     if (container) await refreshGoogleAccountsList(container);
1478   } catch (err) {
1479     console.error('Ошибка сброса статуса аккаунта:', err);
1480     showModelsNotification('Ошибка сброса: ' + err.message, 'd
anger');
1481   }
1482 }
1483
1484 async function testGoogleAccount(name) {
1485   const testCard = document.getElementById('gaccount-test-card
');
1486   const testBody = document.getElementById('gaccount-test-body
');
1487
1488   if (testCard && testBody) {
1489     testCard.style.display = 'block';
1490     testBody.innerHTML = `<span class="spinner-border spinner-
border-sm me-2 text-info"></span>Проверка аутентификации для а
```

```

    ккаунта <strong>${escapeHtml(name)}</strong>...`;
1491   }
1492
1493   try {
1494     const res = await window.api.googleAccounts.test(name);
1495     if (testBody) {
1496       const isSuccess = res.status === 'success';
1497       const isWarning = res.status === 'warning';
1498       const badgeClass = isSuccess ? 'bg-success' : isWarning
1499       ? 'bg-warning text-dark' : 'bg-danger';
1500       testBody.innerHTML = `
1501         <div class="mb-2 d-flex align-items-center gap-2">
1502           <span class="badge ${badgeClass}">${res.status.toUpp
1503           erCase()}</span>
1504           <strong>${escapeHtml(name)}</strong>
1505           </div>
1506           <div class="mb-1">${escapeHtml(res.message || '')}</di
1507           v>
1508           ${res.scopes && res.scopes.length > 0 ? `<div class="t
1509           ext-muted mt-2"><strong>Доступные Scopes:</strong><br>${res.sc
1510           opes.map(s => '• ' + escapeHtml(s)).join('<br>')}</div>` : ''}
1511           `;
1512     }
1513   } catch (err) {
1514     if (testBody) {
1515       testBody.innerHTML = `<span class="text-danger"><i class
1516       ="bi bi-x-circle me-1"></i>Ошибка проверки: ${escapeHtml(err.m
1517       essage)}</span>`;
1518     }
1519   }
1520 }
1521
1522 async function deleteGoogleAccount(name, container) {
1523   if (!confirm(`Вы уверены, что хотите удалить аккаунт "${name}
1524   }" из пула Google Workspace?`)) return;
1525   try {
1526     await window.api.googleAccounts.delete(name);

```



```
1520 |     showModelsNotification(`Аккаунт "${name}" успешно удален`,
      |     'success');
1521 |     if (container) await refreshGoogleAccountsList(container);
1522 | } catch (err) {
1523 |     console.error('Ошибка удаления аккаунта Google:', err);
1524 |     showModelsNotification('Ошибка удаления: ' + err.message,
      |     'danger');
1525 | }
1526 | }
1527 |
1528 | // Export for tab loader
1529 | window.initModelsTab = initModelsTab;
1530 | window.refreshGoogleAccountsList = refreshGoogleAccountsList;
1531 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/network_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/network_tab/index.html"
17     class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/network_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-08 03:08:00
25 =====
26 =====
27 -->
28
29 <!-- Windows Network Terminal Tab (/apps/windows/network) -->
30 <style>
31   .net-container {
32     padding: 0.5rem 0.75rem;
33   }
34   .net-card {
35     background: #0f172a !important;
36     border: 1px solid #334155 !important;
37     border-radius: 8px;
```

```
33 | padding: 0.55rem 0.85rem;
34 | display: flex;
35 | flex-direction: column;
36 | justify-content: space-between;
37 | min-height: 64px;
38 | gap: 0.2rem;
39 | }
40 | .net-label {
41 |     font-size: 0.7rem;
42 |     text-transform: uppercase;
43 |     letter-spacing: 0.05em;
44 |     color: #94a3b8;
45 |     line-height: 1.1;
46 | }
47 | .net-value {
48 |     font-size: 1.25rem;
49 |     font-weight: 700;
50 |     color: #f8fafc;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
52 |     line-height: 1.15;
53 | }
54 | .net-panel {
55 |     background: #0f172a !important;
56 |     border: 1px solid #334155 !important;
57 |     border-radius: 8px;
58 |     overflow: hidden;
59 | }
60 | .net-panel-header {
61 |     background: #1e293b !important;
62 |     border-bottom: 1px solid #334155 !important;
63 |     padding: 0.5rem 0.85rem;
64 |     font-weight: 600;
65 |     font-size: 0.82rem;
66 |     color: #38bdf8;
67 |     display: flex;
68 |     align-items: center;
69 |     justify-content: space-between;
```

```
70 | }
71 | .net-table {
72 |     width: 100%;
73 |     font-size: 0.78rem;
74 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
75 |     border-collapse: collapse;
76 |     margin-bottom: 0;
77 | }
78 | .net-table th {
79 |     background: #1e293b;
80 |     padding: 6px 10px;
81 |     color: #94a3b8;
82 |     font-weight: 600;
83 |     border-bottom: 1px solid #334155;
84 |     position: sticky;
85 |     top: 0;
86 |     z-index: 5;
87 | }
88 | .net-table td {
89 |     padding: 5px 10px;
90 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
91 |     color: #cbd5e1;
92 | }
93 | .proto-badge {
94 |     display: inline-block;
95 |     padding: 1px 6px;
96 |     border-radius: 4px;
97 |     font-size: 0.7rem;
98 |     font-weight: 600;
99 | }
100 | .proto-tcp { background: rgba(59, 130, 246, 0.2); color: #60
    | a5fa; border: 1px solid #3b82f6; }
101 | .proto-udp { background: rgba(168, 85, 247, 0.2); color: #c0
    | 84fc; border: 1px solid #a855f7; }
102 | .proto-tls { background: rgba(16, 185, 129, 0.2); color: #34
    | d399; border: 1px solid #10b981; }
103 | .proto-http { background: rgba(245, 158, 11, 0.2); color: #f
```

```
    bbf24; border: 1px solid #f59e0b; }
104 | .proto-dns { background: rgba(6, 182, 212, 0.2); color: #22d
    3ee; border: 1px solid #06b6d4; }
105 | .proto-other { background: rgba(148, 163, 184, 0.2); color:
    #cbd5e1; border: 1px solid #64748b; }
106 |
107 | .status-badge-up { background: rgba(34, 197, 94, 0.15); colo
    r: #4ade80; border: 1px solid #22c55e; }
108 | .status-badge-down { background: rgba(239, 68, 68, 0.15); co
    lor: #f87171; border: 1px solid #ef4444; }
109 | .status-badge-listen { background: rgba(56, 189, 248, 0.15);
    color: #38bdf8; border: 1px solid #0284c7; }
110 | .status-badge-established { background: rgba(168, 85, 247, 0
    .15); color: #c084fc; border: 1px solid #9333ea; }
111 |
112 | .net-subnav .nav-link {
113 |     font-size: 0.8rem;
114 |     padding: 0.35rem 0.8rem;
115 |     color: #94a3b8;
116 |     border-radius: 6px;
117 |     border: 1px solid transparent;
118 | }
119 | .net-subnav .nav-link.active {
120 |     background: #1e293b;
121 |     color: #38bdf8;
122 |     border: 1px solid #334155;
123 |     font-weight: 600;
124 | }
125 |
126 | .speed-gauge-card {
127 |     background: linear-gradient(145deg, #0f172a 0%, #1e293b 10
    0%);
128 |     border: 1px solid #334155;
129 |     border-radius: 10px;
130 |     padding: 1rem;
131 |     text-align: center;
132 | }
133 | .speed-gauge-val {
```

```
134     font-size: 2.2rem;
135     font-weight: 800;
136     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
137     line-height: 1.1;
138 }
139 </style>
140
141 <div class="net-container">
142     <!-- Header Alert & Status Strip -->
143     <div class="card shadow-sm border-secondary-subtle py-1.5 px-
    -3 mb-2 d-flex justify-content-between align-items-center flex
    -wrap gap-2">
144         <div class="d-flex align-items-center gap-2 flex-wrap">
145             <span class="fs-5">❏ </span>
146             <span class="fw-bold text-white small" style="font-size:
    0.95rem;">Сетевой терминал Windows</span>
147             <span class="badge bg-dark border border-info text-info
    fw-normal" style="font-size: 0.72rem;">apps/windows/network</s
    pan>
148             <span class="text-secondary small d-none d-lg-inline" st
    yle="font-size: 0.76rem;">| Скорость интернета, адаптеры, порт
    ы, соединения и DPI</span>
149         </div>
150         <div class="d-flex align-items-center gap-2 flex-wrap">
151             <span class="badge rounded-pill bg-success-subtle text-s
    uccess border border-success px-2 py-1" style="font-size: 0.72
    rem;" id="net-engine-status">
152                 <i class="bi bi-check-circle-fill me-1"></i> Windows N
    et: Active
153             </span>
154             <span class="badge rounded-pill bg-secondary-subtle text
    -light border border-secondary px-2 py-1" style="font-size: 0.
    72rem;" id="net-tshark-status">
155                 TShark DPI: Проверка...
156             </span>
157             <button class="btn btn-xs btn-outline-info rounded-pill
    px-2.5 py-1" style="font-size: 0.75rem;" id="btn-refresh-netwo
```

```

rk-all">
158     <i class="bi bi-arrow-clockwise me-1"></i> Обновить
159     </button>
160     <button class="btn btn-xs btn-outline-warning rounded-pi
ll px-2 py-1" style="font-size: 0.75rem;" id="btn-network-conf
ig">
161         <i class="bi bi-gear-fill me-1"></i> Настройки
162     </button>
163 </div>
164 </div>
165
166 <!-- Top Metrics KPI Cards -->
167 <div class="row g-2 mb-3">
168     <div class="col-6 col-md-3">
169         <div class="net-card">
170             <span class="net-label">Сетевые адаптеры</span>
171             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
172                 <span class="net-value" id="net-kpi-adapters">--</sp
an>
173                 <span class="small text-success" style="font-size: 0
.68rem;" id="net-kpi-adapters-active">0 активных</span>
174             </div>
175         </div>
176     </div>
177     <div class="col-6 col-md-3">
178         <div class="net-card">
179             <span class="net-label">Активные соединения</span>
180             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
181                 <span class="net-value" id="net-kpi-conns">--</span>
182                 <span class="small text-info" style="font-size: 0.68
rem;" id="net-kpi-listening">0 слушающих</span>
183             </div>
184         </div>
185     </div>
186     <div class="col-6 col-md-3">
187         <div class="net-card">

```

```

188         <span class="net-label">Скорость Интернета</span>
189         <div class="d-flex align-items-baseline justify-content-between mt-1">
190             <span class="net-value text-info" id="net-kpi-speed">-- Mbps</span>
191             <span class="small text-muted" style="font-size: 0.68rem;" id="net-kpi-ping">Ping: -- ms</span>
192         </div>
193     </div>
194 </div>
195 <div class="col-6 col-md-3">
196     <div class="net-card">
197         <span class="net-label">DPI & Безопасность</span>
198         <div class="d-flex align-items-baseline justify-content-between mt-1">
199             <span class="net-value" id="net-kpi-packets" style="color: #38bdf8;">0</span>
200             <span class="small" style="font-size: 0.68rem; color: #4ade80;" id="net-kpi-anomalies">0 аномалий</span>
201         </div>
202     </div>
203 </div>
204 </div>
205
206 <!-- Sub-Navigation Bar -->
207 <ul class="nav net-subnav mb-3 gap-1">
208     <li class="nav-item">
209         <button class="nav-link active" data-bs-toggle="tab" data-bs-target="#tab-net-speedtest">
210             <i class="bi bi-speedometer2 me-1 text-warning"></i> Тест скорости Интернета
211         </button>
212     </li>
213     <li class="nav-item">
214         <button class="nav-link" data-bs-toggle="tab" data-bs-target="#tab-net-adapters">
215             <i class="bi bi-hdd-network me-1"></i> Сетевые адаптеры

```



```

216         </button>
217     </li>
218     <li class="nav-item">
219         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
220 rget="#tab-net-connections">
221             <i class="bi bi-diagram-3 me-1"></i> Соединения и порт
222 ы (<span id="badge-conn-count">0</span>)
223         </button>
224     </li>
225     <li class="nav-item">
226         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
227 rget="#tab-net-capture">
228             <i class="bi bi-broadcast-pin me-1"></i> Захват пакетов
229 в (Live DPI)
230         </button>
231     </li>
232     <li class="nav-item">
233         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
234 rget="#tab-net-pcap">
235             <i class="bi bi-file-earmark-binary me-1"></i> Анализ
236 PCAP & AI
237         </button>
238     </li>
239 </ul>
240 <!-- Tab Content -->
241 <div class="tab-content">
242
243     <!-- SECTION 0: Speedtest & Latency Benchmark -->
244     <div class="tab-pane fade show active" id="tab-net-speedte
245 st">
246
247         <!-- Speedtest Action Control Box -->
248         <div class="net-panel mb-3">
249             <div class="net-panel-header">
250                 <span><i class="bi bi-speedometer2 text-warning me-1
251 "></i> Тестирование пропускной способности и задержки Интернет
252 а</span>
253
254                 <div class="d-flex align-items-center gap-2">

```

```

245 |         <span class="small text-muted" id="speedtest-status-text">Готов к тестированию</span>
246 |         <button class="btn btn-sm btn-warning rounded-pill px-3 fw-bold shadow-sm" id="btn-run-speedtest">
247 |             <i class="bi bi-play-fill me-1"></i> Запустить тест скорости
248 |         </button>
249 |     </div>
250 | </div>
251 | <div class="p-3">
252 |     <!-- 4 Big Gauges -->
253 |     <div class="row g-3 mb-3">
254 |         <div class="col-6 col-lg-3">
255 |             <div class="speed-gauge-card">
256 |                 <div class="text-secondary small text-uppercase fw-semibold mb-1"><i class="bi bi-arrow-down-circle-fill text-success me-1"></i> Скачивание (Download)</div>
257 |                 <div class="speed-gauge-val text-success" id="speed-val-download">--</div>
258 |                 <div class="text-muted small mt-1" id="speed-sub-download">0.0 MB/s</div>
259 |             </div>
260 |         </div>
261 |         <div class="col-6 col-lg-3">
262 |             <div class="speed-gauge-card">
263 |                 <div class="text-secondary small text-uppercase fw-semibold mb-1"><i class="bi bi-arrow-up-circle-fill text-primary me-1"></i> Отдача (Upload)</div>
264 |                 <div class="speed-gauge-val text-primary" id="speed-val-upload">--</div>
265 |                 <div class="text-muted small mt-1" id="speed-sub-upload">0.0 MB/s</div>
266 |             </div>
267 |         </div>
268 |         <div class="col-6 col-lg-3">
269 |             <div class="speed-gauge-card">
270 |                 <div class="text-secondary small text-uppercase fw-semibold mb-1"><i class="bi bi-stopwatch-fill text-info m

```

```

e-1"></i> Задержка (Ping)</div>
271         <div class="speed-gauge-val text-info" id="spe
ed-val-ping">--</div>
272         <div class="text-muted small mt-1" id="speed-s
ub-jitter">Jitter: -- ms</div>
273         </div>
274     </div>
275     <div class="col-6 col-lg-3">
276         <div class="speed-gauge-card">
277             <div class="text-secondary small text-uppercas
e fw-semibold mb-1"><i class="bi bi-shield-check text-warning
me-1"></i> Качество соединения</div>
278             <div class="speed-gauge-val text-warning" id="
speed-val-grade">--</div>
279             <div class="text-muted small text-truncate mt-
1" id="speed-sub-isp">Провайдер: --</div>
280             </div>
281         </div>
282     </div>
283
284     <!-- Quality Summary Bar -->
285     <div class="p-2.5 rounded bg-dark border border-seco
ndary mb-3 d-flex justify-content-between align-items-center f
lex-wrap gap-2">
286         <div class="d-flex align-items-center gap-2">
287             <span class="badge bg-secondary" id="speed-badg
e-ip">Внешний IP: --</span>
288             <span class="badge bg-dark border border-secon
dary" id="speed-badg-location">Локация: --</span>
289         </div>
290         <div class="small text-muted" id="speed-summary-te
xt">Нажмите «Запустить тест скорости», чтобы измерить пропускн
ую способность канала, задержку (Ping) и джиттер.</div>
291     </div>
292
293     <!-- DNS & Edge Servers Latency Table -->
294     <h6 class="small fw-bold text-info mb-2"><i class="b
i bi-globe me-1"></i> Задержка к опорным DNS & CDN серверам:</

```

```

h6>
295     <div class="table-responsive">
296         <table class="net-table">
297             <thead>
298                 <tr>
299                     <th>Сервер / Провайдер</th>
300                     <th style="width: 140px;">Хост / IP</th>
301                     <th style="width: 110px; text-align: right;"
>Средний Ping</th>
302                     <th style="width: 90px; text-align: right;">
Min</th>
303                     <th style="width: 90px; text-align: right;">
Max</th>
304                     <th style="width: 100px; text-align: right;"
>Джиттер</th>
305                     <th style="width: 100px;">Статус</th>
306                 </tr>
307             </thead>
308             <tbody id="speed-servers-tbody">
309                 <tr><td colspan="7" class="text-center py-3 te
xt-muted">Запустите тест скорости для измерения задержки к DNS
-серверам.</td></tr>
310             </tbody>
311         </table>
312     </div>
313 </div>
314 </div>
315 </div>
316
317 <!-- SECTION 1: Network Adapters -->
318 <div class="tab-pane fade" id="tab-net-adapters">
319     <div class="net-panel">
320         <div class="net-panel-header">
321             <span><i class="bi bi-hdd-network me-1"></i> Физичес
кие и виртуальные сетевые интерфейсы Windows</span>
322             <button class="btn btn-xs btn-outline-info rounded-p
ill px-2 py-0" id="btn-refresh-adapters">Обновить адаптеры</bu
tton>

```

```

323 |         </div>
324 |         <div class="table-responsive" style="max-height: 480px
    | ; overflow-y: auto;">
325 |             <table class="net-table">
326 |                 <thead>
327 |                     <tr>
328 |                         <th style="width: 140px;">Адаптер</th>
329 |                         <th>Описание</th>
330 |                         <th style="width: 130px;">IPv4</th>
331 |                         <th style="width: 140px;">MAC адрес</th>
332 |                         <th style="width: 90px;">Скорость</th>
333 |                         <th style="width: 90px;">Статус</th>
334 |                         <th style="width: 130px; text-align: right;">О
    | правлено</th>
335 |                         <th style="width: 130px; text-align: right;">П
    | ринято</th>
336 |                     </tr>
337 |                 </thead>
338 |                 <tbody id="net-adapters-tbody">
339 |                     <tr><td colspan="8" class="text-center py-4 text
    | -muted"><div class="spinner-border spinner-border-sm text-info
    | me-2"></div>Загрузка сетевых адаптеров...</td></tr>
340 |                 </tbody>
341 |             </table>
342 |         </div>
343 |     </div>
344 | </div>
345 |
346 | <!-- SECTION 2: Active Connections & Listening Ports -->
347 | <div class="tab-pane fade" id="tab-net-connections">
348 |     <div class="net-panel">
349 |         <div class="net-panel-header">
350 |             <div class="d-flex align-items-center gap-2 flex-wra
    | p">
351 |                 <span><i class="bi bi-diagram-3 me-1"></i> Таблица
    | сетевых подключений и слушающих портов</span>
352 |                 <div class="btn-group btn-group-sm ms-2" role="gro
    | up">

```

```

353         <button type="button" class="btn btn-outline-sec
ondary active btn-filter-conns" data-filter="all">Все</button>
354         <button type="button" class="btn btn-outline-sec
ondary btn-filter-conns" data-filter="LISTEN">LISTEN</button>
355         <button type="button" class="btn btn-outline-sec
ondary btn-filter-conns" data-filter="ESTABLISHED">ESTABLISHED
</button>
356         <button type="button" class="btn btn-outline-sec
ondary btn-filter-conns" data-filter="UDP">UDP</button>
357     </div>
358 </div>
359 <div class="d-flex align-items-center gap-2">
360     <input type="text" class="form-control form-contro
l-sm bg-dark text-white border-secondary" id="net-conns-filter
" placeholder="Поиск (IP, порт, процесс, PID)..." style="max-w
idth: 250px;">
361     <button class="btn btn-xs btn-outline-info rounded
-pill px-2 py-0" id="btn-refresh-conns">Обновить</button>
362 </div>
363 </div>
364 <div class="table-responsive" style="max-height: 480px
; overflow-y: auto;">
365     <table class="net-table">
366         <thead>
367             <tr>
368                 <th style="width: 70px;">Протокол</th>
369                 <th style="width: 180px;">Локальный адрес</th>
370                 <th style="width: 180px;">Удалённый адрес</th>
371                 <th style="width: 110px;">Статус</th>
372                 <th style="width: 80px;">PID</th>
373                 <th>Процесс / Исполняемый файл</th>
374             </tr>
375         </thead>
376         <tbody id="net-conns-tbody">
377             <tr><td colspan="6" class="text-center py-4 text
-muted"><div class="spinner-border spinner-border-sm text-info
me-2"></div>Загрузка соединений...</td></tr>
378         </tbody>

```

```

379         </table>
380     </div>
381 </div>
382 </div>
383
384 <!-- SECTION 3: Packet Capture & Live DPI -->
385 <div class="tab-pane fade" id="tab-net-capture">
386     <!-- Capture Control Box -->
387     <div class="net-panel mb-3">
388         <div class="net-panel-header">
389             <span><i class="bi bi-broadcast me-1"></i> Настройка
сессии захвата трафика (Live Packet DPI)</span>
390         </div>
391         <div class="p-3">
392             <div class="row g-2 mb-2 align-items-end">
393                 <div class="col-md-5">
394                     <label class="form-label small fw-semibold text-
light mb-1">Сетевой интерфейс для захвата:</label>
395                     <select class="form-select form-select-sm bg-dar
k text-white border-secondary" id="net-interface-select">
396                         <option value="">Загрузка интерфейсов...</opti
on>
397                     </select>
398                 </div>
399                 <div class="col-md-4">
400                     <label class="form-label small fw-semibold text-
light mb-1">Wireshark/BPF фильтр:</label>
401                     <input type="text" class="form-control form-cont
rol-sm bg-dark text-white border-secondary" id="net-capture-fi
lter" placeholder="tcp, udp, port 80, tls">
402                 </div>
403                 <div class="col-md-3 d-flex gap-2">
404                     <button class="btn btn-sm btn-primary rounded-pi
ll px-3 flex-fill" id="btn-start-live-capture">
405                         <i class="bi bi-play-fill me-1"></i> Старт зах
вата
406                     </button>
407                     <button class="btn btn-sm btn-outline-danger rou

```

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    nded-pill px-3 flex-fill" id="btn-stop-live-capture" disabled>
408         <i class="bi bi-stop-fill me-1"></i> Стоп
409     </button>
410 </div>
411 </div>
412 </div>
413 </div>
414
415 <!-- Packets Table -->
416 <div class="net-panel">
417     <div class="net-panel-header">
418         <div class="d-flex align-items-center gap-2">
419             <span><i class="bi bi-reception-4 me-1"></i> Поток
сетевых пакетов (Packet Inspection)</span>
420             <span class="badge bg-secondary" id="net-captured-
badge">0 пакетов</span>
421         </div>
422         <div class="d-flex align-items-center gap-2">
423             <input type="text" class="form-control form-contro
l-sm bg-dark text-white border-secondary" id="net-table-filter
" placeholder="Поиск по IP / протоколу..." style="max-width: 2
20px;">
424             <button class="btn btn-xs btn-outline-secondary ro
unded-pill px-2 py-0" id="btn-clear-packets">Очистить</button>
425         </div>
426     </div>
427     <div class="table-responsive" style="max-height: 400px
; overflow-y: auto;">
428         <table class="net-table">
429             <thead>
430                 <tr>
431                     <th style="width: 50px;">#</th>
432                     <th style="width: 80px;">Время</th>
433                     <th style="width: 140px;">Источник</th>
434                     <th style="width: 140px;">Назначение</th>
435                     <th style="width: 80px;">Протокол</th>
436                     <th style="width: 70px; text-align: right;">Па
змер</th>

```



```

437         <th>Сведения о полезной нагрузке / Инфо</th>
438     </tr>
439 </thead>
440     <tbody id="net-packets-tbody">
441         <tr><td colspan="7" class="text-center py-4 text
| -muted">Нет захваченных пакетов. Выберите интерфейс и нажмите
| "Старт захвата".</td></tr>
442     </tbody>
443 </table>
444 </div>
445 </div>
446 </div>
447
448 <!-- SECTION 4: PCAP Analysis & AI Diagnostics -->
449 <div class="tab-pane fade" id="tab-net-pcap">
450     <div class="row g-3">
451         <div class="col-lg-5">
452             <div class="net-panel h-100">
453                 <div class="net-panel-header">
454                     <span><i class="bi bi-file-earmark-binary me-1">
| </i> Загрузка файла сетевого дампа (.PCAP)</span>
455                 </div>
456                 <div class="p-3">
457                     <div class="mb-3">
458                         <label class="form-label small fw-semibold tex
| t-light">Выберите .pcap, .pcapng или .cap файл:</label>
459                         <input type="file" class="form-control form-co
| ntrol-sm bg-dark text-white border-secondary" id="net-pcap-fil
| e" accept=".pcap,.pcapng,.cap">
460                     </div>
461                     <div class="d-flex justify-content-between align
| -items-center mb-3">
462                         <span class="small text-muted" id="net-upload-
| status">Файл не выбран</span>
463                         <button class="btn btn-sm btn-info rounded-pil
| l px-3 fw-bold" id="btn-analyze-pcap">
464                             <i class="bi bi-cpu me-1"></i> Запустить AI
| Анализ

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```

465 |         </button>
466 |     </div>
467 |     <div class="p-2.5 rounded bg-dark border border-
secondary small text-muted" style="font-size: 0.75rem;">
468 |         <i class="bi bi-info-circle text-info me-1"></
i> PCAP-анализатор проводит глубокую инспекцию заголовков, эвр
истический аудит портов и выявляет аномалии с помощью AI-модел
и.
469 |     </div>
470 | </div>
471 | </div>
472 | </div>
473 |
474 |     <div class="col-lg-7">
475 |         <div class="net-panel h-100">
476 |             <div class="net-panel-header">
477 |                 <span><i class="bi bi-shield-check me-1"></i> Ре
зультаты AI-анализа и безопасности</span>
478 |             </div>
479 |             <div class="p-3">
480 |                 <div class="row g-2 mb-3">
481 |                     <div class="col-6">
482 |                         <div class="p-2 rounded bg-dark border borde
r-secondary">
483 |                             <span class="net-label">AI Health Score</s
pan>
484 |                             <div class="net-value mt-1" id="pcap-healt
h-score">100/100</div>
485 |                         </div>
486 |                     </div>
487 |                     <div class="col-6">
488 |                         <div class="p-2 rounded bg-dark border borde
r-secondary">
489 |                             <span class="net-label">Обнаруженные анома
лии</span>
490 |                             <div class="net-value mt-1 text-success" i
d="pcap-anomalies-count">0</div>
491 |                         </div>

```

```
492 |         </div>
493 |     </div>
494 |
495 |         <h6 class="small fw-bold text-info mb-1.5">Эврис
| тические предупреждения:</h6>
496 |         <div id="pcap-heuristics-list" class="p-2 rounde
| d bg-dark border border-secondary mb-3 small" style="min-heigh
| t: 60px; max-height: 120px; overflow-y: auto;">
497 |             <span class="text-muted">Нет предупреждений. З
| апустите анализ файла.</span>
498 |         </div>
499 |
500 |         <h6 class="small fw-bold text-info mb-1.5">Реком
| ендации AI:</h6>
501 |         <div id="pcap-ai-recommendations" class="p-2 rou
| nded bg-dark border border-secondary small" style="min-height:
| 60px; max-height: 120px; overflow-y: auto;">
502 |             <span class="text-muted">Анализ пока не произв
| одился.</span>
503 |         </div>
504 |     </div>
505 | </div>
506 | </div>
507 | </div>
508 | </div>
509 |
510 | </div>
511 | </div>
512 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Network Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/network_tab/main.js?v=
11  *       20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/network_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21  // =====
22  // Process Name: Windows Network Terminal Tab Client Module
23  // =====
24  // Description:
25  //   Frontend controller for /apps/windows/network.
26  //   Manages internet speedtest, ping/latency benchmark, netwo
27  //   rk adapters,
28  //   active connections, listening ports, live packet capture
29  //   (DPI), PCAP upload,
30  //   and AI diagnostics.
```

```

29 // =====
   =====
30
31 (function() {
32     let isNetInitialized = false;
33     let netEventSource = null;
34     let capturedPackets = [];
35     let allConnections = [];
36     let allAdapters = [];
37     let activeConnFilter = 'all';
38     let isSpeedtestRunning = false;
39
40     // Format bytes to human readable format
41     function formatBytes(bytes, decimals = 1) {
42         if (!bytes || bytes === 0) return '0 B';
43         const k = 1024;
44         const dm = decimals < 0 ? 0 : decimals;
45         const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
46         const i = Math.floor(Math.log(bytes) / Math.log(k));
47         return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
48             ' ' + sizes[i];
49     }
50     // Check overall network status and TShark availability
51     async function fetchNetworkStatus() {
52         const engineBadge = document.getElementById('net-engine-st
53             atus');
54         const tsharkBadge = document.getElementById('net-tshark-st
55             atus');
56         const kpiAdapters = document.getElementById('net-kpi-adapt
57             ers');
58         const kpiAdaptersActive = document.getElementById('net-kpi
59             -adapters-active');
60         const kpiConns = document.getElementById('net-kpi-conns');
61         const kpiListening = document.getElementById('net-kpi-list
62             ening');
63         const kpiPackets = document.getElementById('net-kpi-packet
64             s');

```

```
59 |     const kpiAnomalies = document.getElementById('net-kpi-anom
    | alies');
60 |
61 |     try {
62 |         const res = await fetch('/api/network/status');
63 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
64 |         const data = await res.json();
65 |
66 |         if (engineBadge) {
67 |             engineBadge.className = 'badge rounded-pill bg-success
    | -subtle text-success border border-success px-2 py-1';
68 |             engineBadge.innerHTML = '<i class="bi bi-check-circle-
    | fill me-1"></i> Windows Net: Active';
69 |         }
70 |
71 |         if (tsharkBadge) {
72 |             if (data.tshark_available) {
73 |                 tsharkBadge.className = 'badge rounded-pill bg-succe
    | ss-subtle text-success border border-success px-2 py-1';
74 |                 tsharkBadge.innerText = 'TShark DPI: Доступен';
75 |             } else {
76 |                 tsharkBadge.className = 'badge rounded-pill bg-warni
    | ng-subtle text-warning border border-warning px-2 py-1';
77 |                 tsharkBadge.innerText = 'TShark DPI: Не установлен';
78 |             }
79 |         }
80 |
81 |         if (kpiAdapters) kpiAdapters.innerText = data.total_adap
    | ters ?? '--';
82 |         if (kpiConns) kpiConns.innerText = data.active_connectio
    | ns_count ?? '--';
83 |         if (kpiListening) kpiListening.innerText = `${data.liste
    | ning_ports_count || 0} слушающих`;
84 |         if (kpiPackets) kpiPackets.innerText = data.total_packet
    | s ?? capturedPackets.length;
85 |
86 |         const anomaliesCount = (data.latest_heuristics?.length |
    | | 0) + (data.latest_ai_report?.anomalies?.length || 0);
```

```

87         if (kpiAnomalies) {
88             kpiAnomalies.innerText = `${anomaliesCount} аномалий`;
89             kpiAnomalies.style.color = anomaliesCount > 0 ? '#f871
71' : '#4ade80';
90         }
91     } catch (e) {
92         console.warn('[NetworkTab] Status check error:', e);
93         if (engineBadge) {
94             engineBadge.className = 'badge rounded-pill bg-danger-
subtle text-danger border border-danger px-2 py-1';
95             engineBadge.innerText = 'Windows Net: Ошибка';
96         }
97     }
98 }
99
100 // Load latest or cached speedtest
101 async function fetchLatestSpeedtest() {
102     try {
103         const res = await fetch('/api/network/speedtest/latest')
;
104         if (!res.ok) return;
105         const data = await res.json();
106         if (data && data.download) {
107             renderSpeedtestResults(data);
108         }
109     } catch (e) {
110         console.debug('[NetworkTab] No latest speedtest data:',
e);
111     }
112 }
113
114 // Run full internet speedtest
115 async function runInternetSpeedtest() {
116     if (isSpeedtestRunning) return;
117     isSpeedtestRunning = true;
118
119     const btn = document.getElementById('btn-run-speedtest');
120     const statusText = document.getElementById('speedtest-stat

```

```

us-text');
121     const summaryText = document.getElementById('speed-summary
    -text');
122
123     if (btn) {
124         btn.disabled = true;
125         btn.innerHTML = '<span class="spinner-border spinner-bor
    der-sm me-1"></span> Тестирование...';
126     }
127     if (statusText) statusText.innerText = 'Измерение Ping, Do
    wnload и Upload...';
128     if (summaryText) summaryText.innerText = 'Выполняется подк
    лючение к узлам Cloudflare & DNS серверам для замера скорости.
    ...';
129
130     try {
131         const res = await fetch('/api/network/speedtest/run', {
    method: 'POST' });
132         if (!res.ok) throw new Error(`HTTP ${res.status}`);
133         const data = await res.json();
134
135         renderSpeedtestResults(data);
136         if (statusText) statusText.innerText = `Завершено в ${ne
    w Date().toLocaleTimeString()}`;
137     } catch (e) {
138         console.error('[NetworkTab] Speedtest error:', e);
139         window.showToast?.('Ошибка при выполнении теста скорости
    : ' + e.message, 'danger') || alert('Ошибка при выполнении тес
    та скорости: ' + e.message);
140         if (statusText) statusText.innerText = 'Ошибка теста ско
    рости';
141     } finally {
142         isSpeedtestRunning = false;
143         if (btn) {
144             btn.disabled = false;
145             btn.innerHTML = '<i class="bi bi-play-fill me-1"></i>
    Запустить тест скорости';
146         }

```



```
147     }
148 }
149
150 // Render speedtest report to UI
151 function renderSpeedtestResults(data) {
152     const valDown = document.getElementById('speed-val-downloa
d');
153     const subDown = document.getElementById('speed-sub-downloa
d');
154     const valUp = document.getElementById('speed-val-upload');
155     const subUp = document.getElementById('speed-sub-upload');
156     const valPing = document.getElementById('speed-val-ping');
157     const subJitter = document.getElementById('speed-sub-jitte
r');
158     const valGrade = document.getElementById('speed-val-grade'
);
159     const subIsp = document.getElementById('speed-sub-isp');
160     const badgeIp = document.getElementById('speed-badge-ip');
161     const badgeLoc = document.getElementById('speed-badge-loca
tion');
162     const summaryText = document.getElementById('speed-summary
-text');
163     const kpiSpeed = document.getElementById('net-kpi-speed');
164     const kpiPing = document.getElementById('net-kpi-ping');
165     const serversTbody = document.getElementById('speed-server
s-tbody');
166
167     const downMbps = data.download?.speed_mbps || 0;
168     const downMBs = data.download?.speed_mb_s || 0;
169     const upMbps = data.upload?.speed_mbps || 0;
170     const upMBs = data.upload?.speed_mb_s || 0;
171     const pingMs = data.ping_ms || 0;
172     const jitterMs = data.jitter_ms || 0;
173     const meta = data.meta || {};
174     const quality = data.quality || {};
175
176     if (valDown) valDown.innerHTML = `${downMbps} Mbps`;
177     if (subDown) subDown.innerHTML = `${downMBs} MB/s (${data.
```

```

download?.duration_s || 0}s)`;
178     if (valUp) valUp.innerText = `${upMbps} Mbps`;
179     if (subUp) subUp.innerText = `${upMBs} MB/s (${data.upload
?.duration_s || 0}s)`;
180     if (valPing) valPing.innerText = `${pingMs} ms`;
181     if (subJitter) subJitter.innerText = `Jitter: ${jitterMs}
ms`;
182
183     if (valGrade) {
184         valGrade.innerText = quality.rating || '--';
185         valGrade.style.color = quality.color || '#fbbf24';
186     }
187     if (subIsp) subIsp.innerText = `Провайдер: ${meta.isp || '
Неизвестно'}`;
188
189     if (badgeIp) badgeIp.innerText = `Внешний IP: ${meta.ip ||
'--'}`;
190     if (badgeLoc) badgeLoc.innerText = `Локация: ${meta.city |
| ''} ${meta.country || ''} (${meta.colo || 'Edge'})`.trim();
191     if (summaryText) summaryText.innerText = `${quality.grade
|| 'Готово'}. ${quality.summary || ''}`;
192
193     // Top KPI cards
194     if (kpiSpeed) kpiSpeed.innerText = `${downMbps} Mbps`;
195     if (kpiPing) kpiPing.innerText = `Ping: ${pingMs} ms`;
196
197     // DNS & Edge servers table
198     if (serversTbody && Array.isArray(data.servers)) {
199         serversTbody.innerHTML = data.servers.map(s => {
200             const isOnline = s.avg_ms > 0;
201             const statusBadge = isOnline
202                 ? '<span class="proto-badgе status-badgе-up">Online<
/span>'
203                 : '<span class="proto-badgе status-badgе-down">Offli
ne</span>';
204
205             return `
206                 <tr>

```

```

207         <td class="fw-bold text-white"><i class="bi bi-hdd
| -network me-1 text-info"></i>${s.name}</td>
208         <td class="text-secondary font-monospace">${s.host
| }</td>
209         <td style="text-align: right;" class="text-info fw
| -bold font-monospace">${isOnline ? `${s.avg_ms} ms` : '--'}</t
| d>
210         <td style="text-align: right;" class="text-muted f
| ont-monospace">${isOnline ? `${s.min_ms} ms` : '--'}</td>
211         <td style="text-align: right;" class="text-muted f
| ont-monospace">${isOnline ? `${s.max_ms} ms` : '--'}</td>
212         <td style="text-align: right;" class="text-warning
| font-monospace">${isOnline ? `${s.jitter_ms} ms` : '--'}</td>
213         <td>${statusBadge}</td>
214     </tr>
215     `;
216     })).join('');
217 }
218 }
219
220 // Fetch host network adapters
221 async function fetchAdapters() {
222     const tbody = document.getElementById('net-adapters-tbody'
| );
223     const ifaceSelect = document.getElementById('net-interface
| -select');
224     const kpiActive = document.getElementById('net-kpi-adapter
| s-active');
225
226     try {
227         const res = await fetch('/api/network/interfaces');
228         if (!res.ok) throw new Error(`HTTP ${res.status}`);
229         const adapters = await res.json();
230         allAdapters = Array.isArray(adapters) ? adapters : [];
231
232         let activeCount = 0;
233
234         if (ifaceSelect) {

```

```

235 |         ifaceSelect.innerHTML = allAdapters.map(a => `
236 |             <option value="${a.name || a.id}">${a.name} (${a.des
| cription || 'Ethernet'}) - ${a.status}</option>
237 |         `).join('');
238 |     }
239 |
240 |     if (tbody) {
241 |         if (allAdapters.length === 0) {
242 |             tbody.innerHTML = '<tr><td colspan="8" class="text-c
| enter py-4 text-muted">Сетевые адаптеры не обнаружены.</td></t
| r>';
243 |             return;
244 |         }
245 |
246 |         tbody.innerHTML = allAdapters.map(a => {
247 |             if (a.is_up) activeCount++;
248 |             const statusClass = a.is_up ? 'status-badge-up' : 's
| tatus-badge-down';
249 |             const speedDisplay = a.speed_mbps ? `${a.speed_mbps}
| Mbps` : '--';
250 |             const ipDisplay = a.ipv4 && a.ipv4.length > 0 ? a.ip
| v4.join(', ') : '--';
251 |
252 |             return `
253 |                 <tr>
254 |                     <td class="fw-bold text-white"><i class="bi bi-h
| dd-network me-1 text-info"></i>${a.name}</td>
255 |                     <td class="text-secondary small text-truncate" s
| tyle="max-width: 220px;" title="${a.description}">${a.descript
| ion || '--'}</td>
256 |                     <td class="text-info">${ipDisplay}</td>
257 |                     <td class="text-muted small">${a.mac || '--'}</t
| d>
258 |                     <td><span class="badge bg-dark border border-sec
| ondary text-light">${speedDisplay}</span></td>
259 |                     <td><span class="proto-badge ${statusClass}">${a
| .status || 'Unknown'}</span></td>
260 |                     <td style="text-align: right;" class="text-succe

```

```

ss">${formatBytes(a.bytes_sent)}</td>
261 |         <td style="text-align: right;" class="text-prima
    | ry">${formatBytes(a.bytes_recv)}</td>
262 |     </tr>
263 |     `;
264 |     }).join('');
265 | }
266 |
267 |     if (kpiActive) kpiActive.innerText = `${activeCount} акт
    | ИВНЫХ`;
268 | } catch (e) {
269 |     console.error('[NetworkTab] Failed to fetch adapters:',
    | e);
270 |     if (tbody) tbody.innerHTML = `<tr><td colspan="8" class=
    | "text-center py-3 text-danger">Ошибка загрузки адаптеров: ${e.
    | message}</td></tr>`;
271 | }
272 | }
273 |
274 | // Fetch active connections and listening ports
275 | async function fetchConnections() {
276 |     const tbody = document.getElementById('net-conns-tbody');
277 |     const badgeConn = document.getElementById('badge-conn-coun
    | t');
278 |     const kpiConns = document.getElementById('net-kpi-conns');
279 |     const kpiListening = document.getElementById('net-kpi-list
    | ening');
280 |
281 |     try {
282 |         const res = await fetch('/api/network/connections?limit=
    | 250');
283 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
284 |         const data = await res.json();
285 |
286 |         const conns = Array.isArray(data.connections) ? data.con
    | nections : [];
287 |         const listening = Array.isArray(data.listening_ports) ?
    | data.listening_ports : [];

```

```
288 |         allConnections = [...listening, ...conns];
289 |
290 |         if (badgeConn) badgeConn.innerText = allConnections.leng
| th;
291 |         if (kpiConns) kpiConns.innerText = conns.length;
292 |         if (kpiListening) kpiListening.innerText = `${listening.
| length} слушающих`;
293 |
294 |         renderConnectionsTable();
295 |     } catch (e) {
296 |         console.error('[NetworkTab] Failed to fetch connections:
| ', e);
297 |         if (tbody) tbody.innerHTML = `<tr><td colspan="6" class=
| "text-center py-3 text-danger">Ошибка загрузки соединений: ${e
| .message}</td></tr>`;
298 |     }
299 | }
300 |
301 | // Render connections table with current filter and search
302 | function renderConnectionsTable() {
303 |     const tbody = document.getElementById('net-conns-tbody');
304 |     const searchInput = document.getElementById('net-conns-fil
| ter');
305 |     const query = (searchInput?.value || '').toLowerCase().tri
| m();
306 |
307 |     if (!tbody) return;
308 |
309 |     let filtered = allConnections.filter(c => {
310 |         // Type filter
311 |         if (activeConnFilter === 'LISTEN' && c.status !== 'LISTE
| N') return false;
312 |         if (activeConnFilter === 'ESTABLISHED' && c.status !== '
| ESTABLISHED') return false;
313 |         if (activeConnFilter === 'UDP' && c.protocol !== 'UDP')
| return false;
314 |
315 |         // Text query
```

```
316         if (query) {
317             const text = `${c.protocol} ${c.local_address} ${c.remote_address} ${c.status} ${c.pid} ${c.process_name}`.toLowerCase();
318             return text.includes(query);
319         }
320         return true;
321     });
322
323     if (filtered.length === 0) {
324         tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Соединения, удовлетворяющие критериям, не найдены.</td></tr>';
325         return;
326     }
327
328     tbody.innerHTML = filtered.slice(0, 150).map((c, idx) => {
329         const protoClass = c.protocol === 'TCP' ? 'proto-tcp' : (c.protocol === 'UDP' ? 'proto-udp' : 'proto-other');
330         let statusClass = 'proto-other';
331         if (c.status === 'LISTEN') statusClass = 'status-badge-listen';
332         else if (c.status === 'ESTABLISHED') statusClass = 'status-badge-established';
333
334         return `
335             <tr class="net-conn-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для детальной инспекции процесса и сокета">
336                 <td><span class="proto-badge ${protoClass}">${c.protocol} || 'TCP'</span></td>
337                 <td class="text-info font-monospace">${c.local_address} || '-'</td>
338                 <td class="text-secondary font-monospace">${c.remote_address} || '-'</td>
339                 <td><span class="proto-badge ${statusClass}">${c.status} || '-'</span></td>
340                 <td class="text-muted font-monospace">${c.pid} || '-'
```

```

    }</td>
341 |         <td class="fw-bold text-white"><i class="bi bi-app m
    e-1 text-muted"></i>${c.process_name || 'Неизвестно'}</td>
342 |     </tr>
343 |     `;
344 |     }).join('');
345 |
346 |     // Modal inspection on row click
347 |     tbody.querySelectorAll('.net-conn-row').forEach(row => {
348 |         row.onclick = () => {
349 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
350 |             ;
351 |             const item = filtered[idx];
352 |             if (!item || !window.AITableModal) return;
353 |             window.AITableModal.show({
354 |                 icon: '🔌',
355 |                 title: `Сетевое подключение [${item.protocol || 'TCP
    '}]`,
356 |                 subtitle: `${item.process_name} (PID: ${item.pid ||
    'N/A'})`,
357 |                 tableType: 'network',
358 |                 badges: [
359 |                     { text: item.protocol || 'TCP', class: 'badge bg-p
    rimary' },
360 |                     { text: item.status || 'UNKNOWN', class: 'badge bg
    -info' },
361 |                     { text: `PID ${item.pid || '-'}`, class: 'badge bg
    -secondary' }
362 |                 ],
363 |                 metadata: [
364 |                     { label: 'Процесс', value: item.process_name || '-'
    },
365 |                     { label: 'PID', value: String(item.pid || '-') },
366 |                     { label: 'Протокол', value: item.protocol || '-' }
367 |                 ],
    { label: 'Локальный адрес:Порт', value: item.local
    _address || '-' },

```



```

368 |         { label: 'Удалённый адрес:Порт', value: item.remote_address || '-' },
369 |         { label: 'Статус сокета', value: item.status || '-' }
370 |     ],
371 |     rawTitle: 'Детали сокета',
372 |     rawContent: JSON.stringify(item, null, 2),
373 |     requestData: item
374 | });
375 | };
376 | });
377 | }
378 |
379 | // Render captured packets table
380 | function renderPacketsTable() {
381 |     const tbody = document.getElementById('net-packets-tbody')
382 |     ;
383 |     const badge = document.getElementById('net-captured-badge')
384 |     );
385 |     const filterInput = document.getElementById('net-table-filter');
386 |     const filter = (filterInput?.value || '').toLowerCase().trim();
387 |
388 |     if (!tbody) return;
389 |     if (badge) badge.innerText = `${capturedPackets.length} пакетов`;
390 |
391 |     if (capturedPackets.length === 0) {
392 |         tbody.innerHTML = '<tr><td colspan="7" class="text-center py-4 text-muted">Нет захваченных пакетов. Выберите интерфейс и нажмите "Старт захвата".</td></tr>';
393 |         return;
394 |     }
395 |
396 |     const filtered = capturedPackets.filter(p => {
397 |         if (!filter) return true;
398 |         const str = `${p.number || p.packet_number || ''} ${p.source_address || p.remote_address || ''} ${p.status || ''}`;
399 |         if (filter.length > 0) {
400 |             return str.toLowerCase().indexOf(filter.toLowerCase()) !== -1;
401 |         }
402 |         return true;
403 |     });
404 |
405 |     tbody.innerHTML = '';
406 |     if (filtered.length > 0) {
407 |         filtered.forEach(item => {
408 |             const tr = document.createElement('tr');
409 |             tr.innerHTML = `
410 |                 <td>${item.number || item.packet_number}</td>
411 |                 <td>${item.source_address || item.remote_address}</td>
412 |                 <td>${item.status}</td>
413 |                 <td colspan="4">${JSON.stringify(item, null, 2)}</td>
414 |             `;
415 |             tbody.appendChild(tr);
416 |         });
417 |     }
418 | }
419 |
420 | // Render captured packets table
421 | function renderPacketsTable() {
422 |     const tbody = document.getElementById('net-packets-tbody')
423 |     ;
424 |     const badge = document.getElementById('net-captured-badge')
425 |     );
426 |     const filterInput = document.getElementById('net-table-filter');
427 |     const filter = (filterInput?.value || '').toLowerCase().trim();
428 |
429 |     if (!tbody) return;
430 |     if (badge) badge.innerText = `${capturedPackets.length} пакетов`;
431 |
432 |     if (capturedPackets.length === 0) {
433 |         tbody.innerHTML = '<tr><td colspan="7" class="text-center py-4 text-muted">Нет захваченных пакетов. Выберите интерфейс и нажмите "Старт захвата".</td></tr>';
434 |         return;
435 |     }
436 |
437 |     const filtered = capturedPackets.filter(p => {
438 |         if (!filter) return true;
439 |         const str = `${p.number || p.packet_number || ''} ${p.source_address || p.remote_address || ''} ${p.status || ''}`;
440 |         if (filter.length > 0) {
441 |             return str.toLowerCase().indexOf(filter.toLowerCase()) !== -1;
442 |         }
443 |         return true;
444 |     });
445 |
446 |     tbody.innerHTML = '';
447 |     if (filtered.length > 0) {
448 |         filtered.forEach(item => {
449 |             const tr = document.createElement('tr');
450 |             tr.innerHTML = `
451 |                 <td>${item.number || item.packet_number}</td>
452 |                 <td>${item.source_address || item.remote_address}</td>
453 |                 <td>${item.status}</td>
454 |                 <td colspan="4">${JSON.stringify(item, null, 2)}</td>
455 |             `;
456 |             tbody.appendChild(tr);
457 |         });
458 |     }
459 | }

```

```

    urce || ''} ${p.destination || ''} ${p.protocol || ''} ${p.inf
    o || ''}`.toLowerCase();
397     return str.includes(filter);
398   });
399
400   const pagePackets = filtered.slice(-100);
401   tbody.innerHTML = pagePackets.map((p, idx) => {
402     const proto = (p.protocol || 'OTHER').toUpperCase();
403     let protoClass = 'proto-other';
404     if (proto.includes('TCP')) protoClass = 'proto-tcp';
405     else if (proto.includes('UDP')) protoClass = 'proto-udp'
    ;
406     else if (proto.includes('TLS') || proto.includes('SSL'))
    protoClass = 'proto-tls';
407     else if (proto.includes('HTTP')) protoClass = 'proto-htt
    p';
408     else if (proto.includes('DNS')) protoClass = 'proto-dns'
    ;
409
410     return `
411       <tr class="net-packet-row" data-idx="${idx}" style="cu
    rsor: pointer;" title="Нажмите для детального AI-анализа пакет
    a">
412         <td class="text-muted">${p.number || p.packet_number
    || '---'}</td>
413         <td>${p.timestamp ? (typeof p.timestamp === 'number'
    ? new Date(p.timestamp * 1000).toLocaleTimeString() : p.times
    tamp) : '---'}</td>
414         <td style="color: #38bdf8;">${p.source || '---'}</td>
415         <td style="color: #a855f7;">${p.destination || '---'}
    </td>
416         <td><span class="proto-badge ${protoClass}">${proto}
    </span></td>
417         <td style="text-align: right;">${p.length || 0} B</t
    d>
418         <td class="text-truncate" style="max-width: 320px;">
    ${p.info || '---'}</td>
419       </tr>

```

```

420 |     `;
421 |     }).join('');
422 |
423 |     tbody.querySelectorAll('.net-packet-row').forEach(row => {
424 |         row.onclick = () => {
425 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
426 |             ;
427 |             const p = pagePackets[idx];
428 |             if (!p || !window.AITableModal) return;
429 |
430 |             window.AITableModal.show({
431 |                 icon: '📡',
432 |                 title: `Сетевой пакет #${p.number || p.packet_number
433 | || 'N/A'} [${p.protocol || 'TCP'}]`,
434 |                 subtitle: `${p.source || '0.0.0.0'} → ${p.destination
435 | || '0.0.0.0'}`,
436 |                 tableType: 'network',
437 |                 badges: [
438 |                     { text: (p.protocol || 'TCP').toUpperCase(), class
439 | : 'badge bg-primary' },
440 |                     { text: `${p.length || 0} Bytes`, class: 'badge bg
441 | -secondary' }
442 |                 ],
443 |                 metadata: [
444 |                     { label: 'Номер пакета', value: String(p.number ||
445 | p.packet_number || '-') },
446 |                     { label: 'Протокол', value: (p.protocol || 'TCP').
447 | toUpperCase() },
448 |                     { label: 'Источник', value: p.source || '-' },
449 |                     { label: 'Назначение', value: p.destination || '-'
450 | },
451 |                     { label: 'Размер', value: `${p.length || 0} байт`
452 | },
453 |                     { label: 'Инфо', value: p.info || 'Нет данных', fu
454 | llWidth: true }
455 |                 ],
456 |                 rawTitle: 'Сырые данные пакета',
457 |                 rawContent: JSON.stringify(p, null, 2),

```

```

448         requestData: p
449     });
450 };
451 });
452 }
453
454 // Start live packet capture
455 async function startLiveCapture() {
456     const ifaceSelect = document.getElementById('net-interface
-select');
457     const filterInput = document.getElementById('net-capture-f
ilter');
458     const startBtn = document.getElementById('btn-start-live-c
apture');
459     const stopBtn = document.getElementById('btn-stop-live-cap
ture');
460
461     const iface = ifaceSelect?.value || '1';
462     const filter = filterInput?.value || '';
463
464     try {
465         const res = await fetch(`/api/network/start-capture?inte
rface=${encodeURIComponent(iface)}&display_filter=${encodeURIComponent(filter)}`, {
466             method: 'POST'
467         });
468         if (!res.ok) throw new Error(`HTTP ${res.status}`);
469
470         if (startBtn) startBtn.disabled = true;
471         if (stopBtn) stopBtn.disabled = false;
472
473         // Start SSE event source
474         if (netEventSource) netEventSource.close();
475         netEventSource = new EventSource('/api/network/packets/s
tream');
476         netEventSource.onmessage = async () => {
477             try {
478                 const packRes = await fetch('/api/network/packets?li

```

```

    mit=50');
479         if (packRes.ok) {
480             const data = await packRes.json();
481             if (Array.isArray(data.packets)) {
482                 capturedPackets = data.packets;
483                 renderPacketsTable();
484             }
485         }
486     } catch (e) {
487         console.warn('[NetworkTab] Packet poll error:', e);
488     }
489 };
490 } catch (e) {
491     console.error('[NetworkTab] Live capture error:', e);
492     window.showToast?.('Ошибка запуска захвата пакетов: ' +
    e.message, 'danger') || alert('Ошибка запуска захвата пакетов:
    ' + e.message);
493 }
494 }
495
496 // Stop live packet capture
497 async function stopLiveCapture() {
498     const startBtn = document.getElementById('btn-start-live-c
    apture');
499     const stopBtn = document.getElementById('btn-stop-live-cap
    ture');
500
501     if (netEventSource) {
502         netEventSource.close();
503         netEventSource = null;
504     }
505
506     try {
507         await fetch('/api/network/stop-capture', { method: 'POST
    ' });
508     } catch (e) {
509         console.warn('[NetworkTab] Stop capture error:', e);
510     }

```

```
511 |
512 |     if (startBtn) startBtn.disabled = false;
513 |     if (stopBtn) stopBtn.disabled = true;
514 | }
515 |
516 | // Analyze uploaded PCAP file
517 | async function analyzePcapFile() {
518 |     const fileInput = document.getElementById('net-pcap-file')
519 |     ;
520 |     const statusEl = document.getElementById('net-upload-status');
521 |     const healthEl = document.getElementById('pcap-health-score');
522 |     const anomaliesEl = document.getElementById('pcap-anomalies-count');
523 |     const heuristicsEl = document.getElementById('pcap-heuristics-list');
524 |     const recommendationsEl = document.getElementById('pcap-ai-recommendations');
525 |
526 |     const file = fileInput?.files?.[0];
527 |     if (!file) {
528 |         window.showToast?.(('Пожалуйста, выберите .pcap или .pcapng файл', 'warning')) || alert('Пожалуйста, выберите .pcap или .pcapng файл');
529 |     }
530 |
531 |     const formData = new FormData();
532 |     formData.append('file', file);
533 |     formData.append('max_packets', '500');
534 |
535 |     if (statusEl) statusEl.innerText = 'Анализ пакетов...';
536 |
537 |     try {
538 |         const res = await fetch('/api/network/analyze/pcap', {
539 |             method: 'POST',
540 |             body: formData
```

```
541 |     });
542 |
543 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
544 |     const report = await res.json();
545 |
546 |     if (statusEl) statusEl.innerText = `Анализ завершён: ${r
| eport.stats?.total_packets || 0} пакетов`;
547 |
548 |     if (healthEl) healthEl.innerText = `${report.ai_report?.
| health_score || 100}/100`;
549 |
550 |     const anomaliesCount = (report.heuristics?.length || 0)
| + (report.ai_report?.anomalies?.length || 0);
551 |     if (anomaliesEl) {
552 |         anomaliesEl.innerText = String(anomaliesCount);
553 |         anomaliesEl.className = `net-value mt-1 ${anomaliesCou
| nt > 0 ? 'text-danger' : 'text-success'}`;
554 |     }
555 |
556 |     if (heuristicsEl) {
557 |         if (Array.isArray(report.heuristics) && report.heurist
| ics.length > 0) {
558 |             heuristicsEl.innerHTML = report.heuristics.map(h =>
| `

<i class="bi bi-exclamation-tr
| iangle-fill me-1"></i>${h}</div>`).join('');
559 |         } else {
560 |             heuristicsEl.innerHTML = '<span class="text-success"
| ><i class="bi bi-check-circle-fill me-1"></i>Эвристических угр
| оз не выявлено.</span>';
561 |         }
562 |     }
563 |
564 |     if (recommendationsEl) {
565 |         if (Array.isArray(report.ai_report?.recommended_action
| s) && report.ai_report.recommended_actions.length > 0) {
566 |             recommendationsEl.innerHTML = report.ai_report.recom
| mended_actions.map(r => `

<i class=
| "bi bi-arrow-right-short me-1"></i>${r}</div>`).join('');


```

```
567         } else {
568             recommendationsEl.innerHTML = '<span class="text-mut
ed">Рекомендаций нет, система функционирует штатно.</span>';
569         }
570     }
571
572     if (Array.isArray(report.sample_packets) && report.saml
e_packets.length > 0) {
573         capturedPackets = report.sample_packets;
574         renderPacketsTable();
575     }
576 } catch (e) {
577     console.error('[NetworkTab] PCAP upload error:', e);
578     if (statusEl) statusEl.innerText = 'Ошибка анализа: ' +
e.message;
579     window.showToast?(['Ошибка при анализе PCAP: ' + e.messa
ge, 'danger']) || alert('Ошибка при анализе PCAP: ' + e.message
);
580 }
581 }
582
583 // Main init function for Network Terminal Tab
584 function initNetworkTab() {
585     console.log('[NetworkTab] Initializing Windows Network Ter
minal tab (/apps/windows/network)...');
586     fetchNetworkStatus();
587     fetchLatestSpeedtest();
588     fetchAdapters();
589     fetchConnections();
590
591     if (!isNetInitialized) {
592         // Action buttons
593         const refreshAllBtn = document.getElementById('btn-refre
sh-network-all');
594         const runSpeedtestBtn = document.getElementById('btn-run
-speedtest');
595         const refreshAdaptersBtn = document.getElementById('btn-
refresh-adapters');
```



```
596 |     const refreshConnsBtn = document.getElementById('btn-ref
    | resh-conns');
597 |     const startBtn = document.getElementById('btn-start-live
    | -capture');
598 |     const stopBtn = document.getElementById('btn-stop-live-c
    | apture');
599 |     const analyzePcapBtn = document.getElementById('btn-anal
    | yze-pcap');
600 |     const clearPacketsBtn = document.getElementById('btn-cle
    | ar-packets');
601 |     const connsFilterInput = document.getElementById('net-co
    | nns-filter');
602 |     const tableFilterInput = document.getElementById('net-ta
    | ble-filter');
603 |     const configBtn = document.getElementById('btn-network-c
    | onfig');
604 |
605 |     if (refreshAllBtn) refreshAllBtn.onclick = () => {
606 |         fetchNetworkStatus();
607 |         fetchLatestSpeedtest();
608 |         fetchAdapters();
609 |         fetchConnections();
610 |     };
611 |     if (runSpeedtestBtn) runSpeedtestBtn.onclick = runIntern
    | etSpeedtest;
612 |     if (refreshAdaptersBtn) refreshAdaptersBtn.onclick = fet
    | chAdapters;
613 |     if (refreshConnsBtn) refreshConnsBtn.onclick = fetchConn
    | ections;
614 |     if (startBtn) startBtn.onclick = startLiveCapture;
615 |     if (stopBtn) stopBtn.onclick = stopLiveCapture;
616 |     if (analyzePcapBtn) analyzePcapBtn.onclick = analyzePcap
    | File;
617 |     if (clearPacketsBtn) clearPacketsBtn.onclick = () => {
618 |         capturedPackets = [];
619 |         renderPacketsTable();
620 |     };
621 |
```

```

622         if (connsFilterInput) connsFilterInput.oninput = renderC
onnectionsTable;
623         if (tableFilterInput) tableFilterInput.oninput = renderP
acketsTable;
624
625         // Filter buttons for connections
626         document.querySelectorAll('.btn-filter-conns').forEach(b
tn => {
627             btn.onclick = () => {
628                 document.querySelectorAll('.btn-filter-conns').forEa
ch(b => b.classList.remove('active'));
629                 btn.classList.add('active');
630                 activeConnFilter = btn.getAttribute('data-filter') |
| 'all';
631                 renderConnectionsTable();
632             };
633         });
634
635         if (configBtn) configBtn.onclick = () => {
636             if (typeof window.openAppConfigModal === 'function') {
637                 window.openAppConfigModal('network_terminal', 'Netwo
rk Analyzer Terminal');
638             }
639         };
640
641         isNetInitialized = true;
642     }
643 }
644
645 window.initNetworkTab = initNetworkTab;
646 }));
647

```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/news_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/news_tab/index.html" cl
17     ass="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/news_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- News Tab Component -->
30 <div class="container-fluid p-3" id="news-tab-root">
31   <!-- Top Banner / Digest Generator Bar -->
32   <div class="card border mb-3 shadow-sm">
33     <div class="card-body p-3 d-flex flex-wrap align-items-cen
34     ter justify-content-between gap-3">
35       <div class="d-flex align-items-center gap-3">
36         <div class="p-3 rounded border text-primary display-6
37         d-flex align-items-center justify-content-center" style="width
```

```

: 54px; height: 54px; background: var(--bg-color);">
31     <i class="bi bi-newspaper"></i>
32 </div>
33 <div>
34     <h5 class="mb-1 d-flex align-items-center gap-2">
35         □ Интеллектуальная лента новостей
36         <span class="badge bg-primary rounded-pill small"
id="news-total-badge">0</span>
37     </h5>
38     <p class="mb-0 text-muted small">
39         Только релевантные новости с непрерывным обучением
        под ваши личные интересы и AI-выжимками.
40     </p>
41 </div>
42 </div>
43 <div class="d-flex align-items-center gap-2">
44     <button type="button" class="btn btn-outline-secondary
        btn-sm d-flex align-items-center gap-1" id="btn-refresh-news"
        onclick="window.newsTab?.refreshNews(true)">
45         <i class="bi bi-arrow-repeat"></i> Обновить
46     </button>
47     <button type="button" class="btn btn-outline-info btn-
        sm d-flex align-items-center gap-1 shadow-sm" id="btn-preferen
        ces-news" data-bs-toggle="modal" data-bs-target="#newsPreferen
        cesModal">
48         <i class="bi bi-sliders"></i> Мои интересы
49     </button>
50     <button type="button" class="btn btn-primary btn-sm d-
        flex align-items-center gap-1 shadow-sm" id="btn-generate-dige
        st" onclick="window.newsTab?.generateDigest()">
51         <i class="bi bi-stars"></i> ИИ-дайджест
52     </button>
53 </div>
54 </div>
55 </div>
56
57 <!-- AI Digest Section (Appears when generated) -->
58 <div class="card border-primary mb-3 shadow-sm d-none" id="n

```

```

ews-digest-card">
59     <div class="card-header d-flex justify-content-between align-items-center border-primary py-2">
60         <div class="d-flex align-items-center gap-2">
61             <i class="bi bi-stars text-warning fs-5"></i>
62             <strong id="news-digest-title">Персональный ИИ-дайджест</strong>
63         </div>
64         <div class="d-flex align-items-center gap-2">
65             <button class="btn btn-sm btn-outline-secondary py-0 px-2" id="news-digest-tts-btn" title="Озвучить дайджест" onclick="window.newsTab?.speakDigest()">
66                 <i class="bi bi-volume-up me-1"></i>Слушать
67             </button>
68             <button type="button" class="btn-close" aria-label="Close" onclick="document.getElementById('news-digest-card').classList.add('d-none')"></button>
69         </div>
70     </div>
71     <div class="card-body p-3">
72         <div id="news-digest-content" class="small" style="line-height: 1.6; white-space: pre-line;"></div>
73     </div>
74 </div>
75
76 <!-- Filter & Learning Feedback Bar -->
77 <div class="card mb-3">
78     <div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2">
79         <div class="d-flex align-items-center gap-2 flex-grow-1" style="max-width: 480px;">
80             <div class="input-group input-group-sm">
81                 <span class="input-group-text"><i class="bi bi-search"></i></span>
82                 <input type="text" class="form-control" id="news-search-input" placeholder="Фильтр новостей по тексту..." oninput="window.newsTab?.filterNews()">
83             </div>

```

```
84 |         </div>
85 |
86 |         <div class="d-flex align-items-center gap-2">
87 |             <label class="small text-muted text-nowrap" for="news-
category-filter">Категория:</label>
88 |             <select class="form-select form-select-sm" id="news-ca
tegory-filter" style="width: auto;" onchange="window.newsTab?.
filterNews()">
89 |                 <option value="all">Все категории</option>
90 |                 <option value="ai">ИИ & ML</option>
91 |                 <option value="dev">Разработка</option>
92 |                 <option value="tech">Технологии</option>
93 |                 <option value="science">Наука</option>
94 |                 <option value="world">Мировые</option>
95 |             </select>
96 |
97 |             <label class="small text-muted text-nowrap ms-2" for="
news-min-score">Мин. совпадение:</label>
98 |             <select class="form-select form-select-sm" id="news-mi
n-score" style="width: auto;" onchange="window.newsTab?.filter
News()">
99 |                 <option value="0">Все (0%)</option>
100 |                 <option value="0.25" selected>Релевантные (>= 25%)</
option>
101 |                 <option value="0.5">Высокое (>= 50%)</option>
102 |                 <option value="0.7">Только топ (>= 70%)</option>
103 |             </select>
104 |         </div>
105 |     </div>
106 | </div>
107 |
108 | <!-- News Stream Grid -->
109 | <div class="row g-3" id="news-articles-container">
110 |     <div class="col-12 text-center text-muted p-5" id="news-lo
ading-spinner">
111 |         <div class="spinner-border spinner-border-sm text-primar
y mb-2" role="status"></div>
112 |         <div>Сбор и ранжирование релевантных новостей под ваши и
```

```

    нтересы...</div>
113     </div>
114   </div>
115 </div>
116
117 <!-- Preferences & Topics Modal -->
118 <div class="modal fade" id="newsPreferencesModal" tabindex="-1
    " aria-labelledby="newsPreferencesModalLabel" aria-hidden="tru
    e">
119   <div class="modal-dialog modal-lg modal-dialog-centered">
120     <div class="modal-content shadow">
121       <div class="modal-header border-bottom">
122         <h5 class="modal-title d-flex align-items-center gap-2
    " id="newsPreferencesModalLabel">
123           <i class="bi bi-sliders text-info"></i> Настройка ин
    тересов и профиля обучения
124         </h5>
125         <button type="button" class="btn-close" data-bs-dismis
    s="modal" aria-label="Close"></button>
126       </div>
127       <div class="modal-body p-3">
128         <!-- Learned Weights Section -->
129         <div class="mb-3">
130           <label class="form-label small text-muted fw-semibol
    d">
131             <i class="bi bi-lightning-charge text-warning me-1
    "></i>Накопленные веса интересов (адаптируются на основе лайко
    в):
132           </label>
133           <div class="p-2 border rounded d-flex flex-wrap gap-
    2" id="news-learned-tags-container" style="max-height: 140px;
    overflow-y: auto; background: var(--bg-color);">
134             <!-- Badges rendered via JS -->
135             </div>
136           </div>
137
138         <!-- Preferred Topics Input -->
139         <div class="mb-3">

```

```
140 |         <label class="form-label small fw-semibold" for="new
    | s-pref-topics-input">
141 |             Явные ключевые темы (через запятую):
142 |         </label>
143 |         <input type="text" class="form-control" id="news-pre
    | f-topics-input" placeholder="например: ai, python, robotics, n
    | euroscience, opensource">
144 |         <small class="text-muted">Новости с этими словами по
    | лучат приоритетный буст в вашей ленте.</small>
145 |     </div>
146 |
147 |     <!-- Negative Keywords Input -->
148 |     <div class="mb-3">
149 |         <label class="form-label small fw-semibold" for="new
    | s-neg-topics-input">
150 |             Стоп-слова и нежелательные темы (через запятую):
151 |         </label>
152 |         <input type="text" class="form-control" id="news-neg
    | -topics-input" placeholder="например: криминал, шоу-бизнес, ск
    | андалы">
153 |         <small class="text-muted">Статьи с этими словами буд
    | ут скрываться или получать минимальный скор.</small>
154 |     </div>
155 |
156 |     <!-- Stats & Reset -->
157 |     <div class="d-flex align-items-center justify-content-
    | between p-2 rounded border" style="background: var(--bg-color)
    | ;">
158 |         <span class="small text-muted" id="news-interactions
    | -count-label">Обработано реакций: 0</span>
159 |         <button type="button" class="btn btn-sm btn-outline-
    | danger" onclick="window.newsTab?.resetProfile()">
160 |             <i class="bi bi-arrow-counterclockwise me-1"></i>C
    | бросить обучение
161 |         </button>
162 |     </div>
163 | </div>
164 | <div class="modal-footer border-top">
```



```
165 |         <button type="button" class="btn btn-sm btn-secondary"
    |         data-bs-dismiss="modal">Закрыть</button>
166 |         <button type="button" class="btn btn-sm btn-primary" o
    | nclick="window.newsTab?.savePreferences()">Сохранить интересы<
    | /button>
167 |     </div>
168 | </div>
169 | </div>
170 | </div>
171 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows News Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/news_tab/main.js?v=202
11  * 61001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/news_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // News Tab Controller
22 class NewsTabController {
23   constructor() {
24     this.articles = [];
25     this.profile = null;
26     this.currentDigest = null;
27     this.isLoading = false;
28   }
29
30   async init() {
31     console.log('[NewsTab] Initializing Smart News Feed...');
32     await this.loadProfile();
```

```

33     await this.refreshNews(false);
34 }
35
36 async loadProfile() {
37     try {
38         const res = await fetch('/api/news/profile');
39         if (res.ok) {
40             this.profile = await res.json();
41             this.renderProfileModal();
42         }
43     } catch (e) {
44         console.error('[NewsTab] Error loading profile:', e);
45     }
46 }
47
48 renderProfileModal() {
49     if (!this.profile) return;
50     const container = document.getElementById('news-learned-ta
gs-container');
51     if (container) {
52         container.innerHTML = '';
53         const weights = this.profile.topic_weights || {};
54         const sorted = Object.entries(weights).sort((a, b) => b[
1] - a[1]);
55         if (sorted.length === 0) {
56             container.innerHTML = '<span class="text-muted small">
Пока нет накопленных весов. Оценивайте новости кнопками ⬆ и ⬇
⬆ !</span>';
57         } else {
58             for (const [kw, w] of sorted) {
59                 const badge = document.createElement('span');
60                 badge.className = 'badge bg-secondary border border-
secondary text-light d-flex align-items-center gap-1';
61                 badge.innerHTML = `<span>${kw}</span> <span class="b
adge bg-primary text-white">${w}</span>`;
62                 container.appendChild(badge);
63             }
64         }

```

```
65 |     }
66 |
67 |     const countLabel = document.getElementById('news-interacti
ons-count-label');
68 |     if (countLabel) {
69 |         countLabel.textContent = `Обработано реакций обучения: $
{this.profile.interaction_count || 0}`;
70 |     }
71 |
72 |     const negInput = document.getElementById('news-neg-topics-
input');
73 |     if (negInput && this.profile.negative_keywords) {
74 |         negInput.value = this.profile.negative_keywords.join(',
');
75 |     }
76 | }
77 |
78 | async refreshNews(force = false) {
79 |     if (this.isLoading) return;
80 |     this.isLoading = true;
81 |     const container = document.getElementById('news-articles-c
ontainer');
82 |     const spinner = document.getElementById('news-loading-spin
ner');
83 |     if (spinner) spinner.classList.remove('d-none');
84 |
85 |     try {
86 |         const url = `/api/news/feed?force_refresh=${force}&limit
=35`;
87 |         const res = await fetch(url);
88 |         if (res.ok) {
89 |             this.articles = await res.json();
90 |             this.renderArticles();
91 |             const totalBadge = document.getElementById('news-total
-badge');
92 |             if (totalBadge) totalBadge.textContent = this.articles
.length;
93 |         }
```

```
94 |     } catch (e) {
95 |         console.error('[NewsTab] Error fetching feed:', e);
96 |         if (container) {
97 |             container.innerHTML = `
```

```

126         <i class="bi bi-inbox fs-1 mb-2 d-block"></i>
127         <h6>Нет новостей, подходящих под выбранные фильтры</
h6>
128         <small>Попробуйте уменьшить порог минимального совпа
дения или обновить ленту.</small>
129     </div>
130     `;
131     return;
132 }
133
134     filtered.forEach((art) => {
135         const col = document.createElement('div');
136         col.className = 'col-md-6 col-lg-4';
137
138         const matchPercent = Math.round((art.relevance_score ||
0.5) * 100);
139         let matchBadgeClass = 'bg-secondary';
140         if (matchPercent >= 70) matchBadgeClass = 'bg-success';
141         else if (matchPercent >= 45) matchBadgeClass = 'bg-prima
ry';
142         else if (matchPercent >= 25) matchBadgeClass = 'bg-info
text-dark';
143
144         const isLiked = art.user_interaction === 'like';
145         const isDisliked = art.user_interaction === 'dislike';
146         const isBookmarked = art.user_interaction === 'bookmark'
;
147
148         col.innerHTML = `
149             <div class="card h-100 shadow-sm d-flex flex-column" i
d="news-card-${art.id}">
150                 <div class="card-header d-flex justify-content-betwe
en align-items-center py-2">
151                     <span class="badge bg-secondary-subtle text-second
ary-emphasis border small text-truncate" style="max-width: 130
px;">
152                         ${art.source_name}
153                     </span>

```

```

154 |         <span class="badge ${matchBadgeClass} rounded-pill
      | small" title="Персональная релевантность">
155 |             ${matchPercent}% match
156 |         </span>
157 |     </div>
158 |     <div class="card-body p-3 d-flex flex-column flex-gr
      | ow-1">
159 |         <h6 class="card-title fw-bold mb-2">
160 |             <a href="${art.link}" target="_blank" rel="noope
      | ner noreferrer" class="text-reset text-decoration-none hover-p
      | rimary" onclick="window.newsTab?.onArticleRead('${art.id}', '$
      | {encodeURIComponent(art.title)}')">
161 |                 ${art.title}
162 |             </a>
163 |         </h6>
164 |         <p class="card-text small text-muted flex-grow-1 m
      | b-3">
165 |             ${art.ai_summary || art.summary || ''}
166 |         </p>
167 |         <div class="d-flex align-items-center justify-cont
      | ent-between mt-auto pt-2 border-top border-secondary-subtle">
168 |             <small class="text-secondary" style="font-size:
      | 0.75rem;">
169 |                 <i class="bi bi-clock me-1"></i>${art.publishe
      | d_at || ''}
170 |             </small>
171 |             <div class="btn-group btn-group-sm" role="group"
      | >
172 |                 <button type="button" class="btn ${isLiked ? '
      | btn-success' : 'btn-outline-secondary'} py-0 px-2" title="Поле
      | зно (обучить алгоритм)" onclick="window.newsTab?.sendFeedback(
      | '${art.id}', 'like', '${encodeURIComponent(art.title)}', '${ar
      | t.category}')">
173 |                     <i class="bi bi-hand-thumbs-up"></i>
174 |                 </button>
175 |                 <button type="button" class="btn ${isDisliked
      | ? 'btn-danger' : 'btn-outline-secondary'} py-0 px-2" title="Не
      | интересно (понизить тему)" onclick="window.newsTab?.sendFeedb

```

```

    ack('${art.id}', 'dislike', '${encodeURIComponent(art.title)}'
    , '${art.category}'))">
176         <i class="bi bi-hand-thumbs-down"></i>
177         </button>
178         <button type="button" class="btn ${isBookmarke
    d ? 'btn-warning text-dark' : 'btn-outline-secondary'} py-0 px
    -2" title="Сохранить в закладки" onclick="window.newsTab?.send
    Feedback('${art.id}', 'bookmark', '${encodeURIComponent(art.ti
    tle)}', '${art.category}'))">
179             <i class="bi bi-bookmark"></i>
180             </button>
181             <button type="button" class="btn btn-outline-i
    nfo py-0 px-2" title="Сгенерировать AI-выжимку" onclick="windo
    w.newsTab?.summarizeArticle('${art.id}'))">
182                 <i class="bi bi-magic"></i>
183                 </button>
184             </div>
185         </div>
186     </div>
187 </div>
188 `;
189     container.appendChild(col);
190 });
191 }
192
193 filterNews() {
194     this.renderArticles();
195 }
196
197 async sendFeedback(articleId, action, titleEncoded, category
    ) {
198     const title = decodeURIComponent(titleEncoded || '');
199     try {
200         const res = await fetch('/api/news/feedback', {
201             method: 'POST',
202             headers: { 'Content-Type': 'application/json' },
203             body: JSON.stringify({
204                 article_id: articleId,

```



```

205         action: action,
206         article_title: title,
207         category: category || 'general'
208     })
209 });
210
211 if (res.ok) {
212     // Optimistic UI update
213     const art = this.articles.find((a) => a.id === article
Id);
214     if (art) {
215         art.user_interaction = action;
216     }
217     this.renderArticles();
218     await this.loadProfile();
219     if (window.showNotification) {
220         const msg = action === 'like' ? 'Алгоритм обучен: те
ма получила приоритет ☑ ' :
221             action === 'dislike' ? 'Алгоритм обучен:
интерес к теме снижен ☑ ' :
222             action === 'bookmark' ? 'Добавлено в зак
ладки ☑ ' : 'Действие сохранено';
223         window.showNotification(msg, 'success');
224     }
225 }
226 } catch (e) {
227     console.error('[NewsTab] Error sending feedback:', e);
228 }
229 }
230
231 async onArticleRead(articleId, titleEncoded) {
232     const title = decodeURIComponent(titleEncoded || '');
233     try {
234         await fetch('/api/news/feedback', {
235             method: 'POST',
236             headers: { 'Content-Type': 'application/json' },
237             body: JSON.stringify({
238                 article_id: articleId,

```

```
239         action: 'read',
240         article_title: title
241     })
242 });
243 } catch (_) {}
244 }
245
246 async summarizeArticle(articleId) {
247     const art = this.articles.find((a) => a.id === articleId);
248     if (!art) return;
249
250     const card = document.getElementById(`news-card-${articleId}`);
251     if (card) {
252         const body = card.querySelector('.card-text');
253         if (body) {
254             body.innerHTML = '<span class="spinner-border spinner-border-sm text-info me-1"></span> Генерация ИИ-выжимки...';
255         }
256     }
257
258     try {
259         const res = await fetch('/api/news/summarize-article', {
260             method: 'POST',
261             headers: { 'Content-Type': 'application/json' },
262             body: JSON.stringify(art)
263         });
264         if (res.ok) {
265             const data = await res.json();
266             art.ai_summary = data.ai_summary;
267             this.renderArticles();
268         }
269     } catch (e) {
270         console.error('[NewsTab] Summarize error:', e);
271         this.renderArticles();
272     }
273 }
274
```

```

275 |   async generateDigest() {
276 |       const digestCard = document.getElementById('news-digest-ca
rd');
277 |       const digestContent = document.getElementById('news-digest
-content');
278 |       const digestTitle = document.getElementById('news-digest-t
itle');
279 |
280 |       if (digestCard) digestCard.classList.remove('d-none');
281 |       if (digestContent) {
282 |           digestContent.innerHTML = '<div class="text-center p-3">
<span class="spinner-border spinner-border-sm text-warning me-
2"></span>Формирование персонализированного дайджеста ИИ...</d
iv>';
283 |       }
284 |
285 |       try {
286 |           const res = await fetch('/api/news/digest', { method: 'P
OST' });
287 |           if (res.ok) {
288 |               this.currentDigest = await res.json();
289 |               if (digestTitle) digestTitle.textContent = this.curren
tDigest.title || 'ИИ-Дайджест';
290 |               if (digestContent) {
291 |                   if (window.marked) {
292 |                       digestContent.innerHTML = window.marked.parse(this
.currentDigest.digest_text || '');
293 |                   } else {
294 |                       digestContent.textContent = this.currentDigest.dig
est_text || '';
295 |                   }
296 |               }
297 |           }
298 |       } catch (e) {
299 |           console.error('[NewsTab] Error generating digest:', e);
300 |           if (digestContent) digestContent.textContent = `Ошибка г
енерации дайджеста: ${e.message}`;
301 |       }

```

```
302 | }
303 |
304 | speakDigest() {
305 |     if (!this.currentDigest || !this.currentDigest.digest_text
306 | ) return;
307 |     if (window.chatService && typeof window.chatService.speak
308 | === 'function') {
309 |         window.chatService.speak(this.currentDigest.digest_text)
310 | ;
311 |     } else {
312 |         const utterance = new SpeechSynthesisUtterance(this.curren
313 | tDigest.digest_text);
314 |         utterance.lang = 'ru-RU';
315 |         window.speechSynthesis.speak(utterance);
316 |     }
317 | }
318 |
319 | async savePreferences() {
320 |     const prefInput = document.getElementById('news-pref-topics-
321 | input')?.value || '';
322 |     const negInput = document.getElementById('news-neg-topics-
323 | input')?.value || '';
324 |
325 |     const preferred = prefInput.split(',').map((s) => s.trim())
326 | .filter(Boolean);
327 |     const ignored = negInput.split(',').map((s) => s.trim()).f
328 | ilter(Boolean);
329 |
330 |     try {
331 |         const res = await fetch('/api/news/profile', {
332 |             method: 'POST',
333 |             headers: { 'Content-Type': 'application/json' },
334 |             body: JSON.stringify({
335 |                 preferred_topics: preferred,
336 |                 ignored_topics: ignored
337 |             })
338 |         });
339 |         if (res.ok) {
```

```
332 |         await this.loadProfile();
333 |         await this.refreshNews(true);
334 |         if (window.showNotification) window.showNotification('
Интересы успешно сохранены!', 'success');
335 |         const modalEl = document.getElementById('newsPreferenc
esModal');
336 |         if (modalEl && window.bootstrap) {
337 |             const modalInstance = window.bootstrap.Modal.getInst
ance(modalEl);
338 |             if (modalInstance) modalInstance.hide();
339 |         }
340 |     }
341 | } catch (e) {
342 |     console.error('[NewsTab] Error saving preferences:', e);
343 | }
344 | }
345 |
346 | async resetProfile() {
347 |     if (!confirm('Сбросить накопленные веса интересов и начать
обучение с чистого листа?')) return;
348 |     try {
349 |         const res = await fetch('/api/news/profile', {
350 |             method: 'POST',
351 |             headers: { 'Content-Type': 'application/json' },
352 |             body: JSON.stringify({
353 |                 preferred_topics: [],
354 |                 ignored_topics: []
355 |             })
356 |         });
357 |         if (res.ok) {
358 |             await this.loadProfile();
359 |             await this.refreshNews(true);
360 |             if (window.showNotification) window.showNotification('
Профиль обучения сброшен', 'info');
361 |         }
362 |     } catch (e) {
363 |         console.error('[NewsTab] Error resetting profile:', e);
364 |     }
```

```
365 |     }
366 | }
367 |
368 | window.newsTab = new NewsTabController();
369 |
370 | function initNewsTab() {
371 |     if (window.newsTab) {
372 |         window.newsTab.init();
373 |     }
374 | }
375 |
376 | window.initNewsTab = initNewsTab;
377 |
378 | if (document.getElementById('news-tab-root')) {
379 |     initNewsTab();
380 | }
381 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/ninite_updater_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/ninite_updater_tab/index.html" class="w-100 h-100 border-0"></iframe>
17
18 File: index.html
19 Project: ai-breadboard
20 Package: windows/api/webgui/ninite_updater_tab
21 Author: hypo69
22 Copyright: © 2026 hypo69
23 Updated: 2026-10-01 13:04:40
24 =====
25
26 -->
27
28 <!-- Ninite Auto-Updater & Task Scheduler Management Tab -->
29 <style>
30   .ninite-container {
31     padding: 0.75rem 1rem;
32     max-width: 1200px;
33     margin: 0 auto;
34   }
35   .ninite-card {
36     background: #0f172a !important;
```

```
33     border: 1px solid #334155 !important;
34     border-radius: 10px;
35     padding: 1.25rem;
36     color: #f8fafc;
37     box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.2);
38 }
39 .ninite-header {
40     background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
41 0%);
42     border: 1px solid #334155;
43     border-radius: 10px;
44     padding: 0.85rem 1.25rem;
45 }
46 .ninite-dropzone {
47     border: 2px dashed #38bdf8;
48     background: rgba(56, 189, 248, 0.05);
49     border-radius: 10px;
50     padding: 2rem 1.5rem;
51     text-align: center;
52     cursor: pointer;
53     transition: all 0.2s ease-in-out;
54 }
55 .ninite-dropzone:hover, .ninite-dropzone.dragover {
56     background: rgba(56, 189, 248, 0.12);
57     border-color: #0ea5e9;
58 }
59 .ninite-stat-box {
60     background: #1e293b;
61     border: 1px solid #334155;
62     border-radius: 8px;
63     padding: 0.75rem 1rem;
64 }
65 .ninite-stat-title {
66     font-size: 0.75rem;
67     text-transform: uppercase;
68     letter-spacing: 0.05em;
69     color: #94a3b8;
70 }
```



```

70 | .ninite-stat-val {
71 |     font-size: 1.1rem;
72 |     font-weight: 600;
73 |     color: #f8fafc;
74 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
75 | }
76 | </style>
77 |
78 | <div class="ninite-container">
79 |     <!-- Верхняя информационная панель -->
80 |     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 ninite-header text-white">
81 |         <div class="d-flex align-items-center gap-2.5">
82 |             <span class="fs-4">□ </span>
83 |             <div>
84 |                 <h5 class="mb-0 fw-bold fs-6">Установка и автообновлен
    | ие Ninite</h5>
85 |                 <span class="text-secondary small">Автоматическое полу
    | чение свежих версий ПО через Windows Task Scheduler</span>
86 |             </div>
87 |         </div>
88 |         <div class="d-flex align-items-center gap-2">
89 |             <span class="badge rounded-pill bg-dark border border-se
    | condary text-info px-2.5 py-1.5" id="ninite-status-badge">
90 |                 Загрузка статуса...
91 |             </span>
92 |             <button class="btn btn-sm btn-outline-info rounded-pill
    | px-3" id="btn-ninite-refresh" type="button">
93 |                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
94 |             </button>
95 |         </div>
96 |     </div>
97 |
98 |     <div class="row g-3 mb-3">
99 |         <!-- Левая колонка: Форма загрузки и настройки расписания
    | -->
100 |         <div class="col-12 col-lg-7">

```

```

101         <div class="ninite-card h-100">
102             <h6 class="fw-bold text-info mb-3 d-flex align-items-c
enter gap-2">
103                 <i class="bi bi-cloud-arrow-up-fill"></i> 1. Выбор и
нсталлятора и расписание
104             </h6>
105
106             <form id="ninite-upload-form">
107                 <!-- Dropzone для выбора файла -->
108                 <label for="ninite-file-input" class="ninite-dropzon
e mb-3 d-block text-decoration-none" id="ninite-dropzone">
109                     <i class="bi bi-file-earmark-arrow-up fs-1 text-in
fo mb-2 d-block"></i>
110                     <div class="fw-semibold mb-1 text-white" id="ninit
e-file-label">Перетащите файл установщика Ninite или нажмите д
ля выбора</div>
111                     <div class="text-secondary small">Файл будет сохра
нён и переименован в <code>C:\Program Files\Ninite\ninite.exe<
/code></div>
112                     <input type="file" id="ninite-file-input" class="d
-none" accept=".exe">
113                 </label>
114
115                 <!-- Параметры расписания -->
116                 <div class="row g-2 mb-3">
117                     <div class="col-12 col-sm-6">
118                         <label for="ninite-interval" class="form-label s
mall text-secondary fw-semibold">
119                             <i class="bi bi-calendar-range me-1"></i> Инте
рвал обновления
120                         </label>
121                         <select class="form-select form-select-sm bg-dar
k text-white border-secondary" id="ninite-interval">
122                             <option value="1">Каждую неделю (1 неделя)</op
tion>
123                             <option value="2" selected>Каждые 2 недели (по
умолчанию)</option>
124                             <option value="3">Каждые 3 недели</option>

```

```

125 |             <option value="4">Раз в месяц (4 недели)</opti
| on>
126 |             </select>
127 |         </div>
128 |
129 |         <div class="col-12 col-sm-6">
130 |             <label for="ninite-time" class="form-label small
| text-secondary fw-semibold">
131 |                 <i class="bi bi-clock me-1"></i> Время запуска
132 |             </label>
133 |             <input type="time" class="form-control form-cont
| rol-sm bg-dark text-white border-secondary" id="ninite-time" v
| alue="20:00">
134 |         </div>
135 |     </div>
136 |
137 |     <div class="mb-3">
138 |         <label for="ninite-day" class="form-label small te
| xt-secondary fw-semibold">
139 |             <i class="bi bi-calendar-event me-1"></i> День н
| едели
140 |         </label>
141 |         <select class="form-select form-select-sm bg-dark
| text-white border-secondary" id="ninite-day">
142 |             <option value="Sunday" selected>Воскресенье</opt
| ion>
143 |             <option value="Monday">Понедельник</option>
144 |             <option value="Tuesday">Вторник</option>
145 |             <option value="Wednesday">Среда</option>
146 |             <option value="Thursday">Четверг</option>
147 |             <option value="Friday">Пятница</option>
148 |             <option value="Saturday">Суббота</option>
149 |         </select>
150 |     </div>
151 |
152 |     <!-- Кнопка отправки и сохранения -->
153 |     <button type="submit" class="btn btn-primary w-100 p
| y-2 fw-semibold d-flex align-items-center justify-content-cent

```

```

er gap-2" id="btn-ninite-submit">
154         <i class="bi bi-check-circle-fill"></i> Установить
        в Program Files и активировать Task Scheduler
155         </button>
156     </form>
157
158     <div id="ninite-alert-box" class="mt-3" style="display
: none;"></div>
159     </div>
160 </div>
161
162 <!-- Правая колонка: Текущее состояние и управление -->
163 <div class="col-12 col-lg-5">
164     <div class="ninite-card h-100 d-flex flex-column justify
-content-between">
165         <div>
166             <h6 class="fw-bold text-info mb-3 d-flex align-items
-center gap-2">
167                 <i class="bi bi-cpu-fill"></i> 2. Текущее состояни
е системы
168             </h6>
169
170             <div class="row g-2 mb-3">
171                 <div class="col-12">
172                     <div class="ninite-stat-box">
173                         <div class="ninite-stat-title">Файл Ninite в P
rogram Files</div>
174                         <div class="ninite-stat-val text-truncate" id=
"ninite-file-status">Проверка...</div>
175                         <div class="text-secondary small mt-1" id="nin
ite-file-details">Путь: C:\Program Files\Ninite\ninite.exe</di
v>
176                     </div>
177                 </div>
178
179                 <div class="col-12">
180                     <div class="ninite-stat-box">
181                         <div class="ninite-stat-title">Задача Windows

```

```

Task Scheduler</div>
182 |         <div class="ninite-stat-val text-truncate" id=
| "ninite-task-status">Проверка...</div>
183 |         <div class="text-secondary small mt-1" id="nin
| ite-task-details">Задача: NiniteAutoUpdate</div>
184 |         </div>
185 |     </div>
186 |
187 |     <div class="col-12">
188 |         <div class="ninite-stat-box">
189 |             <div class="ninite-stat-title">Следующий запуск
| по расписанию</div>
190 |             <div class="ninite-stat-val text-info text-tru
| ncate" id="ninite-task-next">--</div>
191 |             </div>
192 |         </div>
193 |     </div>
194 | </div>
195 |
196 | <div>
197 |     <!-- Кнопка восстановления удаленных файлов (R-Studi
| o) -->
198 |     <button class="btn btn-outline-info w-100 py-2 fw-se
| mibold mb-2 d-flex align-items-center justify-content-center g
| ap-2" id="btn-ninite-recovery-launch" type="button" title="Зап
| уск R-Studio Technician Portable для восстановления данных">
199 |         <i class="bi bi-folder-symlink-fill"></i> Восстано
| вить удаленные файлы
200 |     </button>
201 |
202 |     <!-- Кнопка принудительного обновления -->
203 |     <button class="btn btn-outline-warning w-100 py-2 fw
| -semibold mb-2 d-flex align-items-center justify-content-cente
| r gap-2" id="btn-ninite-run-now">
204 |         <i class="bi bi-play-circle-fill"></i> Запустить о
| бновление прямо сейчас
205 |     </button>
206 |

```

```
207 |         <button class="btn btn-sm btn-outline-secondary w-100" id="btn-ninite-update-schedule-only" type="button">
208 |             <i class="bi bi-sliders"></i> Обновить только расписание (без перезагрузки файла)
209 |         </button>
210 |     </div>
211 | </div>
212 | </div>
213 | </div>
214 |
215 | <!-- Журнал выполнения / лог -->
216 | <div class="ninite-card">
217 |     <div class="d-flex justify-content-between align-items-center mb-2">
218 |         <h6 class="fw-bold text-info mb-0 d-flex align-items-center gap-2">
219 |             <i class="bi bi-terminal-fill"></i> Журнал выполнения обновления (Log)
220 |         </h6>
221 |         <span class="text-secondary small" id="ninite-log-timestamp"></span>
222 |     </div>
223 |     <pre class="p-2.5 rounded bg-dark border border-secondary text-light small mb-0" id="ninite-log-viewer" style="max-height: 180px; overflow-y: auto; font-family: ui-monospace, monospace; white-space: pre-wrap;">Логи обновлений пока отсутствуют.</pre>
224 | </div>
225 | </div>
226 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Ninite Updater Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/ninite_updater_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/ninite_updater_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 /**
22  * ninite_updater_tab/main.js – логика управления установкой и
23  *   автообновлением Ninite
24  */
25 (function () {
26   'use strict';
27
28   let selectedFile = null;
29
30   /**
31    * Настройка обработчиков событий
```

```
32 |    */
33 |    function setupEventListeners() {
34 |        const dropzone = document.getElementById('ninite-dropzone'
    |    );
35 |        const fileInput = document.getElementById('ninite-file-inp
    |    ut');
36 |        const form = document.getElementById('ninite-upload-form')
    |    ;
37 |        const refreshBtn = document.getElementById('btn-ninite-ref
    |    resh');
38 |        const runNowBtn = document.getElementById('btn-ninite-run-
    |    now');
39 |        const scheduleOnlyBtn = document.getElementById('btn-ninit
    |    e-update-schedule-only');
40 |
41 |        if (fileInput) {
42 |            fileInput.addEventListener('change', (e) => {
43 |                if (e.target.files && e.target.files[0]) {
44 |                    handleFileSelect(e.target.files[0]);
45 |                }
46 |            });
47 |        }
48 |
49 |        if (dropzone) {
50 |            dropzone.addEventListener('dragover', (e) => {
51 |                e.preventDefault();
52 |                dropzone.classList.add('dragover');
53 |            });
54 |
55 |            dropzone.addEventListener('dragleave', () => {
56 |                dropzone.classList.remove('dragover');
57 |            });
58 |
59 |            dropzone.addEventListener('drop', (e) => {
60 |                e.preventDefault();
61 |                dropzone.classList.remove('dragover');
62 |                if (e.dataTransfer.files && e.dataTransfer.files[0]) {
63 |                    handleFileSelect(e.dataTransfer.files[0]);
```



```
64 |     }
65 |     });
66 | }
67 |
68 | if (form) {
69 |     form.addEventListener('submit', handleFormSubmit);
70 | }
71 |
72 | if (refreshBtn) {
73 |     refreshBtn.addEventListener('click', () => loadStatus())
74 | ;
75 | }
76 |
77 | if (runNowBtn) {
78 |     runNowBtn.addEventListener('click', handleRunNow);
79 | }
80 |
81 | if (scheduleOnlyBtn) {
82 |     scheduleOnlyBtn.addEventListener('click', handleSchedule
83 | Only);
84 | }
85 |
86 | const recoveryBtn = document.getElementById('btn-ninite-re
87 | covery-launch');
88 | if (recoveryBtn) {
89 |     recoveryBtn.addEventListener('click', handleLaunchRecover
90 | y);
91 | }
92 | }
93 |
94 | /**
95 |  * Запуск R-Studio для восстановления удаленных файлов
96 |  */
97 | async function handleLaunchRecovery() {
98 |     const btn = document.getElementById('btn-ninite-recovery-l
99 | aunch');
100 |     if (btn) {
101 |         btn.disabled = true;
```

```

97 |         btn.innerHTML = `<span class="spinner-border spinner-border-sm me-2"></span>Запуск R-Studio...`;
98 |     }
99 |
100 |    try {
101 |        const res = await fetch('/api/recovery/launch', {
102 |            method: 'POST',
103 |            headers: { 'Content-Type': 'application/json' },
104 |        });
105 |        const data = await res.json();
106 |        if (!res.ok) {
107 |            throw new Error(data.detail || `HTTP ${res.status}`);
108 |        }
109 |
110 |        showAlert(`<strong>Успех:</strong> ${data.message || 'Программа восстановления файлов R-Studio запущена!'}`, 'success');
111 |
112 |        if (window.toast) {
113 |            window.toast.success('R-Studio запущена', data.message || 'Окно программы открывается...');
114 |        }
115 |    } catch (err) {
116 |        showAlert(`<strong>Ошибка запуска:</strong> ${err.message}`, 'danger');
117 |
118 |        if (window.toast) {
119 |            window.toast.error('Ошибка запуска R-Studio', err.message);
120 |        }
121 |    } finally {
122 |        if (btn) {
123 |            btn.disabled = false;
124 |            btn.innerHTML = `<i class="bi bi-folder-symlink-fill"></i> Восстановить удаленные файлы`;
125 |        }
126 |    }
127 |    /**

```

```
128 |     * Обработка выбора файла
129 |     */
130 |     function handleFileSelect(file) {
131 |         selectedFile = file;
132 |         const label = document.getElementById('ninite-file-label')
133 |         ;
134 |         if (label) {
135 |             const sizeMb = (file.size / (1024 * 1024)).toFixed(2);
136 |             label.innerHTML = `📄 Выбран файл: <strong class="text-in
137 | fo">${file.name}</strong> (${sizeMb} MB)`;
138 |         }
139 |     }
140 |
141 |     /**
142 |     * Загрузка текущего статуса файла и задачи в Task Scheduler
143 |     */
144 |     async function loadStatus() {
145 |         const statusBadge = document.getElementById('ninite-status
146 | -badge');
147 |         const fileStatus = document.getElementById('ninite-file-st
148 | atus');
149 |         const fileDetails = document.getElementById('ninite-file-d
150 | etails');
151 |         const taskStatus = document.getElementById('ninite-task-st
152 | atus');
153 |         const taskDetails = document.getElementById('ninite-task-d
154 | etails');
155 |         const taskNext = document.getElementById('ninite-task-next
156 | ');
157 |         const logViewer = document.getElementById('ninite-log-view
158 | er');
159 |         const logTimestamp = document.getElementById('ninite-log-t
160 | imESTAMP');
161 |
162 |         try {
163 |             const res = await fetch(`/api/ninite/status?t=${Date.now
164 | ()}`);
165 |             if (!res.ok) throw new Error(`HTTP ${res.status}`);
```

```
155 |         const data = await res.json();
156 |
157 |         if (data.file) {
158 |             if (data.file.installed) {
159 |                 const sizeMb = (data.file.size_bytes / (1024 * 1024)
160 | ).toFixed(2);
161 |                 fileStatus.innerHTML = `<span class="text-success">●
162 | Установлен</span>`;
163 |                 fileDetails.textContent = `Размер: ${sizeMb} MB | Из
164 | менён: ${data.file.modified_time}`;
165 |             } else {
166 |                 fileStatus.innerHTML = `<span class="text-warning">●
167 | Не установлен</span>`;
168 |                 fileDetails.textContent = `Файл ninite.exe отсутству
169 | ет в Program Files`;
170 |             }
171 |         }
172 |
173 |         if (data.task) {
174 |             if (data.task.exists) {
175 |                 taskStatus.innerHTML = `<span class="text-success">●
176 | Активна (${data.task.state})</span>`;
177 |                 taskDetails.textContent = `Задача: ${data.task.task_
178 | name} | Посл. запуск: ${data.task.last_run_time || 'нет'}`;
179 |                 taskNext.textContent = data.task.next_run_time || 'П
180 | о расписанию';
181 |             } else {
182 |                 taskStatus.innerHTML = `<span class="text-secondary"
183 | >● Не запланирована</span>`;
184 |                 taskDetails.textContent = `Задача NiniteAutoUpdate о
185 | тсутствует в Task Scheduler`;
186 |                 taskNext.textContent = '--';
187 |             }
188 |         }
189 |
190 |         if (statusBadge) {
191 |             if (data.file?.installed && data.task?.exists) {
192 |                 statusBadge.className = 'badge rounded-pill bg-succe
```

```

    ss-subtle text-success border border-success px-2.5 py-1.5';
183     statusBadge.textContent = '● Автообновление активно'
    ;
184     } else if (data.file?.installed) {
185         statusBadge.className = 'badge rounded-pill bg-warni
ng-subtle text-warning border border-warning px-2.5 py-1.5';
186         statusBadge.textContent = '● Файл готов, расписание
не задано';
187     } else {
188         statusBadge.className = 'badge rounded-pill bg-secon
dary-subtle text-light border border-secondary px-2.5 py-1.5';
189         statusBadge.textContent = '● Ожидание установки';
190     }
191 }
192
193     if (logViewer && data.log) {
194         logViewer.textContent = data.log;
195         if (logTimestamp) {
196             logTimestamp.textContent = `Обновлено: ${new Date().
toLocaleTimeString()}`;
197         }
198     }
199 } catch (err) {
200     console.error('Ошибка загрузки статуса Ninite:', err);
201     if (statusBadge) {
202         statusBadge.className = 'badge rounded-pill bg-danger-
subtle text-danger border border-danger px-2.5 py-1.5';
203         statusBadge.textContent = '● Ошибка связи с сервером';
204     }
205 }
206 }
207
208 /**
209  * Отправка формы (загрузка файла + настройка Task Scheduler
)
210  */
211 async function handleFormSubmit(e) {
212     e.preventDefault();

```

```
213
214     if (!selectedFile) {
215         showAlert('Пожалуйста, выберите файл инсталлятора Ninite
для загрузки.', 'warning');
216         return;
217     }
218
219     const interval = document.getElementById('ninite-interval'
)?.value || '2';
220     const timeStr = document.getElementById('ninite-time')?.va
lue || '20:00';
221     const submitBtn = document.getElementById('btn-ninite-subm
it');
222
223     const formData = new FormData();
224     formData.append('file', selectedFile);
225     formData.append('interval_weeks', interval);
226     formData.append('time_str', timeStr);
227
228     if (submitBtn) {
229         submitBtn.disabled = true;
230         submitBtn.innerHTML = `
```

```
245     await loadStatus();
246   } catch (err) {
247     showAlert(`Ошибка: ${err.message}`, 'danger');
248   } finally {
249     if (submitBtn) {
250       submitBtn.disabled = false;
251       submitBtn.innerHTML = `<i class="bi bi-check-circle-fi
ll me-1"></i> Установить в Program Files и активировать Task S
cheduler`;
252     }
253   }
254 }
255
256 /**
257  * Обновление только расписания без перезагрузки файла
258  */
259 async function handleScheduleOnly() {
260   const interval = parseInt(document.getElementById('ninite-
interval')?.value || '2', 10);
261   const timeStr = document.getElementById('ninite-time')?.va
lue || '20:00';
262   const day = document.getElementById('ninite-day')?.value |
| 'Sunday';
263
264   try {
265     const res = await fetch('/api/ninite/schedule', {
266       method: 'POST',
267       headers: { 'Content-Type': 'application/json' },
268       body: JSON.stringify({
269         interval_weeks: interval,
270         time_str: timeStr,
271         days_of_week: day,
272       }),
273     });
274
275     const data = await res.json();
276     if (!res.ok) {
277       throw new Error(data.detail || `HTTP ${res.status}`);
```

```

278 |     }
279 |
280 |     showAlert(`Расписание обновлено: каждые ${interval} нед.
    | в ${timeStr}`, 'success');
281 |     await loadStatus();
282 | } catch (err) {
283 |     showAlert(`Ошибка обновления расписания: ${err.message}`
    | , 'danger');
284 | }
285 | }
286 |
287 | /**
288 |  * Ручной запуск обновления сейчас
289 |  */
290 | async function handleRunNow() {
291 |     const btn = document.getElementById('btn-ninite-run-now');
292 |     if (btn) {
293 |         btn.disabled = true;
294 |         btn.innerHTML = `

```



```

311         btn.innerHTML = `<i class="bi bi-play-circle-fill me-1
    "></i> Запустить обновление прямо сейчас`;
312     }
313 }
314 }
315
316 /**
317  * Отображение информационного сообщения
318  */
319 function showAlert(message, type = 'info') {
320     const box = document.getElementById('ninite-alert-box');
321     if (!box) return;
322
323     box.className = `alert alert-${type} alert-dismissible fade show small`;
324     box.innerHTML = `
325         <div>${message}</div>
326         <button type="button" class="btn-close" data-bs-dismiss=
    "alert" aria-label="Заккрыть"></button>
327     `;
328     box.style.display = 'block';
329 }
330
331 /**
332  * Инициализация вкладки Ninite Updater
333  */
334 function initNiniteUpdaterTab() {
335     setupEventListeners();
336     loadStatus();
337 }
338
339 // Экспорт для жизненного цикла tab-core.js и глобального ко
    нтекста
340 window.initNiniteUpdaterTab = initNiniteUpdaterTab;
341 window.initNiniteupdaterTab = initNiniteUpdaterTab;
342 window.initNinite_updaterTab = initNiniteUpdaterTab;
343
344 // Автоматический запуск при автономной загрузке

```

```
345 |   if (document.readyState === 'loading') {
346 |       document.addEventListener('DOMContentLoaded', initNiniteUp
347 |   daterTab);
348 |   } else {
349 |       initNiniteUpdaterTab();
350 |   }
351 | }));
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/performance_tracing_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/performance_tracing_tab
17         /index.html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/performance_tracing_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Performance & ETW Tracing Tab -->
30 <div class="container-fluid py-2 px-3">
31     <!-- Top Stat Cards -->
32     <div class="row g-2 mb-3">
33         <div class="col-6 col-md-3">
34             <div class="card bg-dark text-white border-secondary h-1
35             00 shadow-sm p-3">
36                 <div class="text-muted small text-uppercase fw-semibol
37                 d">ETW Сессии трассировки □ </div>
```

```
31 |         <div class="fs-4 fw-bold mt-1 text-primary" id="pt-ses  
    | sessions-count">--</div>  
32 |         <div class="small text-muted mt-1" id="pt-sessions-sub  
    | ">logman query -ets</div>  
33 |     </div>  
34 | </div>  
35 |     <div class="col-6 col-md-3">  
36 |         <div class="card bg-dark text-white border-secondary h-1  
    | 00 shadow-sm p-3">  
37 |             <div class="text-muted small text-uppercase fw-semibol  
    | d">Счетчики производительности □ </div>  
38 |             <div class="fs-4 fw-bold mt-1 text-info" id="pt-counte  
    | rs-count">--</div>  
39 |             <div class="small text-muted mt-1" id="pt-counters-sub  
    | ">typeperf / relog</div>  
40 |         </div>  
41 |     </div>  
42 |     <div class="col-6 col-md-3">  
43 |         <div class="card bg-dark text-white border-secondary h-1  
    | 00 shadow-sm p-3">  
44 |             <div class="text-muted small text-uppercase fw-semibol  
    | d">Сбор данных (Data Collectors) □ </div>  
45 |             <div class="fs-4 fw-bold mt-1 text-success" id="pt-col  
    | lectors-count">--</div>  
46 |             <div class="small text-muted mt-1" id="pt-collectors-s  
    | ub">Active Collectors</div>  
47 |         </div>  
48 |     </div>  
49 |     <div class="col-6 col-md-3">  
50 |         <div class="card bg-dark text-white border-secondary h-1  
    | 00 shadow-sm p-3">  
51 |             <div class="text-muted small text-uppercase fw-semibol  
    | d">Буфер событий (Lost Events) □ </div>  
52 |             <div class="fs-4 fw-bold mt-1 text-warning" id="pt-los  
    | t-events">0</div>  
53 |             <div class="small text-muted mt-1" id="pt-lost-sub">ET  
    | W Health Status</div>  
54 |         </div>
```

```

55     </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="pt-refresh-btn">
62             <i class="bi bi-arrow-clockwise"></i>
63             <span>Обновить трассировки</span>
64         </button>
65         <button class="btn btn-sm btn-outline-success d-flex align-items-center gap-1.5" id="pt-quick-sample-btn">
66             <i class="bi bi-activity"></i>
67             <span>Замерить CPU % (typeperf)</span>
68         </button>
69     </div>
70     <div class="d-flex align-items-center gap-2">
71         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="pt-search-input" placeholder="Поиск по сессиям ETW..." style="width: 240px;">
72     </div>
73 </div>
74
75 <!-- Main Grid -->
76 <div class="row g-3">
77     <div class="col-12 col-lg-7">
78         <div class="card bg-dark text-white border-secondary shadow-sm">
79             <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
80                 <span class="fw-semibold"><i class="bi bi-activity text-primary me-2"></i>Активные сессии трассировки (ETW Providers)</span>
81                 <span class="badge bg-secondary" id="pt-badge">0 сессий</span>

```

```

82 |         </div>
83 |         <div class="card-body p-0 table-responsive" style="max
| -height: 420px; overflow-y: auto;">
84 |             <table class="table table-dark table-hover table-str
| iped align-middle mb-0 small" id="pt-table">
85 |                 <thead>
86 |                     <tr class="text-secondary">
87 |                         <th>Имя сессии</th>
88 |                         <th>Тип</th>
89 |                         <th>Размер буфера</th>
90 |                         <th>Статус</th>
91 |                         <th class="text-end">Управление</th>
92 |                     </tr>
93 |                 </thead>
94 |                 <tbody id="pt-tbody">
95 |                     <tr><td colspan="5" class="text-center text-mute
| d py-4">0прос сессий ETW через logman...</td></tr>
96 |                     </tbody>
97 |                 </table>
98 |             </div>
99 |         </div>
100 |     </div>
101 |
102 |     <div class="col-12 col-lg-5">
103 |         <div class="card bg-dark text-white border-secondary sha
| dow-sm mb-3">
104 |             <div class="card-header border-secondary d-flex align-
| items-center justify-content-between py-2 px-3">
105 |                 <span class="fw-semibold"><i class="bi bi-sliders te
| xt-warning me-2"></i>Управление трассировкой (logman / typerper
| f)</span>
106 |             </div>
107 |             <div class="card-body p-3">
108 |                 <div class="mb-3">
109 |                     <label class="form-label small text-muted">Действи
| е трассировщика:</label>
110 |                     <select class="form-select form-select-sm bg-dark
| text-white border-secondary" id="pt-action-select">

```

```
111         <option value="logman_query_ets">Опросить сессии
    трассировки (logman query -ets)</option>
112         <option value="typeperf_cpu_sample">Мгновенный с
    рез загрузки CPU (typeperf "\Processor(_Total)\% Processor Tim
    e" -sc 1)</option>
113         <option value="typeperf_memory_sample">Срез дост
    упной памяти (typeperf "\Memory\Available MBytes" -sc 1)</opti
    on>
114         <option value="typeperf_disk_sample">Срез дисков
    ой очереди (typeperf "\PhysicalDisk(_Total)\Current Disk Queue
    Length" -sc 1)</option>
115     </select>
116 </div>
117 <div class="d-flex align-items-center justify-content-between">
118     <div class="form-check form-switch">
119         <input class="form-check-input" type="checkbox"
    id="pt-dry-run-switch" checked>
120         <label class="form-check-label small text-muted"
    for="pt-dry-run-switch">Безопасный режим (Dry Run)</label>
121     </div>
122     <button class="btn btn-sm btn-success px-3" id="pt
    -execute-btn">
123         <i class="bi bi-play-fill"></i> Запустить замер
124     </button>
125 </div>
126 </div>
127 </div>
128
129 <!-- Action Log Output -->
130 <div class="card bg-dark text-white border-secondary sha
    dow-sm">
131     <div class="card-header border-secondary py-2 px-3 d-f
    lex align-items-center justify-content-between">
132         <span class="small fw-semibold text-muted"><i class=
    "bi bi-terminal me-1"></i>Вывод трассировки</span>
133         <button class="btn btn-link btn-sm text-decoration-n
    one text-muted p-0" id="pt-clear-log-btn">Очистить</button>
```

```
134 |         </div>
135 |         <div class="card-body p-2">
136 |             <pre class="mb-0 small bg-black text-light p-2 rounded" id="pt-console-output" style="max-height: 180px; overflow-y: auto; font-family: Consolas, monospace; font-size: 0.78rem;">Трассировщик готов к сбору метрик.</pre>
137 |         </div>
138 |     </div>
139 | </div>
140 | </div>
141 | </div>
142 |
```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Performance Tracing Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/performance_tracing_ta
11  * b/main.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { initPerformanceTracingTab } from '/windows/api
15  * /webgui/performance_tracing_tab/main.js';
16  *
17  * File: main.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/performance_tracing_tab
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-04 11:15:30
23  * =====
24  */
25
26 /**
27  * performance_tracing_tab/main.js – ETW трассировка и счетчик
28  * и производительности (logman, typeperf)
29  * Updated: 2026-10-04 11:15:30
30  */
31
32 const registerTabPoller = window.registerTabPoller || function
33 () {};
```

```
30 |
31 | let isInitialized = false;
32 | let sessionsList = [];
33 |
34 | export async function fetchPerformanceSummary() {
35 |   try {
36 |     const res = await fetch('/api/performance-tracing/summary'
    | );
37 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38 |     const data = await res.json();
39 |     renderPerformance(data);
40 |   } catch (err) {
41 |     console.warn('[PerformanceTracing] Ошибка получения сводки
    | :', err);
42 |   }
43 | }
44 |
45 | function renderPerformance(data) {
46 |   sessionsList = data.sessions || [
47 |     { name: 'EventLog-Application', type: 'Trace', buffer_size
    | : '64 KB', status: 'Running' },
48 |     { name: 'EventLog-System', type: 'Trace', buffer_size: '64
    | KB', status: 'Running' },
49 |     { name: 'Diagtrack-Listener', type: 'Trace', buffer_size:
    | '128 KB', status: 'Running' }
50 |   ];
51 |
52 |   const sessionsCountEl = document.getElementById('pt-sessions
    | -count');
53 |   const countersCountEl = document.getElementById('pt-counters
    | -count');
54 |   const collectorsCountEl = document.getElementById('pt-collec
    | tors-count');
55 |   const tbody = document.getElementById('pt-tbody');
56 |   const badge = document.getElementById('pt-badge');
57 |
58 |   if (sessionsCountEl) sessionsCountEl.textContent = sessionsL
    | ist.length;
```

```

59 |   if (countersCountEl) countersCountEl.textContent = data.coun
    |   ters_count || '120+';
60 |   if (collectorsCountEl) collectorsCountEl.textContent = data.
    |   collectors_count || '4';
61 |   if (badge) badge.textContent = `${sessionsList.length} сессии`;
62 |
63 |   if (!tbody) return;
64 |
65 |   tbody.innerHTML = sessionsList.map(s => `
66 |       <tr>
67 |           <td class="fw-semibold text-light">${s.name}</td>
68 |           <td><span class="badge bg-secondary">${s.type || 'Trace'
    |   }</span></td>
69 |           <td class="font-monospace small text-muted">${s.buffer_s
    |   ize || '64 KB'}</td>
70 |           <td><span class="badge bg-success">${s.status || 'Runnin
    |   g'}</span></td>
71 |           <td class="text-end">
72 |               <button class="btn btn-xs btn-outline-info py-0 px-2"
    |   title="Свойства сессии">
73 |                   <i class="bi bi-info-circle"></i>
74 |               </button>
75 |           </td>
76 |       </tr>
77 |   `).join('');
78 | }
79 |
80 | async function executeTraceAction(customAction = null) {
81 |     const actionSelect = document.getElementById('pt-action-sele
    |   ct');
82 |     const dryRunSwitch = document.getElementById('pt-dry-run-swi
    |   tch');
83 |     const output = document.getElementById('pt-console-output');
84 |
85 |     const action = customAction || (actionSelect ? actionSelect.
    |   value : 'typeperf_cpu_sample');
86 |     const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;

```

```

87
88   if (output) output.textContent = `[${new Date().toLocaleTime
String()}] Запуск трассировки: ${action}...\n`;
89
90   try {
91     const res = await fetch('/api/performance-tracing/action',
    {
92       method: 'POST',
93       headers: { 'Content-Type': 'application/json' },
94       body: JSON.stringify({ action: action, dry_run: dryRun }
    )
95   });
96   const result = await res.json();
97   if (output) output.textContent += JSON.stringify(result, n
ull, 2);
98   if (window.toast) {
99     if (res.ok) window.toast.success('Трассировка', `Замер в
    ыполнен: ${result.status || 'ОК'}`);
100    else window.toast.error('Ошибка', result.detail || 'Сбой
    замера');
101  }
102  } catch (err) {
103    if (output) output.textContent += `\nОшибка запроса: ${err
    .message}`;
104    if (window.toast) window.toast.error('Ошибка сети', err.me
    ssage);
105  }
106 }
107
108 export function initPerformanceTracingTab() {
109   if (isInitialized) return;
110   isInitialized = true;
111
112   const refreshBtn = document.getElementById('pt-refresh-btn')
    ;
113   if (refreshBtn) refreshBtn.addEventListener('click', fetchPe
    rformanceSummary);
114

```

```
115 |   const sampleBtn = document.getElementById('pt-quick-sample-b
    |   tn');
116 |   if (sampleBtn) sampleBtn.addEventListener('click', () => exe
    |   cuteTraceAction('typeperf_cpu_sample'));
117 |
118 |   const execBtn = document.getElementById('pt-execute-btn');
119 |   if (execBtn) execBtn.addEventListener('click', () => execute
    |   TraceAction());
120 |
121 |   const clearBtn = document.getElementById('pt-clear-log-btn')
    |   ;
122 |   if (clearBtn) clearBtn.addEventListener('click', () => {
123 |       const out = document.getElementById('pt-console-output');
124 |       if (out) out.textContent = 'Консоль очищена.';
125 |   });
126 |
127 |   registerTabPoller('tab-performance-tracing', fetchPerformanc
    |   eSummary, 15000, { immediate: true });
128 | }
129 |
130 | window.initPerformanceTracingTab = initPerformanceTracingTab;
131 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 Usage Examples:
9 |Web Route:
10 |/windows/api/webgui/peripherals_tab/index.html
11 |
12 |Tab Frame Integration:
13 |<iframe src="/windows/api/webgui/peripherals_tab/index.h
14 |tml" class="w-100 h-100 border-0"></iframe>
15 |
16 |* File: index.html
17 |* Project: ai-breadboard
18 |* Package: windows/api/webgui/peripherals_tab
19 |* Author: hypo69
20 |* Copyright: © 2026 hypo69
21 |* Updated: 2026-10-08 03:08:00
22 |* =====
23 |=====
24 -->
25
26 <style>
27 |.peripherals-tab-wrapper {
28 |padding: 0.75rem;
29 |color: var(--text-color, #f8fafc);
30 |}
31 |.diag-panel {
32 |background: var(--surface-2, #1e293b);
33 |border: 1px solid var(--border-color, #334155);
34 |border-radius: 8px;
```

```
33     overflow: hidden;
34 }
35 .diag-panel-header {
36     background: rgba(15, 23, 42, 0.6);
37     padding: 8px 14px;
38     border-bottom: 1px solid var(--border-color, #334155);
39     font-weight: 600;
40     font-size: 0.85rem;
41 }
42 .about-sys-spec-table {
43     width: 100%;
44     border-collapse: separate;
45     border-spacing: 0;
46 }
47 .about-sys-spec-table td {
48     padding: 8px 12px;
49     border-bottom: 1px solid rgba(51, 65, 85, 0.35);
50     vertical-align: middle;
51 }
52 .about-sys-spec-key {
53     color: var(--text-muted, #94a3b8);
54     width: 200px;
55     font-size: 0.82rem;
56     font-weight: 500;
57 }
58 .about-sys-spec-val {
59     color: var(--text-color, #f8fafc);
60     font-weight: 500;
61     font-size: 0.82rem;
62 }
63 </style>
64
65 <div class="peripherals-tab-wrapper">
66     <!-- Top Header & Status -->
67     <div class="card shadow-sm border-secondary-subtle mb-3">
68         <div class="card-body p-3">
69             <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
```

```

70         <div class="d-flex align-items-center gap-2">
71             <div class="fs-4">□ </div>
72             <div>
73                 <h5 class="mb-0 fw-bold">5. Сеть и периферия (Peri
pherals & Wireless Network)</h5>
74                 <small class="text-muted">RF-параметры Wi-Fi сетей
, аудио конечные точки и Plug-and-Play USB устройства</small>
75             </div>
76         </div>
77
78         <div class="d-flex align-items-center gap-2 flex-wrap"
>
79             <select class="form-select form-select-sm poll-freq-
select" id="diag-peripherals-poll-freq" data-poll-id="peripher
als" title="Частота опроса: Сеть и периферия">
80                 <option value="start">Только при старте</option>
81                 <option value="manual" selected>По запросу</option>
>
82                 <option value="3">Каждые 3 сек</option>
83                 <option value="5">Каждые 5 сек</option>
84                 <option value="10">Каждые 10 сек</option>
85                 <option value="30">Каждые 30 сек</option>
86                 <option value="60">Каждые 60 сек</option>
87                 <option value="120">Каждые 120 сек</option>
88             </select>
89             <button class="btn btn-sm btn-outline-info rounded-p
ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
"btn-diag-peripherals-refresh">
90                 <i class="bi bi-arrow-clockwise"></i> Обновить
91             </button>
92         </div>
93     </div>
94 </div>
95 </div>
96
97 <div class="row g-2 mb-3">
98     <!-- Wi-Fi Telemetry Panel -->
99     <div class="col-12 col-lg-5">

```



```

100 |     <div class="diag-panel h-100">
101 |         <div class="diag-panel-header d-flex justify-content-b
| etween align-items-center">
102 |             <div class="d-flex align-items-center gap-2">
103 |                 <i class="bi bi-wifi text-info"></i>
104 |                 <span class="text-white">Беспроводная сеть Wi-Fi (
| RF & Security)</span>
105 |             </div>
106 |             <span class="badge bg-success-subtle text-success bo
| rder border-success" id="diag-wifi-status-badge">Wi-Fi Active<
| /span>
107 |         </div>
108 |         <div class="p-3">
109 |             <table class="about-sys-spec-table">
110 |                 <tbody>
111 |                     <tr>
112 |                         <td class="about-sys-spec-key"><i class="bi bi
| -broadcast me-1.5 text-info"></i>Сеть (SSID)</td>
113 |                         <td class="about-sys-spec-val fw-bold text-whi
| te" id="diag-wifi-ssid">--</td>
114 |                     </tr>
115 |                     <tr>
116 |                         <td class="about-sys-spec-key"><i class="bi bi
| -router me-1.5 text-primary"></i>Точка доступа (BSSID)</td>
117 |                         <td class="about-sys-spec-val font-monospace"
| id="diag-wifi-bssid">--</td>
118 |                     </tr>
119 |                     <tr>
120 |                         <td class="about-sys-spec-key"><i class="bi bi
| -reception-4 me-1.5 text-success"></i>Сигнал (RSSI)</td>
121 |                         <td class="about-sys-spec-val" id="diag-wifi-s
| ignal">0% (-- dBm)</td>
122 |                     </tr>
123 |                     <tr>
124 |                         <td class="about-sys-spec-key"><i class="bi bi
| -hash me-1.5 text-warning"></i>Радиоканал</td>
125 |                         <td class="about-sys-spec-val" id="diag-wifi-c
| hannel">--</td>

```

```

126 |             </tr>
127 |             <tr>
128 |                 <td class="about-sys-spec-key"><i class="bi bi
| -cpu me-1.5 text-info"></i>Стандарт связи</td>
129 |                 <td class="about-sys-spec-val font-monospace"
| id="diag-wifi-radio">--</td>
130 |             </tr>
131 |             <tr>
132 |                 <td class="about-sys-spec-key"><i class="bi bi
| -shield-lock me-1.5 text-success"></i>Безопасность</td>
133 |                 <td class="about-sys-spec-val text-truncate" i
| d="diag-wifi-auth">--</td>
134 |             </tr>
135 |         </tbody>
136 |     </table>
137 | </div>
138 | </div>
139 | </div>
140 |
141 | <!-- Audio Endpoints Panel -->
142 | <div class="col-12 col-lg-7">
143 |     <div class="diag-panel h-100">
144 |         <div class="diag-panel-header d-flex justify-content-b
| etween align-items-center">
145 |             <div class="d-flex align-items-center gap-2">
146 |                 <i class="bi bi-volume-up-fill text-warning"></i>
147 |                 <span class="text-white">Аудиоустройства и конечны
| е точки (Audio Endpoints)</span>
148 |             </div>
149 |             <span class="badge bg-secondary" id="diag-audio-coun
| t">0 устройств</span>
150 |         </div>
151 |         <div class="p-2" id="diag-audio-container" style="max-
| height: 260px; overflow-y: auto;">
152 |             <div class="text-center py-4 text-muted small">Опрос
| аудиоустройств...</div>
153 |         </div>
154 |     </div>

```

```

155     </div>
156 </div>
157
158 <!-- PnP USB Devices Table -->
159 <div class="diag-panel">
160     <div class="diag-panel-header d-flex justify-content-between align-items-center">
161         <div class="d-flex align-items-center gap-2">
162             <i class="bi bi-usb-symbol text-info"></i>
163             <span class="text-white">Подключенные PnP USB-устройст
164 ва и концентраторы</span>
165         </div>
166         <span class="badge bg-secondary" id="diag-usb-count">0 у
167 стройств</span>
168     </div>
169     <div class="table-responsive" style="max-height: 480px; ov
170 erflow-y: auto;">
171         <table class="table table-dark table-hover table-sm mb-0
172 " style="font-size: 0.80rem; font-family: ui-monospace, SFMono
173 -Regular, Menlo, Monaco, Consolas, monospace;">
174             <thead>
175                 <tr class="text-secondary border-secondary">
176                     <th>Устройство</th>
177                     <th style="width: 100px;">Vendor ID</th>
178                     <th style="width: 100px;">Product ID</th>
179                     <th style="width: 160px;">Класс устройства</th>
180                     <th style="width: 140px;">Статус</th>
181                     <th>Device Instance ID</th>
182                 </tr>
183             </thead>
184             <tbody id="diag-usb-tbody">
185                 <tr><td colspan="6" class="text-center py-4 text-mut
186 ed">Загрузка USB периферии...</td></tr>
187             </tbody>
188         </table>
189     </div>
190 </div>
191 </div>

```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Peripherals Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/peripherals_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/peripherals_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-08 02:00:00
18  * =====
19  */
20
21 (function () {
22   'use strict';
23
24   let autoRefreshTimer = null;
25
26   function escapeHtml(str) {
27     if (str === null || str === undefined) return '';
28     return String(str)
29       .replace(/&/g, '&amp;')
30       .replace(/</g, '&lt;')
31       .replace(/>/g, '&gt;')
32       .replace(/"/g, '&quot;')
```

```

33 |     .replace(/'/g, '&#039;');
34 | }
35 |
36 | function setText(id, text) {
37 |     const el = document.getElementById(id);
38 |     if (el) el.textContent = text !== null && text !== undefin
    | ed ? String(text) : '--';
39 | }
40 |
41 | async function apiFetch(url) {
42 |     if (window.api && typeof window.api.fetch === 'function')
    | {
43 |         return await window.api.fetch(url);
44 |     }
45 |     const res = await fetch(url);
46 |     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
    | atusText}`);
47 |     return await res.json();
48 | }
49 |
50 | async function fetchPeripheralsNetwork() {
51 |     try {
52 |         const data = await apiFetch('/api/v1/system/diagnostics/
    | peripherals');
53 |         if (!data) return;
54 |
55 |         // Wi-Fi
56 |         if (data.wifi_telemetry) {
57 |             const w = data.wifi_telemetry;
58 |             setText('diag-wifi-ssid', w.ssid || 'Не подключено');
59 |             setText('diag-wifi-bssid', w.bssid || '--');
60 |             setText('diag-wifi-signal', `${w.signal_pct}% (${w.rss
    | i_dbm} dBm)`);
61 |             setText('diag-wifi-channel', w.channel || '--');
62 |             setText('diag-wifi-radio', w.radio_type || '--');
63 |             setText('diag-wifi-auth', w.auth_cipher || '--');
64 |
65 |             const wifiBadge = document.getElementById('diag-wifi-s

```

```

    tatus-badge');
66     if (wifiBadge) {
67         if (w.is_connected) {
68             wifiBadge.className = 'badge bg-success-subtle text-success border border-success';
69             wifiBadge.textContent = 'Wi-Fi Подключено';
70         } else {
71             wifiBadge.className = 'badge bg-secondary';
72             wifiBadge.textContent = 'Wi-Fi Отключено';
73         }
74     }
75 }
76
77 // Audio Endpoints
78 const audioContainer = document.getElementById('diag-audio-container');
79 if (audioContainer && data.audio_endpoints) {
80     setText('diag-audio-count', `${data.audio_endpoints.length} устр.`);
81     if (data.audio_endpoints.length === 0) {
82         audioContainer.innerHTML = '<div class="text-center py-4 text-muted small">Аудиоустройств не обнаружено</div>';
83     } else {
84         audioContainer.innerHTML = data.audio_endpoints.map(
85             a => `
86                 <div class="p-2 mb-1 rounded bg-black bg-opacity-30 border border-secondary-subtle d-flex align-items-center justify-content-between">
87                     <div>
88                         <strong class="text-white"><i class="bi bi-speaker me-1.5 text-warning"></i>${escapeHtml(a.name)}</strong>
89                         <div class="small text-muted">${escapeHtml(a.type)}</div>
90                     </div>
91                     <span class="badge ${a.is_default ? 'bg-info text-dark fw-bold' : 'bg-secondary'}">${a.is_default ? 'По умолчанию' : 'Доступно'}</span>
92                 </div>

```

```

92 |         `).join('');
93 |     }
94 | }
95 |
96 | // USB PnP Table
97 | const usbTbody = document.getElementById('diag-usb-tbody
| ');
98 | if (usbTbody && data.usb_devices) {
99 |     setText('diag-usb-count', `${data.usb_devices.length}
| устройств`);
100 |     if (data.usb_devices.length === 0) {
101 |         usbTbody.innerHTML = '<tr><td colspan="6" class="tex
| t-center py-4 text-muted">USB устройств не обнаружено</td></tr
| >';
102 |     } else {
103 |         usbTbody.innerHTML = data.usb_devices.map(u => `
104 |             <tr>
105 |                 <td class="fw-bold text-white"><i class="bi bi-u
| sb-symbol text-info me-1.5"></i>${escapeHtml(u.name)}</td>
106 |                 <td class="font-monospace text-warning">${escape
| Html(u.vendor_id)}</td>
107 |                 <td class="font-monospace text-info">${escapeHtm
| l(u.product_id)}</td>
108 |                 <td class="text-muted small">${escapeHtml(u.devi
| ce_class || 'USB Device')}</td>
109 |                 <td><span class="badge ${u.has_problem ? 'bg-dan
| ger' : 'bg-success-subtle text-success border border-success'}
| ">${escapeHtml(u.status)}</span></td>
110 |                 <td class="small text-muted font-monospace text-
| truncate" style="max-width: 220px;" title="${escapeHtml(u.devi
| ce_id)}">${escapeHtml(u.device_id)}</td>
111 |             </tr>
112 |         `).join('');
113 |     }
114 | }
115 | } catch (err) {
116 |     console.warn('[PeripheralsTab] fetchPeripheralsNetwork e
| rror:', err);

```



```

117 |     }
118 | }
119 |
120 | const POLL_ID = 'peripherals';
121 |
122 | function getFrequency() {
123 |     try {
124 |         const saved = localStorage.getItem(`poll_freq_${POLL_ID}
125 | `);
126 |         if (saved) return saved;
127 |     } catch (_) {}
128 |     return 'manual';
129 | }
130 |
131 | function setFrequency(freq) {
132 |     try {
133 |         localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
134 |     } catch (_) {}
135 |     applyPoller(freq, false);
136 | }
137 |
138 | function stopPolling() {
139 |     if (window.unregisterTabPoller) {
140 |         window.unregisterTabPoller(`tab-peripherals_${POLL_ID}`)
141 | ;
142 |     }
143 |     if (autoRefreshTimer) {
144 |         clearInterval(autoRefreshTimer);
145 |         autoRefreshTimer = null;
146 |     }
147 | }
148 |
149 | function applyPoller(freq, runInitial = false) {
150 |     stopPolling();
151 |     if (freq === 'start' || freq === 'manual') {
152 |         if (runInitial && (window.isTabActive ? window.isTabActi
153 | ve('tab-peripherals') : true)) {
154 |             fetchPeripheralsNetwork();

```

```
152 |     }
153 |     return;
154 | }
155 |
156 | const intervalSec = parseInt(freq, 10);
157 | if (isNaN(intervalSec) || intervalSec <= 0) return;
158 |
159 | const intervalMs = intervalSec * 1000;
160 | const pollerId = `tab-peripherals_${POLL_ID}`;
161 |
162 | if (window.registerTabPoller) {
163 |     window.registerTabPoller('tab-peripherals', fetchPeripheralsNetwork, intervalMs, { pollerId, immediate: runInitial });
164 | } else {
165 |     if (runInitial && (window.isTabActive ? window.isTabActive('tab-peripherals') : true)) {
166 |         fetchPeripheralsNetwork();
167 |     }
168 |     autoRefreshTimer = setInterval(() => {
169 |         if (window.isTabActive ? window.isTabActive('tab-peripherals') : true) {
170 |             fetchPeripheralsNetwork();
171 |         }
172 |     }, intervalMs);
173 | }
174 | }
175 |
176 | function bindEvents() {
177 |     const btnRefresh = document.getElementById('btn-diag-peripherals-refresh');
178 |     if (btnRefresh) {
179 |         btnRefresh.onclick = () => fetchPeripheralsNetwork();
180 |     }
181 |
182 |     const select = document.getElementById('diag-peripherals-poll-freq');
183 |     if (select) {
184 |         select.onchange = (e) => {
```

```
185 |         const newFreq = e.target.value;
186 |         setFrequency(newFreq);
187 |         if (newFreq !== 'manual' && newFreq !== 'start') {
188 |             fetchPeripheralsNetwork();
189 |         }
190 |         if (window.showToast) {
191 |             const label = select.options[select.selectedIndex]?.
| text || newFreq;
192 |             window.showToast(`Частота опроса периферии: ${label}
| `, 'info');
193 |         }
194 |     };
195 | }
196 | }
197 |
198 | async function initPeripheralsTab() {
199 |     bindEvents();
200 |     const currentFreq = getFrequency();
201 |     const select = document.getElementById('diag-peripherals-p
| oll-freq');
202 |     if (select) select.value = currentFreq;
203 |     applyPoller(currentFreq, true);
204 | }
205 |
206 | window.initPeripheralsTab = initPeripheralsTab;
207 | window.activatePeripheralsTab = () => {
208 |     applyPoller(getFrequency(), true);
209 | };
210 |
211 | if (document.readyState === 'loading') {
212 |     document.addEventListener('DOMContentLoaded', () => {
213 |         if (window.isTabActive ? window.isTabActive('tab-periphe
| rals') : false) {
214 |             initPeripheralsTab();
215 |         } else {
216 |             bindEvents();
217 |         }
218 |     });
```

```
219 |     } else {
220 |         if (window.isTabActive ? window.isTabActive('tab-periphere
    | ls') : false) {
221 |             initPeripheralsTab();
222 |         } else {
223 |             bindEvents();
224 |         }
225 |     }
226 | })();
227 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/pixelrag_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/pixelrag_tab/index.html
15 |        " class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/pixelrag_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!DOCTYPE html>
28 <html lang="ru">
29 <head>
30 |    <meta charset="UTF-8">
31 |    <meta name="viewport" content="width=device-width, initial-s
32 |    cale=1.0">
33 |    <title>PixelRAG</title>
34 </head>
35 <body>
```

```
32 | <div class="container-fluid p-3">
33 |   <h4><i class="bi bi-image-fill text-info"></i> PixelRAG</h
34 |   4>
35 |   <p class="text-muted">Векторный поиск по изображениям и пи
36 |   ксельным данным.</p>
37 |   <div class="card border-secondary mb-3">
38 |     <div class="card-header bg-dark text-white">
39 |       <i class="bi bi-search"></i> Поиск по изображениям
40 |     </div>
41 |     <div class="card-body">
42 |       <p class="text-muted">Функционал PixelRAG будет доступ
43 |       ен после настройки.</p>
44 |     </div>
45 |   </div>
46 | </div>
47 | </body>
48 | </html>
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Pixelrag Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/pixelrag_tab/main.js?v
11  *     =20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/pixelrag_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // PixelRAG Tab Module
22 // =====
23
24 export async function initPixelRagTab() {
25   console.log('[PixelRagTab] Initializing...');
26
27   try {
28     // PixelRAG initialization logic
29     const container = document.getElementById('tab-pixelrag');
30     if (container) {
31       container.innerHTML = `
```

```

32 |         <div class="alert alert-info">
33 |             <i class="bi bi-info-circle"></i>
34 |             <strong>PixelRAG</strong> – векторный поиск по изобра
ажениям и пиксельным данным.
35 |         </div>
36 |         <div class="card border-secondary">
37 |             <div class="card-header bg-dark text-white">
38 |                 <i class="bi bi-image-fill"></i> Настройки PixelRA
G
39 |             </div>
40 |             <div class="card-body">
41 |                 <p class="text-muted">Функционал PixelRAG будет до
ступен после настройки модели и индекса.</p>
42 |                 <button class="btn btn-primary" disabled>Настроить
PixelRAG</button>
43 |             </div>
44 |         </div>
45 |     `;
46 |     }
47 | } catch (err) {
48 |     console.error('[PixelRagTab] Initialization error:', err);
49 | }
50 | }

```


index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/plugins_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/plugins_tab/index.html"
17         class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/plugins_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Plugins Tab -->
30 <div class="container-fluid py-2">
31     <div class="alert alert-primary d-flex flex-wrap align-items
32     -center justify-content-between gap-2 mb-3 shadow-sm">
33         <div class="d-flex align-items-center gap-2">
34             <span class="fs-4">□ </span>
35             <div>
36                 <h5 class="mb-0 fw-bold">Подключатель плагинов и расши
37 рений</h5>
```

```

31 |         <p class="mb-0 small text-muted">Централизованный ката
    | лог модулей, активация и органы динамической настройки</p>
32 |     </div>
33 | </div>
34 | <div class="d-flex gap-2">
35 |     <button class="btn btn-sm btn-outline-primary" id="btn-r
    | efresh-plugins" onclick="loadPluginsList()">
36 |         <i class="bi bi-arrow-clockwise"></i> Обновить список
37 |     </button>
38 | </div>
39 | </div>
40 |
41 | <div class="row g-3">
42 |     <!-- LEFT: Plugin Catalog / List -->
43 |     <div class="col-lg-4 col-md-5">
44 |         <div class="card shadow-sm h-100 border-secondary-subtle
    | ">
45 |             <div class="card-header bg-body-tertiary d-flex justif
    | y-content-between align-items-center py-2">
46 |                 <span class="fw-semibold small text-uppercase">Устан
    | овленные плагины (<span id="plugins-count">0</span>)</span>
47 |                 <span class="badge bg-secondary-subtle text-body" id
    | ="active-plugins-badge">0 активных</span>
48 |             </div>
49 |             <div class="p-2 border-bottom">
50 |                 <div class="input-group input-group-sm">
51 |                     <span class="input-group-text bg-body border-seco
    | ndary-subtle"><i class="bi bi-search"></i></span>
52 |                     <input type="text" class="form-control border-seco
    | ndary-subtle" id="plugin-search-input" placeholder="Поиск плаг
    | ина..." oninput="filterPluginsList()">
53 |                 </div>
54 |             </div>
55 |             <div class="card-body p-2" id="plugins-list-container"
    | style="max-height: 70vh; overflow-y: auto;">
56 |                 <div class="text-center text-muted p-4">
57 |                     <div class="spinner-border spinner-border-sm text-
    | primary mb-2" role="status"></div>

```

```

58         <div>Загрузка плагинов...</div>
59     </div>
60 </div>
61 </div>
62 </div>
63
64 <!-- RIGHT: Selected Plugin Controls & Settings -->
65 <div class="col-lg-8 col-md-7">
66     <div class="card shadow-sm h-100 border-secondary-subtle
67     " id="plugin-details-card">
68         <div class="card-body d-flex flex-column align-items-c
69         enter justify-content-center text-muted p-5" id="plugin-placeh
70         older">
71             <i class="bi bi-puzzle fs-1 mb-2"></i>
72             <h6>Выберите плагин из списка слева для управления и
73             настройки</h6>
74         </div>
75
76         <div class="d-none d-flex flex-column h-100" id="plugi
77         n-content-pane">
78             <!-- Plugin Header -->
79             <div class="card-header bg-body-tertiary d-flex just
80             ify-content-between align-items-center py-3 border-bottom">
81                 <div class="d-flex align-items-center gap-3">
82                     <span class="fs-2" id="plugin-icon">□ </span>
83                     <div>
84                         <div class="d-flex align-items-center gap-2">
85                             <h5 class="mb-0 fw-bold" id="plugin-title">На
86                             звание плагина</h5>
87                             <span class="badge bg-primary-subtle text-pr
88                             imary border border-primary-subtle" id="plugin-version">v1.0.0
89                             </span>
90                             <span class="badge bg-secondary-subtle text-
91                             body border" id="plugin-category">tools</span>
92                             <span class="badge bg-info-subtle text-info-
93                             emphasis border" id="plugin-scope-badge">Системный</span>
94                         </div>
95                         <small class="text-muted" id="plugin-id">id: p

```

```
    lugin_name</small>
85 |         </div>
86 |     </div>
87 |     <div class="form-check form-switch fs-5 mb-0" titl
    e="Включить / отключить плагин">
88 |         <input class="form-check-input" type="checkbox"
    role="switch" id="plugin-toggle-switch" onchange="toggleCurren
    tPlugin(this.checked)">
89 |     </div>
90 | </div>
91 |
92 |     <!-- Description -->
93 |     <div class="p-3 bg-body-secondary border-bottom">
94 |         <p class="mb-0 small text-body" id="plugin-descrip
    tion">Описание возможностей плагина.</p>
95 |     </div>
96 |
97 |     <!-- Body: Settings & Actions -->
98 |     <div class="card-body p-3" style="max-height: 52vh;
    overflow-y: auto;">
99 |         <!-- Actions Section -->
100 |         <div class="mb-4" id="plugin-actions-section">
101 |             <h6 class="fw-bold mb-2 pb-1 border-bottom d-fle
    x align-items-center gap-2">
102 |                 <i class="bi bi-lightning-charge-fill text-war
    ning"></i> Управляющие действия (Actions)
103 |             </h6>
104 |             <div class="d-flex flex-wrap gap-2 pt-1" id="plu
    gin-actions-container">
105 |                 <!-- Action buttons rendered here -->
106 |                 </div>
107 |             </div>
108 |
109 |         <!-- Configuration Section -->
110 |         <div class="mb-4" id="plugin-config-section">
111 |             <h6 class="fw-bold mb-2 pb-1 border-bottom d-fle
    x align-items-center gap-2">
112 |                 <i class="bi bi-sliders text-primary"></i> Нап
```

```

аметры конфигурации
113         </h6>
114         <form id="plugin-config-form" onsubmit="event.pr
eventDefault(); saveCurrentPluginConfig();">
115         <div class="row g-3" id="plugin-fields-contain
er">
116             <!-- Form fields rendered here -->
117         </div>
118         <div class="mt-3 text-end">
119             <button type="submit" class="btn btn-sm btn-
primary px-4" id="btn-save-plugin-config">
120                 <i class="bi bi-check2-circle"></i> Сохран
ить настройки
121             </button>
122         </div>
123     </form>
124 </div>
125
126     <!-- Live Action Console / Terminal -->
127     <div class="mt-3">
128         <div class="d-flex justify-content-between align
-items-center mb-1">
129             <h6 class="fw-bold mb-0 small text-muted d-fle
x align-items-center gap-1">
130                 <i class="bi bi-terminal"></i> Консоль выпол
нения
131             </h6>
132             <button class="btn btn-sm btn-link text-decora
tion-none text-muted py-0" onclick="clearPluginConsole()">Очис
тить</button>
133         </div>
134         <pre class="p-2 rounded bg-dark text-light small
font-monospace mb-0" id="plugin-console-output" style="max-he
ight: 180px; min-height: 60px; overflow-y: auto;">Готов к рабо
те.</pre>
135     </div>
136 </div>
137 </div>

```

```
138 |         </div>
139 |     </div>
140 | </div>
141 | </div>
142 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Plugins Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/plugins_tab/main.js?v=
11 *       20261001_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/plugins_tab
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-01 13:04:40
19 * =====
20 */
21 // — PLUGINS TAB MAIN.JS —————
22
23 let loadedPlugins = [];
24 let selectedPluginName = '';
25
26 // Notification helper
27 function showPluginNotification(message, type = 'info') {
28   if (typeof window.showNotification === 'function') {
29     window.showNotification(message, type);
30     return;
31   }
```

```

32 | console.log(`[Plugin Notification ${type}]: ${message}`);
33 | }
34 |
35 | // Helper to check if current view is user interface
36 | function isRunningInUserInterface() {
37 |     if (document.getElementById('user-interface')) return true;
38 |     if (window.location.pathname.includes('/user')) return true;
39 |     if (!document.getElementById('admin-interface')) {
40 |         if (typeof window.isUserAdmin === 'function') {
41 |             return !window.isUserAdmin();
42 |         }
43 |         return true;
44 |     }
45 |     return false;
46 | }
47 |
48 | // Initialize Plugins Tab
49 | window.initPluginsTab = async function() {
50 |     console.log('Инициализация вкладки плагинов...');
51 |     await loadPluginsList();
52 | };
53 |
54 | // Загрузка списка всех плагинов
55 | async function loadPluginsList() {
56 |     const container = document.getElementById('plugins-list-container');
57 |     if (!container) return;
58 |
59 |     try {
60 |         const isUserUI = isRunningInUserInterface();
61 |         const endpoint = isUserUI ? '/api/plugins?scope=user' : '/api/admin/plugins';
62 |
63 |         let data;
64 |         if (window.api && typeof window.api.fetch === 'function') {
65 |             try {
66 |                 data = await window.api.fetch(endpoint);

```



```

67 |     } catch {
68 |         data = await window.api.fetch('/api/admin/plugins');
69 |     }
70 | } else {
71 |     let resp = await fetch(endpoint);
72 |     if (!resp.ok && isUserUI) {
73 |         resp = await fetch('/api/admin/plugins?scope=user');
74 |     }
75 |     if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
76 |     data = await resp.json();
77 | }
78 | let plugins = data.plugins || [];
79 | if (isUserUI) {
80 |     plugins = plugins.filter(p => !p.is_system && p.scope !=
    = 'system');
81 | }
82 | loadedPlugins = plugins;
83 |
84 | const countEl = document.getElementById('plugins-count');
85 | if (countEl) countEl.textContent = loadedPlugins.length;
86 | const activeCount = loadedPlugins.filter(p => p.enabled).l
    ength;
87 | const badgeEl = document.getElementById('active-plugins-ba
    dge');
88 | if (badgeEl) badgeEl.textContent = `${activeCount} активны
    x`;
89 |
90 | renderPluginsList(loadedPlugins);
91 | syncPluginTabsVisibility(loadedPlugins);
92 |
93 | if (loadedPlugins.length > 0) {
94 |     const toSelect = loadedPlugins.find(p => p.name === sele
    ctedPluginName) || loadedPlugins[0];
95 |     selectPlugin(toSelect.name);
96 | } else {
97 |     showPlaceholder();
98 | }
99 | } catch (ex) {

```

```

100     console.error('Ошибка загрузки плагинов:', ex);
101     container.innerHTML = `
```

```

    nded mb-1 border-0 ${isSelected ? 'active text-white' : ''}"
130     onclick="selectPlugin('${p.name}')"
131     style="cursor: pointer;">
132     <div class="d-flex w-100 justify-content-between align
-items-center mb-1">
133         <div class="d-flex align-items-center gap-2 text-tru
ncate">
134             <span class="fs-5">${p.icon || '☐'}</span>
135             <strong class="text-truncate">${p.title || p.name}
</strong>
136         </div>
137         <div class="d-flex align-items-center gap-1">${typeB
adge}${statusBadge}</div>
138     </div>
139     <div class="small text-truncate ${isSelected ? 'text-w
hite-50' : 'text-muted'}">
140         ${p.description || p.name}
141     </div>
142 </a>
143 `;
144 });
145 html += '</div>';
146 container.innerHTML = html;
147 }
148
149 // Фильтрация списка плагинов
150 function filterPluginsList() {
151     const query = (document.getElementById('plugin-search-input'
)?.value || '').toLowerCase().trim();
152     if (!query) {
153         renderPluginsList(loadedPlugins);
154         return;
155     }
156     const filtered = loadedPlugins.filter(p =>
157         p.name.toLowerCase().includes(query) ||
158         (p.title && p.title.toLowerCase().includes(query)) ||
159         (p.description && p.description.toLowerCase().includes(que
ry))

```

```

160 | );
161 | renderPluginsList(filtered);
162 | }
163 |
164 | // Выбор плагина для отображения органов управления
165 | function selectPlugin(pluginName) {
166 |   if (!pluginName) return;
167 |   selectedPluginName = pluginName;
168 |   const normalized = pluginName.toLowerCase().replace(/-/g, '_');
169 |   const plugin = loadedPlugins.find(p => {
170 |     const pName = (p.name || '').toLowerCase().replace(/-/g, '_');
171 |     const pId = (p.id || '').toLowerCase().replace(/-/g, '_');
172 |     const pTitle = (p.title || '').toLowerCase();
173 |     return pName === normalized || pId === normalized || pTitle === pluginName.toLowerCase();
174 |   });
175 |   if (!plugin) {
176 |     if (!loadedPlugins || loadedPlugins.length === 0) {
177 |       loadPluginsList().then(() => {
178 |         if (loadedPlugins && loadedPlugins.length > 0) {
179 |           const pFound = loadedPlugins.find(p => {
180 |             const pName = (p.name || '').toLowerCase().replace(/-/g, '_');
181 |             const pId = (p.id || '').toLowerCase().replace(/-/g, '_');
182 |             const pTitle = (p.title || '').toLowerCase();
183 |             return pName === normalized || pId === normalized || pTitle === pluginName.toLowerCase();
184 |           });
185 |           if (pFound) selectPlugin(pFound.name);
186 |         }
187 |       });
188 |     }
189 |     return;
190 |   }
191 |

```

```
192 |   renderPluginsList(loadedPlugins);
193 |
194 |   const placeholder = document.getElementById('plugin-placehol
    |   der');
195 |   const contentPane = document.getElementById('plugin-content-
    |   pane');
196 |   if (placeholder) placeholder.classList.add('d-none');
197 |   if (contentPane) contentPane.classList.remove('d-none');
198 |
199 |   // Header info
200 |   document.getElementById('plugin-icon').textContent = plugin.
    |   icon || ' ';
201 |   document.getElementById('plugin-title').textContent = plugin
    |   .title || plugin.name;
202 |   document.getElementById('plugin-version').textContent = `v${
    |   plugin.version || '1.0.0'}`;
203 |   document.getElementById('plugin-category').textContent = plu
    |   gin.category || 'tools';
204 |   document.getElementById('plugin-id').textContent = `id: ${pl
    |   ugin.name}`;
205 |   document.getElementById('plugin-description').textContent =
    |   plugin.description || 'Нет описания';
206 |
207 |   const scopeBadgeEl = document.getElementById('plugin-scope-b
    |   adge');
208 |   if (scopeBadgeEl) {
209 |     if (plugin.is_system) {
210 |       scopeBadgeEl.textContent = 'Системный';
211 |       scopeBadgeEl.className = 'badge bg-secondary-subtle text
    |   -body-secondary border';
212 |     } else {
213 |       scopeBadgeEl.textContent = 'Пользовательский';
214 |       scopeBadgeEl.className = 'badge bg-info-subtle text-info
    |   -emphasis border';
215 |     }
216 |   }
217 |
218 |   const isAdmin = (typeof window.isUserAdmin === 'function') ?
```

```
    window.isUserAdmin() : false;
219   const isLocked = Boolean(plugin.is_system && !isAdmin);
220
221   const toggleSwitch = document.getElementById('plugin-toggle-
switch');
222   if (toggleSwitch) {
223     toggleSwitch.checked = Boolean(plugin.enabled);
224     toggleSwitch.disabled = isLocked;
225     toggleSwitch.title = isLocked
226       ? 'Системный плагин управляется администратором'
227       : 'Включить / отключить плагин';
228   }
229
230   const saveBtn = document.getElementById('btn-save-plugin-con
fig');
231   if (saveBtn) {
232     saveBtn.disabled = isLocked;
233     saveBtn.title = isLocked ? 'Настройки системного плагина и
зменяются только администратором' : '';
234   }
235
236   // Render Actions
237   renderPluginActions(plugin);
238
239   // Render Config Fields
240   renderPluginFields(plugin);
241 }
242
243 // Отрисовка кнопок действий (Actions)
244 function renderPluginActions(plugin) {
245   const container = document.getElementById('plugin-actions-co
ntainer');
246   const section = document.getElementById('plugin-actions-sect
ion');
247   if (!container || !section) return;
248
249   const actions = plugin.actions || [];
250   if (actions.length === 0) {
```

```

251     section.classList.add('d-none');
252     return;
253 }
254
255 section.classList.remove('d-none');
256 let html = '';
257 actions.forEach(act => {
258     const btnColor = act.color || 'primary';
259     const label = act.label || act.name || act.id;
260     const icon = act.icon ? `<span class="me-1">${act.icon}</span>` : '';
261     const safeLabel = label.replace(/'/g, "\\'");
262     html += `
263         <button class="btn btn-sm btn-outline-${btnColor} d-inline-flex align-items-center gap-1 shadow-sm"
264             onclick="executePluginAction('${plugin.name}', '${act.id}', '${safeLabel}')"
265             title="${act.description || ''}"
266             ${icon}<span>${label}</span>
267         </button>
268     `;
269 });
270 container.innerHTML = html;
271 }
272
273 // Отрисовка полей конфигурации
274 function renderPluginFields(plugin) {
275     const container = document.getElementById('plugin-fields-container');
276     const section = document.getElementById('plugin-config-section');
277     if (!container || !section) return;
278
279     const fields = plugin.fields || [];
280     if (fields.length === 0) {
281         section.classList.add('d-none');
282         return;
283     }

```

```

284
285 section.classList.remove('d-none');
286 const currentCfg = plugin.config || {};
287
288 let html = '';
289 fields.forEach(f => {
290     const val = (currentCfg[f.id] !== undefined) ? currentCfg[
291 f.id] : (f.default !== undefined ? f.default : '');
292     const desc = f.description ? `

` : '';
294
295     if (f.type === 'select') {
296         let optionsHtml = '';
297         (f.options || []).forEach(opt => {
298             const isSel = String(opt.value) === String(val) ? 'sel
299 ected' : '';
300             optionsHtml += `


```



```

    field-type="boolean">
314 |         <label class="form-check-label small fw-semibold"
    for="field_{{f.id}}">{{f.label}}</label>
315 |     </div>
316 |     {{desc}}
317 | </div>
318 | `;
319 | } else if (f.type === 'list_string') {
320 |     const listVals = Array.isArray(val) ? val.join(', ') : (
    val || '');
321 |     html += `
322 |         <div class="col-12">
323 |             <label class="form-label small fw-semibold mb-1">{{f
    .label}} (через запятую)</label>
324 |             <input type="text" class="form-control form-control-
    sm font-monospace" name="{{f.id}}" value="{{listVals}}" data-fie
    ld-type="list_string">
325 |                 {{desc}}
326 |             </div>
327 |         `;
328 | } else if (f.type === 'readonly') {
329 |     html += `
330 |         <div class="col-12 col-md-6">
331 |             <label class="form-label small fw-semibold mb-1 text
    -muted">{{f.label}}</label>
332 |             <input type="text" class="form-control form-control-
    sm bg-body-tertiary text-muted" value="{{val}}" readonly>
333 |                 {{desc}}
334 |             </div>
335 |         `;
336 | } else if (f.type === 'number') {
337 |     html += `
338 |         <div class="col-12 col-md-6">
339 |             <label class="form-label small fw-semibold mb-1">{{f
    .label}}</label>
340 |             <input type="number" class="form-control form-contro
    l-sm" name="{{f.id}}" value="{{val}}" data-field-type="number">
341 |                 {{desc}}

```

```

342         </div>
343     `;
344     } else {
345         html += `
346             <div class="col-12 col-md-6">
347                 <label class="form-label small fw-semibold mb-1">${f
348                 .label}</label>
349                 <input type="text" class="form-control form-control-
350                 sm" name="${f.id}" value="${val}" data-field-type="string">
351                 ${desc}
352             </div>
353         `;
354     }
355     container.innerHTML = html;
356 }
357
358 // Переключение состояния плагина (Вкл / Выкл)
359 async function toggleCurrentPlugin(enabled) {
360     if (!selectedPluginName) return;
361
362     try {
363         logToPluginConsole(`Переключение плагина '${selectedPlugin
364         Name}' -> ${enabled ? 'ВКЛ' : 'ВЫКЛ'}...`);
365         const res = await window.api.fetch(`/api/admin/plugins/${s
366         electedPluginName}/toggle`, {
367             method: 'POST',
368             headers: { 'Content-Type': 'application/json' },
369             body: JSON.stringify({ enabled })
370         });
371         const plugin = loadedPlugins.find(p => p.name === selected
372         PluginName);
373         if (plugin) {
374             plugin.enabled = enabled;
375         }
376     }

```

```
375     renderPluginsList(loadedPlugins);
376     syncPluginTabsVisibility(loadedPlugins);
377     showPluginNotification(res.message || 'Статус плагина обновлен', 'success');
378     logToPluginConsole(`    ${res.message || 'Статус успешно обновлен'}`);
379   } catch (ex) {
380     console.error('Ошибка переключения плагина:', ex);
381     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
382     logToPluginConsole(`    Ошибка: ${ex.message}`);
383     // Возвращаем тумблер в исходное состояние
384     const toggleSwitch = document.getElementById('plugin-toggle-switch');
385     if (toggleSwitch) toggleSwitch.checked = !enabled;
386   }
387 }
388
389 // Сохранение настроек текущего плагина
390 async function saveCurrentPluginConfig() {
391   if (!selectedPluginName) return;
392
393   const form = document.getElementById('plugin-config-form');
394   if (!form) return;
395
396   const newConfig = {};
397   const elements = form.querySelectorAll('[data-field-type]');
398
399   elements.forEach(el => {
400     const name = el.getAttribute('name');
401     if (!name) return;
402     const type = el.getAttribute('data-field-type');
403
404     if (type === 'boolean') {
405       newConfig[name] = el.checked;
406     } else if (type === 'number') {
407       newConfig[name] = Number(el.value);
408     } else if (type === 'list_string') {
409       newConfig[name] = el.value.split(',').map(s => s.trim())
```

```

    .filter(Boolean);
410     } else {
411         newConfig[name] = el.value;
412     }
413 });
414
415     try {
416         logToPluginConsole(`Сохранение параметров плагина '${selectedPluginName}'...`);
417         const res = await window.api.fetch(`/api/admin/plugins/${selectedPluginName}/config`, {
418             method: 'POST',
419             headers: { 'Content-Type': 'application/json' },
420             body: JSON.stringify({ config: newConfig })
421         });
422
423         const plugin = loadedPlugins.find(p => p.name === selectedPluginName);
424         if (plugin) {
425             plugin.config = res.config || newConfig;
426         }
427
428         showPluginNotification('Параметры успешно сохранены', 'success');
429         logToPluginConsole(`□ Конфигурация сохранена: ${JSON.stringify(newConfig, null, 2)}`);
430     } catch (ex) {
431         console.error('Ошибка сохранения конфигурации:', ex);
432         showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
433         logToPluginConsole(`□ Ошибка сохранения: ${ex.message}`);
434     }
435 }
436
437 // Выполнение специфического действия (Action) плагина
438 async function executePluginAction(pluginName, actionId, actionLabel) {
439     try {
440         logToPluginConsole(`[ACTION] Запуск '${actionLabel}' для п

```

```

лагина '${pluginName}'...`);
441     showPluginNotification(`Запуск '${actionLabel}'...`, 'info
');
442
443     // Собираем текущие поля формы как параметры действия
444     const form = document.getElementById('plugin-config-form')
;
445     const params = {};
446     if (form) {
447         const elements = form.querySelectorAll('[data-field-type
]');
448         elements.forEach(el => {
449             const name = el.getAttribute('name');
450             if (!name) return;
451             const type = el.getAttribute('data-field-type');
452             if (type === 'boolean') params[name] = el.checked;
453             else if (type === 'number') params[name] = Number(el.v
alue);
454             else if (type === 'list_string') params[name] = el.val
ue.split(',').map(s => s.trim()).filter(Boolean);
455             else params[name] = el.value;
456         });
457     }
458
459     // Проверяем, есть ли обязательные параметры действия, кот
орых нет в форме
460     const plugin = (typeof loadedPlugins !== 'undefined') ? lo
adedPlugins.find(p => p.name === pluginName) : null;
461     const actDef = plugin?.actions?.find(a => a.id === actionI
d);
462     if (actDef && Array.isArray(actDef.parameters)) {
463         for (const param of actDef.parameters) {
464             if (param.required && (params[param.name] === undefine
d || params[param.name] === '')) {
465                 const userVal = prompt(`${param.description || param
.name}:`);
466                 if (userVal === null) {
467                     logToPluginConsole(`[ACTION] Отменено пользователе

```

```

    m.`);
468         return;
469     }
470     params[param.name] = userVal;
471 }
472 }
473 }
474
475     const res = await window.api.fetch(`/api/admin/plugins/${p
pluginName}/action/${actionId}`, {
476         method: 'POST',
477         headers: { 'Content-Type': 'application/json' },
478         body: JSON.stringify({ params })
479     });
480
481     if (res.success || res.status === 'success') {
482         showPluginNotification(res.message || 'Действие успешно
выполнено', 'success');
483         logToPluginConsole(`\n ${res.message || 'Успех'}\n${JSON.
stringify(res.result || res.results || res, null, 2)}`);
484     } else {
485         showPluginNotification(res.message || 'Ошибка выполнения
действия', 'warning');
486         logToPluginConsole(`\n ${res.message || 'Завершено с пред
упреждением'}`);
487     }
488 } catch (ex) {
489     console.error(`Ошибка выполнения действия ${actionId}:`, e
x);
490     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
491     logToPluginConsole(`\n Ошибка выполнения: ${ex.message}`);
492 }
493 }
494
495 // Логирование в консоль плагина
496 function logToPluginConsole(text) {
497     const consoleEl = document.getElementById('plugin-console-ou
tput');

```

```
498 | if (!consoleEl) return;
499 | const time = new Date().toLocaleTimeString();
500 | consoleEl.textContent = `[${time}] ${text}\n` + consoleEl.te
    | xtContent;
501 | }
502 |
503 | // Очистка консоли
504 | function clearPluginConsole() {
505 |     const consoleEl = document.getElementById('plugin-console-ou
    | tput');
506 |     if (consoleEl) consoleEl.textContent = 'Консоль очищена.\n';
507 | }
508 |
509 | function showPlaceholder() {
510 |     const placeholder = document.getElementById('plugin-placehol
    | der');
511 |     const contentPane = document.getElementById('plugin-content-
    | pane');
512 |     if (placeholder) placeholder.classList.remove('d-none');
513 |     if (contentPane) contentPane.classList.add('d-none');
514 | }
515 |
516 | // Переход к конкретному плагину из выпадающего меню
517 | function openPluginFromDropdown(pluginName) {
518 |     if (!pluginName) return;
519 |
520 |     // 1. Закрываем открытые выпадающие списки
521 |     document.querySelectorAll('#mainTabs .dropdown-menu.show').f
    | orEach((m) => {
522 |         m.classList.remove('show');
523 |         m.closest('.dropdown')?.querySelector('.dropdown-toggle')?
    | .classList.remove('show');
524 |     });
525 |
526 |     const normalized = pluginName.toLowerCase().replace(/-/g, '_
    | ');
527 |
528 |     // 2. Если у плагина есть своя выделенная вкладка в DOM, отк
```

```

        ываем ee
529     if (normalized === 'telegram_channel_rag' || normalized ===
        'telegram_rag') {
530         const pane = document.getElementById('tab-telegram-rag') |
        | document.getElementById('tab-telegram_rag');
531         if (pane && typeof window.switchTab === 'function') {
532             window.switchTab('tab-telegram-rag');
533             return;
534         }
535     }
536     if (normalized === 'news_feed' || normalized === 'news' || n
        ormalized === 'smart_news' || normalized === 'smart_feed_news'
        ) {
537         const pane = document.getElementById('tab-news');
538         if (pane && typeof window.switchTab === 'function') {
539             window.switchTab('tab-news');
540             return;
541         }
542     }
543
544     // 3. Иначе переключаемся на вкладку плагинов и выбираем пла
        гин в каталоге
545     if (typeof window.switchTab === 'function') {
546         window.switchTab('tab-plugins');
547     }
548
549     const select = () => {
550         if (typeof window.selectPlugin === 'function') {
551             window.selectPlugin(pluginName);
552         }
553     };
554
555     select();
556     setTimeout(select, 150);
557     setTimeout(select, 400);
558 }
559
560 // Синхронизация видимости вкладок в навбаре с состоянием акти

```


ВНОСТИ ПЛАГИНОВ

```
561 function syncPluginTabsVisibility(plugins) {
562   if (!Array.isArray(plugins)) return;
563   const pluginMap = {};
564   plugins.forEach(p => {
565     pluginMap[p.name] = Boolean(p.enabled);
566   });
567
568   // 1. Обновляем выпадающие списки включенных плагинов в шапк
569   e
570   const enabledContainers = document.querySelectorAll('#nav-enabled-plugins-list, .nav-enabled-plugins-list');
571   const enabledPlugins = plugins.filter(p => p.enabled);
572   enabledContainers.forEach(container => {
573     if (enabledPlugins.length === 0) {
574       container.innerHTML = `
575         <div class="list-group-item dropdown-item text-muted s
576         mall py-2 px-3 fst-italic">
577           Нет активных плагинов
578         </div>
579       `;
580     } else {
581       let html = '';
582       enabledPlugins.forEach(p => {
583         const icon = p.icon || '❏';
584         const title = p.title || p.name;
585         const desc = p.description || title;
586         html += `
587           <button class="list-group-item list-group-item-actio
588           n dropdown-item d-flex align-items-center gap-2 py-2 px-3 text
589           -truncate"
590             type="button"
591             data-plugin="${p.name}"
592             title="${desc}"
593             onclick="openPluginFromDropdown('${p.name}')"
594           ">
595             <span>${icon}</span>
```

```

592 |         <span class="text-truncate">${title}</span>
593 |     </button>
594 |     `;
595 | });
596 |     container.innerHTML = html;
597 | }
598 | });
599 |
600 | // 2. Находим все элементы навигации, привязанные к отдельны
    | м плагинам
601 |     const pluginNavItems = document.querySelectorAll('[data-plug
    | in-tab]');
602 |     pluginNavItems.forEach(item => {
603 |         const pluginName = item.getAttribute('data-plugin-tab');
604 |         const isEnabled = (pluginMap[pluginName] !== undefined) ?
    | pluginMap[pluginName] : false;
605 |
606 |         if (isEnabled) {
607 |             item.classList.remove('d-none');
608 |         } else {
609 |             item.classList.add('d-none');
610 |             // Если отключенная вкладка была активна, переключаемся
    | на вкладку чата
611 |             const button = item.querySelector('.nav-link, .dropdown-
    | item');
612 |             if (button && button.classList.contains('active')) {
613 |                 if (typeof window.switchTab === 'function') {
614 |                     window.switchTab('tab-chat');
615 |                 } else {
616 |                     const chatTab = document.querySelector('[data-bs-tar
    | get="#tab-chat"]');
617 |                     if (chatTab) {
618 |                         const tab = new bootstrap.Tab(chatTab);
619 |                         tab.show();
620 |                     }
621 |                 }
622 |             }
623 |         }

```

```
624 | });
625 | }
626 |
627 | // Make functions globally available for HTML inline handlers
628 | window.loadPluginsList = loadPluginsList;
629 | window.filterPluginsList = filterPluginsList;
630 | window.selectPlugin = selectPlugin;
631 | window.toggleCurrentPlugin = toggleCurrentPlugin;
632 | window.saveCurrentPluginConfig = saveCurrentPluginConfig;
633 | window.executePluginAction = executePluginAction;
634 | window.clearPluginConsole = clearPluginConsole;
635 | window.syncPluginTabsVisibility = syncPluginTabsVisibility;
636 | window.openPluginFromDropdown = openPluginFromDropdown;
637 |
638 | // Auto-run initialization
639 | if (document.readyState === 'loading') {
640 |     document.addEventListener('DOMContentLoaded', () => {
641 |         if (typeof window.initPluginsTab === 'function') window.in
        | itPluginsTab();
642 |     });
643 | } else {
644 |     if (typeof window.initPluginsTab === 'function') window.init
        | PluginsTab();
645 | }
646 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Post-Install Wizard - Index View
5 |=====
6 Description:
7 | Шаблон интерфейса мастера настройки Windows после установки
8 | (Post-Install Wizard).
9 |
10 Usage Examples:
11 | Web Route:
12 | /html/post_install_wizard_tab/index.html
13 |
14 File: index.html
15 Project: ai-breadboard
16 Package: windows/api/webgui/post_install_wizard_tab
17 Author: hypo69
18 Copyright: © 2026 hypo69
19 Updated: 2026-10-08 03:08:00
20 |=====
21 |=====
22 -->
23
24 <style>
25 .piw-container {
26 padding: 0.75rem 1rem;
27 }
28 .piw-header {
29 background: var(--surface-2, #1e293b);
30 border: 1px solid var(--border-color, #334155);
31 border-radius: var(--radius-md, 8px);
32 padding: 0.75rem 1.25rem;
33 }
34 .piw-card {
35 background: #0f172a;
```

```

33     border: 1px solid #334155;
34     border-radius: 8px;
35 }
36 .piw-step-item {
37     background: #1e293b;
38     border: 1px solid #334155;
39     border-radius: 6px;
40     padding: 0.75rem 1rem;
41     transition: all 0.2s ease;
42 }
43 .piw-step-item:hover {
44     border-color: #38bdf8;
45     background: rgba(56, 189, 248, 0.05);
46 }
47 </style>
48
49 <div class="piw-container theme-sysctl">
50     <!-- Top Header Bar -->
51     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 piw-header text-white">
52         <div>
53             <div class="d-flex align-items-center gap-2">
54                 <h5 class="mb-0 fw-bold">
55                     □ Windows Post-Install Wizard
56                 </h5>
57                 <span class="badge rounded-pill bg-warning text-dark px-2 py-1" style="font-size: 0.75rem;">Setup & Hardening</span>
58             </div>
59             <p class="mb-0 small text-light mt-1" >
60                 Пошаговая настройка базовой конфигурации Windows, проф
или оптимизации, отключение телеметрии и настройка параметров
безопасности.
61             </p>
62         </div>
63         <div class="d-flex align-items-center gap-2 flex-wrap">
64             <span class="badge rounded-pill px-3 py-2" id="piw-elevation-badge">
65                 □ Checking privileges...

```

```

66         </span>
67         <button class="btn btn-sm btn-outline-info rounded-pill
px-3" id="btn-piw-refresh">
68             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
69         </button>
70     </div>
71 </div>
72
73 <!-- Main Wizard Card -->
74 <div class="card piw-card">
75     <div class="card-header border-secondary bg-transparent d-
flex justify-content-between align-items-center flex-wrap gap-
2 py-3">
76         <div>
77             <h6 class="mb-0 fw-bold text-warning"><i class="bi bi-
magic me-1"></i> Чек-лист первичной настройки Windows</h6>
78             <span class="small text-muted">Выберите профиль оптими
зации или отметьте необходимые шаги для последовательного прим
енения.</span>
79         </div>
80         <div class="d-flex align-items-center gap-2">
81             <select class="form-select form-select-sm bg-dark text
-white border-secondary" id="piw-profile-select" style="min-wi
dth: 260px;">
82                 <option value="post_install">Windows Post-Install Ba
seline</option>
83                 <option value="secure_hardening">Security Hardening
Profile</option>
84                 <option value="developer_setup">Developer Workstatio
n Profile</option>
85             </select>
86             <button class="btn btn-sm btn-primary px-3 rounded-pil
l" id="btn-piw-apply-profile">
87                 <i class="bi bi-play-fill me-1"></i> Применить профи
ль
88             </button>
89         </div>
90     </div>

```

```
91 |     <div class="card-body p-3">
92 |         <!-- Steps Container -->
93 |         <div id="piw-wizard-steps-container" class="d-flex flex-
column gap-2">
94 |             <div class="text-center text-muted py-4">
95 |                 <div class="spinner-border spinner-border-sm text-in
fo me-2" role="status"></div>
96 |                 Загрузка шагов профиля...
97 |             </div>
98 |         </div>
99 |     </div>
100 |     <div class="card-footer border-secondary bg-transparent d-
flex justify-content-between align-items-center flex-wrap gap-
2">
101 |         <span class="small text-muted" id="piw-status-summary">Г
отово к выполнению</span>
102 |         <div class="d-flex gap-2">
103 |             <button class="btn btn-sm btn-outline-secondary rounde
d-pill px-3" id="btn-piw-select-all">Выбрать все</button>
104 |             <button class="btn btn-sm btn-outline-secondary rounde
d-pill px-3" id="btn-piw-deselect-all">Снять выбор</button>
105 |         </div>
106 |     </div>
107 | </div>
108 </div>
109 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Post-Install Wizard - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт мастера первичной настройки Windows (Post-Install Wizard).
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/html/post_install_wizard_tab/main.js?v=20261006_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/post_install_wizard_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-06 05:55:00
18  * =====
19  */
20
21 (function () {
22   'use strict';
23
24   let isInitialized = false;
25   let activeProfileSteps = [];
26
27   /**
28    * Экранирование HTML-символов для защиты от XSS.
29    *
30    * @param {string} str - Исходная строка.
31    * @returns {string} Экранированная строка.
```



```

32     */
33     function escapeHtml(str) {
34         if (!str) return '';
35         return String(str)
36             .replace(/&/g, '&amp;')
37             .replace(/</g, '&lt;')
38             .replace(/>/g, '&gt;')
39             .replace(/"/g, '&quot;')
40             .replace(/'/g, '&#039;');
41     }
42
43     /**
44     * Проверка прав администратора и обновление бейджа.
45     */
46     async function fetchElevationStatus() {
47         try {
48             const res = await fetch('/api/system-control/status');
49             if (!res.ok) return;
50             const data = await res.json();
51             const elevBadge = document.getElementById('piw-elevation
- badge');
52             if (elevBadge) {
53                 if (data.is_elevated) {
54                     elevBadge.className = 'badge rounded-pill bg-success
- subtle text-success border border-success px-3 py-2';
55                     elevBadge.innerHTML = '👤 Режим: Администратор (Полн
ый доступ)';
56                 } else {
57                     elevBadge.className = 'badge rounded-pill bg-warning
- subtle text-warning border border-warning px-3 py-2';
58                     elevBadge.innerHTML = '👤 Режим: Обычный пользовател
ь (Ограничено)';
59                 }
60             }
61         } catch (e) {
62             console.warn('[PostInstallWizard] Ошибка проверки прав: ', e);
63         }

```

```

64 | }
65 |
66 | /**
67 |  * Загрузка профилей и шагов мастера.
68 |  */
69 | async function loadWizardProfiles() {
70 |     try {
71 |         const res = await fetch('/api/system-control/profiles');
72 |         if (!res.ok) return;
73 |         const data = await res.json();
74 |         const profiles = data.profiles || [];
75 |         const currentProf = profiles[0];
76 |         if (currentProf && currentProf.steps) {
77 |             activeProfileSteps = currentProf.steps;
78 |             renderWizardSteps(currentProf.steps);
79 |         }
80 |     } catch (e) {
81 |         console.error('[PostInstallWizard] Ошибка загрузки профи
лей:', e);
82 |     }
83 | }
84 |
85 | /**
86 |  * Отрисовка элементов шагов мастера.
87 |  *
88 |  * @param {Array} steps - Список шагов.
89 |  */
90 | function renderWizardSteps(steps) {
91 |     const container = document.getElementById('piw-wizard-step
s-container');
92 |     if (!container) return;
93 |
94 |     if (!steps || steps.length === 0) {
95 |         container.innerHTML = '<div class="text-center text-mute
d p-3">Нет доступных шагов для выбранного профиля.</div>';
96 |         return;
97 |     }
98 |

```

```

99     container.innerHTML = steps.map(step => {
100         const badgeClass = step.status === 'SUCCESS' ? 'bg-succe
ss' : step.status === 'RUNNING' ? 'bg-info' : step.status ===
'FAILED' ? 'bg-danger' : 'bg-secondary';
101         return `
102         <div class="piw-step-item d-flex justify-content-between
align-items-center flex-wrap gap-2" id="piw-step-${step.id}">
103             <div class="d-flex align-items-start gap-3">
104                 <input class="form-check-input mt-1 piw-step-checkbo
x" type="checkbox" value="${step.id}" ${step.enabled ? 'checke
d' : ''} style="cursor: pointer;">
105                 <div>
106                     <div class="fw-bold text-white d-flex align-items-
center gap-2">
107                         <span>${escapeHtml(step.title)}</span>
108                         ${step.requires_elevation ? '<span class="badge
bg-warning-subtle text-warning border border-warning px-1.5 py
-0.5" style="font-size: 0.65rem;">Admin</span>' : ''}
109                     </div>
110                     <div class="small text-muted">${escapeHtml(step.de
scription)}</div>
111                 </div>
112             </div>
113             <div class="d-flex align-items-center gap-2">
114                 <span class="badge ${badgeClass} font-monospace px-2
py-1" id="piw-step-status-${step.id}" style="font-size: 0.72r
em;">${step.status || 'READY'}</span>
115                 <button class="btn btn-sm btn-outline-info rounded-p
ill px-3 piw-btn-step-run" data-step-id="${step.id}">
116                     <i class="bi bi-play-circle me-1"></i> Выполнить
117                 </button>
118             </div>
119         </div>
120     `;
121 }).join('');
122
123 // Привязка обработчиков одиночного запуска шагов
124 container.querySelectorAll('.piw-btn-step-run').forEach(bt

```

```

n => {
125     btn.onclick = async () => {
126         const stepId = btn.getAttribute('data-step-id');
127         await executeWizardSteps([stepId]);
128     };
129 });
130 }
131
132 /**
133  * Выполнение списка шагов мастера.
134  *
135  * @param {Array<string>} stepIds - Идентификаторы шагов.
136  */
137 async function executeWizardSteps(stepIds) {
138     if (!stepIds || stepIds.length === 0) {
139         if (window.toast) window.toast.warning('Мастер настройки
140         ', 'Выберите хотя бы один шаг для выполнения.');
```

140 return;

141 }

142

143 const summaryEl = document.getElementById('piw-status-summary');

144 if (summaryEl) summaryEl.innerHTML = `Выполняется \${stepIds.length} шагов...`;

145

146 stepIds.forEach(sid => {

147 const badge = document.getElementById(`piw-step-status-\${sid}`);

148 if (badge) {

149 badge.className = 'badge bg-info font-monospace px-2 py-1';

150 badge.innerHTML = 'RUNNING';

151 }

152 });

153

154 try {

155 const profileSelect = document.getElementById('piw-profile-select');

```
156 |         const profileId = profileSelect ? profileSelect.value :
    | 'post_install';
157 |
158 |         const res = await fetch('/api/system-control/profiles/ap
    | ply', {
159 |             method: 'POST',
160 |             headers: { 'Content-Type': 'application/json' },
161 |             body: JSON.stringify({ profile_id: profileId, step_ids
    | : stepIds }),
162 |         });
163 |
164 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
165 |         const data = await res.json();
166 |
167 |         stepIds.forEach(sid => {
168 |             const badge = document.getElementById(`piw-step-status
    | -${sid}`);
169 |             if (badge) {
170 |                 badge.className = 'badge bg-success font-monospace p
    | x-2 py-1';
171 |                 badge.innerText = 'SUCCESS';
172 |             }
173 |         });
174 |
175 |         if (summaryEl) summaryEl.innerText = 'Профиль успешно пр
    | именен.';
176 |         if (window.toast) window.toast.success('Мастер настройки
    | ', data.message || 'Шаги успешно применены.');
```

```
177 |     } catch (e) {
178 |         console.error('[PostInstallWizard] Ошибка выполнения шаг
    | ов:', e);
179 |         stepIds.forEach(sid => {
180 |             const badge = document.getElementById(`piw-step-status
    | -${sid}`);
181 |             if (badge) {
182 |                 badge.className = 'badge bg-danger font-monospace px
    | -2 py-1';
183 |                 badge.innerText = 'FAILED';
```

```
184         }
185     });
186     if (summaryEl) summaryEl.innerText = `Ошибка применения:
    ${e.message}`;
187     if (window.toast) window.toast.error('Мастер настройки',
    `Ошибка: ${e.message}`);
188     }
189 }
190
191 /**
192  * Инициализация вкладки Post-Install Wizard.
193  */
194 function initPostInstallWizardTab() {
195     fetchElevationStatus();
196     loadWizardProfiles();
197
198     if (isInitialized) return;
199     isInitialized = true;
200
201     const refreshBtn = document.getElementById('btn-piw-refres
h');
202     if (refreshBtn) {
203         refreshBtn.onclick = () => {
204             fetchElevationStatus();
205             loadWizardProfiles();
206         };
207     }
208
209     const applyProfileBtn = document.getElementById('btn-piw-a
pply-profile');
210     if (applyProfileBtn) {
211         applyProfileBtn.onclick = async () => {
212             const checkedBoxes = document.querySelectorAll('.piw-s
tep-checkbox:checked');
213             const stepIds = Array.from(checkedBoxes).map(cb => cb.
value);
214             await executeWizardSteps(stepIds);
215         };
    }
```

```
216 |     }
217 |
218 |     const selectAllBtn = document.getElementById('btn-piw-select-all');
219 |     if (selectAllBtn) {
220 |         selectAllBtn.onclick = () => {
221 |             document.querySelectorAll('.piw-step-checkbox').forEach(
222 |                 h(cb => { cb.checked = true; });
223 |             );
224 |         }
225 |
226 |         const deselectAllBtn = document.getElementById('btn-piw-deselect-all');
227 |         if (deselectAllBtn) {
228 |             deselectAllBtn.onclick = () => {
229 |                 document.querySelectorAll('.piw-step-checkbox').forEach(
230 |                     h(cb => { cb.checked = false; });
231 |                 );
232 |             }
233 |
234 |             window.initPostInstallWizardTab = initPostInstallWizardTab;
235 |         })();
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Power Lifecycle & Boot History Tab
5 |=====
6 |=====
7 Description:
8 |Клиентский веб-интерфейс анализа жизненного цикла питания Windows,
9 |истории перезагрузок (User32 1074, Kernel-Power 41, EventLog 6008/6005/6006),
10 |реконструированных сессий питания (Power Sessions) и корреляции инициаторов.
11 |
12 Usage Examples:
13 |Web Route:
14 |/html/power_lifecycle_tab/index.html
15 |/windows/api/webgui/power_lifecycle_tab/index.html
16 |
17 Tab Frame Integration:
18 |<div id="tab-power-lifecycle" class="tab-pane fade"></div>
19 |
20 File: index.html
21 Project: ai-breadboard
22 Package: windows/api/webgui/power_lifecycle_tab
23 Author: hypo69
24 Copyright: © 2026 hypo69
25 Updated: 2026-10-08 04:25:00
26 |=====
27 |=====
28 -->
29
30 <style>
31 .power-lifecycle-container {
32 color: var(--text-color, #f8fafc);
```



```
30 | }
31 | .pwr-card {
32 |     background: var(--surface-1, #1e293b);
33 |     border: 1px solid var(--border-color, #334155);
34 |     border-radius: var(--radius-md, 8px);
35 |     color: var(--text-color, #f8fafc);
36 | }
37 | .pwr-table {
38 |     width: 100%;
39 |     color: var(--text-color, #f8fafc);
40 |     border-collapse: separate;
41 |     border-spacing: 0;
42 | }
43 | .pwr-table thead th {
44 |     background: var(--surface-2, #0f172a);
45 |     color: var(--text-muted, #94a3b8);
46 |     font-weight: 600;
47 |     font-size: 0.75rem;
48 |     text-transform: uppercase;
49 |     letter-spacing: 0.03em;
50 |     border-bottom: 1.5px solid var(--border-color, #334155);
51 |     padding: 0.55rem 0.75rem;
52 |     white-space: nowrap;
53 | }
54 | .pwr-table td {
55 |     border-bottom: 1px solid var(--border-subtle, rgba(51, 65,
56 | 85, 0.25));
57 |     padding: 0.55rem 0.75rem;
58 |     vertical-align: middle;
59 |     color: var(--text-color, #f8fafc);
60 | }
61 | .pwr-table tbody tr:hover td {
62 |     background-color: var(--nav-hover-bg, rgba(56, 189, 248, 0
63 | .04));
64 | }
65 | .pwr-search-input {
66 |     background: var(--surface-1, #1e293b);
67 |     border: 1px solid var(--border-color, #334155);
```

```
66     color: var(--text-color, #f8fafc);
67 }
68 .pwr-search-input:focus {
69     background: var(--surface-1, #1e293b);
70     border-color: var(--nav-active, #38bdf8);
71     color: var(--text-color, #f8fafc);
72     box-shadow: 0 0 0 0.2rem rgba(56, 189, 248, 0.15);
73 }
74 .pwr-input-addon {
75     background: var(--surface-2, #0f172a);
76     border: 1px solid var(--border-color, #334155);
77     color: var(--text-muted, #94a3b8);
78 }
79 .pwr-nav-pills {
80     border-bottom: 1px solid var(--border-color, #334155);
81 }
82 .pwr-nav-pills .nav-link {
83     color: var(--text-muted, #94a3b8);
84     font-size: 0.84rem;
85     font-weight: 500;
86     border-radius: var(--radius-sm, 6px);
87     padding: 0.4rem 0.9rem;
88     transition: all 0.15s ease;
89 }
90 .pwr-nav-pills .nav-link:hover {
91     color: var(--nav-active, #38bdf8);
92     background: var(--nav-hover-bg, rgba(255, 255, 255, 0.05))
93 ;
94 }
95 .pwr-nav-pills .nav-link.active {
96     color: #fff;
97     background: var(--nav-active, #0284c7);
98     font-weight: 600;
99 }
100 .pwr-detail-card {
101     background: var(--surface-2, #1e293b);
102     border: 1px solid var(--border-color, #334155);
103     border-radius: var(--radius-sm, 6px);
```

```

103 | }
104 | .pwr-xml-pre {
105 |     background: var(--surface-2, #0f172a);
106 |     color: var(--text-color, #f8fafc);
107 |     border: 1px solid var(--border-color, #334155);
108 | }
109 | .pwr-proc-badge {
110 |     background: var(--surface-2, #0f172a);
111 |     color: var(--nav-active, #38bdf8);
112 |     border: 1px solid var(--border-color, #334155);
113 | }
114 | </style>
115 |
116 | <div class="container-fluid py-2 px-3 power-lifecycle-containe
    r">
117 |
118 |     <!-- Верхняя панель метрик и сводки -->
119 |     <div class="row g-2 mb-3">
120 |         <!-- Карточка 1: Текущий Аптайм и Старт ОС -->
121 |         <div class="col-12 col-sm-6 col-xl-3">
122 |             <div class="card pwr-card h-100 shadow-sm p-3 position-r
    elative">
123 |                 <div class="d-flex justify-content-between align-items
    -start">
124 |                     <div class="text-muted small text-uppercase fw-semib
    old">Текущий аптайм </div>
125 |                     <span class="badge bg-success font-monospace" id="pw
    r-uptime-badge">ACTIVE</span>
126 |                 </div>
127 |                 <div class="fs-4 fw-bold mt-1 text-info font-monospace
    " id="pwr-current-uptime">--:--:--</div>
128 |                 <div class="small text-muted mt-1 text-truncate" id="p
    wr-current-boot" style="font-size: 0.75rem;">Загрузка: --</div>
    </div>
129 |                 <div class="mt-2 text-end">
130 |                     <span class="badge bg-secondary-subtle text-secondar
    y border border-secondary-subtle" style="font-size: 0.7rem;">K
    ernel-General 12</span>

```

```

131         </div>
132     </div>
133 </div>
134
135 <!-- Карточка 2: Всего сессий питания -->
136 <div class="col-12 col-sm-6 col-xl-3">
137     <div class="card pwr-card h-100 shadow-sm p-3">
138         <div class="d-flex justify-content-between align-items
139         -start">
140             <div class="text-muted small text-uppercase fw-semib
141             old">Сессии питания □ </div>
142             <span class="badge bg-primary font-monospace" id="pw
143             r-total-sessions-badge">0 Сессий</span>
144         </div>
145         <div class="fs-4 fw-bold mt-1 text-warning font-monosp
146         ace" id="pwr-total-sessions">--</div>
147         <div class="small text-muted mt-1" id="pwr-sessions-de
148         sc">Реконструкция Boot → Shutdown</div>
149         <div class="mt-2">
150             <button class="btn btn-xs btn-outline-warning w-100
151             py-1" id="pwr-rebuild-btn" style="font-size: 0.78rem;">
152                 <i class="bi bi-arrow-repeat me-1"></i><span>Перес
153                 канировать Event Log</span>
154             </button>
155         </div>
156     </div>
157 </div>
158
159 <!-- Карточка 3: Штатные перезагрузки и выключения -->
160 <div class="col-12 col-sm-6 col-xl-3">
161     <div class="card pwr-card h-100 shadow-sm p-3">
162         <div class="d-flex justify-content-between align-items
163         -start">
164             <div class="text-muted small text-uppercase fw-semib
165             old">Штатные завершения □ </div>
166             <span class="badge bg-success-subtle text-success bo
167             rder border-success-subtle font-monospace" id="pwr-clean-ratio
168             ">-- %</span>

```

```

158         </div>
159         <div class="fs-4 fw-bold mt-1 text-success font-monosp
ace" id="pwr-clean-count">--</div>
160         <div class="small text-muted mt-1 text-truncate" id="p
wr-last-initiator" style="font-size: 0.75rem;">Инициатор: --</
div>
161         <div class="mt-2 text-end">
162             <span class="badge bg-secondary-subtle text-secondar
y border border-secondary-subtle" style="font-size: 0.7rem;">U
ser32 1074 / EventLog 6006</span>
163         </div>
164     </div>
165 </div>
166
167 <!-- Карточка 4: Внезапные сбои и BSOD -->
168 <div class="col-12 col-sm-6 col-xl-3">
169     <div class="card pwr-card h-100 shadow-sm p-3">
170         <div class="d-flex justify-content-between align-items
-start">
171             <div class="text-muted small text-uppercase fw-semib
old">Сбои / Потери питания ☐ </div>
172             <span class="badge bg-danger font-monospace" id="pwr
-unexpected-badge">0 Сбоев</span>
173         </div>
174         <div class="fs-4 fw-bold mt-1 text-danger font-monospa
ce" id="pwr-unexpected-count">--</div>
175         <div class="small text-muted mt-1 d-flex justify-conte
nt-between" style="font-size: 0.75rem;">
176             <span id="pwr-bsod-count">BSOD (1001): 0</span>
177             <span>Kernel-Power 41</span>
178         </div>
179         <div class="mt-2">
180             <button class="btn btn-xs btn-outline-primary w-100
py-1" id="pwr-refresh-btn" style="font-size: 0.78rem;">
181                 <i class="bi bi-arrow-clockwise me-1"></i><span>Об
новить данные</span>
182             </button>
183         </div>

```

```
184         </div>
185     </div>
186 </div>
187
188 <!-- Навигация по подразделам: Сессии vs События -->
189 <ul class="nav nav-pills mb-3 pwr-nav-pills pb-2" id="pwrNav
190 Tabs" role="tablist">
191     <li class="nav-item" role="presentation">
192         <button class="nav-link active py-1 px-3" id="pwr-sessio
193 ns-subtab-btn" data-bs-toggle="pill" data-bs-target="#pwr-subt
194 ab-sessions" type="button" role="tab">
195             <i class="bi bi-clock-history me-1.5 text-warning"></i
196 >
197             <span>Сессии питания (Power Sessions)</span>
198         </button>
199     </li>
200     <li class="nav-item" role="presentation">
201         <button class="nav-link py-1 px-3" id="pwr-events-subtab
202 -btn" data-bs-toggle="pill" data-bs-target="#pwr-subtab-events
203 " type="button" role="tab">
204             <i class="bi bi-list-columns-reverse me-1.5 text-info"
205 ></i>
206             <span>Журнал ключевых событий (Event Log)</span>
207         </button>
208     </li>
209 </ul>
210
211 <div class="tab-content" id="pwrTabContent">
212
213     <!-- Вкладка 1: Сессии питания (Power Sessions) -->
214     <div class="tab-pane fade show active" id="pwr-subtab-sess
215 ions" role="tabpanel">
216         <!-- Панель фильтров и поиска -->
217         <div class="card pwr-card mb-3 p-2">
218             <div class="row g-2 align-items-center">
219                 <div class="col-12 col-md-4">
220                     <div class="input-group input-group-sm">
221                         <span class="input-group-text pwr-input-addon"><
```

```

    i class="bi bi-search"></i></span>
214         <input type="text" class="form-control form-cont
rol-sm pwr-search-input" id="pwr-sessions-search" placeholder=
"Поиск по пользователю, процессу, причине...">
215     </div>
216 </div>
217 <div class="col-6 col-md-3">
218     <select class="form-select form-select-sm pwr-sear
ch-input" id="pwr-filter-type">
219         <option value="">Все типы завершения</option>
220         <option value="Active">☐ Активная сессия (Activ
e)</option>
221         <option value="Restart">☐ Перезагрузка (Restart)
</option>
222         <option value="Shutdown">☐ Выключение (Shutdown)
</option>
223         <option value="Unexpected">☐ Внезапное завершени
e (Unexpected)</option>
224         <option value="BSOD">☐ Синий экран (BSOD / BugC
heck)</option>
225     </select>
226 </div>
227 <div class="col-6 col-md-2">
228     <div class="form-check form-switch small mt-1">
229         <input class="form-check-input" type="checkbox"
id="pwr-filter-unexpected-only">
230         <label class="form-check-label text-warning fw-s
emibold" for="pwr-filter-unexpected-only">Только сбои</label>
231     </div>
232 </div>
233 <div class="col-12 col-md-3 text-end">
234     <span class="small text-muted" id="pwr-sessions-co
unter">Найдено: 0</span>
235 </div>
236 </div>
237 </div>
238
239 <!-- Таблица / Список сессий питания -->

```

```

240 |         <div class="card pwr-card shadow-sm">
241 |             <div class="table-responsive">
242 |                 <table class="table table-hover align-middle mb-0 pwr-table" style="font-size: 0.85rem;">
243 |                     <thead>
244 |                         <tr>
245 |                             <th style="width: 140px;">Сессия</th>
246 |                             <th style="width: 160px;">Время старта (Boot)<
247 |                             <th style="width: 160px;">Завершение</th>
248 |                             <th style="width: 130px;">Аптайм</th>
249 |                             <th style="width: 130px;">Статус</th>
250 |                             <th>Инициатор и процесс</th>
251 |                             <th>Причина / Код</th>
252 |                             <th style="width: 80px;" class="text-center">Д
253 |                             етали</th>
254 |                         </tr>
255 |                     </thead>
256 |                     <tbody id="pwr-sessions-tbody">
257 |                         <tr>
258 |                             <td colspan="8" class="text-center py-4 text-m
259 |                             text-info me-2" role="status"></div>
260 |                             Загрузка истории сессий питания...
261 |                         </td>
262 |                     </tr>
263 |                     </tbody>
264 |                 </table>
265 |             </div>
266 |         </div>
267 |
268 |         <!-- Вкладка 2: Журнал ключевых событий (Event Log) -->
269 |         <div class="tab-pane fade" id="pwr-subtab-events" role="ta
270 |         bpanel">
271 |             <!-- Фильтры событий -->
272 |             <div class="card pwr-card mb-3 p-2">

```



```
272 |         <div class="row g-2 align-items-center">
273 |             <div class="col-12 col-md-4">
274 |                 <div class="input-group input-group-sm">
275 |                     <span class="input-group-text pwr-input-addon"><
i class="bi bi-funnel"></i></span>
276 |                     <select class="form-select form-select-sm pwr-se
arch-input" id="pwr-event-id-filter">
277 |                         <option value="">Все Event ID жизненного цикла
</option>
278 |                         <option value="1074">1074 – User32 (Инициатор
shutdown/restart)</option>
279 |                         <option value="41">41 – Kernel-Power (Внезапна
я перезагрузка)</option>
280 |                         <option value="6008">6008 – EventLog (Неожидан
ное выключение)</option>
281 |                         <option value="12">12 – Kernel-General (Запуск
ядра / Boot)</option>
282 |                         <option value="13">13 – Kernel-General (Заверш
ение работы ОС)</option>
283 |                         <option value="6005">6005 – EventLog (Служба E
ventLog запущена)</option>
284 |                         <option value="6006">6006 – EventLog (EventLog
штатно остановлен)</option>
285 |                         <option value="6009">6009 – EventLog (Версия я
дра при загрузке)</option>
286 |                         <option value="1001">1001 – WER (Аварийный сбо
й BSOD / BugCheck)</option>
287 |                         <option value="19">19 – WindowsUpdate (Установ
ка обновлений)</option>
288 |                         <option value="42">42 – Kernel-Power (Переход
в сон)</option>
289 |                         <option value="107">107 – Kernel-Power (Пробуж
дение из сна)</option>
290 |                     </select>
291 |                 </div>
292 |             </div>
293 |             <div class="col-12 col-md-5">
294 |                 <div class="input-group input-group-sm">
```

```

295 |         <span class="input-group-text pwr-input-addon"><
| i class="bi bi-search"></i></span>
296 |         <input type="text" class="form-control form-cont
| rol-sm pwr-search-input" id="pwr-events-search" placeholder="П
| оиск в событиях...">
297 |     </div>
298 | </div>
299 |     <div class="col-12 col-md-3 text-end">
300 |         <span class="small text-muted" id="pwr-events-coun
| ter">Событий: 0</span>
301 |     </div>
302 | </div>
303 | </div>
304 |
305 |     <!-- Таблица сырых событий -->
306 |     <div class="card pwr-card shadow-sm">
307 |         <div class="table-responsive">
308 |             <table class="table table-hover align-middle mb-0 pw
| r-table" style="font-size: 0.85rem;">
309 |                 <thead>
310 |                     <tr>
311 |                         <th style="width: 160px;">Время</th>
312 |                         <th style="width: 90px;">Event ID</th>
313 |                         <th style="width: 140px;">Провайдер</th>
314 |                         <th style="width: 130px;">Тип</th>
315 |                         <th style="width: 180px;">Инициатор / Процесс<
| /th>
316 |                         <th>Описание / Причина</th>
317 |                         <th style="width: 80px;" class="text-center">X
| ML</th>
318 |                     </tr>
319 |                 </thead>
320 |                 <tbody id="pwr-events-tbody">
321 |                     <tr>
322 |                         <td colspan="7" class="text-center py-4 text-m
| uted">
323 |                             <div class="spinner-border spinner-border-sm
| text-info me-2" role="status"></div>

```

```

324 |             Загрузка журнала событий...
325 |         </td>
326 |     </tr>
327 | </tbody>
328 | </table>
329 | </div>
330 | </div>
331 | </div>
332 |
333 | </div>
334 |
335 | <!-- Модальное окно деталей сессии питания -->
336 | <div class="modal fade" id="pwrSessionDetailModal" tabindex=
| "-1" aria-labelledby="pwrSessionDetailModalLabel" aria-hidden=
| "true">
337 |     <div class="modal-dialog modal-lg modal-dialog-centered mo
| dal-dialog-scrollable">
338 |         <div class="modal-content shadow">
339 |             <div class="modal-header border-bottom">
340 |                 <h5 class="modal-title d-flex align-items-center gap
| -2" id="pwrSessionDetailModalLabel">
341 |                     <i class="bi bi-info-circle-fill text-warning"></i
| >
342 |                     <span id="pwr-modal-title">Детали сессии питания</
| span>
343 |                 </h5>
344 |                 <button type="button" class="btn-close" data-bs-dism
| iss="modal" aria-label="Close"></button>
345 |             </div>
346 |             <div class="modal-body p-3" id="pwr-modal-body">
347 |                 <!-- Populated by main.js -->
348 |             </div>
349 |             <div class="modal-footer border-top">
350 |                 <button type="button" class="btn btn-secondary btn-s
| m" data-bs-dismiss="modal">Закрыть</button>
351 |             </div>
352 |         </div>
353 |     </div>

```

```
354 | </div>
355 |
356 | <!-- Модальное окно просмотра сырого XML события -->
357 | <div class="modal fade" id="pwrXmlDetailModal" tabindex="-1"
    | aria-labelledby="pwrXmlDetailModalLabel" aria-hidden="true">
358 |   <div class="modal-dialog modal-lg modal-dialog-centered mo
    | dal-dialog-scrollable">
359 |     <div class="modal-content shadow">
360 |       <div class="modal-header border-bottom">
361 |         <h5 class="modal-title d-flex align-items-center gap
    | -2" id="pwrXmlDetailModalLabel">
362 |           <i class="bi bi-filetype-xml text-info"></i>
363 |           <span>Исходный XML события Windows</span>
364 |         </h5>
365 |         <button type="button" class="btn-close" data-bs-dism
    | iss="modal" aria-label="Close"></button>
366 |       </div>
367 |       <div class="modal-body p-3">
368 |         <pre class="pwr-xml-pre p-3 rounded font-monospace m
    | b-0" id="pwr-xml-content" style="font-size: 0.78rem; max-heigh
    | t: 500px; overflow-y: auto;"></pre>
369 |       </div>
370 |       <div class="modal-footer border-top">
371 |         <button type="button" class="btn btn-secondary btn-s
    | m" data-bs-dismiss="modal">Закрыть</button>
372 |       </div>
373 |     </div>
374 |   </div>
375 | </div>
376 |
377 | </div>
378 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Web Interface - Power Lifecycle Tab S
  cript
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом вкладки Power Li
  fecycle:
7  *   выборка и визуализация реконструированных сессий питания,
  аптайма,
8  *   истории перезагрузок (1074, 41, 6008) и ключевых системны
  х событий.
9  *
10 * Usage Examples:
11 *   HTML Integration:
12 *     <script src="/html/power_lifecycle_tab/main.js?v=202610
  08_v2" type="module"></script>
13 *
14 * File: main.js
15 * Project: ai-breadboard
16 * Package: windows/api/webgui/power_lifecycle_tab
17 * Author: hypo69
18 * Copyright: © 2026 hypo69
19 * Updated: 2026-10-08 04:25:00
20 * =====
21 */
22
23 let allSessions = [];
24 let allEvents = [];
25 let uptimeSeconds = 0;
26 let uptimeInterval = null;
27
28 /**
```

```

29  * Инициализирует вкладку жизненного цикла питания.
30  */
31  export async function initPowerLifecycleTab() {
32      setupEventListeners();
33      await loadPowerData();
34      startUptimeCounter();
35  }
36
37  /**
38   * Устанавливает слушатели событий элементов управления.
39   */
40  function setupEventListeners() {
41      const refreshBtn = document.getElementById('pwr-refresh-btn'
42      );
43      if (refreshBtn) {
44          refreshBtn.addEventListener('click', async () => {
45              refreshBtn.disabled = true;
46              refreshBtn.innerHTML = '<span class="spinner-border spin
47              ner-border-sm me-1"></span>Обновление...';
48              try {
49                  await loadPowerData();
50                  if (window.toast) window.toast.success('Обновлено', 'Д
51                  анные сессий питания успешно обновлены');
52              } finally {
53                  refreshBtn.disabled = false;
54                  refreshBtn.innerHTML = '<i class="bi bi-arrow-clockwis
55                  e me-1"></i><span>Обновить данные</span>';
56              }
57          });
58      }
59
60      const rebuildBtn = document.getElementById('pwr-rebuild-btn'
61      );
62      if (rebuildBtn) {
63          rebuildBtn.addEventListener('click', async () => {
64              rebuildBtn.disabled = true;
65              rebuildBtn.innerHTML = '<span class="spinner-border spin
66              ner-border-sm me-1"></span>Сканирование...';

```

```

61     try {
62         const res = await fetch('/api/v1/power/rebuild', { method: 'POST' });
63         if (!res.ok) throw new Error(`HTTP ${res.status}`);
64         const data = await res.json();
65         renderSummary(data);
66         await loadPowerData();
67         if (window.toast) window.toast.success('Event Log пересканирован', `Реконструировано сессий: ${data.total_sessions_count}`);
68     } catch (err) {
69         console.error('[PowerLifecycle] Ошибка пересканирования:', err);
70         if (window.toast) window.toast.error('Ошибка', 'Не удалось пересканировать журналы: ' + err.message);
71     } finally {
72         rebuildBtn.disabled = false;
73         rebuildBtn.innerHTML = '<i class="bi bi-arrow-repeat me-1"></i><span>Пересканировать Event Log</span>';
74     }
75 });
76 }
77
78 // Фильтры сессий
79 const sessionsSearch = document.getElementById('pwr-sessions-search');
80 if (sessionsSearch) {
81     sessionsSearch.addEventListener('input', () => filterAndRenderSessions());
82 }
83
84 const filterType = document.getElementById('pwr-filter-type');
85 if (filterType) {
86     filterType.addEventListener('change', () => filterAndRenderSessions());
87 }
88

```

```
89 |   const filterUnexpected = document.getElementById('pwr-filter
    | -unexpected-only');
90 |   if (filterUnexpected) {
91 |     filterUnexpected.addEventListener('change', () => filterAn
    | dRenderSessions());
92 |   }
93 |
94 |   // Фильтры событий
95 |   const eventsSearch = document.getElementById('pwr-events-sea
    | rch');
96 |   if (eventsSearch) {
97 |     eventsSearch.addEventListener('input', () => filterAndRend
    | erEvents());
98 |   }
99 |
100 |   const filterEid = document.getElementById('pwr-event-id-filt
    | er');
101 |   if (filterEid) {
102 |     filterEid.addEventListener('change', () => filterAndRender
    | Events());
103 |   }
104 | }
105 |
106 | /**
107 |  * Загружает сводку, сессии и события через API.
108 |  */
109 | async function loadPowerData() {
110 |   try {
111 |     // 1. Сводка
112 |     const sumRes = await fetch('/api/v1/power/summary');
113 |     if (sumRes.ok) {
114 |       const summary = await sumRes.json();
115 |       renderSummary(summary);
116 |     }
117 |
118 |     // 2. Сессии
119 |     const sessRes = await fetch('/api/v1/power/sessions?limit=
    | 100');
```



```

120     if (sessRes.ok) {
121         allSessions = await sessRes.json();
122         filterAndRenderSessions();
123     }
124
125     // 3. События
126     const evRes = await fetch('/api/v1/power/events?limit=200'
127 );
128     if (evRes.ok) {
129         allEvents = await evRes.json();
130         filterAndRenderEvents();
131     } catch (err) {
132         console.error('[PowerLifecycle] Ошибка загрузки данных:',
133 err);
134     }
135 }
136 /**
137  * Отрисовывает верхние карточки сводки.
138  * @param {Object} summary
139  */
140 function renderSummary(summary) {
141     if (!summary) return;
142
143     const curBootEl = document.getElementById('pwr-current-boot'
144 );
145     if (curBootEl) curBootEl.textContent = `Загрузка: ${summary.
146 current_boot_time || '--}`;
147
148     const curUptimeEl = document.getElementById('pwr-current-upt
149 ime');
150     if (curUptimeEl) {
151         curUptimeEl.textContent = summary.current_uptime_human ||
152 '--';
153         uptimeSeconds = summary.current_uptime_seconds || 0;
154     }
155 }

```

```
152 |   const totalSessionsEl = document.getElementById('pwr-total-s
    | sessions');
153 |   if (totalSessionsEl) totalSessionsEl.textContent = summary.t
    | otal_sessions_count;
154 |
155 |   const totalBadge = document.getElementById('pwr-total-sessio
    | ns-badge');
156 |   if (totalBadge) totalBadge.textContent = `${summary.total_se
    | ssions_count} Сессий`;
157 |
158 |   const cleanCountEl = document.getElementById('pwr-clean-coun
    | t');
159 |   if (cleanCountEl) cleanCountEl.textContent = summary.clean_s
    | hutdowns_count;
160 |
161 |   const cleanRatioEl = document.getElementById('pwr-clean-rati
    | o');
162 |   if (cleanRatioEl) {
163 |     const total = summary.total_sessions_count || 1;
164 |     const ratio = Math.round((summary.clean_shutdowns_count /
    | total) * 100);
165 |     cleanRatioEl.textContent = `${ratio} %`;
166 |   }
167 |
168 |   const lastInitEl = document.getElementById('pwr-last-initiat
    | or');
169 |   if (lastInitEl) {
170 |     lastInitEl.textContent = `Инициатор: ${summary.last_initia
    | tor || 'Штатный'}`;
171 |   }
172 |
173 |   const unexpCountEl = document.getElementById('pwr-unexpected
    | -count');
174 |   if (unexpCountEl) unexpCountEl.textContent = summary.unexpec
    | ted_shutdowns_count;
175 |
176 |   const unexpBadge = document.getElementById('pwr-unexpected-b
    | adge');
```

```
177 |   if (unexpBadge) {
178 |       unexpBadge.textContent = `${summary.unexpected_shutdowns_c
|ount} Сбоев`;
179 |       if (summary.unexpected_shutdowns_count > 0) {
180 |           unexpBadge.className = 'badge bg-danger font-monospace';
181 |       } else {
182 |           unexpBadge.className = 'badge bg-success font-monospace'
|;
183 |       }
184 |   }
185 |
186 |   const bsodEl = document.getElementById('pwr-bsod-count');
187 |   if (bsodEl) bsodEl.textContent = `BSOD (1001): ${summary.bso
|d_count || 0}`;
188 | }
189 |
190 | /**
191 |  * Фильтрует и отрисовывает таблицу сессий питания.
192 |  */
193 | function filterAndRenderSessions() {
194 |     const tbody = document.getElementById('pwr-sessions-tbody');
195 |     const counter = document.getElementById('pwr-sessions-counte
|r');
196 |     if (!tbody) return;
197 |
198 |     const searchVal = (document.getElementById('pwr-sessions-sea
|rch')?.value || '').toLowerCase().trim();
199 |     const typeFilter = document.getElementById('pwr-filter-type'
|)?.value || '';
200 |     const unexpOnly = document.getElementById('pwr-filter-unexpe
|cted-only')?.checked || false;
201 |
202 |     const filtered = allSessions.filter(s => {
203 |         if (unexpOnly && !s.unexpected_shutdown) return false;
204 |         if (typeFilter) {
205 |             if (typeFilter === 'Active' && s.shutdown_type !== 'Acti
|ve') return false;
206 |             if (typeFilter === 'Restart' && s.shutdown_type !== 'Res
```

```

    tart') return false;
207     if (typeFilter === 'Shutdown' && s.shutdown_type !== 'Sh
    utdown' && s.shutdown_type !== 'PowerOff') return false;
208     if (typeFilter === 'Unexpected' && !s.unexpected_shutdow
    n) return false;
209     if (typeFilter === 'BSOD' && s.shutdown_type !== 'BSOD'
    && !s.bugcheck) return false;
210     }
211
212     if (searchVal) {
213         const init = (s.initiator || '').toLowerCase();
214         const proc = (s.process || '').toLowerCase();
215         const reason = (s.reason || '').toLowerCase();
216         const code = (s.reason_code || '').toLowerCase();
217         const sId = (s.session_id || '').toLowerCase();
218         if (!init.includes(searchVal) && !proc.includes(searchVa
    l) && !reason.includes(searchVal) && !code.includes(searchVal)
    && !sId.includes(searchVal)) {
219             return false;
220         }
221     }
222     return true;
223 });
224
225 if (counter) counter.textContent = `Найдено: ${filtered.leng
    th}`;
226
227 if (filtered.length === 0) {
228     tbody.innerHTML = `
229         <tr>
230             <td colspan="8" class="text-center py-4 text-muted">
231                 Сессии питания, удовлетворяющие критериям фильтрации
    , не найдены.
232             </td>
233         </tr>
234     `;
235     return;
236 }

```

```

237
238   tbody.innerHTML = filtered.map(s => {
239     let statusBadge = '<span class="badge bg-secondary-subtle
text-secondary border border-secondary-subtle">Неизвестно</spa
n>';
240     if (s.shutdown_type === 'Active') {
241       statusBadge = '<span class="badge bg-success-subtle text
-success border border-success-subtle">⬢ Активная</span>';
242     } else if (s.shutdown_type === 'Restart') {
243       statusBadge = '<span class="badge bg-primary-subtle text
-primary border border-primary-subtle">⬢ Перезагрузка</span>';
244     } else if (s.shutdown_type === 'Shutdown' || s.shutdown_ty
pe === 'PowerOff') {
245       statusBadge = '<span class="badge bg-secondary-subtle te
xt-secondary border border-secondary-subtle">⬢ Выключение</spa
n>';
246     } else if (s.shutdown_type === 'BSOD' || s.bugcheck) {
247       statusBadge = `<span class="badge bg-danger text-white">
⬢ BSOD ${escapeHtml(s.bugcheck || '')}</span>`;
248     } else if (s.unexpected_shutdown) {
249       statusBadge = '<span class="badge bg-danger-subtle text-
danger border border-danger-subtle">⬢ Сбой / 41</span>';
250     }
251
252     let initiatorHtml = '<span class="text-muted">--</span>';
253     if (s.initiator || s.process) {
254       const initName = s.initiator ? `<span class="fw-semibold
text-primary">${escapeHtml(s.initiator)}</span>` : '';
255       const procName = s.process ? `<span class="badge pwr-pro
c-badge font-monospace ms-1">${escapeHtml(s.process)}</span>`
: '';
256       initiatorHtml = `<div class="d-flex align-items-center f
lex-wrap">${initName} ${procName}</div>`;
257       if (s.initiator_chain && s.initiator_chain.length > 1) {
258         initiatorHtml += `<div class="small text-muted font-mo
nospace mt-0.5" style="font-size: 0.72rem;">⬢ ${escapeHtml(s.
initiator_chain.join(' → '))}</div>`;
259     }

```

```

260     }
261
262     let reasonHtml = '<span class="text-muted">--</span>';
263     if (s.reason || s.reason_code) {
264         const codeBadge = s.reason_code ? `<span class="badge bg
-secondary-subtle text-secondary border border-secondary-subtl
e font-monospace me-1">${escapeHtml(s.reason_code)}</span>` :
        '';
265         reasonHtml = `<div>${codeBadge}<span>${escapeHtml(s.reas
on || '')}</span></div>`;
266         if (s.comment) {
267             reasonHtml += `<div class="small text-muted font-itali
c mt-0.5">␣ ${escapeHtml(s.comment)}</div>`;
268         }
269     }
270
271     return `
272         <tr>
273             <td class="font-monospace text-primary small fw-bold">
${escapeHtml(s.session_id)}</td>
274             <td class="font-monospace small">${escapeHtml(s.boot_t
ime)}</td>
275             <td class="font-monospace small text-muted">${s.shutdo
wn_time ? escapeHtml(s.shutdown_time) : '<span class="text-suc
cess fw-bold">Сейчас</span>'}</td>
276             <td class="font-monospace fw-semibold">${escapeHtml(s.
uptime_human || '--')}</td>
277             <td>${statusBadge}</td>
278             <td>${initiatorHtml}</td>
279             <td>${reasonHtml}</td>
280             <td class="text-center">
281                 <button class="btn btn-xs btn-outline-info p-1 px-2
pwr-detail-btn" data-session-id="${escapeHtml(s.session_id)}"
title="Посмотреть подробную хронологию">
282                     <i class="bi bi-eye"></i>
283                 </button>
284             </td>
285         </tr>

```

```

286     `;
287     })).join('');
288
289     // Навешиваем слушатели на кнопки деталей
290     tbody.querySelectorAll('.pwr-detail-btn').forEach(btn => {
291         btn.addEventListener('click', () => {
292             const sId = btn.dataset.sessionId;
293             showSessionDetailModal(sId);
294         });
295     });
296 }
297
298 /**
299  * Фильтрует и отрисовывает таблицу событий Event Log.
300  */
301 function filterAndRenderEvents() {
302     const tbody = document.getElementById('pwr-events-tbody');
303     const counter = document.getElementById('pwr-events-counter'
304     );
305     if (!tbody) return;
306
307     const searchVal = (document.getElementById('pwr-events-search')?.value || '').toLowerCase().trim();
308     const eidFilter = document.getElementById('pwr-event-id-filter')?.value || '';
309
310     const filtered = allEvents.filter(e => {
311         if (eidFilter && String(e.event_id) !== String(eidFilter))
312             return false;
313         if (searchVal) {
314             const prov = (e.provider || '').toLowerCase();
315             const user = (e.user || '').toLowerCase();
316             const proc = (e.process || '').toLowerCase();
317             const reason = (e.reason || '').toLowerCase();
318             const msg = (e.details?.message || '').toLowerCase();
319             if (!prov.includes(searchVal) && !user.includes(searchVal)
320                 && !proc.includes(searchVal) && !reason.includes(searchVal)
321                 && !msg.includes(searchVal)) {

```

```
318         return false;
319     }
320 }
321 return true;
322 });
323
324 if (counter) counter.textContent = `Событий: ${filtered.length}`;
325
326 if (filtered.length === 0) {
327     tbody.innerHTML = `
328         <tr>
329             <td colspan="7" class="text-center py-4 text-muted">
330                 События журнала, удовлетворяющие критериям поиска, не найдены.
331             </td>
332         </tr>
333     `;
334     return;
335 }
336
337 tbody.innerHTML = filtered.map(e => {
338     let eidBadge = `<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle font-monospace">${e.event_id}</span>`;
339     if (e.event_id === 1074) {
340         eidBadge = `<span class="badge bg-warning-subtle text-warning-emphasis border border-warning font-monospace fw-bold">1074 User32</span>`;
341     } else if (e.event_id === 41) {
342         eidBadge = `<span class="badge bg-danger-subtle text-danger border border-danger font-monospace fw-bold">41 Kernel-Power</span>`;
343     } else if (e.event_id === 6008) {
344         eidBadge = `<span class="badge bg-danger-subtle text-danger border border-danger font-monospace fw-bold">6008 EventLog</span>`;
345     } else if (e.event_id === 12 || e.event_id === 6005 || e.e
```



```

vent_id === 6009) {
346     eidBadge = ``;
347 } else if (e.event_id === 13 || e.event_id === 6006) {
348     eidBadge = ``
;
349 } else if (e.event_id === 1001) {
350     eidBadge = ``;
351 }
352
353 let userProc = ``;
354 if (e.user || e.process) {
355     userProc = `

`;
356     if (e.process) userProc += `

`;
357 }
358
359 const desc = e.reason || e.details?.message || '--';
360
361 return `
362     <tr>
363         <td class="font-monospace small">${escapeHtml(e.timest
amp)}</td>
364         <td>${eidBadge}</td>
365         <td class="small text-muted font-monospace">${escapeHt
ml(e.provider)}</td>
366         <td><span class="badge bg-secondary-subtle text-second
ary border border-secondary-subtle">${escapeHtml(e.event_type)
}</span></td>
367         <td>${userProc}</td>
368         <td class="small">${escapeHtml(desc)}</td>
369         <td class="text-center">
370             ${e.raw_xml ? `
371                 <button class="btn btn-xs btn-outline-secondary p-


```

```

1 px-2 pwr-xml-btn" data-xml="${escapeHtml(e.raw_xml)}" title=
"Просмотреть XML">
372     <i class="bi bi-code-square"></i>
373     </button>
374     ` : '<span class="text-muted">-</span>{'
375     </td>
376 </tr>
377 `;
378 }).join('');
379
380 tbody.querySelectorAll('.pwr-xml-btn').forEach(btn => {
381     btn.addEventListener('click', () => {
382         const xml = btn.dataset.xml;
383         showXmlModal(xml);
384     });
385 });
386 }
387
388 /**
389  * Открывает модальное окно деталей конкретной сессии питания.
390  * @param {string} sessionId
391  */
392 function showSessionDetailModal(sessionId) {
393     const session = allSessions.find(s => s.session_id === sessi
onId);
394     if (!session) return;
395
396     const modalTitle = document.getElementById('pwr-modal-title'
);
397     const modalBody = document.getElementById('pwr-modal-body');
398     if (modalTitle) modalTitle.textContent = `Сессия питания: ${
session.session_id}`;
399
400     if (modalBody) {
401         let eventsListHtml = '<div class="text-muted small">Нет св
язанных событий</div>';
402         if (session.events && session.events.length > 0) {
403             eventsListHtml = session.events.map(ev => `

```

```

404         <div class="pwr-detail-card p-2 mb-2">
405             <div class="d-flex justify-content-between align-ite
ms-center mb-1">
406                 <span class="badge bg-primary-subtle text-primary
border border-primary-subtle font-monospace">Event ${ev.event_
id}</span>
407                 <span class="font-monospace small text-muted">${es
capeHtml(ev.timestamp)}</span>
408             </div>
409             <div class="small fw-semibold text-primary">${escape
Html(ev.provider || '')}</div>
410             <div class="small mt-1">${escapeHtml(ev.reason || ev
.details?.message || '')}</div>
411             ${ev.process ? `<div class="small text-muted font-mo
nospace mt-1">Процесс: ${escapeHtml(ev.process)} | Пользовател
ь: ${escapeHtml(ev.user || '--')}</div>` : ''}
412         </div>
413     `).join('');
414 }
415
416 modalBody.innerHTML = `
417     <div class="row g-3 mb-3">
418         <div class="col-6">
419             <div class="pwr-detail-card p-2.5">
420                 <div class="text-muted small">⏱ Время старта (Boo
t):</div>
421                 <div class="font-monospace fw-bold text-info mt-1"
>${escapeHtml(session.boot_time)}</div>
422             </div>
423         </div>
424         <div class="col-6">
425             <div class="pwr-detail-card p-2.5">
426                 <div class="text-muted small">⏱ Завершение работы:
</div>
427                 <div class="font-monospace fw-bold ${session.shutd
own_time ? '' : 'text-success'} mt-1">
428                     ${session.shutdown_time ? escapeHtml(session.shu
tdown_time) : 'Сессия активна'}

```

```

429         </div>
430     </div>
431 </div>
432 </div>
433
434     <div class="pwr-detail-card p-3 mb-3">
435         <h6 class="fw-bold text-warning mb-2"><i class="bi bi-
436 sliders me-1.5"></i>Параметры завершения</h6>
437         <div class="row g-2 small">
438             <div class="col-4 text-muted">Тип завершения:</div>
439             <div class="col-8 fw-semibold">${escapeHtml(session.
440 shutdown_type)}</div>
441             <div class="col-4 text-muted">Аптайм сессии:</div>
442             <div class="col-8 font-monospace fw-bold">${escapeHt
443 ml(session.uptime_human || '--')}</div>
444             <div class="col-4 text-muted">Инициатор:</div>
445             <div class="col-8">${escapeHtml(session.initiator ||
446 'SYSTEM / Hardware')}</div>
447             <div class="col-4 text-muted">Процесс:</div>
448             <div class="col-8 font-monospace text-info">${escape
449 Html(session.process || '--')}</div>
450             <div class="col-4 text-muted">Код причины:</div>
451             <div class="col-8 font-monospace text-secondary">${e
452 scapeHtml(session.reason_code || '--')}</div>
453             <div class="col-4 text-muted">Текст причины:</div>
454             <div class="col-8">${escapeHtml(session.reason || '-
455 -')}</div>
456             <div class="col-4 text-muted">Комментарий:</div>
457             <div class="col-8 font-italic">${escapeHtml(sessio
458 n.comment)}</div>
459         ` : '}`

```

```

459
460         ${session.bugcheck ? `
461         <div class="col-4 text-danger fw-bold">BSOD BugChe
ck:</div>
462         <div class="col-8 text-danger font-monospace fw-bo
ld">${escapeHtml(session.bugcheck)}</div>
463         ` : ''}
464     </div>
465 </div>
466
467     ${session.initiator_chain && session.initiator_chain.len
gth > 0 ? `
468     <div class="pwr-detail-card p-3 mb-3">
469     <h6 class="fw-bold text-info mb-2"><i class="bi bi-d
iagram-3 me-1.5"></i>Корреляция цепочки вызова</h6>
470     <div class="d-flex align-items-center flex-wrap gap-
1 font-monospace small">
471         ${session.initiator_chain.map((step, idx) => `
472         <span class="badge bg-secondary-subtle text-seco
ndary border border-secondary-subtle p-2">${escapeHtml(step)}<
/span>
473         ${idx < session.initiator_chain.length - 1 ? '<i
class="bi bi-arrow-right text-muted"></i>' : ''}
474         `).join('')}
475     </div>
476 </div>
477 ` : ''}
478
479     <div class="pwr-detail-card p-3">
480     <h6 class="fw-bold mb-2"><i class="bi bi-list-nested m
e-1.5"></i>Хронология событий в рамках сессии</h6>
481     <div style="max-height: 250px; overflow-y: auto;">
482         ${eventsListHtml}
483     </div>
484 </div>
485 `;
486
487     const modalEl = document.getElementById('pwrSessionDetailM

```

```

    modal');
488     if (modalEl && window.bootstrap) {
489         const modal = new window.bootstrap.Modal(modalEl);
490         modal.show();
491     }
492 }
493 }
494
495 /**
496  * Открывает модальное окно просмотра XML события.
497  * @param {string} xmlString
498  */
499 function showXmlModal(xmlString) {
500     const pre = document.getElementById('pwr-xml-content');
501     if (pre) pre.textContent = xmlString;
502
503     const modalEl = document.getElementById('pwrXmlDetailModal')
504     ;
505     if (modalEl && window.bootstrap) {
506         const modal = new window.bootstrap.Modal(modalEl);
507         modal.show();
508     }
509 }
510 /**
511  * Запускает локальный секундный таймер для активного аптайма.
512  */
513 function startUptimeCounter() {
514     if (uptimeInterval) clearInterval(uptimeInterval);
515     uptimeInterval = setInterval(() => {
516         uptimeSeconds += 1;
517         const curUptimeEl = document.getElementById('pwr-current-u
518         ptime');
519         if (curUptimeEl && uptimeSeconds > 0) {
520             const days = Math.floor(uptimeSeconds / 86400);
521             const hours = Math.floor((uptimeSeconds % 86400) / 3600)
522             ;
523             const minutes = Math.floor((uptimeSeconds % 3600) / 60);

```

```
522     const sec = Math.floor(uptimeSeconds % 60);
523     const parts = [];
524     if (days > 0) parts.push(`${days}д`);
525     if (hours > 0 || days > 0) parts.push(`${hours}ч`);
526     if (minutes > 0 || hours > 0 || days > 0) parts.push(`${
minutes}м`);
527     parts.push(`${sec}с`);
528     curUptimeEl.textContent = parts.join(' ');
529 }
530 }, 1000);
531 }
532
533 /**
534  * Экранирует HTML строку.
535  * @param {string} str
536  * @returns {string}
537  */
538 function escapeHtml(str) {
539     if (!str) return '';
540     return String(str)
541         .replace(/&/g, '&')
542         .replace(/</g, '<')
543         .replace(/>/g, '>')
544         .replace(/"/g, '"')
545         .replace(/'/g, '&#039;');
546 }
547
548 // Автозапуск при загрузке страницы
549 if (document.readyState === 'loading') {
550     document.addEventListener('DOMContentLoaded', initPowerLifec
ycleTab);
551 } else {
552     initPowerLifecycleTab();
553 }
554
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Processes Load Inspector - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса инспектора активности проц
10  ессов (сетевая активность и мониторинг изменений файлов в реал
11  ьном времени).
12
13 Usage Examples:
14   Web Route:
15   /windows/api/webgui/processes_load_inspector_tab/index.h
16   tml
17
18   Tab Frame Integration:
19   <iframe src="/windows/api/webgui/processes_load_inspecto
20   r_tab/index.html" class="w-100 h-100 border-0"></iframe>
21
22 File: index.html
23 Project: ai-breadboard
24 Package: windows/api/webgui/processes_load_inspector_tab
25 Author: hypo69
26 Copyright: © 2026 hypo69
27 Updated: 2026-10-06 14:24:00
28 =====
29 =====
30 -->
31
32 <!-- Processes Load & Activity Inspector Tab -->
33 <style>
34   .sys-container {
35     padding: 1rem 1.25rem;
36     max-width: 100%;
37     color: var(--text-color);
```



```
30 | }
31 |
32 | .sys-section-divider {
33 |     height: 1px;
34 |     margin: 1.5rem 0;
35 |     background: linear-gradient(90deg, transparent 0%, var(--border-color) 15%, var(--nav-active) 50%, var(--border-color) 85%, transparent 100%);
36 |     position: relative;
37 | }
38 | .sys-section-divider::after {
39 |     content: '';
40 |     position: absolute;
41 |     top: -1.5px;
42 |     left: 50%;
43 |     transform: translateX(-50%);
44 |     width: 60px;
45 |     height: 4px;
46 |     background: linear-gradient(90deg, var(--nav-active), #818cf8);
47 |     border-radius: 4px;
48 |     box-shadow: 0 0 12px rgba(13, 202, 240, 0.4);
49 | }
50 |
51 | .sys-panel {
52 |     background: var(--surface-1);
53 |     border: 1px solid var(--border-color);
54 |     border-radius: 12px;
55 |     box-shadow: var(--shadow-md);
56 |     transition: transform 0.2s ease, box-shadow 0.2s ease, border-color 0.2s ease;
57 |     overflow: hidden;
58 |     color: var(--text-color);
59 | }
60 | .sys-panel:hover {
61 |     border-color: var(--border-strong);
62 |     box-shadow: var(--shadow-lg);
63 | }
```

```
64 | [data-theme="dark"] .sys-panel {
65 |     background: linear-gradient(180deg, #141c2e 0%, #0d1424 10
    0%);
66 |     border: 1px solid rgba(255, 255, 255, 0.14);
67 | }
68 | [data-theme="terminal"] .sys-panel {
69 |     background: #0b120c;
70 |     border: 1px solid #1a4223;
71 | }
72 |
73 | .sys-panel-header {
74 |     padding: 0.65rem 1rem;
75 |     background: var(--surface-2);
76 |     border-bottom: 1px solid var(--border-color);
77 |     display: flex;
78 |     justify-content: space-between;
79 |     align-items: center;
80 |     font-weight: 700;
81 |     font-size: 0.88rem;
82 | }
83 |
84 | .sys-proc-table {
85 |     width: 100%;
86 |     font-size: 0.80rem;
87 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
88 |     border-collapse: collapse;
89 | }
90 | .sys-proc-table th {
91 |     background: var(--surface-2);
92 |     padding: 6px 10px;
93 |     color: var(--text-muted);
94 |     font-weight: 700;
95 |     border-bottom: 1.5px solid var(--border-color);
96 |     position: sticky;
97 |     top: 0;
98 |     z-index: 5;
99 | }
```

```
100 | .sys-proc-table td {
101 |     padding: 5px 10px;
102 |     border-bottom: 1px solid var(--border-subtle);
103 |     vertical-align: middle;
104 | }
105 | .sys-proc-table tbody tr {
106 |     transition: background 0.15s ease;
107 | }
108 | .sys-proc-table tbody tr:hover {
109 |     background: var(--surface-2);
110 | }
111 |
112 | .sys-table-resizer {
113 |     height: 8px;
114 |     cursor: row-resize;
115 |     background: var(--surface-2);
116 |     border-top: 1px solid var(--border-color);
117 |     display: flex;
118 |     align-items: center;
119 |     justify-content: center;
120 |     transition: background-color 0.2s ease;
121 | }
122 | .sys-table-resizer:hover, .sys-table-resizer.resizing {
123 |     background: var(--nav-active);
124 | }
125 | .sys-table-resizer-grip {
126 |     width: 36px;
127 |     height: 3px;
128 |     background: var(--border-strong, #666);
129 |     border-radius: 2px;
130 | }
131 | .badge-cpu-high { color: #dc2626; font-weight: 800; }
132 | .badge-cpu-med  { color: #d97706; font-weight: 700; }
133 | </style>
134 |
135 | <div class="sys-container">
136 |     <!-- Панель 1: Топ процессов хоста (Live Stream) -->
137 |     <div class="row g-2 mb-2">
```

```

138     <div class="col-12">
139         <div class="sys-panel">
140             <div class="sys-panel-header d-flex align-items-center
justify-content-between flex-wrap gap-2">
141                 <div class="d-flex align-items-center gap-2">
142                     <i class="bi bi-list-task text-primary fs-6"></i>
143                     <span class="fw-bold" style="font-size: 0.88rem; c
olor: var(--text-color);">Топ процессов хоста (Live Stream)</s
pan>
144                     <span class="badge bg-primary-subtle text-primary
border border-primary" id="proc-top-count-badge">0 процессов</
span>
145                 </div>
146                 <div class="d-flex align-items-center gap-2 flex-wra
p">
147                     <input type="text" class="form-control form-contro
l-sm" id="sys-proc-search"
148                     placeholder="Фильтр по имени или PID..." st
yle="max-width: 220px; font-size: 0.74rem; padding: 2px 8px;">
149                     <button class="btn btn-xs btn-outline-secondary ro
unded-pill px-2.5 py-0.5" id="btn-sys-pause-proc">Пауза</butto
n>
150                     <button class="btn btn-xs btn-outline-info rounded
-pill px-2 py-0.5" id="btn-sys-refresh-proc" title="Обновить с
писок процессов">
151                         <i class="bi bi-arrow-clockwise"></i>
152                     </button>
153                 </div>
154             </div>
155             <div class="table-responsive" style="height: 380px; mi
n-height: 140px; max-height: 750px; overflow-y: auto;" id="sys
-proc-table-container">
156                 <table class="sys-proc-table">
157                     <thead>
158                         <tr>
159                             <th style="width: 70px;">PID</th>
160                             <th style="width: 200px; max-width: 240px;">Им
я процесса</th>

```

```

161         <th style="width: 90px;">Статус</th>
162         <th style="width: 80px; text-align: right;">CP
    U %</th>
163         <th style="width: 100px; text-align: right;">П
    амять</th>
164         <th style="width: 70px; text-align: right;">По
    токи</th>
165         <th>Пользователь</th>
166     </tr>
167 </thead>
168 <tbody id="sys-proc-tbody">
169     <tr><td colspan="7" class="text-center py-3" sty
    le="color: var(--text-muted);">Ожидание потока процессов...</t
    d></tr>
170     </tbody>
171 </table>
172 </div>
173 <div id="sys-proc-table-resizer" class="sys-table-resi
    zer" title="Потяните для изменения высоты таблицы (двойной кли
    к – сброс)">
174     <div class="sys-table-resizer-grip"></div>
175 </div>
176 </div>
177 </div>
178 </div>
179
180 <!-- Четкий стильный разделитель панелей -->
181 <div class="sys-section-divider"></div>
182
183 <!-- Панель 2: Сетевая активность программ (Интернет, Отправ
    ка и Прием) -->
184 <div class="row g-2">
185     <div class="col-12">
186         <div class="sys-panel">
187             <div class="sys-panel-header d-flex align-items-center
    justify-content-between flex-wrap gap-2">
188                 <div class="d-flex align-items-center gap-2">
189                     <span class="fs-6">📶 </span>

```

```

190         <span class="fw-bold" style="font-size: 0.88rem; c
      olor: var(--text-color);">Сетевая активность программ (Интерне
      т)</span>
191         <span class="badge bg-primary-subtle text-primary
      border border-primary" id="proc-netact-count-badge">0 программ
      </span>
192         <span class="badge bg-dark border border-secondary
      text-info" id="proc-netact-conns-badge">0 сокетов</span>
193     </div>
194     <div class="d-flex align-items-center gap-2 flex-wra
      p">
195         <div class="btn-group btn-group-xs" role="group">
196             <button class="btn btn-xs btn-outline-primary ac
      tive py-0.5 px-2 proc-net-filter-btn" data-filter="internet" id
      ="btn-proc-net-filter-internet">☑ Только Интернет</button>
197             <button class="btn btn-xs btn-outline-secondary
      py-0.5 px-2 proc-net-filter-btn" data-filter="all" id="btn-pro
      c-net-filter-all">Все сокет<
      ы</button>
198             <button class="btn btn-xs btn-outline-secondary
      py-0.5 px-2 proc-net-filter-btn" data-filter="listen" id="btn-
      proc-net-filter-listen">Слушающие (Listen)</button>
199         </div>
200         <input type="text" class="form-control form-contro
      l-sm" id="proc-net-search"
201             placeholder="Поиск по имени, PID или IP..."
      style="max-width: 200px; font-size: 0.74rem; padding: 2px 8px
      ;">
202         <button class="btn btn-xs btn-outline-secondary ro
      unded-pill px-2.5 py-0.5" id="btn-proc-pause-net">Пауза</butto
      n>
203         <button class="btn btn-xs btn-outline-info rounded
      -pill px-2 py-0.5" id="btn-proc-refresh-net" title="Обновить с
      етовые соединения">
204             <i class="bi bi-arrow-clockwise"></i>
205         </button>
206     </div>
207 </div>
208

```

```

209 |         <!-- Network Activity Sub-header & Live Throughput Bar
    |         -->
210 |         <div class="px-3 py-1.5 border-bottom d-flex align-items-center justify-content-between flex-wrap gap-2" style="background: var(--surface-2); border-color: var(--border-color) !important; font-size: 0.76rem;">
211 |             <div class="d-flex align-items-center gap-3 flex-wrap text-muted">
212 |                 <span><i class="bi bi-arrow-up-right text-warning me-1"></i>Отправка: <span class="fw-bold text-light" id="proc-net-send-label">HTTPS запросы, данные, API</span></span>
213 |                 <span class="border-start border-secondary ps-3"><i class="bi bi-arrow-down-left text-success me-1"></i>Прием / Скачано: <span class="fw-bold text-light" id="proc-net-recv-label">Веб-контент, медиа, ответы серверов</span></span>
214 |                 <span class="border-start border-secondary ps-3"><i class="bi bi-shield-lock text-info me-1"></i>Протоколы: <span class="badge bg-dark border border-secondary text-info">TLS / HTTPS / DNS</span></span>
215 |             </div>
216 |             <div class="d-flex align-items-center gap-2 text-muted" style="font-size: 0.74rem;">
217 |                 <span class="badge bg-dark border border-secondary text-warning" title="Измерение трафика за интервал опроса">□
    |                 Период замера: ~2 сек</span>
218 |                 <span title="Автоматическое сопоставление сокетов с процессами ОС">● Live Socket Attribution</span>
219 |             </div>
220 |         </div>
221 |
222 |         <div id="proc-net-table-container" class="p-0 table-responsive" style="height: 360px; min-height: 140px; max-height: 850px; overflow-y: auto;">
223 |             <table class="sys-proc-table">
224 |                 <thead>
225 |                     <tr>
226 |                         <th style="width: 170px;">Программа / PID</th>
227 |                         <th style="width: 220px;">Удаленный сервер (Ин

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```

тернет)</th>
228         <th style="width: 140px;">Служба / Протокол</th>
h>
229         <th style="width: 100px; text-align: center;">
Статус</th>
230         <th style="min-width: 220px;"><i class="bi bi-
arrow-up-right text-warning me-1"></i>Отправка (За период / Вс
его / Данные)</th>
231         <th style="min-width: 220px;"><i class="bi bi-
arrow-down-left text-success me-1"></i>Прием (Скачано за перио
д / Всего / Данные)</th>
232     </tr>
233     </thead>
234     <tbody id="proc-net-tbody">
235         <tr><td colspan="6" class="text-center text-mute
d p-3">0прос активных интернет-соединений программ...</td></tr>
>
236     </tbody>
237 </table>
238 </div>
239 <div id="proc-net-table-resizer" class="sys-table-resi
zer" title="Потяните для изменения высоты таблицы (двойной кли
к – сброс)">
240     <div class="sys-table-resizer-grip"></div>
241 </div>
242 </div>
243 </div>
244 </div>
245
246 <!-- Четкий стильный разделитель панелей -->
247 <div class="sys-section-divider"></div>
248
249 <!-- Панель 2: Изменения файлов в реальном времени (WinAPI R
eadDirectoryChangesW) -->
250 <div class="row g-2 mt-1">
251     <div class="col-12">
252         <div class="sys-panel">
253             <div class="sys-panel-header d-flex align-items-center

```



```

        justify-content-between flex-wrap gap-2">
254         <div class="d-flex align-items-center gap-2">
255             <span><i class="bi bi-activity text-info me-1"></i>
>Изменения файлов в реальном времени</span>
256             <span class="badge bg-info-subtle text-info" id="p
roc-live-badge">ReadDirectoryChangesW</span>
257             <span class="badge bg-dark border border-secondary
text-truncate" id="proc-watch-dir-badge" style="max-width: 28
0px; font-size: 0.72rem; cursor: pointer;" title="Текущая отсл
еживаемая папка. Нажмите для смены.">
258                 □ <span id="proc-watch-dir-path">Загрузка...</s
pan>
259             </span>
260         </div>
261         <div class="d-flex align-items-center gap-1.5">
262             <span class="badge bg-success-subtle text-success
border border-success px-2 py-1" id="proc-watcher-sensor-stat
s" style="font-size: 0.7rem;">□ Штатный режим</span>
263             <button class="btn btn-xs btn-outline-warning roun
ded-pill px-2 py-0.5" style="font-size: 0.72rem;" id="btn-proc
-watch-exclusions" title="Настроить исключения и фильтры монит
оринга">
264                 <i class="bi bi-funnel me-1"></i> Исключения <sp
an class="badge bg-dark border border-secondary text-warning"
id="proc-exclusions-count-badge">0</span>
265             </button>
266             <button class="btn btn-xs btn-outline-light rounde
d-pill px-2 py-0.5" style="font-size: 0.72rem;" id="btn-proc-c
hange-watch-dir" title="Выбрать другую папку для мониторинга">
267                 <i class="bi bi-folder2-open me-1 text-warning">
</i> Папка
268             </button>
269             <button class="btn btn-xs btn-outline-info rounded
-pill px-2 py-0.5" style="font-size: 0.72rem;" id="btn-proc-li
ve-help" title="Подробная справка о работе модуля">
270                 <i class="bi bi-info-circle me-1"></i> Справка
271             </button>
272         </div>

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```

273 |         </div>
274 |
275 |         <!-- Real-Time Watcher Telemetry & Hardware Sensors Ba
r -->
276 |         <div class="px-3 py-2 border-bottom d-flex align-items
-center justify-content-between flex-wrap gap-2" style="backgr
ound: var(--surface-2); border-color: var(--border-color) !imp
ortant; font-size: 0.97rem;">
277 |             <div class="d-flex align-items-center gap-2 flex-wra
p">
278 |                 <span class="text-muted"><i class="bi bi-speedomet
er2 text-info me-1"></i>Темп: <span class="fw-bold text-info"
id="proc-watcher-rate">0.0</span> on/c</span>
279 |                 <span class="text-muted" style="font-size: 0.92rem
;">(<span class="text-success" id="proc-watcher-cr">0</span> c
r / <span class="text-warning" id="proc-watcher-mod">0</span>
mod / <span class="text-danger" id="proc-watcher-del">0</span>
del)</span>
280 |                 <span class="border-start border-secondary ps-2 te
xt-muted" title="Количество отфильтрованных событий">
281 |                     <i class="bi bi-funnel text-warning me-1"></i>Ис
ключено: <span class="text-warning fw-bold" id="proc-watcher-f
iltered-count">0</span>
282 |                 </span>
283 |                 <span class="border-start border-secondary ps-2 te
xt-muted">
284 |                     <i class="bi bi-hdd-network text-primary me-1"><
/i>Диск: <span class="text-light fw-bold" id="proc-watcher-dri
ve">C:</span> (<span class="text-muted" id="proc-watcher-drive
-model">Storage</span>)
285 |                 </span>
286 |                 <span class="badge bg-dark border border-secondary
text-info px-2 py-0.5" id="proc-watcher-temp" style="font-siz
e: 0.92rem;">-- °C</span>
287 |                 <span class="text-muted" style="font-size: 0.92rem
;">R: <span class="text-light" id="proc-watcher-r-kbs">0</span>
> KB/s | W: <span class="text-light" id="proc-watcher-w-kbs">0
</span> KB/s</span>

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288         </div>
289         <div class="d-flex align-items-center gap-1 text-mut
ed" style="font-size: 0.92rem;">
290             <span><i class="bi bi-shield-check text-success me
-1"></i>Audit 4660/4663:</span>
291             <span class="badge bg-dark border border-secondary
text-muted px-2 py-0.5" id="proc-watcher-sec-matches" style="
font-size: 0.92rem;">0</span>
292             <span class="text-muted ms-1" title="Периодическое
логирование в file_watcher_telemetry_polls.csv">● AutoLog Act
ive</span>
293         </div>
294     </div>
295     <div id="proc-watcher-table-container" class="p-0 tabl
e-responsive" style="height: 340px; min-height: 120px; max-hei
ght: 850px; overflow-y: auto;">
296         <table class="sys-proc-table">
297             <thead>
298                 <tr>
299                     <th style="width: 100px;">Время</th>
300                     <th style="width: 120px;">Действие</th>
301                     <th style="width: 170px;">Программа / PID</th>
302                     <th>Полный путь к файлу</th>
303                 </tr>
304             </thead>
305             <tbody id="proc-watcher-tbody">
306                 <tr><td colspan="4" class="text-center text-mute
d p-3">Мониторинг файловых событий...</td></tr>
307             </tbody>
308         </table>
309     </div>
310     <div id="proc-watcher-table-resizer" class="sys-table-
resizer" title="Потяните для изменения высоты таблицы (двойной
клик – сброс)">
311         <div class="sys-table-resizer-grip"></div>
312     </div>
313 </div>
314 </div>

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```

315 |     </div>
316 | </div>
317 |
318 | <!-- Real-Time Watcher Multi-Folder Selection & Exclusions Mod
    | al -->
319 | <div class="modal fade" id="modal-proc-watch-folders" tabindex
    | ="-1" aria-labelledby="modalProcWatchFoldersLabel" aria-hidden
    | ="true">
320 |     <div class="modal-dialog modal-dialog-centered modal-lg moda
    | l-dialog-scrollable">
321 |         <div class="modal-content text-white border-secondary shad
    | ow-lg" style="background: var(--surface-1); border-color: var(
    | --border-color) !important;">
322 |             <div class="modal-header py-2.5 px-3 border-secondary" s
    | tyle="background: var(--surface-2); border-color: var(--border
    | -color) !important;">
323 |                 <h6 class="modal-title d-flex align-items-center gap-2
    | mb-0" id="modalProcWatchFoldersLabel">
324 |                     <i class="bi bi-folder2-open text-warning fs-5"></i>
325 |                     <span class="fw-bold">Отслеживаемые папки и исключен
    | ия (Live Watcher)</span>
326 |                     <span class="badge bg-info-subtle text-info font-mon
    |ospace ms-1" style="font-size: 0.7rem;">ReadDirectoryChangesW<
    | /span>
327 |                 </h6>
328 |                 <button type="button" class="btn-close btn-close-white
    | " data-bs-dismiss="modal" aria-label="Close"></button>
329 |             </div>
330 |             <div class="modal-body p-3" style="max-height: 75vh; ove
    | rflow-y: auto;">
331 |                 <!-- Nav Tabs -->
332 |                 <ul class="nav nav-pills nav-fill mb-3" id="pills-proc
    | -watcher-tabs" role="tablist">
333 |                     <li class="nav-item" role="presentation">
334 |                         <button class="nav-link active py-1 px-3 small fw-
    | bold" id="tab-btn-proc-folders" data-bs-toggle="pill" data-bs-
    | target="#tab-pane-proc-folders" type="button" role="tab">
335 |                             <i class="bi bi-folder2-open me-1.5 text-warning

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    "></i>Папки мониторинга (<span id="modal-proc-dirs-tab-count">
1</span>)
336         </button>
337     </li>
338     <li class="nav-item" role="presentation">
339         <button class="nav-link py-1 px-3 small fw-bold" id="tab-btn-proc-exclusions" data-bs-toggle="pill" data-bs-target="#tab-pane-proc-exclusions" type="button" role="tab">
340             <i class="bi bi-funnel me-1.5 text-warning"></i>
Исключения и фильтры (<span id="modal-proc-exclusions-tab-count">0</span>)
341         </button>
342     </li>
343 </ul>
344
345     <div class="tab-content">
346         <!-- Tab 1: Monitored Folders -->
347         <div class="tab-pane fade show active" id="tab-pane-proc-folders" role="tabpanel">
348             <!-- Alert feedback -->
349             <div id="modal-proc-dirs-alert" class="alert alert-success py-2 px-3 small mb-3 d-none"></div>
350
351             <!-- Section 1: Active Monitored Folders List -->
352             <div class="card border-secondary mb-3" style="background: var(--surface-2); border-color: var(--border-color) !important;">
353                 <div class="card-header py-2 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
354                     <div class="d-flex align-items-center gap-2">
355                         <i class="bi bi-eye text-success"></i>
356                         <span class="small fw-bold text-light">Активные папки мониторинга</span>
357                     <span class="badge bg-primary rounded-pill" id="modal-proc-dirs-count">0</span>
358                     </div>
359                     <span class="small text-muted" style="font-size: 0.8em; margin-left: 10px;">0</span>

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    e: 0.72rem;">События агрегируются в реальном времени</span>
360         </div>
361         <div class="card-body p-2">
362             <div class="d-flex flex-column gap-1.5" id="mo
dal-proc-dirs-active-list" style="max-height: 160px; overflow-
y: auto;">
363                 <!-- Dynamically populated active watched di
rectories -->
364             </div>
365         </div>
366     </div>
367
368     <!-- Section 2: Quick System Presets -->
369     <div class="mb-3">
370         <div class="small fw-bold text-muted mb-1.5"><i
class="bi bi-lightning-charge text-warning me-1"></i>Быстрые с
истемные каталоги:</div>
371         <div class="d-flex flex-wrap gap-1.5" id="modal-
proc-presets-bar">
372             <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 proc-preset-btn" data-preset="proje
ct">
373                 □ Проект AI-Breadboard
374             </button>
375             <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 proc-preset-btn" data-preset="downl
oads">
376                 □ Загрузки (Downloads)
377             </button>
378             <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 proc-preset-btn" data-preset="deskt
op">
379                 □ Рабочий стол (Desktop)
380             </button>
381             <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 proc-preset-btn" data-preset="temp"
>
382                 □ Temp

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383         </button>
384     </div>
385 </div>
386
387     <!-- Section 3: System Folder Explorer -->
388     <div class="card border-secondary" style="background: var(--surface-2); border-color: var(--border-color) !important;">
389         <div class="card-header py-2 px-3 border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2" style="border-color: var(--border-color) !important;">
390             <div class="d-flex align-items-center gap-2">
391                 <i class="bi bi-compass text-info"></i>
392                 <span class="small fw-bold text-light">Проводник файлов и папок</span>
393             </div>
394             <!-- Drives selection bar -->
395             <div class="d-flex align-items-center gap-1" id="modal-proc-drives-bar">
396                 <!-- Drives loaded dynamically -->
397             </div>
398         </div>
399         <div class="card-body p-2.5">
400             <!-- Path bar & Search/Input -->
401             <div class="input-group input-group-sm mb-2">
402                 <button class="btn btn-outline-secondary" type="button" id="btn-modal-proc-folder-up" title="Перейти на уровень выше">
403                     <i class="bi bi-arrow-up-circle"></i> Вверх
404                 </button>
405                 <input type="text" class="form-control bg-dark text-white border-secondary font-monospace" id="modal-proc-folder-path-input" placeholder="C:\..." style="font-size: 0.8rem;">
406                 <button class="btn btn-outline-info" type="button" id="btn-modal-proc-folder-go" title="Открыть указанный путь">

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407 |             <i class="bi bi-arrow-right-circle me-1"><
| /i> Перейти
408 |             </button>
409 |             <button class="btn btn-success" type="button
| " id="btn-modal-proc-folder-add-current" title="Добавить эту п
| апку в мониторинг">
410 |                 <i class="bi bi-plus-circle me-1"></i> + Д
| обавить папку
411 |             </button>
412 |         </div>
413 |
414 |         <!-- Breadcrumbs navigation -->
415 |         <div class="d-flex align-items-center flex-wra
| p gap-1 p-1.5 rounded mb-2 font-monospace small" id="modal-pro
| c-folder-breadcrumbs" style="background: var(--bg-color); font
| -size: 0.75rem;">
416 |             <!-- Breadcrumbs injected dynamically -->
417 |             </div>
418 |
419 |             <!-- Subdirectories list -->
420 |             <div class="border border-secondary rounded p-
| 1" style="background: var(--bg-color); max-height: 200px; over
| flow-y: auto;">
421 |                 <div id="modal-proc-folder-browser-list" cla
| ss="d-flex flex-column gap-1">
422 |                     <div class="text-center text-muted small p
| y-3">Загрузка структуры каталогов...</div>
423 |                 </div>
424 |             </div>
425 |         </div>
426 |     </div>
427 | </div>
428 |
429 |     <!-- Tab 2: Exclusions & Filters -->
430 |     <div class="tab-pane fade" id="tab-pane-proc-exclusi
| ons" role="tabpanel">
431 |         <!-- Alert feedback -->
432 |         <div id="modal-proc-exclusions-alert" class="alert

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    alert-success py-2 px-3 small mb-3 d-none"></div>
433
434         <!-- Master Toggle & Statistics -->
435         <div class="card border-secondary mb-3" style="bac
kground: var(--surface-2); border-color: var(--border-color) !
important;">
436             <div class="card-body p-2.5 d-flex align-items-c
enter justify-content-between flex-wrap gap-2">
437                 <div class="d-flex align-items-center gap-2">
438                     <div class="form-check form-switch mb-0">
439                         <input class="form-check-input" type="chec
kbox" role="switch" id="switch-proc-exclusions-active" checked
style="cursor: pointer;">
440                         <label class="form-check-label fw-bold tex
t-light small" for="switch-proc-exclusions-active">Фильтрация
исключений активна</label>
441                     </div>
442                     <span class="badge bg-success" id="badge-pro
c-exclusions-status">ВКЛ</span>
443                 </div>
444                 <div class="text-muted small" style="font-size
: 0.75rem;">
445                     Отфильтровано спам-событий: <span class="fw-
bold text-warning font-monospace" id="modal-proc-exclusions-fi
ltered-total">0</span>
446                 </div>
447             </div>
448         </div>
449
450         <!-- Quick Exclusions Presets -->
451         <div class="mb-3">
452             <div class="small fw-bold text-muted mb-1.5"><i
class="bi bi-lightning-charge text-warning me-1"></i>Быстрые п
ресеты фильтрации:</div>
453             <div class="d-flex flex-wrap gap-1.5">
454                 <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 proc-ex-preset-btn" data-preset="lo
gs" title="Исключить папки логов, .csv и .log">

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455 |             □ Логи & Телеметрия
456 |             </button>
457 |             <button class="btn btn-xs btn-outline-secondary
| rounded-pill px-2.5 py-1 proc-ex-preset-btn" data-preset="se
| arch" title="Исключить системную службу SearchIndexer.exe и Wi
| ndows.db">
458 |             □ SearchIndexer.exe & Windows.db
459 |             </button>
460 |             <button class="btn btn-xs btn-outline-secondary
| rounded-pill px-2.5 py-1 proc-ex-preset-btn" data-preset="te
| mp" title="Исключить .tmp, ~$* и папку Temp">
461 |             □ Temp & Временные (.tmp)
462 |             </button>
463 |             <button class="btn btn-xs btn-outline-secondary
| rounded-pill px-2.5 py-1 proc-ex-preset-btn" data-preset="de
| v" title="Исключить .git, node_modules, __pycache__">
464 |             □ Разработка (.git, node_modules)
465 |             </button>
466 |         </div>
467 |     </div>
468 |
469 |     <!-- Add Exclusion Rule Form -->
470 |     <div class="card border-secondary mb-3" style="back
| ground: var(--surface-2); border-color: var(--border-color) !
| important;">
471 |         <div class="card-header py-2 px-3 border-seconda
| ry d-flex align-items-center gap-2" style="border-color: var(-
| -border-color) !important;">
472 |             <i class="bi bi-plus-circle text-info"></i>
473 |             <span class="small fw-bold text-light">Добавит
| ь правило исключения</span>
474 |         </div>
475 |         <div class="card-body p-2.5">
476 |             <div class="row g-2 align-items-center">
477 |                 <div class="col-md-4">
478 |                     <select class="form-select form-select-sm
| bg-dark text-white border-secondary" id="select-proc-exclusion
| -category">

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479         <option value="paths">□ Папка / Путь (P
|ath)</option>
480         <option value="extensions">□ Расширение
|файла (.ext)</option>
481         <option value="patterns">□ Шаблон имени
|(*pattern*)</option>
482         <option value="processes">□ Имя программ
|ы (Process.exe)</option>
483     </select>
484 </div>
485     <div class="col-md-6">
486         <input type="text" class="form-control for
|m-control-sm bg-dark text-white border-secondary font-monospac
|e" id="input-proc-exclusion-value" placeholder="Например: AppD
|ata\Local\Temp, .csv, SearchIndexer.exe">
487     </div>
488     <div class="col-md-2">
489         <button class="btn btn-sm btn-info w-100 f
|w-bold rounded-pill" id="btn-proc-add-exclusion">
490             <i class="bi bi-plus-lg me-1"></i> Добав
|ИТЬ
491         </button>
492     </div>
493 </div>
494 </div>
495 </div>
496
497 <!-- Grouped Cards of Active Exclusions -->
498 <div class="row g-2">
499     <!-- Paths -->
500     <div class="col-md-6">
501         <div class="card border-secondary h-100" style
|= "background: var(--surface-2); border-color: var(--border-col
|or) !important;">
502             <div class="card-header py-1.5 px-2.5 border
|-secondary d-flex justify-content-between align-items-center"
|style="border-color: var(--border-color) !important;">
503                 <span class="small fw-bold text-light"><i

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class="bi bi-folder-x text-warning me-1"></i>Папки и пути</spa
n>
504         <span class="badge bg-secondary font-monos
pace" id="badge-proc-ex-paths-count">0</span>
505     </div>
506     <div class="card-body p-2" id="list-proc-ex-
paths" style="max-height: 120px; overflow-y: auto;"></div>
507 </div>
508 </div>
509 <!-- Extensions -->
510 <div class="col-md-6">
511     <div class="card border-secondary h-100" style
="background: var(--surface-2); border-color: var(--border-col
or) !important;">
512         <div class="card-header py-1.5 px-2.5 border
-secondary d-flex justify-content-between align-items-center"
style="border-color: var(--border-color) !important;">
513             <span class="small fw-bold text-light"><i
class="bi bi-file-earmark-x text-info me-1"></i>Расширения фай
лов</span>
514             <span class="badge bg-secondary font-monos
pace" id="badge-proc-ex-extensions-count">0</span>
515         </div>
516         <div class="card-body p-2" id="list-proc-ex-
extensions" style="max-height: 120px; overflow-y: auto;"></div>
>
517     </div>
518 </div>
519 <!-- Patterns -->
520 <div class="col-md-6">
521     <div class="card border-secondary h-100" style
="background: var(--surface-2); border-color: var(--border-col
or) !important;">
522         <div class="card-header py-1.5 px-2.5 border
-secondary d-flex justify-content-between align-items-center"
style="border-color: var(--border-color) !important;">
523             <span class="small fw-bold text-light"><i
class="bi bi-asterisk text-primary me-1"></i>Шаблоны (Wildcard

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s)/span>
524         <span class="badge bg-secondary font-monospace" id="badge-proc-ex-patterns-count">0</span>
525     </div>
526     <div class="card-body p-2" id="list-proc-ex-patterns" style="max-height: 120px; overflow-y: auto;"></div>
527 </div>
528 </div>
529 <!-- Processes -->
530 <div class="col-md-6">
531     <div class="card border-secondary h-100" style="background: var(--surface-2); border-color: var(--border-color) !important;">
532         <div class="card-header py-1.5 px-2.5 border-secondary d-flex justify-content-between align-items-center" style="border-color: var(--border-color) !important;">
533             <span class="small fw-bold text-light"><i class="bi bi-cpu text-danger me-1"></i>Программы (Процессы)</span>
534             <span class="badge bg-secondary font-monospace" id="badge-proc-ex-processes-count">0</span>
535         </div>
536         <div class="card-body p-2" id="list-proc-ex-processes" style="max-height: 120px; overflow-y: auto;"></div>
537     </div>
538 </div>
539 </div>
540 </div>
541 </div>
542 </div>
543
544 <div class="modal-footer py-2 px-3 border-secondary d-flex justify-content-between" style="background: var(--surface-2); border-color: var(--border-color) !important;">
545     <span class="small text-muted" id="modal-proc-dirs-footer-hint">Выберите папки и нажмите «Применить»</span>
546     <div class="d-flex gap-2">
547         <button class="btn btn-sm btn-secondary rounded-pill

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px-3" data-bs-dismiss="modal">Заккрыть</button>
548 |         <button class="btn btn-sm btn-primary rounded-pill p
      | x-4 fw-bold" id="btn-modal-proc-dirs-apply">
549 |         <i class="bi bi-check2-circle me-1"></i> Применить
      | и запустить
550 |         </button>
551 |     </div>
552 | </div>
553 | </div>
554 | </div>
555 | </div>
556 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Processes Load Inspector Tab - Main S
  cript
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля инспектор
  а активности процессов
7   *   (сетевая активность и мониторинг изменений файлов в реаль
  ном времени).
8   *
9   * Usage Examples:
10  *   HTML Integration:
11  *     <script src="/windows/api/webgui/processes_load_inspect
  or_tab/main.js?v=20261006_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/processes_load_inspector_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 04:06:00
19  * =====
20  */
21
22 (function() {
23   let isProcPaused = false;
24   let cachedTopProcesses = [];
25
26   let isNetPaused = false;
27   let currentNetFilter = 'internet';
28   let cachedNetworkActivities = [];
29
```

```

30 | let currentWatchDir = '';
31 | let currentWatchDirs = [];
32 | let stagedWatchDirs = [];
33 | let currentBrowserPath = 'C:\\';
34 | let cachedDrives = [];
35 | let _currentUiRefreshSeconds = 2;
36 |
37 | let currentExclusions = {
38 |     enabled: true,
39 |     paths: [],
40 |     extensions: [],
41 |     patterns: [],
42 |     processes: [],
43 |     filtered_count: 0
44 | };
45 |
46 | async function ensureAITableModal() {
47 |     if (window.AITableModal && typeof window.AITableModal.show
=== 'function') return true;
48 |     if (!document.querySelector('script[src*="ai_table_modal.j
s"]')) {
49 |         await new Promise((resolve) => {
50 |             const script = document.createElement('script');
51 |             script.src = '/html/js/ai_table_modal.js?v=20261008_v1
';
52 |             script.onload = () => resolve(true);
53 |             script.onerror = () => resolve(false);
54 |             document.head.appendChild(script);
55 |         });
56 |     }
57 |     return !(window.AITableModal && typeof window.AITableModa
l.show === 'function');
58 | }
59 |
60 | function escapeHtml(str) {
61 |     if (!str) return '';
62 |     return String(str)
63 |         .replace(/&/g, '&amp;');

```



```

64     .replace(/</g, '&lt;')
65     .replace(/>/g, '&gt;')
66     .replace(/"/g, '&quot;')
67     .replace(/'/g, '&#039;');
68 }
69
70 // =====
71 // 0. Top Host Processes (Live Stream)
72 // =====
73
74 async function fetchTopProcesses() {
75     try {
76         const res = await fetch('/api/v1/system/summary?process_
limit=50');
77         if (res.ok) {
78             const snap = await res.json();
79             const processes = snap.top_processes || snap.processes
|| [];
80             renderProcessTable(processes);
81         }
82     } catch (e) {
83         console.warn('[ProcessesLoadInspectorTab] Failed to fetc
h top processes:', e);
84     }
85 }
86
87 function renderProcessTable(processes) {
88     if (processes) {
89         cachedTopProcesses = processes;
90     }
91     if (isProcPaused) return;
92
93     const filterInput = document.getElementById('sys-proc-sear
ch');
94     const filter = (filterInput?.value || '').toLowerCase().tr
im();

```

```

95     const tbody = document.getElementById('sys-proc-tbody');
96     const countBadge = document.getElementById('proc-top-count
    -badge');
97     if (!tbody) return;
98
99     const rawList = Array.isArray(cachedTopProcesses) ? cached
    TopProcesses : [];
100    const filtered = rawList.filter(p => {
101        if (!filter) return true;
102        return (
103            String(p.pid).includes(filter) ||
104            (p.name || '').toLowerCase().includes(filter) ||
105            (p.username && p.username.toLowerCase().includes(filte
    r))
106        );
107    });
108
109    if (countBadge) {
110        countBadge.textContent = `${filtered.length} процессов`;
111    }
112
113    if (filtered.length === 0) {
114        tbody.innerHTML = '<tr><td colspan="7" class="text-cente
    r py-4 text-muted small">Процессы не найдены</td></tr>';
115        return;
116    }
117
118    tbody.innerHTML = filtered.map((p, idx) => {
119        let cpuClass = '';
120        if (p.cpu_percent > 40) cpuClass = 'badge-cpu-high';
121        else if (p.cpu_percent > 15) cpuClass = 'badge-cpu-med';
122
123        return `
124            <tr class="sys-proc-row" data-idx="${idx}" style="curs
    or: pointer;" title="Нажмите для детальной AI-диагностики проц
    есса">
125                <td class="font-monospace fw-semibold text-info">${p
    .pid}</td>

```

```

126 |         <td style="font-weight: 600; max-width: 200px; overf
| low: hidden; text-overflow: ellipsis; white-space: nowrap; col
| or: var(--text-color);" title="${escapeHtml(p.name || '')}">${
| escapeHtml(p.name || '')}</td>
127 |         <td class="text-muted">${escapeHtml(p.status || 'run
| ning')}</td>
128 |         <td style="text-align: right;" class="${cpuClass}">${
| {Number(p.cpu_percent || 0).toFixed(1)}%</td>
129 |         <td style="text-align: right;" class="font-monospace
| text-success fw-semibold">${Number(p.memory_mb || 0).toFixed(
| 1)} MB</td>
130 |         <td style="text-align: right;" class="font-monospace
| text-warning fw-semibold">${p.num_threads || 1}</td>
131 |         <td class="text-muted">${escapeHtml(p.username || 'S
| YSTEM')}</td>
132 |     </tr>
133 |     `;
134 |     }).join('');
135 |
136 |     tbody.querySelectorAll('.sys-proc-row').forEach(row => {
137 |         row.onclick = async () => {
138 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
139 |             ;
140 |             const p = filtered[idx];
141 |             if (!p) return;
142 |             const isModalReady = await ensureAITableModal();
143 |             if (isModalReady && window.AITableModal) {
144 |                 window.AITableModal.show({
145 |                     icon: '🔍 ',
146 |                     title: p.name,
147 |                     subtitle: `PID: ${p.pid} | ${p.username || 'SYSTEM
| '}` ,
148 |                     tableType: 'process',
149 |                     badges: [
150 |                         { text: `PID ${p.pid}`, class: 'badge bg-info te
| xt-dark' },
151 |                         { text: p.status || 'running', class: 'badge bg-

```

```

    success' }
152         ],
153         metadata: [
154             { label: 'Имя процесса', value: p.name },
155             { label: 'Process ID (PID)', value: String(p.pid
156         ) },
157             { label: 'Пользователь / Учетная запись', value:
158         p.username || 'SYSTEM' },
159             { label: 'Статус', value: p.status || 'Выполняет
160         ся' },
161             { label: 'Загрузка CPU', value: `${Number(p.cpu_
162         percent || 0).toFixed(1)}%` },
163             { label: 'Оперативная память', value: `${Number(
164         p.memory_mb || 0).toFixed(1)} MB` },
165             { label: 'Количество потоков', value: String(p.n
166         um_threads || 1) },
167             { label: 'Исполняемый путь', value: p.exe || p.e
168         xecutable_path || 'Системный процесс Windows', isCode: true, f
169         ullWidth: true }
170         ],
171         rawTitle: 'Команда запуска / Аргументы',
172         rawContent: Array.isArray(p.cmdline) ? p.cmdline.j
173         oin(' ') : (p.cmdline || p.exe || ''),
174         requestData: {
175             pid: p.pid,
176             cpu_percent: p.cpu_percent,
177             memory_mb: p.memory_mb,
178             username: p.username,
179             status: p.status
180         }
181     });
182 }
183 };
184 });
185 }
186
187 // =====
188 =====

```

```

179 // 1. Network Activity of Programs (Интернет & Сокеты)
180 // =====
181
182 async function fetchNetworkActivity() {
183     try {
184         const res = await fetch('/api/v1/system/network-activity
?limit=100');
185         if (res.ok) {
186             const data = await res.json();
187             renderNetworkActivityTable(data);
188         }
189     } catch (e) {
190         console.warn('[ProcessesLoadInspectorTab] Failed to fetch
network activity:', e);
191     }
192 }
193
194 function renderNetworkActivityTable(activities) {
195     if (activities) {
196         cachedNetworkActivities = activities;
197     }
198     if (isNetPaused) return;
199
200     const tbody = document.getElementById('proc-net-tbody');
201     if (!tbody) return;
202
203     const searchInput = document.getElementById('proc-net-search');
204     const filterText = (searchInput?.value || '').toLowerCase().trim();
205     const countBadge = document.getElementById('proc-net-act-count-badge');
206     const connsBadge = document.getElementById('proc-net-act-conns-badge');
207
208     const rawList = Array.isArray(cachedNetworkActivities) ? cachedNetworkActivities : [];

```

```

209
210     let filtered = rawList.filter(item => {
211         if (currentNetFilter === 'internet' && !item.is_internet
212         ) return false;
213         if (currentNetFilter === 'listen' && item.status !== 'LI
214         STEN') return false;
215
216         if (filterText) {
217             return (
218                 String(item.pid).includes(filterText) ||
219                 (item.name || '').toLowerCase().includes(filterText)
220                 ||
221                 (item.user || '').toLowerCase().includes(filterText)
222                 ||
223                 (item.remote_address || '').toLowerCase().includes(f
224                 ilterText) ||
225                 (item.local_address || '').toLowerCase().includes(fi
226                 lterText) ||
227                 (item.service_type || '').toLowerCase().includes(fil
228                 terText) ||
229                 (item.sent_summary || '').toLowerCase().includes(fil
230                 terText) ||
231                 (item.recv_summary || '').toLowerCase().includes(fil
232                 terText)
233             );
234         }
235         return true;
236     });
237
238     const uniqueProcs = new Set(filtered.map(i => i.pid)).size
239     ;
240     if (countBadge) countBadge.textContent = `${uniqueProcs} n
241     porpamm`;
242     if (connsBadge) connsBadge.textContent = `${filtered.lengt
243     h} сокетов`;
244
245     if (filtered.length === 0) {
246         tbody.innerHTML = '<tr><td colspan="6" class="text-cente

```

```

    r py-4 text-muted small">Нет активных сетевых соединений по вы
    бранному фильтру</td></tr>';
235     return;
236   }
237
238   tbody.innerHTML = filtered.map((item, idx) => {
239     const isListen = item.status === 'LISTEN';
240     const statusBadgeClass = isListen
241       ? 'badge bg-secondary-subtle text-light border border-
secondary'
242       : item.status === 'ESTABLISHED'
243       ? 'badge bg-success-subtle text-success border border-
success'
244       : 'badge bg-warning-subtle text-warning border border-
warning';
245
246     const protoBadge = item.protocol === 'UDP'
247       ? '<span class="badge bg-primary text-white" style="fo
nt-size: 0.65rem;">UDP</span>'
248       : '<span class="badge bg-dark border border-secondary
text-info" style="font-size: 0.65rem;">TCP</span>';
249
250     const isExtBadge = item.is_internet
251       ? '<span class="badge bg-primary-subtle text-primary b
order border-primary px-1" style="font-size: 0.62rem;" title="
Внешний сервер в сети Интернет">WAN</span>'
252       : '<span class="badge bg-secondary px-1" style="font-s
ize: 0.62rem;" title="Локальный сокет Loopback">LAN</span>';
253
254     const formatNetKb = (kb) => {
255       if (!kb || kb <= 0) return '0 KB';
256       if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).toF
ixed(2) + ' GB';
257       if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB';
258       return kb.toFixed(1) + ' KB';
259     };
260
261     const formatNetRate = (rate) => {

```

```

262         if (!rate || rate <= 0.05) return '';
263         if (rate >= 1024) return (rate / 1024).toFixed(1) + '
MB/s';
264         return rate.toFixed(1) + ' KB/s';
265     };
266
267     const deltaSentStr = (item.delta_sent_kb && item.delta_s
ent_kb > 0)
268         ? `▲ +${formatNetKb(item.delta_sent_kb)}${item.sent_ra
te_kbs > 0.05 ? ' (' + formatNetRate(item.sent_rate_kbs) + ') '
: ''}`
269         : '▲ 0 KB';
270     const deltaRecvStr = (item.delta_recv_kb && item.delta_r
ecv_kb > 0)
271         ? `▼ +${formatNetKb(item.delta_recv_kb)}${item.recv_ra
te_kbs > 0.05 ? ' (' + formatNetRate(item.recv_rate_kbs) + ') '
: ''}`
272         : '▼ 0 KB';
273
274     const totalSentStr = formatNetKb(item.sent_kb || 0);
275     const totalRecvStr = formatNetKb(item.recv_kb || 0);
276
277     return `
278         <tr class="proc-net-row" data-idx="${idx}" style="curs
or: pointer;" title="Нажмите для детальной диагностики сетевог
о соединения">
279             <td>
280                 <div class="fw-bold text-truncate" style="max-widt
h: 165px; color: var(--text-color);" title="${escapeHtml(item.
name)}">${escapeHtml(item.name)}</div>
281                 <div class="small text-muted" style="font-size: 0.
70rem;">PID: <span class="font-monospace text-info">${item.pid
}</span> ${item.user ? '• ' + escapeHtml(item.user) : ''}</div>
282             </td>
283             <td>
284                 <div class="d-flex align-items-center gap-1">
285                     ${isExtBadge}

```



```

286 |         <span class="font-monospace fw-semibold text-tru
| ncate" style="max-width: 195px; color: var(--text-color);" tit
| le="{escapeHtml(item.remote_address)}">
287 |             ${escapeHtml(item.remote_address !== '-' ? ite
| m.remote_address : item.local_address)}
288 |         </span>
289 |     </div>
290 |     <div class="small text-muted font-monospace" style
| ="font-size: 0.68rem;">Local: ${escapeHtml(item.local_address)
| }</div>
291 | </td>
292 | <td>
293 |     <div class="d-flex align-items-center gap-1 mb-0.5
| ">
294 |         ${protoBadge}
295 |         <span class="fw-semibold text-truncate" style="m
| ax-width: 110px; font-size: 0.74rem; color: var(--nav-active);
| " title="{escapeHtml(item.service_type)}">${escapeHtml(item.s
| ervice_type)}</span>
296 |     </div>
297 | </td>
298 | <td style="text-align: center;">
299 |     <span class="{statusBadgeClass}" style="font-size
| : 0.68rem;">${escapeHtml(item.status)}</span>
300 | </td>
301 | <td>
302 |     <div class="d-flex align-items-center justify-cont
| ent-between gap-1 mb-1">
303 |         <span class="badge bg-warning-subtle text-warnin
| g border border-warning px-1.5 py-0.5" style="font-size: 0.68r
| em;" title="Отправлено за измеряемый период">
304 |             ${deltaSentStr}
305 |         </span>
306 |         <span class="text-muted font-monospace" style="f
| ont-size: 0.66rem;" title="Всего отправлено/записано">
307 |             Σ ${totalSentStr}
308 |         </span>
309 |     </div>

```

```

310         <div class="d-flex align-items-start gap-1">
311             <i class="bi bi-arrow-up-right text-warning mt-0
312 | .5" style="font-size: 0.70rem;"></i>
313             <div class="text-truncate" style="max-width: 250
314 | px; font-size: 0.72rem; color: var(--text-muted);" title="{escapeHtml(item.sent_summary)}">
315                 {escapeHtml(item.sent_summary)}
316             </div>
317         </td>
318         <td>
319             <div class="d-flex align-items-center justify-content-between gap-1 mb-1">
320                 <span class="badge bg-success-subtle text-success border border-success px-1.5 py-0.5" style="font-size: 0.68rem;" title="Скачано/получено за измеряемый период">
321                     {deltaRecvStr}
322                 </span>
323                 <span class="text-muted font-monospace" style="font-size: 0.66rem;" title="Всего скачано/прочитано">
324                     Σ {totalRecvStr}
325                 </span>
326             </div>
327             <div class="d-flex align-items-start gap-1">
328                 <i class="bi bi-arrow-down-left text-success mt-0.5" style="font-size: 0.70rem;"></i>
329                 <div class="text-truncate" style="max-width: 250
330 | px; font-size: 0.72rem; color: var(--text-muted);" title="{escapeHtml(item.recv_summary)}">
331                     {escapeHtml(item.recv_summary)}
332                 </div>
333             </div>
334         </td>
335     </tr>
336 `;
337 }).join('');
338
339 tbody.querySelectorAll('.proc-net-row').forEach(row => {

```

```

338 |         row.onclick = async () => {
339 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
340 |             ;
341 |             const item = filtered[idx];
342 |             if (!item) return;
343 |             const formatNetKb = (kb) => {
344 |                 if (!kb || kb <= 0) return '0 KB';
345 |                 if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).t
346 | oFixed(2) + ' GB';
347 |                 if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB
348 | ';
349 |                 return kb.toFixed(1) + ' KB';
350 |             };
351 |             const formatNetRate = (rate) => {
352 |                 if (!rate || rate <= 0.05) return '';
353 |                 if (rate >= 1024) return (rate / 1024).toFixed(1) +
354 | ' MB/s';
355 |                 return rate.toFixed(1) + ' KB/s';
356 |             };
357 |             const isModalReady = await ensureAITableModal();
358 |             if (isModalReady && window.AITableModal) {
359 |                 window.AITableModal.show({
360 |                     icon: '📶',
361 |                     title: `${item.name} (${item.service_type})`,
362 |                     subtitle: `PID: ${item.pid} | ${item.remote_addres
363 | s}`,
364 |                     tableType: 'network',
365 |                     badges: [
366 |                         { text: `PID ${item.pid}`, class: 'badge bg-info
367 | text-dark' },
368 |                         { text: item.protocol, class: 'badge bg-primary'
369 | },
370 |                         { text: item.status, class: 'badge bg-success' }
371 |                     ],
372 |                     { text: item.is_internet ? 'Интернет (WAN)' : 'Л

```

```

    локально (LAN)', class: item.is_internet ? 'badge bg-warning te
    xt-dark' : 'badge bg-secondary' }
368         ],
369         metadata: [
370             { label: 'Программа / Процесс', value: item.name
            },
371             { label: 'Process ID (PID)', value: String(item.
            pid) },
372             { label: 'Пользователь системы', value: item.use
            r || 'SYSTEM' },
373             { label: 'Удаленный адрес (Remote)', value: item
            .remote_address },
374             { label: 'Локальный сокет (Local)', value: item.
            local_address },
375             { label: 'Протокол / Служба', value: `${item.pro
            tocol} • ${item.service_type}` },
376             { label: 'Статус соединения', value: item.status
            },
377             { label: 'Скачено за период (Прием)', value: `${
            formatNetKb(item.delta_recv_kb || 0)} ${item.recv_rate_kbs > 0
            .05 ? '(' + formatNetRate(item.recv_rate_kbs) + ')' : ''}` },
378             { label: 'Всего скачено / получено', value: form
            atNetKb(item.recv_kb || 0) },
379             { label: 'Отправлено за период', value: `${forma
            tNetKb(item.delta_sent_kb || 0)} ${item.sent_rate_kbs > 0.05 ?
            '(' + formatNetRate(item.sent_rate_kbs) + ')' : ''}` },
380             { label: 'Всего отправлено', value: formatNetKb(
            item.sent_kb || 0) },
381             { label: 'Что шлет (Отправка)', value: item.sent
            _summary, fullWidth: true },
382             { label: 'Что принимает (Прием)', value: item.re
            cv_summary, fullWidth: true }
383         ],
384         rawTitle: 'Сетевой дамп подключения',
385         rawContent: JSON.stringify(item, null, 2),
386         requestData: item
387     });
388 }

```

```

389     };
390   });
391 }
392
393   // =====
394   // 2. Real-Time File Changes (WinAPI ReadDirectoryChangesW)
395   // =====
396
397   async function fetchLiveFileEvents() {
398     try {
399       const [resEvents, resTelem] = await Promise.all([
400         fetch('/api/v1/system/file-audit/live-events?limit=30'
401         ),
402         fetch('/api/v1/system/file-audit/telemetry').catch(()
403         => null)
404       ]);
405
406       if (resTelem && resTelem.ok) {
407         const telem = await resTelem.json();
408         const elRate = document.getElementById('proc-watcher-r
409         ate');
410         const elCr = document.getElementById('proc-watcher-cr'
411         );
412         const elMod = document.getElementById('proc-watcher-mo
413         d');
414         const elDel = document.getElementById('proc-watcher-de
415         l');
416         const elDrive = document.getElementById('proc-watcher-
417         drive');
418         const elDriveModel = document.getElementById('proc-wat
419         cher-drive-model');
420         const elDriveTemp = document.getElementById('proc-watc
421         her-temp');
422         const elDiskR = document.getElementById('proc-watcher-
423         r-kbs');
424         const elDiskW = document.getElementById('proc-watcher-

```

```

w-kbs');
415     const elSecMatches = document.getElementById('proc-wat
cher-sec-matches');
416     const elStatus = document.getElementById('proc-watcher
-sensor-status');
417
418     if (elRate) elRate.textContent = (telem.events_rate_pe
r_sec || 0).toFixed(1);
419     if (elCr) elCr.textContent = (telem.created_rate_per_s
ec || 0).toFixed(1);
420     if (elMod) elMod.textContent = (telem.modified_rate_pe
r_sec || 0).toFixed(1);
421     if (elDel) elDel.textContent = (telem.deleted_rate_per
_sec || 0).toFixed(1);
422
423     if (elDrive) {
424         const drivesStr = (telem.drive_letters && telem.driv
e_letters.length)
425             ? telem.drive_letters.join(', ')
426             : (telem.drive_letter || 'C:');
427         elDrive.textContent = drivesStr;
428     }
429     if (elDriveModel) elDriveModel.textContent = telem.dri
ve_model || 'Storage';
430     if (elDriveTemp) elDriveTemp.textContent = telem.drive
_temperature_c !== null && telem.drive_temperature_c !== undef
ined ? `${telem.drive_temperature_c} °C` : '-- °C';
431     if (elDiskR) elDiskR.textContent = Math.round(telem.di
sk_read_kbs || 0);
432     if (elDiskW) elDiskW.textContent = Math.round(telem.di
sk_write_kbs || 0);
433     if (elSecMatches) elSecMatches.textContent = telem.sec
urity_audit_matched_count || 0;
434
435     if (elStatus) {
436         if (telem.burst_deletions_alert) {
437             elStatus.className = 'badge bg-danger-subtle text-
danger border border-danger px-2 py-1';

```

```
438 |         elStatus.textContent = '⚠ Всплеск удалений';
439 |     } else if (telem.high_activity_alert) {
440 |         elStatus.className = 'badge bg-warning-subtle text
| -warning border border-warning px-2 py-1';
441 |         elStatus.textContent = '⚠ Высокий I/O';
442 |     } else {
443 |         elStatus.className = 'badge bg-success-subtle text
| -success border border-success px-2 py-1';
444 |         elStatus.textContent = '✅ Штатный режим';
445 |     }
446 | }
447 | }
448 |
449 | if (!resEvents || !resEvents.ok) return;
450 | const data = await resEvents.json();
451 | const events = data.events || [];
452 |
453 | if (data.filtered_count !== undefined) {
454 |     const filteredEl = document.getElementById('proc-watch
| er-filtered-count');
455 |     if (filteredEl) filteredEl.textContent = data.filtered
| _count;
456 | }
457 |
458 | currentWatchDirs = data.watch_dirs || (data.watch_dir ?
| [data.watch_dir] : []);
459 | currentWatchDir = currentWatchDirs[0] || data.watch_dir
| || '';
460 |
461 | const dirPathEl = document.getElementById('proc-watch-di
| r-path');
462 | const badge = document.getElementById('proc-watch-dir-ba
| dge');
463 | if (dirPathEl) {
464 |     if (currentWatchDirs.length === 0) {
465 |         dirPathEl.innerText = 'Папки не выбраны';
466 |     } else if (currentWatchDirs.length === 1) {
467 |         const singleName = currentWatchDirs[0].split('\\').p
```

```

    op() || currentWatchDirs[0];
468     dirPathEl.innerText = singleName;
469   } else {
470     const firstNames = currentWatchDirs.slice(0, 2).map(
p => p.split('\\').pop() || p).join(', ');
471     dirPathEl.innerText = `${currentWatchDirs.length} па
пок: ${firstNames}${currentWatchDirs.length > 2 ? '...' : ''}`
;
472   }
473 }
474 if (badge) {
475   badge.title = `Отслеживаемые каталоги (${currentWatchD
irs.length}): \n${currentWatchDirs.join('\n')}\n\n(Нажмите для
настройки и выбора папок)`;
476 }
477
478 const tbody = document.getElementById('proc-watcher-tbod
y');
479 if (tbody) {
480   if (events.length === 0) {
481     const labelDirs = currentWatchDirs.length > 0 ? curr
entWatchDirs.join(', ') : 'проекта';
482     tbody.innerHTML = `|<td colspan="4" class="text-c
enter text-muted p-2">Ожидание изменений в папках <code>${labe
lDirs}</code>...</td></tr>`;
483     return;
484   }
485   tbody.innerHTML = events.map((e, idx) => {
486     const rootDirHint = e.watch_dir ? (e.watch_dir.split
('\\').pop() || e.watch_dir) : '';
487     const procDisplay = e.process_name
488       ? `

|  |

```



```

489         : ``>—</span>`;
490         return `
491         <tr class="proc-live-row" data-idx="${idx}" style=
"cursor: pointer;" title="Нажмите для AI-диагностики события">
492             <td class="font-monospace text-muted small">${e.
timestamp?.slice(11, 19) || ''}</td>
493             <td>
494                 <span class="badge ${e.is_deletion ? 'bg-dange
r' : (e.action === 'Created' ? 'bg-success' : 'bg-secondary')}
">${e.action}</span>
495                 ${rootDirHint && currentWatchDirs.length > 1 ?
`<span class="badge bg-dark border border-secondary text-mute
d ms-1" style="font-size: 0.65rem;" title="Корень: ${escapeHtm
l(e.watch_dir)}">${rootDirHint}</span>` : ''}
496             </td>
497             <td>${procDisplay}</td>
498             <td class="font-monospace small" style="word-bre
ak: break-all; color: var(--text-color);" title="${escapeHtml(
e.path)}">${escapeHtml(e.path)}</td>
499         </tr>
500         `;
501         })).join('');
502
503         tbody.querySelectorAll('.proc-live-row').forEach(row =
> {
504             row.onclick = async () => {
505                 const idx = parseInt(row.getAttribute('data-idx'),
10);
506                 const e = events[idx];
507                 if (!e) return;
508
509                 const isModalReady = await ensureAITableModal();
510                 if (isModalReady && window.AITableModal) {
511                     window.AITableModal.show({
512                         icon: '📁',
513                         title: `Файловое событие: ${e.action}`,
514                         subtitle: `${e.path} | ${e.timestamp}`,

```

```

515         tableType: 'file_event',
516         badges: [
517             { text: e.action, class: e.is_deletion ? 'ba
dge bg-danger' : (e.action === 'Created' ? 'badge bg-success'
: 'badge bg-info text-dark') },
518             { text: e.process_name ? `Программа: ${e.pro
cess_name}` : 'WinAPI ReadDirectoryChangesW', class: 'badge bg
-dark border border-secondary text-info' },
519             { text: 'WinAPI', class: 'badge bg-secondary
' }
520         ],
521         metadata: [
522             { label: 'Действие', value: e.action },
523             { label: 'Полный путь к файлу', value: e.pat
h },
524             { label: 'Программа / Процесс', value: e.pro
cess_name ? `${e.process_name}${e.process_id ? ` (PID: ${e.pro
cess_id})` : ''}` : 'Фоновый процесс / завершен' },
525             { label: 'Время события', value: e.timestamp
},
526             { label: 'Признак удаления', value: e.is_del
etion ? 'Да (Файл удален/переименован)' : 'Нет' },
527             { label: 'Папка события', value: e.watch_dir
|| currentWatchDir },
528             { label: 'Все отслеживаемые папки', value: c
urrentWatchDirs.join('; ') }
529         ],
530         rawTitle: 'Детали события WinAPI & Process Inf
o',
531         rawContent: JSON.stringify(e, null, 2),
532         requestData: e
533     });
534 }
535 };
536 });
537 }
538 } catch (e) {
539     console.error('[ProcessesLoadInspectorTab] Failed to fet

```

```

    ch live file events:', e);
540     }
541   }
542
543   // =====
   =====
544   // 3. Multi-Directory Watcher & Folder Explorer Modal Logic
545   // =====
   =====
546
547   function showWatchDirsAlert(msg, type = 'success') {
548     const alertEl = document.getElementById('modal-proc-dirs-a
alert');
549     if (!alertEl) return;
550     alertEl.className = `alert alert-${type} py-2 px-3 small m
b-3`;
551     alertEl.innerHTML = msg;
552     alertEl.classList.remove('d-none');
553     if (type === 'success') {
554       setTimeout(() => {
555         alertEl.classList.add('d-none');
556       }, 3500);
557     }
558   }
559
560   function renderModalActiveDirsList() {
561     const container = document.getElementById('modal-proc-dirs
-active-list');
562     const countBadge = document.getElementById('modal-proc-dir
s-count');
563     const hintEl = document.getElementById('modal-proc-dirs-fo
oter-hint');
564
565     if (!container) return;
566
567     if (countBadge) countBadge.textContent = String(stagedWatc
hDirs.length);
568     if (hintEl) hintEl.textContent = `Выбрано папок для монито

```

```

    ринга: ${stagedWatchDirs.length}`;
569
570     if (stagedWatchDirs.length === 0) {
571         container.innerHTML = `
572             <div class="text-center text-muted small py-3">
573                 <i class="bi bi-folder-x me-1"></i> Список пуст. Выб
ерите папки в проводнике ниже или воспользуйтесь быстрыми прес
етами.
574             </div>
575         `;
576         return;
577     }
578
579     container.innerHTML = stagedWatchDirs.map((dirPath, idx) =
> {
580         const folderName = dirPath.split('\\').pop() || dirPath;
581         const driveLetter = (dirPath.slice(0, 2)).toUpperCase();
582         return `
583             <div class="p-1.5 px-2 rounded d-flex align-items-cent
er justify-content-between gap-2" style="background: var(--bg-
color); border: 1px solid var(--border-color);">
584                 <div class="d-flex align-items-center gap-2 text-tru
ncate" style="max-width: 82%;">
585                     <i class="bi bi-folder-check text-warning fs-6"></
i>
586                     <span class="badge bg-secondary font-monospace" st
yle="font-size: 0.68rem;">${driveLetter}</span>
587                     <div class="text-truncate">
588                         <span class="fw-bold small" style="color: var(--
text-color);">${escapeHtml(folderName)}</span>
589                         <span class="small text-muted font-monospace d-b
lock text-truncate" style="font-size: 0.72rem;" title="${escap
eHtml(dirPath)}">${escapeHtml(dirPath)}</span>
590                     </div>
591                 </div>
592                 <button class="btn btn-xs btn-outline-danger rounded
-pill px-2 py-0.5" onclick="window._procRemoveStagedWatchDir($
{idx})" title="Удалить из списка мониторинга">

```

```

593         <i class="bi bi-trash3"></i>
594     </button>
595 </div>
596 `;
597 }).join('');
598 }
599
600 window._procRemoveStagedWatchDir = function(index) {
601     if (index >= 0 && index < stagedWatchDirs.length) {
602         const removed = stagedWatchDirs.splice(index, 1)[0];
603         renderModalActiveDirsList();
604         showWatchDirsAlert(`Папка удалена из списка: <code>${esc
apeHtml(removed)}</code>`, 'info');
605     }
606 };
607
608 function addDirToStaged(pathToAdd) {
609     if (!pathToAdd || !pathToAdd.trim()) return;
610     const cleanPath = pathToAdd.trim();
611     if (stagedWatchDirs.includes(cleanPath)) {
612         showWatchDirsAlert(`Папка уже есть в списке: <code>${esc
apeHtml(cleanPath)}</code>`, 'warning');
613         return;
614     }
615     stagedWatchDirs.push(cleanPath);
616     renderModalActiveDirsList();
617     showWatchDirsAlert(`Папка добавлена: <code>${escapeHtml(cle
anPath)}</code>`, 'success');
618 }
619
620 async function loadSystemDrives() {
621     const drivesBar = document.getElementById('modal-proc-driv
es-bar');
622     if (!drivesBar) return;
623
624     try {
625         const res = await fetch('/api/sysadmin/filesystem/drives
');
```

```

626 |         if (res.ok) {
627 |             const data = await res.json();
628 |             cachedDrives = data.drives || [];
629 |         }
630 |     } catch (e) {
631 |         console.warn('[ProcessesLoadInspectorTab] Failed to fetch system drives:', e);
632 |         cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
633 |     }
634 |
635 |     if (cachedDrives.length === 0) {
636 |         cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
637 |     }
638 |
639 |     drivesBar.innerHTML = cachedDrives.map(d => {
640 |         const label = d.mountpoint || 'C:\\';
641 |         const freeTxt = d.free_gb ? `${d.free_gb} GB free` : '';
642 |         return `
643 |             <button class="btn btn-xs btn-outline-secondary rounded-pill px-2 py-0.5 proc-drive-btn font-monospace" data-drive="${
644 |                 escapeHtml(label)}" title="${escapeHtml(label)} ${freeTxt}">
645 |                 <i class="bi bi-hdd me-1"></i>${escapeHtml(label)}
646 |             </button>
647 |         `;
648 |     }).join('');
649 |
650 |     drivesBar.querySelectorAll('.proc-drive-btn').forEach(btn => {
651 |         btn.onclick = () => {
652 |             const drv = btn.getAttribute('data-drive');
653 |             if (drv) loadFolderBrowser(drv);
654 |         };
655 |     });
656 |
657 |     function renderBreadcrumbs(currentPath) {
658 |         const bcContainer = document.getElementById('modal-proc-folder-breadcrumbs');

```

```

659 |     if (!bcContainer) return;
660 |
661 |     const parts = currentPath.replace(/\\+$/, '').split('\\');
662 |     let builtPath = '';
663 |     const crumbsHtml = parts.map((part, idx) => {
664 |         if (idx === 0) {
665 |             builtPath = part + '\\';
666 |         } else {
667 |             builtPath = builtPath + (builtPath.endsWith('\\') ? ''
668 | : '\\') + part;
669 |         }
670 |         const clickPath = builtPath;
671 |         const isLast = idx === parts.length - 1;
672 |         return `
673 |             <span class="d-inline-flex align-items-center">
674 |                 ${idx > 0 ? '<span class="text-muted mx-1">/</span>'
675 | : ''}
676 |                 <a href="#" class="proc-breadcrumb-link text-decoration-
677 | none ${isLast ? 'fw-bold text-info' : ''}" style="color: v
678 | ar(--text-color);" data-path="${escapeHtml(clickPath)}">
679 |                     ${escapeHtml(part || 'Корень')}
680 |                 </a>
681 |             </span>
682 |         `;
683 |     }).join('');
684 |
685 |     bcContainer.innerHTML = crumbsHtml;
686 |
687 |     bcContainer.querySelectorAll('.proc-breadcrumb-link').forEach(link => {
688 |         link.onclick = (e) => {
689 |             e.preventDefault();
690 |             const p = link.getAttribute('data-path');
691 |             if (p) loadFolderBrowser(p);
692 |         };
693 |     });

```

```

692 |   async function loadFolderBrowser(targetPath) {
693 |       const inputPath = document.getElementById('modal-proc-fold
        er-path-input');
694 |       const browserList = document.getElementById('modal-proc-fo
        lder-browser-list');
695 |       if (!targetPath) targetPath = 'C:\\';
696 |       currentBrowserPath = targetPath;
697 |
698 |       if (inputPath) inputPath.value = targetPath;
699 |       renderBreadcrumbs(targetPath);
700 |
701 |       if (!browserList) return;
702 |       browserList.innerHTML = `
```



```

721         return `
722         <div class="p-1 px-2 rounded d-flex align-items-cent
er justify-content-between gap-2 proc-folder-browser-item" sty
le="background: var(--surface-1); border: 1px solid var(--bord
er-color); cursor: pointer;">
723             <div class="d-flex align-items-center gap-2 text-t
runcate flex-grow-1 proc-nav-to-folder" data-path="${escapeHtm
l(d.path)}" title="Нажмите для перехода в папку">
724                 <i class="bi ${d.has_subdirs ? 'bi-folder2 text-
warning' : 'bi-folder text-warning'}"></i>
725                 <span class="small text-light text-truncate">${e
scapeHtml(d.name)}</span>
726                 <span class="small text-muted font-monospace ms-
auto me-2" style="font-size: 0.68rem;">${d.modified || ''}</sp
an>
727             </div>
728             <button class="btn btn-xs ${isAlreadySelected ? 'b
tn-success' : 'btn-outline-info'} rounded-pill px-2 py-0.5 pro
c-add-folder-btn" data-path="${escapeHtml(d.path)}" title="Доб
авить в отслеживаемые">
729                 <i class="bi ${isAlreadySelected ? 'bi-check2' :
'bi-plus-lg'} me-1"></i>${isAlreadySelected ? 'Выбрана' : 'Сл
едить'}
730             </button>
731         </div>
732     `;
733     })).join('');
734
735     browserList.querySelectorAll('.proc-nav-to-folder').forE
ach(el => {
736         el.onclick = () => {
737             const p = el.getAttribute('data-path');
738             if (p) loadFolderBrowser(p);
739         };
740     });
741
742     browserList.querySelectorAll('.proc-add-folder-btn').for
Each(btn => {

```

```

743         btn.onclick = (e) => {
744             e.stopPropagation();
745             const p = btn.getAttribute('data-path');
746             if (p) {
747                 addDirToStaged(p);
748                 btn.className = 'btn btn-xs btn-success rounded-pi
749                 ll px-2 py-0.5 proc-add-folder-btn';
750                 btn.innerHTML = '<i class="bi bi-check2 me-1"></i>
751                 Выбрана';
752             }
753         };
754     });
755     } catch (e) {
756         console.warn('[ProcessesLoadInspectorTab] Failed to brow
757         se folder:', e);
758         browserList.innerHTML = `<div class="text-danger small p
759         y-2 px-2"><i class="bi bi-exclamation-triangle me-1"></i> Ошиб
760         ка доступа: ${escapeHtml(e.message)}</div>`;
761     }
762     }
763     // =====
764     // 4. Exclusions & Filters Management
765     // =====
766     function showExclusionsAlert(msg, type = 'success') {
767         const alertEl = document.getElementById('modal-proc-exclus
768         ions-alert');
769         if (!alertEl) return;
770         alertEl.className = `alert alert-${type} py-2 px-3 small m
771         b-3`;
772         alertEl.innerHTML = msg;
773         alertEl.classList.remove('d-none');
774         if (type === 'success' || type === 'info') {
775             setTimeout(() => {

```

```
772 |         alertEl.classList.add('d-none');
773 |     }, 3500);
774 | }
775 | }
776 |
777 | function renderExclusionsLists() {
778 |     const totalCount = (currentExclusions.paths?.length || 0)
779 | +
780 |     (currentExclusions.extensions?.length || 0) +
781 |     (currentExclusions.patterns?.length || 0) +
782 |     (currentExclusions.processes?.length || 0);
783 |     const tabBadge = document.getElementById('modal-proc-exclu
784 | sions-tab-count');
785 |     const headerBadge = document.getElementById('proc-exclusio
786 | ns-count-badge');
787 |     if (tabBadge) tabBadge.textContent = String(totalCount);
788 |     if (headerBadge) headerBadge.textContent = String(totalCou
789 | nt);
790 |     const switchEl = document.getElementById('switch-proc-excl
791 | usions-active');
792 |     const statusBadge = document.getElementById('badge-proc-ex
793 | clusions-status');
794 |     if (switchEl) switchEl.checked = Boolean(currentExclusions
795 | .enabled);
796 |     if (statusBadge) {
797 |         if (currentExclusions.enabled) {
798 |             statusBadge.className = 'badge bg-success';
799 |             statusBadge.textContent = 'ВКЛ';
800 |         } else {
801 |             statusBadge.className = 'badge bg-danger-subtle text-d
802 | anger border border-danger fw-bold';
803 |             statusBadge.textContent = 'ВыКЛ';
804 |         }
805 |     }
806 | }
807 |
808 | const filteredTotalEl = document.getElementById('modal-pro
```

```

    c-exclusions-filtered-total');
802     if (filteredTotalEl) filteredTotalEl.textContent = String(
currentExclusions.filtered_count || 0);
803
804     // Paths list
805     const pathsContainer = document.getElementById('list-proc-
ex-paths');
806     const pathsBadge = document.getElementById('badge-proc-ex-
paths-count');
807     if (pathsBadge) pathsBadge.textContent = String(currentExc
lusions.paths?.length || 0);
808     if (pathsContainer) {
809         if (!currentExclusions.paths || currentExclusions.paths.
length === 0) {
810             pathsContainer.innerHTML = `<div class="text-muted sma
ll text-center py-2">Нет исключенных папок</div>`;
811         } else {
812             pathsContainer.innerHTML = currentExclusions.paths.map
(p => `
813                 <div class="d-flex align-items-center justify-conten
t-between gap-1.5 p-1 px-2 rounded mb-1" style="background: va
r(--bg-color); border: 1px solid var(--border-color);">
814                 <span class="small font-monospace text-light text-
truncate" style="font-size: 0.72rem;" title="${escapeHtml(p)}"
>${escapeHtml(p)}</span>
815                 <button class="btn btn-xs btn-outline-danger py-0
px-1 rounded-pill" onclick="window._procRemoveExclusionItem('p
aths', '${escapeHtml(p.replace(/\\/g, '\\\\'))}')" title="Удал
ить">❏ </button>
816                 </div>
817             `).join('');
818         }
819     }
820
821     // Extensions list
822     const extsContainer = document.getElementById('list-proc-e
x-extensions');
823     const extsBadge = document.getElementById('badge-proc-ex-e

```

```

xtensions-count');
824     if (extsBadge) extsBadge.textContent = String(currentExclu
sions.extensions?.length || 0);
825     if (extsContainer) {
826         if (!currentExclusions.extensions || currentExclusions.e
xtensions.length === 0) {
827             extsContainer.innerHTML = `
```

```

p gap-1">` + currentExclusions.patterns.map(pat => `
847     <span class="badge bg-dark border border-secondary t
ext-warning d-inline-flex align-items-center gap-1 font-monosp
ace" style="font-size: 0.75rem;">
848         ${escapeHtml(pat)}
849         <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._procRemoveEx
clusionItem('patterns', '${escapeHtml(pat)}')" title="Удалить"
></button>
850     </span>
851     `).join('') + `</div>`;
852 }
853 }
854
855 // Processes list
856 const procsContainer = document.getElementById('list-proc-
ex-processes');
857 const procsBadge = document.getElementById('badge-proc-ex-
processes-count');
858 if (procsBadge) procsBadge.textContent = String(currentExc
lusions.processes?.length || 0);
859 if (procsContainer) {
860     if (!currentExclusions.processes || currentExclusions.pr
ocesses.length === 0) {
861         procsContainer.innerHTML = `<div class="text-muted sma
ll text-center py-2">Нет исключенных программ</div>`;
862     } else {
863         procsContainer.innerHTML = `<div class="d-flex flex-wr
ap gap-1">` + currentExclusions.processes.map(proc => `
864             <span class="badge bg-dark border border-secondary t
ext-danger d-inline-flex align-items-center gap-1 font-monospa
ce" style="font-size: 0.75rem;">
865                 <i class="bi bi-cpu me-0.5"></i>${escapeHtml(proc)
}
866                 <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._procRemoveEx
clusionItem('processes', '${escapeHtml(proc)}')" title="Удалит
ь"></button>

```

```

867         </span>
868     `).join('') + `</div>`;
869     }
870 }
871 }
872
873 async function fetchExclusionsData() {
874     try {
875         const res = await fetch('/api/v1/system/file-audit/exclu
sions');
876         if (res.ok) {
877             currentExclusions = await res.json();
878             renderExclusionsLists();
879         }
880     } catch (e) {
881         console.warn('[ProcessesLoadInspectorTab] Failed to fetc
h exclusions:', e);
882     }
883 }
884
885 async function addExclusionItem(category, value) {
886     if (!value || !value.trim()) return;
887     const cleanVal = value.trim();
888     try {
889         const res = await fetch('/api/v1/system/file-audit/exclu
sions/add', {
890             method: 'POST',
891             headers: { 'Content-Type': 'application/json' },
892             body: JSON.stringify({ category, value: cleanVal })
893         });
894         const data = await res.json();
895         if (res.ok && data.success) {
896             currentExclusions = data.exclusions || currentExclusio
ns;
897             renderExclusionsLists();
898             showExclusionsAlert(`Правило добавлено: <code>${escape
Html(cleanVal)}</code>`, 'success');
899             const inputVal = document.getElementById('input-proc-e

```

```

    xclusion-value');
900     if (inputVal) inputVal.value = '';
901   } else {
902     showExclusionsAlert(data.message || 'Правило уже сущес
твует или невалидно', 'warning');
903   }
904 } catch (e) {
905   showExclusionsAlert(`Ошибка добавления: ${e.message}`, '
danger');
906 }
907 }
908
909 window._procRemoveExclusionItem = async function(category, v
alue) {
910   if (!value) return;
911   try {
912     const res = await fetch('/api/v1/system/file-audit/exclu
sions/remove', {
913       method: 'POST',
914       headers: { 'Content-Type': 'application/json' },
915       body: JSON.stringify({ category, value })
916     });
917     const data = await res.json();
918     if (res.ok && data.success) {
919       currentExclusions = data.exclusions || currentExclusio
ns;
920       renderExclusionsLists();
921       showExclusionsAlert(`Правило удалено: <code>${escapeHt
ml(value)}</code>`, 'info');
922     } else {
923       showExclusionsAlert(data.message || 'Не удалось удалит
ь правило', 'warning');
924     }
925   } catch (e) {
926     showExclusionsAlert(`Ошибка удаления: ${e.message}`, 'da
nger');
927   }
928 };

```



```

929
930   async function toggleExclusionsActive(enabled) {
931     try {
932       const res = await fetch('/api/v1/system/file-audit/exclu
sions/toggle', {
933         method: 'POST',
934         headers: { 'Content-Type': 'application/json' },
935         body: JSON.stringify({ enabled })
936       });
937       const data = await res.json();
938       if (res.ok && data.success) {
939         currentExclusions = data.exclusions || currentExclusio
ns;
940         renderExclusionsLists();
941         showExclusionsAlert(`Фильтрация исключений ${data.enab
led ? 'включена' : 'отключена'}`, 'info');
942       }
943     } catch (e) {
944       showExclusionsAlert(`Ошибка переключения: ${e.message}`,
'danger');
945     }
946   }
947
948   async function applyExclusionPreset(preset) {
949     if (preset === 'logs') {
950       await addExclusionItem('paths', 'AppData\\Roaming\\AI-Br
eadboard\\apps\\windows\\telemetry\\logs');
951       await addExclusionItem('extensions', '.log');
952       await addExclusionItem('extensions', '.csv');
953       await addExclusionItem('patterns', '*system_inspector_po
lls.csv*');
954     } else if (preset === 'search') {
955       await addExclusionItem('processes', 'SearchIndexer.exe')
;
956       await addExclusionItem('patterns', '*Windows.db*');
957       await addExclusionItem('patterns', '*Windows.db-wal*');
958       await addExclusionItem('patterns', '*Windows.db-shm*');
959     } else if (preset === 'temp') {

```

```
960 |     await addExclusionItem('paths', 'AppData\\Local\\Temp');
961 |     await addExclusionItem('extensions', '.tmp');
962 |     await addExclusionItem('patterns', '~$*');
963 | } else if (preset === 'dev') {
964 |     await addExclusionItem('paths', '.git');
965 |     await addExclusionItem('paths', 'node_modules');
966 |     await addExclusionItem('paths', '__pycache__');
967 |     await addExclusionItem('paths', '.pytest_cache');
968 | }
969 | }
970 |
971 | async function openWatchFoldersModal(initialTab = 'folders')
972 | {
973 |     const modalEl = document.getElementById('modal-proc-watch-
974 | folders');
975 |     if (!modalEl) {
976 |         console.warn('[ProcessesLoadInspectorTab] modal-proc-wat
977 | ch-folders element not found');
978 |         return;
979 |     }
980 |     try {
981 |         const res = await fetch('/api/v1/system/file-audit/watch
982 | -dirs');
983 |         if (res.ok) {
984 |             const data = await res.json();
985 |             stagedWatchDirs = data.watch_dirs && data.watch_dirs.l
986 | ength ? [...data.watch_dirs] : [currentWatchDir || 'C:\\'];
987 |         } else {
988 |             stagedWatchDirs = currentWatchDirs.length ? [...curren
989 | tWatchDirs] : [currentWatchDir || 'C:\\'];
990 |         }
991 |     } catch {
992 |         stagedWatchDirs = currentWatchDirs.length ? [...currentW
993 | atchDirs] : [currentWatchDir || 'C:\\'];
994 |     }
995 |     renderModalActiveDirsList();
```

```
991 |     const tabFoldersBadge = document.getElementById('modal-pro
    | c-dirs-tab-count');
992 |     if (tabFoldersBadge) tabFoldersBadge.textContent = String(
    | stagedWatchDirs.length);
993 |
994 |     await loadSystemDrives();
995 |     await fetchExclusionsData();
996 |
997 |     if (initialTab === 'exclusions') {
998 |         const tabExBtn = document.getElementById('tab-btn-proc-e
    | xclusions');
999 |         if (tabExBtn && window.bootstrap?.Tab) {
1000 |             window.bootstrap.Tab.getOrCreateInstance(tabExBtn).sho
    | w();
1001 |         } else if (tabExBtn) {
1002 |             tabExBtn.click();
1003 |         }
1004 |     } else {
1005 |         const tabFoldersBtn = document.getElementById('tab-btn-p
    | roc-folders');
1006 |         if (tabFoldersBtn && window.bootstrap?.Tab) {
1007 |             window.bootstrap.Tab.getOrCreateInstance(tabFoldersBtn
    | ).show();
1008 |         } else if (tabFoldersBtn) {
1009 |             tabFoldersBtn.click();
1010 |         }
1011 |     }
1012 |
1013 |     const startPath = stagedWatchDirs[0] || currentBrowserPath
    | || 'C:\\';
1014 |     loadFolderBrowser(startPath);
1015 |
1016 |     if (window.bootstrap?.Modal) {
1017 |         window.bootstrap.Modal.getOrCreateInstance(modalEl).show
    | ();
1018 |     } else {
1019 |         modalEl.classList.add('show');
1020 |         modalEl.style.display = 'block';
```

```

1021     }
1022 }
1023
1024 async function showLiveWatcherHelpModal() {
1025     const dirsStr = currentWatchDirs.length ? currentWatchDirs
1026     .join('\n- ') : (currentWatchDir || 'Рабочая папка');
1027     const isModalReady = await ensureAITableModal();
1028     if (isModalReady && window.AITableModal) {
1029         window.AITableModal.show({
1030             icon: 'i',
1031             title: 'Справка: Изменения файлов в реальном времени',
1032             subtitle: 'Низкоуровневый мониторинг файловой системы
Windows через WinAPI ReadDirectoryChangesW',
1033             tableType: 'help',
1034             badges: [
1035                 { text: 'WinAPI ReadDirectoryChangesW', class: 'badg
e bg-info text-dark' },
1036                 { text: 'Multi-Directory', class: 'badge bg-primary'
},
1037                 { text: 'Real-Time Streaming', class: 'badge bg-succ
ess' },
1038                 { text: 'Рекурсивно (bWatchSubtree = True)', class:
'badge bg-warning text-dark' }
1039             ],
1040             metadata: [
1041                 { label: 'Технология', value: 'WinAPI ReadDirectoryC
hangesW (kernel32.dll)' },
1042                 { label: 'Активные папки', value: dirsStr, fullWidth
: true },
1043                 { label: 'Автологгер', value: 'file_watcher_telemetr
y_polls.csv' }
1044             ],
1045             rawTitle: 'Архитектурная справка',
1046             rawContent: `Компонент для мгновенного перехвата опера
ций файловой системы в режиме реального времени на базе WinAPI
ReadDirectoryChangesW.\n\nТекущие отслеживаемые каталоги:\n${
dirsStr}`
1047         });

```

```

1047     } else {
1048         alert('Мониторинг файловой системы в реальном времени (R
readDirectoryChangesW).\nОтслеживаемые папки:\n- ' + dirsStr);
1049     }
1050 }
1051
1052 // =====
=====
1053 // 5. Table Resizers
1054 // =====
=====
1055
1056 function initTableResizers() {
1057     // Process table resizer
1058     const procResizer = document.getElementById('sys-proc-table-
resizer');
1059     const procContainer = document.getElementById('sys-proc-table-
container');
1060     if (procResizer && procContainer) {
1061         let startY = 0, startHeight = 0;
1062         const onMouseMove = (e) => {
1063             const delta = e.clientY - startY;
1064             const newH = Math.max(140, Math.min(850, startHeight +
delta));
1065             procContainer.style.height = `${newH}px`;
1066         };
1067         const onMouseUp = () => {
1068             procResizer.classList.remove('resizing');
1069             document.removeEventListener('mousemove', onMouseMove)
;
1070             document.removeEventListener('mouseup', onMouseUp);
1071         };
1072         procResizer.onmousedown = (e) => {
1073             e.preventDefault();
1074             startY = e.clientY;
1075             startHeight = parseInt(window.getComputedStyle(procCon
tainer).height, 10) || 380;
1076             procResizer.classList.add('resizing');

```

```

1077 |         document.addEventListener('mousemove', onMouseMove);
1078 |         document.addEventListener('mouseup', onMouseUp);
1079 |     };
1080 |     procResizer.ondblclick = () => { procContainer.style.height = '380px'; };
1081 |     }
1082 |
1083 |     // Network table resizer
1084 |     const netResizer = document.getElementById('proc-net-table-resizer');
1085 |     const netContainer = document.getElementById('proc-net-table-container');
1086 |     if (netResizer && netContainer) {
1087 |         let startY = 0, startHeight = 0;
1088 |         const onMouseMove = (e) => {
1089 |             const delta = e.clientY - startY;
1090 |             const newH = Math.max(140, Math.min(850, startHeight +
1091 |             delta));
1092 |             netContainer.style.height = `${newH}px`;
1093 |         };
1094 |         const onMouseUp = () => {
1095 |             netResizer.classList.remove('resizing');
1096 |             document.removeEventListener('mousemove', onMouseMove);
1097 |         };
1098 |         netResizer.onmousedown = (e) => {
1099 |             e.preventDefault();
1100 |             startY = e.clientY;
1101 |             startHeight = parseInt(window.getComputedStyle(netContainer).height, 10) || 360;
1102 |             netResizer.classList.add('resizing');
1103 |             document.addEventListener('mousemove', onMouseMove);
1104 |             document.addEventListener('mouseup', onMouseUp);
1105 |         };
1106 |         netResizer.ondblclick = () => { netContainer.style.height = '360px'; };
1107 |     }

```

```

1108
1109     // Watcher table resizer
1110     const watchResizer = document.getElementById('proc-watcher
1111     -table-resizer');
1112     const watchContainer = document.getElementById('proc-watch
1113     er-table-container');
1114     if (watchResizer && watchContainer) {
1115         let startY = 0, startHeight = 0;
1116         const onMouseMove = (e) => {
1117             const delta = e.clientY - startY;
1118             const newH = Math.max(120, Math.min(850, startHeight +
1119             delta));
1120             watchContainer.style.height = `${newH}px`;
1121         };
1122         const onMouseUp = () => {
1123             watchResizer.classList.remove('resizing');
1124             document.removeEventListener('mousemove', onMouseMove)
1125             ;
1126             document.removeEventListener('mouseup', onMouseUp);
1127         };
1128         watchResizer.onmousedown = (e) => {
1129             e.preventDefault();
1130             startY = e.clientY;
1131             startHeight = parseInt(window.getComputedStyle(watchCo
1132             ntainer).height, 10) || 340;
1133             watchResizer.classList.add('resizing');
1134             document.addEventListener('mousemove', onMouseMove);
1135             document.addEventListener('mouseup', onMouseUp);
1136         };
1137         watchResizer.ondblclick = () => { watchContainer.style.h
1138         eight = '340px'; };
1139     }
1140 }
1141
1142 // =====
1143 =====
1144
1145 // 6. Bind Events
1146 // =====

```

```

=====
1139
1140 function bindEvents() {
1141     // Process stream events
1142     const btnPauseProc = document.getElementById('btn-sys-paus
e-proc');
1143     if (btnPauseProc) {
1144         btnPauseProc.onclick = () => {
1145             isProcPaused = !isProcPaused;
1146             btnPauseProc.textContent = isProcPaused ? 'Возобновить
' : 'Пауза';
1147             btnPauseProc.className = isProcPaused
1148                 ? 'btn btn-xs btn-warning rounded-pill px-2.5 py-0.5
'
1149                 : 'btn btn-xs btn-outline-secondary rounded-pill px-
2.5 py-0.5';
1150         };
1151     }
1152
1153     const btnRefreshProc = document.getElementById('btn-sys-re
fresh-proc');
1154     if (btnRefreshProc) {
1155         btnRefreshProc.onclick = () => fetchTopProcesses();
1156     }
1157
1158     const searchProc = document.getElementById('sys-proc-searc
h');
1159     if (searchProc) {
1160         searchProc.oninput = () => renderProcessTable();
1161     }
1162
1163     // Network filter buttons
1164     document.querySelectorAll('.proc-net-filter-btn').forEach(
btn => {
1165         btn.onclick = () => {
1166             document.querySelectorAll('.proc-net-filter-btn').forE
ach(b => {
1167                 b.classList.remove('active', 'btn-outline-primary');

```



```

1168         b.classList.add('btn-outline-secondary');
1169     });
1170     btn.classList.add('active', 'btn-outline-primary');
1171     btn.classList.remove('btn-outline-secondary');
1172     currentNetFilter = btn.getAttribute('data-filter') ||
'all';
1173     renderNetworkActivityTable();
1174 };
1175 });
1176
1177 // Network search
1178 const netSearch = document.getElementById('proc-net-search
');
1179 if (netSearch) {
1180     netSearch.oninput = () => renderNetworkActivityTable();
1181 }
1182
1183 // Network pause
1184 const btnPauseNet = document.getElementById('btn-proc-paus
e-net');
1185 if (btnPauseNet) {
1186     btnPauseNet.onclick = () => {
1187         isNetPaused = !isNetPaused;
1188         btnPauseNet.textContent = isNetPaused ? 'Возобновить'
: 'Пауза';
1189         btnPauseNet.className = isNetPaused
? 'btn btn-xs btn-warning rounded-pill px-2.5 py-0.5
',
1190         : 'btn btn-xs btn-outline-secondary rounded-pill px-
2.5 py-0.5';
1191         if (!isNetPaused) renderNetworkActivityTable();
1192     };
1193 }
1194
1195
1196 // Network refresh
1197 const btnRefreshNet = document.getElementById('btn-proc-re
fresh-net');
1198 if (btnRefreshNet) {

```

```
1199 |         btnRefreshNet.onclick = () => fetchNetworkActivity();
1200 |     }
1201 |
1202 |     // File watcher buttons
1203 |     const btnWatchEx = document.getElementById('btn-proc-watch
| -exclusions');
1204 |     if (btnWatchEx) btnWatchEx.onclick = () => openWatchFolder
| sModal('exclusions');
1205 |
1206 |     const btnChangeDir = document.getElementById('btn-proc-cha
| nge-watch-dir');
1207 |     if (btnChangeDir) btnChangeDir.onclick = () => openWatchFo
| ldersModal('folders');
1208 |
1209 |     const badgeDir = document.getElementById('proc-watch-dir-b
| adge');
1210 |     if (badgeDir) badgeDir.onclick = () => openWatchFoldersMod
| al('folders');
1211 |
1212 |     const btnLiveHelp = document.getElementById('btn-proc-live
| -help');
1213 |     if (btnLiveHelp) btnLiveHelp.onclick = () => showLiveWatch
| erHelpModal();
1214 |
1215 |     // Modal navigation & controls
1216 |     const btnUp = document.getElementById('btn-modal-proc-fold
| er-up');
1217 |     if (btnUp) {
1218 |         btnUp.onclick = () => {
1219 |             const parts = currentBrowserPath.replace(/\\+$/, '').s
| plit('\\');
1220 |             if (parts.length > 1) {
1221 |                 parts.pop();
1222 |                 let parentPath = parts.join('\\');
1223 |                 if (parts.length === 1 && !parentPath.endsWith('\\')
| ) parentPath += '\\';
1224 |                 loadFolderBrowser(parentPath);
1225 |             }
```

```

1226     };
1227 }
1228
1229     const btnGo = document.getElementById('btn-modal-proc-fold
er-go');
1230     const inputPath = document.getElementById('modal-proc-fold
er-path-input');
1231     if (btnGo && inputPath) {
1232         btnGo.onclick = () => {
1233             if (inputPath.value?.trim()) loadFolderBrowser(inputPa
th.value.trim());
1234         };
1235         inputPath.onkeydown = (e) => {
1236             if (e.key === 'Enter') {
1237                 e.preventDefault();
1238                 btnGo.click();
1239             }
1240         };
1241     }
1242
1243     const btnAddCurrent = document.getElementById('btn-modal-p
roc-folder-add-current');
1244     if (btnAddCurrent) {
1245         btnAddCurrent.onclick = () => {
1246             const val = inputPath ? inputPath.value.trim() : curre
ntBrowserPath;
1247             if (val) addDirToStaged(val);
1248         };
1249     }
1250
1251     // Presets
1252     document.querySelectorAll('.proc-preset-btn').forEach(btn
=> {
1253         btn.onclick = () => {
1254             const preset = btn.getAttribute('data-preset');
1255             let target = '';
1256             if (preset === 'project') target = 'C:\\Users\\onela\\
AppData\\Local\\AI-Breadboard';

```

```
1257 |         else if (preset === 'downloads') target = 'C:\\Users\\
| onela\\Downloads';
1258 |         else if (preset === 'desktop') target = 'C:\\Users\\on
| ela\\Desktop';
1259 |         else if (preset === 'temp') target = 'C:\\Users\\onela
| \\AppData\\Local\\Temp';
1260 |         if (target) {
1261 |             addDirToStaged(target);
1262 |             loadFolderBrowser(target);
1263 |         }
1264 |     };
1265 | });
1266 |
1267 |     // Exclusions switch & add
1268 |     const switchExActive = document.getElementById('switch-pro
| c-exclusions-active');
1269 |     if (switchExActive) {
1270 |         switchExActive.onchange = (e) => toggleExclusionsActive(
| e.target.checked);
1271 |     }
1272 |
1273 |     const btnAddEx = document.getElementById('btn-proc-add-exc
| lusion');
1274 |     const selectExCat = document.getElementById('select-proc-e
| xclusion-category');
1275 |     const inputExVal = document.getElementById('input-proc-exc
| lusion-value');
1276 |     if (btnAddEx && selectExCat && inputExVal) {
1277 |         btnAddEx.onclick = () => addExclusionItem(selectExCat.va
| lue, inputExVal.value);
1278 |         inputExVal.onkeydown = (e) => {
1279 |             if (e.key === 'Enter') {
1280 |                 e.preventDefault();
1281 |                 btnAddEx.click();
1282 |             }
1283 |         };
1284 |     }
1285 |
```

```
1286 // Exclusions presets
1287 document.querySelectorAll('.proc-ex-preset-btn').forEach(b
tn => {
1288     btn.onclick = () => {
1289         const preset = btn.getAttribute('data-preset');
1290         if (preset) applyExclusionPreset(preset);
1291     };
1292 });
1293
1294 // Modal apply button
1295 const btnApply = document.getElementById('btn-modal-proc-d
irs-apply');
1296 const modalEl = document.getElementById('modal-proc-watch-
folders');
1297 if (btnApply && modalEl) {
1298     btnApply.onclick = async () => {
1299         if (stagedWatchDirs.length === 0) {
1300             showWatchDirsAlert('Выберите хотя бы одну папку для
мониторинга', 'warning');
1301             return;
1302         }
1303
1304         btnApply.disabled = true;
1305         btnApply.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span> Применение...';
1306
1307         try {
1308             const res = await fetch('/api/v1/system/file-audit/w
atch-dirs', {
1309                 method: 'POST',
1310                 headers: { 'Content-Type': 'application/json' },
1311                 body: JSON.stringify({ paths: stagedWatchDirs })
1312             });
1313             const resData = await res.json();
1314             if (res.ok && resData.success) {
1315                 if (window.bootstrap?.Modal) {
1316                     const modal = window.bootstrap.Modal.getInstance
(modalEl);
```

```

1317         if (modal) modal.hide();
1318     } else {
1319         modalEl.classList.remove('show');
1320         modalEl.style.display = 'none';
1321     }
1322     await fetchLiveFileEvents();
1323 } else {
1324     showWatchDirsAlert(`Ошибка: ${resData.detail} || 'Н
1325 е удалось применить список папок'}`, 'danger');
1326 }
1327 } catch (err) {
1328     showWatchDirsAlert(`Сетевая ошибка: ${err.message}`,
1329 'danger');
1330 } finally {
1331     btnApply.disabled = false;
1332     btnApply.innerHTML = '<i class="bi bi-check2-circle
1333 me-1"></i> Применить и запустить';
1334 }
1335 }
1336 };
1337 }
1338
1339 // =====
1340 // 7. Poller & Tab Lifecycle
1341 // =====
1342
1343 function setupPoller(seconds) {
1344     _currentUiRefreshSeconds = seconds;
1345     const pollHandler = async () => {
1346         await fetchTopProcesses();
1347         await fetchNetworkActivity();
1348         await fetchLiveFileEvents();
1349     };

```

```
1350     if (window.registerTabPoller) {
1351         window.registerTabPoller('tab-processes-load-inspector',
    pollHandler, seconds * 1000, { pollerId: 'proc_load_inspector
    ', immediate: false });
1352     } else {
1353         if (window._procLoadInspectorInterval) {
1354             clearInterval(window._procLoadInspectorInterval);
1355             window._procLoadInspectorInterval = null;
1356         }
1357         window._procLoadInspectorInterval = setInterval(() => {
1358             if (window.isTabActive ? window.isTabActive('tab-proce
    sses-load-inspector') : true) {
1359                 pollHandler();
1360             }
1361             }, seconds * 1000);
1362     }
1363 }
1364
1365 async function initProcessesLoadInspectorTab() {
1366     console.log('[ProcessesLoadInspectorTab] Initializing...')
    ;
1367     bindEvents();
1368     if (window.isTabActive ? window.isTabActive('tab-processes
    -load-inspector') : false) {
1369         await fetchTopProcesses();
1370         await fetchNetworkActivity();
1371         await fetchLiveFileEvents();
1372     }
1373     setupPoller(_currentUiRefreshSeconds || 2);
1374 }
1375
1376 function activateProcessesLoadInspectorTab() {
1377     if (window.isTabActive && !window.isTabActive('tab-process
    es-load-inspector')) return;
1378     console.log('[ProcessesLoadInspectorTab] Tab activated, re
    freshing metrics...');
1379     fetchTopProcesses();
1380     fetchNetworkActivity();
```

```
1381     fetchLiveFileEvents();
1382 }
1383
1384 function deactivateProcessesLoadInspectorTab() {
1385     console.log('[ProcessesLoadInspectorTab] Tab deactivated')
1386     ;
1387 }
1388
1389 window.initProcessesLoadInspectorTab = initProcessesLoadInspectorTab;
1390 window.activateProcessesLoadInspectorTab = activateProcessesLoadInspectorTab;
1391 window.deactivateProcessesLoadInspectorTab = deactivateProcessesLoadInspectorTab;
1392
1393 if (document.getElementById('tab-processes-load-inspector'))
1394 {
1395     initProcessesLoadInspectorTab();
1396 }
```


index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 Usage Examples:
9 |Web Route:
10 |/windows/api/webgui/process_leaks_tab/index.html
11 |
12 |Tab Frame Integration:
13 |<iframe src="/windows/api/webgui/process_leaks_tab/index
14 |.html" class="w-100 h-100 border-0"></iframe>
15 |
16 |* File: index.html
17 |* Project: ai-breadboard
18 |* Package: windows/api/webgui/process_leaks_tab
19 |* Author: hypo69
20 |* Copyright: © 2026 hypo69
21 |* Updated: 2026-10-08 03:08:00
22 |* =====
23 |=====
24 -->
25
26 <style>
27 |.leaks-tab-wrapper {
28 |padding: 0.75rem;
29 |color: var(--text-color, #f8fafc);
30 |}
31 |.diag-meter-card {
32 |background: var(--surface-2, #1e293b);
33 |border: 1px solid var(--border-color, #334155);
34 |border-radius: var(--radius-md, 8px);
```

```
33     padding: 0.75rem 1rem;
34 }
35 .diag-badge-critical {
36     background: rgba(239, 68, 68, 0.2);
37     color: #f87171;
38     border: 1px solid rgba(239, 68, 68, 0.4);
39     font-size: 0.70rem;
40     padding: 2px 6px;
41     border-radius: 4px;
42     font-weight: 600;
43 }
44 .diag-badge-warning {
45     background: rgba(234, 179, 8, 0.2);
46     color: #fde047;
47     border: 1px solid rgba(234, 179, 8, 0.4);
48     font-size: 0.70rem;
49     padding: 2px 6px;
50     border-radius: 4px;
51     font-weight: 600;
52 }
53 .diag-badge-normal {
54     background: rgba(34, 197, 94, 0.15);
55     color: #4ade80;
56     border: 1px solid rgba(34, 197, 94, 0.3);
57     font-size: 0.70rem;
58     padding: 2px 6px;
59     border-radius: 4px;
60     font-weight: 500;
61 }
62 .diag-panel {
63     background: var(--surface-2, #1e293b);
64     border: 1px solid var(--border-color, #334155);
65     border-radius: 8px;
66     overflow: hidden;
67 }
68 .diag-panel-header {
69     background: rgba(15, 23, 42, 0.6);
70     padding: 8px 14px;
```

```

71     border-bottom: 1px solid var(--border-color, #334155);
72     font-weight: 600;
73     font-size: 0.85rem;
74 }
75 </style>
76
77 <div class="leaks-tab-wrapper">
78     <!-- Top Header & Status -->
79     <div class="card shadow-sm border-secondary-subtle mb-3">
80         <div class="card-body p-3">
81             <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
82                 <div class="d-flex align-items-center gap-2">
83                     <div class="fs-4">□ </div>
84                     <div>
85                         <h5 class="mb-0 fw-bold">1. Утечки ресурсов и микр
оффины (Leak Hunter)</h5>
86                         <small class="text-muted">Глубокий Win32 аудит: де
скрипторы Handles, GDI/USER объекты GUI, Hard Page Faults и ри
ски утечек</small>
87                     </div>
88                 </div>
89
90                 <div class="d-flex align-items-center gap-2 flex-wrap"
>
91                     <select class="form-select form-select-sm poll-freq-
select" id="diag-leaks-poll-freq" data-poll-id="process_leaks"
title="Частота опроса: Утечки процессов">
92                         <option value="start">Только при старте</option>
93                         <option value="manual" selected>По запросу</option>
94
95                         <option value="3">Каждые 3 сек</option>
96                         <option value="5">Каждые 5 сек</option>
97                         <option value="10">Каждые 10 сек</option>
98                         <option value="30">Каждые 30 сек</option>
99                         <option value="60">Каждые 60 сек</option>
100                        <option value="120">Каждые 120 сек</option>
</select>

```

```
101         <button class="btn btn-sm btn-outline-info rounded-p  
    ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=  
    "btn-diag-leaks-refresh">  
102             <i class="bi bi-arrow-clockwise"></i> Обновить  
103         </button>  
104     </div>  
105 </div>  
106 </div>  
107 </div>  
108  
109 <!-- Top Leak Hunter KPI Metrics -->  
110 <div class="row g-2 mb-3">  
111     <div class="col-12 col-sm-6 col-lg-3">  
112         <div class="diag-meter-card">  
113             <div class="d-flex justify-content-between align-items  
-center small text-muted">  
114                 <span>Проанализировано процессов</span>  
115                 <i class="bi bi-cpu text-info"></i>  
116             </div>  
117             <div class="fs-4 fw-bold text-info mt-1" id="diag-leak  
s-total-proc">0</div>  
118             <div class="small text-muted">Полнотекстовый срез Win3  
2 API</div>  
119         </div>  
120     </div>  
121     <div class="col-12 col-sm-6 col-lg-3">  
122         <div class="diag-meter-card">  
123             <div class="d-flex justify-content-between align-items  
-center small text-muted">  
124                 <span>С признаками утечек</span>  
125                 <i class="bi bi-exclamation-triangle text-danger"></  
i>  
126             </div>  
127             <div class="fs-4 fw-bold text-danger mt-1" id="diag-le  
aks-suspicious-count">0</div>  
128             <div class="small text-muted">Handles / GDI / Page Fau  
lts</div>  
129         </div>
```

```
130     </div>
131     <div class="col-12 col-sm-6 col-lg-3">
132         <div class="diag-meter-card">
133             <div class="d-flex justify-content-between align-items
134 -center small text-muted">
135                 <span>Пик дескрипторов (Handles)</span>
136                 <i class="bi bi-box text-warning"></i>
137             </div>
138             <div class="fs-4 fw-bold text-warning mt-1" id="diag-l
139 eaks-peak-handles">--</div>
140             <div class="small text-muted text-truncate" id="diag-l
141 eaks-peak-handles-proc">Процесс: --</div>
142         </div>
143     </div>
144     <div class="col-12 col-sm-6 col-lg-3">
145         <div class="diag-meter-card">
146             <div class="d-flex justify-content-between align-items
147 -center small text-muted">
148                 <span>Пик GDI объектов</span>
149                 <i class="bi bi-palette text-primary"></i>
150             </div>
151             <div class="fs-4 fw-bold text-primary mt-1" id="diag-l
152 eaks-peak-gdi">--</div>
153             <div class="small text-muted text-truncate" id="diag-l
154 eaks-peak-gdi-proc">Лимит OS: 10 000</div>
155         </div>
156     </div>
157 </div>
158 <!-- Main Process Leak Hunter Table -->
159 <div class="diag-panel">
160     <div class="diag-panel-header d-flex justify-content-betwe
161 en align-items-center flex-wrap gap-2">
162         <div class="d-flex align-items-center gap-2">
163             <i class="bi bi-shield-shaded text-warning"></i>
164             <span class="text-white">Детектор утечек дескрипторов,
165 GDI и страничных ошибок (Leak Hunter)</span>
166         </div>
167     </div>
```

```

160     <div class="d-flex align-items-center gap-2 flex-wrap">
161         <div class="btn-group btn-group-sm" role="group" id="d
diag-leaks-filter-group">
162             <button type="button" class="btn btn-outline-info ac
tive py-0.5 px-2" data-filter="all" style="font-size: 0.72rem;
">Bce</button>
163             <button type="button" class="btn btn-outline-danger
py-0.5 px-2" data-filter="suspicious" style="font-size: 0.72re
m;">□ Подозрительные</button>
164             <button type="button" class="btn btn-outline-seconda
ry py-0.5 px-2" data-filter="gdi" style="font-size: 0.72rem;">
GDI > 500</button>
165             <button type="button" class="btn btn-outline-seconda
ry py-0.5 px-2" data-filter="handles" style="font-size: 0.72re
m;">Handles > 1500</button>
166         </div>
167         <input type="text" class="form-control form-control-sm
" id="diag-leaks-search" placeholder="Поиск по PID или имени..
." style="max-width: 180px; font-size: 0.74rem; height: 26px;"
>
168     </div>
169 </div>
170 <div class="table-responsive" style="max-height: 540px; ov
erflow-y: auto;">
171     <table class="table table-dark table-hover table-sm mb-0
" style="font-size: 0.78rem; font-family: ui-monospace, SFMono
-Regular, Menlo, Monaco, Consolas, monospace;">
172         <thead>
173             <tr class="text-secondary border-secondary">
174                 <th style="width: 70px;">PID</th>
175                 <th>Имя процесса</th>
176                 <th style="width: 90px; text-align: right;">Handle
s</th>
177                 <th style="width: 90px; text-align: right;">GDI 06
ъекты</th>
178                 <th style="width: 90px; text-align: right;">USER 0
бъекты</th>
179                 <th style="width: 120px; text-align: right;">Page

```

```

Faults</th>
180 |         <th style="width: 90px; text-align: right;">Память
      | (RSS)</th>
181 |         <th style="width: 70px; text-align: right;">Потоки
      | </th>
182 |         <th style="width: 100px;">Статус риска</th>
183 |         <th>Признаки / Аномалии</th>
184 |     </tr>
185 | </thead>
186 | <tbody id="diag-leaks-tbody">
187 |     <tr><td colspan="10" class="text-center py-4 text-mu
      | ted">Загрузка данных глубокой диагностики процессов...</td></t
      | r>
188 |     </tbody>
189 | </table>
190 | </div>
191 | </div>
192 | </div>
193 |

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Process Leaks Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/process_leaks_tab/main
    .js?v=20261001_v1" type="module"></script>
11 *
12 * File: main.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/process_leaks_tab
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-08 02:00:00
18 * =====
19 */
20
21 (function () {
22   'use strict';
23
24   let rawLeakProcesses = [];
25   let leakFilter = 'all';
26   let autoRefreshTimer = null;
27   const POLL_ID = 'process_leaks';
28
29   function escapeHtml(str) {
30     if (str === null || str === undefined) return '';
31     return String(str)
32       .replace(/&/g, '&amp;')
```



```

33 |     .replace(/</g, '&lt;')
34 |     .replace(/>/g, '&gt;')
35 |     .replace(/"/g, '&quot;')
36 |     .replace(/'/g, '&#039;');
37 | }
38 |
39 | function setText(id, text) {
40 |     const el = document.getElementById(id);
41 |     if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
42 | }
43 |
44 | async function apiFetch(url) {
45 |     if (window.api && typeof window.api.fetch === 'function')
{
46 |         return await window.api.fetch(url);
47 |     }
48 |     const res = await fetch(url);
49 |     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
atusText}`);
50 |     return await res.json();
51 | }
52 |
53 | async function fetchProcessLeaks() {
54 |     try {
55 |         const data = await apiFetch('/api/v1/system/diagnostics/
leaks?limit=100');
56 |         if (!data) return;
57 |
58 |         setText('diag-leaks-total-proc', data.total_processes);
59 |         setText('diag-leaks-suspicious-count', data.suspicious_c
ount);
60 |
61 |         if (data.top_handle_hogs && data.top_handle_hogs.length
> 0) {
62 |             const topH = data.top_handle_hogs[0];
63 |             setText('diag-leaks-peak-handles', `${topH.handles_cou
nt.toLocaleString()} шт.`);

```

```

64 |         setText('diag-leaks-peak-handles-proc', `Процесс: ${topH.name} (PID ${topH.pid})`);
65 |     }
66 |     if (data.top_gdi_hogs && data.top_gdi_hogs.length > 0) {
67 |         const topG = data.top_gdi_hogs[0];
68 |         setText('diag-leaks-peak-gdi', `${topG.gdi_objects.toLocaleString()} шт.`);
69 |         setText('diag-leaks-peak-gdi-proc', `Процесс: ${topG.name} (PID ${topG.pid})`);
70 |     }
71 |
72 |     rawLeakProcesses = data.all_processes || [];
73 |     renderProcessLeaksTable(rawLeakProcesses);
74 | } catch (err) {
75 |     console.warn('[ProcessLeaksTab] fetchProcessLeaks error: ', err);
76 | }
77 | }
78 |
79 | function renderProcessLeaksTable(processes) {
80 |     const tbody = document.getElementById('diag-leaks-tbody');
81 |     if (!tbody) return;
82 |
83 |     const query = (document.getElementById('diag-leaks-search')?.value || '').trim().toLowerCase();
84 |
85 |     let filtered = processes.filter(p => {
86 |         if (query && !p.name.toLowerCase().includes(query) && !String(p.pid).includes(query)) {
87 |             return false;
88 |         }
89 |         if (leakFilter === 'suspicious') return p.leak_risk_score !== 'normal';
90 |         if (leakFilter === 'gdi') return p.gdi_objects > 500;
91 |         if (leakFilter === 'handles') return p.handles_count > 1500;
92 |         return true;
93 |     });

```

```

94
95     if (filtered.length === 0) {
96         tbody.innerHTML = '<tr><td colspan="10" class="text-cent
er py-4 text-muted">Процессов по заданным критериям не найдено
</td></tr>';
97         return;
98     }
99
100     tbody.innerHTML = filtered.map(p => {
101         let riskBadge = '<span class="diag-badge-normal">Норма</
span>';
102         if (p.leak_risk_score === 'critical') {
103             riskBadge = '<span class="diag-badge-critical">Критиче
ский</span>';
104         } else if (p.leak_risk_score === 'warning') {
105             riskBadge = '<span class="diag-badge-warning">Внимание
</span>';
106         }
107
108         const reasons = (p.leak_risk_reasons && p.leak_risk_reas
ons.length > 0)
109             ? p.leak_risk_reasons.map(r => `<span class="badge bg-
dark border border-secondary text-warning me-1">${escapeHtml(r
)}</span>`).join('')
110             : '<span class="text-muted small">Штатное поведение</s
pan>';
111
112         return `
113             <tr>
114                 <td><span class="text-info font-monospace">${p.pid}<
/span></td>
115                 <td class="fw-bold text-white text-truncate" style="
max-width: 180px;">${escapeHtml(p.name)}</td>
116                 <td style="text-align: right;" class="${p.handles_co
unt > 1500 ? 'text-danger fw-bold' : 'text-light'}">${p.handle
s_count.toLocaleString()}</td>
117                 <td style="text-align: right;" class="${p.gdi_object
s > 800 ? 'text-danger fw-bold' : 'text-light'}">${p.gdi_objec

```

```

    ts.toLocaleString())</td>
118 |         <td style="text-align: right;" class="{p.user_objec
    ts > 400 ? 'text-warning' : 'text-light'}">{p.user_objects.to
    LocaleString()}</td>
119 |         <td style="text-align: right;" class="text-muted fon
    t-monospace">{p.page_faults_total.toLocaleString()}</td>
120 |         <td style="text-align: right;" class="text-warning">
    {p.memory_mb.toFixed(1)} MB</td>
121 |         <td style="text-align: right;" class="text-muted">{p
    .threads_count}</td>
122 |         <td>{riskBadge}</td>
123 |         <td>{reasons}</td>
124 |     </tr>
125 |     `;
126 |     }).join('');
127 | }
128 |
129 | function getFrequency() {
130 |     try {
131 |         const saved = localStorage.getItem(`poll_freq_${POLL_ID}
    `);
132 |         if (saved) return saved;
133 |     } catch (_) {}
134 |     return 'manual';
135 | }
136 |
137 | function setFrequency(freq) {
138 |     try {
139 |         localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
140 |     } catch (_) {}
141 |     applyPoller(freq, false);
142 | }
143 |
144 | function stopPolling() {
145 |     if (window.unregisterTabPoller) {
146 |         window.unregisterTabPoller(`tab-process-leaks_${POLL_ID}
    `);
147 |     }

```

```
148     if (autoRefreshTimer) {
149         clearInterval(autoRefreshTimer);
150     function applyPoller(freq, runInitial = false) {
151         stopPolling();
152         if (freq === 'start' || freq === 'manual') {
153             if (runInitial && (window.isTabActive ? window.isTabActive('tab-process-leaks') : true)) {
154                 fetchProcessLeaks();
155             }
156             return;
157         }
158
159         const intervalSec = parseInt(freq, 10);
160         if (isNaN(intervalSec) || intervalSec <= 0) return;
161
162         const intervalMs = intervalSec * 1000;
163         const pollerId = `tab-process-leaks_${POLL_ID}`;
164
165         if (window.registerTabPoller) {
166             window.registerTabPoller('tab-process-leaks', fetchProcessLeaks, intervalMs, { pollerId, immediate: runInitial });
167         } else {
168             if (runInitial && (window.isTabActive ? window.isTabActive('tab-process-leaks') : true)) {
169                 fetchProcessLeaks();
170             }
171             autoRefreshTimer = setInterval(() => {
172                 if (window.isTabActive ? window.isTabActive('tab-process-leaks') : true) {
173                     fetchProcessLeaks();
174                 }
175             }, intervalMs);
176         }
177     }
178
179     function bindEvents() {
180         const leakFilters = document.getElementById('diag-leaks-filter-group');
```

```
181 |     if (leakFilters) {
182 |         leakFilters.querySelectorAll('button').forEach(btn => {
183 |             btn.onclick = () => {
184 |                 leakFilters.querySelectorAll('button').forEach(b =>
185 | {
186 |                 b.classList.remove('active', 'btn-info', 'btn-dang
187 | er');
188 |                 b.classList.add('btn-outline-secondary');
189 |                 const f = btn.getAttribute('data-filter') || 'all';
190 |                 btn.classList.add('active', f === 'suspicious' ? 'bt
191 | n-danger' : 'btn-info');
192 |                 btn.classList.remove('btn-outline-secondary');
193 |                 leakFilter = f;
194 |                 renderProcessLeaksTable(rawLeakProcesses);
195 |             };
196 |         });
197 |     }
198 |
199 |     const leakSearch = document.getElementById('diag-leaks-sea
200 | rch');
201 |
202 |     if (leakSearch) {
203 |         leakSearch.oninput = () => renderProcessLeaksTable(rawLe
204 | akProcesses);
205 |     }
206 |
207 |     const btnLeaksRefresh = document.getElementById('btn-diag-
208 | leaks-refresh');
209 |
210 |     if (btnLeaksRefresh) {
211 |         btnLeaksRefresh.onclick = () => fetchProcessLeaks();
212 |     }
213 |
214 |     const select = document.getElementById('diag-leaks-poll-fr
215 | eq');
216 |
217 |     if (select) {
218 |         select.onchange = (e) => {
219 |             const newFreq = e.target.value;
220 |             setFrequency(newFreq);
221 |         }
222 |     }
223 | }
```

```

212         if (newFreq !== 'manual' && newFreq !== 'start') {
213             fetchProcessLeaks();
214         }
215         if (window.showToast) {
216             const label = select.options[select.selectedIndex]?.
text || newFreq;
217             window.showToast(`Частота опроса утечек: ${label}`,
'info');
218         }
219     };
220 }
221 }
222
223 async function initProcessLeaksTab() {
224     bindEvents();
225     const currentFreq = getFrequency();
226     const select = document.getElementById('diag-leaks-poll-fr
eq');
227     if (select) select.value = currentFreq;
228     applyPoller(currentFreq, true);
229 }
230
231 window.initProcessLeaksTab = initProcessLeaksTab;
232 window.activateProcessLeaksTab = () => {
233     applyPoller(getFrequency(), true);
234 };
235
236 if (document.readyState === 'loading') {
237     document.addEventListener('DOMContentLoaded', () => {
238         if (window.isTabActive ? window.isTabActive('tab-process
-leaks') : false) {
239             initProcessLeaksTab();
240         } else {
241             bindEvents();
242         }
243     });
244 } else {
245     if (window.isTabActive ? window.isTabActive('tab-process-l

```

```
    eaks') : false) {  
246 |         initProcessLeaksTab();  
247 |     } else {  
248 |         bindEvents();  
249 |     }  
250 | }  
251 | ))();  
252 |
```


index.html

```
1 <!--
2 =====
3 |
4 | =====
5 | Process Name: Windows Process Manager - Task Manager Index Vie
6 | w
7 | =====
8 | =====
9 | Description:
10 | Клиентский веб-интерфейс диспетчера задач и процессов Window
11 | s с разделением
12 | на категории: Apps (Приложения с окнами), Background process
13 | es (Фоновые процессы)
14 | и Windows processes (Системные процессы), группировкой подпр
15 | оцессов и деревом окон.
16 |
17 | Usage Examples:
18 |     Web Route:
19 |         /windows/api/webgui/process_manager_tab/index.html
20 |
21 |     Tab Frame Integration:
22 |         <iframe src="/windows/api/webgui/process_manager_tab/ind
23 | ex.html?v=20261008_v1" class="w-100 h-100 border-0"></iframe>
24 |
25 | File: index.html
26 | Project: ai-breadboard
27 | Package: windows/api/webgui/process_manager_tab
28 | Author: hypo69
29 | Copyright: © 2026 hypo69
30 | Updated: 2026-10-08 02:10:00
31 | =====
32 | =====
33 | -->
34 |
35 | <style>
36 |     .pm-group-header {
37 |         background: rgba(255, 255, 255, 0.05);
```

```
29 |     cursor: pointer;
30 |     font-weight: 600;
31 |     transition: background 0.15s ease-in-out;
32 | }
33 | .pm-group-header:hover {
34 |     background: rgba(255, 255, 255, 0.1) !important;
35 | }
36 | .pm-chevron {
37 |     display: inline-block;
38 |     transition: transform 0.2s ease;
39 |     width: 14px;
40 |     margin-right: 6px;
41 |     color: var(--bs-secondary, #6c757d);
42 | }
43 | .pm-chevron.expanded {
44 |     transform: rotate(90deg);
45 |     color: var(--bs-primary, #0d6efd);
46 | }
47 | .pm-sub-row {
48 |     background: rgba(0, 0, 0, 0.25);
49 |     font-size: 0.82rem;
50 |     border-left: 3px solid rgba(13, 110, 253, 0.4);
51 | }
52 | .pm-section-badge {
53 |     font-size: 0.85rem;
54 |     font-weight: 700;
55 |     letter-spacing: 0.5px;
56 | }
57 | .pm-type-tag {
58 |     font-size: 0.72rem;
59 |     padding: 2px 6px;
60 |     border-radius: 4px;
61 |     background: rgba(255, 255, 255, 0.07);
62 |     color: #a0aec0;
63 | }
64 | .pm-card-hover {
65 |     transition: transform 0.15s ease, box-shadow 0.15s ease;
66 |     cursor: pointer;
```

```

67 | }
68 | .pm-card-hover:hover {
69 |     transform: translateY(-2px);
70 |     box-shadow: 0 4px 12px rgba(0,0,0,0.3);
71 | }
72 | </style>
73 |
74 | <!-- Process Manager Tab -->
75 | <div class="container-fluid py-2 px-3">
76 |     <!-- Top Stat Cards -->
77 |     <div class="row g-2 mb-3">
78 |         <div class="col-6 col-md-3">
79 |             <div class="card bg-dark text-white border-secondary h-1
00 | 00 shadow-sm p-3 pm-card-hover" id="pm-filter-card-apps">
80 |                 <div class="text-muted small text-uppercase fw-semibol
d d-flex justify-content-between align-items-center">
81 |                     <span>Приложения (Apps)</span>
82 |                     <i class="bi bi-window-stack text-primary"></i>
83 |                 </div>
84 |                 <div class="fs-4 fw-bold mt-1 text-primary" id="pm-app
s-count">--</div>
85 |                 <div class="small text-muted mt-1" id="pm-apps-sub">0к
онные программы</div>
86 |             </div>
87 |         </div>
88 |         <div class="col-6 col-md-3">
89 |             <div class="card bg-dark text-white border-secondary h-1
00 | 00 shadow-sm p-3 pm-card-hover" id="pm-filter-card-bg">
90 |                 <div class="text-muted small text-uppercase fw-semibol
d d-flex justify-content-between align-items-center">
91 |                     <span>Фоновые (Background)</span>
92 |                     <i class="bi bi-gear-wide-connected text-info"></i>
93 |                 </div>
94 |                 <div class="fs-4 fw-bold mt-1 text-info" id="pm-bg-cou
nt">--</div>
95 |                 <div class="small text-muted mt-1" id="pm-bg-sub">Фоно
вые службы и демоны</div>
96 |             </div>

```

```

97     </div>
98     <div class="col-6 col-md-3">
99         <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3 pm-card-hover" id="pm-filter-card-win">
100             <div class="text-muted small text-uppercase fw-semibol
d d-flex justify-content-between align-items-center">
101                 <span>Windows процессы</span>
102                 <i class="bi bi-microsoft text-warning"></i>
103             </div>
104             <div class="fs-4 fw-bold mt-1 text-warning" id="pm-win
-count">--</div>
105             <div class="small text-muted mt-1" id="pm-win-sub">Сис
темные процессы ОС</div>
106         </div>
107     </div>
108     <div class="col-6 col-md-3">
109         <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3 pm-card-hover" id="pm-filter-card-all">
110             <div class="text-muted small text-uppercase fw-semibol
d d-flex justify-content-between align-items-center">
111                 <span>Всего процессов</span>
112                 <i class="bi bi-cpu text-success"></i>
113             </div>
114             <div class="fs-4 fw-bold mt-1 text-success" id="pm-tot
al-count">--</div>
115             <div class="small text-muted mt-1" id="pm-ram-usage">П
амять: --</div>
116         </div>
117     </div>
118 </div>
119
120 <!-- Action Bar & Filter Controls -->
121 <div class="d-flex align-items-center justify-content-between
mb-3 p-2 bg-dark rounded border border-secondary flex-wrap g
ap-2">
122     <div class="d-flex align-items-center gap-2 flex-wrap">
123         <button class="btn btn-sm btn-primary d-flex align-items
-center gap-1" id="pm-refresh-btn">

```

```
124 |         <i class="bi bi-arrow-clockwise"></i>
125 |         <span>Обновить</span>
126 |     </button>
127 |
128 |     <!-- Category Filter Pills -->
129 |     <div class="btn-group btn-group-sm" role="group" id="pm-
category-filter-group">
130 |         <input type="radio" class="btn-check" name="pm-cat-fil
ter" id="pm-filter-all" autocomplete="off" checked value="all"
>
131 |         <label class="btn btn-outline-secondary" for="pm-filte
r-all">Все</label>
132 |
133 |         <input type="radio" class="btn-check" name="pm-cat-fil
ter" id="pm-filter-apps" autocomplete="off" value="app">
134 |         <label class="btn btn-outline-primary" for="pm-filter-
apps">Apps</label>
135 |
136 |         <input type="radio" class="btn-check" name="pm-cat-fil
ter" id="pm-filter-bg" autocomplete="off" value="background">
137 |         <label class="btn btn-outline-info" for="pm-filter-bg"
>Background</label>
138 |
139 |         <input type="radio" class="btn-check" name="pm-cat-fil
ter" id="pm-filter-win-radio" autocomplete="off" value="window
s">
140 |         <label class="btn btn-outline-warning" for="pm-filter-
win-radio">Windows</label>
141 |     </div>
142 |
143 |     <button class="btn btn-sm btn-outline-secondary d-flex a
lign-items-center gap-1" id="pm-toggle-expand-btn" title="Свер
нуть / Развернуть все группы">
144 |         <i class="bi bi-arrows-expand"></i>
145 |         <span id="pm-expand-btn-text">Развернуть все</span>
146 |     </button>
147 | </div>
148 |
```

```

149     <div class="d-flex align-items-center gap-2">
150         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="pm-search-input" placeholder="Поиск приложения, процесса или PID..." style="width: 280px;">
151     </div>
152 </div>
153
154 <!-- Process Table (Task Manager Style) -->
155 <div class="card bg-dark text-white border-secondary shadow-sm">
156     <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
157         <span class="fw-semibold"><i class="bi bi-layout-text-window-reverse text-primary me-2"></i>Диспетчер процессов и приложений Windows</span>
158         <span class="badge bg-secondary" id="pm-table-badge">0 процессов</span>
159     </div>
160     <div class="card-body p-0 table-responsive" style="max-height: 600px; overflow-y: auto;">
161         <table class="table table-dark table-hover align-middle mb-0 small" id="pm-table">
162             <thead class="sticky-top bg-dark border-bottom border-secondary">
163                 <tr class="text-secondary">
164                     <th style="min-width: 280px;">Имя процесса / Приложения</th>
165                     <th style="width: 100px;">PID</th>
166                     <th style="width: 140px;">Тип / Статус</th>
167                     <th style="width: 100px;">CPU %</th>
168                     <th style="width: 120px;">Память (RAM)</th>
169                     <th style="width: 120px;" class="text-end">Действия</th>
170                 </tr>
171             </thead>
172             <tbody id="pm-tbody">
173                 <tr><td colspan="6" class="text-center text-muted py

```

```
173 | -4">Опрос активных приложений и процессов...</td></tr>
174 |     </tbody>
175 | </table>
176 | </div>
177 | </div>
178 | </div>
179 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Process Manager Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт интерактивного диспетчера задач Windows
7   *   (Task Manager)
8   *   с поддержкой группировки Apps, Background processes и Win
9   *   dows processes.
10  *
11  * Usage Examples:
12  *   HTML Integration:
13  *     <script src="/windows/api/webgui/process_manager_tab/main.js?v=20261008_v1" type="module"></script>
14  *
15  *   JavaScript Import:
16  *     import { initProcessManagerTab } from '/windows/api/webgui/process_manager_tab/main.js';
17  *
18  * File: main.js
19  * Project: ai-breadboard
20  * Package: windows/api/webgui/process_manager_tab
21  * Author: hypo69
22  * Copyright: © 2026 hypo69
23  * Updated: 2026-10-08 02:10:00
24  * =====
25  */
26
27 const registerTabPoller = window.registerTabPoller || function
28   () {};
```

```
26
27 let isInitialized = false;
28 let currentReport = null;
```



```

29 let currentCategoryFilter = 'all';
30 let expandedGroups = new Set();
31 let allExpanded = false;
32
33 /**
34  * Загрузка категоризированных процессов с бэкенда
35  */
36 export async function fetchProcessManagerSummary() {
37   try {
38     const res = await fetch('/api/process-manager/categorized'
39   );
40     if (!res.ok) {
41       // Fallback на /summary
42       const fallbackRes = await fetch('/api/process-manager/summary');
43       if (!fallbackRes.ok) throw new Error(`HTTP ${res.status}
44     `);
45     currentReport = await fallbackRes.json();
46   } else {
47     currentReport = await res.json();
48   }
49   renderProcessManager(currentReport);
50 } catch (err) {
51   console.warn('[ProcessManager] Ошибка получения процессов:
52   ', err);
53 }
54 }
55
56 /**
57  * Рендеринг карточек статистики и таблицы процессов
58  */
59 function renderProcessManager(report) {
60   if (!report) return;
61
62   const appsList = report.apps || [];
63   const bgList = report.background_processes || [];
64   const winList = report.windows_processes || [];

```

```

63 | const appsCountEl = document.getElementById('pm-apps-count')
    | ;
64 | const bgCountEl = document.getElementById('pm-bg-count');
65 | const winCountEl = document.getElementById('pm-win-count');
66 | const totalCountEl = document.getElementById('pm-total-count
    | ');
67 | const ramUsageEl = document.getElementById('pm-ram-usage');
68 |
69 | const totalProcs = report.total_processes || (appsList.length
    | h + bgList.length + winList.length);
70 | const totalMemMb = report.total_memory_used_mb || 0;
71 |
72 | if (appsCountEl) appsCountEl.textContent = report.apps_count
    | !== undefined ? report.apps_count : appsList.length;
73 | if (bgCountEl) bgCountEl.textContent = report.background_cou
    | nt !== undefined ? report.background_count : bgList.length;
74 | if (winCountEl) winCountEl.textContent = report.windows_coun
    | t !== undefined ? report.windows_count : winList.length;
75 | if (totalCountEl) totalCountEl.textContent = totalProcs;
76 |
77 | if (ramUsageEl) {
78 |     const memStr = totalMemMb >= 1024 ? `${(totalMemMb / 1024)
    | .toFixed(1)} GB` : `${Math.round(totalMemMb)} MB`;
79 |     ramUsageEl.textContent = `Память: ${memStr}`;
80 | }
81 |
82 | applyProcessFilters();
83 | }
84 |
85 | /**
86 |  * Применение фильтров поиска и категорий
87 |  */
88 | function applyProcessFilters() {
89 |     if (!currentReport) return;
90 |
91 |     const tbody = document.getElementById('pm-tbody');
92 |     const searchInput = document.getElementById('pm-search-input
    | ');

```

```
93 |   const badge = document.getElementById('pm-table-badge');
94 |
95 |   if (!tbody) return;
96 |
97 |   const query = (searchInput ? searchInput.value : '').toLowerCase().trim();
98 |
99 |   const apps = currentReport.apps || [];
100 |   const bg = currentReport.background_processes || [];
101 |   const win = currentReport.windows_processes || [];
102 |
103 |   // Фильтрация групп по строке поиска
104 |   function filterGroup(group) {
105 |     if (!query) return true;
106 |     const nameMatch = (group.name || '').toLowerCase().includes(query);
107 |     const friendlyMatch = (group.friendly_name || '').toLowerCase().includes(query);
108 |     const pidMatch = String(group.main_pid || '').includes(query);
109 |     const titlesMatch = (group.window_titles || []).some(t => t.toLowerCase().includes(query));
110 |     const subMatch = (group.subprocesses || []).some(sub =>
111 |       (sub.name || '').toLowerCase().includes(query) ||
112 |       String(sub.pid || '').includes(query) ||
113 |       (sub.username || '').toLowerCase().includes(query)
114 |     );
115 |     return nameMatch || friendlyMatch || pidMatch || titlesMatch || subMatch;
116 |   }
117 |
118 |   const filteredApps = (currentCategoryFilter === 'all' || currentCategoryFilter === 'app') ? apps.filter(filterGroup) : [];
119 |   const filteredBg = (currentCategoryFilter === 'all' || currentCategoryFilter === 'background') ? bg.filter(filterGroup) : [];
120 |   const filteredWin = (currentCategoryFilter === 'all' || currentCategoryFilter === 'windows') ? win.filter(filterGroup) : []
```

```

    ];
121
122     const totalShown = filteredApps.length + filteredBg.length +
    filteredWin.length;
123     if (badge) {
124         badge.textContent = `${totalShown} групп (${currentReport.
    total_processes || totalShown} процессов)`;
125     }
126
127     if (totalShown === 0) {
128         tbody.innerHTML = '<tr><td colspan="6" class="text-center
    text-muted py-4">Процессы не найдены</td></tr>';
129         return;
130     }
131
132     let html = '';
133
134     // Секция 1: Apps (Интерактивные приложения с окнами)
135     if (filteredApps.length > 0) {
136         html += `
137             <tr class="table-active">
138                 <td colspan="6" class="py-2 px-3 fw-bold text-primary"
    >
139                     <i class="bi bi-window-stack me-1.5"></i> Apps (${fi
    lteredApps.length})
140                 </td>
141             </tr>
142         `;
143         html += filteredApps.map(g => renderGroupRow(g, 'app', 'Ap
    p')).join('');
144     }
145
146     // Секция 2: Background processes (Фоновые процессы)
147     if (filteredBg.length > 0) {
148         html += `
149             <tr class="table-active">
150                 <td colspan="6" class="py-2 px-3 fw-bold text-info">
151                     <i class="bi bi-gear-wide-connected me-1.5"></i> Bac

```

```

    kground processes (${filteredBg.length})
152         </td>
153     </tr>
154     `;
155     html += filteredBg.map(g => renderGroupRow(g, 'background'
    , 'Background process')).join('');
156 }
157
158 // Секция 3: Windows processes (Процессы Windows)
159 if (filteredWin.length > 0) {
160     html += `
161         <tr class="table-active">
162             <td colspan="6" class="py-2 px-3 fw-bold text-warning"
    >
163                 <i class="bi bi-microsoft me-1.5"></i> Windows proce
    sses (${filteredWin.length})
164             </td>
165         </tr>
166     `;
167     html += filteredWin.map(g => renderGroupRow(g, 'windows',
    'Windows process')).join('');
168 }
169
170 tbody.innerHTML = html;
171 attachEventListeners(tbody);
172 }
173
174 /**
175  * Рендеринг строки группы процессов (Task Manager style)
176  */
177 function renderGroupRow(group, category, typeLabel) {
178     const groupId = `grp-${category}-${group.name.replace(/[^\a-z
    A-Z0-9_-]/g, '_')}-${group.main_pid}`;
179     const isExpanded = allExpanded || expandedGroups.has(groupId
    );
180     const count = group.instance_count || (group.subprocesses ?
    group.subprocesses.length : 1);
181     const countBadge = count > 1 ? ` <span class="text-muted sma

```

```

11">(${count})</span>` : '';
182   const chevron = count > 1 || (group.subprocesses && group.subprocesses.length > 0)
183     ? ``
184     : `` : '';
197       return `
198         <tr class="pm-sub-row" data-parent-group="${groupId}">
199           <td class="ps-4 py-1.5">
200             <div class="d-flex align-items-center gap-2">
201               <i class="bi bi-arrow-return-right text-muted sm
                all"></i>
202               <div>
203                 <span class="text-light">${escapeHtml(sub.name
                )}</span>
204                 ${titleInfo}

```

```

205 |             </div>
206 |         </div>
207 |     </td>
208 |     <td class="font-monospace text-warning small py-1.5"
    | >${sub.pid}</td>
209 |     <td class="py-1.5"><span class="pm-type-tag text-mut
    | ed">${typeLabel}</span></td>
210 |     <td class="font-monospace text-light small py-1.5">${
    | sub.cpu_percent.toFixed(1)}%</td>
211 |     <td class="font-monospace text-info small py-1.5">${
    | subMem}</td>
212 |     <td class="text-end py-1.5">
213 |         <button class="btn btn-xs btn-outline-danger py-0
    | px-1.5 pm-kill-btn" data-pid="${sub.pid}" data-name="${sub.nam
    | e}" title="Завершить процесс PID: ${sub.pid}">
214 |             <i class="bi bi-x"></i>
215 |         </button>
216 |     </td>
217 | </tr>
218 | `;
219 | }).join('');
220 | }
221 |
222 | return `
223 |     <tr class="pm-group-header" data-group-id="${groupId}">
224 |         <td class="py-2">
225 |             <div class="d-flex align-items-center">
226 |                 ${chevron}
227 |                 <i class="bi ${iconClass} me-2"></i>
228 |                 <span class="fw-semibold text-white">${escapeHtml(gr
    | oup.friendly_name || group.name)}</span>
229 |                 ${countBadge}
230 |             </div>
231 |         </td>
232 |         <td class="font-monospace text-muted small py-2">${group
    | .main_pid || '-'}</td>
233 |         <td class="py-2"><span class="pm-type-tag">${typeLabel}<
    | /span></td>

```

```

234 |         <td class="font-monospace text-light small py-2">${cpuSt
| r}</td>
235 |         <td class="font-monospace text-success small py-2">${mem
| Str}</td>
236 |         <td class="text-end py-2">
237 |             <button class="btn btn-xs btn-outline-danger py-0 px-2
| pm-kill-btn" data-pid="${group.main_pid}" data-name="${group.
| name}" data-tree="true" title="Завершить дерево процессов (${c
| ount})">
238 |                 <i class="bi bi-x-circle me-1"></i>Завершить
239 |             </button>
240 |         </td>
241 |     </tr>
242 |     ${subRowsHtml}
243 | `;
244 | }
245 |
246 | /**
247 |  * Навешивание обработчиков событий
248 |  */
249 | function attachEventListeners(tbody) {
250 |     // Клик по строке группы для раскрытия
251 |     tbody.querySelectorAll('.pm-group-header').forEach(header =>
| {
252 |         header.addEventListener('click', (e) => {
253 |             if (e.target.closest('.pm-kill-btn')) return;
254 |             const groupId = header.dataset.groupId;
255 |             if (expandedGroups.has(groupId)) {
256 |                 expandedGroups.delete(groupId);
257 |             } else {
258 |                 expandedGroups.add(groupId);
259 |             }
260 |             applyProcessFilters();
261 |         });
262 |     });
263 |
264 |     // Завершение процессов
265 |     tbody.querySelectorAll('.pm-kill-btn').forEach(btn => {

```



```

266     btn.addEventListener('click', (e) => {
267         e.stopPropagation();
268         const pid = btn.dataset.pid;
269         const name = btn.dataset.name;
270         const killTree = btn.dataset.tree === 'true';
271         killProcess(pid, name, killTree);
272     });
273 });
274 }
275
276 /**
277  * Завершение процесса или дерева процессов через API
278  */
279 async function killProcess(pid, name, killTree = false) {
280     const treeMsg = killTree ? ' (и все дочерние процессы)' : ''
281     ;
282     if (!confirm(`Вы действительно хотите завершить ${name} [PID
283 : ${pid}]${treeMsg}?`)) return;
284
285     try {
286         const res = await fetch('/api/process-manager/action', {
287             method: 'POST',
288             headers: { 'Content-Type': 'application/json' },
289             body: JSON.stringify({ pid: parseInt(pid, 10), kill_tree
290 : killTree, dry_run: false, confirmed_by_user: true })
291         });
292         const result = await res.json();
293         if (res.ok && result.status === 'SUCCESS') {
294             if (window.toast) window.toast.success('Процесс завершен
295 ', `${name} (PID: ${pid})`);
296             fetchProcessManagerSummary();
297         } else {
298             if (window.toast) window.toast.error('Ошибка завершения'
299 , result.message || result.detail || 'Отказано в доступе');
300         }
301     } catch (err) {
302         if (window.toast) window.toast.error('Ошибка сети', err.me
303 ssage);

```

```
298 |     }
299 | }
300 |
301 | /**
302 |  * Экранирование спецсимволов HTML
303 |  */
304 | function escapeHtml(str) {
305 |     if (!str) return '';
306 |     return String(str).replace(/&/g, '&amp;').replace(/</g, '&lt;');
307 |     ;').replace(/>/g, '&gt;').replace(/"/g, '&quot;');
308 | }
309 |
310 | /**
311 |  * Инициализация вкладки Process Manager
312 |  */
313 | export function initProcessManagerTab() {
314 |     if (isInitialized) return;
315 |     isInitialized = true;
316 |
317 |     const refreshBtn = document.getElementById('pm-refresh-btn')
318 |     ;
319 |     if (refreshBtn) refreshBtn.addEventListener('click', fetchPr
320 |     ocessManagerSummary);
321 |
322 |     const searchInput = document.getElementById('pm-search-input
323 |     ');
324 |     if (searchInput) searchInput.addEventListener('input', apply
325 |     ProcessFilters);
326 |
327 |     // Фильтры по карточкам
328 |     const cardApps = document.getElementById('pm-filter-card-app
329 |     s');
330 |     const cardBg = document.getElementById('pm-filter-card-bg');
331 |     const cardWin = document.getElementById('pm-filter-card-win'
332 |     );
333 |     const cardAll = document.getElementById('pm-filter-card-all'
334 |     );
```

```

328 |   if (cardApps) cardApps.addEventListener('click', () => setCa
    |   tegoryFilter('app'));
329 |   if (cardBg) cardBg.addEventListener('click', () => setCatego
    |   ryFilter('background'));
330 |   if (cardWin) cardWin.addEventListener('click', () => setCate
    |   goryFilter('windows'));
331 |   if (cardAll) cardAll.addEventListener('click', () => setCate
    |   goryFilter('all'));
332 |
333 |   // Фильтры по радиокнопкам
334 |   document.querySelectorAll('input[name="pm-cat-filter"]').for
    |   Each(radio => {
335 |       radio.addEventListener('change', (e) => {
336 |           currentCategoryFilter = e.target.value;
337 |           applyProcessFilters();
338 |       });
339 |   });
340 |
341 |   // Кнопка развернуть / свернуть все
342 |   const toggleExpandBtn = document.getElementById('pm-toggle-e
    |   xpand-btn');
343 |   const expandText = document.getElementById('pm-expand-btn-te
    |   xt');
344 |   if (toggleExpandBtn) {
345 |       toggleExpandBtn.addEventListener('click', () => {
346 |           allExpanded = !allExpanded;
347 |           if (expandText) expandText.textContent = allExpanded ? '
    |   Свернуть все' : 'Развернуть все';
348 |           applyProcessFilters();
349 |       });
350 |   }
351 |
352 |   registerTabPoller('tab-process-manager', fetchProcessManager
    |   Summary, 5000, { immediate: true });
353 | }
354 |
355 | function setCategoryFilter(cat) {
356 |     currentCategoryFilter = cat;

```

```
357 |   const radio = document.querySelector(`input[name="pm-cat-fil
    |   ter"][value="${cat}"]`);
358 |   if (radio) radio.checked = true;
359 |   applyProcessFilters();
360 | }
361 |
362 | window.initProcessManagerTab = initProcessManagerTab;
363 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/rag_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/rag_tab/index.html" cla
17         ss="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/rag_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- RAG Knowledge Base & Document / Codebase Management Tab -
30 -->
31
32 <div class="container-fluid py-2">
33     <!-- Header Alert with Sub-Navigation -->
34     <div class="alert alert-primary d-flex justify-content-betwe
35     en align-items-center flex-wrap gap-2 mb-3 shadow-sm">
36         <div>
37             <h5 class="mb-1"><i class="bi bi-brain me-2"></i><span d
38             ata-i18n="rag.title">□ База знаний и RAG</span></h5>
```

```

30      <p class="mb-0 small text-secondary" data-i18n="rag.subtitle">Загрузка документов, индексация исходного кода проектов
    и семантический поиск.</p>
31  </div>
32  <div class="d-flex align-items-center gap-2">
33    <!-- Sub-Tabs Navigation -->
34    <ul class="nav nav-pills nav-fill bg-body-secondary p-1
    rounded-pill" id="rag-mode-pills" role="tablist">
35      <li class="nav-item" role="presentation">
36        <button class="nav-link active rounded-pill btn-sm p
    x-3 py-1" id="pills-user-rags-tab" data-bs-toggle="pill" data-
    bs-target="#pills-user-rags" type="button" role="tab">
37          <i class="bi bi-folder2-open me-1"></i> Коллекции
    Workspace
38        </button>
39      </li>
40      <li class="nav-item" role="presentation">
41        <button class="nav-link rounded-pill btn-sm px-3 py-
    1" id="pills-doc-rag-tab" data-bs-toggle="pill" data-bs-target
    ="#pills-doc-rag" type="button" role="tab">
42          <i class="bi bi-file-earmark-text me-1"></i> Докум
    енты
43        </button>
44      </li>
45      <li class="nav-item" role="presentation">
46        <button class="nav-link rounded-pill btn-sm px-3 py-
    1" id="pills-code-rag-tab" data-bs-toggle="pill" data-bs-targe
    t="#pills-code-rag" type="button" role="tab">
47          <i class="bi bi-code-slash me-1"></i> Код проектов
    (AST)
48        </button>
49      </li>
50      <li class="nav-item" role="presentation">
51        <button class="nav-link rounded-pill btn-sm px-3 py-
    1" id="pills-pixel-rag-tab" data-bs-toggle="pill" data-bs-targ
    et="#pills-pixel-rag" type="button" role="tab">
52          <i class="bi bi-image me-1 text-warning"></i> Pixe
    lRAG (Визуальный поиск)

```

```

53 |         </button>
54 |     </li>
55 | </ul>
56 |     <button class="btn btn-outline-primary btn-sm rounded-pi
57 | ll ms-2" id="btn-refresh-rag">
58 |         <i class="bi bi-arrow-clockwise me-1"></i> <span data-
59 | i18n="common.refresh">Обновить</span>
60 |     </button>
61 | </div>
62 | </div>
63 |
64 | <div class="tab-content" id="rag-tab-content">
65 |     <!-- SUB-TAB 0: USER WORKSPACE MULTI-RAG -->
66 |     <div class="tab-pane fade show active" id="pills-user-rags
67 | " role="tabpanel">
68 |         <div class="row g-3">
69 |             <!-- LEFT: Collections List & Create -->
70 |             <div class="col-lg-4 col-md-12">
71 |                 <div class="card shadow-sm border-0 mb-3">
72 |                     <div class="card-header bg-dark text-white d-flex
73 | justify-content-between align-items-center">
74 |                         <span class="fw-semibold"><i class="bi bi-folder
75 | 2-open me-2"></i>Базы знаний (RAG)</span>
76 |                         <button class="btn btn-primary btn-sm rounded-pi
77 | ll px-2 py-0" id="btn-show-create-rag-modal" data-bs-toggle="m
78 | odal" data-bs-target="#modal-create-user-rag">
79 |                             <i class="bi bi-plus-lg me-1"></i> Создать
80 |                         </button>
81 |                     </div>
82 |                     <div class="card-body p-2">
83 |                         <div id="user-rags-list-container" class="d-flex
84 | flex-column gap-2" style="max-height: 480px; overflow-y: auto
85 | ;">
86 |                             <div class="text-center text-muted py-4 small"
87 | >
88 |                                 <i class="bi bi-folder-x fs-3 d-block mb-1 o
89 | pacity-50"></i>

```

```

80 |             <span>Нет созданных коллекций.</span><br>
81 |             <span>Нажмите «+ Создать» выше.</span>
82 |         </div>
83 |     </div>
84 | </div>
85 | </div>
86 | </div>
87 |
88 |     <!-- RIGHT: Selected Collection Details, File Pipeline
    | , & Search -->
89 |     <div class="col-lg-8 col-md-12">
90 |         <div id="user-rag-empty-state" class="card shadow-sm
    | border-0 p-5 text-center text-muted">
91 |             <i class="bi bi-diagram-3 fs-1 mb-2 text-primary o
    | pacity-50"></i>
92 |             <h5>Выберите RAG-коллекцию</h5>
93 |             <p class="small text-secondary mb-3">Выберите суще
    | ствующую базу знаний слева или создайте новую для подготовки и
    | индексации документов.</p>
94 |             <div>
95 |                 <button class="btn btn-outline-primary btn-sm ro
    | unded-pill px-3" data-bs-toggle="modal" data-bs-target="#modal
    | -create-user-rag">
96 |                     <i class="bi bi-plus-lg me-1"></i> Создать пер
    | вую коллекцию
97 |                 </button>
98 |             </div>
99 |         </div>
100 |
101 |     <div id="user-rag-detail-container" class="d-none">
102 |         <!-- Collection Header Card -->
103 |         <div class="card shadow-sm border-0 mb-3">
104 |             <div class="card-header bg-body-tertiary d-flex
    | justify-content-between align-items-center flex-wrap gap-2 py-
    | 2">
105 |                 <div>
106 |                     <h6 class="mb-0 fw-bold d-flex align-items-c
    | enter gap-2">

```



```

107         <i class="bi bi-folder2 text-primary"></i>
108         <span id="active-rag-title">Название колле
109 кции</span>
110         <span id="active-rag-status-badge" class="
111 badge bg-secondary">created</span>
112     </h6>
113     <small class="text-muted" id="active-rag-des
114 с">Описание коллекции</small>
115 </div>
116     <div class="d-flex align-items-center gap-2">
117         <button class="btn btn-outline-danger btn-sm
118 rounded-pill" id="btn-delete-active-rag" title="Удалить колле
119 кцию">
120             <i class="bi bi-trash"></i>
121         </button>
122     </div>
123 </div>
124 <div class="card-body p-3">
125     <!-- Stats Row -->
126     <div class="row g-2 mb-3 text-center">
127         <div class="col-sm-4">
128             <div class="p-2 border rounded bg-body-sec
129 ondary bg-opacity-25">
130                 <div class="small text-muted">Чанков в и
131 ндексе</div>
132                 <div class="fw-bold fs-5 text-success" i
133 d="active-rag-stat-chunks">0</div>

```

```

134 |             <div class="col-sm-4">
135 |                 <div class="p-2 border rounded bg-body-sec
ondary bg-opacity-25">
136 |                     <div class="small text-muted">Обновлено<
/di>
137 |                         <div class="small fw-bold mt-1 text-trun
cate" id="active-rag-stat-updated">--</div>
138 |                     </div>
139 |                 </div>
140 |             </div>
141 |
142 |             <!-- Pipeline & Files Section -->
143 |             <div class="card border border-secondary-subtl
e mb-3">
144 |                 <div class="card-header bg-light d-flex just
ify-content-between align-items-center py-1 px-3">
145 |                     <span class="small fw-bold"><i class="bi b
i-file-earmark-check me-1"></i> Файлы для очистки и сборки (ra
g_cleaner)</span>
146 |                     <button class="btn btn-link btn-sm text-de
coration-none p-0" id="btn-select-all-user-files">Выбрать все<
/button>
147 |                 </div>
148 |                 <div class="card-body p-2">
149 |                     <div id="user-rag-files-checklist" class="
d-flex flex-column gap-1" style="max-height: 140px; overflow-y
: auto;">
150 |                         <div class="text-muted small p-2">Загруз
ка файлов...</div>
151 |                     </div>
152 |                     <div class="d-flex justify-content-between
align-items-center mt-2 pt-2 border-top flex-wrap gap-2">
153 |                         <div class="d-flex align-items-center ga
p-2 flex-wrap">
154 |                             <input type="file" id="user-rag-quick-
upload-input" class="d-none" multiple>
155 |                             <button class="btn btn-outline-seconda
ry btn-sm rounded-pill" type="button" onclick="document.getEle

```

```

mentById('user-rag-quick-upload-input').click();">
156 |             <i class="bi bi-upload me-1"></i> За
    | грузить файлы
157 |         </button>
158 |         <button class="btn btn-outline-info bt
    | n-sm rounded-pill d-flex align-items-center gap-1" id="btn-imp
    | ort-gdocs-rag" type="button" data-bs-toggle="modal" data-bs-ta
    | rget="#modal-import-gdocs">
159 |             <i class="bi bi-google text-danger">
    | </i>
160 |             <span>Google Docs</span>
161 |         </button>
162 |     </div>
163 |     <button class="btn btn-success btn-sm ro
    | unded-pill px-3 fw-semibold shadow-sm" id="btn-build-active-ra
    | g">
164 |         <i class="bi bi-magic me-1"></i> Очист
    | ить и собрать RAG
165 |     </button>
166 | </div>
167 | <div id="user-rag-build-status-alert" clas
    | s="d-none alert p-2 small mt-2 mb-0"></div>
168 | </div>
169 | </div>
170 |
171 | <!-- Q&A and Knowledge Entries Section -->
172 | <div class="card border border-secondary-subtl
    | e mb-3">
173 |     <div class="card-header bg-light d-flex just
    | ify-content-between align-items-center py-1 px-3">
174 |         <span class="small fw-bold"><i class="bi b
    | i-chat-quote me-1 text-primary"></i> Записи и связки «Вопрос-О
    | твет»</span>
175 |         <div class="d-flex align-items-center gap-
    | 2">
176 |             <button class="btn btn-outline-secondary
    | btn-sm py-0 px-2 rounded-pill" id="btn-refresh-rag-entries" t
    | itle="Обновить записи">

```

```

177         <i class="bi bi-arrow-clockwise"></i>
178     </button>
179     <button class="btn btn-primary btn-sm rounded-pill px-2 py-0 fw-semibold" id="btn-show-add-qa-modal" data-bs-toggle="modal" data-bs-target="#modal-add-qa-entry">
180         <i class="bi bi-plus-lg me-1"></i> Добавить Q&A
181     </button>
182 </div>
183 </div>
184 <div class="card-body p-2">
185     <div class="input-group input-group-sm mb-2">
186         <input type="text" class="form-control" id="user-rag-entries-filter" placeholder="Фильтр записей в этой коллекции...">
187         <button class="btn btn-outline-secondary" type="button" id="btn-clear-entries-filter">
188             <i class="bi bi-x-lg"></i>
189         </button>
190     </div>
191     <div id="user-rag-entries-container" class="d-flex flex-column gap-2" style="max-height: 250px; overflow-y: auto;">
192         <div class="text-center text-muted small py-3">Загрузка записей базы знаний...</div>
193     </div>
194 </div>
195 </div>
196
197 <!-- Search Inside Active Collection -->
198 <div class="card border border-secondary-subtle">
199     <div class="card-header bg-light py-1 px-3">
200         <span class="small fw-bold"><i class="bi bi-search me-1"></i> Тестовый поиск по этой коллекции</span>
201     </div>
202     <div class="card-body p-2">

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```

203         <div class="input-group input-group-sm mb-
204         2">
205             <input type="text" class="form-control"
206             id="active-rag-search-query" placeholder="Введите поисковый за
прос по коллекции...">
207             <button class="btn btn-primary px-3" id=
208             "btn-active-rag-search">
209                 <i class="bi bi-search me-1"></i> Поис
210                 к
211             </button>
212         </div>
213         <div id="active-rag-search-results" class=
214         "border rounded p-2 bg-body-tertiary" style="min-height: 100px
; max-height: 200px; overflow-y: auto;">
215             <div class="text-center text-muted small
216             py-3">Введите запрос для поиска по выбранной базе знаний.</di
217             v>
218         </div>
219     </div>
220 </div>
221
222     <!-- Create User RAG Modal -->
223     <div class="modal fade" id="modal-create-user-rag" tabinde
x="-1" aria-labelledby="modalCreateUserRAGLabel" aria-hidden="
true">
224         <div class="modal-dialog modal-dialog-centered">
225             <div class="modal-content shadow">
226                 <div class="modal-header">
227                     <h6 class="modal-title fw-bold" id="modalCreateUse
rRAGLabel"><i class="bi bi-folder-plus text-primary me-2"></i>
Создание новой базы знаний (RAG)</h6>

```

```
228         <button type="button" class="btn-close" data-bs-di  
| smiss="modal" aria-label="Close"></button>  
229     </div>  
230     <div class="modal-body">  
231         <div class="mb-3">  
232             <label for="new-rag-name" class="form-label smal  
| l fw-bold">Имя коллекции:</label>  
233             <input type="text" class="form-control form-cont  
| rol-sm" id="new-rag-name" placeholder="например: movies, codin  
| g, gdocs_knowledge">  
234             <div class="form-text small">Уникальное имя для  
| группировки знаний.</div>  
235         </div>  
236         <div class="mb-3">  
237             <label for="new-rag-desc" class="form-label smal  
| l fw-bold">Описание (опционально):</label>  
238             <textarea class="form-control form-control-sm" i  
| d="new-rag-desc" rows="2" placeholder="Кратко опишите назначен  
| ие этой базы знаний..."></textarea>  
239         </div>  
240         <div class="mb-3">  
241             <label class="form-label small fw-bold">Источник  
| документов:</label>  
242             <div class="form-check">  
243                 <input class="form-check-input" type="radio" n  
| ame="new-rag-source-type" id="rag-src-local" value="local" che  
| cked>  
244                 <label class="form-check-label small" for="rag  
| -src-local">  
245                     <i class="bi bi-hdd me-1 text-primary"></i>Л  
| окальные файлы / Личное хранилище  
246                 </label>  
247             </div>  
248             <div class="form-check">  
249                 <input class="form-check-input" type="radio" n  
| ame="new-rag-source-type" id="rag-src-gdocs" value="gdocs">  
250                 <label class="form-check-label small" for="rag  
| -src-gdocs">
```

```

251 |             <i class="bi bi-google me-1 text-danger"></i
    | >Документы Google Docs / Google Drive
252 |             </label>
253 |         </div>
254 |     </div>
255 |     <div class="row g-2">
256 |         <div class="col-6">
257 |             <label for="new-rag-min-chunk" class="form-label small fw-bold">Мин. длина чанка:</label>
    | <input type="number" class="form-control form-control-sm" id="new-rag-min-chunk" value="20" min="5" max="500"
    | ">
259 |         </div>
260 |         <div class="col-6">
261 |             <label for="new-rag-max-chunk" class="form-label small fw-bold">Макс. длина чанка:</label>
    | <input type="number" class="form-control form-control-sm" id="new-rag-max-chunk" value="1500" min="100" max=
    | "5000">
263 |         </div>
264 |     </div>
265 | </div>
266 | <div class="modal-footer py-2">
267 |     <button type="button" class="btn btn-secondary btn-sm" data-bs-dismiss="modal">Отмена</button>
    | <button type="button" class="btn btn-primary btn-sm fw-semibold" id="btn-confirm-create-rag">Создать</button>
268 | </div>
269 | </div>
270 | </div>
271 | </div>
272 | </div>
273 |
274 | <!-- Import Google Docs Modal -->
275 | <div class="modal fade" id="modal-import-gdocs" tabindex="-1" aria-labelledby="modalImportGDocsLabel" aria-hidden="true"
    | >
276 |     <div class="modal-dialog modal-dialog-centered modal-lg"
    | >

```

```
277 |         <div class="modal-content shadow">
278 |             <div class="modal-header">
279 |                 <h6 class="modal-title fw-bold" id="modalImportGDo
csLabel"><i class="bi bi-google text-danger me-2"></i>Импорт д
окументов из Google Docs & Drive в RAG</h6>
280 |                 <button type="button" class="btn-close" data-bs-di
smis="modal" aria-label="Close"></button>
281 |             </div>
282 |             <div class="modal-body">
283 |                 <div class="alert alert-info py-2 small mb-3">
284 |                     <i class="bi bi-info-circle me-1"></i> Документы
Google Docs автоматически экспортируются, очищаются через <co
de>rag_cleaner</code> и добавляются в векторный индекс коллекц
ии.
285 |                 </div>
286 |                 <div class="row g-2 mb-3">
287 |                     <div class="col-md-6">
288 |                         <label for="import-gdocs-query" class="form-la
bel small fw-bold">Фильтр поиска Google Drive:</label>
289 |                         <input type="text" class="form-control form-co
ntrol-sm font-monospace" id="import-gdocs-query" value="mimeTy
pe = 'application/vnd.google-apps.document' and trashed = fals
e" placeholder="mimeType = '...' or name contains '...'">
290 |                     </div>
291 |                     <div class="col-md-6">
292 |                         <label for="import-gdocs-account" class="form-
label small fw-bold">Google аккаунт (из пула):</label>
293 |                         <input type="text" class="form-control form-co
ntrol-sm" id="import-gdocs-account" placeholder="default (или
имя аккаунта)">
294 |                     </div>
295 |                 </div>
296 |                 <div id="import-gdocs-status-alert" class="alert d
-none py-2 small mb-0"></div>
297 |             </div>
298 |             <div class="modal-footer py-2">
299 |                 <button type="button" class="btn btn-secondary btn
-sm" data-bs-dismiss="modal">Отмена</button>
```



```

300         <button type="button" class="btn btn-primary btn-s
m fw-semibold" id="btn-confirm-import-gdocs">
301             <i class="bi bi-cloud-arrow-down me-1"></i> Синх
ронизировать и собрать RAG
302         </button>
303     </div>
304 </div>
305 </div>
306 </div>
307
308 <!-- Add QA Entry Modal -->
309 <div class="modal fade" id="modal-add-qa-entry" tabindex="
-1" aria-labelledby="modalAddQAEntryLabel" aria-hidden="true">
310     <div class="modal-dialog modal-dialog-centered modal-lg"
>
311         <div class="modal-content shadow">
312             <div class="modal-header">
313                 <h6 class="modal-title fw-bold" id="modalAddQAEntr
yLabel"><i class="bi bi-patch-question text-primary me-2"></i>
Добавление знания / Связки «Вопрос-Ответ»</h6>
314                 <button type="button" class="btn-close" data-bs-di
smis="modal" aria-label="Close"></button>
315             </div>
316             <div class="modal-body">
317                 <div class="mb-3">
318                     <label for="new-qa-question" class="form-label s
mall fw-bold">Вопрос пользователя / Триггерная фраза:</label>
319                     <input type="text" class="form-control form-cont
rol-sm" id="new-qa-question" placeholder="например: Где можно
скачать сериал?">
320                     <div class="form-text small">Фраза или формулиро
вка запроса, при схожести с которой будет применяться этот отв
ет.</div>
321                 </div>
322                 <div class="mb-3">
323                     <label for="new-qa-answer" class="form-label sma
ll fw-bold">Ответ модели / Текст знания:</label>
324                     <textarea class="form-control form-control-sm fo

```

```

nt-monospace" id="new-qa-answer" rows="5" placeholder="Введите
точный ответ, инструкцию или фрагмент знаний..."></textarea>
325     </div>
326     </div>
327     <div class="modal-footer py-2">
328         <button type="button" class="btn btn-secondary btn
-sm" data-bs-dismiss="modal">Отмена</button>
329         <button type="button" class="btn btn-success btn-s
m fw-semibold" id="btn-confirm-add-qa">
330             <i class="bi bi-check2-circle me-1"></i> Сохрани
ть в RAG
331         </button>
332     </div>
333 </div>
334 </div>
335 </div>
336
337
338 <!-- SUB-TAB 1: DOCUMENT RAG -->
339 <div class="tab-pane fade" id="pills-doc-rag" role="tabpan
el">
340     <!-- Top Statistics Summary -->
341     <div class="row g-3 mb-3">
342         <div class="col-md-3 col-sm-6">
343             <div class="card shadow-sm border-0 bg-body-tertiary
">
344                 <div class="card-body p-3 d-flex align-items-cente
r justify-content-between">
345                     <div>
346                         <div class="text-muted small" data-i18n="rag.t
otalDocuments">Документов</div>
347                         <h4 class="mb-0 fw-bold" id="rag-stat-docs">0<
/h4>
348                     </div>
349                     <i class="bi bi-file-earmark-text fs-1 text-prim
ary opacity-50"></i>
350                 </div>
351             </div>

```

```

352 |         </div>
353 |         <div class="col-md-3 col-sm-6">
354 |             <div class="card shadow-sm border-0 bg-body-tertiary
|             ">
355 |                 <div class="card-body p-3 d-flex align-items-cente
|                 r justify-content-between">
356 |                     <div>
357 |                         <div class="text-muted small" data-i18n="rag.t
|                         otalChunks">Чанков в индексе</div>
358 |                         <h4 class="mb-0 fw-bold" id="rag-stat-chunks">
|                         0</h4>
359 |                     </div>
360 |                     <i class="bi bi-diagram-3 fs-1 text-success opac
|                     ity-50"></i>
361 |                 </div>
362 |             </div>
363 |         </div>
364 |         <div class="col-md-3 col-sm-6">
365 |             <div class="card shadow-sm border-0 bg-body-tertiary
|             ">
366 |                 <div class="card-body p-3 d-flex align-items-cente
|                 r justify-content-between">
367 |                     <div>
368 |                         <div class="text-muted small" data-i18n="rag.v
|                         ectorDimension">Размерность вектора</div>
369 |                         <h4 class="mb-0 fw-bold" id="rag-stat-dim">0</
|                         h4>
370 |                     </div>
371 |                     <i class="bi bi-cpu fs-1 text-info opacity-50"><
|                     /i>
372 |                 </div>
373 |             </div>
374 |         </div>
375 |         <div class="col-md-3 col-sm-6">
376 |             <div class="card shadow-sm border-0 bg-body-tertiary
|             ">
377 |                 <div class="card-body p-3 d-flex align-items-cente
|                 r justify-content-between">

```

```

378         <div>
379             <div class="text-muted small" data-i18n="rag.l
astUpdated">Последнее обновление</div>
380             <div class="small fw-bold text-truncate" id="r
ag-stat-updated" style="max-width: 140px;">—</div>
381         </div>
382         <i class="bi bi-clock-history fs-1 text-warning
opacity-50"></i>
383     </div>
384 </div>
385 </div>
386 </div>
387
388 <div class="row g-3">
389     <!-- LEFT COLUMN: File Ingestion & Index Controls -->
390     <div class="col-lg-5 col-md-12">
391         <!-- Upload Dropzone Card -->
392         <div class="card shadow-sm border-0 mb-3">
393             <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
394                 <span class="fw-semibold"><i class="bi bi-cloud-
arrow-up me-2"></i><span data-i18n="rag.uploadTitle">Загрузка
документов</span></span>
395             </div>
396             <div class="card-body">
397                 <div id="rag-dropzone" class="border border-2 bo
rder-dashed rounded p-4 text-center cursor-pointer mb-3 bg-bod
y-secondary bg-opacity-25" style="border-style: dashed !import
ant; transition: all 0.2s;">
398                     <i class="bi bi-file-earmark-arrow-up fs-1 tex
t-primary mb-2 d-block"></i>
399                     <h6 class="fw-bold mb-1" data-i18n="rag.dropzo
neText">Перетащите файлы сюда или нажмите для выбора</h6>
400                     <input type="file" id="rag-file-input" multipl
e class="d-none" accept=".txt,.md,.markdown,.json,.csv,.tsv,.p
y,.js,.ts,.html,.css,.pdf,.log,.sql,.yaml,.yml">
401                     <input type="file" id="rag-folder-input" webki
tdirectory directory multiple class="d-none">

```

```
402 |         <div class="d-flex justify-content-center gap-  
    | 2 mt-2 flex-wrap">  
403 |             <button class="btn btn-outline-primary btn-s  
    | m px-3 rounded-pill" type="button" onclick="document.getElemen  
    | tById('rag-file-input').click();">  
404 |                 <i class="bi bi-file-earmark-plus me-1"></  
    | i> <span data-i18n="rag.browseFiles">Выбрать файлы</span>  
405 |             </button>  
406 |             <button class="btn btn-primary btn-sm px-3 r  
    | ounded-pill" type="button" onclick="document.getElementById('r  
    | ag-folder-input').click();">  
407 |                 <i class="bi bi-folder-plus me-1"></i> <sp  
    | an data-i18n="rag.browseFolder">Выбрать папку целиком</span>  
408 |             </button>  
409 |         </div>  
410 |     </div>  
411 |  
412 |     <div id="rag-upload-progress-container" class="d  
    | -none mb-3">  
413 |         <div class="d-flex justify-content-between sma  
    | ll text-muted mb-1">  
414 |             <span data-i18n="rag.uploading">Загрузка фай  
    | лов...</span>  
415 |             <span id="rag-upload-percent">0%</span>  
416 |         </div>  
417 |         <div class="progress" style="height: 6px;">  
418 |             <div id="rag-upload-progress-bar" class="pro  
    | gress-bar progress-bar-striped progress-bar-animated bg-primar  
    | y" role="progressbar" style="width: 0%"></div>  
419 |         </div>  
420 |     </div>  
421 |  
422 |     <div id="rag-upload-status-alert" class="d-none  
    | alert p-2 small mb-0"></div>  
423 | </div>  
424 | </div>  
425 |  
426 | <!-- Index Settings & Build Card -->
```

```
427 |         <div class="card shadow-sm border-0">
428 |             <div class="card-header bg-dark text-white d-flex
| justify-content-between align-items-center">
429 |                 <span class="fw-semibold"><i class="bi bi-slider
| s me-2"></i><span data-i18n="rag.indexSettings">Параметры инде
| ксации</span></span>
430 |             </div>
431 |             <div class="card-body">
432 |                 <div class="mb-3">
433 |                     <label class="form-label small fw-bold" for="r
| ag-provider-select" data-i18n="rag.providerLabel">Провайдер ве
| кторизации:</label>
434 |                     <select class="form-select form-select-sm" id=
| "rag-provider-select">
435 |                         <option value="auto">Auto (Gemini если досту
| пен ключ, иначе Local TF-IDF)</option>
436 |                         <option value="gemini">Google Gemini Embeddi
| ngs (gemini-embedding-2)</option>
437 |                         <option value="local_tfidf">Local TF-IDF Vec
| torizer (Полностью офлайн)</option>
438 |                     </select>
439 |                 </div>
440 |
441 |                 <div class="mb-3" id="rag-api-key-group">
442 |                     <label class="form-label small fw-bold" for="r
| ag-api-key-input">Gemini API Key (опционально, берется из .env
| ):</label>
443 |                     <input type="password" class="form-control for
| m-control-sm" id="rag-api-key-input" placeholder="AIzaSy... (и
| ли оставить пустым для авто)" autocomplete="new-password" data
| -lignore="true" data-1p-ignore="true">
444 |                     </div>
445 |
446 |                 <div class="row g-2 mb-3">
447 |                     <div class="col-6">
448 |                         <label class="form-label small fw-bold" for=
| "rag-chunk-size" data-i18n="rag.chunkSize">Размер чанка (симво
| лов):</label>
```

```

449         <input type="number" class="form-control form-control-sm" id="rag-chunk-size" value="500" min="100" max="4000" step="50">
450     </div>
451     <div class="col-6">
452         <label class="form-label small fw-bold" for="rag-chunk-overlap" data-i18n="rag.chunkOverlap">Перекрывание (Overlap):</label>
453         <input type="number" class="form-control form-control-sm" id="rag-chunk-overlap" value="50" min="0" max="1000" step="10">
454     </div>
455 </div>
456
457     <div class="d-grid">
458         <button class="btn btn-success btn-sm fw-semibold py-2 shadow-sm" id="btn-build-rag-index">
459             <i class="bi bi-hammer me-1"></i> <span data-i18n="rag.buildIndexBtn">□ Построить / Перестроить индекс</span>
460         </button>
461     </div>
462     <div id="rag-build-status" class="small mt-2 text-center text-muted"></div>
463 </div>
464 </div>
465 </div>
466
467 <!-- RIGHT COLUMN: Documents List & Semantic Search Playground -->
468 <div class="col-lg-7 col-md-12">
469     <!-- Documents Table Card -->
470     <div class="card shadow-sm border-0 mb-3">
471         <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
472             <span class="fw-semibold"><i class="bi bi-files me-2"></i><span data-i18n="rag.documentsTitle">Список документов в базе</span></span>

```

```

473 |         <span class="badge bg-primary" id="rag-docs-badg
| e">0 файлов</span>
474 |     </div>
475 |     <div class="card-body p-0">
476 |         <div class="table-responsive" style="max-height:
| 260px; overflow-y: auto;">
477 |             <table class="table table-hover table-sm align
| -middle mb-0" id="rag-documents-table">
478 |                 <thead class="table-light sticky-top">
479 |                     <tr>
480 |                         <th class="ps-3" data-i18n="rag.fileName
| ">Имя файла</th>
481 |                         <th data-i18n="rag.fileSize">Размер</th>
482 |                         <th data-i18n="rag.chunks">Чанки</th>
483 |                         <th data-i18n="rag.status">Статус</th>
484 |                         <th class="text-end pe-3" data-i18n="com
| mon.actions">Действия</th>
485 |                     </tr>
486 |                 </thead>
487 |                 <tbody id="rag-documents-tbody">
488 |                     <tr>
489 |                         <td colspan="5" class="text-center text-
| muted py-3" data-i18n="rag.noDocuments">
490 |                             Документы не загружены. Загрузите файл
| ы в блоке слева.
491 |                         </td>
492 |                     </tr>
493 |                 </tbody>
494 |             </table>
495 |         </div>
496 |     </div>
497 | </div>
498 |
499 | <!-- Semantic Search Playground Card -->
500 | <div class="card shadow-sm border-0">
501 |     <div class="card-header bg-dark text-white d-flex
| justify-content-between align-items-center">
502 |         <span class="fw-semibold"><i class="bi bi-search

```



```

me-2"></i><span data-i18n="rag.searchPlaygroundTitle">Семанти
ческий поиск по базе</span></span>
503         </div>
504         <div class="card-body">
505             <div class="d-flex align-items-center justify-co
content-between flex-wrap gap-2 mb-2 p-2 bg-body-tertiary rounde
d border small text-muted">
506                 <div class="d-flex align-items-center gap-3">
507                     <div class="d-flex align-items-center gap-1"
>
508                         <label for="rag-top-k" class="text-nowrap
fw-semibold">Top K:</label>
509                         <select class="form-select form-select-sm"
id="rag-top-k" style="width: 75px;">
510                             <option value="3">3</option>
511                             <option value="5" selected>5</option>
512                             <option value="10">10</option>
513                             <option value="20">20</option>
514                         </select>
515                     </div>
516                     <div class="d-flex align-items-center gap-1"
>
517                         <label for="rag-min-score" class="text-now
rap fw-semibold">Min Score:</label>
518                         <input type="number" class="form-control f
orm-control-sm" id="rag-min-score" value="0.0" min="0.0" max="
1.0" step="0.05" style="width: 80px;">
519                     </div>
520                 </div>
521                 <span class="small text-secondary"><i class="b
i bi-info-circle me-1"></i>Фильтры выдачи</span>
522             </div>
523
524             <div class="input-group mb-3">
525                 <input type="text" class="form-control" id="ra
g-search-query" placeholder="Введите поисковый запрос по смысл
у..." data-i18n-placeholder="rag.searchPlaceholder">
526                 <button class="btn btn-primary px-3" id="btn-r

```

```

ag-search">
527         <i class="bi bi-search me-1"></i> <span data
| -i18n="common.search">Поиск</span>
528         </button>
529     </div>
530
531     <div id="rag-search-results" class="border round
| ed p-2 bg-body-tertiary" style="min-height: 180px; max-height:
| 320px; overflow-y: auto;">
532         <div class="text-center text-muted py-4">
533             <i class="bi bi-chat-left-quote fs-3 d-block
| mb-1 opacity-50"></i>
534             <span data-i18n="rag.enterQueryPrompt">Введи
| те запрос и нажмите «Поиск» для проверки RAG.</span>
535         </div>
536     </div>
537 </div>
538 </div>
539 </div>
540 </div>
541 </div>
542
543 <!-- SUB-TAB 2: CODEBASE RAG (AST + Multi-Project) -->
544 <div class="tab-pane fade" id="pills-code-rag" role="tabpa
| nel">
545     <!-- Active Project & Index Selector Bar -->
546     <div class="card shadow-sm border-0 mb-3 bg-body-tertiar
| y">
547         <div class="card-body p-3 d-flex align-items-center ju
| stify-content-between flex-wrap gap-2">
548             <div class="d-flex align-items-center gap-2 flex-gro
| w-1 min-w-0">
549                 <label class="form-label small fw-bold mb-0 text-n
| owrap" for="codebase-active-index-select">
550                     <i class="bi bi-diagram-2 text-primary me-1"></i>
| > АКТИВНЫЙ RAG индекс:
551                 </label>
552                 <select class="form-select form-select-sm" id="cod

```

```

    ebase-active-index-select" style="max-width: 320px;">
553         <option value="codebase">codebase (По умолчанию)
    </option>
554     </select>
555     <button class="btn btn-sm btn-outline-secondary rounded-pill" id="btn-refresh-codebase-indexes" title="Обновить
    список индексов">
556         <i class="bi bi-arrow-clockwise"></i>
557     </button>
558 </div>
559     <div class="d-flex align-items-center gap-2" id="codebase-index-quick-stats">
560         <span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="codebase-stat-chunks">0 чанков</span>
561         <span class="badge bg-success-subtle text-success border border-success-subtle" id="codebase-stat-symbols">0 символов AST</span>
562     </div>
563 </div>
564 </div>
565
566 <div class="row g-3">
567     <!-- LEFT COLUMN: Codebase Index Settings & Trigger -->
568     <div class="col-lg-5 col-md-12">
569         <div class="card shadow-sm border-0">
570             <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
571                 <span class="fw-semibold"><i class="bi bi-folder-2-open me-2"></i>Сборка индекса проекта</span>
572             </div>
573             <div class="card-body">
574                 <div class="mb-3">
575                     <label class="form-label small fw-bold" for="codebase-project-root-input">
576                         Директория проекта (Project Root):
577                     </label>

```

```
578 |         <div class="input-group input-group-sm">
579 |             <span class="input-group-text"><i class="bi
| bi-hdd"></i></span>
580 |             <input type="text" class="form-control" id="
| codebase-project-root-input" value="." placeholder="Например:
| . или C:/Projects/my_app">
581 |         </div>
582 |         <div class="form-text small text-muted">Укажит
| е путь к проекту на диске (абсолютный или относительный).</div
| >
583 |     </div>
584 |
585 |     <div class="mb-3">
586 |         <label class="form-label small fw-bold" for="c
| odebase-index-name-input">
587 |             Имя RAG индекса (Index Name):
588 |         </label>
589 |         <input type="text" class="form-control form-co
| ntrol-sm" id="codebase-index-name-input" value="codebase" plac
| eholder="Например: my_project_v1">
590 |         <div class="form-text small text-muted">Индекс
| ы сохраняются изолированно в вашем персональном хранилище.</di
| v>
591 |     </div>
592 |
593 |     <div class="mb-3">
594 |         <label class="form-label small fw-bold" for="c
| odebase-include-dirs-input">
595 |             Подкаталоги для анализа (через запятую):
596 |         </label>
597 |         <input type="text" class="form-control form-co
| ntrol-sm font-monospace" id="codebase-include-dirs-input" valu
| e="src, plugins, integrations, launchers, scripts, install, do
| cs, prompts">
598 |     </div>
599 |
600 |     <div class="d-grid">
601 |         <button class="btn btn-primary btn-sm fw-semib
```

```

old py-2 shadow-sm d-flex align-items-center justify-content-c
enter gap-2" id="btn-build-codebase-rag">
602         <i class="bi bi-cpu"></i>
603         <span>☐ Собрать Codebase RAG</span>
604     </button>
605 </div>
606
607     <div id="codebase-build-status-alert" class="d-n
one alert p-2 small mt-3 mb-0"></div>
608 </div>
609 </div>
610 </div>
611
612 <!-- RIGHT COLUMN: Code & Symbol Search Playground -->
613 <div class="col-lg-7 col-md-12">
614     <div class="card shadow-sm border-0">
615         <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
616             <ul class="nav nav-tabs card-header-tabs border-
0" id="codebase-search-tabs" role="tablist">
617                 <li class="nav-item" role="presentation">
618                     <button class="nav-link active text-white" i
d="tab-code-semantic-btn" data-bs-toggle="tab" data-bs-target=
"#tab-code-semantic" type="button" role="tab">
619                         <i class="bi bi-search me-1"></i> Семантич
еский поиск по коду
620                     </button>
621                 </li>
622                 <li class="nav-item" role="presentation">
623                     <button class="nav-link text-white-50" id="t
ab-code-symbols-btn" data-bs-toggle="tab" data-bs-target="#tab
-code-symbols" type="button" role="tab">
624                         <i class="bi bi-compass me-1"></i> AST Сим
волы
625                     </button>
626                 </li>
627             </ul>
628         </div>

```

```

629         <div class="card-body">
630             <div class="tab-content" id="codebase-search-tab
        s-content">
631
632                 <!-- Code Semantic Search -->
633                 <div class="tab-pane fade show active" id="tab
        -code-semantic" role="tabpanel">
634                     <div class="input-group mb-3">
635                         <input type="text" class="form-control" id
        ="codebase-search-query" placeholder="Например: Как реализован
        роутинг провайдеров?">
636                         <button class="btn btn-primary px-3" id="b
        tn-codebase-search">
637                             <i class="bi bi-search me-1"></i> Поиск
638                         </button>
639                     </div>
640                     <div id="codebase-search-results" class="bor
        der rounded p-2 bg-body-tertiary" style="min-height: 220px; ma
        x-height: 360px; overflow-y: auto;">
641                         <div class="text-center text-muted py-4">
642                             <i class="bi bi-code-square fs-3 d-block
        mb-1 opacity-50"></i>
643                             <span>Введите вопрос или описание логики
        для поиска по коду.</span>
644                         </div>
645                     </div>
646                 </div>
647
648                 <!-- Symbol Lookup -->
649                 <div class="tab-pane fade" id="tab-code-symbol
        s" role="tabpanel">
650                     <div class="input-group mb-3">
651                         <input type="text" class="form-control" id
        ="codebase-symbol-query" placeholder="Имя класса или метода: н
        апример MediaSearchAgent, search, BasePlugin">
652                         <button class="btn btn-success px-3" id="b
        tn-codebase-symbol-search">
653                             <i class="bi bi-compass me-1"></i> Найти

```

СИМВОЛ

```
654         </button>
655     </div>
656     <div id="codebase-symbol-results" class="border rounded p-2 bg-body-tertiary" style="min-height: 220px; max-height: 360px; overflow-y: auto;">
657         <div class="text-center text-muted py-4">
658             <i class="bi bi-diagram-3 fs-3 d-block mb-1 opacity-50"></i>
659             <span>Введите имя класса, метода или функции для мгновенного AST-поиска.</span>
660         </div>
661     </div>
662 </div>
663
664 </div>
665 </div>
666 </div>
667 </div>
668
669 </div>
670 </div>
671
672 <!-- SUB-TAB 3: PIXELRAG VISUAL HYBRID SEARCH -->
673 <div class="tab-pane fade" id="pills-pixel-rag" role="tabpanel">
674     <!-- Top Metrics Bar -->
675     <div class="row g-3 mb-3">
676         <div class="col-md-3 col-sm-6">
677             <div class="card shadow-sm border-0 bg-body-secondary bg-opacity-25 h-100 p-3 text-center">
678                 <div class="small text-muted mb-1"><i class="bi bi-images text-primary me-1"></i>Всего изображений</div>
679                 <div class="fw-bold fs-4 text-primary" id="pixel-stat-images">0</div>
680                 <div class="small text-muted" id="pixel-stat-size">0.0 MB</div>
681             </div>
```

```
682 |         </div>
683 |         <div class="col-md-3 col-sm-6">
684 |             <div class="card shadow-sm border-0 bg-body-secondar
        y bg-opacity-25 h-100 p-3 text-center">
685 |                 <div class="small text-muted mb-1"><i class="bi bi
        -layers text-success me-1"></i>Чанков в индексе</div>
686 |                 <div class="fw-bold fs-4 text-success" id="pixel-s
        tat-chunks">0</div>
687 |                 <div class="small text-muted">FAISS Hybrid Vector
        Index</div>
688 |             </div>
689 |         </div>
690 |         <div class="col-md-3 col-sm-6">
691 |             <div class="card shadow-sm border-0 bg-body-secondar
        y bg-opacity-25 h-100 p-3 text-center">
692 |                 <div class="small text-muted mb-1"><i class="bi bi
        -robot text-info me-1"></i>Провайдер векторизации</div>
693 |                 <div class="fw-bold fs-6 text-info mt-1" id="pixel
        -stat-provider">FAISS / Visual Embeddings</div>
694 |                 <div class="small text-muted" id="pixel-stat-stat
        us">Статус: Загрузка...</div>
695 |             </div>
696 |         </div>
697 |         <div class="col-md-3 col-sm-6">
698 |             <div class="card shadow-sm border-0 bg-body-secondar
        y bg-opacity-25 h-100 p-3 text-center">
699 |                 <div class="small text-muted mb-1"><i class="bi bi
        -clock-history text-warning me-1"></i>Индекс обновлен</div>
700 |                 <div class="small fw-bold mt-2 text-truncate" id="
        pixel-stat-updated">—</div>
701 |                 <button class="btn btn-link btn-sm text-decoration
        -none p-0 mt-1" id="btn-refresh-pixel-stats">
702 |                     <i class="bi bi-arrow-clockwise"></i> Обновить
703 |                 </button>
704 |             </div>
705 |         </div>
706 |     </div>
707 |
```



```

708 |     <div class="row g-3">
709 |         <!-- LEFT COLUMN: Image Ingestion & Smart Query Router
710 |         -->
711 |         <div class="col-lg-5 col-md-12">
712 |             <!-- Ingest Images Card -->
713 |             <div class="card shadow-sm border-0 mb-3">
714 |                 <div class="card-header bg-dark text-white d-flex
715 | justify-content-between align-items-center">
716 |                     <span class="fw-semibold"><i class="bi bi-cloud-
717 | arrow-up me-2"></i>Загрузка и индексация изображений</span>
718 |                 </div>
719 |                 <div class="card-body">
720 |                     <div id="pixel-dropzone" class="border border-2
721 | border-dashed rounded p-4 text-center cursor-pointer mb-3 bg-b
722 | ody-secondary bg-opacity-25" style="border-style: dashed !impo
723 | rtant; transition: all 0.2s; cursor: pointer;">
724 |                         <i class="bi bi-file-earmark-image fs-1 text-p
725 | rimary mb-2 d-block"></i>
726 |                         <h6 class="fw-bold mb-1">Перетащите изображени
727 | я сюда или нажмите для выбора</h6>
728 |                         <div class="small text-muted mb-3">Поддерживаю
729 | тся .png, .jpg, .jpeg, .webp, .bmp, .gif, .pdf</div>
730 |                         <input type="file" id="pixel-file-input" multi
731 | ple class="d-none" accept="image/*,.pdf">
732 |                         <button class="btn btn-outline-primary btn-sm
733 | px-3 rounded-pill" type="button" onclick="document.getElementB
734 | yId('pixel-file-input').click();">
735 |                             <i class="bi bi-image me-1"></i> Выбрать изо
736 | бражения
737 |                         </button>
738 |                     </div>
739 |                 </div>
740 |                 <div id="pixel-upload-progress" class="d-none mb
741 | -3">
742 |                     <div class="d-flex justify-content-between sma
743 | ll text-muted mb-1">
744 |                         <span>Индексация через PixelRAG...</span>
745 |                         <span id="pixel-upload-percent">0%</span>

```

```

731 |         </div>
732 |         <div class="progress" style="height: 6px;">
733 |             <div id="pixel-upload-progress-bar" class="p
progress-bar progress-bar-striped progress-bar-animated bg-succ
ess" role="progressbar" style="width: 0%"></div>
734 |         </div>
735 |     </div>
736 |
737 |     <div id="pixel-upload-status-alert" class="d-non
e alert p-2 small mb-0"></div>
738 |     </div>
739 | </div>
740 |
741 | <!-- Smart Query Router & Analyzer -->
742 | <div class="card shadow-sm border-0">
743 |     <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
744 |         <span class="fw-semibold"><i class="bi bi-signpo
st-split me-2"></i>Smart Query Router (Анализ запроса)</span>
745 |     </div>
746 |     <div class="card-body">
747 |         <p class="small text-muted mb-2">Автоматическое
определение типа запроса (Text vs Visual vs Hybrid) перед поис
ком.</p>
748 |         <div class="input-group input-group-sm mb-2">
749 |             <input type="text" class="form-control" id="pi
xel-analyze-query-input" placeholder="Например: схема авториза
ции или синий график продаж">
750 |             <button class="btn btn-outline-primary" id="bt
n-pixel-analyze-query">
751 |                 <i class="bi bi-cpu me-1"></i> Анализ
752 |             </button>
753 |         </div>
754 |         <div id="pixel-analyze-result" class="p-2 border
rounded bg-body-tertiary small d-none">
755 |             <div class="d-flex justify-content-between ali
gn-items-center mb-1">
756 |                 <span>Тип маршрутизации: <strong id="pixel-r

```

```

outing-type" class="badge bg-primary">HYBRID</strong></span>
757         <span>Уверенность: <strong id="pixel-routing
-confidence" class="text-success">95%</strong></span>
758         </div>
759         <div class="text-muted" id="pixel-routing-reas
on" style="font-size: 0.8rem;"></div>
760     </div>
761 </div>
762 </div>
763 </div>
764
765 <!-- RIGHT COLUMN: Visual Search Playground -->
766 <div class="col-lg-7 col-md-12">
767     <div class="card shadow-sm border-0">
768         <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
769             <span class="fw-semibold"><i class="bi bi-search
me-2"></i>Интерактивный поиск PixelRAG</span>
770             <div class="form-check form-switch m-0" title="И
спользовать Smart Query Routing">
771                 <input class="form-check-input" type="checkbox"
" id="pixel-search-routing-toggle" checked>
772                 <label class="form-check-label text-light smal
l ms-1" for="pixel-search-routing-toggle">Smart Routing</label
>
773             </div>
774         </div>
775         <div class="card-body">
776             <div class="row g-2 mb-3">
777                 <div class="col-md-8">
778                     <div class="input-group">
779                         <input type="text" class="form-control" id
="pixel-search-query" placeholder="Введите поисковый запрос (т
екст или описание картинки)...>
780                         <button class="btn btn-primary px-3" id="b
tn-pixel-search">
781                             <i class="bi bi-search me-1"></i> Искать
782                             </button>

```

```

783         </div>
784     </div>
785     <div class="col-md-2">
786         <select class="form-select" id="pixel-search
- topk" title="Количество результатов">
787             <option value="3">Top 3</option>
788             <option value="5" selected>Top 5</option>
789             <option value="10">Top 10</option>
790         </select>
791     </div>
792     <div class="col-md-2">
793         <select class="form-select" id="pixel-search
- minscore" title="Минимальный порог">
794             <option value="0.0" selected>Score ≥ 0.0</
option>
795             <option value="0.3">Score ≥ 0.3</option>
796             <option value="0.5">Score ≥ 0.5</option>
797             <option value="0.7">Score ≥ 0.7</option>
798         </select>
799     </div>
800 </div>
801
802     <!-- Search Results List -->
803     <div id="pixel-search-results" class="border rou
nded p-3 bg-body-tertiary" style="min-height: 280px; max-heigh
t: 480px; overflow-y: auto;">
804         <div class="text-center text-muted py-5">
805             <i class="bi bi-images fs-1 d-block mb-2 opa
city-50"></i>
806             <span>Введите запрос для поиска по текстовым
чанкам и визуальным индексам изображений.</span>
807         </div>
808     </div>
809 </div>
810 </div>
811 </div>
812 </div>
813 </div>

```

```
814 |  
815 |   </div>  
816 | </div>  
817 |  
818 |
```

main.js

```
1  /**
2   * =====
3   *
4   * =====
5   * Process Name: Windows Rag Tab - Main Script
6   * =====
7   *
8   * Description:
9   *   Клиентский скрипт управления интерфейсом модуля main.
10  *
11  * Usage Examples:
12  *   HTML Integration:
13  *     <script src="/windows/api/webgui/rag_tab/main.js?v=2026
14  *     1001_v1" type="module"></script>
15  *
16  * File: main.js
17  * Project: ai-breadboard
18  * Package: windows/api/webgui/rag_tab
19  * Author: hypo69
20  * Copyright: © 2026 hypo69
21  * Updated: 2026-10-01 13:04:40
22  * =====
23  */
24
25 import { getAllFilesFromDataTransfer } from './modules/utils.js';
26
27 import {
28   uploadFiles,
29   loadRagStatus,
30   loadRagDocuments,
31   deleteRagDocument,
32   buildRagIndex,
33   executeRagSearch
34 } from './modules/docRag.js';
35
36 import {
37   loadCodebaseIndexes,
```

```

32 |   updateCodebaseStatsDisplay,
33 |   buildCodebaseRag,
34 |   executeCodebaseSearch,
35 |   executeSymbolLookup
36 | } from './modules/codebaseRag.js';
37 | import {
38 |   loadUserRags,
39 |   createUserRag,
40 |   deleteActiveUserRag,
41 |   addQaEntry,
42 |   loadRagEntries
43 | } from './modules/userRagCollections.js';
44 |
45 | import {
46 |   buildActiveUserRag,
47 |   executeActiveUserRagSearch,
48 |   handleQuickUploadToUserStorage,
49 |   syncGoogleDocsIntoActiveRag
50 | } from './modules/userRagPipeline.js';
51 |
52 | import {
53 |   loadPixelRagStatus,
54 |   uploadAndIndexPixelImages,
55 |   analyzePixelQuery,
56 |   executePixelRagSearch
57 | } from './modules/pixelRag.js';
58 |
59 | /**
60 |  * Initialize all RAG tab components, event listeners, and data feeds.
61 |  *
62 |  * @returns {Promise<void>}
63 |  */
64 | export async function initRagTab() {
65 |   setupEventListeners();
66 |   loadUserRags();
67 |   loadRagStatus();
68 |   loadRagDocuments();

```

```

69 | loadCodebaseIndexes();
70 | loadPixelRagStatus();
71 | }
72 |
73 | // Expose globals for external tab orchestrators and inline ha
    | ndlers
74 | window.initRagTab = initRagTab;
75 | window.deleteRagDocument = deleteRagDocument;
76 |
77 | /**
78 |  * Register DOM event listeners for buttons, inputs, dropzones
    | , and keyboard events.
79 |  */
80 | function setupEventListeners() {
81 |     const root = document.getElementById('rag-tab-content');
82 |     if (root) {
83 |         if (root.dataset.listenersBound === 'true') return;
84 |         root.dataset.listenersBound = 'true';
85 |     }
86 |
87 |     const refreshBtn = document.getElementById('btn-refresh-rag'
    | );
88 |     if (refreshBtn) {
89 |         refreshBtn.addEventListener('click', () => {
90 |             loadUserRags();
91 |             loadRagStatus();
92 |             loadRagDocuments();
93 |             loadCodebaseIndexes();
94 |             loadPixelRagStatus();
95 |         });
96 |     }
97 |
98 |     // --- PixelRAG Listeners ---
99 |     const refreshPixelStatsBtn = document.getElementById('btn-re
    | fresh-pixel-stats');
100 |     if (refreshPixelStatsBtn) {
101 |         refreshPixelStatsBtn.addEventListener('click', loadPixelRa
    | gStatus);

```



```
102 | }
103 |
104 | const pixelFileInput = document.getElementById('pixel-file-i
    nput');
105 | if (pixelFileInput) {
106 |     pixelFileInput.addEventListener('change', (e) => {
107 |         if (e.target.files && e.target.files.length > 0) {
108 |             uploadAndIndexPixelImages(Array.from(e.target.files));
109 |             pixelFileInput.value = '';
110 |         }
111 |     });
112 | }
113 |
114 | const pixelDropzone = document.getElementById('pixel-dropzon
    e');
115 | if (pixelDropzone) {
116 |     ['dragenter', 'dragover'].forEach(eventName => {
117 |         pixelDropzone.addEventListener(eventName, (e) => {
118 |             e.preventDefault();
119 |             e.stopPropagation();
120 |             pixelDropzone.classList.add('border-primary', 'bg-prim
    ary', 'bg-opacity-10');
121 |         }, false);
122 |     });
123 |
124 |     ['dragleave', 'drop'].forEach(eventName => {
125 |         pixelDropzone.addEventListener(eventName, (e) => {
126 |             e.preventDefault();
127 |             e.stopPropagation();
128 |             pixelDropzone.classList.remove('border-primary', 'bg-p
    rimary', 'bg-opacity-10');
129 |         }, false);
130 |     });
131 |
132 |     pixelDropzone.addEventListener('drop', async (e) => {
133 |         const dt = e.dataTransfer;
134 |         if (dt) {
135 |             const files = await getAllFilesFromDataTransfer(dt);
```

```
136         if (files && files.length > 0) {
137             uploadAndIndexPixelImages(files);
138         }
139     }
140     }, false);
141 }
142
143 const btnPixelAnalyze = document.getElementById('btn-pixel-a
nalyze-query');
144 const inputPixelAnalyze = document.getElementById('pixel-ana
lyze-query-input');
145 if (btnPixelAnalyze) {
146     btnPixelAnalyze.addEventListener('click', analyzePixelQuer
y);
147 }
148 if (inputPixelAnalyze) {
149     inputPixelAnalyze.addEventListener('keydown', (e) => {
150         if (e.key === 'Enter') analyzePixelQuery();
151     });
152 }
153
154 const btnPixelSearch = document.getElementById('btn-pixel-se
arch');
155 const inputPixelSearch = document.getElementById('pixel-sear
ch-query');
156 if (btnPixelSearch) {
157     btnPixelSearch.addEventListener('click', executePixelRagSe
arch);
158 }
159 if (inputPixelSearch) {
160     inputPixelSearch.addEventListener('keydown', (e) => {
161         if (e.key === 'Enter') executePixelRagSearch();
162     });
163 }
164
165 // --- User Workspace RAG Listeners ---
166 const confirmCreateRagBtn = document.getElementById('btn-con
firm-create-rag');
```

```
167 |   if (confirmCreateRagBtn) {
168 |       confirmCreateRagBtn.addEventListener('click', createUserRa
    | g);
169 |   }
170 |
171 |   const buildActiveRagBtn = document.getElementById('btn-build
    | -active-rag');
172 |   if (buildActiveRagBtn) {
173 |       buildActiveRagBtn.addEventListener('click', buildActiveUse
    | rRag);
174 |   }
175 |
176 |   const deleteActiveRagBtn = document.getElementById('btn-dele
    | te-active-rag');
177 |   if (deleteActiveRagBtn) {
178 |       deleteActiveRagBtn.addEventListener('click', deleteActiveU
    | serRag);
179 |   }
180 |
181 |   const confirmAddQaBtn = document.getElementById('btn-confirm
    | -add-qa');
182 |   if (confirmAddQaBtn) {
183 |       confirmAddQaBtn.addEventListener('click', addQaEntry);
184 |   }
185 |
186 |   const refreshEntriesBtn = document.getElementById('btn-refre
    | sh-rag-entries');
187 |   if (refreshEntriesBtn) {
188 |       refreshEntriesBtn.addEventListener('click', () => loadRagE
    | ntries());
189 |   }
190 |
191 |   const entriesFilterInput = document.getElementById('user-rag
    | -entries-filter');
192 |   if (entriesFilterInput) {
193 |       let filterDebounce = null;
194 |       entriesFilterInput.addEventListener('input', (e) => {
195 |           clearTimeout(filterDebounce);
```

```
196 |         filterDebounce = setTimeout(() => {
197 |             loadRagEntries(undefined, e.target.value.trim());
198 |         }, 300);
199 |     });
200 | }
201 |
202 | const clearFilterBtn = document.getElementById('btn-clear-en
    tries-filter');
203 | if (clearFilterBtn) {
204 |     clearFilterBtn.addEventListener('click', () => {
205 |         if (entriesFilterInput) {
206 |             entriesFilterInput.value = '';
207 |             loadRagEntries(undefined, '');
208 |         }
209 |     });
210 | }
211 |
212 | const searchActiveRagBtn = document.getElementById('btn-acti
    ve-rag-search');
213 |
214 | const searchActiveRagInput = document.getElementById('active
    -rag-search-query');
215 | if (searchActiveRagBtn) {
216 |     searchActiveRagBtn.addEventListener('click', executeActive
    UserRagSearch);
217 | }
218 | if (searchActiveRagInput) {
219 |     searchActiveRagInput.addEventListener('keydown', (e) => {
220 |         if (e.key === 'Enter') executeActiveUserRagSearch();
221 |     });
222 | }
223 |
224 | const quickUploadInput = document.getElementById('user-rag-q
    uick-upload-input');
225 | if (quickUploadInput) {
226 |     quickUploadInput.addEventListener('change', handleQuickUp
    loadToUserStorage);
227 | }
```

```

228
229   const selectAllBtn = document.getElementById('btn-select-all
    -user-files');
230   if (selectAllBtn) {
231       selectAllBtn.addEventListener('click', () => {
232           const checkboxes = document.querySelectorAll('#user-rag-
    files-checklist input[type="checkbox"]');
233           const anyUnchecked = Array.from(checkboxes).some(cb => !
    cb.checked);
234           checkboxes.forEach(cb => cb.checked = anyUnchecked);
235       });
236   }
237
238   const confirmImportGdocsBtn = document.getElementById('btn-c
    onfirm-import-gdocs');
239   if (confirmImportGdocsBtn) {
240       confirmImportGdocsBtn.addEventListener('click', syncGoogle
    DocsIntoActiveRag);
241   }
242
243   // --- Global Document RAG Listeners ---
244   const fileInput = document.getElementById('rag-file-input');
245   const folderInput = document.getElementById('rag-folder-inpu
    t');
246   const dropzone = document.getElementById('rag-dropzone');
247
248   if (fileInput) {
249       fileInput.addEventListener('change', (e) => {
250           if (e.target.files && e.target.files.length > 0) {
251               uploadFiles(Array.from(e.target.files));
252               fileInput.value = '';
253           }
254       });
255   }
256
257   if (folderInput) {
258       folderInput.addEventListener('change', (e) => {
259           if (e.target.files && e.target.files.length > 0) {

```

```

260         uploadFiles(Array.from(e.target.files));
261         folderInput.value = '';
262     }
263 });
264 }
265
266 if (dropzone) {
267     ['dragenter', 'dragover'].forEach(eventName => {
268         dropzone.addEventListener(eventName, (e) => {
269             e.preventDefault();
270             e.stopPropagation();
271             dropzone.classList.add('border-primary', 'bg-primary',
272 'bg-opacity-10');
273         }, false);
274     });
275     ['dragleave', 'drop'].forEach(eventName => {
276         dropzone.addEventListener(eventName, (e) => {
277             e.preventDefault();
278             e.stopPropagation();
279             dropzone.classList.remove('border-primary', 'bg-primar
280 y', 'bg-opacity-10');
281         }, false);
282     });
283     dropzone.addEventListener('drop', async (e) => {
284         const dt = e.dataTransfer;
285         if (dt) {
286             const files = await getAllFilesFromDataTransfer(dt);
287             if (files && files.length > 0) {
288                 uploadFiles(files);
289             }
290         }
291     }, false);
292 }
293
294 const buildBtn = document.getElementById('btn-build-rag-inde
x');

```

```
295 |   if (buildBtn) {
296 |       buildBtn.addEventListener('click', buildRagIndex);
297 |   }
298 |
299 |   const searchBtn = document.getElementById('btn-rag-search');
300 |   const searchInput = document.getElementById('rag-search-quer
    | y');
301 |   if (searchBtn) {
302 |       searchBtn.addEventListener('click', executeRagSearch);
303 |   }
304 |   if (searchInput) {
305 |       searchInput.addEventListener('keydown', (e) => {
306 |           if (e.key === 'Enter') executeRagSearch();
307 |       });
308 |   }
309 |
310 |   // --- Codebase RAG Listeners ---
311 |   const btnRefreshCodebase = document.getElementById('btn-refr
    | esh-codebase-indexes');
312 |   if (btnRefreshCodebase) {
313 |       btnRefreshCodebase.addEventListener('click', loadCodebaseI
    | ndexes);
314 |   }
315 |
316 |   const btnBuildCodebase = document.getElementById('btn-build-
    | codebase-rag');
317 |   if (btnBuildCodebase) {
318 |       btnBuildCodebase.addEventListener('click', buildCodebaseRa
    | g);
319 |   }
320 |
321 |   const btnCodeSearch = document.getElementById('btn-codebase-
    | search');
322 |   const inputCodeSearch = document.getElementById('codebase-se
    | arch-query');
323 |   if (btnCodeSearch) {
324 |       btnCodeSearch.addEventListener('click', executeCodebaseSea
    | rch);
```

```
325 | }
326 | if (inputCodeSearch) {
327 |     inputCodeSearch.addEventListener('keydown', (e) => {
328 |         if (e.key === 'Enter') executeCodebaseSearch();
329 |     });
330 | }
331 |
332 | const btnSymbolSearch = document.getElementById('btn-codebas
333 | e-symbol-search');
334 | const inputSymbolSearch = document.getElementById('codebase-
335 | symbol-query');
336 | if (btnSymbolSearch) {
337 |     btnSymbolSearch.addEventListener('click', executeSymbolLoo
338 | kup);
339 | }
340 | if (inputSymbolSearch) {
341 |     inputSymbolSearch.addEventListener('keydown', (e) => {
342 |         if (e.key === 'Enter') executeSymbolLookup();
343 |     });
344 | }
345 | const activeIndexSelect = document.getElementById('codebase-
346 | active-index-select');
347 | if (activeIndexSelect) {
348 |     activeIndexSelect.addEventListener('change', updateCodebas
349 | eStatsDisplay);
350 | }
351 | // Auto-initialize when loaded or DOM ready
352 | window.initRagTab = initRagTab;
353 | if (document.readyState === 'loading') {
354 |     document.addEventListener('DOMContentLoaded', initRagTab);
355 | } else {
356 |     initRagTab();
357 | }
```


cache.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Cache Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля cache.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/rag_tab/modules/cache.
11  *   js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { CACHE_TAG, updateRagCacheStatus } from '/windo
15  *   ws/api/webgui/rag_tab/modules/cache.js';
16  *
17  * File: cache.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/rag_tab/modules
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-01 13:04:40
23  * =====
24  */
25
26 import { cachedApiFetch, clearCacheByTag } from '../..js/api-
27 cache.js';
28
29 export const CACHE_TAG = 'rag';
30
31 /**
32  * Cached fetch wrapper for RAG tab
33  * @param {string} url - API endpoint
```

```

31 | * @param {object} options - Fetch options
32 | * @returns {Promise<Response>}
33 | */
34 | export async function ragCachedFetch(url, options = {}) {
35 |     // Merge cache tags
36 |     const tags = options.tags || [];
37 |     if (!tags.includes(CACHE_TAG)) {
38 |         tags.push(CACHE_TAG);
39 |     }
40 |
41 |     return cachedApiFetch(url, { ...options, tags });
42 | }
43 |
44 | /**
45 |  * Clear all RAG cache
46 |  * @returns {Promise<void>}
47 |  */
48 | export async function clearRagCache() {
49 |     await clearCacheByTag(CACHE_TAG);
50 |     console.log('RAG cache cleared');
51 | }
52 |
53 | /**
54 |  * Update cache status UI
55 |  */
56 | export function updateRagCacheStatus() {
57 |     const badge = document.getElementById('rag-cache-badge');
58 |     const ratio = document.getElementById('rag-cache-ratio');
59 |     if (!badge || !ratio) return;
60 |
61 |     try {
62 |         const stats = window.apiCacheStats || { hits: 0, total: 0 };
63 |
64 |         const hitRate = stats.total > 0 ? Math.round((stats.hits /
65 | stats.total) * 100) : 0;
66 |         ratio.textContent = `${hitRate}%`;
67 |
68 |         // Color coding

```

```
67 |     badge.className = 'badge cache-badge';
68 |     if (hitRate >= 70) {
69 |         badge.classList.add('bg-success-subtle', 'text-success')
70 |     };
71 |     } else if (hitRate >= 40) {
72 |         badge.classList.add('bg-warning-subtle', 'text-warning')
73 |     };
74 |     } else {
75 |         badge.classList.add('bg-secondary-subtle', 'text-seconda
76 | ry');
77 |     }
78 | }
79 |
```

codebaseRag.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Codebaserag Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля codebaseR
7  *   ag.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/rag_tab/modules/codebaseRag.js?v=20261001_v1" type="module"></script>
11  *
12  *   JavaScript Import:
13  *     import { updateCodebaseStatsDisplay } from './windows/api/webgui/rag_tab/modules/codebaseRag.js';
14  *
15  * File: codebaseRag.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/rag_tab/modules
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-01 13:04:40
21  * =====
22  */
23
24 import { escapeHtml } from './utils.js';
25 import { ragCachedFetch, CACHE_TAG } from './cache.js';
26
27 let cachedCodebaseIndexes = [];
28
29 /**
30  * Fetch list of available codebase AST indexes and populate i
```

```

    ndex selector.
31  *
32  * @returns {Promise<void>}
33  */
34  export async function loadCodebaseIndexes() {
35    const select = document.getElementById('codebase-active-index-select');
36    if (!select) return;
37
38    try {
39      const res = await ragCachedFetch('/api/rag/codebase/indexes', {
40        tags: ['rag-codebase']
41      });
42      if (!res.ok) return;
43      const data = await res.json();
44      cachedCodebaseIndexes = data.indexes || [];
45
46      if (cachedCodebaseIndexes.length === 0) {
47        select.innerHTML = '<option value="codebase">codebase (По умолчанию)</option>';
48      } else {
49        const curVal = select.value;
50        select.innerHTML = cachedCodebaseIndexes.map(idx => {
51          const scopeBadge = idx.scope === 'user' ? '👤' : '👤';
52          const label = `${scopeBadge}${idx.name} (${idx.total_chunks || 0} чанков)`;
53          return `<option value="${escapeHtml(idx.name)}">${escapeHtml(label)}</option>`;
54        }).join('');
55
56        if (curVal && cachedCodebaseIndexes.some(i => i.name === curVal)) {
57          select.value = curVal;
58        }
59      }
60      updateCodebaseStatsDisplay();

```

```

61 | } catch (err) {
62 |     console.error('[Codebase RAG] Error loading indexes:', err
    | );
63 | }
64 | }
65 |
66 | /**
67 |  * Update UI counters for the currently selected codebase inde
    | x.
68 |  */
69 | export function updateCodebaseStatsDisplay() {
70 |     const select = document.getElementById('codebase-active-inde
    | x-select');
71 |     const statChunks = document.getElementById('codebase-stat-ch
    | unks');
72 |     const statSymbols = document.getElementById('codebase-stat-s
    | ymbols');
73 |     if (!select) return;
74 |
75 |     const currentName = select.value;
76 |     const item = cachedCodebaseIndexes.find(i => i.name === curr
    | entName);
77 |     if (item) {
78 |         if (statChunks) statChunks.textContent = `${item.total_chu
    | nks || 0} чанков`;
79 |         if (statSymbols) statSymbols.textContent = `${item.total_s
    | ymbols || 0} символов AST`;
80 |     }
81 | }
82 |
83 | /**
84 |  * Build or rebuild codebase AST and semantic index for a targ
    | et directory.
85 |  *
86 |  * @returns {Promise<void>}
87 |  */
88 | export async function buildCodebaseRag() {
89 |     const btn = document.getElementById('btn-build-codebase-rag'

```

```

);
90 |   const alertEl = document.getElementById('codebase-build-stat
    | us-alert');
91 |   const rootInput = document.getElementById('codebase-project-
    | root-input');
92 |   const nameInput = document.getElementById('codebase-index-na
    | me-input');
93 |   const dirsInput = document.getElementById('codebase-include-
    | dirs-input');
94 |
95 |   const projectRoot = rootInput?.value?.trim() || '.';
96 |   const indexName = nameInput?.value?.trim() || 'codebase';
97 |   const includeDirs = (dirsInput?.value || '').split(',').map(
    | s => s.trim()).filter(Boolean);
98 |
99 |   if (btn) btn.disabled = true;
100 |   if (alertEl) {
101 |     alertEl.className = 'alert alert-info p-2 small mt-3 mb-0'
    | ;
102 |     alertEl.innerHTML = '<span class="spinner-border spinner-b
    | order-sm me-1"></span> Анализ AST и построение индексов...';
103 |     alertEl.classList.remove('d-none');
104 |   }
105 |
106 |   try {
107 |     const res = await fetch('/api/rag/codebase/build', {
108 |       method: 'POST',
109 |       headers: { 'Content-Type': 'application/json' },
110 |       body: JSON.stringify({
111 |         project_root: projectRoot,
112 |         index_name: indexName,
113 |         include_dirs: includeDirs,
114 |       }),
115 |     });
116 |
117 |     const data = await res.json();
118 |     if (!res.ok) throw new Error(data.detail || data.error ||
    | `HTTP ${res.status}`);

```



```

119 |
120 |     const stats = data.result || {};
121 |     if (alertEl) {
122 |         alertEl.className = 'alert alert-success p-2 small mt-3
mb-0';
123 |         alertEl.innerHTML = `
124 |             <strong>□ Индекс «${escapeHtml(indexName)}» успешно со
здан!</strong><br>
125 |             Файлов: ${stats.total_files || 0} | Чанков: ${stats.to
tal_chunks || 0} | Символов AST: ${stats.total_symbols || 0} (
${stats.elapsed_seconds || 0}c)
126 |         `;
127 |     }
128 |
129 |     await loadCodebaseIndexes();
130 |     const select = document.getElementById('codebase-active-in
dex-select');
131 |     if (select) select.value = indexName;
132 |     updateCodebaseStatsDisplay();
133 | } catch (err) {
134 |     console.error('[Codebase RAG] Build error:', err);
135 |     if (alertEl) {
136 |         alertEl.className = 'alert alert-danger p-2 small mt-3 m
b-0';
137 |         alertEl.textContent = `Ошибка сборки: ${err.message}`;
138 |     }
139 | } finally {
140 |     if (btn) btn.disabled = false;
141 | }
142 | }
143 |
144 | /**
145 |  * Execute semantic query against codebase chunks.
146 |  *
147 |  * @returns {Promise<void>}
148 |  */
149 | export async function executeCodebaseSearch() {
150 |     const queryInput = document.getElementById('codebase-search-

```

```

    query');
151   const select = document.getElementById('codebase-active-index-select');
152   const resultsContainer = document.getElementById('codebase-search-results');
153   const btn = document.getElementById('btn-codebase-search');
154
155   const query = queryInput?.value?.trim();
156   if (!query) return;
157
158   const indexName = select?.value || 'codebase';
159   if (btn) btn.disabled = true;
160   if (resultsContainer) {
161     resultsContainer.innerHTML = '<div class="text-center py-4 text-primary"><span class="spinner-border spinner-border-sm me-2"></span> Поиск по коду...</div>';
162   }
163
164   try {
165     const res = await fetch('/api/rag/codebase/search', {
166       method: 'POST',
167       headers: { 'Content-Type': 'application/json' },
168       body: JSON.stringify({
169         query,
170         index_name: indexName,
171         top_k: 5
172       }),
173     });
174
175     const data = await res.json();
176     if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
177
178     const results = data.results || [];
179     if (results.length === 0) {
180       resultsContainer.innerHTML = `<div class="text-center text-muted py-4">По запросу «${escapeHtml(query)}» совпадений не найдено.</div>`;

```

```

181     return;
182 }
183
184 resultsContainer.innerHTML = results.map(r => {
185     return `
186         <div class="card mb-2 border shadow-sm">
187             <div class="card-header py-1 px-2 bg-body-secondary
188 d-flex justify-content-between align-items-center">
189                 <span class="small fw-semibold text-truncate">
190                     <i class="bi bi-code-slash me-1 text-primary"></i>
191                     <span class="small fw-semibold text-truncate">
192                         <i> ${escapeHtml(r.id)}
193                     </span>
194                     <span class="badge bg-info">Score: ${r.similarity_
195 score || 0}</span>
196                 </div>
197                 <div class="card-body p-2 font-monospace small bg-bo
198 dy-tertiary" style="white-space: pre-wrap; font-size: 0.8rem;
199 max-height: 140px; overflow-y: auto;">${escapeHtml(r.text)}</d
200 iv>
201             </div>
202         `;
203     }).join('');
204 } catch (err) {
205     if (resultsContainer) {
206         resultsContainer.innerHTML = `<div class="alert alert-da
207 nger p-2 small mb-0">Ошибка: ${escapeHtml(err.message)}</div>`;
208     };
209 }
210 } finally {
211     if (btn) btn.disabled = false;
212 }
213 }
214
215 /**
216  * Execute symbol lookup in codebase AST symbols table.
217  *
218  * @returns {Promise<void>}
219  */

```

```
211 export async function executeSymbolLookup() {
212   const symbolInput = document.getElementById('codebase-symbol
    -query');
213   const select = document.getElementById('codebase-active-inde
    x-select');
214   const resultsContainer = document.getElementById('codebase-s
    ymbol-results');
215   const btn = document.getElementById('btn-codebase-symbol-sea
    rch');
216
217   const symbol = symbolInput?.value?.trim();
218   if (!symbol) return;
219
220   const indexName = select?.value || 'codebase';
221   if (btn) btn.disabled = true;
222   if (resultsContainer) {
223     resultsContainer.innerHTML = '<div class="text-center py-4
    text-success"><span class="spinner-border spinner-border-sm m
    e-2"></span> Поиск символов в AST...</div>';
224   }
225
226   try {
227     const res = await fetch('/api/rag/codebase/symbols', {
228       method: 'POST',
229       headers: { 'Content-Type': 'application/json' },
230       body: JSON.stringify({
231         symbol,
232         index_name: indexName,
233         limit: 10
234       }),
235     });
236
237     const data = await res.json();
238     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
    atus}`);
239
240     const results = data.results || [];
241     if (results.length === 0) {
```

```

242     resultsContainer.innerHTML = `
```

```
    </div>`;
267 |     }
268 |   } finally {
269 |     if (btn) btn.disabled = false;
270 |   }
271 | }
272 |
```

docRag.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Docrag Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля docRag.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/rag_tab/modules/docRag
11  *     .js?v=20261001_v1" type="module"></script>
12  *
13  * File: docRag.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/rag_tab/modules
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 import { formatBytes, escapeHtml } from './utils.js';
22 import { ragCachedFetch, CACHE_TAG } from './cache.js';
23
24 /**
25  * Upload one or multiple files to the global document RAG storage.
26  *
27  * @param {File[]} files - List of files to upload.
28  * @returns {Promise<void>}
29  */
30 export async function uploadFiles(files) {
31   if (!files || files.length === 0) return;
```

```
32
33   const progressContainer = document.getElementById('rag-uploa
   d-progress-container');
34   const progressBar = document.getElementById('rag-upload-prog
   ress-bar');
35   const percentText = document.getElementById('rag-upload-perc
   ent');
36   const statusAlert = document.getElementById('rag-upload-stat
   us-alert');
37
38   if (progressContainer) progressContainer.classList.remove('d
   -none');
39   if (statusAlert) statusAlert.classList.add('d-none');
40   if (progressBar) progressBar.style.width = '0%';
41   if (percentText) percentText.textContent = '0%';
42
43   const formData = new FormData();
44   files.forEach((file) => {
45     const uploadName = file.customRelativePath || file.webkitR
   elativePath || file.name;
46     formData.append('files', file, uploadName);
47   });
48
49   try {
50     if (progressBar) progressBar.style.width = '50%';
51     if (percentText) percentText.textContent = '50%';
52
53     const res = await fetch('/api/rag/upload', {
54       method: 'POST',
55       body: formData,
56     });
57
58     if (progressBar) progressBar.style.width = '100%';
59     if (percentText) percentText.textContent = '100%';
60
61     const data = await res.json();
62     if (!res.ok) throw new Error(data.detail || `Upload failed
   (HTTP ${res.status})`);
```



```

63 |
64 |     if (statusAlert) {
65 |         statusAlert.className = 'alert alert-success p-2 small m
    | b-0';
66 |         statusAlert.textContent = `Успешно загружено файлов: ${d
    | ata.uploaded?.length || 0}`;
67 |         statusAlert.classList.remove('d-none');
68 |     }
69 |
70 |     await loadRagDocuments();
71 |     await loadRagStatus();
72 | } catch (err) {
73 |     console.error('[RAG] Upload error:', err);
74 |     if (statusAlert) {
75 |         statusAlert.className = 'alert alert-danger p-2 small mb
    | -0';
76 |         statusAlert.textContent = `Ошибка загрузки: ${err.messag
    | e}`;
77 |         statusAlert.classList.remove('d-none');
78 |     }
79 | } finally {
80 |     setTimeout(() => {
81 |         if (progressContainer) progressContainer.classList.add('
    | d-none');
82 |     }, 1500);
83 | }
84 | }
85 |
86 | /**
87 |  * Fetch and display global RAG index status and stats.
88 |  *
89 |  * @returns {Promise<void>}
90 |  */
91 | export async function loadRagStatus() {
92 |     try {
93 |         const res = await ragCachedFetch('/api/rag/status', {
94 |             tags: ['rag-status']
95 |         });

```

```

96     if (!res.ok) return;
97     const data = await res.json();
98     const st = data.data || {};
99
100    const statDocs = document.getElementById('rag-stat-docs');
101    const statChunks = document.getElementById('rag-stat-chunks');
102    const statDim = document.getElementById('rag-stat-dim');
103    const statUpdated = document.getElementById('rag-stat-updated');
104    const providerBadge = document.getElementById('rag-provider-badge');
105
106    if (statDocs) statDocs.textContent = st.total_documents || 0;
107    if (statChunks) statChunks.textContent = st.total_chunks || 0;
108    if (statDim) statDim.textContent = st.vector_dimension || 0;
109    if (statUpdated) {
110        statUpdated.textContent = st.last_built_at
111            ? new Date(st.last_built_at * 1000).toLocaleTimeString()
112            : '-';
113    }
114    if (providerBadge) {
115        const provName = st.provider === 'gemini' ? 'Google Gemini Embeddings' : 'Local TF-IDF';
116        providerBadge.textContent = `Provider: ${provName}`;
117    }
118 } catch (err) {
119     console.error('[RAG] Error loading status:', err);
120 }
121 }
122
123 /**
124  * Fetch and render the list of uploaded global RAG documents.
125  *

```

```

126 | * @returns {Promise<void>}
127 | */
128 | export async function loadRagDocuments() {
129 |     const tbody = document.getElementById('rag-documents-tbody')
130 |     ;
131 |     const badge = document.getElementById('rag-docs-badge');
132 |     if (!tbody) return;
133 |     try {
134 |         const res = await ragCachedFetch('/api/rag/documents', {
135 |             tags: ['rag-documents']
136 |         });
137 |         if (!res.ok) return;
138 |         const data = await res.json();
139 |         const docs = data.documents || [];
140 |
141 |         if (badge) badge.textContent = `${docs.length} файлов`;
142 |         if (docs.length === 0) {
143 |             tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted py-3">Документы не загружены.</td></tr>';
144 |             return;
145 |         }
146 |
147 |         tbody.innerHTML = docs.map((d, idx) => {
148 |             const statusBadge = d.status === 'indexed'
149 |                 ? '<span class="badge bg-success">Индексирован</span>'
150 |                 : (d.status === 'error' ? '<span class="badge bg-danger">Ошибка</span>' : '<span class="badge bg-warning text-dark">Ожидание</span>');
151 |
152 |             return `
153 |                 <tr class="rag-doc-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для AI-анализа документа RAG">
154 |                     <td class="ps-3 text-truncate" style="max-width: 200px;" title="${escapeHtml(d.name)}">
155 |                         <i class="bi bi-file-earmark-code me-1 text-primary"></i> ${escapeHtml(d.name)}
156 |                     </td>

```

```

157 |         <td>${formatBytes(d.size_bytes)}</td>
158 |         <td><span class="badge bg-secondary-subtle text-seco
| ndary">${d.chunks_count || 0}</span></td>
159 |         <td>${statusBadge}</td>
160 |         <td class="text-end pe-3">
161 |             <button class="btn btn-outline-danger btn-sm p-1 p
| y-0 btn-del-rag-doc" data-name="${escapeHtml(d.name)}" title="
| Удалить">
162 |                 <i class="bi bi-trash"></i>
163 |             </button>
164 |         </td>
165 |     </tr>
166 | `;
167 | }).join('');
168 |
169 | tbody.querySelectorAll('.rag-doc-row').forEach(row => {
170 |     row.onclick = (evt) => {
171 |         if (evt.target.closest('.btn-del-rag-doc')) return;
172 |         const idx = parseInt(row.getAttribute('data-idx'), 10)
173 |         ;
174 |         const d = docs[idx];
175 |         if (!d) return;
176 |         if (window.AITableModal) {
177 |             window.AITableModal.show({
178 |                 icon: '🗑',
179 |                 title: d.name,
180 |                 subtitle: `Размер: ${formatBytes(d.size_bytes)} |
| Чанков: ${d.chunks_count || 0}`,
181 |                 tableType: 'rag_doc',
182 |                 badges: [
183 |                     { text: d.status || 'Indexed', class: d.status =
| == 'indexed' ? 'badge bg-success' : 'badge bg-secondary' },
184 |                     { text: `${d.chunks_count || 0} chunks`, class:
| 'badge bg-info text-dark' }
185 |                 ],
186 |                 metadata: [
187 |                     { label: 'Имя документа', value: d.name },

```

```

188         { label: 'Статус индексации', value: d.status ||
  'Indexed' },
189         { label: 'Количество чанков', value: String(d.ch
unks_count || 0) },
190         { label: 'Размер файла', value: formatBytes(d.si
ze_bytes) },
191         { label: 'Дата индексации', value: d.updated_at
|| d.created_at || 'N/A' },
192         { label: 'Путь источника', value: d.source_path
|| d.name, isCode: true, fullWidth: true }
193     ],
194     rawTitle: 'Метаданные документа',
195     rawContent: JSON.stringify(d, null, 2),
196     requestData: {
197         name: d.name,
198         chunks_count: d.chunks_count,
199         size: formatBytes(d.size_bytes),
200         status: d.status
201     }
202     });
203 }
204 };
205 });
206
207 tbody.querySelectorAll('.btn-del-rag-doc').forEach(btn =>
{
208     btn.onclick = (evt) => {
209         evt.stopPropagation();
210         const docName = btn.getAttribute('data-name');
211         deleteRagDocument(docName);
212     };
213 });
214 } catch (err) {
215     console.error('[RAG] Error loading documents:', err);
216 }
217 }
218
219 /**

```

```

220 | * Delete a specific document from global RAG.
221 | *
222 | * @param {string} filename - Name of the document to delete.
223 | * @returns {Promise<void>}
224 | */
225 | export async function deleteRagDocument(filename) {
226 |   if (!confirm(`Удалить документ «${filename}» из базы знаний?`)) return;
227 |   try {
228 |     const res = await fetch(`/api/rag/documents/${encodeURIComponent(filename)}`, {
229 |       method: 'DELETE',
230 |     });
231 |     window.showToast?(`Документ «${filename}» успешно удален`, 'success');
232 |     await loadRagDocuments();
233 |     await loadRagStatus();
234 |   } catch (err) {
235 |     window.showToast?(`Ошибка удаления: ${err.message}`, 'danger') || alert(`Ошибка удаления: ${err.message}`);
236 |   }
237 | }
238 |
239 | /**
240 | * Trigger global RAG index build.
241 | *
242 | * @returns {Promise<void>}
243 | */
244 | export async function buildRagIndex() {
245 |   const buildBtn = document.getElementById('btn-build-rag-index');
246 |   const statusEl = document.getElementById('rag-build-status');
247 |   ;
247 |   const providerSelect = document.getElementById('rag-provider-select');
248 |   const apiKeyInput = document.getElementById('rag-api-key-input');
249 |   const chunkSizeInput = document.getElementById('rag-chunk-si

```

```

    ze');
250   const chunkOverlapInput = document.getElementById('rag-chunk
    -overlap');
251
252   if (buildBtn) buildBtn.disabled = true;
253   if (statusEl) {
254     statusEl.innerHTML = '<span class="spinner-border spinner-
    border-sm me-1"></span> Построение векторного индекса...';
255   }
256
257   try {
258     const res = await fetch('/api/rag/build', {
259       method: 'POST',
260       headers: { 'Content-Type': 'application/json' },
261       body: JSON.stringify({
262         provider: providerSelect?.value || 'auto',
263         api_key: apiKeyInput?.value?.trim() || '',
264         chunk_size: parseInt(chunkSizeInput?.value || '500', 1
0),
265         chunk_overlap: parseInt(chunkOverlapInput?.value || '5
0', 10),
266       }),
267     });
268
269     const data = await res.json();
270     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
271
272     if (statusEl) {
273       statusEl.className = 'small mt-2 text-center text-succes
s fw-semibold';
274       statusEl.textContent = `👉 Индекс успешно построен. Всего
чанков: ${data.result?.total_chunks || 0}`;
275     }
276
277     await loadRagStatus();
278     await loadRagDocuments();
279   } catch (err) {

```

```

280     console.error('[RAG] Build error:', err);
281     if (statusEl) {
282         statusEl.className = 'small mt-2 text-center text-danger
';
283         statusEl.textContent = `❌ Ошибка индексации: ${err.messa
ge}`;
284     }
285     } finally {
286         if (buildBtn) buildBtn.disabled = false;
287     }
288 }
289
290 /**
291  * Execute semantic query against global RAG index.
292  *
293  * @returns {Promise<void>}
294  */
295 export async function executeRagSearch() {
296     const queryInput = document.getElementById('rag-search-query
');
297     const topKSelect = document.getElementById('rag-top-k');
298     const minScoreInput = document.getElementById('rag-min-score
');
299     const resultsContainer = document.getElementById('rag-search
-results');
300     const searchBtn = document.getElementById('btn-rag-search');
301     const apiKeyInput = document.getElementById('rag-api-key-inp
ut');
302
303     const query = queryInput?.value?.trim();
304     if (!query) return;
305
306     if (searchBtn) searchBtn.disabled = true;
307     if (resultsContainer) {
308         resultsContainer.innerHTML = '<div class="text-center py-4
text-primary"><span class="spinner-border spinner-border-sm m
e-2"></span> Идет семантический поиск...</div>';
309     }

```



```

310
311     try {
312         const res = await fetch('/api/rag/search', {
313             method: 'POST',
314             headers: { 'Content-Type': 'application/json' },
315             body: JSON.stringify({
316                 query,
317                 top_k: parseInt(topKSelect?.value || '5', 10),
318                 min_score: parseFloat(minScoreInput?.value || '0.0'),
319                 api_key: apiKeyInput?.value?.trim() || '',
320             }),
321         });
322
323         const data = await res.json();
324         if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
325
326         const results = data.results || [];
327         if (results.length === 0) {
328             resultsContainer.innerHTML = `

<i class="bi bi-search fs-3 d-block mb-1 opacity-50"></i>По запросу «${escapeHtml(query)}» ничего не найдено.</div>`;
329             return;
330         }
331
332         resultsContainer.innerHTML = results.map(r => `
333             <div class="card mb-2 border shadow-sm">
334                 <div class="card-header py-1 px-2 bg-body-secondary d-flex justify-content-between align-items-center">
335                     <span class="small fw-semibold text-truncate" style="max-width: 70%;">
336                         <i class="bi bi-file-earmark-text me-1 text-primary"></i> ${escapeHtml(r.doc_name)} (#${r.chunk_index})
337                     </span>
338                     <span class="badge bg-secondary">Score: ${r.score}</span>
339                 </div>


```

```
340 |         <div class="card-body p-2 font-monospace small bg-body
      | -tertiary" style="white-space: pre-wrap; font-size: 0.8rem; ma
      | x-height: 120px; overflow-y: auto;">${escapeHtml(r.text)}</div
      | >
341 |     </div>
342 |     `).join('');
343 | } catch (err) {
344 |     if (resultsContainer) {
345 |         resultsContainer.innerHTML = `<div class="alert alert-da
      | nger p-2 small mb-0">Ошибка: ${escapeHtml(err.message)}</div>`
      | ;
346 |     }
347 | } finally {
348 |     if (searchBtn) searchBtn.disabled = false;
349 | }
350 | }
351 |
```

pixelRag.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Pixelrag Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля pixelRag.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *   <script src="/windows/api/webgui/rag_tab/modules/pixelR
    ag.js?v=20261001_v1" type="module"></script>
11 *
12 * File: pixelRag.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/rag_tab/modules
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-01 13:04:40
18 * =====
19 */
20
21 import { formatBytes, escapeHtml } from './utils.js';
22
23 /**
24  * Load and display PixelRAG system status, FAISS index metric
    s, and image counts.
25  *
26  * @returns {Promise<void>}
27  */
28 export async function loadPixelRagStatus() {
29   try {
30     const res = await fetch('/api/rag/pixel/status');
31     if (!res.ok) return;
```

```
32 |
33 |     const data = await res.json();
34 |     const imagesEl = document.getElementById('pixel-stat-image
    s');
35 |     const chunksEl = document.getElementById('pixel-stat-chunk
    s');
36 |     const sizeEl = document.getElementById('pixel-stat-size');
37 |     const providerEl = document.getElementById('pixel-stat-pro
    vider');
38 |     const updatedEl = document.getElementById('pixel-stat-upda
    ted');
39 |     const statusEl = document.getElementById('pixel-stat-statu
    s');
40 |
41 |     if (imagesEl) imagesEl.textContent = data.total_images ||
    0;
42 |     if (chunksEl) chunksEl.textContent = data.total_chunks ||
    0;
43 |     if (sizeEl) sizeEl.textContent = `${(data.total_size_mb ||
    0).toFixed(2)} MB`;
44 |     if (providerEl) providerEl.textContent = data.provider ||
    'FAISS / PixelRAG';
45 |     if (statusEl) statusEl.textContent = `Текстовых чанков: ${
    data.sources?.text_chunks || 0} | Картинок: ${data.sources?.im
    ages || 0}`;
46 |
47 |     if (updatedEl) {
48 |         if (data.last_built_at && data.last_built_at > 0) {
49 |             const d = new Date(data.last_built_at * 1000);
50 |             updatedEl.textContent = d.toLocaleString();
51 |         } else {
52 |             updatedEl.textContent = 'Индекс готов';
53 |         }
54 |     }
55 | } catch (err) {
56 |     console.debug('[PixelRAG] Failed to load status:', err);
57 | }
58 | }
```

```
59 |
60 | /**
61 |  * Upload and index image files into PixelRAG.
62 |  *
63 |  * @param {File[]} files - List of image files.
64 |  * @returns {Promise<void>}
65 |  */
66 | export async function uploadAndIndexPixelImages(files) {
67 |   if (!files || files.length === 0) return;
68 |
69 |   const progressContainer = document.getElementById('pixel-upl
oad-progress');
70 |   const progressBar = document.getElementById('pixel-upload-pr
ogress-bar');
71 |   const percentText = document.getElementById('pixel-upload-pe
rcent');
72 |   const statusAlert = document.getElementById('pixel-upload-st
atus-alert');
73 |
74 |   if (progressContainer) progressContainer.classList.remove('d
-none');
75 |   if (statusAlert) statusAlert.classList.add('d-none');
76 |   if (progressBar) progressBar.style.width = '20%';
77 |   if (percentText) percentText.textContent = '20%';
78 |
79 |   const formData = new FormData();
80 |   files.forEach((file) => {
81 |     formData.append('files', file, file.name);
82 |   });
83 |
84 |   try {
85 |     if (progressBar) progressBar.style.width = '60%';
86 |     if (percentText) percentText.textContent = '60%';
87 |
88 |     const res = await fetch('/api/rag/pixel/index', {
89 |       method: 'POST',
90 |       body: formData,
91 |     });
```

```

92
93     if (progressBar) progressBar.style.width = '100%';
94     if (percentText) percentText.textContent = '100%';
95
96     const data = await res.json();
97     if (!res.ok) throw new Error(data.detail || `Индексация не
удалась (HTTP ${res.status})`);
98
99     if (statusAlert) {
100         statusAlert.className = 'alert alert-success p-2 small m
b-0';
101         statusAlert.textContent = `✅ Успешно проиндексировано: $
{data.indexed_count || files.length} файл(ов)`;
102         statusAlert.classList.remove('d-none');
103     }
104
105     await loadPixelRagStatus();
106 } catch (err) {
107     console.error('[PixelRAG] Indexing error:', err);
108     if (statusAlert) {
109         statusAlert.className = 'alert alert-danger p-2 small mb
-0';
110         statusAlert.textContent = `Ошибка индексации: ${err.mess
age}`;
111         statusAlert.classList.remove('d-none');
112     }
113 } finally {
114     setTimeout(() => {
115         if (progressContainer) progressContainer.classList.add('
d-none');
116     }, 2000);
117 }
118 }
119
120 /**
121  * Execute smart query analysis via QueryRouter.
122  *
123  * @returns {Promise<void>}

```

```
124 | */
125 | export async function analyzePixelQuery() {
126 |   const input = document.getElementById('pixel-analyze-query-i
    nput');
127 |   const resultBox = document.getElementById('pixel-analyze-res
    ult');
128 |   const routingTypeEl = document.getElementById('pixel-routing
    -type');
129 |   const confidenceEl = document.getElementById('pixel-routing-
    confidence');
130 |   const reasonEl = document.getElementById('pixel-routing-reas
    on');
131 |
132 |   if (!input || !input.value.trim()) return;
133 |   const query = input.value.trim();
134 |
135 |   try {
136 |     const res = await fetch(`/api/rag/pixel/analyze-query?quer
    y=${encodeURIComponent(query)}`, {
137 |       method: 'POST',
138 |     });
139 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
140 |
141 |     const data = await res.json();
142 |     if (resultBox) resultBox.classList.remove('d-none');
143 |     if (routingTypeEl) {
144 |       routingTypeEl.textContent = data.routing_type?.toUpperCase() || 'HYBRID';
145 |       routingTypeEl.className = data.routing_type === 'visual'
        ? 'badge bg-warning text-dark' : data.routing_type === 'text'
        ? 'badge bg-info' : 'badge bg-primary';
146 |     }
147 |     if (confidenceEl) confidenceEl.textContent = `${Math.round
    ((data.confidence || 1.0) * 100)}%`;
148 |     if (reasonEl) reasonEl.textContent = `Причина: ${data.reas
    on || 'Авто-определение по ключевым признакам'}`;
149 |   } catch (err) {
150 |     console.error('[PixelRAG] Analysis failed:', err);
```

```

151 | }
152 | }
153 |
154 | /**
155 |  * Execute semantic visual search inside PixelRAG.
156 |  *
157 |  * @returns {Promise<void>}
158 |  */
159 | export async function executePixelRagSearch() {
160 |     const queryInput = document.getElementById('pixel-search-que
161 | ry');
162 |     const topKSelect = document.getElementById('pixel-search-top
163 | k');
164 |     const minScoreSelect = document.getElementById('pixel-search
165 | -minscore');
166 |     const routingToggle = document.getElementById('pixel-search-
167 | routing-toggle');
168 |     const resultsContainer = document.getElementById('pixel-sear
169 | ch-results');
170 |
171 |     if (!queryInput || !resultsContainer) return;
172 |     const query = queryInput.value.trim();
173 |     if (!query) {
174 |         queryInput.focus();
175 |         return;
176 |     }
177 |
178 |     const topK = parseInt(topKSelect?.value || '5', 10);
179 |     const minScore = parseFloat(minScoreSelect?.value || '0.0');
180 |     const useRouting = routingToggle ? routingToggle.checked : t
181 | rue;
182 |
183 |     resultsContainer.innerHTML = `
184 |         <div class="text-center py-4 text-muted">
185 |             <div class="spinner-border spinner-border-sm text-primar
186 | y me-2" role="status"></div>
187 |             <span>Поиск по визуальным индексам и текстовым чанкам...
188 | </span>

```



```

181     </div>
182 `;
183
184     try {
185         const res = await fetch('/api/rag/pixel/search', {
186             method: 'POST',
187             headers: { 'Content-Type': 'application/json' },
188             body: JSON.stringify({
189                 query: query,
190                 top_k: topK,
191                 min_score: minScore,
192                 use_routing: useRouting,
193                 version_filter: 'latest',
194             }),
195         });
196
197         if (!res.ok) {
198             const errData = await res.json().catch(() => ({}));
199             throw new Error(errData.detail || `HTTP ${res.status}`);
200         }
201
202         const data = await res.json();
203         const results = data.results || [];
204
205         if (results.length === 0) {
206             resultsContainer.innerHTML = `
207                 <div class="text-center text-muted py-4">
208                     <i class="bi bi-search fs-3 d-block mb-1 opacity-50"
209 ></i>
210                     <span>По запросу «${escapeHtml(query)}» ничего не на
211 йдено.</span>
212                 </div>
213             `;
214             return;
215
216             resultsContainer.innerHTML = `
217                 <div class="d-flex justify-content-between align-items-c

```

```

enter mb-2 pb-2 border-bottom">
217 |         <span class="small text-muted">Найдено совпадений: <strong>${results.length}</strong> (Маршрут: <span class="badge bg-secondary">${data.routing_type}</span>)</span>
218 |         <span class="small text-muted">Уверенность: ${((data.query_confidence * 100).toFixed(0))}%</span>
219 |     </div>
220 |     <div class="d-flex flex-column gap-2">
221 |         ${results.map((r, i) => {
222 |             const scorePercent = Math.min(100, Math.round((r.score || 0) * 100));
223 |             const isVisual = r.source_type === 'pixel' || r.source_type === 'image';
224 |             const typeBadge = isVisual
225 |                 ? '<span class="badge bg-warning text-dark"><i class="bi bi-image me-1"></i>Visual</span>'
226 |                 : '<span class="badge bg-info"><i class="bi bi-file-text me-1"></i>Text</span>';
227 |
228 |             return `
229 |                 <div class="card bg-body border-secondary-subtle p-2">
230 |                     <div class="d-flex justify-content-between align-items-center mb-1">
231 |                         <span class="fw-bold small text-primary text-truncate" style="max-width: 70%;">
232 |                             #${i + 1} ${escapeHtml(r.doc_name || 'Без названия')}
233 |                         </span>
234 |                         <div class="d-flex align-items-center gap-1">
235 |                             ${typeBadge}
236 |                             <span class="badge bg-success-subtle text-success border border-success-subtle">
237 |                                 Score: ${((r.score || 0).toFixed(3))} (${scorePercent}%)
238 |                             </span>
239 |                         </div>
240 |                     </div>

```

```

241 |         <div class="small text-body-secondary font-monospace" style="font-size: 0.8rem; white-space: pre-wrap;">
242 |             ${escapeHtml(r.text || '')}
243 |         </div>
244 |         ${r.source_path ? `<div class="small text-muted mt-1 text-truncate" style="font-size: 0.72rem;"><i class="bi bi-link-45deg"></i> ${escapeHtml(r.source_path)}</div>` : ''}
245 |             </div>
246 |         `;
247 |     }).join('')}
248 | </div>
249 | `;
250 | } catch (err) {
251 |     console.error('[PixelRAG] Search error:', err);
252 |     resultsContainer.innerHTML = `
253 |         <div class="alert alert-danger p-2 small mb-0">
254 |             <i class="bi bi-exclamation-triangle me-1"></i> Ошибка
255 |             поиска: ${escapeHtml(err.message)}
256 |         </div>
257 |     `;
258 | }
259 |

```

userRagCollections.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Userragcollections Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля userRagCo
7  *   llections.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/rag_tab/modules/userRa
11  *   gCollections.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { getActiveUserRagId, setActiveUserRagId, render
15  *   UserRagsList } from './windows/api/webgui/rag_tab/modules/userR
16  *   agCollections.js';
17  *
18  * File: userRagCollections.js
19  * Project: ai-breadboard
20  * Package: windows/api/webgui/rag_tab/modules
21  * Author: hypo69
22  * Copyright: © 2026 hypo69
23  * Updated: 2026-10-01 13:04:40
24  * =====
25  */
26
27 import { escapeHtml } from './utils.js';
28 import { loadUserStorageFiles } from './userRagPipeline.js';
29
```

```
30 /**
31  * Get the currently active collection ID.
32  *
33  * @returns {string|null} Active RAG ID or null.
34  */
35 export function getActiveUserRagId() {
36   return activeUserRagId;
37 }
38
39 /**
40  * Set the active collection ID.
41  *
42  * @param {string|null} id - Target collection ID.
43  */
44 export function setActiveUserRagId(id) {
45   activeUserRagId = id;
46 }
47
48 /**
49  * Fetch all user workspace RAG collections and render sidebar
    list.
50  *
51  * @returns {Promise<void>}
52  */
53 export async function loadUserRags() {
54   const listContainer = document.getElementById('user-rags-lis
    t-container');
55   if (!listContainer) return;
56
57   try {
58     const res = await fetch('/api/user/rags');
59     if (!res.ok) throw new Error(`HTTP ${res.status}`);
60     const data = await res.json();
61     userCollections = data.collections || [];
62     renderUserRagsList(userCollections);
63
64     if (userCollections.length > 0) {
65       if (!activeUserRagId || !userCollections.some(c => c.id
```

```

    === activeUserRagId)) {
66         selectUserRag(userCollections[0].id);
67     } else {
68         selectUserRag(activeUserRagId);
69     }
70 } else {
71     activeUserRagId = null;
72     showUserRagEmptyState();
73 }
74 } catch (err) {
75     listContainer.innerHTML = `<div class="alert alert-danger
p-2 small mb-0">Ошибка загрузки коллекций: ${escapeHtml(err.me
ssage)}</div>`;
76 }
77 }
78
79 /**
80  * Render the collection cards in the left sidebar.
81  *
82  * @param {Array<object>} collections - List of collection sum
mary objects.
83  */
84 export function renderUserRagsList(collections) {
85     const listContainer = document.getElementById('user-rags-lis
t-container');
86     if (!listContainer) return;
87
88     if (collections.length === 0) {
89         listContainer.innerHTML = `
90             <div class="text-center text-muted py-4 small">
91                 <i class="bi bi-folder-x fs-3 d-block mb-1 opacity-50"
></i>
92                 Нет созданных коллекций.<br>Нажмите «+ Создать» выше.
93             </div>
94             `;
95         return;
96     }
97

```

```

98 | listContainer.innerHTML = collections.map(col => {
99 |     const isActive = col.id === activeUserRagId;
100 |     const statusBadgeClass = col.status === 'indexed'
101 |         ? 'bg-success'
102 |         : (col.status === 'cleaning' ? 'bg-warning text-dark' :
103 | (col.status === 'error' ? 'bg-danger' : 'bg-secondary'));
104 |
105 |     return `
106 |         <div class="card p-2 cursor-pointer user-rag-item ${isAc
107 | tive ? 'border-primary bg-primary-subtle' : 'border-secondary-
108 | subtle bg-body-tertiary'}">
109 |             data-rag-id="${escapeHtml(col.id)}" style="cursor:
110 | pointer; transition: all 0.15s ease-in-out;">
111 |             <div class="d-flex justify-content-between align-items
112 | -center mb-1">
113 |                 <strong class="text-truncate small" style="max-width
114 | : 140px;">${escapeHtml(col.name || col.id)}</strong>
115 |                 <span class="badge ${statusBadgeClass} small" style=
116 | "font-size: 0.7rem;">${escapeHtml(col.status || 'created')}</s
117 | pan>
118 |             </div>
119 |             <div class="d-flex justify-content-between align-items
120 | -center small text-muted" style="font-size: 0.75rem;">
121 |                 <span><i class="bi bi-diagram-3 me-1"></i>${col.tota
122 | l_chunks || 0} чанков</span>
123 |                 <span><i class="bi bi-file-earmark me-1"></i>${(col.
124 | files || []).length} файлов</span>
125 |             </div>
126 |         </div>
127 |     `;
128 | }).join('');
129 |
130 | listContainer.querySelectorAll('.user-rag-item').forEach(el
131 | => {
132 |     el.addEventListener('click', () => {
133 |         const ragId = el.getAttribute('data-rag-id');
134 |         if (ragId) selectUserRag(ragId);
135 |     });
136 | });

```

```

124 | });
125 | }
126 |
127 | /**
128 |  * Display placeholder empty state when no collection is selected.
129 |  */
130 | export function showUserRagEmptyState() {
131 |     const emptyState = document.getElementById('user-rag-empty-state');
132 |     const detailContainer = document.getElementById('user-rag-detail-container');
133 |     if (emptyState) emptyState.classList.remove('d-none');
134 |     if (detailContainer) detailContainer.classList.add('d-none');
135 | }
136 |
137 | /**
138 |  * Select a workspace collection and load its metadata and file checklist.
139 |  *
140 |  * @param {string} ragId - ID of the collection to select.
141 |  * @returns {Promise<void>}
142 |  */
143 | export async function selectUserRag(ragId) {
144 |     activeUserRagId = ragId;
145 |     const emptyState = document.getElementById('user-rag-empty-state');
146 |     const detailContainer = document.getElementById('user-rag-detail-container');
147 |     if (emptyState) emptyState.classList.add('d-none');
148 |     if (detailContainer) detailContainer.classList.remove('d-none');
149 |
150 |     document.querySelectorAll('.user-rag-item').forEach(el => {
151 |         if (el.getAttribute('data-rag-id') === ragId) {
152 |             el.classList.add('border-primary', 'bg-primary-subtle');
153 |             el.classList.remove('border-secondary-subtle', 'bg-body-

```



```
    tertiary');
154     } else {
155         el.classList.remove('border-primary', 'bg-primary-subtle
    ');
156         el.classList.add('border-secondary-subtle', 'bg-body-ter
    tiary');
157     }
158 });
159
160 try {
161     const res = await fetch(`/api/user/rags/${encodeURIComponent(ragId)}`);
162     if (!res.ok) throw new Error(`HTTP ${res.status}`);
163     const data = await res.json();
164     const col = data.collection;
165
166     const titleEl = document.getElementById('active-rag-title'
    );
167     const descEl = document.getElementById('active-rag-desc');
168     const statusBadge = document.getElementById('active-rag-st
    atus-badge');
169     const chunksEl = document.getElementById('active-rag-stat-
    chunks');
170     const filesEl = document.getElementById('active-rag-stat-f
    iles');
171     const updatedEl = document.getElementById('active-rag-stat
    -updated');
172
173     if (titleEl) titleEl.textContent = col.name || col.id;
174     if (descEl) descEl.textContent = col.description || 'Без о
    писания';
175     if (statusBadge) {
176         statusBadge.textContent = col.status || 'created';
177         statusBadge.className = `badge ${col.status === 'indexed
    ' ? 'bg-success' : (col.status === 'cleaning' ? 'bg-warning te
    xt-dark' : (col.status === 'error' ? 'bg-danger' : 'bg-seconda
    ry'))}`;
178     }
```

```

179 |     if (chunksEl) chunksEl.textContent = col.total_chunks || 0
    | ;
180 |     if (filesEl) filesEl.textContent = (col.files || []).length;
    |
181 |     if (updatedEl) {
182 |         updatedEl.textContent = col.updated_at ? new Date(col.updated_at * 1000).toLocaleString() : '-';
    |
183 |     }
184 |
185 |     await loadUserStorageFiles(col.files || []);
186 |     await loadRagEntries(ragId);
187 | } catch (err) {
188 |     console.error('Failed to load user RAG details:', err);
189 | }
190 | }
191 |
192 | /**
193 |  * Fetch and render chunk/QA entries for the active collection
    | .
194 |  *
195 |  * @param {string} ragId - Collection identifier.
196 |  * @param {string} query - Optional filter query.
197 |  * @returns {Promise<void>}
198 |  */
199 | export async function loadRagEntries(ragId = activeUserRagId,
    | query = '') {
200 |     const container = document.getElementById('user-rag-entries-
    | container');
201 |     if (!container || !ragId) return;
202 |
203 |     container.innerHTML = '<div class="text-center text-muted small py-2"><div class="spinner-border spinner-border-sm text-primary me-1"></div>Загрузка записей...</div>';
204 |
205 |     try {
206 |         const url = `/api/user/rags/${encodeURIComponent(ragId)}/entries?q=${encodeURIComponent(query)}&limit=100`;
207 |         const res = await fetch(url);

```

```

208     if (!res.ok) throw new Error(`HTTP ${res.status}`);
209     const data = await res.json();
210     renderRagEntries(data.entries || [], data.total || 0);
211   } catch (err) {
212     container.innerHTML = `
```

```

241 |     const badgeClass = isQa ? 'bg-primary' : 'bg-info text-dar
    | k';
242 |     const badgeLabel = isQa ? 'Q&A' : (entry.source_file || 'Ф
    | айл');
243 |
244 |     let displayContent = entry.content || '';
245 |     let questionText = entry.meta?.question || '';
246 |     let answerText = entry.meta?.answer || '';
247 |
248 |     if (!questionText && displayContent.includes('Вопрос:') &&
    | displayContent.includes('Ответ:')) {
249 |         const parts = displayContent.split('\nОтвет:');
250 |         questionText = parts[0].replace('Вопрос:', '').trim();
251 |         answerText = (parts[1] || '').trim();
252 |     }
253 |
254 |     return `
255 |         <div class="card p-2 border border-secondary-subtle bg-b
    | ody-tertiary position-relative rag-entry-card" data-chunk-id="
    | ${escapeHtml(chunkId)}">
256 |             <div class="d-flex justify-content-between align-items
    | -start mb-1">
257 |                 <div class="d-flex align-items-center gap-1 flex-wra
    | p">
258 |                     <span class="badge ${badgeClass} small" style="fon
    | t-size: 0.7rem;">${escapeHtml(badgeLabel)}</span>
259 |                     <small class="text-muted font-monospace" style="fo
    | nt-size: 0.72rem;">#${escapeHtml(chunkId)}</small>
260 |                 </div>
261 |                 <button class="btn btn-outline-danger btn-sm py-0 px
    | -2 rounded-pill btn-delete-entry" data-chunk-id="${escapeHtml(
    | chunkId)}" title="Удалить запись из RAG">
262 |                     <i class="bi bi-trash"></i>
263 |                 </button>
264 |             </div>
265 |             ${questionText ? `
266 |                 <div class="small fw-bold text-primary mb-1"><i clas
    | s="bi bi-question-circle me-1"></i>${escapeHtml(questionText)}

```

```

267 |         <div class="small text-secondary font-monospace bg-b
    | ody p-2 rounded border" style="white-space: pre-wrap; font-siz
    | e: 0.78rem;">${escapeHtml(answerText || displayContent)}</div>
268 |         ` : `
269 |         <div class="small text-secondary font-monospace bg-b
    | ody p-2 rounded border" style="white-space: pre-wrap; font-siz
    | e: 0.78rem;">${escapeHtml(displayContent)}</div>
270 |         `}
271 |     </div>
272 |     `;
273 | }).join('');
274 |
275 | container.querySelectorAll('.btn-delete-entry').forEach(btn
    | => {
276 |     btn.addEventListener('click', (e) => {
277 |         e.stopPropagation();
278 |         const chunkId = btn.getAttribute('data-chunk-id');
279 |         if (chunkId) deleteRagEntry(chunkId);
280 |     });
281 | });
282 | }
283 |
284 | /**
285 |  * Add a new custom Q&A pair entry to the active collection.
286 |  *
287 |  * @returns {Promise<void>}
288 |  */
289 | export async function addQaEntry() {
290 |     if (!activeUserRagId) {
291 |         window.showToast?(['Пожалуйста, выберите коллекцию RAG.',
    | 'warning']) || alert('Пожалуйста, выберите коллекцию RAG.');
```

```
298
299   const question = qInput?.value.trim() || '';
300   const answer = aInput?.value.trim() || '';
301
302   if (!answer) {
303     window.showToast?.('Пожалуйста, введите текст ответа или з
нания.', 'warning') || alert('Пожалуйста, введите текст ответа
или знания.');
```

304 return;

305 }

306

307 if (btn) btn.disabled = true;

308 try {

309 const res = await fetch(`/api/user/rags/\${encodeURIComponent(activeUserRagId)}/entries`, {

310 method: 'POST',

311 headers: { 'Content-Type': 'application/json' },

312 body: JSON.stringify({ question, answer })

313 });

314

315 const data = await res.json();

316 if (!res.ok) throw new Error(data.detail || `HTTP \${res.status}`);

317

318 const modalEl = document.getElementById('modal-add-qa-entry');

319 if (modalEl && window.bootstrap?.Modal) {

320 const modal = bootstrap.Modal.getInstance(modalEl) || new bootstrap.Modal(modalEl);

321 modal.hide();

322 }

323

324 if (qInput) qInput.value = '';

325 if (aInput) aInput.value = '';

326

327 window.showToast?.('Q&A запись успешно добавлена', 'success');

328 await selectUserRag(activeUserRagId);

```

329     await loadUserRags();
330   } catch (err) {
331     window.showToast?(`Ошибка сохранения Q&A записи: ${err.me
ssage}`, 'danger') || alert(`Ошибка сохранения Q&A записи: ${e
rr.message}`);
332   } finally {
333     if (btn) btn.disabled = false;
334   }
335 }
336
337 /**
338  * Delete a specific chunk/QA entry from the active collection
339  *
340  * @param {string} chunkId - Chunk identifier.
341  * @returns {Promise<void>}
342  */
343 export async function deleteRagEntry(chunkId) {
344   if (!activeUserRagId || !chunkId) return;
345
346   const confirmed = confirm(`Удалить эту запись (${chunkId})
из базы знаний`);
347   if (!confirmed) return;
348
349   try {
350     const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/entries/${encodeURIComponent(chunkId)}`,
    {
351       method: 'DELETE'
352     });
353     if (!res.ok) throw new Error(`HTTP ${res.status}`);
354
355     window.showToast?(`Запись успешно удалена`, 'success');
356     await selectUserRag(activeUserRagId);
357     await loadUserRags();
358   } catch (err) {
359     window.showToast?(`Ошибка удаления записи: ${err.message}
`, 'danger') || alert(`Ошибка удаления записи: ${err.message}`

```

```

    );
360 }
361 }
362
363 /**
364  * Create a new user workspace RAG collection via API.
365  *
366  * @returns {Promise<void>}
367  */
368 export async function createUserRag() {
369     const nameInput = document.getElementById('new-rag-name');
370     const descInput = document.getElementById('new-rag-desc');
371     const minChunkInput = document.getElementById('new-rag-min-c
        hunk');
372     const maxChunkInput = document.getElementById('new-rag-max-c
        hunk');
373     const btn = document.getElementById('btn-confirm-create-rag'
        );
374
375     const name = nameInput?.value.trim();
376     if (!name) {
377         window.showToast?.('Пожалуйста, укажите имя коллекции.', '
        warning') || alert('Пожалуйста, укажите имя коллекции.');
```

378 return;

379 }

380

381 if (btn) btn.disabled = true;

382 try {

383 const res = await fetch('/api/user/rags', {

384 method: 'POST',

385 headers: { 'Content-Type': 'application/json' },

386 body: JSON.stringify({

387 name: name,

388 description: descInput?.value.trim() || '',

389 min_chunk_len: parseInt(minChunkInput?.value || '20',

10),

390 max_chunk_len: parseInt(maxChunkInput?.value || '1500'

, 10)


```
391 |     })
392 |   });
393 |
394 |   const data = await res.json();
395 |   if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
396 |
397 |   const srcType = document.querySelector('input[name="new-rag-source-type"]:checked')?.value || 'local';
398 |
399 |   const modalEl = document.getElementById('modal-create-user-rag');
400 |   if (modalEl && window.bootstrap?.Modal) {
401 |     const modal = bootstrap.Modal.getInstance(modalEl) || new bootstrap.Modal(modalEl);
402 |     modal.hide();
403 |   }
404 |
405 |   if (nameInput) nameInput.value = '';
406 |   if (descInput) descInput.value = '';
407 |
408 |   window.showToast?.(`Коллекция '${name}' успешно создана`, 'success');
409 |   await loadUserRags();
410 |   const createdId = data.collection?.id;
411 |   if (createdId) {
412 |     selectUserRag(createdId);
413 |     if (srcType === 'gdocs') {
414 |       const importModalEl = document.getElementById('modal-import-gdocs');
415 |       if (importModalEl && window.bootstrap?.Modal) {
416 |         const importModal = bootstrap.Modal.getInstance(importModalEl) || new bootstrap.Modal(importModalEl);
417 |         importModal.show();
418 |       }
419 |     }
420 |   }
421 | } catch (err) {
```

```
422     window.showToast?(`Ошибка создания коллекции: ${err.messa
    ge}`, 'danger') || alert(`Ошибка создания коллекции: ${err.mes
    sage}`);
423   } finally {
424     if (btn) btn.disabled = false;
425   }
426 }
427
428 /**
429  * Delete the currently active workspace collection.
430  *
431  * @returns {Promise<void>}
432  */
433 export async function deleteActiveUserRag() {
434   if (!activeUserRagId) return;
435
436   const confirmed = confirm(`Вы уверены, что хотите удалить ко
    ллекцию «${activeUserRagId}»?`);
437   if (!confirmed) return;
438
439   try {
440     const res = await fetch(`/api/user/rags/${encodeURIComponent(
    activeUserRagId)}`, {
441       method: 'DELETE'
442     });
443     if (!res.ok) throw new Error(`HTTP ${res.status}`);
444     window.showToast?(`Коллекция '${activeUserRagId}' успешно
    удалена`, 'success');
445     activeUserRagId = null;
446     await loadUserRags();
447   } catch (err) {
448     window.showToast?(`Ошибка удаления коллекции: ${err.messa
    ge}`, 'danger') || alert(`Ошибка удаления коллекции: ${err.mes
    sage}`);
449   }
450 }
451
452
```

userRagPipeline.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Userragpipeline Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля userRagPi
7  *   peline.
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/rag_tab/modules/userRa
11  *     gPipeline.js?v=20261001_v1" type="module"></script>
12  *
13  * File: userRagPipeline.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/rag_tab/modules
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 import { formatBytes, escapeHtml } from './utils.js';
22 import { getActiveUserRagId, loadUserRags, selectUserRag } fro
23   m './userRagCollections.js';
24 /**
25  * Fetch files from user storage and render checkbox selection
26  * list.
27  * @param {string[]} attachedFiles - Array of filenames curren
28  *   tly attached to this collection.
29  * @returns {Promise<void>}
```

```

29  */
30  export async function loadUserStorageFiles(attachedFiles) {
31    const checklistContainer = document.getElementById('user-rag
    -files-checklist');
32    if (!checklistContainer) return;
33
34    try {
35      const res = await fetch('/api/user/files?subfolder=files')
    ;
36      if (!res.ok) throw new Error(`HTTP ${res.status}`);
37      const data = await res.json();
38      const files = data.files || [];
39
40      if (files.length === 0) {
41        checklistContainer.innerHTML = `
42          <div class="text-muted small p-2 text-center">
43            В личном хранилище нет файлов.<br>Нажмите «Загрузить
    файлы» ниже.
44          </div>
45          `;
46        return;
47      }
48
49      checklistContainer.innerHTML = files.map(f => {
50        const isChecked = attachedFiles.length === 0 || attached
    Files.includes(f.name);
51        return `
52          <label class="d-flex align-items-center justify-conten
    t-between p-1 px-2 rounded hover-bg border-bottom border-light
    -subtle small cursor-pointer">
53            <div class="d-flex align-items-center gap-2">
54              <input class="form-check-input mt-0" type="checkbo
    x" value="${escapeHtml(f.name)}" ${isChecked ? 'checked' : ''}
    >
55              <span class="text-truncate font-monospace" style="
    max-width: 280px;">${escapeHtml(f.name)}</span>
56            </div>
57            <span class="text-muted small">${formatBytes(f.size_

```

```

bytes)}}</span>
58 |         </label>
59 |         `;
60 |     }).join('');
61 | } catch (err) {
62 |     checklistContainer.innerHTML = `<div class="alert alert-da
    nger p-2 small mb-0">Ошибка списка файлов: ${escapeHtml(err.me
    ssage)}</div>`;
63 | }
64 | }
65 |
66 | /**
67 |  * Build active collection index through cleaner and TF-IDF pi
    peline.
68 |  *
69 |  * @returns {Promise<void>}
70 |  */
71 | export async function buildActiveUserRag() {
72 |     const activeUserRagId = getActiveUserRagId();
73 |     if (!activeUserRagId) return;
74 |
75 |     const btn = document.getElementById('btn-build-active-rag');
76 |     const statusAlert = document.getElementById('user-rag-build-
    status-alert');
77 |     const checkboxes = document.querySelectorAll('#user-rag-file
    s-checklist input[type="checkbox"]:checked');
78 |     const selectedFiles = Array.from(checkboxes).map(cb => cb.va
    lue);
79 |
80 |     if (btn) btn.disabled = true;
81 |     if (statusAlert) {
82 |         statusAlert.className = 'alert alert-info p-2 small mt-2 m
    b-0 d-block';
83 |         statusAlert.innerHTML = '<span class="spinner-border spinn
    er-border-sm me-2"></span>Выполняется очистка (rag_cleaner) и
    индексация документов...';
84 |     }
85 |

```

```

86 |   try {
87 |       const res = await fetch(`/api/user/rags/${encodeURIComponent(
activeUserRagId)}/build`, {
88 |           method: 'POST',
89 |           headers: { 'Content-Type': 'application/json' },
90 |           body: JSON.stringify({
91 |               files: selectedFiles.length > 0 ? selectedFiles : null
92 |           },
93 |           provider: 'local_tfidf'
94 |       ));
95 |
96 |       const data = await res.json();
97 |       if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
98 |
99 |       if (statusAlert) {
100 |           statusAlert.className = 'alert alert-success p-2 small m
t-2 mb-0 d-block';
101 |           statusAlert.innerHTML = `
Успешно! Создано <strong>${da
ta.chunks_count}</strong> чанков из <strong>${data.processed_f
iles?.length || 0}</strong> файлов.`;
102 |       }
103 |
104 |       await loadUserRags();
105 |   } catch (err) {
106 |       if (statusAlert) {
107 |           statusAlert.className = 'alert alert-danger p-2 small mt
-2 mb-0 d-block';
108 |           statusAlert.textContent = `Ошибка сборки RAG: ${err.mess
age}`;
109 |       }
110 |   } finally {
111 |       if (btn) btn.disabled = false;
112 |   }
113 | }
114 |
115 | /**

```

```

116 | * Perform semantic search in the active workspace collection.
117 | *
118 | * @returns {Promise<void>}
119 | */
120 | export async function executeActiveUserRagSearch() {
121 |     const activeUserRagId = getActiveUserRagId();
122 |     if (!activeUserRagId) return;
123 |
124 |     const queryInput = document.getElementById('active-rag-search-
    h-query');
125 |     const resultsContainer = document.getElementById('active-rag
    -search-results');
126 |     const btn = document.getElementById('btn-active-rag-search')
    ;
127 |     const query = queryInput?.value.trim();
128 |
129 |     if (!query) return;
130 |     if (btn) btn.disabled = true;
131 |     if (resultsContainer) {
132 |         resultsContainer.innerHTML = '<div class="text-center text
    -muted small py-3"><span class="spinner-border spinner-border-
    sm me-1"></span>Поиск...</div>';
133 |     }
134 |
135 |     try {
136 |         const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/search`, {
137 |             method: 'POST',
138 |             headers: { 'Content-Type': 'application/json' },
139 |             body: JSON.stringify({
140 |                 query: query,
141 |                 top_k: 5,
142 |                 min_score: 0.0
143 |             })
144 |         });
145 |
146 |         const data = await res.json();
147 |         if (!res.ok) throw new Error(data.detail || `HTTP ${res.st

```

```

    atus}`);
148
149     const results = data.results || [];
150     if (results.length === 0) {
151         resultsContainer.innerHTML = '<div class="text-center text-muted small py-3">Ничего не найдено по данному запросу.</div>';
152     }
153     return;
154 }
155 resultsContainer.innerHTML = results.map((item) => {
156     const scorePercent = Math.round((item.score || 0) * 100);
157     ;
158     return `
159         <div class="card mb-2 border border-secondary-subtle bg-body">
160             <div class="card-header py-1 px-2 d-flex justify-content-between align-items-center bg-body-tertiary">
161                 <span class="small fw-semibold text-truncate" style="max-width: 70%;">
162                     <i class="bi bi-file-earmark-text text-primary me-1"></i> ${escapeHtml(item.source_file || '')}
163                 </span>
164                 <span class="badge bg-primary-subtle text-primary border border-primary-subtle">${scorePercent}% match</span>
165             </div>
166             <div class="card-body p-2 small">
167                 <div class="text-secondary font-monospace" style="font-size: 0.8rem; white-space: pre-wrap;">${escapeHtml(item.text)}</div>
168             </div>
169         `;
170     }).join('');
171 } catch (err) {
172     if (resultsContainer) {
173         resultsContainer.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка поиска: ${escapeHtml(err.message)}

```



```

    </div>`;
174     }
175   } finally {
176     if (btn) btn.disabled = false;
177   }
178 }
179
180 /**
181  * Handle quick file upload directly into user workspace storage.
182  *
183  * @param {Event} event - Input change event.
184  * @returns {Promise<void>}
185  */
186 export async function handleQuickUploadToUserStorage(event) {
187   const files = event.target.files;
188   if (!files || files.length === 0) return;
189
190   const statusAlert = document.getElementById('user-rag-build-status-alert');
191   if (statusAlert) {
192     statusAlert.className = 'alert alert-info p-2 small mt-2 mb-0 d-block';
193     statusAlert.innerHTML = `<span class="spinner-border spinner-border-sm me-2"></span>Загрузка ${files.length} файл(ов)...`;
194   }
195
196   try {
197     for (const file of files) {
198       const formData = new FormData();
199       formData.append('file', file);
200       formData.append('subfolder', 'files');
201
202       const res = await fetch('/api/user/files/upload', {
203         method: 'POST',
204         body: formData
205       });

```

```

206     if (!res.ok) throw new Error(`Ошибка загрузки ${file.name}
    e}`);
207   }
208
209   if (statusAlert) {
210     statusAlert.className = 'alert alert-success p-2 small m
    t-2 mb-0 d-block';
211     statusAlert.innerHTML = `Загружено <strong>${files.length
    h}</strong> файлов.`;
212   }
213
214   const activeUserRagId = getActiveUserRagId();
215   if (activeUserRagId) {
216     selectUserRag(activeUserRagId);
217   }
218 } catch (err) {
219   if (statusAlert) {
220     statusAlert.className = 'alert alert-danger p-2 small m
    t-2 mb-0 d-block';
221     statusAlert.textContent = err.message;
222   }
223 } finally {
224   event.target.value = '';
225 }
226 }
227
228 /**
229  * Synchronize Google Docs & Drive files directly into the act
    ive collection and build index.
230  *
231  * @returns {Promise<void>}
232  */
233 export async function syncGoogleDocsIntoActiveRag() {
234   const activeUserRagId = getActiveUserRagId();
235   if (!activeUserRagId) {
236     window.showToast?(`Пожалуйста, выберите или создайте колл
    екцию RAG.`, 'warning') || alert('Пожалуйста, выберите или соз
    дайте коллекцию RAG.');
```

```
237     return;
238 }
239
240 const queryInput = document.getElementById('import-gdocs-que
ry');
241 const accountInput = document.getElementById('import-gdocs-a
ccount');
242 const statusAlert = document.getElementById('import-gdocs-st
atus-alert');
243 const btn = document.getElementById('btn-confirm-import-gdoc
s');
244
245 const query = queryInput?.value.trim() || "mimeType = 'appli
cation/vnd.google-apps.document' and trashed = false";
246 const accountName = accountInput?.value.trim() || undefined;
247
248 if (btn) btn.disabled = true;
249 if (statusAlert) {
250     statusAlert.className = 'alert alert-info py-2 small mb-0
d-block';
251     statusAlert.innerHTML = '<span class="spinner-border spinn
er-border-sm me-2"></span>Экспорт Google Docs и построение RAG
индекса...';
252 }
253
254 try {
255     const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/sync-google-docs`, {
256         method: 'POST',
257         headers: { 'Content-Type': 'application/json' },
258         body: JSON.stringify({
259             query: query,
260             account_name: accountName
261         })
262     });
263
264     const data = await res.json();
265     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
```

```

    atus}`);
266
267     if (statusAlert) {
268         statusAlert.className = 'alert alert-success py-2 small
mb-0 d-block';
269         statusAlert.innerHTML = `
Синхронизировано! Загружено д
окументов: <strong>${data.downloaded_files?.length || 0}</stro
ng>, создано чанков: <strong>${data.chunks_count || 0}</strong
>.`;
270     }
271
272     await loadUserRags();
273     await selectUserRag(activeUserRagId);
274
275     setTimeout(() => {
276         const modalEl = document.getElementById('modal-import-gd
ocs');
277         if (modalEl && window.bootstrap?.Modal) {
278             const modal = bootstrap.Modal.getInstance(modalEl);
279             if (modal) modal.hide();
280         }
281     }, 1500);
282
283 } catch (err) {
284     if (statusAlert) {
285         statusAlert.className = 'alert alert-danger py-2 small m
b-0 d-block';
286         statusAlert.textContent = `Ошибка синхронизации: ${err.m
essage}`;
287     }
288 } finally {
289     if (btn) btn.disabled = false;
290 }
291 }
292
293

```

utils.js

```
1 /**
2  * =====
3  * Process Name: Windows Modules - Utils Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля utils.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/rag_tab/modules/utils.
11  *js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { formatBytes, escapeHtml, traverseEntry } from
15  *'/windows/api/webgui/rag_tab/modules/utils.js';
16  *
17  * File: utils.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/rag_tab/modules
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-01 13:04:40
23  * =====
24  */
25
26 /**
27  * Format bytes into human-readable string (B, KB, MB, GB).
28  *
29  * @param {number} bytes - Size in bytes.
30  * @returns {string} Formatted size string.
31  */
32 export function formatBytes(bytes) {
33   if (!bytes || bytes === 0) return '0 B';
```

```

32 |   const k = 1024;
33 |   const sizes = ['B', 'KB', 'MB', 'GB'];
34 |   const i = Math.floor(Math.log(bytes) / Math.log(k));
35 |   return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + ' '
    |   + sizes[i];
36 | }
37 |
38 | /**
39 |  * Escape HTML special characters to prevent XSS.
40 |  *
41 |  * @param {string} text - Raw string.
42 |  * @returns {string} Escaped HTML string.
43 |  */
44 | export function escapeHtml(text) {
45 |   if (!text) return '';
46 |   return String(text)
47 |     .replace(/&/g, '&amp;')
48 |     .replace(/</g, '&lt;')
49 |     .replace(/>/g, '&gt;')
50 |     .replace(/"/g, '&quot;')
51 |     .replace(/'/g, '&#039;');
52 | }
53 |
54 | /**
55 |  * Recursively traverse a file system entry.
56 |  *
57 |  * @param {FileSystemEntry} entry - FileSystem entry.
58 |  * @param {string} path - Current relative path.
59 |  * @returns {Promise<File[]>} Array of files with relative paths.
60 |  */
61 | export function traverseEntry(entry, path = '') {
62 |   return new Promise((resolve) => {
63 |     if (entry.isFile) {
64 |       entry.file((file) => {
65 |         const relPath = path ? `${path}/${file.name}` : file.name;
66 |         Object.defineProperty(file, 'customRelativePath', {

```

```

67         value: relPath,
68         writable: true
69     });
70     resolve([file]);
71 }, () => resolve([]));
72 } else if (entry.isDirectory) {
73     const dirReader = entry.createReader();
74     const entries = [];
75     const readEntries = () => {
76         dirReader.readEntries(async (result) => {
77             if (!result.length) {
78                 const currentPath = path ? `${path}/${entry.name}`
79 : entry.name;
80                 const promises = entries.map(e => traverseEntry(e,
81 currentPath));
82                 const nestedFiles = await Promise.all(promises);
83                 resolve(nestedFiles.flat());
84             } else {
85                 entries.push(...result);
86                 readEntries();
87             }
88         }, () => resolve([]));
89     };
90     readEntries();
91 } else {
92     resolve([]);
93 }
94
95 /**
96  * Extract all files from a DataTransfer object recursively.
97  *
98  * @param {DataTransfer} dataTransfer - Drag and drop DataTransfer object.
99  * @returns {Promise<File[]>} Array of files.
100  */
101 export async function getAllFilesFromDataTransfer(dataTransfer

```

```
    ) {
102 |   if (!dataTransfer) return [];
103 |   const items = dataTransfer.items;
104 |   if (items && items.length > 0 && items[0].webkitGetAsEntry)
    {
105 |     const queue = [];
106 |     for (let i = 0; i < items.length; i++) {
107 |       const entry = items[i].webkitGetAsEntry();
108 |       if (entry) queue.push(traverseEntry(entry));
109 |     }
110 |     const results = await Promise.all(queue);
111 |     return results.flat();
112 |   }
113 |   if (dataTransfer.files && dataTransfer.files.length > 0) {
114 |     return Array.from(dataTransfer.files);
115 |   }
116 |   return [];
117 | }
118 |
```


index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/rc/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/rc/index.html" class="w
15 |        -100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/rc
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!DOCTYPE html>
28 <html lang="ru">
29 <head>
30 |    <meta charset="UTF-8">
31 |    <meta name="viewport" content="width=device-width, initial-s
32 |    cale=1.0, maximum-scale=1.0, user-scalable=no">
33 |    <title>Медиа Плейер - Пульт ДУ</title>
34 |
35 |    <!-- Bootstrap 5 CSS -->
```

```
32 | <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
    | t/css/bootstrap.min.css" rel="stylesheet">
33 | <!-- Bootstrap Icons -->
34 | <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    | 0.0/font/bootstrap-icons.css" rel="stylesheet">
35 |
36 | <!-- Telegram Web Apps SDK -->
37 | <script src="https://telegram.org/js/telegram-web-app.js"></
    | script>
38 | <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
    | js"></script>
39 |
40 | <style>
41 |     :root {
42 |         --tg-theme-bg-color: #0d1117;
43 |         --tg-theme-text-color: #c9d1d9;
44 |         --tg-theme-hint-color: #8b949e;
45 |         --tg-theme-secondary-bg-color: #161b22;
46 |         --primary-gradient: linear-gradient(135deg, #1f6feb 0%,
    | #094cb5 100%);
47 |         --accent-gradient: linear-gradient(135deg, #ff7b00 0%, #
    | ffae00 100%);
48 |         --success-gradient: linear-gradient(135deg, #238636 0%,
    | #2ea043 100%);
49 |     }
50 |
51 |     body {
52 |         background: var(--tg-theme-bg-color);
53 |         color: var(--tg-theme-text-color);
54 |         font-family: system-ui, sans-serif;
55 |         min-height: 100vh;
56 |         display: flex;
57 |         flex-direction: column;
58 |         justify-content: center;
59 |         align-items: center;
60 |         padding: 10px;
61 |     }
62 |
```

```
63 | .rc-container, .auth-container, .scan-container {
64 |     width: 100%;
65 |     max-width: 480px;
66 |     padding: 2px;
67 | }
68 |
69 | .card-rc, .card-auth, .card-scan {
70 |     background-color: var(--tg-theme-secondary-bg-color);
71 |     border: 1px solid #30363d;
72 |     border-radius: 16px;
73 |     padding: 1px;
74 |     box-shadow: 0 8px 32px rgba(0,0,0,0.6);
75 |     backdrop-filter: blur(8px);
76 | }
77 |
78 | .btn-circle-lg {
79 |     width: 64px;
80 |     height: 64px;
81 |     border-radius: 50%;
82 |     display: flex;
83 |     align-items: center;
84 |     justify-content: center;
85 | }
86 |
87 | .btn-circle-xl {
88 |     width: 80px;
89 |     height: 80px;
90 |     border-radius: 50%;
91 |     display: flex;
92 |     align-items: center;
93 |     justify-content: center;
94 | }
95 |
96 | .track-title {
97 |     font-size: 1.15rem;
98 |     font-weight: 600;
99 |     color: #58a6ff;
100 | }
```

```
101
102 .playlist-box {
103     max-height: 200px;
104     overflow-y: auto;
105     border-radius: 8px;
106     background: #0d1117;
107 }
108
109 .list-group-item-action:hover {
110     background-color: #21262d !important;
111 }
112
113 .btn-carousel {
114     background: var(--accent-gradient);
115     border: none;
116     color: #000;
117     font-weight: bold;
118     transition: all 0.3s ease;
119     box-shadow: 0 4px 15px rgba(255, 123, 0, 0.3);
120 }
121 .btn-carousel:hover {
122     transform: translateY(-2px);
123     box-shadow: 0 6px 20px rgba(255, 123, 0, 0.5);
124     background: linear-gradient(135deg, #ffae00 0%, #ff7b00
125 100%);
126     color: #000;
127 }
128 .btn-carousel:active {
129     transform: translateY(1px);
130 }
131 .btn-carousel:hover i {
132     display: inline-block;
133     animation: spin 1s linear infinite;
134 }
135 @keyframes spin {
136     100% { transform: rotate(360deg); }
137 }
```

```
138 | /* Chat bubbles styling */
139 | .chat-msg {
140 |     display: flex;
141 |     flex-direction: column;
142 |     max-width: 85%;
143 |     margin-bottom: 8px;
144 | }
145 | .chat-msg.user {
146 |     align-self: flex-end;
147 | }
148 | .chat-msg.ai, .chat-msg.system {
149 |     align-self: flex-start;
150 | }
151 | .chat-msg.user .msg-bubble {
152 |     background: var(--primary-gradient);
153 |     color: #fff;
154 |     border-radius: 12px 12px 0 12px;
155 |     box-shadow: 0 2px 8px rgba(31, 111, 235, 0.2);
156 | }
157 | .chat-msg.ai .msg-bubble {
158 |     background-color: #21262d;
159 |     color: #c9d1d9;
160 |     border-radius: 12px 12px 12px 0;
161 |     border: 1px solid #30363d;
162 | }
163 | .chat-msg.system .msg-bubble {
164 |     background-color: rgba(255, 255, 255, 0.05);
165 |     color: #8b949e;
166 |     border-radius: 8px;
167 |     border-left: 3px solid #58a6ff;
168 | }
169 | .msg-bubble {
170 |     padding: 8px 12px;
171 |     font-size: 0.9rem;
172 |     word-break: break-word;
173 | }
174 | .msg-time {
175 |     font-size: 0.7rem;
```

```
176 |     color: #8b949e;
177 |     margin-top: 2px;
178 |     padding: 0 4px;
179 | }
180 | .chat-msg.user .msg-time {
181 |     text-align: right;
182 | }
183 | .animate-pulse {
184 |     animation: pulse 1.5s infinite;
185 | }
186 | @keyframes pulse {
187 |     0% { transform: scale(1); opacity: 1; }
188 |     50% { transform: scale(1.15); opacity: 0.7; }
189 |     100% { transform: scale(1); opacity: 1; }
190 | }
191 |
192 | .search-result-card {
193 |     background: rgba(88, 166, 255, 0.07);
194 |     border: 1px solid rgba(88, 166, 255, 0.2);
195 |     border-radius: 8px;
196 |     transition: all 0.2s ease;
197 | }
198 | .search-result-card:hover {
199 |     background: rgba(88, 166, 255, 0.12);
200 |     border-color: rgba(88, 166, 255, 0.4);
201 | }
202 |
203 | /* Auth panel specific */
204 | .form-control {
205 |     background: #0d1117;
206 |     border: 1px solid #30363d;
207 |     color: #c9d1d9;
208 | }
209 | .form-control:focus {
210 |     background: #0d1117;
211 |     border-color: #388bfd;
212 |     color: #c9d1d9;
213 |     box-shadow: 0 0 0 2px rgba(56,139,253,0.3);
```

```
214 |     }
215 |     .btn-auth-primary {
216 |         background: var(--primary-gradient);
217 |         border: none;
218 |         color: #fff;
219 |     }
220 |     .btn-auth-success {
221 |         background: var(--success-gradient);
222 |         border: none;
223 |         color: #fff;
224 |     }
225 |
226 |     /* Scanning styles */
227 |     .scan-radar {
228 |         position: relative;
229 |         width: 100px;
230 |         height: 100px;
231 |         margin: 0 auto 20px;
232 |         border-radius: 50%;
233 |         border: 2px solid rgba(88, 166, 255, 0.3);
234 |         display: flex;
235 |         align-items: center;
236 |         justify-content: center;
237 |     }
238 |     .scan-radar::after {
239 |         content: '';
240 |         position: absolute;
241 |         width: 100%;
242 |         height: 100%;
243 |         border-radius: 50%;
244 |         border: 2px solid #58a6ff;
245 |         animation: radar-ping 2s infinite linear;
246 |     }
247 |     @keyframes radar-ping {
248 |         0% { transform: scale(1); opacity: 1; }
249 |         100% { transform: scale(1.6); opacity: 0; }
250 |     }
251 |     .player-list-item {
```

```
252     background: #161b22;
253     border: 1px solid #30363d;
254     border-radius: 8px;
255     padding: 12px;
256     margin-bottom: 8px;
257     display: flex;
258     justify-content: space-between;
259     align-items: center;
260     transition: all 0.2s ease;
261 }
262 .player-list-item:hover {
263     border-color: #58a6ff;
264     background: rgba(88, 166, 255, 0.05);
265 }
266
267 /* Theme toggle button */
268 .theme-toggle-btn {
269     position: absolute;
270     top: 10px;
271     right: 10px;
272     background: rgba(255, 255, 255, 0.1);
273     border: 1px solid rgba(255, 255, 255, 0.2);
274     color: #fff;
275     width: 36px;
276     height: 36px;
277     border-radius: 50%;
278     cursor: pointer;
279     display: flex;
280     align-items: center;
281     justify-content: center;
282     font-size: 16px;
283     z-index: 1000;
284 }
285 .theme-toggle-btn:hover {
286     background: rgba(255, 255, 255, 0.2);
287 }
288
289 /* Light theme styles */
```



```
290 | .theme-light .card-rc,  
291 | .theme-light .card-auth,  
292 | .theme-light .card-scan {  
293 |     background-color: #ffffff;  
294 |     border-color: #d0d7de;  
295 | }  
296 | .theme-light .track-title {  
297 |     color: #0969da;  
298 | }  
299 | .theme-light .form-control {  
300 |     background: #ffffff;  
301 |     border-color: #d0d7de;  
302 |     color: #24292f;  
303 | }  
304 | .theme-light .form-control:focus {  
305 |     background: #ffffff;  
306 |     border-color: #0969da;  
307 | }  
308 | .theme-light .btn-primary {  
309 |     background: #2da44e;  
310 |     border-color: #2da44e;  
311 | }  
312 | .theme-light .btn-primary:hover {  
313 |     background: #2da44e;  
314 | }  
315 | .theme-light .btn-outline-light {  
316 |     color: #24292f;  
317 |     border-color: #d0d7de;  
318 | }  
319 | .theme-light .btn-outline-light:hover {  
320 |     background: #f6f8fa;  
321 | }  
322 | .theme-light .btn-circle-lg {  
323 |     background: #ffffff;  
324 |     border-color: #d0d7de;  
325 |     color: #24292f;  
326 | }  
327 | .theme-light .btn-circle-lg:hover {
```

```
328     background: #f6f8fa;
329 }
330 .theme-light .text-muted {
331     color: #6e7781 !important;
332 }
333 .theme-light .chat-msg.user .msg-bubble {
334     background: #0969da;
335 }
336 .theme-light .chat-msg.ai .msg-bubble {
337     background-color: #f6f8fa;
338     color: #24292f;
339 }
340 .theme-light .chat-msg.system .msg-bubble {
341     background-color: #f6f8fa;
342     color: #6e7781;
343 }
344 .theme-light .player-list-item {
345     background: #ffffff;
346     border-color: #d0d7de;
347 }
348 .theme-light .player-list-item:hover {
349     background: #f6f8fa;
350 }
351 .theme-light .list-group-item {
352     background: #ffffff;
353     border-color: #d0d7de;
354     color: #24292f;
355 }
356 .theme-light .list-group-item:hover {
357     background: #f6f8fa;
358 }
359 .theme-light .form-range {
360     color-scheme: light;
361 }
362 </style>
363 </head>
364 <body>
365
```

```
366 | <!-- 1. AUTHENTICATION PANEL -->
367 | <div class="auth-container" id="auth-panel">
368 |     <div class="card-auth">
369 |         <div class="text-center mb-4">
370 |             <i class="bi bi-shield-lock-fill fs-1 text-primary"></i>
371 |             <h4 class="mt-2 text-white">Вход в пульт ДУ</h4>
372 |             <p class="text-muted small">Пожалуйста, войдите в аккаунт для продолжения</p>
373 |         </div>
374 |
375 |         <div id="auth-alert" class="alert alert-danger d-none" role="alert"></div>
376 |
377 |         <!-- Login Form -->
378 |         <div id="auth-form-login">
379 |             <form id="form-login">
380 |                 <div class="mb-3">
381 |                     <label for="login-email" class="form-label text-muted small">Email</label>
382 |                     <input type="email" class="form-control" id="login-email" required>
383 |                 </div>
384 |                 <div class="mb-3">
385 |                     <label for="login-password" class="form-label text-muted small">Пароль</label>
386 |                     <input type="password" class="form-control" id="login-password" required>
387 |                 </div>
388 |                 <button type="submit" class="btn btn-auth-primary w-100 py-2.5 rounded-pill mb-3">Войти</button>
389 |                 <p class="text-center small mb-0">Нет аккаунта? <a href="#" id="go-to-register" class="text-info">Зарегистрироваться</a></p>
390 |             </form>
391 |         </div>
392 |
393 |         <!-- Register Form -->
```

```

394 |     <div id="auth-form-register" class="d-none">
395 |         <form id="form-register">
396 |             <div class="mb-3">
397 |                 <label for="register-name" class="form-label text-
| muted small">Имя</label>
398 |                 <input type="text" class="form-control" id="regist
| er-name" required>
399 |             </div>
400 |             <div class="mb-3">
401 |                 <label for="register-email" class="form-label text
| -muted small">Email</label>
402 |                 <input type="email" class="form-control" id="regis
| ter-email" required>
403 |             </div>
404 |             <div class="mb-3">
405 |                 <label for="register-password" class="form-label t
| ext-muted small">Пароль</label>
406 |                 <input type="password" class="form-control" id="re
| gister-password" required>
407 |             </div>
408 |             <button type="submit" class="btn btn-auth-success w-
| 100 py-2.5 rounded-pill mb-3">Зарегистрироваться</button>
409 |             <p class="text-center small mb-0">Уже есть аккаунт?
| <a href="#" id="go-to-login" class="text-info">Войти</a></p>
410 |         </form>
411 |     </div>
412 |
413 |     <!-- Verify Form -->
414 |     <div id="auth-form-verify" class="d-none">
415 |         <p class="text-center small">На вашу почту отправлен 6
| -значный код подтверждения. Пожалуйста, введите его ниже.</p>
416 |         <form id="form-verify">
417 |             <div class="mb-3">
418 |                 <label for="verify-code" class="form-label text-mu
| ted small">Код подтверждения</label>
419 |                 <input type="text" class="form-control text-center
| fs-3 py-2" id="verify-code" maxlength="6" placeholder="XXXXXX
| " required>

```

```

420         </div>
421         <button type="submit" class="btn btn-info w-100 py-2
422         .5 text-white rounded-pill mb-3">Подтвердить</button>
423         <p class="text-center small mb-0"><a href="#" id="ve
424         rify-cancel" class="text-info">Назад к входу</a></p>
425         </form>
426     </div>
427
428     <div class="my-4 text-center position-relative">
429         <hr class="border-secondary">
430         <span class="px-2 text-muted position-absolute top-50
431         start-50 translate-middle small" style="background-color: var(
432         --tg-theme-secondary-bg-color);">Или войти через</span>
433     </div>
434
435     <div class="d-flex flex-column gap-2 align-items-center"
436     >
437         <button id="google-signin-btn" class="btn btn-outline-
438         light w-100 rounded-pill d-flex align-items-center justify-con
439         tent-center gap-2 py-2">
440             <i class="bi bi-google text-danger"></i>
441             <span>Google</span>
442         </button>
443     </div>
444 </div>
445
446 <!-- 2. COMPUTER DISCOVERY SCANNER PANEL -->
447 <div class="scan-container d-none" id="scan-panel">
448     <div class="card-scan text-center">
449         <div class="scan-radar">
450             <i class="bi bi-laptop fs-2 text-primary"></i>
451         </div>
452         <h4 class="text-white" id="scan-title">Поиск открытых ко
453         мпьютеров</h4>
454         <p class="text-muted small" id="scan-status">Сканируем с
455         еть на наличие открытых сессий davidka.net...</p>
456     </div>

```

```

449         <div id="discovered-players-list" class="mt-4 text-start
    "></div>
450
451         <button id="btn-retry-scan" class="btn btn-outline-prima
    ry w-100 py-2.5 rounded-pill mt-3 d-none">
452             <i class="bi bi-arrow-clockwise"></i> Обновить поиск
453         </button>
454
455         <button id="btn-scan-logout" class="btn btn-sm btn-outli
    ne-danger w-100 mt-2 py-2 border-0 small">
456             <i class="bi bi-box-arrow-right"></i> Выйти из аккаунт
    а
457         </button>
458     </div>
459 </div>
460
461 <!-- 3. REMOTE CONTROL MAIN PANEL -->
462 <div class="rc-container d-none" id="rc-app-container">
463     <div class="card-rc">
464         <!-- Status Badge & User info -->
465         <div class="d-flex justify-content-between align-items-c
    enter mb-3">
466             <span class="badge bg-secondary" id="ws-status"><i cla
    ss="bi bi-wifi-off"></i> Подключение...</span>
467             <div class="d-flex gap-2 align-items-center">
468                 <button class="theme-toggle-btn" id="theme-toggle-bt
    n" title="Переключить тему">
469                     <i class="bi bi-brightness-high"></i>
470                 </button>
471                 <button class="btn btn-sm btn-outline-secondary py-1
    px-2 border-secondary small" id="btn-change-player" title="См
    енить компьютер">
472                     <i class="bi bi-arrow-left-right"></i> <span id="r
    oom-info">ПК: default</span>
473                 </button>
474             </div>
475         </div>
476

```

```

477 |         <!-- Now Playing Info -->
478 |         <div class="text-center my-3 p-3 rounded bg-dark bg-opac
| ity-25">
479 |             <i class="bi bi-music-note-beamed fs-1 text-primary mb
| -3 d-block"></i>
480 |             <div id="track-title" class="track-title mb-1 text-tru
| ncate">Файл не выбран</div>
481 |             <small id="time-display" class="text-muted">00:00 / 00
| :00</small>
482 |
483 |         <!-- Progress Bar -->
484 |         <div class="mt-3">
485 |             <input type="range" class="form-range" id="seek-slid
| er" min="0" max="100" value="0">
486 |             </div>
487 |         </div>
488 |
489 |         <!-- Central Controls -->
490 |         <div class="d-flex justify-content-center align-items-ce
| nter gap-3 my-4">
491 |             <button class="btn btn-outline-light btn-circle-lg" id
| ="btn-prev" title="Предыдущий"><i class="bi bi-skip-start-fill
| fs-3"></i></button>
492 |             <button class="btn btn-primary btn-circle-xl" id="btn-
| playtoggle" title="Воспроизведение/Пауза"><i class="bi bi-play
| -fill fs-1" id="play-icon"></i></button>
493 |             <button class="btn btn-outline-light btn-circle-lg" id
| ="btn-next" title="Следующий"><i class="bi bi-skip-end-fill fs
| -3"></i></button>
494 |         </div>
495 |
496 |         <!-- Carousel / Random Movie Control -->
497 |         <div class="my-3 px-2">
498 |             <button class="btn btn-carousel w-100 py-2.5 rounded-p
| ill d-flex align-items-center justify-content-center gap-2" id
| ="btn-carousel" title="Случайный фильм">
499 |                 <i class="bi bi-arrow-repeat fs-5"></i>
500 |                 <span>Карусель</span>

```

```
501         </button>
502     </div>
503
504     <!-- Quick Google Workspace & Social Control Panel Butto
ns -->
505     <div class="row g-2 px-2 my-2">
506         <div class="col-6 col-sm-3">
507             <button class="btn btn-outline-primary w-100 py-2 ro
unded-3 d-flex align-items-center justify-content-center gap-1
small" id="rc-btn-check-mail" title="Проверить почту Gmail">
508                 <i class="bi bi-envelope-check-fill text-primary">
</i>
509                 <span>Почта</span>
510             </button>
511         </div>
512         <div class="col-6 col-sm-3">
513             <button class="btn btn-outline-info w-100 py-2 round
ed-3 d-flex align-items-center justify-content-center gap-1 sm
all" id="rc-btn-search-gdocs" title="Поиск в Google Docs">
514                 <i class="bi bi-file-earmark-text-fill text-info">
</i>
515                 <span>Docs</span>
516             </button>
517         </div>
518         <div class="col-6 col-sm-3">
519             <button class="btn btn-outline-success w-100 py-2 ro
unded-3 d-flex align-items-center justify-content-center gap-1
small" id="rc-btn-gsheets" title="Работа с Google Sheets">
520                 <i class="bi bi-file-earmark-spreadsheet-fill text
-success"></i>
521                 <span>Таблицы</span>
522             </button>
523         </div>
524         <div class="col-6 col-sm-3">
525             <button class="btn btn-outline-primary w-100 py-2 ro
unded-3 d-flex align-items-center justify-content-center gap-1
small" id="rc-btn-facebook" title="Опубликовать в Facebook">
526                 <i class="bi bi-facebook text-primary"></i>
```



```
527         <span>Facebook</span>
528     </button>
529 </div>
530 </div>
531
532 <!-- Volume Controls -->
533 <div class="d-flex align-items-center gap-3 my-3 px-2">
534     <i class="bi bi-volume-down-fill text-muted fs-4"></i>
535     <input type="range" class="form-range" id="volume-slider" min="0" max="1" step="0.05" value="1">
536     <i class="bi bi-volume-up-fill text-muted fs-4"></i>
537 </div>
538
539 <!-- Voice Search & AI Chat -->
540 <div class="mt-4 p-3 rounded border border-secondary bg-dark bg-opacity-25">
541     <h6 class="text-center mb-3"><i class="bi bi-mic"></i>
542     Голосовой Ассистент</h6>
543
544     <!-- Status -->
545     <div class="text-muted small text-center mb-3" id="voice-status">Готов к работе</div>
546
547     <!-- Control buttons -->
548     <div class="d-flex flex-column gap-3 align-items-center">
549         <button class="btn btn-outline-danger btn-circle-xl"
550             id="btn-mic" title="Нажмите, чтобы говорить">
551             <i class="bi bi-mic-fill fs-1" id="mic-icon"></i>
552         </button>
553
554         <!-- Language Selection -->
555         <div class="btn-group w-100 mt-2" role="group">
556             <input type="radio" class="btn-check" name="lang-radio" id="lang-ru" value="ru-RU" autocomplete="off" checked>
557             <label class="btn btn-outline-secondary" for="lang-ru">RU</label>
```

```

557         <input type="radio" class="btn-check" name="lang-r
      | adio" id="lang-en" value="en-US" autocomplete="off">
558         <label class="btn btn-outline-secondary" for="lang
      | -en">EN</label>
559
560         <input type="radio" class="btn-check" name="lang-r
      | adio" id="lang-he" value="he-IL" autocomplete="off">
561         <label class="btn btn-outline-secondary" for="lang
      | -he">HE</label>
562     </div>
563 </div>
564 </div>
565
566 <!-- Playlist Picker -->
567 <div class="mt-4">
568     <h6 class="border-bottom border-secondary pb-2"><i cla
      | ss="bi bi-collection-play"></i> Список воспроизведения плеера<
      | /h6>
569     <div class="list-group list-group-flush playlist-box"
      | id="playlist-items">
570         <div class="text-muted small p-3 text-center">Ожидан
      | ие подключения к плееру...</div>
571     </div>
572 </div>
573
574 <!-- User Manual (Read & Listen) -->
575 <div class="mt-4 p-3 rounded border border-secondary bg-
      | dark bg-opacity-25">
576     <div class="d-flex justify-content-between align-items
      | -center border-bottom border-secondary pb-2 mb-2">
577         <h6 class="m-0"><i class="bi bi-info-circle"></i> Ин
      | струкция пользователя</h6>
578         <button class="btn btn-outline-info btn-sm d-flex al
      | ign-items-center gap-1" id="btn-read-manual" title="Прослушать
      | инструкцию">
579             <i class="bi bi-volume-up-fill" id="read-manual-ic
      | on"></i> <span>Прослушать</span>
580         </button>

```

```
581 |         </div>
582 |         <div id="user-manual-text" class="small text-muted" st
    | yle="line-height: 1.5; text-align: justify;">
583 |             Для дистанционного управления сначала откройте Медиа
    | плеер на вашем компьютере или телевизоре.
584 |             Данный пульт автоматически подключится к нему через
    | сервер.
585 |             Вы можете нажимать кнопки для запуска файлов, переключе
    | ния треков, перемотки или изменения громкости.
586 |             Также доступно голосовое управление кнопкой микрофон
    | а.
587 |             Простые команды: «пауза», «плей», «громче», «тише»,
    | «вперед» и «назад» сработают мгновенно.
588 |             А любые другие фразы будут отправлены вашему AI-асси
    | стенту Gemini для помощи в выборе фильма или ответа на вопросы
    | .
589 |         </div>
590 |     </div>
591 | </div>
592 | </div>
593 |
594 | <script>
595 |     // API Module
596 |     const api = {
597 |         async fetch(url, options = {}) {
598 |             const response = await fetch(url, options);
599 |             if (!response.ok) {
600 |                 let msg = response.statusText;
601 |                 try {
602 |                     const data = await response.json();
603 |                     msg = data.detail || msg;
604 |                 } catch {}
605 |                 throw new Error(msg);
606 |             }
607 |             return response.json();
608 |         },
609 |         auth: {
610 |             async check() {
```

```

611         try {
612             const response = await fetch('/auth/check');
613             return await response.json();
614         } catch {
615             return { authenticated: false };
616         }
617     },
618     async register(email, password, name) {
619         return api.fetch('/auth/register', {
620             method: 'POST',
621             headers: { 'Content-Type': 'application/json' },
622             body: JSON.stringify({ email, password, name })
623         });
624     },
625     async verify(email, code) {
626         return api.fetch('/auth/verify', {
627             method: 'POST',
628             headers: { 'Content-Type': 'application/json' },
629             body: JSON.stringify({ email, code })
630         });
631     },
632     async loginEmail(email, password) {
633         return api.fetch('/auth/login/email', {
634             method: 'POST',
635             headers: { 'Content-Type': 'application/json' },
636             body: JSON.stringify({ email, password })
637         });
638     },
639     async logout() {
640         return api.fetch('/auth/logout', { method: 'POST' });
641     };
642 }
643 };
644
645 // Panels state
646 const authPanel = document.getElementById('auth-panel');
647 const scanPanel = document.getElementById('scan-panel');

```

```
648     const rcAppContainer = document.getElementById('rc-app-con
tainer');
649
650     const authFormLogin = document.getElementById('auth-form-l
ogin');
651     const authFormRegister = document.getElementById('auth-for
m-register');
652     const authFormVerify = document.getElementById('auth-form-
verify');
653     const authAlert = document.getElementById('auth-alert');
654
655     let currentVerifyEmail = '';
656     let currentRoomName = '';
657     let ws = null;
658     let isSeeking = false;
659     let playerState = { duration: 0, currentTime: 0, isPlaying
: false, volume: 1 };
660
661     // Toggle panels helpers
662     function showPanel(panelId) {
663         authPanel.classList.add('d-none');
664         scanPanel.classList.add('d-none');
665         rcAppContainer.classList.add('d-none');
666
667         if (panelId === 'auth') authPanel.classList.remove('d-no
ne');
668         else if (panelId === 'scan') scanPanel.classList.remove(
'd-none');
669         else if (panelId === 'rc') rcAppContainer.classList.remo
ve('d-none');
670     }
671
672     // — AUTH PANEL LOGIC —————
673
674     document.getElementById('go-to-register').addEventListener
('click', (e) => {
675         e.preventDefault();
676         authAlert.classList.add('d-none');
```

```
676     authFormLogin.classList.add('d-none');
677     authFormRegister.classList.remove('d-none');
678 });
679
680     document.getElementById('go-to-login').addEventListener('c
lick', (e) => {
681         e.preventDefault();
682         authAlert.classList.add('d-none');
683         authFormRegister.classList.add('d-none');
684         authFormLogin.classList.remove('d-none');
685     });
686
687     document.getElementById('verify-cancel').addEventListener(
'click', (e) => {
688         e.preventDefault();
689         authAlert.classList.add('d-none');
690         authFormVerify.classList.add('d-none');
691         authFormLogin.classList.remove('d-none');
692     });
693
694     document.getElementById('google-signin-btn').addEventListe
ner('click', () => {
695         window.location.href = '/auth/google?next=' + encodeURICom
ponent(window.location.pathname + window.location.search);
696     });
697
698     // Login Form Submit
699     document.getElementById('form-login').addEventListener('su
bmit', async (e) => {
700         e.preventDefault();
701         authAlert.classList.add('d-none');
702         const email = document.getElementById('login-email').val
ue;
703         const pass = document.getElementById('login-password').v
alue;
704
705         try {
706             await api.auth.loginEmail(email, pass);
```

```
707 |         checkAndInit();
708 |     } catch (err) {
709 |         authAlert.textContent = err.message || 'Ошибка входа';
710 |         authAlert.classList.remove('d-none');
711 |     }
712 | });
713 |
714 | // Register Form Submit
715 | document.getElementById('form-register').addEventListener(
716 | 'submit', async (e) => {
717 |     e.preventDefault();
718 |     authAlert.classList.add('d-none');
719 |     const name = document.getElementById('register-name').va
720 | lue;
721 |     const email = document.getElementById('register-email').
722 | value;
723 |     const pass = document.getElementById('register-password'
724 | ).value;
725 |
726 |     try {
727 |         const res = await api.auth.register(email, pass, name)
728 | ;
729 |         currentVerifyEmail = email;
730 |         authFormRegister.classList.add('d-none');
731 |         authFormVerify.classList.remove('d-none');
732 |     } catch (err) {
733 |         authAlert.textContent = err.message || 'Ошибка регистр
734 | ации';
735 |         authAlert.classList.remove('d-none');
736 |     }
737 | });
738 |
739 | // Verify Form Submit
740 | document.getElementById('form-verify').addEventListener('s
741 | ubmit', async (e) => {
742 |     e.preventDefault();
743 |     authAlert.classList.add('d-none');
744 |     const code = document.getElementById('verify-code').valu
```

```

e;
738
739     try {
740         await api.auth.verify(currentVerifyEmail, code);
741         // After verify, try to log in
742         authFormVerify.classList.add('d-none');
743         authFormLogin.classList.remove('d-none');
744         authAlert.className = 'alert alert-success';
745         authAlert.textContent = 'Почта успешно подтверждена. Т
еперь войдите.';
746         authAlert.classList.remove('d-none');
747     } catch (err) {
748         authAlert.textContent = err.message || 'Неверный код';
749         authAlert.classList.remove('d-none');
750     }
751 });
752
753     // — COMPUTER DISCOVERY LOGIC —————
754
755     async function scanForPlayers() {
756         showPanel('scan');
757         document.getElementById('scan-title').textContent = 'Пои
ск открытых компьютеров';
758         document.getElementById('scan-status').textContent = 'Ск
анируем сеть на наличие открытых сессий davidka.net...';
759         document.getElementById('btn-retry-scan').classList.add(
'd-none');
760
761         const listDiv = document.getElementById('discovered-play
ers-list');
762         listDiv.innerHTML = '<div class="text-center p-3"><span
class="spinner-border spinner-border-sm text-primary"></span><
/div>';
763
764         try {
765             const response = await fetch('/api/control/active_play
ers');
766             const data = await response.json();

```



```

766 |     const players = data.players || [];
767 |     const currentUserEmail = data.user_email;
768 |
769 |     const mappedPlayers = players.map(p => {
770 |         if (p.includes('@')) {
771 |             return { room: p, name: `davidka.net (${p})` };
772 |         }
773 |         return { room: p, name: p === 'default' ? 'Основной
компьютер (davidka.net)' : `ПК: ${p}` };
774 |     });
775 |
776 |     if (mappedPlayers.length === 0) {
777 |         listDiv.innerHTML = `
778 |             <div class="alert alert-warning text-center small
mb-0">
779 |                 <i class="bi bi-exclamation-circle-fill"></i> От
крытые компьютеры не обнаружены.<br>
780 |                 Пожалуйста, откройте davidka.net на вашем компью
тере.
781 |             </div>`;
782 |         document.getElementById('scan-title').textContent =
'Компьютеры не найдены';
783 |         document.getElementById('btn-retry-scan').classList.
remove('d-none');
784 |     } else if (mappedPlayers.length === 1) {
785 |         // Exactly one computer found - connect automaticall
y
786 |         listDiv.innerHTML = `<div class="alert alert-success
text-center small">Обнаружен компьютер "${mappedPlayers[0].na
me}". Подключаемся...</div>`;
787 |         setTimeout(() => {
788 |             initStream(mappedPlayers[0].room);
789 |         }, 800);
790 |     } else {
791 |         // Multiple computers found - ask user to select one
792 |         document.getElementById('scan-title').textContent =
'Куда подключиться?';
793 |         document.getElementById('scan-status').textContent =

```

```

      'Обнаружено несколько открытых компьютеров. Выберите один: ';
794
795      listDiv.innerHTML = mappedPlayers.map(p => `
796          <div class="player-list-item">
797              <div>
798                  <i class="bi bi-laptop-fill text-primary me-2"
></i>
799                      <strong>${p.name}</strong>
800                  </div>
801                  <button class="btn btn-sm btn-primary rounded-pi
ll px-3" onclick="selectPlayer('${p.room}')">Выбрать</button>
802              </div>
803          `).join('');
804      document.getElementById('btn-retry-scan').classList.
remove('d-none');
805      }
806      } catch (err) {
807          listDiv.innerHTML = `<div class="alert alert-danger te
xt-center small">Ошибка сканирования: ${err.message}</div>`;
808          document.getElementById('btn-retry-scan').classList.re
move('d-none');
809      }
810      }
811
812      window.selectPlayer = function(room) {
813          initStream(room);
814      };
815
816      document.getElementById('btn-retry-scan').addEventListener
('click', scanForPlayers);
817
818      document.getElementById('btn-change-player').addEventListe
ner('click', () => {
819          // Disconnect current stream
820          if (ws) {
821              ws.close();
822              ws = null;
823          }

```

```

824     scanForPlayers();
825 });
826
827     document.getElementById('btn-scan-logout').addEventListener('click', async () => {
828         await api.auth.logout();
829         showPanel('auth');
830     });
831
832     // — MAIN TELEGRAM / WEB APP ROOM SETUP —————
833     function getRoomNameFromQuery() {
834         const urlParams = new URLSearchParams(window.location.search);
835         return urlParams.get('room') || '';
836     }
837
838     // Initialize Telegram WebApp if available
839     if (window.Telegram && window.Telegram.WebApp) {
840         const tg = window.Telegram.WebApp;
841         tg.ready();
842         tg.expand();
843
844         // Apply theme
845         function applyTgTheme() {
846             if (tg.themeParams.bg_color) {
847                 document.documentElement.style.setProperty('--tg-theme-bg-color', tg.themeParams.bg_color);
848             }
849             if (tg.themeParams.text_color) {
850                 document.documentElement.style.setProperty('--tg-theme-text-color', tg.themeParams.text_color);
851             }
852             if (tg.themeParams.secondary_bg_color) {
853                 document.documentElement.style.setProperty('--tg-theme-secondary-bg-color', tg.themeParams.secondary_bg_color);
854             }
855             if (tg.themeParams.hint_color) {

```

```

856 |         document.documentElement.style.setProperty('--tg-theme-hint-color', tg.themeParams.hint_color);
857 |     }
858 | }
859 | applyTgTheme();
860 | tg.onEvent('themeChanged', applyTgTheme);
861 |
862 | // Haptic feedback function
863 | window.triggerTgHaptic = function() {
864 |     if (tg.HapticFeedback) {
865 |         tg.HapticFeedback.impactOccurred('light');
866 |     }
867 | };
868 | } else {
869 |     window.triggerTgHaptic = function() {};
870 | }
871 |
872 | const statusBadge = document.getElementById('ws-status');
873 | const trackTitle = document.getElementById('track-title');
874 | const timeDisplay = document.getElementById('time-display'
875 | );
876 | const seekSlider = document.getElementById('seek-slider');
877 | const volumeSlider = document.getElementById('volume-slider');
878 | const playIcon = document.getElementById('play-icon');
879 | const playlistContainer = document.getElementById('playlist-items');
880 |
881 | function formatTime(seconds) {
882 |     if (!seconds || isNaN(seconds)) return '00:00';
883 |     const m = Math.floor(seconds / 60);
884 |     const s = Math.floor(seconds % 60);
885 |     return `${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`;
886 | }
887 |
888 | // Connect to WebSocket with Selected Room (Strictly One Stream)

```

```

888     function initStream(room) {
889         currentRoomName = room;
890         showPanel('rc');
891
892         const roomInfoElement = document.getElementById('room-in
fo');
893         if (roomInfoElement) {
894             const displayRoom = room.includes('@') ? 'kino.davidka
.net.tv' : room;
895             roomInfoElement.textContent = `Плеер: ${displayRoom}`;
896         }
897
898         // Close previous connection to satisfy "Strictly One St
ream" requirement
899         if (ws) {
900             ws.close();
901         }
902
903         const protocol = window.location.protocol === 'https:' ?
'wss:' : 'ws:';
904         const wsUrl = `${protocol}://${window.location.host}/api/
control/ws?role=remote&room=${encodeURIComponent(currentRoomNa
me)}`;
905
906         ws = new WebSocket(wsUrl);
907
908         ws.onopen = () => {
909             statusBadge.className = 'badge bg-success';
910             statusBadge.innerHTML = '<i class="bi bi-wifi"></i> По
дключено';
911         };
912
913         ws.onmessage = (event) => {
914             try {
915                 const msg = JSON.parse(event.data);
916                 if (msg.event === 'status_update') {
917                     updateUI(msg);
918                 } else if (msg.event === 'playlist_update') {

```

```

919         updatePlaylist(msg.files || []);
920     }
921 } catch (e) {
922     console.error(e);
923 }
924 };
925
926 ws.onclose = (e) => {
927     // Stop speech queue when connection drops
928     if (currentSpeechQueue) {
929         try {
930             currentSpeechQueue.stop();
931             currentSpeechQueue = null;
932         } catch(err) {}
933     }
934     if (currentSpeechAudio) {
935         try {
936             currentSpeechAudio.pause();
937             currentSpeechAudio.src = "";
938             currentSpeechAudio = null;
939         } catch(err) {}
940     }
941
942     // If we deliberately closed it to switch player, don'
t auto-reconnect
943     if (currentRoomName !== room) return;
944
945     statusBadge.className = 'badge bg-warning text-dark';
946     statusBadge.innerHTML = '<i class="bi bi-arrow-clockwi
se"></i> Переподключение...';
947     setTimeout(() => {
948         if (currentRoomName === room) {
949             initStream(room);
950         }
951     }, 3000);
952 };
953 }
954

```

```

955 |     function sendCommand(action, payload = {}) {
956 |         window.triggerTgHaptic();
957 |         if (ws && ws.readyState === WebSocket.OPEN) {
958 |             ws.send(JSON.stringify({ action, ...payload }));
959 |         }
960 |     }
961 |
962 |     function updateUI(data) {
963 |         playerState = { ...playerState, ...data };
964 |         trackTitle.textContent = data.fileName || 'Файл не выбра
965 | н';
966 |         timeDisplay.textContent = `${formatTime(data.currentTime
967 | )} / ${formatTime(data.duration)}`;
968 |
969 |         if (!isSeeking && data.duration > 0) {
970 |             seekSlider.max = Math.floor(data.duration);
971 |             seekSlider.value = Math.floor(data.currentTime);
972 |         }
973 |
974 |         if (typeof data.volume === 'number') {
975 |             volumeSlider.value = data.volume;
976 |         }
977 |
978 |         if (data.isPlaying) {
979 |             playIcon.className = 'bi bi-pause-fill fs-1';
980 |         } else {
981 |             playIcon.className = 'bi bi-play-fill fs-1';
982 |         }
983 |     }
984 |
985 |     function updatePlaylist(files) {
986 |         window.playlistFiles = files;
987 |         if (!files.length) {
988 |             playlistContainer.innerHTML = '<div class="text-muted
989 | small p-3 text-center">Плейлист пуст</div>';

```

```

990     playlistContainer.innerHTML = files.map((f, i) => `
991         <button class="list-group-item list-group-item-action
    bg-transparent text-white border-secondary py-2 px-3 small d-f
    lex justify-content-between align-items-center" onclick="playF
    ile(${i})">
992             <span class="text-truncate">${i + 1}. ${f.name}</spa
    n>
993             <span class="badge bg-secondary ms-2">${f.type === '
    series' ? 'Сериал' : 'Фильм'}</span>
994         </button>
995     `).join('');
996 }
997
998 window.playFile = (index) => {
999     sendCommand('play_file', { index });
1000 };
1001
1002 // — VOICE CONTROL & SEARCH CHAT MODULE —————
    _____
1003 let recognition = null;
1004 let isListening = false;
1005 let rcAbortController = null;
1006 let rcHistory = [];
1007
1008
1009
1010 function tryParseMovieCard(text) {
1011     const trimmed = text.trim();
1012     if (trimmed.startsWith('{') && trimmed.endsWith('}')) {
1013         try {
1014             const data = JSON.parse(trimmed);
1015             if (data.title || data.title_ru) return data;
1016         } catch (e) {}
1017     }
1018     const match = text.match(/\\{[\\s\\S]*?\\}/);
1019     if (match) {
1020         try {
1021             const data = JSON.parse(match[0]);

```



```

1022         if (data.title || data.title_ru) return data;
1023     } catch (e) {}
1024 }
1025 return null;
1026 }
1027
1028 function renderMovieCardHtml(card) {
1029     const title = card.title_ru || card.title;
1030     const subtitle = card.title_orig || '';
1031     const year = card.year || '';
1032     const category = card.main_category || '';
1033     const genres = card.genres ? card.genres.join(', ') : ''
1034     ;
1035     const directors = card.directors ? card.directors.join('
1036     , ') : '';
1037     const cast = card.cast ? card.cast.slice(0, 5).join(', '
1038     ) : '';
1039     const plot = card.plot || '';
1040     const whyWatch = card.why_watch || '';
1041
1042     let ratingHtml = '';
1043     if (card.rating) {
1044         if (card.rating.imdb) ratingHtml += `<span class="badg
1045         e bg-warning text-dark me-1">IMDb: ${card.rating.imdb}</span>`;
1046         ;
1047         if (card.rating.кинопоиск) ratingHtml += `<span class=
1048         "badge bg-danger text-white me-1">КП: ${card.rating.кинопоиск}
1049         </span>`;
1050         if (card.rating.tmbd) ratingHtml += `<span class="badg
1051         e bg-primary text-white">TMDB: ${card.rating.tmbd}</span>`;
1052     }
1053
1054     return `
1055     <div class="p-3 text-white" style="border-radius: 10px
1056     ; background: rgba(88, 166, 255, 0.08); border: 1px solid rgba
1057     (88, 166, 255, 0.2);">
1058     <h5 class="mb-1 text-info">${title}</h5>
1059     <div class="small text-muted mb-2">${subtitle} ${yea

```

```

r ? `(${year})` : ''</div>
1050     <div class="mb-2">${ratingHtml}</div>
1051     ${category ? `<div class="small text-muted mb-1" sty
le="font-size: 0.8rem;"><strong>Категория:</strong> ${category
}</div>` : ''}
1052     ${genres ? `<div class="small text-muted mb-1" style
="font-size: 0.8rem;"><strong>Жанры:</strong> ${genres}</div>`
: ''}
1053     ${directors ? `<div class="small text-muted mb-1" st
yle="font-size: 0.8rem;"><strong>Режиссер:</strong> ${director
s}</div>` : ''}
1054     ${cast ? `<div class="small text-muted mb-2" style="
font-size: 0.8rem;"><strong>В ролях:</strong> ${cast}...</div>
` : ''}
1055     ${plot ? `<p class="small text-secondary mb-2" style
="font-size: 0.85rem; line-height: 1.4;">${plot}</p>` : ''}
1056     ${whyWatch ? `<div class="small text-info mb-2" styl
e="font-size: 0.8rem;"><strong>Рекомендация:</strong> ${whyWat
ch}</div>` : ''}
1057     <div class="d-flex justify-content-end mt-2">
1058     <button class="btn btn-sm btn-primary rounded-pill
px-3" onclick="playMovieDirect('${title.replace(/'/g, "\\')}
')">
1059         <i class="bi bi-play-fill"></i> Запустить фильм
1060     </button>
1061 </div>
1062 </div>
1063 `;
1064 }
1065
1066 window.playMovieDirect = (title) => {
1067     if (!window.playlistFiles) return;
1068     const cleanT = title.toLowerCase();
1069     const foundIdx = window.playlistFiles.findIndex(f => (f.
name || '').toLowerCase().includes(cleanT));
1070     if (foundIdx !== -1) {
1071         playFile(foundIdx);
1072         speakText(`Запускаю ${window.playlistFiles[foundIdx].n

```

```

    ame}``);
1073     } else {
1074         speakText(`Фильм не найден в плейлисте`);
1075     }
1076 };
1077
1078 async function processChatMessage(text) {
1079     if (!text || !text.trim()) return;
1080     const query = text.trim();
1081
1082     stopAudio(); // <--- STOP AUDIO IMMEDIATELY
1083
1084     // 0. Transmit transcript to open computers via WebSocke
t
1085     if (ws && ws.readyState === WebSocket.OPEN) {
1086         sendCommand('transcript', { text: query });
1087     }
1088
1089     // 1. Check local control commands first
1090     if (handleLocalVoiceCommand(query.toLowerCase())) {
1091         speakText('Команда выполнена');
1092         return;
1093     }
1094
1095     // 2. Perform search in playlist
1096     const cleanQ = query.toLowerCase().replace(/^(найди|поис
к|включи|запусти|покажи|фильм|сериал)\s+/i, '').trim();
1097     let matches = [];
1098     if (window.playlistFiles && cleanQ.length > 1) {
1099         matches = window.playlistFiles.map((f, idx) => ({ ...f
, index: idx })).filter(f => {
1100             const name = (f.name || '').toLowerCase();
1101             const type = (f.type || '').toLowerCase();
1102             return name.includes(cleanQ) || (f.year && f.year.to
String().includes(cleanQ));
1103         });
1104     }
1105

```

```
1106 |         if (matches.length > 0) {
1107 |             speakText(`Найдено совпадений: ${matches.length}`);
1108 |
1109 |             const queryLower = query.toLowerCase();
1110 |             if (matches.length === 1 && (queryLower.includes('включи') || queryLower.includes('запусти') || queryLower.includes('плей') || queryLower.includes('играй')))) {
1111 |                 playFile(matches[0].index);
1112 |                 speakText(`Запускаю ${matches[0].name}`);
1113 |             }
1114 |             return;
1115 |         }
1116 |
1117 |         // 3. Forward to AI Chat API with Streaming
1118 |         if (rcAbortController) {
1119 |             try { rcAbortController.abort(); } catch(e) {}
1120 |         }
1121 |         rcAbortController = new AbortController();
1122 |
1123 |         const voiceStatus = document.getElementById('voice-status');
1124 |         voiceStatus.textContent = 'Обработка запроса моделью...';
1125 |
1126 |         let fullReply = '';
1127 |         let started = false;
1128 |
1129 |         try {
1130 |             const payload = {
1131 |                 message: query,
1132 |                 history: rcHistory,
1133 |                 generation_config: { is_rc: true }
1134 |             };
1135 |
1136 |             const response = await fetch('/api/chat', {
1137 |                 method: 'POST',
1138 |                 headers: { 'Content-Type': 'application/json' },
1139 |                 body: JSON.stringify(payload),
```

```
1140         signal: rcAbortController.signal
1141     });
1142
1143     if (!response.ok) {
1144         throw new Error(`Ошибка HTTP ${response.status}`);
1145     }
1146
1147     const reader = response.body.getReader();
1148     const decoder = new TextDecoder('utf-8');
1149     let buffer = '';
1150     let voiceReply = '';
1151
1152     while (true) {
1153         const { value, done } = await reader.read();
1154         if (done) break;
1155
1156         buffer += decoder.decode(value, { stream: true });
1157         const lines = buffer.split('\n');
1158         buffer = lines.pop();
1159
1160         for (const line of lines) {
1161             const trimmed = line.trim();
1162             if (!trimmed) continue;
1163             if (trimmed.startsWith('data: ')) {
1164                 try {
1165                     const data = JSON.parse(trimmed.substring(6));
1166                     if (data.error) {
1167                         throw new Error(data.error);
1168                     }
1169                     if (data.status) {
1170                         voiceStatus.textContent = data.status;
1171                     }
1172                     if (data.voice) {
1173                         voiceReply += data.voice;
1174                     }
1175                     if (data.text) {
1176                         fullReply += data.text;
1177                     }
1178                 } catch (error) {
1179                     console.error('Error parsing JSON:', error);
1180                 }
1181             }
1182         }
1183     }
1184 }
```

```

1178         } catch (e) {
1179             console.error(e);
1180         }
1181     }
1182 }
1183 }
1184
1185 voiceStatus.textContent = 'Готов к работе';
1186
1187 // — Обработка тегов <film> из ответа AI —————
1188
1189 const filmMatches = [...fullReply.matchAll(/<film>(.*?
)<\</film>/gi)];
1190
1191 if (filmMatches.length > 0) {
1192     const filmTitle = filmMatches[0][1].trim();
1193     try {
1194         const mediaResp = await fetch('/api/media/by-title
', {
1195             method: 'POST',
1196             headers: { 'Content-Type': 'application/json' },
1197             body: JSON.stringify({ title: filmTitle })
1198         });
1199         if (mediaResp.ok) {
1200             const mediaData = await mediaResp.json();
1201             const filmPath = mediaData.path;
1202             if (filmPath) {
1203                 sendCommand('play_file_by_path', { path: filmP
ath });
1204                 voiceStatus.textContent = `▶ Запускаю: ${film
Title}`;
1205             }
1206         }
1207     } catch (e) {
1208         console.error('Ошибка поиска пути фильма:', e);
1209     }
1210 }
1211 // —————

```

```
1210 |
1211 |     const card = tryParseMovieCard(fullReply);
1212 |     if (card) {
1213 |         speakText(`Найдена карточка фильма ${card.title_ru |
| card.title}. ${card.why_watch || card.plot || ''}`);
1214 |     } else {
1215 |         speakText(voiceReply || fullReply);
1216 |     }
1217 |
1218 |     rcHistory.push({ role: 'user', parts: [query] });
1219 |     rcHistory.push({ role: 'model', parts: [fullReply] });
1220 |     if (rcHistory.length > 20) {
1221 |         rcHistory = rcHistory.slice(-20);
1222 |     }
1223 |
1224 |     } catch (err) {
1225 |         if (err.name === 'AbortError') return;
1226 |         voiceStatus.textContent = 'Ошибка AI: ' + err.message;
1227 |     }
1228 | }
1229 |
1230 | function initVoiceControl() {
1231 |     const btnMic = document.getElementById('btn-mic');
1232 |     const micIcon = document.getElementById('mic-icon');
1233 |     const voiceStatus = document.getElementById('voice-status');
1234 |
1235 |     const SpeechRecognition = window.SpeechRecognition || window.webkitSpeechRecognition;
1236 |     if (!SpeechRecognition) {
1237 |         voiceStatus.textContent = 'Голосовой ввод не поддерживается вашим браузером.';
1238 |         if (btnMic) btnMic.disabled = true;
1239 |     } else {
1240 |         recognition = new SpeechRecognition();
1241 |
1242 |         // Dynamically get selected language
1243 |         const getSelectedLang = () => {
```

```
1244 |         const selected = document.querySelector('input[name=
| "lang-radio"]:checked');
1245 |         return selected ? selected.value : 'ru-RU';
1246 |     };
1247 |
1248 |     recognition.lang = getSelectedLang();
1249 |
1250 |     // Update language if radio button changes
1251 |     document.querySelectorAll('input[name="lang-radio"]').
|     forEach(radio => {
1252 |         radio.addEventListener('change', () => {
1253 |             recognition.lang = getSelectedLang();
1254 |         });
1255 |     });
1256 |
1257 |     recognition.continuous = false;
1258 |     recognition.interimResults = false;
1259 |
1260 |     recognition.onstart = () => {
1261 |         isListening = true;
1262 |         if (btnMic) btnMic.className = 'btn btn-danger btn-c
| ircle-lg flex-shrink-0';
1263 |         if (micIcon) micIcon.className = 'bi bi-mic-fill fs-
| 3 animate-pulse';
1264 |         voiceStatus.textContent = 'Слушаю...';
1265 |     };
1266 |
1267 |     recognition.onend = () => {
1268 |         isListening = false;
1269 |         if (btnMic) btnMic.className = 'btn btn-outline-dang
| er btn-circle-lg flex-shrink-0';
1270 |         if (micIcon) micIcon.className = 'bi bi-mic-fill fs-
| 3';
1271 |         voiceStatus.textContent = 'Готов к работе';
1272 |     };
1273 |
1274 |     recognition.onerror = (e) => {
1275 |         voiceStatus.textContent = `Ошибка ввода: ${e.error}`
```



```

1276 |         };
1277 |
1278 |         recognition.onresult = async (event) => {
1279 |             const text = event.results[0][0].transcript;
1280 |             if (text) {
1281 |                 processChatMessage(text);
1282 |             }
1283 |         };
1284 |
1285 |         if (btnMic) {
1286 |             btnMic.addEventListener('click', () => {
1287 |                 window.triggerTgHaptic();
1288 |                 // Unlock SpeechSynthesis for mobile devices
1289 |                 if ('speechSynthesis' in window) {
1290 |                     const silentUtterance = new SpeechSynthesisUtter
1291 |                 }
1292 |                 silentUtterance.volume = 0;
1293 |                 window.speechSynthesis.speak(silentUtterance);
1294 |             }
1295 |             if (isListening) {
1296 |                 recognition.stop();
1297 |             } else {
1298 |                 recognition.start();
1299 |             }
1300 |         });
1301 |     }
1302 | }
1303 |
1304 | let currentSpeechAudio = null;
1305 | let currentSpeechQueue = null;
1306 |
1307 | class AudioQueuePlayer {
1308 |     constructor(chunks, userSettings) {
1309 |         this.chunks = chunks;
1310 |         this.userSettings = userSettings;
1311 |         this.currentIndex = 0;

```

```
1312         this.isPlaying = false;
1313         this.audioElements = new Map();
1314         this.activeAudio = null;
1315         this._prefetch(0);
1316         this._prefetch(1);
1317     }
1318
1319     _prefetch(index) {
1320         if (index >= this.chunks.length || this.audioElements.
has(index)) return;
1321         const text = this.chunks[index];
1322         const queryParams = new URLSearchParams({
1323             text: text,
1324             system: this.userSettings.tts_system,
1325             voice: this.userSettings.tts_voice
1326         });
1327         const audioUrl = `/api/tts/synthesize?${queryParams.to
String()}`;
1328         const audio = new Audio(audioUrl);
1329         audio.load();
1330         this.audioElements.set(index, audio);
1331     }
1332
1333     async play() {
1334         this.isPlaying = true;
1335         while (this.currentIndex < this.chunks.length && this.
isPlaying) {
1336             this._prefetch(this.currentIndex);
1337             this._prefetch(this.currentIndex + 1);
1338             this._prefetch(this.currentIndex + 2);
1339
1340             const audio = this.audioElements.get(this.currentInd
ex);
1341             if (!audio) {
1342                 this.currentIndex++;
1343                 continue;
1344             }
1345
```

```
1346         this.activeAudio = audio;
1347         currentSpeechAudio = audio;
1348
1349         const playPromise = new Promise((resolve) => {
1350             audio.onended = () => resolve();
1351             audio.onerror = (e) => {
1352                 console.error(`Audio error for chunk ${this.cu
1353 | rrentIndex}:`, e);
1354                 this.stop(); // Останавливаем всю очередь при
1355 | ошибке загрузки
1356                 resolve();
1357             };
1358             });
1359
1360         try {
1361             await audio.play();
1362             await playPromise;
1363         } catch (err) {
1364             console.error("Audio playback interrupted/failed:"
1365 | , err);
1366         }
1367         this.currentIndex++;
1368     }
1369     this.isPlaying = false;
1370 }
1371
1372 stop() {
1373     this.isPlaying = false;
1374     if (this.activeAudio) {
1375         try {
1376             this.activeAudio.pause();
1377             this.activeAudio.src = "";
1378         } catch (e) {}
1379     }
1380     this.audioElements.clear();
1381 }
```

```

1381     function stopAudio() {
1382         if (currentSpeechQueue) {
1383             try {
1384                 currentSpeechQueue.stop();
1385                 currentSpeechQueue = null;
1386             } catch (e) {
1387                 console.error("Error stopping currentSpeechQueue:",
1388                     e);
1389             }
1390         }
1391         if (currentSpeechAudio) {
1392             try {
1393                 currentSpeechAudio.pause();
1394                 currentSpeechAudio.src = "";
1395                 currentSpeechAudio = null;
1396             } catch (e) {
1397                 console.error("Error stopping currentSpeechAudio:",
1398                     e);
1399             }
1400         }
1401         if ('speechSynthesis' in window) {
1402             try {
1403                 window.speechSynthesis.cancel();
1404             } catch (e) {
1405                 console.error("Error cancelling SpeechSynthesis:", e
1406                     );
1407             }
1408         }
1409     }
1410
1411     async function speakText(text) {
1412         if (!text) return;
1413
1414         // Stop any currently playing audio
1415         stopAudio();
1416
1417         // Remove markdown or HTML tags if any from voice output
1418         const cleanText = text.replace(/<[^\>]*>/g, '').replace(/
1419             [*_`#]/g, '').trim();

```

```

1415 |         if (!cleanText) return;
1416 |
1417 |         // Try backend TTS first
1418 |         try {
1419 |             const response = await fetch('/auth/settings');
1420 |             if (response.ok) {
1421 |                 const userSettings = await response.json();
1422 |                 if (userSettings.tts_enabled === 1 && userSettings.t
1423 | ts_system && userSettings.tts_system !== 'browser') {
1424 |                     const paragraphs = cleanText
1425 |                         .split(/\n+/)
1426 |                         .map(p => p.trim())
1427 |                         .filter(p => p.length > 0);
1428 |
1429 |                     let chunks = [];
1430 |                     for (const para of paragraphs) {
1431 |                         const sentences = para.match(/[^.!?]+[.!?]*\/g) |
1432 | | [para];
1433 |                         let currentChunk = "";
1434 |                         for (const sentence of sentences) {
1435 |                             const trimmedSentence = sentence.trim();
1436 |                             if (!trimmedSentence) continue;
1437 |                             if ((currentChunk + " " + trimmedSentence).len
1438 | gth > 200) {
1439 |                                 chunks.push(currentChunk.trim());
1440 |                                 currentChunk = trimmedSentence;
1441 |                             } else {
1442 |                                 currentChunk = (currentChunk + " " + trimmed
1443 | Sentence).trim();
1444 |                             }
1445 |                         }
1446 |                         if (currentChunk) {
1447 |                             chunks.push(currentChunk.trim());
1448 |                         }
1449 |                     }
1450 |                     chunks = chunks.filter(c => c.length > 0);
1451 |
1452 |                     if (chunks.length > 0) {

```

```
1449         currentSpeechQueue = new AudioQueuePlayer(chunks
1450         , userSettings);
1451         await currentSpeechQueue.play();
1452     }
1453     return;
1454 }
1455 } catch (e) {
1456     console.error('Failed to speak using backend TTS, fall
1457     ing back to browser:', e);
1458 }
1459 // Fallback to browser SpeechSynthesis
1460 if ('speechSynthesis' in window) {
1461     window.currentUtterance = new SpeechSynthesisUtterance
1462     (cleanText);
1463     window.currentUtterance.lang = 'ru-RU';
1464     window.speechSynthesis.speak(window.currentUtterance);
1465 }
1466
1467 function handleLocalVoiceCommand(text) {
1468     const t = text.toLowerCase();
1469     if (t.includes('пауз') || t.includes('стоп')) {
1470         sendCommand('pause');
1471         return true;
1472     }
1473     if (t.includes('продолж') || t.includes('плей') || t.inc
1474     ludes('играй') || t.includes('старт')) {
1475         sendCommand('play');
1476         return true;
1477     }
1478     if (t.includes('громче') || t.includes('прибавь звук'))
1479     {
1480         const targetVol = Math.min(1, playerState.volume + 0.2
1481         );
1482         sendCommand('set_volume', { level: targetVol });
1483         return true;
1484     }
1485 }
```

```

1481 |     }
1482 |     if (t.includes('тише') || t.includes('убавь звук')) {
1483 |         const targetVol = Math.max(0, playerState.volume - 0.2
1484 | );
1485 |         sendCommand('set_volume', { level: targetVol });
1486 |         return true;
1487 |     }
1488 |     if (t.includes('следующ') || t.includes('вперед')) {
1489 |         sendCommand('next');
1490 |         return true;
1491 |     }
1492 |     if (t.includes('предыдущ') || t.includes('назад')) {
1493 |         sendCommand('prev');
1494 |         return true;
1495 |     }
1496 |     if (t.includes('карусель') || t.includes('случайн') || t
1497 | .includes('рандом')) {
1498 |         sendCommand('play_random_movie');
1499 |         return true;
1500 |     }
1501 |     return false;
1502 | }
1503 |
1504 | async function sendToAiModel(message) {
1505 |     const response = await fetch('/api/chat', {
1506 |         method: 'POST',
1507 |         headers: { 'Content-Type': 'application/json' },
1508 |         body: JSON.stringify({ message, generation_config: { i
1509 | s_rc: true } })
1510 |     });
1511 |     if (!response.ok) {
1512 |         throw new Error('Не удалось получить ответ от AI');
1513 |     }
1514 |     const data = await response.json();
1515 |     return data.response || data.message || 'Модель не ответ

```

```
1515 // Set listeners
1516 document.getElementById('btn-playtoggle').addEventListener
('click', () => sendCommand('toggle_play'));
1517 document.getElementById('btn-prev').addEventListener('click',
() => sendCommand('prev'));
1518 document.getElementById('btn-next').addEventListener('click',
() => sendCommand('next'));
1519 document.getElementById('btn-carousel').addEventListener('
click', () => sendCommand('play_random_movie'));
1520
1521 const rcCheckMailBtn = document.getElementById('rc-btn-check-mail');
1522 if (rcCheckMailBtn) {
1523     rcCheckMailBtn.addEventListener('click', () => {
1524         processChatMessage('Проверь мою почту Gmail и покажи п
оследние непрочитанные письма.');
```

1525 });

1526 }

1527

1528 const rcSearchGdocsBtn = document.getElementById('rc-btn-search-gdocs');

1529 if (rcSearchGdocsBtn) {

1530 rcSearchGdocsBtn.addEventListener('click', () => {

1531 const query = prompt('Введите запрос для поиска в Google Docs / Drive:');

1532 if (query && query.trim()) {

1533 processChatMessage(`Найди в Google Docs / Google Drive документы по запросу: "\${query.trim()}".`);

1534 }

1535 });

1536 }

1537

1538 const rcGsheetsBtn = document.getElementById('rc-btn-gsheets');

1539 if (rcGsheetsBtn) {

1540 rcGsheetsBtn.addEventListener('click', () => {

1541 const query = prompt('Введите ID таблицы или команду для Google Sheets (например: "Прочитай диапазон A1:D10 из табли


```

    цы ID"):');
1542     if (query && query.trim()) {
1543         processChatMessage(`Google Sheets: ${query.trim()}`)
1544     };
1545     });
1546 }
1547
1548 const rcFacebookBtn = document.getElementById('rc-btn-face
book');
1549 if (rcFacebookBtn) {
1550     rcFacebookBtn.addEventListener('click', () => {
1551         const text = prompt('Введите текст для публикации в Fa
cebook:');
1552         if (text && text.trim()) {
1553             processChatMessage(`Опубликуй пост в Facebook с текс
том: "${text.trim()}"`);
1554         }
1555     });
1556 }
1557
1558 seekSlider.addEventListener('mousedown', () => { isSeeking
= true; });
1559 seekSlider.addEventListener('touchstart', () => { isSeekin
g = true; });
1560 seekSlider.addEventListener('change', (e) => {
1561     isSeeking = false;
1562     sendCommand('seek', { seconds: parseFloat(e.target.value
) });
1563 });
1564
1565 volumeSlider.addEventListener('change', (e) => {
1566     sendCommand('set_volume', { level: parseFloat(e.target.v
alue) });
1567 });
1568
1569 // — USER MANUAL AUDIO READER —————

```

```
1570 |     let isSpeakingManual = false;
1571 |     const btnRead = document.getElementById('btn-read-manual')
    |     ;
1572 |     const readIcon = document.getElementById('read-manual-icon
    |     ');
1573 |     const readTextSpan = btnRead ? btnRead.querySelector('span
    |     ') : null;
1574 |
1575 |     if (btnRead) {
1576 |         btnRead.addEventListener('click', () => {
1577 |             window.triggerTgHaptic();
1578 |             const text = document.getElementById('user-manual-text
    |             ').innerText;
1579 |
1580 |             if (isSpeakingManual) {
1581 |                 window.speechSynthesis.cancel();
1582 |                 isSpeakingManual = false;
1583 |                 if (readIcon) readIcon.className = 'bi bi-volume-up-
    |                 fill';
1584 |                 if (readTextSpan) readTextSpan.textContent = 'Прослы
    |                 шать';
1585 |                 btnRead.className = 'btn btn-outline-info btn-sm d-f
    |                 lex align-items-center gap-1';
1586 |             } else {
1587 |                 window.speechSynthesis.cancel();
1588 |                 const utterance = new SpeechSynthesisUtterance(text)
    |                 ;
1589 |                 utterance.lang = 'ru-RU';
1590 |
1591 |                 utterance.onend = () => {
1592 |                     isSpeakingManual = false;
1593 |                     if (readIcon) readIcon.className = 'bi bi-volume-u
    |                     p-fill';
1594 |                     if (readTextSpan) readTextSpan.textContent = 'Прос
    |                     лушать';
1595 |                     btnRead.className = 'btn btn-outline-info btn-sm d
    |                     -flex align-items-center gap-1';
1596 |                 };
```

```
1597 |
1598 |         utterance.onerror = () => {
1599 |             isSpeakingManual = false;
1600 |             if (readIcon) readIcon.className = 'bi bi-volume-u
| p-fill';
1601 |             if (readTextSpan) readTextSpan.textContent = 'Прос
| лушать';
1602 |             btnRead.className = 'btn btn-outline-info btn-sm d
| -flex align-items-center gap-1';
1603 |             };
1604 |
1605 |             window.speechSynthesis.speak(utterance);
1606 |             isSpeakingManual = true;
1607 |             if (readIcon) readIcon.className = 'bi bi-stop-fill'
| ;
1608 |             if (readTextSpan) readTextSpan.textContent = 'Остано
| вить';
1609 |             btnRead.className = 'btn btn-info btn-sm text-dark d
| -flex align-items-center gap-1';
1610 |             }
1611 |         });
1612 |     }
1613 |
1614 |     // Theme toggle function
1615 |     function toggleTheme() {
1616 |         const body = document.body;
1617 |         const currentTheme = body.classList.contains('theme-ligh
| t') ? 'light' : 'dark';
1618 |         const newTheme = currentTheme === 'light' ? 'dark' : 'li
| ght';
1619 |
1620 |         if (newTheme === 'light') {
1621 |             body.classList.add('theme-light');
1622 |         } else {
1623 |             body.classList.remove('theme-light');
1624 |         }
1625 |     }
1626 |
```

```
1627 |     // Set theme toggle listener
1628 |     document.getElementById('theme-toggle-btn')?.addEventListener('click', toggleTheme);
1629 |
1630 |     // — AUTH CHECK INITIALIZER —————
1631 |     async function checkAndInit() {
1632 |         const authState = await api.auth.check();
1633 |         if (authState.authenticated) {
1634 |             // If a room parameter is present in URL, connect directly to it
1635 |             const queryRoom = getRoomNameFromQuery();
1636 |             if (queryRoom) {
1637 |                 initStream(queryRoom);
1638 |             } else {
1639 |                 // Otherwise, scan for open players
1640 |                 scanForPlayers();
1641 |             }
1642 |         } else {
1643 |             showPanel('auth');
1644 |         }
1645 |     }
1646 |
1647 |     // Run on startup
1648 |     document.addEventListener('DOMContentLoaded', () => {
1649 |         checkAndInit();
1650 |         initVoiceControl();
1651 |     });
1652 |
1653 | </script>
1654 | </body>
1655 | </html>
1656 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/registry_viewer_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/registry_viewer_tab/index.html" class="w-100 h-100 border-0"></iframe>
17
18 File: index.html
19 Project: ai-breadboard
20 Package: windows/api/webgui/registry_viewer_tab
21 Author: hypo69
22 Copyright: © 2026 hypo69
23 Updated: 2026-10-08 03:08:00
24 =====
25
26 -->
27
28 <!-- Windows Registry Viewer Tab -->
29 <style>
30   .reg-container {
31     padding: 1rem;
32   }
33   .reg-card {
34     background: #0f172a !important;
35     border: 1px solid #334155 !important;
36     border-radius: 10px;
```

```
33 | }
34 | .reg-path-bar {
35 |     background: #1e293b !important;
36 |     border: 1px solid #334155 !important;
37 |     border-radius: 8px;
38 |     padding: 0.5rem 0.75rem;
39 | }
40 | .reg-tree-pane {
41 |     background: #0b1329;
42 |     border: 1px solid #334155;
43 |     border-radius: 8px;
44 |     height: 580px;
45 |     overflow-y: auto;
46 |     font-size: 0.82rem;
47 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
48 | }
49 | .reg-tree-item {
50 |     padding: 4px 8px;
51 |     cursor: pointer;
52 |     border-radius: 4px;
53 |     display: flex;
54 |     align-items: center;
55 |     gap: 6px;
56 |     color: #cbd5e1;
57 |     white-space: nowrap;
58 |     overflow: hidden;
59 |     text-overflow: ellipsis;
60 | }
61 | .reg-tree-item:hover {
62 |     background: rgba(56, 189, 248, 0.1);
63 |     color: #38bdf8;
64 | }
65 | .reg-tree-item.active {
66 |     background: rgba(56, 189, 248, 0.25);
67 |     color: #38bdf8;
68 |     font-weight: 600;
69 | }
```

```
70 | .reg-values-pane {
71 |     background: #0f172a;
72 |     border: 1px solid #334155;
73 |     border-radius: 8px;
74 |     height: 580px;
75 |     overflow-y: auto;
76 | }
77 | .reg-table {
78 |     width: 100%;
79 |     font-size: 0.8rem;
80 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
81 |     border-collapse: collapse;
82 | }
83 | .reg-table th {
84 |     background: #1e293b;
85 |     padding: 8px 12px;
86 |     color: #94a3b8;
87 |     font-weight: 600;
88 |     border-bottom: 1px solid #334155;
89 |     position: sticky;
90 |     top: 0;
91 |     z-index: 2;
92 | }
93 | .reg-table td {
94 |     padding: 6px 12px;
95 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
96 |     vertical-align: top;
97 | }
98 | .reg-table tbody tr:hover {
99 |     background: rgba(56, 189, 248, 0.05);
100 | }
101 | .reg-type-badge {
102 |     font-size: 0.68rem;
103 |     padding: 2px 6px;
104 |     border-radius: 4px;
105 |     background: #1e293b;
106 |     border: 1px solid #475569;
```

```

107     color: #38bdf8;
108 }
109 .reg-bookmark-btn {
110     font-size: 0.75rem;
111     padding: 4px 10px;
112     border-radius: 6px;
113     background: #1e293b;
114     border: 1px solid #334155;
115     color: #cbd5e1;
116     transition: all 0.2s ease;
117 }
118 .reg-bookmark-btn:hover {
119     background: #334155;
120     color: #38bdf8;
121     border-color: #38bdf8;
122 }
123 </style>
124
125 <div class="reg-container">
126     <!-- Header Alert & Status -->
127     <div class="card shadow-sm border-secondary-subtle p-3 mb-3
128         d-flex justify-content-between align-items-center flex-wrap ga
129         p-3">
130         <div>
131             <h5 class="mb-1 fw-bold d-flex align-items-center gap-2"
132             >
133                 <span class="fs-4">🔍 </span> Просмотр реестра (Windows
134                 Registry Viewer)
135             </h5>
136             <p class="mb-0 small text-muted" >
137                 Безопасный навигатор по системному реестру Windows (HK
138                 LM, HKCU, HKCR, HKU, HKCC), быстрый переход к ключевым веткам
139                 автозагрузки, служб и политик.
140             </p>
141         </div>
142         <div class="d-flex align-items-center gap-2 flex-wrap">
143             <span class="badge rounded-pill bg-success-subtle text-s
144             uccess border border-success px-3 py-2" id="reg-status-badge">

```



```
138         • Реестр: Подключено
139     </span>
140     <button class="btn btn-sm btn-outline-info rounded-pill
px-3" id="btn-reg-refresh">
141         <i class="bi bi-arrow-clockwise me-1"></i> Обновить
142     </button>
143     <button class="btn btn-sm btn-outline-light rounded-pill
px-3" id="btn-reg-copy-path">
144         <i class="bi bi-clipboard me-1"></i> Копировать путь
145     </button>
146     <button class="btn btn-sm btn-warning rounded-pill px-3"
id="btn-reg-backups-modal">
147         <i class="bi bi-shield-check me-1"></i> Бэкапы и откат
148     </button>
149 </div>
150 </div>
151
152 <!-- Bookmarks Quick Bar -->
153 <div class="mb-3 d-flex align-items-center gap-1 flex-wrap p
-2 rounded bg-dark border border-secondary" id="reg-bookmarks-
container">
154     <span class="text-muted small me-2"><i class="bi bi-bookma
rk-star-fill text-warning me-1"></i> Закладки:</span>
155     <!-- Bookmarks will be loaded here -->
156 </div>
157
158 <!-- Navigation Path Bar -->
159 <div class="reg-path-bar mb-3">
160     <div class="row g-2 align-items-center">
161         <div class="col-md-3 col-sm-4">
162             <select class="form-select form-select-sm bg-dark text
-white border-secondary font-monospace" id="reg-hive-select">
163                 <option value="HKEY_LOCAL_MACHINE">HKEY_LOCAL_MACHIN
E (HKLM)</option>
164                 <option value="HKEY_CURRENT_USER">HKEY_CURRENT_USER
(HKCU)</option>
165                 <option value="HKEY_CLASSES_ROOT">HKEY_CLASSES_ROOT
(HKCR)</option>
```

```

166         <option value="HKEY_USERS">HKEY_USERS (HKU)</option>
167         <option value="HKEY_CURRENT_CONFIG">HKEY_CURRENT_CON
FIG (HKCC)</option>
168     </select>
169 </div>
170 <div class="col-md-7 col-sm-6">
171     <div class="input-group input-group-sm">
172         <button class="btn btn-outline-secondary" id="btn-re
g-up" title="На уровень выше">
173             <i class="bi bi-arrow-up-short fs-6"></i>
174         </button>
175         <input type="text" class="form-control bg-dark text-
white border-secondary font-monospace" id="reg-path-input" pla
ceholder="SOFTWARE\Microsoft\Windows..." value="SOFTWARE">
176         <button class="btn btn-primary" id="btn-reg-go">
177             Перейти
178         </button>
179     </div>
180 </div>
181 <div class="col-md-2 col-sm-2 text-end">
182     <div class="input-group input-group-sm">
183         <input type="text" class="form-control bg-dark text-
white border-secondary" id="reg-filter-input" placeholder="Фил
ьтр параметров...">
184     </div>
185 </div>
186 </div>
187 </div>
188
189 <!-- Main Explorer View (Subkeys Tree + Values Table) -->
190 <div class="row g-3">
191     <!-- Subkeys Tree Pane (Left) -->
192     <div class="col-md-4">
193         <div class="reg-tree-pane p-2">
194             <div class="d-flex justify-content-between align-items
-center mb-2 pb-1 border-bottom border-secondary">
195                 <span class="text-muted small fw-bold"><i class="bi
bi-folder2-open me-1"></i> Подразделы (Subkeys)</span>

```

```

196         <div class="d-flex gap-1 align-items-center">
197             <span class="badge bg-secondary" id="reg-subkeys-c
ount-badge">0</span>
198             <button class="btn btn-xs btn-outline-success py-0
px-2" id="btn-reg-add-key" title="Создать подраздел" style="f
ont-size: 0.75rem;">
199                 <i class="bi bi-plus-lg"></i>
200             </button>
201             <button class="btn btn-xs btn-outline-danger py-0
px-2" id="btn-reg-delete-current-key" title="Удалить текущий р
аздел" style="font-size: 0.75rem;">
202                 <i class="bi bi-trash"></i>
203             </button>
204         </div>
205     </div>
206     <div id="reg-subkeys-list">
207         <div class="text-muted text-center py-4 small">Загру
зка ключей...</div>
208     </div>
209 </div>
210 </div>
211
212 <!-- Values & Data Pane (Right) -->
213 <div class="col-md-8">
214     <div class="reg-values-pane p-0">
215         <div class="d-flex justify-content-between align-items
-center p-2 border-bottom border-secondary bg-dark">
216             <span class="text-muted small fw-bold"><i class="bi
bi-card-list me-1"></i> Параметры и значения (Values)</span>
217             <div class="d-flex gap-2 align-items-center">
218                 <span class="badge bg-secondary" id="reg-values-co
unt-badge">0</span>
219                 <button class="btn btn-sm btn-success py-1 px-3" i
d="btn-reg-add-value">
220                     <i class="bi bi-plus-circle me-1"></i> Создать п
араметр
221                 </button>
222             </div>

```

```

223         </div>
224         <div class="table-responsive" style="max-height: 535px
    ; overflow-y: auto;">
225             <table class="reg-table">
226                 <thead>
227                     <tr>
228                         <th style="width: 28%;">Имя параметра</th>
229                         <th style="width: 18%;">Тип</th>
230                         <th style="width: 42%;">Значение</th>
231                         <th style="width: 12%; text-align: center;">Де
йствия</th>
232                     </tr>
233                 </thead>
234                 <tbody id="reg-values-tbody">
235                     <tr><td colspan="4" class="text-center text-mute
d p-4">Выберите ключ реестра для просмотра параметров</td></tr>
    <tr>
236                     </tbody>
237                 </table>
238             </div>
239         </div>
240     </div>
241 </div>
242
243 <!-- Modal: Редактировать / Создать параметр -->
244 <div class="modal fade" id="regValueModal" tabindex="-1" ari
a-hidden="true">
245     <div class="modal-dialog modal-dialog-centered">
246         <div class="modal-content bg-dark text-white border-seco
ndary">
247             <div class="modal-header border-secondary">
248                 <h5 class="modal-title" id="regValueModalTitle">□ Pe
дактирование параметра</h5>
249                 <button type="button" class="btn-close btn-close-whi
te" data-bs-dismiss="modal" aria-label="Close"></button>
250             </div>
251             <div class="modal-body">
252                 <form id="regValueForm">

```

```
253 |         <div class="mb-3">
254 |             <label class="form-label small text-muted">Имя п
| арамeтpа</label>
255 |             <input type="text" class="form-control bg-second
| ary text-white border-dark font-monospace" id="regModalValueNa
| me" placeholder="Оставьте пустым для (По умолчанию)">
256 |         </div>
257 |         <div class="mb-3">
258 |             <label class="form-label small text-muted">Тип д
| анных</label>
259 |             <select class="form-select bg-secondary text-whi
| te border-dark font-monospace" id="regModalValueType">
260 |                 <option value="REG_SZ">REG_SZ (Строка)</option
| >
261 |                 <option value="REG_DWORD">REG_DWORD (Число 32-
| бит)</option>
262 |                 <option value="REG_QWORD">REG_QWORD (Число 64-
| бит)</option>
263 |                 <option value="REG_EXPAND_SZ">REG_EXPAND_SZ (П
| асшиpямая строка)</option>
264 |                 <option value="REG_MULTI_SZ">REG_MULTI_SZ (Мно
| гостpочный текст)</option>
265 |                 <option value="REG_BINARY">REG_BINARY (Двоичны
| е данные / HEX)</option>
266 |             </select>
267 |         </div>
268 |         <div class="mb-3">
269 |             <label class="form-label small text-muted">Значе
| ние</label>
270 |             <textarea class="form-control bg-secondary text-
| white border-dark font-monospace" id="regModalValueData" rows=
| "4" placeholder="Введите значение параметра"></textarea>
271 |             <div class="form-text text-muted small">Для REG_
| MULTI_SZ разделяйте строки новой строкой. Для REG_BINARY вводи
| те HEX байты.</div>
272 |         </div>
273 |         <div class="form-check form-switch mb-2">
274 |             <input class="form-check-input" type="checkbox">
```

```

    id="regModalBackupCheck" checked>
275         <label class="form-check-label small text-light"
    for="regModalBackupCheck">
276             ☐ Создать резервную копию ветки перед сохране
нием
277         </label>
278     </div>
279 </form>
280 </div>
281 <div class="modal-footer border-secondary">
282     <button type="button" class="btn btn-secondary" data
-bs-dismiss="modal">Отмена</button>
283     <button type="button" class="btn btn-primary" id="bt
n-reg-modal-save">Сохранить</button>
284 </div>
285 </div>
286 </div>
287 </div>
288
289 <!-- Modal: Создать подраздел -->
290 <div class="modal fade" id="regKeyModal" tabindex="-1" aria-
hidden="true">
291     <div class="modal-dialog modal-dialog-centered">
292         <div class="modal-content bg-dark text-white border-seco
ndary">
293             <div class="modal-header border-secondary">
294                 <h5 class="modal-title"><input type="checkbox"/> Новый подраздел реестра</
h5>
295                 <button type="button" class="btn-close btn-close-whi
te" data-bs-dismiss="modal" aria-label="Close"></button>
296             </div>
297             <div class="modal-body">
298                 <div class="mb-3">
299                     <label class="form-label small text-muted">Имя нов
ого подраздела</label>
300                     <input type="text" class="form-control bg-secondar
y text-white border-dark font-monospace" id="regModalNewKeyName"
placeholder="MyNewSubkey">

```

```

301         </div>
302         <p class="small text-muted mb-0">Ключ будет создан в
нутри текущего пути.</p>
303     </div>
304     <div class="modal-footer border-secondary">
305         <button type="button" class="btn btn-secondary" data
-bs-dismiss="modal">Отмена</button>
306         <button type="button" class="btn btn-success" id="bt
n-reg-modal-create-key">Создать</button>
307     </div>
308 </div>
309 </div>
310 </div>
311
312 <!-- Modal: Резервные копии и откат -->
313 <div class="modal fade" id="regBackupsModal" tabindex="-1" a
ria-hidden="true">
314     <div class="modal-dialog modal-lg modal-dialog-centered">
315         <div class="modal-content bg-dark text-white border-seco
ndary">
316             <div class="modal-header border-secondary">
317                 <h5 class="modal-title">□ Точки восстановления и ис
тория бэкапов</h5>
318                 <button type="button" class="btn-close btn-close-whi
te" data-bs-dismiss="modal" aria-label="Close"></button>
319             </div>
320             <div class="modal-body p-0">
321                 <div class="table-responsive" style="max-height: 400
px; overflow-y: auto;">
322                     <table class="table table-dark table-hover mb-0 fo
nt-monospace small">
323                         <thead>
324                             <tr>
325                                 <th>Время</th>
326                                 <th>Ветка и путь</th>
327                                 <th>Операция</th>
328                                 <th>Действие</th>
329                             </tr>

```

```
330 |         </thead>
331 |         <tbody id="reg-backups-tbody">
332 |             <tr><td colspan="4" class="text-center py-4 text-muted">Загрузка списка бэкапов...</td></tr>
333 |         </tbody>
334 |     </table>
335 | </div>
336 | </div>
337 | <div class="modal-footer border-secondary">
338 |     <button type="button" class="btn btn-secondary" data-
    | -bs-dismiss="modal">Закрыть</button>
339 | </div>
340 | </div>
341 | </div>
342 | </div>
343 | </div>
344 |
345 |
```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Registry Viewer Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/registry_viewer_tab/main.js?v=20261001_v1" type="module"></script>
11 *
12 * File: main.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/registry_viewer_tab
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-01 13:04:40
18 * =====
19 */
20
21 // Windows Registry Viewer Tab JavaScript Module
22 (function() {
23   let currentHive = 'HKEY_LOCAL_MACHINE';
24   let currentPath = 'SOFTWARE';
25   let currentKeyData = null;
26   let isInitialized = false;
27
28   async function loadBookmarks() {
29     const container = document.getElementById('reg-bookmarks-container');
30     if (!container) return;
31
```

```

32 |     try {
33 |         const res = await fetch('/api/registry/bookmarks');
34 |         if (!res.ok) throw new Error('Bookmarks API error');
35 |         const data = await res.json();
36 |         const bookmarks = data.bookmarks || [];
37 |
38 |         container.innerHTML = `
39 |             <span class="text-muted small me-2"><i class="bi bi-bo
okmark-star-fill text-warning me-1"></i> Закладки:</span>
40 |             ` + bookmarks.map(b => `
41 |                 <button class="reg-bookmark-btn" data-hive="${b.hive}"
data-path="${b.path}" title="${b.description || ''}">
42 |                     <span>${b.icon || '□'}</span> ${b.title}
43 |                 </button>
44 |             `).join('');
45 |
46 |         container.querySelectorAll('.reg-bookmark-btn').forEach(
btn => {
47 |             btn.onclick = () => {
48 |                 const h = btn.getAttribute('data-hive');
49 |                 const p = btn.getAttribute('data-path');
50 |                 navigateTo(h, p);
51 |             };
52 |         });
53 |     } catch (e) {
54 |         console.warn('[RegistryViewerTab] Failed to load bookmar
ks:', e);
55 |     }
56 | }
57 |
58 | async function navigateTo(hive, path) {
59 |     currentHive = hive;
60 |     currentPath = path.trim().replace(/^[\\\/]+|[\\\/]+$/g, ''
);
61 |
62 |     const hiveSelect = document.getElementById('reg-hive-selec
t');
63 |     const pathInput = document.getElementById('reg-path-input'

```

```

    );
64     if (hiveSelect) hiveSelect.value = currentHive;
65     if (pathInput) pathInput.value = currentPath;
66
67     await loadKey(currentHive, currentPath);
68 }
69
70 async function loadKey(hive, path) {
71     const subkeysList = document.getElementById('reg-subkeys-l
ist');
72     const valuesTbody = document.getElementById('reg-values-tb
ody');
73     const badge = document.getElementById('reg-status-badge');
74     const subkeysCountBadge = document.getElementById('reg-sub
keys-count-badge');
75     const valuesCountBadge = document.getElementById('reg-valu
es-count-badge');
76
77     if (badge) badge.innerText = `• Чтение ${hive}...`;
78
79     try {
80         const url = `/api/registry/key?hive=${encodeURIComponent
(hive)}&path=${encodeURIComponent(path)}`;
81         const res = await fetch(url);
82         if (!res.ok) {
83             const errJson = await res.json().catch(() => ({}));
84             throw new Error(errJson.detail || `HTTP ${res.status}`
);
85         }
86
87         currentKeyData = await res.json();
88
89         // Рендер подразделов (subkeys)
90         const subkeys = currentKeyData.subkeys || [];
91         if (subkeysCountBadge) subkeysCountBadge.innerText = sub
keys.length;
92
93         if (subkeys.length === 0) {

```

```

94 |         subkeysList.innerHTML = '<div class="text-muted text-c
enter py-4 small">Нет подразделов</div>';
95 |     } else {
96 |         subkeysList.innerHTML = subkeys.map(sk => `
97 |             <div class="reg-tree-item" data-subkey="${sk}" title
="${sk}">
98 |                 <i class="bi bi-folder-fill text-warning"></i>
99 |                 <span class="text-truncate">${sk}</span>
100 |             </div>
101 |         `).join('');
102 |
103 |         subkeysList.querySelectorAll('.reg-tree-item').forEach
(item => {
104 |             item.onclick = () => {
105 |                 const subkey = item.getAttribute('data-subkey');
106 |                 const newPath = currentPath ? `${currentPath}\\${s
ubkey}` : subkey;
107 |                 navigateTo(currentHive, newPath);
108 |             };
109 |         });
110 |     }
111 |
112 |     // Рендер параметров (values)
113 |     renderValues();
114 |
115 |     if (badge) {
116 |         badge.className = 'badge rounded-pill bg-success-subtl
e text-success border border-success px-3 py-2';
117 |         badge.innerText = `• Реестр: ${hive}`;
118 |     }
119 | } catch (err) {
120 |     console.error('[RegistryViewerTab] Error loading key:',
err);
121 |     if (valuesTbody) {
122 |         valuesTbody.innerHTML = `<tr><td colspan="3" class="te
xt-center text-danger p-4"><i class="bi bi-exclamation-triangl
e-fill me-1"></i> Ошибка чтения ветки: ${err.message}</td></tr>
>`;

```

```
123     }
124     if (subkeysList) {
125         subkeysList.innerHTML = `
```

```
151 |     return;
152 | }
153 |
154 | valuesTbody.innerHTML = filtered.map((v, idx) => {
155 |     let dataDisplay = v.data;
156 |     if (dataDisplay === null || dataDisplay === undefined) {
157 |         dataDisplay = '<span class="text-muted">(Значение не з
| адано)</span>';
158 |     } else if (typeof dataDisplay === 'object') {
159 |         dataDisplay = `<pre class="mb-0 text-info" style="font
| -size: 0.75rem;">${JSON.stringify(dataDisplay, null, 2)}</pre>
| `;
160 |     } else {
161 |         dataDisplay = `<span class="text-light text-break">${e
| scapeHtml(String(dataDisplay))}</span>`;
162 |     }
163 |
164 |     const isDefault = v.name === '(Default / По умолчанию)'
| || v.name === '(Default)';
165 |     const rawNameAttr = escapeHtml(isDefault ? '' : v.name);
166 |
167 |     return `
168 |         <tr class="reg-val-row" data-idx="${idx}" style="curso
| r: pointer;" title="Нажмите для AI-объяснения параметра реестр
| а">
169 |             <td>
170 |                 <div class="fw-bold text-white text-truncate" styl
| e="max-width: 220px;" title="${v.name}">
171 |                     <i class="bi bi-file-earmark-binary text-seconda
| ry me-1"></i>
172 |                         ${escapeHtml(v.name)}
173 |                     </div>
174 |                 </td>
175 |                 <td>
176 |                     <span class="reg-type-badge">${v.type_name}</span>
177 |                 </td>
178 |                 <td>
179 |                     ${dataDisplay}
```

```

180         </td>
181         <td class="text-center">
182             <button class="btn btn-xs btn-outline-info py-0 px
-2 me-1 btn-edit-val" data-idx="${idx}" title="Редактировать">
183                 <i class="bi bi-pencil"></i>
184             </button>
185             <button class="btn btn-xs btn-outline-danger py-0
px-2 btn-del-val" data-name="${rawNameAttr}" title="Удалить">
186                 <i class="bi bi-trash"></i>
187             </button>
188         </td>
189     </tr>
190     `;
191     }).join('');
192
193     // Привязка кликабельности всей строки к AITableModal
194     valuesTbody.querySelectorAll('.reg-val-row').forEach(row =
> {
195         row.onclick = (evt) => {
196             if (evt.target.closest('.btn-edit-val') || evt.target.
closest('.btn-del-val')) return;
197             const idx = parseInt(row.getAttribute('data-idx'), 10)
;
198             const v = filtered[idx];
199             if (!v) return;
200
201             if (window.AITableModal) {
202                 window.AITableModal.show({
203                     icon: '□ ',
204                     title: v.name || '(Default)',
205                     subtitle: `${currentHive}\\${currentPath}`,
206                     tableType: 'registry',
207                     badges: [
208                         { text: v.type_name || 'REG_SZ', class: 'badge b
g-info text-dark' },
209                         { text: currentHive, class: 'badge bg-secondary'
}
210                     ],

```

```

211         metadata: [
212             { label: 'Параметр', value: v.name || '(По умолч
анию)' },
213             { label: 'Тип данных', value: v.type_name },
214             { label: 'Ветка реестра (Hive)', value: currentH
ive },
215             { label: 'Путь к разделу', value: currentPath, i
sCode: true, fullWidth: true },
216             { label: 'Значение', value: String(v.data ?? '')
, isCode: true, fullWidth: true }
217         ],
218         rawTitle: 'Значение параметра',
219         rawContent: typeof v.data === 'object' ? JSON.stri
ngify(v.data, null, 2) : String(v.data ?? ''),
220         requestData: {
221             hive: currentHive,
222             path: currentPath,
223             name: v.name,
224             type: v.type_name,
225             data: v.data
226         }
227     });
228 }
229 };
230 });
231
232 // Привязка событий кнопок редактирования и удаления парам
етров
233 valuesTbody.querySelectorAll('.btn-edit-val').forEach(btn
=> {
234     btn.onclick = (evt) => {
235         evt.stopPropagation();
236         const idx = parseInt(btn.getAttribute('data-idx'), 10)
;
237         const valObj = filtered[idx];
238         if (valObj) openEditValueModal(valObj);
239     };
240 });

```



```

241
242     valuesTbody.querySelectorAll('.btn-del-val').forEach(btn =
    > {
243         btn.onclick = (evt) => {
244             evt.stopPropagation();
245             const valName = btn.getAttribute('data-name');
246             confirmDeleteValue(valName);
247         };
248     });
249 }
250
251 function openEditValueModal(valObj) {
252     const isDefault = !valObj.name || valObj.name === '(Default / По умолчанию)' || valObj.name === '(Default)';
253     const nameInput = document.getElementById('regModalValueName');
254     const typeSelect = document.getElementById('regModalValueType');
255     const dataInput = document.getElementById('regModalValueData');
256     const title = document.getElementById('regValueModalTitle');
257
258     if (title) title.innerText = '✎ Редактирование параметра';
259     if (nameInput) {
260         nameInput.value = isDefault ? '' : valObj.name;
261         nameInput.disabled = true; // имя существующего параметра не меняем
262     }
263     if (typeSelect) typeSelect.value = valObj.type_name || 'REG_SZ';
264     if (dataInput) {
265         if (Array.isArray(valObj.data)) {
266             dataInput.value = valObj.data.join('\n');
267         } else {
268             dataInput.value = valObj.data !== null && valObj.data !== undefined ? String(valObj.data) : '';
269         }

```

```
270 |     }
271 |
272 |     const modalEl = document.getElementById('regValueModal');
273 |     if (modalEl && window.bootstrap) {
274 |         const modal = bootstrap.Modal.getOrCreateInstance(modalEl
275 | 1);
276 |         modal.show();
277 |     }
278 |
279 |     function openCreateValueModal() {
280 |         const nameInput = document.getElementById('regModalValueNa
281 | me');
282 |         const typeSelect = document.getElementById('regModalValueT
283 | ype');
284 |         const dataInput = document.getElementById('regModalValueDa
285 | ta');
286 |         const title = document.getElementById('regValueModalTitle'
287 | );
288 |         if (title) title.innerText = '□ Создание нового параметра'
289 | ;
290 |         if (nameInput) {
291 |             nameInput.value = '';
292 |             nameInput.disabled = false;
293 |         }
294 |         if (typeSelect) typeSelect.value = 'REG_SZ';
295 |         if (dataInput) dataInput.value = '';
296 |
297 |         const modalEl = document.getElementById('regValueModal');
298 |         if (modalEl && window.bootstrap) {
299 |             const modal = bootstrap.Modal.getOrCreateInstance(modalEl
300 | 1);
301 |             modal.show();
302 |         }
303 |     }
304 |
305 |     async function saveModalValue() {
```

```
301     const nameInput = document.getElementById('regModalValueNa
me');
302     const typeSelect = document.getElementById('regModalValueT
ype');
303     const dataInput = document.getElementById('regModalValueDa
ta');
304     const backupCheck = document.getElementById('regModalBacku
pCheck');
305
306     const name = nameInput ? nameInput.value.trim() : '';
307     const type_name = typeSelect ? typeSelect.value : 'REG_SZ'
;
308     const rawData = dataInput ? dataInput.value : '';
309     const create_backup = backupCheck ? backupCheck.checked :
true;
310
311     try {
312         const payload = {
313             hive: currentHive,
314             path: currentPath,
315             name: name,
316             type_name: type_name,
317             data: rawData,
318             create_backup: create_backup
319         };
320
321         const res = await fetch('/api/registry/value', {
322             method: 'POST',
323             headers: { 'Content-Type': 'application/json' },
324             body: JSON.stringify(payload)
325         });
326
327         if (!res.ok) {
328             const errJson = await res.json().catch(() => ({}));
329             throw new Error(errJson.detail || `HTTP ${res.status}`
);
330         }
331
```

```
332 |     const modalEl = document.getElementById('regValueModal')
    | ;
333 |     if (modalEl && window.bootstrap) {
334 |         const modal = bootstrap.Modal.getInstance(modalEl);
335 |         if (modal) modal.hide();
336 |     }
337 |
338 |     await loadKey(currentHive, currentPath);
339 | } catch (e) {
340 |     window.showToast?(`Ошибка сохранения параметра: ${e.message}`, 'danger') || alert(`Ошибка сохранения параметра: ${e.message}`);
341 | }
342 | }
343 |
344 | async function confirmDeleteValue(name) {
345 |     const displayName = name ? `${name}` : '(По умолчанию)';
346 |     if (!confirm(`Вы действительно хотите удалить параметр ${displayName}? Перед удалением будет автоматически создан бэкап.`)) {
347 |         return;
348 |     }
349 |
350 |     try {
351 |         const res = await fetch('/api/registry/value', {
352 |             method: 'DELETE',
353 |             headers: { 'Content-Type': 'application/json' },
354 |             body: JSON.stringify({
355 |                 hive: currentHive,
356 |                 path: currentPath,
357 |                 name: name,
358 |                 create_backup: true
359 |             })
360 |         });
361 |
362 |         if (!res.ok) {
363 |             const errJson = await res.json().catch(() => ({}));
364 |             throw new Error(errJson.detail || `HTTP ${res.status}`
```

```

    );
365     }
366
367     window.showToast?(`Параметр ${displayName} удален`, 'su
ccess');
368     await loadKey(currentHive, currentPath);
369   } catch (e) {
370     window.showToast?(`Ошибка удаления параметра: ${e.messa
ge}`, 'danger') || alert(`Ошибка удаления параметра: ${e.messa
ge}`);
371   }
372 }
373
374 function openCreateKeyModal() {
375   const keyNameInput = document.getElementById('regModalNewK
eyName');
376   if (keyNameInput) keyNameInput.value = '';
377
378   const modalEl = document.getElementById('regKeyModal');
379   if (modalEl && window.bootstrap) {
380     const modal = bootstrap.Modal.getOrCreateInstance(modalE
l);
381     modal.show();
382   }
383 }
384
385 async function saveModalKey() {
386   const keyNameInput = document.getElementById('regModalNewK
eyName');
387   const newKeyName = keyNameInput ? keyNameInput.value.trim(
) : '';
388   if (!newKeyName) {
389     window.showToast?(`Введите имя подраздела`, 'warning')
|| alert(`Введите имя подраздела`);
390     return;
391   }
392
393   const newPath = currentPath ? `${currentPath}\\${newKeyNam

```

```

    e}` : newKeyName;
394
395     try {
396         const res = await fetch('/api/registry/key', {
397             method: 'POST',
398             headers: { 'Content-Type': 'application/json' },
399             body: JSON.stringify({
400                 hive: currentHive,
401                 path: newPath
402             })
403         });
404
405         if (!res.ok) {
406             const errJson = await res.json().catch(() => ({}));
407             throw new Error(errJson.detail || `HTTP ${res.status}`
408         );
409
410         const modalEl = document.getElementById('regKeyModal');
411         if (modalEl && window.bootstrap) {
412             const modal = bootstrap.Modal.getInstance(modalEl);
413             if (modal) modal.hide();
414         }
415
416         window.showToast?.(`Подраздел '${newKeyName}' успешно со
417     здан`, 'success');
418         await loadKey(currentHive, currentPath);
419     } catch (e) {
420         window.showToast?.(`Ошибка создания ключа: ${e.message}`
421     , 'danger') || alert(`Ошибка создания ключа: ${e.message}`);
422     }
423
424     async function deleteCurrentKey() {
425         if (!currentPath) {
426             window.showToast?.('Нельзя удалить корневой раздел', 'wa
427     rning') || alert('Нельзя удалить корневой раздел');
428         return;

```

```
427 |     }
428 |
429 |     if (!confirm(`ВНИМАНИЕ: Вы действительно хотите удалить ра
    | здел '${currentHive}\\${currentPath}' со всеми подразделами и
    | значениями? Будет создан снимок резервной копии.`)) {
430 |         return;
431 |     }
432 |
433 |     try {
434 |         const res = await fetch('/api/registry/key', {
435 |             method: 'DELETE',
436 |             headers: { 'Content-Type': 'application/json' },
437 |             body: JSON.stringify({
438 |                 hive: currentHive,
439 |                 path: currentPath,
440 |                 recursive: true,
441 |                 create_backup: true
442 |             })
443 |         });
444 |
445 |         if (!res.ok) {
446 |             const errJson = await res.json().catch(() => ({}));
447 |             throw new Error(errJson.detail || `HTTP ${res.status}`
    | );
448 |         }
449 |
450 |         window.showToast?(`Раздел реестра успешно удален`, 'suc
    | cess');
451 |         goUp();
452 |     } catch (e) {
453 |         window.showToast?(`Ошибка удаления раздела: ${e.message
    | }`, 'danger') || alert(`Ошибка удаления раздела: ${e.message}`
    | );
454 |     }
455 | }
456 |
457 | async function openBackupsModal() {
458 |     const modalEl = document.getElementById('regBackupsModal')
```

```
;
459   const tbody = document.getElementById('reg-backups-tbody')
;
460
461   if (modalEl && window.bootstrap) {
462     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
463     modal.show();
464   }
465
466   if (tbody) tbody.innerHTML = '<tr><td colspan="4" class="text-center py-3 text-muted">Загрузка истории снимков...</td></tr>';
467
468   try {
469     const res = await fetch('/api/registry/backups');
470     if (!res.ok) throw new Error('Ошибка загрузки бэкапов');
471     const data = await res.json();
472     const backups = data.backups || [];
473
474     if (backups.length === 0) {
475       tbody.innerHTML = '<tr><td colspan="4" class="text-center py-4 text-muted">Точки восстановления отсутствуют</td></tr>';
476       return;
477     }
478
479     tbody.innerHTML = backups.map(b => `
480       <tr>
481         <td><small class="text-light">${new Date(b.timestamp).toLocaleString()}</small></td>
482         <td class="text-truncate" style="max-width: 250px;" title="${b.hive}\\${b.path}">
483           <span class="text-warning">${b.hive}</span>\\${escapeHtml(b.path)}
484         </td>
485         <td><span class="badge bg-secondary">${escapeHtml(b.operation)}</span></td>
```



```

486 |         <td>
487 |             <button class="btn btn-xs btn-outline-warning py-0
      | px-2 btn-restore-backup" data-id="${b.backup_id}">
488 |                 <i class="bi bi-arrow-counterclockwise me-1"></i
      | > Откатить
489 |             </button>
490 |         </td>
491 |     </tr>
492 |     `).join('');
493 |
494 |     tbody.querySelectorAll('.btn-restore-backup').forEach(bt
      | n => {
495 |         btn.onclick = async () => {
496 |             const backupId = btn.getAttribute('data-id');
497 |             if (!confirm(`Восстановить реестр из снимка ${backup
      | Id}?`)) return;
498 |
499 |             try {
500 |                 const rRes = await fetch(`/api/registry/restore?ba
      | ckup_id=${encodeURIComponent(backupId)}`, {
501 |                     method: 'POST'
502 |                 });
503 |                 if (!rRes.ok) {
504 |                     const err = await rRes.json().catch(() => ({}));
505 |                     throw new Error(err.detail || `HTTP ${rRes.statu
      | s}`);
506 |                 }
507 |                 window.showToast?(`Реестр успешно восстановлен из
      | резервной копии!`, 'success');
508 |                 await loadKey(currentHive, currentPath);
509 |             } catch (err) {
510 |                 window.showToast?(`Ошибка восстановления: ${err.m
      | essage}`, 'danger') || alert(`Ошибка восстановления: ${err.mes
      | sage}`);
511 |             }
512 |         };
513 |     });
514 | } catch (e) {

```

```

515 |         if (tbody) tbody.innerHTML = `|<td colspan="4" class=
| "text-center py-3 text-danger">${e.message}</td></tr>`;
516 |     }
517 | }
518 |
519 | function escapeHtml(text) {
520 |     return text
521 |         .replace(/&/g, "&amp;")
522 |         .replace(/</g, "&lt;")
523 |         .replace(/>/g, "&gt;")
524 |         .replace(/"/g, "&quot;")
525 |         .replace(/'/g, "&#039;");
526 | }
527 |
528 | function goUp() {
529 |     if (!currentPath) return;
530 |     const parts = currentPath.split('\\');
531 |     parts.pop();
532 |     const newPath = parts.join('\\');
533 |     navigateTo(currentHive, newPath);
534 | }
535 |
536 | function copyCurrentPath() {
537 |     const full = `${currentHive}\\${currentPath}`.replace(/\\+
| $/, '');
538 |     if (navigator.clipboard) {
539 |         navigator.clipboard.writeText(full).then(() => {
540 |             const btn = document.getElementById('btn-reg-copy-path
| ');
541 |             if (btn) {
542 |                 const orig = btn.innerHTML;
543 |                 btn.innerHTML = '<i class="bi bi-check me-1"></i> Ск
| опировано!';
544 |                 setTimeout(() => { btn.innerHTML = orig; }, 1500);
545 |             }
546 |         });
547 |     }
548 | }

|  |

```

```
549 |
550 |   window.initRegistryViewerTab = async function() {
551 |     console.log('[RegistryViewerTab] Initializing Registry Vie
552 | wer & Editor tab...');
553 |     const hiveSelect = document.getElementById('reg-hive-selec
554 | t');
555 |     const pathInput = document.getElementById('reg-path-input'
556 | );
557 |     const goBtn = document.getElementById('btn-reg-go');
558 |     const upBtn = document.getElementById('btn-reg-up');
559 |     const filterInput = document.getElementById('reg-filter-in
560 | put');
561 |     const refreshBtn = document.getElementById('btn-reg-refres
562 | h');
563 |     const copyBtn = document.getElementById('btn-reg-copy-path
564 | ');
565 |     const backupsBtn = document.getElementById('btn-reg-backup
566 | s-modal');
567 |     const addValBtn = document.getElementById('btn-reg-add-val
568 | ue');
569 |     const addKeyBtn = document.getElementById('btn-reg-add-key
570 | ');
571 |     const delKeyBtn = document.getElementById('btn-reg-delete-
572 | current-key');
573 |     const saveValModalBtn = document.getElementById('btn-reg-m
574 | odal-save');
575 |     const saveKeyModalBtn = document.getElementById('btn-reg-m
576 | odal-create-key');
577 |
578 |     if (hiveSelect) {
579 |       hiveSelect.onChange = () => {
580 |         navigateTo(hiveSelect.value, '');
581 |       };
582 |     }
583 |     if (goBtn && pathInput) {
584 |       goBtn.onclick = () => {
585 |         navigateTo(hiveSelect.value, pathInput.value);
586 |       };
587 |     }
588 |   };
589 | }
```

```

575 |     };
576 | }
577 | if (pathInput) {
578 |     pathInput.onkeydown = (e) => {
579 |         if (e.key === 'Enter') {
580 |             navigateTo(hiveSelect.value, pathInput.value);
581 |         }
582 |     };
583 | }
584 | if (upBtn) upBtn.onclick = goUp;
585 | if (filterInput) filterInput.oninput = renderValues;
586 | if (refreshBtn) refreshBtn.onclick = () => loadKey(current
Hive, currentPath);
587 | if (copyBtn) copyBtn.onclick = copyCurrentPath;
588 | if (backupsBtn) backupsBtn.onclick = openBackupsModal;
589 | if (addValBtn) addValBtn.onclick = openCreateValueModal;
590 | if (addKeyBtn) addKeyBtn.onclick = openCreateKeyModal;
591 | if (delKeyBtn) delKeyBtn.onclick = deleteCurrentKey;
592 | if (saveValModalBtn) saveValModalBtn.onclick = saveModalVa
lue;
593 | if (saveKeyModalBtn) saveKeyModalBtn.onclick = saveModalKe
y;
594 |
595 |     await loadBookmarks();
596 |     await navigateTo(currentHive, currentPath);
597 |     isInitialized = true;
598 | };
599 | })();
600 |
601 |

```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/remote_mic/index.html
15
16   Tab Frame Integration:
17     <iframe src="/windows/api/webgui/remote_mic/index.html"
18     class="w-100 h-100 border-0"></iframe>
19
20 File: index.html
21 Project: ai-breadboard
22 Package: windows/api/webgui/remote_mic
23 Author: hypo69
24 Copyright: © 2026 hypo69
25 Updated: 2026-10-01 13:04:40
26 =====
27 =====
28 -->
29
30 <!DOCTYPE html>
31 <html lang="ru">
32 <head>
33   <meta charset="UTF-8">
34   <meta name="viewport" content="width=device-width, initial-s
35   cale=1.0, maximum-scale=1.0, user-scalable=no">
36   <title>Голосовой Пульт - AI Breadboard</title>
37
38   <!-- Bootstrap 5 CSS -->
```

```
32 | <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
    | t/css/bootstrap.min.css" rel="stylesheet">
33 | <!-- Bootstrap Icons -->
34 | <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    | 0.0/font/bootstrap-icons.css" rel="stylesheet">
35 |
36 | <!-- Telegram Web Apps SDK & Marked.js -->
37 | <script src="https://telegram.org/js/telegram-web-app.js"></
    | script>
38 | <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
    | js"></script>
39 |
40 | <style>
41 |     :root {
42 |         --tg-theme-bg-color: #0b0f19;
43 |         --tg-theme-text-color: #f1f5f9;
44 |         --tg-theme-hint-color: #94a3b8;
45 |         --tg-theme-secondary-bg-color: #151c2e;
46 |         --primary-gradient: linear-gradient(135deg, #3b82f6 0%,
    | #1d4ed8 100%);
47 |         --mic-idle-gradient: linear-gradient(135deg, #ef4444 0%,
    | #b91c1c 100%);
48 |         --mic-listening-gradient: linear-gradient(135deg, #10b98
    | 1 0%, #047857 100%);
49 |         --mic-thinking-gradient: linear-gradient(135deg, #8b5cf6
    | 0%, #6d28d9 100%);
50 |         --accent-color: #38bdf8;
51 |     }
52 |
53 |     body {
54 |         background: var(--tg-theme-bg-color);
55 |         color: var(--tg-theme-text-color);
56 |         font-family: -apple-system, BlinkMacSystemFont, "Segoe U
    | I", Roboto, "Helvetica Neue", Arial, sans-serif;
57 |         min-height: 100vh;
58 |         display: flex;
59 |         flex-direction: column;
60 |         justify-content: flex-start;
```

```
61 |     align-items: center;
62 |     padding: 12px;
63 |     margin: 0;
64 |     overflow-x: hidden;
65 | }
66 |
67 | .app-container {
68 |     width: 100%;
69 |     max-width: 520px;
70 |     display: flex;
71 |     flex-direction: column;
72 |     height: 100%;
73 |     min-height: 94vh;
74 | }
75 |
76 | .card-panel {
77 |     background-color: var(--tg-theme-secondary-bg-color);
78 |     border: 1px solid rgba(255, 255, 255, 0.08);
79 |     border-radius: 20px;
80 |     padding: 18px;
81 |     box-shadow: 0 10px 30px rgba(0, 0, 0, 0.4);
82 |     backdrop-filter: blur(12px);
83 | }
84 |
85 | /* Giant Mic Button */
86 | .mic-centerpiece {
87 |     display: flex;
88 |     flex-direction: column;
89 |     align-items: center;
90 |     justify-content: center;
91 |     padding: 16px 0 12px;
92 |     position: relative;
93 | }
94 |
95 | .mic-button-wrapper {
96 |     position: relative;
97 |     display: flex;
98 |     align-items: center;
```

```
99 |     justify-content: center;
100 | }
101 |
102 | .btn-mic-giant {
103 |     width: 110px;
104 |     height: 110px;
105 |     border-radius: 50%;
106 |     border: none;
107 |     outline: none;
108 |     background: var(--mic-idle-gradient);
109 |     color: #ffffff;
110 |     display: flex;
111 |     align-items: center;
112 |     justify-content: center;
113 |     font-size: 44px;
114 |     box-shadow: 0 8px 25px rgba(239, 68, 68, 0.4);
115 |     cursor: pointer;
116 |     transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
117 |     position: relative;
118 |     z-index: 2;
119 | }
120 |
121 | .btn-mic-giant:hover {
122 |     transform: scale(1.05);
123 |     box-shadow: 0 12px 30px rgba(239, 68, 68, 0.6);
124 | }
125 |
126 | .btn-mic-giant:active {
127 |     transform: scale(0.95);
128 | }
129 |
130 | .btn-mic-giant.listening {
131 |     background: var(--mic-listening-gradient);
132 |     box-shadow: 0 8px 30px rgba(16, 185, 129, 0.5);
133 |     animation: mic-listening-pulse 1.8s infinite;
134 | }
135 |
136 | .btn-mic-giant.thinking {
```



```

137     background: var(--mic-thinking-gradient);
138     box-shadow: 0 8px 30px rgba(139, 92, 246, 0.5);
139     animation: mic-thinking-pulse 1.2s infinite ease-in-out;
140 }
141
142 @keyframes mic-listening-pulse {
143     0% { transform: scale(1); box-shadow: 0 0 0 0 rgba(16, 1
144 85, 129, 0.7); }
145     70% { transform: scale(1.06); box-shadow: 0 0 0 22px rgb
146 a(16, 185, 129, 0); }
147     100% { transform: scale(1); box-shadow: 0 0 0 0 rgba(16,
148 185, 129, 0); }
149 }
150
151 @keyframes mic-thinking-pulse {
152     0% { transform: rotate(0deg) scale(1); }
153     50% { transform: rotate(180deg) scale(1.05); }
154     100% { transform: rotate(360deg) scale(1); }
155 }
156
157 /* Wave visualizer dots */
158 .wave-bars {
159     display: flex;
160     align-items: center;
161     justify-content: center;
162     gap: 5px;
163     height: 20px;
164     margin-top: 8px;
165 }
166
167 .wave-bar {
168     width: 4px;
169     height: 6px;
170     background: #ef4444;
171     border-radius: 2px;
172     transition: height 0.15s ease;
173 }

```

```
172 |     .wave-bars.active .wave-bar {
173 |         background: #10b981;
174 |         animation: wave-anim 0.8s ease-in-out infinite alternate
    | ;
175 |     }
176 |
177 |     .wave-bars.active .wave-bar:nth-child(2) { animation-delay
    | : 0.15s; }
178 |     .wave-bars.active .wave-bar:nth-child(3) { animation-delay
    | : 0.3s; }
179 |     .wave-bars.active .wave-bar:nth-child(4) { animation-delay
    | : 0.45s; }
180 |     .wave-bars.active .wave-bar:nth-child(5) { animation-delay
    | : 0.2s; }
181 |
182 |     @keyframes wave-anim {
183 |         0% { height: 6px; }
184 |         100% { height: 20px; }
185 |     }
186 |
187 |     /* Media Actions Bar */
188 |     .media-actions-bar {
189 |         display: flex;
190 |         align-items: center;
191 |         justify-content: center;
192 |         gap: 8px;
193 |         margin-top: 10px;
194 |         width: 100%;
195 |     }
196 |
197 |     .btn-media-action {
198 |         background: rgba(255, 255, 255, 0.06);
199 |         border: 1px solid rgba(255, 255, 255, 0.12);
200 |         color: #e2e8f0;
201 |         font-size: 0.82rem;
202 |         font-weight: 500;
203 |         padding: 6px 14px;
204 |         border-radius: 20px;
```

```
205 |     display: inline-flex;
206 |     align-items: center;
207 |     gap: 6px;
208 |     transition: all 0.2s ease;
209 |     cursor: pointer;
210 | }
211 |
212 | .btn-media-action:hover {
213 |     background: rgba(255, 255, 255, 0.14);
214 |     color: #ffffff;
215 |     border-color: rgba(255, 255, 255, 0.25);
216 | }
217 |
218 | .btn-media-action.active-audio {
219 |     border-color: #38bdf8;
220 |     color: #38bdf8;
221 |     background: rgba(56, 189, 248, 0.1);
222 | }
223 |
224 | .btn-media-action.active-media {
225 |     border-color: #a855f7;
226 |     color: #c084fc;
227 |     background: rgba(168, 85, 247, 0.1);
228 | }
229 |
230 | /* Chat Messages Container */
231 | .chat-container {
232 |     flex: 1;
233 |     overflow-y: auto;
234 |     border-radius: 14px;
235 |     background: rgba(0, 0, 0, 0.2);
236 |     border: 1px solid rgba(255, 255, 255, 0.05);
237 |     padding: 12px;
238 |     display: flex;
239 |     flex-direction: column;
240 |     gap: 10px;
241 |     max-height: 44vh;
242 |     min-height: 180px;
```

```
243 |     scroll-behavior: smooth;
244 | }
245 |
246 | .chat-container::-webkit-scrollbar {
247 |     width: 5px;
248 | }
249 | .chat-container::-webkit-scrollbar-thumb {
250 |     background: rgba(255, 255, 255, 0.15);
251 |     border-radius: 10px;
252 | }
253 |
254 | .chat-bubble {
255 |     max-width: 90%;
256 |     padding: 10px 14px;
257 |     border-radius: 14px;
258 |     font-size: 0.92rem;
259 |     line-height: 1.45;
260 |     word-break: break-word;
261 |     position: relative;
262 | }
263 |
264 | .chat-bubble.user {
265 |     align-self: flex-end;
266 |     background: var(--primary-gradient);
267 |     color: #ffffff;
268 |     border-bottom-right-radius: 4px;
269 | }
270 |
271 | .chat-bubble.ai {
272 |     align-self: flex-start;
273 |     background: #1e293b;
274 |     color: #f1f5f9;
275 |     border: 1px solid rgba(255, 255, 255, 0.08);
276 |     border-bottom-left-radius: 4px;
277 | }
278 |
279 | .chat-bubble.ai p:last-child {
280 |     margin-bottom: 0;
```

```
281 |     }
282 |
283 |     .chat-meta {
284 |         display: flex;
285 |         align-items: center;
286 |         justify-content: space-between;
287 |         gap: 8px;
288 |         margin-top: 4px;
289 |         font-size: 0.72rem;
290 |         opacity: 0.75;
291 |     }
292 |
293 |     .btn-speak-again {
294 |         background: none;
295 |         border: none;
296 |         color: inherit;
297 |         padding: 0;
298 |         cursor: pointer;
299 |         font-size: 0.85rem;
300 |     }
301 |
302 |     /* Media Attachment Previews */
303 |     .media-preview-card {
304 |         background: rgba(0, 0, 0, 0.25);
305 |         border: 1px solid rgba(255, 255, 255, 0.1);
306 |         border-radius: 10px;
307 |         padding: 8px 10px;
308 |         margin-bottom: 6px;
309 |         display: flex;
310 |         align-items: center;
311 |         gap: 8px;
312 |     }
313 |
314 |     .media-player-widget {
315 |         width: 100%;
316 |         margin-top: 6px;
317 |         border-radius: 8px;
318 |         background: rgba(0, 0, 0, 0.2);
```

```
319 |     }
320 |
321 |     .media-thumb-img {
322 |         max-width: 100%;
323 |         max-height: 200px;
324 |         border-radius: 8px;
325 |         margin-top: 6px;
326 |         object-fit: cover;
327 |     }
328 |
329 |     /* Language and Theme Switchers */
330 |     .btn-lang-pill {
331 |         font-size: 0.75rem;
332 |         font-weight: 600;
333 |         padding: 3px 9px;
334 |         border-radius: 20px;
335 |     }
336 |
337 |     .theme-toggle-btn {
338 |         background: rgba(255, 255, 255, 0.08);
339 |         border: 1px solid rgba(255, 255, 255, 0.15);
340 |         color: #ffffff;
341 |         width: 34px;
342 |         height: 34px;
343 |         border-radius: 50%;
344 |         display: flex;
345 |         align-items: center;
346 |         justify-content: center;
347 |         cursor: pointer;
348 |         transition: all 0.2s;
349 |     }
350 |     .theme-toggle-btn:hover {
351 |         background: rgba(255, 255, 255, 0.18);
352 |     }
353 |
354 |     /* Input text area fallback */
355 |     .form-control-custom {
356 |         background: rgba(0, 0, 0, 0.25);
```

```
357 |     border: 1px solid rgba(255, 255, 255, 0.12);
358 |     color: #ffffff;
359 |     border-radius: 12px;
360 | }
361 | .form-control-custom:focus {
362 |     background: rgba(0, 0, 0, 0.35);
363 |     border-color: #3b82f6;
364 |     color: #ffffff;
365 |     box-shadow: 0 0 0 2px rgba(59, 130, 246, 0.3);
366 | }
367 |
368 | /* Light Theme */
369 | body.theme-light {
370 |     --tg-theme-bg-color: #f8fafc;
371 |     --tg-theme-text-color: #0f172a;
372 |     --tg-theme-hint-color: #64748b;
373 |     --tg-theme-secondary-bg-color: #ffffff;
374 | }
375 | body.theme-light .card-panel {
376 |     border-color: #e2e8f0;
377 |     box-shadow: 0 8px 24px rgba(0, 0, 0, 0.06);
378 | }
379 | body.theme-light .chat-container {
380 |     background: #f1f5f9;
381 |     border-color: #e2e8f0;
382 | }
383 | body.theme-light .chat-bubble.ai {
384 |     background: #ffffff;
385 |     color: #0f172a;
386 |     border-color: #e2e8f0;
387 | }
388 | body.theme-light .theme-toggle-btn {
389 |     background: #e2e8f0;
390 |     color: #0f172a;
391 |     border-color: #cbd5e1;
392 | }
393 | body.theme-light .btn-media-action {
394 |     background: #f1f5f9;
```

```

395 |         color: #334155;
396 |         border-color: #cbd5e1;
397 |     }
398 |     body.theme-light .form-control-custom {
399 |         background: #ffffff;
400 |         color: #0f172a;
401 |         border-color: #cbd5e1;
402 |     }
403 | </style>
404 | </head>
405 | <body>
406 |
407 |     <!-- AUTHENTICATION PANEL -->
408 |     <div class="app-container" id="auth-panel">
409 |         <div class="card-panel my-auto">
410 |             <div class="text-center mb-4">
411 |                 <i class="bi bi-mic-fill fs-1 text-primary"></i>
412 |                 <h4 class="mt-2 text-white fw-bold">Голосовой Пульт</h
413 | 4>
414 |                 <p class="text-muted small">Войдите в систему для пере
415 | дачи голосовых сообщений и медиа в чат</p>
416 |             </div>
417 |
418 |             <div id="auth-alert" class="alert alert-danger d-none" r
419 | ole="alert"></div>
420 |
421 |             <!-- Login Form -->
422 |             <div id="auth-form-login">
423 |                 <form id="form-login">
424 |                     <div class="mb-3">
425 |                         <label for="login-email" class="form-label text-mu
426 | ted small">Email</label>
427 |                         <input type="email" class="form-control form-contr
428 | ol-custom" id="login-email" required>
429 |                     </div>
430 |                     <div class="mb-3">
431 |                         <label for="login-password" class="form-label text
432 | -muted small">Пароль</label>

```



```
427 |         <input type="password" class="form-control form-co
| ntrol-custom" id="login-password" required>
428 |     </div>
429 |     <button type="submit" class="btn btn-primary w-100 p
| y-2.5 rounded-pill mb-3">Войти</button>
430 |     <p class="text-center small mb-0">Нет аккаунта? <a h
| ref="#" id="go-to-register" class="text-info">Регистрация</a><
| /p>
431 | </form>
432 | </div>
433 |
434 | <!-- Register Form -->
435 | <div id="auth-form-register" class="d-none">
436 |     <form id="form-register">
437 |         <div class="mb-3">
438 |             <label for="register-name" class="form-label text-
| muted small">Имя</label>
439 |             <input type="text" class="form-control form-contr
| ol-custom" id="register-name" required>
440 |         </div>
441 |         <div class="mb-3">
442 |             <label for="register-email" class="form-label text
| -muted small">Email</label>
443 |             <input type="email" class="form-control form-contr
| ol-custom" id="register-email" required>
444 |         </div>
445 |         <div class="mb-3">
446 |             <label for="register-password" class="form-label t
| ext-muted small">Пароль</label>
447 |             <input type="password" class="form-control form-co
| ntrol-custom" id="register-password" required>
448 |         </div>
449 |         <button type="submit" class="btn btn-success w-100 p
| y-2.5 rounded-pill mb-3">Зарегистрироваться</button>
450 |         <p class="text-center small mb-0">Уже есть аккаунт?
| <a href="#" id="go-to-login" class="text-info">Войти</a></p>
451 |     </form>
452 | </div>
```

```
453
454     <!-- Verify Form -->
455     <div id="auth-form-verify" class="d-none">
456         <p class="text-center small">Код подтверждения отправл
ен на почту:</p>
457         <form id="form-verify">
458             <div class="mb-3">
459                 <input type="text" class="form-control form-contro
l-custom text-center fs-3 py-2" id="verify-code" maxlength="6"
placeholder="XXXXXX" required>
460             </div>
461             <button type="submit" class="btn btn-info w-100 py-2
.5 text-white rounded-pill mb-3">Подтвердить</button>
462             <p class="text-center small mb-0"><a href="#" id="ve
rify-cancel" class="text-info">Назад</a></p>
463         </form>
464     </div>
465
466     <div class="my-4 text-center position-relative">
467         <hr class="border-secondary">
468         <span class="px-2 text-muted position-absolute top-50
start-50 translate-middle small bg-dark">Или</span>
469     </div>
470
471     <button id="google-signin-btn" class="btn btn-outline-li
ght w-100 rounded-pill d-flex align-items-center justify-conte
nt-center gap-2 py-2">
472         <i class="bi bi-google text-danger"></i>
473         <span>Войти через Google</span>
474     </button>
475 </div>
476 </div>
477
478 <!-- MAIN VOICE REMOTE PANEL -->
479 <div class="app-container d-none" id="main-panel">
480     <div class="card-panel d-flex flex-column h-100">
481
482         <!-- Top Bar -->
```

```
483 |         <div class="d-flex justify-content-between align-items-c
| enter mb-2">
484 |             <div class="d-flex align-items-center gap-2">
485 |                 <span class="badge bg-success rounded-pill px-2.5 py
| -1.5" id="connection-status">
486 |                     <i class="bi bi-broadcast"></i> Онлайн
487 |                 </span>
488 |                 <span class="small text-muted text-truncate" style="
| max-width: 120px;" id="user-display-name"></span>
489 |             </div>
490 |
491 |             <div class="d-flex align-items-center gap-2">
492 |                 <!-- Language Selector -->
493 |                 <div class="btn-group btn-group-sm" role="group">
494 |                     <input type="radio" class="btn-check" name="lang-r
| adio" id="lang-ru" value="ru-RU" checked>
495 |                     <label class="btn btn-outline-secondary btn-lang-p
| ill" for="lang-ru">RU</label>
496 |
497 |                     <input type="radio" class="btn-check" name="lang-r
| adio" id="lang-en" value="en-US">
498 |                     <label class="btn btn-outline-secondary btn-lang-p
| ill" for="lang-en">EN</label>
499 |
500 |                     <input type="radio" class="btn-check" name="lang-r
| adio" id="lang-he" value="he-IL">
501 |                     <label class="btn btn-outline-secondary btn-lang-p
| ill" for="lang-he">HE</label>
502 |                 </div>
503 |
504 |                 <!-- TTS Mute/Unmute Toggle -->
505 |                 <button class="theme-toggle-btn" id="tts-toggle-btn"
| title="Озвучивание ответов">
506 |                     <i class="bi bi-volume-up-fill" id="tts-icon"></i>
507 |                 </button>
508 |
509 |                 <!-- Theme Toggle -->
510 |                 <button class="theme-toggle-btn" id="theme-toggle-bt
```

```

n" title="Сменить тему">
511     <i class="bi bi-moon-stars-fill" id="theme-icon"><
    /i>
512     </button>
513 </div>
514 </div>
515
516 <!-- Big Microphone Centerpiece -->
517 <div class="mic-centerpiece">
518     <div class="mic-button-wrapper">
519         <button class="btn-mic-giant" id="btn-mic" title="На
Жмите для голосового ввода">
520             <i class="bi bi-mic-fill" id="mic-icon"></i>
521         </button>
522     </div>
523
524     <div class="wave-bars" id="wave-bars">
525         <div class="wave-bar"></div>
526         <div class="wave-bar"></div>
527         <div class="wave-bar"></div>
528         <div class="wave-bar"></div>
529         <div class="wave-bar"></div>
530     </div>
531
532     <div class="fw-semibold mt-2 text-center" id="voice-st
atus-text" style="font-size: 0.95rem;">
533         Нажмите кнопку и говорите
534     </div>
535     <div class="small text-muted text-center" id="voice-su
btext" style="font-size: 0.78rem;">
536         Голос будет распознан и отправлен в чат
537     </div>
538
539 <!-- Quick Media & Audio Upload Actions -->
540 <div class="media-actions-bar">
541     <button class="btn-media-action" id="btn-upload-audi
o" title="Загрузить аудиозапись или диктофон">
542         <i class="bi bi-file-earmark-music-fill text-info"

```

```

></i>
543         <span>Аудиофайл</span>
544     </button>
545     <button class="btn-media-action" id="btn-upload-media" title="Загрузить фото, видео или документ">
546         <i class="bi bi-cloud-arrow-up-fill text-primary">
547     </i>
548         <span>Медиа</span>
549     </button>
550 </div>
551 <!-- Hidden File Inputs -->
552 <input type="file" id="audio-file-input" accept="audio
/*,.mp3,.wav,.ogg,.m4a,.flac,.aac,.webm,.opus" class="d-none">
553 <input type="file" id="media-file-input" accept="image
/*,video/*,audio/*,.pdf,.doc,.docx,.txt" class="d-none">
554 </div>
555
556 <!-- Chat Stream Feed -->
557 <div class="d-flex justify-content-between align-items-center mt-1 mb-1 px-1">
558     <span class="small text-muted fw-bold text-uppercase" style="letter-spacing: 0.5px; font-size: 0.75rem;">Диалог</span>
559     <button class="btn btn-sm btn-link text-muted p-0 text-decoration-none small" id="btn-clear-chat" title="Очистить диалог">
560         <i class="bi bi-trash"></i> Очистить
561     </button>
562 </div>
563
564 <div class="chat-container" id="chat-feed">
565     <div class="text-muted small text-center my-auto p-3" id="empty-feed-hint">
566         <i class="bi bi-chat-dots fs-3 d-block mb-1 opacity-50"></i>
567         Голосовые команды и медиафайлы отобразятся здесь
568     </div>

```

```
569 |     </div>
570 |
571 |     <!-- Fallback Quick Text Bar & Attach Clip -->
572 |     <div class="mt-2">
573 |         <form id="text-send-form" class="d-flex gap-2">
574 |             <button type="button" class="btn btn-outline-seconda
ry btn-sm px-2.5 rounded-3 d-flex align-items-center justify-c
content-center" id="btn-clip-attachment" title="Прикрепить файл
">
575 |                 <i class="bi bi-paperclip fs-6"></i>
576 |             </button>
577 |             <input type="text" class="form-control form-control-
custom form-control-sm py-2" id="text-input" placeholder="Или
введите сообщение текстом..." autocomplete="off">
578 |             <button type="submit" class="btn btn-primary btn-sm
px-3 rounded-3" id="btn-send-text" title="Отправить">
579 |                 <i class="bi bi-send-fill"></i>
580 |             </button>
581 |         </form>
582 |     </div>
583 |
584 | </div>
585 | </div>
586 |
587 | <script>
588 |     // Global State
589 |     let recognition = null;
590 |     let isListening = false;
591 |     let isTtsEnabled = true;
592 |     let chatHistory = [];
593 |     let abortController = null;
594 |     let currentSpeechAudio = null;
595 |     let currentSpeechQueue = null;
596 |
597 |     // Elements
598 |     const authPanel = document.getElementById('auth-panel');
599 |     const mainPanel = document.getElementById('main-panel');
600 |     const authAlert = document.getElementById('auth-alert');
```

```
601     const authFormLogin = document.getElementById('auth-form-l  
    ogin');  
602     const authFormRegister = document.getElementById('auth-for  
    m-register');  
603     const authFormVerify = document.getElementById('auth-form-  
    verify');  
604  
605     const btnMic = document.getElementById('btn-mic');  
606     const micIcon = document.getElementById('mic-icon');  
607     const voiceStatusText = document.getElementById('voice-sta  
    tus-text');  
608     const voiceSubtext = document.getElementById('voice-subtex  
    t');  
609     const waveBars = document.getElementById('wave-bars');  
610     const chatFeed = document.getElementById('chat-feed');  
611     const emptyFeedHint = document.getElementById('empty-feed-  
    hint');  
612     const userDisplayName = document.getElementById('user-disp  
    lay-name');  
613  
614     const btnUploadAudio = document.getElementById('btn-upload  
    -audio');  
615     const btnUploadMedia = document.getElementById('btn-upload  
    -media');  
616     const btnClipAttachment = document.getElementById('btn-cli  
    p-attachment');  
617     const audioFileInput = document.getElementById('audio-file  
    -input');  
618     const mediaFileInput = document.getElementById('media-file  
    -input');  
619  
620     const textSendForm = document.getElementById('text-send-fo  
    rm');  
621     const textInput = document.getElementById('text-input');  
622     const ttsToggleBtn = document.getElementById('tts-toggle-b  
    tn');  
623     const ttsIcon = document.getElementById('tts-icon');  
624     const btnClearChat = document.getElementById('btn-clear-ch
```

```

    at');
625
626 // Telegram Web App Integration
627 function initTelegram() {
628     if (window.Telegram && window.Telegram.WebApp) {
629         const tg = window.Telegram.WebApp;
630         tg.ready();
631         tg.expand();
632
633         window.triggerHaptic = function(type = 'light') {
634             if (tg.HapticFeedback) {
635                 tg.HapticFeedback.impactOccurred(type);
636             }
637         };
638     } else {
639         window.triggerHaptic = function() {};
640     }
641 }
642
643 // Auth API
644 const api = {
645     async fetch(url, options = {}) {
646         const res = await fetch(url, options);
647         if (!res.ok) {
648             let msg = res.statusText;
649             try {
650                 const data = await res.json();
651                 msg = data.detail || msg;
652             } catch {}
653             throw new Error(msg);
654         }
655         return res.json();
656     },
657     auth: {
658         async check() {
659             try {
660                 const res = await fetch('/auth/check');
661                 return await res.json();

```



```
662         } catch {
663             return { authenticated: false };
664         }
665     },
666     async login(email, password) {
667         return api.fetch('/auth/login/email', {
668             method: 'POST',
669             headers: { 'Content-Type': 'application/json' },
670             body: JSON.stringify({ email, password })
671         });
672     },
673     async register(email, password, name) {
674         return api.fetch('/auth/register', {
675             method: 'POST',
676             headers: { 'Content-Type': 'application/json' },
677             body: JSON.stringify({ email, password, name })
678         });
679     },
680     async verify(email, code) {
681         return api.fetch('/auth/verify', {
682             method: 'POST',
683             headers: { 'Content-Type': 'application/json' },
684             body: JSON.stringify({ email, code })
685         });
686     }
687 }
688 };
689
690 // Panel Switching
691 function showPanel(panel) {
692     if (panel === 'auth') {
693         authPanel.classList.remove('d-none');
694         mainPanel.classList.add('d-none');
695     } else {
696         authPanel.classList.add('d-none');
697         mainPanel.classList.remove('d-none');
698     }
699 }
```

```
700
701 // Auth Handlers
702 document.getElementById('go-to-register').addEventListener
  ('click', (e) => {
703     e.preventDefault();
704     authAlert.classList.add('d-none');
705     authFormLogin.classList.add('d-none');
706     authFormRegister.classList.remove('d-none');
707 });
708
709 document.getElementById('go-to-login').addEventListener('c
lick', (e) => {
710     e.preventDefault();
711     authAlert.classList.add('d-none');
712     authFormRegister.classList.add('d-none');
713     authFormLogin.classList.remove('d-none');
714 });
715
716 let currentVerifyEmail = '';
717 document.getElementById('form-register').addEventListener(
  'submit', async (e) => {
718     e.preventDefault();
719     authAlert.classList.add('d-none');
720     const name = document.getElementById('register-name').va
lue;
721     const email = document.getElementById('register-email').
value;
722     const password = document.getElementById('register-passw
ord').value;
723     try {
724         await api.auth.register(email, password, name);
725         currentVerifyEmail = email;
726         authFormRegister.classList.add('d-none');
727         authFormVerify.classList.remove('d-none');
728     } catch (err) {
729         authAlert.textContent = err.message || 'Ошибка регистр
ации';
730         authAlert.classList.remove('d-none');
```

```
731 |     }
732 |   });
733 |
734 |   document.getElementById('form-verify').addEventListener('submit', async (e) => {
735 |     e.preventDefault();
736 |     authAlert.classList.add('d-none');
737 |     const code = document.getElementById('verify-code').value;
738 |     try {
739 |       await api.auth.verify(currentVerifyEmail, code);
740 |       authFormVerify.classList.add('d-none');
741 |       authFormLogin.classList.remove('d-none');
742 |       authAlert.className = 'alert alert-success';
743 |       authAlert.textContent = 'Почта подтверждена! Войдите в аккаунт.';
744 |       authAlert.classList.remove('d-none');
745 |     } catch (err) {
746 |       authAlert.textContent = err.message || 'Неверный код';
747 |       authAlert.classList.remove('d-none');
748 |     }
749 |   });
750 |
751 |   document.getElementById('form-login').addEventListener('submit', async (e) => {
752 |     e.preventDefault();
753 |     authAlert.classList.add('d-none');
754 |     const email = document.getElementById('login-email').value;
755 |     const password = document.getElementById('login-password').value;
756 |     try {
757 |       await api.auth.login(email, password);
758 |       await initApp();
759 |     } catch (err) {
760 |       authAlert.textContent = err.message || 'Ошибка входа';
761 |       authAlert.classList.remove('d-none');
762 |     }
```

```
763     });
764
765     document.getElementById('google-signin-btn').addEventListener('click', () => {
766         window.location.href = '/auth/google?next=' + encodeURIComponent(window.location.pathname);
767     });
768
769     // Voice & Speech Recognition Module
770     function initSpeechRecognition() {
771         const SpeechRecognition = window.SpeechRecognition || window.webkitSpeechRecognition;
772         if (!SpeechRecognition) {
773             voiceStatusText.textContent = 'Голосовой ввод не поддерживается';
774             voiceSubtext.textContent = 'Используйте загрузку аудио файлов или ввод ниже';
775             btnMic.disabled = true;
776             btnMic.style.opacity = '0.5';
777             return;
778         }
779
780         recognition = new SpeechRecognition();
781         recognition.continuous = false;
782         recognition.interimResults = false;
783
784         const getSelectedLang = () => {
785             const checked = document.querySelector('input[name="lang-radio"]:checked');
786             return checked ? checked.value : 'ru-RU';
787         };
788
789         recognition.lang = getSelectedLang();
790         document.querySelectorAll('input[name="lang-radio"]').forEach((radio) => {
791             radio.addEventListener('change', () => {
792                 recognition.lang = getSelectedLang();
793             });
794         });
795     }
796 }
```

```
794 |     });
795 |
796 |     recognition.onstart = () => {
797 |         isListening = true;
798 |         btnMic.className = 'btn-mic-giant listening';
799 |         micIcon.className = 'bi bi-mic-fill';
800 |         waveBars.classList.add('active');
801 |         voiceStatusText.textContent = 'Слушаю вас...';
802 |         voiceSubtext.textContent = 'Говорите команду или вопро
| с';
803 |     };
804 |
805 |     recognition.onend = () => {
806 |         isListening = false;
807 |         waveBars.classList.remove('active');
808 |         if (!btnMic.classList.contains('thinking')) {
809 |             btnMic.className = 'btn-mic-giant';
810 |             micIcon.className = 'bi bi-mic-fill';
811 |             voiceStatusText.textContent = 'Нажмите кнопку и гово
| рите';
812 |             voiceSubtext.textContent = 'Голос будет распознан и
| отправлен в чат';
813 |         }
814 |     };
815 |
816 |     recognition.onerror = (e) => {
817 |         isListening = false;
818 |         waveBars.classList.remove('active');
819 |         btnMic.className = 'btn-mic-giant';
820 |         voiceStatusText.textContent = 'Ошибка ввода: ' + (e.er
| ror || 'попробуйте снова');
821 |         voiceSubtext.textContent = 'Нажмите микрофон для повто
| рной попытки';
822 |     };
823 |
824 |     recognition.onresult = (event) => {
825 |         const transcript = event.results[0][0].transcript;
826 |         if (transcript && transcript.trim()) {
```

```

827         handleUserMessage(transcript.trim());
828     }
829 };
830
831 btnMic.addEventListener('click', () => {
832     window.triggerHaptic('medium');
833     stopAudio();
834
835     // Unlock audio context for mobile iOS/Safari/Chrome
836     if ('speechSynthesis' in window) {
837         const silent = new SpeechSynthesisUtterance(' ');
838         silent.volume = 0;
839         window.speechSynthesis.speak(silent);
840     }
841
842     if (isListening) {
843         recognition.stop();
844     } else {
845         try {
846             recognition.start();
847         } catch (e) {
848             console.warn('Recognition start error:', e);
849         }
850     }
851 });
852 }
853
854 // Audio Queue Player for Backend TTS
855 class AudioQueuePlayer {
856     constructor(chunks) {
857         this.chunks = chunks;
858         this.currentIndex = 0;
859         this.isPlaying = false;
860         this.audioElements = new Map();
861         this.activeAudio = null;
862         this._prefetch(0);
863         this._prefetch(1);
864     }

```

```

865 |
866 |     _prefetch(index) {
867 |         if (index >= this.chunks.length || this.audioElements.
has(index)) return;
868 |         const text = this.chunks[index];
869 |         const queryParams = new URLSearchParams({ text });
870 |         const audioUrl = `/api/tts/synthesize?${queryParams.to
String()}`;
871 |         const audio = new Audio(audioUrl);
872 |         audio.load();
873 |         this.audioElements.set(index, audio);
874 |     }
875 |
876 |     async play() {
877 |         this.isPlaying = true;
878 |         while (this.currentIndex < this.chunks.length && this.
isPlaying) {
879 |             this._prefetch(this.currentIndex + 1);
880 |             this._prefetch(this.currentIndex + 2);
881 |
882 |             const audio = this.audioElements.get(this.currentInd
ex);
883 |             if (!audio) {
884 |                 this.currentIndex++;
885 |                 continue;
886 |             }
887 |
888 |             this.activeAudio = audio;
889 |             currentSpeechAudio = audio;
890 |
891 |             const playPromise = new Promise((resolve) => {
892 |                 audio.onended = () => resolve();
893 |                 audio.onerror = () => {
894 |                     this.stop();
895 |                     resolve();
896 |                 };
897 |             });
898 |

```

```
899 |         try {
900 |             await audio.play();
901 |             await playPromise;
902 |         } catch (err) {
903 |             console.error('Audio playback interrupted:', err);
904 |         }
905 |         this.currentIndex++;
906 |     }
907 |     this.isPlaying = false;
908 | }
909 |
910 | stop() {
911 |     this.isPlaying = false;
912 |     if (this.activeAudio) {
913 |         try {
914 |             this.activeAudio.pause();
915 |             this.activeAudio.src = '';
916 |         } catch (e) {}
917 |     }
918 |     this.audioElements.clear();
919 | }
920 |
921 |
922 | function stopAudio() {
923 |     if (currentSpeechQueue) {
924 |         try { currentSpeechQueue.stop(); } catch (e) {}
925 |         currentSpeechQueue = null;
926 |     }
927 |     if (currentSpeechAudio) {
928 |         try {
929 |             currentSpeechAudio.pause();
930 |             currentSpeechAudio.src = '';
931 |         } catch (e) {}
932 |         currentSpeechAudio = null;
933 |     }
934 |     if ('speechSynthesis' in window) {
935 |         try { window.speechSynthesis.cancel(); } catch (e) {}
936 |     }
```



```

937 |     }
938 |
939 |     async function speakText(text) {
940 |         if (!isTtsEnabled || !text) return;
941 |         stopAudio();
942 |
943 |         const cleanText = text.replace(/<[^>]*>/g, '').replace(/
944 |         [*_`#]/g, '').trim();
945 |         if (!cleanText) return;
946 |
947 |         // Try Backend TTS
948 |         try {
949 |             const paragraphs = cleanText.split(/\n+/).map(p => p.t
950 |             rim()).filter(Boolean);
951 |             let chunks = [];
952 |             for (const para of paragraphs) {
953 |                 const sentences = para.match(/^[^.!?]+[.!?]*$/g) || [p
954 |                 ara];
955 |                 let currentChunk = '';
956 |                 for (const s of sentences) {
957 |                     const tr = s.trim();
958 |                     if (!tr) continue;
959 |                     if ((currentChunk + ' ' + tr).length > 200) {
960 |                         chunks.push(currentChunk.trim());
961 |                         currentChunk = tr;
962 |                     } else {
963 |                         currentChunk = (currentChunk + ' ' + tr).trim();
964 |                     }
965 |                 }
966 |                 if (currentChunk) chunks.push(currentChunk.trim());
967 |             }
968 |             chunks = chunks.filter(Boolean);
969 |
970 |             if (chunks.length > 0) {
971 |                 currentSpeechQueue = new AudioQueuePlayer(chunks);
972 |                 await currentSpeechQueue.play();
973 |                 return;
974 |             }

```

```

972 |         } catch (e) {
973 |             console.warn('Backend TTS failed, falling back to Web
| Speech Synthesis:', e);
974 |         }
975 |
976 |         // Fallback: Browser SpeechSynthesis
977 |         if ('speechSynthesis' in window) {
978 |             const utterance = new SpeechSynthesisUtterance(cleanTe
| xt);
979 |             const selectedLang = document.querySelector('input[nam
| e="lang-radio"]:checked')?.value || 'ru-RU';
980 |             utterance.lang = selectedLang;
981 |             window.speechSynthesis.speak(utterance);
982 |         }
983 |     }
984 |
985 |     // Message Rendering Helpers
986 |     function appendMessage(role, text, isStreaming = false) {
987 |         if (emptyFeedHint) {
988 |             emptyFeedHint.remove();
989 |         }
990 |
991 |         const bubble = document.createElement('div');
992 |         bubble.className = `chat-bubble ${role}`;
993 |
994 |         const timeStr = new Date().toLocaleTimeString([], { hour
| : '2-digit', minute: '2-digit' });
995 |
996 |         if (role === 'user') {
997 |             bubble.innerHTML = `<div>${escapeHtml(text)}</div><div
| class="chat-meta"><span>${timeStr}</span></div>`;
998 |         } else {
999 |             bubble.innerHTML = `
1000 |                 <div class="ai-content">${window.marked ? marked.par
| se(text) : escapeHtml(text)}</div>
1001 |                 <div class="chat-meta">
1002 |                     <span>${timeStr}</span>
1003 |                     <button class="btn-speak-again" title="Озвучить"><

```

```

        i class="bi bi-volume-up-fill"></i></button>
1004         </div>
1005         `;
1006         const speakBtn = bubble.querySelector('.btn-speak-again');
1007         if (speakBtn) {
1008             speakBtn.addEventListener('click', () => speakText(bubble.querySelector('.ai-content').innerText));
1009         }
1010     }
1011
1012     chatFeed.appendChild(bubble);
1013     chatFeed.scrollTop = chatFeed.scrollHeight;
1014     return bubble;
1015 }
1016
1017 function appendMediaUserMessage(file, previewUrl) {
1018     if (emptyFeedHint) {
1019         emptyFeedHint.remove();
1020     }
1021
1022     const bubble = document.createElement('div');
1023     bubble.className = 'chat-bubble user';
1024     const timeStr = new Date().toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
1025
1026     const sizeFormatted = (file.size / (1024 * 1024)).toFixed(2) + ' MB';
1027     const isAudio = file.type.startsWith('audio/') || /\. (mp3|wav|ogg|m4a|flac|aac|webm)$/i.test(file.name);
1028     const isImage = file.type.startsWith('image/');
1029     const isVideo = file.type.startsWith('video/');
1030
1031     let mediaSnippet = '';
1032     if (isAudio && previewUrl) {
1033         mediaSnippet = `<audio class="media-player-widget" controls src="${previewUrl}"></audio>`;
1034     } else if (isImage && previewUrl) {

```

```
1035     mediaSnippet = ``;
1036     } else if (isVideo && previewUrl) {
1037     mediaSnippet = `<video class="media-player-widget" con
| trols src="${previewUrl}"></video>`;
1038     }
1039
1040     const iconClass = isAudio ? 'bi-file-earmark-music' : (i
| sImage ? 'bi-image' : (isVideo ? 'bi-camera-video' : 'bi-file-
| earmark-text'));
1041
1042     bubble.innerHTML = `
1043     <div class="media-preview-card">
1044     <i class="bi ${iconClass} fs-4 text-info"></i>
1045     <div class="text-truncate">
1046     <div class="fw-semibold text-truncate small">${esc
| apeHtml(file.name)}</div>
1047     <div class="text-muted" style="font-size: 0.72rem;
| ">${sizeFormatted}</div>
1048     </div>
1049     </div>
1050     ${mediaSnippet}
1051     <div class="chat-meta"><span>${timeStr}</span></div>
1052     `;
1053
1054     chatFeed.appendChild(bubble);
1055     chatFeed.scrollTop = chatFeed.scrollHeight;
1056     return bubble;
1057 }
1058
1059 function escapeHtml(text) {
1060     const div = document.createElement('div');
1061     div.textContent = text;
1062     return div.innerHTML;
1063 }
1064
1065 // Send Voice/Text Query to AI Chat API
1066 async function handleUserMessage(query) {
```

```
1067 |         if (!query || !query.trim()) return;
1068 |
1069 |         appendMessage('user', query);
1070 |         stopAudio();
1071 |
1072 |         btnMic.className = 'btn-mic-giant thinking';
1073 |         micIcon.className = 'bi bi-arrow-repeat';
1074 |         voiceStatusText.textContent = 'Обработка ответа AI...';
1075 |         voiceSubtext.textContent = 'Генерация ответа моделью';
1076 |
1077 |         if (abortController) {
1078 |             try { abortController.abort(); } catch (e) {}
1079 |         }
1080 |         abortController = new AbortController();
1081 |
1082 |         const aiBubble = appendMessage('ai', '...');
1083 |         const aiContentDiv = aiBubble.querySelector('.ai-content
1084 | ');
1085 |
1086 |         let fullReply = '';
1087 |         let voiceReply = '';
1088 |
1089 |         try {
1090 |             const payload = {
1091 |                 message: query,
1092 |                 history: chatHistory,
1093 |                 generation_config: { is_rc: true }
1094 |             };
1095 |
1096 |             const response = await fetch('/api/chat', {
1097 |                 method: 'POST',
1098 |                 headers: { 'Content-Type': 'application/json' },
1099 |                 body: JSON.stringify(payload),
1100 |                 signal: abortController.signal
1101 |             });
1102 |
1103 |             if (!response.ok) {
1104 |                 throw new Error(`Ошибка сервера: ${response.status}`
```

```

    );
1104     }
1105
1106     const reader = response.body.getReader();
1107     const decoder = new TextDecoder('utf-8');
1108     let buffer = '';
1109
1110     while (true) {
1111         const { value, done } = await reader.read();
1112         if (done) break;
1113
1114         buffer += decoder.decode(value, { stream: true });
1115         const lines = buffer.split('\n');
1116         buffer = lines.pop();
1117
1118         for (const line of lines) {
1119             const tr = line.trim();
1120             if (!tr) continue;
1121             if (tr.startsWith('data: ')) {
1122                 try {
1123                     const data = JSON.parse(tr.substring(6));
1124                     if (data.error) throw new Error(data.error);
1125                     if (data.voice) voiceReply += data.voice;
1126                     if (data.text) {
1127                         fullReply += data.text;
1128                         aiContentDiv.innerHTML = window.marked ? mar
1129 ked.parse(fullReply) : escapeHtml(fullReply);
1130                         chatFeed.scrollTop = chatFeed.scrollHeight;
1131                     }
1132                 } catch (e) {
1133                     console.error(e);
1134                 }
1135             }
1136         }
1137
1138         btnMic.className = 'btn-mic-giant';
1139         micIcon.className = 'bi bi-mic-fill';

```

```
1140 |         voiceStatusText.textContent = 'Нажмите кнопку и говори
      | те';
1141 |         voiceSubtext.textContent = 'Готов к следующему вопросу
      | ';
1142 |
1143 |         // Update history
1144 |         chatHistory.push({ role: 'user', parts: [query] });
1145 |         chatHistory.push({ role: 'model', parts: [fullReply] }
      | );
1146 |         if (chatHistory.length > 20) chatHistory = chatHistory
      | .slice(-20);
1147 |
1148 |         // Speak response
1149 |         speakText(voiceReply || fullReply);
1150 |
1151 |     } catch (err) {
1152 |         if (err.name === 'AbortError') return;
1153 |         aiContentDiv.innerHTML = `<span class="text-danger">Ош
      | ибка: ${escapeHtml(err.message)}</span>`;
1154 |         btnMic.className = 'btn-mic-giant';
1155 |         micIcon.className = 'bi bi-mic-fill';
1156 |         voiceStatusText.textContent = 'Ошибка выполнения';
1157 |         voiceSubtext.textContent = 'Попробуйте нажать кнопку е
      | ще раз';
1158 |     }
1159 | }
1160 |
1161 | // Process Uploaded Audio / Media File
1162 | async function processUploadedFile(file, isAudioOnly = fal
      | se) {
1163 |     if (!file) return;
1164 |     window.triggerHaptic('medium');
1165 |     stopAudio();
1166 |
1167 |     const previewUrl = URL.createObjectURL(file);
1168 |     appendMediaUserMessage(file, previewUrl);
1169 |
1170 |     btnMic.className = 'btn-mic-giant thinking';
```

```
1171 micIcon.className = 'bi bi-arrow-repeat';
1172 voiceStatusText.textContent = 'Загрузка и анализ медиа..
    .';
1173 voiceSubtext.textContent = file.name;
1174
1175 const aiBubble = appendMessage('ai', '□ Загрузка и обраб
отка файла...');
1176 const aiContentDiv = aiBubble.querySelector('.ai-content
    ');
1177
1178 const isAudio = isAudioOnly || file.type.startsWith('aud
io/') || /\.(mp3|wav|ogg|m4a|flac|aac|webm)$/i.test(file.name)
    ;
1179
1180 try {
1181     if (isAudio) {
1182         // Send to Audio Diarization & Transcription service
1183         const formData = new FormData();
1184         formData.append('file', file);
1185         const selectedLang = document.querySelector('input[n
ame="lang-radio"]:checked').value || 'ru-RU';
1186         formData.append('language', selectedLang.split('-')[
0]);
1187
1188         const res = await fetch('/api/audio/diarize', {
1189             method: 'POST',
1190             body: formData
1191         });
1192
1193         if (!res.ok) {
1194             const errData = await res.json().catch(() => ({}))
        ;
1195             throw new Error(errData.detail || `Ошибка сервера:
        ${res.status}`);
1196         }
1197
1198         const result = await res.json();
1199         const diarizationData = result.data || {};
```



```

1200 |
1201 |         let formattedOutput = '';
1202 |         if (diarizationData.summary) {
1203 |             formattedOutput += `### □ Краткая сводка\n${diari
1204 | zationData.summary}\n\n`;
1205 |         }
1206 |         if (diarizationData.key_points && diarizationData.ke
1207 | y_points.length > 0) {
1208 |             formattedOutput += `### □ Ключевые тезисы\n` + di
1209 | arizationData.key_points.map(kp => `- ${kp}`).join('\n') + `\n
1210 | \n`;
1211 |         }
1212 |         if (diarizationData.action_items && diarizationData.
1213 | action_items.length > 0) {
1214 |             formattedOutput += `### □ Задачи и решения\n` + di
1215 | arizationData.action_items.map(ai => `- [ ] ${ai}`).join('\n')
1216 | + `\n\n`;
1217 |         }
1218 |         if (diarizationData.transcript && diarizationData.tr
1219 | anscript.length > 0) {
1220 |             formattedOutput += `### □ Стенограмма диалога\n`;
1221 |             for (const item of diarizationData.transcript) {
1222 |                 const speaker = item.speaker || 'Собеседник';
1223 |                 const text = item.text || '';
1224 |                 formattedOutput += `**${speaker}:** ${text}\n\n`
1225 | ;
1226 |             }
1227 |         }
1228 |         if (!formattedOutput) {
1229 |             formattedOutput = diarizationData.markdown_report
1230 | || 'Аудиозапись успешно обработана.';
1231 |         }
1232 |         aiContentDiv.innerHTML = window.marked ? marked.pars

```

```

    e(formattedOutput) : escapeHtml(formattedOutput);
1228     chatFeed.scrollTop = chatFeed.scrollHeight;
1229
1230     chatHistory.push({ role: 'user', parts: [ `[Аудиозапи
    съ ${file.name}]` ] });
1231     chatHistory.push({ role: 'model', parts: [formattedO
    utput] });
1232     if (chatHistory.length > 20) chatHistory = chatHisto
    ry.slice(-20);
1233
1234     // Speak the executive summary if available
1235     if (diarizationData.summary) {
1236         speakText(diarizationData.summary);
1237     }
1238
1239     } else {
1240         // General Media Upload to storage
1241         const formData = new FormData();
1242         formData.append('file', file);
1243
1244         const res = await fetch('/api/user/storage/upload',
    {
1245             method: 'POST',
1246             body: formData
1247         });
1248
1249         if (!res.ok) {
1250             const errData = await res.json().catch(() => ({}))
    ;
1251             throw new Error(errData.detail || `Ошибка загрузки
    : ${res.status}`);
1252         }
1253
1254         const resData = await res.json();
1255         const fileName = resData.filename || file.name;
1256
1257         // Request AI analysis of the uploaded document/medi
    a

```

```
1258 |         const prompt = `Пользователь загрузил файл ${fileName}
      |     (${file.type} || 'документ')). Подтверди получение и опиши д
      |     альнейшие действия.`;
1259 |
1260 |         const chatRes = await fetch('/api/chat', {
1261 |             method: 'POST',
1262 |             headers: { 'Content-Type': 'application/json' },
1263 |             body: JSON.stringify({
1264 |                 message: prompt,
1265 |                 history: chatHistory,
1266 |                 generation_config: { is_rc: true }
1267 |             })
1268 |         });
1269 |
1270 |         let modelReply = `Файл **${escapeHtml(fileName)}** у
      |     спешно загружен и сохранен в вашем хранилище.`;
1271 |         if (chatRes.ok) {
1272 |             // Read stream or fallback
1273 |             const reader = chatRes.body.getReader();
1274 |             const decoder = new TextDecoder('utf-8');
1275 |             let buffer = '';
1276 |             let streamReply = '';
1277 |
1278 |             while (true) {
1279 |                 const { value, done } = await reader.read();
1280 |                 if (done) break;
1281 |                 buffer += decoder.decode(value, { stream: true }
      |     );
1282 |
1283 |                 const lines = buffer.split('\n');
1284 |                 buffer = lines.pop();
1285 |
1286 |                 for (const line of lines) {
1287 |                     const tr = line.trim();
1288 |                     if (tr.startsWith('data: ')) {
1289 |                         try {
1290 |                             const data = JSON.parse(tr.substring(6));
1291 |                             if (data.text) streamReply += data.text;
```

```

1292         }
1293     }
1294 }
1295     if (streamReply) modelReply = streamReply;
1296 }
1297
1298     aiContentDiv.innerHTML = window.marked ? marked.parse(modelReply) : escapeHtml(modelReply);
1299     chatFeed.scrollTop = chatFeed.scrollHeight;
1300
1301     chatHistory.push({ role: 'user', parts: [`[Загружен
1302 файл ${file.name}]`] });
1303     chatHistory.push({ role: 'model', parts: [modelReply
1304 ] });
1305     if (chatHistory.length > 20) chatHistory = chatHistory.slice(-20);
1306
1307     speakText(modelReply);
1308 }
1309
1310     btnMic.className = 'btn-mic-giant';
1311     micIcon.className = 'bi bi-mic-fill';
1312     voiceStatusText.textContent = 'Нажмите кнопку и говорите';
1313     voiceSubtext.textContent = 'Готов к следующему вопросу или загрузке';
1314
1315 } catch (err) {
1316     aiContentDiv.innerHTML = `Ошибка обработки: ${escapeHtml(err.message)}</span>`;
1317     btnMic.className = 'btn-mic-giant';
1318     micIcon.className = 'bi bi-mic-fill';
1319     voiceStatusText.textContent = 'Ошибка загрузки';
1320     voiceSubtext.textContent = 'Попробуйте загрузить файл повторно';
1321 }
1322 }

```

```
1322 // Media Buttons Event Listeners
1323 btnUploadAudio.addEventListener('click', () => {
1324     window.triggerHaptic('light');
1325     audioFileInput.click();
1326 });
1327
1328 btnUploadMedia.addEventListener('click', () => {
1329     window.triggerHaptic('light');
1330     mediaFileInput.click();
1331 });
1332
1333 btnClipAttachment.addEventListener('click', () => {
1334     window.triggerHaptic('light');
1335     mediaFileInput.click();
1336 });
1337
1338 audioFileInput.addEventListener('change', (e) => {
1339     const file = e.target.files && e.target.files[0];
1340     if (file) {
1341         processUploadedFile(file, true);
1342         audioFileInput.value = '';
1343     }
1344 });
1345
1346 mediaFileInput.addEventListener('change', (e) => {
1347     const file = e.target.files && e.target.files[0];
1348     if (file) {
1349         processUploadedFile(file, false);
1350         mediaFileInput.value = '';
1351     }
1352 });
1353
1354 // Text Fallback Form
1355 textSendForm.addEventListener('submit', (e) => {
1356     e.preventDefault();
1357     const val = textInput.value.trim();
1358     if (val) {
1359         textInput.value = '';
```

```

1360     handleUserMessage(val);
1361 }
1362 });
1363
1364 // TTS Toggle Button
1365 ttsToggleBtn.addEventListener('click', () => {
1366     window.triggerHaptic('light');
1367     isTtsEnabled = !isTtsEnabled;
1368     if (isTtsEnabled) {
1369         ttsIcon.className = 'bi bi-volume-up-fill';
1370         ttsToggleBtn.classList.remove('opacity-50');
1371     } else {
1372         stopAudio();
1373         ttsIcon.className = 'bi bi-volume-mute-fill';
1374         ttsToggleBtn.classList.add('opacity-50');
1375     }
1376 });
1377
1378 // Clear Chat Button
1379 btnClearChat.addEventListener('click', () => {
1380     window.triggerHaptic('light');
1381     stopAudio();
1382     chatHistory = [];
1383     chatFeed.innerHTML = `
1384         <div class="text-muted small text-center my-auto p-3"
1385         id="empty-feed-hint">
1386             <i class="bi bi-chat-dots fs-3 d-block mb-1 opacity-
1387             50"></i>
1388                 Голосовые команды и медиафайлы отобразятся здесь
1389             </div>
1390         `;
1391 });
1392
1393 // Theme Toggle
1394 document.getElementById('theme-toggle-btn').addEventListener('click', () => {
1395     window.triggerHaptic('light');
1396     const isLight = document.body.classList.toggle('theme-li

```

```

    ght');
1395     const icon = document.getElementById('theme-icon');
1396     icon.className = isLight ? 'bi bi-sun-fill' : 'bi bi-moon-stars-fill';
1397   });
1398
1399   // App Initialization
1400   async function initApp() {
1401     const state = await api.auth.check();
1402     if (state.authenticated) {
1403       if (userDisplayName && state.user) {
1404         userDisplayName.textContent = state.user.name || state.user.email || 'Пользователь';
1405       }
1406       showPanel('main');
1407       initSpeechRecognition();
1408     } else {
1409       showPanel('auth');
1410     }
1411   }
1412
1413   document.addEventListener('DOMContentLoaded', () => {
1414     initTelegram();
1415     initApp();
1416   });
1417 </script>
1418 </body>
1419 </html>
1420

```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/scenarios_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/scenarios_tab/index.htm
17         l" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/scenarios_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-06 10:38:00
25 =====
26 =====
27 -->
28
29 <!-- Scenarios & Automated Diagnostics Tab Template -->
30 <div class="container-fluid p-0 scenarios-tab-wrapper">
31
32     <!-- Header & Quick Actions -->
33     <div class="card border-secondary-subtle mb-3 shadow-sm">
34         <div class="card-body p-3">
35             <div class="d-flex flex-wrap align-items-center justify-
36             content-between gap-3">
37                 <div class="d-flex align-items-center gap-2">
```



```

32 |         <div class="fs-2">□ </div>
33 |         <div>
34 |             <h5 class="mb-0 fw-bold d-flex align-items-center
gap-2">
35 |                 <span>Сценарии тестирования и автоматизации</spa
n>
36 |                 <span class="badge bg-primary-subtle text-primar
y border border-primary-subtle small fs-6">Test Computer (/tc)
</span>
37 |             </h5>
38 |             <div class="text-muted small">
39 |                 Центр управления: экспресс-сценарии проверки хос
та, аудит логов, диагностика оборудования и интеллектуальный а
ссистент навыков.
40 |             </div>
41 |         </div>
42 |     </div>
43 |     <div class="d-flex align-items-center gap-2 flex-wrap"
>
44 |         <button class="btn btn-outline-info btn-sm d-flex al
ign-items-center gap-1 shadow-sm" id="btn-scenario-refresh-lis
t" title="Обновить список сценариев">
45 |             <i class="bi bi-arrow-clockwise"></i>
46 |             <span>Обновить</span>
47 |         </button>
48 |         <button class="btn btn-success btn-sm d-flex align-i
tems-center gap-1 shadow-sm fw-semibold" id="btn-run-quick-che
ck" title="Запустить экспресс-проверку окружения">
49 |             <span>□ Быстрая проверка</span>
50 |             <span class="badge bg-white text-dark ms-1">Рекоме
ндуется</span>
51 |         </button>
52 |         <button class="btn btn-outline-primary btn-sm d-flex
align-items-center gap-1 shadow-sm" id="btn-run-all-scenarios
" title="Последовательно выполнить все доступные сценарии">
53 |             <i class="bi bi-play-fill"></i>
54 |             <span>Запустить все</span>
55 |         </button>

```

```

56 |         <button class="btn btn-primary btn-sm d-flex align-i
| items-center gap-1.5 shadow-sm fw-semibold" id="btn-open-scenar
| ios-drawer" data-bs-toggle="offcanvas" data-bs-target="#scenar
| iosOffcanvas" aria-controls="scenariosOffcanvas" title="Открыт
| ь выдвижную панель со сценариями">
57 |             <i class="bi bi-layout-sidebar-reverse"></i>
58 |             <span>Сценарии</span>
59 |             <span class="badge bg-white text-dark ms-1" id="sc
| enarios-count-badge-header">6</span>
60 |         </button>
61 |     </div>
62 | </div>
63 | </div>
64 | </div>
65 |
66 | <!-- Main View: AI Assistant Mini-Chat (expanded to full pan
| el width and >= 80% height) -->
67 | <div class="row g-3">
68 |     <div class="col-12">
69 |         <div class="card scenario-chat-card border-secondary-sub
| tle shadow-sm d-flex flex-column" style="min-height: 80vh; hei
| ght: calc(100vh - 170px);">
70 |
71 |             <!-- Chat Header -->
72 |             <div class="card-header border-secondary-subtle py-2 p
| x-3 d-flex align-items-center justify-content-between flex-wra
| p gap-2">
73 |                 <div class="d-flex align-items-center gap-2">
74 |                     <span class="fs-5">📄 </span>
75 |                     <div>
76 |                         <h6 class="mb-0 fw-bold">AI Ассистент сценариев
| & Навыков</h6>
77 |                         <div class="text-muted small" style="font-size:
| 0.75rem;">Спросите что угодно о компьютере – система проверит
| и создаст навык</div>
78 |                     </div>
79 |                 </div>
80 |                 <div class="d-flex align-items-center gap-2">

```

```

81 |         <button class="btn btn-outline-info btn-sm py-0.5
    | px-2 text-info border-info border-opacity-50 d-flex align-item
    | s-center gap-1" data-bs-toggle="offcanvas" data-bs-target="#sc
    | enariosOffcanvas" aria-controls="scenariosOffcanvas" title="От
    | крыть выдвижную панель со сценариями" style="font-size: 0.75re
    | m;">
82 |             <i class="bi bi-layout-sidebar-reverse"></i>
83 |             <span>Сценарии</span>
84 |         </button>
85 |         <button class="btn btn-outline-secondary btn-sm py
    | -0.5 px-2 border-secondary" id="btn-scenario-new-chat" title="
    | Очистить контекст и начать новую сессию диалога" style="font-s
    | ize: 0.75rem;">
86 |             <i class="bi bi-arrow-counterclockwise me-1"></i>
    | >Новая сессия
87 |         </button>
88 |         <button class="btn btn-outline-danger btn-sm d-fle
    | x align-items-center gap-1 ms-2" id="btn-clear-dialog" title="
    | Очистить диалог и сбросить историю" style="font-size: 0.75rem;
    | ">
89 |             <i class="bi bi-trash"></i>Очистить диалог
90 |         </button>
91 |         <span class="badge bg-success-subtle text-success
    | border border-success-subtle small">Online</span>
92 |     </div>
93 | </div>
94 |
95 |     <!-- Mini-Chat Suggestions / Prompt Pills -->
96 |     <div class="p-2 border-bottom border-secondary-subtle
    | bg-body-secondary bg-opacity-50">
97 |         <div class="small text-muted mb-1" style="font-size:
    | 0.72rem;">Быстрые запросы:</div>
98 |         <div class="d-flex flex-wrap gap-1" id="chat-quick-p
    | rompts">
99 |             <button class="btn btn-outline-secondary btn-sm py
    | -0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
100 |                 □ Какие мыши были подключены?
101 |             </button>

```

```

102 |         <button class="btn btn-outline-secondary btn-sm py
      | -0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
103 |             □ Покажи USB устройства
104 |         </button>
105 |         <button class="btn btn-outline-secondary btn-sm py
      | -0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
106 |             □ Проверь ошибки в логах
107 |         </button>
108 |         <button class="btn btn-outline-secondary btn-sm py
      | -0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
109 |             □ Сделай быструю проверку
110 |         </button>
111 |     </div>
112 | </div>
113 |
114 |     <!-- Chat Messages History Container -->
115 |     <div class="card-body p-3 overflow-y-auto flex-grow-1
      | scenario-chat-history" id="scenario-chat-history" style="min-h
      | eight: 480px; max-height: none;">
116 |         <div class="chat-message-row bot-row mb-3">
117 |             <div class="chat-msg-avatar">□ </div>
118 |             <div class="chat-msg-bot">
119 |                 <div class="mb-2">
120 |                     Привет! Я интеллектуальный ассистент Test Comp
      | uter. Вы можете задать мне любой вопрос о конфигурации хоста,
      | например:
121 |                 </div>
122 |                 <div id="chat-greeting-prompts" class="d-flex fl
      | ex-column gap-1 my-2">
123 |                     <div>• <a href="#" class="chat-prompt-link tex
      | t-decoration-none border-bottom border-primary border-opacity-
      | 50" data-prompt="Какие мыши были подключены к этому компьютеру
      | ?">«Какие мыши были подключены к этому компьютеру?»</a></div>
124 |                     <div>• <a href="#" class="chat-prompt-link tex
      | t-decoration-none border-bottom border-primary border-opacity-
      | 50" data-prompt="Покажи список сетевых портов и активных соеди
      | нений">«Покажи список сетевых портов и активных соединений»</a
      | ></div>

```

```
125 |         <div>• <a href="#" class="chat-prompt-link tex
    | t-decoration-none border-bottom border-primary border-opacity-
    | 50" data-prompt="Проверь автозапуск и службы Windows">«Проверь
    | автозапуск и службы Windows»</a></div>
126 |         </div>
127 |         <div class="text-muted small mt-2">
128 |             Если для вопроса нет готового сценария, я опро
    | шу оборудование через PowerShell/WMI, сформирую для вас ответ
    | и <strong>автоматически сохраню новый навык</strong> в каталог
    | <code>.skills/</code>.
129 |         </div>
130 |     </div>
131 | </div>
132 | </div>
133 |
134 | <!-- Chat Input Form -->
135 | <div class="card-footer border-secondary-subtle p-2 fl
    | ex-shrink-0">
136 |     <form id="scenario-chat-form" class="d-flex flex-col
    | umn gap-2">
137 |         <div class="input-group input-group-sm">
138 |             <input type="text" class="form-control" id="scen
    | ario-chat-input" placeholder="Задайте вопрос или введите /skil
    | ls list..." autocomplete="off">
139 |             <button class="btn btn-primary d-flex align-item
    | s-center gap-1 px-3" type="submit" id="btn-scenario-chat-send"
    | >
140 |                 <span>Отправить</span>
141 |                 <i class="bi bi-send-fill"></i>
142 |             </button>
143 |         </div>
144 |         <div class="d-flex align-items-center justify-cont
    | ent-between px-1 flex-wrap gap-2">
145 |             <span class="text-muted" style="font-size: 0.72r
    | em;" id="chat-status-indicator">Готов к работе</span>
146 |             <div class="d-flex align-items-center gap-2">
147 |                 <div class="form-check form-switch m-0 d-flex
    | align-items-center gap-1">
```

```

148 |         <input class="form-check-input mt-0" type="checkbox" role="switch" id="check-scenario-keep-context" title="Хранить историю диалога в контексте запроса">
149 |         <label class="form-check-label text-muted user-select-none" for="check-scenario-keep-context" style="font-size: 0.75rem;" title="Хранить историю диалога в контексте запроса">
150 |             <i class="bi bi-clock-history text-secondary me-1"></i>Хранить историю
151 |         </label>
152 |     </div>
153 |     <div class="form-check form-switch m-0 d-flex align-items-center gap-1">
154 |         <input class="form-check-input mt-0" type="checkbox" role="switch" id="check-scenario-rag" checked="" title="Использовать базу знаний (RAG)">
155 |         <label class="form-check-label text-muted user-select-none" for="check-scenario-rag" style="font-size: 0.75rem;" title="Использовать базу знаний (RAG)">
156 |             <i class="bi bi-database-check text-info me-1"></i>RAG
157 |         </label>
158 |     </div>
159 | </div>
160 | </div>
161 | </form>
162 | </div>
163 |
164 | </div>
165 | </div>
166 | </div>
167 |
168 | <!-- Right Slide-out Drawer (Offcanvas): Scenarios List & Execution Results -->
169 | <div class="offcanvas offcanvas-end text-bg-dark border-start border-secondary-subtle shadow-lg" tabindex="-1" id="scenariosOffcanvas" aria-labelledby="scenariosOffcanvasLabel" style="width: 560px; max-width: 95vw;">

```

```
170 |     <div class="offcanvas-header border-bottom border-secondary
    | y-subtle py-3 px-3">
171 |         <div class="d-flex align-items-center gap-2">
172 |             <span class="fs-4">❏ </span>
173 |             <div>
174 |                 <h5 class="offcanvas-title fw-bold text-white mb-0"
    | id="scenariosOffcanvasLabel">Сценарии тестирования</h5>
175 |                 <div class="text-muted small" style="font-size: 0.75
    | rem;">Экспресс-диагностика и системные проверки</div>
176 |             </div>
177 |         </div>
178 |         <div class="d-flex align-items-center gap-2">
179 |             <button type="button" class="btn-close btn-close-white
    | " data-bs-dismiss="offcanvas" aria-label="Заккрыть"></button>
180 |         </div>
181 |     </div>
182 |
183 |     <div class="offcanvas-body d-flex flex-column gap-3 p-3">
184 |
185 |         <!-- Filter Bar & Count Badge -->
186 |         <div class="d-flex align-items-center justify-content-be
    | tween gap-2 flex-wrap">
187 |             <div class="btn-group btn-group-sm" role="group" id="s
    | cenario-category-filters">
188 |                 <button type="button" class="btn btn-outline-seconda
    | ry active" data-filter="all">Все (6)</button>
189 |                 <button type="button" class="btn btn-outline-seconda
    | ry" data-filter="quick">❏ Экспресс</button>
190 |                 <button type="button" class="btn btn-outline-seconda
    | ry" data-filter="logs">❏ Логи</button>
191 |                 <button type="button" class="btn btn-outline-seconda
    | ry" data-filter="diagnostics">❏ Система</button>
192 |                 <button type="button" class="btn btn-outline-seconda
    | ry" data-filter="security">❏ Безопасность</button>
193 |             </div>
194 |             <span class="text-muted small" id="scenarios-count-bad
    | ge">Доступно: 6</span>
195 |         </div>
```

```
196 |
197 |     <!-- Execution Console & Results (shown on run) -->
198 |     <div class="card bg-dark bg-opacity-75 border-secondary-
    subtle shadow-sm" id="scenario-result-panel" style="display: n
    one;">
199 |         <div class="card-header bg-transparent border-secondar
    y-subtle d-flex align-items-center justify-content-between py-
    2 px-3">
200 |             <div class="d-flex align-items-center gap-2">
201 |                 <span id="result-status-icon" class="fs-5">❏ </span
    >
202 |                 <div>
203 |                     <strong id="result-scenario-title" class="text-w
    hite">Выполнение сценария...</strong>
204 |                     <div class="text-muted small" id="result-scenari
    o-meta">Инициализация шагов...</div>
205 |                 </div>
206 |             </div>
207 |             <div class="d-flex align-items-center gap-2">
208 |                 <span class="badge bg-secondary" id="result-durati
    on-badge">0.0 ms</span>
209 |                 <button class="btn btn-sm btn-outline-secondary py
    -0 px-2" id="btn-close-result-panel" title="Скрыть панель резу
    льтатов">❏ </button>
210 |             </div>
211 |         </div>
212 |         <div class="card-body p-3">
213 |             <!-- Overall Summary Banner -->
214 |             <div class="alert alert-secondary py-2 px-3 mb-3 sma
    ll d-flex align-items-start gap-2" id="result-summary-banner">
215 |                 <i class="bi bi-info-circle-fill mt-1"></i>
216 |                 <span id="result-summary-text">Результат проверки
    будет отображен здесь...</span>
217 |             </div>
218 |
219 |             <!-- Progress Bar -->
220 |             <div class="progress mb-3" style="height: 6px;" id="
    scenario-progress-container">
```



```
221 |         <div class="progress-bar progress-bar-striped prog  
    | ress-bar-animated bg-primary" role="progressbar" style="width:  
    | 100%" id="scenario-progress-bar"></div>  
222 |     </div>  
223 |  
224 |     <!-- Step Results List -->  
225 |     <div class="d-flex flex-column gap-2" id="scenario-s  
    | teps-container">  
226 |         <!-- Dynamic step rows injected here -->  
227 |     </div>  
228 | </div>  
229 | </div>  
230 |  
231 | <!-- Scenarios Cards Container -->  
232 | <div class="row g-2" id="scenarios-cards-container">  
233 |     <!-- Injected dynamically by main.js -->  
234 |     <div class="col-12 text-center py-5 text-muted small">  
235 |         <div class="spinner-border spinner-border-sm text-pr  
    | imary mb-2" role="status"></div>  
236 |         <div>Загрузка сценариев тестирования...</div>  
237 |     </div>  
238 | </div>  
239 |  
240 | </div>  
241 | </div>  
242 |  
243 | </div>  
244 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Scenarios Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/scenarios_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/scenarios_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-06 10:38:00
18  * =====
19  */
20
21 // Scenarios and Test Computer Automated Diagnostics Module (/
22 tc)
23 (function () {
24   'use strict';
25
26   let scenariosList = [];
27   let currentFilter = 'all';
28   let isRunning = false;
29
30   // Icons and badges helper
31   const categoryIcons = {
32     quick: '⚡',
```

```

32 |     logs: '🔍 ',
33 |     diagnostics: '🔍 ',
34 |     security: '🔍 ',
35 |     network: '🔍 ',
36 |     ai: '🔍 ',
37 | };
38 |
39 | const statusIcons = {
40 |     ok: '✅ ',
41 |     warn: '⚠️ ',
42 |     error: '❌ ',
43 |     skipped: '⏸️ ',
44 | };
45 |
46 | // Helper for safe HTML escaping
47 | function escapeHtml(str) {
48 |     if (!str) return '';
49 |     return String(str)
50 |         .replace(/&/g, '&amp;')
51 |         .replace(/</g, '&lt;')
52 |         .replace(/>/g, '&gt;')
53 |         .replace(/"/g, '&quot;')
54 |         .replace(/'/g, '&#039;');
55 | }
56 |
57 | // Simple Markdown to HTML formatter for mini-chat
58 | function formatMarkdown(text) {
59 |     if (!text) return '';
60 |     let html = escapeHtml(text);
61 |     // Bold
62 |     html = html.replace(/\*\*(.*?)\*\*/g, '<strong>$1</strong>');
63 |     // Italic
64 |     html = html.replace(/\*(.*?)\*/g, '<em>$1</em>');
65 |     // Inline code
66 |     html = html.replace(/`(.*)`/g, '<code class="text-primary bg-body-secondary px-1 rounded">$1</code>');
67 |     // Blockquote

```

```

68 |     html = html.replace(/^\>\s+(.*)/gim, '<div class="border-start border-primary border-3 ps-2 my-1 text-body bg-body-secondary py-1 rounded-end">$1</div>');
69 |     // Headers
70 |     html = html.replace(/^###\s+(.*)/gim, '<h6 class="fw-bold text-primary mt-2 mb-1">$1</h6>');
71 |     html = html.replace(/^##\s+(.*)/gim, '<h5 class="fw-bold text-body mt-2 mb-1">$1</h5>');
72 |     // Line breaks
73 |     html = html.replace(/\n/g, '<br>');
74 |     return html;
75 | }
76 |
77 | // Load scenarios from API
78 | async function loadScenarios() {
79 |     const container = document.getElementById('scenarios-cards-container');
80 |     if (!container) return;
81 |
82 |     try {
83 |         const res = await fetch('/api/v1/scenarios');
84 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
85 |         scenariosList = await res.json();
86 |         renderScenariosGrid();
87 |     } catch (err) {
88 |         console.warn('[ScenariosTab] Error fetching /api/v1/scenarios, using local fallback:', err);
89 |         scenariosList = [
90 |             {
91 |                 id: 'quick_check',
92 |                 title: 'Быстрая проверка (Smoke Test)',
93 |                 description: 'Экспресс-проверка целостности окружения, доступности ядра FastAPI, AI-провайдеров, свободного места и ключевых логов. Рекомендуется после установки.',
94 |                 category: 'quick',
95 |                 icon: '🔍',
96 |                 recommended: true,
97 |                 estimated_duration_sec: 2,

```

```
98         },
99         {
100             id: 'log_audit',
101             title: 'Запустить аудит логов',
102             description: 'Комплексный анализ файлов логов (info.
log, errors.log, app.log) и журналов событий Windows. Поиск кр
итических ошибок и аномалий.',
103             category: 'logs',
104             icon: '🔍',
105             recommended: false,
106             estimated_duration_sec: 3,
107         },
108         {
109             id: 'system_inspector',
110             title: 'Диагностика системы и оборудования',
111             description: 'Срез телеметрии: загрузка ядер процесс
ора, температура, оперативная память, свободное место на диска
х и фоновые процессы.',
112             category: 'diagnostics',
113             icon: '🔧',
114             recommended: false,
115             estimated_duration_sec: 3,
116         },
117         {
118             id: 'windows_admin',
119             title: 'Аудит автозапуска и служб Windows',
120             description: 'Проверка записей реестра Run/RunOnce,
папки автозагрузки, системных служб Windows и прав текущего пр
оцесса.',
121             category: 'security',
122             icon: '🔒',
123             recommended: false,
124             estimated_duration_sec: 4,
125         },
126         {
127             id: 'network_test',
128             title: 'Проверка сетевой доступности и портов',
129             description: 'Тестирование локального шлюза, разреше
```

```

        ния DNS, открытых серверных портов (8000) и доступности внешни
        х AI-сервисов.',
130         category: 'network',
131         icon: '🌐 ',
132         recommended: false,
133         estimated_duration_sec: 4,
134     },
135     {
136         id: 'ai_providers_check',
137         title: 'Проверка статуса AI-провайдеров',
138         description: 'Опрос сконфигурированных AI-моделей (G
        oogle Gemini, AGY, Ollama, Foundry) и замер времени отклика ин
        ференса.',
139         category: 'ai',
140         icon: '🤖 ',
141         recommended: false,
142         estimated_duration_sec: 5,
143     },
144 ];
145 renderScenariosGrid();
146 }
147 }
148
149 // Render cards grid based on active filter
150 function renderScenariosGrid() {
151     const container = document.getElementById('scenarios-cards
        -container');
152     if (!container) return;
153
154     const filtered = scenariosList.filter(sc => {
155         if (currentFilter === 'all') return true;
156         return sc.category === currentFilter;
157     });
158
159     const badgeEl = document.getElementById('scenarios-count-b
        adge');
160     if (badgeEl) {
161         badgeEl.innerText = `Доступно: ${filtered.length}`;

```

```

162     }
163     const headerBadgeEl = document.getElementById('scenarios-c
ount-badge-header');
164     if (headerBadgeEl) {
165         headerBadgeEl.innerText = `${scenariosList.length}`;
166     }
167
168     if (filtered.length === 0) {
169         container.innerHTML = '<div class="col-12 text-center py
-4 text-muted small">В этой категории нет сценариев.</div>';
170         return;
171     }
172
173     container.innerHTML = filtered.map(sc => {
174         const isRecommended = Boolean(sc.recommended);
175         const borderClass = isRecommended ? 'border-success bord
er-opacity-50 shadow-sm' : 'border-secondary-subtle';
176         const bgHeader = isRecommended ? 'bg-success bg-opacity-
10' : 'bg-transparent';
177         const btnClass = isRecommended ? 'btn-success' : 'btn-ou
tline-primary';
178
179         return `
180             <div class="col-12">
181                 <div class="card ${borderClass} transition-all hover
-shadow">
182                     <div class="card-header ${bgHeader} border-seconda
ry-subtle py-2 px-3 d-flex align-items-center justify-content-
between">
183                         <div class="d-flex align-items-center gap-2">
184                             <span class="fs-5">${sc.icon || categoryIcons[
sc.category] || '❏ '}</span>
185                             <strong class="small">${escapeHtml(sc.title)}<
/strong>
186                         </div>
187                     <div>
188                         ${isRecommended ? '<span class="badge bg-succe
ss small">Рекомендуется</span>' : '<span class="badge bg-secon

```

```

    dary small">${sc.estimated_duration_sec || 3} сек</span>`}
189         </div>
190     </div>
191     <div class="card-body p-3 d-flex flex-column justify-content-between">
192         <p class="text-muted small mb-2" style="font-size: 0.8rem; line-height: 1.35;">
193             ${escapeHtml(sc.description)}
194         </p>
195         <div class="d-flex align-items-center justify-content-between gap-2 mt-auto pt-2 border-top border-secondary-subtle">
196             <span class="text-secondary small" style="font-size: 0.72rem;">
197                 ID: <code>${escapeHtml(sc.id)}</code>
198             </span>
199             <button class="btn ${btnClass} btn-sm py-1 px-3 d-flex align-items-center gap-1 btn-run-scenario-item" data-scenario-id="${escapeHtml(sc.id)}" ${isRunning ? 'disabled' : ''}>
200                 <i class="bi bi-play-fill"></i>
201                 <span>Запустить</span>
202             </button>
203         </div>
204     </div>
205 </div>
206 </div>
207 `;
208 }).join('');
209
210 // Attach listeners to individual run buttons
211 container.querySelectorAll('.btn-run-scenario-item').forEach(btn => {
212     btn.onclick = () => {
213         const scId = btn.getAttribute('data-scenario-id');
214         if (scId) runScenario(scId);
215     };
216 });

```



```
217 | }
218 |
219 | // Execute scenario via API
220 | async function runScenario(scenarioId) {
221 |     if (isRunning) return;
222 |     isRunning = true;
223 |
224 |     // Automatically open the scenarios drawer if closed
225 |     const offcanvasEl = document.getElementById('scenariosOffc
226 | anvas');
227 |     if (offcanvasEl && typeof bootstrap !== 'undefined' && boo
228 | tstrap.Offcanvas) {
229 |         try {
230 |             const offcanvasInstance = bootstrap.Offcanvas.getOrCre
231 | ateInstance(offcanvasEl);
232 |             offcanvasInstance.show();
233 |         } catch (e) {
234 |             console.warn('[ScenariosTab] Offcanvas show error:', e
235 | );
236 |         }
237 |     }
238 |
239 |     const resultPanel = document.getElementById('scenario-resu
240 | lt-panel');
241 |     const titleEl = document.getElementById('result-scenario-t
242 | itle');
243 |     const metaEl = document.getElementById('result-scenario-me
244 | ta');
245 |     const iconEl = document.getElementById('result-status-icon
246 | ');
247 |     const durationBadge = document.getElementById('result-dura
248 | tion-badge');
249 |     const summaryBanner = document.getElementById('result-summ
250 | ary-banner');
251 |     const summaryText = document.getElementById('result-summar
252 | y-text');
253 |     const stepsContainer = document.getElementById('scenario-s
254 | teps-container');
```

```
243     const progressBar = document.getElementById('scenario-prog
ress-bar');
244
245     if (resultPanel) {
246         resultPanel.style.display = '';
247         resultPanel.scrollIntoView({ behavior: 'smooth', block:
'nearest' });
248     }
249     if (titleEl) titleEl.innerText = `Выполнение сценария [{s
cenarioId}]...`;
250     if (metaEl) metaEl.innerText = 'Опрос системных подсистем
и выполнение тестов...';
251     if (iconEl) iconEl.innerText = '❏ ';
252     if (durationBadge) durationBadge.innerText = 'Выполняется.
..';
253     if (summaryBanner) summaryBanner.className = 'alert alert-
info py-2 px-3 mb-3 small d-flex align-items-start gap-2';
254     if (summaryText) summaryText.innerText = 'Идет тестировани
е компонентов... Пожалуйста, подождите.';
255     if (stepsContainer) stepsContainer.innerHTML = '<div class
="text-center py-3 text-muted small"><div class="spinner-borde
r spinner-border-sm text-primary mb-2"></div><div>Тестирование
запущено...</div></div>';
256     if (progressBar) {
257         progressBar.style.width = '100%';
258         progressBar.className = 'progress-bar progress-bar-strip
ed progress-bar-animated bg-primary';
259     }
260
261     renderScenariosGrid();
262
263     const tStart = performance.now();
264     try {
265         const response = await fetch('/api/v1/scenarios/run', {
266             method: 'POST',
267             headers: { 'Content-Type': 'application/json' },
268             body: JSON.stringify({ scenario_id: scenarioId }),
269         });
```

```
270 |
271 |     if (!response.ok) {
272 |         throw new Error(`Ошибка сервера: HTTP ${response.statu
    | s}`);
273 |     }
274 |
275 |     const res = await response.json();
276 |     displayScenarioResults(res);
277 | } catch (err) {
278 |     console.error('[ScenariosTab] Error running scenario:',
    | err);
279 |     const totalDur = Math.round(performance.now() - tStart);
280 |     displayScenarioResults({
281 |         scenario_id: scenarioId,
282 |         title: scenarioId,
283 |         status: 'error',
284 |         duration_ms: totalDur,
285 |         total_steps: 1,
286 |         passed_steps: 0,
287 |         warn_steps: 0,
288 |         failed_steps: 1,
289 |         steps: [
290 |             {
291 |                 name: 'Выполнение запроса сценария',
292 |                 status: 'error',
293 |                 duration_ms: totalDur,
294 |                 details: `Ошибка: ${err.message}`,
295 |                 recommendation: 'Проверьте соединение с бэкенд сер
    | вером FastAPI.',
296 |             },
297 |         ],
298 |         summary: `Сбой при выполнении сценария: ${err.message}
    | `,
299 |     });
300 | } finally {
301 |     isRunning = false;
302 |     renderScenariosGrid();
303 | }
```

```

304 | }
305 |
306 | // Display structured results in panel
307 | function displayScenarioResults(res) {
308 |     const titleEl = document.getElementById('result-scenario-t
| title');
309 |     const metaEl = document.getElementById('result-scenario-me
| ta');
310 |     const iconEl = document.getElementById('result-status-icon
| ');
311 |     const durationBadge = document.getElementById('result-dura
| tion-badge');
312 |     const summaryBanner = document.getElementById('result-summ
| ary-banner');
313 |     const summaryText = document.getElementById('result-summar
| y-text');
314 |     const stepsContainer = document.getElementById('scenario-s
| teps-container');
315 |     const progressBar = document.getElementById('scenario-prog
| ress-bar');
316 |
317 |     if (titleEl) titleEl.innerText = res.title || res.scenario
| _id;
318 |     if (metaEl) metaEl.innerText = `Шагов: ${res.total_steps |
| | 0} (Пройдено: ${res.passed_steps || 0}, Замечаний: ${res.war
| n_steps || 0}, Ошибок: ${res.failed_steps || 0})`;
319 |     if (durationBadge) durationBadge.innerText = `${res.durati
| on_ms || 0} ms`;
320 |
321 |     // Overall status styling
322 |     if (res.status === 'ok') {
323 |         if (iconEl) iconEl.innerText = '✅';
324 |         if (summaryBanner) summaryBanner.className = 'alert aler
| t-success py-2 px-3 mb-3 small d-flex align-items-start gap-2'
| ;
325 |         if (progressBar) progressBar.className = 'progress-bar b
| g-success';
326 |     } else if (res.status === 'warn') {

```

```

327 |         if (iconEl) iconEl.innerText = '⚠';
328 |         if (summaryBanner) summaryBanner.className = 'alert alert-warning py-2 px-3 mb-3 small d-flex align-items-start gap-2';
329 |         if (progressBar) progressBar.className = 'progress-bar bg-warning';
330 |     } else {
331 |         if (iconEl) iconEl.innerText = '⚠';
332 |         if (summaryBanner) summaryBanner.className = 'alert alert-danger py-2 px-3 mb-3 small d-flex align-items-start gap-2';
333 |         if (progressBar) progressBar.className = 'progress-bar bg-danger';
334 |     }
335 |
336 |     if (summaryText) summaryText.innerText = res.summary || 'Погон завершен.';
337 |
338 |     // Steps rendering
339 |     if (stepsContainer && Array.isArray(res.steps)) {
340 |         stepsContainer.innerHTML = res.steps.map((step, idx) =>
341 |         {
342 |             const icon = statusIcons[step.status] || '⚠';
343 |             const badgeBg = step.status === 'ok' ? 'bg-success' :
344 |             (step.status === 'warn' ? 'bg-warning text-dark' : 'bg-danger');
345 |
346 |             const hasData = step.data && Object.keys(step.data).length > 0;
347 |             const dataId = `step-data-${idx}`;
348 |
349 |             return `
350 |                 <div class="card border border-secondary-subtle p-2 rounded">
351 |                     <div class="d-flex align-items-center justify-content-between gap-2 flex-wrap">
352 |                         <div class="d-flex align-items-center gap-2">

```

```

    </strong>
353         </div>
354         <div class="d-flex align-items-center gap-2">
355             <span class="badge ${badgeBg} small" style="font-size: 0.68rem;">${step.status.toUpperCase()}</span>
356             <span class="text-secondary small" style="font-size: 0.7rem;">${step.duration_ms || 0} ms</span>
357         </div>
358     </div>
359     <div class="text-muted small mt-1 ps-4" style="font-size: 0.78rem;">
360         ${escapeHtml(step.details)}
361     </div>
362     ${step.recommendation ? `
363         <div class="mt-1 ps-4 small text-warning" style="font-size: 0.75rem;">
364              <strong>Рекомендация:</strong> ${escapeHtml(step.recommendation)}
365         </div>
366         ` : ''}
367     ${hasData ? `
368         <div class="ps-4 mt-1">
369             <button class="btn btn-link btn-sm p-0 text-primary text-decoration-none small" style="font-size: 0.72rem;" onclick="const el=document.getElementById('${dataId}'); if(el) el.classList.toggle('d-none')">
370                  Показать детали JSON
371             </button>
372             <div id="${dataId}" class="d-none mt-1 p-2 bg-body-secondary rounded border border-secondary font-monospace small overflow-auto" style="max-height: 140px; font-size: 0.7rem;">
373                 <pre class="m-0">${escapeHtml(JSON.stringify(step.data, null, 2))}</pre>
374             </div>
375         </div>
376         ` : ''}
377     </div>

```

```

378 |         `;
379 |     }).join('');
380 | }
381 | }
382 |
383 | let isChatSubmitting = false;
384 | let isEventsSetup = false;
385 |
386 | // Extract [CHAT] and [VOICE] blocks from response and clean
    embedded action blocks
387 | function parseChatAndVoice(rawText) {
388 |     if (!rawText) return { chatText: '', voiceText: '' };
389 |     let text = rawText;
390 |     let voiceText = '';
391 |
392 |     if (text.includes('[CHAT]') || text.includes('[VOICE]')) {
393 |         const voiceMatch = text.match(/\[VOICE\](\[s\S]*?)(?:\[\/
    \[VOICE\]|(?:\[CHAT\])|$/i);
394 |         if (voiceMatch) {
395 |             voiceText = voiceMatch[1].trim();
396 |         }
397 |         const chatMatch = text.match(/\[CHAT\](\[s\S]*?)(?:\[\/C
    HAT\]|(?:\[VOICE\])|$/i);
398 |         if (chatMatch) {
399 |             text = chatMatch[1].trim();
400 |         } else {
401 |             text = text.replace(/\[VOICE\](\[s\S]*?)(?:\[\/VOICE\]|(
    ?\[CHAT\])|$/gi, '').trim();
402 |         }
403 |     }
404 |
405 |     // Clean action blocks from chat display text
406 |     text = text.replace(/```(?:action|json:action)[s\S]*?```/
    gi, '').trim();
407 |     text = text.replace(/\[ACTION:[s\S]*?\]/gi, '').trim();
408 |
409 |     // Clean remaining tags
410 |     text = text.replace(/\[\/?CHAT\]/gi, '').replace(/\[\/?VOI

```

```

    CE\]/gi, '').trim();
411     return { chatText: text, voiceText: voiceText || text };
412 }
413
414 // Dynamic pool of questions & quick buttons loaded from external file
415 let sampleQuestionsPool = [];
416 let quickButtonsPool = [];
417
418 // Asynchronously load questions and quick buttons from external file / API
419 async function loadQuestionsFromConfig() {
420     try {
421         const res = await fetch('/api/v1/scenarios/questions');
422         if (res.ok) {
423             const data = await res.json();
424             if (Array.isArray(data.prompts) && data.prompts.length > 0) {
425                 sampleQuestionsPool = data.prompts;
426             }
427             if (Array.isArray(data.quick_buttons) && data.quick_buttons.length > 0) {
428                 quickButtonsPool = data.quick_buttons;
429                 renderQuickButtons();
430             }
431         }
432     } catch (err) {
433         console.warn('[ScenariosTab] Falling back to direct JSON load for questions:', err);
434         try {
435             const fallbackRes = await fetch('/html/scenarios_tab/questions.json');
436             if (fallbackRes.ok) {
437                 const fbData = await fallbackRes.json();
438                 if (Array.isArray(fbData.prompts)) sampleQuestionsPool = fbData.prompts;
439                 if (Array.isArray(fbData.quick_buttons)) {
440                     quickButtonsPool = fbData.quick_buttons;

```



```

441         renderQuickButtons();
442     }
443 }
444 } catch {}
445 }
446 renderDynamicGreetingPrompts();
447 }
448
449 function renderQuickButtons() {
450     const container = document.getElementById('chat-quick-prom
pts');
451     if (!container || !quickButtonsPool.length) return;
452
453     container.innerHTML = quickButtonsPool.map(btn => `
454         <button class="btn btn-outline-secondary btn-sm py-0 px-
2 quick-prompt-btn" style="font-size: 0.75rem;" data-prompt="$
455         ${escapeHtml(btn.icon || '🔍')} ${escapeHtml(btn.text)}
456         </button>
457     `).join('');
458
459     // Re-bind click handlers
460     container.querySelectorAll('.quick-prompt-btn').forEach(bt
n => {
461         btn.onclick = () => {
462             if (isChatSubmitting) return;
463             const text = btn.dataset.prompt || btn.innerText.repla
ce(/^[^a-zA-Za-яA-ЯëË]+/, '').trim();
464             const chatInput = document.getElementById('scenario-ch
at-input');
465             if (chatInput) {
466                 chatInput.value = text;
467                 submitChat();
468             }
469         };
470     });
471 }
472

```

```
473 function renderDynamicGreetingPrompts() {
474     const greetingContainer = document.getElementById('chat-gr
eeting-prompts');
475     if (!greetingContainer) return;
476     if (!sampleQuestionsPool.length) return;
477
478     // Pick 3 random distinct questions from external config
479     const shuffled = [...sampleQuestionsPool].sort(() => 0.5 -
Math.random());
480     const selected = shuffled.slice(0, 3);
481
482     greetingContainer.innerHTML = selected.map(q => `
483         <div>• <a href="#" class="chat-prompt-link text-decorati
on-none border-bottom border-primary border-opacity-50" data-p
rompt="${escapeHtml(q)}">«${escapeHtml(q)}»</a></div>
484     `).join('');
485 }
486
487 function getGreetingHtml() {
488     const shuffled = sampleQuestionsPool.length ? [...sampleQu
estionsPool].sort(() => 0.5 - Math.random()) : [];
489     const selected = shuffled.slice(0, 3);
490     const promptsHtml = selected.length ? selected.map(q => `
491         <div>• <a href="#" class="chat-prompt-link text-decorati
on-none border-bottom border-primary border-opacity-50" data-p
rompt="${escapeHtml(q)}">«${escapeHtml(q)}»</a></div>
492     `).join('') : `
493         <div>• <a href="#" class="chat-prompt-link text-decorati
on-none border-bottom border-primary border-opacity-50" data-p
rompt="Какие мыши были подключены к этому компьютеру?">«Какие
мыши были подключены к этому компьютеру?»</a></div>
494         <div>• <a href="#" class="chat-prompt-link text-decorati
on-none border-bottom border-primary border-opacity-50" data-p
rompt="Покажи список сетевых портов и активных соединений">«По
кажи список сетевых портов и активных соединений»</a></div>
495         <div>• <a href="#" class="chat-prompt-link text-decorati
on-none border-bottom border-primary border-opacity-50" data-p
rompt="Проверь автозапуск и службы Windows">«Проверь автозапуск
```

```

    к и службы Windows»</a></div>
496     `;
497
498     return `
499         <div class="chat-message-row bot-row mb-3">
500             <div class="chat-msg-avatar">□ </div>
501             <div class="chat-msg-bot">
502                 <div class="mb-2">
503                     Привет! Я интеллектуальный ассистент Test Computer
                    . Вы можете задать мне любой вопрос о конфигурации хоста, напр
                    имер:
504                 </div>
505                 <div id="chat-greeting-prompts" class="d-flex flex-c
                    olumn gap-1 my-2">
506                     ${promptsHtml}
507                 </div>
508                 <div class="text-muted small mt-2">
509                     Если для вопроса нет готового сценария, я опрошу о
                    борудование через PowerShell/WMI, сформирую для вас ответ и <s
                    trong>автоматически сохраню новый навык</strong> в каталог <co
                    de>.skills/</code>.
510                 </div>
511             </div>
512         </div>
513     `;
514 }
515
516 // Handle Mini-Chat interaction
517 function setupMiniChat() {
518     const chatForm = document.getElementById('scenario-chat-fo
    rm');
519     const chatInput = document.getElementById('scenario-chat-i
    nput');
520     const chatHistory = document.getElementById('scenario-chat
    -history');
521     const sendBtn = document.getElementById('btn-scenario-chat
    -send');
522     const statusInd = document.getElementById('chat-status-ind

```

```

icator');
523     const ragCheck = document.getElementById('check-scenario-r
rag');
524     const keepContextCheck = document.getElementById('check-sc
enario-keep-context');
525
526     if (!chatForm || !chatInput || !chatHistory) return;
527
528     // Fetch and render questions dynamically from external co
nfiguration
529     loadQuestionsFromConfig();
530
531     if (keepContextCheck) {
532         const savedKeepContext = localStorage.getItem('tc_scnar
io_keep_context');
533         if (savedKeepContext !== null) {
534             keepContextCheck.checked = savedKeepContext === 'true'
;
535         } else {
536             keepContextCheck.checked = false;
537         }
538         keepContextCheck.addEventListener('change', (e) => {
539             localStorage.setItem('tc_scenario_keep_context', e.tar
get.checked);
540         });
541     }
542
543     if (ragCheck) {
544         const savedRagState = localStorage.getItem('tc_scenario_
use_rag');
545         if (savedRagState !== null) {
546             ragCheck.checked = savedRagState === 'true';
547         }
548         ragCheck.addEventListener('change', (e) => {
549             localStorage.setItem('tc_scenario_use_rag', e.target.c
hecked);
550         });
551     }

```

```

552
553     let currentConversationId = `scenario-chat-${Date.now()}`;
554
555     // New Session button handler
556     const btnNewChat = document.getElementById('btn-scenario-n
ew-chat');
557     if (btnNewChat) {
558         btnNewChat.onclick = async () => {
559             try {
560                 await fetch('/api/v1/scenarios/chat/clear', {
561                     method: 'POST',
562                     headers: { 'Content-Type': 'application/json' },
563                     body: JSON.stringify({ conversation_id: currentCon
versationId, message: 'clear' }),
564                 });
565             } catch (e) {
566                 console.warn('[ScenariosTab] Clear session error:',
e);
567             }
568             currentConversationId = `scenario-chat-${Date.now()}`;
569             chatHistory.innerHTML = getGreetingHtml();
570             if (statusInd) statusInd.innerText = 'Новая сессия нач
ата';
571             chatInput.focus();
572         };
573     }
574
575     // Quick prompt buttons & clickable prompt links
576     document.querySelectorAll('.quick-prompt-btn').forEach(btn
=> {
577         btn.onclick = () => {
578             if (isChatSubmitting) return;
579             const text = btn.innerText.replace(/^[^a-zA-Za-яA-ЯёЁ]
+/, '').trim();
580             chatInput.value = text;
581             submitChat();
582         };
583     });

```

```

584
585     // Delegated click handler for inline prompt links in chat
    history
586     chatHistory.onclick = (e) => {
587         const link = e.target.closest('.chat-prompt-link');
588         if (link) {
589             e.preventDefault();
590             e.stopPropagation();
591             if (isChatSubmitting) return;
592             const promptText = link.dataset.prompt || link.innerHTMLTe
xt.replace(/^[«"']|['»"]$/g, '').trim();
593             if (promptText) {
594                 chatInput.value = promptText;
595                 submitChat();
596             }
597         }
598     };
599
600     async function submitChat() {
601         const message = chatInput.value.trim();
602         if (!message || isChatSubmitting) return;
603
604         isChatSubmitting = true;
605         chatInput.value = '';
606         chatInput.disabled = true;
607         if (sendBtn) sendBtn.disabled = true;
608         if (statusInd) statusInd.innerText = '□ Инициализация с
истемного зонда...';
609
610         // Append User message
611         const userMsgHtml = `
612             <div class="chat-message-row user-row mb-2">
613                 <div class="chat-msg-user">
614                     ${escapeHtml(message)}
615                 </div>
616             </div>
617         `;
618         chatHistory.insertAdjacentHTML('beforeend', userMsgHtml)

```

```

;
619     chatHistory.scrollTop = chatHistory.scrollHeight;
620
621     // Handle client-side slash commands (/skills list, /ski
lls, /help)
622     const cleanCmd = message.trim().toLowerCase();
623     if (cleanCmd === '/skills' || cleanCmd === '/skills list
' || cleanCmd.startsWith('/skills search')) {
624         try {
625             const searchParam = cleanCmd.startsWith('/skills sea
rch') ? cleanCmd.replace('/skills search', '').trim() : '';
626             const url = searchParam ? `/api/admin/skills?q=${enc
odeURIComponent(searchParam)}` : '/api/admin/skills';
627             const res = await fetch(url);
628             if (!res.ok) throw new Error(`HTTP ${res.status}`);
629             const data = await res.json();
630             const skills = data.skills || [];
631
632             let replyMd = `### □ Каталог зарегистрированных нав
ыков (.skills)\n\n`;
633             if (skills.length === 0) {
634                 replyMd += `*Навыки не найдены.*`;
635             } else {
636                 replyMd += `Всего доступно навыков: **${skills.len
gth}**\n\n`;
637                 skills.forEach((s, idx) => {
638                     const pathStr = s.root_relative || s.path || `.s
kills/${s.name}/SKILL.md`;
639                     const desc = s.description_ru || s.description |
| 'Без описания';
640                     replyMd += `${idx + 1}. □ **${escapeHtml(s.name
)}** – **${escapeHtml(desc)}* (\`${escapeHtml(pathStr)}\`)\n\n`;
641                 });
642             }
643
644             const assistantMsgHtml = `
645                 <div class="chat-message-row bot-row mb-3">
646                     <div class="chat-msg-avatar">□ </div>

```

```

647         <div class="chat-msg-bot">
648             <div>${formatMarkdown(replyMd)}</div>
649         </div>
650     </div>
651     `;
652     chatHistory.insertAdjacentHTML('beforeend', assistantMsgHtml);
653     chatHistory.scrollTop = chatHistory.scrollHeight;
654     if (statusInd) statusInd.innerText = 'Список навыков
выведен';
655     } catch (err) {
656         const errorMsgHtml = `
657             <div class="chat-message-row bot-row mb-3">
658                 <div class="chat-msg-avatar">❌ </div>
659                 <div class="chat-msg-bot border-danger text-dang
er">
660                     Ошибка получения списка навыков: ${escapeHtml(
err.message)}
661                 </div>
662             </div>
663         `;
664         chatHistory.insertAdjacentHTML('beforeend', errorMsg
Html);
665         chatHistory.scrollTop = chatHistory.scrollHeight;
666         if (statusInd) statusInd.innerText = 'Ошибка запроса
';
667     } finally {
668         isChatSubmitting = false;
669         chatInput.disabled = false;
670         if (sendBtn) sendBtn.disabled = false;
671         chatInput.focus();
672     }
673     return;
674 }
675
676     // Append Streaming Assistant placeholder with multi-ste
p live tracker
677     const msgId = `msg-${Date.now()}-${Math.random().toStrin

```



```

    g(36).substr(2, 6)}`;
678     const stagesLog = [];
679
680     const renderStagesHtml = (stages, isFinal = false) => {
681         if (!stages || stages.length === 0) return '';
682         const currentCount = stages.length;
683         const isComplete = isFinal;
684         const headerIcon = isComplete
685             ? '<i class="bi bi-check-circle-fill text-success"><
        /i>'
686             : '<div class="spinner-border spinner-border-sm text
        -primary" role="status" style="width:0.75rem;height:0.75rem;">
        </div>';
687         const headerText = isComplete
688             ? `Диагностика хоста завершена (${currentCount} шаго
        в)`
689             : `Диагностический конвейер хоста (${currentCount} ш
        агов)...`;
690         const headerClass = isComplete ? 'text-success' : 'tex
        t-primary';
691         const collapseClass = isComplete ? 'd-none' : '';
692         const chevronIcon = isComplete ? 'bi-chevron-down' : '
        bi-chevron-up';
693         const lastTitle = stages[stages.length - 1]?.title ||
        'Зонд';
694
695         return `
696         <div class="chat-stage-pipeline mb-2 shadow-sm" id="
        ${msgId}-stages-wrapper">
697             <div class="d-flex align-items-center justify-cont
        ent-between flex-wrap gap-1" onclick="const b=document.getElem
        entById('${msgId}-stages-content'); const c=document.getElemen
        tById('${msgId}-stages-chevron'); if(b){b.classList.toggle('d-
        none'); if(c){c.classList.toggle('bi-chevron-down'); c.classLi
        st.toggle('bi-chevron-up');}" style="cursor:pointer;" title="
        Нажмите, чтобы развернуть/свернуть детали этапов">
698                 <span class="${headerClass} small fw-semibold d-
        flex align-items-center gap-1.5" style="font-size:0.75rem;">

```

```

699         ${headerIcon}
700         <span>${escapeHtml(headerText)}</span>
701     </span>
702     <div class="d-flex align-items-center gap-1">
703         <span class="badge bg-body-secondary text-body
border border-secondary-subtle font-monospace" style="font-si
ze:0.63rem;">${escapeHtml(lastTitle)}</span>
704         <i class="bi ${chevronIcon} text-muted" id="${
msgId}-stages-chevron" style="font-size:0.65rem;"></i>
705     </div>
706 </div>
707     <div class="mt-1.5 pt-1.5 border-top border-second
ary border-opacity-25 d-flex flex-column gap-1.5 ${collapseCla
ss}" id="${msgId}-stages-content">
708         ${stages.map((st, sIdx) => {
709             const isLatest = sIdx === stages.length - 1 &&
!isComplete;
710             const icon = isLatest
711                 ? '<div class="spinner-grow spinner-grow-sm
text-primary" style="width: 0.55rem; height: 0.55rem;" role="s
tatus"></div>'
712                 : '<span class="text-success small" style="f
ont-size:0.7rem;">◻ </span>';
713             const textClass = isLatest ? 'text-body fw-med
ium' : 'text-muted';
714             return `
715                 <div class="small ${textClass} d-flex flex-c
olumn ps-0.5 stage-item ${isLatest ? 'active' : ''}" style="fo
nt-size: 0.74rem; line-height: 1.35;">
716                     <div class="d-flex align-items-center gap-
1.5">
717                         <span>${icon}</span>
718                         <span>${escapeHtml(st.message || '')}</s
pan>
719                     </div>
720                     ${st.details ? `<div class="text-muted ps-
3 font-monospace" style="font-size: 0.68rem; word-break: break
-all;">◻ ${escapeHtml(st.details)}</div>` : ''}

```

```

721         ${st.generated_prompt ? `
722         <div class="mt-1 ps-3">
723             <button class="btn btn-sm btn-outline-
warning py-0 px-2 d-inline-flex align-items-center gap-1 font-
monospace" type="button" onclick="event.stopPropagation(); con
st p=document.getElementById('${msgId}-stage-prompt-${sIdx}');
const ic=document.getElementById('${msgId}-stage-prompt-ic-${
sIdx}'); if(p){p.classList.toggle('d-none'); if(ic){ic.classLi
st.toggle('bi-chevron-down'); ic.classList.toggle('bi-chevron-
up');}}" style="font-size: 0.68rem;">
724                 <i class="bi bi-cpu-fill"></i>
725                 <span>Показать отправленный промт</
span>
726                 <i class="bi bi-chevron-down text-wa
rning" id="${msgId}-stage-prompt-ic-${sIdx}" style="font-size:
0.6rem;"></i>
727             </button>
728             <div id="${msgId}-stage-prompt-${sIdx}
" class="d-none mt-1.5 p-2 bg-dark bg-opacity-25 border border
-warning border-opacity-40 rounded text-start" onclick="event.
stopPropagation();">
729                 <div class="d-flex justify-content-b
etween align-items-center text-warning mb-1" style="font-size:
0.68rem;">
730                     <span><i class="bi bi-terminal me-
1"></i>Точный текст промта, отправленный модели:</span>
731                     <button class="btn btn-dark btn-sm
py-0 px-1.5 text-secondary border border-secondary-subtle" ty
pe="button" onclick="navigator.clipboard.writeText(this.closes
t('#${msgId}-stage-prompt-${sIdx}').querySelector('pre')).inne
rText || ''); this.innerText='Скопировано!'; setTimeout(()=>th
is.innerText='Копировать', 1500);" style="font-size: 0.65rem;"
>Копировать</button>
732                 </div>
733                 <pre class="m-0 p-1.5 bg-body-tertia
ry text-body border border-secondary border-opacity-30 rounded
font-monospace" style="font-size: 0.70rem; line-height: 1.35;
max-height: 220px; overflow-y: auto; white-space: pre-wrap; w

```

```

ord-break: break-word;">${escapeHtml(st.generated_prompt)}</pre>
e>
734         </div>
735     </div>
736     ` : ''}
737 </div>
738 `;
739     }).join('')}
740 </div>
741 </div>
742 `;
743 };
744
745     const initialStage = {
746         stage: 'init',
747         title: 'Инициализация',
748         message: '□ Подключение к локальной подсистеме Window
s API и сервисам хоста...',
749         details: 'Подготовка системных интерфейсов WMI/PnP, ко
ллекторов и контекста сессии',
750     };
751     stagesLog.push(initialStage);
752
753     const loadingHtml = `
754         <div class="chat-message-row bot-row mb-3" id="${msgId
}-container">
755             <div class="chat-msg-avatar">□ </div>
756             <div class="chat-msg-bot shadow-sm" id="${msgId}-bod
y">
757                 ${renderStagesHtml(stagesLog, false)}
758             </div>
759         </div>
760     `;
761     chatHistory.insertAdjacentHTML('beforeend', loadingHtml)
;
762     chatHistory.scrollTop = chatHistory.scrollHeight;
763
764     try {

```

```

765 |         const useRag = ragCheck ? ragCheck.checked : true;
766 |         const keepContext = keepContextCheck ? keepContextChec
| k.checked : false;
767 |
768 |         const res = await fetch('/api/v1/scenarios/chat/stream
| ', {
769 |             method: 'POST',
770 |             headers: { 'Content-Type': 'application/json' },
771 |             body: JSON.stringify({
772 |                 message: message,
773 |                 conversation_id: currentConversationId,
774 |                 auto_create_skill: false,
775 |                 keep_context: keepContext,
776 |                 use_rag: useRag,
777 |             }),
778 |         });
779 |
780 |         if (!res.ok) {
781 |             throw new Error(`Ошибка ответа сервера: HTTP ${res.s
| tatus}`);
782 |         }
783 |
784 |         const reader = res.body.getReader();
785 |         const decoder = new TextDecoder('utf-8');
786 |         let buffer = '';
787 |         let finalData = null;
788 |         let streamedReplyText = '';
789 |
790 |         while (true) {
791 |             const { done, value } = await reader.read();
792 |             if (done) break;
793 |             buffer += decoder.decode(value, { stream: true });
794 |
795 |             const lines = buffer.split('\n\n');
796 |             buffer = lines.pop(); // Оставляем неполный хвост в
| буфере
797 |
798 |             for (const line of lines) {

```

```

799 |         const cleanLine = line.trim();
800 |         if (!cleanLine.startsWith('data:')) continue;
801 |         let evt = null;
802 |         try {
803 |             evt = JSON.parse(cleanLine.substring(5).trim());
804 |         } catch (pErr) {
805 |             console.warn('[ScenariosTab] SSE parse warning: '
      | , pErr);
806 |             continue;
807 |         }
808 |         if (!evt) continue;
809 |
810 |         if (evt.type === 'stage') {
811 |             const lastSt = stagesLog[stagesLog.length - 1];
812 |             if (!lastSt || lastSt.stage !== evt.stage || las
      | tSt.message !== evt.message) {
813 |                 stagesLog.push(evt);
814 |             }
815 |             if (statusInd) statusInd.innerText = evt.message
      | || 'Обработка...';
816 |
817 |             const bodyElem = document.getElementById(`${msgI
      | d}-body`);
818 |             if (bodyElem) {
819 |                 const { chatText } = parseChatAndVoice(streame
      | dReplyText);
820 |                 bodyElem.innerHTML = `
821 |                     ${renderStagesHtml(stagesLog, false)}
822 |                     ${streamedReplyText ? `<div class="chat-stre
      | aming-content mt-1.5">${formatMarkdown(chatText)}<span class="
      | streaming-cursor">█</span></div>` : ''}
823 |                 `;
824 |                 chatHistory.scrollTop = chatHistory.scrollHeight;
825 |             }
826 |         } else if (evt.type === 'chunk') {
827 |             streamedReplyText += (evt.content || '');
828 |             const bodyElem = document.getElementById(`${msgI

```

```

    d}-body`);
829         if (bodyElem) {
830             const { chatText } = parseChatAndVoice(streamedReplyText);
            bodyElem.innerHTML = `
831                 ${renderStagesHtml(stagesLog, false)}
832                 <div class="chat-streaming-content mt-1.5">${
833                     {formatMarkdown(chatText)}<span class="streaming-cursor">█</span></div>
834             `;
835             chatHistory.scrollTop = chatHistory.scrollHeight;
836         }
837     } else if (evt.type === 'done') {
838         finalData = evt;
839     } else if (evt.type === 'error') {
840         const streamErr = new Error(evt.error || 'Ошибка
            в потоке');
841         streamErr.details = evt.details || '';
842         streamErr.errorType = evt.error_type || 'StreamError';
843         streamErr.rawEvent = evt;
844         throw streamErr;
845     }
846 }
847 }
848
849 if (!finalData) {
850     throw new Error('Не удалось получить завершающий ответ от сервера.');
```

```

856
857     let remediationHtml = '';
858     if (Array.isArray(finalData.remediation_actions) && fi
nalData.remediation_actions.length > 0) {
859         const actionsList = finalData.remediation_actions;
860         remediationHtml = `
861             <div class="mt-2.5 p-2.5 bg-body-secondary border
border-warning border-opacity-50 rounded" id="remediation-cont
ainer-${msgId}">
862                 <div class="d-flex align-items-center justify-co
ntent-between mb-2">
863                     <span class="fw-semibold text-warning small d-
flex align-items-center gap-1">
864                         <i class="bi bi-shield-check text-warning"><
/i> <span>Предложенные действия и исправления:</span>
865                         </span>
866                         <span class="badge bg-warning text-dark small"
style="font-size: 0.68rem;">SafeOps (${actionsList.length})</
span>
867                     </div>
868                     <div class="d-flex flex-column gap-1.5" id="reme
diation-buttons-${msgId}">
869                         ${actionsList.map((act, aIdx) => {
870                             const isCaution = (act.risk || '').toLowerCase() === 'caution' || (act.risk || '').toLowerCase() === 'critical';
871                             const badgeRisk = isCaution ? '<span class="
badge bg-danger-subtle text-danger border border-danger-subtle
ms-1" style="font-size:0.65rem;">Требуется подтверждения</span>
' : '<span class="badge bg-info-subtle text-info border border
-info-subtle ms-1" style="font-size:0.65rem;">Безопасно</span>
';
872                             return `
873                                 <div class="d-flex align-items-center just
ify-content-between p-2 bg-body rounded border border-secondar
y border-opacity-25 gap-2 flex-wrap" id="act-row-${msgId}-${aI
dx}">
874                                     <div class="small d-flex flex-column" st

```



```

yle="font-size: 0.78rem;">
875 |             <div><strong>${escapeHtml(act.title ||
|             act.action_id)}</strong>${badgeRisk}</div>
876 |             ${act.description ? `<span class="text
|             -muted mt-0.5" style="font-size: 0.72rem;">${escapeHtml(act.de
|             scription)}</span>` : ''}
877 |             ${act.execution_command ? `<code class
|             ="text-warning-emphasis bg-body-secondary px-1 py-0.5 rounded
|             mt-1 font-monospace" style="font-size: 0.68rem; word-break: br
|             eak-all;">${escapeHtml(act.execution_command)}</code>` : ''}
878 |             </div>
879 |             <button class="btn btn-warning btn-sm py
|             -1 px-3 d-flex align-items-center gap-1.5 shadow-sm fw-semibol
|             d btn-exec-action"
880 |                 id="btn-act-${msgId}-${aIdx}"
881 |                 data-msg-id="${msgId}"
882 |                 data-action-id="${escapeHtml(act.actio
|             n_id)}"
883 |                 data-action-type="${escapeHtml(act.act
|             ion_type)}"
884 |                 data-title="${escapeHtml(act.title)}"
885 |                 data-description="${escapeHtml(act.des
|             cription)}"
886 |                 data-target="${escapeHtml(act.target)}
|             "
887 |                 data-risk="${escapeHtml(act.risk)}"
888 |                 data-command="${escapeHtml(act.executi
|             on_command)}"
889 |                 style="font-size: 0.78rem;">
890 |                 <i class="bi bi-play-circle-fill"></i>
891 |                 <span>Подтвердить</span>
892 |             </button>
893 |         </div>
894 |     `;
895 |     })).join('')}
896 | </div>
897 | </div>
898 | `;

```

```

899         }
900
901         let skillActionHtml = '';
902         if (finalData.created_skill) {
903             const isNew = finalData.created_skill.is_new !== false;
904             const badgeTitle = isNew ? ' Автоматически  
создан навык:</strong>' : ' Использован существующий  
навык:</strong>';
905             const badgeClass = isNew ? 'bg-success bg-opacity-10  
border-success border-opacity-50 text-success' : 'bg-info bg-  
opacity-10 border-info border-opacity-50 text-info';
906             skillActionHtml = `
907                 <div class="mt-2 p-2 ${badgeClass} border rounded  
small" id="skill-action-container-${msgId}">
908                     ${badgeTitle} <code>${escapeHtml(finalData.creat  
ed_skill.path)}</code>
909                     <div class="text-muted small mt-1">${escapeHtml(  
finalData.created_skill.description_ru)}</div>
910                 </div>
911             `;
912         } else if (finalData.tool_plan) {
913             skillActionHtml = `
914                 <div class="mt-2 pt-2 border-top border-secondary  
border-opacity-25 d-flex align-items-center gap-2 flex-wrap" i  
d="skill-action-container-${msgId}">
915                     <button class="btn btn-outline-success btn-sm d-  
flex align-items-center gap-1.5 py-1 px-2.5 shadow-sm" id="btn  
-save-skill-${msgId}" type="button">
916                         <i class="bi bi-bookmark-check-fill text-succe  
ss"></i>
917                         <span>Ответ правильный, запомнить как навык</s  
pan>
918                     </button>
919                     <button class="btn btn-outline-secondary btn-sm  
d-flex align-items-center gap-1.5 py-1 px-2 shadow-sm" id="btn  
-edit-skill-${msgId}" type="button">
920                         <i class="bi bi-pencil-square"></i>

```

```
921         <span>Скорректировать</span>
922     </button>
923 </div>
924     <div id="skill-edit-form-{$msgId}" class="mt-2 p-2
    .5 bg-body-secondary border border-secondary-subtle rounded sm
    all d-none">
925         <div class="fw-semibold text-warning mb-2"><i cl
    ass="bi bi-sliders me-1"></i>Корректировка навыка перед сохран
    ением</div>
926         <div class="mb-2">
927             <label class="form-label text-muted mb-1" styl
    e="font-size:0.75rem;">Заголовок навыка:</label>
928             <input type="text" class="form-control form-co
    ntrol-sm border-secondary" id="edit-title-{$msgId}" value="{$e
    scapeHtml(finalData.tool_plan.tool_title || '')}">
929         </div>
930         <div class="mb-2">
931             <label class="form-label text-muted mb-1" styl
    e="font-size:0.75rem;">Описание (русский язык):</label>
932             <textarea class="form-control form-control-sm
    border-secondary" id="edit-desc-{$msgId}" rows="2">{$escapeHtm
    l(finalData.tool_plan.description_ru || '')}</textarea>
933         </div>
934         <div class="mb-2">
935             <label class="form-label text-muted mb-1" styl
    e="font-size:0.75rem;">Команда проверки / скрипт:</label>
936             <input type="text" class="form-control form-co
    ntrol-sm border-secondary font-monospace" id="edit-cmd-{$msgId
    }" value="{$escapeHtml(finalData.command_executed || finalData
    .tool_plan.probe_script || '')}">
937         </div>
938         <div class="d-flex gap-2 justify-content-end mt-
    2">
939             <button class="btn btn-secondary btn-sm py-1 p
    x-2" id="btn-cancel-edit-{$msgId}" type="button">Отмена</butto
    n>
940             <button class="btn btn-success btn-sm py-1 px-
    2.5" id="btn-save-edited-{$msgId}" type="button">
```

```

941 |         <i class="bi bi-check-circle-fill me-1"></i>
    | Сохранить скорректированный навык
942 |         </button>
943 |     </div>
944 | </div>
945 | `;
946 | }
947 |
948 | let metaInfoHtml = '';
949 | const agentsList = finalData.agents_used || [];
950 | const knowledgeList = finalData.knowledge_used || [];
951 |
952 | if (agentsList.length > 0 || knowledgeList.length > 0)
    | {
953 |     const totalCount = agentsList.length + knowledgeList
    | .length;
954 |     const metaCollapseId = `meta-details-${msgId}`;
955 |     metaInfoHtml = `
956 |         <div class="mt-2.5 pt-2 border-top border-secondar
    | y border-opacity-25" id="meta-container-${msgId}">
957 |             <div class="d-flex align-items-center justify-co
    | ntent-between flex-wrap gap-1">
958 |                 <button class="btn btn-sm btn-link p-0 text-pr
    | imary text-decoration-none d-flex align-items-center gap-1.5"
    | type="button" onclick="const el=document.getElementById('${met
    | aCollapseId}'); const ic=document.getElementById('meta-icon-${
    | msgId}'); if(el){el.classList.toggle('d-none'); if(ic) { ic.cl
    | assList.toggle('bi-chevron-down'); ic.classList.toggle('bi-che
    | vron-up'); }}" style="font-size: 0.75rem;">
959 |                 <i class="bi bi-diagram-3-fill text-primary"
    | ></i>
960 |                 <span><strong>Задействованные знания и агент
    | ы</strong> (${totalCount})</span>
961 |                 <i class="bi bi-chevron-down text-muted" id=
    | "meta-icon-${msgId}" style="font-size: 0.7rem;"></i>
962 |                 </button>
963 |                 <div class="d-flex gap-1 flex-wrap">
964 |                     ${agentsList.map(a => `<span class="badge bg

```

```

-primary bg-opacity-10 text-primary border border-primary border-opacity-25" style="font-size: 0.65rem;" title="${escapeHtml(a.role || '')}">${escapeHtml(a.icon || '🤖')} ${escapeHtml(a.name)}</span>`).slice(0, 2).join('')
965         </div>
966     </div>
967
968     <div id="${metaCollapseId}" class="d-none mt-2 p-2 bg-body-secondary border border-secondary border-opacity-50 rounded small">
969         ${agentsList.length > 0 ? `
970             <div class="mb-2">
971                 <div class="text-warning fw-semibold mb-1"
972                 style="font-size: 0.73rem;">
973                     <i class="bi bi-robot me-1"></i>Задействованные агенты и модули:
974                     <div class="d-flex flex-column gap-1 ps-1">
975                         >
976                             ${agentsList.map(ag => `
977                                 <div class="d-flex align-items-start gap-1.5" style="font-size: 0.73rem;">
978                                     <span>${escapeHtml(ag.icon || '🤖')}
979                                     </span>
980                                     <div>
981                                         <strong>${escapeHtml(ag.name)}</strong>
982                                         <span class="text-muted ms-1">— ${escapeHtml(ag.role || 'Системный агент')}</span>
983                                     </div>
984                                     </div>
985                                     `).join('')}
986                                 </div>
987                             </div>
988                             ` : ''}
989
990             ${knowledgeList.length > 0 ? `
991                 <div>

```

```

990 |         <div class="text-primary fw-semibold mb-1"
    |         style="font-size: 0.73rem;">
991 |             <i class="bi bi-book-half me-1"></i>Испо
    | льзованные источники знаний и артефакты:
992 |         </div>
993 |         <div class="d-flex flex-column gap-1 ps-1"
    | >
994 |             ${knowledgeList.map(kn => `
995 |                 <div class="d-flex align-items-start g
    | ap-1.5" style="font-size: 0.73rem;">
996 |                     <span class="badge bg-info bg-opacit
    | y-25 text-info border border-info border-opacity-25" style="fo
    | nt-size: 0.65rem;">${escapeHtml(kn.badge || 'База')}</span>
997 |                     <div>
998 |                         <strong>${escapeHtml(kn.title)}</s
    | trong>
999 |                         <span class="text-muted ms-1">— ${
    | escapeHtml(kn.description || '')}</span>
1000 |                     </div>
1001 |                 </div>
1002 |             `).join('')}
1003 |         </div>
1004 |     </div>
1005 |     ` : ''}
1006 | </div>
1007 | </div>
1008 | `;
1009 | }
1010 |
1011 | let promptInfoHtml = '';
1012 | const generatedPromptText = finalData.generated_prompt
    | || (stagesLog.find(s => s.stage === 'synthesizing' && s.gener
    | ated_prompt)?.generated_prompt);
1013 | if (generatedPromptText) {
1014 |     const promptCollapseId = `prompt-details-${msgId}`;
1015 |     const promptCodeId = `prompt-code-${msgId}`;
1016 |     promptInfoHtml = `
1017 |         <div class="mt-2.5 pt-2 border-top border-secondar

```

```

y border-opacity-25" id="prompt-container-${msgId}">
1018     <div class="d-flex align-items-center justify-co
content-between flex-wrap gap-1">
1019         <button class="btn btn-sm btn-link p-0 text-wa
arning text-decoration-none d-flex align-items-center gap-1.5"
type="button" onclick="const el=document.getElementById('${pro
mptCollapseId}'); const ic=document.getElementById('prompt-ico
n-${msgId}'); if(el){el.classList.toggle('d-none'); if(ic) { i
c.classList.toggle('bi-chevron-down'); ic.classList.toggle('bi
-chevron-up'); }}" style="font-size: 0.75rem;">
1020             <i class="bi bi-cpu-fill text-warning"></i>
1021             <span><strong>Сформированный промпт модели</
strong></span>
1022             <i class="bi bi-chevron-down text-muted" id=
"prompt-icon-${msgId}" style="font-size: 0.7rem;"></i>
1023         </button>
1024         <div class="d-flex align-items-center gap-1">
1025             <button class="btn btn-outline-secondary btn
-sm py-0 px-2 d-flex align-items-center gap-1" type="button" o
nclick="navigator.clipboard.writeText(document.getElementById(
'${promptCodeId}')?.innerText || ''); const s=this.querySelector(
'span'); if(s){s.innerText='Скопировано!'; setTimeout(()=>s
.innerText='Копировать', 1800);} " style="font-size: 0.68rem;"
title="Скопировать отправленный промпт">
1026                 <i class="bi bi-clipboard"></i>
1027                 <span>Копировать</span>
1028             </button>
1029         </div>
1030     </div>
1031
1032     <div id="${promptCollapseId}" class="d-none mt-2
p-2.5 bg-body-secondary border border-warning border-opacity-
40 rounded">
1033         <div class="d-flex align-items-center justify-
content-between text-warning small mb-1.5" style="font-size: 0
.72rem;">
1034             <span><i class="bi bi-terminal me-1"></i>Точ
ный текст запроса, отправленный языковой модели:</span>

```

```

1035 |         <span class="badge bg-warning bg-opacity-10
      | text-warning border border-warning border-opacity-25 font-mono
      | space" style="font-size: 0.65rem;">${generatedPromptText.length}
      | симв.</span>
1036 |     </div>
1037 |     <pre class="m-0 p-2 bg-body text-body border b
      | order-secondary border-opacity-30 rounded font-monospace small
      | " id="${promptCodeId}" style="font-size: 0.72rem; line-height:
      | 1.4; max-height: 280px; overflow-y: auto; white-space: pre-wr
      | ap; word-break: break-word;">${escapeHtml(generatedPromptText)
      | }</pre>
1038 |     </div>
1039 | </div>
1040 | `;
1041 | }
1042 |
1043 | const trainingActionHtml = `
1044 |     <div class="mt-2.5 pt-2 border-top border-secondary
      | border-opacity-25 d-flex align-items-center justify-content-be
      | tween flex-wrap gap-2" id="training-save-container-${msgId}">
1045 |         <button class="btn btn-outline-primary btn-sm d-fl
      | ex align-items-center gap-1.5 py-1 px-2.5 shadow-sm btn-save-t
      | raining-qa" id="btn-save-training-${msgId}" type="button" titl
      | e="Сохранить пару вопрос-ответ в базу данных Test Computer для
      | RAG и последующего обучения (тюнинга) модели">
1046 |             <i class="bi bi-mortarboard-fill text-primary"><
      | /i>
1047 |                 <span>Сохранить ответ для обучения модели</span>
1048 |             </button>
1049 |             <span class="text-muted" style="font-size: 0.68rem
      | ;"><i class="bi bi-database me-1"></i>data/tc/approved_respons
      | es</span>
1050 |         </div>
1051 |     `;
1052 |
1053 | const bodyElem = document.getElementById(`${msgId}-bod
      | y`);
1054 | if (bodyElem) {

```



```

1055 |         bodyElem.innerHTML = `
1056 |             ${renderStagesHtml(stagesLog, true)}
1057 |             <div>${formattedReply}</div>
1058 |             ${remediationHtml}
1059 |             ${skillActionHtml}
1060 |             ${trainingActionHtml}
1061 |             ${metaInfoHtml}
1062 |             ${promptInfoHtml}
1063 |         `;
1064 |     }
1065 |     chatHistory.scrollTop = chatHistory.scrollHeight;
1066 |
1067 |     // Wire up save response for training & RAG button
1068 |     const btnSaveTraining = document.getElementById(`btn-s
ave-training-${msgId}`);
1069 |     if (btnSaveTraining) {
1070 |         btnSaveTraining.onclick = async () => {
1071 |             btnSaveTraining.disabled = true;
1072 |             btnSaveTraining.innerHTML = `<div class="spinner-b
order spinner-border-sm text-info me-1" role="status"></div><s
pan>Сохранение...</span>`;
1073 |             try {
1074 |                 const res = await fetch('/api/v1/scenarios/save-
approved-response', {
1075 |                     method: 'POST',
1076 |                     headers: { 'Content-Type': 'application/json'
},
1077 |                     body: JSON.stringify({
1078 |                         user_id: 'tc_admin',
1079 |                         query: message,
1080 |                         chat_text: chatText,
1081 |                         voice_text: voiceText || '',
1082 |                         system_context: {
1083 |                             command_executed: finalData.command_execut
ed || '',
1084 |                             tool_plan: finalData.tool_plan || null,
1085 |                             platform: 'Windows'
1086 |                         },

```

```

1087         tags: ['tc', 'diagnostics', 'training']
1088     })),
1089     });
1090     if (!res.ok) throw new Error(`HTTP ${res.status}
1091 `);
1092     const resData = await res.json();
1093     const container = document.getElementById(`train
1094 ing-save-container-${msgId}`);
1095     if (container) {
1096         container.innerHTML = `
1097         <div class="p-1.5 px-2.5 bg-success bg-opaci
1098 ty-10 border border-success border-opacity-50 text-success rou
1099 nded small d-flex align-items-center gap-1.5 w-100 shadow-sm">
1100         <i class="bi bi-check-circle-fill text-suc
1101 cess"></i>
1102         <span><strong>Успешно:</strong> ответ сохр
1103 анён для обучения модели и RAG (<code>data/tc/approved_respons
1104 es</code>)</span>
1105         </div>
1106         `;
1107     }
1108     if (statusInd) statusInd.innerText = 'Ответ сохр
1109 анён в базу для обучения модели!';
1110     } catch (saveErr) {
1111         console.error('[ScenariosTab] Error saving train
1112 ing QA:', saveErr);
1113         btnSaveTraining.disabled = false;
1114         btnSaveTraining.innerHTML = `<i class="bi bi-exc
1115 lamation-triangle-fill text-danger me-1"></i><span>Ошибка сохр
1116 анения. Повторить</span>`;
1117     }
1118 };
1119 }
1120
1121 // Wire up manual save & inline edit logic
1122 if (!finalData.created_skill && finalData.tool_plan) {
1123     const saveBtn = document.getElementById(`btn-save-sk

```

```

    ill-${msgId}`);
1114     const editBtn = document.getElementById(`btn-edit-sk
    ill-${msgId}`);
1115     const editForm = document.getElementById(`skill-edit
    -form-${msgId}`);
1116     const cancelEditBtn = document.getElementById(`btn-c
    ancel-edit-${msgId}`);
1117     const saveEditedBtn = document.getElementById(`btn-s
    ave-edited-${msgId}`);
1118
1119     const executeSave = async (payload) => {
1120         const container = document.getElementById(`skill-a
    ction-container-${msgId}`);
1121         if (saveBtn) saveBtn.disabled = true;
1122         if (saveEditedBtn) saveEditedBtn.disabled = true;
1123         if (saveBtn) saveBtn.innerHTML = `
```

```

    eHtml(savedData.description_ru)}</div>
1138         </div>
1139         `;
1140     }
1141     if (editForm) editForm.classList.add('d-none');
1142     if (statusInd) statusInd.innerText = `Навык '${s
savedData.name}' сохранён!`;
1143     } catch (saveErr) {
1144         console.error('[ScenariosTab] Error saving skill
: ', saveErr);
1145         if (saveBtn) {
1146             saveBtn.disabled = false;
1147             saveBtn.innerHTML = `<i class="bi bi-exclamati
on-triangle-fill text-danger me-1"></i><span>Ошибка. Повторить
</span>`;
1148         }
1149         if (saveEditedBtn) saveEditedBtn.disabled = fals
e;
1150     }
1151 };
1152
1153 if (saveBtn) {
1154     saveBtn.addEventListener('click', () => {
1155         executeSave({
1156             tool_name: finalData.tool_plan.tool_name,
1157             tool_title: finalData.tool_plan.tool_title,
1158             description_ru: finalData.tool_plan.descriptio
n_ru,
1159             probe_type: finalData.tool_plan.probe_type ||
'powershell',
1160             collector_name: finalData.tool_plan.collector_
name,
1161             probe_script: finalData.tool_plan.probe_script
,
1162             instructions: finalData.tool_plan.instructions
,
1163             command_executed: finalData.command_executed,
1164         });

```

```
1165         });
1166     }
1167
1168     if (editBtn && editForm) {
1169         editBtn.addEventListener('click', () => {
1170             editForm.classList.toggle('d-none');
1171         });
1172     }
1173
1174     if (cancelEditBtn && editForm) {
1175         cancelEditBtn.addEventListener('click', () => {
1176             editForm.classList.add('d-none');
1177         });
1178     }
1179
1180     if (saveEditedBtn) {
1181         saveEditedBtn.addEventListener('click', () => {
1182             const editedTitle = document.getElementById(`edit-
1183 t-title-${msgId}`)?.value || finalData.tool_plan.tool_title;
1184             const editedDesc = document.getElementById(`edit-
1185 -desc-${msgId}`)?.value || finalData.tool_plan.description_ru;
1186             const editedCmd = document.getElementById(`edit-
1187 cmd-${msgId}`)?.value || finalData.command_executed;
1188
1189             executeSave({
1190                 tool_name: finalData.tool_plan.tool_name,
1191                 tool_title: editedTitle,
1192                 description_ru: editedDesc,
1193                 probe_type: finalData.tool_plan.probe_type ||
1194 'powershell',
1195                 collector_name: finalData.tool_plan.collector_
1196 name,
1197                 probe_script: editedCmd,
1198                 instructions: finalData.tool_plan.instructions
1199             },
1200             {
1201                 command_executed: editedCmd,
1202             });
1203         });
1204     }
1205 }
```

```

1197 |         }
1198 |     }
1199 |
1200 |     // Wire up remediation action execution buttons
1201 |     if (Array.isArray(finalData.remediation_actions) && fi
1202 | nalData.remediation_actions.length > 0) {
1203 |         const remContainer = document.getElementById(`remedi
1204 | ation-container-${msgId}`);
1205 |         if (remContainer) {
1206 |             remContainer.querySelectorAll('.btn-exec-action').
1207 | forEach(btn => {
1208 |                 btn.addEventListener('click', async () => {
1209 |                     const actionId = btn.dataset.actionId;
1210 |                     const actionType = btn.dataset.actionType;
1211 |                     const title = btn.dataset.title;
1212 |                     const description = btn.dataset.description;
1213 |                     const target = btn.dataset.target;
1214 |                     const risk = btn.dataset.risk;
1215 |                     const command = btn.dataset.command;
1216 |
1217 |                     // Confirm if risk is caution or critical
1218 |                     const isRisky = (risk || '').toLowerCase() ===
1219 | 'caution' || (risk || '').toLowerCase() === 'critical';
1220 |                     if (isRisky) {
1221 |                         const ok = confirm(`Подтвердите применение и
1222 | справления:\n\n«${title}»\n\nКоманда: ${command || target}`);
1223 |                         if (!ok) return;
1224 |                     }
1225 |
1226 |                     btn.disabled = true;
1227 |                     btn.innerHTML = `<div class="spinner-border sp
1228 | inner-border-sm text-warning" role="status"></div><span>Примен
1229 | ение...</span>`;
1230 |                     if (statusInd) statusInd.innerText = `Применен
1231 | ие '${title}'...`;
1232 |
1233 |                     try {
1234 |                         const execRes = await fetch(`/api/v1/scenari

```

```

    os/execute-fix', {
1227         method: 'POST',
1228         headers: { 'Content-Type': 'application/js
on' },
1229         body: JSON.stringify({
1230             action_id: actionId,
1231             action_type: actionType,
1232             title: title,
1233             description: description,
1234             target: target,
1235             risk: risk,
1236             execution_command: command,
1237             confirmed_by_user: true,
1238         }),
1239     });
1240
1241     if (!execRes.ok) throw new Error(`HTTP ${exe
cRes.status}`);
1242     const resData = await execRes.json();
1243     if (resData.success) {
1244         btn.className = 'btn btn-success btn-sm py-
1 px-3 d-flex align-items-center gap-1.5 shadow-sm';
1245         btn.innerHTML = `

```

```

1255 |             console.error('[ScenariosTab] Error executin
| g fix:', exErr);
1256 |             btn.disabled = false;
1257 |             btn.className = 'btn btn-outline-danger btn-
| sm py-0.5 px-2.5 d-flex align-items-center gap-1 shadow-sm';
1258 |             btn.innerHTML = `
```



```

xt-danger"></i>
1279         <span>Ошибка: ${escapeHtml(err.message || 'Н
известная ошибка выполнения')}</span>
1280     </span>
1281     ${errorTypeBadge}
1282 </div>
1283     ${errDetail && errDetail.trim() !== (err.message
|| '').trim() ? `
1284         <div class="mt-2 pt-1.5 border-top border-dang
er border-opacity-25">
1285             <div class="text-muted small mb-1" style="fo
nt-size: 0.72rem;"><i class="bi bi-code-slash me-1"></i>Параме
тры и стек ошибки:</div>
1286             <pre class="m-0 p-2 bg-body-secondary text-d
anger border border-danger border-opacity-25 rounded font-mono
space small" style="font-size: 0.72rem; line-height: 1.35; max
-height: 220px; overflow-y: auto; white-space: pre-wrap; word-
break: break-all;">${escapeHtml(errDetail)}</pre>
1287         </div>
1288         ` : ''}
1289     </div>
1290     `;
1291 }
1292 chatHistory.scrollTop = chatHistory.scrollHeight;
1293 if (statusInd) statusInd.innerText = `Ошибка: ${err.me
ssage || 'Сбой'}`;
1294 } finally {
1295     isChatSubmitting = false;
1296     chatInput.disabled = false;
1297     if (sendBtn) sendBtn.disabled = false;
1298     chatInput.focus();
1299 }
1300 }
1301
1302 chatForm.onSubmit = (e) => {
1303     e.preventDefault();
1304     submitChat();
1305 };

```

```
1306 | }
1307 |
1308 | // Setup UI event handlers
1309 | function setupEvents() {
1310 |     if (isEventsSetup) return;
1311 |     isEventsSetup = true;
1312 |
1313 |     // Quick Check Button
1314 |     const btnQuick = document.getElementById('btn-run-quick-check');
1315 |     if (btnQuick) {
1316 |         btnQuick.onclick = () => runScenario('quick_check');
1317 |     }
1318 |
1319 |     // Run All Button
1320 |     const btnAll = document.getElementById('btn-run-all-scenarios');
1321 |     if (btnAll) {
1322 |         btnAll.onclick = () => runScenario('all');
1323 |     }
1324 |
1325 |     // Refresh List Button
1326 |     const btnRefresh = document.getElementById('btn-scenario-refresh-list');
1327 |     if (btnRefresh) {
1328 |         btnRefresh.onclick = () => loadScenarios();
1329 |     }
1330 |
1331 |     // Close Results Button
1332 |     const btnCloseRes = document.getElementById('btn-close-result-panel');
1333 |     if (btnCloseRes) {
1334 |         btnCloseRes.onclick = () => {
1335 |             const panel = document.getElementById('scenario-result-panel');
1336 |             if (panel) panel.style.display = 'none';
1337 |         };
1338 |     }
```

```

1339 |
1340 |     // Category Filter Buttons
1341 |     const filterContainer = document.getElementById('scenario-
category-filters');
1342 |     if (filterContainer) {
1343 |         filterContainer.querySelectorAll('button').forEach(btn =
> {
1344 |             btn.onclick = () => {
1345 |                 filterContainer.querySelectorAll('button').forEach(b
=> b.classList.remove('active'));
1346 |                 btn.classList.add('active');
1347 |                 currentFilter = btn.getAttribute('data-filter') || '
all';
1348 |                 renderScenariosGrid();
1349 |             };
1350 |         });
1351 |     }
1352 |
1353 |     setupMiniChat();
1354 | }
1355 |
1356 | // Public initialization entrypoint
1357 | window.initScenariosTab = async function () {
1358 |     console.log('[ScenariosTab] Initializing Scenarios & Test
Computer Tab...');
1359 |     setupEvents();
1360 |     loadQuestionsFromConfig();
1361 |     await loadScenarios();
1362 |     console.log('[ScenariosTab] Scenarios tab ready.');
```

```

1363 | };
1364 | window.initChatTab = window.initScenariosTab;
1365 |
1366 | // Auto-init if container exists
1367 | if (document.getElementById('scenarios-cards-container') ||
document.getElementById('scenario-chat-history')) {
1368 |     window.initScenariosTab();
1369 | }
1370 | })();

```


index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/search_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/search_tab/index.html"
17     class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/search_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Web Search MCP Providers Management Tab -->
30 <div class="container-fluid py-3">
31   <!-- Header -->
32   <div class="alert alert-info d-flex justify-content-between
33   align-items-center flex-wrap gap-2 mb-3 shadow-sm">
34     <div>
35       <h5 class="mb-1"><i class="bi bi-globe2 me-2"></i> Про
36       вайдеры веб-поиска (MCP)</h5>
37     <p class="mb-0 small text-secondary">Управление серверам
```

и интернет-поиска, алгоритмами извлечения и тестирование выдачи.
и.

```
31     </div>
32     <div class="input-group input-group-sm" style="max-width:
33         440px;">
34         <span class="input-group-text fw-bold">Активный движок п
35         оиска:</span>
36         <select class="form-select" id="search-tab-engine-select
37         or">
38             <option value="playwright">□ Playwright MCP</option>
39             <option value="langchain">□ LangChain MCP</option>
40             <option value="gemini">□ Google Gemini (Grounding)</op
41             tion>
42             <option value="gemini_cli">□ Google Gemini CLI</optio
43             n>
44             <option value="agy">□ Antigravity (AGY)</option>
45         </select>
46         <button class="btn btn-primary" id="btn-save-search-tab-
47         engine">
48             <i class="bi bi-check-circle me-1"></i> Сохранить
49         </button>
50     </div>
51 </div>
52
53 <!-- Reference Info Block (Collapse, matching RAG tab style)
54 -->
55 <div class="card mb-4 border-0 shadow-sm bg-light">
56     <div class="card-body p-3">
57         <div class="d-flex align-items-center justify-content-be
58         tween">
59             <div class="d-flex align-items-center gap-2">
60                 <i class="bi bi-info-circle-fill text-primary fs-5">
61                 </i>
62                 <span class="fw-bold">Справка: архитектура и принцип
63                 ы работы 5 серверов веб-поиска</span>
64             </div>
65             <button class="btn btn-sm btn-link text-decoration-non
66             e p-0 text-secondary" type="button" data-bs-toggle="collapse"
```

```

data-bs-target="#searchArchCollapse" aria-expanded="true" aria-
-controls="searchArchCollapse">
56     <i class="bi bi-chevron-down"></i> Развернуть / Скры
    ть
57     </button>
58 </div>
59 <div class="collapse show" id="searchArchCollapse">
60     <div class="row g-3 mt-2 pt-2 border-top">
61         <!-- 1. Playwright -->
62         <div class="col-xl col-md-6">
63             <div class="p-3 border rounded bg-white h-100 shad
                ow-sm">
64                 <div class="fw-bold text-primary mb-1">
65                     □ Playwright MCP
66                 </div>
67                 <div class="small text-muted mb-2">
68                     Прямой headless-браузер (Chromium).
69                 </div>
70                 <ul class="small text-secondary mb-0 ps-3">
71                     <li>Поиск в DuckDuckGo, Bing и Google.</li>
72                     <li>Глубокий сбор DOM-структуры и текста сайта
73                     в.</li>
74                     <li><strong>Плюс:</strong> не расходует токены
75                     ИИ на сбор данных.</li>
76                     <li><strong>Особенность:</strong> требует запу
77                     щенного Playwright.</li>
78                 </ul>
79                 </div>
80             </div>
81             <!-- 2. LangChain -->
82             <div class="col-xl col-md-6">
83                 <div class="p-3 border rounded bg-white h-100 shad
84                 ow-sm">
85                     <div class="fw-bold text-success mb-1">
86                         □ LangChain MCP
87                     </div>
88                     <div class="small text-muted mb-2">

```

```

86 |             Автономный ReAct-агент Breadboard.
87 |         </div>
88 |         <ul class="small text-secondary mb-0 ps-3">
89 |             <li>Инструменты: веб-поиск, RAG, Python-вычисл
| ения.</li>
90 |             <li>Агент сам принимает решение о вызове инстр
| ументов.</li>
91 |             <li><strong>Плюс:</strong> гибкое решение комп
| лексных задач.</li>
92 |             <li><strong>Особенность:</strong> многошаговый
| reasoning-цикл.</li>
93 |         </ul>
94 |     </div>
95 | </div>
96 |
97 | <!-- 3. Gemini Grounding -->
98 | <div class="col-xl col-md-6">
99 |     <div class="p-3 border rounded bg-white h-100 shad
| ow-sm">
100 |         <div class="fw-bold text-info mb-1">
101 |             □ Google Gemini Grounding
102 |         </div>
103 |         <div class="small text-muted mb-2">
104 |             Официальный Google Search Grounding.
105 |         </div>
106 |         <ul class="small text-secondary mb-0 ps-3">
107 |             <li>Прямой поиск в реальном индексе Google Sea
| rch.</li>
108 |             <li>Автоматическая ротация пула ключей при оши
| бках 429.</li>
109 |             <li><strong>Плюс:</strong> проверенная точност
| ь фактов и цитаты.</li>
110 |             <li><strong>Особенность:</strong> требует вали
| дных API-ключей.</li>
111 |         </ul>
112 |     </div>
113 | </div>
114 |

```



```

115 |         <!-- 4. Gemini CLI -->
116 |         <div class="col-xl col-md-6">
117 |             <div class="p-3 border rounded bg-white h-100 shadow-sm">
118 |                 <div class="fw-bold text-primary mb-1">
119 |                     □ Google Gemini CLI
120 |                 </div>
121 |                 <div class="small text-muted mb-2">
122 |                     Локальный терминальный агент Google.
123 |                 </div>
124 |                 <ul class="small text-secondary mb-0 ps-3">
125 |                     <li>Прямое выполнение через CLI подпроцесс.</li>
126 |                     <li>Встроенные агентные инструменты веб-поиска
127 |                     .</li>
128 |                     <li><strong>Плюс:</strong> автономный агент, модель по умолчанию flash-lite.</li>
129 |                     <li><strong>Особенность:</strong> требует установленной утилиты gemini CLI.</li>
130 |                 </ul>
131 |             </div>
132 |
133 |         <!-- 5. Antigravity AGY -->
134 |         <div class="col-xl col-md-6">
135 |             <div class="p-3 border rounded bg-white h-100 shadow-sm">
136 |                 <div class="fw-bold text-warning mb-1">
137 |                     □ Antigravity (AGY)
138 |                 </div>
139 |                 <div class="small text-muted mb-2">
140 |                     Мультиагентный поиск платформы Google.
141 |                 </div>
142 |                 <ul class="small text-secondary mb-0 ps-3">
143 |                     <li>Инструменты <code>SEARCH_WEB</code> и <code>READ_URL_CONTENT</code>.</li>
144 |                     <li>Глубокий ресерч и синтез сложных многоходовок.</li>

```

```

145 |             <li><strong>Плюс:</strong> автономный агентный
      | цикл.</li>
146 |             <li><strong>Особенность:</strong> использует A
      | ntigravity SDK / CLI.</li>
147 |         </ul>
148 |     </div>
149 | </div>
150 | </div>
151 | </div>
152 | </div>
153 | </div>
154 |
155 | <!-- Detailed Provider Settings & Live Search Tester -->
156 | <div class="row g-4 mb-4">
157 |     <!-- LEFT: Provider Detailed Settings -->
158 |     <div class="col-lg-6">
159 |         <div class="card shadow-sm border-0 h-100">
160 |             <div class="card-header bg-dark text-white fw-bold">
161 |                 <i class="bi bi-sliders me-1"></i> Параметры поиско
      | ых модулей
162 |             </div>
163 |             <div class="card-body">
164 |                 <!-- Gemini Grounding Settings -->
165 |                 <div class="mb-4 p-3 border rounded bg-light">
166 |                     <h6 class="fw-bold text-info mb-2"><i class="bi bi
      | -stars me-1"></i> Gemini Grounding</h6>
167 |                     <div class="mb-2">
168 |                         <label class="form-label small text-muted fw-bol
      | d">Модель для поиска Grounding (динамически из SDK):</label>
169 |                         <select class="form-select form-select-sm" id="s
      | earch-gemini-model">
170 |                             <option value="">Загрузка доступных моделей че
      | рез Google GenAI SDK...</option>
171 |                         </select>
172 |                     </div>
173 |                     <div class="small text-muted">
174 |                         Ключи автоматически считываются из вкладки «Моде
      | ли и API» и ротируются по кругу.

```

```
175         </div>
176     </div>
177
178     <!-- Gemini CLI Settings -->
179     <div class="mb-4 p-3 border rounded bg-light">
180         <h6 class="fw-bold text-primary mb-2"><i class="bi
181         bi-terminal-fill me-1"></i> Google Gemini CLI</h6>
182         <div class="mb-2">
183             <label class="form-label small text-muted fw-bol
184             d">Модель Gemini CLI (динамически из инструмента):</label>
185             <select class="form-select form-select-sm" id="s
186             earch-gemini-cli-model">
187                 <option value="">Загрузка доступных моделей че
188                 рез Gemini CLI...</option>
189             </select>
190         </div>
191         <div class="small text-muted">
192             Использует локальную утилиту <code>gemini</code>
193             с автоматическим вызовом поисковых инструментов.
194         </div>
195     </div>
196
197     <!-- Antigravity AGY Settings -->
198     <div class="mb-4 p-3 border rounded bg-light">
199         <h6 class="fw-bold text-warning mb-2"><i class="bi
200         bi-rocket-takeoff me-1"></i> Antigravity (AGY) Search</h6>
201         <div class="mb-2">
202             <label class="form-label small text-muted fw-bol
203             d">Модель агента Antigravity (динамически из SDK):</label>
204             <select class="form-select form-select-sm" id="s
205             earch-agy-model">
206                 <option value="">Загрузка доступных моделей че
207                 рез Antigravity SDK...</option>
208             </select>
209         </div>
210         <div class="small text-muted">
211             Использует встроенные агентные инструменты <code>
212             >SEARCH_WEB</code> и <code>READ_URL_CONTENT</code>.
```

```

203 |         </div>
204 |     </div>
205 |
206 |     <div class="d-grid">
207 |         <button class="btn btn-success fw-bold" id="btn-sa
ve-search-params">
208 |             <i class="bi bi-floppy me-1"></i> Сохранить наст
ройки модулей
209 |         </button>
210 |     </div>
211 | </div>
212 | </div>
213 | </div>
214 |
215 | <!-- RIGHT: Live Interactive Search Tester -->
216 | <div class="col-lg-6">
217 |     <div class="card shadow-sm border-0 h-100">
218 |         <div class="card-header bg-dark text-white d-flex just
ify-content-between align-items-center">
219 |             <span class="fw-bold"><i class="bi bi-terminal me-1"
></i> Интерактивный тест поиска</span>
220 |             <span class="badge bg-primary" id="test-search-engin
e-badge">playwright</span>
221 |         </div>
222 |         <div class="card-body d-flex flex-column">
223 |             <div class="mb-3">
224 |                 <label class="form-label small text-muted fw-bold"
>Тестовый поисковый запрос:</label>
225 |                 <div class="input-group">
226 |                     <input type="text" class="form-control form-cont
rol-sm" id="test-search-query-input" placeholder="Например: по
следние новости марсохода 2026">
227 |                     <button class="btn btn-primary btn-sm px-3" id="
btn-run-test-search">
228 |                         <i class="bi bi-search me-1"></i> Искать
229 |                     </button>
230 |                 </div>
231 |             </div>

```

```
232 |
233 |         <div class="flex-grow-1 d-flex flex-column">
234 |             <label class="form-label small text-muted fw-bold"
|>Результат работы провайдера:</label>
235 |             <div id="test-search-output-container" class="bord
|er rounded p-3 bg-light flex-grow-1 font-monospace" style="min
|height: 230px; max-height: 350px; overflow-y: auto; font-size
|: 0.82rem; white-space: pre-wrap;">
236 |                 <span class="text-muted">Введите запрос и нажмит
|е «Искать» для проверки выбранного провайдера.</span>
237 |             </div>
238 |         </div>
239 |     </div>
240 | </div>
241 | </div>
242 | </div>
243 | </div>
244 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Search Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/search_tab/main.js?v=2
11  *       0261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/search_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Search Tab Logic - Web Search MCP Providers Management
22
23 async function initSearchTab() {
24   const engineSelect = document.getElementById('search-tab-engine-selector');
25   const btnSaveEngine = document.getElementById('btn-save-search-tab-engine');
26   const btnSaveParams = document.getElementById('btn-save-search-params');
27   const geminiModelSelect = document.getElementById('search-gemini-model');
28   const geminiCliModelSelect = document.getElementById('search
```

```

    -gemini-cli-model');
29   const agyModelSelect = document.getElementById('search-agy-m
    odel');
30   const engineBadge = document.getElementById('test-search-eng
    ine-badge');
31   const btnRunTest = document.getElementById('btn-run-test-sea
    rch');
32   const testQueryInput = document.getElementById('test-search-
    query-input');
33   const testOutputContainer = document.getElementById('test-se
    arch-output-container');
34
35   if (!engineSelect) return;
36
37   // --- 1. Load Dynamic Models via SDK / Tools ---
38   async function loadModels() {
39     try {
40       const modelsData = await window.api.fetch('/api/chat/mod
        els');
41       const modelsGrouped = modelsData.models || {};
42       const geminiList = modelsGrouped.gemini || [];
43       const geminiCliList = modelsGrouped.gemini_cli || ['gemi
        ni-3.1-flash-lite', 'gemini-3.1-flash', 'gemini-2.5-pro'];
44       const agyList = modelsGrouped.agy || [];
45
46       if (geminiModelSelect && geminiList.length > 0) {
47         const curVal = geminiModelSelect.value;
48         geminiModelSelect.innerHTML = '';
49         geminiList.forEach(m => {
50           const opt = document.createElement('option');
51           opt.value = m;
52           opt.textContent = m;
53           geminiModelSelect.appendChild(opt);
54         });
55         if (curVal && geminiList.includes(curVal)) {
56           geminiModelSelect.value = curVal;
57         }
58       }

```

```
59 |
60 |     if (geminiCliModelSelect && geminiCliList.length > 0) {
61 |         const curVal = geminiCliModelSelect.value;
62 |         geminiCliModelSelect.innerHTML = '';
63 |         geminiCliList.forEach(m => {
64 |             const opt = document.createElement('option');
65 |             opt.value = m;
66 |             opt.textContent = m;
67 |             geminiCliModelSelect.appendChild(opt);
68 |         });
69 |         if (curVal && geminiCliList.includes(curVal)) {
70 |             geminiCliModelSelect.value = curVal;
71 |         } else if (geminiCliList.includes('gemini-3.1-flash-li
72 | te')) {
73 |             geminiCliModelSelect.value = 'gemini-3.1-flash-lite'
74 | ;
75 |     }
76 | }
77 |
78 | if (agyModelSelect && agyList.length > 0) {
79 |     const curVal = agyModelSelect.value;
80 |     agyModelSelect.innerHTML = '';
81 |     agyList.forEach(m => {
82 |         const opt = document.createElement('option');
83 |         opt.value = m;
84 |         opt.textContent = m;
85 |         agyModelSelect.appendChild(opt);
86 |     });
87 |     if (curVal && agyList.includes(curVal)) {
88 |         agyModelSelect.value = curVal;
89 |     }
90 | } catch (err) {
91 |     console.error('[SearchTab] Ошибка загрузки динамических
92 | моделей SDK/CLI:', err);
93 | }
```



```

94 | // --- 2. Load Current Config ---
95 | async function loadConfig() {
96 |     try {
97 |         await loadModels();
98 |         const data = await window.api.fetch('/api/admin/web-search/config');
99 |         if (data) {
100 |             if (data.engine && engineSelect) {
101 |                 engineSelect.value = data.engine;
102 |                 if (engineBadge) engineBadge.textContent = data.engine;
103 |             }
104 |             if (data.gemini_model && geminiModelSelect) {
105 |                 geminiModelSelect.value = data.gemini_model;
106 |             }
107 |             if (data.gemini_cli_model && geminiCliModelSelect) {
108 |                 geminiCliModelSelect.value = data.gemini_cli_model;
109 |             }
110 |             if (data.agy_model && agyModelSelect) {
111 |                 agyModelSelect.value = data.agy_model;
112 |             }
113 |         }
114 |     } catch (e) {
115 |         console.error('[SearchTab] Ошибка загрузки конфигурации поиска:', e);
116 |         notifySearch('Ошибка загрузки настроек поиска: ' + e.message, 'danger');
117 |     }
118 | }
119 |
120 | // --- 3. Save Config Helper ---
121 | async function saveConfig() {
122 |     const engine = engineSelect ? engineSelect.value : 'playwright';
123 |     const gemini_model = geminiModelSelect ? geminiModelSelect.value : 'gemini-flash-lite-latest';
124 |     const gemini_cli_model = geminiCliModelSelect ? geminiCliModelSelect.value : 'gemini-3.1-flash-lite';

```

```
125     const agy_model = agyModelSelect ? agyModelSelect.value :
    'agy-flash';
126
127     try {
128         await window.api.fetch('/api/admin/web-search/config', {
129             method: 'POST',
130             headers: { 'Content-Type': 'application/json' },
131             body: JSON.stringify({ engine, gemini_model, gemini_cli_model, agy_model })
132         });
133         if (engineBadge) engineBadge.textContent = engine;
134         notifySearch(`Настройки веб-поиска сохранены (активен: ${engine})`, 'success');
135
136         let compoundSearch = engine;
137         if (engine === 'gemini') compoundSearch = `${engine}:${gemini_model}`;
138         if (engine === 'gemini_cli') compoundSearch = `${engine}:${gemini_cli_model}`;
139         if (engine === 'agy') compoundSearch = `${engine}:${agy_model}`;
140
141         if (typeof window.updateChatBadges === 'function') {
142             window.updateChatBadges(undefined, compoundSearch);
143         }
144     } catch (e) {
145         console.error('[SearchTab] Ошибка сохранения:', e);
146         notifySearch(`Ошибка сохранения настроек: ` + e.message, 'danger');
147     }
148 }
149
150 if (btnSaveEngine) btnSaveEngine.onclick = saveConfig;
151 if (btnSaveParams) btnSaveParams.onclick = saveConfig;
152
153 if (engineSelect) {
154     engineSelect.onChange = () => {
155         if (engineBadge) engineBadge.textContent = engineSelect.
```

```

    value;
156     };
157   }
158
159   // --- 3. Interactive Search Tester ---
160   if (btnRunTest && testQueryInput && testOutputContainer) {
161     const runTest = async () => {
162       const query = testQueryInput.value.trim();
163       if (!query) {
164         notifySearch('Введите поисковый запрос для теста!', 'w
arning');
165         testQueryInput.focus();
166         return;
167       }
168
169       const engine = engineSelect ? engineSelect.value : 'play
wright';
170       btnRunTest.disabled = true;
171       const originalText = btnRunTest.innerHTML;
172       btnRunTest.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1" role="status"></span> Поиск...';
173       testOutputContainer.innerHTML = `<span class="text-info"
>□ Выполнение поиска через <strong>${engine}</strong> по запро
су: "${query}"...</span>`;
174
175       try {
176         const response = await window.api.fetch('/api/admin/we
b-search/test', {
177           method: 'POST',
178           headers: { 'Content-Type': 'application/json' },
179           body: JSON.stringify({ query, engine })
180         });
181
182         if (response.status === 'ok') {
183           testOutputContainer.textContent = response.result ||
'Пустой ответ.';
184         } else {
185           testOutputContainer.innerHTML = `<span class="text-d

```

```

    anger">❑ Ошибка: ${response.message || 'Неизвестная ошибка'}</
    span>`;
186     }
187     } catch (err) {
188         console.error('[SearchTab] Ошибка теста:', err);
189         testOutputContainer.innerHTML = `<span class="text-dan
    ger">❑ Ошибка запроса: ${err.message}</span>`;
190     } finally {
191         btnRunTest.disabled = false;
192         btnRunTest.innerHTML = originalText;
193     }
194 };
195
196 btnRunTest.onclick = runTest;
197 testQueryInput.onkeydown = (e) => {
198     if (e.key === 'Enter') {
199         e.preventDefault();
200         runTest();
201     }
202 };
203 }
204
205 // Helper notification
206 function notifySearch(msg, type = 'info') {
207     if (typeof showNotification === 'function') {
208         showNotification(msg, type);
209     } else if (typeof showModelsNotification === 'function') {
210         showModelsNotification(msg, type);
211     } else {
212         console.log(`[${type}] ${msg}`);
213     }
214 }
215
216 await loadConfig();
217 }
218
219 window.initSearchTab = initSearchTab;
220

```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/security_acl_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/security_acl_tab/index.
17     html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/security_acl_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Security & ACL Management Tab -->
30 <div class="container-fluid py-2 px-3">
31   <!-- Top Stat Cards -->
32   <div class="row g-2 mb-3">
33     <div class="col-6 col-md-3">
34       <div class="card bg-dark text-white border-secondary h-1
35       00 shadow-sm p-3">
36         <div class="text-muted small text-uppercase fw-semibol
37         d">Владелец объекта □ </div>
```

```
31 |         <div class="fs-5 fw-bold mt-1 text-primary text-trunca
    | te" id="acl-owner-val">--</div>
32 |         <div class="small text-muted mt-1" id="acl-owner-sub">
    | takeown.exe</div>
33 |     </div>
34 | </div>
35 | <div class="col-6 col-md-3">
36 |     <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
37 |         <div class="text-muted small text-uppercase fw-semibol
    | d">Наследование прав □ </div>
38 |         <div class="fs-4 fw-bold mt-1 text-info" id="acl-inher
    | it-val">Включено</div>
39 |         <div class="small text-muted mt-1" id="acl-inherit-sub
    | ">Inheritance Mode</div>
40 |     </div>
41 | </div>
42 | <div class="col-6 col-md-3">
43 |     <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
44 |         <div class="text-muted small text-uppercase fw-semibol
    | d">Количество ACE □ </div>
45 |         <div class="fs-4 fw-bold mt-1 text-success" id="acl-ac
    | e-count">--</div>
46 |         <div class="small text-muted mt-1" id="acl-ace-sub">ic
    | acls entries</div>
47 |     </div>
48 | </div>
49 | <div class="col-6 col-md-3">
50 |     <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
51 |         <div class="text-muted small text-uppercase fw-semibol
    | d">Целостность ACL □ </div>
52 |         <div class="fs-4 fw-bold mt-1 text-warning" id="acl-in
    | tegrity-val">Normal</div>
53 |         <div class="small text-muted mt-1" id="acl-integrity-s
    | ub">ACL Health</div>
54 |     </div>
```

```

55     </div>
56 </div>
57
58 <!-- Path & Inspect Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2 flex-grow-1">
61         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="acl-path-input" placeholder="Путь к файлу или папке (например: C:\Windows\System32 или C:\Data)" value="C:\Windows\System32">
62         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5 flex-shrink-0" id="acl-inspect-btn">
63             <i class="bi bi-search"></i>
64             <span>Инспектировать ACL</span>
65         </button>
66         <button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5 flex-shrink-0" id="acl-takeown-btn">
67             <i class="bi bi-person-check"></i>
68             <span>Взять владение (takeown)</span>
69         </button>
70     </div>
71 </div>
72
73 <!-- Main Grid -->
74 <div class="row g-3">
75     <div class="col-12 col-lg-7">
76         <div class="card bg-dark text-white border-secondary shadow-sm">
77             <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
78                 <span class="fw-semibold"><i class="bi bi-shield-lock text-primary me-2"></i>Записи контроля доступа (ACE List)</span>
79                 <span class="badge bg-secondary" id="acl-badge">0 записей</span>
80             </div>

```

```

81 |         <div class="card-body p-0 table-responsive" style="max
| -height: 420px; overflow-y: auto;">
82 |             <table class="table table-dark table-hover table-str
| iped align-middle mb-0 small" id="acl-table">
83 |                 <thead>
84 |                     <tr class="text-secondary">
85 |                         <th>Пользователь / Группа (Principal)</th>
86 |                         <th>Права доступа</th>
87 |                         <th>Тип</th>
88 |                         <th>Наследование</th>
89 |                     </tr>
90 |                 </thead>
91 |                 <tbody id="acl-tbody">
92 |                     <tr><td colspan="4" class="text-center text-mute
| d py-4">Введите путь и нажмите «Инспектировать ACL»</td></tr>
93 |                 </tbody>
94 |             </table>
95 |         </div>
96 |     </div>
97 | </div>
98 |
99 |     <div class="col-12 col-lg-5">
100 |         <div class="card bg-dark text-white border-secondary sha
| dow-sm mb-3">
101 |             <div class="card-header border-secondary d-flex align-
| items-center justify-content-between py-2 px-3">
102 |                 <span class="fw-semibold"><i class="bi bi-person-gear
| text-warning me-2"></i>Изменение разрешений (icacls)</span>
103 |             </div>
104 |             <div class="card-body p-3">
105 |                 <div class="mb-2">
106 |                     <label class="form-label small text-muted">Пользов
| атель / Субъект:</label>
107 |                     <input type="text" class="form-control form-control-sm
| bg-dark text-white border-secondary" id="acl-user-input"
| placeholder="Administrators или Users" value="Administrators">
108 |                 </div>
109 |                 <div class="mb-3">

```



```

110         <label class="form-label small text-muted">Уровень
      | разрешений:</label>
111         <select class="form-select form-select-sm bg-dark
      | text-white border-secondary" id="acl-perm-select">
112             <option value="F">Полный доступ (Full Control -
      | :F)</option>
113             <option value="M">Изменение (Modify - :M)</optio
      | n>
114             <option value="RX">Чтение и выполнение (Read/Exe
      | cute - :RX)</option>
115             <option value="R">Только чтение (Read - :R)</opt
      | ion>
116             <option value="W">Только запись (Write - :W)</op
      | tion>
117         </select>
118     </div>
119     <div class="d-flex align-items-center justify-content-between">
120         <div class="form-check form-switch">
121             <input class="form-check-input" type="checkbox"
      | id="acl-dry-run-switch" checked>
122             <label class="form-check-label small text-muted"
      | for="acl-dry-run-switch">Безопасный режим (Dry Run)</label>
123         </div>
124         <button class="btn btn-sm btn-success px-3" id="ac
      | l-grant-btn">
125             <i class="bi bi-check-lg"></i> Применить права
126         </button>
127     </div>
128 </div>
129 </div>
130
131     <!-- Action Log Output -->
132     <div class="card bg-dark text-white border-secondary sha
      | dow-sm">
133         <div class="card-header border-secondary py-2 px-3 d-f
      | lex align-items-center justify-content-between">
134             <span class="small fw-semibold text-muted"><i class=

```

```
135 | "bi bi-terminal me-1"></i>Консольный вывод icacIs</span>
136 |     <button class="btn btn-link btn-sm text-decoration-n
137 | one text-muted p-0" id="acl-clear-log-btn">Очистить</button>
138 |     </div>
139 |     <div class="card-body p-2">
140 |         <pre class="mb-0 small bg-black text-light p-2 round
141 | ed" id="acl-console-output" style="max-height: 180px; overflow
142 | -y: auto; font-family: Consolas, monospace; font-size: 0.78rem
143 | ;">Готов к инспекции прав доступа.</pre>
144 |     </div>
145 | </div>
146 | </div>
147 | </div>
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Security Acl Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *   <script src="/windows/api/webgui/security_acl_tab/main.
11 *   js?v=20261001_v1" type="module"></script>
12 *
13 *   JavaScript Import:
14 *   import { initSecurityAclTab } from '/windows/api/webgui
15 *   /security_acl_tab/main.js';
16 *
17 * File: main.js
18 * Project: ai-breadboard
19 * Package: windows/api/webgui/security_acl_tab
20 * Author: hypo69
21 * Copyright: © 2026 hypo69
22 * Updated: 2026-10-04 11:15:30
23 * =====
24 */
25
26 * security_acl_tab/main.js – Управление правами доступа и дес
27 * крипторами безопасности (icaccls, takeown)
28 * Updated: 2026-10-04 11:15:30
29 */
30
31 const registerTabPoller = window.registerTabPoller || function
32 () {};
```

```
30
31 let isInitialized = false;
32
33 export async function inspectPathAcl() {
34   const pathInput = document.getElementById('acl-path-input');
35   const path = pathInput ? pathInput.value.trim() : 'C:\\Windows\\System32';
36
37   try {
38     const res = await fetch(`/api/security-acl/summary?path=${encodeURIComponent(path)}`);
39     if (!res.ok) throw new Error(`HTTP ${res.status}`);
40     const data = await res.json();
41     renderAclSummary(data);
42   } catch (err) {
43     console.warn('[SecurityACL] Ошибка получения ACL:', err);
44   }
45 }
46
47 function renderAclSummary(data) {
48   const ownerEl = document.getElementById('acl-owner-val');
49   const inheritEl = document.getElementById('acl-inherit-val');
50   ;
51   const aceCountEl = document.getElementById('acl-ace-count');
52   const tbody = document.getElementById('acl-tbody');
53   const badge = document.getElementById('acl-badge');
54
55   if (ownerEl) ownerEl.textContent = data.owner || 'NT AUTHORITY\\SYSTEM';
56
57   if (inheritEl) inheritEl.textContent = data.inheritance_enabled ? 'Включено' : 'Отключено';
58
59   const aces = data.aces || [
60     { principal: 'NT AUTHORITY\\SYSTEM', permissions: 'Full Control', type: 'Allow', inheritance: 'Inherited' },
61     { principal: 'BUILTIN\\Administrators', permissions: 'Full Control', type: 'Allow', inheritance: 'Inherited' },
62     { principal: 'BUILTIN\\Users', permissions: 'Read & Execut
```

```

    e', type: 'Allow', inheritance: 'Inherited' }
61   ];
62
63   if (aceCountEl) aceCountEl.textContent = aces.length;
64   if (badge) badge.textContent = `${aces.length} записей`;
65
66   if (!tbody) return;
67
68   tbody.innerHTML = aces.map(a => `
69     <tr>
70       <td class="fw-semibold text-light">${a.principal || '-'}
    </td>
71       <td class="text-info font-monospace small">${a.permissions || 'Read'}</td>
72       <td><span class="badge ${a.type === 'Allow' ? 'bg-success' : 'bg-danger'}">${a.type || 'Allow'}</span></td>
73       <td class="small text-muted">${a.inheritance || 'Direct'}</td>
74     </tr>
75   `).join('');
76 }
77
78 async function executeAclAction(customAction = null) {
79   const pathInput = document.getElementById('acl-path-input');
80   const userInput = document.getElementById('acl-user-input');
81   const permSelect = document.getElementById('acl-perm-select');
82   const dryRunSwitch = document.getElementById('acl-dry-run-switch');
83   const output = document.getElementById('acl-console-output');
84   ;
85   const path = pathInput ? pathInput.value.trim() : 'C:\\Windows\\System32';
86   const user = userInput ? userInput.value.trim() : 'Administrators';
87   const perm = permSelect ? permSelect.value : 'F';
88   const action = customAction || 'icacls_grant';

```

```
89 |   const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;
90 |
91 |   if (output) output.textContent = `[${new Date().toLocaleTime
String()}] Выполнение ${action} для "${path}"...\n`;
92 |
93 |   try {
94 |     const res = await fetch('/api/security-acl/action', {
95 |       method: 'POST',
96 |       headers: { 'Content-Type': 'application/json' },
97 |       body: JSON.stringify({ path: path, user: user, permissio
n: perm, action: action, dry_run: dryRun })
98 |     });
99 |     const result = await res.json();
100 |     if (output) output.textContent += JSON.stringify(result, n
ull, 2);
101 |     if (window.toast) {
102 |       if (res.ok) window.toast.success('Права обновлены', `Ста
тус: ${result.status || 'OK'}`);
103 |       else window.toast.error('Ошибка', result.detail || 'Сбой
выполнения icacls');
104 |     }
105 |   } catch (err) {
106 |     if (output) output.textContent += `\nОшибка запроса: ${err
.message}`;
107 |     if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
108 |   }
109 | }
110 |
111 | export function initSecurityAclTab() {
112 |   if (isInitialized) return;
113 |   isInitialized = true;
114 |
115 |   const inspectBtn = document.getElementById('acl-inspect-btn'
);
116 |   if (inspectBtn) inspectBtn.addEventListener('click', inspect
PathAcl);
117 |
```

```
118 |   const takeownBtn = document.getElementById('acl-takeown-btn'  
    |   );  
119 |   if (takeownBtn) takeownBtn.addEventListener('click', () => e  
    |   xecuteAclAction('takeown'));  
120 |  
121 |   const grantBtn = document.getElementById('acl-grant-btn');  
122 |   if (grantBtn) grantBtn.addEventListener('click', () => execu  
    |   teAclAction('icacls_grant'));  
123 |  
124 |   const clearBtn = document.getElementById('acl-clear-log-btn'  
    |   );  
125 |   if (clearBtn) clearBtn.addEventListener('click', () => {  
126 |       const out = document.getElementById('acl-console-output');  
127 |       if (out) out.textContent = 'Консоль очищена.';  
128 |   });  
129 |  
130 |   registerTabPoller('tab-security-acl', inspectPathAcl, 20000,  
    |   { immediate: true });  
131 | }  
132 |  
133 | window.initSecurityAclTab = initSecurityAclTab;  
134 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Services Manager View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса диспетчера служб Windows.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/services_manager_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/services_manager_tab/in
15 |dex.html" class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/services_manager_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-06 11:24:00
23 |=====
24 |=====
25 -->
26
27 <style>
28 |.svc-manager-wrapper {
29 |    color: var(--text-color);
30 |}
31 |.svc-card {
32 |    background: var(--surface-1);
33 |    border: 1px solid var(--border-color);
34 |    color: var(--text-color);
35 |}
```



```

33 | .svc-card-header {
34 |     background: var(--surface-2);
35 |     border-bottom: 1px solid var(--border-color);
36 |     color: var(--text-color);
37 | }
38 | #svc-table tbody tr {
39 |     border-bottom: 1px solid var(--border-color);
40 | }
41 | #svc-table tbody tr:hover {
42 |     background-color: var(--surface-2, rgba(56, 189, 248, 0.06
43 | )) !important;
44 | }
45 | #svc-table td {
46 |     background: transparent;
47 |     color: var(--text-color);
48 | }
49 | </style>
50 | <!-- Services Manager Tab -->
51 | <div class="container-fluid py-2 px-3 svc-manager-wrapper">
52 |     <!-- Top Stat Cards -->
53 |     <div class="row g-2 mb-3">
54 |         <div class="col-6 col-md-3">
55 |             <div class="card svc-card h-100 shadow-sm p-3">
56 |                 <div class="text-muted small text-uppercase fw-semibol
57 | d">Всего служб ☐ </div>
58 |                 <div class="fs-4 fw-bold mt-1 text-primary" id="svc-to
59 | tal-count">--</div>
60 |                 <div class="small text-muted mt-1" id="svc-total-sub">
61 | sc query state= all</div>
62 |             </div>
63 |         </div>
64 |         <div class="col-6 col-md-3">
65 |             <div class="card svc-card h-100 shadow-sm p-3">
66 |                 <div class="text-muted small text-uppercase fw-semibol
67 | d">Работают (Running) ☐ </div>
68 |                 <div class="fs-4 fw-bold mt-1 text-success" id="svc-ru
69 | nning-count">--</div>

```

```

65 |         <div class="small text-muted mt-1" id="svc-running-sub
    | ">Активные процессы</div>
66 |     </div>
67 | </div>
68 | <div class="col-6 col-md-3">
69 |     <div class="card svc-card h-100 shadow-sm p-3">
70 |         <div class="text-muted small text-uppercase fw-semibol
    | d">Остановлены (Stopped) □ </div>
71 |         <div class="fs-4 fw-bold mt-1 text-secondary" id="svc-
    | stopped-count">--</div>
72 |         <div class="small text-muted mt-1" id="svc-stopped-sub
    | ">В режиме ожидания</div>
73 |     </div>
74 | </div>
75 | <div class="col-6 col-md-3">
76 |     <div class="card svc-card h-100 shadow-sm p-3">
77 |         <div class="text-muted small text-uppercase fw-semibol
    | d">С автозапуском (Auto) □ </div>
78 |         <div class="fs-4 fw-bold mt-1 text-info" id="svc-auto-
    | count">--</div>
79 |         <div class="small text-muted mt-1" id="svc-auto-sub">C
    | истемный старт</div>
80 |     </div>
81 | </div>
82 | </div>
83 |
84 | <!-- Action Bar -->
85 | <div class="d-flex align-items-center justify-content-between
    | mb-3 p-2 rounded flex-wrap gap-2 svc-card">
86 |     <div class="d-flex align-items-center gap-2">
87 |         <button class="btn btn-sm btn-primary d-flex align-items
    | -center gap-1.5" id="svc-refresh-btn">
88 |             <i class="bi bi-arrow-clockwise"></i>
89 |             <span>Обновить список</span>
90 |         </button>
91 |         <div class="btn-group btn-group-sm">
92 |             <button class="btn btn-sm btn-outline-secondary active
    | " id="svc-filter-all">Все</button>

```

```

93 |         <button class="btn btn-sm btn-outline-secondary" id="s
|vc-filter-running">Работающие</button>
94 |         <button class="btn btn-sm btn-outline-secondary" id="s
|vc-filter-stopped">Остановленные</button>
95 |     </div>
96 | </div>
97 | <div class="d-flex align-items-center gap-2">
98 |     <input type="text" class="form-control form-control-sm"
|id="svc-search-input" placeholder="Поиск службы (имя или отобр
|ажение)..." style="width: 260px; background: var(--surface-2);
|color: var(--text-color); border-color: var(--border-color);"
|>
99 | </div>
100 | </div>
101 |
102 | <!-- Services Table -->
103 | <div class="card svc-card shadow-sm">
104 |     <div class="card-header svc-card-header d-flex align-items
|center justify-content-between py-2 px-3">
105 |         <span class="fw-semibold"><i class="bi bi-gear-wide-conn
|ected text-primary me-2"></i>Службы Windows (Service Control M
|anager)</span>
106 |         <span class="badge bg-secondary-subtle text-secondary bo
|rder border-secondary" id="svc-table-badge">0 служб</span>
107 |     </div>
108 |     <div class="card-body p-0 table-responsive" style="max-hei
|ght: 560px; overflow-y: auto;">
109 |         <table class="table table-hover align-middle mb-0 small"
|id="svc-table">
110 |             <thead class="sticky-top" style="background: var(--sur
|face-2); border-bottom: 2px solid var(--border-color);">
111 |                 <tr>
112 |                     <th style="background: var(--surface-2); color: va
|r(--text-color);">Имя службы</th>
113 |                     <th style="background: var(--surface-2); color: va
|r(--text-color);">Отображаемое имя</th>
114 |                     <th style="background: var(--surface-2); color: va
|r(--text-color);">Состояние</th>

```

```
115 |         <th style="background: var(--surface-2); color: va  
| r(--text-color);">Тип запуска</th>  
116 |         <th style="background: var(--surface-2); color: va  
| r(--text-color);">PID</th>  
117 |         <th class="text-end" style="background: var(--surf  
| ace-2); color: var(--text-color);">Управление</th>  
118 |     </tr>  
119 | </thead>  
120 |     <tbody id="svc-tbody">  
121 |         <tr><td colspan="6" class="text-center text-muted py  
| -4">Опрос диспетчера служб Windows...</td></tr>  
122 |     </tbody>  
123 | </table>  
124 | </div>  
125 | </div>  
126 | </div>  
127 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Services Manager Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом диспетчера служб
   Windows.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/services_manager_tab/main.js?v=20261006_v1" type="module"></script>
11  *
12  *   JavaScript Import:
13  *     import { initServicesManagerTab } from '/windows/api/webgui/services_manager_tab/main.js';
14  *
15  * File: main.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/services_manager_tab
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-06 11:24:00
21  * =====
22  */
23
24 const registerTabPoller = window.registerTabPoller || function
   () {};
25
26 let isInitialized = false;
27 let servicesList = [];
28 let currentFilter = 'all';
29
```

```
30 function escapeHtml(str) {
31   if (str === null || str === undefined) return '';
32   return String(str)
33     .replace(/&/g, '&amp;')
34     .replace(/</g, '&lt;')
35     .replace(/>/g, '&gt;')
36     .replace(/"/g, '&quot;')
37     .replace(/'/g, '&#039;');
38 }
39
40 export async function fetchServicesSummary() {
41   const refreshBtn = document.getElementById('svc-refresh-btn'
42   );
43   if (refreshBtn) {
44     const icon = refreshBtn.querySelector('i');
45     if (icon) icon.classList.add('spin-animation');
46   }
47   try {
48     const res = await fetch('/api/services-manager/summary');
49     if (!res.ok) throw new Error(`HTTP ${res.status}`);
50     const data = await res.json();
51     renderServices(data);
52   } catch (err) {
53     console.warn('[ServicesManager] Ошибка получения списка сл
54     ужб:', err);
55   } finally {
56     if (refreshBtn) {
57       const icon = refreshBtn.querySelector('i');
58       if (icon) icon.classList.remove('spin-animation');
59     }
60   }
61
62   function renderServices(data) {
63     servicesList = data.services || [];
64
65     const totalEl = document.getElementById('svc-total-count');
```

```

66 |   const runningEl = document.getElementById('svc-running-count
    | ');
67 |   const stoppedEl = document.getElementById('svc-stopped-count
    | ');
68 |   const autoEl = document.getElementById('svc-auto-count');
69 |
70 |   const runningCount = servicesList.filter(s => (s.state || ''
    | ).toLowerCase().includes('running') || s.state === '4')).length
    | ;
71 |   const stoppedCount = servicesList.filter(s => (s.state || ''
    | ).toLowerCase().includes('stopped') || s.state === '1')).length
    | ;
72 |   const autoCount = servicesList.filter(s => (s.start_type ||
    | '').toLowerCase().includes('auto')).length;
73 |
74 |   if (totalEl) totalEl.textContent = servicesList.length;
75 |   if (runningEl) runningEl.textContent = runningCount;
76 |   if (stoppedEl) stoppedEl.textContent = stoppedCount;
77 |   if (autoEl) autoEl.textContent = autoCount;
78 |
79 |   applyServiceFilters();
80 | }
81 |
82 | function applyServiceFilters() {
83 |   const tbody = document.getElementById('svc-tbody');
84 |   const searchInput = document.getElementById('svc-search-inpu
    | t');
85 |   const badge = document.getElementById('svc-table-badge');
86 |
87 |   if (!tbody) return;
88 |
89 |   const query = (searchInput ? searchInput.value : '').toLower
    | Case().trim();
90 |
91 |   const filtered = servicesList.filter(s => {
92 |     const isRunning = (s.state || '').toLowerCase().includes('
    | running') || s.state === '4';
93 |     if (currentFilter === 'running' && !isRunning) return fals

```

```

    e;
94     if (currentFilter === 'stopped' && isRunning) return false
    ;
95
96     if (query) {
97         const matchName = (s.name || '').toLowerCase().includes(
query);
98         const matchDisp = (s.display_name || '').toLowerCase().i
ncludes(query);
99         return matchName || matchDisp;
100     }
101     return true;
102 });
103
104     if (badge) badge.textContent = `${filtered.length} из ${serv
icesList.length} служб`;
105
106     if (filtered.length === 0) {
107         tbody.innerHTML = '<tr><td colspan="6" class="text-center
text-muted py-4">Службы не найдены</td></tr>';
108         return;
109     }
110
111     tbody.innerHTML = filtered.slice(0, 150).map(s => {
112         const isRunning = (s.state || '').toLowerCase().includes('
running') || s.state === '4';
113         const badgeClass = isRunning
114             ? 'badge bg-success-subtle text-success border border-su
ccess'
115             : 'badge bg-secondary-subtle text-secondary border borde
r-secondary';
116         const statusText = isRunning ? 'Running' : 'Stopped';
117
118         return `
119             <tr>
120                 <td class="fw-bold font-monospace" style="color: var(--
accent-color, #0ea5e9);">${escapeHtml(s.name)}</td>
121                 <td style="color: var(--text-color);">${escapeHtml(s.d

```



```

    display_name || s.name))</td>
122     <td>
123         <span class="${badgeClass}">
124             ${statusText}
125         </span>
126     </td>
127     <td class="small text-muted">${escapeHtml(s.start_type
    || 'Manual')}</td>
128     <td class="font-monospace small text-muted">${escapeHt
    ml(String(s.pid || '-'))}</td>
129     <td class="text-end">
130         <button class="btn btn-xs ${isRunning ? 'btn-outline
    -danger' : 'btn-outline-success'} py-0 px-2 svc-action-btn"
131             data-service="${escapeHtml(s.name)}" data-ac
    tion="${isRunning ? 'sc_stop' : 'sc_start'}">
132             <i class="bi ${isRunning ? 'bi-stop-fill' : 'bi-pl
    ay-fill'}"></i> ${isRunning ? 'Стон' : 'Старт'}
133         </button>
134     </td>
135 </tr>
136 `;
137 }).join('');
138
139 tbody.querySelectorAll('.svc-action-btn').forEach(btn => {
140     btn.addEventListener('click', () => {
141         const sName = btn.dataset.service;
142         const sAct = btn.dataset.action;
143         executeServiceCmd(sName, sAct);
144     });
145 });
146 }
147
148 async function executeServiceCmd(serviceName, action) {
149     if (window.showToast) window.showToast(`Команда отправлена:
    ${action} ${serviceName}`, 'info');
150     else if (window.toast) window.toast.info('Команда отправлена
    ', `${action}: ${serviceName}`);
151

```

```
152 |   try {
153 |     const res = await fetch('/api/services-manager/action', {
154 |       method: 'POST',
155 |       headers: { 'Content-Type': 'application/json' },
156 |       body: JSON.stringify({ service_name: serviceName, action
    : action, dry_run: false })
157 |     });
158 |     const result = await res.json();
159 |     if (res.ok) {
160 |       if (window.showToast) window.showToast(`☑  ${serviceName}
    : команда выполнена`, 'success');
161 |       else if (window.toast) window.toast.success('Успех', `${
    serviceName}: команда выполнена`);
162 |       fetchServicesSummary();
163 |     } else {
164 |       const errDetail = result.detail || result.message || 'Сб
    ой операции';
165 |       if (window.showToast) window.showToast(`Ошибка: ${errDet
    ail}`, 'danger');
166 |       else if (window.toast) window.toast.error('Ошибка', errD
    etail);
167 |     }
168 |   } catch (err) {
169 |     const errMsg = `Ошибка сети: ${err.message}`;
170 |     if (window.showToast) window.showToast(errMsg, 'danger');
171 |     else if (window.toast) window.toast.error('Ошибка сети', e
    rr.message);
172 |   }
173 | }
174 |
175 | export function initServicesManagerTab() {
176 |   if (isInitialized) return;
177 |   isInitialized = true;
178 |
179 |   const refreshBtn = document.getElementById('svc-refresh-btn'
    );
180 |   if (refreshBtn) refreshBtn.addEventListener('click', fetchSe
    rvicesSummary);
```

```
181
182   const searchInput = document.getElementById('svc-search-input');
183   if (searchInput) searchInput.addEventListener('input', applyServiceFilters);
184
185   const filterAll = document.getElementById('svc-filter-all');
186   const filterRunning = document.getElementById('svc-filter-running');
187   const filterStopped = document.getElementById('svc-filter-stopped');
188
189   [filterAll, filterRunning, filterStopped].forEach(btn => {
190     if (!btn) return;
191     btn.addEventListener('click', () => {
192       [filterAll, filterRunning, filterStopped].forEach(b => {
193         if (!b) return;
194         b.classList.remove('active', 'btn-primary');
195         b.classList.add('btn-outline-secondary');
196       });
197       btn.classList.add('active', 'btn-primary');
198       btn.classList.remove('btn-outline-secondary');
199       currentFilter = btn.id.replace('svc-filter-', '');
200       applyServiceFilters();
201     });
202   });
203
204   registerTabPoller('tab-services-manager', fetchServicesSummary, 10000, { immediate: true });
205 }
206
207 window.initServicesManagerTab = initServicesManagerTab;
208
209 // Автозапуск если вкладка уже в DOM
210 if (document.getElementById('svc-table') || document.getElementById('tab-services-manager')) {
211   if (document.readyState === 'loading') {
212     document.addEventListener('DOMContentLoaded', () => initSe
```

```
    rvicesManagerTab());  
213 | } else {  
214 |     setTimeout(() => initServicesManagerTab(), 10);  
215 | }  
216 | }  
217 |  
218 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/servicing_integrity_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/servicing_integrity_tab
17 /index.html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/servicing_integrity_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Servicing & Component Integrity Tab -->
30 <div class="container-fluid py-2 px-3">
31   <!-- Top Stat Cards -->
32   <div class="row g-2 mb-3">
33     <div class="col-6 col-md-3">
34       <div class="card bg-dark text-white border-secondary h-1
35 00 shadow-sm p-3">
36         <div class="text-muted small text-uppercase fw-semibol
37 d">Хранилище компонентов WinSxS □ </div>
```

```
31 |         <div class="fs-4 fw-bold mt-1 text-info" id="si-winsxs
    | -size">--</div>
32 |         <div class="small text-muted mt-1" id="si-winsxs-sub">
    | DISM AnalyzeComponentStore</div>
33 |     </div>
34 | </div>
35 |     <div class="col-6 col-md-3">
36 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
37 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Рекомендуется очистка □ </div>
38 |             <div class="fs-4 fw-bold mt-1 text-warning" id="si-cle
    | anup-recommended">--</div>
39 |             <div class="small text-muted mt-1" id="si-cleanup-sub"
    | >StartComponentCleanup</div>
40 |         </div>
41 |     </div>
42 |     <div class="col-6 col-md-3">
43 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
44 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Целостность файлов SFC □ </div>
45 |             <div class="fs-4 fw-bold mt-1 text-success" id="si-sfc
    | -status">OK</div>
46 |             <div class="small text-muted mt-1" id="si-sfc-sub">sfc
    | /scannow</div>
47 |         </div>
48 |     </div>
49 |     <div class="col-6 col-md-3">
50 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
51 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Состояние образа DISM □ </div>
52 |             <div class="fs-4 fw-bold mt-1 text-primary" id="si-dis
    | m-status">Healthy</div>
53 |             <div class="small text-muted mt-1" id="si-dism-sub">Ch
    | eckHealth / ScanHealth</div>
54 |         </div>
```

```

55     </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="si-refresh-btn">
62             <i class="bi bi-arrow-clockwise"></i>
63             <span>Обновить состояние</span>
64         </button>
65         <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5" id="si-dism-check-btn">
66             <i class="bi bi-heart-pulse"></i>
67             <span>DISM CheckHealth</span>
68         </button>
69         <button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5" id="si-sfc-scan-btn">
70             <i class="bi bi-shield-check"></i>
71             <span>SFC Scannow</span>
72         </button>
73         <button class="btn btn-sm btn-outline-danger d-flex align-items-center gap-1.5" id="si-winsxs-clean-btn">
74             <i class="bi bi-trash"></i>
75             <span>Очистить WinSxS</span>
76         </button>
77     </div>
78 </div>
79
80 <!-- Main Grid -->
81 <div class="row g-3">
82     <div class="col-12 col-lg-6">
83         <div class="card bg-dark text-white border-secondary shadow-sm mb-3">
84             <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">

```

```

85 |         <span class="fw-semibold"><i class="bi bi-tools text
    | -primary me-2"></i>Параметры обслуживания образа</span>
86 |     </div>
87 |     <div class="card-body p-3">
88 |         <div class="mb-3">
89 |             <label class="form-label small text-muted">Выберит
    | е операцию обслуживания:</label>
90 |             <select class="form-select form-select-sm bg-dark
    | text-white border-secondary" id="si-action-select">
91 |                 <option value="dism_check_health">DISM: Быстрая
    | проверка повреждений (/CheckHealth)</option>
92 |                 <option value="dism_scan_health">DISM: Полное ск
    | анирование целостности (/ScanHealth)</option>
93 |                 <option value="dism_restore_health">DISM: Восста
    | новление поврежденного образа (/RestoreHealth)</option>
94 |                 <option value="dism_analyze_component_store">DIS
    | M: Анализ размера хранилища WinSxS (/AnalyzeComponentStore)</o
    | ption>
95 |                 <option value="dism_start_component_cleanup">DIS
    | M: Очистка старых версий компонентов (/StartComponentCleanup)<
    | /option>
96 |                 <option value="dism_reset_base">DISM: Фиксация б
    | азы обновлений (/ResetBase)</option>
97 |                 <option value="sfc_scannow">SFC: Проверка и восс
    | тановление системных файлов (sfc /scannow)</option>
98 |                 <option value="sfc_verifyonly">SFC: Только прове
    |рка без восстановления (sfc /verifyonly)</option>
99 |             </select>
100 |         </div>
101 |         <div class="d-flex align-items-center justify-conten
    | t-between">
102 |             <div class="form-check form-switch">
103 |                 <input class="form-check-input" type="checkbox"
    | id="si-dry-run-switch" checked>
104 |                 <label class="form-check-label small text-muted"
    | for="si-dry-run-switch">Безопасный режим (Dry Run)</label>
105 |             </div>
106 |             <button class="btn btn-sm btn-success px-3" id="si

```



```

-execute-btn">
107         <i class="bi bi-play-fill"></i> Запустить операц
ию
108     </button>
109 </div>
110 </div>
111 </div>
112
113 <!-- Quick Info Card -->
114 <div class="card bg-dark text-white border-secondary sha
dow-sm">
115     <div class="card-header border-secondary py-2 px-3">
116         <span class="small fw-semibold text-info"><i class="
bi bi-info-circle me-1"></i>Справочная информация</span>
117     </div>
118     <div class="card-body p-3 small text-muted">
119         <p class="mb-2"><strong class="text-light">DISM</str
ong> (Deployment Image Servicing and Management) обслуживает к
омпоненты Windows и исправит ошибки хранилища манифестов.</p>
120         <p class="mb-0"><strong class="text-light">SFC</stro
ng> (System File Checker) сверяет контрольные суммы системных
dll и exe файлов с резервными копиями в WinSxS.</p>
121     </div>
122 </div>
123 </div>
124
125 <div class="col-12 col-lg-6">
126     <div class="card bg-dark text-white border-secondary sha
dow-sm h-100">
127         <div class="card-header border-secondary py-2 px-3 d-f
lex align-items-center justify-content-between">
128             <span class="small fw-semibold text-muted"><i class=
"bi bi-terminal me-1"></i>Журнал выполнения обслуживания</span>
>
129             <button class="btn btn-link btn-sm text-decoration-n
one text-muted p-0" id="si-clear-log-btn">Очистить</button>
130         </div>
131         <div class="card-body p-2 d-flex flex-column">

```

```
132 |         <pre class="mb-0 small bg-black text-light p-2 rounded flex-grow-1" id="si-console-output" style="min-height: 280px; max-height: 480px; overflow-y: auto; font-family: Consolas, monospace; font-size: 0.78rem;">Готов к диагностике компонентов.</pre>
133 |     </div>
134 | </div>
135 | </div>
136 | </div>
137 | </div>
138 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Servicing Integrity Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/servicing_integrity_ta
11  * b/main.js?v=20261001_v1" type="module"></script>
12  *
13  *   JavaScript Import:
14  *     import { initServicingIntegrityTab } from '/windows/api
15  * /webgui/servicing_integrity_tab/main.js';
16  *
17  * File: main.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/servicing_integrity_tab
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-04 11:15:30
23  * =====
24  */
25
26 /**
27  * servicing_integrity_tab/main.js – Обслуживание образов DISM
28  * и целостность SFC
29  * Updated: 2026-10-04 11:15:30
30  */
31
32 const registerTabPoller = window.registerTabPoller || function
33 () {};
```

```
30
31 let isInitialized = false;
32
33 export async function fetchServicingSummary() {
34   try {
35     const res = await fetch('/api/servicing-integrity/summary'
36   );
37     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38     const data = await res.json();
39     renderServicingSummary(data);
40   } catch (err) {
41     console.warn('[ServicingIntegrity] Ошибка получения сводки
42   :', err);
43   }
44 }
45
46 function renderServicingSummary(data) {
47   const winsxsSizeEl = document.getElementById('si-winsxs-size
48   ');
49   const cleanupRecommendedEl = document.getElementById('si-cle
50   anup-recommended');
51   const sfcStatusEl = document.getElementById('si-sfc-status')
52   ;
53   const dismStatusEl = document.getElementById('si-dism-status
54   ');
55
56   if (winsxsSizeEl) winsxsSizeEl.textContent = data.winsxs_siz
57   e || '8.4 GB';
58   if (cleanupRecommendedEl) cleanupRecommendedEl.textContent =
59   data.cleanup_recommended ? 'Да' : 'Нет';
60   if (sfcStatusEl) sfcStatusEl.textContent = data.sfc_status |
61   | 'OK';
62   if (dismStatusEl) dismStatusEl.textContent = data.dism_healt
63   h || 'Healthy';
64 }
65
66 async function executeServicingAction(customAction = null) {
67   const actionSelect = document.getElementById('si-action-sele
```

```

ct');
58   const dryRunSwitch = document.getElementById('si-dry-run-switch');
59   const output = document.getElementById('si-console-output');
60
61   const op = customAction || (actionSelect ? actionSelect.value : 'dism_check_health');
62   const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;
63
64   if (output) output.textContent = `[${new Date().toLocaleTimeString()}] Запуск операции обслуживания: ${op} (DryRun: ${dryRun})...\nПожалуйста, подождите...\n`;
65
66   try {
67     const res = await fetch('/api/servicing-integrity/action',
68     {
69       method: 'POST',
70       headers: { 'Content-Type': 'application/json' },
71       body: JSON.stringify({ action: op, dry_run: dryRun })
72     });
73     const result = await res.json();
74     if (output) output.textContent += JSON.stringify(result, null, 2);
75     if (window.toast) {
76       if (res.ok) window.toast.success('Операция завершена', `Статус: ${result.status} || 'OK'`);
77       else window.toast.error('Ошибка', result.detail || 'Сбой обслуживания');
78     }
79   } catch (err) {
80     if (output) output.textContent += `\nОшибка запроса: ${err.message}`;
81     if (window.toast) window.toast.error('Ошибка сети', err.message);
82   }
83
84 export function initServicingIntegrityTab() {

```

```

85 |   if (isInitialized) return;
86 |   isInitialized = true;
87 |
88 |   const refreshBtn = document.getElementById('si-refresh-btn')
89 |   ;
90 |   if (refreshBtn) refreshBtn.addEventListener('click', fetchSe
91 |   rvicingSummary);
92 |
93 |   const dismCheckBtn = document.getElementById('si-dism-check-
94 |   btn');
95 |   if (dismCheckBtn) dismCheckBtn.addEventListener('click', ()
96 |   => executeServicingAction('dism_check_health'));
97 |
98 |   const sfcScanBtn = document.getElementById('si-sfc-scan-btn'
99 |   );
100 |   if (sfcScanBtn) sfcScanBtn.addEventListener('click', () => e
101 |   xecuteServicingAction('sfc_scannow'));
102 |
103 |   const winsxsCleanBtn = document.getElementById('si-winsxs-cl
104 |   ean-btn');
105 |   if (winsxsCleanBtn) winsxsCleanBtn.addEventListener('click',
106 |   () => executeServicingAction('dism_start_component_cleanup'))
107 |   ;
108 |
109 |   const execBtn = document.getElementById('si-execute-btn');
110 |   if (execBtn) execBtn.addEventListener('click', () => execute
    ServicingAction());
111 |
112 |   const clearBtn = document.getElementById('si-clear-log-btn')
113 |   ;
114 |   if (clearBtn) clearBtn.addEventListener('click', () => {
115 |       const out = document.getElementById('si-console-output');
116 |       if (out) out.textContent = 'Журнал очищен.';
117 |   });
118 |
119 |   registerTabPoller('tab-servicing-integrity', fetchServicingS
120 |   ummary, 20000, { immediate: true });
121 | }

```

```
111 |  
112 | window.initServicingIntegrityTab = initServicingIntegrityTab;  
113 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/skills_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/skills_tab/index.html"
15 |        class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/skills_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Skills Tab Component -->
28 <div class="container-fluid p-3" id="skills-tab-root">
29 |<!-- Top Banner / Summary -->
30 |<div class="alert alert-dark border-secondary bg-dark text-l
31 |ight d-flex flex-wrap align-items-center justify-content-betwe
32 |en gap-3 shadow-sm">
33 |    <div class="d-flex align-items-center gap-3">
34 |        <div class="p-3 bg-black rounded border border-secondary
35 |text-warning display-6 d-flex align-items-center justify-cont
```



```

ent-center" style="width: 54px; height: 54px;">
30     <i class="bi bi-lightning-charge-fill"></i>
31     </div>
32     <div>
33         <h5 class="mb-1 text-white d-flex align-items-center gap-2">
34             □ Управление навыками (Agent Skills)
35             <span class="badge bg-primary rounded-pill small" id="skills-total-badge">0</span>
36         </h5>
37         <p class="mb-0 text-muted small">
38             Просмотр, создание, настройка и упаковка переиспользуемых навыков для AI-агентов (Antigravity, Gemini CLI, Breadboard).
39         </p>
40     </div>
41 </div>
42 <div class="d-flex align-items-center gap-2">
43     <button type="button" class="btn btn-outline-light btn-sm d-flex align-items-center gap-1" id="btn-refresh-skills" onclick="window.skillsTab?.refresh()">
44         <i class="bi bi-arrow-clockwise"></i> Обновить
45     </button>
46     <button type="button" class="btn btn-primary btn-sm d-flex align-items-center gap-1 shadow-sm" data-bs-toggle="modal" data-bs-target="#createSkillModal">
47         <i class="bi bi-plus-circle-fill"></i> Новый навык
48     </button>
49 </div>
50 </div>
51
52 <!-- Statistics Row -->
53 <div class="row g-2 mb-3" id="skills-stats-row">
54     <div class="col-6 col-md-3">
55         <div class="p-2 bg-dark rounded border border-secondary text-center">
56             <div class="h4 mb-0 text-white" id="stat-total-skills">0</div>

```

```

57 |         <div class="small text-muted">Всего навыков</div>
58 |     </div>
59 | </div>
60 |     <div class="col-6 col-md-3">
61 |         <div class="p-2 bg-dark rounded border border-secondary
text-center">
62 |             <div class="h4 mb-0 text-info" id="stat-scripts-skills
">0</div>
63 |             <div class="small text-muted">С исполняемыми скриптами
</div>
64 |         </div>
65 |     </div>
66 |     <div class="col-6 col-md-3">
67 |         <div class="p-2 bg-body-tertiary rounded border text-cen
ter">
68 |             <div class="h4 mb-0 text-warning" id="stat-refs-skills
">0</div>
69 |             <div class="small text-muted">Со справочниками</div>
70 |         </div>
71 |     </div>
72 |     <div class="col-6 col-md-3">
73 |         <div class="p-2 bg-body-tertiary rounded border text-cen
ter">
74 |             <div class="h4 mb-0 text-success" id="stat-packaged-sk
ills">0</div>
75 |             <div class="small text-muted">Скомпилировано (.skill)<
/div>
76 |         </div>
77 |     </div>
78 | </div>
79 |
80 | <!-- Search & View Controls -->
81 | <div class="card bg-body-tertiary border mb-3 shadow-sm">
82 |     <div class="card-body p-2 d-flex flex-wrap align-items-cen
ter justify-content-between gap-2">
83 |         <div class="input-group input-group-sm flex-grow-1" styl
e="max-width: 450px;">
84 |             <span class="input-group-text"><i class="bi bi-search"

```

```

></i></span>
85     <input type="text" class="form-control" id="skills-sea
rch-input" placeholder="Поиск по названию или описанию навыка.
..">
86     <button class="btn btn-outline-secondary" type="button
" id="skills-search-clear" onclick="window.skillsTab?.clearSea
rch()">🔍 </button>
87     </div>
88
89     <div class="d-flex align-items-center gap-2 flex-wrap">
90     <select class="form-select form-select-sm" id="skills-
filter-select" style="width: auto;" onchange="window.skillsTab
?.applyFilters()">
91         <option value="all">Все навыки</option>
92         <option value="scripts">Только со скриптами</option>
93         <option value="references">Только со справочниками</
option>
94         <option value="packaged">Собранные (.skill)</option>
95     </select>
96
97     <div class="btn-group btn-group-sm" role="group">
98         <button type="button" class="btn btn-outline-seconda
ry active" id="btn-view-cards" onclick="window.skillsTab?.setV
iewMode('cards')">
99             <i class="bi bi-grid-fill"></i> Карточки
100         </button>
101         <button type="button" class="btn btn-outline-seconda
ry" id="btn-view-table" onclick="window.skillsTab?.setViewMode
('table')">
102             <i class="bi bi-table"></i> Таблица
103         </button>
104     </div>
105 </div>
106 </div>
107 </div>
108
109 <!-- Loading & Empty Indicators -->
110 <div id="skills-loading" class="text-center py-5 text-muted

```

```

d-none">
111     <div class="spinner-border text-primary mb-2" role="status
"></div>
112     <div>Загрузка навыков...</div>
113 </div>
114
115 <div id="skills-empty" class="text-center py-5 text-muted d-
none">
116     <i class="bi bi-folder-x display-4 d-block mb-2 text-secon
dary"></i>
117     <h5>Навыки не найдены</h5>
118     <p class="small">Попробуйте изменить поисковый запрос или
создайте новый навык.</p>
119 </div>
120
121 <!-- Cards Container -->
122 <div class="row g-3" id="skills-cards-container"></div>
123
124 <!-- Table Container -->
125 <div class="card shadow-sm border d-none" id="skills-table-c
ard">
126     <div class="card-body p-0">
127         <div class="table-responsive">
128             <table class="table table-hover align-middle mb-0">
129                 <thead class="table-light text-uppercase small">
130                     <tr>
131                         <th scope="col" style="width: 25%;">Название нав
ыка</th>
132                         <th scope="col" style="width: 35%;">Описание</th>
133                         <th scope="col" style="width: 15%;">Расположение
</th>
134                         <th scope="col" style="width: 10%;">Компоненты</
th>
135                         <th scope="col" style="width: 15%; text-align: r
ight;">Действия</th>
136                     </tr>
137                 </thead>

```

```

138         <tbody id="skills-table-body"></tbody>
139     </table>
140 </div>
141 </div>
142 </div>
143 </div>
144
145 <!-- Modal: Create New Skill Wizard -->
146 <div class="modal fade" id="createSkillModal" tabindex="-1" ar
    ia-labelledby="createSkillModalLabel" aria-hidden="true">
147     <div class="modal-dialog modal-xl modal-dialog-centered moda
    l-dialog-scrollable">
148         <div class="modal-content bg-dark text-white border-second
    ary shadow-lg">
149             <div class="modal-header border-secondary py-2 px-3 bg-b
    lack bg-opacity-40">
150                 <div class="d-flex align-items-center gap-2">
151                     <span class="fs-5 text-warning">⚠ </span>
152                 </div>
153                 <h5 class="modal-title mb-0 fs-6 fw-bold" id="crea
    teSkillModalLabel">Мастер создания навыков (Skill Wizard)</h5>
154                 <small class="text-muted">Создавайте переиспользуе
    мые навыки для AI Breadboard, Gemini CLI и Antigravity</small>
155             </div>
156         </div>
157         <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="modal" aria-label="Close"></button>
158     </div>
159
160     <!-- Mode Selector Pills -->
161     <div class="px-3 pt-3 pb-2 bg-black bg-opacity-25 border
    -bottom border-secondary">
162         <ul class="nav nav-pills nav-fill gap-2" id="skillCrea
    teModes" role="tablist">
163             <li class="nav-item" role="presentation">
164                 <button class="nav-link active py-2 d-flex align-i
    tems-center justify-content-center gap-2" id="mode-wizard-tab"
    data-bs-toggle="pill" data-bs-target="#mode-wizard-pane" type

```

```

    ="button" role="tab" onclick="window.skillsTab?.switchCreateMo
de('wizard')">
165         <i class="bi bi-magic text-primary"></i> <strong>
    > ⚡ Пошаговый мастер</strong>
166     </button>
167 </li>
168     <li class="nav-item" role="presentation">
169         <button class="nav-link py-2 d-flex align-items-ce
nter justify-content-center gap-2" id="mode-ai-tab" data-bs-to
ogle="pill" data-bs-target="#mode-ai-pane" type="button" role=
"tab" onclick="window.skillsTab?.switchCreateMode('ai')">
170             <i class="bi bi-robot text-info"></i> <strong>⚡
    AI-Ассистент генерации</strong>
171         </button>
172     </li>
173     <li class="nav-item" role="presentation">
174         <button class="nav-link py-2 d-flex align-items-ce
nter justify-content-center gap-2" id="mode-presets-tab" data-
bs-toggle="pill" data-bs-target="#mode-presets-pane" type="but
ton" role="tab" onclick="window.skillsTab?.switchCreateMode('p
resets')">
175             <i class="bi bi-collection text-warning"></i> <s
trong>⚡ Готовые шаблоны</strong>
176         </button>
177     </li>
178 </ul>
179 </div>
180
181     <div class="modal-body p-3">
182         <div class="tab-content" id="skillCreateModesContent">
183
184             <!-- MODE 1: STEP-BY-STEP WIZARD -->
185             <div class="tab-pane fade show active" id="mode-wiza
rd-pane" role="tabpanel">
186                 <!-- Steps Stepper Header -->
187                 <div class="d-flex justify-content-between align-i
tems-center mb-3 px-2">
188                     <div class="d-flex align-items-center gap-2 curs

```

```
or-pointer" onclick="window.skillsTab?.goToStep(1)">
189         <span class="badge rounded-circle p-2 bg-prim
ry step-indicator" id="step-badge-1">1</span>
190         <span class="small fw-semibold" id="step-label
-1">Основное</span>
191     </div>
192     <div class="flex-grow-1 mx-2 border-top border-s
econdary"></div>
193     <div class="d-flex align-items-center gap-2 curs
or-pointer" onclick="window.skillsTab?.goToStep(2)">
194         <span class="badge rounded-circle p-2 bg-secon
dary step-indicator" id="step-badge-2">2</span>
195         <span class="small text-muted" id="step-label-
2">Инструкции</span>
196     </div>
197     <div class="flex-grow-1 mx-2 border-top border-s
econdary"></div>
198     <div class="d-flex align-items-center gap-2 curs
or-pointer" onclick="window.skillsTab?.goToStep(3)">
199         <span class="badge rounded-circle p-2 bg-secon
dary step-indicator" id="step-badge-3">3</span>
200         <span class="small text-muted" id="step-label-
3">Скрипты и файлы</span>
201     </div>
202     <div class="flex-grow-1 mx-2 border-top border-s
econdary"></div>
203     <div class="d-flex align-items-center gap-2 curs
or-pointer" onclick="window.skillsTab?.goToStep(4)">
204         <span class="badge rounded-circle p-2 bg-secon
dary step-indicator" id="step-badge-4">4</span>
205         <span class="small text-muted" id="step-label-
4">Финализация</span>
206     </div>
207 </div>
208
209 <!-- Wizard Step 1: Basic Info -->
210 <div id="wizard-step-1" class="wizard-step">
211     <div class="card bg-black bg-opacity-50 border-s
```

```

secondary mb-3">
212         <div class="card-body p-3">
213             <h6 class="text-primary mb-3"><i class="bi bi
i-info-circle"></i> Шаг 1: Название, назначение и расположение
</h6>
214
215             <div class="row g-3">
216                 <div class="col-12 col-md-6">
217                     <label for="wizard-skill-name" class="fo
rm-label small fw-semibold">
218                         Идентификатор навыка (kebab-case) <spa
n class="text-danger">*</span>
219                     </label>
220                     <div class="input-group input-group-sm">
221                         <span class="input-group-text bg-dark
border-secondary text-warning">□ </span>
222                         <input type="text" class="form-control
bg-dark text-white border-secondary" id="wizard-skill-name" p
laceholder="напр. media-downloader или log-parser" required>
223                         <button class="btn btn-outline-seconda
ry" type="button" title="Сгенерировать имя из описания" onclik
k="window.skillsTab?.generateSlugFromName()">
224                             <i class="bi bi-magic"></i>
225                         </button>
226                     </div>
227                     <small class="text-muted">Латиница, цифр
ы, дефис и подчеркивания (используется агентом для вызова).</s
mall>
228                 </div>
229
230                 <div class="col-12 col-md-6">
231                     <label for="wizard-skill-dir" class="for
m-label small fw-semibold">Целевой каталог</label>
232                     <select class="form-select form-select-s
m bg-dark text-white border-secondary" id="wizard-skill-dir">
233                         <option value=".agents/skills" selecte
d>.agents/skills (Проектный навык AI Breadboard)</option>
234                         <option value="~/gemini/skills">~/ge

```



```

mini/skills (Пользовательский – Gemini CLI / Antigravity)</opt
ion>
235         <option value="~/agents/skills">~/agents/skills (Глобальный пользовательский навык)</option>
236         <option value="skills">skills (Локальный каталог)</option>
237     </select>
238     <small class="text-muted">Определяет область видимости навыка в системе.</small>
239 </div>
240
241     <div class="col-12 col-md-6">
242         <label for="wizard-skill-category" class="form-label small fw-semibold">Категория</label>
243         <select class="form-select form-select-sm bg-dark text-white border-secondary" id="wizard-skill-category" onchange="window.skillsTab?.onCategoryChanged()">
244             <option value="tools">☐ Инструмент и автоматизация (CLI / Script)</option>
245             <option value="api">☐ Интеграция с API и сервисами</option>
246             <option value="data">☐ Анализ данных и парсинг</option>
247             <option value="persona">☐ AI-Персона / Специализированный эксперт</option>
248             <option value="monitoring">☐ Мониторинг и нотификации</option>
249             <option value="docgen">☐ Генератор документов и отчетов</option>
250             <option value="general" selected>☐ Общего назначения</option>
251         </select>
252     </div>
253
254     <div class="col-12 col-md-6">
255         <label for="wizard-skill-desc-ru" class="form-label small fw-semibold">Назначение на русском</label>
256         <input type="text" class="form-control f

```

```

orm-control-sm bg-dark text-white border-secondary" id="wizard
-skill-desc-ru" placeholder="Понятное описание для пользовател
я...">
257         </div>
258
259         <div class="col-12">
260             <label for="wizard-skill-desc-en" class=
"form-label small fw-semibold">Описание для AI-агента (English
description)</label>
261             <input type="text" class="form-control f
orm-control-sm bg-dark text-white border-secondary" id="wizard
-skill-desc-en" placeholder="Concise description in English us
ed by agent router...">
262             <small class="text-muted">Агент использу
ет это описание при маршрутизации и выборе навыка.</small>
263         </div>
264     </div>
265 </div>
266 </div>
267 </div>
268
269 <!-- Wizard Step 2: Instructions (SKILL.md) -->
270 <div id="wizard-step-2" class="wizard-step d-none"
>
271     <div class="card bg-black bg-opacity-50 border-s
econdary mb-3">
272         <div class="card-body p-3">
273             <div class="d-flex justify-content-between a
lign-items-center mb-2">
274                 <h6 class="text-primary mb-0"><i class="bi
bi-file-text"></i> Шаг 2: Структура и логика инструкций (SKIL
L.md)</h6>
275                 <button type="button" class="btn btn-outli
ne-info btn-sm" onclick="window.skillsTab?.fillStandardInstruc
tions()">
276                     <i class="bi bi-arrow-repeat"></i> Запол
нить стандартным шаблоном
277                 </button>

```

```
278 |         </div>
279 |
280 |         <div class="row g-2 mb-3">
281 |             <div class="col-12 col-md-6">
282 |                 <label for="wizard-inst-purpose" class="
form-label small fw-semibold">□ Цель (Purpose):</label>
283 |                 <textarea class="form-control form-contr
ol-sm bg-dark text-white border-secondary" id="wizard-inst-pur
pose" rows="2" placeholder="Опишите главную цель работы навыка
..."></textarea>
284 |             </div>
285 |             <div class="col-12 col-md-6">
286 |                 <label for="wizard-inst-triggers" class="
"form-label small fw-semibold">□ Условия запуска (Triggers):<
/label>
287 |                 <textarea class="form-control form-contr
ol-sm bg-dark text-white border-secondary" id="wizard-inst-tri
ggers" rows="2" placeholder="Когда агент должен активировать э
тот навык..."></textarea>
288 |             </div>
289 |             <div class="col-12 col-md-6">
290 |                 <label for="wizard-inst-protocol" class="
"form-label small fw-semibold">□ Протокол действий (Execution
Protocol):</label>
291 |                 <textarea class="form-control form-contr
ol-sm bg-dark text-white border-secondary" id="wizard-inst-pro
tocol" rows="4" placeholder="1. Проверить входные данные&#10;2
. Запустить скрипт или обработку&#10;3. Сформировать ответ"></
textarea>
292 |             </div>
293 |             <div class="col-12 col-md-6">
294 |                 <label for="wizard-inst-rules" class="fo
rm-label small fw-semibold">□ Правила и ограничения (Rules & C
onstraints):</label>
295 |                 <textarea class="form-control form-contr
ol-sm bg-dark text-white border-secondary" id="wizard-inst-rul
es" rows="4" placeholder="- Обработка ошибок без падений&#1
0;- Возвращать четкий статус выполнения"></textarea>
```

```

296         </div>
297     </div>
298
299     <label for="wizard-skill-instructions" class
="form-label small fw-semibold">Полный текст Markdown для SKIL
L.md:</label>
300     <textarea class="form-control bg-dark text-w
hite border-secondary font-monospace" id="wizard-skill-instruc
tions" rows="6" style="font-size: 0.85rem;" placeholder="Автом
атически генерируется из блоков выше..."></textarea>
301     </div>
302 </div>
303 </div>
304
305 <!-- Wizard Step 3: Scripts & Files -->
306 <div id="wizard-step-3" class="wizard-step d-none"
>
307     <div class="card bg-black bg-opacity-50 border-s
econdary mb-3">
308         <div class="card-body p-3">
309             <h6 class="text-primary mb-3"><i class="bi b
i-folder-plus"></i> Шаг 3: Дополнительные скрипты, документац
ия и справочники</h6>
310
311             <div class="row g-3">
312                 <div class="col-12">
313                     <div class="form-check form-switch mb-2"
>
314                         <input class="form-check-input" type="
checkbox" id="wizard-enable-script" onchange="window.skillsTab
?.toggleScriptEditor()">
315                         <label class="form-check-label fw-semi
bold small" for="wizard-enable-script">
316                             Добавить исполняемый Python-скрипт (
<code>scripts/main.py</code>)
317                         </label>
318                     </div>
319                     <div id="wizard-script-container" class=

```

```

320 |         <textarea class="form-control bg-dark
| text-info border-secondary font-monospace" id="wizard-script-c
| ode" rows="6" style="font-size: 0.85rem;" spellcheck="false"><
| /textarea>
321 |             </div>
322 |         </div>
323 |
324 |         <div class="col-12">
325 |             <div class="form-check form-switch mb-2"
| >
326 |                 <input class="form-check-input" type="
| checkbox" id="wizard-enable-ref" onchange="window.skillsTab?.t
| oggleRefEditor()">
327 |                 <label class="form-check-label fw-semi
| bold small" for="wizard-enable-ref">
328 |                     Добавить справочный документ (<code>
| references/guide.md</code>)
329 |                 </label>
330 |             </div>
331 |             <div id="wizard-ref-container" class="d-
| none mt-2">
332 |                 <textarea class="form-control bg-dark
| text-light border-secondary font-monospace" id="wizard-ref-con
| tent" rows="4" style="font-size: 0.85rem;" spellcheck="false">
| </textarea>
333 |             </div>
334 |         </div>
335 |
336 |         <div class="col-12">
337 |             <label for="wizard-skill-readme" class="
| form-label small fw-semibold">Документация <code>README.md</co
| de> (на английском языке):</label>
338 |             <textarea class="form-control bg-dark te
| xt-light border-secondary font-monospace" id="wizard-skill-rea
| dme" rows="4" style="font-size: 0.85rem;" spellcheck="false"><
| /textarea>
339 |         </div>

```

```

340         </div>
341     </div>
342 </div>
343 </div>
344
345     <!-- Wizard Step 4: Finalize & Preview -->
346     <div id="wizard-step-4" class="wizard-step d-none"
347 >
348         <div class="card bg-black bg-opacity-50 border-s
349 econdary mb-3">
350             <div class="card-body p-3">
351                 <h6 class="text-success mb-3"><i class="bi b
352 i-check2-circle"></i> Шаг 4: Предпросмотр создаваемого навыка<
353 /h6>
354
355                 <div class="row g-2 mb-3">
356                     <div class="col-12 col-md-4">
357                         <div class="p-2 bg-dark rounded border b
358 order-secondary">
359                             <span class="text-muted small d-block"
360 >Имя:</span>
361                             <strong class="text-warning font-monos
362 pace" id="preview-skill-name">—</strong>
363                         </div>
364                     </div>
365                     <div class="col-12 col-md-4">
366                         <div class="p-2 bg-dark rounded border b
367 order-secondary">
368                             <span class="text-muted small d-block"
369 >Путь:</span>
370                             <code class="text-info" id="preview-sk
371 ill-path">—</code>
372                         </div>
373                     </div>
374                     <div class="col-12 col-md-4">
375                         <div class="p-2 bg-dark rounded border b
376 order-secondary">
377                             <span class="text-muted small d-block"

```

```

|>Компоненты:</span>
367|         <span class="badge bg-primary me-1">SK
|ILL.md</span>
368|         <span class="badge bg-secondary me-1">
|README.md</span>
369|         <span class="badge bg-info d-none" id=
|"preview-badge-script">scripts/</span>
370|         <span class="badge bg-warning d-none t
|ext-dark" id="preview-badge-ref">references/</span>
371|         </div>
372|     </div>
373| </div>
374|
375|     <label class="form-label small fw-semibold">
|Результирующий <code>SKILL.md</code> (Frontmatter + Markdown):
|</label>
376|     <pre class="bg-dark p-3 rounded border borde
|r-secondary text-info font-monospace" id="preview-skillmd-full
|" style="max-height: 220px; overflow-y: auto; font-size: 0.85r
|em;"></pre>
377|         </div>
378|     </div>
379| </div>
380|
381| <!-- Wizard Navigation Bar -->
382| <div class="d-flex justify-content-between align-i
|tems-center mt-3 pt-2 border-top border-secondary">
383|     <button type="button" class="btn btn-outline-sec
|ondary btn-sm" id="btn-wizard-prev" onclick="window.skillsTab?
|.wizardPrevStep()" disabled>
384|         <i class="bi bi-arrow-left"></i> Назад
385|     </button>
386|
387|     <div class="d-flex gap-2">
388|         <button type="button" class="btn btn-primary b
|tn-sm" id="btn-wizard-next" onclick="window.skillsTab?.wizardN
|extStep()">
389|             Далее <i class="bi bi-arrow-right"></i>

```

```

390         </button>
391         <button type="button" class="btn btn-success b
tn-sm d-none" id="btn-wizard-submit" onclick="window.skillsTab
?.createSkillFromWizard()">
392             <span class="spinner-border spinner-border-s
m d-none" id="wizard-submit-spinner" role="status"></span>
393             <span><i class="bi bi-rocket-takeoff-fill"><
/i> Создать навык</span>
394         </button>
395     </div>
396 </div>
397 </div>
398
399 <!-- MODE 2: AI ASSISTANT GENERATION -->
400 <div class="tab-pane fade" id="mode-ai-pane" role="t
abpanel">
401     <div class="card bg-black bg-opacity-50 border-sec
ondary mb-3">
402         <div class="card-body p-3">
403             <div class="d-flex align-items-center gap-2 mb
-2 text-info">
404                 <i class="bi bi-robot fs-5"></i>
405                 <h6 class="mb-0">Генерация навыка с помощью
ИИ (Natural Language to Skill)</h6>
406             </div>
407             <p class="text-muted small mb-3">
408                 Просто опишите задачу или идею навыка обычны
ми словами на русском или английском. ИИ автоматически сформиру
ет имя, структуру инструкций, описание, контракт и вспомога
тельные скрипты.
409             </p>
410
411             <div class="mb-3">
412                 <label for="ai-skill-prompt" class="form-lab
el small fw-semibold">Описание задачи навыка <span class="text
-danger">*</span></label>
413                 <textarea class="form-control bg-dark text-w
hite border-secondary" id="ai-skill-prompt" rows="3" placehold

```



```
er="например: Навык для автоматического анализа логов сервера
Nginx, выявления подозрительных IP-адресов и выгрузки отчета в
формате CSV"></textarea>
```

```
414         </div>
415
416         <div class="row g-2 mb-3">
417             <div class="col-12 col-md-6">
418                 <label for="ai-skill-category" class="form-
-label small fw-semibold">Категория</label>
419                 <select class="form-select form-select-sm
bg-dark text-white border-secondary" id="ai-skill-category">
420                     <option value="tools">☐ Инструмент и ск
рипт</option>
421                     <option value="api">☐ API и интеграция<
/option>
422                     <option value="data">☐ Анализ данных</o
ption>
423                     <option value="persona">☐ AI-Эксперт</o
ption>
424                     <option value="monitoring">☐ Мониторинг
</option>
425                     <option value="general" selected>☐ Общий
навык</option>
426                 </select>
427             </div>
428             <div class="col-12 col-md-6">
429                 <label for="ai-skill-dir" class="form-labe
l small fw-semibold">Куда сохранить</label>
430                 <select class="form-select form-select-sm
bg-dark text-white border-secondary" id="ai-skill-dir">
431                     <option value=".agents/skills" selected>
.agents/skills (Проектный)</option>
432                     <option value="~/ .gemini/skills">~/ .gemi
ni/skills (Gemini / Antigravity)</option>
433                     <option value="~/ .agents/skills">~/ .agen
ts/skills (Глобальный)</option>
434                 </select>
435             </div>
```

```
436 |         </div>
437 |
438 |         <button type="button" class="btn btn-info btn-
      sm d-flex align-items-center gap-2" id="btn-generate-ai" oncli
      ck="window.skillsTab?.generateSkillWithAi()">
439 |             <span class="spinner-border spinner-border-s
      m d-none" id="ai-gen-spinner" role="status"></span>
440 |             <i class="bi bi-sparkles"></i> <strong>Сгене
      рировать навык</strong>
441 |         </button>
442 |
443 |         <!-- AI Result Box -->
444 |         <div id="ai-result-box" class="d-none mt-3 pt-
      3 border-top border-secondary">
445 |             <div class="alert alert-success py-2 d-flex
      align-items-center justify-content-between">
446 |                 <span><i class="bi bi-check-circle-fill me
      -1"></i> Навык успешно сгенерирован!</span>
447 |                 <span class="badge bg-dark font-monospace"
      id="ai-res-slug"></span>
448 |             </div>
449 |
450 |             <div class="mb-2">
451 |                 <label class="form-label small fw-semibold
      text-muted">Сгенерированные инструкции (SKILL.md):</label>
452 |                 <pre class="bg-dark p-3 rounded border bor
      der-secondary text-info font-monospace" id="ai-res-preview" st
      yle="max-height: 180px; overflow-y: auto; font-size: 0.85rem;"
      ></pre>
453 |             </div>
454 |
455 |             <div class="d-flex justify-content-end gap-2
      ">
456 |                 <button type="button" class="btn btn-outli
      ne-primary btn-sm" onclick="window.skillsTab?.applyAiResultToW
      izard()">
457 |                     <i class="bi bi-sliders"></i> Настроить
      в мастере
```

```

458         </button>
459         <button type="button" class="btn btn-succe
ss btn-sm" onclick="window.skillsTab?.saveAiSkillDirectly()">
460             <i class="bi bi-plus-circle"></i> Создат
ь сразу
461         </button>
462     </div>
463 </div>
464 </div>
465 </div>
466 </div>
467
468 <!-- MODE 3: PRESETS LIBRARY -->
469 <div class="tab-pane fade" id="mode-presets-pane" ro
le="tabpanel">
470     <div class="row g-3" id="presets-cards-container">
471         <!-- Preset Cards populated dynamically -->
472     </div>
473 </div>
474
475 </div>
476
477 <div id="create-skill-alert" class="alert d-none py-2
mt-3 mb-0"></div>
478 </div>
479
480 <div class="modal-footer border-secondary py-2 px-3 bg-b
lack bg-opacity-40">
481     <button type="button" class="btn btn-outline-secondary
btn-sm" data-bs-dismiss="modal">Закрыть</button>
482 </div>
483 </div>
484 </div>
485 </div>
486
487 <!-- Modal / Inspector: View, Edit & Live Test Skill -->
488 <div class="modal fade" id="skillDetailsModal" tabindex="-1" a
ria-labelledby="skillDetailsModalLabel" aria-hidden="true">

```

```
489 | <div class="modal-dialog modal-xl modal-dialog-scrollable">
490 |   <div class="modal-content bg-dark text-white border-second
    | ary">
491 |     <div class="modal-header border-secondary d-flex justify
    | -content-between align-items-center">
492 |       <div>
493 |         <h5 class="modal-title d-flex align-items-center gap
    | -2 mb-0" id="skillDetailsModalLabel">
494 |           <i class="bi bi-lightning-charge-fill text-warning
    | "></i>
495 |             <span id="detail-skill-title">Навык</span>
496 |           </h5>
497 |           <small class="text-muted" id="detail-skill-path">—</
    | small>
498 |         </div>
499 |         <div class="d-flex align-items-center gap-2">
500 |           <button type="button" class="btn btn-outline-info bt
    | n-sm d-flex align-items-center gap-1" id="btn-package-modal" o
    | nclick="window.skillsTab?.packageCurrentSkill()">
501 |             <i class="bi bi-box-seam"></i> Собрать .skill
502 |           </button>
503 |           <button type="button" class="btn btn-outline-danger
    | btn-sm d-flex align-items-center gap-1" id="btn-delete-modal"
    | onclick="window.skillsTab?.deleteCurrentSkill()">
504 |             <i class="bi bi-trash"></i> Удалить
505 |           </button>
506 |           <button type="button" class="btn-close btn-close-whi
    | te" data-bs-dismiss="modal" aria-label="Close"></button>
507 |         </div>
508 |       </div>
509 |     <div class="modal-body p-3">
510 |       <!-- Navigation tabs in inspector -->
511 |       <ul class="nav nav-pills mb-3" id="skillDetailTabs" ro
    | le="tablist">
512 |         <li class="nav-item" role="presentation">
513 |           <button class="nav-link active" id="detail-tab-ski
    | llmd" data-bs-toggle="pill" data-bs-target="#detail-pane-skill
    | md" type="button" role="tab">
```

```

514 |         □ SKILL.md (Инструкции)
515 |     </button>
516 | </li>
517 |     <li class="nav-item" role="presentation">
518 |         <button class="nav-link" id="detail-tab-sandbox" data-
    | ata-bs-toggle="pill" data-bs-target="#detail-pane-sandbox" type=
    | e="button" role="tab">
519 |             □ Тестирование (Sandbox)
520 |         </button>
521 |     </li>
522 |     <li class="nav-item" role="presentation">
523 |         <button class="nav-link" id="detail-tab-readme" data-
    | ta-bs-toggle="pill" data-bs-target="#detail-pane-readme" type=
    | "button" role="tab">
524 |             □ README.md
525 |         </button>
526 |     </li>
527 |     <li class="nav-item" role="presentation">
528 |         <button class="nav-link" id="detail-tab-files" data-
    | a-bs-toggle="pill" data-bs-target="#detail-pane-files" type="b
    | utton" role="tab">
529 |             □ Файлы и структура
530 |         </button>
531 |     </li>
532 |     <li class="nav-item" role="presentation">
533 |         <button class="nav-link" id="detail-tab-contract"
    | data-bs-toggle="pill" data-bs-target="#detail-pane-contract" t
    | ype="button" role="tab">
534 |             □ JSON Contract / Метаданные
535 |         </button>
536 |     </li>
537 | </ul>
538 |
539 | <div class="tab-content" id="skillDetailTabContent">
540 |     <!-- SKILL.md Tab Pane -->
541 |     <div class="tab-pane fade show active" id="detail-pa
    | ne-skillmd" role="tabpanel">
542 |         <div class="row g-2 mb-3">

```

```
543 |         <div class="col-12 col-md-6">
544 |             <label for="detail-skill-desc-input" class="fo
| rm-label fw-semibold small text-muted">English Description (Fr
| ontmatter):</label>
545 |             <input type="text" class="form-control bg-blac
| k text-white border-secondary form-control-sm" id="detail-skil
| l-desc-input">
546 |         </div>
547 |         <div class="col-12 col-md-6">
548 |             <label for="detail-skill-desc-ru-input" class=
| "form-label fw-semibold small text-muted">Описание на русском
| (RU):</label>
549 |             <input type="text" class="form-control bg-blac
| k text-white border-secondary form-control-sm" id="detail-skil
| l-desc-ru-input">
550 |         </div>
551 |     </div>
552 |     <div class="mb-2 d-flex justify-content-between al
| ign-items-center">
553 |         <label for="detail-skill-instructions" class="fo
| rm-label fw-semibold small text-muted mb-0">Тело инструкции (M
| arkdown):</label>
554 |         <small class="text-muted">Редактируемый текст</s
| mall>
555 |     </div>
556 |     <textarea class="form-control bg-black text-white
| border-secondary font-monospace" id="detail-skill-instructions
| " rows="16" style="font-size: 0.9rem;" spellcheck="false"></te
| xtaarea>
557 | </div>
558 |
559 |     <!-- Live Sandbox Tester Tab Pane -->
560 |     <div class="tab-pane fade" id="detail-pane-sandbox"
| role="tabpanel">
561 |         <div class="p-3 bg-black rounded border border-sec
| ondary mb-3">
562 |             <h6 class="text-warning mb-2"><i class="bi bi-te
| rminal"></i> Интерактивная песочница для тестирования навыка</
```

```

h6>
563         <p class="text-muted small mb-3">
564             Введите тестовый запрос, чтобы проверить, как
агент интерпретирует инструкции данного навыка и выполняет зап
рос.
565         </p>
566         <div class="mb-3">
567             <label for="sandbox-test-prompt" class="form-l
abel small fw-semibold">Тестовый запрос агенту:</label>
568             <div class="input-group">
569                 <input type="text" class="form-control bg-da
rk text-white border-secondary" id="sandbox-test-prompt" place
holder="например: Выполни анализ данных или проверь состояние.
..">
570                 <button class="btn btn-warning text-dark fw-
bold d-flex align-items-center gap-1" id="btn-run-sandbox" onc
lick="window.skillsTab?.runSandboxTest()">
571                     <span class="spinner-border spinner-border
-sm d-none" id="sandbox-spinner" role="status"></span>
572                     <i class="bi bi-play-fill"></i> Запустить
573                 </button>
574             </div>
575         </div>
576
577         <label class="form-label small fw-semibold text-
muted">Результат выполнения (Output):</label>
578         <div class="p-3 bg-dark rounded border border-se
condary text-light font-monospace" id="sandbox-output-box" sty
le="min-height: 140px; max-height: 320px; overflow-y: auto; wh
ite-space: pre-wrap; font-size: 0.85rem;">
579 Готов к тестированию. Введите запрос и нажмите «Запустить».
580         </div>
581     </div>
582 </div>
583
584     <!-- README.md Tab Pane -->
585     <div class="tab-pane fade" id="detail-pane-readme" r
ole="tabpanel">

```

```
586 |         <div class="mb-2 d-flex justify-content-between align-items-center">
587 |             <label for="detail-skill-readme" class="form-label fw-semibold small text-muted mb-0">README.md документация:</label>
588 |             <small class="text-muted">Англоязычная документация пакета</small>
589 |         </div>
590 |         <textarea class="form-control bg-black text-white border-secondary font-monospace" id="detail-skill-readme" rows="18" style="font-size: 0.9rem;" spellcheck="false"></textarea>
591 |     </div>
592 |
593 |     <!-- Files Structure Tab Pane -->
594 |     <div class="tab-pane fade" id="detail-pane-files" role="tabpanel">
595 |         <div class="p-3 bg-black rounded border border-secondary">
596 |             <h6 class="text-info mb-2"><i class="bi bi-folder2-open"></i> Содержимое папки навыка:</h6>
597 |             <div id="detail-files-list" class="list-group list-group-flush bg-transparent"></div>
598 |         </div>
599 |     </div>
600 |
601 |     <!-- JSON Contract Tab Pane -->
602 |     <div class="tab-pane fade" id="detail-pane-contract" role="tabpanel">
603 |         <pre class="bg-black p-3 rounded border border-secondary text-info mb-0" id="detail-json-contract" style="font-size: 0.85rem; max-height: 400px; overflow-y: auto;"></pre>
604 |     </div>
605 | </div>
606 |
607 | <div id="detail-skill-alert" class="alert d-none py-2 mt-3 mb-0"></div>
608 | </div>
```



```
609 |         <div class="modal-footer border-secondary d-flex justify
      | -content-between">
610 |             <button type="button" class="btn btn-outline-secondary
      | " data-bs-dismiss="modal">Заккрыть</button>
611 |             <button type="button" class="btn btn-primary d-flex al
      | ign-items-center gap-1" id="btn-save-skill" onclick="window.sk
      | illsTab?.saveCurrentSkill()">
612 |                 <span class="spinner-border spinner-border-sm d-none
      | " id="save-skill-spinner" role="status"></span>
613 |                 <span><i class="bi bi-save"></i> Сохранить изменения
      | </span>
614 |             </button>
615 |         </div>
616 |     </div>
617 | </div>
618 | </div>
619 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Skills Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/skills_tab/main.js?v=2
11  *     0261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/skills_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Skills Tab Logic for AI Breadboard Admin Interface
22 // Supports 3 Creation Modes (Step-by-Step Wizard, AI Assistant, Presets Library) and Live Sandbox Testing
23
24 class SkillsTabManager {
25   constructor() {
26     this.skills = [];
27     this.filteredSkills = [];
28     this.currentSkill = null;
29     this.viewMode = 'cards'; // 'cards' | 'table'
30     this.searchQuery = '';
31     this.activeFilter = 'all'; // 'all' | 'scripts' | 'referen
```

```

ces' | 'packaged'
32     this.isInitialized = false;
33
34     // Wizard state
35     this.currentWizardStep = 1;
36     this.createMode = 'wizard'; // 'wizard' | 'ai' | 'presets'
37     this.aiGeneratedData = null;
38
39     // Presets catalog
40     this.presets = [
41         {
42             id: 'tool-automation',
43             title: '☐ CLI & Python Tool',
44             category: 'tools',
45             badge: 'Automation',
46             description: 'Исполняемый навык с Python-скриптом авто-
матизации в папке scripts/. Подходит для утилит, конвертеров и
системных задач.',
47             name: 'custom-cli-tool',
48             descRu: 'Автоматизированная CLI утилита для выполнения
задач через скрипты.',
49             descEn: 'Automated CLI utility providing executable sc
ripts for system routines.',
50             purpose: 'Execute specialized automation commands via
internal scripts.',
51             triggers: 'Triggered when the user requests executing
a custom tool or batch script task.',
52             protocol: '1. Validate arguments and environment.\n2.
Execute python script in `scripts/main.py`.\n3. Return formatt
ed stdout/stderr status.',
53             rules: '- Ensure non-zero exit codes are logged and re
ported.\n- Maintain English logging.',
54             hasScript: true,
55             scriptCode: `# -*- coding: utf-8 -*-\n"""\nCLI tool auto
mation script.\n\nimport sys\n\ndef main():\n    print("Execu
ting automated tool task...")\n    return 0\n\nif __name__ ==
"__main__":\n    sys.exit(main())\n`,
56             hasRef: false,

```

```

57 |     },
58 |     {
59 |         id: 'api-connector',
60 |         title: '⬢ REST API Connector',
61 |         category: 'api',
62 |         badge: 'Integration',
63 |         description: 'Интеграция с внешним REST/JSON API. Шабл
он с проверкой авторизации, обработкой таймаутов и структуриро
ванием ответов.',
64 |         name: 'rest-api-connector',
65 |         descRu: 'Интеграция с внешним сервисом через REST API.
',
66 |         descEn: 'Connector skill for external REST API service
s and endpoints.',
67 |         purpose: 'Query external HTTP REST API endpoints and p
rocess JSON payloads.',
68 |         triggers: 'Triggered when user asks to fetch data from
or post payloads to remote APIs.',
69 |         protocol: '1. Parse endpoint parameters and auth token
s.\n2. Send HTTP request with exponential retry.\n3. Parse JSO
N response and summarize.',
70 |         rules: '- Protect sensitive secrets and auth tokens.\n
- Handle rate limits and connection timeouts gracefully.',
71 |         hasScript: true,
72 |         scriptCode: `# -*- coding: utf-8 -*-\n""REST API clie
nt script.""\nimport os, urllib.request, json\n\ndef query_ap
i(url: str):\n    req = urllib.request.Request(url, headers={"
User-Agent": "AI-Breadboard/1.0"})\n    with urllib.request.ur
lopen(req, timeout=10) as resp:\n        return json.loads(res
p.read().decode("utf-8"))\n`,
73 |         hasRef: true,
74 |         refContent: `# REST API Protocol Guidelines\n\n- Stand
ard timeout: 10s\n- Required headers: User-Agent, Accept\n`,
75 |     },
76 |     {
77 |         id: 'data-parser',
78 |         title: '⬢ Data Analyzer & Parser',
79 |         category: 'data',

```

```
80 |         badge: 'Analytics',
81 |         description: 'Обработка и анализ структурированных дан
ных (CSV, JSON, XML, таблицы, логи). Извлечение инсайтов и фор
матирование результатов.',
82 |         name: 'data-analyzer',
83 |         descRu: 'Анализ структурированных данных, логов и табл
иц с выводом статистики.',
84 |         descEn: 'Analyze structured data, logs, CSV/JSON datas
ets, and generate statistics.',
85 |         purpose: 'Parse input data files, compute aggregated m
etrics, and present visual summaries.',
86 |         triggers: 'Triggered when datasets, spreadsheets or da
ta files need parsing and reporting.',
87 |         protocol: '1. Identify input format (CSV/JSON/Log).\n2
. Parse schema and clean invalid rows.\n3. Compute totals, dis
tributions, and outliers.\n4. Output Markdown summary tables.'
,
88 |         rules: '- Avoid memory leaks on large files.\n- Strict
ly English docstrings.',
89 |         hasScript: true,
90 |         scriptCode: `# -*- coding: utf-8 -*-\n"""Data parsing
helper."""\nimport csv, json\n\ndef parse_records(filepath: str):\n
    # Add parser logic here\n    pass\n`,
91 |         hasRef: false,
92 |     },
93 |     {
94 |         id: 'ai-persona',
95 |         title: 'AI Expert & Persona',
96 |         category: 'persona',
97 |         badge: 'Agent Role',
98 |         description: 'Специализированная роль и экспертные инс
трукции (Архитектор, Аналитик безопасности, Code Reviewer, Кон
сультант).',
99 |         name: 'domain-expert',
100 |         descRu: 'Экспертная роль и правила поведения для специ
ализированных консультаций.',
101 |         descEn: 'Specialized domain expert persona with strict
evaluation guidelines.',
```

```
102 |         purpose: 'Act as a senior domain expert offering high-
103 |         triggers: 'Triggered whenever specialized architectura
104 |         protocol: '1. Analyze problem from the perspective of
105 |         rules: '- Be concise, objective, and reference industr
106 |         hasScript: false,
107 |         hasRef: true,
108 |         refContent: `# Domain Expertise Standards\n\nKey metho
109 |     },
110 |     {
111 |         id: 'monitoring-notifier',
112 |         title: '□ Monitor & Notifier',
113 |         category: 'monitoring',
114 |         badge: 'Observability',
115 |         description: 'Периодическая проверка сервисов, мониторинг системных параметров, выявление сбоев и отправка нотификаций.',
116 |         name: 'health-monitor',
117 |         descRu: 'Мониторинг доступности сервисов и формирование алертов при сбоях.',
118 |         descEn: 'Health monitoring and alerting skill for infrastructure and services.',
119 |         purpose: 'Continuously verify target service endpoints and notify on anomalies.',
120 |         triggers: 'Triggered by scheduled cron jobs or on-demand health check requests.',
121 |         protocol: '1. Ping healthcheck endpoints.\n2. Measure latency and status codes.\n3. If failure detected, format diagnostic alert.',
122 |         rules: '- Fail-fast on network unreachability.\n- Include timestamps and severity level.',
123 |         hasScript: true,
124 |         scriptCode: `# -*- coding: utf-8 -*-\n""Health check`
```

```

monitor."""\nimport time\n\ndef check_service(host: str, port:
    int):\n    # Health probe\n    return True\n`,
125     hasRef: false,
126     },
127     {
128         id: 'doc-generator',
129         title: '□ Report & Doc Generator',
130         category: 'docgen',
131         badge: 'Documentation',
132         description: 'Автоматическая генерация документации, о
тчетов в Markdown/HTML и сводок по проекту на основе входных д
анных.',
133         name: 'report-generator',
134         descRu: 'Генератор красивых отчетов, проектной докумен
тации и сводок.',
135         descEn: 'Automated documentation and structured report
generation skill.',
136         purpose: 'Generate comprehensive Markdown reports, cha
nge logs, and release notes.',
137         triggers: 'Triggered when documentation, release summa
ries or audit reports are requested.',
138         protocol: '1. Collect source items and metadata.\n2. A
pply template sections and headings.\n3. Format tables and Git
Hub alerts.\n4. Save output file.',
139         rules: '- Follow documentation style standards.\n- Ens
ure all links and tables are valid.',
140         hasScript: false,
141         hasRef: true,
142         refContent: `# Document Generation Templates\n\nTempla
te blocks, header standards, and formatting guidelines.\n`,
143     }
144 ];
145 }
146
147 async init() {
148     if (this.isInitialized) {
149         await this.refresh();
150     }
    return;

```

```

151     }
152     this.bindEvents();
153     this.renderPresets();
154     this.isInitialized = true;
155     await this.refresh();
156 }
157
158 bindEvents() {
159     const searchInput = document.getElementById('skills-search
-input');
160     if (searchInput) {
161         searchInput.addEventListener('input', (e) => {
162             this.searchQuery = e.target.value.trim().toLowerCase()
;
163             this.applyFilters();
164         });
165     }
166
167     const wizardNameInput = document.getElementById('wizard-sk
ill-name');
168     if (wizardNameInput) {
169         wizardNameInput.addEventListener('input', (e) => {
170             const val = e.target.value;
171             const normalized = val.toLowerCase().replace(/[^a-z0-9
_-]/g, '-');
172             if (val !== normalized) {
173                 e.target.value = normalized;
174             }
175             this.syncWizardPreview();
176         });
177     }
178
179     // Step 2 field listeners for live markdown generator
180     ['wizard-inst-purpose', 'wizard-inst-triggers', 'wizard-in
st-protocol', 'wizard-inst-rules'].forEach(id => {
181         const el = document.getElementById(id);
182         if (el) {
183             el.addEventListener('input', () => this.syncInstructio

```



```

    nsFromFields());
184     }
185   });
186
187   ['wizard-skill-desc-ru', 'wizard-skill-desc-en', 'wizard-s
kill-dir'].forEach(id => {
188     const el = document.getElementById(id);
189     if (el) {
190       el.addEventListener('input', () => this.syncWizardPrev
iew());
191       el.addEventListener('change', () => this.syncWizardPre
view());
192     }
193   });
194 }
195
196 async apiFetch(endpoint, options = {}) {
197   let url = endpoint;
198   let res = await fetch(url, options);
199   if (!res.ok && url.startsWith('/api/skills')) {
200     const fallbackUrl = url.replace('/api/skills', '/api/adm
in/skills');
201     const fallbackRes = await fetch(fallbackUrl, options);
202     if (fallbackRes.ok) {
203       return fallbackRes;
204     }
205   }
206   return res;
207 }
208
209 async refresh() {
210   this.showLoading(true);
211   try {
212     const response = await this.apiFetch('/api/skills');
213     if (!response.ok) {
214       throw new Error(`HTTP ${response.status}: ${response.s
tatusText}`);
215     }

```

```

216 |     const data = await response.json();
217 |     this.skills = data.skills || [];
218 |     this.updateStats();
219 |     this.applyFilters();
220 | } catch (err) {
221 |     console.error('[SkillsTab] Load error:', err);
222 |     const cardsContainer = document.getElementById('skills-c
ards-container');
223 |     if (cardsContainer) {
224 |         cardsContainer.innerHTML = `
225 |             <div class="col-12">
226 |                 <div class="alert alert-danger d-flex align-items-
center justify-content-between p-3 shadow-sm">
227 |                     <div><i class="bi bi-exclamation-triangle-fill m
e-2"></i><strong>Ошибка загрузки навыков:</strong> ${err.messa
ge}</div>
228 |                     <button class="btn btn-sm btn-outline-light" onc
lick="window.skillsTab?.refresh()">Повторить</button>
229 |                 </div>
230 |             </div>
231 |         `;
232 |     }
233 | } finally {
234 |     this.showLoading(false);
235 | }
236 | }
237 |
238 | updateStats() {
239 |     const total = this.skills.length;
240 |     const withScripts = this.skills.filter(s => s.has_scripts)
.length;
241 |     const withRefs = this.skills.filter(s => s.has_references)
.length;
242 |     const packaged = this.skills.filter(s => s.has_dist).lengt
h;
243 |
244 |     const setVal = (id, val) => {
245 |         const el = document.getElementById(id);

```

```

246         if (el) el.textContent = val;
247     };
248
249     setVal('skills-total-badge', total);
250     setVal('stat-total-skills', total);
251     setVal('stat-scripts-skills', withScripts);
252     setVal('stat-refs-skills', withRefs);
253     setVal('stat-packaged-skills', packaged);
254 }
255
256 applyFilters() {
257     const filterSelect = document.getElementById('skills-filte
r-select');
258     this.activeFilter = filterSelect ? filterSelect.value : 'a
ll';
259
260     this.filteredSkills = this.skills.filter(skill => {
261         if (this.searchQuery) {
262             const nameMatch = (skill.name || '').toLowerCase().inc
ludes(this.searchQuery);
263             const descMatch = (skill.description || '').toLowerCas
e().includes(this.searchQuery);
264             const i18nMatch = Object.values(skill.descriptions_i18
n || {}).some(d => (d || '').toLowerCase().includes(this.searc
hQuery));
265             if (!nameMatch && !descMatch && !i18nMatch) return fal
se;
266         }
267
268         if (this.activeFilter === 'scripts' && !skill.has_script
s) return false;
269         if (this.activeFilter === 'references' && !skill.has_ref
erences) return false;
270         if (this.activeFilter === 'packaged' && !skill.has_dist)
return false;
271
272         return true;
273     });

```

```
274 |
275 |     this.render();
276 | }
277 |
278 | clearSearch() {
279 |     const searchInput = document.getElementById('skills-search
    -input');
280 |     if (searchInput) searchInput.value = '';
281 |     this.searchQuery = '';
282 |     this.applyFilters();
283 | }
284 |
285 | setViewMode(mode) {
286 |     this.viewMode = mode;
287 |     const btnCards = document.getElementById('btn-view-cards')
    ;
288 |     const btnTable = document.getElementById('btn-view-table')
    ;
289 |
290 |     if (mode === 'cards') {
291 |         btnCards?.classList.add('active');
292 |         btnTable?.classList.remove('active');
293 |     } else {
294 |         btnCards?.classList.remove('active');
295 |         btnTable?.classList.add('active');
296 |     }
297 |     this.render();
298 | }
299 |
300 | render() {
301 |     const cardsContainer = document.getElementById('skills-car
    ds-container');
302 |     const tableCard = document.getElementById('skills-table-ca
    rd');
303 |     const emptyNotice = document.getElementById('skills-empty'
    );
304 |
305 |     if (!cardsContainer || !tableCard) return;
```

```

306
307     if (this.filteredSkills.length === 0) {
308         cardsContainer.innerHTML = '';
309         tableCard.classList.add('d-none');
310         emptyNotice?.classList.remove('d-none');
311         return;
312     }
313
314     emptyNotice?.classList.add('d-none');
315
316     if (this.viewMode === 'cards') {
317         tableCard.classList.add('d-none');
318         cardsContainer.classList.remove('d-none');
319         cardsContainer.innerHTML = this.filteredSkills.map(s =>
this.renderCard(s)).join('');
320     } else {
321         cardsContainer.classList.add('d-none');
322         tableCard.classList.remove('d-none');
323         const tableBody = document.getElementById('skills-table-
body');
324         if (tableBody) {
325             tableBody.innerHTML = this.filteredSkills.map(s => thi
s.renderTableRow(s)).join('');
326         }
327     }
328 }
329
330 renderCard(skill) {
331     const escapeHtml = str => String(str || '').replace(/[<>
"]/g, tag => ({
332         '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&#39;', "'
': '&quot;';
333     }[tag] || tag));
334
335     const badges = [];
336     if (skill.has_scripts) badges.push('<span class="badge bg-
info-subtle text-info border border-info-subtle small me-1"><i
class="bi bi-file-earmark-code"></i> scripts</span>');

```

```

337     if (skill.has_references) badges.push('<span class="badge
bg-warning-subtle text-warning border border-warning-subtle sm
all me-1"><i class="bi bi-journal-text"></i> references</span>
');
338     if (skill.has_assets) badges.push('<span class="badge bg-s
econdary border border-secondary small me-1"><i class="bi bi-i
mages"></i> assets</span>');
339     if (skill.has_dist) badges.push('<span class="badge bg-suc
cess-subtle text-success border border-success-subtle small me
-1"><i class="bi bi-box-seam"></i> .skill</span>');
340
341     const descRu = skill.descriptions_i18n?.ru || '';
342     const descDisplay = descRu ? `${descRu}` : (skill.descript
ion || 'Описание отсутствует');
343
344     return `
345         <div class="col-12 col-md-6 col-lg-4">
346             <div class="card bg-dark border-secondary text-white h
-100 shadow-sm skill-card">
347                 <div class="card-header border-secondary d-flex just
ify-content-between align-items-center bg-black bg-opacity-25
py-2">
348                     <span class="badge bg-primary font-monospace fs-6
text-truncate" style="max-width: 70%;" title="${escapeHtml(ski
ll.name)}">
349                         □ ${escapeHtml(skill.name)}
350                     </span>
351                     <span class="text-muted small" title="Количество ф
айлов">
352                         <i class="bi bi-files"></i> ${skill.files_count
|| 1}
353                     </span>
354                 </div>
355                 <div class="card-body py-2">
356                     <p class="card-text text-light small mb-2 text-tru
ncate-3" style="min-height: 40px; display: -webkit-box; -webki
t-line-clamp: 2; -webkit-box-orient: vertical; overflow: hidde
n;" title="${escapeHtml(descDisplay)}">

```

```

357         ${escapeHtml(descDisplay)}
358     </p>
359     <div class="mb-2 text-truncate">
360         <small class="text-secondary font-monospace"><i
class="bi bi-folder"></i> ${escapeHtml(skill.relative_path)}</
small>
361     </div>
362     <div class="d-flex flex-wrap gap-1">
363         ${badges.join('')}
364     </div>
365 </div>
366     <div class="card-footer border-secondary bg-black bg
-opacity-25 py-2 d-flex justify-content-between align-items-ce
nter">
367         <button type="button" class="btn btn-sm btn-outlin
e-primary d-flex align-items-center gap-1" onclick="window.ski
llsTab?.openSkillDetails('${escapeHtml(skill.name)}')">
368             <i class="bi bi-sliders"></i> Детали
369         </button>
370         <div class="btn-group btn-group-sm">
371             <button type="button" class="btn btn-outline-war
ning" title="Тестировать в Sandbox" onclick="window.skillsTab?
.openSkillSandbox('${escapeHtml(skill.name)}')">
372                 <i class="bi bi-play-circle"></i>
373             </button>
374             <button type="button" class="btn btn-outline-inf
o" title="Собрать .skill архив" onclick="window.skillsTab?.pac
kageSkill('${escapeHtml(skill.name)}')">
375                 <i class="bi bi-box-seam"></i>
376             </button>
377             <button type="button" class="btn btn-outline-dan
ger" title="Удалить навык" onclick="window.skillsTab?.deleteSk
ill('${escapeHtml(skill.name)}')">
378                 <i class="bi bi-trash"></i>
379             </button>
380         </div>
381     </div>
382 </div>

```

```

383     </div>
384     `;
385 }
386
387 renderTableRow(skill) {
388     const escapeHtml = str => String(str || '').replace(/[<>'
389     '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&#39;', "'
390     ': '&quot;';
391     }[tag] || tag));
392     const badges = [];
393     if (skill.has_scripts) badges.push('<span class="badge bg-
394     info-subtle text-info border border-info-subtle small me-1">sc
395     ripts</span>');
396     if (skill.has_references) badges.push('<span class="badge
397     bg-warning-subtle text-warning border border-warning-subtle sm
398     all me-1">refs</span>');
399     if (skill.has_dist) badges.push('<span class="badge bg-suc
400     cess-subtle text-success border border-success-subtle small me
401     -1">.skill</span>');
402
403     const descRu = skill.descriptions_i18n?.ru || '';
404     const descDisplay = descRu ? `${descRu}` : (skill.descript
405     ion || '-');
406
407     return `
408     <tr>
409     <td>
410         <span class="badge bg-primary font-monospace fs-6">
411         ${escapeHtml(skill.name)}</span>
412     </td>
413     <td>
414         <div class="small text-light text-truncate" style="m
415         ax-width: 350px;" title="${escapeHtml(descDisplay)}">
416         ${escapeHtml(descDisplay)}
417     </div>
418     </td>

```



```

410         <td>
411             <code class="small text-secondary">${escapeHtml(skill
412 | l.relative_path)}</code>
413         </td>
414         <td>
415             ${badges.length ? badges.join('') : '<span class="te
416 | xt-muted small">—</span>'}
417         </td>
418         <td style="text-align: right;">
419             <div class="btn-group btn-group-sm">
420                 <button type="button" class="btn btn-outline-prima
421 | ry" title="Открыть детали" onclick="window.skillsTab?.openSkill
422 | lDetails('${escapeHtml(skill.name)}')">
423                     <i class="bi bi-sliders"></i>
424                 </button>
425                 <button type="button" class="btn btn-outline-warni
426 | ng" title="Тестировать в Sandbox" onclick="window.skillsTab?.o
427 | penSkillSandbox('${escapeHtml(skill.name)}')">
428                     <i class="bi bi-play-circle"></i>
429                 </button>
430                 <button type="button" class="btn btn-outline-info"
431 | title="Собрать архив" onclick="window.skillsTab?.packageSkill
432 | ('${escapeHtml(skill.name)}')">
433                     <i class="bi bi-box-seam"></i>
434                 </button>
435                 <button type="button" class="btn btn-outline-dange
436 | r" title="Удалить" onclick="window.skillsTab?.deleteSkill('${e
437 | scapeHtml(skill.name)}')">
438                     <i class="bi bi-trash"></i>
439                 </button>
440             </div>
441         </td>
442     </tr>
443 `;
444 }
445
446 // =====
=====

```

```

437 // CREATION WIZARD LOGIC (3 Modes: Wizard, AI, Presets)
438 // =====
    =====
439
440 switchCreateMode(mode) {
441     this.createMode = mode;
442 }
443
444 goToStep(step) {
445     if (step < 1 || step > 4) return;
446     this.currentWizardStep = step;
447
448     for (let i = 1; i <= 4; i++) {
449         const stepEl = document.getElementById(`wizard-step-${i}
450 `);
451         const badgeEl = document.getElementById(`step-badge-${i}
452 `);
453         const labelEl = document.getElementById(`step-label-${i}
454 `);
455
456         if (stepEl) {
457             if (i === step) stepEl.classList.remove('d-none');
458             else stepEl.classList.add('d-none');
459         }
460
461         if (badgeEl && labelEl) {
462             if (i === step) {
463                 badgeEl.className = 'badge rounded-circle p-2 bg-pri
464 mary step-indicator';
465                 labelEl.className = 'small fw-semibold text-white';
466             } else if (i < step) {
467                 badgeEl.className = 'badge rounded-circle p-2 bg-suc
468 cess step-indicator';
469                 labelEl.className = 'small text-success';
470             } else {
471                 badgeEl.className = 'badge rounded-circle p-2 bg-sec
472 ondary step-indicator';
473                 labelEl.className = 'small text-muted';

```

```
468 |     }
469 |   }
470 | }
471 |
472 |   const prevBtn = document.getElementById('btn-wizard-prev')
    | ;
473 |   const nextBtn = document.getElementById('btn-wizard-next')
    | ;
474 |   const submitBtn = document.getElementById('btn-wizard-submit');
475 |
476 |   if (prevBtn) prevBtn.disabled = (step === 1);
477 |   if (nextBtn) {
478 |     if (step === 4) nextBtn.classList.add('d-none');
479 |     else nextBtn.classList.remove('d-none');
480 |   }
481 |   if (submitBtn) {
482 |     if (step === 4) submitBtn.classList.remove('d-none');
483 |     else submitBtn.classList.add('d-none');
484 |   }
485 |
486 |   if (step === 4) {
487 |     this.syncWizardPreview();
488 |   }
489 | }
490 |
491 | wizardNextStep() {
492 |   if (this.currentWizardStep === 1) {
493 |     const name = document.getElementById('wizard-skill-name')?.value.trim();
494 |     if (!name) {
495 |       this.showCreateAlert('Укажите имя навыка (kebab-case)', 'danger');
496 |       return;
497 |     }
498 |     this.clearCreateAlert();
499 |   }
500 |   this.goToStep(this.currentWizardStep + 1);
```

```

501 | }
502 |
503 | wizardPrevStep() {
504 |     this.goToStep(this.currentWizardStep - 1);
505 | }
506 |
507 | generateSlugFromName() {
508 |     const descRu = document.getElementById('wizard-skill-desc-
509 | ru')?.value.trim();
510 |     const descEn = document.getElementById('wizard-skill-desc-
511 | en')?.value.trim();
512 |     const source = descEn || descRu || 'custom-skill';
513 |
514 |     // Transliterate or simple slugify
515 |     let slug = source.toLowerCase()
516 |         .replace(/[а-яё]/g, char => {
517 |             const tr = {'а': 'a', 'б': 'b', 'в': 'v', 'г': 'g', 'д': 'd', 'е': 'e', 'ё': 'yo', 'ж': 'zh', 'з': 'z', 'и': 'i', 'й': 'y', 'к': 'k', 'л': 'l', 'м': 'm', 'н': 'n', 'о': 'o', 'п': 'p', 'р': 'r', 'с': 's', 'т': 't', 'у': 'u', 'ф': 'f', 'х': 'h', 'ц': 'ts', 'ч': 'ch', 'ш': 'sh', 'щ': 'sch', 'ъ': ' ', 'ы': 'y', 'ь': ' ', 'э': 'e', 'ю': 'yu', 'я': 'ya'};
518 |             return tr[char] || char;
519 |         })
520 |         .replace(/^[a-z0-9_-]/g, '-')
521 |         .replace(/-+/g, '-')
522 |         .replace(/^-|-$/g, '')
523 |         .slice(0, 32);
524 |
525 |     if (slug) {
526 |         const nameInput = document.getElementById('wizard-skill-
527 | name');
528 |         if (nameInput) nameInput.value = slug;
529 |         this.syncWizardPreview();
530 |     }
531 | }
532 |
533 | onCategoryChanged() {
534 |     const cat = document.getElementById('wizard-skill-category

```

```

    ')??.value;
532     const descRuInput = document.getElementById('wizard-skill-
desc-ru');
533     const descEnInput = document.getElementById('wizard-skill-
desc-en');
534
535     if (cat && (!descRuInput?.value || !descEnInput?.value)) {
536         const preset = this.presets.find(p => p.category === cat
);
537         if (preset) {
538             if (!descRuInput.value) descRuInput.value = preset.des
cRu;
539             if (!descEnInput.value) descEnInput.value = preset.des
cEn;
540         }
541     }
542     this.syncWizardPreview();
543 }
544
545     fillStandardInstructions() {
546         const name = document.getElementById('wizard-skill-name')?
.value.trim() || 'custom-skill';
547         const descRu = document.getElementById('wizard-skill-desc-
ru')?.value.trim() || 'Выполнение специализированных задач';
548         const descEn = document.getElementById('wizard-skill-desc-
en')?.value.trim() || 'Specialized agent task execution';
549
550         document.getElementById('wizard-inst-purpose').value = des
cEn;
551         document.getElementById('wizard-inst-triggers').value = `A
ctivate when the user asks to perform tasks matching: ${descRu
}`;
552         document.getElementById('wizard-inst-protocol').value = `1
. Validate input query and preconditions.\n2. Execute core log
ic and inspect helper scripts in scripts/.\n3. Return structur
ed markdown response.`;
553         document.getElementById('wizard-inst-rules').value = `- En
sure English logging via src.logger.logger.\n- Gracefully catc

```

```

    h and report all execution errors.`;
554
555     this.syncInstructionsFromFields();
556   }
557
558   syncInstructionsFromFields() {
559     const name = document.getElementById('wizard-skill-name')?
    .value.trim() || 'custom-skill';
560     const title = name.replace('-', ' ').replace('_', ' ').rep
    lace(/\b\w/g, l => l.toUpperCase());
561     const purpose = document.getElementById('wizard-inst-purpo
    se')?.value.trim() || 'Provide specialized agent capability.';
562     const triggers = document.getElementById('wizard-inst-trig
    gers')?.value.trim() || 'Activate upon relevant user requests.
    ';
563     const protocol = document.getElementById('wizard-inst-prot
    ocol')?.value.trim() || '1. Parse input.\n2. Execute action.\n
    3. Output result.';
564     const rules = document.getElementById('wizard-inst-rules')
    ?.value.trim() || '- Maintain fail-fast validation.\n- Report
    status clearly.';
565
566     const md = `# ${title}
567
568 ## □ Purpose
569 ${purpose}
570
571 ## □ When to Use & Triggers
572 ${triggers}
573
574 ## □ Execution Protocol
575 ${protocol}
576
577 ## □ Rules & Constraints
578 ${rules}
579 `;
580     const instEl = document.getElementById('wizard-skill-instr
    uctions');

```

```

581     if (instEl) instEl.value = md;
582     this.syncWizardPreview();
583 }
584
585 toggleScriptEditor() {
586     const chk = document.getElementById('wizard-enable-script'
);
587     const container = document.getElementById('wizard-script-c
ontainer');
588     const scriptCode = document.getElementById('wizard-script-
code');
589
590     if (chk?.checked) {
591         container?.classList.remove('d-none');
592         if (!scriptCode.value.trim()) {
593             const name = document.getElementById('wizard-skill-nam
e')?.value.trim() || 'custom-skill';
594             scriptCode.value = `# -*- coding: utf-8 -*-\n""Execut
able tool helper for ${name}.""\n\nimport sys\n\ndef main():\n
    print("Running helper script...")\n    return 0\n\nif __n
ame__ == "__main__":\n    sys.exit(main())\n`;
595         }
596     } else {
597         container?.classList.add('d-none');
598     }
599     this.syncWizardPreview();
600 }
601
602 toggleRefEditor() {
603     const chk = document.getElementById('wizard-enable-ref');
604     const container = document.getElementById('wizard-ref-cont
ainer');
605     const refContent = document.getElementById('wizard-ref-con
tent');
606
607     if (chk?.checked) {
608         container?.classList.remove('d-none');
609         if (!refContent.value.trim()) {

```

```

610         const name = document.getElementById('wizard-skill-name')?.value.trim() || 'custom-skill';
611         refContent.value = `# ${name.replace('-', ' ').toUpperCase()} Specification\n\nDetailed operational rules and reference data.\n`;
612     }
613     } else {
614         container?.classList.add('d-none');
615     }
616     this.syncWizardPreview();
617 }
618
619 syncWizardPreview() {
620     const name = document.getElementById('wizard-skill-name')?.value.trim() || 'custom-skill';
621     const dir = document.getElementById('wizard-skill-dir')?.value || '.agents/skills';
622     const descEn = document.getElementById('wizard-skill-desc-en')?.value.trim() || `Agent skill for ${name}.`;
623     const descRu = document.getElementById('wizard-skill-desc-ru')?.value.trim() || descEn;
624     const instructions = document.getElementById('wizard-skill-instructions')?.value.trim() || `# ${name}\n\n## Purpose\n${descEn}\n`;
625
626     const hasScript = document.getElementById('wizard-enable-script')?.checked;
627     const hasRef = document.getElementById('wizard-enable-ref')?.checked;
628
629     // Badges in step 4
630     const nameEl = document.getElementById('preview-skill-name');
631     const pathEl = document.getElementById('preview-skill-path');
632     const badgeScript = document.getElementById('preview-badge-script');
633     const badgeRef = document.getElementById('preview-badge-re

```



```

    f');
634     const previewFull = document.getElementById('preview-skill
md-full');
635
636     if (nameEl) nameEl.textContent = name;
637     if (pathEl) pathEl.textContent = `${dir}/${name}`;
638     if (badgeScript) {
639         if (hasScript) badgeScript.classList.remove('d-none');
640         else badgeScript.classList.add('d-none');
641     }
642     if (badgeRef) {
643         if (hasRef) badgeRef.classList.remove('d-none');
644         else badgeRef.classList.add('d-none');
645     }
646
647     const fullSkillMd = `---
648 name: ${name}
649 description: ${descEn}
650 description_i18n:
651   en: ${descEn}
652   ru: ${descRu}
653 ---
654
655 ${instructions}`;
656
657     if (previewFull) previewFull.textContent = fullSkillMd;
658
659     // Sync README default
660     const readmeEl = document.getElementById('wizard-skill-rea
dme');
661     if (readmeEl && !readmeEl.value.trim()) {
662         readmeEl.value = `# ${name.replace('-', ' ').replace(/\b
\w/g, l => l.toUpperCase())}\n\n## Overview\n${descEn}\n\n## L
ocation\n`${dir}/${name}/${name}`;
663     }
664 }
665
666 async createSkillFromWizard() {

```

```
667     const name = document.getElementById('wizard-skill-name')?
        .value.trim().toLowerCase();
668     if (!name) {
669         this.goToStep(1);
670         this.showCreateAlert('Укажите имя навыка', 'danger');
671         return;
672     }
673
674     const spinner = document.getElementById('wizard-submit-spi
        nner');
675     spinner?.classList.remove('d-none');
676
677     const descEn = document.getElementById('wizard-skill-desc-
        en')?.value.trim() || `Agent skill for ${name}.`;
678     const descRu = document.getElementById('wizard-skill-desc-
        ru')?.value.trim() || descEn;
679     const targetDir = document.getElementById('wizard-skill-di
        r')?.value || '.agents/skills';
680     const category = document.getElementById('wizard-skill-cat
        egor')?.value || 'general';
681     const instructions = document.getElementById('wizard-skill
        -instructions')?.value.trim() || '';
682     const readme = document.getElementById('wizard-skill-readm
        e')?.value.trim() || '';
683
684     const scripts = {};
685     if (document.getElementById('wizard-enable-script')?.check
        ed) {
686         scripts['main.py'] = document.getElementById('wizard-scr
        ipt-code')?.value || '';
687     }
688
689     const references = {};
690     if (document.getElementById('wizard-enable-ref')?.checked)
        {
691         references['guide.md'] = document.getElementById('wizard
        -ref-content')?.value || '';
692     }
```

```
693 |
694 |     try {
695 |         const res = await this.apiFetch('/api/skills', {
696 |             method: 'POST',
697 |             headers: { 'Content-Type': 'application/json' },
698 |             body: JSON.stringify({
699 |                 name,
700 |                 description: descEn,
701 |                 description_ru: descRu,
702 |                 category,
703 |                 target_dir: targetDir,
704 |                 instructions,
705 |                 readme,
706 |                 scripts,
707 |                 references
708 |             })
709 |         });
710 |
711 |         if (!res.ok) {
712 |             const errData = await res.json().catch(() => ({}));
713 |             throw new Error(errData.detail || res.statusText);
714 |         }
715 |
716 |         // Close modal
717 |         const modalEl = document.getElementById('createSkillModal');
718 |         const modal = bootstrap.Modal.getInstance(modalEl);
719 |         modal?.hide();
720 |
721 |         await this.refresh();
722 |         await this.openSkillDetails(name);
723 |     } catch (err) {
724 |         console.error('[SkillsTab] Create error:', err);
725 |         this.showCreateAlert(`Ошибка создания навыка: ${err.message}`, 'danger');
726 |     } finally {
727 |         spinner?.classList.add('d-none');
728 |     }
```

```

729 | }
730 |
731 | // =====
732 | // AI GENERATOR MODE
733 | // =====
734 |
735 | async generateSkillWithAi() {
736 |     const prompt = document.getElementById('ai-skill-prompt')?
737 |     .value.trim();
738 |     if (!prompt) {
739 |         this.showCreateAlert('Введите описание задачи для генера
740 |         ции навыка', 'warning');
741 |         return;
742 |     }
743 |     const spinner = document.getElementById('ai-gen-spinner');
744 |     const resBox = document.getElementById('ai-result-box');
745 |     const resSlug = document.getElementById('ai-res-slug');
746 |     const resPreview = document.getElementById('ai-res-preview
747 |     ');
748 |     spinner?.classList.remove('d-none');
749 |     this.clearCreateAlert();
750 |     try {
751 |         const category = document.getElementById('ai-skill-categ
752 |         ory')?.value || 'general';
753 |         const targetDir = document.getElementById('ai-skill-dir'
754 |         )?.value || '.agents/skills';
755 |         const res = await this.apiFetch('/api/skills/generate-ai
756 |         ', {
757 |             method: 'POST',
758 |             headers: { 'Content-Type': 'application/json' },
759 |             body: JSON.stringify({ prompt, category, target_dir: t
760 |             argetDir })

```

```

758         });
759
760         if (!res.ok) {
761             const errData = await res.json().catch(() => ({}));
762             throw new Error(errData.detail || res.statusText);
763         }
764
765         const data = await res.json();
766         this.aiGeneratedData = data.generated;
767
768         if (resSlug) resSlug.textContent = ` ${this.aiGenerated
Data.name}`;
769         if (resPreview) resPreview.textContent = this.aiGenerate
dData.instructions;
770         resBox?.classList.remove('d-none');
771     } catch (err) {
772         console.error('[SkillsTab] AI generate error:', err);
773         this.showCreateAlert(`Ошибка генерации навыка: ${err.mes
sage}`, 'danger');
774     } finally {
775         spinner?.classList.add('d-none');
776     }
777 }
778
779 applyAiResultToWizard() {
780     if (!this.aiGeneratedData) return;
781     const g = this.aiGeneratedData;
782
783     document.getElementById('wizard-skill-name').value = g.nam
e || '';
784     document.getElementById('wizard-skill-desc-en').value = g.
description || '';
785     document.getElementById('wizard-skill-desc-ru').value = g.
description_ru || '';
786     document.getElementById('wizard-skill-category').value = g
.category || 'general';
787     document.getElementById('wizard-skill-dir').value = g.targ
et_dir || '.agents/skills';

```

```
788     document.getElementById('wizard-skill-instructions').value
      = g.instructions || '';
789     document.getElementById('wizard-skill-readme').value = g.r
      eadme || '';
790
791     if (g.scripts && g.scripts['main.py']) {
792         document.getElementById('wizard-enable-script').checked
      = true;
793         document.getElementById('wizard-script-container').class
      List.remove('d-none');
794         document.getElementById('wizard-script-code').value = g.
      scripts['main.py'];
795     }
796
797     if (g.references && g.references['guide.md']) {
798         document.getElementById('wizard-enable-ref').checked = t
      rue;
799         document.getElementById('wizard-ref-container').classLis
      t.remove('d-none');
800         document.getElementById('wizard-ref-content').value = g.
      references['guide.md'];
801     }
802
803     // Switch to wizard tab
804     const wizardTabBtn = document.getElementById('mode-wizard-
      tab');
805     if (wizardTabBtn) {
806         const tab = new bootstrap.Tab(wizardTabBtn);
807         tab.show();
808     }
809     this.goToStep(1);
810     this.syncWizardPreview();
811 }
812
813 async saveAiSkilllDirectly() {
814     if (!this.aiGeneratedData) return;
815     const g = this.aiGeneratedData;
816
```

```

817 |     try {
818 |         const res = await this.apiFetch('/api/skills', {
819 |             method: 'POST',
820 |             headers: { 'Content-Type': 'application/json' },
821 |             body: JSON.stringify({
822 |                 name: g.name,
823 |                 description: g.description,
824 |                 description_ru: g.description_ru,
825 |                 category: g.category,
826 |                 target_dir: g.target_dir,
827 |                 instructions: g.instructions,
828 |                 readme: g.readme,
829 |                 scripts: g.scripts || {},
830 |                 references: g.references || {}
831 |             })
832 |         });
833 |
834 |         if (!res.ok) {
835 |             const errData = await res.json().catch(() => ({}));
836 |             throw new Error(errData.detail || res.statusText);
837 |         }
838 |
839 |         const modalEl = document.getElementById('createSkillModal');
840 |         const modal = bootstrap.Modal.getInstance(modalEl);
841 |         modal?.hide();
842 |
843 |         await this.refresh();
844 |         await this.openSkillDetails(g.name);
845 |     } catch (err) {
846 |         console.error('[SkillsTab] AI Direct Save error:', err);
847 |         this.showCreateAlert(`Ошибка сохранения: ${err.message}`
848 |             , 'danger');
849 |     }
850 |
851 |     // =====
      |     =====

```

```

852 // PRESETS LIBRARY
853 // =====
854
855 renderPresets() {
856     const container = document.getElementById('presets-cards-c
ontainer');
857     if (!container) return;
858
859     container.innerHTML = this.presets.map(p => `
860         <div class="col-12 col-md-6 col-lg-4">
861             <div class="card bg-dark border-secondary text-white h
-100 shadow-sm">
862                 <div class="card-header border-secondary d-flex just
ify-content-between align-items-center bg-black bg-opacity-25
py-2">
863                     <span class="fw-bold small text-white">${p.title}<
/span>
864                     <span class="badge bg-primary-subtle text-primary
border border-primary-subtle small">${p.badge}</span>
865                 </div>
866                 <div class="card-body py-2">
867                     <p class="card-text text-muted small mb-2" style="
min-height: 48px;">
868                         ${p.description}
869                     </p>
870                     <div class="small font-monospace text-secondary mb
-2">
871                         <code>❏  ${p.name}</code>
872                     </div>
873                 </div>
874                 <div class="card-footer border-secondary bg-black bg
-opacity-25 py-2">
875                     <button type="button" class="btn btn-outline-prima
ry btn-sm w-100 d-flex align-items-center justify-content-cent
er gap-1" onclick="window.skillsTab?.applyPreset('${p.id}')">
876                         <i class="bi bi-box-arrow-in-right"></i> Исполни
вать шаблон

```



```
877         </button>
878     </div>
879 </div>
880 </div>
881 `).join('');
882 }
883
884 applyPreset(presetId) {
885     const preset = this.presets.find(p => p.id === presetId);
886     if (!preset) return;
887
888     document.getElementById('wizard-skill-name').value = prese
t.name;
889     document.getElementById('wizard-skill-desc-en').value = pr
eset.descEn;
890     document.getElementById('wizard-skill-desc-ru').value = pr
eset.descRu;
891     document.getElementById('wizard-skill-category').value = p
reset.category;
892     document.getElementById('wizard-skill-dir').value = '.agen
ts/skills';
893
894     document.getElementById('wizard-inst-purpose').value = pre
set.purpose;
895     document.getElementById('wizard-inst-triggers').value = pr
eset.triggers;
896     document.getElementById('wizard-inst-protocol').value = pr
eset.protocol;
897     document.getElementById('wizard-inst-rules').value = prese
t.rules;
898     this.syncInstructionsFromFields();
899
900     // Scripts
901     const scriptChk = document.getElementById('wizard-enable-s
cript');
902     if (scriptChk) {
903         scriptChk.checked = preset.hasScript;
904         if (preset.hasScript) {
```

```

905 |         document.getElementById('wizard-script-container')?.classList.remove('d-none');
906 |         document.getElementById('wizard-script-code').value =
| preset.scriptCode || '';
907 |     } else {
908 |         document.getElementById('wizard-script-container')?.classList.add('d-none');
909 |     }
910 | }
911 |
912 | // References
913 | const refChk = document.getElementById('wizard-enable-ref');
| );
914 | if (refChk) {
915 |     refChk.checked = preset.hasRef;
916 |     if (preset.hasRef) {
917 |         document.getElementById('wizard-ref-container')?.classList.remove('d-none');
918 |         document.getElementById('wizard-ref-content').value =
| preset.refContent || '';
919 |     } else {
920 |         document.getElementById('wizard-ref-container')?.classList.add('d-none');
921 |     }
922 | }
923 |
924 | // Switch to wizard tab
925 | const wizardTabBtn = document.getElementById('mode-wizard-tab');
| tab');
926 | if (wizardTabBtn) {
927 |     const tab = new bootstrap.Tab(wizardTabBtn);
928 |     tab.show();
929 | }
930 | this.goToStep(1);
931 | this.syncWizardPreview();
932 | }
933 |
934 | showCreateAlert(msg, type = 'danger') {

```

```

935     const alertEl = document.getElementById('create-skill-aler
t');
936     if (alertEl) {
937         alertEl.className = `alert alert-${type} py-2 mt-3 mb-0`
;
938         alertEl.textContent = msg;
939         alertEl.classList.remove('d-none');
940     }
941 }
942
943 clearCreateAlert() {
944     const alertEl = document.getElementById('create-skill-aler
t');
945     if (alertEl) alertEl.classList.add('d-none');
946 }
947
948 // =====
=====
949 // SKILL DETAILS & SANDBOX TESTER
950 // =====
=====
951
952 async openSkillDetails(name, defaultTab = 'detail-tab-skillm
d') {
953     try {
954         const res = await this.apiFetch(`/api/skills/${encodeURIComponent(name)}`);
955         if (!res.ok) throw new Error(`Failed to load skill detai
ls: ${res.statusText}`);
956         const data = await res.json();
957         this.currentSkill = data.skill;
958
959         // Populate modal fields
960         document.getElementById('detail-skill-title').textConten
t = this.currentSkill.name;
961         document.getElementById('detail-skill-path').textContent
= this.currentSkill.relative_path;
962         document.getElementById('detail-skill-desc-input').value

```

```

    = this.currentSkill.description || '';
963     document.getElementById('detail-skill-desc-ru-input').va
lue = this.currentSkill.descriptions_i18n?.ru || '';
964     document.getElementById('detail-skill-instructions').val
ue = this.currentSkill.instructions || '';
965     document.getElementById('detail-skill-readme').value = t
his.currentSkill.readme_raw || '';
966
967     // Populate files list
968     const filesContainer = document.getElementById('detail-f
iles-list');
969     if (filesContainer) {
970         if (this.currentSkill.files && this.currentSkill.files
.length > 0) {
971             filesContainer.innerHTML = this.currentSkill.files.m
ap(f => `
972                 <div class="list-group-item bg-transparent text-li
ght border-secondary d-flex justify-content-between align-item
s-center py-1 small">
973                     <span><i class="bi bi-file-earmark text-secondar
y"></i> ${f.rel_path}</span>
974                     <span class="badge bg-secondary font-monospace">
${this.formatBytes(f.size)}</span>
975                 </div>
976             `).join('');
977         } else {
978             filesContainer.innerHTML = '<div class="text-muted s
mall py-2">Файлы отсутствуют</div>';
979         }
980     }
981
982     // Populate JSON contract
983     const contractEl = document.getElementById('detail-json-
contract');
984     if (contractEl) {
985         const contractObj = {
986             name: this.currentSkill.name,
987             description: this.currentSkill.description,

```

```
988 |         descriptions_i18n: this.currentSkill.descriptions_i1
    | 8n,
989 |         metadata: this.currentSkill.metadata,
990 |         manifest: this.currentSkill.manifest,
991 |     };
992 |     contractEl.textContent = JSON.stringify(contractObj, null, 2);
993 |     }
994 |
995 |     // Reset sandbox box
996 |     const sandboxOutput = document.getElementById('sandbox-output-box');
997 |     if (sandboxOutput) {
998 |         sandboxOutput.textContent = `Готов к тестированию навыка '${this.currentSkill.name}'. Введите тестовый запрос и нажмите «Запустить».`;
999 |     }
1000 |
1001 |     // Hide alerts
1002 |     const alertEl = document.getElementById('detail-skill-alert');
1003 |     if (alertEl) alertEl.classList.add('d-none');
1004 |
1005 |     // Show modal
1006 |     const modalEl = document.getElementById('skillDetailsModal');
1007 |     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
1008 |     modal.show();
1009 |
1010 |     // Switch to requested tab
1011 |     const targetTabBtn = document.getElementById(defaultTab);
    | ;
1012 |     if (targetTabBtn) {
1013 |         const tab = new bootstrap.Tab(targetTabBtn);
1014 |         tab.show();
1015 |     }
1016 | } catch (err) {
```

```

1017 |         console.error('[SkillsTab] Details error:', err);
1018 |         window.showToast?.(`Ошибка открытия деталей навыка: ${err.message}`, 'danger') || alert(`Ошибка открытия деталей навыка: ${err.message}`);
1019 |     }
1020 | }
1021 |
1022 | async openSkillSandbox(name) {
1023 |     await this.openSkillDetails(name, 'detail-tab-sandbox');
1024 | }
1025 |
1026 | async runSandboxTest() {
1027 |     if (!this.currentSkill) return;
1028 |     const promptInput = document.getElementById('sandbox-test-prompt');
1029 |     const outputBox = document.getElementById('sandbox-output-box');
1030 |     const spinner = document.getElementById('sandbox-spinner');
1031 |
1032 |     const prompt = promptInput?.value.trim() || `Test execute skill ${this.currentSkill.name}`;
1033 |     spinner?.classList.remove('d-none');
1034 |     if (outputBox) outputBox.textContent = `⏳ Отправка запроса к модели для навыка '${this.currentSkill.name}'...\n`;
1035 |
1036 |     try {
1037 |         const res = await this.apiFetch(`/api/skills/${encodeURIComponent(this.currentSkill.name)}/test`, {
1038 |             method: 'POST',
1039 |             headers: { 'Content-Type': 'application/json' },
1040 |             body: JSON.stringify({ prompt })
1041 |         });
1042 |
1043 |         if (!res.ok) {
1044 |             const errData = await res.json().catch(() => ({}));
1045 |             throw new Error(errData.detail || res.statusText);
1046 |         }

```

```
1047 |
1048 |     const data = await res.json();
1049 |     if (outputBox) {
1050 |         outputBox.textContent = `[Model: ${data.model_used}]\n
| \n${data.response}`;
1051 |     }
1052 | } catch (err) {
1053 |     console.error('[SkillsTab] Sandbox test error:', err);
1054 |     if (outputBox) outputBox.textContent = `❌ Ошибка тестиро
| вания: ${err.message}`;
1055 | } finally {
1056 |     spinner?.classList.add('d-none');
1057 | }
1058 | }
1059 |
1060 | async saveCurrentSkill() {
1061 |     if (!this.currentSkill) return;
1062 |     const spinner = document.getElementById('save-skill-spinne
| r');
1063 |     const alertEl = document.getElementById('detail-skill-aler
| t');
1064 |
1065 |     spinner?.classList.remove('d-none');
1066 |     try {
1067 |         const description = document.getElementById('detail-skil
| l-desc-input').value.trim();
1068 |         const description_ru = document.getElementById('detail-s
| kill-desc-ru-input').value.trim();
1069 |         const instructions = document.getElementById('detail-ski
| ll-instructions').value;
1070 |         const readme = document.getElementById('detail-skill-rea
| dme').value;
1071 |
1072 |         const res = await this.apiFetch(`/api/skills/${encodeURIComponent(
| Component(this.currentSkill.name))}`, {
1073 |             method: 'PUT',
1074 |             headers: { 'Content-Type': 'application/json' },
1075 |             body: JSON.stringify({ description, description_ru, in
```

```

        instructions, readme })
1076 |         });
1077 |
1078 |         if (!res.ok) {
1079 |             const errData = await res.json().catch(() => ({}));
1080 |             throw new Error(errData.detail || res.statusText);
1081 |         }
1082 |
1083 |         if (alertEl) {
1084 |             alertEl.className = 'alert alert-success py-2 mt-3 mb-
0';
1085 |             alertEl.textContent = 'Изменения успешно сохранены';
1086 |             alertEl.classList.remove('d-none');
1087 |         }
1088 |         window.showToast?.('Изменения навыка успешно сохранены',
'success');
1089 |
1090 |         await this.refresh();
1091 |     } catch (err) {
1092 |         console.error('[SkillsTab] Save error:', err);
1093 |         if (alertEl) {
1094 |             alertEl.className = 'alert alert-danger py-2 mt-3 mb-0
';
1095 |             alertEl.textContent = `Ошибка сохранения: ${err.messag
e}`;
1096 |             alertEl.classList.remove('d-none');
1097 |         }
1098 |         window.showToast?.(`Ошибка сохранения: ${err.message}`,
'danger');
1099 |     } finally {
1100 |         spinner?.classList.add('d-none');
1101 |     }
1102 | }
1103 |
1104 | async packageSkill(name) {
1105 |     try {
1106 |         const res = await this.apiFetch(`/api/skills/${encodeURIComponent
Component(name)}/package`, {

```



```

1107         method: 'POST'
1108     });
1109     if (!res.ok) {
1110         const errData = await res.json().catch(() => ({}));
1111         throw new Error(errData.detail || res.statusText);
1112     }
1113     const data = await res.json();
1114     window.showToast?(`📦 Навык '${name}' успешно упакован в
архив:\n${data.archive?.path} (${this.formatBytes(data.archive?.size)})`, 'success');
1115     await this.refresh();
1116 } catch (err) {
1117     console.error('[SkillsTab] Package error:', err);
1118     window.showToast?(`Ошибка сборки архива: ${err.message}
`, 'danger') || alert(`Ошибка сборки архива: ${err.message}`);
1119 }
1120 }
1121
1122 async packageCurrentSkill() {
1123     if (!this.currentSkill) return;
1124     await this.packageSkill(this.currentSkill.name);
1125     await this.openSkillDetails(this.currentSkill.name);
1126 }
1127
1128 async deleteSkill(name) {
1129     if (!confirm(`Вы действительно хотите удалить навык '${name}' и все его файлы? Это действие необратимо.`)) {
1130         return;
1131     }
1132
1133     try {
1134         const res = await this.apiFetch(`/api/skills/${encodeURIComponent(name)}`, {
1135             method: 'DELETE'
1136         });
1137         if (!res.ok) {
1138             const errData = await res.json().catch(() => ({}));
1139             throw new Error(errData.detail || res.statusText);

```

```
1140     }
1141
1142     const modalEl = document.getElementById('skillDetailsModal');
1143     const modal = bootstrap.Modal.getInstance(modalEl);
1144     modal?.hide();
1145
1146     window.showToast?(`Навык '${name}' успешно удален`, 'success');
1147     await this.refresh();
1148   } catch (err) {
1149     console.error('[SkillsTab] Delete error:', err);
1150     window.showToast?(`Ошибка удаления навыка: ${err.message}`, 'danger') || alert(`Ошибка удаления навыка: ${err.message}`);
1151   }
1152 }
1153
1154 async deleteCurrentSkill() {
1155   if (!this.currentSkill) return;
1156   await this.deleteSkill(this.currentSkill.name);
1157 }
1158
1159 showLoading(show) {
1160   const loadingEl = document.getElementById('skills-loading');
1161   if (loadingEl) {
1162     if (show) loadingEl.classList.remove('d-none');
1163     else loadingEl.classList.add('d-none');
1164   }
1165 }
1166
1167 formatBytes(bytes) {
1168   if (!bytes || bytes === 0) return '0 B';
1169   const k = 1024;
1170   const sizes = ['B', 'KB', 'MB', 'GB'];
1171   const i = Math.floor(Math.log(bytes) / Math.log(k));
1172   return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + ' ';
```

```
    ' + sizes[i];
1173 }
1174 }
1175
1176 // Global initialization hook
1177 window.SkillsTabManager = SkillsTabManager;
1178 if (!window.skillsTab) {
1179     window.skillsTab = new SkillsTabManager();
1180 }
1181 window.initSkillsTab = function() {
1182     if (window.skillsTab) {
1183         window.skillsTab.init();
1184     }
1185 };
1186
1187 // Auto-init on script load if container is present
1188 if (document.getElementById('skills-tab-root')) {
1189     window.skillsTab.init();
1190 }
1191
1192
1193
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/software_audit_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/software_audit_tab/index.html" class="w-100 h-100 border-0"></iframe>
17
18 File: index.html
19 Project: ai-breadboard
20 Package: windows/api/webgui/software_audit_tab
21 Author: hypo69
22 Copyright: © 2026 hypo69
23 Updated: 2026-10-08 03:08:00
24 =====
25
26 -->
27
28 <!-- Software Audit Application Tab -->
29 <style>
30   .sw-audit-container {
31     padding: 0.5rem 0.75rem;
32   }
33   .sw-audit-card {
34     background: #0f172a !important;
35     border: 1px solid #334155 !important;
36     border-radius: 8px;
```

```
33 | padding: 0.45rem 0.75rem;
34 | display: flex;
35 | flex-direction: column;
36 | justify-content: space-between;
37 | min-height: 58px;
38 | transition: border-color 0.15s ease;
39 | }
40 | .sw-audit-card:hover {
41 |   border-color: #38bdf8 !important;
42 | }
43 | .sw-audit-label {
44 |   font-size: 0.68rem;
45 |   text-transform: uppercase;
46 |   letter-spacing: 0.05em;
47 |   color: #94a3b8;
48 |   line-height: 1.1;
49 | }
50 | .sw-audit-value {
51 |   font-size: 1.25rem;
52 |   font-weight: 700;
53 |   color: #f8fafc;
54 |   font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
55 |   line-height: 1.15;
56 | }
57 | .sw-audit-panel {
58 |   background: #0f172a !important;
59 |   border: 1px solid #334155 !important;
60 |   border-radius: 8px;
61 |   overflow: hidden;
62 | }
63 | .sw-audit-panel-header {
64 |   background: #1e293b !important;
65 |   border-bottom: 1px solid #334155 !important;
66 |   padding: 0.45rem 0.85rem;
67 |   font-weight: 600;
68 |   font-size: 0.82rem;
69 |   color: #38bdf8;
```

```
70 |     display: flex;
71 |     align-items: center;
72 |     justify-content: space-between;
73 |     flex-wrap: wrap;
74 |     gap: 0.35rem;
75 | }
76 | .sw-audit-table {
77 |     width: 100%;
78 |     font-size: 0.78rem;
79 |     border-collapse: collapse;
80 | }
81 | .sw-audit-table th {
82 |     background: #1e293b;
83 |     padding: 6px 10px;
84 |     color: #94a3b8;
85 |     font-weight: 600;
86 |     border-bottom: 1px solid #334155;
87 |     position: sticky;
88 |     top: 0;
89 |     z-index: 2;
90 | }
91 | .sw-audit-table td {
92 |     padding: 6px 10px;
93 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
94 |     vertical-align: middle;
95 | }
96 | .sw-audit-table tbody tr:hover {
97 |     background: rgba(56, 189, 248, 0.05);
98 | }
99 | .sw-badge-category {
100 |     font-size: 0.7rem;
101 |     padding: 2px 6px;
102 |     border-radius: 4px;
103 |     background: #1e293b;
104 |     border: 1px solid #475569;
105 |     color: #e2e8f0;
106 | }
107 | .sw-badge-active {
```

```

108     background: rgba(34, 197, 94, 0.15);
109     border: 1px solid rgba(34, 197, 94, 0.4);
110     color: #4ade80;
111 }
112 .sw-badge-dormant {
113     background: rgba(245, 158, 11, 0.15);
114     border: 1px solid rgba(245, 158, 11, 0.4);
115     color: #fbbf24;
116 }
117 </style>
118
119 <div class="sw-audit-container">
120     <!-- Header Alert & Controls (Compact Strip) -->
121     <div class="card shadow-sm border-secondary-subtle py-1.5 px-3 mb-2 d-flex justify-content-between align-items-center flex-wrap gap-2">
122         <div class="d-flex align-items-center gap-2 flex-wrap">
123             <span class="fs-5">❏ </span>
124             <span class="fw-bold text-white small" style="font-size: 0.92rem;">Аудит программного обеспечения</span>
125             <span class="badge bg-dark border border-secondary text-info fw-normal" style="font-size: 0.7rem;">Software Audit</span>
126             <span class="text-secondary small d-none d-lg-inline" style="font-size: 0.76rem;">| Инвентаризация ПО, UserAssist & Prefetch</span>
127         </div>
128         <div class="d-flex align-items-center gap-1.5 flex-wrap">
129             <span class="badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1" style="font-size: 0.72rem;" id="sw-audit-status-badge">
130                 • Готов
131             </span>
132             <button class="btn btn-xs btn-outline-info rounded-pill px-2.5 py-1" style="font-size: 0.75rem;" id="btn-sw-audit-refresh">
133                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
134             </button>

```

```

135 |         <button class="btn btn-xs btn-outline-light rounded-pill
      | px-2 py-1" style="font-size: 0.75rem;" id="btn-sw-audit-expor
      | t">
136 |             <i class="bi bi-download me-1"></i> Экспорт JSON
137 |         </button>
138 |     </div>
139 </div>
140
141 <!-- Metric Statistics Cards (Compact Density Grid) -->
142 <div class="row g-2 mb-2">
143     <div class="col-6 col-lg-3">
144         <div class="sw-audit-card">
145             <span class="sw-audit-label">Всего программ</span>
146             <div class="d-flex align-items-baseline justify-conten
      | t-between mt-1">
147                 <span class="sw-audit-value text-info" id="sw-metric
      | -total">0</span>
148                 <small class="text-muted" style="font-size: 0.7rem;"
      | >В реестре</small>
149             </div>
150         </div>
151     </div>
152     <div class="col-6 col-lg-3">
153         <div class="sw-audit-card">
154             <span class="sw-audit-label">Активно используются</spa
      | n>
155             <div class="d-flex align-items-baseline justify-conten
      | t-between mt-1">
156                 <span class="sw-audit-value text-success" id="sw-met
      | ric-active">0</span>
157                 <small class="text-muted" style="font-size: 0.7rem;"
      | >Недавние запуски</small>
158             </div>
159         </div>
160     </div>
161     <div class="col-6 col-lg-3">
162         <div class="sw-audit-card">
163             <span class="sw-audit-label">Неиспользуемые (>90 дн)</

```



```

span>
164     <div class="d-flex align-items-baseline justify-content-between mt-1">
165         <span class="sw-audit-value text-warning" id="sw-metric-dormant">0</span>
166         <small class="text-muted" style="font-size: 0.7rem;">Кандидаты на удаление</small>
167     </div>
168 </div>
169 </div>
170 <div class="col-6 col-lg-3">
171     <div class="sw-audit-card">
172         <span class="sw-audit-label">Категоризировано</span>
173         <div class="d-flex align-items-baseline justify-content-between mt-1">
174             <span class="sw-audit-value text-info" id="sw-metric-categorized">0%</span>
175             <small class="text-muted" style="font-size: 0.7rem;">AI эвристика</small>
176         </div>
177     </div>
178 </div>
179 </div>
180
181 <!-- Filters & Controls Bar -->
182 <div class="card bg-dark border-secondary p-2 mb-2" style="border-radius: 8px;">
183     <div class="row g-2 align-items-center">
184         <div class="col-md-5">
185             <div class="input-group input-group-sm">
186                 <span class="input-group-text bg-dark border-secondary text-light"><i class="bi bi-search"></i></span>
187                 <input type="text" class="form-control bg-dark text-white border-secondary" id="sw-search-input" placeholder="Поиск по названию, издателю или назначению...">
188             </div>
189         </div>
190         <div class="col-md-3">

```

```

191 |         <select class="form-select form-select-sm bg-dark text
      | -white border-secondary" id="sw-category-filter">
192 |             <option value="">Все категории</option>
193 |         </select>
194 |     </div>
195 |     <div class="col-md-2">
196 |         <select class="form-select form-select-sm bg-dark text
      | -white border-secondary" id="sw-status-filter">
197 |             <option value="all">Все программы</option>
198 |             <option value="active">Только активные</option>
199 |             <option value="dormant">Только неиспользуемые</optio
      | n>
200 |         </select>
201 |     </div>
202 |     <div class="col-md-2 text-end">
203 |         <span class="badge bg-secondary" id="sw-filtered-count
      | -badge">0 / 0</span>
204 |     </div>
205 | </div>
206 | </div>
207 |
208 | <!-- Main Table Panel -->
209 | <div class="sw-audit-panel">
210 |     <div class="sw-audit-panel-header">
211 |         <span><i class="bi bi-table me-1"></i> Список установлен
      | ных программных продуктов</span>
212 |         <span class="small text-muted" id="sw-last-scanned-time"
      | >Сканирование...</span>
213 |     </div>
214 |     <div class="p-0 table-responsive" style="max-height: 540px
      | ; overflow-y: auto;">
215 |         <table class="sw-audit-table">
216 |             <thead>
217 |                 <tr>
218 |                     <th>Программа</th>
219 |                     <th>Категория</th>
220 |                     <th>Издатель / Версия</th>
221 |                     <th>Последний запуск</th>

```

```

222         <th>Запусков</th>
223         <th>Время работы</th>
224         <th>Размер / Аpx</th>
225         <th style="text-align: right;">Действия</th>
226     </tr>
227 </thead>
228 <tbody id="sw-audit-tbody">
229     <tr><td colspan="8" class="text-center text-muted p-
4"><i class="bi bi-arrow-repeat spin me-1"></i> Загрузка аудит
а программного обеспечения...</td></tr>
230 </tbody>
231 </table>
232 </div>
233 </div>
234 </div>
235
236 <!-- Modal: App Details -->
237 <div class="modal fade" id="swAppDetailModal" tabindex="-1" ar
ia-labelledby="swAppDetailModalLabel" aria-hidden="true">
238     <div class="modal-dialog modal-lg modal-dialog-centered">
239         <div class="modal-content bg-dark text-white border-second
ary">
240             <div class="modal-header border-secondary">
241                 <h5 class="modal-title d-flex align-items-center gap-2
" id="swAppDetailModalLabel">
242                     <span class="fs-5">□ </span> <span id="sw-modal-app-
name">Детали программы</span>
243                 </h5>
244                 <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
245             </div>
246             <div class="modal-body p-3" id="sw-modal-app-body">
247                 <!-- Content will be injected by main.js -->
248             </div>
249             <div class="modal-footer border-secondary">
250                 <button type="button" class="btn btn-secondary btn-sm"
data-bs-dismiss="modal">Заккрыть</button>
251             </div>

```

```
252 |     </div>
253 | </div>
254 | </div>
255 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Software Audit Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/software_audit_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/software_audit_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 // Software Audit Tab JavaScript Module
22 (function() {
23   let allApps = [];
24   let auditReport = null;
25   let isInitialized = false;
26
27   function formatRelativeTime(dateStr) {
28     if (!dateStr) return '<span class="text-muted">Нет данных</span>';
29     const d = new Date(dateStr);
30     if (isNaN(d.getTime())) return '<span class="text-muted">-</span>';
```

```
31 |     const now = new Date();
32 |     const diffMs = now - d;
33 |     const diffSec = Math.floor(diffMs / 1000);
34 |     const diffMin = Math.floor(diffSec / 60);
35 |     const diffHrs = Math.floor(diffMin / 60);
36 |     const diffDays = Math.floor(diffHrs / 24);
37 |
38 |     if (diffDays > 90) {
39 |         return `<span class="text-warning" title="${d.toLocaleSt
ring()}">${diffDays} дн назад</span>`;
40 |     } else if (diffDays > 0) {
41 |         return `<span class="text-success" title="${d.toLocaleSt
ring()}">${diffDays} дн назад</span>`;
42 |     } else if (diffHrs > 0) {
43 |         return `<span class="text-info" title="${d.toLocaleStrin
g()}">${diffHrs} ч назад</span>`;
44 |     } else if (diffMin > 0) {
45 |         return `<span class="text-info" title="${d.toLocaleStrin
g()}">${diffMin} мин назад</span>`;
46 |     } else {
47 |         return `<span class="text-info">Только что</span>`;
48 |     }
49 | }
50 |
51 | function formatDuration(sec) {
52 |     if (!sec || sec <= 0) return '<span class="text-muted">-</
span>';
53 |     const hrs = Math.floor(sec / 3600);
54 |     const mins = Math.floor((sec % 3600) / 60);
55 |     if (hrs > 0) return `${hrs}ч ${mins}м`;
56 |     if (mins > 0) return `${mins} мин`;
57 |     return `${sec} сек`;
58 | }
59 |
60 | async function loadSoftwareAudit() {
61 |     const tbody = document.getElementById('sw-audit-tbody');
62 |     const badge = document.getElementById('sw-audit-status-bad
ge');
```

```
63 |     if (badge) badge.innerText = '● Сканирование реестра...';
64 |
65 |     try {
66 |         // 1. Пытаемся получить сводный аналитический отчет
67 |         let reportRes = await fetch('/api/windows/software/audit
    ');
68 |         if (!reportRes.ok) {
69 |             reportRes = await fetch('/api/v1/windows-admin/software/audit');
70 |         }
71 |         if (reportRes.ok) {
72 |             auditReport = await reportRes.json();
73 |         }
74 |
75 |         // 2. Получаем полный список установленных приложений
76 |         let appsRes = await fetch('/api/windows/software?limit=5
    00');
77 |         if (!appsRes.ok) {
78 |             // Fallback на router_windows_admin
79 |             appsRes = await fetch('/api/v1/windows-admin/software?
    limit=500');
80 |         }
81 |
82 |         if (appsRes.ok) {
83 |             allApps = await appsRes.json();
84 |         } else {
85 |             throw new Error(`HTTP ${appsRes.status}`);
86 |         }
87 |
88 |         // Обновляем метрики
89 |         updateMetrics();
90 |         // Заполняем выпадающий список категорий
91 |         populateCategories();
92 |         // Рендерим таблицу
93 |         renderTable();
94 |
95 |         if (badge) {
96 |             badge.className = 'badge rounded-pill bg-success-subtl
```

```

    e text-success border border-success px-3 py-2';
97     badge.innerText = `• Аудит: Завершено (${allApps.length
h} программ)`;
98     }
99
100     const timeEl = document.getElementById('sw-last-scanned-
time');
101     if (timeEl) {
102         timeEl.innerText = `Обновлено: ${new Date().toLocaleTi
meString()}`;
103     }
104     } catch (err) {
105         console.error('[SoftwareAuditTab] Failed to load softwar
e audit:', err);
106         if (tbody) {
107             tbody.innerHTML = `<tr><td colspan="8" class="text-cen
ter text-danger p-4">Ошибка сбора сведений об установленном ПО
: ${err.message}</td></tr>`;
108         }
109         if (badge) {
110             badge.className = 'badge rounded-pill bg-danger-subtle
text-danger border border-danger px-3 py-2';
111             badge.innerText = '• Ошибка аудита';
112         }
113     }
114 }
115
116 function updateMetrics() {
117     const totalEl = document.getElementById('sw-metric-total')
;
118     const activeEl = document.getElementById('sw-metric-active
');
119     const dormantEl = document.getElementById('sw-metric-dorma
nt');
120     const catEl = document.getElementById('sw-metric-categorie
s');
121
122     if (auditReport) {

```



```
123     if (totalEl) totalEl.innerText = auditReport.total_apps
    || allApps.length;
124     if (activeEl) activeEl.innerText = auditReport.active_ap
    ps_count || 0;
125     if (dormantEl) dormantEl.innerText = auditReport.never_l
    aunched_or_dormant_count || 0;
126     if (catEl) catEl.innerText = Object.keys(auditReport.cat
    egories_breakdown || {}).length;
127   } else {
128     const activeCount = allApps.filter(a => a.execution_info
    && a.execution_info.last_run_time).length;
129     if (totalEl) totalEl.innerText = allApps.length;
130     if (activeEl) activeEl.innerText = activeCount;
131     if (dormantEl) dormantEl.innerText = allApps.length - ac
    tiveCount;
132     const cats = new Set(allApps.map(a => a.category).filter
    (Boolean));
133     if (catEl) catEl.innerText = cats.size;
134   }
135 }
136
137 function populateCategories() {
138   const sel = document.getElementById('sw-category-filter');
139   if (!sel) return;
140
141   const currentVal = sel.value;
142   const cats = Array.from(new Set(allApps.map(a => a.categor
    y).filter(Boolean))).sort();
143
144   sel.innerHTML = '<option value="">Все категории</option>'
    + cats.map(c =>
145     `<option value="${c}" ${c === currentVal ? 'selected' :
    ''}>${c}</option>`
146   ).join('');
147 }
148
149 function getFilteredApps() {
150   const search = (document.getElementById('sw-search-input'))
```

```
?.value || '').toLowerCase().trim();
151     const category = document.getElementById('sw-category-filter')?.value || '';
152     const status = document.getElementById('sw-status-filter')?.value || 'all';
153
154     const dormantThresholdMs = 90 * 24 * 3600 * 1000;
155     const now = Date.now();
156
157     return allApps.filter(app => {
158         // Поиск по ключевым полям
159         if (search) {
160             const text = `${app.display_name || ''} ${app.name || ''} ${app.publisher || ''} ${app.purpose_description || ''}`.toLowerCase();
161             if (!text.includes(search)) return false;
162         }
163
164         // Фильтр по категории
165         if (category && app.category !== category) {
166             return false;
167         }
168
169         // Фильтр по статусу использования
170         const lastRun = app.execution_info?.last_run_time ? new Date(app.execution_info.last_run_time).getTime() : null;
171         if (status === 'active') {
172             if (!lastRun || (now - lastRun > dormantThresholdMs))
173                 return false;
174             if (lastRun && (now - lastRun <= dormantThresholdMs))
175                 return false;
176         }
177         return true;
178     });
179 }
```

```

181 | function renderTable() {
182 |     const tbody = document.getElementById('sw-audit-tbody');
183 |     const badge = document.getElementById('sw-filtered-count-b
    | adge');
184 |     if (!tbody) return;
185 |
186 |     const filtered = getFilteredApps();
187 |     if (badge) {
188 |         badge.innerText = `${filtered.length} / ${allApps.length
    | }`;
189 |     }
190 |
191 |     if (filtered.length === 0) {
192 |         tbody.innerHTML = '<tr><td colspan="8" class="text-cente
    | r text-muted p-4">Программы по заданным критериям не найдены</
    | td></tr>';
193 |         return;
194 |     }
195 |
196 |     tbody.innerHTML = filtered.map((app, idx) => {
197 |         const exec = app.execution_info;
198 |         const lastRunFormatted = exec ? formatRelativeTime(exec.
    | last_run_time) : '<span class="text-muted">Никогда</span>';
199 |         const runCount = exec && exec.run_count ? '<span class="
    | badge bg-secondary">${exec.run_count}</span>' : '<span class="
    | text-muted">0</span>';
200 |         const focusTime = exec ? formatDuration(exec.focus_time_
    | seconds) : '<span class="text-muted">-</span>';
201 |         const sizeStr = app.size_mb > 0 ? `${app.size_mb} MB` :
    | '<span class="text-muted">-</span>';
202 |         const archBadge = app.architecture === 'x64' ? '<span cl
    | ass="badge bg-primary-subtle text-primary" style="font-size: 0
    | .65rem;">x64</span>' : '<span class="badge bg-secondary-subtle
    | text-secondary" style="font-size: 0.65rem;">x86</span>';
203 |
204 |         return `
205 |             <tr class="sw-row-item" data-idx="${idx}" style="curso
    | r: pointer;" title="Нажмите для AI-диагностики и подробных све

```

```

    дений">
206         <td>
207             <div class="fw-bold text-white text-truncate" styl
e="max-width: 260px;" title="{app.display_name || app.name}">
208                 {app.display_name || app.name}
209             </div>
210             <div class="small text-muted text-truncate" style=
"max-width: 260px; font-size: 0.72rem;" title="{app.purpose_d
escription || ''}">
211                 {app.purpose_description || 'Прикладное ПО'}
212             </div>
213         </td>
214         <td>
215             <span class="sw-badge-category">{app.category ||
'Прочее'}</span>
216         </td>
217         <td>
218             <div class="text-truncate text-light" style="max-w
idth: 180px;" title="{app.publisher || 'Неизвестен'}">{app.p
ublisher || '<span class="text-muted">-</span>'}</div>
219             <div class="small text-muted" style="font-size: 0.
72rem;">{app.version || '-'}</div>
220         </td>
221         <td>{lastRunFormatted}</td>
222         <td>{runCount}</td>
223         <td class="small text-light">{focusTime}</td>
224         <td class="small">
225             <div>{sizeStr}</div>
226             <div>{archBadge}</div>
227         </td>
228         <td style="text-align: right;">
229             <button class="btn btn-sm btn-outline-info p-1 px-
2 btn-sw-detail" data-idx="{idx}" title="Подробности">
230                 <i class="bi bi-info-circle"></i>
231             </button>
232         </td>
233     </tr>
234     `;

```

```
235 |     }).join('');
236 |
237 |     // Привязка кликабельности всей строки
238 |     tbody.querySelectorAll('.sw-row-item').forEach(row => {
239 |         row.onclick = (evt) => {
240 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
241 |             ;
242 |             showAppDetails(filtered[idx]);
243 |         };
244 |     });
245 |
246 |     // Привязка обработчиков кнопки деталей
247 |     tbody.querySelectorAll('.btn-sw-detail').forEach(btn => {
248 |         btn.onclick = (evt) => {
249 |             evt.stopPropagation();
250 |             const idx = parseInt(btn.getAttribute('data-idx'), 10)
251 |             ;
252 |             showAppDetails(filtered[idx]);
253 |         };
254 |     });
255 | }
256 |
257 | function showAppDetails(app) {
258 |     if (!app) return;
259 |     const exec = app.execution_info;
260 |
261 |     if (window.AITableModal) {
262 |         window.AITableModal.show({
263 |             icon: '□ ',
264 |             title: app.display_name || app.name,
265 |             subtitle: app.publisher || 'Неизвестный разработчик',
266 |             tableType: 'software',
267 |             badges: [
268 |                 { text: app.category || 'ПО', class: 'badge bg-info' },
269 |                 { text: app.architecture || 'x64', class: 'badge bg-secondary' }
270 |             ],
271 |         });
272 |     }
273 | }
```

```

269         metadata: [
270             { label: 'Продукт', value: app.display_name || app.n
ame },
271             { label: 'Издатель', value: app.publisher || 'Не ука
зан' },
272             { label: 'Версия', value: app.version || 'Не указана
' },
273             { label: 'Категория', value: app.category || 'Прочее
' },
274             { label: 'Размер на диске', value: app.size_mb > 0 ?
`${app.size_mb} MB` : 'Не указан' },
275             { label: 'Дата установки', value: app.install_date |
| 'Неизвестно' },
276             { label: 'Запусков (UserAssist)', value: exec?.run_c
ount ? `${exec.run_count} раз` : '0' },
277             { label: 'Время в фокусе', value: exec ? formatDurat
ion(exec.focus_time_seconds) : '-' },
278             { label: 'Последний запуск', value: exec?.last_run_t
ime ? new Date(exec.last_run_time).toLocaleString() : 'Нет дан
ных', fullWidth: true },
279             { label: 'Путь установки', value: app.install_locati
on || 'Не указан', isCode: true, fullWidth: true }
280         ],
281         rawTitle: 'Команда удаления / Путь',
282         rawContent: app.uninstall_string || app.install_locati
on || '',
283         requestData: {
284             install_location: app.install_location,
285             uninstall_string: app.uninstall_string,
286             version: app.version,
287             size_mb: app.size_mb,
288             run_count: exec?.run_count
289         }
290     });
291     return;
292 }
293 }
294

```

```

295 | function exportJson() {
296 |     const dataStr = "data:text/json;charset=utf-8," + encodeUR
    | IComponent(JSON.stringify(allApps, null, 2));
297 |     const dlAnchor = document.createElement('a');
298 |     dlAnchor.setAttribute("href", dataStr);
299 |     dlAnchor.setAttribute("download", `software_audit_${new Da
    | te().toISOString().slice(0, 10)}.json`);
300 |     document.body.appendChild(dlAnchor);
301 |     dlAnchor.click();
302 |     dlAnchor.remove();
303 | }
304 |
305 | window.initSoftwareAuditTab = async function() {
306 |     console.log('[SoftwareAuditTab] Initializing Software Audi
    | t tab...');
307 |
308 |     // Привязываем контролы поиска и фильтрации
309 |     const searchInput = document.getElementById('sw-search-inp
    | ut');
310 |     const catSelect = document.getElementById('sw-category-fil
    | ter');
311 |     const statusSelect = document.getElementById('sw-status-fi
    | lter');
312 |     const refreshBtn = document.getElementById('btn-sw-audit-r
    | efresh');
313 |     const exportBtn = document.getElementById('btn-sw-audit-ex
    | port');
314 |
315 |     if (searchInput) searchInput.oninput = renderTable;
316 |     if (catSelect) catSelect.onchange = renderTable;
317 |     if (statusSelect) statusSelect.onchange = renderTable;
318 |     if (refreshBtn) refreshBtn.onclick = loadSoftwareAudit;
319 |     if (exportBtn) exportBtn.onclick = exportJson;
320 |
321 |     await loadSoftwareAudit();
322 |     isInitialized = true;
323 | };
324 | })();

```


index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/software_manager_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/software_manager_tab/in
17     dex.html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/software_manager_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Windows Software Manager Tab -->
30 <div class="container-fluid py-2 px-3">
31   <!-- Top Stat Cards -->
32   <div class="row g-2 mb-3">
33     <div class="col-6 col-md-3">
34       <div class="card bg-dark text-white border-secondary h-1
35       00 shadow-sm p-3">
36         <div class="text-muted small text-uppercase fw-semibol
37         d">Установлено пакетов 0 </div>
```

```
31 |         <div class="fs-4 fw-bold mt-1 text-primary" id="sw-ins
    | talled-count">--</div>
32 |         <div class="small text-muted mt-1" id="sw-installed-su
    | b">winget list / MSI</div>
33 |     </div>
34 | </div>
35 |     <div class="col-6 col-md-3">
36 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
37 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Доступно обновлений 0 </div>
38 |             <div class="fs-4 fw-bold mt-1 text-warning" id="sw-upg
    | rades-count">--</div>
39 |             <div class="small text-muted mt-1" id="sw-upgrades-sub
    | ">winget upgrade</div>
40 |         </div>
41 |     </div>
42 |     <div class="col-6 col-md-3">
43 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
44 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Источники пакетов (Sources) 0 </div>
45 |             <div class="fs-4 fw-bold mt-1 text-info" id="sw-source
    | s-count">--</div>
46 |             <div class="small text-muted mt-1" id="sw-sources-sub"
    | >winget / msstore</div>
47 |         </div>
48 |     </div>
49 |     <div class="col-6 col-md-3">
50 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
51 |             <div class="text-muted small text-uppercase fw-semibol
    | d">MSI Инсталлятор 0 </div>
52 |             <div class="fs-4 fw-bold mt-1 text-success" id="sw-msi
    | -status">Готов</div>
53 |             <div class="small text-muted mt-1" id="sw-msi-sub">msi
    | exec.exe engine</div>
54 |         </div>
```

```

55     </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="sw-refresh-btn">
62             <i class="bi bi-arrow-clockwise"></i>
63             <span>Обновить список</span>
64         </button>
65         <button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5" id="sw-check-upgrades-btn">
66             <i class="bi bi-arrow-up-circle"></i>
67             <span>Проверить обновления</span>
68         </button>
69     </div>
70     <div class="d-flex align-items-center gap-2">
71         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="sw-search-input" placeholder="Поиск пакета по имени или ID..." style="width: 260px;">
72     </div>
73 </div>
74
75 <!-- Software Table -->
76 <div class="card bg-dark text-white border-secondary shadow-sm">
77     <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
78         <span class="fw-semibold"><i class="bi bi-box-seam text-primary me-2"></i>Установленные приложения и пакеты (Winget & MSI)</span>
79         <span class="badge bg-secondary" id="sw-table-badge">0 пакетов</span>
80     </div>

```

```
81 |     <div class="card-body p-0 table-responsive" style="max-hei
| ght: 520px; overflow-y: auto;">
82 |         <table class="table table-dark table-hover table-striped
| align-middle mb-0 small" id="sw-table">
83 |             <thead class="sticky-top bg-dark">
84 |                 <tr class="text-secondary">
85 |                     <th>Имя приложения</th>
86 |                     <th>Идентификатор (ID)</th>
87 |                     <th>Версия</th>
88 |                     <th>Доступная версия</th>
89 |                     <th>Источник</th>
90 |                     <th class="text-end">Действия</th>
91 |                 </tr>
92 |             </thead>
93 |             <tbody id="sw-tbody">
94 |                 <tr><td colspan="6" class="text-center text-muted py
| -4">Опрос установленных программ через winget...</td></tr>
95 |             </tbody>
96 |         </table>
97 |     </div>
98 | </div>
99 | </div>
100 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Software Manager Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/software_manager_tab/main.js?v=20261004_v1" type="module"></script>
11 *
12 *   JavaScript Import:
13 *     import { initSoftwareManagerTab } from '/windows/api/webgui/software_manager_tab/main.js';
14 *
15 * File: main.js
16 * Project: ai-breadboard
17 * Package: windows/api/webgui/software_manager_tab
18 * Author: hypo69
19 * Copyright: © 2026 hypo69
20 * Updated: 2026-10-04 12:40:00
21 * =====
22 */
23
24 /**
25 * software_manager_tab/main.js – Управление пакетами и программами (winget, msixexec)
26 * Updated: 2026-10-04 12:40:00
27 */
28
29 const registerTabPoller = window.registerTabPoller || function
  () {};
```

```

30
31 let isInitialized = false;
32 let packagesList = [];
33
34 export async function fetchSoftwareSummary() {
35   try {
36     const res = await fetch('/api/software-manager/summary');
37     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38     const data = await res.json();
39     renderSoftware(data);
40   } catch (err) {
41     console.warn('[SoftwareManager] Ошибка получения списка ПО
: ', err);
42     const tbody = document.getElementById('sw-tbody');
43     if (tbody && packagesList.length === 0) {
44       tbody.innerHTML = `<tr><td colspan="6" class="text-cente
r text-danger py-4">Ошибка загрузки данных: ${err.message}</td
></tr>`;
45     }
46   }
47 }
48
49 function renderSoftware(data) {
50   packagesList = data.packages || [];
51
52   const installedEl = document.getElementById('sw-installed-co
unt');
53   const upgradesEl = document.getElementById('sw-upgrades-coun
t');
54   const sourcesEl = document.getElementById('sw-sources-count'
);
55
56   if (installedEl) installedEl.textContent = data.total_packag
es ?? packagesList.length;
57   if (upgradesEl) upgradesEl.textContent = data.updates_availa
ble_count ?? data.upgrades_count ?? 0;
58   if (sourcesEl) sourcesEl.textContent = data.sources_count ??
(data.winget_available ? 'winget' : 'MSI');

```

```
59 |
60 |   applySoftwareFilters();
61 | }
62 |
63 | function applySoftwareFilters() {
64 |   const tbody = document.getElementById('sw-tbody');
65 |   const searchInput = document.getElementById('sw-search-input
    ');
66 |   const badge = document.getElementById('sw-table-badge');
67 |
68 |   if (!tbody) return;
69 |
70 |   const query = (searchInput ? searchInput.value : '').toLower
    Case().trim();
71 |
72 |   const filtered = packagesList.filter(p => {
73 |     if (!query) return true;
74 |     const nameMatch = (p.name || '').toLowerCase().includes(qu
    ery);
75 |     const idMatch = (p.package_id || p.id || '').toLowerCase()
    .includes(query);
76 |     return nameMatch || idMatch;
77 |   });
78 |
79 |   if (badge) badge.textContent = `${filtered.length} из ${pack
    agesList.length} пакетов`;
80 |
81 |   if (filtered.length === 0) {
82 |     tbody.innerHTML = '<tr><td colspan="6" class="text-center
    text-muted py-4">Пакеты не найдены</td></tr>';
83 |     return;
84 |   }
85 |
86 |   tbody.innerHTML = filtered.slice(0, 150).map(p => {
87 |     const name = p.name || 'Application';
88 |     const id = p.package_id || p.id || '-';
89 |     const ver = p.version || '-';
90 |     const avail = p.available_version || p.latest_version || '
```

```

-';
91     const source = p.source || 'winget';
92     const hasUpgrade = avail && avail !== '-' && avail !== ver
    ;
93
94     return `
95         <tr>
96             <td class="fw-semibold text-light">${name}</td>
97             <td class="font-monospace text-info small">${id}</td>
98             <td class="text-muted small">${ver}</td>
99             <td>
100                 <span class="badge ${hasUpgrade ? 'bg-warning text-d
ark' : 'bg-secondary'}">
101                     ${avail}
102                 </span>
103             </td>
104             <td class="small text-muted">${source}</td>
105             <td class="text-end">
106                 ${hasUpgrade ? `
107                     <button class="btn btn-xs btn-outline-warning py-0
px-2 sw-action-btn" data-id="${id}" data-action="upgrade" tit
le="Обновить пакет">
108                         <i class="bi bi-arrow-up-circle"></i> Обновить
109                     </button>
110                 ` : `
111                     <button class="btn btn-xs btn-outline-danger py-0
px-2 sw-action-btn" data-id="${id}" data-action="uninstall" ti
tle="Удалить пакет">
112                         <i class="bi bi-trash"></i>
113                     </button>
114                 `}
115             </td>
116         </tr>
117     `;
118 }).join('');
119
120 tbody.querySelectorAll('.sw-action-btn').forEach(btn => {
121     btn.addEventListener('click', () => {

```



```
122     const pkgId = btn.dataset.id;
123     const act = btn.dataset.action;
124     executeSoftwareAction(pkgId, act);
125   });
126 });
127 }
128
129 async function executeSoftwareAction(packageId, action) {
130   if (window.toast) window.toast.info('Пакетный менеджер', `${
131     action}: ${packageId}`);
132   try {
133     const res = await fetch('/api/software-manager/action', {
134       method: 'POST',
135       headers: { 'Content-Type': 'application/json' },
136       body: JSON.stringify({ package_id: packageId, action: ac
137         tion, dry_run: false, confirmed_by_user: true })
138     });
139     const result = await res.json();
140     if (res.ok) {
141       if (window.toast) window.toast.success('Успех', `${packa
142         geId}: выполнено`);
143       fetchSoftwareSummary();
144     } else {
145       if (window.toast) window.toast.error('Ошибка', result.de
146         tail || result.message || 'Сбой операции');
147     }
148   } catch (err) {
149     if (window.toast) window.toast.error('Ошибка сети', err.me
150       ssage);
151   }
152 }
153
154 export function initSoftwareManagerTab() {
155   if (isInitialized) {
156     fetchSoftwareSummary();
157     return;
158   }
159   isInitialized = true;
```

```
155 |
156 |     const refreshBtn = document.getElementById('sw-refresh-btn')
    |     ;
157 |     if (refreshBtn) refreshBtn.addEventListener('click', fetchSo
    |     ftwareSummary);
158 |
159 |     const checkUpgradesBtn = document.getElementById('sw-check-u
    |     pgrades-btn');
160 |     if (checkUpgradesBtn) checkUpgradesBtn.addEventListener('cli
    |     ck', () => {
161 |         if (window.toast) window.toast.info('Проверка', 'Поиск дос
    |         тупных обновлений...');
162 |         fetchSoftwareSummary();
163 |     });
164 |
165 |     const searchInput = document.getElementById('sw-search-input
    |     ');
166 |     if (searchInput) searchInput.addEventListener('input', apply
    |     SoftwareFilters);
167 |
168 |     registerTabPoller('tab-software-manager', fetchSoftwareSumma
    |     ry, 30000, { immediate: true });
169 |     fetchSoftwareSummary();
170 | }
171 |
172 | window.initSoftwareManagerTab = initSoftwareManagerTab;
173 | window.fetchSoftwareSummary = fetchSoftwareSummary;
174 |
175 | if (document.getElementById('tab-software-manager') && documen
    |     t.getElementById('sw-tbody')) {
176 |     initSoftwareManagerTab();
177 | }
178 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/software_transparency_tab/index.html
15
16   Tab Frame Integration:
17     <iframe src="/windows/api/webgui/software_transparency_t
18 ab/index.html" class="w-100 h-100 border-0"></iframe>
19
20 File: index.html
21 Project: ai-breadboard
22 Package: windows/api/webgui/software_transparency_tab
23 Author: hypo69
24 Copyright: © 2026 hypo69
25 Updated: 2026-10-01 13:04:40
26 =====
27 =====
28 -->
29
30 <!-- AI Software Transparency Scanner Tab UI -->
31 <div class="container-fluid p-2" id="st-scanner-container">
32   <!-- Верхняя карточка с метриками и кнопками управления -->
33   <div class="card bg-dark text-white border-secondary mb-3 sh
34 adow-sm">
35     <div class="card-body p-3">
36       <div class="d-flex align-items-center justify-content-be
37 tween flex-wrap gap-2 mb-3">
38         <div class="d-flex align-items-center gap-2">
```

```

31 |         <span class="fs-4">□ </span>
32 |         <div>
33 |             <h5 class="mb-0 fw-bold">AI Software Transparency
Scanner</h5>
34 |             <div class="text-muted small">Интеллектуальный инв
ентаризатор установленного ПО, конфигураций, хранилищ и сетевое
й активности</div>
35 |         </div>
36 |     </div>
37 |     <div class="d-flex align-items-center gap-2">
38 |         <span id="st-status-badge" class="badge rounded-pill
bg-success-subtle text-success border border-success px-3 py-
2">
39 |             • Готов к сканированию
40 |         </span>
41 |         <button id="st-btn-refresh" class="btn btn-sm btn-pr
imary d-flex align-items-center gap-1 shadow-sm">
42 |             <i class="bi bi-arrow-clockwise"></i> Сканировать
систему
43 |         </button>
44 |     </div>
45 | </div>
46 |
47 | <!-- Сводные метрики -->
48 | <div class="row g-2">
49 |     <div class="col-6 col-md-3">
50 |         <div class="p-2 bg-secondary bg-opacity-10 border bo
rder-secondary rounded">
51 |             <div class="text-muted small text-uppercase">Устан
овлено ПО</div>
52 |             <div class="fs-4 fw-bold text-info" id="st-metric-
total-apps">0</div>
53 |         </div>
54 |     </div>
55 |     <div class="col-6 col-md-3">
56 |         <div class="p-2 bg-secondary bg-opacity-10 border bo
rder-secondary rounded">
57 |             <div class="text-muted small text-uppercase">Файло

```

```

    в КОНФИГОВ</div>
58         <div class="fs-4 fw-bold text-warning" id="st-metric-total-configs">0</div>
59     </div>
60 </div>
61     <div class="col-6 col-md-3">
62         <div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
63             <div class="text-muted small text-uppercase">Сетевых узлов / Доменов</div>
64             <div class="fs-4 fw-bold text-success" id="st-metric-total-domains">0</div>
65         </div>
66     </div>
67     <div class="col-6 col-md-3">
68         <div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
69             <div class="text-muted small text-uppercase">Время сканирования</div>
70             <div class="fs-4 fw-bold text-light" id="st-metric-scan-duration">0.0 с</div>
71         </div>
72     </div>
73 </div>
74 </div>
75 </div>
76
77 <!-- Фильтры и строка поиска -->
78 <div class="d-flex align-items-center justify-content-between mb-3 flex-wrap gap-2">
79     <div class="d-flex align-items-center gap-2 flex-grow-1" style="max-width: 450px;">
80         <div class="input-group input-group-sm">
81             <span class="input-group-text bg-dark text-secondary border-secondary"><i class="bi bi-search"></i></span>
82             <input type="text" id="st-search-input" class="form-control bg-dark text-white border-secondary" placeholder="Поиск по названию программы или издателю...">

```

```

83     </div>
84 </div>
85 <div class="d-flex align-items-center gap-2 small text-mut
ed">
86     <span>Легенда достоверности:</span>
87     <span class="badge bg-success-subtle text-success border
border-success">□ Локально</span>
88     <span class="badge bg-info-subtle text-info border borde
r-info">□ Документация</span>
89     <span class="badge bg-warning-subtle text-warning border
border-warning">□ Предположение AI</span>
90     <span class="badge bg-secondary-subtle text-secondary bo
rder border-secondary">□ Неизвестно</span>
91 </div>
92 </div>
93
94 <!-- Таблица программ -->
95 <div class="card bg-dark text-white border-secondary shadow-
sm">
96     <div class="table-responsive">
97         <table class="table table-dark table-hover mb-0 align-mi
ddle">
98             <thead class="table-dark text-secondary small border-s
econdary">
99                 <tr>
100                     <th style="width: 25%;">Программа</th>
101                     <th style="width: 15%;">Издатель / Версия</th>
102                     <th style="width: 20%;">Каталоги и хранилище</th>
103                     <th style="width: 15%;">Конфиги & Секреты</th>
104                     <th style="width: 15%;">Сетевые домены</th>
105                     <th style="width: 10%; text-align: right;">AI Анал
из</th>
106                 </tr>
107             </thead>
108             <tbody id="st-table-body">
109                 <tr>
110                     <td colspan="6" class="text-center py-4 text-muted
">

```

```

111 |             Нажмите кнопку «Сканировать систему» для сбора с
112 |             ведений об установленном ПО.
113 |         </td>
114 |     </tr>
115 | </tbody>
116 | </table>
117 | </div>
118 | </div>
119 |
120 | <!-- Модальное окно просмотра очищенного конфигурационного фай
121 | ла -->
122 | <div class="modal fade" id="st-config-modal" tabindex="-1" ari
123 | a-hidden="true">
124 |     <div class="modal-dialog modal-lg modal-dialog-scrollable">
125 |         <div class="modal-content bg-dark text-white border-second
126 | ary">
127 |             <div class="modal-header border-secondary">
128 |                 <h6 class="modal-title fw-bold" id="st-config-modal-ti
129 | tle">Конфигурационный файл</h6>
130 |                 <button type="button" class="btn-close btn-close-white
131 | " data-bs-dismiss="modal" aria-label="Close"></button>
132 |             </div>
133 |             <div class="modal-body">
134 |                 <div class="alert alert-warning py-2 small mb-3">
135 |                     <i class="bi bi-shield-lock-fill me-1"></i> Конфиден
136 | циальные токены, пароли и ключи безопасности автоматически мас
137 | кированы (<code>[REDACTED_SECRET]</code>).
138 |                 </div>
139 |                 <div class="small text-muted mb-2" id="st-config-modal
140 | -path"></div>
141 |                 <pre class="p-3 bg-black rounded border border-seconda
142 | ry text-success small font-monospace overflow-auto" style="max
143 | -height: 400px;" id="st-config-modal-content"></pre>
144 |             </div>
145 |         </div>
146 |     </div>
147 | </div>
148 | </div>

```

```
138 |
139 | <!-- Модальное окно подробного AI-исследования Gemini -->
140 | <div class="modal fade" id="st-research-modal" tabindex="-1" a
    | ria-hidden="true">
141 |     <div class="modal-dialog modal-lg modal-dialog-scrollable">
142 |         <div class="modal-content bg-dark text-white border-second
    | ary">
143 |             <div class="modal-header border-secondary">
144 |                 <h6 class="modal-title fw-bold" id="st-research-modal-
    | title">AI Исследование Gemini</h6>
145 |                 <button type="button" class="btn-close btn-close-white
    | " data-bs-dismiss="modal" aria-label="Close"></button>
146 |             </div>
147 |             <div class="modal-body" id="st-research-modal-body">
148 |                 <div class="text-center py-4">
149 |                     <div class="spinner-border spinner-border-sm text-pr
    | imary me-2"></div>
150 |                     Выполняется аналитическое исследование программы...
151 |                 </div>
152 |             </div>
153 |         </div>
154 |     </div>
155 | </div>
156 |
```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Software Transparency Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/software_transparency_
11  *   tab/main.js?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/software_transparency_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Software Transparency Scanner script
22 // Frontend Controller
23 (function () {
24   let allApps = [];
25   let isScanning = false;
26
27   async function initSoftwareTransparencyTab() {
28     console.log('[TransparencyScanner] Initializing tab controller...');
29     bindEvents();
30     await loadScannerData(false);
```

```

31 | }
32 |
33 | function bindEvents() {
34 |     const btnRefresh = document.getElementById('st-btn-refresh
    ');
35 |     if (btnRefresh) {
36 |         btnRefresh.onclick = () => loadScannerData(true);
37 |     }
38 |
39 |     const searchInput = document.getElementById('st-search-inp
    ut');
40 |     if (searchInput) {
41 |         searchInput.oninput = () => renderTable();
42 |     }
43 | }
44 |
45 | async function loadScannerData(forceRefresh = false) {
46 |     if (isScanning) return;
47 |     isScanning = true;
48 |
49 |     const badge = document.getElementById('st-status-badge');
50 |     const tbody = document.getElementById('st-table-body');
51 |
52 |     if (badge) {
53 |         badge.className = 'badge rounded-pill bg-warning-subtle
    text-warning border border-warning px-3 py-2';
54 |         badge.innerText = '• Идет сбор сведений о ПО и сети...';
55 |     }
56 |
57 |     if (tbody && (!allApps || allApps.length === 0)) {
58 |         tbody.innerHTML = '<tr><td colspan="6" class="text-cen
    te
    r py-4 text-muted"><div class="spinner-border spinner-border-s
    m text-primary me-2" role="status"></div>Выполняется инвентари
    зация установленных программ, конфигураций и сетевых сокетов..
    .</td></tr>';
59 |     }
60 |
61 |     try {

```

```

62 |     const url = '/api/v1/software-scanner/scan?force_refresh
    | =' + (forceRefresh ? 'true' : 'false');
63 |     const res = await fetch(url);
64 |     if (!res.ok) throw new Error('HTTP ' + res.status);
65 |
66 |     const data = await res.json();
67 |     allApps = data.apps || [];
68 |     const summary = data.summary || {};
69 |
70 |     document.getElementById('st-metric-total-apps').innerTex
    | t = summary.total_apps || allApps.length;
71 |     document.getElementById('st-metric-total-configs').inner
    | Text = summary.total_configs_found || 0;
72 |     document.getElementById('st-metric-total-domains').inner
    | Text = summary.total_network_domains || 0;
73 |     document.getElementById('st-metric-scan-duration').inner
    | Text = (summary.scan_duration_sec || 0) + ' c.';
74 |
75 |     if (badge) {
76 |         badge.className = 'badge rounded-pill bg-success-subtl
    | e text-success border border-success px-3 py-2';
77 |         badge.innerText = '● Обнаружено: ' + allApps.length +
    | ' программ';
78 |     }
79 |
80 |     renderTable();
81 | } catch (err) {
82 |     console.error('[TransparencyScanner] Scan error:', err);
83 |     if (badge) {
84 |         badge.className = 'badge rounded-pill bg-danger-subtle
    | text-danger border border-danger px-3 py-2';
85 |         badge.innerText = '● Ошибка сканирования';
86 |     }
87 |     if (tbody) {
88 |         tbody.innerHTML = '<tr><td colspan="6" class="text-cen
    | ter text-danger py-4">Ошибка: ' + err.message + '</td></tr>';
89 |     }
90 | } finally {

```

```
91     isScanning = false;
92   }
93 }
94
95 function renderTable() {
96   const tbody = document.getElementById('st-table-body');
97   const searchInput = document.getElementById('st-search-input');
98   const searchVal = (searchInput ? searchInput.value : '').toLowerCase().trim();
99
100   if (!tbody) return;
101
102   const filtered = allApps.filter((a) => {
103     if (!searchVal) return true;
104     return (
105       a.name.toLowerCase().includes(searchVal) ||
106       (a.publisher && a.publisher.toLowerCase().includes(searchVal)) ||
107       (a.install_location && a.install_location.toLowerCase().includes(searchVal))
108     );
109   });
110
111   if (filtered.length === 0) {
112     tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Ничего не найдено по фильтру «' + searchVal + '»</td></tr>';
113     return;
114   }
115
116   tbody.innerHTML = filtered.map((app) => renderAppRow(app)).join('');
117   bindRowActions();
118 }
119
120 function renderAppRow(app) {
121   const dirs = (app.data_directories || []).map(d => `<div c
```

```

    class="text-truncate small text-muted" title="${d.path}">📁 ${d
    .path} (${(d.total_size_bytes / (1024*1024)).toFixed(1)} MB)</
    div>`).join('') || '<span class="text-muted small">—</span>';
122     const cfgs = (app.config_files || []).map(c => `<button cl
    ass="btn btn-link btn-sm p-0 text-warning text-decoration-none
    text-truncate d-block view-config-btn" data-app-id="${app.id}
    " data-cfg-path="${c.path}" title="${c.path}">📄 ${c.filename}<
    /button>`).join('') || '<span class="text-muted small">—</span
    >';
123     const nets = (app.network_endpoints || []).map(n => `<span
    class="badge bg-secondary me-1 mb-1" title="${n.endpoint}">${
    n.domain || n.endpoint}</span>`).join('') || '<span class="tex
    t-muted small">—</span>';
124
125     return `
126         <tr>
127             <td>
128                 <div class="fw-bold">${escapeHtml(app.name)}</div>
129                 <div class="small text-muted text-truncate" style="m
    ax-width: 250px;" title="${app.install_location || ''}">${esca
    peHtml(app.install_location || '')}</div>
130             </td>
131             <td>
132                 <div>${escapeHtml(app.publisher || '-')}</div>
133                 <div class="small text-muted">${escapeHtml(app.versi
    on || '')}</div>
134             </td>
135             <td>${dirs}</td>
136             <td>${cfgs}</td>
137             <td>${nets}</td>
138             <td class="text-end">
139                 <button class="btn btn-sm btn-outline-info research-
    app-btn" data-app-id="${app.id}" title="AI Исследование">
140                     <i class="bi bi-robot"></i> Анализ
141                 </button>
142             </td>
143         </tr>
144     `;

```

```
145 | }
146 |
147 | function bindRowActions() {
148 |     document.querySelectorAll('.view-config-btn').forEach(btn
=> {
149 |         btn.onclick = () => {
150 |             const appId = btn.getAttribute('data-app-id');
151 |             const cfgPath = btn.getAttribute('data-cfg-path');
152 |             openConfigModal(appId, cfgPath);
153 |         };
154 |     });
155 |
156 |     document.querySelectorAll('.research-app-btn').forEach(btn
=> {
157 |         btn.onclick = () => {
158 |             const appId = btn.getAttribute('data-app-id');
159 |             openResearchModal(appId);
160 |         };
161 |     });
162 | }
163 |
164 | function openConfigModal(appId, cfgPath) {
165 |     const app = allApps.find(a => a.id === appId);
166 |     const cfg = app ? (app.config_files || []).find(c => c.pat
h === cfgPath) : null;
167 |
168 |     document.getElementById('st-config-modal-path').innerText
= cfgPath;
169 |     document.getElementById('st-config-modal-content').innerTe
xt = cfg ? (cfg.content_preview || '[Пустой файл]') : '[Файл н
е найден]';
170 |
171 |     const modalEl = document.getElementById('st-config-modal')
;
172 |     if (modalEl && window.bootstrap) {
173 |         new bootstrap.Modal(modalEl).show();
174 |     }
175 | }
```

```
176
177   async function openResearchModal(appId) {
178     const app = allApps.find(a => a.id === appId);
179     const titleEl = document.getElementById('st-research-modal-
    -title');
180     const bodyEl = document.getElementById('st-research-modal-
    body');
181
182     if (titleEl) titleEl.innerText = 'AI Исследование: ' + (ap
    p ? app.name : appId);
183     if (bodyEl) {
184       bodyEl.innerHTML = '<div class="text-center py-4"><div c
    lass="spinner-border spinner-border-sm text-primary me-2"></di
    v>Выполняется аналитическое исследование программы через Gemin
    i...</div>';
185     }
186
187     const modalEl = document.getElementById('st-research-modal
    ');
188     if (modalEl && window.bootstrap) {
189       new bootstrap.Modal(modalEl).show();
190     }
191
192     try {
193       const res = await fetch('/api/v1/software-scanner/resear
    ch', {
194         method: 'POST',
195         headers: { 'Content-Type': 'application/json' },
196         body: JSON.stringify({ app_id: appId })
197       });
198       if (!res.ok) throw new Error('HTTP ' + res.status);
199       const data = await res.json();
200
201       if (bodyEl) {
202         bodyEl.innerHTML = `
203           <h6>□ Описание программы:</h6>
204           <p>${escapeHtml(data.summary || data.description ||
    'Нет данных')}</p>
```

```

205 |         ${data.risk_score !== undefined ? `<div class="mb-2"
    | ><strong>Оценка риска:</strong> <span class="badge bg-${data.r
    | isk_score > 5 ? 'danger' : 'success'}">${data.risk_score}/10</
    | span></div>` : ''}
206 |         ${data.recommendations ? `<h6>📌 Рекомендации:</h6><
    | p>${escapeHtml(data.recommendations)}</p>` : ''}
207 |         `;
208 |     }
209 |     } catch (err) {
210 |         if (bodyEl) {
211 |             bodyEl.innerHTML = `<div class="alert alert-danger">❗
    | Ошибка анализа: ${err.message}</div>`;
212 |         }
213 |     }
214 | }
215 |
216 | function escapeHtml(text) {
217 |     if (!text) return '';
218 |     const div = document.createElement('div');
219 |     div.innerText = text;
220 |     return div.innerHTML;
221 | }
222 |
223 | window.initSoftwareTransparencyTab = initSoftwareTransparenc
    | yTab;
224 |
225 | if (document.readyState === 'loading') {
226 |     document.addEventListener('DOMContentLoaded', initSoftware
    | TransparencyTab);
227 | } else {
228 |     initSoftwareTransparencyTab();
229 | }
230 | })();
231 |

```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Sources - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/sources/main.js?v=2026
11  *     1001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/sources
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * sources_tab/main.js
23  * Логика вкладки редактирования источников.
24  */
25
26 let currentSourcesObj = {};
27
28 async function loadSourcesRaw() {
29     const editor = document.getElementById('sources-json-editor');
30     const tbody = document.getElementById('sources-list-body')
31     ;
```

```

31
32     if (editor) {
33         editor.disabled = true;
34         editor.value = 'Загрузка...';
35     }
36     if (tbody) {
37         tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Загрузка источников...</td></tr>';
38     }
39
40     try {
41         const response = await window.api.fetch('/api/admin/sources/raw');
42         const rawContent = response.content || '{}';
43
44         if (editor) editor.value = rawContent;
45
46         try {
47             currentSourcesObj = JSON.parse(rawContent);
48             renderSourcesTable();
49         } catch (e) {
50             console.error('Ошибка парсинга JSON для таблицы:',
51                 e);
52             if (tbody) {
53                 tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-danger">Ошибка формата JSON. Таблица недоступна.</td></tr>';
54             }
55
56         } catch (err) {
57             console.error('Ошибка загрузки источников:', err);
58             if (editor) editor.value = 'Ошибка загрузки:\n' + err.message;
59             if (tbody) tbody.innerHTML = `<tr><td colspan="6" class="text-center py-4 text-danger">Ошибка загрузки: ${err.message}</td></tr>`;
60         } finally {

```

```

61         if (editor) editor.disabled = false;
62     }
63 }
64
65 async function saveSourcesRaw(content) {
66     const editor = document.getElementById('sources-json-editor');
67
68     // Проверка синтаксиса на клиенте перед отправкой
69     try {
70         JSON.parse(content);
71     } catch (e) {
72         if (typeof showNotification === 'function') {
73             showNotification('Синтаксическая ошибка JSON: ' +
74 e.message, 'danger');
75         } else {
76             alert('Синтаксическая ошибка JSON: ' + e.message);
77         }
78         return false;
79     }
80
81     try {
82         if (editor) editor.disabled = true;
83         const response = await window.api.fetch('/api/admin/sources/raw', {
84             method: 'POST',
85             headers: { 'Content-Type': 'application/json' },
86             body: JSON.stringify({ content: content })
87         });
88
89         if (response.status === 'ok') {
90             if (typeof showNotification === 'function') {
91                 showNotification('Источники успешно сохранены!', 'success');
92             }
93             return true;
94         } catch (err) {

```

```
95 |         console.error('Ошибка сохранения источников:', err);
96 |         if (typeof showNotification === 'function') {
97 |             showNotification('Ошибка сохранения: ' + err.messa
| ge, 'danger');
98 |         }
99 |         return false;
100 |     } finally {
101 |         if (editor) editor.disabled = false;
102 |     }
103 | }
104 |
105 | function renderSourcesTable() {
106 |     const tbody = document.getElementById('sources-list-body')
| ;
107 |     const filterSelect = document.getElementById('sources-cate
| gory-filter');
108 |     if (!tbody) return;
109 |
110 |     const filterVal = filterSelect ? filterSelect.value : 'all
| ';
111 |     tbody.innerHTML = '';
112 |
113 |     let hasItems = false;
114 |
115 |     for (const [category, items] of Object.entries(currentSour
| cesObj)) {
116 |         if (!Array.isArray(items)) continue; // Пропускаем мет
| аданные самого файла (например, "metadata": {...})
117 |
118 |         // Маппинг категорий
119 |         let catForFilter = 'other';
120 |         if (category === 'metadata_apis') catForFilter = 'meta
| data';
121 |         else if (category === 'streaming_search') catForFilter
| = 'streaming_search';
122 |         else if (category.includes('streaming') || category ==
| = 'direct_sites') catForFilter = 'streaming';
123 |         else if (category === 'iframe_players') catForFilter =
```

```

    'embed_player';
124     else if (category === 'torrent_trackers') catForFilter
    = 'torrent';
125     else if (category === 'video_search') catForFilter = '
search';
126
127     if (filterVal !== 'all' && catForFilter !== filterVal)
    continue;
128
129     items.forEach((item, index) => {
130         hasItems = true;
131         const tr = document.createElement('tr');
132
133         const isEnabled = item.enabled !== false; // по ум
    олчанию true
134
135         tr.innerHTML = `
136             <td>
137                 <div class="form-check form-switch">
138                     <input class="form-check-input toggle-
source-btn" type="checkbox" role="switch"
139                         data-category="${category}" data-i
    ndex="${index}" ${isEnabled ? 'checked' : ''}>
140                 </div>
141             </td>
142             <td class="fw-bold">${item.name || item.id ||
    'Без названия'}</td>
143             <td><a href="${item.url || '#'}" target="_blan
    k" class="small text-truncate d-inline-block" style="max-width
    : 150px;">${item.url || '-'}</a></td>
144             <td><span class="badge bg-secondary">${categor
    y}</span></td>
145             <td class="small text-muted">${item.descriptio
    n || '-'}</td>
146             <td class="text-end">
147                 <button class="btn btn-sm btn-outline-dang
    er delete-source-btn" data-category="${category}" data-index="
    ${index}" title="Удалить">

```

```

148         <i class="bi bi-trash"></i>
149     </button>
150 </td>
151 `;
152     tbody.appendChild(tr);
153 });
154 }
155
156 if (!hasItems) {
157     tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Нет источников для отображения</td></tr>';
158 }
159
160 // Привязываем обработчики к сгенерированным кнопкам
161 document.querySelectorAll('.toggle-source-btn').forEach(btn => {
162     btn.addEventListener('change', handleToggleSource);
163 });
164
165 document.querySelectorAll('.delete-source-btn').forEach(btn => {
166     btn.addEventListener('click', handleDeleteSource);
167 });
168 }
169
170 async function handleToggleSource(e) {
171     const category = e.target.getAttribute('data-category');
172     const index = parseInt(e.target.getAttribute('data-index'));
173
174     if (currentSourcesObj[category] && currentSourcesObj[category][index]) {
175         currentSourcesObj[category][index].enabled = e.target.checked;
176         const newJson = JSON.stringify(currentSourcesObj, null, 2);
177     }

```

```

178 |         // Обновляем текстовое поле и сервер
179 |         const editor = document.getElementById('sources-json-e
ditor');
180 |         if (editor) editor.value = newJson;
181 |
182 |         await saveSourcesRaw(newJson);
183 |     }
184 | }
185 |
186 | async function handleDeleteSource(e) {
187 |     const btn = e.target.closest('button');
188 |     const category = btn.getAttribute('data-category');
189 |     const index = parseInt(btn.getAttribute('data-index'));
190 |
191 |     if (confirm('Вы уверены, что хотите удалить этот источник?
')) {
192 |         if (currentSourcesObj[category] && currentSourcesObj[c
ategory][index]) {
193 |             currentSourcesObj[category].splice(index, 1);
194 |
195 |             const newJson = JSON.stringify(currentSourcesObj,
null, 2);
196 |             const editor = document.getElementById('sources-js
on-editor');
197 |             if (editor) editor.value = newJson;
198 |
199 |             const success = await saveSourcesRaw(newJson);
200 |             if (success) renderSourcesTable();
201 |         }
202 |     }
203 | }
204 |
205 | async function handleAddSource() {
206 |     const id = document.getElementById('new-source-id')?.value
.trim();
207 |     const name = document.getElementById('new-source-name')?.v
alue.trim();
208 |     const url = document.getElementById('new-source-url')?.val

```

```
ue.trim();
209     const filterCat = document.getElementById('new-source-cate
gory')?.value;
210     const desc = document.getElementById('new-source-descripti
on')?.value.trim();
211
212     if (!id || !name || !url) {
213         window.showToast?.('Заполните обязательные поля: ID, Н
азвание и URL', 'warning') || alert('Заполните обязательные по
ля: ID, Название и URL');
214         return;
215     }
216
217     // Определяем в какую секцию JSON положить новый элемент
218     let targetCategory = 'other';
219     if (filterCat === 'metadata') targetCategory = 'metadata_a
pis';
220     else if (filterCat === 'streaming_search') targetCategory
= 'streaming_search';
221     else if (filterCat === 'streaming') targetCategory = 'dire
ct_sites';
222     else if (filterCat === 'embed_player') targetCategory = 'i
frame_players';
223     else if (filterCat === 'torrent') targetCategory = 'torren
t_trackers';
224     else if (filterCat === 'search') targetCategory = 'video_s
earch';
225
226     if (!currentSourcesObj[targetCategory]) {
227         currentSourcesObj[targetCategory] = [];
228     }
229
230     const newItem = {
231         id: id,
232         name: name,
233         url: url,
234         description: desc,
235         enabled: true
```



```
236 |     };
237 |
238 |     currentSourcesObj[targetCategory].push(newItem);
239 |
240 |     const newJson = JSON.stringify(currentSourcesObj, null, 2)
241 |     ;
242 |     const editor = document.getElementById('sources-json-editor');
243 |     if (editor) editor.value = newJson;
244 |
245 |     const success = await saveSourcesRaw(newJson);
246 |     if (success) {
247 |         renderSourcesTable();
248 |         // Очищаем форму
249 |         document.getElementById('form-add-source').reset();
250 |     }
251 | }
252 | // Инициализация вкладки
253 | window.initSourcesTab = function() {
254 |     console.log('Инициализация вкладки Источники...');
255 |
256 |     const btnRefresh = document.getElementById('btn-refresh-sources-json');
257 |     const btnSave = document.getElementById('btn-save-sources-json');
258 |     const btnSaveAll = document.getElementById('btn-save-all-sources');
259 |     const btnAdd = document.getElementById('btn-add-source');
260 |     const filterSelect = document.getElementById('sources-category-filter');
261 |     const editor = document.getElementById('sources-json-editor');
262 |
263 |     if (btnRefresh) btnRefresh.onclick = loadSourcesRaw;
264 |
265 |     if (btnSave) {
266 |         btnSave.onclick = () => {
```

```

267         if (editor) saveSourcesRaw(editor.value).then(() =
    > {
268             // Обновляем таблицу после ручного изменения J
    JSON
269             try {
270                 currentSourcesObj = JSON.parse(editor.valu
    e);
271                 renderSourcesTable();
272             } catch(e){}
273         });
274     };
275 }
276
277 if (btnSaveAll) {
278     btnSaveAll.onclick = btnSave.onclick;
279 }
280
281 if (btnAdd) {
282     btnAdd.onclick = handleAddSource;
283 }
284
285 if (filterSelect) {
286     filterSelect.addEventListener('change', renderSourcesT
    able);
287 }
288
289 // Также обновляем таблицу при потере фокуса редактором, е
    сли JSON валидный
290 if (editor) {
291     editor.addEventListener('blur', () => {
292         try {
293             currentSourcesObj = JSON.parse(editor.value);
294             renderSourcesTable();
295         } catch(e){}
296     });
297 }
298
299 // Загружаем данные при открытии

```

```
300 |     loadSourcesRaw();  
301 | };  
302 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/sources_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/sources_tab/index.html"
15 |       class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/sources_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Sources Registry Tab -->
28 <div class="container-fluid">
29 |   <div class="alert alert-info d-flex justify-content-between
30 |     align-items-center flex-wrap gap-2 mb-3 shadow-sm">
31 |     <div>
32 |       <h5 class="mb-1"><i class="bi bi-collection-play me-2"><
33 |         /i> ☐ Реестр источников данных</h5>
34 |       <p class="mb-0 small text-secondary">Управление списком
35 |         сайтов, стримингов и парсеров для медиатеки.</p>
```

```

30     </div>
31     <div>
32         <button class="btn btn-outline-primary btn-sm rounded-pi
ll" onclick="const t = document.querySelector('[data-bs-target
=\#tab-search\']'); if(t) t.click();">
33             <i class="bi bi-globe2 me-1"></i> Настройки веб-поиска
(MCP) →
34         </button>
35     </div>
36 </div>
37
38 <div class="row g-3">
39     <!-- LEFT: Add new source -->
40     <div class="col-4">
41         <div class="card mb-3">
42             <div class="card-header bg-dark text-white">
43                 <span>□ Добавить источник</span>
44             </div>
45             <div class="card-body">
46                 <form id="form-add-source" onsubmit="event.preventDe
fault();">
47                     <div class="mb-2">
48                         <label for="new-source-id" class="form-label sma
ll fw-bold">ID (уникальный):</label>
49                         <input type="text" class="form-control form-cont
rol-sm" id="new-source-id"
50                             placeholder="Например: my_site" required>
51                     </div>
52                     <div class="mb-2">
53                         <label for="new-source-name" class="form-label s
mall fw-bold">Название:</label>
54                         <input type="text" class="form-control form-cont
rol-sm" id="new-source-name"
55                             placeholder="Например: My Site" required>
56                     </div>
57                     <div class="mb-2">
58                         <label for="new-source-url" class="form-label sm
all fw-bold">URL:</label>

```

```

59 |         <input type="url" class="form-control form-control-sm" id="new-source-url"
60 |             placeholder="https://example.com" required>
61 |     </div>
62 |     <div class="mb-2">
63 |         <label for="new-source-category" class="form-label small fw-bold">Категория:</label>
64 |         <select class="form-select form-select-sm" id="new-source-category">
65 |             <option value="streaming">streaming – стриминг
66 |             <option value="streaming_search">streaming_search – поиск стримингов</option>
67 |             <option value="metadata">metadata – метаданные
68 |             <option value="embed_player">embed_player – встроенный плеер</option>
69 |             <option value="torrent">torrent – торрент-трекер</option>
70 |             <option value="search">search – поисковик</option>
71 |             <option value="other">other – другое</option>
72 |         </select>
73 |     </div>
74 |     <div class="mb-3">
75 |         <label for="new-source-description" class="form-label small fw-bold">Описание:</label>
76 |         <textarea class="form-control form-control-sm" id="new-source-description" rows="2"
77 |             placeholder="Краткое описание источника"></textarea>
78 |     </div>
79 |     <div class="d-grid">
80 |         <button type="button" class="btn btn-success btn-sm" id="btn-add-source">
81 |             □ Добавить
82 |     </button>

```

```
83 |         </div>
84 |     </form>
85 | </div>
86 | </div>
87 |
88 | <!-- JSON Raw Editor -->
89 | <div class="card">
90 |     <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
91 |         <span>🔄 JSON-редактор</span>
92 |         <div class="d-flex gap-2">
93 |             <button class="btn btn-outline-light btn-sm" id="btn-refresh-sources-json">
94 |                 <i class="bi bi-arrow-clockwise"></i>
95 |             </button>
96 |             <button class="btn btn-success btn-sm" id="btn-save-sources-json">
97 |                 <i class="bi bi-floppy"></i> Сохранить JSON
98 |             </button>
99 |         </div>
100 |     </div>
101 |     <div class="card-body p-2">
102 |         <p class="small text-muted mb-2">Прямое редактирование JSON. Используйте с осторожностью.</p>
103 |         <textarea
104 |             id="sources-json-editor"
105 |             class="form-control font-monospace"
106 |             style="min-height: 250px; max-height: 250px; resize: none; font-size: 0.78rem;"
107 |             spellcheck="false"
108 |         ></textarea>
109 |     </div>
110 | </div>
111 | </div>
112 |
113 | <!-- RIGHT: Sources Table -->
114 | <div class="col-8">
115 |     <div class="card">
```

```

116 |         <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
117 |             <span>☰ Список источников</span>
118 |             <button class="btn btn-outline-light btn-sm" id="btn-refresh-sources">
119 |                 <i class="bi bi-arrow-clockwise"></i> Обновить
120 |             </button>
121 |         </div>
122 |         <div class="card-body p-0">
123 |             <!-- Category filter -->
124 |             <div class="p-2 border-bottom d-flex gap-2 flex-wrap align-items-center">
125 |                 <span class="small text-muted">Фильтр:</span>
126 |                 <select class="form-select form-select-sm" id="sources-category-filter" style="width: auto;">
127 |                     <option value="all">Все категории</option>
128 |                     <option value="streaming">Стриминг</option>
129 |                     <option value="streaming_search">Поиск стримингов</option>
130 |                     <option value="metadata">Метаданные</option>
131 |                     <option value="embed_player">Встроенные плееры</option>
132 |                     <option value="torrent">Торренты</option>
133 |                     <option value="search">Поисковики</option>
134 |                     <option value="other">Другое</option>
135 |                 </select>
136 |             </div>
137 |             <!-- Table container with vertical scroll -->
138 |             <div style="height: 500px; overflow-y: scroll; overflow-x: auto;">
139 |                 <table class="table table-striped table-hover align-middle mb-0" id="sources-table" style="min-width: 600px;">
140 |                     <thead class="table-dark">
141 |                         <tr>
142 |                             <th style="width: 5%;">On</th>
143 |                             <th style="width: 15%;">Название</th>
144 |                             <th style="width: 25%;">URL</th>
145 |                             <th style="width: 15%;">Категория</th>

```



```
146 |             <th style="width: 25%;">Описание</th>
147 |             <th style="width: 15%;" class="text-end">Дей
    | ствия</th>
148 |         </tr>
149 |     </thead>
150 |     <tbody id="sources-list-body" style="word-wrap:
    | break-word; overflow-wrap: break-word;">
151 |         <tr>
152 |             <td colspan="6" class="text-center py-4 text
    | -muted">Загрузка источников...</td>
153 |         </tr>
154 |     </tbody>
155 | </table>
156 | </div>
157 | </div>
158 | </div>
159 | </div>
160 | </div>
161 |
162 | <!-- Save all button at bottom -->
163 | <div class="mt-3 text-end">
164 |     <button class="btn btn-primary" id="btn-save-all-sources">
165 |         <i class="bi bi-floppy"></i> Сохранить все изменения
166 |     </button>
167 | </div>
168 | </div>
169 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Sources Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/sources_tab/main.js?v=
11  *       20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/sources_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * sources_tab/main.js
23  * Логика вкладки редактирования источников.
24  */
25
26 let currentSourcesObj = {};
27
28 async function loadSourcesRaw() {
29     const editor = document.getElementById('sources-json-editor');
30     const tbody = document.getElementById('sources-list-body')
31     ;
```

```
31 |
32 |     if (editor) {
33 |         editor.disabled = true;
34 |         editor.value = 'Загрузка...';
35 |     }
36 |     if (tbody) {
37 |         tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Загрузка источников...</td></tr>';
38 |     }
39 |
40 |     try {
41 |         const response = await window.api.fetch('/api/admin/sources/raw');
42 |         const rawContent = response.content || '{}';
43 |
44 |         if (editor) editor.value = rawContent;
45 |
46 |         try {
47 |             currentSourcesObj = JSON.parse(rawContent);
48 |             renderSourcesTable();
49 |         } catch (e) {
50 |             console.error('Ошибка парсинга JSON для таблицы:',
51 |                 e);
52 |             if (tbody) {
53 |                 tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-danger">Ошибка формата JSON. Таблица недоступна.</td></tr>';
54 |             }
55 |
56 |         } catch (err) {
57 |             console.error('Ошибка загрузки источников:', err);
58 |             if (editor) editor.value = 'Ошибка загрузки:\n' + err.message;
59 |             if (tbody) tbody.innerHTML = `<tr><td colspan="6" class="text-center py-4 text-danger">Ошибка загрузки: ${err.message}</td></tr>`;
60 |         } finally {
```

```

61         if (editor) editor.disabled = false;
62     }
63 }
64
65 async function saveSourcesRaw(content) {
66     const editor = document.getElementById('sources-json-editor');
67
68     // Проверка синтаксиса на клиенте перед отправкой
69     try {
70         JSON.parse(content);
71     } catch (e) {
72         if (typeof showNotification === 'function') {
73             showNotification('Синтаксическая ошибка JSON: ' +
74 e.message, 'danger');
75         } else {
76             alert('Синтаксическая ошибка JSON: ' + e.message);
77         }
78         return false;
79     }
80
81     try {
82         if (editor) editor.disabled = true;
83         const response = await window.api.fetch('/api/admin/sources/raw', {
84             method: 'POST',
85             headers: { 'Content-Type': 'application/json' },
86             body: JSON.stringify({ content: content })
87         });
88
89         if (response.status === 'ok') {
90             if (typeof showNotification === 'function') {
91                 showNotification('Источники успешно сохранены!', 'success');
92             }
93             return true;
94         } catch (err) {

```

```

95 |         console.error('Ошибка сохранения источников:', err);
96 |         if (typeof showNotification === 'function') {
97 |             showNotification('Ошибка сохранения: ' + err.messa
| ge, 'danger');
98 |         }
99 |         return false;
100 |     } finally {
101 |         if (editor) editor.disabled = false;
102 |     }
103 | }
104 |
105 | function renderSourcesTable() {
106 |     const tbody = document.getElementById('sources-list-body')
| ;
107 |     const filterSelect = document.getElementById('sources-cate
| gory-filter');
108 |     if (!tbody) return;
109 |
110 |     const filterVal = filterSelect ? filterSelect.value : 'all
| ';
111 |     tbody.innerHTML = '';
112 |
113 |     let hasItems = false;
114 |
115 |     for (const [category, items] of Object.entries(currentSour
| cesObj)) {
116 |         if (!Array.isArray(items)) continue; // Пропускаем мет
| аданные самого файла (например, "metadata": {...})
117 |
118 |         // Маппинг категорий
119 |         let catForFilter = 'other';
120 |         if (category === 'metadata_apis') catForFilter = 'meta
| data';
121 |         else if (category === 'streaming_search') catForFilter
| = 'streaming_search';
122 |         else if (category.includes('streaming') || category ==
| = 'direct_sites') catForFilter = 'streaming';
123 |         else if (category === 'iframe_players') catForFilter =

```

```

    'embed_player';
124     else if (category === 'torrent_trackers') catForFilter
    = 'torrent';
125     else if (category === 'video_search') catForFilter = '
search';
126
127     if (filterVal !== 'all' && catForFilter !== filterVal)
    continue;
128
129     items.forEach((item, index) => {
130         hasItems = true;
131         const tr = document.createElement('tr');
132
133         const isEnabled = item.enabled !== false; // по ум
    олчанию true
134
135         tr.innerHTML = `
136             <td>
137                 <div class="form-check form-switch">
138                     <input class="form-check-input toggle-
source-btn" type="checkbox" role="switch"
139                         data-category="${category}" data-i
    ndex="${index}" ${isEnabled ? 'checked' : ''}>
140                 </div>
141             </td>
142             <td class="fw-bold">${item.name || item.id ||
    'Без названия'}</td>
143             <td><a href="${item.url || '#'}" target="_blan
    k" class="small text-truncate d-inline-block" style="max-width
    : 150px;">${item.url || '-'}</a></td>
144             <td><span class="badge bg-secondary">${categor
    y}</span></td>
145             <td class="small text-muted">${item.descriptio
    n || '-'}</td>
146             <td class="text-end">
147                 <button class="btn btn-sm btn-outline-dang
    er delete-source-btn" data-category="${category}" data-index="
    ${index}" title="Удалить">

```

```

148         <i class="bi bi-trash"></i>
149     </button>
150 </td>
151 `;
152     tbody.appendChild(tr);
153 });
154 }
155
156 if (!hasItems) {
157     tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Нет источников для отображения</td></tr>';
158 }
159
160 // Привязываем обработчики к сгенерированным кнопкам
161 document.querySelectorAll('.toggle-source-btn').forEach(btn => {
162     btn.addEventListener('change', handleToggleSource);
163 });
164
165 document.querySelectorAll('.delete-source-btn').forEach(btn => {
166     btn.addEventListener('click', handleDeleteSource);
167 });
168 }
169
170 async function handleToggleSource(e) {
171     const category = e.target.getAttribute('data-category');
172     const index = parseInt(e.target.getAttribute('data-index'));
173
174     if (currentSourcesObj[category] && currentSourcesObj[category][index]) {
175         currentSourcesObj[category][index].enabled = e.target.checked;
176         const newJson = JSON.stringify(currentSourcesObj, null, 2);
177     }

```

```
178 |         // Обновляем текстовое поле и сервер
179 |         const editor = document.getElementById('sources-json-e
ditor');
180 |         if (editor) editor.value = newJson;
181 |
182 |         await saveSourcesRaw(newJson);
183 |     }
184 | }
185 |
186 | async function handleDeleteSource(e) {
187 |     const btn = e.target.closest('button');
188 |     const category = btn.getAttribute('data-category');
189 |     const index = parseInt(btn.getAttribute('data-index'));
190 |
191 |     if (confirm('Вы уверены, что хотите удалить этот источник?
')) {
192 |         if (currentSourcesObj[category] && currentSourcesObj[c
ategory][index]) {
193 |             currentSourcesObj[category].splice(index, 1);
194 |
195 |             const newJson = JSON.stringify(currentSourcesObj,
null, 2);
196 |             const editor = document.getElementById('sources-js
on-editor');
197 |             if (editor) editor.value = newJson;
198 |
199 |             const success = await saveSourcesRaw(newJson);
200 |             if (success) renderSourcesTable();
201 |         }
202 |     }
203 | }
204 |
205 | async function handleAddSource() {
206 |     const id = document.getElementById('new-source-id')?.value
.trim();
207 |     const name = document.getElementById('new-source-name')?.v
alue.trim();
208 |     const url = document.getElementById('new-source-url')?.val
```



```
ue.trim();
209     const filterCat = document.getElementById('new-source-cate
gory')?.value;
210     const desc = document.getElementById('new-source-descripti
on')?.value.trim();
211
212     if (!id || !name || !url) {
213         window.showToast?.('Заполните обязательные поля: ID, Н
азвание и URL', 'warning') || alert('Заполните обязательные по
ля: ID, Название и URL');
214         return;
215     }
216
217     // Определяем в какую секцию JSON положить новый элемент
218     let targetCategory = 'other';
219     if (filterCat === 'metadata') targetCategory = 'metadata_a
pis';
220     else if (filterCat === 'streaming_search') targetCategory
= 'streaming_search';
221     else if (filterCat === 'streaming') targetCategory = 'dire
ct_sites';
222     else if (filterCat === 'embed_player') targetCategory = 'i
frame_players';
223     else if (filterCat === 'torrent') targetCategory = 'torren
t_trackers';
224     else if (filterCat === 'search') targetCategory = 'video_s
earch';
225
226     if (!currentSourcesObj[targetCategory]) {
227         currentSourcesObj[targetCategory] = [];
228     }
229
230     const newItem = {
231         id: id,
232         name: name,
233         url: url,
234         description: desc,
235         enabled: true
```

```
236 |     };
237 |
238 |     currentSourcesObj[targetCategory].push(newItem);
239 |
240 |     const newJson = JSON.stringify(currentSourcesObj, null, 2)
241 |     ;
242 |     const editor = document.getElementById('sources-json-editor');
243 |     if (editor) editor.value = newJson;
244 |
245 |     const success = await saveSourcesRaw(newJson);
246 |     if (success) {
247 |         renderSourcesTable();
248 |         // Очищаем форму
249 |         document.getElementById('form-add-source').reset();
250 |     }
251 | }
252 | // Инициализация вкладки
253 | window.initSourcesTab = function() {
254 |     console.log('Инициализация вкладки Источники...');
255 |
256 |     const btnRefresh = document.getElementById('btn-refresh-sources-json');
257 |     const btnSave = document.getElementById('btn-save-sources-json');
258 |     const btnSaveAll = document.getElementById('btn-save-all-sources');
259 |     const btnAdd = document.getElementById('btn-add-source');
260 |     const filterSelect = document.getElementById('sources-category-filter');
261 |     const editor = document.getElementById('sources-json-editor');
262 |
263 |     if (btnRefresh) btnRefresh.onclick = loadSourcesRaw;
264 |
265 |     if (btnSave) {
266 |         btnSave.onclick = () => {
```

```

267         if (editor) saveSourcesRaw(editor.value).then(() =
    > {
268             // Обновляем таблицу после ручного изменения J
    JSON
269             try {
270                 currentSourcesObj = JSON.parse(editor.valu
    e);
271                 renderSourcesTable();
272             } catch(e){}
273         });
274     };
275 }
276
277 if (btnSaveAll) {
278     btnSaveAll.onclick = btnSave.onclick;
279 }
280
281 if (btnAdd) {
282     btnAdd.onclick = handleAddSource;
283 }
284
285 if (filterSelect) {
286     filterSelect.addEventListener('change', renderSourcesT
    able);
287 }
288
289 // Также обновляем таблицу при потере фокуса редактором, е
    сли JSON валидный
290 if (editor) {
291     editor.addEventListener('blur', () => {
292         try {
293             currentSourcesObj = JSON.parse(editor.value);
294             renderSourcesTable();
295         } catch(e){}
296     });
297 }
298
299 // Загружаем данные при открытии

```

```
300 |     loadSourcesRaw();
301 | };
302 |
303 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 =====
9 Description:
10   Шаблон клиентского веб-интерфейса аудита автозапуска Windows
11   (Startup Auditor).
12   Полная поддержка светлых и тёмных тем с использованием дизай
13   н-токенов CSS.
14
15 Usage Examples:
16   Web Route:
17   /windows/api/webgui/startup_auditor_tab/index.html
18
19   Tab Frame Integration:
20   <iframe src="/windows/api/webgui/startup_auditor_tab/ind
21   ex.html" class="w-100 h-100 border-0"></iframe>
22
23 File: index.html
24 Project: ai-breadboard
25 Package: windows/api/webgui/startup_auditor_tab
26 Author: hypo69
27 Copyright: © 2026 hypo69
28 Updated: 2026-10-06 14:05:00
29 =====
30 =====
31 -->
32
33 <!-- Startup Auditor Tab Content -->
34 <style>
35   .sa-container {
36     padding: 0.5rem 0.75rem;
37     color: var(--text-color);
38   }
```

```
31 | .sa-card {
32 |     background: var(--surface-1) !important;
33 |     border: 1px solid var(--border-color) !important;
34 |     border-radius: 8px;
35 |     padding: 0.45rem 0.75rem;
36 |     display: flex;
37 |     flex-direction: column;
38 |     justify-content: space-between;
39 |     height: 100%;
40 |     min-height: 58px;
41 |     box-shadow: var(--shadow-sm);
42 |     transition: border-color 0.15s ease, box-shadow 0.15s ease
    | ;
43 | }
44 | .sa-card:hover {
45 |     border-color: var(--nav-active) !important;
46 |     box-shadow: var(--shadow-md);
47 | }
48 | .sa-card-label {
49 |     font-size: 0.68rem;
50 |     text-transform: uppercase;
51 |     letter-spacing: 0.05em;
52 |     color: var(--text-muted);
53 |     font-weight: 600;
54 |     display: flex;
55 |     justify-content: space-between;
56 |     align-items: center;
57 |     line-height: 1.1;
58 | }
59 | .sa-card-value {
60 |     font-size: 1.25rem;
61 |     font-weight: 700;
62 |     font-family: var(--font-mono, ui-monospace, SFMono-Regular
    | , Menlo, Monaco, Consolas, monospace);
63 |     line-height: 1.15;
64 | }
65 | .sa-card-sub {
66 |     font-size: 0.7rem;
```

```
67 |     color: var(--text-muted);
68 |     line-height: 1.1;
69 | }
70 |
71 | .sa-panel {
72 |     background: var(--surface-1) !important;
73 |     border: 1px solid var(--border-color) !important;
74 |     border-radius: 8px;
75 |     overflow: hidden;
76 |     box-shadow: var(--shadow-sm);
77 | }
78 | .sa-panel-header {
79 |     background: var(--surface-2) !important;
80 |     border-bottom: 1px solid var(--border-color) !important;
81 |     padding: 0.45rem 0.85rem;
82 |     font-weight: 600;
83 |     font-size: 0.82rem;
84 |     color: var(--nav-active);
85 |     display: flex;
86 |     align-items: center;
87 |     justify-content: space-between;
88 |     flex-wrap: wrap;
89 |     gap: 0.35rem;
90 | }
91 |
92 | .sa-table {
93 |     width: 100%;
94 |     font-size: 0.78rem;
95 |     border-collapse: collapse;
96 |     color: var(--text-color);
97 |     background: var(--surface-1);
98 | }
99 | .sa-table thead th {
100 |     background: var(--surface-2) !important;
101 |     padding: 6px 10px;
102 |     color: var(--text-muted);
103 |     font-weight: 600;
104 |     font-size: 0.72rem;
```

```
105 |     text-transform: uppercase;
106 |     letter-spacing: 0.04em;
107 |     border-bottom: 1px solid var(--border-color);
108 |     white-space: nowrap;
109 |     position: sticky;
110 |     top: 0;
111 |     z-index: 2;
112 | }
113 | .sa-table tbody tr {
114 |     background: var(--surface-1) !important;
115 |     transition: background-color 0.1s ease;
116 | }
117 | .sa-table tbody tr:nth-child(even) {
118 |     background: var(--surface-2) !important;
119 | }
120 | .sa-table tbody tr:hover {
121 |     background: var(--nav-hover-bg, rgba(0, 0, 0, 0.04)) !impo
122 | rtant;
123 | }
124 | .sa-table td {
125 |     padding: 6px 10px;
126 |     border-bottom: 1px solid var(--border-subtle, var(--border
127 | -color));
128 |     vertical-align: middle;
129 |     color: var(--text-color);
130 | }
131 | /* Badge styles */
132 | .sa-badge {
133 |     font-size: 0.7rem;
134 |     font-weight: 600;
135 |     padding: 2px 6px;
136 |     border-radius: 4px;
137 |     display: inline-flex;
138 |     align-items: center;
139 |     gap: 3px;
140 |     letter-spacing: 0.02em;
141 |     white-space: nowrap;
```



```
141 | }
142 | .sa-badge-enabled {
143 |     background: var(--status-success-bg, rgba(34, 197, 94, 0.1
144 | 5));
145 |     border: 1px solid var(--status-success-border, rgba(34, 19
146 | 7, 94, 0.45));
147 |     color: var(--status-success-text, #185223);
148 | }
149 | .sa-badge-disabled {
150 |     background: var(--status-danger-bg, rgba(239, 68, 68, 0.18
151 | ));
152 |     border: 1px solid var(--status-danger-border, rgba(239, 68
153 | , 68, 0.45));
154 |     color: var(--status-danger-text, #7e1f1f);
155 |     font-weight: bold;
156 | }
157 | .sa-badge-loc {
158 |     background: var(--surface-2);
159 |     border: 1px solid var(--border-color);
160 |     color: var(--nav-active);
161 |     font-family: var(--font-mono, monospace);
162 | }
163 | .sa-badge-cat {
164 |     background: var(--surface-2);
165 |     border: 1px solid var(--border-color);
166 |     color: var(--text-muted);
167 | }
168 |
169 | /* Risk Badges */
170 | .sa-risk-critical {
171 |     background: var(--status-danger-bg, rgba(239, 68, 68, 0.22
172 | ));
173 |     border: 1px solid var(--status-danger-border, rgba(239, 68
174 | , 68, 0.6));
175 |     color: var(--status-danger-text, #7e1f1f);
176 | }
177 | .sa-risk-suspicious {
178 |     background: var(--status-warning-bg, rgba(249, 115, 22, 0.
```

```
22));
173     border: 1px solid var(--status-warning-border, rgba(249, 1
174     15, 22, 0.6));
175     color: var(--status-warning-text, #704705);
176 }
177 .sa-risk-warning {
178     background: var(--status-warning-bg, rgba(234, 179, 8, 0.2
179     2));
180     border: 1px solid var(--status-warning-border, rgba(234, 1
181     79, 8, 0.6));
182     color: var(--status-warning-text, #704705);
183 }
184 .sa-risk-notice {
185     background: var(--status-info-bg, rgba(56, 189, 248, 0.18)
186     );
187     border: 1px solid var(--status-info-border, rgba(56, 189,
188     248, 0.5));
189     color: var(--status-info-text, #184674);
190 }
191 .sa-risk-clean {
192     background: var(--status-success-bg, rgba(34, 197, 94, 0.1
193     6));
194     border: 1px solid var(--status-success-border, rgba(34, 19
195     7, 94, 0.45));
196     color: var(--status-success-text, #185223);
197 }
198
199 /* Toggle Switch button */
200 .sa-toggle-btn {
201     background: transparent;
202     border: none;
203     cursor: pointer;
204     padding: 1px 4px;
205     border-radius: 4px;
206     font-size: 1.15rem;
207     line-height: 1;
208     transition: transform 0.15s ease, opacity 0.15s ease;
209     display: inline-flex;
```

```

203     align-items: center;
204     justify-content: center;
205 }
206 .sa-toggle-btn:hover {
207     transform: scale(1.15);
208 }
209 .sa-toggle-btn:active {
210     transform: scale(0.95);
211 }
212
213 .sa-input {
214     background: var(--surface-1) !important;
215     border: 1px solid var(--border-color) !important;
216     color: var(--text-color) !important;
217     font-size: 0.8rem;
218     padding: 0.25rem 0.5rem;
219 }
220 .sa-input:focus {
221     border-color: var(--nav-active) !important;
222     box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.2) !important;
223 }
224 .sa-input option {
225     background: var(--surface-1);
226     color: var(--text-color);
227 }
228 </style>
229
230 <div class="sa-container" id="startup-auditor-container">
231
232     <!-- Header Alert Banner (Compact Strip) -->
233     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 sa-header-banner"
234         style="background: var(--surface-1); border: 1px solid var(--border-color); color: var(--text-color);">
235         <div class="d-flex align-items-center gap-2 flex-wrap">
236             <span class="fs-5">□ </span>
237             <span class="fw-bold small" style="font-size: 0.92rem; color: var(--text-color);">Аудит автозапуска Windows</span>

```

```
237 |         <span class="badge border fw-normal" style="font-size: 0
| .7rem; background: var(--surface-2); border-color: var(--borde
| r-color) !important; color: var(--nav-active);">Startup Audito
| r</span>
238 |         <span class="badge border fw-normal d-none d-sm-inline-f
| lex align-items-center gap-1" style="font-size: 0.7rem; backgr
| ound: var(--surface-2); border-color: var(--border-color) !imp
| ortant; color: var(--text-muted);" id="sa-telemetry-status-pil
| l" title="Статус снимка автозапуска в базе телеметрии">
239 |         <i class="bi bi-database-check text-success"></i> <spa
| n id="sa-telemetry-last-snap">Телеметрия: синхронизировано</sp
| an>
240 |         </span>
241 |         <span class="text-muted small d-none d-lg-inline" style=
| "font-size: 0.76rem;">| Run/RunOnce, папки автозагрузки, IFE0
| и службы</span>
242 |     </div>
243 |     <div class="d-flex align-items-center gap-1.5 flex-wrap">
244 |         <button class="btn btn-xs btn-primary rounded-pill px-3
| py-1 d-flex align-items-center gap-1.5 fw-semibold shadow-sm"
| style="font-size: 0.76rem;" id="sa-btn-refresh-telemetry" titl
| e="Асинхронно перезапустить сбор автозапуска и зафиксировать о
| тличия в телеметрии">
245 |             <i class="bi bi-arrow-repeat" id="sa-refresh-spinner">
| </i> <span>Обновить</span>
246 |         </button>
247 |         <button class="btn btn-xs btn-outline-secondary rounded-
| pill px-2.5 py-1 d-flex align-items-center gap-1" style="font-
| size: 0.75rem;" id="sa-btn-view-changes" title="Просмотр зафик
| сированных изменений между снимками телеметрии">
248 |             <i class="bi bi-clock-history"></i> <span>Изменения</s
| pan>
249 |             <span class="badge bg-warning text-dark px-1.5 py-0.5
| rounded-pill d-none" id="sa-changes-badge" style="font-size: 0
| .65rem;">0</span>
250 |         </button>
251 |         <button class="btn btn-xs btn-outline-secondary rounded-
| pill px-2 py-1" style="font-size: 0.75rem;" id="sa-btn-export-
```

```

    json" title="Экспорт в JSON">
252     <i class="bi bi-file-earmark-code me-1"></i> JSON
253     </button>
254     <button class="btn btn-xs btn-outline-secondary rounded-
pill px-2 py-1" style="font-size: 0.75rem;" id="sa-btn-export-
csv" title="Экспорт в CSV">
255     <i class="bi bi-file-earmark-spreadsheet me-1"></i> CS
V
256     </button>
257 </div>
258 </div>
259
260 <!-- Top Metric & Health Score Cards (Compact Density Grid)
-->
261 <div class="row g-2 mb-2">
262     <!-- Health Score Card -->
263     <div class="col-6 col-md-3">
264         <div class="sa-card">
265             <div class="sa-card-label">
266                 <span>Индекс чистоты</span>
267                 <i class="bi bi-shield-check text-success"></i>
268             </div>
269             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
270                 <div class="d-flex align-items-baseline gap-1">
271                     <div class="sa-card-value text-success" id="sa-hea
lth-score">--</div>
272                     <span class="text-muted small" style="font-size: 0
.75rem;">/ 100</span>
273                 </div>
274                 <div class="sa-card-sub text-end" id="sa-health-stat
us">Оценка безопасности</div>
275             </div>
276         </div>
277     </div>
278
279     <!-- Active & Total Items -->
280     <div class="col-6 col-md-3">

```

```

281 |     <div class="sa-card">
282 |         <div class="sa-card-label">
283 |             <span>Элементы автозапуска</span>
284 |             <i class="bi bi-play-circle text-info"></i>
285 |         </div>
286 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
287 |             <div class="d-flex align-items-baseline gap-1">
288 |                 <div class="sa-card-value text-info" id="sa-active-count">--</div>
289 |                 <span class="text-muted small" style="font-size: 0.75rem;">из <span id="sa-total-count" class="fw-bold" style="color: var(--text-color);">--</span></span>
290 |             </div>
291 |             <div class="sa-card-sub text-end">Откл: <span id="sa-disabled-count" class="text-danger fw-bold">--</span></div>
292 |         </div>
293 |     </div>
294 | </div>
295 |
296 | <!-- Broken Entries Card -->
297 | <div class="col-6 col-md-3">
298 |     <div class="sa-card">
299 |         <div class="sa-card-label">
300 |             <span>Битые ссылки</span>
301 |             <i class="bi bi-exclamation-triangle text-warning"></i>
302 |         </div>
303 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
304 |             <div class="d-flex align-items-baseline gap-1">
305 |                 <div class="sa-card-value text-warning" id="sa-broken-count">--</div>
306 |                 <span class="text-muted small" style="font-size: 0.75rem;">записей</span>
307 |             </div>
308 |             <div class="sa-card-sub text-end">Файлы не найдены</div>

```

```

309         </div>
310     </div>
311 </div>
312
313 <!-- Threats & Critical Alerts -->
314 <div class="col-6 col-md-3">
315     <div class="sa-card">
316         <div class="sa-card-label">
317             <span>Угрозы и аномалии</span>
318             <i class="bi bi-shield-slash text-danger"></i>
319         </div>
320         <div class="d-flex align-items-baseline justify-content-between mt-1">
321             <div class="d-flex align-items-baseline gap-1">
322                 <div class="sa-card-value text-danger" id="sa-threats-count">--</div>
323                 <span class="text-muted small" style="font-size: 0.75rem;">предупреждений</span>
324             </div>
325             <div class="sa-card-sub text-end">IFE0 и скрипты</div>
326         </div>
327     </div>
328 </div>
329 </div>
330
331 <!-- Recommendations Alert Banner (Compact Sleek Strip) -->
332 <div class="py-1.5 px-3 mb-2 rounded-2 d-flex align-items-center justify-content-between" id="sa-recommendations-box" style="background: var(--surface-1); border: 1px solid var(--border-color); border-left: 3px solid #eab308; font-size: 0.78rem;">
333     <div class="d-flex align-items-center gap-2">
334         <span class="text-warning">⚠ </span>
335         <div class="d-flex align-items-center gap-2 flex-wrap">
336             <span class="fw-semibold" style="color: var(--text-color);" id="sa-rec-title">Рекомендации:</span>
337             <span class="text-muted" id="sa-rec-desc">Загрузка рек

```

```

омендаций по оптимизации...</span>
338     </div>
339   </div>
340 </div>
341
342 <!-- Search & Filter Controls (Compact Toolbar) -->
343 <div class="card p-2 mb-2 sa-panel">
344   <div class="row g-2 align-items-center">
345     <!-- Search Input -->
346     <div class="col-12 col-md-4">
347       <div class="input-group input-group-sm">
348         <span class="input-group-text sa-input text-muted py
-1" style="border-right: none;"><i class="bi bi-search"></i></
span>
349         <input type="text" class="form-control form-control-
sm sa-input" id="sa-search-input" placeholder="Поиск по имени,
команде или пути...">
350       </div>
351     </div>
352
353     <!-- Risk Filter -->
354     <div class="col-6 col-md-3">
355       <select class="form-select form-select-sm sa-input" id
="sa-filter-risk">
356         <option value="">Все уровни риска</option>
357         <option value="critical">⚠ Critical (Критические)</
option>
358         <option value="suspicious">⚠ Suspicious (Подозритель
ные)</option>
359         <option value="warning">⚠ Warning (Предупреждения)</
option>
360         <option value="notice">ℹ Notice (Инфо)</option>
361         <option value="clean">✅ Clean (Безопасные)</option>
362       </select>
363     </div>
364
365     <!-- Location Filter -->
366     <div class="col-6 col-md-3">

```



```

367 |         <select class="form-select form-select-sm sa-input" id
    | ="sa-filter-location">
368 |             <option value="">Все точки автозапуска</option>
369 |             <option value="registry_run">Реестр: Run</option>
370 |             <option value="registry_runonce">Реестр: RunOnce</op
    | tion>
371 |             <option value="startup_folder_user">Папка автозагруз
    | ки (User)</option>
372 |             <option value="startup_folder_common">Папка автозагр
    | узки (Common)</option>
373 |             <option value="winlogon">Winlogon</option>
374 |             <option value="ifeo">IFE0 Перехваты</option>
375 |             <option value="scheduled_task">Планировщик задач</op
    | tion>
376 |             <option value="windows_service">Службы Windows</opti
    | on>
377 |         </select>
378 |     </div>
379 |
380 |     <!-- State Filter -->
381 |     <div class="col-12 col-md-2">
382 |         <select class="form-select form-select-sm sa-input" id
    | ="sa-filter-state">
383 |             <option value="">Все статусы</option>
384 |             <option value="enabled">Только включенные</option>
385 |             <option value="disabled">Только отключенные</option>
386 |         </select>
387 |     </div>
388 | </div>
389 | </div>
390 |
391 | <!-- Entries Data Table Panel -->
392 | <div class="sa-panel">
393 |     <div class="sa-panel-header">
394 |         <span><i class="bi bi-table me-1"></i> Список зарегистри
    | рованных элементов автозапуска</span>
395 |         <span class="small text-muted" id="sa-table-count">Загру
    |зка...</span>

```

```

396     </div>
397     <div class="table-responsive" style="max-height: calc(100v
h - 245px); overflow-y: auto;">
398         <table class="sa-table" id="sa-entries-table">
399             <thead>
400                 <tr>
401                     <th style="width: 80px;" class="text-center">Стату
c</th>
402                     <th style="width: 230px;">Имя / Программа</th>
403                     <th style="width: 170px;">Точка автозапуска</th>
404                     <th style="width: 140px;">Категория</th>
405                     <th style="width: 110px;" class="text-center">Риск
</th>
406                     <th>Исполняемый путь / Аргументы</th>
407                     <th style="width: 110px;" class="text-center">Влия
ние</th>
408                     <th style="width: 80px;" class="text-center">Дейст
вие</th>
409                 </tr>
410             </thead>
411             <tbody id="sa-table-body">
412                 <tr>
413                     <td colspan="8" class="text-center py-4 text-muted
">
414                         <div class="spinner-border spinner-border-sm tex
t-primary me-2" role="status"></div>
415                         Выполняется сканирование точек автозапуска Windo
ws...
416                     </td>
417                 </tr>
418             </tbody>
419         </table>
420     </div>
421 </div>
422
423 <!-- Program Details & AI Context Modal -->
424 <div class="modal fade" id="sa-entry-detail-modal" tabindex=
"-1" aria-hidden="true">

```

```
425 |     <div class="modal-dialog modal-lg modal-dialog-centered mo
    | dal-dialog-scrollable">
426 |         <div class="modal-content shadow-lg">
427 |             <div class="modal-header py-2 px-3">
428 |                 <div class="d-flex align-items-center gap-2">
429 |                     <span class="fs-5 text-primary">□ </span>
430 |                     <h5 class="modal-title fw-bold text-primary mb-0"
    | id="sa-modal-title">Детали программы</h5>
431 |                     <span id="sa-modal-status-badge"></span>
432 |                     <span id="sa-modal-risk-badge"></span>
433 |                 </div>
434 |                 <button type="button" class="btn-close" data-bs-dism
    | iss="modal" aria-label="Close"></button>
435 |             </div>
436 |             <div class="modal-body py-3 px-3">
437 |                 <!-- Metadata Grid -->
438 |                 <div class="row g-2 mb-3 small">
439 |                     <div class="col-sm-6"><strong class="text-muted">И
    | мя / Программа:</strong> <span id="sa-modal-name" class="fw-se
    | mibold"></span></div>
440 |                     <div class="col-sm-6"><strong class="text-muted">И
    | здатель:</strong> <span id="sa-modal-publisher"></span></div>
441 |                     <div class="col-sm-6"><strong class="text-muted">Т
    | очка автозапуска:</strong> <span id="sa-modal-location-type" c
    | lass="sa-badge sa-badge-loc"></span></div>
442 |                     <div class="col-sm-6"><strong class="text-muted">К
    | атегория:</strong> <span id="sa-modal-category" class="sa-badg
    | e sa-badge-cat"></span></div>
443 |                     <div class="col-sm-6"><strong class="text-muted">В
    | лияние на запуск:</strong> <span id="sa-modal-boot-impact"></s
    | pan></div>
444 |                     <div class="col-sm-6"><strong class="text-muted">Ц
    | ифровая подпись:</strong> <span id="sa-modal-is-signed"></span
    | ></div>
445 |                     <div class="col-sm-6"><strong class="text-muted">С
    | татус файла:</strong> <span id="sa-modal-file-exists"></span><
    | /div>
446 |                     <div class="col-sm-6"><strong class="text-muted">P
```

```

    азмёр файла:</strong> <span id="sa-modal-file-size"></span></div>
  </div>
447     <div class="col-sm-12"><strong class="text-muted">
    Расположение / Ключ:</strong> <span id="sa-modal-location-path
    " class="font-monospace text-muted small" style="word-break: b
    reak-all;"></span></div>
448     </div>
449
450     <!-- Command & Path Section -->
451     <div class="card p-3 mb-3" style="background: var(--
    surface-2); border: 1px solid var(--border-color);">
452         <div class="d-flex justify-content-between align-i
    tems-center mb-1">
453             <h6 class="small text-uppercase text-muted fw-bo
    ld mb-0">Команда и исполняемый файл</h6>
454         </div>
455         <div class="small font-monospace mb-2 text-primary
    " id="sa-modal-executable-path" style="word-break: break-all;"
    ></div>
456         <div class="small font-monospace text-muted" id="s
    a-modal-command" style="white-space: pre-wrap; word-break: bre
    ak-all;"></div>
457     </div>
458
459     <!-- AI Diagnostic & Explanation Section -->
460     <div class="card p-3" id="sa-modal-ai-section" style
    ="background: var(--surface-1); border: 1px solid var(--border
    -color);">
461         <div class="d-flex justify-content-between align-i
    tems-center mb-2">
462             <h6 class="fw-bold text-primary mb-0"><i class="
    bi bi-robot me-1"></i> AI Contextual Explanation</h6>
463             <button class="btn btn-sm btn-outline-primary py
    -0 px-2 rounded-pill" id="btn-sa-modal-diagnose">
464                 <i class="bi bi-magic me-1"></i>Анализ контекс
    та
465             </button>
466         </div>

```

```

467         <div class="small text-muted" id="sa-modal-ai-expl
| anation">
468             Нажмите «Анализ контекста», чтобы получить экспе
| ртное объяснение назначения программы от языковой модели, оцен
| ку рисков и рекомендации по оптимизации.
469         </div>
470     </div>
471 </div>
472     <div class="modal-footer py-2 px-3 d-flex justify-cont
| ent-between">
473         <div>
474             <button type="button" class="btn btn-sm btn-outlin
| e-warning me-1" id="btn-sa-modal-toggle">
475                 <i class="bi bi-toggle2-on me-1"></i>Переключить
| автозапуск
476             </button>
477             <button type="button" class="btn btn-sm btn-outlin
| e-secondary" id="btn-sa-modal-copy">
478                 <i class="bi bi-clipboard me-1"></i>Копировать п
| уть
479             </button>
480         </div>
481         <button type="button" class="btn btn-sm btn-secondar
| y" data-bs-dismiss="modal">Заккрыть</button>
482     </div>
483 </div>
484 </div>
485 </div>
486
487 <!-- Telemetry Diff / Changes History Modal -->
488 <div class="modal fade" id="sa-changes-modal" tabindex="-1"
| aria-hidden="true">
489     <div class="modal-dialog modal-xl modal-dialog-centered mo
| dal-dialog-scrollable">
490         <div class="modal-content shadow-lg">
491             <div class="modal-header py-2 px-3">
492                 <div class="d-flex align-items-center gap-2">
493                     <span class="fs-5 text-warning">⚠ </span>

```

```

494         <h5 class="modal-title fw-bold text-warning mb-0">
История изменений автозапуска (Телеметрия)</h5>
495         <span class="badge border bg-surface-2 text-muted
fw-normal" id="sa-changes-modal-count">0 записей</span>
496     </div>
497     <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Close"></button>
498 </div>
499     <div class="modal-body py-3 px-3">
500         <div class="d-flex justify-content-between align-ite
ms-center mb-2 flex-wrap gap-2">
501             <div class="small text-muted">
502                 Каждый запуск телеметрии и ручное обновление («О
бновить») фиксирует снимок автозапуска в SQLite и вычисляет ди
фференциальные изменения.
503             </div>
504             <button class="btn btn-xs btn-outline-primary roun
ded-pill px-2.5 py-1" id="sa-btn-refresh-changes-modal">
505                 <i class="bi bi-arrow-clockwise me-1"></i>Обнови
ть историю
506             </button>
507         </div>
508         <div class="table-responsive" style="max-height: 55v
h;">
509             <table class="sa-table" id="sa-changes-table">
510                 <thead>
511                     <tr>
512                         <th style="width: 140px;">Время фиксации</th>
513                         <th style="width: 125px;" class="text-center
">Тип изменения</th>
514                         <th style="width: 190px;">Программа</th>
515                         <th style="width: 150px;">Точка автозапуска<
/th>
516                         <th>Детали изменения</th>
517                     </tr>
518                 </thead>
519                 <tbody id="sa-changes-table-body">

```

```
520 |             <tr>
521 |                 <td colspan="5" class="text-center py-4 text
| -muted">
522 |                     Загрузка истории изменений...
523 |                 </td>
524 |             </tr>
525 |         </tbody>
526 |     </table>
527 | </div>
528 | </div>
529 | <div class="modal-footer py-2 px-3">
530 |     <button type="button" class="btn btn-sm btn-secondar
| y" data-bs-dismiss="modal">Закрыть</button>
531 | </div>
532 | </div>
533 | </div>
534 | </div>
535 | </div>
536 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Startup Auditor Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main (Startup Auditor).
7   *   Обеспечивает загрузку, фильтрацию, переключение состояния
8   *   , AI-диагностику,
9   *   асинхронное обновление снимков телеметрии и аудит дифференциальных изменений.
10  *
11  * Usage Examples:
12  *   HTML Integration:
13  *     <script src="/windows/api/webgui/startup_auditor_tab/main.js?v=20261006_v15" type="module"></script>
14  *
15  * File: main.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/startup_auditor_tab
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-06 14:05:00
21  * =====
22  */
23 // Windows Startup Auditor Frontend Logic
24 (function () {
25   let allEntries = [];
26   let currentSummary = null;
27
28   async function initStartupAuditorTab() {
29     console.log('[StartupAuditor] Initializing tab...');
```



```
30 |     bindEvents();
31 |     await loadAuditData();
32 |     checkRecentChangesBadge().catch(() => {});
33 | }
34 |
35 | function bindEvents() {
36 |     const btnRefreshTelemetry = document.getElementById('sa-bt
n-refresh-telemetry');
37 |     if (btnRefreshTelemetry) {
38 |         btnRefreshTelemetry.onclick = () => refreshStartupTeleme
try();
39 |     }
40 |
41 |     const btnRefreshLegacy = document.getElementById('sa-btn-r
efresh');
42 |     if (btnRefreshLegacy) {
43 |         btnRefreshLegacy.onclick = () => refreshStartupTelemetry
();
44 |     }
45 |
46 |     const btnViewChanges = document.getElementById('sa-btn-vie
w-changes');
47 |     if (btnViewChanges) {
48 |         btnViewChanges.onclick = () => showChangesModal();
49 |     }
50 |
51 |     const btnRefreshChangesModal = document.getElementById('sa
-btn-refresh-changes-modal');
52 |     if (btnRefreshChangesModal) {
53 |         btnRefreshChangesModal.onclick = () => loadChangesData()
;
54 |     }
55 |
56 |     const searchInput = document.getElementById('sa-search-inp
ut');
57 |     if (searchInput) {
58 |         searchInput.oninput = () => renderFilteredTable();
59 |     }
```

```
60
61     const filterRisk = document.getElementById('sa-filter-risk
    ');
62     if (filterRisk) {
63         filterRisk.onchange = () => renderFilteredTable();
64     }
65
66     const filterLocation = document.getElementById('sa-filter-
location');
67     if (filterLocation) {
68         filterLocation.onchange = () => renderFilteredTable();
69     }
70
71     const filterState = document.getElementById('sa-filter-sta
te');
72     if (filterState) {
73         filterState.onchange = () => renderFilteredTable();
74     }
75
76     const btnExportJson = document.getElementById('sa-btn-expo
rt-json');
77     if (btnExportJson) {
78         btnExportJson.onclick = () => {
79             window.open('/api/v1/startup-auditor/export?format=jso
n', '_blank');
80         };
81     }
82
83     const btnExportCsv = document.getElementById('sa-btn-expor
t-csv');
84     if (btnExportCsv) {
85         btnExportCsv.onclick = () => {
86             window.open('/api/v1/startup-auditor/export?format=csv
', '_blank');
87         };
88     }
89 }
90
```

```

91 |  /**
92 |   * Асинхронно перезапускает сбор данных автозапуска через AP
   | I телеметрии,
93 |   * регистрирует снимок в SQLite и сохраняет дифференциальные
   | изменения.
94 |   */
95 |   async function refreshStartupTelemetry() {
96 |       const btn = document.getElementById('sa-btn-refresh-teleme
   | try');
97 |       const spinner = document.getElementById('sa-refresh-spinne
   | r');
98 |       const statusText = document.getElementById('sa-telemetry-l
   | ast-snap');
99 |
100 |       if (btn) btn.disabled = true;
101 |       if (spinner) spinner.classList.add('spinner-border', 'spin
   | ner-border-sm');
102 |
103 |       try {
104 |           if (statusText) statusText.textContent = 'Телеметрия: сб
   | ор снимка...';
105 |
106 |           const res = await window.api.fetch('/api/v1/startup-audi
   | tor/refresh', {
107 |               method: 'POST',
108 |               headers: { 'Content-Type': 'application/json' }
109 |           });
110 |
111 |           if (res && res.report) {
112 |               allEntries = res.report.entries || [];
113 |               currentSummary = res.report.summary || {};
114 |               renderSummaryCards(res.report);
115 |               renderFilteredTable();
116 |           }
117 |
118 |           const count = res.changes_count || (res.changes ? res.ch
   | anges.length : 0);
119 |           const timeStr = new Date().toLocaleTimeString('ru-RU', {

```

```
    hour: '2-digit', minute: '2-digit', second: '2-digit' });
120
121     if (statusText) {
122         statusText.textContent = count > 0
123             ? `Телеметрия: снимок ${timeStr} (изменений: ${count
124             : `Телеметрия: снимок ${timeStr} (без изменений)`;
125     }
126
127     const badge = document.getElementById('sa-changes-badge'
128 );
129     if (badge) {
130         if (count > 0) {
131             badge.textContent = count;
132             badge.classList.remove('d-none');
133         }
134     }
135
136     if (count > 0) {
137         window.showToast?.(`Снимок сохранен! Зафиксировано изм
138 енений: ${count}`, 'warning');
139     } else {
140         window.showToast?.('Снимок телеметрии успешно зафиксир
141 ован (изменений нет)', 'success');
142     }
143
144     } catch (err) {
145         console.error('[StartupAuditor] Error refreshing telemet
146 ry snapshot:', err);
147         if (statusText) statusText.textContent = 'Телеметрия: ош
148 ибка сбора';
149         window.showToast?.(`Ошибка обновления телеметрии: ${err.
150 message}`, 'danger');
151     } finally {
152         if (btn) btn.disabled = false;
153         if (spinner) spinner.classList.remove('spinner-border',
154 'spinner-border-sm');
155     }
156 }
```

```
149
150 async function checkRecentChangesBadge() {
151     try {
152         const data = await window.api.fetch('/api/v1/startup-aud
itor/changes?limit=10');
153         const badge = document.getElementById('sa-changes-badge'
);
154         if (badge && data && data.changes && data.changes.length
> 0) {
155             badge.textContent = data.changes.length;
156             badge.classList.remove('d-none');
157         }
158     } catch {
159         // Игнорируем фоновые ошибки бейджа
160     }
161 }
162
163 async function showChangesModal() {
164     const modalEl = document.getElementById('sa-changes-modal'
);
165     if (!modalEl) return;
166
167     if (window.bootstrap && window.bootstrap.Modal) {
168         const bsModal = window.bootstrap.Modal.getOrCreateInstan
ce(modalEl);
169         bsModal.show();
170     }
171     await loadChangesData();
172 }
173
174 async function loadChangesData() {
175     const tbody = document.getElementById('sa-changes-table-bo
dy');
176     const countEl = document.getElementById('sa-changes-modal-
count');
177
178     if (tbody) {
179         tbody.innerHTML = `
```

```

180         <tr>
181             <td colspan="5" class="text-center py-4 text-muted">
182                 <div class="spinner-border spinner-border-sm text-
warning me-2" role="status"></div>
183                 Загрузка истории изменений автозапуска из телеметр
ии...
184             </td>
185         </tr>
186     `;
187 }
188
189     try {
190         const data = await window.api.fetch('/api/v1/startup-aud
itor/changes?limit=100');
191         const changes = data.changes || [];
192
193         if (countEl) {
194             countEl.textContent = `${changes.length} записей`;
195         }
196
197         if (changes.length === 0) {
198             if (tbody) {
199                 tbody.innerHTML = `
200                     <tr>
201                         <td colspan="5" class="text-center py-4 text-mut
ed">
202                             <i class="bi bi-shield-check text-success fs-4
d-block mb-1"></i>
203                             Изменений в конфигурации автозапуска между сн
мками телеметрии не зафиксировано.
204                         </td>
205                     </tr>
206                 `;
207             }
208             return;
209         }
210
211         if (tbody) {

```

```

212         tbody.innerHTML = changes.map(ch => {
213             const changeBadge = getChangeTypeBadge(ch.change_type);
214             const timeFormatted = ch.created_at ? ch.created_at.replace('T', ' ').substring(0, 19) : '-';
215
216             let detailsHtml = '';
217             if (ch.change_type === 'added') {
218                 detailsHtml = `Состояние: <strong class="text-danger">${escapeHtml(ch.old_value)}</strong> <i class="bi bi-arrow-right"></i> <strong class="text-success">${escapeHtml(ch.new_value)}</strong></div>`;
223             } else if (ch.change_type === 'risk_changed') {
224                 detailsHtml = `

Уровень риска: <strong>${escapeHtml(ch.old_value)}</strong> <i class="bi bi-arrow-right"></i> <strong class="text-warning">${escapeHtml(ch.new_value)}</strong></div>`;
225             } else {
226                 detailsHtml = `
227                 <div class="small font-monospace text-muted text-truncate" style="max-width: 420px;" title="Было: ${escapeHtml(ch.old_value)}">Было: ${escapeHtml(ch.old_value)} || '-')</div>
228
229                 <div class="small font-monospace text-primary text-truncate" style="max-width: 420px;" title="Стало: ${escapeHtml(ch.new_value)}">Стало: ${escapeHtml(ch.new_value)} || '-')</div>
230
231                 `;


```

```

230         }
231
232         return `
233             <tr>
234                 <td class="font-monospace small text-muted">${ti
meFormatted}</td>
235                 <td class="text-center">${changeBadge}</td>
236                 <td>
237                     <div class="fw-semibold text-truncate" style="
max-width: 190px;" title="${escapeHtml(ch.item_name)}">${escap
eHtml(ch.item_name)}</div>
238                 </td>
239                 <td><span class="sa-badge sa-badge-loc">${escape
Html(ch.location_type || 'registry_run')}</span></td>
240                 <td>${detailsHtml}</td>
241             </tr>
242         `;
243     })).join('');
244 }
245 } catch (err) {
246     console.error('[StartupAuditor] Error loading changes hi
story:', err);
247     if (tbody) {
248         tbody.innerHTML = `
249             <tr>
250                 <td colspan="5" class="text-center py-4 text-dange
r">
251                     Ошибка загрузки истории изменений: ${escapeHtml(
err.message)}
252                 </td>
253             </tr>
254         `;
255     }
256 }
257 }
258
259 function getChangeTypeBadge(changeType) {
260     switch (changeType) {

```



```

261         case 'added':
262             return '<span class="sa-badge sa-badge-enabled"><i cla
ss="bi bi-plus-circle"></i> Добавлен</span>';
263         case 'removed':
264             return '<span class="sa-badge sa-badge-disabled"><i cl
ass="bi bi-dash-circle"></i> Удален</span>';
265         case 'state_changed':
266             return '<span class="sa-badge sa-risk-warning"><i clas
s="bi bi-toggle2-on"></i> Статус</span>';
267         case 'path_changed':
268             return '<span class="sa-badge sa-risk-suspicious"><i c
lass="bi bi-pencil"></i> Путь</span>';
269         case 'risk_changed':
270             return '<span class="sa-badge sa-risk-critical"><i cla
ss="bi bi-shield-exclamation"></i> Риск</span>';
271         default:
272             return '<span class="sa-badge sa-badge-cat">${escapeHt
ml(changeType)}</span>`;
273     }
274 }
275
276 async function loadAuditData() {
277     const tbody = document.getElementById('sa-table-body');
278     if (tbody) {
279         tbody.innerHTML = `
280             <tr>
281                 <td colspan="8" class="text-center py-4 text-muted">
282                     <div class="spinner-border spinner-border-sm text-
primary me-2" role="status"></div>
283                     Выполняется сканирование точек автозапуска Windows
...
284                 </td>
285             </tr>
286         `;
287     }
288
289     try {
290         const report = await window.api.fetch('/api/v1/startup-a

```

```

    uditor/audit');
291     allEntries = report.entries || [];
292     currentSummary = report.summary || {};
293
294     renderSummaryCards(report);
295     renderFilteredTable();
296 } catch (err) {
297     console.error('[StartupAuditor] Error loading audit repo
rt:', err);
298     if (tbody) {
299         tbody.innerHTML = `
300             <tr>
301                 <td colspan="8" class="text-center py-4 text-dange
r">
302                     Ошибка загрузки данных аудита: ${err.message}
303                 </td>
304             </tr>
305         `;
306     }
307 }
308 }
309
310 function renderSummaryCards(report) {
311     const summary = report.summary || {};
312
313     // Health Score
314     const scoreEl = document.getElementById('sa-health-score')
315     ;
316     if (scoreEl) {
317         scoreEl.textContent = summary.health_score ?? '--';
318         scoreEl.className = 'sa-card-value ' + (
319             summary.health_score >= 80 ? 'text-success' : (summary
320             .health_score >= 60 ? 'text-warning' : 'text-danger')
321         );
322     }
323
324     // Counters
325     const activeEl = document.getElementById('sa-active-count'

```

```

    );
324     if (activeEl) activeEl.textContent = summary.active_entries ?? '--';
325
326     const totalEl = document.getElementById('sa-total-count');
327     if (totalEl) totalEl.textContent = summary.total_entries ?
    ? '--';
328
329     const disabledEl = document.getElementById('sa-disabled-count');
330     if (disabledEl) disabledEl.textContent = summary.disabled_entries ?? '0';
331
332     const brokenEl = document.getElementById('sa-broken-count');
333     if (brokenEl) brokenEl.textContent = summary.broken_entries ?? '0';
334
335     const threatsEl = document.getElementById('sa-threats-count');
336     if (threatsEl) threatsEl.textContent = (summary.critical_count || 0) + (summary.suspicious_count || 0);
337
338     // Recommendations
339     const recDesc = document.getElementById('sa-rec-desc');
340     if (recDesc) {
341         if (report.recommendations && report.recommendations.length > 0) {
342             recDesc.innerHTML = report.recommendations.map(r => `<div>${r}</div>`).join('');
343         } else {
344             recDesc.textContent = 'Конфигурация автозапуска оптимальна, критических замечаний нет.';
345         }
346     }
347 }
348
349 function renderFilteredTable() {

```

```
350     const tbody = document.getElementById('sa-table-body');
351     const tableCount = document.getElementById('sa-table-count
    ');
352     if (!tbody) return;
353
354     const query = (document.getElementById('sa-search-input')?
    .value || '').toLowerCase().trim();
355     const riskFilter = (document.getElementById('sa-filter-ris
    k')?.value || '').toLowerCase();
356     const locFilter = (document.getElementById('sa-filter-loca
    tion')?.value || '').toLowerCase();
357     const stateFilter = (document.getElementById('sa-filter-st
    ate')?.value || '').toLowerCase();
358
359     const filtered = allEntries.filter(e => {
360         if (query) {
361             const fullText = `${e.name || ''} ${e.command || ''} $
    {e.executable_path || ''} ${e.publisher || ''} ${e.category ||
    ''}`.toLowerCase();
362             if (!fullText.includes(query)) return false;
363         }
364         if (riskFilter && (e.risk_level || '').toLowerCase() !==
    riskFilter) {
365             return false;
366         }
367         if (locFilter && (e.location_type || '').toLowerCase() !
    == locFilter) {
368             return false;
369         }
370         if (stateFilter === 'enabled' && !e.is_enabled) return f
    else;
371         if (stateFilter === 'disabled' && e.is_enabled) return f
    else;
372
373         return true;
374     });
375
376     if (tableCount) {
```

```

377 |         tableCount.textContent = `Показано: ${filtered.length} и
378 |         з ${allEntries.length}`;
379 |     }
380 |     if (filtered.length === 0) {
381 |         tbody.innerHTML = `
382 |             <tr>
383 |                 <td colspan="8" class="text-center py-5 text-muted">
384 |                     <div class="fs-4 mb-2">❌ </div>
385 |                     <div>Элементы автозапуска по заданным критериям не
386 |                     найдены</div>
387 |                 </td>
388 |             </tr>
389 |         `;
390 |         return;
391 |     }
392 |     tbody.innerHTML = filtered.map(e => {
393 |         const isEnabled = e.is_enabled;
394 |         const statusBadge = isEnabled
395 |             ? '<span class="sa-badge sa-badge-enabled"><i class="b
396 | i bi-check-circle-fill"></i> ВКЛ</span>'
397 |             : '<span class="sa-badge sa-badge-disabled"><i class="
398 | bi bi-dash-circle"></i> ОТКЛ</span>';
399 |
400 |         const riskBadge = getRiskBadge(e.risk_level);
401 |         const isBroken = !e.file_exists && e.executable_path;
402 |
403 |         const pathDisplay = isBroken
404 |             ? `<div class="d-flex align-items-center gap-1 text-da
405 | nger small font-monospace">
406 |                 <i class="bi bi-x-octagon-fill text-danger flex-s
407 | hrink-0"></i>
408 |                 <span class="text-truncate fw-semibold" style="ma
409 | x-width: 480px;" title="${e.executable_path || e.command}">[Файл не найден] ${e.executable_path || e.command}</span>
410 |             </div>`
411 |             : `<div class="font-monospace small text-truncate" sty

```

```

le="max-width: 480px; color: var(--text-color);" title="${e.co
mmand || e.executable_path}">
407     ${e.executable_path || e.command}
408 </div>`;
409
410     const argsDisplay = e.arguments ? `
411     <div class="d-flex align-items-center gap-1 font-monos
pace text-truncate mt-1 text-muted" style="max-width: 480px; f
ont-size: 0.73rem;" title="${e.arguments}">
412     <span class="badge bg-secondary-subtle text-secondar
y-emphasis border border-secondary" style="font-size: 0.68rem;
padding: 1px 4px;">ARG</span>
413     <span class="text-truncate">${e.arguments}</span>
414     </div>` : '';
415
416     const toggleIcon = isEnabled
417     ? '<i class="bi bi-toggle2-on text-success" style="fon
t-size: 1.4rem;"></i>'
418     : '<i class="bi bi-toggle2-off text-muted" style="font
-size: 1.4rem;"></i>';
419     const toggleTitle = isEnabled ? 'Отключить элемент автоз
апуска' : 'Включить элемент автозапуска';
420
421     let impactBadge = '';
422     if (!isEnabled) {
423         impactBadge = '<span class="sa-badge sa-badge-disabled
">Отключено</span>';
424     } else if (e.boot_impact === 'Высокое') {
425         impactBadge = '<span class="sa-badge sa-risk-critical"
>Высокое</span>';
426     } else if (e.boot_impact === 'Среднее') {
427         impactBadge = '<span class="sa-badge sa-risk-warning">
Среднее</span>';
428     } else {
429         impactBadge = '<span class="sa-badge sa-badge-cat">Низ
кое</span>';
430     }
431

```

```

432         return `
433         <tr class="sa-row-item" data-id="${e.id}" style="curso
r: pointer;" title="Нажмите для подробного AI-аудита">
434             <td class="text-center">${statusBadge}</td>
435             <td>
436                 <div class="fw-semibold text-truncate" style="max-
width: 230px; color: var(--text-color);" title="${e.name}">${e
.name}</div>
437                 <div class="small text-truncate text-muted" style=
"max-width: 230px; font-size: 0.73rem;" title="${e.publisher |
| 'Неизвестен'}">${e.publisher || 'Неизвестный издатель'}</div
>
438             </td>
439             <td><span class="sa-badge sa-badge-loc">${e.location
_type}</span></td>
440             <td><span class="sa-badge sa-badge-cat">${e.category
|| 'Неизвестно'}</span></td>
441             <td class="text-center">${riskBadge}</td>
442             <td>
443                 ${pathDisplay}
444                 ${argsDisplay}
445             </td>
446             <td class="text-center">${impactBadge}</td>
447             <td class="text-center">
448                 <button class="sa-toggle-btn" data-id="${e.id}" da
ta-enabled="${isEnabled}" title="${toggleTitle}">
449                     ${toggleIcon}
450                 </button>
451             </td>
452         </tr>
453     `;
454     }).join('');
455
456     // Attach row click listeners for modal details
457     tbody.querySelectorAll('.sa-row-item').forEach(row => {
458         row.onclick = (evt) => {
459             // Prevent modal open when clicking directly on the to
toggle button

```

```

460         if (evt.target.closest('.sa-toggle-btn')) {
461             return;
462         }
463         const entryId = row.getAttribute('data-id');
464         const entry = allEntries.find(item => item.id === entr
yId);
465         if (entry) {
466             showEntryModal(entry);
467         }
468     };
469 });
470
471 // Attach toggle listeners
472 tbody.querySelectorAll('.sa-toggle-btn').forEach(btn => {
473     btn.onclick = async (evt) => {
474         evt.stopPropagation();
475         const entryId = btn.getAttribute('data-id');
476         const currentState = btn.getAttribute('data-enabled')
=== 'true';
477         const newState = !currentState;
478         try {
479             btn.disabled = true;
480             btn.style.opacity = '0.5';
481             const res = await window.api.fetch('/api/v1/startup-
auditor/toggle', {
482                 method: 'POST',
483                 headers: { 'Content-Type': 'application/json' },
484                 body: JSON.stringify({ entry_id: entryId, enable:
newState })
485             });
486             if (res.success) {
487                 window.showToast?(`Автозагрузка ${newState ? 'вкл
ючена' : 'отключена'}`, 'success');
488                 await loadAuditData();
489             } else {
490                 window.showToast?(`Ошибка переключения: ${res.mes
sage}`, 'danger') || alert(`Ошибка переключения: ${res.message
}`);

```



```
491 |         }
492 |     } catch (err) {
493 |         window.showToast?(`Ошибка: ${err.message}`, 'danger
    | `') || alert(`Ошибка: ${err.message}`);
494 |     } finally {
495 |         btn.disabled = false;
496 |         btn.style.opacity = '1';
497 |     }
498 | };
499 | });
500 | }
501 |
502 | function showEntryModal(entry) {
503 |     if (!entry) return;
504 |
505 |     const modalEl = document.getElementById('sa-entry-detail-m
    | odal');
506 |     if (!modalEl) return;
507 |
508 |     // Header info
509 |     const titleEl = document.getElementById('sa-modal-title');
510 |     if (titleEl) titleEl.textContent = entry.name || 'Детали п
    | рограммы';
511 |
512 |     const statusBadgeEl = document.getElementById('sa-modal-st
    | atus-badge');
513 |     if (statusBadgeEl) {
514 |         statusBadgeEl.innerHTML = entry.is_enabled
515 |             ? '<span class="sa-badge sa-badge-enabled"><i class="b
    | i bi-check-circle-fill"></i> ВКЛ</span>'
516 |             : '<span class="sa-badge sa-badge-disabled"><i class="
    | bi bi-dash-circle"></i> ОТКЛ</span>';
517 |     }
518 |
519 |     const riskBadgeEl = document.getElementById('sa-modal-risk
    | -badge');
520 |     if (riskBadgeEl) {
521 |         riskBadgeEl.innerHTML = getRiskBadge(entry.risk_level);
```

```
522     }
523
524     // Metadata Fields
525     const setTxt = (id, val) => {
526         const el = document.getElementById(id);
527         if (el) el.textContent = val ?? '-';
528     };
529     const setHtml = (id, html) => {
530         const el = document.getElementById(id);
531         if (el) el.innerHTML = html;
532     };
533
534     setTxt('sa-modal-name', entry.name);
535     setTxt('sa-modal-publisher', entry.publisher || 'Неизвестн
536    ый разработчик');
537     setTxt('sa-modal-location-type', entry.location_type || 'r
538    egistry_run');
539     setTxt('sa-modal-category', entry.category || 'Неизвестно'
540    );
541     setTxt('sa-modal-boot-impact', entry.boot_impact || 'Низко
542    е');
543     setTxt('sa-modal-location-path', entry.location_path || 'Н
544    е указан');
545
546     setHtml('sa-modal-is-signed', entry.is_signed
547     ? '<span class="text-success"><i class="bi bi-patch-chec
548    k me-1"></i>Подписан (Valid)</span>'
549     : '<span class="text-muted"><i class="bi bi-patch-questi
550    on me-1"></i>Без подписи / Неизвестно</span>'
551     );
552
553     setHtml('sa-modal-file-exists', entry.file_exists
554     ? '<span class="text-success"><i class="bi bi-file-earma
555    rk-check me-1"></i>Файл существует</span>'
556     : '<span class="text-danger fw-bold"><i class="bi bi-fil
557    e-earmark-x me-1"></i>Файл не найден (битая ссылка)</span>'
558     );
559
560 }
```

```

551     setTxt('sa-modal-file-size', entry.file_size_kb > 0 ? `${e
| ntry.file_size_kb.toFixed(1)} KB` : '-');
552     setTxt('sa-modal-executable-path', entry.executable_path |
| entry.command || 'Путь не указан');
553     setTxt('sa-modal-command', entry.command || '-');
554
555     // Reset AI Diagnostic section
556     const aiExplanation = document.getElementById('sa-modal-ai
| -explanation');
557     if (aiExplanation) {
558         aiExplanation.innerHTML = `Нажмите «Анализ контекста», ч
| тобы получить экспертное объяснение назначения программы от яз
| ыковой модели, оценку рисков и рекомендации по оптимизации.`;
559     }
560
561     // AI Diagnose button
562     const diagnoseBtn = document.getElementById('btn-sa-modal-
| diagnose');
563     if (diagnoseBtn) {
564         diagnoseBtn.onclick = async () => {
565             aiExplanation.innerHTML = `
566                 <div class="d-flex align-items-center gap-2 py-2 tex
| t-primary">
567                     <div class="spinner-border spinner-border-sm" role
| ="status"></div>
568                     <span>Генерация AI-диагностики и анализа рисков дл
| я программы «${escapeHtml(entry.name)}»...</span>
569                 </div>
570             `;
571             const activeModel = window.activeModelName
572                 || window.userSettings?.model
573                 || document.getElementById('model-select')?.value
574                 || document.getElementById('header-model-select')?.v
| alue
575                 || '';
576             try {
577                 const res = await window.api.fetch('/api/v1/startup-
| auditor/explain', {

```

```

578         method: 'POST',
579         headers: { 'Content-Type': 'application/json' },
580         body: JSON.stringify({
581             entry_id: entry.id,
582             name: entry.name,
583             publisher: entry.publisher,
584             executable_path: entry.executable_path,
585             command: entry.command,
586             arguments: entry.arguments,
587             location_type: entry.location_type,
588             location_path: entry.location_path,
589             risk_level: entry.risk_level,
590             is_enabled: entry.is_enabled,
591             file_exists: entry.file_exists,
592             is_signed: entry.is_signed,
593             boot_impact: entry.boot_impact,
594             model: activeModel || undefined
595         })
596     });
597
598     aiExplanation.innerHTML = `
599         <div class="mb-2"><strong class="text-primary"><i
600         class="bi bi-card-text me-1"></i>Назначение:</strong> ${escape
        Html(res.summary)}</div>
601         <div class="mb-2"><strong class="text-muted"><i cl
        ass="bi bi-building me-1"></i>Разработчик / Категория:</strong>
        > ${escapeHtml(res.developer || entry.publisher || 'Неизвестен
        ')} (${escapeHtml(res.category || 'Приложение')})</div>
602         <div class="mb-2"><strong class="text-warning"><i
        class="bi bi-shield-lock me-1"></i>Оценка безопасности:</stron
        g> ${escapeHtml(res.security_verdict)}</div>
603         <div class="mb-2"><strong class="text-info"><i cla
        ss="bi bi-speedometer2 me-1"></i>Влияние на запуск:</strong> $
        {escapeHtml(res.boot_impact_analysis)}</div>
604         <div class="p-2 mb-2 rounded alert alert-warning">
        <strong class="text-warning"><i class="bi bi-lig
        htbulb me-1"></i>Рекомендация:</strong> ${escapeHtml(res.start
        up_recommendation)}

```

```

605 |         </div>
606 |         ${res.action_steps && res.action_steps.length > 0
| ? `
607 |             <h6 class="small text-uppercase text-muted fw-bo
| ld mb-1">Рекомендуемые действия:</h6>
608 |             <ul class="mb-0 ps-3">
609 |                 ${res.action_steps.map(s => `<li>${escapeHtml(
| s)}</li>`).join('')}
610 |             </ul>
611 |             ` : ''}
612 |         `;
613 |     } catch (err) {
614 |         aiExplanation.innerHTML = `<div class="text-danger p
| y-2"><i class="bi bi-exclamation-octagon me-1"></i>Ошибка полу
| чения AI-анализа: ${escapeHtml(err.message)}</div>`;
615 |     }
616 | };
617 | }
618 |
619 | // Modal Footer Toggle Button
620 | const modalToggleBtn = document.getElementById('btn-sa-mod
| al-toggle');
621 | if (modalToggleBtn) {
622 |     modalToggleBtn.innerHTML = entry.is_enabled
623 |     ? '<i class="bi bi-toggle2-off me-1 text-warning"></i>
| Отключить автозапуск'
624 |     : '<i class="bi bi-toggle2-on me-1 text-success"></i>B
| ключить автозапуск';
625 |     modalToggleBtn.onclick = async () => {
626 |         const newState = !entry.is_enabled;
627 |         try {
628 |             modalToggleBtn.disabled = true;
629 |             const res = await window.api.fetch('/api/v1/startup-
| auditor/toggle', {
630 |                 method: 'POST',
631 |                 headers: { 'Content-Type': 'application/json' },
632 |                 body: JSON.stringify({ entry_id: entry.id, enable:
| newState })

```

```

633         });
634         if (res.success) {
635             entry.is_enabled = newState;
636             showEntryModal(entry);
637             window.showToast?.`Автозагрузка ${newState ? 'вкл
        ючена' : 'отключена'}` , 'success');
638             await loadAuditData();
639         } else {
640             window.showToast?.`Ошибка переключения: ${res.mes
        sage}` , 'danger') || alert(`Ошибка переключения: ${res.message
        }`);
641         }
642     } catch (err) {
643         window.showToast?.`Ошибка: ${err.message}` , 'danger
        ') || alert(`Ошибка: ${err.message}`);
644     } finally {
645         modalToggleBtn.disabled = false;
646     }
647 };
648 }
649
650 // Modal Footer Copy Button
651 const modalCopyBtn = document.getElementById('btn-sa-modal
        -copy');
652 if (modalCopyBtn) {
653     modalCopyBtn.onclick = () => {
654         const textToCopy = entry.executable_path || entry.comm
        and || '';
655         navigator.clipboard.writeText(textToCopy).then(() => {
656             const orig = modalCopyBtn.innerHTML;
657             modalCopyBtn.innerHTML = '<i class="bi bi-check2 me-
        1 text-success"></i>Скопировано!';
658             setTimeout(() => { modalCopyBtn.innerHTML = orig; },
        1800);
659         });
660     };
661 }
662

```

```
663 // Show modal via Bootstrap
664 if (window.bootstrap && window.bootstrap.Modal) {
665     const bsModal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
666     bsModal.show();
667 }
668 }
669
670 function getRiskBadge(level) {
671     switch ((level || '').toLowerCase()) {
672         case 'critical':
673             return '<span class="sa-badge sa-risk-critical"><i class="bi bi-shield-x"></i> CRITICAL</span>';
674         case 'suspicious':
675             return '<span class="sa-badge sa-risk-suspicious"><i class="bi bi-exclamation-octagon"></i> SUSPICIOUS</span>';
676         case 'warning':
677             return '<span class="sa-badge sa-risk-warning"><i class="bi bi-exclamation-triangle"></i> WARNING</span>';
678         case 'notice':
679             return '<span class="sa-badge sa-risk-notice"><i class="bi bi-info-circle"></i> NOTICE</span>';
680         default:
681             return '<span class="sa-badge sa-risk-clean"><i class="bi bi-shield-check"></i> CLEAN</span>';
682     }
683 }
684
685 function escapeHtml(str) {
686     if (!str) return '';
687     return String(str)
688         .replace(/&/g, '&amp;')
689         .replace(/</g, '&lt;')
690         .replace(/>/g, '&gt;')
691         .replace(/"/g, '&quot;')
692         .replace(/'/g, '&#039;');
693 }
694
```

```
695 | // Expose globally for Apps Hub
696 | window.initStartupAuditorTab = initStartupAuditorTab;
697 |})();
698 |
```


index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/storage_manager_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/storage_manager_tab/index.html" class="w-100 h-100 border-0"></iframe>
17
18 * File: index.html
19 * Project: ai-breadboard
20 * Package: windows/api/webgui/storage_manager_tab
21 * Author: hypo69
22 * Copyright: © 2026 hypo69
23 * Updated: 2026-10-06 00:20:00
24 * =====
25 =====
26 -->
27
28 <!-- Storage & Disk Volumes Management Tab -->
29 <div class="container-fluid py-2 px-3">
30   <!-- Top Stat Cards -->
31   <div class="row g-2 mb-3">
32     <div class="col-6 col-md-3">
33       <div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3">
34         <div class="text-muted small text-uppercase fw-semibold">Физические диски □ </div>
```

```

31 |         <div class="fs-4 fw-bold mt-1 text-info" id="sm-disks-
    | count">--</div>
32 |         <div class="small text-muted mt-1" id="sm-disks-sub">0
    | прос накопителей...</div>
33 |     </div>
34 | </div>
35 |     <div class="col-6 col-md-3">
36 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
37 |             <div class="text-muted small text-uppercase fw-semibol
    | d">Логические тома □ </div>
38 |             <div class="fs-4 fw-bold mt-1 text-primary" id="sm-vol
    | umes-count">--</div>
39 |             <div class="small text-muted mt-1" id="sm-volumes-sub"
    | >NTFS / ReFS / FAT32</div>
40 |         </div>
41 |     </div>
42 |     <div class="col-6 col-md-3">
43 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
44 |             <div class="text-muted small text-uppercase fw-semibol
    | d">BitLocker Защита □ </div>
45 |             <div class="fs-4 fw-bold mt-1 text-success" id="sm-bit
    | locker-val">--</div>
46 |             <div class="small text-muted mt-1" id="sm-bitlocker-su
    | b">Статус шифрования</div>
47 |         </div>
48 |     </div>
49 |     <div class="col-6 col-md-3">
50 |         <div class="card bg-dark text-white border-secondary h-1
    | 00 shadow-sm p-3">
51 |             <div class="text-muted small text-uppercase fw-semibol
    | d">NTFS Сжатие / VHD □ </div>
52 |             <div class="fs-4 fw-bold mt-1 text-warning" id="sm-com
    | pression-val">--</div>
53 |             <div class="small text-muted mt-1" id="sm-compression-
    | sub">Компактность дисков</div>
54 |         </div>

```

```

55     </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <select class="form-select form-select-sm poll-freq-select" id="sm-poll-freq" data-poll-id="storage_manager" title="Частота опроса: Хранилище">
62             <option value="start">Только при старте</option>
63             <option value="manual" selected>По запросу</option>
64             <option value="3">Каждые 3 сек</option>
65             <option value="5">Каждые 5 сек</option>
66             <option value="10">Каждые 10 сек</option>
67             <option value="30">Каждые 30 сек</option>
68             <option value="60">Каждые 60 сек</option>
69             <option value="120">Каждые 120 сек</option>
70         </select>
71         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="sm-refresh-btn">
72             <i class="bi bi-arrow-clockwise"></i>
73             <span>Обновить состояние</span>
74         </button>
75         <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5" id="sm-quick-rescan-btn">
76             <i class="bi bi-search"></i>
77             <span>Rescan (diskpart)</span>
78         </button>
79         <button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5" id="sm-defrag-btn">
80             <i class="bi bi-speedometer"></i>
81             <span>Дефрагментация / TRIM</span>
82         </button>
83     </div>
84     <div class="d-flex align-items-center gap-2">
85         <input type="text" class="form-control form-control-sm b

```

```

    g-dark text-white border-secondary" id="sm-filter-input" place
holder="Фильтр по томам/дискам..." style="width: 220px;">
86     </div>
87 </div>
88
89 <!-- Tables Section -->
90 <div class="row g-3">
91     <div class="col-12 col-lg-7">
92         <div class="card bg-dark text-white border-secondary sha
dow-sm">
93             <div class="card-header border-secondary d-flex align-
items-center justify-content-between py-2 px-3">
94                 <span class="fw-semibold"><i class="bi bi-hdd-stack
text-primary me-2"></i>Томы и разделы дисков</span>
95                 <span class="badge bg-secondary" id="sm-volumes-badg
e">0 томов</span>
96             </div>
97             <div class="card-body p-0 table-responsive" style="max
-height: 420px; overflow-y: auto;">
98                 <table class="table table-dark table-hover table-str
iped align-middle mb-0 small" id="sm-volumes-table">
99                     <thead>
100                         <tr class="text-secondary">
101                             <th>Том / Буква</th>
102                             <th>Метка</th>
103                             <th>ФС</th>
104                             <th>Размер</th>
105                             <th>Свободно</th>
106                             <th>BitLocker</th>
107                             <th class="text-end">Действия</th>
108                         </tr>
109                     </thead>
110                     <tbody id="sm-volumes-tbody">
111                         <tr><td colspan="7" class="text-center text-mute
d py-3">Загрузка данных о томах...</td></tr>
112                     </tbody>
113                 </table>
114             </div>

```

```

115 |         </div>
116 |     </div>
117 |
118 |     <div class="col-12 col-lg-5">
119 |         <div class="card bg-dark text-white border-secondary sha
120 |         <div class="card-header border-secondary d-flex align-
121 |         <span class="fw-semibold"><i class="bi bi-terminal t
122 |         </div>
123 |         <div class="card-body p-3">
124 |             <div class="mb-3">
125 |                 <label class="form-label small text-muted">Выбор о
126 |                 <select class="form-select form-select-sm bg-dark
127 |                 <option value="diskpart_rescan">Обновить конфигу
128 |                 <option value="bitlocker_status">Проверить стату
129 |                 <option value="compact_query">Запрос сжатия ОС (
130 |                 <option value="defrag_analyze">Анализ оптимизаци
131 |                 </select>
132 |             </div>
133 |             <div class="mb-3">
134 |                 <label class="form-label small text-muted">Целевой
135 |                 <input type="text" class="form-control form-contro
136 |                 </div>
137 |                 <div class="d-flex align-items-center justify-conten
138 |                 <div class="form-check form-switch">
139 |                     <input class="form-check-input" type="checkbox"

```

```
id="sm-dry-run-switch" checked>
140     <label class="form-check-label small text-muted"
    for="sm-dry-run-switch">Безопасный режим (Dry Run)</label>
141     </div>
142     <button class="btn btn-sm btn-success px-3" id="sm
    -execute-btn">
143         <i class="bi bi-play-fill"></i> Выполнить
144     </button>
145 </div>
146 </div>
147 </div>
148
149 <!-- Action Log Output -->
150 <div class="card bg-dark text-white border-secondary sha
    dow-sm">
151     <div class="card-header border-secondary py-2 px-3 d-f
    lex align-items-center justify-content-between">
152         <span class="small fw-semibold text-muted"><i class=
    "bi bi-code-square me-1"></i>Консольный вывод</span>
153         <button class="btn btn-link btn-sm text-decoration-n
    one text-muted p-0" id="sm-clear-log-btn">Очистить</button>
154     </div>
155     <div class="card-body p-2">
156         <pre class="mb-0 small bg-black text-light p-2 round
    ed" id="sm-console-output" style="max-height: 180px; overflow-
    y: auto; font-family: Consolas, monospace; font-size: 0.78rem;
    ">Готов к операциям.</pre>
157     </div>
158 </div>
159 </div>
160 </div>
161 </div>
162
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Storage Manager Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/storage_manager_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  *   JavaScript Import:
13  *     import { initStorageManagerTab } from '/windows/api/webgui/storage_manager_tab/main.js';
14  *
15  * File: main.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/storage_manager_tab
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-08 02:00:00
21  * =====
22  */
23
24 /**
25  * storage_manager_tab/main.js – Управление накопителями, разделами и томами
26  * Updated: 2026-10-04 11:15:30
27  */
28
29 const registerTabPoller = window.registerTabPoller || function
  () {};
```

```
30 |
31 | let isInitialized = false;
32 |
33 | export async function fetchStorageSummary() {
34 |   try {
35 |     const res = await fetch('/api/storage-manager/summary');
36 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
37 |     const data = await res.json();
38 |     renderStorageSummary(data);
39 |   } catch (err) {
40 |     console.warn('[StorageManager] Ошибка получения сводки:',
41 | err);
42 |   }
43 | }
44 | function renderStorageSummary(data) {
45 |   const disksCountEl = document.getElementById('sm-disks-count
46 | ');
47 |   const volumesCountEl = document.getElementById('sm-volumes-c
48 | ount');
49 |   const bitlockerEl = document.getElementById('sm-bitlocker-va
50 | l');
51 |   const compressionEl = document.getElementById('sm-compressio
52 | n-val');
53 |   const tbody = document.getElementById('sm-volumes-tbody');
54 |   const badge = document.getElementById('sm-volumes-badge');
55 |   if (disksCountEl) disksCountEl.textContent = data.disks_coun
56 | t ?? '--';
57 |   if (volumesCountEl) volumesCountEl.textContent = data.volume
58 | s ? data.volumes.length : '--';
59 |   if (bitlockerEl) bitlockerEl.textContent = data.bitlocker_st
60 | atus || 'Готово';
61 |   if (compressionEl) compressionEl.textContent = data.compact_
62 | os_status || 'CompactOS';
63 |   if (!tbody) return;
```



```

59 |   const volumes = data.volumes || [
60 |     { drive: 'C:', label: 'System', fs: 'NTFS', size: '476 GB'
    |     , free: '180 GB', bitlocker: 'Включен' },
61 |     { drive: 'D:', label: 'Data', fs: 'NTFS', size: '931 GB',
    |     free: '420 GB', bitlocker: 'Выключен' }
62 |   ];
63 |
64 |   if (badge) badge.textContent = `${volumes.length} томов`;
65 |
66 |   tbody.innerHTML = volumes.map(v => `
67 |     <tr>
68 |       <td class="fw-bold text-info">${v.drive} || v.name || 'То
    | м'</td>
69 |       <td>${v.label} || '- '</td>
70 |       <td><span class="badge bg-secondary">${v.fs} || 'NTFS'</
    | span></td>
71 |       <td>${v.size} || '- '</td>
72 |       <td class="text-success">${v.free} || '- '</td>
73 |       <td><span class="badge ${v.bitlocker === 'Включен' ? 'bg
    | -success' : 'bg-warning text-dark'}">${v.bitlocker} || 'N/A'</
    | span></td>
74 |       <td class="text-end">
75 |         <button class="btn btn-xs btn-outline-primary py-0 px-
    | 1 sm-btn-action" data-drive="${v.drive} || 'C:'" title="Анализ
    | оптимизации">
76 |           <i class="bi bi-gear"></i>
77 |         </button>
78 |       </td>
79 |     </tr>
80 |   `).join('');
81 | }
82 |
83 | async function executeStorageAction() {
84 |   const actionSelect = document.getElementById('sm-action-sele
    | ct');
85 |   const targetInput = document.getElementById('sm-target-input
    | ');
86 |   const dryRunSwitch = document.getElementById('sm-dry-run-swi

```

```

tch');
87   const output = document.getElementById('sm-console-output');
88
89   if (!actionSelect || !output) return;
90
91   const op = actionSelect.value;
92   const target = targetInput ? targetInput.value.trim() : '';
93   const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;
94
95   output.textContent = `[${new Date().toLocaleTimeString()}] 3
апуск операции: ${op} (DryRun: ${dryRun})...\n`;
96
97   try {
98     const res = await fetch('/api/storage-manager/action', {
99       method: 'POST',
100       headers: { 'Content-Type': 'application/json' },
101       body: JSON.stringify({ action: op, target: target, dry_r
un: dryRun })
102     });
103     const result = await res.json();
104     output.textContent += JSON.stringify(result, null, 2);
105     if (window.toast) {
106       if (res.ok) window.toast.success('Операция выполнена', `
Код: ${result.status} || 'OK'`);
107       else window.toast.error('Ошибка', result.detail || 'Сбой
операции');
108     }
109   } catch (err) {
110     output.textContent += `\nОшибка запроса: ${err.message}`;
111     if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
112   }
113 }
114
115 const POLL_ID = 'storage_manager';
116 let timerId = null;
117
118 function getFrequency() {

```

```
119 |   try {
120 |     const saved = localStorage.getItem(`poll_freq_${POLL_ID}`)
    | ;
121 |     if (saved) return saved;
122 |   } catch (_) {}
123 |   return 'manual';
124 | }
125 |
126 | function setFrequency(freq) {
127 |   try {
128 |     localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
129 |   } catch (_) {}
130 |   applyPoller(freq, false);
131 | }
132 |
133 | function stopPolling() {
134 |   if (window.unregisterTabPoller) {
135 |     window.unregisterTabPoller(`tab-storage-manager_${POLL_ID}
    | `);
136 |   }
137 |   if (timerId) {
138 |     clearInterval(timerId);
139 |     timerId = null;
140 |   }
141 | }
142 |
143 | function applyPoller(freq, runInitial = false) {
144 |   stopPolling();
145 |   if (freq === 'start' || freq === 'manual') {
146 |     if (runInitial && (window.isTabActive ? window.isTabActive
    | ('tab-storage-manager') : true)) {
147 |       fetchStorageSummary();
148 |     }
149 |     return;
150 |   }
151 |
152 |   const intervalSec = parseInt(freq, 10);
153 |   if (isNaN(intervalSec) || intervalSec <= 0) return;
```

```
154
155   const intervalMs = intervalSec * 1000;
156   const pollerId = `tab-storage-manager_${POLL_ID}`;
157
158   if (window.registerTabPoller) {
159     window.registerTabPoller('tab-storage-manager', fetchStorageSummary, intervalMs, { pollerId, immediate: runInitial });
160   } else {
161     if (runInitial && (window.isTabActive ? window.isTabActive('tab-storage-manager') : true)) {
162       fetchStorageSummary();
163     }
164     timerId = setInterval(() => {
165       if (window.isTabActive ? window.isTabActive('tab-storage-manager') : true) {
166         fetchStorageSummary();
167       }
168     }, intervalMs);
169   }
170 }
171
172 export function initStorageManagerTab() {
173   if (isInitialized) return;
174   isInitialized = true;
175
176   const refreshBtn = document.getElementById('sm-refresh-btn');
177   ;
178   if (refreshBtn) refreshBtn.addEventListener('click', fetchStorageSummary);
179
180   const execBtn = document.getElementById('sm-execute-btn');
181   if (execBtn) execBtn.addEventListener('click', executeStorageAction);
182
183   const clearBtn = document.getElementById('sm-clear-log-btn');
184   ;
185   if (clearBtn) clearBtn.addEventListener('click', () => {
186     const out = document.getElementById('sm-console-output');
```

```
185     if (out) out.textContent = 'Журнал очищен.';
186   });
187
188   const select = document.getElementById('sm-poll-freq');
189   if (select) {
190     const currentFreq = getFrequency();
191     select.value = currentFreq;
192     select.onchange = (e) => {
193       const newFreq = e.target.value;
194       setFrequency(newFreq);
195       if (newFreq !== 'manual' && newFreq !== 'start') {
196         fetchStorageSummary();
197       }
198       if (window.showToast) {
199         const label = select.options[select.selectedIndex]?.text || newFreq;
200         window.showToast(`Частота опроса хранилища: ${label}`,
201           'info');
202       }
203       applyPoller(currentFreq, window.isTabActive ? window.isTabActive('tab-storage-manager') : true);
204     } else {
205       applyPoller(getFrequency(), window.isTabActive ? window.isTabActive('tab-storage-manager') : true);
206     }
207   }
208
209   export function activateStorageManagerTab() {
210     applyPoller(getFrequency(), true);
211   }
212
213   window.initStorageManagerTab = initStorageManagerTab;
214   window.activateStorageManagerTab = activateStorageManagerTab;
215
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/storage_wear_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/storage_wear_tab/index.
15 |     html" class="w-100 h-100 border-0"></iframe>
16 |
17 | * File: index.html
18 | * Project: ai-breadboard
19 | * Package: windows/api/webgui/storage_wear_tab
20 | * Author: hypo69
21 | * Copyright: © 2026 hypo69
22 | * Updated: 2026-10-08 03:08:00
23 | * =====
24 |=====
25 -->
26
27 <style>
28   .wear-tab-wrapper {
29     padding: 0.75rem;
30     color: var(--text-color, #f8fafc);
31   }
32   .diag-panel {
33     background: var(--surface-2, #1e293b);
34     border: 1px solid var(--border-color, #334155);
35     border-radius: 8px;
```

```
33     overflow: hidden;
34 }
35 .diag-panel-header {
36     background: rgba(15, 23, 42, 0.6);
37     padding: 8px 14px;
38     border-bottom: 1px solid var(--border-color, #334155);
39     font-weight: 600;
40     font-size: 0.85rem;
41 }
42 .about-sys-progress-track {
43     background: #0f172a;
44     border-radius: 4px;
45     height: 8px;
46     overflow: hidden;
47     min-width: 80px;
48 }
49 .about-sys-progress-bar {
50     height: 100%;
51     border-radius: 4px;
52     transition: width 0.3s ease;
53 }
54 .about-sys-spec-table {
55     width: 100%;
56     border-collapse: separate;
57     border-spacing: 0;
58 }
59 .about-sys-spec-table td {
60     padding: 8px 12px;
61     border-bottom: 1px solid rgba(51, 65, 85, 0.35);
62     vertical-align: middle;
63 }
64 .about-sys-spec-key {
65     color: var(--text-muted, #94a3b8);
66     width: 200px;
67     font-size: 0.82rem;
68     font-weight: 500;
69 }
70 .about-sys-spec-val {
```

```

71 |     color: var(--text-color, #f8fafc);
72 |     font-weight: 500;
73 |     font-size: 0.82rem;
74 | }
75 | </style>
76 |
77 | <div class="wear-tab-wrapper">
78 |   <!-- Top Header & Status -->
79 |   <div class="card shadow-sm border-secondary-subtle mb-3">
80 |     <div class="card-body p-3">
81 |       <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
82 |         <div class="d-flex align-items-center gap-2">
83 |           <div class="fs-4">□ </div>
84 |           <div>
85 |             <h5 class="mb-0 fw-bold">4. Износ накопителей и ба
тареи (Storage & Battery Health)</h5>
86 |             <small class="text-muted">Мониторинг ресурса SSD/N
VMe, файловых систем, SMART и деградации аккумулятора</small>
87 |           </div>
88 |         </div>
89 |
90 |         <div class="d-flex align-items-center gap-2 flex-wrap"
>
91 |           <select class="form-select form-select-sm poll-freq-
select" id="diag-wear-poll-freq" data-poll-id="storage_wear" t
itle="Частота опроса: Износ накопителей">
92 |             <option value="start">Только при старте</option>
93 |             <option value="manual" selected>По запросу</option>
94 |             <option value="3">Каждые 3 сек</option>
95 |             <option value="5">Каждые 5 сек</option>
96 |             <option value="10">Каждые 10 сек</option>
97 |             <option value="30">Каждые 30 сек</option>
98 |             <option value="60">Каждые 60 сек</option>
99 |             <option value="120">Каждые 120 сек</option>
100 |           </select>
101 |           <button class="btn btn-sm btn-outline-info rounded-p

```



```

    ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
    "btn-diag-wear-refresh">
102         <i class="bi bi-arrow-clockwise"></i> Обновить
103     </button>
104 </div>
105 </div>
106 </div>
107 </div>
108
109 <div class="row g-2 mb-3">
110     <!-- Disks Health & Wear Panel -->
111     <div class="col-12 col-xl-8 col-lg-8">
112         <div class="diag-panel h-100">
113             <div class="diag-panel-header d-flex justify-content-b
between align-items-center">
114                 <div class="d-flex align-items-center gap-2">
115                     <i class="bi bi-hdd-fill text-primary"></i>
116                     <span class="text-white">Здоровье накопителей SSD/
NVMe и износ (Wear & SMART)</span>
117                 </div>
118                 <span class="badge bg-secondary" id="diag-wear-disks
-count">0 дисков</span>
119             </div>
120             <div class="table-responsive" style="max-height: 480px
; overflow-y: auto;">
121                 <table class="table table-dark table-hover table-sm
mb-0" style="font-size: 0.82rem; font-family: ui-monospace, SF
Mono-Regular, Menlo, Monaco, Consolas, monospace;">
122                     <thead>
123                         <tr class="text-secondary border-secondary">
124                             <th>Физический накопитель</th>
125                             <th>Тип / Шина</th>
126                             <th>Разделы</th>
127                             <th style="text-align: right;">Объем</th>
128                             <th style="text-align: right;">Свободно</th>
129                             <th>Наработка / Старт</th>
130                             <th style="text-align: right;">Ввод / Вывод (I
/ O)</th>

```

```

131 |             <th style="width: 130px;">Здоровье / Износ</th>
132 |         >
133 |             <th>Статус</th>
134 |         </tr>
135 |     </thead>
136 |     <tbody id="diag-wear-disks-tbody">
137 |         <tr><td colspan="9" class="text-center py-4 text
138 | -muted">Опрос контроллеров накопителей...</td></tr>
139 |     </tbody>
140 | </table>
141 | </div>
142 |
143 | <!-- Battery Deep Telemetry -->
144 | <div class="col-12 col-xl-4 col-lg-4">
145 |     <div class="diag-panel h-100">
146 |         <div class="diag-panel-header d-flex justify-content-b
147 | etween align-items-center">
148 |             <div class="d-flex align-items-center gap-2">
149 |                 <i class="bi bi-battery-charging text-warning"></i>
150 |             >
151 |                 <span class="text-white">Деградация и состояние ба
152 | тарей (Battery Degradation)</span>
153 |             </div>
154 |             <span class="badge bg-dark border border-secondary t
155 | ext-info" id="diag-battery-source-badge">AC Mains</span>
156 |         </div>
157 |         <div class="p-3" id="diag-battery-details-container">
158 |             <div class="text-center py-4 text-muted small">
159 |                 <i class="bi bi-plug-fill fs-4 text-info d-block m
160 | b-1"></i>
161 |                 Стационарный компьютер – питание напрямую от элект
162 | росети
163 |             </div>
164 |         </div>
165 |     </div>
166 | </div>

```

```
161 |     </div>
```

```
162 | </div>
```

```
163 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Storage Wear Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/storage_wear_tab/main.
11  *     js?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/storage_wear_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 02:00:00
19  * =====
20  */
21 (function () {
22   'use strict';
23
24   let autoRefreshTimer = null;
25
26   function escapeHtml(str) {
27     if (str === null || str === undefined) return '';
28     return String(str)
29       .replace(/&/g, '&amp;')
30       .replace(/</g, '&lt;')
31       .replace(/>/g, '&gt;')
32       .replace(/"/g, '&quot;')
```

```

33     .replace(/'/g, '&#039;');
34 }
35
36 function setText(id, text) {
37     const el = document.getElementById(id);
38     if (el) el.textContent = text !== null && text !== undefined ? String(text) : '--';
39 }
40
41 async function apiFetch(url) {
42     if (window.api && typeof window.api.fetch === 'function')
43     {
44         return await window.api.fetch(url);
45     }
46     const res = await fetch(url);
47     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.statusText}`);
48     return await res.json();
49 }
50
51 async function fetchStorageBatteryWear() {
52     try {
53         const data = await apiFetch('/api/v1/system/diagnostics/storage-battery');
54         if (!data) return;
55
56         // Disks
57         const disksTbody = document.getElementById('diag-wear-disks-tbody');
58         if (disksTbody && data.disks_wear) {
59             setText('diag-wear-disks-count', `${data.disks_wear.length} дисков`);
60             if (data.disks_wear.length === 0) {
61                 disksTbody.innerHTML = '<tr><td colspan="9" class="text-center py-4 text-muted">Физических накопителей не обнаружено</td></tr>';
62             } else {
63                 const formatBytesLocal = (bytes) => {

```

```

63 |         if (bytes == null || bytes === 0) return '0 B';
64 |         const k = 1024;
65 |         const sizes = ['B', 'KB', 'MB', 'GB', 'TB', 'PB'];
66 |         const i = Math.floor(Math.log(bytes) / Math.log(k)
    | );
67 |         return `${parseFloat((bytes / Math.pow(k, i)).toFixed(1)))} ${sizes[i]}`;
68 |     };
69 |
70 |     const formatPohLocal = (hours) => {
71 |         if (hours == null || hours <= 0) return null;
72 |         const days = Math.floor(hours / 24);
73 |         const years = (hours / (24 * 365.25)).toFixed(1);
74 |         if (days >= 365) return `${hours.toLocaleString()}
    | ч (≈ ${years} г)`;
75 |         if (days >= 1) return `${hours.toLocaleString()} ч
    | (${days} д)`;
76 |         return `${hours.toLocaleString()} ч`;
77 |     };
78 |
79 |     disksTbody.innerHTML = data.disks_wear.map(d => {
80 |         const diskName = d.name || d.model || d.device_id
    | || 'Физический диск';
81 |         const devId = d.device_id ? `<span class="badge bg
    | -secondary-subtle text-light border border-secondary me-1.5">${
    | escapeHtml(d.device_id)}</span>` : '';
82 |         const isSsd = (d.media_type && d.media_type.toUpperCase().includes('SSD')) || (d.bus_type && d.bus_type.toUpperCase().includes('NVME'));
83 |         const diskIcon = isSsd ? 'bi-device-ssd text-info'
    | : 'bi-hdd text-primary';
84 |         const typeBus = [d.media_type, d.bus_type].filter(
    | Boolean).join(' / ') || 'Storage';
85 |         const partsBadge = d.partitions && d.partitions !==
    | = '_'
86 |         ? `<span class="badge bg-primary-subtle text-info border border-info-subtle font-monospace">${escapeHtml(d.partitions)}</span>`

```

```

87         : '<span class="text-muted">—</span>';
88         const freeGbStr = d.free_gb != null ? `${d.free_gb
} GB` : '<span class="text-muted">—</span>';
89         const healthPct = d.health_pct != null ? d.health_
pct : 100;
90         const healthColor = healthPct >= 80 ? 'success' :
(healthPct >= 50 ? 'warning' : 'danger');
91
92         // Наработка и дата первого включения
93         const pohFormatted = formatPohLocal(d.power_on_hou
rs);
94         const pohHtml = pohFormatted
95         ? '<div class="font-monospace text-info">${pohFo
rmatted}</div><div class="small text-muted font-monospace mt-0
.5">Старт: ${escapeHtml(d.first_power_on || '-')}</div>'
96         : '<span class="text-muted font-monospace small"
>—</span>';
97
98         // Объемы ввода/вывода (запись и чтение)
99         const ioHtml = `
100         <div class="font-monospace text-light" style="wh
ite-space: nowrap;" title="Записано за сессию">
101         <i class="bi bi-arrow-up-circle text-warning m
e-1"></i>Записано: ${formatBytesLocal(d.bytes_written)}
102         </div>
103         <div class="small font-monospace text-info mt-0.
5" style="white-space: nowrap;" title="Прочитано за сессию">
104         <i class="bi bi-arrow-down-circle text-info me
-1"></i>Прочитано: ${formatBytesLocal(d.bytes_read)}
105         </div>
106         `;
107
108         return `
109         <tr>
110         <td class="fw-bold text-white">
111         <div class="d-flex align-items-center">
112         <i class="bi ${diskIcon} me-2 fs-6"></i>
113         <div>

```

```

114 |         <div class="text-truncate" style="max-widt
      | h: 250px;" title="{escapeHtml(diskName)}">{escapeHtml(diskNa
      | me)}</div>
115 |         <div class="small text-muted font-monospac
      | e mt-0.5">{devId}{d.serial_number && d.serial_number !== 'N/
      | A' ? `<span class="text-secondary">S/N: {escapeHtml(d.serial_
      | number)}</span>` : ''}</div>
116 |         </div>
117 |     </div>
118 | </td>
119 |     <td class="text-muted font-monospace"><span clas
      | s="badge bg-dark border border-secondary text-light">{escapeH
      | tml(typeBus)}</span></td>
120 |     <td>{partsBadge}</td>
121 |     <td style="text-align: right;" class="text-light
      | font-monospace">{d.total_gb} GB</td>
122 |     <td style="text-align: right;" class="text-succe
      | ss font-monospace">{freeGbStr}</td>
123 |     <td>{pohHtml}</td>
124 |     <td style="text-align: right;">{ioHtml}</td>
125 |     <td>
126 |         <div class="d-flex align-items-center gap-1.5"
      | >
127 |             <div class="about-sys-progress-track flex-gr
      | ow-1" style="height: 6px; min-width: 55px;">
128 |                 <div class="about-sys-progress-bar bg-{$he
      | althColor}" style="width: {healthPct}%;"></div>
129 |             </div>
130 |             <span class="small font-monospace text-{$hea
      | lthColor} ms-1">{healthPct}%</span>
131 |         </div>
132 |     </td>
133 |     <td><span class="badge bg-{$healthColor}-subtle
      | text-{$healthColor} border border-{$healthColor}">{escapeHtml
      | (d.status || 'OK')}</span></td>
134 | </tr>
135 | `;)).join('');
136 | }

```



```

137 |     }
138 |
139 |     // Battery
140 |     const batContainer = document.getElementById('diag-batte
ry-details-container');
141 |     const batSourceBadge = document.getElementById('diag-bat
tery-source-badge');
142 |     if (batContainer && data.battery_wear) {
143 |         const b = data.battery_wear;
144 |         if (batSourceBadge) {
145 |             batSourceBadge.textContent = b.power_source || 'AC M
ains';
146 |         }
147 |         if (!b.has_battery) {
148 |             batContainer.innerHTML = `
149 |                 <div class="text-center py-4 text-muted small">
150 |                     <i class="bi bi-plug-fill fs-4 text-info d-block
mb-1"></i>
151 |                         Стационарный компьютер – питание напрямую от эле
ктросети
152 |                 </div>
153 |             `;
154 |         } else {
155 |             batContainer.innerHTML = `
156 |                 <table class="about-sys-spec-table">
157 |                     <tbody>
158 |                         <tr>
159 |                             <td class="about-sys-spec-key"><i class="bi
bi-battery-charging me-1.5 text-warning"></i>Уровень заряда</t
d>
160 |                             <td class="about-sys-spec-val fw-bold text-w
hite">${b.percent}% (${b.is_charging ? 'Заряжается' : 'Разряд'
})</td>
161 |                         </tr>
162 |                         <tr>
163 |                             <td class="about-sys-spec-key"><i class="bi
bi-shield-shaded me-1.5 text-info"></i>Заводская емкость</td>
164 |                             <td class="about-sys-spec-val font-monospace

```

```

    ">${b.design_capacity_mwh ? b.design_capacity_mwh + ' mWh' : '
    --'}</td>
165 |             </tr>
166 |             <tr>
167 |                 <td class="about-sys-spec-key"><i class="bi
    bi-battery-full me-1.5 text-success"></i>Текущая емкость</td>
168 |                 <td class="about-sys-spec-val font-monospace
    ">${b.full_charge_capacity_mwh ? b.full_charge_capacity_mwh +
    ' mWh' : '--'}</td>
169 |             </tr>
170 |             <tr>
171 |                 <td class="about-sys-spec-key"><i class="bi
    bi-heart-pulse me-1.5 text-danger"></i>Деградация (Износ)</td>
172 |                 <td class="about-sys-spec-val ${b.wear_level
    _pct > 20 ? 'text-danger fw-bold' : 'text-success'}">
173 |                     ${b.wear_level_pct}% износа
174 |                 </td>
175 |             </tr>
176 |         </tbody>
177 |     </table>
178 |     `;
179 |     }
180 | }
181 | } catch (err) {
182 |     console.warn('[StorageWearTab] fetchStorageBatteryWear e
    rror:', err);
183 | }
184 | }
185 |
186 | const POLL_ID = 'storage_wear';
187 |
188 | function getFrequency() {
189 |     try {
190 |         const saved = localStorage.getItem(`poll_freq_${POLL_ID}
    `);
191 |         if (saved) return saved;
192 |     } catch (_) {}
193 |     return 'manual';

```

```

194 | }
195 |
196 | function setFrequency(freq) {
197 |     try {
198 |         localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
199 |     } catch (_) {}
200 |     applyPoller(freq, false);
201 | }
202 |
203 | function stopPolling() {
204 |     if (window.unregisterTabPoller) {
205 |         window.unregisterTabPoller(`tab-storage-wear_${POLL_ID}`
206 | );
207 |     }
208 |     if (autoRefreshTimer) {
209 |         clearInterval(autoRefreshTimer);
210 |         autoRefreshTimer = null;
211 |     }
212 | }
213 |
214 | function applyPoller(freq, runInitial = false) {
215 |     stopPolling();
216 |     if (freq === 'start' || freq === 'manual') {
217 |         if (runInitial && (window.isTabActive ? window.isTabActi
218 | ve('tab-storage-wear') : true)) {
219 |             fetchStorageBatteryWear();
220 |         }
221 |         return;
222 |     }
223 |     const intervalSec = parseInt(freq, 10);
224 |     if (isNaN(intervalSec) || intervalSec <= 0) return;
225 |
226 |     const intervalMs = intervalSec * 1000;
227 |     const pollerId = `tab-storage-wear_${POLL_ID}`;
228 |
229 |     if (window.registerTabPoller) {
230 |         window.registerTabPoller('tab-storage-wear', fetchStorag

```

```

    eBatteryWear, intervalMs, { pollerId, immediate: runInitial })
;
230     } else {
231         if (runInitial && (window.isTabActive ? window.isTabActive('tab-storage-wear') : true)) {
232             fetchStorageBatteryWear();
233         }
234         autoRefreshTimer = setInterval(() => {
235             if (window.isTabActive ? window.isTabActive('tab-storage-wear') : true) {
236                 fetchStorageBatteryWear();
237             }
238         }, intervalMs);
239     }
240 }
241
242 function bindEvents() {
243     const btnRefresh = document.getElementById('btn-diag-wear-refresh');
244     if (btnRefresh) {
245         btnRefresh.onclick = () => fetchStorageBatteryWear();
246     }
247
248     const select = document.getElementById('diag-wear-poll-frequency');
249     if (select) {
250         select.onchange = (e) => {
251             const newFreq = e.target.value;
252             setFrequency(newFreq);
253             if (newFreq !== 'manual' && newFreq !== 'start') {
254                 fetchStorageBatteryWear();
255             }
256             if (window.showToast) {
257                 const label = select.options[select.selectedIndex]?.text || newFreq;
258                 window.showToast(`Частота опроса износа: ${label}`, 'info');
259             }

```

```

260     };
261   }
262 }
263
264 async function initStorageWearTab() {
265   bindEvents();
266   const currentFreq = getFrequency();
267   const select = document.getElementById('diag-wear-poll-fre
q');
268   if (select) select.value = currentFreq;
269   applyPoller(currentFreq, true);
270 }
271
272 window.initStorageWearTab = initStorageWearTab;
273 window.activateStorageWearTab = () => {
274   applyPoller(getFrequency(), true);
275 };
276
277 if (document.readyState === 'loading') {
278   document.addEventListener('DOMContentLoaded', () => {
279     if (window.isTabActive ? window.isTabActive('tab-storage
-wear') : false) {
280       initStorageWearTab();
281     } else {
282       bindEvents();
283     }
284   });
285 } else {
286   if (window.isTabActive ? window.isTabActive('tab-storage-w
ear') : false) {
287     initStorageWearTab();
288   } else {
289     bindEvents();
290   }
291 }
292 }));
293

```

index.html

```
1 <!--
2 =====
3 |
4 |
5 |
6 | Process Name: Windows Web Interface - System32 Commands & Control Catalog
7 |
8 | =====
9 |
10 | Description:
11 |     Информационный каталог штатных средств Windows (%SystemRoot%\System32),
12 |     пояснения по использованию, эквиваленты Control Plane, дерево подкоманд
13 |     и интеллектуальный пульт подбора и запуска команд по свободному тексту.
14 |
15 | Usage Examples:
16 |     Web Route:
17 |         /windows/api/webgui/system32_commands_tab/index.html
18 |
19 |     Tab Frame Integration:
20 |         <iframe src="/windows/api/webgui/system32_commands_tab/index.html" class="w-100 h-100 border-0"></iframe>
21 |
22 | File: index.html
23 | Project: ai-breadboard
24 | Package: windows/api/webgui/system32_commands_tab
25 | Author: hypo69
26 | Copyright: © 2026 hypo69
27 | Updated: 2026-10-08 03:08:00
28 | =====
29 |
30 | -->
31 |
32 | <div class="container-fluid p-0 system32-tab-wrapper d-flex flex-column" style="min-height: 100vh;">
```

```

28 | <!-- Header Card & Summary Stats -->
29 | <div class="card border-secondary-subtle mb-3 shadow-sm flex
    | -shrink-0">
30 |     <div class="card-body p-3">
31 |         <div class="d-flex flex-wrap align-items-center justify-
    | content-between gap-3">
32 |             <div class="d-flex align-items-center gap-2">
33 |                 <div class="fs-2">□ </div>
34 |                 <div>
35 |                     <h5 class="mb-0 fw-bold d-flex align-items-center
    | gap-2">
36 |                         <span>Каталог штатных средств Windows (System32)
    | </span>
37 |                         <span class="badge bg-primary-subtle text-primar
    | y border border-primary-subtle small fs-6">Control Plane & AIT
    | elemetry</span>
38 |                     </h5>
39 |                     <div class="text-muted small">
40 |                         66 функциональных категорий, Telemetry Tiers, эк
    | виваленты Win32 API / PowerShell / WMI / COM, USN Journal и за
    | пуск по свободному тексту.
41 |                     </div>
42 |                 </div>
43 |             </div>
44 |             <div class="d-flex align-items-center gap-2 flex-wrap"
    | >
45 |                 <button class="btn btn-outline-info btn-sm d-flex al
    | ign-items-center gap-1 shadow-sm" id="btn-refresh-system32-cat
    | alog" title="Обновить каталог">
46 |                     <i class="bi bi-arrow-clockwise"></i>
47 |                     <span>Обновить</span>
48 |                 </button>
49 |                 <button class="btn btn-outline-secondary btn-sm d-fl
    | ex align-items-center gap-1 shadow-sm" id="btn-open-inventory-
    | modal" data-bs-toggle="modal" data-bs-target="#inventoryModal"
    | title="Проверить наличие на диске">
50 |                     <i class="bi bi-hdd-stack"></i>
51 |                     <span>Инвентарь хоста</span>

```

```

52 |         </button>
53 |         <button class="btn btn-primary btn-sm d-flex align-i
    | tems-center gap-1.5 shadow-sm fw-semibold" id="btn-open-guide-
    | modal" data-bs-toggle="modal" data-bs-target="#guideModal" tit
    | le="Руководство и архитектура">
54 |             <i class="bi bi-book"></i>
55 |             <span>Руководство</span>
56 |         </button>
57 |     </div>
58 | </div>
59 |
60 |     <!-- Quick Metrics Bar -->
61 |     <div class="row g-2 mt-2 pt-2 border-top border-secondar
    | y-subtle" id="system32-summary-bar">
62 |         <div class="col-6 col-md-2">
63 |             <div class="p-2 bg-body-tertiary rounded text-center
    | ">
64 |                 <div class="text-muted small" style="font-size: 0.
    | 72rem;">Всего средств</div>
65 |                 <div class="fw-bold fs-6 text-primary" id="stat-to
    | tal-tools">...</div>
66 |             </div>
67 |         </div>
68 |         <div class="col-6 col-md-2">
69 |             <div class="p-2 bg-body-tertiary rounded text-center
    | ">
70 |                 <div class="text-muted small" style="font-size: 0.
    | 72rem;">Категорий</div>
71 |                 <div class="fw-bold fs-6 text-info" id="stat-total
    | -categories">...</div>
72 |             </div>
73 |         </div>
74 |         <div class="col-6 col-md-2">
75 |             <div class="p-2 bg-body-tertiary rounded text-center
    | ">
76 |                 <div class="text-muted small" style="font-size: 0.
    | 72rem;">▯ OBSERVE (Телеметрия)</div>
77 |                 <div class="fw-bold fs-6 text-success" id="stat-ob

```



```

    serve-tools">...</div>
78         </div>
79     </div>
80     <div class="col-6 col-md-2">
81         <div class="p-2 bg-body-tertiary rounded text-center
">
82             <div class="text-muted small" style="font-size: 0.
72rem;"> ETW Конвейер</div>
83             <div class="fw-bold fs-6 text-warning" id="stat-et
w-tools">...</div>
84         </div>
85     </div>
86     <div class="col-6 col-md-2">
87         <div class="p-2 bg-body-tertiary rounded text-center
">
88             <div class="text-muted small" style="font-size: 0.
72rem;"> USN Journal</div>
89             <div class="fw-bold fs-6 text-info" id="stat-usn-t
ools">...</div>
90         </div>
91     </div>
92     <div class="col-6 col-md-2">
93         <div class="p-2 bg-body-tertiary rounded text-center
">
94             <div class="text-muted small" style="font-size: 0.
72rem;"> Оснастки MMC/CPL</div>
95             <div class="fw-bold fs-6 text-secondary" id="stat-
gui-tools">...</div>
96         </div>
97     </div>
98 </div>
99 </div>
100 </div>
101
102 <!-- Filters & Search Toolbar -->
103 <div class="card border-secondary-subtle mb-3 shadow-sm flex
-shrink-0">
104     <div class="card-body p-2 d-flex flex-wrap align-items-cen

```

```

ter justify-content-between gap-2">
105     <div class="d-flex align-items-center gap-2 flex-wrap">
106         <span class="small fw-semibold text-muted">AITelemetry
Tier:</span>
107         <div class="btn-group btn-group-sm" role="group" id="t
ier-filters">
108             <button type="button" class="btn btn-outline-seconda
ry active" data-tier="all">Bce</button>
109             <button type="button" class="btn btn-outline-success
" data-tier="OBSERVE">⏏ OBSERVE</button>
110             <button type="button" class="btn btn-outline-warning
" data-tier="DIAGNOSE">⏏ DIAGNOSE</button>
111             <button type="button" class="btn btn-outline-info" d
ata-tier="CONTROL">⏏ CONTROL</button>
112             <button type="button" class="btn btn-outline-danger"
data-tier="ADMIN">⏏ ADMIN</button>
113             <button type="button" class="btn btn-outline-dark" d
ata-tier="DESTRUCTIVE">⏏ ⏏ DESTRUCTIVE</button>
114             <button type="button" class="btn btn-outline-primary
" data-tier="RECOVERY">⏏ RECOVERY</button>
115         </div>
116     </div>
117
118     <div class="d-flex align-items-center gap-2 flex-grow-1
flex-md-grow-0" style="min-width: 260px;">
119         <div class="input-group input-group-sm">
120             <span class="input-group-text bg-body-tertiary"><i c
lass="bi bi-search"></i></span>
121             <input type="text" class="form-control" id="tool-sea
rch-input" placeholder="Поиск по имени, назначению или тегам..
.">
122             <button class="btn btn-outline-secondary" type="butt
on" id="btn-clear-search" title="Очистить"><i class="bi bi-x">
</i></button>
123         </div>
124     </div>
125 </div>
126 </div>

```

```

127
128 <!-- Tool Cards Grid / Main Catalog Display -->
129 <div class="flex-grow-1 overflow-auto mb-3" style="max-height:
    calc(100vh - 440px); min-height: 280px;">
130     <div class="row g-2" id="system32-cards-container">
131         <div class="col-12 text-center py-5 text-muted">
132             <div class="spinner-border spinner-border-sm text-primary me-2"></div>
133             <span>Загрузка каталога штатных средств Windows...</span>
134         </div>
135     </div>
136 </div>
137
138 <!-- Bottom AI Command Execution & Interpretation Panel (Identical in Style to tab-scenarios) -->
139 <div class="card border-primary border-opacity-50 shadow-lg mt-auto flex-shrink-0" >
140
141     <!-- Quick Prompts / Pills -->
142     <div class="p-2 border-bottom border-secondary-subtle bg-body-secondary bg-opacity-50">
143         <div class="d-flex align-items-center justify-content-between mb-1">
144             <span class="small text-muted fw-semibold" style="font-size: 0.75rem;">□ Экспресс-команды свободным текстом:</span>
145             <span class="text-muted small" style="font-size: 0.72rem;">Модель подберет программу и сформирует точный синтаксис</span>
146         </div>
147         <div class="d-flex flex-wrap gap-1" id="system32-quick-prompts">
148             <button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Проверь статус TRIM для SSD">
149                 □ Проверить статус TRIM для SSD
150             </button>
151             <button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Очисти локальный кэш сопостав

```

```

        ителя DNS">
152         □ Сбросить кэш DNS (flushdns)
153     </button>
154     <button class="btn btn-outline-secondary btn-sm py-0 p
x-2 quick-cmd-pill" data-prompt="Покажи все активные сетевые п
орты и PID процессов">
155         □ Активные сетевые порты (netstat)
156     </button>
157     <button class="btn btn-outline-secondary btn-sm py-0 p
x-2 quick-cmd-pill" data-prompt="Проверь статус шифрования Bit
Locker на дисках">
158         □ Статус шифрования BitLocker
159     </button>
160     <button class="btn btn-outline-secondary btn-sm py-0 p
x-2 quick-cmd-pill" data-prompt="Покажи активные сессии ETW тр
ассировки">
161         □ Активные ETW сессии (logman)
162     </button>
163     <button class="btn btn-outline-secondary btn-sm py-0 p
x-2 quick-cmd-pill" data-prompt="Покажи метаданные NTFS USN Jo
urnal тома C">
164         □ NTFS USN Journal
165     </button>
166     <button class="btn btn-outline-secondary btn-sm py-0 p
x-2 quick-cmd-pill" data-prompt="Кто текущий пользователь и ка
кие у него привилегии">
167         □ Кто я и мои привилегии (whoami)
168     </button>
169 </div>
170 </div>
171
172 <!-- Live Execution Results Console (Shown on run) -->
173 <div class="p-3 border-bottom border-secondary-subtle bg-d
ark text-white" id="ai-command-result-box" style="display: non
e;">
174     <div class="d-flex align-items-center justify-content-be
tween mb-2">
175         <div class="d-flex align-items-center gap-2">

```

```

176         <span class="fs-5" id="res-status-icon">␣ </span>
177         <div>
178             <span class="fw-bold text-info" id="res-tool-name"
179             >...</span>
180             <span class="badge bg-secondary ms-2" id="res-tier
181             -badge">OBSERVE</span>
182             <span class="badge bg-danger ms-1" id="res-danger-
183             badge" style="display: none;">HIGH</span>
184         </div>
185     </div>
186     <div class="d-flex align-items-center gap-2">
187         <span class="text-muted small" id="res-duration-badg
188         e">0 ms</span>
189         <button class="btn btn-outline-light btn-sm py-0 px-
190         2" id="btn-copy-command-output" title="Скопировать вывод">
191             <i class="bi bi-clipboard me-1"></i>Копировать
192         </button>
193         <button class="btn btn-outline-danger btn-sm py-0 px
194         -1.5" id="btn-close-result-box" title="Закрыть панель">
195             <i class="bi bi-x"></i>
196         </button>
197     </div>
198 </div>
199
200     <div class="mb-2 small text-light bg-secondary bg-opacit
201     y-25 p-2 rounded">
202         <div><strong>␣ Сформированная команда:</strong> <code
203         class="text-warning" id="res-cmd-line">...</code></div>
204         <div class="mt-1"><strong>␣ Назначение:</strong> <spa
205         n id="res-explanation">...</span></div>
206         <div class="mt-1 d-flex gap-3 flex-wrap text-muted" st
207         yle="font-size: 0.75rem;">
208             <span id="res-ps-equiv"></span>
209             <span id="res-win32-equiv"></span>
210         </div>
211     </div>
212
213 <!-- Output Console -->

```

```

204     <pre class="p-2 bg-black rounded text-success small mb-0
      font-monospace" style="max-height: 200px; overflow-y: auto; w
      hite-space: pre-wrap;" id="res-output-pre">Ожидание выполнения
      ...</pre>
205   </div>
206
207   <!-- Bottom Input Form -->
208   <div class="p-2">
209     <form id="system32-ai-cmd-form" class="d-flex flex-colum
      n gap-2">
210       <div class="input-group">
211         <span class="input-group-text bg-primary text-white"
      ><i class="bi bi-terminal-fill"></i></span>
212         <input type="text" class="form-control form-control-
      lg fs-6" id="system32-cmd-input" placeholder="Введите задачу с
      вободным текстом (например: 'проверь статус trim для ssd', 'по
      кажи порты netstat', 'сбрось кэш dns')..." autocomplete="off">
213         <button class="btn btn-primary d-flex align-items-ce
      nter gap-2 px-3 fw-semibold" type="submit" id="btn-system32-cm
      d-send">
214           <span>Подобрать и запустить</span>
215           <i class="bi bi-play-circle-fill"></i>
216         </button>
217       </div>
218
219       <div class="d-flex align-items-center justify-content-
      between px-1 flex-wrap gap-2">
220         <span class="text-muted small" id="ai-cmd-status-ind
      icator">Готов к приему команд</span>
221         <div class="d-flex align-items-center gap-3">
222           <div class="form-check form-switch m-0 d-flex alig
      n-items-center gap-1">
223             <input class="form-check-input mt-0" type="check
      box" role="switch" id="check-cmd-dry-run">
224             <label class="form-check-label text-muted user-s
      elect-none small" for="check-cmd-dry-run" title="Только симули
      ровать и показать подобранную команду">
225               Симуляция (Dry-Run)

```

```

226         </label>
227     </div>
228     <div class="form-check form-switch m-0 d-flex align-items-center gap-1">
229         <input class="form-check-input mt-0" type="checkbox" role="switch" id="check-cmd-confirm-admin" checked>
230         <label class="form-check-label text-muted user-select-none small" for="check-cmd-confirm-admin" title="Защита SafeOps: запрашивать подтверждение для HIGH/CRITICAL">
231             Подтверждать опасные операции
232         </label>
233     </div>
234 </div>
235 </div>
236 </form>
237 </div>
238
239 </div>
240
241 <!-- Tool Details Modal -->
242 <div class="modal fade" id="toolDetailsModal" tabindex="-1"
243     aria-labelledby="toolDetailsModalLabel" aria-hidden="true">
244     <div class="modal-dialog modal-lg modal-dialog-scrollable">
245     <div class="modal-content">
246         <div class="modal-header py-2 px-3">
247             <h6 class="modal-title fw-bold d-flex align-items-center gap-2" id="toolDetailsModalLabel">
248                 <span class="fs-5" id="modal-tool-icon">❏ </span>
249                 <span id="modal-tool-title">Сведения об инструменте</span>
250             </h6>
251             <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Закрыть"></button>
252         </div>
253         <div class="modal-body p-3" id="modal-tool-body">
254             <!-- Populated dynamically -->

```

```

255 |         <div class="modal-footer py-2 px-3">
256 |             <button type="button" class="btn btn-secondary btn-s
m" data-bs-dismiss="modal">Закрыть</button>
257 |             <button type="button" class="btn btn-primary btn-sm"
id="btn-modal-paste-command">Вставить команду в пульт</button
>
258 |         </div>
259 |     </div>
260 | </div>
261 | </div>
262 |
263 | <!-- Host Inventory Modal -->
264 | <div class="modal fade" id="inventoryModal" tabindex="-1" ar
ia-labelledby="inventoryModalLabel" aria-hidden="true">
265 |     <div class="modal-dialog modal-lg modal-dialog-scrollable"
>
266 |         <div class="modal-content">
267 |             <div class="modal-header py-2 px-3">
268 |                 <h6 class="modal-title fw-bold" id="inventoryModallLa
bel">□ Инвентарь файлов %SystemRoot%\System32</h6>
269 |                 <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Закрыть"></button>
270 |             </div>
271 |             <div class="modal-body p-3" id="modal-inventory-body">
272 |                 <div class="text-center py-4 text-muted">
273 |                     <div class="spinner-border spinner-border-sm text-
primary mb-2"></div>
274 |                     <div>Сверка с файловой системой хоста...</div>
275 |                 </div>
276 |             </div>
277 |         </div>
278 |     </div>
279 | </div>
280 |
281 | <!-- Architecture & Usage Guide Modal -->
282 | <div class="modal fade" id="guideModal" tabindex="-1" aria-l
abelledby="guideModalLabel" aria-hidden="true">
283 |     <div class="modal-dialog modal-xl modal-dialog-scrollable"

```



```

>
284     <div class="modal-content">
285         <div class="modal-header py-2 px-3">
286             <h6 class="modal-title fw-bold" id="guideModalLabel"
>□ Руководство по штатным средствам Windows 10/11</h6>
287             <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Закрыть"></button>
288         </div>
289         <div class="modal-body p-3">
290             <h6>□ 1. Слои Windows Control Plane</h6>
291             <p class="small text-muted">
292                 Каждая операция в Windows может выполняться не тол
ько через <code>System32/*.exe</code>, но и через соответствую
щие эквиваленты:
293                 PowerShell Cmdlets, WMI/CIM классы, COM интерфейсы
, прямые вызовы Win32 API в системных DLL, события ETW и канал
ы Windows Event Log.
294             </p>
295
296             <h6 class="mt-3">□ 2. Градация AITelemetry Tiers</h
6>
297             <ul class="small text-muted">
298                 <li><strong>□ OBSERVE:</strong> Безопасное считыв
ание телеметрии (<code>systeminfo</code>, <code>tasklist</code>
>, <code>netstat</code>, <code>fsutil usn</code>, <code>logman
query</code>, <code>whoami</code>).</li>
299                 <li><strong>□ DIAGNOSE:</strong> Трассировка, диа
гностика задержек, сетевые снифферы (<code>ping</code>, <code>
tracert</code>, <code>pathping</code>, <code>pktmon</code>, <c
ode>sxstrace</code>).</li>
300                 <li><strong>□ CONTROL:</strong> Управление служба
ми, расписанием, некритичными параметрами (<code>sc start/stop
</code>, <code>schtasks /run</code>, <code>tzutil</code>).</li>
>
301                 <li><strong>□ ADMIN:</strong> Изменение configura
ции системы, брандмауэра, политик безопасности (<code>reg add<
/code>, <code>sc config</code>, <code>netsh advfirewall</code>
, <code>dism</code>).</li>

```

```

302 |         <li><strong>☐ DESTRUCTIVE:</strong> Деструктивн
|   ые операции (<code>diskpart clean/format</code>, <code>format.
|   com</code>, <code>bcdedit /delete</code>).</li>
303 |         <li><strong>☐ RECOVERY:</strong> Среда восстановл
|   ения и ремонт ОС (<code>reagentc</code>, <code>bcdboot</code>,
|   <code>bootrec</code>, <code>wbadmin</code>, <code>sfc</code>)
|   .</li>
304 |     </ul>
305 |
306 |     <h6 class="mt-3">☐ 3. Сквозной конвейер ETW</h6>
307 |     <p class="small text-muted">
308 |         <code>Sensors → ETW / Performance Counters (logman
|   , pktmon, typeperf) → AITelemetry Pipeline → SQLite (telemetry
|   .db)</code>
309 |     </p>
310 |
311 |     <h6 class="mt-3">☐ 4. Наблюдение через NTFS USN Jou
|   rnal</h6>
312 |     <p class="small text-muted">
313 |         Команда <code>fsutil usn queryjournal C:</code> по
|   зволяет отслеживать изменения файлов в реальном времени без ре
|   сурсоемкого сканирования всего диска.
314 |     </p>
315 |     </div>
316 |   </div>
317 | </div>
318 | </div>
319 |
320 | </div>
321 |

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Web Interface - System32 Commands Tab
4  * Main Script
5  * =====
6  * Description:
7  *   Клиентский контроллер вкладки tab-system32-commands: отображение каталога
8  *   штатных средств Windows, слоев Control Plane, AITelemetry Tiers, инвентаря
9  *   и интеллектуального пульта подбора и запуска команд по свободному тексту.
10 *
11 * Usage Examples:
12 *   HTML Integration:
13 *     <script src="/windows/api/webgui/system32_commands_tab/main.js?v=20261006_v1" type="module"></script>
14 *
15 * File: main.js
16 * Project: ai-breadboard
17 * Package: windows/api/webgui/system32_commands_tab
18 * Author: hypo69
19 * Copyright: © 2026 hypo69
20 * Updated: 2026-10-06 15:20:00
21 * =====
22 */
23 (function () {
24   'use strict';
25
26   let toolsList = [];
27   let currentTierFilter = 'all';
28   let searchQuery = '';
```

```
29 | let activeSelectedTool = null;
30 | let isExecuting = false;
31 |
32 | const tierColors = {
33 |     OBSERVE: { bg: 'bg-success-subtle', text: 'text-success',
border: 'border-success-subtle', badge: 'bg-success' },
34 |     DIAGNOSE: { bg: 'bg-warning-subtle', text: 'text-warning',
border: 'border-warning-subtle', badge: 'bg-warning text-dark
' },
35 |     CONTROL: { bg: 'bg-info-subtle', text: 'text-info', border
: 'border-info-subtle', badge: 'bg-info text-dark' },
36 |     ADMIN: { bg: 'bg-danger-subtle', text: 'text-danger', bord
er: 'border-danger-subtle', badge: 'bg-danger' },
37 |     DESTRUCTIVE: { bg: 'bg-dark-subtle', text: 'text-light', b
order: 'border-dark', badge: 'bg-dark text-danger fw-bold' },
38 |     RECOVERY: { bg: 'bg-primary-subtle', text: 'text-primary',
border: 'border-primary-subtle', badge: 'bg-primary' },
39 | };
40 |
41 | const planeIcons = {
42 |     CLI: '□ ',
43 |     POWERSHELL: '□ ',
44 |     WMI_CIM: '□ ',
45 |     COM: '□ ',
46 |     WIN32_API: '□ ',
47 |     NATIVE_NT_API: '□ ',
48 |     ETW: '□ ',
49 |     EVENT_LOG: '□ ',
50 |     REGISTRY: '□ ',
51 |     GUI_MMC_CPL: '□ ',
52 | };
53 |
54 | function escapeHtml(str) {
55 |     if (!str) return '';
56 |     return String(str)
57 |         .replace(/&/g, '&amp;')
58 |         .replace(/</g, '&lt;')
59 |         .replace(/>/g, '&gt;')
```

```

60     .replace(/"/g, '&quot;');
61     .replace(/'/g, '&#039;');
62 }
63
64 // Load summary metrics and full catalog from backend
65 async function loadCatalog() {
66     const container = document.getElementById('system32-cards-
67     container');
68     if (!container) return;
69
70     try {
71         const [catRes, sumRes] = await Promise.all([
72             fetch('/api/v1/system32/catalog'),
73             fetch('/api/v1/system32/summary')
74         ]);
75
76         if (!catRes.ok) throw new Error(`HTTP ${catRes.status}`)
77         ;
78
79         const catData = await catRes.json();
80         toolsList = catData.tools || [];
81
82         if (sumRes.ok) {
83             const sumData = await sumRes.json();
84             updateSummaryMetrics(sumData);
85         }
86
87         renderToolsGrid();
88     } catch (err) {
89         console.error('[System32CommandsTab] Error loading catal
90         og:', err);
91         if (container) {
92             container.innerHTML = `
93                 <div class="col-12 text-center py-4 text-danger">
94                     <i class="bi bi-exclamation-triangle fs-3 mb-2"></
95                 i>
96
97                 <div>Не удалось загрузить каталог System32: ${esca
98                 peHtml(err.message)}</div>
99                 <button class="btn btn-outline-primary btn-sm mt-2

```

```

    " onclick="location.reload()">Повторить</button>
93     </div>
94     `;
95     }
96     }
97     }
98
99     function updateSummaryMetrics(sum) {
100         const elTotal = document.getElementById('stat-total-tools'
    );
101         const elCats = document.getElementById('stat-total-categor
    ies');
102         const elObs = document.getElementById('stat-observe-tools'
    );
103         const elEtw = document.getElementById('stat-etw-tools');
104         const elUsn = document.getElementById('stat-usn-tools');
105         const elGui = document.getElementById('stat-gui-tools');
106
107         if (elTotal) elTotal.innerText = sum.total_tools || toolsL
    ist.length;
108         if (elCats) elCats.innerText = sum.categories_count || '66
    ';
109         if (elObs) elObs.innerText = sum.by_telemetry_tier?.OBSERV
    E || sum.read_tools_count || '0';
110         if (elEtw) elEtw.innerText = sum.etw_tools_count || '0';
111         if (elUsn) elUsn.innerText = sum.usn_journal_tools_count |
    | '1';
112         if (elGui) elGui.innerText = (sum.gui_tools_count || 0) +
    (sum.msc_consoles_count || 0);
113     }
114
115     function getFilteredTools() {
116         let list = [...toolsList];
117         if (currentTierFilter !== 'all') {
118             list = list.filter(t => t.telemetry_tier === currentTier
    Filter);
119         }
120         if (searchQuery) {

```

```

121 |     const q = searchQuery.toLowerCase().trim();
122 |     list = list.filter(t =>
123 |         t.executable.toLowerCase().includes(q) ||
124 |         t.purpose.toLowerCase().includes(q) ||
125 |         t.category_code.toLowerCase().includes(q) ||
126 |         (t.tags && t.tags.some(tag => tag.toLowerCase().includ
    | es(q)))
127 |     );
128 | }
129 | return list;
130 | }
131 |
132 | function renderToolsGrid() {
133 |     const container = document.getElementById('system32-cards-
    | container');
134 |     if (!container) return;
135 |
136 |     const filtered = getFilteredTools();
137 |
138 |     if (filtered.length === 0) {
139 |         container.innerHTML = `
140 |             <div class="col-12 text-center py-5 text-muted">
141 |                 <i class="bi bi-search fs-2 mb-2"></i>
142 |                 <div>Инструменты не найдены по заданному фильтру</di
    | v>
143 |             </div>
144 |         `;
145 |         return;
146 |     }
147 |
148 |     container.innerHTML = filtered.map((tool, idx) => {
149 |         const tierStyle = tierColors[tool.telemetry_tier] || { b
    | adge: 'bg-secondary' };
150 |         const planeIcon = planeIcons[tool.primary_control_plane]
    | || '□';
151 |         const firstTemplate = tool.command_templates && tool.com
    | mand_templates.length > 0 ? tool.command_templates[0] : tool.e
    | xecutable;

```

```

152 |
153 |     return `
154 |         <div class="col-12 col-md-6 col-lg-4 col-xl-3">
155 |             <div class="card h-100 border-secondary-subtle shadow-sm hover-shadow transition">
156 |                 <div class="card-header bg-transparent border-secondary-subtle py-2 px-3 d-flex align-items-center justify-content-between">
157 |                     <div class="d-flex align-items-center gap-1.5 overflow-hidden text-truncate">
158 |                         <span class="fs-5">${planeIcon}</span>
159 |                         <span class="fw-bold font-monospace text-truncate text-primary" title="${escapeHtml(tool.executable)}">${escapeHtml(tool.executable)}</span>
160 |                     </div>
161 |                     <span class="badge ${tierStyle.badge} small" style="font-size: 0.68rem;">${escapeHtml(tool.telemetry_tier)}</span>
162 |                 </div>
163 |                 <div class="card-body p-2 d-flex flex-column justify-content-between">
164 |                     <div>
165 |                         <div class="text-muted small mb-1 text-truncate" style="font-size: 0.72rem;" title="${escapeHtml(tool.category_code)}">
166 |                             □ ${escapeHtml(tool.category_code)}
167 |                         </div>
168 |                         <p class="card-text small text-body mb-2" style="font-size: 0.78rem; display: -webkit-box; -webkit-line-clamp: 2; -webkit-box-orient: vertical; overflow: hidden;" title="${escapeHtml(tool.purpose)}">
169 |                             ${escapeHtml(tool.purpose)}
170 |                         </p>
171 |                     </div>
172 |                     <div class="pt-2 border-top border-secondary-subtle d-flex align-items-center justify-content-between gap-1">
173 |                         <button class="btn btn-outline-info btn-sm py-0.5 px-2 btn-open-tool-details" data-idx="${idx}" style="font-

```



```

size: 0.75rem;" title="Посмотреть эквиваленты и руководство">
174 |         □ Инструкция
175 |     </button>
176 |     <button class="btn btn-outline-primary btn-sm
py-0.5 px-2 btn-paste-cmd" data-cmd="{escapeHtml(firstTemplat
e)}" style="font-size: 0.75rem;" title="Вставить шаблон в пуль
т">
177 |         □ В пульт
178 |     </button>
179 | </div>
180 | </div>
181 | </div>
182 | </div>
183 | `;
184 | }).join('');
185 |
186 | // Attach click events
187 | container.querySelectorAll('.btn-open-tool-details').forEa
ch(btn => {
188 |     btn.onclick = () => {
189 |         const idx = parseInt(btn.getAttribute('data-idx'), 10)
190 |         ;
191 |         if (!isNaN(idx) && filtered[idx]) {
192 |             showToolDetailsModal(filtered[idx]);
193 |         }
194 |     };
195 | });
196 | container.querySelectorAll('.btn-paste-cmd').forEach(btn =
> {
197 |     btn.onclick = () => {
198 |         const cmd = btn.getAttribute('data-cmd');
199 |         const input = document.getElementById('system32-cmd-in
put');
200 |         if (input && cmd) {
201 |             input.value = cmd;
202 |             input.focus();
203 |         }

```

```

204     };
205   });
206 }
207
208 function showToolDetailsModal(tool) {
209   activeSelectedTool = tool;
210   const modalTitle = document.getElementById('modal-tool-tit
le');
211   const modalIcon = document.getElementById('modal-tool-icon
');
212   const modalBody = document.getElementById('modal-tool-body
');
213
214   if (modalTitle) modalTitle.innerText = `${tool.executable}
(${tool.category_code})`;
215   if (modalIcon) modalIcon.innerText = planeIcons[tool.prima
ry_control_plane] || '□';
216
217   let subcommandsHtml = '';
218   if (tool.subcommands_info && Object.keys(tool.subcommands_
info).length > 0) {
219     subcommandsHtml = `
220       <h6 class="fw-bold mt-3 mb-2">□ Подкоманды и контексты
:</h6>
221       <div class="list-group list-group-flush border rounded
small">
222         ${Object.entries(tool.subcommands_info).map(([sub, d
esc]) => `
223           <div class="list-group-item py-1.5 px-2 d-flex jus
tify-content-between align-items-center">
224             <code class="text-primary font-monospace">${esca
peHtml(sub)}</code>
225             <span class="text-muted">${escapeHtml(desc)}</sp
an>
226           </div>
227           `).join('')}
228       </div>
229     `;

```

```

230     }
231
232     let templatesHtml = '';
233     if (tool.command_templates && tool.command_templates.length
234         h > 0) {
235         templatesHtml = `
236             <h6 class="fw-bold mt-3 mb-2">📄 Шаблоны команд:</h6>
237             <div class="d-flex flex-column gap-1">
238                 ${tool.command_templates.map(tmpl => `
239                     <div class="d-flex align-items-center justify-content-between p-2 bg-body-secondary rounded font-monospace small">
240                         <code class="text-warning">${escapeHtml(tmpl)}</code>
241                         <button class="btn btn-sm btn-outline-secondary py-0 px-1.5 btn-copy-inline" data-clip="${escapeHtml(tmpl)}" title="Копировать">
242                             <i class="bi bi-clipboard"></i>
243                         </button>
244                     </div>
245                 `).join('')}
246             </div>
247         `;
248
249     if (modalBody) {
250         modalBody.innerHTML = `
251             <div class="mb-3">
252                 <div class="text-muted small mb-1">Назначение:</div>
253                 <div class="p-2 bg-body-tertiary rounded small fw-medium">${escapeHtml(tool.purpose)}</div>
254             </div>
255
256             <div class="row g-2 mb-3">
257                 <div class="col-6 col-md-3">
258                     <div class="p-2 bg-body-tertiary rounded text-center small">
259                         <span class="text-muted d-block" style="font-siz

```

```

e: 0.7rem;">Tier:</span>
260         <span class="badge ${tierColors[tool.telemetry_t
ier]}.badge || 'bg-secondary'}">${escapeHtml(tool.telemetry_ti
er)}</span>
261     </div>
262 </div>
263     <div class="col-6 col-md-3">
264         <div class="p-2 bg-body-tertiary rounded text-cent
er small">
265             <span class="text-muted d-block" style="font-siz
e: 0.7rem;">Опасность:</span>
266             <span class="fw-bold ${tool.danger_level === 'CR
ITICAL' ? 'text-danger' : 'text-body'}">${escapeHtml(tool.dang
er_level)}</span>
267         </div>
268     </div>
269     <div class="col-6 col-md-3">
270         <div class="p-2 bg-body-tertiary rounded text-cent
er small">
271             <span class="text-muted d-block" style="font-siz
e: 0.7rem;">Привилегии:</span>
272             <span class="fw-semibold">${escapeHtml(tool.requ
ired_privileges)}</span>
273         </div>
274     </div>
275     <div class="col-6 col-md-3">
276         <div class="p-2 bg-body-tertiary rounded text-cent
er small">
277             <span class="text-muted d-block" style="font-siz
e: 0.7rem;">Control Plane:</span>
278             <span class="badge bg-secondary">${escapeHtml(to
ol.primary_control_plane)}</span>
279         </div>
280     </div>
281 </div>
282
283     <h6 class="fw-bold mb-2">□ Эквиваленты в Control Plan
e:</h6>

```

```

284 |         <div class="table-responsive">
285 |             <table class="table table-sm table-bordered small mb
| -0">
286 |                 <tbody>
287 |                     <tr>
288 |                         <th style="width: 25%;">PowerShell</th>
289 |                         <td><code>${escapeHtml(tool.powershell_equival
| ent || '-' )}</code></td>
290 |                     </tr>
291 |                     <tr>
292 |                         <th>Win32 API</th>
293 |                         <td><code>${escapeHtml(tool.native_api_equival
| ent || '-' )}</code></td>
294 |                     </tr>
295 |                     <tr>
296 |                         <th>WMI / CIM</th>
297 |                         <td><code>${escapeHtml(tool.wmi_cim_equivalent
| || '-' )}</code></td>
298 |                     </tr>
299 |                     <tr>
300 |                         <th>COM</th>
301 |                         <td><code>${escapeHtml(tool.com_equivalent ||
| '-' )}</code></td>
302 |                     </tr>
303 |                     ${tool.etw_pipeline_enabled ? '<tr><th>ETW Pipel
| ine</th><td><span class="badge bg-success">Интегрирован в AITe
| lemetry</span></td></tr>' : ''}
304 |                     ${tool.usn_journal_enabled ? '<tr><th>USN Journa
| l</th><td><span class="badge bg-info">Поддержка прямого чтения
| NTFS USN</span></td></tr>' : ''}
305 |                 </tbody>
306 |             </table>
307 |         </div>
308 |
309 |         ${subcommandsHtml}
310 |         ${templatesHtml}
311 |     `;
312 |

```

```

313 |         // Copy buttons in modal
314 |         modalBody.querySelectorAll('.btn-copy-inline').forEach(b
    | tn => {
315 |             btn.onclick = () => {
316 |                 const txt = btn.getAttribute('data-clip');
317 |                 if (txt) navigator.clipboard.writeText(txt);
318 |             };
319 |         });
320 |     }
321 |
322 |     const modalEl = document.getElementById('toolDetailsModal'
    | );
323 |     if (modalEl && typeof bootstrap !== 'undefined' && bootstr
    | ap.Modal) {
324 |         const modal = bootstrap.Modal.getOrCreateInstance(modalE
    | l);
325 |         modal.show();
326 |     }
327 | }
328 |
329 | // Load and display host inventory
330 | async function loadHostInventory() {
331 |     const body = document.getElementById('modal-inventory-body
    | ');
332 |     if (!body) return;
333 |
334 |     try {
335 |         const res = await fetch('/api/v1/system32/inventory');
336 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
337 |         const data = await res.json();
338 |
339 |         body.innerHTML = `
340 |             <div class="d-flex align-items-center justify-content-
    | between mb-3 p-2 bg-body-tertiary rounded small">
341 |                 <div>📦 Каталог: <code>${escapeHtml(data.system32_pa
    | th)}</code></div>
342 |                 <div>🔍 Найдено: <strong>${data.total_installed}</str
    | ong> / <strong>${data.total_registered}</strong></div>

```

```

343 |         </div>
344 |         <div class="table-responsive" style="max-height: 400px
    | ; overflow-y: auto;">
345 |             <table class="table table-sm table-hover table-strip
    | ed small">
346 |                 <thead>
347 |                     <tr>
348 |                         <th>Инструмент</th>
349 |                         <th>Категория</th>
350 |                         <th>Статус</th>
351 |                         <th>Путь / Источник</th>
352 |                     </tr>
353 |                 </thead>
354 |                 <tbody>
355 |                     ${data.installed.map(item => `
356 |                         <tr>
357 |                             <td class="font-monospace fw-bold text-succe
    | ss">${escapeHtml(item.executable)}</td>
358 |                             <td class="text-muted">${escapeHtml(item.cat
    | egory_code || item.category)}</td>
359 |                             <td><span class="badge bg-success-subtle tex
    | t-success border border-success-subtle">Доступен</span></td>
360 |                             <td class="font-monospace text-muted" style=
    | "font-size: 0.72rem;">${escapeHtml(item.path)}</td>
361 |                         </tr>
362 |                     `).join('')}</tbody>
363 |                     ${data.missing.map(item => `
364 |                         <tr class="table-warning">
365 |                             <td class="font-monospace text-muted">${esca
    | peHtml(item.executable)}</td>
366 |                             <td class="text-muted">${escapeHtml(item.cat
    | egory_code || item.category)}</td>
367 |                             <td><span class="badge bg-secondary-subtle t
    | ext-secondary">Не найден</span></td>
368 |                             <td class="text-muted" style="font-size: 0.7
    | 2rem;">${escapeHtml(item.source)}</td>
369 |                         </tr>
370 |                     `).join('')}</tbody>

```

```
371         </tbody>
372     </table>
373 </div>
374 `;
375 } catch (err) {
376     body.innerHTML = `<div class="text-danger p-3 text-center">Ошибка инвентаризации: ${escapeHtml(err.message)}</div>`;
377 }
378 }
379
380 // Setup Bottom AI Command Input Bar & Execution
381 function setupAiCommandRunner() {
382     const form = document.getElementById('system32-ai-cmd-form');
383     const input = document.getElementById('system32-cmd-input');
384     const sendBtn = document.getElementById('btn-system32-cmd-send');
385     const statusInd = document.getElementById('ai-cmd-status-indicator');
386     const resultBox = document.getElementById('ai-command-result-box');
387     const dryRunCheck = document.getElementById('check-cmd-dry-run');
388     const confirmAdminCheck = document.getElementById('check-cmd-confirm-admin');
389
390     const resToolName = document.getElementById('res-tool-name');
391     const resTierBadge = document.getElementById('res-tier-badge');
392     const resDangerBadge = document.getElementById('res-danger-badge');
393     const resDuration = document.getElementById('res-duration-badge');
394     const resCmdLine = document.getElementById('res-cmd-line');
395     const resExplanation = document.getElementById('res-explan
```



```

    ation');
396     const resPsEquiv = document.getElementById('res-ps-equiv')
    ;
397     const resWin32Equiv = document.getElementById('res-win32-e
    quiv');
398     const resOutput = document.getElementById('res-output-pre'
    );
399     const btnCopyOutput = document.getElementById('btn-copy-co
    mmand-output');
400     const btnCloseBox = document.getElementById('btn-close-res
    ult-box');
401
402     if (btnCloseBox && resultBox) {
403         btnCloseBox.onclick = () => { resultBox.style.display =
    'none'; };
404     }
405
406     if (btnCopyOutput && resOutput) {
407         btnCopyOutput.onclick = () => {
408             navigator.clipboard.writeText(resOutput.innerText);
409             btnCopyOutput.innerHTML = '<i class="bi bi-check2 me-1
    "></i>Скопировано';
410             setTimeout(() => {
411                 btnCopyOutput.innerHTML = '<i class="bi bi-clipboard
    me-1"></i>Копировать';
412             }, 1800);
413         };
414     }
415
416     // Modal paste command button
417     const btnModalPaste = document.getElementById('btn-modal-p
    aste-command');
418     if (btnModalPaste && input) {
419         btnModalPaste.onclick = () => {
420             if (activeSelectedTool) {
421                 const tpl = activeSelectedTool.command_templates?.[
    0] || activeSelectedTool.executable;
422                 input.value = tpl;

```

```
423         input.focus();
424         const modalEl = document.getElementById('toolDetails
Modal');
425         if (modalEl && typeof bootstrap !== 'undefined' && b
ootstrap.Modal) {
426             bootstrap.Modal.getInstance(modalEl)?.hide();
427         }
428     }
429 };
430 }
431
432 // Quick command pills
433 document.querySelectorAll('.quick-cmd-pill').forEach(btn =
> {
434     btn.onclick = () => {
435         if (isExecuting) return;
436         const prompt = btn.getAttribute('data-prompt');
437         if (prompt && input) {
438             input.value = prompt;
439             executePrompt(prompt);
440         }
441     };
442 });
443
444 if (form && input) {
445     form.onsubmit = (e) => {
446         e.preventDefault();
447         const prompt = input.value.trim();
448         if (prompt) executePrompt(prompt);
449     };
450 }
451
452 async function executePrompt(promptText) {
453     if (isExecuting) return;
454     isExecuting = true;
455
456     input.disabled = true;
457     if (sendBtn) sendBtn.disabled = true;
```

```

458         if (statusInd) statusInd.innerText = '□ Подбор инструме
| нта и формирование команды...';
459
460         if (resultBox) {
461             resultBox.style.display = '';
462             resultBox.scrollToView({ behavior: 'smooth', block:
| 'nearest' });
463         }
464         if (resToolName) resToolName.innerText = 'Анализ намерен
| ия...';
465         if (resCmdLine) resCmdLine.innerText = '...';
466         if (resExplanation) resExplanation.innerText = 'Подбор ш
| татной утилиты System32...';
467         if (resOutput) {
468             resOutput.className = 'p-2 bg-black rounded text-info
| small mb-0 font-monospace';
469             resOutput.innerText = 'Выполнение системной команды...
| Пожалуйста, подождите.';
470         }
471
472         const isDryRun = dryRunCheck ? dryRunCheck.checked : fal
| se;
473         const confirmAdmin = confirmAdminCheck ? confirmAdminChe
| ck.checked : true;
474
475         try {
476             const response = await fetch('/api/v1/system32/interpr
| et-and-run', {
477                 method: 'POST',
478                 headers: { 'Content-Type': 'application/json' },
479                 body: JSON.stringify({
480                     prompt: promptText,
481                     execute: !isDryRun,
482                     confirmed_by_user: !confirmAdmin,
483                 }),
484             });
485
486             if (!response.ok) throw new Error(`HTTP ${response.sta

```

```

tus}`);
487         const data = await response.json();
488
489         if (resToolName) resToolName.innerText = data.matched_
executable;
490         if (resTierBadge) {
491             resTierBadge.className = `badge ${tierColors[data.te
lemetry_tier]?.badge || 'bg-secondary'} ms-2`;
492             resTierBadge.innerText = data.telemetry_tier;
493         }
494         if (resDangerBadge) {
495             if (data.danger_level === 'HIGH' || data.danger_leve
l === 'CRITICAL') {
496                 resDangerBadge.style.display = '';
497                 resDangerBadge.innerText = data.danger_level;
498             } else {
499                 resDangerBadge.style.display = 'none';
500             }
501         }
502         if (resDuration) resDuration.innerText = `${data.execu
tion_time_ms} ms`;
503         if (resCmdLine) resCmdLine.innerText = data.command_li
ne;
504         if (resExplanation) resExplanation.innerText = data.ex
planation;
505         if (resPsEquiv) resPsEquiv.innerHTML = data.powershell
_equivalent ? `⌨ PowerShell: <code>${escapeHtml(data.powershel
l_equivalent)}</code>` : '';
506         if (resWin32Equiv) resWin32Equiv.innerHTML = data.win3
2_api_equivalent ? `⌨ Win32 API: <code>${escapeHtml(data.win32
_api_equivalent)}</code>` : '';
507
508         if (resOutput) {
509             if (data.status === 'success') {
510                 resOutput.className = 'p-2 bg-black rounded text-s
uccess small mb-0 font-monospace';
511             } else if (data.status === 'warn' || data.status ===
'requires_confirmation') {

```

```

512         resOutput.className = 'p-2 bg-black rounded text-w
arning small mb-0 font-monoospace';
513     } else {
514         resOutput.className = 'p-2 bg-black rounded text-d
anger small mb-0 font-monoospace';
515     }
516     resOutput.innerText = data.execution_output || 'Кома
нда выполнена успешно.';
517 }
518
519     if (statusInd) statusInd.innerText = `Готово: ${data.m
atched_executable} (${data.status})`;
520 } catch (err) {
521     console.error('[System32CommandsTab] Execution error:'
, err);
522     if (resOutput) {
523         resOutput.className = 'p-2 bg-black rounded text-dan
ger small mb-0 font-monoospace';
524         resOutput.innerText = `Сбой запроса: ${err.message}`
;
525     }
526     if (statusInd) statusInd.innerText = 'Ошибка выполнени
я';
527 } finally {
528     isExecuting = false;
529     input.disabled = false;
530     if (sendBtn) sendBtn.disabled = false;
531     input.focus();
532 }
533 }
534 }
535
536 // Setup toolbar filters & search
537 function setupFilters() {
538     const tierGroup = document.getElementById('tier-filters');
539     if (tierGroup) {
540         tierGroup.querySelectorAll('button').forEach(btn => {
541             btn.onclick = () => {

```

```
542 |         tierGroup.querySelectorAll('button').forEach(b => b.
| classList.remove('active'));
543 |         btn.classList.add('active');
544 |         currentTierFilter = btn.getAttribute('data-tier') ||
| 'all';
545 |         renderToolsGrid();
546 |     };
547 | });
548 | }
549 |
550 | const searchInput = document.getElementById('tool-search-i
| nput');
551 | const clearBtn = document.getElementById('btn-clear-search
| ');
552 | if (searchInput) {
553 |     searchInput.oninput = () => {
554 |         searchQuery = searchInput.value;
555 |         renderToolsGrid();
556 |     };
557 | }
558 | if (clearBtn && searchInput) {
559 |     clearBtn.onclick = () => {
560 |         searchInput.value = '';
561 |         searchQuery = '';
562 |         renderToolsGrid();
563 |     };
564 | }
565 |
566 | const refreshBtn = document.getElementById('btn-refresh-sy
| stem32-catalog');
567 | if (refreshBtn) {
568 |     refreshBtn.onclick = () => { loadCatalog(); };
569 | }
570 |
571 | const invBtn = document.getElementById('btn-open-inventory
| -modal');
572 | if (invBtn) {
573 |     invBtn.onclick = () => { loadHostInventory(); };

```

```
574 |     }
575 | }
576 |
577 | // Module initialization
578 | function init() {
579 |     setupFilters();
580 |     setupAiCommandRunner();
581 |     loadCatalog();
582 | }
583 |
584 | if (document.readyState === 'loading') {
585 |     document.addEventListener('DOMContentLoaded', init);
586 | } else {
587 |     init();
588 | }
589 |})();
590 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows System Logs & Activity - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса журналов событий и активно
8 | сти системы Windows.
9 | Оформлен на базе стандартных стилей About System и дизайн-то
10 | кенов темы.
11 |
12 Usage Examples:
13 | Web Route:
14 | /html/system_control_tab/index.html
15 |
16 File: index.html
17 Project: ai-breadboard
18 Package: windows/api/webgui/system_control_tab
19 Author: hypo69
20 Copyright: © 2026 hypo69
21 Updated: 2026-10-08 05:15:00
22 |=====
23 |=====
24 -->
25
26 <!-- Windows System Logs & Activity Center Web Tab -->
27 <style>
28 | .scc-container {
29 |     padding: 0.75rem;
30 |     color: var(--text-color, #18181b);
31 | }
32 | .scc-header {
33 |     background: var(--surface-2);
34 |     border: 1px solid var(--border-color);
35 |     border-radius: var(--radius-md, 8px);
```



```
32 | padding: 0.75rem 1rem;
33 | margin-bottom: 0.75rem;
34 | color: var(--text-color);
35 | }
36 | .slc-compact-header {
37 |   background: var(--surface-2);
38 |   border: 1px solid var(--border-color);
39 |   border-radius: var(--radius-md, 8px);
40 |   padding: 0.5rem 0.75rem;
41 |   margin-bottom: 0.5rem;
42 |   color: var(--text-color);
43 | }
44 | .slc-metric-pill {
45 |   background: var(--surface-1);
46 |   border: 1px solid var(--border-color);
47 |   border-radius: 6px;
48 |   padding: 0.25rem 0.6rem;
49 |   font-size: 0.75rem;
50 |   display: inline-flex;
51 |   align-items: center;
52 |   gap: 0.35rem;
53 |   color: var(--text-color);
54 | }
55 | .slc-subtabs-nav {
56 |   background: var(--surface-2);
57 |   border: 1px solid var(--border-color);
58 |   border-radius: var(--radius-md, 8px);
59 |   padding: 0.35rem 0.5rem;
60 | }
61 | .slc-subtab-btn {
62 |   transition: all 0.18s ease;
63 |   font-size: 0.78rem;
64 |   color: var(--text-color);
65 |   border: 1px solid transparent;
66 |   border-radius: 6px;
67 | }
68 | .slc-subtab-btn:hover {
69 |   background: rgba(56, 189, 248, 0.08);
```

```
70 |     color: var(--text-color);
71 | }
72 | .slc-subtab-btn.active {
73 |     background: #0ea5e9 !important;
74 |     color: #0f172a !important;
75 |     border-color: #0ea5e9 !important;
76 |     font-weight: 700;
77 | }
78 | .slc-panel {
79 |     background: var(--surface-1);
80 |     border: 1px solid var(--border-color);
81 |     border-radius: var(--radius-md, 8px);
82 |     overflow: hidden;
83 | }
84 | .slc-panel-header {
85 |     background: var(--surface-2);
86 |     border-bottom: 1px solid var(--border-color);
87 |     padding: 0.5rem 0.9rem;
88 |     font-size: 0.85rem;
89 |     font-weight: 600;
90 |     color: var(--text-color);
91 |     display: flex;
92 |     align-items: center;
93 |     justify-content: space-between;
94 | }
95 | .slc-tree-list {
96 |     max-height: 560px;
97 |     overflow-y: auto;
98 |     font-size: 0.82rem;
99 | }
100 | .slc-tree-item {
101 |     padding: 0.45rem 0.75rem;
102 |     cursor: pointer;
103 |     border-radius: 6px;
104 |     margin-bottom: 2px;
105 |     color: var(--text-color) !important;
106 |     display: flex;
107 |     justify-content: space-between;
```

```
108 | align-items: center;
109 | transition: background 0.15s ease;
110 | border: 1px solid transparent;
111 | }
112 | .slc-tree-item span {
113 |     color: inherit !important;
114 | }
115 | .slc-tree-item:hover {
116 |     background: rgba(56, 189, 248, 0.08);
117 |     color: var(--text-color) !important;
118 | }
119 | .slc-tree-item.active {
120 |     background: #0ea5e9 !important;
121 |     color: #0f172a !important;
122 |     font-weight: 700;
123 |     border-color: #0ea5e9 !important;
124 | }
125 | .slc-tree-item.active span {
126 |     color: #0f172a !important;
127 | }
128 | .slc-tree-item.active .badge {
129 |     background: #0f172a !important;
130 |     color: #ffffff !important;
131 | }
132 | .slc-table {
133 |     width: 100%;
134 |     border-collapse: separate;
135 |     border-spacing: 0;
136 |     margin-bottom: 0;
137 |     font-size: 0.82rem;
138 |     color: var(--text-color) !important;
139 | }
140 | .slc-table th {
141 |     padding: 8px 12px;
142 |     font-size: 0.75rem;
143 |     font-weight: 600;
144 |     color: var(--text-muted) !important;
145 |     text-transform: uppercase;
```

```
146 |     letter-spacing: 0.04em;
147 |     border-bottom: 1px solid var(--border-color);
148 |     background: var(--surface-2) !important;
149 |     vertical-align: middle;
150 | }
151 | .slc-table td {
152 |     padding: 8px 12px;
153 |     border-bottom: 1px solid var(--border-subtle, rgba(0, 0, 0
154 | , 0.06));
155 |     vertical-align: middle;
156 |     color: var(--text-color) !important;
157 |     background: transparent;
158 |     line-height: 1.4;
159 | }
160 | .slc-table tr:hover td {
161 |     background: rgba(56, 189, 248, 0.06) !important;
162 | }
163 | .slc-table tr.table-active td {
164 |     background: rgba(14, 165, 233, 0.18) !important;
165 |     font-weight: 600;
166 |     color: var(--text-color) !important;
167 | }
168 | .slc-table tr.table-danger-subtle td {
169 |     background: rgba(239, 68, 68, 0.08);
170 | }
171 | .scc-container .form-control,
172 | .scc-container .form-select {
173 |     background-color: var(--surface-1);
174 |     color: var(--text-color);
175 |     border: 1px solid var(--border-color);
176 | }
177 | .scc-container .input-group-text {
178 |     background-color: var(--surface-2);
179 |     color: var(--text-muted);
180 |     border: 1px solid var(--border-color);
181 | }
182 | .slc-tooltip-popup {
183 |     position: fixed;
```

```
183 |     z-index: 99999;
184 |     max-width: 320px;
185 |     background: var(--surface-2);
186 |     border: 1.5px solid var(--border-color);
187 |     border-radius: 8px;
188 |     padding: 0.65rem 0.85rem;
189 |     box-shadow: 0 4px 16px rgba(0, 0, 0, 0.15);
190 |     pointer-events: none;
191 |     transition: opacity 0.15s ease, transform 0.15s ease;
192 |     opacity: 0;
193 |     transform: translateY(4px);
194 |     font-size: 0.78rem;
195 |     line-height: 1.4;
196 |     color: var(--text-color);
197 | }
198 | .slc-tooltip-popup.show {
199 |     opacity: 1;
200 |     transform: translateY(0);
201 | }
202 | .slc-card {
203 |     background: var(--surface-1);
204 |     border: 1px solid var(--border-color);
205 |     border-radius: var(--radius-md, 8px);
206 | }
207 | .slc-card-header {
208 |     background: var(--surface-2);
209 |     border-bottom: 1px solid var(--border-color);
210 |     color: var(--text-color);
211 |     padding: 0.5rem 0.85rem;
212 |     font-weight: 600;
213 | }
214 | .badge-crit { background: #dc2626 !important; color: #fff !i
215 | mportant; }
216 | .badge-err { background: #ef4444 !important; color: #fff !im
217 | portant; }
218 | .badge-warn { background: #f59e0b !important; color: #000 !i
219 | mportant; }
220 | .badge-inf { background: #10b981 !important; color: #fff !im
```

```
    portant; }
218 | .badge-verb { background: #64748b !important; color: #fff !i
    mportant; }
219 |
220 | .slc-table-resizer {
221 |     height: 8px;
222 |     cursor: row-resize;
223 |     background: var(--surface-2);
224 |     border-top: 1px solid var(--border-color);
225 |     display: flex;
226 |     align-items: center;
227 |     justify-content: center;
228 |     transition: background-color 0.2s ease;
229 | }
230 | .slc-table-resizer:hover, .slc-table-resizer.resizing {
231 |     background: #0ea5e9 !important;
232 | }
233 | .slc-table-resizer-grip {
234 |     width: 36px;
235 |     height: 3px;
236 |     background: var(--border-strong, #666);
237 |     border-radius: 2px;
238 | }
239 | .pwr-detail-card {
240 |     background: var(--surface-2);
241 |     border: 1px solid var(--border-color);
242 |     border-radius: 6px;
243 | }
244 | .pwr-xml-pre {
245 |     background: var(--surface-2);
246 |     color: var(--text-color);
247 |     border: 1px solid var(--border-color);
248 | }
249 | .pwr-proc-badge {
250 |     background: var(--surface-2);
251 |     color: #38bdf8;
252 |     border: 1px solid var(--border-color);
253 | }
```

```

254 | </style>
255 |
256 | <div class="scc-container">
257 |   <!-- Top Header Bar -->
258 |   <div class="scc-header d-flex justify-content-between align-
      items-center flex-wrap gap-3">
259 |     <div>
260 |       <div class="d-flex align-items-center gap-2">
261 |         <h5 class="mb-0 fw-bold" data-i18n="systemControl.head
      erTitle">
262 |           □ Windows System Logs & Activity Center
263 |         </h5>
264 |         <span class="badge rounded-pill bg-primary px-2 py-1"
      style="font-size: 0.75rem;" data-i18n="systemControl.badgeLogs
      Activity">Logs & Activity</span>
265 |       </div>
266 |       <div class="small text-muted mt-1" data-i18n="systemCont
      rol.headerSubtitle">
267 |         Комплексный мониторинг журналов событий Windows, семан
      тический RAG-поиск, аудит аномалий и история активности систем
      ы.
268 |       </div>
269 |     </div>
270 |     <div class="d-flex align-items-center gap-2 flex-wrap">
271 |       <span class="badge rounded-pill px-3 py-2" id="scc-eleva
      tion-badge" data-i18n="systemControl.elevChecking">
272 |         □ Проверка прав доступа...
273 |       </span>
274 |       <button class="btn btn-sm btn-outline-info rounded-pill
      px-3" id="btn-scc-refresh">
275 |         <i class="bi bi-arrow-clockwise me-1"></i> <span data-
      i18n="systemControl.btnRefresh">Обновить</span>
276 |       </button>
277 |     </div>
278 |   </div>
279 |
280 |   <!-- Main System Logs Section -->
281 |   <div>

```

```

282 |     <!-- Compact Header Bar with Inline Stats & Actions -->
283 |     <div class="slc-compact-header d-flex justify-content-between align-items-center flex-wrap gap-2">
284 |         <!-- Title & Inline Metrics -->
285 |         <div class="d-flex align-items-center gap-2 flex-wrap">
286 |             <div class="d-flex align-items-center gap-1.5 me-1">
287 |                 <span class="fs-6">□ </span>
288 |                 <span class="fw-bold small" data-i18n="systemControl
| .centerTitle">Windows OS & System Log Center</span>
289 |             </div>
290 |
291 |             <!-- Quick Metrics Pills -->
292 |             <div class="slc-metric-pill" title="Источники журналов
| " data-i18n-title="systemControl.statSourcesTitle">
293 |                 <span class="text-muted small"><i class="bi bi-diagram-3 me-1"></i><span data-i18n="systemControl.statSources">Ист
| очники:</span></span>
294 |                 <span class="fw-bold text-info font-monospace" id="s
| lc-stat-sources">1,247</span>
295 |             </div>
296 |             <div class="slc-metric-pill" title="Активный канал лог
| ов" data-i18n-title="systemControl.statChannelTitle">
297 |                 <span class="text-muted small"><i class="bi bi-folder-fill me-1"></i><span data-i18n="systemControl.statChannel">К
| анал:</span></span>
298 |                 <span class="fw-bold text-warning font-monospace tex
| t-truncate" id="slc-stat-channel" style="max-width: 140px;">Sy
| stem</span>
299 |             </div>
300 |             <div class="slc-metric-pill" title="Загружено событий"
| data-i18n-title="systemControl.statLoadedTitle">
301 |                 <span class="text-muted small"><i class="bi bi-list-ol me-1"></i><span data-i18n="systemControl.statLoaded">Загруж
| ено:</span></span>
302 |                 <span class="fw-bold text-primary font-monospace" id
| ="slc-stat-events">0</span>
303 |             </div>
304 |             <div class="slc-metric-pill" title="Ошибки и инциденты

```



```

" data-i18n-title="systemControl.statErrorsTitle">
305     <span class="text-muted small"><i class="bi bi-exclamation-triangle-fill me-1"></i><span data-i18n="systemControl.statErrors">Ошибки:</span></span>
306     <span class="fw-bold text-danger font-monospace" id="slc-stat-incidents">0</span>
307 </div>
308 </div>
309
310 <!-- Actions Toolbar -->
311 <div class="d-flex align-items-center gap-2 flex-wrap">
312     <button class="btn btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-scc-open-event-logs" style="font-size: 0.78rem;" title="Открыть встроенную панель журналов событий">
313         <i class="bi bi-journal-text"></i> <span data-i18n="systemControl.modeEventLogs">Журналы событий</span>
314     </button>
315     <button class="btn btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-scc-open-power-lifecycle" style="font-size: 0.78rem;" title="Открыть панель жизненного цикла питания и аптайма">
316         <i class="bi bi-lightning-charge"></i> <span data-i18n="systemControl.modePowerLifecycle">Power Lifecycle</span>
317     </button>
318     <button class="btn btn-outline-info rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-slc-scan" style="font-size: 0.78rem;">
319         <i class="bi bi-radar"></i> <span data-i18n="systemControl.btnScan">Сканировать</span>
320     </button>
321     <div class="form-check form-switch mb-0 d-inline-flex align-items-center" style="font-size: 0.78rem; min-height: auto;">
322         <input class="form-check-input me-1" type="checkbox" role="switch" id="slc-live-switch" style="cursor: pointer;">
323         <label class="form-check-label text-muted small user-select-none" for="slc-live-switch" data-i18n="systemControl.l

```

```

ive">Live</label>
324     </div>
325     <button class="btn btn-primary rounded-pill px-2.5 py-
1 small d-flex align-items-center gap-1" id="btn-slc-refresh"
style="font-size: 0.78rem;">
326         <i class="bi bi-arrow-clockwise"></i> <span data-i18
n="systemControl.btnRefresh">Обновить</span>
327     </button>
328     <div class="dropdown">
329         <button class="btn btn-sm btn-outline-secondary drop
down-toggle rounded-pill px-2.5 py-1 small d-flex align-items-
center gap-1" type="button" data-bs-toggle="dropdown" style="f
ont-size: 0.78rem;">
330             <i class="bi bi-download"></i> <span data-i18n="sy
stemControl.btnExport">Экспорт</span>
331         </button>
332         <ul class="dropdown-menu dropdown-menu-end shadow" s
tyle="font-size: 0.8rem;">
333             <li><a class="dropdown-item" href="#" id="btn-slc-
export-json"><i class="bi bi-filetype-json me-2"></i><span dat
a-i18n="systemControl.exportJson">JSON</span></a></li>
334             <li><a class="dropdown-item" href="#" id="btn-slc-
export-csv"><i class="bi bi-filetype-csv me-2"></i><span data-
i18n="systemControl.exportCsv">CSV</span></a></li>
335         </ul>
336     </div>
337 </div>
338 </div>
339
340 <!-- Search & View Mode Navigation (Compact) -->
341 <div class="slc-panel mb-2 p-2">
342     <div class="d-flex justify-content-between align-items-c
enter flex-wrap gap-1 mb-2 pb-1 border-bottom" style="border-c
olor: var(--border-color) !important;">
343         <!-- Mode Tabs -->
344         <ul class="nav nav-pills gap-1" id="slc-mode-tabs">
345             <li class="nav-item">
346                 <button class="nav-link slc-subtab-btn active py-0

```

```
    .5 px-2.5 small" data-mode="live" data-i18n="systemControl.mod
eLive">● Live Events</button>
347         </li>
348         <li class="nav-item">
349             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="audit" data-i18n="systemControl.modeAudit
">□ Аудит (Smart Digest)</button>
350         </li>
351         <li class="nav-item">
352             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="rag" data-i18n="systemControl.modeRag">□
RAG Поиск</button>
353         </li>
354         <li class="nav-item">
355             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="errors" data-i18n="systemControl.modeErro
rs">□ Errors Only</button>
356         </li>
357         <li class="nav-item">
358             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="incidents" data-i18n="systemControl.modeI
ncidents">□ Incidents</button>
359         </li>
360         <li class="nav-item">
361             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="timeline" data-i18n="systemControl.modeTi
meline">□ Timeline</button>
362         </li>
363         <li class="nav-item">
364             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="raw" data-i18n="systemControl.modeRaw">□
Raw Record</button>
365         </li>
366         <li class="nav-item">
367             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small" data-mode="activity" data-i18n="systemControl.modeAc
tivity">□ App Activity Log</button>
368         </li>
```

```

369         <li class="nav-item">
370             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small fw-semibold text-warning" data-mode="event-logs" id="
subtab-btn-event-logs" data-i18n="systemControl.modeEventLogs"
>📖 Журналы событий (Event Logs)</button>
371         </li>
372         <li class="nav-item">
373             <button class="nav-link slc-subtab-btn py-0.5 px-2
.5 small fw-semibold text-warning" data-mode="power-lifecycle"
id="subtab-btn-power-lifecycle" data-i18n="systemControl.mode
PowerLifecycle">🔌 Жизненный цикл питания (Power Lifecycle)</bu
tton>
374         </li>
375     </ul>
376
377     <!-- Action AI Button -->
378     <div class="d-flex align-items-center gap-2">
379         <button class="btn btn-sm btn-outline-warning rounde
d-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="
btn-slc-ask-ai" style="font-size: 0.78rem;">
380             <i class="bi bi-robot"></i> <span data-i18n="syste
mControl.btnAskAi">Спросить ИИ</span>
381         </button>
382     </div>
383 </div>
384
385     <!-- Filter Inputs Grid -->
386     <div class="row g-1.5 align-items-center" id="slc-filter
-row">
387         <div class="col-lg-4 col-md-5">
388             <div class="input-group input-group-sm">
389                 <span class="input-group-text py-0 px-2"><i class=
"bi bi-search" style="font-size: 0.75rem;"></i></span>
390                 <input type="text" class="form-control form-contro
l-sm py-1" id="slc-search-input" placeholder="Поиск по тексту,
источнику..." data-i18n-placeholder="systemControl.searchPlac
eholder" style="font-size: 0.8rem;">
391             </div>

```

```
392 |         </div>
393 |         <div class="col-lg-2 col-md-2 col-6">
394 |             <select class="form-select form-select-sm py-1" id="
| slc-level-select" style="font-size: 0.8rem;">
395 |                 <option value="" data-i18n="systemControl.allLevel
| s">Все уровни</option>
396 |                 <option value="Critical">Critical</option>
397 |                 <option value="Error">Error</option>
398 |                 <option value="Warning">Warning</option>
399 |                 <option value="Information">Information</option>
400 |             </select>
401 |         </div>
402 |         <div class="col-lg-2 col-md-2 col-6">
403 |             <select class="form-select form-select-sm py-1" id="
| slc-hours-select" style="font-size: 0.8rem;">
404 |                 <option value="1" data-i18n="systemControl.hours1"
| >За последний 1 час</option>
405 |                 <option value="6" data-i18n="systemControl.hours6"
| >За последние 6 часов</option>
406 |                 <option value="24" data-i18n="systemControl.hours2
| 4" selected>За последние 24 часа</option>
407 |                 <option value="168" data-i18n="systemControl.hours
| 168">За последние 7 дней</option>
408 |                 <option value="0" data-i18n="systemControl.hoursAl
| l">За все доступное время</option>
409 |             </select>
410 |         </div>
411 |         <div class="col-lg-2 col-md-1.5 col-6">
412 |             <input type="number" class="form-control form-contro
| l-sm py-1" id="slc-eventid-input" placeholder="Event ID" data-
| i18n-placeholder="systemControl.eventIdPlaceholder" style="fon
| t-size: 0.8rem;">
413 |         </div>
414 |         <div class="col-lg-2 col-md-1.5 col-6">
415 |             <select class="form-select form-select-sm py-1" id="
| slc-limit-select" style="font-size: 0.8rem;">
416 |                 <option value="50" data-i18n="systemControl.limit5
| 0">50 записей</option>
```

```
417 |         <option value="100" data-i18n="systemControl.limit
    | 100" selected>100 записей</option>
418 |         <option value="250" data-i18n="systemControl.limit
    | 250">250 записей</option>
419 |         <option value="500" data-i18n="systemControl.limit
    | 500">500 записей</option>
420 |     </select>
421 | </div>
422 | </div>
423 | </div>
424 |
425 | <!-- Main Multi-Column View -->
426 | <div class="row g-3" id="slc-main-row">
427 |     <!-- Channel & Source Tree Sidebar -->
428 |     <div class="col-lg-3" id="slc-channel-tree-sidebar">
429 |         <div class="slc-panel h-100">
430 |             <div class="slc-panel-header">
431 |                 <span><i class="bi bi-folder2-open me-2 text-prima
    | ry"></i><span data-i18n="systemControl.channelsTitle">Каналы с
    | обытий</span></span>
432 |                 <span class="badge bg-secondary" id="slc-channel-c
    | ount-badge">0</span>
433 |             </div>
434 |             <div class="p-2 border-bottom" style="border-color:
    | var(--border-color) !important;">
435 |                 <input type="text" class="form-control form-contro
    | l-sm" id="slc-channel-filter-input" placeholder="Фильтр канало
    | в..." data-i18n-placeholder="systemControl.filterChannelsPlace
    | holder">
436 |             </div>
437 |             <div class="p-2 slc-tree-list" id="slc-channel-tree-
    | container">
438 |                 <!-- Dynamic channels injected here -->
439 |             </div>
440 |         </div>
441 |     </div>
442 |
443 | <!-- Events / Incidents Main Stream -->
```

```

444 |     <div class="col-lg-9" id="slc-main-stream-column">
445 |         <div class="slc-panel h-100">
446 |             <div class="slc-panel-header">
447 |                 <span id="slc-main-view-title"><i class="bi bi-lis
| t-columns-reverse me-2 text-primary"></i><span data-i18n="syst
| emControl.mainViewTitle">Поток событий Windows</span></span>
448 |                 <div class="d-flex align-items-center gap-2">
449 |                     <button class="btn btn-sm btn-info rounded-pill
| px-3 fw-bold text-dark" id="btn-slc-audit" title="Провести ауд
| ит текущего журнала логов" data-i18n-title="systemControl.btnA
| uditTitle">
450 |                         <i class="bi bi-shield-check me-1"></i> <span
| data-i18n="systemControl.btnAudit">Аудит</span>
451 |                     </button>
452 |                     <span class="badge bg-secondary" id="slc-stream-
| count-badge">0 записей</span>
453 |                 </div>
454 |             </div>
455 |
456 |             <!-- Table View Container -->
457 |             <div class="table-responsive" style="min-height: 140
| px; max-height: 560px; overflow-y: auto;" id="slc-table-contai
| ner">
458 |                 <table class="slc-table">
459 |                     <thead>
460 |                         <tr>
461 |                             <th style="width: 155px;" data-i18n="systemC
| ontrol.colTime">Время</th>
462 |                             <th style="width: 90px;" data-i18n="systemCo
| ntrol.colLevel">Уровень</th>
463 |                             <th style="width: 65px;" data-i18n="systemCo
| ntrol.colId">ID</th>
464 |                             <th style="width: 160px;" data-i18n="systemC
| ontrol.colProvider">Поставщик</th>
465 |                             <th style="width: 65px;" data-i18n="systemCo
| ntrol.colPid">PID</th>
466 |                             <th data-i18n="systemControl.colMessage">Coo
| бщение</th>

```

```

467         </tr>
468     </thead>
469     <tbody id="slc-table-body">
470         <tr>
471             <td colspan="6" class="text-center text-mute
d py-4">
472                 <div class="spinner-border spinner-border-
sm text-info me-2" role="status"></div>
473                 <span data-i18n="systemControl.loadingEven
ts">Загрузка событий Windows...</span>
474             </td>
475         </tr>
476     </tbody>
477 </table>
478 </div>
479     <div id="slc-table-resizer" class="slc-table-resizer
" title="Потяните для изменения высоты таблицы (двойной клик –
сброс)">
480         <div class="slc-table-resizer-grip"></div>
481     </div>
482
483     <!-- Audit Digest View Container -->
484     <div class="p-3 d-none" id="slc-audit-container" sty
le="max-height: 560px; overflow-y: auto;">
485         <div id="slc-audit-content"></div>
486     </div>
487
488     <!-- Dynamic Log RAG Search Container -->
489     <div class="p-3 d-none" id="slc-rag-container" style
="max-height: 560px; overflow-y: auto;">
490         <div class="slc-card mb-3 shadow-sm">
491             <div class="slc-card-header p-2.5 d-flex justify
-content-between align-items-center">
492                 <span class="fw-bold text-primary"><i class="b
i bi-search me-2"></i><span data-i18n="systemControl.ragTitle"
>Семантический RAG-поиск по логам</span></span>
493                 <button class="btn btn-sm btn-outline-info rou
nded-pill px-3" id="btn-slc-rag-rebuild">

```



```
494 |         <i class="bi bi-arrow-repeat me-1"></i> <spa
    | n data-i18n="systemControl.btnRagRebuild">Перестроить RAG-инде
    | кс</span>
495 |     </button>
496 | </div>
497 |     <div class="p-3">
498 |         <div class="input-group mb-2">
499 |             <span class="input-group-text"><i class="bi
    | bi-chat-left-dots text-primary"></i></span>
500 |             <input type="text" class="form-control" id="
    | slc-rag-input" placeholder="Введите запрос (например: 'ошибки
    | службы DCOM', 'сбои сети', 'Disk warning')..." data-i18n-place
    | holder="systemControl.ragInputPlaceholder">
501 |             <button class="btn btn-info px-4 fw-bold tex
    | t-dark" id="btn-slc-rag-search" data-i18n="systemControl.btnRa
    | gSearch">Найти</button>
502 |         </div>
503 |         <div class="small text-muted" data-i18n="syste
    | mControl.ragHint">
504 |             Поиск использует динамический индекс кластер
    | ов и аномалий.
505 |         </div>
506 |     </div>
507 | </div>
508 |
509 |     <div id="slc-rag-results">
510 |         <div class="slc-card p-4 text-center text-muted"
    | data-i18n="systemControl.ragEmptyPrompt">
511 |             Введите запрос для поиска по динамическому RAG
    | -индексу логов или нажмите «Перестроить RAG-индекс».
512 |         </div>
513 |     </div>
514 | </div>
515 |
516 |     <!-- Incidents View Container -->
517 |     <div class="p-3 d-none" id="slc-incidents-container"
    | style="max-height: 560px; overflow-y: auto;">
518 |         <div id="slc-incidents-list"></div>
```

```

519         </div>
520
521         <!-- Timeline Histogram View Container -->
522         <div class="p-3 d-none" id="slc-timeline-container"
style="max-height: 560px; overflow-y: auto;">
523             <div id="slc-timeline-list"></div>
524         </div>
525
526         <!-- Raw Record Container -->
527         <div class="p-3 d-none" id="slc-raw-container" style
="max-height: 560px; overflow-y: auto;">
528             <pre class="p-3 font-monospace small rounded borde
r" style="background: var(--surface-2); color: var(--text-colo
r); border-color: var(--border-color) !important;" id="slc-raw
-text" data-i18n="systemControl.rawSelectPrompt">Выберите собы
тие из списка для просмотра данных.</pre>
529         </div>
530
531         <!-- Application Activity Log View Container -->
532         <div class="p-3 d-none" id="slc-activity-container"
style="max-height: 560px; overflow-y: auto;">
533             <div class="d-flex justify-content-between align-i
tems-center mb-2">
534                 <span class="fw-bold"><i class="bi bi-journal-te
xt me-1 text-primary"></i> <span data-i18n="systemControl.acti
vityTitle">Журнал активности</span></span>
535                 <button class="btn btn-sm btn-outline-secondary
rounded-pill px-3" id="btn-scc-refresh-logs">
536                     <i class="bi bi-arrow-clockwise me-1"></i> <sp
an data-i18n="systemControl.btnRefresh">Обновить</span>
537                 </button>
538             </div>
539             <div class="table-responsive" style="min-height: 1
40px; max-height: 480px; overflow-y: auto;" id="scc-activity-t
able-container">
540                 <table class="slc-table">
541                     <thead>
542                         <tr>

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543         <th style="width: 180px;" data-i18n="systemControl.colTime">Время</th>
544         <th style="width: 160px;" data-i18n="systemControl.colAction">Действие</th>
545         <th data-i18n="systemControl.colTarget">Объект</th>
546         <th style="width: 120px;" data-i18n="systemControl.colStatus">Статус</th>
547         <th style="width: 30%;" data-i18n="systemControl.colDetails">Подробности</th>
548     </tr>
549 </thead>
550 <tbody id="scc-logs-tbody">
551     <tr><td colspan="5" class="text-center text-muted p-3" data-i18n="systemControl.loadingActivity">Загрузка
журнала активности...</td></tr>
552 </tbody>
553 </table>
554 </div>
555 <div id="scc-activity-table-resizer" class="slc-table-resizer" title="Потяните для изменения высоты таблицы (двойной клик – сброс)">
556     <div class="slc-table-resizer-grip"></div>
557 </div>
558 </div>
559
560 <!-- Event Logs (WevtAPI & Log Intelligence) Container Panel -->
561 <div class="p-3 d-none" id="slc-event-logs-container">
562     <!-- Top Stat Cards -->
563     <div class="row g-2 mb-3">
564         <div class="col-6 col-md-3">
565             <div class="slc-card h-100 shadow-sm p-3">
566                 <div class="text-muted small text-uppercase fw-semibold" data-i18n="systemControl.elStatCritical">Критические
ошибки (Critical) </div>
567                 <div class="fs-4 fw-bold mt-1 text-danger fo

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nt-monospace" id="el-critical-count">--</div>
568 |         <div class="small text-muted mt-1" id="el-cr
critical-sub" data-i18n="systemControl.elStatCriticalSub">Уровен
ь 1 (WevtAPI)</div>
569 |         </div>
570 |     </div>
571 |     <div class="col-6 col-md-3">
572 |         <div class="slc-card h-100 shadow-sm p-3">
573 |             <div class="text-muted small text-uppercase
fw-semibold" data-i18n="systemControl.elStatErrors">Ошибки (Er
rors) □ </div>
574 |                 <div class="fs-4 fw-bold mt-1 text-warning f
ont-monospace" id="el-error-count">--</div>
575 |                 <div class="small text-muted mt-1" id="el-er
ror-sub" data-i18n="systemControl.elStatErrorsSub">Уровень 2 (
WevtAPI)</div>
576 |                 </div>
577 |             </div>
578 |         <div class="col-6 col-md-3">
579 |             <div class="slc-card h-100 shadow-sm p-3">
580 |                 <div class="text-muted small text-uppercase
fw-semibold" data-i18n="systemControl.elStatHealth">Здоровье 0
C (Health Score) □ </div>
581 |                 <div class="fs-4 fw-bold mt-1 text-success f
ont-monospace" id="el-health-score">--</div>
582 |                 <div class="small text-muted mt-1" id="el-he
alth-sub" data-i18n="systemControl.elStatHealthSub">Data Resea
rcher EDA</div>
583 |                 </div>
584 |             </div>
585 |         <div class="col-6 col-md-3">
586 |             <div class="slc-card h-100 shadow-sm p-3">
587 |                 <div class="text-muted small text-uppercase
fw-semibold" data-i18n="systemControl.elStatRedundancy">Избыто
чность / Шум (Rdup) □ </div>
588 |                 <div class="fs-4 fw-bold mt-1 text-info font
-monospace" id="el-redundancy-pct">--</div>
589 |                 <div class="small text-muted mt-1" id="el-re

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    dundancy-sub" data-i18n="systemControl.elStatRedundancySub">Де
    дупликация шаблонов</div>
590         </div>
591     </div>
592 </div>
593
594     <!-- Action & Intelligence Bar -->
595     <div class="slc-panel mb-3 p-2 d-flex align-items-
center justify-content-between flex-wrap gap-2">
596         <div class="d-flex align-items-center gap-2 flex
-wrap">
597             <button class="btn btn-sm btn-primary d-flex a
align-items-center gap-1.5" id="el-refresh-btn">
598                 <i class="bi bi-arrow-clockwise"></i>
599                 <span data-i18n="systemControl.btnRefresh">О
бновить</span>
600             </button>
601             <select class="form-select form-select-sm" id=
"el-channel-select" style="width: 180px;">
602                 <option value="System">Журнал System</option
>
603                 <option value="Application">Журнал Applicati
on</option>
604                 <option value="Security">Журнал Security</op
tion>
605             </select>
606             <button class="btn btn-sm btn-outline-info d-f
lex align-items-center gap-1.5" id="el-intel-analyze-btn">
607                 <i class="bi bi-cpu"></i>
608                 <span data-i18n="systemControl.elBtnIntel">L
og Intelligence (EDA)</span>
609             </button>
610             <button class="btn btn-sm btn-outline-warning
d-flex align-items-center gap-1.5" id="el-intel-audit-btn">
611                 <i class="bi bi-shield-shaded"></i>
612                 <span data-i18n="systemControl.elBtnAudit">П
олный аудит</span>
613             </button>

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614 |         </div>
615 |         <div class="d-flex align-items-center gap-2 flex
| -wrap">
616 |             <input type="text" class="form-control form-co
| ntrol-sm" id="el-rag-query-input" placeholder="RAG поиск (напр
| : сбой сети)..." data-i18n-placeholder="systemControl.elRagPla
| ceholder" style="width: 240px;">
617 |             <button class="btn btn-sm btn-info text-dark f
| w-semibold" id="el-rag-search-btn" title="Искать в RAG">
618 |                 <i class="bi bi-search"></i>
619 |             </button>
620 |             <input type="text" class="form-control form-co
| ntrol-sm" id="el-search-input" placeholder="Фильтр таблицы..."
| data-i18n-placeholder="systemControl.elFilterPlaceholder" sty
| le="width: 170px;">
621 |         </div>
622 |     </div>
623 |
624 |     <!-- Intelligence Digest Banner (Hidden by default
| ) -->
625 |     <div id="el-intel-banner" class="alert alert-dark
| border-info shadow-sm mb-3 d-none" style="background: var(--su
| rface-2); color: var(--text-color); border-color: var(--border
| -color) !important;">
626 |         <div class="d-flex justify-content-between align
| -items-start flex-wrap gap-2">
627 |             <div>
628 |                 <div class="d-flex align-items-center gap-2
| mb-1">
629 |                     <span class="badge bg-info text-dark fw-bo
| ld font-monospace" id="el-strategy-badge">STRATEGY</span>
630 |                     <strong class="fs-6" id="el-intel-title" s
| tyle="color: var(--text-color);">Аналитический вердикт Log Int
| elligence</strong>
631 |                 </div>
632 |                 <div class="small font-monospace text-muted"
| id="el-intel-rationale">--</div>
633 |                 <div class="small text-warning mt-1" id="el-
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intel-recommendation"></div>
634         </div>
635         <div class="text-end">
636             <span class="badge bg-secondary font-monospa
ce" id="el-chunks-badge">0 RAG chunks</span>
637             <button class="btn-close ms-2" id="el-intel-
close-btn" aria-label="Close"></button>
638         </div>
639     </div>
640 </div>
641
642     <!-- RAG Search Results Panel (Hidden by default)
-->
643     <div id="el-rag-results-card" class="slc-card shad
ow-sm mb-3 d-none border-info">
644         <div class="slc-card-header d-flex justify-conte
nt-between align-items-center py-2 px-3">
645             <span class="fw-semibold text-info"><i class="
bi bi-brain me-2"></i><span data-i18n="systemControl.elRagResu
ltsTitle">Результаты Adaptive Log RAG</span></span>
646             <button class="btn btn-sm btn-outline-secondar
y py-0 px-2" id="el-rag-close-btn" data-i18n="systemControl.el
RagClose">Закрыть</button>
647         </div>
648         <div class="card-body p-2" id="el-rag-results-bo
dy">
649             <!-- Dynamic Search Items -->
650         </div>
651     </div>
652
653     <!-- Event Logs Table -->
654     <div class="slc-card shadow-sm">
655         <div class="slc-card-header d-flex align-items-c
enter justify-content-between py-2 px-3">
656             <span class="fw-semibold"><i class="bi bi-jour
nal-text text-primary me-2"></i><span data-i18n="systemControl
.elViewerTitle">Записи системного журнала (Event Viewer)</span
></span>

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657 |         <span class="badge bg-secondary font-monospace
      | " id="el-table-badge">0 событий</span>
658 |     </div>
659 |     <div class="table-responsive" style="min-height:
      | 140px; max-height: 520px; overflow-y: auto;" id="el-table-con
      | tainer">
660 |         <table class="slc-table" id="el-table">
661 |             <thead class="sticky-top">
662 |                 <tr>
663 |                     <th style="width: 100px;" data-i18n="sys
      | temControl.elColLevel">Уровень</th>
664 |                     <th style="width: 160px;" data-i18n="sys
      | temControl.elColTime">Время</th>
665 |                     <th style="width: 180px;" data-i18n="sys
      | temControl.elColProvider">Источник (Provider)</th>
666 |                     <th style="width: 80px;" data-i18n="syst
      | emControl.elColEventId">Event ID</th>
667 |                     <th data-i18n="systemControl.elColDescri
      | ption">Описание события</th>
668 |                 </tr>
669 |             </thead>
670 |             <tbody id="el-tbody">
671 |                 <tr><td colspan="5" class="text-center tex
      | t-muted py-4" data-i18n="systemControl.elLoading">Чтение журна
      | ла событий через WevtAPI...</td></tr>
672 |             </tbody>
673 |         </table>
674 |     </div>
675 |     <div id="el-table-resizer" class="slc-table-resi
      | zer" title="Потяните для изменения высоты таблицы (двойной кли
      | к – сброс)">
676 |         <div class="slc-table-resizer-grip"></div>
677 |     </div>
678 | </div>
679 | </div>
680 |
681 | <!-- Power Lifecycle (Uptime, Sessions, Reboots & Ev
      | entLog 1074/41/6008) Container Panel -->

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682 |         <div class="p-3 d-none" id="slc-power-lifecycle-cont
      | ainer">
683 |             <!-- Верхняя панель метрик и сводки -->
684 |             <div class="row g-2 mb-3">
685 |                 <!-- Карточка 1: Текущий Аптайм и Старт ОС -->
686 |                 <div class="col-12 col-sm-6 col-xl-3">
687 |                     <div class="slc-card h-100 shadow-sm p-3 posit
      | ion-relative">
688 |                         <div class="d-flex justify-content-between a
      | lign-items-start">
689 |                             <div class="text-muted small text-uppercas
      | e fw-semibold">Текущий аптайм □ </div>
690 |                             <span class="badge bg-success font-monospa
      | ce" id="pwr-uptime-badge">ACTIVE</span>
691 |                         </div>
692 |                         <div class="fs-4 fw-bold mt-1 text-info font
      | -monospace" id="pwr-current-uptime">--:--:--</div>
693 |                         <div class="small text-muted mt-1 text-trunc
      | ate" id="pwr-current-boot" style="font-size: 0.75rem;">Загрузка:
      | --</div>
694 |                         <div class="mt-2 text-end">
695 |                             <span class="badge bg-secondary-subtle tex
      | t-secondary border border-secondary-subtle" style="font-size:
      | 0.7rem;">Kernel-General 12</span>
696 |                         </div>
697 |                     </div>
698 |                 </div>
699 |
700 |             <!-- Карточка 2: Всего сессий питания -->
701 |             <div class="col-12 col-sm-6 col-xl-3">
702 |                 <div class="slc-card h-100 shadow-sm p-3">
703 |                     <div class="d-flex justify-content-between a
      | lign-items-start">
704 |                         <div class="text-muted small text-uppercas
      | e fw-semibold">Сессии питания □ </div>
705 |                         <span class="badge bg-primary font-monospa
      | ce" id="pwr-total-sessions-badge">0 Сессий</span>
706 |                     </div>

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707 |         <div class="fs-4 fw-bold mt-1 text-warning font-monospace" id="pwr-total-sessions">--</div>
708 |         <div class="small text-muted mt-1" id="pwr-sessions-desc">Реконструкция Boot → Shutdown</div>
709 |         <div class="mt-2">
710 |             <button class="btn btn-xs btn-outline-warning w-100 py-1" id="pwr-rebuild-btn" style="font-size: 0.78rem;">
711 |                 <i class="bi bi-arrow-repeat me-1"></i><span>Пересканировать Event Log</span>
712 |             </button>
713 |         </div>
714 |     </div>
715 | </div>
716 |
717 |     <!-- Карточка 3: Штатные перезагрузки и включения -->
718 |     <div class="col-12 col-sm-6 col-xl-3">
719 |         <div class="slc-card h-100 shadow-sm p-3">
720 |             <div class="d-flex justify-content-between align-items-start">
721 |                 <div class="text-muted small text-uppercase fw-semibold">Штатные завершения □ </div>
722 |                 <span class="badge bg-success-subtle text-success border border-success-subtle font-monospace" id="pwr-clean-ratio">-- %</span>
723 |             </div>
724 |             <div class="fs-4 fw-bold mt-1 text-success font-monospace" id="pwr-clean-count">--</div>
725 |             <div class="small text-muted mt-1 text-truncate" id="pwr-last-initiator" style="font-size: 0.75rem;">Инициатор: --</div>
726 |             <div class="mt-2 text-end">
727 |                 <span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle" style="font-size: 0.7rem;">User32 1074 / EventLog 6006</span>
728 |             </div>
729 |         </div>

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730         </div>
731
732         <!-- Карточка 4: Внезапные сбои и BSOD -->
733         <div class="col-12 col-sm-6 col-xl-3">
734             <div class="slc-card h-100 shadow-sm p-3">
735                 <div class="d-flex justify-content-between align-items-start">
736                     <div class="text-muted small text-uppercase fw-semibold">Сбои / Потери питания </div>
737                     <span class="badge bg-danger font-monospace" id="pwr-unexpected-badge">0 Сбоев</span>
738                 </div>
739                 <div class="fs-4 fw-bold mt-1 text-danger font-monospace" id="pwr-unexpected-count">--</div>
740                 <div class="small text-muted mt-1 d-flex justify-content-between" style="font-size: 0.75rem;">
741                     <span id="pwr-bsod-count">BSOD (1001): 0</span>
742                     <span>Kernel-Power 41</span>
743                 </div>
744                 <div class="mt-2">
745                     <button class="btn btn-xs btn-outline-primary w-100 py-1" id="pwr-refresh-btn" style="font-size: 0.78rem;">
746                         <i class="bi bi-arrow-clockwise me-1"></i><span>Обновить данные</span>
747                     </button>
748                 </div>
749             </div>
750         </div>
751     </div>
752
753     <!-- Навигация по подразделам: Сессии vs События -->
754     <ul class="nav nav-pills mb-3 slc-subtabs-nav pb-1" id="pwrNavTabs" role="tablist">
755         <li class="nav-item" role="presentation">
756             <button class="nav-link active py-1 px-3 small

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" id="pwr-sessions-subtab-btn" data-bs-toggle="pill" data-bs-t
arget="#pwr-subtab-sessions" type="button" role="tab">
757         <i class="bi bi-clock-history me-1.5 text-wa
rning"></i>
758         <span>Сессии питания (Power Sessions)</span>
759     </button>
760 </li>
761     <li class="nav-item" role="presentation">
762         <button class="nav-link py-1 px-3 small" id="p
wr-events-subtab-btn" data-bs-toggle="pill" data-bs-target="#p
wr-subtab-events" type="button" role="tab">
763         <i class="bi bi-list-columns-reverse me-1.5
text-info"></i>
764         <span>Журнал ключевых событий (Event Log)</s
pan>
765         </button>
766     </li>
767 </ul>
768
769     <div class="tab-content" id="pwrTabContent">
770         <!-- Вкладка 1: Сессии питания (Power Sessions)
-->
771         <div class="tab-pane fade show active" id="pwr-s
ubtab-sessions" role="tabpanel">
772             <div class="slc-panel mb-3 p-2">
773                 <div class="row g-2 align-items-center">
774                     <div class="col-12 col-md-4">
775                         <div class="input-group input-group-sm">
776                             <span class="input-group-text"><i clas
s="bi bi-search"></i></span>
777                             <input type="text" class="form-control
form-control-sm" id="pwr-sessions-search" placeholder="Поиск
по пользователю, процессу, причине...">
778                         </div>
779                     </div>
780                     <div class="col-6 col-md-3">
781                         <select class="form-select form-select-s
m" id="pwr-filter-type">

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782 |             <option value="">Все типы завершения</
| option>
783 |             <option value="Active">☐ Активная сес
| сия (Active)</option>
784 |             <option value="Restart">☐ Перезагрузка
| (Restart)</option>
785 |             <option value="Shutdown">☐ Выключение
| (Shutdown)</option>
786 |             <option value="Unexpected">☐ Внезапное
| завершение (Unexpected)</option>
787 |             <option value="BSOD">☐ Синий экран (B
| SOD / BugCheck)</option>
788 |         </select>
789 |     </div>
790 |     <div class="col-6 col-md-2">
791 |         <div class="form-check form-switch small
| mt-1">
792 |             <input class="form-check-input" type="
| checkbox" id="pwr-filter-unexpected-only">
793 |             <label class="form-check-label text-wa
| rning fw-semibold" for="pwr-filter-unexpected-only">Только сбо
| и</label>
794 |         </div>
795 |     </div>
796 |     <div class="col-12 col-md-3 text-end">
797 |         <span class="small text-muted" id="pwr-s
| essions-counter">Найдено: 0</span>
798 |     </div>
799 | </div>
800 | </div>
801 |
802 |     <div class="slc-card shadow-sm">
803 |         <div class="table-responsive" style="min-hei
| ght: 140px; max-height: 520px; overflow-y: auto;" id="pwr-sess
| ions-table-container">
804 |             <table class="slc-table" style="font-size:
| 0.85rem;">
805 |                 <thead class="sticky-top">

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806 |         <tr>
807 |             <th style="width: 140px;">Сессия</th>
808 |         >
809 |             <th style="width: 160px;">Время стар
та (Boot)</th>
810 |             <th style="width: 160px;">Завершение
</th>
811 |             <th style="width: 130px;">Аптайм</th>
812 |             >
813 |             <th style="width: 130px;">Статус</th>
814 |             >
815 |             <th>Инициатор и процесс</th>
816 |             <th>Причина / Код</th>
817 |             <th style="width: 80px;" class="text
-center">Детали</th>
818 |         </tr>
819 |     </thead>
820 |     <tbody id="pwr-sessions-tbody">
821 |         <tr>
822 |             <td colspan="8" class="text-center p
y-4 text-muted">
823 |                 <div class="spinner-border spinner
-border-sm text-info me-2" role="status"></div>
824 |                 Загрузка истории сессий питания...
825 |             </td>
826 |         </tr>
827 |     </tbody>
828 | </table>
829 | </div>
830 | <div id="pwr-sessions-table-resizer" class="
slc-table-resizer" title="Потяните для изменения высоты таблиц
ы (двойной клик – сброс)">
831 |     <div class="slc-table-resizer-grip"></div>
832 | </div>
833 | </div>
834 | <!-- Вкладка 2: Журнал ключевых событий (Event L

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og) -->
834         <div class="tab-pane fade" id="pwr-subtab-events
" role="tabpanel">
835             <div class="slc-panel mb-3 p-2">
836                 <div class="row g-2 align-items-center">
837                     <div class="col-12 col-md-4">
838                         <div class="input-group input-group-sm">
839                             <span class="input-group-text"><i clas
s="bi bi-funnel"></i></span>
840                             <select class="form-select form-select
-sm" id="pwr-event-id-filter">
841                                 <option value="">Все Event ID жизнен
ного цикла</option>
842                                 <option value="1074">1074 – User32 (
Инициатор shutdown/restart)</option>
843                                 <option value="41">41 – Kernel-Power
(Внезапная перезагрузка)</option>
844                                 <option value="6008">6008 – EventLog
(Неожиданное выключение)</option>
845                                 <option value="12">12 – Kernel-Gener
al (Запуск ядра / Boot)</option>
846                                 <option value="13">13 – Kernel-Gener
al (Завершение работы ОС)</option>
847                                 <option value="6005">6005 – EventLog
(Служба EventLog запущена)</option>
848                                 <option value="6006">6006 – EventLog
(EventLog штатно остановлен)</option>
849                                 <option value="6009">6009 – EventLog
(Версия ядра при загрузке)</option>
850                                 <option value="1001">1001 – WER (Ава
рийный сбой BSOD / BugCheck)</option>
851                                 <option value="19">19 – WindowsUpdat
e (Установка обновлений)</option>
852                                 <option value="42">42 – Kernel-Power
(Переход в сон)</option>
853                                 <option value="107">107 – Kernel-Pow
er (Пробуждение из сна)</option>
854                                 </select>

```

```

855 |         </div>
856 |     </div>
857 |     <div class="col-12 col-md-5">
858 |         <div class="input-group input-group-sm">
859 |             <span class="input-group-text"><i clas
s="bi bi-search"></i></span>
860 |             <input type="text" class="form-control
form-control-sm" id="pwr-events-search" placeholder="Поиск в
событиях...">
861 |         </div>
862 |     </div>
863 |     <div class="col-12 col-md-3 text-end">
864 |         <span class="small text-muted" id="pwr-e
vents-counter">Событий: 0</span>
865 |     </div>
866 | </div>
867 | </div>
868 |
869 |     <div class="slc-card shadow-sm">
870 |         <div class="table-responsive" style="min-hei
ght: 140px; max-height: 520px; overflow-y: auto;" id="pwr-even
ts-table-container">
871 |             <table class="slc-table" style="font-size:
0.85rem;">
872 |                 <thead class="sticky-top">
873 |                     <tr>
874 |                         <th style="width: 160px;">Время</th>
875 |                         <th style="width: 90px;">Event ID</t
h>
876 |                         <th style="width: 140px;">Провайдер<
/th>
877 |                         <th style="width: 130px;">Тип</th>
878 |                         <th style="width: 180px;">Инициатор
/ Процесс</th>
879 |                         <th>Описание / Причина</th>
880 |                         <th style="width: 80px;" class="text
-center">XML</th>
881 |                     </tr>

```



```

882         </thead>
883         <tbody id="pwr-events-tbody">
884             <tr>
885                 <td colspan="7" class="text-center p
y-4 text-muted">
886                     <div class="spinner-border spinner
-border-sm text-info me-2" role="status"></div>
887                     Загрузка журнала событий...
888                 </td>
889             </tr>
890         </tbody>
891     </table>
892 </div>
893     <div id="pwr-events-table-resizer" class="sl
c-table-resizer" title="Потяните для изменения высоты таблицы
(двойной клик – сброс)">
894         <div class="slc-table-resizer-grip"></div>
895     </div>
896 </div>
897 </div>
898 </div>
899 </div>
900
901 </div>
902 </div>
903 </div>
904 </div>
905
906 <!-- Modal: Event Details & AI Diagnostics -->
907 <div class="modal fade" id="slc-event-detail-modal" tabindex
="-1">
908     <div class="modal-dialog modal-lg modal-dialog-centered">
909         <div class="modal-content" style="background: var(--surf
ace-1); color: var(--text-color); border: 1px solid var(--bord
er-color);">
910             <div class="modal-header" style="border-bottom: 1px so
lid var(--border-color);">
911                 <h5 class="modal-title fw-bold text-primary" id="slc

```

```

    -modal-title" data-i18n="systemControl.modalTitle">Детали собы
тия Windows</h5>
912     <button type="button" class="btn-close" data-bs-dism
iss="modal"></button>
913     </div>
914     <div class="modal-body">
915         <div class="row g-2 mb-3 small font-monospace">
916             <div class="col-sm-6"><strong class="text-muted" d
ata-i18n="systemControl.modalTime">Время:</strong> <span id="s
lc-modal-time"></span></div>
917             <div class="col-sm-6"><strong class="text-muted" d
ata-i18n="systemControl.modalLevel">Уровень:</strong> <span id
="slc-modal-level"></span></div>
918             <div class="col-sm-6"><strong class="text-muted" d
ata-i18n="systemControl.modalEventId">Event ID:</strong> <span
id="slc-modal-id"></span></div>
919             <div class="col-sm-6"><strong class="text-muted" d
ata-i18n="systemControl.modalProvider">Поставщик:</strong> <sp
an id="slc-modal-provider"></span></div>
920             <div class="col-sm-6"><strong class="text-muted" d
ata-i18n="systemControl.modalComputer">Компьютер:</strong> <sp
an id="slc-modal-computer"></span></div>
921             <div class="col-sm-6"><strong class="text-muted" d
ata-i18n="systemControl.modalPid">PID:</strong> <span id="slc-
modal-pid"></span></div>
922             <div class="col-sm-12"><strong class="text-muted"
data-i18n="systemControl.modalChannel">Канал:</strong> <span i
d="slc-modal-channel"></span></div>
923         </div>
924
925         <div class="p-3 mb-3 rounded border" style="backgrou
nd: var(--surface-2); border-color: var(--border-color) !impor
tant;">
926             <h6 class="small text-uppercase text-muted fw-bold
mb-2" data-i18n="systemControl.modalMessageHeader">Текст собы
тия</h6>
927             <div class="small font-monospace" id="slc-modal-me
ssage" style="white-space: pre-wrap; word-break: break-word; c

```

```

    color: var(--text-color);"></div>
928         </div>
929
930         <!-- AI Diagnostic Response Section -->
931         <div class="p-3 rounded border" style="background: v
ar(--surface-2); border-color: var(--border-color) !important;
" id="slc-modal-ai-section">
932             <div class="d-flex justify-content-between align-i
tems-center mb-2">
933                 <h6 class="fw-bold text-primary mb-0"><i class="
bi bi-robot me-1"></i> <span data-i18n="systemControl.modalAiH
eader">Контекстный анализ ИИ</span></h6>
934                 <button class="btn btn-sm btn-outline-info py-0
px-2 rounded-pill" id="btn-slc-modal-diagnose" data-i18n="syst
emControl.modalBtnDiagnose">Анализ контекста</button>
935             </div>
936             <div class="small text-muted" id="slc-modal-ai-exp
lanation" data-i18n="systemControl.modalAiPrompt">
937                 Нажмите «Анализ контекста», чтобы провести диагн
остику выбранной записи.
938             </div>
939         </div>
940     </div>
941     <div class="modal-footer" style="border-top: 1px solid
var(--border-color);">
942         <button type="button" class="btn btn-sm btn-secondar
y" data-bs-dismiss="modal" data-i18n="systemControl.modalBtnCl
ose">Закрыть</button>
943     </div>
944 </div>
945 </div>
946 </div>
947
948 <!-- Модальное окно деталей сессии питания -->
949 <div class="modal fade" id="pwrSessionDetailModal" tabindex=
"-1" aria-labelledby="pwrSessionDetailModalLabel" aria-hidden=
"true">
950     <div class="modal-dialog modal-lg modal-dialog-centered mo

```

```

dal-dialog-scrollable">
951     <div class="modal-content shadow" style="background: var
    (--surface-1); color: var(--text-color); border: 1px solid var
    (--border-color);">
952     <div class="modal-header border-bottom" style="border-
    color: var(--border-color) !important;">
953     <h5 class="modal-title d-flex align-items-center gap
    -2" id="pwrSessionDetailModalLabel">
954     <i class="bi bi-info-circle-fill text-warning"></i
    >
955     <span id="pwr-modal-title">Детали сессии питания</
    span>
956     </h5>
957     <button type="button" class="btn-close" data-bs-dism
    iss="modal" aria-label="Close"></button>
958     </div>
959     <div class="modal-body p-3" id="pwr-modal-body">
960     <!-- Populated by main.js -->
961     </div>
962     <div class="modal-footer border-top" style="border-col
    or: var(--border-color) !important;">
963     <button type="button" class="btn btn-secondary btn-s
    m" data-bs-dismiss="modal">Закрыть</button>
964     </div>
965     </div>
966     </div>
967 </div>
968
969 <!-- Модальное окно просмотра сырого XML события -->
970 <div class="modal fade" id="pwrXmlDetailModal" tabindex="-1"
    aria-labelledby="pwrXmlDetailModalLabel" aria-hidden="true">
971     <div class="modal-dialog modal-lg modal-dialog-centered mo
    dal-dialog-scrollable">
972     <div class="modal-content shadow" style="background: var
    (--surface-1); color: var(--text-color); border: 1px solid var
    (--border-color);">
973     <div class="modal-header border-bottom" style="border-
    color: var(--border-color) !important;">

```

```
974 |         <h5 class="modal-title d-flex align-items-center gap
    | -2" id="pwrXmlDetailModalLabel">
975 |             <i class="bi bi-filetype-xml text-info"></i>
976 |             <span>Исходный XML события Windows</span>
977 |         </h5>
978 |         <button type="button" class="btn-close" data-bs-dism
    | iss="modal" aria-label="Close"></button>
979 |     </div>
980 |     <div class="modal-body p-3">
981 |         <pre class="pwr-xml-pre p-3 rounded font-monospace m
    | b-0" id="pwr-xml-content" style="font-size: 0.78rem; max-heigh
    | t: 500px; overflow-y: auto;"></pre>
982 |     </div>
983 |     <div class="modal-footer border-top" style="border-col
    | or: var(--border-color) !important;">
984 |         <button type="button" class="btn btn-secondary btn-s
    | m" data-bs-dismiss="modal">Заккрыть</button>
985 |     </div>
986 | </div>
987 | </div>
988 | </div>
989 |
990 | </div>
991 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows System Logs & Activity - Main Script
4   * =====
5   * Description:
6   *   Клиентский веб-скрипт модуля журналов событий и активност
   и системы Windows.
7   *   Полная поддержка интернационализации (i18n), светлой и тёмной темы.
8   *
9   * Usage Examples:
10  *   HTML Integration:
11  *     <script src="/html/system_control_tab/main.js?v=20261006_v14" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/system_control_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 05:15:00
19  * =====
20  */
21
22 (function () {
23   'use strict';
24
25   let isSCCInitialized = false;
26
27   // Состояния центра журналов событий
28   let currentChannel = 'System';
29   let currentFilePath = '';
30   let currentMode = 'live';
```

```

31 | let cachedEvents = [];
32 | let cachedChannels = [];
33 | let selectedEvent = null;
34 | let liveIntervalTimer = null;
35 | let isLogRequestInProgress = false;
36 | let globalTooltipEl = null;
37 |
38 | /**
39 |  * Локализация строки через i18next с fallback.
40 |  *
41 |  * @param {string} key - Ключ локализации.
42 |  * @param {string} fallback - Значение по умолчанию.
43 |  * @returns {string} Локализованная строка.
44 |  */
45 | function t(key, fallback) {
46 |     if (window.i18next && typeof window.i18next.t === 'function' && window.i18next.isInitialized) {
47 |         const res = window.i18next.t(key);
48 |         if (res && res !== key) return res;
49 |     }
50 |     return fallback !== undefined ? fallback : key;
51 | }
52 |
53 | /**
54 |  * Экранирование HTML-символов для защиты от XSS.
55 |  *
56 |  * @param {string} str - Исходная строка.
57 |  * @returns {string} Экранированная строка.
58 |  */
59 | function escapeHtml(str) {
60 |     if (!str) return '';
61 |     return String(str)
62 |         .replace(/&/g, '&amp;')
63 |         .replace(/</g, '&lt;')
64 |         .replace(/>/g, '&gt;')
65 |         .replace(/"/g, '&quot;')
66 |         .replace(/'/g, '&#039;');
67 | }

```

```

68 |
69 | /**
70 |  * Форматирование ISO-времени.
71 |  *
72 |  * @param {string} isoStr - Строка ISO Timestamp.
73 |  * @returns {string} Отформатированное время.
74 |  */
75 | function formatTime(isoStr) {
76 |     if (!isoStr) return '-';
77 |     try {
78 |         return new Date(isoStr).toLocaleTimeString();
79 |     } catch {
80 |         return isoStr.slice(11, 19);
81 |     }
82 | }
83 |
84 | /**
85 |  * Определение CSS-класса для уровня важности события.
86 |  *
87 |  * @param {string} level - Уровень важности.
88 |  * @returns {string} Имя CSS-класса.
89 |  */
90 | function getSeverityClass(level) {
91 |     const lvl = (level || '').toLowerCase();
92 |     if (lvl.includes('crit')) return 'badge-crit';
93 |     if (lvl.includes('err')) return 'badge-err';
94 |     if (lvl.includes('warn')) return 'badge-warn';
95 |     if (lvl.includes('verb')) return 'badge-verb';
96 |     return 'badge-inf';
97 | }
98 |
99 | /**
100 |  * Проверка прав администратора и обновление бейджа.
101 |  */
102 | async function fetchStatus() {
103 |     try {
104 |         const res = await fetch('/api/system-control/status');
105 |         if (!res.ok) return;

```



```

106         const data = await res.json();
107
108         const elevBadge = document.getElementById('scc-elevation
| -badge');
109         if (elevBadge) {
110             if (data.is_elevated) {
111                 elevBadge.className = 'badge rounded-pill bg-success
| -subtle text-success border border-success px-3 py-2';
112                 elevBadge.innerHTML = t('systemControl.elevAdmin', '
| □ Режим: Администратор (Полный доступ)');
113             } else {
114                 elevBadge.className = 'badge rounded-pill bg-warning
| -subtle text-warning border border-warning px-3 py-2';
115                 elevBadge.innerHTML = t('systemControl.elevUser', '□
| □ Режим: Обычный пользователь (Ограничено)');
116             }
117         }
118     } catch (e) {
119         console.warn('[SystemLogs] Ошибка проверки прав:', e);
120     }
121 }
122
123 // =====
| =====
124 // ЖУРНАЛ АКТИВНОСТИ И ОПЕРАЦИЙ
125 // =====
| =====
126
127 async function loadSccActivityLogs() {
128     const tbody = document.getElementById('scc-logs-tbody');
129     if (!tbody) return;
130
131     try {
132         const res = await fetch('/api/system-control/logs');
133         if (!res.ok) return;
134         const data = await res.json();
135         const logs = data.logs || [];
136

```

```

137         if (logs.length === 0) {
138             tbody.innerHTML = `<tr><td colspan="5" class="text-center text-muted p-3">${t('systemControl.emptyActivity', 'Журнал операций пуст.')}</td></tr>`;
139             return;
140         }
141
142         tbody.innerHTML = logs.map(l => `
143             <tr>
144                 <td class="font-monospace small" style="color: var(--text-muted);">${escapeHtml(l.timestamp)}</td>
145                 <td><span class="badge bg-secondary font-monospace">${escapeHtml(l.action)}</span></td>
146                 <td style="color: var(--text-color);">${escapeHtml(l.target)}</td>
147                 <td><span class="badge ${l.status === 'SUCCESS' || l.status === 'OK' ? 'bg-success' : 'bg-warning text-dark'}">${escapeHtml(l.status)}</span></td>
148                 <td class="small text-truncate" style="max-width: 280px; color: var(--text-muted);" title="${escapeHtml(l.details)}">${escapeHtml(l.details)}</td>
149             </tr>
150         `).join('');
151     } catch (e) {
152         console.warn('[SystemLogs] Ошибка загрузки логов активно сти:', e);
153     }
154 }
155
156 // =====
157 // WINDOWS OS SYSTEM LOG CENTER (ПОТОК, КАНАЛЫ, АУДИТ, RAG, ИНЦИДЕНТЫ)
158 // =====
159
160 function createGlobalChannelTooltip() {
161     if (globalTooltipEl) return;

```

```

162     globalTooltipEl = document.createElement('div');
163     globalTooltipEl.className = 'slc-tooltip-popup';
164     globalTooltipEl.id = 'slc-global-tooltip';
165     document.body.appendChild(globalTooltipEl);
166 }
167
168 function showChannelTooltip(channelData, mouseEvent) {
169     if (!globalTooltipEl) createGlobalChannelTooltip();
170     const isFile = !(channelData.file_path || channelData.location || (channelData.source_type && channelData.source_type !== 'channel'));
171     const icon = isFile ? '📁' : '📄';
172     const rawTitle = channelData.display_name || channelData.channel_name || channelData.name || 'Channel';
173     const chanTitle = rawTitle.replace(/^evt:/, '');
174     const typeLabel = (channelData.source_type === 'channel' || !isFile) ? t('systemControl.tooltipTypeLive', 'Live Windows Channel') : t('systemControl.tooltipTypeArchive', 'EVTX File Archive');
175     const countVal = channelData.record_count !== undefined ? channelData.record_count : (channelData.records || 'N/A');
176     const recCount = typeof countVal === 'number' ? countVal.toLocaleString() : String(countVal);
177     const sizeStr = channelData.file_size_mb ? `${channelData.file_size_mb} MB` : (channelData.size_bytes ? `${(channelData.size_bytes / 1024 / 1024).toFixed(1)} MB` : 'N/A');
178
179     globalTooltipEl.innerHTML = `
180         <div class="fw-bold mb-1 d-flex align-items-center gap-1.5" style="color: var(--text-color);">
181             <span>${icon}</span>
182             <span class="text-truncate">${escapeHtml(chanTitle)}</span>
183         </div>
184         <div class="mb-2" style="font-size: 0.76rem; color: var(--text-muted) !important;">
185             ${escapeHtml(channelData.description || t('systemControl.tooltipDefaultDesc', 'Стандартный системный канал событий W

```

```

indows.'))}
186     </div>
187     <div class="d-flex justify-content-between align-items-c
enter pt-1 border-top" style="border-color: var(--border-color
) !important; font-size: 0.74rem;">
188         <span class="text-muted">${t('systemControl.tooltipTyp
e', 'Тип:')} <strong style="color: var(--text-color);">${typeL
abel}</strong></span>
189         <span class="text-muted">${t('systemControl.tooltipRec
ords', 'Записей:')} <strong class="text-info">${recCount}</str
ong></span>
190     </div>
191     <div class="d-flex justify-content-between align-items-c
enter pt-1" style="font-size: 0.74rem;">
192         <span class="text-muted">${t('systemControl.tooltipSiz
e', 'Размер:')} <strong style="color: var(--text-color);">${si
zeStr}</strong></span>
193         <span class="text-success fw-bold">${t('systemControl.
tooltipAvailable', 'Доступен')}</span>
194     </div>
195     `;
196
197     const pad = 15;
198     let left = mouseEvent.clientX + pad;
199     let top = mouseEvent.clientY + pad;
200     if (left + 330 > window.innerWidth) left = mouseEvent.clie
ntX - 335;
201     if (top + 140 > window.innerHeight) top = mouseEvent.clie
ntY - 145;
202
203     globalTooltipEl.style.left = `${Math.max(10, left)}px`;
204     globalTooltipEl.style.top = `${Math.max(10, top)}px`;
205     globalTooltipEl.classList.add('show');
206 }
207
208 function hideChannelTooltip() {
209     if (globalTooltipEl) {
210         globalTooltipEl.classList.remove('show');

```

```

211     }
212 }
213
214 async function scanChannels(force = false) {
215     const treeEl = document.getElementById('slc-channel-tree-c
216     ontainer');
217     const countBadge = document.getElementById('slc-channel-co
218     unt-badge');
219     const sourcesStat = document.getElementById('slc-stat-sour
220     ces');
221     if (treeEl && (!cachedChannels || cachedChannels.length ==
222     = 0)) {
223         treeEl.innerHTML = `
```

```
238
239     renderChannelTree(cachedChannels);
240 } catch (err) {
241     if (treeEl) treeEl.innerHTML = `
```

```

267 |         const icon = isFile ? 'bi-file-earmark-text text-seconda
      | ry' : 'bi-folder-fill text-warning';
268 |
269 |         let sizeBadge = '';
270 |         if (ch.record_count !== undefined && ch.record_count > 0
      | ) {
271 |             const countText = ch.record_count >= 1000 ? `${(ch.rec
      | ord_count / 1000).toFixed(1)}k` : String(ch.record_count);
272 |             sizeBadge = `${countText}</span>`;
273 |         } else if (ch.file_size_mb) {
274 |             sizeBadge = `${ch.file_size_mb} MB</span
      | >`;
275 |         }
276 |
277 |         return `
278 |             <div class="slc-tree-item ${isActive ? 'active' : ''}"
      | data-channel="${escapeHtml(sysName)}" data-display-name="${es
      | capeHtml(disName)}" data-path="${escapeHtml(chPath)}">
279 |                 <div class="d-flex align-items-center gap-2 text-tru
      | ncate me-2" style="min-width: 0; flex: 1 1 auto;">
280 |                     <i class="bi ${icon} flex-shrink-0"></i>
281 |                     <span class="text-truncate fw-medium" title="${esc
      | apeHtml(disName)}">${escapeHtml(disName)}</span>
282 |                 </div>
283 |                 <div class="d-flex align-items-center flex-shrink-0"
      | >
284 |                     ${sizeBadge}
285 |                 </div>
286 |             </div>
287 |         `;
288 |     })).join('');
289 |
290 |     treeEl.querySelectorAll('.slc-tree-item').forEach((el, ind
      | ex) => {
291 |         const chData = channels[index];
292 |         el.addEventListener('mouseenter', (e) => {

```

```
293 |         if (chData) showChannelTooltip(chData, e);
294 |     });
295 |     el.addEventListener('mousemove', (e) => {
296 |         if (chData) showChannelTooltip(chData, e);
297 |     });
298 |     el.addEventListener('mouseleave', () => {
299 |         hideChannelTooltip();
300 |     });
301 |
302 |     el.addEventListener('click', () => {
303 |         hideChannelTooltip();
304 |         treeEl.querySelectorAll('.slc-tree-item').forEach(i =>
305 | i.classList.remove('active'));
306 |         el.classList.add('active');
307 |
308 |         currentChannel = (el.getAttribute('data-channel') || e
309 | l.getAttribute('data-name') || 'System').replace(/^evt:/, '').
310 | split(' (')[0].trim();
311 |         currentFilePath = el.getAttribute('data-path') || '';
312 |         const dispName = el.getAttribute('data-display-name')
313 | || currentChannel;
314 |
315 |         const statChan = document.getElementById('slc-stat-cha
316 | nnel');
317 |         if (statChan) statChan.textContent = dispName;
318 |
319 |         const mainTitle = document.getElementById('slc-main-vi
320 | ew-title');
321 |         if (mainTitle) mainTitle.innerHTML = `
```



```
321         for (let opt of elSelect.options) {
322             if (opt.value.toLowerCase() === currentChannel.t
oLowerCase()) {
323                 elSelect.value = opt.value;
324                 found = true;
325                 break;
326             }
327         }
328         if (!found && !currentFilePath) {
329             const opt = document.createElement('option');
330             opt.value = currentChannel;
331             opt.textContent = `Журнал ${dispName}`;
332             elSelect.appendChild(opt);
333             elSelect.value = currentChannel;
334         }
335     }
336     fetchEventLogsSummary();
337 } else {
338     loadEvents();
339 }
340 });
341 });
342 }
343
344 async function loadEvents() {
345     if (isLogRequestInProgress) return;
346     isLogRequestInProgress = true;
347
348     const tbody = document.getElementById('slc-table-body');
349     const statEvents = document.getElementById('slc-stat-event
s');
350     const statIncidents = document.getElementById('slc-stat-in
cidents');
351     const streamBadge = document.getElementById('slc-stream-co
unt-badge');
352
353     const searchInput = document.getElementById('slc-search-in
put');
```

```

354     const levelSelect = document.getElementById('slc-level-select');
355     const hoursSelect = document.getElementById('slc-hours-select');
356     const limitSelect = document.getElementById('slc-limit-select');
357     const eventIdInput = document.getElementById('slc-eventid-input');
358
359     const query = searchInput ? searchInput.value.trim() : '';
360     const level = levelSelect ? levelSelect.value : '';
361     const hours = hoursSelect ? hoursSelect.value : '24';
362     const limit = limitSelect ? limitSelect.value : '100';
363     const eventId = eventIdInput ? eventIdInput.value.trim() :
        '';
364
365     const cleanChan = currentChannel.replace(/^evt:/, '').split(' (')[0].trim();
366     const params = new URLSearchParams({
367         channel: cleanChan,
368         limit: limit,
369         hours: hours,
370     });
371
372     if (query) {
373         params.append('search', query);
374         params.append('query', query);
375     }
376     if (level) params.append('level', level);
377     if (eventId) params.append('event_id', eventId);
378     if (currentFilePath) params.append('file_path', currentFilePath);
379
380     let url = `/api/v1/system_logs/events?${params.toString()}`;
381
382     if (currentMode === 'errors') {
383         params.set('level', 'Error');
384         url = `/api/v1/system_logs/events?${params.toString()}`;

```

```

384 |     }
385 |
386 |     try {
387 |         const res = await fetch(url);
388 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
389 |         const data = await res.json();
390 |         cachedEvents = data.events || data.entries || [];
391 |
392 |         if (statEvents) statEvents.textContent = cachedEvents.length.toLocaleString();
393 |         if (streamBadge) streamBadge.textContent = `${cachedEvents.length} ${t('systemControl.records', 'записей')}`;
394 |
395 |         const faults = cachedEvents.filter(e => {
396 |             const l = (e.level || '').toLowerCase();
397 |             return l.includes('err') || l.includes('crit');
398 |         }).length;
399 |         if (statIncidents) statIncidents.textContent = faults;
400 |
401 |         renderEventsTable(cachedEvents);
402 |     } catch (err) {
403 |         if (tbody) tbody.innerHTML = `<tr><td colspan="6" class="text-center text-danger py-4">${t('systemControl.errorLoadingEvents', 'Ошибка загрузки событий:')} ${escapeHtml(err.message)}</td></tr>`;
404 |     } finally {
405 |         isLogRequestInProgress = false;
406 |     }
407 | }
408 |
409 | function renderEventsTable(events) {
410 |     const tbody = document.getElementById('slc-table-body');
411 |     if (!tbody) return;
412 |
413 |     if (!events || events.length === 0) {
414 |         tbody.innerHTML = `<tr><td colspan="6" class="text-center text-muted py-4">${t('systemControl.noEvents', 'Нет записей, удовлетворяющих заданным фильтрам.')}</td></tr>`;

```

```

415     return;
416 }
417
418 tbody.innerHTML = events.map((e, idx) => {
419     const timeFormatted = formatTime(e.timestamp);
420     const isErr = (e.level || '').toLowerCase().includes('err') || (e.level || '').toLowerCase().includes('crit');
421     const provName = e.provider || e.source || '-';
422     const msgText = e.message || e.raw_data || '-';
423
424     return `
425         <tr data-index="${idx}" class="${isErr ? 'table-danger
426         -subtle' : ''}">
427             <td class="font-monospace small" style="color: var(--text-muted);">${escapeHtml(timeFormatted)}</td>
428             <td><span class="badge ${getSeverityClass(e.level)}
429             text-uppercase" style="font-size: 0.68rem;">${escapeHtml(e.level || 'Info')}</span></td>
430             <td class="font-monospace small fw-bold" style="color: #0284c7;">${e.event_id > 0 ? e.event_id : '-'}</td>
431             <td class="text-truncate small" style="max-width: 160px; color: var(--text-color);" title="${escapeHtml(provName)}">${escapeHtml(provName)}</td>
432             <td class="font-monospace small" style="color: var(--text-muted);">${e.process_id || '-'}</td>
433             <td class="text-truncate" style="max-width: 420px; color: var(--text-color);" title="${escapeHtml(msgText)}">${escapeHtml(msgText)}</td>
434         </tr>
435     `;
436 }).join('');
437
438 tbody.querySelectorAll('tr').forEach(row => {
439     row.onclick = () => {
440         tbody.querySelectorAll('tr').forEach(r => r.classList.remove('table-active'));
441         row.classList.add('table-active');
442         const idx = parseInt(row.getAttribute('data-index'), 1

```

```

0);
441     selectedEvent = events[idx];
442
443     if (currentMode === 'raw') {
444         const rawEl = document.getElementById('slc-raw-text'
);
445         if (rawEl && selectedEvent) rawEl.textContent = JSON
.stringify(selectedEvent, null, 2);
446     } else {
447         showEventDetails(selectedEvent);
448     }
449 };
450 });
451 }
452
453 async function loadAuditDigest() {
454     const contentEl = document.getElementById('slc-audit-conte
nt');
455     if (!contentEl) return;
456     contentEl.innerHTML = `<div class="text-center text-muted
py-4"><div class="spinner-border spinner-border-sm text-info m
e-2"></div>${escapeHtml(t('systemControl.auditSummaryGeneratin
g', 'Формирование сводки аудита и дедупликации...'))}</div>`;
457
458     try {
459         const hoursSelect = document.getElementById('slc-hours-s
elect');
460         const hours = hoursSelect ? hoursSelect.value : '24';
461         const cleanChan = currentChannel.replace(/^evt:/i, '');
462         const res = await fetch(`/api/v1/system_logs/audit?chann
el=${encodeURIComponent(cleanChan)}&hours=${hours}&limit=300&f
ile_path=${encodeURIComponent(currentFilePath)}`);
463         if (!res.ok) throw new Error(`HTTP ${res.status}`);
464         const data = await res.json();
465
466         const totalEvents = data.total_events || 0;
467         const critSum = data.severity_summary?.Critical || 0;
468         const errSum = data.severity_summary?.Error || 0;

```

```

469         const warnSum = data.severity_summary?.Warning || 0;
470
471         let anomaliesHtml = `
```

```
491 |         <td class="font-monospace small" style="color: var
    | (-text-muted);">${escapeHtml(formatTime(c.first_seen))}</td>
492 |     </tr>
493 |     `).join('');
494 |     } else {
495 |         clustersHtml = `<tr><td colspan="7" class="text-center
    | text-muted py-3">${escapeHtml(t('systemControl.noClusters', '
    | Кластеры не сформированы.'))}</td></tr>`;
496 |     }
497 |
498 |     contentEl.innerHTML = `
499 |         <div class="row g-2 mb-3">
500 |             <div class="col-sm-3 col-6">
501 |                 <div class="p-2 rounded border" style="background:
    | var(--surface-1); border-color: var(--border-color) !important;
    | ">
502 |                     <div class="small text-muted" style="font-size:
    | 0.72rem;">${escapeHtml(t('systemControl.totalAnalyzed', 'Всего
    | проанализировано'))}</div>
503 |                     <div class="fs-5 fw-bold text-info font-monospac
    | e">${totalEvents}</div>
504 |                 </div>
505 |             </div>
506 |             <div class="col-sm-3 col-6">
507 |                 <div class="p-2 rounded border" style="background:
    | var(--surface-1); border-color: var(--border-color) !important;
    | ">
508 |                     <div class="small text-muted" style="font-size:
    | 0.72rem;">${escapeHtml(t('systemControl.critFaults', 'Критичес
    | ких сбоев'))}</div>
509 |                     <div class="fs-5 fw-bold text-danger font-monospac
    | ace">${critSum}</div>
510 |                 </div>
511 |             </div>
512 |             <div class="col-sm-3 col-6">
513 |                 <div class="p-2 rounded border" style="background:
    | var(--surface-1); border-color: var(--border-color) !important;
    | ">
```

```
514 |         <div class="small text-muted" style="font-size:
    | 0.72rem;">${escapeHtml(t('systemControl.errorsCount', 'Ошибок'
    | ))}</div>
515 |         <div class="fs-5 fw-bold text-danger font-monosp
    | ace">${errSum}</div>
516 |     </div>
517 | </div>
518 |     <div class="col-sm-3 col-6">
519 |         <div class="p-2 rounded border" style="background:
    | var(--surface-1); border-color: var(--border-color) !importan
    | t;">
520 |             <div class="small text-muted" style="font-size:
    | 0.72rem;">${escapeHtml(t('systemControl.warningsCount', 'Преду
    | преждений'))}</div>
521 |             <div class="fs-5 fw-bold text-warning font-monos
    | pace">${warnSum}</div>
522 |         </div>
523 |     </div>
524 | </div>
525 |
526 |     <div class="slc-card mb-2 shadow-sm">
527 |         <div class="slc-card-header text-warning fw-bold py-
    | 1.5 px-3 small">
528 |             <i class="bi bi-exclamation-triangle me-1"></i>${e
    | scapeHtml(t('systemControl.anomaliesTitle', 'Обнаруженные аном
    | алии и всплески'))}
529 |         </div>
530 |         <div class="p-2">
531 |             ${anomaliesHtml}
532 |         </div>
533 |     </div>
534 |
535 |     <div class="slc-card shadow-sm">
536 |         <div class="slc-card-header fw-bold d-flex justify-c
    | ontent-between align-items-center py-1.5 px-3">
537 |             <span><i class="bi bi-collection me-2 text-primary
    | "></i>${escapeHtml(t('systemControl.patternsTitle', 'Топ шабло
    | нов событий (Дедупликация'))}</span>
```



```

538         <span class="small text-muted">${escapeHtml(t('sys
temControl.top15', 'Топ 15 кластеров'))}</span>
539     </div>
540     <div class="table-responsive">
541         <table class="slc-table mb-0">
542             <thead>
543                 <tr>
544                     <th style="width: 40px;">${escapeHtml(t('sys
temControl.colNum', '#'))}</th>
545                     <th style="width: 80px;">${escapeHtml(t('sys
temControl.colLevel', 'Уровень'))}</th>
546                     <th style="width: 140px;">${escapeHtml(t('sy
stemControl.colProvider', 'Поставщик'))}</th>
547                     <th style="width: 60px;">${escapeHtml(t('sys
temControl.colId', 'ID'))}</th>
548                     <th>${escapeHtml(t('systemControl.colPattern
', 'Шаблон сообщения (Masked)'))}</th>
549                     <th style="width: 70px;" class="text-center"
>${escapeHtml(t('systemControl.colCount', 'Кол-во'))}</th>
550                     <th style="width: 140px;">${escapeHtml(t('sy
stemControl.colFirstSeen', 'Впервые'))}</th>
551                 </tr>
552             </thead>
553             <tbody>
554                 ${clustersHtml}
555             </tbody>
556         </table>
557     </div>
558 </div>
559 `;
560 } catch (err) {
561     contentEl.innerHTML = `<div class="alert alert-danger">${
escapeHtml(t('systemControl.errorAudit', 'Ошибка проведения а
удита:'))} ${escapeHtml(err.message)}</div>`;
562 }
563 }
564
565 async function runRagSearch() {

```

```

566     const inputEl = document.getElementById('slc-rag-input');
567     const resultsEl = document.getElementById('slc-rag-results
    ');
568     if (!inputEl || !resultsEl) return;
569     const query = inputEl.value.trim();
570     if (!query) return;
571
572     resultsEl.innerHTML = `
```

```

e">Relevance: ${Math.round((c.score || 0) * 100)}%</span>
594         <span class="small font-monospace" style="color: v
ar(--text-muted);">${escapeHtml(c.metadata?.timestamp || '')}<
/span>
595     </div>
596     <div class="small font-monospace" style="white-space
: pre-wrap; color: var(--text-color);">${escapeHtml(c.text ||
c.content)}</div>
597 </div>
598 `).join('');
599 } catch (err) {
600     resultsEl.innerHTML = `<div class="alert alert-danger">$
{escapeHtml(t('systemControl.errorRagSearch', 'Ошибка поиска R
AG:'))} ${escapeHtml(err.message)}</div>`;
601 }
602 }
603
604 async function rebuildRagIndex() {
605     const resultsEl = document.getElementById('slc-rag-results
');
606     if (!resultsEl) return;
607     resultsEl.innerHTML = `<div class="text-center text-info p
y-4"><div class="spinner-border spinner-border-sm text-info me
-2"></div>${escapeHtml(t('systemControl.ragRebuilding', 'Перес
троение семантического RAG-индекса...'))}</div>`;
608
609     try {
610         const cleanChan = currentChannel.replace(/^evt:/i, '');
611         const res = await fetch(`/api/v1/system_logs/rag/build?c
hannel=${encodeURIComponent(cleanChan)}&limit=500&hours=24&fil
e_path=${encodeURIComponent(currentFilePath)}`, {
612             method: 'POST',
613         });
614         if (!res.ok) throw new Error(`HTTP ${res.status}`);
615         const data = await res.json();
616         const summaryMsg = t('systemControl.ragAddedChunks', 'До
бавлено {{added}} чанков (Всего в индексе: {{total}}).')
617         .replace('{{added}}', String(data.chunks_added ?? 0))

```

```

618 |         .replace('{{total}}', String(data.total_index_chunks ?
    | ? 0));
619 |         resultsEl.innerHTML = `
620 |             <div class="alert alert-success d-flex align-items-cen
    | ter gap-2">
621 |                 <i class="bi bi-check-circle-fill fs-5"></i>
622 |                 <div>
623 |                     <strong>${escapeHtml(t('systemControl.ragRebuildSu
    | ccess', 'RAG-индекс успешно обновлен!'))}</strong> ${escapeHtm
    | l(summaryMsg)}
624 |                     <div class="small mt-1 text-muted">${escapeHtml(da
    | ta.executive_summary || ''})</div>
625 |                 </div>
626 |             </div>
627 |         `;
628 |     } catch (err) {
629 |         resultsEl.innerHTML = `<div class="alert alert-danger">${
    | escapeHtml(t('systemControl.errorRagRebuild', 'Ошибка обновле
    | ния RAG:'))} ${escapeHtml(err.message)}</div>`;
630 |     }
631 | }
632 |
633 | function showEventDetails(entry) {
634 |     if (!entry) return;
635 |     const modalEl = document.getElementById('slc-event-detail-
    | modal');
636 |     if (!modalEl) return;
637 |
638 |     const setElTxt = (id, val) => {
639 |         const el = document.getElementById(id);
640 |         if (el) el.textContent = val || '';
641 |     };
642 |
643 |     setElTxt('slc-modal-time', entry.timestamp);
644 |     const lvlEl = document.getElementById('slc-modal-level');
645 |     if (lvlEl) lvlEl.innerHTML = `<span class="badge ${getSeve
    | rityClass(entry.level)}">${escapeHtml(entry.level)}</span>`;
646 |     setElTxt('slc-modal-id', entry.event_id > 0 ? String(entry

```

```

        .event_id) : '-');
647     setElTxt('slc-modal-provider', entry.provider || entry.sou
    rce || '-');
648     setElTxt('slc-modal-computer', entry.computer || '-');
649     setElTxt('slc-modal-pid', entry.process_id > 0 ? String(en
    try.process_id) : '-');
650     setElTxt('slc-modal-channel', (entry.channel || currentCha
    nnel).replace(/^evt:/i, ''));
651     setElTxt('slc-modal-message', entry.message);
652
653     const aiExplanation = document.getElementById('slc-modal-a
    i-explanation');
654     if (aiExplanation) {
655         aiExplanation.innerHTML = escapeHtml(t('systemControl.mo
    dalAiPrompt', 'Нажмите «Анализ контекста», чтобы провести диаг
    ностику выбранной записи.'));
656     }
657
658     const diagnoseBtn = document.getElementById('btn-slc-modal
    -diagnose');
659     if (diagnoseBtn) {
660         diagnoseBtn.onclick = async () => {
661             aiExplanation.innerHTML = `
```

```

673         message: entry.message || '',
674         target_entry: entry,
675         query_text: `Event diagnostic ${entry.provider |
| entry.source} (ID ${entry.event_id || 0}): ${entry.message |
| ''}` ,
676     )),
677     ));
678     if (!res.ok) throw new Error(`HTTP ${res.status}`);
679     const report = await res.json();
680     aiExplanation.innerHTML = `
681         <div class="mb-2"><strong class="text-info">${esca
| peHtml(t('systemControl.modalAiSummary', 'Сводка:'))}</strong>
| ${escapeHtml(report.summary)}</div>
682         <div class="mb-2"><strong class="text-warning">${e
| scapeHtml(t('systemControl.modalAiRootCause', 'Причина / Анали
| з:'))}</strong> ${escapeHtml(report.root_cause)}</div>
683         <h6 class="small text-uppercase text-muted fw-bold
| mb-1">${escapeHtml(t('systemControl.modalAiRecs', 'Рекомендац
| ии:'))}</h6>
684         <ul class="mb-0 ps-3">
685             ${((report.recommendations || []).map(r => `<li>$
| {escapeHtml(r)}</li>`).join(''))}
686         </ul>
687     `;
688     } catch (e) {
689         aiExplanation.innerHTML = `<div class="text-danger">
| ${escapeHtml(t('systemControl.errorDiagnose', 'Ошибка диагност
| ики:'))} ${escapeHtml(e.message)}</div>`;
690     }
691     };
692     }
693
694     if (window.bootstrap?.Modal) {
695         const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl);
696         bsModal.show();
697     }
698 }

```

```
699
700   async function loadIncidents() {
701     const listEl = document.getElementById('slc-incidents-list
');
702     if (!listEl) return;
703     listEl.innerHTML = `
```

```

724 |         <div class="small font-monospace text-muted">${esc
|apeHtml(t('systemControl.timeWindow', 'Временное окно:'))} ${e
|scapeHtml(inc.time_window || '')}</div>
725 |     </div>
726 | </div>
727 | `).join('');
728 | } catch (e) {
729 |     listEl.innerHTML = `<div class="alert alert-danger">${es
|capeHtml(t('systemControl.errorIncidents', 'Ошибка загрузки ин
|цидентов:'))} ${escapeHtml(e.message)}</div>`;
730 | }
731 | }
732 |
733 | async function loadTimeline() {
734 |     const listEl = document.getElementById('slc-timeline-list'
|);
735 |     if (!listEl) return;
736 |     listEl.innerHTML = `<div class="text-center text-muted py-
|4"><div class="spinner-border spinner-border-sm text-info me-2
|"></div>${escapeHtml(t('systemControl.formingTimeline', 'Форми
|рование временной шкалы...'))}</div>`;
737 |
738 |     try {
739 |         const res = await fetch('/api/v1/system_logs/timeline');
740 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
741 |         const data = await res.json();
742 |         const points = data.timeline || [];
743 |
744 |         if (points.length === 0) {
745 |             listEl.innerHTML = `<div class="slc-card p-3 text-cent
|er text-muted">${escapeHtml(t('systemControl.noTimeline', 'Дан
|ные временной шкалы отсутствуют.'))}</div>`;
746 |             return;
747 |         }
748 |
749 |         listEl.innerHTML = points.map(p => `
750 |             <div class="d-flex align-items-center gap-2 mb-1">
751 |                 <span class="font-monospace small" style="width: 120

```



```

px; color: var(--text-muted);">${escapeHtml(p.time_slot)}</span>
752     <div class="progress flex-grow-1" style="height: 12px; background: var(--surface-2);">
753         <div class="progress-bar bg-info" style="width: ${Math.min(100, p.count * 4)}%;"></div>
754     </div>
755     <span class="font-monospace small fw-bold" style="width: 40px; color: #0284c7;">${p.count}</span>
756 </div>
757     `).join('');
758 } catch (e) {
759     listEl.innerHTML = `<div class="alert alert-danger">${escapeHtml(t('systemControl.errorTimeline', 'Ошибка таймлайна:'))} ${escapeHtml(e.message)}</div>`;
760 }
761 }
762
763 // =====
764 // ПАНЕЛЬ ЖУРНАЛОВ СОБЫТИЙ WINDOWS (WEVTAPI & LOG INTELLIGENCE)
765 // =====
766
767 let elEventsList = [];
768 let isElInitialized = false;
769
770 async function fetchEventLogsSummary() {
771     const channelSelect = document.getElementById('el-channel-select');
772     const channel = channelSelect ? channelSelect.value : 'System';
773
774     try {
775         const res = await fetch(`/api/event-logs/events?channel=${encodeURIComponent(channel)}&limit=100&hours=24`);
776         if (!res.ok) throw new Error(`HTTP ${res.status}`);

```

```

777     const entries = await res.json();
778     renderElEvents(entries);
779     fetchIntelligenceProfile(channel, false);
780 } catch (err) {
781     console.warn('[EventLogs] Ошибка получения событий:', er
r);
782 }
783 }
784
785 async function fetchIntelligenceProfile(channel, showBanner
= true) {
786     try {
787         const res = await fetch(`/api/event-logs/intelligence/pr
ofile?channel=${encodeURIComponent(channel)}&hours=24&limit=10
0`);
788         if (!res.ok) return;
789         const data = await res.json();
790
791         const healthEl = document.getElementById('el-health-scor
e');
792         const redEl = document.getElementById('el-redundancy-pct
');
793
794         if (data.profile) {
795             if (healthEl) {
796                 const score = Math.round(data.profile.health_score ?
? 100);
797                 healthEl.textContent = `${score}/100`;
798                 healthEl.className = `fs-4 fw-bold mt-1 font-monospa
ce ${score >= 80 ? 'text-success' : (score >= 50 ? 'text-warni
ng' : 'text-danger')}`;
799             }
800             if (redEl) {
801                 const rPct = (data.profile.redundancy_ratio_pct ?? 0
).toFixed(1);
802                 redEl.textContent = `${rPct}%`;
803             }
804         }

```

```
805 |
806 |     if (showBanner && data.decision) {
807 |         renderIntelligenceBanner(data);
808 |     }
809 | } catch (err) {
810 |     console.debug('[EventLogs] Фоновый сбор аналитики:', err
811 | );
812 | }
813 |
814 | function renderIntelligenceBanner(data) {
815 |     const banner = document.getElementById('el-intel-banner');
816 |     const title = document.getElementById('el-intel-title');
817 |     const strat = document.getElementById('el-strategy-badge')
818 | ;
819 |     const rationale = document.getElementById('el-intel-ration
820 | ale');
821 |     const rec = document.getElementById('el-intel-recommendati
822 | on');
823 |     const chunks = document.getElementById('el-chunks-badge');
824 |
825 |     if (!banner) return;
826 |
827 |     if (title) title.textContent = `Log Intelligence: Канал ${
828 | data.channel || 'System'}`;
829 |     if (strat) strat.textContent = (data.decision.strategy ||
830 | 'SNAPSHOT').toUpperCase();
831 |     if (rationale) rationale.textContent = data.decision.ratio
832 | nale || '';
833 |     if (rec) rec.innerHTML = `
```

```
833 |   async function runElFullAudit() {
834 |     const channelSelect = document.getElementById('el-channel-
      | select');
835 |     const channel = channelSelect ? channelSelect.value : 'Sys
      | tem';
836 |
837 |     try {
838 |       const res = await fetch(`/api/event-logs/intelligence/au
      | dit?channel=${encodeURIComponent(channel)}&hours=24&limit=200`
      | );
839 |       if (!res.ok) throw new Error(`HTTP ${res.status}`);
840 |       const data = await res.json();
841 |       renderIntelligenceBanner({
842 |         channel: data.channel,
843 |         decision: {
844 |           strategy: data.strategy,
845 |           rationale: data.strategy_rationale,
846 |           recommended_llm_action: data.recommended_llm_action,
847 |           chunks_generated: data.top_clusters ? data.top_clust
      | ers.length : 0,
848 |         }
849 |       });
850 |     } catch (err) {
851 |       console.warn('[EventLogs] Ошибка аудита:', err);
852 |     }
853 |   }
854 |
855 |   async function runElRAGSearch() {
856 |     const input = document.getElementById('el-rag-query-input'
      | );
857 |     const channelSelect = document.getElementById('el-channel-
      | select');
858 |     const resultsCard = document.getElementById('el-rag-result
      | s-card');
859 |     const resultsBody = document.getElementById('el-rag-result
      | s-body');
860 |
861 |     if (!input || !resultsCard || !resultsBody) return;
```

```

862     const query = input.value.trim();
863     if (!query) return;
864
865     resultsBody.innerHTML = `
```

```

    || 'Adaptive')) • ${escapeHtml(r.created_at || '')}</div>
887     </div>
888     `).join('');
889     } catch (err) {
890         resultsBody.innerHTML = `<div class="text-danger small p
-2">Ошибка поиска: ${escapeHtml(err.message)}</div>`;
891     }
892 }
893
894 function renderElEvents(entries) {
895     elEventsList = entries || [];
896
897     const critEl = document.getElementById('el-critical-count'
);
898     const errEl = document.getElementById('el-error-count');
899
900     let crits = 0;
901     let errs = 0;
902
903     elEventsList.forEach(e => {
904         const l = (e.level || '').toLowerCase();
905         if (l.includes('crit')) crits++;
906         else if (l.includes('err')) errs++;
907     });
908
909     if (critEl) critEl.textContent = crits;
910     if (errEl) errEl.textContent = errs;
911
912     applyElFilters();
913 }
914
915 function applyElFilters() {
916     const tbody = document.getElementById('el-tbody');
917     const searchInput = document.getElementById('el-search-inp
ut');
918     const badge = document.getElementById('el-table-badge');
919
920     if (!tbody) return;

```

```
921 |
922 |     const query = (searchInput ? searchInput.value : '').toLowerCase().trim();
923 |
924 |     const filtered = elEventsList.filter(e => {
925 |         if (!query) return true;
926 |         const msgMatch = (e.message || '').toLowerCase().includes(query);
927 |         const provMatch = (e.provider_name || e.provider || '').toLowerCase().includes(query);
928 |         const idMatch = String(e.event_id || e.id || '').includes(query);
929 |         return msgMatch || provMatch || idMatch;
930 |     });
931 |
932 |     if (badge) badge.textContent = `${filtered.length} из ${elEventsList.length} событий`;
933 |
934 |     if (filtered.length === 0) {
935 |         tbody.innerHTML = `<tr><td colspan="5" class="text-center text-muted py-4">${escapeHtml(t('systemControl.noEvents', 'События не найдены'))}</td></tr>`;
936 |         return;
937 |     }
938 |
939 |     const itemsToRender = filtered.slice(0, 150);
940 |     tbody.innerHTML = itemsToRender.map((e, idx) => {
941 |         const level = (e.level || 'Information').toLowerCase();
942 |         let badgeClass = 'bg-secondary';
943 |         if (level.includes('crit') || level.includes('err')) badgeClass = 'bg-danger text-white';
944 |         else if (level.includes('warn')) badgeClass = 'bg-warning text-dark';
945 |         else if (level.includes('info')) badgeClass = 'bg-info text-dark';
946 |
947 |         return `
948 |             <tr class="slc-el-table-row" data-idx="${idx}" style="`
```

```

        cursor: pointer;" title="Нажмите для просмотра подробностей со
        бытия">
949         <td><span class="badge ${badgeClass}">${escapeHtml(e
        .level || 'Info')}</span></td>
950         <td class="text-muted small font-monospace">${escape
        Html(e.time_created || e.timestamp || '-')}</td>
951         <td class="fw-semibold small" style="color: var(--te
        xt-color);">${escapeHtml(e.provider_name || e.provider || 'Win
        dows')}</td>
952         <td class="font-monospace text-warning small">${esca
        peHtml(String(e.event_id || e.id || '-'))}</td>
953         <td class="small text-truncate" style="max-width: 50
        0px; color: var(--text-color);">${escapeHtml(e.message || '-')}
        </td>
954     </tr>
955     `;
956     }).join('');
957
958     // Привязываем клики по строкам для просмотра деталей и ИИ
    -диагностики
959     tbody.querySelectorAll('.slc-el-table-row').forEach(row =>
    {
960         row.onclick = () => {
961             const idx = parseInt(row.getAttribute('data-idx'), 10)
962             ;
963             const item = itemsToRender[idx];
964             if (item) {
965                 showEventDetails({
966                     timestamp: item.time_created || item.timestamp,
967                     level: item.level,
968                     event_id: item.event_id || item.id,
969                     provider: item.provider_name || item.provider,
970                     computer: item.computer || item.machine_name || ''
971                     ,
972                     process_id: item.process_id || item.pid,
973                     channel: item.channel || (document.getElementById(
974                     'el-channel-select')?.value || 'System'),
975                     message: item.message,

```



```
973         });
974     }
975 };
976 });
977 }
978
979 function initEventLogsPanel() {
980     if (isElInitialized) return;
981     isElInitialized = true;
982
983     const refreshBtn = document.getElementById('el-refresh-btn
984 ');
985     if (refreshBtn) refreshBtn.addEventListener('click', fetch
986 EventLogsSummary);
987
988     const channelSelect = document.getElementById('el-channel-
989 select');
990     if (channelSelect) channelSelect.addEventListener('change'
991 , fetchEventLogsSummary);
992
993     const searchInput = document.getElementById('el-search-inp
994 ut');
995     if (searchInput) searchInput.addEventListener('input', app
996 lyElFilters);
997
998     const analyzeBtn = document.getElementById('el-intel-analy
999 ze-btn');
1000     if (analyzeBtn) analyzeBtn.addEventListener('click', () =>
1001 {
1002     const chan = channelSelect ? channelSelect.value : 'Syst
1003 em';
1004     fetchIntelligenceProfile(chan, true);
1005 });
1006
1007     const auditBtn = document.getElementById('el-intel-audit-b
1008 tn');
1009     if (auditBtn) auditBtn.addEventListener('click', runElFull
1010 Audit);
```

```

1000
1001     const ragBtn = document.getElementById('el-rag-search-btn'
1002     );
1003     if (ragBtn) ragBtn.addEventListener('click', runElRAGSearch
1004     h);
1005     const ragInput = document.getElementById('el-rag-query-inp
1006     ut');
1007     if (ragInput) ragInput.addEventListener('keydown', (e) =>
1008     {
1009         if (e.key === 'Enter') runElRAGSearch();
1010     });
1011     const bannerCloseBtn = document.getElementById('el-intel-c
1012     lose-btn');
1013     if (bannerCloseBtn) bannerCloseBtn.addEventListener('click
1014     ', () => {
1015         const banner = document.getElementById('el-intel-banner'
1016         );
1017         if (banner) banner.classList.add('d-none');
1018     });
1019     const ragCloseBtn = document.getElementById('el-rag-close-
1020     btn');
1021     if (ragCloseBtn) ragCloseBtn.addEventListener('click', ()
1022     => {
1023         const card = document.getElementById('el-rag-results-car
1024         d');
1025         if (card) card.classList.add('d-none');
1026     });
1027     initTableResizers();
1028 }
1029
1030 // =====
1031 // ПАНЕЛЬ ЖИЗНЕННОГО ЦИКЛА ПИТАНИЯ (POWER LIFECYCLE & BOOT H
1032 ISTORY)

```

```

1026 // =====
1027 =====
1028 let allPowerSessions = [];
1029 let allPowerEvents = [];
1030 let powerUptimeSeconds = 0;
1031 let powerUptimeInterval = null;
1032 let isPowerInitialized = false;
1033
1034 function initPowerLifecyclePanel() {
1035     if (isPowerInitialized) return;
1036     isPowerInitialized = true;
1037
1038     const refreshBtn = document.getElementById('pwr-refresh-bt
n');
1039     if (refreshBtn) {
1040         refreshBtn.addEventListener('click', async () => {
1041             refreshBtn.disabled = true;
1042             refreshBtn.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span>Обновление...';
1043             try {
1044                 await loadPowerData();
1045                 if (window.toast) window.toast.success('Обновлено',
'Dанные сессий питания успешно обновлены');
1046             } finally {
1047                 refreshBtn.disabled = false;
1048                 refreshBtn.innerHTML = '<i class="bi bi-arrow-clockw
ise me-1"></i><span>Обновить данные</span>';
1049             }
1050         });
1051     }
1052
1053     const rebuildBtn = document.getElementById('pwr-rebuild-bt
n');
1054     if (rebuildBtn) {
1055         rebuildBtn.addEventListener('click', async () => {
1056             rebuildBtn.disabled = true;
1057             rebuildBtn.innerHTML = '<span class="spinner-border sp

```

```
inner-border-sm me-1"></span>Сканирование...';
1058     try {
1059         const res = await fetch('/api/v1/power/rebuild', { m
method: 'POST' });
1060         if (!res.ok) throw new Error(`HTTP ${res.status}`);
1061         const data = await res.json();
1062         renderPowerSummary(data);
1063         await loadPowerData();
1064         if (window.toast) window.toast.success('Event Log пе
ресканирован', `Реконструировано сессий: ${data.total_sessions
_count}`);
1065     } catch (err) {
1066         console.error('[PowerLifecycle] Ошибка пересканирова
ния:', err);
1067         if (window.toast) window.toast.error('Ошибка', 'Не у
далось пересканировать журналы: ' + err.message);
1068     } finally {
1069         rebuildBtn.disabled = false;
1070         rebuildBtn.innerHTML = '<i class="bi bi-arrow-repeat
me-1"></i><span>Пересканировать Event Log</span>';
1071     }
1072 });
1073 }
1074
1075 // Фильтры сессий
1076 const sessionsSearch = document.getElementById('pwr-sessio
ns-search');
1077 if (sessionsSearch) {
1078     sessionsSearch.addEventListener('input', () => filterAnd
RenderPowerSessions());
1079 }
1080
1081 const filterType = document.getElementById('pwr-filter-typ
e');
1082 if (filterType) {
1083     filterType.addEventListener('change', () => filterAndRen
derPowerSessions());
1084 }
```

```
1085 |
1086 |     const filterUnexpected = document.getElementById('pwr-filt
1087 | er-unexpected-only');
1088 |     if (filterUnexpected) {
1089 |         filterUnexpected.addEventListener('change', () => filter
1090 | AndRenderPowerSessions());
1091 |     }
1092 |
1093 |     // Фильтры событий
1094 |     const eventsSearch = document.getElementById('pwr-events-s
1095 | earch');
1096 |     if (eventsSearch) {
1097 |         eventsSearch.addEventListener('input', () => filterAndRe
1098 | nderPowerEvents());
1099 |     }
1100 |
1101 |     const filterEid = document.getElementById('pwr-event-id-fi
1102 | lter');
1103 |     if (filterEid) {
1104 |         filterEid.addEventListener('change', () => filterAndRend
1105 | erPowerEvents());
1106 |     }
1107 |
1108 |     startPowerUptimeCounter();
1109 |     initTableResizers();
1110 | }
1111 |
1112 | async function loadPowerData() {
1113 |     try {
1114 |         // 1. Сводка
1115 |         const sumRes = await fetch('/api/v1/power/summary');
1116 |         if (sumRes.ok) {
1117 |             const summary = await sumRes.json();
1118 |             renderPowerSummary(summary);
1119 |         }
1120 |
1121 |         // 2. Сессии
1122 |         const sessRes = await fetch('/api/v1/power/sessions?limi
```

```
t=100');
1117     if (sessRes.ok) {
1118         allPowerSessions = await sessRes.json();
1119         filterAndRenderPowerSessions();
1120     }
1121
1122     // 3. События
1123     const evRes = await fetch('/api/v1/power/events?limit=20
0');
1124     if (evRes.ok) {
1125         allPowerEvents = await evRes.json();
1126         filterAndRenderPowerEvents();
1127     }
1128 } catch (err) {
1129     console.error('[PowerLifecycle] Ошибка загрузки данных: '
, err);
1130 }
1131 }
1132
1133 function renderPowerSummary(summary) {
1134     if (!summary) return;
1135
1136     const curBootEl = document.getElementById('pwr-current-boo
t');
1137     if (curBootEl) curBootEl.textContent = `Загрузка: ${summar
y.current_boot_time || '--'}`;
1138
1139     const curUptimeEl = document.getElementById('pwr-current-u
ptime');
1140     if (curUptimeEl) {
1141         curUptimeEl.textContent = summary.current_uptime_human |
| '--';
1142         powerUptimeSeconds = summary.current_uptime_seconds || 0
;
1143     }
1144
1145     const totalSessionsEl = document.getElementById('pwr-total
-sessions');
```

```
1146     if (totalSessionsEl) totalSessionsEl.textContent = summary
      |.total_sessions_count;
1147
1148     const totalBadge = document.getElementById('pwr-total-sess
      |ions-badge');
1149     if (totalBadge) totalBadge.textContent = `${summary.total_
      |sessions_count} Сессий`;
1150
1151     const cleanCountEl = document.getElementById('pwr-clean-co
      |unt');
1152     if (cleanCountEl) cleanCountEl.textContent = summary.clean
      |_shutdowns_count;
1153
1154     const cleanRatioEl = document.getElementById('pwr-clean-ra
      |tio');
1155     if (cleanRatioEl) {
1156         const total = summary.total_sessions_count || 1;
1157         const ratio = Math.round((summary.clean_shutdowns_count
      | / total) * 100);
1158         cleanRatioEl.textContent = `${ratio} %`;
1159     }
1160
1161     const lastInitEl = document.getElementById('pwr-last-initi
      |ator');
1162     if (lastInitEl) {
1163         lastInitEl.textContent = `Инициатор: ${summary.last_init
      |iator || 'Штатный'}`;
1164     }
1165
1166     const unexpCountEl = document.getElementById('pwr-unexpect
      |ed-count');
1167     if (unexpCountEl) unexpCountEl.textContent = summary.unexp
      |ected_shutdowns_count;
1168
1169     const unexpBadge = document.getElementById('pwr-unexpected
      |-badge');
1170     if (unexpBadge) {
1171         unexpBadge.textContent = `${summary.unexpected_shutdowns
```

```

    _count} C60eB`;
1172     if (summary.unexpected_shutdowns_count > 0) {
1173         unexpBadge.className = 'badge bg-danger font-monospace
';
1174     } else {
1175         unexpBadge.className = 'badge bg-success font-monospac
e';
1176     }
1177 }
1178
1179     const bsodEl = document.getElementById('pwr-bsod-count');
1180     if (bsodEl) bsodEl.textContent = `BSOD (1001): ${summary.b
sod_count || 0}`;
1181 }
1182
1183     function filterAndRenderPowerSessions() {
1184         const tbody = document.getElementById('pwr-sessions-tbody'
);
1185         const counter = document.getElementById('pwr-sessions-coun
ter');
1186         if (!tbody) return;
1187
1188         const searchVal = (document.getElementById('pwr-sessions-s
earch')?.value || '').toLowerCase().trim();
1189         const typeFilter = document.getElementById('pwr-filter-typ
e')?.value || '';
1190         const unexpOnly = document.getElementById('pwr-filter-unex
pected-only')?.checked || false;
1191
1192         const filtered = allPowerSessions.filter(s => {
1193             if (unexpOnly && !s.unexpected_shutdown) return false;
1194             if (typeFilter) {
1195                 if (typeFilter === 'Active' && s.shutdown_type !== 'Ac
tive') return false;
1196                 if (typeFilter === 'Restart' && s.shutdown_type !== 'R
estart') return false;
1197                 if (typeFilter === 'Shutdown' && s.shutdown_type !== '
Shutdown' && s.shutdown_type !== 'PowerOff') return false;

```



```
1198 |         if (typeFilter === 'Unexpected' && !s.unexpected_shutdown) return false;
1199 |         if (typeFilter === 'BSOD' && s.shutdown_type !== 'BSOD' && !s.bugcheck) return false;
1200 |     }
1201 |
1202 |     if (searchVal) {
1203 |         const init = (s.initiator || '').toLowerCase();
1204 |         const proc = (s.process || '').toLowerCase();
1205 |         const reason = (s.reason || '').toLowerCase();
1206 |         const code = (s.reason_code || '').toLowerCase();
1207 |         const sId = (s.session_id || '').toLowerCase();
1208 |         if (!init.includes(searchVal) && !proc.includes(searchVal) && !reason.includes(searchVal) && !code.includes(searchVal) && !sId.includes(searchVal)) {
1209 |             return false;
1210 |         }
1211 |     }
1212 |     return true;
1213 | });
1214 |
1215 | if (counter) counter.textContent = `Найдено: ${filtered.length}`;
1216 |
1217 | if (filtered.length === 0) {
1218 |     tbody.innerHTML = `
1219 |         <tr>
1220 |             <td colspan="8" class="text-center py-4 text-muted">
1221 |                 Сессии питания, удовлетворяющие критериям фильтрации, не найдены.
1222 |             </td>
1223 |         </tr>
1224 |     `;
1225 |     return;
1226 | }
1227 |
1228 | tbody.innerHTML = filtered.map(s => {
1229 |     let statusBadge = '<span class="badge bg-secondary-subtl
```

```

    e text-secondary border border-secondary-subtle">Неизвестно</span>';
1230     if (s.shutdown_type === 'Active') {
1231         statusBadge = '<span class="badge bg-success-subtle text-success border border-success-subtle">⬢ Активная</span>';
1232     } else if (s.shutdown_type === 'Restart') {
1233         statusBadge = '<span class="badge bg-primary-subtle text-primary border border-primary-subtle">⬢ Перезагрузка</span>';
1234     } else if (s.shutdown_type === 'Shutdown' || s.shutdown_type === 'PowerOff') {
1235         statusBadge = '<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle">⬢ Выключение</span>';
1236     } else if (s.shutdown_type === 'BSOD' || s.bugcheck) {
1237         statusBadge = '<span class="badge bg-danger text-white">⬢ BSOD ${escapeHtml(s.bugcheck || '')}</span>';
1238     } else if (s.unexpected_shutdown) {
1239         statusBadge = '<span class="badge bg-danger-subtle text-danger border border-danger-subtle">⬢ Сбой / 41</span>';
1240     }
1241
1242     let initiatorHtml = '<span class="text-muted">--</span>';
1243
1244     if (s.initiator || s.process) {
1245         const initName = s.initiator ? '<span class="fw-semibold text-primary">${escapeHtml(s.initiator)}</span>' : '';
1246         const procName = s.process ? '<span class="badge pwr-proc-badge font-monospace ms-1">${escapeHtml(s.process)}</span>' : '';
1247         initiatorHtml = '<div class="d-flex align-items-center flex-wrap">${initName} ${procName}</div>';
1248         if (s.initiator_chain && s.initiator_chain.length > 1) {
1249             initiatorHtml += '<div class="small text-muted font-monospace mt-0.5" style="font-size: 0.72rem;">⬢ ${escapeHtml(s.initiator_chain.join(' → '))}</div>';
1250         }

```

```

1250     }
1251
1252     let reasonHtml = '<span class="text-muted">--</span>';
1253     if (s.reason || s.reason_code) {
1254         const codeBadge = s.reason_code ? `<span class="badge
bg-secondary-subtle text-secondary border border-secondary-sub
tle font-monospace me-1">${escapeHtml(s.reason_code)}</span>`
: '';
1255         reasonHtml = `<div>${codeBadge}<span>${escapeHtml(s.re
ason || '')}</span></div>`;
1256         if (s.comment) {
1257             reasonHtml += `<div class="small text-muted font-ita
lic mt-0.5">␣ ${escapeHtml(s.comment)}</div>`;
1258         }
1259     }
1260
1261     return `
1262         <tr>
1263             <td class="font-monospace text-primary small fw-bold
">${escapeHtml(s.session_id)}</td>
1264             <td class="font-monospace small">${escapeHtml(s.boot
_time)}</td>
1265             <td class="font-monospace small text-muted">${s.shut
down_time ? escapeHtml(s.shutdown_time) : '<span class="text-s
uccess fw-bold">Сейчас</span>'}</td>
1266             <td class="font-monospace fw-semibold">${escapeHtml(
s.uptime_human || '--')}</td>
1267             <td>${statusBadge}</td>
1268             <td>${initiatorHtml}</td>
1269             <td>${reasonHtml}</td>
1270             <td class="text-center">
1271                 <button class="btn btn-xs btn-outline-info p-1 px-
2 pwr-detail-btn" data-session-id="${escapeHtml(s.session_id)}
" title="Посмотреть подробную хронологию">
1272                     <i class="bi bi-eye"></i>
1273                 </button>
1274             </td>
1275         </tr>

```

```

1276     `;
1277     })).join('');
1278
1279     tbody.querySelectorAll('.pwr-detail-btn').forEach(btn => {
1280         btn.addEventListener('click', () => {
1281             const sId = btn.dataset.sessionId;
1282             showPowerSessionDetailModal(sId);
1283         });
1284     });
1285 }
1286
1287 function filterAndRenderPowerEvents() {
1288     const tbody = document.getElementById('pwr-events-tbody');
1289     const counter = document.getElementById('pwr-events-counte
1290 r');
1291     if (!tbody) return;
1292
1293     const searchVal = (document.getElementById('pwr-events-sea
1294 rch')?.value || '').toLowerCase().trim();
1295     const eidFilter = document.getElementById('pwr-event-id-fi
1296 lter')?.value || '';
1297
1298     const filtered = allPowerEvents.filter(e => {
1299         if (eidFilter && String(e.event_id) !== String(eidFilter
1300 )) return false;
1301         if (searchVal) {
1302             const prov = (e.provider || '').toLowerCase();
1303             const user = (e.user || '').toLowerCase();
1304             const proc = (e.process || '').toLowerCase();
1305             const reason = (e.reason || '').toLowerCase();
1306             const msg = (e.details?.message || '').toLowerCase();
1307             if (!prov.includes(searchVal) && !user.includes(search
1308 Val) && !proc.includes(searchVal) && !reason.includes(searchVa
1309 l) && !msg.includes(searchVal)) {
1310                 return false;
1311             }
1312         }
1313     });
1314     return true;

```

```
1308     });
1309
1310     if (counter) counter.textContent = `Событий: ${filtered.length}`;
1311
1312     if (filtered.length === 0) {
1313         tbody.innerHTML = `
1314             <tr>
1315                 <td colspan="7" class="text-center py-4 text-muted">
1316                     События журнала, удовлетворяющие критериям поиска,
1317                     не найдены.
1318                 </td>
1319             </tr>
1320         `;
1321         return;
1322     }
1323
1324     tbody.innerHTML = filtered.map(e => {
1325         let eidBadge = `<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle font-monospace">${e.event_id}</span>`;
1326
1327         if (e.event_id === 1074) {
1328             eidBadge = `<span class="badge bg-warning-subtle text-warning-emphasis border border-warning font-monospace fw-bold">1074 User32</span>`;
1329         } else if (e.event_id === 41) {
1330             eidBadge = `<span class="badge bg-danger-subtle text-danger border border-danger font-monospace fw-bold">41 Kernel-Power</span>`;
1331         } else if (e.event_id === 6008) {
1332             eidBadge = `<span class="badge bg-danger-subtle text-danger border border-danger font-monospace fw-bold">6008 EventLog</span>`;
1333         } else if (e.event_id === 12 || e.event_id === 6005 || e.event_id === 6009) {
1334             eidBadge = `<span class="badge bg-success-subtle text-success border border-success font-monospace">${e.event_id} Boot</span>`;
1335         }
1336     });
```

```

1333     } else if (e.event_id === 13 || e.event_id === 6006) {
1334         eidBadge = `<--</span>`;
1340     if (e.user || e.process) {
1341         userProc = `
```

```

1360         ` : '<span class="text-muted">-</span>'}
1361     </td>
1362 </tr>
1363 `;
1364 }).join('');
1365
1366 tbody.querySelectorAll('.pwr-xml-btn').forEach(btn => {
1367     btn.addEventListener('click', () => {
1368         const xml = btn.dataset.xml;
1369         showPowerXmlModal(xml);
1370     });
1371 });
1372 }
1373
1374 function showPowerSessionDetailModal(sessionId) {
1375     const session = allPowerSessions.find(s => s.session_id ==
= sessionId);
1376     if (!session) return;
1377
1378     const modalTitle = document.getElementById('pwr-modal-titl
e');
1379     const modalBody = document.getElementById('pwr-modal-body'
);
1380     if (modalTitle) modalTitle.textContent = `Сессия питания:
${session.session_id}`;
1381
1382     if (modalBody) {
1383         let eventsListHtml = '<div class="text-muted small">Нет
связанных событий</div>';
1384         if (session.events && session.events.length > 0) {
1385             eventsListHtml = session.events.map(ev => `
1386                 <div class="pwr-detail-card p-2 mb-2">
1387                     <div class="d-flex justify-content-between align-i
tems-center mb-1">
1388                         <span class="badge bg-primary-subtle text-primar
y border border-primary-subtle font-monospace">Event ${ev.even
t_id}</span>
1389                         <span class="font-monospace small text-muted">${

```

```

    escapeHtml(ev.timestamp)}</span>
1390         </div>
1391         <div class="small fw-semibold text-primary">${escape
    peHtml(ev.provider || ' ')}</div>
1392         <div class="small mt-1">${escapeHtml(ev.reason ||
    ev.details?.message || ' ')}</div>
1393         ${ev.process ? `<div class="small text-muted font-
    monospace mt-1">Процесс: ${escapeHtml(ev.process)} | Пользоват
    ель: ${escapeHtml(ev.user || '--')}</div>` : ''}
1394         </div>
1395         `).join('');
1396     }
1397
1398     modalBody.innerHTML = `
1399         <div class="row g-3 mb-3">
1400             <div class="col-6">
1401                 <div class="pwr-detail-card p-2.5">
1402                     <div class="text-muted small">⏱ Время старта (B
    oot):</div>
1403                     <div class="font-monospace fw-bold text-info mt-
    1">${escapeHtml(session.boot_time)}</div>
1404                 </div>
1405             </div>
1406             <div class="col-6">
1407                 <div class="pwr-detail-card p-2.5">
1408                     <div class="text-muted small">⏱ Завершение работ
    ы:</div>
1409                     <div class="font-monospace fw-bold ${session.shu
    tdown_time ? '' : 'text-success'} mt-1">
1410                         ${session.shutdown_time ? escapeHtml(session.s
    hutdown_time) : 'Сессия активна'}
1411                     </div>
1412                 </div>
1413             </div>
1414         </div>
1415
1416         <div class="pwr-detail-card p-3 mb-3">
1417             <h6 class="fw-bold text-warning mb-2"><i class="bi b

```



```

1418         <div class="row g-2 small">
1419             <div class="col-4 text-muted">Тип завершения:</div>
1420             <div class="col-8 fw-semibold">${escapeHtml(session.shutdown_type)}</div>
1421             <div class="col-4 text-muted">Аптайм сессии:</div>
1422             <div class="col-8 font-monospace fw-bold">${escapeHtml(session.uptime_human || '--')}</div>
1423             <div class="col-4 text-muted">Инициатор:</div>
1424             <div class="col-8">${escapeHtml(session.initiator || 'SYSTEM / Hardware')}</div>
1425             <div class="col-4 text-muted">Процесс:</div>
1426             <div class="col-8 font-monospace text-info">${escapeHtml(session.process || '--')}</div>
1427             <div class="col-4 text-muted">Код причины:</div>
1428             <div class="col-8 font-monospace text-secondary">${escapeHtml(session.reason_code || '--')}</div>
1429             <div class="col-4 text-muted">Текст причины:</div>
1430             <div class="col-8">${escapeHtml(session.reason || '--')}</div>
1431             ${session.comment ? `
1432             <div class="col-4 text-muted">Комментарий:</div>
1433             <div class="col-8 font-italic">${escapeHtml(session.comment)}</div>
1434             ` : ''}
1435             ${session.bugcheck ? `
1436             <div class="col-4 text-danger fw-bold">BSOD BugC
1437             heck:</div>
1438             <div class="col-8 text-danger font-monospace fw-
1439             bold">${escapeHtml(session.bugcheck)}</div>

```

```

1445 |         ` : ''}
1446 |     </div>
1447 | </div>
1448 |
1449 |     ${session.initiator_chain && session.initiator_chain.l
length > 0 ? `
1450 |         <div class="pwr-detail-card p-3 mb-3">
1451 |             <h6 class="fw-bold text-info mb-2"><i class="bi bi
-diagram-3 me-1.5"></i>Корреляция цепочки вызова</h6>
1452 |             <div class="d-flex align-items-center flex-wrap ga
p-1 font-monospace small">
1453 |                 ${session.initiator_chain.map((step, idx) => `
1454 |                     <span class="badge bg-secondary-subtle text-se
condary border border-secondary-subtle p-2">${escapeHtml(step)
}</span>
1455 |                     ${idx < session.initiator_chain.length - 1 ? '
<i class="bi bi-arrow-right text-muted"></i>' : ''}
1456 |                     `).join('')}
1457 |                 </div>
1458 |             </div>
1459 |         ` : ''}
1460 |
1461 |     <div class="pwr-detail-card p-3">
1462 |         <h6 class="fw-bold mb-2"><i class="bi bi-list-nested
me-1.5"></i>Хронология событий в рамках сессии</h6>
1463 |         <div style="max-height: 250px; overflow-y: auto;">
1464 |             ${eventsListHtml}
1465 |         </div>
1466 |     </div>
1467 | `;
1468 |
1469 |     const modalEl = document.getElementById('pwrSessionDetailModal');
1470 |     if (modalEl && window.bootstrap) {
1471 |         const modal = new window.bootstrap.Modal(modalEl);
1472 |         modal.show();
1473 |     }
1474 | }

```

```
1475 | }
1476 |
1477 | function showPowerXmlModal(xmlString) {
1478 |     const pre = document.getElementById('pwr-xml-content');
1479 |     if (pre) pre.textContent = xmlString;
1480 |
1481 |     const modalEl = document.getElementById('pwrXmlDetailModal
1482 | ');
1483 |     if (modalEl && window.bootstrap) {
1484 |         const modal = new window.bootstrap.Modal(modalEl);
1485 |         modal.show();
1486 |     }
1487 |
1488 | function startPowerUptimeCounter() {
1489 |     if (powerUptimeInterval) clearInterval(powerUptimeInterval
1490 | );
1491 |     powerUptimeInterval = setInterval(() => {
1492 |         powerUptimeSeconds += 1;
1493 |         const curUptimeEl = document.getElementById('pwr-current
1494 | -uptime');
1495 |         if (curUptimeEl && powerUptimeSeconds > 0) {
1496 |             const days = Math.floor(powerUptimeSeconds / 86400);
1497 |             const hours = Math.floor((powerUptimeSeconds % 86400)
1498 | / 3600);
1499 |             const minutes = Math.floor((powerUptimeSeconds % 3600)
1500 | / 60);
1501 |             const sec = Math.floor(powerUptimeSeconds % 60);
1502 |             const parts = [];
1503 |             if (days > 0) parts.push(`${days}д`);
1504 |             if (hours > 0 || days > 0) parts.push(`${hours}ч`);
1505 |             if (minutes > 0 || hours > 0 || days > 0) parts.push(`
1506 | ${minutes}м`);
1507 |             parts.push(`${sec}с`);
1508 |             curUptimeEl.textContent = parts.join(' ');
1509 |         }
1510 |     }, 1000);
1511 | }
```

```

1507
1508  /**
1509   * Инициализация ползунков изменения высоты таблиц журнала и
      событий.
1510   */
1511   function initTableResizers() {
1512       function setupResizer(resizerId, containerId, defaultMaxHeight = 520, minH = 140, maxH = 1500) {
1513           const resizer = document.getElementById(resizerId);
1514           const container = document.getElementById(containerId);
1515           if (!resizer || !container || resizer._resizerBound) return;
1516           resizer._resizerBound = true;
1517
1518           let startY = 0, startHeight = 0;
1519           const onMouseMove = (e) => {
1520               const delta = e.clientY - startY;
1521               const newH = Math.max(minH, Math.min(maxH, startHeight + delta));
1522               container.style.maxHeight = 'none';
1523               container.style.height = `${newH}px`;
1524           };
1525           const onMouseUp = () => {
1526               resizer.classList.remove('resizing');
1527               document.removeEventListener('mousemove', onMouseMove);
1528               document.removeEventListener('mouseup', onMouseUp);
1529           };
1530           resizer.onmousedown = (e) => {
1531               e.preventDefault();
1532               startY = e.clientY;
1533               startHeight = container.getBoundingClientRect().height || parseInt(window.getComputedStyle(container).height, 10) || defaultMaxHeight;
1534               resizer.classList.add('resizing');
1535               document.addEventListener('mousemove', onMouseMove);
1536               document.addEventListener('mouseup', onMouseUp);
1537           };

```

```
1538     resizer.ondblclick = () => {
1539         container.style.height = '';
1540         container.style.maxHeight = `${defaultMaxHeight}px`;
1541     };
1542 }
1543
1544     setupResizer('el-table-resizer', 'el-table-container', 520
1545     , 140, 1500);
1546     setupResizer('slc-table-resizer', 'slc-table-container', 5
1547     60, 140, 1500);
1548     setupResizer('scc-activity-table-resizer', 'scc-activity-t
1549     able-container', 480, 140, 1500);
1550     setupResizer('pwr-sessions-table-resizer', 'pwr-sessions-t
1551     able-container', 520, 140, 1500);
1552     setupResizer('pwr-events-table-resizer', 'pwr-events-table
1553     -container', 520, 140, 1500);
1554 }
1555
1556 function setMode(mode) {
1557     currentMode = mode;
1558     const tableContainer = document.getElementById('slc-table-
1559     container');
1560     const auditContainer = document.getElementById('slc-audit-
1561     container');
1562     const ragContainer = document.getElementById('slc-rag-cont
1563     ainer');
1564     const incidentsContainer = document.getElementById('slc-in
1565     cidents-container');
1566     const timelineContainer = document.getElementById('slc-tim
1567     eline-container');
1568     const rawContainer = document.getElementById('slc-raw-cont
1569     ainer');
1570     const activityContainer = document.getElementById('slc-act
1571     ivity-container');
1572     const eventLogsContainer = document.getElementById('slc-ev
1573     ent-logs-container');
1574     const powerLifecycleContainer = document.getElementById('s
1575     lc-power-lifecycle-container');
```

```
1562     const filterRow = document.getElementById('slc-filter-row'
    );
1563     const sidebar = document.getElementById('slc-channel-tree-
    sidebar');
1564     const mainStreamCol = document.getElementById('slc-main-st
    ream-column');
1565     const mainStreamHeader = document.querySelector('#slc-main
    -stream-column .slc-panel-header');
1566
1567     if (tableContainer) tableContainer.classList.add('d-none')
    ;
1568     if (auditContainer) auditContainer.classList.add('d-none')
    ;
1569     if (ragContainer) ragContainer.classList.add('d-none');
1570     if (incidentsContainer) incidentsContainer.classList.add('
    d-none');
1571     if (timelineContainer) timelineContainer.classList.add('d-
    none');
1572     if (rawContainer) rawContainer.classList.add('d-none');
1573     if (activityContainer) activityContainer.classList.add('d-
    none');
1574     if (eventLogsContainer) eventLogsContainer.classList.add('
    d-none');
1575     if (powerLifecycleContainer) powerLifecycleContainer.class
    List.add('d-none');
1576     if (filterRow) filterRow.classList.remove('d-none');
1577
1578     // Управление шириной колонок и сайдбаром каналов
1579     if (mode === 'event-logs' || mode === 'power-lifecycle') {
1580         if (sidebar) sidebar.classList.add('d-none');
1581         if (mainStreamCol) {
1582             mainStreamCol.classList.remove('col-lg-9');
1583             mainStreamCol.classList.add('col-12');
1584         }
1585         if (mainStreamHeader) mainStreamHeader.classList.add('d-
    none');
1586     } else {
1587         if (sidebar) sidebar.classList.remove('d-none');
```

```

1588         if (mainStreamCol) {
1589             mainStreamCol.classList.remove('col-12');
1590             mainStreamCol.classList.add('col-lg-9');
1591         }
1592         if (mainStreamHeader) mainStreamHeader.classList.remove(
1593             'd-none');
1594     }
1595     document.querySelectorAll('#slc-mode-tabs button').forEach
1596     (btn => {
1597         btn.classList.toggle('active', btn.getAttribute('data-mo-
1598         de') === mode);
1599     });
1600     if (mode === 'live' || mode === 'errors') {
1601         if (tableContainer) tableContainer.classList.remove('d-n-
1602         one');
1603         loadEvents();
1604     } else if (mode === 'audit') {
1605         if (auditContainer) auditContainer.classList.remove('d-n-
1606         one');
1607         loadAuditDigest();
1608     } else if (mode === 'rag') {
1609         if (ragContainer) ragContainer.classList.remove('d-none'
1610         );
1611         if (filterRow) filterRow.classList.add('d-none');
1612     } else if (mode === 'incidents') {
1613         if (incidentsContainer) incidentsContainer.classList.rem-
1614         ove('d-none');
1615         loadIncidents();
1616     } else if (mode === 'timeline') {
1617         if (timelineContainer) timelineContainer.classList.remov-
1618         e('d-none');
1619         loadTimeline();
1620     } else if (mode === 'raw') {
1621         if (rawContainer) rawContainer.classList.remove('d-none'
1622         );
1623         if (tableContainer) tableContainer.classList.remove('d-n

```

```

one');
1617     } else if (mode === 'activity') {
1618         if (activityContainer) activityContainer.classList.remove('d-none');
1619         if (filterRow) filterRow.classList.add('d-none');
1620         loadSccActivityLogs();
1621     } else if (mode === 'event-logs') {
1622         if (eventLogsContainer) eventLogsContainer.classList.remove('d-none');
1623         if (filterRow) filterRow.classList.add('d-none');
1624         initEventLogsPanel();
1625         fetchEventLogsSummary();
1626     } else if (mode === 'power-lifecycle') {
1627         if (powerLifecycleContainer) powerLifecycleContainer.classList.remove('d-none');
1628         if (filterRow) filterRow.classList.add('d-none');
1629         initPowerLifecyclePanel();
1630         loadPowerData();
1631     }
1632 }
1633
1634 /**
1635  * Экспорт текущих событий.
1636  */
1637 function exportEvents(format = 'json') {
1638     if (!cachedEvents || cachedEvents.length === 0) {
1639         if (window.toast) {
1640             window.toast.warning(
1641                 t('systemControl.btnExport', 'Экспорт'),
1642                 t('systemControl.exportNoEvents', 'Нет загруженных событий для экспорта.')
1643             );
1644         }
1645         return;
1646     }
1647     let content = '';
1648     let mime = 'application/json';
1649     let ext = 'json';

```



```

1650
1651     if (format === 'json') {
1652         content = JSON.stringify(cachedEvents, null, 2);
1653     } else {
1654         mime = 'text/csv';
1655         ext = 'csv';
1656         const headers = ['timestamp', 'level', 'event_id', 'prov
ider', 'process_id', 'message'];
1657         const rows = cachedEvents.map(e => headers.map(h => `"$${
String(e[h] || '').replace(/"/g, '"')}"`).join(','));
1658         content = [headers.join(','), ...rows].join('\n');
1659     }
1660
1661     const cleanChan = currentChannel.replace(/^evt:/i, '').rep
lace(/[a-zA-Z0-9_-]/g, '_');
1662     const blob = new Blob([content], { type: mime });
1663     const url = URL.createObjectURL(blob);
1664     const a = document.createElement('a');
1665     a.href = url;
1666     a.download = `system_logs_${cleanChan}_${Date.now()}.${ext}
`;
1667     a.click();
1668     URL.revokeObjectURL(url);
1669 }
1670
1671 /**
1672  * Инициализация вкладки System Logs & Activity Center.
1673  */
1674 function initSystemControlTab() {
1675     window.applyTranslations?.();
1676     if (window.isTabActive ? window.isTabActive('tab-system-co
ntrol') : true) {
1677         fetchStatus();
1678         scanChannels();
1679         loadEvents();
1680     }
1681
1682     if (window.registerTabPoller) {

```

```
1683     window.registerTabPoller('tab-system-control', loadEvent
1684     s, 4000, { pollerId: 'tab-system-control_logs', immediate: tru
1685     e });
1686     }
1687     // Слушатель глобального изменения языка в приложении
1688     window.addEventListener('languageChanged', () => {
1689         window.applyTranslations?();
1690         fetchStatus();
1691         if (cachedChannels.length > 0) {
1692             renderChannelTree(cachedChannels);
1693         }
1694         if (cachedEvents.length > 0) {
1695             renderEventsTable(cachedEvents);
1696         }
1697         if (currentMode === 'audit') loadAuditDigest();
1698         else if (currentMode === 'incidents') loadIncidents();
1699         else if (currentMode === 'timeline') loadTimeline();
1700         else if (currentMode === 'activity') loadSccActivityLogs
1701         ();
1702         else if (currentMode === 'event-logs') fetchEventLogsSum
1703         mary();
1704         else if (currentMode === 'power-lifecycle') loadPowerDat
1705         a();
1706     });
1707     initEventLogsPanel();
1708     initTableResizers();
1709     if (isSCCInitialized) return;
1710     isSCCInitialized = true;
1711     // Главная кнопка обновления шапки
1712     const refreshBtn = document.getElementById('btn-scc-refres
1713     h');
1714     if (refreshBtn) {
1715         refreshBtn.onclick = () => {
1716             fetchStatus();
1717         }
1718     }
```

```
1715 |         if (currentMode === 'event-logs') {
1716 |             fetchEventLogsSummary();
1717 |         } else if (currentMode === 'power-lifecycle') {
1718 |             loadPowerData();
1719 |         } else {
1720 |             loadEvents();
1721 |         }
1722 |     };
1723 | }
1724 |
1725 | // Кнопка обновления потока логов
1726 | const slcRefreshBtn = document.getElementById('btn-slc-ref
resh');
1727 | if (slcRefreshBtn) {
1728 |     slcRefreshBtn.onclick = () => {
1729 |         if (currentMode === 'event-logs') {
1730 |             fetchEventLogsSummary();
1731 |         } else if (currentMode === 'power-lifecycle') {
1732 |             loadPowerData();
1733 |         } else {
1734 |             loadEvents();
1735 |         }
1736 |     };
1737 | }
1738 |
1739 | // Кнопка сканирования источников
1740 | const slcScanBtn = document.getElementById('btn-slc-scan')
;
1741 | if (slcScanBtn) {
1742 |     slcScanBtn.onclick = () => {
1743 |         scanChannels(true);
1744 |     };
1745 | }
1746 |
1747 | // Кнопка быстрого открытия встроенной панели журналов соб
ытий
1748 | const openEventLogsBtn = document.getElementById('btn-scc-
open-event-logs');
```

```
1749     if (openEventLogsBtn) {
1750         openEventLogsBtn.onclick = () => {
1751             setMode('event-logs');
1752         };
1753     }
1754
1755     // Кнопка быстрого открытия встроенной панели жизненного ц
    икла питания
1756     const openPowerLifecycleBtn = document.getElementById('btn
    -scc-open-power-lifecycle');
1757     if (openPowerLifecycleBtn) {
1758         openPowerLifecycleBtn.onclick = () => {
1759             setMode('power-lifecycle');
1760         };
1761     }
1762
1763     // Кнопка аудита
1764     const auditBtn = document.getElementById('btn-slc-audit');
1765     if (auditBtn) {
1766         auditBtn.onclick = () => {
1767             setMode('audit');
1768         };
1769     }
1770
1771     // Авто-активация режима по хэшу/параметру
1772     if (location.hash.includes('power-lifecycle') || location.
    search.includes('power-lifecycle') || location.search.includes
    ('panel=power-lifecycle')) {
1773         setMode('power-lifecycle');
1774     } else if (location.hash.includes('event-logs') || locatio
    n.search.includes('event-logs') || location.search.includes('p
    anel=event-logs')) {
1775         setMode('event-logs');
1776     }
1777
1778     // Фильтр каналов в дереве
1779     const channelFilterInput = document.getElementById('slc-ch
    annel-filter-input');
```

```
1780     if (channelFilterInput) {
1781         channelFilterInput.oninput = (e) => {
1782             const query = e.target.value.toLowerCase();
1783             const filtered = cachedChannels.filter(c => {
1784                 const name = (c.channel_name || c.display_name || c.
name || '').toLowerCase();
1785                 const desc = (c.description || '').toLowerCase();
1786                 return name.includes(query) || desc.includes(query);
1787             });
1788             renderChannelTree(filtered);
1789         };
1790     }
1791
1792     // Переключение режимов работы (Live, Audit, RAG, Incident
s, Timeline, Raw, Activity, Event Logs, Power Lifecycle)
1793     document.querySelectorAll('#slc-mode-tabs button').forEach
(btn => {
1794         btn.onclick = () => {
1795             const mode = btn.getAttribute('data-mode') || 'live';
1796             setMode(mode);
1797         };
1798     });
1799
1800     // Фильтры потока
1801     const searchInput = document.getElementById('slc-search-in
put');
1802     if (searchInput) {
1803         let debounceTimer = null;
1804         searchInput.oninput = () => {
1805             clearTimeout(debounceTimer);
1806             debounceTimer = setTimeout(loadEvents, 300);
1807         };
1808     }
1809
1810     const levelSelect = document.getElementById('slc-level-sel
ect');
1811     if (levelSelect) levelSelect.onchange = loadEvents;
1812
```

```
1813     const hoursSelect = document.getElementById('slc-hours-select');
1814     if (hoursSelect) hoursSelect.onchange = loadEvents;
1815
1816     const limitSelect = document.getElementById('slc-limit-select');
1817     if (limitSelect) limitSelect.onchange = loadEvents;
1818
1819     const eventIdInput = document.getElementById('slc-eventid-input');
1820     if (eventIdInput) {
1821         let debounceTimer = null;
1822         eventIdInput.oninput = () => {
1823             clearTimeout(debounceTimer);
1824             debounceTimer = setTimeout(loadEvents, 300);
1825         };
1826     }
1827
1828     // Live Auto-refresh switch
1829     const liveSwitch = document.getElementById('slc-live-switch');
1830     if (liveSwitch) {
1831         liveSwitch.onchange = (e) => {
1832             if (e.target.checked) {
1833                 if (window.registerTabPoller) {
1834                     window.registerTabPoller('tab-system-control', loadEvents, 3000, { pollerId: 'tab-system-control_live', immediate: true });
1835                 } else {
1836                     if (liveIntervalTimer) clearInterval(liveIntervalTimer);
1837                     liveIntervalTimer = setInterval(loadEvents, 3000);
1838                 }
1839             } else {
1840                 if (window.unregisterTabPoller) {
1841                     window.unregisterTabPoller('tab-system-control_live');
1842                 }
1843             }
1844         };
1845     }
```

```
1843         if (liveIntervalTimer) {
1844             clearInterval(liveIntervalTimer);
1845             liveIntervalTimer = null;
1846         }
1847     }
1848 };
1849 }
1850
1851 // Экспорт событий
1852 const exportJson = document.getElementById('btn-slc-export
1853 -json');
1854 if (exportJson) {
1855     exportJson.onclick = (e) => {
1856         e.preventDefault();
1857         exportEvents('json');
1858     };
1859 }
1860
1861 const exportCsv = document.getElementById('btn-slc-export-
1862 csv');
1863 if (exportCsv) {
1864     exportCsv.onclick = (e) => {
1865         e.preventDefault();
1866         exportEvents('csv');
1867     };
1868 }
1869
1870 // RAG Search actions
1871 const ragSearchBtn = document.getElementById('btn-slc-rag-
1872 search');
1873 if (ragSearchBtn) ragSearchBtn.onclick = runRagSearch;
1874
1875 const ragInput = document.getElementById('slc-rag-input');
1876 if (ragInput) {
1877     ragInput.onkeydown = (e) => {
1878         if (e.key === 'Enter') runRagSearch();
1879     };
1880 }
```

```
1878
1879     const ragRebuildBtn = document.getElementById('btn-slc-rag
-rebuild');
1880     if (ragRebuildBtn) ragRebuildBtn.onclick = rebuildRagIndex
;
1881
1882     // AI Ask Action
1883     const askAiBtn = document.getElementById('btn-slc-ask-ai')
;
1884     if (askAiBtn) {
1885         askAiBtn.onclick = () => {
1886             const topEvents = cachedEvents.slice(0, 10).map(e => `[${e.level}] ${e.provider}: ${e.message}`).join('\n');
1887             const cleanChan = currentChannel.replace(/^evt:/i, '')
;
1888             const prompt = `${t('systemControl.headerTitle', 'Wind
ows System Logs')} (${cleanChan}):\n${topEvents}`;
1889             if (window.sendChatMessage) {
1890                 window.sendChatMessage(prompt);
1891             } else if (window.toast) {
1892                 window.toast.info(
1893                     t('systemControl.aiToastTitle', 'ИИ Анализ'),
1894                     t('systemControl.aiToastBody', 'Запрос направлен в
чат-ассистент.')
1895                 );
1896             }
1897         };
1898     }
1899
1900     // Application activity refresh
1901     const refreshActivityBtn = document.getElementById('btn-sc
c-refresh-logs');
1902     if (refreshActivityBtn) refreshActivityBtn.onclick = loadS
ccActivityLogs;
1903 }
1904
1905 // Экспорт для динамического загрузчика
1906 window.initSystemControlTab = initSystemControlTab;
```



```
1907 |
1908 | // Автозапуск при готовности DOM
1909 | if (document.readyState === 'loading') {
1910 |     document.addEventListener('DOMContentLoaded', initSystemCo
1911 | ntrolTab);
1912 | } else {
1913 |     initSystemControlTab();
1914 | }
1915 | }());
```

system_inspector.html

```
1 | <!--
2 | =====
3 | Process Name: Windows Web Interface - System Inspector View
4 | =====
5 | Description:
6 |   Шаблон клиентского веб-интерфейса system_inspector.
7 |
8 | Usage Examples:
9 |   Web Route:
10 |     /windows/api/webgui/system_inspector.html
11 |
12 |   Tab Frame Integration:
13 |     <iframe src="/windows/api/webgui/system_inspector.html"
14 |     class="w-100 h-100 border-0"></iframe>
15 | File: system_inspector.html
16 | Project: ai-breadboard
17 | Package: windows/api/webgui
18 | Author: hypo69
19 | Copyright: © 2026 hypo69
20 | Updated: 2026-10-01 13:04:40
21 | =====
22 | -->
23 |
24 | <!DOCTYPE html>
25 | <html lang="en">
26 | <head>
27 |   <meta charset="UTF-8">
28 |   <meta name="viewport" content="width=device-width, initial
29 |   -scale=1.0">
30 |   <title>AI-Breadboard | System & Hardware Inspector</title>
31 |   <style>
32 |     :root {
```

```
32 |         --bg-base: #0f172a;
33 |         --bg-card: #1e293b;
34 |         --bg-hover: #334155;
35 |         --text-main: #f8fafc;
36 |         --text-muted: #94a3b8;
37 |         --accent-cyan: #06b6d4;
38 |         --accent-green: #10b981;
39 |         --accent-yellow: #f59e0b;
40 |         --accent-red: #ef4444;
41 |         --accent-purple: #a855f7;
42 |         --border-color: #334155;
43 |     }
44 |
45 |     * {
46 |         box-sizing: border-box;
47 |         margin: 0;
48 |         padding: 0;
49 |         font-family: -apple-system, BlinkMacSystemFont, "S
egoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
50 |     }
51 |
52 |     body {
53 |         background-color: var(--bg-base);
54 |         color: var(--text-main);
55 |         height: 100vh;
56 |         display: flex;
57 |         flex-direction: column;
58 |         overflow: hidden;
59 |     }
60 |
61 |     header {
62 |         background-color: var(--bg-card);
63 |         border-bottom: 1px solid var(--border-color);
64 |         padding: 12px 24px;
65 |         display: flex;
66 |         align-items: center;
67 |         justify-content: space-between;
68 |     }
```

```
69 |
70 |     .header-title {
71 |         display: flex;
72 |         align-items: center;
73 |         gap: 12px;
74 |         font-size: 1.15rem;
75 |         font-weight: 700;
76 |         color: var(--accent-cyan);
77 |     }
78 |
79 |     .status-badge {
80 |         font-size: 0.75rem;
81 |         padding: 4px 8px;
82 |         border-radius: 9999px;
83 |         background: rgba(16, 185, 129, 0.2);
84 |         color: var(--accent-green);
85 |         border: 1px solid var(--accent-green);
86 |         display: flex;
87 |         align-items: center;
88 |         gap: 6px;
89 |     }
90 |
91 |     .pulse-dot {
92 |         width: 8px;
93 |         height: 8px;
94 |         border-radius: 50%;
95 |         background: var(--accent-green);
96 |         animation: pulse 1.5s infinite;
97 |     }
98 |
99 |     @keyframes pulse {
100 |         0% { opacity: 1; transform: scale(1); }
101 |         50% { opacity: 0.4; transform: scale(0.8); }
102 |         100% { opacity: 1; transform: scale(1); }
103 |     }
104 |
105 |     /* Top Metric Cards */
106 |     .metric-cards {
```

```
107 |         display: grid;
108 |         grid-template-columns: repeat(auto-fit, minmax(200
    | px, 1fr));
109 |         gap: 12px;
110 |         padding: 12px 24px;
111 |         background-color: #0b1120;
112 |         border-bottom: 1px solid var(--border-color);
113 |     }
114 |
115 |     .card {
116 |         background: var(--bg-card);
117 |         border: 1px solid var(--border-color);
118 |         border-radius: 8px;
119 |         padding: 12px 16px;
120 |         display: flex;
121 |         flex-direction: column;
122 |         gap: 6px;
123 |     }
124 |
125 |     .card-label {
126 |         font-size: 0.75rem;
127 |         text-transform: uppercase;
128 |         letter-spacing: 0.05em;
129 |         color: var(--text-muted);
130 |     }
131 |
132 |     .card-value {
133 |         font-size: 1.4rem;
134 |         font-weight: 700;
135 |         color: var(--text-main);
136 |     }
137 |
138 |     .card-sub {
139 |         font-size: 0.75rem;
140 |         color: var(--text-muted);
141 |     }
142 |
143 |     .progress-bar {
```

```
144         height: 4px;
145         background: #334155;
146         border-radius: 2px;
147         overflow: hidden;
148         margin-top: 4px;
149     }
150
151     .progress-fill {
152         height: 100%;
153         background: var(--accent-cyan);
154         transition: width 0.3s ease;
155     }
156
157     /* Main Workspace Layout */
158     .main-workspace {
159         display: grid;
160         grid-template-columns: 380px 1fr;
161         flex: 1;
162         overflow: hidden;
163     }
164
165     /* Left Side: AIDA64 Hardware Tree */
166     .left-panel {
167         background: #131d31;
168         border-right: 1px solid var(--border-color);
169         display: flex;
170         flex-direction: column;
171         overflow: hidden;
172     }
173
174     .panel-header {
175         padding: 12px 16px;
176         font-size: 0.85rem;
177         font-weight: 600;
178         text-transform: uppercase;
179         letter-spacing: 0.05em;
180         color: var(--accent-cyan);
181         border-bottom: 1px solid var(--border-color);
```

```
182         background: rgba(0, 0, 0, 0.2);
183         display: flex;
184         justify-content: space-between;
185         align-items: center;
186     }
187
188     .tree-container {
189         flex: 1;
190         overflow-y: auto;
191         padding: 12px;
192     }
193
194     .tree-node {
195         margin-bottom: 10px;
196         background: var(--bg-card);
197         border: 1px solid var(--border-color);
198         border-radius: 6px;
199         overflow: hidden;
200     }
201
202     .tree-node-title {
203         padding: 8px 12px;
204         font-size: 0.85rem;
205         font-weight: 600;
206         cursor: pointer;
207         background: rgba(255, 255, 255, 0.03);
208         display: flex;
209         align-items: center;
210         justify-content: space-between;
211         color: #e2e8f0;
212     }
213
214     .tree-node-title:hover {
215         background: rgba(255, 255, 255, 0.08);
216     }
217
218     .tree-node-body {
219         padding: 8px 12px;
```

```
220         font-size: 0.8rem;
221         border-top: 1px solid var(--border-color);
222         background: #0f172a;
223     }
224
225     .prop-row {
226         display: flex;
227         justify-content: space-between;
228         margin-bottom: 4px;
229     }
230
231     .prop-key {
232         color: var(--text-muted);
233     }
234
235     .prop-val {
236         font-weight: 500;
237         color: #f1f5f9;
238     }
239
240     /* Right Side: Wireshark Process Stream + AI Copilot *
241 /
242     .right-panel {
243         display: flex;
244         flex-direction: column;
245         overflow: hidden;
246     }
247
248     .stream-toolbar {
249         padding: 10px 16px;
250         background: var(--bg-card);
251         border-bottom: 1px solid var(--border-color);
252         display: flex;
253         align-items: center;
254         gap: 12px;
255     }
256
257     .search-box {
```



```
257 |         flex: 1;
258 |         background: #0f172a;
259 |         border: 1px solid var(--border-color);
260 |         border-radius: 4px;
261 |         padding: 6px 12px;
262 |         color: var(--text-main);
263 |         font-size: 0.85rem;
264 |         outline: none;
265 |     }
266 |
267 |     .btn {
268 |         padding: 6px 14px;
269 |         border-radius: 4px;
270 |         font-size: 0.8rem;
271 |         font-weight: 600;
272 |         cursor: pointer;
273 |         border: none;
274 |         transition: all 0.2s;
275 |     }
276 |
277 |     .btn-primary {
278 |         background: var(--accent-cyan);
279 |         color: #0f172a;
280 |     }
281 |
282 |     .btn-primary:hover {
283 |         opacity: 0.9;
284 |     }
285 |
286 |     .btn-ai {
287 |         background: linear-gradient(135deg, #a855f7, #6366
f1);
288 |         color: #ffffff;
289 |     }
290 |
291 |     .table-container {
292 |         flex: 1;
293 |         overflow-y: auto;
```

```
294 |         position: relative;
295 |     }
296 |
297 |     table {
298 |         width: 100%;
299 |         border-collapse: collapse;
300 |         font-size: 0.8rem;
301 |         text-align: left;
302 |     }
303 |
304 |     th {
305 |         background: #1e293b;
306 |         position: sticky;
307 |         top: 0;
308 |         padding: 8px 12px;
309 |         color: var(--text-muted);
310 |         font-weight: 600;
311 |         border-bottom: 1px solid var(--border-color);
312 |         z-index: 10;
313 |     }
314 |
315 |     td {
316 |         padding: 8px 12px;
317 |         border-bottom: 1px solid rgba(51, 65, 85, 0.4);
318 |         font-family: ui-monospace, SFMono-Regular, Menlo,
319 | Monaco, Consolas, monospace;
320 |     }
321 |
322 |     tr:hover {
323 |         background: rgba(255, 255, 255, 0.04);
324 |     }
325 |
326 |     .badge-cpu-high {
327 |         color: var(--accent-red);
328 |         font-weight: 700;
329 |     }
330 |
331 |     .badge-cpu-med {
```

```
331         color: var(--accent-yellow);
332         font-weight: 600;
333     }
334
335     /* Bottom AI Copilot Drawer */
336     .ai-drawer {
337         background: #111827;
338         border-top: 1px solid var(--border-color);
339         padding: 12px 20px;
340         display: flex;
341         flex-direction: column;
342         gap: 8px;
343         max-height: 180px;
344         overflow-y: auto;
345     }
346
347     .ai-header {
348         display: flex;
349         align-items: center;
350         justify-content: space-between;
351     }
352
353     .ai-title {
354         font-size: 0.85rem;
355         font-weight: 700;
356         color: var(--accent-purple);
357         display: flex;
358         align-items: center;
359         gap: 8px;
360     }
361
362     .ai-content {
363         font-size: 0.82rem;
364         color: #cbd5e1;
365         line-height: 1.4;
366     }
367
368     .anomaly-pill {
```

```

369         display: inline-block;
370         font-size: 0.75rem;
371         padding: 2px 8px;
372         border-radius: 4px;
373         background: rgba(239, 68, 68, 0.2);
374         color: #f87171;
375         border: 1px solid rgba(239, 68, 68, 0.4);
376         margin-right: 6px;
377         margin-top: 4px;
378     }
379 </style>
380 </head>
381 <body>
382     <header>
383         <div class="header-title">
384             <span>📡 AI-BREADBOARD</span>
385             <span style="color: var(--text-muted); font-weight
: 400;">|</span>
386             <span>System & Hardware Inspector</span>
387         </div>
388         <div style="display: flex; align-items: center; gap: 1
6px;">
389             <div id="conn-status" class="status-badge">
390                 <span class="pulse-dot"></span>
391                 <span>TELEMETRY STREAMING</span>
392             </div>
393         </div>
394     </header>
395
396     <!-- Top Telemetry Metrics -->
397     <div class="metric-cards">
398         <div class="card">
399             <span class="card-label">CPU Load</span>
400             <span id="metric-cpu-val" class="card-value">0.0%<
/span>
401             <div class="progress-bar"><div id="metric-cpu-fill
" class="progress-fill" style="width: 0%"></div></div>
402             <span id="metric-cpu-sub" class="card-sub">-- Core

```

```

    s</span>
403         </div>
404         <div class="card">
405             <span class="card-label">Physical RAM</span>
406             <span id="metric-ram-val" class="card-value">0.0 G
    B</span>
407             <div class="progress-bar"><div id="metric-ram-fill
    " class="progress-fill" style="width: 0%; background: var(--ac
    cent-green);"></div></div>
408             <span id="metric-ram-sub" class="card-sub">0% Used
    </span>
409         </div>
410         <div class="card">
411             <span class="card-label">GPU Accelerator</span>
412             <span id="metric-gpu-val" class="card-value">--</s
    pan>
413             <span id="metric-gpu-sub" class="card-sub">VRAM: -
    - / -- GB</span>
414         </div>
415         <div class="card">
416             <span class="card-label">Disk Throughput</span>
417             <span id="metric-disk-val" class="card-value">0.0
    MB/s</span>
418             <span id="metric-disk-sub" class="card-sub">R: 0 K
    B/s | W: 0 KB/s</span>
419         </div>
420     </div>
421
422     <!-- Main Workspace -->
423     <div class="main-workspace">
424         <!-- Left Pane: Hardware Tree -->
425         <div class="left-panel">
426             <div class="panel-header">
427                 <span>📁 Hardware Specification</span>
428                 <button class="btn btn-primary" style="font-si
    ze: 0.7rem; padding: 2px 8px;" onclick="fetchHardwareTree()">R
    efresh</button>
429             </div>

```

```

430         <div id="hardware-tree" class="tree-container">
431             <div style="color: var(--text-muted); font-size: 0.8rem; text-align: center; padding: 20px;">Loading hardware tree...</div>
432         </div>
433     </div>
434
435     <!-- Right Pane: Wireshark Process Stream -->
436     <div class="right-panel">
437         <div class="stream-toolbar">
438             <input id="process-search" type="text" class="search-box" placeholder="Filter processes by name, PID, or use r..." oninput="renderProcessTable()">
439             <button id="pause-btn" class="btn btn-primary" onclick="togglePause()">Pause Stream</button>
440             <button class="btn btn-ai" onclick="triggerAiAudit()"> Run AI Audit</button>
441         </div>
442         <div class="table-container">
443             <table>
444                 <thead>
445                     <tr>
446                         <th style="width: 70px;">PID</th>
447                         <th>Process Name</th>
448                         <th style="width: 90px;">Status</th>
449                         <th style="width: 90px; text-align: right;">CPU %</th>
450                         <th style="width: 100px; text-align: right;">Memory</th>
451                         <th style="width: 80px; text-align: right;">Threads</th>
452                         <th style="width: 120px;">User</th>
453                     </tr>
454                 </thead>
455                 <tbody id="process-tbody">
456                     <tr><td colspan="7" style="text-align:

```

```

        center; color: var(--text-muted);">Waiting for telemetry stre
am...</td></tr>
457 |         </tbody>
458 |     </table>
459 | </div>
460 |
461 |     <!-- Bottom AI Copilot -->
462 |     <div class="ai-drawer">
463 |         <div class="ai-header">
464 |             <div class="ai-title">
465 |                 <span> AI Telemetry & Bottleneck Cop
466 | ilot</span>
467 |                 <span id="ai-health-badge" style="font
468 | -size: 0.75rem; color: var(--accent-green); font-weight: norma
469 | l;">[Health: 100/100]</span>
470 |             </div>
471 |             <span id="ai-engine-tag" style="font-size:
472 | 0.7rem; color: var(--text-muted);">Heuristic Engine</span>
473 |             </div>
474 |             <div id="ai-anomalies-container"></div>
475 |             <div id="ai-summary-text" class="ai-content">
476 |                 Telemetry monitor active. Running automate
477 | d heuristic checks over CPU, memory, and hardware thermals.
478 |             </div>
479 |         </div>
480 |     </div>
481 |
482 |     <script>
483 |         let isPaused = false;
484 |         let latestSnapshot = null;
485 |         let ws = null;
486 |
487 |         let wsReconnectTimer = null;
488 |
489 |         function connectWebSocket() {
490 |             if (wsReconnectTimer) {
491 |                 clearTimeout(wsReconnectTimer);

```

```

488         wsReconnectTimer = null;
489     }
490     if (ws) {
491         try {
492             ws.onclose = null;
493             ws.onerror = null;
494             ws.close();
495         } catch {}
496     }
497     const proto = window.location.protocol === 'https:
' ? 'wss:' : 'ws:';
498     const wsUrl = `${proto}://${window.location.host}/a
pi/v1/system/stream`;
499     ws = new WebSocket(wsUrl);
500
501     ws.onopen = () => {
502         const connStatus = document.getElementById('co
nn-status');
503         if (connStatus) {
504             connStatus.innerHTML = '<span class="pulse
-dot"></span><span>TELEMETRY STREAMING</span>';
505             connStatus.style.borderColor = 'var(--acce
nt-green)';
506         }
507     };
508
509     ws.onmessage = (event) => {
510         if (isPaused) return;
511         try {
512             const data = JSON.parse(event.data);
513             latestSnapshot = data;
514             updateDashboard(data);
515         } catch (e) {
516             console.error("Error parsing telemetry sna
pshot:", e);
517         }
518     };
519

```



```

520         const scheduleReconnect = () => {
521             if (wsReconnectTimer) return;
522             const connStatus = document.getElementById('co
nn-status');
523             if (connStatus) {
524                 connStatus.innerHTML = '<span style="color
: var(--accent-red);">●</span><span>RECONNECTING...</span>';
525                 connStatus.style.borderColor = 'var(--acce
nt-red)';
526             }
527             wsReconnectTimer = setTimeout(() => {
528                 wsReconnectTimer = null;
529                 connectWebSocket();
530             }, 3000);
531         };
532
533         ws.onclose = scheduleReconnect;
534         ws.onerror = scheduleReconnect;
535     }
536
537     function updateDashboard(snap) {
538         // Update Top Cards
539         const cpu = snap.cpu;
540         document.getElementById('metric-cpu-val').innerTex
t = `${cpu.total_percent.toFixed(1)}%`;
541         document.getElementById('metric-cpu-fill').style.w
idth = `${Math.min(100, cpu.total_percent)}%`;
542         document.getElementById('metric-cpu-sub').innerTex
t = `${cpu.physical_cores} Physical / ${cpu.logical_cores} Thr
eads`;
543
544         const mem = snap.memory;
545         document.getElementById('metric-ram-val').innerTex
t = `${mem.used_gb.toFixed(1)} / ${mem.total_gb.toFixed(1)} GB
`;
546         document.getElementById('metric-ram-fill').style.w
idth = `${mem.percent}%`;
547         document.getElementById('metric-ram-sub').innerTex

```

```

t = `${mem.percent}% Utilized (${mem.available_gb.toFixed(1)}
GB free)`;
548
549         if (snap.gpus && snap.gpus.length > 0) {
550             const g = snap.gpus[0];
551             document.getElementById('metric-gpu-val').inne
rText = g.name;
552             document.getElementById('metric-gpu-sub').inne
rText = `VRAM: ${g.memory_total_gb.toFixed(1)} GB | CUDA: ${g.
has_cuda ? 'Yes' : 'No'}`;
553         }
554
555         const io = snap.disk_io;
556         const totalMb = ((io.read_bytes_per_sec + io.write
_bytes_per_sec) / (1024 * 1024)).toFixed(2);
557         const rKb = (io.read_bytes_per_sec / 1024).toFixed
(0);
558         const wKb = (io.write_bytes_per_sec / 1024).toFixe
d(0);
559         document.getElementById('metric-disk-val').innerTe
xt = `${totalMb} MB/s`;
560         document.getElementById('metric-disk-sub').innerTe
xt = `R: ${rKb} KB/s | W: ${wKb} KB/s`;
561
562         renderProcessTable();
563     }
564
565     function renderProcessTable() {
566         if (!latestSnapshot) return;
567         const filter = document.getElementById('process-se
arch').value.toLowerCase();
568         const tbody = document.getElementById('process-tbo
dy');
569
570         const filtered = latestSnapshot.top_processes.filt
er(p => {
571             return !filter || p.name.toLowerCase().include
s(filter) || String(p.pid).includes(filter) || (p.username &&

```

```

    p.username.toLowerCase().includes(filter));
572         });
573
574         if (filtered.length === 0) {
575             tbody.innerHTML = '<tr><td colspan="7" style="
text-align: center; color: var(--text-muted);">No matching pro
cesses found.</td></tr>';
576             return;
577         }
578
579         tbody.innerHTML = filtered.map(p => {
580             let cpuClass = '';
581             if (p.cpu_percent > 40) cpuClass = 'badge-cpu-
high';
582             else if (p.cpu_percent > 15) cpuClass = 'badge
-cpu-med';
583
584             return `
585                 <tr>
586                     <td style="color: var(--accent-cyan);"
>${p.pid}</td>
587                     <td style="font-weight: 600;">${p.name
}</td>
588                     <td style="color: var(--text-muted);">
${p.status}</td>
589                     <td style="text-align: right;" class="
${cpuClass}">${p.cpu_percent.toFixed(1)}%</td>
590                     <td style="text-align: right; color: v
ar(--accent-green);">${p.memory_mb.toFixed(1)} MB</td>
591                     <td style="text-align: right; color: v
ar(--accent-purple);">${p.num_threads}</td>
592                     <td style="color: var(--text-muted);">
${p.username || 'SYSTEM'}</td>
593                 </tr>
594             `;
595         }).join('');
596     }
597

```

```

598     function togglePause() {
599         isPaused = !isPaused;
600         const btn = document.getElementById('pause-btn');
601         btn.innerText = isPaused ? 'Resume Stream' : 'Pause Stream';
602         btn.style.background = isPaused ? 'var(--accent-yellow)' : 'var(--accent-cyan)';
603     }
604
605     async function fetchHardwareTree() {
606         try {
607             const res = await fetch('/api/v1/system/hardware');
608             if (!res.ok) return;
609             const nodes = await res.json();
610             const container = document.getElementById('hardware-tree');
611
612             container.innerHTML = nodes.map(node => {
613                 const propsHtml = Object.entries(node.properties).map(([k, v]) => `
614                     <div class="prop-row">
615                         <span class="prop-key">${k}</span>
616                         <span class="prop-val">${v}</span>
617                     </div>
618                 `).join('');
619
620                 return `
621                     <div class="tree-node">
622                         <div class="tree-node-title" onclick="this.nextElementSibling.classList.toggle('hidden')">
623                             <span>${node.category}: ${node.name}</span>
624                             <span style="font-size: 0.7rem; color: var(--accent-cyan);">▼</span>
625                         </div>
626                         <div class="tree-node-body">
627                             ${propsHtml}

```

```

628 |         </div>
629 |     </div>
630 |     `;
631 |     }).join('');
632 | } catch (e) {
633 |     console.error("Failed to load hardware tree:",
634 | e);
635 | }
636 |
637 | async function triggerAiAudit() {
638 |     const summaryEl = document.getElementById('ai-summ
639 | ary-text');
640 |     const badgeEl = document.getElementById('ai-health
641 | -badge');
642 |     const anomalyEl = document.getElementById('ai-anom
643 | alies-container');
644 |
645 |     summaryEl.innerText = "Running AI telemetry diagno
646 | stic analysis...";
647 |     try {
648 |         const res = await fetch('/api/v1/system/diagno
649 | se', { method: 'POST' });
650 |         const report = await res.json();
651 |
652 |         badgeEl.innerText = `[Health: ${report.health_
653 | score}/100]`;
654 |
655 |         badgeEl.style.color = report.health_score >= 8
656 | 0 ? 'var(--accent-green)' : (report.health_score >= 60 ? 'var(
657 | --accent-yellow)' : 'var(--accent-red)');
658 |
659 |         document.getElementById('ai-engine-tag').inner
660 | Text = report.ai_model_used;
661 |
662 |         if (report.anomalies && report.anomalies.lengt
663 | h > 0) {
664 |             anomalyEl.innerHTML = report.anomalies.map
665 | (a => `<span class="anomaly-pill">⬢ [${a.subsystem}] ${a.title
666 | }</span>`).join('');

```

```
653 |             } else {
654 |                 anomalyEl.innerHTML = '<span style="color:
    var(--accent-green); font-size: 0.75rem;">␣ No performance an
    omalies detected</span>';
655 |             }
656 |
657 |             summaryEl.innerText = `${report.summary}\nReco
    mmendations: ${report.recommendations.join(', ')}`;
658 |         } catch (e) {
659 |             summaryEl.innerText = `Diagnostic error: ${e.m
    essage}`;
660 |         }
661 |     }
662 |
663 |     // Initialize on load
664 |     window.addEventListener('DOMContentLoaded', () => {
665 |         connectWebSocket();
666 |         fetchHardwareTree();
667 |     });
668 | </script>
669 | </body>
670 | </html>
671 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/system_inspector_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/system_inspector_tab/in
17         dex.html" class="w-100 h-100 border-0"></iframe>
18
19 * File: index.html
20 * Project: ai-breadboard
21 * Package: windows/api/webgui/system_inspector_tab
22 * Author: hypo69
23 * Copyright: © 2026 hypo69
24 * Updated: 2026-10-08 03:51:00
25 =====
26 =====
27 -->
28
29 <!-- System & Hardware Inspector Application Tab -->
30 <style>
31     /* Адаптивные стили системного инспектора через CSS-переменн
32     ые */
33     .sys-container {
34         padding: 1rem 1.25rem;
35         max-width: 100%;
36         color: var(--text-color);
37     }
```

```
32 |
33 | /* Адаптация text-light и бейджей для светлой темы */
34 | .sys-container .text-light {
35 |     color: var(--text-color) !important;
36 | }
37 | [data-theme="dark"] .sys-container .text-light,
38 | [data-theme="terminal"] .sys-container .text-light {
39 |     color: #e2e8f0 !important;
40 | }
41 |
42 | .sys-container .badge.bg-dark {
43 |     background: var(--surface-2) !important;
44 |     color: var(--text-color) !important;
45 |     border: 1px solid var(--border-color) !important;
46 | }
47 | [data-theme="dark"] .sys-container .badge.bg-dark {
48 |     background: #0f172a !important;
49 |     color: #e2e8f0 !important;
50 |     border-color: rgba(255, 255, 255, 0.18) !important;
51 | }
52 | [data-theme="terminal"] .sys-container .badge.bg-dark {
53 |     background: #0b120c !important;
54 |     color: #22c55e !important;
55 |     border-color: #1a4223 !important;
56 | }
57 |
58 | /* Стильные градиентные разделители между основными панелями
59 | */
60 | .sys-section-divider {
61 |     height: 1px;
62 |     margin: 1.75rem 0;
63 |     background: linear-gradient(90deg, transparent 0%, var(--border-color) 15%, var(--nav-active) 50%, var(--border-color) 85%, transparent 100%);
64 |     position: relative;
65 | }
66 | .sys-section-divider::after {
67 |     content: '';
```



```

67 |     position: absolute;
68 |     top: -1.5px;
69 |     left: 50%;
70 |     transform: translateX(-50%);
71 |     width: 60px;
72 |     height: 4px;
73 |     background: linear-gradient(90deg, var(--nav-active), #818
cf8);
74 |     border-radius: 4px;
75 |     box-shadow: 0 0 12px rgba(13, 202, 240, 0.4);
76 | }
77 |
78 | /* Карточки основных разделов (CPU, GPU, RAM, Net, Storage)
*/
79 | .sys-card, .sys-panel {
80 |     background: var(--surface-1);
81 |     border: 1px solid var(--border-color);
82 |     border-radius: 12px;
83 |     box-shadow: var(--shadow-md);
84 |     transition: transform 0.2s ease, box-shadow 0.2s ease, bor
der-color 0.2s ease;
85 |     overflow: hidden;
86 |     color: var(--text-color);
87 | }
88 | .sys-card:hover {
89 |     border-color: var(--border-strong);
90 |     box-shadow: var(--shadow-lg);
91 | }
92 | [data-theme="dark"] .sys-card,
93 | [data-theme="dark"] .sys-panel {
94 |     background: linear-gradient(180deg, #141c2e 0%, #0d1424 10
0%);
95 |     border: 1px solid rgba(255, 255, 255, 0.14);
96 | }
97 | [data-theme="terminal"] .sys-card,
98 | [data-theme="terminal"] .sys-panel {
99 |     background: #0b120c;
100 |     border: 1px solid #1a4223;

```

```
101 | }
102 |
103 | .sys-card-cpu { border-top: 4px solid #0284c7; }
104 | .sys-card-gpu { border-top: 4px solid #d97706; }
105 | .sys-card-ram { border-top: 4px solid #16a34a; }
106 | .sys-card-net { border-top: 4px solid #6366f1; }
107 | .sys-card-storage { border-top: 4px solid #0891b2; }
108 |
109 | /* Заголовки секций с повышенной контрастностью и интерактив
    | ностью */
110 | .sys-core-section-title {
111 |     font-size: 0.85rem !important;
112 |     color: var(--text-color) !important;
113 |     font-weight: 700 !important;
114 |     padding-bottom: 6px;
115 |     border-bottom: 1.5px solid var(--border-subtle);
116 |     margin-top: 14px;
117 |     margin-bottom: 12px !important;
118 |     letter-spacing: 0.3px;
119 | }
120 |
121 | /* Шапка карточек панелей для четкого разделения и иерархии
    | */
122 | .sys-card-header-bar {
123 |     padding: 0.6rem 0.85rem;
124 |     margin: -1rem -1rem 1rem -1rem;
125 |     background: var(--surface-2);
126 |     border-bottom: 1px solid var(--border-color);
127 |     display: flex;
128 |     justify-content: space-between;
129 |     align-items: center;
130 |     border-top-left-radius: 11px;
131 |     border-top-right-radius: 11px;
132 | }
133 |
134 | /* Компактный режим: отображаются только верхние блоки панел
    | ей */
135 | .sys-compact-mode .sys-full-only,
```

```
136 | .sys-compact-mode .sys-detailed-sensors,
137 | .sys-compact-mode .sys-card-header-bar {
138 |     display: none !important;
139 | }
140 | .sys-compact-mode .sys-card {
141 |     padding: 0.6rem 0.75rem !important;
142 | }
143 | .sys-compact-mode .sys-card-top-banner {
144 |     margin-bottom: 0 !important;
145 |     padding: 0.6rem 0.85rem !important;
146 | }
147 | .sys-compact-mode .sys-section-divider {
148 |     margin: 0.65rem 0 !important;
149 | }
150 | .sys-compact-mode .row.mb-3 {
151 |     margin-bottom: 0.5rem !important;
152 | }
153 | .sys-compact-mode #sys-gpu-panels-container {
154 |     gap: 0.5rem !important;
155 |     margin-bottom: 0.5rem !important;
156 | }
157 |
158 | /* Принудительное разворачивание карточки компонента по клику
    | на значок дрели (Drill-down) */
159 | .sys-compact-mode .sys-card.sys-card-drill-expanded {
160 |     padding: 1rem !important;
161 |     border-color: var(--nav-active) !important;
162 |     box-shadow: 0 0 16px rgba(13, 202, 240, 0.25) !important;
163 | }
164 | .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-car
    | d-header-bar {
165 |     display: flex !important;
166 | }
167 | .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-ful
    | l-only {
168 |     display: block !important;
169 | }
170 | .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-ful
```

```
l-only.row {
171     display: flex !important;
172 }
173 .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-det
ailed-sensors {
174     display: block !important;
175 }
176 .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-car
d-top-banner {
177     margin-bottom: 1rem !important;
178 }
179
180 /* Интерактивная кнопка со значком дрели (Drill-down) слева
от названия компонента */
181 .sys-drill-btn {
182     display: inline-flex;
183     align-items: center;
184     justify-content: center;
185     width: 28px;
186     height: 28px;
187     min-width: 28px;
188     min-height: 28px;
189     padding: 0;
190     margin-right: 8px;
191     border-radius: 7px;
192     border: 1.5px solid var(--border-color);
193     background: var(--surface-1);
194     cursor: pointer;
195     transition: all 0.22s cubic-bezier(0.4, 0, 0.2, 1);
196     box-shadow: var(--shadow-xs);
197     flex-shrink: 0;
198     vertical-align: middle;
199 }
200 .sys-drill-btn img {
201     width: 18px;
202     height: 18px;
203     object-fit: contain;
204     transition: transform 0.22s ease, opacity 0.22s ease;
```

```

205     display: block;
206 }
207 .sys-drill-btn:hover {
208     border-color: var(--nav-active) !important;
209     transform: scale(1.12);
210     box-shadow: 0 0 10px rgba(239, 68, 68, 0.45);
211 }
212 .sys-drill-btn:hover img {
213     transform: rotate(12deg);
214 }
215 .sys-drill-btn.active {
216     box-shadow: 0 0 14px rgba(239, 68, 68, 0.65) !important;
217     border-color: #ef4444 !important;
218     transform: rotate(-8deg) scale(1.06);
219 }
220 [data-theme="dark"] .sys-drill-btn {
221     background: rgba(15, 23, 42, 0.85);
222     border-color: rgba(255, 255, 255, 0.2);
223 }
224 [data-theme="terminal"] .sys-drill-btn {
225     background: #0b120c;
226     border-color: #1a4223;
227 }
228
229 /* Верхний баннер панелей: строгое выравнивание колонок чере
3 CSS Grid */
230 .sys-card-top-banner {
231     display: grid !important;
232     grid-template-columns: minmax(280px, 1fr) 280px 330px !imp
ortant;
233     align-items: center;
234     gap: 1.25rem;
235     width: 100%;
236     background: var(--surface-2);
237     border: 1px solid var(--border-color);
238 }
239 [data-theme="dark"] .sys-card-top-banner {
240     background: rgba(0, 0, 0, 0.45);

```

```
241     border: 1.5px solid rgba(255, 255, 255, 0.18);
242 }
243 [data-theme="terminal"] .sys-card-top-banner {
244     background: #0b120c;
245     border: 1px solid #1a4223;
246 }
247 .sys-banner-col-info {
248     min-width: 0;
249     width: 100%;
250 }
251 .sys-banner-col-gauge {
252     width: 280px;
253     min-width: 280px;
254     max-width: 280px;
255     justify-self: center;
256 }
257 .sys-banner-col-specs {
258     width: 330px;
259     min-width: 330px;
260     max-width: 330px;
261     justify-self: end;
262 }
263
264 /* Панель графиков истории */
265 .sys-history-panel {
266     background: var(--surface-2);
267     border: 1px solid var(--border-color);
268     position: relative;
269 }
270 [data-theme="dark"] .sys-history-panel {
271     background: rgba(0, 0, 0, 0.4);
272     border: 1.5px solid rgba(255, 255, 255, 0.15);
273 }
274 [data-theme="terminal"] .sys-history-panel {
275     background: #0b120c;
276     border: 1px solid #1a4223;
277 }
278
```

```
279 | .sys-spark-tooltip {
280 |     position: absolute;
281 |     top: 6px;
282 |     pointer-events: none;
283 |     background: var(--surface-1);
284 |     color: var(--text-color);
285 |     border: 1px solid var(--nav-active);
286 |     border-radius: 6px;
287 |     padding: 4px 8px;
288 |     font-size: 0.70rem;
289 |     z-index: 10;
290 |     box-shadow: var(--shadow-md);
291 |     white-space: nowrap;
292 | }
293 | [data-theme="dark"] .sys-spark-tooltip {
294 |     background: rgba(15, 23, 42, 0.95);
295 |     box-shadow: 0 4px 12px rgba(0, 0, 0, 0.6);
296 | }
297 | [data-theme="terminal"] .sys-spark-tooltip {
298 |     background: #0b120c;
299 |     border-color: #1a4223;
300 |     color: #22c55e;
301 | }
302 |
303 | @media (max-width: 1280px) {
304 |     .sys-card-top-banner {
305 |         grid-template-columns: 1fr 280px !important;
306 |     }
307 |     .sys-banner-col-specs {
308 |         grid-column: span 2;
309 |         justify-self: stretch;
310 |         width: 100%;
311 |         max-width: 100%;
312 |     }
313 | }
314 |
315 | @media (max-width: 860px) {
316 |     .sys-card-top-banner {
```

```
317     grid-template-columns: 1fr !important;
318 }
319 .sys-banner-col-gauge,
320 .sys-banner-col-specs {
321     grid-column: span 1;
322     justify-self: stretch;
323     width: 100%;
324     max-width: 100%;
325 }
326 }
327
328 /* Сетка и плашки ключевых экспресс-метрик (KPI Pills) в цен
тре верхних баннеров */
329 .sys-banner-kpi-grid {
330     display: grid;
331     grid-template-columns: repeat(auto-fit, minmax(115px, 1fr)
);
332     gap: 6px;
333     width: 100%;
334     max-width: 580px;
335     margin-top: 4px;
336 }
337 .sys-kpi-pill {
338     background: var(--surface-2);
339     border: 1px solid var(--border-color);
340     border-radius: 7px;
341     padding: 5px 8px;
342     display: flex;
343     flex-direction: column;
344     justify-content: center;
345     gap: 1px;
346     transition: all 0.2s ease;
347     box-shadow: var(--shadow-xs);
348 }
349 .sys-kpi-pill:hover {
350     border-color: var(--nav-active);
351     transform: translateY(-1px);
352     box-shadow: var(--shadow-sm);
```



```
353 | }
354 | [data-theme="dark"] .sys-kpi-pill {
355 |     background: rgba(15, 23, 42, 0.65);
356 |     border: 1px solid rgba(255, 255, 255, 0.12);
357 | }
358 | [data-theme="terminal"] .sys-kpi-pill {
359 |     background: #0b120c;
360 |     border: 1px solid #1a4223;
361 | }
362 | .sys-kpi-label {
363 |     font-size: 0.68rem;
364 |     font-weight: 600;
365 |     color: var(--text-muted);
366 |     white-space: nowrap;
367 |     overflow: hidden;
368 |     text-overflow: ellipsis;
369 |     display: flex;
370 |     align-items: center;
371 | }
372 | .sys-kpi-value {
373 |     font-size: 0.80rem;
374 |     font-weight: 800;
375 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
376 | Consolas, monospace;
377 |     white-space: nowrap;
378 |     overflow: hidden;
379 |     text-overflow: ellipsis;
380 | }
381 | /* Сенсоры LHM */
382 | .sys-sensor-group {
383 |     margin-bottom: 8px;
384 |     background: var(--surface-2);
385 |     border: 1.5px solid var(--border-color);
386 |     border-radius: 6px;
387 |     overflow: hidden;
388 | }
389 | .sys-sensor-group-title {
```

```
390 padding: 6px 10px;
391 font-size: 0.80rem;
392 font-weight: 700;
393 background: var(--surface-1);
394 display: flex;
395 align-items: center;
396 justify-content: space-between;
397 color: var(--nav-active);
398 border-bottom: 1px solid var(--border-color);
399 }
400 .sys-sensor-row {
401 display: flex;
402 justify-content: space-between;
403 align-items: center;
404 padding: 4px 10px;
405 font-size: 0.78rem;
406 border-bottom: 1px solid var(--border-subtle);
407 background: var(--surface-2);
408 }
409 .sys-sensor-row:last-child {
410 border-bottom: none;
411 }
412 .sys-sensor-name {
413 color: var(--text-color);
414 font-weight: 600;
415 max-width: 55%;
416 overflow: hidden;
417 text-overflow: ellipsis;
418 white-space: nowrap;
419 }
420 .sys-sensor-value {
421 font-weight: 700;
422 font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
423 Consolas, monospace;
424 font-size: 0.80rem;
425 text-align: right;
426 color: var(--text-color);
427 }
```

```
427
428  /* Таблица процессов */
429  .sys-proc-table {
430      width: 100%;
431      font-size: 0.80rem;
432      font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
433      Consolas, monospace;
434      border-collapse: collapse;
435  }
436  .sys-proc-table th {
437      background: var(--surface-2);
438      padding: 6px 10px;
439      color: var(--text-muted);
440      font-weight: 700;
441      border-bottom: 1.5px solid var(--border-color);
442      position: sticky;
443      top: 0;
444      z-index: 5;
445  }
446  .sys-proc-table td {
447      padding: 5px 10px;
448      border-bottom: 1px solid var(--border-subtle);
449      color: var(--text-color);
450  }
451
452  /* Статусы загрузки */
453  .badge-cpu-high { color: #dc2626; font-weight: 800; }
454  .badge-cpu-med  { color: #d97706; font-weight: 700; }
455
456  /* Аномалии */
457  .anomaly-tag {
458      display: inline-block;
459      font-size: 0.74rem;
460      padding: 2px 7px;
461      border-radius: 4px;
462      background: var(--status-danger-bg);
463      color: var(--status-danger-text);
464      border: 1.5px solid var(--status-danger-border);
```

```
464     margin-right: 4px;
465     margin-top: 2px;
466     font-weight: 700;
467 }
468
469 /* Интерактивный ресайзер для таблицы */
470 .sys-table-resizer {
471     height: 10px;
472     background: var(--surface-2);
473     border-top: 1px solid var(--border-color);
474     cursor: ns-resize;
475     display: flex;
476     align-items: center;
477     justify-content: center;
478     transition: background-color 0.15s ease;
479     user-select: none;
480     -webkit-user-select: none;
481     touch-action: none;
482 }
483 .sys-table-resizer:hover,
484 .sys-table-resizer.resizing {
485     background: var(--nav-hover-bg);
486 }
487 .sys-table-resizer-grip {
488     width: 42px;
489     height: 3px;
490     background: var(--border-strong);
491     border-radius: 3px;
492     pointer-events: none;
493     transition: background-color 0.15s ease, width 0.15s ease;
494 }
495 .sys-table-resizer:hover .sys-table-resizer-grip,
496 .sys-table-resizer.resizing .sys-table-resizer-grip {
497     background: var(--nav-active);
498     width: 56px;
499 }
500
501 /* Карточки ядер CPU и блоков GPU, полукруги и термо-слайдер
```

```
51  */
502 .sys-core-card {
503     background: var(--surface-2);
504     border: 1.5px solid var(--border-color);
505     border-radius: 8px;
506     padding: 10px 12px;
507     transition: all 0.2s ease;
508     box-shadow: var(--shadow-xs);
509     display: flex;
510     flex-direction: column;
511     justify-content: space-between;
512     min-height: 148px;
513     color: var(--text-color);
514 }
515 .sys-core-card:hover {
516     border-color: var(--nav-active);
517     transform: translateY(-2px);
518     box-shadow: var(--shadow-sm);
519 }
520 [data-theme="dark"] .sys-core-card {
521     background: #0f172a;
522     border: 1.5px solid rgba(255, 255, 255, 0.22);
523     box-shadow: 0 4px 12px rgba(0, 0, 0, 0.5);
524 }
525 [data-theme="terminal"] .sys-core-card {
526     background: #0b120c;
527     border: 1.5px solid #1a4223;
528 }
529
530 .sys-core-title {
531     font-size: 0.88rem;
532     font-weight: 800;
533     color: var(--text-color);
534     letter-spacing: 0.3px;
535 }
536 .sys-core-load-badge {
537     font-size: 0.82rem;
538     font-weight: 800;
```

```
539 | padding: 2px 7px;
540 | border-radius: 5px;
541 | background: var(--status-info-bg);
542 | color: var(--status-info-text);
543 | border: 1.5px solid var(--status-info-border);
544 | }
545 |
546 | .sys-temp-slider-wrap {
547 |     margin-top: 5px;
548 | }
549 | .sys-temp-slider-track {
550 |     position: relative;
551 |     height: 8px;
552 |     background: var(--border-color);
553 |     border-radius: 4px;
554 |     overflow: visible;
555 | }
556 | .sys-temp-slider-bar {
557 |     height: 100%;
558 |     border-radius: 4px;
559 |     transition: width 0.4s ease, background-color 0.4s ease;
560 | }
561 | .sys-temp-slider-thumb {
562 |     position: absolute;
563 |     top: 50%;
564 |     width: 14px;
565 |     height: 14px;
566 |     border-radius: 50%;
567 |     border: 2px solid var(--surface-1);
568 |     transform: translate(-50%, -50%);
569 |     transition: left 0.4s ease, background-color 0.4s ease;
570 |     box-shadow: 0 2px 6px rgba(0,0,0,0.5);
571 | }
572 |
573 | /* Таблица параметров ядра CPU внутри карточки */
574 | .sys-core-param-table {
575 |     display: flex;
576 |     flex-direction: column;
```

```
577     gap: 3px;
578     width: 100%;
579 }
580 .sys-core-param-row {
581     display: flex;
582     justify-content: space-between;
583     align-items: center;
584     padding: 2px 0;
585     border-bottom: 1px dashed var(--border-subtle);
586     font-size: 0.74rem;
587     line-height: 1.25;
588 }
589 .sys-core-param-row:last-child {
590     border-bottom: none;
591 }
592 .sys-core-param-key {
593     color: var(--text-muted);
594     font-size: 0.72rem;
595     font-weight: 600;
596     white-space: nowrap;
597     overflow: hidden;
598     text-overflow: ellipsis;
599     max-width: 58%;
600 }
601 .sys-core-param-val {
602     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
603     Consolas, monospace;
604     font-weight: 800;
605     font-size: 0.76rem;
606     text-align: right;
607     white-space: nowrap;
608     color: var(--text-color);
609 }
610 /* Крупные слайдеры для главного заголовка */
611 .sys-temp-slider-lg .sys-temp-slider-track {
612     height: 12px;
613     border-radius: 6px;
```

```
614     background: var(--border-color);
615 }
616 .sys-temp-slider-lg .sys-temp-slider-bar {
617     border-radius: 6px;
618 }
619 .sys-temp-slider-lg .sys-temp-slider-thumb {
620     width: 20px;
621     height: 20px;
622     border: 2.5px solid var(--surface-1);
623     box-shadow: 0 2px 8px rgba(0,0,0,0.5);
624 }
625 .sys-temp-slider-empty {
626     font-size: 0.80rem;
627     color: var(--text-muted);
628     text-align: center;
629     font-weight: 600;
630 }
631
632 /* Спецификации CPU: интерактивная карточка */
633 .sys-cpu-specs-card,
634 .sys-gpu-specs-card {
635     transition: all 0.2s ease;
636     cursor: pointer;
637     position: relative;
638     user-select: none;
639     background: var(--surface-2);
640     border: 1px solid var(--border-color);
641     color: var(--text-color);
642 }
643 .sys-cpu-specs-card:hover,
644 .sys-gpu-specs-card:hover {
645     border-color: var(--nav-active) !important;
646     box-shadow: var(--shadow-sm) !important;
647     transform: translateY(-1px);
648 }
649
650 /* Сворачиваемая интерактивная плашка детализации CPU */
651 .sys-cpu-details-accordion-btn,
```



```
652 | .sys-gpu-details-accordion-btn {
653 |     cursor: pointer;
654 |     user-select: none;
655 |     padding: 8px 14px;
656 |     background: var(--surface-2);
657 |     border-radius: 8px;
658 |     border: 1.5px solid var(--border-color);
659 |     transition: all 0.25s ease;
660 |     box-shadow: var(--shadow-xs);
661 |     color: var(--text-color);
662 | }
663 | .sys-cpu-details-accordion-btn:hover,
664 | .sys-gpu-details-accordion-btn:hover {
665 |     border-color: var(--nav-active);
666 |     box-shadow: var(--shadow-sm);
667 |     transform: translateY(-1px);
668 | }
669 |
670 | /* Сенсорные виджеты */
671 | .sys-sensor-widget-item {
672 |     background: var(--surface-2) !important;
673 |     border: 1px solid var(--border-color) !important;
674 |     box-shadow: var(--shadow-xs);
675 |     color: var(--text-color);
676 | }
677 | [data-theme="dark"] .sys-sensor-widget-item {
678 |     background: #0f172a !important;
679 |     border: 1.5px solid rgba(255, 255, 255, 0.18) !important;
680 | }
681 | [data-theme="terminal"] .sys-sensor-widget-item {
682 |     background: #0b120c !important;
683 |     border: 1px solid #1a4223 !important;
684 | }
685 |
686 | /* Полоса заголовка инспектора */
687 | .sys-header-strip {
688 |     background: var(--surface-2);
689 |     border: 1px solid var(--border-color);
```

```

690     box-shadow: var(--shadow-xs);
691 }
692 [data-theme="dark"] .sys-header-strip {
693     background: linear-gradient(180deg, #182238 0%, #0f172a 10
0%) !important;
694     border: 1px solid rgba(255, 255, 255, 0.18) !important;
695     box-shadow: 0 4px 16px rgba(0,0,0,0.45) !important;
696 }
697 [data-theme="terminal"] .sys-header-strip {
698     background: #0b120c !important;
699     border: 1px solid #1a4223 !important;
700 }
701 </style>
702
703 <div class="sys-container theme-inspector sys-compact-mode">
704     <!-- Header Alert & Status (Compact Strip) -->
705     <div class="d-flex justify-content-between align-items-cente
r flex-wrap gap-2 mb-3.5 py-2 px-3 rounded-3 sys-header-strip"
>
706         <div class="d-flex align-items-center gap-1.5 flex-wrap">
707             <button type="button" class="panel-toggle-drill collapse
d me-2" id="btn-sys-toggle-compact" title="Развернуть все / Св
ернуть все панели">
708                 
709             </button>
710         </div>
711         <div class="d-flex align-items-center gap-2 flex-wrap">
712             <span class="fs-5">□ </span>
713             <span class="fw-bold" style="font-size: 0.95rem; color:
var(--text-color);">Телеметрия хоста и оборудования</span>
714             <span class="badge bg-secondary fw-bold" style="font-siz
e: 0.72rem;">Live Telemetry</span>
715             <span class="d-none d-lg-inline text-muted" style="font-
size: 0.78rem;">| Комплексный мониторинг CPU, RAM, GPU, дисков
, питания и процессов</span>
716         </div>
717         <div class="d-flex align-items-center gap-1.5 flex-wrap">

```

```

718 |         <span class="badge rounded-pill bg-success text-white bo
      | rder border-light px-2.5 py-1" style="font-size: 0.74rem; font
      | -weight: 700;" id="sys-conn-status">
719 |             • АКТИВНА
720 |         </span>
721 |     </div>
722 | </div>
723 |
724 | <!-- 1. Панель Загрузка CPU -->
725 | <div class="row mb-3">
726 |     <div class="col-12">
727 |         <div class="sys-card sys-card-cpu h-100 p-3" id="sys-car
      | d-cpu">
728 |             <div class="sys-card-header-bar">
729 |                 <div class="d-flex align-items-center gap-2">
730 |                     <span class="sys-label text-info" style="font-size
      | : 1.05rem; font-weight: 800;"><i class="bi bi-cpu text-info me
      | -1.5"></i>Загрузка CPU</span>
731 |                 </div>
732 |                 <span class="small badge bg-dark border border-secon
      | dary text-info font-monospace" style="font-size: 0.84rem;" id=
      | "sys-metric-cpu-sub">-- Ядер</span>
733 |             </div>
734 |             <!-- Верхний баннер: Название слева, Спидометр и датчи
      | к температуры по центру, Сводная таблица CPU справа -->
735 |             <div class="sys-card-top-banner p-3 rounded-3 mb-3">
736 |                 <!-- Левая колонка: Модель процессора и экспресс-мет
      |рики KPI -->
737 |                 <div class="sys-banner-col-info d-flex flex-column a
      | lign-items-start justify-content-center">
738 |                     <div class="text-start mb-1.5 d-flex align-items-c
      | enter">
739 |                         <button type="button" class="sys-drill-btn" id="
      | sys-drill-btn-cpu" onclick="window.toggleSysCardDrill && windo
      | w.toggleSysCardDrill('sys-card-cpu')" title="Drill-down: Разве
      | рнуть/скрыть все скрытые панели и датчики CPU">
740 |                         

```

```
741 |         </button>
742 |         <div>
743 |             <span class="fw-bold text-info" style="font-size: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-shadow: 0 0 14px rgba(13, 202, 240, 0.4);" id="sys-metric-cpu-model">--</span>
744 |             <div class="d-flex align-items-center gap-1.5 mt-0.5">
745 |                 <span class="badge bg-dark border border-secondary text-light font-monospace" id="sys-metric-cpu-vendor-badge" style="font-size: 0.70rem;">Intel Corporation · 64-bit</span>
746 |                 <span class="badge bg-info-subtle text-info border border-info font-monospace" id="sys-cpu-status-badge" style="font-size: 0.70rem;">● Active</span>
747 |             </div>
748 |         </div>
749 |     </div>
750 |     <div class="sys-banner-kpi-grid">
751 |         <div class="sys-kpi-pill">
752 |             <span class="sys-kpi-label"><i class="bi bi-lightning-charge text-warning me-1"></i>Мощность</span>
753 |             <span class="sys-kpi-value text-warning" id="sys-cpu-kpi-power">-- W</span>
754 |         </div>
755 |         <div class="sys-kpi-pill">
756 |             <span class="sys-kpi-label"><i class="bi bi-thermometer-half text-danger me-1"></i>Пакет t°</span>
757 |             <span class="sys-kpi-value text-info" id="sys-cpu-kpi-temp">-- °C</span>
758 |         </div>
759 |         <div class="sys-kpi-pill">
760 |             <span class="sys-kpi-label"><i class="bi bi-cpu text-info me-1"></i>Ядра/Потоки</span>
761 |             <span class="sys-kpi-value text-light" id="sys-cpu-kpi-cores">-- / --</span>
762 |         </div>
763 |     <div class="sys-kpi-pill">
```

```

764 |         <span class="sys-kpi-label"><i class="bi bi-sp
| eedometer2 text-primary me-1"></i>Частота</span>
765 |         <span class="sys-kpi-value text-primary" id="s
| ys-cpu-kpi-freq">-- GHz</span>
766 |         </div>
767 |     </div>
768 | </div>
769 |
770 |     <!-- Центральная колонка: Спидометр загрузки, справа
| от него значение с динамическим оттенком, а СНИЗУ – датчик те
| мпературы -->
771 |     <div class="sys-banner-col-gauge d-flex flex-column
| justify-content-between gap-1.5">
772 |         <div class="d-flex align-items-center gap-2.5">
773 |             <div id="sys-metric-cpu-gauge" style="width: 120
| px; height: 68px; flex-shrink: 0;"></div>
774 |             <div>
775 |                 <div class="sys-value text-info" id="sys-metri
| c-cpu-val" style="font-size: 2.1rem; line-height: 1; font-weig
| ht: 800; transition: color 0.3s ease, text-shadow 0.3s ease;">
| 0.0%</div>
776 |                 <div class="text-light mt-1 font-monospace" st
| yle="font-size: 0.82rem; font-weight: 700;" id="sys-metric-cpu
| -package-temp">Пакет: -- °C</div>
777 |                 </div>
778 |             </div>
779 |             <!-- Датчик температуры Package расположен ПОД спи
| дометром загрузки -->
780 |             <div id="sys-metric-cpu-package-slider" style="wid
| th: 100%;"></div>
781 |             </div>
782 |
783 |     <!-- Правая колонка: Сводная таблица о процессоре (к
| ликабельна для запуска AI-аудита) -->
784 |     <div class="sys-banner-col-specs">
785 |         <div class="sys-cpu-specs-card p-2 rounded-2 d-fle
| x flex-column justify-content-between h-100"
786 |             id="sys-cpu-specs-card"

```

```

787         onclick="window.inspectCpuSpecsWithAi && wind
      ow.inspectCpuSpecsWithAi()"
788         title="Нажмите для вызова AI-аудита и эксперт
      ного анализа архитектуры процессора">
789         <div class="d-flex justify-content-between align
      -items-center mb-1 pb-1 border-bottom" style="border-color: va
      r(--border-subtle) !important;">
790             <span class="text-light fw-bold d-flex align-i
      tems-center gap-1" style="font-size: 0.78rem;">
791                 <i class="bi bi-info-circle text-info"></i>Х
      арактеристики CPU
792                 <span class="badge bg-primary-subtle text-pr
      imary border border-primary-subtle font-monospace ms-1" style=
      "font-size: 0.62rem;"><i class="bi bi-stars me-0.5"></i>AI</sp
      an>
793             </span>
794             <span class="badge bg-dark border border-secon
      dary text-info font-monospace" style="font-size: 0.68rem;" id=
      "sys-cpu-spec-arch">x86_64</span>
795         </div>
796         <div class="sys-cpu-specs-grid" style="display:
      grid; grid-template-columns: 1fr 1fr; gap: 4px 10px; font-size
      : 0.74rem;">
797             <div class="d-flex justify-content-between"><s
      pan class="text-muted">Сокет:</span><span class="text-light fo
      nt-monospace fw-semibold" id="sys-cpu-spec-socket">--</span></
      div>
798             <div class="d-flex justify-content-between"><s
      pan class="text-muted">Ядра/Потоки:</span><span class="text-li
      ght font-monospace fw-semibold" id="sys-cpu-spec-cores">--</sp
      an></div>
799             <div class="d-flex justify-content-between"><s
      pan class="text-muted">Базовая част.:</span><span class="text-
      info font-monospace fw-semibold" id="sys-cpu-spec-base-freq">-
      -</span></div>
800             <div class="d-flex justify-content-between"><s
      pan class="text-muted">Макс. част.:</span><span class="text-in
      fo font-monospace fw-semibold" id="sys-cpu-spec-max-freq">--</

```

```

span></div>
801         <div class="d-flex justify-content-between" st
yle="grid-column: span 2;"><span class="text-muted">Кэш L2 / L
3:</span><span class="text-warning font-monospace fw-semibold"
id="sys-cpu-spec-cache">--</span></div>
802         </div>
803     </div>
804 </div>
805 </div>
806
807     <!-- График истории CPU (Загрузка %, Температура °C, М
ощность W) с логарифмической/линейной шкалой -->
808     <div class="sys-full-only sys-history-panel p-2.5 mb-2
.5 rounded-2">
809         <div class="d-flex justify-content-between align-ite
ms-center mb-1.5 flex-wrap gap-2">
810             <div class="d-flex align-items-center gap-2">
811                 <span class="small text-light" style="font-size:
0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-inf
o me-1"></i>История CPU</span>
812                 <div class="btn-group btn-group-sm" role="group"
style="height: 22px;">
813                     <button type="button" class="btn btn-xs btn-ou
tline-info active py-0 px-2 fw-bold" id="btn-cpu-spark-log" st
yle="font-size: 0.68rem;" onclick="window.setCpuSparkScale &&
window.setCpuSparkScale('log')">Лог. шкала (Log)</button>
814                     <button type="button" class="btn btn-xs btn-ou
tline-secondary py-0 px-2" id="btn-cpu-spark-lin" style="font-
size: 0.68rem;" onclick="window.setCpuSparkScale && window.set
CpuSparkScale('linear')">Линейная (Lin)</button>
815                 </div>
816             </div>
817             <div class="d-flex align-items-center gap-2.5" sty
le="font-size: 0.74rem; font-weight: 600;">
818                 <span class="text-info" style="text-shadow: 0 0
8px rgba(13,202,240,0.5);" title="Основная шкала: Загрузка яде
p CPU (0-100%)">● Нагрузка %</span>
819                 <span class="text-danger" style="text-shadow: 0

```

```

0 8px rgba(239,68,68,0.5);" title="Компактная шкала: Температу
ра пакета CPU (°C)">● Температура °C</span>
820     <span class="text-warning" style="text-shadow: 0
0 8px rgba(245,158,11,0.5);" title="Компактная шкала: Энергоп
отребление пакета CPU (W)">● Мощность W</span>
821     <span class="text-muted font-monospace ms-2" id=
"sys-cpu-spark-live-cursor" style="font-size: 0.70rem;">Live</
span>
822         </div>
823     </div>
824     <!-- Контейнер графика и интерактивного тултипа -->
825     <div style="position: relative; width: 100%; height:
90px;" id="sys-cpu-spark-container">
826         <div id="sys-cpu-spark" style="width: 100%; height
: 100%;"></div>
827         <div id="sys-cpu-spark-tooltip" class="sys-spark-t
ooltip" style="display: none;"></div>
828     </div>
829 </div>
830
831     <!-- Сворачиваемый блок детализации CPU: Ядра, потоки
и датчики системы (расположен под графиком истории) -->
832     <div class="sys-full-only sys-detailed-sensors mt-2">
833         <div class="sys-cpu-details-accordion-btn d-flex jus
tify-content-between align-items-center flex-wrap gap-2"
834             id="sys-cpu-details-toggle-btn"
835             title="Нажмите, чтобы развернуть или скрыть дет
ализацию по ядрам, потокам и сенсорам CPU"
836             onclick="window.toggleSysCpuDetails && window.t
oggleSysCpuDetails()">
837             <div class="d-flex align-items-center gap-2 flex-w
rap">
838                 <button type="button" class="panel-toggle-drill
collapsed me-1" id="sys-cpu-details-drill-btn" tabindex="-1" s
tyle="pointer-events: none;">
839                     
840                 </button>

```



```

841         <span class="fw-bold" style="font-size: 0.88rem;
      color: var(--text-color);">Детализация: Ядра, потоки и сенсор
      ы CPU</span>
842         <span class="badge bg-dark border border-seconda
      ry text-info font-monospace ms-1" style="font-size: 0.72rem;"
      id="sys-cores-count-badge">0 ядер</span>
843         <span class="badge bg-dark border border-seconda
      ry text-warning font-monospace" style="font-size: 0.72rem;" id
      ="sys-cpu-sensors-count-badge">0 параметров</span>
844     </div>
845 </div>
846
847 <!-- Скрытый по умолчанию общий контейнер деталей CP
      U -->
848 <div id="sys-cpu-details-collapse" style="display: n
      one; flex-direction: column; gap: 14px; margin-top: 10px;">
849     <!-- 1. Нагрузка и температура по ядрам и потокам
      -->
850     <div>
851         <div class="sys-core-section-title d-flex justif
      y-content-between align-items-center mb-2" style="font-size: 0
      .82rem;">
852             <span><i class="bi bi-cpu text-info me-1"></i>
      Нагрузка и температура по ядрам и потокам</span>
853         </div>
854         <div id="sys-metric-cpu-cores" style="display: g
      rid; grid-template-columns: repeat(auto-fill, minmax(300px, 1f
      r)); gap: 10px; font-size: 0.78rem;"></div>
855     </div>
856
857     <!-- 2. Общие показатели CPU (датчики системы) -->
858     <div>
859         <div class="sys-core-section-title d-flex justif
      y-content-between align-items-center mb-2" style="font-size: 0
      .82rem;">
860             <span><i class="bi bi-lightning-charge text-wa
      rning me-1"></i>Общие показатели CPU (датчики системы)</span>
861         </div>

```

```

862 |         <div id="sys-metric-cpu-sensors" style="display:
      | grid; grid-template-columns: repeat(auto-fill, minmax(180px,
      | 1fr)); gap: 8px; font-size: 0.78rem;">
863 |             <div class="text-muted small text-center py-2
      | col-12">Опрос сенсоров CPU...</div>
864 |         </div>
865 |     </div>
866 | </div>
867 | </div>
868 | </div>
869 | </div>
870 | </div>
871 |
872 | <!-- Четкий стильный разделитель панелей -->
873 | <div class="sys-section-divider"></div>
874 |
875 | <!-- 2. Панели Загрузки GPU (Для каждого найденного GPU созд
      | ается отдельная полноразмерная панель) -->
876 | <div id="sys-gpu-panels-container" class="d-flex flex-column
      | gap-3 mb-3">
877 |     <!-- GPU карточки динамически монтируются и обновляются из
      | main.js -->
878 | </div>
879 |
880 | <!-- Четкий стильный разделитель панелей -->
881 | <div class="sys-section-divider"></div>
882 |
883 | <!-- 3. Панель Память (RAM) и Дисковый ввод-вывод -->
884 | <div class="row mb-3">
885 |     <div class="col-12">
886 |         <div class="sys-card sys-card-ram p-3" id="sys-card-ram"
      | >
887 |             <div class="sys-card-header-bar">
888 |                 <div class="d-flex align-items-center gap-2">
889 |                     <span class="sys-label text-success" style="font-s
      | ize: 1.05rem; font-weight: 800;"><i class="bi bi-memory text-s
      | uccess me-1.5"></i>Память (RAM) и Дисковый ввод-вывод</span>
890 |                 </div>

```

```

891 |         <span class="small badge bg-dark border border-secon
      | dary text-light font-monospace" style="font-size: 0.84rem;" id
      | ="sys-memio-ts">--</span>
892 |     </div>
893 |
894 |     <!-- Верхний баннер: Модель слева, Gauge RAM по центру
      | , Слайдер файла подкачки справа -->
895 |     <div class="sys-card-top-banner p-3 rounded-3 mb-3">
896 |         <!-- Левая колонка: Модель RAM и экспресс-метрики KP
      | I -->
897 |         <div class="sys-banner-col-info d-flex flex-column a
      | lign-items-start justify-content-center">
898 |             <div class="text-start mb-1.5 d-flex align-items-c
      | enter">
899 |                 <button type="button" class="sys-drill-btn" id="
      | sys-drill-btn-ram" onclick="window.toggleSysCardDrill && windo
      | w.toggleSysCardDrill('sys-card-ram')" title="Drill-down: Разве
      | рнуть/скрыть все скрытые панели и датчики RAM">
900 |                     
901 |                 </button>
902 |                 <div>
903 |                     <span class="fw-bold text-success" style="font
      | -size: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-
      | shadow: 0 0 14px rgba(16, 185, 129, 0.4);" id="sys-metric-ram-
      | model">-- GB RAM</span>
904 |                     <div class="d-flex align-items-center gap-1.5
      | mt-0.5">
905 |                         <span class="badge bg-dark border border-sec
      | ondary text-light font-monospace" id="sys-ram-type-badge" styl
      | e="font-size: 0.70rem;">DDR4 · System Memory</span>
906 |                         <span class="badge bg-success-subtle text-su
      | ccess border border-success font-monospace" id="sys-ram-status
      | -badge" style="font-size: 0.70rem;">● Dual Channel</span>
907 |                     </div>
908 |                 </div>
909 |             </div>
910 |             <div class="sys-banner-kpi-grid">

```

```

911         <div class="sys-kpi-pill">
912             <span class="sys-kpi-label"><i class="bi bi-pi
e-chart text-success me-1"></i>Занято RAM</span>
913             <span class="sys-kpi-value text-success" id="s
ys-ram-kpi-used">-- GB</span>
914         </div>
915         <div class="sys-kpi-pill">
916             <span class="sys-kpi-label"><i class="bi bi-ch
eck2-circle text-info me-1"></i>Свободно</span>
917             <span class="sys-kpi-value text-info" id="sys-
ram-kpi-free">-- GB</span>
918         </div>
919         <div class="sys-kpi-pill">
920             <span class="sys-kpi-label"><i class="bi bi-ar
row-repeat text-warning me-1"></i>Swap Подкачка</span>
921             <span class="sys-kpi-value text-warning" id="s
ys-ram-kpi-swap">--%</span>
922         </div>
923         <div class="sys-kpi-pill">
924             <span class="sys-kpi-label"><i class="bi bi-hd
d-stack text-primary me-1"></i>Диск I/O</span>
925             <span class="sys-kpi-value text-primary" id="s
ys-ram-kpi-io">-- MB/s</span>
926         </div>
927     </div>
928 </div>
929 <!-- Центральная колонка: Большой Gauge RAM -->
930 <div class="sys-banner-col-gauge d-flex align-items-
center gap-2.5">
931     <div id="sys-metric-ram-gauge" style="width: 120px
; height: 68px; flex-shrink: 0;"></div>
932     <div>
933         <div class="sys-value text-success" id="sys-memi
o-pct" style="font-size: 2.1rem; line-height: 1; font-weight:
800;">0.0%</div>
934         <div class="text-light mt-1 font-monospace" styl
e="font-size: 0.82rem; font-weight: 700;" id="sys-memio-used">
0.0 / 0.0 GB</div>

```

```

935 |         </div>
936 |     </div>
937 |     <!-- Правая колонка: Файл подкачки со слайдером -->
938 |     <div class="sys-banner-col-specs d-flex flex-column
| justify-content-center">
939 |         <span class="text-light mb-1 font-monospace" style
| ="font-size: 0.80rem; font-weight: 700;" id="sys-memio-swap">Ф
| айл подкачки: 0%</span>
940 |         <div id="sys-metric-ram-slider" style="width: 100%
| ;"></div>
941 |     </div>
942 | </div>
943 |
944 | <!-- Сетка 4 блоков: RAM, Swap, Disk Read, Disk Write
| -->
945 | <div class="row g-2 mb-2 sys-full-only">
946 |     <div class="col-6 col-md-3">
947 |         <div class="sys-core-card h-100" title="Физическая
| оперативная память">
948 |             <div class="d-flex justify-content-between align
| -items-center mb-1">
949 |                 <span class="sys-core-title">Физическая RAM</s
| pan>
950 |                 <span class="sys-core-load-badge text-success
| border-success" id="sys-ram-sub-badge">0%</span>
951 |             </div>
952 |             <div class="py-1" id="sys-ram-sub-gauge"></div>
953 |             <div id="sys-ram-sub-slider"></div>
954 |         </div>
955 |     </div>
956 |
957 |     <div class="col-6 col-md-3">
958 |         <div class="sys-core-card h-100" title="Виртуальна
| я память / файл подкачки">
959 |             <div class="d-flex justify-content-between align
| -items-center mb-1">
960 |                 <span class="sys-core-title">Подкачка (Swap)</
| span>

```

```
961 |         <span class="sys-core-load-badgе text-warning
| border-warning" id="sys-swap-sub-badgе">0%</span>
962 |     </div>
963 |     <div class="py-1" id="sys-swap-sub-gaugе"></div>
964 |     <div id="sys-swap-sub-slider"></div>
965 | </div>
966 | </div>
967 |
968 |     <div class="col-6 col-md-3">
969 |         <div class="sys-core-card h-100" title="Скорость ч
| тения с диска">
970 |             <div class="d-flex justify-content-between align
| -items-center mb-1">
971 |                 <span class="sys-core-title">Чтение (Read)</sp
| an>
972 |                 <span class="sys-core-load-badgе text-info bor
| der-info" id="sys-read-sub-badgе">0 B/s</span>
973 |             </div>
974 |             <div class="py-1" id="sys-read-sub-gaugе"></div>
975 |             <div class="sys-temp-slider-wrap">
976 |                 <div class="d-flex justify-content-between ali
| gn-items-center mb-1" style="font-size: 0.78rem;">
977 |                     <span class="text-light fw-semibold"><i clas
| s="bi bi-arrow-down-circle me-1 text-info"></i>Чтение</span>
978 |                     <span class="font-monospace text-info fw-bol
| d" id="sys-memio-read">0 B/s</span>
979 |                 </div>
980 |                 <div class="sys-temp-slider-track">
981 |                     <div class="sys-temp-slider-bar" id="sys-rea
| d-sub-bar" style="width: 0%; background: #0dcaf0;"></div>
982 |                     <div class="sys-temp-slider-thumb" id="sys-r
| ead-sub-thumb" style="left: 0%; background: #0dcaf0;"></div>
983 |                 </div>
984 |             </div>
985 |         </div>
986 |     </div>
987 |
988 |     <div class="col-6 col-md-3">
```

```

989 |         <div class="sys-core-card h-100" title="Скорость з
| аписи на диск">
990 |             <div class="d-flex justify-content-between align
| -items-center mb-1">
991 |                 <span class="sys-core-title">Запись (Write)</s
| pan>
992 |                 <span class="sys-core-load-badge text-warning
| border-warning" id="sys-write-sub-badge">0 B/s</span>
993 |             </div>
994 |             <div class="py-1" id="sys-write-sub-gauge"></div
| >
995 |             <div class="sys-temp-slider-wrap">
996 |                 <div class="d-flex justify-content-between ali
| gn-items-center mb-1" style="font-size: 0.78rem;">
997 |                     <span class="text-light fw-semibold"><i clas
| s="bi bi-arrow-up-circle me-1 text-warning"></i>Запись</span>
998 |                     <span class="font-monospace text-warning fw-
| bold" id="sys-memio-write">0 B/s</span>
999 |                 </div>
1000 |                 <div class="sys-temp-slider-track">
1001 |                     <div class="sys-temp-slider-bar" id="sys-wri
| te-sub-bar" style="width: 0%; background: #f59e0b;"></div>
1002 |                     <div class="sys-temp-slider-thumb" id="sys-w
| rite-sub-thumb" style="left: 0%; background: #f59e0b;"></div>
1003 |                 </div>
1004 |             </div>
1005 |         </div>
1006 |     </div>
1007 | </div>
1008 |
1009 |     <!-- Параметры, тайминги и сенсоры RAM (Скрывается в к
| ompактном режиме) -->
1010 |     <div class="sys-full-only sys-detailed-sensors">
1011 |         <div class="sys-core-section-title d-flex justify-co
| ntent-between align-items-center mt-3">
1012 |             <span><i class="bi bi-lightning-charge text-warnin
| g me-1.5"></i>Параметры, тайминги и сенсоры RAM</span>
1013 |             <span class="badge bg-secondary font-monospace" st

```

```

yle="font-size: 0.76rem;" id="sys-ram-sensors-count-badge">0 п
а параметров</span>
1014     </div>
1015     <div id="sys-metric-ram-sensors" class="mb-2" style=
"display: grid; grid-template-columns: repeat(auto-fill, minma
x(180px, 1fr)); gap: 8px; font-size: 0.78rem;">
1016         <div class="text-muted small text-center py-2 col-
12">Опрос параметров RAM...</div>
1017     </div>
1018 </div>
1019
1020 <!-- График истории RAM -->
1021 <div class="sys-full-only sys-history-panel p-2.5 mt-3
rounded-2">
1022     <div class="d-flex justify-content-between align-ite
ms-center mb-1">
1023         <span class="small text-light" style="font-size: 0
.78rem; font-weight: 700;"><i class="bi bi-graph-up text-prima
ry me-1"></i>История загрузки (RAM % · Чтение · Запись)</span>
1024         <div class="d-flex align-items-center gap-2" style
="font-size: 0.72rem; font-weight: 600;">
1025             <span class="text-success">● RAM</span>
1026             <span class="text-info">● Read</span>
1027             <span class="text-warning">● Write</span>
1028         </div>
1029     </div>
1030     <div id="sys-memio-spark" style="height: 60px;"></di
v>
1031 </div>
1032 </div>
1033 </div>
1034 </div>
1035
1036 <!-- Четкий стильный разделитель панелей -->
1037 <div class="sys-section-divider"></div>
1038
1039 <!-- 4. Панель Сетевые адаптеры и беспроводная связь (Ethern
et / Wi-Fi / Bluetooth) -->

```



```

1040 | <div class="row mb-3">
1041 |     <div class="col-12">
1042 |         <div class="sys-card sys-card-net p-3" id="sys-card-net"
      | >
1043 |             <div class="sys-card-header-bar">
1044 |                 <div class="d-flex align-items-center gap-2">
1045 |                     <span class="sys-label text-info" style="font-size
      | : 1.05rem; font-weight: 800;"><i class="bi bi-hdd-network text
      | -info me-1.5"></i>Сетевые адаптеры и беспроводная связь</span>
1046 |                 </div>
1047 |                 <span class="small badge bg-dark border border-secon
      | dary text-light font-monospace" style="font-size: 0.84rem;" id
      | ="sys-net-ts">--</span>
1048 |             </div>
1049 |
1050 |             <!-- Верхний баннер: Модель слева, Gauge трафика по це
      | нтру, Слайдер линка справа -->
1051 |             <div class="sys-card-top-banner p-3 rounded-3 mb-3">
1052 |                 <!-- Левая колонка: Модель сети и экспресс-метрики K
      | PI -->
1053 |                 <div class="sys-banner-col-info d-flex flex-column a
      | lign-items-start justify-content-center">
1054 |                     <div class="text-start mb-1.5 d-flex align-items-c
      | enter">
1055 |                         <button type="button" class="sys-drill-btn" id="
      | sys-drill-btn-net" onclick="window.toggleSysCardDrill && windo
      | w.toggleSysCardDrill('sys-card-net')" title="Drill-down: Разве
      | рнуть/скрыть все скрытые панели и адаптеры сети">
1056 |                             
1057 |                         </button>
1058 |                         <div>
1059 |                             <span class="fw-bold text-info" style="font-si
      | ze: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-sha
      | dow: 0 0 14px rgba(13, 202, 240, 0.4);" id="sys-metric-net-mod
      | el">Ethernet (1000 Mbps) · Wi-Fi / BT</span>
1060 |                             <div class="d-flex align-items-center gap-1.5
      | mt-0.5">

```

```

1061 |         <span class="badge bg-dark border border-sec
      | ondary text-light font-monospace" id="sys-net-speed-badge" sty
      | le="font-size: 0.70rem;">1.0 Gbps Full Duplex</span>
1062 |         <span class="badge bg-info-subtle text-info
      | border border-info font-monospace" id="sys-net-status-badge" s
      | tyle="font-size: 0.70rem;">● Link Active</span>
1063 |     </div>
1064 | </div>
1065 | </div>
1066 | <div class="sys-banner-kpi-grid">
1067 |     <div class="sys-kpi-pill">
1068 |         <span class="sys-kpi-label"><i class="bi bi-ar
      | row-down-circle text-info me-1"></i>Входящий (Rx)</span>
1069 |         <span class="sys-kpi-value text-info" id="sys-
      | net-kpi-rx">0 B/s</span>
1070 |     </div>
1071 |     <div class="sys-kpi-pill">
1072 |         <span class="sys-kpi-label"><i class="bi bi-ar
      | row-up-circle text-warning me-1"></i>Исходящий (Tx)</span>
1073 |         <span class="sys-kpi-value text-warning" id="s
      | ys-net-kpi-tx">0 B/s</span>
1074 |     </div>
1075 |     <div class="sys-kpi-pill">
1076 |         <span class="sys-kpi-label"><i class="bi bi-hd
      | d-network text-primary me-1"></i>Всего сессия</span>
1077 |         <span class="sys-kpi-value text-primary" id="s
      | ys-net-kpi-total">0 B</span>
1078 |     </div>
1079 |     <div class="sys-kpi-pill">
1080 |         <span class="sys-kpi-label"><i class="bi bi-ac
      | tivity text-success me-1"></i>Утилизация</span>
1081 |         <span class="sys-kpi-value text-success" id="s
      | ys-net-kpi-util">0.0%</span>
1082 |     </div>
1083 | </div>
1084 | </div>
1085 | <!-- Центральная колонка: Большой Gauge трафика -->
1086 | <div class="sys-banner-col-gauge d-flex align-items-

```

```

center gap-2.5">
1087     <div id="sys-metric-net-gauge" style="width: 120px
; height: 68px; flex-shrink: 0;"></div>
1088     <div>
1089         <div class="sys-value text-info" id="sys-net-spe
ed-val" style="font-size: 2.1rem; line-height: 1; font-weight:
800;">0 B/s</div>
1090         <div class="text-light mt-1 font-monospace text-
truncate" style="font-size: 0.82rem; font-weight: 700; max-wid
th: 150px;" id="sys-net-sub-details">□ 0 B/s · □ 0 B/s</div>
1091     </div>
1092 </div>
1093 <!-- Правая колонка: Загрузка канала со слайдером --
>
1094     <div class="sys-banner-col-specs d-flex flex-column
justify-content-center">
1095         <span class="text-light mb-1 font-monospace" style
="font-size: 0.80rem; font-weight: 700;" id="sys-net-util-labe
l">Загрузка канала: 0%</span>
1096         <div id="sys-metric-net-slider" style="width: 100%
;"></div>
1097     </div>
1098 </div>
1099
1100 <!-- Сетка 4 блоков: Download, Upload, Wi-Fi, Bluetoot
h -->
1101     <div class="row g-2 mb-2 sys-full-only">
1102         <div class="col-6 col-md-3">
1103             <div class="sys-core-card h-100" title="Входящий с
етевой трафик (Download / Rx)">
1104                 <div class="d-flex justify-content-between align
-items-center mb-1">
1105                     <span class="sys-core-title">Входящий (Rx)</sp
an>
1106                     <span class="sys-core-load-badge text-info bor
der-info" id="sys-rx-sub-badge">0 B/s</span>
1107                 </div>
1108                 <div class="py-1" id="sys-rx-sub-gauge"></div>

```

```
1109         <div class="sys-temp-slider-wrap">
1110             <div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
1111                 <span class="text-light fw-semibold"><i class="bi bi-arrow-down-circle me-1 text-info"></i>Скачивание</span>
1112                 <span class="font-monospace text-info fw-bold text-truncate" style="max-width: 110px;" id="sys-rx-total-text">0 B</span>
1113             </div>
1114             <div class="sys-temp-slider-track">
1115                 <div class="sys-temp-slider-bar" id="sys-rx-sub-bar" style="width: 0%; background: #0dcaf0;"></div>
1116                 <div class="sys-temp-slider-thumb" id="sys-rx-sub-thumb" style="left: 0%; background: #0dcaf0;"></div>
1117             </div>
1118         </div>
1119     </div>
1120 </div>
1121
1122     <div class="col-6 col-md-3">
1123         <div class="sys-core-card h-100" title="Исходящий сетевой трафик (Upload / Tx)">
1124             <div class="d-flex justify-content-between align-items-center mb-1">
1125                 <span class="sys-core-title">Исходящий (Tx)</span>
1126                 <span class="sys-core-load-badge text-warning border-warning" id="sys-tx-sub-badge">0 B/s</span>
1127             </div>
1128             <div class="py-1" id="sys-tx-sub-gauge"></div>
1129             <div class="sys-temp-slider-wrap">
1130                 <div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
1131                     <span class="text-light fw-semibold"><i class="bi bi-arrow-up-circle me-1 text-warning"></i>Отдача</span>
1132                     <span class="font-monospace text-warning fw-bold text-truncate" style="max-width: 110px;" id="sys-tx-total
```

```

    -text">0 B</span>
1133         </div>
1134         <div class="sys-temp-slider-track">
1135             <div class="sys-temp-slider-bar" id="sys-tx-
sub-bar" style="width: 0%; background: #f59e0b;"></div>
1136             <div class="sys-temp-slider-thumb" id="sys-t
x-sub-thumb" style="left: 0%; background: #f59e0b;"></div>
1137         </div>
1138     </div>
1139 </div>
1140 </div>
1141
1142     <div class="col-6 col-md-3">
1143         <div class="sys-core-card h-100" title="Беспроводн
ой адаптер Wi-Fi">
1144             <div class="d-flex justify-content-between align
-items-center mb-1">
1145                 <span class="sys-core-title">Wi-Fi Адаптер</sp
an>
1146                 <span class="sys-core-load-badge text-primary
border-primary" id="sys-wifi-sub-badge">0n</span>
1147             </div>
1148             <div class="py-1" id="sys-wifi-sub-gauge"></div>
1149             <div class="sys-temp-slider-wrap">
1150                 <div class="d-flex justify-content-between ali
gn-items-center mb-1" style="font-size: 0.78rem;">
1151                     <span class="text-light fw-semibold"><i clas
s="bi bi-wifi me-1 text-primary"></i>Статус</span>
1152                     <span class="font-monospace text-primary fw-
bold text-truncate" style="max-width: 110px;" id="sys-wifi-sta
tus-text">Disconnected</span>
1153                 </div>
1154                 <div class="sys-temp-slider-track">
1155                     <div class="sys-temp-slider-bar" id="sys-wif
i-sub-bar" style="width: 0%; background: #38bdf8;"></div>
1156                     <div class="sys-temp-slider-thumb" id="sys-w
ifi-sub-thumb" style="left: 0%; background: #38bdf8;"></div>
1157                 </div>

```

```

1158         </div>
1159     </div>
1160 </div>
1161
1162     <div class="col-6 col-md-3">
1163         <div class="sys-core-card h-100" title="Bluetooth
1164 | контроллер и устройства">
1165             <div class="d-flex justify-content-between align
1166 | -items-center mb-1">
1167                 <span class="sys-core-title">Bluetooth (BT)</s
1168 | pan>
1169                 <span class="sys-core-load-badge text-success
1170 | border-success" id="sys-bt-sub-badge">Active</span>
1171             </div>
1172             <div class="py-1" id="sys-bt-sub-gauge"></div>
1173             <div class="sys-temp-slider-wrap">
1174                 <div class="d-flex justify-content-between ali
1175 | gn-items-center mb-1" style="font-size: 0.78rem;">
1176                     <span class="text-light fw-semibold"><i clas
1177 | s="bi bi-bluetooth me-1 text-success"></i>Устройства</span>
1178                     <span class="font-monospace text-success fw-
1179 | bold text-truncate" style="max-width: 110px;" id="sys-bt-devic
1180 | es-text">0 устр.</span>
1181                 </div>
1182                 <div class="sys-temp-slider-track">
1183                     <div class="sys-temp-slider-bar" id="sys-bt-
1184 | sub-bar" style="width: 0%; background: #10b981;"></div>
1185                     <div class="sys-temp-slider-thumb" id="sys-b
1186 | t-sub-thumb" style="left: 0%; background: #10b981;"></div>
1187                 </div>
1188             </div>
1189         </div>
1190     </div>
1191
1192     <!-- Параметры, адаптеры и сенсоры сети (Скрывается в
1193 | компактном режиме) -->
1194     <div class="sys-full-only sys-detailed-sensors">

```

```
1185 |         <div class="sys-core-section-title d-flex justify-co
| ntent-between align-items-center mt-3">
1186 |             <span><i class="bi bi-lightning-charge text-warnin
| g me-1.5"></i>Параметры, адаптеры и сенсоры сети / Wi-Fi / BT<
| /span>
1187 |             <span class="badge bg-secondary font-monospace" st
| yle="font-size: 0.76rem;" id="sys-net-sensors-count-badge">0 п
| араметров</span>
1188 |         </div>
1189 |         <div id="sys-metric-net-sensors" class="mb-2" style=
| "display: grid; grid-template-columns: repeat(auto-fill, minma
| x(180px, 1fr)); gap: 8px; font-size: 0.78rem;">
1190 |             <div class="text-muted small text-center py-2 col-
| 12">Опрос сетевых интерфейсов...</div>
1191 |         </div>
1192 |     </div>
1193 |
1194 |     <!-- График истории сети -->
1195 |     <div class="sys-full-only sys-history-panel p-2.5 mt-3
| rounded-2">
1196 |         <div class="d-flex justify-content-between align-ite
| ms-center mb-1">
1197 |             <span class="small text-light" style="font-size: 0
| .78rem; font-weight: 700;"><i class="bi bi-graph-up text-info
| me-1"></i>История сети (▢ Download · ▢ Upload)</span>
1198 |             <div class="d-flex align-items-center gap-2" style
| ="font-size: 0.72rem; font-weight: 600;">
1199 |                 <span class="text-info">● Download</span>
1200 |                 <span class="text-warning">● Upload</span>
1201 |             </div>
1202 |         </div>
1203 |         <div id="sys-net-spark" style="height: 60px;"></div>
1204 |     </div>
1205 | </div>
1206 | </div>
1207 | </div>
1208 |
1209 | <!-- Четкий стильный разделитель панелей -->
```

```
1210 | <div class="sys-section-divider"></div>
1211 |
1212 | <!-- 5. Панель Параметры накопителей (Диски SSD / HDD / NVMe
    | ) -->
1213 | <div class="row mb-3">
1214 |     <div class="col-12">
1215 |         <div class="sys-card sys-card-storage p-3" id="sys-card-
    | storage">
1216 |             <div class="sys-card-header-bar">
1217 |                 <div class="d-flex align-items-center gap-2">
1218 |                     <span class="sys-label text-info" style="font-size
    | : 1.05rem; font-weight: 800;"><i class="bi bi-hdd-fill text-in
    | fo me-1.5"></i>Параметры накопителей (Диски SSD / HDD / NVMe)<
    | /span>
1219 |                     <span class="badge bg-info-subtle text-info border
    | border-info" style="font-size: 0.74rem;" id="sys-storage-driv
    | es-badge">0 дисков</span>
1220 |                     <span class="badge bg-dark border border-secondary
    | text-light" style="font-size: 0.74rem;" id="sys-storage-parts
    | -badge">0 разделов</span>
1221 |                 </div>
1222 |                 <span class="small badge bg-dark border border-secon
    | dary text-light font-monospace" style="font-size: 0.84rem;" id
    | ="sys-storage-ts">Live</span>
1223 |             </div>
1224 |
1225 | <!-- Верхний баннер: Модель слева, Gauge общего объема
    | по центру, Слайдер макс t° справа -->
1226 | <div class="sys-card-top-banner p-3 rounded-3 mb-3">
1227 |     <!-- Левая колонка: Модель хранилища и экспресс-метр
    | ики KPI -->
1228 |         <div class="sys-banner-col-info d-flex flex-column a
    | lign-items-start justify-content-center">
1229 |             <div class="text-start mb-1.5 d-flex align-items-c
    | enter">
1230 |                 <button type="button" class="sys-drill-btn" id="
    | sys-drill-btn-storage" onclick="window.toggleSysCardDrill && w
    | indow.toggleSysCardDrill('sys-card-storage')" title="Drill-dow
```



```
n: Развернуть/скрыть все скрытые панели накопителей">
1231         
1233         </button>
1234         <div>
1235             <span class="fw-bold text-info" style="font-si
1236             ze: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-sha
1237             dow: 0 0 14px rgba(13, 202, 240, 0.4);" id="sys-metric-storage
1238             -model">Дисковая подсистема и накопители</span>
1239             <div class="d-flex align-items-center gap-1.5
1240             mt-0.5">
1241                 <span class="badge bg-dark border border-sec
1242                 ondary text-light font-monospace" id="sys-storage-summary-badg
1243                 e-top" style="font-size: 0.70rem;">NVMe / SSD / HDD</span>
1244                 <span class="badge bg-success-subtle text-su
1245                 ccess border border-success font-monospace" id="sys-storage-st
1246                 atus-badge" style="font-size: 0.70rem;">● SMART OK</span>
1247             </div>
1248         </div>
1249     </div>
1250     <div class="sys-banner-kpi-grid">
1251         <div class="sys-kpi-pill">
1252             <span class="sys-kpi-label"><i class="bi bi-pi
1253             e-chart text-info me-1"></i>Занято всего</span>
1254             <span class="sys-kpi-value text-info" id="sys-
1255             storage-kpi-used">0.0 GB</span>
1256         </div>
1257         <div class="sys-kpi-pill">
1258             <span class="sys-kpi-label"><i class="bi bi-ch
1259             eck2-circle text-success me-1"></i>Свободно</span>
1260             <span class="sys-kpi-value text-success" id="s
1261             ys-storage-kpi-free">0.0 GB</span>
1262         </div>
1263         <div class="sys-kpi-pill">
1264             <span class="sys-kpi-label"><i class="bi bi-hd
1265             d text-primary me-1"></i>Диски/Разделы</span>
1266             <span class="sys-kpi-value text-primary" id="s
1267             ys-storage-kpi-drives">-- / --</span>
```

```
1253 |         </div>
1254 |         <div class="sys-kpi-pill">
1255 |             <span class="sys-kpi-label"><i class="bi bi-thermometer-half text-danger me-1"></i>Макс. t°</span>
1256 |             <span class="sys-kpi-value text-warning" id="sys-storage-kpi-temp">-- °C</span>
1257 |         </div>
1258 |     </div>
1259 | </div>
1260 | <!-- Центральная колонка: Большой Gauge общего объема -->
1261 | <div class="sys-banner-col-gauge d-flex align-items-center gap-2.5">
1262 |     <div id="sys-metric-storage-gauge" style="width: 120px; height: 68px; flex-shrink: 0;"></div>
1263 |     <div>
1264 |         <div class="sys-value text-info" id="sys-storage-pct" style="font-size: 2.1rem; line-height: 1; font-weight: 800;">0.0%</div>
1265 |         <div class="text-light mt-1 font-monospace text-truncate" style="font-size: 0.82rem; font-weight: 700; max-width: 150px;" id="sys-storage-used">0.0 / 0.0 GB</div>
1266 |     </div>
1267 | </div>
1268 | <!-- Правая колонка: Слайдер максимальной t° -->
1269 | <div class="sys-banner-col-specs d-flex flex-column justify-content-center">
1270 |     <span class="text-light mb-1 font-monospace" style="font-size: 0.80rem; font-weight: 700;" id="sys-storage-temp-label">Макс. температура: -- °C</span>
1271 |     <div id="sys-metric-storage-temp-slider" style="width: 100%;"></div>
1272 | </div>
1273 | </div>
1274 |
1275 | <!-- Сетка карточек накопителей и разделов -->
1276 | <div class="sys-full-only">
1277 |     <div class="sys-core-section-title d-flex justify-co
```

```

ntent-between align-items-center">
1278     <span><i class="bi bi-hdd-network text-info me-1.5
"></i>Физические накопители и разделы</span>
1279     <span class="badge bg-secondary font-monospace" st
yle="font-size: 0.76rem;" id="sys-storage-summary-badg
e">Автом
атический опрос</span>
1280     </div>
1281     <div id="sys-metric-storage-cards" style="display: g
rid; grid-template-columns: repeat(auto-fill, minmax(240px, 1f
r)); gap: 10px; font-size: 0.78rem;">
1282         <div class="text-muted small text-center py-3 col-
12">Опрос накопителей выполняется...</div>
1283     </div>
1284 </div>
1285 </div>
1286 </div>
1287 </div>
1288
1289 <!-- Четкий стильный разделитель панелей -->
1290 <div class="sys-section-divider sys-full-only"></div>
1291
1292 <!-- Extended Hardware & Health Status Strip -->
1293 <div class="row g-2 mb-2 sys-full-only">
1294     <!-- Physical Storage & NVMe Health -->
1295     <div class="col-md-6 col-lg-3">
1296         <div class="sys-panel p-2 h-100">
1297             <div class="d-flex justify-content-between align-items
-center mb-1">
1298                 <span class="small fw-bold" style="font-size: 0.76re
m; color: var(--text-color);"><i class="bi bi-hdd-network text
-info me-1"></i>Здоровье дисков (NVMe/SSD)</span>
1299                 <span class="badge bg-success-subtle text-success bo
rder border-success" style="font-size: 0.65rem;" id="sys-disk-
health-badg
e">SMART OK</span>
1300             </div>
1301             <div id="sys-physical-disks-container" class="small te
xt-muted" style="font-size: 0.72rem; line-height: 1.3; max-hei
ght: 120px; overflow-y: auto;">

```

```

1302         Опрос состояния физических дисков...
1303     </div>
1304 </div>
1305 </div>
1306
1307 <!-- Power & Battery -->
1308 <div class="col-md-6 col-lg-3">
1309     <div class="sys-panel p-2 h-100">
1310         <div class="d-flex justify-content-between align-items
1311         -center mb-1">
1312             <span class="small fw-bold" style="font-size: 0.76re
1313             m; color: var(--text-color);"><i class="bi bi-battery-charging
1314             text-warning me-1"></i> Питание и Энергия</span>
1315             <span class="badge bg-secondary" style="font-size: 0
1316             .65rem;" id="sys-power-status-badge">AC Mains</span>
1317         </div>
1318         <div id="sys-power-info-container" class="small text-m
1319         uted" style="font-size: 0.72rem; line-height: 1.3;">
1320             Питание от электросети (Стационарный ПК)
1321         </div>
1322     </div>
1323 </div>
1324
1325 <!-- RAM SPD Sticks -->
1326 <div class="col-md-6 col-lg-3">
1327     <div class="sys-panel p-2 h-100">
1328         <div class="d-flex justify-content-between align-items
1329         -center mb-1">
1330             <span class="small fw-bold" style="font-size: 0.76re
1331             m; color: var(--text-color);"><i class="bi bi-memory text-prim
1332             ary me-1"></i> Модули RAM (SPD)</span>
1333             <span class="badge bg-dark border border-secondary t
1334             ext-info" style="font-size: 0.65rem;" id="sys-ram-sticks-badg
1335             e">Слоты</span>
1336         </div>
1337         <div id="sys-ram-sticks-container" class="small text-m
1338         uted" style="font-size: 0.72rem; line-height: 1.3;">
1339             Опрос конфигурации планок памяти...

```

```
1329 |         </div>
1330 |     </div>
1331 | </div>
1332 |
1333 | <!-- Reliability & Open Ports -->
1334 | <div class="col-md-6 col-lg-3">
1335 |     <div class="sys-panel p-2 h-100">
1336 |         <div class="d-flex justify-content-between align-items
1337 | -center mb-1">
1338 |             <span class="small fw-bold" style="font-size: 0.76re
1339 | m; color: var(--text-color);"><i class="bi bi-shield-check tex
1340 | t-success me-1"></i> Стабильность и Сеть</span>
1341 |             <span class="badge bg-info-subtle text-info border b
1342 | order-info" style="font-size: 0.65rem;" id="sys-alerts-badge">
1343 | Стабильно</span>
1344 |         </div>
1345 |         <div id="sys-alerts-ports-container" class="small text
1346 | -muted" style="font-size: 0.72rem; line-height: 1.3;">
1347 |             Службы активны. Ожидание телеметрии портов.
1348 |         </div>
1349 |     </div>
1350 | </div>
1351 | </div>
1352 | <!-- Telemetry & Polling Intervals Editor Modal (config.json)
1353 | -->
1354 | <div class="modal fade" id="sysIntervalsModal" tabindex="-1" a
1355 | ria-labelledby="sysIntervalsModalLabel" aria-hidden="true">
1356 |     <div class="modal-dialog modal-dialog-centered modal-lg moda
1357 | l-dialog-scrollable">
1358 |         <div class="modal-content text-white border-secondary shad
1359 | ow-lg" style="background: var(--surface-1); border-color: var(
1360 | --border-color) !important;">
1361 |             <div class="modal-header py-2.5 px-3 border-secondary" s
1362 | tyle="background: var(--surface-2); border-color: var(--border
1363 | -color) !important;">
1364 |                 <h6 class="modal-title d-flex align-items-center gap-2
```

```

    mb-0" id="sysIntervalsModalLabel">
1354         <i class="bi bi-stopwatch text-primary"></i>
1355         <span class="fw-bold">Редактор интервалов опроса ре
        урсов</span>
1356         <span class="badge bg-secondary font-monospace" id="
        modal-intervals-cfg-file">config.json</span>
1357     </h6>
1358     <button type="button" class="btn-close btn-close-white
        " data-bs-dismiss="modal" aria-label="Close"></button>
1359 </div>
1360
1361     <div class="modal-body p-3" style="max-height: 75vh; ove
        rflow-y: auto;">
1362         <!-- Alert message -->
1363         <div id="modal-intervals-alert" class="alert d-none py
        -2 px-3 small mb-3"></div>
1364
1365         <!-- Engine Status Strip -->
1366         <div class="p-2.5 rounded mb-3 d-flex align-items-cent
        er justify-content-between flex-wrap gap-2" style="background:
        var(--surface-2); border: 1px solid var(--border-color);">
1367             <div class="d-flex align-items-center gap-2">
1368                 <span class="fs-4">⏱ </span>
1369             </div>
1370             <div class="fw-bold small" style="color: var(--t
        ext-color);">Служба автологгирования и опроса телеметрии</div>
1371             <div class="small text-muted" style="font-size:
        0.73rem;">Интервалы сохраняются напрямую в активный файл конфи
        гурации и применяются без перезапуска сервера</div>
1372         </div>
1373     </div>
1374     <div class="d-flex align-items-center gap-2">
1375         <span class="badge bg-success" id="modal-intervals
        -engine-status">AutoLog Active</span>
1376         <div class="form-check form-switch m-0">
1377             <input class="form-check-input" type="checkbox"
        id="modal-intervals-global-enable" checked>
1378             <label class="form-check-label small fw-semibold

```

```

" for="modal-intervals-global-enable">Автологгирование</label>
1379     </div>
1380     </div>
1381 </div>
1382
1383     <!-- Section 1: Live UI Refresh Interval -->
1384     <div class="card border-secondary mb-3" style="backgro
und: var(--surface-2); border-color: var(--border-color) !impo
rtant;">
1385         <div class="card-header py-1.5 px-3 border-secondary
d-flex align-items-center justify-content-between" style="bor
der-color: var(--border-color) !important;">
1386             <span class="small fw-bold text-info"><i class="bi
bi-display me-1"></i> Интервал обновления Live-интерфейса вкл
адки (UI)</span>
1387             <span class="badge bg-dark text-info border border
-secondary" id="modal-ui-refresh-badge">5 сек</span>
1388         </div>
1389         <div class="card-body p-3">
1390             <div class="row g-2 align-items-center">
1391                 <div class="col-sm-7">
1392                     <label class="form-label small text-muted mb-0
">Частота опроса сенсоров и оборудования в открытом окне брауз
ера:</label>
1393                 </div>
1394                 <div class="col-sm-5">
1395                     <select class="form-select form-select-sm bg-d
ark text-white border-secondary" id="modal-ui-refresh-select">
1396                         <option value="1">1 секунда (Высокая частота
)</option>
1397                         <option value="2">2 секунды</option>
1398                         <option value="3">3 секунды</option>
1399                         <option value="5" selected>5 секунд (По умол
чанию)</option>
1400                         <option value="10">10 секунд (Экономный режи
м)</option>
1401                         <option value="30">30 секунд</option>
1402                     </select>

```

```
1403         </div>
1404     </div>
1405 </div>
1406 </div>
1407
1408     <!-- Section 2: Core Resource Loggers (config.json) --
>
1409     <div class="card border-secondary mb-3" style="background: var(--surface-2); border-color: var(--border-color) !important;">
1410         <div class="card-header py-1.5 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
1411             <span class="small fw-bold text-primary"><i class="bi bi-hdd-stack me-1"></i> Интервалы опроса основных сенсоров и песчпсов (config.json)</span>
1412             <span class="badge bg-secondary font-monospace" style="font-size: 0.68rem;">logging.loggers</span>
1413         </div>
1414         <div class="card-body p-3">
1415             <div class="d-flex flex-column gap-2.5" id="modal-core-loggers-container">
1416                 <!-- Dynamically populated core loggers -->
1417             </div>
1418         </div>
1419     </div>
1420
1421     <!-- Section 3: Other Services Loggers Collapsible -->
1422     <div class="card border-secondary mb-3" style="background: var(--surface-2); border-color: var(--border-color) !important;">
1423         <div class="card-header py-1.5 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
1424             <a class="small fw-bold text-light text-decoration-none d-flex align-items-center gap-1" data-bs-toggle="collapse" href="#modal-other-loggers-collapse" role="button" aria-expanded="false">
```



```

1425         <i class="bi bi-chevron-down"></i> Все остальные
        логиеры и службы системы (<span id="modal-other-loggers-count
        ">0</span>)
1426     </a>
1427     <span class="small text-muted" style="font-size: 0
        .72rem;">Windows SysAdmin, Defender, StartUp, Net...</span>
1428 </div>
1429 <div class="collapse" id="modal-other-loggers-collap
        se">
1430     <div class="card-body p-3 border-top border-second
        ary" style="border-color: var(--border-color) !important;">
1431         <div class="d-flex flex-column gap-2" id="modal-
        other-loggers-container">
1432             <!-- Dynamically populated other loggers -->
1433         </div>
1434     </div>
1435 </div>
1436 </div>
1437
1438 <!-- Section 4: Default fallback interval -->
1439 <div class="p-2.5 rounded d-flex align-items-center ju
        stify-content-between flex-wrap gap-2" style="background: var(
        --bg-color); border: 1px solid var(--border-color);">
1440     <div class="d-flex align-items-center gap-2">
1441         <i class="bi bi-clock-history text-muted"></i>
1442         <span class="small" style="color: var(--text-color
        );">Интервал по умолчанию (default_interval):</span>
1443     </div>
1444     <div style="min-width: 170px;">
1445         <select class="form-select form-select-sm bg-dark
        text-white border-secondary" id="modal-intervals-default-selec
        t">
1446             <option value="5 seconds">5 seconds (5 сек)</opt
        ion>
1447             <option value="10 seconds">10 seconds (10 сек)</
        option>
1448             <option value="30 seconds">30 seconds (30 сек)</
        option>

```

```
1449 |         <option value="1 minute" selected>1 minute (1 ми  
      | н)</option>  
1450 |         <option value="5 minutes">5 minutes (5 мин)</opt  
      | ion>  
1451 |         <option value="10 minutes">10 minutes (10 мин)</  
      | option>  
1452 |         <option value="1 hour">1 hour (1 час)</option>  
1453 |     </select>  
1454 | </div>  
1455 | </div>  
1456 | </div>  
1457 |  
1458 |     <div class="modal-footer py-2 px-3 border-secondary d-fl  
      | ex justify-content-between" style="background: var(--surface-2  
      | ); border-color: var(--border-color) !important;">  
1459 |         <button class="btn btn-sm btn-outline-secondary rounde  
      | d-pill px-3" id="btn-modal-intervals-refresh" title="Перезагру  
      | зить текущую конфигурацию из файла">  
1460 |             <i class="bi bi-arrow-clockwise me-1"></i> Обновить  
1461 |         </button>  
1462 |         <div class="d-flex gap-2">  
1463 |             <button class="btn btn-sm btn-secondary rounded-pill  
      | px-3" data-bs-dismiss="modal">Отмена</button>  
1464 |             <button class="btn btn-sm btn-primary rounded-pill p  
      | x-4 fw-bold" id="btn-modal-intervals-save">  
1465 |                 <i class="bi bi-check2-circle me-1"></i> Сохранить  
      | в config.json  
1466 |             </button>  
1467 |         </div>  
1468 |     </div>  
1469 | </div>  
1470 | </div>  
1471 | </div>  
1472 |  
1473 |  
1474 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows System Inspector Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/system_inspector_tab/main.js?v=20261001_v1" type="module"></script>
11 *
12 * File: main.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/system_inspector_tab
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-08 03:51:00
18 * =====
19 */
20
21 // System & Hardware Inspector Tab JS Module
22 (function() {
23   let sysWs = null;
24   let isSysInitialized = false;
25   let isSysPaused = false;
26   let isNetPaused = false;
27   let currentNetFilter = 'internet';
28   let cachedNetworkActivities = [];
29   let latestTelemetrySnapshot = null;
30   let cachedSensors = [];
31   let currentSensorCategory = 'all';
32   let isLhmRunning = false;
```

```

33 | let cachedHardwareSensorsData = null;
34 | let lastCpuCoresData = [];
35 | let lastCpuSensorsData = [];
36 | let lastCpuData = null;
37 | let lastGpuData = null;
38 | let lastGpuDevices = [];
39 | const _gpuSparkHistories = {};
40 | const currentGpuSparkScales = {};
41 | const lastGpuSparkDataMap = {};
42 |
43 | async function fetchLhmSensors() {
44 |     const badgeCount = document.getElementById('sys-lhm-sensor
| s-count');
45 |     const badgeStatus = document.getElementById('sys-lhm-statu
| s-badge');
46 |     const container = document.getElementById('sys-lhm-sensors
| -container');
47 |     const maxTempLabel = document.getElementById('sys-sensors-
| max-temp');
48 |     const warningsBadge = document.getElementById('sys-sensors
| -warnings-badge');
49 |     const dbTsBadge = document.getElementById('sys-sensors-db-
| ts');
50 |
51 |     try {
52 |         const res = await fetch('/api/v1/panel/hardware-sensors'
| );
53 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
54 |         const data = await res.json();
55 |         cachedHardwareSensorsData = data;
56 |         cachedSensors = data.all_sensors || [];
57 |
58 |         const summary = data.summary || {};
59 |
60 |         if (badgeCount) {
61 |             badgeCount.textContent = `${summary.total_sensors || c
| achedSensors.length} шт. (БД)`;
62 |         }

```

```

63 |         if (badgeStatus) {
64 |             badgeStatus.textContent = '● telemetry.db';
65 |             badgeStatus.className = 'badge bg-success';
66 |             if (summary.timestamp) {
67 |                 const tsStr = new Date(summary.timestamp).toLocaleTi
meString();
68 |                 badgeStatus.title = `Данные из базы telemetry.db: ${
tsStr}`;
69 |                 if (dbTsBadge) dbTsBadge.textContent = tsStr;
70 |             }
71 |         }
72 |         if (maxTempLabel) {
73 |             maxTempLabel.textContent = summary.max_temperature_c !
= null ? `${summary.max_temperature_c.toFixed(0)} °C` : '-- °C
';
74 |             if (summary.max_temperature_c && summary.max_temperatu
re_c >= 75) {
75 |                 maxTempLabel.className = 'fw-bold text-danger font-m
onospace';
76 |             } else if (summary.max_temperature_c && summary.max_te
mperature_c >= 60) {
77 |                 maxTempLabel.className = 'fw-bold text-warning font-
monospace';
78 |             } else {
79 |                 maxTempLabel.className = 'fw-bold text-info font-mon
ospace';
80 |             }
81 |         }
82 |         if (warningsBadge) {
83 |             warningsBadge.textContent = summary.warnings_count ||
'0';
84 |             warningsBadge.className = summary.warnings_count > 0 ?
'badge bg-danger text-white' : 'badge bg-dark border border-s
econdary text-success';
85 |         }
86 |
87 |         renderSensors();
88 |     } catch (e) {

```

```

89     console.warn('[SystemInspectorTab] Failed to fetch hardw
are sensors from DB:', e);
90     if (container) {
91         container.innerHTML = `<div class="text-center py-4 te
xt-muted small">Датчики опрашиваются из БД... (${e.message})</
div>`;
92     }
93 }
94 }
95
96 function createSensorWidgetHtml(s) {
97     const cat = (s.sensor_category || '').toLowerCase();
98     const val = Number(s.value || 0);
99     const valRaw = s.value_raw || `${val.toFixed(1)} ${s.unit
|| ''}`.trim();
100     const name = escapeHtml(s.sensor_name);
101
102     if (cat.includes('temp')) {
103         const tempSlider = createTempSliderHtml(val, 25, 95, fal
se);
104         const gaugeSvg = createGaugeSvg(Math.min(100, Math.max(0
, ((val - 25) / 70) * 100)), 64, 38);
105         return `
106         <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-
1" style="background: rgba(255,255,255,0.02); border: 1px soli
d var(--border-subtle, rgba(255,255,255,0.06));">
107             <div class="d-flex justify-content-between align-ite
ms-center mb-1">
108                 <span class="sys-sensor-name text-truncate" title=
"${name}" style="font-size: 0.74rem; font-weight: 600;">${name
}</span>
109                 <span class="badge ${val >= 80 ? 'bg-danger' : val
>= 65 ? 'bg-warning text-dark' : 'bg-info-subtle text-info'}"
style="font-size: 0.65rem;">${val.toFixed(0)}°C</span>
110             </div>
111             <div class="d-flex align-items-center gap-2">
112                 <div style="width: 64px; height: 38px; flex-shrink
: 0;">${gaugeSvg}</div>

```

```

113         <div class="flex-grow-1">${tempSlider}</div>
114     </div>
115 </div>
116 `;
117 }
118
119 if (cat.includes('load')) {
120     const pctSlider = createPercentSliderHtml(val, false, `
121     {val.toFixed(0)}%`);
122     const gaugeSvg = createGaugeSvg(val, 64, 38);
123     return `
124     <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-
125     1" style="background: rgba(255,255,255,0.02); border: 1px soli
126     d var(--border-subtle, rgba(255,255,255,0.06));">
127         <div class="d-flex justify-content-between align-ite
128         ms-center mb-1">
129             <span class="sys-sensor-name text-truncate" title=
130             "${name}" style="font-size: 0.74rem; font-weight: 600;">${name
131             }</span>
132             <span class="badge ${val >= 85 ? 'bg-danger' : val
133             >= 60 ? 'bg-warning text-dark' : 'bg-success-subtle text-succ
134             ess'}" style="font-size: 0.65rem;">${val.toFixed(0)}%</span>
135         </div>
136         <div class="d-flex align-items-center gap-2">
137             <div style="width: 64px; height: 38px; flex-shrink
138             : 0;">${gaugeSvg}</div>
139             <div class="flex-grow-1">${pctSlider}</div>
140         </div>
141     </div>
142 `;
143 }
144
145 if (cat.includes('fan')) {
146     const pct = Math.min(100, Math.max(0, (val / 2800) * 100
147     ));
148     return `
149     <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-
150     1" style="background: rgba(255,255,255,0.02); border: 1px soli

```

```

    d var(--border-subtle, rgba(255,255,255,0.06));">
140         <div class="d-flex justify-content-between align-items-center mb-1">
141             <span class="sys-sensor-name text-truncate" title=
"$${name}" style="font-size: 0.74rem; font-weight: 600;"><i class="bi bi-fan text-info me-1"></i>${name}</span>
142             <span class="font-monospace text-info fw-bold" style="font-size: 0.76rem;">${val.toFixed(0)} RPM</span>
143         </div>
144         <div class="progress" style="height: 5px; background: rgba(255,255,255,0.1); border-radius: 3px;">
145             <div class="progress-bar bg-info" style="width: ${pct}%;"></div>
146         </div>
147     </div>
148     `;
149 }
150
151     if (cat.includes('volt') || cat.includes('power')) {
152         const isPower = cat.includes('power') || (s.unit || '').toLowerCase() === 'w';
153         return `
154             <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-1" style="background: rgba(255,255,255,0.02); border: 1px solid var(--border-subtle, rgba(255,255,255,0.06));">
155                 <div class="d-flex justify-content-between align-items-center">
156                     <span class="sys-sensor-name text-truncate" title=
"$${name}" style="font-size: 0.74rem; font-weight: 600;"><i class="bi bi-lightning-charge text-warning me-1"></i>${name}</span>
157                     <span class="badge ${isPower ? 'bg-warning-subtle text-warning' : 'bg-primary-subtle text-primary'} font-monospace" style="font-size: 0.72rem;">${valRaw}</span>
158                 </div>
159             </div>
160         `;
161     }

```



```

162
163     if (cat.includes('clock')) {
164         return `
165             <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-
166             1" style="background: rgba(255,255,255,0.02); border: 1px solid
167             var(--border-subtle, rgba(255,255,255,0.06));">
168                 <div class="d-flex justify-content-between align-items-center">
169                     <span class="sys-sensor-name text-truncate" title=
170                     "${name}" style="font-size: 0.74rem; font-weight: 600;"><i class="bi bi-speedometer2 text-primary me-1"></i>${name}</span>
171                     <span class="font-monospace text-primary fw-bold"
172                     style="font-size: 0.74rem;">${valRaw}</span>
173                 </div>
174             </div>
175         `;
176     }
177
178     return `
179         <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-1"
180         style="background: rgba(255,255,255,0.02); border: 1px solid
181         var(--border-subtle, rgba(255,255,255,0.06));">
182             <div class="d-flex justify-content-between align-items-center">
183                 <span class="sys-sensor-name text-truncate" title="${
184                 name}" style="font-size: 0.74rem; font-weight: 600;">${name}<
185                 /span>
186                 <span class="font-monospace text-light fw-bold" style="font-size: 0.74rem;">${valRaw}</span>
187             </div>
188         </div>
189     `;
190 }
191
192 function renderSensors() {
193     const container = document.getElementById('sys-lhm-sensors-container');
194     if (!container) return;

```

```

187
188     const searchInput = document.getElementById('sys-lhm-searc
h');
189     const filterText = (searchInput ? searchInput.value : '').
trim().toLowerCase();
190
191     if (!Array.isArray(cachedSensors) || cachedSensors.length
=== 0) {
192         container.innerHTML = `
193             <div class="text-center py-4 text-muted small">
194                 <div>В базе данных telemetry.db сенсоры не обнаружен
ы</div>
195                 <div class="mt-2 text-muted" style="font-size: 0.72r
ем;">Ожидание сбора и сохранения телеметрии сенсоров...</div>
196             </div>
197         `;
198         return;
199     }
200
201     const filtered = cachedSensors.filter(s => {
202         // Category filter
203         if (currentSensorCategory !== 'all') {
204             const catLower = (s.sensor_category || '').toLowerCase
();
205             if (currentSensorCategory === 'temperature' && !catLow
er.includes('temp')) return false;
206             if (currentSensorCategory === 'load' && !catLower.incl
udes('load')) return false;
207             if (currentSensorCategory === 'fan' && !catLower.inclu
des('fan') && !catLower.includes('control')) return false;
208             if (currentSensorCategory === 'voltage' && !catLower.i
ncludes('volt') && !catLower.includes('power')) return false;
209             if (currentSensorCategory === 'clock' && !catLower.inc
ludes('clock')) return false;
210             if (currentSensorCategory === 'storage' && !catLower.i
ncludes('data') && !catLower.includes('throughput')) return fa
lse;
211         }

```

```

212
213     // Text search
214     if (filterText) {
215         const hName = (s.hardware_name || '').toLowerCase();
216         const sName = (s.sensor_name || '').toLowerCase();
217         const sCat = (s.sensor_category || '').toLowerCase();
218         const valStr = String(s.value_raw || s.value || '').to
LowerCase();
219         return hName.includes(filterText) || sName.includes(fi
lterText) || sCat.includes(filterText) || valStr.includes(filt
erText);
220     }
221     return true;
222 });
223
224     if (filtered.length === 0) {
225         container.innerHTML = `<div class="text-center py-4 text
-muted small">По выбранному фильтру сенсоров не найдено</div>`;
226         return;
227     }
228
229     // Group sensors by Hardware Name
230     const groups = {};
231     filtered.forEach(s => {
232         const grpKey = s.hardware_name || 'Оборудование';
233         if (!groups[grpKey]) groups[grpKey] = [];
234         groups[grpKey].push(s);
235     });
236
237     container.innerHTML = Object.entries(groups).map(([grpName
, items]) => {
238         const type = (items[0].hardware_type || '').toLowerCase(
);
239         let icon = 'bi bi-cpu';
240         let iconColor = 'text-info';
241         if (type.includes('gpu')) { icon = 'bi bi-gpu-card'; ico
nColor = 'text-warning'; }

```

```

242     else if (type.includes('storage') || type.includes('disk
| ')) { icon = 'bi bi-hdd'; iconColor = 'text-success'; }
243     else if (type.includes('memory') || type.includes('ram')
| ) { icon = 'bi bi-memory'; iconColor = 'text-primary'; }
244     else if (type.includes('network') || type.includes('net'
| )) { icon = 'bi bi-globe'; iconColor = 'text-info'; }
245     else if (type.includes('motherboard') || type.includes('
| mainboard')) { icon = 'bi bi-motherboard'; iconColor = 'text-s
| econdary'; }
246
247     const temps = items.filter(i => (i.sensor_category || ''
| ).toLowerCase().includes('temp')).map(i => Number(i.value || 0
| ));
248     const maxT = temps.length ? Math.max(...temps) : null;
249     const tempBadge = maxT !== null ? `` : '';
250
251     const widgetsHtml = items.map(s => createSensorWidgetHtm
| l(s)).join('');
252
253     return `
254     <div class="sys-core-card mb-2 p-2" style="min-height:
| auto; background: var(--surface-2, rgba(255,255,255,0.02)); b
| order: 1px solid var(--border-color, rgba(255,255,255,0.08));"
| >
255         <div class="d-flex justify-content-between align-ite
| ms-center mb-1.5 pb-1 border-bottom" style="border-color: var(
| --border-subtle, rgba(255,255,255,0.06)) !important;">
256             <div class="d-flex align-items-center gap-1.5 text
| -truncate" style="max-width: 70%;">
257                 <i class="${icon} ${iconColor}"></i>
258                 <span class="sys-core-title text-truncate" style
| ="font-size: 0.82rem;" title="${escapeHtml(grpName)}">${escape
| Html(grpName)}</span>
259             </div>
260             <div class="d-flex align-items-center gap-1">

```

```

261         ${tempBadge}
262         <span class="badge bg-dark border border-seconda
ry text-muted" style="font-size: 0.65rem;">${items.length} шт.
</span>
263     </div>
264 </div>
265 <div>
266     ${widgetsHtml}
267 </div>
268 </div>
269 `;
270 }).join('');
271
272 // Также обновляем блоки сенсоров в целевых панелях CPU, G
PU и RAM
273 updateAllComponentSensors();
274 if (lastCpuCoresData && lastCpuCoresData.length) {
275     renderCpuCores(lastCpuCoresData);
276 }
277 }
278
279 function getSensorSortPriority(s) {
280     const cat = (s.sensor_category || '').toLowerCase();
281     const unit = (s.unit || '').toLowerCase();
282     if (cat.includes('power') || unit === 'w') return 1;
283     if (cat.includes('volt') || unit === 'v') return 2;
284     if (cat.includes('clock') || unit.includes('hz')) return 3
;
285     if (cat.includes('fan') || unit.includes('rpm')) return 4;
286     if (cat.includes('timing') || cat.includes('factor')) retu
rn 5;
287     if (cat.includes('temp') || unit.includes('°c')) return 6;
288     return 7;
289 }
290
291 function renderComponentSensors(containerId, countBadgeId, s
ensors) {
292     const container = document.getElementById(containerId);

```

```

293 |     const countBadge = document.getElementById(countBadgeId);
294 |     if (!container) return;
295 |
296 |     if (sensors && sensors.length > 0) {
297 |         if (countBadge) {
298 |             countBadge.textContent = `${sensors.length} параметров
    | `;
299 |         }
300 |     } else {
301 |         // Если элементы уже отрендерены, не перетираем их заглу
    | шкой при временной задержке поллинга
302 |         if (container.children.length > 0 && !container.querySel
    | ector('.sys-no-sensors-msg')) {
303 |             return;
304 |         }
305 |         if (countBadge) {
306 |             countBadge.textContent = '0 параметров';
307 |         }
308 |         container.innerHTML = `<div class="sys-no-sensors-msg te
    | xt-muted small text-center py-2 col-12" style="font-size: 0.74
    | rem;">Нет дополнительных данных сенсоров</div>`;
309 |         return;
310 |     }
311 |
312 |     const sorted = [...sensors].sort((a, b) => {
313 |         const pA = getSensorSortPriority(a);
314 |         const pB = getSensorSortPriority(b);
315 |         if (pA !== pB) return pA - pB;
316 |         return (a.sensor_name || '').localeCompare(b.sensor_name
    | || '');
317 |     });
318 |
319 |     container.innerHTML = sorted.map(s => {
320 |         const rawName = s.sensor_name || s.name || '';
321 |         const name = escapeHtml((!rawName || rawName.toLowerCase
    | () === 'unknown') ? (s.id ? `Sensor #${s.id}` : 'Sensor') : ra
    | wName);
322 |         const cat = (s.sensor_category || '').toLowerCase();

```

```

323 |     const val = Number(s.value || 0);
324 |     const unit = (s.unit || '').trim();
325 |     const valRaw = s.value_raw || `${val.toFixed(1)} ${unit}
    |     `.trim();
326 |
327 |     let icon = 'bi bi-lightning-charge';
328 |     let iconColor = 'text-warning';
329 |     let badgeClass = 'bg-warning-subtle text-warning border
    | border-warning';
330 |
331 |     if (cat.includes('volt') || unit.toLowerCase() === 'v')
    | {
332 |         icon = 'bi bi-lightning-charge';
333 |         iconColor = 'text-primary';
334 |         badgeClass = 'bg-primary-subtle text-primary border bo
    | rder-primary';
335 |     } else if (cat.includes('power') || unit.toLowerCase() =
    | == 'w') {
336 |         icon = 'bi bi-lightning-charge';
337 |         iconColor = 'text-warning';
338 |         badgeClass = 'bg-warning-subtle text-warning border bo
    | rder-warning';
339 |     } else if (cat.includes('clock') || unit.toLowerCase().i
    | ncludes('hz')) {
340 |         icon = 'bi bi-speedometer2';
341 |         iconColor = 'text-info';
342 |         badgeClass = 'bg-info-subtle text-info border border-i
    | nfo';
343 |     } else if (cat.includes('fan') || unit.toLowerCase().inc
    | ludes('rpm')) {
344 |         icon = 'bi bi-fan';
345 |         iconColor = 'text-info';
346 |         badgeClass = 'bg-dark border border-secondary text-inf
    | o';
347 |     } else if (cat.includes('timing') || cat.includes('facto
    | r')) {
348 |         icon = 'bi bi-clock-history';
349 |         iconColor = 'text-secondary';

```

```

350 |         badgeClass = 'bg-dark border border-secondary text-light';
351 |     } else if (cat.includes('data') || unit.toLowerCase().includes('mb') || unit.toLowerCase().includes('gb')) {
352 |         icon = 'bi bi-hdd-fill';
353 |         iconColor = 'text-success';
354 |         badgeClass = 'bg-success-subtle text-success border border-success';
355 |     } else if (cat.includes('temp') || unit.toLowerCase().includes('°c')) {
356 |         icon = 'bi bi-thermometer-half';
357 |         iconColor = val >= 75 ? 'text-danger' : val >= 60 ? 'text-warning' : 'text-info';
358 |         badgeClass = val >= 75 ? 'bg-danger text-white' : val >= 60 ? 'bg-warning text-dark' : 'bg-info-subtle text-info';
359 |     }
360 |
361 |     return `
362 |         <div class="sys-sensor-widget-item p-1.5 rounded-2" style="background: var(--surface-2, rgba(255,255,255,0.03)); border: 1px solid var(--border-color, rgba(255,255,255,0.08));">
363 |             <div class="d-flex justify-content-between align-items-center">
364 |                 <span class="sys-sensor-name text-truncate" title="${name}" style="font-size: 0.74rem; font-weight: 600;">
365 |                     <i class="${icon} ${iconColor} me-1"></i>${name}
366 |                 </span>
367 |                 <span class="badge ${badgeClass} font-monospace" style="font-size: 0.72rem;">${valRaw}</span>
368 |             </div>
369 |         </div>
370 |     `;
371 | }).join('');
372 | }
373 |
374 | function toggleSysCpuDetails() {
375 |     const box = document.getElementById('sys-cpu-details-collapse');

```



```
376     const drillBtn = document.getElementById('sys-cpu-details-  
drill-btn');  
377     const parentBtn = document.getElementById('sys-cpu-details  
-toggle-btn');  
378     if (!box) return;  
379     const isHidden = box.style.display === 'none' || getComput  
edStyle(box).display === 'none';  
380     if (isHidden) {  
381         box.style.display = 'flex';  
382         if (drillBtn) drillBtn.classList.remove('collapsed');  
383         if (parentBtn) parentBtn.classList.add('expanded');  
384     } else {  
385         box.style.display = 'none';  
386         if (drillBtn) drillBtn.classList.add('collapsed');  
387         if (parentBtn) parentBtn.classList.remove('expanded');  
388     }  
389 }  
390 window.toggleSysCpuDetails = toggleSysCpuDetails;  
391 window.toggleSysCpuSensors = toggleSysCpuDetails;  
392  
393 function toggleSysGpuDetails(idx = 0) {  
394     const box = document.getElementById(`sys-gpu-details-colla  
pse-${idx}`) || document.getElementById('sys-gpu-details-colla  
pse');  
395     const drillBtn = document.getElementById(`sys-gpu-details-  
drill-btn-${idx}`) || document.getElementById('sys-gpu-details  
-drill-btn');  
396     const parentBtn = document.getElementById(`sys-gpu-details  
-toggle-btn-${idx}`) || document.getElementById('sys-gpu-detai  
ls-toggle-btn');  
397     if (!box) return;  
398     const isHidden = box.style.display === 'none' || getComput  
edStyle(box).display === 'none';  
399     if (isHidden) {  
400         box.style.display = 'flex';  
401         if (drillBtn) drillBtn.classList.remove('collapsed');  
402         if (parentBtn) parentBtn.classList.add('expanded');  
403     } else {
```

```
404     box.style.display = 'none';
405     if (drillBtn) drillBtn.classList.add('collapsed');
406     if (parentBtn) parentBtn.classList.remove('expanded');
407 }
408 }
409 window.toggleSysGpuDetails = toggleSysGpuDetails;
410 window.toggleSysGpuSensors = toggleSysGpuDetails;
411
412 /**
413  * Управление режимом Drill-down по нажатию на значок дрели
414  * слева от названия компонента.
415  *
416  * @param {string|HTMLElement} cardIdOrEl - Идентификатор ка
417  * рточки или элемент кнопки
418  * @param {number} [idx] - Индекс устройства (для GPU)
419  */
419 function toggleSysCardDrill(cardIdOrEl, idx) {
420     let card = null;
421     let btn = null;
422     if (typeof cardIdOrEl === 'string') {
423         card = document.getElementById(cardIdOrEl);
424     } else if (cardIdOrEl instanceof HTMLElement) {
425         btn = cardIdOrEl;
426         card = btn.closest('.sys-card');
427     }
428     if (!card) return;
429     if (!btn) {
430         btn = card.querySelector('.sys-drill-btn');
431     }
432
433     const isExpanded = card.classList.contains('sys-card-drill
434 -expanded');
435     if (isExpanded) {
436         card.classList.remove('sys-card-drill-expanded');
437         if (btn) {
438             btn.classList.remove('active');
```

```
438 |         btn.title = 'Drill-down: Развернуть/скрыть все скрытые
    | панели и датчики';
439 |     }
440 | } else {
441 |     card.classList.add('sys-card-drill-expanded');
442 |     if (btn) {
443 |         btn.classList.add('active');
444 |         btn.title = 'Drill-down: Свернуть скрытые панели';
445 |     }
446 |
447 |     // Если это CPU - также открываем вложенный аккордеон де
    | талей ядер и сенсоров
448 |     if (card.classList.contains('sys-card-cpu') || card.id =
    | == 'sys-card-cpu') {
449 |         const box = document.getElementById('sys-cpu-details-c
    | ollapse');
450 |         const icon = document.getElementById('sys-cpu-details-
    | toggle-icon');
451 |         const text = document.getElementById('sys-cpu-details-
    | toggle-text');
452 |         const actionBtn = document.getElementById('sys-cpu-det
    | ails-action-btn');
453 |         const parentBtn = document.getElementById('sys-cpu-det
    | ails-toggle-btn');
454 |         if (box) box.style.display = 'flex';
455 |         if (icon) icon.className = 'bi bi-chevron-up';
456 |         if (text) text.textContent = 'Свернуть детализацию';
457 |         if (actionBtn) actionBtn.className = 'btn btn-xs btn-i
    | nfo text-dark rounded-pill px-3 py-1 fw-bold d-flex align-item
    | s-center gap-1.5';
458 |         if (parentBtn) parentBtn.classList.add('expanded');
459 |         if (typeof renderCpuSpark === 'function' && Array.isAr
    | ray(_cpuSparkHistory)) {
460 |             renderCpuSpark(_cpuSparkHistory);
461 |         }
462 |     }
463 |
464 |     // Если это GPU - также открываем вложенный аккордеон де
```

тaley GPU

```
465     if (card.classList.contains('sys-card-gpu') || (card.id
    && card.id.startsWith('sys-card-gpu-'))) {
466         const cardIdx = idx != null ? idx : (card.id ? card.id
    .replace('sys-card-gpu-', '') : 0);
467         const box = document.getElementById(`sys-gpu-details-c
    ollapse-${cardIdx}`) || document.getElementById('sys-gpu-detai
    ls-collapse');
468         const icon = document.getElementById(`sys-gpu-details-
    toggle-icon-${cardIdx}`) || document.getElementById('sys-gpu-d
    etails-toggle-icon');
469         const text = document.getElementById(`sys-gpu-details-
    toggle-text-${cardIdx}`) || document.getElementById('sys-gpu-d
    etails-toggle-text');
470         const actionBtn = document.getElementById(`sys-gpu-det
    ails-action-btn-${cardIdx}`) || document.getElementById('sys-g
    pu-details-action-btn');
471         const parentBtn = document.getElementById(`sys-gpu-det
    ails-toggle-btn-${cardIdx}`) || document.getElementById('sys-g
    pu-details-toggle-btn');
472         if (box) box.style.display = 'flex';
473         if (icon) icon.className = 'bi bi-chevron-up';
474         if (text) text.textContent = 'Свернуть детализацию';
475         if (actionBtn) actionBtn.className = 'btn btn-xs btn-w
    arning text-dark rounded-pill px-3 py-1 fw-bold d-flex align-i
    tems-center gap-1.5';
476         if (parentBtn) parentBtn.classList.add('expanded');
477         if (typeof renderGpuSpark === 'function') {
478             renderGpuSpark(_gpuSparkHistories[cardIdx] || [], ca
    rdIdx);
479         }
480     }
481
482     // RAM график
483     if (card.classList.contains('sys-card-ram') || card.id =
    == 'sys-card-ram') {
484         if (typeof renderMemIoSpark === 'function' && Array.is
    Array(_memIoSparkHistory)) {
```

```
485         renderMemIoSpark(_memIoSparkHistory);
486     }
487 }
488
489     // Net график
490     if (card.classList.contains('sys-card-net') || card.id == 'sys-card-net') {
491         if (typeof renderNetSpark === 'function' && Array.isArray(_netSparkHistory)) {
492             renderNetSpark(_netSparkHistory);
493         }
494     }
495 }
496 }
497 window.toggleSysCardDrill = toggleSysCardDrill;
498
499 function updateAllComponentSensors() {
500     if (!Array.isArray(cachedSensors) || cachedSensors.length === 0) return;
501
502     // 1. CPU Sensors (электрические параметры, мощности, вольтажи, шины, TjMax)
503     const cpuSensors = cachedSensors.filter(s => {
504         const sid = String(s.id || s.sensor_id || '').toLowerCase();
505         const hwType = (s.hardware_type || '').toLowerCase();
506         const hwName = (s.hardware_name || '').toLowerCase();
507         const sName = (s.sensor_name || '').toLowerCase();
508         const cat = (s.sensor_category || '').toLowerCase();
509
510         // Строго отсекаем дисковые метрики и не-CPU компоненты
511         if (sid.startsWith('disk_speed_') || hwType.includes('storage') || cat.includes('storage')) return false;
512
513         const isCpu = hwType === 'cpu' || hwName.includes('cpu') || hwName.includes('intel') || hwName.includes('amd');
514         if (!isCpu) return false;
515     });
```

```
516 |         // Исключаем по-ядерную загрузку и температуру отдельных
      | ядер (они уже в карточках ядер)
517 |         if (cat.includes('load') && (sName.includes('core #') ||
      | sName === 'cpu total')) return false;
518 |         if (cat.includes('temp') && sName.includes('core #') &&
      | !sName.includes('tjmax')) return false;
519 |         if (cat.includes('temp') && sName === 'cpu package') ret
      | urn false; // показана в главном слайдере Package
520 |
521 |         return true;
522 |     });
523 |     if (cpuSensors.length > 0) {
524 |         lastCpuSensorsData = cpuSensors;
525 |         renderComponentSensors('sys-metric-cpu-sensors', 'sys-cp
      | u-sensors-count-badge', cpuSensors);
526 |     } else if (lastCpuSensorsData && lastCpuSensorsData.length
      | > 0) {
527 |         renderComponentSensors('sys-metric-cpu-sensors', 'sys-cp
      | u-sensors-count-badge', lastCpuSensorsData);
528 |     } else {
529 |         renderComponentSensors('sys-metric-cpu-sensors', 'sys-cp
      | u-sensors-count-badge', []);
530 |     }
531 |
532 |     // 2. GPU Sensors (мощности, вольтажи, частоты, вентилятор
      | ы, память GPU)
533 |     const gpuSensors = cachedSensors.filter(s => {
534 |         const hwType = (s.hardware_type || '').toLowerCase();
535 |         const hwName = (s.hardware_name || '').toLowerCase();
536 |         const sName = (s.sensor_name || '').toLowerCase();
537 |         const cat = (s.sensor_category || '').toLowerCase();
538 |
539 |         const isGpu = hwType === 'gpu' || hwName.includes('gpu')
      | || hwName.includes('nvidia') || hwName.includes('geforce') ||
      | hwName.includes('radeon') || hwName.includes('intel arc');
540 |         if (!isGpu) return false;
541 |
542 |         // Исключаем общую загрузку/температуру GPU Core (они в
```

```
главном Gauge/Slider)
543     if (cat.includes('load') && sName === 'gpu core') return
      false;
544     if (cat.includes('temp') && sName === 'gpu core') return
      false;
545
546     return true;
547   });
548   renderComponentSensors('sys-metric-gpu-sensors', 'sys-gpu-
sensors-count-badge', gpuSensors);
549
550   // 3. RAM Sensors (мощности, вольтажи, тайминги, доступный
      объем)
551   const ramSensors = cachedSensors.filter(s => {
552     const hwType = (s.hardware_type || '').toLowerCase();
553     const hwName = (s.hardware_name || '').toLowerCase();
554     const sName = (s.sensor_name || '').toLowerCase();
555
556     const isRam = hwType === 'memory' || hwType === 'ram' ||
      hwName.includes('memory') || hwName.includes('ram') || sName.
includes('cpu memory');
557     return isRam;
558   });
559   renderComponentSensors('sys-metric-ram-sensors', 'sys-ram-
sensors-count-badge', ramSensors);
560
561   // 4. Network Sensors (трафик, сетевые интерфейсы, Wi-Fi,
      Bluetooth)
562   const netSensors = cachedSensors.filter(s => {
563     const hwType = (s.hardware_type || '').toLowerCase();
564     const hwName = (s.hardware_name || '').toLowerCase();
565     const isNet = hwType === 'network' || hwType === 'net' |
      | hwName.includes('net') || hwName.includes('ethernet') || hwN
ame.includes('wi-fi') || hwName.includes('wifi') || hwName.inc
cludes('bluetooth');
566     return isNet;
567   });
568   renderComponentSensors('sys-metric-net-sensors', 'sys-net-
```

```

sensors-count-badge', netSensors);
569   }
570
571   let lastAnalysisReportText = '';
572
573   async function analyzeSensorsAndReport() {
574     const modalEl = document.getElementById('sysSensorAnalysis
Modal');
575     let modal = null;
576     if (modalEl && window.bootstrap && window.bootstrap.Modal)
    {
577       modal = window.bootstrap.Modal.getOrCreateInstance(modal
El);
578       modal.show();
579     }
580
581     // Set Loading state in modal
582     const titleEl = document.getElementById('modal-sensor-stat
us-title');
583     const subEl = document.getElementById('modal-sensor-status
-sub');
584     const aiCompEl = document.getElementById('modal-sensor-ai-
comparison');
585     const devListEl = document.getElementById('modal-sensor-de
vices-list');
586     const devCountEl = document.getElementById('modal-sensor-d
evices-count');
587     const modelBadge = document.getElementById('modal-sensor-m
odel-badge');
588
589     if (titleEl) titleEl.textContent = 'Выполняется AI-аудит з
аложенных сенсоров...';
590     if (subEl) subEl.textContent = 'Чтение CSV-логов, усреднен
ие значений и сравнение со спецификациями железа';
591     if (aiCompEl) {
592       aiCompEl.innerHTML = '<div class="d-flex align-items-cen
ter gap-2 py-3"><div class="spinner-border spinner-border-sm t
ext-primary" role="status"></div><span>Запрос к модели AI и ср

```



```

        авнительный анализ номинальных параметров железа...</span></div>
        v>';
593     }
594     if (devListEl) {
595         devListEl.innerHTML = '<span class="badge bg-dark border
        border-secondary text-muted"><span class="spinner-border spin
        ner-border-sm me-1" style="width:0.6rem;height:0.6rem;"></span
        >Определение устройств...</span>';
596     }
597
598     let report = null;
599     try {
600         let res = await fetch('/api/v1/system/lhm-audit', { meth
        od: 'POST' });
601         if (!res.ok) {
602             res = await fetch('/api/v1/lhm/audit', { method: 'POST
        ' });
603         }
604         if (res.ok) {
605             report = await res.json();
606         }
607     } catch (err) {
608         console.warn('[SystemInspectorTab] LHM Audit API error:'
        , err);
609     }
610
611     // Extract metrics from report or fallback to cachedSensor
        s
612     const sensorsList = (report && report.aggregated_sensors)
        ? report.aggregated_sensors : (cachedSensors || []);
613     const devices = (report && report.devices) ? report.device
        s : [];
614     let healthScore = report ? (report.health_score || 100) :
        100;
615     const recommendations = [];
616
617     let maxCpuTemp = null;
618     let maxGpuTemp = null;

```

```
619 |     let maxMbTemp = null;
620 |     let maxStorageTemp = null;
621 |     let fanSpeeds = [];
622 |     let voltages = [];
623 |     let cpuLoads = [];
624 |     let gpuLoads = [];
625 |
626 |     sensorsList.forEach(s => {
627 |         const cat = ((s.category || s.sensor_category) || '').to
LowerCase();
628 |         const hw = ((s.hardware || s.hardware_name) || '').toLow
erCase();
629 |         const sName = (s.sensor_name || '').toLowerCase();
630 |         const num = s.avg !== undefined ? s.avg : s.value_numeri
c;
631 |
632 |         if (num === null || num === undefined || isNaN(num)) ret
urn;
633 |
634 |         if (cat.includes('temp')) {
635 |             if (hw.includes('cpu') || sName.includes('cpu') || sNa
me.includes('core')) {
636 |                 if (maxCpuTemp === null || num > maxCpuTemp) maxCpuT
emp = num;
637 |             } else if (hw.includes('gpu') || hw.includes('nvidia')
|| sName.includes('gpu')) {
638 |                 if (maxGpuTemp === null || num > maxGpuTemp) maxGpuT
emp = num;
639 |             } else if (hw.includes('storage') || hw.includes('nvme
') || hw.includes('ssd') || hw.includes('hdd') || sName.includ
es('drive')) {
640 |                 if (maxStorageTemp === null || num > maxStorageTemp)
maxStorageTemp = num;
641 |             } else {
642 |                 if (maxMbTemp === null || num > maxMbTemp) maxMbTemp
= num;
643 |             }
644 |         } else if (cat.includes('fan')) {
```

```
645 |         fanSpeeds.push({ name: s.sensor_name, value: num, unit
    | : s.unit || 'RPM' });
646 |     } else if (cat.includes('volt')) {
647 |         voltages.push({ name: s.sensor_name, value: num, unit:
    | s.unit || 'V' });
648 |     } else if (cat.includes('load')) {
649 |         if (hw.includes('cpu') || sName.includes('cpu')) cpuLo
    | ads.push(num);
650 |         else if (hw.includes('gpu') || sName.includes('gpu'))
    | gpuLoads.push(num);
651 |     }
652 | });
653 |
654 | const overallMaxTemp = Math.max(
655 |     maxCpuTemp || 0,
656 |     maxGpuTemp || 0,
657 |     maxMbTemp || 0,
658 |     maxStorageTemp || 0
659 | );
660 |
661 | // Update Model Badge
662 | if (modelBadge && report) {
663 |     modelBadge.textContent = report.ai_model_used || 'Heuris
    | tic Engine';
664 | }
665 |
666 | // Render Devices List
667 | if (devCountEl) {
668 |     devCountEl.textContent = `${devices.length} устр.`;
669 | }
670 | if (devListEl) {
671 |     if (devices.length > 0) {
672 |         devListEl.innerHTML = devices.map(d => {
673 |             let icon = 'bi-hdd-network';
674 |             const low = d.toLowerCase();
675 |             if (low.includes('core') || low.includes('cpu') || l
    | ow.includes('intel') || low.includes('amd')) icon = 'bi-cpu';
676 |             else if (low.includes('geforce') || low.includes('gp
```

```

u') || low.includes('nvidia') || low.includes('radeon')) icon
= 'bi-gpu-card';
677     else if (low.includes('ssd') || low.includes('hd') |
| low.includes('wdc') || low.includes('toshiba') || low.includ
es('st3500')) icon = 'bi-device-hdd';
678     else if (low.includes('memory') || low.includes('ram
')) icon = 'bi-memory';
679     else if (low.includes('ethernet') || low.includes('w
i-fi') || low.includes('network')) icon = 'bi-wifi';
680
681     return `

```

```
702 |         elCpuBadge.textContent = `${Math.round(maxCpuTemp)} °C`;  
703 |         elCpuBadge.className = `badge ${maxCpuTemp > 82 ? 'bg-danger' : maxCpuTemp > 70 ? 'bg-warning text-dark' : 'bg-success'}`;  
704 |     }  
705 |     if (maxCpuTemp > 82) {  
706 |         if (elCpuText) elCpuText.textContent = `Критический нагрев (${Math.round(maxCpuTemp)}°C). Риск теплового троттлинга.`;  
707 |         recommendations.push({  
708 |             type: 'danger',  
709 |             icon: 'bi-exclamation-triangle-fill',  
710 |             title: 'Высокая температура процессора',  
711 |             text: `Пиковая/средняя температура CPU достигает ${Math.round(maxCpuTemp)}°C. Проверьте плотность прижима кулера, термопасту и запыленность радиатора.`  
712 |         });  
713 |     } else if (maxCpuTemp > 70) {  
714 |         if (elCpuText) elCpuText.textContent = `Повышенная температура (${Math.round(maxCpuTemp)}°C) под нагрузкой.`;  
715 |         recommendations.push({  
716 |             type: 'warning',  
717 |             icon: 'bi-thermometer-high',  
718 |             title: 'Повышенный нагрев CPU',  
719 |             text: `Температура CPU ${Math.round(maxCpuTemp)}°C выше оптимального порога. Рекомендуется настроить кривую оборотов вентилятора.`  
720 |         });  
721 |     } else {  
722 |         if (elCpuText) elCpuText.textContent = `Штатный температурный режим (${Math.round(maxCpuTemp)}°C). Троттлинг отсутствует.`;  
723 |         recommendations.push({  
724 |             type: 'success',  
725 |             icon: 'bi-check-circle-fill',  
726 |             title: 'Тепловой режим процессора оптимален',  
727 |             text: `Температура процессора ${Math.round(maxCpuTemp)}°C`
```

```

    p))°C находится в безопасной зоне номинальных характеристик.`
728         });
729     }
730     } else {
731         if (elCpuBadge) elCpuBadge.textContent = 'N/A';
732         if (elCpuText) elCpuText.textContent = 'Температурные да
тчики CPU не предоставили данных.';
733     }
734
735     // Analyze GPU
736     const elGpuBadge = document.getElementById('modal-gpu-temp
-badge');
737     const elGpuText = document.getElementById('modal-gpu-analy
sis-text');
738     if (maxGpuTemp !== null) {
739         if (elGpuBadge) {
740             elGpuBadge.textContent = `${Math.round(maxGpuTemp)} °C
`;
741             elGpuBadge.className = `badge ${maxGpuTemp > 80 ? 'bg-
danger' : maxGpuTemp > 72 ? 'bg-warning text-dark' : 'bg-succe
ss'}`;
742         }
743         if (maxGpuTemp > 80) {
744             if (elGpuText) elGpuText.textContent = `Критический на
грев GPU (${Math.round(maxGpuTemp)}°C).`;
745             recommendations.push({
746                 type: 'danger',
747                 icon: 'bi-gpu-card',
748                 title: 'Высокая температура видеокарты',
749                 text: `Графический чип нагревается до ${Math.round(m
axGpuTemp)}°C. Проверьте циркуляцию воздуха в корпусе ПК.`
750             });
751         } else {
752             if (elGpuText) elGpuText.textContent = `Температура GP
U ${Math.round(maxGpuTemp)}°C в норме.`;
753             recommendations.push({
754                 type: 'success',
755                 icon: 'bi-check-circle-fill',

```

```
756 |         title: 'Видеокарта работает штатно',
757 |         text: `Температурные показатели GPU (${Math.round(maxGpuTemp)}°C) соответствуют номиналу.`
758 |     });
759 | }
760 | } else {
761 |     if (elGpuBadge) elGpuBadge.textContent = 'N/A';
762 |     if (elGpuText) elGpuText.textContent = 'Дискретный GPU в
режиме энергосбережения или данные отсутствуют.';
763 | }
764 |
765 | // Analyze Cooling
766 | const elFanBadge = document.getElementById('modal-fan-stat
us-badge');
767 | const elFanText = document.getElementById('modal-fan-analy
sis-text');
768 | if (fanSpeeds.length > 0) {
769 |     const activeFans = fanSpeeds.filter(f => f.value > 0);
770 |     if (elFanBadge) {
771 |         elFanBadge.textContent = `${activeFans.length} акт. ку
леров`;
772 |         elFanBadge.className = 'badge bg-success';
773 |     }
774 |     if (elFanText) {
775 |         elFanText.textContent = `Опрошено ${fanSpeeds.length}
кулеров. Обороты стабильны.`;
776 |     }
777 | } else {
778 |     if (elFanBadge) elFanBadge.textContent = 'Пассив / WMI';
779 |     if (elFanText) elFanText.textContent = 'Управление венти
ляторами через BIOS или пассивное охлаждение.';
780 | }
781 |
782 | // Additional recommendations from backend report
783 | if (report && Array.isArray(report.recommendations)) {
784 |     report.recommendations.forEach(r => {
785 |         if (!recommendations.some(ex => ex.text.includes(r)))
{
```

```
786         recommendations.push({
787             type: 'info',
788             icon: 'bi-info-circle',
789             title: 'Рекомендация по оборудованию',
790             text: r
791         });
792     }
793 });
794 }
795
796 // Update Header & Badge
797 const badgeHealth = document.getElementById('modal-sensor-
health-badge');
798 if (badgeHealth) {
799     badgeHealth.textContent = `Health: ${healthScore}/100`;
800     badgeHealth.className = `badge ms-1 ${healthScore >= 85
? 'bg-success' : healthScore >= 65 ? 'bg-warning text-dark' :
'bg-danger'}`;
801 }
802
803 const countStat = document.getElementById('modal-sensor-co
unt-stat');
804 if (countStat) {
805     const totalSmpls = report ? report.total_samples : senso
rsList.length;
806     countStat.textContent = `${sensorsList.length} шт. (${to
talSmpls} замеров)`;
807 }
808
809 const maxTempStat = document.getElementById('modal-sensor-
max-temp');
810 if (maxTempStat) maxTempStat.textContent = overallMaxTemp
> 0 ? `${Math.round(overallMaxTemp)} °C` : 'N/A';
811
812 const statusTitle = document.getElementById('modal-sensor-
status-title');
813 const statusSub = document.getElementById('modal-sensor-st
atus-sub');
```



```

814     const statusIcon = document.getElementById('modal-sensor-s
tatus-icon');
815
816     if (healthScore >= 85) {
817         if (statusIcon) statusIcon.textContent = '✅';
818         if (statusTitle) statusTitle.textContent = 'Все аппаратн
ые подсистемы работают в идеальном режиме';
819         if (statusSub) statusSub.textContent = `Усредненные пока
затели соответствуют спецификациям реального железа (макс. ${M
ath.round(overallMaxTemp)}°C).`;
820     } else if (healthScore >= 65) {
821         if (statusIcon) statusIcon.textContent = '⚠️';
822         if (statusTitle) statusTitle.textContent = 'Обнаружены п
араметры оборудования, требующие внимания';
823         if (statusSub) statusSub.textContent = `Пиковая температ
ура ${Math.round(overallMaxTemp)}°C. Ознакомьтесь с AI-заключе
нием и рекомендациями.`;
824     } else {
825         if (statusIcon) statusIcon.textContent = '❌';
826         if (statusTitle) statusTitle.textContent = 'Внимание: Кр
итические параметры оборудования!';
827         if (statusSub) statusSub.textContent = `Зафиксирован опа
сный нагрев или высокая перегрузка компонентов.`;
828     }
829
830     // Render Recommendations List
831     const recList = document.getElementById('modal-sensor-reco
mmendations-list');
832     if (recList) {
833         recList.innerHTML = recommendations.map(r => `
834             <div class="p-2.5 rounded border d-flex align-items-st
art gap-2.5" style="background: var(--surface-2); border-color
: var(--border-color) !important;">
835                 <div class="mt-0.5">
836                     <span class="badge ${r.type === 'danger' ? 'bg-dan
ger' : r.type === 'warning' ? 'bg-warning text-dark' : r.type
=== 'info' ? 'bg-info text-dark' : 'bg-success'} rounded-circl
e p-1.5 d-flex align-items-center justify-content-center" styl

```

```

e="width: 24px; height: 24px;">
837 |         <i class="bi ${r.icon}" style="font-size: 0.75rem;"></i>
838 |         </span>
839 |     </div>
840 |     <div class="flex-grow-1">
841 |         <div class="fw-bold text-white mb-0.5" style="font-size: 0.82rem;">${escapeHtml(r.title)}</div>
842 |         <div class="text-muted" style="font-size: 0.76rem; line-height: 1.4;">${escapeHtml(r.text)}</div>
843 |     </div>
844 | </div>
845 | `).join('');
846 | }
847 |
848 | // Prepare Clipboard text
849 | lastAnalysisReportText = `=== AI-Breadboard: Отчет AI-аудита сенсоров и оборудования ===\n` +
850 |     `Дата: ${new Date().toLocaleString('ru-RU')}\n` +
851 |     `Индекс здоровья системы: ${healthScore}/100\n` +
852 |     `AI-модель: ${report ? report.ai_model_used : 'Heuristic'}\n` +
853 |     `Распознанные устройства (${devices.length} шт.): ${devices.join(', ')}\n` +
854 |     `Сенсоров в аудите: ${sensorsList.length}\n` +
855 |     `Максимальная температура: ${overallMaxTemp > 0 ? Math.round(overallMaxTemp) + ' °C' : 'N/A'}\n\n` +
856 |     `=== AI СРАВНЕНИЕ СО СПЕЦИФИКАЦИЯМИ РЕАЛЬНОГО ЖЕЛЕЗА ===\n` +
857 |     `${(report && report.comparison_report) ? report.comparison_report : 'Параметры в норме.'}\n\n` +
858 |     `=== РЕКОМЕНДАЦИИ И ВЫВОДЫ ===\n` +
859 |     recommendations.map((r, i) => `${i + 1}. [${r.title}] ${r.text}`).join('\n') +
860 |     `\n=====
=`;
861 | }
862 |

```

```

863 | /**
864 |  * Генерация полукруглого спидометра (Gauge) со стрелкой.
865 |  * @param {number|null} percent - Значение нагрузки (0-100)
866 |  * @param {number} width - Ширина SVG контейнера
867 |  * @param {number} height - Высота SVG контейнера
868 |  * @returns {string} SVG разметка
869 |  */
870 | function createGaugeSvg(percent, width = 100, height = 56) {
871 |     const val = percent == null ? 0 : Math.min(100, Math.max(0
872 | , Number(percent)));
873 |     // Угол поворота стрелки: 0% -> -90deg (влево), 50% -> 0deg
874 |     // (вверх), 100% -> +90deg (вправо)
875 |     const angle = -90 + (val / 100) * 180;
876 |     const needleColor = val > 80 ? '#ef4444' : val > 50 ? '#f5
877 | 9e0b' : '#38bdf8';
878 |
879 |     return `
880 |         <svg width="${width}" height="${height}" viewBox="0 0 10
881 | 0 56" preserveAspectRatio="xMidYMid meet" style="display:block
882 | ;margin:0 auto;overflow:visible;">
883 |             <defs>
884 |                 <linearGradient id="sysGaugeGrad" x1="0%" y1="0%" x2
885 | ="100%" y2="0%">
886 |                     <stop offset="0%" stop-color="#10b981"/>
887 |                     <stop offset="50%" stop-color="#f59e0b"/>
888 |                     <stop offset="100%" stop-color="#ef4444"/>
889 |                 </linearGradient>
890 |             </defs>
891 |             <!-- Фоновая дуга -->
892 |             <path d="M 12 50 A 38 38 0 0 1 88 50" fill="none" stro
893 | ke="var(--border-color, rgba(128,128,128,0.25))" stroke-width=
894 | "9" stroke-linecap="round"/>
895 |             <!-- Градиентная активная шкала -->
896 |             <path d="M 12 50 A 38 38 0 0 1 88 50" fill="none" stro
897 | ke="url(#sysGaugeGrad)" stroke-width="9" stroke-linecap="round
898 | " opacity="0.9"/>
899 |             <!-- Стрелка прибора -->
900 |             <g transform="rotate(${angle.toFixed(1)}, 50, 50)" sty

```

```

le="transition: transform 0.4s cubic-bezier(0.4, 0, 0.2, 1);">
891     <polygon points="48,50 50,14 52,50" fill="{needleCo
lor}" filter="drop-shadow(0 2px 3px rgba(0,0,0,0.6))"/>
892     <line x1="50" y1="50" x2="50" y2="14" stroke="#ffffff
f" stroke-width="1.2" opacity="0.9"/>
893     </g>
894     <!-- Центральный шарнир -->
895     <circle cx="50" cy="50" r="6" fill="#38bdf8"/>
896     <circle cx="50" cy="50" r="2.5" fill="var(--surface-1,
#0f172a)"/>
897 </svg>
898 `;
899 }
900
901 /**
902  * Генерация цветного термо-слайдера с бегунком.
903  * @param {number|null} tempC - Температура в градусах Цельс
ия
904  * @param {number} minTemp - Мин. шкала (по умолчанию 30°C)
905  * @param {number} maxTemp - Макс. шкала (по умолчанию 95°C)
906  * @param {boolean} isLarge - Флаг увеличенного отображения
для шапки
907  * @returns {string} HTML разметка
908  */
909 function createTempSliderHtml(tempC, minTemp = 30, maxTemp =
95, isLarge = false) {
910     if (tempC == null) {
911         return `<div class="sys-temp-slider-empty">-- °C</div>`;
912     }
913     const val = Number(tempC);
914     const pct = Math.min(100, Math.max(0, ((val - minTemp) / (
maxTemp - minTemp)) * 100));
915
916     let barColor = '#10b981'; // Зеленый (<50°C)
917     let badgeClass = 'text-success';
918     if (val >= 80) {
919         barColor = '#ef4444'; // Красный (>=80°C)
920         badgeClass = 'text-danger fw-bold';

```

```

921     } else if (val >= 68) {
922         barColor = '#f97316'; // Оранжевый (68-80°C)
923         badgeClass = 'text-warning fw-bold';
924     } else if (val >= 50) {
925         barColor = '#eab308'; // Желтый (50-68°C)
926         badgeClass = 'text-warning';
927     }
928
929     const wrapClass = isLarge ? 'sys-temp-slider-wrap sys-temp
-slider-lg' : 'sys-temp-slider-wrap';
930     const fontSize = isLarge ? 'font-size: 0.85rem;' : 'font-s
ize: 0.78rem;';
931
932     return `
933         <div class="${wrapClass}">
934             <div class="d-flex justify-content-between align-items
-center mb-1" style="${fontSize}">
935                 <span class="text-muted"><i class="bi bi-thermometer
-half me-1"></i>t°</span>
936                 <span class="font-monospace ${badgeClass}" style="fo
nt-weight: 700;">${val.toFixed(0)}°C</span>
937             </div>
938             <div class="sys-temp-slider-track">
939                 <div class="sys-temp-slider-bar" style="width: ${pct
.toFixed(1)}%; background: ${barColor};"></div>
940                 <div class="sys-temp-slider-thumb" style="left: ${pc
t.toFixed(1)}%; background: ${barColor};"></div>
941             </div>
942         </div>
943     `;
944 }
945
946 /**
947  * Извлечение и сопоставление параметров сенсоров конкретнo
o ядра CPU.
948  * @param {number} coreIdx - Индекс ядра (0-based)
949  * @param {number} totalCores - Общее количество ядер/потоко
В

```

```

950 |     * @param {Array} sensors - Массив объектов сенсоров из tele
    |     metry.db / LHM
951 |     * @returns {Array<{icon: string, label: string, value: stri
    |     ng, badgeClass: string}>}
952 |     */
953 |     function extractCoreParameters(coreIdx, totalCores, sensors)
    |     {
954 |         if (!Array.isArray(sensors) || sensors.length === 0) {
955 |             return [];
956 |         }
957 |
958 |         const cpuSensors = sensors.filter(s => {
959 |             const hwType = (s.hardware_type || '').toLowerCase();
960 |             const hwName = (s.hardware_name || '').toLowerCase();
961 |             return hwType === 'cpu' || hwName.includes('cpu') || hwN
    |             ame.includes('intel') || hwName.includes('amd');
962 |         });
963 |
964 |         if (cpuSensors.length === 0) return [];
965 |
966 |         const idx1 = coreIdx + 1;
967 |         const physIdx1 = Math.floor(coreIdx / 2) + 1;
968 |
969 |         const params = [];
970 |
971 |         // 1. Вольтаж ядра (Voltage)
972 |         const voltSensor = cpuSensors.find(s => {
973 |             const cat = (s.sensor_category || '').toLowerCase();
974 |             const unit = (s.unit || '').toLowerCase();
975 |             const name = (s.sensor_name || '').toLowerCase();
976 |             if (!cat.includes('volt') && unit !== 'v') return false;
977 |             return name.includes(`core #${idx1}`) || name.includes(`
    |             core #${physIdx1}`) || name.includes(`core #${coreIdx}`);
978 |         }) || cpuSensors.find(s => {
979 |             const cat = (s.sensor_category || '').toLowerCase();
980 |             const unit = (s.unit || '').toLowerCase();
981 |             const name = (s.sensor_name || '').toLowerCase();
982 |             return (cat.includes('volt') || unit === 'v') && (name.i

```

```
includes('cpu core') || name.includes('vcore')));
983     });
984
985     if (voltSensor) {
986         params.push({
987             icon: 'bi bi-lightning-charge text-primary',
988             label: 'Вольтаж',
989             value: voltSensor.value_raw || `${Number(voltSensor.va
990 lue || 0).toFixed(2)} V`,
991             badgeClass: 'text-primary'
992         });
993     }
994
995     // 2. Частота ядра (Clock / Frequency)
996     const clockSensor = cpuSensors.find(s => {
997         const cat = (s.sensor_category || '').toLowerCase();
998         const unit = (s.unit || '').toLowerCase();
999         const name = (s.sensor_name || '').toLowerCase();
1000         if (!cat.includes('clock') && !unit.includes('hz')) retu
1001 rn false;
1002         return name.includes(`core #${idx1}`) || name.includes(`
1003 core #${physIdx1}`) || name.includes(`core #${coreIdx}`);
1004     }) || cpuSensors.find(s => {
1005         const cat = (s.sensor_category || '').toLowerCase();
1006         const unit = (s.unit || '').toLowerCase();
1007         const name = (s.sensor_name || '').toLowerCase();
1008         return (cat.includes('clock') || unit.includes('hz')) &&
1009 (name.includes('core freq') || name.includes('cpu core'));
1010     });
1011
1012     if (clockSensor) {
1013         params.push({
1014             icon: 'bi bi-speedometer2 text-info',
1015             label: 'Частота',
1016             value: clockSensor.value_raw || `${Number(clockSensor.
1017 value || 0).toFixed(0)} MHz`,
1018             badgeClass: 'text-info'
1019         });
1020     }
```

```

1015 |     }
1016 |
1017 |     // 3. Расстояние до TjMax (Distance to TjMax)
1018 |     const tjmaxSensor = cpuSensors.find(s => {
1019 |         const name = (s.sensor_name || '').toLowerCase();
1020 |         if (!name.includes('tjmax')) return false;
1021 |         return name.includes(`core #${idx1}`) || name.includes(`
1022 | core #${physIdx1}`) || name.includes(`core #${coreIdx}`);
1023 |     }) || cpuSensors.find(s => (s.sensor_name || '').toLowerCase().includes('tjmax'));
1024 |
1025 |     if (tjmaxSensor) {
1026 |         params.push({
1027 |             icon: 'bi bi-shield-check text-success',
1028 |             label: 'До TjMax',
1029 |             value: tjmaxSensor.value_raw || `${Number(tjmaxSensor.
1030 | value || 0).toFixed(0)} °C`,
1031 |             badgeClass: 'text-success'
1032 |         });
1033 |     }
1034 |
1035 |     // 4. Мощность (Power / Cores Power)
1036 |     const powerSensor = cpuSensors.find(s => {
1037 |         const cat = (s.sensor_category || '').toLowerCase();
1038 |         const unit = (s.unit || '').toLowerCase();
1039 |         const name = (s.sensor_name || '').toLowerCase();
1040 |         if (!cat.includes('power') && unit !== 'w') return false
1041 |         ;
1042 |         return name.includes(`core #${idx1}`) || name.includes(`
1043 | core #${physIdx1}`);
1044 |     }) || cpuSensors.find(s => {
1045 |         const cat = (s.sensor_category || '').toLowerCase();
1046 |         const unit = (s.unit || '').toLowerCase();
1047 |         const name = (s.sensor_name || '').toLowerCase();
1048 |         return (cat.includes('power') || unit === 'w') && (name.
1049 | includes('cpu cores') || name.includes('cpu core'));
1050 |     });

```



```
1047 |     if (powerSensor) {
1048 |         params.push({
1049 |             icon: 'bi bi-lightning text-warning',
1050 |             label: 'Мощность',
1051 |             value: powerSensor.value_raw || `${Number(powerSensor.
value || 0).toFixed(1)} W`,
1052 |             badgeClass: 'text-warning'
1053 |         });
1054 |     }
1055 |
1056 |     // 5. Пиковая загрузка / Core Max / Шина (Bus Speed)
1057 |     const maxSensor = cpuSensors.find(s => {
1058 |         const name = (s.sensor_name || '').toLowerCase();
1059 |         return name.includes('core max') || name.includes('cpu c
ore max');
1060 |     }) || cpuSensors.find(s => {
1061 |         const name = (s.sensor_name || '').toLowerCase();
1062 |         return name.includes('bus speed');
1063 |     });
1064 |
1065 |     if (maxSensor) {
1066 |         const isBus = (maxSensor.sensor_name || '').toLowerCase(
).includes('bus');
1067 |         params.push({
1068 |             icon: isBus ? 'bi bi-hdd-network text-secondary' : 'bi
bi-graph-up-arrow text-danger',
1069 |             label: isBus ? 'Шина' : 'Пик Max',
1070 |             value: maxSensor.value_raw || `${Number(maxSensor.valu
e || 0).toFixed(1)} ${maxSensor.unit || ''}`.trim(),
1071 |             badgeClass: isBus ? 'text-secondary' : 'text-danger'
1072 |         });
1073 |     }
1074 |
1075 |     return params;
1076 | }
1077 |
1078 | function renderCpuCores(cores, totalThreadsCount) {
1079 |     const box = document.getElementById('sys-metric-cpu-cores'
```

```

);
1080     const badgeCount = document.getElementById('sys-cores-count-badge');
1081     if (!box) return;
1082
1083     if (Array.isArray(cores)) {
1084         lastCpuCoresData = cores;
1085     } else if (lastCpuCoresData && lastCpuCoresData.length) {
1086         cores = lastCpuCoresData;
1087     }
1088
1089     const cCount = cores ? cores.length : 0;
1090     const thCount = totalThreadsCount || (cores ? cores.reduce(
        ((acc, c) => acc + (c.threads ? c.threads.length : 0), 0) : 0)
        ;
1091
1092     if (badgeCount) {
1093         badgeCount.textContent = cCount ? `${cCount} ядер • ${th
        Count} потоков` : '0 ядер';
1094     }
1095
1096     if (!Array.isArray(cores) || cores.length === 0) {
1097         box.innerHTML = `<div class="text-muted small text-center
        r py-3 col-12">Опрос ядер CPU выполняется...</div>`;
1098         return;
1099     }
1100
1101     box.innerHTML = cores.map(c => {
1102         const load = c.load_percent == null ? null : Number(c.lo
        ad_percent);
1103         const temp = c.temperature_c == null ? null : Number(c.t
        emperature_c);
1104         const freqStr = c.frequency_str || (c.frequency_mhz ? (c
        .frequency_mhz >= 1000 ? (c.frequency_mhz / 1000).toFixed(2) +
        ' GHz' : c.frequency_mhz.toFixed(0) + ' MHz') : '--');
1105         const powerStr = c.power_w != null ? `${Number(c.power_w
        ).toFixed(1)} W` : (c.power_str || '--');
1106         const loadStr = load == null ? '--' : `${load.toFixed(0)}

```

```

    }%`;
1107     const tempStr = temp != null ? `${temp.toFixed(0)} °C` :
    '-- °C';
1108     const gaugeSvg = createGaugeSvg(load, 100, 54);
1109     const tempHtml = createTempSliderHtml(temp, 30, 95, fals
    e);
1110
1111     const threadsList = Array.isArray(c.threads) ? c.threads
    : [];
1112     const threadsHtml = threadsList.map(th => {
1113         const thLoad = Number(th.load_percent || 0);
1114         const barClass = thLoad >= 80 ? 'bg-danger' : thLoad >
    = 60 ? 'bg-warning' : 'bg-info';
1115         return `
1116             <div class="d-flex align-items-center justify-conten
    t-between gap-1 mb-1" style="font-size: 0.72rem;">
1117                 <span class="text-truncate" style="max-width: 65px
    ; color: var(--text-color);" title="${escapeHtml(th.name || 'T
    hread')}">
1118                     ${escapeHtml(th.name || 'Thread')}
1119                 </span>
1120                 <div class="progress flex-grow-1" style="height: 5
    px; background: var(--border-color); border-radius: 3px;">
1121                     <div class="progress-bar ${barClass}" style="wid
    th: ${Math.min(100, thLoad)}%; border-radius: 3px;"></div>
1122                 </div>
1123                 <span class="font-monospace fw-bold" style="width:
    32px; text-align: right; font-size: 0.69rem; color: var(--tex
    t-color);">${thLoad.toFixed(0)}%</span>
1124             </div>
1125         `;
1126     }).join('');
1127
1128     return `
1129         <div class="sys-core-card p-2 rounded-2" style="backgr
    ound: var(--surface-2, rgba(255,255,255,0.03)); border: 1px so
    lid var(--border-color, rgba(255,255,255,0.08));" title="Ядро
    #${c.index}: Нагрузка ${loadStr}, Температура ${tempStr}">

```

```

1130         <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom" style="border-color: var(--border-subtle, rgba(255,255,255,0.06)) !important;">
1131             <div class="d-flex align-items-center gap-1.5">
1132                 <i class="bi bi-cpu text-info"></i>
1133                 <span class="sys-core-title fw-bold" style="font-size: 0.82rem;">Core #${c.index}</span>
1134             </div>
1135             <span class="badge ${load != null && load >= 80 ? 'bg-danger' : 'bg-info-subtle text-info border border-info'}" style="font-size: 0.70rem;">${loadStr}</span>
1136         </div>
1137
1138         <div class="d-flex align-items-center gap-2 mb-2">
1139             <div class="d-flex flex-column align-items-center justify-content-center" style="width: 96px; flex-shrink: 0;">
1140                 <div style="width: 96px; height: 52px;">
1141                     ${gaugeSvg}
1142                 </div>
1143                 <div class="w-100 mt-1">
1144                     ${tempHtml}
1145                 </div>
1146             </div>
1147             <div class="flex-grow-1 ps-1" style="min-width: 0;">
1148                 <div class="sys-core-param-table">
1149                     <div class="sys-core-param-row">
1150                         <span class="sys-core-param-key"><i class="bi bi-activity text-info me-1"></i>Нагрузка</span>
1151                         <span class="sys-core-param-val text-info">${loadStr}</span>
1152                     </div>
1153                     <div class="sys-core-param-row">
1154                         <span class="sys-core-param-key"><i class="bi bi-thermometer-half text-danger me-1"></i>Температура</span>
1155                         <span class="sys-core-param-val ${temp != null && temp >= 75 ? 'text-danger' : 'text-warning'}">${tempStr}</span>

```

```

1156         </div>
1157         <div class="sys-core-param-row">
1158             <span class="sys-core-param-key"><i class="b
1159 i bi-speedometer2 text-info me-1"></i>Частота</span>
1160             <span class="sys-core-param-val text-info">$
1161 {escapeHtml(freqStr)}</span>
1162         </div>
1163         <div class="sys-core-param-row">
1164             <span class="sys-core-param-key"><i class="b
1165 i bi-lightning text-warning me-1"></i>Мощность</span>
1166             <span class="sys-core-param-val text-warning
1167 ">${escapeHtml(powerStr)}</span>
1168         </div>
1169     </div>
1170     </div>
1171     </div>
1172 </div>
1173
1174     <div class="pt-1 border-top" style="border-color: va
1175 r(--border-subtle, rgba(255,255,255,0.06)) !important;">
1176         <div class="text-muted small mb-1" style="font-siz
1177 e: 0.68rem; font-weight: 600;">Потоки:</div>
1178         <div class="sys-core-threads-list">
1179             ${threadsHtml || '<div class="text-muted small">
1180 1 поток</div>'}
1181         </div>
1182     </div>
1183 </div>
1184 `;
1185 }).join('');
1186 }
1187
1188 function formatBytes(v) {
1189     const n = Number(v) || 0;
1190     const units = ['B', 'KB', 'MB', 'GB', 'TB'];
1191     let i = 0;
1192     let x = n;
1193     while (x >= 1024 && i < units.length - 1) { x /= 1024; i++
1194 ; }

```

```
1186     return `${x.toFixed(i === 0 ? 0 : 1)} ${units[i]}`;
1187 }
1188
1189 function formatBytesPerSec(v) {
1190     const n = Number(v) || 0;
1191     const units = ['B/s', 'KB/s', 'MB/s', 'GB/s'];
1192     let i = 0;
1193     let x = n;
1194     while (x >= 1024 && i < units.length - 1) { x /= 1024; i++
1195 ; }
1196     return `${x.toFixed(i === 0 ? 0 : 1)} ${units[i]}`;
1197 }
1198
1199 const _cpuSparkHistory = [];
1200 const _gpuSparkHistory = [];
1201
1202 let currentCpuSparkScale = 'log';
1203 let lastCpuSparkData = [];
1204
1205 function setCpuSparkScale(scale) {
1206     currentCpuSparkScale = scale;
1207     const btnLog = document.getElementById('btn-cpu-spark-log'
1208 );
1209     const btnLin = document.getElementById('btn-cpu-spark-lin'
1210 );
1211     if (btnLog && btnLin) {
1212         if (scale === 'log') {
1213             btnLog.className = 'btn btn-xs btn-outline-info active
1214 py-0 px-2 fw-bold';
1215             btnLin.className = 'btn btn-xs btn-outline-secondary p
1216 y-0 px-2';
1217         } else {
1218             btnLin.className = 'btn btn-xs btn-outline-info active
1219 py-0 px-2 fw-bold';
1220             btnLog.className = 'btn btn-xs btn-outline-secondary p
1221 y-0 px-2';
1222         }
1223     }
1224 }
```

```
1217     if (lastCpuSparkData && lastCpuSparkData.length) {
1218         renderCpuSpark(lastCpuSparkData);
1219     }
1220 }
1221 window.setCpuSparkScale = setCpuSparkScale;
1222
1223 function buildSmoothSvgPath(pts) {
1224     if (!pts || pts.length === 0) return '';
1225     if (pts.length === 1) return `M ${pts[0].x.toFixed(1)} ${p
ts[0].y.toFixed(1)}`;
1226     let d = `M ${pts[0].x.toFixed(1)} ${pts[0].y.toFixed(1)}`;
1227     for (let i = 0; i < pts.length - 1; i++) {
1228         const p0 = i > 0 ? pts[i - 1] : pts[i];
1229         const p1 = pts[i];
1230         const p2 = pts[i + 1];
1231         const p3 = i !== pts.length - 2 ? pts[i + 2] : p2;
1232
1233         const cp1x = p1.x + (p2.x - p0.x) / 6;
1234         const cp1y = p1.y + (p2.y - p0.y) / 6;
1235         const cp2x = p2.x - (p3.x - p1.x) / 6;
1236         const cp2y = p2.y - (p3.y - p1.y) / 6;
1237
1238         d += ` C ${cp1x.toFixed(1)} ${cp1y.toFixed(1)}, ${cp2x.t
oFixed(1)} ${cp2y.toFixed(1)}, ${p2.x.toFixed(1)} ${p2.y.toFix
ed(1)}`;
1239     }
1240     return d;
1241 }
1242
1243 function renderCpuSpark(history) {
1244     const box = document.getElementById('sys-cpu-spark');
1245     const container = document.getElementById('sys-cpu-spark-c
ontainer');
1246     const tooltip = document.getElementById('sys-cpu-spark-too
ltip');
1247     if (!box) return;
1248
1249     if (Array.isArray(history) && history.length >= 2) {
```

```
1250     lastCpuSparkData = history;
1251   } else if (lastCpuSparkData && lastCpuSparkData.length >=
1252   2) {
1253     history = lastCpuSparkData;
1254   } else {
1255     return;
1256   }
1257   box.replaceChildren();
1258   const ns = 'http://www.w3.org/2000/svg';
1259   const W = 600, H = 84;
1260   const padTop = 6, padBottom = 6;
1261   const innerH = H - padTop - padBottom;
1262
1263   const svg = document.createElementNS(ns, 'svg');
1264   svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
1265   svg.setAttribute('preserveAspectRatio', 'none');
1266   svg.style.cssText = 'width:100%;height:100%;display:block;
1267   cursor:crosshair;';
1268
1269   // 1. Defs с градиентами и фильтром свечения
1270   const defs = document.createElementNS(ns, 'defs');
1271   defs.innerHTML = `
1272     <linearGradient id="cpuGradLoad" x1="0%" y1="0%" x2="0%"
1273     y2="100%">
1274       <stop offset="0%" stop-color="#0dcaf0" stop-opacity="0
1275       .38"/>
1276       <stop offset="60%" stop-color="#0dcaf0" stop-opacity="
1277       0.12"/>
1278       <stop offset="100%" stop-color="#0dcaf0" stop-opacity=
1279       "0.0"/>
1280     </linearGradient>
1281     <linearGradient id="cpuGradTemp" x1="0%" y1="0%" x2="0%"
1282     y2="100%">
1283       <stop offset="0%" stop-color="#ef4444" stop-opacity="0
1284       .28"/>
1285       <stop offset="100%" stop-color="#ef4444" stop-opacity=
1286       "0.0"/>
1287     </linearGradient>
1288   `;
```



```

1279         </linearGradient>
1280         <linearGradient id="cpuGradPower" x1="0%" y1="0%" x2="0%
1281         " y2="100%">
1282             <stop offset="0%" stop-color="#f59e0b" stop-opacity="0
1283             .22"/>
1284             <stop offset="100%" stop-color="#f59e0b" stop-opacity=
1285             "0.0"/>
1286         </linearGradient>
1287         `;
1288         svg.appendChild(defs);
1289
1290         // 2. Горизонтальная сетка (Grid Lines)
1291         const gridG = document.createElementNS(ns, 'g');
1292         gridG.setAttribute('opacity', '0.2');
1293         [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
1294             const y = H - paddingBottom - ratio * innerH;
1295             const gl = document.createElementNS(ns, 'line');
1296             gl.setAttribute('x1', '0');
1297             gl.setAttribute('y1', y.toFixed(1));
1298             gl.setAttribute('x2', String(W));
1299             gl.setAttribute('y2', y.toFixed(1));
1300             gl.setAttribute('stroke', '#ffffff');
1301             gl.setAttribute('stroke-dasharray', ratio === 0.0 || ratio
1302             === 1.0 ? 'none' : '3,3');
1303             gl.setAttribute('stroke-width', '1');
1304             gridG.appendChild(gl);
1305         });
1306         svg.appendChild(gridG);
1307
1308         // 3. Расчет динамических диапазонов
1309         const count = history.length;
1310         const isLog = currentCpuSparkScale === 'log';
1311
1312         // Температурный локальный диапазон (для выявления микроко
1313         лебаний)
1314         const rawTemps = history.map(h => Number(h.temperature_c !
1315         = null ? h.temperature_c : (h.temp || 0))).filter(t => t > 0);
1316         const tMin = rawTemps.length ? Math.min(...rawTemps) : 30;

```

```
1311     const tMax = rawTemps.length ? Math.max(...rawTemps) : 90;
1312     const tDelta = Math.max(6, tMax - tMin);
1313
1314     // Диапазон мощности (для выявления микроколебаний)
1315     const rawPowers = history.map(h => Number(h.power_w != null ? h.power_w : (h.power || 0))).filter(p => p > 0);
1316     const pMin = rawPowers.length ? Math.min(...rawPowers) : 5
1317     ;
1318     const pMax = rawPowers.length ? Math.max(...rawPowers) : 1
1319     00;
1320     const pDelta = Math.max(8, pMax - pMin);
1321
1322     // Главная шкала: Загрузка CPU (0..100% высоты холста)
1323     const transformLoad = (v) => {
1324         const cl = Math.min(100, Math.max(0, Number(v || 0)));
1325         if (!isLog) return cl / 100;
1326         return Math.log10(1 + 9 * (cl / 100)); // Log-10 mapping
1327         : 10% -> 0.28, 50% -> 0.74, 100% -> 1.0
1328     };
1329
1330     // Вторичная компактная шкала: Температура (занимает 10% -
1331     38% высоты)
1332     const transformTemp = (v) => {
1333         if (v == null) return 0.12;
1334         const frac = Math.min(1, Math.max(0, (Number(v) - (tMin
1335         - 1)) / (tDelta + 2)));
1336         return 0.10 + frac * 0.28;
1337     };
1338
1339     // Вторичная компактная шкала: Мощность (занимает 4% - 28%
1340     высоты)
1341     const transformPower = (v) => {
1342         if (v == null) return 0.08;
1343         const frac = Math.min(1, Math.max(0, (Number(v) - (pMin
1344         - 1.5)) / (pDelta + 3)));
1345         return 0.04 + frac * 0.24;
1346     };
1347
```

```

1341 // Точки графиков
1342 const loadPts = [];
1343 const tempPts = [];
1344 const powerPts = [];
1345
1346 history.forEach((h, i) => {
1347     const x = (i / (count - 1)) * W;
1348
1349     const lVal = Number(h.load_percent != null ? h.load_percent : (h.load || 0));
1350     const tVal = h.temperature_c != null ? Number(h.temperature_c) : (h.temp != null ? Number(h.temp) : null);
1351     const pVal = h.power_w != null ? Number(h.power_w) : (h.power != null ? Number(h.power) : null);
1352
1353     const yLoad = H - paddingBottom - transformLoad(lVal) * innerH;
1354     const yTemp = H - paddingBottom - transformTemp(tVal) * innerH;
1355     const yPower = H - paddingBottom - transformPower(pVal) * innerH;
1356
1357     loadPts.push({ x, y: yLoad, raw: lVal, time: h.time_label || h.timestamp });
1358     tempPts.push({ x, y: yTemp, raw: tVal });
1359     powerPts.push({ x, y: yPower, raw: pVal });
1360 });
1361
1362 // 4. Отрисовка площадей и кривых (Area & Spline)
1363 const drawSeries = (pts, strokeColor, fillColor, strokeWidth = 1.8) => {
1364     if (pts.length < 2) return;
1365     const linePath = buildSmoothSvgPath(pts);
1366
1367     // Заливка градиентом
1368     if (fillColor) {
1369         const areaPath = `${linePath} L ${pts[pts.length - 1].x.toFixed(1)} ${H - paddingBottom} L ${pts[0].x.toFixed(1)} ${H -

```

```

    padBottom} Z`;
1370     const aEl = document.createElementNS(ns, 'path');
1371     aEl.setAttribute('d', areaPath);
1372     aEl.setAttribute('fill', fillColor);
1373     svg.appendChild(aEl);
1374 }
1375
1376 // Линия кривой
1377 const pEl = document.createElementNS(ns, 'path');
1378 pEl.setAttribute('d', linePath);
1379 pEl.setAttribute('fill', 'none');
1380 pEl.setAttribute('stroke', strokeColor);
1381 pEl.setAttribute('stroke-width', String(strokeWidth));
1382 pEl.setAttribute('stroke-linejoin', 'round');
1383 pEl.setAttribute('stroke-linecap', 'round');
1384 svg.appendChild(pEl);
1385 };
1386
1387 // Рисуем: Тонкие тренды температуры и мощности на компакт
    ной шкале + Основной график загрузки с градиентом
1388 drawSeries(powerPts, '#f59e0b', null, 1.2);
1389 drawSeries(tempPts, '#ef4444', null, 1.3);
1390 drawSeries(loadPts, '#0dcaf0', 'url(#cpuGradLoad)', 2.0);
1391
1392 // 5. Интерактивный Hover-курсор и Тултип
1393 const crosshair = document.createElementNS(ns, 'line');
1394 crosshair.setAttribute('y1', '0');
1395 crosshair.setAttribute('y2', String(H));
1396 crosshair.setAttribute('stroke', 'var(--text-muted, rgba(1
    28,128,128,0.5))');
1397 crosshair.setAttribute('stroke-dasharray', '2,2');
1398 crosshair.setAttribute('stroke-width', '1');
1399 crosshair.style.display = 'none';
1400 svg.appendChild(crosshair);
1401
1402 svg.addEventListener('mousemove', (evt) => {
1403     const rect = svg.getBoundingClientRect();
1404     const mouseX = Math.max(0, Math.min(rect.width, evt.clie

```

```

    ntX - rect.left));
1405     const ratio = mouseX / rect.width;
1406     const idx = Math.min(count - 1, Math.max(0, Math.round(r
    atio * (count - 1))));
1407     const pt = loadPts[idx];
1408     const hItem = history[idx];
1409
1410     if (pt && tooltip) {
1411         crosshair.setAttribute('x1', pt.x.toFixed(1));
1412         crosshair.setAttribute('x2', pt.x.toFixed(1));
1413         crosshair.style.display = 'block';
1414
1415         const tStr = hItem.time_label || (hItem.timestamp ? ne
    w Date(hItem.timestamp).toLocaleTimeString() : (hItem.time ? n
    ew Date(hItem.time).toLocaleTimeString() : '--:--:--'));
1416         const lStr = `${pt.raw.toFixed(1)}%`;
1417         const tempVal = tempPts[idx] && tempPts[idx].raw != nu
    ll ? `${tempPts[idx].raw.toFixed(0)} °C` : '--';
1418         const powVal = powerPts[idx] && powerPts[idx].raw != n
    ull ? `${powerPts[idx].raw.toFixed(1)} W` : '--';
1419
1420         tooltip.innerHTML = `
1421         <div class="d-flex align-items-center justify-conten
    t-between gap-2 border-bottom pb-0.5 mb-1" style="border-color
    : rgba(255,255,255,0.1) !important;">
1422             <span class="text-muted"><i class="bi bi-clock me-
    1"></i>${escapeHtml(tStr)}</span>
1423             <span class="badge ${isLog ? 'bg-info-subtle text-
    info' : 'bg-secondary text-light'}" style="font-size:0.62rem;"
    >${isLog ? 'Log' : 'Lin'}</span>
1424         </div>
1425         <div class="d-flex gap-2 font-monospace">
1426             <span class="text-info fw-bold"><i class="bi bi-ac
    tivity me-0.5"></i>${lStr}</span>
1427             <span class="text-danger fw-bold"><i class="bi bi-
    thermometer-half me-0.5"></i>${tempVal}</span>
1428             <span class="text-warning fw-bold"><i class="bi bi
    -lightning me-0.5"></i>${powVal}</span>

```

```
1429         </div>
1430     `;
1431     tooltip.style.display = 'block';
1432
1433     // Позиционирование тултипа
1434     const tipX = Math.min(rect.width - 150, Math.max(10, mouseX - 60));
1435     tooltip.style.left = `${tipX}px`;
1436     }
1437     });
1438
1439     svg.addEventListener('mouseleave', () => {
1440         crosshair.style.display = 'none';
1441         if (tooltip) tooltip.style.display = 'none';
1442     });
1443
1444     box.appendChild(svg);
1445     }
1446
1447     function setGpuSparkScale(scale, idx = 0) {
1448         currentGpuSparkScales[idx] = scale;
1449         const btnLog = document.getElementById(`btn-gpu-spark-log-${idx}`) || document.getElementById('btn-gpu-spark-log');
1450         const btnLin = document.getElementById(`btn-gpu-spark-lin-${idx}`) || document.getElementById('btn-gpu-spark-lin');
1451         if (btnLog && btnLin) {
1452             if (scale === 'log') {
1453                 btnLog.className = 'btn btn-xs btn-outline-warning active py-0 px-2 fw-bold';
1454                 btnLin.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
1455             } else {
1456                 btnLin.className = 'btn btn-xs btn-outline-warning active py-0 px-2 fw-bold';
1457                 btnLog.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
1458             }
1459         }
1460     }
```

```
1460     if (lastGpuSparkDataMap[idx] && lastGpuSparkDataMap[idx].length) {
1461         renderGpuSpark(lastGpuSparkDataMap[idx], idx);
1462     }
1463 }
1464 window.setGpuSparkScale = setGpuSparkScale;
1465
1466 function renderGpuSpark(history, idx = 0) {
1467     const box = document.getElementById(`sys-gpu-spark-${idx}`) || document.getElementById('sys-gpu-spark');
1468     const container = document.getElementById(`sys-gpu-spark-container-${idx}`) || document.getElementById('sys-gpu-spark-container');
1469     const tooltip = document.getElementById(`sys-gpu-spark-tooltip-${idx}`) || document.getElementById('sys-gpu-spark-tooltip');
1470     if (!box) return;
1471
1472     if (Array.isArray(history) && history.length >= 2) {
1473         lastGpuSparkDataMap[idx] = history;
1474     } else if (lastGpuSparkDataMap[idx] && lastGpuSparkDataMap[idx].length >= 2) {
1475         history = lastGpuSparkDataMap[idx];
1476     } else {
1477         return;
1478     }
1479
1480     box.replaceChildren();
1481     const ns = 'http://www.w3.org/2000/svg';
1482     const W = 600, H = 84;
1483     const padTop = 6, padBottom = 6;
1484     const innerH = H - padTop - padBottom;
1485
1486     const svg = document.createElementNS(ns, 'svg');
1487     svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
1488     svg.setAttribute('preserveAspectRatio', 'none');
1489     svg.style.cssText = 'width:100%;height:100%;display:block;cursor:crosshair;';
```

```

1490
1491 // 1. Defs с уникальным ID градиента для каждого GPU
1492 const defs = document.createElementNS(ns, 'defs');
1493 const gradId = `gpuGradLoad_${idx}`;
1494 defs.innerHTML = `
1495     <linearGradient id="${gradId}" x1="0%" y1="0%" x2="0%" y
1496     2="100%">
1497         <stop offset="0%" stop-color="#f59e0b" stop-opacity="0
1498         .38"/>
1499         <stop offset="60%" stop-color="#f59e0b" stop-opacity="
1500         0.12"/>
1501         <stop offset="100%" stop-color="#f59e0b" stop-opacity=
1502         "0.0"/>
1503     </linearGradient>
1504 `;
1505 svg.appendChild(defs);
1506
1507 // 2. Горизонтальная сетка (Grid Lines)
1508 const gridG = document.createElementNS(ns, 'g');
1509 gridG.setAttribute('opacity', '0.2');
1510 [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
1511     const y = H - paddingBottom - ratio * innerH;
1512     const gl = document.createElementNS(ns, 'line');
1513     gl.setAttribute('x1', '0');
1514     gl.setAttribute('y1', y.toFixed(1));
1515     gl.setAttribute('x2', String(W));
1516     gl.setAttribute('y2', y.toFixed(1));
1517     gl.setAttribute('stroke', '#ffffff');
1518     gl.setAttribute('stroke-dasharray', ratio === 0.0 || rat
1519     io === 1.0 ? 'none' : '3,3');
1520     gl.setAttribute('stroke-width', '1');
1521     gridG.appendChild(gl);
1522 });
1523 svg.appendChild(gridG);
1524
1525 // 3. Расчет динамических диапазонов
1526 const count = history.length;
1527 const isLog = currentGpuSparkScales[idx] === 'log' || curr

```



```

entGpuSparkScales[idx] === undefined;
1523
1524     // Температурный локальный диапазон
1525     const rawTemps = history.map(h => Number(h.temperature_c !
= null ? h.temperature_c : (h.temp || 0))).filter(t => t > 0);
1526     const tMin = rawTemps.length ? Math.min(...rawTemps) : 30;
1527     const tMax = rawTemps.length ? Math.max(...rawTemps) : 90;
1528     const tDelta = Math.max(6, tMax - tMin);
1529
1530     // Диапазон мощности / памяти
1531     const rawPowers = history.map(h => Number(h.power_w != nul
l ? h.power_w : (h.power || h.vram_percent || 0))).filter(p =>
p > 0);
1532     const pMin = rawPowers.length ? Math.min(...rawPowers) : 5
;
1533     const pMax = rawPowers.length ? Math.max(...rawPowers) : 1
00;
1534     const pDelta = Math.max(8, pMax - pMin);
1535
1536     // Главная шкала: Загрузка GPU Core (0..100% высоты холста
)
1537     const transformLoad = (v) => {
1538         const cl = Math.min(100, Math.max(0, Number(v || 0)));
1539         if (!isLog) return cl / 100;
1540         return Math.log10(1 + 9 * (cl / 100));
1541     };
1542
1543     // Вторичная компактная шкала: Температура (занимает 10% -
38% высоты)
1544     const transformTemp = (v) => {
1545         if (v == null) return 0.12;
1546         const frac = Math.min(1, Math.max(0, (Number(v) - (tMin
- 1)) / (tDelta + 2)));
1547         return 0.10 + frac * 0.28;
1548     };
1549
1550     // Вторичная компактная шкала: Мощность/Память (занимает 4
% - 28% высоты)

```

```
1551     const transformPower = (v) => {
1552         if (v == null) return 0.08;
1553         const frac = Math.min(1, Math.max(0, (Number(v) - (pMin
1554 - 1.5)) / (pDelta + 3)));
1555         return 0.04 + frac * 0.24;
1556     };
1557     // Точки графиков
1558     const loadPts = [];
1559     const tempPts = [];
1560     const powerPts = [];
1561
1562     history.forEach((h, i) => {
1563         const x = (i / (count - 1)) * W;
1564
1565         const lVal = Number(h.load_percent != null ? h.load_perc
1566 ent : (h.load || 0));
1567         const tVal = h.temperature_c != null ? Number(h.temperat
1568 ure_c) : (h.temp != null ? Number(h.temp) : null);
1569         const pVal = h.power_w != null ? Number(h.power_w) : (h.
1570 power != null ? Number(h.power) : (h.vram_percent != null ? Nu
1571 mber(h.vram_percent) : null));
1572
1573         const yLoad = H - paddingBottom - transformLoad(lVal) * inne
1574 rH;
1575         const yTemp = H - paddingBottom - transformTemp(tVal) * inne
1576 rH;
1577         const yPower = H - paddingBottom - transformPower(pVal) * in
1578 nerH;
1579
1580         loadPts.push({ x, y: yLoad, raw: lVal, time: h.time_labe
1581 l || h.timestamp });
1582         tempPts.push({ x, y: yTemp, raw: tVal });
1583         powerPts.push({ x, y: yPower, raw: pVal });
1584     });
1585
1586     // 4. Отрисовка площадей и кривых (Area & Spline)
1587     const drawSeries = (pts, strokeColor, fillColor, strokeWid
```

```

th = 1.8) => {
1580     if (pts.length < 2) return;
1581     const linePath = buildSmoothSvgPath(pts);
1582
1583     // Заливка градиентом
1584     if (fillColor) {
1585         const areaPath = `${linePath} L ${pts[pts.length - 1].
x.toFixed(1)} ${H - paddingBottom} L ${pts[0].x.toFixed(1)} ${H -
paddingBottom} Z`;
1586         const aEl = document.createElementNS(ns, 'path');
1587         aEl.setAttribute('d', areaPath);
1588         aEl.setAttribute('fill', fillColor);
1589         svg.appendChild(aEl);
1590     }
1591
1592     // Линия кривой
1593     const pEl = document.createElementNS(ns, 'path');
1594     pEl.setAttribute('d', linePath);
1595     pEl.setAttribute('fill', 'none');
1596     pEl.setAttribute('stroke', strokeColor);
1597     pEl.setAttribute('stroke-width', String(strokeWidth));
1598     pEl.setAttribute('stroke-linejoin', 'round');
1599     pEl.setAttribute('stroke-linecap', 'round');
1600     svg.appendChild(pEl);
1601 };
1602
1603 drawSeries(powerPts, '#0dcaf0', null, 1.2);
1604 drawSeries(tempPts, '#ef4444', null, 1.3);
1605 drawSeries(loadPts, '#f59e0b', `url(#${gradId})`, 2.0);
1606
1607 // 5. Интерактивный Hover-курсор и Тултип
1608 const crosshair = document.createElementNS(ns, 'line');
1609 crosshair.setAttribute('y1', '0');
1610 crosshair.setAttribute('y2', String(H));
1611 crosshair.setAttribute('stroke', 'var(--text-muted, rgba(1
28,128,128,0.5))');
1612 crosshair.setAttribute('stroke-dasharray', '2,2');
1613 crosshair.setAttribute('stroke-width', '1');

```

```

1614 |     crosshair.style.display = 'none';
1615 |     svg.appendChild(crosshair);
1616 |
1617 |     svg.addEventListener('mousemove', (evt) => {
1618 |         const rect = svg.getBoundingClientRect();
1619 |         const mouseX = Math.max(0, Math.min(rect.width, evt.clientX - rect.left));
1620 |         const ratio = mouseX / rect.width;
1621 |         const curIndex = Math.min(count - 1, Math.max(0, Math.round(ratio * (count - 1))));
1622 |         const pt = loadPts[curIndex];
1623 |         const hItem = history[curIndex];
1624 |
1625 |         if (pt && tooltip) {
1626 |             crosshair.setAttribute('x1', pt.x.toFixed(1));
1627 |             crosshair.setAttribute('x2', pt.x.toFixed(1));
1628 |             crosshair.style.display = 'block';
1629 |
1630 |             const tStr = hItem.time_label || (hItem.timestamp ? new Date(hItem.timestamp).toLocaleTimeString() : (hItem.time ? new Date(hItem.time).toLocaleTimeString() : '--:--:--'));
1631 |             const lStr = `${pt.raw.toFixed(1)}%`;
1632 |             const tempVal = tempPts[curIndex] && tempPts[curIndex].raw != null ? `${tempPts[curIndex].raw.toFixed(0)} °C` : '--';
1633 |             const powVal = powerPts[curIndex] && powerPts[curIndex].raw != null ? (hItem.power_w != null ? `${powerPts[curIndex].raw.toFixed(1)} W` : `${powerPts[curIndex].raw.toFixed(0)}%`) : '--';
1634 |
1635 |             tooltip.innerHTML = `
1636 |                 <div class="d-flex align-items-center justify-content-between gap-2 border-bottom pb-0.5 mb-1" style="border-color: rgba(255,255,255,0.1) !important;">
1637 |                     <span class="text-muted"><i class="bi bi-clock me-1"></i>${escapeHtml(tStr)}</span>
1638 |                     <span class="badge ${isLog ? 'bg-warning-subtle text-warning' : 'bg-secondary text-light'}" style="font-size:0.6

```

```

2rem;">${isLog ? 'Log' : 'Lin'}</span>
1639     </div>
1640     <div class="d-flex gap-2 font-monospace">
1641         <span class="text-warning fw-bold"><i class="bi bi-
-activity me-0.5"></i>${lStr}</span>
1642         <span class="text-danger fw-bold"><i class="bi bi-
thermometer-half me-0.5"></i>${tempVal}</span>
1643         <span class="text-info fw-bold"><i class="bi bi-li
ghtning me-0.5"></i>${powVal}</span>
1644     </div>
1645     `;
1646     tooltip.style.display = 'block';
1647
1648     const tipX = Math.min(rect.width - 150, Math.max(10, m
ouseX - 60));
1649     tooltip.style.left = `${tipX}px`;
1650     }
1651     });
1652
1653     svg.addEventListener('mouseleave', () => {
1654         crosshair.style.display = 'none';
1655         if (tooltip) tooltip.style.display = 'none';
1656     });
1657
1658     box.appendChild(svg);
1659 }
1660
1661 function renderMemIoSpark(history) {
1662     const box = document.getElementById('sys-memio-spark');
1663     if (!box) return;
1664     box.replaceChildren();
1665     if (!Array.isArray(history) || history.length < 2) return;
1666     const ns = 'http://www.w3.org/2000/svg';
1667     const W = 200, H = 56;
1668     const svg = document.createElementNS(ns, 'svg');
1669     svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
1670     svg.setAttribute('preserveAspectRatio', 'none');
1671     svg.style.cssText = 'width:100%;height:100%;';

```

```

1672     const maxIo = Math.max(1, ...history.map(h => Math.max(h.r
ead_bytes_sec, h.write_bytes_sec)));
1673     const line = (getY, color) => {
1674         const pts = history.map((h, i) => `${(i / (history.length
h - 1) * W).toFixed(1)}${(H - getY(h) * (H - 2) - 1).toFixed(
1)} `);
1675         const pl = document.createElementNS(ns, 'polyline');
1676         pl.setAttribute('points', pts.join(' '));
1677         pl.setAttribute('fill', 'none');
1678         pl.setAttribute('stroke', color);
1679         pl.setAttribute('stroke-width', '1.2');
1680         svg.appendChild(pl);
1681     };
1682     line(h => Math.min(100, h.memory_percent) / 100, '#198754'
);
1683     line(h => h.read_bytes_sec / maxIo, '#0dcaf0');
1684     line(h => h.write_bytes_sec / maxIo, '#ffc107');
1685     box.appendChild(svg);
1686 }
1687
1688 /**
1689  * Генерация цветного процентного слайдера (для памяти, RAM,
Swap).
1690  * @param {number|null} percent - Значение в % (0-100)
1691  * @param {boolean} isLarge - Флаг увеличенного отображения
для шапки
1692  * @param {string} labelText - Подпись бейджа справа
1693  * @returns {string} HTML разметка
1694  */
1695 function createPercentSliderHtml(percent, isLarge = false, l
abelText = '') {
1696     const val = percent == null ? 0 : Math.min(100, Math.max(0
, Number(percent)));
1697     let barColor = '#10b981'; // Зеленый (<60%)
1698     let badgeClass = 'text-success';
1699     if (val >= 85) {
1700         barColor = '#ef4444'; // Красный
1701         badgeClass = 'text-danger fw-bold';

```

```
1702     } else if (val >= 70) {
1703         barColor = '#f97316'; // Оранжевый
1704         badgeClass = 'text-warning fw-bold';
1705     } else if (val >= 50) {
1706         barColor = '#eab308'; // Желтый
1707         badgeClass = 'text-warning';
1708     }
1709
1710     const wrapClass = isLarge ? 'sys-temp-slider-wrap sys-temp
1711     -slider-lg' : 'sys-temp-slider-wrap';
1712     const fontSize = isLarge ? 'font-size: 0.85rem;' : 'font-s
1713     ize: 0.78rem;';
1714     const displayLabel = labelText || `${val.toFixed(0)}%`;
1715
1716     return `
1717         <div class="${wrapClass}">
1718             <div class="d-flex justify-content-between align-items
1719             -center mb-1" style="${fontSize}">
1720                 <span class="text-muted"><i class="bi bi-memory me-1
1721                 "></i>Занято</span>
1722                 <span class="font-monospace ${badgeClass}" style="fo
1723                 nt-weight: 700;">${displayLabel}</span>
1724             </div>
1725             <div class="sys-temp-slider-track">
1726                 <div class="sys-temp-slider-bar" style="width: ${val
1727                 .toFixed(1)}%; background: ${barColor};"></div>
1728                 <div class="sys-temp-slider-thumb" style="left: ${va
1729                 l.toFixed(1)}%; background: ${barColor};"></div>
1730             </div>
1731         </div>
1732     `;
1733 }
1734
1735 async function fetchMemoryIoFromApi() {
1736     try {
1737         const res = await fetch('/api/v1/panel/memory-io?limit=6
1738         0');
1739         if (!res.ok) return;
```

```
1732 |         const data = await res.json();
1733 |         const set = (id, text) => { const el = document.getEleme
1734 | ntById(id); if (el) el.innerText = text; };
1735 |         const m = data.memory || {};
1736 |         const io = data.disk_io || {};
1737 |         const pct = Number(m.percent || 0);
1738 |         const swapPct = Number(m.swap_percent || 0);
1739 |         const usedGb = Number(m.used_gb || 0);
1740 |         const totalGb = Number(m.total_gb || 0);
1741 |         const freeGb = Number(m.free_gb || 0);
1742 |         // 1. Главный баннер RAM
1743 |         const ramModel = document.getElementById('sys-metric-ram
1744 | -model');
1745 |         if (ramModel) {
1746 |             ramModel.innerText = data.name || (m.name || (m.total_
1747 | gb ? `${Math.round(m.total_gb)} GB RAM` : 'RAM'));
1748 |         }
1749 |         set('sys-memio-pct', `${pct.toFixed(1)}%`);
1750 |         set('sys-memio-used', `${usedGb.toFixed(1)} / ${totalGb.
1751 | toFixed(1)} GB (свободно ${freeGb.toFixed(1)} GB)`);
1752 |         set('sys-memio-swap', `Файл подкачки: ${swapPct.toFixed(
1753 | 1)}%`);
1754 |         set('sys-memio-ts', data.timestamp ? new Date(data.times
1755 | tamp).toLocaleTimeString() : '--');
1756 |
1757 |         const ramGauge = document.getElementById('sys-metric-ram
1758 | -gauge');
1759 |         if (ramGauge) {
1760 |             ramGauge.innerHTML = createGaugeSvg(pct, 130, 74);
1761 |         }
1762 |
1763 |         const ramSlider = document.getElementById('sys-metric-ra
1764 | m-slider');
1765 |         if (ramSlider) {
1766 |             ramSlider.innerHTML = createPercentSliderHtml(pct, tru
1767 | e, `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)} GB`);
1768 |         }
1769 |     }
1770 | }
```



```
1761
1762     // 2. Блок 1: Физическая RAM
1763     set('sys-ram-sub-badge', `${pct.toFixed(0)}%`);
1764     const ramSubGauge = document.getElementById('sys-ram-sub
-gauge');
1765     if (ramSubGauge) {
1766         ramSubGauge.innerHTML = createGaugeSvg(pct, 100, 56);
1767     }
1768     const ramSubSlider = document.getElementById('sys-ram-su
b-slider');
1769     if (ramSubSlider) {
1770         ramSubSlider.innerHTML = createPercentSliderHtml(pct,
false, `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)} GB`);
1771     }
1772
1773     // 3. Блок 2: Файл подкачки (Swap)
1774     set('sys-swap-sub-badge', `${swapPct.toFixed(1)}%`);
1775     const swapSubGauge = document.getElementById('sys-swap-s
ub-gauge');
1776     if (swapSubGauge) {
1777         swapSubGauge.innerHTML = createGaugeSvg(swapPct, 100,
56);
1778     }
1779     const swapSubSlider = document.getElementById('sys-swap-
sub-slider');
1780     if (swapSubSlider) {
1781         swapSubSlider.innerHTML = createPercentSliderHtml(swap
Pct, false, `Подкачка ${swapPct.toFixed(1)}%`);
1782     }
1783
1784     // 4. Блок 3 & 4: Чтение и запись на диск (IO)
1785     const rBps = Number(io.read_bytes_sec || 0);
1786     const wBps = Number(io.write_bytes_sec || 0);
1787     const readFormatted = formatBytesPerSec(rBps);
1788     const writeFormatted = formatBytesPerSec(wBps);
1789
1790     set('sys-memio-read', readFormatted);
1791     set('sys-read-sub-badge', readFormatted);
```

```
1792 |         set('sys-memio-write', writeFormatted);
1793 |         set('sys-write-sub-badge', writeFormatted);
1794 |
1795 |         // Вычисление процента скорости относительно шкалы 50 MB
1796 |         /s
1797 |         const maxIoScale = 50 * 1024 * 1024; // 50 MB/s
1798 |         const readPct = Math.min(100, (rBps / maxIoScale) * 100)
1799 |         ;
1800 |         const writePct = Math.min(100, (wBps / maxIoScale) * 100
1801 |         );
1802 |         const readSubGauge = document.getElementById('sys-read-s
1803 |         ub-gauge');
1804 |         if (readSubGauge) {
1805 |             readSubGauge.innerHTML = createGaugeSvg(readPct, 100,
1806 |             56);
1807 |         }
1808 |         const readSubBar = document.getElementById('sys-read-sub
1809 |         -bar');
1810 |         const readSubThumb = document.getElementById('sys-read-s
1811 |         ub-thumb');
1812 |         if (readSubBar) readSubBar.style.width = `${readPct.toFi
1813 |         xed(1)}%`;
1814 |         if (readSubThumb) readSubThumb.style.left = `${readPct.t
1815 |         oFixed(1)}%`;
1816 |
1817 |         const writeSubGauge = document.getElementById('sys-write
1818 |         -sub-gauge');
1819 |         if (writeSubGauge) {
1820 |             writeSubGauge.innerHTML = createGaugeSvg(writePct, 100
1821 |             , 56);
1822 |         }
1823 |         const writeSubBar = document.getElementById('sys-write-s
1824 |         ub-bar');
1825 |         const writeSubThumb = document.getElementById('sys-write
1826 |         -sub-thumb');
1827 |         if (writeSubBar) writeSubBar.style.width = `${writePct.t
1828 |         oFixed(1)}%`;
```

```
1816         if (writeSubThumb) writeSubThumb.style.left = `${writePc
|t.toFixed(1)}%`;
1817
1818         // 5. График спарклайна
1819         renderMemIoSpark(data.history);
1820     } catch (e) {
1821         console.warn('[SystemInspectorTab] Ошибка получения памя
|ти/IO из API:', e);
1822     }
1823 }
1824
1825 const _netSparkHistory = [];
1826
1827 function renderNetSpark(history) {
1828     const box = document.getElementById('sys-net-spark');
1829     if (!box) return;
1830     box.replaceChildren();
1831     if (!Array.isArray(history) || history.length < 2) return;
1832     const ns = 'http://www.w3.org/2000/svg';
1833     const W = 200, H = 56;
1834     const svg = document.createElementNS(ns, 'svg');
1835     svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
1836     svg.setAttribute('preserveAspectRatio', 'none');
1837     svg.style.cssText = 'width:100%;height:100%;';
1838     const maxSpeed = Math.max(1024, ...history.map(h => Math.m
|ax(h.download_bytes_sec || 0, h.upload_bytes_sec || 0)));
1839     const line = (getY, color) => {
1840         const pts = history.map((h, i) => `${(i / (history.lengt
|h - 1) * W).toFixed(1)},${(H - getY(h) * (H - 4) - 2).toFixed(
|1)} `);
1841         const pl = document.createElementNS(ns, 'polyline');
1842         pl.setAttribute('points', pts.join(' '));
1843         pl.setAttribute('fill', 'none');
1844         pl.setAttribute('stroke', color);
1845         pl.setAttribute('stroke-width', '1.4');
1846         svg.appendChild(pl);
1847     };
1848     line(h => Math.min(1, Math.max(0, Number(h.download_bytes_
```

```
sec || 0) / maxSpeed)), '#0dcaf0');
1849     line(h => Math.min(1, Math.max(0, Number(h.upload_bytes_se
c || 0) / maxSpeed)), '#f59e0b');
1850     box.appendChild(svg);
1851   }
1852
1853   async function fetchNetworkLoadFromApi() {
1854     try {
1855       const res = await fetch('/api/v1/panel/network-load');
1856       if (!res.ok) return;
1857       const data = await res.json();
1858       const set = (id, text) => {
1859         const el = document.getElementById(id);
1860         if (el) el.innerText = text;
1861       };
1862
1863       const rxBps = Number(data.download_bytes_sec || 0);
1864       const txBps = Number(data.upload_bytes_sec || 0);
1865       const totalBps = rxBps + txBps;
1866       const utilPct = Number(data.utilization_percent || 0);
1867
1868       const rxFormatted = formatBytesPerSec(rxBps);
1869       const txFormatted = formatBytesPerSec(txBps);
1870       const totalFormatted = formatBytesPerSec(totalBps);
1871
1872       const rxTotalFormatted = formatBytes(data.download_total
_bytes || 0);
1873       const txTotalFormatted = formatBytes(data.upload_total_b
ytes || 0);
1874       const sumTotalFormatted = formatBytes((data.download_tot
al_bytes || 0) + (data.upload_total_bytes || 0));
1875
1876       // 1. Главный баннер Сети
1877       const netModel = document.getElementById('sys-metric-net
-model');
1878       if (netModel) {
1879         netModel.innerText = data.name || 'Сетевые адаптеры (E
thernet / Wi-Fi / BT)';
```

```
1880 |     }
1881 |     set('sys-net-ts', data.timestamp || 'Live');
1882 |     set('sys-net-speed-val', totalFormatted);
1883 |     set('sys-net-sub-details', `❏ ${rxFormatted} · ❏ ${txFor
1884 | matted} (Всего: ${sumTotalFormatted})`);
1885 |     set('sys-net-util-label', `Загрузка канала: ${utilPct.to
1886 | Fixed(1)}% (Линк: ${data.link_speed_mbps || 1000} Mbps)`);
1887 |     set('sys-net-speed-badge', `${data.link_speed_mbps || 10
1888 | 00} Mbps Full Duplex`);
1889 |
1890 |     // Экспресс-метрики KPI в центре баннера сети
1891 |     set('sys-net-kpi-rx', rxFormatted);
1892 |     set('sys-net-kpi-tx', txFormatted);
1893 |     set('sys-net-kpi-total', sumTotalFormatted);
1894 |     set('sys-net-kpi-util', `${utilPct.toFixed(1)}%`);
1895 |
1896 |     const netGauge = document.getElementById('sys-metric-net
1897 | -gauge');
1898 |     if (netGauge) {
1899 |         netGauge.innerHTML = createGaugeSvg(utilPct, 130, 74);
1900 |     }
1901 |
1902 |     const netSlider = document.getElementById('sys-metric-ne
1903 | t-slider');
1904 |     if (netSlider) {
1905 |         netSlider.innerHTML = createPercentSliderHtml(utilPct,
1906 | true, `Канал ${utilPct.toFixed(1)}%`);
1907 |     }
1908 |
1909 |     // 2. Блок 1: Входящий (Rx)
1910 |     set('sys-rx-sub-badge', rxFormatted);
1911 |     set('sys-rx-total-text', `${rxTotalFormatted} всего`);
1912 |     const maxSpeedScale = 12.5 * 1024 * 1024; // 100 Mbps =
1913 | 12.5 MB/s
1914 |     const rxPct = Math.min(100, (rxBps / maxSpeedScale) * 10
1915 | 0);
1916 |     const rxSubGauge = document.getElementById('sys-rx-sub-g
1917 | auge');
```

```
1909 |         if (rxSubGauge) rxSubGauge.innerHTML = createGaugeSvg(rx
| Pct, 100, 56);
1910 |         const rxSubBar = document.getElementById('sys-rx-sub-bar
| ');
1911 |         const rxSubThumb = document.getElementById('sys-rx-sub-t
| humb');
1912 |         if (rxSubBar) rxSubBar.style.width = `${rxPct.toFixed(1)
| }%`;
1913 |         if (rxSubThumb) rxSubThumb.style.left = `${rxPct.toFixed
| (1)}%`;
1914 |
1915 |         // 3. Блок 2: Исходящий (Tx)
1916 |         set('sys-tx-sub-badge', txFormatted);
1917 |         set('sys-tx-total-text', `${txTotalFormatted} всего`);
1918 |         const txPct = Math.min(100, (txBps / maxSpeedScale) * 10
| 0);
1919 |         const txSubGauge = document.getElementById('sys-tx-sub-g
| auge');
1920 |         if (txSubGauge) txSubGauge.innerHTML = createGaugeSvg(tx
| Pct, 100, 56);
1921 |         const txSubBar = document.getElementById('sys-tx-sub-bar
| ');
1922 |         const txSubThumb = document.getElementById('sys-tx-sub-t
| humb');
1923 |         if (txSubBar) txSubBar.style.width = `${txPct.toFixed(1)
| }%`;
1924 |         if (txSubThumb) txSubThumb.style.left = `${txPct.toFixed
| (1)}%`;
1925 |
1926 |         // 4. Блок 3: Wi-Fi
1927 |         const wifi = data.wifi || {};
1928 |         const wifiState = wifi.state || 'Disconnected';
1929 |         const wifiSignal = wifi.signal_percent !== null ? Number(
| wifi.signal_percent) : (wifi.radio_status && wifi.radio_status
| .includes('On') ? 100 : 0);
1930 |         set('sys-wifi-sub-badge', wifi.ssid ? `SSID: ${wifi.ssid
| }` : (wifi.radio_status || 'On'));
1931 |         set('sys-wifi-status-text', wifi.ssid ? `${wifi.ssid} ($
```

```

    {wifiSignal}%)` : wifiState);
1932     const wifiSubGauge = document.getElementById('sys-wifi-s
ub-gauge');
1933     if (wifiSubGauge) wifiSubGauge.innerHTML = createGaugeSv
g(wifiSignal, 100, 56);
1934     const wifiSubBar = document.getElementById('sys-wifi-sub
-bar');
1935     const wifiSubThumb = document.getElementById('sys-wifi-s
ub-thumb');
1936     if (wifiSubBar) wifiSubBar.style.width = `${wifiSignal}%
`;
1937     if (wifiSubThumb) wifiSubThumb.style.left = `${wifiSigna
l}%`;
1938
1939     // 5. Блок 4: Bluetooth
1940     const bt = data.bluetooth || {};
1941     const btDevCount = bt.devices_count || 0;
1942     set('sys-bt-sub-badge', bt.status || 'Active');
1943     set('sys-bt-devices-text', btDevCount > 0 ? `${btDevCoun
t} устр. (${bt.devices ? bt.devices[0] : ''})` : 'Активен (0 у
стр.)');
1944     const btPct = Math.min(100, btDevCount * 25);
1945     const btSubGauge = document.getElementById('sys-bt-sub-g
auge');
1946     if (btSubGauge) btSubGauge.innerHTML = createGaugeSvg(bt
Pct > 0 ? btPct : 10, 100, 56);
1947     const btSubBar = document.getElementById('sys-bt-sub-bar
');
1948     const btSubThumb = document.getElementById('sys-bt-sub-t
humb');
1949     if (btSubBar) btSubBar.style.width = `${btPct > 0 ? btPc
t : 10}%`;
1950     if (btSubThumb) btSubThumb.style.left = `${btPct > 0 ? b
tPct : 10}%`;
1951
1952     // 6. График истории
1953     if (Array.isArray(data.history) && data.history.length >
0) {

```

```
1954         renderNetSpark(data.history);
1955     }
1956 } catch (e) {
1957     console.warn('[SystemInspectorTab] Ошибка получения сете
вой нагрузки из API:', e);
1958 }
1959 }
1960
1961 function renderStorageCards(data) {
1962     const box = document.getElementById('sys-metric-storage-ca
rds');
1963     const badgeDrives = document.getElementById('sys-storage-d
rives-badge');
1964     const badgeParts = document.getElementById('sys-storage-pa
rts-badge');
1965     const badgeSummary = document.getElementById('sys-storage-
summary-badge');
1966     if (!box) return;
1967
1968     const drives = Array.isArray(data.drives) ? data.drives :
[];
1969     const partitions = Array.isArray(data.partitions) ? data.p
artitions : [];
1970
1971     if (badgeDrives) badgeDrives.textContent = `${drives.lengt
h} дисков`;
1972     if (badgeParts) badgeParts.textContent = `${partitions.len
gth} разделов`;
1973     if (badgeSummary) {
1974         const summary = data.summary || {};
1975         badgeSummary.textContent = `${summary.used_gb || 0} / ${
summary.total_gb || 0} GB (${summary.used_percent || 0}%)`;
1976     }
1977
1978     if (drives.length === 0 && partitions.length === 0) {
1979         box.innerHTML = `<div class="text-muted small text-cen
ter py-3 col-12">Опрос накопителей выполняется...</div>`;
1980     }
    return;
```



```

1981     }
1982
1983     if (drives.length > 0) {
1984         box.innerHTML = drives.map(d => {
1985             const pct = d.used_percent != null ? Number(d.used_per
cent) : 0;
1986             const temp = d.temperature_c != null ? Number(d.temper
ature_c) : null;
1987             const totGb = d.total_gb != null ? Number(d.total_gb)
: 0;
1988             const usedGb = d.used_gb != null ? Number(d.used_gb) :
0;
1989             const gaugeSvg = createGaugeSvg(pct, 100, 56);
1990             const percentSlider = createPercentSliderHtml(pct, fal
se, `${usedGb.toFixed(0)} / ${totGb.toFixed(0)} GB`);
1991             const tempHtml = temp != null ? createTempSliderHtml(t
emp, 25, 75, false) : '';
1992             const mediaBadgeClass = d.media_type === 'SSD' || d.me
dia_type === 'NVMe' ? 'bg-primary-subtle text-primary border-p
rimary' : 'bg-warning-subtle text-warning border-warning';
1993
1994             let subInfo = '';
1995             if (d.read_rate_raw || d.write_rate_raw) {
1996                 subInfo = `
1997                     <div class="d-flex justify-content-between text-mu
ted mt-1" style="font-size: 0.72rem;">
1998                         <span><i class="bi bi-arrow-down text-info"></i>
${escapeHtml(d.read_rate_raw || '0 B/s')}</span>
1999                         <span><i class="bi bi-arrow-up text-warning"></i>
> ${escapeHtml(d.write_rate_raw || '0 B/s')}</span>
2000                     </div>
2001                 `;
2002             } else if (d.activity_percent != null) {
2003                 subInfo = `<div class="text-muted mt-1" style="font-
size: 0.72rem;">Активность: <span class="text-light fw-bold">${
{d.activity_percent}%</span></div>`;
2004             }
2005

```

```

2006         return `
2007         <div class="sys-core-card" title="${escapeHtml(d.name)}: Занято ${pct.toFixed(1)}% (${usedGb.toFixed(1)} GB / ${totalGb.toFixed(1)} GB)${temp != null ? ', Температура ' + temp.toFixed(0) + '°C' : ''}">
2008             <div class="d-flex justify-content-between align-items-center mb-1">
2009                 <span class="sys-core-title text-truncate" style="max-width: 140px;" title="${escapeHtml(d.name)}">${escapeHtml(d.name)}</span>
2010                 <span class="badge ${mediaBadgeClass} border" style="font-size: 0.65rem;">${escapeHtml(d.media_type)}</span>
2011             </div>
2012             <div class="py-1">
2013                 ${gaugeSvg}
2014             </div>
2015             <div class="mb-1">
2016                 ${percentSlider}
2017             </div>
2018             ${tempHtml ? `<div class="mb-1">${tempHtml}</div>`
2019             : ''}
2019             ${subInfo}
2020         </div>
2021         `;
2022     }).join('');
2023 } else {
2024     box.innerHTML = partitions.map(p => {
2025         const pct = Number(p.used_percent || 0);
2026         const gaugeSvg = createGaugeSvg(pct, 100, 56);
2027         const percentSlider = createPercentSliderHtml(pct, false, `${p.used_gb.toFixed(0)} / ${p.total_gb.toFixed(0)} GB`);
2028
2029         return `
2030         <div class="sys-core-card" title="Раздел ${escapeHtml(p.device)} (${escapeHtml(p.fstype)}): Занято ${pct.toFixed(1)}%">
2031             <div class="d-flex justify-content-between align-items-center mb-1">

```

```

2032 |         <span class="sys-core-title">${escapeHtml(p.device)} (${escapeHtml(p.fstype))}</span>
2033 |         <span class="badge bg-secondary border" style="font-size: 0.65rem;">${p.free_gb.toFixed(0)} GB free</span>
2034 |     </div>
2035 |     <div class="py-1">
2036 |         ${gaugeSvg}
2037 |     </div>
2038 |     <div>
2039 |         ${percentSlider}
2040 |     </div>
2041 | </div>
2042 | `;
2043 | }).join('');
2044 | }
2045 | }
2046 |
2047 | async function fetchStorageLoadFromApi() {
2048 |     try {
2049 |         const res = await fetch('/api/v1/panel/storage-load');
2050 |         if (!res.ok) return;
2051 |         const data = await res.json();
2052 |         const set = (id, text) => { const el = document.getElementByIdById(id); if (el) el.innerText = text; };
2053 |         const s = data.summary || {};
2054 |         const pct = Number(s.used_percent || 0);
2055 |         const usedGb = Number(s.used_gb || 0);
2056 |         const totalGb = Number(s.total_gb || 0);
2057 |         const freeGb = Number(s.free_gb || 0);
2058 |         const maxTemp = s.max_temperature_c !== null ? Number(s.max_temperature_c) : null;
2059 |
2060 |         // 1. Главный баннер хранилища
2061 |         set('sys-storage-pct', `${pct.toFixed(1)}%`);
2062 |         set('sys-storage-used', `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)} GB (свободно ${freeGb.toFixed(1)} GB)`);
2063 |         set('sys-storage-temp-label', maxTemp !== null ? `Макс. температура: ${maxTemp.toFixed(0)} °C` : 'Макс. температура: --

```

```

    °C');
2064     set('sys-storage-ts', new Date().toLocaleTimeString());
2065
2066     const stModel = document.getElementById('sys-metric-storage-model');
2067     if (stModel && totalGb > 0) {
2068         const totalTb = totalGb >= 1000 ? `${(totalGb / 1024).toFixed(2)} TB` : `${totalGb.toFixed(0)} GB`;
2069         stModel.innerText = `Дисковая подсистема (${totalTb})`;
2070     }
2071
2072     // Экспресс-метрики KPI в центре баннера хранилища
2073     set('sys-storage-kpi-used', `${usedGb.toFixed(1)} GB`);
2074     set('sys-storage-kpi-free', `${freeGb.toFixed(1)} GB`);
2075     const dCount = s.drives_count || (Array.isArray(data.drives) ? data.drives.length : '--');
2076     const pCount = s.partitions_count || (Array.isArray(data.partitions) ? data.partitions.length : '--');
2077     set('sys-storage-kpi-drives', `${dCount} диск. / ${pCount} разд.`);
2078     const stKpiTemp = document.getElementById('sys-storage-kpi-temp');
2079     if (stKpiTemp) {
2080         stKpiTemp.textContent = maxTemp != null ? `${maxTemp.toFixed(0)} °C` : '-- °C';
2081         if (maxTemp != null) {
2082             stKpiTemp.className = maxTemp >= 60 ? 'sys-kpi-value text-danger' : maxTemp >= 48 ? 'sys-kpi-value text-warning' : 'sys-kpi-value text-info';
2083         }
2084     }
2085
2086     const stGauge = document.getElementById('sys-metric-storage-gauge');
2087     if (stGauge) {
2088         stGauge.innerHTML = createGaugeSvg(pct, 130, 74);
2089     }

```

```
2090 |
2091 |     const stSlider = document.getElementById('sys-metric-storage-temp-slider');
2092 |     if (stSlider) {
2093 |         stSlider.innerHTML = createTempSliderHtml(maxTemp, 25,
2094 |             75, true);
2095 |     }
2096 |     // 2. Карточки дисков
2097 |     renderStorageCards(data);
2098 | } catch (e) {
2099 |     console.warn('[SystemInspectorTab] Ошибка получения параметров дисков из API:', e);
2100 | }
2101 | }
2102 |
2103 | async function fetchCpuLoadFromApi() {
2104 |     try {
2105 |         const res = await fetch('/api/v1/panel/cpu-load');
2106 |         if (!res.ok) return;
2107 |         const data = await res.json();
2108 |         lastCpuData = data;
2109 |         const cpuVal = document.getElementById('sys-metric-cpu-val');
2110 |         const cpuFill = document.getElementById('sys-metric-cpu-fill');
2111 |         const cpuSub = document.getElementById('sys-metric-cpu-sub');
2112 |         const cpuModel = document.getElementById('sys-metric-cpu-model');
2113 |         const cpuGauge = document.getElementById('sys-metric-cpu-gauge');
2114 |         const cpuPkgSlider = document.getElementById('sys-metric-cpu-package-slider');
2115 |         const cpuPkgTemp = document.getElementById('sys-metric-cpu-package-temp');
2116 |
2117 |         const pct = Number(data.total_percent || 0);
```

```
2118     const cores = Array.isArray(data.cores) ? data.cores : [
2119     ];
2119     const threads = Array.isArray(data.threads) ? data.threads : [];
2120     const sensors = Array.isArray(data.sensors) ? data.sensors : [];
2121     const history = Array.isArray(data.history) ? data.history : [];
2122
2123     if (cpuVal) {
2124         cpuVal.innerText = `${pct.toFixed(1)}%`;
2125         // Динамический оттенок значения загрузки
2126         if (pct >= 85) {
2127             cpuVal.style.color = '#ef4444'; // Красный (критическая нагрузка)
2128             cpuVal.style.textShadow = '0 0 14px rgba(239, 68, 68, 0.6)';
2129         } else if (pct >= 70) {
2130             cpuVal.style.color = '#f97316'; // Оранжевый (высокая нагрузка)
2131             cpuVal.style.textShadow = '0 0 12px rgba(249, 115, 22, 0.45)';
2132         } else if (pct >= 50) {
2133             cpuVal.style.color = '#eab308'; // Желтый / янтарный (умеренная нагрузка)
2134             cpuVal.style.textShadow = '0 0 10px rgba(234, 179, 8, 0.4)';
2135         } else {
2136             cpuVal.style.color = '#0dcaf0'; // Голубой (штатная нагрузка)
2137             cpuVal.style.textShadow = '0 0 10px rgba(13, 202, 240, 0.35)';
2138         }
2139     }
2140     if (cpuFill) cpuFill.style.width = `${Math.min(100, pct)}%`;
2141
2142     if (cpuModel && data.name) {
```

```

2143         cpuModel.innerText = data.name;
2144         cpuModel.title = data.name;
2145     }
2146
2147     const pkgTemp = data.package_temperature_c != null ? Num
ber(data.package_temperature_c) : null;
2148     const pkgPower = data.package_power_w != null ? Number(d
ata.package_power_w) : null;
2149     if (cpuSub) {
2150         const t = pkgTemp == null ? '' : ` · ${pkgTemp.toFixed
(0)} °C`;
2151         const p = pkgPower == null ? '' : ` · ${pkgPower.toFix
ed(0)} W`;
2152         const cCount = data.cores_count || cores.length;
2153         const thCount = data.threads_count || threads.length;
2154         cpuSub.innerText = cCount ? `${cCount} ядер • ${thCoun
t} потоков${t}${p}` : '-- Ядер';
2155     }
2156     if (cpuPkgTemp) {
2157         const pStr = pkgPower != null ? ` · ${pkgPower.toFixed
(0)} W` : '';
2158         cpuPkgTemp.innerText = pkgTemp != null ? `Пакет: ${pkg
Temp.toFixed(0)} °C${pStr}` : 'Пакет: -- °C';
2159         if (pkgTemp != null) {
2160             if (pkgTemp >= 80) {
2161                 cpuPkgTemp.className = 'text-danger fw-bold font-m
onospace';
2162             } else if (pkgTemp >= 65) {
2163                 cpuPkgTemp.className = 'text-warning fw-bold font-
monospace';
2164             } else {
2165                 cpuPkgTemp.className = 'text-info font-monospace';
2166             }
2167         }
2168     }
2169
2170     // 1. Заполнение сводной таблицы характеристик CPU (Spec
s)

```

```
2171     const specs = data.specs || {};
2172     const setSpec = (id, val) => { const el = document.getEl
ementById(id); if (el) el.innerText = val; };
2173     if (specs.socket || specs.physical_cores) {
2174         setSpec('sys-cpu-spec-arch', specs.architecture || 'x8
6_64');
2175         setSpec('sys-cpu-spec-socket', specs.socket || 'LGA120
0');
2176         setSpec('sys-cpu-spec-cores', `${specs.physical_cores
|| cores.length}C / ${specs.logical_cores || threads.length}T`
);
2177         setSpec('sys-cpu-spec-base-freq', specs.base_frequency
_str || '--');
2178         setSpec('sys-cpu-spec-max-freq', specs.max_frequency_s
tr || '--');
2179         setSpec('sys-cpu-spec-cache', specs.cache_combined_str
|| '--');
2180     }
2181
2182     // Экспресс-метрики KPI в центре баннера CPU
2183     const kpiPower = document.getElementById('sys-cpu-kpi-po
wer');
2184     const kpiTemp = document.getElementById('sys-cpu-kpi-tem
p');
2185     const kpiCores = document.getElementById('sys-cpu-kpi-co
res');
2186     const kpiFreq = document.getElementById('sys-cpu-kpi-fre
q');
2187     const cCount = data.cores_count || (specs.physical_cores
|| cores.length);
2188     const thCount = data.threads_count || (specs.logical_cor
es || threads.length);
2189
2190     if (kpiPower) kpiPower.textContent = pkgPower != null ?
`${pkgPower.toFixed(0)} W` : (data.package_power_w != null ? `
${Number(data.package_power_w).toFixed(0)} W` : '-- W');
2191     if (kpiTemp) {
2192         kpiTemp.textContent = pkgTemp != null ? `${pkgTemp.toF
```



```

    ixed(0)} °C` : '-- °C';
2193     if (pkgTemp !== null) {
2194         kpiTemp.className = pkgTemp >= 80 ? 'sys-kpi-value t
ext-danger' : pkgTemp >= 65 ? 'sys-kpi-value text-warning' : '
sys-kpi-value text-info';
2195     }
2196 }
2197     if (kpiCores) kpiCores.textContent = cCount ? `${cCount}
C / ${thCount}T` : '-- / --';
2198     if (kpiFreq) {
2199         const fMax = specs.max_frequency_str || (specs.max_fre
quency_ghz ? `${specs.max_frequency_ghz} GHz` : null);
2200         const fBase = specs.base_frequency_str || (specs.base_
frequency_ghz ? `${specs.base_frequency_ghz} GHz` : null);
2201         kpiFreq.textContent = fMax || fBase || '-- GHz';
2202     }
2203
2204     const vendorBadge = document.getElementById('sys-metric-
cpu-vendor-badge');
2205     if (vendorBadge) {
2206         const cpus = Array.isArray(data.cpus) && data.cpus.len
gth > 0 ? data.cpus : [];
2207         if (cpus.length > 1) {
2208             vendorBadge.innerHTML = cpus.map((c, i) => `<div cla
ss="text-truncate font-monospace" style="font-size: 0.74rem;">
CPU #${c.id ?? i}: ${escapeHtml(c.description || `${c.raw_desc
riptor} || c.name} (${c.logical_cores || 12} logical cores)`)<
/div>`).join('');
2209         } else if (data.description || specs.description) {
2210             vendorBadge.innerText = data.description || specs.de
scription;
2211         } else {
2212             const rawDesc = specs.raw_descriptor || 'Intel64 Fam
ily 6 Model 165 Stepping 3, GenuineIntel';
2213             const logCores = data.threads_count || specs.logical
_cores || (threads ? threads.length : 12) || 12;
2214             vendorBadge.innerText = `${rawDesc} (${logCores} log
ical cores)`;

```

```
2215 |     }
2216 | }
2217 |
2218 | // Gauge полукруг со стрелкой для общего CPU
2219 | if (cpuGauge) {
2220 |     cpuGauge.innerHTML = createGaugeSvg(pct, 120, 68);
2221 | }
2222 |
2223 | // Слайдер температуры Package (расположен прямо под спи
дометром)
2224 |     if (cpuPkgSlider) {
2225 |         cpuPkgSlider.innerHTML = createTempSliderHtml(pkgTemp,
30, 95, false);
2226 |     }
2227 |
2228 | // 2. Физические ядра со своими потоками
2229 | renderCpuCores(cores, data.threads_count || threads.leng
th);
2230 |
2231 | // 3. Общие показатели CPU
2232 | if (sensors && sensors.length > 0) {
2233 |     lastCpuSensorsData = sensors;
2234 |     renderComponentSensors('sys-metric-cpu-sensors', 'sys-
cpu-sensors-count-badge', sensors);
2235 | }
2236 |
2237 | // 4. График истории CPU (Загрузка %, Температура °C, Мо
щность W)
2238 | if (history && history.length > 0) {
2239 |     renderCpuSpark(history);
2240 | } else {
2241 |     _cpuSparkHistory.push({ load: pct, temp: pkgTemp, powe
r: pkgPower, time: new Date() });
2242 |     if (_cpuSparkHistory.length > 60) _cpuSparkHistory.shi
ft();
2243 |     renderCpuSpark(_cpuSparkHistory);
2244 | }
2245 | } catch (e) {
```

```
2246     console.warn('[SystemInspectorTab] Ошибка получения загр
узки CPU из API:', e);
2247   }
2248 }
2249
2250 /**
2251  * Запуск экспертного AI-аудита и анализа характеристик CPU
через Universal AITableModal.
2252  */
2253 function inspectCpuSpecsWithAi() {
2254   if (!lastCpuData) {
2255     console.warn('[SystemInspectorTab] Нет загруженных данны
х CPU для AI-инспекции');
2256     return;
2257   }
2258   const specs = lastCpuData.specs || {};
2259   const title = lastCpuData.name || 'Центральный процессор (
CPU)';
2260   const subtitle = `${specs.vendor || 'Intel/AMD'} · ${specs
.socket || 'Socket'} · ${specs.architecture || 'x86_64'}`;
2261   const metadata = {
2262     'Сокет': specs.socket || 'N/A',
2263     'Ядра / Поток': `${specs.physical_cores || lastCpuData.
cores_count || '--'}C / ${specs.logical_cores || lastCpuData.t
hread_count || '--'}T`,
2264     'Базовая частота': specs.base_frequency_str || '--',
2265     'Макс. частота': specs.max_frequency_str || '--',
2266     'Кэш L2 / L3': specs.cache_combined_str || '--',
2267     'Архитектура': specs.architecture || 'x86_64',
2268     'Текущая загрузка': `${(lastCpuData.total_percent || 0).
toFixed(1)}%`,
2269     'Температура пакета': lastCpuData.package_temperature_c
!= null ? `${lastCpuData.package_temperature_c.toFixed(0)} °C`
: 'N/A',
2270     'Энергопотребление': lastCpuData.package_power_w != null
? `${lastCpuData.package_power_w.toFixed(1)} W` : 'N/A'
2271   };
2272
```

```
2273     if (window.AITableModal && typeof window.AITableModal.show
    === 'function') {
2274         window.AITableModal.show({
2275             icon: '□ ',
2276             title: title,
2277             subtitle: subtitle,
2278             badges: [
2279                 { text: specs.architecture || 'x86_64', class: 'bg-p
rimary-subtle text-primary border border-primary' },
2280                 { text: specs.socket || 'Socket', class: 'bg-info-su
btle text-info border border-info' },
2281                 { text: `${specs.physical_cores || lastCpuData.cores
_count || '--'} Cores`, class: 'bg-success-subtle text-success
border border-success' }
2282             ],
2283             metadata: metadata,
2284             rawTitle: 'CPU Inventory Telemetry & Specs',
2285             rawContent: JSON.stringify({
2286                 name: lastCpuData.name,
2287                 specs: specs,
2288                 metrics: {
2289                     total_percent: lastCpuData.total_percent,
2290                     package_temperature_c: lastCpuData.package_tempera
ture_c,
2291                     package_power_w: lastCpuData.package_power_w
2292                 },
2293                 cores_count: (lastCpuData.cores || []).length,
2294                 threads_count: (lastCpuData.threads || []).length
2295             }, null, 2),
2296             tableType: 'cpu',
2297             actions: []
2298         });
2299     } else {
2300         console.warn('[SystemInspectorTab] Universal AITableModa
l недоступен в DOM');
2301     }
2302 }
2303 window.inspectCpuSpecsWithAi = inspectCpuSpecsWithAi;
```

```
2304
2305  /**
2306   * Запуск экспертного AI-аудита и анализа характеристик GPU
    через Universal AITableModal.
2307   */
2308   function inspectGpuSpecsWithAi(idx = 0) {
2309     const gpu = (lastGpuDevices && lastGpuDevices[idx]) || las
    tGpuData;
2310     if (!gpu) {
2311       console.warn('[SystemInspectorTab] Нет загруженных данны
    х GPU для AI-инспекции');
2312       return;
2313     }
2314     const specs = gpu.specs || {};
2315     const title = gpu.name || `Графический процессор (GPU #{i
    dx})`;
2316     const subtitle = `${specs.vendor || gpu.vendor || 'GPU'} ·
    ${specs.pci_bus || 'PCIe'} · ${specs.directx || 'DirectX 12'}
    `;
2317     const metadata = {
2318       'Производитель': specs.vendor || gpu.vendor || 'NVIDIA/I
    ntel/AMD',
2319       'VRAM Объем': specs.vram_str || (gpu.memory && gpu.memor
    y.total_mb ? `${(gpu.memory.total_mb/1024).toFixed(1)} GB` : '
    --'),
2320       'Использование VRAM': specs.vram_used_str || (gpu.memory
    && gpu.memory.used_percent != null ? `${gpu.memory.used_perce
    nt.toFixed(0)}%` : '--'),
2321       'Частота ядра': specs.core_clock_str || (gpu.clocks && g
    pu.clocks.core_mhz ? `${Math.round(gpu.clocks.core_mhz)} MHz`
    : '--'),
2322       'Частота памяти': specs.memory_clock_str || (gpu.clocks
    && gpu.clocks.memory_mhz ? `${Math.round(gpu.clocks.memory_mhz
    )} MHz` : '--'),
2323       'Версия драйвера': specs.driver_version || '--',
2324       'Шина': specs.pci_bus || 'PCIe',
2325       'DirectX API': specs.directx || 'DirectX 12',
2326       'Текущая загрузка': `${(gpu.core_load_percent || 0).toFi
```

```

    xed(1)}%`,
2327     'Температура GPU': gpu.core_temperature_c != null ? `${g
    pu.core_temperature_c.toFixed(0)} °C` : 'N/A',
2328     'Энергопотребление': gpu.power_w != null ? `${gpu.power_
    w.toFixed(1)} W` : 'N/A'
2329     };
2330
2331     if (window.AITableModal && typeof window.AITableModal.show
    === 'function') {
2332         window.AITableModal.show({
2333             icon: '⚠',
2334             title: title,
2335             subtitle: subtitle,
2336             badges: [
2337                 { text: specs.vendor || gpu.vendor || 'GPU', class:
    'bg-warning-subtle text-warning border border-warning' },
2338                 { text: specs.vram_str || (gpu.memory && gpu.memory.
    total_mb ? `${(gpu.memory.total_mb/1024).toFixed(1)} GB` : 'VR
    AM'), class: 'bg-info-subtle text-info border border-info' },
2339                 { text: specs.directx || 'DirectX 12', class: 'bg-su
    ccess-subtle text-success border border-success' }
2340             ],
2341             metadata: metadata,
2342             rawTitle: `GPU #${gpu.index != null ? gpu.index : idx}
    Inventory Telemetry & Specs`,
2343             rawContent: JSON.stringify({
2344                 index: gpu.index != null ? gpu.index : idx,
2345                 name: gpu.name,
2346                 vendor: gpu.vendor,
2347                 specs: specs,
2348                 metrics: {
2349                     core_load_percent: gpu.core_load_percent,
2350                     core_temperature_c: gpu.core_temperature_c,
2351                     hotspot_temperature_c: gpu.hotspot_temperature_c,
2352                     power_w: gpu.power_w,
2353                     memory: gpu.memory,
2354                     clocks: gpu.clocks
2355                 },

```

```
2356 |         engines: gpu.engines || [],
2357 |         sensors_count: (gpu.sensors || []).length
2358 |     }, null, 2),
2359 |     tableType: 'gpu',
2360 |     actions: []
2361 | });
2362 | } else {
2363 |     console.warn('[SystemInspectorTab] Universal AITableModa
2364 | l недоступен в DOM');
2365 | }
2366 | window.inspectGpuSpecsWithAi = inspectGpuSpecsWithAi;
2367 |
2368 | function renderGpuEngines(engines, idx = 0) {
2369 |     const box = document.getElementById(`sys-metric-gpu-engine
2370 | s-${idx}`) || document.getElementById('sys-metric-gpu-engines'
2371 | );
2372 |     const badgeCount = document.getElementById(`sys-gpu-engine
2373 | s-count-badge-${idx}`) || document.getElementById('sys-gpu-eng
2374 | ines-count-badge');
2375 |     if (!box) return;
2376 |
2377 |     if (badgeCount) {
2378 |         badgeCount.textContent = engines && engines.length ? `${
2379 | engines.length} блоков` : '0 блоков';
2380 |     }
2381 |
2382 |     if (!Array.isArray(engines) || engines.length === 0) {
2383 |         box.innerHTML = `
```

```

2385     const loadStr = load == null ? '--' : `${load.toFixed(0)}%`;
2386     const gaugeSvg = createGaugeSvg(load, 110, 62);
2387     const tempHtml = createTempSliderHtml(temp, 30, 95, false);
2388
2389     return `
2390         <div class="sys-core-card p-2.5" style="min-height: 156px; display: flex; flex-direction: column; justify-content: space-between;" title="${escapeHtml(e.name)}: Нагрузка ${loadStr}${temp != null ? ', Температура ' + temp.toFixed(0) + '°C' : ''}">
2391             <div class="d-flex justify-content-between align-items-start gap-1.5 mb-1" style="min-height: 32px;">
2392                 <span class="sys-core-title" style="font-size: 0.84rem; font-weight: 800; color: var(--text-color); line-height: 1.25; word-break: break-word;" title="${escapeHtml(e.name)}">${escapeHtml(e.name)}</span>
2393                 <span class="sys-core-load-badge font-monospace flex-shrink-0" style="font-size: 0.76rem; padding: 2px 6px;">${loadStr}</span>
2394             </div>
2395             <div class="py-1 d-flex justify-content-center">
2396                 ${gaugeSvg}
2397             </div>
2398             <div class="mt-auto">
2399                 ${tempHtml}
2400             </div>
2401         </div>
2402     `;
2403     }).join('');
2404 }
2405
2406 function renderGpuCardHtml(gpu, idx, totalDevices) {
2407     const gpuName = escapeHtml(gpu.name || `GPU #${idx}`);
2408     const vendor = escapeHtml(gpu.vendor || 'GPU');
2409     const headerTitle = totalDevices > 1 ? `Загрузка GPU #${gpu.index != null ? gpu.index : idx}: ${gpuName}` : `Загрузка GP

```



```

    U: ${gpuName}`;
2410
2411     return `
2412         <div class="sys-card sys-card-gpu h-100 p-3" id="sys-car
d-gpu-${idx}">
2413             <div class="sys-card-header-bar">
2414                 <div class="d-flex align-items-center gap-2">
2415                     <span class="sys-label text-warning" style="font-s
ize: 1.05rem; font-weight: 800;">
2416                         <i class="bi bi-gpu-card text-warning me-1.5"></
i>${headerTitle}
2417                     </span>
2418                     <span class="badge bg-dark border border-secondary
text-info font-monospace ms-1" style="font-size: 0.74rem;">
2419                         ${vendor}
2420                     </span>
2421                 </div>
2422                 <span class="small text-truncate font-monospace badg
e bg-dark border border-secondary text-warning" style="font-si
ze: 0.84rem; max-width: 320px;" id="sys-metric-gpu-sub-${idx}"
>--</span>
2423             </div>
2424
2425             <!-- Верхний баннер: Название слева, Спидометр и датчи
к температуры по центру, Сводная спецификация GPU справа -->
2426             <div class="sys-card-top-banner p-3 rounded-3 mb-3">
2427                 <!-- Левая колонка: Модель графического процессора и
экспресс-метрики KPI -->
2428                 <div class="sys-banner-col-info d-flex flex-column a
lign-items-start justify-content-center">
2429                     <div class="text-start mb-1.5 d-flex align-items-c
enter">
2430                         <button type="button" class="sys-drill-btn" id="
sys-drill-btn-gpu-${idx}" onclick="window.toggleSysCardDrill &
& window.toggleSysCardDrill('sys-card-gpu-${idx}', ${idx})" ti
tle="Drill-down: Развернуть/скрыть все скрытые панели и датчик
и GPU">
2431                         
2432         </button>
2433         <div>
2434             <span class="fw-bold text-warning" style="font
-size: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-
shadow: 0 0 14px rgba(245, 158, 11, 0.4);" id="sys-metric-gpu-
model-{$idx}">{$gpuName}</span>
2435             <div class="d-flex align-items-center gap-1.5
mt-0.5">
2436                 <span class="badge bg-dark border border-sec
ondary text-light font-monospace" id="sys-metric-gpu-vendor-ba
dge-{$idx}" style="font-size: 0.70rem;">{$vendor} · PCIe</span
>
2437                 <span class="badge bg-warning-subtle text-wa
rning border border-warning font-monospace" id="sys-gpu-status
-badge-{$idx}" style="font-size: 0.70rem;">● Ready</span>
2438             </div>
2439         </div>
2440     </div>
2441     <div class="sys-banner-kpi-grid">
2442         <div class="sys-kpi-pill">
2443             <span class="sys-kpi-label"><i class="bi bi-me
mory text-info me-1"></i>VRAM Память</span>
2444             <span class="sys-kpi-value text-info" id="sys-
gpu-kpi-vram-{$idx}">-- / -- GB</span>
2445         </div>
2446         <div class="sys-kpi-pill">
2447             <span class="sys-kpi-label"><i class="bi bi-li
ghtning-charge text-warning me-1"></i>Мощность</span>
2448             <span class="sys-kpi-value text-warning" id="s
ys-gpu-kpi-power-{$idx}">-- W</span>
2449         </div>
2450         <div class="sys-kpi-pill">
2451             <span class="sys-kpi-label"><i class="bi bi-sp
eedometer2 text-primary me-1"></i>Частота Core</span>
2452             <span class="sys-kpi-value text-primary" id="s
ys-gpu-kpi-clock-{$idx}">-- MHz</span>
2453         </div>

```

```

2454 |         <div class="sys-kpi-pill">
2455 |             <span class="sys-kpi-label"><i class="bi bi-thermometer-half text-danger me-1"></i>GPU t°</span>
2456 |             <span class="sys-kpi-value text-warning" id="sys-gpu-kpi-temp-${idx}">-- °C</span>
2457 |         </div>
2458 |     </div>
2459 | </div>
2460 |
2461 |     <!-- Центральная колонка: Спидометр загрузки, справа
2462 | от него значение с динамическим оттенком, а СНИЗУ – датчик температуры -->
2463 |     <div class="sys-banner-col-gauge d-flex flex-column justify-content-between gap-1.5">
2464 |         <div class="d-flex align-items-center gap-2.5">
2465 |             <div id="sys-metric-gpu-gauge-${idx}" style="width: 120px; height: 68px; flex-shrink: 0;"></div>
2466 |             <div>
2467 |                 <div class="sys-value text-warning" id="sys-metric-gpu-val-${idx}" style="font-size: 2.1rem; line-height: 1; font-weight: 800; transition: color 0.3s ease, text-shadow 0.3s ease;">0.0%</div>
2468 |                 <div class="text-light mt-1 font-monospace" style="font-size: 0.82rem; font-weight: 700;" id="sys-metric-gpu-core-temp-${idx}">GPU: -- °C</div>
2469 |             </div>
2470 |         </div>
2471 |         <!-- Датчик температуры GPU Core расположен ПОД спидометром загрузки -->
2472 |         <div id="sys-metric-gpu-temp-slider-${idx}" style="width: 100%;"></div>
2473 |     </div>
2474 |
2475 |     <!-- Правая колонка: Сводная таблица о GPU (кликабельна для запуска AI-аудита) -->
2476 |     <div class="sys-banner-col-specs">
2477 |         <div class="sys-gpu-specs-card p-2 rounded-2 d-flex flex-column justify-content-between h-100">

```

```
2477         id="sys-gpu-specs-card-{$idx}"
2478         onclick="window.inspectGpuSpecsWithAi && wind
2479         ow.inspectGpuSpecsWithAi({$idx})"
2480         title="Нажмите для вызова AI-аудита и эксперт
2481         ного анализа графического процессора">
2482         <div class="d-flex justify-content-between align
2483         -items-center mb-1 pb-1 border-bottom" style="border-color: va
2484         r(--border-subtle) !important;">
2485         <span class="text-light fw-bold d-flex align-i
2486         tems-center gap-1" style="font-size: 0.78rem;">
2487         <i class="bi bi-info-circle text-warning"></
2488         i>Характеристики GPU
2489         <span class="badge bg-primary-subtle text-pr
2490         imary border border-primary-subtle font-monospace ms-1" style=
2491         "font-size: 0.62rem;"><i class="bi bi-stars me-0.5"></i>AI</sp
2492         an>
2493         </span>
2494         <span class="badge bg-dark border border-secon
2495         dary text-warning font-monospace" style="font-size: 0.68rem;"
2496         id="sys-gpu-spec-directx-{$idx}">DirectX 12</span>
2497         </div>
2498         <div class="sys-gpu-specs-grid" style="display:
2499         grid; grid-template-columns: 1fr 1fr; gap: 4px 10px; font-size
2500         : 0.74rem;">
2501         <div class="d-flex justify-content-between"><s
2502         pan class="text-muted">VRAM Объем:</span><span class="text-lig
2503         ht font-monospace fw-semibold" id="sys-gpu-spec-vram-{$idx}">-
2504         -</span></div>
2505         <div class="d-flex justify-content-between"><s
2506         pan class="text-muted">Исп. VRAM:</span><span class="text-warn
2507         ing font-monospace fw-semibold" id="sys-gpu-spec-vram-used-{$i
2508         dx}">--</span></div>
2509         <div class="d-flex justify-content-between"><s
2510         pan class="text-muted">Частота Core:</span><span class="text-w
2511         arning font-monospace fw-semibold" id="sys-gpu-spec-core-clock
2512         -{$idx}">--</span></div>
2513         <div class="d-flex justify-content-between"><s
2514         pan class="text-muted">Частота Mem:</span><span class="text-wa
```

```

ning font-monospace fw-semibold" id="sys-gpu-spec-mem-clock-  

{idx}">--</span></div>
2492         <div class="d-flex justify-content-between" st  

yle="grid-column: span 2;"><span class="text-muted">Драйвер /  

API:</span><span class="text-info font-monospace fw-semibold"  

id="sys-gpu-spec-driver-{$idx}">--</span></div>
2493         </div>
2494     </div>
2495 </div>
2496 </div>
2497
2498     <!-- График истории GPU (Загрузка %, Температура °C, М  

ощность / Память %) с логарифмической/линейной шкалой -->
2499     <div class="sys-full-only sys-history-panel p-2.5 mb-2  

.5 rounded-2">
2500         <div class="d-flex justify-content-between align-ite  

ms-center mb-1.5 flex-wrap gap-2">
2501             <div class="d-flex align-items-center gap-2">
2502                 <span class="small text-light" style="font-size:  

0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-war  

ning me-1"></i>История GPU</span>
2503                 <div class="btn-group btn-group-sm" role="group"  

style="height: 22px;">
2504                     <button type="button" class="btn btn-xs btn-ou  

tline-warning active py-0 px-2 fw-bold" id="btn-gpu-spark-log-  

{$idx}" style="font-size: 0.68rem;" onclick="window.setGpuSpar  

kScale && window.setGpuSparkScale('log', {$idx})">Лог. шкала (  

Log)</button>
2505                     <button type="button" class="btn btn-xs btn-ou  

tline-secondary py-0 px-2" id="btn-gpu-spark-lin-{$idx}" style  

="font-size: 0.68rem;" onclick="window.setGpuSparkScale && win  

dow.setGpuSparkScale('linear', {$idx})">Линейная (Lin)</button  

>
2506                 </div>
2507             </div>
2508             <div class="d-flex align-items-center gap-2.5" sty  

le="font-size: 0.74rem; font-weight: 600;">
2509                 <span class="text-warning" style="text-shadow: 0

```

```

    0 8px rgba(245,158,11,0.5);" title="Основная шкала: Загрузка
GPU Core (0-100%)">● Нагрузка Core %</span>
2510      <span class="text-danger" style="text-shadow: 0
0 8px rgba(239,68,68,0.5);" title="Компактная шкала: Температу
ра GPU Core (°C)">● Температура °C</span>
2511      <span class="text-info" style="text-shadow: 0 0
8px rgba(13,202,240,0.5);" title="Компактная шкала: Мощность W
/ VRAM %">● Мощность / Память %</span>
2512      <span class="text-muted font-monospace ms-2" id=
"sys-gpu-spark-live-cursor-{$idx}" style="font-size: 0.70rem;"
>Live</span>
2513    </div>
2514  </div>
2515  <!-- Контейнер графика и интерактивного тултипа -->
2516  <div style="position: relative; width: 100%; height:
90px;" id="sys-gpu-spark-container-{$idx}">
2517    <div id="sys-gpu-spark-{$idx}" style="width: 100%;
height: 100%;"></div>
2518    <div id="sys-gpu-spark-tooltip-{$idx}" class="sys-
spark-tooltip" style="display: none;"></div>
2519  </div>
2520 </div>
2521
2522  <!-- Сворачиваемый блок детализации GPU: Блоки, подсисте
мы и сенсоры питания/напряжения/кулеров -->
2523  <div class="sys-full-only sys-detailed-sensors mt-2">
2524    <div class="sys-gpu-details-accordion-btn d-flex jus
tify-content-between align-items-center flex-wrap gap-2"
2525      id="sys-gpu-details-toggle-btn-{$idx}"
2526      title="Нажмите, чтобы развернуть или скрыть дет
ализацию по подсистемам и сенсорам GPU"
2527      onclick="window.toggleSysGpuDetails && window.t
oggleSysGpuDetails({$idx})">
2528      <div class="d-flex align-items-center gap-2 flex-w
rap">
2529        <button type="button" class="panel-toggle-drill
collapsed me-1" id="sys-gpu-details-drill-btn-{$idx}" tabindex
="-1" style="pointer-events: none;">

```

```
2530 |         
2531 |     </button>
2532 |     <span class="fw-bold" style="font-size: 0.88rem; color: var(--text-color);">Детализация: Блоки, подсистемы и сенсоры GPU</span>
2533 |     <span class="badge bg-dark border border-secondary text-warning font-monospace ms-1" style="font-size: 0.72rem;" id="sys-gpu-engines-count-badge-${idx}">0 блоков</span>
2534 |     <span class="badge bg-dark border border-secondary text-danger font-monospace" style="font-size: 0.72rem;" id="sys-gpu-sensors-count-badge-${idx}">0 параметров</span>
2535 | </div>
2536 | </div>
2537 |
2538 | <!-- Скрытый по умолчанию общий контейнер деталей GPU -->
2539 | <div id="sys-gpu-details-collapse-${idx}" style="display: none; flex-direction: column; gap: 14px; margin-top: 10px;">
2540 |     <!-- 1. Блоки и подсистемы GPU -->
2541 |     <div>
2542 |         <div class="sys-core-section-title d-flex justify-content-between align-items-center mb-2" style="font-size: 0.82rem;">
2543 |             <span><i class="bi bi-grid-3x3-gap-fill text-warning me-1"></i>Блоки и подсистемы GPU</span>
2544 |         </div>
2545 |         <div id="sys-metric-gpu-engines-${idx}" style="display: grid; grid-template-columns: repeat(auto-fill, minmax(175px, 1fr)); gap: 10px; font-size: 0.78rem;"></div>
2546 |     </div>
2547 |
2548 |     <!-- 2. Энергопотребление, напряжение, частоты и сенсоры GPU -->
2549 |     <div>
2550 |         <div class="sys-core-section-title d-flex justify-content-between align-items-center mb-2" style="font-size: 0
```

```

.82rem;">
2551         <span><i class="bi bi-lightning-charge text-wa
rning me-1"></i>Энергопотребление, напряжение, вентиляторы и ч
астоты GPU</span>
2552         </div>
2553         <div id="sys-metric-gpu-sensors-`${idx}" style="d
isplay: grid; grid-template-columns: repeat(auto-fill, minmax(
180px, 1fr)); gap: 8px; font-size: 0.78rem;">
2554             <div class="text-muted small text-center py-2
col-12">Опрос сенсоров GPU...</div>
2555             </div>
2556         </div>
2557     </div>
2558 </div>
2559 </div>
2560 `;
2561 }
2562
2563 function updateSingleGpuCard(gpu, idx, totalDevices) {
2564     const pct = Number(gpu.core_load_percent || 0);
2565     const coreTemp = gpu.core_temperature_c !== null ? Number(g
pu.core_temperature_c) : null;
2566     const hotspotTemp = gpu.hotspot_temperature_c !== null ? Nu
mber(gpu.hotspot_temperature_c) : null;
2567     const powerW = gpu.power_w !== null ? Number(gpu.power_w) :
null;
2568     const clocks = gpu.clocks || {};
2569     const memory = gpu.memory || {};
2570     const specs = gpu.specs || {};
2571     const engines = Array.isArray(gpu.engines) ? gpu.engines :
[];
2572     const sensorsList = Array.isArray(gpu.sensors) ? gpu.senso
rs : [];
2573
2574     const gpuVal = document.getElementById(`sys-metric-gpu-val
-${idx}`);
2575     const gpuModel = document.getElementById(`sys-metric-gpu-m
odel-${idx}`);

```



```
2576     const gpuVendorBadge = document.getElementById(`sys-metric
-gpu-vendor-badge-${idx}`);
2577     const gpuSub = document.getElementById(`sys-metric-gpu-sub
-${idx}`);
2578     const gpuGauge = document.getElementById(`sys-metric-gpu-g
auge-${idx}`);
2579     const gpuTempSlider = document.getElementById(`sys-metric-
gpu-temp-slider-${idx}`);
2580     const gpuCoreTemp = document.getElementById(`sys-metric-gp
u-core-temp-${idx}`);
2581
2582     if (gpuVal) {
2583         gpuVal.innerText = `${pct.toFixed(1)}%`;
2584         if (pct >= 85) {
2585             gpuVal.style.color = '#ef4444';
2586             gpuVal.style.textShadow = '0 0 16px rgba(239, 68, 68,
0.7)';
2587         } else if (pct >= 60) {
2588             gpuVal.style.color = '#f59e0b';
2589             gpuVal.style.textShadow = '0 0 14px rgba(245, 158, 11,
0.6)';
2590         } else if (pct >= 25) {
2591             gpuVal.style.color = '#eab308';
2592             gpuVal.style.textShadow = '0 0 12px rgba(234, 179, 8,
0.4)';
2593         } else {
2594             gpuVal.style.color = '#10b981';
2595             gpuVal.style.textShadow = '0 0 10px rgba(16, 185, 129,
0.3)';
2596         }
2597     }
2598
2599     if (gpuModel) {
2600         gpuModel.innerText = gpu.name || 'GPU';
2601         gpuModel.title = gpu.name || 'GPU';
2602     }
2603
2604     if (gpuVendorBadge) {
```

```

2605     const v = specs.vendor || gpu.vendor || 'GPU';
2606     const bus = specs.pci_bus || 'PCIe x16';
2607     gpuVendorBadge.textContent = `${v} Corporation · ${bus}`
2608     ;
2609
2610     if (gpuCoreTemp) {
2611         let tempText = coreTemp != null ? `GPU: ${coreTemp.toFixed(0)} °C` : 'GPU: -- °C';
2612         if (hotspotTemp != null) {
2613             tempText += ` (HotSpot: ${hotspotTemp.toFixed(0)}°C)`;
2614         }
2615         gpuCoreTemp.innerText = tempText;
2616     }
2617
2618     if (gpuSub) {
2619         let subText = '';
2620         if (clocks && clocks.core_mhz) {
2621             subText += `${Math.round(clocks.core_mhz)} MHz`;
2622         }
2623         if (memory && memory.total_mb) {
2624             const totalGb = (memory.total_mb / 1024).toFixed(1);
2625             subText += (subText ? ' · ' : '') + `${totalGb} GB VRAM`;
2626         }
2627         gpuSub.innerText = subText || (gpu.name || 'GPU');
2628         gpuSub.title = subText || (gpu.name || 'GPU');
2629     }
2630
2631     // Спецификации
2632     const specVram = document.getElementById(`sys-gpu-spec-vram-${idx}`);
2633     const specVramUsed = document.getElementById(`sys-gpu-spec-vram-used-${idx}`);
2634     const specCoreClock = document.getElementById(`sys-gpu-spec-core-clock-${idx}`);
2635     const specMemClock = document.getElementById(`sys-gpu-spec-mem-clock-${idx}`);

```

```
2636     const specDriver = document.getElementById(`sys-gpu-spec-d
river-${idx}`);
2637     const specDirectx = document.getElementById(`sys-gpu-spec-
directx-${idx}`);
2638
2639     if (specVram) specVram.textContent = specs.vram_str || (me
memory.total_mb ? `${(memory.total_mb/1024).toFixed(1)} GB` : '-
-');
2640     if (specVramUsed) specVramUsed.textContent = specs.vram_us
ed_str || (memory.used_percent !== null ? `${memory.used_percen
t.toFixed(0)}%` : '--');
2641     if (specCoreClock) specCoreClock.textContent = specs.core_
clock_str || (clocks.core_mhz ? `${Math.round(clocks.core_mhz)
} MHz` : '--');
2642     if (specMemClock) specMemClock.textContent = specs.memory_
clock_str || (clocks.memory_mhz ? `${Math.round(clocks.memory_
mhz)} MHz` : '--');
2643     if (specDriver) specDriver.textContent = specs.driver_vers
ion ? `${specs.driver_version} (DirectX 12)` : 'WDDM 2.7 (Dire
ctX 12)';
2644     if (specDirectx) specDirectx.textContent = specs.directx |
| 'DirectX 12';
2645
2646     // Экспресс-метрики KPI в центре баннера GPU
2647     const kpiVram = document.getElementById(`sys-gpu-kpi-vram-
${idx}`);
2648     const kpiPower = document.getElementById(`sys-gpu-kpi-powe
r-${idx}`);
2649     const kpiClock = document.getElementById(`sys-gpu-kpi-cloc
k-${idx}`);
2650     const kpiTemp = document.getElementById(`sys-gpu-kpi-temp-
${idx}`);
2651     if (kpiVram) {
2652         if (memory && memory.total_mb) {
2653             const uGb = ((memory.used_mb || 0) / 1024).toFixed(1);
2654             const tGb = (memory.total_mb / 1024).toFixed(1);
2655             kpiVram.textContent = `${uGb} / ${tGb} GB`;
2656         } else {
```

```
2657         kpiVram.textContent = '-- / -- GB';
2658     }
2659 }
2660 if (kpiPower) kpiPower.textContent = powerW != null ? `${p
owerW.toFixed(0)} W` : (gpu.power_w ? `${gpu.power_w} W` : '--
W');
2661 if (kpiClock) kpiClock.textContent = clocks && clocks.core
_mhz ? `${Math.round(clocks.core_mhz)} MHz` : '-- MHz';
2662 if (kpiTemp) {
2663     kpiTemp.textContent = coreTemp != null ? `${coreTemp.toF
ixed(0)} °C` : '-- °C';
2664     if (coreTemp != null) {
2665         kpiTemp.className = coreTemp >= 80 ? 'sys-kpi-value te
xt-danger' : coreTemp >= 65 ? 'sys-kpi-value text-warning' : '
sys-kpi-value text-info';
2666     }
2667 }
2668
2669 // Gauge SVG
2670 if (gpuGauge) {
2671     gpuGauge.innerHTML = createGaugeSvg(pct, 120, 68);
2672 }
2673
2674 // Temp slider
2675 if (gpuTempSlider) {
2676     gpuTempSlider.innerHTML = createTempSliderHtml(coreTemp,
30, 95, true);
2677 }
2678
2679 // Engines
2680 renderGpuEngines(engines, idx);
2681
2682 // Sensors
2683 const gpuSensorsContainer = document.getElementById(`sys-m
etric-gpu-sensors-${idx}`);
2684 const gpuSensorsCountBadge = document.getElementById(`sys-
gpu-sensors-count-badge-${idx}`);
2685 if (gpuSensorsCountBadge) {
```

```

2686         gpuSensorsCountBadge.textContent = `${sensorsList.length
    } параметров`;
2687     }
2688     if (gpuSensorsContainer) {
2689         if (sensorsList.length === 0) {
2690             gpuSensorsContainer.innerHTML = `
```

```
2706 |         }).join('');
2707 |     }
2708 | }
2709 |
2710 | // Sparkline
2711 | if (Array.isArray(gpu.history) && gpu.history.length > 0)
2712 | {
2713 |     renderGpuSpark(gpu.history, idx);
2714 | } else {
2715 |     if (!_gpuSparkHistories[idx]) _gpuSparkHistories[idx] =
2716 |     [];
2717 |     _gpuSparkHistories[idx].push({ load: pct, temp: coreTemp
2718 | , power: powerW, time: new Date() });
2719 |     if (_gpuSparkHistories[idx].length > 60) _gpuSparkHistor
2720 | ies[idx].shift();
2721 |     renderGpuSpark(_gpuSparkHistories[idx], idx);
2722 | }
2723 | }
2724 |
2725 | async function fetchGpuLoadFromApi() {
2726 |     try {
2727 |         const res = await fetch('/api/v1/panel/gpu-load');
2728 |         if (!res.ok) return;
2729 |         const data = await res.json();
2730 |         lastGpuData = data;
2731 |
2732 |         const devices = Array.isArray(data.devices) && data.devi
2733 | ces.length > 0 ? data.devices : [data];
2734 |         lastGpuDevices = devices;
2735 |
2736 |         const container = document.getElementById('sys-gpu-panel
2737 | s-container');
2738 |         if (!container) return;
2739 |
2740 |         const currentCards = container.querySelectorAll('.sys-ca
2741 | rd-gpu');
2742 |         if (currentCards.length !== devices.length) {
2743 |             container.innerHTML = devices.map((gpu, idx) => render
```

```

    GpuCardHtml(gpu, idx, devices.length)).join('');
2737     }
2738
2739     devices.forEach((gpu, idx) => {
2740         updateSingleGpuCard(gpu, idx, devices.length);
2741     });
2742     } catch (e) {
2743         console.warn('[SystemInspectorTab] Ошибка получения загрузки GPU из API:', e);
2744     }
2745 }
2746
2747 function updateTelemetryDashboard(snap) {
2748     if (!snap) return;
2749
2750     // CPU
2751     if (snap.cpu) {
2752         const cpuVal = document.getElementById('sys-metric-cpu-val');
2753         const cpuModel = document.getElementById('sys-metric-cpu-model');
2754         const cpuFill = document.getElementById('sys-metric-cpu-fill');
2755         const cpuSub = document.getElementById('sys-metric-cpu-sub');
2756         const cpuGauge = document.getElementById('sys-metric-cpu-gauge');
2757
2758         const pct = Number(snap.cpu.total_percent || 0);
2759         if (cpuVal) cpuVal.innerText = `${pct.toFixed(1)}%`;
2760         if (cpuModel && snap.cpu.model) cpuModel.innerText = snap.cpu.model;
2761         if (cpuFill) cpuFill.style.width = `${Math.min(100, pct)}%`;
2762         if (cpuSub) cpuSub.innerText = `${snap.cpu.physical_cores || '--'} Физических / ${snap.cpu.logical_cores || '--'} Поток
ков`;
2763         if (cpuGauge) cpuGauge.innerHTML = createGaugeSvg(pct, 1

```

```

30, 74);
2764     }
2765
2766     // RAM
2767     if (snap.memory) {
2768         const ramVal = document.getElementById('sys-metric-ram-v
al');
2769         const ramFill = document.getElementById('sys-metric-ram-
fill');
2770         const ramSub = document.getElementById('sys-metric-ram-s
ub');
2771         const ramModel = document.getElementById('sys-metric-ram
-model');
2772
2773         const usedGb = Number(snap.memory.used_gb || 0);
2774         const totalGb = Number(snap.memory.total_gb || 0);
2775         const pct = Number(snap.memory.percent || 0);
2776         const swapPct = Number(snap.memory.swap_percent || 0);
2777
2778         if (ramModel && totalGb > 0) ramModel.innerText = `${Mat
h.round(totalGb)} GB RAM`;
2779         if (ramVal) ramVal.innerText = `${usedGb.toFixed(1)} / $
{totalGb.toFixed(1)} GB`;
2780         if (ramFill) ramFill.style.width = `${pct}%`;
2781         if (ramSub) ramSub.innerText = `${pct}% занято (${Number
(snap.memory.available_gb || 0).toFixed(1)} GB свободно)`;
2782
2783         const memioPct = document.getElementById('sys-memio-pct'
);
2784         if (memioPct) memioPct.innerText = `${pct.toFixed(1)}%`;
2785         const memioUsed = document.getElementById('sys-memio-use
d');
2786         if (memioUsed) memioUsed.innerText = `${usedGb.toFixed(1
)} / ${totalGb.toFixed(1)} GB (свободно ${Number(snap.memory.a
vailable_gb || (totalGb - usedGb)).toFixed(1)} GB)`;
2787         const memioSwap = document.getElementById('sys-memio-swa
p');
2788         if (memioSwap) memioSwap.innerText = `Файл подкачки: ${s

```



```

    wapPct.toFixed(1)}}%`;
2789     const memioTs = document.getElementById('sys-memio-ts');
2790     if (memioTs) memioTs.innerHTML = new Date().toLocaleTime
String();
2791
2792     const ramGauge = document.getElementById('sys-metric-ram
-gauge');
2793     if (ramGauge) ramGauge.innerHTML = createGaugeSvg(pct, 1
30, 74);
2794     const ramSlider = document.getElementById('sys-metric-ra
m-slider');
2795     if (ramSlider) ramSlider.innerHTML = createPercentSlider
Html(pct, true, `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)}
GB`);
2796     const ramSubGauge = document.getElementById('sys-ram-sub
-gauge');
2797     if (ramSubGauge) ramSubGauge.innerHTML = createGaugeSvg(
pct, 100, 56);
2798     const ramSubSlider = document.getElementById('sys-ram-su
b-slider');
2799     if (ramSubSlider) ramSubSlider.innerHTML = createPercent
SliderHtml(pct, false, `${usedGb.toFixed(1)} / ${totalGb.toFix
ed(1)} GB`);
2800     const ramSubBadge = document.getElementById('sys-ram-sub
-badge');
2801     if (ramSubBadge) ramSubBadge.textContent = `${pct.toFixe
d(0)}}%`;
2802
2803     // Экспресс-метрики KPI в центре баннера RAM
2804     const ramKpiUsed = document.getElementById('sys-ram-kpi-
used');
2805     const ramKpiFree = document.getElementById('sys-ram-kpi-
free');
2806     const ramKpiSwap = document.getElementById('sys-ram-kpi-
swap');
2807     if (ramKpiUsed) ramKpiUsed.textContent = `${usedGb.toFix
ed(1)} GB`;
2808     if (ramKpiFree) ramKpiFree.textContent = `${Number(snap.

```

```

memory.available_gb || (totalGb - usedGb)).toFixed(1)} GB`;
2809     if (ramKpiSwap) ramKpiSwap.textContent = `${swapPct.toFixed(1)}%`;
2810
2811     // Swap
2812     const swapSubBadge = document.getElementById('sys-swap-sub-badge');
2813     if (swapSubBadge) swapSubBadge.textContent = `${swapPct.toFixed(1)}%`;
2814     const swapSubGauge = document.getElementById('sys-swap-sub-gauge');
2815     if (swapSubGauge) swapSubGauge.innerHTML = createGaugeSvg(swapPct, 100, 56);
2816     const swapSubSlider = document.getElementById('sys-swap-sub-slider');
2817     if (swapSubSlider) swapSubSlider.innerHTML = createPercentSliderHtml(swapPct, false, `Подкачка ${swapPct.toFixed(1)}%`);
2818 }
2819
2820 // GPU: данные для независимых карточек берутся строго из
fetchGpuLoadFromApi()
2821 if (!lastGpuData) {
2822     fetchGpuLoadFromApi();
2823 }
2824
2825 // Disk I/O
2826 if (snap.disk_io) {
2827     const diskVal = document.getElementById('sys-metric-disk-val');
2828     const diskSub = document.getElementById('sys-metric-disk-sub');
2829
2830     const rBps = Number(snap.disk_io.read_bytes_per_sec || 0);
2831     const wBps = Number(snap.disk_io.write_bytes_per_sec || 0);
2832     const totalMb = ((rBps + wBps) / (1024 * 1024)).toFixed(

```

```
2);
2833     const rKb = (rBps / 1024).toFixed(0);
2834     const wKb = (wBps / 1024).toFixed(0);
2835
2836     if (diskVal) diskVal.innerText = `${totalMb} MB/s`;
2837     if (diskSub) diskSub.innerText = `Чтение: ${rKb} KB/s |
Запись: ${wKb} KB/s`;
2838     const ramKpiIo = document.getElementById('sys-ram-kpi-io
');
2839     if (ramKpiIo) ramKpiIo.textContent = `${totalMb} MB/s`;
2840
2841     const readFormatted = formatBytesPerSec(rBps);
2842     const writeFormatted = formatBytesPerSec(wBps);
2843
2844     const memioRead = document.getElementById('sys-memio-rea
d');
2845     if (memioRead) memioRead.innerText = readFormatted;
2846     const readSubBadge = document.getElementById('sys-read-s
ub-badge');
2847     if (readSubBadge) readSubBadge.innerText = readFormatted
;
2848
2849     const memioWrite = document.getElementById('sys-memio-wr
ite');
2850     if (memioWrite) memioWrite.innerText = writeFormatted;
2851     const writeSubBadge = document.getElementById('sys-write
-sub-badge');
2852     if (writeSubBadge) writeSubBadge.innerText = writeFormat
ted;
2853
2854     const maxIoScale = 50 * 1024 * 1024;
2855     const readPct = Math.min(100, (rBps / maxIoScale) * 100)
;
2856     const writePct = Math.min(100, (wBps / maxIoScale) * 100
);
2857
2858     const readSubGauge = document.getElementById('sys-read-s
ub-gauge');
```

```
2859 |         if (readSubGauge) readSubGauge.innerHTML = createGaugeSv
| g(readPct, 100, 56);
2860 |         const readSubBar = document.getElementById('sys-read-sub
| -bar');
2861 |         const readSubThumb = document.getElementById('sys-read-s
| ub-thumb');
2862 |         if (readSubBar) readSubBar.style.width = `${readPct.toFi
| xed(1)}%`;
2863 |         if (readSubThumb) readSubThumb.style.left = `${readPct.t
| oFixed(1)}%`;
2864 |
2865 |         const writeSubGauge = document.getElementById('sys-write
| -sub-gauge');
2866 |         if (writeSubGauge) writeSubGauge.innerHTML = createGauge
| Svg(writePct, 100, 56);
2867 |         const writeSubBar = document.getElementById('sys-write-s
| ub-bar');
2868 |         const writeSubThumb = document.getElementById('sys-write
| -sub-thumb');
2869 |         if (writeSubBar) writeSubBar.style.width = `${writePct.t
| oFixed(1)}%`;
2870 |         if (writeSubThumb) writeSubThumb.style.left = `${writePc
| t.toFixed(1)}%`;
2871 |     }
2872 |
2873 |     // Physical Disks SMART Health
2874 |     const diskCont = document.getElementById('sys-physical-dis
| ks-container');
2875 |     const diskBadge = document.getElementById('sys-disk-health
| -badge');
2876 |     if (diskCont && Array.isArray(snap.physical_disks) && snap
| .physical_disks.length > 0) {
2877 |         const allHealthy = snap.physical_disks.every(d => d.heal
| th_status === 'Healthy');
2878 |         if (diskBadge) {
2879 |             diskBadge.textContent = allHealthy ? 'SMART OK' : 'Вни
| мание';
2880 |             diskBadge.className = allHealthy ? 'badge bg-success-s
```

```

        subtle text-success border border-success' : 'badge bg-warning-
        subtle text-warning border border-warning';
2881     }
2882     diskCont.innerHTML = snap.physical_disks.map(d => `
2883         <div class="d-flex justify-content-between align-items
        -center py-0.5 border-bottom border-dark-subtle" style="border
        -bottom-style: dashed !important;">
2884             <span class="text-truncate me-2" style="max-width: 1
        60px;" title="${escapeHtml(d.model)}">${escapeHtml(d.model)}</
        span>
2885             <span class="badge bg-secondary font-monospace" styl
        e="font-size: 0.65rem;">${d.size_gb} GB ${escapeHtml(d.media_t
        ype || '')}</span>
2886         </div>
2887     `).join('');
2888 }
2889
2890 // Battery & Power
2891 const powerCont = document.getElementById('sys-power-info-
        container');
2892 const powerBadge = document.getElementById('sys-power-stat
        us-badge');
2893 if (powerCont && snap.battery) {
2894     if (snap.battery.has_battery && snap.battery.percent !==
        null) {
2895         if (powerBadge) {
2896             powerBadge.textContent = `${snap.battery.percent}% $
        {snap.battery.power_plugged ? '⏻ Зарядка' : '🔋 Батарея'}`;
2897             powerBadge.className = snap.battery.power_plugged ?
        'badge bg-success-subtle text-success border border-success' :
        'badge bg-warning-subtle text-warning border border-warning';
2898         }
2899         const minsLeft = snap.battery.secs_left ? Math.round(s
        nap.battery.secs_left / 60) : null;
2900         powerCont.innerHTML = `
2901             <div>Статус: ${snap.battery.power_plugged ? 'Подключ
        ено к сети' : 'Работа от батареи'}</div>
2902             <div>${minsLeft ? `Осталось ~${minsLeft} мин.` : 'Пр

```

```

    2903 |     'филь: ' + (snap.battery.power_profile || 'Balanced'))</div>
2904 |     `;
2905 |     } else {
2906 |         if (powerBadge) {
2907 |             powerBadge.textContent = 'AC Mains';
2908 |             powerBadge.className = 'badge bg-secondary';
2909 |         }
2910 |         powerCont.innerHTML = `
2911 |             <div>Питание: Стационарная электросеть 220V</div>
2912 |             <div class="text-muted">Профиль: ${escapeHtml(snap.b
2913 | attery.power_profile || 'High Performance'))</div>
2914 |             `;
2915 |         }
2916 |     }
2917 |
2918 |     // RAM SPD Modules
2919 |     const ramSticksCont = document.getElementById('sys-ram-sti
2920 | cks-container');
2921 |     const ramSticksBadge = document.getElementById('sys-ram-st
2922 | icks-badge');
2923 |     if (ramSticksCont && Array.isArray(snap.ram_sticks) && sna
2924 | p.ram_sticks.length > 0) {
2925 |         if (ramSticksBadge) {
2926 |             ramSticksBadge.textContent = `${snap.ram_sticks.length
2927 | } модулей`;
2928 |         }
2929 |         ramSticksCont.innerHTML = snap.ram_sticks.slice(0, 2).ma
2930 | p(m => `
    <div class="d-flex justify-content-between align-items
    -center py-0.5">
        <span>${escapeHtml(m.bank_label)}: ${escapeHtml(m.ma
        nufacturer)}</span>
        <span class="font-monospace text-primary">${m.capaci
        ty_gb} GB @ ${m.speed_mhz} MT/s</span>
    </div>
    `).join('');
    } else if (ramSticksCont) {
    ramSticksCont.innerHTML = `<div>RAM: ${snap.memory ? sna

```

```

    p.memory.total_gb : '--'} GB физической памяти</div>`;
2931     }
2932
2933     // Reliability & Open Ports
2934     const alertsCont = document.getElementById('sys-alerts-por
ts-container');
2935     const alertsBadge = document.getElementById('sys-alerts-ba
dge');
2936     if (alertsCont) {
2937         const portCount = Array.isArray(snap.listening_ports) ?
snap.listening_ports.length : 0;
2938         const isReboot = snap.alerts && snap.alerts.reboot_pendi
ng;
2939         if (alertsBadge) {
2940             alertsBadge.textContent = isReboot ? 'Перезагрузка' :
'Стабильно';
2941             alertsBadge.className = isReboot ? 'badge bg-warning t
ext-dark' : 'badge bg-info-subtle text-info border border-info
';
2942         }
2943         alertsCont.innerHTML = `
2944             <div>Сетевые сокетy: <span class="fw-bold" style="colo
r: var(--text-color);">${portCount} портов LISTEN</span></div>
2945             <div class="text-truncate" title="${escapeHtml(snap.al
erts?.latest_alert || '')}">${escapeHtml(snap.alerts?.latest_a
lert || 'Система стабильна')}</div>
2946         `;
2947     }
2948 }
2949
2950 async function fetchNetworkActivity() {
2951     try {
2952         const res = await fetch('/api/v1/system/network-activity
?limit=100');
2953         if (res.ok) {
2954             const data = await res.json();
2955             renderNetworkActivityTable(data);
2956         }

```

```
2957     } catch (e) {
2958         console.warn('[SystemInspectorTab] Failed to fetch network activity:', e);
2959     }
2960 }
2961
2962 function renderNetworkActivityTable(activities) {
2963     if (activities) {
2964         cachedNetworkActivities = activities;
2965     }
2966     if (isNetPaused) return;
2967
2968     const tbody = document.getElementById('sys-net-tbody');
2969     if (!tbody) return;
2970
2971     const searchInput = document.getElementById('sys-net-search');
2972     const filterText = (searchInput?.value || '').toLowerCase().trim();
2973     const countBadge = document.getElementById('sys-netact-count-badge');
2974     const connsBadge = document.getElementById('sys-netact-conns-badge');
2975
2976     const rawList = Array.isArray(cachedNetworkActivities) ? cachedNetworkActivities : [];
2977
2978     // Filter by category and search query
2979     let filtered = rawList.filter(item => {
2980         if (currentNetFilter === 'internet' && !item.is_internet)
2981             return false;
2982         if (currentNetFilter === 'listen' && item.status !== 'LISTEN')
2983             return false;
2984
2985         if (filterText) {
2986             return (String(item.pid).includes(filterText) ||
2987                 (item.name || '').toLowerCase().includes(filterText))
2988         }
2989     });
```



```

2987 |         (item.user || '').toLowerCase().includes(filterText)
2988 |         (item.remote_address || '').toLowerCase().includes(f
filterText) ||
2989 |         (item.local_address || '').toLowerCase().includes(fi
lterText) ||
2990 |         (item.service_type || '').toLowerCase().includes(fil
terText) ||
2991 |         (item.sent_summary || '').toLowerCase().includes(fil
terText) ||
2992 |         (item.recv_summary || '').toLowerCase().includes(fil
terText)
2993 |     );
2994 | }
2995 |     return true;
2996 | });
2997 |
2998 |     const uniqueProcs = new Set(filtered.map(i => i.pid)).size
;
2999 |     if (countBadge) countBadge.textContent = `${uniqueProcs} н
программ`;
3000 |     if (connsBadge) connsBadge.textContent = `${filtered.lengt
h} сокетов`;
3001 |
3002 |     if (filtered.length === 0) {
3003 |         tbody.innerHTML = '<tr><td colspan="6" class="text-cen
ter py-4 text-muted small">Нет активных сетевых соединений по вы
бранному фильтру</td></tr>';
3004 |         return;
3005 |     }
3006 |
3007 |     tbody.innerHTML = filtered.map((item, idx) => {
3008 |         const isListen = item.status === 'LISTEN';
3009 |         const statusBadgeClass = isListen
3010 |             ? 'badge bg-secondary-subtle text-light border border-
secondary'
3011 |             : item.status === 'ESTABLISHED'

```

```

3012 |         ? 'badge bg-success-subtle text-success border border-
      | success'
3013 |         : 'badge bg-warning-subtle text-warning border border-
      | warning';
3014 |
3015 |         const protoBadge = item.protocol === 'UDP'
3016 |         ? '<span class="badge bg-primary text-white" style="fo
      | nt-size: 0.65rem;">UDP</span>'
3017 |         : '<span class="badge bg-dark border border-secondary
      | text-info" style="font-size: 0.65rem;">TCP</span>';
3018 |
3019 |         const isExtBadge = item.is_internet
3020 |         ? '<span class="badge bg-primary-subtle text-primary b
      | order border-primary px-1" style="font-size: 0.62rem;" title="
      | Внешний сервер в сети Интернет">WAN</span>'
3021 |         : '<span class="badge bg-secondary px-1" style="font-s
      | ize: 0.62rem;" title="Локальный сокет Loopback">LAN</span>';
3022 |
3023 |         const formatNetKb = (kb) => {
3024 |             if (!kb || kb <= 0) return '0 KB';
3025 |             if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).toF
      | ixed(2) + ' GB';
3026 |             if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB';
3027 |             return kb.toFixed(1) + ' KB';
3028 |         };
3029 |
3030 |         const formatNetRate = (rate) => {
3031 |             if (!rate || rate <= 0.05) return '';
3032 |             if (rate >= 1024) return (rate / 1024).toFixed(1) + '
      | MB/s';
3033 |             return rate.toFixed(1) + ' KB/s';
3034 |         };
3035 |
3036 |         const deltaSentStr = (item.delta_sent_kb && item.delta_s
      | ent_kb > 0)
3037 |         ? `▲ +${formatNetKb(item.delta_sent_kb)}${item.sent_ra
      | te_kbs > 0.05 ? ' (' + formatNetRate(item.sent_rate_kbs) + ')
      | ': ''}`

```

```

3038 |         : '▲ 0 KB';
3039 |         const deltaRecvStr = (item.delta_recv_kb && item.delta_r
|         ecv_kb > 0)
3040 |         ? `▼ +${formatNetKb(item.delta_recv_kb)}${item.recv_ra
|         te_kbs > 0.05 ? ' (' + formatNetRate(item.recv_rate_kbs) + ') '
|         : ''}`
3041 |         : '▼ 0 KB';
3042 |
3043 |         const totalSentStr = formatNetKb(item.sent_kb || 0);
3044 |         const totalRecvStr = formatNetKb(item.recv_kb || 0);
3045 |
3046 |         return `
3047 |             <tr class="sys-net-row" data-idx="${idx}" style="curso
|             r: pointer;" title="Нажмите для детальной диагностики сетевого
|             соединения">
3048 |                 <td>
3049 |                     <div class="fw-bold text-truncate" style="max-widt
|                     h: 165px; color: var(--text-color);" title="${escapeHtml(item.
|                     name)}">${escapeHtml(item.name)}</div>
3050 |                     <div class="small text-muted" style="font-size: 0.
|                     70rem;">PID: <span class="font-monospace text-info">${item.pid
|                     }</span> ${item.user ? '• ' + escapeHtml(item.user) : ''}</div
|                     >
3051 |                 </td>
3052 |                 <td>
3053 |                     <div class="d-flex align-items-center gap-1">
3054 |                         ${isExtBadge}
3055 |                         <span class="font-monospace fw-semibold text-tru
|                         ncate" style="max-width: 195px; color: var(--text-color);" tit
|                         le="${escapeHtml(item.remote_address)}">
3056 |                             ${escapeHtml(item.remote_address !== '-' ? ite
|                             m.remote_address : item.local_address)}
3057 |                         </span>
3058 |                     </div>
3059 |                     <div class="small text-muted font-monospace" style
|                     ="font-size: 0.68rem;">Local: ${escapeHtml(item.local_address)
|                     }</div>
3060 |                 </td>

```

```

3061         <td>
3062             <div class="d-flex align-items-center gap-1 mb-0.5
3063             ">
3064                 ${protoBadge}
3065                 <span class="fw-semibold text-truncate" style="m
3066                 ax-width: 110px; font-size: 0.74rem; color: var(--nav-active);
3067                 " title="${escapeHtml(item.service_type)}">${escapeHtml(item.s
3068                 ervice_type)}</span>
3069             </div>
3070         </td>
3071         <td style="text-align: center;">
3072             <span class="${statusBadgeClass}" style="font-size
3073             : 0.68rem;">${escapeHtml(item.status)}</span>
3074         </td>
3075         <td>
3076             <div class="d-flex align-items-center justify-cont
3077             ent-between gap-1 mb-1">
3078                 <span class="badge bg-warning-subtle text-warnin
3079                 g border border-warning px-1.5 py-0.5" style="font-size: 0.68r
3080                 em;" title="Отправлено за измеряемый период">
3081                     ${deltaSentStr}
3082                 </span>
3083                 <span class="text-muted font-monospace" style="f
3084                 ont-size: 0.66rem;" title="Всего отправлено/записано">
3085                     Σ ${totalSentStr}
3086                 </span>
3087             </div>
3088             <div class="d-flex align-items-start gap-1">
3089                 <i class="bi bi-arrow-up-right text-warning mt-0
3090                 .5" style="font-size: 0.70rem;"></i>
3091                 <div class="text-truncate" style="max-width: 250
3092                 px; font-size: 0.72rem; color: var(--text-muted);" title="${es
3093                 capeHtml(item.sent_summary)}">
3094                     ${escapeHtml(item.sent_summary)}
3095                 </div>
3096             </div>
3097         </td>
3098         <td>

```

```

3087 |         <div class="d-flex align-items-center justify-content-between gap-1 mb-1">
3088 |             <span class="badge bg-success-subtle text-success border border-success px-1.5 py-0.5" style="font-size: 0.68rem;" title="Скачано/получено за измеряемый период">
3089 |                 ${deltaRecvStr}
3090 |             </span>
3091 |             <span class="text-muted font-monospace" style="font-size: 0.66rem;" title="Всего скачано/прочитано">
3092 |                 Σ ${totalRecvStr}
3093 |             </span>
3094 |         </div>
3095 |         <div class="d-flex align-items-start gap-1">
3096 |             <i class="bi bi-arrow-down-left text-success mt-0.5" style="font-size: 0.70rem;"></i>
3097 |             <div class="text-truncate" style="max-width: 250px; font-size: 0.72rem; color: var(--text-muted);" title="${escapeHtml(item.recv_summary)}">
3098 |                 ${escapeHtml(item.recv_summary)}
3099 |             </div>
3100 |         </div>
3101 |     </td>
3102 | </tr>
3103 | `;
3104 | }).join('');
3105 |
3106 | tbody.querySelectorAll('.sys-net-row').forEach(row => {
3107 |     row.onclick = () => {
3108 |         const idx = parseInt(row.getAttribute('data-idx'), 10)
3109 |         ;
3110 |         const item = filtered[idx];
3111 |         if (!item) return;
3112 |         const formatNetKb = (kb) => {
3113 |             if (!kb || kb <= 0) return '0 KB';
3114 |             if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).toFixed(2) + ' GB';
3115 |             if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB

```

```

';
3116         return kb.toFixed(1) + ' KB';
3117     };
3118
3119     const formatNetRate = (rate) => {
3120         if (!rate || rate <= 0.05) return '';
3121         if (rate >= 1024) return (rate / 1024).toFixed(1) +
3122         ' MB/s';
3123         return rate.toFixed(1) + ' KB/s';
3124     };
3125
3126     if (window.AITableModal) {
3127         window.AITableModal.show({
3128             icon: '📡',
3129             title: `${item.name} (${item.service_type})`,
3130             subtitle: `PID: ${item.pid} | ${item.remote_address}`,
3131             tableType: 'network',
3132             badges: [
3133                 { text: `PID ${item.pid}`, class: 'badge bg-info text-dark' },
3134                 { text: item.protocol, class: 'badge bg-primary' },
3135                 { text: item.status, class: 'badge bg-success' },
3136                 { text: item.is_internet ? 'Интернет (WAN)' : 'Локально (LAN)', class: item.is_internet ? 'badge bg-warning text-dark' : 'badge bg-secondary' },
3137             ],
3138             metadata: [
3139                 { label: 'Программа / Процесс', value: item.name },
3140                 { label: 'Process ID (PID)', value: String(item.pid) },
3141                 { label: 'Пользователь системы', value: item.user || 'SYSTEM' },
3142                 { label: 'Удаленный адрес (Remote)', value: item.remote_address },

```

```

3142         { label: 'Локальный сокет (Local)', value: item.
local_address },
3143         { label: 'Протокол / Служба', value: `${item.pro
tocol} • ${item.service_type}` },
3144         { label: 'Статус соединения', value: item.status
},
3145         { label: 'Скачано за период (Прием)', value: `${
formatNetKb(item.delta_recv_kb || 0)} ${item.recv_rate_kbs > 0
.05 ? '(' + formatNetRate(item.recv_rate_kbs) + ')' : ''}` },
3146         { label: 'Всего скачано / получено', value: form
atNetKb(item.recv_kb || 0) },
3147         { label: 'Отправлено за период', value: `${forma
tNetKb(item.delta_sent_kb || 0)} ${item.sent_rate_kbs > 0.05 ?
 '(' + formatNetRate(item.sent_rate_kbs) + ')' : ''}` },
3148         { label: 'Всего отправлено', value: formatNetKb(
item.sent_kb || 0) },
3149         { label: 'Что шлет (Отправка)', value: item.sent
_summary, fullWidth: true },
3150         { label: 'Что принимает (Прием)', value: item.re
cv_summary, fullWidth: true }
3151     ],
3152     rawTitle: 'Сетевой дамп подключения',
3153     rawContent: JSON.stringify(item, null, 2),
3154     requestData: item
3155     });
3156 }
3157 };
3158 });
3159 }
3160
3161 async function runAiDiagnostics() {
3162     const summaryEl = document.getElementById('sys-ai-summary-
text');
3163     const badgeEl = document.getElementById('sys-ai-health-bad
ge');
3164     const anomaliesEl = document.getElementById('sys-ai-anomal
ies-container');
3165     const engineTagEl = document.getElementById('sys-ai-model-

```

```

tag');
3166
3167     if (summaryEl) summaryEl.innerText = 'Запуск глубокого AI-
аудита системы и оборудования...';
3168
3169     try {
3170         const res = await fetch('/api/v1/system/diagnose', { met
hod: 'POST' });
3171         if (!res.ok) throw new Error('AI Diagnosis error');
3172         const report = await res.json();
3173
3174         if (badgeEl) {
3175             badgeEl.innerText = `Health: ${report.health_score ||
100}/100`;
3176             badgeEl.className = `badge ${report.health_score >= 80
? 'bg-success' : report.health_score >= 60 ? 'bg-warning' : '
bg-danger'}`;
3177         }
3178
3179         if (engineTagEl) {
3180             engineTagEl.innerText = report.ai_model_used || 'Heuri
stic Engine';
3181         }
3182
3183         if (anomaliesEl) {
3184             if (Array.isArray(report.anomalies) && report.anomalie
s.length > 0) {
3185                 anomaliesEl.innerHTML = report.anomalies.map(a => `<
span class="anomaly-tag"> [${a.subsystem}] ${escapeHtml(a.tit
le)}</span>`).join('');
3186             } else {
3187                 anomaliesEl.innerHTML = '<span style="color: #4ade80
; font-size: 0.75rem;"><i class="bi bi-check-circle me-1"></i>
Аномалий в работе оборудования не обнаружено</span>';
3188             }
3189         }
3190
3191         if (summaryEl) {

```



```
3192 |         const recs = Array.isArray(report.recommendations) &&
| report.recommendations.length > 0
3193 |         ? `\\nРекомендации: ${report.recommendations.join(',
| ')} `
3194 |         : '';
3195 |         summaryEl.innerHTML = `${report.summary || 'Телеметрия
| в норме.'}${recs}`;
3196 |     }
3197 |     } catch (e) {
3198 |         console.error('[SystemInspectorTab] AI Diagnose error:',
| e);
3199 |         if (summaryEl) summaryEl.innerHTML = 'Ошибка выполнения
| AI-диагностики: ' + e.message;
3200 |     }
3201 | }
3202 |
3203 | let sysWsReconnectTimer = null;
3204 |
3205 | function connectSystemWebSocket() {
3206 |     if (sysWsReconnectTimer) {
3207 |         clearTimeout(sysWsReconnectTimer);
3208 |         sysWsReconnectTimer = null;
3209 |     }
3210 |     if (sysWs) {
3211 |         try {
3212 |             sysWs.onclose = null;
3213 |             sysWs.onerror = null;
3214 |             sysWs.close();
3215 |         } catch {}
3216 |         sysWs = null;
3217 |     }
3218 |
3219 |     if (window.isTabActive && !window.isTabActive('tab-hardwar
| e-load-inspector') && !window.isTabActive('tab-system-load-ins
| pector') && !window.isTabActive('tab-system-inspector')) {
3220 |         const statusBadge = document.getElementById('sys-conn-st
| atus');
3221 |         if (statusBadge) {
```

```
3222 |         statusBadge.className = 'badge rounded-pill bg-seconda
      | ry text-light px-3 py-2';
3223 |         statusBadge.innerText = '○ Поток приостановлен';
3224 |     }
3225 |     return;
3226 | }
3227 |
3228 |     const proto = window.location.protocol === 'https:' ? 'wss
      | :' : 'ws: ';
3229 |     const wsUrl = `${proto}://${window.location.host}/api/v1/sy
      | stem/stream`;
3230 |
3231 |     try {
3232 |         sysWs = new WebSocket(wsUrl);
3233 |         const statusBadge = document.getElementById('sys-conn-st
      | atus');
3234 |
3235 |         sysWs.onopen = () => {
3236 |             console.info('[SystemInspectorTab] WebSocket соединени
      | е с телеметрией установлено. ');
3237 |             if (statusBadge) {
3238 |                 statusBadge.className = 'badge rounded-pill bg-succe
      | ss-subtle text-success border border-success px-3 py-2';
3239 |                 statusBadge.innerText = '● Телеметрия активна';
3240 |             }
3241 |         };
3242 |
3243 |         sysWs.onmessage = (evt) => {
3244 |             try {
3245 |                 const snap = JSON.parse(evt.data);
3246 |                 latestTelemetrySnapshot = snap;
3247 |                 updateTelemetryDashboard(snap);
3248 |             } catch (e) {
3249 |                 console.error('[SystemInspectorTab] Ошибка разбора с
      | ообщения телеметрии:', e);
3250 |             }
3251 |         };
3252 |     }
```

```
3253 |         const scheduleReconnect = (reason) => {
3254 |             if (sysWsReconnectTimer) return;
3255 |             if (window.isTabActive && !window.isTabActive('tab-har
dware-load-inspector') && !window.isTabActive('tab-system-load
-inspector') && !window.isTabActive('tab-system-inspector')) r
eturn;
3256 |             console.warn(`[SystemInspectorTab] WebSocket отключен
(${reason}). Автопереподключение через 3 сек...`);
3257 |             if (statusBadge) {
3258 |                 statusBadge.className = 'badge rounded-pill bg-warni
ng-subtle text-warning border border-warning px-3 py-2';
3259 |                 statusBadge.innerText = '○ Переподключение...';
3260 |             }
3261 |             sysWsReconnectTimer = setTimeout(() => {
3262 |                 sysWsReconnectTimer = null;
3263 |                 if (!window.isTabActive || window.isTabActive('tab-h
ardware-load-inspector') || window.isTabActive('tab-system-loa
d-inspector') || window.isTabActive('tab-system-inspector')) {
3264 |                     connectSystemWebSocket();
3265 |                 }
3266 |             }, 3000);
3267 |         };
3268 |
3269 |         sysWs.onclose = (evt) => {
3270 |             scheduleReconnect(`код закрытия: ${evt.code}`);
3271 |         };
3272 |
3273 |         sysWs.onerror = (err) => {
3274 |             console.warn('[SystemInspectorTab] Ошибка соединения W
ebSocket:', err);
3275 |             scheduleReconnect('ошибка связи');
3276 |         };
3277 |     } catch (err) {
3278 |         console.error('[SystemInspectorTab] Ошибка инициализации
WebSocket:', err);
3279 |         sysWsReconnectTimer = setTimeout(() => {
3280 |             sysWsReconnectTimer = null;
3281 |             if (!window.isTabActive || window.isTabActive('tab-har
```

```

    'dware-load-inspector') || window.isTabActive('tab-system-load-
inspector') || window.isTabActive('tab-system-inspector')) {
3282         connectSystemWebSocket();
3283     }
3284     }, 5000);
3285 }
3286 }
3287
3288 function disconnectSystemWebSocket() {
3289     if (sysWsReconnectTimer) {
3290         clearTimeout(sysWsReconnectTimer);
3291         sysWsReconnectTimer = null;
3292     }
3293     if (sysWs) {
3294         try {
3295             sysWs.onclose = null;
3296             sysWs.onerror = null;
3297             sysWs.close();
3298         } catch {}
3299         sysWs = null;
3300     }
3301     const statusBadge = document.getElementById('sys-conn-stat
us');
3302     if (statusBadge) {
3303         statusBadge.className = 'badge rounded-pill bg-secondary
text-light px-3 py-2';
3304         statusBadge.innerText = '○ Поток приостановлен';
3305     }
3306 }
3307
3308 function bindTabEvents() {
3309     const btnRefreshSensors = document.getElementById('btn-sys
-refresh-sensors');
3310     if (btnRefreshSensors) {
3311         btnRefreshSensors.onclick = () => fetchLhmSensors();
3312     }
3313
3314     const btnRefreshLhm = document.getElementById('btn-sys-ref

```

```
    resh-lhm');
3315     if (btnRefreshLhm) {
3316         btnRefreshLhm.onclick = () => fetchLhmSensors();
3317     }
3318
3319     const btnLaunchLhm = document.getElementById('btn-sys-laun
ch-lhm');
3320     if (btnLaunchLhm) {
3321         btnLaunchLhm.onclick = async () => {
3322             btnLaunchLhm.disabled = true;
3323             btnLaunchLhm.innerHTML = '<span class="spinner-border
spinner-border-sm me-1"></span> Заныск...';
3324             try {
3325                 const res = await fetch('/api/v1/lhm/launch', { meth
od: 'POST' });
3326                 if (res.ok) {
3327                     setTimeout(fetchLhmSensors, 2500);
3328                 }
3329             } catch (e) {
3330                 console.error('[SystemInspectorTab] Failed to launch
LHM:', e);
3331             } finally {
3332                 btnLaunchLhm.disabled = false;
3333             }
3334         };
3335     }
3336
3337     const searchSensors = document.getElementById('sys-lhm-sea
rch');
3338     if (searchSensors) {
3339         searchSensors.oninput = () => renderSensors();
3340     }
3341
3342     // Category pills
3343     document.querySelectorAll('.sys-lhm-cat-btn').forEach(btn
=> {
3344         btn.onclick = () => {
3345             document.querySelectorAll('.sys-lhm-cat-btn').forEach(
```

```

      b => {
3346         b.classList.remove('active', 'btn-outline-primary');
3347         b.classList.add('btn-outline-secondary');
3348     });
3349     btn.classList.remove('btn-outline-secondary');
3350     btn.classList.add('active', 'btn-outline-primary');
3351     currentSensorCategory = btn.getAttribute('data-cat') |
    | 'all';
3352     renderSensors();
3353     };
3354     });
3355
3356     // Analyze Button (Sensors)
3357     const btnAnalyze = document.getElementById('btn-sys-sensor
-analyze');
3358     if (btnAnalyze) {
3359         btnAnalyze.onclick = () => analyzeSensorsAndReport();
3360     }
3361
3362     // Modal Re-analyze Button
3363     const btnReAnalyze = document.getElementById('btn-re-analy
ze-sensors');
3364     if (btnReAnalyze) {
3365         btnReAnalyze.onclick = async () => {
3366             btnReAnalyze.disabled = true;
3367             const icon = btnReAnalyze.querySelector('i');
3368             if (icon) icon.classList.add('spin-animation');
3369             await fetchLhmSensors();
3370             await analyzeSensorsAndReport();
3371             if (icon) icon.classList.remove('spin-animation');
3372             btnReAnalyze.disabled = false;
3373         };
3374     }
3375
3376     // Modal Copy Report Button
3377     const btnCopyReport = document.getElementById('btn-copy-se
nsor-report');
3378     if (btnCopyReport) {

```

```
3379     btnCopyReport.onclick = async () => {
3380         if (!lastAnalysisReportText) return;
3381         try {
3382             await navigator.clipboard.writeText(lastAnalysisReportText);
3383             const origHtml = btnCopyReport.innerHTML;
3384             btnCopyReport.innerHTML = '<i class="bi bi-check2 text-success me-1"></i> Скопировано!';
3385             setTimeout(() => {
3386                 btnCopyReport.innerHTML = origHtml;
3387             }, 2000);
3388         } catch (e) {
3389             console.warn('[SystemInspectorTab] Clipboard copy failed:', e);
3390         }
3391     };
3392 }
3393
3394 const btnAudit = document.getElementById('btn-sys-run-audit');
3395 if (btnAudit) {
3396     btnAudit.onclick = () => runAiDiagnostics();
3397 }
3398
3399 const btnCompact = document.getElementById('btn-sys-toggle-compact');
3400 const sysContainer = document.querySelector('.sys-container');
3401 const compactText = document.getElementById('sys-compact-button-text');
3402
3403 const updateCompactUI = (isCompact) => {
3404     if (!sysContainer) return;
3405     if (isCompact) {
3406         sysContainer.classList.add('sys-compact-mode');
3407         if (btnCompact) {
3408             btnCompact.classList.add('collapsed');
3409             btnCompact.title = 'Развернуть все панели';
```

```
3410         }
3411     } else {
3412         sysContainer.classList.remove('sys-compact-mode');
3413         if (btnCompact) {
3414             btnCompact.classList.remove('collapsed');
3415             btnCompact.title = 'Свернуть все панели';
3416         }
3417     }
3418 };
3419
3420     if (btnCompact && sysContainer) {
3421         const savedCompact = localStorage.getItem('sys_inspector_
3422         _compact_mode');
3423         const isCompact = savedCompact === null ? true : savedCo
3424         mpact === 'true';
3425         updateCompactUI(isCompact);
3426         btnCompact.onclick = () => {
3427             const nextState = !sysContainer.classList.contains('sy
3428             s-compact-mode');
3429             updateCompactUI(nextState);
3430             try { localStorage.setItem('sys_inspector_compact_mode
3431             ', String(nextState)); } catch (e) {}
3432         };
3433     }
3434
3435     const btnConfig = document.getElementById('btn-sys-config'
3436     );
3437     if (btnConfig) {
3438         btnConfig.onclick = () => {
3439             if (window.appConfigEditor) {
3440                 window.appConfigEditor.open('system_inspector', 'Нас
3441                 тройки телеметрии и инспектора');
3442             } else {
3443                 console.info('[SystemInspectorTab] Config editor not
3444                 available');
3445             }
3446         };
3447     }
```



```
3441 |
3442 |     // Interval editor modal buttons
3443 |     const btnIntervalsTop = document.getElementById('btn-sys-i
|ntervals');
3444 |     if (btnIntervalsTop) {
3445 |         btnIntervalsTop.onclick = () => openSysIntervalsModal();
3446 |     }
3447 |
3448 |     const btnIntervalsLhm = document.getElementById('btn-sys-i
|ntervals-lhm');
3449 |     if (btnIntervalsLhm) {
3450 |         btnIntervalsLhm.onclick = () => openSysIntervalsModal();
3451 |     }
3452 |
3453 |     const btnRefreshIntervals = document.getElementById('btn-m
|odal-intervals-refresh');
3454 |     if (btnRefreshIntervals) {
3455 |         btnRefreshIntervals.onclick = () => loadSysIntervalsConf
|ig(true);
3456 |     }
3457 |
3458 |     const btnSaveIntervals = document.getElementById('btn-moda
|l-intervals-save');
3459 |     if (btnSaveIntervals) {
3460 |         btnSaveIntervals.onclick = () => saveSysIntervalsConfig(
|);
3461 |     }
3462 |
3463 |     const selUiRefresh = document.getElementById('modal-ui-ref
|resh-select');
3464 |     if (selUiRefresh) {
3465 |         selUiRefresh.onChange = (e) => {
3466 |             const sec = parseInt(e.target.value, 10) || 5;
3467 |             setupSysSensorInterval(sec);
3468 |             const badge = document.getElementById('modal-ui-refres
|h-badge');
3469 |             if (badge) badge.textContent = `${sec} сек`;
3470 |         };
```

```
3471 |     }
3472 |
3473 |     // Изменения файлов в реальном времени controls
3474 |     const changeWatchDirBtn = document.getElementById('btn-sys
-change-watch-dir');
3475 |     if (changeWatchDirBtn) {
3476 |         changeWatchDirBtn.onclick = () => openWatchFoldersModal(
'folders');
3477 |     }
3478 |     const exclusionsBtn = document.getElementById('btn-sys-wat
ch-exclusions');
3479 |     if (exclusionsBtn) {
3480 |         exclusionsBtn.onclick = () => openWatchFoldersModal('exc
lusions');
3481 |     }
3482 |     const watchDirBadge = document.getElementById('sys-watch-d
ir-badge');
3483 |     if (watchDirBadge) {
3484 |         watchDirBadge.onclick = () => openWatchFoldersModal('fol
ders');
3485 |     }
3486 |     const liveHelpBtn = document.getElementById('btn-sys-live-
help');
3487 |     if (liveHelpBtn) {
3488 |         liveHelpBtn.onclick = showLiveWatcherHelpModal;
3489 |     }
3490 |
3491 |     // Network Activity Table controls
3492 |     const searchNet = document.getElementById('sys-net-search'
);
3493 |     if (searchNet) {
3494 |         searchNet.oninput = () => renderNetworkActivityTable();
3495 |     }
3496 |
3497 |     const btnPauseNet = document.getElementById('btn-sys-pause
-net');
3498 |     if (btnPauseNet) {
3499 |         btnPauseNet.onclick = () => {
```

```

3500         isNetPaused = !isNetPaused;
3501         btnPauseNet.innerText = isNetPaused ? 'Возобновить' :
'Пауза';
3502         btnPauseNet.className = isNetPaused ? 'btn btn-xs btn-
warning rounded-pill px-2.5 py-0.5' : 'btn btn-xs btn-outline-
secondary rounded-pill px-2.5 py-0.5';
3503         if (!isNetPaused) renderNetworkActivityTable();
3504     };
3505 }
3506
3507     const btnRefreshNet = document.getElementById('btn-sys-ref
resh-net');
3508     if (btnRefreshNet) {
3509         btnRefreshNet.onclick = () => fetchNetworkActivity();
3510     }
3511
3512     document.querySelectorAll('.sys-net-filter-btn').forEach(b
tn => {
3513         btn.onclick = () => {
3514             document.querySelectorAll('.sys-net-filter-btn').forEa
ch(b => {
3515                 b.classList.remove('active', 'btn-outline-primary');
3516                 b.classList.add('btn-outline-secondary');
3517             });
3518             btn.classList.remove('btn-outline-secondary');
3519             btn.classList.add('active', 'btn-outline-primary');
3520             currentNetFilter = btn.getAttribute('data-filter') ||
'internet';
3521             renderNetworkActivityTable();
3522         };
3523     });
3524
3525     // Инициализация интерактивных ресайзеров таблиц
3526     initNetTableResizer();
3527     initWatcherTableResizer();
3528 }
3529
3530 function initNetTableResizer() {

```

```
3531     const resizer = document.getElementById('sys-net-table-res
izer');
3532     const tableContainer = document.getElementById('sys-net-ta
ble-container');
3533     if (!resizer || !tableContainer) return;
3534
3535     if (resizer._resizerInitialized) return;
3536     resizer._resizerInitialized = true;
3537
3538     try {
3539         const savedHeight = localStorage.getItem('sys_inspector_
net_height');
3540         if (savedHeight) {
3541             const parsed = parseInt(savedHeight, 10);
3542             if (!isNaN(parsed) && parsed >= 140 && parsed <= 1200)
{
3543                 tableContainer.style.height = `${parsed}px`;
3544             }
3545         }
3546     } catch {}
3547
3548     let startY = 0;
3549     let startHeight = 0;
3550     let isDragging = false;
3551
3552     function onMouseMove(e) {
3553         if (!isDragging) return;
3554         const clientY = e.clientY ?? (e.touches && e.touches[0]
? e.touches[0].clientY : null);
3555         if (clientY === null) return;
3556
3557         const deltaY = clientY - startY;
3558         let newHeight = startHeight + deltaY;
3559         if (newHeight < 140) newHeight = 140;
3560         if (newHeight > 1200) newHeight = 1200;
3561
3562         tableContainer.style.height = `${newHeight}px`;
3563     }
```

```
3564 |
3565 |     function onMouseUp() {
3566 |         if (!isDragging) return;
3567 |         isDragging = false;
3568 |         resizer.classList.remove('resizing');
3569 |         document.body.style.cursor = '';
3570 |         document.body.style.userSelect = '';
3571 |
3572 |         window.removeEventListener('mousemove', onMouseMove);
3573 |         window.removeEventListener('mouseup', onMouseUp);
3574 |         window.removeEventListener('touchmove', onMouseMove);
3575 |         window.removeEventListener('touchend', onMouseUp);
3576 |
3577 |         const currentH = parseInt(tableContainer.style.height, 1
0);
3578 |         if (!isNaN(currentH)) {
3579 |             try {
3580 |                 localStorage.setItem('sys_inspector_net_height', Str
ing(currentH));
3581 |             } catch {}
3582 |         }
3583 |     }
3584 |
3585 |     function onMouseDown(e) {
3586 |         isDragging = true;
3587 |         startY = e.clientY ?? (e.touches && e.touches[0] ? e.tou
ches[0].clientY : 0);
3588 |         startHeight = tableContainer.getBoundingClientRect().hei
ght;
3589 |         resizer.classList.add('resizing');
3590 |         document.body.style.cursor = 'ns-resize';
3591 |         document.body.style.userSelect = 'none';
3592 |
3593 |         window.addEventListener('mousemove', onMouseMove, { pass
ive: false });
3594 |         window.addEventListener('mouseup', onMouseUp);
3595 |         window.addEventListener('touchmove', onMouseMove, { pass
ive: false });
```

```
3596     window.addEventListener('touchend', onMouseUp);
3597     e.preventDefault();
3598 }
3599
3600     resizer.addEventListener('mousedown', onMouseDown);
3601     resizer.addEventListener('touchstart', onMouseDown, { passive: false });
3602
3603     resizer.addEventListener('dblclick', () => {
3604         tableContainer.style.height = '300px';
3605         try {
3606             localStorage.removeItem('sys_inspector_net_height');
3607         } catch {}
3608     });
3609 }
3610
3611 function initWatcherTableResizer() {
3612     const resizer = document.getElementById('sys-watcher-table-resizer');
3613     const tableContainer = document.getElementById('sys-watcher-table-container');
3614     if (!resizer || !tableContainer) return;
3615
3616     if (resizer._resizerInitialized) return;
3617     resizer._resizerInitialized = true;
3618
3619     // Восстановление сохранённой пользователем высоты
3620     try {
3621         const savedHeight = localStorage.getItem('sys_inspector_watcher_height');
3622         if (savedHeight) {
3623             const parsed = parseInt(savedHeight, 10);
3624             if (!isNaN(parsed) && parsed >= 120 && parsed <= 1200)
3625             {
3626                 tableContainer.style.height = `${parsed}px`;
3627             }
3628         } catch {}
3629     }
```

```
3629 |
3630 |     let startY = 0;
3631 |     let startHeight = 0;
3632 |     let isDragging = false;
3633 |
3634 |     function onMouseMove(e) {
3635 |         if (!isDragging) return;
3636 |         const clientY = e.clientY ?? (e.touches && e.touches[0]
3637 | ? e.touches[0].clientY : null);
3638 |         if (clientY === null) return;
3639 |
3640 |         const deltaY = clientY - startY;
3641 |         let newHeight = startHeight + deltaY;
3642 |         if (newHeight < 120) newHeight = 120;
3643 |         if (newHeight > 1200) newHeight = 1200;
3644 |
3645 |         tableContainer.style.height = `${newHeight}px`;
3646 |     }
3647 |
3648 |     function onMouseUp() {
3649 |         if (!isDragging) return;
3650 |         isDragging = false;
3651 |         resizer.classList.remove('resizing');
3652 |         document.body.style.cursor = '';
3653 |         document.body.style.userSelect = '';
3654 |
3655 |         window.removeEventListener('mousemove', onMouseMove);
3656 |         window.removeEventListener('mouseup', onMouseUp);
3657 |         window.removeEventListener('touchmove', onMouseMove);
3658 |         window.removeEventListener('touchend', onMouseUp);
3659 |
3660 |         const currentH = parseInt(tableContainer.style.height, 1
3661 | 0);
3662 |         if (!isNaN(currentH)) {
3663 |             try {
3664 |                 localStorage.setItem('sys_inspector_watcher_height',
3665 | String(currentH));
3666 |             } catch {}
```

```

3664     }
3665 }
3666
3667 function onMouseDown(e) {
3668     isDragging = true;
3669     startY = e.clientY ?? (e.touches && e.touches[0] ? e.tou
ches[0].clientY : 0);
3670     startHeight = tableContainer.getBoundingClientRect().hei
ght;
3671     resizer.classList.add('resizing');
3672     document.body.style.cursor = 'ns-resize';
3673     document.body.style.userSelect = 'none';
3674
3675     window.addEventListener('mousemove', onMouseMove, { pass
ive: false });
3676     window.addEventListener('mouseup', onMouseUp);
3677     window.addEventListener('touchmove', onMouseMove, { pass
ive: false });
3678     window.addEventListener('touchend', onMouseUp);
3679     e.preventDefault();
3680 }
3681
3682 resizer.addEventListener('mousedown', onMouseDown);
3683 resizer.addEventListener('touchstart', onMouseDown, { pass
ive: false });
3684
3685 // Сброс по двойному клику
3686 resizer.addEventListener('dblclick', () => {
3687     tableContainer.style.height = '280px';
3688     try {
3689         localStorage.removeItem('sys_inspector_watcher_height'
);
3690     } catch {}
3691 });
3692 }
3693
3694 // =====
=====

```



```

3695 // Telemetry & Polling Intervals Editor Controller (config.j
    son)
3696 // =====
    =====
3697
3698 const INTERVAL_PRESETS = [
3699     { value: '1 second', label: '1 сек' },
3700     { value: '2 seconds', label: '2 сек' },
3701     { value: '3 seconds', label: '3 сек' },
3702     { value: '5 seconds', label: '5 сек' },
3703     { value: '10 seconds', label: '10 сек' },
3704     { value: '30 seconds', label: '30 сек' },
3705     { value: '1 minute', label: '1 мин' },
3706     { value: '5 minutes', label: '5 мин' },
3707     { value: '10 minutes', label: '10 мин' },
3708     { value: '30 minutes', label: '30 мин' },
3709     { value: '1 hour', label: '1 час' },
3710     { value: '6 hours', label: '6 часов' },
3711     { value: '24 hours', label: '24 часа' },
3712 ];
3713
3714 const CORE_RESOURCE_LOGGERS = [
3715     'system_inspector',
3716     'hardware_monitor'
3717 ];
3718
3719 const LOGGER_META = {
3720     system_inspector: {
3721         icon: '□ ',
3722         title: 'System Inspector',
3723         desc: 'Комплексный снимок: загрузка CPU, RAM, дисковый в
вод-вывод, процессы хоста',
3724     },
3725     hardware_monitor: {
3726         icon: '□ ',
3727         title: 'Hardware Monitor',
3728         desc: 'Аппаратные датчики WMI, GPU SMI (NVIDIA/AMD/Intel
) и физические шины',

```

```
3729 | },
3730 | windows_sysadmin: {
3731 |     icon: '☐ ',
3732 |     title: 'Windows SysAdmin',
3733 |     desc: 'Системные службы Windows, пользователи, локальные
      | группы и сетевые порты',
3734 | },
3735 | windows_defender: {
3736 |     icon: '☐ ',
3737 |     title: 'Windows Defender',
3738 |     desc: 'Статус антивирусной защиты, сигнатуры, обнаруженн
      | ые угрозы и карантин',
3739 | },
3740 | windows_startup_auditor: {
3741 |     icon: '☐ ',
3742 |     title: 'StartUp Auditor',
3743 |     desc: 'Автозагрузка программ, реестр Run/RunOnce, сервис
      | ы и планировщик задач',
3744 | },
3745 | windows_backup_manager: {
3746 |     icon: '☐ ',
3747 |     title: 'Backup Manager',
3748 |     desc: 'Резервные копии баз данных SQLite, конфигураций и
      | снапшотов системы',
3749 | },
3750 | website_monitor: {
3751 |     icon: '☐ ',
3752 |     title: 'Website Monitor',
3753 |     desc: 'Проверка доступности веб-сервисов, задержка ответ
      | ов HTTP/HTTPS',
3754 | },
3755 | gcloud_monitor: {
3756 |     icon: '☐ ',
3757 |     title: 'GCloud Monitor',
3758 |     desc: 'Телеметрия виртуальных машин, квоты и мониторинг
      | Google Cloud',
3759 | },
3760 | cloudflared_monitor: {
```

```
3761         icon: '☐ ',
3762         title: 'Cloudflare Tunnel',
3763         desc: 'Статус защищенных туннелей cloudflared и исходящи
x подключений',
3764     },
3765     user_assistant: {
3766         icon: '☐ ',
3767         title: 'User Assistant',
3768         desc: 'Фоновые периодические напоминания, календарь и ув
едомления',
3769     },
3770     trading_terminal: {
3771         icon: '☐ ',
3772         title: 'Trading Terminal',
3773         desc: 'Торговые котировки, балансы криптобирж и открытые
ордера',
3774     },
3775     registry_viewer: {
3776         icon: '☐ ',
3777         title: 'Registry Viewer',
3778         desc: 'Мониторинг изменений системного реестра Windows',
3779     },
3780     software_audit: {
3781         icon: '☐ ',
3782         title: 'Software Audit',
3783         desc: 'Аудит неиспользуемого и редко запускаемого ПО (Us
erAssist/Prefetch)',
3784     },
3785     helpdesk: {
3786         icon: '☐ ',
3787         title: 'Helpdesk',
3788         desc: 'Мониторинг тикетов техподдержки и уведомлений',
3789     },
3790 };
3791
3792 let _sysIntervalsConfigData = null;
3793 let _currentUiRefreshSeconds = 5;
3794
```

```
3795 | function showAlertInModal(msg, type = 'success') {
3796 |     const alertEl = document.getElementById('modal-intervals-a
| lert');
3797 |     if (!alertEl) return;
3798 |     alertEl.className = `alert alert-${type} py-2 px-3 small m
| b-3`;
3799 |     alertEl.innerHTML = msg;
3800 |     alertEl.classList.remove('d-none');
3801 |     if (type === 'success') {
3802 |         setTimeout(() => {
3803 |             alertEl.classList.add('d-none');
3804 |         }, 4000);
3805 |     }
3806 | }
3807 |
3808 | async function loadSysIntervalsConfig(notify = false) {
3809 |     const btnRefresh = document.getElementById('btn-modal-inte
| rvals-refresh');
3810 |     const icon = btnRefresh ? btnRefresh.querySelector('i') :
| null;
3811 |     if (icon) icon.classList.add('spin-animation');
3812 |
3813 |     try {
3814 |         let res = await fetch('/api/autolog/config');
3815 |         if (!res.ok) {
3816 |             res = await fetch('/sysautologging/config');
3817 |         }
3818 |         if (!res.ok) {
3819 |             throw new Error(`HTTP ${res.status}`);
3820 |         }
3821 |
3822 |         _sysIntervalsConfigData = await res.json();
3823 |         renderSysIntervalsForm(_sysIntervalsConfigData);
3824 |
3825 |         if (notify) {
3826 |             showAlertInModal('<i class="bi bi-check2 me-1"></i> Ко
| нфигурация успешно загружена из файла', 'success');
3827 |         }
```

```
3828     } catch (e) {
3829         console.warn('[SystemInspectorTab] Failed to load intervals config:', e);
3830         showAlertInModal(`<i class="bi bi-exclamation-triangle me-1"></i> Ошибка загрузки конфигурации: ${escapeHtml(e.message)}</i>`, 'danger');
3831     } finally {
3832         if (icon) icon.classList.remove('spin-animation');
3833     }
3834 }
3835
3836 function buildIntervalSelectHtml(loggerName, currentInterval) {
3837     const val = currentInterval || '1 minute';
3838     const isCustom = !INTERVAL_PRESETS.some(p => p.value === val);
3839
3840     let html = `
```

```
    cfg-file');
3857     if (cfgBadge && data.config_file) {
3858         cfgBadge.textContent = data.config_file;
3859     }
3860
3861     // 2. Engine status & global switch
3862     const engineStatusBadge = document.getElementById('modal-intervals-engine-status');
3863     if (engineStatusBadge) {
3864         if (data.is_running) {
3865             engineStatusBadge.textContent = 'AutoLog Active';
3866             engineStatusBadge.className = 'badge bg-success';
3867         } else {
3868             engineStatusBadge.textContent = 'AutoLog Paused';
3869             engineStatusBadge.className = 'badge bg-danger-subtle text-danger border border-danger fw-bold';
3870         }
3871     }
3872
3873     const globalEnable = document.getElementById('modal-intervals-global-enable');
3874     if (globalEnable) {
3875         globalEnable.checked = data.enable_autolog !== false;
3876     }
3877
3878     // 3. Default fallback interval select
3879     const defSelect = document.getElementById('modal-intervals-default-select');
3880     if (defSelect && data.default_interval) {
3881         defSelect.value = data.default_interval;
3882     }
3883
3884     // 4. Populate Core resource loggers & Other loggers
3885     const coreContainer = document.getElementById('modal-core-loggers-container');
3886     const otherContainer = document.getElementById('modal-other-loggers-container');
3887     const otherCountEl = document.getElementById('modal-other-
```

```

    loggers-count');
3888
3889     if (!coreContainer || !otherContainer) return;
3890
3891     const loggers = data.loggers || {};
3892     let coreHtml = '';
3893     let otherHtml = '';
3894     let otherCount = 0;
3895
3896     // Process core loggers
3897     CORE_RESOURCE_LOGGERS.forEach(name => {
3898         const cfg = loggers[name] || { interval: '5 seconds', en
3899         abled: true };
3900         const meta = LOGGER_META[name] || { icon: '⚙', title: na
3901         me, desc: 'Системный сбор метрик' };
3902         const isEnabled = cfg.enabled !== false;
3903         const intervalVal = cfg.interval || '5 seconds';
3904
3905         coreHtml += `
3906         <div class="p-2 rounded d-flex align-items-center just
3907         ify-content-between flex-wrap gap-2" style="background: var(--
3908         bg-color); border: 1px solid var(--border-color);">
3909         <div class="d-flex align-items-center gap-2" style="
3910         max-width: 58%;">
3911             <span class="fs-5">${meta.icon}</span>
3912             <div>
3913                 <div class="fw-bold small" style="color: var(--t
3914                 ext-color);">${meta.title}</div>
3915                 <div class="small text-muted" style="font-size:
3916                 0.72rem; line-height: 1.2;">${meta.desc}</div>
3917             </div>
3918             </div>
3919             <div class="d-flex align-items-center gap-2 ms-auto"
3920             >
3921                 <div class="form-check form-switch m-0" title="Вкл
3922                 ючить/отключить сбор метрик для этого сервиса">
3923                 <input class="form-check-input sys-int-enable" t
3924                 ype="checkbox" data-logger="${name}" ${isEnabled ? 'checked' :

```

```

    ''}>
3915         </div>
3916         ${buildIntervalSelectHtml(name, intervalVal)}
3917         <button class="btn btn-xs btn-outline-secondary rounded px-1.5 py-0.5 sys-int-custom-btn" data-logger="${name}"
title="Ввести произвольный интервал вручную">
3918             <i class="bi bi-pencil"></i>
3919         </button>
3920     </div>
3921 </div>
3922 `;
3923 });
3924
3925 // Process remaining loggers
3926 const allLoggerNames = Object.keys(loggers).length > 0
3927   ? Object.keys(loggers)
3928   : Object.keys(LOGGER_META);
3929
3930 allLoggerNames.forEach(name => {
3931     if (CORE_RESOURCE_LOGGERS.includes(name)) return;
3932     otherCount++;
3933     const cfg = loggers[name] || { interval: data.default_in
3934     terval || '1 minute', enabled: true };
3935     const meta = LOGGER_META[name] || { icon: '□ ', title: n
3936     ame, desc: 'Фоновый опрос службы' };
3937     const isEnabled = cfg.enabled !== false;
3938     const intervalVal = cfg.interval || '1 minute';
3939
3940     otherHtml += `
3941         <div class="p-2 rounded d-flex align-items-center justify-content-between flex-wrap gap-2" style="background: var(--bg-color); border: 1px solid var(--border-color);">
3942             <div class="d-flex align-items-center gap-2" style="max-width: 58%;">
3943                 <span class="fs-6">${meta.icon}</span>
3944                 <div>
3945                     <div class="fw-bold small" style="color: var(--text-color); font-size: 0.78rem;">${meta.title}</div>

```



```

3944 |         <div class="small text-muted" style="font-size:
      | 0.7rem; line-height: 1.2;">${meta.desc}</div>
3945 |     </div>
3946 | </div>
3947 |     <div class="d-flex align-items-center gap-2 ms-auto"
      | >
3948 |         <div class="form-check form-switch m-0" title="Вкл
      | ючить/выключить">
3949 |             <input class="form-check-input sys-int-enable" t
      | ype="checkbox" data-logger="${name}" ${isEnabled ? 'checked' :
      | ''}>
3950 |         </div>
3951 |         ${buildIntervalSelectHtml(name, intervalVal)}
3952 |         <button class="btn btn-xs btn-outline-secondary ro
      | unded px-1.5 py-0.5 sys-int-custom-btn" data-logger="${name}"
      | title="Ввести произвольный интервал вручную">
3953 |             <i class="bi bi-pencil"></i>
3954 |         </button>
3955 |     </div>
3956 | </div>
3957 | `;
3958 | });
3959 |
3960 | coreContainer.innerHTML = coreHtml;
3961 | otherContainer.innerHTML = otherHtml;
3962 | if (otherCountEl) otherCountEl.textContent = String(otherC
      | ount);
3963 |
3964 | // Bind custom interval prompt buttons
3965 | document.querySelectorAll('.sys-int-custom-btn').forEach(b
      | tn => {
3966 |     btn.onclick = () => {
3967 |         const lName = btn.getAttribute('data-logger');
3968 |         const sel = document.querySelector(`.sys-int-select[da
      | ta-logger="${lName}"]`);
3969 |         const currentVal = sel ? sel.value : '1 minute';
3970 |         const custom = prompt(`Введите интервал опроса для ${l
      | Name} (например: "2 seconds", "15 seconds", "10 minutes", "1 h

```

```

        our"):`, currentVal);
3971         if (custom && custom.trim()) {
3972             const trimmed = custom.trim();
3973             let opt = sel.querySelector(`option[value="${trimmed
        }"]`);
3974             if (!opt) {
3975                 opt = document.createElement('option');
3976                 opt.value = trimmed;
3977                 opt.textContent = `${trimmed} (пользовательский)`;
3978                 sel.appendChild(opt);
3979             }
3980             sel.value = trimmed;
3981         }
3982     };
3983 });
3984 }
3985
3986 async function saveSysIntervalsConfig() {
3987     const btnSave = document.getElementById('btn-modal-interva
        ls-save');
3988     if (btnSave) {
3989         btnSave.disabled = true;
3990         btnSave.innerHTML = '<span class="spinner-border spinner
        -border-sm me-1" role="status"></span> Сохранение...';
3991     }
3992
3993     try {
3994         const globalEnable = document.getElementById('modal-inte
        rvals-global-enable');
3995         const defSelect = document.getElementById('modal-interva
        ls-default-select');
3996
3997         const payload = {
3998             enable_autolog: globalEnable ? globalEnable.checked :
        true,
3999             default_interval: defSelect ? defSelect.value : '1 min
        ute',
4000             loggers: {}

```

```

4001     };
4002
4003     // Collect values from all interval selects & switches
4004     document.querySelectorAll('.sys-int-select').forEach(sel
=> {
4005         const lName = sel.getAttribute('data-logger');
4006         if (!lName) return;
4007         const sw = document.querySelector(`.sys-int-enable[dat
a-logger="${lName}"]`);
4008         payload.loggers[lName] = {
4009             interval: sel.value,
4010             enabled: sw ? sw.checked : true
4011         };
4012     });
4013
4014     let saveRes = await fetch('/api/autolog/config', {
4015         method: 'POST',
4016         headers: { 'Content-Type': 'application/json' },
4017         body: JSON.stringify(payload)
4018     });
4019
4020     if (!saveRes.ok) {
4021         saveRes = await fetch('/sysautologging/config', {
4022             method: 'POST',
4023             headers: { 'Content-Type': 'application/json' },
4024             body: JSON.stringify(payload)
4025         });
4026     }
4027
4028     if (!saveRes.ok) {
4029         const errJson = await saveRes.json().catch(() => ({}))
;
4030         throw new Error(errJson.detail || `HTTP ${saveRes.stat
us}`);
4031     }
4032
4033     const result = await saveRes.json();
4034     showAlertInModal(

```

```
4035 |         `<i class="bi bi-check-circle-fill text-success me-1">
      | </i> ${result.message || 'Интервалы успешно сохранены в config
      | .json и применены!'}`,
4036 |         'success'
4037 |     );
4038 |
4039 |     // Refresh engine badge
4040 |     const engineStatusBadge = document.getElementById('modal
      | -intervals-engine-status');
4041 |     if (engineStatusBadge) {
4042 |         if (result.is_running) {
4043 |             engineStatusBadge.textContent = 'AutoLog Active';
4044 |             engineStatusBadge.className = 'badge bg-success';
4045 |         } else {
4046 |             engineStatusBadge.textContent = 'AutoLog Paused';
4047 |             engineStatusBadge.className = 'badge bg-danger-subtl
      | e text-danger border border-danger fw-bold';
4048 |         }
4049 |     }
4050 | } catch (e) {
4051 |     console.error('[SystemInspectorTab] Save intervals confi
      | g error:', e);
4052 |     showAlertInModal(`<i class="bi bi-exclamation-triangle-f
      | ill text-danger me-1"></i> Не удалось сохранить интервалы: ${e
      | scapeHtml(e.message)}`, 'danger');
4053 | } finally {
4054 |     if (btnSave) {
4055 |         btnSave.disabled = false;
4056 |         btnSave.innerHTML = '<i class="bi bi-check2-circle me-
      | 1"></i> Сохранить в config.json';
4057 |     }
4058 | }
4059 | }
4060 |
4061 | let currentWatchDirs = [];
4062 | let currentWatchDir = '';
4063 | let stagedWatchDirs = [];
4064 | let currentBrowserPath = 'C:\\';
```

```
4065 | let cachedDrives = [];
4066 |
4067 | async function fetchLiveFileEvents() {
4068 |     try {
4069 |         const [resEvents, resTelem] = await Promise.all([
4070 |             fetch('/api/v1/system/file-audit/live-events?limit=30'
4071 | ),
4072 |             fetch('/api/v1/system/file-audit/telemetry').catch(()
4073 | => null)
4074 |         ]));
4075 |
4076 |         if (resTelem && resTelem.ok) {
4077 |             const telem = await resTelem.json();
4078 |             const elRate = document.getElementById('sys-watcher-ra
4079 | te');
4080 |             const elCr = document.getElementById('sys-watcher-cr')
4081 | ;
4082 |             const elMod = document.getElementById('sys-watcher-mod
4083 | ');
4084 |             const elDel = document.getElementById('sys-watcher-del
4085 | ');
4086 |             const elDrive = document.getElementById('sys-watcher-d
4087 | rive');
4088 |             const elDriveModel = document.getElementById('sys-watc
4089 | her-drive-model');
4090 |             const elDriveTemp = document.getElementById('sys-watch
4091 | er-temp');
4092 |             const elDiskR = document.getElementById('sys-watcher-r
4093 | -kbs');
4094 |             const elDiskW = document.getElementById('sys-watcher-w
4095 | -kbs');
4096 |             const elSecMatches = document.getElementById('sys-watc
4097 | her-sec-matches');
4098 |             const elStatus = document.getElementById('sys-watcher-
4099 | sensor-status');
4100 |
4101 |             if (elRate) elRate.textContent = (telem.events_rate_pe
4102 | r_sec || 0).toFixed(1);
```

```
4089 |         if (elCr) elCr.textContent = (telem.created_rate_per_s
|         ec || 0).toFixed(1);
4090 |         if (elMod) elMod.textContent = (telem.modified_rate_pe
|         r_sec || 0).toFixed(1);
4091 |         if (elDel) elDel.textContent = (telem.deleted_rate_per
|         _sec || 0).toFixed(1);
4092 |
4093 |         if (elDrive) {
4094 |             const drivesStr = (telem.drive_letters && telem.driv
|         e_letters.length)
4095 |                 ? telem.drive_letters.join(', ')
4096 |                 : (telem.drive_letter || 'C:');
4097 |             elDrive.textContent = drivesStr;
4098 |         }
4099 |         if (elDriveModel) elDriveModel.textContent = telem.dri
|         ve_model || 'Storage';
4100 |         if (elDriveTemp) elDriveTemp.textContent = telem.drive
|         _temperature_c !== null && telem.drive_temperature_c !== undef
|         ined ? `${telem.drive_temperature_c} °C` : '-- °C';
4101 |         if (elDiskR) elDiskR.textContent = Math.round(telem.di
|         sk_read_kbs || 0);
4102 |         if (elDiskW) elDiskW.textContent = Math.round(telem.di
|         sk_write_kbs || 0);
4103 |         if (elSecMatches) elSecMatches.textContent = telem.sec
|         urity_audit_matched_count || 0;
4104 |
4105 |         if (elStatus) {
4106 |             if (telem.burst_deletions_alert) {
4107 |                 elStatus.className = 'badge bg-danger-subtle text-
|         danger border border-danger px-2 py-1';
4108 |                 elStatus.textContent = '⚠ Всплеск удалений';
4109 |             } else if (telem.high_activity_alert) {
4110 |                 elStatus.className = 'badge bg-warning-subtle text
|         -warning border border-warning px-2 py-1';
4111 |                 elStatus.textContent = '⚠ Высокий I/O';
4112 |             } else {
4113 |                 elStatus.className = 'badge bg-success-subtle text
|         -success border border-success px-2 py-1';
```

```
4114 |         elStatus.textContent = 'Штатный режим';
4115 |     }
4116 | }
4117 | }
4118 |
4119 | if (!resEvents || !resEvents.ok) return;
4120 | const data = await resEvents.json();
4121 | const events = data.events || [];
4122 |
4123 | if (data.filtered_count !== undefined) {
4124 |     const filteredEl = document.getElementById('sys-watche
r-filtered-count');
4125 |     if (filteredEl) filteredEl.textContent = data.filtered
_count;
4126 | }
4127 |
4128 | currentWatchDirs = data.watch_dirs || (data.watch_dir ?
[data.watch_dir] : []);
4129 | currentWatchDir = currentWatchDirs[0] || data.watch_dir
|| '';
4130 |
4131 | const dirPathEl = document.getElementById('sys-watch-dir
-path');
4132 | const badge = document.getElementById('sys-watch-dir-bad
ge');
4133 | if (dirPathEl) {
4134 |     if (currentWatchDirs.length === 0) {
4135 |         dirPathEl.innerText = 'Папки не выбраны';
4136 |     } else if (currentWatchDirs.length === 1) {
4137 |         const singleName = currentWatchDirs[0].split('\\').p
op() || currentWatchDirs[0];
4138 |         dirPathEl.innerText = singleName;
4139 |     } else {
4140 |         const firstNames = currentWatchDirs.slice(0, 2).map(
p => p.split('\\').pop() || p).join(', ');
4141 |         dirPathEl.innerText = `${currentWatchDirs.length} па
нок: ${firstNames}${currentWatchDirs.length > 2 ? '...' : ''}`
;

```

```

4142     }
4143     }
4144     if (badge) {
4145         badge.title = `Отслеживаемые каталоги (${currentWatchD
irs.length}): \n${currentWatchDirs.join('\n')} \n\n(Нажмите для
настройки и выбора папок)`;
4146     }
4147
4148     const tbody = document.getElementById('sys-watcher-tbody
');
4149     if (tbody) {
4150         if (events.length === 0) {
4151             const labelDirs = currentWatchDirs.length > 0 ? curr
entWatchDirs.join(', ') : 'проекта';
4152             tbody.innerHTML = `|<td colspan="4" class="text-c
enter text-muted p-2">Ожидание изменений в папках <code>${labe
lDirs}</code>...</td></tr>`;
4153             return;
4154         }
4155         tbody.innerHTML = events.map((e, idx) => {
4156             const rootDirHint = e.watch_dir ? (e.watch_dir.split
('\').pop() || e.watch_dir) : '';
4157             const procDisplay = e.process_name
? `

|  |

```



```

4164         <span class="badge ${e.is_deletion ? 'bg-dange
r' : (e.action === 'Created' ? 'bg-success' : 'bg-secondary')}
">${e.action}</span>
4165         ${rootDirHint && currentWatchDirs.length > 1 ?
`<span class="badge bg-dark border border-secondary text-mute
d ms-1" style="font-size: 0.65rem;" title="Корень: ${escapeHtm
l(e.watch_dir)}">${rootDirHint}</span>` : ''}
4166     </td>
4167     <td>${procDisplay}</td>
4168     <td class="font-monospace small" style="word-bre
ak: break-all; color: var(--text-color);" title="${escapeHtml(
e.path)}">${escapeHtml(e.path)}</td>
4169 </tr>
4170 `;
4171 }).join('');
4172
4173 tbody.querySelectorAll('.sys-live-row').forEach(row =>
{
4174     row.onclick = () => {
4175         const idx = parseInt(row.getAttribute('data-idx'),
10);
4176         const e = events[idx];
4177         if (!e) return;
4178
4179         if (window.AITableModal) {
4180             window.AITableModal.show({
4181                 icon: '📁',
4182                 title: `Файловое событие: ${e.action}`,
4183                 subtitle: `${e.path} | ${e.timestamp}`,
4184                 tableType: 'file_event',
4185                 badges: [
4186                     { text: e.action, class: e.is_deletion ? 'ba
dge bg-danger' : (e.action === 'Created' ? 'badge bg-success'
: 'badge bg-info text-dark') },
4187                     { text: e.process_name ? `Программа: ${e.pro
cess_name}` : 'WinAPI ReadDirectoryChangesW', class: 'badge bg
-dark border border-secondary text-info' },
4188                     { text: 'WinAPI', class: 'badge bg-secondary

```

```

    ' }
4189         ],
4190         metadata: [
4191             { label: 'Действие', value: e.action },
4192             { label: 'Полный путь к файлу', value: e.path },
4193             { label: 'Программа / Процесс', value: e.process_name ? `${e.process_name}${e.process_id ? ` (PID: ${e.process_id})` : ''}` : 'Фоновый процесс / завершен' },
4194             { label: 'Время события', value: e.timestamp },
4195             { label: 'Признак удаления', value: e.is_deletion ? 'Да (Файл удален/переименован)' : 'Нет' },
4196             { label: 'Папка события', value: e.watch_dir || currentWatchDir },
4197             { label: 'Все отслеживаемые папки', value: currentWatchDirs.join('; ') }
4198         ],
4199         rawTitle: 'Детали события WinAPI & Process Info',
4200         rawContent: JSON.stringify(e, null, 2),
4201         requestData: e
4202     });
4203 }
4204 };
4205 });
4206 }
4207 } catch (e) {
4208     console.error('[SystemInspectorTab] Failed to fetch live file events:', e);
4209 }
4210 }
4211
4212 // =====
4213 // Multi-Directory Watcher Manager & Folder Explorer Modal Logic
4214 // =====

```

=====

```
4215
4216   function showWatchDirsAlert(msg, type = 'success') {
4217       const alertEl = document.getElementById('modal-watch-dirs-
alert');
4218       if (!alertEl) return;
4219       alertEl.className = `alert alert-${type} py-2 px-3 small m
b-3`;
4220       alertEl.innerHTML = msg;
4221       alertEl.classList.remove('d-none');
4222       if (type === 'success') {
4223           setTimeout(() => {
4224               alertEl.classList.add('d-none');
4225           }, 3500);
4226       }
4227   }
4228
4229   function renderModalActiveDirsList() {
4230       const container = document.getElementById('modal-watch-dir
s-active-list');
4231       const countBadge = document.getElementById('modal-watch-di
rs-count');
4232       const hintEl = document.getElementById('modal-watch-dirs-f
ooter-hint');
4233
4234       if (!container) return;
4235
4236       if (countBadge) countBadge.textContent = String(stagedWatc
hDirs.length);
4237       if (hintEl) hintEl.textContent = `Выбрано папок для монито
ринга: ${stagedWatchDirs.length}`;
4238
4239       if (stagedWatchDirs.length === 0) {
4240           container.innerHTML = `
4241               <div class="text-center text-muted small py-3">
4242                   <i class="bi bi-folder-x me-1"></i> Список пуст. Выб
ерите папки в проводнике ниже или воспользуйтесь быстрыми прес
етами.
```

```

4243         </div>
4244     `;
4245     return;
4246 }
4247
4248     container.innerHTML = stagedWatchDirs.map((dirPath, idx) =
> {
4249         const folderName = dirPath.split('\\').pop() || dirPath;
4250         const driveLetter = (dirPath.slice(0, 2)).toUpperCase();
4251         return `
4252             <div class="p-1.5 px-2 rounded d-flex align-items-cent
er justify-content-between gap-2" style="background: var(--bg-
color); border: 1px solid var(--border-color);">
4253                 <div class="d-flex align-items-center gap-2 text-tru
ncate" style="max-width: 82%;">
4254                     <i class="bi bi-folder-check text-warning fs-6"></
i>
4255                     <span class="badge bg-secondary font-monospace" st
yle="font-size: 0.68rem;">${driveLetter}</span>
4256                     <div class="text-truncate">
4257                         <span class="fw-bold small" style="color: var(--
text-color);">${escapeHtml(folderName)}</span>
4258                         <span class="small text-muted font-monospace d-b
lock text-truncate" style="font-size: 0.72rem;" title="${escap
eHtml(dirPath)}">${escapeHtml(dirPath)}</span>
4259                     </div>
4260                 </div>
4261                 <button class="btn btn-xs btn-outline-danger rounded
-pill px-2 py-0.5" onclick="window._removeStagedWatchDir(${idx
})" title="Удалить из списка мониторинга">
4262                     <i class="bi bi-trash3"></i>
4263                 </button>
4264             </div>
4265         `;
4266     }).join('');
4267 }
4268
4269 window._removeStagedWatchDir = function(index) {

```

```
4270     if (index >= 0 && index < stagedWatchDirs.length) {
4271         const removed = stagedWatchDirs.splice(index, 1)[0];
4272         renderModalActiveDirsList();
4273         showWatchDirsAlert(`Папка удалена из списка: <code>${escapeHtml(removed)}</code>`, 'info');
4274     }
4275 };
4276
4277 function addDirToStaged(pathToAdd) {
4278     if (!pathToAdd || !pathToAdd.trim()) return;
4279     const cleanPath = pathToAdd.trim();
4280     if (stagedWatchDirs.includes(cleanPath)) {
4281         showWatchDirsAlert(`Папка уже есть в списке: <code>${escapeHtml(cleanPath)}</code>`, 'warning');
4282         return;
4283     }
4284     stagedWatchDirs.push(cleanPath);
4285     renderModalActiveDirsList();
4286     showWatchDirsAlert(`Папка добавлена: <code>${escapeHtml(cleanPath)}</code>`, 'success');
4287 }
4288
4289 async function loadSystemDrives() {
4290     const drivesBar = document.getElementById('modal-watch-drives-bar');
4291     if (!drivesBar) return;
4292
4293     try {
4294         const res = await fetch('/api/sysadmin/filesystem/drives');
4295         if (res.ok) {
4296             const data = await res.json();
4297             cachedDrives = data.drives || [];
4298         }
4299     } catch (e) {
4300         console.warn('[SystemInspectorTab] Failed to fetch system drives:', e);
4301         cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];

```

```
4302     }
4303
4304     if (cachedDrives.length === 0) {
4305         cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
4306     }
4307
4308     drivesBar.innerHTML = cachedDrives.map(d => {
4309         const label = d.mountpoint || 'C:\\';
4310         const freeTxt = d.free_gb ? `${d.free_gb} GB free` : '';
4311         return `
4312             <button class="btn btn-xs btn-outline-secondary rounded-pill px-2 py-0.5 sys-drive-btn font-monospace" data-drive="${
4313                 {escapeHtml(label)}" title="${{escapeHtml(label)} ${freeTxt}">
4314                 <i class="bi bi-hdd me-1"></i>${escapeHtml(label)}
4315             </button>
4316         `;
4317     }).join('');
4318
4319     drivesBar.querySelectorAll('.sys-drive-btn').forEach(btn => {
4320         btn.onclick = () => {
4321             const drv = btn.getAttribute('data-drive');
4322             if (drv) loadFolderBrowser(drv);
4323         };
4324     });
4325
4326     function renderBreadcrumbs(currentPath) {
4327         const bcContainer = document.getElementById('modal-folder-breadcrumbs');
4328         if (!bcContainer) return;
4329
4330         const parts = currentPath.replace(/\\+$/, '').split('\\');
4331         let builtPath = '';
4332         const crumbsHtml = parts.map((part, idx) => {
4333             if (idx === 0) {
4334                 builtPath = part + '\\';
4335             } else {
```

```

4336         builtPath = builtPath + (builtPath.endsWith('\\') ? ''
      : '\\') + part;
4337     }
4338     const clickPath = builtPath;
4339     const isLast = idx === parts.length - 1;
4340     return `
4341         <span class="d-inline-flex align-items-center">
4342             ${idx > 0 ? '<span class="text-muted mx-1">/</span>'
      : ''}
4343             <a href="#" class="sys-breadcrumb-link text-decoration-
      none ${isLast ? 'fw-bold text-info' : ''}" style="color: va
      r(--text-color);" data-path="${escapeHtml(clickPath)}">
4344                 ${escapeHtml(part || 'Корень')}
4345             </a>
4346         </span>
4347     `;
4348     })).join('');
4349
4350     bcContainer.innerHTML = crumbsHtml;
4351
4352     bcContainer.querySelectorAll('.sys-breadcrumb-link').forEa
      ch(link => {
4353         link.onclick = (e) => {
4354             e.preventDefault();
4355             const p = link.getAttribute('data-path');
4356             if (p) loadFolderBrowser(p);
4357         };
4358     });
4359 }
4360
4361 async function loadFolderBrowser(targetPath) {
4362     const inputPath = document.getElementById('modal-folder-pa
      th-input');
4363     const browserList = document.getElementById('modal-folder-
      browser-list');
4364     if (!targetPath) targetPath = 'C:\\';
4365     currentBrowserPath = targetPath;
4366

```

```

4367     if (inputPath) inputPath.value = targetPath;
4368     renderBreadcrumbs(targetPath);
4369
4370     if (!browserList) return;
4371     browserList.innerHTML = `
```



```

4393 |         <i class="bi ${d.has_subdirs ? 'bi-folder2 text-
| warning' : 'bi-folder text-warning'}"></i>
4394 |         <span class="small text-light text-truncate">${e
| scapeHtml(d.name)}</span>
4395 |         <span class="small text-muted font-monospace ms-
| auto me-2" style="font-size: 0.68rem;">${d.modified || ''}</sp
| an>
4396 |     </div>
4397 |     <button class="btn btn-xs ${isAlreadySelected ? 'b
| tn-success' : 'btn-outline-info'} rounded-pill px-2 py-0.5 sys
| -add-folder-btn" data-path="${escapeHtml(d.path)}" title="Доба
| вить в отслеживаемые">
4398 |         <i class="bi ${isAlreadySelected ? 'bi-check2' :
| 'bi-plus-lg'} me-1"></i>${isAlreadySelected ? 'Выбрана' : 'Сл
| едить'}
4399 |     </button>
4400 | </div>
4401 | `;
4402 | }).join('');
4403 |
4404 | // Bind folder navigation
4405 | browserList.querySelectorAll('.sys-nav-to-folder').forEa
| ch(el => {
4406 |     el.onclick = () => {
4407 |         const p = el.getAttribute('data-path');
4408 |         if (p) loadFolderBrowser(p);
4409 |     };
4410 | });
4411 |
4412 | // Bind Add buttons
4413 | browserList.querySelectorAll('.sys-add-folder-btn').forE
| ach(btn => {
4414 |     btn.onclick = (e) => {
4415 |         e.stopPropagation();
4416 |         const p = btn.getAttribute('data-path');
4417 |         if (p) {
4418 |             addDirToStaged(p);
4419 |             btn.className = 'btn btn-xs btn-success rounded-pi

```

```

    ll px-2 py-0.5 sys-add-folder-btn';
4420         btn.innerHTML = '<i class="bi bi-check2 me-1"></i>
        Выбрана';
4421     }
4422 };
4423 });
4424
4425 } catch (e) {
4426     console.warn('[SystemInspectorTab] Failed to browse fold
        er:', e);
4427     browserList.innerHTML = `<div class="text-danger small p
        y-2 px-2"><i class="bi bi-exclamation-triangle me-1"></i> Ошиб
        ка доступа: ${escapeHtml(e.message)}</div>`;
4428 }
4429 }
4430
4431 let currentExclusions = {
4432     enabled: true,
4433     paths: [],
4434     extensions: [],
4435     patterns: [],
4436     processes: [],
4437     filtered_count: 0
4438 };
4439
4440 function showExclusionsAlert(msg, type = 'success') {
4441     const alertEl = document.getElementById('modal-exclusions-
        alert');
4442     if (!alertEl) return;
4443     alertEl.className = `alert alert-${type} py-2 px-3 small m
        b-3`;
4444     alertEl.innerHTML = msg;
4445     alertEl.classList.remove('d-none');
4446     if (type === 'success' || type === 'info') {
4447         setTimeout(() => {
4448             alertEl.classList.add('d-none');
4449         }, 3500);
4450     }

```

```
4451 | }
4452 |
4453 | function renderExclusionsLists() {
4454 |     const totalCount = (currentExclusions.paths?.length || 0)
4455 | +
4456 |     (currentExclusions.extensions?.length || 0) +
4457 |     (currentExclusions.patterns?.length || 0) +
4458 |     (currentExclusions.processes?.length || 0);
4459 |     const tabBadge = document.getElementById('modal-watch-excl
4460 | usions-tab-count');
4461 |     const headerBadge = document.getElementById('sys-exclusion
4462 | s-count-badge');
4463 |     if (tabBadge) tabBadge.textContent = String(totalCount);
4464 |     if (headerBadge) headerBadge.textContent = String(totalCou
4465 | nt);
4466 |
4467 |     const switchEl = document.getElementById('switch-exclusion
4468 | s-active');
4469 |     const statusBadge = document.getElementById('badge-exclusi
4470 | ons-status');
4471 |     if (switchEl) switchEl.checked = Boolean(currentExclusions
4472 | .enabled);
4473 |     if (statusBadge) {
4474 |         if (currentExclusions.enabled) {
4475 |             statusBadge.className = 'badge bg-success';
4476 |             statusBadge.textContent = 'ВКЛ';
4477 |         } else {
4478 |             statusBadge.className = 'badge bg-danger-subtle text-d
4479 | anger border border-danger fw-bold';
4480 |             statusBadge.textContent = 'ВыКЛ';
4481 |         }
4482 |     }
4483 |
4484 |     const filteredTotalEl = document.getElementById('modal-exc
4485 | lusions-filtered-total');
4486 |     if (filteredTotalEl) filteredTotalEl.textContent = String(
4487 | currentExclusions.filtered_count || 0);
```

```
4479 |
4480 |     // 1. Paths list
4481 |     const pathsContainer = document.getElementById('list-ex-pa
|ths');
4482 |     const pathsBadge = document.getElementById('badge-ex-paths
|-count');
4483 |     if (pathsBadge) pathsBadge.textContent = String(currentExc
|lusions.paths?.length || 0);
4484 |     if (pathsContainer) {
4485 |         if (!currentExclusions.paths || currentExclusions.paths.
|length === 0) {
4486 |             pathsContainer.innerHTML = `
```

```
4501     if (extsContainer) {
4502         if (!currentExclusions.extensions || currentExclusions.e
xtensions.length === 0) {
4503             extsContainer.innerHTML = `
```

```

ace" style="font-size: 0.75rem;">
4524     ${escapeHtml(pat)}
4525     <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._removeExclus
ionItem('patterns', '${escapeHtml(pat)}')" title="Удалить"></b
utton>
4526     </span>
4527     `).join('') + `</div>`;
4528     }
4529     }
4530
4531     // 4. Processes list
4532     const procsContainer = document.getElementById('list-ex-pr
ocesses');
4533     const procsBadge = document.getElementById('badge-ex-proce
sses-count');
4534     if (procsBadge) procsBadge.textContent = String(currentExc
lusions.processes?.length || 0);
4535     if (procsContainer) {
4536         if (!currentExclusions.processes || currentExclusions.pr
ocesses.length === 0) {
4537             procsContainer.innerHTML = `<div class="text-muted sma
ll text-center py-2">Нет исключенных программ</div>`;
4538         } else {
4539             procsContainer.innerHTML = `<div class="d-flex flex-wr
ap gap-1">` + currentExclusions.processes.map(proc => `
4540             <span class="badge bg-dark border border-secondary t
ext-danger d-inline-flex align-items-center gap-1 font-monospa
ce" style="font-size: 0.75rem;">
4541                 <i class="bi bi-cpu me-0.5"></i>${escapeHtml(proc)
}
4542                 <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._removeExclus
ionItem('processes', '${escapeHtml(proc)}')" title="Удалить"><
/button>
4543             </span>
4544             `).join('') + `</div>`;
4545         }

```

```
4546     }
4547 }
4548
4549 async function fetchExclusionsData() {
4550     try {
4551         const res = await fetch('/api/v1/system/file-audit/exclu
sions');
4552         if (res.ok) {
4553             currentExclusions = await res.json();
4554             renderExclusionsLists();
4555         }
4556     } catch (e) {
4557         console.warn('[SystemInspectorTab] Failed to fetch exclu
sions:', e);
4558     }
4559 }
4560
4561 async function addExclusionItem(category, value) {
4562     if (!value || !value.trim()) return;
4563     const cleanVal = value.trim();
4564     try {
4565         const res = await fetch('/api/v1/system/file-audit/exclu
sions/add', {
4566             method: 'POST',
4567             headers: { 'Content-Type': 'application/json' },
4568             body: JSON.stringify({ category, value: cleanVal })
4569         });
4570         const data = await res.json();
4571         if (res.ok && data.success) {
4572             currentExclusions = data.exclusions || currentExclusio
ns;
4573             renderExclusionsLists();
4574             showExclusionsAlert(`Правило добавлено: <code>${escape
Html(cleanVal)}</code>`, 'success');
4575             const inputVal = document.getElementById('input-exclus
ion-value');
4576             if (inputVal) inputVal.value = '';
4577         } else {
```

```
4578         showExclusionsAlert(data.message || 'Правило уже суцес
| твует или невалидно', 'warning');
4579     }
4580     } catch (e) {
4581         showExclusionsAlert(`Ошибка добавления: ${e.message}`, '
| danger');
4582     }
4583 }
4584
4585 window._removeExclusionItem = async function(category, value
| ) {
4586     if (!value) return;
4587     try {
4588         const res = await fetch('/api/v1/system/file-audit/exclu
| sions/remove', {
4589             method: 'POST',
4590             headers: { 'Content-Type': 'application/json' },
4591             body: JSON.stringify({ category, value })
4592         });
4593         const data = await res.json();
4594         if (res.ok && data.success) {
4595             currentExclusions = data.exclusions || currentExclusio
| ns;
4596             renderExclusionsLists();
4597             showExclusionsAlert(`Правило удалено: <code>${escapeHt
| ml(value)}</code>`, 'info');
4598         } else {
4599             showExclusionsAlert(data.message || 'Не удалось удалит
| ь правило', 'warning');
4600         }
4601     } catch (e) {
4602         showExclusionsAlert(`Ошибка удаления: ${e.message}`, 'da
| nger');
4603     }
4604 };
4605
4606 async function toggleExclusionsActive(enabled) {
4607     try {
```



```
4608     const res = await fetch('/api/v1/system/file-audit/exclu
sions/toggle', {
4609         method: 'POST',
4610         headers: { 'Content-Type': 'application/json' },
4611         body: JSON.stringify({ enabled })
4612     });
4613     const data = await res.json();
4614     if (res.ok && data.success) {
4615         currentExclusions = data.exclusions || currentExclusio
ns;
4616         renderExclusionsLists();
4617         showExclusionsAlert(`Фильтрация исключений ${data.enab
led ? 'включена' : 'отключена'}`, 'info');
4618     }
4619 } catch (e) {
4620     showExclusionsAlert(`Ошибка переключения: ${e.message}`,
'danger');
4621 }
4622 }
4623
4624 async function applyExclusionPreset(preset) {
4625     if (preset === 'logs') {
4626         await addExclusionItem('paths', 'AppData\\Roaming\\AI-Br
eadboard\\apps\\windows\\telemetry\\logs');
4627         await addExclusionItem('extensions', '.log');
4628         await addExclusionItem('extensions', '.csv');
4629         await addExclusionItem('patterns', '*system_inspector_po
lls.csv*');
4630     } else if (preset === 'search') {
4631         await addExclusionItem('processes', 'SearchIndexer.exe')
;
4632         await addExclusionItem('patterns', '*Windows.db*');
4633         await addExclusionItem('patterns', '*Windows.db-wal*');
4634         await addExclusionItem('patterns', '*Windows.db-shm*');
4635     } else if (preset === 'temp') {
4636         await addExclusionItem('paths', 'AppData\\Local\\Temp');
4637         await addExclusionItem('extensions', '.tmp');
4638         await addExclusionItem('patterns', '~$*');
```

```
4639     } else if (preset === 'dev') {
4640         await addExclusionItem('paths', '.git');
4641         await addExclusionItem('paths', 'node_modules');
4642         await addExclusionItem('paths', '__pycache__');
4643         await addExclusionItem('paths', '.pytest_cache');
4644     }
4645 }
4646
4647 async function openWatchFoldersModal(initialTab = 'folders')
4648 {
4649     const modalEl = document.getElementById('modal-sys-watch-f
4650     olders');
4651     if (!modalEl) {
4652         console.warn('[SystemInspectorTab] modal-sys-watch-folde
4653         rs element not found');
4654         return;
4655     }
4656     try {
4657         const res = await fetch('/api/v1/system/file-audit/watch
4658         -dirs');
4659         if (res.ok) {
4660             const data = await res.json();
4661             stagedWatchDirs = data.watch_dirs && data.watch_dirs.l
4662             ength ? [...data.watch_dirs] : [currentWatchDir || 'C:\\'];
4663         } else {
4664             stagedWatchDirs = currentWatchDirs.length ? [...curren
4665             tWatchDirs] : [currentWatchDir || 'C:\\'];
4666         }
4667     } catch {
4668         stagedWatchDirs = currentWatchDirs.length ? [...currentW
4669         atchDirs] : [currentWatchDir || 'C:\\'];
4670     }
4671     renderModalActiveDirsList();
4672     const tabFoldersBadge = document.getElementById('modal-wat
4673     ch-dirs-tab-count');
4674     if (tabFoldersBadge) tabFoldersBadge.textContent = String(
```

```

    stagedWatchDirs.length);
4669
4670     await loadSystemDrives();
4671     await fetchExclusionsData();
4672
4673     // Tab activation
4674     if (initialTab === 'exclusions') {
4675         const tabExBtn = document.getElementById('tab-btn-watch-
exclusions');
4676         if (tabExBtn && window.bootstrap && window.bootstrap.Tab
) {
4677             const tab = window.bootstrap.Tab.getOrCreateInstance(t
abExBtn);
4678             tab.show();
4679         } else if (tabExBtn) {
4680             tabExBtn.click();
4681         }
4682     } else {
4683         const tabFoldersBtn = document.getElementById('tab-btn-w
atch-folders');
4684         if (tabFoldersBtn && window.bootstrap && window.bootstra
p.Tab) {
4685             const tab = window.bootstrap.Tab.getOrCreateInstance(t
abFoldersBtn);
4686             tab.show();
4687         } else if (tabFoldersBtn) {
4688             tabFoldersBtn.click();
4689         }
4690     }
4691
4692     const startPath = stagedWatchDirs[0] || currentBrowserPath
|| 'C:\\';
4693     loadFolderBrowser(startPath);
4694
4695     // Bind navigation buttons
4696     const btnUp = document.getElementById('btn-modal-folder-up
');
4697     if (btnUp) {

```

```
4698 |         btnUp.onclick = () => {
4699 |             const parts = currentBrowserPath.replace(/\\+$/, '').s
|             plit('\\\\');
4700 |             if (parts.length > 1) {
4701 |                 parts.pop();
4702 |                 let parentPath = parts.join('\\\\');
4703 |                 if (parts.length === 1 && !parentPath.endsWith('\\\\'))
|             ) parentPath += '\\\\';
4704 |                 loadFolderBrowser(parentPath);
4705 |             }
4706 |         };
4707 |     }
4708 |
4709 |     const btnGo = document.getElementById('btn-modal-folder-go
|     ');
4710 |     const inputPath = document.getElementById('modal-folder-pa
|     th-input');
4711 |     if (btnGo && inputPath) {
4712 |         btnGo.onclick = () => {
4713 |             if (inputPath.value && inputPath.value.trim()) {
4714 |                 loadFolderBrowser(inputPath.value.trim());
4715 |             }
4716 |         };
4717 |         inputPath.onkeydown = (e) => {
4718 |             if (e.key === 'Enter') {
4719 |                 e.preventDefault();
4720 |                 btnGo.click();
4721 |             }
4722 |         };
4723 |     }
4724 |
4725 |     const btnAddCurrent = document.getElementById('btn-modal-f
|     older-add-current');
4726 |     if (btnAddCurrent) {
4727 |         btnAddCurrent.onclick = () => {
4728 |             const val = inputPath ? inputPath.value.trim() : curre
|             ntBrowserPath;
4729 |             if (val) addDirToStaged(val);
```

```

4730     };
4731 }
4732
4733 // Presets buttons (folders)
4734 document.querySelectorAll('.sys-preset-btn').forEach(btn =
4735 > {
4736     btn.onclick = () => {
4737         const preset = btn.getAttribute('data-preset');
4738         let target = '';
4739         if (preset === 'project') target = 'C:\\Users\\onela\\
AppData\\Local\\AI-Breadboard';
4740         else if (preset === 'downloads') target = 'C:\\Users\\
onela\\Downloads';
4741         else if (preset === 'desktop') target = 'C:\\Users\\on
ela\\Desktop';
4742         else if (preset === 'temp') target = 'C:\\Users\\onela
\\AppData\\Local\\Temp';
4743
4744         if (target) {
4745             addDirToStaged(target);
4746             loadFolderBrowser(target);
4747         }
4748     };
4749 });
4750
4751 // Exclusions switch and add buttons binding
4752 const switchExActive = document.getElementById('switch-exc
lusions-active');
4753 if (switchExActive) {
4754     switchExActive.onchange = (e) => {
4755         toggleExclusionsActive(e.target.checked);
4756     };
4757 }
4758
4759 const btnAddEx = document.getElementById('btn-add-exclusio
n');
4760 const selectExCat = document.getElementById('select-exclus
ion-category');

```

```

4760     const inputExVal = document.getElementById('input-exclusio
n-value');
4761     if (btnAddEx && selectExCat && inputExVal) {
4762         btnAddEx.onclick = () => {
4763             addExclusionItem(selectExCat.value, inputExVal.value);
4764         };
4765         inputExVal.onkeydown = (e) => {
4766             if (e.key === 'Enter') {
4767                 e.preventDefault();
4768                 btnAddEx.click();
4769             }
4770         };
4771     }
4772
4773     // Exclusions preset buttons
4774     document.querySelectorAll('.sys-ex-preset-btn').forEach(bt
n => {
4775         btn.onclick = () => {
4776             const preset = btn.getAttribute('data-preset');
4777             if (preset) applyExclusionPreset(preset);
4778         };
4779     });
4780
4781     // Apply button
4782     const btnApply = document.getElementById('btn-modal-watch-
dirs-apply');
4783     if (btnApply) {
4784         btnApply.onclick = async () => {
4785             if (stagedWatchDirs.length === 0) {
4786                 showWatchDirsAlert('Выберите хотя бы одну папку для
мониторинга', 'warning');
4787                 return;
4788             }
4789
4790             btnApply.disabled = true;
4791             btnApply.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span> Применение...';
4792

```

```
4793 |         try {
4794 |             const res = await fetch('/api/v1/system/file-audit/watch-dirs', {
4795 |                 method: 'POST',
4796 |                 headers: { 'Content-Type': 'application/json' },
4797 |                 body: JSON.stringify({ paths: stagedWatchDirs })
4798 |             });
4799 |             const resData = await res.json();
4800 |             if (res.ok && resData.success) {
4801 |                 if (window.bootstrap && window.bootstrap.Modal) {
4802 |                     const modal = window.bootstrap.Modal.getInstance
4803 | (modalEl);
4804 |                     if (modal) modal.hide();
4805 |                     } else {
4806 |                         modalEl.classList.remove('show');
4807 |                         modalEl.style.display = 'none';
4808 |                     }
4809 |                     await fetchLiveFileEvents();
4810 |                 } else {
4811 |                     showWatchDirsAlert(`Ошибка: ${resData.detail} || 'Не
4812 | удалось применить список папок'`, 'danger');
4813 |                 }
4814 |             } catch (err) {
4815 |                 showWatchDirsAlert(`Сетевая ошибка: ${err.message}`,
4816 | 'danger');
4817 |             } finally {
4818 |                 btnApply.disabled = false;
4819 |                 btnApply.innerHTML = '<i class="bi bi-check2-circle
4820 | me-1"></i> Применить и запустить';
4821 |             }
4822 |         };
4823 |     }
4824 | }

if (window.bootstrap && window.bootstrap.Modal) {
    const modal = window.bootstrap.Modal.getOrCreateInstance
(modalEl);
    modal.show();
} else {
```

```
4825     modalEl.classList.add('show');
4826     modalEl.style.display = 'block';
4827 }
4828 }
4829
4830 function showLiveWatcherHelpModal() {
4831     const dirsStr = currentWatchDirs.length ? currentWatchDirs
4832     .join('\n- ') : (currentWatchDir || 'Рабочая папка');
4833     if (window.AITableModal) {
4834         window.AITableModal.show({
4835             icon: 'i',
4836             title: 'Справка:Изменения файлов в реальном времени (M
4837             ulti-Directory)',
4838             subtitle: 'Низкоуровневый мониторинг файловой системы
4839             Windows через WinAPI ReadDirectoryChangesW',
4840             tableType: 'help',
4841             badges: [
4842                 { text: 'WinAPI ReadDirectoryChangesW', class: 'badg
4843                 e bg-info text-dark' },
4844                 { text: 'Multi-Directory', class: 'badge bg-primary'
4845                 },
4846                 { text: 'Real-Time Streaming', class: 'badge bg-succ
4847                 ess' },
4848                 { text: 'Рекурсивно (bWatchSubtree = True)', class:
4849                 'badge bg-warning text-dark' }
4850             ],
4851             metadata: [
4852                 { label: 'Технология', value: 'WinAPI ReadDirectoryC
4853                 hangesW (нативный вызов ядра Windows kernel32.dll)' },
4854                 { label: 'Режим работы', value: 'Множественные поток
4855                 и мониторинга с агрегацией в единый кольцевой буфер' },
4856                 { label: 'Отслеживаемые папки', value: currentWatchD
4857                 irs.join('; ') || 'Рабочая папка проекта' },
4858                 { label: 'Хранение настроек', value: 'apps/windows_s
4859                 yadmin/config.json (ключ watch_directories)' }
4860             ],
4861             rawTitle: 'Подробное руководство по панели мониторинга
4862             ',
```



```
4851 |         rawContent: `# ПанельИзменения файлов в реальном време
      | ни
4852 |
4853 | ### 1. Что это такое?
4854 | Компонент для мгновенного перехвата операций файловой системы
      | в режиме реального времени на базе WinAPI ReadDirectoryChanges
      | W с поддержкой одновременного наблюдения за произвольным колич
      | еством папок.
4855 |
4856 | ### 2. Типы отслеживаемых действий:
4857 | - Created: Создание нового файла или папки (FILE_ACTION_ADDED)
      | .
4858 | - Modified: Модификация содержимого, атрибутов или размера фай
      | ла (FILE_ACTION_MODIFIED).
4859 | - Deleted: Удаление файла или папки с диска (FILE_ACTION_REMOV
      | ED).
4860 | - Renamed: Переименование объекта (старое и новое имя).
4861 |
4862 | ### 3. Как выбрать или добавить папки?
4863 | 1. Нажмите кнопку «Папка» в заголовке панели.
4864 | 2. Откроется модальное окно с проводником файловой системы.
4865 | 3. Выберите логический диск (C:, D:, E:), перейдите в нужную п
      | апку или вставьте путь вручную.
4866 | 4. Нажмите «+ Добавить папку» или кнопку «Следить» напротив лю
      | бой подпапки.
4867 | 5. Нажмите «Применить и запустить». Список сохранится в config
      | .json и мониторинг стартует автоматически!`,
4868 |         requestData: {
4869 |             current_watch_dirs: currentWatchDirs,
4870 |             engine: 'ReadDirectoryChangesW (Multi-Directory)'
4871 |         }
4872 |     });
4873 | } else {
4874 |     alert('Мониторинг файловой системыИзменения файлов в реа
      | льном времени (Multi-Directory).\nОтслеживаемые папки:\n- ' +
      | dirsStr);
4875 | }
4876 | }
```

```
4877 |
4878 | function setupSysSensorInterval(seconds) {
4879 |     _currentUiRefreshSeconds = seconds;
4880 |     const pollHandler = async () => {
4881 |         await fetchCpuLoadFromApi();
4882 |         await fetchGpuLoadFromApi();
4883 |         await fetchMemoryIoFromApi();
4884 |         await fetchNetworkLoadFromApi();
4885 |         await fetchStorageLoadFromApi();
4886 |     };
4887 |     if (window.registerTabPoller) {
4888 |         window.registerTabPoller('tab-hardware-load-inspector',
pollHandler, seconds * 1000, { pollerId: 'hw_load_inspector',
immediate: false });
4889 |     } else {
4890 |         if (window._sysSensorInterval) {
4891 |             clearInterval(window._sysSensorInterval);
4892 |             window._sysSensorInterval = null;
4893 |         }
4894 |         window._sysSensorInterval = setInterval(() => {
4895 |             if (window.isTabActive ? (window.isTabActive('tab-hard
ware-load-inspector') || window.isTabActive('tab-system-load-i
nspector') || window.isTabActive('tab-system-inspector')) : tr
ue) {
4896 |                 pollHandler();
4897 |             }
4898 |             }, seconds * 1000);
4899 |     }
4900 |     console.log(`[SystemInspectorTab] UI sensor refresh interv
al set to ${seconds}s`);
4901 | }
4902 |
4903 | function openSysIntervalsModal() {
4904 |     const modalEl = document.getElementById('sysIntervalsModal
');
4905 |     if (!modalEl) {
4906 |         console.warn('[SystemInspectorTab] sysIntervalsModal ele
ment not found');
```

```
4907     return;
4908 }
4909
4910 // Set UI select to current seconds
4911 const selUi = document.getElementById('modal-ui-refresh-se
lect');
4912 if (selUi) {
4913     selUi.value = String(_currentUiRefreshSeconds);
4914 }
4915 const uiBadge = document.getElementById('modal-ui-refresh-
badge');
4916 if (uiBadge) {
4917     uiBadge.textContent = `${_currentUiRefreshSeconds} cek`;
4918 }
4919
4920 loadSysIntervalsConfig(false);
4921
4922 if (window.bootstrap && window.bootstrap.Modal) {
4923     const modal = window.bootstrap.Modal.getOrCreateInstance
(modalEl);
4924     modal.show();
4925 } else {
4926     modalEl.classList.add('show');
4927     modalEl.style.display = 'block';
4928 }
4929 }
4930
4931 function escapeHtml(str) {
4932     if (!str) return '';
4933     return String(str)
4934         .replace(/&/g, '&amp;')
4935         .replace(/</g, '&lt;')
4936         .replace(/>/g, '&gt;')
4937         .replace(/"/g, '&quot;')
4938         .replace(/'/g, '&#039;');
4939 }
4940
4941 async function initSystemInspectorTab() {
```

```

4942     console.log('[SystemInspectorTab] Initializing...');
4943     bindTabEvents();
4944     if (window.isTabActive ? (window.isTabActive('tab-hardware
-load-inspector') || window.isTabActive('tab-system-load-inspe
ctor') || window.isTabActive('tab-system-inspector')) : false)
    {
4945         await fetchCpuLoadFromApi();
4946         await fetchGpuLoadFromApi();
4947         await fetchMemoryIoFromApi();
4948         await fetchNetworkLoadFromApi();
4949         await fetchStorageLoadFromApi();
4950
4951         try {
4952             const snapRes = await fetch('/api/v1/system/summary');
4953             if (snapRes.ok) {
4954                 latestTelemetrySnapshot = await snapRes.json();
4955                 updateTelemetryDashboard(latestTelemetrySnapshot);
4956             }
4957         } catch (e) {
4958             console.warn('[SystemInspectorTab] Initial snapshot fe
tch failed:', e);
4959         }
4960
4961         if (typeof connectSystemWebSocket === 'function') {
4962             try { connectSystemWebSocket(); } catch (err) { consol
e.debug(err); }
4963         }
4964     }
4965
4966     // Periodic sensor refresh
4967     setupSysSensorInterval(_currentUiRefreshSeconds || 5);
4968 }
4969
4970 function activateSystemInspectorTab() {
4971     if (window.isTabActive && !window.isTabActive('tab-hardwar
e-load-inspector') && !window.isTabActive('tab-system-load-ins
pector') && !window.isTabActive('tab-system-inspector')) retur
n;

```

```
4972     console.log('[SystemInspectorTab] Tab activated, refreshin
|g metrics...');
4973     fetchCpuLoadFromApi();
4974     fetchGpuLoadFromApi();
4975     fetchMemoryIoFromApi();
4976     fetchNetworkLoadFromApi();
4977     fetchStorageLoadFromApi();
4978     if (typeof connectSystemWebSocket === 'function') {
4979         try { connectSystemWebSocket(); } catch (err) { console.
|debug(err); }
4980     }
4981 }
4982
4983 function deactivateSystemInspectorTab() {
4984     console.log('[SystemInspectorTab] Tab deactivated, pausing
|telemetry stream...');
4985     if (typeof disconnectSystemWebSocket === 'function') {
4986         try { disconnectSystemWebSocket(); } catch (err) { conso
|le.debug(err); }
4987     }
4988 }
4989
4990 window.initSystemInspectorTab = initSystemInspectorTab;
4991 window.initSystemLoadInspectorTab = initSystemInspectorTab;
4992 window.initHardwareLoadInspectorTab = initSystemInspectorTab
|;
4993 window.activateSystemInspectorTab = activateSystemInspectorT
|ab;
4994 window.activateSystemLoadInspectorTab = activateSystemInspec
|torTab;
4995 window.activateHardwareLoadInspectorTab = activateSystemInsp
|ectorTab;
4996 window.deactivateSystemInspectorTab = deactivateSystemInspec
|torTab;
4997 window.deactivateSystemLoadInspectorTab = deactivateSystemIn
|spectatorTab;
4998 window.deactivateHardwareLoadInspectorTab = deactivateSystem
|InspectorTab;
```

```
4999 | window.openSysIntervalsModal = openSysIntervalsModal;
5000 |
5001 | // Auto-init if tab is already in DOM and active or open
5002 | if (document.getElementById('tab-hardware-load-inspector'))
    | {
5003 |     initSystemInspectorTab();
5004 | }
5005 |})();
5006 |
5007 |
```

index.html

```
1 <!--
2 =====
3 |
4 |=====
5 | Process Name: Windows Web Interface - Index View
6 |=====
7 |
8 | Description:
9 |   Шаблон клиентского веб-интерфейса index.
10 |
11 | Usage Examples:
12 |   Web Route:
13 |     /windows/api/webgui/system_logs_tab/index.html
14 |
15 |   Tab Frame Integration:
16 |     <iframe src="/windows/api/webgui/system_logs_tab/index.h
17 | tml" class="w-100 h-100 border-0"></iframe>
18 |
19 | File: index.html
20 | Project: ai-breadboard
21 | Package: windows/api/webgui/system_logs_tab
22 | Author: hypo69
23 | Copyright: © 2026 hypo69
24 | Updated: 2026-10-01 13:04:40
25 |=====
26 |=====
27 | -->
28 |
29 | <style>
30 |   .slc-container {
31 |     padding: 0.25rem 0.5rem;
32 |   }
33 |   .slc-compact-header {
34 |     background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
35 | 0%);
36 |     border: 1px solid #334155;
37 |     border-radius: 8px;
```

```
32 |     padding: 0.4rem 0.75rem;
33 |     margin-bottom: 0.5rem;
34 | }
35 | .slc-metric-pill {
36 |     background: rgba(15, 23, 42, 0.8);
37 |     border: 1px solid #334155;
38 |     border-radius: 6px;
39 |     padding: 0.2rem 0.5rem;
40 |     font-size: 0.75rem;
41 |     display: inline-flex;
42 |     align-items: center;
43 |     gap: 0.35rem;
44 | }
45 | .slc-card {
46 |     background: #0f172a !important;
47 |     border: 1px solid #334155 !important;
48 |     border-radius: 8px;
49 |     padding: 0.5rem 0.75rem;
50 |     display: flex;
51 |     flex-direction: column;
52 |     gap: 0.1rem;
53 | }
54 | .slc-label {
55 |     font-size: 0.68rem;
56 |     text-transform: uppercase;
57 |     letter-spacing: 0.04em;
58 |     color: #94a3b8;
59 | }
60 | .slc-value {
61 |     font-size: 1.05rem;
62 |     font-weight: 700;
63 |     color: #f8fafc;
64 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
65 |     Consolas, monospace;
66 | }
67 | .slc-panel {
68 |     background: #0f172a !important;
69 |     border: 1px solid #334155 !important;
```



```
69 |     border-radius: 8px;
70 |     overflow: hidden;
71 | }
72 | .slc-panel-header {
73 |     background: #1e293b !important;
74 |     border-bottom: 1px solid #334155 !important;
75 |     padding: 0.6rem 1rem;
76 |     font-weight: 600;
77 |     font-size: 0.85rem;
78 |     color: #38bdf8;
79 |     display: flex;
80 |     align-items: center;
81 |     justify-content: space-between;
82 | }
83 | .slc-tree-list {
84 |     max-height: 580px;
85 |     overflow-y: auto;
86 |     font-size: 0.8rem;
87 | }
88 | .slc-tree-item {
89 |     padding: 0.4rem 0.75rem;
90 |     cursor: pointer;
91 |     border-radius: 6px;
92 |     margin-bottom: 2px;
93 |     color: #cbd5e1;
94 |     display: flex;
95 |     justify-content: space-between;
96 |     align-items: center;
97 |     transition: background 0.15s ease;
98 | }
99 | .slc-tree-item:hover {
100 |     background: rgba(56, 189, 248, 0.1);
101 |     color: #38bdf8;
102 | }
103 | .slc-tree-item.active {
104 |     background: #0284c7;
105 |     color: #fff;
106 |     font-weight: 600;
```

```
107 | }
108 | .slc-table {
109 |     width: 100%;
110 |     margin-bottom: 0;
111 |     color: #e2e8f0;
112 |     font-size: 0.8rem;
113 | }
114 | .slc-table th {
115 |     background: #1e293b;
116 |     color: #94a3b8;
117 |     border-bottom: 1px solid #334155;
118 |     font-weight: 600;
119 |     text-transform: uppercase;
120 |     font-size: 0.7rem;
121 |     letter-spacing: 0.05em;
122 |     padding: 0.5rem 0.75rem;
123 | }
124 | .slc-table td {
125 |     border-bottom: 1px solid #1e293b;
126 |     padding: 0.45rem 0.75rem;
127 |     vertical-align: middle;
128 | }
129 | .slc-table tr:hover td {
130 |     background: rgba(56, 189, 248, 0.06);
131 |     cursor: pointer;
132 | }
133 | .slc-table tr.selected td,
134 | .slc-table tr.table-active td {
135 |     background: rgba(56, 189, 248, 0.2) !important;
136 |     color: #fff !important;
137 | }
138 | .badge-crit { background: #dc2626 !important; color: #fff !i
139 | mportant; }
140 | .badge-err { background: #ef4444 !important; color: #fff !im
141 | portant; }
142 | .badge-warn { background: #f59e0b !important; color: #000 !i
143 | mportant; }
144 | .badge-inf { background: #10b981 !important; color: #fff !im
```

```
    portant; }
142 | .badge-verb { background: #64748b !important; color: #fff !i
    mportant; }
143 |
144 | /* Interactive Channel Description Popover Tooltip */
145 | .slc-tooltip-popup {
146 |     position: fixed;
147 |     z-index: 99999;
148 |     max-width: 320px;
149 |     background: #0f172a;
150 |     border: 1px solid #38bdf8;
151 |     border-radius: 8px;
152 |     padding: 0.6rem 0.8rem;
153 |     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.7), 0 0 15px
    rgba(56, 189, 248, 0.2);
154 |     pointer-events: none;
155 |     transition: opacity 0.15s ease, transform 0.15s ease;
156 |     opacity: 0;
157 |     transform: translateY(4px);
158 |     font-size: 0.75rem;
159 |     line-height: 1.35;
160 |     color: #e2e8f0;
161 | }
162 | .slc-tooltip-popup.show {
163 |     opacity: 1;
164 |     transform: translateY(0);
165 | }
166 | .slc-tooltip-title {
167 |     font-weight: 700;
168 |     color: #38bdf8;
169 |     margin-bottom: 0.25rem;
170 |     display: flex;
171 |     align-items: center;
172 |     gap: 0.4rem;
173 | }
174 | .slc-tooltip-meta {
175 |     color: #94a3b8;
176 |     font-size: 0.7rem;
```

```

177     border-top: 1px solid #334155;
178     padding-top: 0.35rem;
179     margin-top: 0.35rem;
180     display: flex;
181     justify-content: space-between;
182 }
183 </style>
184
185 <div class="slc-container">
186     <!-- Compact Header Bar with Inline Stats & Actions -->
187     <div class="slc-compact-header d-flex justify-content-between align-items-center flex-wrap gap-2">
188         <!-- Title & Inline Metrics -->
189         <div class="d-flex align-items-center gap-2 flex-wrap">
190             <div class="d-flex align-items-center gap-1.5 me-1">
191                 <span class="fs-6">□ </span>
192                 <span class="fw-bold text-white small">Windows System
193             </div>
194             <span class="fw-bold text-white small">Log Center</span>
195         </div>
196         <!-- Quick Metrics Pills -->
197         <div class="slc-metric-pill" title="Discovered Log Sources">
198             <span class="text-secondary small"><i class="bi bi-diagram-3 me-1"></i>Sources:</span>
199             <span class="fw-bold text-info font-monospace" id="slc-stat-sources">1,247</span>
200         </div>
201         <div class="slc-metric-pill" title="Active Log Channel">
202             <span class="text-secondary small"><i class="bi bi-folder-fill me-1"></i>Channel:</span>
203             <span class="fw-bold text-warning font-monospace text-truncate" id="slc-stat-channel" style="max-width: 140px;">System</span>
204         </div>
205         <div class="slc-metric-pill" title="Events Loaded in View">
206             <span class="text-secondary small"><i class="bi bi-list"></i>

```

```

t-ol me-1"></i>Loaded:</span>
206     <span class="fw-bold text-light font-monospace" id="sl
c-stat-events">0</span>
207     </div>
208     <div class="slc-metric-pill" title="Errors and Incidents
Detected">
209         <span class="text-secondary small"><i class="bi bi-exc
lamation-triangle-fill me-1"></i>Faults:</span>
210         <span class="fw-bold text-danger font-monospace" id="s
lc-stat-incidents">0</span>
211     </div>
212 </div>
213
214 <!-- Actions Toolbar -->
215 <div class="d-flex align-items-center gap-2 flex-wrap">
216     <button class="btn btn-outline-info rounded-pill px-2.5
py-1 small d-flex align-items-center gap-1" id="btn-slc-scan"
style="font-size: 0.78rem;">
217         <i class="bi bi-radar"></i> <span>Scan</span>
218     </button>
219     <div class="form-check form-switch text-light mb-0 d-inl
ine-flex align-items-center" style="font-size: 0.78rem; min-he
ight: auto;">
220         <input class="form-check-input me-1" type="checkbox" r
ole="switch" id="slc-live-switch" style="cursor: pointer;">
221         <label class="form-check-label text-secondary small us
er-select-none" for="slc-live-switch">Live</label>
222     </div>
223     <button class="btn btn-primary rounded-pill px-2.5 py-1
small d-flex align-items-center gap-1" id="btn-slc-refresh" st
yle="font-size: 0.78rem;">
224         <i class="bi bi-arrow-clockwise"></i> <span>Refresh</s
pan>
225     </button>
226     <div class="dropdown">
227         <button class="btn btn-xs btn-outline-secondary dropdo
wn-toggle rounded-pill px-2.5 py-1 text-light small d-flex ali
gn-items-center gap-1" type="button" data-bs-toggle="dropdown"

```

```

        style="font-size: 0.78rem;">
228         <i class="bi bi-download"></i> <span>Export</span>
229     </button>
230     <ul class="dropdown-menu dropdown-menu-dark dropdown-m
enu-end shadow" style="font-size: 0.8rem;">
231         <li><a class="dropdown-item" href="#" id="btn-slc-ex
port-json"><i class="bi bi-filetype-json me-2"></i>JSON</a></li>
232         <li><a class="dropdown-item" href="#" id="btn-slc-ex
port-csv"><i class="bi bi-filetype-csv me-2"></i>CSV</a></li>
233     </ul>
234 </div>
235 </div>
236 </div>
237
238 <!-- Search & View Mode Navigation (Compact) -->
239 <div class="slc-panel mb-2 p-2">
240     <div class="d-flex justify-content-between align-items-cen
ter flex-wrap gap-1 mb-2 pb-1 border-bottom border-secondary">
241         <!-- Mode Tabs -->
242         <ul class="nav nav-pills gap-1" id="slc-mode-tabs">
243             <li class="nav-item">
244                 <button class="nav-link active py-0.5 px-2.5 small r
ounded-pill" data-mode="live" style="font-size: 0.78rem;">● Li
ve Events</button>
245             </li>
246             <li class="nav-item">
247                 <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="audit" style="font-size: 0.78rem;"
>□ Аудит (Smart Digest)</button>
248             </li>
249             <li class="nav-item">
250                 <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="rag" style="font-size: 0.78rem;">□
□ RAG Поиск</button>
251             </li>
252             <li class="nav-item">
253                 <button class="nav-link py-0.5 px-2.5 small rounded-

```

```

    pill text-light" data-mode="errors" style="font-size: 0.78rem;
">❏ Errors Only</button>
254     </li>
255     <li class="nav-item">
256         <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="incidents" style="font-size: 0.78r
em;">❏ Incidents</button>
257     </li>
258     <li class="nav-item">
259         <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="timeline" style="font-size: 0.78re
m;">❏ Timeline</button>
260     </li>
261     <li class="nav-item">
262         <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="raw" style="font-size: 0.78rem;">❏
❏ Raw Record</button>
263     </li>
264 </ul>
265
266 <!-- Action AI Button -->
267 <div class="d-flex align-items-center gap-2">
268     <button class="btn btn-xs btn-outline-warning rounded-
pill px-2.5 py-1 small d-flex align-items-center gap-1" id="bt
n-slc-ask-ai" style="font-size: 0.78rem;">
269         <i class="bi bi-robot"></i> <span>Ask AI</span>
270     </button>
271 </div>
272 </div>
273
274 <!-- Filter Inputs Grid -->
275 <div class="row g-1.5 align-items-center">
276     <div class="col-lg-4 col-md-5">
277         <div class="input-group input-group-sm">
278             <span class="input-group-text bg-dark text-secondary
border-secondary py-0 px-2"><i class="bi bi-search" style="fo
nt-size: 0.75rem;"></i></span>
279             <input type="text" class="form-control form-control-

```

```

sm bg-dark text-light border-secondary py-1" id="slc-search-in
put" placeholder="Search message, provider..." style="font-siz
e: 0.8rem;">
280     </div>
281 </div>
282 <div class="col-lg-2 col-md-2 col-6">
283     <select class="form-select form-select-sm bg-dark text
-light border-secondary py-1" id="slc-level-select" style="fon
t-size: 0.8rem;">
284         <option value="">All Severities</option>
285         <option value="Critical">Critical</option>
286         <option value="Error">Error</option>
287         <option value="Warning">Warning</option>
288         <option value="Information">Information</option>
289     </select>
290 </div>
291 <div class="col-lg-2 col-md-2 col-6">
292     <select class="form-select form-select-sm bg-dark text
-light border-secondary py-1" id="slc-hours-select" style="fon
t-size: 0.8rem;">
293         <option value="1">Last 1 hour</option>
294         <option value="6">Last 6 hours</option>
295         <option value="24" selected>Last 24 hours</option>
296         <option value="168">Last 7 days</option>
297         <option value="0">All available</option>
298     </select>
299 </div>
300 <div class="col-lg-2 col-md-1.5 col-6">
301     <input type="number" class="form-control form-control-
sm bg-dark text-light border-secondary py-1" id="slc-eventid-i
nput" placeholder="Event ID" style="font-size: 0.8rem;">
302 </div>
303 <div class="col-lg-2 col-md-1.5 col-6">
304     <select class="form-select form-select-sm bg-dark text
-light border-secondary py-1" id="slc-limit-select" style="fon
t-size: 0.8rem;">
305         <option value="50">50 records</option>
306         <option value="100" selected>100 records</option>

```



```

307         <option value="250">250 records</option>
308         <option value="500">500 records</option>
309     </select>
310 </div>
311 </div>
312 </div>
313
314 <!-- Main Multi-Column View -->
315 <div class="row g-3">
316     <!-- Channel & Source Tree Sidebar -->
317     <div class="col-lg-3">
318         <div class="slc-panel h-100">
319             <div class="slc-panel-header">
320                 <span><i class="bi bi-folder2-open me-2"></i>Log Cha
nnels Tree</span>
321                 <span class="badge bg-secondary" id="slc-channel-cou
nt-badge">0</span>
322             </div>
323             <div class="p-2 border-bottom border-secondary">
324                 <input type="text" class="form-control form-control-
sm bg-dark text-light border-secondary" id="slc-channel-filter
-input" placeholder="Filter channels...">
325             </div>
326             <div class="p-2 slc-tree-list" id="slc-channel-tree-co
ntainer">
327                 <!-- Dynamic channels injected here -->
328             </div>
329         </div>
330     </div>
331
332 <!-- Events / Incidents Main Stream -->
333 <div class="col-lg-9">
334     <div class="slc-panel h-100">
335         <div class="slc-panel-header">
336             <span id="slc-main-view-title"><i class="bi bi-list-
columns-reverse me-2"></i>Live Windows Event Stream</span>
337             <div class="d-flex align-items-center gap-2">
338                 <button class="btn btn-sm btn-info rounded-pill px

```

```

-3 fw-bold text-dark" id="btn-slc-audit" title="Провести аудит
текущего журнала логов">
339         <i class="bi bi-shield-check me-1"></i> Аудит
340     </button>
341     <span class="badge bg-secondary" id="slc-stream-co
unt-badge">0 entries</span>
342 </div>
343 </div>
344
345 <!-- Table View Container -->
346 <div class="table-responsive" style="max-height: 580px
; overflow-y: auto;" id="slc-table-container">
347     <table class="table slc-table">
348         <thead>
349             <tr>
350                 <th style="width: 155px;">Timestamp</th>
351                 <th style="width: 90px;">Level</th>
352                 <th style="width: 65px;">ID</th>
353                 <th style="width: 160px;">Provider</th>
354                 <th style="width: 65px;">PID</th>
355                 <th>Message</th>
356             </tr>
357         </thead>
358         <tbody id="slc-table-body">
359             <tr>
360                 <td colspan="6" class="text-center text-muted
py-4">
361                     <div class="spinner-border spinner-border-sm
text-info me-2" role="status"></div>
362                     Loading system events...
363                 </td>
364             </tr>
365         </tbody>
366     </table>
367 </div>
368
369 <!-- Audit Digest View Container (Hidden by default) -
->

```

```

370         <div class="p-3 d-none" id="slc-audit-container" style
    ="max-height: 580px; overflow-y: auto;">
371             <div id="slc-audit-content"></div>
372         </div>
373
374         <!-- Dynamic Log RAG Search Container (Hidden by default) -->
375         <div class="p-3 d-none" id="slc-rag-container" style="
    max-height: 580px; overflow-y: auto;">
376             <div class="card bg-dark border-info mb-3 shadow">
377                 <div class="card-header bg-black d-flex justify-content-between align-items-center">
378                     <span class="text-info fw-bold"><i class="bi bi-search me-2"></i>Семантический RAG-поиск по логам</span>
379                     <button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-slc-rag-rebuild">
380                         <i class="bi bi-arrow-repeat me-1"></i> Перестроить RAG-индекс
381                     </button>
382                 </div>
383                 <div class="card-body">
384                     <div class="input-group mb-2">
385                         <span class="input-group-text bg-black text-info border-secondary"><i class="bi bi-chat-left-dots"></i></span>
386                         <input type="text" class="form-control bg-dark text-light border-secondary" id="slc-rag-input" placeholder="
    Введите естественный запрос (например: 'почему завис BITS', 'ошибки DCOM 10016', 'сетевые сбои')...">
387                     <button class="btn btn-info px-4 fw-bold text-dark" id="btn-slc-rag-search">Найти</button>
388                 </div>
389                 <div class="small text-muted">
390                     Поиск использует динамический TF-IDF/BM25 индекс кластеров, аномалий и сводок с нулевым расходом токенов LLM
391                     .
392                 </div>
393             </div>
394         </div>

```

```

393         </div>
394
395         <div id="slc-rag-results">
396             <div class="card bg-dark border-secondary p-4 text
- center text-muted">
397                 Введите запрос для поиска по динамическому RAG-и
ндексу логов или нажмите «Перестроить RAG-индекс».
398             </div>
399         </div>
400     </div>
401
402     <!-- Incidents View Container (Hidden by default) -->
403     <div class="p-3 d-none" id="slc-incidents-container" s
tyle="max-height: 580px; overflow-y: auto;">
404         <div id="slc-incidents-list"></div>
405     </div>
406
407     <!-- Timeline Histogram View Container (Hidden by defa
ult) -->
408     <div class="p-3 d-none" id="slc-timeline-container" st
yle="max-height: 580px; overflow-y: auto;">
409         <div id="slc-timeline-list"></div>
410     </div>
411
412     <!-- Raw Record Container (Hidden by default) -->
413     <div class="p-3 d-none" id="slc-raw-container" style="
max-height: 580px; overflow-y: auto;">
414         <pre class="bg-black text-info p-3 rounded font-mono
space small" id="slc-raw-text">Select an event from the list t
o view raw structured XML/JSON data.</pre>
415     </div>
416 </div>
417 </div>
418 </div>
419 </div>
420
421 <!-- Modal: Event Details & AI Diagnostics -->
422 <div class="modal fade" id="slc-event-detail-modal" tabindex="

```

```

-1">
423 <div class="modal-dialog modal-lg modal-dialog-centered">
424 <div class="modal-content bg-dark text-light border-second
ary">
425 <div class="modal-header border-secondary">
426 <h5 class="modal-title fw-bold text-info" id="slc-moda
l-title">Log Event Details</h5>
427 <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal"></button>
428 </div>
429 <div class="modal-body">
430 <div class="row g-2 mb-3 small font-monospace">
431 <div class="col-sm-6"><strong class="text-secondary"
>Timestamp:</strong> <span id="slc-modal-time"></span></div>
432 <div class="col-sm-6"><strong class="text-secondary"
>Severity:</strong> <span id="slc-modal-level"></span></div>
433 <div class="col-sm-6"><strong class="text-secondary"
>Event ID:</strong> <span id="slc-modal-id"></span></div>
434 <div class="col-sm-6"><strong class="text-secondary"
>Provider:</strong> <span id="slc-modal-provider"></span></div>
>
435 <div class="col-sm-6"><strong class="text-secondary"
>Computer:</strong> <span id="slc-modal-computer"></span></div>
>
436 <div class="col-sm-6"><strong class="text-secondary"
>Process ID:</strong> <span id="slc-modal-pid"></span></div>
437 <div class="col-sm-12"><strong class="text-secondary
">Channel:</strong> <span id="slc-modal-channel"></span></div>
438 </div>
439
440 <div class="card bg-black border-secondary p-3 mb-3">
441 <h6 class="small text-uppercase text-muted fw-bold m
b-2">Message Body</h6>
442 <div class="small font-monospace" id="slc-modal-mess
age" style="white-space: pre-wrap; word-break: break-word;"></
div>
443 </div>
444

```

```
445 |         <!-- AI Diagnostic Response Section -->
446 |         <div class="card border-info bg-dark p-3" id="slc-modal-ai-section">
447 |             <div class="d-flex justify-content-between align-items-center mb-2">
448 |                 <h6 class="fw-bold text-info mb-0"><i class="bi bi-robot me-1"></i> AI Contextual Explanation</h6>
449 |                 <button class="btn btn-sm btn-outline-info py-0 px-2 rounded-pill" id="btn-slc-modal-diagnose">Analyze Context</button>
450 |             </div>
451 |             <div class="small" id="slc-modal-ai-explanation">
452 |                 Click "Analyze Context" to inspect surrounding events and identify the root cause of this incident.
453 |             </div>
454 |         </div>
455 |     </div>
456 |     <div class="modal-footer border-secondary">
457 |         <button type="button" class="btn btn-sm btn-secondary" data-bs-dismiss="modal">Close</button>
458 |     </div>
459 | </div>
460 | </div>
461 | </div>
462 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows System Logs Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/system_logs_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/system_logs_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-08 02:00:00
18  * =====
19  */
20
21 (function () {
22   'use strict';
23
24   let currentChannel = 'System';
25   let currentFilePath = '';
26   let currentMode = 'live';
27   let cachedEvents = [];
28   let cachedChannels = [];
29   let selectedEvent = null;
30   let liveIntervalTimer = null;
31   let isRequestInProgress = false;
32 }
```

```
33 | /**
34 |  * Escape HTML entities to protect against XSS injection.
35 |  *
36 |  * @param {string} str - Raw input text.
37 |  * @returns {string} Sanitized string.
38 |  */
39 | function escapeHtml(str) {
40 |     if (!str) return '';
41 |     return String(str)
42 |         .replace(/&/g, '&amp;')
43 |         .replace(/</g, '&lt;')
44 |         .replace(/>/g, '&gt;')
45 |         .replace(/"/g, '&quot;')
46 |         .replace(/'/g, '&#039;');
47 | }
48 |
49 | /**
50 |  * Map log severity to badge styling class.
51 |  *
52 |  * @param {string} level - Severity level.
53 |  * @returns {string} CSS class name.
54 |  */
55 | function getSeverityClass(level) {
56 |     const lvl = (level || '').toLowerCase();
57 |     if (lvl.includes('crit')) return 'badge-crit';
58 |     if (lvl.includes('err')) return 'badge-err';
59 |     if (lvl.includes('warn')) return 'badge-warn';
60 |     if (lvl.includes('verb')) return 'badge-verb';
61 |     return 'badge-inf';
62 | }
63 |
64 | /**
65 |  * Perform dynamic system scan and populate channels tree.
66 |  *
67 |  * @param {boolean} force - Force refresh flag.
68 |  */
69 | async function performSystemScan(force = false) {
70 |     try {
```



```

71 |     const response = await fetch(`/api/v1/system_logs/scan?f
    | orce=${force}`);
72 |     if (!response.ok) return;
73 |     const data = await response.json();
74 |
75 |     cachedChannels = data.channels || [];
76 |     const activeChannels = cachedChannels.filter((c) => (c.r
    | ecord_count || 0) > 0);
77 |     const sourcesBadge = document.getElementById('slc-stat-s
    | ources');
78 |     const countBadge = document.getElementById('slc-channel-
    | count-badge');
79 |
80 |     if (sourcesBadge) {
81 |         sourcesBadge.textContent = `${activeChannels.length.to
    | LocaleString()} active / ${data.total_sources.toLocaleString()
    | } sources`;
82 |     }
83 |     if (countBadge) {
84 |         countBadge.textContent = String(activeChannels.length)
    | ;
85 |     }
86 |
87 |     renderChannelsTree();
88 | } catch (err) {
89 |     console.warn('Failed to perform system discovery scan:',
    | err);
90 | }
91 | }
92 |
93 | // Singleton global popup tooltip element
94 | let globalTooltipEl = null;
95 |
96 | function getOrCreateTooltipEl() {
97 |     if (!globalTooltipEl) {
98 |         globalTooltipEl = document.createElement('div');
99 |         globalTooltipEl.className = 'slc-tooltip-popup';
100 |         document.body.appendChild(globalTooltipEl);

```

```
101     }
102     return globalTooltipEl;
103 }
104
105 function showChannelTooltip(e, channel) {
106     const tip = getOrCreateTooltipEl();
107     const countFormatted = (channel.record_count || 0).toLocaleString();
108     const descText = channel.description || 'Служебный журнал операционной системы Windows.';
109     const typeLabel = channel.source_type === 'channel' ? 'Windows Event Channel' : 'Log File';
110
111     tip.innerHTML = `
112         <div class="slc-tooltip-title">
113             <i class="bi bi-info-circle text-info"></i>
114             <span>${escapeHtml(channel.display_name)}</span>
115         </div>
116         <div style="color: #cbd5e1; margin-bottom: 0.35rem;">${escapeHtml(descText)}</div>
117         <div class="slc-tooltip-meta">
118             <span><strong class="text-info">${countFormatted}</strong> записей</span>
119             <span class="text-muted font-monospace">${typeLabel}</span>
120         </div>
121     `;
122
123     tip.classList.add('show');
124     positionTooltip(e, tip);
125 }
126
127 function hideChannelTooltip() {
128     if (globalTooltipEl) {
129         globalTooltipEl.classList.remove('show');
130     }
131 }
132
```

```

133 | function positionTooltip(e, tip) {
134 |     const margin = 12;
135 |     let x = e.clientX + margin;
136 |     let y = e.clientY + margin;
137 |
138 |     const tipRect = tip.getBoundingClientRect();
139 |     if (x + tipRect.width > window.innerWidth) {
140 |         x = e.clientX - tipRect.width - margin;
141 |     }
142 |     if (y + tipRect.height > window.innerHeight) {
143 |         y = e.clientY - tipRect.height - margin;
144 |     }
145 |
146 |     tip.style.left = `${Math.max(10, x)}px`;
147 |     tip.style.top = `${Math.max(10, y)}px`;
148 | }
149 |
150 | /**
151 |  * Render channels tree list in sidebar with category groupi
152 |  */
153 | function renderChannelsTree() {
154 |     const container = document.getElementById('slc-channel-tre
155 |     const filterInput = document.getElementById('slc-channel-f
156 |     const filterText = filterInput ? filterInput.value.trim().
157 |     toLowerCase() : '';
158 |
159 |     if (!container) return;
160 |     container.innerHTML = '';
161 |
162 |     const filtered = cachedChannels.filter((c) => {
163 |         // Не выводить каналы и источники, в которых 0 записей
164 |         if (c.record_count === 0) return false;
165 |
166 |         if (!filterText) return true;
167 |         return (

```

```

167 |         c.channel_name.toLowerCase().includes(filterText) ||
168 |         c.display_name.toLowerCase().includes(filterText) ||
169 |         (c.description && c.description.toLowerCase().includes
| (filterText)) ||
170 |         (c.category && c.category.toLowerCase().includes(filte
| rText))
171 |     );
172 | });
173 |
174 | // Group by category
175 | const grouped = {};
176 | filtered.forEach((c) => {
177 |     const cat = c.category || 'Windows Event Log';
178 |     if (!grouped[cat]) grouped[cat] = [];
179 |     grouped[cat].push(c);
180 | });
181 |
182 | const categoryIcons = {
183 |     'Windows Event Log': '📄 ',
184 |     'Windows OS Logs': '📄 ',
185 |     'Application Logs': '📄 ',
186 |     'AI-Breadboard Logs': '📄 ',
187 | };
188 |
189 | Object.keys(grouped).forEach((catName) => {
190 |     const catHeader = document.createElement('div');
191 |     catHeader.className = 'd-flex justify-content-between al
| ign-items-center px-2 py-1 mt-2 mb-1 rounded bg-dark border bo
| rder-secondary text-info small fw-bold';
192 |     const icon = categoryIcons[catName] || '📄 ';
193 |     catHeader.innerHTML = `
194 |         <span>${icon} ${escapeHtml(catName)}</span>
195 |         <span class="badge bg-secondary font-monospace">${grou
| ped[catName].length}</span>
196 |     `;
197 |     container.appendChild(catHeader);
198 |
199 |     grouped[catName].slice(0, 400).forEach((c) => {

```

```

200 |         const item = document.createElement('div');
201 |         const isActive = (c.channel_name === currentChannel &&
    | !currentFilePath) || (currentFilePath && currentFilePath ===
    | c.location);
202 |         item.className = `slc-tree-item ${isActive ? 'active'
    | : ''}`;
203 |
204 |         let subBadge = '0';
205 |         let badgeStyle = 'bg-dark text-secondary';
206 |         if (c.record_count > 0) {
207 |             subBadge = c.record_count >= 1000 ? `${(c.record_cou
    | nt / 1000).toFixed(1)}k` : String(c.record_count);
208 |             badgeStyle = c.record_count > 1000 ? 'bg-primary tex
    | t-white' : 'bg-dark text-info border border-secondary';
209 |         } else if (c.source_type !== 'channel') {
210 |             subBadge = 'FILE';
211 |         }
212 |
213 |         item.innerHTML = `
214 |             <span class="text-truncate me-2" style="max-width: 1
    | 75px;">${escapeHtml(c.display_name)}</span>
215 |             <span class="badge ${badgeStyle} font-monospace" sty
    | le="font-size: 0.72rem; min-width: 32px; text-align: right;">${
    | {subBadge}</span>
216 |         `;
217 |
218 |         // Tooltip events
219 |         item.onmouseenter = (e) => showChannelTooltip(e, c);
220 |         item.onmousemove = (e) => {
221 |             const tip = getOrCreateTooltipEl();
222 |             if (tip.classList.contains('show')) positionTooltip(
    | e, tip);
223 |         };
224 |         item.onmouseleave = () => hideChannelTooltip();
225 |
226 |         item.onclick = () => {
227 |             hideChannelTooltip();
228 |             if (c.source_type === 'channel') {

```

```
229         currentChannel = c.channel_name;
230         currentFilePath = '';
231     } else {
232         currentChannel = c.display_name;
233         currentFilePath = c.location;
234     }
235     const statChan = document.getElementById('slc-stat-c
hannel');
236     if (statChan) statChan.textContent = c.display_name;
237     renderChannelsTree();
238     loadEvents();
239 };
240 container.appendChild(item);
241 });
242 });
243 }
244
245 /**
246  * Fetch events matching current channel, filters, and activ
e view mode.
247  */
248 async function loadEvents() {
249     if (isRequestInProgress) return;
250     isRequestInProgress = true;
251
252     const levelSelect = document.getElementById('slc-level-sel
ect');
253     const hoursSelect = document.getElementById('slc-hours-sel
ect');
254     const searchInput = document.getElementById('slc-search-in
put');
255     const eventIdInput = document.getElementById('slc-eventid-
input');
256     const limitSelect = document.getElementById('slc-limit-sel
ect');
257     const tableBody = document.getElementById('slc-table-body'
);
258     const eventsBadge = document.getElementById('slc-stat-even
```

```

    ts');
259     const faultsBadge = document.getElementById('slc-stat-incidents');
260     const streamBadge = document.getElementById('slc-stream-count-badge');
261
262     const level = levelSelect ? levelSelect.value : '';
263     const hours = hoursSelect ? hoursSelect.value : '24';
264     const search = searchInput ? searchInput.value.trim() : ''
    ;
265     const eventId = eventIdInput && eventIdInput.value.trim()
    ? eventIdInput.value.trim() : '0';
266     const limit = limitSelect ? limitSelect.value : '100';
267
268     try {
269         const url = `/api/v1/system_logs/events?channel=${encodeURIComponent(currentChannel)}&limit=${limit}&level=${encodeURIComponent(level)}&search=${encodeURIComponent(search)}&hours=${hours}&event_id=${eventId}&file_path=${encodeURIComponent(currentFilePath)}`;
270
271         const res = await fetch(url);
272         if (!res.ok) {
273             if (tableBody) tableBody.innerHTML = `<tr><td colspan=
    "6" class="text-center text-danger py-3">Error reading channel
    (HTTP ${res.status})</td></tr>`;
274             isRequestInProgress = false;
275             return;
276         }
277
278         const data = await res.json();
279         cachedEvents = data.entries || [];
280
281         if (eventsBadge) eventsBadge.textContent = String(cachedEvents.length);
282         if (streamBadge) streamBadge.textContent = `${cachedEvents.length} records`;
283

```

```

284 |     let errCount = 0;
285 |     let critCount = 0;
286 |     cachedEvents.forEach((e) => {
287 |         const lvl = (e.level || '').toLowerCase();
288 |         if (lvl.includes('crit')) critCount++;
289 |         if (lvl.includes('err')) errCount++;
290 |     });
291 |
292 |     if (faultsBadge) {
293 |         faultsBadge.textContent = `${errCount + critCount} faults`;
294 |     }
295 |
296 |     renderCurrentModeView();
297 | } catch (err) {
298 |     console.warn('Failed to load events:', err);
299 |     if (tableBody) tableBody.innerHTML = `<tr><td colspan="6
    " class="text-center text-danger py-3">Network error loading logs.</td></tr>`;
300 | } finally {
301 |     isRequestInProgress = false;
302 | }
303 | }
304 |
305 | /**
306 |  * Render view corresponding to active mode (Live, Errors, Incidents, Timeline, Raw).
307 |  */
308 | function renderCurrentModeView() {
309 |     const tableContainer = document.getElementById('slc-table-container');
310 |     const incidentsContainer = document.getElementById('slc-incidents-container');
311 |     const timelineContainer = document.getElementById('slc-timeline-container');
312 |     const rawContainer = document.getElementById('slc-raw-container');
313 |     const auditContainer = document.getElementById('slc-audit-

```



```
container');
314     const tableBody = document.getElementById('slc-table-body'
);
315     const viewTitle = document.getElementById('slc-main-view-t
itle');
316
317     if (!tableContainer || !tableBody) return;
318
319     // Reset visibility
320     tableContainer.classList.add('d-none');
321     if (incidentsContainer) incidentsContainer.classList.add('
d-none');
322     if (timelineContainer) timelineContainer.classList.add('d-
none');
323     if (rawContainer) rawContainer.classList.add('d-none');
324     if (auditContainer) auditContainer.classList.add('d-none')
;
325
326     if (currentMode === 'audit') {
327         if (viewTitle) viewTitle.innerHTML = `
```

```

341 |         incidentsContainer.classList.remove('d-none');
342 |         loadIncidentsView();
343 |     }
344 | } else if (currentMode === 'timeline') {
345 |     if (viewTitle) viewTitle.innerHTML = `

```

```

371 |         const l = (e.level || '').toLowerCase();
372 |         return l.includes('err') || l.includes('crit') || l.
includes('warn');
373 |     });
374 | }
375 |
376 |     if (entriesToRender.length === 0) {
377 |         tableBody.innerHTML = `|  |  |  |  |  |  |
| --- | --- | --- | --- | --- | --- |
|  | | | | | |

```

```

    ${escapeHtml(pidStr)}</td>
397         <td class="font-monospace text-light" style="word-
break: break-word;">${escapeHtml(e.message)}</td>
398     </tr>
399     `;
400     }).join('');
401
402     // Attach row click listeners for row selection and deta
il inspection
403     tableBody.querySelectorAll('tr').forEach((row) => {
404         row.onclick = () => {
405             const idx = parseInt(row.getAttribute('data-index')
|| '0', 10);
406             selectedEvent = entriesToRender[idx];
407             tableBody.querySelectorAll('tr').forEach(r => r.clas
sList.remove('selected', 'table-active'));
408             row.classList.add('selected', 'table-active');
409             showEventModal(selectedEvent);
410         };
411     });
412 }
413 }
414
415 /**
416  * Fetch and render correlated incident clusters.
417  */
418 async function loadIncidentsView() {
419     const listEl = document.getElementById('slc-incidents-list
');
420     if (!listEl) return;
421     listEl.innerHTML = `<div class="text-center py-4 text-mute
d"><div class="spinner-border spinner-border-sm text-info me-2
"></div>Correlating incident cascades...</div>`;
422
423     try {
424         const res = await fetch('/api/v1/system_logs/incidents?w
indow_seconds=25.0');
425         if (!res.ok) return;

```

```

426 |     const data = await res.json();
427 |     const incidents = data.incidents || [];
428 |
429 |     if (incidents.length === 0) {
430 |         listEl.innerHTML = `
```

```

451     } catch (err) {
452         listEl.innerHTML = `
```

```

    ngs}</span>
481         <span class="badge badge-info">Info: ${h.info}</span>
    n>
482     </div>
483 </div>
484     `).join('');
485     } catch (err) {
486         listEl.innerHTML = `<div class="alert alert-danger">Error
loading timeline: ${err.message}</div>`;
487     }
488 }
489
490 /**
491  * Fetch and render multi-level log audit digest (Noise filtering,
clustering, anomalies).
492  */
493 async function loadAuditView() {
494     const contentEl = document.getElementById('slc-audit-content');
495     if (!contentEl) return;
496
497     const hoursSelect = document.getElementById('slc-hours-select');
498     const limitSelect = document.getElementById('slc-limit-select');
499     const hours = hoursSelect ? hoursSelect.value : '24';
500     const limit = limitSelect ? limitSelect.value : '500';
501
502     contentEl.innerHTML = `
503         <div class="text-center py-5 text-muted">
504             <div class="spinner-border text-info mb-3" role="status"></div>
505             <div class="fw-bold text-light">Выполняется многоуровневый аудит массива логов...</div>
506             <div class="small text-secondary mt-1">Фильтрация шума
• Шаблонизация • Детекция аномалий • Сжатие данных</div>
507         </div>
508     `;

```

```

509
510     try {
511         const url = `/api/v1/system_logs/audit?channel=${encodeURIComponentComponent(currentChannel)}&limit=${limit}&hours=${hours}&file_path=${encodeURIComponentComponent(currentFilePath)}`;
512         const res = await fetch(url);
513         if (!res.ok) throw new Error(`HTTP ${res.status}`);
514         const data = await res.json();
515
516         const anomaliesHtml = (data.anomalies || []).map((a) =>
517         {
518             const badgeClass = getSeverityClass(a.level);
519             return `
520                 <div class="p-2 mb-2 rounded bg-black border border-secondary d-flex justify-content-between align-items-center flex-wrap gap-2">
521                     <div>
522                         <span class="badge ${badgeClass} text-uppercase me-2">${escapeHtml(a.level)}</span>
523                         <strong class="text-info">${escapeHtml(a.provider)}</strong>
524                         <span class="text-muted font-monospace small">(ID ${a.event_id})</span>
525                         <div class="small text-light mt-1 font-monospace" style="word-break: break-word;">${escapeHtml(a.sample)}</div>
526                         <div class="small text-secondary mt-1"><i class="bi bi-clock me-1"></i>Окно: ${escapeHtml(a.time_window)}</div>
527                     </div>
528                     <div class="text-end">
529                         <span class="badge bg-danger rounded-pill px-3 py-2 fs-6">${a.count}x</span>
530                     </div>
531                 `;
532             }).join('') || `<div class="text-success small p-2"><i class="bi bi-check-circle me-1"></i>Аномалий и критических сбое

```



```

    в не обнаружено.</div>`;
533
534     const clustersHtml = (data.top_clusters || []).map((cl,
535     idx) => {
536         const badgeClass = getSeverityClass(cl.level);
537         return `
538             <tr class="small">
539                 <td class="text-center font-monospace">${idx + 1}<
540 /td>
541                 <td><span class="badge ${badgeClass}">${escapeHtml
542 (cl.level)}</span></td>
543                 <td class="text-info text-truncate" style="max-wid
544 th: 140px;" title="${escapeHtml(cl.provider)}">${escapeHtml(cl
545 .provider)}</td>
546                 <td class="text-center font-monospace">${cl.event_
547 id || '-'}</td>
548                 <td class="font-monospace text-light" style="max-w
549 idth: 320px; word-break: break-word;" title="${escapeHtml(cl.t
550 emplate)}">${escapeHtml(cl.template)}</td>
551                 <td class="text-center"><span class="badge bg-seco
552 ndary font-monospace">${cl.count}</span></td>
553                 <td class="small text-secondary font-monospace tex
554 t-nowrap">${escapeHtml(cl.first_seen)}</td>
555             </tr>
556         `;
557     }).join('');
558
559     const totalScanned = data.total_analyzed ?? data.total_s
560 canned ?? 0;
561     const uniqueCount = data.unique_patterns_count ?? (data.
562 top_clusters ? data.top_clusters.length : 0);
563     const redundancyPct = data.redundancy_pct !== undefined
564 ? `${data.redundancy_pct.toFixed(1)}%` : (data.compression_rat
565 io || '0%');
566     const errSum = (data.error_count || 0) + (data.critical_
567 count || 0);
568     const warnSum = data.warning_count || 0;
569     const summaryText = data.strategy_rationale || data.exec

```

```
utive_summary || `Стратегия обработки: ${data.strategy || 'Стандартная'}. Найдено ${uniqueCount} ключевых шаблонов.`;
```

```
555 |
556 |     contentEl.innerHTML = `
557 |         <!-- Executive Summary Card (Compact) -->
558 |         <div class="card bg-dark border-info mb-2 shadow-sm">
559 |             <div class="card-header bg-black text-info fw-bold p
y-1 px-3 d-flex justify-content-between align-items-center sma
ll">
560 |                 <span><i class="bi bi-cpu me-1"></i>Сводка аудита:
${escapeHtml(data.channel)}</span>
561 |                 <span class="badge bg-info text-dark font-monospac
e">Дублирование: ${redundancyPct}</span>
562 |             </div>
563 |             <div class="card-body p-2">
564 |                 <div class="row g-2 text-center mb-2">
565 |                     <div class="col-sm-3 col-6">
566 |                         <div class="p-1 rounded bg-black border border
-secondary">
567 |                             <div class="small text-muted" style="font-si
ze: 0.72rem;">Просканировано</div>
568 |                             <div class="fs-5 fw-bold text-light font-mon
ospace">${totalScanned}</div>
569 |                         </div>
570 |                     </div>
571 |                     <div class="col-sm-3 col-6">
572 |                         <div class="p-1 rounded bg-black border border
-secondary">
573 |                             <div class="small text-muted" style="font-si
ze: 0.72rem;">Шаблонов</div>
574 |                             <div class="fs-5 fw-bold text-info font-mono
space">${uniqueCount}</div>
575 |                         </div>
576 |                     </div>
577 |                     <div class="col-sm-3 col-6">
578 |                         <div class="p-1 rounded bg-black border border
-secondary">
579 |                             <div class="small text-muted" style="font-si
```

```

ze: 0.72rem;">Ошибок / Сбоев</div>
580         <div class="fs-5 fw-bold text-danger font-mo
nospace">${errSum}</div>
581         </div>
582     </div>
583     <div class="col-sm-3 col-6">
584         <div class="p-1 rounded bg-black border border
-secondary">
585             <div class="small text-muted" style="font-si
ze: 0.72rem;">Предупреждений</div>
586             <div class="fs-5 fw-bold text-warning font-m
onospace">${warnSum}</div>
587             </div>
588         </div>
589     </div>
590     <p class="mb-0 text-light small"><i class="bi bi-i
nfo-circle text-info me-1"></i>${escapeHtml(summaryText)}</p>
591     </div>
592 </div>
593
594 <!-- Detected Anomalies Section -->
595 <div class="card bg-dark border-secondary mb-2 shadow-
sm">
596     <div class="card-header bg-black text-warning fw-bol
d py-1 px-3 small">
597         <i class="bi bi-exclamation-triangle me-1"></i>Обн
аруженные аномалии и всплески
598     </div>
599     <div class="card-body p-2">
600         ${anomaliesHtml}
601     </div>
602 </div>
603
604 <!-- Top Clustered Patterns Table -->
605 <div class="card bg-dark border-secondary">
606     <div class="card-header bg-black text-light fw-bold
d-flex justify-content-between align-items-center">
607         <span><i class="bi bi-collection me-2"></i>Топ шаб

```

```

        лонов событий (Дедупликация)</span>
608         <span class="small text-muted">Топ 10 кластеров</s
        pan>
609     </div>
610     <div class="table-responsive">
611         <table class="table slc-table mb-0">
612             <thead>
613                 <tr>
614                     <th style="width: 40px;">#</th>
615                     <th style="width: 80px;">Уровень</th>
616                     <th style="width: 140px;">Поставщик</th>
617                     <th style="width: 60px;">ID</th>
618                     <th>Шаблон сообщения (Masked)</th>
619                     <th style="width: 70px;" class="text-center"
        >Кол-во</th>
620                     <th style="width: 140px;">Впервые</th>
621                 </tr>
622             </thead>
623             <tbody>
624                 ${clustersHtml}
625             </tbody>
626         </table>
627     </div>
628 </div>
629 `;
630 } catch (err) {
631     contentEl.innerHTML = `<div class="alert alert-danger">0
        шибка проведения аудита: ${escapeHtml(err.message)}</div>`;
632 }
633 }
634
635 /**
636  * Execute dynamic RAG search over indexed patterns, anomal
        ies, and digests.
637  */
638 async function performRAGSearch() {
639     const inputEl = document.getElementById('slc-rag-input');
640     const resultsEl = document.getElementById('slc-rag-results

```

```

    ');
641     if (!inputEl || !resultsEl) return;
642
643     const query = inputEl.value.trim();
644     if (!query) return;
645
646     resultsEl.innerHTML = `<div class="text-center py-4 text-m
    uted"><div class="spinner-border spinner-border-sm text-info m
    e-2"></div>Поиск по динамическому RAG-индексу...</div>`;
647
648     try {
649         const res = await fetch('/api/v1/system_logs/rag/query',
        {
650             method: 'POST',
651             headers: { 'Content-Type': 'application/json' },
652             body: JSON.stringify({ query: query, channel: currentC
        hannel, top_k: 6 }),
653         });
654         if (!res.ok) throw new Error(`HTTP ${res.status}`);
655         const data = await res.json();
656         const results = data.results || [];
657
658         if (results.length === 0) {
659             resultsEl.innerHTML = `
660                 <div class="card bg-dark border-secondary p-4 text-c
        enter text-muted">
661                     По запросу «${escapeHtml(query)}» ничего не найден
        о в RAG-индексе. Попробуйте нажать «Перестроить RAG-индекс».
662                 </div>
663                 `;
664             return;
665         }
666
667         resultsEl.innerHTML = results.map((r) => {
668             const badgeClass = getSeverityClass(r.meta?.level || '
        info');
669             const scorePct = Math.min(100, Math.round((r.relevance
        _score || 0) * 40));

```

```

670         return `
671             <div class="card bg-dark border-secondary mb-3 shado
w-sm">
672                 <div class="card-header bg-black d-flex justify-co
ntent-between align-items-center">
673                     <span class="fw-bold text-info"><i class="bi bi-
file-earmark-text me-2"></i>${escapeHtml(r.title)}</span>
674                     <div class="d-flex align-items-center gap-2">
675                         <span class="badge ${badgeClass}">${escapeHtml
(r.meta?.level || r.type)}</span>
676                         <span class="badge bg-secondary font-monospace
">Score: ${r.relevance_score}</span>
677                     </div>
678                 </div>
679                 <div class="card-body">
680                     <pre class="bg-black text-light p-3 rounded font
-monospace small mb-0" style="white-space: pre-wrap; word-brea
k: break-word;">${escapeHtml(r.text)}</pre>
681                 </div>
682             </div>
683         `;
684     }).join('');
685 } catch (err) {
686     resultsEl.innerHTML = `<div class="alert alert-danger">0
шибка поиска RAG: ${escapeHtml(err.message)}</div>`;
687 }
688 }
689
690 /**
691  * Trigger building or rebuilding dynamic RAG index for curr
ent channel.
692  */
693 async function rebuildRAGIndex() {
694     const resultsEl = document.getElementById('slc-rag-results
');
695     if (!resultsEl) return;
696     resultsEl.innerHTML = `<div class="text-center py-4 text-m
uted"><div class="spinner-border spinner-border-sm text-info m

```

```

e-2"></div>Перестроение динамического RAG-индекса для «${escapeHtml(currentChannel)}»...</div>`;
697
698     try {
699         const res = await fetch(`/api/v1/system_logs/rag/build?channel=${encodeURIComponent(currentChannel)}&limit=500&hours=24&file_path=${encodeURIComponent(currentFilePath)}`, {
700             method: 'POST',
701         });
702         if (!res.ok) throw new Error(`HTTP ${res.status}`);
703         const data = await res.json();
704         resultsEl.innerHTML = `
705             <div class="alert alert-success d-flex align-items-center gap-2">
706                 <i class="bi bi-check-circle-fill fs-5"></i>
707                 <div>
708                     <strong>RAG-индекс успешно обновлен!</strong> Добавлено ${data.chunks_added} чанков (Всего в индексе: ${data.total_index_chunks}).
709                     <div class="small mt-1 text-light">${escapeHtml(data.executive_summary)}</div>
710                 </div>
711             </div>
712         `;
713     } catch (err) {
714         resultsEl.innerHTML = `<div class="alert alert-danger">Ошибка перестроения RAG: ${escapeHtml(err.message)}</div>`;
715     }
716 }
717
718 /**
719  * Display event details modal with AI diagnostic action.
720  *
721  * @param {object} entry - Event object.
722  */
723 function showEventModal(entry) {
724     if (!entry) return;
725     selectedEvent = entry;

```

```
726
727     const modalEl = document.getElementById('slc-event-detail-
modal');
728     if (!modalEl) return;
729
730     document.getElementById('slc-modal-time').textContent = en
try.timestamp;
731     document.getElementById('slc-modal-level').innerHTML = `
```



```

748     try {
749         const res = await fetch('/api/v1/system_logs/explain
    ', {
750             method: 'POST',
751             headers: { 'Content-Type': 'application/json' },
752             body: JSON.stringify({
753                 target_timestamp: entry.timestamp,
754                 window_minutes: 5,
755                 channel: entry.channel || currentChannel,
756                 event_id: entry.event_id || 0,
757                 provider: entry.provider || entry.source || '',
758                 level: entry.level || '',
759                 message: entry.message || '',
760                 target_entry: entry,
761                 query_text: `Диагностика события ${entry.provider
    || entry.source} (ID ${entry.event_id || 0}): ${entry.message
    || ''}`,
762             }),
763         });
764         if (!res.ok) throw new Error(`HTTP ${res.status}`);
765         const report = await res.json();
766         aiExplanation.innerHTML = `
767             <div class="mb-2"><strong class="text-info">Сводка
    :</strong> ${escapeHtml(report.summary)}</div>
768             <div class="mb-2"><strong class="text-warning">При
    чина / Анализ:</strong> ${escapeHtml(report.root_cause)}</div>
769             <h6 class="small text-uppercase text-muted fw-bold
    mb-1">Рекомендации:</h6>
770             <ul class="mb-0 ps-3">
771                 ${report.recommendations || []}.map(r => `<li>${
    escapeHtml(r)}</li>`).join('')
772             </ul>
773             `;
774         } catch (err) {
775             aiExplanation.innerHTML = `<div class="text-danger">
    Ошибка диагностики: ${escapeHtml(err.message)}</div>`;
776         }
777     };

```

```

778     }
779
780     const bsModal = new bootstrap.Modal(modalEl);
781     bsModal.show();
782 }
783
784 /**
785  * Export logs as downloadable file.
786  *
787  * @param {string} format - "json" or "csv".
788  */
789 function exportLogs(format) {
790     const level = (document.getElementById('slc-level-select')
791     || {}).value || '';
792     const search = (document.getElementById('slc-search-input'
793     ) || {}).value || '';
794     const limit = (document.getElementById('slc-limit-select')
795     || {}).value || '200';
796     const hours = (document.getElementById('slc-hours-select')
797     || {}).value || '24';
798
799     const url = `/api/v1/system_logs/export?format=${format}&c
800 hannel=${encodeURIComponent(currentChannel)}&level=${encodeURIComponent(level)}&search=${encodeURIComponent(search)}&hours=${
801 {hours}&limit=${limit}`;
802     window.open(url, '_blank');
803 }
804
805 /**
806  * Main initializer for System Log Center tab.
807  */
808 function initSystemLogsTab() {
809     if (window.isTabActive ? window.isTabActive('tab-system-lo
810 gs') : false) {
811         performSystemScan().then(() => {
812             loadEvents();
813         });
814     }
815 }

```

```
808
809     const scanBtn = document.getElementById('btn-slc-scan');
810     if (scanBtn) {
811         scanBtn.onclick = () => performSystemScan(true);
812     }
813
814     const refreshBtn = document.getElementById('btn-slc-refres
h');
815     if (refreshBtn) {
816         refreshBtn.onclick = loadEvents;
817     }
818
819     const jsonExportBtn = document.getElementById('btn-slc-exp
ort-json');
820     if (jsonExportBtn) {
821         jsonExportBtn.onclick = (e) => {
822             e.preventDefault();
823             exportLogs('json');
824         };
825     }
826
827     const csvExportBtn = document.getElementById('btn-slc-expo
rt-csv');
828     if (csvExportBtn) {
829         csvExportBtn.onclick = (e) => {
830             e.preventDefault();
831             exportLogs('csv');
832         };
833     }
834
835     const modeTabs = document.querySelectorAll('#slc-mode-tabs
.nav-link');
836     modeTabs.forEach((tab) => {
837         tab.onclick = () => {
838             modeTabs.forEach(t => t.classList.remove('active'));
839             tab.classList.add('active');
840             currentMode = tab.getAttribute('data-mode') || 'live';
841             renderCurrentModeView();
```

```
842 |     };
843 | });
844 |
845 | const auditBtn = document.getElementById('btn-slc-audit');
846 | if (auditBtn) {
847 |     auditBtn.onclick = () => {
848 |         modeTabs.forEach(t => {
849 |             if (t.getAttribute('data-mode') === 'audit') {
850 |                 t.classList.add('active');
851 |             } else {
852 |                 t.classList.remove('active');
853 |             }
854 |         });
855 |         currentMode = 'audit';
856 |         renderCurrentModeView();
857 |     };
858 | }
859 |
860 | const ragSearchBtn = document.getElementById('btn-slc-rag-
search');
861 | if (ragSearchBtn) {
862 |     ragSearchBtn.onclick = performRAGSearch;
863 | }
864 |
865 | const ragRebuildBtn = document.getElementById('btn-slc-rag
-rebuild');
866 | if (ragRebuildBtn) {
867 |     ragRebuildBtn.onclick = rebuildRAGIndex;
868 | }
869 |
870 | const ragInput = document.getElementById('slc-rag-input');
871 | if (ragInput) {
872 |     ragInput.onkeydown = (e) => {
873 |         if (e.key === 'Enter') {
874 |             e.preventDefault();
875 |             performRAGSearch();
876 |         }
877 |     };

```

```
878 |     }
879 |
880 |     const askAiBtn = document.getElementById('btn-slc-ask-ai')
      | ;
881 |     if (askAiBtn) {
882 |         askAiBtn.onclick = () => {
883 |             if (selectedEvent) {
884 |                 showEventModal(selectedEvent);
885 |             } else if (cachedEvents.length > 0) {
886 |                 selectedEvent = cachedEvents[0];
887 |                 showEventModal(selectedEvent);
888 |             }
889 |         };
890 |     }
891 |
892 |     const levelSelect = document.getElementById('slc-level-select');
893 |     if (levelSelect) levelSelect.onchange = loadEvents;
894 |
895 |     const hoursSelect = document.getElementById('slc-hours-select');
896 |     if (hoursSelect) hoursSelect.onchange = loadEvents;
897 |
898 |     const limitSelect = document.getElementById('slc-limit-select');
899 |     if (limitSelect) limitSelect.onchange = loadEvents;
900 |
901 |     const eventIdInput = document.getElementById('slc-eventid-input');
902 |     if (eventIdInput) eventIdInput.onchange = loadEvents;
903 |
904 |     let searchTimer = null;
905 |     const searchInput = document.getElementById('slc-search-input');
906 |     if (searchInput) {
907 |         searchInput.oninput = () => {
908 |             clearTimeout(searchTimer);
909 |             searchTimer = setTimeout(loadEvents, 350);
```

```

910     };
911 }
912
913     const channelFilterInput = document.getElementById('slc-channel-filter-input');
914     if (channelFilterInput) {
915         channelFilterInput.oninput = renderChannelsTree;
916     }
917
918     const liveSwitch = document.getElementById('slc-live-switch');
919     if (liveSwitch) {
920         liveSwitch.onChange = () => {
921             if (window.registerTabPoller && window.unregisterTabPoller) {
922                 if (liveSwitch.checked) {
923                     window.registerTabPoller('tab-system-logs', loadEvents, 4000, { pollerId: 'tab-system-logs_live', immediate: true });
924                 } else {
925                     window.unregisterTabPoller('tab-system-logs_live');
926                 }
927             } else {
928                 if (liveSwitch.checked) {
929                     liveIntervalTimer = setInterval(loadEvents, 4000);
930                 } else {
931                     clearInterval(liveIntervalTimer);
932                     liveIntervalTimer = null;
933                 }
934             }
935         };
936     }
937 }
938
939 window.activateSystemLogsTab = () => {
940     performSystemScan().then(() => loadEvents());
941 };

```

```
942 |  
943 | // Export initializer for admin UI  
944 | window.initSystemLogsTab = initSystemLogsTab;  
945 | }());  
946 |
```

index.html

```
1 <!--
2 =====
3 |
4 |
5 |
6 | Process Name: Windows Web Interface - Taskbar & Window Managem
7 | ent Tab
8 | =====
9 |
10 | Description:
11 |   Интерактивный пульт управления панелью задач Windows 10/11,
12 |   окнами рабочего
13 |   стола, закрепленными приложениями, ITaskbarList3 UX, Command
14 |   Registry
15 |   и историей изменений в telemetry.db с поддержкой отката (Roll
16 |   back).
17 |
18 | Usage Examples:
19 |   Web Route:
20 |     /windows/api/webgui/taskbar_tab/index.html
21 |
22 |   Tab Frame Integration:
23 |     <iframe src="/windows/api/webgui/taskbar_tab/index.html"
24 |     class="w-100 h-100 border-0"></iframe>
25 |
26 | File: index.html
27 | Project: ai-breadboard
28 | Package: windows/api/webgui/taskbar_tab
29 | Author: hypo69
30 | Copyright: © 2026 hypo69
31 | Updated: 2026-10-06 17:58:00
32 | =====
33 |
34 | -->
35 |
36 | <div class="container-fluid p-0 taskbar-tab-wrapper d-flex fle
37 | x-column" style="min-height: 100vh;">
```



```

28 | <!-- Header Card & Summary Stats -->
29 | <div class="card border-secondary-subtle mb-3 shadow-sm flex
    | -shrink-0">
30 |     <div class="card-body p-3">
31 |         <div class="d-flex flex-wrap align-items-center justify-
    | content-between gap-3">
32 |             <div class="d-flex align-items-center gap-2">
33 |                 <div class="fs-2">□ </div>
34 |                 <div>
35 |                     <h5 class="mb-0 fw-bold d-flex align-items-center
    | gap-2">
36 |                         <span>Панель задач и Окна (Windows Taskbar & Win
    | dows)</span>
37 |                         <span class="badge bg-primary-subtle text-primar
    | y border border-primary-subtle small fs-6" id="tb-os-badge">Wi
    | n32 / COM / Registry</span>
38 |                         <span class="badge bg-success-subtle text-succes
    | s border border-success-subtle small fs-6">telemetry.db Audit
    | & Rollback</span>
39 |                     </h5>
40 |                     <div class="text-muted small">
41 |                         Управление положением, автоскрытием, значками по
    | иска/виджетов, окнами рабочего стола, ITaskbarList3, историей
    | в telemetry.db и откатом.
42 |                     </div>
43 |                 </div>
44 |             </div>
45 |             <div class="d-flex align-items-center gap-2 flex-wrap"
    | >
46 |                 <button class="btn btn-outline-warning btn-sm d-flex
    | align-items-center gap-1 shadow-sm" id="btn-rollback-last-qui
    | ck" title="Откатить последнее действие">
47 |                     <i class="bi bi-arrow-counterclockwise"></i>
48 |                     <span>Откат последнего действия</span>
49 |                 </button>
50 |                 <button class="btn btn-outline-info btn-sm d-flex al
    | ign-items-center gap-1 shadow-sm" id="btn-refresh-taskbar-data
    | " title="Обновить состояние">

```

```

51         <i class="bi bi-arrow-clockwise"></i>
52         <span>Обновить</span>
53     </button>
54     <button class="btn btn-outline-secondary btn-sm d-fl
ex align-items-center gap-1 shadow-sm" id="btn-restart-explor
er-quick" title="Перезапустить Explorer">
55         <i class="bi bi-arrow-repeat"></i>
56         <span>Перезапуск Explorer</span>
57     </button>
58 </div>
59 </div>
60
61 <!-- Quick Metrics Bar -->
62 <div class="row g-2 mt-2 pt-2 border-top border-secondar
y-subtle small">
63     <div class="col-6 col-md-3">
64         <div class="p-2 rounded bg-body-tertiary border bord
er-secondary-subtle text-center">
65             <div class="text-muted small">Габариты панели</div
>
66             <div class="fw-bold text-info" id="metric-tb-rect"
>--</div>
67         </div>
68     </div>
69     <div class="col-6 col-md-3">
70         <div class="p-2 rounded bg-body-tertiary border bord
er-secondary-subtle text-center">
71             <div class="text-muted small">Выравнивание значков
</div>
72             <div class="fw-bold text-success" id="metric-tb-al
ign">--</div>
73         </div>
74     </div>
75     <div class="col-6 col-md-3">
76         <div class="p-2 rounded bg-body-tertiary border bord
er-secondary-subtle text-center">
77             <div class="text-muted small">Открытых окон</div>
78             <div class="fw-bold text-primary" id="metric-tb-wi

```

```

ndows">0</div>
79         </div>
80     </div>
81     <div class="col-6 col-md-3">
82         <div class="p-2 rounded bg-body-tertiary border border-secondary-subtle text-center">
83             <div class="text-muted small">Активное окно (Foreground)</div>
84             <div class="fw-bold text-warning text-truncate" id="metric-tb-fg">--</div>
85         </div>
86     </div>
87 </div>
88 </div>
89 </div>
90
91 <!-- Navigation Subtabs -->
92 <ul class="nav nav-pills mb-3 gap-2 flex-shrink-0" id="taskbarSubTabs" role="tablist">
93     <li class="nav-item" role="presentation">
94         <button class="nav-link active btn-sm" id="tab-btn-windows" data-bs-toggle="pill" data-bs-target="#panel-windows" type="button" role="tab">
95             <i class="bi bi-window-stack me-1"></i> Окона рабочего стола (<span id="badge-windows-count">0</span>)
96         </button>
97     </li>
98     <li class="nav-item" role="presentation">
99         <button class="nav-link btn-sm" id="tab-btn-settings" data-bs-toggle="pill" data-bs-target="#panel-settings" type="button" role="tab">
100             <i class="bi bi-sliders me-1"></i> Настройки панели задач
101         </button>
102     </li>
103     <li class="nav-item" role="presentation">
104         <button class="nav-link btn-sm" id="tab-btn-apps" data-bs-toggle="pill" data-bs-target="#panel-apps" type="button" rol

```

```

e="tab">
105     <i class="bi bi-pin-angle me-1"></i> Закрепленные и За
пуск
106     </button>
107     </li>
108     <li class="nav-item" role="presentation">
109         <button class="nav-link btn-sm" id="tab-btn-ux" data-bs-
toggle="pill" data-bs-target="#panel-ux" type="button" role="t
ab">
110             <i class="bi bi-bar-chart-steps me-1"></i> Индикаторы
ITaskbarList3
111             </button>
112         </li>
113         <li class="nav-item" role="presentation">
114             <button class="nav-link btn-sm" id="tab-btn-history" dat
a-bs-toggle="pill" data-bs-target="#panel-history" type="butto
n" role="tab">
115                 <i class="bi bi-clock-history me-1"></i> История в tel
emetry.db (<span id="badge-history-count">0</span>)
116                 </button>
117             </li>
118             <li class="nav-item" role="presentation">
119                 <button class="nav-link btn-sm" id="tab-btn-commands" da
ta-bs-toggle="pill" data-bs-target="#panel-commands" type="but
ton" role="tab">
120                     <i class="bi bi-terminal me-1"></i> Command Registry (
260)
121                 </button>
122             </li>
123     </ul>
124
125     <!-- Subtabs Content -->
126     <div class="tab-content flex-grow-1" id="taskbarSubTabsConte
nt">
127
128         <!-- 1. WINDOWS MANAGEMENT -->
129         <div class="tab-pane fade show active" id="panel-windows"
role="tabpanel">

```

```

130 |         <div class="card border-secondary-subtle shadow-sm h-100
    | ">
131 |             <div class="card-header bg-body-tertiary d-flex flex-w
    | rap align-items-center justify-content-between gap-2 p-2.5">
132 |                 <!-- Filters -->
133 |                 <div class="d-flex align-items-center gap-2 flex-gro
    | w-1" style="max-width: 500px;">
134 |                     <div class="input-group input-group-sm">
135 |                         <span class="input-group-text"><i class="bi bi-s
    | earch"></i></span>
136 |                         <input type="text" class="form-control" id="inpu
    | t-search-windows" placeholder="Фильтр по заголовку или процесс
    | у...">
137 |                     </div>
138 |                 </div>
139 |                 <!-- Batch Actions -->
140 |                 <div class="d-flex align-items-center gap-1.5 flex-w
    | rap">
141 |                     <button class="btn btn-sm btn-outline-secondary" i
    | d="btn-batch-minimize-all" title="Свернуть все окна (Показать
    | рабочий стол)">
142 |                         <i class="bi bi-dash-square me-1"></i> Свернуть
    | все
143 |                     </button>
144 |                     <button class="btn btn-sm btn-outline-secondary" i
    | d="btn-batch-restore-all" title="Восстановить все окна">
145 |                         <i class="bi bi-window me-1"></i> Развернуть все
146 |                     </button>
147 |                 </div>
148 |             </div>
149 |
150 |             <div class="card-body p-0 table-responsive" style="max
    | -height: 60vh;">
151 |                 <table class="table table-hover table-sm align-middl
    | e mb-0" id="table-windows-list">
152 |                     <thead class="table-dark sticky-top small">
153 |                         <tr>
154 |                             <th style="width: 90px;">HWND</th>

```

```

155 |         <th>Заголовок окна</th>
156 |         <th style="width: 150px;">Процесс (PID)</th>
157 |         <th style="width: 140px;">Размеры (WxH)</th>
158 |         <th style="width: 100px;">Состояние</th>
159 |         <th style="width: 220px;" class="text-end">Дей
    | ствия</th>
160 |     </tr>
161 | </thead>
162 | <tbody id="tbody-windows-list" class="small">
163 |     <tr>
164 |         <td colspan="6" class="text-center text-muted
    | p-4">
165 |             <div class="spinner-border spinner-border-sm
    | text-primary me-2"></div> Загрузка списка открытых окон...
166 |         </td>
167 |     </tr>
168 | </tbody>
169 | </table>
170 | </div>
171 | </div>
172 | </div>
173 |
174 | <!-- 2. TASKBAR SETTINGS -->
175 | <div class="tab-pane fade" id="panel-settings" role="tabpa
    | nel">
176 |     <div class="card border-secondary-subtle shadow-sm">
177 |         <div class="card-header bg-body-tertiary fw-semibold">
178 |             <i class="bi bi-sliders me-1 text-primary"></i> Сист
    | емные настройки и внешний вид панели задач (с фиксацией в tele
    | metry.db)
179 |         </div>
180 |         <div class="card-body p-3">
181 |             <form id="form-taskbar-settings">
182 |                 <div class="row g-3">
183 |                     <!-- Alignment -->
184 |                     <div class="col-md-6">
185 |                         <label class="form-label fw-semibold small">Вы
    | равнивание значков (Windows 11)</label>

```

```

186 |         <div class="btn-group w-100" role="group">
187 |             <input type="radio" class="btn-check" name="
| tb-alignment" id="align-left" value="0">
188 |             <label class="btn btn-outline-primary btn-sm
| " for="align-left">
189 |                 <i class="bi bi-text-left me-1"></i> По ле
| вому краю
190 |             </label>
191 |             <input type="radio" class="btn-check" name="
| tb-alignment" id="align-center" value="1">
192 |             <label class="btn btn-outline-primary btn-sm
| " for="align-center">
193 |                 <i class="bi bi-text-center me-1"></i> По
| центру
194 |             </label>
195 |         </div>
196 |     </div>
197 |
198 |     <!-- Search mode -->
199 |     <div class="col-md-6">
200 |         <label class="form-label fw-semibold small" fo
| r="select-search-mode">Режим строки поиска</label>
201 |         <select class="form-select form-select-sm" id=
| "select-search-mode">
202 |             <option value="0">Скрыт</option>
203 |             <option value="1">Только значок поиска</opti
| on>
204 |             <option value="2">Поле поиска</option>
205 |             <option value="3">Кнопка поиска</option>
206 |         </select>
207 |     </div>
208 |
209 |     <!-- Switches -->
210 |     <div class="col-md-4">
211 |         <div class="form-check form-switch mt-2">
212 |             <input class="form-check-input" type="checkb
| ox" id="switch-auto-hide">
213 |             <label class="form-check-label small fw-semi

```

```

    bold" for="switch-auto-hide">Автоматически скрывать панель</la
    bel>
214         </div>
215     </div>
216
217         <div class="col-md-4">
218             <div class="form-check form-switch mt-2">
219                 <input class="form-check-input" type="checkb
    ox" id="switch-widgets">
220                 <label class="form-check-label small fw-semi
    bold" for="switch-widgets">Показывать виджеты (Widgets)</label
    >
221             </div>
222         </div>
223
224         <div class="col-md-4">
225             <div class="form-check form-switch mt-2">
226                 <input class="form-check-input" type="checkb
    ox" id="switch-task-view">
227                 <label class="form-check-label small fw-semi
    bold" for="switch-task-view">Кнопка Task View</label>
228             </div>
229         </div>
230
231         <div class="col-md-4">
232             <div class="form-check form-switch">
233                 <input class="form-check-input" type="checkb
    ox" id="switch-copilot">
234                 <label class="form-check-label small fw-semi
    bold" for="switch-copilot">Кнопка Copilot</label>
235             </div>
236         </div>
237
238         <div class="col-md-4">
239             <div class="form-check form-switch">
240                 <input class="form-check-input" type="checkb
    ox" id="switch-badges">
241                 <label class="form-check-label small fw-semi

```



```

    bold" for="switch-badges">Бейджи уведомлений на значках</label
  >
242         </div>
243     </div>
244
245         <div class="col-md-4">
246             <div class="form-check form-switch">
247                 <input class="form-check-input" type="checkbox" id="switch-restart-explorer">
248                 <label class="form-check-label small fw-semi
bold text-warning" for="switch-restart-explorer">Перезапустить
Explorer после сохранения</label>
249             </div>
250         </div>
251     </div>
252
253         <div class="mt-4 pt-3 border-top d-flex justify-co
content-end gap-2">
254             <button type="submit" class="btn btn-primary btn
-sm px-3 shadow-sm">
255                 <i class="bi bi-check2-circle me-1"></i> Приме
нить настройки
256             </button>
257         </div>
258     </form>
259 </div>
260 </div>
261 </div>
262
263 <!-- 3. PINNED APPS & LAUNCH -->
264 <div class="tab-pane fade" id="panel-apps" role="tabpanel"
>
265     <div class="row g-3">
266         <!-- Pinned List -->
267         <div class="col-md-7">
268             <div class="card border-secondary-subtle shadow-sm h
-100">
269                 <div class="card-header bg-body-tertiary fw-semibo

```

```

ld d-flex justify-content-between align-items-center">
270     <span><i class="bi bi-pin-angle me-1 text-primar
y"></i> Закрепленные на панели задачи ярлыки</span>
271     <button class="btn btn-sm btn-outline-secondary
py-0 px-2" id="btn-refresh-pinned">
272         <i class="bi bi-arrow-clockwise"></i>
273     </button>
274 </div>
275 <div class="card-body p-2" id="pinned-apps-contain
er">
276     <div class="text-muted small p-3 text-center">За
грузка закрепленных приложений...</div>
277 </div>
278 </div>
279 </div>
280
281 <!-- App Launcher -->
282 <div class="col-md-5">
283     <div class="card border-secondary-subtle shadow-sm h
-100">
284         <div class="card-header bg-body-tertiary fw-semibo
ld">
285             <i class="bi bi-play-circle me-1 text-success"><
/i> Быстрый запуск приложения
286         </div>
287         <div class="card-body p-3">
288             <form id="form-launch-app">
289                 <div class="mb-3">
290                     <label class="form-label small fw-semibold">
Путь к файлу / Команда</label>
291                     <input type="text" class="form-control form-
control-sm" id="launch-app-path" placeholder="notepad.exe, mse
dge, explorer..." required>
292                 </div>
293                 <div class="mb-3">
294                     <label class="form-label small fw-semibold">
Аргументы</label>
295                     <input type="text" class="form-control form-

```

```

control-sm" id="launch-app-args" placeholder="--incognito, pat
h...">
296         </div>
297         <div class="mb-3 form-check form-switch">
298             <input class="form-check-input" type="checkb
ox" id="launch-app-admin">
299             <label class="form-check-label small fw-semi
bold" for="launch-app-admin">Запуск от имени администратора (U
AC)</label>
300         </div>
301         <button type="submit" class="btn btn-success b
tn-sm w-100">
302             <i class="bi bi-rocket-takeoff me-1"></i> За
пустить
303         </button>
304     </form>
305 </div>
306 </div>
307 </div>
308 </div>
309 </div>
310
311 <!-- 4. TASKBAR UX & ITASKBARLIST3 -->
312 <div class="tab-pane fade" id="panel-ux" role="tabpanel">
313     <div class="card border-secondary-subtle shadow-sm">
314         <div class="card-header bg-body-tertiary fw-semibold">
315             <i class="bi bi-bar-chart-steps me-1 text-primary"><
/i> Интерактивный эмулятор ITaskbarList3 (Индикаторы прогресса
и оверлеи)
316         </div>
317         <div class="card-body p-3">
318             <div class="row g-3">
319                 <div class="col-md-6">
320                     <label class="form-label small fw-semibold">Целе
вое окно (HWND)</label>
321                     <select class="form-select form-select-sm" id="s
elect-ux-target-window"></select>
322                 </div>

```

```

323 |
324 |         <div class="col-md-6">
325 |             <label class="form-label small fw-semibold">Режи
326 | м индикатора прогресса</label>
327 |             <select class="form-select form-select-sm" id="s
328 | elect-ux-progress-state">
329 |                 <option value="normal">Обычный (Зеленый / Norm
330 | al)</option>
331 |                 <option value="indeterminate">Неопределенный (
332 | Бегущий / Indeterminate)</option>
333 |                 <option value="error">Ошибка (Красный / Error)
334 | </option>
335 |                 <option value="paused">Пауза (Желтый / Paused)
336 | </option>
337 |                 <option value="no_progress">Сбросить (No Progr
338 | ess)</option>
339 |             </select>
340 |         </div>
341 |
342 |         <div class="col-md-6">
343 |             <label class="form-label small fw-semibold">Прогр
344 | есс выполнения (%): <span id="label-progress-val">50%</span><
345 | /label>
346 |             <input type="range" class="form-range" id="range
347 | -ux-progress" min="0" max="100" value="50">
348 |             </div>
349 |
350 |         <div class="col-md-6 d-flex align-items-end">
351 |             <button class="btn btn-primary btn-sm w-100" id=
352 | "btn-apply-ux-progress">
353 |                 <i class="bi bi-send me-1"></i> Передать прогр
354 | есс на панель задач
355 |             </button>
356 |         </div>
357 |     </div>
358 | </div>

```

```

349 |
350 |     <!-- 5. HISTORY & ROLLBACK (telemetry.db) -->
351 |     <div class="tab-pane fade" id="panel-history" role="tabpanel"
    | el">
352 |         <div class="card border-secondary-subtle shadow-sm">
353 |             <div class="card-header bg-body-tertiary d-flex flex-wrap align-items-center justify-content-between gap-2 p-2.5">
354 |                 <div class="d-flex align-items-center gap-2">
355 |                     <i class="bi bi-clock-history me-1 text-primary"><
    | /i>
356 |                     <span class="fw-semibold">Журнал изменений в SQLite telemetry.db</span>
357 |                 </div>
358 |                 <div class="d-flex align-items-center gap-2">
359 |                     <button class="btn btn-outline-info btn-sm" id="btn-refresh-history">
360 |                         <i class="bi bi-arrow-clockwise me-1"></i> Обновить историю
361 |                     </button>
362 |                 </div>
363 |             </div>
364 |
365 |             <div class="card-body p-0 table-responsive" style="max-height: 55vh;">
366 |                 <table class="table table-hover table-sm align-middle mb-0" id="table-history-list">
367 |                     <thead class="table-dark sticky-top small">
368 |                         <tr>
369 |                             <th style="width: 60px;">ID</th>
370 |                             <th style="width: 160px;">Время</th>
371 |                             <th style="width: 180px;">Команда</th>
372 |                             <th>Детали и состояния</th>
373 |                             <th style="width: 110px;">Статус</th>
374 |                             <th style="width: 120px;" class="text-end">Откат</th>
375 |                         </tr>
376 |                     </thead>
377 |                     <tbody id="tbody-history-list" class="small">

```

```

378         <tr>
379             <td colspan="6" class="text-center text-muted
p-4">Загрузка журнала изменений из telemetry.db...</td>
380         </tr>
381     </tbody>
382 </table>
383 </div>
384 </div>
385 </div>
386
387 <!-- 6. COMMAND REGISTRY (260) -->
388 <div class="tab-pane fade" id="panel-commands" role="tabpa
nel">
389     <div class="card border-secondary-subtle shadow-sm">
390         <div class="card-header bg-body-tertiary d-flex flex-w
rap align-items-center justify-content-between gap-2 p-2.5">
391             <div class="d-flex align-items-center gap-2">
392                 <i class="bi bi-terminal me-1 text-primary"></i>
393                 <span class="fw-semibold">Каталог команд Control P
lane</span>
394                 <span class="badge bg-secondary-subtle text-second
ary small" id="badge-commands-count">0</span>
395             </div>
396             <div class="d-flex align-items-center gap-2">
397                 <select class="form-select form-select-sm" id="fil
ter-cmd-category" style="width: 140px;">
398                     <option value="">Все категории</option>
399                     <option value="window">window</option>
400                     <option value="taskbar">taskbar</option>
401                     <option value="shell">shell</option>
402                     <option value="policy">policy</option>
403                 </select>
404                 <select class="form-select form-select-sm" id="fil
ter-cmd-risk" style="width: 130px;">
405                     <option value="">Любой риск</option>
406                     <option value="safe">□ safe</option>
407                     <option value="caution">□ caution</option>
408                     <option value="admin">□ admin</option>

```

```

409         <option value="high">□ high</option>
410     </select>
411 </div>
412 </div>
413
414     <div class="card-body p-0 table-responsive" style="max
-height: 55vh;">
415         <table class="table table-hover table-sm align-middl
e mb-0" id="table-commands-list">
416             <thead class="table-dark sticky-top small">
417                 <tr>
418                     <th style="width: 220px;">ID Команды</th>
419                     <th style="width: 100px;">Класс</th>
420                     <th style="width: 90px;">Риск</th>
421                     <th>Описание</th>
422                     <th>API / Механизм</th>
423                     <th style="width: 100px;" class="text-end">Дей
ствие</th>
424                 </tr>
425             </thead>
426             <tbody id="tbody-commands-list" class="small">
427                 <tr>
428                     <td colspan="6" class="text-center text-muted
p-4">Загрузка каталога команд...</td>
429                 </tr>
430             </tbody>
431         </table>
432     </div>
433 </div>
434 </div>
435
436 </div>
437 </div>
438

```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Web Interface - Taskbar Tab Controller
4   * =====
5   * Description:
6   *   Клиентский контроллер интерактивного управления панелью задач, окнами
7   *   и историей изменений в telemetry.db с поддержкой Rollback
8   *
9   * Usage Examples:
10  *   HTML Integration:
11  *     <script src="/windows/api/webgui/taskbar_tab/main.js?v=20261006_v2" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/taskbar_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-06 17:58:00
19  * =====
20  */
21
22 let _windowsCache = [];
23 let _commandsCache = [];
24 let _historyCache = [];
25
26 export async function init() {
27   bindEvents();
28   await refreshAll();
29 }
```



```
30
31 function bindEvents() {
32   document.getElementById('btn-refresh-taskbar-data')?.addEventListener('click', () => refreshAll());
33   document.getElementById('btn-restart-explorer-quick')?.addEventListener('click', () => restartExplorer());
34   document.getElementById('btn-rollback-last-quick')?.addEventListener('click', () => rollbackLast());
35   document.getElementById('btn-refresh-history')?.addEventListener('click', () => loadHistory());
36   document.getElementById('input-search-windows')?.addEventListener('input', () => renderWindowsTable());
37   document.getElementById('btn-batch-minimize-all')?.addEventListener('click', () => executeBatchWindowAction('minimize_all'
));
38   document.getElementById('btn-batch-restore-all')?.addEventListener('click', () => executeBatchWindowAction('restore_all'
));
39   document.getElementById('form-taskbar-settings')?.addEventListener('submit', (e) => saveTaskbarSettings(e));
40   document.getElementById('form-launch-app')?.addEventListener('submit', (e) => handleLaunchApp(e));
41   document.getElementById('btn-refresh-pinned')?.addEventListener('click', () => loadPinnedApps());
42
43   const range = document.getElementById('range-ux-progress');
44   const label = document.getElementById('label-progress-val');
45   if (range && label) {
46     range.addEventListener('input', () => { label.textContent
= `${range.value}%`; });
47   }
48   document.getElementById('btn-apply-ux-progress')?.addEventListener('click', () => handleApplyUxProgress());
49
50   document.getElementById('filter-cmd-category')?.addEventListener('change', () => renderCommandsTable());
51   document.getElementById('filter-cmd-risk')?.addEventListener('change', () => renderCommandsTable());
```

```

52 | }
53 |
54 | export async function refreshAll() {
55 |     await Promise.all([
56 |         loadSummary(),
57 |         loadWindows(),
58 |         loadSettings(),
59 |         loadPinnedApps(),
60 |         loadHistory(),
61 |         loadCommandCatalog(),
62 |     ]);
63 | }
64 |
65 | async function loadSummary() {
66 |     try {
67 |         const res = await fetch('/api/v1/taskbar/summary');
68 |         if (!res.ok) return;
69 |         const data = await res.json();
70 |
71 |         const rectEl = document.getElementById('metric-tb-rect');
72 |         if (rectEl) {
73 |             rectEl.textContent = `${data.taskbar_rect.width}x${data.
74 | taskbar_rect.height} px`;
75 |         }
76 |         const alignEl = document.getElementById('metric-tb-align')
77 | ;
78 |         if (alignEl) {
79 |             alignEl.textContent = data.settings.alignment === 1 ? 'П
80 | о центру' : 'По левому краю';
81 |         }
82 |         const winEl = document.getElementById('metric-tb-windows')
83 | ;
84 |         if (winEl) {
85 |             winEl.textContent = `${data.visible_windows_count} / ${d
86 | ata.windows_count}`;
87 |         }

```

```

85 |
86 |     const fgEl = document.getElementById('metric-tb-fg');
87 |     if (fgEl) {
88 |         fgEl.textContent = data.foreground_window?.title || '-';
89 |         fgEl.title = data.foreground_window?.title || '';
90 |     }
91 |
92 |     const osBadge = document.getElementById('tb-os-badge');
93 |     if (osBadge) {
94 |         osBadge.textContent = data.is_win11 ? 'Windows 11 (Cente
95 | r/Left Taskbar)' : 'Windows 10 / Classic';
96 |     }
97 | } catch (err) {
98 |     console.error('Ошибка загрузки summary:', err);
99 | }
100 |
101 | async function loadWindows() {
102 |     const tbody = document.getElementById('tbody-windows-list');
103 |     try {
104 |         const res = await fetch('/api/v1/taskbar/windows?only_visi
105 | ble=false&only_taskbar=false');
106 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
107 |         _windowsCache = await res.json();
108 |         renderWindowsTable();
109 |         populateWindowSelects();
110 |     } catch (err) {
111 |         if (tbody) {
112 |             tbody.innerHTML = `<tr><td colspan="6" class="text-dange
113 | r p-3 text-center">Ошибка: ${err.message}</td></tr>`;
114 |         }
115 |     }
116 | }
117 |
118 | function renderWindowsTable() {
119 |     const tbody = document.getElementById('tbody-windows-list');
120 |     const filter = (document.getElementById('input-search-window
121 | s')?.value || '').toLowerCase();

```

```
119 |   if (!tbody) return;
120 |
121 |   const filtered = _windowsCache.filter(w => {
122 |     if (!filter) return w.title;
123 |     return w.title.toLowerCase().includes(filter) || w.process
124 | _name.toLowerCase().includes(filter) || String(w.hwnd).include
125 | s(filter);
126 |   });
127 |
128 |   const badge = document.getElementById('badge-windows-count')
129 | ;
130 |   if (badge) badge.textContent = filtered.length;
131 |
132 |   if (filtered.length === 0) {
133 |     tbody.innerHTML = `<tr><td colspan="6" class="text-muted p
134 | -4 text-center">Окна по фильтру не найдены</td></tr>`;
135 |     return;
136 |   }
137 |
138 |   tbody.innerHTML = filtered.map(w => {
139 |     const isFg = w.is_foreground ? '<span class="badge bg-warn
140 | ing text-dark me-1">Фокус</span>' : '';
141 |     const stateBadge = w.is_minimized
142 |       ? '<span class="badge bg-secondary">Свернуто</span>'
143 |       : w.is_maximized
144 |       ? '<span class="badge bg-success">Развернуто</span>'
145 |       : '<span class="badge bg-info text-dark">Нормальное</s
146 | pan>';
147 |
148 |     return `
149 |       <tr class="${w.is_foreground ? 'table-active' : ''}">
150 |         <td><code class="user-select-all">${w.hwnd}</code></td>
151 |
152 |         <td class="text-truncate" style="max-width: 250px;" ti
153 | tle="${escapeHtml(w.title)}">
154 |           ${isFg}<strong>${escapeHtml(w.title || '(Без названи
155 | я)')}</strong>
156 |         </td>
```

```

148         <td>
149             <div class="fw-semibold text-truncate" style="max-wi
150 dth: 140px;">${escapeHtml(w.process_name || '-')}</div>
151             <div class="text-muted small">PID: ${w.process_id}</
152 div>
153         </td>
154         <td><small class="text-muted">${w.rect.width} × ${w.re
155 ct.height}</small></td>
156         <td>${stateBadge}</td>
157         <td class="text-end">
158             <div class="btn-group btn-group-sm">
159                 <button class="btn btn-outline-primary" onclick="w
160 indow._taskbarActions.activate(${w.hwnd})" title="Активировать
161 "><i class="bi bi-lightning-fill"></i></button>
162                 <button class="btn btn-outline-secondary" onclick=
163 "window._taskbarActions.minimize(${w.hwnd})" title="Свернуть">
164 <i class="bi bi-dash"></i></button>
165                 <button class="btn btn-outline-secondary" onclick=
166 "window._taskbarActions.maximize(${w.hwnd})" title="Развернуть
167 "><i class="bi bi-square"></i></button>
168                 <button class="btn btn-outline-secondary" onclick=
169 "window._taskbarActions.restore(${w.hwnd})" title="Восстановит
170 ь"><i class="bi bi-window"></i></button>
171                 <button class="btn btn-outline-danger" onclick="wi
172 ndow._taskbarActions.close(${w.hwnd})" title="Закрыть"><i clas
173 s="bi bi-x-lg"></i></button>
174             </div>
175         </td>
176     </tr>
177 `;
178 }).join('');
179 }
180
181 function populateWindowSelects() {
182     const sel = document.getElementById('select-ux-target-window
183 ');
184     if (!sel) return;
185     const named = _windowsCache.filter(w => w.title);

```

```
172 |   sel.innerHTML = named.map(w => `<option value="${w.hwnd}">${
    | w.hwnd} - ${escapeHtml(w.title.slice(0, 40))} (${w.process_name})</option>`).join('');
173 | }
174 |
175 | // Window Actions
176 | window._taskbarActions = {
177 |   activate: async (hwnd) => {
178 |     await fetch(`/api/v1/taskbar/windows/${hwnd}/activate`, {
    | method: 'POST' });
179 |     await loadWindows();
180 |     await loadHistory();
181 |   },
182 |   minimize: async (hwnd) => {
183 |     await fetch(`/api/v1/taskbar/windows/${hwnd}/minimize`, {
    | method: 'POST' });
184 |     await loadWindows();
185 |     await loadHistory();
186 |   },
187 |   maximize: async (hwnd) => {
188 |     await fetch(`/api/v1/taskbar/windows/${hwnd}/maximize`, {
    | method: 'POST' });
189 |     await loadWindows();
190 |     await loadHistory();
191 |   },
192 |   restore: async (hwnd) => {
193 |     await fetch(`/api/v1/taskbar/windows/${hwnd}/restore`, { method: '
    | POST' });
194 |     await loadWindows();
195 |     await loadHistory();
196 |   },
197 |   close: async (hwnd) => {
198 |     await fetch(`/api/v1/taskbar/windows/${hwnd}`, { method: '
    | DELETE' });
199 |     await loadWindows();
200 |     await loadHistory();
201 |   }
202 | };
```

```
203
204 async function executeBatchWindowAction(action) {
205   try {
206     const res = await fetch('/api/v1/taskbar/windows/batch', {
207       method: 'POST',
208       headers: { 'Content-Type': 'application/json' },
209       body: JSON.stringify({ action })
210     });
211     if (res.ok) {
212       if (window.toast) window.toast.success('Действие выполне
но', `Пакетное действие: ${action}`);
213       await loadWindows();
214       await loadHistory();
215     }
216   } catch (err) {
217     if (window.toast) window.toast.error('Ошибка', err.message
);
218   }
219 }
220
221 async function loadSettings() {
222   try {
223     const res = await fetch('/api/v1/taskbar/settings');
224     if (!res.ok) return;
225     const s = await res.json();
226
227     const alignVal = s.alignment === 1 ? '1' : '0';
228     const alignRadio = document.querySelector(`input[name="tb-
alignment"][value="${alignVal}"]`);
229     if (alignRadio) alignRadio.checked = true;
230
231     const selSearch = document.getElementById('select-search-m
ode');
232     if (selSearch) selSearch.value = String(s.search_mode);
233
234     setCheck('switch-auto-hide', s.auto_hide);
235     setCheck('switch-widgets', s.widgets_visible);
236     setCheck('switch-task-view', s.task_view_visible);
```

```
237     setCheck('switch-copilot', s.copilot_visible);
238     setCheck('switch-badges', s.badges_enabled);
239 } catch (err) {
240     console.error('Ошибка настроек:', err);
241 }
242 }
243
244 function setCheck(id, val) {
245     const el = document.getElementById(id);
246     if (el) el.checked = Boolean(val);
247 }
248
249 async function saveTaskbarSettings(e) {
250     e.preventDefault();
251     const alignRadio = document.querySelector('input[name="tb-al
252     alignment"]:checked');
253     const alignment = alignRadio ? parseInt(alignRadio.value, 10
254     ) : 1;
255     const search_mode = parseInt(document.getElementById('select
256     -search-mode')?.value || '1', 10);
257     const auto_hide = document.getElementById('switch-auto-hide'
258     )?.checked || false;
259     const widgets_visible = document.getElementById('switch-widg
260     ets')?.checked || false;
261     const task_view_visible = document.getElementById('switch-ta
262     sk-view')?.checked || false;
263     const copilot_visible = document.getElementById('switch-copi
264     lot')?.checked || false;
265     const badges_enabled = document.getElementById('switch-badg
266     es')?.checked || false;
267     const restart_explorer = document.getElementById('switch-res
268     tart-explorer')?.checked || false;
269
270     try {
271         const res = await fetch('/api/v1/taskbar/settings', {
272             method: 'PUT',
273             headers: { 'Content-Type': 'application/json' },
274             body: JSON.stringify({
```



```
266 |         alignment,
267 |         search_mode,
268 |         auto_hide,
269 |         widgets_visible,
270 |         task_view_visible,
271 |         copilot_visible,
272 |         badges_enabled,
273 |         restart_explorer,
274 |     })
275 | });
276 | if (res.ok) {
277 |     if (window.toast) window.toast.success('Настройки сохран
278 | ены', 'Параметры панели задач успешно зафиксированы в telemetr
279 | y.db');
280 |     await refreshAll();
281 | } catch (err) {
282 |     if (window.toast) window.toast.error('Ошибка сохранения',
283 | err.message);
284 | }
285 | async function restartExplorer() {
286 |     if (!confirm('Перезапустить процесс Explorer? Экран на 1-2 с
287 | екунды обновится.')) return;
288 |     try {
289 |         const res = await fetch('/api/v1/taskbar/settings', {
290 |             method: 'PUT',
291 |             headers: { 'Content-Type': 'application/json' },
292 |             body: JSON.stringify({ restart_explorer: true })
293 |         });
294 |         if (res.ok && window.toast) {
295 |             window.toast.success('Explorer перезапущен', 'Оболочка W
296 | indows успешно перезапущена');
297 |         } catch (err) {
298 |             if (window.toast) window.toast.error('Ошибка', err.message
299 | );
```

```

298 |     }
299 | }
300 |
301 | async function loadPinnedApps() {
302 |     const container = document.getElementById('pinned-apps-container');
303 |     try {
304 |         const res = await fetch('/api/v1/taskbar/apps');
305 |         if (!res.ok) return;
306 |         const apps = await res.json();
307 |         if (!container) return;
308 |
309 |         if (apps.length === 0) {
310 |             container.innerHTML = `<div class="text-muted small p-3 text-center">Нет закрепленных ярлыков</div>`;
311 |             return;
312 |         }
313 |
314 |         container.innerHTML = `<div class="list-group list-group-flush">${apps.map(a => `
315 |             <div class="list-group-item d-flex justify-content-between align-items-center p-2">
316 |                 <div class="d-flex align-items-center gap-2">
317 |                     <i class="bi bi-app text-primary"></i>
318 |                     <div>
319 |                         <div class="fw-semibold small">${escapeHtml(a.name)}</div>
320 |                         <div class="text-muted font-monospace" style="font-size: 0.72rem;">${escapeHtml(a.target_path || a.link_path)}</div>
321 |                     </div>
322 |                 </div>
323 |             </div>
324 |         `).join('')}</div>`;
325 |     } catch (err) {
326 |         if (container) container.innerHTML = `<div class="text-danger small p-2">Ошибка: ${err.message}</div>`;
327 |     }

```

```

328 }
329
330 async function handleLaunchApp(e) {
331     e.preventDefault();
332     const path = document.getElementById('launch-app-path')?.value || '';
333     const args = document.getElementById('launch-app-args')?.value || '';
334     const admin = document.getElementById('launch-app-admin')?.checked || false;
335
336     try {
337         const res = await fetch('/api/v1/taskbar/apps/launch', {
338             method: 'POST',
339             headers: { 'Content-Type': 'application/json' },
340             body: JSON.stringify({ app_path: path, arguments: args |
341 | null, admin })
342         });
343         const data = await res.json();
344         if (res.ok) {
345             if (window.toast) window.toast.success('Приложение запущено', data.message);
346             document.getElementById('launch-app-path').value = '';
347             setTimeout(() => { loadWindows(); loadHistory(); }, 1000);
348         } else {
349             if (window.toast) window.toast.error('Ошибка запуска', data.detail || 'Сбой');
350         }
351     } catch (err) {
352         if (window.toast) window.toast.error('Ошибка сети', err.message);
353     }
354
355     async function handleApplyUxProgress() {
356         const hwnd = parseInt(document.getElementById('select-ux-target-window')?.value || '0', 10);

```

```

357 |   const state = document.getElementById('select-ux-progress-st
    |   ate')?.value || 'normal';
358 |   const val = parseInt(document.getElementById('range-ux-progr
    |   ess')?.value || '50', 10);
359 |
360 |   if (!hwnd) {
361 |     if (window.toast) window.toast.warning('Внимание', 'Выбери
    |   те целевое окно');
362 |     return;
363 |   }
364 |
365 |   try {
366 |     const res = await fetch('/api/v1/taskbar/ux/progress', {
367 |       method: 'POST',
368 |       headers: { 'Content-Type': 'application/json' },
369 |       body: JSON.stringify({ hwnd, state, completed: val, tota
    |   l: 100 })
370 |     });
371 |     if (res.ok && window.toast) {
372 |       window.toast.success('ITaskbarList3', `Прогресс ${val}%
    |   передан окну HWND ${hwnd}`);
373 |       await loadHistory();
374 |     }
375 |   } catch (err) {
376 |     if (window.toast) window.toast.error('Ошибка', err.message
    |   );
377 |   }
378 | }
379 |
380 | // History & Rollback
381 | async function loadHistory() {
382 |   const tbody = document.getElementById('tbody-history-list');
383 |   try {
384 |     const res = await fetch('/api/v1/taskbar/history?limit=50'
    |   );
385 |     if (!res.ok) return;
386 |     _historyCache = await res.json();
387 |     renderHistoryTable();

```

```

388 |   } catch (err) {
389 |     if (tbody) {
390 |       tbody.innerHTML = `|<td colspan="6" class="text-dange
r p-3 text-center">Ошибка: ${err.message}</td></tr>`;
391 |     }
392 |   }
393 | }
394 |
395 | function renderHistoryTable() {
396 |   const tbody = document.getElementById('tbody-history-list');
397 |   const badge = document.getElementById('badge-history-count')
    ;
398 |   if (badge) badge.textContent = _historyCache.length;
399 |   if (!tbody) return;
400 |
401 |   if (_historyCache.length === 0) {
402 |     tbody.innerHTML = `|<td colspan="6" class="text-muted p
-4 text-center">История в telemetry.db пуста</td></tr>`;
403 |     return;
404 |   }
405 |
406 |   tbody.innerHTML = _historyCache.map(h => {
407 |     const rolledBadge = h.is_rolled_back
408 |       ? '<span class="badge bg-secondary">Откачено</span>'
409 |       : '<span class="badge bg-success-subtle text-success bor
der border-success-subtle">Активно</span>';
410 |
411 |     const canRollback = !h.is_rolled_back && h.previous_state;
412 |     const timeShort = (h.timestamp || '').replace('T', ' ').sl
ice(0, 19);
413 |
414 |     let detailsStr = '';
415 |     if (h.previous_state || h.new_state) {
416 |       detailsStr = `

|  |

|  |

```

```

    N.stringify(h.details).slice(0, 60))}</small>`;
419     }
420
421     return `
422         <tr class="${h.is_rolled_back ? 'opacity-75' : ''}">
423             <td><code>#${h.id}</code></td>
424             <td><small>${escapeHtml(timeShort)}</small></td>
425             <td>
426                 <div class="fw-semibold"><code>${escapeHtml(h.command_id)}</code></div>
427                 <small class="text-muted">${escapeHtml(h.action_type)} ${h.target_hwnd ? `(HWND: ${h.target_hwnd})` : ''}</small>
428             </td>
429             <td>${detailsStr}</td>
430             <td>${rolledBadge}</td>
431             <td class="text-end">
432                 ${canRollback ? `
433                     <button class="btn btn-outline-warning btn-sm py-0
434                     px-2" onclick="window._rollbackAction(${h.id})" title="Откати
435                     ть это изменение">
436                         <i class="bi bi-arrow-counterclockwise me-1"></i>
437                     > Откат
438                     </button>
439                     ` : `<span class="text-muted small">—</span>`}
440             </td>
441         </tr>
442     `;
443     })).join('');
444 }
445
446 window._rollbackAction = async (historyId) => {
447     if (!confirm(`Откатить действие #${historyId}? Предыдущее со
448     стояние будет восстановлено.`)) return;
449     try {
450         const res = await fetch(`/api/v1/taskbar/history/${history
451         Id}/rollback`, { method: 'POST' });
452         const data = await res.json();
453         if (res.ok) {

```

```
449     if (window.toast) window.toast.success('Откат выполнен',
    data.message);
450     await refreshAll();
451   } else {
452     if (window.toast) window.toast.error('Ошибка отката', da
    ta.detail || 'Сбой');
453   }
454 } catch (err) {
455   if (window.toast) window.toast.error('Ошибка', err.message
    );
456 }
457 };
458
459 async function rollbackLast() {
460   try {
461     const res = await fetch('/api/v1/taskbar/history/rollback-
    last', { method: 'POST' });
462     const data = await res.json();
463     if (res.ok) {
464       if (window.toast) window.toast.success('Откат выполнен',
    data.message);
465       await refreshAll();
466     } else {
467       if (window.toast) window.toast.error('Ошибка отката', da
    ta.detail || 'Сбой');
468     }
469   } catch (err) {
470     if (window.toast) window.toast.error('Ошибка', err.message
    );
471   }
472 }
473
474 async function loadCommandCatalog() {
475   try {
476     const res = await fetch('/api/v1/taskbar/commands');
477     if (!res.ok) return;
478     _commandsCache = await res.json();
479     renderCommandsTable();
```

```
480 | } catch (err) {
481 |     console.error('Ошибка каталога команд:', err);
482 | }
483 | }
484 |
485 | function renderCommandsTable() {
486 |     const tbody = document.getElementById('tbody-commands-list')
487 |     ;
488 |     const cat = document.getElementById('filter-cmd-category')?.
489 |     value || '';
490 |     const risk = document.getElementById('filter-cmd-risk')?.val
491 |     ue || '';
492 |     if (!tbody) return;
493 |
494 |     const filtered = _commandsCache.filter(c => {
495 |         if (cat && c.category.toLowerCase() !== cat.toLowerCase())
496 |         return false;
497 |         if (risk && c.risk.toLowerCase() !== risk.toLowerCase()) r
498 |         eturn false;
499 |         return true;
500 |     });
501 |
502 |     const badge = document.getElementById('badge-commands-count'
503 |     );
504 |     if (badge) badge.textContent = filtered.length;
505 |
506 |     tbody.innerHTML = filtered.map(c => {
507 |         const riskBadge = c.risk === 'safe'
508 |         ? '<span class="badge bg-success-subtle text-success bor
509 |         der border-success-subtle">safe</span>'
510 |         : c.risk === 'caution'
511 |         ? '<span class="badge bg-warning-subtle text-warning b
512 |         order border-warning-subtle">caution</span>'
513 |         : c.risk === 'admin'
514 |         ? '<span class="badge bg-danger-subtle text-danger b
515 |         order border-danger-subtle">admin</span>'
516 |         : '<span class="badge bg-danger text-white">high</sp
517 |         an>';
```



```

508
509     const apiStr = c.api.win32 || c.api.com || c.api.registry
    || c.api.powershell || '-';
510
511     return `
512         <tr>
513             <td><code>${escapeHtml(c.id)}</code></td>
514             <td><span class="badge bg-secondary-subtle text-second
    ary">${escapeHtml(c.execution_class)}</span></td>
515             <td>${riskBadge}</td>
516             <td>${escapeHtml(c.description)}</td>
517             <td class="font-monospace small text-muted text-trunca
    te" style="max-width: 200px;" title="${escapeHtml(apiStr)}">${
    escapeHtml(apiStr)}</td>
518             <td class="text-end">
519                 <button class="btn btn-outline-primary btn-sm py-0 p
    x-2" onclick="window._executeCommandDirect('${c.id}')" title="
    Выполнить">
520                     <i class="bi bi-play"></i>
521                 </button>
522             </td>
523         </tr>
524     `;
525     }).join('');
526 }
527
528 window._executeCommandDirect = async (cmdId) => {
529     const meta = _commandsCache.find(c => c.id === cmdId);
530     if (meta?.requires_confirmation) {
531         if (!confirm(`Команда ${cmdId} имеет повышенный риск (${me
    ta.risk}). Подтвердить запуск?`)) return;
532     }
533     try {
534         const res = await fetch('/api/v1/taskbar/execute', {
535             method: 'POST',
536             headers: { 'Content-Type': 'application/json' },
537             body: JSON.stringify({ command: cmdId, params: {}, confi
    rmed_by_user: true })

```

```
538 |     });
539 |     const data = await res.json();
540 |     if (res.ok && window.toast) {
541 |         window.toast.success(`Команда ${cmdId}`, data.message ||
|         'Выполнено успешно');
542 |         await loadHistory();
543 |     }
544 | } catch (err) {
545 |     if (window.toast) window.toast.error('Ошибка', err.message
| );
546 | }
547 | };
548 |
549 | function escapeHtml(str) {
550 |     if (!str) return '';
551 |     return String(str)
552 |         .replace(/&/g, '&amp;')
553 |         .replace(/</g, '&lt;')
554 |         .replace(/>/g, '&gt;')
555 |         .replace(/"/g, '&quot;');
556 | }
557 |
558 | if (document.readyState === 'loading') {
559 |     document.addEventListener('DOMContentLoaded', () => init());
560 | } else {
561 |     init();
562 | }
563 |
```

index.html

```
1 <!--
2 =====
3 |
4 | =====
5 | Process Name: Windows Web Interface - Task Scheduler View
6 | =====
7 | =====
8 | Description:
9 |   Шаблон клиентского веб-интерфейса планировщика задач Windows
10 |   .
11 |
12 | Usage Examples:
13 |   Web Route:
14 |     /windows/api/webgui/task_scheduler_tab/index.html
15 |
16 |   Tab Frame Integration:
17 |     <iframe src="/windows/api/webgui/task_scheduler_tab/index.html" class="w-100 h-100 border-0"></iframe>
18 |
19 | File: index.html
20 | Project: ai-breadboard
21 | Package: windows/api/webgui/task_scheduler_tab
22 | Author: hypo69
23 | Copyright: © 2026 hypo69
24 | Updated: 2026-10-06 12:35:00
25 | =====
26 | =====
27 | -->
28 |
29 | <!-- Task Scheduler Tab -->
30 | <div class="container-fluid py-2 px-3">
31 |   <!-- Top Stat Cards -->
32 |   <div class="row g-2 mb-3">
33 |     <div class="col-6 col-md-3">
34 |       <div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3">
35 |         <div class="text-muted small text-uppercase fw-semibol
```

```

d">Всего задач □ </div>
31     <div class="fs-4 fw-bold mt-1 text-primary" id="ts-tot
al-count">--</div>
32     <div class="small text-muted mt-1" id="ts-total-sub">s
chtasks.exe /query</div>
33     </div>
34 </div>
35 <div class="col-6 col-md-3">
36     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
37         <div class="text-muted small text-uppercase fw-semibol
d">Готовы к запуску (Ready) □ </div>
38         <div class="fs-4 fw-bold mt-1 text-success" id="ts-rea
dy-count">--</div>
39         <div class="small text-muted mt-1" id="ts-ready-sub">0
ждуют расписания</div>
40     </div>
41 </div>
42 <div class="col-6 col-md-3">
43     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
44         <div class="text-muted small text-uppercase fw-semibol
d">Выполняются сейчас □ </div>
45         <div class="fs-4 fw-bold mt-1 text-info" id="ts-runnin
g-count">--</div>
46         <div class="small text-muted mt-1" id="ts-running-sub"
>Активные задачи</div>
47     </div>
48 </div>
49 <div class="col-6 col-md-3">
50     <div class="card bg-dark text-white border-secondary h-1
00 shadow-sm p-3">
51         <div class="text-muted small text-uppercase fw-semibol
d">Отключены (Disabled) □ </div>
52         <div class="fs-4 fw-bold mt-1 text-danger" id="ts-disa
bled-count">--</div>
53         <div class="small text-muted mt-1" id="ts-disabled-sub
">Деактивированы</div>

```

```

54     </div>
55 </div>
56 </div>
57
58 <!-- Action Bar -->
59 <div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
60     <div class="d-flex align-items-center gap-2">
61         <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="ts-refresh-btn">
62             <i class="bi bi-arrow-clockwise"></i>
63             <span>Обновить задачи</span>
64         </button>
65     </div>
66     <div class="d-flex align-items-center gap-2">
67         <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary" id="ts-search-input" placeholder="Поиск по имени задачи..." style="width: 260px;">
68     </div>
69 </div>
70
71 <!-- Tasks Table -->
72 <div class="card bg-dark text-white border-secondary shadow-sm">
73     <div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
74         <span class="fw-semibold"><i class="bi bi-calendar-check text-primary me-2"></i>Задачи Планировщика Windows (Task Scheduler)</span>
75         <span class="badge bg-secondary" id="ts-table-badge">0 задач</span>
76     </div>
77     <div class="card-body p-0 table-responsive" style="max-height: 520px; overflow-y: auto;">
78         <table class="table table-dark table-hover table-striped align-middle mb-0 small" id="ts-table">
79             <thead class="sticky-top bg-dark">

```

```
80 |         <tr class="text-secondary">
81 |             <th>Имя / Путь задачи</th>
82 |             <th>Время след. запуска</th>
83 |             <th>Статус</th>
84 |             <th>Режим расписания</th>
85 |             <th class="text-end">Действия</th>
86 |         </tr>
87 |     </thead>
88 |     <tbody id="ts-tbody">
89 |         <tr><td colspan="5" class="text-center text-muted py
90 | -4">Опрос планировщика задач...</td></tr>
91 |     </tbody>
92 | </table>
93 | </div>
94 | </div>
95 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Task Scheduler Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля Планировщ
7   *   ика задач Windows.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/task_scheduler_tab/mai
11  *     n.js?v=20261006_v2" type="module"></script>
12  *   JavaScript Import:
13  *     import { initTaskSchedulerTab } from '/windows/api/webg
14  *     ui/task_scheduler_tab/main.js';
15  * File: main.js
16  * Project: ai-breadboard
17  * Package: windows/api/webgui/task_scheduler_tab
18  * Author: hypo69
19  * Copyright: © 2026 hypo69
20  * Updated: 2026-10-06 12:35:00
21  * =====
22  */
23
24 /**
25  * task_scheduler_tab/main.js – Управление задачами Windows Ta
26  * sk Scheduler
27  * Updated: 2026-10-06 12:35:00
28  */
29 const registerTabPoller = window.registerTabPoller || function
```

```

    () {};
```

30

```

31 let isInitialized = false;
32 let tasksList = [];
33
34 function escapeHtml(str) {
35     if (str === null || str === undefined) return '';
36     return String(str)
37         .replace(/&/g, '&amp;')
38         .replace(/</g, '&lt;')
39         .replace(/>/g, '&gt;')
40         .replace(/"/g, '&quot;')
41         .replace(/'/g, '&#039;');
42 }
43
44 export async function fetchTaskSchedulerSummary() {
45     const refreshBtn = document.getElementById('ts-refresh-btn')
46     ;
47     if (refreshBtn) {
48         const icon = refreshBtn.querySelector('i');
49         if (icon) icon.classList.add('spin-animation');
50     }
51     try {
52         const res = await fetch('/api/task-scheduler/summary');
53         if (!res.ok) throw new Error(`HTTP ${res.status}`);
54         const data = await res.json();
55         renderTasks(data);
56     } catch (err) {
57         console.warn('[TaskScheduler] Ошибка получения списка задач:', err);
58         const tbody = document.getElementById('ts-tbody');
59         if (tbody && tasksList.length === 0) {
60             tbody.innerHTML = `<tr><td colspan="5" class="text-center text-danger py-4">Ошибка загрузки задач: ${escapeHtml(err.message)}</td></tr>`;
61         }
62     } finally {
```



```
63 |     if (refreshBtn) {
64 |         const icon = refreshBtn.querySelector('i');
65 |         if (icon) icon.classList.remove('spin-animation');
66 |     }
67 | }
68 | }
69 |
70 | function renderTasks(data) {
71 |     tasksList = data.tasks || [];
72 |
73 |     const totalEl = document.getElementById('ts-total-count');
74 |     const readyEl = document.getElementById('ts-ready-count');
75 |     const runningEl = document.getElementById('ts-running-count'
    | );
76 |     const disabledEl = document.getElementById('ts-disabled-coun
    | t');
77 |
78 |     const readyCount = data.ready_tasks !== undefined
79 |         ? data.ready_tasks
80 |         : tasksList.filter(t => {
81 |             const st = (t.state || t.status || '').toLowerCase();
82 |             return st.includes('ready') || st.includes('готов');
83 |         }).length;
84 |
85 |     const runningCount = data.running_tasks !== undefined
86 |         ? data.running_tasks
87 |         : tasksList.filter(t => {
88 |             const st = (t.state || t.status || '').toLowerCase();
89 |             return st.includes('running') || st.includes('работ')
    | || st.includes('выполн');
90 |         }).length;
91 |
92 |     const disabledCount = data.disabled_tasks !== undefined
93 |         ? data.disabled_tasks
94 |         : tasksList.filter(t => {
95 |             const st = (t.state || t.status || '').toLowerCase();
96 |             return st.includes('disabled') || st.includes('отключ'
    | );
```

```
97     }).length;
98
99     if (totalEl) totalEl.textContent = tasksList.length;
100    if (readyEl) readyEl.textContent = readyCount;
101    if (runningEl) runningEl.textContent = runningCount;
102    if (disabledEl) disabledEl.textContent = disabledCount;
103
104    applyTaskFilters();
105 }
106
107 function applyTaskFilters() {
108     const tbody = document.getElementById('ts-tbody');
109     const searchInput = document.getElementById('ts-search-input');
110     const badge = document.getElementById('ts-table-badge');
111
112     if (!tbody) return;
113
114     const query = (searchInput ? searchInput.value : '').toLowerCase().trim();
115
116     const filtered = tasksList.filter(t => {
117         if (!query) return true;
118         const name = (t.name || t.task_name || '').toLowerCase();
119         const path = (t.task_path || '').toLowerCase();
120         const act = (t.action || '').toLowerCase();
121         return name.includes(query) || path.includes(query) || act
122             .includes(query);
123     });
124
125     if (badge) badge.textContent = `${filtered.length} из ${tasksList.length} задач`;
126
127     if (filtered.length === 0) {
128         tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted py-4">Задачи не найдены</td></tr>';
129         return;
130     }
```

```
130
131   tbody.innerHTML = filtered.slice(0, 300).map(t => {
132     const tName = t.name || t.task_name || (t.task_path ? t.task_path.split('\\').pop() : 'Task');
133     const tPath = t.task_path || tName;
134     const status = t.state || t.status || 'Ready';
135     const statusLower = status.toLowerCase();
136     const isReady = statusLower.includes('ready') || statusLower.includes('готов');
137     const isRunning = statusLower.includes('running') || statusLower.includes('работ') || statusLower.includes('выполн');
138     const isDisabled = statusLower.includes('disabled') || statusLower.includes('отключ');
139
140     let badgeClass = 'bg-secondary';
141     if (isRunning) badgeClass = 'bg-info text-dark';
142     else if (isReady) badgeClass = 'bg-success';
143     else if (isDisabled) badgeClass = 'bg-danger-subtle text-danger border border-danger fw-bold';
144
145     const scheduleType = t.schedule_type || 'Custom';
146     const nextRun = t.next_run_time || '-';
147
148     return `
149       <tr>
150         <td class="fw-semibold text-light font-monospace small" title="${escapeHtml(tPath)}">
151           <div>${escapeHtml(tName)}</div>
152           ${tPath !== tName ? `<div class="text-muted" style="font-size: 0.75rem;">${escapeHtml(tPath)}</div>` : ''}
153         </td>
154         <td class="text-muted small">${escapeHtml(nextRun)}</td>
155         <td><span class="badge ${badgeClass}">${escapeHtml(status)}</span></td>
156         <td class="small text-muted">${escapeHtml(scheduleType)}</td>
157         <td class="text-end">
```

```

158 |         <button class="btn btn-xs btn-outline-primary py-0 p
      | x-2 ts-action-btn" data-task="${escapeHtml(tPath)}" data-actio
      | n="schtasks_run" title="Запустить немедленно">
159 |             <i class="bi bi-play-fill"></i>
160 |         </button>
161 |         <button class="btn btn-xs ${isDisabled ? 'btn-outlin
      | e-success' : 'btn-outline-warning'} py-0 px-2 ts-action-btn" d
      | ata-task="${escapeHtml(tPath)}" data-action="${isDisabled ? 's
      | chtasks_enable' : 'schtasks_disable'}" title="${isDisabled ? '
      | Включить' : 'Отключить'}">
162 |             <i class="bi ${isDisabled ? 'bi-check-circle' : 'b
      | i-pause-fill'}"></i>
163 |         </button>
164 |     </td>
165 | </tr>
166 | `;
167 | }).join('');
168 |
169 | tbody.querySelector('.ts-action-btn').forEach(btn => {
170 |     btn.addEventListener('click', () => {
171 |         const taskTarget = btn.dataset.task;
172 |         const act = btn.dataset.action;
173 |         executeTaskAction(taskTarget, act);
174 |     });
175 | });
176 | }
177 |
178 | async function executeTaskAction(taskTarget, action) {
179 |     if (window.toast) window.toast.info('Планировщик', `${action
      | }: ${taskTarget}`);
180 |     try {
181 |         const res = await fetch('/api/task-scheduler/action', {
182 |             method: 'POST',
183 |             headers: { 'Content-Type': 'application/json' },
184 |             body: JSON.stringify({
185 |                 task_path: taskTarget,
186 |                 task_name: taskTarget,
187 |                 action: action,

```

```
188         dry_run: false,
189         confirmed_by_user: true
190     })
191 });
192 const result = await res.json();
193 if (res.ok && (result.status === 'SUCCESS' || result.status === 'DRY_RUN_SUCCESS')) {
194     if (window.toast) window.toast.success('Успех', `${taskTarget}: ${result.message || 'выполнено'}`);
195     fetchTaskSchedulerSummary();
196 } else {
197     if (window.toast) window.toast.error('Ошибка', result.message || result.detail || 'Сбой операции');
198 }
199 } catch (err) {
200     if (window.toast) window.toast.error('Ошибка сети', err.message);
201 }
202 }
203
204 export function initTaskSchedulerTab() {
205     if (!isInitialized) {
206         isInitialized = true;
207
208         const refreshBtn = document.getElementById('ts-refresh-btn');
209         if (refreshBtn) refreshBtn.addEventListener('click', fetchTaskSchedulerSummary);
210
211         const searchInput = document.getElementById('ts-search-input');
212         if (searchInput) searchInput.addEventListener('input', applyTaskFilters);
213
214         registerTabPoller('tab-task-scheduler', fetchTaskSchedulerSummary, 15000, { immediate: true });
215     }
216 }
```

```
217 |   fetchTaskSchedulerSummary();
218 | }
219 |
220 | window.initTaskSchedulerTab = initTaskSchedulerTab;
221 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/telegram_rag_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/telegram_rag_tab/index.
15 | html" class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/telegram_rag_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Telegram Channel & Group RAG Search Tab -->
28 <div class="container-fluid py-2">
29 | <!-- Top Banner / Header -->
30 | <div class="alert alert-primary d-flex flex-wrap align-items
31 | -center justify-content-between gap-2 mb-3 shadow-sm">
```

руппам (RAG)</h5>

32 <p class="mb-0 small text-muted">Индексация публичных
каналов, поиск по сообщениям с точными ссылками и мгновенным о
тветом</p>

33 </div>

34 </div>

35 <div class="d-flex align-items-center gap-2">

36 <span class="badge bg-primary-subtle text-primary border
border-primary-subtle px-3 py-2" id="tgram-stats-badge">

37 Загрузка каналов...

38

39 <button class="btn btn-sm btn-outline-primary" id="btn-r
efresh-tgram-channels" onclick="loadTelegramChannels()">

40 <i class="bi bi-arrow-clockwise"></i> Обновить

41 </button>

42 </div>

43 </div>

44

45 <div class="row g-3">

46 <!-- LEFT: Channel Management & Subscriptions -->

47 <div class="col-lg-4 col-md-5">

48 <!-- Add / Subscribe Card -->

49 <div class="card shadow-sm border-secondary-subtle mb-3"
>

50 <div class="card-header bg-body-tertiary py-2">

51 <h6 class="mb-0 fw-bold small text-uppercase d-flex
align-items-center gap-2">

52 <i class="bi bi-plus-circle-fill text-primary"></i>
> Добавить канал / группу

53 </h6>

54 </div>

55 <div class="card-body p-3">

56 <form id="tgram-add-channel-form" onsubmit="event.pr
eventDefault(); handleAddTelegramChannel();">

57 <div class="mb-2">

58 <label class="form-label small fw-semibold mb-1"
>Имя или ссылка на канал</label>

59 <div class="input-group input-group-sm">


```

60 |             <span class="input-group-text bg-body-secondar
| y">@</span>
61 |             <input type="text" class="form-control" id="tg
| ram-channel-input" placeholder="canozrimb или t.me/canozrimb"
| required>
62 |         </div>
63 |         <div class="form-text text-muted small" style="f
| ont-size: 0.75rem;">
64 |             Поддерживаются публичные каналы и веб-превью T
| elegram
65 |         </div>
66 |     </div>
67 |
68 |     <div class="mb-3">
69 |         <label class="form-label small fw-semibold mb-1"
| >Лимит сообщений для сбора</label>
70 |         <select class="form-select form-select-sm" id="t
| gram-max-msgs-select">
71 |             <option value="500" selected>500 сообщений (бы
| стро)</option>
72 |             <option value="1000">1000 сообщений</option>
73 |             <option value="3000">3000 сообщений</option>
74 |             <option value="5000">5000 сообщений (глубокий
| RAG)</option>
75 |         </select>
76 |     </div>
77 |
78 |     <button type="submit" class="btn btn-sm btn-primar
| y w-100 d-flex align-items-center justify-content-center gap-2
| shadow-sm" id="btn-tgram-subscribe">
79 |         <i class="bi bi-cloud-arrow-down-fill"></i>
80 |         <span>Подключить и индексировать</span>
81 |     </button>
82 | </form>
83 | </div>
84 | </div>
85 |
86 | <!-- Channels List Card -->

```

```

87 |         <div class="card shadow-sm border-secondary-subtle">
88 |             <div class="card-header bg-body-tertiary d-flex justify-content-between align-items-center py-2">
89 |                 <span class="fw-semibold small text-uppercase">
90 |                     Подключенные каналы (<span id="tgram-channels-count">0</span>)
91 |                 </span>
92 |                 <span class="badge bg-secondary-subtle text-body" id="tgram-active-count-badge">0 в пуле</span>
93 |             </div>
94 |             <div class="card-body p-2" id="tgram-channels-list-container" style="max-height: 55vh; overflow-y: auto;">
95 |                 <div class="text-center text-muted p-4">
96 |                     <div class="spinner-border spinner-border-sm text-primary mb-2" role="status"></div>
97 |                     <div>Загрузка списка каналов...</div>
98 |                 </div>
99 |             </div>
100 |         </div>
101 |     </div>
102 |
103 |     <!-- RIGHT: Search Interface & Results -->
104 |     <div class="col-lg-8 col-md-7">
105 |         <div class="card shadow-sm border-secondary-subtle h-100 d-flex flex-column">
106 |             <!-- Search Input Bar & Settings Header -->
107 |             <div class="card-header bg-body-tertiary p-3 border-bottom">
108 |                 <!-- Main Search Input -->
109 |                 <form id="tgram-search-form" onsubmit="event.preventDefault(); executeTelegramSearch();" class="mb-3">
110 |                     <div class="input-group shadow-sm">
111 |                         <span class="input-group-text bg-body text-primary"><i class="bi bi-search"></i></span>
112 |                         <input type="text" class="form-control form-control-lg fs-6" id="tgram-search-input" placeholder="Введите поисковый запрос, тему или вопрос..." required autofocus>
113 |                         <button class="btn btn-outline-secondary" type="

```

```
button" title="Очистить" onclick="document.getElementById('tgram-search-input').value='';">
114     <i class="bi bi-x-lg"></i>
115     </button>
116     <button class="btn btn-primary px-4 fw-semibold"
type="submit" id="btn-tgram-do-search">
117         <i class="bi bi-search me-1"></i> Найти
118     </button>
119 </div>
120 </form>
121
122 <!-- Search Settings & Scope Filters -->
123 <div class="row g-2 align-items-center bg-body p-2 rounded border border-secondary-subtle small">
124     <!-- Channel Filter -->
125     <div class="col-md-5 col-12">
126         <label class="form-label text-muted mb-1 fw-semibold" style="font-size: 0.75rem;">
127             <i class="bi bi-funnel-fill text-primary"></i>
Область поиска:
128         </label>
129         <select class="form-select form-select-sm" id="tgram-filter-channel">
130             <option value="all" selected>☐ Все подключенные каналы (Общий пул)</option>
131         </select>
132     </div>
133
134     <!-- Search Mode -->
135     <div class="col-md-4 col-12">
136         <label class="form-label text-muted mb-1 fw-semibold" style="font-size: 0.75rem;">
137             <i class="bi bi-diagram-2-fill text-info"></i>
Режим индексов:
138         </label>
139         <select class="form-select form-select-sm" id="tgram-search-mode">
140             <option value="unified" selected>Общий RAG пул
```

```

    (Unified)</option>
141         <option value="individual">Раздельно по канала
    м</option>
142     </select>
143 </div>
144
145     <!-- Top K -->
146     <div class="col-md-3 col-12">
147         <label class="form-label text-muted mb-1 fw-semi
    bold" style="font-size: 0.75rem;">
148             <i class="bi bi-list-ol text-warning"></i> Лим
    ит выдачи:
149         </label>
150         <select class="form-select form-select-sm" id="t
    gram-topk-select">
151             <option value="5" selected>5 сообщений</option
    >
152             <option value="10">10 сообщений</option>
153             <option value="20">20 сообщений</option>
154             <option value="50">50 сообщений</option>
155         </select>
156     </div>
157 </div>
158 </div>
159
160     <!-- Search Results Body -->
161     <div class="card-body p-3 flex-grow-1" style="max-heig
    ht: 60vh; overflow-y: auto;" id="tgram-results-container">
162         <!-- Initial Placeholder -->
163         <div class="d-flex flex-column align-items-center ju
    stify-content-center text-muted p-5 text-center" id="tgram-sea
    rch-placeholder">
164             <span class="fs-1 mb-2">□ </span>
165             <h6 class="fw-bold">Поиск по базе знаний Telegram<
    /h6>
166             <p class="small text-muted mb-0" style="max-width:
    480px;">
167                 Введите ключевые слова или вопрос выше, чтобы на

```

йти точные сообщения с ссылками на первоисточники в Telegram.

168 | </p>

169 | </div>

170 | </div>

171 | </div>

172 | </div>

173 | </div>

174 | </div>

175 |

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Telegram Rag Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/telegram_rag_tab/main.
11 *       js?v=20261001_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/telegram_rag_tab
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-01 13:04:40
19 * =====
20 */
21 // — TELEGRAM CHANNEL & GROUP RAG TAB MAIN.JS —————
22
23 let tgramSubscribedChannels = [];
24 let tgramAvailableChannels = [];
25
26 // Initialize Telegram RAG Tab
27 window.initTelegramRagTab = async function() {
28   console.log('[TelegramRAG] Initializing Telegram RAG Tab...'
29   );
29   await loadTelegramChannels();
30 };
```

```

31 |
32 | // Notification helper
33 | function showTgramNotification(message, type = 'info') {
34 |     if (typeof window.showNotification === 'function') {
35 |         window.showNotification(message, type);
36 |         return;
37 |     }
38 |     console.log(`[TelegramRAG Notification ${type}]: ${message}`
39 | );
40 | }
41 |
42 | // Load channels from backend API
43 | async function loadTelegramChannels() {
44 |     const container = document.getElementById('tgram-channels-li
45 | st-container');
46 |     const statsBadge = document.getElementById('tgram-stats-badg
47 | e');
48 |     const countEl = document.getElementById('tgram-channels-coun
49 | t');
50 |     const activeCountBadge = document.getElementById('tgram-acti
51 | ve-count-badge');
52 |     const filterSelect = document.getElementById('tgram-filter-c
53 | hannel');
54 |
55 |     try {
56 |         const res = await fetch('/api/telegram_rag/channels');
57 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
58 |         const data = await res.json();
59 |
60 |         tgramSubscribedChannels = data.user_subscriptions || [];
61 |         tgramAvailableChannels = data.available_channels || [];
62 |
63 |         // Calculate total messages
64 |         const totalMsgs = tgramAvailableChannels.reduce((sum, c) =
65 | > sum + (c.messages_count || 0), 0);
66 |
67 |         if (statsBadge) {
68 |             statsBadge.textContent = `${tgramSubscribedChannels.leng

```

```

th} подписок • ${totalMsgs} сообщений`;
62     }
63     if (countEl) countEl.textContent = tgramSubscribedChannels
.length;
64     if (activeCountBadge) activeCountBadge.textContent = `${tg
ramAvailableChannels.length} в пуле`;
65
66     // Render channels list
67     renderTelegramChannelsList(tgramSubscribedChannels, tgramA
vailableChannels);
68
69     // Update filter dropdown
70     if (filterSelect) {
71         const currentVal = filterSelect.value || 'all';
72         let optsHtml = '<option value="all">☐ Все подключенные
каналы (Общий пул)</option>';
73         tgramSubscribedChannels.forEach(ch => {
74             const isSel = ch === currentVal ? 'selected' : '';
75             const found = tgramAvailableChannels.find(a => a.chann
el === ch);
76             const countText = found ? ` (${found.messages_count} с
ообщ.)` : '';
77             optsHtml += `<option value="${ch}" ${isSel}>@${ch}${co
untText}</option>`;
78         });
79         filterSelect.innerHTML = optsHtml;
80     }
81 } catch (err) {
82     console.error('Error loading Telegram channels:', err);
83     if (container) {
84         container.innerHTML = `<div class="alert alert-danger m-
2 small">Ошибка загрузки каналов: ${err.message}</div>`;
85     }
86 }
87 }
88
89 // Render Channels List
90 function renderTelegramChannelsList(userSubs, availablePool) {

```



```

91 |   const container = document.getElementById('tgram-channels-li
    | st-container');
92 |   if (!container) return;
93 |
94 |   if (userSubs.length === 0) {
95 |     container.innerHTML = `
96 |       <div class="text-center text-muted p-4 small">
97 |         <i class="bi bi-chat-left-dots fs-3 d-block mb-2 text-
    | secondary"></i>
98 |         Нет подключенных каналов.<br>Добавьте канал выше для и
    | ндексации.
99 |       </div>
100 |     `;
101 |     return;
102 |   }
103 |
104 |   let html = '<div class="list-group list-group-flush">';
105 |   userSubs.forEach(channelName => {
106 |     const info = availablePool.find(p => p.channel === channel
    | Name) || {
107 |       channel: channelName,
108 |       messages_count: 0,
109 |       is_indexed: false,
110 |     };
111 |
112 |     const statusBadge = info.is_indexed
113 |       ? `<span class="badge bg-success-subtle text-success bor
    | der border-success-subtle">${info.messages_count} сообщ.</span
    | >`
114 |       : `<span class="badge bg-warning-subtle text-warning bor
    | der">Не индексирован</span>`;
115 |
116 |     html += `
117 |       <div class="list-group-item p-2 rounded mb-1 border-0 bg
    | -body-tertiary shadow-sm">
118 |         <div class="d-flex justify-content-between align-items
    | -center mb-1">
119 |           <div class="d-flex align-items-center gap-2 text-tru

```

```

ncate">
120         <span class="fs-5 text-primary">□ </span>
121         <a href="https://t.me/${channelName}" target="_bla
nk" rel="noopener noreferrer" class="fw-bold text-decoration-n
one text-truncate" title="Открыть в Telegram">
122             @${channelName}
123             <i class="bi bi-box-arrow-up-right ms-1 text-mut
ed" style="font-size: 0.75rem;"></i>
124         </a>
125     </div>
126     <div>${statusBadge}</div>
127 </div>
128 <div class="d-flex justify-content-between align-items
-center mt-2 pt-1 border-top border-secondary-subtle">
129     <button class="btn btn-sm btn-outline-secondary py-0
px-2 small" onclick="filterByChannel('${channelName}')" title
="Искать только в этом канале">
130         <i class="bi bi-search"></i> Искать здесь
131     </button>
132     <div class="btn-group btn-group-sm">
133         <button class="btn btn-outline-primary py-0 px-2"
onclick="handleReindexTelegramChannel('${channelName}')" title
="Переиндексировать канал">
134             <i class="bi bi-arrow-clockwise"></i>
135         </button>
136         <button class="btn btn-outline-danger py-0 px-2" o
nclick="handleUnsubscribeTelegramChannel('${channelName}')" ti
tle="Отключить канал">
137             <i class="bi bi-trash"></i>
138         </button>
139     </div>
140 </div>
141 </div>
142 `;
143 });
144 html += '</div>';
145 container.innerHTML = html;
146 }

```

```
147 |
148 | // Quick filter helper from channel list
149 | function filterByChannel(channelName) {
150 |     const filterSelect = document.getElementById('tgram-filter-c
    hannel');
151 |     if (filterSelect) {
152 |         filterSelect.value = channelName;
153 |     }
154 |     const searchInput = document.getElementById('tgram-search-in
    put');
155 |     if (searchInput) {
156 |         searchInput.focus();
157 |     }
158 | }
159 |
160 | // Add / Subscribe Channel Handler
161 | async function handleAddTelegramChannel() {
162 |     const input = document.getElementById('tgram-channel-input')
    ;
163 |     const maxMsgsSelect = document.getElementById('tgram-max-msg
    s-select');
164 |     const btn = document.getElementById('btn-tgram-subscribe');
165 |
166 |     if (!input || !input.value.trim()) return;
167 |     const channel = input.value.trim();
168 |     const maxMessages = parseInt(maxMsgsSelect?.value || '500',
    10);
169 |
170 |     try {
171 |         if (btn) {
172 |             btn.disabled = true;
173 |             btn.innerHTML = '<span class="spinner-border spinner-bor
    der-sm me-1" role="status"></span> Индексация...';
174 |         }
175 |
176 |         showTgramNotification(`Подключение канала @${channel}...`,
    'info');
177 |         const resp = await fetch('/api/telegram_rag/subscribe', {
```

```

178     method: 'POST',
179     headers: { 'Content-Type': 'application/json' },
180     body: JSON.stringify({ channel, max_messages: maxMessage
    s })),
181   });
182
183   if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
184   const data = await resp.json();
185
186   showTgramNotification(data.message || `Канал @${channel} у
    спешно подключен`, 'success');
187   input.value = '';
188   await loadTelegramChannels();
189 } catch (err) {
190   console.error('Error adding Telegram channel:', err);
191   showTgramNotification(`Ошибка: ${err.message}`, 'danger');
192 } finally {
193   if (btn) {
194     btn.disabled = false;
195     btn.innerHTML = '<i class="bi bi-cloud-arrow-down-fill m
    e-1"></i> Подключить и индексировать';
196   }
197 }
198 }
199
200 // Unsubscribe Channel Handler
201 async function handleUnsubscribeTelegramChannel(channel) {
202   if (!confirm(`Вы действительно хотите отключить канал @${cha
    nnel}?`)) return;
203
204   try {
205     const resp = await fetch('/api/telegram_rag/unsubscribe',
    {
206       method: 'POST',
207       headers: { 'Content-Type': 'application/json' },
208       body: JSON.stringify({ channel }),
209     });
210     if (!resp.ok) throw new Error(`HTTP ${resp.status}`);

```

```

211     const data = await resp.json();
212     showTgramNotification(data.message || `Канал @${channel} о
    тключен`, 'info');
213     await loadTelegramChannels();
214 } catch (err) {
215     console.error('Error unsubscribing channel:', err);
216     showTgramNotification(`Ошибка: ${err.message}`, 'danger');
217 }
218 }
219
220 // Reindex Channel Handler
221 async function handleReindexTelegramChannel(channel) {
222     const maxMsgs = parseInt(document.getElementById('tgram-max-
    msgs-select')?.value || '500', 10);
223     try {
224         showTgramNotification(`Обновление индекса @${channel}...`,
    'info');
225         const resp = await fetch('/api/telegram_rag/reindex', {
226             method: 'POST',
227             headers: { 'Content-Type': 'application/json' },
228             body: JSON.stringify({ channel, max_messages: maxMsgs })
    ,
229         });
230         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
231         const data = await resp.json();
232         showTgramNotification(data.message || `Индекс @${channel}
    успешно обновлен`, 'success');
233         await loadTelegramChannels();
234     } catch (err) {
235         console.error('Error reindexing channel:', err);
236         showTgramNotification(`Ошибка переиндексации: ${err.messag
    e}`, 'danger');
237     }
238 }
239
240 // Highlight keywords in text
241 function highlightKeywords(text, query) {
242     if (!text || !query) return text || '';

```

```
243 |   const words = query.split(/\s+/).filter(w => w.length > 2);
244 |   if (words.length === 0) return text;
245 |
246 |   let escaped = text
247 |     .replace(/&/g, '&amp;')
248 |     .replace(/</g, '&lt;')
249 |     .replace(/>/g, '&gt;');
250 |
251 |   words.forEach(w => {
252 |     const regex = new RegExp(`(${w.replace(/[\.*+?^${}()|[\]\$\\"/g, '\\$&')})`, 'gi');
253 |     escaped = escaped.replace(regex, '<mark class="bg-warning-subtle text-body px-1 rounded">$1</mark>');
254 |   });
255 |   return escaped;
256 | }
257 |
258 | // Execute Telegram RAG Search
259 | async function executeTelegramSearch() {
260 |   const searchInput = document.getElementById('tgram-search-input');
261 |   const filterSelect = document.getElementById('tgram-filter-channel');
262 |   const modeSelect = document.getElementById('tgram-search-mode');
263 |   const topkSelect = document.getElementById('tgram-topk-select');
264 |   const resultsContainer = document.getElementById('tgram-results-container');
265 |   const searchBtn = document.getElementById('btn-tgram-do-search');
266 |
267 |   if (!searchInput || !searchInput.value.trim()) return;
268 |   const query = searchInput.value.trim();
269 |   const channel = filterSelect ? filterSelect.value : 'all';
270 |   const searchMode = modeSelect ? modeSelect.value : 'unified';
271 |   const topK = parseInt(topkSelect?.value || '5', 10);
```

```

272
273   if (resultsContainer) {
274       resultsContainer.innerHTML = `
275           <div class="text-center text-muted p-5">
276               <div class="spinner-border text-primary mb-3" role="st
atus"></div>
277               <h6>Поиск в базе знаний Telegram...</h6>
278               <div class="small text-muted">Запрос: «${query}»</div>
279           </div>
280       `;
281   }
282
283   if (searchBtn) searchBtn.disabled = true;
284
285   try {
286       const startTime = performance.now();
287       const resp = await fetch('/api/telegram_rag/search', {
288           method: 'POST',
289           headers: { 'Content-Type': 'application/json' },
290           body: JSON.stringify({
291               query,
292               channel: channel === 'all' ? null : channel,
293               search_mode: searchMode,
294               top_k: topK,
295           }),
296       });
297
298       if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
299       const data = await resp.json();
300       const duration = Math.round(performance.now() - startTime)
301       ;
302       renderTelegramSearchResults(data.results || [], query, dur
ation);
303   } catch (err) {
304       console.error('Error executing Telegram search:', err);
305       if (resultsContainer) {
306           resultsContainer.innerHTML = `

```

```

307 |         <div class="alert alert-danger shadow-sm">
308 |             <h6><i class="bi bi-exclamation-triangle-fill me-2">
| </i> Ошибка поиска</h6>
309 |             <p class="mb-0 small">${err.message}</p>
310 |         </div>
311 |     `;
312 | }
313 | } finally {
314 |     if (searchBtn) searchBtn.disabled = false;
315 | }
316 | }
317 |
318 | // Render Search Results Cards
319 | function renderTelegramSearchResults(results, query, durationM
| s) {
320 |     const container = document.getElementById('tgram-results-con
| tainer');
321 |     if (!container) return;
322 |
323 |     if (results.length === 0) {
324 |         container.innerHTML = `
325 |             <div class="text-center text-muted p-5">
326 |                 <span class="fs-1 mb-2">❌ </span>
327 |                 <h6 class="fw-bold">Ничего не найдено</h6>
328 |                 <p class="small text-muted mb-0">По запросу «${query}»
| в индексах Telegram сообщений не обнаружено. Попробуйте измен
| ить формулировку или подключить дополнительные каналы.</p>
329 |             </div>
330 |         `;
331 |         return;
332 |     }
333 |
334 |     let html = `
335 |         <div class="d-flex justify-content-between align-items-cen
| ter mb-3 pb-2 border-bottom">
336 |             <div class="small text-muted">
337 |                 Найдено результатов: <strong class="text-primary">${re
| sults.length}</strong> (${durationMs} мс)

```



```

338         </div>
339         <span class="badge bg-secondary-subtle text-body border"
>RAG Ranker</span>
340     </div>
341     <div class="d-flex flex-column gap-3">
342     `;
343
344     results.forEach((r, idx) => {
345         const channelName = r.channel || 'unknown';
346         const author = r.author || 'Telegram';
347         const dateStr = r.date || '';
348         const msgUrl = r.url || `https://t.me/${channelName}/${r.i
d || ''}`;
349         const highlightedSnippet = highlightKeywords(r.snippet ||
r.text || '', query);
350         const scoreBadge = r.score !== undefined
351             ? `<span class="badge bg-info-subtle text-info border">P
елевантность: ${Math.round(r.score * 100)}%</span>`
352             : '';
353
354         html += `
355             <div class="card border shadow-sm rounded-3">
356                 <div class="card-header bg-body-tertiary d-flex justif
y-content-between align-items-center py-2">
357                     <div class="d-flex align-items-center gap-2">
358                         <span class="badge bg-primary text-white">#${idx +
1}</span>
359                         <a href="https://t.me/${channelName}" target="_bla
nk" rel="noopener noreferrer" class="fw-bold text-decoration-n
one text-primary">
360                             @${channelName}
361                         </a>
362                         <span class="text-muted small">• ${author}</span>
363                         ${dateStr ? `<span class="text-muted small">• ${da
teStr}</span>` : ''}
364                     </div>
365                     <div>${scoreBadge}</div>
366                 </div>

```

```

367 |         <div class="card-body p-3">
368 |             <p class="mb-0 small text-body font-monospace" style
| ="white-space: pre-wrap; word-break: break-word;">
369 |                 ${highlightedSnippet}
370 |             </p>
371 |         </div>
372 |         <div class="card-footer bg-body-secondary py-2 d-flex
| justify-content-between align-items-center flex-wrap gap-2">
373 |             <a href="${msgUrl}" target="_blank" rel="noopener no
| referrer" class="btn btn-sm btn-outline-primary d-inline-flex
| align-items-center gap-1">
374 |                 <i class="bi bi-box-arrow-up-right"></i>
375 |                 <span>Открыть в Telegram</span>
376 |             </a>
377 |             <div class="btn-group btn-group-sm">
378 |                 <button class="btn btn-outline-secondary" onclick=
| "copyTelegramLink('${msgUrl}')" title="Скопировать ссылку на с
| ообщение">
379 |                     <i class="bi bi-clipboard"></i> Ссылка
380 |                 </button>
381 |                 <button class="btn btn-outline-success" onclick="s
| endTelegramResultToChat('${msgUrl}', '${encodeURIComponent(r.s
| nippet || '')}')" title="Вставить цитату в чат">
382 |                     <i class="bi bi-chat-quote-fill"></i> В чат
383 |                 </button>
384 |             </div>
385 |         </div>
386 |     </div>
387 |     `;
388 | });
389 |
390 | html += '</div>';
391 | container.innerHTML = html;
392 | }
393 |
394 | // Copy link to clipboard
395 | function copyTelegramLink(url) {
396 |     navigator.clipboard.writeText(url).then(() => {

```

```
397     showTgramNotification('Ссылка скопирована в буфер обмена',
    'success');
398   }).catch(() => {
399     prompt('Скопируйте ссылку вручную:', url);
400   });
401 }
402
403 // Insert result snippet into main chat
404 function sendTelegramResultToChat(url, encodedSnippet) {
405   const snippet = decodeURIComponent(encodedSnippet);
406   const textToInsert = `[Сообщение Telegram](${url}):> ${snippet.replace(/\n/g, '\n> ')}`;
407
408   if (typeof window.switchTab === 'function') {
409     window.switchTab('tab-chat');
410   }
411
412   const msgInput = document.getElementById('message-input');
413   if (msgInput) {
414     msgInput.value = (msgInput.value ? msgInput.value + '\n\n' : '') + textToInsert;
415     msgInput.focus();
416     showTgramNotification('Цитата вставлена в поле ввода чата', 'success');
417   }
418 }
419
420 // Make globally accessible
421 window.loadTelegramChannels = loadTelegramChannels;
422 window.handleAddTelegramChannel = handleAddTelegramChannel;
423 window.handleUnsubscribeTelegramChannel = handleUnsubscribeTelegramChannel;
424 window.handleReindexTelegramChannel = handleReindexTelegramChannel;
425 window.executeTelegramSearch = executeTelegramSearch;
426 window.filterByChannel = filterByChannel;
427 window.copyTelegramLink = copyTelegramLink;
428 window.sendTelegramResultToChat = sendTelegramResultToChat;
```


index.html

```
1 <!--
2 =====
3 |
4 |=====
5 Process Name: Windows Web Interface - Telemetry Config Tab Vie
6 |w
7 |=====
8 |=====
9 Description:
10 |Клиентский веб-интерфейс редактора файла конфигурации телеме
11 |трии
12 |(%APPDATA%\AI-Breadboard\apps\windows\telemetry\config.json)
13 |с поддержкой
14 |визуального и текстового редактирования и применения на лету
15 |.
16 |
17 Usage Examples:
18 |Web Route:
19 |/windows/api/webgui/telemetry_config_tab/index.html
20 |
21 File: index.html
22 Project: ai-breadboard
23 Package: windows/api/webgui/telemetry_config_tab
24 Author: hypo69
25 Copyright: © 2026 hypo69
26 Updated: 2026-10-08 03:08:00
27 |=====
28 |=====
29 -->
30
31 <div class="container-fluid p-2" id="telemetry-config-tab-cont
32 |ainer">
33 |
34 |<!-- Header & Status Card -->
35 |<div class="card shadow-sm border-secondary-subtle mb-3">
36 |<div class="card-body p-3">
37 |<div class="d-flex flex-wrap align-items-center justify-
```

```

content-between gap-2">
29     <div class="d-flex align-items-center gap-3">
30         <div class="fs-2">□ </div>
31         <div>
32             <h5 class="mb-0 fw-bold d-flex align-items-center
gap-2">
33                 Редактор конфигурации телеметрии
34                 <span class="badge bg-success-subtle text-succes
s border border-success-subtle fs-7">Hot Reload Активен</span>
35             </h5>
36             <small class="text-secondary d-block mt-1">
37                 Управление параметрами сбора, режимами реального
времени (ТС), интервалами опроса датчиков и SQLite БД с мгновенным
перечитыванием на каждом тике
38             </small>
39         </div>
40     </div>
41
42     <!-- Toolbar Buttons -->
43     <div class="d-flex align-items-center gap-2 flex-wrap"
>
44         <button id="tc-cfg-btn-reload" class="btn btn-sm btn
-outline-info d-flex align-items-center gap-1 shadow-sm" title
="Перечитать конфигурацию с диска">
45             <i class="bi bi-arrow-clockwise"></i>
46             <span>Перечитать</span>
47         </button>
48         <button id="tc-cfg-btn-reset" class="btn btn-sm btn-
outline-warning d-flex align-items-center gap-1 shadow-sm" tit
le="Сбросить к исходному шаблону по умолчанию">
49             <i class="bi bi-arrow-counterclockwise"></i>
50             <span>По умолчанию</span>
51         </button>
52         <button id="tc-cfg-btn-save" class="btn btn-sm btn-p
rimary d-flex align-items-center gap-1 shadow-sm px-3" title="
Сохранить изменения и применить на лету">
53             <i class="bi bi-check2-circle"></i>
54             <span class="fw-semibold">Сохранить и применить</s

```

```

pan>
55         </button>
56     </div>
57 </div>
58
59     <!-- File Location & Meta Bar -->
60     <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2 mt-3 pt-2 border-top border-secondary bo
rder-opacity-25 fs-8 text-secondary">
61         <div class="d-flex align-items-center gap-2">
62             <i class="bi bi-file-earmark-code text-info"></i>
63             <span>Файл: <code class="text-info bg-dark px-2 py-0
.5 rounded border border-secondary border-opacity-25" id="tc-c
fg-file-path">%APPDATA%\AI-Breadboard\apps\windows\telemetry\c
onfig.json</code></span>
64             <button class="btn btn-link btn-sm p-0 text-muted ho
ver-info" id="tc-cfg-copy-path-btn" title="Скопировать путь">
65                 <i class="bi bi-clipboard"></i>
66             </button>
67         </div>
68         <div class="d-flex align-items-center gap-3">
69             <span id="tc-cfg-mtime-wrap"><i class="bi bi-clock-h
istory me-1"></i>Изменен: <span id="tc-cfg-mtime" class="text-
white">-</span></span>
70             <span class="badge bg-primary-subtle text-primary bo
rder border-primary-subtle" id="tc-cfg-active-mode-badge">Режи
м: TELEMETRY</span>
71         </div>
72     </div>
73 </div>
74 </div>
75
76 <!-- Alert / Toast Banner -->
77 <div id="tc-cfg-alert" class="alert alert-dismissible fade s
how d-none mb-3 py-2 px-3" role="alert">
78     <div class="d-flex align-items-center gap-2">
79         <i class="bi bi-info-circle fs-5" id="tc-cfg-alert-icon"
></i>

```

```
80     <span id="tc-cfg-alert-msg">Сообщение</span>
81 </div>
82     <button type="button" class="btn-close py-2" id="tc-cfg-al
ert-close" aria-label="Заккрыть"></button>
83 </div>
84
85 <!-- Editor Mode Navigation Tabs -->
86 <ul class="nav nav-tabs nav-tabs-bordered mb-3" id="tc-cfg-v
iew-tabs" role="tablist">
87     <li class="nav-item" role="presentation">
88         <button class="nav-link active d-flex align-items-center
gap-2" id="tc-tab-gui-btn" data-bs-toggle="tab" data-bs-targe
t="#tc-tab-gui" type="button" role="tab">
89             <i class="bi bi-sliders"></i>
90             <span>Визуальный конфигуратор</span>
91         </button>
92     </li>
93     <li class="nav-item" role="presentation">
94         <button class="nav-link d-flex align-items-center gap-2"
id="tc-tab-json-btn" data-bs-toggle="tab" data-bs-target="#tc
-tab-json" type="button" role="tab">
95             <i class="bi bi-code-square"></i>
96             <span>Исходный JSON</span>
97         </button>
98     </li>
99 </ul>
100
101 <!-- Tab Content -->
102 <div class="tab-content" id="tc-cfg-tab-content">
103
104     <!-- 1. GUI FORM TAB -->
105     <div class="tab-pane fade show active" id="tc-tab-gui" rol
e="tabpanel">
106         <form id="tc-cfg-form" onsubmit="return false;">
107
108             <div class="row g-3">
109                 <!-- Column 1: Core Telemetry & DB Options -->
110                 <div class="col-lg-6">
```



```

111 |
112 |         <!-- Section 0: Telemetry Mode & Flush (Standard v
      | s TC Realtime) -->
113 |         <div class="card bg-dark bg-opacity-50 border-prim
      | ary border-opacity-50 shadow-sm mb-3">
114 |             <div class="card-header bg-dark bg-opacity-75 py
      | -2 px-3 border-primary border-opacity-25 d-flex align-items-ce
      | nter justify-content-between">
115 |                 <div class="d-flex align-items-center gap-2">
116 |                     <i class="bi bi-broadcast text-primary"></i>
117 |                     <h6 class="mb-0 fw-bold text-white fs-7">Реж
      | им работы телеметрии и Flush в БД</h6>
118 |                 </div>
119 |                 <span class="badge bg-primary-subtle text-prim
      | ary border border-primary-subtle" id="tc-current-mode-status">
      | TELEMETRY</span>
120 |             </div>
121 |             <div class="card-body p-3">
122 |                 <div class="row g-3">
123 |                     <div class="col-12">
124 |                         <label class="form-label text-muted fs-8 m
      | b-1">Режим телеметрии (telemetry_mode):</label>
125 |                         <select class="form-select form-select-sm
      | bg-dark text-white border-primary" id="tc-field-telemetry-mode
      | ">
126 |                             <option value="telemetry">telemetry – Ст
      | андартный режим (Flush каждые 30с или при 100МБ)</option>
127 |                             <option value="tc">tc – Режим реального
      | времени TC (Flush каждые 5с, автовозврат через 5 мин)</option>
128 |                         </select>
129 |                     </div>
130 |
131 |                     <div class="col-sm-6">
132 |                         <label class="form-label text-muted fs-8 m
      | b-1">Период Flush в стандартном режиме:</label>
133 |                         <div class="input-group input-group-sm">
134 |                             <input type="number" step="1" min="1" cl
      | ass="form-control bg-dark text-white border-secondary" id="tc-

```

```

field-standard-flush-interval" value="30.0">
135         <span class="input-group-text bg-dark text-muted border-secondary">c</span>
136     </div>
137 </div>
138
139     <div class="col-sm-6">
140         <label class="form-label text-muted fs-8 mb-1">Период Flush в режиме TC (реальное время):</label>
141         <div class="input-group input-group-sm">
142             <input type="number" step="0.5" min="0.5"
143             " class="form-control bg-dark text-white border-secondary" id="
144             "tc-field-tc-flush-interval" value="5.0">
145             <span class="input-group-text bg-dark text-muted border-secondary">c</span>
146         </div>
147     </div>
148
149     <div class="col-sm-6">
150         <label class="form-label text-muted fs-8 mb-1">Таймаут возврата из режима TC в стандартный:</label>
151         <div class="input-group input-group-sm">
152             <input type="number" step="10" min="10"
153             class="form-control bg-dark text-white border-secondary" id="t
154             c-field-tc-mode-timeout" value="300.0">
155             <span class="input-group-text bg-dark text-muted border-secondary">c</span>
156         </div>
157     </div>
158
159     <div class="col-sm-6">
160         <label class="form-label text-muted fs-8 mb-1">Макс. размер базы данных SQLite:</label>
161         <div class="input-group input-group-sm">
162             <input type="number" step="5" min="1" class="form-control bg-dark text-white border-secondary" id="tc-field-max-db-size" value="100.0">
163             <span class="input-group-text bg-dark text-muted border-secondary">c</span>
164         </div>
165     </div>
166 </div>

```

```

xt-muted border-secondary">МБ</span>
160         </div>
161     </div>
162 </div>
163 </div>
164 </div>
165
166     <!-- Section 1: Main Engine Parameters -->
167     <div class="card bg-dark bg-opacity-50 border-seco
168 ndary border-opacity-25 shadow-sm mb-3">
169         <div class="card-header bg-dark bg-opacity-75 py
170 -2 px-3 border-secondary border-opacity-25 d-flex align-items-
171 center gap-2">
172             <i class="bi bi-speedometer2 text-info"></i>
173             <h6 class="mb-0 fw-bold text-white fs-7">Парам
174 етры сбора сенсоров и нагрузки</h6>
175         </div>
176         <div class="card-body p-3">
177             <div class="row g-3">
178
179                 <div class="col-sm-6">
180                     <label class="form-label text-muted fs-8 m
181 b-1">Профиль сбора (mode):</label>
182                     <select class="form-select form-select-sm
183 bg-dark text-white border-secondary" id="tc-field-mode">
184                         <option value="minimal">Minimal (Легкове
185 сный CPU/RAM/Диск)</option>
186                         <option value="hybrid" selected>Hybrid (
187 Базовый + тяжелые периодически)</option>
188                         <option value="full">Full (Полный непрер
189 ывный стек)</option>
190                     </select>
191                 </div>
192
193                 <div class="col-sm-6">
194                     <label class="form-label text-muted fs-8 m
195 b-1">Режим фильтрации процессов:</label>
196                     <select class="form-select form-select-sm

```

```

    bg-dark text-white border-secondary" id="tc-field-process-mode
">
187         <option value="top_n" selected>Top-N про
цессов (по нагрузке)</option>
188         <option value="all">Все запущенные проце
ссы</option>
189     </select>
190 </div>
191
192     <div class="col-sm-6">
193         <label class="form-label text-muted fs-8 m
b-1">Быстрый базовый интервал (сек):</label>
194         <div class="input-group input-group-sm">
195             <input type="number" step="0.1" min="0.1
" class="form-control bg-dark text-white border-secondary" id=
"tc-field-interval" value="5.0">
196             <span class="input-group-text bg-dark te
xt-muted border-secondary">c</span>
197         </div>
198     </div>
199
200     <div class="col-sm-6">
201         <label class="form-label text-muted fs-8 m
b-1">Интервал тяжелых сенсоров (сек):</label>
202         <div class="input-group input-group-sm">
203             <input type="number" step="1" min="1" cl
ass="form-control bg-dark text-white border-secondary" id="tc-
field-heavy-interval" value="60.0">
204             <span class="input-group-text bg-dark te
xt-muted border-secondary">c</span>
205         </div>
206     </div>
207
208     <div class="col-sm-6">
209         <label class="form-label text-muted fs-8 m
b-1">Количество Top-процессов:</label>
210         <input type="number" step="1" min="1" max=
"500" class="form-control form-control-sm bg-dark text-white b

```

```

order-secondary" id="tc-field-top-processes" value="10">
211         </div>
212
213         <div class="col-sm-6">
214             <label class="form-label text-muted fs-8 m
b-1">Интервал агрегации (Rollup, сек):</label>
215             <div class="input-group input-group-sm">
216                 <input type="number" step="1" min="10" c
lass="form-control bg-dark text-white border-secondary" id="tc
-field-agg-interval" value="3600">
217                 <span class="input-group-text bg-dark te
xt-muted border-secondary">c</span>
218             </div>
219         </div>
220
221         <div class="col-12 pt-2 border-top border-se
condary border-opacity-25">
222             <div class="form-check form-switch mb-1">
223                 <input class="form-check-input" type="ch
eckbox" role="switch" id="tc-field-low-priority" checked>
224                 <label class="form-check-label text-whit
e fs-8" for="tc-field-low-priority">Пониженный приоритет проце
сса телеметрии (BELOW_NORMAL_PRIORITY_CLASS)</label>
225             </div>
226             <div class="form-check form-switch">
227                 <input class="form-check-input" type="ch
eckbox" role="switch" id="tc-field-heavy-auto-switch" checked>
228                 <label class="form-check-label text-whit
e fs-8" for="tc-field-heavy-auto-switch">Авто-переключение тяж
елого режима в гибридный по истечении лимита</label>
229             </div>
230         </div>
231
232     </div>
233 </div>
234 </div>
235
236 <!-- Section 2: SQLite DB & Buffer Management -->

```

```

237 |         <div class="card bg-dark bg-opacity-50 border-seco
      | ndary border-opacity-25 shadow-sm mb-3">
238 |             <div class="card-header bg-dark bg-opacity-75 py
      | -2 px-3 border-secondary border-opacity-25 d-flex align-items-
      | center gap-2">
239 |                 <i class="bi bi-database text-info"></i>
240 |                 <h6 class="mb-0 fw-bold text-white fs-7">База
      | данных SQLite и буферизация</h6>
241 |             </div>
242 |             <div class="card-body p-3">
243 |                 <div class="row g-3">
244 |
245 |                     <div class="col-sm-6">
246 |                         <label class="form-label text-muted fs-8 m
      | b-1">Срок хранения записей (дней):</label>
247 |                         <div class="input-group input-group-sm">
248 |                             <input type="number" step="1" min="1" cl
      | ass="form-control bg-dark text-white border-secondary" id="tc-
      | field-retention-days" value="7">
249 |                             <span class="input-group-text bg-dark te
      | xt-muted border-secondary">дн</span>
250 |                         </div>
251 |                     </div>
252 |
253 |                     <div class="col-sm-6">
254 |                         <label class="form-label text-muted fs-8 m
      | b-1">Интервал автоочистки БД (сек):</label>
255 |                         <div class="input-group input-group-sm">
256 |                             <input type="number" step="1" min="5" cl
      | ass="form-control bg-dark text-white border-secondary" id="tc-
      | field-db-cleanup-interval" value="300.0">
257 |                             <span class="input-group-text bg-dark te
      | xt-muted border-secondary">с</span>
258 |                         </div>
259 |                     </div>
260 |
261 |                     <div class="col-sm-6">
262 |                         <label class="form-label text-muted fs-8 m

```

```

b-1">Период сброса буфера (сек):</label>
263         <div class="input-group input-group-sm">
264             <input type="number" step="1" min="1" class="form-control bg-dark text-white border-secondary" id="tc-field-flush-interval" value="5.0">
265             <span class="input-group-text bg-dark text-muted border-secondary">c</span>
266         </div>
267     </div>
268
269     <div class="col-sm-6">
270         <label class="form-label text-muted fs-8 mb-1">Режим буферизации:</label>
271         <select class="form-select form-select-sm bg-dark text-white border-secondary" id="tc-field-buffer-mode">
272             <option value="memory" selected>Memory (Память с периодическим сбросом)</option>
273             <option value="direct">Direct (Прямая запись без буфера)</option>
274             <option value="file">File (JSONL аварийный буфер)</option>
275         </select>
276     </div>
277
278     <div class="col-sm-6">
279         <label class="form-label text-muted fs-8 mb-1">Размер буфера (записей):</label>
280         <input type="number" step="1" min="1" class="form-control form-control-sm bg-dark text-white border-secondary" id="tc-field-buffer-size" value="50">
281     </div>
282
283     <div class="col-12 pt-2 border-top border-secondary border-opacity-25">
284         <div class="form-check form-switch">
285             <input class="form-check-input" type="checkbox" role="switch" id="tc-field-auto-vacuum" checked>

```

```

286         <label class="form-check-label text-whit
e fs-8" for="tc-field-auto-vacuum">Автоматический VACUUM SQLit
e базы данных при очистке</label>
287         </div>
288     </div>
289
290     </div>
291 </div>
292 </div>
293
294     <!-- Section 3: Heavy Collectors -->
295     <div class="card bg-dark bg-opacity-50 border-seco
ndary border-opacity-25 shadow-sm mb-3">
296         <div class="card-header bg-dark bg-opacity-75 py
-2 px-3 border-secondary border-opacity-25 d-flex align-items-
center gap-2">
297             <i class="bi bi-cpu text-warning"></i>
298             <h6 class="mb-0 fw-bold text-white fs-7">Тяжел
ые периодические коллекторы (Heavy Collectors)</h6>
299         </div>
300         <div class="card-body p-3">
301             <div class="row g-2">
302                 <div class="col-sm-6">
303                     <div class="form-check form-switch">
304                         <input class="form-check-input" type="ch
eckbox" role="switch" id="tc-hc-hw-sensors" checked>
305                         <label class="form-check-label text-whit
e fs-8" for="tc-hc-hw-sensors">Аппаратные датчики (LHM)</label
>
306                     </div>
307                 </div>
308                 <div class="col-sm-6">
309                     <div class="form-check form-switch">
310                         <input class="form-check-input" type="ch
eckbox" role="switch" id="tc-hc-smart" checked>
311                         <label class="form-check-label text-whit
e fs-8" for="tc-hc-smart">SMART накопителей</label>
312                     </div>

```



```

313 |         </div>
314 |         <div class="col-sm-6">
315 |             <div class="form-check form-switch">
316 |                 <input class="form-check-input" type="checkbox"
| eckbox" role="switch" id="tc-hc-ping" checked>
317 |                 <label class="form-check-label text-white fs-8"
| for="tc-hc-ping">Сетевой пинг и Fast.com</label>
318 |             </div>
319 |         </div>
320 |         <div class="col-sm-6">
321 |             <div class="form-check form-switch">
322 |                 <input class="form-check-input" type="checkbox"
| eckbox" role="switch" id="tc-hc-wmi" checked>
323 |                 <label class="form-check-label text-white fs-8"
| for="tc-hc-wmi">WMI инвентарь оборудования</label>
324 |             </div>
325 |         </div>
326 |     </div>
327 | </div>
328 | </div>
329 |
330 | </div>
331 |
332 | <!-- Column 2: Sensors Matrix & W64/ETW Monitoring -
| ->
333 |     <div class="col-lg-6">
334 |
335 |         <!-- Section 4: Individual Sensors Grid -->
336 |         <div class="card bg-dark bg-opacity-50 border-secondary
| border-opacity-25 shadow-sm mb-3">
337 |             <div class="card-header bg-dark bg-opacity-75 py-2
| px-3 border-secondary border-opacity-25 d-flex align-items-
| center justify-content-between">
338 |                 <div class="d-flex align-items-center gap-2">
339 |                     <i class="bi bi-diagram-3 text-success"></i>
340 |                     <h6 class="mb-0 fw-bold text-white fs-7">Индивидуальные
| сенсоры и интервалы</h6>
341 |                 </div>

```

```

342 |         <span class="badge bg-secondary-subtle text-se
    | condary fs-8" id="tc-sensors-count-badge">9 сенсоров</span>
343 |     </div>
344 |     <div class="card-body p-3">
345 |         <div class="table-responsive">
346 |             <table class="table table-dark table-sm tabl
    | e-hover align-middle mb-0 fs-8 border-secondary border-opacity
    | -25" id="tc-sensors-table">
347 |                 <thead>
348 |                     <tr class="text-secondary border-bottom
    | border-secondary border-opacity-25">
349 |                         <th style="width: 40px;">Вкл</th>
350 |                         <th>Сенсор</th>
351 |                         <th style="width: 130px;">Интервал (се
    | к)</th>
352 |                         <th>Метрики</th>
353 |                     </tr>
354 |                 </thead>
355 |                 <tbody id="tc-sensors-tbody">
356 |                     <!-- Rendered dynamically by main.js -->
357 |                 </tbody>
358 |             </table>
359 |         </div>
360 |     </div>
361 | </div>
362 |
363 | <!-- Section 5: W64 & ETW Events -->
364 | <div class="card bg-dark bg-opacity-50 border-seco
    | ndary border-opacity-25 shadow-sm mb-3">
365 |     <div class="card-header bg-dark bg-opacity-75 py
    | -2 px-3 border-secondary border-opacity-25 d-flex align-items-
    | center gap-2">
366 |         <i class="bi bi-shield-check text-danger"></i>
367 |         <h6 class="mb-0 fw-bold text-white fs-7">W64 и
    | ETW мониторинг системных событий</h6>
368 |     </div>
369 |     <div class="card-body p-3">
370 |         <div class="row g-2">

```

```
371 |         <div class="col-sm-6">
372 |             <div class="form-check form-switch">
373 |                 <input class="form-check-input" type="checkbox"
374 | e fs-8" role="switch" id="tc-w64-enabled" checked>
375 |                 <label class="form-check-label text-white fs-8" for="tc-w64-enabled">W64 сборщик активен</label>
376 |             </div>
377 |         </div>
378 |         <div class="col-sm-6">
379 |             <div class="form-check form-switch">
380 |                 <input class="form-check-input" type="checkbox"
381 | e fs-8" role="switch" id="tc-w64-file" checked>
382 |                 <label class="form-check-label text-white fs-8" for="tc-w64-file">Мониторинг файлов</label>
383 |             </div>
384 |         </div>
385 |         <div class="col-sm-6">
386 |             <div class="form-check form-switch">
387 |                 <input class="form-check-input" type="checkbox"
388 | e fs-8" role="switch" id="tc-w64-proc" checked>
389 |                 <label class="form-check-label text-white fs-8" for="tc-w64-proc">Мониторинг процессов</label>
390 |             </div>
391 |         </div>
392 |         <div class="col-sm-6">
393 |             <div class="form-check form-switch">
394 |                 <input class="form-check-input" type="checkbox"
395 | e fs-8" role="switch" id="tc-w64-reg" checked>
396 |                 <label class="form-check-label text-white fs-8" for="tc-w64-reg">Мониторинг реестра</label>
397 |             </div>
398 |         </div>
399 |         <div class="col-sm-6">
400 |             <div class="form-check form-switch">
401 |                 <input class="form-check-input" type="checkbox"
402 | e fs-8" role="switch" id="tc-w64-net" checked>
403 |                 <label class="form-check-label text-white fs-8" for="tc-w64-net">Сетевые события</label>
```

```
399         </div>
400     </div>
401     <div class="col-sm-6">
402         <div class="form-check form-switch">
403             <input class="form-check-input" type="checkbox"
404             eckbox" role="switch" id="tc-w64-evlog" checked>
405             <label class="form-check-label text-white fs-8" for="tc-w64-evlog">Журнал событий Windows</label>
406         </div>
407     </div>
408     <div class="col-sm-6">
409         <div class="form-check form-switch">
410             <input class="form-check-input" type="checkbox"
411             eckbox" role="switch" id="tc-w64-etw-proc" checked>
412             <label class="form-check-label text-white fs-8" for="tc-w64-etw-proc">ETW трассировка процессов</label>
413         </div>
414     </div>
415     <div class="col-sm-6">
416         <div class="form-check form-switch">
417             <input class="form-check-input" type="checkbox"
418             eckbox" role="switch" id="tc-w64-etw-disk" checked>
419             <label class="form-check-label text-white fs-8" for="tc-w64-etw-disk">ETW трассировка диска</label>
420         </div>
421     </div>
422 </div>
423 </div>
424 </div>
425
426 </form>
427 </div>
428
429 <!-- 2. RAW JSON EDITOR TAB -->
```

```
430 |     <div class="tab-pane fade" id="tc-tab-json" role="tabpanel
    | ">
431 |         <div class="card bg-dark bg-opacity-50 border-secondary
    | border-opacity-25 shadow-sm">
432 |             <div class="card-header bg-dark bg-opacity-75 py-2 px-
    | 3 border-secondary border-opacity-25 d-flex align-items-center
    | justify-content-between flex-wrap gap-2">
433 |                 <div class="d-flex align-items-center gap-2">
434 |                     <i class="bi bi-filetype-json text-warning"></i>
435 |                     <h6 class="mb-0 fw-bold text-white fs-7">Прямое пе
    | дактирование JSON</h6>
436 |                     <span class="badge bg-secondary-subtle text-second
    | ary fs-8" id="tc-json-status">Готов</span>
437 |                 </div>
438 |                 <div class="d-flex align-items-center gap-2">
439 |                     <button class="btn btn-sm btn-outline-secondary d-
    | flex align-items-center gap-1" id="tc-btn-format-json" title="
    | Авто-форматирование с отступами">
440 |                         <i class="bi bi-text-indent-left"></i>
441 |                         <span>Форматировать</span>
442 |                     </button>
443 |                     <button class="btn btn-sm btn-outline-secondary d-
    | flex align-items-center gap-1" id="tc-btn-copy-json" title="Ск
    | опировать весь JSON в буфер">
444 |                         <i class="bi bi-clipboard"></i>
445 |                         <span>Копировать</span>
446 |                     </button>
447 |                 </div>
448 |             </div>
449 |             <div class="card-body p-2">
450 |                 <textarea class="form-control font-monospace bg-dark
    | text-info border-secondary border-opacity-25 fs-8" id="tc-raw
    | -json-editor" rows="22" spellcheck="false" style="tab-size: 2;
    | line-height: 1.45;"></textarea>
451 |                 <div id="tc-json-error" class="text-danger fs-8 mt-2
    | d-none d-flex align-items-center gap-1">
452 |                     <i class="bi bi-exclamation-triangle-fill"></i>
453 |                     <span id="tc-json-error-text">Ошибка парсинга</spa
```

```
      n>
454      </div>
455      </div>
456      </div>
457      </div>
458
459      </div>
460
461 </div>
462
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Telemetry Config Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт редактора файла конфигурации телеметрии
7   *   Windows.
8   *   Обеспечивает интерактивное визуальное и raw-JSON редактир
9   *   ование параметров
10  *   с мгновенной синхронизацией и применением на лету.
11  *
12  * Usage Examples:
13  *   HTML Integration:
14  *     <script src="/windows/api/webgui/telemetry_config_tab/m
15  *     ain.js?v=20261006_v1" type="module"></script>
16  *
17  * File: main.js
18  * Project: ai-breadboard
19  * Package: windows/api/webgui/telemetry_config_tab
20  * Author: hypo69
21  * Copyright: © 2026 hypo69
22  * Updated: 2026-10-06 03:32:00
23  * =====
24  */
25
26 (function () {
27   'use strict';
28
29   const API_BASE = '/api/windows/telemetry/config';
30   let currentConfig = {};
```

```
31 | const SENSOR_META = {
32 |   cpu: { name: 'Процессор (CPU)', icon: 'bi-cpu', defaultInterval: 5.0, defaultMetrics: ['temperature', 'load', 'clocks'] },
33 |   gpu: { name: 'Видеокарта (GPU)', icon: 'bi-gpu-card', defaultInterval: 10.0, defaultMetrics: ['temperature', 'load', 'memory', 'power'] },
34 |   ram: { name: 'Оперативная память (RAM)', icon: 'bi-memory', defaultInterval: 10.0, defaultMetrics: ['usage', 'swap'] },
35 |   disk: { name: 'Диски и разделы (Disk)', icon: 'bi-hdd', defaultInterval: 30.0, defaultMetrics: ['usage', 'io'] },
36 |   network: { name: 'Сетевые адаптеры (Network)', icon: 'bi-ethernet', defaultInterval: 10.0, defaultMetrics: ['throughput', 'connections'] },
37 |   sensors: { name: 'LHM Сенсоры и вентиляторы', icon: 'bi-fan', defaultInterval: 10.0, defaultMetrics: ['temperature', 'fan', 'voltage'] },
38 |   internet: { name: 'Скорость Fast.com / Latency', icon: 'bi-globe2', defaultInterval: 3600.0, defaultMetrics: ['ping', 'download', 'upload', 'bufferbloat', 'latency_loaded'] },
39 |   storage: { name: 'SMART здоровье накопителей', icon: 'bi-hdd-network', defaultInterval: 32000.0, defaultMetrics: ['smart_attributes', 'temperature', 'wear_level'] },
40 |   device_flapping: { name: 'Детектор PnP Flapping', icon: 'bi-usb-symbol', defaultInterval: 30.0, defaultMetrics: ['connect_events', 'disconnect_events', 'flapping_count'] },
41 | };
42 |
43 | /**
44 |  * Инициализация при монтировании вкладки
45 |  */
46 | async function init() {
47 |   bindEvents();
48 |   await loadConfig();
49 | }
50 |
51 | /**
52 |  * Привязка обработчиков событий
```



```
53 |    */
54 |    function bindEvents() {
55 |        // Кнопка сохранения
56 |        const btnSave = document.getElementById('tc-cfg-btn-save')
57 |    ;
58 |        if (btnSave) {
59 |            btnSave.addEventListener('click', saveActiveConfig);
60 |        }
61 |        // Кнопка перечитывания
62 |        const btnReload = document.getElementById('tc-cfg-btn-reload');
63 |        if (btnReload) {
64 |            btnReload.addEventListener('click', async () => {
65 |                showAlert('Перечитывание конфигурации с диска...', 'info');
66 |                await loadConfig();
67 |            });
68 |        }
69 |
70 |        // Кнопка сброса к значениям по умолчанию
71 |        const btnReset = document.getElementById('tc-cfg-btn-reset');
72 |        if (btnReset) {
73 |            btnReset.addEventListener('click', async () => {
74 |                if (confirm('Сбросить конфигурацию телеметрии к исходным настройкам по умолчанию?')) {
75 |                    await resetConfigToDefault();
76 |                }
77 |            });
78 |        }
79 |
80 |        // Кнопка копирования пути к файлу
81 |        const btnCopyPath = document.getElementById('tc-cfg-copy-path-btn');
82 |        if (btnCopyPath) {
83 |            btnCopyPath.addEventListener('click', () => {
84 |                const pathEl = document.getElementById('tc-cfg-file-path');
```

```

th');
85     if (pathEl) {
86         navigator.clipboard.writeText(pathEl.innerText.trim(
        )).then(() => {
87             showAlert('Путь к файлу скопирован в буфер обмена'
            , 'success', 2500);
88         });
89     }
90 });
91 }
92
93 // Переключение вкладок GUI / JSON
94 const tabGuiBtn = document.getElementById('tc-tab-gui-btn'
    );
95 const tabJsonBtn = document.getElementById('tc-tab-json-bt
    n');
96 if (tabGuiBtn && tabJsonBtn) {
97     tabGuiBtn.addEventListener('shown.bs.tab', () => {
98         activeTab = 'gui';
99         syncJsonToGui();
100     });
101     tabJsonBtn.addEventListener('shown.bs.tab', () => {
102         activeTab = 'json';
103         syncGuiToJson();
104     });
105 }
106
107 // Форматирование JSON
108 const btnFormatJson = document.getElementById('tc-btn-form
    at-json');
109 if (btnFormatJson) {
110     btnFormatJson.addEventListener('click', () => {
111         const editor = document.getElementById('tc-raw-json-ed
            itor');
112         if (!editor) return;
113         try {
114             const parsed = JSON.parse(editor.value);
115             editor.value = JSON.stringify(parsed, null, 2);

```

```
116 |         hideJsonError();
117 |         showAlert('JSON отформатирован', 'info', 2000);
118 |     } catch (e) {
119 |         showJsonError(e.message);
120 |     }
121 | });
122 | }
123 |
124 | // Копирование JSON
125 | const btnCopyJson = document.getElementById('tc-btn-copy-j
son');
126 | if (btnCopyJson) {
127 |     btnCopyJson.addEventListener('click', () => {
128 |         const editor = document.getElementById('tc-raw-json-ed
itor');
129 |         if (editor) {
130 |             navigator.clipboard.writeText(editor.value).then(()
=> {
131 |                 showAlert('JSON скопирован в буфер обмена', 'succe
ss', 2500);
132 |             });
133 |         }
134 |     });
135 | }
136 |
137 | // Live-валидация JSON при вводе
138 | const jsonEditor = document.getElementById('tc-raw-json-ed
itor');
139 | if (jsonEditor) {
140 |     jsonEditor.addEventListener('input', () => {
141 |         try {
142 |             JSON.parse(jsonEditor.value);
143 |             hideJsonError();
144 |         } catch (e) {
145 |             showJsonError(e.message);
146 |         }
147 |     });
148 | }
```

```
149 |
150 |     // Зкрытие алерта
151 |     const alertCloseBtn = document.getElementById('tc-cfg-aler
    t-close');
152 |     if (alertCloseBtn) {
153 |         alertCloseBtn.addEventListener('click', () => {
154 |             const alertBox = document.getElementById('tc-cfg-alert
    ');
155 |             if (alertBox) alertBox.classList.add('d-none');
156 |         });
157 |     }
158 | }
159 |
160 | /**
161 |  * Загрузка конфигурации с бэкенда
162 |  */
163 | async function loadConfig() {
164 |     try {
165 |         const response = await fetch(API_BASE, { cache: 'no-cach
    e' });
166 |         if (!response.ok) {
167 |             throw new Error(`HTTP error ${response.status}`);
168 |         }
169 |         const data = await response.json();
170 |         if (data.success) {
171 |             currentConfig = data.config || {};
172 |             updateMeta(data.config_path, data.last_modified, curre
    ntConfig.mode, currentConfig.telemetry_mode);
173 |             renderGuiForm(currentConfig);
174 |             renderRawJson(currentConfig);
175 |             showAlert('Конфигурация успешно загружена', 'success',
    2500);
176 |         } else {
177 |             showAlert(data.message || 'Ошибка загрузки конфигураци
    и', 'danger');
178 |         }
179 |     } catch (err) {
180 |         console.error('Ошибка загрузки конфигурации телеметрии:')
```

```

    , err);
181     showAlert(`Не удалось загрузить конфигурацию: ${err.mess
age}`, 'danger');
182     }
183     }
184
185     /**
186     * Обновление метаинформации о файле
187     */
188     function updateMeta(path, mtime, mode, telemetryMode) {
189         const pathEl = document.getElementById('tc-cfg-file-path')
190         ;
191         if (pathEl && path) pathEl.innerText = path;
192
193         const mtimeEl = document.getElementById('tc-cfg-mtime');
194         if (mtimeEl && mtime) {
195             try {
196                 const d = new Date(mtime);
197                 mtimeEl.innerText = d.toLocaleString('ru-RU');
198             } catch (_) {
199                 mtimeEl.innerText = mtime;
200             }
201
202             const tMode = (telemetryMode || 'telemetry').toUpperCase()
203             ;
204             const modeBadge = document.getElementById('tc-cfg-active-m
ode-badge');
205             if (modeBadge) {
206                 modeBadge.innerText = `Режим: ${tMode}`;
207                 modeBadge.className = `badge border fs-8 ${
208                     tMode === 'TC' ? 'bg-warning-subtle text-warning borde
r-warning-subtle' :
209                     'bg-primary-subtle text-primary border-primary-subtle'
210                 }`;
211             }
212
213             const currentModeStatus = document.getElementById('tc-curr

```

```

ent-mode-status');
213     if (currentModeStatus) {
214         currentModeStatus.innerText = tMode === 'TC' ? 'TC (РЕАЛ
215         БНОЕ ВРЕМЯ)' : 'СТАНДАРТНЫЙ (TELEMETRY)';
216         currentModeStatus.className = `badge border ${tMode ===
217         'TC' ? 'bg-warning-subtle text-warning border-warning-subtle'
218         : 'bg-primary-subtle text-primary border-primary-subtle'}`;
219     }
220 }
221
222 /**
223  * Отрисовка значений в элементах GUI формы
224  */
225 function renderGuiForm(cfg) {
226     setVal('tc-field-telemetry-mode', cfg.telemetry_mode || 't
227     elemetry');
228     setVal('tc-field-standard-flush-interval', cfg.standard_fl
229     ush_interval_seconds ?? 30.0);
230     setVal('tc-field-tc-flush-interval', cfg.tc_flush_interval
231     _seconds ?? 5.0);
232     setVal('tc-field-tc-mode-timeout', cfg.tc_mode_timeout_sec
233     onds ?? 300.0);
234     setVal('tc-field-max-db-size', cfg.max_db_size_mb ?? 100.0
235     );
236     setVal('tc-field-mode', cfg.mode || 'hybrid');
237     setVal('tc-field-process-mode', cfg.process_mode || 'top_n
238     ');
239     setVal('tc-field-interval', cfg.interval_seconds ?? 5.0);
240     setVal('tc-field-heavy-interval', cfg.heavy_interval_secon
241     ds ?? 60.0);
242     setVal('tc-field-top-processes', cfg.top_processes ?? 10);
243     setVal('tc-field-agg-interval', cfg.aggregation_interval_s
244     econds ?? 3600);
245     setChecked('tc-field-low-priority', cfg.low_priority !== f
246     alse);
247     setChecked('tc-field-heavy-auto-switch', cfg.heavy_mode_au
248     to_switch_enabled !== false);

```

```
237 |
238 |     setVal('tc-field-retention-days', cfg.retention_days ?? 7)
    | ;
239 |     setVal('tc-field-db-cleanup-interval', cfg.db_cleanup_inte
    | rval_seconds ?? 300.0);
240 |     setVal('tc-field-flush-interval', cfg.flush_interval_secon
    | ds ?? 5.0);
241 |     setVal('tc-field-buffer-mode', cfg.buffer_mode || 'memory'
    | );
242 |     setVal('tc-field-buffer-size', cfg.buffer_size ?? 50);
243 |     setChecked('tc-field-auto-vacuum', cfg.auto_vacuum_enabled
    | !== false);
244 |
245 |     const hc = cfg.heavy_collectors || {};
246 |     setChecked('tc-hc-hw-sensors', hc.hardware_sensors !== fal
    | se);
247 |     setChecked('tc-hc-smart', hc.storage_smart !== false);
248 |     setChecked('tc-hc-ping', hc.network_ping !== false);
249 |     setChecked('tc-hc-wmi', hc.inventory_wmi === true);
250 |
251 |     const w64 = cfg.w64_collector || {};
252 |     setChecked('tc-w64-enabled', w64.enabled !== false);
253 |     setChecked('tc-w64-file', w64.enable_file_monitoring !== f
    | alse);
254 |     setChecked('tc-w64-proc', w64.enable_process_monitoring !=
    | = false);
255 |     setChecked('tc-w64-reg', w64.enable_registry_monitoring !=
    | = false);
256 |     setChecked('tc-w64-net', w64.enable_network_monitoring !=
    | false);
257 |     setChecked('tc-w64-evlog', w64.enable_event_log_monitoring
    | !== false);
258 |     setChecked('tc-w64-etw-proc', w64.enable_process_trace !=
    | false);
259 |     setChecked('tc-w64-etw-disk', w64.enable_disk_trace !== fa
    | lse);
260 |
261 |     renderSensorsTable(cfg.sensors || {});
```

```

262 | }
263 |
264 | /**
265 |  * Отрисовка таблицы сенсоров
266 |  */
267 | function renderSensorsTable(sensorsConfig) {
268 |     const tbody = document.getElementById('tc-sensors-tbody');
269 |     if (!tbody) return;
270 |
271 |     tbody.innerHTML = '';
272 |     const sensorKeys = Object.keys(SENSOR_META);
273 |
274 |     // Добавляем также сенсоры, которые есть в конфиге, но не
275 |     в SENSOR_META
276 |     Object.keys(sensorsConfig).forEach((k) => {
277 |         if (!sensorKeys.includes(k)) sensorKeys.push(k);
278 |     });
279 |
280 |     const countBadge = document.getElementById('tc-sensors-count-badge');
281 |     if (countBadge) countBadge.innerText = `${sensorKeys.length} сенсоров`;
282 |
283 |     sensorKeys.forEach((key) => {
284 |         const sMeta = SENSOR_META[key] || {
285 |             name: key,
286 |             icon: 'bi-activity',
287 |             defaultInterval: 10.0,
288 |             defaultMetrics: [],
289 |         };
290 |         const sConf = sensorsConfig[key] || {
291 |             enabled: true,
292 |             interval_seconds: sMeta.defaultInterval,
293 |             metrics: sMeta.defaultMetrics,
294 |         };
295 |
296 |         const tr = document.createElement('tr');
297 |         tr.className = 'border-bottom border-secondary border-op

```



```

    acity-10';
297
298     const isEnabled = sConf.enabled !== false;
299     const intervalSec = sConf.interval_seconds ?? sMeta.defaultInterval;
300     const metricsList = Array.isArray(sConf.metrics) ? sConf.metrics.join(', ') : '';
301
302     tr.innerHTML = `
303         <td class="text-center">
304             <div class="form-check form-switch mb-0">
305                 <input class="form-check-input tc-sensor-enabled"
306                 type="checkbox" role="switch" data-sensor="${key}" ${isEnabled
307                 ? 'checked' : ''}>
308             </div>
309         </td>
310         <td>
311             <div class="d-flex align-items-center gap-2">
312                 <i class="bi ${sMeta.icon} text-info"></i>
313                 <div>
314                     <span class="fw-semibold text-white">${sMeta.name}</span>
315                     <span class="text-secondary d-block fs-9 font-monospace">${key}</span>
316                 </div>
317             </div>
318         </td>
319         <td>
320             <div class="input-group input-group-sm" style="max-width: 120px;">
321                 <input type="number" step="0.5" min="0.1" class="form-control form-control-sm bg-dark text-white border-secondary tc-sensor-interval" data-sensor="${key}" value="${intervalSec}">
322                 <span class="input-group-text bg-dark text-muted border-secondary fs-9">c</span>
323             </div>
324         </td>

```

```

323         <td>
324             <span class="text-secondary fs-9 font-monospace" tit
le="${metricsList}">${metricsList || '-'}</span>
325         </td>
326     `;
327
328     tbody.appendChild(tr);
329 });
330 }
331
332 /**
333  * Сбор параметров из элементов GUI формы в объект
334  */
335 function gatherFormConfig() {
336     const cfg = JSON.parse(JSON.stringify(currentConfig));
337
338     cfg.telemetry_mode = getVal('tc-field-telemetry-mode') ||
'telemetry';
339     cfg.standard_flush_interval_seconds = parseFloat(getVal('t
c-field-standard-flush-interval')) || 30.0;
340     cfg.tc_flush_interval_seconds = parseFloat(getVal('tc-fiel
d-tc-flush-interval')) || 5.0;
341     cfg.tc_mode_timeout_seconds = parseFloat(getVal('tc-field-
tc-mode-timeout')) || 300.0;
342     cfg.max_db_size_mb = parseFloat(getVal('tc-field-max-db-si
ze')) || 100.0;
343
344     cfg.mode = getVal('tc-field-mode') || 'hybrid';
345     cfg.process_mode = getVal('tc-field-process-mode') || 'top
_n';
346     cfg.interval_seconds = parseFloat(getVal('tc-field-interva
l')) || 5.0;
347     cfg.heavy_interval_seconds = parseFloat(getVal('tc-field-h
eavy-interval')) || 60.0;
348     cfg.top_processes = parseInt(getVal('tc-field-top-processe
s'), 10) || 10;
349     cfg.aggregation_interval_seconds = parseInt(getVal('tc-fie
ld-agg-interval'), 10) || 3600;

```

```
350     cfg.low_priority = getChecked('tc-field-low-priority');
351     cfg.heavy_mode_auto_switch_enabled = getChecked('tc-field-
heavy-auto-switch');
352
353     cfg.retention_days = parseInt(getVal('tc-field-retention-d
ays'), 10) || 7;
354     cfg.db_cleanup_interval_seconds = parseFloat(getVal('tc-fi
eld-db-cleanup-interval')) || 300.0;
355     cfg.flush_interval_seconds = parseFloat(getVal('tc-field-f
lush-interval')) || 5.0;
356     cfg.buffer_mode = getVal('tc-field-buffer-mode') || 'memor
y';
357     cfg.buffer_size = parseInt(getVal('tc-field-buffer-size'),
10) || 50;
358     cfg.auto_vacuum_enabled = getChecked('tc-field-auto-vacuum
');
359
360     cfg.heavy_collectors = cfg.heavy_collectors || {};
361     cfg.heavy_collectors.hardware_sensors = getChecked('tc-hc-
hw-sensors');
362     cfg.heavy_collectors.storage_smart = getChecked('tc-hc-sma
rt');
363     cfg.heavy_collectors.network_ping = getChecked('tc-hc-ping
');
364     cfg.heavy_collectors.inventory_wmi = getChecked('tc-hc-wmi
');
365
366     cfg.w64_collector = cfg.w64_collector || {};
367     cfg.w64_collector.enabled = getChecked('tc-w64-enabled');
368     cfg.w64_collector.enable_file_monitoring = getChecked('tc-
w64-file');
369     cfg.w64_collector.enable_process_monitoring = getChecked('
tc-w64-proc');
370     cfg.w64_collector.enable_registry_monitoring = getChecked(
'tc-w64-reg');
371     cfg.w64_collector.enable_network_monitoring = getChecked('
tc-w64-net');
372     cfg.w64_collector.enable_event_log_monitoring = getChecked
```

```

    ('tc-w64-evlog');
373     cfg.w64_collector.enable_process_trace = getChecked('tc-w6
    4-etw-proc');
374     cfg.w64_collector.enable_disk_trace = getChecked('tc-w64-e
    tw-disk');
375
376     cfg.sensors = cfg.sensors || {};
377     document.querySelectorAll('.tc-sensor-enabled').forEach((c
    b) => {
378         const sKey = cb.getAttribute('data-sensor');
379         if (!sKey) return;
380         cfg.sensors[sKey] = cfg.sensors[sKey] || {};
381         cfg.sensors[sKey].enabled = cb.checked;
382     });
383
384     document.querySelectorAll('.tc-sensor-interval').forEach((
    inp) => {
385         const sKey = inp.getAttribute('data-sensor');
386         if (!sKey) return;
387         cfg.sensors[sKey] = cfg.sensors[sKey] || {};
388         cfg.sensors[sKey].interval_seconds = parseFloat(inp.valu
    e) || 5.0;
389     });
390
391     return cfg;
392 }
393
394 /**
395  * Отрисовка текста в редакторе сырого JSON
396  */
397 function renderRawJson(cfg) {
398     const editor = document.getElementById('tc-raw-json-editor
    ');
399     if (editor) {
400         editor.value = JSON.stringify(cfg, null, 2);
401         hideJsonError();
402     }
403 }

```

```

404
405  /**
406   * Синхронизация формы в JSON при переходе на вкладку JSON
407   */
408  function syncGuiToJson() {
409    try {
410      const updated = gatherFormConfig();
411      currentConfig = updated;
412      renderRawJson(updated);
413    } catch (e) {
414      console.warn('Ошибка сбора формы:', e);
415    }
416  }
417
418  /**
419   * Синхронизация JSON в форму при переходе на вкладку GUI
420   */
421  function syncJsonToGui() {
422    const editor = document.getElementById('tc-raw-json-editor
423  ');
424    if (!editor) return;
425    try {
426      const parsed = JSON.parse(editor.value);
427      currentConfig = parsed;
428      renderGuiForm(parsed);
429      hideJsonError();
430    } catch (e) {
431      showJsonError(`Невозможно отобразить визуальную форму: $
432      {e.message}`);
433    }
434  }
435
436  /**
437   * Сохранение активной конфигурации (из GUI или JSON)
438   */
439  async function saveActiveConfig() {
440    const btnSave = document.getElementById('tc-cfg-btn-save')
441  ;

```

```
439     const originalText = btnSave ? btnSave.innerHTML : '';
440     if (btnSave) {
441         btnSave.disabled = true;
442         btnSave.innerHTML = '<span class="spinner-border spinner
-border-sm me-1" role="status"></span><span>Сохранение...</spa
n>';
443     }
444
445     try {
446         if (activeTab === 'json') {
447             const editor = document.getElementById('tc-raw-json-ed
itor');
448             if (!editor) throw new Error('Редактор JSON не найден'
);
449             const rawJson = editor.value;
450
451             // Предварительная валидация
452             JSON.parse(rawJson);
453
454             const response = await fetch(`${API_BASE}/raw`, {
455                 method: 'POST',
456                 headers: { 'Content-Type': 'application/json' },
457                 body: JSON.stringify({ raw_json: rawJson }),
458             });
459
460             const resData = await response.json();
461             if (!response.ok || !resData.success) {
462                 throw new Error(resData.detail || resData.message ||
`Ошибка сервера ${response.status}`);
463             }
464
465             currentConfig = resData.config;
466             renderGuiForm(currentConfig);
467             updateMeta(resData.config_path, new Date().toISOString
(), currentConfig.mode, currentConfig.telemetry_mode);
468             showAlert('☑ Конфигурация телеметрии успешно сохранена
и применена на лету!', 'success');
469         } else {
```

```

470     const updatedConfig = gatherFormConfig();
471     const response = await fetch(API_BASE, {
472         method: 'POST',
473         headers: { 'Content-Type': 'application/json' },
474         body: JSON.stringify({ config: updatedConfig }),
475     });
476
477     const resData = await response.json();
478     if (!response.ok || !resData.success) {
479         throw new Error(resData.detail || resData.message ||
`Ошибка сервера ${response.status}`);
480     }
481
482     currentConfig = resData.config;
483     renderRawJson(currentConfig);
484     updateMeta(resData.config_path, new Date().toISOString
(), currentConfig.mode, currentConfig.telemetry_mode);
485     showAlert('✅ Параметры телеметрии успешно сохранены и
применены на лету!', 'success');
486 }
487 } catch (err) {
488     console.error('Ошибка сохранения конфигурации:', err);
489     showAlert(`❌ Ошибка сохранения: ${err.message}`, 'danger
');
490     if (activeTab === 'json') {
491         showJsonError(err.message);
492     }
493 } finally {
494     if (btnSave) {
495         btnSave.disabled = false;
496         btnSave.innerHTML = originalText;
497     }
498 }
499 }
500
501 /**
502  * Сброс конфигурации к значениям по умолчанию
503  */

```

```

504 | async function resetConfigToDefault() {
505 |     try {
506 |         const response = await fetch(`${API_BASE}/reset`, {
507 |             method: 'POST',
508 |             headers: { 'Content-Type': 'application/json' },
509 |         });
510 |         const data = await response.json();
511 |         if (data.success) {
512 |             currentConfig = data.config || {};
513 |             updateMeta(data.config_path, new Date().toISOString(),
514 | currentConfig.mode, currentConfig.telemetry_mode);
515 |             renderGuiForm(currentConfig);
516 |             renderRawJson(currentConfig);
517 |             showAlert('⚠ Конфигурация телеметрии сброшена к значен
иям по умолчанию!', 'warning');
518 |         } else {
519 |             showAlert(data.message || 'Ошибка сброса конфигурации'
, 'danger');
520 |         }
521 |     } catch (err) {
522 |         showAlert(`Не удалось сбросить конфигурацию: ${err.messa
ge}`, 'danger');
523 |     }
524 |
525 |     /**
526 |      * Показать плашку ошибки валидации JSON
527 |      */
528 |     function showJsonError(msg) {
529 |         const errBox = document.getElementById('tc-json-error');
530 |         const errText = document.getElementById('tc-json-error-tex
t');
531 |         const statusBadge = document.getElementById('tc-json-stat
s');
532 |         if (errBox && errText) {
533 |             errText.innerText = msg;
534 |             errBox.classList.remove('d-none');
535 |         }

```



```

536 |     if (statusBadge) {
537 |         statusBadge.innerText = 'Ошибка синтаксиса';
538 |         statusBadge.className = 'badge bg-danger-subtle text-dan
    | ger border border-danger-subtle fs-8';
539 |     }
540 | }
541 |
542 | /**
543 |  * Скрыть плашку ошибки валидации JSON
544 |  */
545 | function hideJsonError() {
546 |     const errBox = document.getElementById('tc-json-error');
547 |     const statusBadge = document.getElementById('tc-json-statu
    | s');
548 |     if (errBox) errBox.classList.add('d-none');
549 |     if (statusBadge) {
550 |         statusBadge.innerText = 'Синтаксис корректен';
551 |         statusBadge.className = 'badge bg-success-subtle text-su
    | ccess border border-success-subtle fs-8';
552 |     }
553 | }
554 |
555 | /**
556 |  * Отображение Toast / Alert сообщения
557 |  */
558 | function showAlert(msg, type = 'info', autoHideMs = 4000) {
559 |     const alertBox = document.getElementById('tc-cfg-alert');
560 |     const alertMsg = document.getElementById('tc-cfg-alert-msg
    | ');
561 |     const alertIcon = document.getElementById('tc-cfg-alert-ic
    | on');
562 |     if (!alertBox || !alertMsg) return;
563 |
564 |     alertMsg.innerText = msg;
565 |     alertBox.className = `alert alert-${type} alert-dismissibl
    | e fade show mb-3 py-2 px-3 shadow-sm border`;
566 |
567 |     if (alertIcon) {

```

```
568 |         alertIcon.className =
569 |             type === 'success' ? 'bi bi-check-circle-fill text-suc
| ccess fs-5' :
570 |             type === 'danger' ? 'bi bi-exclamation-triangle-fill t
| ext-danger fs-5' :
571 |             type === 'warning' ? 'bi bi-exclamation-circle-fill te
| xt-warning fs-5' :
572 |             'bi bi-info-circle-fill text-info fs-5';
573 |     }
574 |
575 |     alertBox.classList.remove('d-none');
576 |
577 |     if (autoHideMs > 0) {
578 |         setTimeout(() => {
579 |             if (!alertBox.classList.contains('d-none')) {
580 |                 alertBox.classList.add('d-none');
581 |             }
582 |             }, autoHideMs);
583 |     }
584 | }
585 |
586 | // Вспомогательные утилиты для работы с элементами DOM
587 | function setVal(id, val) {
588 |     const el = document.getElementById(id);
589 |     if (el) el.value = val;
590 | }
591 |
592 | function getVal(id) {
593 |     const el = document.getElementById(id);
594 |     return el ? el.value : '';
595 | }
596 |
597 | function setChecked(id, val) {
598 |     const el = document.getElementById(id);
599 |     if (el) el.checked = Boolean(val);
600 | }
601 |
602 | function getChecked(id) {
```

```
603 |     const el = document.getElementById(id);
604 |     return el ? Boolean(el.checked) : false;
605 | }
606 |
607 | // Запуск при загрузке DOM
608 | if (document.readyState === 'loading') {
609 |     document.addEventListener('DOMContentLoaded', init);
610 | } else {
611 |     init();
612 | }
613 |})();
614 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/telemetry_research_tab/index.html
15
16   Tab Frame Integration:
17     <iframe src="/windows/api/webgui/telemetry_research_tab/
18 index.html" class="w-100 h-100 border-0"></iframe>
19
20
21 File: index.html
22 Project: ai-breadboard
23 Package: windows/api/webgui/telemetry_research_tab
24 Author: hypo69
25 Copyright: © 2026 hypo69
26 Updated: 2026-10-08 03:08:00
27 =====
28 =====
29 -->
30
31 <div class="container-fluid p-2" id="telemetry-research-tab-co
32 ntainer">
33
34   <!-- Header & Control Toolbar -->
35   <div class="card shadow-sm border-secondary-subtle mb-3">
36     <div class="card-body p-3">
37       <div class="d-flex flex-wrap align-items-center justify-
38 content-between gap-2 mb-3">
39         <div class="d-flex align-items-center gap-2">
```

```

31         <div class="fs-2">□ </div>
32     <div>
33         <h5 class="mb-0 fw-bold">Исследования телеметрии и
| оборудование</h5>
34         <small class="text-light text-opacity-75">Глубокий
| анализ корреляций, проверка гипотез производительности, анома
| лии и Health Score</small>
35     </div>
36 </div>
37
38     <div class="d-flex align-items-center gap-2 flex-wrap"
| >
39         <div class="input-group input-group-sm" style="width
| : auto;">
40             <span class="input-group-text bg-dark text-muted b
| order-secondary"><i class="bi bi-folder2-open"></i></span>
41             <select class="form-select form-select-sm bg-dark
| text-white border-secondary" id="tr-select-source" style="max-
| width: 260px;">
42                 <option value="">Все доступные логи телеметрии</
| option>
43             </select>
44         </div>
45
46         <button id="tr-btn-run" class="btn btn-sm btn-primar
| y d-flex align-items-center gap-1 shadow-sm" title="Запустить
| полное исследование">
47             <i class="bi bi-play-circle-fill"></i>
48             <span>Исследовать</span>
49         </button>
50
51         <button id="tr-btn-dashboard" class="btn btn-sm btn-
| outline-success d-flex align-items-center gap-1 shadow-sm" tit
| le="Открыть автономный HTML-дашборд в новой вкладке">
52             <i class="bi bi-file-earmark-bar-graph"></i>
53             <span>HTML-дашборд</span>
54         </button>
55

```

```

56 |         <button id="tr-btn-json" class="btn btn-sm btn-outli
    | ne-secondary d-flex align-items-center gap-1 shadow-sm" title=
    | "Скачать JSON отчет">
57 |             <i class="bi bi-download"></i>
58 |             <span>JSON</span>
59 |         </button>
60 |     </div>
61 | </div>
62 |
63 | <!-- KPI Metrics Cards -->
64 | <div class="row g-2 text-white">
65 |     <!-- Health Score -->
66 |     <div class="col-6 col-md-3">
67 |         <div class="p-2 rounded bg-dark bg-opacity-50 border
    | border-secondary d-flex align-items-center gap-3">
68 |             <div class="fs-3 text-info"><i class="bi bi-heart-
    | pulse-fill"></i></div>
69 |             <div class="flex-grow-1">
70 |                 <div class="text-muted small">Индекс здоровья</d
    | iv>
71 |                 <div class="d-flex align-items-baseline gap-2">
72 |                     <span class="h5 mb-0 fw-bold" id="tr-val-healt
    | h">--</span>
73 |                     <span class="small text-muted">/100</span>
74 |                 </div>
75 |                 <div class="progress mt-1" style="height: 4px;">
76 |                     <div id="tr-progress-health" class="progress-b
    | ar bg-success" style="width: 0%;"></div>
77 |                 </div>
78 |             </div>
79 |         </div>
80 |     </div>
81 |
82 | <!-- Records Analyzed -->
83 | <div class="col-6 col-md-3">
84 |     <div class="p-2 rounded bg-dark bg-opacity-50 border
    | border-secondary d-flex align-items-center gap-3">
85 |         <div class="fs-3 text-primary"><i class="bi bi-dat

```

```

abase-check"></i></div>
86         <div>
87             <div class="text-muted small">Записей исследован
o</div>
88             <div class="h5 mb-0 fw-bold" id="tr-val-records"
>0</div>
89             <div class="small text-muted" id="tr-val-timespa
n">Временной срез</div>
90         </div>
91     </div>
92 </div>
93
94     <!-- Hypotheses Confirmed -->
95     <div class="col-6 col-md-3">
96         <div class="p-2 rounded bg-dark bg-opacity-50 border
border-secondary d-flex align-items-center gap-3">
97             <div class="fs-3 text-warning"><i class="bi bi-lig
htbulb-fill"></i></div>
98             <div>
99                 <div class="text-muted small">Проблемных гипотез
</div>
100                 <div class="h5 mb-0 fw-bold" id="tr-val-hypothes
es">0</div>
101                 <div class="small text-muted">Из проверенных</di
v>
102             </div>
103         </div>
104     </div>
105
106     <!-- Anomalies & Incidents -->
107     <div class="col-6 col-md-3">
108         <div class="p-2 rounded bg-dark bg-opacity-50 border
border-secondary d-flex align-items-center gap-3">
109             <div class="fs-3 text-danger"><i class="bi bi-excl
amation-triangle-fill"></i></div>
110             <div>
111                 <div class="text-muted small">Аномалии и сбои</d
iv>

```

```

112 |         <div class="h5 mb-0 fw-bold" id="tr-val-anomalie
    | s">0</div>
113 |         <div class="small text-muted" id="tr-val-devices
    | ">Устройства в норме</div>
114 |     </div>
115 | </div>
116 | </div>
117 | </div>
118 | </div>
119 | </div>
120 |
121 | <!-- Summary Alert Banner -->
122 | <div id="tr-summary-banner" class="alert alert-info border-i
    | nfo d-flex align-items-center gap-2 mb-3 py-2 px-3 small d-non
    | e">
123 |     <i class="bi bi-info-circle-fill fs-5"></i>
124 |     <div id="tr-summary-text" class="flex-grow-1"></div>
125 | </div>
126 |
127 | <!-- Tabs Navigation -->
128 | <ul class="nav nav-tabs border-secondary mb-3" id="trResearc
    | hTabs" role="tablist">
129 |     <li class="nav-item" role="presentation">
130 |         <button class="nav-link active fw-medium py-2 px-3" id="
    | tr-tab-hypotheses-btn" data-bs-toggle="tab" data-bs-target="#t
    | r-tab-hypotheses" type="button" role="tab">
131 |             □ Проверка гипотез
132 |         </button>
133 |     </li>
134 |     <li class="nav-item" role="presentation">
135 |         <button class="nav-link fw-medium py-2 px-3" id="tr-tab-
    | correlations-btn" data-bs-toggle="tab" data-bs-target="#tr-tab
    | -correlations" type="button" role="tab">
136 |             □ Матрица корреляций
137 |         </button>
138 |     </li>
139 |     <li class="nav-item" role="presentation">
140 |         <button class="nav-link fw-medium py-2 px-3" id="tr-tab-

```



```

charts-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-chart
s" type="button" role="tab">
141     □ Графики метрик
142     </button>
143 </li>
144 <li class="nav-item" role="presentation">
145     <button class="nav-link fw-medium py-2 px-3" id="tr-tab-
anomalies-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-an
omalies" type="button" role="tab">
146     □ Аномалии и устройства
147     </button>
148 </li>
149 <li class="nav-item" role="presentation">
150     <button class="nav-link fw-medium py-2 px-3" id="tr-tab-
records-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-reco
rds" type="button" role="tab">
151     □ Записи телеметрии
152     </button>
153 </li>
154 </ul>
155
156 <!-- Tab Contents -->
157 <div class="tab-content" id="trResearchTabContent">
158
159 <!-- 1. Hypotheses Tab -->
160 <div class="tab-pane fade show active" id="tr-tab-hypothes
es" role="tabpanel">
161     <div class="row g-3" id="tr-hypotheses-list">
162         <div class="col-12 text-center text-muted py-4">Нажмит
е «Исследовать» для проверки системных гипотез...</div>
163     </div>
164 </div>
165
166 <!-- 2. Correlations Tab -->
167 <div class="tab-pane fade" id="tr-tab-correlations" role="
tabpanel">
168     <div class="card bg-dark border-secondary">
169         <div class="card-header border-secondary d-flex align-

```

```

items-center justify-content-between">
170     <span class="fw-bold small text-light">Взаимосвязи м
    ежду системными метриками (Пирсон)</span>
171     <span class="badge bg-secondary" id="tr-correlations
    -count">0 пар</span>
172     </div>
173     <div class="table-responsive">
174         <table class="table table-dark table-hover table-str
    iped mb-0 align-middle small">
175             <thead>
176                 <tr class="text-muted border-secondary">
177                     <th>Метрика А</th>
178                     <th>Метрика Б</th>
179                     <th>Коэффициент (r)</th>
180                     <th>Выборка</th>
181                     <th>Интерпретация</th>
182                 </tr>
183             </thead>
184             <tbody id="tr-correlations-tbody">
185                 <tr><td colspan="5" class="text-center text-mute
    d py-3">Нет данных корреляций</td></tr>
186             </tbody>
187         </table>
188     </div>
189 </div>
190 </div>
191
192 <!-- 3. Charts Tab -->
193 <div class="tab-pane fade" id="tr-tab-charts" role="tabpan
    el">
194     <div class="row g-3" id="tr-charts-container">
195         <div class="col-12 text-center text-muted py-4">График
    и появятся после запуска исследования...</div>
196     </div>
197 </div>
198
199 <!-- 4. Anomalies & Devices Tab -->
200 <div class="tab-pane fade" id="tr-tab-anomalies" role="tab

```

```

panel">
201     <div class="row g-3">
202         <!-- Anomalies Table -->
203         <div class="col-12 col-lg-7">
204             <div class="card bg-dark border-secondary h-100">
205                 <div class="card-header border-secondary fw-bold s
mall text-light d-flex justify-content-between">
206                     <span>Зафиксированные аномалии</span>
207                     <span class="badge bg-danger" id="tr-anomalies-b
adge">0</span>
208                 </div>
209                 <div class="table-responsive">
210                     <table class="table table-dark table-hover table
-striped mb-0 align-middle small">
211                         <thead>
212                             <tr class="text-muted border-secondary">
213                                 <th>Время</th>
214                                 <th>Метрика</th>
215                                 <th>Уровень</th>
216                                 <th>Значение / Порог</th>
217                                 <th>Описание</th>
218                             </tr>
219                         </thead>
220                         <tbody id="tr-anomalies-tbody">
221                             <tr><td colspan="5" class="text-center text-
muted py-3">Аномалий не обнаружено</td></tr>
222                         </tbody>
223                     </table>
224                 </div>
225             </div>
226         </div>
227
228         <!-- Devices Summary -->
229         <div class="col-12 col-lg-5">
230             <div class="card bg-dark border-secondary h-100">
231                 <div class="card-header border-secondary fw-bold s
mall text-light d-flex justify-content-between">
232                     <span>Состояние оборудования</span>

```

```

233 |         <span class="badge bg-info" id="tr-devices-badge
    | ">OK</span>
234 |     </div>
235 |     <div class="card-body p-3 small" id="tr-devices-co
    | ntent">
236 |         <p class="text-muted">Данные об оборудовании заг
    | ружаются...</p>
237 |     </div>
238 | </div>
239 | </div>
240 | </div>
241 | </div>
242 |
243 | <!-- 5. Records Tab -->
244 | <div class="tab-pane fade" id="tr-tab-records" role="tabpa
    | nel">
245 |     <div class="card bg-dark border-secondary">
246 |         <div class="card-header border-secondary d-flex align-
    | items-center justify-content-between gap-2 flex-wrap">
247 |             <div class="d-flex align-items-center gap-2">
248 |                 <input type="text" id="tr-records-search" class="f
    | orm-control form-control-sm bg-dark text-white border-secondar
    | y" placeholder="Поиск по записям..." style="width: 250px;">
249 |                 <button id="tr-records-search-btn" class="btn btn-
    | sm btn-outline-secondary"><i class="bi bi-search"></i></button
    | >
250 |             </div>
251 |             <div class="d-flex align-items-center gap-2">
252 |                 <span class="small text-muted" id="tr-records-pagi
    | nation-info">Стр. 1 из 1</span>
253 |                 <div class="btn-group btn-group-sm">
254 |                     <button class="btn btn-outline-secondary" id="tr
    | -records-prev-btn"><i class="bi bi-chevron-left"></i></button>
255 |                     <button class="btn btn-outline-secondary" id="tr
    | -records-next-btn"><i class="bi bi-chevron-right"></i></button
    | >
256 |                 </div>
257 |             </div>

```

```

258 |         </div>
259 |         <div class="table-responsive" style="max-height: 500px
    |         ;">
260 |             <table class="table table-dark table-hover table-str
    |             iped mb-0 align-middle small font-monospace">
261 |                 <thead>
262 |                     <tr class="text-muted border-secondary">
263 |                         <th>#</th>
264 |                         <th>Timestamp</th>
265 |                         <th>CPU %</th>
266 |                         <th>RAM %</th>
267 |                         <th>GPU %</th>
268 |                         <th>Диск (R/W)</th>
269 |                         <th>Событие / Детали</th>
270 |                     </tr>
271 |                 </thead>
272 |                 <tbody id="tr-records-tbody">
273 |                     <tr><td colspan="7" class="text-center text-mute
    |                     d py-3">Записи не загружены</td></tr>
274 |                 </tbody>
275 |             </table>
276 |         </div>
277 |     </div>
278 | </div>
279 |
280 | </div>
281 |
282 | </div>
283 |

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Telemetry Research Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/telemetry_research_tab
11  * /main.js?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/telemetry_research_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 let trCurrentReport = null;
22 let trChartsInstances = [];
23 let trCurrentPage = 1;
24 const trPageSize = 25;
25 let trTotalPages = 1;
26 let trSearchQuery = '';
27
28 /**
29  * Инициализация вкладки исследования телеметрии.
30  */
31 export async function initTelemetryResearchTab() {
32   bindEvents();
```

```
33 |   await loadSources();
34 |   await runResearch();
35 | }
36 |
37 | /**
38 |  * Привязка событий элементов управления.
39 |  */
40 | function bindEvents() {
41 |   const btnRun = document.getElementById('tr-btn-run');
42 |   if (btnRun) {
43 |     btnRun.onclick = () => runResearch();
44 |   }
45 |
46 |   const btnDashboard = document.getElementById('tr-btn-dashboa
  rd');
47 |   if (btnDashboard) {
48 |     btnDashboard.onclick = () => openHtmlDashboard();
49 |   }
50 |
51 |   const btnJson = document.getElementById('tr-btn-json');
52 |   if (btnJson) {
53 |     btnJson.onclick = () => downloadJsonReport();
54 |   }
55 |
56 |   const selectSource = document.getElementById('tr-select-sour
  ce');
57 |   if (selectSource) {
58 |     selectSource.onchange = () => runResearch();
59 |   }
60 |
61 |   const searchBtn = document.getElementById('tr-records-search
  -btn');
62 |   const searchInput = document.getElementById('tr-records-sear
  ch');
63 |   if (searchBtn && searchInput) {
64 |     searchBtn.onclick = () => {
65 |       trSearchQuery = searchInput.value.trim();
66 |       trCurrentPage = 1;
```

```
67 |     loadRecords();
68 | };
69 | searchInput.onkeydown = (e) => {
70 |     if (e.key === 'Enter') {
71 |         trSearchQuery = searchInput.value.trim();
72 |         trCurrentPage = 1;
73 |         loadRecords();
74 |     }
75 | };
76 | }
77 |
78 | const prevBtn = document.getElementById('tr-records-prev-btn
    ');
79 | const nextBtn = document.getElementById('tr-records-next-btn
    ');
80 | if (prevBtn) {
81 |     prevBtn.onclick = () => {
82 |         if (trCurrentPage > 1) {
83 |             trCurrentPage--;
84 |             loadRecords();
85 |         }
86 |     };
87 | }
88 | if (nextBtn) {
89 |     nextBtn.onclick = () => {
90 |         if (trCurrentPage < trTotalPages) {
91 |             trCurrentPage++;
92 |             loadRecords();
93 |         }
94 |     };
95 | }
96 |
97 | const recordsTabBtn = document.getElementById('tr-tab-record
    s-btn');
98 | if (recordsTabBtn) {
99 |     recordsTabBtn.addEventListener('shown.bs.tab', () => {
100 |         loadRecords();
101 |     });
```



```

102 | }
103 | }
104 |
105 | /**
106 |  * Загрузка списка доступных источников телеметрии.
107 |  */
108 | async function loadSources() {
109 |     const select = document.getElementById('tr-select-source');
110 |     if (!select) return;
111 |
112 |     try {
113 |         const res = await fetch('/apps/telemetry_research/sources'
114 | );
115 |         if (!res.ok) return;
116 |         const files = await res.json();
117 |         select.innerHTML = '<option value="">Все доступные логи те
118 | леметрии</option>';
119 |         files.forEach((f) => {
120 |             const opt = document.createElement('option');
121 |             opt.value = f.path;
122 |             const sizeKb = (f.size_bytes / 1024).toFixed(1);
123 |             opt.textContent = `${f.name} (${sizeKb} KB)`;
124 |             select.appendChild(opt);
125 |         });
126 |     } catch (err) {
127 |         console.warn('[TelemetryResearch] Ошибка загрузки источник
128 | ов:', err);
129 |     }
130 | }
131 |
132 | /**
133 |  * Запуск глубокого исследования телеметрии.
134 |  */
135 | async function runResearch() {
136 |     const btnRun = document.getElementById('tr-btn-run');
137 |     const selectSource = document.getElementById('tr-select-sour
138 | ce');
139 |     const sourcePath = selectSource ? selectSource.value : null;

```

```
136 |
137 |   if (btnRun) {
138 |     btnRun.disabled = true;
139 |     btnRun.innerHTML = '<span class="spinner-border spinner-bo
140 | rder-sm" role="status"></span> Анализ...';
141 |   }
142 |   try {
143 |     const res = await fetch('/apps/telemetry_research/run-rese
144 | arch', {
145 |       method: 'POST',
146 |       headers: { 'Content-Type': 'application/json' },
147 |       body: JSON.stringify({ source_path: sourcePath || null }
148 | ),
149 |     });
150 |     if (!res.ok) {
151 |       throw new Error(`Ошибка сервера: ${res.status}`);
152 |     }
153 |     const report = await res.json();
154 |     trCurrentReport = report;
155 |
156 |     renderKPI(report);
157 |     renderHypotheses(report.hypotheses);
158 |     renderCorrelations(report.correlations);
159 |     renderCharts(report.base_report.charts);
160 |     renderAnomalies(report.base_report.anomalies, report.base_
161 | report.device_summary);
162 |
163 |     if (window.toast) {
164 |       window.toast.success('Исследование завершено', `Проанали
165 | зировано ${report.base_report.records_analyzed} записей.`);
166 |     }
167 |   } catch (err) {
168 |     console.error('[TelemetryResearch] Ошибка анализа:', err);
169 |     if (window.toast) {
170 |       window.toast.error('Сбой исследования', err.message);
171 |     }
172 |   }
173 | }
```

```
169 |     }
170 | } finally {
171 |     if (btnRun) {
172 |         btnRun.disabled = false;
173 |         btnRun.innerHTML = '<i class="bi bi-play-circle-fill"></i> <span>Исследовать</span>';
174 |     }
175 | }
176 | }
177 |
178 | /**
179 |  * Отрисовка сводных KPI карточек.
180 |  */
181 | function renderKPI(report) {
182 |     const base = report.base_report;
183 |     const valHealth = document.getElementById('tr-val-health');
184 |     const progHealth = document.getElementById('tr-progress-health');
185 |     const valRecords = document.getElementById('tr-val-records');
186 |     const valTimespan = document.getElementById('tr-val-timespan');
187 |     const valHypotheses = document.getElementById('tr-val-hypotheses');
188 |     const valAnomalies = document.getElementById('tr-val-anomalies');
189 |     const valDevices = document.getElementById('tr-val-devices');
190 |
191 |     if (valHealth) valHealth.textContent = `${base.health_score}`;
192 |     if (progHealth) {
193 |         progHealth.style.width = `${base.health_score}%`;
194 |         progHealth.className = `progress-bar ${
195 |             base.health_score >= 80 ? 'bg-success' : base.health_score >= 60 ? 'bg-warning' : 'bg-danger'
196 |         }`;
197 |     }
```

```
198
199   if (valRecords) valRecords.textContent = `${base.records_analyzed}`;
200   if (valTimespan && base.time_range) {
201     valTimespan.textContent = `${base.time_range.start ? base.time_range.start.slice(11, 19) : ''} - ${base.time_range.end ? base.time_range.end.slice(11, 19) : ''}`;
202   }
203 }
204
205
206 const confirmedCount = (report.hypotheses || []).filter((h) => h.confirmed).length;
207 if (valHypotheses) valHypotheses.textContent = `${confirmedCount} / ${report.hypotheses ? report.hypotheses.length : 0}`;
208
209 if (valAnomalies) valAnomalies.textContent = `${(base.anomalies || []).length}`;
210 if (valDevices) {
211   valDevices.textContent = base.device_summary.error_count > 0
212     ? `Ошибок устройств: ${base.device_summary.error_count}`
213     : 'Оборудование в норме';
214 }
215
216 // Summary alert banner
217 const banner = document.getElementById('tr-summary-banner');
218 const bannerText = document.getElementById('tr-summary-text');
219 if (banner && bannerText && report.investigation_summary) {
220   banner.classList.remove('d-none');
221   bannerText.textContent = report.investigation_summary;
222 }
223 }
224
225 /**
226  * Отрисовка списка гипотез.
227  */
```

```

228 function renderHypotheses(hypotheses) {
229   const container = document.getElementById('tr-hypotheses-list');
230   if (!container) return;
231
232   if (!hypotheses || hypotheses.length === 0) {
233     container.innerHTML = '<div class="col-12 text-center text-muted py-4">Гипотезы не проверены</div>';
234     return;
235   }
236
237   container.innerHTML = '';
238   hypotheses.forEach((h) => {
239     const col = document.createElement('div');
240     col.className = 'col-12 col-md-6';
241
242     const cardBorder = h.confirmed ? 'border-warning' : 'border-success-subtle';
243     const statusBadge = h.confirmed
244       ? '<span class="badge bg-warning text-dark"><i class="bi bi-exclamation-triangle"></i> Подтверждена</span>'
245       : '<span class="badge bg-success"><i class="bi bi-check-circle"></i> В норме</span>';
246
247     const evidenceItems = h.evidence.map((ev) => `<li>${ev}</li>`).join('');
248
249     col.innerHTML = `
250       <div class="card bg-dark ${cardBorder} h-100 shadow-sm">
251         <div class="card-header bg-transparent border-secondary d-flex justify-content-between align-items-center">
252           <span class="fw-bold text-light">${h.title}</span>
253           ${statusBadge}
254         </div>
255         <div class="card-body p-3 small">
256           <p class="text-muted mb-2">${h.description}</p>
257           <div class="mb-2">
258             <strong class="text-secondary">Факты телеметрии:</

```

```

    strong>
259         <ul class="text-light text-opacity-75 ps-3 mb-1">${
    {evidenceItems}</ul>
260     </div>
261     ${
262         h.recommendation
263         ? `<div class="p-2 rounded bg-body-tertiary bg-o
    pacity-10 text-info mt-2">
264             <strong>□ Рекомендация:</strong> ${h.recom
    mendation}
265             </div>`
266         : ''
267     }
268 </div>
269 <div class="card-footer bg-transparent border-secondar
    y text-muted small d-flex justify-content-between">
270     <span>ID: <code>${h.hypothesis_id}</code></span>
271     <span>Уверенность: <strong>${Math.round(h.confidence
    * 100)}%</strong></span>
272 </div>
273 </div>
274 `;
275 container.appendChild(col);
276 });
277 }
278
279 /**
280  * Отрисовка матрицы корреляций.
281  */
282 function renderCorrelations(correlations) {
283     const tbody = document.getElementById('tr-correlations-tbody
    ');
284     const countBadge = document.getElementById('tr-correlations-
    count');
285     if (!tbody) return;
286
287     if (!correlations || correlations.length === 0) {
288         tbody.innerHTML = '<tr><td colspan="5" class="text-center

```

```

    text-muted py-3">Корреляции не рассчитаны</td></tr>';
289     if (countBadge) countBadge.textContent = '0 nap';
290     return;
291   }
292
293   if (countBadge) countBadge.textContent = `${correlations.length} nap`;
294
295   tbody.innerHTML = correlations
296     .map((c) => {
297       const absVal = Math.abs(c.coefficient);
298       let coeffBadgeClass = 'bg-secondary';
299       if (absVal >= 0.7) coeffBadgeClass = 'bg-danger';
300       else if (absVal >= 0.5) coeffBadgeClass = 'bg-warning text-dark';
301       else if (absVal >= 0.3) coeffBadgeClass = 'bg-info text-dark';
302
303       return `
304         <tr>
305           <td><code class="text-info">${c.metric_a}</code></td>
306           <td><code class="text-info">${c.metric_b}</code></td>
307           <td><span class="badge ${coeffBadgeClass} font-monospace">${c.coefficient > 0 ? '+' : ''}${c.coefficient.toFixed(3)}</span></td>
308           <td>${c.sample_size} точек</td>
309           <td class="text-light text-opacity-75">${c.interpretation}</td>
310         </tr>
311       `;
312     })
313     .join('');
314 }
315
316 /**
317  * Отрисовка интерактивных графиков Chart.js.

```

```

318 |  */
319 | function renderCharts(charts) {
320 |     const container = document.getElementById('tr-charts-contain
    er');
321 |     if (!container) return;
322 |
323 |     // Очищаем старые инстансы графиков
324 |     trChartsInstances.forEach((inst) => {
325 |         try {
326 |             inst.destroy();
327 |         } catch {}
328 |     });
329 |     trChartsInstances = [];
330 |
331 |     if (!charts || charts.length === 0) {
332 |         container.innerHTML = '<div class="col-12 text-center text
    -muted py-4">Нет доступных графиков</div>';
333 |         return;
334 |     }
335 |
336 |     container.innerHTML = '';
337 |
338 |     charts.forEach((ch, idx) => {
339 |         const col = document.createElement('div');
340 |         col.className = 'col-12 col-xl-6';
341 |
342 |         const canvasId = `tr-chart-canvas-${idx}`;
343 |         col.innerHTML = `
344 |             <div class="card bg-dark border-secondary h-100 shadow-s
    m">
345 |                 <div class="card-header bg-transparent border-secondar
    y d-flex justify-content-between align-items-center">
346 |                     <span class="fw-bold text-light">${ch.title}</span>
347 |                     <span class="badge bg-secondary small">${ch.chart_ty
    pe}</span>
348 |                 </div>
349 |                 <div class="card-body p-2" style="position: relative;
    height: 280px;">

```



```

350         <canvas id="${canvasId}"></canvas>
351     </div>
352 </div>
353 `;
354 container.appendChild(col);
355
356 const ctx = document.getElementById(canvasId);
357 if (ctx && typeof Chart !== 'undefined') {
358     const labels = ch.labels.map((l) => (l.length > 19 ? l.s
lice(11, 19) : l));
359     const datasets = ch.datasets.map((ds) => ({
360         label: ds.label,
361         data: ds.data,
362         borderColor: ds.color || '#3b82f6',
363         backgroundColor: (ds.color || '#3b82f6') + '22',
364         borderWidth: 2,
365         fill: ch.chart_type === 'area',
366         tension: 0.25,
367         pointRadius: ds.data.length > 50 ? 0 : 3,
368     }));
369
370     const newInst = new Chart(ctx, {
371         type: ch.chart_type === 'bar' ? 'bar' : 'line',
372         data: { labels, datasets },
373         options: {
374             responsive: true,
375             maintainAspectRatio: false,
376             animation: { duration: 400 },
377             scales: {
378                 x: {
379                     grid: { color: 'rgba(255,255,255,0.06)' },
380                     ticks: { color: '#9ca3af', font: { size: 10 } },
381                 },
382                 y: {
383                     grid: { color: 'rgba(255,255,255,0.06)' },
384                     ticks: { color: '#9ca3af', font: { size: 10 } },
385                 },
386             },

```

```

387         plugins: {
388             legend: {
389                 labels: { color: '#e5e7eb', boxWidth: 12, font:
| { size: 11 } },
390             },
391         },
392     },
393 });
394     trChartsInstances.push(newInst);
395 }
396 });
397 }
398
399 /**
400  * Отрисовка аномалий и состояния устройств.
401  */
402 function renderAnomalies(anomalies, deviceSummary) {
403     const tbody = document.getElementById('tr-anomalies-tbody');
404     const anomBadge = document.getElementById('tr-anomalies-badg
| e');
405     const devBadge = document.getElementById('tr-devices-badge')
| ;
406     const devContent = document.getElementById('tr-devices-conte
| nt');
407
408     if (anomBadge) anomBadge.textContent = anomalies ? `${anomal
| ies.length}` : '0';
409
410     if (tbody) {
411         if (!anomalies || anomalies.length === 0) {
412             tbody.innerHTML = '<tr><td colspan="5" class="text-cen
| ter text-success py-3"><i class="bi bi-check-circle"></i> Аномал
| ий в метриках не зафиксировано</td></tr>';
413         } else {
414             tbody.innerHTML = anomalies
415                 .map((a) => {
416                 const sevClass = a.severity === 'critical' ? 'bg-dan
| ger' : a.severity === 'high' ? 'bg-warning text-dark' : 'bg-in

```

```

fo text-dark';
417         return `
418         <tr>
419             <td class="text-muted">${a.timestamp ? a.timestamp.slice(11, 19) : '--'}</td>
420             <td><strong>${a.metric}</strong></td>
421             <td><span class="badge ${sevClass}">${a.severity}</span></td>
422             <td><code class="text-warning">${a.value}</code>
/ ${a.threshold}</td>
423             <td class="text-light text-opacity-75">${a.message}</td>
424         </tr>
425     `;
426     })
427     .join('');
428 }
429 }
430
431 if (deviceSummary && devContent) {
432     const hasIssues = deviceSummary.error_count > 0 || deviceSummary.flapping_devices.length > 0;
433     if (devBadge) {
434         devBadge.className = `badge ${hasIssues ? 'bg-danger' : 'bg-success'}`;
435         devBadge.textContent = hasIssues ? 'Внимание' : 'Норма';
436     }
437
438     let flapHtml = '<span class="text-success">Нет</span>';
439     if (deviceSummary.flapping_devices && deviceSummary.flapping_devices.length > 0) {
440         flapHtml = deviceSummary.flapping_devices.map((d) => `<span class="badge bg-warning text-dark me-1">${d}</span>`).join('');
441     }
442
443     devContent.innerHTML = `
444         <div class="mb-2"><strong>Всего инцидентов:</strong> ${d

```

```

    deviceSummary.total_events}</div>
445     <div class="mb-2"><strong>Ошибок и сбоев:</strong> <span
    class="{deviceSummary.error_count > 0 ? 'text-danger fw-bold
    ' : 'text-success'}">${deviceSummary.error_count}</span></div>
446     <div class="mb-2"><strong>Флаппирующие устройства:</stro
    ng> ${flapHtml}</div>
447     `;
448   }
449 }
450
451 /**
452  * Загрузка записей телеметрии с пагинацией.
453  */
454 async function loadRecords() {
455   const tbody = document.getElementById('tr-records-tbody');
456   const infoEl = document.getElementById('tr-records-paginat
    ion-info');
457   const selectSource = document.getElementById('tr-select-sour
    ce');
458   const sourcePath = selectSource ? selectSource.value : null;
459
460   if (tbody) {
461     tbody.innerHTML = '<tr><td colspan="7" class="text-center
    text-muted py-3"><span class="spinner-border spinner-border-sm
    "></span> Загрузка...</td></tr>';
462   }
463
464   try {
465     const url = new URL('/apps/telemetry_research/records', wi
    ndow.location.origin);
466     if (sourcePath) url.searchParams.set('source_path', source
    Path);
467     if (trSearchQuery) url.searchParams.set('query', trSearchQ
    uery);
468     url.searchParams.set('page', trCurrentPage);
469     url.searchParams.set('page_size', trPageSize);
470
471     const res = await fetch(url.toString());

```

```

472     if (!res.ok) throw new Error(`HTTP ${res.status}`);
473     const data = await res.json();
474
475     trTotalPages = data.total_pages || 1;
476     if (infoEl) {
477         infoEl.textContent = `Стр. ${data.page} из ${trTotalPage
s} (всего ${data.total})`;
478     }
479
480     if (!data.items || data.items.length === 0) {
481         tbody.innerHTML = '<tr><td colspan="7" class="text-cente
r text-muted py-3">Записей не найдено</td></tr>';
482         return;
483     }
484
485     tbody.innerHTML = data.items
486         .map((r, i) => {
487             const num = (trCurrentPage - 1) * trPageSize + i + 1;
488             const ts = r.timestamp || r.time || '--';
489             const cpu = r.cpu ? `${r.cpu.total_percent || r.cpu.lo
ad_percent || '--'}%` : '--';
490             const ram = r.memory ? `${r.memory.percent || '--'}%`
: '--';
491             const gpu = r.gpu ? `${r.gpu.load_percent || '--'}%` :
'--';
492             let disk = '--';
493             if (r.disk_io) {
494                 const rMb = ((r.disk_io.read_bytes_per_sec || 0) / 1
048576).toFixed(1);
495                 const wMb = ((r.disk_io.write_bytes_per_sec || 0) /
1048576).toFixed(1);
496                 disk = `${rMb} / ${wMb} MB/s`;
497             }
498             const eventInfo = r.event_type || r.friendly_name || J
SON.stringify(r).slice(0, 50);
499
500             return `
501                 <tr>

```

```

502         <td>${num}</td>
503         <td class="text-muted">${ts}</td>
504         <td>${cpu}</td>
505         <td>${ram}</td>
506         <td>${gpu}</td>
507         <td>${disk}</td>
508         <td class="text-truncate" style="max-width: 250px;
" title="${eventInfo}">${eventInfo}</td>
509     </tr>
510     `;
511     })
512     .join('');
513 } catch (err) {
514     if (tbody) {
515         tbody.innerHTML = `<tr><td colspan="7" class="text-cen
r text-danger py-3">Ошибка: ${err.message}</td></tr>`;
516     }
517 }
518 }
519
520 /**
521  * Открытие автономного HTML дашборда.
522  */
523 function openHtmlDashboard() {
524     const selectSource = document.getElementById('tr-select-sour
ce');
525     const sourcePath = selectSource ? selectSource.value : '';
526     const url = sourcePath
527         ? `/apps/telemetry_research/dashboard?source_path=${encode
URIComponent(sourcePath)}`
528         : '/apps/telemetry_research/dashboard';
529     window.open(url, '_blank');
530 }
531
532 /**
533  * Экспорт JSON отчета.
534  */
535 function downloadJsonReport() {

```

```
536 |     if (!trCurrentReport) {
537 |         if (window.toast) window.toast.warning('Нет данных', 'Снач
538 | ала выполните исследование.');
```

```
539 |         return;
540 |     }
541 |     const blob = new Blob([JSON.stringify(trCurrentReport, null,
542 | 2)], { type: 'application/json' });
543 |     const url = URL.createObjectURL(blob);
544 |     const a = document.createElement('a');
545 |     a.href = url;
546 |     a.download = `telemetry_research_${Date.now()}.json`;
547 |     a.click();
548 |     URL.revokeObjectURL(url);
549 | }
550 |
551 | // Автозапуск при подключении модуля
552 | if (document.readyState === 'loading') {
553 |     document.addEventListener('DOMContentLoaded', initTelemetryR
554 | esearchTab);
555 | } else {
556 |     initTelemetryResearchTab();
557 | }
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/throttling_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/throttling_tab/index.ht
17 ml" class="w-100 h-100 border-0"></iframe>
18
19 * File: index.html
20 * Project: ai-breadboard
21 * Package: windows/api/webgui/throttling_tab
22 * Author: hypo69
23 * Copyright: © 2026 hypo69
24 * Updated: 2026-10-08 03:08:00
25 * =====
26 =====
27 -->
28
29 <style>
30   .throttling-tab-wrapper {
31     padding: 0.75rem;
32     color: var(--text-color, #f8fafc);
33   }
34
35   .diag-meter-card {
36     background: var(--surface-2, #1e293b);
37     border: 1px solid var(--border-color, #334155);
38     border-radius: var(--radius-md, 8px);
```



```
33     padding: 0.75rem 1rem;
34 }
35 .diag-panel {
36     background: var(--surface-2, #1e293b);
37     border: 1px solid var(--border-color, #334155);
38     border-radius: 8px;
39     overflow: hidden;
40 }
41 .diag-panel-header {
42     background: rgba(15, 23, 42, 0.6);
43     padding: 8px 14px;
44     border-bottom: 1px solid var(--border-color, #334155);
45     font-weight: 600;
46     font-size: 0.85rem;
47 }
48 .about-sys-spec-table {
49     width: 100%;
50     border-collapse: separate;
51     border-spacing: 0;
52 }
53 .about-sys-spec-table td {
54     padding: 8px 12px;
55     border-bottom: 1px solid rgba(51, 65, 85, 0.35);
56     vertical-align: middle;
57 }
58 .about-sys-spec-key {
59     color: var(--text-muted, #94a3b8);
60     width: 220px;
61     font-size: 0.82rem;
62     font-weight: 500;
63 }
64 .about-sys-spec-val {
65     color: var(--text-color, #f8fafc);
66     font-weight: 500;
67     font-size: 0.82rem;
68 }
69 </style>
70
```

```
71 <div class="throttling-tab-wrapper">
72   <!-- Top Header & Status -->
73   <div class="card shadow-sm border-secondary-subtle mb-3">
74     <div class="card-body p-3">
75       <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
76         <div class="d-flex align-items-center gap-2">
77           <div class="fs-4">□ </div>
78           <div>
79             <h5 class="mb-0 fw-bold">3. Качество ядра и Троттл
инг (Kernel Quality & Hardware Health)</h5>
80             <small class="text-muted">DPC/ISR задержки драйвер
ов, аппаратный сброс частот PROCHOT/Power Limit, PCIe линии и
аварии BSOD</small>
81           </div>
82         </div>
83
84         <div class="d-flex align-items-center gap-2 flex-wrap"
>
85           <select class="form-select form-select-sm poll-freq-
select" id="diag-throttling-poll-freq" data-poll-id="throttlin
g" title="Частота опроса: Троттлинг и ядро">
86             <option value="start">Только при старте</option>
87             <option value="manual" selected>По запросу</option>
88             <option value="3">Каждые 3 сек</option>
89             <option value="5">Каждые 5 сек</option>
90             <option value="10">Каждые 10 сек</option>
91             <option value="30">Каждые 30 сек</option>
92             <option value="60">Каждые 60 сек</option>
93             <option value="120">Каждые 120 сек</option>
94           </select>
95           <button class="btn btn-sm btn-outline-info rounded-p
ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
"btn-diag-throttling-refresh">
96             <i class="bi bi-arrow-clockwise"></i> Обновить
97           </button>
98         </div>
```

```
99 |         </div>
100 |     </div>
101 | </div>
102 |
103 | <!-- KPI Latency and Status Meters -->
104 | <div class="row g-2 mb-3">
105 |     <div class="col-12 col-sm-6 col-lg-3">
106 |         <div class="diag-meter-card">
107 |             <div class="d-flex justify-content-between align-items
108 | -center small text-muted">
109 |                 <span>DPC Latency (Отложенные вызовы)</span>
110 |                 <i class="bi bi-activity text-info"></i>
111 |             </div>
112 |             <div class="fs-4 fw-bold mt-1" id="diag-dpc-latency-val">0.0%</div>
113 |             <div class="small" id="diag-dpc-latency-sub"><span class="badge bg-success-subtle text-success border border-success">Optimal</span></div>
114 |         </div>
115 |     </div>
116 |     <div class="col-12 col-sm-6 col-lg-3">
117 |         <div class="diag-meter-card">
118 |             <div class="d-flex justify-content-between align-items
119 | -center small text-muted">
120 |                 <span>ISR Latency (Прерывания драйверов)</span>
121 |                 <i class="bi bi-cpu text-warning"></i>
122 |             </div>
123 |             <div class="fs-4 fw-bold text-warning mt-1" id="diag-isr-latency-val">0.0%</div>
124 |             <div class="small text-muted">Нагрузка драйверов на ЦП</div>
125 |         </div>
126 |     </div>
127 |     <div class="col-12 col-sm-6 col-lg-3">
128 |         <div class="diag-meter-card">
129 |             <div class="d-flex justify-content-between align-items
130 | -center small text-muted">
131 |                 <span>Аппаратный троттлинг (PROCHOT)</span>
```

```

129         <i class="bi bi-thermometer-sun text-danger"></i>
130     </div>
131     <div class="fs-4 fw-bold mt-1" id="diag-throttling-sta
tus-val">Inactive</div>
132     <div class="small text-muted" id="diag-throttling-stat
us-sub">Терморежим в норме</div>
133 </div>
134 </div>
135 <div class="col-12 col-sm-6 col-lg-3">
136     <div class="diag-meter-card">
137         <div class="d-flex justify-content-between align-items
-center small text-muted">
138             <span>Время непрерывной работы</span>
139             <i class="bi bi-clock text-primary"></i>
140         </div>
141         <div class="fs-4 fw-bold text-white mt-1" id="diag-upt
ime-val">--</div>
142         <div class="small text-muted">System Uptime</div>
143     </div>
144 </div>
145 </div>
146
147 <!-- GPU PCIe Bus & Kernel BSOD Crash Dumps -->
148 <div class="row g-2">
149     <div class="col-12 col-lg-6">
150         <div class="diag-panel h-100">
151             <div class="diag-panel-header d-flex justify-content-b
etween align-items-center">
152                 <div class="d-flex align-items-center gap-2">
153                     <i class="bi bi-gpu-card text-success"></i>
154                     <span class="text-white">Режим шины PCIe графическ
ого ускорителя (GPU Link)</span>
155                 </div>
156                 <span class="badge bg-success-subtle text-success bo
rder border-success">PCIe Prober</span>
157             </div>
158             <div class="p-3">
159                 <table class="about-sys-spec-table">

```

```

160         <tbody>
161             <tr>
162                 <td class="about-sys-spec-key"><i class="bi bi
- gpu-card me-1.5 text-success"></i>Видеокарта</td>
163                 <td class="about-sys-spec-val text-info" id="d
iag-gpu-pcie-name">GeForce GPU</td>
164             </tr>
165             <tr>
166                 <td class="about-sys-spec-key"><i class="bi bi
- speedometer2 me-1.5 text-warning"></i>Скорость шины (Link Spe
ed)</td>
167                 <td class="about-sys-spec-val" id="diag-gpu-pc
ie-speed">PCIe 3.0 / 4.0 (8.0 GT/s)</td>
168             </tr>
169             <tr>
170                 <td class="about-sys-spec-key"><i class="bi bi
- arrows-expand me-1.5 text-primary"></i>Ширина линий (Link Wid
th)</td>
171                 <td class="about-sys-spec-val" id="diag-gpu-pc
ie-width">x16 Lanes (Full Width)</td>
172             </tr>
173             <tr>
174                 <td class="about-sys-spec-key"><i class="bi bi
- shield-check me-1.5 text-success"></i>Статус пропускной спосо
бности</td>
175                 <td class="about-sys-spec-val text-success" id
="diag-gpu-pcie-status">Штатный режим без деградации</td>
176             </tr>
177         </tbody>
178     </table>
179 </div>
180 </div>
181 </div>
182
183 <div class="col-12 col-lg-6">
184     <div class="diag-panel h-100">
185         <div class="diag-panel-header d-flex justify-content-b
etween align-items-center">

```

```
186 |         <div class="d-flex align-items-center gap-2">
187 |             <i class="bi bi-bug-fill text-danger"></i>
188 |             <span class="text-white">Журнал аварийных дампов я
дра (Minidump / BSOD)</span>
189 |         </div>
190 |         <span class="badge bg-secondary" id="diag-bsod-count
">0 сбоев</span>
191 |     </div>
192 |     <div class="p-2" id="diag-bsod-container" style="max-h
eight: 260px; overflow-y: auto;">
193 |         <div class="text-center py-4 text-success small">
194 |             <i class="bi bi-check-circle-fill fs-5 d-block mb-
1 text-success"></i>
195 |             Аварийных дампов падения ядра (BSOD) в Minidump не
обнаружено
196 |         </div>
197 |     </div>
198 | </div>
199 | </div>
200 | </div>
201 | </div>
202 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Throttling Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/throttling_tab/main.js
11  *     ?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/throttling_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-08 02:00:00
19  * =====
20  */
21 (function () {
22   'use strict';
23
24   let autoRefreshTimer = null;
25
26   function escapeHtml(str) {
27     if (str === null || str === undefined) return '';
28     return String(str)
29       .replace(/&/g, '&amp;')
30       .replace(/</g, '&lt;')
31       .replace(/>/g, '&gt;')
32       .replace(/"/g, '&quot;')
```

```

33     .replace(/'/g, '&#039;');
34 }
35
36 function setText(id, text) {
37     const el = document.getElementById(id);
38     if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
39 }
40
41 async function apiFetch(url) {
42     if (window.api && typeof window.api.fetch === 'function')
{
43         return await window.api.fetch(url);
44     }
45     const res = await fetch(url);
46     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
atusText}`);
47     return await res.json();
48 }
49
50 async function fetchKernelThrottling() {
51     try {
52         const data = await apiFetch('/api/v1/system/diagnostics/
throttling');
53         if (!data) return;
54
55         setText('diag-dpc-latency-val', `${data.dpc_latency_pct.
toFixed(2)}%`);
56         const dpcSub = document.getElementById('diag-dpc-latency
-sub');
57         if (dpcSub) {
58             if (data.dpc_status === 'severe') {
59                 dpcSub.innerHTML = '<span class="badge bg-danger-sub
tle text-danger border border-danger">Критическая латентность
DPC</span>';
60             } else if (data.dpc_status === 'elevated') {
61                 dpcSub.innerHTML = '<span class="badge bg-warning-su
btle text-warning border border-warning">Повышенная задержка D

```



```

PC</span>';
62         } else {
63             dpcSub.innerHTML = '<span class="badge bg-success-subtle text-success border border-success">Оптимально (< 1.5%)</span>';
64         }
65     }
66
67     setText('diag-isr-latency-val', `${data.interrupt_latency_pct.toFixed(2)}%`);
68     setText('diag-uptime-val', data.uptime_formatted || '--');
69
70     const thVal = document.getElementById('diag-throttling-status-val');
71     const thSub = document.getElementById('diag-throttling-status-sub');
72     if (thVal && thSub) {
73         if (data.thermal_throttling_detected) {
74             thVal.textContent = 'PROCHOT Active';
75             thVal.className = 'fs-4 fw-bold text-danger mt-1';
76             thSub.textContent = 'Обнаружен термический сброс частот CPU';
77         } else if (data.power_limit_throttling_detected) {
78             thVal.textContent = 'Power Limit PL1/PL2';
79             thVal.className = 'fs-4 fw-bold text-warning mt-1';
80             thSub.textContent = 'Ограничение по энергопотреблению';
81         } else {
82             thVal.textContent = 'Inactive (Норма)';
83             thVal.className = 'fs-4 fw-bold text-success mt-1';
84             thSub.textContent = 'Терморезим и питание в норме';
85         }
86     }
87
88     // GPU PCIe Link
89     if (data.gpu_pcie_link) {
90         setText('diag-gpu-pcie-name', data.gpu_pcie_link.gpu_n

```

```

    ame || 'GPU Accelerator');
91     setText('diag-gpu-pcie-speed', data.gpu_pcie_link.curr
ent_link_speed || 'PCIe 3.0 / 4.0');
92     setText('diag-gpu-pcie-width', data.gpu_pcie_link.curr
ent_link_width || 'x16 Lanes');
93     setText('diag-gpu-pcie-status', data.gpu_pcie_link.sta
tus || 'Штатный режим без деградации');
94     }
95
96     // BSOD Minidump
97     const bsodContainer = document.getElementById('diag-bsod
-container');
98     if (bsodContainer && data.last_bsod_crashes) {
99         setText('diag-bsod-count', `${data.last_bsod_crashes.l
ength} сбоев`);
100         if (data.last_bsod_crashes.length === 0) {
101             bsodContainer.innerHTML = `
102                 <div class="text-center py-4 text-success small">
103                     <i class="bi bi-check-circle-fill fs-5 d-block m
b-1 text-success"></i>
104                     Аварийных дампов падения ядра (BSOD) в Minidump
не обнаружено
105                 </div>
106                 `;
107             } else {
108                 bsodContainer.innerHTML = data.last_bsod_crashes.map
(c => `
109                     <div class="p-2 mb-1.5 rounded bg-black bg-opacity
-40 border border-danger-subtle d-flex align-items-center just
ify-content-between">
110                         <div>
111                             <strong class="text-danger"><i class="bi bi-fi
le-earmark-binary me-1"></i>${escapeHtml(c.file_name)}</strong>
112                             <div class="small text-muted font-monospace">${
escapeHtml(c.timestamp)} (${c.size_kb} KB)</div>
113                         </div>
114                         <span class="badge bg-danger">Crash Dump</span>

```

```
115 |         </div>
116 |         `).join('');
117 |     }
118 | }
119 | } catch (err) {
120 |     console.warn('[ThrottlingTab] fetchKernelThrottling error:', err);
121 | }
122 | }
123 |
124 | const POLL_ID = 'throttling';
125 |
126 | function getFrequency() {
127 |     try {
128 |         const saved = localStorage.getItem(`poll_freq_${POLL_ID}`);
129 |         if (saved) return saved;
130 |     } catch (_) {}
131 |     return 'manual';
132 | }
133 |
134 | function setFrequency(freq) {
135 |     try {
136 |         localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
137 |     } catch (_) {}
138 |     applyPoller(freq, false);
139 | }
140 |
141 | function stopPolling() {
142 |     if (window.unregisterTabPoller) {
143 |         window.unregisterTabPoller(`tab-throttling_${POLL_ID}`);
144 |     }
145 |     if (autoRefreshTimer) {
146 |         clearInterval(autoRefreshTimer);
147 |         autoRefreshTimer = null;
148 |     }
149 | }
150 |
```

```

151 | function applyPoller(freq, runInitial = false) {
152 |     stopPolling();
153 |     if (freq === 'start' || freq === 'manual') {
154 |         if (runInitial && (window.isTabActive ? window.isTabActive('tab-throttling') : true)) {
155 |             fetchKernelThrottling();
156 |         }
157 |         return;
158 |     }
159 |
160 |     const intervalSec = parseInt(freq, 10);
161 |     if (isNaN(intervalSec) || intervalSec <= 0) return;
162 |
163 |     const intervalMs = intervalSec * 1000;
164 |     const pollerId = `tab-throttling_${POLL_ID}`;
165 |
166 |     if (window.registerTabPoller) {
167 |         window.registerTabPoller('tab-throttling', fetchKernelThrottling, intervalMs, { pollerId, immediate: runInitial });
168 |     } else {
169 |         if (runInitial && (window.isTabActive ? window.isTabActive('tab-throttling') : true)) {
170 |             fetchKernelThrottling();
171 |         }
172 |         autoRefreshTimer = setInterval(() => {
173 |             if (window.isTabActive ? window.isTabActive('tab-throttling') : true) {
174 |                 fetchKernelThrottling();
175 |             }
176 |         }, intervalMs);
177 |     }
178 | }
179 |
180 | function bindEvents() {
181 |     const btnRefresh = document.getElementById('btn-diag-throttling-refresh');
182 |     if (btnRefresh) {
183 |         btnRefresh.onclick = () => fetchKernelThrottling();

```

```

184     }
185
186     const select = document.getElementById('diag-throttling-po
ll-freq');
187     if (select) {
188         select.onchange = (e) => {
189             const newFreq = e.target.value;
190             setFrequency(newFreq);
191             if (newFreq !== 'manual' && newFreq !== 'start') {
192                 fetchKernelThrottling();
193             }
194             if (window.showToast) {
195                 const label = select.options[select.selectedIndex]?.
text || newFreq;
196                 window.showToast(`Частота опроса троттлинга: ${label
}`, 'info');
197             }
198         };
199     }
200 }
201
202 async function initThrottlingTab() {
203     bindEvents();
204     const currentFreq = getFrequency();
205     const select = document.getElementById('diag-throttling-po
ll-freq');
206     if (select) select.value = currentFreq;
207     applyPoller(currentFreq, true);
208 }
209
210 window.initThrottlingTab = initThrottlingTab;
211 window.activateThrottlingTab = () => {
212     applyPoller(getFrequency(), true);
213 };
214
215 if (document.readyState === 'loading') {
216     document.addEventListener('DOMContentLoaded', () => {
217         if (window.isTabActive ? window.isTabActive('tab-throttling-

```

```
ing') : false) {
218 |         initThrottlingTab();
219 |     } else {
220 |         bindEvents();
221 |     }
222 | });
223 | } else {
224 |     if (window.isTabActive ? window.isTabActive('tab-throttlin
    | g') : false) {
225 |         initThrottlingTab();
226 |     } else {
227 |         bindEvents();
228 |     }
229 | }
230 | })();
231 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/trading_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/trading_tab/index.html"
17     class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/trading_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-08 03:08:00
25 =====
26 =====
27 -->
28
29 <!-- Trading Terminal Application Tab -->
30 <style>
31   .trading-container {
32     padding: 1rem;
33   }
34   .trading-metric-card {
35     background: #0f172a !important;
36     border: 1px solid #334155 !important;
37     border-radius: 10px;
```

```
33 | padding: 1rem;
34 | display: flex;
35 | flex-direction: column;
36 | gap: 0.25rem;
37 | }
38 | .trading-metric-label {
39 |   font-size: 0.75rem;
40 |   text-transform: uppercase;
41 |   letter-spacing: 0.05em;
42 |   color: #94a3b8;
43 | }
44 | .trading-metric-value {
45 |   font-size: 1.4rem;
46 |   font-weight: 700;
47 |   color: #f8fafc;
48 |   font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
49 | }
50 | .trading-panel {
51 |   background: #0f172a !important;
52 |   border: 1px solid #334155 !important;
53 |   border-radius: 10px;
54 |   overflow: hidden;
55 | }
56 | .trading-panel-header {
57 |   background: #1e293b !important;
58 |   border-bottom: 1px solid #334155 !important;
59 |   padding: 0.75rem 1rem;
60 |   font-weight: 600;
61 |   font-size: 0.88rem;
62 |   color: #38bdf8;
63 |   display: flex;
64 |   align-items: center;
65 |   justify-content: space-between;
66 | }
67 | .trading-orderbook-table {
68 |   width: 100%;
69 |   font-size: 0.8rem;
```



```
70     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,  
    Consolas, monospace;  
71     border-collapse: collapse;  
72 }  
73 .trading-orderbook-table th {  
74     background: #1e293b;  
75     padding: 6px 12px;  
76     color: #94a3b8;  
77     font-weight: 600;  
78     border-bottom: 1px solid #334155;  
79 }  
80 .trading-orderbook-table td {  
81     padding: 5px 12px;  
82     border-bottom: 1px solid rgba(51, 65, 85, 0.4);  
83 }  
84 .trading-ask-row {  
85     color: #f87171;  
86     background: rgba(239, 68, 68, 0.05);  
87 }  
88 .trading-bid-row {  
89     color: #4ade80;  
90     background: rgba(34, 197, 94, 0.05);  
91 }  
92 .btn-kill-switch {  
93     background: linear-gradient(135deg, #ef4444 0%, #b91c1c 10  
    0%) !important;  
94     border: 1px solid #f87171 !important;  
95     color: #ffffff !important;  
96     font-weight: 700;  
97     box-shadow: 0 0 12px rgba(239, 68, 68, 0.4);  
98 }  
99 .btn-kill-switch:hover {  
100     background: linear-gradient(135deg, #dc2626 0%, #991b1b 10  
    0%) !important;  
101     box-shadow: 0 0 18px rgba(239, 68, 68, 0.7);  
102 }  
103 </style>  
104
```

```

105 <div class="trading-container">
106   <!-- Header Alert & Status -->
107   <div class="card shadow-sm border-secondary-subtle p-3 mb-3
      d-flex justify-content-between align-items-center flex-wrap ga
      p-3">
108     <div>
109       <h5 class="mb-1 fw-bold d-flex align-items-center gap-2"
      >
110         <span class="fs-4">☐ </span> Торговый терминал (Exchan
      ge Trading Desk)
111       </h5>
112       <p class="mb-0 small text-muted" >
113         Автономный торговый терминал с обновлением стакана цен
      L2, расчётом PnL и кнопкой экстренной ликвидации (Kill-Switch
      ).
114       </p>
115     </div>
116     <div class="d-flex align-items-center gap-2">
117       <span class="badge rounded-pill bg-success-subtle text-s
      uccess border border-success px-3 py-2" id="trading-stream-sta
      tus">
118         • Стрим активен
119       </span>
120       <button class="btn btn-sm btn-outline-info rounded-pill
      px-3" id="btn-refresh-trading">
121         <i class="bi bi-arrow-clockwise me-1"></i> Обновить
122       </button>
123       <button class="btn btn-sm btn-outline-warning rounded-pi
      ll px-3" id="btn-trading-config">
124         <i class="bi bi-gear-fill me-1"></i> Настройки (config
      .json)
125       </button>
126       <button class="btn btn-sm btn-kill-switch rounded-pill p
      x-3" id="btn-trigger-kill-switch">
127         ☐ Kill-Switch (Ликвидация)
128       </button>
129     </div>
130 </div>

```

```
131 |
132 | <!-- Top Metrics Bar -->
133 | <div class="row g-3 mb-3">
134 |     <div class="col-md-3 col-sm-6">
135 |         <div class="trading-metric-card">
136 |             <span class="trading-metric-label">Торговая пара</span>
137 |             <span class="trading-metric-value" style="color: #38bdf8;" id="trading-symbol">BTC/USDT</span>
138 |             <span class="small text-muted" id="trading-spread-info">Спред: --</span>
139 |         </div>
140 |     </div>
141 |     <div class="col-md-3 col-sm-6">
142 |         <div class="trading-metric-card">
143 |             <span class="trading-metric-label">Текущая цена</span>
144 |             <span class="trading-metric-value" id="trading-current-price">$0.00</span>
145 |             <span class="small" id="trading-price-delta" style="color: #4ade80;">Bid: $0 | Ask: $0</span>
146 |         </div>
147 |     </div>
148 |     <div class="col-md-3 col-sm-6">
149 |         <div class="trading-metric-card">
150 |             <span class="trading-metric-label">Баланс USDT / Эквити</span>
151 |             <span class="trading-metric-value" id="trading-total-equity">$0.00</span>
152 |             <span class="small text-muted" id="trading-cash-balance">Доступно: $0.00</span>
153 |         </div>
154 |     </div>
155 |     <div class="col-md-3 col-sm-6">
156 |         <div class="trading-metric-card">
157 |             <span class="trading-metric-label">Позиция / PnL</span>
158 |             <span class="trading-metric-value" id="trading-pnl-value" style="color: #4ade80;">+$0.00</span>
```

```

159 |         <span class="small text-muted" id="trading-pos-size">P
      | азмeр позиции: 0.0 BTC</span>
160 |     </div>
161 | </div>
162 | </div>
163 |
164 | <!-- Main Trading Desk Grid -->
165 | <div class="row g-3">
166 |     <!-- Left Column: Orderbook (L2 Depth) -->
167 |     <div class="col-lg-5">
168 |         <div class="trading-panel h-100">
169 |             <div class="trading-panel-header">
170 |                 <span><i class="bi bi-bar-chart-fill me-1"></i> Стак
      | ан котировок (L2 Orderbook)</span>
171 |                 <span class="badge bg-secondary" id="trading-orderbo
      | ok-depth">Глубина: 6</span>
172 |             </div>
173 |             <div class="p-0">
174 |                 <table class="trading-orderbook-table">
175 |                     <thead>
176 |                         <tr>
177 |                             <th>Цена (USDT)</th>
178 |                             <th style="text-align: right;">Объём</th>
179 |                             <th style="text-align: right;">Сумма</th>
180 |                         </tr>
181 |                     </thead>
182 |                     <tbody id="trading-asks-tbody">
183 |                         <tr><td colspan="3" class="text-center py-2 text
      | -muted">Загрузка заявок на продажу...</td></tr>
184 |                     </tbody>
185 |                     <tbody >
186 |                         <tr >
187 |                             <td colspan="3" class="text-center py-2 fw-bol
      | d" id="trading-orderbook-midprice" style="font-size: 0.95rem;
      | color: #f8fafc;">
188 |                                 $0.00
189 |                             </td>
190 |                         </tr>

```

```

191         </tbody>
192         <tbody id="trading-bids-tbody">
193             <tr><td colspan="3" class="text-center py-2 text
-muted">Загрузка заявок на покупку...</td></tr>
194         </tbody>
195     </table>
196 </div>
197 </div>
198 </div>
199
200 <!-- Right Column: Order Placement & Recent Orders -->
201 <div class="col-lg-7 d-flex flex-column gap-3">
202     <!-- Order Form -->
203     <div class="trading-panel">
204         <div class="trading-panel-header">
205             <span><i class="bi bi-lightning-charge-fill me-1"></
i> Выставление ордера</span>
206             <span class="badge bg-primary" id="trading-mode-badg
e">Direct Execution</span>
207         </div>
208         <div class="p-3">
209             <form id="trading-order-form">
210                 <div class="row g-3">
211                     <div class="col-md-4">
212                         <label class="form-label small fw-semibold tex
t-light">Тип ордера:</label>
213                         <select class="form-select form-select-sm bg-d
ark text-white border-secondary" id="trading-order-type">
214                             <option value="MARKET">Рыночный (Market)</op
tion>
215                             <option value="LIMIT">Лимитный (Limit)</opti
on>
216                         </select>
217                     </div>
218                     <div class="col-md-4">
219                         <label class="form-label small fw-semibold tex
t-light">Количество:</label>
220                         <div class="input-group input-group-sm">

```

```

221 |         <input type="number" step="0.01" min="0.001"
      |         class="form-control bg-dark text-white border-secondary" id="
      | trading-order-amount" value="0.1" required>
222 |         <span class="input-group-text bg-secondary t
      | ext-white border-secondary">BTC</span>
223 |     </div>
224 | </div>
225 |     <div class="col-md-4">
226 |         <label class="form-label small fw-semibold tex
      | t-light">Цена (для Limit):</label>
227 |         <div class="input-group input-group-sm">
228 |             <span class="input-group-text bg-secondary t
      | ext-white border-secondary">$</span>
229 |             <input type="number" step="1" class="form-co
      | ntrol bg-dark text-white border-secondary" id="trading-order-p
      | rice" placeholder="По рынку">
230 |         </div>
231 |     </div>
232 | </div>
233 |
234 |     <div class="d-flex justify-content-end gap-3 mt-3"
      | >
235 |         <button type="button" class="btn btn-success fw-
      | bold px-4 rounded-pill" id="btn-order-buy">
236 |             <i class="bi bi-arrow-up-circle me-1"></i> Куп
      | ить (BUY)
237 |         </button>
238 |         <button type="button" class="btn btn-danger fw-b
      | old px-4 rounded-pill" id="btn-order-sell">
239 |             <i class="bi bi-arrow-down-circle me-1"></i> П
      | родать (SELL)
240 |         </button>
241 |     </div>
242 | </form>
243 | </div>
244 | </div>
245 |
246 | <!-- Recent Orders Table -->

```

```

247 |     <div class="trading-panel flex-grow-1">
248 |         <div class="trading-panel-header">
249 |             <span><i class="bi bi-clock-history me-1"></i> Истоп
            ия ордеров и исполнений</span>
250 |             <span class="badge bg-secondary" id="trading-orders-
            count">0 ордеров</span>
251 |         </div>
252 |         <div class="table-responsive" style="max-height: 260px
            ; overflow-y: auto;">
253 |             <table class="table table-dark table-hover table-sm
            mb-0 small font-monospace">
254 |                 <thead>
255 |                     <tr class="table-secondary text-dark">
256 |                         <th>Время</th>
257 |                         <th>ID</th>
258 |                         <th>Направление</th>
259 |                         <th>Тип</th>
260 |                         <th style="text-align: right;">Объём</th>
261 |                         <th style="text-align: right;">Цена исполнения
            </th>
262 |                         <th style="text-align: right;">Статус</th>
263 |                     </tr>
264 |                 </thead>
265 |                 <tbody id="trading-orders-tbody">
266 |                     <tr><td colspan="7" class="text-center py-3 text
            -muted">История сделок пуста</td></tr>
267 |                 </tbody>
268 |             </table>
269 |         </div>
270 |     </div>
271 | </div>
272 | </div>
273 | </div>
274 |

```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Trading Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/trading_tab/main.js?v=
11  *       20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/trading_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Trading Terminal Tab JS Module
22 (function() {
23   let tradingWs = null;
24   let isTradingInitialized = false;
25
26   async function fetchTradingState() {
27     try {
28       const res = await fetch('/api/v1/trading/status');
29       if (!res.ok) return;
30       const data = await res.json();
31       updateTradingUI(data);
32     } catch (e) {
```



```

33 |     console.error('[TradingTab] Failed to fetch state:', e);
34 | }
35 | }
36 |
37 | async function fetchOrderbook() {
38 |     try {
39 |         const res = await fetch('/api/v1/trading/orderbook?depth
| =6');
40 |         if (!res.ok) return;
41 |         const data = await res.json();
42 |         renderOrderbook(data);
43 |     } catch (e) {
44 |         console.error('[TradingTab] Failed to fetch orderbook:',
| e);
45 |     }
46 | }
47 |
48 | async function fetchOrdersList() {
49 |     try {
50 |         const res = await fetch('/api/v1/trading/orders');
51 |         if (!res.ok) return;
52 |         const orders = await res.json();
53 |         renderOrders(orders);
54 |     } catch (e) {
55 |         console.error('[TradingTab] Failed to fetch orders:', e)
| ;
56 |     }
57 | }
58 |
59 | function updateTradingUI(state) {
60 |     if (!state) return;
61 |     const symEl = document.getElementById('trading-symbol');
62 |     const priceEl = document.getElementById('trading-current-p
| rice');
63 |     const equityEl = document.getElementById('trading-total-eq
| uity');
64 |     const balanceEl = document.getElementById('trading-cash-ba
| lance');

```

```
65 |     const pnlEl = document.getElementById('trading-pnl-val');
66 |     const posEl = document.getElementById('trading-pos-size');
67 |
68 |     if (symEl) symEl.innerText = state.symbol || 'BTC/USDT';
69 |     if (priceEl) priceEl.innerText = `$$${Number(state.current_
price || 0).toLocaleString(undefined, { minimumFractionDigits:
2, maximumFractionDigits: 2 })}`;
70 |     if (equityEl) equityEl.innerText = `$$${Number(state.total_
equity || 0).toLocaleString(undefined, { minimumFractionDigits
: 2, maximumFractionDigits: 2 })}`;
71 |     if (balanceEl) balanceEl.innerText = `Доступно: $$${Number(
state.balance || 0).toLocaleString(undefined, { minimumFractio
nDigits: 2, maximumFractionDigits: 2 })}`;
72 |
73 |     if (pnlEl) {
74 |         const pnl = Number(state.unrealized_pnl || 0);
75 |         const sign = pnl >= 0 ? '+' : '';
76 |         pnlEl.innerText = `${sign}$$${pnl.toLocaleString(undefine
d, { minimumFractionDigits: 2, maximumFractionDigits: 2 })}`;
77 |         pnlEl.style.color = pnl >= 0 ? '#4ade80' : '#f87171';
78 |     }
79 |
80 |     if (posEl) {
81 |         posEl.innerText = `Размер позиции: ${state.position_size
|| 0} ${state.symbol?.split('/')[0] || 'BTC'}`;
82 |     }
83 | }
84 |
85 | function renderOrderbook(ob) {
86 |     if (!ob) return;
87 |     const asksTbody = document.getElementById('trading-asks-tb
ody');
88 |     const bidsTbody = document.getElementById('trading-bids-tb
ody');
89 |     const midEl = document.getElementById('trading-orderbook-m
idprice');
90 |     const spreadEl = document.getElementById('trading-spread-i
nfo');
```

```

91
92     if (midEl && ob.mid_price) {
93         midEl.innerText = `$$${Number(ob.mid_price).toLocaleString(
          undefined, { minimumFractionDigits: 2, maximumFractionDigits
          : 2 })}}`;
94     }
95     if (spreadEl && ob.spread !== undefined) {
96         spreadEl.innerText = `Сред: $$${Number(ob.spread).toFixed(
          2)}}`;
97     }
98
99     if (asksTbody && Array.isArray(ob.asks)) {
100         // Show asks reversed (highest on top)
101         const reversedAsks = [...ob.asks].reverse();
102         asksTbody.innerHTML = reversedAsks.map(a => `
103             <tr class="trading-ask-row">
104                 <td>$$${Number(a.price).toLocaleString(undefined, { m
          inimumFractionDigits: 2, maximumFractionDigits: 2 })}</td>
105                 <td style="text-align: right;">${Number(a.amount).to
          Fixed(4)}</td>
106                 <td style="text-align: right;">$$${Number(a.total).to
          LocaleString(undefined, { minimumFractionDigits: 2, maximumFra
          ctionDigits: 2 })}</td>
107             </tr>
108         `).join('');
109     }
110
111     if (bidsTbody && Array.isArray(ob.bids)) {
112         bidsTbody.innerHTML = ob.bids.map(b => `
113             <tr class="trading-bid-row">
114                 <td>$$${Number(b.price).toLocaleString(undefined, { m
          inimumFractionDigits: 2, maximumFractionDigits: 2 })}</td>
115                 <td style="text-align: right;">${Number(b.amount).to
          Fixed(4)}</td>
116                 <td style="text-align: right;">$$${Number(b.total).to
          LocaleString(undefined, { minimumFractionDigits: 2, maximumFra
          ctionDigits: 2 })}</td>
117             </tr>

```

```

118 |     `).join('');
119 | }
120 | }
121 |
122 | function renderOrders(orders) {
123 |     const tbody = document.getElementById('trading-orders-tbod
124 | y');
125 |     const countEl = document.getElementById('trading-orders-co
126 | unt');
127 |     if (!tbody) return;
128 |     if (!Array.isArray(orders) || orders.length === 0) {
129 |         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
130 | ter py-3 text-muted">История сделок пуста</td></tr>';
131 |         if (countEl) countEl.innerText = '0 ордеров';
132 |         return;
133 |     }
134 |     if (countEl) countEl.innerText = `${orders.length} ордеров
135 | `;
136 |     tbody.innerHTML = orders.map(o => {
137 |         const isBuy = (o.side || '').toUpperCase() === 'BUY';
138 |         const sideBadge = isBuy
139 |             ? '<span class="badge bg-success">BUY</span>'
140 |             : '<span class="badge bg-danger">SELL</span>';
141 |         return `
142 |             <tr>
143 |                 <td>${new Date(o.timestamp * 1000).toLocaleTimeStrin
144 | g()}</td>
145 |                 <td class="text-muted">${o.order_id}</td>
146 |                 <td>${sideBadge}</td>
147 |                 <td>${o.order_type}</td>
148 |                 <td style="text-align: right;">${Number(o.amount).to
149 | Fixed(4)}</td>
150 |                 <td style="text-align: right;">$$${Number(o.price).to
151 | LocaleString(undefined, { minimumFractionDigits: 2, maximumFra
152 | ctionDigits: 2 })}</td>

```

```
148 |         <td style="text-align: right;"><span class="badge bg  
    | -secondary">${o.status}</span></td>  
149 |     </tr>  
150 |     `;  
151 |     }).join('');  
152 | }  
153 |  
154 | async function placeOrder(side) {  
155 |     const amountInput = document.getElementById('trading-order-  
    | -amount');  
156 |     const priceInput = document.getElementById('trading-order-  
    | price');  
157 |     const typeSelect = document.getElementById('trading-order-  
    | type');  
158 |     const amount = parseFloat(amountInput?.value || '0');  
159 |     const orderType = typeSelect?.value || 'MARKET';  
160 |     const price = priceInput?.value ? parseFloat(priceInput.va  
    | lue) : null;  
161 |  
162 |     if (!amount || amount <= 0) {  
163 |         alert('Укажите корректный объём ордера');  
164 |         return;  
165 |     }  
166 |  
167 |     try {  
168 |         const payload = {  
169 |             symbol: 'BTC/USDT',  
170 |             side: side,  
171 |             amount: amount,  
172 |             order_type: orderType,  
173 |             price: price  
174 |         };  
175 |  
176 |         const res = await fetch('/api/v1/trading/order', {  
177 |             method: 'POST',  
178 |             headers: { 'Content-Type': 'application/json' },  
179 |             body: JSON.stringify(payload)  
180 |         });
```

```
181
182     if (!res.ok) {
183         const err = await res.json().catch(() => ({ detail: 'О
184        шибка выполнения' }));
185         alert('Ошибка выставления ордера: ' + (err.detail || r
186         es.statusText));
187         return;
188     }
189     await fetchTradingState();
190     await fetchOrdersList();
191     await fetchOrderbook();
192 } catch (e) {
193     console.error('[TradingTab] Order error:', e);
194     alert('Ошибка соединения при отправке ордера: ' + e.mess
195     age);
196 }
197
198 async function triggerKillSwitch() {
199     if (!confirm('⚠ ВНИМАНИЕ: Вы действительно хотите немедлен
200     но закрыть все позиции по рыночной цене (Kill-Switch)?')) {
201         return;
202     }
203     try {
204         const res = await fetch('/api/v1/trading/kill-switch', {
205             method: 'POST' });
206         if (!res.ok) throw new Error('Ошибка вызова Kill-Switch'
207         );
208         const data = await res.json();
209         alert(`Ликвидация завершена: ${data.message || 'Позиции
210         закрыты'}`);
211         await fetchTradingState();
212         await fetchOrdersList();
213     } catch (e) {
214         alert('Ошибка при вызове Kill-Switch: ' + e.message);
215     }
216 }
```

```

212 | }
213 |
214 | function isTradingTabActive() {
215 |     const el = document.getElementById('tab-trading');
216 |     if (!el) return false;
217 |     return el.classList.contains('active') || el.classList.con
218 | tains('show') || (el.offsetWidth > 0 && el.offsetHeight > 0);
219 | }
220 |
221 | function connectTradingWebSocket() {
222 |     if (tradingWs) {
223 |         try { tradingWs.close(); } catch {}
224 |     }
225 |     const proto = window.location.protocol === 'https:' ? 'wss
226 | : ' : 'ws:';
227 |     const wsUrl = `${proto}://${window.location.host}/api/v1/tr
228 | ading/ws/stream`;
229 |
230 |     try {
231 |         tradingWs = new WebSocket(wsUrl);
232 |         const statusBadge = document.getElementById('trading-str
233 | eam-status');
234 |
235 |         tradingWs.onopen = () => {
236 |             if (statusBadge) {
237 |                 statusBadge.className = 'badge rounded-pill bg-succe
238 | ss-subtle text-success border border-success px-3 py-2';
239 |                 statusBadge.innerText = '● Стрим активен';
240 |             }
241 |         };
242 |
243 |         tradingWs.onmessage = (evt) => {
244 |             try {
245 |                 const state = JSON.parse(evt.data);
246 |                 updateTradingUI(state);
247 |                 if (isTradingTabActive()) {
248 |                     fetchOrderbook();
249 |                     fetchOrdersList();

```

```
245 |         }
246 |     } catch (e) {
247 |         console.error('[TradingTab] WS message parse error:'
    | , e);
248 |     }
249 | };
250 |
251 | tradingWs.onclose = () => {
252 |     if (statusBadge) {
253 |         statusBadge.className = 'badge rounded-pill bg-dange
    | r-subtle text-danger border border-danger px-3 py-2';
254 |         statusBadge.innerText = '● Отключено';
255 |     }
256 | };
257 | } catch (e) {
258 |     console.error('[TradingTab] WS connection error:', e);
259 | }
260 | }
261 |
262 | function initTradingTab(force = false) {
263 |     if (!isTradingInitialized) {
264 |         const buyBtn = document.getElementById('btn-order-buy');
265 |         const sellBtn = document.getElementById('btn-order-sell'
    | );
266 |         const refreshBtn = document.getElementById('btn-refresh-
    | trading');
267 |         const killBtn = document.getElementById('btn-trigger-kil
    | l-switch');
268 |
269 |         const configBtn = document.getElementById('btn-trading-c
    | onfig');
270 |
271 |         if (buyBtn) buyBtn.onclick = () => placeOrder('BUY');
272 |         if (sellBtn) sellBtn.onclick = () => placeOrder('SELL');
273 |         if (refreshBtn) refreshBtn.onclick = () => {
274 |             fetchTradingState();
275 |             fetchOrderbook();
276 |             fetchOrdersList();
```



```

277 |     };
278 |     if (killBtn) killBtn.onclick = triggerKillSwitch;
279 |     if (configBtn) configBtn.onclick = () => {
280 |         if (typeof window.openAppConfigModal === 'function') {
281 |             window.openAppConfigModal('trading_terminal', 'Excha
nge Trading Terminal');
282 |         }
283 |     };
284 |
285 |     isTradingInitialized = true;
286 | }
287 |
288 | if (!force && !isTradingTabActive()) {
289 |     console.log('[TradingTab] Tab is not active; skipping ba
ckground network polling and WebSocket connection. ');
290 |     return;
291 | }
292 |
293 | console.log('[TradingTab] Initializing active Trading Term
inal network polling & WebSocket... ');
294 | fetchTradingState();
295 | fetchOrderbook();
296 | fetchOrdersList();
297 | connectTradingWebSocket();
298 | }
299 |
300 | window.initTradingTab = initTradingTab;
301 | })();
302 |

```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/tts_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/tts_tab/index.html" cla
17     ss="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/tts_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- TTS Settings Tab -->
30 <div class="container-fluid">
31   <div class="alert alert-info">
32     <h5 data-i18n="tts.title">□ Настройка TTS (Синтез речи)</
33     h5>
34     <p data-i18n="tts.description">Установите голос и движок п
35     о умолчанию для озвучки описания медиафайлов, а также протести
36     руйте их звучание.</p>
37   </div>
```

```

30
31 <div class="row g-3">
32     <!-- LEFT: Settings Configuration -->
33     <div class="col-5">
34         <div class="card mb-3">
35             <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
36                 <span>□ Параметры озвучки</span>
37             </div>
38             <div class="card-body">
39                 <form id="admin-tts-form" onsubmit="event.preventDefault();">
40                     <!-- Active Voice Badge -->
41                     <div id="admin-active-voice-container" class="mb-3" style="display: none;">
42                         <span class="badge bg-success d-inline-flex align-items-center gap-1 p-2">
43                             <i class="bi bi-check-circle-fill"></i>
44                             <span>Активный голос: <strong id="admin-active-voice-name">...</strong></span>
45                         </span>
46                     </div>
47
48                     <!-- Engine Selection -->
49                     <div class="mb-3">
50                         <label for="admin-tts-engine" class="form-label small fw-bold">Система синтеза:</label>
51                         <select id="admin-tts-engine" class="form-select form-select-sm">
52                             <option value="browser">Web Speech API (Браузерная)</option>
53                             <option value="gtts">Google TTS (Базовая)</option>
54                             <option value="edge-tts" selected>Microsoft Edge TTS (Нейросетевая)</option>
55                             <option value="silero">Silero TTS (Локальная оффлайн)</option>
56                             </select>

```

```

57 |         </div>
58 |
59 |         <!-- Voice Selection -->
60 |         <div class="mb-3" id="admin-voice-select-container
61 |         ">
62 |             <label for="admin-tts-voice" class="form-label s
63 |             mall fw-bold">Голос / Спикер:</label>
64 |             <select id="admin-tts-voice" class="form-select
65 |             form-select-sm">
66 |                 <!-- Dynamically populated -->
67 |                 </select>
68 |             </div>
69 |
70 |             <!-- Save Default Button -->
71 |             <div class="d-grid">
72 |                 <button type="button" class="btn btn-success btn
73 |                 -sm" id="btn-admin-tts-save">
74 |                     <i class="bi bi-floppy"></i> Применить по умол
75 |                     чанию
76 |                 </button>
77 |             </div>
78 |         </form>
79 |     </div>
80 | </div>
81 |
82 | <!-- RIGHT: Text to Speech Playground -->
83 | <div class="col-7">
84 |     <div class="card">
85 |         <div class="card-header bg-dark text-white">
86 |             <span>🔊 Тестирование озвучки</span>
87 |         </div>
88 |         <div class="card-body">
89 |             <div class="mb-3">
90 |                 <label for="admin-tts-text" class="form-label smal
91 |                 l fw-bold">Текст для проверки:</label>
92 |                 <textarea id="admin-tts-text" class="form-control
93 |                 form-control-sm" rows="5">Привет! Это проверка работы модульно

```

й системы синтеза речи. Выберите систему и голос слева, затем нажмите кнопку «Озвучить» для воспроизведения.</textarea>

88 </div>

89

90 <div class="d-flex align-items-center gap-3">

91 <button type="button" class="btn btn-primary btn-sm px-4" id="btn-admin-tts-speak">

92 <i class="bi bi-play-fill"></i> Озвучить текст

93 </button>

94

95 <!-- Loading Spinner -->

96 <div class="status-badge small text-muted d-none" id="admin-tts-status">

97 <div class="spinner-border spinner-border-sm text-primary" role="status"></div>

98 Генерация аудио...

99 </div>

100 </div>

101

102 <!-- Audio Player Container -->

103 <div class="mt-4 border-top pt-3 d-none" id="admin-tts-player-container">

104 <label class="form-label small fw-bold">Результат генерации:</label>

105 <audio id="admin-tts-audio-player" class="w-100" controls></audio>

106 </div>

107 </div>

108 </div>

109 </div>

110 </div>

111 </div>

112

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Tts Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/tts_tab/main.js?v=2026
11  *     1001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/tts_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // TTS Tab Configuration and Testing Logic
22 // Called via window.initTtsTab() when tab is loaded
23
24 'use strict';
25
26 function ttsFetch(url, options = {}) {
27   return window.api
28     ? window.api.fetch(url, options)
29     : fetch(url, options).then(r => {
30       if (!r.ok) throw new Error(`HTTP ${r.status}`);
31       return r.json();
32     });
```

```

33 | }
34 |
35 | function ttsNotify(msg, type = 'info') {
36 |     if (typeof showNotification === 'function') {
37 |         showNotification(msg, type);
38 |     } else {
39 |         console.log(`[${type}] ${msg}`);
40 |     }
41 | }
42 |
43 | // Available voices dictionary
44 | const ADMIN_VOICE_MAP = {
45 |     'browser': [], // Populated dynamically from window.speechSynthesis
46 |     'gtts': [
47 |         { value: 'ru', label: 'Русский (Google TTS)' },
48 |         { value: 'en', label: 'English (Google TTS)' },
49 |         { value: 'he', label: 'עברית (Google TTS)' },
50 |         { value: 'de', label: 'Deutsch (Google TTS)' },
51 |         { value: 'fr', label: 'Français (Google TTS)' },
52 |         { value: 'es', label: 'Español (Google TTS)' }
53 |     ],
54 |     'edge-tts': [
55 |         { value: 'ru-RU-DmitryNeural', label: 'Дмитрий (RU - Мужской)' },
56 |         { value: 'ru-RU-SvetlanaNeural', label: 'Светлана (RU - Женский)' },
57 |         { value: 'en-US-JennyNeural', label: 'Jenny (US - Female)' },
58 |         { value: 'en-US-GuyNeural', label: 'Guy (US - Male)' },
59 |         { value: 'en-US-AriaNeural', label: 'Aria (US - Female)' },
60 |         { value: 'en-GB-SoniaNeural', label: 'Sonia (UK - Female)' },
61 |         { value: 'en-GB-RyanNeural', label: 'Ryan (UK - Male)' },
62 |         { value: 'he-IL-AvriNeural', label: 'Avri (HE - עברי)' },
63 |         { value: 'he-IL-HilaNeural', label: 'Hila (HE - עברית)' },
64 |         { value: 'de-DE-KatjaNeural', label: 'Katja (DE - Female)' }

```

```

    },
65     { value: 'de-DE-ConradNeural', label: 'Conrad (DE - Male)'
    },
66     { value: 'fr-FR-DeniseNeural', label: 'Denise (FR - Female
    )' },
67     { value: 'fr-FR-HenriNeural', label: 'Henri (FR - Male)' }
    ,
68     { value: 'es-ES-ElviraNeural', label: 'Elvira (ES - Female
    )' },
69     { value: 'es-ES-AlvaroNeural', label: 'Alvaro (ES - Male)'
    }
70 ],
71 'silero': [
72     { value: 'eugene', label: 'Евгений (Silero RU)' },
73     { value: 'aidar', label: 'Айдар (Silero RU)' },
74     { value: 'baya', label: 'Бая (Silero RU)' },
75     { value: 'kseniya', label: 'Ксения (Silero RU)' },
76     { value: 'xenia', label: 'Ксения v2 (Silero RU)' },
77     { value: 'random', label: 'Рандомный (Silero RU)' }
78 ]
79 };
80
81 function getFriendlyVoiceName(system, voice) {
82     if (system === 'edge-tts') {
83         const found = ADMIN_VOICE_MAP['edge-tts'].find(v => v.value
84         === voice);
85         return found ? `Edge: ${found.label}` : `Edge (${voice})`;
86     }
87     if (system === 'silero') {
88         const found = ADMIN_VOICE_MAP['silero'].find(v => v.value
89         === voice);
90         return found ? found.label : `Silero (${voice})`;
91     }
92     if (system === 'gtts') {
93         const found = ADMIN_VOICE_MAP['gtts'].find(v => v.value ==
94         = voice);
95         return found ? found.label : `Google TTS (${voice})`;
96     }
97 }

```



```
94 |   if (system === 'browser') {
95 |       return `Браузерный (${voice} || 'По умолчанию')`;
96 |   }
97 |   return `${system} – ${voice}`;
98 | }
99 |
100 | async function initTtsTab() {
101 |     const engineSelect = document.getElementById('admin-tts-engine');
102 |     const voiceSelect = document.getElementById('admin-tts-voice');
103 |     const voiceContainer = document.getElementById('admin-voice-select-container');
104 |     const btnSpeak = document.getElementById('btn-admin-tts-speak');
105 |     const btnSave = document.getElementById('btn-admin-tts-save');
106 |     const ttsStatus = document.getElementById('admin-tts-status');
107 |     const statusText = document.getElementById('admin-status-text');
108 |     const playerContainer = document.getElementById('admin-tts-player-container');
109 |     const audioPlayer = document.getElementById('admin-tts-audio-player');
110 |     const activeVoiceContainer = document.getElementById('admin-active-voice-container');
111 |     const activeVoiceName = document.getElementById('admin-active-voice-name');
112 |
113 |     if (!engineSelect || !voiceSelect) {
114 |         console.error('TTS Settings Tab DOM elements not found.');
```

return;

```
116 |     }
117 |
118 |     function updateActiveBadge(system, voice) {
119 |         if (system && voice) {
120 |             activeVoiceName.textContent = getFriendlyVoiceName(system, voice);
```

```

    m, voice);
121     activeVoiceContainer.style.display = 'block';
122 } else {
123     activeVoiceContainer.style.display = 'none';
124 }
125 }
126
127 // Populate voices based on selected engine
128 function populateVoices() {
129     const engine = engineSelect.value;
130     const voices = ADMIN_VOICE_MAP[engine] || [];
131
132     voiceSelect.innerHTML = '';
133
134     if (voices.length === 0) {
135         voiceContainer.style.display = 'none';
136     } else {
137         voiceContainer.style.display = 'block';
138         voices.forEach(voice => {
139             const opt = document.createElement('option');
140             opt.value = voice.value;
141             opt.textContent = voice.label;
142             voiceSelect.appendChild(opt);
143         });
144     }
145 }
146
147 // Load browser speech synthesis voices if available
148 function loadBrowserVoices() {
149     if ('speechSynthesis' in window) {
150         const voices = window.speechSynthesis.getVoices();
151         ADMIN_VOICE_MAP['browser'] = voices
152             .map(v => ({ value: v.name, label: `[${v.lang}] ${v.name}` }));
153
154         if (ADMIN_VOICE_MAP['browser'].length === 0) {
155             ADMIN_VOICE_MAP['browser'].push({ value: 'default', label: 'Браузер по умолчанию' });

```

```

156 |     }
157 |
158 |     if (engineSelect.value === 'browser') {
159 |         populateVoices();
160 |     }
161 | }
162 | }
163 |
164 | if ('speechSynthesis' in window) {
165 |     window.speechSynthesis.onvoiceschanged = loadBrowserVoices
166 | ;
167 |     loadBrowserVoices();
168 | }
169 | engineSelect.addEventListener('change', populateVoices);
170 | populateVoices(); // Initial run
171 |
172 | // Fetch current user settings
173 | try {
174 |     const settings = await ttsFetch('/auth/settings');
175 |     if (settings) {
176 |         if (settings.tts_system) {
177 |             engineSelect.value = settings.tts_system;
178 |             populateVoices();
179 |         }
180 |         if (settings.tts_voice) {
181 |             voiceSelect.value = settings.tts_voice;
182 |         }
183 |         updateActiveBadge(settings.tts_system, settings.tts_voic
184 | e);
185 |     } catch (e) {
186 |         console.error('Failed to load active TTS settings:', e);
187 |     }
188 |
189 | // Speak button handler
190 | btnSpeak.addEventListener('click', async () => {
191 |     const text = document.getElementById('admin-tts-text').val

```

```
ue.trim();
192     if (!text) {
193         ttsNotify('Пожалуйста, введите текст для озвучки', 'warn
ing');
194         return;
195     }
196
197     const engine = engineSelect.value;
198     const voice = voiceSelect.value;
199
200     if (engine === 'browser') {
201         if ('speechSynthesis' in window) {
202             window.speechSynthesis.cancel();
203             const cleanText = text.replace(/<[^>]*>/g, '').replace
(/[*_`#]/g, '');
204             const utterance = new SpeechSynthesisUtterance(cleanTe
xt);
205             const voices = window.speechSynthesis.getVoices();
206             const selectedVoice = voices.find(v => v.name === voic
e);
207             if (selectedVoice) {
208                 utterance.voice = selectedVoice;
209                 if (selectedVoice.lang) {
210                     utterance.lang = selectedVoice.lang;
211                 }
212             }
213
214             ttsStatus.classList.remove('d-none');
215             statusText.textContent = 'Воспроизведение браузером...
';
216
217             utterance.onend = () => ttsStatus.classList.add('d-non
e');
218             utterance.onerror = () => ttsStatus.classList.add('d-n
one');
219
220             window.speechSynthesis.speak(utterance);
221             playerContainer.classList.add('d-none');
```

```
222     } else {
223         ttsNotify('Ваш браузер не поддерживает Web Speech API'
224         , 'danger');
225     }
226     return;
227 }
228 // Server-side synthesis
229 ttsStatus.classList.remove('d-none');
230 statusText.textContent = 'Синтез аудиофайла на сервере...'
231 ;
232 btnSpeak.disabled = true;
233 playerContainer.classList.add('d-none');
234 try {
235     const queryParams = new URLSearchParams({
236         text: text,
237         system: engine,
238         voice: voice
239     });
240
241     const audioUrl = `/api/tts/synthesize?${queryParams.toString()}`;
242
243     audioPlayer.src = audioUrl;
244     audioPlayer.load();
245
246     audioPlayer.oncanplaythrough = () => {
247         ttsStatus.classList.add('d-none');
248         btnSpeak.disabled = false;
249         playerContainer.classList.remove('d-none');
250         audioPlayer.play().catch(err => console.log('Audio autoplay blocked or failed:', err));
251     };
252
253     audioPlayer.onerror = (err) => {
254         console.error('Audio element error:', err);
255         statusText.textContent = 'Ошибка загрузки аудиофайла';
```

```

256 |         btnSpeak.disabled = false;
257 |         setTimeout(() => ttsStatus.classList.add('d-none'), 30
    | 00);
258 |     };
259 |
260 | } catch (err) {
261 |     console.error(err);
262 |     statusText.textContent = 'Ошибка сети';
263 |     btnSpeak.disabled = false;
264 |     setTimeout(() => ttsStatus.classList.add('d-none'), 3000
    | );
265 | }
266 | });
267 |
268 | // Save Settings handler
269 | btnSave.addEventListener('click', async () => {
270 |     const engine = engineSelect.value;
271 |     const voice = voiceSelect.value;
272 |
273 |     btnSave.disabled = true;
274 |     const originalText = btnSave.innerHTML;
275 |     btnSave.innerHTML = '<span class="spinner-border spinner-b
    | order-sm" role="status" aria-hidden="true"></span> Сохранение.
    | ..';
276 |
277 |     try {
278 |         const payload = {
279 |             tts_enabled: 1,
280 |             tts_system: engine,
281 |             tts_voice: voice
282 |         };
283 |
284 |         const response = await fetch('/auth/settings', {
285 |             method: 'POST',
286 |             headers: { 'Content-Type': 'application/json' },
287 |             body: JSON.stringify(payload)
288 |         });
289 |

```

```
290         if (response.ok) {
291             updateActiveBadge(engine, voice);
292             ttsNotify('Настройки TTS успешно применены по умолчанию!', 'success');
293         } else {
294             const errData = await response.json().catch(() => ({}))
295             );
296             throw new Error(errData.detail || 'Ошибка сохранения настроек');
297         }
298     } catch (err) {
299         console.error(err);
300         ttsNotify('Не удалось сохранить настройки: ' + err.message, 'danger');
301     } finally {
302         btnSave.disabled = false;
303         btnSave.innerHTML = originalText;
304     }
305 });
306
307 window.initTtsTab = initTtsTab;
308
```

google_signin.html

```
1 <!--
2 =====
3 |
4 |=====
5 | Process Name: Windows Web Interface - Google Signin View
6 |=====
7 |
8 | Description:
9 |   Шаблон клиентского веб-интерфейса google_signin.
10 |
11 | Usage Examples:
12 |   Web Route:
13 |     /windows/api/webgui/user/components/google_signin.html
14 |
15 |   Tab Frame Integration:
16 |     <iframe src="/windows/api/webgui/user/components/google_
17 |signin.html" class="w-100 h-100 border-0"></iframe>
18 |
19 | File: google_signin.html
20 | Project: ai-breadboard
21 | Package: windows/api/webgui/user/components
22 | Author: hypo69
23 | Copyright: © 2026 hypo69
24 | Updated: 2026-10-01 13:04:40
25 |=====
26 |=====
27 | -->
28 |
29 |<!-- Google Sign-In Component -->
30 |<div id="google-signin-container" class="d-flex align-items-ce
31 |nter gap-2">
32 |  <button id="google-signin-btn" class="btn btn-outline-light
33 |btn-sm">
34 |    <i class="bi bi-google"></i>
35 |    <span data-i18n="auth.googleSignIn">Войти через Google</sp
36 |an>
37 |  </button>
```



```

30 | <div id="user-profile" class="d-none flex-grow-1">
31 |   <div class="d-flex align-items-center gap-2">
32 |     <img id="user-avatar" src="" class="rounded-circle" width
33 | h="32" height="32">
34 |     <span id="user-name" class="text-muted small"></span>
35 |     <button id="logout-btn" class="btn btn-sm btn-danger">
36 |       <i class="bi bi-box-arrow-right"></i> Выйти
37 |     </button>
38 |   </div>
39 </div>
40
41 <script>
42 // Google Sign-In Component Logic
43 (async () => {
44   const googleSignInBtn = document.getElementById('google-sign
45 in-btn');
46   const logoutBtn = document.getElementById('logout-btn');
47   const userProfile = document.getElementById('user-profile');
48   const userAvatar = document.getElementById('user-avatar');
49   const userName = document.getElementById('user-name');
50
51   // Check authentication status
52   async function checkAuthStatus() {
53     try {
54       const response = await fetch('/auth/check');
55       const data = await response.json();
56
57       if (data.authenticated) {
58         // User is authenticated
59         userProfile.classList.remove('d-none');
60         googleSignInBtn.classList.add('d-none');
61
62         if (data.name) {
63           userName.textContent = data.name;
64         }
65         if (data.email) {
66           userName.textContent = data.email.split('@')[0];

```

```
66 |     }
67 |     if (data.picture) {
68 |         userAvatar.src = data.picture;
69 |     }
70 | } else {
71 |     // User is not authenticated
72 |     userProfile.classList.add('d-none');
73 |     googleSignInBtn.classList.remove('d-none');
74 | }
75 | } catch (error) {
76 |     console.error('Auth check failed:', error);
77 |     userProfile.classList.add('d-none');
78 |     googleSignInBtn.classList.remove('d-none');
79 | }
80 | }
81 |
82 | // Handle Google Sign-In click
83 | googleSignInBtn?.addEventListener('click', () => {
84 |     window.location.href = '/auth/google?next=' + encodeURIComponent(
window.location.pathname + window.location.search);
85 | });
86 |
87 | // Handle logout click
88 | logoutBtn?.addEventListener('click', async () => {
89 |     try {
90 |         await fetch('/auth/logout', { method: 'POST' });
91 |         // Reload page to update UI
92 |         window.location.reload();
93 |     } catch (error) {
94 |         console.error('Logout failed:', error);
95 |     }
96 | });
97 |
98 | // Initial auth check
99 | checkAuthStatus();
100 | })();
101 | </script>
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/user/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/user/index.html" class=
15 | "w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/user
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!DOCTYPE html>
28 <html lang="ru">
29 <head>
30 |   <meta charset="UTF-8">
31 |   <meta name="viewport" content="width=device-width, initial-s
32 | cale=1.0">
33 |   <title>AI Assistant</title>
34 |   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
35 | t/css/bootstrap.min.css" rel="stylesheet">
```

```

31 <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    0.0/font/bootstrap-icons.css" rel="stylesheet">
32 <link href="/html/css/main.css?v=20260924_v1" rel="styleshee
    t">
33 <script src="/html/js/toast.js?v=20260924_v1"></script>
34 <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
    8next.min.js"></script>
35 <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
    js"></script>
36 </head>
37 <body>
38 <div class="container-fluid mt-2" id="user-interface">
39
40 <div class="d-flex align-items-center justify-content-betwee
    n mb-3 main-nav-container rounded px-2 py-1">
41 <ul class="nav nav-tabs flex-nowrap flex-grow-1 border-0 m
    e-2 align-items-center" id="mainTabs">
42 <!-- Группа: Стандартные -->
43 <li class="nav-item dropdown">
44 <button class="nav-link dropdown-toggle active d-flex
    align-items-center gap-1" type="button" id="standardTabsDropdo
    wn">
45 <i class="bi bi-grid-fill text-primary me-1"></i>
46 <span data-i18n="tabs.groupStandard">Стандартные</sp
    an>
47 </button>
48 <div class="dropdown-menu shadow border-secondary-subtl
    e">
49 <button class="dropdown-item active d-flex align-ite
    ms-center gap-2" data-tab="tab-chat" data-bs-target="#tab-chat
    " type="button" data-i18n="tabs.chat">
50 <span>Чат
51 </button>
52 <button class="dropdown-item d-flex align-items-cent
    er gap-2" data-tab="tab-voice" data-bs-target="#tab-voice" typ
    e="button" data-i18n="tabs.voice">
53 <span>Голос и аудио
54 </button>

```

```
55 |         </div>
56 |     </li>
57 |
58 |     <!-- Разделитель -->
59 |     <li class="nav-item d-flex align-items-center">
60 |         <div class="nav-group-divider"></div>
61 |     </li>
62 |
63 |     <!-- Группа: Расширенные -->
64 |     <li class="nav-item dropdown">
65 |         <button class="nav-link dropdown-toggle d-flex align-i
66 | items-center gap-1" type="button" id="extendedTabsDropdown">
67 |             <i class="bi bi-sliders2 text-info me-1"></i>
68 |             <span data-i18n="tabs.groupExtended">Расширенные</sp
69 | an>
70 |         </button>
71 |         <div class="dropdown-menu shadow border-secondary-subt
72 | le">
73 |             <button class="dropdown-item d-flex align-items-cent
74 | er gap-2" data-tab="tab-rag" data-bs-target="#tab-rag" type="b
75 | utton" data-i18n="tabs.rag">
76 |                 □ RAG
77 |             </button>
78 |             <button class="dropdown-item d-flex align-items-cent
79 | er gap-2" data-tab="tab-tts" data-bs-target="#tab-tts" type="b
80 | utton" data-i18n="tabs.tts">
81 |                 □ Настройка TTS
82 |             </button>
83 |             <button class="dropdown-item d-flex align-items-cent
84 | er gap-2" data-tab="tab-skills" data-bs-target="#tab-skills" t
85 | ype="button" data-i18n="tabs.skills">
86 |                 □ Навыки
87 |             </button>
88 |             <button class="dropdown-item d-flex align-items-cent
89 | er gap-2" data-tab="tab-news" data-bs-target="#tab-news" type=
90 | "button" data-i18n="tabs.news">
91 |                 □ Новости
92 |             </button>
```

```

82 |         <button class="dropdown-item d-flex align-items-cent
| er gap-2" data-tab="tab-user-directories" data-bs-target="#tab
| -user-directories" type="button" data-i18n="tabs.userDirectori
| es">
83 |             □ Директории и файлы
84 |         </button>
85 |     </div>
86 | </li>
87 |
88 | <!-- Разделитель -->
89 | <li class="nav-item d-flex align-items-center">
90 |     <div class="nav-group-divider"></div>
91 | </li>
92 |
93 | <!-- Группа: Плагины -->
94 | <li class="nav-item dropdown" id="pluginsNavDropdownWrap
| per">
95 |     <button class="nav-link dropdown-toggle d-flex align-i
| tems-center gap-1" type="button" id="pluginsTabsDropdown">
96 |         <i class="bi bi-puzzle-fill text-warning me-1"></i>
97 |         <span data-i18n="tabs.plugins">Плагины</span>
98 |     </button>
99 |     <div class="dropdown-menu shadow border-secondary-subt
| le" id="nav-plugins-dropdown-menu">
100 |         <div id="nav-enabled-plugins-list" class="nav-enable
| d-plugins-list">
101 |             <!-- Dynamic enabled plugins will be injected here
| -->
102 |         </div>
103 |         <hr class="dropdown-divider my-1">
104 |         <button class="dropdown-item d-flex align-items-cent
| er gap-2" data-tab="tab-plugins" data-bs-target="#tab-plugins"
| type="button" data-i18n="tabs.managePlugins">
105 |             □ Управление
106 |         </button>
107 |     </div>
108 | </li>
109 | <!-- Разделитель -->

```

```
110 |     <li class="nav-item d-flex align-items-center">
111 |         <div class="nav-group-divider"></div>
112 |     </li>
113 |
114 |     <!-- Группа: Сервисы -->
115 |     <li class="nav-item dropdown">
116 |         <button class="nav-link dropdown-toggle d-flex align-i
117 | items-center gap-1" type="button" id="userServicesDropdown">
118 |             <i class="bi bi-grid-3x3-gap-fill text-success me-1"
119 | ></i>
120 |             <span data-i18n="tabs.groupUserServices">Сервисы</sp
121 | an>
122 |         </button>
123 |         <div class="dropdown-menu shadow border-secondary-subt
124 | le">
125 |             <button class="dropdown-item d-flex align-items-cent
126 | er gap-2 py-1" id="nav-btn-check-mail" type="button">
127 |                 <i class="bi bi-envelope-check text-primary"></i>
128 |                 <span>Почта</span>
129 |             </button>
130 |             <button class="dropdown-item d-flex align-items-cent
131 | er gap-2 py-1" id="nav-btn-search-gdocs" type="button">
132 |                 <i class="bi bi-file-earmark-text text-info"></i>
133 |                 <span>Google Docs</span>
134 |             </button>
135 |             <button class="dropdown-item d-flex align-items-cent
136 | er gap-2 py-1" id="nav-btn-gsheets" type="button">
137 |                 <i class="bi bi-file-earmark-spreadsheet text-succ
138 | ess"></i>
139 |                 <span>Google Таблицы</span>
140 |             </button>
141 |             <button class="dropdown-item d-flex align-items-cent
142 | er gap-2 py-1" id="nav-btn-news-digest" type="button">
143 |                 <i class="bi bi-newspaper text-warning"></i>
144 |                 <span>Новости</span>
145 |             </button>
146 |             <button class="dropdown-item d-flex align-items-cent
147 | er gap-2 py-1" id="nav-btn-facebook-post" type="button">
```

```

138         <i class="bi bi-facebook text-primary"></i>
139         <span>Facebook</span>
140     </button>
141 </div>
142 </li>
143 </ul>
144
145     <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto pb-1" id="nav-auth-container">
146         <!-- Google OAuth Register / Sign-in Button (Shown when
not logged in) -->
147         <button class="btn btn-sm btn-light text-dark fw-semibol
d d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-
pill btn-google-login" id="nav-google-btn" title="Войти или за
регистрироваться через Google" data-i18n-title="auth.googleSig
nInTitle" style="border: 1px solid rgba(0,0,0,0.15);">
148             <svg width="18" height="18" viewBox="0 0 24 24"><path
fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h
5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.
74 3.28-8.09z"/><path fill="#34A853" d="M12 23c2.97 0 5.46-.98
7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-
1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/><path fill
="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.
35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.
85-2.22.81-.63z"/><path fill="#EA4335" d="M12 5.38c1.62 0 3.06
.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.4
7 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/></svg>
149         <span data-i18n="auth.googleSignUp">Регистрация через
Google</span>
150     </button>
151
152     <!-- Authenticated User Menu Dropdown (Shown when logged
in) -->
153     <div class="dropdown d-none" id="user-menu-dropdown-wrap
per">
154         <button class="btn btn-sm btn-outline-secondary d-flex
align-items-center gap-2 px-2 py-1 rounded-pill dropdown-togg
le" id="user-settings-btn" type="button" data-bs-toggle="drop

```



```

own" aria-expanded="false" title="Пользовательское меню">
155     <img id="nav-user-avatar" src="" class="rounded-circle d-none" width="22" height="22" style="object-fit: cover;">
156     <i class="bi bi-person-circle" id="nav-user-icon"></i>
157     <span id="nav-user-label" class="small fw-semibold" data-i18n="userSettings.button">Пользователь</span>
158 </button>
159 <ul class="dropdown-menu dropdown-menu-end shadow border-secondary-subtle py-1" aria-labelledby="user-settings-btn" style="min-width: 220px;">
160     <li class="px-3 py-2 border-bottom border-secondary-subtle">
161         <div class="fw-bold text-truncate" id="user-menu-display-name">Пользователь</div>
162         <div class="small text-muted text-truncate" id="user-menu-display-email" style="font-size: 0.75rem;">user@gmail.com</div>
163     </li>
164     <li><h6 class="dropdown-header small text-uppercase text-muted fw-semibold mt-1">Сервисы</h6></li>
165     <li>
166         <button class="dropdown-item d-flex align-items-center gap-2 py-1" type="button" id="user-menu-btn-mail">
167             <i class="bi bi-envelope-check text-primary"></i>
168             <span>Почта</span>
169         </button>
170     </li>
171     <li>
172         <button class="dropdown-item d-flex align-items-center gap-2 py-1" type="button" id="user-menu-btn-docs">
173             <i class="bi bi-file-earmark-text text-info"></i>
174             <span>Google Docs</span>
175         </button>
176     </li>
177     <li>

```

```
178         <button class="dropdown-item d-flex align-items-ce
| nter gap-2 py-1" type="button" id="user-menu-btn-sheets">
179         <i class="bi bi-file-earmark-spreadsheet text-su
| ccess"></i>
180         <span>Google Таблицы</span>
181     </button>
182 </li>
183 <li>
184     <button class="dropdown-item d-flex align-items-ce
| nter gap-2 py-1" type="button" id="user-menu-btn-news">
185     <i class="bi bi-newspaper text-warning"></i>
186     <span>Новости</span>
187 </button>
188 </li>
189 <li>
190     <button class="dropdown-item d-flex align-items-ce
| nter gap-2 py-1" type="button" id="user-menu-btn-facebook">
191     <i class="bi bi-facebook text-primary"></i>
192     <span>Facebook</span>
193 </button>
194 </li>
195 <li><hr class="dropdown-divider my-1"></li>
196 <li>
197     <button class="dropdown-item d-flex align-items-ce
| nter gap-2 py-1" type="button" data-bs-toggle="modal" data-bs-
| target="#userSettingsModal">
198     <i class="bi bi-gear-fill text-secondary"></i>
199     <span data-i18n="userSettings.title">Настройки п
| рофиля</span>
200 </button>
201 </li>
202 <li>
203     <button class="dropdown-item d-flex align-items-ce
| nter gap-2 py-1 text-danger btn-user-logout" type="button">
204     <i class="bi bi-box-arrow-right"></i>
205     <span>Выйти</span>
206 </button>
207 </li>
```

```
208         </ul>
209     </div>
210
211
212     <!-- Quick Settings Gear Button -->
213     <button class="btn btn-sm btn-outline-secondary d-flex align-items-center justify-content-center p-0 rounded-circle" id="user-settings-gear-btn" data-bs-toggle="modal" data-bs-target="#userSettingsModal" title="Настройки" style="width: 32px; height: 32px;" data-i18n-title="userSettings.title">
214         <i class="bi bi-gear-fill"></i>
215     </button>
216 </div>
217 </div>
218
219 <div class="tab-content" id="mainTabContent">
220     <!-- Chat Tab -->
221     <div id="tab-chat" class="tab-pane fade show active"></div>
222 >
223     <!-- RAG Tab -->
224     <div id="tab-rag" class="tab-pane fade"></div>
225
226     <!-- Telegram RAG Tab -->
227     <div id="tab-telegram-rag" class="tab-pane fade"></div>
228
229     <!-- News Tab -->
230     <div id="tab-news" class="tab-pane fade"></div>
231
232     <!-- TTS Tab -->
233     <div id="tab-tts" class="tab-pane fade"></div>
234
235     <!-- Voice Tab -->
236     <div id="tab-voice" class="tab-pane fade"></div>
237
238     <!-- Skills Tab -->
239     <div id="tab-skills" class="tab-pane fade"></div>
240
```

```

241     <!-- Plugins Tab -->
242     <div id="tab-plugins" class="tab-pane fade"></div>
243
244     <!-- User Directories Tab -->
245     <div id="tab-user-directories" class="tab-pane fade"></div>
246 >
247 </div>
248
249 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    /js/bootstrap.bundle.min.js"></script>
250 <script src="/html/js/chatService.js?v=20260924_v1"></script>
251 <script type="module" src="/html/user/main.js?v=20260924_v1"><
    /script>
252
253 <!-- Chat Logic & RAG Architecture Modal -->
254 <div class="modal fade" id="chat-logic-modal" tabindex="-1" ar
    ia-labelledby="chat-logic-modal-label" aria-hidden="true">
255     <div class="modal-dialog modal-xl modal-dialog-scrollable">
256         <div class="modal-content bg-dark text-white border-second
    ary">
257             <div class="modal-header border-secondary d-flex justify
    -content-between align-items-center">
258                 <h5 class="modal-title d-flex align-items-center gap-2
    " id="chat-logic-modal-label">
259                     <i class="bi bi-diagram-3-fill text-info"></i> □ Ap
    хитектура и алгоритм диспетчеризации ответов чата
260                 </h5>
261                 <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="modal" aria-label="Close"></button>
262             </div>
263             <div class="modal-body p-3">
264                 <div class="p-2 border border-secondary rounded bg-bla
    ck d-flex justify-content-center align-items-center" style="ov
    erflow-x: auto; min-height: 520px;">
265                     

```

```

266         </div>
267     </div>
268     <div class="modal-footer border-secondary">
269         <button type="button" class="btn btn-sm btn-secondary"
270         data-bs-dismiss="modal">Закрыть</button>
271     </div>
272 </div>
273 </div>
274
275 <!-- User Settings Modal -->
276 <div class="modal fade" id="userSettingsModal" tabindex="-1" a
277     ria-labelledby="userSettingsModalLabel" aria-hidden="true">
278     <div class="modal-dialog modal-dialog-centered">
279         <div class="modal-content shadow border-secondary-subtle">
280             <div class="modal-header border-secondary-subtle">
281                 <h5 class="modal-title d-flex align-items-center gap-2
282                 " id="userSettingsModalLabel">
283                     <i class="bi bi-person-gear text-primary"></i>
284                     <span data-i18n="userSettings.title">Пользовательски
285                     е настройки</span>
286                 </h5>
287                 <button type="button" class="btn-close" data-bs-dismis
288                 s="modal" aria-label="Close"></button>
289             </div>
290             <div class="modal-body p-3">
291                 <!-- User Account & Google OAuth Card -->
292                 <div class="card bg-body-tertiary border-secondary-sub
293                 tle mb-3 p-3" id="user-settings-auth-card">
294                     <div class="d-flex align-items-center gap-3">
295                         <img id="user-settings-avatar" src="" class="round
296                         ed-circle border border-2 border-primary" width="48" height="4
297                         8" style="object-fit: cover;">
298                         <div class="flex-grow-1 min-w-0">
299                             <div class="d-flex align-items-center gap-2">
300                                 <strong id="user-settings-name" class="text-tr
301                                 uncate">Пользователь</strong>
302                                 <span id="user-settings-role-badge" class="bad

```

```

    ge bg-primary">User</span>
295         </div>
296         <div id="user-settings-email" class="small text-
muted text-truncate">email@gmail.com</div>
297         <div id="user-settings-google-badge" class="small
mt-1 text-muted">
298             <i class="bi bi-google text-danger me-1"></i>G
oogle OAuth
299         </div>
300     </div>
301     <button type="button" class="btn btn-outline-dange
r btn-sm btn-user-logout" title="Выйти">
302         <i class="bi bi-box-arrow-right"></i>
303     </button>
304 </div>
305 </div>
306
307 <!-- Guest / Connect Google OAuth Card -->
308 <div class="card bg-body-tertiary border-secondary-sub
tle mb-3 p-3 text-center d-none" id="user-settings-guest-card"
>
309     <div class="text-muted small mb-2">
310         <i class="bi bi-shield-lock me-1"></i> Авторизация
и синхронизация аккаунта
311     </div>
312     <button type="button" class="btn btn-outline-danger
btn-sm w-100 d-flex align-items-center justify-content-center
gap-2 btn-google-login">
313         <i class="bi bi-google"></i>
314         <span>Войти через Google (OAuth)</span>
315     </button>
316     <div class="small text-muted mt-2" style="font-size:
0.75rem;">
317         Синхронизация с Google Drive, Docs, Calendar и Con
tacts
318     </div>
319 </div>
320

```

```
321 |         <!-- Language Selection -->
322 |         <div class="mb-3">
323 |             <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-lang-selector">
324 |                 <i class="bi bi-translate text-info"></i>
325 |                 <span data-i18n="userSettings.language">Язык интерфейса</span>
326 |             </label>
327 |             <select class="form-select lang-selector" id="modal-lang-selector">
328 |                 <option value="ru">Русский</option>
329 |                 <option value="en">English</option>
330 |                 <option value="he">עברית</option>
331 |             </select>
332 |         </div>
333 |
334 |         <!-- Theme Selection -->
335 |         <div class="mb-2">
336 |             <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-theme-selector">
337 |                 <i class="bi bi-palette text-warning"></i>
338 |                 <span data-i18n="userSettings.theme">Тема оформления</span>
339 |             </label>
340 |             <select class="form-select theme-selector" id="modal-theme-selector">
341 |                 <option value="light" data-i18n="userSettings.themeLight">☀ Светлая</option>
342 |                 <option value="dark" data-i18n="userSettings.themeDark">☿ Тёмная</option>
343 |                 <option value="system" data-i18n="userSettings.themeSystem">☒ Системная</option>
344 |             </select>
345 |         </div>
346 |     </div>
347 |     <div class="modal-footer border-secondary-subtle py-2">
348 |         <button type="button" class="btn btn-secondary btn-sm" data-bs-dismiss="modal" data-i18n="userSettings.close">Закрыт
```

```
        ь</button>
349 |         </div>
350 |     </div>
351 | </div>
352 </div>
353 |
354 </body>
355 </html>
```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows User - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/user/main.js?v=2026100
11 *     1_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/user
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-06 22:15:00
19 * =====
20 */
21 // User Interface Main JS
22 import { initI18n, switchLang, applyTranslations } from '../js/i18n.js';
23 import { initTheme, setTheme, getThemeMode, getResolvedTheme } from '../js/theme.js';
24 import { initUserSettings, refreshUserProfile } from '../js/userSettings.js';
25
26 // Make switchLang and theme functions available globally
27 window.switchLang = switchLang;
28 window.applyTranslations = applyTranslations;
29 window.setTheme = setTheme;
```

```

30 window.getThemeMode = getThemeMode;
31 window.getResolvedTheme = getResolvedTheme;
32 window.initUserSettings = initUserSettings;
33 window.refreshUserProfile = refreshUserProfile;
34
35 // Helper to open plugin from top navbar dropdown
36 window.openPluginFromDropdown = function(pluginName) {
37     if (!pluginName) return;
38     document.querySelectorAll('#mainTabs .dropdown-menu.show').f
forEach((m) => {
39         m.classList.remove('show');
40         m.closest('.dropdown')?.querySelector('.dropdown-toggle')?
.classList.remove('show');
41     });
42
43     const normalized = pluginName.toLowerCase().replace(/-/g, '_
');
44
45     if (normalized === 'telegram_channel_rag' || normalized ===
'telegram_rag') {
46         const pane = document.getElementById('tab-telegram-rag') |
| document.getElementById('tab-telegram_rag');
47         if (pane && typeof window.switchTab === 'function') {
48             window.switchTab('tab-telegram-rag');
49             return;
50         }
51     }
52
53     if (normalized === 'news_feed' || normalized === 'news' || n
ormalized === 'smart_news' || normalized === 'smart_feed_news'
) {
54         const pane = document.getElementById('tab-news');
55         if (pane && typeof window.switchTab === 'function') {
56             window.switchTab('tab-news');
57             return;
58         }
59     }
60

```

```

61 | if (typeof window.switchTab === 'function') {
62 |     window.switchTab('tab-plugins');
63 | }
64 |
65 | const select = () => {
66 |     if (typeof window.selectPlugin === 'function') {
67 |         window.selectPlugin(pluginName);
68 |     }
69 | };
70 |
71 | select();
72 | setTimeout(select, 150);
73 | setTimeout(select, 400);
74 | };
75 |
76 | // Global API module
77 | window.api = {
78 |     async fetch(url, options = {}) {
79 |         const response = await fetch(url, options);
80 |         if (!response.ok) {
81 |             let msg = response.statusText;
82 |             try {
83 |                 const data = await response.json();
84 |                 if (data && data.detail) {
85 |                     if (typeof data.detail === 'string') {
86 |                         msg = data.detail;
87 |                     } else if (Array.isArray(data.detail)) {
88 |                         msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
89 |                     } else {
90 |                         msg = JSON.stringify(data.detail);
91 |                     }
92 |                 }
93 |             } catch {}
94 |             throw new Error(`${response.status} ${msg}`);
95 |         }
96 |         return response.json();
97 |     }

```

```

98 | };
99 |
100 | // UI Notifications
101 | function showNotification(message, type = 'info') {
102 |     const notification = document.createElement('div');
103 |     notification.className = `alert alert-${type} position-fixed
    |     top-0 end-0 m-3 shadow`;
104 |     notification.style.zIndex = '9999';
105 |     notification.style.maxWidth = '400px';
106 |     notification.textContent = message;
107 |     document.body.appendChild(notification);
108 |
109 |     setTimeout(() => {
110 |         notification.remove();
111 |     }, 5000);
112 | }
113 | window.showNotification = showNotification;
114 |
115 | // Tab Switch Handler
116 | function onTabSwitched(targetId) {
117 |     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
    |     : targetId;
118 |     if (cleanId === 'tab-chat') {
119 |         const msgInput = document.getElementById('message-input');
120 |         if (msgInput) msgInput.focus();
121 |     } else if (cleanId === 'tab-rag' && typeof window.initRagTab
    |     === 'function') {
122 |         console.log('[UserInterface] Switching to RAG tab...');
123 |         window.initRagTab();
124 |     } else if ((cleanId === 'tab-telegram-rag' || cleanId === 't
    |     ab-telegram_rag') && typeof window.initTelegramRagTab === 'fun
    |     ction') {
125 |         console.log('[UserInterface] Switching to Telegram RAG tab
    |         ...');
126 |         window.initTelegramRagTab();
127 |     } else if (cleanId === 'tab-tts' && typeof window.initTtsTab
    |     === 'function') {
128 |         console.log('[UserInterface] Switching to TTS tab...');

```

```
129     window.initTtsTab();
130   } else if (cleanId === 'tab-voice' && typeof window.initVoiceTab === 'function') {
131     console.log('[UserInterface] Switching to Voice tab...');
132     window.initVoiceTab();
133   } else if (cleanId === 'tab-skills' && typeof window.initSkillsTab === 'function') {
134     console.log('[UserInterface] Switching to Skills tab...');
135     window.initSkillsTab();
136   } else if (cleanId === 'tab-news' && typeof window.initNewsTab === 'function') {
137     console.log('[UserInterface] Switching to News tab...');
138     window.initNewsTab();
139   } else if (cleanId === 'tab-plugins' && typeof window.initPluginsTab === 'function') {
140     console.log('[UserInterface] Switching to Plugins tab...');
141     ;
142     window.initPluginsTab();
143   } else if (cleanId === 'tab-user-directories' && typeof window.initUserDirectoriesTab === 'function') {
144     console.log('[UserInterface] Switching to User Directories tab...');
145     window.initUserDirectoriesTab();
146   }
147 }
148 // Карта конфигураций вкладок
149 const USER_TAB_DEFS = {
150   'chat': { html: '/html/chat/index.html', js: '/html/chat/main.js' },
151   'rag': { html: '/html/rag_tab/index.html', js: '/html/rag_tab/main.js' },
152   'telegram-rag': { html: '/html/telegram_rag_tab/index.html', js: '/html/telegram_rag_tab/main.js' },
153   'news': { html: '/html/news_tab/index.html', js: '/html/news_tab/main.js' },
154   'tts': { html: '/html/tts_tab/index.html', js: '/html/tts_tab/main.js' },
```

```

155 |   'voice': { html: '/html/voice_tab/index.html', js: '/html/vo
      ice_tab/main.js' },
156 |   'skills': { html: '/html/skills_tab/index.html', js: '/html/
      skills_tab/main.js' },
157 |   'plugins': { html: '/html/plugins_tab/index.html', js: '/htm
      l/plugins_tab/main.js' },
158 |   'user-directories': { html: '/html/user_directories_tab/inde
      x.html', js: '/html/user_directories_tab/main.js' },
159 | };
160 |
161 | const loadedUserTabs = new Set();
162 | const loadingUserTabs = new Map();
163 |
164 | async function ensureTabLoaded(tabName) {
165 |   if (loadedUserTabs.has(tabName)) return true;
166 |   if (loadingUserTabs.has(tabName)) return loadingUserTabs.get
      (tabName);
167 |
168 |   const def = USER_TAB_DEFS[tabName];
169 |   if (!def) return false;
170 |
171 |   const cb = Date.now();
172 |   const appendCb = (u, cb) => u ? (u.includes('?') ? `${u}&t=
      ${cb}` : `${u}?v=${cb}`) : '';
173 |   const promise = (async () => {
174 |     try {
175 |       await loadTabContent(tabName, appendCb(def.html, cb), ap
      pendCb(def.js, cb));
176 |       loadedUserTabs.add(tabName);
177 |       return true;
178 |     } finally {
179 |       loadingUserTabs.delete(tabName);
180 |     }
181 |   })();
182 |
183 |   loadingUserTabs.set(tabName, promise);
184 |   return promise;
185 | }

```

```

186
187 // Programmatic tab switcher
188 async function switchTab(targetId) {
189   if (!targetId) return;
190   const cleanId = targetId.startsWith('#') ? targetId.slice(1)
191     : targetId;
192   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
cleanId}`;
193
194   const tabName = tabId.replace(/^tab-/, '');
195
196   // 1. Update active state on dropdown items & toggles
197   document.querySelectorAll('#mainTabs .dropdown-item').forEac
h((item) => {
198     const itemTarget = item.getAttribute('data-tab') || item.g
etAttribute('data-bs-target')?.replace('#', '');
199     if (itemTarget === tabId || itemTarget === cleanId) {
200       item.classList.add('active');
201     } else {
202       item.classList.remove('active');
203     }
204   });
205
206   document.querySelectorAll('#mainTabs .dropdown').forEach((dr
opdown) => {
207     const toggle = dropdown.querySelector('.dropdown-toggle');
208     const hasActiveChild = dropdown.querySelector('.dropdown-i
tem.active');
209     if (toggle) {
210       if (hasActiveChild) {
211         toggle.classList.add('active');
212       } else {
213         toggle.classList.remove('active');
214       }
215     }
216   });
217
218   // 2. Switch tab-pane (scoped to top-level container to pres
erve nested subtabs)

```

```
217 |   const mainTabContent = document.getElementById('mainTabConte
    | nt');
218 |   if (mainTabContent) {
219 |     Array.from(mainTabContent.children).forEach((pane) => {
220 |       if (pane.classList.contains('tab-pane')) {
221 |         pane.classList.remove('show', 'active');
222 |       }
223 |     });
224 |   } else {
225 |     document.querySelectorAll('#mainTabContent > .tab-pane, bo
    | dy > .container-fluid > .tab-content > .tab-pane').forEach((pa
    | ne) => {
226 |       pane.classList.remove('show', 'active');
227 |     });
228 |   }
229 |   const targetPane = document.getElementById(tabId) || documen
    | t.getElementById(cleanId);
230 |   if (targetPane) {
231 |     targetPane.classList.add('show', 'active');
232 |   }
233 |
234 |   // 3. Загружаем содержимое вкладки по требованию, если ещё н
    | е загружено
235 |   await ensureTabLoaded(tabName);
236 |
237 |   // 4. Notify lifecycle callback
238 |   onTabSwitched(tabId);
239 | }
240 | window.switchTab = switchTab;
241 | window.switchToTab = switchTab;
242 |
243 | // Setup dropdowns navigation
244 | function setupDropdownTabs() {
245 |   const mainTabs = document.getElementById('mainTabs');
246 |   if (!mainTabs) return;
247 |
248 |   // 1. Dropdown Toggle Buttons
249 |   mainTabs.querySelectorAll('.dropdown-toggle').forEach((btn)
```



```

=> {
250   btn.onclick = (e) => {
251     e.preventDefault();
252     e.stopPropagation();
253
254     const dropdown = btn.closest('.dropdown');
255     const menu = dropdown?.querySelector('.dropdown-menu');
256     const isAlreadyOpen = menu?.classList.contains('show');
257
258     // Close all dropdowns
259     document.querySelectorAll('#mainTabs .dropdown-menu.show
').forEach((m) => {
260       m.classList.remove('show');
261       m.closest('.dropdown')?.querySelector('.dropdown-toggl
e')?.classList.remove('show');
262     });
263
264     // Toggle clicked dropdown
265     if (!isAlreadyOpen && menu) {
266       menu.classList.add('show');
267       btn.classList.add('show');
268     }
269   };
270 });
271
272 // 2. Dropdown Item Buttons (Tab Switchers & Action Buttons)
273 mainTabs.querySelectorAll('.dropdown-item').forEach((item) =
> {
274   item.onclick = async (e) => {
275     const targetId = item.getAttribute('data-tab') || item.g
etAttribute('data-bs-target')?.replace('#', '');
276     const pluginName = item.getAttribute('data-plugin');
277
278     // Close dropdown menu
279     item.closest('.dropdown-menu')?.classList.remove('show')
;
280     item.closest('.dropdown')?.querySelector('.dropdown-togg
le')?.classList.remove('show');

```

```

281
282     if (pluginName && typeof window.openPluginFromDropdown =
    == 'function') {
283         e.preventDefault();
284         e.stopPropagation();
285         window.openPluginFromDropdown(pluginName);
286     } else if (targetId) {
287         e.preventDefault();
288         e.stopPropagation();
289         await switchTab(targetId);
290     }
291 };
292 });
293
294 // 3. Document Click to Close Dropdowns
295 if (!document.body.dataset.dropdownOutsideBound) {
296     document.body.dataset.dropdownOutsideBound = 'true';
297     document.addEventListener('click', (e) => {
298         if (!e.target.closest('#mainTabs .dropdown')) {
299             document.querySelectorAll('#mainTabs .dropdown-menu.sh
ow').forEach((menu) => {
300                 menu.classList.remove('show');
301                 menu.closest('.dropdown')?.querySelector('.dropdown-
toggle')?.classList.remove('show');
302             });
303         }
304     });
305 }
306 }
307
308 async function startUserInterface() {
309     console.log('User interface initializing...');
310     try {
311         initTheme();
312         setupDropdownTabs();
313         const el = document.getElementById('user-interface');
314         if (el) el.style.display = 'block';
315         await initInterface();

```

```
316     console.log('User interface ready');
317   } catch (err) {
318     console.error('Error during startUserInterface initializat
ion:', err);
319     const el = document.getElementById('user-interface');
320     if (el) el.style.display = 'block';
321   }
322 }
323
324 if (document.readyState === 'loading') {
325   document.addEventListener('DOMContentLoaded', () => {
326     setupDropdownTabs();
327     startUserInterface();
328   });
329 } else {
330   setupDropdownTabs();
331   startUserInterface();
332 }
333
334 async function initInterface() {
335   // Initialize theme
336   initTheme();
337
338   // Initialize i18n
339   await initI18n();
340
341   // Initialize User Settings & Google OAuth
342   await initUserSettings();
343
344   // Определение и загрузка ТОЛЬКО активной вкладки
345   const hash = location.hash.replace('#', '');
346   const initialTabId = hash || 'tab-chat';
347   const initialTabName = initialTabId.replace(/^tab-/, '');
348
349   console.log(`[UserInterface] Loading initial active tab: ${i
nitialTabName}`);
350   await ensureTabLoaded(initialTabName);
351   await switchTab(initialTabId);
```

```
352 |
353 | // Apply translations
354 | applyTranslations();
355 |
356 | // Synchronize plugin tabs visibility
357 | try {
358 |     let pluginsResp = await fetch('/api/plugins?scope=user');
359 |     if (!pluginsResp.ok) {
360 |         pluginsResp = await fetch('/api/admin/plugins?scope=user
361 | ');
362 |     }
363 |     if (pluginsResp.ok) {
364 |         const pluginsData = await pluginsResp.json();
365 |         if (window.syncPluginTabsVisibility && pluginsData && pl
366 | uuginsData.plugins) {
367 |             window.syncPluginTabsVisibility(pluginsData.plugins);
368 |         }
369 |     } catch (err) {
370 |         console.error('Ошибка синхронизации видимости плагинов:',
371 | err);
372 |     }
373 | }
374 |
375 | async function loadTabContent(tabName, url, jsOverrideSrc) {
376 |     try {
377 |         const container = document.getElementById(`tab-${tabName}`
378 | );
379 |         if (container && !container.innerHTML.trim()) {
380 |             container.innerHTML = `
381 |                 <div class="d-flex justify-content-center align-items-
382 | center" style="height: 200px;">
383 |                     <div class="spinner-border text-primary" role="statu
384 | s">
385 |                         <span class="visually-hidden">Загрузка...</span>
386 |                     </div>
```

```

384         </div>
385     `;
386 }
387
388     const response = await fetch(url);
389     if (!response.ok) throw new Error(`HTTP ${response.status}
    `);
390     const html = await response.text();
391     if (!container) return;
392     container.innerHTML = html;
393
394     // Load JS for the tab if provided or present
395     const cacheBuster = Date.now();
396     const scriptSrc = jsOverrideSrc || `/html/${tabName}/main.
    js?v=${cacheBuster}`;
397
398     await new Promise((resolve) => {
399         const script = document.createElement('script');
400         if (tabName === 'instructions' || tabName === 'rag') {
401             script.type = 'module';
402         }
403         script.src = scriptSrc;
404         script.onload = () => {
405             console.log(`[User] Loaded JS for ${tabName}`);
406             resolve();
407         };
408         script.onerror = (err) => {
409             console.debug(`[User] Note: No JS loaded for ${tabName}
    } from ${scriptSrc}`);
410             resolve();
411         };
412         document.body.appendChild(script);
413     });
414
415     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
    .toUpperCase());
416     const initFuncName = 'init' + camelName.charAt(0).toUpperC
    ase() + camelName.slice(1) + 'Tab';

```

```
417 |     if (typeof window[initFuncName] === 'function') {
418 |         console.log(`[User] Calling ${initFuncName} for ${tabName}
    | e`);
419 |         try {
420 |             await window[initFuncName]();
421 |         } catch (err) {
422 |             console.error(`[User] Error executing ${initFuncName}:
    | `, err);
423 |         }
424 |     }
425 |     applyTranslations();
426 | } catch (e) {
427 |     console.error(`[User] Error loading tab ${tabName}:`, e);
428 |     const container = document.getElementById(`tab-${tabName}`
    | );
429 |     if (container) {
430 |         container.innerHTML = `<div class="alert alert-danger">0
    | ошибка загрузки: ${e.message}</div>`;
431 |     }
432 | }
433 | }
```

debounce-patch.js

```
1  /**
2   * =====
3   * Process Name: Windows Users Tab - Debounce-Patch Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля debounce-
7   *   patch.
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/users_tab/debounce-pat
11  *   ch.js?v=20261001_v1" type="module"></script>
12  *
13  * File: debounce-patch.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/users_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 /**
22  * Users Tab Debounce Patch
23  *
24  * Применяет улучшенный дебаунс к поиску пользователей,
25  * используя RequestManager для снижения нагрузки на сервер
26  */
27
28 import { createUserSearchDebounce } from '../js/debounce-integ
29 ration.js';
30
31 // Ждем загрузки основного скрипта
```

```
31 | if (window.initUsersTab) {
32 |   const originalInitUsersTab = window.initUsersTab;
33 |
34 |   window.initUsersTab = async function() {
35 |     // Вызываем оригинальную функцию
36 |     await originalInitUsersTab.apply(this, arguments);
37 |
38 |     console.log('[UsersDebouncePatch] Patching users tab search...');
39 |
40 |     // После инициализации, патчим поиск
41 |     setTimeout(() => {
42 |       const searchInput = document.getElementById('users-search-input');
43 |
44 |       if (searchInput && window.globalRequestManager) {
45 |         // Сохраняем оригинальный обработчик
46 |         const listeners = searchInput._eventListeners?.input || [];
47 |
48 |         // Удаляем старые обработчики
49 |         searchInput.replaceWith(searchInput.cloneNode(true));
50 |         const newSearchInput = document.getElementById('users-search-input');
51 |
52 |         // Добавляем новый дебаунсированный обработчик
53 |         newSearchInput.addEventListener('input', (e) => {
54 |           const val = e.target.value;
55 |           const searchClear = document.getElementById('users-search-clear');
56 |
57 |           if (searchClear) {
58 |             searchClear.classList.toggle('d-none', !val);
59 |           }
60 |
61 |           // Используем улучшенный дебаунс
62 |           window.globalRequestManager.debounce(
63 |             'user-search-optimized',
```



```

64         async () => {
65             // Обновляем state поиска (симуляция)
66             if (window.updateUserSearch) {
67                 window.updateUserSearch(val.trim());
68             }
69         },
70         500
71     ).catch(e => console.error('Search debounce error:',
    e));
72     });
73
74     // Обновляем кнопку очистки
75     const searchClear = document.getElementById('users-search-clear');
76     if (searchClear) {
77         searchClear.onclick = () => {
78             newSearchInput.value = '';
79             searchClear.classList.add('d-none');
80             if (window.updateUserSearch) {
81                 window.updateUserSearch('');
82             }
83         };
84     }
85
86     console.log('[UsersDebouncePatch] Users tab search patched with RequestManager debounce');
87     }
88     }, 500);
89 };
90 }
91
92 // Альтернативный подход: Если users tab уже загружена, патчим сразу
93 if (document.getElementById('users-search-input')) {
94     console.log('[UsersDebouncePatch] Users tab already loaded, applying patch immediately...');
95
96     const searchInput = document.getElementById('users-search-in

```

```
    put');
97   const searchClear = document.getElementById('users-search-clear');
98
99   // Создаем новый дебаунсированный обработчик
100  const debouncedSearch = createUserSearchDebounce(async (query) => {
101    console.log(`[UsersDebouncePatch] Executing debounced search for: ${query}`);
102    // Обновляем UI через существующую функцию если она есть
103    if (window.updateUserSearch) {
104      await window.updateUserSearch(query);
105    }
106  });
107
108  // Заменяем обработчик
109  searchInput.replaceWith(searchInput.cloneNode(true));
110  const newSearchInput = document.getElementById('users-search-input');
111
112  newSearchInput.addEventListener('input', (e) => {
113    const val = e.target.value;
114
115    if (searchClear) {
116      searchClear.classList.toggle('d-none', !val);
117    }
118
119    debouncedSearch(val.trim());
120  });
121
122  if (searchClear) {
123    searchClear.addEventListener('click', () => {
124      newSearchInput.value = '';
125      searchClear.classList.add('d-none');
126      debouncedSearch('');
127    });
128  }
129
```

```
130 | console.log('[UsersDebouncePatch] Search debouncing initiali  
    | zed');  
131 | }  
132 |  
133 | export {};  
134 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/users_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/users_tab/index.html" c
17     lass="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/users_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-01 13:04:40
25 =====
26 =====
27 -->
28
29 <!-- Users Management Tab -->
30 <div class="container-fluid py-2" id="users-tab-root">
31   <!-- Top Banner & Stats -->
32   <div class="d-flex flex-wrap align-items-center justify-cont
33   ent-between gap-3 mb-3 p-3 bg-body-tertiary border rounded sha
34   dow-sm">
35     <div>
36       <h5 class="mb-1 text-primary d-flex align-items-center g
37       ap-2">
```

```

30         <i class="bi bi-people-fill text-primary"></i>
31         <span>Управление пользователями</span>
32     </h5>
33     <p class="mb-0 text-muted small">Управление учетными за-
34     писями, ролями, доступом к системе и сбросом паролей.</p>
35 </div>
36 <div class="d-flex flex-wrap gap-2">
37     <button class="btn btn-outline-warning btn-sm d-flex ali-
38     gn-items-center gap-1" id="btn-check-orphaned-dirs" title="Про-
39     верить мертвые директории пользователей, которые уже удалены и
40     з системы">
41         <i class="bi bi-folder-x"></i> Проверить мертвые дирек-
42         тории
43     </button>
44     <button class="btn btn-primary btn-sm d-flex align-items-
45     center gap-1" id="btn-open-create-user">
46         <i class="bi bi-person-plus-fill"></i> Добавить пользо-
47         вателя
48     </button>
49     <button class="btn btn-outline-secondary btn-sm d-flex a-
50     lign-items-center gap-1" id="btn-refresh-users" title="Обновит
51     ь список">
52         <i class="bi bi-arrow-clockwise"></i> Обновить
53     </button>
54 </div>
55 </div>
56 <!-- Summary Metric Cards -->
57 <div class="row g-3 mb-3">
58     <div class="col-6 col-md-3">
59         <div class="card bg-body-tertiary border h-100 shadow-sm
60         ">
61             <div class="card-body p-3 d-flex align-items-center ju-
62             stify-content-between">
63                 <div>
64                     <div class="text-muted small">Всего пользователей<
65                     /div>
66                     <h4 class="mb-0 fw-bold" id="stat-total-users">0</

```

```

h4>
56         </div>
57         <div class="bg-secondary bg-opacity-25 rounded p-2 t
ext-primary">
58             <i class="bi bi-people fs-4"></i>
59         </div>
60     </div>
61 </div>
62 </div>
63 <div class="col-6 col-md-3">
64     <div class="card bg-body-tertiary border h-100 shadow-sm
">
65         <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
66             <div>
67                 <div class="text-muted small">Активные аккаунты</d
iv>
68                 <h4 class="mb-0 text-success fw-bold" id="stat-act
ive-users">0</h4>
69             </div>
70             <div class="bg-success bg-opacity-25 rounded p-2 tex
t-success">
71                 <i class="bi bi-check-circle fs-4"></i>
72             </div>
73         </div>
74     </div>
75 </div>
76 <div class="col-6 col-md-3">
77     <div class="card bg-body-tertiary border h-100 shadow-sm
">
78         <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
79             <div>
80                 <div class="text-muted small">Администраторы</div>
81                 <h4 class="mb-0 text-warning fw-bold" id="stat-adm
in-users">0</h4>
82             </div>
83             <div class="bg-warning bg-opacity-25 rounded p-2 tex

```

```

t-warning">
84         <i class="bi bi-shield-lock fs-4"></i>
85     </div>
86 </div>
87 </div>
88 </div>
89 <div class="col-6 col-md-3">
90     <div class="card bg-body-tertiary border h-100 shadow-sm
">
91         <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
92             <div>
93                 <div class="text-muted small">Telegram привязано</
div>
94                 <h4 class="mb-0 text-info fw-bold" id="stat-tg-use
rs"></h4>
95             </div>
96             <div class="bg-info bg-opacity-25 rounded p-2 text-i
nfo">
97                 <i class="bi bi-telegram fs-4"></i>
98             </div>
99         </div>
100     </div>
101 </div>
102 </div>
103
104 <!-- Search & Filter Controls -->
105 <div class="card bg-body-tertiary border mb-3 shadow-sm">
106     <div class="card-body p-3">
107         <div class="row g-2 align-items-center">
108             <div class="col-12 col-md-6 col-lg-5">
109                 <div class="input-group input-group-sm">
110                     <span class="input-group-text"><i class="bi bi-sea
rch"></i></span>
111                     <input type="text" class="form-control" id="users-
search-input" placeholder="Поиск по имени, email, Telegram..."
>
112                 <button class="btn btn-outline-secondary d-none" t

```

```

    type="button" id="users-search-clear">❌ </button>
113         </div>
114     </div>
115     <div class="col-6 col-md-3 col-lg-3">
116         <select class="form-select form-select-sm" id="users
- filter-role">
117             <option value="">Все роли</option>
118             <option value="admin">Администраторы (admin)</opti
on>
119             <option value="user">Пользователи (user)</option>
120             <option value="guest">Гости (guest)</option>
121         </select>
122     </div>
123     <div class="col-6 col-md-3 col-lg-2">
124         <select class="form-select form-select-sm" id="users
- filter-status">
125             <option value="">Все статусы</option>
126             <option value="active">Только активные</option>
127             <option value="inactive">Заблокированные</option>
128         </select>
129     </div>
130     <div class="col-12 col-lg-2 text-end text-muted small"
id="users-count-badge">
131         Загрузка...
132     </div>
133 </div>
134 </div>
135 </div>
136
137 <!-- User Alert/Status message -->
138 <div id="users-alert-box" class="d-none alert alert-dismissi
ble fade show" role="alert">
139     <span id="users-alert-message"></span>
140     <button type="button" class="btn-close" aria-label="Close"
onclick="document.getElementById('users-alert-box').classList
.add('d-none')"></button>
141 </div>
142

```



```

143 | <!-- Users Table Card -->
144 | <div class="card shadow-sm border">
145 |     <div class="card-body p-0">
146 |         <div class="table-responsive">
147 |             <table class="table table-hover mb-0 align-middle">
148 |                 <thead class="table-light small text-uppercase">
149 |                     <tr>
150 |                         <th style="width: 50px;" class="text-center">ID<
151 | /th>
152 |                         <th>Пользователь</th>
153 |                         <th style="width: 140px;" class="text-center">По
154 | ль</th>
155 |                         <th style="width: 130px;" class="text-center">Ст
156 | атус</th>
157 |                         <th style="width: 150px;">Telegram</th>
158 |                         <th style="width: 120px;" class="text-center">Па
159 | роль</th>
160 |                         <th style="width: 160px;">Регистрация / Вход</th>
161 | >
162 |                         <th style="width: 170px;" class="text-center">Де
163 | йствия</th>
164 |                     </tr>
165 |                 </thead>
166 |                 <tbody id="users-table-body">
167 |                     <tr>
168 |                         <td colspan="8" class="text-center py-4 text-mut
169 | ed">
170 |                             <div class="spinner-border spinner-border-sm t
171 | ext-primary me-2" role="status"></div>
172 |                             Загрузка списка пользователей...
173 |                         </td>
174 |                     </tr>
175 |                 </tbody>
176 |             </table>
177 |         </div>
178 |     </div>
179 | </div>

```

```

173
174 <!-- =====
    ===== -->
175 <!-- MODAL: Create User -->
176 <!-- =====
    ===== -->
177 <div class="modal fade" id="modal-create-user" tabindex="-1" a
    ria-labelledby="modal-create-user-title" aria-hidden="true">
178   <div class="modal-dialog modal-dialog-centered">
179     <div class="modal-content shadow-lg">
180       <div class="modal-header">
181         <h5 class="modal-title d-flex align-items-center gap-2
    " id="modal-create-user-title">
182           <i class="bi bi-person-plus text-primary"></i> Добав
    ить нового пользователя
183         </h5>
184         <button type="button" class="btn-close" data-bs-dismis
    s="modal" aria-label="Close"></button>
185       </div>
186       <form id="form-create-user">
187         <div class="modal-body p-3">
188           <div class="mb-3">
189             <label class="form-label small fw-bold">Email <spa
    n class="text-danger">*</span></label>
190             <input type="email" class="form-control form-contr
    ol-sm" id="create-user-email" required placeholder="user@examp
    le.com">
191           </div>
192           <div class="mb-3">
193             <label class="form-label small fw-bold">Отображаем
    ое имя <span class="text-danger">*</span></label>
194             <input type="text" class="form-control form-contro
    l-sm" id="create-user-name" required placeholder="Иван Иванов"
    >
195           </div>
196           <div class="mb-3">
197             <label class="form-label small fw-bold">Пароль (оп
    ционально)</label>

```

```
198         <div class="input-group input-group-sm">
199             <input type="text" class="form-control font-mono
| space" id="create-user-password" placeholder="Оставьте пустым
| или задайте пароль">
200             <button class="btn btn-outline-secondary" type="
| button" id="btn-gen-create-password" title="Сгенерировать безо
| пасный пароль">
201                 <i class="bi bi-dice-5"></i>
202             </button>
203         </div>
204         <div class="form-text text-muted small">Если парол
| ь не задан, пользователь сможет авторизоваться через Google ил
| и Telegram.</div>
205     </div>
206     <div class="mb-3 p-3 bg-body-tertiary border rounded
| ">
207         <div class="form-check form-switch d-flex align-it
| ems-center justify-content-between p-0 m-0">
208             <label class="form-check-label fw-bold" for="cre
| ate-user-is-admin">
209                 <i class="bi bi-shield-lock-fill text-warning
| me-1"></i> Права Администратора
210             </label>
211             <input class="form-check-input ms-0 fs-5" type="
| checkbox" role="switch" id="create-user-is-admin">
212         </div>
213         <div class="small text-muted mt-1" id="create-role
| -hint">Роль: <strong class="text-info" id="create-role-label">
| Пользователь (user)</strong></div>
214     </div>
215     <div class="row g-2 mb-2">
216         <div class="col-6">
217             <div class="form-check form-switch">
218                 <input class="form-check-input" type="checkbox"
| " role="switch" id="create-user-is-active" checked>
219                 <label class="form-check-label small" for="cre
| ate-user-is-active">Активен</label>
220             </div>
```

```

221         </div>
222         <div class="col-6">
223             <div class="form-check form-switch">
224                 <input class="form-check-input" type="checkbox"
225                 " role="switch" id="create-user-verified" checked>
226                 <label class="form-check-label small" for="cre
227                 ate-user-verified">Email подтвержден</label>
228             </div>
229         </div>
230     <div class="modal-footer">
231         <button type="button" class="btn btn-secondary btn-s
232         m rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
233         <button type="submit" class="btn btn-primary btn-sm
234         rounded-pill px-3" id="btn-submit-create-user">Создать пользо
235         вателя</button>
236     </div>
237 </div>
238
239 <!-- =====
240 ===== -->
241 <!-- MODAL: Edit User -->
242 <!-- =====
243 ===== -->
244 <div class="modal fade" id="modal-edit-user" tabindex="-1" ari
245     a-labelledby="modal-edit-user-title" aria-hidden="true">
246     <div class="modal-dialog modal-dialog-centered">
247         <div class="modal-content shadow-lg">
248             <div class="modal-header">
249                 <h5 class="modal-title d-flex align-items-center gap-2
250                 " id="modal-edit-user-title">
251                     <i class="bi bi-pencil-square text-warning"></i> Ред
252                     активирование пользователя #<span id="edit-user-id-badge"></span
253                     >

```

```
248         </h5>
249         <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
250     </div>
251     <form id="form-edit-user">
252         <input type="hidden" id="edit-user-id">
253         <div class="modal-body p-3">
254             <div class="mb-3">
255                 <label class="form-label small fw-bold">Email <span class="text-danger">*</span></label>
256                 <input type="email" class="form-control form-control-sm" id="edit-user-email" required>
257             </div>
258             <div class="mb-3">
259                 <label class="form-label small fw-bold">Имя <span class="text-danger">*</span></label>
260                 <input type="text" class="form-control form-control-sm" id="edit-user-name" required>
261             </div>
262             <div class="mb-3 p-3 bg-body-tertiary border rounded">
263                 <div class="form-check form-switch d-flex align-items-center justify-content-between p-0 m-0">
264                     <label class="form-check-label fw-bold" for="edit-user-is-admin">
265                         <i class="bi bi-shield-lock-fill text-warning me-1"></i> Права Администратора
266                     </label>
267                     <input class="form-check-input ms-0 fs-5" type="checkbox" role="switch" id="edit-user-is-admin">
268                 </div>
269                 <div class="small text-muted mt-1" id="edit-role-hint">Роль: <strong class="text-info" id="edit-role-label">Пользователь (user)</strong></div>
270             </div>
271             <div class="row g-2 mb-2">
272                 <div class="col-6">
273                     <div class="form-check form-switch">
```

```

274 |         <input class="form-check-input" type="checkbox"
    | " role="switch" id="edit-user-is-active">
275 |         <label class="form-check-label small" for="edi
    | t-user-is-active">Активен</label>
276 |     </div>
277 | </div>
278 |     <div class="col-6">
279 |         <div class="form-check form-switch">
280 |             <input class="form-check-input" type="checkbox"
    | " role="switch" id="edit-user-verified">
281 |             <label class="form-check-label small" for="edi
    | t-user-verified">Email подтвержден</label>
282 |         </div>
283 |     </div>
284 | </div>
285 | </div>
286 |     <div class="modal-footer">
287 |         <button type="button" class="btn btn-secondary btn-s
    | m rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
288 |         <button type="submit" class="btn btn-warning btn-sm
    | rounded-pill px-3" id="btn-submit-edit-user">Сохранить изменен
    | ия</button>
289 |     </div>
290 | </form>
291 | </div>
292 | </div>
293 | </div>
294 |
295 | <!-- =====
    | ===== -->
296 | <!-- MODAL: Set / Reset Password -->
297 | <!-- =====
    | ===== -->
298 | <div class="modal fade" id="modal-password-user" tabindex="-1"
    | aria-labelledby="modal-password-user-title" aria-hidden="true
    | ">
299 |     <div class="modal-dialog modal-dialog-centered">
300 |         <div class="modal-content shadow-lg">

```

```
301 |     <div class="modal-header">
302 |         <h5 class="modal-title d-flex align-items-center gap-2
    | " id="modal-password-user-title">
303 |             <i class="bi bi-key-fill text-info"></i> Смена пароля
    | пользователя
304 |         </h5>
305 |         <button type="button" class="btn-close" data-bs-dismiss="modal"
    | aria-label="Close"></button>
306 |     </div>
307 |     <form id="form-password-user">
308 |         <input type="hidden" id="pwd-user-id">
309 |         <div class="modal-body p-3">
310 |             <p class="small text-muted mb-3">
311 |                 Установка нового пароля для <strong id="pwd-user-target"></strong>
    | </p>
312 |             <div class="mb-3">
313 |                 <label class="form-label small fw-bold">Новый пароль
    | <span class="text-danger">*</span></label>
314 |                 <div class="input-group">
315 |                     <input type="text" class="form-control font-monospace"
    | id="pwd-input-val" required minlength="4" placeholder="Введите пароль">
316 |                     <button class="btn btn-outline-info" type="button"
    | id="btn-gen-pwd-val" title="Сгенерировать безопасный пароль">
317 |                         <i class="bi bi-dice-5"></i> Сгенерировать
318 |                     </button>
319 |                 </div>
320 |             </div>
321 |         </div>
322 |     </div>
323 |     <div class="modal-footer">
324 |         <button type="button" class="btn btn-secondary btn-sm rounded-pill
    | px-3" data-bs-dismiss="modal">Отмена</button>
325 |         <button type="submit" class="btn btn-info btn-sm rounded-pill
    | px-3" id="btn-submit-pwd-user">Установить пароль</button>
326 |     </div>
```

```

327 |         </form>
328 |     </div>
329 | </div>
330 </div>
331
332 <!-- =====
    ===== -->
333 <!-- MODAL: User Details & Profile Settings -->
334 <!-- =====
    ===== -->
335 <div class="modal fade" id="modal-details-user" tabindex="-1"
    aria-labelledby="modal-details-user-title" aria-hidden="true">
336     <div class="modal-dialog modal-dialog-centered modal-lg">
337         <div class="modal-content shadow-lg">
338             <div class="modal-header">
339                 <h5 class="modal-title d-flex align-items-center gap-2
    " id="modal-details-user-title">
340                     <i class="bi bi-person-badge text-primary"></i> Проф
    иль и настройки пользователя
341                 </h5>
342                 <button type="button" class="btn-close" data-bs-dismiss="modal"
    aria-label="Close"></button>
343             </div>
344             <div class="modal-body p-3" id="details-user-content">
345                 <div class="text-center py-3 text-muted">Загрузка данн
    ых...</div>
346             </div>
347             <div class="modal-footer">
348                 <button type="button" class="btn btn-secondary btn-sm
    rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
349             </div>
350         </div>
351     </div>
352 </div>
353
354 <!-- =====
    ===== -->
355 <!-- MODAL: Delete Confirmation -->

```



```

356 | <!-- =====
      | ===== -->
357 | <div class="modal fade" id="modal-delete-user" tabindex="-1" a
      | ria-labelledby="modal-delete-user-title" aria-hidden="true">
358 |   <div class="modal-dialog modal-dialog-centered">
359 |     <div class="modal-content border-danger shadow-lg">
360 |       <div class="modal-header border-danger">
361 |         <h5 class="modal-title text-danger d-flex align-items-
      | center gap-2" id="modal-delete-user-title">
362 |           <i class="bi bi-exclamation-triangle-fill"></i> Подт
      | верждение удаления
363 |         </h5>
364 |         <button type="button" class="btn-close" data-bs-dismis
      | s="modal" aria-label="Close"></button>
365 |       </div>
366 |       <div class="modal-body p-3">
367 |         <p class="mb-2">Вы действительно хотите удалить пользо
      | вателя <strong class="text-danger" id="delete-user-name-target
      | "></strong>?</p>
368 |         <p class="small text-muted mb-0">Это действие удалит у
      | четную запись, настройки и сессии пользователя. Операция необр
      | атима.</p>
369 |         <input type="hidden" id="delete-user-id">
370 |       </div>
371 |       <div class="modal-footer">
372 |         <button type="button" class="btn btn-secondary btn-sm
      | rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
373 |         <button type="button" class="btn btn-danger btn-sm rou
      | nded-pill px-3" id="btn-confirm-delete-user">Удалить пользоват
      | еля</button>
374 |       </div>
375 |     </div>
376 |   </div>
377 | </div>
378 |
379 | <!-- =====
      | ===== -->
380 | <!-- MODAL: Orphaned User Directories Management -->

```

```

381 | <!-- =====
      | ===== -->
382 | <div class="modal fade" id="modal-orphaned-dirs" tabindex="-1"
      |   aria-labelledby="modal-orphaned-dirs-title" aria-hidden="true
      |   ">
383 |     <div class="modal-dialog modal-dialog-centered modal-lg">
384 |       <div class="modal-content shadow-lg border-warning">
385 |         <div class="modal-header border-warning">
386 |           <h5 class="modal-title text-warning d-flex align-items
      | -center gap-2" id="modal-orphaned-dirs-title">
387 |             <i class="bi bi-folder-x"></i> Мертвые директории по
      | льзователей
388 |           </h5>
389 |           <button type="button" class="btn-close" data-bs-dismis
      | s="modal" aria-label="Close"></button>
390 |         </div>
391 |         <div class="modal-body p-3">
392 |           <div class="alert alert-info border-0 small mb-3">
393 |             <i class="bi bi-info-circle text-info me-1"></i>
394 |             Обнаружены директории в хранилище <code>data/users/<
      | /code>, которые не привязаны ни к одному активному пользовател
      | ю в базе данных или остались после тестов.
395 |           </div>
396 |
397 |           <!-- Summary Stats Banner -->
398 |           <div class="d-flex align-items-center justify-content-
      | between p-2 mb-3 bg-body-tertiary border rounded">
399 |             <div class="small">
400 |               <span class="text-muted">Найдено директорий:</span>
      | <strong id="orphaned-dirs-count" class="text-warning">0</str
      | ong>
401 |               <span class="mx-2 text-secondary">|</span>
402 |               <span class="text-muted">Занято на диске:</span> <
      | strong id="orphaned-dirs-size" class="text-info">0 B</strong>
403 |               <span class="mx-2 text-secondary">|</span>
404 |               <span class="text-muted">Файлов:</span> <strong id
      | ="orphaned-dirs-files">0</strong>
405 |             </div>

```

```

406         <button class="btn btn-outline-secondary btn-sm py-0
      px-2 rounded-pill" id="btn-refresh-orphaned-dirs" title="Пере
      сканировать">
407             <i class="bi bi-arrow-clockwise"></i> Обновить
408         </button>
409     </div>
410
411     <div id="orphaned-dirs-table-wrapper" class="table-res
      ponsive" style="max-height: 380px; overflow-y: auto;">
412         <table class="table table-hover table-sm align-middl
      e mb-0">
413             <thead class="table-light small text-uppercase sti
      cky-top">
414                 <tr>
415                     <th>Папка / ID</th>
416                     <th>Причина</th>
417                     <th class="text-center" style="width: 80px;">Ф
      айлов</th>
418                     <th class="text-center" style="width: 100px;">
      Размер</th>
419                     <th style="width: 140px;">Изменено</th>
420                     <th class="text-center" style="width: 90px;">Д
      ействие</th>
421                 </tr>
422             </thead>
423             <tbody id="orphaned-dirs-tbody">
424                 <tr>
425                     <td colspan="6" class="text-center py-4 text-m
      uted">
426                         <div class="spinner-border spinner-border-sm
      text-warning me-2" role="status"></div>
427                         Сканирование хранилища...
428                     </td>
429                 </tr>
430             </tbody>
431         </table>
432     </div>
433 </div>

```

```
434 |         <div class="modal-footer justify-content-between">
435 |             <button type="button" class="btn btn-secondary btn-sm
436 | rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
437 |             <button type="button" class="btn btn-danger btn-sm rou
438 | nded-pill px-3 d-flex align-items-center gap-1" id="btn-cleanu
439 | p-all-orphaned">
440 |                 <i class="bi bi-trash3-fill"></i> Удалить все мертвы
441 | е директории
442 |             </button>
443 |         </div>
444 |     </div>
445 | </div>
```

main.js

```
1  /**
2   * =====
3   *
4   * =====
5   * Process Name: Windows Users Tab - Main Script
6   * =====
7   *
8   * Description:
9   *   Клиентский скрипт управления интерфейсом модуля main.
10  *
11  * Usage Examples:
12  *   HTML Integration:
13  *     <script src="/windows/api/webgui/users_tab/main.js?v=20
14  *     261001_v1" type="module"></script>
15  *
16  * File: main.js
17  * Project: ai-breadboard
18  * Package: windows/api/webgui/users_tab
19  * Author: hypo69
20  * Copyright: © 2026 hypo69
21  * Updated: 2026-10-01 13:04:40
22  * =====
23  */
24
25 // =====
26 // Webinterface: Users Tab Logic
27 // Module: webinterface/users_tab/main.js
28 // Author: hypo69
29 // Copyright: © 2026 hypo69
30 // =====
31
32 'use strict';
33
34 (function () {
```

```

31 | const state = {
32 |     users: [],
33 |     stats: {
34 |         total: 0,
35 |         active: 0,
36 |         admins: 0,
37 |         telegram: 0
38 |     },
39 |     filterRole: '',
40 |     filterStatus: '',
41 |     searchQuery: '',
42 |     editingUserId: null,
43 |     savingUserId: null,
44 |     isLoading: false,
45 |     initialized: false
46 | };
47 |
48 | // Helper API fetch
49 | async function apiFetch(url, options = {}) {
50 |     if (window.api && typeof window.api.fetch === 'function')
51 |     {
52 |         return window.api.fetch(url, options);
53 |     }
54 |     const res = await fetch(url, options);
55 |     if (!res.ok) {
56 |         let errMsg = `HTTP ${res.status}`;
57 |         try {
58 |             const errData = await res.json();
59 |             if (errData && errData.detail) {
60 |                 errMsg = typeof errData.detail === 'string' ? errData
61 |                 .detail : JSON.stringify(errData.detail);
62 |             } catch (_) {}
63 |             throw new Error(errMsg);
64 |         }
65 |         return res.json();
66 |     }

```

```

67 | // Show notification
68 | function showStatusAlert(msg, type = 'info') {
69 |     const alertBox = document.getElementById('users-alert-box'
    | );
70 |     const alertMsg = document.getElementById('users-alert-mess
    | age');
71 |     if (!alertBox || !alertMsg) return;
72 |
73 |     alertBox.className = `alert alert-${type} alert-dismissible
    | e fade show mb-3`;
74 |     alertMsg.innerHTML = msg;
75 |     alertBox.classList.remove('d-none');
76 |
77 |     setTimeout(() => {
78 |         alertBox.classList.add('d-none');
79 |     }, 5000);
80 | }
81 |
82 | // Generate secure password
83 | function generatePassword(length = 12) {
84 |     const charset = 'abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ
    | STUVWXYZ23456789!@#%&*';
85 |     let pwd = '';
86 |     const randomValues = new Uint32Array(length);
87 |     window.crypto.getRandomValues(randomValues);
88 |     for (let i = 0; i < length; i++) {
89 |         pwd += charset[randomValues[i] % charset.length];
90 |     }
91 |     return pwd;
92 | }
93 |
94 | // Escape HTML
95 | function escapeHtml(str) {
96 |     if (!str) return '';
97 |     return String(str)
98 |         .replace(/&/g, '&amp;')
99 |         .replace(/</g, '&lt;')
100 |         .replace(/>/g, '&gt;')

```

```

101     .replace(/"/g, '&quot;');
102     .replace(/'/g, '&#039;');
103 }
104
105 // Format datetime
106 function formatDate(isoStr) {
107     if (!isoStr) return '<span class="text-muted">–</span>';
108     try {
109         const d = new Date(isoStr);
110         if (isNaN(d.getTime())) return escapeHtml(isoStr);
111         return `<span title="${escapeHtml(isoStr)}">${d.toLocale
DateString('ru-RU')} <small class="text-muted">${d.toLocaleTim
eString('ru-RU', { hour: '2-digit', minute: '2-digit' })}</sma
ll></span>`;
112     } catch (_) {
113         return escapeHtml(isoStr);
114     }
115 }
116
117 // Load Users List from Backend
118 async function loadUsers() {
119     state.isLoading = true;
120     const tableBody = document.getElementById('users-table-bod
y');
121     const countBadge = document.getElementById('users-count-ba
dge');
122
123     if (tableBody && (!state.users || state.users.length === 0
)) {
124         tableBody.innerHTML = `
125             <tr>
126                 <td colspan="8" class="text-center py-4 text-muted">
127                     <div class="spinner-border spinner-border-sm text-
primary me-2" role="status"></div>
128                     Загрузка списка пользователей...
129                 </td>
130             </tr>`;
131     }

```



```
132 |
133 |     try {
134 |         const params = new URLSearchParams();
135 |         if (state.searchQuery) params.append('q', state.searchQu
136 |         if (state.filterRole) params.append('role', state.filter
137 |         if (state.filterStatus) params.append('status', state.fi
138 |
139 |         const url = `/api/admin/users?${params.toString()}`;
140 |         const data = await apiFetch(url);
141 |
142 |         state.users = data.users || [];
143 |         if (data.stats) {
144 |             state.stats = data.stats;
145 |         }
146 |
147 |         renderStats();
148 |         renderTable();
149 |     } catch (err) {
150 |         console.error('[UsersTab] Error loading users:', err);
151 |         showStatusAlert(`Ошибка загрузки пользователей: ${err.me
152 |         if (tableBody) {
153 |             tableBody.innerHTML = `
154 |                 <tr>
155 |                     <td colspan="8" class="text-center py-4 text-dange
156 |                         <i class="bi bi-exclamation-octagon fs-4 d-block
157 |                         Не удалось загрузить пользователей: ${escapeHtml
158 |                     </td>
159 |                 </tr>`;
160 |             }
161 |         } finally {
162 |             state.isLoading = false;
```

```
163     }
164   }
165
166   // Render Metric Cards
167   function renderStats() {
168     const totalEl = document.getElementById('stat-total-users'
169   );
170     const activeEl = document.getElementById('stat-active-user
171   s');
172     const adminEl = document.getElementById('stat-admin-users'
173   );
174     const tgEl = document.getElementById('stat-tg-users');
175
176     if (totalEl) totalEl.textContent = state.stats.total || 0;
177     if (activeEl) activeEl.textContent = state.stats.active ||
178   0;
179     if (adminEl) adminEl.textContent = state.stats.admins || 0
180   ;
181     if (tgEl) tgEl.textContent = state.stats.telegram || 0;
182   }
183
184   // Render Users Table
185   function renderTable() {
186     const tableBody = document.getElementById('users-table-bod
187   y');
188     const countBadge = document.getElementById('users-count-ba
189   dge');
190     if (!tableBody) return;
191
192     if (countBadge) {
193       countBadge.textContent = `Показано: ${state.users.length
194   } из ${state.stats.total || state.users.length}`;
195     }
196
197     if (state.users.length === 0) {
198       tableBody.innerHTML = `
199       <tr>
200         <td colspan="8" class="text-center py-5 text-muted">
```

```

193 |         <i class="bi bi-person-x fs-2 d-block mb-2 text-se
condary"></i>
194 |         Пользователи не найдены.
195 |     </td>
196 | </tr>`;
197 |     return;
198 | }
199 |
200 | tableBody.innerHTML = state.users.map((user) => {
201 |     const isRoot = user.id === 1;
202 |     const isAdmin = Boolean(user.is_admin || user.role === '
admin');
203 |     const isActive = Boolean(user.is_active);
204 |     const isEmailVerified = Boolean(user.is_email_verified);
205 |     const hasPassword = Boolean(user.has_password);
206 |     const isRowEditing = (state.editingUserId === user.id);
207 |     const isSaving = (state.savingUserId === user.id);
208 |
209 |     // User Initials or Avatar
210 |     let avatarHtml = '';
211 |     if (user.picture) {
212 |         avatarHtml = ``;
213 |     } else {
214 |         const initial = (user.name || user.email || 'U').charA
t(0).toUpperCase();
215 |         const bgClass = isAdmin ? 'bg-warning text-dark' : 'bg
-primary text-white';
216 |         avatarHtml = `<div class="rounded-circle ${bgClass} d-
flex align-items-center justify-content-center me-2 flex-shrin
k-0 fw-bold" style="width:36px;height:36px;font-size:14px;">\${
escapeHtml(initial)}</div>`;
217 |     }
218 |
219 |     // Password column
220 |     const pwdHtml = hasPassword
221 |         ? `<span class="badge bg-success-subtle text-success b

```

```

    order border-success-subtle" title="Пароль установлен"><i clas
s="bi bi-key-fill"></i> Задан</span>`
222      : `<span class="badge bg-warning-subtle text-warning b
order border-warning-subtle" title="Пароль не установлен (OAut
h/TG)"><i class="bi bi-dash-circle"></i> Нет</span>`;
223
224      // Email verified badge
225      const emailVerifiedBadge = isEmailVerified
226      ? `<i class="bi bi-patch-check-fill text-success ms-1"
title="Email подтвержден"></i>`
227      : `<i class="bi bi-question-circle text-muted ms-1" ti
tle="Email не подтвержден"></i>`;
228
229      // If this row is in inline-edit mode:
230      if (isRowEditing) {
231          return `
232              <tr data-user-id="${user.id}" class="table-active bo
rder-primary">
233                  <td class="text-center text-muted fw-bold align-mi
ddle">${user.id}</td>
234                  <td class="align-middle">
235                      <div class="d-flex align-items-center">
236                          ${avatarHtml}
237                          <div class="w-100">
238                              <div class="input-group input-group-sm mb-1"
>
239                                  <span class="input-group-text text-muted">
<i class="bi bi-person"></i></span>
240                                  <input type="text" class="form-control inl
ine-edit-name" value="${escapeHtml(user.name || '')}" placehol
der="Имя пользователя">
241                                  </div>
242                                  <div class="input-group input-group-sm">
243                                      <span class="input-group-text text-muted">
<i class="bi bi-envelope"></i></span>
244                                      <input type="email" class="form-control in
line-edit-email" value="${escapeHtml(user.email || '')}" place
holder="Email">

```

```

245 |         </div>
246 |     </div>
247 | </div>
248 | </td>
249 | <td class="text-center align-middle">
250 |     <select class="form-select form-select-sm inline
-edit-role" ${isRoot ? 'disabled' : ''}>
251 |         <option value="user" ${user.role === 'user' ?
'selected' : ''}>□ User</option>
252 |         <option value="admin" ${user.role === 'admin'
? 'selected' : ''}>□ Admin</option>
253 |         <option value="guest" ${user.role === 'guest'
? 'selected' : ''}>□ Guest</option>
254 |     </select>
255 | </td>
256 | <td class="text-center align-middle">
257 |     <select class="form-select form-select-sm inline
-edit-status" ${isRoot ? 'disabled' : ''}>
258 |         <option value="1" ${isActive ? 'selected' : ''
}>□ Активен</option>
259 |         <option value="0" ${!isActive ? 'selected' : ''
}>□ Блок</option>
260 |     </select>
261 | </td>
262 | <td class="align-middle">
263 |     <div class="input-group input-group-sm">
264 |         <span class="input-group-text text-info"><i cl
ass="bi bi-telegram"></i></span>
265 |         <input type="text" class="form-control inline-
edit-tg" value="${escapeHtml(user.telegram_username || '')}" p
laceholder="Username">
266 |     </div>
267 | </td>
268 | <td class="text-center align-middle">${pwdHtml}</t
d>
269 | <td class="small align-middle">
270 |     <div><span class="text-muted">Создан:</span> ${f
ormatDate(user.created_at)}</div>

```

```

271 |         </td>
272 |         <td class="text-center align-middle">
273 |             <div class="d-flex justify-content-center gap-1"
    | >
274 |                 <button class="btn btn-success btn-sm btn-save
    | -inline" data-id="{user.id}" title="Сохранить изменения (Ente
    | r)" {isSaving ? 'disabled' : ''}>
275 |                     {isSaving ? '<span class="spinner-border sp
    | inner-border-sm"></span>' : '<i class="bi bi-check-lg"></i>'}
276 |                 </button>
277 |                 <button class="btn btn-outline-secondary btn-s
    | m btn-cancel-inline" data-id="{user.id}" title="Отмена (Esc)"
    | {isSaving ? 'disabled' : ''}>
278 |                     <i class="bi bi-x-lg"></i>
279 |                 </button>
280 |             </div>
281 |         </td>
282 |     </tr>
283 |     `;
284 | }
285 |
286 | // Normal view mode with DIRECT editable table fields:
287 | const roleSelectHtml = `
288 |     <select class="form-select form-select-sm bg-dark bord
    | er-secondary user-table-role-select text-center {isAdmin ? 't
    | ext-warning fw-bold' : 'text-info'}"
289 |         data-user-id="{user.id}"
290 |         {isRoot ? 'disabled title="Нельзя изменить роль Roo
    | t"' : 'title="Изменить роль прямо в таблице"}>
291 |         <option value="user" {user.role === 'user' ? 'selec
    | ted' : ''} class="text-info">☐ User</option>
292 |         <option value="admin" {isAdmin ? 'selected' : ''} c
    | lass="text-warning">☐ Admin</option>
293 |         <option value="guest" {user.role === 'guest' ? 'sel
    | ected' : ''} class="text-secondary">☐ Guest</option>
294 |     </select>
295 |     `;
296 |

```

```

297 |         const statusSelectHtml = `
298 |             <select class="form-select form-select-sm bg-dark bord
| er-secondary user-table-status-select text-center ${isActive ?
| 'text-success' : 'text-danger'}"
299 |                 data-user-id="${user.id}"
300 |                 ${isRoot ? 'disabled title="Нельзя деактивировать Ro
| ot"' : 'title="Изменить статус прямо в таблице"'}>
301 |                 <option value="1" ${isActive ? 'selected' : ''} clas
| s="text-success">☑ Активен</option>
302 |                 <option value="0" ${!isActive ? 'selected' : ''} cla
| ss="text-danger">☐ Блок</option>
303 |             </select>
304 |         `;
305 |
306 |         // Telegram column
307 |         let tgHtml = '<span class="text-muted small">—</span>';
308 |         if (user.telegram_username) {
309 |             tgHtml = `<a href="https://t.me/${escapeHtml(user.tele
| gram_username)}" target="_blank" class="text-info text-decorat
| ion-none small d-flex align-items-center gap-1" title="Открыть
| профиль Telegram">
310 |                 <i class="bi bi-telegram"></i> @${escapeHtml(user.te
| legram_username)}
311 |                 </a>`;
312 |         } else if (user.telegram_id) {
313 |             tgHtml = `<span class="text-muted small" title="Telegr
| am ID"><i class="bi bi-telegram text-info"></i> ID: ${escapeHt
| ml(user.telegram_id)}</span>`;
314 |         }
315 |
316 |         return `
317 |             <tr data-user-id="${user.id}">
318 |                 <td class="text-center text-muted fw-bold align-midd
| le">${user.id}</td>
319 |                 <td class="align-middle">
320 |                     <div class="d-flex align-items-center justify-cont
| ent-between">
321 |                         <div class="d-flex align-items-center">

```

```

322 |         ${avatarHtml}
323 |         <div>
324 |             <div class="fw-bold text-white user-cell-name" title="Дважды кликните для быстрого редактирования" style="
| cursor: pointer;">
325 |                 ${escapeHtml(user.name || 'Без имени')}
326 |                 <i class="bi bi-pencil-fill text-muted ms-
| 1 opacity-25 hover-opacity-100" style="font-size:10px;"></i>
327 |             </div>
328 |             <div class="small text-muted d-flex align-items-center user-cell-email" title="Дважды кликните для быстрого
| редактирования" style="cursor: pointer;">
329 |                 ${escapeHtml(user.email)} ${emailVerifiedB
| adge}
330 |             </div>
331 |         </div>
332 |     </div>
333 | </div>
334 | </td>
335 |     <td class="text-center align-middle" style="min-width: 130px;">${roleSelectHtml}</td>
336 |     <td class="text-center align-middle" style="min-width: 125px;">${statusSelectHtml}</td>
337 |     <td class="align-middle" style="min-width: 140px;">
338 |         <div class="d-flex align-items-center justify-content-between">
339 |             <div class="user-cell-tg" title="Дважды кликните
| для быстрого редактирования" style="cursor: pointer;">${tgHtml}</div>
340 |         </div>
341 |     </td>
342 |     <td class="text-center align-middle">${pwdHtml}</td>
343 |     <td class="small align-middle">
344 |         <div><span class="text-muted">Создан:</span> ${formatDate(user.created_at)}</div>
345 |         ${user.last_login ? `<div><span class="text-muted">Вход:</span> ${formatDate(user.last_login)}</div>` : ''}
346 |     </td>

```



```

347         <td class="text-center align-middle">
348             <div class="btn-group btn-group-sm" role="group">
349                 <button class="btn btn-outline-warning btn-inlin
e-edit" data-id="${user.id}" title="Редактировать поля строки
прямо в таблице">
350                     <i class="bi bi-pencil-square"></i>
351                 </button>
352                 <button class="btn btn-outline-info btn-pwd-user
" data-id="${user.id}" data-name="${escapeHtml(user.name || us
er.email)}" title="Сменить пароль">
353                     <i class="bi bi-key-fill"></i>
354                 </button>
355                 <button class="btn btn-outline-light btn-details
-user" data-id="${user.id}" title="Профиль и настройки">
356                     <i class="bi bi-info-circle-fill"></i>
357                 </button>
358                 <button class="btn btn-outline-secondary btn-edi
t-user-modal" data-id="${user.id}" title="Редактировать в мода
льном окне">
359                     <i class="bi bi-sliders"></i>
360                 </button>
361                 <button class="btn btn-outline-danger btn-delete
-user" data-id="${user.id}" data-name="${escapeHtml(user.name
|| user.email)}" ${isRoot ? 'disabled title="Нельзя удалить Ro
ot"' : 'title="Удалить"'}>
362                     <i class="bi bi-trash-fill"></i>
363                 </button>
364             </div>
365         </td>
366     </tr>`;
367     }).join('');
368
369     attachTableEvents();
370 }
371
372 // Attach Table Action Button Events
373 function attachTableEvents() {
374     // Direct Table Role Selector Change

```

```

375 |     document.querySelectorAll('.user-table-role-select').forEa
| ch(sel => {
376 |         sel.addEventListener('change', async (e) => {
377 |             const userId = parseInt(e.target.getAttribute('data-us
| er-id'), 10);
378 |             const newRole = e.target.value;
379 |             const isAdmin = (newRole === 'admin') ? 1 : 0;
380 |             await handleUpdateField(userId, { role: newRole, is_ad
| min: isAdmin }, `Роль изменена на "${newRole}"`);
381 |         });
382 |     });
383 |
384 |     // Direct Table Status Selector Change
385 |     document.querySelectorAll('.user-table-status-select').for
| Each(sel => {
386 |         sel.addEventListener('change', async (e) => {
387 |             const userId = parseInt(e.target.getAttribute('data-us
| er-id'), 10);
388 |             const newStatus = parseInt(e.target.value, 10);
389 |             await handleUpdateField(userId, { is_active: newStatus
| }, `Статус изменен на "${newStatus ? 'Активен' : 'Заблокирова
| н'}"`);
390 |         });
391 |     });
392 |
393 |     // Double-click row or name/email/tg to enter inline-edit
| mode
394 |     document.querySelectorAll('#users-table-body tr').forEach(
| row => {
395 |         const userId = parseInt(row.getAttribute('data-user-id')
| , 10);
396 |         if (!userId) return;
397 |
398 |         const nameEl = row.querySelector('.user-cell-name');
399 |         const emailEl = row.querySelector('.user-cell-email');
400 |         const tgEl = row.querySelector('.user-cell-tg');
401 |
402 |         [nameEl, emailEl, tgEl].forEach(el => {

```

```

403         if (el) {
404             el.addEventListener('dblclick', () => {
405                 state.editingUserId = userId;
406                 renderTable();
407             });
408         }
409     });
410 });
411
412 // Inline Edit Button
413 document.querySelectorAll('.btn-inline-edit').forEach(btn
=> {
414     btn.addEventListener('click', (e) => {
415         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
416         state.editingUserId = userId;
417         renderTable();
418     });
419 });
420
421 // Inline Cancel Button
422 document.querySelectorAll('.btn-cancel-inline').forEach(bt
n => {
423     btn.addEventListener('click', () => {
424         state.editingUserId = null;
425         renderTable();
426     });
427 });
428
429 // Inline Save Button
430 document.querySelectorAll('.btn-save-inline').forEach(btn
=> {
431     btn.addEventListener('click', async (e) => {
432         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
433         await saveInlineRow(userId);
434     });
435 });

```

```

436
437     // Enter & Escape key handling in inline inputs
438     document.querySelectorAll('.inline-edit-name, .inline-edit
    -email, .inline-edit-tg').forEach(inp => {
439         inp.addEventListener('keydown', async (e) => {
440             const tr = inp.closest('tr');
441             const userId = tr ? parseInt(tr.getAttribute('data-use
    r-id'), 10) : null;
442             if (!userId) return;
443
444             if (e.key === 'Enter') {
445                 e.preventDefault();
446                 await saveInlineRow(userId);
447             } else if (e.key === 'Escape') {
448                 e.preventDefault();
449                 state.editingUserId = null;
450                 renderTable();
451             }
452         });
453     });
454
455     // Modal Edit User Button
456     document.querySelectorAll('.btn-edit-user-modal').forEach(
    btn => {
457         btn.addEventListener('click', (e) => {
458             const userId = parseInt(e.currentTarget.getAttribute('
    data-id'), 10);
459             openEditModal(userId);
460         });
461     });
462
463     // Change Password Button
464     document.querySelectorAll('.btn-pwd-user').forEach(btn =>
    {
465         btn.addEventListener('click', (e) => {
466             const userId = parseInt(e.currentTarget.getAttribute('
    data-id'), 10);
467             const userName = e.currentTarget.getAttribute('data-na

```

```

    me');
468         openPasswordModal(userId, userName);
469     });
470 });
471
472 // View Details Button
473 document.querySelectorAll('.btn-details-user').forEach(btn
    => {
474     btn.addEventListener('click', (e) => {
475         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
476         openDetailsModal(userId);
477     });
478 });
479
480 // Delete User Button
481 document.querySelectorAll('.btn-delete-user').forEach(btn
    => {
482     btn.addEventListener('click', (e) => {
483         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
484         const userName = e.currentTarget.getAttribute('data-na
me');
485         openDeleteModal(userId, userName);
486     });
487 });
488 }
489
490 // Save Inline Row Changes
491 async function saveInlineRow(userId) {
492     const row = document.querySelector(`tr[data-user-id="${use
rId}"]`);
493     if (!row) return;
494
495     const nameInput = row.querySelector('.inline-edit-name');
496     const emailInput = row.querySelector('.inline-edit-email')
;
497     const roleSelect = row.querySelector('.inline-edit-role');

```

```
498     const statusSelect = row.querySelector('.inline-edit-status');
499     const tgInput = row.querySelector('.inline-edit-tg');
500
501     const name = nameInput ? nameInput.value.trim() : '';
502     const email = emailInput ? emailInput.value.trim() : '';
503     const role = roleSelect ? roleSelect.value : 'user';
504     const isActive = statusSelect ? parseInt(statusSelect.value, 10) : 1;
505     const tg = tgInput ? tgInput.value.trim().replace(/^@/, '') : '';
506
507     if (!email) {
508         showStatusAlert('Email не может быть пустым', 'warning')
509     };
510     return;
511     if (!name) {
512         showStatusAlert('Имя пользователя не может быть пустым', 'warning');
513     }
514     return;
515
516     const payload = {
517         name: name,
518         email: email,
519         role: role,
520         is_admin: (role === 'admin') ? 1 : 0,
521         is_active: isActive,
522         telegram_username: tg
523     };
524
525     state.savingUserId = userId;
526     const saveBtn = row.querySelector('.btn-save-inline');
527     if (saveBtn) saveBtn.disabled = true;
528
529     try {
530         const res = await apiFetch(`/api/admin/users/${userId}`,
```

```

    {
531         method: 'PATCH',
532         headers: { 'Content-Type': 'application/json' },
533         body: JSON.stringify(payload)
534     });
535
536     state.editingUserId = null;
537     showStatusAlert(`Пользователь #${userId} (<strong>${escapeHtml(name)}</strong>) успешно сохранён`, 'success');
538     loadUsers();
539 } catch (err) {
540     showStatusAlert(`Ошибка сохранения: ${err.message}`, 'danger');
541 } finally {
542     state.savingUserId = null;
543     if (saveBtn) saveBtn.disabled = false;
544 }
545 }
546
547 // Update a Single Field or Subset of Fields
548 async function handleUpdateField(userId, payload, successMsg = 'Данные сохранены') {
549     try {
550         const res = await apiFetch(`/api/admin/users/${userId}`,
551         {
552             method: 'PATCH',
553             headers: { 'Content-Type': 'application/json' },
554             body: JSON.stringify(payload)
555         });
556
557         // Update local state user
558         const user = state.users.find(u => u.id === userId);
559         if (user && res.user) {
560             Object.assign(user, res.user);
561         }
562         showStatusAlert(`Пользователь #${userId}: ${successMsg}`, 'success');
563         loadUsers();

```

```
563     } catch (err) {
564         showStatusAlert(`Ошибка обновления: ${err.message}`, 'da
nger');
565         loadUsers();
566     }
567 }
568
569 // Open Edit Modal
570 function openEditModal(userId) {
571     const user = state.users.find(u => u.id === userId);
572     if (!user) return;
573
574     const isEditAdmin = Boolean(user.is_admin || user.role ===
'admin');
575     document.getElementById('edit-user-id').value = user.id;
576     document.getElementById('edit-user-id-badge').textContent
= user.id;
577     document.getElementById('edit-user-email').value = user.em
ail || '';
578     document.getElementById('edit-user-name').value = user.nam
e || '';
579
580     const editAdminSw = document.getElementById('edit-user-is-
admin');
581     const editRoleLabel = document.getElementById('edit-role-l
abel');
582     if (editAdminSw) {
583         editAdminSw.checked = isEditAdmin;
584         editAdminSw.disabled = (userId === 1);
585         if (editRoleLabel) {
586             editRoleLabel.textContent = isEditAdmin ? 'Администрат
оп (admin)' : 'Пользователь (user)';
587             editRoleLabel.className = isEditAdmin ? 'text-warning'
: 'text-info';
588         }
589     }
590
591     document.getElementById('edit-user-is-active').checked = B
```



```

        boolean(user.is_active);
592     document.getElementById('edit-user-verified').checked = Bo
        olean(user.is_email_verified);
593
594     const modalEl = document.getElementById('modal-edit-user')
        ;
595     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
        ;
596     modal.show();
597 }
598
599 // Open Password Modal
600 function openPasswordModal(userId, userName) {
601     document.getElementById('pwd-user-id').value = userId;
602     document.getElementById('pwd-user-target').textContent = `
        ${userName} (ID #${userId})`;
603     document.getElementById('pwd-input-val').value = '';
604
605     const modalEl = document.getElementById('modal-password-us
        er');
606     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
        ;
607     modal.show();
608 }
609
610 // Open Details Modal
611 async function openDetailsModal(userId) {
612     const modalEl = document.getElementById('modal-details-use
        r');
613     const contentEl = document.getElementById('details-user-co
        ntent');
614     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
        ;
615     modal.show();
616
617     contentEl.innerHTML = `
618         <div class="text-center py-4 text-muted">
619             <div class="spinner-border spinner-border-sm text-prim

```

```

    ary me-2" role="status"></div>
620     Загрузка профиля пользователя...
621     </div>`;
622
623     try {
624         const data = await apiFetch(`/api/admin/users/${userId}`
        );
625         const u = data.user || {};
626         const s = data.settings || {};
627         const perms = data.permissions || [];
628
629         contentEl.innerHTML = `
630             <div class="row g-3">
631                 <div class="col-md-6">
632                     <h6 class="text-primary border-bottom pb-1 mb-2">□
        □ Основные данные</h6>
633                     <table class="table table-sm table-borderless smal
        l mb-0">
634                         <tr><td class="text-muted" style="width:120px;">
        ID:</td><td class="fw-bold">${u.id}</td></tr>
635                         <tr><td class="text-muted">Email:</td><td>${esca
        peHtml(u.email)} ${u.is_email_verified ? '<span class="badge b
        g-success">Verified</span>' : '<span class="badge bg-secondary
        ">Unverified</span>'}</td></tr>
636                         <tr><td class="text-muted">Имя:</td><td>${escape
        Html(u.name || '-')}</td></tr>
637                         <tr><td class="text-muted">Роль:</td><td><span c
        lass="badge bg-info text-dark">${escapeHtml(u.role || 'user')}
        </span> ${u.is_admin ? '<span class="badge bg-warning text-dar
        k">Admin</span>' : ''}</td></tr>
638                         <tr><td class="text-muted">Статус:</td><td>${u.i
        s_active ? '<span class="text-success">Активен</span>' : '<spa
        n class="text-danger">Заблокирован</span>'}</td></tr>
639                         <tr><td class="text-muted">Пароль:</td><td>${u.h
        as_password ? '<span class="text-success">Установлен</span>' :
        '<span class="text-muted">Не задан</span>'}</td></tr>
640                         <tr><td class="text-muted">Создан:</td><td>${esc
        apeHtml(u.created_at || '-')}</td></tr>

```

```

641         <tr><td class="text-muted">Последний вход:</td><
        |td>${escapeHtml(u.last_login || '-')}</td></tr>
642     </table>
643 </div>
644 <div class="col-md-6">
645     <h6 class="text-info border-bottom pb-1 mb-2">Telegram & Настройки</h6>
646     <table class="table table-sm table-borderless small mb-0">
647         <tr><td class="text-muted" style="width:130px;">Telegram ID:</td><td>${u.telegram_id ? escapeHtml(u.telegram_id) : '<span class="text-muted">Не привязан</span>'}</td></tr>
648         <tr><td class="text-muted">TG Username:</td><td>
        |${u.telegram_username ? '@${escapeHtml(u.telegram_username)}' : '<span class="text-muted">—</span>'}</td></tr>
649         <tr><td class="text-muted">Тема UI:</td><td>${escapeHtml(s.theme || 'dark')}</td></tr>
650         <tr><td class="text-muted">Язык:</td><td>${escapeHtml(s.language || 'ru')}</td></tr>
651         <tr><td class="text-muted">TTS голос:</td><td>${escapeHtml(s.tts_voice || 'ru-RU-DmitryNeural')} (${escapeHtml(s.tts_system || 'edge-tts')})</td></tr>
652         <tr><td class="text-muted">Модель чата:</td><td>
        |${escapeHtml(s.model || 'По умолчанию')}</td></tr>
653     </table>
654 </div>
655 <div class="col-12 mt-3">
656     <h6 class="text-warning border-bottom pb-1 mb-2">Разрешения системы</h6>
657     <div class="d-flex flex-wrap gap-1">
658         ${perms.length > 0 ? perms.map(p => '<span class="badge bg-secondary">${escapeHtml(p)}</span>').join('')} : '<span class="text-muted small">Нет явных разрешений</span>'}
659     </div>
660 </div>
661 ${s.system_instruction ? `
662 <div class="col-12 mt-2">
663     <h6 class="border-bottom pb-1 mb-1">Пользователь

```

ьская системная инструкция</h6>

```
664      <pre class="bg-body-tertiary p-2 rounded border sm
all font-monospace" style="max-height:120px;overflow-y:auto;">
${escapeHtml(s.system_instruction)}</pre>
665      </div>` : ''}
666      </div>`;
667    } catch (err) {
668      contentEl.innerHTML = `<div class="text-danger py-3">Оши
бка загрузки данных: ${escapeHtml(err.message)}</div>`;
669    }
670  }
671
672  // Open Delete Modal
673  function openDeleteModal(userId, userName) {
674    document.getElementById('delete-user-id').value = userId;
675    document.getElementById('delete-user-name-target').textCon
tent = `${userName} (ID #${userId})`;
676
677    const modalEl = document.getElementById('modal-delete-user
');
678    const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
;
679    modal.show();
680  }
681
682  // Initialize Event Listeners
683  function initListeners() {
684    // Open Create User Modal
685    const btnOpenCreate = document.getElementById('btn-open-cr
eate-user');
686    if (btnOpenCreate) {
687      btnOpenCreate.addEventListener('click', () => {
688        const form = document.getElementById('form-create-user
');
689        if (form) form.reset();
690        document.getElementById('create-user-is-admin').checke
d = false;
691        const createRoleLabel = document.getElementById('creat
```

```

    e-role-label');
692     if (createRoleLabel) {
693         createRoleLabel.textContent = 'Пользователь (user)';
694         createRoleLabel.className = 'text-info';
695     }
696     document.getElementById('create-user-is-active').checked = true;
697     document.getElementById('create-user-verified').checked = true;
698     const modalEl = document.getElementById('modal-create-user');
699     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
700     modal.show();
701 });
702 }
703
704 // Dynamic Switch Hints
705 const createAdminSw = document.getElementById('create-user-is-admin');
706 const createRoleLabel = document.getElementById('create-role-label');
707 if (createAdminSw && createRoleLabel) {
708     createAdminSw.addEventListener('change', (e) => {
709         createRoleLabel.textContent = e.target.checked ? 'Администратор (admin)' : 'Пользователь (user)';
710         createRoleLabel.className = e.target.checked ? 'text-warning' : 'text-info';
711     });
712 }
713
714 const editAdminSw = document.getElementById('edit-user-is-admin');
715 const editRoleLabel = document.getElementById('edit-role-label');
716 if (editAdminSw && editRoleLabel) {
717     editAdminSw.addEventListener('change', (e) => {
718         editRoleLabel.textContent = e.target.checked ? 'Админи

```

```
    стратор (admin)' : 'Пользователь (user)';
719     editRoleLabel.className = e.target.checked ? 'text-war
ning' : 'text-info';
720   });
721   }
722
723   // Refresh Users Button
724   const btnRefresh = document.getElementById('btn-refresh-us
ers');
725   if (btnRefresh) {
726     btnRefresh.addEventListener('click', () => {
727       loadUsers();
728     });
729   }
730
731   // Search Input with Debounce
732   const searchInput = document.getElementById('users-search-
input');
733   const searchClear = document.getElementById('users-search-
clear');
734   let debounceTimeout = null;
735
736   if (searchInput) {
737     searchInput.addEventListener('input', (e) => {
738       const val = e.target.value;
739       if (searchClear) {
740         searchClear.classList.toggle('d-none', !val);
741       }
742       clearTimeout(debounceTimeout);
743       debounceTimeout = setTimeout(() => {
744         state.searchQuery = val.trim();
745         loadUsers();
746       }, 300);
747     });
748   }
749
750   if (searchClear) {
751     searchClear.addEventListener('click', () => {
```

```
752         if (searchInput) searchInput.value = '';
753         searchClear.classList.add('d-none');
754         state.searchQuery = '';
755         loadUsers();
756     });
757 }
758
759 // Role Filter
760 const filterRole = document.getElementById('users-filter-role');
761 if (filterRole) {
762     filterRole.addEventListener('change', (e) => {
763         state.filterRole = e.target.value;
764         loadUsers();
765     });
766 }
767
768 // Status Filter
769 const filterStatus = document.getElementById('users-filter-status');
770 if (filterStatus) {
771     filterStatus.addEventListener('change', (e) => {
772         state.filterStatus = e.target.value;
773         loadUsers();
774     });
775 }
776
777 // Generator button for create modal
778 const btnGenCreatePwd = document.getElementById('btn-gen-create-password');
779 if (btnGenCreatePwd) {
780     btnGenCreatePwd.addEventListener('click', () => {
781         const pwdInput = document.getElementById('create-user-password');
782         if (pwdInput) pwdInput.value = generatePassword(12);
783     });
784 }
785
```

```
786 // Generator button for password change modal
787 const btnGenPwdVal = document.getElementById('btn-gen-pwd-
val');
788 if (btnGenPwdVal) {
789     btnGenPwdVal.addEventListener('click', () => {
790         const pwdInput = document.getElementById('pwd-input-va
l');
791         if (pwdInput) pwdInput.value = generatePassword(12);
792     });
793 }
794
795 // Create User Form Submit
796 const formCreate = document.getElementById('form-create-us
er');
797 if (formCreate) {
798     formCreate.addEventListener('submit', async (e) => {
799         e.preventDefault();
800         const submitBtn = document.getElementById('btn-submit-
create-user');
801         if (submitBtn) submitBtn.disabled = true;
802
803         const isAdmin = document.getElementById('create-user-i
s-admin').checked ? 1 : 0;
804         const payload = {
805             email: document.getElementById('create-user-email').
value.trim(),
806             name: document.getElementById('create-user-name').va
lue.trim(),
807             password: document.getElementById('create-user-passw
ord').value.trim(),
808             role: isAdmin ? 'admin' : 'user',
809             is_admin: isAdmin,
810             is_active: document.getElementById('create-user-is-a
ctive').checked ? 1 : 0,
811             is_email_verified: document.getElementById('create-u
ser-verified').checked ? 1 : 0
812         };
813
```



```
814 |         try {
815 |             await apiFetch('/api/admin/users', {
816 |                 method: 'POST',
817 |                 headers: { 'Content-Type': 'application/json' },
818 |                 body: JSON.stringify(payload)
819 |             });
820 |             const modalEl = document.getElementById('modal-creat
e-user');
821 |             const modal = bootstrap.Modal.getInstance(modalEl);
822 |             if (modal) modal.hide();
823 |
824 |             showStatusAlert(`Пользователь ${escapeHtml(payload.n
ame)} успешно создан`, 'success');
825 |             loadUsers();
826 |         } catch (err) {
827 |             showStatusAlert(`Ошибка создания пользователя: ${err
.message}`, 'danger');
828 |         } finally {
829 |             if (submitBtn) submitBtn.disabled = false;
830 |         }
831 |     });
832 | }
833 |
834 | // Edit User Form Submit (Modal)
835 | const formEdit = document.getElementById('form-edit-user')
;
836 | if (formEdit) {
837 |     formEdit.addEventListener('submit', async (e) => {
838 |         e.preventDefault();
839 |         const submitBtn = document.getElementById('btn-submit-
edit-user');
840 |         if (submitBtn) submitBtn.disabled = true;
841 |
842 |         const userId = parseInt(document.getElementById('edit-
user-id').value, 10);
843 |         const isAdmin = document.getElementById('edit-user-is-
admin').checked ? 1 : 0;
844 |         const payload = {
```

```
845 |         email: document.getElementById('edit-user-email').va
      | lue.trim(),
846 |         name: document.getElementById('edit-user-name').valu
      | e.trim(),
847 |         role: isAdmin ? 'admin' : 'user',
848 |         is_admin: isAdmin,
849 |         is_active: document.getElementById('edit-user-is-act
      | ive').checked ? 1 : 0,
850 |         is_email_verified: document.getElementById('edit-use
      | r-verified').checked ? 1 : 0
851 |     };
852 |
853 |     try {
854 |         await apiFetch(`/api/admin/users/${userId}`, {
855 |             method: 'PUT',
856 |             headers: { 'Content-Type': 'application/json' },
857 |             body: JSON.stringify(payload)
858 |         });
859 |         const modalEl = document.getElementById('modal-edit-
      | user');
860 |         const modal = bootstrap.Modal.getInstance(modalEl);
861 |         if (modal) modal.hide();
862 |
863 |         showStatusAlert(`Данные пользователя #${userId} успе
      | шно обновлены!`, 'success');
864 |         loadUsers();
865 |     } catch (err) {
866 |         showStatusAlert(`Ошибка обновления: ${err.message}`,
      | 'danger');
867 |     } finally {
868 |         if (submitBtn) submitBtn.disabled = false;
869 |     }
870 | });
871 | }
872 |
873 | // Change Password Form Submit
874 | const formPwd = document.getElementById('form-password-use
      | r');
```

```

875 |     if (formPwd) {
876 |         formPwd.addEventListener('submit', async (e) => {
877 |             e.preventDefault();
878 |             const submitBtn = document.getElementById('btn-submit-
pwd-user');
879 |             if (submitBtn) submitBtn.disabled = true;
880 |
881 |             const userId = parseInt(document.getElementById('pwd-u
ser-id').value, 10);
882 |             const newPassword = document.getElementById('pwd-input
-val').value.trim();
883 |
884 |             try {
885 |                 await apiFetch(`/api/admin/users/${userId}/password`
, {
886 |                     method: 'POST',
887 |                     headers: { 'Content-Type': 'application/json' },
888 |                     body: JSON.stringify({ password: newPassword })
889 |                 });
890 |                 const modalEl = document.getElementById('modal-passw
ord-user');
891 |                 const modal = bootstrap.Modal.getInstance(modalEl);
892 |                 if (modal) modal.hide();
893 |
894 |                 showStatusAlert(`Пароль для пользователя #${userId}
успешно установлен`, 'success');
895 |                 loadUsers();
896 |             } catch (err) {
897 |                 showStatusAlert(`Ошибка установки пароля: ${err.mess
age}`, 'danger');
898 |             } finally {
899 |                 if (submitBtn) submitBtn.disabled = false;
900 |             }
901 |         });
902 |     }
903 |
904 |     // Delete User Confirm Button
905 |     const btnConfirmDelete = document.getElementById('btn-conf

```

```

    irm-delete-user');
906     if (btnConfirmDelete) {
907         btnConfirmDelete.addEventListener('click', async () => {
908             btnConfirmDelete.disabled = true;
909             const userId = parseInt(document.getElementById('delete-user-id').value, 10);
910
911             try {
912                 await apiFetch(`/api/admin/users/${userId}`, { method: 'DELETE' });
913                 const modalEl = document.getElementById('modal-delete-user');
914                 const modal = bootstrap.Modal.getInstance(modalEl);
915                 if (modal) modal.hide();
916
917                 showStatusAlert(`Пользователь #${userId} успешно удалён.`, 'success');
918                 loadUsers();
919             } catch (err) {
920                 showStatusAlert(`Ошибка удаления пользователя: ${err.message}`, 'danger');
921             } finally {
922                 btnConfirmDelete.disabled = false;
923             }
924         });
925     }
926
927     // Orphaned Directories Check Button
928     const btnCheckOrphaned = document.getElementById('btn-check-orphaned-dirs');
929     if (btnCheckOrphaned) {
930         btnCheckOrphaned.addEventListener('click', () => {
931             const modalEl = document.getElementById('modal-orphaned-dirs');
932             if (modalEl) {
933                 const modal = new bootstrap.Modal(modalEl);
934                 modal.show();
935                 loadOrphanedDirs();

```

```
936 |         }
937 |     });
938 | }
939 |
940 | // Orphaned Directories Refresh Button
941 | const btnRefreshOrphaned = document.getElementById('btn-re
fresh-orphaned-dirs');
942 | if (btnRefreshOrphaned) {
943 |     btnRefreshOrphaned.addEventListener('click', () => {
944 |         loadOrphanedDirs();
945 |     });
946 | }
947 |
948 | // Cleanup All Orphaned Directories Button
949 | const btnCleanupAll = document.getElementById('btn-cleanup
-all-orphaned');
950 | if (btnCleanupAll) {
951 |     btnCleanupAll.addEventListener('click', async () => {
952 |         if (!confirm('Вы уверены, что хотите удалить ВСЕ обнар
уженные мертвые директории пользователей? Это действие необрат
имо.')) {
953 |             return;
954 |         }
955 |         btnCleanupAll.disabled = true;
956 |         try {
957 |             const res = await apiFetch('/api/admin/users/orphane
d-dirs/clean', {
958 |                 method: 'POST',
959 |                 headers: { 'Content-Type': 'application/json' },
960 |                 body: JSON.stringify({})
961 |             });
962 |             showStatusAlert(`Успешно удалено мертвых директорий:
${res.total_deleted} (освобождено: ${formatBytes(res.freed_by
tes)})`, 'success');
963 |             loadOrphanedDirs();
964 |         } catch (err) {
965 |             showStatusAlert(`Ошибка очистки мертвых директорий:
${err.message}`, 'danger');
```

```

966         } finally {
967             btnCleanupAll.disabled = false;
968         }
969     });
970 }
971 }
972
973 // Format bytes helper
974 function formatBytes(bytes, decimals = 1) {
975     if (!bytes || bytes <= 0) return '0 B';
976     const k = 1024;
977     const dm = decimals < 0 ? 0 : decimals;
978     const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
979     const i = Math.floor(Math.log(bytes) / Math.log(k));
980     return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
    ' ' + sizes[i];
981 }
982
983 // Load Orphaned Directories from Backend
984 async function loadOrphanedDirs() {
985     const tbody = document.getElementById('orphaned-dirs-tbody
    ');
986     const countEl = document.getElementById('orphaned-dirs-cou
    nt');
987     const sizeEl = document.getElementById('orphaned-dirs-size
    ');
988     const filesEl = document.getElementById('orphaned-dirs-fil
    es');
989     const btnCleanupAll = document.getElementById('btn-cleanup
    -all-orphaned');
990
991     if (tbody) {
992         tbody.innerHTML = `
993             <tr>
994                 <td colspan="6" class="text-center py-4 text-muted">
995                     <div class="spinner-border spinner-border-sm text-
    warning me-2" role="status"></div>
996                     Сканирование хранилища data/users/...

```

```

997         </td>
998     </tr>`;
999     }
1000
1001     try {
1002         const data = await apiFetch('/api/admin/users/orphaned-d
1003         irects');
1004
1005         const dirs = data.orphaned_dirs || [];
1006
1007         if (countEl) countEl.textContent = data.total || 0;
1008         if (sizeEl) sizeEl.textContent = formatBytes(data.total_
1009         size_bytes || 0);
1010         if (filesEl) filesEl.textContent = data.total_files || 0
1011         ;
1012
1013         if (btnCleanupAll) {
1014             btnCleanupAll.disabled = (dirs.length === 0);
1015         }
1016
1017         if (!tbody) return;
1018
1019         if (dirs.length === 0) {
1020             tbody.innerHTML = `
1021                 <tr>
1022                     <td colspan="6" class="text-center py-4 text-succe
1023                     ss">
1024                         <i class="bi bi-check-circle-fill fs-4 d-block m
1025                         b-1"></i>
1026                             Мертвых директорий не обнаружено. Все папки соот
1027                             ветствуют активным пользователям.
1028                 </td>
1029             </tr>`;
1030             return;
1031         }
1032
1033         tbody.innerHTML = dirs.map(d => {
1034             const badgeClass = d.is_test_dir ? 'bg-secondary' : 'b
1035             g-warning text-dark';

```

```

1028         return `
1029             <tr data-dir-name="${escapeHtml(d.name)}">
1030                 <td class="font-monospace fw-bold text-light">
1031                     <i class="bi bi-folder text-warning me-1"></i>${
escapeHtml(d.name)}
1032                 </td>
1033                 <td>
1034                     <span class="badge ${badgeClass} small">${escape
Html(d.reason)}</span>
1035                 </td>
1036                 <td class="text-center">${d.files_count}</td>
1037                 <td class="text-center">${formatBytes(d.size_bytes
)}}</td>
1038                 <td class="small">${formatDate(d.modified_at)}</td>
1039                 <td class="text-center">
1040                     <button class="btn btn-outline-danger btn-sm py-
0 px-2 btn-delete-single-orphaned" data-dir="${escapeHtml(d.na
me)}" title="Удалить директорию">
1041                         <i class="bi bi-trash"></i>
1042                     </button>
1043                 </td>
1044             </tr>`;
1045         })).join('');
1046
1047         // Add click listeners to single delete buttons
1048         tbody.querySelectorAll('.btn-delete-single-orphaned').fo
rEach(btn => {
1049             btn.addEventListener('click', async () => {
1050                 const dirName = btn.getAttribute('data-dir');
1051                 if (!dirName) return;
1052                 if (!confirm(`Удалить мертвую директорию "${dirName}
"?`)) return;
1053
1054                 btn.disabled = true;
1055                 try {
1056                     const res = await apiFetch('/api/admin/users/orpha
ned-dirs/clean', {

```



```

1057         method: 'POST',
1058         headers: { 'Content-Type': 'application/json' },
1059         body: JSON.stringify({ dirs: [dirName] })
1060     });
1061     showStatusAlert(`Директория "${dirName}" успешно у
далена.` , 'success');
1062     loadOrphanedDirs();
1063     } catch (err) {
1064         showStatusAlert(`Ошибка удаления директории ${dirN
ame}: ${err.message}`, 'danger');
1065         btn.disabled = false;
1066     }
1067     });
1068     });
1069     } catch (err) {
1070         console.error('[UsersTab] Error loading orphaned dirs:',
err);
1071         if (tbody) {
1072             tbody.innerHTML = `
1073                 <tr>
1074                     <td colspan="6" class="text-center py-4 text-dange
r">
1075                         <i class="bi bi-exclamation-octagon fs-4 d-block
mb-1"></i>
1076                         Ошибка сканирования: ${escapeHtml(err.message)}
1077                     </td>
1078                 </tr>`;
1079         }
1080     }
1081 }
1082
1083 // Global Init Function
1084 function initUsersTab() {
1085     console.log('[UsersTab] Initializing user management tab w
ith inline editing...');
1086     if (!state.initialized) {
1087         initListeners();
1088         state.initialized = true;

```

```
1089 |     }
1090 |     loadUsers();
1091 | }
1092 |
1093 | // Register globally
1094 | window.initUsersTab = initUsersTab;
1095 | ))();
1096 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/user_assistant_tab/index.html
15
16   Tab Frame Integration:
17     <iframe src="/windows/api/webgui/user_assistant_tab/index.html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/user_assistant_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-08 03:08:00
25 =====
26 =====
27 -->
28
29 <!-- User Assistant Application Tab -->
30 <style>
31   .assistant-container {
32     padding: 1rem;
33   }
34   .assistant-card {
35     background: #0f172a !important;
36     border: 1px solid #334155 !important;
37     border-radius: 10px;
```

```
33 | padding: 1rem;
34 | display: flex;
35 | flex-direction: column;
36 | gap: 0.25rem;
37 | }
38 | .assistant-label {
39 |   font-size: 0.75rem;
40 |   text-transform: uppercase;
41 |   letter-spacing: 0.05em;
42 |   color: #94a3b8;
43 | }
44 | .assistant-value {
45 |   font-size: 1.35rem;
46 |   font-weight: 700;
47 |   color: #f8fafc;
48 |   font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
  | Consolas, monospace;
49 | }
50 | .assistant-panel {
51 |   background: #0f172a !important;
52 |   border: 1px solid #334155 !important;
53 |   border-radius: 10px;
54 |   overflow: hidden;
55 | }
56 | .assistant-panel-header {
57 |   background: #1e293b !important;
58 |   border-bottom: 1px solid #334155 !important;
59 |   padding: 0.75rem 1rem;
60 |   font-weight: 600;
61 |   font-size: 0.88rem;
62 |   color: #38bdf8;
63 |   display: flex;
64 |   align-items: center;
65 |   justify-content: space-between;
66 | }
67 | .assistant-list-item {
68 |   padding: 0.75rem 1rem;
69 |   border-bottom: 1px solid rgba(51, 65, 85, 0.4);
```

```

70     transition: background 0.15s ease;
71 }
72 .assistant-list-item:hover {
73     background: rgba(30, 41, 59, 0.5);
74 }
75 </style>
76
77 <div class="assistant-container">
78     <!-- Header Alert & Status -->
79     <div class="card shadow-sm border-secondary-subtle p-3 mb-3
80         d-flex justify-content-between align-items-center flex-wrap ga
81         p-3">
82         <div>
83             <h5 class="mb-1 fw-bold d-flex align-items-center gap-2"
84             >
85                 <span class="fs-4">☑ </span> User Assistant (Agenda, M
86                 ail & Calendar)
87             </h5>
88             <p class="mb-0 small text-muted" >
89                 Персональный ассистент продуктивности: расписание дня,
90                 сортировка входящих писем, события календаря и поиск документ
91                 ов.
92             </p>
93             </div>
94             <div class="d-flex align-items-center gap-2">
95                 <span class="badge rounded-pill bg-success-subtle text-s
96                 uccess border border-success px-3 py-2" id="assistant-status-b
97                 adge">
98                     • Готов к работе
99                 </span>
100                 <button class="btn btn-sm btn-outline-info rounded-pill
101                 px-3" id="btn-assistant-refresh">
102                     <i class="bi bi-arrow-clockwise me-1"></i> Обновить
103                 </button>
104                 <button class="btn btn-sm btn-outline-warning rounded-pi
105                 ll px-3" id="btn-assistant-config">
106                     <i class="bi bi-gear-fill me-1"></i> Настройки (config
107                     .json)

```

```
97 |         </button>
98 |     </div>
99 | </div>
100 |
101 | <!-- Top Metrics Bar -->
102 | <div class="row g-3 mb-3">
103 |     <div class="col-md-3 col-sm-6">
104 |         <div class="assistant-card">
105 |             <span class="assistant-label">Событий на сегодня</span
106 |             <span class="assistant-value text-info" id="assistant-
107 |             events-count">0</span>
108 |         </div>
109 |     </div>
110 |     <div class="col-md-3 col-sm-6">
111 |         <div class="assistant-card">
112 |             <span class="assistant-label">Непрочитанных писем</spa
113 |             n>
114 |             <span class="assistant-value text-warning" id="assista
115 |             nt-unread-count">0</span>
116 |         </div>
117 |     </div>
118 |     <div class="col-md-3 col-sm-6">
119 |         <div class="assistant-card">
120 |             <span class="assistant-label">Недавних файлов</span>
121 |             <span class="assistant-value text-success" id="assista
122 |             nt-files-count">0</span>
123 |         </div>
124 |     </div>
125 |     <div class="col-md-3 col-sm-6">
126 |         <div class="assistant-card">
127 |             <span class="assistant-label">Рабочее пространство</sp
128 |             an>
129 |             <span class="assistant-value text-light" style="font-s
130 |             ize: 1.05rem;" id="assistant-workspace-status">Активно</span>
131 |         </div>
132 |     </div>
133 | </div>
```

```
128
129 <!-- Main Content Panels Grid -->
130 <div class="row g-3">
131   <!-- Calendar & Agenda Panel -->
132   <div class="col-lg-6">
133     <div class="assistant-panel h-100">
134       <div class="assistant-panel-header">
135         <span>📅 Повестка дня и ближайшие события</span>
136         <span class="badge bg-secondary" id="assistant-event
s-badge">0 событий</span>
137       </div>
138       <div class="p-0" id="assistant-events-list" style="max
-height: 450px; overflow-y: auto;">
139         <div class="p-3 text-muted text-center small">Загруз
ка повестки дня...</div>
140       </div>
141     </div>
142   </div>
143
144   <!-- Email Triage Panel -->
145   <div class="col-lg-6">
146     <div class="assistant-panel h-100">
147       <div class="assistant-panel-header">
148         <span>✉ Непрочитанные сообщения</span>
149         <span class="badge bg-secondary" id="assistant-mail-
badge">0 писем</span>
150       </div>
151       <div class="p-0" id="assistant-mail-list" style="max-h
eight: 450px; overflow-y: auto;">
152         <div class="p-3 text-muted text-center small">Загруз
ка почты...</div>
153       </div>
154     </div>
155   </div>
156 </div>
157 </div>
158
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows User Assistant Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/user_assistant_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/user_assistant_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 // =====
22 // Process Name: User Assistant Web Tab Module
23 // =====
24 // Description:
25 //   Client-side JavaScript controller for the User Assistant
26 //   administrative web interface tab.
27 //
28 // File: main.js
29 // Project: ai-breadboard
30 // Package: src.api.webinterface.user_assistant_tab
```



```

31 // Author: hypo69
32 // Copyright: © 2026 hypo69
33 // =====
34 |=====
35 (function() {
36     let isAssistantInitialized = false;
37
38     async function fetchAgenda() {
39         try {
40             const res = await fetch('/api/v1/assistant/agenda');
41             if (!res.ok) return;
42             const data = await res.json();
43
44             const eventsCountEl = document.getElementById('assistant
45 |-events-count');
46             const unreadCountEl = document.getElementById('assistant
47 |unread-count');
48             const filesCountEl = document.getElementById('assistant-
49 |files-count');
50             const eventsBadge = document.getElementById('assistant-e
51 |vents-badge');
52             const mailBadge = document.getElementById('assistant-mai
53 |l-badge');
54             const eventsList = document.getElementById('assistant-ev
55 |ents-list');
56             const mailList = document.getElementById('assistant-mail
57 |-list');
58
59             const events = data.events || [];
60             const emails = data.unread_emails || [];
61             const files = data.recent_files || [];
62
63             if (eventsCountEl) eventsCountEl.innerText = events.leng
64 |th;
65             if (unreadCountEl) unreadCountEl.innerText = emails.leng
66 |th;
67             if (filesCountEl) filesCountEl.innerText = files.length;

```

```

59 |         if (eventsBadge) eventsBadge.innerText = `${events.length
    | h} событий`;
60 |         if (mailBadge) mailBadge.innerText = `${emails.length} н
    | исем`;
61 |
62 |         if (eventsList) {
63 |             if (events.length === 0) {
64 |                 eventsList.innerHTML = '<div class="p-3 text-muted t
    | ext-center small">Нет запланированных событий на сегодня.</div
    | >';
65 |             } else {
66 |                 eventsList.innerHTML = events.map(e => `
67 |                     <div class="assistant-list-item d-flex justify-con
    | tent-between align-items-center">
68 |                         <div>
69 |                             <div class="fw-semibold text-light">${e.summar
    | y || 'Без названия'}</div>
70 |                             <div class="text-secondary small">📅 ${e.start
    | || 'Весь день'} ${e.location ? '📍 ' + e.location : ''}</div>
71 |                         </div>
72 |                     </div>
73 |                 `).join('');
74 |             }
75 |         }
76 |
77 |         if (maillist) {
78 |             if (emails.length === 0) {
79 |                 maillist.innerHTML = '<div class="p-3 text-muted tex
    | t-center small">Нет новых непрочитанных сообщений.</div>';
80 |             } else {
81 |                 maillist.innerHTML = emails.map(m => `
82 |                     <div class="assistant-list-item d-flex justify-con
    | tent-between align-items-center">
83 |                         <div>
84 |                             <div class="fw-semibold text-light">${m.subjec
    | t || 'Без темы'}</div>
85 |                             <div class="text-secondary small">📧 ${m.from
    | || 'Неизвестный'} &bull; ${m.snippet || ''}</div>

```

```

86 |         </div>
87 |     </div>
88 |     `).join('');
89 |     }
90 | }
91 | } catch (e) {
92 |     console.error('[UserAssistantTab] Failed to fetch agenda
    | :', e);
93 | }
94 | }
95 |
96 | function initUserAssistantTab() {
97 |     fetchAgenda();
98 |     if (!isAssistantInitialized) {
99 |         const refreshBtn = document.getElementById('btn-assistan
    | t-refresh');
100 |         if (refreshBtn) refreshBtn.addEventListener('click', fet
    | chAgenda);
101 |
102 |         const configBtn = document.getElementById('btn-assistant
    | -config');
103 |         if (configBtn) {
104 |             configBtn.addEventListener('click', () => {
105 |                 if (typeof window.openAppConfigModal === 'function')
    | {
106 |                     window.openAppConfigModal('user_assistant');
107 |                 }
108 |             });
109 |         }
110 |         isAssistantInitialized = true;
111 |     }
112 | }
113 |
114 | window.initUserAssistantTab = initUserAssistantTab;
115 | })();
116 |

```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/user_directories_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/user_directories_tab/index.html" class="w-100 h-100 border-0"></iframe>
15 |
16 File: index.html
17 Project: ai-breadboard
18 Package: windows/api/webgui/user_directories_tab
19 Author: hypo69
20 Copyright: © 2026 hypo69
21 Updated: 2026-10-01 13:04:40
22 |=====
23 |=====
24 -->
25 <!-- User Directories and Workspace Storage Tab -->
26 <div class="container-fluid py-2" id="user-directories-tab-root">
27 |
28 |   <!-- Top Banner & Quick Controls -->
29 |   <div class="d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3 p-3 bg-body-tertiary border rounded shadow-sm">
30 |     <div>
31 |       <h5 class="mb-1 text-primary d-flex align-items-center g
```

```

ap-2">
30     <i class="bi bi-folder2-open text-primary"></i>
31     <span>Пользовательские директории и хранилище</span>
32 </h5>
33     <p class="mb-0 text-muted small">Мониторинг файловых хра
нилиц пользователей (<code>data/users/&lt;user_id&gt;/</code>)
, анализ дисковых квот, аудит деревьев файлов и очистка осирот
евших каталогов.</p>
34 </div>
35 <div class="d-flex flex-wrap gap-2 align-items-center">
36     <button class="btn btn-outline-warning btn-sm d-flex ali
gn-items-center gap-1.5 rounded-pill px-3" id="btn-open-orphan
ed-dirs-modal" title="Проверить и удалить неиспользуемые папки
">
37         <i class="bi bi-folder-x"></i> Мертвые директории (<sp
an id="badge-orphaned-count">0</span>)
38     </button>
39     <button class="btn btn-primary btn-sm d-flex align-items
-center gap-1.5 rounded-pill px-3" id="btn-refresh-user-dirs"
title="Обновить данные хранилища">
40         <i class="bi bi-arrow-clockwise"></i> Обновить
41     </button>
42 </div>
43 </div>
44
45 <!-- Metric Cards -->
46 <div class="row g-3 mb-3">
47     <div class="col-6 col-md-3">
48         <div class="card bg-body-tertiary border h-100 shadow-sm
">
49             <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
50                 <div>
51                     <div class="text-muted small">Пользователей с хран
илищем</div>
52                     <h4 class="mb-0 fw-bold text-primary" id="metric-t
otal-users">0</h4>
53                 </div>

```

```

54 |         <div class="bg-primary bg-opacity-25 rounded p-2 tex
    | t-primary">
55 |             <i class="bi bi-people-fill fs-4"></i>
56 |         </div>
57 |     </div>
58 | </div>
59 | </div>
60 | <div class="col-6 col-md-3">
61 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
62 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
63 |             <div>
64 |                 <div class="text-muted small">Занято на диске</div
    | >
65 |                 <h4 class="mb-0 fw-bold text-info" id="metric-tota
    | l-storage">0 B</h4>
66 |             </div>
67 |             <div class="bg-info bg-opacity-25 rounded p-2 text-i
    | nfo">
68 |                 <i class="bi bi-hdd-fill fs-4"></i>
69 |             </div>
70 |         </div>
71 |     </div>
72 | </div>
73 | <div class="col-6 col-md-3">
74 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
75 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
76 |             <div>
77 |                 <div class="text-muted small">Всего файлов</div>
78 |                 <h4 class="mb-0 fw-bold text-success" id="metric-t
    | otal-files">0</h4>
79 |             </div>
80 |             <div class="bg-success bg-opacity-25 rounded p-2 tex
    | t-success">
81 |                 <i class="bi bi-file-earmark-text-fill fs-4"></i>

```

```

82         </div>
83     </div>
84 </div>
85 </div>
86 <div class="col-6 col-md-3">
87     <div class="card bg-body-tertiary border h-100 shadow-sm
">
88         <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
89             <div>
90                 <div class="text-muted small">Осиротевшие каталоги
</div>
91                 <h4 class="mb-0 fw-bold text-warning" id="metric-o
rphaned-size">0 B</h4>
92             </div>
93             <div class="bg-warning bg-opacity-25 rounded p-2 tex
t-warning">
94                 <i class="bi bi-trash3-fill fs-4"></i>
95             </div>
96         </div>
97     </div>
98 </div>
99 </div>
100
101 <!-- Search & Filter Controls -->
102 <div class="card bg-body-tertiary border mb-3 shadow-sm">
103     <div class="card-body p-2.5">
104         <div class="row g-2 align-items-center">
105             <div class="col-12 col-md-5">
106                 <div class="input-group input-group-sm">
107                     <span class="input-group-text"><i class="bi bi-sea
rch"></i></span>
108                     <input type="text" class="form-control" id="user-d
irs-search-input" placeholder="Поиск по имени пользователя, em
ail или имени файла...">
109                     <button class="btn btn-outline-secondary d-none" t
ype="button" id="user-dirs-search-clear">⌫ </button>
110                 </div>

```

```

111 |         </div>
112 |         <div class="col-6 col-md-3">
113 |             <select class="form-select form-select-sm" id="user-
114 | dirs-ext-filter">
115 |                 <option value="">Все типы файлов</option>
116 |                 <option value="pdf">Документы PDF (.pdf)</option>
117 |                 <option value="txt">Текстовые (.txt)</option>
118 |                 <option value="md">Markdown (.md)</option>
119 |                 <option value="json">Данные JSON (.json)</option>
120 |                 <option value="csv">Таблицы CSV (.csv)</option>
121 |                 <option value="png">Изображения (.png, .jpg)</opti
122 | on>
123 |                 <option value="mp3">Аудио (.mp3, .wav)</option>
124 |             </select>
125 |         </div>
126 |         <div class="col-6 col-md-4 text-end text-muted small"
127 | id="user-dirs-status-label">
128 |             Загрузка...
129 |         </div>
130 |     </div>
131 |     <!-- Alert Message Box -->
132 |     <div id="user-dirs-alert-box" class="d-none alert alert-dism
133 | issible fade show" role="alert">
134 |         <span id="user-dirs-alert-message"></span>
135 |         <button type="button" class="btn-close" aria-label="Close"
136 | onclick="document.getElementById('user-dirs-alert-box').class
137 | List.add('d-none')"></button>
138 |     </div>
139 |     <!-- Two-Panel Split Layout (Users List & Files Explorer) --
140 | >
141 |     <div class="row g-3">
142 |         <!-- Left Column: Users List & Storage Quotas -->
143 |         <div class="col-12 col-lg-4">
144 |             <div class="card border shadow-sm h-100">

```



```

142 |         <div class="card-header bg-body-tertiary d-flex align-
      | items-center justify-content-between py-2 px-3">
143 |             <span class="fw-semibold small"><i class="bi bi-peop
      | le me-1 text-primary"></i> Пользователи</span>
144 |             <span class="badge bg-secondary rounded-pill" id="us
      | ers-badge-count">0</span>
145 |         </div>
146 |         <div class="card-body p-0" style="max-height: 540px; o
      | verflow-y: auto;">
147 |             <div class="list-group list-group-flush" id="user-di
      | rs-users-list">
148 |                 <div class="text-center py-4 text-muted small">
149 |                     <div class="spinner-border spinner-border-sm tex
      | t-primary me-2" role="status"></div>
150 |                     Загрузка каталогов пользователей...
151 |                 </div>
152 |             </div>
153 |         </div>
154 |     </div>
155 | </div>
156 |
157 | <!-- Right Column: Selected User File Tree & Inspector -->
158 | <div class="col-12 col-lg-8">
159 |     <div class="card border shadow-sm h-100">
160 |         <div class="card-header bg-body-tertiary d-flex align-
      | items-center justify-content-between py-2 px-3 flex-wrap gap-2
      | ">
161 |             <div class="d-flex align-items-center gap-2">
162 |                 <span class="fw-semibold text-truncate small" id="
      | explorer-user-title">
163 |                     <i class="bi bi-folder-fill text-warning me-1"><
      | /i> Выберите пользователя слева
164 |                 </span>
165 |                 <span class="badge bg-info-subtle text-info border
      | border-info-subtle d-none" id="explorer-user-path-badge"></sp
      | an>
166 |             </div>
167 |

```

```

168         <!-- Subfolder Filters Buttons -->
169         <div class="btn-group btn-group-sm" role="group" id=
"subfolder-filter-group">
170             <button type="button" class="btn btn-outline-prima
ry active" data-subfolder="">Все папки</button>
171             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="files">📁 files</button>
172             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="rag">📁 rag</button>
173             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="profile">📁 profile</button>
174             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="temp">📁 temp</button>
175         </div>
176     </div>
177
178     <!-- Files Table -->
179     <div class="card-body p-0" style="max-height: 480px; o
verflow-y: auto;">
180         <div class="table-responsive">
181             <table class="table table-hover table-sm align-mid
dle mb-0" style="font-size: 0.84rem;">
182                 <thead class="table-light small text-uppercase s
ticky-top">
183                     <tr>
184                         <th style="width: 35px;"></th>
185                         <th>Имя файла / Путь</th>
186                         <th style="width: 90px;" class="text-center"
>Раздел</th>
187                         <th style="width: 95px;" class="text-center"
>Размер</th>
188                         <th style="width: 140px;">Изменен</th>
189                         <th style="width: 110px;" class="text-center
">Действия</th>
190                     </tr>
191                 </thead>
192                 <tbody id="user-files-table-body">
193                     <tr>

```

```

194         <td colspan="6" class="text-center py-5 text
    -muted">
195             <i class="bi bi-arrow-left-circle fs-3 d-b
    lock mb-2 text-secondary"></i>
196             Выберите пользователя из списка слева для
    просмотра файлов и директорий
197         </td>
198     </tr>
199 </tbody>
200 </table>
201 </div>
202 </div>
203
204 <!-- Explorer Footer Stats -->
205 <div class="card-footer bg-body-tertiary d-flex align-
    items-center justify-content-between py-2 px-3 small text-mute
    d">
206     <div id="explorer-footer-stats">
207         Файлов: <strong>0</strong> | Общий размер: <strong
    >0 B</strong>
208     </div>
209     <div id="explorer-footer-quota">
210         Квота: <strong>-</strong>
211     </div>
212 </div>
213 </div>
214 </div>
215 </div>
216 </div>
217
218 <!-- =====
    ===== -->
219 <!-- MODAL: File Preview -->
220 <!-- =====
    ===== -->
221 <div class="modal fade" id="modal-file-preview" tabindex="-1"
    aria-labelledby="modal-file-preview-title" aria-hidden="true">
222     <div class="modal-dialog modal-dialog-centered modal-lg">

```

```
223     <div class="modal-content shadow-lg border-primary">
224         <div class="modal-header border-primary py-2.5">
225             <h6 class="modal-title d-flex align-items-center gap-2
text-truncate" id="modal-file-preview-title">
226                 <i class="bi bi-file-earmark-text text-primary"></i>
Предпросмотр файла
227             </h6>
228             <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
229         </div>
230         <div class="modal-body p-3">
231             <!-- File Info Bar -->
232             <div class="d-flex align-items-center justify-content-between p-2 mb-2 bg-body-tertiary border rounded small">
233                 <div>
234                     <span class="text-muted">Путь:</span> <code id="preview-file-path"></code>
235                 </div>
236                 <div>
237                     <span class="badge bg-secondary me-1" id="preview-file-size"></span>
238                     <span class="badge bg-dark border" id="preview-file-type"></span>
239                 </div>
240             </div>
241
242             <!-- File Content Container -->
243             <div id="preview-content-wrapper" class="bg-dark text-white rounded p-3 font-monospace" style="max-height: 440px; overflow-y: auto; font-size: 0.82rem; white-space: pre-wrap; word-break: break-all;">
244                 <div class="text-center py-4 text-muted">
245                     <div class="spinner-border spinner-border-sm text-primary me-2"></div> Загрузка содержимого...
246                 </div>
247             </div>
248         </div>
249         <div class="modal-footer justify-content-between py-2">
```

```

250         <button type="button" class="btn btn-outline-danger btn-sm rounded-pill px-3" id="btn-preview-delete-file">
251             <i class="bi bi-trash3 me-1"></i> Удалить файл
252         </button>
253         <div class="d-flex gap-2">
254             <button type="button" class="btn btn-outline-secondary btn-sm rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
255             <a href="#" class="btn btn-primary btn-sm rounded-pill px-3" id="btn-preview-download-file" target="_blank" download>
256                 <i class="bi bi-download me-1"></i> Скачать
257             </a>
258         </div>
259     </div>
260 </div>
261 </div>
262 </div>
263
264 <!-- =====
265 ===== -->
266 <!-- MODAL: Orphaned User Directories Management -->
267 <!-- =====
268 ===== -->
267 <div class="modal fade" id="modal-orphaned-user-dirs" tabindex="-1" aria-labelledby="modal-orphaned-user-dirs-title" aria-hidden="true">
268     <div class="modal-dialog modal-dialog-centered modal-lg">
269         <div class="modal-content shadow-lg border-warning">
270             <div class="modal-header border-warning py-2.5">
271                 <h6 class="modal-title text-warning d-flex align-items-center gap-2" id="modal-orphaned-user-dirs-title">
272                     <i class="bi bi-folder-x"></i> Осиротевшие (мертвые)
273                     директории пользователей
274                 </h6>
275                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
276             </div>

```

```

276 |         <div class="modal-body p-3">
277 |             <div class="alert alert-info border-0 small mb-3">
278 |                 <i class="bi bi-info-circle text-info me-1"></i>
279 |                 Директории в папке <code>data/users/</code>, которые
|                 не принадлежат ни одной активной учетной записи в базе данных
|                 .
280 |             </div>
281 |
282 |             <div class="d-flex align-items-center justify-content-
| between p-2 mb-3 bg-body-tertiary border rounded small">
283 |                 <div>
284 |                     Найдено папок: <strong id="modal-orphaned-count" c
| lass="text-warning">0</strong> |
285 |                     Занято места: <strong id="modal-orphaned-size" cla
| ss="text-info">0 В</strong> |
286 |                     Файлов: <strong id="modal-orphaned-files">0</stron
| g>
287 |                 </div>
288 |                 <button class="btn btn-outline-secondary btn-xs roun
| ded-pill px-2 py-0.5" id="btn-modal-refresh-orphaned">
289 |                     <i class="bi bi-arrow-clockwise"></i> Обновить
290 |                 </button>
291 |             </div>
292 |
293 |             <div class="table-responsive" style="max-height: 360px
| ; overflow-y: auto;">
294 |                 <table class="table table-hover table-sm align-middl
| e mb-0" style="font-size: 0.84rem;">
295 |                     <thead class="table-light small text-uppercase sti
| cky-top">
296 |                         <tr>
297 |                             <th>Каталог / ID</th>
298 |                             <th>Причина</th>
299 |                             <th class="text-center" style="width: 80px;">Ф
| айл</th>
300 |                             <th class="text-center" style="width: 95px;">P
| аметр</th>
301 |                             <th style="width: 140px;">Изменен</th>

```

```

302 |         <th class="text-center" style="width: 85px;">Д
    | действие</th>
303 |     </tr>
304 | </thead>
305 | <tbody id="modal-orphaned-dirs-tbody">
306 |     <tr>
307 |         <td colspan="6" class="text-center py-4 text-m
    | uted">
308 |             <div class="spinner-border spinner-border-sm
    | text-warning me-2"></div> Сканирование...
309 |         </td>
310 |     </tr>
311 | </tbody>
312 | </table>
313 | </div>
314 | </div>
315 | <div class="modal-footer justify-content-between py-2">
316 |     <button type="button" class="btn btn-secondary btn-sm
    | rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
317 |     <button type="button" class="btn btn-danger btn-sm rou
    | nded-pill px-3 d-flex align-items-center gap-1.5" id="btn-moda
    | l-clean-all-orphaned">
318 |         <i class="bi bi-trash3-fill"></i> Удалить все мертвы
    | е директории
319 |     </button>
320 | </div>
321 | </div>
322 | </div>
323 | </div>
324 |

```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows User Directories Tab - Main Script
4   * =====
5   * Description:
6   *   Клиентский скрипт управления интерфейсом модуля main.
7   *
8   * Usage Examples:
9   *   HTML Integration:
10  *     <script src="/windows/api/webgui/user_directories_tab/main.js?v=20261001_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/user_directories_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-01 13:04:40
18  * =====
19  */
20
21 /**
22  * user_directories_tab/main.js – Модуль управления пользовательскими директориями и хранилищем
23  * AI Breadboard
24  */
25
26 let state = {
27   users: [],
28   selectedUserId: null,
29   selectedUserName: '',
30   currentSubfolder: '',
31   searchQuery: '',
```



```
32 |   extFilter: '',
33 |   files: [],
34 |   previewFile: null,
35 |   orphanedDirs: [],
36 | };
37 |
38 | /**
39 |  * Инициализация вкладки пользовательских директорий
40 |  */
41 | export async function initUserDirectoriesTab() {
42 |   const root = document.getElementById('user-directories-tab-root');
43 |   if (!root) return;
44 |
45 |   setupEventListeners();
46 |   await refreshData();
47 | }
48 |
49 | /**
50 |  * Настройка обработчиков событий элементов управления
51 |  */
52 | function setupEventListeners() {
53 |   // Кнопка обновления
54 |   const refreshBtn = document.getElementById('btn-refresh-user-dirs');
55 |   if (refreshBtn) {
56 |     refreshBtn.onclick = () => refreshData();
57 |   }
58 |
59 |   // Поиск с дебаунсом
60 |   const searchInput = document.getElementById('user-dirs-search-input');
61 |   const searchClear = document.getElementById('user-dirs-search-clear');
62 |   if (searchInput) {
63 |     let timeout = null;
64 |     searchInput.oninput = () => {
65 |       clearTimeout(timeout);
```

```

66     state.searchQuery = searchInput.value.trim();
67     if (searchClear) {
68         searchClear.classList.toggle('d-none', !state.searchQu
        ery);
69     }
70     timeout = setTimeout(() => {
71         applyFilters();
72     }, 300);
73 };
74 }
75
76 if (searchClear && searchInput) {
77     searchClear.onclick = () => {
78         searchInput.value = '';
79         state.searchQuery = '';
80         searchClear.classList.add('d-none');
81         applyFilters();
82     };
83 }
84
85 // Фильтр по расширениям
86 const extSelect = document.getElementById('user-dirs-ext-fil
    ter');
87 if (extSelect) {
88     extSelect.onchange = () => {
89         state.extFilter = extSelect.value;
90         if (state.selectedUserId) {
91             loadUserFiles(state.selectedUserId);
92         }
93     };
94 }
95
96 // Фильтры подпапок
97 const subfolderGroup = document.getElementById('subfolder-fi
    lter-group');
98 if (subfolderGroup) {
99     subfolderGroup.addEventListener('click', (e) => {
100         const btn = e.target.closest('button[data-subfolder]');

```

```
101     if (!btn) return;
102     subfolderGroup.querySelectorAll('button').forEach(b => b
    .classList.remove('active'));
103     btn.classList.add('active');
104     state.currentSubfolder = btn.dataset.subfolder || '';
105     if (state.selectedUserId) {
106         loadUserFiles(state.selectedUserId);
107     }
108 });
109 }
110
111 // Кнопки модального окна осиротевших каталогов
112 const openOrphanedBtn = document.getElementById('btn-open-or
    phaned-dirs-modal');
113 if (openOrphanedBtn) {
114     openOrphanedBtn.onclick = () => openOrphanedModal();
115 }
116
117 const refreshOrphanedBtn = document.getElementById('btn-moda
    l-refresh-orphaned');
118 if (refreshOrphanedBtn) {
119     refreshOrphanedBtn.onclick = () => loadOrphanedDirs();
120 }
121
122 const cleanAllOrphanedBtn = document.getElementById('btn-mod
    al-clean-all-orphaned');
123 if (cleanAllOrphanedBtn) {
124     cleanAllOrphanedBtn.onclick = () => cleanOrphanedDirs();
125 }
126
127 // Кнопка удаления файла из предпросмотра
128 const previewDeleteBtn = document.getElementById('btn-previe
    w-delete-file');
129 if (previewDeleteBtn) {
130     previewDeleteBtn.onclick = () => {
131         if (state.previewFile && state.selectedUserId) {
132             deleteFile(state.selectedUserId, state.previewFile.rel
                ative_path);
```

```
133     }
134   };
135 }
136 }
137
138 /**
139  * Полное обновление данных: сводка и список пользователей
140  */
141 async function refreshData() {
142   await Promise.all([
143     loadSummary(),
144     loadUsersList(),
145     loadOrphanedBadge(),
146   ]);
147 }
148
149 /**
150  * Загрузка сводной статистики хранилища
151  */
152 async function loadSummary() {
153   try {
154     const res = await fetch('/api/admin/user-directories/summary');
155     if (!res.ok) return;
156     const data = await res.json();
157     if (data.status === 'ok' && data.summary) {
158       const s = data.summary;
159       const elUsers = document.getElementById('metric-total-users');
160       const elStorage = document.getElementById('metric-total-storage');
161       const elFiles = document.getElementById('metric-total-files');
162       const elOrphaned = document.getElementById('metric-orphaned-size');
163
164       if (elUsers) elUsers.textContent = s.total_users || 0;
165       if (elStorage) elStorage.textContent = s.total_storage_f
```

```

    formatted || '0 B';
166     if (elFiles) elFiles.textContent = s.total_files_count |
    | 0;
167     if (elOrphaned) elOrphaned.textContent = s.orphaned_form
    atted || '0 B';
168     }
169   } catch (err) {
170     console.warn('Ошибка загрузки сводки хранилища:', err);
171   }
172 }
173
174 /**
175  * Загрузка бейджа осиротевших каталогов
176  */
177 async function loadOrphanedBadge() {
178   try {
179     const res = await fetch('/api/admin/user-directories/orpha
    ned');
180     if (!res.ok) return;
181     const data = await res.json();
182     if (data.status === 'ok') {
183       const badge = document.getElementById('badge-orphaned-co
    unt');
184       if (badge) badge.textContent = data.total || 0;
185     }
186   } catch (err) {
187     console.warn('Ошибка проверки осиротевших каталогов:', err
    );
188   }
189 }
190
191 /**
192  * Загрузка списка пользователей с метриками хранилища
193  */
194 async function loadUsersList() {
195   const userListEl = document.getElementById('user-dirs-users
    -list');
196   const countBadge = document.getElementById('users-badge-coun

```

```

    t');
197   const statusLabel = document.getElementById('user-dirs-status-label');
198
199   try {
200     if (statusLabel) statusLabel.textContent = 'Обновление списка пользователей...';
201     const res = await fetch('/api/admin/user-directories/users');
202     if (!res.ok) throw new Error(`HTTP ${res.status}`);
203     const data = await res.json();
204
205     state.users = data.users || [];
206     if (countBadge) countBadge.textContent = state.users.length;
207     if (statusLabel) statusLabel.textContent = `Загружено пользователей: ${state.users.length}`;
208
209     renderUsersList();
210
211     // Если уже был выбран пользователь – перезагрузим его файлы, иначе выберем первого
212     if (state.selectedUserId && state.users.some(u => u.id === state.selectedUserId)) {
213       selectUser(state.selectedUserId);
214     } else if (state.users.length > 0) {
215       selectUser(state.users[0].id);
216     }
217   } catch (err) {
218     console.error('Ошибка загрузки пользователей:', err);
219     if (usersListEl) {
220       usersListEl.innerHTML = `
221         <div class="text-center py-4 text-danger small">
222           <i class="bi bi-exclamation-triangle fs-4 d-block mb-1"></i>
223           Не удалось загрузить список пользователей: ${err.message}
224         </div>

```

```

225 |     `;
226 |     }
227 | }
228 | }
229 |
230 | /**
231 |  * Рендеринг списка пользователей в левой колонке
232 |  */
233 | function renderUsersList() {
234 |     const userListEl = document.getElementById('user-dirs-users
235 |     -list');
236 |     if (!userListEl) return;
237 |
238 |     const query = state.searchQuery.toLowerCase();
239 |     const filteredUsers = state.users.filter(u => {
240 |         if (!query) return true;
241 |         return (
242 |             (u.name && u.name.toLowerCase().includes(query)) ||
243 |             (u.email && u.email.toLowerCase().includes(query)) ||
244 |             String(u.id).includes(query)
245 |         );
246 |     });
247 |
248 |     if (filteredUsers.length === 0) {
249 |         userListEl.innerHTML = `
250 |             <div class="text-center py-4 text-muted small">
251 |                 Пользователи не найдены
252 |             </div>
253 |         `;
254 |         return;
255 |     }
256 |
257 |     userListEl.innerHTML = filteredUsers.map(u => {
258 |         const isSelected = u.id === state.selectedUserId;
259 |         const activeClass = isSelected ? 'active border-primary' :
260 |         '';
261 |         const roleBadge = u.is_admin
262 |         ? '<span class="badge bg-warning text-dark me-1">admin</

```

```

span> '
261      : '<span class="badge bg-secondary me-1">user</span>';
262
263      const diskWarning = u.quota_percent > 85 ? 'bg-danger' : u
      .quota_percent > 60 ? 'bg-warning' : 'bg-primary';
264
265      return `
266          <a href="javascript:void(0)"
267              class="list-group-item list-group-item-action p-2.5 u
      ser-select-item ${activeClass}"
268              data-user-id="${u.id}">
269              <div class="d-flex w-100 justify-content-between align
      -items-center mb-1">
270                  <div class="d-flex align-items-center gap-1.5 text-t
      runcate">
271                      <span class="badge bg-dark border text-info" style
      ="font-size: 0.72rem;">#${u.id}</span>
272                      <strong class="text-truncate small ${isSelected ?
      'text-white' : ''}">${escapeHtml(u.name)}</strong>
273                  </div>
274                  <div>${roleBadge}</div>
275              </div>
276              <div class="text-truncate small text-muted mb-1.5" sty
      le="font-size: 0.76rem;">
277                  ${escapeHtml(u.email || '-')}
278              </div>
279              <div class="d-flex justify-content-between align-items
      -center small mb-1" style="font-size: 0.74rem;">
280                  <span class="text-muted"><i class="bi bi-files me-1"
      ></i>${u.files_count} файлов</span>
281                  <span class="fw-semibold text-info">${u.size_formatt
      ed}</span>
282              </div>
283              <div class="progress" style="height: 4px;" title="Испо
      льзовано квоты: ${u.quota_percent}%">
284                  <div class="progress-bar ${diskWarning}" role="progr
      essbar" style="width: ${u.quota_percent}%;"></div>
285              </div>

```



```
286     </a>
287     `;
288   }).join('');
289
290   // Обработчики клика по элементам пользователей
291   userListEl.querySelectorAll('.user-select-item').forEach(it
  em => {
292     item.onclick = () => {
293       const uid = parseInt(item.dataset.userId, 10);
294       selectUser(uid);
295     };
296   });
297 }
298
299 /**
300  * Выбор пользователя и загрузка его файлового проводника
301  */
302 function selectUser(userId) {
303   state.selectedUserId = userId;
304   const user = state.users.find(u => u.id === userId);
305   state.selectedUserName = user ? user.name : `Пользователь #${
     {userId}`;
306
307   // Обновление подсветки в списке слева
308   const userListEl = document.getElementById('user-dirs-users
    -list');
309   if (userListEl) {
310     userListEl.querySelectorAll('.user-select-item').forEach(
      item => {
311       const uid = parseInt(item.dataset.userId, 10);
312       item.classList.toggle('active', uid === userId);
313       item.classList.toggle('border-primary', uid === userId);
314     });
315   }
316
317   // Обновление заголовка проводника
318   const titleEl = document.getElementById('explorer-user-title
    ');
```

```

319 |   const pathBadge = document.getElementById('explorer-user-pat
    | h-badge');
320 |   if (titleEl) {
321 |       titleEl.innerHTML = `

```

```

350 |   try {
351 |     let url = `/api/admin/user-directories/users/${userId}/tree?`;
352 |     const params = new URLSearchParams();
353 |     if (state.currentSubfolder) params.append('subfolder', state.currentSubfolder);
354 |     if (state.searchQuery) params.append('q', state.searchQuery);
355 |     if (state.extFilter) params.append('extension', state.extFilter);
356 |
357 |     const res = await fetch(url + params.toString());
358 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
359 |     const data = await res.json();
360 |
361 |     state.files = data.items || [];
362 |
363 |     if (footerStats) {
364 |       footerStats.innerHTML = `Файлов: <strong>${data.total_items || 0}</strong> | Общий размер: <strong>${data.total_size_formatted || '0 B'}</strong>`;
365 |     }
366 |
367 |     const user = state.users.find(u => u.id === userId);
368 |     if (footerQuota && user) {
369 |       footerQuota.innerHTML = `Квота: <strong>${user.size_formatted} / ${user.quota_formatted}</strong> (${user.quota_percent}%)`;
370 |     }
371 |
372 |     renderUserFilesTable();
373 |   } catch (err) {
374 |     console.error('Ошибка загрузки файлов:', err);
375 |     if (tbody) {
376 |       tbody.innerHTML = `
377 |         <tr>
378 |           <td colspan="6" class="text-center py-4 text-danger small">

```

```

379 |             Ошибка загрузки файлов: ${err.message}
380 |         </td>
381 |     </tr>
382 |     `;
383 | }
384 | }
385 | }
386 |
387 | /**
388 |  * Рендеринг таблицы файлов пользователя
389 |  */
390 | function renderUserFilesTable() {
391 |     const tbody = document.getElementById('user-files-table-body
392 |     ');
393 |     if (!tbody) return;
394 |     if (state.files.length === 0) {
395 |         tbody.innerHTML = `
396 |             <tr>
397 |                 <td colspan="6" class="text-center py-5 text-muted sma
398 |                 ll">
399 |                     <i class="bi bi-inbox fs-3 d-block mb-1 text-seconda
400 |                     ry"></i>
401 |                     В выбранном разделе нет файлов
402 |                 </td>
403 |             </tr>
404 |             `;
405 |     }
406 |     return;
407 | }
408 |
409 | tbody.innerHTML = state.files.map(f => {
410 |     const isDir = f.is_dir;
411 |     const icon = isDir
412 |         ? '<i class="bi bi-folder-fill text-warning fs-6"></i>'
413 |         : getFileIcon(f.extension);
414 |
415 |     const subfolderBadge = f.subfolder === 'root'
416 |         ? '<span class="badge bg-secondary-subtle text-secondary

```

```

border">root</span>'
414         : `<span class="badge bg-info-subtle text-info border bo
border-info-subtle">${escapeHtml(f.subfolder)}</span>`;
415
416     return `
417         <tr>
418             <td class="text-center">${icon}</td>
419             <td>
420                 <div class="fw-semibold text-truncate text-white" st
style="max-width: 260px;" title="${escapeHtml(f.relative_path)}"
>
421                     ${escapeHtml(f.name)}
422                 </div>
423                 <div class="text-muted small font-monospace" style="
font-size: 0.72rem;">
424                     ${escapeHtml(f.relative_path)}
425                 </div>
426             </td>
427             <td class="text-center">${subfolderBadge}</td>
428             <td class="text-center font-monospace small">${f.size_
formatted}</td>
429             <td class="small text-muted font-monospace" style="fon
t-size: 0.76rem;">${f.modified_formatted}</td>
430             <td class="text-center">
431                 <div class="btn-group btn-group-sm" role="group">
432                     ${!isDir ? `
433                         <button class="btn btn-xs btn-outline-info btn-p
review-file"
434                             data-path="${escapeHtml(f.relative_path)
}"
435                             title="Предпросмотр файла">
436                             <i class="bi bi-eye"></i>
437                         </button>
438                         <a href="/api/admin/user-directories/users/${sta
te.selectedUserId}/file/download?path=${encodeURIComponent(f.r
elative_path)}"
439                             class="btn btn-xs btn-outline-success"
440                             title="Скачать файл"

```

```

441         download>
442         <i class="bi bi-download"></i>
443     </a>
444     ` : `'}
445     <button class="btn btn-xs btn-outline-danger btn-d
delete-file"
446         data-path="${escapeHtml(f.relative_path)}"
447         data-is-dir="${isDir}"
448         title="Удалить">
449         <i class="bi bi-trash3"></i>
450     </button>
451 </div>
452 </td>
453 </tr>
454 `;
455 }).join('');
456
457 // Навешивание событий предпросмотра
458 tbody.querySelectorAll('.btn-preview-file').forEach(btn => {
459     btn.onclick = () => {
460         const path = btn.dataset.path;
461         openFilePreview(state.selectedUserId, path);
462     };
463 });
464
465 // Навешивание событий удаления
466 tbody.querySelectorAll('.btn-delete-file').forEach(btn => {
467     btn.onclick = () => {
468         const path = btn.dataset.path;
469         const isDir = btn.dataset.isDir === 'true';
470         const promptText = isDir
471             ? `Вы уверены, что хотите удалить директорию "${path}"
со всем содержимым?`
472             : `Вы уверены, что хотите удалить файл "${path}"?`;
473         if (confirm(promptText)) {
474             deleteFile(state.selectedUserId, path);
475         }

```

```
476     };
477   });
478 }
479
480 /**
481  * Иконка по расширению файла
482  */
483 function getFileIcon(ext) {
484   const e = (ext || '').toLowerCase();
485   switch (e) {
486     case 'pdf':
487       return '<i class="bi bi-file-earmark-pdf-fill text-dange
r fs-6"></i>';
488     case 'txt':
489     case 'md':
490     case 'log':
491       return '<i class="bi bi-file-earmark-text-fill text-info
fs-6"></i>';
492     case 'json':
493     case 'xml':
494     case 'yaml':
495     case 'yml':
496       return '<i class="bi bi-file-earmark-code-fill text-warn
ing fs-6"></i>';
497     case 'csv':
498     case 'tsv':
499     case 'xlsx':
500       return '<i class="bi bi-file-earmark-spreadsheet-fill te
xt-success fs-6"></i>';
501     case 'png':
502     case 'jpg':
503     case 'jpeg':
504     case 'webp':
505     case 'gif':
506     case 'svg':
507       return '<i class="bi bi-file-earmark-image-fill text-pri
mary fs-6"></i>';
508     case 'mp3':
```

```

509     case 'wav':
510     case 'ogg':
511         return '<i class="bi bi-file-earmark-music-fill text-pur
ple fs-6"></i>';
512     case 'py':
513     case 'js':
514     case 'sh':
515     case 'ps1':
516         return '<i class="bi bi-filetype-py text-warning fs-6"><
/i>';
517     default:
518         return '<i class="bi bi-file-earmark-fill text-secondary
fs-6"></i>';
519     }
520 }
521
522 /**
523  * Открытие модального окна предпросмотра файла
524  */
525 async function openFilePreview(userId, path) {
526     const modalEl = document.getElementById('modal-file-preview'
);
527     if (!modalEl) return;
528
529     const titleEl = document.getElementById('modal-file-preview-
title');
530     const pathEl = document.getElementById('preview-file-path');
531     const sizeEl = document.getElementById('preview-file-size');
532     const typeEl = document.getElementById('preview-file-type');
533     const contentWrapper = document.getElementById('preview-cont
ent-wrapper');
534     const downloadLink = document.getElementById('btn-preview-do
wnload-file');
535
536     if (titleEl) titleEl.textContent = `Предпросмотр: ${path.spl
it('/').pop()}`;
537     if (pathEl) pathEl.textContent = `data/users/${userId}/${pat
h}`;

```



```
538 |   if (sizeEl) sizeEl.textContent = '...';
539 |   if (typeEl) typeEl.textContent = '...';
540 |   if (contentWrapper) {
541 |     contentWrapper.innerHTML = `
542 |       <div class="text-center py-4 text-muted">
543 |         <div class="spinner-border spinner-border-sm text-primary me-2"></div> Загрузка содержимого...
544 |       </div>
545 |     `;
546 |   }
547 |   if (downloadLink) {
548 |     downloadLink.href = `/api/admin/user-directories/users/${userId}/file/download?path=${encodeURIComponent(path)}`;
549 |   }
550 |
551 |   const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
552 |   modal.show();
553 |
554 |   try {
555 |     const res = await fetch(`/api/admin/user-directories/users/${userId}/file/preview?path=${encodeURIComponent(path)}`);
556 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
557 |     const data = await res.json();
558 |     state.previewFile = data;
559 |
560 |     if (sizeEl) sizeEl.textContent = data.size_formatted;
561 |     if (typeEl) typeEl.textContent = data.mime_type || data.extension;
562 |
563 |     if (contentWrapper) {
564 |       if (data.is_image) {
565 |         contentWrapper.innerHTML = `
566 |           <div class="text-center p-2">
567 |             
568 |           </div>
569 |         `;
```

```

570         } else {
571             contentWrapper.textContent = data.content || '(Файл ну
ст)';
572         }
573     }
574 } catch (err) {
575     console.error('Ошибка предпросмотра файла:', err);
576     if (contentWrapper) {
577         contentWrapper.innerHTML = `<div class="text-danger p-3"
>Ошибка загрузки содержимого: ${err.message}</div>`;
578     }
579 }
580 }
581
582 /**
583  * Удаление файла или папки
584  */
585 async function deleteFile(userId, path) {
586     try {
587         const res = await fetch(`/api/admin/user-directories/users
/${userId}/file?path=${encodeURIComponent(path)}`, {
588             method: 'DELETE',
589         });
590         if (!res.ok) {
591             const errData = await res.json().catch(() => ({}));
592             throw new Error(errData.detail || `HTTP ${res.status}`);
593         }
594
595         showAlert(`Файл "${path}" успешно удален`, 'success');
596
597         // Закрываем модальное окно предпросмотра, если оно открыт
о
598         const previewModalEl = document.getElementById('modal-file
-preview');
599         if (previewModalEl) {
600             const modal = bootstrap.Modal.getInstance(previewModalEl
);
601             if (modal) modal.hide();

```

```
602     }
603
604     // Перезагружаем файлы и сводку
605     await loadUserFiles(userId);
606     await loadSummary();
607     await loadUsersList();
608 } catch (err) {
609     console.error('Ошибка удаления файла:', err);
610     showAlert(`Ошибка удаления файла: ${err.message}`, 'danger
    ');
611 }
612 }
613
614 /**
615  * Открытие модального окна управления осиротевшими директория
    ми
616  */
617 async function openOrphanedModal() {
618     const modalEl = document.getElementById('modal-orphaned-user
    -dirs');
619     if (!modalEl) return;
620     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
621     modal.show();
622     await loadOrphanedDirs();
623 }
624
625 /**
626  * Загрузка списка осиротевших директорий
627  */
628 async function loadOrphanedDirs() {
629     const tbody = document.getElementById('modal-orphaned-dirs-t
    body');
630     const countEl = document.getElementById('modal-orphaned-coun
    t');
631     const sizeEl = document.getElementById('modal-orphaned-size'
    );
632     const filesEl = document.getElementById('modal-orphaned-file
    s');
```

```
633 |
634 |   if (tbody) {
635 |     tbody.innerHTML = `
636 |       <tr>
637 |         <td colspan="6" class="text-center py-4 text-muted">
638 |           <div class="spinner-border spinner-border-sm text-wa
639 |             rning me-2"></div> Сканирование...
640 |         </td>
641 |       </tr>
642 |     `;
643 |   }
644 |   try {
645 |     const res = await fetch('/api/admin/user-directories/orpha
646 |       ned');
647 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
648 |     const data = await res.json();
649 |
650 |     state.orphanedDirs = data.orphaned_dirs || [];
651 |
652 |     if (countEl) countEl.textContent = data.total || 0;
653 |     if (sizeEl) sizeEl.textContent = data.total_size_formatted
654 |       || '0 B';
655 |     if (filesEl) filesEl.textContent = data.total_files || 0;
656 |
657 |     const badge = document.getElementById('badge-orphaned-coun
658 |       t');
659 |     if (badge) badge.textContent = data.total || 0;
660 |
661 |     renderOrphanedDirsTable();
662 |   } catch (err) {
663 |     console.error('Ошибка загрузки осиротевших директорий:', e
664 |       rr);
665 |     if (tbody) {
666 |       tbody.innerHTML = `
667 |         <tr>
668 |           <td colspan="6" class="text-center py-3 text-danger
669 |             small">
```

```

665 |             Ошибка сканирования: ${err.message}
666 |         </td>
667 |     </tr>
668 |     `;
669 | }
670 | }
671 | }
672 |
673 | /**
674 |  * Рендеринг таблицы осиротевших каталогов
675 |  */
676 | function renderOrphanedDirsTable() {
677 |     const tbody = document.getElementById('modal-orphaned-dirs-t
678 |     body');
679 |     if (!tbody) return;
680 |     if (state.orphanedDirs.length === 0) {
681 |         tbody.innerHTML = `
682 |             <tr>
683 |                 <td colspan="6" class="text-center py-4 text-success s
684 |                 mall">
685 |                     <i class="bi bi-check-circle fs-4 d-block mb-1"></i>
686 |                     Осиротевших каталогов не обнаружено. Все директории
687 |                     привязаны к активным пользователям!
688 |                 </td>
689 |             </tr>
690 |             `;
691 |         return;
692 |     }
693 |     tbody.innerHTML = state.orphanedDirs.map(d => `
694 |         <tr>
695 |             <td>
696 |                 <div class="fw-bold font-monospace text-warning small"
697 |                 >${escapeHtml(d.dir_name)}</div>
698 |                 <div class="text-muted small" style="font-size: 0.72re
699 |                 m;">${escapeHtml(d.full_path)}</div>
700 |             </td>

```

```

698         <td><span class="badge bg-secondary-subtle text-secondary border">${escapeHtml(d.reason)}</span></td>
699         <td class="text-center font-monospace">${d.files_count}<
700         /td>
701         <td class="text-center font-monospace small">${d.size_formatted}</td>
702         <td class="small text-muted font-monospace" style="font-size: 0.76rem;">${d.modified_formatted}</td>
703         <td class="text-center">
704             <button class="btn btn-xs btn-outline-danger btn-clean-single-orphaned"
705                 data-dir="${escapeHtml(d.dir_name)}"
706                 title="Удалить этот каталог">
707                 <i class="bi bi-trash3"></i>
708             </button>
709         </td>
710     </tr>
711     `).join('');
712     tbody.querySelectorAll('.btn-clean-single-orphaned').forEach(
713     (btn => {
714         btn.onclick = () => {
715             const dirName = btn.dataset.dir;
716             if (confirm(`Удалить осиротевший каталог "${dirName}"?`)) {
717                 cleanOrphanedDirs([dirName]);
718             }
719         });
720     });
721 }
722 /**
723  * Очистка осиротевших каталогов (конкретных или всех)
724  */
725 async function cleanOrphanedDirs(dirsList = null) {
726     if (!dirsList && !confirm('Вы уверены, что хотите удалить ВСЕ осиротевшие каталоги пользователей? Это действие необратимо.')) {

```

```
727     return;
728 }
729
730 try {
731     const payload = dirsList ? { dirs: dirsList } : {};
732     const res = await fetch('/api/admin/user-directories/orpha
733     ned/clean', {
734         method: 'POST',
735         headers: { 'Content-Type': 'application/json' },
736         body: JSON.stringify(payload),
737     });
738     if (!res.ok) throw new Error(`HTTP ${res.status}`);
739     const data = await res.json();
740
741     showAlert(`Удалено каталогов: ${data.deleted_count}, освоб
742     ождено места: ${data.freed_formatted}`, 'success');
743
744     await loadOrphanedDirs();
745     await loadSummary();
746 } catch (err) {
747     console.error('Ошибка очистки осиротевших каталогов:', err
748     );
749     showAlert(`Ошибка очистки каталогов: ${err.message}`, 'dan
750     ger');
751 }
752 }
753
754 /**
755  * Применение клиентских фильтров поиска
756  */
757 function applyFilters() {
758     renderUsersList();
759     if (state.selectedUserId) {
760         loadUserFiles(state.selectedUserId);
761     }
762 }
```

```

761 /**
762  * Вывод сообщения alert
763  */
764 function showAlert(message, type = 'info') {
765   const alertBox = document.getElementById('user-dirs-alert-bo
x');
766   const alertMsg = document.getElementById('user-dirs-alert-me
ssage');
767   if (!alertBox || !alertMsg) return;
768
769   alertBox.className = `alert alert-${type} alert-dismissible
fade show mb-3 shadow-sm`;
770   alertMsg.textContent = message;
771   alertBox.classList.remove('d-none');
772
773   setTimeout(() => {
774     alertBox.classList.add('d-none');
775   }, 6000);
776 }
777
778 /**
779  * Экранирование HTML
780  */
781 function escapeHtml(text) {
782   if (!text) return '';
783   const map = {
784     '&': '&',
785     '<': '<',
786     '>': '>',
787     '"': '"',
788     "'": '&#039;',
789   };
790   return String(text).replace(/[<>"']/g, m => map[m]);
791 }
792
793 // Экспорт по умолчанию
794 export default initUserDirectoriesTab;
795

```


index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/user_tts/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/user_tts/index.html" class="w-100 h-100 border-0"></iframe>
15 |
16 File: index.html
17 Project: ai-breadboard
18 Package: windows/api/webgui/user_tts
19 Author: hypo69
20 Copyright: © 2026 hypo69
21 Updated: 2026-10-01 13:04:40
22 |=====
23 |=====
24 -->
25
26 <!DOCTYPE html>
27 <html lang="ru">
28 <head>
29 |    <meta charset="UTF-8">
30 |    <meta name="viewport" content="width=device-width, initial-scale=1.0">
31 |    <title data-i18n="title">Эксперименты с TTS (Синтез речи)</title>
32 |    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
```

```
t/css/bootstrap.min.css" rel="stylesheet">
31 <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
    0.0/font/bootstrap-icons.css" rel="stylesheet">
32 <link href="/html/css/main.css?v=20260915_light_v2" rel="sty
    lesheet">
33 <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
    8next.min.js"></script>
34 <style>
35     /* Custom styles for TTS page */
36     :root {
37         --tts-nav-bg: #161b22;
38     }
39
40     body {
41         background: #0d1117;
42         color: #c9d1d9;
43         min-height: 100vh;
44     }
45
46     .tts-container {
47         max-width: 800px;
48         margin: 2rem auto;
49         padding: 0 1rem;
50     }
51
52     .tts-card {
53         background: #161b22;
54         border: 1px solid #30363d;
55         border-radius: 12px;
56         padding: 2rem;
57         box-shadow: 0 4px 12px rgba(0, 0, 0, 0.3);
58     }
59
60     h1 {
61         color: #c9d1d9;
62         font-weight: 600;
63         margin-bottom: 0.5rem;
64     }
```

```
65 |
66 | .subtitle {
67 |     color: #8b949e;
68 |     font-size: 0.95rem;
69 |     margin-bottom: 2rem;
70 | }
71 |
72 | .form-label {
73 |     color: #8b949e;
74 |     font-weight: 500;
75 |     margin-bottom: 0.5rem;
76 | }
77 |
78 | .form-control, .form-select {
79 |     background: #0d1117;
80 |     border: 1px solid #30363d;
81 |     color: #c9d1d9;
82 |     border-radius: 8px;
83 |     padding: 0.75rem 1rem;
84 | }
85 |
86 | .form-control:focus, .form-select:focus {
87 |     background: #0d1117;
88 |     border-color: #58a6ff;
89 |     color: #c9d1d9;
90 |     box-shadow: 0 0 0 2px rgba(88, 166, 255, 0.25);
91 | }
92 |
93 | textarea.form-control {
94 |     resize: vertical;
95 |     min-height: 120px;
96 |     font-family: system-ui, -apple-system, sans-serif;
97 | }
98 |
99 | .btn-tts {
100 |     background: #1f6feb;
101 |     color: #fff;
102 |     border: none;
```

```
103 |     border-radius: 8px;
104 |     padding: 0.85rem 1.5rem;
105 |     font-weight: 500;
106 |     transition: background-color 0.2s;
107 | }
108 |
109 | .btn-tts:hover {
110 |     background: #388bfd;
111 |     color: #fff;
112 | }
113 |
114 | .btn-tts:disabled {
115 |     opacity: 0.6;
116 |     cursor: not-allowed;
117 | }
118 |
119 | .btn-apply {
120 |     background: #238636;
121 |     color: #fff;
122 |     border: none;
123 |     border-radius: 8px;
124 |     padding: 0.85rem 1.5rem;
125 |     font-weight: 500;
126 |     transition: background-color 0.2s;
127 | }
128 |
129 | .btn-apply:hover {
130 |     background: #2ea043;
131 |     color: #fff;
132 | }
133 |
134 | .btn-apply:disabled {
135 |     opacity: 0.6;
136 |     cursor: not-allowed;
137 | }
138 |
139 | .active-badge {
140 |     background: rgba(35, 134, 54, 0.15);
```

```
141 |     border: 1px solid rgba(35, 134, 54, 0.3);
142 |     color: #56d364;
143 |     padding: 0.5rem 1rem;
144 |     border-radius: 20px;
145 |     font-size: 0.85rem;
146 |     display: inline-flex;
147 |     align-items: center;
148 |     gap: 0.5rem;
149 |     margin-bottom: 1.5rem;
150 | }
151 |
152 | .audio-player-container {
153 |     margin-top: 2rem;
154 |     padding-top: 2rem;
155 |     border-top: 1px solid #30363d;
156 |     display: none;
157 | }
158 |
159 | audio {
160 |     width: 100%;
161 |     border-radius: 8px;
162 | }
163 |
164 | .status-badge {
165 |     display: inline-flex;
166 |     align-items: center;
167 |     gap: 0.5rem;
168 |     font-size: 0.9rem;
169 |     color: #8b949e;
170 |     margin-top: 1rem;
171 | }
172 |
173 | .spinner-border {
174 |     width: 1rem;
175 |     height: 1rem;
176 |     border-width: 0.15em;
177 | }
178 |
```

```
179 | .back-link {
180 |     display: inline-flex;
181 |     align-items: center;
182 |     gap: 0.5rem;
183 |     color: #8b949e;
184 |     text-decoration: none;
185 |     font-size: 0.95rem;
186 |     margin-top: 2rem;
187 |     transition: color 0.2s;
188 | }
189 |
190 | .back-link:hover {
191 |     color: #58a6ff;
192 | }
193 |
194 | /* Dark theme audio player controls */
195 | audio::-webkit-media-controls-panel {
196 |     background-color: rgba(13, 17, 23, 0.9);
197 | }
198 |
199 | audio::-webkit-media-controls-current-time-display,
200 | audio::-webkit-media-controls-time-remaining-display {
201 |     color: #c9d1d9;
202 | }
203 | </style>
204 | </head>
205 | <body>
206 |
207 | <div class="tts-container">
208 |     <div class="tts-card">
209 |         <h1><i class="bi bi-mic"></i> <span data-i18n="title">Эксп
210 | ерименты с TTS (Синтез речи)</span></h1>
211 |         <p class="subtitle" data-i18n="subtitle">Сравните качество
212 | синтеза речи различных моделей: Web Speech, Google, Edge и Si
213 | lero.</p>
214 |
215 |         <div id="active-voice-container" class="mb-3" style="display: none;">
```

```
213 |     <span class="active-badge">
214 |         <i class="bi bi-check-circle-fill"></i>
215 |         <span><span data-i18n="label_active_voice">Активный го
    | лос:</span> <strong id="active-voice-name">...</strong></span>
216 |     </span>
217 | </div>
218 |
219 | <form id="tts-form">
220 |     <div class="mb-4">
221 |         <label for="tts-text" class="form-label" data-i18n="fo
    | rm.label_text">Текст для озвучки</label>
222 |         <textarea id="tts-text" class="form-control" placehold
    | er data-i18n-placeholder="form.placeholder">Привет! Это демонс
    | трация работы различных систем синтеза речи. Какая из них звуч
    | ит лучше всего?</textarea>
223 |     </div>
224 |
225 |     <div class="row">
226 |         <div class="col-md-6 mb-4">
227 |             <label for="tts-engine" class="form-label" data-i18n
    | ="form.label_engine">Система синтеза</label>
228 |             <select id="tts-engine" class="form-select">
229 |                 <option value="browser" data-i18n="options.browser
    | ">Web Speech API (Браузерная)</option>
230 |                 <option value="gtts" data-i18n="options.gtts">Goog
    | le TTS (Базовая)</option>
231 |                 <option value="edge-tts" selected data-i18n="optio
    | ns.edge-tts">Microsoft Edge TTS (Нейросетевая)</option>
232 |                 <option value="silero" data-i18n="options.silero">
    | Silero TTS (Локальная оффлайн)</option>
233 |             </select>
234 |         </div>
235 |
236 |         <div class="col-md-6 mb-4" id="voice-select-container"
    | >
237 |             <label for="tts-voice" class="form-label" data-i18n=
    | "form.label_voice">Голос / Спикер</label>
238 |             <select id="tts-voice" class="form-select">
```

```
239 |         <!-- Populated dynamically via JS -->
240 |         </select>
241 |     </div>
242 | </div>
243 |
244 |     <div class="d-flex gap-2">
245 |         <button type="submit" class="btn btn-tts flex-grow-1"
  id="btn-speak" data-i18n="button_speak">Озвучить текст</button
  >
246 |         <button type="button" class="btn btn-apply" id="btn-sa
  ve-default" data-i18n="button_apply">Применить изменения</butt
  on>
247 |     </div>
248 |
249 |     <div class="status-badge" id="tts-status" style="display
  : none;">
250 |         <div class="spinner-border text-success" role="status"
  ></div>
251 |         <span id="status-text" data-i18n="status_generating">Г
  енерация аудиофайла на сервере...</span>
252 |     </div>
253 | </form>
254 |
255 |     <div class="audio-player-container" id="player-container">
256 |         <label class="form-label" data-i18n="label_result">Резул
  ьтат генерации</label>
257 |         <audio id="audio-player" controls></audio>
258 |     </div>
259 | </div>
260 |
261 |     <a href="/" class="back-link">
262 |         <i class="bi bi-arrow-left"></i>
263 |         <span data-i18n="back_link">Вернуться в медиатеку</span>
264 |     </a>
265 | </div>
266 |
267 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
  /js/bootstrap.bundle.min.js"></script>
```



```

268 <script type="module">
269   import { initI18n, switchLang, applyTranslations } from '/ht
    ml/js/i18n.js';
270
271   window.switchLang = switchLang;
272   window.applyTranslations = applyTranslations;
273
274   // Initialize i18n
275   const savedLang = localStorage.getItem('app_language') || 'r
    u';
276   await initI18n(savedLang);
277
278   const engineSelect = document.getElementById('tts-engine');
279   const voiceSelect = document.getElementById('tts-voice');
280   const voiceContainer = document.getElementById('voice-select
    -container');
281   const ttsForm = document.getElementById('tts-form');
282   const btnSpeak = document.getElementById('btn-speak');
283   const ttsStatus = document.getElementById('tts-status');
284   const statusText = document.getElementById('status-text');
285   const playerContainer = document.getElementById('player-cont
    ainer');
286   const audioPlayer = document.getElementById('audio-player');
287   const activeVoiceContainer = document.getElementById('active
    -voice-container');
288   const activeVoiceName = document.getElementById('active-voic
    e-name');
289
290   function getFriendlyVoiceName(system, voice) {
291     if (system === 'edge-tts') {
292       const found = voiceMap['edge-tts'].find(v => v.value ===
        voice);
293       return found ? translateVoiceLabel(found.labelKey) : `Ed
        ge (${voice})`;
294     }
295     if (system === 'silero') {
296       const found = voiceMap['silero'].find(v => v.value === v
        oice);

```

```

297     return found ? translateVoiceLabel(found.labelKey) : `Si
    lero (${voice})`;
298 }
299 if (system === 'gtts') {
300     const found = voiceMap['gtts'].find(v => v.value === voi
    ce);
301     return found ? translateVoiceLabel(found.labelKey) : `Go
    ogle TTS (${voice})`;
302 }
303 if (system === 'browser') {
304     return `Браузерный (${voice} || 'По умолчанию')`;
305 }
306 return `${system} – ${voice}`;
307 }
308
309 function updateActiveBadge(system, voice) {
310     if (system && voice) {
311         activeVoiceName.textContent = getFriendlyVoiceName(system, voice);
312         activeVoiceContainer.style.display = 'block';
313     } else {
314         activeVoiceContainer.style.display = 'none';
315     }
316 }
317
318 // Available voices dictionary with translated labels
319 const voiceMap = {
320     'browser': [], // Populated dynamically from window.speech
    Synthesis
321     'gtts': [
322         { value: 'ru', labelKey: 'voices.ru_standard' },
323         { value: 'en', labelKey: 'voices.en_standard' },
324         { value: 'he', labelKey: 'voices.he_standard' },
325         { value: 'de', labelKey: 'voices.de_standard' },
326         { value: 'fr', labelKey: 'voices.fr_standard' },
327         { value: 'es', labelKey: 'voices.es_standard' }
328     ],
329     'edge-tts': [

```

```

330     { value: 'ru-RU-DmitryNeural', labelKey: 'voices.dmitry'
      },
331     { value: 'ru-RU-SvetlanaNeural', labelKey: 'voices.svetl
      ana' },
332     { value: 'en-US-JennyNeural', labelKey: 'voices.jenny' }
      ,
333     { value: 'en-US-GuyNeural', labelKey: 'voices.guy' },
334     { value: 'en-US-AriaNeural', labelKey: 'voices.aria' },
335     { value: 'en-GB-SoniaNeural', labelKey: 'voices.sonia' }
      ,
336     { value: 'en-GB-RyanNeural', labelKey: 'voices.ryan' },
337     { value: 'he-IL-AvriNeural', labelKey: 'voices.avri' },
338     { value: 'he-IL-HilaNeural', labelKey: 'voices.hila' },
339     { value: 'de-DE-KatjaNeural', labelKey: 'voices.katja' }
      ,
340     { value: 'de-DE-ConradNeural', labelKey: 'voices.conrad'
      },
341     { value: 'fr-FR-DeniseNeural', labelKey: 'voices.denise'
      },
342     { value: 'fr-FR-HenriNeural', labelKey: 'voices.henri' }
      ,
343     { value: 'es-ES-ElviraNeural', labelKey: 'voices.elvira'
      },
344     { value: 'es-ES-AlvaroNeural', labelKey: 'voices.alvaro'
      }
345 ],
346 'silero': [
347     { value: 'eugene', labelKey: 'voices.eugene' },
348     { value: 'aidar', labelKey: 'voices.aidar' },
349     { value: 'baya', labelKey: 'voices.baya' },
350     { value: 'kseniya', labelKey: 'voices.kseniya' },
351     { value: 'xenia', labelKey: 'voices.xenia' },
352     { value: 'random', labelKey: 'voices.random' }
353 ],
354 };
355
356 // Translate voice labels
357 function translateVoiceLabel(labelKey) {

```

```
358     return i18next?.t(labelKey) || labelKey;
359 }
360
361 // Load browser native voices
362 function loadBrowserVoices() {
363     if ('speechSynthesis' in window) {
364         const voices = window.speechSynthesis.getVoices();
365         voiceMap['browser'] = voices
366             .map(v => ({ value: v.name, labelKey: `[${v.lang}] ${v
.name}` }));
367
368         // Fallback if no voices found in browser
369         if (voiceMap['browser'].length === 0) {
370             voiceMap['browser'].push({ value: 'default', labelKey:
'voices.default' });
371         }
372
373         if (engineSelect.value === 'browser') {
374             updateVoiceSelect();
375         }
376     }
377 }
378
379 if ('speechSynthesis' in window) {
380     window.speechSynthesis.onvoiceschanged = loadBrowserVoices
;
381     loadBrowserVoices();
382 }
383
384 function updateVoiceSelect() {
385     const engine = engineSelect.value;
386     const voices = voiceMap[engine] || [];
387
388     voiceSelect.innerHTML = '';
389
390     if (voices.length === 0) {
391         voiceContainer.style.display = 'none';
392     } else {
```

```
393 |     voiceContainer.style.display = 'block';
394 |     voices.forEach(voice => {
395 |         const opt = document.createElement('option');
396 |         opt.value = voice.value;
397 |         opt.textContent = translateVoiceLabel(voice.labelKey);
398 |         voiceSelect.appendChild(opt);
399 |     });
400 | }
401 | }
402 |
403 | engineSelect.addEventListener('change', updateVoiceSelect);
404 | updateVoiceSelect(); // Initial populate
405 |
406 | // On page load, fetch current settings to pre-select saved
    voice/engine
407 | try {
408 |     const settingsResponse = await fetch('/auth/settings');
409 |     if (settingsResponse.ok) {
410 |         const settings = await settingsResponse.json();
411 |         if (settings.tts_system) {
412 |             engineSelect.value = settings.tts_system;
413 |             updateVoiceSelect();
414 |         }
415 |         if (settings.tts_voice) {
416 |             voiceSelect.value = settings.tts_voice;
417 |         }
418 |         updateActiveBadge(settings.tts_system, settings.tts_voic
    e);
419 |     }
420 | } catch (e) {
421 |     console.error('Failed to load user settings:', e);
422 | }
423 |
424 | // Update translations when language changes
425 | const originalSwitchLang = window.switchLang;
426 | window.switchLang = async function(lang) {
427 |     await originalSwitchLang(lang);
428 |     updateVoiceSelect();
```

```

429 | };
430 |
431 | ttsForm.addEventListener('submit', async (e) => {
432 |     e.preventDefault();
433 |     const text = document.getElementById('tts-text').value.trim();
434 |     if (!text) return;
435 |
436 |     const engine = engineSelect.value;
437 |     const voice = voiceSelect.value;
438 |
439 |     // 1. Browser Web Speech synthesis
440 |     if (engine === 'browser') {
441 |         if ('speechSynthesis' in window) {
442 |             window.speechSynthesis.cancel();
443 |             const cleanText = text.replace(/<[^>]*>/g, '').replace(
444 | (/[*_`#]/g, ''));
445 |             const utterance = new SpeechSynthesisUtterance(cleanText);
446 |             const voices = window.speechSynthesis.getVoices();
447 |             const selectedVoice = voices.find(v => v.name === voice);
448 |             if (selectedVoice) {
449 |                 utterance.voice = selectedVoice;
450 |                 if (selectedVoice.lang) {
451 |                     utterance.lang = selectedVoice.lang;
452 |                 }
453 |             }
454 |
455 |             ttsStatus.style.display = 'inline-flex';
456 |             statusText.textContent = i18next.t('status_generating') || 'Воспроизведение речи браузером...';
457 |
458 |             utterance.onend = () => {
459 |                 ttsStatus.style.display = 'none';
460 |             };
461 |             utterance.onerror = () => {
462 |                 ttsStatus.style.display = 'none';

```

```
462         };
463
464         window.speechSynthesis.speak(utterance);
465         playerContainer.style.display = 'none';
466     } else {
467         const msg = i18next?.t('errors.no_api') || 'Ваш браузер
    не поддерживает Web Speech API.';
468         window.showToast?.(msg, 'warning') || alert(msg);
469     }
470     return;
471 }
472
473 // 2. Server-side TTS synthesis (gtts, edge-tts, silero)
474 ttsStatus.style.display = 'inline-flex';
475 statusText.textContent = i18next?.t('status_generating') |
    | 'Генерация аудиофайла на сервере...';
476 btnSpeak.disabled = true;
477 playerContainer.style.display = 'none';
478
479 try {
480     const queryParams = new URLSearchParams({
481         text: text,
482         system: engine,
483         voice: voice
484     });
485
486     // Load audio stream from server
487     const audioUrl = `/api/tts/synthesize?${queryParams.toSt
    ring()}`;
488
489     // Force preload of the audio
490     audioPlayer.src = audioUrl;
491     audioPlayer.load();
492
493     audioPlayer.oncanplaythrough = () => {
494         ttsStatus.style.display = 'none';
495         btnSpeak.disabled = false;
496         playerContainer.style.display = 'block';
```

```
497 |         audioPlayer.play();
498 |     };
499 |
500 |     audioPlayer.onerror = (err) => {
501 |         console.error(err);
502 |         statusText.textContent = i18next?.t('errors.server_err
or') || 'Ошибка синтеза речи на сервере';
503 |         btnSpeak.disabled = false;
504 |         setTimeout(() => { ttsStatus.style.display = 'none'; }
, 3000);
505 |     };
506 |
507 |     } catch (err) {
508 |         console.error(err);
509 |         statusText.textContent = i18next?.t('errors.network_err
or') || 'Сбой сети при запросе к серверу';
510 |         btnSpeak.disabled = false;
511 |         setTimeout(() => { ttsStatus.style.display = 'none'; },
3000);
512 |     }
513 | });
514 |
515 | const btnSaveDefault = document.getElementById('btn-save-def
ault');
516 | btnSaveDefault.addEventListener('click', async () => {
517 |     const engine = engineSelect.value;
518 |     const voice = voiceSelect.value;
519 |
520 |     btnSaveDefault.disabled = true;
521 |     try {
522 |         const payload = {
523 |             tts_enabled: 1,
524 |             tts_system: engine,
525 |             tts_voice: voice
526 |         };
527 |
528 |         const response = await fetch('/auth/settings', {
529 |             method: 'POST',
```



```
530 |         headers: { 'Content-Type': 'application/json' },
531 |         body: JSON.stringify(payload)
532 |     });
533 |
534 |     if (response.ok) {
535 |         updateActiveBadge(engine, voice);
536 |         alert('Изменения успешно применены! Этот голос теперь
| установлен по умолчанию.');
```

537 | } else {

538 | alert('Не удалось сохранить настройки. Убедитесь, что

539 | вы вошли в систему.');

540 | } catch (err) {

541 | console.error(err);

542 | alert('Ошибка подключения к серверу при сохранении настр

543 | } finally {

544 | btnSaveDefault.disabled = false;

545 | }

546 | });

547 | </script>

548 | </body>

549 | </html>

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11
12 Usage Examples:
13     Web Route:
14         /windows/api/webgui/voice_tab/index.html
15
16     Tab Frame Integration:
17         <iframe src="/windows/api/webgui/voice_tab/index.html" c
18         lass="w-100 h-100 border-0"></iframe>
19
20 File: index.html
21 Project: ai-breadboard
22 Package: windows/api/webgui/voice_tab
23 Author: hypo69
24 Copyright: © 2026 hypo69
25 Updated: 2026-10-01 13:04:40
26 =====
27 =====
28 -->
29
30 <!-- Voice Diarization & Audio Analyzer Tab -->
31 <div class="container-fluid py-2">
32     <div class="alert alert-danger d-flex justify-content-betwee
33     n align-items-center flex-wrap gap-2 mb-3 shadow-sm">
34         <div>
35             <h5 class="mb-1"><i class="bi bi-mic-fill me-2"></i><spa
36             n data-i18n="voice.title">□ Разбор голосовых сообщений и диар
37             изация</span></h5>
38             <p class="mb-0 small text-secondary" data-i18n="voice.su
```

```
title">Запись с микрофона или загрузка аудиофайла: разделение
по собеседникам, стенограмма и сводка встреч.</p>
```

```
</div>
```

```
<div class="d-flex align-items-center gap-2">
```

```
<span class="badge bg-dark font-monospace p-2">Gemini Mu
ltimodal Audio</span>
```

```
</div>
```

```
</div>
```

```
<div class="row g-3">
```

```
<!-- LEFT COLUMN: Recording & Audio Upload Controls -->
```

```
<div class="col-lg-5 col-md-12">
```

```
<!-- Microphone Live Recording Card -->
```

```
<div class="card shadow-sm border-0 mb-3">
```

```
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
```

```
<span class="fw-semibold"><i class="bi bi-broadcast-me-2 text-danger"></i><span data-i18n="voice.recordTitle">Запись с микрофона</span></span>
```

```
<span class="badge bg-danger font-monospace d-none" id="vtab-record-timer">00:00</span>
```

```
</div>
```

```
<div class="card-body text-center p-4">
```

```
<div id="vtab-mic-visual" class="my-3">
```

```
<div class="d-inline-flex align-items-center justify-content-center rounded-circle p-4 border border-3 border-danger shadow" style="width: 100px; height: 100px; transition: all 0.3s;" id="vtab-mic-circle">
```

```
<i class="bi bi-mic-fill fs-1 text-danger" id="vtab-mic-icon"></i>
```

```
</div>
```

```
</div>
```

```
<h6 class="fw-bold mb-1" id="vtab-record-status-label" data-i18n="voice.clickToRecord">Нажмите для начала записи</h6>
```

```
<p class="text-muted small mb-3">Говорите в микрофон – система автоматически распознает разных спикеров.</p>
```

```

54         <div class="d-flex justify-content-center gap-2">
55             <button class="btn btn-danger px-4 py-2 rounded-pi
ll fw-bold shadow-sm" id="btn-vtab-record-toggle">
56                 <i class="bi bi-record-fill me-1"></i> <span id=
"vtab-record-btn-text" data-i18n="voice.startRecord">Начать за
пись</span>
57             </button>
58             <button class="btn btn-outline-secondary px-3 py-2
rounded-pill d-none" id="btn-vtab-record-cancel">
59                 Отмена
60             </button>
61         </div>
62     </div>
63 </div>
64
65     <!-- File Upload Card -->
66     <div class="card shadow-sm border-0 mb-3">
67         <div class="card-header bg-dark text-white d-flex just
ify-content-between align-items-center">
68             <span class="fw-semibold"><i class="bi bi-cloud-arro
w-up me-2 text-info"></i><span data-i18n="voice.uploadTitle">3
агрузка аудиофайла</span></span>
69         </div>
70         <div class="card-body">
71             <div id="vtab-dropzone" class="border border-2 borde
r-dashed rounded p-4 text-center cursor-pointer mb-2 bg-body-s
econdary bg-opacity-25" style="border-style: dashed !important
;">
72                 <i class="bi bi-file-earmark-music fs-1 text-info
mb-2 d-block"></i>
73                 <h6 class="fw-bold mb-1" data-i18n="voice.dropzone
Text">Перетащите аудиофайл сюда или нажмите для выбора</h6>
74                 <p class="text-muted small mb-2">Форматы: .mp3, .w
av, .ogg, .m4a, .webm, .aac</p>
75                 <input type="file" id="vtab-file-input" class="d-n
one" accept="audio/*, .mp3, .wav, .ogg, .m4a, .webm, .aac">
76                 <button class="btn btn-outline-info btn-sm px-3 ro
unded-pill mt-1" type="button" onclick="document.getElementByI

```

```

d('vtab-file-input').click();">
77 |         <i class="bi bi-folder2-open me-1"></i> <span da
| ta-i18n="voice.browseAudio">Выбрать аудиофайл</span>
78 |         </button>
79 |     </div>
80 | </div>
81 | </div>
82 |
83 | <!-- Audio Analysis Settings -->
84 | <div class="card shadow-sm border-0">
85 |     <div class="card-header bg-dark text-white">
86 |         <span class="fw-semibold"><i class="bi bi-sliders me
| -2"></i>Параметры распознавания</span>
87 |     </div>
88 |     <div class="card-body">
89 |         <div class="mb-2">
90 |             <label class="form-label small fw-bold" for="vtab-
| model-select">Модель:</label>
91 |             <select class="form-select form-select-sm" id="vta
| b-model-select">
92 |                 <option value="gemini-3.1-flash" selected>gemini
| -3.1-flash (Рекомендуется для аудио)</option>
93 |                 <option value="gemini-2.0-flash">gemini-2.0-flas
| h</option>
94 |                 <option value="gemini-1.5-flash">gemini-1.5-flas
| h</option>
95 |                 <option value="gemini-1.5-pro">gemini-1.5-pro</o
| ption>
96 |             </select>
97 |         </div>
98 |         <div class="mb-2">
99 |             <label class="form-label small fw-bold" for="vtab-
| lang-select">Язык отчёта и сводки:</label>
100 |             <select class="form-select form-select-sm" id="vta
| b-lang-select">
101 |                 <option value="ru" selected>Русский (Russian)</o
| ption>
102 |                 <option value="en">English</option>

```

```
103 |         <option value="he">□ □ □ □ □ (Hebrew)</option>
104 |     </select>
105 | </div>
106 | <div>
107 |     <label class="form-label small fw-bold" for="vtab-
    api-key">Gemini API Key (опционально):</label>
108 |     <input type="password" class="form-control form-co
    ntrol-sm" id="vtab-api-key" placeholder="Из .env по умолчанию"
    autocomplete="new-password" data-lpignore="true" data-lp-igno
    re="true">
109 | </div>
110 | </div>
111 | </div>
112 | </div>
113 |
114 | <!-- RIGHT COLUMN: Audio Playback & Diarization Result -->
115 | <div class="col-lg-7 col-md-12">
116 |     <div class="card shadow-sm border-0 mb-3" id="vtab-resul
    t-card">
117 |         <div class="card-header bg-dark text-white d-flex just
    ify-content-between align-items-center">
118 |             <span class="fw-semibold"><i class="bi bi-file-text
    me-2 text-warning"></i><span data-i18n="voice.resultTitle">Рез
    ультат анализа и стенограмма</span></span>
119 |             <div class="d-flex align-items-center gap-2">
120 |                 <button class="btn btn-success btn-sm d-none" id="
    vtab-btn-save-rag">
121 |                     <i class="bi bi-floppy me-1"></i> <span data-i18
    n="voice.saveToRag">Сохранить в RAG</span>
122 |                 </button>
123 |             </div>
124 |         </div>
125 |         <div class="card-body p-3" id="vtab-result-body">
126 |             <!-- Audio Player Preview -->
127 |             <div id="vtab-player-container" class="mb-3 d-none p
    -2 bg-body-tertiary border rounded">
128 |                 <div class="small fw-semibold mb-1 text-muted"><i
    class="bi bi-play-circle me-1 text-danger"></i>Исходная аудиоз
```

```

    апись: <span id="vtab-current-filename" class="text-body font-
monospace"></span></div>
129         <audio id="vtab-audio-element" controls class="w-1
00" style="max-height: 38px;"></audio>
130     </div>
131
132     <!-- Loading State -->
133     <div id="vtab-loading" class="d-none text-center py-
5">
134         <div class="spinner-border text-danger mb-3" role=
"status" style="width: 3rem; height: 3rem;"></div>
135         <h6 class="fw-bold text-primary">Идет распознавани
е и диаризация...</h6>
136         <p class="text-muted small">Gemini слушает аудио,
разделяет собеседников и готовит сводку.</p>
137     </div>
138
139     <!-- Empty Initial State -->
140     <div id="vtab-empty-state" class="text-center py-5 t
ext-muted">
141         <i class="bi bi-soundwave fs-1 d-block mb-2 opacit
y-50"></i>
142         <h6 class="fw-bold" data-i18n="voice.emptyStateTit
le">Запишите голос или загрузите аудиофайл</h6>
143         <p class="small mb-0">Здесь появится краткая сводк
а, список ключевых решений и реплики каждого собеседника.</p>
144     </div>
145
146     <!-- Analysis Content Container -->
147     <div id="vtab-analysis-content" class="d-none"></div>
>
148     </div>
149 </div>
150 </div>
151 </div>
152 </div>
153

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Voice Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/voice_tab/main.js?v=20
11 *     261001_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/voice_tab
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-01 13:04:40
19 * =====
20 */
21 // — VOICE_TAB MAIN.JS —————
22
23 (function () {
24   let isInitialized = false;
25   let tabMediaRecorder = null;
26   let tabAudioChunks = [];
27   let tabRecordingStartTime = 0;
28   let tabTimerInterval = null;
29   let lastAnalysisData = null;
30   let lastAudioFilename = 'voice_recording';
31 }
```



```
32 | function initVoiceTab() {
33 |     const recordBtn = document.getElementById('btn-vtab-record
    | -toggle');
34 |     if (!recordBtn) return;
35 |     if (isInitialized) return;
36 |     isInitialized = true;
37 |     setupEventListeners();
38 | }
39 |
40 | window.initVoiceTab = initVoiceTab;
41 |
42 | function setupEventListeners() {
43 |     const recordBtn = document.getElementById('btn-vtab-record
    | -toggle');
44 |     const cancelBtn = document.getElementById('btn-vtab-record
    | -cancel');
45 |     const fileInput = document.getElementById('vtab-file-input
    | ');
46 |     const dropzone = document.getElementById('vtab-dropzone');
47 |     const saveRagBtn = document.getElementById('vtab-btn-save-
    | rag');
48 |
49 |     if (recordBtn) {
50 |         recordBtn.addEventListener('click', toggleRecording);
51 |     }
52 |     if (cancelBtn) {
53 |         cancelBtn.addEventListener('click', cancelRecording);
54 |     }
55 |     if (fileInput) {
56 |         fileInput.addEventListener('change', (e) => {
57 |             if (e.target.files && e.target.files.length > 0) {
58 |                 const file = e.target.files[0];
59 |                 analyzeAudioBlob(file, file.name);
60 |                 fileInput.value = '';
61 |             }
62 |         });
63 |     }
64 |     if (dropzone) {
```

```

65 |     ['dragenter', 'dragover'].forEach(eventName => {
66 |         dropzone.addEventListener(eventName, (e) => {
67 |             e.preventDefault();
68 |             e.stopPropagation();
69 |             dropzone.classList.add('border-info', 'bg-info', 'bg
| -opacity-10');
70 |             }, false);
71 |         });
72 |
73 |     ['dragleave', 'drop'].forEach(eventName => {
74 |         dropzone.addEventListener(eventName, (e) => {
75 |             e.preventDefault();
76 |             e.stopPropagation();
77 |             dropzone.classList.remove('border-info', 'bg-info',
| 'bg-opacity-10');
78 |             }, false);
79 |         });
80 |
81 |     dropzone.addEventListener('drop', (e) => {
82 |         const dt = e.dataTransfer;
83 |         if (dt && dt.files && dt.files.length > 0) {
84 |             const file = dt.files[0];
85 |             analyzeAudioBlob(file, file.name);
86 |         }
87 |     }, false);
88 | }
89 | if (saveRagBtn) {
90 |     saveRagBtn.addEventListener('click', saveResultToRag);
91 | }
92 | }
93 |
94 | async function toggleRecording() {
95 |     if (tabMediaRecorder && tabMediaRecorder.state === 'record
| ing') {
96 |         stopRecording();
97 |     } else {
98 |         await startRecording();
99 |     }

```

```
100 | }
101 |
102 | async function startRecording() {
103 |     if (!navigator.mediaDevices || !navigator.mediaDevices.get
104 | UserMedia) {
105 |         alert('Доступ к микрофону не поддерживается в вашем брау
106 | зере. ');
107 |         return;
108 |     }
109 |     try {
110 |         const stream = await navigator.mediaDevices.getUserMedia
111 | ({ audio: true });
112 |         tabAudioChunks = [];
113 |         let options = {};
114 |         if (MediaRecorder.isTypeSupported('audio/webm;codecs=opus
115 | s')) {
116 |             options = { mimeType: 'audio/webm;codecs=opus' };
117 |         } else if (MediaRecorder.isTypeSupported('audio/ogg;code
118 | cs=opus')) {
119 |             options = { mimeType: 'audio/ogg;codecs=opus' };
120 |         }
121 |         tabMediaRecorder = new MediaRecorder(stream, options);
122 |         tabMediaRecorder.ondataavailable = (e) => {
123 |             if (e.data && e.data.size > 0) {
124 |                 tabAudioChunks.push(e.data);
125 |             }
126 |         };
127 |         tabMediaRecorder.onstop = () => {
128 |             stream.getTracks().forEach(t => t.stop());
129 |             if (tabAudioChunks.length > 0) {
130 |                 const mime = tabMediaRecorder.mimeType || 'audio/web
131 | m';
132 |                 const blob = new Blob(tabAudioChunks, { type: mime })
```

```

);
132 |         const fname = `voice_${new Date().toISOString().slice(0, 19).replace(/[:T]/g, '_')}.webm`;
133 |         analyzeAudioBlob(blob, fname);
134 |     }
135 | };
136 |
137 |     tabMediaRecorder.start(250);
138 |     tabRecordingStartTime = Date.now();
139 |
140 |     // UI state update
141 |     const recordBtn = document.getElementById('btn-vtab-record-toggle');
142 |     const cancelBtn = document.getElementById('btn-vtab-record-cancel');
143 |     const btnText = document.getElementById('vtab-record-btn-text');
144 |     const statusLabel = document.getElementById('vtab-record-status-label');
145 |     const timerBadge = document.getElementById('vtab-record-timer');
146 |     const micCircle = document.getElementById('vtab-mic-circle');
147 |
148 |     if (recordBtn) {
149 |         recordBtn.className = 'btn btn-danger px-4 py-2 rounded-pill fw-bold shadow-sm';
150 |     }
151 |     if (btnText) btnText.textContent = 'Остановить и разобрать';
152 |     if (cancelBtn) cancelBtn.classList.remove('d-none');
153 |     if (statusLabel) statusLabel.textContent = 'Идет запись речи...';
154 |     if (timerBadge) {
155 |         timerBadge.classList.remove('d-none');
156 |         timerBadge.textContent = '00:00';
157 |     }
158 |     if (micCircle) {

```

```
159 |         micCircle.classList.add('bg-danger', 'bg-opacity-25');
160 |         micCircle.style.transform = 'scale(1.1)';
161 |     }
162 |
163 |     clearInterval(tabTimerInterval);
164 |     tabTimerInterval = setInterval(() => {
165 |         const elapsedSec = Math.floor((Date.now() - tabRecordi
ngStartTime) / 1000);
166 |         const m = String(Math.floor(elapsedSec / 60)).padStart
(2, '0');
167 |         const s = String(elapsedSec % 60).padStart(2, '0');
168 |         if (timerBadge) timerBadge.textContent = `${m}:${s}`;
169 |     }, 500);
170 |
171 | } catch (err) {
172 |     console.error('Microphone error:', err);
173 |     alert(`Не удалось включить микрофон: ${err.message}`);
174 | }
175 | }
176 |
177 | function stopRecording() {
178 |     resetRecordingUi();
179 |     if (tabMediaRecorder && tabMediaRecorder.state === 'record
ing') {
180 |         tabMediaRecorder.stop();
181 |     }
182 | }
183 |
184 | function cancelRecording() {
185 |     resetRecordingUi();
186 |     if (tabMediaRecorder) {
187 |         if (tabMediaRecorder.state === 'recording') {
188 |             tabMediaRecorder.onstop = null;
189 |             tabMediaRecorder.stop();
190 |         }
191 |         tabAudioChunks = [];
192 |     }
193 | }
```

```
194
195 function resetRecordingUi() {
196     clearInterval(tabTimerInterval);
197     const cancelBtn = document.getElementById('btn-vtab-record
-cancel');
198     const btnText = document.getElementById('vtab-record-btn-t
ext');
199     const statusLabel = document.getElementById('vtab-record-s
tatus-label');
200     const timerBadge = document.getElementById('vtab-record-ti
mer');
201     const micCircle = document.getElementById('vtab-mic-circle
');
202
203     if (cancelBtn) cancelBtn.classList.add('d-none');
204     if (btnText) btnText.textContent = 'Начать запись';
205     if (statusLabel) statusLabel.textContent = 'Нажмите для на
чала записи';
206     if (timerBadge) timerBadge.classList.add('d-none');
207     if (micCircle) {
208         micCircle.classList.remove('bg-danger', 'bg-opacity-25')
;
209         micCircle.style.transform = 'scale(1)';
210     }
211 }
212
213 async function analyzeAudioBlob(audioBlob, filename) {
214     lastAudioFilename = filename;
215     const playerContainer = document.getElementById('vtab-play
er-container');
216     const audioElement = document.getElementById('vtab-audio-e
lement');
217     const filenameLabel = document.getElementById('vtab-curren
t-filename');
218     const loadingState = document.getElementById('vtab-loading
');
219     const emptyState = document.getElementById('vtab-empty-sta
te');
```

```
220     const analysisContent = document.getElementById('vtab-anal
ysis-content');
221     const saveRagBtn = document.getElementById('vtab-btn-save-
rag');
222
223     // Show audio player
224     if (playerContainer && audioElement) {
225         const audioUrl = URL.createObjectURL(audioBlob);
226         audioElement.src = audioUrl;
227         if (filenameLabel) filenameLabel.textContent = filename;
228         playerContainer.classList.remove('d-none');
229     }
230
231     if (emptyState) emptyState.classList.add('d-none');
232     if (analysisContent) analysisContent.classList.add('d-none
');
233     if (loadingState) loadingState.classList.remove('d-none');
234     if (saveRagBtn) saveRagBtn.classList.add('d-none');
235
236     const model = document.getElementById('vtab-model-select')
?.value || 'gemini-3.1-flash';
237     const lang = document.getElementById('vtab-lang-select')?.
value || 'ru';
238     const apiKey = document.getElementById('vtab-api-key')?.va
lue || '';
239
240     const formData = new FormData();
241     formData.append('file', audioBlob, filename);
242     formData.append('model', model);
243     formData.append('language', lang);
244     if (apiKey) formData.append('api_key', apiKey);
245
246     try {
247         const res = await fetch('/api/audio/diarize', {
248             method: 'POST',
249             body: formData,
250         });
251
```

```

252     const json = await res.json();
253     if (!res.ok) throw new Error(json.detail || 'Analysis er
ror');
254
255     lastAnalysisData = json.data || {};
256     renderResults(lastAnalysisData);
257     if (saveRagBtn) saveRagBtn.classList.remove('d-none');
258 } catch (err) {
259     console.error('[VoiceTab] Diarization failed:', err);
260     if (analysisContent) {
261         analysisContent.innerHTML = `
262             <div class="alert alert-danger p-3 mb-0">
263                 <h6 class="fw-bold mb-1"><i class="bi bi-exclamati
on-triangle-fill me-2"></i>Ошибка анализа аудио</h6>
264                 <p class="mb-0 small">${escapeHtml(err.message)}</
p>
265             </div>
266         `;
267         analysisContent.classList.remove('d-none');
268     }
269 } finally {
270     if (loadingState) loadingState.classList.add('d-none');
271 }
272 }
273
274 function renderResults(data) {
275     const analysisContent = document.getElementById('vtab-anal
ysis-content');
276     if (!analysisContent) return;
277
278     const summary = data.summary || '';
279     const keyPoints = data.key_points || [];
280     const actionItems = data.action_items || [];
281     const transcript = data.transcript || [];
282     const speakers = data.speakers || [];
283
284     const speakerColors = [
285         '#3b82f6', '#10b981', '#f59e0b', '#ec4899', '#8b5cf6', '

```



```

#06b6d4'
286     ];
287
288     let keyPointsHtml = '';
289     if (keyPoints.length > 0) {
290         keyPointsHtml = `
291             <div class="card bg-body-tertiary border-0 mb-3 shadow
- sm">
292                 <div class="card-body p-3">
293                     <h6 class="fw-bold text-info mb-2"><i class="bi bi
- key-fill me-2"></i>Ключевые темы и решения</h6>
294                     <ul class="mb-0 ps-3 small text-body-secondary">
295                         ${keyPoints.map(kp => `<li class="mb-1">${escape
Html(kp)}</li>`).join('')}
296                     </ul>
297                 </div>
298             </div>
299         `;
300     }
301
302     let actionItemsHtml = '';
303     if (actionItems.length > 0) {
304         actionItemsHtml = `
305             <div class="card bg-body-tertiary border-0 mb-3 shadow
- sm">
306                 <div class="card-body p-3">
307                     <h6 class="fw-bold text-success mb-2"><i class="bi
bi-check2-square me-2"></i>Задачи и договоренности (Action It
ems)</h6>
308                     <ul class="list-unstyled mb-0 small">
309                         ${actionItems.map(ai => `
310                             <li class="d-flex align-items-start gap-2 mb-1
">
311                             <span class="badge bg-success bg-opacity-25
text-success font-monospace">TODO</span>
312                             <span>${escapeHtml(ai)}</span>
313                             </li>
314                         `).join('')}

```

```

315 |         </ul>
316 |     </div>
317 | </div>
318 |     `;
319 | }
320 |
321 | let transcriptHtml = '';
322 | if (transcript.length > 0) {
323 |     transcriptHtml = `
324 |         <div class="card bg-body-tertiary border-0 shadow-sm">
325 |             <div class="card-header bg-dark text-white d-flex ju
326 | stify-content-between align-items-center py-2">
327 |                 <span class="fw-semibold small"><i class="bi bi-ch
328 | at-quote-fill me-2 text-warning"></i>Стенограмма по собеседник
329 | ам (${speakers.length || 2} участников)</span>
330 |             </div>
331 |             <div class="card-body p-3 d-flex flex-column gap-2"
332 | style="max-height: 400px; overflow-y: auto;">
333 |                 ${transcript.map((t, idx) => {
334 |                     const sp = t.speaker || `Собеседник ${idx % 2 +
335 | 1}`;
336 |                     const col = speakerColors[idx % speakerColors.le
337 | ngth];
338 |                     const ts = t.timestamp ? `<span class="badge bg-
339 | dark text-muted font-monospace">${escapeHtml(t.timestamp)}</sp
340 | an>` : '';
341 |                     return `
342 |                         <div class="p-2 rounded bg-body border-start b
343 | order-3 shadow-sm" style="border-left-color: ${col} !important
344 | ;">
345 |                             <div class="d-flex align-items-center justif
346 | y-content-between mb-1">
347 |                                 <span class="fw-bold small" style="color:
348 | ${col};"><i class="bi bi-person-fill me-1"></i>${escapeHtml(sp
349 | )}</span>
350 |
351 |                                 ${ts}
352 |                             </div>
353 |                             <div class="small text-body">${escapeHtml(t.

```

```

    text)}}</div>
340         </div>
341         `;
342         }).join('')}
343     </div>
344 </div>
345 `;
346 }
347
348 analysisContent.innerHTML = `
349     ${summary ? `
350         <div class="alert alert-primary mb-3 shadow-sm">
351             <h6 class="fw-bold mb-1"><i class="bi bi-card-text m
e-2"></i>Краткая сводка (Executive Summary)</h6>
352             <p class="mb-0 small">${escapeHtml(summary)}</p>
353         </div>
354         ` : ''}
355         ${keyPointsHtml}
356         ${actionItemsHtml}
357         ${transcriptHtml}
358     `;
359
360 analysisContent.classList.remove('d-none');
361 }
362
363 async function saveResultToRag() {
364     if (!lastAnalysisData) return;
365
366     try {
367         const res = await fetch('/api/audio/save-to-rag', {
368             method: 'POST',
369             headers: { 'Content-Type': 'application/json' },
370             body: JSON.stringify({
371                 title: `Голосовая запись: ${lastAudioFilename}`,
372                 summary: lastAnalysisData.summary || '',
373                 transcript: lastAnalysisData.transcript || [],
374                 key_points: lastAnalysisData.key_points || [],
375                 action_items: lastAnalysisData.action_items || [],

```

```
376 |         markdown_report: lastAnalysisData.markdown_report ||
377 |         '',
378 |         });
379 |
380 |         const json = await res.json();
381 |         if (!res.ok) throw new Error(json.detail || 'Save error'
382 | );
383 |         alert(`❌ ${json.message} || 'Разговор успешно сохранен в
384 | базу знаний RAG!'`);
385 |     } catch (err) {
386 |         console.error('[VoiceTab] Failed to save in RAG:', err);
387 |         alert(`Ошибка сохранения в RAG: ${err.message}`);
388 |     }
389 |
390 | function escapeHtml(text) {
391 |     if (!text) return '';
392 |     return String(text)
393 |         .replace(/&/g, '&amp;')
394 |         .replace(/</g, '&lt;')
395 |         .replace(/>/g, '&gt;')
396 |         .replace(/"/g, '&quot;')
397 |         .replace(/'/g, '&#039;');
398 | }
399 |
400 | if (document.readyState === 'loading') {
401 |     document.addEventListener('DOMContentLoaded', initVoiceTab
402 | );
403 | } else {
404 |     initVoiceTab();
405 | }
406 | }());
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9     Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12     Web Route:
13         /windows/api/webgui/website_monitor_tab/index.html
14
15     Tab Frame Integration:
16         <iframe src="/windows/api/webgui/website_monitor_tab/index.html" class="w-100 h-100 border-0"></iframe>
17
18 File: index.html
19 Project: ai-breadboard
20 Package: windows/api/webgui/website_monitor_tab
21 Author: hypo69
22 Copyright: © 2026 hypo69
23 Updated: 2026-10-08 03:08:00
24 =====
25
26 -->
27
28 <!-- Website Intelligence Monitor Application Tab -->
29 <style>
30     .webmon-container {
31         padding: 0.5rem 0.75rem;
32     }
33     .webmon-card {
34         background: #0f172a !important;
35         border: 1px solid #334155 !important;
36         border-radius: 8px;
```

```
33 | padding: 0.45rem 0.75rem;
34 | display: flex;
35 | flex-direction: column;
36 | justify-content: space-between;
37 | min-height: 58px;
38 | gap: 0.2rem;
39 | }
40 | .webmon-label {
41 |     font-size: 0.68rem;
42 |     text-transform: uppercase;
43 |     letter-spacing: 0.05em;
44 |     color: #94a3b8;
45 |     line-height: 1.1;
46 | }
47 | .webmon-value {
48 |     font-size: 1.25rem;
49 |     font-weight: 700;
50 |     color: #f8fafc;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
52 |     line-height: 1.15;
53 | }
54 | .webmon-panel {
55 |     background: #0f172a !important;
56 |     border: 1px solid #334155 !important;
57 |     border-radius: 8px;
58 |     overflow: hidden;
59 | }
60 | .webmon-panel-header {
61 |     background: #1e293b !important;
62 |     border-bottom: 1px solid #334155 !important;
63 |     padding: 0.45rem 0.85rem;
64 |     font-weight: 600;
65 |     font-size: 0.82rem;
66 |     color: #38bdf8;
67 |     display: flex;
68 |     align-items: center;
69 |     justify-content: space-between;
```

```

70 | }
71 | .webmon-table {
72 |     width: 100%;
73 |     font-size: 0.78rem;
74 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
75 |     border-collapse: collapse;
76 | }
77 | .webmon-table th {
78 |     background: #1e293b;
79 |     padding: 5px 8px;
80 |     color: #94a3b8;
81 |     font-weight: 600;
82 |     border-bottom: 1px solid #334155;
83 | }
84 | .webmon-table td {
85 |     padding: 4px 8px;
86 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
87 | }
88 | </style>
89 |
90 | <div class="webmon-container">
91 |     <!-- Header Alert & Status (Compact Strip) -->
92 |     <div class="card shadow-sm border-secondary-subtle py-1.5 px
    | -3 mb-2 d-flex justify-content-between align-items-center flex
    | -wrap gap-2">
93 |         <div class="d-flex align-items-center gap-2 flex-wrap">
94 |             <span class="fs-5">❏ </span>
95 |             <span class="fw-bold text-white small" style="font-size:
    | 0.92rem;">Website Intelligence</span>
96 |             <span class="badge bg-dark border border-secondary text-
    | info fw-normal" style="font-size: 0.7rem;">Traffic Monitor</sp
    | an>
97 |             <span class="text-secondary small d-none d-lg-inline" st
    | yle="font-size: 0.76rem;">| GA4 Realtime, Google Search Consol
    | e CTR & аномалии</span>
98 |         </div>
99 |         <div class="d-flex align-items-center gap-1.5 flex-wrap">

```

```

100      <span class="badge rounded-pill bg-success-subtle text-s
      uccess border border-success px-2 py-1" style="font-size: 0.72
      rem;" id="webmon-status-badge">
101          • АКТИВЕН
102      </span>
103      <button class="btn btn-xs btn-outline-info rounded-pill
      px-2.5 py-1" style="font-size: 0.75rem;" id="btn-webmon-refres
      h">
104          <i class="bi bi-arrow-clockwise me-1"></i> Обновить
105      </button>
106      <button class="btn btn-xs btn-outline-warning rounded-pi
      ll px-2 py-1" style="font-size: 0.75rem;" id="btn-webmon-confi
      g">
107          <i class="bi bi-gear-fill me-1"></i> Настройки
108      </button>
109  </div>
110 </div>
111
112 <!-- Top Metrics Bar (Compact Density Grid) -->
113 <div class="row g-2 mb-2">
114     <div class="col-6 col-md-3">
115         <div class="webmon-card">
116             <span class="webmon-label">Онлайн (Realtime)</span>
117             <div class="d-flex align-items-baseline justify-conten
      t-between mt-1">
118                 <span class="webmon-value text-success" id="webmon-r
      ealtime-users">0</span>
119                 <small class="text-muted" style="font-size: 0.68rem;
      ">Пользователей</small>
120             </div>
121         </div>
122     </div>
123     <div class="col-6 col-md-3">
124         <div class="webmon-card">
125             <span class="webmon-label">Доступность / Ping</span>
126             <div class="d-flex align-items-baseline justify-conten
      t-between mt-1">
127                 <span class="webmon-value text-info" style="font-siz

```



```

    e: 0.95rem;" id="webmon-availability">100% (24ms)</span>
128     </div>
129   </div>
130 </div>
131 <div class="col-6 col-md-3">
132   <div class="webmon-card">
133     <span class="webmon-label">GSC CTR</span>
134     <div class="d-flex align-items-baseline justify-content-between mt-1">
135       <span class="webmon-value text-warning" id="webmon-ctr-score">4.2%</span>
136       <small class="text-muted" style="font-size: 0.68rem;">Search Console</small>
137     </div>
138   </div>
139 </div>
140 <div class="col-6 col-md-3">
141   <div class="webmon-card">
142     <span class="webmon-label">Индекс здоровья</span>
143     <div class="d-flex align-items-baseline justify-content-between mt-1">
144       <span class="webmon-value text-light" id="webmon-health-score">98/100</span>
145     </div>
146   </div>
147 </div>
148 </div>
149
150 <!-- Main Content Panels Grid -->
151 <div class="row g-3">
152   <!-- Top Pages Panel -->
153   <div class="col-lg-6">
154     <div class="webmon-panel h-100">
155       <div class="webmon-panel-header">
156         <span>📄 Популярные страницы (GA4)</span>
157         <span class="badge bg-secondary" id="webmon-pages-badge">Топ-10</span>
158       </div>

```

```

159 |         <div class="table-responsive">
160 |             <table class="webmon-table">
161 |                 <thead>
162 |                     <tr>
163 |                         <th>Путь страницы</th>
164 |                         <th class="text-end">Просмотры</th>
165 |                         <th class="text-end">Пользователи</th>
166 |                     </tr>
167 |                 </thead>
168 |                 <tbody id="webmon-pages-table-body">
169 |                     <tr><td colspan="3" class="text-center text-mute
170 | d p-3">Загрузка популярных страниц...</td></tr>
171 |                     </tbody>
172 |                 </table>
173 |             </div>
174 |         </div>
175 |
176 |         <!-- Traffic Sources & Diagnostics Panel -->
177 |         <div class="col-lg-6">
178 |             <div class="webmon-panel h-100">
179 |                 <div class="webmon-panel-header">
180 |                     <span>▯ Каналы трафика и AI Диагностика</span>
181 |                     <span class="badge bg-info text-dark">GA4 & Anomalie
182 | s</span>
183 |                 </div>
184 |                 <div class="p-3">
185 |                     <div id="webmon-traffic-channels" class="mb-3">
186 |                         <div class="text-muted small text-center">Загрузка
187 | источников...</div>
188 |                     </div>
189 |                     <hr class="border-secondary my-2">
190 |                     <h6 class="fw-bold text-info mb-1" id="webmon-diag-t
191 | itle">Анализ трендов трафика...</h6>
192 |                     <p class="small text-light mb-0" id="webmon-diag-sum
193 | mary">Ожидание нормализованного отчёта...</p>
194 |                     </div>
195 |                 </div>
196 |             </div>
197 |         </div>

```

```
192 |     </div>
193 | </div>
194 | </div>
195 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Website Monitor Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/website_monitor_tab/main.js?v=20261001_v1" type="module"></script>
11 *
12 * File: main.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/website_monitor_tab
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-01 13:04:40
18 * =====
19 */
20
21 // =====
22 // Process Name: Website Intelligence Monitor Web Tab Module
23 // =====
24 // Description:
25 //   Client-side JavaScript controller for the Website Intelligence Monitor
26 //   administrative web interface tab.
27 //
28 // File: main.js
29 // Project: ai-breadboard
```

```

30 // Package: src.api.webinterface.website_monitor_tab
31 // Author: hypo69
32 // Copyright: © 2026 hypo69
33 // =====
34 |=====
35 (function() {
36     let isWebmonInitialized = false;
37
38     async function fetchRealtime() {
39         try {
40             const res = await fetch('/api/v1/website-monitor/realtime');
41             if (!res.ok) return;
42             const data = await res.json();
43
44             const usersEl = document.getElementById('webmon-realtime-users');
45             if (usersEl) usersEl.innerText = data.active_users || 0;
46         } catch (e) {
47             console.error('[WebsiteMonitorTab] Failed to fetch realtime:', e);
48         }
49     }
50
51     async function fetchSummary() {
52         try {
53             const res = await fetch('/api/v1/website-monitor/summary');
54             if (!res.ok) return;
55             const data = await res.json();
56
57             const availEl = document.getElementById('webmon-availability');
58             const ctrEl = document.getElementById('webmon-ctr-score');
59             const healthEl = document.getElementById('webmon-health-score');

```

```
60 |     const diagTitle = document.getElementById('webmon-diag-t
    | itle');
61 |     const diagSummary = document.getElementById('webmon-diag
    | -summary');
62 |
63 |     if (data.technical) {
64 |         if (availEl) availEl.innerText = `${data.technical.ava
    | ilability_pct || 100}% (${data.technical.avg_latency_ms || 0}
    | ms)`;
65 |     }
66 |     if (data.search_console) {
67 |         if (ctrEl) ctrEl.innerText = `${data.search_console.av
    | g_ctr || 0}%`;
68 |     }
69 |     if (healthEl) healthEl.innerText = `${data.health_score
    | || 100}/100`;
70 |
71 |     if (diagTitle) diagTitle.innerText = `Статус: ${data.sta
    | tus || 'OK'}`;
72 |     if (diagSummary) diagSummary.innerText = data.summary ||
    | 'Все веб-сервисы и каналы трафика функционируют стабильно.';
73 |     } catch (e) {
74 |         console.error('[WebsiteMonitorTab] Failed to fetch summa
    | ry:', e);
75 |     }
76 | }
77 |
78 | async function fetchPages() {
79 |     try {
80 |         const res = await fetch('/api/v1/website-monitor/pages?l
    | imit=8');
81 |         if (!res.ok) return;
82 |         const pages = await res.json();
83 |
84 |         const tbody = document.getElementById('webmon-pages-tabl
    | e-body');
85 |         if (tbody) {
86 |             if (!Array.isArray(pages) || pages.length === 0) {
```

```

87 |         tbody.innerHTML = '<tr><td colspan="3" class="text-c
    | enter text-muted p-3">Нет данных о просмотрах.</td></tr>';
88 |         return;
89 |     }
90 |     tbody.innerHTML = pages.map((p, idx) => `
91 |         <tr class="webmon-page-row" data-idx="${idx}" style=
    | "cursor: pointer;" title="Нажмите для AI-анализа страницы">
92 |             <td class="text-info text-truncate" style="max-wid
    | th: 240px;" title="${p.page_path || '/'}">${p.page_path || '/'
    | }</td>
93 |             <td class="text-end text-light">${p.screen_page_vi
    | ews || p.views || 0}</td>
94 |             <td class="text-end text-secondary">${p.active_use
    | rs || p.users || 0}</td>
95 |         </tr>
96 |     `).join('');
97 |
98 |     tbody.querySelectorAll('.webmon-page-row').forEach(row
    | => {
99 |         row.onclick = () => {
100 |             const idx = parseInt(row.getAttribute('data-idx'),
    | 10);
101 |             const p = pages[idx];
102 |             if (!p) return;
103 |
104 |             if (window.AITableModal) {
105 |                 window.AITableModal.show({
106 |                     icon: '🌐',
107 |                     title: `Веб-страница ${p.page_path || '/'}`,
108 |                     subtitle: `Просмотров: ${p.screen_page_views |
    | p.views || 0} | Пользователей: ${p.active_users || p.users |
    | 0}`,
109 |                     tableType: 'website',
110 |                     badges: [
111 |                         { text: `${p.screen_page_views || 0} views`,
    | class: 'badge bg-info text-dark' },
112 |                         { text: `${p.active_users || 0} users`, clas
    | s: 'badge bg-success' }

```

```

113         ],
114         metadata: [
115             { label: 'URL / Путь страницы', value: p.page_path || '/' },
116             { label: 'Просмотры страниц', value: String(
117 p.screen_page_views || p.views || 0) },
117             { label: 'Активные пользователи', value: String(
118 p.active_users || p.users || 0) },
118             { label: 'Показатель отказов', value: p.bounce_rate ? `${p.bounce_rate}%` : 'N/A' },
119             { label: 'Средняя длительность', value: p.avg_session_duration ? `${p.avg_session_duration} сек` : 'N/A' }
120         ],
121         rawTitle: 'Аналитика веб-страницы',
122         rawContent: JSON.stringify(p, null, 2),
123         requestData: {
124             url: p.page_path,
125             views: p.screen_page_views || p.views,
126             users: p.active_users || p.users
127         }
128     });
129 }
130 };
131 });
132 }
133 } catch (e) {
134     console.error('[WebsiteMonitorTab] Failed to fetch pages
135 :', e);
136 }
137
138 async function fetchTrafficSources() {
139     try {
140         const res = await fetch('/api/v1/website-monitor/traffic
141 -sources');
142         if (!res.ok) return;
143         const channels = await res.json();

```



```

144     const container = document.getElementById('webmon-traffic-channels');
145     if (container && Array.isArray(channels)) {
146         container.innerHTML = channels.slice(0, 5).map(c => `
147             <div class="d-flex justify-content-between align-items-center py-1 border-bottom border-secondary-subtle small">
148                 <span class="text-light">${c.channel_group || c.name || 'Organic'}</span>
149                 <span class="badge bg-secondary">${c.sessions || c.users || 0} сессий</span>
150             </div>
151         `).join('');
152     }
153     } catch (e) {
154         console.error('[WebsiteMonitorTab] Failed to fetch traffic sources:', e);
155     }
156 }
157
158 function initWebsiteMonitorTab() {
159     fetchRealtime();
160     fetchSummary();
161     fetchPages();
162     fetchTrafficSources();
163
164     if (!isWebmonInitialized) {
165         const refreshBtn = document.getElementById('btn-webmon-refresh');
166         if (refreshBtn) {
167             refreshBtn.addEventListener('click', () => {
168                 fetchRealtime();
169                 fetchSummary();
170                 fetchPages();
171                 fetchTrafficSources();
172             });
173         }
174
175         const configBtn = document.getElementById('btn-webmon-co

```

```
    nfig');
176 |         if (configBtn) {
177 |             configBtn.addEventListener('click', () => {
178 |                 if (typeof window.openAppConfigModal === 'function')
179 |                 {
180 |                     window.openAppConfigModal('website_monitor');
181 |                 }
182 |             });
183 |             isWebmonInitialized = true;
184 |         }
185 |     }
186 |
187 |     window.initWebsiteMonitorTab = initWebsiteMonitorTab;
188 | })();
189 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 | Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |   Web Route:
11 |     /windows/api/webgui/wikipedia_research_tab/index.html
12 |
13 |   Tab Frame Integration:
14 |     <iframe src="/windows/api/webgui/wikipedia_research_tab/
15 | index.html" class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/wikipedia_research_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!-- Wikipedia Research & Multi-Model Benchmark Laboratory Tab
28 | -->
29 <style>
30 | .wiki-lab-container {
31 |   padding: 1rem;
32 | }
33 | .wiki-card {
34 |   background: #0f172a !important;
35 |   border: 1px solid #334155 !important;
```

```
32 |     border-radius: 10px;
33 |     padding: 1.25rem;
34 |     margin-bottom: 1rem;
35 | }
36 | .wiki-card-title {
37 |     font-size: 1rem;
38 |     font-weight: 700;
39 |     color: #38bdf8;
40 |     margin-bottom: 0.75rem;
41 |     display: flex;
42 |     align-items: center;
43 |     gap: 0.5rem;
44 | }
45 | .wiki-badge-pos {
46 |     background: rgba(34, 197, 94, 0.2);
47 |     color: #4ade80;
48 |     border: 1px solid rgba(34, 197, 94, 0.4);
49 |     padding: 2px 8px;
50 |     border-radius: 4px;
51 |     font-weight: 600;
52 | }
53 | .wiki-badge-neg {
54 |     background: rgba(239, 68, 68, 0.2);
55 |     color: #f87171;
56 |     border: 1px solid rgba(239, 68, 68, 0.4);
57 |     padding: 2px 8px;
58 |     border-radius: 4px;
59 |     font-weight: 600;
60 | }
61 | .wiki-badge-neu {
62 |     background: rgba(234, 179, 8, 0.2);
63 |     color: #facc15;
64 |     border: 1px solid rgba(234, 179, 8, 0.4);
65 |     padding: 2px 8px;
66 |     border-radius: 4px;
67 |     font-weight: 600;
68 | }
69 | .wiki-table {
```

```

70     width: 100%;
71     font-size: 0.85rem;
72     border-collapse: collapse;
73     color: #f1f5f9;
74 }
75 .wiki-table th {
76     background: #1e293b;
77     padding: 8px 12px;
78     color: #94a3b8;
79     font-weight: 600;
80     border-bottom: 1px solid #334155;
81     text-align: left;
82 }
83 .wiki-table td {
84     padding: 8px 12px;
85     border-bottom: 1px solid #1e293b;
86 }
87 .wiki-table tr:hover {
88     background: rgba(51, 65, 85, 0.3);
89 }
90 .wiki-checkbox-grid {
91     display: grid;
92     grid-template-columns: repeat(auto-fill, minmax(130px, 1fr
93 ));
94     gap: 0.5rem;
95 }
96 </style>
97 <div class="wiki-lab-container">
98     <!-- Header -->
99     <div class="d-flex justify-content-between align-items-cente
100 r mb-3">
101         <div>
102             <h4 class="fw-bold text-white mb-0">□ Wikipedia Researc
103 h & Model Laboratory</h4>
104             <small class="text-muted">Кросс-языковой сравнительный а
105 нализ контента Википедии и бенчмарк интерпретации моделей ИИ</
106 small>

```

```
103     </div>
104     <div class="d-flex gap-2">
105         <button class="btn btn-outline-info btn-sm" id="btn-wiki
-health" title="Проверить статус API">
106             <i class="bi bi-heart-pulse"></i> Статус
107         </button>
108     </div>
109 </div>
110
111 <!-- Experiment Configuration Panel -->
112 <div class="row g-3 mb-3">
113     <div class="col-lg-8">
114         <div class="wiki-card">
115             <div class="wiki-card-title">
116                 <i class="bi bi-sliders"></i> Параметры исследования
117             </div>
118
119             <!-- Topic Search -->
120             <div class="mb-3">
121                 <label class="form-label text-light small fw-bold">Т
ема для исследования (Wikipedia Topic):</label>
122                 <div class="input-group">
123                     <span class="input-group-text bg-dark border-secon
dary text-info"><i class="bi bi-search"></i></span>
124                     <input type="text" class="form-control bg-dark tex
t-white border-secondary" id="wiki-topic-input" placeholder="На
пример: Israel-Gaza war, Quantum computing, Artificial intell
igence" value="Israel-Gaza war">
125                     <button class="btn btn-primary px-3" id="btn-run-e
xp-a" title="Запустить кросс-языковое сравнение одной моделью"
>
126                         <i class="bi bi-translate"></i> Exp A: Языки
127                     </button>
128                     <button class="btn btn-success px-3" id="btn-run-e
xp-b" title="Запустить кросс-модельный бенчмарк">
129                         <i class="bi bi-cpu"></i> Exp B: Модели
130                     </button>
131                 </div>
```

```
132 |         </div>
133 |
134 |         <!-- Languages Selector -->
135 |         <div class="mb-3">
136 |             <label class="form-label text-light small fw-bold">Я
137 | языковые разделы Википедии:</label>
138 |             <div class="wiki-checkbox-grid" id="wiki-languages-l
139 | ist">
140 |                 <div class="form-check">
141 |                     <input class="form-check-input wiki-lang-check"
142 | type="checkbox" value="en" id="lang-en" checked>
143 |                     <label class="form-check-label text-white small"
144 | for="lang-en">English (en)</label>
145 |                 </div>
146 |                 <div class="form-check">
147 |                     <input class="form-check-input wiki-lang-check"
148 | type="checkbox" value="ru" id="lang-ru" checked>
149 |                     <label class="form-check-label text-white small"
150 | for="lang-ru">Русский (ru)</label>
151 |                 </div>
152 |                 <div class="form-check">
153 |                     <input class="form-check-input wiki-lang-check"
154 | type="checkbox" value="he" id="lang-he" checked>
155 |                     <label class="form-check-label text-white small"
156 | for="lang-he">עברית (he)</label>
157 |                 </div>
158 |                 <div class="form-check">
159 |                     <input class="form-check-input wiki-lang-check"
160 | type="checkbox" value="de" id="lang-de" checked>
161 |                     <label class="form-check-label text-white small"
162 | for="lang-de">Deutsch (de)</label>
163 |                 </div>
164 |                 <div class="form-check">
165 |                     <input class="form-check-input wiki-lang-check"
166 | type="checkbox" value="fr" id="lang-fr" checked>
167 |                     <label class="form-check-label text-white small"
168 | for="lang-fr">Français (fr)</label>
169 |                 </div>
```

```
158         <div class="form-check">
159             <input class="form-check-input wiki-lang-check"
type="checkbox" value="uk" id="lang-uk">
160             <label class="form-check-label text-white small"
for="lang-uk">Українська (uk)</label>
161         </div>
162     </div>
163 </div>
164
165     <!-- Models & Analysis Dimensions -->
166     <div class="row">
167         <div class="col-md-6">
168             <label class="form-label text-light small fw-bold"
>AI Моделі (Experiment B):</label>
169             <div class="d-flex flex-wrap gap-2" id="wiki-model
s-list">
170                 <div class="form-check form-check-inline">
171                     <input class="form-check-input wiki-model-chec
k" type="checkbox" value="gemini" id="mod-gemini" checked>
172                     <label class="form-check-label text-white smal
l" for="mod-gemini">Gemini</label>
173                 </div>
174                 <div class="form-check form-check-inline">
175                     <input class="form-check-input wiki-model-chec
k" type="checkbox" value="foundry" id="mod-foundry" checked>
176                     <label class="form-check-label text-white smal
l" for="mod-foundry">Foundry Local</label>
177                 </div>
178                 <div class="form-check form-check-inline">
179                     <input class="form-check-input wiki-model-chec
k" type="checkbox" value="ollama" id="mod-ollama" checked>
180                     <label class="form-check-label text-white smal
l" for="mod-ollama">Ollama</label>
181                 </div>
182                 <div class="form-check form-check-inline">
183                     <input class="form-check-input wiki-model-chec
k" type="checkbox" value="onnx" id="mod-onnx">
184                     <label class="form-check-label text-white smal
```



```

1" for="mod-onnx">ONNX DirectML</label>
185     </div>
186     </div>
187 </div>
188     <div class="col-md-6">
189         <label class="form-label text-light small fw-bold"
>Базовая модель для Exp A:</label>
190         <select class="form-select form-select-sm bg-dark
text-white border-secondary" id="wiki-primary-model">
191             <option value="gemini" selected>Google Gemini</o
ption>
192             <option value="foundry">Microsoft Foundry Local<
/option>
193             <option value="ollama">Ollama Local LLM</option>
194             <option value="onnx">ONNX DirectML</option>
195         </select>
196     </div>
197 </div>
198 </div>
199 </div>
200
201 <!-- Quick Insights & Stats -->
202 <div class="col-lg-4">
203     <div class="wiki-card h-100">
204         <div class="wiki-card-title">
205             <i class="bi bi-lightbulb"></i> Аналитические инсайт
ы
206         </div>
207         <div id="wiki-insights-container" class="text-secondar
y small">
208             <em>Запустите Experiment A или Experiment B для отоб
ражения выводов и кросс-сравнения...</em>
209         </div>
210     </div>
211 </div>
212 </div>
213
214 <!-- Loading State -->

```

```
215 | <div id="wiki-loading-spinner" class="text-center my-4 d-non
    | e">
216 |     <div class="spinner-border text-primary" role="status"></d
    | iv>
217 |     <div class="text-light mt-2 small" id="wiki-loading-text">
    | Выполняется сбор статей и AI-анализ...</div>
218 | </div>
219 |
220 | <!-- Results Section -->
221 | <div id="wiki-results-section" class="d-none">
222 |     <div class="wiki-card">
223 |         <div class="wiki-card-title" id="wiki-results-title">
224 |             <i class="bi bi-bar-chart-fill"></i> Результаты экспер
    | имента
225 |         </div>
226 |
227 |         <div class="table-responsive">
228 |             <div id="wiki-results-table-container"></div>
229 |         </div>
230 |     </div>
231 | </div>
232 | </div>
233 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Wikipedia Research Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/wikipedia_research_tab
11 | /main.js?v=20261001_v1" type="module"></script>
12 *
13 * File: main.js
14 * Project: ai-breadboard
15 * Package: windows/api/webgui/wikipedia_research_tab
16 * Author: hypo69
17 * Copyright: © 2026 hypo69
18 * Updated: 2026-10-01 13:04:40
19 * =====
20 */
21 // Wikipedia Research & Multi-Model Benchmark Laboratory WebGU
22 I JS
23 (function () {
24   'use strict';
25   console.log('[WikiLab] Initializing Wikipedia Research Tab J
26   S...');
27   async function checkHealth() {
28     try {
29       const res = await fetch('/api/v1/wikipedia-research/health');
30     }
31   }
32 }
```

```
30     if (res.ok) {
31         const data = await res.json();
32         alert(`Wikipedia Research API статус: ${data.status.toUpperCase()} (приложение: ${data.app})`);
33     } else {
34         alert(`Ошибка API: HTTP ${res.status}`);
35     }
36 } catch (e) {
37     alert(`Не удалось связаться с API: ${e.message}`);
38 }
39 }
40
41 function getSelectedLanguages() {
42     const checkboxes = document.querySelectorAll('.wiki-lang-check:checked');
43     const langs = [];
44     checkboxes.forEach((cb) => langs.push(cb.value));
45     return langs.length > 0 ? langs : ['en', 'ru'];
46 }
47
48 function getSelectedModels() {
49     const checkboxes = document.querySelectorAll('.wiki-model-check:checked');
50     const models = [];
51     checkboxes.forEach((cb) => models.push(cb.value));
52     return models.length > 0 ? models : ['gemini', 'foundry'];
53 }
54
55 function formatSentimentBadge(val) {
56     if (val === undefined || val === null) return '<span class="text-muted">n/a</span>';
57     const num = parseFloat(val);
58     if (num > 0.05) {
59         return `<span class="wiki-badge-pos">+${num.toFixed(2)}</span>`;
60     } else if (num < -0.05) {
61         return `<span class="wiki-badge-neg">${num.toFixed(2)}</span>`;
62     }
63 }
```

```

62     }
63     return `${num.toFixed(2)}</span>`;
64 }
65
66 function setLoading/loading, text = 'Выполняется сбор и анализ...') {
67     const spinner = document.getElementById('wiki-loading-spinner');
68     const loadingText = document.getElementById('wiki-loading-text');
69     const resultsSec = document.getElementById('wiki-results-section');
70     if (spinner) {
71         if (loading) {
72             spinner.classList.remove('d-none');
73             if (loadingText) loadingText.textContent = text;
74             if (resultsSec) resultsSec.classList.add('d-none');
75         } else {
76             spinner.classList.add('d-none');
77             if (resultsSec) resultsSec.classList.remove('d-none');
78         }
79     }
80 }
81
82 function renderInsights/insights) {
83     const container = document.getElementById('wiki-insights-container');
84     if (!container) return;
85     if (!insights || insights.length === 0) {
86         container.innerHTML = '<div class="text-muted small">Нет выявленных значимых расхождений.</div>';
87         return;
88     }
89     let html = '<ul class="list-unstyled mb-0 d-flex flex-column gap-2">';
90     insights.forEach((ins) => {
91         html += '<li class="d-flex align-items-start gap-2 text-

```

```

    light"><i class="bi bi-check-circle-fill text-info mt-1"></i>
    <span>${ins}</span></li>`;
92     });
93     html += '</ul>';
94     container.innerHTML = html;
95   }
96
97   async function runExperimentA() {
98     const topicInput = document.getElementById('wiki-topic-input');
99     const topic = topicInput ? topicInput.value.trim() : 'Israel-Gaza war';
100    const languages = getSelectedLanguages();
101    const model = document.getElementById('wiki-primary-model')?.value || 'gemini';
102
103    setLoading(true, `Эксперимент A: сбор статей по теме "${topic}" и анализ моделью ${model}...`);
104
105    try {
106      const response = await fetch('/api/v1/wikipedia-research/experiment/languages', {
107        method: 'POST',
108        headers: { 'Content-Type': 'application/json' },
109        body: JSON.stringify({ topic, languages, model }),
110      });
111
112      if (!response.ok) {
113        throw new Error(`HTTP ${response.status}: ${await response.text()}`);
114      }
115
116      const report = await response.json();
117      renderExperimentAResults(report);
118    } catch (err) {
119      alert(`Ошибка выполнения Эксперимента A: ${err.message}`);
120    } finally {

```

```

121 |     setLoading(false);
122 |   }
123 | }
124 |
125 | function renderExperimentAResults(report) {
126 |   const titleEl = document.getElementById('wiki-results-titl
127 | e');
128 |   if (titleEl) {
129 |     titleEl.innerHTML = `</i> Exp
130 | eriment A: Кросс-языковое сравнение темы «${report.topic}» (Мо
131 | дель: ${report.model_used})`;
132 |   }
133 |
134 |   renderInsights(report.cross_language_insights);
135 |
136 |   const tableContainer = document.getElementById('wiki-resul
137 | ts-table-container');
138 |   if (!tableContainer) return;
139 |
140 |   let html = `
141 |     <table class="wiki-table">
142 |       <thead>
143 |         <tr>
144 |           <th>Язык</th>
145 |           <th>Статья</th>
146 |           <th>Тональность</th>
147 |           <th>Субъективность</th>
148 |           <th>Критика</th>
149 |           <th>Похвала</th>
150 |           <th>Неопределенность</th>
151 |           <th>Спорные тезисы</th>
152 |           <th>Фрейминг</th>
153 |           <th>Слов</th>
154 |         </tr>
155 |       </thead>
156 |       <tbody>

```

```

155 |     (report.metrics_summary_table || []).forEach((row) => {
156 |         html += `
157 |             <tr>
158 |                 <td><strong class="text-info">${row.lang.toUpperCase
|    ()}</strong></td>
159 |                 <td><a href="${row.url}" target="_blank" class="text
|    -light text-decoration-underline">${row.title}</a></td>
160 |                 <td>${formatSentimentBadge(row.sentiment)}</td>
161 |                 <td><span class="text-warning">${row.subjectivity.to
|    Fixed(2)}</span></td>
162 |                 <td><span class="text-danger">${row.criticism.toFixe
|    d(2)}</span></td>
163 |                 <td><span class="text-success">${row.praise.toFixed(
|    2)}</span></td>
164 |                 <td><span class="text-primary">${row.uncertainty.toF
|    ixed(2)}</span></td>
165 |                 <td class="text-center"><span class="badge bg-second
|    ary">${row.claims_count}</span></td>
166 |                 <td><span class="badge bg-dark border border-seconda
|    ry text-info">${row.framing}</span></td>
167 |                 <td class="text-muted small">${row.word_count.toLoca
|    leString()}</td>
168 |             </tr>
169 |         `;
170 |     });
171 |
172 |     html += `</tbody></table>`;
173 |     tableContainer.innerHTML = html;
174 | }
175 |
176 | async function runExperimentB() {
177 |     const topicInput = document.getElementById('wiki-topic-inp
|    ut');
178 |     const topic = topicInput ? topicInput.value.trim() : 'Isra
|    el-Gaza war';
179 |     const languages = getSelectedLanguages();
180 |     const models = getSelectedModels();
181 |

```



```

182     setLoading(true, `Эксперимент В: сопоставление интерпретац
    ий моделями [${models.join(', ')}]...`);
183
184     try {
185         const response = await fetch('/api/v1/wikipedia-research
    /experiment/models', {
186             method: 'POST',
187             headers: { 'Content-Type': 'application/json' },
188             body: JSON.stringify({ topic, languages, models }),
189         });
190
191         if (!response.ok) {
192             throw new Error(`HTTP ${response.status}: ${await resp
    onse.text()}`);
193         }
194
195         const report = await response.json();
196         renderExperimentBResults(report);
197     } catch (err) {
198         alert(`Ошибка выполнения Эксперимента В: ${err.message}`
    );
199     } finally {
200         setLoading(false);
201     }
202 }
203
204 function renderExperimentBResults(report) {
205     const titleEl = document.getElementById('wiki-results-titl
    e');
206     if (titleEl) {
207         titleEl.innerHTML = `</i> Experimen
    t В: Мульти-модельный бенчмарк интерпретации «${report.topic}»
    `;
208     }
209
210     renderInsights(report.model_alignment_insights);
211
212     const tableContainer = document.getElementById('wiki-resul

```

```

    ts-table-container');
213     if (!tableContainer) return;
214
215     let html = `
216         <table class="wiki-table">
217             <thead>
218                 <tr>
219                     <th>Языковой раздел</th>
220             `;
221
222     report.models.forEach((mod) => {
223         html += `<th>Модель: ${mod}</th>`;
224     });
225
226     html += `
227         </tr>
228         </thead>
229         <tbody>
230     `;
231
232     report.languages.forEach((lang) => {
233         html += `<tr><td><strong class="text-warning">${lang.toUpperCase()}</strong></td>`;
234         report.models.forEach((mod) => {
235             const sentiment = report.sentiment_grid?.[lang]?.[mod]
236             ;
237             const res = report.matrix?.[lang]?.[mod];
238             const framing = res?.metrics?.framing_tone || 'n/a';
239             html += `
240                 <td>
241                     <div class="d-flex flex-column gap-1">
242                         <div>${formatSentimentBadge(sentiment)}</div>
243                         <small class="text-muted" style="font-size: 0.75rem;">(${framing})</small>
244                     </div>
245                 </td>
246             `;
        });
    });

```

```
247 |     html += `</tr>`;
248 | });
249 |
250 |     html += `</tbody></table>`;
251 |     tableContainer.innerHTML = html;
252 | }
253 |
254 | function bindEvents() {
255 |     const btnHealth = document.getElementById('btn-wiki-health
256 | ');
257 |     if (btnHealth) btnHealth.onclick = checkHealth;
258 |
259 |     const btnExpA = document.getElementById('btn-run-exp-a');
260 |     if (btnExpA) btnExpA.onclick = runExperimentA;
261 |
262 |     const btnExpB = document.getElementById('btn-run-exp-b');
263 |     if (btnExpB) btnExpB.onclick = runExperimentB;
264 | }
265 |
266 | window.initWikipediaResearchTab = function () {
267 |     console.log('[WikiLab] initWikipediaResearchTab triggered'
268 | );
269 |     bindEvents();
270 | };
271 |
272 | if (document.readyState === 'loading') {
273 |     document.addEventListener('DOMContentLoaded', bindEvents);
274 | } else {
275 |     bindEvents();
276 | }
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон клиентского веб-интерфейса index.
10
11 Usage Examples:
12   Web Route:
13     /windows/api/webgui/windows_admin_tab/index.html
14
15   Tab Frame Integration:
16     <iframe src="/windows/api/webgui/windows_admin_tab/index
17     .html" class="w-100 h-100 border-0"></iframe>
18
19 File: index.html
20 Project: ai-breadboard
21 Package: windows/api/webgui/windows_admin_tab
22 Author: hypo69
23 Copyright: © 2026 hypo69
24 Updated: 2026-10-06 11:38:46
25 =====
26 =====
27 -->
28
29 <!-- Windows System Administrator Application Tab -->
30 <style>
31   .winadmin-container {
32     padding: 0.5rem 0.75rem;
33   }
34   .winadmin-table {
35     width: 100%;
36     font-size: 0.78rem;
37     font-family: var(--font-mono, monospace);
```

```
33     border-collapse: collapse;
34 }
35 .winadmin-table th {
36     background: var(--surface-2);
37     padding: 5px 8px;
38     color: var(--text-muted);
39     font-weight: 600;
40     border-bottom: 1px solid var(--border-color);
41 }
42 .winadmin-table td {
43     padding: 4px 8px;
44     border-bottom: 1px solid var(--border-subtle, rgba(51, 65,
85, 0.4));
45 }
46 .winadmin-filter-btn.active {
47     background-color: #3b82f6 !important;
48     color: #fff !important;
49     border-color: #3b82f6 !important;
50 }
51 .badge-hidden {
52     background-color: #6b21a8;
53     color: #e9d5ff;
54     border: 1px solid #9333ea;
55 }
56 .badge-admin {
57     background-color: #9a3412;
58     color: #ffedd5;
59     border: 1px solid #ea580c;
60 }
61 .badge-disabled {
62     background-color: rgba(239, 68, 68, 0.2);
63     color: #f87171;
64     border: 1px solid rgba(239, 68, 68, 0.5);
65     font-weight: bold;
66 }
67 </style>
68
69 <div class="winadmin-container theme-sysadmin">
```

```
70 | <!-- Top Metrics Bar (Compact Density Grid) -->
71 | <div class="row g-2 mb-2">
72 |     <div class="col-6 col-md-3">
73 |         <div class="winadmin-card">
74 |             <span class="winadmin-label">Хост / Домен</span>
75 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
76 |                 <span class="winadmin-value text-truncate" id="winadmin-hostname" style="font-size: 0.95rem;">-</span>
77 |             </div>
78 |         </div>
79 |     </div>
80 |     <div class="col-6 col-md-3">
81 |         <div class="winadmin-card">
82 |             <span class="winadmin-label">Active Directory</span>
83 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
84 |                 <span class="winadmin-value text-info text-truncate" id="winadmin-ad-status" style="font-size: 0.9rem;">Локальный хост</span>
85 |             </div>
86 |         </div>
87 |     </div>
88 |     <div class="col-6 col-md-3">
89 |         <div class="winadmin-card">
90 |             <span class="winadmin-label">Пользователи (Все / Онлайн)</span>
91 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
92 |                 <span class="winadmin-value text-success" id="winadmin-user-count">0</span>
93 |                 <span class="badge bg-dark border border-secondary text-muted small" id="winadmin-user-breakdown" style="font-size: 0.7rem;">0 скрытых</span>
94 |             </div>
95 |         </div>
96 |     </div>
97 | </div class="col-6 col-md-3">
```

```

98         <div class="winadmin-card">
99             <span class="winadmin-label">События безопасности</spa
n>
100         <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
101             <span class="winadmin-value text-warning" id="winadm
in-events-count">0</span>
102         </div>
103     </div>
104 </div>
105 </div>
106
107 <!-- Main Panels Row 1: Users & Security Log -->
108 <div class="row g-3 mb-3">
109     <!-- User Accounts & Sessions Panel -->
110     <div class="col-lg-7">
111         <div class="winadmin-panel h-100">
112             <div class="winadmin-panel-header d-flex justify-conte
nt-between align-items-center flex-wrap gap-2">
113                 <div class="d-flex align-items-center gap-2">
114                     <span><i class="bi bi-people-fill me-1"></i> Учетн
ые записи и сессии Windows</span>
115                     <span class="badge bg-secondary" id="winadmin-user
s-badge">0 аккаунтов</span>
116                 </div>
117                 <!-- Filter Buttons Toolbar -->
118                 <div class="btn-group btn-group-sm" role="group" id=
"winadmin-user-filters">
119                     <button type="button" class="btn btn-outline-secon
dary py-0 px-2 winadmin-filter-btn active" data-filter="all">В
се</button>
120                     <button type="button" class="btn btn-outline-secon
dary py-0 px-2 winadmin-filter-btn" data-filter="active">В сер
ии</button>
121                     <button type="button" class="btn btn-outline-secon
dary py-0 px-2 winadmin-filter-btn" data-filter="hidden">Скрыт
ые</button>
122                     <button type="button" class="btn btn-outline-secon

```

```

123         </div>
124     </div>
125     <div class="p-0 table-responsive" style="max-height: 3
126         <table class="winadmin-table">
127             <thead>
128                 <tr>
129                     <th>Учетная запись</th>
130                     <th>Группы / SID</th>
131                     <th>Профиль / Диск</th>
132                     <th>Ресурсы</th>
133                     <th>Статус</th>
134                     <th style="text-align: right;">Действие</th>
135                 </tr>
136             </thead>
137             <tbody id="winadmin-users-tbody">
138                 <tr><td colspan="6" class="text-center text-mute
139                     d p-3">Загрузка пользователей и метрик...</td></tr>
140             </tbody>
141         </table>
142     </div>
143 </div>
144
145 <!-- Security Events Audit Log Panel -->
146 <div class="col-lg-5">
147     <div class="winadmin-panel h-100">
148         <div class="winadmin-panel-header">
149             <span><i class="bi bi-shield-exclamation me-1"></i>
150             Общий журнал безопасности</span>
151             <span class="badge bg-secondary" id="winadmin-events
152             -badge">0 записей</span>
153         </div>
154         <div class="p-0 table-responsive" style="max-height: 3
155             80px; overflow-y: auto;">
156             <table class="winadmin-table">

```



```
154 |         <thead>
155 |             <tr>
156 |                 <th>Время</th>
157 |                 <th>Event ID</th>
158 |                 <th>Уровень</th>
159 |                 <th>Источник / Описание</th>
160 |             </tr>
161 |         </thead>
162 |         <tbody id="winadmin-events-tbody">
163 |             <tr><td colspan="4" class="text-center text-mute
164 |             </tbody>
165 |         </table>
166 |     </div>
167 | </div>
168 | </div>
169 | </div>
170 | </div>
171 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Windows Admin Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10 *     <script src="/windows/api/webgui/windows_admin_tab/main
    .js?v=20261001_v1" type="module"></script>
11 *
12 * File: main.js
13 * Project: ai-breadboard
14 * Package: windows/api/webgui/windows_admin_tab
15 * Author: hypo69
16 * Copyright: © 2026 hypo69
17 * Updated: 2026-10-04 07:09:00
18 * =====
19 */
20
21 // =====
22 // Process Name: Windows System Administrator Web Tab Module
23 // =====
24 // Description:
25 //   Клиентский контроллер вкладки Windows System Administrator:
26 //   - Отображение всех локальных и доменных учетных записей W
    indows
27 //   - Обнаружение скрытых пользователей (SpecialAccounts User
    List)
```

```

28 // - Подробная телеметрия ресурсов (память, CPU, процессы, п
   | рофили на диске)
29 // - Фильтрация (Все, В сети, Скрытые, Администраторы)
30 // - Интерактивное досье с поддержкой AI-аудита прав и безоп
   | асности
31 //
32 // File: main.js
33 // Project: ai-breadboard
34 // Package: src.api.webgui.windows_admin_tab
35 // Author: hypo69
36 // Copyright: © 2026 hypo69
37 // =====
   | =====
38
39 (function() {
40     let isWinAdminInitialized = false;
41     let currentUserFilter = 'all';
42
43     async function fetchStatus() {
44         try {
45             const res = await fetch('/api/sysadmin/status');
46             if (!res.ok) return;
47             const data = await res.json();
48
49             const hostEl = document.getElementById('winadmin-hostnam
   | e');
50             const userCountEl = document.getElementById('winadmin-us
   | er-count');
51             const userBreakdownEl = document.getElementById('winadmi
   | n-user-breakdown');
52             const adStatusEl = document.getElementById('winadmin-ad-
   | status');
53             const eventsCountEl = document.getElementById('winadmin-
   | events-count');
54
55             if (hostEl) hostEl.innerText = `${data.hostname || 'LOCA
   | L'} / ${data.domain || 'WORKGROUP'}`;
56             if (userCountEl) {

```

```

57 |         userCountEl.innerText = `${data.total_accounts || data
    | .user_count || 0} (${data.active_users || 0} онлайн)`;
58 |     }
59 |     if (userBreakdownEl) {
60 |         userBreakdownEl.innerText = `${data.hidden_users || 0}
    | скрытых`;
61 |     }
62 |     if (adStatusEl) adStatusEl.innerText = data.ad_connected
    | ? 'Подключен' : 'Локальный хост';
63 |     if (eventsCountEl) eventsCountEl.innerText = data.event_
    | count || 0;
64 |     } catch (e) {
65 |         console.error('[WinAdminTab] Failed to fetch status:', e
    | );
66 |     }
67 | }
68 |
69 | async function fetchAccounts(filterType = currentUserFilter)
    | {
70 |     try {
71 |         const res = await fetch(`/api/sysadmin/accounts?filter_t
    | ype=${encodeURIComponent(filterType)}`);
72 |         if (!res.ok) return;
73 |         const data = await res.json();
74 |         const accounts = data.accounts || [];
75 |
76 |         const tbody = document.getElementById('winadmin-users-tb
    | ody');
77 |         const badge = document.getElementById('winadmin-users-ba
    | dge');
78 |         if (badge) badge.innerText = `${accounts.length} аккаунт
    | ов (${data.total || accounts.length} всего)`;
79 |
80 |         if (tbody) {
81 |             if (accounts.length === 0) {
82 |                 tbody.innerHTML = '<tr><td colspan="6" class="text-c
    | enter text-muted p-3">Учетные записи по выбранному фильтру не
    | найдены</td></tr>';

```

```

83         return;
84     }
85
86     tbody.innerHTML = accounts.map((u, idx) => {
87         // Вычисление бейджей
88         const badges = [];
89         if (u.is_admin) badges.push('<span class="badge badge-
e-admin me-1">Admin</span>');
90         if (u.is_hidden) badges.push('<span class="badge badge-
ge-hidden me-1">Скрытый</span>');
91         if (!u.enabled) badges.push('<span class="badge badge-
e-disabled me-1">Отключен</span>');
92
93         // Статус
94         let statusBadge = '<span class="badge bg-secondary">
Офлайн</span>';
95         if (u.is_logged_in) {
96             statusBadge = '<span class="badge bg-success-subtl
e text-success border border-success">● В сети</span>';
97         } else if (!u.enabled) {
98             statusBadge = '<span class="badge badge-disabled m
e-1">Отключен</span>';
99         }
100
101         // Ресурсы
102         const resInfo = u.process_count > 0
103             ? `<span class="text-info">${u.process_count} проц
.</span> <span class="text-muted">(${u.memory_rss_mb} MB)</spa
n>`
104             : '<span class="text-muted">-</span>';
105
106         // Профиль
107         const profInfo = u.profile_size_mb > 0
108             ? `<span title="${u.profile_path}">${u.profile_siz
e_mb} MB</span>`
109             : (u.profile_path ? `<span class="text-muted" titl
e="${u.profile_path}">Локальный</span>` : '<span class="text-m
uted">-</span>');

```

```

110 |
111 |         // Группы
112 |         const groupsStr = (u.groups && u.groups.length > 0)
113 |         ? u.groups.join(', ') : 'Users';
114 |
115 |         return `
116 |             <tr class="winadmin-user-row" data-idx="${idx}" st
117 |             yle="cursor: pointer;" title="Нажмите для просмотра полного до
118 |             сье пользователя и AI-аудита">
119 |                 <td>
120 |                     <div class="d-flex align-items-center gap-1.5"
121 |                     >
122 |                         <i class="bi bi-person-badge text-info"></i>
123 |                         <div>
124 |                             <span class="fw-semibold text-white">${u.n
125 |                             ame}</span>
126 |                             <div class="mt-0.5">${badges.join('')}</di
127 |                             v>
128 |                         </div>
129 |                     </div>
130 |                 </td>
131 |                 <td>
132 |                     <div class="text-truncate text-secondary" styl
133 |                     e="max-width: 140px;" title="${groupsStr}">${groupsStr}</div>
134 |                     <div class="font-monospace text-muted small te
135 |                     xt-truncate" style="max-width: 130px;" title="${u.sid}">${u.si
136 |                     d.slice(-10) || '-'}</div>
137 |                 </td>
138 |                 <td class="small">
139 |                     ${profInfo}
140 |                 </td>
141 |                 <td class="small font-monospace">
142 |                     ${resInfo}
143 |                 </td>
144 |                 <td>${statusBadge}</td>
145 |                 <td style="text-align: right;">
146 |                     <button class="btn btn-xs btn-outline-info py-
147 |                     0 px-2 btn-inspect-user me-1" data-user="${u.name}" title="Дос

```

```

    ые и метрики">
138         Досье
139     </button>
140     ${u.is_logged_in ? `
141         <button class="btn btn-xs btn-outline-danger
py-0 px-1.5 btn-disconnect-user" data-user="${u.name}" title=
"Отключить сессию">
142             Выход
143         </button>
144         ` : ''}
145     </td>
146 </tr>
147 `;
148 }).join('');
149
150 // Клик по строке для открытия досье
151 tbody.querySelectorAll('.winadmin-user-row').forEach(r
ow => {
152     row.onclick = (evt) => {
153         if (evt.target.closest('.btn-disconnect-user')) re
turn;
154         const idx = parseInt(row.getAttribute('data-idx'),
10);
155         const u = accounts[idx];
156         if (!u) return;
157
158         openUserDossier(u);
159     };
160 });
161
162 // Кнопки "Досье"
163 tbody.querySelectorAll('.btn-inspect-user').forEach(bt
n => {
164     btn.onclick = (evt) => {
165         evt.stopPropagation();
166         const username = btn.getAttribute('data-user');
167         const u = accounts.find(a => a.name === username);
168         if (u) openUserDossier(u);

```

```

169         };
170     });
171
172     // Кнопки "Выход"
173     tbody.querySelectorAll('.btn-disconnect-user').forEach
174     (btn => {
175         btn.onclick = async (evt) => {
176             evt.stopPropagation();
177             const user = btn.getAttribute('data-user');
178             if (confirm(`Отключить сессию пользователя ${user}
179             ?`)) {
180                 await fetch(`/api/sysadmin/users/${user}/disconn
181                 ect`, { method: 'POST' });
182                 fetchAccounts();
183                 fetchStatus();
184             }
185         };
186     });
187 }
188 }
189
190 function openUserDossier(u) {
191     if (!window.AITableModal) return;
192
193     const metadata = [
194         { label: 'Имя пользователя', value: u.name },
195         { label: 'Полное имя', value: u.full_name || 'Не указано
196         ' },
197         { label: 'Описание', value: u.description || 'Отсутствует' },
198         { label: 'SID пользователя', value: u.sid || 'N/A' },
199         { label: 'Тип аккаунта', value: `${u.account_type} (${u.
200         is_admin ? 'Администратор' : 'Обычный'})` },
201         { label: 'Скрытая учетная запись', value: u.is_hidden ?

```



```

    'Да (SpecialAccounts / Built-in)' : 'Нет' },
200     { label: 'Учетная запись включена', value: u.enabled ? '
Да' : 'Отключена' },
201     { label: 'Членство в группах', value: (u.groups && u.gro
ups.length > 0) ? u.groups.join(', ') : 'Users' },
202     { label: 'Путь к профилю', value: u.profile_path || 'Не
создан' },
203     { label: 'Размер профиля на диске', value: `${u.profile_
size_mb} MB` },
204     { label: 'Последняя смена пароля', value: u.password_las
t_set || 'N/A' },
205     { label: 'Неверных попыток пароля', value: String(u.bad_
password_count || 0) },
206     { label: 'Последний вход в систему', value: u.last_logon
|| 'Не зафиксирован' },
207     { label: 'Активных процессов', value: `${u.process_count
} (RAM: ${u.memory_rss_mb} MB, CPU: ${u.cpu_percent}%)` }
208 ];
209
210     const badges = [
211       { text: u.is_logged_in ? 'Online' : 'Offline', class: u.
is_logged_in ? 'badge bg-success' : 'badge bg-secondary' }
212     ];
213     if (u.is_admin) badges.push({ text: 'Администратор', class
: 'badge bg-warning text-dark' });
214     if (u.is_hidden) badges.push({ text: 'Скрытый', class: 'ba
dge bg-purple text-white' });
215     if (!u.enabled) badges.push({ text: 'Отключен', class: 'ba
dge bg-danger' });
216
217     window.AITableModal.show({
218       icon: '👤',
219       title: `Учетная запись: ${u.name}`,
220       subtitle: `${u.full_name ? u.full_name + ' | ' : ''}SID:
${u.sid}`,
221       tableType: 'user_account',
222       badges: badges,
223       metadata: metadata,

```

```
224 |     rawTitle: 'Полная конфигурация и метрики пользователя',
225 |     rawContent: JSON.stringify(u, null, 2),
226 |     requestData: {
227 |         username: u.name,
228 |         is_admin: u.is_admin,
229 |         is_hidden: u.is_hidden,
230 |         groups: u.groups,
231 |         process_count: u.process_count,
232 |         memory_rss_mb: u.memory_rss_mb,
233 |         profile_path: u.profile_path
234 |     }
235 | });
236 | }
237 |
238 | async function fetchEvents() {
239 |     try {
240 |         const res = await fetch('/api/sysadmin/events?hours=24')
241 |         ;
242 |         if (!res.ok) return;
243 |         const data = await res.json();
244 |         const events = data.events || [];
245 |         const tbody = document.getElementById('winadmin-events-t
246 |         body');
247 |         const badge = document.getElementById('winadmin-events-b
248 |         adge');
249 |         if (badge) badge.innerText = `${events.length} записей`;
250 |
251 |         if (tbody) {
252 |             if (events.length === 0) {
253 |                 tbody.innerHTML = '<tr><td colspan="4" class="text-c
254 |                 enter text-muted p-3">События не зафиксированы</td></tr>';
255 |                 return;
256 |             }
257 |             const displayEvents = events.slice(0, 30);
258 |             tbody.innerHTML = displayEvents.map((e, idx) => `
259 |                 <tr class="winadmin-event-row" data-idx="${idx}" sty
260 |                 le="cursor: pointer;" title="Нажмите для AI-диагностики событи
```

```

я">
257         <td class="font-monospace text-muted small">${e.timestamp?.slice(11, 19) || ''}</td>
258         <td class="font-monospace fw-semibold text-info">${e.event_id}</td>
259         <td><span class="badge ${e.level === 'Warning' ? 'bg-warning-subtle text-warning' : (e.level === 'Error' ? 'bg-danger-subtle text-danger' : 'bg-info-subtle text-info')}">${e.level}</span></td>
260         <td class="small text-light">${e.source}: ${e.description}</td>
261     </tr>
262     `).join('');
263
264     tbody.querySelectorAll('.winadmin-event-row').forEach(
row => {
265         row.onclick = () => {
266             const idx = parseInt(row.getAttribute('data-idx'),
10);
267             const e = displayEvents[idx];
268             if (!e) return;
269
270             if (window.AITableModal) {
271                 window.AITableModal.show({
272                     icon: '⚠',
273                     title: `Событие #${e.event_id} (${e.source})`,
274                     subtitle: `Уровень: ${e.level} | ${e.timestamp}
|| ''`,
275                     tableType: 'generic',
276                     badges: [
277                         { text: e.level || 'Info', class: 'badge bg-info text-dark' }
278                     ],
279                     metadata: [
280                         { label: 'Event ID', value: String(e.event_id) },
281                         { label: 'Источник / Provider', value: e.source },

```

```

282 |             { label: 'Уровень важности', value: e.level
    | },
283 |             { label: 'Время регистрации', value: e.times
    | tamp }
284 |         ],
285 |         rawTitle: 'Текст сообщения журнала',
286 |         rawContent: e.description || '',
287 |         requestData: {
288 |             event_id: e.event_id,
289 |             source: e.source,
290 |             level: e.level,
291 |             description: e.description
292 |         }
293 |     });
294 | }
295 | };
296 | });
297 | }
298 | } catch (e) {
299 |     console.error('[WinAdminTab] Failed to fetch events:', e
    | );
300 | }
301 | }
302 |
303 | function initWindowsAdminTab() {
304 |     console.log('[WinAdminTab] Initializing Windows Sysadmin U
    | ser Metrics & Security tab...');
305 |     fetchStatus();
306 |     fetchAccounts('all');
307 |     fetchEvents();
308 |
309 |     if (!isWinAdminInitialized) {
310 |         const refreshBtn = document.getElementById('btn-winadmin
    | -refresh');
311 |         const configBtn = document.getElementById('btn-winadmin-
    | config');
312 |
313 |         if (refreshBtn) {

```

```
314 |         refreshBtn.onclick = () => {
315 |             fetchStatus();
316 |             fetchAccounts(currentUserFilter);
317 |             fetchEvents();
318 |         };
319 |     }
320 |
321 |     // Фильтры пользователей
322 |     const filterGroup = document.getElementById('winadmin-us
er-filters');
323 |     if (filterGroup) {
324 |         filterGroup.querySelectorAll('.winadmin-filter-btn').f
orEach(btn => {
325 |             btn.onclick = () => {
326 |                 filterGroup.querySelectorAll('.winadmin-filter-btn
').forEach(b => b.classList.remove('active'));
327 |                 btn.classList.add('active');
328 |                 currentUserFilter = btn.getAttribute('data-filter'
) || 'all';
329 |                 fetchAccounts(currentUserFilter);
330 |             };
331 |         });
332 |     }
333 |
334 |     if (configBtn) {
335 |         configBtn.onclick = () => {
336 |             if (typeof window.openAppConfigModal === 'function')
{
337 |                 window.openAppConfigModal('windows_sysadmin', 'Win
dows System Administrator');
338 |             }
339 |         };
340 |     }
341 |
342 |     isWinAdminInitialized = true;
343 | }
344 | }
345 |
```

```
346 | window.initWindowsAdminTab = initWindowsAdminTab;  
347 | }>();  
348 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Web Interface - Index View
6 =====
7
8 Description:
9   Шаблон веб-интерфейса модуля резервного копирования и управл
10  ения данными Windows.
11
12 Usage Examples:
13   Web Route:
14     /windows/api/webgui/windows_backup_tab/index.html
15
16 File: index.html
17 Project: ai-breadboard
18 Package: windows/api/webgui/windows_backup_tab
19 Author: hypo69
20 Copyright: © 2026 hypo69
21 Updated: 2026-10-04 11:20:00
22 =====
23 =====
24 -->
25
26 <!-- Windows Backup & Storage Management Tab -->
27 <style>
28   .wb-container {
29     padding: 0.65rem 0.85rem;
30   }
31   .wb-card {
32     background: #0f172a !important;
33     border: 1px solid #334155 !important;
34     border-radius: 8px;
35     padding: 0.5rem 0.8rem;
36     display: flex;
37     flex-direction: column;
```

```
33 |     justify-content: space-between;
34 |     height: 100%;
35 |     min-height: 64px;
36 |     transition: border-color 0.15s ease, box-shadow 0.15s ease
37 | ;
38 | }
39 | .wb-card:hover {
40 |     border-color: #38bdf8 !important;
41 |     box-shadow: 0 2px 10px rgba(0, 0, 0, 0.3);
42 | }
43 | .wb-card-label {
44 |     font-size: 0.7rem;
45 |     text-transform: uppercase;
46 |     letter-spacing: 0.05em;
47 |     color: #94a3b8;
48 |     font-weight: 600;
49 |     display: flex;
50 |     justify-content: space-between;
51 |     align-items: center;
52 |     line-height: 1.1;
53 | }
54 | .wb-card-value {
55 |     font-size: 1.3rem;
56 |     font-weight: 700;
57 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
58 | Consolas, monospace;
59 |     line-height: 1.2;
60 | }
61 | .wb-card-sub {
62 |     font-size: 0.72rem;
63 |     color: #64748b;
64 |     line-height: 1.1;
65 | }
66 | .wb-panel {
67 |     background: #0f172a !important;
68 |     border: 1px solid #334155 !important;
69 |     border-radius: 8px;
```



```
69 |     overflow: hidden;
70 |     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.2);
71 |     margin-bottom: 0.75rem;
72 | }
73 | .wb-panel-header {
74 |     background: #1e293b !important;
75 |     border-bottom: 1px solid #334155 !important;
76 |     padding: 0.45rem 0.85rem;
77 |     font-weight: 600;
78 |     font-size: 0.82rem;
79 |     color: #38bdf8;
80 |     display: flex;
81 |     align-items: center;
82 |     justify-content: space-between;
83 | }
84 | .wb-table {
85 |     margin-bottom: 0;
86 |     font-size: 0.8rem;
87 | }
88 | .wb-table th {
89 |     background: #1e293b;
90 |     color: #94a3b8;
91 |     font-weight: 600;
92 |     font-size: 0.72rem;
93 |     text-transform: uppercase;
94 |     border-bottom: 1px solid #334155;
95 |     padding: 0.45rem 0.65rem;
96 | }
97 | .wb-table td {
98 |     border-bottom: 1px solid #1e293b;
99 |     padding: 0.45rem 0.65rem;
100 |     vertical-align: middle;
101 | }
102 | .wb-nav-tabs {
103 |     border-bottom: 1px solid #334155;
104 | }
105 | .wb-nav-tabs .nav-link {
106 |     color: #94a3b8;
```

```

107     font-size: 0.82rem;
108     font-weight: 500;
109     border: 1px solid transparent;
110     border-top-left-radius: 6px;
111     border-top-right-radius: 6px;
112     padding: 0.45rem 1rem;
113     transition: all 0.15s ease;
114 }
115 .wb-nav-tabs .nav-link:hover {
116     color: #38bdf8;
117     border-color: #334155 #334155 transparent;
118 }
119 .wb-nav-tabs .nav-link.active {
120     color: #38bdf8;
121     background-color: #0f172a;
122     border-color: #334155 #334155 transparent;
123     font-weight: 600;
124 }
125 </style>
126
127 <div class="wb-container">
128     <!-- Top Header Bar -->
129     <div class="d-flex align-items-center justify-content-between mb-2.5 flex-wrap gap-2">
130         <div>
131             <h6 class="mb-0 fw-bold text-white d-flex align-items-center gap-2">
132                 <span>⏮ </span> Резервное копирование и управление данными Windows
133             </h6>
134             <div class="small text-muted" style="font-size: 0.72rem;">
135                 Пользовательские папки, системные библиотеки (.library-ms), служба File History, теньевые копии томов (VSS), аудит удаления файлов и RAG-поиск по архивам
136             </div>
137         </div>
138     <div class="d-flex align-items-center gap-2">

```

```
139 |         <button class="btn btn-sm btn-outline-info d-flex align-
      | items-center gap-1" id="wb-btn-open-create-lib" title="Создать
      | новую системную библиотеку Windows">
140 |             <i class="bi bi-folder-plus"></i>
141 |             <span>Создать библиотеку</span>
142 |         </button>
143 |         <button class="btn btn-sm btn-primary d-flex align-items
      | -center gap-1" id="wb-btn-trigger-backup" title="Принудительны
      | й запуск архивации через fhexec">
144 |             <i class="bi bi-play-fill"></i>
145 |             <span>Запустить бэкап</span>
146 |         </button>
147 |         <button class="btn btn-sm btn-outline-secondary d-flex a
      | lign-items-center gap-1" id="wb-btn-refresh" title="Обновить в
      | се данные вкладки">
148 |             <i class="bi bi-arrow-clockwise"></i>
149 |             <span>Обновить</span>
150 |         </button>
151 |     </div>
152 | </div>
153 |
154 | <!-- KPI Stats Row -->
155 | <div class="row g-2 mb-2.5">
156 |     <div class="col-6 col-md-3">
157 |         <div class="wb-card">
158 |             <div class="wb-card-label">
159 |                 <span>Готовность защиты</span>
160 |                 <span class="badge bg-secondary" id="wb-badge-status
      | ">-</span>
161 |             </div>
162 |             <div class="wb-card-value text-info" id="wb-val-health
      | ">- / 100</div>
163 |             <div class="wb-card-sub" id="wb-sub-health">Готовность
      | защиты данных</div>
164 |         </div>
165 |     </div>
166 |     <div class="col-6 col-md-3">
167 |         <div class="wb-card">
```

```

168         <div class="wb-card-label">
169             <span>Служба File History</span>
170             <span>□ </span>
171         </div>
172         <div class="wb-card-value text-white" id="wb-val-servi
ce">-</div>
173         <div class="wb-card-sub text-truncate" id="wb-sub-serv
ice">fhsvc статус</div>
174     </div>
175 </div>
176 <div class="col-6 col-md-3">
177     <div class="wb-card">
178         <div class="wb-card-label">
179             <span>Библиотеки / VSS</span>
180             <span>□ </span>
181         </div>
182         <div class="wb-card-value text-success" id="wb-val-lib
raries">-</div>
183         <div class="wb-card-sub" id="wb-sub-libraries">Библиот
ек / VSS снимков</div>
184     </div>
185 </div>
186 <div class="col-6 col-md-3">
187     <div class="wb-card">
188         <div class="wb-card-label">
189             <span>Аудит удаления (24ч)</span>
190             <button class="btn btn-xs btn-outline-primary py-0 p
x-1" id="wb-btn-enable-auditpol" title="Включить auditpol /sub
category:File System" style="font-size: 0.65rem;">Вкл</button>
191         </div>
192         <div class="wb-card-value text-danger" id="wb-val-dele
tions">0</div>
193         <div class="wb-card-sub" id="wb-sub-deletions">Event 4
663 / 4660</div>
194     </div>
195 </div>
196 </div>
197

```

```

198 | <!-- Sub Navigation Tabs -->
199 | <ul class="nav wb-nav-tabs mb-2" id="wbSubTabs" role="tablis
    | t">
200 |     <li class="nav-item" role="presentation">
201 |         <button class="nav-link active" id="wb-tab-storage-btn"
    | data-bs-toggle="tab" data-bs-target="#wb-pane-storage" type="b
    | utton" role="tab">
202 |             <i class="bi bi-folder2-open me-1 text-info"></i>Папки
    | и библиотеки
203 |         </button>
204 |     </li>
205 |     <li class="nav-item" role="presentation">
206 |         <button class="nav-link" id="wb-tab-backup-btn" data-bs-
    | toggle="tab" data-bs-target="#wb-pane-backup" type="button" ro
    | le="tab">
207 |             <i class="bi bi-shield-lock me-1 text-warning"></i>Рез
    | ервное копирование и VSS
208 |         </button>
209 |     </li>
210 |     <li class="nav-item" role="presentation">
211 |         <button class="nav-link" id="wb-tab-audit-btn" data-bs-t
    | oggle="tab" data-bs-target="#wb-pane-audit" type="button" role
    | ="tab">
212 |             <i class="bi bi-trash3 me-1 text-danger"></i>Аудит уда
    | ления файлов
213 |         </button>
214 |     </li>
215 | </ul>
216 |
217 | <!-- Sub Panes Content -->
218 | <div class="tab-content" id="wbTabContent">
219 |
220 |     <!-- ===== -->
221 |     <!-- Tab 1: Пользовательские папки и библиотеки -->
222 |     <!-- ===== -->
223 |     <div class="tab-pane fade show active" id="wb-pane-storage
    | " role="tabpanel">
224 |         <!-- User Shell Folders Storage & Relocation Management

```

```

-->
225     <div class="wb-panel">
226         <div class="wb-panel-header">
227             <div class="d-flex align-items-center gap-2">
228                 <i class="bi bi-hdd-network-fill text-info"></i>
229                 <span>Пользовательские директории и перенос на вто
ричные диски</span>
230             </div>
231             <div class="d-flex align-items-center gap-2">
232                 <span class="badge bg-primary-subtle text-primary
border border-primary-subtle" id="wb-total-user-size">Объем: 0
ГБ</span>
233                 <button class="btn btn-xs btn-outline-info py-0 px
-2" id="wb-btn-refresh-user-folders" title="Пересчитать размер
пользовательских папок" style="font-size: 0.72rem;">
234                 <i class="bi bi-arrow-repeat me-1"></i>Пересчита
ТЬ
235                 </button>
236             </div>
237         </div>
238
239     <div class="p-2">
240         <!-- Disks summary row -->
241         <div class="mb-2 p-2 rounded bg-dark border border-s
econdary" style="font-size: 0.76rem;" id="wb-drives-summary">
242             <div class="text-muted small mb-1 fw-bold">□ Оба
руженные физические накопители:</div>
243             <div class="d-flex flex-wrap gap-2" id="wb-drives-
list">
244                 <span class="text-muted">Загрузка информации о д
исках...</span>
245             </div>
246         </div>
247
248         <div class="table-responsive" style="max-height: 280
px; overflow-y: auto;">
249             <table class="table table-dark table-hover wb-tabl
e">

```

```

250         <thead>
251             <tr>
252                 <th>Директория</th>
253                 <th>Текущий путь</th>
254                 <th>Диск</th>
255                 <th>Размер</th>
256                 <th>Файлов</th>
257                 <th class="text-end">Действие</th>
258             </tr>
259         </thead>
260         <tbody id="wb-user-folders-tbody">
261             <tr>
262                 <td colspan="6" class="text-center text-mute
d p-3">Подсчет размера пользовательских папок...</td>
263             </tr>
264         </tbody>
265     </table>
266 </div>
267 </div>
268 </div>
269
270 <!-- System Libraries (.library-ms) -->
271 <div class="wb-panel mb-0">
272     <div class="wb-panel-header">
273         <div class="d-flex align-items-center gap-2">
274             <i class="bi bi-folder-symlink text-info"></i>
275             <span>Системные библиотеки Windows (.library-ms)</
span>
276         </div>
277         <div class="d-flex align-items-center gap-2">
278             <button class="btn btn-xs btn-outline-info py-0 px
-2" id="wb-btn-quick-add-folder" style="font-size: 0.72rem;" t
itle="Добавить папку в существующую библиотеку">
279                 <i class="bi bi-plus-circle me-1"></i>Добавить п
апку
280             </button>
281             <span class="badge bg-dark border border-secondary
text-muted" id="wb-lib-count">0 шт.</span>

```

```

282 |         </div>
283 |     </div>
284 |     <div class="table-responsive" style="max-height: 260px
    | ; overflow-y: auto;">
285 |         <table class="table table-dark table-hover wb-table"
    | >
286 |             <thead>
287 |                 <tr>
288 |                     <th>Библиотека</th>
289 |                     <th>Папки / Источники</th>
290 |                     <th>Папка сохранения</th>
291 |                     <th>Закреплена</th>
292 |                 </tr>
293 |             </thead>
294 |             <tbody id="wb-libraries-body">
295 |                 <tr>
296 |                     <td colspan="4" class="text-center py-3 text-m
    | uted">Загрузка библиотек...</td>
297 |                 </tr>
298 |             </tbody>
299 |         </table>
300 |     </div>
301 | </div>
302 | </div>
303 |
304 | <!-- ===== -->
305 | <!-- Tab 2: Резервное копирование и VSS -->
306 | <!-- ===== -->
307 | <div class="tab-pane fade" id="wb-pane-backup" role="tabpa
    | nel">
308 |     <div class="row g-2">
309 |         <!-- Left Column: File History Configuration & Recomme
    | ndations -->
310 |         <div class="col-12 col-lg-6">
311 |             <!-- File History Config -->
312 |             <div class="wb-panel">
313 |                 <div class="wb-panel-header">
314 |                     <div class="d-flex align-items-center gap-2">

```



```
315 |         <i class="bi bi-clock-history text-primary"></i>
316 |         <span>Настройка File History (История файлов)</span>
317 |     </div>
318 | </div>
319 |     <div class="p-2.5" style="font-size: 0.8rem;" id="
wb-fh-details">
320 |         <div class="mb-2">
321 |             <div class="text-muted small">Целевой каталог
бэкапа:</div>
322 |             <div class="fw-semibold text-break font-monosp
ace small text-info" id="wb-fh-target">Не настроено</div>
323 |             </div>
324 |             <div class="mb-2">
325 |                 <div class="text-muted small">Файл configuraц
и:</div>
326 |                 <div class="text-break font-monospace small te
xt-muted" id="wb-fh-config">-</div>
327 |                 </div>
328 |                 <div class="row g-2 mb-1">
329 |                     <div class="col-6">
330 |                         <div class="text-muted small">Интервал архив
ации:</div>
331 |                         <div class="fw-bold text-white" id="wb-fh-in
terval">-</div>
332 |                         </div>
333 |                         <div class="col-6">
334 |                             <div class="text-muted small">Срок хранения
версий:</div>
335 |                             <div class="fw-bold text-white" id="wb-fh-re
tention">-</div>
336 |                             </div>
337 |                         </div>
338 |                         <div class="mt-2 pt-2 border-top border-secondar
y">
339 |                             <div class="text-muted small">Последняя успешн
ая архивация:</div>
```

```

340 |         <div class="fw-semibold text-warning" id="wb-f
| h-last-backup">-</div>
341 |         </div>
342 |     </div>
343 | </div>
344 |
345 | <!-- Recommendations and Health Checklist -->
346 | <div class="wb-panel mb-0">
347 |     <div class="wb-panel-header">
348 |         <div class="d-flex align-items-center gap-2">
349 |             <i class="bi bi-shield-check text-success"></i
| >
350 |             <span>Рекомендации безопасности данных</span>
351 |         </div>
352 |     </div>
353 |     <div class="p-2.5" style="max-height: 200px; overf
| low-y: auto;">
354 |         <ul class="list-unstyled mb-0" id="wb-recommenda
| tions-list" style="font-size: 0.78rem;">
355 |             <li class="text-muted text-center py-2">Загруз
| ка рекомендаций...</li>
356 |         </ul>
357 |     </div>
358 | </div>
359 | </div>
360 |
361 | <!-- Right Column: VSS Snapshots & RAG Search -->
362 | <div class="col-12 col-lg-6">
363 |     <!-- VSS Snapshots -->
364 |     <div class="wb-panel">
365 |         <div class="wb-panel-header">
366 |             <div class="d-flex align-items-center gap-2">
367 |                 <i class="bi bi-camera text-warning"></i>
368 |                 <span>Теневые копии томов VSS (Volume Shadow C
| opies)</span>
369 |             </div>
370 |             <span class="badge bg-dark border border-seconda
| ry text-muted" id="wb-vss-count">0 шт.</span>

```

```

371 |         </div>
372 |         <div class="table-responsive" style="max-height: 1
373 | 80px; overflow-y: auto;">
374 |             <table class="table table-dark table-hover wb-ta
375 | ble">
376 |                 <thead>
377 |                     <tr>
378 |                         <th>ID снимка</th>
379 |                         <th>Исходный том</th>
380 |                         <th>Время создания</th>
381 |                         <th>Состояние</th>
382 |                     </tr>
383 |                 </thead>
384 |                 <tbody id="wb-vss-body">
385 |                     <tr>
386 |                         <td colspan="4" class="text-center py-3 te
387 | xt-muted">Загрузка VSS снимков...</td>
388 |                     </tr>
389 |                 </tbody>
390 |             </table>
391 |         </div>
392 |     </div>
393 |
394 | <!-- RAG Search in File History Archive -->
395 | <div class="wb-panel mb-0">
396 |     <div class="wb-panel-header">
397 |         <div class="d-flex align-items-center gap-2">
398 |             <i class="bi bi-search-heart text-info"></i>
399 |             <span>RAG поиск по архивам бэкапа</span>
400 |         </div>
401 |         <button class="btn btn-xs btn-outline-primary py
402 | -0 px-2" id="wb-btn-rag-sync" title="Индексировать новые верси
и файлов" style="font-size: 0.72rem;">
403 |             <i class="bi bi-arrow-repeat me-1"></i>Синхрон
изация
404 |         </button>
405 |     </div>
406 |     <div class="p-2" style="font-size: 0.8rem;">

```

```

403         <div class="input-group input-group-sm mb-2">
404             <input type="text" class="form-control bg-dark
text-white border-secondary" id="wb-rag-query" placeholder="П
иск документа или фрагмента текста..." />
405             <button class="btn btn-primary" id="wb-btn-rag
-search"><i class="bi bi-search"></i></button>
406         </div>
407         <div id="wb-rag-results" style="max-height: 160p
x; overflow-y: auto;">
408             <div class="text-muted text-center py-2 small"
>Введите запрос для поиска по архивным копиям</div>
409         </div>
410     </div>
411 </div>
412 </div>
413 </div>
414 </div>
415
416 <!-- ===== -->
417 <!-- Tab 3: Аудит удаления файлов -->
418 <!-- ===== -->
419 <div class="tab-pane fade" id="wb-pane-audit" role="tabpan
el">
420     <div class="wb-panel mb-0">
421         <div class="wb-panel-header">
422             <div class="d-flex align-items-center gap-2">
423                 <i class="bi bi-trash3-fill text-danger"></i>
424                 <span>Журнал аудита изменения и удаления файлов (E
vent 4663 / 4660)</span>
425             </div>
426             <div class="d-flex align-items-center gap-2">
427                 <button class="btn btn-xs btn-outline-info py-0 px
-2" id="wb-btn-configure-sacl" title="Настроить SACL Delete дл
я папки" style="font-size: 0.72rem;">
428                     <i class="bi bi-folder-check me-1"></i>Настроить
SACL папки
429                 </button>
430                 <span class="badge bg-danger-subtle text-danger bo

```

```

    rder border-danger-subtle" id="wb-deletions-badge">0 удалений<
    /span>
431         </div>
432     </div>
433     <div class="p-0 table-responsive" style="max-height: 3
434         80px; overflow-y: auto;">
435         <table class="table table-dark table-hover wb-table"
436         >
437             <thead>
438                 <tr>
439                     <th>Время</th>
440                     <th>Event ID</th>
441                     <th>Файл / Папка</th>
442                     <th>Процесс-инициатор</th>
443                     <th>Пользователь</th>
444                 </tr>
445             </thead>
446             <tbody id="wb-deletions-tbody">
447                 <tr><td colspan="5" class="text-center text-mute
448                 d p-3">Поиск событий удаления файлов...</td></tr>
449             </tbody>
450         </table>
451     </div>
452 </div>
453
454 <!-- Modal for Relocating Folder -->
455 <div class="modal fade" id="wb-relocate-modal" tabindex="-1"
456     aria-hidden="true">
457     <div class="modal-dialog modal-dialog-centered">
458         <div class="modal-content bg-dark text-white border-seco
459         ndary">
460             <div class="modal-header border-secondary py-2">
461                 <h6 class="modal-title fw-bold text-info d-flex alig
462                 n-items-center gap-2">
463                     <i class="bi bi-box-arrow-right"></i>

```

```

461         <span>Перенос пользовательской директории</span>
462     </h6>
463     <button type="button" class="btn-close btn-close-whi
te" data-bs-dismiss="modal" aria-label="Close"></button>
464 </div>
465     <div class="modal-body" style="font-size: 0.85rem;">
466         <input type="hidden" id="wb-relocate-folder-id" />
467         <div class="mb-2">
468             <label class="text-muted small">Переносимая папка:
</label>
469             <div class="fw-bold text-white fs-6" id="wb-reloca
te-folder-name">-</div>
470             <div class="font-monospace small text-muted text-b
reak" id="wb-relocate-folder-path">-</div>
471         </div>
472         <div class="mb-2">
473             <label class="text-muted small">Текущий размер:</l
abel>
474             <div class="fw-semibold text-warning" id="wb-reloc
ate-folder-size">-</div>
475         </div>
476         <div class="mb-2">
477             <label class="text-muted small mb-1" for="wb-reloc
ate-target-drive">Целевой диск для переноса:</label>
478             <select class="form-select form-select-sm bg-secon
dary text-white border-0" id="wb-relocate-target-drive">
479                 <option value="">Выберите диск с достаточным мес
том...</option>
480             </select>
481             <div class="form-text text-muted small" id="wb-rel
ocate-drive-hint"></div>
482         </div>
483         <div class="mb-3">
484             <label class="text-muted small mb-1" for="wb-reloc
ate-target-path">Целевая директория (папка):</label>
485             <div class="input-group input-group-sm">
486                 <input type="text" class="form-control form-cont
rol-sm bg-secondary text-white border-0 font-monospace" id="wb

```

```

    -relocate-target-path" placeholder="D:\Users\Username\Pictures
    " />
487         <button class="btn btn-outline-info d-flex align
    -items-center gap-1" type="button" id="wb-btn-browse-folder" t
    itle="Выбрать целевую папку через системный проводник Windows"
    >
488             <i class="bi bi-folder2-open"></i>
489             <span>Обзор...</span>
490         </button>
491     </div>
492     <div class="form-text text-muted small">Можно выбр
    ать готовую папку через проводник или отредактировать путь в ру
    чную.</div>
493 </div>
494 <div class="form-check mb-2">
495     <input class="form-check-input" type="checkbox" id
    ="wb-relocate-delete-source" />
496     <label class="form-check-label text-white small" f
    or="wb-relocate-delete-source">
497         Очистить исходную папку после успешного копирова
    ния и перенастройки
498     </label>
499 </div>
500 <div class="alert alert-info py-2 px-3 small mb-0">
501     <i class="bi bi-info-circle me-1"></i>
502     Перенос скопирует файлы, обновит реестр Windows (U
    ser Shell Folders) и системную библиотеку Windows, благодаря ч
    ему служба File History и Проводник продолжат работать без сбо
    ев.
503 </div>
504 </div>
505 <div class="modal-footer border-secondary py-1.5">
506     <button type="button" class="btn btn-sm btn-outline-
    secondary" data-bs-dismiss="modal">Отмена</button>
507     <button type="button" class="btn btn-sm btn-primary
    d-flex align-items-center gap-1" id="wb-btn-execute-relocate">
508         <i class="bi bi-check-circle"></i>
509         <span>Выполнить перенос</span>

```

```

510         </button>
511     </div>
512 </div>
513 </div>
514 </div>
515
516 <!-- Modal for Creating Library -->
517 <div class="modal fade" id="wb-create-lib-modal" tabindex="-
1" aria-hidden="true">
518     <div class="modal-dialog modal-dialog-centered">
519         <div class="modal-content bg-dark text-white border-seco
ndary">
520             <div class="modal-header border-secondary py-2">
521                 <h6 class="modal-title fw-bold text-info d-flex align-items-center gap-2">
522                     <i class="bi bi-folder-plus"></i>
523                     <span>Создать библиотеку Windows (.library-ms)</span>
524                 </h6>
525                 <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
526             </div>
527             <div class="modal-body" style="font-size: 0.85rem;">
528                 <div class="mb-2">
529                     <label class="text-muted small" for="wb-create-lib-name">Название библиотеки:</label>
530                     <input type="text" class="form-control form-control-sm bg-secondary text-white border-0" id="wb-create-lib-name" placeholder="Например: Архивы Проектов" />
531                 </div>
532                 <div class="mb-2">
533                     <label class="text-muted small" for="wb-create-lib-folders">Пути к папкам (по одной строке на папку):</label>
534                     <textarea class="form-control form-control-sm bg-secondary text-white border-0 font-monospace" id="wb-create-lib-folders" rows="3" placeholder="C:\Data\Backups&#10;D:\Projects"></textarea>
535                 </div>

```



```

536 |         <div class="form-check mb-2">
537 |             <input class="form-check-input" type="checkbox" id
| ="wb-create-lib-pinned" checked />
538 |             <label class="form-check-label text-white small" f
| or="wb-create-lib-pinned">
539 |                 Закрепить в панели навигации Проводника
540 |             </label>
541 |         </div>
542 |     </div>
543 |     <div class="modal-footer border-secondary py-1.5">
544 |         <button type="button" class="btn btn-sm btn-outline-
| secondary" data-bs-dismiss="modal">Отмена</button>
545 |         <button type="button" class="btn btn-sm btn-info d-f
| lex align-items-center gap-1" id="wb-btn-execute-create-lib">
546 |             <i class="bi bi-check-lg"></i>
547 |             <span>Создать</span>
548 |         </button>
549 |     </div>
550 | </div>
551 | </div>
552 | </div>
553 |
554 | <!-- Modal for Adding Folder to Library -->
555 | <div class="modal fade" id="wb-add-folder-modal" tabindex="-
| 1" aria-hidden="true">
556 |     <div class="modal-dialog modal-dialog-centered">
557 |         <div class="modal-content bg-dark text-white border-seco
| ndary">
558 |             <div class="modal-header border-secondary py-2">
559 |                 <h6 class="modal-title fw-bold text-info d-flex align-
| n-items-center gap-2">
560 |                     <i class="bi bi-folder-plus"></i>
561 |                     <span>Добавить папку в библиотеку</span>
562 |                 </h6>
563 |                 <button type="button" class="btn-close btn-close-whi
| te" data-bs-dismiss="modal" aria-label="Close"></button>
564 |             </div>
565 |             <div class="modal-body" style="font-size: 0.85rem;">

```

```

566         <div class="mb-2">
567             <label class="text-muted small" for="wb-add-folder
- lib-select">Выберите целевую библиотеку:</label>
568             <select class="form-select form-select-sm bg-secon
dary text-white border-0" id="wb-add-folder-lib-select">
569                 <option value="">Загрузка библиотек...</option>
570             </select>
571         </div>
572         <div class="mb-2">
573             <label class="text-muted small" for="wb-add-folder
- path">Полный путь к физической папке:</label>
574             <input type="text" class="form-control form-contro
l-sm bg-secondary text-white border-0 font-monospace" id="wb-a
dd-folder-path" placeholder="D:\BackupFolder" />
575         </div>
576         <div class="form-check mb-2">
577             <input class="form-check-input" type="checkbox" id
="wb-add-folder-default-save" />
578             <label class="form-check-label text-white small" f
or="wb-add-folder-default-save">
579                 Сделать основной папкой сохранения по умолчанию
580             </label>
581         </div>
582     </div>
583     <div class="modal-footer border-secondary py-1.5">
584         <button type="button" class="btn btn-sm btn-outline-
secondary" data-bs-dismiss="modal">Отмена</button>
585         <button type="button" class="btn btn-sm btn-primary
d-flex align-items-center gap-1" id="wb-btn-execute-add-folder
">
586             <i class="bi bi-check-lg"></i>
587             <span>Добавить</span>
588         </button>
589     </div>
590 </div>
591 </div>
592 </div>
593 </div>

```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Windows Backup Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля резервног
   о копирования и данных Windows.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/windows_backup_tab/mai
   n.js?v=20261004_v1" type="module"></script>
11  *
12  * File: main.js
13  * Project: ai-breadboard
14  * Package: windows/api/webgui/windows_backup_tab
15  * Author: hypo69
16  * Copyright: © 2026 hypo69
17  * Updated: 2026-10-08 03:45:00
18  * =====
19  */
20
21 // Windows Backup & Storage Manager Frontend Logic
22 (function () {
23   let _userFoldersData = null;
24   let _librariesData = [];
25
26   async function initWindowsBackupTab() {
27     console.log('[WindowsBackup] Initializing tab...');
28     bindEvents();
29     await loadBackupData();
30   }
31   window.initWindowsBackupTab = initWindowsBackupTab;
```

```
32 |
33 | function bindEvents() {
34 |     // Поддержка переключения подвкладок
35 |     const subTabButtons = document.querySelectorAll('#wbSubTab
36 | s button[data-bs-toggle="tab"]');
37 |     subTabButtons.forEach(btn => {
38 |         btn.onclick = (e) => {
39 |             e.preventDefault();
40 |             const targetSelector = btn.getAttribute('data-bs-targe
41 | t');
42 |             if (!targetSelector) return;
43 |             subTabButtons.forEach(b => b.classList.remove('active'
44 | ));
45 |             btn.classList.add('active');
46 |             const panes = document.querySelectorAll('#wbTabContent
47 | .tab-pane');
48 |             panes.forEach(p => p.classList.remove('show', 'active'
49 | ));
50 |             const targetPane = document.querySelector(targetSelect
51 | or);
52 |             if (targetPane) {
53 |                 targetPane.classList.add('show', 'active');
54 |             }
55 |         };
56 |     });
57 |     const btnRefresh = document.getElementById('wb-btn-refresh
58 | ');
59 |     if (btnRefresh) {
60 |         btnRefresh.onclick = () => loadBackupData();
61 |     }
62 |     const btnTrigger = document.getElementById('wb-btn-trigger
63 | -backup');
64 |     if (btnTrigger) {
```

```

62 |         btnTrigger.onclick = async () => {
63 |             btnTrigger.disabled = true;
64 |             btnTrigger.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span> Запуск...';
65 |             try {
66 |                 const res = await fetch('/api/v1/windows-backup/file
-history/trigger', { method: 'POST' });
67 |                 const data = await res.json();
68 |                 if (res.ok) {
69 |                     alert('Архивация File History успешно запущена: '
+ (data.message || 'OK'));
70 |                 } else {
71 |                     alert('Ошибка запуска: ' + (data.detail || JSON.st
ringify(data)));
72 |                 }
73 |             } catch (e) {
74 |                 alert('Сетевая ошибка при запуске архивации: ' + e.m
essage);
75 |             } finally {
76 |                 btnTrigger.disabled = false;
77 |                 btnTrigger.innerHTML = '<i class="bi bi-play-fill"><
/i> <span>Запустить бэкап</span>';
78 |                 await loadBackupData();
79 |             }
80 |         };
81 |     }
82 |
83 |     // Modal Create Library
84 |     const btnOpenCreateLib = document.getElementById('wb-btn-o
pen-create-lib');
85 |     if (btnOpenCreateLib) {
86 |         btnOpenCreateLib.onclick = () => {
87 |             const modalEl = document.getElementById('wb-create-lib
-modal');
88 |             if (modalEl) {
89 |                 if (window.bootstrap?.Modal) {
90 |                     new window.bootstrap.Modal(modalEl).show();
91 |                 } else {

```

```

92 |         modalEl.classList.add('show');
93 |         modalEl.style.display = 'block';
94 |     }
95 | }
96 | };
97 | }
98 |
99 |     const btnExecCreateLib = document.getElementById('wb-btn-e
xecute-create-lib');
100 |     if (btnExecCreateLib) {
101 |         btnExecCreateLib.onclick = async () => {
102 |             const name = document.getElementById('wb-create-lib-na
me')?.value?.trim();
103 |             const foldersRaw = document.getElementById('wb-create-
lib-folders')?.value || '';
104 |             const isPinned = document.getElementById('wb-create-li
b-pinned')?.checked ?? true;
105 |
106 |             if (!name) {
107 |                 alert('Укажите название библиотеки');
108 |                 return;
109 |             }
110 |
111 |             const folders = foldersRaw
112 |                 .split('\n')
113 |                 .map(f => f.trim())
114 |                 .filter(f => f.length > 0);
115 |
116 |             btnExecCreateLib.disabled = true;
117 |             btnExecCreateLib.innerHTML = '<span class="spinner-bor
der spinner-border-sm me-1"></span> Создание...';
118 |
119 |             try {
120 |                 const res = await fetch('/api/v1/windows-backup/libr
aries', {
121 |                     method: 'POST',
122 |                     headers: { 'Content-Type': 'application/json' },
123 |                     body: JSON.stringify({ name, folders, is_pinned: i

```

```

    sPinned })
124         });
125         const data = await res.json();
126         if (res.ok) {
127             alert(`Библиотека "${name}" успешно создана!`);
128             const modalEl = document.getElementById('wb-create
-lib-modal');
129             if (modalEl && window.bootstrap?.Modal) {
130                 const modalInst = window.bootstrap.Modal.getInst
ance(modalEl);
131                 if (modalInst) modalInst.hide();
132             }
133             await loadLibraries();
134         } else {
135             alert('Ошибка: ' + (data.detail || JSON.stringify(
data)));
136         }
137     } catch (e) {
138         alert('Ошибка сети: ' + e.message);
139     } finally {
140         btnExecCreateLib.disabled = false;
141         btnExecCreateLib.innerHTML = '<i class="bi bi-check-
lg"></i> <span>Создать</span>';
142     }
143 };
144 }
145
146 // Modal Add Folder to Library
147 const btnQuickAddFolder = document.getElementById('wb-btn-
quick-add-folder');
148 if (btnQuickAddFolder) {
149     btnQuickAddFolder.onclick = () => {
150         const select = document.getElementById('wb-add-folder-
lib-select');
151         if (select) {
152             select.innerHTML = _librariesData.length > 0
? _librariesData.map(l => `<option value="${escape
Html(l.name)}">${escapeHtml(l.name)} (${l.folder_count || 0} n

```



```

        апок)</option>`).join('')
154         : '<option value="">Нет доступных библиотек</optio
        n>';
155     }
156     const modalEl = document.getElementById('wb-add-folder
        -modal');
157     if (modalEl) {
158         if (window.bootstrap?.Modal) {
159             new window.bootstrap.Modal(modalEl).show();
160         } else {
161             modalEl.classList.add('show');
162             modalEl.style.display = 'block';
163         }
164     }
165 };
166 }
167
168     const btnExecAddFolder = document.getElementById('wb-btn-e
        xecute-add-folder');
169     if (btnExecAddFolder) {
170         btnExecAddFolder.onclick = async () => {
171             const libName = document.getElementById('wb-add-folder
        -lib-select')?.value;
172             const folderPath = document.getElementById('wb-add-fol
        der-path')?.value?.trim();
173             const isDefault = document.getElementById('wb-add-fold
        er-default-save')?.checked ?? false;
174
175             if (!libName || !folderPath) {
176                 alert('Выберите библиотеку и укажите путь к папке');
177                 return;
178             }
179
180             btnExecAddFolder.disabled = true;
181             btnExecAddFolder.innerHTML = '<span class="spinner-bor
        der spinner-border-sm me-1"></span> Добавление...';
182
183             try {

```

```

184         const res = await fetch(`/api/v1/windows-backup/libr
aries/${encodeURIComponent(libName)}/folders`, {
185             method: 'POST',
186             headers: { 'Content-Type': 'application/json' },
187             body: JSON.stringify({ folder_path: folderPath, is
_default_save: isDefault })
188         });
189         const data = await res.json();
190         if (res.ok) {
191             alert(`Папка успешно добавлена в библиотеку "${lib
Name}"!`);
192             const modalEl = document.getElementById('wb-add-fo
lder-modal');
193             if (modalEl && window.bootstrap?.Modal) {
194                 const modalInst = window.bootstrap.Modal.getInst
ance(modalEl);
195                 if (modalInst) modalInst.hide();
196             }
197             await loadLibraries();
198         } else {
199             alert('Ошибка: ' + (data.detail || JSON.stringify(
data)));
200         }
201     } catch (e) {
202         alert('Ошибка сети: ' + e.message);
203     } finally {
204         btnExecAddFolder.disabled = false;
205         btnExecAddFolder.innerHTML = '<i class="bi bi-check-
lg"></i> <span>Добавить</span>';
206     }
207 };
208 }
209
210 // RAG Search & Sync
211 const btnRagSync = document.getElementById('wb-btn-rag-syn
c');
212 if (btnRagSync) {
213     btnRagSync.onclick = async () => {

```

```
214 |         btnRagSync.disabled = true;
215 |         btnRagSync.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span> Синхр...';
216 |         try {
217 |             const res = await fetch('/api/v1/windows-backup/file
-history/rag/sync', {
218 |                 method: 'POST',
219 |                 headers: { 'Content-Type': 'application/json' },
220 |                 body: JSON.stringify({ force_rebuild: false })
221 |             });
222 |             const data = await res.json();
223 |             if (res.ok) {
224 |                 alert(`Синхронизация RAG завершена!\nПроиндексиров
ано: ${data.total_indexed || 0} версий файлов.`);
225 |             } else {
226 |                 alert('Ошибка синхронизации: ' + (data.detail || J
SON.stringify(data)));
227 |             }
228 |         } catch (e) {
229 |             alert('Ошибка сети: ' + e.message);
230 |         } finally {
231 |             btnRagSync.disabled = false;
232 |             btnRagSync.innerHTML = '<i class="bi bi-arrow-repeat
me-1"></i>Синхронизация';
233 |         }
234 |     };
235 | }
236 |
237 |     const btnRagSearch = document.getElementById('wb-btn-rag-s
earch');
238 |     const inputRagQuery = document.getElementById('wb-rag-quer
y');
239 |     const performRagSearch = async () => {
240 |         const query = inputRagQuery?.value?.trim();
241 |         if (!query) return;
242 |
243 |         const resultsContainer = document.getElementById('wb-rag
-results');
```

```

244     if (resultsContainer) {
245         resultsContainer.innerHTML = '<div class="text-center
text-muted py-2"><span class="spinner-border spinner-border-sm
me-1"></span>Поиск в архиве...</div>';
246     }
247
248     try {
249         const res = await fetch('/api/v1/windows-backup/file-h
istory/rag/search', {
250             method: 'POST',
251             headers: { 'Content-Type': 'application/json' },
252             body: JSON.stringify({ query, top_k: 5 })
253         });
254         const items = await res.json();
255         if (resultsContainer) {
256             if (!items || items.length === 0) {
257                 resultsContainer.innerHTML = '<div class="text-mut
ed text-center py-2 small">Ничего не найдено по запросу</div>'
;
258                 return;
259             }
260             resultsContainer.innerHTML = items.map(item => `
261                 <div class="p-1.5 mb-1 rounded bg-dark border bord
er-secondary" style="font-size: 0.75rem;">
262                     <div class="d-flex justify-content-between align
-items-center mb-0.5">
263                         <span class="fw-bold text-info text-truncate"
title="${escapeHtml(item.file_name)}">${escapeHtml(item.file_n
ame)}</span>
264                         <span class="badge bg-primary-subtle text-prim
ary">${(item.score * 100).toFixed(0)}%</span>
265                     </div>
266                     <div class="text-muted font-monospace small text
-truncate" title="${escapeHtml(item.original_path)}">${escapeH
tml(item.original_path)}</div>
267                     ${item.snippet ? `<div class="text-white-50 mt-1
small text-truncate">${escapeHtml(item.snippet)}</div>` : ''}
268                 </div>

```

```

269 |         `).join('');
270 |     }
271 | } catch (e) {
272 |     if (resultsContainer) {
273 |         resultsContainer.innerHTML = `
```

```

301         await loadBackupData();
302     } catch (err) {
303         alert(`Ошибка выполнения: ${err.message}`);
304     }
305 }
306 };
307 }
308
309 const btnConfigureSacl = document.getElementById('wb-btn-c
onfigure-sacl');
310 if (btnConfigureSacl) {
311     btnConfigureSacl.onclick = async () => {
312         const targetPath = prompt('Укажите путь к папке для на
стройки SAcl аудита удаления:', 'c:\\Users\\onela\\AppData\\Lo
cal\\AI-Breadboard');
313         if (targetPath) {
314             try {
315                 const res = await fetch('/api/v1/system/file-audit
/folder-sacl', {
316                     method: 'POST',
317                     headers: { 'Content-Type': 'application/json' },
318                     body: JSON.stringify({ path: targetPath, princip
al: 'Everyone', enable: true })
319                 });
320                 const data = await res.json();
321                 alert(data.Success ? `SAcl успешно настроен для ${
targetPath}!` : `Ошибка: ${data.Error || 'Сбой'}`);
322                 await loadFileDeletions();
323             } catch (err) {
324                 alert(`Ошибка: ${err.message}`);
325             }
326         }
327     };
328 }
329
330 const btnRefreshFolders = document.getElementById('wb-btn-
refresh-user-folders');
331 if (btnRefreshFolders) {

```

```

332 |         btnRefreshFolders.onclick = () => loadUserFoldersOverview();
333 |     }
334 |
335 |     // Modal Relocate Target Drive Change
336 |     const selectRelocateDrive = document.getElementById('wb-relocate-target-drive');
337 |     if (selectRelocateDrive) {
338 |         selectRelocateDrive.onchange = () => {
339 |             const driveVal = selectRelocateDrive.value;
340 |             const folderId = document.getElementById('wb-relocate-folder-id')?.value;
341 |             const folder = (_userFoldersData?.folders || []).find(
342 |                 f => f.folder_id === folderId);
343 |             const inputPath = document.getElementById('wb-relocate-target-path');
344 |             if (driveVal && folder && inputPath) {
345 |                 const d = driveVal.replace('/', '\\').replace(/\\+$/, '\\');
346 |                 const parts = folder.current_path.split('\\');
347 |                 const folderSubname = parts[parts.length - 1] || folder.name;
348 |                 const userMatch = folder.current_path.match(/Users\\([^\s]+)/i);
349 |                 const username = userMatch ? userMatch[1] : 'User';
350 |                 inputPath.value = `${d}\\Users\\${username}\\${folderSubname}`;
351 |             }
352 |         };
353 |     }
354 |
355 |     // Modal Relocate Browse Folder Button
356 |     const btnBrowseFolder = document.getElementById('wb-btn-browse-folder');
357 |     if (btnBrowseFolder) {
358 |         btnBrowseFolder.onclick = async () => {
359 |             const currentPathVal = document.getElementById('wb-relocate-target-path')?.value || '';

```

```

359 |         const currentDriveVal = document.getElementById('wb-re
| locate-target-drive)?.value || '';
360 |         const initialPath = currentPathVal || (currentDriveVal
| ? currentDriveVal + '\\\\' : '');
361 |
362 |         btnBrowseFolder.disabled = true;
363 |         btnBrowseFolder.innerHTML = '<span class="spinner-bord
| er spinner-border-sm me-1"></span> Выбор...';
364 |
365 |         try {
366 |             const res = await fetch(`/api/v1/windows-backup/brow
| se-folder?initial_path=${encodeURIComponent(initialPath)}`, {
367 |                 method: 'POST'
368 |             });
369 |             const data = await res.json();
370 |             if (res.ok && data.selected_path) {
371 |                 const chosenPath = data.selected_path;
372 |                 const inputPath = document.getElementById('wb-relo
| cate-target-path');
373 |                 if (inputPath) inputPath.value = chosenPath;
374 |
375 |                 // Синхронизируем выпадающий список дисков
376 |                 const driveLetter = chosenPath.match(/^([a-zA-Z]:)
| ?.[0]??.toUpperCase());
377 |                 if (driveLetter && selectRelocateDrive) {
378 |                     for (let opt of selectRelocateDrive.options) {
379 |                         if (opt.value && opt.value.toUpperCase().start
| sWith(driveLetter)) {
380 |                             selectRelocateDrive.value = opt.value;
381 |                             break;
382 |                         }
383 |                     }
384 |                 }
385 |             }
386 |         } catch (err) {
387 |             console.error('[WindowsBackup] Browse folder error:'
| , err);
388 |         } finally {

```



```

389         btnBrowseFolder.disabled = false;
390         btnBrowseFolder.innerHTML = '<i class="bi bi-folder2
-open"></i> <span>Обзор...</span>';
391     }
392 };
393 }
394
395 // Modal Relocate Execution
396 const btnExecRelocate = document.getElementById('wb-btn-ex
ecute-relocate');
397 if (btnExecRelocate) {
398     btnExecRelocate.onclick = async () => {
399         const folderId = document.getElementById('wb-relocate-
folder-id')?.value;
400         const targetDrive = document.getElementById('wb-reloca
te-target-drive')?.value;
401         const targetPath = document.getElementById('wb-relocat
e-target-path')?.value?.trim();
402         const deleteSource = document.getElementById('wb-reloc
ate-delete-source')?.checked || false;
403
404         if (!folderId || (!targetDrive && !targetPath)) {
405             alert('Пожалуйста, выберите целевой диск или укажите
целевую папку для переноса.');
```

406 return;

407 }

408

409 const destDisplay = targetPath || targetDrive;

410 if (!confirm(`Подтвердите перенос папки в "\${destDispl
ay}".\nВсе файлы будут скопированы, а системные пути и библиот
еки перенастроены.`)) {

411 return;

412 }

413

414 btnExecRelocate.disabled = true;

415 btnExecRelocate.innerHTML = '<span class="spinner-bord
er spinner-border-sm me-1"> Перенос...';

416

```
417 |         try {
418 |             const res = await fetch('/api/v1/windows-backup/user
| -folders/relocate', {
419 |                 method: 'POST',
420 |                 headers: { 'Content-Type': 'application/json' },
421 |                 body: JSON.stringify({
422 |                     folder_id: folderId,
423 |                     target_drive_letter: targetDrive || null,
424 |                     target_path: targetPath || null,
425 |                     delete_source_after: deleteSource
426 |                 })
427 |             });
428 |
429 |             const data = await res.json();
430 |             if (res.ok && data.success) {
431 |                 alert(data.message || 'Перенос успешно завершен!')
| ;
432 |                 const modalEl = document.getElementById('wb-reloca
| te-modal');
433 |                 if (modalEl && window.bootstrap?.Modal) {
434 |                     const modalInstance = window.bootstrap.Modal.get
| Instance(modalEl);
435 |                     if (modalInstance) modalInstance.hide();
436 |                 }
437 |                 await loadBackupData();
438 |             } else {
439 |                 alert('Ошибка переноса: ' + (data.detail || data.m
| essage || JSON.stringify(data)));
440 |             }
441 |         } catch (e) {
442 |             alert('Сетевая ошибка при переносе: ' + e.message);
443 |         } finally {
444 |             btnExecRelocate.disabled = false;
445 |             btnExecRelocate.innerHTML = '<i class="bi bi-check-c
| ircle"></i> <span>Выполнить перенос</span>';
446 |         }
447 |     };
448 | }
```

```
449 | }
450 |
451 | async function loadBackupData() {
452 |     await Promise.allSettled([
453 |         loadHealthReport(),
454 |         loadLibraries(),
455 |         loadFileHistoryStatus(),
456 |         loadUserFoldersOverview(),
457 |         loadVssSnapshots(),
458 |         loadFileDeletions()
459 |     ]);
460 | }
461 |
462 | async function loadHealthReport() {
463 |     try {
464 |         const res = await fetch('/api/v1/windows-backup/health')
465 |         ;
466 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
467 |         const data = await res.json();
468 |         const elHealth = document.getElementById('wb-val-health'
469 |         );
470 |         const elBadge = document.getElementById('wb-badge-status
471 |         ');
472 |         const elSub = document.getElementById('wb-sub-health');
473 |         const score = data.health_score ?? 0;
474 |         if (elHealth) elHealth.textContent = `${score} / 100`;
475 |         if (elBadge) {
476 |             elBadge.textContent = score >= 80 ? 'Готов' : score >=
477 |             50 ? 'Внимание' : 'Критично';
478 |             elBadge.className = `badge bg-${score >= 80 ? 'success
479 |             ' : score >= 50 ? 'warning' : 'danger'}`;
480 |         }
481 |         if (elSub) {
482 |             elSub.textContent = data.service_running ? 'Служба акт
483 |             ивна' : 'Служба остановлена';
484 |         }
485 |     } catch (error) {
486 |         console.error('Error loading health report:', error);
487 |     }
488 | }
```

```

481
482     const recList = document.getElementById('wb-recommendati
ons-list');
483     if (recList && Array.isArray(data.recommendations)) {
484         if (data.recommendations.length === 0) {
485             recList.innerHTML = '<li class="text-success small">
<i class="bi bi-check-circle me-1"></i>Все проверки безопаснос
ти данных пройдены успешно.</li>';
486         } else {
487             recList.innerHTML = data.recommendations.map(r => `
488                 <li class="mb-1.5 text-warning d-flex align-items-
start gap-1">
489                     <i class="bi bi-exclamation-triangle-fill text-w
arning flex-shrink-0 mt-0.5"></i>
490                     <span>${escapeHtml(r)}</span>
491                 </li>
492             `).join('');
493         }
494     }
495 } catch (e) {
496     console.warn('[WindowsBackup] Error loading health:', e)
;
497 }
498 }
499
500 async function loadLibraries() {
501     const tbody = document.getElementById('wb-libraries-body')
;
502     const countBadge = document.getElementById('wb-lib-count')
;
503     const valLibs = document.getElementById('wb-val-libraries'
);
504
505     try {
506         const res = await fetch('/api/v1/windows-backup/librarie
s');
507         if (!res.ok) throw new Error(`HTTP ${res.status}`);
508         const libs = await res.json();

```

```

509 |     _librariesData = libs || [];
510 |
511 |     if (countBadge) countBadge.textContent = `${libs.length}
шт.`;
512 |     if (vallibs) vallibs.textContent = `${libs.length}`;
513 |
514 |     if (!tbody) return;
515 |     if (!libs || libs.length === 0) {
516 |         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Библиотеки не найдены</td></tr>';
517 |         return;
518 |     }
519 |
520 |     tbody.innerHTML = libs.map(lib => {
521 |         const foldersHtml = (lib.folders || []).map(f => {
522 |             const path = typeof f === 'string' ? f : f.folder_pa
th;
523 |             const exists = typeof f === 'object' ? f.exists : tr
ue;
524 |             return `
525 |                 <div class="text-break font-monospace small d-flex
align-items-center gap-1 ${exists ? 'text-white' : 'text-dang
er'}">
526 |                     <i class="bi ${exists ? 'bi-folder-check text-in
fo' : 'bi-folder-x text-danger'}"></i>
527 |                     <span>${escapeHtml(path)}</span>
528 |                     ${!exists ? '<span class="badge bg-danger-subtle
text-danger py-0 px-1" style="font-size:0.6rem;">Недоступна</
span>' : ''}
529 |                 </div>
530 |             `;
531 |         }).join('') || '<span class="text-muted small">Нет пап
ок</span>';
532 |
533 |     return `
534 |         <tr>
535 |             <td class="fw-bold text-white">
536 |                 <i class="bi bi-collection me-1 text-primary"></

```

```

        i>${escapeHtml(lib.name)}
537         </td>
538         <td>${foldersHtml}</td>
539         <td class="text-break font-monospace small text-in
fo">
540             ${lib.default_save_folder ? escapeHtml(lib.defau
lt_save_folder) : '<span class="text-muted">—</span>'}
541         </td>
542         <td>
543             <span class="badge ${lib.is_pinned ? 'bg-primary
' : 'bg-secondary'}">${lib.is_pinned ? 'Да' : 'Нет'}</span>
544         </td>
545     </tr>
546     `;
547     }).join('');
548     } catch (e) {
549         if (tbody) tbody.innerHTML = `<tr><td colspan="4" class=
"text-center py-3 text-danger">Ошибка: ${escapeHtml(e.message)
}</td></tr>`;
550     }
551 }
552
553 async function loadFileHistoryStatus() {
554     try {
555         const res = await fetch('/api/v1/windows-backup/file-his
tory/status');
556         if (!res.ok) throw new Error(`HTTP ${res.status}`);
557         const data = await res.json();
558
559         const elService = document.getElementById('wb-val-servic
e');
560         const elSubService = document.getElementById('wb-sub-ser
vice');
561         const elTarget = document.getElementById('wb-fh-target')
;
562         const elConfig = document.getElementById('wb-fh-config')
;
563         const elInterval = document.getElementById('wb-fh-interv

```

```
al');
564     const elRetention = document.getElementById('wb-fh-retention');
565     const elLast = document.getElementById('wb-fh-last-backup');
566
567     const isRunning = data.is_service_running;
568     if (elService) {
569         elService.textContent = isRunning ? 'Работает' : 'Остановлена';
570         elService.className = `wb-card-value ${isRunning ? 'text-success' : 'text-danger'}`;
571     }
572     if (elSubService) {
573         elSubService.textContent = `Запуск: ${data.service_start_type || 'Manual'}`;
574     }
575
576     const cfg = data.config;
577     if (cfg) {
578         if (elTarget) elTarget.textContent = cfg.target_path || cfg.target_url || 'Не настроено';
579         if (elConfig) elConfig.textContent = cfg.config_file_path || 'Стандартный профиль';
580         if (elInterval) elInterval.textContent = cfg.backup_interval_minutes ? `${cfg.backup_interval_minutes} мин.` : 'По умолчанию (1 ч)';
581         if (elRetention) elRetention.textContent = cfg.retention_policy || 'Всегда';
582         if (elLast) elLast.textContent = cfg.last_backup_time || 'Нет записей';
583     } else {
584         if (elTarget) elTarget.textContent = 'Конфигурация отсутствует';
585         if (elConfig) elConfig.textContent = '-';
586         if (elInterval) elInterval.textContent = '-';
587         if (elRetention) elRetention.textContent = '-';
588         if (elLast) elLast.textContent = '-';
```

```

589 |     }
590 |   } catch (e) {
591 |     console.warn('[WindowsBackup] Error loading File History
status:', e);
592 |   }
593 | }
594 |
595 | async function loadVssSnapshots() {
596 |   const tbody = document.getElementById('wb-vss-body');
597 |   const countBadge = document.getElementById('wb-vss-count')
;
598 |
599 |   try {
600 |     const res = await fetch('/api/v1/windows-backup/vss/snap
shots');
601 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
602 |     const snapshots = await res.json();
603 |
604 |     if (countBadge) countBadge.textContent = `${snapshots.le
ngth} шт.`;
605 |
606 |     if (!tbody) return;
607 |     if (!snapshots || snapshots.length === 0) {
608 |       tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Теневые копии (VSS) не найдены</td></tr>'
;
609 |       return;
610 |     }
611 |
612 |     tbody.innerHTML = snapshots.map(s => `
613 |       <tr>
614 |         <td class="font-monospace small text-truncate" style
="max-width: 150px;" title="${escapeHtml(s.snapshot_id || '')}
">
615 |           ${escapeHtml(s.snapshot_id || '-')}
616 |         </td>
617 |         <td class="fw-semibold text-info">${escapeHtml(s.ori
ginal_volume || '-')}</td>

```



```

618 |         <td class="small text-muted">${escapeHtml(s.creation
|_time || '-')}}</td>
619 |         <td><span class="badge bg-success">${escapeHtml(s.st
|atus || 'OK')}}</span></td>
620 |     </tr>
621 |     `).join('');
622 | } catch (e) {
623 |     if (tbody) tbody.innerHTML = `<tr><td colspan="4" class=
|"text-center py-3 text-danger">Ошибка: ${escapeHtml(e.message)
|}</td></tr>`;
624 | }
625 | }
626 |
627 | async function loadFileDeletions() {
628 |     try {
629 |         const res = await fetch('/api/v1/system/file-audit/delet
|ions?hours=24&limit=100&deletions_only=true');
630 |         if (!res.ok) return;
631 |         const data = await res.json();
632 |         const events = data.events || [];
633 |
634 |         const tbody = document.getElementById('wb-deletions-tbod
|y');
635 |         const badge = document.getElementById('wb-deletions-badg
|e');
636 |         const valDeletions = document.getElementById('wb-val-del
|etions');
637 |
638 |         if (badge) badge.innerText = `${events.length} удалений`
|;
639 |         if (valDeletions) valDeletions.innerText = events.length
|;
640 |
641 |         if (tbody) {
642 |             if (events.length === 0) {
643 |                 tbody.innerHTML = '<tr><td colspan="5" class="text-c
|enter text-muted p-3">События удаления (4663/4660) не зафиксир
|ованы или аудит не включен</td></tr>';

```

```

644         return;
645     }
646     tbody.innerHTML = events.slice(0, 30).map((e, idx) =>
        `
647         <tr class="wb-deletion-row" data-idx="${idx}" style=
        "cursor: pointer;" title="Нажмите для детального анализа удале
        ния">
648             <td class="font-monospace text-muted small">${e.ti
        mestamp?.slice(11, 19) || ''}</td>
649             <td class="font-monospace fw-semibold ${e.event_id
        === 4660 ? 'text-danger' : 'text-warning'}">${e.event_id}</td
        >
650             <td class="font-monospace text-truncate text-white
        " style="max-width: 260px;" title="${escapeHtml(e.object_name)
        }">${escapeHtml(e.object_name.split('\\').pop() || e.object_na
        me)}</td>
651             <td class="small text-info text-truncate" style="m
        ax-width: 140px;" title="${escapeHtml(e.process_name)}">${esca
        peHtml(e.process_name ? e.process_name.split('\\').pop() : 'N/
        A')}</td>
652             <td class="small text-muted">${escapeHtml(e.subjec
        t_user_name || 'SYSTEM')}</td>
653         </tr>
654     `).join('');
655
656     tbody.querySelectorAll('.wb-deletion-row').forEach(row
    => {
657         row.onclick = () => {
658             const idx = parseInt(row.getAttribute('data-idx'),
        10);
659             const e = events[idx];
660             if (!e) return;
661
662             if (window.AITableModal) {
663                 window.AITableModal.show({
664                     icon: '🗑',
665                     title: `Удаление: ${e.object_name.split('\\').
        pop() || e.object_name}`,

```

```

666         subtitle: `Event ID: ${e.event_id} | ${e.times
|tamp}``,
667         tableType: 'security_event',
668         badges: [
669             { text: `Event ${e.event_id}`, class: e.even
|t_id === 4660 ? 'badge bg-danger' : 'badge bg-warning text-dar
|k' },
670             { text: e.status || 'Success', class: 'badge
|bg-success' }
671         ],
672         metadata: [
673             { label: 'Целевой объект', value: e.object_n
|ame },
674             { label: 'Тип объекта', value: e.object_type
| || 'File' },
675             { label: 'Процесс-инициатор', value: e.proce
|ss_name || 'Не указан' },
676             { label: 'PID процесса', value: e.process_id
| ? String(e.process_id) : 'N/A' },
677             { label: 'Пользователь', value: `${e.subject
|_domain_name}\\${e.subject_user_name}` },
678             { label: 'Дескриптор HandleId', value: e.han
|dle_id || 'N/A' },
679             { label: 'Маска доступа', value: e.access_ma
|sk || 'DELETE (0x10000)' }
680         ],
681         rawTitle: 'Полное сообщение Security Log',
682         rawContent: e.raw_message || JSON.stringify(e,
| null, 2),
683         requestData: e
684     });
685 }
686 };
687 });
688 }
689 } catch (e) {
690     console.error('[WindowsBackup] Failed to fetch deletion
|events:', e);

```

```

691     }
692 }
693
694 async function loadUserFoldersOverview() {
695     const tbody = document.getElementById('wb-user-folders-tbo
dy');
696     const totalSizeBadge = document.getElementById('wb-total-u
ser-size');
697     const drivesList = document.getElementById('wb-drives-list
');
698
699     try {
700         const res = await fetch('/api/v1/windows-backup/user-fol
ders/overview');
701         if (!res.ok) throw new Error(`HTTP ${res.status}`);
702         const data = await res.json();
703         _userFoldersData = data;
704
705         if (totalSizeBadge) {
706             totalSizeBadge.textContent = `Общий объем: ${data.tota
l_user_size_gb} ГБ (${data.total_user_size_mb} МБ)`;
707         }
708
709         if (drivesList) {
710             if (!data.drives || data.drives.length === 0) {
711                 drivesList.innerHTML = '<span class="text-muted">Дис
ки не обнаружены</span>';
712             } else {
713                 drivesList.innerHTML = data.drives.map(d => `
714                     <div class="px-2 py-1 rounded border ${d.is_system
_drive ? 'border-info bg-dark' : 'border-secondary bg-dark'} d
-flex align-items-center gap-1.5">
715                         <i class="bi ${d.is_system_drive ? 'bi-hdd-fill
text-info' : 'bi-hdd text-success'}"></i>
716                         <span class="fw-bold">${escapeHtml(d.drive_lette
r)}</span>
717                         <span class="text-muted">(${escapeHtml(d.fstype)
})</span>:

```

```

718 |         <span class="text-success fw-semibold">${d.free_
| space_gb} ГБ своб.</span>
719 |         <span class="text-muted small">из ${d.total_spac
| e_gb} ГБ</span>
720 |         ${d.is_system_drive ? '<span class="badge bg-inf
| o-subtle text-info py-0 px-1" style="font-size: 0.65rem;">Syst
| em</span>' : ''}
721 |     </div>
722 |     `).join('');
723 | }
724 | }
725 |
726 | if (!tbody) return;
727 | if (!data.folders || data.folders.length === 0) {
728 |     tbody.innerHTML = '<tr><td colspan="6" class="text-cen
| ter py-3 text-muted">Пользовательские папки не найдены</td></t
| r>';
729 |     return;
730 | }
731 |
732 | tbody.innerHTML = data.folders.map(f => {
733 |     const suitableDrives = (data.available_target_drives |
| []).filter(d => {
734 |         const folderDrive = f.drive_letter.replace('\\', '')
| ;
735 |         const targetDrive = d.drive_letter.replace('\\', '')
| ;
736 |         return folderDrive !== targetDrive && d.free_space_g
| b >= (f.size_gb + 1.0);
737 |     });
738 |
739 |     const canRelocate = suitableDrives.length > 0;
740 |
741 |     return `
742 |         <tr>
743 |             <td class="fw-bold text-white">
744 |                 <i class="bi bi-folder2-open me-1 text-primary">
| </i>${escapeHtml(f.name)}

```

```

745 |         </td>
746 |         <td class="font-monospace text-muted small text-tr
uncate" style="max-width: 250px;" title="{escapeHtml(f.curren
t_path)}">
747 |             {escapeHtml(f.current_path)}
748 |         </td>
749 |         <td>
750 |             <span class="badge {f.drive_letter.startsWith('
C') ? 'bg-secondary' : 'bg-primary'}">{escapeHtml(f.drive_let
ter)}</span>
751 |         </td>
752 |         <td class="fw-semibold font-monospace {f.size_gb
> 1.0 ? 'text-warning' : 'text-info'}">
753 |             {f.size_gb >= 0.01 ? `{f.size_gb} ГБ` : `{f.s
ize_mb} МБ`}
754 |         </td>
755 |         <td class="small text-muted font-monospace">{f.fi
le_count}</td>
756 |         <td class="text-end">
757 |             <button class="btn btn-xs {canRelocate ? 'btn-o
utline-warning' : 'btn-outline-secondary'} py-0 px-2 wb-btn-re
locate-modal"
758 |                 data-folder-id="{escapeHtml(f.folder_id
)}"
759 |                 {!canRelocate ? 'title="Нет подходящих
дисков с достаточным местом"' : 'title="Перенести на другой ди
ск"' }
760 |                 style="font-size: 0.72rem;">
761 |                 <i class="bi bi-box-arrow-right me-1"></i>Пеpe
нести
762 |             </button>
763 |         </td>
764 |     </tr>
765 |     `;
766 |     }).join('');
767 |
768 |     tbody.querySelectorAll('.wb-btn-relocate-modal').forEach
(btn => {

```

```
769         btn.onclick = () => {
770             const fid = btn.getAttribute('data-folder-id');
771             openRelocateModal(fid);
772         };
773     });
774
775     } catch (e) {
776         if (tbody) tbody.innerHTML = `|<td colspan="6" class=
"text-center py-3 text-danger">Ошибка загрузки: ${escapeHtml(e
.message)}</td></tr>`;
777     }
778 }
779
780 function openRelocateModal(folderId) {
781     if (!_userFoldersData) return;
782     const folder = (_userFoldersData.folders || []).find(f =>
f.folder_id === folderId);
783     if (!folder) return;
784
785     const elId = document.getElementById('wb-relocate-folder-i
d');
786     const elName = document.getElementById('wb-relocate-folder
-name');
787     const elPath = document.getElementById('wb-relocate-folder
-path');
788     const elSize = document.getElementById('wb-relocate-folder
-size');
789     const selectDrive = document.getElementById('wb-relocate-t
arget-drive');
790     const elHint = document.getElementById('wb-relocate-drive-
hint');
791
792     if (elId) elId.value = folder.folder_id;
793     if (elName) elName.textContent = folder.name;
794     if (elPath) elPath.textContent = folder.current_path;
795     if (elSize) elSize.textContent = `${folder.size_gb >= 0.01
? `${folder.size_gb} ГБ` : `${folder.size_mb} МБ`} (${folder.
file_count} файлов)`;
|  |

```

```

796
797     const inputPath = document.getElementById('wb-relocate-tar
get-path');
798     if (inputPath) inputPath.value = '';
799
800     if (selectDrive) {
801         selectDrive.innerHTML = '<option value="">-- Выберите ди
ск --</option>';
802         const folderDrive = folder.drive_letter.replace('\\', '
').toUpperCase();
803         let firstSuitableDrive = null;
804
805         (_userFoldersData.drives || []).forEach(d => {
806             const dLetter = d.drive_letter.replace('\\', '
').toUpp
erCase();
807             if (dLetter === folderDrive) return;
808
809             const isEnough = d.free_space_gb >= (folder.size_gb +
1.0);
810             const opt = document.createElement('option');
811             opt.value = d.drive_letter;
812             opt.textContent = `${d.drive_letter} (${d.fstype}) – С
вободно: ${d.free_space_gb} ГБ из ${d.total_space_gb} ГБ ${isE
nough ? '☐ Достаточно места' : '☐ Мало места'}`;
813             if (!isEnough) {
814                 opt.disabled = true;
815             } else if (!firstSuitableDrive) {
816                 firstSuitableDrive = d.drive_letter;
817             }
818             selectDrive.appendChild(opt);
819         });
820
821         if (firstSuitableDrive) {
822             selectDrive.value = firstSuitableDrive;
823             if (inputPath) {
824                 const d = firstSuitableDrive.replace('/', '\\').repl
ace(/\\+$/, '');
825                 const parts = folder.current_path.split('\\');

```



```

826 |         const folderSubname = parts[parts.length - 1] || folder.name;
827 |         const userMatch = folder.current_path.match(/Users\\([^\s\\]+)/i);
828 |         const username = userMatch ? userMatch[1] : 'User';
829 |         inputPath.value = `${d}\\Users\\${username}\\${folderSubname}`;
830 |     }
831 | }
832 | }
833 |
834 | if (elHint) {
835 |     elHint.textContent = 'Требуется свободного места: минимум ' + (folder.size_gb + 1.0).toFixed(2) + ' ГБ (размер + 1 ГБ резерва).';
836 | }
837 |
838 | const modalEl = document.getElementById('wb-relocate-modal');
839 | if (modalEl) {
840 |     if (window.bootstrap?.Modal) {
841 |         new window.bootstrap.Modal(modalEl).show();
842 |     } else {
843 |         modalEl.classList.add('show');
844 |         modalEl.style.display = 'block';
845 |     }
846 | }
847 | }
848 |
849 | function escapeHtml(str) {
850 |     if (str === null || str === undefined) return '';
851 |     return String(str)
852 |         .replace(/&/g, '&')
853 |         .replace(/</g, '<')
854 |         .replace(/>/g, '>')
855 |         .replace(/"/g, '"')
856 |         .replace(/'/g, '&#039;');
857 | }

```

858 | `})();`

859 |

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Web Interface - Index View
5 |=====
6 Description:
7 |Шаблон клиентского веб-интерфейса index.
8 |
9 Usage Examples:
10 |    Web Route:
11 |        /windows/api/webgui/windows_users_tab/index.html
12 |
13 |    Tab Frame Integration:
14 |        <iframe src="/windows/api/webgui/windows_users_tab/index
15 |.html" class="w-100 h-100 border-0"></iframe>
16 |
17 File: index.html
18 Project: ai-breadboard
19 Package: windows/api/webgui/windows_users_tab
20 Author: hypo69
21 Copyright: © 2026 hypo69
22 Updated: 2026-10-01 13:04:40
23 |=====
24 |=====
25 -->
26
27 <!DOCTYPE html>
28 <html lang="ru">
29 <head>
30 |    <meta charset="UTF-8">
31 |    <meta name="viewport" content="width=device-width, initial-s
32 |cale=1.0">
33 |    <title>Пользователи Windows</title>
34 </head>
35 <body>
```

```
32 | <div class="container-fluid p-3">
33 |   <h4><i class="bi bi-people-fill text-primary"></i> Пользов
   | атели Windows</h4>
34 |   <p class="text-muted">Управление локальными учетными запис
   | ями операционной системы Windows.</p>
35 |
36 |   <div class="card border-secondary mb-3">
37 |     <div class="card-header bg-dark text-white">
38 |       <i class="bi bi-list-ul"></i> Список пользователей
39 |     </div>
40 |     <div class="card-body">
41 |       <div id="windows-users-list">
42 |         <p class="text-muted">Загрузка...</p>
43 |       </div>
44 |     </div>
45 |   </div>
46 |
47 |   <div class="card border-secondary">
48 |     <div class="card-header bg-dark text-white">
49 |       <i class="bi bi-people-fill"></i> Управление группами
50 |     </div>
51 |     <div class="card-body">
52 |       <div id="windows-groups-list">
53 |         <p class="text-muted">Загрузка...</p>
54 |       </div>
55 |     </div>
56 |   </div>
57 | </div>
58 |
59 | <script type="module">
60 |   import { initWindowsUsersTab } from '/html/windows_users_t
   | ab/main.js?v=20260920_v1';
61 |   window.initWindowsUsersTab = initWindowsUsersTab;
62 | </script>
63 | </body>
64 | </html>
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Windows Users Tab - Main Script
4  * =====
5  * Description:
6  *   Клиентский скрипт управления интерфейсом модуля main.
7  *
8  * Usage Examples:
9  *   HTML Integration:
10  *     <script src="/windows/api/webgui/windows_users_tab/main
11  *     .js?v=20261001_v1" type="module"></script>
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: windows/api/webgui/windows_users_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * Updated: 2026-10-01 13:04:40
19  * =====
20  */
21 // Windows Users Tab Module
22 // =====
23
24 export async function initWindowsUsersTab() {
25   console.log('[WindowsUsersTab] Initializing...');
26
27   try {
28     // Load Windows users
29     const usersContainer = document.getElementById('windows-us
30     ers-list');
31     const groupsContainer = document.getElementById('windows-g
```

```

    rous-list');
31
32     if (usersContainer) {
33         usersContainer.innerHTML = '<div class="spinner-border text-
    ext-primary" role="status"><span class="visually-hidden">Loadi
    ng...</span></div>';
34     }
35
36     if (groupsContainer) {
37         groupsContainer.innerHTML = '<div class="spinner-border
    text-primary" role="status"><span class="visually-hidden">Load
    ing...</span></div>';
38     }
39
40     // Fetch Windows users
41     try {
42         const response = await fetch('/api/windows-users');
43         if (response.ok) {
44             const data = await response.json();
45             if (data.status === 'ok' && data.users) {
46                 if (usersContainer) {
47                     usersContainer.innerHTML = `
48                     <div class="table-responsive">
49                     <table class="table table-dark table-sm table-
    bordered">
50                         <thead>
51                             <tr>
52                                 <th>Имя</th>
53                                 <th>Enabled</th>
54                                 <th>Описание</th>
55                                 <th>SID</th>
56                             </tr>
57                         </thead>
58                         <tbody>
59                             ${data.users.map(u => `
60                             <tr>
61                                 <td><strong>${escapeHtml(u.Name || '')}
    }</strong></td>

```

```

62 |                 <td>${u.Enabled ? '<span class="badge
|bg-success">Да</span>' : '<span class="badge bg-danger">Нет</s
|pan>'}</td>
63 |                 <td>${escapeHtml(u.Description || '')}
|</td>
64 |                 <td class="text-muted small">${escapeH
|tml(u.SID || '')}</td>
65 |             </tr>
66 |             `).join('')`
67 |         </tbody>
68 |     </table>
69 | </div>
70 |     `;
71 |     }
72 | }
73 | }
74 | } catch (err) {
75 |     console.error('[WindowsUsersTab] Error loading users:',
|err);
76 |     if (usersContainer) {
77 |         usersContainer.innerHTML = `<div class="alert alert-da
|nger">Ошибка загрузки: ${escapeHtml(err.message)}</div>`;
78 |     }
79 | }
80 |
81 | // Fetch Windows groups
82 | try {
83 |     const response = await fetch('/api/windows-users/groups'
|);
84 |     if (response.ok) {
85 |         const data = await response.json();
86 |         if (data.status === 'ok' && data.groups) {
87 |             if (groupsContainer) {
88 |                 groupsContainer.innerHTML = `
89 |                     <div class="table-responsive">
90 |                         <table class="table table-dark table-sm table-
|bordered">
91 |                             <thead>

```

```

92         <tr>
93             <th>Группа</th>
94             <th>SID</th>
95             <th>Описание</th>
96         </tr>
97     </thead>
98     <tbody>
99         ${data.groups.map(g => `
100             <tr>
101                 <td><strong>${escapeHtml(g.Name || '')}
102             </strong></td>
103                 <td class="text-muted small">${escapeH
104             tml(g.SID || '')}</td>
105                 <td>${escapeHtml(g.Description || '')}
106             </td>
107             </tr>
108             `).join('')}
109         </tbody>
110     </table>
111 </div>
112 `;
113     }
114     }
115     } catch (err) {
116         console.error('[WindowsUsersTab] Error loading groups:',
117             err);
118         if (groupsContainer) {
119             groupsContainer.innerHTML = `<div class="alert alert-d
120             anger">Ошибка загрузки групп: ${escapeHtml(err.message)}</div>
121             `;
122         }
123     }
124     } catch (err) {
125         console.error('[WindowsUsersTab] Initialization error:', e
126             rr);
127     }

```



```
123 | }
124 |
125 | function escapeHtml(text) {
126 |     if (!text) return '';
127 |     return text
128 |         .replace(/&/g, '&amp;')
129 |         .replace(/</g, '&lt;')
130 |         .replace(/>/g, '&gt;')
131 |         .replace(/"/g, '&quot;')
132 |         .replace(/'/g, '&#039;');
133 | }
```

__init__.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api - Init
4 # =====
5 # Description:
6 #     Внутренний FastAPI-сервис подсистемы Windows.
7 #
8 # Usage Examples:
9 #
10 #
11 # File: __init__.py
12 # Project: ai-breadboard
13 # Package: apps.windows.api
14 # Author: hypo69
15 # Copyright: © 2026 hypo69
16 # Updated: 2026-10-01 13:28:28
17 # =====
18
19 """Внутренний FastAPI-сервис подсистемы Windows."""
20
21 from .internal_app import create_internal_app, load_config
22
23 __all__ = [
24     'create_internal_app',
25     'load_config',
26 ]
```

__main__.py

```
1 # -*- coding: utf-8 -*-
2 # =====
3 # Process Name: AI-Breadboard Apps Windows Api - Main
4 # =====
5 # Description:
6 #   Точка входа и запуск внутреннего FastAPI сервера Windows S
7 #   ystem API (TC)
8 #   через uvicorn с поддержкой форматирования логов с временны
9 #   ми метками.
10 #
11 # Usage Examples:
12 #   CLI:
13 #       python -m apps.windows.api.__main__
14 #   Python API:
15 #       from apps.windows.api.__main__ import main
16 #
17 #       res = main()
18 #
19 # File: __main__.py
20 # Project: ai-breadboard
21 # Package: apps.windows.api
22 # Author: hypo69
23 # Copyright: © 2026 hypo69
24 # Updated: 2026-10-08 01:30:00
25 # =====
26
27
28 from __future__ import annotations
29 """Точка входа внутреннего FastAPI-сервиса Windows System API.
30 """
31
32 import argparse
33 import copy
34 import sys
```

```

31 from pathlib import Path
32 from typing import Any, Dict
33
34 # Добавляем корень проекта в sys.path, чтобы работали импорты
   src.*, apps.*
35 _ROOT = Path(__file__).resolve().parents[3]
36 if str(_ROOT) not in sys.path:
37     sys.path.insert(0, str(_ROOT))
38
39
40 def _build_log_config() -> Dict[str, Any]:
41     """Формирует конфигурацию логирования uvicorn с временными
   метками.
42
43     Returns:
44         Dict[str, Any]: Конфигурация логирования uvicorn с фор
   матом даты и времени.
45     """
46     import uvicorn.config
47
48     log_config: Dict[str, Any] = copy.deepcopy(uvicorn.config.
   LOGGING_CONFIG)
49     log_config['formatters']['default']['fmt'] = '%(asctime)s
   %(levelname)s %(message)s'
50     log_config['formatters']['default']['datefmt'] = '%Y-%m-%d
   %H:%M:%S'
51     log_config['formatters']['access']['fmt'] = (
52         '%(asctime)s %(levelname)s %(client_addr)s - "%(requ
   est_line)s" %(status_code)s'
53     )
54     log_config['formatters']['access']['datefmt'] = '%Y-%m-%d
   %H:%M:%S'
55     return log_config
56
57
58 def _parse_args() -> argparse.Namespace:
59     """Разбирает аргументы командной строки.
60

```

```

61 | Returns:
62 |     argparse.Namespace: Распарсенные аргументы.
63 |     """
64 |     parser = argparse.ArgumentParser(
65 |         description='AI-Breadboard Windows Internal API Server
66 |         ',
67 |         formatter_class=argparse.ArgumentDefaultsHelpFormatter
68 |     )
69 |     parser.add_argument(
70 |         '--host',
71 |         default='127.0.0.1',
72 |         help='IP-адрес для прослушивания (только localhost)',
73 |     )
74 |     parser.add_argument(
75 |         '--port',
76 |         type=int,
77 |         default=8001,
78 |         help='Порт внутреннего сервиса',
79 |     )
80 |     parser.add_argument(
81 |         '--reload',
82 |         action='store_true',
83 |         default=False,
84 |         help='Включить hot-reload (только для разработки)',
85 |     )
86 |     parser.add_argument(
87 |         '--log-level',
88 |         default='info',
89 |         choices=['critical', 'error', 'warning', 'info', 'debug', 'trace'],
90 |         help='Уровень логирования uvicorn',
91 |     )
92 |     return parser.parse_args()
93 |
94 | def main() -> None:
95 |     """Запускает uvicorn с внутренним FastAPI-сервисом.

```

```
96 |
97 |     Принудительно ограничивает bind только на localhost.
98 |     """
99 |     args = _parse_args()
100 |
101 |     # Защита: запрещаем bind на 0.0.0.0 или внешние адреса
102 |     if args.host not in ('127.0.0.1', 'localhost', '::1'):
103 |         print(
104 |             f'[WARN] Внутренний сервис не может слушать на {ar
105 | gs.host}. '
106 |             'Принудительно используется 127.0.0.1.',
107 |             file=sys.stderr,
108 |         )
109 |         args.host = '127.0.0.1'
110 |
111 |     try:
112 |         import uvicorn
113 |     except ImportError:
114 |         print('[ERROR] uvicorn не установлен. Установите: pip
115 | install uvicorn', file=sys.stderr)
116 |         sys.exit(1)
117 |
118 |     print(f' [Windows Internal API] Запуск на http://{args.ho
119 | st}:{args.port}')
120 |     print(f' [Windows Internal API] TC UI: http://{args.host}
121 | :{args.port}/tc')
122 |     print(f' [Windows Internal API] Health: http://{args.host}
123 | :{args.port}/health')
124 |     print(f' [Windows Internal API] Docs: http://{args.host}
125 | :{args.port}/docs')
126 |
127 |     uvicorn.run(
128 |         'apps.windows.api.internal_app:create_internal_app',
129 |         factory=True,
130 |         host=args.host,
131 |         port=args.port,
132 |         reload=args.reload,
133 |         log_level=args.log_level,
```

```
128 |         log_config=_build_log_config(),
129 |         # Заголовок сервера не раскрываем
130 |         server_header=False,
131 |         access_log=True,
132 |     )
133 |
134 |
135 | if __name__ == '__main__':
136 |     main()
137 |
138 |
```